

Medical Dictionary by Specialty

A Comprehensive Reference of Medical Terminology

Chaman Singh Verma

September 12, 2026

Preface

This dictionary is a concise reference to medical terms, diseases, investigations, treatments, anatomy, physiology, and related clinical concepts. It is intended to help students, healthcare professionals, educators, and general readers understand unfamiliar medical language.

The entries are arranged alphabetically from A to Z, with each letter presented in its own chapter. This arrangement allows a reader to locate a known term quickly while maintaining a consistent structure throughout the book. Related terms, recognized common names, and alternative names are included when they improve clarity or aid searching.

Definitions are written in simple, formal language and are intended to be clear, accurate, and useful for reference. Medical terminology changes as scientific knowledge and clinical guidance develop; therefore, important information should be checked against current textbooks, primary research, professional guidance, and local clinical protocols.

This book is an educational reference, not a substitute for professional medical advice, diagnosis, or treatment. Readers should seek advice from a qualified healthcare professional when making decisions about health.

Chaman Singh Verma

Medical Disclaimer

Educational and Reference Purpose Only: This publication, *Medical Dictionary by Specialty*, is prepared and compiled strictly for educational, academic, historical, and reference purposes. It is intended for medical students, healthcare professionals, educators, and researchers.

No Medical Advice: The information contained herein does not constitute professional medical advice, clinical diagnosis, treatment protocols, or formal therapeutic recommendations. It should not be used as a substitute for the professional judgment of qualified physicians, surgeons, or certified healthcare providers.

Clinical Practice Guidelines: Healthcare practitioners must always consult primary evidence-based medical literature, institutional protocols, manufacturer prescribing information, and national clinical guidelines (such as WHO, CDC, NICE, or national medical councils) when making diagnostic or therapeutic decisions for individual patients.

Limitation of Liability: Neither the author nor the publishers accept any legal responsibility or liability for any errors, omissions, or clinical outcomes resulting from the use or application of any information presented in this work.

A

5-Alpha Reductase

An enzyme that changes testosterone into dihydrotestosterone. DHT supports male sexual development and contributes to prostate growth and male-pattern hair loss. Medicines that block this enzyme are used for an enlarged prostate and some types of hair loss.

A & E Medicine

Accident and emergency (A&E) medicine provides rapid assessment, stabilization, diagnosis, and initial treatment for sudden illness and injury.

A Fib

Alternate name; see *Atrial fibrillation*.

A-A Gradient

The difference between the oxygen level in the air sacs of the lungs and the oxygen level in arterial blood. It helps find the cause of low blood oxygen. The gradient is often increased when oxygen transfer is impaired, but may remain normal when low oxygen is caused by slow or shallow breathing or low oxygen in the inhaled air.

A-T

The adenine–thymine base pair in DNA. The two bases join by hydrogen bonds and help form the structure of DNA.

AAA Repair

Surgical or endovascular treatment of an abdominal aortic aneurysm to prevent rupture or treat a rupture.

Aagenaes Syndrome

A rare inherited disorder, also called cholestasis-lymphedema syndrome, that causes cholestasis beginning in infancy and persistent lymphedema, usually of the lower limbs. Liver disease may become intermittent with age, but some people develop fat-soluble vitamin deficiency, poor growth, or cirrhosis.

Aarskog-Scott Syndrome

A rare inherited developmental disorder, often related to FGD1 variants, characterized by distinctive facial features, short stature, skeletal differences, and genital anomalies.

Alternate Names:

Aarskog Syndrome

Aase Syndrome

A rare congenital disorder that causes hypoplastic anemia, meaning reduced production of red blood cells, and skeletal abnormalities, especially thumbs with three bones (triphalangeal thumbs). It is often considered part of the Diamond–Blackfan anemia spectrum; not all cases are known to be inherited.

Aase-Smith Syndrome I

A rare congenital-malformation syndrome associated with hydrocephalus, cleft palate, severe joint contractures, distinctive hand and facial findings, and sometimes heart defects.

Aase-Smith Syndrome II

A rare inherited disorder characterized by hypoplastic anemia and triphalangeal thumbs, in which the thumbs contain three phalanges rather than the usual two.

AATD

See *Alpha-1-antitrypsin deficiency*.

Abandoned Lead

A disconnected or no-longer-used wire from an implanted cardiac or other electrical device that remains in the body.

Alternate Names:

Retained cardiac device lead

Abasia

Inability to walk despite preserved strength and the ability to move the legs in other circumstances. It usually reflects a neurologic or functional gait disorder rather than paralysis.

Abate

To become less intense or to reduce in severity, frequency, or amount; symptoms, inflammation, pain, or fever may abate.

Abatement

A reduction in the severity, intensity, frequency, or amount of a symptom, disease process, exposure, or other condition.

ABCDE Assessment

A systematic emergency assessment of airway, breathing, circulation, disability, and exposure. Life-threatening abnormalities are treated as they are identified, with repeated reassessment.

ABCDE Rule For Melanoma

A screening mnemonic for suspicious pigmented lesions: asymmetry, border irregularity, color variation, diameter greater than about 6 mm, and evolution. It does not diagnose melanoma; suspicious lesions require clinical assessment.

Abdomen

The body region between the diaphragm and pelvis. It contains most digestive organs, the kidneys, spleen, pancreas, and major blood vessels and is described clinically by four quadrants or nine regions.

Abdomen - Swollen

An abdomen that is larger than usual or feels full and tight, also called abdominal distention. It may be caused by gas, constipation, overeating, fluid buildup, pregnancy, weight gain, bowel blockage, or enlarged organs. Severe or persistent swelling, especially with pain, vomiting, fever, or bloody stools, needs medical assessment.

Abdomen, Acute

Acute abdomen is sudden or severe abdominal pain with possible tenderness, guarding, vomiting, distention, or altered bowel function that may indicate a surgical or other urgent condition.

Abdomen, Diseases Of

Diseases of the abdomen affect abdominal organs, blood vessels, lining, or abdominal wall and may cause pain, vomiting, altered bowel function, bleeding, or distention.

Abdominal

Relating to the abdomen, the body region between the chest and pelvis that contains organs of the digestive, urinary, and reproductive systems.

Abdominal Abscess

A localized collection of pus within the abdomen, usually caused by bacterial infection, inflammation, surgery, or organ perforation. It may cause fever and abdominal pain and often requires antibiotics and drainage.

Abdominal Angina

Pain in the abdomen after eating, caused by reduced blood flow through the arteries supplying the intestines. It is usually due to hardening and narrowing of these arteries.

Abdominal Aorta

The portion of the aorta below the diaphragm that extends to its bifurcation into the common iliac arteries. It supplies the abdominal organs, pelvic organs, and lower limbs through its branches.

Abdominal Aortic Aneurysm

A localized enlargement of the abdominal aorta, usually defined as an aortic diameter of at least 3 cm or more than 50 percent above normal. Enlargement can progress to rupture.

Alternate Names:

AAA (Abdominal Aortic Aneurysm)

Abdominal Aortic Aneurysm Repair - Open

Open surgery to repair an enlarged abdominal aorta using an artificial graft. The surgeon reaches the aorta through the abdominal cavity or behind the abdominal lining. It may be needed for a large, painful, or rapidly growing aneurysm when repair from inside the blood vessel is not possible.

Abdominal Aortic Aneurysm Screening

Ultrasound examination used to detect an abdominal aortic aneurysm before symptoms occur. Screening recommendations depend on age, sex, smoking history, and clinical risk; follow-up intervals depend on aneurysm size.

Abdominal Aortic Ultrasound

Ultrasound imaging of the abdominal aorta used to detect or monitor an abdominal aortic aneurysm.

Abdominal Binder

A wide supportive garment placed around the abdomen to provide gentle compression after surgery, injury, or selected abdominal-wall conditions.

Abdominal Bloating

A sensation of fullness or visible enlargement of the abdomen, commonly caused by intestinal gas, constipation, food intolerance, or functional gastrointestinal disorders. Persistent or severe bloating may require evaluation for organic disease.

Abdominal Bruit

An abnormal blowing or whooshing sound heard over the abdomen with a stethoscope, usually caused by turbulent blood flow through a narrowed or enlarged artery. It may occur with renal-artery stenosis or an abdominal aortic aneurysm. Further assessment depends on symptoms and vascular risk factors.

Abdominal Cavity

The cavity between the diaphragm and pelvis containing the stomach, liver, gallbladder, spleen, pancreas, kidneys, and intestines.

Abdominal Circumference

The distance around the abdomen. In pregnancy, fetal abdominal circumference is measured by ultrasound at a standardized level that includes the stomach, portal sinus, and spine; it is used with other measurements to estimate fetal size and assess growth.

Abdominal Colic

Cramping abdominal pain caused by contractions or obstruction of a hollow organ, such as the intestine, ureter, or bile duct. Pain often comes in waves and may be accompanied by nausea, vomiting, sweating, or altered bowel or urinary function. Severe or persistent pain requires prompt medical assessment.

Abdominal Compartment Syndrome

A dangerous rise in pressure inside the abdomen, usually above 20 millimeters of mercury, that causes new problems with organ function. Causes include severe fluid buildup, an overly swollen or blocked intestine, internal bleeding, and some conditions after abdominal surgery.

Abdominal CT

Computed tomography of the abdomen used to evaluate selected causes of abdominal pain, trauma, bowel obstruction, inflammation, bleeding, and masses. The protocol and use of contrast depend on the suspected condition and the patient's clinical status.

Abdominal CT Scan

A computed tomography study that produces cross-sectional images of the abdomen using X-rays. It can identify structural abnormalities, injury, inflammation, bleeding, or masses.

Abdominal Distention

Visible or felt enlargement, tightness, or fullness of the abdomen caused by gas, fluid, stool, pregnancy, organ enlargement, a mass, or bowel blockage. Sudden distention with pain, vomiting, fever, faintness, bleeding, or signs of irritation of the abdominal lining requires urgent evaluation.

Abdominal Exploration

Surgical inspection of the abdominal cavity to identify or treat a suspected source of pain, bleeding, infection, obstruction, or other disease.

Abdominal Fluid

Fluid within the abdominal cavity; excessive accumulation is called ascites.

Alternate Names:

Peritoneal fluid

Abdominal Girth

The distance around the abdomen, measured at a specified level. It may be used to monitor abdominal distention, fetal growth, obesity, ascites, or changes in intra-abdominal volume.

Abdominal Guarding

Tensing of the abdominal muscles when the abdomen is touched. It may occur when the lining around the abdominal organs is irritated or inflamed.

Abdominal Hernia

Protrusion of an organ or tissue through a weakness or opening in the abdominal wall. Common types include incisional, umbilical, epigastric, and ventral hernias.

Abdominal Hysterectomy

Surgical removal of the uterus through an incision in the abdominal wall; the cervix, ovaries, and fallopian tubes may or may not also be removed.

Abdominal Mass

A palpable or imaging-detected abnormal collection, enlargement, or lesion within the abdomen.

Causes include distended bowel or bladder, organ enlargement, hernia, inflammatory collection, benign tumour, and malignancy; assessment depends on location, mobility, tenderness, systemic features, and appropriate imaging.

Abdominal Migraine

A migraine-related disorder characterized mainly by recurrent episodes of moderate to severe midline abdominal pain, often with nausea or vomiting and little or no headache.

Abdominal MRI Scan

A magnetic resonance imaging study that produces detailed images of the abdomen using magnetic fields and radiofrequency energy.

Abdominal Muscles

The muscles forming the front and sides of the abdominal wall. They support the trunk, help with posture and breathing, and increase intra-abdominal pressure during coughing, lifting, and defecation.

Alternate Names:

Abdominal wall muscles.

Abdominal Neoplasm

An abnormal tissue growth arising in an abdominal organ or structure; it may be benign or malignant.

Alternate Names:

Abdominal tumor

Abdominal Pain

Discomfort within the abdomen classified as visceral (dull, midline, poorly localized), parietal (sharp, well localized, worsened by movement), or referred from a distant source.

Alternate Names:

Stomach pain

Abdominal Pregnancy

An ectopic pregnancy implanted in the peritoneal cavity rather than the uterus, fallopian tubes, or ovaries. It is rare and can cause severe internal bleeding. Diagnosis requires specialist imaging and management.

Abdominal Radiography

A regular X-ray of the abdomen. It may be taken while a person is lying down, standing, or lying on their side.

Abdominal Reflex

Contraction of the abdominal muscles after stimulation of the abdominal skin. An absent or asymmetric reflex may indicate neurologic disease, although it can be absent normally in some people.

Abdominal Regions

The nine named areas used to describe locations on the abdominal wall: the right and left hypochondriac, epigastric, right and left lumbar, umbilical, right and left iliac, and hypogastric regions.

Abdominal Rigidity

Involuntary tensing of the abdominal wall, usually caused by irritation of the peritoneum. It may occur with peritonitis, perforation, severe infection, or intra-abdominal bleeding and requires clinical assessment.

Abdominal Sounds

Sounds produced by movement of gas and fluid through the intestines; changes in frequency or character may support, but do not establish, a diagnosis.

Abdominal Swelling

Visible or felt enlargement of the abdomen caused by gas, constipation, pregnancy, fluid accumulation, organ enlargement, mass, or abdominal-wall changes. Persistent, progressive, painful, or sudden swelling—especially with vomiting, fever, bleeding, or inability to pass stool or gas—requires medical evaluation.

Abdominal Tap

A procedure in which a needle or catheter is inserted into the abdominal cavity to remove or test ascitic fluid. It is also called paracentesis.

Abdominal Thrusts

First-aid maneuver used for severe choking in a

conscious person, intended to expel an obstructing object from the airway.

Abdominal Ultrasound

An ultrasound examination that uses sound waves to produce images of the abdomen and assess its structure and blood flow when applicable.

Abdominal Wall

The layered muscular, fascial, and skin-covered boundary surrounding the abdominal cavity. It supports and protects abdominal organs and helps with posture, breathing, coughing, and straining. Weakness or defects may produce abdominal or incisional hernias; trauma, infection, and tumors may also affect the wall.

Abdominal Wall Fat Pad Biopsy

Removal of a small sample of abdominal fat to look for amyloid, usually with a special Congo-red stain. It is less invasive than biopsy of an affected organ, but a negative result does not rule out amyloidosis.

Abdominal Wall Hernia

Protrusion of abdominal tissue through a weak area or opening in the abdominal wall. Common types include inguinal, umbilical, femoral, and incisional hernias. A painful, firm, or non-reducible bulge with vomiting or abdominal distension may indicate incarceration or strangulation and requires urgent assessment.

Abdominal Wall Surgery

Surgical repair of defects in the abdominal wall, most commonly for hernias using open or laparoscopic techniques with mesh reinforcement. Indications include inguinal, umbilical, femoral, or incisional hernias, and the approach depends on defect size, location, and patient factors.

Abdominal X-Ray

A plain radiograph of the abdomen used to assess selected causes of pain, obstruction, perforation, foreign bodies, or urinary stones. It has limited sensitivity for many abdominal disorders.

Abdominocentesis

Inserting a needle through the abdominal wall into the space around the abdominal organs to remove or test extra fluid, called ascites. It is also called paracentesis.

Abdominopelvic Cavity

The continuous body cavity below the diaphragm that contains the abdominal and pelvic organs.

Abdominoperineal Resection

Surgery removing the rectum and anus, usually with a permanent colostomy.

Alternate Names:

APR

Abdominoplasty

A surgical procedure that removes excess abdominal skin and may tighten the abdominal wall. It is commonly called a tummy tuck.

Alternate Names:

a tummy tuck.

Abducens Nerve

The sixth cranial nerve. It controls the muscle that moves the eye outward. Damage to this nerve can prevent outward eye movement and cause double vision.

Abducent Nerve

Another name for the abducens nerve, the sixth cranial nerve, which supplies the lateral rectus muscle and moves the eye outward.

Abduct

To abduct a limb or body part is to move it away from the body's midline; the opposite movement is adduction.

Abduction

Movement of a limb or other body part away from the median plane of the body.

Abductor Muscle

A muscle that moves a body part away from the

midline or, for a digit, away from the axis of the hand or foot.

Aberration

A deviation from normal structure, function, or appearance; in optics, it is an imperfection that prevents light from forming a sharply focused image.

Abetalipoproteinemia

A rare inherited disorder in which the body cannot produce normal apolipoprotein-B-containing lipoproteins. It causes fat malabsorption, deficiencies of fat-soluble vitamins, abnormal red cells, and possible neurologic or retinal disease.

Abfraction

Loss of tooth structure near the gum line, possibly related to mechanical stress. It may cause tooth sensitivity and may require dental treatment.

Abiotic

Nonliving; in biology and medicine, the term describes physical or chemical environmental factors rather than living organisms.

Abiotrophy

Progressive degeneration or loss of function in a tissue or organ, often due to an inherited or developmental defect rather than an acquired injury. The process may reflect a genetically determined failure of cellular maintenance, metabolism, or structural integrity. Clinical manifestations vary with the affected organ and can include progressive weakness, sensory loss, ataxia, or organ-specific dysfunction. Diagnosis is guided by the clinical course, examination, laboratory studies, imaging, and genetic or pathologic evaluation when indicated.

Ablate

To remove or destroy tissue, usually by surgery, heat, cold, chemicals, or another energy source.

Ablation

Destruction or removal of a small area of abnormal tissue using methods such as heat, cold, chemicals,

or electrical energy.

Ablative Surgery

Surgery that destroys or removes abnormal tissue, often using excision, heat, cold, electrical energy, or other tissue-destructive methods.

Ablepharon

A congenital absence or severe underdevelopment of an eyelid.

Abnormal

Not normal or not what is expected for a person's age or situation. It may be harmless or may point to disease, so it must be considered with other findings.

Abnormal Color

A color change in skin, mucosa, secretions, or another tissue that differs from the expected appearance.

Alternate Names:

Discoloration

Abnormal Fissure

An unusual crack, groove, or cleft in a tissue or organ.

Abnormal Hemoglobins Testing

A laboratory test that identifies variant or abnormal hemoglobin molecules using hemoglobin electrophoresis, high-performance liquid chromatography, or molecular methods. It is used to diagnose hemoglobinopathies such as sickle cell disease, thalassemia, and other inherited hemoglobin disorders.

Abnormal Uterine Bleeding

Uterine bleeding that is abnormal in frequency, regularity, duration, or volume, occurring outside of normal menstruation or in women who are not pregnant.

Abnormal Virilization

Development of masculinizing physical features in a person whose expected sex development does not include them, often from excess androgens.

Alternate Names:

Virilization

Abnormality

A structural, functional, or developmental difference from the usual state.

Alternate Names:

Anomaly

Abnormally Dark Or Light Skin

Skin that has become darker or lighter than a person's usual skin tone. Darkening is called hyperpigmentation and lightening is called hypopigmentation. Causes include inflammation or injury, sun exposure, medicines, infections, hormonal disorders, and genetic conditions.

Abnormally Excessive Sweating

See *Hyperhidrosis*.

ABO Blood Group System

The major red-cell blood-group system based on whether A and/or B antigens are present on the red-cell surface and the corresponding antibodies are present in plasma. The four principal phenotypes are A, B, AB, and O; correct ABO matching is critical for safe transfusion.

Alternate Names:

ABO Blood Group ABO Blood Groups ABO Blood Types ABO System

ABO Blood Typing

A laboratory test that determines a person's ABO and Rh blood groups. It is performed before transfusion, transplantation, and some obstetric procedures.

ABO Incompatibility

An immune reaction caused by transfusion or pregnancy involving incompatible A, B, or O blood-group antigens. It can cause acute hemolysis, transfusion reactions, or hemolytic disease of the newborn. See also Hemolytic Disease of the Newborn.

Abomasum

The fourth and glandular compartment of the stomach in ruminant animals.

Alternate Names:

True stomach

Abortion

The intentional ending of an ongoing pregnancy using medication or a medical procedure. It may be induced for personal or medical reasons and is distinct from spontaneous abortion, also called miscarriage. Laws and access vary by jurisdiction.

Abortion - Procedure

Surgical evacuation of the uterus to end a pregnancy, most commonly by vacuum aspiration under local or general anesthesia. It is performed for unwanted pregnancy, fetal demise, or incomplete miscarriage; complications include bleeding, infection, and uterine perforation.

Abortive

Incomplete, unsuccessful, or stopped before full development; an abortive infection produces limited disease without the usual complete clinical course.

Abortive Polio

A mild, nonparalytic poliovirus infection causing nonspecific fever, sore throat, headache, nausea, or malaise without the neurologic findings of paralytic polio.

Above The Knee Amputation

Amputation of the lower limb through the thigh above the knee joint, usually performed for severe vascular disease, infection, trauma, or cancer.

ABR Test

Auditory brainstem response testing records electrical activity from the auditory nerve and brainstem through scalp electrodes to assess hearing and auditory-pathway function, especially in infants or people unable to provide reliable behavioral responses.

Abrade

To wear away tissue by friction or erosion; an abrasion is the resulting scrape or superficial loss of skin.

Abrasion

A superficial injury in which the skin is scraped or rubbed away, usually by friction. In dentistry, abrasion means mechanical wearing away of tooth structure by an external force.

Abreaction

Abreaction is the expression and re-experiencing of intense emotion linked to a distressing memory; it is a historical psychological term and is not itself a diagnosis.

Abrin

A highly poisonous protein found in rosary pea seeds. It stops cells from making proteins, causing them to die. Swallowing chewed seeds or exposure to abrin powder can cause vomiting, diarrhea, dehydration, breathing problems, organ failure, and death. Suspected exposure requires urgent medical care; there is no routine antidote.

Abruptio Placenta

Abruptio placenta, more commonly called placental abruption, is premature separation of the placenta from the uterine wall before delivery, which can cause maternal bleeding and fetal compromise.

Abruptio Placentae

Premature separation of the placenta from the uterine wall before delivery. It can cause vaginal bleeding, abdominal pain, uterine tenderness, fetal distress, and maternal shock and requires urgent obstetric assessment.

Abs

An informal term for the abdominal muscles, particularly the anterior abdominal-wall muscles.

Abscess

A pocket of pus that forms in tissue, an organ, or a

body cavity, usually because of an infection. It may cause pain, swelling, warmth, and fever; treatment often requires drainage and sometimes antibiotics.

Abscess - Abdomen Or Pelvis

A localized collection of pus within the abdominal or pelvic cavity, usually caused by infection or leakage from an injured or inflamed organ. It may cause pain, fever, and sepsis and often requires antimicrobial treatment and drainage.

Abscess, Bartholin'S

See *Bartholin's cyst*.

Abscessed Tooth

A collection of pus caused by bacterial infection of the tooth or surrounding tissues, usually arising from untreated dental decay, a cracked tooth, or periodontal disease. Pain, facial swelling, fever, and sensitivity may occur. Dental treatment and drainage are usually required; swelling that affects swallowing or breathing is an emergency.

Abscesses Skin (Boils)

Alternate index label for Boils.

Abscission

Removal of tissue by cutting it away, as in a surgical procedure.

Absence Epilepsy

A generalized epilepsy characterized by brief episodes of impaired awareness, often with staring and subtle movements.

Alternate Names:

Absence seizures

Absence Of Sweating

Anhidrosis is diminished or absent sweating, which may be focal or generalized. It impairs thermoregulation and can result from autonomic dysfunction, peripheral neuropathy, skin disease, medications, or congenital conditions. Severe generalized anhidrosis can cause heat intolerance and exertional hyperthermia.

Absence Of The Breast

Congenital absence of breast development, called amastia; it may affect one or both sides and can occur with other developmental anomalies.

Absence Of The Nipple

Athelia: a rare congenital absence of one or both nipples, which may occur with hereditary disorders, chromosomal abnormalities, or developmental syndromes.

Absence Seizure

A brief generalized seizure causing sudden impaired awareness and behavioral arrest, usually followed by rapid recovery. Electroencephalography commonly shows generalized three-hertz spike-and-wave activity.

Alternate Names:

Petit Mal Seizure Seizure, Absence Seizure, Petit Mal

Absent Breath Sounds

Lack of audible air movement over part or all of a lung. Causes include pneumothorax, pleural effusion, or airway obstruction; sudden onset requires urgent assessment.

Absent Eye

Anophthalmia, a congenital absence or severe underdevelopment of the eye globe and ocular tissue; it may affect one or both eyes and may occur with other birth defects.

Absent Menstrual Periods - Primary

Primary amenorrhea, meaning failure to have the first menstrual period by the expected age. Causes include chromosomal or reproductive-tract abnormalities, endocrine disorders, chronic disease, and inadequate energy availability.

Absent Menstrual Periods - Secondary

Secondary amenorrhea, meaning absence of menstrual periods for an extended interval in a person who previously menstruated. Pregnancy is considered first; other causes include endocrine disease,

medications, stress, weight change, and ovarian dysfunction.

Absent Pulmonary Valve

Absent pulmonary valve is a congenital heart defect in which the pulmonary valve is missing or severely underdeveloped, often causing pulmonary-artery enlargement and respiratory symptoms.

Absent Reflex

Failure of an expected tendon or neurologic reflex to occur when stimulated.

Absinthism

A historical disorder attributed to habitual absinthe abuse, associated with alcohol dependence and reported hallucinations, insomnia, tremor, and seizures; the specific contribution of absinthe constituents was uncertain.

Absolute Neutrophil Count

The number of circulating neutrophils calculated from the white-cell count and differential. A low ANC indicates neutropenia; severe neutropenia greatly increases the risk of serious infection.

Alternate Names:

ANC

Absorb

To take in a substance, such as through the skin or intestine, or to reduce the intensity of radiation or light by taking up its energy.

Absorption

The process by which a drug enters the bloodstream from its site of administration, such as the digestive tract, skin, or muscle.

Abstinence

Deliberate avoidance of a substance or behavior, especially refraining from alcohol, drugs, tobacco, or sexual activity.

Abstract

In medical literature, an abstract is a brief summary

of a study or report describing its purpose, methods, main results, and conclusions.

Abulia

Marked reduction in motivation, initiative, and spontaneous activity, usually caused by dysfunction of frontal-subcortical brain circuits or related neurologic disease.

Abusive Head Trauma

Injury to an infant or young child caused by inflicted blunt impact, violent shaking, or both. Findings may include subdural hemorrhage, retinal hemorrhage, encephalopathy, fractures, or other injuries; diagnosis requires careful multidisciplinary assessment because no single finding is diagnostic.

Alternate Names:

Shaken Baby Syndrome (Abusive Head Trauma)

Abutment

A tooth or implant that supports and anchors a dental prosthesis.

AC Joint

The acromioclavicular joint, a gliding joint between the acromion of the scapula and the clavicle at the top of the shoulder.

Acalculia

Loss of the ability to do or understand numbers and calculations that a person could previously manage. It may result from injury or disease affecting areas of the brain involved in calculation.

Acalculous Cholecystitis

Inflammation of the gallbladder without gallstones. It usually occurs suddenly in people who are critically ill, injured, recovering from surgery, unable to eat, or in shock. Reduced bile flow and poor blood supply can contribute, and severe cases may cause tissue death or a hole in the gallbladder. Fever and pain in the upper-right abdomen may occur; ultrasound is commonly used for evaluation.

Acanthamoeba

A free-living amoeba found in soil and water that can cause keratitis, skin infection, or rarely granulomatous amoebic encephalitis, especially in people with impaired immunity.

Acanthamoeba Keratitis

A rare but potentially sight-threatening infection of the cornea caused by the free-living amoeba *Acanthamoeba*. It is strongly associated with contaminated contact-lens water or poor lens hygiene and may cause severe eye pain, redness, photophobia, blurred vision, and a ring-shaped corneal infiltrate. Prompt ophthalmic assessment and prolonged anti-amoebic treatment are required.

Acanthocyte

An abnormally shaped red blood cell with irregular thorn-like projections.

Alternate Names:

Spur cell

Acanthocytes (Spur Cells)

Abnormal red blood cells with irregular, spiky, thorn-like membrane projections of varying length and width. Caused by alterations in membrane lipid composition (increased cholesterol-to-phospholipid ratio).

Alternate Names:

Spur Cells

Acantholysis

Loss of attachment between cells in the outer layer of the skin, causing the cells to separate.

Acanthoma

A benign growth composed of thickened squamous epidermal cells.

Acanthosis Nigricans

A skin change causing darker, thicker, velvety patches, usually in skin folds such as the neck, armpits, or groin. It is often associated with insulin resistance, obesity, type 2 diabetes, or metabolic syndrome; less commonly, it is related to medicines

or other medical conditions.

Acapnia

An unusually low carbon-dioxide level in arterial blood, usually caused by breathing too quickly or deeply. It may cause light-headedness, tingling, or a change in the blood's acid level.

Acardia

Absence or severe malformation of the fetal heart, usually occurring in the setting of a rare twin perfusion sequence. The affected fetus cannot sustain independent circulation.

Acariasis

Infestation or disease caused by mites or ticks. Clinical manifestations vary with the organism and may include dermatitis, pruritus, tissue invasion, or transmission of an associated pathogen.

Acaricide

A pesticide or other agent used to kill mites and related arachnids such as ticks.

Acarus

Acarus is a mite genus; mites and related arthropods can cause skin infestation, allergy, or occupational exposure depending on the species.

Acatalasia

A rare inherited deficiency of catalase, an enzyme that breaks down hydrogen peroxide.

Alternate Names:

Catalase deficiency

ACC

Adenoid cystic carcinoma, a malignant tumor that most often arises in glandular tissues such as the salivary glands and may grow slowly but invade nerves or recur.

Accelerated Growth

Growth occurring faster than expected for a person's age, tissue, or developmental stage.

Accelerometry

Measurement of movement or acceleration, often used to assess activity, gait, or sleep.

Acceptance-Based Therapies

Psychotherapeutic approaches that help a person notice and accept thoughts, feelings, or bodily sensations while reducing unhelpful attempts to control or avoid them.

Alternate Names:

Acceptance-based psychotherapy; acceptance-focused therapy.

Access

In health care, the opportunity to obtain care; in dialysis, vascular access is the site where a needle or catheter enters the bloodstream.

Access Needle (Interventional)

A needle used to enter a vessel, cavity, or tissue under clinical or image guidance for sampling, drainage, infusion, or placement of another device.

Alternate Names:

Interventional

Accessory

Additional or supplementary; in anatomy, an accessory structure is an extra part or a structure that assists the usual organ or system.

Accessory Digestive Organ

An organ that supports digestion but is not part of the alimentary canal, such as the liver, pancreas, gallbladder, or salivary glands.

Accessory Dwelling Unit

A secondary, independent residential unit on the same property as a primary home; it is a housing and community-health term rather than a disease or body structure.

Accessory Muscles

Muscles recruited to assist the primary muscles of respiration or another movement when the normal workload is increased; use of respiratory accessory

muscles may indicate respiratory distress.

Accessory Nerve

The eleventh cranial nerve (CN XI), a primarily motor nerve whose spinal component supplies the sternocleidomastoid and trapezius muscles. It enables head rotation and shoulder elevation; injury may cause weak head rotation, impaired shoulder shrug, and shoulder droop.

Alternate Names:

Cranial nerve XI; spinal accessory nerve

Accessory Pathway

An abnormal electrical connection between the atria and ventricles that can bypass the atrioventricular node and cause pre-excitation or tachyarrhythmias, including atrioventricular reentrant tachycardia.

Accessory Placenta

An extra placental lobe separate from the main placenta but connected to it by fetal blood vessels; it can increase the risk of retained placental tissue or vessel injury during delivery.

Accident And Emergency Medicine

Accident and emergency medicine is the specialty concerned with immediate evaluation and treatment of acute illness, injury, poisoning, and life-threatening emergencies.

Accidental Death

Death caused by an unintentional injury or event rather than disease, suicide, or homicide. The cause is determined through medical, scene, and sometimes legal investigation.

Alternate Names:

Unintentional death

Accidental Fall

An unintentional descent to a lower level or the ground.

Alternate Names:

Fall injury

Accommodation

The ability of the eye to change the shape of its lens so that near objects can be focused clearly.

Accommodation Reflex

The coordinated adjustment of the lens, ciliary muscle, and pupils that shifts the eye from distance to near focus; it includes lens thickening, convergence, and pupillary constriction.

Accommodative Insufficiency

Inability of the eye to increase lens power adequately for near focusing, causing eyestrain, blurred near vision, headache, or difficulty reading.

Accouchement

Accouchement is the process of childbirth, including labor, delivery of the infant, and expulsion of the placenta.

Accretion

Accretion is gradual accumulation or growth of material; in medicine it may describe deposits such as dental calculus or mineral buildup around a lesion or device.

ACE Blood Test

ACE blood test measures angiotensin-converting enzyme in serum; it may support evaluation of sarcoidosis and selected granulomatous or liver disorders but is not diagnostic alone.

ACE Inhibitors

ACE inhibitors block formation of angiotensin II and are used for hypertension, heart failure, diabetic kidney protection, and selected cardiovascular conditions.

Acebutolol

Acebutolol is a beta-adrenergic blocker with intrinsic sympathomimetic activity used in some settings for hypertension and cardiac rhythm disorders; it can slow heart rate and conduction.

Acellular

Lacking cells; an acellular material may consist of extracellular matrix, mineral, fluid, or other components without intact living cells.

Acentric Chromosome

A chromosome fragment that lacks a centromere and therefore usually cannot attach normally to the spindle during cell division.

Acephalostomia

A congenital anomaly involving absence of the head and mouth.

Acetabular Labrum

A fibrocartilaginous rim attached to the margin of the acetabulum that deepens the hip socket, helps maintain a fluid seal, and contributes to hip-joint stability.

Acetabular Notch

The indentation in the inferior margin of the acetabulum, bridged by the transverse ligament of the acetabulum and forming a passage for structures entering the hip joint.

Acetabuloplasty

A surgical procedure that reshapes or augments the acetabular rim or socket to improve coverage of the femoral head in selected structural hip disorders.

Acetabulum

The cup-shaped socket of the hip formed by the fused ilium, ischium, and pubis. It articulates with the femoral head and is deepened by the acetabular labrum.

Acetaldehyde

A chemical produced when the body metabolizes alcohol. Accumulation may contribute to flushing, nausea, and vomiting.

Acetaminophen

A non-opioid analgesic and antipyretic, also called paracetamol, used to relieve mild-to-moderate pain

and reduce fever. It has little clinically useful anti-inflammatory activity. Excessive doses can cause severe, potentially fatal hepatic injury, especially with alcohol misuse, fasting, liver disease, or concurrent use of other acetaminophen-containing products; dosing must therefore follow the product label or a clinician's advice.

Acetaminophen (Paracetamol) Toxicity

Liver injury caused by an excessive dose of acetaminophen. Its toxic metabolite depletes glutathione and causes hepatic necrosis. N-acetylcysteine is the antidote and should be given promptly when toxicity is suspected.

Alternate Names:

Paracetamol

Acetaminophen And Codeine Overdose

Toxic effects caused by excessive exposure to Acetaminophen and codeine. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Acetaminophen Dosing For Children

Weight-based dosing guidance for acetaminophen in children; the total daily dose must account for all combination products to avoid liver toxicity.

Acetaminophen Overdose

Toxic effects caused by excessive exposure to Acetaminophen. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Acetate

The salt or ester of acetic acid; acetate is also a two-carbon metabolic substrate that can be converted to acetyl-CoA.

Acetazolamide

Acetazolamide inhibits carbonic anhydrase and promotes bicarbonate loss in urine; it is used for glaucoma, altitude illness, selected edema, and some acid-base or neurologic disorders.

Acetone

A volatile ketone body produced during increased fat metabolism. It may contribute to the characteristic breath odor of ketosis and ketoacidosis.

Acetone Body

A ketone body produced during fatty-acid metabolism; acetone is one of the three major ketone bodies and may contribute to the characteristic odor of ketosis.

Acetone Poisoning

Toxic effects caused by excessive exposure to acetone. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Acetonuria

The presence of acetone or other ketone bodies in the urine, which may occur with fasting, vomiting, uncontrolled diabetes, or other states of increased fat metabolism.

Acetyl-CoA Carboxylase

An enzyme that converts acetyl-CoA to malonyl-CoA, the first committed step in fatty-acid synthesis. Its activity is regulated by cellular energy status and hormones.

Acetyl-Coenzyme A

A central metabolic intermediate consisting of an acetyl group linked to coenzyme A by a high-energy thioester bond. It enters the citric acid cycle and provides carbon for fatty-acid, cholesterol, and ketone-body synthesis.

Acetylcholine

A major neurotransmitter in both the central and peripheral nervous systems. In the peripheral nervous system, it is released at the neuromuscular junction (nicotinic receptors) to excite skeletal muscle, at all autonomic ganglia (nicotinic receptors), and at parasympathetic postganglionic terminals (muscarinic receptors).

Acetylcholine Receptor Antibody

Acetylcholine receptor antibody testing detects antibodies associated with autoimmune myasthenia gravis and is interpreted with the neurologic examination and other studies.

Acetylcysteine

A cysteine derivative used as a mucolytic and as the antidote for acetaminophen poisoning because it replenishes glutathione and helps detoxify its harmful metabolite.

Acetylsalicyclic Acid

Acetylsalicyclic acid is the commonly misspelled name for acetylsalicylic acid, or aspirin, a salicylate that reduces pain, fever, and inflammation and inhibits platelet aggregation.

Acetylsalicylic Acid

The chemical name for aspirin, a salicylate drug with analgesic, antipyretic, anti-inflammatory, and antiplatelet effects.

ACh (Acetylcholine)

Acetylcholine, a neurotransmitter released at the neuromuscular junction and in autonomic and central nervous systems; it acts through nicotinic and muscarinic receptors.

Achalasia

A disorder in which the lower esophageal sphincter fails to relax normally and esophageal peristalsis is impaired, causing difficulty swallowing and regurgitation.

Alternate Names:

Cardiospasm (Achalasia)

Achalasia Of The Cardia

Achalasia of the cardia is failure of the lower esophageal sphincter to relax with loss of esophageal peristalsis, causing progressive swallowing difficulty, regurgitation, and food retention.

AChE (Acetylcholinesterase)

An enzyme that rapidly breaks down acetylcholine

into acetate and choline, terminating cholinergic signaling at synapses and neuromuscular junctions.

Achenbach'S Syndrome

A benign condition of recurrent, painful, spontaneous hematoma formation in the pulp of a finger or thumb, typically affecting middle-aged women. It presents with sudden onset of throbbing pain and blue-black discoloration of the fingertip, resolving spontaneously over one to two weeks without treatment.

Aches And Pains During Pregnancy

Musculoskeletal discomfort occurring during pregnancy, commonly affecting the lower back, pelvis, and joints. Mechanisms include hormonal ligamentous laxity, postural changes, and weight gain. Persistent or severe pain, especially with bleeding or fever, requires obstetric evaluation to exclude complications.

Aches, Pain, Fever

Aches, pain, and fever are common symptoms of infection, inflammation, injury, medication effects, or other illness. The pattern, duration, associated symptoms, and temperature help determine whether evaluation or urgent treatment is needed.

Achilles Tendinitis

Painful inflammation or degeneration of the Achilles tendon, usually caused by repeated loading.

Alternate Names:

Tendinitis, Achilles

Achilles Tendinopathy

Pain and structural disorder of the Achilles tendon, usually caused by repeated loading. It causes posterior heel pain and stiffness and is managed with activity modification, rehabilitation, and treatment of contributing factors.

Achilles Tendon

The strong tendon connecting the calf muscles to the heel bone. It can become inflamed or rupture.

Achilles Tendon (Calcaneal Tendon)

The thickest and strongest tendon in the human body, connecting the gastrocnemius and soleus (triceps surae) muscles to the calcaneus (heel bone). It plays a crucial role in plantarflexion of the foot, enabling walking, running, and jumping.

Alternate Names:

Calcaneal Tendon

Achilles Tendon Repair

Achilles tendon repair reconnects a ruptured Achilles tendon, followed by immobilization and graduated rehabilitation to restore plantar-flexion strength.

Achilles Tendon Rupture

A partial or complete tear of the Achilles tendon, usually caused by sudden force. It causes posterior ankle pain, weakness in plantar flexion, and difficulty walking or standing on tiptoe.

Achilles Tendonitis

A commonly used term for pain and inflammation or degeneration of the Achilles tendon. It may affect the tendon at its heel insertion or its mid-portion and is often related to repetitive loading.

Achillobursitis

Inflammation of a bursa near the Achilles tendon, causing pain, tenderness, swelling, or friction-related discomfort at the back of the heel.

Achlorhydria

Absence of hydrochloric acid in gastric secretions. It may result from autoimmune gastritis, long-term acid suppression, or loss of acid-producing stomach cells.

Achondrogenesis

A group of severe inherited disorders in which cartilage and bone development are profoundly impaired.

Achondroplasia

Autosomal dominant skeletal dysplasia (FGFR3 gene). Most common form of dwarfism. Features: short limbs, macrocephaly, frontal bossing, trident

hand, spinal stenosis. Homozygous form lethal. Normal intelligence.

Alternate Names:

Dwarfism (Achondroplasia)

Achoo Syndrome

The photic sneeze reflex: repeated sneezing triggered by sudden exposure to bright light, especially after entering sunlight from darkness.

Achromatopsia

An inherited disorder causing absent or greatly reduced color vision, often with light sensitivity and poor visual acuity.

Alternate Names:

Complete color blindness

Achromia

Achromia is loss or marked reduction of pigment in skin, hair, eye, or another tissue; causes include genetic disorders, autoimmune depigmentation, injury, and chemical exposure.

Aciclovir

Aciclovir, also called acyclovir, is an antiviral drug that inhibits herpesvirus DNA polymerase and is used for infections such as herpes simplex and varicella-zoster.

Acid

A substance that donates hydrogen ions or accepts an electron pair. In water, an acid increases hydrogen-ion concentration and lowers pH.

Acid Base Balance

The body's control of acid levels in its fluids, keeping arterial blood pH usually between 7.35 and 7.45.

Acid Indigestion

Dyspepsia or upper-abdominal discomfort that may include burning, fullness, belching, or nausea; it can accompany reflux, gastritis, ulcer disease, or other conditions.

Acid Loading Test

A test of the kidney's ability to excrete acid after an acid load, used in evaluating renal tubular acidification disorders.

Acid Mucopolysaccharides

An older term for glycosaminoglycans, long carbohydrate chains in connective-tissue substances such as cartilage, skin, and synovial fluid. Abnormal degradation or storage can cause mucopolysaccharidoses.

Acid Phosphatase

An enzyme that works best in acidic conditions and is found in several tissues, including prostate and blood cells.

Alternate Names:

ACP

Acid Reflux

Backward flow of stomach contents into the esophagus, which may cause heartburn, regurgitation, or irritation.

Acid Soldering Flux Poisoning

Toxic effects caused by excessive exposure to acid soldering flux. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Acid-Base Analysis

Evaluation of blood pH, bicarbonate, and carbon dioxide to identify metabolic or respiratory disturbances, including those caused by poisoning.

Acid-Base Disorder

Disturbance of the body's acid-base balance, classified by primary disturbance and compensation. Respiratory acidosis: hypoventilation, high PaCO₂, low pH. Respiratory alkalosis: hyperventilation, low PaCO₂, high pH.

Acid-Fast Bacillus Smear

Microscopic examination of a specimen, often sputum, for acid-fast organisms using a special stain. A positive result supports mycobacterial infection but

does not by itself identify the species or establish drug susceptibility; culture or molecular testing is also required.

Acid-Fast Stain

Acid-fast stain uses dyes that remain in the cell wall of mycobacteria and some related organisms, helping identify them in clinical specimens.

Acidophil

A cell or tissue component that stains readily with acidic dyes, especially eosin; acidophilic cells are recognized in histology and include selected endocrine and pituitary cell populations.

Acidophilus

Lactobacillus acidophilus, a lactic-acid-producing bacterium found in the intestinal and genital microbiota and used in some fermented foods and probiotic products.

Acidosis

An acid-base disturbance in which the concentration of hydrogen ions increases, tending to lower arterial blood pH below 7.35.

Aciduria

Aciduria means an abnormally acidic urine or the presence of excess organic acids in urine; it may reflect diet, dehydration, medication, or an inherited metabolic disorder.

Acinar Cell

A secretory cell arranged in a cluster, such as a pancreatic cell that releases digestive enzymes.

Acinetobacter Baumannii Resistance

Acinetobacter baumannii is a gram-negative coccobacillus that has emerged as a significant cause of healthcare-associated infections, particularly in intensive care units. The organism has intrinsic and acquired resistance mechanisms, including beta-lactamases, efflux pumps, and porin mutations.

Acinus

An acinus is a small functional secretory unit of a gland, such as a pancreatic or salivary acinus, whose cells release their product into a duct.

Acinus, Pulmonary

The portion of a lung supplied by a terminal bronchiole and containing respiratory bronchioles, alveolar ducts, and alveoli where gas exchange occurs.

ACL Injury

A sprain or tear of the anterior cruciate ligament in the knee.

ACL Reconstruction

Surgical reconstruction of the anterior cruciate ligament following rupture, one of the most common knee ligament injuries. The ACL is a primary restraint to anterior tibial translation and rotatory instability, and its rupture frequently occurs during non-contact pivoting sports activities (skiing, soccer, basketball).

Acne

A chronic inflammatory disorder of hair follicles and sebaceous glands. It causes comedones, papules, pustules, or nodules, most often on the face, chest, and back.

Alternate Names:

Pimples Pimples (Acne)

Acne Inversa

See *Hidradenitis suppurativa*.

Acne Keloid

A chronic inflammatory condition causing firm scar-like bumps around hair follicles, usually on the back of the neck.

Alternate Names:

Acne keloidalis nuchae

Acne Rosacea

Acne rosacea is a chronic inflammatory facial skin disorder causing flushing, persistent redness, visible vessels, papules, pustules, or eye symptoms; it is

distinct from acne vulgaris.

Acne Vulgaris

The common form of acne caused by follicular plugging, increased sebum production, microbial activity, and inflammation. It usually affects the face, trunk, and upper arms.

Acoustic

Relating to sound, hearing, or the physical properties and transmission of sound waves.

Acoustic Nerve

The acoustic nerve, more properly called the auditory or vestibulocochlear nerve (cranial nerve VIII), carries hearing and balance information from the inner ear to the brainstem.

Acoustic Neuroma

A benign, slow-growing tumor of the vestibular portion of the vestibulocochlear nerve. It may cause hearing loss, tinnitus, or balance problems.

Acoustic Neuroma (Vestibular Schwannoma)

A usually benign, slow-growing tumor of Schwann cells involving the vestibular portion of cranial nerve VIII. It may cause unilateral sensorineural hearing loss, tinnitus, imbalance, or facial sensory symptoms; MRI of the internal auditory canals is the preferred diagnostic study.

Alternate Names:

Vestibular Schwannoma Neuroma, Acoustic

Acoustic Reflex

An involuntary contraction of the stapedius muscle in response to a loud sound. It can be measured with immittance testing and helps assess the middle-ear muscles, auditory pathways, and facial nerve.

Alternate Names:

Stapedial reflex; stapedius reflex; middle-ear muscle reflex.

Acoustic Shock

Auditory and other symptoms after exposure to a

sudden or intense sound, potentially including ear pain, tinnitus, hyperacusis, dizziness, or temporary hearing change.

Acoustic Trauma

Acute damage to the auditory system from a brief, intense sound such as an explosion or firearm. It may cause tinnitus, temporary or permanent sensorineural hearing loss, and tympanic or ossicular injury; urgent audiologic assessment is indicated when hearing loss is sudden.

Acquired

Developed after birth rather than being present at birth or inherited; an acquired condition may result from infection, exposure, injury, aging, or another later event.

Acquired Feminization

Development of typically female secondary sexual characteristics after birth in a person not expected to develop them, often from hormonal change.

Acquired Generalized And Inherited Lipodystrophy

Lipodystrophy is abnormal loss or absence of adipose tissue. Acquired generalized forms develop after an immune or other trigger, while inherited forms result from gene variants; both can cause severe insulin resistance, diabetes, high triglycerides, and fatty liver.

Alternate Names:

Lipodystrophy (Acquired Generalized And Inherited Lipodystrophy)

Acquired Immunity

Immune protection that develops after infection, vaccination, or transfer of antibodies, involving adaptive immune responses directed toward specific antigens.

Acquired Immunodeficiency Syndrome

See *HIV/AIDS*.

Acquired Immunodeficiency Syndrome

Alternate index label for AIDS.

Acquired Platelet Function Defect

Impaired platelet adhesion or aggregation that develops after birth. Causes include antiplatelet medicines, kidney failure, hematologic disease, and other systemic disorders; platelet counts may remain normal, but mucocutaneous bleeding can occur.

Acquisition

The process by which new information or a skill is learned and initially encoded, especially in discussions of memory and learning.

Alternate Names:

Learning acquisition; memory acquisition.

Acquisition Duration

The time required to obtain or record a measurement, image, or physiologic signal.

Acquisition Time

The time during which a medical imaging system collects data. The required time depends on the imaging method, body region, protocol, and patient movement.

Acral Lentiginous Melanoma

A subtype of melanoma arising on acral skin (palms, soles, or under nails), appearing as a dark, irregularly pigmented patch or plaque. It is more common in people with darker skin tones and often presents at a later stage due to less frequent surveillance of these sites. Prognosis depends on thickness and stage at diagnosis.

Acro-Osteolysis

Resorption or destruction of the ends of the fingers or toes.

Acrocentric Chromosome

A chromosome whose centromere lies close to one end, producing a very short p arm and a longer q arm; human acrocentric chromosomes include 13, 14, 15, 21, and 22.

Acrochordon

A benign, soft, flesh-colored skin tag, usually occurring in skin folds such as the neck, axilla, or groin.

Acrochordon (Skin Tag)

Alternate index label for Skin Tag.

Acrocyanosis

Persistent, usually painless bluish discoloration and coolness of the hands or feet caused by reduced peripheral blood flow, often worsened by cold exposure.

Acrodermatitis

Inflammation affecting the skin of the hands, feet, or other distal body regions.

Acrodermatitis Enteropathica

A zinc-deficiency disorder causing periorificial and acral dermatitis, alopecia, diarrhoea, poor wound healing, and increased susceptibility to infection. It may be inherited from impaired intestinal zinc absorption or acquired through malnutrition, malabsorption, or prolonged parenteral nutrition.

Acrodynia

A painful condition with red, swollen, sweaty hands and feet, classically associated with mercury poisoning and historically called pink disease in children.

Acrodysostosis

A rare genetic skeletal dysplasia characterized by short, broad fingers and toes, nasal and midface hypoplasia, short stature, and variable intellectual disability. Some patients have resistance to parathyroid or other hormones.

Acromegaly

Chronic growth hormone excess in adulthood, typically from a pituitary somatotroph adenoma. Complications may include diabetes, hypertension, heart disease, sleep apnea, joint disease, nerve compression, visual problems, and other organ complications.

Acromioclavicular Joint

A plane synovial joint between the acromion of the scapula and the lateral end of the clavicle. It permits gliding movements and is stabilized primarily by the acromioclavicular and coracoclavicular ligaments.

Acromioclavicular Joint Injuries

Sprains or dislocations of the joint between the acromion and clavicle, usually caused by a fall or direct shoulder impact. Severity ranges from ligament sprain to complete displacement and is assessed clinically and with imaging.

Acromion

Bony process forming the shoulder's highest point, extending laterally from the scapular spine. Articulates with the clavicle at the acromioclavicular joint and arches over the glenohumeral joint. Attachment for deltoid and trapezius. The subacromial space contains the supraspinatus tendon and subacromial bursa.

Acroparaesthesia

Acroparaesthesia is tingling, pins-and-needles, burning, or altered sensation in the hands or feet, commonly related to nerve compression, neuropathy, vascular disease, or metabolic causes.

Acrophobia

An excessive, persistent fear of heights that causes marked anxiety, avoidance, or impairment beyond the ordinary caution associated with high places.

Acrosin

An enzyme in sperm that helps it penetrate the egg's surrounding layers.

Acrosome

A cap-like structure on the head of a sperm cell containing enzymes needed for fertilization.

Acrosome Reaction

Release of acrosomal enzymes that enables a sperm cell to penetrate the egg's outer layers.

ACTH Blood Test

A blood test that measures the level of ACTH, a hormone made by the pituitary gland. It is usually checked with a cortisol test to help find problems of the adrenal or pituitary glands.

ACTH Stimulation Test

A test that shows whether the adrenal glands can make cortisol. A synthetic form of ACTH is given, and cortisol levels are measured before and after it is given. A small rise may suggest adrenal insufficiency.

Actigraphy

Monitoring of movement over time to estimate sleep-wake patterns and activity.

Alternate Names:

Actigraphic monitoring

Actin

A structural and contractile protein found in many cells. In muscle, actin interacts with myosin to produce contraction; in other cells, it supports shape, movement, and intracellular transport.

Alternate Names:

Actin filament; microfilament protein.

Actin Myofilament

A thin muscle filament made mainly of actin that interacts with myosin during contraction.

Alternate Names:

Thin filament

Actin-Binding Protein

A protein that regulates, anchors, or cross-links actin filaments.

Acting Out

Expressing emotional conflict or distress through behavior rather than words.

Actinic

Relating to ultraviolet radiation from sunlight or UV lamps; chronic actinic exposure can cause sunburn, photoaging, and actinic keratoses.

Actinic Cheilitis

A precancerous sun-damage condition of the lips, usually the lower lip, caused by long-term ultraviolet exposure. It may cause persistent dryness, scaling, whitening, thickening, or blurring of the lip border and can progress to squamous cell carcinoma.

Actinic Granuloma

A benign granulomatous skin condition arising on sun-damaged skin, presenting as annular or ring-shaped plaques with central atrophy and an elevated, erythematous border. It is thought to represent a foreign-body reaction to degenerated elastic tissue from chronic ultraviolet exposure.

Actinic Keratosis

A rough, scaly skin lesion caused by chronic ultraviolet exposure, usually on sun-exposed skin. It can progress to squamous-cell carcinoma, so persistent, thickened, or changing lesions require clinical assessment.

Alternate Names:

Keratosis, Actinic Solar Keratosis Solar Keratosis (Actinic Keratosis)

Actinic Keratosis (Solar Keratosis)

A rough, scaly precancerous skin lesion caused by long-term ultraviolet exposure and capable of progressing to squamous cell carcinoma.

Alternate Names:

Solar Keratosis

Actinic Keratosis Management

Actinic keratosis management requires field-directed and lesion-directed therapy. Lesion-directed: cryotherapy (liquid nitrogen, 5 to 10 second freeze-thaw cycle), curettage, electrodesiccation.

Actinic Purpura

Bruising caused by fragile, sun-damaged skin and connective tissue, usually on the backs of the hands and forearms in older adults. The purple patches may appear after minor trauma and are generally harmless, although medicines such as anticoagulants can worsen them. New, widespread, or unexplained bruising warrants assessment.

Actinomycetoma

A chronic, progressive granulomatous infection of the skin and subcutaneous tissue caused by aerobic actinomycetes, characterized by swelling, draining sinuses, and dark granules in the discharge. It primarily affects the feet and is endemic in tropical regions. Diagnosis is by culture and histopathology.

Actinomycin D

Actinomycin D, or dactinomycin, is an antineoplastic antibiotic that blocks DNA-dependent RNA synthesis and is used for selected cancers; myelosuppression and tissue injury are important risks.

Actinomycosis

A chronic infection caused by branching, gram-positive anaerobic bacteria of the genus *Actinomyces*, which normally colonize the mouth, gastrointestinal tract, and genital tract. It can form infiltrative masses and draining sinuses containing sulfur granules.

Action Potential

A rapid, all-or-nothing depolarization of the cell membrane that propagates along excitable membranes (neurons, muscle, some endocrine cells).

Activated Charcoal

A porous form of carbon that adsorbs some substances in the gastrointestinal tract and can reduce their absorption after selected poisonings. It is not effective for every toxin and can cause aspiration or other complications.

Activated Partial Thromboplastin Time

A blood test that measures clotting through the intrinsic and common coagulation pathways. It may be prolonged by unfractionated heparin, clotting-factor deficiencies, inhibitors, or some liver and systemic disorders.

Active Caries

Tooth decay that is continuing to progress. The affected surface is often soft, rough, or chalky and requires dental assessment and management.

Active Management Of Third Stage

A set of measures used after birth to reduce postpartum hemorrhage, usually including a uterotonic medicine, controlled cord traction when appropriate, and assessment of uterine tone and bleeding.

Active Phase Of Labor

The part of the first stage of labor in which the cervix dilates progressively, generally beginning at about 6 cm and continuing to full dilation. Progress varies between individuals.

Active Site

The specific three-dimensional region of an enzyme where substrate binding and catalysis occur. Formed by amino acid residues that may be distant in primary sequence but brought together by protein folding.

Active Surveillance

Close, structured monitoring of a cancer or precancerous condition without immediate treatment. It includes scheduled examinations, laboratory tests, imaging, or biopsies, with treatment started if evidence of progression develops.

Active Transport

Movement of substances across cell membranes against their concentration gradient, requiring energy input, typically from ATP hydrolysis.

Activities Of Daily Living

Routine tasks required for personal care and independent living, including basic activities and more complex instrumental activities. Performance is assessed to describe functional status and support care planning.

Acupressure

Application of pressure to selected body points, usually to relieve symptoms such as pain or nausea.

Acupuncture

A treatment in which thin needles are inserted into selected points of the body.

Acute

Beginning suddenly or developing over a short period. The term may describe symptoms, illnesses, or complications.

Alternate Names:

Bronchitis (Acute)

Acute Abdomen

Sudden or severe abdominal pain with clinical features suggesting a potentially serious intra-abdominal disorder that may require urgent investigation or surgery.

Acute Acalculous Cholecystitis

Acute gallbladder inflammation without gallstones, usually occurring in critically ill, injured, postoperative, or fasting patients. Ischaemia, bile stasis, and infection may contribute; it can progress rapidly to necrosis or perforation and requires urgent assessment.

Acute Adrenal Crisis

Acute adrenal crisis is life-threatening cortisol deficiency causing hypotension, vomiting, abdominal pain, electrolyte disturbance, hypoglycemia, or shock; urgent glucocorticoid and fluid treatment is required.

Acute Agitation

A state of heightened motor activity, emotional tension, irritability, and impaired behavioral control that may threaten the patient or others. Management begins with de-escalation and assessment for delirium, intoxication, withdrawal, medical illness, and psychosis.

Acute Alcohol Intoxication

Neurobehavioral and physiologic impairment caused by recent ethanol exposure. Severity varies with concentration, tolerance, co-ingestants, and comorbidity; dangerous effects include respiratory depression, aspiration, hypoglycemia, hypothermia, trauma, and coma.

Acute And Chronic Bursitis

Bursitis is inflammation of a fluid-filled bursa that cushions a joint. Acute bursitis may follow trauma, infection, or sudden overuse; chronic bursitis reflects repeated pressure or inflammatory disease and causes localized pain, swelling, and tenderness with movement.

Alternate Names:

Bursa Inflammation (Acute And Chronic Bursitis)

Acute Angle-Closure Glaucoma

An ophthalmic emergency in which sudden closure of the anterior-chamber angle causes a rapid rise in intraocular pressure. Severe eye pain, blurred vision, halos, headache, nausea, vomiting, and a red eye may occur; urgent ophthalmic treatment is required to protect vision.

Acute Aortic Syndrome

A spectrum of life-threatening conditions involving the thoracic aorta, including aortic dissection, intramural hematoma (IMH), and penetrating aortic ulcer (PAU). See acute aortic syndromes.

Acute Arterial Occlusion - Kidney

Acute arterial occlusion of the kidney abruptly reduces renal blood flow and may cause flank pain, hematuria, hypertension, or acute kidney injury.

Acute Bacterial Rhinosinusitis

Bacterial infection of the paranasal sinuses suspected when symptoms persist without improvement for at least 10 days, become substantially worse after initial improvement, or are severe with high fever and purulent discharge. Treatment decisions depend on severity, complications, resistance risk, and local guidance.

Acute Bacterial Prostatitis

An acute, severe bacterial infection of the prostate gland, classified as NIH Category I prostatitis, most commonly caused by uropathogens such as *Escherichia coli*, *Proteus*, or *Klebsiella*. Clinically presents with sudden high fever, chills, perineal and low-back pain, dysuria, and acute urinary retention. Vigorous digital rectal examination is avoided due

to the risk of bacteremia and septic shock.

Acute Bilirubin Encephalopathy

The acute, potentially reversible neurologic syndrome caused by unconjugated bilirubin toxicity in the basal ganglia, hippocampus, and brainstem nuclei of neonates with severe hyperbilirubinemia. Early signs include poor feeding, lethargy, and hypotonia; intermediate progression brings irritability, high-pitched cry, and retrocollis; advanced stages manifest with stupor or coma, persistent retrocollis, opisthotonos, bicycling movements, and seizures. Urgent exchange transfusion and intensive phototherapy are mandatory to prevent progression to permanent kernicterus.

Alternate Names:

Bilirubin Neurotoxicity; Bilirubin Encephalopathy (Acute)

Acute Blood Loss

Rapid loss of circulating blood volume that can cause hypovolemic shock, tissue hypoxia, tachycardia, hypotension, and acute anemia.

Acute Bronchitis

Acute inflammation of the tracheobronchial tree, usually caused by a viral respiratory infection and characterized by cough, with or without sputum production, lasting up to several weeks.

Acute Bronchitis (Bronchitis (Acute))

Alternate index label for Bronchitis (Acute).

Acute Caries

Tooth decay that progresses rapidly, often producing a soft lesion or cavity. Risk is increased by factors such as frequent sugar exposure, reduced saliva, or poor oral hygiene.

Acute Cerebellar Ataxia

Sudden or rapidly developing incoordination caused by cerebellar dysfunction. In children it may follow an infection, but stroke, intoxication, inflammation, tumor, and other urgent causes must be considered.

Acute Chest Syndrome

A potentially life-threatening pulmonary complication of sickle cell disease characterized by a new lung infiltrate plus fever or respiratory symptoms such as chest pain, cough, hypoxemia, or breathing difficulty.

Acute Cholecystitis

Acute inflammation of the gallbladder, most often caused by persistent cystic-duct obstruction by a gallstone. It commonly causes sustained right-upper-quadrant pain, fever, nausea, and a positive Murphy sign; ultrasound and timely antimicrobial and surgical assessment are standard.

Acute Coronary Syndrome

A clinical spectrum caused by acute myocardial ischemia from plaque disruption, thrombosis, or less commonly coronary spasm. It includes unstable angina, NSTEMI, and STEMI and requires rapid assessment with symptoms, serial ECGs, and cardiac troponin testing.

Acute Coronary Syndromes

A spectrum of acute myocardial ischemia comprising unstable angina, non-ST-elevation myocardial infarction, and ST-elevation myocardial infarction, usually caused by acute coronary plaque disruption and thrombosis.

Acute Cough

A cough lasting less than three weeks. Common causes include viral respiratory infection, acute bronchitis, and pneumonia.

Acute Dacryocystitis

Acute infection and inflammation of the lacrimal sac, most commonly caused by obstruction of the nasolacrimal duct leading to stasis and secondary bacterial infection. It presents with painful swelling, redness, and tenderness at the medial canthus with purulent discharge.

Acute Decompensated Heart Failure

A rapid onset or worsening of heart-failure symptoms and signs caused by increased cardiac filling

pressures and/or inadequate cardiac output. It commonly presents with breathlessness, pulmonary or systemic congestion, edema, or fatigue and requires prompt assessment for precipitating causes and urgent treatment when respiratory failure, hypotension, or organ dysfunction is present.

Acute Diarrhea

Passage of loose or watery stools, usually three or more times daily, lasting less than two weeks. Infection, food-related illness, medicines, and other intestinal disorders can cause it. Oral rehydration is important; blood in stool, severe dehydration, persistent vomiting, or severe pain requires prompt assessment.

Acute Disease

A condition that begins suddenly and generally lasts a limited time.

Alternate Names:

Acute illness

Acute Disseminated Encephalomyelitis

An acute, immune-mediated demyelinating disorder, usually occurring days to weeks after an infection and less often after vaccination. It commonly causes encephalopathy with multifocal neurologic deficits, ataxia, or seizures, particularly in children.

Acute Endometritis

Acute inflammation of the endometrium, usually caused by ascending infection or retained tissue after pregnancy-related events or uterine instrumentation; fever, pelvic pain, and abnormal bleeding may occur.

Acute Esophageal Necrosis

Ischemic necrosis of the esophageal mucosa, most commonly caused by low-flow states such as shock, sepsis, or thromboembolism, presenting with hematemesis and chest pain. Endoscopy shows circumferential black necrotic mucosa, typically in the distal esophagus.

Acute Exacerbation

A rapid worsening of a chronic pulmonary condition, with increased breathlessness, cough, oxygen need,

or other symptoms; infection, aspiration, embolism, and other triggers may contribute.

Acute Exacerbation Of Asthma

A progressive increase in dyspnea, cough, wheezing, or chest tightness with reduced expiratory airflow compared with the patient's baseline. Severe attacks may cause exhaustion, silent chest, hypercapnia, respiratory failure, and need for urgent treatment.

Acute Fatty Liver Of Pregnancy

A rare, potentially life-threatening complication of late pregnancy involving liver dysfunction and microvesicular fat accumulation. It may cause nausea, abdominal pain, jaundice, low glucose, coagulopathy, or organ failure.

Acute Febrile Illness

A sudden illness characterized by fever, often with headache, chills, muscle aches, gastrointestinal symptoms, or other signs of infection; the cause may be infectious or noninfectious.

Acute Flaccid Myelitis

A rare but serious inflammatory disorder of the spinal cord, usually affecting the gray matter and causing sudden flaccid limb weakness, reduced reflexes, and sometimes cranial-nerve or respiratory involvement. MRI and cerebrospinal-fluid studies support the diagnosis; urgent neurologic and respiratory monitoring is required.

Acute Flaccid Myelitis

A rare neurologic condition causing sudden weakness and loss of muscle tone, usually affecting the arms or legs and sometimes the face or breathing muscles.

Acute Generalized Exanthematous Pustulosis

An acute drug reaction characterised by dozens to thousands of small, sterile, nonfollicular pustules arising on an erythematous base, often with fever and neutrophilia. It usually begins rapidly after exposure to an antibiotic or other medication and resolves after withdrawal of the trigger.

Acute Gastritis

Sudden, transient inflammation of the gastric mucosa, characterized by mucosal edema, hemorrhage, and neutrophil infiltration. Common causes include nonsteroidal anti-inflammatory drugs (NSAIDs), heavy alcohol consumption, severe physiologic stress (such as Curling or Cushing ulcers), and acute infection.

Alternate Names:

Acute erosive gastritis

Acute Glomerulonephritis

Acute inflammation of the renal glomeruli causing hematuria, proteinuria, reduced glomerular filtration, edema, and often hypertension.

Acute Granulocytic Leukemia

See *Acute myelogenous leukemia*.

Acute Hepatitis

Acute inflammation and injury of hepatocytes, usually developing over days to months and characterized by elevated aminotransferases, with or without jaundice, nausea, right-upper-quadrant discomfort, or constitutional symptoms.

Acute HIV Syndrome

The acute retroviral syndrome occurring 2-6 weeks after primary HIV infection, corresponding to high-level viremia and immune activation. Presents with fever (most common), lymphadenopathy, pharyngitis, rash (maculopapular, trunk/face), myalgia, arthralgia, headache, nausea, diarrhea, and oral ulcers.

Acute Hypercapnic Respiratory Failure

Inability to eliminate carbon dioxide, producing elevated arterial PaCO₂ and often respiratory acidemia. Causes include COPD exacerbation, severe asthma, neuromuscular weakness, central respiratory depression, and chest-wall restriction.

Acute Hypoxemic Respiratory Failure

Inability of the respiratory system to maintain adequate arterial oxygenation, usually manifested by

low arterial oxygen or oxygen saturation despite appropriate inspired oxygen. Causes include pneumonia, pulmonary edema, ARDS, atelectasis, and pulmonary embolism.

Acute Idiopathic Polyneuritis

Guillain-Barré syndrome, an acute immune-mediated disorder of peripheral nerves that usually causes progressive symmetric weakness and reduced reflexes and may impair breathing.

Acute Illness

A disease or health problem that begins suddenly or over a short period and generally lasts a limited time, although its severity may range from mild to life-threatening.

Acute Inflammation

A rapid, predominantly non-specific immune response to tissue injury or infection that typically lasts hours to days and involves vascular and cellular reactions. The cardinal signs are redness, heat, swelling, pain, and loss of function, resulting from increased blood flow, vascular permeability, and cellular infiltration.

Acute Inflammatory Demyelinating Polyneuropathy

See *Guillain-Barre syndrome*.

Acute Inflammatory Demyelinating Polyradiculoneuropathy

The most common form of Guillain-Barré syndrome, in which immune-mediated demyelination of peripheral nerves causes progressive, usually symmetric weakness and reduced reflexes.

Acute Interstitial Nephritis

Inflammatory infiltrate in the renal interstitium and tubules most commonly caused by drug reactions including NSAIDs, antibiotics particularly penicillins and cephalosporins, proton pump inhibitors, and diuretics. Also caused by infections and autoimmune diseases.

Acute Ischemic Stroke

Sudden neurologic dysfunction caused by focal cerebral, spinal, or retinal ischemia. Emergency evaluation distinguishes hemorrhage from ischemia and determines eligibility for reperfusion therapy while supporting airway, glucose, temperature, and blood pressure management.

Acute Kidney Failure

A sudden decline in kidney function that causes waste products and fluid to accumulate in the body.

Acute Kidney Injury

A sudden decline in kidney function over hours or days, causing reduced filtration and possible fluid, electrolyte, or acid-base disturbances. Causes include reduced blood flow, direct kidney injury, and urinary obstruction.

Alternate Names:
AKI

Acute Kidney Injury Staging

A standardized classification based on changes in serum creatinine and urine output. KDIGO stages 1 through 3 quantify the severity of abrupt kidney-function decline and support prognosis, monitoring, and treatment decisions.

Acute Leukemia

A leukemia characterized by a rapid onset, clonal proliferation of immature blast cells (>20% of bone marrow cells), and rapid progression without treatment. Subtypes: acute lymphoblastic leukemia (ALL, B-cell or T-cell, most common childhood cancer, peak age 2-5) and acute myeloid leukemia (AML, more common in adults, median age 65-70).

Acute Limb Ischemia

Sudden reduction in arterial blood flow to a limb that threatens tissue viability, usually caused by embolism, thrombosis, or graft occlusion. The six classic findings are pain, pallor, pulselessness, paresthesia, paralysis, and poikilothermia; urgent vascular assessment and revascularization are required.

Acute Limb Ischemia Six Ps

The classic findings of acute arterial perfusion loss: pain, pallor, pulselessness, paresthesia, paralysis, and poikilothermia. Motor or sensory deficit indicates threatened limb and requires urgent vascular assessment and revascularization planning.

Acute Liver Failure

Rapid deterioration of liver function in a previously healthy individual without prior liver disease. Causes include acetaminophen toxicity, viral hepatitis, drug reactions, autoimmune hepatitis, and Wilson disease. Features include coagulopathy, encephalopathy, and multiorgan failure. Presents with jaundice, confusion, and bleeding.

Alternate Names:
Liver Failure, Acute

Acute Lymphangitis

Acute inflammation of lymphatic vessels, typically resulting from streptococcal or staphylococcal infection spreading from a distal wound. It presents with red, tender streaks extending proximally from the site of infection, along with fever, malaise, and regional lymphadenopathy. Treatment requires systemic antibiotics.

Acute Lymphoblastic Leukemia

A clonal malignancy of lymphoid progenitor cells, most common in children (peak age 2-5 years) but also occurring in adults with a worse prognosis with increasing age.

Acute Lymphoblastic Leukemia

Acute lymphoblastic leukemia is a rapidly progressive cancer of immature lymphoid precursor cells that crowds the bone marrow and can involve blood, lymph nodes, or the central nervous system.

Acute Lymphocytic Leukemia

A malignant proliferation of immature lymphoid precursors in the bone marrow and blood, commonly presenting with cytopenias, fever, bone pain, or lymphadenopathy.

Alternate Names:
Leukemia, Acute Lymphocytic

Acute Lymphoid Leukemia

See *Acute lymphocytic leukemia*.

Acute Megakaryoblastic Leukemia

A rare subtype of acute myeloid leukemia characterized by the proliferation of megakaryoblasts in the bone marrow and peripheral blood.

Acute Mesenteric Ischemia

Occlusion or reduced flow in the mesenteric arteries causing bowel ischemia, most commonly from embolism or atherosclerosis. Presents with severe abdominal pain disproportionate to physical findings. Delayed treatment leads to bowel necrosis, perforation, and high mortality.

Acute Mountain Sickness

A mild altitude illness caused by rapid ascent, usually producing headache with nausea, fatigue, dizziness, or sleep disturbance. Prevention and treatment include gradual acclimatization, descent when symptoms worsen, oxygen, and appropriate medical therapy.

Acute Mountain Sickness

A high-altitude illness caused by insufficient acclimatization, typically producing headache with nausea, dizziness, fatigue, or sleep disturbance after ascent.

Acute Myeloblastic Leukemia

See *Acute myelogenous leukemia*.

Acute Myelogenous Leukemia

A fast-growing cancer of immature myeloid blood cells in the bone marrow and blood; also called acute myeloid leukemia.

Alternate Names:

acute myeloid leukemia. Leukemia, Acute Myelogenous

Acute Myeloid Leukemia

A clonal hematologic malignancy characterized by proliferation of immature myeloid blasts that suppress normal hematopoiesis. Diagnosis requires the appropriate blast burden or defining genetic abnor-

malities, and presentation commonly includes anemia, infection, and bleeding from marrow failure.

Acute Myeloid Leukemia - Children

Childhood acute myeloid leukemia is a marrow and blood cancer caused by uncontrolled proliferation of immature myeloid cells, producing anemia, infection, and bleeding risk.

Acute Myeloid Leukemia - Adult

Adult acute myeloid leukemia is an aggressive malignancy of myeloid blood-cell precursors that impairs normal blood-cell production.

Acute Myocardial Infarction

Sudden death of heart muscle caused by prolonged inadequate coronary blood flow, usually from an acutely obstructed coronary artery; it is a medical emergency.

Acute Necrotizing Ulcerative Gingivitis

A painful, rapidly developing infection of the gums characterized by punched-out gingival papillae, ulceration, spontaneous bleeding, gray necrotic tissue, and foul breath. It is associated with anaerobic oral bacteria and risk factors such as poor oral hygiene, smoking, stress, malnutrition, or immunosuppression; urgent dental evaluation, cleaning, and treatment are needed.

Acute Nephritic Syndrome

A clinical syndrome of glomerular inflammation marked by hematuria, variable proteinuria, edema, hypertension, and reduced kidney function.

Acute Nonlymphocytic Leukemia

See *Acute myelogenous leukemia*.

Acute Obstructive Uropathy

Sudden impairment of urinary flow caused by obstruction of the urinary tract, potentially producing hydronephrosis, pain, infection, and acute kidney injury.

Acute Onset Pain

Pain that begins suddenly, often over seconds, minutes, or hours.

Alternate Names:

Sudden-onset pain

Acute Otitis Media

Infection of the middle ear space with acute onset of symptoms and signs of middle ear inflammation. Diagnosis requires middle ear effusion plus acute symptoms (ear pain, fever, irritability) or signs (bulging tympanic membrane, erythema). Most common pathogen: *Streptococcus pneumoniae*.

Acute Pain

Pain that begins suddenly and usually lasts for a limited period, often because of injury, illness, or a procedure.

Acute Pancreatitis

Sudden inflammation of the pancreas, most often caused by gallstones or alcohol. It causes upper-abdominal pain and may cause vomiting or systemic complications; severity ranges from mild to life-threatening.

Acute Promyelocytic Leukemia

A distinct subtype of acute myeloid leukemia characterized by the t(15;17)(q24;q21) translocation, which fuses the promyelocytic leukemia (PML) gene on chromosome 15 with the retinoic acid receptor alpha (RARA) gene on chromosome 17.

Acute Pyelonephritis

An acute, severe bacterial infection of the renal parenchyma and pelvicalyceal system, typically arising from an ascending lower urinary tract infection (most commonly *Escherichia coli*). Clinically presents with flank pain, costovertebral angle tenderness, high fever, rigors, and urinary symptoms.

Acute Radiation Sickness

Illness caused by a high dose of penetrating radiation received over a short time.

Acute Radiation Syndrome

See *Radiation sickness*.

Acute Rehabilitation

Multidisciplinary rehabilitation provided during or soon after hospitalization for conditions such as stroke, spinal-cord injury, traumatic brain injury, or major trauma. Treatment is tailored to functional needs.

Acute Renal Failure

See *Acute kidney injury*.

Acute Respiratory Distress Syndrome

A severe inflammatory lung injury causing acute hypoxemic respiratory failure and diffuse lung opacities not fully explained by heart failure. Common triggers include sepsis, pneumonia, aspiration, trauma, and pancreatitis.

Acute Respiratory Distress Syndrome

ARDS is acute inflammatory lung injury causing severe hypoxemia and diffuse pulmonary opacities that are not fully explained by heart failure or fluid overload.

Acute Respiratory Failure

A sudden inability of the respiratory system to maintain adequate oxygenation, carbon-dioxide removal, or both. It may result from lung disease, airway obstruction, neuromuscular weakness, or impaired respiratory drive and requires urgent evaluation.

Acute Retinal Necrosis

A rapidly progressive necrotizing viral retinitis, usually caused by herpes viruses, with vitreous inflammation, peripheral retinal necrosis, occlusive vasculitis, and risk of retinal detachment.

Acute Retroviral Syndrome

The acute illness that may occur shortly after HIV infection, commonly involving fever, rash, sore throat, lymph-node enlargement, muscle aches, or headache.

Acute Rhinitis

Sudden inflammation of the nasal lining, commonly caused by a viral infection or an irritant, producing congestion, nasal discharge, and sneezing.

Acute Rhinosinusitis

See *Acute sinusitis*.

Acute Sinusitis

Acute inflammation of the mucosa of one or more paranasal sinuses, usually following a viral upper-respiratory infection. It causes nasal obstruction, purulent or discoloured discharge, facial pressure or pain, and reduced smell; persistent, severe, or worsening symptoms may indicate acute bacterial rhinosinusitis or a complication.

Alternate Names:

Sinusitis, Acute

Acute Stress Disorder

A trauma-related mental-health condition beginning after exposure to actual or threatened death, serious injury, or sexual violence and lasting from three days to one month. Symptoms may include intrusive memories, avoidance, negative mood, dissociation, and increased arousal; professional assessment and support are recommended.

Acute Symptomatic Seizures Secondary To Autoimmune Encephalitis

See *Autoimmune epilepsy*.

Acute Thrombocytopenic Purpura

An acute disorder with a markedly reduced platelet count and bleeding or purpura, historically used for immune thrombocytopenic purpura, often following an infection in children.

Acute Transverse Myelitis

Inflammation across a segment of the spinal cord that can cause rapidly developing weakness, sensory changes, and bladder or bowel dysfunction.

Acute Tubular Necrosis

Most common cause of intrinsic AKI resulting from

ischemic or nephrotoxic injury to tubular epithelial cells. Ischemic ATN develops from prolonged prerenal states. Nephrotoxic ATN results from direct injury by aminoglycosides, amphotericin B, cisplatin, contrast, and myoglobin.

Acute Urinary Retention

Sudden inability to empty the bladder, often causing painful lower-abdominal distension and requiring urgent assessment.

Acute Viral Illness

A short-duration illness caused by a virus. Symptoms depend on the affected organ system and may include fever, fatigue, cough, vomiting, diarrhea, or rash.

Acute-On-Chronic Liver Failure

Acute deterioration of pre-existing chronic liver disease associated with organ failure and high short-term mortality. It is commonly precipitated by infection, alcohol-related hepatitis, bleeding, or other acute insults and requires urgent multidisciplinary management.

Acyclovir

An antiviral drug that inhibits herpesvirus DNA polymerase and is used for infections caused by herpes simplex virus and varicella-zoster virus.

Ad Lib

A prescription or clinical instruction meaning as desired or as tolerated, from the Latin *ad libitum*.

ADA Deficiency

An inherited lack of adenosine deaminase activity that causes toxic purine metabolites to accumulate and can produce severe combined immunodeficiency affecting both T- and B-cell function.

Adactyly

Adactyly is congenital absence of one or more fingers or toes, ranging from loss of a digit to more extensive limb malformation.

Adam'S Apple

The common name for the laryngeal prominence, the visible projection at the front of the neck formed by the angle where the two laminae of the thyroid cartilage meet.

Adamantine

Hard or resembling diamond; in dentistry, adamantine refers to enamel or enamel-like tissue.

Adaptation

Adaptation is the adjustment of a cell, organism, or person to a changing internal or external condition; in medicine it may involve physiologic compensation, behavior, or rehabilitation.

Adaptive Design

A clinical-study design that permits planned changes in response to accumulating data while preserving validity.

Adaptive Equipment

Devices that help individuals with disabilities perform daily activities independently. Examples: reachers, dressing sticks, button hooks, ergonomic utensils, grab bars, shower chairs, wheelchairs, walkers, canes. Selection based on individual needs and goals. Occupational therapy assessment recommended.

Adaptive Immunity

Antigen-specific immunity mediated principally by B and T lymphocytes. It develops after antigen exposure, generates immunologic memory, and includes humoral responses from B-cell antibodies and cellular responses from T cells.

Addiction

A commonly used, non-diagnostic term for compulsive substance use or recurrent behavior despite harm, with impaired control, craving, or difficulty stopping. In clinical practice, the specific diagnosis is usually named as a substance-use disorder or a recognized disorder due to addictive behavior; physical dependence alone does not establish addiction.

Addiction Sexual (Sexual Addiction)

Alternate index label; see *Compulsive Sexual Behavior Disorder*.

Addiction, Alcohol

Legacy label; see *Alcohol Use Disorder*.

Addiction, Gambling

Legacy label; see *Gambling Disorder*.

Addiction, Nicotine

Legacy label; see *Nicotine Dependence*.

Addison Anemia

An older name for pernicious anemia, a vitamin B12-deficiency megaloblastic anemia caused by inadequate intrinsic factor and impaired B12 absorption.

Addison Disease

Primary adrenal insufficiency resulting from destruction or dysfunction of the adrenal cortex, leading to deficiency of cortisol, aldosterone, and adrenal androgens.

Addison'S Disease

A disorder in which the adrenal glands produce too little cortisol and, often, too little aldosterone.

Alternate Names:

Chronic Adrenal Insufficiency Primary Adrenal Insufficiency

Addisonian Anemia

Another historical term for pernicious anemia, in which autoimmune loss of gastric parietal-cell function or intrinsic factor causes vitamin B12 deficiency.

Addisonian Crisis

Life-threatening emergency resulting from acute cortisol deficiency in patients with adrenal insufficiency. Precipitated by infection, trauma, surgery, or abrupt discontinuation of chronic glucocorticoid therapy.

Addisons Anemia (Pernicious Anemia)

Alternate index label for Pernicious Anemia.

Additive Genetic Effects

The contributions of individual alleles that combine approximately additively to influence a measurable trait, without requiring interaction between those alleles.

Adduct

To adduct a limb or body part is to move it toward the body's midline; the opposite movement is abduction.

Adduction

Movement of a body part toward the body's midline.

Adductor Muscle

A muscle that moves a body part toward the body's midline or, for a digit, toward the axis of the hand or foot.

Adductor Muscles

A group of muscles on the medial thigh that adduct (pull toward the midline) the hip: adductor longus, adductor brevis, adductor magnus, gracilis, and pectineus. Innervated by the obturator nerve (L2-L4); adductor magnus also receives tibial nerve (sciatic). Adductor longus is a common site of athletic injury ('groin strain').

Adenine (A)

In genetics, A denotes adenine, a nucleobase that pairs with thymine in DNA and with uracil in RNA.

Adenitis

Inflammation of a gland or lymph node, usually caused by infection or an inflammatory disorder. Lymphadenitis is a common form.

Adenitis, Mesenteric

See *Mesenteric lymphadenitis*.

Adenocarcinoma

A malignant tumor arising from glandular epithelial tissue, forming gland-like structures. The most common type of carcinoma, occurring in organs such as the colon, breast, lung, pancreas, and prostate.

Adenocarcinoma In Situ Of Cervix

A noninvasive glandular precursor lesion of the cervix, usually associated with high-risk human papillomavirus; abnormal glandular cells remain confined to the epithelium but may progress to invasive adenocarcinoma.

Adenocarcinoma Of The Colon

A malignant gland-forming epithelial tumor of the colon, commonly presenting with altered bowel habits, occult bleeding, anemia, or an obstructing mass.

Adenocarcinoma Of The Lung

The most common histologic subtype of non-small cell lung cancer, originating from glandular epithelial cells of the peripheral lung and bronchial submucosal glands. Often associated with molecular driver mutations (such as EGFR, ALK, or KRAS) and commonly diagnosed in both smokers and non-smokers.

Alternate Names:

Lung adenocarcinoma

Adenohypophysis

The anterior lobe of the pituitary gland, which produces and secretes hormones including adrenocorticotrophic hormone, thyroid-stimulating hormone, growth hormone, prolactin, and gonadotropins.

Adenoid Basal Carcinoma

A rare, usually indolent cervical carcinoma composed of small basaloid nests, often associated with squamous intraepithelial lesions and requiring distinction from more aggressive tumors.

Adenoid Cystic Carcinoma

A rare malignant tumor of glandular tissue, most commonly of the salivary glands, characterized by slow growth, perineural spread, and late recurrence.

Adenoid Cystic Carcinoma Of The Vulva

A rare malignant tumor of vulvar Bartholin-type glands, often causing pain because of perineural invasion. It is usually slow-growing but may recur

locally or metastasize after long intervals.

Adenoid Removal

Surgical excision of the pharyngeal tonsils (adenoids) from the nasopharynx, typically performed in children for chronic nasal obstruction, recurrent otitis media, or obstructive sleep apnea unresponsive to medical management.

Adenoidectomy

The surgical removal of the pharyngeal tonsils (adenoids), which are lymphoid tissue structures located in the nasopharynx, and it is commonly performed in children for obstructive sleep apnea, chronic nasal obstruction, recurrent otitis media, and chronic rhinosinusitis that is refractory to medical management.

Adenoiditis

Inflammation or infection of the adenoids, the pharyngeal tonsil behind the nose, which may cause nasal obstruction, mouth breathing, snoring, or ear problems.

Adenoids

Lymphoid tissue in the posterior wall and roof of the nasopharynx, part of Waldeyer ring of pharyngeal lymphoid tissue. Enlarged adenoids (adenoidal hypertrophy) in children cause nasal obstruction, mouth breathing, snoring, obstructive sleep apnea, serous otitis media (eustachian tube obstruction), and facial changes ('adenoid facies').

Adenoids And Tonsils

The adenoids and tonsils are lymphoid tissues in the upper airway that participate in immune surveillance. Enlargement or recurrent infection can cause mouth breathing, snoring, sleep apnea, swallowing problems, or recurrent throat disease and may require medical or surgical evaluation.

Alternate Names:

Tonsil Inflammation (Adenoids And Tonsils)

Adenolipoma

An adenolipoma is a usually benign tumor containing both glandular tissue and mature fatty tissue.

Adenolymphoma

A benign salivary-gland tumor, also called Warthin tumor, composed of oncocytic epithelium and lymphoid stroma and occurring most often in the parotid gland.

Adenoma

A benign tumour arising from glandular or gland-like epithelial tissue. Some adenomas, particularly certain colorectal adenomas, can develop dysplasia and progress to carcinoma.

Adenoma Malignum

A rare, deceptively bland, deeply infiltrative gastric-type adenocarcinoma of the cervix with minimal cytologic atypia; persistent watery discharge may occur and diagnosis requires expert histopathology.

Adenoma-Carcinoma Sequence

Established model of stepwise colorectal carcinogenesis progressing from normal epithelium through adenomas to carcinoma over ten to fifteen years.

Adenomatosis

Adenomatosis is the presence of multiple adenomas or diffuse adenomatous overgrowth in an organ or tissue; its implications depend on the site and underlying syndrome.

Adenomatous Polyp

A benign gland-forming polyp, most commonly in the colon, that may contain dysplasia and can progress through the adenoma–carcinoma sequence. Tubular, villous, and tubulovillous types are recognized.

Adenomyoepithelioma Of The Breast

A rare biphasic breast tumor composed of proliferating epithelial and myoepithelial cells. Most are benign, but local recurrence and malignant transformation can occur, so complete excision is usually recommended.

Adenomyoma

A benign uterine lesion composed of endometrial

glands and stroma within a proliferation of smooth muscle; it may cause abnormal uterine bleeding or mimic other uterine tumors.

Adenomyosis

A condition in which endometrial glands and stroma are present within the myometrium (uterine muscle). Most common in multiparous women in their 40s and 50s. Presents with heavy menstrual bleeding (menorrhagia), severe dysmenorrhea, chronic pelvic pain, and uterine enlargement.

Adenopathy

Disease or enlargement of a gland, most often used for lymph-node enlargement or abnormal lymph-node involvement.

Adenosine

A purine nucleoside made of adenine linked to ribose; it participates in cellular energy transfer and signaling and can transiently slow atrioventricular-node conduction when given medically.

Adenosine Deaminase Deficiency

An inherited purine-metabolism disorder in which toxic deoxyadenosine metabolites accumulate and damage lymphocytes, commonly causing severe combined immunodeficiency.

Adenosine Diphosphate

A nucleotide composed of adenine, ribose, and two phosphate groups. Formed by hydrolysis of ATP ($\text{ATP} \rightarrow \text{ADP} + \text{P}_i$, $\Delta G^\circ = -30.5 \text{ kJ/mol}$). Serves as the immediate precursor for ATP synthesis via oxidative phosphorylation (mitochondrial ATP synthase), substrate-level phosphorylation (glycolysis, TCA cycle), and photophosphorylation.

Alternate Names:

Adenosine Diphosphate

Adenosine Monophosphate

A nucleotide composed of adenine, ribose, and a single phosphate group. Formed by hydrolysis of ADP ($\text{ADP} \rightarrow \text{AMP} + \text{P}_i$) or directly from ATP via adenylate kinase ($2 \text{ ADP} \rightleftharpoons \text{ATP} + \text{AMP}$).

Alternate Names:

Adenosine Monophosphate

Adenosine Triphosphate

A molecule that stores and transfers energy within cells. It is commonly abbreviated ATP.

Adenosine Triphosphate

The primary energy currency of the cell, composed of adenine, ribose, and three phosphate groups. Hydrolysis of the terminal phosphoanhydride bond releases $\Delta G^\circ = -30.5 \text{ kJ/mol}$ (-7.3 kcal/mol), driving endergonic cellular processes such as muscle contraction, active transport, and biosynthesis.

Alternate Names:

Adenosine Triphosphate

Adenosis

A benign or abnormal increase in the number or size of glands or glandular tissue; the clinical meaning depends on the organ, such as breast adenosis or pulmonary adenosis.

Adenovirus

A non-enveloped DNA virus that causes respiratory infections, conjunctivitis, gastroenteritis, and cystitis. Adenoviruses are highly stable in the environment and can cause outbreaks in closed populations. The virus has over 50 serotypes with different tissue tropisms. Diagnosis is made by PCR, antigen detection, or viral culture.

Adenovirus Infection

Infection with an adenovirus, commonly causing sore throat, cough, fever, conjunctivitis, diarrhea, or vomiting. Most infections are self-limited, but severe respiratory or disseminated disease can occur in infants and people with weakened immunity. Treatment is usually supportive, with medical assessment for breathing difficulty, dehydration, or severe illness.

Adenoviruses

Adenoviruses are non-enveloped DNA viruses that can cause respiratory, eye, gastrointestinal, urinary, or other infections, usually self-limited but some-

times severe in vulnerable people.

Adequate Intake

A recommended average daily nutrient intake based on observed or experimentally determined levels in healthy people when evidence is insufficient to establish a recommended dietary allowance.

Alternate Names:

AI; adequate nutrient intake.

Adherence

The extent to which a person follows an agreed treatment plan, including prescribed medicines and other care. Adherence may be affected by cost, adverse effects, treatment complexity, health beliefs, or difficulty accessing care.

Adherence (Geriatric)

Adherence to an agreed treatment plan in an older adult, including medicine use, appointments, and recommended self-care. Cognitive impairment, polypharmacy, adverse effects, cost, and support needs may affect adherence.

Alternate Names:

Geriatric

Adherens Junction

A cell-to-cell connection that uses cadherin proteins to link neighboring cells through actin filaments.

Adhesiolysis

Surgical division or removal of abnormal fibrous adhesions that bind tissues or organs together, often after inflammation, infection, surgery, or trauma.

Adhesion

A band of scar-like tissue that abnormally connects surfaces or organs inside the body.

Adhesion Molecules

Cell-surface proteins that mediate attachment between cells or between cells and the extracellular matrix; they are important in leukocyte trafficking, inflammation, tissue organization, and metastasis.

Adhesive Capsulitis

A condition causing progressive shoulder pain and stiffness due to inflammation and fibrosis of the glenohumeral joint capsule. See Frozen Shoulder.

Adhesive Capsulitis Of Shoulder (Frozen Shoulder)

Alternate index label for Frozen Shoulder.

Adhesive Tape

A material used to secure dressings, devices, or tubing to the skin. It is available in paper, cloth, plastic, and other forms.

ADI

Acceptable daily intake: the estimated amount of a substance that can be consumed every day over a lifetime without appreciable health risk, based on available evidence and safety factors.

Adie Tonic Pupil

Uncommon cause of anisocoria, a benign parasympathetic denervation of the iris sphincter and ciliary muscle, typically unilateral in young women, idiopathic or associated with diabetes, herpes zoster, or autoimmune neuropathy.

Adipocere (Grave Wax)

Waxy, grayish-white substance formed postmortem in warm, moist, anaerobic environments (water, damp soil) by the hydrolysis and hydrogenation of body fat into free fatty acids; retards putrefaction and preserves anatomical details for months to years.

Alternate Names:

Grave Wax

Adipocyte

A specialized mesenchymal cell responsible for the storage and mobilization of energy in the form of neutral triglycerides. Classified into white adipocytes (unilocular lipid droplets, specialized for long-term caloric storage and endocrine secretion of leptin and adiponectin), brown adipocytes (multilocular with abundant mitochondria express-

ing uncoupling protein 1 [UCP1/thermogenin] for non-shivering thermogenesis), and beige/brite adipocytes (inducible within white fat depots upon cold exposure or adrenergic stimulation). Pathologic hypertrophy and hyperplasia of white adipocytes promote chronic low-grade systemic inflammation, macrophage recruitment (crown-like structures), and insulin resistance in obesity and metabolic syndrome.

Alternate Names:

Fat cell

Adipogenesis

The biological differentiation process by which multipotent mesenchymal stem cells (MSCs) commit to the adipocyte lineage and mature into functional lipid-storing adipocytes. Driven by a coordinated transcriptional cascade in which CCAAT/enhancer-binding proteins (C/EBP- β and C/EBP- δ) induce the master regulators peroxisome proliferator-activated receptor gamma (PPAR- γ) and C/EBP- α , which activate downstream lipogenic enzymes and adipokine production. Targeted pharmacologically by thiazolidinediones (e.g., pioglitazone), which act as synthetic PPAR- γ agonists to promote adipocyte differentiation, enhance subcutaneous lipid storage capacity, and improve peripheral insulin sensitivity.

Adipokine

A signaling molecule released by adipose tissue that influences metabolism or inflammation.

Adiponectin

A hormone produced mainly by adipose tissue that influences fatty-acid metabolism, insulin sensitivity, and inflammatory responses.

Adipose

Relating to body fat or composed of adipose tissue, which stores energy and also functions as an endocrine and immune-active organ.

Adipose Tissue

A specialized connective tissue composed mainly of adipocytes that store energy as triglycerides. It provides thermal insulation, cushioning for organs,

and endocrine regulation through hormones and signaling molecules such as leptin and adiponectin.

Adiposis Dolorosa

Adiposis dolorosa, or Dercum disease, is a rare disorder characterized by painful fatty deposits, often with obesity, fatigue, and other systemic symptoms.

Adiposity

The amount or distribution of body fat; excess adiposity is associated with increased risk of metabolic, cardiovascular, and mechanical disease.

Adipsia

Absent or markedly reduced thirst, which can impair normal fluid intake and lead to dehydration or hypernatremia.

Adjacent

Next to, close to, or sharing a boundary with another structure, tissue, organ, or lesion.

Adjustment Disorder

Emotional or behavioral symptoms in response to an identifiable stressor occurring within three months of the onset of the stressor. The symptoms are disproportionate to the severity of the stressor and cause significant distress or impairment. It resolves within six months of the termination of the stressor.

Adjustment Disorder With Anxiety

An emotional or behavioral response to an identifiable stressor that causes distress or impairment but does not meet criteria for another mental disorder.

Adjustment Disorders

Emotional or behavioral symptoms occurring in response to an identifiable stressor and causing significant distress or impairment.

Alternate Names:

Adjustment disorder

Adjuvant

An added treatment or substance that enhances the effect of a primary treatment; in vaccines, an

adjuvant strengthens the immune response to an antigen.

Adjuvant Therapy

Additional treatment after primary treatment to reduce recurrence risk. Examples: chemotherapy after surgery.

ADL Training

Training for activities of daily living: dressing, bathing, grooming, feeding, toileting. Adaptive techniques, equipment.

ADLs (Activities Of Daily Living)

Basic self-care activities such as bathing, dressing, toileting, transferring, eating, and maintaining continence; ADL ability is used to assess functional independence.

ADME

Absorption, distribution, metabolism, and excretion: the processes governing a substance's movement through the body.

Alternate Names:

Pharmacokinetics

Administered Activity

The measured amount of radioactive material given to a patient for imaging or therapy, usually expressed in becquerels. It should be prescribed and recorded with the radionuclide, radiopharmaceutical, route, date, and time of administration.

Administration Procedure

A procedure used to deliver a medicine, fluid, vaccine, or other intervention to a patient.

Admission

The process of accepting a patient into a hospital or other facility for evaluation, treatment, monitoring, or care.

Adnexa

Structures next to an organ. In gynecology, the uterine adnexa are the ovaries, fallopian tubes, and

supporting ligaments and tissues around the uterus.

Adnexal Dysplasia

Abnormal precancerous or developmental cellular changes in tissue beside the uterus, such as the ovary or tube.

Adnexal Mass

A growth or enlargement involving structures beside the uterus, such as an ovary or fallopian tube.

Adnexal Tumors And Masses

Abnormal growths or masses involving the ovary, fallopian tube, or nearby supporting tissues; they may be benign or malignant.

Adnexitis

Inflammation of an adnexal structure, usually the fallopian tube or ovary.

Alternate Names:

Salpingo-oophoritis

Adolescence

The developmental period between childhood and adulthood, involving physical maturation, puberty, and major psychological and social changes.

Adolescent Confidentiality And Consent

The principles governing private healthcare discussions with adolescents and their ability to participate in medical decisions. The exact rules for confidentiality, consent, and parental involvement depend on age, capacity, local law, and safeguarding concerns.

Adolescent Development

The physical, cognitive, emotional, and social changes that occur as a person develops from childhood into adulthood, including puberty and the development of greater independence.

Adolescent Health

The health and wellbeing of young people (10-19 years), addressing the physical, psychological, and social transitions of puberty and adolescence.

Adolescent Medicine

A medical specialty focused on the physical, mental, reproductive, and social health needs of adolescents and young adults, including prevention, risk behavior, chronic disease, and confidentiality-sensitive care.

Adolescent Obesity

Alternate index label; see *Pediatric Obesity*.

Adolescent Schizophrenia

See *Childhood schizophrenia*.

Adolescent Test Or Procedure Preparation

Age-appropriate preparation of an adolescent for a medical test or procedure, including explanation of the procedure, expectations, sensation, and coping strategies. It aims to reduce anxiety, improve cooperation, and support informed assent.

Adoption

In healthcare and genetics, adoption may be relevant to family history, consent, psychosocial care, and assessment when biological family information is unavailable or differs from the caregiving family.

Adoptive Cell Therapy

A personalized cellular immunotherapy in which a patient's own lymphocytes are expanded or engineered ex vivo and reinfused to attack the tumor; examples include tumor-infiltrating lymphocyte (TIL) therapy and chimeric antigen receptor (CAR) T-cell therapy.

Adoptive Immunotherapy

Treatment in which immune cells are collected, modified or activated, and returned to the patient.

ADP-Ribosylation

Addition of an ADP-ribose group to a protein or other molecule, altering its activity or interactions.

Adrenal Cancer

A malignant neoplasm of the adrenal gland, most

importantly adrenocortical carcinoma, a rare, aggressive malignancy of the adrenal cortex with a bimodal age distribution (early childhood and 40–50 years). Approximately 50–60% of tumors are functional, presenting with acute endocrine excess syndromes: hypercortisolemia (rapid-onset Cushing syndrome), virilization and hirsutism from excess adrenal androgens, or rarely primary aldosteronism. Non-functioning tumors present late with abdominal or flank pain and a palpable mass. Suspicion on CT or MRI is indicated by diameter > 4 cm, high unenhanced attenuation (> 10 Hounsfield units), heterogeneous necrosis, and delayed contrast washout (< 50%). Complete surgical resection (adrenalectomy with regional lymphadenectomy) offers the only potential cure; adjuvant therapy incorporates the adrenolytic agent mitotane with or without combination chemotherapy.

Alternate Names:

Adrenocortical cancer

Adrenal Cortex

The outer part of an adrenal gland that produces steroid hormones, including cortisol and aldosterone.

Adrenal Cortex Physiology

The adrenal cortex produces steroid hormones in three zones: the zona glomerulosa produces mineralocorticoids, mainly aldosterone; the zona fasciculata produces glucocorticoids, mainly cortisol; and the zona reticularis produces adrenal androgens. Secretion is regulated by the renin-angiotensin system, plasma potassium, and the hypothalamic-pituitary-adrenal axis.

Adrenal Cortical Carcinoma

A rare malignant tumor arising from the adrenal cortex. It may be functional, producing excess adrenal hormones, or nonfunctional.

Adrenal Crisis

An acute, life-threatening manifestation of severe adrenal insufficiency. Causes: primary (Addison disease), secondary (pituitary failure), or tertiary (exogenous corticosteroid withdrawal). Precipi-

tants: systemic infection, trauma, surgery, or abrupt steroid cessation.

Adrenal Failure

Inadequate production of adrenal hormones, especially cortisol and sometimes aldosterone; severe acute deficiency can cause adrenal crisis with hypotension and shock.

Adrenal Gland

Paired endocrine glands atop each kidney (suprarenal), composed of cortex and medulla. Cortex (three zones): zona glomerulosa (aldosterone, mineralocorticoids), zona fasciculata (cortisol, glucocorticoids), zona reticularis (androgens). Medulla: chromaffin cells secrete epinephrine and norepinephrine (catecholamines).

Adrenal Gland Removal

Surgical excision of an adrenal gland, performed for tumors, hyperplasia, or other adrenal disorders. Approaches include open, laparoscopic, or retroperitoneal; perioperative management requires assessment of hormone production and potential adrenal insufficiency.

Adrenal Glands

Adrenal glands sit above the kidneys and produce cortisol, aldosterone, adrenal androgens, and catecholamines that regulate stress, blood pressure, salt balance, and metabolism.

Adrenal Hemorrhage

Bleeding into the adrenal gland most commonly occurring in critically ill patients with sepsis, coagulopathy, or anticoagulation therapy. Bilateral adrenal hemorrhage may cause acute adrenal insufficiency with circulatory collapse. Waterhouse-Friderichsen syndrome: adrenal hemorrhage associated with meningococcal sepsis.

Adrenal Hyperplasia

Enlargement or overgrowth of adrenal tissue that may increase production of adrenal hormones. Causes include congenital adrenal hyperplasia, chronic stimulation by ACTH, and some adrenal

tumors. Symptoms depend on the hormone affected and may include abnormal genital development, early puberty, excess body hair, high blood pressure, or electrolyte disturbances.

Adrenal Insufficiency

Deficient production of cortisol by the adrenal cortex classified as primary adrenal insufficiency or Addison disease from adrenal gland destruction, secondary from pituitary ACTH deficiency, and tertiary from chronic exogenous glucocorticoid suppression of the hypothalamic-pituitary-adrenal axis.

Adrenal Mass

See *Benign adrenal tumors*.

Adrenal Medulla

The inner part of the adrenal gland that produces epinephrine and norepinephrine.

Adrenal Medulla Physiology

The adrenal medulla produces catecholamines primarily epinephrine and norepinephrine in response to sympathetic nervous system activation. Epinephrine accounts for approximately eighty percent and norepinephrine approximately twenty percent of catecholamine output.

Adrenal Myelolipoma

A usually benign adrenal tumor composed of mature fat and blood-forming tissue. It is often found incidentally on imaging and causes no symptoms, but larger lesions may produce mass effect or bleeding. Management depends on size, symptoms, growth, and diagnostic certainty.

Adrenal Tumors

Tumors arising in the adrenal cortex or medulla may be benign or malignant and may or may not produce hormones. Evaluation uses clinical assessment, hormone testing when indicated, and imaging to assess size and appearance; treatment depends on the diagnosis, hormone secretion, symptoms, growth, and risk of malignancy.

Adrenal Venous Sampling

Interventional radiology procedure for lateralizing aldosterone secretion in primary hyperaldosteronism. Catheterization of both adrenal veins allows measurement of aldosterone and cortisol levels to calculate selectivity index confirming successful cannulation and lateralization ratio.

Adrenalectomy

Surgical removal of one or both adrenal glands, performed for selected adrenal tumors, severe hormone excess, or other localized disease. Removal of one gland may leave adequate hormone production; bilateral removal causes permanent adrenal insufficiency and requires lifelong hormone replacement and emergency-steroid planning.

Adrenaline

Epinephrine, a hormone and neurotransmitter released during stress that increases heart rate and contractility, constricts some blood vessels, and relaxes bronchial muscle.

Adrenergic Agent

A substance that affects receptors responsive to adrenaline-like neurotransmitters.

Adrenergic Agonist

A drug or other substance that binds to and activates alpha- or beta-adrenergic receptors, producing effects that depend on the receptor subtype and tissue. Examples include epinephrine, norepinephrine, phenylephrine, and dobutamine.

Alternate Names:

Sympathomimetic

Adrenergic Antagonists

Substances that block the effects of adrenaline-like signaling at adrenergic receptors.

Alternate Names:

Adrenergic blockers

Adrenergic Bronchodilator Overdose

Toxic effects caused by excessive exposure to Adrenergic bronchodilator. Severity depends on the dose,

route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Adrenergic Receptor

A cell-surface receptor activated by adrenaline (epinephrine), noradrenaline (norepinephrine), or related medicines. Alpha and beta receptors help control heart rate, blood pressure, airway size, pupil size, and other body responses.

Adrenergic Receptors

Cell-surface receptors activated by adrenaline and noradrenaline. Alpha and beta subtypes regulate cardiovascular, respiratory, metabolic, and other physiological responses.

Adrenocortical Carcinoma

A rare malignant tumor of the adrenal cortex that may secrete cortisol, androgens, or aldosterone or be non-functional. Often large at diagnosis; treatment is en bloc resection, with mitotane for advanced disease.

Adrenocorticotrophic Hormone

A hormone made by the pituitary gland that tells the adrenal glands to make cortisol. Cortisol helps the body respond to stress and helps control blood pressure, blood sugar, and inflammation.

Alternate Names:

Adrenocorticotrophic Hormone (ACTH); ACTH; Corticotropin

Adrenogenital Syndrome

A group of disorders, usually caused by congenital adrenal hyperplasia, in which excess adrenal androgens cause atypical genital development or early puberty.

Adrenoleukodystrophy

An inherited disorder causing abnormal buildup of very-long-chain fatty acids and progressive nervous-system or adrenal damage.

Alternate Names:

ALD

Adrenomedullin

A peptide hormone that promotes blood-vessel dilation and helps regulate circulation and fluid balance.

Adsorption

Attachment of molecules to the surface of a solid or liquid rather than absorption into its interior.

ADU

Accessory dwelling unit, a secondary independent residential unit located on the same property as a primary home.

Adult Day Services

Community-based daytime programs providing supervision, social activities, health-related support, or assistance with daily activities for adults who cannot safely remain alone throughout the day.

Alternate Names:

Adult day care; adult day program.

Adult Polycystic Kidney Disease

An inherited disorder, usually autosomal dominant polycystic kidney disease, in which multiple kidney cysts enlarge over time and may cause hypertension, pain, hematuria, and chronic kidney failure.

Adult Primary Liver Cancer

Cancer that begins in the liver of an adult, most commonly hepatocellular carcinoma or intrahepatic cholangiocarcinoma, rather than cancer that has spread to the liver from another organ.

Adult Rickets

Poor mineralization of adult bone, more accurately called osteomalacia. It may cause bone pain, muscle weakness, fractures, or difficulty walking and can result from vitamin-D deficiency, poor absorption, kidney disease, or phosphate loss.

Alternate Names:

Osteomalacia

Adult Soft Tissue Sarcoma

A malignant tumor arising from connective or supporting tissues, such as muscle, fat, fibrous tissue,

blood vessels, or peripheral nerves. It often presents as a painless enlarging mass and requires imaging and biopsy for diagnosis.

Adult Spinal Deformity

Abnormal curvature or alignment of the spine in an adult, including scoliosis, kyphosis, or loss of normal sagittal balance. It may cause back pain, nerve compression, difficulty walking, or functional disability.

Adult Still Disease

Alternate name; see *Adult-Onset Still Disease*.

Adult Still'S Disease

Alternate spelling; see *Adult-Onset Still Disease*.

Adult-Onset Asthma

Adult-onset asthma is variable airway inflammation and narrowing that begins in adulthood, producing episodic wheezing, cough, chest tightness, or shortness of breath. Diagnosis uses symptoms and lung-function testing, and treatment reduces triggers and airway inflammation.

Alternate Names:

Asthma Adult-Onset (Adult-Onset Asthma)

Adult-Onset Still Disease

A rare inflammatory disease that starts in adulthood. It can cause high fevers, joint pain or swelling, a salmon-colored rash, sore throat, and inflammation in organs such as the liver, heart, or lungs.

Adults Autism (Autism And Communication)

Alternate index label for Autism and Communication.

Advance Directive

Legal document expressing healthcare wishes. Components: living will (treatment preferences), durable power of attorney for healthcare (surrogate decision-maker). Should be reviewed periodically, discussed with surrogate and healthcare team.

Advance Directives

Written instructions about a person's future medical-care preferences and/or designation of a health-care proxy if the person later cannot make or communicate decisions.

Advance Directives (Advance Medical Directives)

Alternate index label for Advance Medical Directives.

Advance Medical Directives

Advance medical directives record a person's preferences for future medical care and identify a health-care proxy when the person cannot decide. They may address resuscitation, life support, artificial nutrition, and other treatments, and should be reviewed as circumstances change.

Alternate Names:

Advance Directives (Advance Medical Directives)

Advance Statements About Medical Treatment

Advance statements about medical treatment record a person's preferences, values, and wishes for future care if illness later prevents effective communication; legal status varies by jurisdiction.

Advanced Cancer

Cancer that has grown beyond its original site. It may be locally advanced when it has invaded nearby tissues or metastatic when it has spread to distant organs.

Advanced Cardiac Life Support

A structured set of interventions for cardiac arrest and other life-threatening cardiovascular emergencies, including high-quality CPR, rhythm recognition, defibrillation, airway management, and medications.

Advanced Cardiovascular Life Support

A standardized set of advanced resuscitation measures for cardiac arrest and other life-threatening cardiovascular emergencies. It includes high-quality

CPR, defibrillation when indicated, airway management, medicines, and treatment of reversible causes.

Advanced Maternal Age

Maternal age of 35 years or older at the estimated time of delivery. It is associated with increased risks of some chromosomal conditions and pregnancy complications.

Advanced Sleep-Wake Phase Disorder

A circadian rhythm sleep-wake disorder in which a person becomes sleepy and falls asleep earlier than desired and wakes earlier than desired; symptoms persist despite an opportunity for a preferred schedule.

Adventitia

The outer connective-tissue layer of a blood vessel or hollow organ.

Alternate Names:

Tunica adventitia

Adventitious

Abnormal, unexpected, or arising from an unusual location or process; adventitious lung sounds include crackles, wheezes, and other added breath sounds.

Adverse Effect

An unintended and harmful or bothersome effect of a medicine or other treatment given at normal doses or for an intended purpose.

Adverse Event

Any untoward medical occurrence during or after a treatment or intervention, whether or not it is considered causally related to that treatment. Severity, seriousness, expectedness, and relatedness are assessed separately.

Adverse Reaction

An unintended, harmful response to a medicine, treatment, or other exposure.

Alternate Names:

Adverse drug reaction

Adverse Reactions To Drugs

Adverse drug reactions are harmful, unintended responses to medicines taken at normal doses, ranging from predictable side effects to allergic, organ-toxic, or idiosyncratic reactions.

Aedes Aegypti

Aedes aegypti is a mosquito that can transmit dengue, yellow fever, chikungunya, Zika, and other viruses.

Aegophony

Aegophony is an altered “E” sound heard through a stethoscope as an “A,” classically suggesting lung consolidation or fluid-related change when present over the chest.

Aerobe

A microorganism that requires molecular oxygen for growth or uses oxygen efficiently for energy production. Obligate aerobes are unable to grow without oxygen, whereas facultative organisms can grow with or without it.

Aerobic

Requiring or involving molecular oxygen. In microbiology, aerobic organisms use oxygen for growth or energy production, either obligatorily or as one option among several; in physiology, aerobic metabolism uses oxidative phosphorylation in mitochondria to generate ATP from nutrients.

Aerobic Bacteria

Bacteria that grow using oxygen or tolerate oxygen, although individual species differ in their exact oxygen requirements.

Aerobic Bacterium

An aerobic bacterium grows using oxygen, although some species can tolerate or use alternative metabolic pathways; oxygen requirements help guide culture and infection assessment.

Aerobic Exercise

Physical activity that uses sustained oxygen-

dependent energy production, such as walking or cycling.

Alternate Names:

Cardio exercise

Aerobiosis

A state or process requiring the presence of oxygen.

Aerophagia

Excessive swallowing of air, which may cause belching, abdominal bloating, distention, or increased intestinal gas.

Aerophagy

Excessive swallowing of air, also called aerophagia, which can lead to frequent belching, abdominal bloating, distension, and discomfort.

Alternate Names:

aerophagia, which can lead to frequent.

Aerophobia

An excessive fear of flying or air travel that causes significant anxiety, avoidance, or impairment.

Aerospace Medicine

The field of medicine concerned with maintaining the health, safety, and performance of people who travel or work in aircraft and spacecraft. It addresses effects such as changes in pressure, acceleration, low oxygen, radiation, isolation, and microgravity.

Aerosol

A suspension of fine solid particles or liquid droplets in a gas.

Aerotitis

Ear pain or middle-ear injury caused by failure to equalize pressure during rapid changes in altitude, such as during aircraft ascent or descent.

Aesculapius

The Roman name for Asclepius, the Greek god of healing; the name is associated with medicine and the rod-and-serpent medical symbol.

Aesthetic Dentistry

Dental care intended to improve the appearance of the teeth, gums, or smile. Procedures may include tooth-colored restorations, veneers, whitening, or selected crowns.

Aesthetic Medicine

Medical care intended to improve appearance or reduce visible signs of aging through procedures, devices, medicines, or skin care; appropriate assessment includes risks, contraindications, and realistic expectations.

Aesthetician

A trained skin-care professional who provides non-medical cosmetic skin treatments, such as cleansing, superficial exfoliation, and selected skin-care procedures within the scope of local regulation.

Alternate Names:

Esthetician; skin-care specialist.

Aetiology

The cause or causes of a disease, injury, symptom, or other medical condition; also spelled etiology.

Aetiology (Etiology)

The cause or group of causes underlying a disease, injury, or other medical condition.

Alternate Names:

Etiology

Affect

The observable expression of a person's emotional state, including facial expression, voice, posture, and emotional responsiveness.

Affective Flattening

Markedly reduced outward expression of emotion, including limited facial expression, voice modulation, and gestures. It may occur in schizophrenia and other psychiatric or neurologic conditions.

Afferent

Carrying signals toward a central organ or structure; afferent nerves transmit sensory information from

the body toward the brain and spinal cord.

Afferent Loop Syndrome

Obstruction or impaired emptying of the afferent intestinal limb created by some gastric operations, causing bile and pancreatic-secretions to accumulate with pain, nausea, vomiting, or malabsorption.

Afferent Neuron

A sensory nerve cell that carries impulses from receptors toward the central nervous system.

Afferent Pathway

A sensory nerve route carrying information from the body toward the brain or spinal cord.

Alternate Names:

Sensory pathway

Affinity

The strength of attraction or binding between two molecules, such as an antibody and antigen, receptor and ligand, or enzyme and substrate.

Affinity Chromatography

A laboratory separation method that isolates a molecule through its specific binding to an immobilized partner.

Afib (Atrial Fibrillation (AFib) Warning Symptoms)

Alternate index label for Atrial Fibrillation (AFib) Warning Symptoms.

Afib (Atrial Fibrillation)

Alternate index label for Atrial Fibrillation.

Afibrinogenaemia

Afibrinogenaemia is severe or complete deficiency of fibrinogen, an essential clotting protein, causing a bleeding tendency; it is usually inherited but may be acquired.

Afibrinogenemia

A rare inherited or acquired absence of fibrinogen, causing impaired blood clotting.

Aflatoxin

Aflatoxin is a toxin produced by certain *Aspergillus* molds in contaminated foods; chronic exposure can injure the liver and increase hepatocellular-carcinoma risk.

African Trypanosomiasis

A parasitic infection transmitted by tsetse flies and affecting the blood and nervous system.

Alternate Names:

African sleeping sickness

After-Cataract

Clouding of the posterior lens capsule that develops after cataract surgery and may cause blurred or hazy vision. It is also called posterior capsule opacification.

Alternate Names:

Posterior capsule opacification; secondary cataract.

Afterbirth

The placenta and fetal membranes expelled from the uterus after delivery of the baby.

Afterload

The resistance against which the ventricles must eject blood during systole. Left ventricular afterload is primarily determined by systemic vascular resistance (SVR); right ventricular afterload is determined by pulmonary vascular resistance (PVR). Afterload is estimated by blood pressure or calculated as wall stress.

Afterpains

Intermittent uterine cramps occurring after childbirth as the uterus contracts and returns toward its pre-pregnancy size. They may be more noticeable during breastfeeding.

Aftershave Poisoning

Toxic effects caused by excessive exposure to aftershave products. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Agammaglobulinaemia

Agammaglobulinaemia is profound deficiency of immunoglobulins and mature B cells, leading to recurrent bacterial infections; X-linked disease is a classic form.

Agammaglobulinemia

A severe deficiency or absence of immunoglobulins, causing recurrent infections.

Agar

A gelatinous polysaccharide from red algae used to solidify culture media. Melting near 85°C and solidifying at 35–40°C, it remains solid at incubation temperatures and is largely undegraded by bacteria, enabling isolation of pure colonies on media such as blood, MacConkey, and Sabouraud agars.

Agatston Score

A standardized semi-automated radiologic scoring system used to quantify the burden of coronary artery calcification on non-contrast, electrocardiographically gated cardiac computed tomography. Calculated by multiplying the area of each calcified lesion (defined by a CT attenuation threshold ≥ 130 Hounsfield units) by a peak density weighting factor (1 to 4). Total scores are stratified into established prognostic risk tiers (0: no identifiable plaque, extremely low 10-year cardiovascular risk; 1–99: mild plaque burden; 100–399: moderate plaque; ≥ 400 : severe calcification and high risk of major adverse cardiovascular events), guiding shared decision-making regarding statin initiation and preventive cardiovascular therapy.

Alternate Names:

Coronary Artery Calcium (CAC) Score; Agatston Calcium Score

Age By Decade

Classification of a person's age into ten-year groups, such as the twenties, thirties, or seventies, for demographic description or analysis.

Age Spots (Liver Spots)

Flat brown, gray, or black skin spots caused by long-term ultraviolet exposure, usually appearing

on sun-exposed areas.

Alternate Names:

Lentigines, Solar Solar Lentigines

Age-Related Cognitive Decline

Small changes in memory, thinking speed, or other mental abilities that can happen with aging. These changes do not usually interfere greatly with daily independence.

Alternate Names:

Age-associated cognitive decline; normal cognitive aging.

Age-Related Hearing Loss

Gradual hearing loss in both ears caused by aging. It often affects high-pitched sounds first and can make speech harder to understand.

Age-Related Macular Degeneration

A disease that damages the macula, the part of the retina needed for clear central vision. It can cause blurred or missing central vision. Dry AMD usually worsens slowly; wet AMD can cause faster vision loss because abnormal blood vessels leak under the retina. Smoking and family history increase the risk.

Age-Related Macular Degeneration, Dry

See *Dry macular degeneration*.

Age-Related Macular Degeneration, Wet

See *Wet macular degeneration*.

Age-Standardized Rate

A rate adjusted to a specified standard population age distribution so that populations with different age structures can be compared more fairly. It is commonly used for mortality, cancer, and disease-incidence comparisons.

Ageing

Ageing is the progressive biological and functional change over time, including altered repair, immunity, metabolism, organ reserve, and susceptibility to disease.

Ageism

Stereotyping, prejudice, or discrimination directed at people because of their age, which can impair access to appropriate medical care.

Agenesis

Complete failure of an organ or body structure to develop, usually because of a congenital developmental abnormality.

Agenesis Of The Corpus Callosum

Complete or partial absence of the corpus callosum, the major commissural fiber bundle connecting the cerebral hemispheres. May be isolated or associated with other CNS anomalies (Dandy-Walker, Chiari, heterotopias) and genetic syndromes (Aicardi, Andermann, acrocallosal syndromes).

Ageusia

Complete loss of the sense of taste. Causes include viral infections, oral or nasal disease, head injury, damage to the facial or glossopharyngeal nerves, medications, chemotherapy, zinc deficiency, and neurologic disorders.

Agglutination

The visible clumping of cells or particles caused by antibodies or other binding molecules linking antigens on their surfaces. It is used in blood-grouping and crossmatching, infectious-disease serology, and some laboratory identification tests.

Agglutination Test

A laboratory test in which particles visibly clump when a specific antibody or antigen is present.

Agglutinin

An antibody or other substance that causes cells or particles to clump together.

Agglutininogen

An agglutininogen is an antigen, often on a red blood cell, that can trigger production of a corresponding agglutinin and cause visible cell clumping in a compatible reaction.

Aggregation, Platelet

The process by which activated platelets adhere to one another to form a primary platelet plug at a site of vascular injury. Platelet agonists such as ADP, thromboxane A2, and thrombin activate glycoprotein IIb/IIIa, which binds fibrinogen and links adjacent platelets.

Aggression

Verbal or physical behavior intended to threaten, harm, or dominate another person or object. It may occur with mania, psychosis, personality disorders, dementia, delirium, substance intoxication or withdrawal, traumatic brain injury, or severe frustration.

Aggressive

Characterized by forceful or hostile behavior, or by rapid growth and invasive or destructive clinical behavior, depending on context.

Aggressive Angiomyxoma

A locally infiltrative, usually slow-growing mesenchymal tumor of the pelvis and perineum, predominantly affecting reproductive-age women. It may be mistaken for a Bartholin cyst and has a tendency to recur.

Aggressive Cancer

A cancer that grows, invades, or spreads rapidly and may require prompt diagnosis and treatment. Aggressiveness depends on tumour biology, grade, stage, and response to therapy.

Agility

The ability to move or change direction quickly while maintaining control and balance.

Aging

Biological process of growing older, involving cumulative changes at molecular, cellular, tissue, organ levels.

Aging Spots - Should You Be Concerned?

A topic about age-related lentigines, which are usu-

ally flat brown skin patches caused by cumulative sun exposure. A changing or bleeding lesion requires assessment.

Agita

An informal term for heartburn, indigestion, or an uneasy feeling of stomach discomfort; its meaning can vary by region and context.

Agitation

A state of excessive motor activity, emotional excitement, and restlessness. In psychiatric settings, it may be a symptom of mania, psychosis, delirium, substance intoxication, or withdrawal.

Agnathia

A rare congenital absence or severe underdevelopment of the mandible, maxilla, or both. It is usually part of a craniofacial developmental disorder and may cause profound facial deformity, airway obstruction, feeding difficulty, and impaired dentition.

Agnogenic Myeloid Metaplasia

See *Myelofibrosis*.

Agnosia

A higher-order neurological deficit characterized by the inability to recognize or process sensory stimuli (visual, auditory, tactile) despite intact primary sensory organs and pathways. Caused by focal lesions in specific cortical association areas (parietal, temporal, or occipital lobes).

Agonist

A substance that binds to a receptor and activates it, producing a biological response. A full agonist can produce the receptor's maximum response, whereas a partial agonist produces a lower maximum response even when it occupies all available receptors.

Agoraphobia

Anxiety disorder characterized by fear and avoidance of situations where escape might be difficult or help unavailable during a panic-like episode. Common feared situations include public transportation,

open spaces, enclosed spaces, standing in line, being in crowds, and being outside the home alone.

Agranulocytes

White blood cells without prominent cytoplasmic granules; the main types are lymphocytes and monocytes.

Agranulocytosis

A severe reduction or near-absence of circulating neutrophils, usually defined as an absolute neutrophil count below $0.5 \times 10^9/\text{L}$. It may be caused by medications, chemotherapy, autoimmune disease, infections, or bone-marrow disorders and presents with fever, sore throat, oral ulcers, or recurrent serious infections.

Agraphesthesia

Inability to recognize numbers or letters traced on the skin, indicating a lesion of the contralateral parietal lobe (somatosensory association cortex, post-central gyrus). Tested with a blunt object on the palm or fingertip while the patient's eyes are closed.

Agraphia

An acquired inability to write or express language in written form despite adequate education and motor ability. It may result from dominant-parietal or language-network lesions, especially stroke, traumatic brain injury, dementia, or brain tumors, and may occur with aphasia, alexia, or acalculia.

Ague

A fever, often associated with malaria, marked by recurrent episodes of chills, fever, and sweating; the term can also mean a fit of shivering.

Atherosclerotic Heart Disease

A condition in which fatty deposits called plaques narrow or block the arteries that supply blood to the heart. It is a form of coronary artery disease and may cause chest pain, heart attack, or other heart problems.

AI

In medicine, AI means artificial intelligence: computer methods that analyze data or recognize patterns to support research, diagnosis, prediction, treatment, or clinical workflow; they require validation and oversight.

Aicardi Syndrome

Aicardi Syndrome is a rare developmental brain disorder classically involving agenesis of the corpus callosum, chorioretinal lacunae, and infantile seizures.

Aicardi-Goutieres Syndrome

Rare autosomal recessive inflammatory disorder mimicking congenital viral infection. Characterized by cerebral calcifications, white matter disease, cerebral atrophy, and cerebrospinal fluid lymphocytosis. Presents with severe neurodevelopmental impairment, seizures, and spasticity.

AIDS

Acquired immunodeficiency syndrome: the advanced stage of HIV infection, defined by CD4 count <200 cells/ μL or the presence of AIDS-defining illnesses.

Alternate Names:

Acquired Immunodeficiency Syndrome (AIDS)

AIDS Dementia

A disorder of thinking, memory, behavior, or movement caused by advanced HIV-related disease affecting the brain.

AIDS-Associated Nephropathy

Kidney disease linked to untreated or advanced HIV infection. It can cause large amounts of protein in the urine, swelling, and falling kidney function; antiretroviral treatment and kidney assessment are important.

Alternate Names:

HIV-associated nephropathy

AIDS/HIV

See *HIV/AIDS*.

Ainhum

A rare disorder causing a constricting groove around a toe that may eventually lead to autoamputation.

Alternate Names:

Dactylolysis spontanea

Air Abrasion

Minimally invasive caries removal using abrasive particles. Alternative to drilling for small lesions.

Air Bag

An inflatable vehicle safety device that rapidly deploys during a collision to reduce impact with the steering wheel, dashboard, or window; it supplements seat belts.

Air Bronchogram

A radiographic sign showing air-filled bronchi visible against opacified lung parenchyma, indicating that the airways remain patent while the surrounding alveoli are filled with fluid, pus, or cells.

Air Contaminants

Solid, liquid, or gaseous substances in air that may harm health, such as particulate matter, fumes, gases, allergens, or microorganisms. Effects depend on the contaminant, concentration, and duration of exposure.

Air Embolism

Entry of gas into the venous or arterial circulation, where it can obstruct blood flow and injure tissues. It may occur during medical procedures, trauma, or rapid changes in environmental pressure. Symptoms and treatment depend on the amount and location of gas and require urgent clinical assessment when significant embolism is suspected.

Air Passages

The air passages are the conducting airways from the nose and mouth through the pharynx, larynx, trachea, bronchi, and bronchioles; obstruction can impair ventilation.

Air Polishing

Removal of stains using air-abrasive device. Glycine or erythritol powder.

Air Quality

The condition of air as determined by the concentration of pollutants such as particulate matter, ozone, nitrogen dioxide, and sulfur dioxide.

Air Sac

A cavity or thin-walled space that contains air, such as a pulmonary alveolus or bird air sac.

Alternate Names:

Air space

Air Trapping

Retention of air in the lungs because it cannot be fully exhaled, often from airway obstruction.

Air-Conditioner Lung

Hypersensitivity pneumonitis caused by inhaled mold, fungus, or other contaminants from air-conditioning or humidifying systems.

Air-Sickness

Air-sickness is motion sickness triggered by aircraft movement or conflicting visual and vestibular signals, causing nausea, dizziness, sweating, and sometimes vomiting.

Airborne Disease

An infection that can spread through small suspended particles or aerosols under suitable conditions. Transmission depends on the organism, environment, ventilation, and duration of exposure.

Airborne Precautions

Infection-control measures for infections that can remain suspended in the air and spread over a distance. They include an airborne infection isolation room with negative pressure, limiting patient movement, and appropriate respiratory protection, often a fit-tested N95 or higher-level respirator. They are used for tuberculosis, measles, chickenpox, and disseminated shingles.

Airplane Ear

Ear pain, pressure, or temporary hearing loss caused by unequal air pressure across the eardrum during flight.

Alternate Names:

Barotitis Media

Airway

A passage through which air moves between the outside environment and the lungs, including the nose, mouth, pharynx, larynx, trachea, and bronchi.

Airway Clearance Techniques

Physical or mechanical methods used to mobilize and remove airway secretions. Examples include directed coughing, breathing exercises, postural drainage, positive expiratory pressure, and high-frequency chest-wall oscillation. Selection depends on the disease, secretion burden, strength, and ability to cooperate.

Airway Management

Airway management encompasses techniques and devices used to maintain ventilation and oxygenation in patients with compromised airways.

Airway Obstruction

Partial or complete blockage of airflow through the upper or lower respiratory tract. It may result from a foreign body, secretions, swelling, trauma, tumor, or loss of airway muscle tone and can cause stridor, wheeze, hypoxaemia, respiratory distress, or arrest.

Airway Resistance

The opposition to airflow through the conducting airways, determined largely by airway radius and flow pattern. Bronchoconstriction, mucus, edema, and reduced elastic traction increase resistance; because resistance varies strongly with radius, small changes can markedly affect work of breathing.

AK

Actinic keratosis, a rough, scaly precancerous skin lesion caused by chronic ultraviolet exposure.

Akathisia

Subjective feeling of inner restlessness accompanied by a compulsive need to move, most commonly an extrapyramidal side effect of antipsychotic drugs (typical and atypical neuroleptics), also seen with dopamine antagonists (metoclopramide), selective serotonin reuptake inhibitor withdrawal, and in Parkinson disease.

Akinesia

Impairment or absence of voluntary movement, classically a hypokinetic feature of basal ganglia disorders such as Parkinson disease and related parkinsonian syndromes.

Akinetic

Characterized by little or no voluntary movement; akinesia may result from neurologic disease, medication effects, or severe motor impairment.

Akinetic Mutism

A state of profound lack of movement and speech despite wakefulness and preserved basic alertness.

Alagille Syndrome

An inherited disorder of Notch signaling that commonly affects the liver, heart, skeleton, eyes, and characteristic facial features. It is usually caused by a pathogenic variant in JAG1 or, less often, NOTCH2.

Alanine

An amino acid used in protein synthesis and glucose metabolism.

Alanine Aminotransferase

An enzyme found mainly in liver cells. When these cells are damaged, ALT enters the blood; high levels may indicate liver injury or disease. It is usually interpreted with AST and other liver tests.

Alternate Names:

ALT

Alanine Transaminase Blood Test

A laboratory test that measures serum ALT activity,

a hepatocellular enzyme released during liver injury. Elevated ALT is more specific for hepatocellular damage than AST and is used to evaluate hepatitis, drug toxicity, and other liver diseases.

ALARA (As Low As Is Reasonably Achievable) Principle

A radiation-safety principle that aims to keep exposure to ionizing radiation as low as practical while still achieving the needed diagnostic or treatment benefit. It uses measures such as justification, optimization, shielding, shorter exposure time, and greater distance.

Alarmin

A molecule released by damaged or stressed cells that activates innate immune responses.

Alastrim

Alastrim is a historical name for a generally milder form of smallpox, caused by variola minor; smallpox has been eradicated worldwide.

Albendazole

Albendazole is a benzimidazole antiparasitic drug that disrupts parasite microtubules and is used for several intestinal and tissue helminth infections.

Albers-Schönberg Disease

An autosomal dominant form of osteopetrosis (specifically Autosomal Dominant Osteopetrosis Type II [ADO II]) caused by heterozygous loss-of-function mutations in the *CLCN7* gene encoding the chloride channel 7 required for osteoclast lysosomal acidification and bone resorption. Characterized by generalized osteosclerosis, cortical bone thickening, and excessive bone brittleness leading to multiple pathologic fractures, cranial nerve entrapment (causing facial palsy and optic atrophy), and pathognomonic radiological findings including “sandwich vertebrae” (sclerosis of vertebral endplates) and “bone-in-bone” appearance of the pelvis and tubular bones.

Alternate Names:

Autosomal Dominant Osteopetrosis Type II (ADO II); Marble Bone Disease

Albinism

An inherited condition involving reduced or absent melanin production, affecting the skin, hair, and eyes.

Alternate Names:

Oculocutaneous Albinism

Albinism (Oculocutaneous Albinism)

A group of inherited disorders that reduce melanin production in the skin, hair, and eyes. It may cause light pigmentation, reduced visual acuity, nystagmus, and increased sensitivity to sunlight; the severity depends on the genetic subtype.

Alternate Names:

Oculocutaneous Albinism

Albinism, Oculocutaneous

An inherited disorder of melanin production affecting the skin, hair, and eyes, often causing light sensitivity, nystagmus, reduced visual acuity, and increased sun sensitivity.

Albino

A person or animal with albinism; person-first language such as “person with albinism” is generally preferred.

Albright’S Hereditary Osteodystrophy

A group of inherited features associated with abnormalities of GNAS signaling, including short metacarpal bones, a round facial appearance, short stature, and subcutaneous ossification. Some associated disorders also cause resistance to parathyroid hormone or other hormones.

Albumen

The white of an egg, composed largely of water and proteins; it is distinct from albumin, the major protein in blood plasma.

Albumin

The main protein in blood plasma, produced by the liver. It helps maintain oncotic pressure and transports various substances, including hormones, fatty acids, and medications.

Albumin Blood (Serum) Test

A laboratory test that measures serum albumin concentration, the principal plasma protein synthesized by the liver. Low levels indicate hepatic insufficiency, nephrotic syndrome, malnutrition, or chronic inflammation; it is also used to assess nutritional status and prognosis.

Albumin Clearance

The removal rate of albumin from blood or another body compartment, used in selected physiologic studies.

Albumin-To-Creatinine Ratio

The urinary albumin concentration standardized to urinary creatinine concentration, usually measured in a spot urine sample. Persistent albuminuria is a marker of kidney damage and is categorized as A1 (<30 mg/g), A2 (30–300 mg/g), or A3 (>300 mg/g) in commonly used CKD classification.

Albumins

Albumins are water-soluble proteins, especially serum albumin made by the liver, that maintain oncotic pressure and transport hormones, fatty acids, drugs, and other substances.

Albuminuria

Abnormally increased excretion of albumin in the urine, indicating disruption of the glomerular filtration barrier or another form of kidney disease. It is commonly detected with a urine albumin-to-creatinine ratio and classified as moderately or severely increased according to the quantity and persistence of excretion.

Alcohol

Ethanol, a psychoactive substance that can impair judgment and coordination and cause intoxication or dependence.

Alcohol Abuse

Legacy, nonpreferred label; see *Alcohol Use Disorder*.

Alcohol Abuse And Alcoholism

Legacy index label; see *Alcohol Use Disorder*.

Alcohol Addiction

Legacy label; see *Alcohol Use Disorder*.

Alcohol And Pregnancy

Alcohol crosses the placenta and can cause fetal alcohol spectrum disorders; no known amount or timing during pregnancy is established as safe.

Alcohol And Teens

Alcohol exposure during adolescence can impair judgment, increase injury and unsafe-sex risk, and affect developing brain systems involved in learning and impulse control. Early screening, supportive communication, treatment of coexisting mental-health problems, and limiting access reduce harm.

Alcohol Dehydrogenase

An enzyme, found mainly in the liver, that catalyzes the first major step in ethanol metabolism by converting alcohol to acetaldehyde.

Alternate Names:

ADH; ethanol dehydrogenase.

Alcohol Dependence

Legacy diagnostic label; see *Alcohol Use Disorder*.

Alcohol Dependence (Alcohol Abuse And Alcoholism)

Legacy index label; see *Alcohol Use Disorder*.

Alcohol Dependency

Legacy label; see *Alcohol Use Disorder*.

Alcohol Intolerance

An unpleasant reaction to alcohol caused by difficulty metabolizing alcohol or one of its components.

Alcohol Intoxication

Acute impairment caused by ethanol, which may produce disinhibition, impaired coordination, slurred speech, vomiting, reduced consciousness, or respiratory depression. Severe intoxication can be

life-threatening and requires assessment of breathing, circulation, glucose, trauma, and other substances.

Alcohol Poisoning

A life-threatening condition caused by drinking enough alcohol to impair breathing, consciousness, and other vital functions.

Alcohol Pregnancy (Fetal Alcohol Syndrome)

Alternate index label for Fetal Alcohol Syndrome.

Alcohol Swab

A single-use pad containing alcohol for cleaning intact skin before selected procedures. It should be used according to the product instructions and allowed to dry.

Alcohol Use

Consumption of alcoholic beverages. Health effects depend on amount, frequency, pattern, and individual risk.

Alcohol Use And Safe Drinking

Guidance on reducing alcohol-related harm, including standard-drink awareness, lower-risk limits, and situations in which alcohol should be avoided.

Alcohol Use Disorder

A problematic pattern of alcohol use leading to significant impairment or distress. Severity is classified as mild (2–3 criteria), moderate (4–5 criteria), or severe (6 or more criteria).

Alcohol Withdrawal

A syndrome that may occur after abrupt cessation or substantial reduction of prolonged heavy alcohol use. Symptoms can include tremor, anxiety, sweating, nausea, hallucinations, seizures, or delirium; onset and severity vary, and severe withdrawal requires urgent medical care.

Alcohol Withdrawal Syndrome

Alternate label; see *Alcohol Withdrawal*.

Alcohol-Associated Hepatitis

Alternate label; see *Alcoholic Hepatitis*.

Alcoholic Cardiomyopathy

Dilated cardiomyopathy caused by chronic excessive alcohol consumption, one of the most common reversible causes of dilated cardiomyopathy. Direct toxic effect of ethanol and its metabolite acetaldehyde on cardiomyocytes. Presentation is identical to other dilated cardiomyopathies.

Alcoholic Hepatitis

An acute inflammation and injury of the liver caused by heavy alcohol use. It commonly causes jaundice, fatigue, fever, nausea, and abdominal pain and can become severe enough to cause liver failure.

Alcoholic Hepatitis (Alcohol-Associated Hepatitis)

Alternate index label; see *Alcoholic Hepatitis*.

Alcoholic Intoxication

Alternate label; see *Alcohol Intoxication*.

Alcoholic Ketoacidosis

A high-anion-gap metabolic acidosis caused by prolonged heavy alcohol use, poor nutritional intake, and vomiting. It typically presents with abdominal pain, nausea, vomiting, dehydration, and tachypnea, often with a normal or low blood glucose concentration.

Alcoholic Liver Disease

Alcoholic liver disease includes steatosis, alcohol-associated hepatitis, and cirrhosis caused by sustained alcohol exposure.

Alcoholic Myopathy

Skeletal muscle damage and dysfunction directly resulting from acute or chronic toxic exposure to ethanol and its metabolite acetaldehyde. Acute alcoholic myopathy presents following an episode of heavy binge drinking with abrupt, painful muscle swelling, tenderness, severe weakness, and acute rhabdomyolysis with massive creatine kinase eleva-

tion and myoglobinuria. Chronic alcoholic myopathy is far more prevalent, manifesting as an insidious, painless, progressive weakness and wasting of proximal limb-girdle muscles (predominantly pelvic girdle) due to impaired muscle protein synthesis, oxidative stress, and selective atrophy of fast-twitch glycolytic type IIb muscle fibers, which reverses with abstinence and nutritional repletion.

Alternate Names:

Alcohol-Induced Myopathy; Ethanol-Induced Myopathy

Alcoholic Neuropathy

A length-dependent peripheral neuropathy associated with chronic excessive alcohol use and nutritional deficiency, especially thiamine deficiency. It causes burning pain, numbness, paraesthesia, and weakness, most prominently in the feet and lower legs.

Alcoholism

Legacy, nonpreferred label; see *Alcohol Use Disorder*.

Alcuronium

Alcuronium is a nondepolarizing neuromuscular-blocking drug that causes skeletal-muscle paralysis during anesthesia; it requires airway and respiratory support.

Aldolase Blood Test

A laboratory test that measures serum aldolase activity, a glycolytic enzyme present in skeletal muscle, liver, and brain. Elevated levels suggest muscle disease such as dermatomyositis or muscular dystrophy, and may complement creatine kinase in evaluating myopathy.

Aldosterone

A mineralocorticoid hormone produced by the zona glomerulosa of the adrenal cortex, the final effector of the renin-angiotensin-aldosterone system (RAAS).

Aldosterone Blood Test

A laboratory test that measures serum aldosterone

concentration, the principal mineralocorticoid hormone produced by the adrenal cortex. It is used to evaluate primary aldosteronism, secondary hypertension, and adrenal insufficiency, often interpreted with plasma renin activity.

Alembic

A distillation apparatus historically used in chemistry and laboratory work to separate or purify substances by differences in volatility.

Alexander Disease

A rare, progressive fatal leukodystrophy inherited in an autosomal dominant manner, caused in almost all cases by de novo heterozygous gain-of-function mutations in the glial fibrillary acidic protein (*GFAP*) gene on chromosome 17q21. Pathologically defined by the widespread accumulation of astrocytic cytoplasmic inclusions known as Rosenthal fibers. Primarily manifests in infancy with megalencephaly (progressive head enlargement), frontal-predominant white matter demyelination on neuroimaging, severe psychomotor retardation, spastic quadriparesis, bulbar signs, and intractable seizures.

Alternate Names:

Fibrinoid Leukodystrophy; GFAP Leukodystrophy

Alexia

An acquired inability to comprehend written language despite intact vision and otherwise preserved language function, usually caused by stroke or other lesions affecting the dominant-hemisphere reading networks. When reading is impaired but writing remains possible, the pattern is called alexia without agraphia.

Alexithymia

Difficulty identifying, describing, or distinguishing one's emotional experiences.

Alfacalcidol

Alfacalcidol is a vitamin-D analogue converted in the liver to an active form that increases calcium and phosphate absorption and is used in selected bone and mineral disorders.

Algia

A word ending meaning pain, as in neuralgia, myalgia, and arthralgia.

Alglucerase

Alglucerase is a modified form of the lysosomal enzyme glucocerebrosidase formerly used as enzyme-replacement therapy for Gaucher disease; newer preparations are more commonly used.

Algor Mortis

Post-mortem cooling of the body following death until it reaches ambient environmental temperature. Serves as one of the physical estimators of the post-mortem interval (time since death).

Algorithm

A medical algorithm is a defined sequence of rules or calculations used to support diagnosis, risk assessment, treatment, triage, or data analysis; it does not replace clinical judgment.

Aliasing Artifact

An ultrasound display artifact in which blood flow exceeds the selected velocity scale and appears to reverse direction. It can be reduced by increasing the scale or changing the imaging settings.

Alimentary

Relating to food, nourishment, or the organs and processes of digestion.

Alimentary Canal

The continuous muscular tube from the mouth to the anus through which food passes; it includes the pharynx, esophagus, stomach, small intestine, and large intestine.

Alkali

An alkali is a basic substance that accepts hydrogen ions or produces hydroxide in solution; caustic alkalis can cause severe chemical burns, while controlled bases have medical and laboratory uses.

Alkaline Phosphatase

A group of enzymes that catalyze the hydrolysis of phosphate esters at alkaline pH, found in high concentrations in the liver, bone, placenta, and intestinal mucosa. Elevated serum levels occur in obstructive liver disease, bone disorders including Paget disease and fractures, pregnancy, and certain malignancies.

Alkaloid

A naturally occurring nitrogen-containing organic compound, often produced by plants, with potentially potent physiologic or pharmacologic effects.

Alkaloids

Alkaloids are nitrogen-containing plant or microbial compounds with diverse biologic effects; examples include morphine, quinine, nicotine, and atropine.

Alkalosis

A metabolic (elevated bicarbonate) or respiratory (low PaCO₂) disturbance that raises arterial pH above 7.45. Metabolic alkalosis: vomiting/NG suction (HCl loss), diuretics, mineralocorticoid excess, hypokalemia, and alkali ingestion—compensated by hypoventilation; chloride-responsive vs. chloride-resistant distinguishes causes.

Alkaptonuria

An inherited disorder of tyrosine metabolism causing homogentisic-acid accumulation and darkening of urine or tissues.

Alkylating Agents

Alkylating agents are drugs or chemicals that add alkyl groups to DNA or other molecules; anticancer agents exploit this to inhibit replication, while normal-tissue toxicity includes myelosuppression and mutagenic risk.

Allantoin

Allantoin is a compound used in some topical products for its skin-protective, moisturizing, and keratolytic properties; formulation and indication determine its clinical role.

Allantois

The allantois is an embryonic membrane involved in early waste handling, blood-vessel development, and formation of structures related to the umbilicus and urinary bladder.

Allele

Alternative form of a gene at a particular locus. Individuals inherit two alleles for each gene (one from each parent). Alleles can be dominant, recessive, or codominant.

Allelic

Relating to an allele, one of two or more alternative forms of a gene at a particular genomic location.

Allen Test

A bedside test of collateral blood flow to the hand through the radial and ulnar arteries, commonly performed before radial-artery puncture or cannulation.

Allergen

A substance capable of causing an allergic immune response in a susceptible person. Common allergens include proteins in foods, pollens, medicines, animal dander, insect venom, and natural rubber.

Allergen Immunotherapy

Treatment that gives gradually increasing doses of a specific allergen by injection or under the tongue to reduce allergic sensitivity. It may be used for selected allergic rhinitis, conjunctivitis, asthma, or insect-venom allergy.

Allergic Bronchopulmonary Aspergillosis

A hypersensitivity reaction to *Aspergillus fumigatus* in asthma or CF patients. Causes central bronchiectasis, elevated total IgE, and eosinophilia. Treatment includes corticosteroids and itraconazole.

Allergic Cascade

The allergic cascade is the sequence in which an allergen activates immune cells, including IgE-bearing mast cells, causing release of histamine and other

mediators. This produces symptoms such as itching, swelling, mucus, wheezing, hives, or anaphylaxis.

Allergic Conjunctivitis

An inflammatory condition of the conjunctiva mediated by IgE-dependent mast cell degranulation in response to environmental allergens, characterized by bilateral ocular itching, tearing, conjunctival injection, and eyelid edema.

Allergic Contact Dermatitis

A delayed, T-cell-mediated skin reaction after contact with a substance to which the person has become sensitized, causing itching, redness, swelling, and sometimes blisters or scaling.

Allergic Dermatitis

An inflammatory skin reaction caused by an immune response to an external allergen, most commonly allergic contact dermatitis. It is a delayed, T-cell-mediated hypersensitivity reaction that typically develops after prior sensitization.

Allergic Granulomatosis

See *EGPA (formerly Churg-Strauss syndrome)*.

Allergic Granulomatosis And Angiitis

See *EGPA (formerly Churg-Strauss syndrome)*.

Allergic Reaction

An immune response to an otherwise harmless substance, ranging from mild symptoms to anaphylaxis.

Alternate Names:

Allergy

Allergic Reactions

Immune reactions to substances that are usually harmless, such as foods, medicines, pollens, or insect venom. They may be immediate or delayed and can range from localized symptoms to anaphylaxis.

Allergic Rhinitis

An IgE-mediated inflammatory response of the nasal mucosa to inhaled allergens, causing sneezing, nasal congestion, clear rhinorrhea, nasal itching, and of-

ten conjunctival symptoms. Management may include allergen avoidance, intranasal corticosteroids, antihistamines, and allergen immunotherapy when indicated.

Allergic Rhinitis (Hay Fever)

Allergies occur when the immune system overreacts to harmless environmental substances such as pollen, dust mites, mold, or pet dander. Hay fever (allergic rhinitis) specifically involves nasal symptoms like sneezing, congestion, and itchy eyes triggered by airborne allergens.

Alternate Names:

Hay Fever

Allergies

Immune reactions to normally harmless substances such as foods, pollens, medicines, insect venom, or animal dander. Reactions range from localized rhinitis or dermatitis to life-threatening anaphylaxis.

Allergies (Allergy)

Alternate index label for Allergy.

Allergies, Asthma, And Dust

Dust can carry allergens such as dust-mite particles that trigger allergic rhinitis or asthma in susceptible people. Symptoms may include sneezing, congestion, wheezing, cough, and shortness of breath.

Allergies, Asthma, And Molds

Mold spores and fragments can trigger allergic rhinitis, conjunctivitis, or asthma in sensitized people. Reducing exposure and treating the specific allergic or airway condition are guided by clinical assessment.

Allergies, Asthma, And Pollen

Pollen is a seasonal airborne allergen that can trigger allergic rhinitis, conjunctivitis, or asthma. Typical symptoms include sneezing, itchy eyes, nasal congestion, cough, and wheezing.

Allergy

An allergy is an inappropriate immune response

to a normally harmless substance such as pollen, food, medicine, insect venom, or animal proteins. Symptoms range from localized rhinitis or hives to life-threatening anaphylaxis.

Alternate Names:

Allergies (Allergy)

Allergy Diaper (Intertrigo)

Alternate index label for Intertrigo.

Allergy Eczema (Atopic Dermatitis)

Alternate index label for Atopic Dermatitis.

Allergy Eye (Eye Allergy)

Alternate index label for Eye Allergy.

Allergy Food (Food Allergy)

Alternate index label for Food Allergy.

Allergy Insect (Insect Sting Allergies)

Alternate index label for Insect Sting Allergies.

Allergy Latex (Latex Allergy)

Alternate index label for Latex Allergy.

Allergy Plant Contact (Poison Ivy, Oak, And Sumac)

Alternate index label for Poison Ivy, Oak, and Sumac.

Allergy Rash (Rash)

Alternate index label for Rash.

Allergy Shots

Subcutaneous allergen immunotherapy involving gradually increasing doses of selected allergens to reduce allergic sensitivity; administration requires monitoring for systemic reactions.

Allergy Testing

Identification of IgE-mediated allergic sensitization. Skin prick test: allergen extract applied to skin via lancet, read after 15 min (wheal >3mm = positive). Intradermal test: more sensitive, used when SPT

negative but suspicion remains.

Allergy Testing - Skin

A diagnostic test in which small amounts of allergen extracts are applied to the skin by prick or intradermal injection to detect IgE-mediated allergic sensitization. A wheal-and-flare reaction at the test site indicates sensitization to the corresponding allergen.

Allergy, Dust Mite

See *Dust mite allergy*.

Allergy, Egg

See *Egg allergy*.

Allergy, Food

See *Food allergy*.

Allergy, Latex

See *Latex allergy*.

Allergy, Milk

See *Milk allergy*.

Allergy, Mold

See *Mold allergy*.

Allergy, Nickel

See *Nickel allergy*.

Allergy, Peanut

See *Peanut allergy*.

Allergy, Penicillin

See *Penicillin allergy*.

Allergy, Pet

See *Pet allergy*.

Allergy, Shellfish

See *Shellfish allergy*.

Allergy, Wheat

See *Wheat allergy*.

Allis Clamp

A toothed surgical clamp used to grasp and hold firm tissue during an operation.

Allodynia

Pain caused by a stimulus that is normally not painful, such as light touch or gentle pressure. It may occur with neuropathic, nociplastic, or other pain conditions.

Allogeneic

Derived from a genetically different individual of the same species, as in an allogeneic blood, tissue, stem-cell, or organ transplant.

Allograft

Tissue transplanted between genetically different individuals of the same species. Allografts may provoke immune rejection and require appropriate donor screening, processing, and, in some settings, immunosuppression.

Allopathy

A historical label for conventional medical treatment, often contrasted with homeopathy; modern clinical practice is more accurately described by its specific methods and evidence.

Allosteric Regulation

The regulation of enzyme activity by the binding of an effector molecule at a site other than the active site (allosteric site), causing a conformational change that alters the enzyme's activity. Allosteric enzymes typically have multiple subunits with cooperative binding kinetics (sigmoidal rather than hyperbolic velocity curves).

Aloe

A plant-derived substance used in some topical products and supplements; topical preparations may soothe minor irritation, while oral latex-containing products can cause cramps, diarrhea, and electrolyte

problems.

Alogia

Poverty of speech or poverty of content of speech, a negative symptom of schizophrenia. It manifests as brief, empty replies to questions, reduced spontaneous speech, and lack of elaboration. It reflects dysfunction in the mesocortical dopamine pathway and is less responsive to antipsychotic medications than positive symptoms.

Alongshan Virus

A tick-associated segmented RNA virus reported to cause a febrile illness, often with headache, fatigue, nausea, or neurologic symptoms; its epidemiology and clinical significance remain under study.

Alopecia

The general medical term for loss of hair from the scalp or body, rather than a single specific disease. Causes include disorders of the hair follicle, immune disease, hormones, medicines, systemic illness, or physical injury. Androgenetic alopecia (male- or female-pattern hair loss) is common and is associated with genetic susceptibility and androgen, particularly dihydrotestosterone, effects. Alopecia areata is an autoimmune, usually non-scarring, patchy hair-loss disorder that may progress to alopecia totalis (entire scalp) or universalis (whole body).

Alopecia (Hair Loss)

Alternate index label for Hair Loss.

Alopecia Areata

An autoimmune condition in which the immune system attacks hair follicles, causing sudden, usually non-scarring hair loss. It often begins as smooth, round or oval patches on the scalp but may affect the beard, eyebrows, eyelashes, or other body hair. Some people develop nail changes or more extensive hair loss, called alopecia totalis or alopecia universalis. Hair may regrow, but episodes can recur.

Alternate Names:

AA (Alopecia Areata) Patchy Hair Loss (Alopecia Areata)

Alopecia Totalis

Complete loss of hair from the scalp, usually caused by an autoimmune disorder.

Alopecia Universalis

Loss of hair from the entire body, including the scalp, eyebrows, eyelashes, and other hair-bearing areas.

ALP - Blood Test

A laboratory test that measures serum alkaline phosphatase activity, an enzyme present in liver, bone, intestine, and placenta. Elevated levels may indicate hepatobiliary disease, bone turnover, or pregnancy; isoenzyme testing helps distinguish the source.

ALP Isoenzyme Test

A laboratory test that identifies the tissue source of elevated alkaline phosphatase by separating liver, bone, intestinal, and placental isoenzymes. It helps determine whether elevated ALP arises from hepatobiliary disease or bone disorders.

Alpha 1

See *Alpha-1-antitrypsin deficiency*.

Alpha Blockers

Medicines that block alpha-adrenergic receptors and may lower blood pressure or improve urine flow in prostate enlargement.

Alpha Cells

Endocrine cells in the pancreatic islets that produce and release glucagon, a hormone that raises blood glucose during fasting or low-glucose states.

Alternate Names:

Pancreatic alpha cells; glucagon-secreting cells.

Alpha Hydroxy Acids

A group of organic acids used in dermatology and cosmetics to exfoliate the outer skin layer and improve texture or pigmentation. Examples include glycolic and lactic acid.

Alternate Names:

AHAs; fruit acids.

Alpha Thalassemia

An inherited blood disorder caused by reduced production of alpha-globin chains of hemoglobin. Severity ranges from mild microcytosis or anemia to severe fetal or newborn disease.

Alpha Waves

Rhythmic electroencephalographic waves, typically around 8–13 Hz, most prominent when a relaxed, awake person has the eyes closed.

Alternate Names:

Alpha rhythm; posterior dominant rhythm.

Alpha-1 Antitrypsin Blood Test

A laboratory test that measures serum alpha-1 antitrypsin concentration, a protease inhibitor synthesized by the liver. Low levels suggest alpha-1 antitrypsin deficiency, a genetic disorder predisposing to early emphysema and liver disease; phenotyping or genotyping confirms the diagnosis.

Alpha-1 Antitrypsin Deficiency

An autosomal codominant genetic disorder caused by mutations in the SERPINA1 gene. It predisposes to early-onset panlobular emphysema (lower lobes), liver disease, and panniculitis. PiZZ genotype is most severe. Treatment includes AAT augmentation therapy and smoking avoidance.

Alpha-Agonists

Medicines that stimulate alpha-adrenergic receptors. Depending on the receptor and tissue, they can constrict blood vessels, alter autonomic activity, or reduce aqueous-humor production in the eye. Their uses and adverse effects depend on the specific medicine and route.

Alpha-Delta Sleep

A sleep pattern in which alpha-frequency activity intrudes into delta-wave deep sleep and may be associated with nonrestorative sleep or chronic pain.

Alternate Names:

Alpha-delta intrusion; alpha-delta rhythm.

Alpha-Fetoprotein

A protein produced mainly by the fetal liver; maternal or blood levels can help assess certain fetal and adult conditions.

Alternate Names:

AFP; alpha fetoprotein (alternate spelling).

Alpha-Fetoprotein

A protein produced mainly by the fetal liver. Maternal blood AFP is used in prenatal screening, and serum AFP may help evaluate selected liver and germ-cell tumors. Results are not diagnostic by themselves.

Alternate Names:

Alpha-Fetoprotein

Alpha-Gal Syndrome

An allergy to the carbohydrate alpha-gal found in most nonhuman mammalian meat and some products derived from mammals.

Alpha-Globulin

A group of blood proteins that includes transport and acute-phase proteins.

Alpha-Glucosidase Inhibitors

Oral glucose-lowering medications including acarbose, miglitol, and voglibose that delay carbohydrate digestion and glucose absorption in the small intestine by inhibiting brush border alpha-glucoside enzymes. Reduce postprandial glucose excursions without affecting fasting glucose significantly.

Alpha-Ketoglutarate

A five-carbon intermediate of the citric acid cycle formed from isocitrate and converted to succinyl-CoA. It also serves as a carbon skeleton for amino-acid metabolism and participates in oxidative deamination and cellular signaling.

Alpha-Synuclein

A neuronal protein that can accumulate abnormally in disorders such as Parkinson disease and Lewy-body dementia.

Alpha-Thalassemia

An inherited disorder in which reduced alpha-globin production impairs hemoglobin formation.

Alport Syndrome

Hereditary nephritis caused by mutations in type four collagen genes leading to progressive sensorineural hearing loss, ocular anomalies, and renal failure. X-linked form from COL4A5 mutations is the most common. Autosomal recessive and dominant forms exist.

Alprazolam

A short- to intermediate-acting benzodiazepine used mainly for anxiety and panic disorder; it can cause sedation, dependence, withdrawal, and dangerous respiratory depression with opioids or other depressants.

ALS

Amyotrophic lateral sclerosis (motor neuron disease; Lou Gehrig disease) is the progressive neurodegenerative disease characterized by degeneration of upper and lower motor neurons controlling voluntary movement. It causes progressive weakness, muscle wasting, fasciculations, spasticity, dysarthria, or dysphagia; sensation is usually relatively preserved, but respiratory muscles may eventually be affected.

Alstrom Syndrome

A rare autosomal-recessive disorder caused by ALMS1 variants, associated with cone-rod dystrophy, hearing loss, obesity, insulin resistance, cardiomyopathy, and progressive kidney or liver disease.

Alternative Therapy For Multiple Sclerosis

Complementary approaches for multiple sclerosis may support symptoms such as fatigue, pain, stress, or mobility but do not replace disease-modifying treatment. Exercise, rehabilitation, mindfulness, and selected supplements should be discussed with clinicians because interactions and unproven claims are common.

Alternate Names:

MS Alternative And Complementary Therapies (Alternative Therapy For Multiple Sclerosis)

Alternative Treatments For Hot Flashes

Nonhormonal approaches to hot flashes include cooling strategies, layered clothing, sleep and stress measures, cognitive behavioral therapy, and selected prescription medicines. Herbal products have variable evidence and can interact with medicines; treatment choice depends on symptoms and medical history.

Altitude Illness

Acute mountain sickness: headache + 1 symptom (nausea, fatigue, dizziness, insomnia) at >2500m. HACE: AMS + ataxia/altered mental status (cerebral edema). HAPE: dyspnea at rest, cough, frothy sputum, crackles, hypoxemia (noncardiogenic pulmonary edema).

Altitude Sickness

Illness caused by rapid ascent to high altitude without adequate acclimatization, ranging from acute mountain sickness to life-threatening pulmonary or cerebral edema.

Aluminum Phosphide

A highly toxic fumigant that releases phosphine gas when exposed to moisture or stomach acid. Poisoning can cause severe cardiovascular, respiratory, gastrointestinal, and metabolic effects and requires immediate emergency treatment and supportive intensive care.

Alvarado Score

Clinical scoring system for acute appendicitis. Components: Migration of pain (1), Anorexia (1), Nausea/vomiting (1), Tenderness in RLQ (2), Rebound pain (1), Elevated temperature (1), Leukocytosis (2), Left shift (1).

Alveolar

Relating to an alveolus, especially one of the tiny air sacs in the lungs where oxygen and carbon dioxide are exchanged.

Alveolar Abnormalities

Structural or functional changes in the lung alveoli that can impair gas exchange and occur with infection, edema, emphysema, interstitial disease, or developmental disorders.

Alveolar Bone

Bone surrounding and supporting tooth roots. Resorbs with tooth loss. Maintained by function. Bone grafting may be needed for implants.

Alveolar Gas Equation

$PAO_2 = (FiO_2 \times (P_{atm} - PH_2O)) - (PaCO_2 / R)$.
 $P_{atm} = 760$ mmHg, $PH_2O = 47$ mmHg, $R = 0.8$.
Allows calculation of A-a gradient.

Alveolar Hydatid Disease

A serious parasitic infection caused by the larval stage of *Echinococcus multilocularis*, usually producing an infiltrative, tumor-like lesion in the liver that can spread to other organs.

Alveolar Macrophage

An immune cell in the lung air sacs that removes particles, microbes, and cellular debris.

Alternate Names:

Dust cell

Alveolar Osteitis

See *Dry socket*.

Alveolar Osteitis (Dry Socket Overview)

Alternate index label for Dry Socket Overview.

Alveolar Rhabdomyosarcoma

A malignant small-round-cell tumor showing skeletal-muscle differentiation, usually arising in the extremities, trunk, or head and neck of children and adolescents. It may disseminate early through lymphatic and hematogenous routes and often carries characteristic PAX3-FOXO1 or PAX7-FOXO1 fusions. Diagnosis requires biopsy with immunophenotyping and molecular testing, followed by staging for local and distant disease. Treatment is multimodal and commonly includes systemic chemotherapy

with surgery and/or radiotherapy according to risk group.

Alveolar Ventilation

The volume of fresh air reaching the alveoli per minute, representing the portion of minute ventilation that participates in gas exchange. Calculated as (tidal volume - dead space volume) x respiratory rate. Normal alveolar ventilation is approximately 4-5 L/min.

Alveolar Ventilation Equation

The relationship between alveolar ventilation and arterial carbon-dioxide tension, approximately $PaCO_2 = 0.863$ times carbon-dioxide production divided by alveolar ventilation. It explains why reduced effective ventilation causes hypercapnia.

Alveoli

Microscopic air sacs in the lungs where oxygen enters the blood and carbon dioxide leaves it.

Alveolitis

Inflammation of the pulmonary alveoli, usually causing cough and breathlessness, or inflammation of a tooth socket after extraction, called alveolar osteitis or dry socket; the organ involved distinguishes the two meanings.

Alveolus

The terminal air sac (200-300 microns) where gas exchange occurs. Approximately 300-500 million per lung, providing 70 m² surface area. Wall: type I pneumocytes (95% surface, thin for diffusion), type II pneumocytes (5%, secrete pulmonary surfactant reducing surface tension), alveolar macrophages (dust cells, phagocytosis).

Alzheimer Disease

A progressive neurodegenerative disorder characterized pathologically by amyloid-beta plaques, tau neurofibrillary pathology, and neuronal loss. It most commonly presents with insidious impairment of episodic memory followed by broader cognitive and functional decline.

Alzheimer'S

A progressive neurodegenerative disease causing memory loss and decline in thinking and daily function.

Alternate Names:

Alzheimer disease

Alzheimer'S Disease

A progressive brain disorder causing deterioration of memory, thinking, behavior, and the ability to perform daily activities.

Alzheimer'S Disease Causes, Stages, And Symptoms

Alzheimer disease is a progressive neurodegenerative disorder causing decline in memory and other thinking abilities, followed by increasing difficulty with communication, judgment, movement, and self-care. Progression varies, and evaluation looks for reversible contributors and other dementias.

Alternate Names:

Am I Depressed

Depression is more than temporary sadness: it involves persistent low mood or loss of interest with changes in sleep, appetite, energy, concentration, movement, self-worth, or hope. Thoughts of self-harm require immediate emergency or crisis support.

Amalgam

Restorative material composed of mercury alloyed with silver, tin, copper, and other metals. Properties: strong, durable, cost-effective. Concerns: mercury toxicity debates, aesthetic limitations. Placement: requires cavity preparation, condensation, carving. Removal: sectioning, cooling to prevent mercury vapor.

Amastia

Congenital absence of one or both breasts, sometimes accompanied by absence of the nipple or pectoral muscles and other developmental anomalies.

Amaurosis Fugax

Transient monocular vision loss lasting seconds to minutes, classically described as a curtain descending over the field, caused by retinal or optic nerve ischemia.

Ambidextrous

Able to use the right and left hands with approximately equal skill or preference.

Ambient

Surrounding or present in the immediate environment, as in ambient temperature, light, noise, or air.

Ambiguous Genitalia

External genital anatomy that does not have typical male or female features, usually because of a difference in sex development. Evaluation includes examination, chromosome and hormone testing, pelvic or abdominal imaging, and assessment for congenital adrenal hyperplasia.

Amblyopia

A developmental reduction in vision caused by abnormal visual input during early childhood, without a sufficient structural explanation. It commonly results from misalignment, unequal focusing power, or obstruction of vision in one eye.

Amblyopia Lazy Eye

Amblyopia is reduced vision in an eye that did not develop normal visual processing during childhood, often because of unequal focusing power, misalignment, or obstruction. Early treatment may include glasses, patching or blurring the stronger eye, and correction of the underlying problem.

Ambu Bag

A self-inflating manual resuscitation bag used with a mask or airway device to provide positive-pressure ventilation.

Ambulance

A vehicle equipped to transport and provide emer-

gency medical care to ill or injured people.

Ambulatory Blood Pressure Monitoring

Measurement of blood pressure at regular intervals, usually over twenty-four hours, with a portable cuff worn during normal activities and sleep. It detects white-coat and masked hypertension, evaluates nighttime blood-pressure patterns, and helps confirm hypertension diagnosed in the clinic.

Amebiasis

Infection with the protozoan parasite *Entamoeba histolytica*, acquired mainly by ingestion of faecally contaminated food or water. It may be asymptomatic or cause amoebic colitis with abdominal pain, diarrhoea, or dysentery, and can spread through the portal circulation to produce a liver abscess.

Amebiasis *Entamoeba Histolytica* Infection

Amebiasis is infection with *Entamoeba histolytica*, acquired through fecally contaminated food, water, or hands. It may cause diarrhea or dysentery and can spread to the liver; stool testing and appropriate tissue-active and luminal treatment are used.

Alternate Names:

Amoebic Dysentery (*Amebiasis Entamoeba Histolytica* Infection)

Amebic Colitis

Inflammation of the colon caused by infection with *Entamoeba histolytica*.

Alternate Names:

Amoebic colitis

Amebic Liver Abscess

A focal collection of necrotic material in the liver caused by *Entamoeba histolytica*, usually following intestinal infection. It commonly causes fever, right-upper-quadrant pain, and tender hepatomegaly; diagnosis is supported by imaging and serology.

Amelanotic Melanoma

A variant of melanoma that lacks significant melanin

production, presenting as a pink, red, or flesh-coloured papule or nodule rather than a typical dark lesion. Its paucity of pigment can delay diagnosis, making it a diagnostic challenge. Histopathology with immunohistochemical markers confirms the diagnosis.

Amelia

A rare congenital anomaly characterized by complete absence of one or more limbs. It may occur alone or as part of a genetic or developmental syndrome. Prenatal imaging can identify the anomaly and help assess associated abnormalities.

Amelioration

Improvement or reduction in the severity of a symptom, disease, impairment, or other adverse condition.

Ameloblastoma

Locally aggressive odontogenic tumor. Multicystic radiolucency in posterior mandible. High recurrence rate. Treatment: surgical resection with margins.

Amelogenesis

The formation and maturation of tooth enamel by ameloblasts during tooth development.

Amelogenesis Imperfecta

A group of inherited disorders in which enamel formation is abnormal, causing enamel that may be thin, poorly mineralized, weak, or easily worn on most or all teeth.

Amenorrhea

Amenorrhea is the absence of menstrual periods. Primary amenorrhea is failure to experience menarche by age 15 despite normal secondary sexual development, or by age 13 when secondary sexual development is absent. Secondary amenorrhea is the absence of menstruation for 3 months in someone with previously regular cycles or 6 months in someone with irregular cycles. It may be physiological or caused by pregnancy, endocrine disorders, reproductive-system abnormalities, medications, illness, inadequate energy availability, or excessive

exercise.

American Trypanosomiasis

See *Chagas disease*.

AMI

Acute myocardial infarction, the sudden death of heart muscle caused by prolonged inadequate coronary blood flow.

Amino Acid

An organic molecule containing an amino group, a carboxyl group, a hydrogen atom, and a variable side chain attached to a central carbon. Amino acids are the building blocks of proteins; cells use some also in energy metabolism and other biochemical pathways.

Amino Acid Catabolism

The breakdown of amino acids after protein digestion, with the amino group removed and the carbon skeleton metabolized for energy or gluconeogenesis/ketogenesis. Transamination: amino groups are transferred to alpha-ketoglutarate, forming glutamate, by aminotransferases requiring pyridoxal phosphate (PLP, vitamin B6) as a cofactor.

Amino Acid Symbols

Standard one-letter and three-letter abbreviations used to identify the 20 common protein-building amino acids in biochemical and genetic notation.

Amino Acids

Amino acids are organic molecules that form proteins and also serve as precursors or fuels in human metabolism.

Aminoaciduria

Aminoaciduria is abnormal urinary loss of amino acids caused by renal tubular transport defects, inherited metabolic disease, or generalized tubular injury.

Aminoglycoside Nephrotoxicity

Kidney damage caused by aminoglycoside an-

tibiotics, characterized by acute tubular necrosis. Nephrotoxicity is caused by accumulation of aminoglycosides in the proximal tubular cells. The risk is increased with prolonged therapy, high doses, concurrent nephrotoxic drugs, and renal impairment.

Aminoglycoside Ototoxicity

Damage to the inner ear caused by aminoglycoside antibiotics, resulting in hearing loss and vestibular dysfunction. Ototoxicity is caused by accumulation of aminoglycosides in the inner ear, damaging hair cells. The risk is increased with prolonged therapy, high doses, concurrent ototoxic drugs, and renal impairment.

Aminoglycosides

A class of bactericidal antibiotics that bind irreversibly to the 30S ribosomal subunit, causing misreading of messenger RNA and inhibiting protein synthesis. They include gentamicin, tobramycin, amikacin, and streptomycin.

Aminophylline Overdose

Toxic effects caused by excessive exposure to Aminophylline. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Aminotransferase

An enzyme that transfers an amino group between an amino acid and an alpha-keto acid; clinically measured examples include ALT and AST.

Amiodarone-Induced Skin Discoloration

A slate-grey to bluish hyperpigmentation of sun-exposed skin caused by long-term amiodarone therapy, due to deposition of the drug and its metabolites in dermal macrophages. It is dose-related, more pronounced in photodistributed areas, and may be partially reversible upon drug cessation.

Amiodarone-Induced Thyroid Dysfunction

Thyroid abnormalities occurring in approximately fifteen to twenty percent of patients on amiodarone due to the high iodine content of the molecule. Type

one results from iodine-induced increased thyroid hormone synthesis in patients with underlying thyroid disease. Type two results from destructive thyroiditis.

Amiodarone-Induced Vortex Keratopathy

Whorl-like corneal epithelial deposits caused by amiodarone accumulation, appearing as golden-brown verticillata on slit-lamp examination, usually asymptomatic but reversible upon drug discontinuation.

Amitriptyline

A tricyclic antidepressant that affects serotonin and norepinephrine signaling and is also used for selected chronic pain, migraine-prevention, and sleep-related indications.

Amitriptyline And Perphenazine Overdose

Toxic effects caused by excessive exposure to Amitriptyline and perphenazine. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Amitriptyline Hydrochloride Overdose

Toxic effects caused by excessive exposure to Amitriptyline hydrochloride. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

AML

Acute myeloid leukemia, an aggressive cancer of immature myeloid blood-cell precursors that crowds the bone marrow and impairs normal blood-cell production.

Ammonia

A nitrogen-containing waste product produced during protein metabolism and normally converted to urea by the liver. Elevated blood ammonia may occur with liver failure, urea-cycle disorders, some medicines, or other metabolic conditions.

Ammonia Blood Test

A laboratory test that measures venous or arterial blood ammonia concentration, a nitrogenous waste product of protein metabolism normally cleared by the liver. Elevated levels may indicate hepatic failure, urea-cycle disorders, or valproic acid toxicity, and correlate with risk of encephalopathy.

Ammonia Poisoning

Toxic effects caused by excessive exposure to ammonia. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Ammonium Hydroxide Poisoning

Toxic effects caused by excessive exposure to ammonium hydroxide. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Amnesia

Loss or impairment of memory that is greater than ordinary forgetting. It may affect the ability to form new memories (anterograde amnesia), recall past events (retrograde amnesia), or both.

Amnesia, Transient Global

See *Transient global amnesia*.

Amnestic Syndrome

See *Amnesia*.

Amniocentesis

A procedure in which a needle is used to obtain amniotic fluid for genetic, chromosomal, infectious, or other prenatal testing. It is usually performed after the first trimester when clinically indicated; risks and benefits depend on the reason for testing and gestational age.

Amnioinfusion

Infusion of sterile fluid into the amniotic sac, often to reduce umbilical-cord compression during labor.

Amnion

The innermost fetal membrane, lining the amniotic cavity and containing the amniotic fluid. A thin, avascular membrane composed of a single layer of cuboidal epithelium overlying a basement membrane and connective tissue. The amnion produces and resorbs amniotic fluid. It fuses with the chorion by 12-14 weeks.

Amnion Nodosum

Multiple small nodules of vernix or amniotic debris attached to the amnion, usually associated with reduced or absent fetal urine contribution to amniotic fluid and oligohydramnios.

Amnioscope

An instrument used to inspect amniotic fluid and the membranes through the cervix.

Amniotic Band

A strand of amniotic tissue that can entangle or constrict developing fetal body parts.

Alternate Names:

Amniotic band sequence

Amniotic Band Sequence

Amniotic band sequence results when fibrous amniotic strands entangle developing fetal parts, producing constriction rings, limb defects, or craniofacial and body-wall abnormalities.

Amniotic Band Syndrome

A spectrum of congenital anomalies caused by strands of a ruptured amniotic membrane that constrict or entangle developing fetal parts. Effects range from ring-like limb constrictions, swelling, or loss of digits to facial and body-wall disruptions; severity varies widely, and management depends on the structures involved.

Amniotic Constriction Bands

Alternate descriptive label; see *Amniotic band sequence*.

Amniotic Fluid

The clear to pale-yellow fluid surrounding the fetus within the amniotic sac. Early in pregnancy it is derived largely from maternal plasma; later, fetal urine and lung fluid are major sources, and the fetus continuously swallows and recycles it.

Amniotic Sac

The fluid-filled sac in the uterus that surrounds and protects the developing baby. It contains the amniotic fluid and is also called the bag of waters.

Amniotic Fluid Embolism

A rare obstetric emergency involving sudden respiratory or cardiovascular collapse and often severe coagulopathy during labor, delivery, or shortly afterward. It requires immediate supportive and multidisciplinary treatment.

Amniotic Fluid Index

An ultrasound estimate of amniotic-fluid volume obtained by dividing the uterus into four quadrants, measuring the deepest unobstructed vertical pocket in each quadrant, and adding the measurements. Interpretation uses gestational-age and practice-specific thresholds to identify unusually low or high fluid volume.

Amoebiasis

Infection caused by the protozoan parasite *Entamoeba histolytica*, transmitted via the fecal-oral route (contaminated food or water).

Amoebic Dysentery (Amebiasis *Entamoeba Histolytica* Infection)

Alternate index label for Amebiasis *Entamoeba histolytica* Infection.

Amok

A historical culture-related term for a sudden episode of uncontrolled, often violent behavior followed by exhaustion or amnesia; modern assessment considers psychiatric, neurologic, substance-related, and cultural factors.

Amoxicillin Rash In Infectious Mononucleosis

A maculopapular, non-allergic rash that develops in up to 90 percent of patients with Epstein-Barr virus infectious mononucleosis who receive amoxicillin or ampicillin. The rash is often widespread and pruritic, and its presence does not indicate true penicillin allergy.

Amphetamine

A central nervous system stimulant that increases catecholamine signaling and is used in selected attention-deficit/hyperactivity and sleep disorders; misuse can cause dependence and cardiovascular or psychiatric harm.

AMPK

AMP-activated protein kinase. Energy sensor. Activated by AMP. Inhibits anabolism, stimulates catabolism.

Amplified Musculoskeletal Pain Syndrome

A chronic pain condition, often beginning in childhood or adolescence, in which pain is unusually intense or widespread and may be accompanied by sensitivity, fatigue, or functional limitation despite no single structural explanation; it is also called AMPS.

Ampulla

A widened portion of a tubular structure, such as the ampulla of the bile and pancreatic ducts or the ampulla of a fallopian tube.

Ampullary Cancer

Cancer arising in the ampulla of Vater, where the bile duct and pancreatic duct empty into the small intestine.

Ampullary Carcinoma

A malignant epithelial tumor arising in or around the ampulla of Vater, where the bile duct and pancreatic duct enter the duodenum. It commonly presents with obstructive jaundice, abnormal liver tests, pancreatitis, or gastrointestinal bleeding. His-

tologic subtype and local extension influence prognosis, staging, and treatment selection. Diagnosis uses endoscopy, cross-sectional imaging, and tissue sampling, with surgery considered for resectable disease.

Amputation

Surgical removal of a limb or part of a limb. The level may involve a digit, foot, lower leg, thigh, or hip, or a corresponding part of the upper limb. The level is selected according to tissue viability, function, and potential for rehabilitation.

Amputation Stump

The remaining part of a limb after amputation. It must be monitored for wound healing, infection, poor circulation, neuroma pain, phantom-limb pain, and skin problems caused by prosthetic use. Care includes hygiene, compression or shaping when advised, progressive strengthening, and professional prosthetic and rehabilitation assessment.

Amsler Grid

A simple chart used to detect or monitor distortions and missing areas in central vision, especially in macular disease.

Amusia

An acquired or developmental inability to perceive, recognize, produce, or understand music despite otherwise adequate hearing and language ability.

Amygdala

Almond-shaped nucleus cluster in the anterior medial temporal lobe, part of the limbic system, primarily processing emotions (especially fear/anxiety). Receives thalamic/cortical sensory input; projects to hypothalamus, brainstem, and cortex for autonomic/behavioral fear responses.

Amygdaloid Nucleus

A group of nuclei in the medial temporal lobe involved in processing emotional significance, fear responses, learning, memory, and autonomic responses.

Amylase

An enzyme that breaks down starch and is produced mainly by the salivary glands and pancreas. Blood levels may rise in several conditions, including some pancreatic disorders.

Amylase - Blood

A blood test measuring amylase, an enzyme produced mainly by the pancreas and salivary glands; elevated values may occur in pancreatitis and other conditions.

Amylase - Urine

A urine test measuring amylase excretion, sometimes used with blood amylase when evaluating pancreatic or salivary-gland disease.

Amylase And Lipase

Digestive enzymes measured in blood when pancreatic disease is suspected. Lipase is generally more specific for pancreatic injury and remains elevated longer than amylase, but neither test alone diagnoses acute pancreatitis; results must be interpreted with symptoms and imaging.

Amylin Analogues

Medicines that mimic amylin, a hormone released with insulin by pancreatic beta cells. Pramlintide slows gastric emptying, reduces post-meal glucagon secretion, and may be used with insulin in selected patients with diabetes.

Amyloid

An abnormal protein that can accumulate in tissues and impair their function in several diseases.

Amyloid Disease

See *Amyloidosis*.

Amyloid Disease (Amyloidosis)

Alternate index label for Amyloidosis.

Amyloid Neuropathy

Peripheral nerve damage caused by deposition of abnormal amyloid protein.

Amyloid PET

A positron-emission tomography scan using a tracer that binds to amyloid deposits in the brain. It can support evaluation of selected cognitive disorders but is interpreted with clinical assessment and other tests.

Amyloid Precursor Protein

A normal transmembrane protein that can be processed into amyloid-beta peptides, which may accumulate in the brain in Alzheimer disease.

Alternate Names:

APP; amyloid-beta precursor protein.

Amyloid Protein

A normally soluble protein that has misfolded into insoluble fibrils and may accumulate in tissues.

Alternate Names:

Amyloid

Amyloid Purpura

Purpuric discoloration of the eyelids and periorbital skin (raccoon eyes) caused by cutaneous amyloid deposition, which weakens vessel walls and leads to extravasation of erythrocytes. It may be a sign of systemic amyloidosis, particularly AL amyloidosis, and warrants further workup including serum free light chains and tissue biopsy.

Amyloidosis

A group of diseases in which misfolded proteins form insoluble deposits called amyloid in tissues and organs. The deposits may be limited to one site or spread throughout the body and can impair organs such as the kidneys, heart, nerves, liver, or digestive tract. On biopsy, amyloid typically binds Congo red dye and shows apple-green birefringence under polarized light.

Alternate Names:

Amyloid Disease (Amyloidosis)

Amyotrophic Lateral Sclerosis

Progressive neurodegenerative disorder affecting both upper and lower motor neurons, causing weakness, atrophy, fasciculations, and eventually respira-

tory failure. Most cases are sporadic (90 percent); 10 percent are familial. Average age of onset 55-60 years.

Amyotrophic Lateral Sclerosis

A progressive neurodegenerative disease that damages motor neurons, causing muscle weakness, loss of movement, speech or swallowing difficulty, and eventually respiratory failure.

Alternate Names:

Lou Gehrig's Disease

ANA

Antinuclear antibody, an autoantibody directed against components of cell nuclei; ANA testing can support evaluation of autoimmune disease but is not diagnostic by itself.

Anabolic Steroid Induced Hypogonadism

A form of secondary (hypogonadotropic) hypogonadism caused by the supra-physiologic use of exogenous anabolic-androgenic steroids. Exogenous androgens trigger profound negative feedback on the hypothalamic-pituitary-gonadal axis, shutting down pulsatile gonadotropin-releasing hormone (GnRH), luteinizing hormone, and follicle-stimulating hormone secretion. Clinically manifests as severe testicular atrophy, azoospermia or severe oligozoospermia, erectile dysfunction, loss of libido, depression, and prolonged withdrawal hypogonadism upon cessation of steroid use, often requiring medical endocrine therapy (such as human chorionic gonadotropin or selective estrogen receptor modulators) to restore endogenous spermatogenesis and testosterone synthesis.

Alternate Names:

ASIH; Anabolic-Androgenic Steroid-Induced Hypogonadism

Anabolic Steroids

Synthetic derivatives of testosterone used in selected medical conditions and misused for muscle building or performance enhancement. Nonmedical use can cause acne, infertility, testicular shrinkage, breast enlargement, mood disturbance, and cardiovascular disease, and sudden cessation may produce symp-

toms of hormone deficiency.

Anabolism

The set of metabolic pathways that construct complex molecules from simpler precursors, requiring energy input (ATP, reducing equivalents NADPH). Examples: protein synthesis from amino acids, glycogen synthesis (glycogenesis), fatty acid synthesis (lipogenesis), cholesterol synthesis, nucleotide synthesis, gluconeogenesis.

Anaemia

A condition in which the number of red blood cells or the hemoglobin concentration is below the normal range, reducing the oxygen-carrying capacity of the blood.

Anaerobe

A microorganism that grows without molecular oxygen. Obligate anaerobes may be injured or killed by oxygen, while aerotolerant anaerobes tolerate oxygen but do not use it for energy production.

Anaerobic

Able to live, grow, or function without oxygen. Anaerobic bacteria can cause infections in deep wounds, abscesses, the mouth, abdomen, or pelvis and may need antibiotics that work without oxygen.

Anaerobic Bacteria

Bacteria that grow without oxygen; some are harmed by oxygen.

Anaesthesia

Reversible loss of sensation, with or without loss of consciousness, produced by medicines or other techniques; also spelled anesthesia.

Anagen

The active, proliferative growth phase of the physiological hair follicle cycle, during which matrix cells in the hair bulb undergo rapid mitotic division and keratinization to produce the hair shaft. Lasts approximately 2 to 7 years on the human scalp, determining the maximum length of scalp hair, with

approximately 85–90% of scalp follicles in anagen under normal conditions. Characterized by high metabolic activity that makes anagen matrix cells acutely vulnerable to cytotoxic chemotherapy and ionizing radiation, precipitating rapid-onset hair breakage and diffuse shedding termed anagen effluvium.

Anal Abscess

Collection of pus in the perianal or ischiorectal space usually from infected anal glands. Presents with throbbing anal pain, swelling, fever, and difficulty sitting. Diagnosed clinically with tenderness and fluctuance. Treatment is incision and drainage under local anesthesia. Antibiotics are not routinely required for simple abscesses.

Anal Canal

The terminal 3–4 cm of the gastrointestinal tract, extending from the anorectal junction (puborectalis/pectinate line) to the anal verge. Above the pectinate line: lined by columnar epithelium, innervated by autonomic nerves, draining to portal system, supplied by superior rectal artery.

Anal Cancer

Malignant tumor of the anal canal or margin, most commonly squamous cell carcinoma associated with high-risk human papillomavirus infection. Presents with bleeding, pain, or a mass; primary treatment is chemoradiation, with surgery reserved for residual or recurrent disease.

Anal Carcinoma

A malignant neoplasm arising from the epithelial lining of the anal canal or anal verge, overwhelmingly represented by squamous cell carcinoma (SCC; > 85% of cases). Strongly etiologically linked to persistent high-risk human papillomavirus infection (particularly HPV-16 and HPV-18), with elevated risk among immunocompromised individuals (HIV infection, solid organ transplant recipients) and individuals with a history of receptive anal intercourse. Presents clinically with rectal bleeding, anal pain, a palpable perianal mass, or pruritus. Unlike rectal adenocarcinoma, localized anal canal squamous

cell carcinoma is primarily managed with definitive organ-preserving chemoradiation (the Nigro protocol, combining 5-fluorouracil, mitomycin-C, and radiation therapy), reserving abdominoperineal resection for persistent or recurrent disease.

Alternate Names:

Anal cancer

Anal Condyloma

A benign wart-like squamous lesion of the anus or perianal skin, usually caused by low-risk HPV types. Multiple or recurrent lesions may coexist with high-grade intraepithelial neoplasia.

Anal Fissure

Linear tear in the distal anal mucosa usually posterior in the midline. Presents with painful defecation and bright red blood on wiping or in the toilet. Associated with constipation, hard stools, and chronic diarrhea. The internal anal sphincter spasm perpetuates the fissure by reducing blood flow to the posterior midline.

Alternate Names:

Anal Ulcer (Anal Fissure)

Anal Fistula

Abnormal tract connecting the anal canal to the perianal skin usually following a drained abscess. Presents with recurrent drainage, pain, and pruritus. Parks classification categorizes fistulae by relationship to the sphincter muscles. Diagnosed by examination under anesthesia and MRI.

Anal Intercourse

The insertion of the penis into the anus and rectum for sexual pleasure. The anal canal is rich in nerve endings, making it a sensitive erogenous zone. The rectum lacks natural lubrication, so artificial lubricants are essential to prevent mucosal trauma.

Anal Intraepithelial Neoplasia

A premalignant squamous lesion of the anal canal or perianal skin, usually related to persistent high-risk HPV infection. It ranges from low- to high-grade disease and may progress to anal squamous cell carcinoma.

Anal Itching

Pruritus around the anus. Common causes include irritation, moisture, dermatitis, hemorrhoids, infections, and occasionally pinworms or other disease. Gentle cleansing, keeping the area dry, and avoiding irritants may help; persistent, severe, bleeding, or painful symptoms require medical evaluation.

Alternate Names:

Itching Anal (Anal Itching) Pruritis Ani Rectal Itching

Anal Mucositis

Inflammation and ulceration of the anal mucous membrane, often after radiation or chemotherapy.

Anal Necrosis

Pathologic death and sloughing of perianal, anal canal, or perineal tissues. Most critically represented by Fournier gangrene, a synergistic polymicrobial necrotizing fasciitis involving aerobic and anaerobic flora (including *Escherichia coli*, *Bacteroides*, and *Streptococcus* species) commonly arising in immunocompromised, diabetic, or malnourished patients. Other etiologies include severe pelvic ischemic compromise, strangulated grade IV hemorrhoids with infarction, caustic chemical injury, and late-stage radiation necrosis. Characterized by severe perianal pain disproportionate to physical findings, violaceous skin discoloration, cutaneous bullae, subcutaneous crepitus from gas-forming organisms, and rapid progression to septic shock. Mandates emergent, radical surgical debridement of non-viable tissue, broad-spectrum parenteral antibiotics, and frequent surgical re-exploration.

Anal Pain

Pain localized to the anal verge, anal canal, or perineum (proctalgia). The clinical differential diagnosis is structured by onset, duration, and relationship to defecation: severe, sharp tearing pain during defecation followed by a lingering burn is pathognomonic for an acute anal fissure; constant, throbbing, worsening pain accompanied by fever and tender induration indicates an acute perianal or ischiorectal abscess requiring prompt incision and drainage; acute exquisite pain with a firm, violaceous sub-

cutaneous nodule indicates a thrombosed external hemorrhoid; sudden, severe, transient episodes of deep rectal spasm occurring without organic pathology characterize proctalgia fugax; and persistent pain associated with bleeding, tenesmus, or an indurated ulcer necessitates biopsy to rule out anal squamous cell carcinoma.

Alternate Names:

Proctalgia

Anal Papilla

A small normal projection of mucosa at the anal canal's dentate line, where columnar and squamous epithelium meet.

Anal Papillae

Multiple small mucosal projections at the dentate line of the anal canal; they are usually normal but may become enlarged or symptomatic.

Anal Sphincter

A muscular structure that controls passage of stool and gas from the anal canal. It includes the involuntary internal anal sphincter and the voluntary external anal sphincter.

Anal Ulcer (Anal Fissure)

Alternate index label for Anal Fissure.

Analgesia

Absence or reduction of pain without necessarily causing loss of consciousness.

Analog

A structure, substance, or process that resembles another and performs a similar function but is not necessarily chemically or structurally identical.

Analogous

Similar in function or purpose to another structure or process, despite having a different origin or structure.

Analysis

Systematic examination or interpretation of a spec-

imen, measurement, dataset, symptom pattern, or clinical problem.

Analysis Of Urine

Laboratory examination of urine for cells, blood, protein, glucose, microorganisms, and other substances.

Analytic Sensitivity

The smallest amount or concentration of an analyte that a laboratory method can reliably detect. It concerns measurement performance and differs from clinical sensitivity, which describes detection of disease in affected patients.

Anaphia

Loss or absence of the sense of touch or tactile sensation.

Anaphylactic Shock

Anaphylactic shock is severe anaphylaxis causing distributive shock, profound hypotension, and inadequate tissue perfusion. It may include airway swelling, bronchospasm, skin or mucosal findings, vomiting, abdominal pain, dizziness, collapse, or cardiac arrest. It is a medical emergency requiring immediate treatment.

Anaphylactoid Purpura

An older term for IgA vasculitis, an immune-mediated small-vessel vasculitis causing palpable purpura, often with joint, abdominal, or kidney involvement.

Anaphylaxis

A severe, potentially life-threatening systemic hypersensitivity reaction characterized by rapid onset of multisystem involvement, primarily affecting the skin, respiratory tract, cardiovascular system, and gastrointestinal tract.

Anaphylaxis During Anesthesia

A life-threatening systemic hypersensitivity reaction that may cause sudden hypotension, bronchospasm, cardiovascular collapse, and sometimes urticaria

or flushing. Common perioperative triggers include neuromuscular-blocking agents, antibiotics, chlorhexidine, and latex.

Anaplasia

A marked loss of normal cellular differentiation, often with pleomorphic cells, abnormal nuclei, disorganized architecture, and loss of specialized tissue function. It is a feature of some high-grade malignant tumors rather than a diagnosis by itself.

Anaplastic Astrocytoma

A high-grade astrocytic brain tumor with rapid growth and infiltrative behavior.

Anaplastic Ependymoma

An aggressive, high-grade tumor arising from cells lining the brain's ventricles or spinal canal.

Anaplastic Large Cell Lymphoma

A rare T-cell non-Hodgkin lymphoma characterized by large abnormal lymphocytes and expression of the CD30 marker; it may be systemic or skin-limited.

Alternate Names:

ALCL

Anaplastic Thyroid Cancer

A rare, rapidly growing, and highly aggressive thyroid cancer that often presents as a large fixed neck mass with dysphagia or airway compromise. Prognosis is poor; management is usually palliative.

Anaptic

Lacking a normal sense of touch or tactile sensation; the term is rarely used in modern clinical writing.

Anasarca

Severe, generalized edema involving large areas of the body, often associated with heart failure, kidney disease, liver disease, or profound hypoalbuminemia.

Anastomosis

A natural or surgically created connection between two tubular structures, such as blood vessels or

segments of the intestine.

Alternate Names:

Anastomosis, Intestinal

Anastomosis (Vascular)

A surgical or natural connection between blood vessels that permits blood to flow from one vessel to another.

Alternate Names:

Vascular

Anastomosis, Intestinal

Alternate index label; see *Anastomosis*.

Anastomotic Leak

Breakdown of a surgical anastomosis in the gastrointestinal tract causing spillage of luminal contents into the peritoneal cavity. Presents with fever, tachycardia, abdominal pain, leukocytosis, and peritonitis. Diagnosed by CT with oral or rectal contrast showing extraluminal contrast or fluid collection.

Anastrozole

An aromatase inhibitor that lowers estrogen production and is used mainly in hormone-receptor-positive breast cancer, particularly after menopause.

Anatomic Orientation Terms

Standard words used to describe the position and relationships of body structures, including superior, inferior, anterior, posterior, medial, lateral, proximal, distal, superficial, and deep.

Anatomic Snuffbox

A triangular depression on the lateral wrist that becomes visible when the thumb is extended. Tenderness there may suggest a scaphoid fracture.

Anatomical Axis

An imaginary straight line around which a body part moves or rotates.

Anatomical Conduit

A natural or surgically created passage that carries fluid, air, tissue, or another structure.

Anatomical Orifice

A natural opening into or through a body structure.

Alternate Names:

Body opening

Anatomical Planes

Hypothetical flat surfaces that pass through the body in anatomical position, used to slice the body for anatomical descriptions or medical imaging.

Anatomical Position

The standard reference position of the body used to describe anatomy, defined as standing erect, facing forward, with arms at the sides and palms facing forward.

Anatomical Snuffbox

Triangular depression on the dorsal wrist, bounded by extensor pollicis longus medially and extensor pollicis brevis/abductor pollicis longus laterally. Floor contains the scaphoid, trapezium, and radial artery. Tenderness after a fall on an outstretched hand suggests scaphoid fracture. The radial artery pulse is palpable here.

Anatomist

A specialist who studies the structure of the body.

Anatomy

Study of body structure. Regions: head, neck, thorax, abdomen, pelvis, extremities. Systems: skeletal, muscular, nervous, circulatory, respiratory, digestive, urinary, reproductive, endocrine, immune.

Anatomy Survey (Level II Ultrasound)

A comprehensive fetal ultrasound performed at 18-22 weeks to evaluate fetal anatomy and screen for structural anomalies.

Alternate Names:

Level II ultrasound; anatomy scan.

Anatomy, Gross

The study of body structures visible without a microscope, including organs, regions, bones, muscles, vessels, and their relationships.

ANCA-Associated Vasculitis

A group of small-vessel vasculitides associated with antineutrophil cytoplasmic antibodies. Includes granulomatosis with polyangiitis (c-ANCA/PR3), microscopic polyangiitis (p-ANCA/MPO), and eosinophilic granulomatosis with polyangiitis (p-ANCA/MPO with asthma/eosinophilia).

Ancillary Test (Pathology)

A test performed in addition to routine morphology to refine a pathologic diagnosis. Examples include IHC, flow cytometry, cytogenetics, molecular testing, special stains, and electron microscopy.

Alternate Names:

Pathology

Ancylostoma (Hookworm)

Nematode infestation by *Ancylostoma duodenale* (Old World) or *Necator americanus* (New World), common in tropical and subtropical regions where sanitation is poor.

Alternate Names:

Hookworm

Andersen-Tawil Syndrome

A rare autosomal dominant channelopathy (Periodic Paralysis Type 3) caused in approximately 60% of cases by loss-of-function mutations in the *KCNJ2* gene on chromosome 17q24, which encodes the inward-rectifier potassium channel Kir2.1. Clinically characterized by a pathognomonic cardinal triad: (1) episodic flaccid muscle weakness and periodic paralysis (hypokalemic, hyperkalemic, or normokalemic); (2) cardiac electrical instability with prolonged QTc intervals, prominent U waves, and frequent ventricular arrhythmias (particularly bidirectional or polymorphic ventricular tachycardia, which is often surprisingly well tolerated); and (3) characteristic physical dysmorphisms including low-set ears, ocular hypertelorism, broad forehead, micrognathia, mandibular hypoplasia, clinodactyly, syndactyly, and short stature.

Alternate Names:

ATS; Periodic Paralysis Type 3; Andersen Syndrome

Anderson Disease

An exceptionally rare, severe autosomal recessive disorder of glycogen metabolism (Glycogen Storage Disease Type IV [GSD IV]) caused by pathogenic mutations in the *GBE1* gene, encoding 1,4-alpha-glucan branching enzyme. Deficiency leads to the intracellular accumulation of abnormal, poorly soluble glycogen with few branching points that resembles amylopectin (polyglucosan bodies). Typically manifests in early infancy with failure to thrive, hepatomegaly, and progressive hepatic fibrosis rapidly culminating in biliary cirrhosis, portal hypertension, and liver failure, often requiring early liver transplantation.

Alternate Names:

Glycogen Storage Disease Type IV (GSD IV); Amylopectinosis

Androgen

A sex hormone, such as testosterone, that contributes to male reproductive development and other body functions.

Androgen Deficiency

Hypogonadism in males: reduced testosterone production causing symptoms and signs. Causes: primary (testicular—Klinefelter, mumps orchitis, chemotherapy, trauma) and secondary (pituitary/hypothalamic—prolactinoma, iron overload, pituitary lesions, opioids, obesity, aging).

Androgen Excess

Abnormally high androgen activity or concentration, which can cause acne, excess hair, or menstrual changes.

Alternate Names:

Hyperandrogenism

Androgen Insensitivity Syndrome

A difference of sex development in which the cells of a genetically male fetus cannot respond to androgens because of a defect in the androgen receptor. Affected individuals have typical female external genitalia and breast development but no uterus, and testes that may lie in the groin or abdomen; diagnosis and care require specialist endocrine and

genetic assessment.

Androgen Insufficiency

Inadequate androgen production or activity for a person's physiologic needs.

Alternate Names:

Androgen deficiency

Androgen Receptor

A cellular protein that binds testosterone or related hormones and mediates their effects.

Androgen Suppression

Reduction of androgen production or action, used in selected hormone-sensitive conditions.

Alternate Names:

Androgen deprivation

Androgenetic Alopecia

The most common cause of hair loss, caused by progressive miniaturization of genetically susceptible hair follicles under androgen influence. Male pattern commonly affects the frontal hairline and vertex; female pattern commonly causes diffuse crown thinning with relative preservation of the frontal hairline.

Androgenic

Relating to androgens or producing masculine sexual development and other androgen-mediated effects.

Android Pelvis

A relatively heart-shaped or funnel-shaped pelvic configuration with a narrower midpelvis, which may increase the risk of obstructed labor compared with a gynecoid pelvis.

Andropause

Age-related testosterone decline in men. Symptoms: fatigue, decreased libido, depression. Treatment controversial.

Anejaculation

Inability to release semen during orgasm or sexual

activity. Causes include nerve or spinal disease, pelvic surgery, medicines, diabetes, hormonal problems, and psychological factors; it may affect fertility even when erections and orgasm are preserved.

Anemia

A reduction in the oxygen-carrying capacity of the blood, defined by a hemoglobin concentration below the normal range for age and sex: less than 13.5 g/dL in adult males, less than 12.0 g/dL in adult females.

Alternate Names:

Blood Low Red Cell Count (Anemia)

Anemia (Cancer)

Anemia occurring in a person with cancer, caused by the cancer itself, bleeding, inflammation, nutritional deficiency, or treatment. Evaluation identifies the cause; management may include treating the underlying problem, iron or other replacement, transfusion, or selected medicines.

Alternate Names:

Cancer

Anemia Caused By Low Iron - Children

Iron-deficiency anemia in a child caused by insufficient iron intake, increased requirements, blood loss, or impaired absorption. It reduces hemoglobin production and may cause pallor, fatigue, irritability, or impaired development.

Anemia Caused By Low Iron - Infants And Toddlers

Iron-deficiency anemia in an infant or toddler, commonly related to inadequate dietary iron, excessive cow's milk intake, increased growth requirements, or blood loss. It may cause pallor, fatigue, irritability, and developmental effects.

Anemia In Pregnancy

Anemia occurring during pregnancy, commonly because of iron deficiency or the normal plasma-volume expansion of pregnancy. Assessment considers hemoglobin, mean corpuscular volume, iron status, folate, vitamin B12, bleeding, and other causes.

Alternate Names:

Gestational anemia; pregnancy-associated anemia.

Anemia Of Chronic Disease

Anemia of inflammation. Normocytic or microcytic. Low iron, low TIBC, normal/high ferritin.

Anemia Of Inflammation

An anemia caused by chronic infection, inflammatory disease, or cancer, involving impaired iron availability, reduced erythropoietin response, and shortened red-cell survival.

Anemic

Having anemia, meaning reduced red-cell mass or hemoglobin and therefore reduced blood oxygen-carrying capacity.

Anencephaly

A serious birth defect in which much of the brain and skull do not develop. It happens when the upper part of the neural tube does not close early in pregnancy. It is usually found before birth and is almost always fatal shortly before or after birth.

Anergy

State of T-lymphocyte non-responsiveness to nominal antigen despite prior sensitization, reflecting functional inactivation rather than cell death.

Anesthesia

The use of medications to prevent sensation, particularly pain, during medical procedures or surgery.

Anesthesia For Coagulopathy

Coagulopathy can increase bleeding risk during an operation or invasive procedure. Anesthetic planning considers the cause, bleeding history, platelet count, coagulation tests, fibrinogen, the procedure, and available correction or replacement therapies.

Anesthesia For Epilepsy

Patients with epilepsy generally require continuation of antiseizure medication perioperatively because missed doses increase seizure risk. Propofol

and benzodiazepines are anticonvulsant; high concentrations of some volatile agents may produce epileptiform activity.

Anesthesia For Hemophilia

Hemophilia A results from factor VIII deficiency and hemophilia B from factor IX deficiency. Perioperative care is coordinated with hematology and uses factor replacement or, in selected mild hemophilia A cases, desmopressin.

Anesthesia For Neurosurgery

Neuroanesthesia aims to maintain cerebral perfusion pressure ($CPP = \text{mean arterial pressure} - \text{intracranial pressure}$), optimize brain relaxation, and minimize secondary brain injury. Induction agents (propofol, etomidate) reduce cerebral metabolic rate and intracranial pressure.

Anesthesia For Pheochromocytoma

Anesthetic care for pheochromocytoma requires control of catecholamine effects before surgery, careful management of blood pressure and heart rate, and preparation for rapid hemodynamic changes.

Anesthesia For Porphyria

Anesthetic care in acute porphyria requires avoidance of medicines known to precipitate attacks and attention to nutrition, pain, and metabolic stability. The specific drug plan should be checked against a current porphyria-safety reference.

Anesthesia Gas Analyzers

Anesthesia gas analyzers measure delivered and exhaled concentrations of oxygen, carbon dioxide, and volatile anesthetic agents. The oxygen analyzer measures inspired oxygen concentration; pulse oximetry measures peripheral hemoglobin oxygen saturation.

Anesthesia Informatics

Anesthesia informatics integrates electronic health records, decision support systems, and data analytics into clinical practice. Automated documentation reduces charting time. Clinical decision support alerts for drug interactions and dosing errors. Big data analytics identify patterns in anesthesia out-

comes.

Anesthesia Machine

The anesthesia machine delivers precise mixtures of oxygen, air, and volatile anesthetics to the patient.

Anesthesia Monitoring

Continuous assessment of vital functions during anesthesia, typically including oxygenation, ventilation, circulation, temperature, and depth of anesthesia as clinically indicated.

Anesthesia Pharmacogenomics

Anesthesia pharmacogenomics studies genetic variations affecting drug response. Pseudocholinesterase deficiency causes prolonged succinylcholine paralysis. CYP2D6 polymorphisms affect codeine metabolism to morphine. Malignant hyperthermia susceptibility is linked to RYR1 gene mutations.

Anesthesiologist

A physician who provides anesthesia and manages pain and vital functions during procedures.

Anesthesiology

The medical specialty concerned with anesthesia, perioperative care, and pain management.

Aneuploidy

The presence of an abnormal number of chromosomes in a cell, with either extra or missing copies of whole chromosomes, such as trisomy 21. It results from errors in cell division and is a major cause of miscarriage, congenital syndromes, and chromosomal disorders identified by prenatal testing.

Aneurysm

A localized weakening and enlargement of a blood vessel. Aneurysms may occur in arteries or, rarely, veins and may rupture, dissect, or form emboli depending on their location and cause.

Aneurysm Clip

Surgical clip occluding aneurysm neck. Materials: titanium, stainless steel. Indications: intracranial

aneurysm. Placed via craniotomy. Complications: clip migration, vessel occlusion, rebleeding.

Aneurysm Coils

Flexible coils, often made of platinum, placed inside an aneurysm to promote clotting and reduce blood flow into the aneurysm.

Aneurysm In The Brain

A brain aneurysm is a focal weakness and ballooning of a cerebral-artery wall; rupture causes subarachnoid hemorrhage and sudden severe headache.

Aneurysm Repair

Surgical or catheter-based treatment that excludes or replaces a weakened, dilated section of a blood vessel when the risk of rupture or other complications justifies intervention.

Aneurysms

Abnormal bulges or enlargements in weakened areas of blood-vessel walls that may enlarge, clot, dissect, or rupture.

Angelman Syndrome

A genetic disorder caused by loss of function of the UBE3A gene on chromosome 15 (maternal deletion or paternal uniparental disomy). Features: severe intellectual disability, absent speech, seizures, ataxia, and frequent laughter. Characteristic happy demeanor with hand-flapping movements.

Anger

An emotional response involving displeasure or perceived threat, ranging from mild irritation to intense hostility; persistent or harmful anger may require clinical assessment.

Angiitis

See *Vasculitis*.

Angiitis (Vasculitis)

Alternate index label for *Vasculitis*.

Angina

Chest discomfort caused by inadequate oxygen-rich blood flow to the heart muscle, usually from coronary artery disease.

Angina - When You Have Chest Pain

Angina with chest pain reflects transient myocardial ischemia; new, severe, prolonged, or rest pain may signal acute coronary syndrome and requires emergency assessment.

Angina Pectoris

Chest discomfort caused by myocardial ischemia without infarction. Stable angina occurs predictably with exertion and resolves with rest. Unstable angina occurs at rest or with minimal exertion and is an acute coronary syndrome. Prinzmetal (variant) angina occurs at rest due to coronary vasospasm.

Angina Symptoms

Angina is chest discomfort caused by inadequate blood flow to heart muscle, commonly from coronary artery disease. Pressure or squeezing may spread to the arm, jaw, back, or shoulder and can occur with exertion; new, severe, or resting symptoms require emergency assessment.

Alternate Names:

Ischemic Chest Pain (Angina Symptoms)

Angioblastoma

See *Hemangioblastoma*.

Angiocardiology

Imaging of the heart and major vessels after injection of contrast material.

Angiodysplasia

Dilated, fragile blood vessels in the gastrointestinal mucosa or submucosa, often occurring in the right colon of older adults. They may cause intermittent or recurrent gastrointestinal bleeding and anemia.

Angiodysplasia Of The Colon

A degenerative abnormality of small blood vessels in the colon that can cause intermittent or recurrent

lower gastrointestinal bleeding, often without pain. Diagnosis and treatment depend on the bleeding pattern and endoscopic findings.

Angioedema

Rapid swelling of the deep dermis/subcutaneous and submucosal tissues (lips, eyelids, tongue, throat, hands, GI tract) due to mast cell mediator release (histamine—allergic/idiopathic, urticaria associated) or bradykinin-mediated (hereditary angioedema—C1 esterase inhibitor deficiency; ACE-inhibitor-induced).

Angioedema (Hives)

Alternate index label for Hives.

Angioedema And Hives

See *Hives and angioedema*.

Angioedema Hereditary (Hereditary Angioedema HAE)

Alternate index label for Hereditary Angioedema HAE.

Angiofibroma

A benign tumor composed of blood vessels and fibrous tissue.

Angiofollicular Lymph Node Hyperplasia

See *Castleman disease*.

Angiogenesis

Formation of new blood vessels. Tumors induce angiogenesis for growth. Targeted by anti-angiogenic drugs (bevacizumab).

Angiogenesis Inhibition

Suppression of new blood-vessel formation, a strategy used in some cancer treatments.

Alternate Names:

Antiangiogenic therapy

Angiogram

An image of blood vessels, usually obtained after contrast material is injected, used to identify narrow-

ing, blockage, aneurysm, malformation, or bleeding.

Angiography

Invasive imaging of blood vessels by injecting contrast material under fluoroscopy. Coronary angiography (cardiac catheterization): assesses coronary artery stenosis, guides PCI. Cerebral angiography: evaluates aneurysms, AVMs, and vasculitis. Peripheral angiography: evaluates PAD and guides intervention.

Angioid Streak

Breaks in a tissue layer beneath the retina that appear as irregular dark lines.

Angiomyofibrolastoma

A benign mesenchymal tumor of the vulvovaginal region, usually presenting as a small, well-circumscribed painless mass in reproductive-age or middle-aged women.

Angiomyolipoma

A benign renal tumor composed of blood vessels, smooth muscle, and fat; it is associated with tuberous sclerosis and may cause retroperitoneal hemorrhage.

Angioneurotic Edema, Hereditary

Hereditary angioedema, an inherited bradykinin-mediated disorder causing recurrent swelling of the skin, gastrointestinal tract, or airway, usually without hives.

Angiopathy

Disease or abnormality of blood vessels, including changes caused by diabetes, inflammation, atherosclerosis, trauma, or inherited disorders.

Angioplasty

A procedure that widens a narrowed blood vessel, usually with a balloon and sometimes a stent. It is used most often for coronary or peripheral arterial disease.

Angioplasty And Stent Placement - Heart

A catheter-based procedure that widens a narrowed coronary artery, usually with an inflatable balloon, and may place a stent to help keep the artery open and improve blood flow to the heart muscle.

Angioplasty And Stent Placement - Peripheral Arteries

A catheter-based procedure that widens a narrowed peripheral artery, usually with an inflatable balloon, and may place a stent to improve blood flow to an affected limb or organ.

Angioplasty Balloon (Vascular)

An inflatable catheter balloon used to widen a narrowed blood vessel or expand a stent.

Alternate Names:

Vascular

Angiosarcoma

A rare, aggressive malignant tumor of blood vessel or lymphatic endothelial cells. Common sites include skin, breast, liver, and bone. Treatment typically involves wide surgical excision with chemotherapy and radiation.

Angiosarcoma Of The Breast

A malignant vascular tumor of the breast that may arise primarily or, more rarely, after radiation or chronic lymphoedema. It often presents as an enlarging mass or bruise-like skin discoloration and requires oncologic management.

Angiosarcoma Of The Heart

A rare aggressive primary cardiac sarcoma, most often arising in the right atrium and potentially causing obstruction, pericardial effusion, arrhythmia, or metastases. Diagnosis requires imaging and tissue evaluation.

Angiotensin

A family of peptides in the renin–angiotensin system that regulate blood pressure and fluid balance, with angiotensin II causing vasoconstriction and stimulating aldosterone release.

Angiotensin Converting Enzyme

An enzyme that converts angiotensin I to vasoconstricting angiotensin II and also breaks down bradykinin; it is inhibited by ACE-inhibitor medicines.

Angiotensin I

An inactive precursor that is converted to angiotensin II in the renin-angiotensin system.

Angiotensin II

A hormone that narrows blood vessels and promotes sodium retention, thereby increasing blood pressure.

Angiotensin Receptor Blockers

A class of cardiovascular drugs that selectively block the angiotensin II type 1 receptor, preventing the binding of angiotensin II and its effects including vasoconstriction, aldosterone secretion, and sympathetic activation.

Angle-Closure Glaucoma

Glaucoma caused by narrowing or closure of the drainage angle, which can abruptly raise eye pressure and threaten vision; acute attacks require emergency treatment.

Anhedonia

The inability to experience pleasure from activities that were previously enjoyable. It is a core symptom of major depressive disorder and a negative symptom of schizophrenia. It is mediated by dysfunction in the brain's reward circuitry, particularly the mesolimbic dopamine system.

Anhidrosis

Absent or greatly reduced sweating. It may result from nerve damage, skin disease, dehydration, medications, or inherited conditions. Inability to sweat can cause dangerous overheating, especially during exercise or hot weather.

Animal-Assisted Therapy

A structured intervention in which trained animals are incorporated into care to support defined thera-

peutic goals. It may be used to support engagement, mood, communication, or activity when appropriate safeguards are in place.

Anion

Anion is an ion with a net negative electrical charge, such as chloride, bicarbonate, or phosphate in body fluids.

Anion Gap

A calculated estimate of unmeasured serum anions, commonly sodium minus chloride and bicarbonate. An increased anion gap supports the presence of unmeasured acids in metabolic acidosis, but interpretation must account for albumin concentration and laboratory method.

Anisocoria

Unequal pupil size. It may be physiologic or caused by disorders of the eye, autonomic nerves, or brain; sudden anisocoria with headache, eye pain, or neurologic symptoms requires urgent assessment.

Anisocytosis

Increased variation in red blood cell size on a peripheral blood film (e.g., macrocytes with microcytes together), not a diagnosis in itself. Derived from the Greek "anisokytosis" (unequal cells). Quantified by an elevated red cell distribution width (RDW).

Anisometropia

A difference in the refractive power of the two eyes, which can cause unequal image focus, blurred vision, or amblyopia.

Ankle

The joint connecting the lower leg to the foot.

Ankle Arthroplasty

Surgical replacement of the ankle joint with a prosthesis, usually for severe symptomatic ankle arthritis. Suitability depends on deformity, bone quality, infection risk, activity level, and the condition of the surrounding joints.

Ankle Arthroscopy

A minimally invasive surgical technique using small incisions and an endoscope to visualize and treat intra-articular ankle pathology, including loose bodies, osteochondral lesions, synovitis, and impingement. It offers faster recovery compared with open arthrotomy.

Ankle Bone

A bone forming the ankle joint, especially the talus or the distal tibia and fibula that form the ankle mortise.

Ankle Dislocated (Dislocated Ankle (Ankle Dislocation))

Alternate index label for Dislocated Ankle (Ankle Dislocation).

Ankle Dislocation

Complete displacement of the ankle joint surfaces, usually resulting from a high-energy injury and often accompanied by a fracture, ligament damage, or neurovascular compromise. It causes severe pain, deformity, and inability to bear weight. The joint requires urgent reduction, imaging, and immobilization, with surgical management when instability or associated fractures demand it.

Alternate Names:

Dislocated Ankle (Ankle Dislocation)

Ankle Fracture

A break in one or more bones forming the ankle joint.

Ankle Fractures

Ankle fractures: Weber classification (fibular fracture relative to syndesmosis): A (below syndesmosis, stable), B (at syndesmosis, variable stability), C (above syndesmosis, unstable). Lauge-Hansen: supination-external rotation (SER, most common), supination-adduction, pronation-external rotation, pronation-abduction.

Ankle Joint

Uniaxial hinge synovial joint between distal tibia,

lateral malleolus, and superior talus, allowing dorsiflexion/plantarflexion. Medial (deltoid) ligament resists eversion; lateral complex resists inversion. The ankle mortise provides bony stability.

Ankle Pain

Pain in or around the ankle, commonly caused by sprain, fracture, tendinitis, arthritis, bursitis, or infection. Inability to bear weight, deformity, marked swelling, or fever requires prompt assessment.

Ankle Pain And Tendinitis

Ankle pain may result from sprain, fracture, arthritis, tendon inflammation, nerve disease, infection, or referred pain. Tendinitis causes pain along an affected tendon and often worsens with loading; swelling, deformity, inability to bear weight, fever, or numbness warrants evaluation.

Ankle Replacement

Total ankle arthroplasty, the surgical replacement of the ankle joint with a prosthetic implant for severe arthritis when joint-sparing options are insufficient. It aims to preserve motion and reduce pain; outcomes depend on deformity correction, alignment, and implant selection.

Ankle Sprain

An injury to the ligaments surrounding the ankle joint, most commonly involving the lateral ligament complex (anterior talofibular ligament [ATFL, most commonly injured], calcaneofibular ligament [CFL], and posterior talofibular ligament [PTFL]) from an inversion mechanism.

Ankle Swelling

Enlargement around the ankle caused by injury, inflammation, infection, venous disease, lymphatic problems, or fluid retention. One-sided sudden swelling with pain, warmth, or shortness of breath may indicate a blood clot or infection and requires urgent assessment; swelling after trauma may require evaluation for fracture.

Ankle Swollen (Swollen Ankles And

Swollen Feet)

Alternate index label for Swollen Ankles and Swollen Feet.

Ankle-Brachial Index

The ratio of ankle systolic blood pressure to brachial systolic blood pressure. A low result supports peripheral arterial disease, while a very high result may indicate noncompressible arteries.

Ankle-Foot Orthosis

A brace that supports the ankle and foot and helps control alignment, stability, or foot drop.

Ankle-Foot Orthosis

A brace that supports the ankle and foot, helps control alignment and motion, and may improve walking in people with weakness, spasticity, instability, or foot drop.

Ankyloglossia

An abnormally short or tight lingual frenulum that restricts tongue movement.

Alternate Names:

Tongue-tie

Ankyloglossia (Tongue-Tie)

Short or thick lingual frenulum restricting tongue movement. May cause breastfeeding difficulties, speech problems, oral hygiene issues. Treatment: frenotomy (simple release) or frenuloplasty.

Alternate Names:

Tongue-Tie

Ankylose

To become stiff or immobile at a joint, sometimes because the joint surfaces become fused.

Ankylosing

Causing or involving progressive stiffness and fixation of a joint, often through fibrous or bony fusion.

Ankylosing Spondylitis

A long-term form of arthritis that mainly affects the joints between the spine and pelvis and the spine

itself. It causes back pain and stiffness. Over time, new bone may form around the spine, causing some spinal bones to join together and making the back less flexible. It can also affect other joints, the eyes, skin, or bowel.

Alternate Names:

Arthritis Ankylosing Spondylitis (Ankylosing Spondylitis)

Ankylosis

Abnormal stiffness or fusion of a joint. In dentistry, it may refer to fusion of a tooth root to the surrounding bone.

Ankyrin

A structural protein that links membrane proteins to the cell's internal cytoskeleton.

Annihilation

Positron-electron annihilation producing two 511 keV photons in opposite directions.

Annular Pancreas

A congenital anomaly where a ring of pancreatic tissue completely or partially encircles the second portion of the duodenum, causing duodenal obstruction. It results from failure of the ventral pancreatic bud to completely rotate around the duodenum during embryogenesis.

Annulus

A ring-shaped structure or band; examples include the annulus fibrosus around an intervertebral disk and the fibrous rings around heart valves.

ANOCA

Angina with non-obstructive coronary arteries: angina symptoms without a flow-limiting obstructive stenosis in the epicardial coronary arteries. Causes may include coronary microvascular dysfunction or vasospastic angina, and further evaluation can help identify the mechanism.

Anodontia

Complete absence of teeth. Rare, associated with

ectodermal dysplasia.

Anogenital Tract

The connected anatomic region comprising the anus, perianal tissues, genital structures, and related mucosal surfaces.

Anomalous Anatomy

A structural variation or congenital abnormality in the expected arrangement, number, origin, or course of an organ, vessel, duct, or other body structure. It may be incidental or clinically important during diagnosis, surgery, or interventional procedures.

Anomalous Aortic Origin Of A Coronary Artery

A high-risk congenital coronary anomaly in which a coronary artery arises from the inappropriate opposite or non-coronary sinus of Valsalva and traverses between the ascending aorta and the main pulmonary artery (interarterial course). Compression of the anomalous vessel between the great arteries during exercise, an acute takeoff angle, or an intramural slit-like lumen can produce myocardial ischemia, exertional syncope, and sudden cardiac arrest in children and young athletes; diagnosis is established with coronary CT angiography or cardiac MRI, and surgical unroofing or reimplantation is indicated for high-risk anatomy.

Alternate Names:

AAOCA; Anomalous Coronary Artery from Opposite Sinus; ACAOS

Anomalous Left Coronary Artery From The Pulmonary Artery

A rare congenital anomaly in which the left coronary artery arises from the pulmonary artery instead of the aorta. It can cause myocardial ischemia, heart failure, arrhythmia, or sudden death, particularly in infancy or exertion.

Anomaly

A deviation from the usual structure, development, function, or measurement of the body; it may be harmless or clinically significant.

Anomaly Scan

A detailed routine structural ultrasound performed between 18 and 22 weeks of gestation to assess fetal anatomical development, placental localization, amniotic fluid volume, and screening for major congenital anomalies, including neural tube defects, congenital cardiac anomalies, abdominal wall defects, and renal tract abnormalities.

Alternate Names:

Mid-Pregnancy Ultrasound; 20-Week Scan; Fetal Anomaly Scan

Anomia

Difficulty retrieving or producing a known word, especially the name of a familiar object or person, despite knowing its meaning. It is a common language symptom of aphasia and may occur after stroke or with other neurologic disorders.

Anophthalmia

A birth defect in which one or both eyes do not develop. It may occur with other birth defects and can affect the growth of the eye socket and face. Care may include an eye specialist, socket treatment, and a prosthetic eye.

Anotia

A birth defect in which the outer ear is completely absent. The ear canal may also be absent, and hearing may be reduced.

Anorchia

Complete absence of testicular tissue, present from birth or caused by loss of previously formed testes. It results in absent or severely reduced testosterone production and infertility unless treated with appropriate hormone replacement.

Anorectal Abscess

A localized collection of pus in the tissues around the anus or rectum, usually caused by infection of an anal gland. It commonly causes severe anal pain, swelling, fever, and difficulty sitting. Prompt incision and drainage is the standard treatment.

Anorectal Fistula

An abnormal channel connecting the anus or rectum with nearby skin or another organ.

Alternate Names:

Anal fistula

Anorectal Infection

Infection of the anus, rectum, or nearby tissues. Causes include sexually transmitted infections, abscesses, bowel bacteria, and inflammatory disease; pain, swelling, discharge, fever, or difficulty passing stool may occur and an abscess may need drainage.

Anorectal Malformation

A congenital abnormality in which the anus or rectum does not develop normally or does not connect appropriately to the perineum. Types vary in anatomy and severity and usually require specialist surgical assessment.

Anorectal Manometry

Measurement of pressures in the anal canal and rectum to assess sphincter function and rectal sensation. A pressure-sensing catheter is inserted into the rectum and slowly withdrawn through the anal canal. Measures resting and squeeze pressures reflecting internal and external sphincter function respectively.

Anorexia

Loss of appetite, distinct from anorexia nervosa (the eating disorder).

Anorexia Nervosa

An eating disorder characterized by restriction of energy intake, intense fear of weight gain, and distorted body image. Highest mortality of any psychiatric disorder. Features: emaciation, lanugo, bradycardia, hypotension, amenorrhea, and electrolyte abnormalities. Diagnosis by DSM-5 criteria.

Alternate Names:

Eating Disorders, Anorexia

Anorgasmia

Persistent or recurrent delay in, or absence of, or-

gasm following a normal sexual excitement phase.

Anorgasmia In Women

Persistent or recurrent difficulty achieving orgasm despite adequate sexual stimulation, causing personal distress.

Anorgasmy

Inability to achieve orgasm despite adequate sexual stimulation; possible contributors include medications, hormonal or neurologic disorders, psychological factors, and relationship difficulties.

Anoscopy

Examination of the anal canal with a short lighted instrument called an anoscope. It is used to evaluate rectal bleeding, hemorrhoids, fissures, fistulas, and other anorectal disorders.

Anoscopy And High-Resolution Anoscopy

An examination of the anal canal and lower rectum using an anoscope; high-resolution anoscopy provides magnified views for evaluation of abnormal tissue.

Anosmia

Complete loss of the sense of smell. Causes include post-viral olfactory dysfunction, chronic rhinosinusitis with polyps, head trauma, neurodegenerative disease, congenital disorders such as Kallmann syndrome, medications, and toxic exposure.

Anosognosia

Denial or unawareness of one's own neurologic deficit, classically hemiplegia following a nondominant (right) parietal lobe lesion. Associated with right middle cerebral artery strokes and sometimes with neglect of the left side of space.

Anovulation

Failure to release an egg from the ovary during the menstrual cycle. It may cause irregular or absent periods, infertility, or abnormal bleeding and can result from hormonal disorders, excessive exercise, low body weight, stress, or polycystic ovary syndrome.

Anoxia

Complete absence of oxygen supply to tissues, distinguished from hypoxia (reduced oxygen). Anoxia causes rapid cell death, particularly in highly aerobic tissues (brain, heart, kidneys).

Antacids

Medications that neutralise gastric acid, providing rapid symptomatic relief in dyspepsia, gastroesophageal reflux, and peptic ulcer disease. Commonly contain aluminium hydroxide, magnesium hydroxide, calcium carbonate, or sodium bicarbonate.

Antagonist

A substance, drug, muscle, or other factor that opposes or blocks the action of another; a receptor antagonist binds without activating the receptor.

Antecubital Fossa

Triangular depression on the anterior elbow, bounded laterally by brachioradialis, medially by pronator teres, and superiorly by a line connecting the epicondyles. Floor formed by brachialis and supinator. Contains biceps tendon, brachial artery, and median nerve. Primary site for venipuncture and blood pressure measurement.

Antegrade

Moving or occurring in the normal forward direction of flow or conduction, such as blood flow away from the heart or electrical conduction through the usual pathway.

Antemortem Injury

Injury inflicted upon a body prior to death, characterized microscopically and macroscopically by physiological tissue responses including hemorrhage, blood clotting, inflammatory cell infiltration, tissue edema, and active healing reactions (differentiating it from postmortem artifact).

Antenatal

The period before birth, also called prenatal. The medical supervision and screening provided to pregnant women to monitor the health of the mother

and fetus, detect complications early, and provide health education.

Alternate Names:
prenatal.

Antepartum Fetal Surveillance

The monitoring of fetal well-being in the third trimester for high-risk pregnancies. Methods: fetal kick counts (daily maternal counting of fetal movements), non-stress test (NST), biophysical profile (BPP, AFI, fetal tone, movement, breathing), and Doppler (umbilical artery, MCA).

Antepartum Hemorrhage

Bleeding from the genital tract after the threshold of fetal viability and before birth. Important causes include placenta previa, placental abruption, vasa previa, and local cervical or vaginal lesions; maternal stabilization and fetal assessment are priorities.

Anterior

Relating to the front of the body or positioned closer to the front.

Anterior Chamber

The fluid-filled space between the cornea and iris of the eye.

Anterior Chamber Angle

The anterior chamber angle is the anatomical region where the cornea and iris meet, and it contains the trabecular meshwork, the primary drainage structure for aqueous humor from the eye.

Anterior Cord Syndrome

A spinal-cord injury syndrome causing impaired motor function and loss of pain and temperature sensation below the lesion, with relative preservation of vibration and position sense.

Anterior Cruciate Ligament

A strong ligament inside the knee that runs from the femur to the tibia and limits forward movement and rotation of the tibia.

Anterior Cruciate Ligament Injury

A sprain or tear of the ligament that limits forward movement and rotation of the tibia relative to the femur. It commonly occurs during sudden stopping, pivoting, or landing and may cause a pop, rapid swelling, pain, and knee instability.

Anterior Cruciate Ligament Injury

Tear of the ACL, a common sports injury (non-contact deceleration, pivoting, or hyperextension). Presents with a pop, immediate swelling (hemarthrosis), and giving way of the knee. Positive Lachman test (most sensitive), anterior drawer test, and pivot shift test.

Anterior Cruciate Ligament Tear

ACL tear: common knee injury, often non-contact pivoting sports. Symptoms: popping sound, swelling, instability, inability to continue play. Diagnosis: Lachman test, anterior drawer, MRI. Treatment: surgical reconstruction (autograft/allograft), rehabilitation.

Anterior Knee Pain

Pain localized to the front of the knee around or behind the patella, often worsened by stair climbing, squatting, or prolonged sitting. Common causes include patellofemoral pain syndrome, chondromalacia patellae, patellar tendinopathy, and plica syndrome. Evaluation includes physical examination, imaging, and assessment of biomechanical factors.

Anterior Myocardial Infarction

A myocardial infarction involving the anterior wall of the left ventricle, usually because of obstruction in the left anterior descending coronary artery.

Alternate Names:

Anterior heart attack; anterior MI.

Anterior Pituitary

The glandular portion of the pituitary that secretes ACTH, TSH, growth hormone, prolactin, FSH, and LH in response to hypothalamic releasing hormones. Its hormones regulate adrenal, thyroid, growth, lactation, and reproductive function.

Anterior Tibial Artery

Arises from the popliteal artery, passes through the anterior intermuscular septum to the anterior compartment, continuing to the ankle as the dorsalis pedis artery. Supplies anterior leg and dorsal foot muscles. Palpable anterior to the lateral malleolus as the dorsalis pedis pulse.

Anterior Uveitis

Anterior uveitis, also known as iritis or iridocyclitis when the ciliary body is involved, is inflammation of the anterior segment of the eye involving the iris and/or ciliary body, and it is the most common form of uveitis.

Alternate Names:

iritis or iridocyclitis when the ciliary body is.

Anterior Vaginal Prolapse

See *Anterior vaginal prolapse (cystocele)*.

Anterior Vaginal Prolapse (Cystocele)

Bulging of the front wall of the vagina caused by weakening of the tissue supporting the bladder, sometimes producing urinary symptoms.

Alternate Names:

Bladder Prolapse Prolapsed Bladder

Anterior Vaginal Wall Repair

Surgical repair of the anterior vaginal wall to correct cystocele (bladder prolapse), in which the bladder bulges into the vaginal canal. The procedure restores pelvic-floor support and may be performed with or without mesh depending on the defect and surgeon judgment.

Anterior/Posterior Drawer Test

A clinical test of knee stability. Anterior displacement assesses the anterior cruciate ligament, and posterior displacement assesses the posterior cruciate ligament.

Anterolisthesis

Forward displacement of one vertebral body relative to the vertebra below it, usually in the spine.

Anteroposterior

Directed from front to back; an anteroposterior radiographic view passes through the body from the anterior surface toward the posterior surface.

Anteroposterior

Directed from the front of the body toward the back; an AP radiograph is made with the X-ray beam entering anteriorly and exiting posteriorly.

Anthrax

A serious infectious disease caused by the gram-positive bacterium *Bacillus anthracis*, which forms highly resistant spores that can persist in the environment for decades, and it is primarily a zoonotic disease of herbivores (cattle, sheep, goats) that can be transmitted to humans through contact with infected animals or their products.

Anthrax Blood Test

A laboratory test that detects *Bacillus anthracis* antigens, antibodies, or nucleic acid in blood or other specimens. It is used to confirm suspected anthrax infection, which may present as cutaneous, inhalational, or gastrointestinal disease with potential for severe toxemia.

Anthropometry

Body measurements: weight, height, BMI, waist, hip, skinfold thickness. Assessment of nutritional status and body composition.

Anti-Anxiety Agents

Medicines or other treatments used to reduce clinically significant anxiety.

Alternate Names:

Anxiolytics

Anti-CCP

Alternate index label; see *Anti-Cyclic Citrullinated Peptide (Anti-CCP) Antibody*.

Anti-CCP Antibody

Alternate index label; see *Anti-Cyclic Citrullinated Peptide (Anti-CCP) Antibody*.

Anti-Citrullinated Peptide Antibody

Alternate index label; see *Anti-Cyclic Citrullinated Peptide (Anti-CCP) Antibody*.

Anti-Cyclic Citrullinated Peptide (Anti-CCP) Antibody

An autoantibody measured in a blood test that targets citrullinated proteins. A positive result supports rheumatoid arthritis, especially when joint symptoms are present, but it does not diagnose the disease by itself.

Alternate Names:

Anti-CCP; Anti-CCP antibody; Anti-Citrullinated Peptide Antibody; Anti-Cyclic Citrullinated Peptide; cyclic citrullinated peptide antibody.

Anti-D

An injection of anti-D immunoglobulin given to RhD-negative pregnant people to prevent antibodies forming against RhD-positive fetal blood cells.

Anti-DNase B Blood Test

A blood test that measures antibodies against deoxyribonuclease B (DNase B), a protein made by group A *Streptococcus*. It helps show a recent past strep infection and may help evaluate rheumatic fever or kidney inflammation that develops after a strep infection.

Anti-Double-Stranded DNA (Anti-dsDNA) Antibody

An autoantibody that targets double-stranded DNA. A blood test for it helps evaluate systemic lupus erythematosus (lupus); high levels may be associated with active disease, especially kidney involvement, but results must be interpreted with symptoms and other tests.

Alternate Names:

Anti-DsDNA Antibody; Anti-Double-Stranded DNA Antibody.

Anti-GBM Disease

Autoimmune disease with antibodies against alpha-3 chain of type IV collagen in glomerular and alveolar basement membranes. Presents with crescentic GN and pulmonary hemorrhage (Goodpasture

syndrome). Linear IgG deposits on immunofluorescence. Treatment requires emergent plasmapheresis, cyclophosphamide, and corticosteroids.

Anti-Glomerular Basement Membrane Blood Test

A laboratory test that detects circulating anti-glomerular basement membrane antibodies, which target type IV collagen in renal and sometimes pulmonary basement membranes. Positive results support the diagnosis of anti-GBM disease, including Goodpasture syndrome.

Anti-Glomerular Basement Membrane Disease

An autoimmune disease in which antibodies target basement-membrane components in the kidney and sometimes the lungs. It can cause rapidly progressive glomerulonephritis, pulmonary hemorrhage, or both and requires urgent treatment.

Anti-HMGCR Myopathy

A severe subtype of immune-mediated necrotizing myopathy triggered in a subset of patients by exposure to 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase inhibitors (statins), but continuing to progress autonomously after statin cessation. Characterized by autoantibodies directed against the HMGCR enzyme, presenting with subacute or insidious, progressive, symmetrical proximal muscle weakness, severe muscle atrophy, and massive serum creatine kinase elevation (typically > 5,000–10,000 U/L). Muscle biopsy reveals widespread, scattered myofiber necrosis, regeneration, and myophagocytosis with remarkably minimal lymphocytic inflammation and sarcolemmal upregulation of MHC class I, requiring aggressive combination immunotherapy (glucocorticoids, intravenous immunoglobulin, and steroid-sparing immunosuppressants).

Alternate Names:

Statin-Induced Immune-Mediated Necrotizing Myopathy; Anti-HMG-CoA Reductase Myopathy

Anti-IgE Therapy

Omalizumab is a monoclonal antibody that binds

free IgE, preventing it from binding to mast cells and basophils. It reduces allergic inflammation and is approved for moderate-to-severe persistent allergic asthma. It reduces exacerbations and the need for oral corticosteroids. It is administered by subcutaneous injection every 2–4 weeks.

Anti-IL4 Receptor Therapy

Dupilumab is a monoclonal antibody blocking the IL-4 receptor alpha subunit, inhibiting both IL-4 and IL-13 signaling. It is approved for moderate-to-severe asthma with eosinophilic phenotype or oral corticosteroid dependence. It reduces exacerbations and improves lung function.

Anti-IL5 Therapy

Treatment with monoclonal antibodies that block interleukin-5 signaling or its receptor, thereby reducing eosinophil production or survival. It is used for selected eosinophilic diseases, including some forms of severe asthma.

Anti-Insulin Antibody Test

A laboratory test that detects autoantibodies directed against insulin, which may cause hypoglycemia by binding and releasing insulin unpredictably. It is used to evaluate factitious hypoglycemia and insulin autoimmune syndrome.

Anti-MDA5 Dermatomyositis

A clinically distinct subtype of dermatomyositis associated with anti-melanoma differentiation-associated gene 5 antibodies, characterised by skin-predominant disease, mechanic's hands, and a high risk of rapidly progressive interstitial lung disease.

Anti-Mullerian Hormone

A hormone produced by granulosa cells of ovarian follicles. It is measured in selected fertility and reproductive evaluations, but it does not by itself determine fertility.

Anti-NMDA Receptor Encephalitis

A severe autoimmune encephalitis mediated by IgG antibodies directed against the GluN1 subunit of the N-methyl-D-aspartate receptor in the central

nervous system. Most commonly affects children and young women, frequently associated with an underlying ovarian teratoma. Classic progression involves a viral prodrome followed by acute psychosis, memory deficits, catatonia, autonomic instability, choreoathetoid or orofacial dyskinesias, hypoventilation, and refractory seizures.

Alternate Names:

Anti-N-methyl-D-aspartate receptor encephalitis; NMDA receptor encephalitis

Anti-Reflux Surgery

An operation intended to reduce reflux of stomach contents into the esophagus, usually by reinforcing the gastroesophageal junction. It may be considered for selected patients with gastroesophageal reflux disease or a hiatal hernia.

Anti-Reflux Surgery - Children

Surgical procedure to correct gastroesophageal reflux in children, typically by fundoplication, in which the gastric fundus is wrapped around the lower esophagus. It is considered for severe refractory reflux, aspiration, failure to thrive, or esophageal strictures.

Anti-Rust Product Poisoning

Toxic effects caused by excessive exposure to anti-rust products. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Anti-Smooth Muscle Antibody

An autoantibody associated especially with autoimmune hepatitis, although low levels can occur in other autoimmune or infectious conditions.

Anti-SRP Myopathy

A particularly severe, aggressive form of immune-mediated necrotizing myopathy associated with high-titer autoantibodies targeting the 54-kDa subunit of the signal recognition particle (*SRP*), a ribonucleoprotein complex vital for translocating secretory and membrane proteins across the endoplasmic reticulum. Manifests with acute or subacute, rapidly progressive, profound symmetrical proxi-

mal limb weakness, prominent neck flexor weakness, severe dysphagia, and marked functional disability accompanied by extreme serum creatine kinase elevations ($> 10,000$ U/L). Muscle biopsy shows prominent myofiber necrosis, sparse inflammatory infiltrates, and membrane attack complex (C5b-9) deposition on non-necrotic sarcolemma, typically exhibiting resistance to glucocorticoid monotherapy and necessitating early intensive B-cell-depleting therapy or plasma exchange.

Alternate Names:

Anti-Signal Recognition Particle Myopathy; SRP-Associated Necrotizing Myopathy

Antianginal

A class of medications used to prevent or relieve angina pectoris (chest pain from myocardial ischemia).

Antibacterial

Destroying bacteria or inhibiting their growth; the term may describe a drug, disinfectant, surface treatment, or immune defense.

Antibiogram

A periodic summary of antimicrobial susceptibility patterns of bacterial isolates from a hospital or laboratory. Guides empiric antibiotic therapy by showing local resistance patterns (e.g., % of *E. coli* susceptible to TMP-SMX). Updates annually.

Antibodies

Immunoglobulin proteins produced by B cells that bind to specific antigens. There are five classes: IgG (most abundant), IgM (first produced), IgA (mucosal immunity), IgE (allergic responses), and IgD. Antibodies neutralise pathogens, opsonise for phagocytosis, activate complement, and mediate antibody-dependent cellular cytotoxicity.

Antibody

A protein produced by the immune system that specifically binds to a foreign or abnormal substance.

Antibody Therapy

Treatment using antibodies to block, activate, or

target a specific molecule or cell.

Alternate Names:

Immunotherapy

Antibody Titer Blood Test

A laboratory test that measures the concentration of specific antibodies in serum, expressed as the highest dilution that produces a detectable reaction. Rising titers between acute and convalescent specimens indicate recent or active infection; high single titers suggest prior exposure or immunity.

Antibody-Dependent Cell-Mediated Cytotox.

Antibody-dependent cellular cytotoxicity, in which immune effector cells such as natural killer cells recognize antibodies attached to a target cell and destroy it.

Antibody Engineering

Designing or changing antibodies to make them bind more accurately, last longer, cause fewer unwanted immune reactions, or work better as medicines. Scientists may make antibodies more human-like, improve their binding, change how they activate immune cells, create smaller or bispecific antibodies, or attach drugs to them.

Anticholinergic

Blocking the effects of acetylcholine, commonly at muscarinic receptors; these drugs can reduce secretions and smooth-muscle activity but may cause dry mouth, blurred vision, constipation, urinary retention, or confusion.

Anticholinergic Burden

The cumulative effect of taking multiple medications with anticholinergic properties. It can cause dry mouth, constipation, urinary retention, blurred vision, confusion, and delirium, especially in elderly patients. Many psychiatric medications (TCAs, first-generation antipsychotics, some antihistamines) have anticholinergic effects.

Anticholinergic Toxidrome

A poisoning pattern caused by blockade of mus-

carinic acetylcholine receptors, commonly producing dry skin and mouth, dilated pupils, rapid heart rate, urinary retention, reduced bowel sounds, overheating, agitation, or delirium.

Anticoagulation

Treatment that reduces blood clot formation by inhibiting the coagulation system. It is used for conditions such as venous thromboembolism and selected cardiac or vascular disorders.

Anticodon

A three-nucleotide sequence at one end of transfer RNA that base-pairs with a complementary messenger RNA codon during translation, ensuring the correct amino acid is incorporated into the growing polypeptide.

Anticonvulsants

Medications used to prevent or control seizures in epilepsy and other seizure disorders.

Antideformity Position

A therapeutic positioning method that places a limb or joint in an alignment that helps prevent or limit a contracture or deformity.

Antidepressants

Medicines used mainly to treat depressive disorders; different classes alter serotonin, norepinephrine, dopamine, or other signaling and may also treat anxiety, pain, or related conditions.

Antidiabetic

A class of medications used to manage diabetes mellitus by lowering blood glucose levels through various mechanisms.

Antidiabetics-Oral

Oral antidiabetic medicines lower blood glucose through different mechanisms, including increasing insulin release, improving insulin sensitivity, reducing hepatic glucose production, or increasing urinary glucose loss. Selection considers kidney function, hypoglycemia risk, weight, cardiovascular disease,

and cost.

Alternate Names:

Medications Type 2 Diabetes (Antidiabetics-Oral)

Antidiuretic Hormone

A hypothalamic hormone released from the posterior pituitary in response to increased plasma osmolality or reduced effective circulating volume. It increases collecting-duct water permeability through V2 receptors and promotes vasoconstriction through V1 receptors.

Antidiuretic Hormone

Vasopressin, a hormone released from the posterior pituitary that increases water reabsorption in the kidney and, at higher concentrations, constricts blood vessels.

Antidiuretic Hormone Blood Test

A laboratory test that measures plasma vasopressin concentration, the hormone responsible for water reabsorption in the kidney collecting ducts. It is used to evaluate diabetes insipidus, syndrome of inappropriate ADH secretion, and disorders of water balance.

Antidiuretic Hormone Physiology

Antidiuretic hormone also known as vasopressin is synthesized in the supraoptic and paraventricular nuclei of the hypothalamus and stored in the posterior pituitary. Released in response to increased serum osmolality detected by hypothalamic osmoreceptors and decreased blood volume detected by baroreceptors.

Alternate Names:

vasopressin is synthesized in the supraoptic and.

Antidiuretic Hormone Resistance

Impaired renal response to vasopressin, causing excessive production of dilute urine and polydipsia. Causes include lithium, hypercalcemia, hypokalemia, chronic kidney disease, and inherited defects of the vasopressin receptor or aquaporin pathway.

Antidote

Substance that counteracts poison toxicity. Specific: naloxone (opioids), flumazenil (benzodiazepines), N-acetylcysteine (acetaminophen), digoxin immune Fab (digoxin), chelators (EDTA, deferoxamine, dimercaprol), atropine + pralidoxime (organophosphates), methylene blue (methemoglobinemia), cyanide antidote kit (nitrite + thiosulfate).

Antiemetics

Antiemetics are medicines used to prevent or treat nausea and vomiting. Drug classes act on serotonin, dopamine, histamine, acetylcholine, or neurokinin pathways; choice depends on the cause, age, pregnancy, other medicines, heart rhythm risk, and hydration status.

Antiepileptic Drugs

Medications used to prevent and control seizures by modulating neuronal excitability through various mechanisms. They can be classified by their primary mechanism of action.

Antifreeze Poisoning

Toxic effects caused by excessive exposure to antifreeze, usually products containing ethylene glycol or propylene glycol. Severity depends on the substance, dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Antifungal Prophylaxis

The use of antifungal agents to prevent invasive fungal infections in high-risk patients. Common indications include prolonged neutropenia, hematopoietic stem cell transplantation, and advanced HIV infection. Fluconazole, posaconazole, and micafungin are used for prophylaxis.

Antifungal Resistance

The ability of fungi to withstand the effects of antifungal agents. Resistance can be intrinsic or acquired. Azole resistance occurs through efflux pumps and target mutations. Echinocandin resistance occurs through FKS gene mutations. Amphotericin B resistance is rare.

Antigen

A molecule or molecular structure specifically recognized by an antibody, B-cell receptor, or T-cell receptor. An immunogen is an antigen capable of inducing an immune response; antigenicity and immunogenicity are related but not synonymous properties.

Antigen Detection Test

A diagnostic assay that identifies a pathogen-derived antigen in a clinical specimen using a specific antibody or related binding reagent. It is generally rapid but may be less sensitive than nucleic-acid amplification, especially when organism burden is low.

Antigen Presentation

Display of antigen fragments on cell surfaces for recognition by T cells.

Antigen Receptor

A receptor on an immune cell that specifically recognizes an antigen.

Antigen-Presenting Cell

Specialized immune cells that process protein antigens and display peptide fragments bound to Major Histocompatibility Complex (MHC) molecules on their surface for recognition by T-cell receptors. Professional APCs: dendritic cells (most potent for naive T cells), macrophages, and B cells.

Alternate Names:

APC (antigen-presenting cell)

Antigenic Drift

The gradual accumulation of point mutations in viral surface antigens (hemagglutinin, neuraminidase) during replication, allowing influenza viruses to evade pre-existing antibodies and cause recurrent seasonal epidemics. Drift is continuous and selected by host immunity; it is why the seasonal vaccine is updated annually.

Antigenic Shift

A major, abrupt change in the antigenic structure

of a virus (especially influenza A) resulting from genetic reassortment between different strains (e.g., human and avian viruses) infecting the same cell.

Antigenic Variation

The process by which pathogens change their surface proteins to evade immune recognition. Antigenic drift: gradual changes in surface proteins. Antigenic shift: major changes through reassortment of genome segments. Influenza virus demonstrates both mechanisms. Trypanosomes and Plasmodium also use antigenic variation.

Antigout

A class of medications used to treat gout by reducing uric acid levels or controlling acute inflammation.

Antihistamines

Medicines that block histamine receptors; H1 antihistamines relieve allergy symptoms and some cause sedation, while H2 antihistamines reduce gastric acid secretion.

Antihistamines For Allergies

Medicines that block histamine effects and reduce symptoms such as itching, sneezing, and runny nose; sedation and interactions vary by agent.

Antihypertensives In Pregnancy

Medicines used to lower high blood pressure during pregnancy. Common options include labetalol, nifedipine, and methyldopa; severe hypertension needs urgent treatment, and ACE inhibitors and angiotensin-receptor blockers are generally avoided during pregnancy.

Antileukotriene

A drug class that blocks leukotriene-mediated airway inflammation or bronchoconstriction and is used in selected allergic or respiratory conditions.

Alternate Names:

Leukotriene modifier; leukotriene pathway inhibitor.

Antimetabolite

A compound resembling a substance needed for

metabolism or DNA synthesis that disrupts cellular processes; many antimetabolites are used as anti-cancer or immunosuppressive drugs.

Antibiotics

Medicines that kill bacteria or stop them from multiplying. They treat susceptible bacterial infections but do not treat viral infections such as colds or influenza.

Antimicrobial

A general term for antibiotics and other drugs that fight microscopic organisms in the body, such as bacteria, viruses, fungi, and parasites.

Antimicrobial Adverse Effects

Side effects associated with antimicrobial agents. Beta-lactams can cause allergic reactions. Fluoroquinolones can cause tendon rupture and QT prolongation. Aminoglycosides can cause nephrotoxicity and ototoxicity. Azole antifungals can cause hepatotoxicity.

Antimicrobial Agent

A substance that kills microorganisms or inhibits their growth, including antibacterial, antiviral, antifungal, and antiparasitic agents.

Antimicrobial Desensitization

A procedure that administers gradually increasing doses of an antimicrobial to temporarily induce tolerance in a person with a documented or strongly suspected immediate allergy. It may be considered when that antimicrobial is clinically necessary and suitable alternatives are not available.

Antimicrobial Dosing Optimization

Strategies to ensure antimicrobial agents achieve therapeutic concentrations at the site of infection. Optimization includes appropriate drug selection, dosing interval, route of administration, and duration of therapy. Extended or continuous infusion of beta-lactams may improve outcomes. Weight-based dosing is important for many agents.

Antimicrobial Hypersensitivity Reactions

Immune-mediated reactions to antimicrobial agents that may range from a mild rash to life-threatening anaphylaxis. The reaction depends on the drug, the patient, and the type of immune response.

Antimicrobial Pharmacodynamics

The relationship between antimicrobial drug concentrations and their effect on microorganisms. Time-dependent antibiotics (beta-lactams) are most effective when concentrations remain above the MIC. Concentration-dependent antibiotics (aminoglycosides) are most effective with high peak concentrations.

Antimicrobial Pharmacokinetics

The absorption, distribution, metabolism, and elimination of antimicrobial agents. Pharmacokinetics varies by drug class, patient factors, and disease state. Renal impairment affects elimination of many antibiotics. Hepatic impairment affects metabolism of many antifungals.

Antimicrobial Resistance

The ability of a microorganism to survive or grow despite a medicine that would normally kill it or stop its growth. It includes resistance in bacteria, viruses, fungi, and parasites; genetic changes and spread between organisms can cause it, and unnecessary or incorrect antimicrobial use can accelerate it.

Alternate Names:

AMR; Antimicrobial Resistance

Antimicrobial Resistance Mechanisms

Biological processes that allow microorganisms to survive antimicrobial exposure, including drug inactivation, modification of the drug target, reduced permeability, and active drug efflux.

Antimicrobial Resistance Surveillance

The systematic monitoring of antimicrobial resistance patterns in clinical isolates. Surveillance programmes track resistance rates over time and guide empiric therapy. National and international networks share resistance data.

Antimicrobial Sensitivity Test

A laboratory test that determines which antimicrobial medicines inhibit or kill a microorganism isolated from a patient.

Antimicrobial Stewardship

Coordinated actions that promote appropriate antimicrobial selection, dose, route, and duration to improve patient outcomes and limit resistance, adverse effects, and unnecessary cost.

Antimicrobial Susceptibility Testing

Laboratory methods used to estimate how susceptible a microorganism is to antimicrobial agents. Methods include disk diffusion, broth microdilution, and automated systems. Results are interpreted using standardized breakpoints and can help guide antimicrobial therapy.

Antimicrobial Therapy

The use of antibacterial, antiviral, antifungal, or antiparasitic medicines to prevent or treat infection. Selection depends on the suspected or identified organism, infection site, severity, resistance patterns, and patient factors.

Antimitochondrial Antibody

An autoantibody strongly associated with primary biliary cholangitis, interpreted with cholestatic liver tests and clinical findings.

Antimony

A metallic chemical element whose compounds can be toxic; selected antimony compounds, such as the pentavalent agents, have been used to treat leishmaniasis.

Antinuclear Antibody

ANA: autoantibodies against nuclear components (DNA, histones, ribonucleoproteins, etc.), detected by immunofluorescence (HEp-2 cells), reported with pattern (homogeneous, speckled, nucleolar, centromere) and titer.

Alternate Names:

Antinuclear Antibodies

Antinuclear Antibody Panel

A laboratory test that measures multiple antinuclear antibody specificities, including anti-dsDNA, anti-Smith, anti-RNP, anti-Ro, anti-La, and anti-histone antibodies. It is used to evaluate suspected autoimmune connective tissue diseases such as systemic lupus erythematosus when ANA screening is positive.

Antinuclear Autoantibody

An autoantibody directed against components of a cell nucleus.

Alternate Names:

ANA

Antioxidant

A substance that limits oxidation by reacting with oxidizing molecules or free radicals.

Antiparietal Cell Antibody Test

A laboratory test that detects autoantibodies against gastric parietal cells, which produce intrinsic factor and hydrochloric acid. Positive results are found in pernicious anemia and autoimmune gastritis and may indicate risk of vitamin B12 deficiency.

Antiparkinson

A class of medications used to manage Parkinson disease symptoms by restoring dopaminergic activity or modulating other neurotransmitter systems.

Antiphospholipid Antibodies

Include lupus anticoagulant (prolongs phospholipid-dependent coagulation tests), anticardiolipin antibodies (IgG/IgM), and anti-beta-2-glycoprotein I antibodies. Detected on ≥ 2 occasions 12 weeks apart. Clinical criteria: vascular thrombosis or pregnancy morbidity. Can be transient (infection, drugs) or persistent.

Antiphospholipid Antibody

An autoantibody directed against phospholipid-binding proteins. Persistent antibodies may be associated with thrombosis or pregnancy complications, particularly in antiphospholipid syndrome.

Alternate Names:

APA (antiphospholipid antibody)

Antiphospholipid Antibody Syndrome

An autoimmune disorder in which antiphospholipid antibodies are associated with abnormal blood clotting, recurrent pregnancy loss, thrombocytopenia, and sometimes migraine or heart-valve abnormalities.

Antiphospholipid Syndrome

A systemic autoimmune thrombophilia characterized by arterial and venous thrombosis, pregnancy morbidity, and persistent antiphospholipid antibody positivity.

Alternate Names:

APS (Antiphospholipid Syndrome)

Antiphospholipid Syndrome

Alternate index label for Antiphospholipid Syndrome.

Antiphospholipid Syndrome - APS

Antiphospholipid syndrome is an autoimmune thrombophilia associated with venous or arterial clots and pregnancy morbidity in the presence of persistent antiphospholipid antibodies.

Antiplatelet Agents

Medicines that reduce platelet activation or aggregation and thereby lower the risk of some blood clots.

Antiplatelet Medicines - P2Y12 Inhibitors

Antiplatelet drugs such as clopidogrel, prasugrel, and ticagrelor that reduce ADP-mediated platelet activation and thrombosis risk, with bleeding as a major adverse effect.

Antiplatelet Therapy

Treatment with medicines that reduce platelet activation or aggregation, such as aspirin or clopidogrel, when clinically indicated to reduce arterial clotting.

Antiprotozoal

Destroying protozoa or inhibiting their growth; antiprotozoal medicines treat infections such as malaria, giardiasis, or amoebiasis.

Antiretroviral

Active against retroviruses; antiretroviral medicines inhibit different stages of HIV replication and are generally used in combination therapy.

Antiretroviral Therapy

Treatment of HIV infection with combinations of antiretroviral agents. ART targets multiple steps in the viral life cycle including entry, reverse transcription, integration, and maturation. Combination ART suppresses viral replication and restores immune function. ART is lifelong therapy.

Antiretroviral Therapy

Combination treatment with antiretroviral medicines that suppress HIV replication, preserve immune function, and reduce the risk of HIV transmission and disease progression.

Antisepsis

Use of chemical methods to reduce or eliminate harmful microorganisms on living tissue.

Antiseptic

Substances used on wounds to prevent or treat infection; they kill or slow the growth of diseasecausing organisms, such as bacteria, on the surface of the body.

Antiserum

Blood serum containing antibodies against a particular antigen or toxin.

Antisocial Personality

A persistent pattern of disregard for other people's rights, social rules, and responsibilities.

Alternate Names:

Antisocial personality disorder

Antisocial Personality Disorder

A pattern of disregard for and violation of the rights of others occurring since age 15 years. At least three of seven criteria: failure to conform to social norms, deceitfulness, impulsivity, irritability and aggressiveness, reckless disregard for safety, consistent irresponsibility, and lack of remorse.

Alternate Names:

Disorder Antisocial Personality (Antisocial Personality Disorder) Personality Disorder, Antisocial Personality Disorder Antisocial (Antisocial Personality Disorder)

Antispasmodic Drugs

Medicines that relax smooth muscle and reduce cramping or spasms, most often in the digestive tract. They may also be used for bladder or bile-duct spasms; the choice depends on the cause and the medicine's risks.

Alternate Names:

Antispasmodics; spasmolytics.

Antistreptolysin O Titer

A blood test that measures antibodies against streptolysin O, a toxin produced by group A streptococci. Rising or elevated titers support recent streptococcal infection and help evaluate suspected rheumatic fever or poststreptococcal glomerulonephritis, although a single value does not prove active infection.

Antisynthetase Syndrome

An autoimmune disorder associated with antibodies against aminoacyl-tRNA synthetases, especially anti-Jo-1. It may cause interstitial lung disease, inflammatory muscle disease, arthritis, fever, Raynaud phenomenon, or mechanic's hands.

Antithrombin

Natural anticoagulant. Inhibits thrombin, factor Xa. Deficiency causes thrombosis.

Antithrombin Deficiency

A rare (1 in 2,000-5,000), strongly thrombophilic inherited disorder (autosomal dominant). Antithrombin (previously antithrombin III) is a serine protease

inhibitor that inactivates thrombin and factors Xa, IXa, XIa, and XIIa. Heterozygous: 10-50 fold increased VTE risk, more severe than factor V Leiden.

Antithrombin III Blood Test

A laboratory test that measures antithrombin activity or antigen level, a natural anticoagulant that inactivates thrombin and factor Xa. Deficiency is a strong risk factor for venous thromboembolism; acquired decreases occur with liver disease, nephrotic syndrome, or heparin therapy.

Antithrombin III Deficiency

An inherited or acquired thrombophilia caused by reduced antithrombin activity, leading to increased risk of venous thromboembolism.

Antithymocyte Globulin

A preparation of antibodies directed against human T lymphocytes, used as immunosuppressive therapy for aplastic anemia, solid-organ transplantation, and selected immune-mediated disorders. Important risks include infusion reactions, serum sickness, cytopenias, and infection.

Alternate Names:

Rabbit Antithymocyte Globulin rATG Thymoglobulin

Antithyroglobulin Antibody Test

A laboratory test that measures antibodies against thyroglobulin, a thyroid hormone precursor protein. Elevated levels are found in Hashimoto thyroiditis, Graves disease, and other autoimmune thyroid conditions, and help evaluate thyroid dysfunction of autoimmune origin.

Antitoxin

An antibody or antibody preparation that binds to and neutralizes a toxin. Antitoxins may be made from human or animal antibodies and are used for selected poisonings or toxin-producing infections.

Antitubercular Agents

Medicines used to treat tuberculosis caused by *Mycobacterium tuberculosis*. Treatment uses several drugs together for months to prevent resistance,

with monitoring for liver injury, nerve problems, vision changes, and other adverse effects.

Antral Follicle Count

A standardized transvaginal ultrasound biometric assessment performed during the early follicular phase (cycle days 2–4) to enumerate resting follicles measuring 2–10 mm in diameter across both ovaries. Serves as a primary clinical biomarker of functional ovarian reserve, predictive of ovarian response to exogenous gonadotropin stimulation in assisted reproductive technology, but not of natural fecundability or cycle-specific pregnancy rates.

Alternate Names:

AFC; Basal Antral Follicle Count

Antrochoanal Polyp

A benign, single nasal polyp originating from the maxillary sinus mucosa and extending through the accessory ostium into the choana and nasopharynx. It is most common in children and young adults, presenting with unilateral nasal obstruction and rhinorrhea. Treatment is endoscopic surgical removal.

Antrum

A cavity or chamber in an organ, especially the lower part of the stomach near the pylorus or the air-filled maxillary sinus.

Anuria

Near absence of urine production, often defined in adults as less than about 100 milliliters in 24 hours.

Anus

The terminal opening of the gastrointestinal tract through which feces are expelled. It is surrounded by the internal smooth-muscle sphincter and external skeletal-muscle sphincter.

Anxiety

A normal emotional response to perceived danger, but anxiety disorders involve excessive, persistent worry that interferes with daily life. Symptoms can include restlessness, rapid heart rate, muscle tension, and avoidance behaviors.

Anxiety Disorder

A group of psychiatric conditions characterized by excessive fear, anxiety, and related behavioral disturbances. Includes generalized anxiety disorder (GAD), panic disorder, social anxiety disorder, specific phobias, and agoraphobia.

Anxiety Disorder, Generalized

See *Generalized anxiety disorder*.

Anxiety Disorder, Social

See *Social anxiety disorder (social phobia)*.

Anxiety Disorders

Mental health disorders involving excessive fear, worry, or anxiety that is persistent, difficult to control, and disruptive to daily life.

Anxiety Separation (Separation Anxiety)

Separation anxiety is excessive fear or distress when separated from an attachment figure, beyond what is expected for developmental stage. It can cause avoidance, physical complaints, sleep difficulty, and school impairment and is treated with behavioral therapy and, when needed, medication.

Aorta

Largest artery, from the left ventricle distributing oxygenated blood systemically. Approximately 30 cm long, 2.5 cm diameter. Four segments: ascending aorta, aortic arch (brachiocephalic trunk, left common carotid, left subclavian), descending thoracic aorta, descending abdominal aorta (bifurcating into common iliacs at L4).

Aortal

Relating to the aorta, the large artery that carries blood from the left ventricle to the systemic circulation.

Aortic

Relating to the aorta, the main artery carrying blood from the heart to the body.

Aortic Aneurysm

A permanent, localized enlargement of the aorta caused by weakening of its wall. It may occur in the thoracic or abdominal aorta and can enlarge, dissect, or rupture; management depends on its site, size, growth, symptoms, and the patient's risk.

Aortic Aneurysm Repair - Endovascular

Endovascular aneurysm repair is a minimally invasive technique using a stent-graft deployed through femoral artery access to exclude an aortic aneurysm from circulation. It is indicated for suitable anatomy and offers faster recovery than open repair.

Aortic Aneurysm Rupture

Full-thickness failure of an aortic aneurysm with bleeding into a body cavity or surrounding tissue. Ruptured aneurysm commonly causes sudden pain, shock, and collapse and requires immediate resuscitation and operative or endovascular repair.

Aortic Aneurysm Screening

Evaluation of selected people for an aortic aneurysm before symptoms develop. Screening may use abdominal ultrasonography or other imaging according to the person's risk.

Aortic Angiography

An imaging study of blood vessels that uses contrast material to identify narrowing, blockage, aneurysm, or abnormal blood flow.

Aortic Arch

Curves from second right sternocostal joint superiorly and left, then posteriorly and inferiorly to T4. From right to left: brachiocephalic trunk, left common carotid, left subclavian. Related to left recurrent laryngeal nerve (hooking under it), trachea, esophagus, left phrenic nerve. Arch aneurysms may cause hoarseness and SVC syndrome.

Aortic Arch Syndrome

Symptoms and signs caused by narrowing or blockage of the aorta or its major branches. Causes include large-vessel vasculitis, atherosclerosis, thrombosis, and congenital disease; findings may include

unequal pulses, limb claudication, or neurologic symptoms.

Aortic Body

A small group of chemoreceptor cells near the aortic arch that helps sense blood oxygen and carbon dioxide.

Aortic Coarctation

Congenital narrowing of the aorta, usually near the insertion of the ductus arteriosus. It causes higher blood pressure and stronger pulses in the arms than in the legs, with a radio-femoral delay; severe neonatal disease may present with shock when the ductus closes.

Aortic Disease

Any disorder affecting the aorta, including aneurysm, dissection, narrowing, or inflammation.

Aortic Dissection

A tear in the inner layer of the aorta that allows blood to enter the wall and form a false channel. Stanford type A involves the ascending aorta; type B does not. Treatment depends on the type, symptoms, complications, and overall clinical condition.

Alternate Names:

Dissecting Aneurysm Dissection Aorta (Aortic Dissection)

Aortic Embolism

Obstruction of a systemic artery by material originating in the aorta, such as thrombus or atheromatous debris, causing distal ischemia.

Aortic Graft Infection

Infection involving a prosthetic or endovascular aortic graft, potentially causing sepsis, pseudoaneurysm, hemorrhage, or aortoenteric fistula. Management usually requires prolonged antimicrobial treatment and individualized surgical or endovascular source control.

Aortic Injury

Damage to the aorta, often caused by severe trauma

or a medical procedure.

Aortic Insufficiency

See *Aortic valve regurgitation*.

Aortic Regurgitation

Incomplete closure of the aortic valve allowing blood to flow backward from the aorta to the left ventricle during diastole. Causes include bicuspid aortic valve, endocarditis, aortic root dilation, hypertension, and connective tissue disorders. Murmur: early diastolic decrescendo murmur at the left sternal border.

Aortic Regurgitation Murmur

Early diastolic decrescendo murmur heard best at the left sternal border with the patient sitting up and leaning forward in end-expiration. Wide pulse pressure, water-hammer pulses (Corrigan pulse), de Musset sign (head bobbing with each pulse), and Duroziez sign (femoral bruit with compression).

Aortic Rupture

A life-threatening tear through the wall of the aorta causing internal bleeding.

Aortic Stenosis

Narrowing of the aortic valve opening, obstructing left ventricular outflow. It is the most common valvular disease in developed countries, usually caused by calcific degeneration. Symptoms include angina, syncope, and heart failure (SAD triad).

Aortic Stenosis Grading

Severity grading: mild (valve area >1.5 cm², mean gradient <20 mmHg), moderate (valve area 1.0-1.5 cm², mean gradient 20-40 mmHg), severe (valve area <1.0 cm², mean gradient >40 mmHg). Very severe (valve area <0.6 cm², mean gradient >60 mmHg) indicates critical stenosis. Grading guides timing of intervention.

Aortic Valve

A semilunar valve between the left ventricle and aorta, normally composed of right, left, and non-

coronary cusps. It opens during systole and closes during diastole to direct blood flow. Each cusp forms a sinus of Valsalva from which coronary arteries originate. Opens during systole; closes during diastole.

Aortic Valve Disease

Abnormal narrowing, leakage, or structural change of the valve between the left ventricle and aorta. It may cause chest pain, breathlessness, fatigue, fainting, or no symptoms initially. Evaluation commonly includes examination and echocardiography; severe disease may require specialist treatment.

Aortic Valve Insufficiency

Leakage of the aortic valve that allows blood to flow back into the left ventricle.

Aortic Valve Regurgitation

Backward leakage of blood through an incompletely closing aortic valve, which can make the heart work harder and eventually cause heart failure.

Alternate Names:

Regurgitation, Aortic Valve

Aortic Valve Stenosis

Narrowing of the aortic valve that obstructs blood flow from the left ventricle into the aorta.

Alternate Names:

Stenosis, Aortic Valve

Aortic Valve Surgery - Minimally Invasive

Aortic valve replacement performed through a small incision or port access, avoiding median sternotomy. It may be surgical or transcatheter and is used for aortic stenosis or regurgitation when a less invasive approach is appropriate.

Aortic Valve Surgery - Open

Open surgical replacement or repair of the aortic valve through median sternotomy, typically using a mechanical or bioprosthetic valve. It is indicated for severe aortic stenosis or regurgitation causing symptoms or left ventricular dysfunction.

Aortitis

Infectious or non-infectious inflammatory disease involving the aortic wall. Non-infectious etiologies are primarily large-vessel vasculitides, including giant cell arteritis (predominantly in patients older than 50 years) and Takayasu arteritis (predominantly in women younger than 40 years, historically termed pulseless disease), as well as IgG4-related systemic disease and ankylosing spondylitis. Infectious aortitis (mycotic aneurysm) results from bacteremic seeding (*Salmonella*, *Staphylococcus aureus*) or tertiary syphilis, which causes obliterative endarteritis of the vasa vasorum leading to ascending aortic dilation, tree-bark intimal wrinkling, and aortic root dilation. Evaluated using CT angiography, MR angiography, or FDG-PET; major complications include aortic regurgitation, pseudoaneurysm, dissection, and catastrophic rupture.

Aortography

Contrast imaging of aorta. Evaluates aneurysm, dissection, occlusion. Largely replaced by CTA/MRA.

Aortopulmonary Window

A congenital opening between the ascending aorta and pulmonary artery.

AP, X-Ray

An anteroposterior radiographic projection in which the X-ray beam travels from the front of the body toward the back.

Alkaline Phosphatase Activity

A measurement of alkaline phosphatase enzyme activity in the blood, used to help evaluate liver, bile-duct, or bone disorders. Results are interpreted with other liver tests and clinical findings.

APACHE II Score

Severity scoring system for critically ill patients incorporating physiological variables, age, and chronic health status to generate scores from zero to seventy-one. Higher scores indicate greater severity and higher mortality. In acute pancreatitis, scores of eight or greater indicate severe disease.

Apathy

Reduced motivation, interest, or emotional response. It may occur with depression, dementia, Parkinson disease, brain injury, other neurologic illness, or medicines and is different from simple tiredness or sadness.

Argon Plasma Coagulation

A procedure that uses ionized argon gas and electrical energy to coagulate tissue and control bleeding or treat some superficial abnormal tissue, often during endoscopy.

Alternate Names:

APC (argon plasma coagulation)

APC Tumor Suppressor

The adenomatous polyposis coli protein is a large multidomain tumor suppressor that functions as a critical negative regulator of the canonical Wnt signaling pathway.

Apert Syndrome

An autosomal dominant craniosynostosis syndrome caused by FGFR2 mutations (most commonly Ser252Trp or Pro253Arg).

Apex

The tip or pointed end of a structure; examples include the apex of the heart, formed mainly by the left ventricle, and the upper tip of a lung.

Apgar Score

A structured assessment of a newborn's appearance, pulse, grimace, activity, and respiration, usually recorded at 1 and 5 minutes after birth. It describes immediate adaptation after delivery and helps guide resuscitation; it is not a standalone diagnosis or long-term prognostic test.

Alternate Names:

Apgar assessment; newborn Apgar score.

Aphagia

Complete inability to swallow, which may result from neurologic disease, obstruction, severe pain, or loss of swallowing-muscle function.

Aphasia

Impairment of language comprehension or production resulting from brain damage, most commonly from stroke affecting the left hemisphere (dominant) language areas.

Aphasia, Primary Progressive

See *Primary progressive aphasia*.

Apheresis

A procedure that removes a selected blood component, such as plasma, platelets, or abnormal cells, and returns the remaining components to the patient or donor.

Apheresis Donation

A blood donation in which a machine collects one needed component, such as platelets or plasma, and returns the remaining blood components to the donor.

Aphonia

Complete loss of the ability to produce a voice. A person may still be able to whisper. Causes include vocal-cord paralysis, laryngitis, injury to the larynx, growths on the vocal cords, or psychological factors.

Aphrasia

Loss or impairment of the ability to speak, often used as an older or less precise term related to aphasia.

Aphthous Stomatitis

Recurrent painful ulcers of the oral mucosa with a shallow base and erythematous border, commonly known as canker sores.

Aphthous Ulcer

A recurrent, painful ulcer of the oral mucosa. See Canker Sore.

Apical

Relating to or located at the apex, or tip, of a structure such as the lung, heart, or tooth.

Apical Ballooning Syndrome

See *Broken heart syndrome*.

Apical Pulse

The heart rate measured by listening over the apex of the heart, usually at the left fifth intercostal space near the midclavicular line.

Apicoectomy

Surgical removal of root tip and periapical lesion. Indication: failed root canal therapy.

Aplasia

Complete failure of development of an organ, tissue, or cell line, resulting in absence or severe deficiency of the expected structure. It may arise from congenital developmental abnormalities, inherited disorders, injury to a precursor cell population, or severe marrow suppression. Examples include aplastic anemia, renal aplasia, and aplasia of selected hematopoietic or lymphoid cell lines. Clinical effects depend on the affected structure and may range from an incidental anatomic finding to life-threatening organ failure.

Aplastic

Describing failure of a tissue, especially bone marrow, to produce adequate cells; aplastic anemia causes pancytopenia from marrow failure.

Aplastic Anemia

A bone marrow failure syndrome characterized by pancytopenia (anemia, leukopenia, thrombocytopenia) and a hypocellular bone marrow with replacement by fat cells, resulting from immune-mediated destruction of hematopoietic stem cells.

Aplastic Crisis

An abrupt temporary cessation of red-cell production causing a rapid fall in hemoglobin and reticulocyte count. It most often follows Parvovirus B19 infection in people with shortened red-cell survival, such as those with sickle-cell disease or hereditary spherocytosis. Symptoms may include pallor, fatigue, tachycardia, and worsening anemia without the usual increase in reticulocytes. Diagnosis relies on blood counts, reticulocyte measurement, and

targeted viral testing; treatment is supportive and directed at the underlying disorder.

Apnea

Temporary cessation of breathing. Clinically important apnea may cause low oxygen, carbon-dioxide retention, or bradycardia.

Apnea Of Prematurity

Recurrent pauses in breathing in a premature infant, sometimes accompanied by bradycardia or oxygen desaturation. It may be central, obstructive, or mixed. Management includes monitoring and, when needed, respiratory stimulation or support.

Apnea Sleep (Sleep Apnea)

Sleep apnea consists of repeated pauses or reductions in breathing during sleep. Obstructive apnea results from upper-airway collapse, while central apnea reflects reduced respiratory drive; untreated disease can cause daytime sleepiness, impaired concentration, hypertension, and cardiovascular risk.

Apnea-Hypopnea Index

Number of apneas plus hypopneas per hour of sleep. Apnea = airflow cessation > 10 seconds. Hypopnea = partial obstruction with desaturation or arousal. AHI < 5 normal, 5–14 mild, 15–29 moderate, ≥ 30 severe.

Apneic

Relating to or characterized by apnea, meaning a temporary cessation of breathing.

Apneustic Breathing

A breathing pattern characterized by a prolonged inspiratory pause, seen in brainstem lesions (particularly the pons). It indicates damage to the apneustic center and is a poor prognostic sign in neurological injury. It may require mechanical ventilation.

Apnoea

Temporary cessation of breathing, defined as complete cessation of airflow for at least 10 seconds. Obstructive apnea: airway collapse despite respira-

tory effort (common in obstructive sleep apnea).

Apoenzyme

The protein component of an enzyme without its required cofactor or coenzyme. Inactive or minimally active until the cofactor binds. The complete, catalytically active complex of apoenzyme + cofactor is called the holoenzyme.

Apolipoprotein B100

The structural apolipoprotein on atherogenic particles such as LDL and VLDL; its concentration reflects the number of potentially atherogenic lipoprotein particles.

Apolipoprotein CII

A cofactor required to activate lipoprotein lipase, the enzyme that hydrolyzes triglycerides in circulating lipoproteins.

Apolipoproteins

Proteins associated with lipids in the blood that help transport and metabolize cholesterol and triglycerides.

Aponeurosis

A broad, flat sheet of connective tissue that attaches muscle to bone or other tissues.

Apophenia

The perception of meaningful patterns or connections in unrelated or random events; marked apophenia may occur in some psychiatric or neurologic conditions.

Apoplexy

An older term for sudden loss of neurologic function, historically referring to stroke and sometimes used for hemorrhage or infarction in an organ.

Apoptosis

Programmed cell death. Dysregulated in cancer (avoiding death). Pro-apoptotic genes: BAX, P53. Anti-apoptotic: BCL-2.

Apoptosis Extrinsic Pathway

The extrinsic pathway of apoptosis is triggered by extracellular ligands binding to death receptors belonging to the tumor necrosis factor receptor superfamily, including Fas (CD95), TNFR1, and TRAIL receptors DR4 and DR5.

Apoptosis Intrinsic Pathway

The intrinsic, or mitochondrial, pathway of apoptosis is initiated by intracellular stress signals including DNA damage, growth factor withdrawal, endoplasmic reticulum stress, and hypoxia.

Apoptotic Body

A membrane-bound fragment of a cell undergoing programmed cell death.

Apparent Mineralocorticoid Excess

Rare autosomal recessive disorder caused by deficiency of 11-beta-hydroxysteroid dehydrogenase type 2 normally converting cortisol to cortisone in the collecting duct. Cortisol activates mineralocorticoid receptors causing severe hypertension, hypokalemia, metabolic alkalosis, and low renin with low aldosterone.

Appendageal

Relating to skin appendages or structures derived from the skin, including hair follicles, sweat glands, and sebaceous glands.

Appendectomy

Surgical removal of the appendix, the most common emergency surgical procedure performed worldwide. Appendectomy is indicated for acute appendicitis, the most common surgical emergency with a lifetime risk of 7-8 percent and peak incidence between 10 and 30 years.

Appendiceal

Relating to the vermiform appendix, a narrow pouch projecting from the beginning of the large intestine.

Appendiceal Adenocarcinoma

A malignant gland-forming epithelial tumor of the

appendix. It may present with appendicitis or be discovered incidentally, and its management depends on histologic type, grade, invasion, and peritoneal spread.

Appendiceal Cancer

A malignant tumor arising in the appendix; types include neuroendocrine tumors and epithelial neoplasms, which can rupture or spread mucous-producing tumor cells within the abdomen.

Appendiceal Rupture

Tearing or perforation of an inflamed appendix, allowing intestinal contents to enter the abdomen and potentially causing peritonitis or an abscess.

Appendiceal Tumors

Neoplasms arising in the appendix, including neuroendocrine tumors, mucinous neoplasms, and adenocarcinoma. Some are found incidentally, whereas others cause pain, obstruction, or abdominal distention. Evaluation and treatment depend on the tumor type, size, spread, and pathologic features.

Appendicitis

Inflammation of the vermiform appendix, often caused by obstruction of its lumen. It may cause abdominal pain that moves to the right lower quadrant, nausea, loss of appetite, and fever; diagnosis and treatment depend on clinical assessment, laboratory tests, imaging, and complications.

Alternate Names:

Pain Lower Right Abdomen (Appendicitis)

Appendix

A blind-ended tubular diverticulum of the cecum, typically 6-10 cm long, rich in lymphoid tissue. Its position is variable (retrocecal, pelvic, or retroileal); the base usually projects near McBurney's point, one-third of the distance from the anterior superior iliac spine to the umbilicus.

Appendix Cancer

Cancer that begins in the appendix, including neuroendocrine tumors, mucinous tumors, and adenocarcinoma.

Appetite

The coordinated neuroendocrine drive and psychological desire to consume food, distinct from physiological hunger (the biological requirement for energy). Regulated in the arcuate nucleus of the hypothalamus by balancing orexigenic (appetite-stimulating) mediators—notably gastric ghrelin, neuropeptide Y, and agouti-related peptide (AgRP)—with anorexigenic (satiety-inducing) peptides—including adipose-derived leptin, pro-opiomelanocortin, glucagon-like peptide-1, peptide YY, and cholecystikinin. Pathologic extremes include anorexia (loss of appetite seen in cancer cachexia, chronic kidney disease, depression, and anorexia nervosa) and polyphagia or hyperphagia (excessive appetite in Prader-Willi syndrome, uncontrolled diabetes mellitus, and hypothalamic injury). Modulated pharmacologically with GLP-1 receptor agonists for obesity or appetite stimulants (mirtazapine, megestrol acetate) for cachexia.

Appetite - Decreased

Anorexia, or reduced desire to eat, may result from depression, infection, malignancy, chronic disease, medications, or psychological stress. Persistent unexplained appetite loss can cause unintentional weight loss and malnutrition. Evaluation considers duration, associated symptoms, and underlying etiology.

Appetite - Increased

Hyperphagia is abnormally increased hunger or food intake. It may result from endocrine disorders such as hyperthyroidism or diabetes mellitus, hypothalamic lesions, medications, psychological conditions, or recovery from illness. Persistent unexplained increase in appetite warrants clinical evaluation.

Appetite Regulation

Complex control by hormones (leptin, ghrelin, insulin, CCK), neural signals, psychological factors. Leptin signals satiety, ghrelin signals hunger.

Applied Behavior Analysis

A behavioral intervention based on learning principles that uses observation, measurement, and rein-

forcement to increase useful behaviors and reduce behaviors that interfere with functioning. It may be used in autism care and other settings, with goals individualized to the person.

Apposition

Placement of two structures next to or against each other; in wound care, good apposition means that tissue edges are accurately aligned.

Appropriate For Gestational Age

Describing a fetus or newborn whose estimated weight or size falls within the expected reference range for the gestational age, commonly between the 10th and 90th percentiles.

Apraxia

Inability to perform learned, skilled motor movements despite intact motor function, sensation, coordination, and comprehension.

Aquagenic Keratoderma

Transient, painful whitish thickening and wrinkling of the palms and soles occurring within minutes of water immersion, accompanied by a prickling or burning sensation. It is often associated with cystic fibrosis or other sweat gland disorders but can also be idiopathic or drug-induced. Symptoms resolve shortly after drying the skin.

Aquaporin

A membrane water-channel protein that permits rapid movement of water across cells; aquaporin-2 in kidney collecting ducts is regulated by vasopressin.

Aquatic Therapy

Exercise or rehabilitation performed in water, using buoyancy and resistance to support movement and conditioning.

Aqueous Humor

A clear, watery fluid that fills the anterior and posterior chambers of the eye, maintaining intraocular pressure and providing nutrients and oxygen to the avascular cornea and lens while removing metabolic

waste products.

Arachnodactyly

Abnormally long and slender fingers or toes. It may occur as an isolated trait or as a feature of connective-tissue disorders such as Marfan syndrome.

Arachnoid Cyst

A congenital, benign, cerebrospinal fluid-filled cyst within the arachnoid membrane, most commonly found in the middle cranial fossa. Most are asymptomatic and discovered incidentally on imaging, but large cysts may cause headache, seizures, or focal neurological deficits.

Arachnoid Mater

The middle of the three meninges, a delicate avascular membrane lying between the dura mater and pia mater. The subarachnoid space between arachnoid and pia contains cerebrospinal fluid (CSF) and major vessels. Arachnoid granulations (villi) project into the dural venous sinuses and mediate CSF resorption.

Arachnoiditis

Inflammation and scarring of the arachnoid membrane and nearby spinal nerves, often causing chronic pain or neurological symptoms.

Arbovirus

A virus transmitted by mosquitoes or other member of the arthropod phylum.

ARBs

Block angiotensin II AT1 receptors providing renoprotection similar to ACE inhibitors. Alternative for ACE inhibitor cough. Do not combine ACE inhibitor and ARB due to increased adverse effects without benefit. Similar renoprotective effects to ACE inhibitors.

Arciform

Having a curved or arch-like shape, used to describe the configuration of a skin lesion, rash, or other

anatomical finding.

Arcuate Uterus

A mild congenital uterine cavity variation characterized by a smooth, broad-based, saddle-shaped concavity of the fundal endometrium protruding < 1 cm into the uterine cavity, with a normal, convex external fundal contour. Arising from near-complete resorption of the uterovaginal septum, it is considered an anatomic variant with minimal or no adverse impact on reproductive outcomes or pregnancy loss.

Arcus Senilis (Corneal Arcus)

A white, gray, or bluish opaque ring in the peripheral corneal stroma separated from the limbus by a clear zone (lucid interval of Vogt). Caused by lipid (cholesterol) deposition in the peripheral cornea. Common and benign in elderly individuals (>60 years).

Alternate Names:

Corneal Arcus

Acute Respiratory Disease

A general term for a respiratory illness that begins suddenly or over a short period. It is not a specific diagnosis and may include infections or other conditions affecting breathing; it is different from acute respiratory distress syndrome (ARDS).

Alternate Names:

ARD

Area Under The Concentration-Time Curve

The integral of a drug's plasma concentration over time. AUC estimates total systemic exposure and is used to compare bioavailability, guide dosing for selected medicines, and characterize pharmacokinetic interactions.

Arenaviruses

See *Viral hemorrhagic fevers*.

Areola

The pigmented circular area surrounding the nipple.

It contains glands that help lubricate and protect the skin during breastfeeding.

Argentine Hemorrhagic Fever

See *Viral hemorrhagic fevers*.

Argininosuccinic Aciduria

Autosomal recessive urea cycle disorder caused by deficiency of argininosuccinate lyase (ASL gene, 7cen-q11.2). Biochemical hallmark: accumulation of argininosuccinic acid in blood, urine, and CSF with hyperammonemia, low citrulline, and elevated argininosuccinic acid.

Argyll Robertson Pupil

A small, irregular pupil that constricts during near focusing but responds poorly or not at all to light. It is classically associated with neurosyphilis but can have other causes; the finding requires clinical evaluation.

Arias-Stella Reaction

A hormonally induced atypical change in endometrial or endocervical glands during pregnancy or exposure to progestins; it is benign and can mimic glandular neoplasia histologically.

Arm

The upper limb between the shoulder and elbow; in everyday use, the term may also include the entire upper extremity.

Arm CT Scan

A computed tomography study that produces cross-sectional images of the arm using X-rays. It can identify structural abnormalities, injury, inflammation, bleeding, or masses.

Arm Fracture

See *Broken arm*.

Arm MRI Scan

A magnetic resonance imaging study that produces detailed images of the arm using magnetic fields and radiofrequency energy.

Arm Pain

Pain in the shoulder, upper arm, elbow, forearm, or related tissues. Causes include muscle or tendon injury, nerve compression, arthritis, vascular disease, and referred pain; sudden severe pain with chest discomfort, weakness, or shortness of breath requires urgent assessment.

Alternate Names:

Pain in the arm; upper-limb pain.

Armpit Lump

A lump or swelling in the axilla, commonly caused by an enlarged lymph node, skin infection, cyst, or benign growth. Persistent, hard, fixed, enlarging, or unexplained lumps require medical assessment.

Arnold-Chiari Malformation

Herniation of cerebellar tonsils through the foramen magnum into the upper cervical spinal canal. Type II (myelomeningocele-associated) is the most common form in pediatrics. Associated with hydrocephalus, syringomyelia, and spinal cord dysfunction. Symptoms include headache, neck pain, and lower cranial nerve palsies.

Aromatase

An enzyme that converts androgens into estrogens in tissues such as the ovaries, adipose tissue, and breast; it is inhibited by drugs such as anastrozole.

Aromatase Excess Syndrome

Rare autosomal dominant condition caused by mutations in the CYP19A1 gene leading to overexpression of aromatase enzyme and excessive estrogen production. Presents in males with gynecomastia, advanced bone age, and accelerated linear growth with early epiphyseal closure. In females with precocious puberty, macroclitoris, and virilization.

Aromatherapy

Use of essential oils or other plant-derived scents, usually by inhalation or topical application, for perceived relaxation or symptom relief; evidence and safety vary by oil and use.

Arousal Index

The number of brief awakenings or arousals from sleep per hour of sleep, usually measured during polysomnography; an elevated value may indicate sleep fragmentation.

Arrector Pili

A small involuntary smooth muscle attached to a hair follicle that contracts to raise the hair and produce gooseflesh.

Arrector Pili Muscle

A small band of involuntary smooth muscle attached to a hair follicle and the superficial dermis. Sympathetic stimulation causes it to contract, pulling the hair upright and producing temporary piloerection, commonly called goosebumps. Contraction also compresses nearby sebaceous glands and may help release sebum onto the skin.

Alternate Names:
goosebumps.

Arrest Cardiac (Sudden Cardiac Death)

Alternate index label for Sudden Cardiac Death.

Arrested Caries

A carious lesion that is no longer progressing, often because the surface has become hard, smooth, and cleansable. Arrested lesions may be monitored and remineralized; treatment depends on activity, cavitation, cleansability, and patient risk.

Arrhythmia

Any abnormality of cardiac rate, rhythm, or site of impulse origin, arising through enhanced automaticity, reentry, or triggered activity. Classified by rate as tachyarrhythmia or bradyarrhythmia and by site as supraventricular or ventricular. Diagnosis rests on the 12-lead ECG and ambulatory monitoring when intermittent.

Arrhythmias

Abnormal heart rhythms caused by altered impulse formation or conduction; they range from harmless premature beats to rhythms that impair circulation.

Arrhythmogenic Right Ventricular Cardiomyopathy

A genetic cardiomyopathy characterized by fibrofatty replacement of the right ventricular myocardium, predisposing to ventricular arrhythmias and sudden cardiac death, particularly during exercise. ECG shows T-wave inversions in V1-V3 and epsilon waves; diagnosis uses echocardiography, cardiac MRI, and genetic testing.

Arsenic

A naturally occurring metalloid that is toxic in many forms; exposure can cause gastrointestinal, neurologic, cardiovascular, and skin effects, while arsenic trioxide has specific therapeutic uses.

Arsenic Poisoning

Arsenic poisoning results from acute or chronic exposure to inorganic or organic arsenic. It can cause gastrointestinal illness, cardiovascular instability, neuropathy, characteristic skin changes, anemia, and increased cancer risk; suspected exposure requires poison-control or emergency guidance.

Alternate Names:
Poisoning Arsenic (Arsenic Poisoning)

Arsenic Toxicity

Systemic poisoning from arsenic exposure causing gastrointestinal distress, cardiac arrhythmias, peripheral neuropathy, and characteristic Mees lines on nails, requiring chelation therapy with dimercaprol or succimer.

Art Therapy

Use of art-making for therapeutic purposes. Emotional expression, coping, self-awareness.

Artemisinin

A plant-derived antimalarial compound from *Artemisia annua*; artemisinin derivatives are used in combination regimens, especially for falciparum malaria.

Arteria Femoralis

The femoral artery, the major artery of the thigh

and continuation of the external iliac artery below the inguinal ligament.

Arterial Blood Gas

Measurement of arterial pH, carbon-dioxide tension, oxygen tension, bicarbonate, and related acid-base variables. It helps assess ventilation, oxygenation, and metabolic or respiratory acid-base disorders.

Arterial Blood Gas

A test of arterial blood that measures pH, oxygen, carbon dioxide, and usually bicarbonate or related values to assess oxygenation, ventilation, and acid-base balance.

Arterial Blood Gas Analyzer

An analyzer that measures arterial pH, partial pressures of oxygen and carbon dioxide, and usually bicarbonate or calculated acid-base variables. It is used to assess ventilation, oxygenation, and metabolic or respiratory acid-base disorders.

Arterial Catheter

A catheter placed in an artery, commonly the radial artery, for continuous blood-pressure monitoring and repeated arterial blood sampling. Complications include hemorrhage, thrombosis, infection, and distal ischemia.

Arterial Catheterization

Placement of a catheter in an artery, usually the radial artery, for continuous blood-pressure measurement and repeated arterial blood sampling. It is used when rapid hemodynamic or gas-exchange assessment is clinically necessary and carries risks including bleeding, infection, thrombosis, and ischemia.

Arterial Duplex

Ultrasound technique combining B-mode imaging and Doppler flow assessment. Used for evaluating arterial stenosis, aneurysm, graft patency. Non-invasive, widely available.

Arterial Embolism

Sudden blockage of an artery by material carried through the bloodstream, usually a blood clot. It can cause acute limb, brain, kidney, intestinal, or other organ ischemia and requires urgent assessment.

Alternate Names:

Arterial Thromboembolism

Arterial Embolization

Intentional blockage of an artery, often to control bleeding or reduce blood supply to a tumor.

Arterial Injury

Damage to an artery that may cause bleeding, narrowing, clotting, or impaired blood flow.

Arterial Insufficiency

Reduced arterial blood flow to a tissue or limb, causing ischemic symptoms such as claudication, rest pain, cool skin, delayed capillary refill, and diminished pulses. Most commonly caused by atherosclerosis; assessed by ankle-brachial index and duplex ultrasound.

Arterial Line

A catheter inserted into a peripheral artery (most commonly radial, also femoral, brachial, dorsalis pedis) for continuous blood pressure monitoring and arterial blood gas sampling. Allen test: assesses ulnar collateral flow before radial artery cannulation.

Arterial Resistance

The opposition to blood flow produced mainly by the diameter and tone of small arteries and arterioles. It contributes to systemic vascular resistance and blood pressure.

Alternate Names:

Vascular resistance; systemic vascular resistance.

Arterial Stenosis

Pathologic narrowing of an arterial lumen that compromises downstream perfusion and tissue oxygenation. Most commonly caused by atheromatous plaque formation, followed by fibromuscular dysplasia.

sia (classically affecting renal and carotid arteries in young women with a string-of-beads angiographic appearance), vasculitis, radiation arteriopathy, or post-interventional neointimal hyperplasia. Hemodynamically significant stenosis typically exceeds 50–70% luminal diameter reduction, manifesting clinically as myocardial ischemia (angina pectoris), cerebral hypoperfusion (transient ischemic attack or stroke), renovascular hypertension (renal artery stenosis), or lower-extremity claudication and tissue loss in peripheral artery disease. Diagnosed via duplex Doppler ultrasonography, CT or MR angiography, or digital subtraction angiography; managed with medical risk-factor modification, antiplatelet therapy, percutaneous transluminal angioplasty with stenting, or surgical revascularization.

Arterial Stick

Needle puncture of an artery, usually to obtain blood for arterial blood-gas or related testing; bleeding and arterial injury are possible complications.

Arterial Switch Operation

The definitive anatomical surgical repair for complete D-transposition of the great arteries (D-TGA) with an intact ventricular septum or ventricular septal defect, routinely performed in the neonatal period (typically within the first 1 to 2 weeks of life before left ventricular deconditioning occurs). Involves transecting both great arteries above the sinuses of Valsalva, transposing them to their anatomically appropriate ventricles (connecting the neo-aorta to the left ventricle and the neopulmonary artery to the right ventricle), translocating the pulmonary artery confluence anterior to the neo-aorta via the Lecompte maneuver, and delicately excising and reimplanting the coronary artery buttons into the neo-aortic root.

Alternate Names:

Jatene Procedure; ASO

Arterial Tension

The pressure or partial pressure measured in arterial blood; depending on context, it may refer to blood pressure or to the tension of an arterial gas such as oxygen or carbon dioxide.

Arterial Thromboembolism

Alternate index label; see *Arterial embolism*.

Arterial Thrombosis

The formation of an occlusive or mural blood clot within an arterial lumen, most commonly triggered by atherosclerotic plaque rupture or erosion exposing subendothelial collagen and tissue factor. Under high arterial shear conditions, this leads to rapid platelet activation and aggregation, yielding a platelet-rich white clot. Represents an acute, organ-threatening emergency resulting in tissue infarction: acute coronary syndromes (STEMI/NSTEMI), ischemic stroke, acute mesenteric ischemia, or acute limb ischemia characterized by the classic 6 Ps (pain, pallor, pulselessness, paresthesias, paralysis, and poikilothermia). Treatment requires immediate systemic anticoagulation with intravenous unfractionated heparin, antiplatelet therapy, and emergency revascularization via percutaneous catheter-directed thrombectomy, catheter-directed thrombolysis, or surgical thromboembolectomy.

Arterial Ulcer

A chronic wound caused by arterial insufficiency (peripheral artery disease). Located on distal extremities: toes, heels, lateral malleolus, and sites of pressure. Presents with punched-out, deep, well-demarcated wound with pale or necrotic base, minimal exudate, and cool, hairless, shiny skin.

Arteries

Blood vessels that carry blood away from the heart. Most arteries carry oxygenated blood, except the pulmonary and umbilical arteries; their walls contain elastic tissue and smooth muscle to withstand and regulate pressure.

Alternate Names:

arterial vessels

Arteriogram

An imaging study of blood vessels that uses contrast material to identify narrowing, blockage, aneurysm, or abnormal blood flow.

Arteriole

A small artery that delivers blood to a capillary bed. Arteriolar smooth muscle regulates vascular resistance and controls tissue perfusion and systemic blood pressure.

Arterioles

Small branches of arteries that regulate blood flow into capillary beds through contraction and relaxation of their smooth muscle.

Alternate Names:

Resistance vessels; small arteries.

Arteriosclerosis

Chronic inflammatory disease of medium and large arteries characterized by accumulation of lipids, inflammatory cells, and fibrous tissue within the arterial wall, forming atherosclerotic plaques. See Atherosclerosis.

Arteriosclerosis / Atherosclerosis

Arteriosclerosis is hardening and thickening of arteries; atherosclerosis is plaque buildup within artery walls that can restrict blood flow.

Arteriosclerotic Retinopathy

Retinal vascular changes associated with thickening or hardening of small arteries, often occurring with chronic hypertension or atherosclerotic disease.

Arteriovenous Fistula

An abnormal connection between an artery and a vein. It may be present from birth, caused by an injury or procedure, or created for hemodialysis. A large fistula can put extra strain on the heart.

Arteriovenous Graft

Synthetic conduit connecting artery to vein for dialysis. Used when veins inadequate for fistula. Thrombosis and infection more common than AVF. Can be used earlier than fistula. Dec clotting with thrombectomy or angioplasty.

Arteriovenous Malformation

Abnormal connection between artery and vein with-

out capillary bed. Can bleed or cause symptoms.

Arteriovenous Malformation

An abnormal tangle of arteries and veins connected without the normal capillary bed; an AVM can bleed, cause seizures or neurologic symptoms, or produce high-flow complications.

Arteritis

Inflammation of the arterial wall occurring as part of primary systemic vasculitides, secondary autoimmune disorders, or localized infections. Classified by vessel caliber: large-vessel (giant cell arteritis, Takayasu arteritis), medium-vessel (polyarteritis nodosa, Kawasaki disease), and small-vessel (ANCA-associated vasculitides, IgA vasculitis). Manifestations include constitutional symptoms (fever, weight loss), localized ischemic pain (claudication, jaw claudication), absent pulses, and ischemic tissue infarction. Diagnosis relies on inflammatory markers, vessel imaging (CTA, MRA, catheter angiography), and definitive histopathologic tissue biopsy (such as temporal artery biopsy demonstrating granulomatous inflammation); managed acutely with high-dose corticosteroids, disease-modifying antirheumatic drugs (DMARDs), or targeted biologics (e.g., tocilizumab).

Arteritis, Giant Cell

See *Giant cell arteritis*.

Arteritis, Takayasu's

See *Takayasu's arteritis*.

Artery

A blood vessel that carries oxygenated blood away from the heart to the tissues, except for the pulmonary arteries which carry deoxygenated blood to the lungs. Arteries have thick, elastic, muscular walls that withstand high pressure and distribute blood throughout the body. The aorta is the largest artery.

Artery Occlusion

A blockage in an artery, often caused by a blood clot, fatty buildup, or material that traveled from

elsewhere. It can cut off blood flow; sudden pain, pale or cold skin, numbness, weakness, or no pulse is an emergency.

Arthralgia

Pain in one or more joints without objective evidence of joint inflammation such as swelling, warmth, or structural damage. Common causes include viral infections, overuse, autoimmune disease, and early degenerative change; persistent, severe, or multifocal pain warrants clinical evaluation.

Arthritis

Inflammation of one or more joints, characterized by joint pain, swelling, warmth, and stiffness.

Alternate Names:

Joint Inflammation (Arthritis)

Arthritis Ankylosing Spondylitis (Ankylosing Spondylitis)

Alternate index label for Ankylosing Spondylitis.

Arthritis Degenerative (Osteoarthritis)

Alternate index label for Osteoarthritis.

Arthritis Gout (Gout (Gouty Arthritis))

Alternate index label for Gout (Gouty Arthritis).

Arthritis Infectious (Septic Arthritis)

Alternate index label for Septic Arthritis.

Arthritis Juvenile (Juvenile Arthritis)

Alternate index label for Juvenile Arthritis.

Arthritis Lyme (Lyme Disease)

Alternate index label for Lyme Disease.

Arthritis MCTD (Mixed Connective Tissue Disease)

Alternate index label for Mixed Connective Tissue Disease.

Arthritis Plant Thorn (Plant Thorn

Synovitis)

Alternate index label for Plant Thorn Synovitis.

Arthritis Pseudogout (Pseudogout)

Alternate index label for Pseudogout.

Arthritis Psoriatic (Psoriatic Arthritis)

Alternate index label for Psoriatic Arthritis.

Arthritis Quackery (Quackery Arthritis)

Alternate index label for Quackery Arthritis.

Arthritis Reactive (Reactive Arthritis)

Alternate index label for Reactive Arthritis.

Arthritis Reiters (Reactive Arthritis)

Alternate index label for Reactive Arthritis.

Arthritis Sarcoid (Sarcoidosis)

Alternate index label for Sarcoidosis.

Arthritis Scleroderma (Scleroderma)

Alternate index label for Scleroderma.

Arthritis Sjogrens Syndrome (Sjogren's Syndrome)

Alternate index label for Sjogren's Syndrome.

Arthritis SLE (Systemic Lupus)

Alternate index label for Systemic Lupus.

Arthritis Still (Still's Disease)

Alternate name; see *Still Disease*.

Arthritis, Basal Joint

See *Thumb arthritis*.

Arthritis, Degenerative

See *Osteoarthritis*.

Arthritis, Gouty

See *Gout*.

Arthritis, Infectious

See *Septic arthritis*.

Arthritis, Juvenile Idiopathic

See *Juvenile idiopathic arthritis*.

Arthritis, Osteoarthritis

See *Osteoarthritis*.

Arthritis, Psoriatic

See *Psoriatic arthritis*.

Arthritis, Reactive

See *Reactive arthritis*.

Arthritis, Rheumatoid

See *Rheumatoid arthritis*.

Arthritis, Septic

See *Septic arthritis*.

Arthritis, Thumb

See *Thumb arthritis*.

Arthrocentesis

Aspiration of joint fluid for diagnostic or therapeutic purposes. Indications: septic arthritis (urgent, crystals), inflammatory arthritis (gout, pseudogout), hemorrhagic effusion (trauma, coagulopathy), and therapeutic drainage for symptomatic relief.

Arthrodesis

Surgical fusion of a joint by removing or preparing articular surfaces and stabilizing the bones until osseous union occurs. It relieves pain and provides stability at the cost of joint motion.

Arthrolysis

A procedure that releases scar tissue, adhesions, or other restrictions around a joint to improve movement. It may be performed through an arthroscope or by open surgery, depending on the joint, cause of stiffness, and clinical circumstances.

Arthrogram

An image of a joint obtained after contrast material is introduced into or around the joint, used to evaluate structures such as cartilage, ligaments, or the joint capsule.

Arthrography

Contrast injected into joint for imaging. Used for shoulder, knee, hip evaluation. MR arthrography more common now.

Arthrogryposis

A group of congenital disorders characterized by multiple joint contractures present at birth, caused by decreased fetal movement (fetal akinesia) during development.

Arthrogryposis Multiplex Congenita

A group of congenital conditions characterized by multiple joint contractures present at birth, often with muscle weakness or abnormal development of nerves, muscles, or connective tissue.

Arthropathy

Any pathologic disorder or structural disease affecting a synovial joint or articulation. Broadly categorized into inflammatory arthropathies (rheumatoid arthritis, seronegative spondyloarthropathies, crystal-induced arthropathies like gout and pseudogout), non-inflammatory degenerative arthropathies (osteoarthritis / degenerative joint disease), neuropathic arthropathies (Charcot joint secondary to sensory denervation in diabetic neuropathy, tabes dorsalis, or syringomyelia), and hemophilic arthropathy (resulting from recurrent intra-articular hemarthroses). Differentiated clinically through detailed joint examination (swelling, warmth, range of motion), synovial fluid analysis (clarity, cell count, crystals, Gram stain), and radiography.

Alternate Names:

Joint disease

Arthroplasty

Surgical reconstruction or replacement of a joint to relieve pain, restore alignment, or improve function. It may be partial or total and is commonly

performed for advanced degenerative joint disease, fracture, inflammatory arthritis, or other destructive joint conditions.

Arthropod Vector

An arthropod, such as a mosquito or tick, that transmits an infectious agent between hosts.

Alternate Names:

Disease vector

Arthroscope

A small illuminated instrument used to view the inside of a joint.

Arthroscopy

Minimally invasive joint visualization and intervention via a rigid arthroscope inserted through small portals, used for meniscal and ligamentous repair, chondroplasty, loose body removal, and synovectomy.

Arthrosis

Degenerative disease of a joint, characterized by progressive cartilage loss and changes in the underlying bone; osteoarthritis is the most common example.

Arthus Reaction

A localized immune-complex reaction causing inflammation at the site of antigen exposure.

Alternate Names:

AI

Articular Cartilage

Smooth cartilage covering the ends of bones within a joint, allowing low-friction movement and distributing load.

Articular Cartilage Injuries

Chondral/osteochondral lesions: Outerbridge grading (I: softening, II: fibrillation <50%, III: fibrillation >50%, IV: subchondral bone exposed). OCD (osteochondritis dissecans): subchondral bone necrosis, fragment separation (knee: lateral femoral condyle, elbow: capitellum, ankle: talar dome).

Articular Ligament

A band of connective tissue that stabilizes a joint by connecting bone to bone.

Alternate Names:

Joint ligament

Articulation

A joint or connection between bones; the term also refers to the production and clarity of speech sounds by the vocal organs.

Artifacts

Image distortions not representing true anatomy. Types: motion, beam hardening, metal, aliasing, partial volume. Recognition important for accurate interpretation.

Artificial Abortion

An induced termination of pregnancy using medication, a procedure, or both, rather than a spontaneous miscarriage.

Artificial Heart

A mechanical device that replaces or supports the pumping function of the heart.

Artificial Insemination

Placement of sperm into the reproductive tract by a medical procedure to assist conception.

Alternate Names:

Intrauterine insemination

Artificial Insemination By Donor

Intrauterine insemination using sperm from a donor other than the intended mother's partner to attempt fertilization and pregnancy.

Artificial Insemination By Husband

Therapeutic placement of washed, motile sperm obtained from a woman's husband or male partner directly into the uterine cavity (intrauterine insemination [IUI]) to facilitate conception in mild male-factor subfertility, cervical mucus hostility, or unexplained infertility; also termed homologous insemination.

Artificial Pacemaker

An implanted device that delivers electrical impulses to maintain an appropriate heart rate when the heart's natural conduction system is inadequate.

Artificial Rupture Of Membranes

The intentional rupture of the amniotic sac with a small hook-like instrument (Amnihook) during a cervical exam to induce or augment labor. Indications: induction or augmentation of labor, placement of internal monitors (fetal scalp electrode, intrauterine pressure catheter), and assessment of amniotic fluid (meconium staining).

Artificial Saliva

A product used to moisten the mouth and relieve symptoms of dry mouth.

Alternate Names:

Saliva substitute

Artificial Skin

A biological or synthetic material used to cover wounds and support skin regeneration.

Alternate Names:

Skin substitute

Artificial Urinary Sphincter

An implanted device that restores urinary continence by allowing controlled compression and release of the urethra. It is used mainly for selected cases of stress urinary incontinence and requires surgical placement and follow-up.

Artificial Ventricle (Total Artificial Heart)

A pulsatile, pneumatically or electrically driven biventricular replacement that removes the native ventricles. Indications: end-stage heart failure bridging to transplant or destination therapy. Requirements: surgical implantation, external power source, anticoagulation.

Alternate Names:

Total Artificial Heart

Arytenoid Cartilage

The arytenoid cartilages are a pair of small, pyramid-shaped cartilages located on the posterior aspect of the larynx, articulating with the superior border of the cricoid cartilage lamina, and they serve as the attachment points for the vocal cords and the intrinsic laryngeal muscles that control vocal cord movement.

ASA Physical Status Classification System

A system developed by the American Society of Anesthesiologists to assess and communicate a patient's pre-operative physical status and comorbidities. It ranges from ASA I (a normal healthy patient) to ASA VI (a declared brain-dead patient whose organs are being removed for donor purposes).

Asbestos-Related Disorders

Asbestos exposure can cause pleural plaques, diffuse pleural thickening, asbestosis, lung cancer, and mesothelioma after a long latency. Risk increases with cumulative exposure and smoking, and evaluation may include occupational history, imaging, and lung-function testing.

Asbestosis

Diffuse pulmonary fibrosis caused by prolonged inhalation of asbestos fibers. It may cause progressive shortness of breath and increases the risk of lung cancer and mesothelioma.

Ascariasis

Infection with the soil-transmitted roundworm *Ascaris lumbricoides*, usually acquired by swallowing eggs from contaminated soil, food, or water. It may cause no symptoms, abdominal problems, intestinal blockage, or cough during larval migration and is treated with antiparasitic medicine.

Ascending Aorta

From left ventricle at the aortic valve to the sternal angle (approximately 5 cm), enclosed in pericardium. Gives rise to right/left coronary arteries at the aortic sinuses. Most common site of aneurysm and dissection, especially in Marfan syndrome, bicuspid

aortic valve, and hypertension.

Ascending Aortic Aneurysm

Abnormal enlargement of the ascending portion of the aorta. It may be associated with a bicuspid aortic valve, connective-tissue disease, hypertension, or atherosclerosis.

Ascending Cholangitis

An acute, life-threatening infection of the biliary tract caused by the combination of biliary outflow obstruction (often by choledocholithiasis or stricture) and bacterial superinfection. Classic presentation includes Charcot's triad (fever, jaundice, right upper quadrant pain), which may progress to Reynolds' pentad with hypotension and altered mental status.

Alternate Names:

Acute cholangitis; Acute suppurative cholangitis

Ascending Colon

Extends from cecum to hepatic flexure below the liver, approximately 15-20 cm, retroperitoneal (fixed to posterior abdominal wall). Anteriorly: abdominal wall. Posteriorly: right kidney. Absorbs water/electrolytes and compacts fecal matter. Tumors may present as a palpable right-sided mass.

Ascites

Abnormal buildup of fluid in the abdominal cavity, which can cause abdominal swelling. It is most often caused by high blood pressure in the liver's portal veins due to cirrhosis, but it may also result from cancer, heart failure, kidney disease, or pancreatitis.

Ascitic Fluid

Fluid that collects in the space around the abdominal organs, called ascites. Common causes include cirrhosis, cancer, heart failure, and infection; new or painful fluid, especially with fever or confusion, may need testing with paracentesis.

Alternate Names:

Ascites

Ascorbic Acid

Vitamin C, a water-soluble vitamin needed for collagen formation, wound healing, antioxidant activity, and normal immune and connective-tissue function.

Ascus

An abbreviation for atypical squamous cells of undetermined significance, a Pap-test finding in which cervical squamous cells are abnormal but not clearly precancerous.

Aseptic

Free from disease-causing microorganisms, or performed using methods intended to prevent microbial contamination; it does not necessarily mean sterile.

Aseptic Loosening

Loss of fixation of an implant without active infection, commonly caused by mechanical stress and a biologic response to wear particles. It may cause pain, radiographic lucency, instability, and need for revision surgery.

Aseptic Meningitis

Inflammation of the meninges without evidence of typical bacterial meningitis, often caused by viruses or medicines.

Aseptic Necrosis

See *Avascular necrosis (osteonecrosis)*.

Alternate Names:

Avascular Necrosis (Aseptic Necrosis)

Aseptic Technique

A set of laboratory practices preventing contamination of sterile sites, cultures, and specimens by unwanted microorganisms. Key elements include hand hygiene, sterile fields and instruments, surface disinfection, and minimal exposure of sterile items to air and touch, forming the basis of reliable clinical and culture work.

Asherman Syndrome

Intrauterine adhesions or scar tissue that partially or completely obliterate the uterine cavity, most

often after uterine instrumentation or pregnancy-related curettage. It may cause hypomenorrhea or amenorrhea, infertility, recurrent miscarriage, or abnormal placentation.

Ashman Phenomenon

A physiological, rate-dependent form of aberrant intraventricular conduction (most frequently displaying a right bundle branch block morphology, owing to the inherently longer refractory period of the right bundle branch) that occurs when a short R-R interval immediately follows a long R-R interval. Classically observed in atrial fibrillation or atrial premature complexes, it occurs because the refractory period of the His-Purkinje system is directly proportional to the duration of the preceding cardiac cycle length (the preceding long cycle prolongs bundle refractoriness, causing the early impulse ending the short cycle to find one bundle branch still refractory). Differentiated from ventricular premature complexes by normal initial QRS vector and absence of a compensatory pause.

Alternate Names:

Ashman's Phenomenon; Aberrant Intraventricular Conduction

ASL

American Sign Language, a visual language used by many Deaf and hard-of-hearing people.

ASOT

Antistreptolysin O titer, a blood test measuring antibodies to streptolysin O produced by group A *Streptococcus*.

Asp

The standard amino-acid abbreviation for aspartic acid, an acidic amino acid incorporated into proteins and involved in cellular metabolism.

Aspartame

A low-calorie artificial sweetener made from two amino-acid components; people with phenylketonuria must avoid it because it provides phenylalanine.

Aspartate Aminotransferase

An enzyme found mainly in the liver, heart, and skeletal muscle; increased blood AST can indicate injury to these tissues, especially liver injury.

Aspartate Aminotransferase Blood Test

A laboratory test that measures serum AST activity, an enzyme present in liver, heart, skeletal muscle, and red blood cells. Elevated AST indicates tissue injury and, when interpreted with ALT, helps evaluate hepatocellular damage, myocardial infarction, or muscle disease.

Aspartic Acid

A nonessential amino acid incorporated into proteins and involved in the urea cycle, nucleotide synthesis, and other metabolic pathways. It is not the same as aspartame, an artificial sweetener that contains aspartic acid.

Asperger Syndrome

A former diagnostic term for a neurodevelopmental condition characterized by persistent social-communication difficulties and restricted or repetitive interests or behaviors, without the intellectual or language impairment traditionally associated with classic autism. Current diagnostic systems generally classify it within autism spectrum disorder; support is individualized.

Alternate Names:

Disorder Asperger (Asperger Syndrome) Syndrome
Asperger (Asperger Syndrome)

Aspergillosis

A spectrum of diseases caused by the fungus *Aspergillus* (most commonly *A. fumigatus*). Allergic forms: allergic bronchopulmonary aspergillosis (ABPA, in asthma or cystic fibrosis – wheezing, eosinophilia, pulmonary infiltrates), allergic *Aspergillus* sinusitis.

Alternate Names:

Aspergillus Infection (Aspergillosis)

Aspergillosis Precipitin

An antibody test that may support evaluation of exposure or immune response to *Aspergillus*; results

must be interpreted with symptoms, imaging, and other tests.

Aspergillus Infection (Aspergillosis)

Alternate index label for Aspergillosis.

Asphalt Cement Poisoning

Toxic effects caused by excessive exposure to asphalt cement products. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Asphyxia

A life-threatening lack of oxygen due to drowning, choking, or an obstruction of the airways. Management: remove obstruction, rescue breathing, CPR if needed. Foreign body airway obstruction: Heimlich maneuver (abdominal thrusts) for conscious adults; back blows and chest thrusts for infants.

Asphyxia Neonatorum

Failure of a newborn to establish adequate breathing at or shortly after birth.

Alternate Names:

Birth asphyxia

Aspirate

To withdraw fluid, cells, or tissue with suction, usually through a needle; to aspirate can also mean to inhale material into the airway.

Aspirate (Substance)

Material withdrawn from a body cavity, lesion, or airway by suction.

Alternate Names:

Aspirated material

Aspiration

The inhalation of foreign material (oropharyngeal secretions, gastric contents, food, liquid, or blood) into the larynx and lower respiratory tract. Aspiration pneumonitis (Mendelson syndrome) is a chemical lung injury caused by acidic gastric fluid.

Aspiration Pneumonia

Infectious pneumonia from aspiration of oropharyngeal secretions. Risk factors: dysphagia, impaired consciousness, enteral feeding. Common pathogens: anaerobes, gram-negative bacilli. Treatment: antibiotics covering oral flora.

Aspiration Pneumonitis

Chemical lung injury caused by inhalation of gastric contents, characterized by bronchospasm, hypoxemia, pulmonary edema, and atelectasis. Risk factors include full stomach, emergency surgery, obesity, pregnancy, gastroesophageal reflux, bowel obstruction, and decreased consciousness.

Aspiration Risk

The likelihood that food, liquid, saliva, or gastric contents will enter the larynx or lower respiratory tract. Risk rises with dysphagia, delirium, stroke, neuromuscular disease, reflux, poor dentition, and reduced consciousness; assessment and prevention are multidisciplinary.

Aspirin (Salicylate) Toxicity

Mixed respiratory alkalosis and metabolic acidosis. Early: tinnitus, hyperventilation, respiratory alkalosis. Late: metabolic acidosis, fever, seizures, coma. Treatment: activated charcoal, urine alkalinization (sodium bicarbonate), hemodialysis if severe (renal failure, pulmonary edema, severe acidosis, very high levels).

Alternate Names:

Salicylate

Aspirin And Heart Disease

Aspirin inhibits platelet aggregation and may prevent recurrent cardiovascular events in selected patients; routine primary prevention requires individualized assessment of benefit and bleeding risk.

Aspirin Overdose

Toxic effects caused by excessive exposure to Aspirin. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Aspirin-Exacerbated Respiratory Disease

A triad of asthma, nasal polyps, and aspirin sensitivity due to COX-1 inhibition. It involves overproduction of cysteinyl leukotrienes. Leukotriene receptor antagonists are specifically effective.

Assay

A laboratory procedure used to detect or measure a substance, activity, or biological response.

Assist-Control Ventilation

Ventilator delivers a set tidal volume at a set rate. Every breath receives the full set volume. Guarantees minimum rate and minute ventilation. Risk of respiratory alkalosis and auto-PEEP.

Assisted Breech Delivery

Vaginal breech delivery with operator assistance after spontaneous delivery to the umbilicus.

Assisted Circulation

Mechanical or procedural support used to maintain blood flow when the heart cannot pump adequately.

Alternate Names:

Mechanical circulatory support

Assisted Delivery With Forceps

Obstetric application of metal forceps to the fetal head during the second stage of labor to facilitate delivery when progress is arrested. Indications include prolonged second stage, fetal distress, or maternal exhaustion; complications include maternal lacerations and fetal facial nerve palsy.

Assisted Living

Residential setting with assistance for ADLs. Less intensive than nursing home.

Assisted Reproductive Technology

ART: fertility treatments involving manipulation of oocytes/sperm/embryos: in vitro fertilization (IVF), intracytoplasmic sperm injection (ICSI), intrauterine insemination (IUI, some definitions), embryo cryopreservation, gamete donation, and surrogacy.

Assistive Device

Equipment that helps a person perform activities, move safely, communicate, or maintain independence.

Assistive Device Training

Instruction in selecting, fitting, using, and maintaining devices such as walkers, canes, or crutches, including safe walking and stair techniques.

Association

A relationship or co-occurrence between two variables, findings, or conditions; association alone does not prove that one causes the other.

Association Cortex

Cerebral cortex involved in integrating information and supporting memory, language, judgment, and complex behavior.

Association Study

Comparison of allele frequencies between cases and controls. Genome-wide association studies (GWAS) identify common variants with small effects on complex diseases.

Astereognosis

Inability to identify an object by touch alone despite intact primary somatosensory modalities (touch, vibration, proprioception), indicating dysfunction of the contralateral parietal association cortex (typically the postcentral gyrus region or the dominant inferior parietal lobule).

Asterixis

Brief, irregular, non-rhythmic lapse of sustained posture causing a flapping movement, best elicited by asking the patient to hold the arms extended with wrists dorsiflexed ("hold arms up") for 30 seconds.

Asthenia

Abnormal lack of physical strength or energy. It is a symptom that may occur with illness, malnutrition, or medication effects.

Asthenic

Characterized by reduced strength or a slender, lightly built body habitus; in clinical descriptions it may refer to physical weakness or, less precisely, an asthenic constitution and should be distinguished from asthenia as a symptom.

Asthenopia

Eye strain or visual fatigue, often experienced as aching, burning, blurred vision, headache, or difficulty focusing after prolonged visual tasks. Refractive errors, dry eye, poor lighting, and extended screen use may contribute.

Asthenozoospermia

Reduced sperm motility, which may contribute to difficulty conceiving.

Alternate Names:

Asthenospermia

Asthma

A chronic inflammatory disorder of the airways characterized by recurrent episodes of wheezing, breathlessness, chest tightness, and cough. Airflow limitation is often reversible. It involves eosinophilic inflammation, mucus hypersecretion, and smooth muscle hypertrophy.

Asthma - Control Drugs

Long-term asthma medicines, especially inhaled corticosteroids, reduce airway inflammation and prevent exacerbations; they do not replace a rapid-relief medicine for acute symptoms.

Asthma - Quick-Relief Drugs

Rapid-relief asthma medicines, commonly inhaled short-acting beta-2 agonists, relax bronchial smooth muscle for prompt symptom relief; frequent use signals inadequate control.

Asthma Adult-Onset (Adult-Onset Asthma)

Alternate index label for Adult-Onset Asthma.

Asthma And School

School-based asthma care includes an action plan, access to prescribed rescue medicine, trigger reduction, and staff training to recognize worsening breathing.

Asthma Attack

A sudden worsening of asthma caused by increased airway narrowing and inflammation.

Asthma Complexities

Asthma varies in trigger, severity, airway inflammation, and response to treatment. Difficult-to-control symptoms may reflect poor inhaler technique, ongoing exposure, incorrect diagnosis, comorbid disease, or severe asthma requiring specialist assessment.

Asthma Exacerbation

An acute or subacute worsening of asthma symptoms and airflow limitation, typically causing increased wheeze, cough, chest tightness, or breathlessness.

Asthma Medications

Asthma medicines include controller treatments that reduce airway inflammation and reliever treatments that quickly relax airway muscle. The regimen is guided by symptom frequency, lung function, exacerbation risk, age, pregnancy, technique, and response.

Asthma Overview

Asthma is a chronic disorder of variable airway inflammation and narrowing that causes wheezing, cough, chest tightness, or breathlessness. Symptoms fluctuate with triggers, and management combines an individualized action plan, inhaled medicines, trigger reduction, and monitoring.

Alternate Names:

Bronchial Asthma (Asthma Overview)

Asthma, Exercise-Induced

See *Exercise-induced asthma*.

Asthma, Occupational

See *Occupational asthma*.

Asthma-COPD Overlap

Features of both asthma and COPD in the same patient. Characterized by persistent airflow limitation with features of both diseases: airway inflammation (eosinophilic and neutrophilic), bronchial hyperresponsiveness, and partially reversible obstruction. Treatment: inhaled corticosteroids plus long-acting bronchodilators.

Asthma: Over The Counter Treatment

Over-the-counter products do not replace prescription asthma evaluation or a rescue plan. Some may temporarily relieve symptoms but can delay treatment of dangerous airway obstruction; worsening breathlessness, difficulty speaking, cyanosis, or poor response to a reliever is an emergency.

Alternate Names:

Asthma OTC Medication (Asthma: Over The Counter Treatment)

Astigmatism

A refractive error caused by an irregular curvature of the cornea or lens, resulting in light rays being focused at multiple points rather than a single point on the retina, leading to blurred or distorted vision at all distances.

Astigmatism Overview

Astigmatism is an irregular curvature of the cornea or lens that prevents light from focusing at one retinal point. It can cause blurred or distorted vision, eyestrain, and headaches and is corrected with glasses, contact lenses, or selected procedures.

Astringent

A substance that contracts tissue and reduces secretions or minor bleeding.

Astrocyte

A star-shaped glial cell that supports neurons, maintains the blood-brain barrier, and regulates the nervous-system environment.

Astrocytoma

A glioma arising from star-shaped cells called astrocytes. Graded I-IV by the WHO; grade IV (glioblastoma) is the most aggressive and common primary malignant brain tumor in adults.

Asymmetrical

Unequal in shape, size, distribution, or appearance between corresponding sides; asymmetry of a pigmented lesion can be a warning feature requiring assessment.

Asymptomatic

Having a disease, infection, or abnormal finding without noticeable symptoms. A person may still have detectable disease, develop complications, or transmit an infection; asymptomatic is different from presymptomatic, when symptoms develop later.

Asymptomatic Adults

Adults who report no symptoms attributable to the condition or disease being assessed. Asymptomatic status does not exclude subclinical disease, risk factors, or early pathology. Evaluation may involve history, examination, screening tests, or targeted investigations according to age, risk, and clinical guidelines. The term is used in screening, preventive medicine, epidemiology, and studies of disease detected before symptoms develop.

Asymptomatic Bacteriuria

Bacteria detected in urine without urinary symptoms; treatment is selective, such as during pregnancy or before certain urologic procedures.

Asymptomatic Bacteriuria In Pregnancy

The presence of significant bacterial growth in a pregnant person's urine without urinary symptoms. It is identified by urine culture and is treated because it increases the risk of symptomatic urinary infection and kidney infection during pregnancy.

Asymptomatic Carrier

A person who harbors a pathogen and can potentially transmit it despite having no symptoms of

the associated infection.

Asymptomatic HIV Infection

HIV infection without current symptoms; ongoing viral replication can damage immunity, so diagnosis, monitoring, and antiretroviral treatment remain important.

Asymptomatic Infections

Infections in which the person does not notice or develop symptoms, although the pathogen may still be detectable and may sometimes cause complications or be transmitted to others. This differs from presymptomatic infection, in which symptoms develop later.

Alternate Names:

Subclinical infections

Asystole

Complete cessation of cardiac electrical activity, presenting as a flat line on the ECG with no QRS complexes, P waves, or other organized electrical activity.

Ataxia

Lack of voluntary coordination of muscle movements, resulting from dysfunction of the cerebellum, posterior columns, or peripheral nerves. Types: cerebellar (gait and limb ataxia, dysarthria, nystagmus), sensory (impaired proprioception causing sensory ataxia, positive Romberg sign), and vestibular (vertigo, nystagmus, nausea).

Ataxia Telangiectasia

An inherited disorder causing progressive coordination problems, visible small blood vessels, and immune deficiency.

Alternate Names:

A-T

Ataxia-Telangiectasia

An autosomal-recessive disorder usually caused by biallelic ATM variants, characterized by progressive ataxia, telangiectasias, immune deficiency, and increased cancer risk.

Ataxic Breathing

An irregular, unpredictable breathing pattern with random variations in rate and depth, seen in medullary lesions. It is the most dangerous respiratory pattern and often precedes respiratory arrest. It requires mechanical ventilation.

Atelectasis

Complete or partial collapse of lung tissue, resulting in loss of aeration. It may be caused by airway obstruction (resorption atelectasis), pleural disease (compressive atelectasis), surfactant dysfunction (dissolution atelectasis), or scarring (cicatrizatio atelectasis). It impairs gas exchange and may predispose to infection.

Alternate Names:

Lung Collapse

Athelia

Congenital or acquired absence of one or both nipples.

Atherectomy

Mechanical removal of atherosclerotic plaque. Types: directional, laser, rotational, orbital. Used for peripheral arterial disease.

Atherectomy (Vascular)

Removal or modification of atherosclerotic plaque from inside a blood vessel, usually with a catheter-based device.

Alternate Names:

Vascular

Atheroembolic Renal Disease

Cholesterol crystal embolization to kidneys after vascular procedures. AKI with livedo reticularis and peripheral eosinophilia. Renal biopsy shows cholesterol clefts. Poor prognosis with progressive renal failure.

Atheroma

A localized accumulation of lipid, inflammatory cells, and fibrous tissue within an arterial wall, forming part of an atherosclerotic plaque.

Alternate Names:

Atheromatous plaque; arterial plaque.

Atherosclerosis

Chronic inflammatory disease of medium and large arteries characterized by accumulation of lipids, inflammatory cells, and fibrous tissue within the arterial wall, forming atherosclerotic plaques. Initiating event is endothelial dysfunction from risk factors (hypertension, smoking, diabetes, dyslipidemia).

Atherosclerotic

Relating to atherosclerosis, the buildup of lipid-rich plaque and fibrous tissue in arterial walls that can narrow arteries or promote thrombosis.

Atherosclerotic Plaque

An atherosclerotic plaque is a lesion within the arterial intima composed of a necrotic lipid core covered by a fibrous cap, representing the pathological substrate of atherosclerotic cardiovascular disease.

Atherothrombotic Stroke

An ischemic stroke caused by thrombosis or artery-to-artery embolism arising from atherosclerotic disease in a large artery supplying the brain.

Alternate Names:

Large-artery atherosclerotic stroke; atherothrombotic ischemic stroke.

Athetosis

Slow, involuntary, writhing, sinuous movements predominantly of the distal extremities, sometimes also affecting the face, neck, and trunk, arising from dysfunction of the basal ganglia (putamen and globus pallidus).

Athlete Foot

Tinea pedis: dermatophyte infection (*Trichophyton rubrum*, *T. mentagrophytes*) of the feet, common in those who wear occlusive footwear or use communal showers. Forms: interdigital (scaly, fissured, macerated between toes—most common), moccasin (diffuse plantar scaling), vesicular.

Athlete'S Foot

A superficial fungal infection of the feet, usually affecting the spaces between the toes and causing itching, scaling, maceration, or fissures. It is also called tinea pedis.

Alternate Names:

tinea. Athletes Foot (Athlete's Foot) Ringworm Of The Foot Tinea Pedis (Athlete's Foot)

Athletes Foot (Athlete'S Foot)

Alternate index label for Athlete's Foot.

Athlete'S Heart

A constellation of benign, physiological cardiovascular adaptations resulting from prolonged, intense physical conditioning in endurance and strength athletes. Characterized morphologically by balanced, biventricular eccentric or concentric hypertrophy, increased left ventricular end-diastolic chamber dimensions, resting sinus bradycardia, sinus pauses, and physiologic flow murmurs. It is distinguished from hypertrophic cardiomyopathy and dilated cardiomyopathy by normal or supranormal diastolic filling, preserved left ventricular ejection fraction, regression after detraining, and absence of late gadolinium enhancement on cardiac MRI.

Alternate Names:

Athletic Heart Syndrome; Athlete Heart Remodeling

Atlanto-Occipital Dislocation

Traumatic separation of the atlas from the occipital condyles, a highly unstable injury with high mortality often associated with spinal cord injury and requiring emergent immobilization.

Atlas

First cervical vertebra articulating with occipital condyles at the atlanto-occipital joint for nodding movement. Lacks a body and spinous process; has anterior/posterior arches and lateral masses. Transverse foramina transmit vertebral arteries. The anterior arch has a facet for the axis dens.

Atlas And Axis Joint

The atlantoaxial joint between the first cervical

vertebra, the atlas, and the second, the axis; it permits much of the head's rotation.

Atom

The smallest unit of an element that retains that element's chemical properties, consisting of a nucleus surrounded by electrons.

Atonic

Lacking normal muscle tone or contractile strength; an atonic bladder, for example, does not contract effectively to empty urine.

Atonic Seizure

A generalized seizure characterized by a sudden loss of muscle tone lasting one to two seconds, frequently causing the patient to collapse abruptly to the floor (drop attack). Common in childhood-onset epileptic encephalopathies, particularly Lennox–Gastaut syndrome.

Alternate Names:

Drop attack; Akinetic seizure

Atopic

Relating to an inherited or acquired tendency to develop allergic reactions such as eczema, allergic rhinitis, or asthma.

Atopic Dermatitis

A chronic, relapsing inflammatory skin disease characterized by pruritus, xerosis, and eczematous lesions. It reflects epidermal-barrier dysfunction and immune dysregulation, with flares triggered by irritants, allergens, infection, stress, or climate.

Alternate Names:

Allergy Eczema (Atopic Dermatitis)

Atopic Dermatitis (Eczema)

A chronic inflammatory skin condition causing itchy, dry, irritated, or rash-covered skin, often associated with an atopic tendency.

Alternate Names:

Dermatitis, Atopic

Atopic Dermatitis - Children - Homecare

Home management of childhood eczema using regular emollients, gentle skin cleansers, and topical anti-inflammatory agents as prescribed. Trigger avoidance, infection prevention, and appropriate clothing help reduce flares and improve quality of life.

Atopic Eruption Of Pregnancy

Pruritic eczematous skin eruptions occurring in pregnancy, presenting as eczematous patches or pruritic urticarial papules and plaques, typically resolving after delivery.

Atopic Rhinitis

Allergic inflammation of the nasal lining causing sneezing, itching, congestion, or discharge.

Alternate Names:

Allergic rhinitis

Atopy

The genetic predisposition to develop immediate (IgE-mediated, type I) hypersensitivity reactions to common environmental allergens (pollen, dust mites, foods, pet dander).

Atovaquone–Proguanil

An antimalarial combination used for prevention and treatment of *Plasmodium falciparum* malaria and selected travel-related malaria regimens.

ATP

Adenosine triphosphate, the principal energy-transfer molecule in cells; hydrolysis of its phosphate bonds powers many cellular processes.

ATP Synthase

Complex V of electron transport chain. Uses proton gradient to synthesize ATP. Rotational catalysis mechanism.

Atransferrinemia

Very rare autosomal recessive disorder of iron transport caused by mutations in the TF gene (3q22.1) leading to absence or severe deficiency of transferrin.

Atresia

Congenital absence, closure, or severe narrowing of a normal body opening or passage, such as the intestinal, biliary, or pulmonary outflow tract.

Atresia Of Bile Ducts

See *Biliary atresia*.

Atria

The two upper chambers of the heart, which receive blood from the systemic and pulmonary veins and pass it to the ventricles.

Atrial

Relating to an atrium, especially one of the two upper chambers of the heart.

Atrial Ablation

A procedure that destroys selected heart tissue to prevent or control an atrial arrhythmia.

Alternate Names:

Cardiac ablation

Atrial Arrhythmia

An abnormal heart rhythm that starts in the atria, the heart's upper chambers. Examples include atrial fibrillation, atrial flutter, and some supraventricular tachycardias; symptoms may include palpitations, breathlessness, dizziness, or fainting.

Atrial Fib

Atrial fibrillation, an irregular and often rapid atrial rhythm that can reduce coordinated heart contraction and increase the risk of blood clots and stroke.

Atrial Fibrillation

The most common sustained cardiac arrhythmia, characterized by rapid, disorganized atrial activity with an irregular ventricular response. ECG typically shows absent organized P waves and irregularly irregular RR intervals. Associated conditions include hypertension, ischemic or valvular heart disease, cardiomyopathy, atrial septal defect, and thyroid dysfunction, especially hyperthyroidism. Alcohol, myocarditis, pericarditis, pulmonary embolism,

pneumonia, and cardiac surgery can precipitate AF; sick sinus syndrome may coexist as tachycardia-bradycardia syndrome. Some cases have no identified cause (formerly called lone atrial fibrillation).

Alternate Names:

AF (Atrial Fibrillation) Afib (Atrial Fibrillation)

Atrial Fibrillation

Alternate index label for Atrial Fibrillation.

Atrial Fibrillation (AFib) Warning Symptoms

Alternate index label for AFib) Warning Symptom.

Atrial Fibrillation Ablation

A catheter-based procedure that creates lesions, usually around the pulmonary veins, to reduce the electrical triggers of atrial fibrillation. It may be considered for symptomatic patients when medicines are ineffective, not tolerated, or not preferred. Results and the need for repeat treatment vary with the type of atrial fibrillation and the person's clinical features.

Atrial Fibrillation AFib Treatment Drugs

Medicines for atrial fibrillation may slow the ventricular rate, restore or maintain rhythm, or reduce stroke risk with anticoagulation. Choice depends on symptoms, heart structure, kidney function, bleeding risk, other diseases, and whether rhythm control is appropriate.

Atrial Fibrillation And Atrial Flutter

Atrial fibrillation produces chaotic atrial activity, while atrial flutter produces a rapid organized circuit; both can cause palpitations and increase stroke risk.

Atrial Flutter

A rapid, regular atrial rhythm (typically 250-350 bpm) caused by a macro-reentrant circuit, most commonly in the right atrium. ECG shows sawtooth flutter waves best seen in leads II, III, and aVF, with regular or irregular ventricular response depending on AV conduction ratio (commonly 2:1 or 4:1).

Atrial Myxoma

A usually benign primary heart tumor, most often in the left atrium, that may obstruct blood flow or produce emboli and constitutional symptoms.

Atrial Natriuretic Peptide

A peptide hormone secreted by atrial myocytes in response to atrial stretch from volume expansion. ANP promotes natriuresis and diuresis by increasing glomerular filtration rate, inhibiting sodium reabsorption in the collecting ducts, and reducing renin and aldosterone secretion.

Atrial Septal Defect

Congenital opening in the interatrial secundum allowing left-to-right shunting of blood. Small defects may close spontaneously. Larger defects cause right heart volume overload, right atrial and ventricular dilation, pulmonary hypertension, and eventually right heart failure. Paradoxical embolism may occur with right-to-left shunting.

Atrial Septal Defect

A congenital opening between the heart's upper chambers that permits abnormal blood flow between the atria.

Atrial Septum

The wall separating the right and left atria; an atrial septal defect is an abnormal opening in this wall.

Atrial Standstill

Absence of normal electrical and mechanical activity in the atria.

Atrial Switch Operation

A historical physiological reparative surgical operation for complete D-transposition of the great arteries (D-TGA), developed prior to the modern arterial switch operation. Involves extensive intra-atrial baffle reconstruction utilizing resected atrial septal flaps and autologous pericardium (the Mustard procedure) or atrial wall flaps (the Senning procedure) to redirect systemic and pulmonary venous return at the atrial level without transposing the great arteries: systemic venous blood is directed

across the mitral valve into the left ventricle and out the pulmonary artery to the lungs, while oxygenated pulmonary venous blood is directed across the tricuspid valve into the right ventricle and out the aorta. Over decades, patients frequently develop systemic right ventricular dysfunction, tricuspid regurgitation, sinus node dysfunction, and baffle leak or stenosis.

Alternate Names:

Mustard Procedure; Senning Procedure

Atrial Tachycardia

Supraventricular tachycardia arising from an atrial focus outside the sinus node, often focal with rates 100-250 beats per minute with discrete P waves of different morphology from sinus P waves.

Atrial Thrombosis

Formation of a blood clot within an atrium of the heart.

Atrioventricular

Pertaining anatomically and electrophysiologically to both the atria and ventricles of the heart, specifically encompassing the atrioventricular valves (mitral and tricuspid) and the specialized junctional conduction system (AV node and bundle of His) that coordinates the timed delay between atrial and ventricular contraction.

Atrioventricular Block

Delayed or interrupted electrical conduction between the atria and ventricles of the heart.

Atrioventricular Bundle

Conducting fibers that carry electrical impulses from the atrioventricular node to the ventricles.

Alternate Names:

Bundle of His

Atrioventricular Canal Defect

A congenital heart defect involving abnormal development of the septum and atrioventricular valves; also called an atrioventricular septal defect.

Alternate Names:

an atrioventricular septal defect. Septal Defect, Atrioventricular

Atrioventricular Nodal Reentrant Tachycardia

The most common type of paroxysmal regular supraventricular tachycardia, caused by a functional reentrant circuit within the atrioventricular node utilizing distinct fast and slow conduction pathways. Characterized on electrocardiogram by narrow QRS complexes at rates of 140–220 beats per minute with retrograde P waves embedded within or just following the QRS.

Alternate Names:

AV nodal reentrant tachycardia; AVNRT

Atrioventricular Nodal Reentry Tachycardia

A rapid regular heart rhythm caused by a reentry circuit near the atrioventricular node; commonly abbreviated AVNRT.

Atrioventricular Nodal Reentry Tachycardia

A rapid regular heart rhythm caused by a reentrant electrical circuit involving or near the atrioventricular node.

Atrioventricular Node

Located in the posterior-inferior region of the interatrial septum near the coronary sinus. It receives impulses from the SA node via internodal pathways and delays them by approximately 120-200 milliseconds before transmitting them to the ventricles. This delay allows atrial contraction to complete before ventricular contraction.

Atrioventricular Valves

The heart valves between the atria and ventricles: the tricuspid valve on the right and the mitral valve on the left.

Atrium

One of the two upper chambers of the heart that receive blood returning to the heart. The right atrium receives deoxygenated blood from the superior and

inferior venae cavae and the coronary sinus; the left atrium receives oxygenated blood from the four pulmonary veins.

Atrophoderma Maculatum

A rare skin condition causing localized, depressed patches of altered pigmentation and tissue thinning.

Atrophy

Reduction in the size and functional capacity of a tissue, organ, or muscle caused by loss or shrinkage of its cells. It may result from disuse, denervation, inadequate blood supply, malnutrition, endocrine deficiency, chronic disease, or aging. Atrophy can be reversible when the underlying cause is corrected, but prolonged injury may produce permanent structural loss and fibrosis. Clinical assessment identifies the distribution, time course, associated weakness or dysfunction, and the underlying neurologic, vascular, nutritional, or systemic cause.

Atrophy Vaginal (Vaginal Dryness And Vaginal Atrophy)

Alternate index label for Vaginal Dryness and Vaginal Atrophy.

Atropine

An anticholinergic drug that blocks muscarinic acetylcholine receptors; it can increase heart rate, reduce secretions, dilate the pupils, and treat selected poisonings or bradycardia.

Atropine In Organophosphate Poisoning

A muscarinic antagonist used to reverse bronchorrhea, bronchospasm, bradycardia, and other muscarinic effects of organophosphate toxicity. Treatment is titrated to drying of respiratory secretions and improvement in ventilation, not to pupil size alone.

Attachment (Dental)

Retentive device connecting denture to implant or tooth. Precision attachments preferred over bar attachments.

Alternate Names:

Dental

Attachment Disorder

See *Reactive attachment disorder*.

Attack Rate

The proportion of an at-risk population that develops a specified illness during a defined outbreak or exposure period. It is a cumulative incidence measure commonly used for foodborne and other localized outbreaks.

Attention

The ability to select, focus on, and sustain mental processing of relevant information while limiting distraction.

Attention-Deficit/Hyperactivity Disorder

A neurodevelopmental disorder involving persistent inattention and/or hyperactivity-impulsivity that begins in childhood and impairs functioning in more than one setting.

Attenuated

Weakened or reduced in force, severity, concentration, or biological activity; an attenuated vaccine contains a weakened form of a microorganism.

Attenuation Correction

Compensation for photon absorption in tissue. CT-based (PET/CT), transmission source (SPECT), or MR-based (PET/MRI).

Attributable Risk

Difference in risk between exposed and unexposed. $AR = \text{incidence exposed} - \text{incidence unexposed}$.

Atypia

Cytologic or architectural abnormality that deviates from the expected appearance of a cell or tissue. Atypia may be reactive, degenerative, premalignant, or malignant and must be interpreted with architecture, clinical context, and ancillary studies.

Atypia Morphology

Abnormal cellular appearance or architecture that may be benign, reactive, precancerous, or cancerous

depending on context.

Alternate Names:

Cytologic atypia

Atypical

Not typical of the usual appearance, behavior, structure, or course of a condition; atypical does not by itself indicate whether a finding is benign or malignant.

Atypical Antipsychotics

A class of modern psychiatric medications used to treat schizophrenia and bipolar disorder. They work by balancing dopamine and serotonin levels in the brain, which helps reduce symptoms like hallucinations and delusions with fewer movement-related side effects than older antipsychotics.

Atypical Endometrial Hyperplasia

A premalignant proliferation of endometrial glands with cytologic atypia, now commonly classified as endometrial intraepithelial neoplasia; it carries a risk of concurrent or subsequent endometrial carcinoma.

Atypical Facial Pain

Chronic facial pain without identifiable cause. Psychological factors may contribute.

Atypical Features

A specifier for depressive episodes characterized by mood reactivity, significant weight gain or increased appetite, hypersomnia, leaden paralysis, and a long-standing pattern of interpersonal rejection sensitivity. Atypical depression often presents in younger patients and may respond preferentially to MAOIs or SSRIs rather than TCAs.

Atypical Fibroxanthoma

A superficial, usually low-grade malignant fibrohistiocytic tumor of sun-damaged skin in older adults; it may resemble squamous-cell carcinoma or melanoma and is treated by complete excision.

Atypical Genitalia

External genital anatomy that does not have typical

male or female features, often because of differences in sex development.

Atypical Hemolytic Uremic Syndrome

A rare, life-threatening thrombotic microangiopathy caused by uncontrolled dysregulation and hyperactivation of the alternative complement pathway. Clinically characterized by microangiopathic hemolytic anemia, severe thrombocytopenia, and progressive acute kidney injury without preceding Shiga toxin-producing bacterial infection.

Alternate Names:

aHUS; Complement-mediated HUS

Atypical Hyperplasia Of The Breast

An overgrowth of abnormal cells in breast tissue that is not cancer but may increase future breast-cancer risk.

Atypical Lymphocyte

A lymphocyte with an unusual appearance, often reflecting immune activation or infection.

Atypical Measles Syndrome

A rare severe illness formerly reported after exposure to measles virus in people who had received an inactivated measles vaccine; it can cause unusual rash, high fever, and pneumonia.

Atypical Pneumonia

Pneumonia with a gradual onset, relatively mild chest findings, and prominent systemic or extrapulmonary symptoms. Common causes include *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, and respiratory viruses.

Atypical Spitz Nevus

A melanocytic proliferation with Spitz-like features and atypia that may resemble melanoma; diagnosis relies on clinicopathologic correlation and, when needed, expert histopathologic review.

Atypical Teratoid Rhabdoid Tumor

A rare, aggressive cancer of the brain and spinal cord that occurs mainly in infants and young children.

Alternate Names:

ATRT Atypical Teratoid Rhabdoid Tumor Atypical Teratoid/Rhabdoid Tumor Brain Cancer

Audiogram

A chart that shows a person's ability to hear at different pitches or frequencies.

Audiologist

A health professional who assesses hearing and fits hearing aids.

Audiology

The science and clinical evaluation of hearing, balance, and related disorders.

Audiometry

Measurement of hearing sensitivity at different sound frequencies and intensities, usually using headphones or insert earphones.

Auditory Acuity

The sensitivity or sharpness of hearing, commonly assessed by determining the softest sounds a person can detect at different frequencies.

Auditory Area

A region of the cerebral cortex that processes sound information.

Alternate Names:

Auditory cortex

Auditory Fatigue

Temporary reduction in hearing sensitivity or listening performance after prolonged or intense sound exposure.

Auditory Information Processing Disorder

See *Auditory processing disorder*.

Auditory Nerve

The portion of the vestibulocochlear nerve that carries hearing signals from the inner ear to the brain.

Auditory Ossicle

One of the three small middle-ear bones that transmit sound vibrations.

Alternate Names:

Ear ossicle

Auditory Pathway

The neural route carrying sound information from the inner ear to the auditory cortex.

Auditory Perception

The brain's process of detecting, recognizing, and interpreting sounds received through the auditory system.

Auditory Processing Disorder

Difficulty recognizing or interpreting sounds despite normal or near-normal hearing sensitivity.

Alternate Names:

APD

Auditory Threshold

The quietest sound intensity a person can detect at a specified frequency.

Alternate Names:

Hearing threshold

Auer Rods

Rod-shaped inclusions formed from fused azurophilic granules in myeloid precursor cells. They strongly support a myeloid neoplasm, especially acute myeloid leukemia, but diagnosis requires immunophenotyping, genetic studies, and clinical correlation.

Auger Electrons

Low-energy electrons emitted during electron capture decay. Short range (nanometers), high linear energy transfer. Potential for targeted radiotherapy if localized near DNA.

Augmentation Mammoplasty

Surgical enlargement of the breasts using implants or, in selected procedures, transferred fat; risks, expected results, and suitability depend on anatomy

and the chosen technique.

Augmentation Mammoplasty

Surgery to increase breast size or alter breast shape, commonly using implants or fat transfer.

Alternate Names:

Breast augmentation

Augmentation Of Labor

The stimulation of spontaneous contractions that are judged to be inadequate to achieve progressive cervical dilatation. Most commonly uses oxytocin (Pitocin) infusion. Also includes artificial rupture of membranes (AROM, amniotomy).

Aura

A temporary sensory, visual, or other neurological symptom that may precede a migraine or seizure.

Aural Fullness

A sensation of pressure, blockage, or plugged feeling in the ear. Causes include eustachian-tube dysfunction, middle-ear fluid, cerumen impaction, inner-ear disorders, and temporomandibular-joint disease.

Alternate Names:

Ear pressure; plugged-ear sensation; blocked-ear sensation.

Aural Polyps

Abnormal tissue projections in the ear canal or middle ear, often associated with chronic infection, inflammation, or cholesteatoma and requiring otologic evaluation.

Auricle

The visible external portion of the ear that projects from the side of the head, designed to collect and channel sound waves into the ear canal.

Auricular Hematoma

A collection of blood between the cartilage of the external ear and its perichondrium, usually after blunt trauma. Without prompt drainage and compression, it may cause cartilage necrosis and permanent cauliflower-ear deformity.

Alternate Names:

Pinna hematoma; subperichondrial auricular hematoma.

Auscultation

The act of listening to sounds produced within the body, typically using a stethoscope. Essential for assessing heart sounds, lung sounds, bowel sounds, and vascular bruits. Normal findings: S1 and S2 heart sounds, vesicular breath sounds, normal bowel sounds. Abnormal: murmurs, crackles, wheezes, rubs, bruits.

Auscultatory Gap

A transient silence between the first and second sounds of blood pressure measurement, sometimes heard in hypertensive patients. Can lead to underestimation of systolic pressure if not recognized.

Austin Flint Murmur

A mid-diastolic, low-pitched rumbling murmur heard at the apex, caused by the regurgitant jet from severe aortic regurgitation striking the anterior mitral leaflet.

Autism And Communication

Autism can affect social communication, language development, nonverbal interaction, and the use of communication for shared attention or relationships. Individualized speech-language therapy, visual supports, augmentative communication, and respectful environmental accommodations can improve participation.

Alternate Names:

Adults Autism (Autism And Communication)

Autism Spectrum Disorder

Neurodevelopmental disorder characterized by persistent deficits in social communication and social interaction across multiple contexts, restricted and repetitive patterns of behavior, interests, or activities. Symptoms present in early developmental period and cause clinically significant impairment.

Alternate Names:

Autism

Autism Spectrum Disorder - Asperger Syndrome

A former diagnostic label for autism spectrum disorder in a person without intellectual disability or clinically significant early language delay. Current diagnostic systems classify this presentation under autism spectrum disorder rather than as a separate disorder.

Autism Spectrum Disorder - Childhood Disintegrative Disorder

Childhood disintegrative disorder, also called Heller syndrome, is a rare condition involving marked loss of previously acquired language, social, and adaptive skills after apparently typical early development. Modern classifications place it within autism spectrum disorder, and evaluation should exclude other neurologic causes.

Autistic Savant

A person with autism who has an unusually advanced, specific ability—such as in music, art, calculation, or memory—relative to overall functioning.

Autoantibodies

Antibodies that mistakenly react with the body's own cells or tissues.

Autoantibody

An immunoglobulin that recognizes and binds an endogenous antigen. Autoantibodies may be pathogenic, as in autoimmune hemolytic anemia, or may serve as laboratory markers of autoimmune disease.

Autoantigen

A normal body molecule mistakenly targeted by the immune system in an autoimmune disease.

Autoclave

Sterilization device using high-pressure steam. Parameters: 121°C, 15 psi, 15-30 minutes. Essential for infection control.

Autoerythrocyte Sensitivity

A rare syndrome of painful bruising or purpura attributed to an abnormal reaction to a person's own red-cell components. The diagnosis is one of exclusion after bleeding disorders, medication effects, and other causes of bruising are assessed.

Autogenic Drainage

A breathing technique using controlled breathing at different lung volumes to mobilize secretions from peripheral to central airways. It involves breathing at low lung volumes (collect), mid lung volumes (loosen), and high lung volumes (evacuate). It requires patient training and cooperation.

Autograft

Tissue transplanted from one site to another within the same individual. Autografts are genetically identical to the recipient and generally avoid alloimmune rejection, although donor-site morbidity may occur.

Autoimmune

Relating to an immune response directed against the body's own cells, tissues, or molecules.

Autoimmune Associated Epilepsy

See *Autoimmune epilepsy*.

Autoimmune Disease

Autoimmune diseases result from loss of immunological self-tolerance, leading to immune-mediated attack on the body's own tissues.

Autoimmune Diseases

Autoimmune diseases arise when immune responses target the body's own tissues, producing organ-specific or systemic inflammation and damage.

Autoimmune Disorder

A disease caused by an abnormal immune response against the body's own cells, tissues, or organs. Autoimmune disorders may be organ-specific or systemic and can produce inflammation, tissue injury, functional impairment, or circulating autoantibodies.

Autoimmune Encephalitis

Brain inflammation caused by antibodies or other immune responses directed against neuronal proteins.

Autoimmune Encephalopathy

See *Autoimmune Encephalitis*.

Autoimmune Enteropathy

Rare immune-mediated disease causing persistent diarrhoea and small-intestinal mucosal injury, usually with villous atrophy and crypt hyperplasia. It may occur in children or adults and can be associated with other autoimmune disorders or immune dysregulation.

Autoimmune Epilepsy

Recurrent seizures caused by an immune attack on the brain or by inflammation associated with an autoimmune disorder.

Autoimmune Gastritis

Chronic atrophic gastritis caused by antibodies against parietal cells leading to achlorhydria and vitamin B12 deficiency. Associated with pernicious anemia and increased risk of gastric carcinoid tumors and type 1 gastric neuroendocrine tumors. Anti-parietal cell antibodies are present in approximately ninety percent of patients.

Autoimmune Hemolytic Anemia

AIHA: hemolytic anemia due to antibodies directed against red blood cells. Warm AIHA (IgG, 37C—idiopathic, SLE, CLL, drugs: methyldopa, penicillin): splenic sequestration, spherocytes, positive direct antiglobulin (DAT).

Autoimmune Hepatitis

Chronic hepatitis caused by immune-mediated destruction of hepatocytes characterized by interface hepatitis with lymphoplasmacytic infiltrate, elevated transaminases, hypergammaglobulinemia, and autoantibodies.

Autoimmune Keratitis

Corneal inflammation associated with an immune-mediated disorder, sometimes occurring with rheumatoid arthritis, lupus, or granulomatosis with polyangiitis. It may cause pain, redness, corneal opacity, and reduced vision and requires urgent ophthalmic assessment. Treatment is directed at the cause and may include anti-inflammatory or immunosuppressive therapy.

Autoimmune Liver Disease Panel

A laboratory test that measures autoantibodies associated with autoimmune hepatitis and other autoimmune liver disorders, including antinuclear antibody, anti-smooth muscle antibody, anti-liver kidney microsomal antibody, and anti-mitochondrial antibody. Results help distinguish autoimmune hepatitis from primary biliary cholangitis and primary sclerosing cholangitis.

Autoimmune Pancreatitis

Chronic inflammatory condition characterized by dense lymphoplasmacytic infiltrate with IgG4-positive plasma cells representing the pancreatic manifestation of IgG4-related disease. Type 1 is more common affecting older men with elevated serum IgG4 and extrapancreatic involvement.

Autoimmune Polyendocrine Syndromes

APS-1 (APECED, AIRE mutation): autosomal recessive, chronic mucocutaneous candidiasis, hypoparathyroidism, adrenal insufficiency (Addison's), plus type 1 diabetes, autoimmune thyroiditis, vitiligo, alopecia, pernicious anemia, hepatitis, asplenia.

Autoimmune Polyendocrinopathy

An inherited autoimmune syndrome, also called autoimmune polyendocrine syndrome type 1, characterized by chronic mucocutaneous candidiasis, hypoparathyroidism, adrenal insufficiency, and other autoimmune disorders.

Autoimmune Polyglandular Syndrome

A group of disorders in which autoimmune disease affects multiple endocrine glands; manifestations

vary by type and may include adrenal insufficiency, hypoparathyroidism, and thyroid disease.

Autoimmune Thyroiditis

Immune-mediated inflammation and damage of the thyroid gland. See *Hashimoto Thyroiditis*.

Autoimmunity

An immune response against self-antigens; it can be harmless or can cause autoimmune disease when it damages tissues or disrupts their function.

Autoimmunity: Mechanisms

Loss of tolerance: central (thymic negative selection defects, AIRE mutation → APS-1) and peripheral (Treg dysfunction, anergy failure, ignorance loss). Molecular mimicry: cross-reactivity between pathogen and self (rheumatic fever: Strep M protein + cardiac myosin; Guillain-Barre: *Campylobacter* + GM1 ganglioside; reactive arthritis).

Autoinflammatory Diseases

Disorders of innate immune regulation that cause recurrent or persistent inflammation without the disease-specific autoantibodies typical of autoimmune disorders. They may cause fever, rash, joint inflammation, or organ damage and can be inherited or acquired.

Autoinoculation

Transfer of an infectious agent from one body site to another on the same person, often by touching or scratching.

Autologous

Derived from or involving the same individual who will receive it, such as an autologous blood transfusion or stem-cell transplant.

Autologous Blood Donation

Autologous blood donation involves preoperative collection and storage of the patient's own blood for intraoperative or postoperative transfusion. It eliminates risks of alloimmune reactions, transfusion-transmitted infections, and immune modulation.

Autologous Fat Transplant

A procedure that removes a person's own fat from one body area and transfers it to another area for soft-tissue volume restoration or contouring.

Alternate Names:

Autologous fat grafting; fat transfer; lipofilling.

Autolysis

Self-digestion of cells or tissues by their own enzymes, commonly occurring after cell death and contributing to decomposition.

Automated External Defibrillator

A portable device that analyzes heart rhythm and delivers an electrical shock when it detects a shockable cardiac arrest rhythm such as ventricular fibrillation.

Automatic Dishwasher Soap Poisoning

Toxic effects caused by excessive exposure to automatic dishwasher detergents. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Autonomic Disorder

Dysfunction of the autonomic nervous system, which regulates involuntary cardiovascular, gastrointestinal, sudomotor, urinary, sexual, and pupillary functions. It may be primary or secondary to conditions such as diabetes, neurodegenerative disease, autoimmune neuropathy, infection, or medication effects, and may cause orthostatic hypotension, abnormal heart-rate responses, sweating abnormalities, gastrointestinal dysmotility, bladder dysfunction, or sexual dysfunction. Evaluation and treatment depend on the underlying cause and the organs affected.

Autonomic Dysreflexia

Acute, potentially life-threatening sympathetic discharge in a person with spinal-cord injury at or above approximately T6, triggered by a noxious stimulus below the lesion. It causes sudden hypertension, headache, flushing or sweating above the lesion, and reflex bradycardia or other heart-rate

changes.

Autonomic Ganglion

A cluster of nerve-cell bodies in the autonomic nervous system that relays signals to organs.

Autonomic Nervous System

The part of the peripheral nervous system that regulates involuntary functions, including heart rate, vascular tone, digestion, sweating, pupil size, and bladder activity.

Autonomic Neuropathy

Damage to nerves that control involuntary functions such as heart rate, blood pressure, digestion, sweating, bladder function, and sexual function. Diabetes is a common cause, but other diseases, toxins, and medicines may contribute. Symptoms depend on the affected organs and require evaluation of the underlying cause.

Alternate Names:

Neuropathy, Autonomic

Autonomic Neuropathy Diabetic (Diabetic Neuropathy)

Alternate index label for Diabetic Neuropathy.

Autonomy, Patient

A patient's right and capacity to make informed, voluntary decisions about medical care, including accepting or refusing treatment.

Autophagy

Cellular self-degradation process delivering cytoplasmic components to lysosomes for recycling. Macroautophagy: phagophore forms, expands to autophagosome (LC3-II on membrane), fuses with lysosome (autolysosome) for degradation. Selective autophagy: mitophagy (damaged mitochondria), xenophagy (pathogens), and aggrephagy (protein aggregates).

Autopsy

A postmortem examination of a body to determine cause and manner of death, performed by a foren-

sic pathologist (medical examiner/coroner) or, for hospital autopsies, a pathologist.

Autopsy (Medicolegal)

Systematic external examination and surgical dissection of a deceased body performed under legal authority by a forensic pathologist to determine the Cause, Manner, and Mechanism of Death, document injuries, collect evidence, and assist judicial investigations.

Alternate Names:

Medicolegal

Autopsy (Post-Mortem Examination)

Specialized surgical procedure performed on a deceased body to determine the cause of death, mechanism of death, manner of death (natural, accidental, suicide, homicide, or undetermined), state of health prior to death, and to document injuries or disease processes.

Alternate Names:

Post-Mortem Examination

Autoradiography

An imaging technique that uses radioactive labels to show the location of substances in tissue.

Autosomal Dominant

Autosomal dominant inheritance occurs when one altered copy of a gene can produce a trait or disorder; an affected heterozygous parent may transmit it to about half of offspring.

Autosomal Dominant Inheritance

Inheritance in which one pathogenic variant on a non-sex chromosome can cause a trait or disorder. An affected heterozygous parent may transmit it to about half of the children.

Autosomal Dominant PKD

Most common hereditary kidney disease affecting 1 in 400 to 1000. PKD1 chromosome 16 more severe with earlier ESRD than PKD2 chromosome 4. Progressive cyst growth causes hypertension, pain, hematuria, infections, and eventually ESRD. Screen

family members with ultrasound. Tolvaptan may slow cyst growth.

Autosomal Dominant Polycystic Kidney Disease

An inherited disorder causing progressive growth of numerous renal cysts that enlarge the kidneys and destroy parenchyma, leading to hypertension, flank pain, and eventual kidney failure. Associated with hepatic cysts and intracranial aneurysms.

Alternate Names:

ADPKD

Autosomal Dominant Tubulointerstitial Kidney Disease

An inherited kidney disorder caused by pathogenic variants in genes such as UMOD, REN, or MUC1. It causes progressive loss of kidney function with relatively bland urine findings; some subtypes cause early gout or anemia.

Autosomal Inheritance

Transmission of a genetic trait through one of the 22 non-sex chromosomes. Autosomal dominant traits may appear in successive generations, whereas autosomal recessive disease generally requires pathogenic variants on both homologous chromosomes.

Autosomal Recessive

Autosomal recessive inheritance generally requires pathogenic variants in both copies of a gene; unaffected carrier parents can have an affected child.

Autosomal Recessive Disorders

Disorders caused when a person inherits pathogenic variants in both copies of a gene, usually one from each parent; carriers with one altered copy are often unaffected.

Autosomal Recessive Inheritance

Inheritance in which a disorder generally requires pathogenic variants in both copies of a gene. Carrier parents typically have a one-in-four chance of an affected child in each pregnancy.

Autosomal Recessive PKD

Rare childhood disorder from PKHD1 mutation. Bilateral enlarged kidneys with fusiform dilation of collecting ducts. Associated with congenital hepatic fibrosis. Severe forms cause pulmonary hypoplasia in utero. Variable severity from perinatal lethal to presentation in childhood or adolescence.

Autosomal Recessive Polycystic Kidney Disease

An inherited cystic kidney disorder, usually presenting in infancy or childhood, caused by pathogenic variants commonly affecting PKHD1; it may cause enlarged echogenic kidneys, impaired kidney function, liver fibrosis, and hypertension.

Alternate Names:

ARPKD

Autosome

Any nuclear chromosome that is not a sex chromosome (allosome). In humans, normal diploid somatic cells contain 22 pairs of homologous autosomes (numbered 1 through 22 in descending order of physical size) and one pair of sex chromosomes (XX in females, XY in males), totaling 46 chromosomes. Genes located on autosomes exhibit autosomal inheritance patterns, classified as autosomal dominant (where a single mutant allele on an autosome produces the clinical phenotype, such as in Marfan syndrome and Huntington disease) or autosomal recessive (where biallelic mutations are required, such as in cystic fibrosis and sickle cell disease), affecting males and females with equal probability.

Alternate Names:

Non-sex chromosome

Avascular Necrosis

Death of bone tissue caused by impaired blood supply. It commonly affects the hip or shoulder and may cause pain, collapse of the joint surface, and arthritis. Risk factors include corticosteroid exposure, alcohol use, trauma, sickle-cell disease, and systemic lupus erythematosus. Treatment depends on the site and stage and may include observation, joint-preserving surgery, or joint replacement.

Avascular Necrosis (Aseptic Necrosis)

Alternate index label for Aseptic Necrosis.

Avascular Necrosis (Osteonecrosis)

Death of bone tissue caused by impaired blood supply. It commonly affects the femoral head and may lead to subchondral collapse, secondary arthritis, and need for joint-preserving surgery or arthroplasty.

Alternate Names:

Osteonecrosis Ischemic Bone Necrosis

Avian Flu (Avian Influenza (Bird Flu))

Avian influenza is infection with influenza A viruses adapted primarily to birds. Human infection is uncommon but can be severe after contact with infected birds or mammals; exposure history, respiratory symptoms, and public-health testing guide evaluation and treatment.

Avian Influenza

Influenza caused by avian-adapted influenza A viruses, acquired mainly through contact with infected birds or contaminated environments. Human infection can cause severe pneumonia, respiratory failure, and occasionally systemic disease.

Avian Influenza (Bird Flu)

Avian influenza is infection with influenza A viruses adapted primarily to birds. Human infection is uncommon but can be severe after contact with infected birds or mammals; exposure history, respiratory symptoms, and public-health testing guide evaluation and treatment.

AVN

See *Avascular necrosis (osteonecrosis)*.

Avoidance Behavior

Deliberate behavior intended to prevent contact with a feared, distressing, or threatening situation.

Avoidant Personality Disorder

A pattern of social inhibition, feelings of inadequacy, and hypersensitivity to negative evaluation.

Avolition

A severe lack of motivation to initiate and perform self-directed, purposeful activities. It is a negative symptom of schizophrenia and is seen in severe depression. It manifests as neglect of personal hygiene, inability to complete tasks, and social withdrawal.

Avulsed Tooth

A tooth completely displaced from its socket by trauma. It is a dental emergency; management depends on patient age, extraoral dry time, storage medium, root development, and periodontal-ligament viability, and may require urgent replantation by a dental professional.

Avulsion

An injury in which tissue is forcibly torn from the body by shearing or pulling. Common forms include dental, fingertip, nail, and degloving avulsions. Management focuses on hemorrhage control, debridement, tissue preservation, and urgent consultation.

Avulsion Fracture

A fracture in which a fragment of bone is pulled away by the tendon or ligament attached to it, usually during sudden forceful contraction or injury. Common sites include the pelvis, ankle, elbow, and fingers; evaluation considers displacement, joint involvement, skeletal maturity, and associated soft-tissue injury.

Avuncular

Relating to an uncle or resembling the kindly, familiar manner traditionally associated with an uncle.

Awake Fiberoptic Intubation

An intubation technique performed while the patient remains responsive and breathing spontaneously, using topical airway anesthesia and a flexible bronchoscope. It is one option when difficult airway management is anticipated and maintaining oxygenation is important.

Awareness

Conscious recognition of oneself, surroundings,

thoughts, or sensations. In clinical examination, awareness is assessed along with alertness, attention, memory, and orientation.

Alternate Names:

Consciousness

Awareness Under Anesthesia

Unintended awareness or recall of events during general anesthesia. It may involve hearing, sensation, or pain and can be associated with significant psychological distress after the procedure.

Axial Skeleton

The bones of the head, neck, and trunk, including the skull, spine, ribs, and sternum.

Axial Spondyloarthritis

An inflammatory rheumatic disease primarily affecting the sacroiliac joints and spine, causing chronic inflammatory back pain and stiffness; it includes radiographic and non-radiographic forms.

Axial Tomography, Computerized

Computed tomography, or CT, an imaging method that uses X-rays and computer processing to create cross-sectional images of the body.

Axilla

Pyramidal space between the upper lateral thorax and medial arm. Bounded by anterior fold (pectoralis major/minor), posterior fold (latissimus dorsi/teres major), medial wall (ribs/serratus anterior), and lateral wall (humerus). Contains axillary artery/vein, brachial plexus cords, and lymph nodes.

Axillary

Relating to the armpit or to structures located in the axilla, including axillary lymph nodes and vessels.

Alternate Names:

Armpit-related; pertaining to the axilla.

Axillary Artery

The continuation of the subclavian artery, running from the outer border of the first rib to the lower border of teres major, where it becomes the brachial

artery. Divided into three parts by the pectoralis minor.

Axillary Vein

A major vein of the upper limb formed by the union of the brachial and basilic veins. It continues as the subclavian vein and drains the arm and axilla.

Axillary Lymphadenectomy

Surgery to remove lymph nodes from the armpit, usually to stage or treat breast cancer or melanoma. Possible complications include pain, numbness, reduced shoulder movement, fluid collection, and arm lymphedema.

Alternate Names:

Axillary node dissection

Axillary Nerve

From posterior cord (C5-C6), passing through the quadrangular space with the posterior circumflex humeral artery. Innervates deltoid and teres minor; provides sensation to the regimental badge area (lateral deltoid skin). Vulnerable in anterior shoulder dislocations and surgical neck humeral fractures.

Axillary Nerve Dysfunction

Impairment of the axillary nerve, often from shoulder trauma, dislocation, compression, or surgery. It can cause weakness of shoulder abduction, deltoid wasting, and numbness over the lateral shoulder.

Axis

Second cervical vertebra with the dens (odontoid process) projecting superiorly, serving as a pivot for atlas rotation, allowing approximately 50 percent of cervical rotation. Has a large spinous process and flat superior articular facets.

Axis Vertebra

The second cervical vertebra, which supports rotation of the first cervical vertebra and head.

Alternate Names:

C2; axis

Axolemma

The specialized trilaminar plasma membrane enclosing an axon of a neuron. Essential for the generation and propagation of electrical action potentials through its high density of voltage-gated ion channels, notably voltage-gated sodium (Na_v) and potassium (K_v) channels. In unmyelinated fibers, action potential conduction occurs continuously along the entire axolemma; in myelinated fibers, the axolemma is insulated by myelin sheaths (formed by Schwann cells in the PNS and oligodendrocytes in the CNS), confining voltage-gated sodium channels to uninsulated gaps termed nodes of Ranvier to facilitate rapid, energy-efficient saltatory conduction.

Axon

The single long process of a neuron that conducts action potentials away from the cell body toward the axon terminal for synaptic transmission.

Axon Fasciculation

The bundling of developing axons into organized nerve tracts during nervous-system development. Guidance signals and cell-adhesion molecules help axons grow together toward their targets.

Axon Guidance

The developmental process directing growing axons toward their correct targets.

Axon Hillock

The region of a neuron where action potentials are usually initiated.

Axon Terminal

The end of a nerve-cell axon that releases chemical messengers to communicate with another cell.

Axonal Transport

Movement of proteins, organelles, and other materials along an axon.

Axoneme

The internal microtubule-based structure of a cilium or flagellum.

Ayurveda

A traditional system of medicine from India that emphasizes balance among bodily principles and uses individualized diet, lifestyle practices, and herbal or other treatments.

Azole Antifungals

A class of antifungal agents that inhibit the enzyme lanosterol 14-alpha-demethylase, a cytochrome P450 enzyme essential for ergosterol synthesis in fungal cell membranes. The depletion of ergosterol and accumulation of toxic methylated sterols disrupt membrane integrity and function.

Azoospermia

Absence of spermatozoa in the ejaculate after appropriate centrifugation and microscopic examination. It may be obstructive, with sperm production preserved but outflow blocked, or nonobstructive, with impaired spermatogenesis.

Azotaemia

Azotaemia is excess nitrogen-containing waste, especially urea and creatinine, in the blood, usually reflecting reduced kidney filtration, increased production, or altered renal blood flow.

Azotemia

Accumulation of nitrogen-containing waste products in the blood, usually because the kidneys are not filtering adequately.

Azygos Vein

A vein that drains the posterior intercostal veins and empties into the superior vena cava. It forms an important collateral pathway when the SVC is obstructed. On chest imaging, enlargement of the azygos vein may indicate SVC obstruction, heart failure, or portal hypertension.

Azygous Vein

A large vein that drains the posterior thoracic wall and mediastinum, ascending along the right side of the vertebral column to empty into the superior vena cava. It provides collateral venous return when the superior or inferior vena cava is obstructed.

Labor: Signs And Symptoms

Labor is the process of regular uterine contractions that produces progressive cervical effacement and dilation and leads to birth. Possible signs include increasingly regular and painful contractions, rupture of membranes, bloody show, pelvic pressure, and back pain; urgent assessment is needed for heavy bleeding, decreased fetal movement, severe pain, or suspected complications.

Primary Clear Cell Adenocarcinoma Of The Vulva

A rare primary vulvar clear-cell adenocarcinoma, usually showing Müllerian differentiation. Diagnosis requires histopathologic examination and immunohistochemical correlation while excluding metastatic clear-cell carcinoma from sites such as the kidney or gynecologic tract.

B

B And T Cell Screen

A blood test that identifies and measures B cells and T cells, two major types of immune cells. It can help evaluate immune deficiencies, leukemia, and lymphoma.

B And T Lymphocytes

B lymphocytes produce antibodies, while T lymphocytes coordinate immune responses or kill infected and abnormal cells; both are adaptive immune cells.

B Cell

A lymphocyte that develops in the bone marrow and can differentiate into antibody-producing plasma cells or memory B cells after activation.

Alternate Names:

B lymphocyte; B-cell.

B Cell (B Lymphocyte)

An immune cell that matures in the bone marrow. After recognizing a foreign substance and receiving signals from other immune cells, it multiplies and develops into cells that make antibodies or remember the substance for future defense.

Alternate Names:

B Lymphocyte

B Chromosome

A supernumerary chromosome that is not part of the standard chromosome complement and is not required for normal development. It may occur in certain plants, animals, or tumors and varies in number between cells.

B Hemophilia (Hemophilia)

Alternate index label for Hemophilia.

B Virus

Herpes B virus, also called Macacine alphaherpesvirus 1, a macaque virus that can cause severe and sometimes fatal encephalomyelitis in humans after exposure to infected animals or tissues.

B-Cell Activation

The process by which a B lymphocyte is stimulated by antigen and other immune signals to multiply and develop into antibody-producing plasma cells or memory B cells.

B-Cell Leukemia/Lymphoma Panel

A laboratory test that uses cell analysis and antibody markers to identify abnormal B-cell groups in blood, bone marrow, or tissue.

B-Cell Lymphoma

A group of lymphatic cancers arising from B lymphocytes.

B-Cell Receptor

A protein on the surface of a B cell that recognizes foreign substances. Signals from this receptor help the B cell multiply, mature, and make antibodies.

B-Lymphocyte

A B lymphocyte is an immune cell that can mature into an antibody-producing plasma cell or a memory B cell.

B-Lynch Suture

A surgical stitch placed around the uterus to control heavy bleeding after childbirth when the uterus has not contracted and medicines have not worked. The stitch compresses the uterus and reduces bleeding.

B-Mode Ultrasound

A standard two-dimensional ultrasound image made from sound waves that bounce back from tissues. It is used to examine many organs and soft tissues.

Ba

The chemical symbol for barium, a metal used in compounds such as barium sulfate for contrast studies of the digestive tract.

Babcock Clamp

An atraumatic surgical clamp with fenestrated jaws used to hold delicate tubular tissue.

Babesia

A group of parasites spread mainly by Ixodes ticks. They infect red blood cells and can cause fever, anemia from red-cell destruction, and a low platelet count. Severe disease is more likely in people without a spleen or with a weakened immune system.

Babesia Divergens

A species of Babesia, an intraerythrocytic protozoan parasite transmitted by Ixodes ticks, primarily causing babesiosis in Europe. It can cause severe hemolytic anemia, particularly in asplenic or immunocompromised individuals.

Babesia Microti

The most common cause of babesiosis in North America, an intraerythrocytic protozoan parasite transmitted by Ixodes scapularis ticks. Infection can range from asymptomatic to severe hemolytic anemia with fever, jaundice, and organ failure in immunocompromised patients.

Babesiosis

A tick-borne infection caused by Babesia parasites. The parasites infect red blood cells and can cause

fever, tiredness, anemia, a low platelet count, jaundice, or blood pigment in the urine. Severe disease is more likely in people without a spleen or with a weakened immune system.

Babinski Reflex

A normal reflex in infants up to about 2 years of age. Stroking the outer sole makes the big toe point upward and the other toes spread. The reflex normally changes as the nervous system develops; persistence in older children or adults may indicate a nervous-system problem.

Babinski Response

Upward movement of the big toe and spreading of the other toes when the sole is stroked. It is normal in infants but may indicate a nervous-system problem in older children or adults.

Babinski Sign

Upward movement of the big toe and spreading of the other toes when the outer sole is stroked. This can indicate a problem in the brain or spinal cord in older children and adults; a normal response is downward curling of the toes.

Baby Acne

Small red or white bumps on an infant's skin, usually caused by temporary hormonal effects and often resolving without treatment.

Baby Blues

A common, short-lived period of tearfulness, irritability, anxiety, or mood lability during the first days after childbirth, usually resolving within about two weeks.

Babyhood

Babyhood is the period of infancy, when rapid growth, brain development, motor skills, feeding, and immune maturation occur.

Bacillaceae

A family of Gram-positive, spore-forming rod-shaped bacteria in the order Bacillales. It includes

medically important genera such as *Bacillus* and *Clostridium*, some of which cause anthrax, botulism, or gas gangrene.

Bacillary Angiomatosis

A vascular proliferative disease caused by *Bartonella henselae* or *B. quintana*, occurring in immunocompromised patients (HIV/AIDS, transplant). Presents with erythematous papules, nodules, or pedunculated lesions resembling Kaposi sarcoma or pyogenic granuloma. Can involve skin, liver (peliosis hepatis), spleen, bone, and lymph nodes.

Bacillary Dysentery

Acute inflammatory diarrhea caused by *Shigella* species, typically producing fever, abdominal cramps, and frequent stools that may contain blood or mucus.

Bacille Calmette-Guérin

Bacille Calmette-Guérin is a live attenuated mycobacterial vaccine used mainly against tuberculosis and, in some settings, as bladder-cancer immunotherapy.

Bacilluria

The presence of bacilli in the urine, usually referring to rod-shaped bacteria detected by microscopy or culture. Its clinical significance depends on symptoms, quantitative culture, and the organism identified.

Bacillus

A rod-shaped bacterium. The term may refer broadly to bacterial shape or specifically to members of the genus *Bacillus*, some of which form resistant spores and cause human disease.

Bacillus Anthracis

The spore-forming bacterium that causes anthrax. Infection may be cutaneous, gastrointestinal, or inhalational.

Bacillus Cereus

A spore-forming bacterium that can cause food poi-

soning, usually through a rapid-onset vomiting syndrome or a diarrheal syndrome after ingestion of contaminated food.

Bacillus Lentus

A Gram-positive, spore-forming rod bacterium in the genus *Bacillus*, commonly found in soil. It is generally of low virulence but may cause opportunistic infections in immunocompromised individuals.

Bacillus Licheniformis

A Gram-positive, spore-forming rod bacterium found in soil and plant surfaces. It is an occasional cause of food contamination and has been rarely associated with opportunistic infections in immunocompromised patients.

Bacillus Megaterium

A large, Gram-positive, spore-forming rod bacterium found in soil and water. It is of low pathogenicity but has been reported as an opportunistic cause of bacteremia and endocarditis in rare cases.

Bacillus Mycoides

A Gram-positive, spore-forming soil bacterium recognized by its distinctive rhizoid colony morphology. It is not considered a significant human pathogen but is studied as an environmental indicator organism.

Bacillus Pumilus

A Gram-positive, spore-forming rod bacterium commonly found in soil and on skin. It has been associated with rare opportunistic infections, including endophthalmitis and bacteremia, particularly in immunocompromised individuals.

Bacillus Subtilis

A Gram-positive, spore-forming rod bacterium widely distributed in soil and vegetation. Generally considered non-pathogenic, it is used as a probiotic and in biotechnology; rare opportunistic infections have been reported.

Bacitracin

Bacitracin is a topical polypeptide antibiotic that interferes with bacterial cell-wall synthesis and is used for selected superficial infections.

Bacitracin Overdose

Toxic effects from excessive exposure to bacitracin. Topical exposure usually causes local irritation, while large, ingested, or systemic exposures may cause gastrointestinal symptoms or kidney injury; assessment depends on the dose and route.

Bacitracin Zinc Overdose

Toxic effects from excessive exposure to bacitracin zinc. Severity depends on the dose and route; large exposures may cause gastrointestinal symptoms, allergic reactions, or kidney injury and require prompt assessment.

Back Injury

Damage to the muscles, ligaments, joints, discs, vertebrae, or nerves of the back caused by trauma, lifting, or repetitive strain. Pain and stiffness are common; weakness, numbness, loss of bladder or bowel control, or severe pain after major trauma requires urgent assessment.

Back Pain

One of the most common medical complaints, affecting up to 80 percent of the population at some point in life, and the leading cause of disability worldwide. Back pain is classified by duration: acute (less than 4 weeks), subacute (4-12 weeks), and chronic (greater than 12 weeks).

Alternate Names:

Pain, Back

Back Pressure

Back pressure is pressure exerted in the reverse direction of flow; in medicine it may result from obstruction in an airway, vessel, duct, or organ.

Back-Blows

Back-blows are firm blows between the shoulder blades used as a first-aid maneuver for severe chok-

ing in a responsive person.

Backache

Pain or discomfort located anywhere in the posterior trunk from neck to buttocks, a symptom rather than a disease. Causes include musculoskeletal strain (most common), disc herniation, spinal stenosis, facet arthropathy, sacroiliac joint dysfunction, and referred pain from renal, pancreatic, or vascular sources.

Baclofen

A muscle relaxant that activates GABA-B receptors and is used mainly to reduce spasticity. Drowsiness, weakness, confusion, and dangerous withdrawal after abrupt stopping can occur.

Bacteraemia

Bacteraemia is the presence of viable bacteria in the bloodstream; it may be transient or lead to sepsis and infection of distant sites.

Bacteremia

The presence of viable bacteria in the bloodstream, a serious condition that can be transient (after dental procedures, toothbrushing), intermittent (undrained abscess), or continuous (endovascular infection like endocarditis). Bacteremia triggers systemic inflammatory responses ranging from fever and leukocytosis to septic shock.

Bacteria

Single-celled microscopic organisms lacking a nucleus, which can exist independently or in colonies; some are beneficial to health (microbiome) while others cause infectious disease.

Bacterial Adherence

The process by which bacteria attach to the body's cells, often an early step in infection. Bacterial surface structures, including hair-like pili and fimbriae, help them attach to specific sites on human cells.

Bacterial Biofilms

Groups of bacteria attached to a surface and embed-

ded in a protective substance they make themselves. Biofilms can cause long-lasting infections, especially on catheters, artificial joints, and heart valves.

Bacterial Capsule

A sugar-rich outer layer around some bacteria that helps protect them from the body's defenses. It can help bacteria cause disease; examples include *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Klebsiella pneumoniae*.

Bacterial Cell Wall

The rigid outer layer of bacteria that provides structural integrity and determines cell shape. The cell wall is composed primarily of peptidoglycan, a polymer of sugars cross-linked by short peptides.

Bacterial Classification

The organisation of bacteria into taxonomic groups based on genetic, phenotypic, and evolutionary relationships. Bacteria are classified using a hierarchical system: domain, phylum, class, order, family, genus, and species.

Bacterial Conjugation

The transfer of DNA from one bacterium to another through direct contact. A donor bacterium uses a tiny connecting structure to pass DNA, often including genes for antibiotic resistance.

Bacterial Conjunctivitis

A contagious infection of the conjunctiva, often caused by staphylococci or streptococci. It commonly causes eye redness, irritation, a gritty sensation, and mucopurulent discharge that can mat the eyelids, especially after sleep. Symptoms overlap with viral and allergic conjunctivitis, and most uncomplicated cases improve within 1–2 weeks; topical antibiotics may shorten the illness. Contact-lens wearers with pain, light sensitivity, or reduced vision need prompt eye assessment.

Bacterial Dermatitis

Inflammation of the skin caused by bacterial infection, typically presenting with erythema, edema, warmth, pain, and purulent discharge. Common

causative organisms include *Staphylococcus aureus* and *Streptococcus pyogenes*, and the condition may arise de novo or as a superinfection of pre-existing dermatitis.

Bacterial Dormancy

A state of reduced metabolic activity that allows bacteria to survive unfavourable conditions. Dormant forms include spores, persisters, and viable but non-culturable cells. Spores are highly resistant structures formed by *Bacillus* and *Clostridium*. Persisters are phenotypic variants that are tolerant to antibiotics.

Bacterial Endocarditis

Infection of the heart's inner lining or valves by bacteria, usually producing fever and a new or changing heart murmur. It can cause valve damage and embolic complications.

Bacterial Endocarditis Vegetations

Clumps of bacteria, clotting material, and platelets that form on a heart valve or the heart's inner lining during bacterial endocarditis. They can develop when bacteria in the blood attach to a damaged or abnormal heart surface.

Bacterial Endospores

Highly resistant dormant structures formed by certain gram-positive bacteria including *Bacillus* and *Clostridium* species. Endospores form during nutrient deprivation and can survive for decades. They are resistant to heat, radiation, chemicals, and desiccation.

Bacterial Flagella

Thread-like appendages composed of flagellin protein that provide motility to bacteria. Flagella are composed of three parts: the filament, hook, and basal body. The basal body is embedded in the cell wall and functions as a rotary motor.

Bacterial Gastroenteritis

Acute inflammation of the stomach and intestines caused by pathogenic bacteria such as *Salmonella*, *Campylobacter*, *Shigella*, or pathogenic *Escherichia*

coli. It commonly causes diarrhea, abdominal cramps, fever, and sometimes bloody stools or vomiting.

Bacterial Genetics

The study of genes and heredity in bacteria. Most bacteria have one circular chromosome and may also have small DNA rings called plasmids. Groups of bacterial genes may be controlled together.

Bacterial Growth Phases

The stages of bacterial growth in culture. The lag phase is when bacteria adapt to their environment. The log (exponential) phase is when bacteria multiply at their maximum rate. The stationary phase is when nutrient depletion and waste accumulation limit growth. The death phase is when bacteria die.

Bacterial Infections

Illnesses caused by pathogenic bacteria proliferating in the body, ranging from mild sore throats and ear infections to life-threatening sepsis and meningitis. They are diagnosed through cultures, PCR testing, or other microbiological methods.

Bacterial Meningitis

Infection of the meninges by bacteria, a medical emergency. In neonates: Group B Streptococcus, *E. coli*, *Listeria*. In older children: *Neisseria meningitidis*, *Streptococcus pneumoniae*. Presents with fever, headache, neck stiffness, altered consciousness, and Kernig/Brudzinski signs.

Bacterial Metabolism

The chemical processes that occur within bacteria to maintain life. Bacteria have diverse metabolic capabilities including aerobic and anaerobic respiration, fermentation, and photosynthesis. Metabolic pathways are targets for antimicrobial drugs.

Bacterial Morphology

The shape and arrangement of bacteria seen under a microscope. Common shapes include spheres, rods, and spirals. Spherical bacteria may occur in pairs, chains, clusters, or groups of four.

Bacterial Overgrowth

Excessive proliferation of bacteria in the small intestine, commonly called small-intestinal bacterial overgrowth. It may result from impaired intestinal motility, anatomical stasis, blind loops, strictures, diverticula, or loss of the normal ileocaecal barrier.

Alternate Names:

small-intestinal bacterial overgrowth.

Bacterial Pathogenesis

The mechanisms by which bacteria cause disease. Pathogenesis involves adhesion, invasion, toxin production, and immune evasion. Adhesins mediate attachment to host cells. Invasins allow penetration of tissues. Toxins damage host cells.

Bacterial Peritonitis

Infection of the space inside the abdomen around the organs. It may occur in people with extra abdominal fluid or after bowel perforation, surgery, or another abdominal infection. It requires prompt evaluation and antibiotic treatment.

Bacterial Pharyngitis

Bacterial infection and inflammation of the throat, most commonly caused by group A Streptococcus. Symptoms may include sore throat, fever, painful swallowing, and tender lymph nodes in the neck.

Bacterial Pili

Hair-like appendages on the bacterial surface composed of pilin protein. Pili are shorter and thicker than flagella. Type I pili mediate adherence to host cells and are important virulence factors. Type IV pili are involved in twitching motility and DNA uptake during transformation.

Bacterial Pneumonia

Infection of the lung tissue by bacteria, causing inflammation and air-space filling. It may cause fever, cough, sputum, chest pain, and shortness of breath.

Bacterial Protein

A protein produced by a bacterium that may form

part of its structure, support its metabolism, help it attach to host cells, or contribute to disease.

Bacterial Spore

A metabolically dormant, highly resistant structure produced by certain bacteria, notably *Bacillus* and *Clostridioides* species, in response to environmental stress. Spores resist desiccation, heat, and many disinfectants and require validated sterilization for reliable destruction.

Bacterial Toxemia

The presence or systemic effects of bacterial toxins in the bloodstream. It may occur during severe bacterial infection and can contribute to shock or organ injury.

Bacterial Toxin

A harmful substance produced or released by bacteria. Toxins may damage cells directly or disrupt normal body functions and include secreted exotoxins and components such as endotoxin.

Bacterial Toxins

Substances produced by bacteria that cause damage to host cells. There are two major categories: exotoxins and endotoxins. Exotoxins are proteins secreted by bacteria that have specific toxic effects. Endotoxins are lipopolysaccharide components of gram-negative bacterial outer membranes released during cell lysis.

Bacterial Tracheitis

Bacterial infection and inflammation of the trachea, often causing fever, a harsh cough, noisy breathing, and potentially rapid airway obstruction.

Bacterial Transduction

The transfer of bacterial DNA from one bacterium to another through bacteriophages (bacterial viruses).

Bacterial Transformation

The uptake of loose DNA from the environment by bacteria that are able to absorb it. This process occurs naturally in some species, including *Strep-*

tococcus pneumoniae, *Haemophilus influenzae*, and *Neisseria*.

Bacterial Typing

Laboratory methods used to distinguish bacterial strains within the same species, especially for tracking transmission, investigating outbreaks, or linking an isolate to a source.

Bacterial Vaginosis

BV: replacement of normal vaginal lactobacilli by an overgrowth of mixed anaerobes (*Gardnerella vaginalis*, *Mobiluncus*, etc.), the most common cause of vaginal discharge in reproductive-age women. Risk: new/multiple partners, douching, smoking.

Bacterial Vaginosis (Causes, Symptoms, Treatment)

Alternate index label for Causes, Symptoms, Treatment.

Bacterial Virulence Factors

Molecules produced by bacteria that contribute to their ability to cause disease. Virulence factors include adhesins, invasins, toxins, capsules, and enzymes. Adhesins mediate attachment to host cells. Toxins damage host cells directly. Capsules protect against phagocytosis.

Bactericide

An agent that kills bacteria rather than merely inhibiting their growth (bacteriostatic). Bactericidal antibiotics include beta-lactams (penicillins, cephalosporins, carbapenems), aminoglycosides, fluoroquinolones, metronidazole, and vancomycin.

Bacteriocin

An antibacterial protein or peptide produced by some bacteria that inhibits or kills susceptible bacteria, often closely related strains.

Bacteriology

The branch of microbiology concerned with the study of bacteria, including their morphology, genetics, physiology, ecology, and clinical significance.

Bacteriophage

A virus that infects bacteria and uses them to make more virus particles. Phages are studied or used in some research and experimental treatments for antibiotic-resistant bacterial infections.

Bacteriophage Lambda

A virus that infects *Escherichia coli*. It can replicate by destroying the bacterial cell or integrate its genetic material into the bacterial chromosome as a prophage.

Bacteriophage T4

A virus that infects *Escherichia coli* and usually replicates by producing new phage particles and breaking open the bacterial cell.

Bacteriophage Therapy

The investigational use of bacteriophages, viruses that infect bacteria, to treat selected bacterial infections. Their activity is usually limited to susceptible bacterial strains.

Bacteriophages

Viruses that infect bacteria and use bacterial cells to reproduce. They may destroy the host cell or insert their genetic material into the bacterial genome.

Bacteriorhodopsin

A light-sensitive protein in some microorganisms that uses light energy to pump protons across the cell membrane and help generate chemical energy.

Bacteriostasis

Inhibition of bacterial growth and reproduction without necessarily killing the bacteria.

Bacteriostatic

Describing an agent or condition that inhibits bacterial growth rather than directly killing bacteria.

Bacterium

A single-celled organism without a nucleus or specialized internal compartments. Most bacteria have one circular chromosome and a firm outer cell wall.

Bacteriuria

The presence of bacteria in urine, detected by urinalysis or culture. It may occur with urinary symptoms, as in a urinary-tract infection, or without symptoms, as in asymptomatic bacteriuria; interpretation depends on specimen quality, bacterial quantity, patient context, and the presence of inflammation or symptoms.

Bacteroidaceae

A family of mostly Gram-negative anaerobic bacteria that includes *Bacteroides* and commonly forms part of the normal intestinal microbiota. Some species can cause opportunistic infection when they enter sterile tissues.

Bacteroides

A genus of mostly Gram-negative anaerobic bacteria that commonly live in the intestine. They are usually harmless in the gut but can cause abscesses and other infections when displaced into sterile sites.

Bacteroides Caccae

An anaerobic Gram-negative bacterial species found mainly in the intestine and occasionally recovered from polymicrobial abdominal or bloodstream infections.

Bacteroides Dorei

An anaerobic Gram-negative bacterial species found in the human intestine and usually considered part of the normal gut microbiota.

Bacteroides Eggerthii

An anaerobic Gram-negative bacterial species found in the human intestine that is usually a commensal organism but can rarely be associated with opportunistic infection.

Bacteroides Faecis

An anaerobic Gram-negative bacterial species associated with the human intestinal microbiota and occasionally recovered from polymicrobial infections.

Bacteroides Fragilis

An anaerobic Gram-negative bacterium that is part of the intestinal microbiota and is an important cause of intra-abdominal abscesses and other infections after bowel disruption.

Bacteroides Gingivalis

An older name for Porphyromonas gingivalis, an anaerobic oral bacterium associated with dental plaque and periodontal disease.

Bacteroides Heparinolyticus

An anaerobic Gram-negative bacterial species that can be found in human-associated microbial communities and may rarely be recovered from clinical infections.

Bacteroides Intestinalis

An anaerobic Gram-negative bacterial species found in the human intestinal microbiota; it is usually a commensal organism but may be recovered from polymicrobial infections.

Bacteroides Nordii

An anaerobic Gram-negative bacterial species found in the intestine and occasionally identified in polymicrobial clinical specimens.

Bacteroides Ovatus

An anaerobic Gram-negative bacterium that is part of the intestinal microbiota and can contribute to opportunistic infection if it enters damaged or normally sterile tissue.

Bacteroides Pyogenes

An anaerobic Gram-negative bacterial species that may occur in human-associated microbiota and has occasionally been recovered from polymicrobial infections.

Bacteroides Salyersiae

An anaerobic Gram-negative bacterial species associated with the human intestine and occasionally implicated in opportunistic infection.

Bacteroides Stercoris

An anaerobic Gram-negative bacterial species commonly associated with intestinal microbiota and only rarely reported in clinical infection.

Bacteroides Thetaiotaomicon

An anaerobic Gram-negative bacterium that is an important member of the human gut microbiota and helps break down complex dietary carbohydrates; it can cause opportunistic infection outside the intestine.

Bacteroides Uniformis

An anaerobic Gram-negative bacterial species commonly found in the human intestine and usually regarded as a commensal organism.

Bacteroides Ureolyticus

A bacterial name historically applied to an anaerobic, urease-producing organism now generally classified as Campylobacter ureolyticus, which can be associated with wound or bloodstream infection.

Bacteroides Vulgatus

An anaerobic Gram-negative bacterium commonly found in the human intestine that can contribute to infection when the intestinal barrier is disrupted.

Bacteroides Xylanisolvens

An anaerobic Gram-negative bacterial species found in the human intestine that can break down plant-derived carbohydrates and is usually a commensal organism.

Bacteroidete

A member of the bacterial group now commonly called Bacteroidota, which includes many Gram-negative organisms found in soil, water, and the human intestine.

Baculoviridae

Baculoviridae is a virus or viral group classified by its genetic and biologic features; clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Bad Breath

An unpleasant odor from the mouth, medically called halitosis. It most often results from bacteria, food debris, gum disease, tooth decay, dry mouth, or tonsil problems; less commonly, it may be related to nose, throat, digestive, or other medical conditions.

Alternate Names:

Breath Bad (Bad Breath)

Bad Protein

Bad protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

BAER - Brainstem Auditory Evoked Response

BAER records electrical responses from the auditory nerve and brainstem after sound stimulation, helping assess hearing pathways when behavioral testing is difficult.

Bag-Mask Ventilation

Manual ventilation of a non-breathing or inadequately breathing patient using a self-inflating bag (AMBU) attached to a mask sealed over the face. Effective technique: correct mask size, jaw thrust/head extension, two-person technique (two-hand seal), adequate tidal volumes with ETCO₂ monitoring.

Bagassosis

An occupational hypersensitivity pneumonitis (extrinsic allergic alveolitis) caused by inhalation of dust from moldy bagasse—the fibrous residue of sugarcane after extraction. Caused by thermophilic actinomycetes (*Saccharopolyspora rectivirgula*) that colonize stored bagasse.

Bags Under Eyes

Mild swelling or fullness beneath the eyes, commonly related to aging, fluid retention, or inherited anatomy.

Alternate Names:

Puffy Eyes

Bainbridge Reflex

An increase in heart rate produced by stretch of the atrial walls, especially during increased venous return. The reflex helps limit venous congestion by adjusting cardiac output.

Bak Protein

Bak protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Baker Cyst

A fluid-filled cyst in the popliteal fossa, resulting from herniation of the synovial membrane through the posterior knee joint capsule or from a communication between the knee joint and the gastrocnemius-semimembranosus bursa.

Alternate Names:

Bakers Cyst (Baker Cyst) Popliteal Cyst (Baker Cyst)

Baker'S Cyst

A fluid-filled synovial cyst arising from the gastrocnemius-semimembranosus bursa posterior to the knee, typically caused by chronic knee joint effusion from conditions such as osteoarthritis, rheumatoid arthritis, or meniscal tears. It presents as a palpable, tender mass in the popliteal fossa that may rupture causing calf swelling.

Bakers Cyst (Baker Cyst)

Alternate index label for Baker Cyst.

Baking Powder Overdose

Toxic effects from ingesting a large amount of baking powder, which may cause abdominal distention, vomiting, excess sodium, electrolyte disturbances, or metabolic alkalosis. Severity depends on the amount and ingredients.

Baking Soda Overdose

Toxic effects from ingesting excessive sodium bicarbonate, which may cause vomiting, abdominal distention, high blood sodium, metabolic alkalo-

sis, or fluid disturbances. Severe exposure requires prompt assessment.

Balamuthia

A biomedical term referring to balamuthia; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Balamuthia Mandrillaris

A biomedical term referring to balamuthia mandrillaris; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Balamuthia Mandrillaris Infection

A rare, usually fatal infection caused by the free-living amoeba *Balamuthia mandrillaris*. It may begin as a skin lesion and progress to granulomatous amoebic encephalitis with headache, confusion, seizures, or focal neurologic deficits. Diagnosis is difficult and requires specialized testing; urgent infectious-disease and neurologic evaluation is essential.

Balance

The ability to maintain the body's center of mass over its base of support, relying on input from the vestibular system, vision, and proprioception. It is essential for walking, standing, and performing daily activities without falling. Age-related decline, medications, and neurological disorders can impair balance, increasing fall risk.

Balance Assessment

An evaluation of steadiness and the ability to maintain safe posture during standing and movement. It may include standardized tests and helps identify mobility or fall risk.

Balance Disorders And Vertigo

Balance disorders encompass conditions that cause dizziness, unsteadiness, or a sensation of motion, often arising from dysfunction in the inner ear, brain, or sensory pathways.

Balance Impairment

The ability to maintain the body's center of mass over its base of support, relying on input from the vestibular system, vision, and proprioception. See balance.

Balance Problems

Difficulty maintaining body stability while standing or moving, often caused by disorders of the inner ear, nervous system, vision, or muscles.

Balance Training

Exercises improving postural stability and preventing falls. Components: static balance, dynamic balance, reactive balance. Techniques: weight shifting, tandem stance, single-leg stance, perturbations, balance boards. Essential for elderly, neurological conditions.

Balanced Study

A clinical study design in which participant characteristics are evenly distributed between comparison groups at baseline, either by design or by randomization, to reduce confounding and allow fair comparison of interventions.

Balanitis

Inflammation of the glans penis, often involving the prepuce (balanoposthitis). Causes include poor hygiene, irritants, infection, and dermatologic conditions such as lichen planus, psoriasis, and lichen sclerosus. It may cause erythema, swelling, discharge, pruritus, and discomfort; treatment depends on the cause.

Balanitis (Penis Disorder)

Alternate index label for Penis Disorder.

Balanitis Xerotica Obliterans

A chronic, progressive inflammatory mucocutaneous disease of the male external genitalia, representing the penile manifestation of lichen sclerosus. Characterized by porcelain-white, sclerotic, atrophic plaques that predominantly involve the prepuce, glans penis, frenulum, and urethral meatus. Progression causes non-retractile phimosis, painful erec-

tions, meatal stenosis with urinary flow obstruction, and an elevated risk of developing invasive penile squamous cell carcinoma. Management includes ultra-potent topical corticosteroids, circumcision, and meatal reconstruction when stricturing occurs.

Alternate Names:

BZO; Penile Lichen Sclerosus

Balanoposthitis

Inflammation of both the glans penis and the prepuce. It may cause pain, erythema, swelling, discharge, or difficulty retracting the foreskin and can result from infection, irritation, or inflammatory skin disease.

Balantidiasis

Balantidiasis is intestinal infection caused by the protozoan *Balantidium coli*, producing diarrhea, abdominal pain, or colitis, especially after fecal exposure.

Balantidium

A biomedical term referring to *balantidium*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Balantidium Coli

A biomedical term referring to *balantidium coli*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Balbani Body

A specialized aggregate of organelles, including mitochondria and messenger RNA, near the nucleus of a developing oocyte; it helps organize maternal determinants during early embryonic development.

Baldness

Commonly used term for substantial hair loss, especially androgenetic alopecia, also called pattern hair loss. In this condition, genetically susceptible hair follicles gradually become smaller under the influence of androgens.

Alternate Names:

pattern.

Balint's Syndrome

Balint's syndrome is a neurologic disorder caused by bilateral parietal-occipital injury, characterized by impaired visual attention, reaching under visual guidance, and simultaneous perception.

Ballard Score

Clinical assessment of gestational age in newborns using 6 physical and 6 neuromuscular maturity criteria. Physical parameters: skin texture, ear form and recoil, eye appearance, breast tissue, and genitalia. Neuromuscular: posture, arm and leg recoil, popliteal angle, scarf sign, and heel-to-ear.

Ballism

A hyperkinetic movement disorder characterized by sudden, involuntary, large-amplitude, flinging movements of a limb or limbs. It is often related to dysfunction of the subthalamic region or connected basal-ganglia circuits.

Ballistocardiography

The graphic recording of repetitive body movements produced by the ejection of blood with each heartbeat. It has been used to study cardiac mechanical activity and circulation.

Balloon (Vascular)

An inflatable catheter component used to widen a narrowed blood vessel or expand a vascular stent.

Alternate Names:

Vascular

Balloon Angioplasty

A procedure that uses an inflatable balloon catheter to widen a narrowed artery, often followed by stent placement.

Balloon Aortic Valvuloplasty

Catheter-based procedure using balloon inflation to dilate a stenotic aortic valve. Provides temporary hemodynamic improvement in severe AS but restenosis is common within 6-12 months. Used as

a bridge to TAVR or surgical AVR in hemodynamically unstable patients, or as palliation in patients not eligible for definitive valve replacement.

Balloon Brachytherapy

A form of intracavitary radiation therapy in which a balloon catheter is placed in a body cavity, such as the breast lumpectomy cavity or uterine cavity, and filled with radioactive material to deliver localized radiation to surrounding tissue.

Balloon Catheter

A catheter with an inflatable tip used to widen a narrowed vessel, expand a stent, or dilate another body passage.

Balloon Dilation

A procedure that widens a narrowed body passage or vessel by inflating a balloon at the affected site.

Alternate Names:

Balloon dilatation; balloon angioplasty when performed in a blood vessel.

Balloon Tamponade

Temporary control of bleeding or obstruction using an inflatable balloon catheter to apply pressure or occlude a passage. Examples include tamponade of variceal bleeding, postpartum hemorrhage, or selected vascular injuries.

Balneology

Balneology is the study or therapeutic use of mineral waters, baths, and spa treatments; evidence and safety depend on the condition and treatment.

Balneotherapy

Balneotherapy is the therapeutic use of mineral-rich thermal waters, baths, muds, or related spa treatments for selected medical conditions; evidence, contraindications, and safety depend on the indication and treatment.

Balsam

A balsam is a resinous plant-derived substance used in some traditional remedies, fragrances, and topical

preparations; it may cause contact allergy.

Band Cell

An immature neutrophil with a curved, unsegmented nucleus. An increased number in blood may occur during acute infection or inflammation.

Band Keratopathy

A horizontal band of calcium deposition in the superficial cornea, typically across the interpalpebral area. It may occur with chronic ocular inflammation, hypercalcemia, renal disease, or longstanding eye disease and can impair vision.

Bandage

Material applied to cover, protect, compress, or support an injured body part. Examples include gauze, adhesive, elastic, and tubular bandages.

BANF

Alternate index label; see *NF2-related schwannomatosis (neurofibromatosis type 2)*.

Bankart Lesion

A tear of the anterior-inferior glenoid labrum, usually caused by anterior shoulder dislocation and associated with recurrent instability.

Bannayan-Riley-Ruvalcaba Syndrome

Autosomal dominant PTEN hamartoma tumor syndrome caused by loss-of-function mutations in PTEN (10q23), allelic to Cowden syndrome.

Banti Syndrome

A historical term for portal hypertension with splenomegaly and hypersplenism. Modern evaluation usually identifies an underlying hepatic, portal-venous, or hematologic disorder.

Alternate Names:

Banti disease; Banti's syndrome

Bar Graph

A chart that represents values with rectangular bars whose lengths or heights correspond to the quantities being compared.

Barakat Syndrome

A clinical disorder or finding related to barakat syndrome; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Barber Itch

An inflammatory infection of hair follicles in the beard area, often caused by bacteria or dermatophyte fungi and associated with itchy bumps or pustules.

Alternate Names:

Barber's itch

Barber Say Syndrome

A rare congenital disorder characterized by excess, lax facial skin, distinctive facial features, sparse hair and eyelashes, and eyelid abnormalities such as ectropion.

Barbiturate

A class of CNS depressants derived from barbituric acid that enhance GABA-A receptor activity by increasing the duration of chloride channel opening (as opposed to benzodiazepines, which increase the frequency). At low doses: anxiolysis and sedation. At moderate doses: hypnosis and anesthesia.

Barbiturate Intoxication And Overdose

Toxic effects caused by excessive exposure to a barbiturate medicine. Severe poisoning may cause central nervous system depression, respiratory failure, hypotension, or coma and requires emergency treatment.

Barbiturates

A class of central nervous system depressants derived from barbituric acid, acting as positive allosteric modulators of GABA-A receptors. They range from short-acting (thiopental) to long-acting (phenobarbital) and are used for anesthesia, seizure management, and, historically, sedation; dependence and overdose risk limit current use.

Bardet-Biedl Syndrome

A rare inherited disorder that may cause retinal degeneration, obesity, extra fingers or toes, kidney disease, and reproductive abnormalities.

Bardet-Biedl Syndrome

A rare inherited disorder that may cause retinal degeneration, obesity, extra fingers or toes, kidney disease, and developmental or reproductive problems.

Alternate Names:

BBS

Bare Stent (Vascular)

A metal mesh tube placed inside a narrowed vessel without a drug coating to keep the vessel open.

Alternate Names:

Vascular

Bariatric Surgery

Weight loss procedures and surgery, collectively known as bariatric surgery, are invasive interventions for severe obesity that produce substantial and sustained weight loss by altering gastrointestinal anatomy and physiology.

Bariatrics

The branch of medicine concerned with the prevention and treatment of obesity and its associated diseases, including lifestyle, medication, and bariatric surgery.

Barium

A metallic element (atomic number 56, symbol Ba) whose sulfate salt (barium sulfate) is used as a radiocontrast agent due to its high atomic number, which effectively absorbs X-rays.

Barium Enema

An X-ray examination of the large intestine after barium is put into the rectum. Sometimes air is added to show the bowel lining more clearly. It may be used to examine blockage, diverticula, or changes after surgery; other tests are now preferred for routine colon-cancer screening.

Barium Meal

A barium meal is an upper-gastrointestinal contrast study in which swallowed barium outlines the esophagus, stomach, and duodenum on radiographs or fluoroscopy.

Barium Study

An X-ray examination that uses barium to outline parts of the digestive tract. Examples include a barium swallow, barium enema, and upper-GI series. It can show the shape and movement of the digestive tract.

Barium Sulfate

An insoluble radiopaque contrast agent used for radiographic examinations of the gastrointestinal tract; aspiration or leakage from the bowel can cause serious complications.

Barium Swallow

An X-ray examination of the swallowing tube after a person drinks barium. It can show narrowing, abnormal shape, or problems with movement. A barium follow-through continues the examination into the small intestine.

Barium-133

Calibration source. Half-life 10.5 years. Used for energy calibration.

Barlow Test

A physical examination maneuver for detecting hip instability in infants. With the hip flexed, gentle pressure is applied to see whether the femoral head can be displaced out of the socket, suggesting developmental dysplasia of the hip.

Barlow's Disease

Barlow's disease is an older name for severe vitamin-C deficiency, or scurvy, causing bleeding gums, bruising, poor wound healing, and bone or joint symptoms.

Barognosis

The ability to recognize and compare the weight of

objects by holding them. Loss of this ability may indicate a problem in areas of the brain that process sensation.

Baroreceptor

A pressure sensor in major arteries, especially in the neck and chest, that detects changes in blood pressure and helps the body adjust heart rate and blood-vessel tightness.

Baroreceptor Reflex

The rapid neural mechanism for short-term blood pressure regulation, involving baroreceptors (mechanoreceptors) in the carotid sinus (innervated by CN IX) and aortic arch (innervated by CN X). When blood pressure rises, baroreceptors increase their firing rate, signaling the cardiovascular center in the medulla.

Baroreceptors

Baroreceptors are stretch-sensitive nerve endings in major arteries that detect blood-pressure changes and help regulate heart rate and vascular tone through reflexes.

Baroreflex

A rapid reflex that helps regulate blood pressure by adjusting heart rate and blood-vessel tone in response to changes in arterial pressure.

Barotitis Media

Alternate index label; see *Airplane ear*.

Barotrauma

Lung injury from excessive pressure during ventilation, causing alveolar rupture (pneumothorax, pneumomediastinum). Risk factors: high pressures, high PEEP. Prevented by lung-protective ventilation.

Barr Body

A Barr body is the condensed, inactive X chromosome visible in the nucleus of many cells with more than one X chromosome.

Barraquer-Simons Disease

Barraquer-Simons disease is acquired partial lipodystrophy with loss of subcutaneous fat from the face and upper body and relative fat accumulation in the lower body.

Barrel Chest

A rounded, enlarged appearance of the chest caused by increased lung volume or changes in the chest wall. It may occur with chronic obstructive pulmonary disease, severe asthma, or other long-standing lung conditions.

Barrett Esophagus

A change in the lower lining of the swallowing tube caused by long-term acid reflux. The changed lining can increase the risk of precancerous changes and cancer.

Barrett Esophagus Surveillance

Periodic endoscopic evaluation of Barrett esophagus to detect dysplasia or early esophageal adenocarcinoma. The interval and biopsy protocol depend on the presence and grade of dysplasia and current clinical guidelines.

Barrett'S Esophagus

A change in the lower esophageal lining caused by chronic reflux and associated with an increased risk of esophageal adenocarcinoma.

Alternate Names:

Barretts Esophagus (Barrett's Esophagus) Columnar Epithelium Lined Lower Esophagus

Barrett'S Oesophagus

Barrett's oesophagus is replacement of normal lower-esophageal lining by specialized intestinal-type epithelium after chronic reflux, increasing the risk of dysplasia and adenocarcinoma.

Barretts Esophagus (Barrett'S Esophagus)

Alternate index label for Barrett's Esophagus.

Barrier Creams

Topical preparations applied to the skin to create a physical protective layer against irritants, moisture, and pathogens. Common formulations include zinc oxide, petrolatum, dimethicone, and titanium dioxide.

Barrier Nursing (Isolation Nursing)

Nursing care arranged to reduce the spread of infection between a patient, healthcare workers, and other patients by using appropriate hand hygiene, protective equipment, and isolation measures.

Alternate Names:

Isolation Nursing

Bart'S Hemoglobinopathy

A severe alpha-thalassaemia syndrome caused by deletion or inactivation of all four alpha-globin genes, producing haemoglobin Bart's (gamma4); it usually causes hydrops fetalis and is commonly fatal before or shortly after birth.

Barthel Index

A standardized scale that measures assistance needed for activities of daily living, including feeding, bathing, dressing, toileting, transfers, mobility, and stair use. Scores are interpreted according to the version of the scale and the clinical setting.

Bartholin Abscess

An acute, painful, fluctuant purulent infection of the greater vestibular (Bartholin) gland, occurring when ductal occlusion leads to secondary polymicrobial or opportunistic bacterial infection. It presents with severe vulvar throbbing pain, localized swelling, erythema, and difficulty sitting or walking. Management requires urgent incision and drainage with Word catheter placement, marsupialization, and appropriate antimicrobial therapy.

Alternate Names:

Infected Bartholin Cyst; Bartholin Gland Abscess

Bartholin Cyst

A cyst of the Bartholin gland (paired mucus-secreting glands at the posterolateral introitus) due to obstruction of the duct; if infected, a Bartholin

abscess.

Bartholin Cyst Or Abscess

Bartholin cyst is a blocked gland duct near the vaginal opening; infection can produce a painful abscess requiring drainage and sometimes antibiotics.

Bartholin Gland

One of two small glands near the vaginal opening that produce lubricating mucus. Blockage of its duct can cause a painless cyst; infection can produce a painful abscess with swelling, redness, and difficulty walking or sitting. Recurrent or atypical masses require gynecologic assessment.

Bartholin Gland Hyperplasia

Enlargement and proliferation of Bartholin gland tissue, which may present as a vulvar mass and can mimic a cyst or neoplasm; persistent or atypical lesions require clinical assessment and biopsy.

Bartholin Gland Mucinous Cyst

A mucus-filled cyst arising from obstruction of a Bartholin gland duct, usually presenting as a fluctuant mass in the posterolateral vaginal introitus; infection may produce an abscess.

Bartholin's Cyst

A fluid-filled cyst caused by blockage of a Bartholin gland near the vaginal opening.

Alternate Names:

Cyst, Bartholin's

Bartholin's Glands

Bartholin's glands are paired glands near the vaginal opening that secrete mucus for lubrication through ducts opening into the vestibule.

Bartholinitis

Bartholinitis is inflammation or infection of a Bartholin gland, often causing painful swelling or an abscess near the vaginal opening.

Bartonella

A genus of bacteria that includes organisms caus-

ing cat-scratch disease, trench fever, bacillary angiomatosis, and other infections.

Bartonella Elizabethae

A biomedical term referring to bartonella elizabethae; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Bartonella Henselae

A small gram-negative bacterium that causes cat-scratch disease and can produce regional lymphadenitis, fever, and other systemic or atypical infections.

Bartonella Infections

A biomedical term referring to bartonella infections; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Bartonella Quintana

A biomedical term referring to bartonella quintana; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Bartonellaceae

A family of small gram-negative bacteria that includes Bartonella species, which can cause cat-scratch disease, bacillary angiomatosis, endocarditis, and relapsing fever.

Bartonellosis

A group of infections caused by Bartonella species, including cat-scratch disease, bacillary angiomatosis, and, in some regions, bloodstream infection with fever and anemia. The clinical features depend on the species and the patient's immune status.

Bartter Disease

A group of inherited kidney disorders that cause excessive loss of salt and potassium in the urine.

Bartter Syndrome

A group of inherited kidney disorders that cause the body to lose too much salt and potassium in the urine. They can lead to dehydration, muscle weakness, frequent urination, excessive thirst, and poor growth in children.

Basal

Basal means at the base, lowest level, or resting state, such as basal metabolism, basal cells, or basal body temperature.

Basal Body

A microtubule-organizing structure at the base of a cilium or flagellum, derived from a centriole, that anchors the cilium and initiates its axoneme.

Basal Cell

A cell in the deepest layer of an epithelial tissue, where it can divide and produce new cells. In the epidermis, basal cells help renew the skin and contain keratinocyte stem and progenitor cells.

Basal Cell Carcinoma

The most common malignant skin tumor, arising from basal keratinocyte-lineage cells and associated with cumulative ultraviolet exposure and Hedgehog-pathway abnormalities. It commonly presents as a pearly papule or nodule with telangiectasia, ulceration, or a rolled border.

Basal Cell Skin Cancer

Basal cell skin cancer is a usually slow-growing skin malignancy arising from basal keratinocytes; it rarely metastasizes but can destroy nearby tissue if untreated.

Basal Ganglia

Deep brain structures that help control movement, habits, learning, and emotions. Problems in these structures can cause tremor, stiffness, slowed movement, or unwanted movements.

Basal Ganglia Dysfunction

Abnormal function of the deep cerebral nuclei that

regulate movement, cognition, and procedural learning. It may cause tremor, rigidity, bradykinesia, chorea, dystonia, or abnormal involuntary movements, as in Parkinson disease and Huntington disease.

Basal Joint Arthritis

Alternate index label; see *Thumb arthritis*.

Basal Lamina

A thin supporting layer beneath cells that line body surfaces and organs. It helps anchor these cells to the underlying tissue.

Basal Metabolic Rate

Energy expenditure at rest for basic physiological functions. Accounts for 60-75% of total energy expenditure. Affected by age, sex, body composition, thyroid status, genetics. Measured by indirect calorimetry.

Basal Metabolic Rate

Basal metabolic rate is the energy the body uses at rest to maintain essential functions such as breathing, circulation, temperature, and cellular activity.

Basal Nuclei

The basal nuclei are deep cerebral structures that help select and regulate movement, habits, learning, and other motor and cognitive functions.

Basal Tear

A basal tear is a tear occurring at the base of a structure; in the eye, a basal tear layer helps lubricate and protect the ocular surface.

Basal Temperature

The body's resting temperature, measured immediately after waking and before activity; a sustained small rise can occur after ovulation.

Base

A substance that accepts protons (H⁺) or donates electron pairs (Lewis base). In aqueous solution, bases increase OH⁻ concentration (pH >7). Strong

bases (NaOH, KOH) dissociate completely; weak bases (NH₃, HCO₃⁻, protein side chains) partially.

Base Excision Repair

The repair pathway for small, non-helix-distorting base lesions including oxidized bases (8-oxoguanine), deaminated bases (uracil from cytosine deamination), alkylated bases, and abasic (AP) sites.

Base Pair

Two complementary nucleobases joined by hydrogen bonds in nucleic acids: adenine pairs with thymine (or uracil in RNA), and cytosine pairs with guanine.

Base Pairing

The specific hydrogen-bonded association of complementary nucleobases that stabilizes nucleic-acid structure and permits accurate DNA replication, transcription, and RNA hybridization.

Base Sequence

The ordered sequence of nucleotides in a DNA or RNA molecule; it encodes genetic information and can be analysed to identify variants, pathogens, or inherited disorders.

Basedow's Disease

Basedow's disease is Graves disease, an autoimmune disorder in which thyroid-stimulating antibodies cause hyperthyroidism, goiter, and sometimes eye or skin manifestations.

Baseline

A reference measurement or clinical state obtained before an intervention, illness, or period of observation, used for comparison with subsequent findings.

Baseline EKG

An electrocardiogram obtained for comparison with later tracings or before a treatment, procedure, or change in clinical status.

Alternate Names:

Baseline ECG; reference electrocardiogram.

Basement Membrane

A thin, delicate layer of extracellular matrix that anchors epithelial tissue to the underlying connective tissue.

Basic First Aid

Basic first aid is immediate care for illness or injury, including scene safety, emergency activation, bleeding control, resuscitation, choking care, and monitoring until help arrives.

Basic Fuchsin

Basic fuchsin is a magenta laboratory dye used in stains and microbiology, including methods for detecting selected bacteria.

Basic Life Support

Immediate care for a person with cardiac arrest or respiratory failure, including recognition, emergency activation, chest compressions, ventilation, and defibrillation when indicated.

Basic Metabolic Panel

A panel of blood tests measuring serum glucose, sodium, potassium, chloride, bicarbonate, BUN, and creatinine. Also includes calcium in some institutions. Uses: assess hydration status, renal function, acid-base balance, and glucose control.

Basic Reproduction Number

The expected number of secondary infections generated by one typical infectious person in a completely susceptible population, under specified conditions. It is a population and context parameter, not a fixed biological constant.

Basic Research

Scientific investigation aimed at understanding fundamental biological, chemical, or physical mechanisms underlying health and disease, without immediate direct clinical application. It provides the foundation for translational and clinical research.

Basidiomycota

A major fungal group that reproduces by forming

spores on club-shaped structures called basidia. It includes mushrooms, rusts, smuts, and some fungi that can cause human disease.

Basilar

Relating to or located at the base of a structure, such as the base of the skull or the basilar artery.

Basilar Artery

The artery formed by the union of the vertebral arteries that supplies the brainstem, cerebellum, and posterior brain.

Basilar Artery Aneurysm

Dilation of the basilar artery, most commonly fusiform, often associated with atherosclerosis or hypertension. May compress cranial nerves or brainstem structures causing facial pain, double vision, or hemiparesis. Rupture causes subarachnoid hemorrhage with high morbidity.

Basilar Membrane

A flexible membrane in the cochlea that supports the organ of Corti and vibrates in response to sound. Different parts respond best to different sound frequencies.

Basilic Vein

A superficial vein of the upper limb that ascends along the medial side of the arm and joins the brachial veins to form the axillary vein. It is used for venous access in selected settings.

Brachial Vein

Deep veins accompanying the brachial artery that drain the arm and join the basilic vein to form the axillary vein.

Basiliximab

A monoclonal antibody that blocks the interleukin-2 receptor alpha chain on activated T cells. It is used as induction immunosuppression, especially in kidney transplantation, and increases susceptibility to infection.

Basis Pontis

The basis pontis is the front portion of the pons containing descending motor fibers and pontine nuclei that relay signals to the cerebellum.

Basket Cell

A basket cell is an inhibitory interneuron that forms synaptic contacts around the cell body or axon initial segment of another neuron.

Basophil

A rare (<1 percent of leukocytes) granulocyte with large, basophilic granules containing histamine, heparin, and leukotrienes. Basophils bear high-affinity IgE receptors (Fc-epsilon-RI) and, like mast cells, release histamine and mediators during immediate hypersensitivity (allergic) reactions.

Basophil Adenoma

A basophil adenoma is a usually benign pituitary tumor composed of basophilic hormone-producing cells and may cause hormone excess or pressure symptoms.

Basophilia

An abnormally increased number of basophils in the blood. It may occur with myeloproliferative disorders, allergic or inflammatory conditions, and some infections.

Basophilic Focus

A basophilic focus is an area that stains blue or purple with basic dyes, reflecting particular cellular or tissue components.

Basosquamous Carcinoma

A neoplastic process affecting basosquamous carcinoma; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Bassen-Kornzweig Syndrome

Bassen-Kornzweig syndrome, or abetalipoproteinemia, is an inherited disorder of lipoprotein assembly causing fat malabsorption, acanthocytosis, vitamin E deficiency, and neurologic disease.

Batabulin

Batabulin is an investigational vascular-disrupting antineoplastic agent studied for effects on tumor blood vessels and cancer growth.

Bath Anoaesthesia

Bath an(a)esthesia is a historical term for loss of sensation associated with bathing or immersion; the wording is uncommon and its meaning depends on the source.

Bath Salts Abuse And Addiction

Misuse of synthetic cathinones, stimulant drugs that may cause agitation, paranoia, hallucinations, seizures, dangerously high temperature, cardiovascular complications, and dependence. Severe agitation, chest pain, seizure, hyperthermia, or altered consciousness requires emergency care. Treatment is supportive and may require addiction and mental-health services.

Bather'S Itch

Bather's itch is an itchy rash caused by skin penetration of cercarial larvae released by freshwater or marine snails, usually resolving without systemic illness.

Bathing A Patient In Bed

Safe bathing care involves maintaining water temperature, preventing falls or drowning, protecting skin, and providing assistance appropriate to the person's age and mobility.

Bathing An Infant

Safe bathing care involves maintaining water temperature, preventing falls or drowning, protecting skin, and providing assistance appropriate to the person's age and mobility.

Batimastat

Batimastat is an investigational matrix-metalloproteinase inhibitor studied to limit tumor invasion and tissue breakdown.

Batrachotoxin

A highly potent steroidal alkaloid toxin found in some poison frogs and birds that persistently activates voltage-gated sodium channels, causing neuronal and cardiac depolarization, arrhythmias, paralysis, and potentially death.

Batracylin

Batracylin is an investigational anticancer compound studied for DNA damage and topoisomerase-related cytotoxic effects.

Batroxobin

Batroxobin is a snake-venom-derived thrombin-like enzyme that converts fibrinogen to fibrin and has been studied or used in selected clotting and bleeding settings.

Batten Disease

An inherited group of neuronal ceroid lipofuscinoses causing progressive neurodegeneration, seizures, vision loss, and cognitive or motor decline, usually beginning in childhood.

Battered Child Syndrome

A pattern of repeated, non-accidental physical injury inflicted on a child by a caregiver, formally described by Kempe in 1962. Features include fractures of differing ages, subdural and retinal hemorrhages, inconsistent histories, and delayed presentation; mandatory reporting is required in most jurisdictions.

Battery

A battery is a collection of cells or tests used together; in healthcare it may also refer to a power source for an implanted device or to harmful battery ingestion.

Batley Bacillus

An older name for *Mycobacterium avium* complex, a group of nontuberculous mycobacteria that cause pulmonary disease, lymphadenitis, and disseminated infection in immunocompromised patients, particularly those with advanced HIV/AIDS.

Bauhin Valve

The ileocecal valve, a mucosal sphincter at the junction of the terminal ileum and cecum that helps regulate passage of intestinal contents and limits reflux from the colon.

Bauxite-Workers' Lung

Bauxite-workers' lung is an occupational pneumoconiosis associated with inhalation of bauxite dust, potentially causing chronic airway or interstitial lung disease.

Bax Protein

Bax protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Bayes Theorem

A rule for updating the probability of a condition after new evidence, using the prior probability and the likelihood of that evidence in people with and without the condition.

Bazin'S Disease

Bazin's disease is erythema induratum, a nodular inflammation of subcutaneous fat on the calves, classically associated with tuberculosis but having other causes.

Bcgosis

BCGosis is infection or disseminated disease caused by *Bacille Calmette–Guérin* organisms, usually after intravesical BCG treatment or in immune deficiency.

Bdellotomy

Bdellotomy is the surgical incision or removal of a leech attached to tissue, a historical procedure used to control bleeding or prevent retained mouthparts.

Beach Chair Position

A semisitting position in which the patient's back is elevated and the hips and knees are flexed; it is commonly used for shoulder surgery.

Beacopp Regimen

A combination chemotherapy regimen used mainly to treat Hodgkin lymphoma; it includes bleomycin, etoposide, doxorubicin, cyclophosphamide, vincristine, procarbazine, and prednisone, with dosing and supportive care determined by the oncology team.

Beals Syndrome

A congenital connective-tissue disorder characterized by congenital contractures, arachnodactyly, kyphoscoliosis, and crumpled ears. It is usually associated with pathogenic variants affecting fibrillin-2.

Beam Hardening Artifact

CT artifact from dense structures. Reduced by beam hardening correction algorithms.

Beat Number

Beat number is the count of heartbeats or other rhythmic events over a specified interval and helps describe rate or rhythm.

Beats, Cardiac

Cardiac beats are the repeated cycles of atrial and ventricular electrical activation and contraction that propel blood through the circulation.

Beau Lines

Transverse depressions or grooves across the nail plate caused by temporary interruption of nail-matrix growth. They may follow severe illness, fever, trauma, malnutrition, chemotherapy, or systemic inflammatory disease.

Beau'S Lines

Transverse grooves or depressions across a fingernail or toenail caused by temporary interruption of nail-matrix growth. They may follow severe illness, injury, high fever, nutritional stress, or local trauma; the timing of the groove can help estimate when the event occurred.

Becaplermin

Becaplermin is a recombinant platelet-derived

growth factor gel used in selected diabetic foot ulcers with appropriate wound care.

Becatecarin

Becatecarin is an investigational topoisomerase-I inhibitor studied for anticancer activity.

Bechterew Disease

Historical eponym for ankylosing spondylitis (also known as Marie-Strümpell disease), describing the chronic inflammatory arthritis of the axial skeleton first reported by Vladimir Bekhterev (Russian, 1892) and Adolf Strümpell (German, 1884).

Alternate Names:

Marie-Strümpell disease),.

Beck Triad

Classic diagnostic clinical triad of Cardiac Tamponade: (1) Hypotension, (2) Distended neck veins (elevated JVP), and (3) Muffled / distant heart sounds. See Cardiac Tamponade.

Becker Muscular Dystrophy

An X-linked recessive neuromuscular disorder caused by mutations in the dystrophin gene, similar to Duchenne muscular dystrophy (DMD), but with in-frame mutations that allow production of some functional dystrophin protein, albeit reduced in quantity or abnormal in size. The result is a milder clinical phenotype compared to DMD.

Becker Myotonia

The autosomal recessive form of myotonia congenita caused by loss-of-function mutations in the *CLCN1* gene on chromosome 7q35, which encodes the skeletal muscle voltage-gated chloride channel ClC-1. Characterized by reduced chloride conductance that causes sarcolemmal electrical hyperexcitability and delayed muscle relaxation following voluntary contraction. Typically manifests in late childhood with generalized muscle stiffness, transient weakness during sudden vigorous movements, prominent muscle hypertrophy (giving patients a bodybuilder or “Herculean” appearance), and classic percussion and action myotonia that improves with repetitive activity (the warm-up phenomenon); clinical severity

is significantly greater than in autosomal dominant Thomsen disease.

Alternate Names:

Autosomal Recessive Myotonia Congenita; Recessive Generalized Myotonia

Becker’S Muscular Dystrophy

Becker muscular dystrophy is an X-linked dystrophinopathy causing progressive skeletal-muscle weakness, usually later and more slowly than Duchenne muscular dystrophy.

Becker’S Nevus

A benign hamartomatous lesion appearing as a unilateral, hyperpigmented, often hairy patch, typically developing in adolescence on the shoulder or upper trunk of males. It results from a localized androgen-dependent increase in dermal smooth muscle and pigmentation. It is cosmetically concerning but carries no malignant potential.

Beckwith-Wiedemann Syndrome

An overgrowth disorder associated with 11p15.5 imprinting defects. Features: macrosomia, macroglossia, omphalocele, visceromegaly, and neonatal hypoglycemia. Increased risk of embryonal tumors (Wilms tumor, hepatoblastoma).

Beclomethasone

Beclomethasone is a corticosteroid used by inhalation, nasal spray, or topical routes to reduce inflammation in asthma, rhinitis, or skin disease.

Becquerel

A becquerel is the SI unit of radioactivity equal to one nuclear decay per second.

Becquerel (Bq)

The SI unit of radioactive activity, equal to one nuclear transformation per second. Administered activity is commonly reported in megabecquerels or gigabecquerels in nuclear medicine.

Alternate Names:

Bq

Bed Rest

Prescribed or self-imposed restriction of physical activity to a bed or reclining position, used therapeutically for selected conditions such as high-risk pregnancy or acute back pain. Prolonged bed rest can cause deconditioning, thromboembolism, and pressure injury.

Bed Rest During Pregnancy

Bed rest during pregnancy is a medical recommendation that restricts activity for a specific indication. Routine bed rest is not generally used because it can cause deconditioning and thromboembolism.

Bedbug

A bedbug is a blood-feeding *Cimex* insect that causes itchy bites, skin reactions, anxiety, and occasionally secondary infection but is not known to transmit human disease routinely.

Bedbug Bites

Cutaneous reactions to bites of *Cimex lectularius* (bedbugs), nocturnal blood-feeding insects. Bites are painless but cause erythematous, pruritic papules, often in a linear arrangement (breakfast, lunch, dinner sign) or clusters on exposed skin (face, neck, arms, hands). Delayed hypersensitivity develops over days.

Bedbugs

Small blood-feeding insects that cause itchy bites and skin reactions and can spread through bedding, clothing, and furniture.

Bedbugs Bite (Bed Bugs)

Alternate index label for Bed Bugs.

Bednar Aphthae

Traumatic ulcers on the posterior palate of an infant, often associated with pressure or irritation during feeding.

Bedpan

A receptacle used by a person who cannot safely leave bed to urinate or pass stool.

Bedsore

Pressure ulcer (decubitus): localized skin/soft tissue injury from prolonged pressure over bony prominences (sacrum, heels, occiput, trochanters), usually in immobile patients.

Bedsore (Pressure Ulcers)

Injuries to skin and underlying tissue caused by prolonged pressure, usually over bony areas in people with limited mobility.

Bedtime Habits For Infants And Children

Healthy bedtime habits for infants and children include a consistent routine, a safe sleep environment, and age-appropriate sleep duration.

Bedwetting

Involuntary urination during sleep after the age at which nighttime bladder control is usually established. It may be primary or secondary to urinary infection, constipation, diabetes, sleep disorders, stress, or other medical conditions and should be assessed when persistent, new, or accompanied by daytime symptoms.

Bee And Wasp Sting

Bee and wasp stings inject venom and usually cause immediate localized pain, redness, swelling, and itching. Widespread hives, throat or tongue swelling, breathing difficulty, dizziness, or vomiting may indicate anaphylaxis and require emergency treatment.

Bee Sting

A skin injury caused by a bee's stinger that may cause local pain and swelling or, in allergic people, a serious systemic reaction.

Alternate Names:

Stings, Bee

Bee Stings

Bee stings inject venom and can cause localized pain and swelling, large local reactions, or life-threatening anaphylaxis in sensitized people.

Bee, Wasp, Hornet, Or Yellow Jacket

Sting

A clinical topic about bee, wasp, hornet, or yellow jacket sting. It includes the purpose, common findings, risks, and appropriate clinical follow-up.

Beer-Drinker'S Heart

Beer-drinker's heart is an older term for alcohol-related dilated cardiomyopathy, in which chronic heavy alcohol use weakens and enlarges the heart.

Beers Criteria

A regularly updated list of medicines that may pose increased risks for some older adults. It supports medication review but does not replace individualized clinical judgment.

Beeswax Poisoning

Illness after ingesting or inhaling a large amount of beeswax. Beeswax is generally of low toxicity, but large ingestions may cause gastrointestinal blockage or aspiration, especially in children.

Behavior

The observable actions, responses, and activities of an organism, influenced by genetic, neurobiological, psychological, and environmental factors. In medicine, behavioral assessment helps diagnose psychiatric, developmental, and neurological conditions.

Behavior Disorder

A clinical disorder or finding related to behavior disorder; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Behavior Test

A structured assessment of behavior, mood, attention, thinking, or social functioning. It may use observation, interviews, questionnaires, or standardized tasks to help identify developmental, psychiatric, or neurologic concerns.

Behavior Therapy

A form of psychotherapy based on learning principles

that uses reinforcement, extinction, and other behavioral techniques to modify maladaptive behaviors. It is effective for anxiety disorders, phobias, obsessive-compulsive disorder, and habit disorders.

Behavioral Abnormality

A deviation from typical patterns of behavior that may indicate a psychiatric, neurodevelopmental, or neurologic condition. It is assessed by history, observation, standardized measures, and developmental norms to determine clinical significance.

Behavioral Genetics

Behavioral genetics is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

Behavioral Rehabilitation

Behavioral interventions for chronic conditions: pain management, addiction, adjustment.

Behcet Disease

A chronic, relapsing systemic vasculitis characterized by recurrent oral ulcers, genital ulcers, and ocular inflammation, with a distribution along the ancient Silk Road from the Mediterranean to East Asia.

Behcet Syndrome

A chronic inflammatory disorder that can cause recurrent mouth and genital ulcers, eye inflammation, and other organ involvement.

Behcet'S Syndrome

Behcet syndrome is a relapsing inflammatory disorder that can cause recurrent mouth and genital ulcers, eye inflammation, skin lesions, and vascular, neurologic, or gastrointestinal disease. Treatment is tailored to the organs involved and may include immunosuppression.

Alternate Names:

Silk Road Disease (Behcet's Syndrome)

Behçet'S Syndrome

Behçet's syndrome is a relapsing inflammatory disorder causing recurrent oral and genital ulcers, eye inflammation, skin lesions, and sometimes vascular, neurologic, or gastrointestinal disease.

Behenic Acid

Behenic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Behr Syndrome

A clinical disorder or finding related to behr syndrome; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Beijerinck, Martinus W.

A Dutch microbiologist who helped establish the concept of viruses as infectious agents that reproduce only in living cells and contributed to the development of enrichment culture methods.

Being Active When You Have Heart Disease

Physical activity after a heart attack or with heart disease should be gradual and individualized, usually with clinician guidance or cardiac rehabilitation.

Being Good Partner To Someone With Chronic Illness

Supporting a partner with chronic illness involves listening, respecting autonomy, sharing practical tasks by agreement, and recognizing fluctuating symptoms and treatment demands. Clear communication, respite, boundaries, and professional or peer support help protect both partners' well-being.

Bejel

Bejel is endemic treponematosi s, a chronic infection caused by *Treponema pallidum* pertenue or related treponemes and spread mainly by close nonsexual contact in endemic areas.

Bekam

A Malay name for cupping, a practice in which cups create suction on the skin; wet cupping may also involve superficial skin incisions.

Belching

The release of swallowed air or gas from the stomach through the mouth. Frequent belching may occur with aerophagia, gastroesophageal reflux, functional dyspepsia, or carbonated-drink consumption.

Belinostat

Belinostat is a histone-deacetylase inhibitor used for selected relapsed or refractory peripheral T-cell lymphomas.

Bell-Clapper Deformity

A congenital anatomical variation of the scrotum and testis in which the visceral and parietal layers of the tunica vaginalis insert abnormally high on the spermatic cord rather than posterolaterally on the testis. This failure of normal testicular posterior fixation allows the testis to hang horizontally and mobile within the tunica vaginalis cavity like a bell clapper within a bell. The deformity is frequently bilateral and is the primary anatomical predisposing risk factor for intravaginal testicular torsion.

Alternate Names:

Bell-Clapper Anomaly; High Tunica Vaginalis Insertion

Bell Palsy

Acute, idiopathic, unilateral facial nerve (CN VII) paralysis of lower motor neuron type, with an incidence of approximately 20-30 per 100,000 person-years.

Bell'S Palsy

Acute peripheral facial-nerve weakness of uncertain cause, usually affecting one side of the face. Eye protection and timely clinical assessment are important; treatment is individualized.

Alternate Names:

Palsy, Bell's Palsy, Facial

Belladonna

Atropa belladonna, the deadly nightshade plant, whose tropane alkaloids such as atropine and scopolamine can cause anticholinergic poisoning.

Bells Palsy (Facial Nerve Problems)

Alternate index label for Facial Nerve Problems.

Belly

The abdomen, the region of the body between the chest and pelvis containing much of the digestive, urinary, and reproductive organs.

Belotecan Hydrochloride

Belotecan hydrochloride is an investigational topoisomerase-I inhibitor studied as an anticancer drug.

Bence Jones Protein

Monoclonal immunoglobulin light chains (free kappa or lambda) detectable in urine of patients with monoclonal gammopathies, most characteristically multiple myeloma but also Waldenstrom macroglobulinemia, MGUS, and primary amyloidosis (AL).

Bence-Jones Protein

A monoclonal immunoglobulin light chain found in urine or blood, classically associated with plasma-cell disorders such as multiple myeloma. Detection contributes to evaluation of monoclonal gammopathy and kidney injury.

Benchmarking

Benchmarking compares performance, outcomes, quality, or processes with a reference standard or peer group to identify improvement opportunities.

Bendamustine

Bendamustine is an alkylating antineoplastic drug used for selected lymphoid cancers and chronic lymphocytic leukemia, with marrow and infection risks.

Bendamustine Hydrochloride

Bendamustine hydrochloride is the hydrochloride salt formulation of the alkylating antineoplastic drug

bendamustine.

Bendiocarb

Bendiocarb is a carbamate insecticide that inhibits acetylcholinesterase and can cause cholinergic poisoning after exposure.

Bendroflumethiazide

Bendroflumethiazide is a thiazide diuretic used for hypertension or fluid retention and can lower blood pressure, potassium, and sodium.

Bends

A manifestation of decompression sickness caused by gas bubbles forming in tissues or blood after a rapid reduction in ambient pressure. It commonly produces deep joint or muscle pain but may also involve the nervous system or lungs.

Benedikt'S Syndrome

Benedikt's syndrome is a midbrain lesion causing ipsilateral oculomotor palsy with contralateral tremor, ataxia, or involuntary movements.

Beneficence And Non-Maleficence

Core biomedical ethical principles guiding clinical practice and medical research. Beneficence requires healthcare providers to act in the best interest of the patient, taking positive steps to promote health, prevent harm, and relieve suffering.

Benefits And Risks Of Taking Amino Acid Supplement

Amino-acid supplements may be useful for a documented deficiency or a specific clinical indication, but most people obtain adequate amino acids from protein-containing foods. Excessive or unnecessary use can cause gastrointestinal symptoms, metabolic effects, interactions, or kidney and liver concerns.

Benfuracarb

Benfuracarb is a carbamate insecticide that inhibits acetylcholinesterase and can cause cholinergic toxicity after exposure.

Benign

Not malignant and not expected to invade nearby tissue or spread to distant sites, although a benign lesion can still cause problems because of its size or location.

Alternate Names:

Nonmalignant; noncancerous.

Benign Adrenal Tumors

Noncancerous growths of an adrenal gland that may be discovered incidentally or may produce excess hormones.

Benign Ameloblastoma

A neoplastic process affecting benign ameloblastoma; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Benign Brain Tumor

A noncancerous growth in or around the brain that can still cause symptoms by pressing on brain tissue, nerves, or fluid pathways.

Benign Chondroblastoma

A benign cartilage-producing bone tumor, usually arising in the epiphysis of adolescents and young adults and often causing joint-region pain.

Benign CNS Cyst

A non-neoplastic fluid-filled lesion within or around the central nervous system, such as an arachnoid, epidermoid, or colloid cyst. Clinical significance depends on location, size, growth, and obstruction of cerebrospinal-fluid pathways.

Benign Ear Cyst Or Tumor

A benign ear cyst or tumor is a noncancerous mass involving the external, middle, or inner ear. Symptoms and treatment depend on location and size.

Benign Epithelioma

A benign epithelioma is a noncancerous epithelial tumor or growth that does not invade or metastasize but may require removal if symptomatic or uncertain.

Benign Hemangiopericytoma

A benign hemangiopericytoma is a usually nonmalignant tumor around small blood vessels; modern classification often groups related lesions as solitary fibrous tumors.

Benign Intracranial Hypertension

An older term for idiopathic intracranial hypertension, a disorder of raised intracranial pressure without an intracranial mass or hydrocephalus. Headache and papilledema are typical, and untreated disease may threaten vision.

Benign Melanoma

A nonstandard or contradictory phrase: melanoma is a malignant melanocytic tumor, so a benign pigmented lesion should generally be described by its specific benign diagnosis rather than as benign melanoma.

Benign Meningioma

A benign meningioma is a usually slow-growing tumor arising from the membranes around the brain and spinal cord and may cause symptoms by compression.

Benign Mesenchymoma

A benign mesenchymoma is a rare noncancerous tumor containing more than one type of mesenchymal tissue, such as fat, cartilage, or bone.

Benign Migratory Glossitis

Alternate index label; see *Geographic tongue*.

Benign Mixed Tumor

A benign salivary-gland tumor containing epithelial and connective-tissue components, usually called a pleomorphic adenoma.

Benign Mullerian Papilloma

A rare benign papillary lesion of the vulvovaginal region showing Müllerian-type epithelial differentiation, usually presenting as a small polypoid lesion.

Benign Myoepithelioma

A benign myoepithelioma is a noncancerous tumor of myoepithelial cells, commonly arising in a salivary gland or other glandular tissue.

Benign Neoplasm

Benign neoplasms are proliferations of cells that remain confined to their site of origin, grow by expansion rather than invasion, and do not metastasize to distant sites. They typically grow slowly, pushing aside adjacent normal tissue and often becoming enclosed by a fibrous pseudocapsule formed from compressed connective tissue.

Benign Orgasmic Headache

A sudden severe headache occurring during sexual activity or at orgasm after dangerous secondary causes have been excluded.

Alternate Names:

Primary headache associated with sexual activity; sexual-activity headache.

Benign Paraganglioma

A benign paraganglioma is a usually slow-growing neuroendocrine tumor arising from paraganglion cells and may secrete catecholamines or compress nearby structures.

Benign Paroxysmal Positional Vertigo

A common peripheral vertigo from dislodged otoconia (canalithiasis) within the semicircular canals, most often the posterior canal. Brief episodes of rotational vertigo lasting under one minute provoked by changes in head position.

Benign Paroxysmal Positional Vertigo

Brief episodes of spinning dizziness triggered by changes in head position, usually caused by displaced inner-ear crystals.

Benign Peripheral Nerve Tumor

A noncancerous tumor arising from or near a peripheral nerve, potentially causing a lump, pain, tingling, numbness, or weakness.

Benign Positional Vertigo

Brief episodes of spinning dizziness triggered by changes in head position, usually caused by displaced calcium carbonate crystals in a semicircular canal. It is also called benign paroxysmal positional vertigo or BPPV.

Benign Prostate Hyperplasia

A biomedical term referring to benign prostate hyperplasia; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Benign Prostatic Hyperplasia

Noncancerous enlargement of the prostate that may obstruct urine flow and cause hesitancy, weak stream, urgency, frequency, nocturia, or incomplete emptying. Assessment uses the symptoms, examination, and selected tests.

Alternate Names:

BPH (Benign Prostatic Hyperplasia)

Benign Prostatic Hyperplasia

Noncancerous enlargement of the prostate that can obstruct urine flow and cause urinary frequency, urgency, weak stream, or incomplete emptying.

Alternate Names:

Prostate Gland Enlargement

Benign Prostatic Obstruction

Bladder-outlet obstruction caused by benign prostatic enlargement or increased prostatic smooth-muscle tone. It can produce weak stream, hesitancy, intermittency, straining, incomplete emptying, retention, infection, and upper-tract deterioration.

Benign Rolandic Epilepsy

The most common childhood epilepsy syndrome (also known as self-limited epilepsy with centrotemporal spikes, SeLECTS), typically presenting between ages 3 and 12 years. Characterized clinically by brief focal sensorimotor seizures involving the face, mouth, and tongue during drowsiness or sleep, presenting with unilateral facial twitching, perioral paresthesias, drooling, and speech arrest (anarthria). Electroencephalography shows characteristic high-

voltage centrottemporal sharp-and-slow-wave complexes that increase during non-REM sleep. The condition has an excellent prognosis and almost always resolves spontaneously by adolescence.

Alternate Names:

Self-Limited Epilepsy with Centrottemporal Spikes; SeLECTS; Benign Childhood Epilepsy with Centrottemporal Spikes; Rolandic Epilepsy

Benign Tumor

Non-cancerous growth, slow-growing, well-differentiated, localized, does not invade or metastasize. Examples: lipoma, fibroma, adenoma. May cause problems by compression or hormone secretion.

Benserazide

Benserazide inhibits peripheral dopa decarboxylase and is combined with levodopa to increase brain delivery and reduce peripheral adverse effects in Parkinson disease.

Bent Penis

Curvature of the penis that may be congenital or acquired, often from Peyronie disease; it can cause pain or difficulty with sexual intercourse.

Bent Spine Syndrome

A marked, usually reducible forward flexion of the trunk caused by weakness or abnormal contraction of axial extensor muscles. It is also called camptocormia and may accompany neuromuscular or movement disorders.

Alternate Names:

camptocormia and may accompany.

Bentall Procedure

A classic cardiovascular surgical operation for combined disease of the aortic valve, aortic root, and ascending aorta (such as severe aortic regurgitation with an ascending aortic aneurysm, annuloaortic ectasia in Marfan syndrome, or acute Stanford type A aortic dissection). The diseased aortic valve, root, and ascending aorta are resected en bloc and replaced with a composite prosthetic conduit containing a mechanical or tissue valve inside a woven

Dacron tube graft, with direct reimplantation of the mobilized coronary artery buttons into the side of the graft.

Alternate Names:

Bentall Operation; Composite Aortic Root Replacement

Benzaldehyde

An aromatic aldehyde used mainly as a flavouring, fragrance, and chemical intermediate; concentrated exposure may irritate the eyes, skin, and respiratory tract and it is not a routine therapeutic medicine.

Benzazepines

A family of chemical compounds containing a benzene ring fused to a seven-membered azepine ring; some derivatives have pharmacological activity, but the clinical effects depend on the individual compound.

Benzene

Solvent, industrial chemical. Chronic exposure: aplastic anemia, leukemia (especially AML). Acute: CNS depression, arrhythmias. Prevention: substitution, ventilation, monitoring.

Benzene Poisoning

Toxic effects from exposure to benzene, including dizziness and central nervous system depression after high short-term exposure and bone-marrow injury with repeated exposure. Long-term exposure increases the risk of leukemia.

Benzhexol

Benzhexol, or trihexyphenidyl, is an anticholinergic drug used mainly to reduce tremor and rigidity in Parkinson disease or drug-induced extrapyramidal symptoms.

Benzo(A)Pyrene

Polycyclic aromatic hydrocarbon from combustion. Carcinogenic (lung, skin cancer). Found in tobacco smoke, charred meat, air pollution.

Alternate Names:

a

Benzoates

Salts or esters of benzoic acid, used as preservatives and, in selected preparations, as pharmaceutical excipients or metabolic agents; their safety depends on the specific compound and exposure.

Benzocaine

A topical ester local anesthetic used to numb skin or mucous membranes. Rarely, especially after excessive exposure, it can cause methemoglobinemia.

Benzodiazepine Use Disorder

A problematic pattern of benzodiazepine use leading to significant impairment. Benzodiazepines include alprazolam, clonazepam, diazepam, and lorazepam. Dependence can develop rapidly with regular use. Withdrawal can be life-threatening, presenting with anxiety, insomnia, tremor, seizures, and psychosis.

Benzodiazepine Withdrawal

Symptoms from cessation or reduction of prolonged benzodiazepine use. Early symptoms include anxiety, insomnia, tremor, and diaphoresis. Severe withdrawal can include seizures, psychosis, and delirium. Withdrawal severity depends on the half-life of the benzodiazepine, duration of use, and dose.

Benzopyran

A fused benzene-pyran chemical ring system found in several natural products and drug classes; it is a structural chemical term rather than one specific medicine.

Benzopyrene

A polycyclic aromatic hydrocarbon formed during incomplete combustion and present in tobacco smoke and some charred foods; benzo[a]pyrene is mutagenic and carcinogenic after metabolic activation.

Benzopyrenes

Polycyclic aromatic hydrocarbons containing a benzopyrene structure, produced by incomplete combustion; several are environmental carcinogens and may damage DNA after metabolic activation.

Benzoyl Peroxide

Benzoyl peroxide is a defined chemical or biologic substance involved in benzoyl peroxide and studied in physiology, pharmacology, laboratory medicine, or treatment.

Benzoylphenylurea

A class of insecticides that disrupt chitin formation in developing insects; these compounds are used for pest control rather than as human medicines.

Benzphetamine

A sympathomimetic amine and stimulant formerly used in some countries as a short-term adjunct for weight reduction; it can cause dependence, cardiovascular effects, and insomnia and is restricted or unavailable in many jurisdictions.

Benztropine

An antimuscarinic medicine used mainly to reduce drug-induced parkinsonism and acute dystonia; adverse effects include dry mouth, blurred vision, constipation, urinary retention, confusion, and worsening of glaucoma.

Benzydamine

A topical non-steroidal anti-inflammatory analgesic used for painful inflammation of the mouth, throat, or skin; local irritation, numbness, or allergic reactions may occur.

Benzyl Benzoate

Benzyl benzoate is a topical acaricide and insecticide used for scabies and sometimes lice; skin irritation and careful application are important.

Benzylamines

Organic compounds containing a benzyl group attached to an amino group; some derivatives are pharmacologically active, while others are chemical intermediates or industrial compounds.

Benzylpenicillin

Benzylpenicillin, or penicillin G, is a beta-lactam antibiotic used for susceptible serious infections,

including some streptococcal and syphilis infections.

BEP Regimen

A combination chemotherapy regimen containing bleomycin, etoposide, and cisplatin, commonly used for germ-cell tumors such as testicular cancer; treatment requires monitoring for organ toxicity and complications.

Berberine

Berberine is a plant alkaloid sold in supplements and studied for metabolic and antimicrobial effects; interactions and adverse effects are possible.

Bereavement

Natural grief response to the death of a loved one, marked by sadness, yearning, and preoccupation with the deceased, experienced in waves that diminish over weeks to months with preserved self-esteem. Distinct from major depression by its wave-like quality and preserved self-worth.

Bereavement Reaction

A normal emotional response to the death of a loved one, involving grief that generally follows a variable course without necessarily indicating a mental disorder.

Berger Disease

Historical name for IgA nephropathy named after Jean Berger who first characterized mesangial IgA deposition in 1968. Characterized by recurrent gross hematuria triggered by mucosal infections with intervals of microscopic hematuria and proteinuria. Generally benign course in most patients but subset progresses to CKD.

Berger'S Disease

Alternate index label; see *IgA nephropathy (Berger disease)*.

Bergeyella

Bergeyella is a genus of Gram-negative bacteria found in animals and the environment; human infection is uncommon and depends on the clinical

isolate.

Bergeyella Zoohelcum

Bergeyella zoohelcum is an uncommon zoonotic Gram-negative bacterium reported after animal exposure, especially in people with underlying illness.

Beriberi

A nutritional deficiency disease caused by inadequate thiamine (vitamin B1), occurring in two forms. Wet beriberi presents with high-output heart failure, peripheral vasodilation, edema, and tachycardia due to impaired cardiac energy metabolism.

Berkeley'S Sign

Berkeley's sign is an eponym with varying historical use; the exact clinical finding should be confirmed from the source in which it appears.

Bernard-Soulier Disease

Bernard-Soulier disease is a rare inherited platelet disorder caused by defects in the platelet glycoprotein Ib-IX-V complex. Platelets are unusually large and do not adhere normally, producing mucocutaneous bleeding, easy bruising, and prolonged bleeding after procedures.

Bernard–Soulier Syndrome

An inherited platelet-adhesion disorder caused by deficiency or dysfunction of the glycoprotein Ib-IX-V complex, producing mucocutaneous bleeding and giant platelets.

Bernstein Test

A historical provocative test in which acid is introduced into the esophagus to reproduce heartburn symptoms; it has largely been replaced by more direct evaluation of reflux.

Alternate Names:

Acid perfusion test; Bernstein acid perfusion test.

Berry Aneurysm

A small, rounded saccular aneurysm arising at an arterial branch point in the circle of Willis. Rupture is a major cause of nontraumatic subarachnoid

hemorrhage.

Berylliosis

A granulomatous lung disease from beryllium exposure. Chronic berylliosis resembles sarcoidosis. Diagnosis requires beryllium lymphocyte proliferation test.

Beryllium

Beryllium is a metal whose dust or fumes can cause sensitization, chronic granulomatous lung disease, and cancer risk after occupational exposure.

Beta 2-Microglobulin

A small protein associated with the class I major histocompatibility complex on nucleated cells; its blood or urine concentration may rise with inflammation, lymphoproliferative disease, or impaired renal handling.

Beta Amyloid

Peptides generated from amyloid precursor protein that can accumulate as plaques in the brain in Alzheimer disease.

Alternate Names:

Amyloid-beta; A-beta peptide.

Beta Carotene

A carotenoid pigment found in plants that can be converted by the body into vitamin A.

Alternate Names:

Beta-carotene; provitamin A carotenoid.

Beta Cell

An endocrine cell in the pancreatic islets that produces and releases insulin in response to increased blood glucose.

Beta Cells

Cells in clusters called pancreatic islets that make and release insulin, the hormone that lowers blood glucose. They also release amylin, which helps control the rise in glucose after meals; loss or failure of beta cells contributes to diabetes.

Beta Globulin

A group of serum proteins that migrate in the beta region during protein electrophoresis, including transferrin, complement proteins, and some lipoproteins; abnormal levels can reflect inflammation, liver disease, or plasma-cell disorders.

Beta Particle

A high-energy electron or positron emitted during radioactive beta decay; beta radiation can damage tissue and is used in selected medical imaging and radiotherapy applications.

Beta Radiation

Ionizing radiation consisting of high-energy electrons or positrons emitted by unstable nuclei; it has moderate tissue penetration and can cause radiation injury or be used therapeutically.

Beta Thalassemia

An inherited blood disorder in which reduced production of beta-globin causes anemia.

Beta Tocopherol

One of the naturally occurring forms of vitamin E, a lipid-soluble antioxidant; it is less biologically active than alpha-tocopherol in humans.

Beta-Adrenoceptor-Blocking Drugs

Beta-adrenoceptor-blocking drugs reduce beta-adrenergic effects on the heart and other tissues and are used for hypertension, angina, arrhythmias, heart failure, and other indications.

Beta-Blockers

Beta-adrenergic blocking agents, commonly known as beta-blockers, are topical medications used in the treatment of glaucoma that lower intraocular pressure by reducing the production of aqueous humor by the ciliary body, primarily through blockade of beta-2 adrenergic receptors.

Beta-Blockers Overdose

Toxic effects from excessive beta-blocker exposure, which may cause slow heart rate, low blood pres-

sure, conduction abnormalities, low blood glucose, seizures, or shock. Severe poisoning requires emergency assessment.

Beta-Carotene Blood Test

A laboratory test that measures serum beta-carotene concentration, a provitamin A carotenoid obtained from dietary sources. Low levels may indicate malabsorption or nutritional deficiency; elevated levels can cause carotenoderma, a harmless yellow-orange skin discoloration.

Beta-Chemokines

A group of chemokines, generally in the CC family, that attract monocytes, lymphocytes, eosinophils, and other leukocytes through specific chemokine receptors during immune and inflammatory responses.

Beta-Endorphin

An endogenous opioid peptide derived from proopiomelanocortin in the pituitary and hypothalamus; it binds opioid receptors and contributes to modulation of pain, stress, and neuroendocrine function.

Beta-Glucans

Polysaccharides made of glucose units linked by beta bonds, found in cereals, fungi, and yeast; their physiological effects vary by structure and may include effects on fibre intake and innate immune signalling.

Beta-Haemolytic Streptococcus

A beta-haemolytic streptococcus completely lyses red cells around colonies on blood agar; important species include group A and group B streptococci.

Beta-Human Chorionic Gonadotrophin

Beta-human chorionic gonadotrophin is the beta subunit of hCG, measured in blood or urine to detect pregnancy and to assess selected tumors or pregnancy problems.

Beta-Lactam Antibiotics

A large class of antimicrobial agents characterized by a beta-lactam ring in their molecular structure, which is essential for their bactericidal activity.

Beta-Lactamase

A bacterial enzyme that hydrolyses the beta-lactam ring of susceptible penicillins, cephalosporins, or related antibiotics, producing antimicrobial resistance; inhibitors may restore activity against selected organisms.

Beta-Mannosidosis

Beta-mannosidosis is an inherited lysosomal storage disorder caused by beta-mannosidase deficiency and can produce developmental, hearing, immune, or skeletal problems.

Beta-Oxidation

The catabolic process by which fatty acid molecules are broken down in the mitochondria (and peroxisomes) to generate acetyl-CoA, NADH, and FADH₂.

Beta-Sheet

A protein secondary structure formed by extended polypeptide strands stabilized by hydrogen bonds; abnormal beta-sheet-rich aggregates occur in disorders such as amyloidosis and some neurodegenerative diseases.

Beta-Thalassemia

An inherited hemoglobin disorder with deficient or absent beta-globin chain synthesis, ranging from mild trait to transfusion-dependent major disease with iron overload, extramedullary hematopoiesis, and skeletal changes.

Beta-Thalassemia Intermedia

A hereditary hemoglobinopathy characterized by moderate reduction in beta-globin chain synthesis, resulting in anemia of intermediate severity between mild beta thalassemia trait and transfusion-dependent beta thalassemia major. Patients present with moderate microcytic anemia, hepatosplenomegaly, and bone changes, but typically do not require lifelong regular blood transfusions.

Beta-Thalassemia Minor

The heterozygous carrier state for beta thalassemia caused by a mutation in a single beta-globin gene allele. Typically asymptomatic or presenting with

mild, well-tolerated microcytic hypochromic anemia, an elevated red blood cell count, and characteristic elevation of hemoglobin A2 (usually above 3.5%) on hemoglobin electrophoresis.

Alternate Names:

Beta-thalassemia trait; Cooley trait

Betacoronavirus

Betacoronavirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Betaherpesvirinae

A subfamily of herpesviruses that includes cytomegalovirus and human herpesviruses 6 and 7. These viruses establish lifelong latent infection and may cause severe disease in immunocompromised people.

Betaine

A naturally occurring zwitterionic compound that acts as a methyl-group donor in homocysteine metabolism and helps maintain cellular osmotic balance; inherited defects in its pathway can cause homocystinuria.

Betamethasone

A potent synthetic glucocorticoid used topically or systemically to suppress inflammation and immune responses; prolonged or extensive use can cause skin atrophy, infection, adrenal suppression, and other corticosteroid effects.

Betamethasone Valerate

A topical corticosteroid ester used for inflammatory skin disorders such as eczema and psoriasis; prolonged, occlusive, or extensive application increases the risk of local skin atrophy and systemic corticosteroid absorption.

Betaretrovirus

Betaretrovirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Betaxolol

A relatively beta-1-selective blocker used orally for some cardiovascular conditions and topically to lower intraocular pressure in glaucoma. It may cause bradycardia, hypotension, or bronchospasm.

Betazole

A histamine H2-receptor agonist formerly used to stimulate gastric acid secretion during diagnostic testing; it is rarely used in current practice.

Bethanechol

Bethanechol is a muscarinic cholinergic agonist that stimulates bladder and gastrointestinal smooth muscle, used selectively for urinary retention or impaired motility.

Bethesda System For Reporting Thyroid Cytopathology

A six-tier standardized system for reporting thyroid fine-needle aspiration results, from nondiagnostic and benign categories to follicular-pattern, suspicious, and malignant categories; each tier guides risk assessment and management.

Bethlem Myopathy

A slowly progressive, benign inherited skeletal muscle disorder within the spectrum of collagen VI-related myopathies, transmitted in an autosomal dominant (or rarely autosomal recessive) pattern and caused by mutations in *COL6A1*, *COL6A2*, or *COL6A3*. Characterized by proximal muscle weakness and wasting with onset in childhood or early adulthood, combined with a unique, pathognomonic pattern of joint contractures: prominent flexion contractures of the interphalangeal joints of the fingers, elbows, and ankles, juxtaposed with marked hyperextensibility and hyperlaxity of the wrists, metacarpophalangeal joints, and ankles. Additional findings include follicular hyperkeratosis, keloid formation, and preserved ambulation into late adulthood.

Alternate Names:

Collagen VI-Related Myopathy (Mild Form); Bethlem Distal Myopathy

Bevacizumab

A monoclonal antibody that blocks vascular endothelial growth factor and inhibits new blood-vessel formation. It is used for selected cancers and some eye diseases; important risks include hypertension, bleeding, thrombosis, proteinuria, impaired wound healing, and gastrointestinal perforation.

Bexarotene

An oral retinoid X-receptor agonist used mainly for cutaneous T-cell lymphoma; it commonly causes hypertriglyceridaemia, hypothyroidism, and liver abnormalities and is teratogenic.

Bezafibrate

Bezafibrate is a fibrate lipid-lowering drug that activates PPAR-alpha and lowers triglycerides; muscle and liver effects require monitoring.

Bezoar

A mass of indigestible material retained in the gastrointestinal tract, commonly composed of hair, plant fiber, or medication material.

Breast Imaging-Reporting And Data System

Breast Imaging-Reporting and Data System standardizes breast-imaging reports and categories to guide follow-up and biopsy recommendations. Categories: 0 (incomplete), 1 (negative), 2 (benign), 3 (probably benign), 4 (suspicious), 5 (highly suggestive of malignancy), 6 (known biopsy-proven malignancy).

Biacetyl

Biacetyl, or diacetyl, is a diketone compound with a buttery odor used in flavoring and industry; inhalational exposure can injure the small airways.

Bias

A systematic deviation that distorts a measurement, study result, clinical judgment, or decision away from the truth or from an intended fair comparison.

Bible Cyst (Ganglion)

Alternate index label for Ganglion.

Bibliotherapy

The use of reading materials as a therapeutic intervention to support mental health, educate patients about conditions, or promote behavioral change. It may be used alone or alongside other treatments for depression, anxiety, or chronic disease self-management.

Bicalutamide

A non-steroidal androgen-receptor antagonist used in prostate cancer, usually with androgen-deprivation therapy; adverse effects include hot flashes, breast enlargement, sexual dysfunction, and possible liver injury.

Bicarbonate

The principal extracellular buffer anion, formed from carbon dioxide and water and regulated chiefly by the lungs and kidneys. The bicarbonate-carbonic-acid system helps maintain arterial pH; serum bicarbonate is commonly used to assess metabolic components of acid-base disorders.

Biceps

The biceps is a two-headed muscle, especially the biceps brachii of the upper arm, that flexes the elbow and supinates the forearm.

Biceps Brachii Muscle

Two-headed anterior arm muscle. Long head from supraglenoid tubercle, short head from coracoid process, both inserting on the radial tuberosity. Innervated by musculocutaneous nerve (C5-C6). Most powerful elbow flexor in supination; supinates forearm; weakly flexes shoulder. Distal rupture causes Popeye deformity.

Biceps Reflex

A tendon reflex produced by tapping the biceps tendon, causing elbow flexion through the musculocutaneous nerve and predominantly C5-C6 spinal segments. It helps assess the corresponding peripheral and central pathways.

Bichat'S Membrane

Bichat's membrane is an older anatomical term generally referring to a serous or connective-tissue membrane; its exact usage depends on the historical source.

Bicipital

Bicipital means relating to a two-headed muscle, especially the biceps tendon, groove, or muscle of the upper arm.

Bicipital Groove

The intertubercular groove of the humerus that transmits the tendon of the long head of the biceps brachii, with the transverse humeral ligament and adjacent rotator-cuff structures contributing to tendon stability.

Bicipital Tuberosity

The radial tuberosity, a roughened prominence on the proximal radius where the biceps brachii tendon inserts and produces flexion and supination of the forearm.

Bicolor

Bicolor means having two colors; in medicine it may describe a two-toned lesion, specimen, iris, or other structure and requires context.

Biconcave

Biconcave means concave on both sides; the normal red blood cell is a classic biconcave disc, a shape that supports flexibility and gas exchange.

Biconvex

Biconvex means convex on both sides; the term may describe a lens, implant, or radiologic shape.

Bicornuate

Having two horn-shaped cavities or projections; a bicornuate uterus is a congenital uterine anomaly with two partially separated upper cavities.

Bicornuate Uterus

A bicornuate uterus is a congenital uterine shape

with two upper cavities caused by incomplete fusion of the embryonic Müllerian ducts; it may affect pregnancy outcomes.

Bicuculline

Bicuculline is a research compound that blocks GABA-A receptors and can provoke seizures in experimental systems.

Bicuspid

Bicuspid means having two cusps, as in a bicuspid aortic valve or a premolar tooth with two main points.

Bicuspid Aortic Valve

A congenital aortic valve with two leaflets instead of the usual three. It can lead to aortic stenosis, aortic regurgitation, enlargement of the aorta, or other congenital heart abnormalities.

Bicuspid Valve

A heart valve with two leaflets rather than the usual three. The term most often refers to congenital bicuspid aortic valve, which can cause aortic stenosis, regurgitation, infective endocarditis, or ascending-aortic disease.

Bicycled Foot

Bicycled foot is an uncommon descriptive term for a foot deformity or position associated with cycling or repetitive mechanical stress; exact meaning depends on the source.

Bid Protein

Bid protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Biermer'S Anemia

Pernicious anemia, a vitamin B12-deficiency anemia caused by impaired absorption, usually because autoimmune loss of intrinsic factor prevents normal B12 uptake.

Bietti'S Crystalline Dystrophy

A rare autosomal recessive chorioretinal dystrophy characterized by tiny crystalline deposits in the retina and cornea with progressive chorioretinal atrophy, presenting with night blindness and progressive visual field loss beginning in early adulthood.

Bifidobacterium

A genus of anaerobic, Gram-positive bacteria commonly found in the human intestine. Some species are used as probiotics, while rare bloodstream infections have been reported, mainly in people with serious underlying illness.

Bifidobacterium Adolescentis

An anaerobic bacterium commonly found in the human intestine and used in some probiotic preparations; it is usually harmless but can rarely cause opportunistic infection.

Bifidobacterium Animalis

An anaerobic intestinal bacterium used in some fermented foods and probiotic preparations. It is generally regarded as low risk, although rare opportunistic infections can occur.

Bifidobacterium Bifidum

An anaerobic bacterium of the human intestine, especially common early in life, and a species used in some probiotic preparations.

Bifidobacterium Breve

An anaerobic intestinal bacterium commonly found in infants and used in some probiotic preparations.

Bifidobacterium Dentium

An anaerobic bacterium found in the mouth and intestine that may contribute to dental caries and, rarely, opportunistic infection.

Bifidobacterium Longum

An anaerobic intestinal bacterium commonly found in humans and used in some probiotic preparations; it is usually harmless but can rarely cause opportunistic infection.

Bifocal

Bifocal means having two focal points; bifocal spectacles correct near and distance vision, while bifocal imaging or lenses have two optical zones.

Bifurcation

Bifurcation is the division of one structure, vessel, airway, nerve, or pathway into two branches.

Big Toe Sign

Upward extension of the great toe, often with fanning of the other toes, when the sole is stimulated; in adults it may indicate corticospinal-tract disease.

Bigeminy

Bigeminy is a cardiac rhythm pattern in which a normal beat alternates regularly with a premature beat, often a premature ventricular contraction.

Biglycan

Biglycan is a small proteoglycan of the extracellular matrix involved in collagen organization, inflammation, and tissue repair.

Biguanide

A class of medicines that lower blood glucose by reducing hepatic glucose production and improving insulin sensitivity. Metformin is the principal biguanide in current clinical use; its use depends on kidney function, tolerability, and the person's treatment goals.

Biguanides

A class of glucose-lowering medicines that reduce hepatic glucose production and improve insulin sensitivity; metformin is the principal example.

Alternate Names:

Biguanide drugs; biguanide antidiabetics.

Bilateral

Bilateral means affecting or involving both sides of the body or a paired structure.

Bilateral Acoustic Neurofibromatosis

Alternate index label; see *NF2-related schwanno-*

matosis (neurofibromatosis type 2).

Bilateral Gynecomastia

Enlargement of breast glandular tissue affecting both breasts in a male patient; causes include hormonal imbalance, medicines, obesity, liver or kidney disease, and rarely malignancy.

Bilateral Hydronephrosis

Bilateral hydronephrosis is enlargement of both renal collecting systems from impaired urine drainage, with causes including obstruction, reflux, stones, or bladder dysfunction.

Bilateral Microphthalmos

Bilateral microphthalmos is abnormally small size of both eyes, a congenital condition that can impair vision and occur with other developmental abnormalities.

Bilateral Oophorectomy

Surgical removal of both ovaries, performed for conditions such as ovarian cancer, endometriosis, or prophylaxis in high-risk genetic mutation carriers. It induces immediate surgical menopause and requires hormone replacement consideration in premenopausal women.

Bilateral Peritonsillar Abscesses

Suppurative infection in the peritonsillar space on both sides, a rare complication of acute tonsillitis presenting with severe sore throat, dysphagia, trismus, and muffled hot potato voice. Bilateral involvement risks acute upper airway obstruction and requires urgent needle aspiration or incision and drainage.

Bilateral Pneumonia

Pneumonia involving both lungs; severity ranges from limited infection to diffuse respiratory failure depending on the cause and host factors.

Bilateral Tonic-Clonic Seizure

A bilateral tonic-clonic seizure causes loss of consciousness followed by stiffening and rhythmic jerk-

ing of both sides of the body.

Bile

A fluid produced by the liver that helps digest and absorb dietary fats.

Bile Acid Sequestrants For Cholesterol

Medicines that bind bile acids in the intestine and lower low-density lipoprotein cholesterol by increasing hepatic conversion of cholesterol into bile acids.

Bile Acids

Substances produced by the liver that aid digestion and absorption of fats and fat-soluble vitamins. They are also measured when a hepatobiliary disorder, including intrahepatic cholestasis of pregnancy, is suspected.

Bile Culture

A laboratory culture of bile to identify bacteria or fungi in suspected biliary infection.

Bile Duct

A channel that carries bile from the liver and gallbladder to the duodenum. The right and left hepatic ducts unite as the common hepatic duct; union with the cystic duct forms the common bile duct, which usually joins the pancreatic duct before entering the duodenum at the major duodenal papilla.

Bile Duct Cancer

Cancer arising in the bile ducts; also called cholangiocarcinoma.

Alternate Names:

cholangiocarcinoma.

Bile Duct Cancer (Cholangiocarcinoma)

Alternate index label for Cholangiocarcinoma.

Bile Duct Obstruction

Impaired flow of bile through the biliary tract, causing jaundice, dark urine, pale stools, itching, abdominal pain, or infection.

Bile Duct Stricture

Narrowing of a bile duct that impairs bile flow and may cause jaundice, abdominal pain, cholangitis, or abnormal liver tests.

Bile Ducts

Bile ducts are channels that carry bile from the liver and gallbladder to the duodenum; obstruction can cause jaundice, pain, infection, or pancreatitis.

Bile Pigment

A pigment derived largely from haem breakdown, chiefly bilirubin, that is processed by the liver and excreted in bile; accumulation in blood and tissues causes jaundice.

Bile Reflux

Reflux of duodenal contents (bile acids, lysolecithin, pancreatic enzymes) into the stomach and esophagus, caused by incompetence of the pyloric sphincter, prior gastric surgery (Billroth II, Roux-en-Y, vagotomy with pyloroplasty), cholecystectomy, or functional gastroparesis.

Alternate Names:

Reflux, Bile

Bile Salt Emulsification

Bile salts (derived from cholesterol in the liver) are amphipathic molecules with hydrophobic and hydrophilic regions that act as biological detergents, emulsifying dietary lipids into smaller droplets (micelles). This increases the surface area of lipid substrate available for pancreatic lipase action.

Bile Salt Malabsorption

Condition characterized by impaired reabsorption of bile acids in the terminal ileum leading to excessive bile acids entering the colon where they stimulate water and electrolyte secretion causing chronic watery diarrhea.

Bile Salts

Bile salts are bile-derived detergent molecules that emulsify dietary fats and help absorb lipids and fat-soluble vitamins in the intestine.

Bilevel Positive Airway Pressure

Noninvasive ventilation that provides different pressures during inhalation and exhalation to support breathing.

Bilharzia

A parasitic infection caused by *Schistosoma* species. See Schistosomiasis.

Bilharzia (Schistosomiasis)

Alternate index label for Schistosomiasis.

Bili Lights

Phototherapy lights used to treat neonatal jaundice by converting bilirubin into forms that can be eliminated more easily.

Biliary

Biliary means relating to bile, the gallbladder, or bile ducts that transport bile from the liver to the intestine.

Biliary Atresia

A progressive, obliterative cholangiopathy of the extrahepatic bile ducts in neonates, leading to biliary obstruction, cholestasis, and ultimately biliary cirrhosis if untreated.

Alternate Names:

Congenital Biliary Atresia Extrahepatic Biliary Atresia

Biliary Calculi

Biliary calculi are gallstones or stones in the biliary tract formed from crystallized bile components and capable of causing pain, obstruction, or inflammation.

Biliary Cirrhosis

Chronic liver scarring associated with long-standing injury to the bile ducts. The term is often used for primary biliary cholangitis, an autoimmune small-duct disease.

Biliary Cirrhosis Primary (Primary

Biliary Cirrhosis)

Alternate index label for Primary Biliary Cirrhosis.

Biliary Colic

Episodic right-upper-quadrant or epigastric pain caused by transient obstruction of the cystic duct, usually by a gallstone. Pain is typically steady, may follow meals, and resolves when the obstruction passes; fever and persistent inflammatory findings suggest a complication.

Biliary Decompression

Drainage of obstructed bile ducts, using methods such as endoscopic stenting, nasobiliary drainage, or percutaneous catheter drainage, to restore bile flow.

Biliary Dyskinesia

A functional gallbladder disorder causing biliary-type pain without gallstones or another clear structural cause. A hepatobiliary scan may show abnormal gallbladder emptying, but no single ejection-fraction cutoff establishes the diagnosis; other causes must be considered.

Biliary Fever

Fever associated with infection or inflammation of the biliary tract, classically acute ascending cholangitis. It may occur with right-upper-quadrant pain and jaundice and can progress rapidly to sepsis.

Biliary Fistula

A biliary fistula is an abnormal connection allowing bile to leak from the biliary system into another organ, the abdomen, or the skin.

Biliary Sand

Fine particulate material or thick sludge in bile, usually composed of cholesterol crystals, calcium salts, and pigment, which can contribute to gallstones or duct obstruction.

Biliary Stent

A tube placed in a bile duct to maintain drainage when the duct is narrowed or blocked. Stents may

be plastic or metal, and complications include migration, blockage, infection, and pancreatitis.

Biliary System

The network of liver, gallbladder, and bile ducts that produces, stores, and transports bile to the intestine.

Biliary Tract

The network of ducts that carries bile from the liver and gallbladder to the small intestine. Blockage or inflammation can cause abdominal pain, jaundice, infection, or pancreatitis.

Biliary Tract Injuries

Damage to the gallbladder, bile ducts, or adjacent biliary structures, often occurring after abdominal surgery or trauma. It may cause bile leakage, obstruction, jaundice, infection, or later biliary narrowing.

Bilious

Containing or relating to bile; bilious vomiting is typically yellow-green because bile has entered the stomach contents.

Bilious Vomiting

Vomiting of green/yellow, bilious material in the newborn or infant, defined as emerald green or yellow vomitus that contains bile.

Biliousness

An older nonspecific term for nausea, indigestion, or vomiting thought to be associated with bile or liver and gallbladder dysfunction.

Bilirubin

A yellow pigment formed mainly from heme breakdown. Unconjugated bilirubin is transported bound to albumin to the liver, where it is conjugated with glucuronic acid and excreted in bile; accumulation causes jaundice.

Bilirubin (Bilirubin And Bilirubin Blood

Test)

Alternate index label for Bilirubin and Bilirubin Blood Test.

Bilirubin - Urine

A urine test for conjugated bilirubin. Its presence may indicate cholestasis or hepatocellular disease because unconjugated bilirubin is not normally excreted in urine.

Bilirubin And Bilirubin Blood Test

Bilirubin is a yellow pigment produced when red blood cells are broken down. A blood test measures total and sometimes direct and indirect bilirubin to help evaluate liver disease, bile-duct obstruction, hemolysis, and newborn jaundice.

Alternate Names:

Bilirubin (Bilirubin And Bilirubin Blood Test)

Bilirubin Blood Test

A laboratory test that measures total and direct (conjugated) bilirubin in serum, a breakdown product of heme catabolism. Elevated bilirubin causes jaundice and may indicate hemolysis, hepatocellular disease, or biliary obstruction; neonatal testing identifies risk for kernicterus.

Bilirubin Encephalopathy

Neurologic injury caused by deposition of unconjugated bilirubin in the brain, especially the basal ganglia. Severe neonatal hyperbilirubinemia can produce acute lethargy, abnormal tone, feeding difficulty, seizures, and permanent kernicterus.

Bilirubin Metabolism

Bilirubin is the breakdown product of heme from senescent red blood cells, metabolized primarily by the liver through a series of enzymatic steps. Heme is converted to biliverdin by heme oxygenase, then to unconjugated bilirubin by biliverdin reductase. Unconjugated bilirubin is lipophilic and transported bound to albumin to the liver.

Bilirubinuria

The presence of conjugated bilirubin in urine. It

indicates that water-soluble bilirubin has entered the circulation, often because of hepatocellular disease or biliary obstruction.

Biliverdin

Biliverdin is a green bile pigment formed during heme breakdown and normally converted to bilirubin by biliverdin reductase.

Billroth'S Operation

Billroth's operations are historical partial gastrectomy procedures, especially Billroth I and II, that reconnect the remaining stomach to the duodenum or jejunum.

Bilophila

A genus of anaerobic Gram-negative rod-shaped bacteria found in the human gastrointestinal tract. *Bilophila wadsworthia* is the primary species and has been associated with appendicitis, intra-abdominal infections, and bacteremia.

Bilophila Wadsworthia

A bile-tolerant anaerobic Gram-negative rod found in the human colon. It is a minor constituent of normal gut flora but has been implicated in appendicitis, peritonitis, and wound infections, particularly following bowel surgery or in immunocompromised patients.

Bimanual

Bimanual means performed with both hands, as in a bimanual pelvic examination or two-handed surgical technique.

Bimatoprost

Bimatoprost is a prostaglandin analogue used as eye drops to lower intraocular pressure and, in some formulations, promote eyelash growth.

Bimedial

Relating to or positioned on the medial side of two structures, or involving both medial aspects; the exact anatomical reference must be specified by the structure being described.

Bimellar

Bimellar means composed of two lamellae or layers, such as a double-layered membrane or tissue structure.

Binary Fission

Binary fission is asexual reproduction in which one cell or organism divides into two genetically similar descendants, common among bacteria and protozoa.

Binder Excipient

A binder excipient is an inactive pharmaceutical ingredient that helps powder particles adhere during tablet formation.

Binding Protein

Binding protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Binding Site

A specific region of a receptor, enzyme, antibody, nucleic acid, or other molecule to which another molecule binds through chemical and physical interactions, influencing biological activity or measurement.

Binet-Simon Scale

An early developmental assessment used to estimate intellectual functioning in children by comparing performance with age-based tasks. Modern evaluation uses standardized, validated developmental and cognitive instruments.

Binge Drinking

A pattern of consuming enough alcohol over a short period to produce a high blood alcohol concentration and increased risk of injury or acute toxicity.

Alternate Names:

Episodic heavy drinking; heavy episodic drinking.

Binge Eating Disorder

Recurrent episodes of eating large quantities of food

with a sense of lack of control, without regular compensatory behavior. Associated with obesity, depression, and anxiety. Diagnosis by DSM-5 criteria. Treatment: cognitive behavioral therapy, interpersonal therapy, and SSRIs. Weight management with dietary counseling and exercise.

Binge-Eating Disorder

A disorder marked by recurrent episodes of eating unusually large amounts of food with a sense of loss of control and without regular compensatory behaviors.

Alternate Names:

Compulsive Overeating Eating Disorders, Binge Eating

Binocular Vision

Vision that uses both eyes together to produce a single image and support depth perception.

Alternate Names:

Two-eyed vision; binocular visual function.

Binocular Visual Loss

Reduction in vision affecting both eyes simultaneously, typically resulting from bilateral optic neuropathy, chiasmal lesions, or diffuse bilateral retinal disease rather than monocular ocular pathology. Evaluation requires neuroimaging and visual field testing to localise the lesion along the visual pathway.

Binswanger Disease

A clinical disorder or finding related to Binswanger disease; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Binswanger's Dementia

A form of vascular cognitive impairment caused by diffuse small-vessel disease affecting the brain's deep white matter.

Alternate Names:

Subcortical ischemic vascular disease; subcortical arteriosclerotic encephalopathy.

Bio-Informatics

The use of computational methods to store, analyse, and interpret biological data such as DNA sequences, gene expression, protein structures, and clinical genomic information.

Bioabsorbable Stent (Vascular)

A vascular scaffold designed to provide temporary support and gradually break down after its function is completed.

Alternate Names:

Vascular

Bioaccumulation

Bioaccumulation is a biochemical reaction, pathway, or chemical process in which molecules are transformed; its rate and effects depend on substrates, enzymes, and cellular conditions.

Bioactive

Having an effect on living cells, tissues, or organisms, such as a bioactive drug, hormone, nutrient, or signaling molecule.

Bioassay

A test that measures the effect or activity of a substance by observing its action on living cells, tissues, organisms, or a biological system. It can be used to estimate potency, toxicity, or biologic activity.

Bioavailability

The fraction of an administered drug that reaches the systemic circulation in its unchanged active form. It is expressed as a percentage and is determined by the route of administration and the extent of first-pass metabolism. For intravenous administration, bioavailability is 100 percent by definition.

Bioburden

The number and type of viable microorganisms contaminating a surface, medical device, tissue, or pharmaceutical product; a measure of contamination. Bioburden testing assesses the level of microbial contamination (total viable count) before steriliza-

tion or in quality control.

Biochanin A

A naturally occurring isoflavone found mainly in legumes and other plants. It is studied for biologic effects but is not established as a treatment for disease.

Alternate Names:

Biochanin A isoflavone; 5,7-dihydroxy-4'-methoxyisoflavone.

Biochemical

Relating to the chemical substances and reactions that occur in living organisms, including metabolism and cellular signaling.

Biochemical Assessment

Blood tests: albumin, prealbumin, transferrin, electrolytes, vitamins, minerals, glucose, lipids. Reflect nutritional status and metabolic function.

Biochemical Reaction

Biochemical reaction is a biochemical reaction, pathway, or chemical process in which molecules are transformed; its rate and effects depend on substrates, enzymes, and cellular conditions.

Biochemical Recurrence

A rise in a disease-specific laboratory marker after treatment, suggesting recurrent or persistent disease before clinical or imaging findings become apparent.

Alternate Names:

Biochemical relapse; biochemical failure.

Biochemistry

The study of the chemical substances and reactions occurring in living organisms. Clinical biochemistry uses laboratory measurements to assess health and disease.

Biocompatibility

Ability of material to function without adverse response. ISO 10993 standards for biocompatibility testing.

Biodegradable Implant

An implant designed to perform a temporary function and then degrade into products that can be metabolized or eliminated by the body. Degradation rate, local tissue response, and mechanical strength must be matched to the intended clinical use.

Biodistribution

The time-dependent distribution of a radiopharmaceutical among organs, tissues, blood, and excretory pathways. It determines image quality, target-to-background contrast, therapeutic delivery, and radiation exposure to normal organs.

Bioengineering

The application of engineering principles to biology and medicine, including the design of medical devices, biomaterials, tissue-engineering systems, prostheses, and diagnostic technologies.

Bioequivalence

The property of two drug products (brand-name and generic) that have the same rate and extent of absorption when administered at the same molar dose under similar conditions, resulting in comparable bioavailability and therapeutic effect.

Bioethics

Interdisciplinary study of the ethical, legal, philosophical, and social implications of advancements in medicine, biology, healthcare technology, and biotechnology.

Bioethics (Geriatric)

Ethical considerations in elderly care: autonomy vs. paternalism, informed consent capacity, end-of-life decisions, resource allocation, elder abuse. Principles: respect for autonomy, beneficence, non-maleficence, justice. Advance directives: living will, durable power of attorney for healthcare.

Alternate Names:

Geriatric

Biofeedback

A technique that provides information about a phys-

iological process, such as muscle activity, heart rate, or breathing, to help a person learn voluntary control of it.

Biofilm

An organized community of microorganisms attached to a surface and embedded in a self-produced extracellular polymeric matrix (polysaccharides, proteins, DNA).

Biofilm-Associated Infections

Infections caused by bacteria growing in biofilms on medical devices or tissues. Biofilms are structured communities of bacteria embedded in extracellular polymeric substances. Bacteria in biofilms are more resistant to antibiotics and host immune responses.

Biofluid

A fluid produced by or present in a living organism, such as blood, urine, saliva, cerebrospinal fluid, or synovial fluid.

Biologic Therapy

Treatment with biologic agents targeting specific immune pathways, used in autoimmune and inflammatory disease.

Biologic Therapy In Dermatology

Treatment with targeted biological agents, usually monoclonal antibodies or receptor proteins, that inhibit selected immune pathways. Biologics are used for diseases such as psoriasis, atopic dermatitis, hidradenitis suppurativa, and urticaria; infection risk, vaccination status, screening, and monitoring must be considered.

Biologic Variation

Natural fluctuation in a measurable analyte within an individual over time and variation between individuals. It should be considered when interpreting serial results and deciding whether a change is clinically meaningful.

Biological

Relating to living organisms, their structures, func-

tions, processes, or products; in medicine, a biological product is a therapeutic or diagnostic substance made from or resembling a biological source.

Biological Availability

Biological availability is a pharmacologic, physiologic, or clinical-care concept used to describe a measurable effect, intervention, or patient-management issue.

Biological Clock

An internal timing system that coordinates recurring biological rhythms, especially the circadian rhythm, through networks involving the suprachiasmatic nucleus, hormones, and environmental cues.

Biological Half-Life

Time required for the human body to eliminate half the administered quantity of a drug or radiopharmaceutical through metabolic and excretory processes. Different from physical half-life (radioactive decay) and effective half-life (combined biological and physical elimination). Formula: $1/T_{\text{effective}} = 1/T_{\text{physical}} + 1/T_{\text{biological}}$.

Biological Model

A cell, tissue, organism, computational system, or other representation used to study biological mechanisms, disease, or treatment responses.

Biological Process

A coordinated series of molecular, cellular, or physiological events in a living organism, such as metabolism, cell division, inflammation, or tissue repair.

Biological Relative

A person related by shared genetic ancestry, in contrast to a relative related solely by marriage or legal adoption; family history uses this distinction when assessing inherited risk.

Biological Therapy

Treatment using biological agents, including monoclonal antibodies, cytokines, vaccines, and cell-based

therapies, to modify immune responses or target specific molecular pathways in cancer, autoimmune diseases, and other conditions.

Biological Transport

The movement of ions, molecules, cells, or fluids within or across biological membranes and tissues, by processes such as diffusion, filtration, channels, pumps, and vesicular transport.

Biological Variability

Natural variation in a measured biologic quantity within the same person over time or between people, independent of measurement error.

Alternate Names:

Biologic variation; physiologic variability.

Biological Warfare

The deliberate use of microorganisms, toxins, or other biological agents to cause disease or death in humans, animals, or plants.

Bioluminescence

Light produced by a living organism through a chemical reaction, commonly involving luciferin and luciferase; it is used in biological research and diagnostic assays.

Biomarker

Measurable indicator of disease. Cancer biomarkers: diagnostic, prognostic, predictive.

Biomarker Testing

Laboratory analysis of biological features, such as genes, proteins, or metabolites, that may help diagnose disease, estimate prognosis, or guide treatment.

Biomaterial

A natural or synthetic material designed to interact with a biological system, for example in implants, prostheses, drug delivery, wound dressings, or tissue engineering.

Biomechanical Compliance

The change in volume or displacement produced by

a given change in pressure or force in a biological structure; examples include lung, vascular, and joint compliance.

Biomechanics

Study of movement mechanics. Gait analysis, joint forces, muscle activation. Applied to rehabilitation device design.

Biomedical Cement

Biomedical cement is a material used to fill, stabilize, anchor, or repair bone or other tissues in medical procedures.

Biomedical Engineering

Biomedical engineering applies engineering, materials science, computing, and design to medical devices, diagnostics, prostheses, imaging, and health systems.

Biomedical Research

Scientific investigation aimed at understanding disease mechanisms, developing diagnostic tools, and creating new treatments through laboratory experiments, animal models, and clinical trials.

Biomedical Tape

Biomedical tape is adhesive material used to secure dressings, devices, sensors, or skin edges; its adhesive and backing determine skin tolerance.

Biomedicine

Medicine grounded in the biological sciences, including the study of disease mechanisms, diagnosis, prevention, and treatment through laboratory and clinical research.

Biometry

The measurement of biological structures or functions. In eye care, optical biometry measures the eye's dimensions to help calculate the power of an intraocular lens before cataract surgery.

Biomicroscopy

Biomicroscopy is microscopic examination of liv-

ing tissue, especially slit-lamp examination of the eye's cornea, anterior chamber, lens, and related structures.

Biomimetics

The design of materials, devices, or processes that imitate biological structures or functions, such as tissue-like scaffolds or gecko-inspired adhesives.

Biomolecule

Any organic molecule produced by living organisms, essential for structure, function, and regulation of biological processes.

Biomonitoring

Measurement of substances in body. Examples: blood glucose, drug levels, toxicants.

Biomphalaria

A genus of freshwater snails that serves as an intermediate host for *Schistosoma mansoni*, the parasite causing intestinal schistosomiasis in humans.

Bionics

Bionics uses electronic or mechanical systems to reproduce or augment biological functions, such as hearing, movement, or sensory feedback.

Biophysical Profile

A non-stress test combined with ultrasound assessment of five fetal parameters, each scored 0 or 2 for a total of 0-10. Parameters include non-stress test (fetal heart rate reactivity), fetal breathing movements, fetal tone, fetal movement, and amniotic fluid volume.

Biopolymer

A polymer produced by living organisms, such as proteins, nucleic acids, polysaccharides, and natural polyesters; biopolymers provide biological structure and function and are also used in biomaterials.

Biopsy

Removal of cells or tissue for examination by a pathologist to diagnose or characterize disease.

Common approaches include fine-needle aspiration, core-needle, incisional, excisional, punch, and endoscopic biopsy. Depending on the site and technique, complications may include pain, bleeding, infection, or—infrequently—tumor-tract seeding.

Alternate Names:

Biopsies

Biopsy - Biliary Tract

Sampling of a bile-duct or gallbladder lesion, usually endoscopically with forceps, brushings, or needle guidance, for cytology or histopathology. It may evaluate suspected malignancy or selected inflammatory and infectious disorders.

Biopsy Lung

Removal of lung tissue by needle, bronchoscopy, or surgery for evaluation of infection, inflammation, interstitial disease, or cancer. The route is selected according to lesion location, diagnostic yield, and procedural risk.

Biopsy Needle

A needle designed to obtain cells or tissue for cytologic or histopathologic examination. Fine-needle aspiration retrieves cells, whereas core biopsy obtains a tissue cylinder that preserves architecture.

Biopsy Site

The body location from which tissue or cells are removed for biopsy. The site is selected to sample the abnormal area while minimizing injury to nearby structures.

Biopsy Specimen

Tissue or cells removed from the body for examination. A laboratory may evaluate the specimen with microscopy, special stains, immunostains, or molecular tests to identify disease.

Bioreactor

A controlled vessel or system in which cells, tissues, microorganisms, or enzymes are cultured or used to produce biological materials, medicines, or engineered tissues.

Biorhythm

A recurring biological pattern, such as the circadian sleep-wake cycle, hormonal rhythms, or seasonal changes. Popular fixed-cycle theories claiming that biorhythms can predict health, mood, or performance are not supported by scientific evidence.

Alternate Names:

Biorhythms

Biosafety

The practices, equipment, and containment procedures used to prevent accidental exposure to or release of biological agents and toxins.

Biosensor

A device that combines a biological recognition element with a physical or chemical transducer to detect and measure an analyte. Medical examples include sensors for glucose, pathogens, hormones, and cardiac biomarkers.

Biosimilar

A biologic medicine highly similar to an approved reference biologic, with no clinically meaningful differences in safety, purity, or potency for its approved uses. Because biologics are complex, biosimilarity is established through a comparative evidence program rather than simple chemical identity.

Biospecimen

A sample of human tissue, cells, blood, urine, or another body fluid collected for clinical testing or research. Proper collection, labeling, storage, and handling help preserve its usefulness.

Biostatistics

Biostatistics applies statistical methods to biomedical data, including study design, estimation, hypothesis testing, epidemiology, and interpretation of clinical evidence.

Biosynthesis

Biosynthesis is the enzyme-directed production of complex molecules by living cells from simpler precursors.

Biot Breathing

An irregular breathing pattern with random periods of normal breathing interspersed with apneic episodes, seen in brainstem lesions. It is less organized than Cheyne-Stokes breathing and indicates medullary damage.

Biot's Breathing

Biot's breathing is an abnormal pattern of irregular breaths interrupted by unpredictable periods of apnea, associated with disruption of respiratory control.

Biotechnology

The use of living cells, organisms, or biological molecules to develop products and processes, including vaccines, recombinant medicines, molecular diagnostics, and gene therapies.

Biotelemetry

The remote measurement and transmission of physiologic data from a person or animal. Clinical applications include ambulatory cardiac rhythm monitoring, temperature monitoring, and intensive-care observation.

Bioterrorism

Bioterrorism is the deliberate release of biological agents to cause illness, death, or disruption. Preparedness includes surveillance, laboratory identification, public-health communication, vaccination or prophylaxis when indicated, and rapid isolation or treatment of exposed people.

Bioterrorism Agents

Category A: anthrax (*Bacillus anthracis*), smallpox (*variola*), botulism (*Clostridium botulinum*), plague (*Yersinia pestis*), tularemia (*Francisella tularensis*), viral hemorrhagic fevers (Ebola, Marburg, Lassa, Machupo). Anthrax: inhalational (mediastinal widening, hemorrhagic mediastinitis), cutaneous (eschar), GI.

Biotherapy

Treatment that uses living organisms, cells, genes, or biological substances to prevent or treat disease;

immunotherapy is one form of biotherapy.

Biotin

A water-soluble B vitamin that acts as a cofactor for carboxylase enzymes involved in fatty-acid synthesis, gluconeogenesis, and amino-acid metabolism; deficiency is uncommon but can cause dermatitis, alopecia, and neurological symptoms.

Biotinidase Deficiency

An autosomal recessive inborn error of biotin recycling caused by BTB gene mutations, resulting in inability to cleave biotin from biocytin (proteolytic degradation product of biotin-dependent carboxylases).

Biotinylation

The attachment of biotin to another molecule, usually a protein, so it can be found or separated in a laboratory test. The biotin acts as a tag that binds to special substances, making the target easier to detect or collect. Biotinylation can occur naturally or be done artificially.

Biotransformation

The natural process by which the body changes drugs and other foreign substances, mainly through enzymes in the liver but also in other tissues. It often makes them less active and easier to remove. Phase I reactions modify the substance; phase II reactions may attach another molecule to help eliminate it. These steps are not always sequential, and the process can produce an active or harmful substance instead.

Biotrauma

Inflammatory response to ventilator-induced lung injury with systemic cytokine release. Can lead to multi-organ dysfunction. Minimized by lung-protective ventilation strategies.

BiPAP

Bilevel Positive Airway Pressure (BiPAP), a form of noninvasive breathing support delivered through a mask. It provides higher pressure during inhalation and lower pressure during exhalation. It may

be used for conditions such as hypercapnic COPD exacerbation, sleep-related breathing disorders, and cardiogenic pulmonary edema.

Alternate Names:

Bilevel positive airway pressure

BiPAP Machine

The device that delivers BiPAP treatment through a mask by providing separate inhalation and exhalation pressures. The machine's settings are prescribed and adjusted according to the patient's condition and response.

Biparental

Involving both biological parents. In genetics, biparental inheritance means that offspring receive genetic material for a trait from both parents, unlike maternal or paternal inheritance in which the relevant material comes mainly from one parent.

Biparietal Diameter

Biparietal diameter is the width of a baby's head measured from one side to the other during an ultrasound. It helps estimate the length of the pregnancy and check the baby's growth. Doctors use it with other measurements because head shape can affect the result.

Bipartite Ossification

Formation or persistence of two separate ossification centers in a bone. If they do not fuse, a bipartite bone, such as a bipartite patella, may result. This is usually an incidental developmental variant but can become painful after injury or repetitive stress.

Biphenyl Compounds

Chemicals made of two linked benzene rings. Polychlorinated biphenyls (PCBs) are long-lasting industrial pollutants; exposure can harm the liver, immune system, and hormone system and may increase cancer risk.

Bisphosphonates

Medicines that slow the breakdown of bone. They are used mainly to prevent fractures in osteoporosis and to treat Paget disease and some bone problems

caused by cancer.

Alternate Names:

Bisphosphonate

Bipolar

Alternate index label; see *Bipolar disorder*.

Bipolar Affective Disorder

Alternate name for bipolar disorder; see *Bipolar disorder*.

Bipolar Disorder

A mental health condition with episodes of mania or hypomania and often depression. Bipolar I includes at least one manic episode; bipolar II includes hypomania and major depression but no manic episode. Episodes can affect mood, sleep, energy, judgment, and daily functioning.

Alternate Names:

Manic-Depressive Illness

Bipolar I Disorder

A mood disorder defined by at least one manic episode lasting at least seven days or requiring hospitalization. A manic episode is a distinct period of abnormally elevated, expansive, or irritable mood and increased goal-directed activity or energy.

Bipolar II Disorder

A mood disorder characterized by at least one hypomanic episode and at least one major depressive episode, with no history of a full manic episode. The depressive episodes in bipolar II tend to be more frequent and longer-lasting than in bipolar I, leading to greater cumulative disability.

Bipolar Neuron

A bipolar neuron has one dendrite and one axon extending from opposite sides of its cell body, as found in the retina and sensory pathways.

Bipolaris

A biomedical term referring to bipolaris; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Bird Fancier'S Lung

A form of hypersensitivity pneumonitis caused by breathing proteins from bird feathers or droppings. It can cause cough and shortness of breath; repeated exposure may lead to permanent lung scarring.

Bird Flu

Infection with avian influenza A viruses that primarily circulate in birds. People rarely become infected, usually after close, unprotected exposure to infected animals or contaminated environments. Illness may range from eye or upper-respiratory symptoms to severe pneumonia; evaluation depends on exposure history and public-health testing.

Bird Flu (Avian Influenza)

Influenza caused by avian influenza viruses, which primarily infect birds but can occasionally infect humans and cause severe respiratory illness.

Alternate Names:

Influenza, Avian

Birt-Hogg-Dubé Syndrome

An inherited condition caused by a change in the FLCN gene. It can cause small, usually harmless skin growths, cysts in the lungs with repeated collapsed lung, and an increased risk of kidney tumors. It is autosomal dominant, meaning one altered copy of the gene can cause the condition.

Birth

Birth is delivery of a fetus from the uterus, involving labor, delivery of the infant, and separation or delivery of the placenta.

Birth Asphyxia

Inadequate oxygenation or ventilation around the time of birth, potentially causing respiratory failure, metabolic acidosis, organ injury, or hypoxic-ischaemic encephalopathy. It requires immediate neonatal assessment and resuscitation.

Birth Control

Methods used to prevent pregnancy, including barrier, hormonal, intrauterine, behavioral, and perma-

nent methods.

Birth Control - Slow Release Methods

Information about contraceptive methods, including effectiveness, use, adverse effects, contraindications, and reproductive planning.

Birth Control And Family Planning

Information about contraceptive methods, including effectiveness, use, adverse effects, contraindications, and reproductive planning.

Birth Control Methods

Birth-control methods include condoms, pills, implants, injections, intrauterine devices, patches, rings, fertility-awareness approaches, and permanent contraception. Effectiveness, sexually transmitted infection protection, side effects, reversibility, and medical eligibility vary by method.

Birth Control Pill Overdose

Toxic effects from ingesting an excessive amount of oral contraceptive medication. Most exposures cause nausea, vomiting, or drowsiness, but the product, dose, co-ingestants, and symptoms determine the need for assessment.

Birth Control Pills

Information about contraceptive methods, including effectiveness, use, adverse effects, contraindications, and reproductive planning.

Birth Control Pills - Combination

Information about contraceptive methods, including effectiveness, use, adverse effects, contraindications, and reproductive planning.

Birth Control Pills - Progestin Only

Information about contraceptive methods, including effectiveness, use, adverse effects, contraindications, and reproductive planning.

Birth Defect

A structural, functional, or biochemical abnormality present at birth, caused by genetic, environmental,

developmental, or often unknown factors.

Birth Defect (Congenital Anomaly)

A structural, functional, or metabolic abnormality present at birth, caused by genetic, environmental, nutritional, infectious, or multifactorial factors.

Alternate Names:

Congenital Anomaly

Birth Injury

A clinical disorder or finding related to birth injury; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Birth Order

The sequence in which siblings are born within a family. While birth order has been studied for psychological effects, its direct medical significance is limited to obstetric considerations such as risks associated with grand multiparity.

Birth Pains

Birth pains are painful uterine contractions and related pressure during labor as the cervix dilates and the baby moves through the birth canal.

Birth Rate

The number of live births in a population during a specified period, usually expressed per 1,000 people per year.

Birth Weight

The weight of a newborn measured within the first hour of life. Low birth weight (less than 2,500 g) is associated with increased risk of neonatal morbidity and mortality; very low birth weight (less than 1,500 g) identifies high-risk infants requiring specialized care.

Birthmark

A localized skin or vascular mark present at birth or appearing soon afterward, such as a nevus, hemangioma, or port-wine stain.

Birthmarks

Congenital or early-appearing marks on the skin, including vascular lesions such as hemangiomas and port-wine stains and pigmented lesions such as nevi and café-au-lait macules. Most are benign, but changes in appearance warrant assessment.

Birthmarks - Pigmented

Pigmented birthmarks are localized areas of increased or altered skin pigmentation present at birth or appearing early in life. Changes in size, color, or shape require assessment.

Birthmarks And Other Skin Pigmentation Problems

Birthmarks and pigment changes may reflect vascular lesions, melanocytic cells, post-inflammatory change, medication, hormonal conditions, or genetic disease. A lesion that changes rapidly, bleeds, ulcerates, or differs from others should be examined.

Bis In Die

Latin for twice a day, traditionally abbreviated b.i.d.; medication directions should state the intended dosing times clearly.

Bisacodyl

Bisacodyl is a stimulant laxative that increases bowel movement and is used for short-term constipation or bowel preparation.

Bisbenzimidazole

A class of fluorescent compounds that bind DNA and are used mainly as laboratory stains, including Hoechst dyes.

Bisexual

Bisexual describes sexual or romantic attraction to more than one gender; it is an identity or orientation, not a disease or disorder.

Bisexuality

Bisexuality is a sexual orientation involving attraction to more than one gender; it is not a disease or disorder.

Bishop Score

A clinical score used before labor induction to describe cervical readiness. It incorporates cervical dilation, effacement, consistency, position, and fetal-head station; a more favorable cervix generally indicates a greater likelihood of successful induction.

Bismuth

Bismuth is a metal used in some medicines and industrial products; excessive exposure or certain compounds can cause toxicity.

Bisoprolol

A beta-1-selective blocker used to treat hypertension, angina, and selected patients with heart failure. It lowers heart rate and blood pressure and may cause bradycardia, hypotension, or fatigue.

Bisphenol A

Endocrine disruptor in plastics. Estrogenic effects, associated with reproductive disorders, obesity, diabetes. Avoid BPA-containing products, especially for children.

Bisphosphonate

A medicine that reduces bone breakdown and is used to prevent or treat osteoporosis and some other bone disorders.

Bistoury

A bistoury is a small surgical knife, historically used for precise incision or opening of tissue.

Bite Chigger (Chiggers (Bites))

Alternate index label for Chiggers (Bites).

Bite Dog (Dog Bite Treatment)

Alternate index label for Dog Bite Treatment.

Bite Snake (Snake Bite)

Alternate index label for Snake Bite.

Bitemporal Hemianopia

Loss of vision in the temporal (lateral) half of both visual fields due to lesions that cross the optic chi-

asm, damaging the decussating fibers from the nasal retina of each eye (which carry temporal field information).

Bites Flea (Flea Bites In Humans)

Alternate index label for Flea Bites in Humans.

Bitewing

A dental radiograph that records the crowns of the upper and lower teeth and the adjacent alveolar bone in one image. It is particularly useful for detecting interproximal caries and assessing crestal bone levels.

Bitewing Radiograph

An intraoral radiograph showing the crowns of the maxillary and mandibular posterior teeth and the alveolar crest. It is particularly useful for detecting interproximal caries and assessing crestal bone levels.

Bitot'S Spots

Foamy, whitish, triangular patches on the conjunctiva, usually near the temporal limbus, associated classically with vitamin A deficiency. They may accompany night blindness and xerosis of the conjunctiva or cornea.

Bitters

Bitters are intensely flavored plant preparations traditionally used as digestive remedies; evidence, composition, interactions, and safety vary by product.

Bituminosis

Bituminosis is an occupational lung condition caused by inhalation of bitumen or asphalt fumes and dust, potentially producing airway irritation or chronic respiratory disease.

Biuret Test

The biuret test detects peptide bonds by producing a violet color with proteins in alkaline copper solution; it is a laboratory chemistry test.

Bivalve

Bivalve means having two shells or valves; bivalve mollusks may be relevant to food allergy, poisoning, or parasitic exposure.

Bizarre Parosteal Osteochondromatous Proliferation

A rare benign but locally recurrent bone-surface lesion, also called Nora Lesion, composed of cartilage, bone, and spindle cells; it most often affects the hands or feet.

Bizelesin

Bizelesin is an investigational DNA-alkylating anti-neoplastic compound studied for cytotoxic effects.

BK Virus

BK virus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

BK Virus Nephropathy

Polyomavirus infection in transplant recipients. Viral cytopathic changes on biopsy. Reduce immunosuppression as primary treatment. Can lead to graft loss. Monitor viral loads in blood for early detection.

Below-Knee Amputation

Below-knee amputation is surgical removal of the lower leg while preserving the knee joint.

Black Death

The Black Death was the historic pandemic of plague, primarily caused by *Yersinia pestis*, with severe mortality in the fourteenth century.

Black Eye

Bruising and swelling around the eye, usually caused by blunt trauma. Vision may remain normal, but eye pain, double vision, reduced vision, an irregular pupil, severe headache, vomiting, or loss of consciousness can indicate an eye or orbital injury and requires urgent assessment.

Alternate Names:

Eye Black (Black Eye)

Black Fly

A black fly is a biting Simuliidae insect that can cause local reactions and, in endemic regions, transmit *Onchocerca volvulus*.

Black Hair, Hairy Tongue

Black hairy tongue is benign elongation and discoloration of the tongue's filiform papillae, associated with smoking, antibiotics, poor oral hygiene, and some foods or medicines.

Black Hairy Tongue

A benign elongation and dark discoloration of the filiform papillae on the dorsal tongue caused by impaired desquamation and trapping of debris or pigment. Predisposing factors include smoking, poor oral hygiene, dry mouth, and some exposures.

Black Lung

Coal workers' pneumoconiosis caused by prolonged inhalation of coal-mine dust. It ranges from simple anthracosis to progressive massive fibrosis and may cause cough, breathlessness, and permanent lung impairment.

Black Lung Disease

Black lung disease is coal workers' pneumoconiosis caused by inhaled coal dust, ranging from simple nodules to progressive massive fibrosis and respiratory impairment.

Black Nightshade Poisoning

Poisoning from eating black nightshade or its berries, which contain toxic glycoalkaloids. Symptoms may include nausea, vomiting, abdominal pain, confusion, dilated pupils, seizures, or abnormal heart rhythm.

Black Or Tarry Stools

Black or tarry stools may indicate upper gastrointestinal bleeding, but can also result from iron or bismuth. New or persistent findings require medical

assessment.

Black Stain

Dark discoloration on the tooth surface, often near the gumline, caused by external deposits or chromogenic bacteria. Dental assessment determines the cause and appropriate removal.

Black Triangle

Triangular space between teeth at gumline. Caused by papilla loss, tooth shape, bone level. Treatment: orthodontics, restorative, papilla reconstruction.

Black Widow Spider

The black widow spider is a venomous spider whose bite can cause severe muscle pain, abdominal cramping, sweating, hypertension, and autonomic symptoms.

Blackflesh

Blackflesh is a descriptive term for dark or blackened tissue, potentially reflecting bruising, necrosis, gangrene, pigmentation, or contamination; clinical examination determines the cause.

Blackhead

A blackhead is an open comedone in which sebum and keratin plug a hair follicle and darken at the surface through oxidation.

Blackheads

Open comedones formed when a hair follicle becomes blocked with sebum and keratin and the surface material oxidizes. They are a common non-inflammatory lesion of acne, especially on the face, chest, and back.

Blackout

A period of partial or complete memory loss, commonly associated with alcohol or other causes of impaired consciousness.

Alternate Names:

Amnestic episode; alcohol-related amnesia when caused by alcohol.

Blackwater Fever

A severe form of malaria, classically associated with *Plasmodium falciparum*, characterized by intravascular hemolysis, hemoglobinuria, anemia, and possible kidney failure.

Bladder

A hollow muscular organ that stores urine received from the kidneys and expels it through the urethra during urination.

Bladder Biopsy

Removal of bladder-mucosa tissue, usually during cystoscopy, for histopathologic evaluation of a visible lesion, unexplained hematuria, inflammation, or suspected urothelial cancer.

Bladder Calculi

Alternate index label; see *Bladder stones*.

Bladder Cancer

A malignant tumor arising in the urinary bladder, most often from the urothelial lining. Blood in the urine is a common presenting feature, although symptoms vary.

Alternate Names:

Bladder Carcinoma (Bladder Cancer)

Bladder Carcinoma

A malignant tumor arising in the urinary bladder, most often from the urothelial lining. Blood in the urine is a common presenting sign; smoking and certain occupational chemical exposures increase risk.

Bladder Carcinoma (Bladder Cancer)

Alternate index label for Bladder Cancer.

Bladder Control, Loss Of

Alternate index label; see *Urinary incontinence*.

Bladder Diverticula

Outpouchings of the bladder mucosa through the muscular wall, often acquired from chronic bladder outlet obstruction or congenital in origin. Acquired

diverticula are associated with benign prostatic hyperplasia and may cause urinary stasis, recurrent infections, and stone formation.

Bladder Diverticulum

Outpouching of bladder wall. Congenital or acquired (obstruction). Asymptomatic or causes urinary stasis, stones, infection. Treatment: address obstruction, diverticulectomy if symptomatic.

Bladder Exstrophy

A rare congenital abdominal wall defect where the bladder is exposed on the outside of the body due to failure of the lower abdominal wall and bladder to close during embryonic development. Part of the exstrophy-epispadias complex. Associated with pubic bone diastasis, external genital anomalies, and potential bladder dysfunction.

Bladder Exstrophy Repair

Bladder exstrophy repair reconstructs the exposed bladder and lower abdominal and genital structures present in this congenital defect, often through staged surgery.

Bladder Fistula

An abnormal connection between the bladder and another organ or the skin, most often the vagina, bowel, or uterus. It may cause continuous urine leakage, recurrent infections, gas or stool in the urine, or unusual vaginal wetness. Treatment depends on the cause and may require catheter drainage and surgical repair.

Bladder Health And Disorders

Bladder health involves the proper storage and voluntary release of urine through coordinated function of the detrusor muscle and sphincters. Common disorders include urinary tract infections, overactive bladder, stress incontinence, and interstitial cystitis.

Bladder Infection

Alternate index label; see *Cystitis*.

Bladder Infection (Urinary Tract

Infection)

Alternate index label for Urinary Tract Infection.

Bladder Inflammation

Alternate index label; see *Interstitial cystitis*.

Bladder Neck

The narrow lower part of the urinary bladder where it connects to the urethra and contributes to urinary continence.

Alternate Names:

Vesical neck; bladder outlet.

Bladder Neoplasm

A neoplastic process affecting bladder neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Bladder Outlet Obstruction

Blockage of urine flow at the bladder neck or urethra, commonly caused by benign prostatic enlargement, urethral stricture, stones, tumors, or pelvic-organ prolapse. It may cause weak stream, hesitancy, retention, hydronephrosis, infection, and renal impairment.

Bladder Outlet Obstruction

Blockage or narrowing at the bladder neck or urethra that restricts urine flow and can cause hesitancy, weak stream, retention, hydronephrosis, or kidney injury.

Bladder Pain

Pain perceived in the lower abdomen, pelvis, or bladder region, often related to urinary infection, bladder pain syndrome, stones, inflammation, or malignancy; assessment depends on urinary and systemic symptoms.

Bladder Pain Syndrome

Chronic bladder-related pain, pressure, or discomfort associated with urinary symptoms in the absence of infection or another identifiable cause. It is a diagnosis of exclusion and is managed with multimodal, patient-centered treatment.

Bladder Perforation

A tear or hole in the bladder wall caused by trauma, pelvic surgery, instrumentation, severe inflammation, or obstruction. Urine may leak into surrounding tissues, causing abdominal pain, distention, blood in the urine, or infection. It requires urgent medical assessment and may need catheter drainage or surgical repair.

Bladder Problems

A range of conditions affecting urine storage and voiding, including incontinence, overactive bladder, urinary retention, and recurrent infections. Causes may include neurological damage, pelvic floor weakness, prostate enlargement, or bladder outlet obstruction.

Bladder Prolapse

Alternate index label; see *Anterior vaginal prolapse (cystocele)*.

Bladder Retraining

Scheduled voiding and gradual increases in the time between voids to improve bladder control.

Bladder Spasm

An involuntary contraction of the bladder muscle that can cause sudden urgency, cramping, urine leakage, or pain. It may occur with infection, catheter use, bladder irritation, neurological conditions, or after surgery. Treatment addresses the cause and may include bladder training or medicines.

Bladder Stone

A mineral deposit formed in or entering the urinary bladder. It may cause painful urination, blood in the urine, interrupted flow, or recurrent infection; treatment addresses the stone and its cause.

Bladder Stones

Hard mineral deposits that form in the bladder and may cause painful urination, blood in urine, urinary frequency, or difficulty emptying the bladder.

Alternate Names:

Bladder Calculi

Bladder, Diseases Of

Diseases of the bladder include infection, inflammation, stones, obstruction, neurogenic dysfunction, bleeding, and tumors, often causing urinary or pelvic symptoms.

Bladder, Urinary

The urinary bladder is a muscular reservoir that stores urine and contracts to expel it through the urethra.

Blalock-Taussig Operation

A surgical systemic-to-pulmonary shunt that increases pulmonary blood flow in selected cyanotic congenital heart disease. The modern modified procedure commonly uses a prosthetic conduit rather than direct sacrifice of a subclavian artery.

Bland Diet

A bland diet consists of foods that are soft, mildly seasoned, and low in irritants. It may be used temporarily for selected gastrointestinal symptoms.

Blaschko Lines

Invisible developmental skin-cell lines that may become visible when certain congenital or acquired dermatoses follow their characteristic linear or whorled patterns. They do not usually correspond to nerves or blood vessels.

BLAST

An immature blood-forming cell, also called a blast; an increased blast percentage in blood or bone marrow can indicate an acute leukemia.

Blast Cell

An immature precursor cell, especially an immature blood-forming cell in bone marrow. An abnormally high proportion of blasts may indicate an acute leukemia or another marrow disorder.

Blast Crisis

Aggressive, terminal phase of myeloid (chiefly CML) or, less commonly, lymphoid malignancies in which immature blasts dominate the marrow and periph-

eral blood, behaving like acute leukemia. CML blast crisis is defined by 20% blasts in peripheral blood or bone marrow or by the presence of extramedullary blastic infiltrates (chloromas).

Blast Injury

Tissue injury caused by an explosion, classified as primary injury from the pressure wave, secondary injury from projectiles, tertiary injury from displacement or impact, and quaternary injury from burns, inhalation, toxins, or crush. Multiple mechanisms may coexist.

Blast Phase

Blast phase is an advanced stage of a myeloproliferative blood cancer marked by a high proportion of immature blast cells and aggressive disease.

Blastic Plasmacytoid Dendritic-Cell Neoplasm

A rare, aggressive hematologic malignancy derived from plasmacytoid dendritic cells. It often involves skin, bone marrow, lymph nodes, and central nervous system, with a poor prognosis.

Blastocyst

An early stage in embryonic development, consisting of a hollow sphere of cells that implants into the uterine wall.

Blastocyst Transfer

An assisted reproductive technology procedure in which embryos cultured in vitro to day 5 or day 6 (blastocyst stage, characterized by an inner cell mass and an outer trophoctoderm surrounding a fluid-filled blastocoel) are transferred into the uterine cavity. Offers physiological synchronization with the receptive endometrium and higher implantation rates compared to cleavage-stage transfers, facilitating elective single-embryo transfer (eSET) to minimize multiple gestations.

Blastocystis

A biomedical term referring to blastocystis; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Blastocystis Hominis

A microscopic organism that can inhabit the intestinal tract; its role in causing gastrointestinal symptoms is variable and sometimes uncertain.

Alternate Names:

Blastocystosis

Blastocystosis

Alternate index label; see *Blastocystis hominis*.

Blastoma

Generic descriptor for embryonal (primitive) malignant tumors arising from precursor ("blast") cells that fail to differentiate normally.

Blastomyces Dermatitidis

A dimorphic fungus that causes blastomycosis, a systemic pyogranulomatous infection acquired by inhalation of conidia from soil in endemic regions such as the Ohio and Mississippi River valleys. Cutaneous manifestations include verrucous or ulcerated plaques, and pulmonary involvement is common.

Blastomycosis

A fungal infection caused by *Blastomyces dermatitidis*, endemic to the Ohio and Mississippi River valleys and Great Lakes region. Inhalation of conidia causes pulmonary infection. Pulmonary: acute pneumonia (often misdiagnosed as bacterial), chronic pneumonia with nodules and cavitation.

Blasts

Immature precursor cells in the blood-forming tissues; excess blasts may be seen in acute leukemia or other marrow disorders.

Blautia Producta

Blautia producta is an anaerobic bacterium found in the human gut; its clinical significance depends on the specimen and compatible illness.

Bleb

A thin-walled, blister-like structure, generally larger than 5 mm and containing fluid or air; a fluid-filled bleb is also called a bulla.

Bleeder'S Disease

Bleeder's disease is an older name for hemophilia, an inherited clotting-factor deficiency causing prolonged or recurrent bleeding.

Bleeding

The escape of blood from the vascular system, which may be external (visible hemorrhage from a wound, orifice, or natural opening) or internal (bleeding into a body cavity, organ, or tissue).

Postmenopausal Bleeding

Vaginal bleeding after menopause. Even light or one-time bleeding needs medical evaluation because causes include tissue thinning, polyps, hormone effects, and cancer.

Bleeding Disorders

A heterogeneous group of conditions characterized by abnormal hemostasis, resulting from defects in blood vessels, platelets, or coagulation factors.

Bleeding During Cancer Treatment

A clinical topic about bleeding during cancer treatment. It includes the purpose, common findings, risks, and appropriate clinical follow-up.

Bleeding During Pregnancy

Vaginal bleeding during pregnancy; its significance varies with gestational age and amount. Urgent evaluation is needed for heavy bleeding, pain, dizziness, or bleeding later in pregnancy.

Bleeding Esophageal Varices

Life-threatening hemorrhage from dilated submucosal veins in the lower esophagus, occurring as a complication of portal hypertension, most commonly from cirrhosis. Varices form when the portal venous pressure gradient exceeds 10 mmHg (the threshold for variceal development).

Bleeding Gums

Bleeding from the gum tissue, often caused by plaque-related gingivitis, periodontitis, trauma, or vigorous brushing. It can also occur with blood

disorders or medicines that affect clotting. Persistent bleeding, swelling, loose teeth, or spontaneous bleeding requires dental or medical evaluation.

Bleeding Into A Thyroid Nodule

Hemorrhage within a thyroid nodule, presenting with sudden onset of neck pain and rapid enlargement of a pre-existing or new thyroid mass. It may mimic thyroid malignancy clinically and on imaging. Ultrasound shows a complex cystic lesion with internal echoes. Fine needle aspiration confirms the benign nature of the hemorrhagic cyst.

Bleeding Into The Skin

A clinical topic about bleeding into the skin. It includes the purpose, common findings, risks, and appropriate clinical follow-up.

Bleeding Tendency

An increased susceptibility to bleeding, which may result from platelet disorders, coagulation factor deficiencies, vascular abnormalities, or anticoagulant medications. Evaluation includes personal and family history, complete blood count, and coagulation studies.

Bleeding Time

A bedside test that measures the time required for a standardized skin puncture to stop bleeding, primarily reflecting platelet function and vascular integrity. It is largely replaced by more specific platelet function assays in current practice.

Bleeding Varices

Varices are enlarged, fragile veins, commonly in the esophagus or stomach when portal pressure is high. Rupture causes life-threatening gastrointestinal bleeding with vomiting blood, black stool, weakness, or shock and requires emergency treatment.

Blennorrhea

A mucopurulent discharge from a mucous membrane, especially the urethra or conjunctiva. The finding requires clinical assessment for infectious and noninfectious causes rather than treatment based on the symptom alone.

Bleomycin

An antitumor antibiotic that damages DNA and is used in selected cancer treatments. Pulmonary toxicity, skin changes, fever, and allergic reactions are important adverse effects.

Bleomycin Sulfate

The sulfate salt formulation of bleomycin, an antitumor antibiotic that damages DNA. Its most important serious toxicity is potentially irreversible pulmonary injury.

Blepharitis

A common, usually chronic inflammation of the eyelid margins that can cause itchy or burning eyelids, redness, tearing, dry-eye symptoms, and crusting at the base of the eyelashes. Anterior blepharitis affects the skin, eyelash follicles, and lash bases; posterior blepharitis affects the meibomian oil glands. It is associated with bacterial overgrowth, dandruff, rosacea, allergies, or gland dysfunction and is not usually contagious.

Alternate Names:

Eyelid Inflammation Eyelid Inflammation (Blepharitis)

Blepharoconjunctivitis

Inflammation involving the eyelid margins and conjunctiva. It may cause crusting, itching, redness, tearing, irritation, and recurrent symptoms associated with infection, allergy, or eyelid-gland dysfunction.

Blepharophimosis

Blepharophimosis is abnormally short horizontal eyelid openings, often with drooping lids and other features in blepharophimosis-ptosis syndromes.

Blepharoplasty

Surgical reconstruction or reshaping of the eyelids, often involving removal or repositioning of excess skin, muscle, or orbital fat. It may be functional, cosmetic, or reconstructive.

Blepharoptosis

Blepharoptosis is drooping of the upper eyelid caused by muscle, nerve, tendon, congenital, aging, or mechanical problems.

Blepharospasm

Involuntary spasms of the muscles around the eyes that cause repeated or sustained eyelid closure. It may be primary or associated with eye irritation, medications, or a neurologic disorder.

Blind

Having severe or complete loss of vision that substantially limits visual function; the degree and cause of visual impairment should be specified.

Blind Loop Syndrome

Malabsorption syndrome due to stasis and bacterial overgrowth in a non-propulsive ("blind") segment of small bowel, classically caused by surgical bypass loops (Billroth II, end-to-side anastomosis), strictures (Crohn disease, radiation, adhesions), diverticulosis, or impaired motility (diabetes, scleroderma).

Blind Spot

A blind spot is an area of absent vision, especially the optic-disc region where the optic nerve exits the retina and photoreceptors are absent.

Blinded

Blinded describes a research participant, clinician, or assessor who does not know the assigned treatment or relevant result, reducing expectation bias.

Blinding

Concealing group assignment. Single-blind: participants. Double-blind: participants and investigators. Triple-blind: participants, investigators, analysts.

Blindness

Severe or complete loss of vision in one or both eyes, caused by disease or injury affecting the cornea, lens, retina, optic nerve, visual pathways, or visual cortex. It may be congenital or acquired, sudden or progressive, and reversible or permanent; sudden

visual loss is an ophthalmic emergency.

Blindness And Vision Loss

Blindness or vision loss is reduced visual function that may result from eye disease, optic-nerve disease, brain disease, injury, or congenital conditions.

Blister

A raised, fluid-filled pocket in or under the outer layer of skin. Blisters commonly form from rubbing, heat, burns, infections, allergic reactions, or skin diseases, and often appear on the hands or feet.

Alternate Names:

Bulla; vesicle when small.

Bloating

A subjective sensation of abdominal fullness or distention, commonly caused by gas, functional gastrointestinal disorders, constipation, or dietary factors. Persistent bloating with weight loss, pelvic pain, or early satiety warrants medical evaluation.

Block

A block is obstruction or interruption of flow, conduction, movement, or communication, such as an airway block, nerve block, or heart block.

Blockage Of Upper Airway

Upper-airway obstruction prevents normal airflow through the nose, mouth, pharynx, or larynx and can rapidly cause respiratory failure.

Blocked Tear Duct

Partial or complete blockage of the passage that drains tears from the eye into the nose.

Blocked Ureter

Alternate index label; see *Ureteral obstruction*.

Blocking Agent

A substance that prevents or reduces a biologic process by blocking a receptor, channel, enzyme, or other target.

Alternate Names:

Blocker; antagonist when it acts by receptor antag-

onism.

Blocking Agents

Blocking agents inhibit a receptor, enzyme, ion channel, nerve signal, or physiologic pathway; their effects and risks depend on the target and drug.

Blocking Antibody

An antibody that prevents a ligand from binding to its receptor or interferes with a biological interaction; blocking antibodies can be therapeutic, protective, or involved in autoimmune disease.

Blood

A fluid connective tissue consisting of plasma, red blood cells, white blood cells, and platelets; it transports gases, nutrients, hormones, and waste and participates in immunity, coagulation, and temperature regulation.

Blood Thinner

A common name for an anticoagulant medicine that lowers the blood's ability to form harmful clots. It does not actually make the blood thinner. These medicines help prevent or treat conditions such as deep vein thrombosis, pulmonary embolism, and stroke, but can increase bleeding risk.

Alternate Names:

Anticoagulant

Blood Alcohol Concentration

The concentration of ethanol in blood, commonly expressed as mass per volume or as a percentage. Interpretation depends on sampling, timing, metabolism, tolerance, co-ingestants, and local standards; clinical impairment cannot be inferred from a single value without context.

Blood Alcohol Level

Measurement of ethanol concentration in blood. The clinical effects depend on the concentration, timing, tolerance, co-ingestants, and individual factors; a single level does not by itself determine impairment or prognosis.

Blood Bank

A service that collects, tests, processes, stores, and distributes blood and blood components for transfusion, with compatibility testing and safeguards against infection and transfusion reactions.

Blood Banking

The collection, testing, processing, storage, and distribution of blood and blood components for transfusion.

Blood Cancer

A cancer that affects the blood, bone marrow, or lymphatic system. It develops when abnormal cells grow without control and interfere with normal blood-cell or immune-system function.

Blood Blister

A raised, fluid-filled blister containing blood, usually caused by friction, pinching, pressure, or other injury that damages small blood vessels.

Blood Brain Barrier

The blood-brain barrier is a selective system of brain capillary cells and supporting structures that limits movement of substances from blood into nervous tissue.

Blood Cancer (Leukemia)

Alternate index label for Leukemia.

Blood Cell

A formed element of blood, including red blood cells, white blood cells, and platelets, produced mainly in the bone marrow.

Blood Circulation

The continuous movement of blood through the pulmonary and systemic vascular circuits, driven by the heart, to deliver oxygen and nutrients and remove carbon dioxide and metabolic waste.

Blood Cleaner

A pathogen-inactivation process for donated blood components that uses chemicals, light, or other

methods to reduce infectious microorganisms while preserving the blood product.

Blood Clot

A mass of coagulated blood within a blood vessel or body cavity. A clot may be protective when it stops bleeding, but an abnormal thrombus can obstruct blood flow or embolize to the lungs, brain, heart, or other organs.

Blood Clot In The Lung (Pulmonary Embolism)

Alternate index label for Pulmonary Embolism.

Blood Clots

Clumps of blood formed during clotting; abnormal clots can obstruct veins or arteries and cause deep vein thrombosis, pulmonary embolism, stroke, or other ischemia.

Alternate Names:

Clot Blood (Blood Clots)

Blood Coagulation

The cascade of enzymatic reactions converting circulating fibrinogen into an insoluble fibrin clot.

Blood Coagulation Disorder

A disorder in which clot formation is deficient, excessive, or inappropriately regulated, causing abnormal bleeding or thrombosis. Causes include inherited factor deficiencies, platelet disorders, liver disease, vitamin deficiencies, immune conditions, and acquired systemic disease.

Blood Count

A blood count measures numbers and characteristics of red cells, white cells, and platelets to assess anemia, infection, inflammation, marrow disease, and bleeding risk.

Blood Culture

A laboratory test in which blood is placed in culture bottles to detect bacteria or fungi in the bloodstream. It is used when bloodstream infection, sepsis, or infective endocarditis is suspected.

Blood Culture Contamination

Introduction of skin or environmental microorganisms into a blood-culture bottle during collection or handling rather than true bloodstream infection. It can lead to unnecessary antimicrobial treatment and should be assessed using the organism, number of positive sets, time to positivity, and clinical findings.

Blood Differential Test

A laboratory test that quantifies the relative and absolute numbers of each white blood cell type—neutrophils, lymphocytes, monocytes, eosinophils, and basophils—in peripheral blood. It helps identify infection, inflammation, allergy, leukemia, and bone marrow disorders.

Blood Donation Before Surgery

A medical procedure involving blood donation before surgery. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Blood Donor

A person who voluntarily gives blood or blood components for transfusion or medical use. Donors are screened to protect both donor health and recipient safety.

Blood Draw

Collection of blood, usually by inserting a needle into a vein, for laboratory testing, donation, or other medical use.

Blood Film (Peripheral Blood Smear)

A stained microscopic preparation of peripheral blood used to assess erythrocyte, leukocyte, and platelet morphology and to detect abnormal cells, parasites, blasts, and red-cell fragments.

Alternate Names:

Peripheral Blood Smear

Blood Flow

The volume of blood passing through a vessel, organ, or the systemic circulation per unit time, typically

expressed in mL/min. Governed by the hemodynamic equivalent of Ohm's law: flow = pressure gradient / resistance. Total cardiac output (5 L/min at rest) equals systemic blood flow.

Blood Gas (Venous)

Sampling of venous blood (typically from a peripheral vein or central line) for pH, PvCO₂, HCO₃, and base excess. Less painful than ABG, avoids arterial puncture risks. Limitations: no PaO₂ or SpO₂ data; peripheral venous gases in shock may not reflect arterial pH. VBG pH is typically 0.03-0.05 lower than ABG; PvCO₂ is 3-8 mmHg higher.

Alternate Names:

Venous

Blood Gas Analysis

Blood gas analysis measures oxygen, carbon dioxide, pH, and related values to assess ventilation, oxygenation, and acid-base status.

Blood Gases

Laboratory measurements of oxygen, carbon dioxide, acidity, and related values in blood to assess ventilation, oxygenation, and acid-base balance.

Blood Glucose

The concentration of glucose in blood, maintained by dietary absorption, hepatic production, insulin, and counter-regulatory hormones; abnormal levels indicate hypoglycaemia or hyperglycaemia.

Blood Glucose Monitoring

Blood sugar monitoring involves measuring glucose levels in the blood to manage diabetes, typically using a glucometer with finger-stick testing or continuous glucose monitors that track interstitial glucose in real time.

Blood Lancet

Alternate index label; see *Lancet*.

Blood Group

The classification of blood based on the presence or absence of specific antigens (agglutinogens) on the

surface of red blood cells. The ABO system has four groups: A (A antigen), B (B antigen), AB (both A and B), and O (neither A nor B). The Rh system: RhD-positive (85% of population) or RhD-negative (15%).

Blood Group, ABO

The blood-group system defined by A and B antigens on red blood cells and corresponding antibodies in plasma, producing type A, B, AB, or O.

Blood Groups

Blood groups classify inherited red-cell antigens, such as ABO and Rh, to guide compatible transfusion, pregnancy care, and transplantation.

Blood In Semen

Hemospermia, or the presence of blood in the ejaculate. It is often temporary and related to inflammation, infection, recent urologic procedures, or trauma, but persistent or recurrent bleeding, pain, fever, urinary symptoms, or bleeding in older men warrants medical evaluation.

Alternate Names:

Semen Blood (Blood In Semen)

Blood In The Eye

Bleeding within or around the eye, such as a subconjunctival hemorrhage on the eye surface or a hyphema in the front chamber; the location and cause determine urgency.

Blood In The Semen

Blood in semen, called hemospermia, is usually benign and self-limited but may result from infection, inflammation, trauma, or a urologic disorder.

Blood In The Stool (Rectal Bleeding)

Alternate index label for Rectal Bleeding.

Blood In The Urine

Hematuria, the presence of red blood cells or blood products in urine; it may be visible or detected only by urinalysis and requires evaluation for urinary-tract causes.

Blood In Urine

Blood in the urine, or hematuria, may arise from infection, stones, prostate disease, kidney disease, trauma, medicines, or urinary-tract cancer. Visible blood, clots, pain, fever, inability to urinate, or persistent microscopic blood requires timely evaluation.

Blood In Urine (Hematuria)

Visible or microscopic blood in the urine, which may result from infection, stones, injury, kidney disease, medicines, or urinary tract cancer.

Alternate Names:

Microscopic Hematuria

Blood Loss

The volume of blood lost during surgery or trauma, estimated by weighing swabs, suction contents, and hemodynamic monitoring, guiding fluid and transfusion management.

Blood Low Red Cell Count (Anemia)

Alternate index label for Anemia.

Blood PH

A measure of blood acidity or alkalinity, regulated mainly by the lungs and kidneys; normal arterial blood pH is approximately 7.35–7.45.

Blood Plasma

Blood plasma is the liquid component of blood containing water, electrolytes, proteins, nutrients, hormones, gases, and waste products.

Blood Poisoning

Blood poisoning is a nontechnical term usually referring to sepsis or bloodstream infection, a potentially life-threatening dysregulated response to infection.

Blood Poisoning (Septicaemia)

An older term commonly used for serious bloodstream infection or sepsis. Sepsis is life-threatening organ dysfunction caused by a dysregulated response to infection.

Alternate Names:

Septicaemia

Blood Preservation

The collection, anticoagulation, processing, refrigeration, freezing, and quality control used to maintain donated blood or blood components until transfusion.

Blood Pressure

The force exerted by blood against arterial walls, measured in millimeters of mercury (mmHg). Systolic pressure (peak during ventricular ejection, normally approximately 120 mmHg) and diastolic pressure (minimum during ventricular relaxation, normally approximately 80 mmHg).

Blood Pressure Chart: Reading By Age

Blood-pressure interpretation depends on repeated measurements, technique, age, pregnancy, comorbidities, and the guideline used; age alone does not define a safe target. Very high readings with chest pain, neurologic symptoms, shortness of breath, or confusion are emergencies.

Blood Pressure Low (Low Blood Pressure)

Low blood pressure can be normal or can result from dehydration, bleeding, infection, heart disease, endocrine disease, or medicines. Symptoms such as fainting, confusion, chest pain, severe weakness, or shock require urgent assessment.

Blood Pressure Measurement

Measurement of the pressure exerted by circulating blood on arterial walls, recorded as systolic and diastolic pressure.

Blood Pressure Monitors For Home

Devices used to measure blood pressure outside a clinic. Accurate cuff size, positioning, and repeated measurements improve reliability.

Blood Product

A therapeutic component prepared from donated blood, such as red-cell concentrates, platelets, plasma, or cryoprecipitate, selected according to the patient's clinical need.

Blood Products

Components prepared from donated blood for transfusion, including red blood cells, platelets, plasma, and cryoprecipitate. Each component has specific indications, benefits, and risks; selection depends on the patient's clinical condition and laboratory findings.

Blood Protein

Blood protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Blood Results

Blood results are laboratory measurements from a blood sample, interpreted using the test method, reference interval, symptoms, and clinical context.

Blood Sedimentation

The settling of red blood cells in anticoagulated blood; the erythrocyte sedimentation rate is a non-specific marker that may increase with inflammation, infection, anaemia, or other conditions.

Blood Smear

Peripheral blood film: a thin blood smear on a slide stained (Wright/Giemsa) and examined microscopically for cell morphology, aiding diagnosis.

Blood Spatter Analysis

The forensic interpretation of bloodstain patterns to reconstruct events surrounding a violent incident. Types: passive (drops, pools, flows), transfer (contact patterns), and impact/expired spatter.

Blood Substitutes

Fluids or artificial products intended to restore circulating volume or, in some cases, oxygen-carrying capacity when donated blood is unavailable or unsuitable; most currently used substitutes provide volume rather than durable oxygen transport.

Blood Sugar Test

Measurement of blood glucose to detect or monitor diabetes and other disorders of glucose regulation. Testing may be performed fasting, randomly, or after a glucose load, using a laboratory sample or a point-of-care meter.

Blood Test

A blood test analyzes blood cells, chemicals, proteins, hormones, antibodies, genes, or other markers to investigate health or disease.

Blood Transfusion

The administration of whole blood or a specific blood component, such as red blood cells, platelets, plasma, or cryoprecipitate, to treat blood loss, anaemia, thrombocytopenia, or a clotting-factor deficiency. Red cells improve oxygen-carrying capacity; platelets, plasma, and cryoprecipitate support haemostasis. ABO/Rh typing, antibody screening, and compatibility testing help reduce transfusion reactions. Risks include febrile or allergic reactions, acute haemolysis, circulatory overload, and rare transfusion-transmitted infection; the decision depends on the patient's symptoms, laboratory results, bleeding, and overall clinical context.

Blood Treatment

The diagnosis-specific use of medicines, transfusion, apheresis, replacement factors, or other interventions to treat disorders of blood cells, coagulation, plasma proteins, or blood-borne infection.

Blood Typing

Laboratory determination of ABO and Rh blood-group antigens to support safe transfusion and pregnancy-related care.

Blood Urea Nitrogen

BUN: serum urea, the end product of protein catabolism, filtered and partially reabsorbed by the tubules. Used with creatinine to assess renal function and volume status.

Blood Urea Nitrogen Test

A blood test that measures urea nitrogen and helps

assess kidney function and hydration.

Blood Vessel

A tube that carries blood through the body. Arteries carry blood away from the heart, veins return it, and capillaries allow exchange with tissues.

Blood Viscosity

The internal friction of blood that resists flow, determined primarily by hematocrit (red cell concentration), with smaller contributions from plasma proteins and temperature. Normal whole-blood viscosity is approximately 3-4 times that of water.

Blood Volume

The total amount of blood in the circulation, comprising plasma and cellular components; it varies with body size and fluid balance, and abnormal reductions or increases contribute to shock or circulatory strain.

Blood-Borne Disease

An infection that can spread through contact with infected blood or other body fluids. Examples include HIV and hepatitis B and C.

Blood-Borne Pathogens

Infectious microorganisms transmitted through blood or certain body fluids, including viruses such as HIV and hepatitis B and C; prevention requires appropriate barrier precautions and safe handling of sharps.

Blood-Brain Barrier

The blood-brain barrier is a highly selective semipermeable membrane that separates the circulating blood from the brain extracellular fluid, consisting of endothelial cells connected by tight junctions, astrocyte foot processes, and a basement membrane.

Bloodborne Pathogens

Bloodborne pathogens are infectious agents transmitted through blood or certain body fluids, including HIV, hepatitis B virus, and hepatitis C virus.

Bloom Syndrome

Rare autosomal recessive chromosomal instability syndrome caused by biallelic mutations in BLM (15q26.1) encoding a RecQ helicase.

Blount Disease

A growth disorder of the proximal tibia causing progressive bowing of the legs, usually involving the medial tibial physis. It may occur in infants or adolescents and can lead to gait abnormality, limb-length discrepancy, and early knee degeneration.

Blue Baby

Generic term for an infant with cyanosis (bluish skin, lips, nail beds) from arterial oxygen desaturation.

Blue Baby Operation

A historical term for palliative surgery used to improve oxygenation in an infant with cyanotic congenital heart disease, often by increasing pulmonary blood flow.

Blue Bloater

A clinical phenotype of chronic bronchitis in COPD. Patients present with chronic cough, sputum production, cyanosis, right heart failure, and airway-centric inflammation. They tend to be younger, have more hypercapnia, and less emphysema on imaging.

Blue Bottle

Blue bottle is a common name for several organisms or objects; in medical contexts it may refer to a bluebottle jellyfish whose sting causes pain and skin inflammation.

Blue Discoloration Of The Skin

Bluish discoloration of the skin or mucous membranes, called cyanosis, may indicate reduced oxygenation or abnormal blood flow.

Blue Nevus

A blue nevus is a usually benign blue-gray skin mole caused by pigment cells located deeper in the dermis.

Blue Nightshade Poisoning

Poisoning from eating blue nightshade or related Solanum plants containing toxic glycoalkaloids. It may cause gastrointestinal symptoms, confusion, weakness, seizures, or abnormal heart rhythm.

Blue Nævus

A blue naevus is a usually benign melanocytic skin lesion containing pigment deep in the dermis, giving it a blue or blue-black appearance.

Blue Sclera

A bluish appearance of the normally white outer coat of the eye, often caused by scleral thinning and classically associated with osteogenesis imperfecta.

Blue-Green Algae

Blue-green algae are cyanobacteria; some produce toxins that can contaminate water or food and cause gastrointestinal, neurologic, liver, or skin illness.

Blunt Abdominal Trauma

Trauma to the abdomen without penetrating injury. Mechanisms: motor vehicle collisions (most common), falls, assaults. Commonly injured organs: spleen (most common, left upper quadrant), liver (right upper quadrant), kidneys (flank), bowel (seatbelt sign, delayed perforation), and pancreas (epigastric).

Blunt Force Trauma

Injury produced when a blunt object strikes the body, or the body strikes a blunt surface, without a sharp edge penetrating the skin. Patterns include abrasions, contusions, lacerations with tissue bridging, and fractures; patterned injuries may reproduce the impacting object.

Blunted Affect

A restricted range of emotional expression observed in psychiatric or neurologic conditions. The individual shows diminished responsiveness to emotional stimuli, with reduced facial expression, vocal tone, and body language; it is commonly seen in schizophrenia, depression, and traumatic brain injury.

Blurred Vision

Impaired visual acuity in which objects appear out of focus or hazy. Causes range from refractive errors and dry eye to retinal disease, optic neuritis, increased intracranial pressure, and acute vascular events; sudden onset warrants urgent evaluation.

Blush

Temporary reddening or warmth of the face caused by increased superficial blood flow, which may result from emotion, heat, exertion, substances, or medical conditions.

Blushing

Blushing is temporary facial redness caused by increased cutaneous blood flow, triggered by emotion, heat, exertion, hormones, medicines, or some diseases.

Bone Mineral Density

Bone mineral density—the amount of calcium and other minerals in a specific area of bone. It is commonly measured by a dual-energy X-ray absorptiometry (DXA) scan to help diagnose osteoporosis, estimate fracture risk, and monitor treatment. Results must be interpreted with the person's age and other clinical risk factors.

Alternate Names:

BMD; bone mineral density; bone density.

Board-Like Rigidity

Board-like rigidity is marked involuntary abdominal-wall stiffness, classically suggesting peritoneal irritation and a possible surgical abdomen.

Boas' Sign

Boas' sign is referred pain or skin hypersensitivity below the right shoulder blade described in association with gallbladder disease; it is not sufficiently specific alone for diagnosis.

Bobbing, Ocular

Ocular bobbing is a rapid downward movement of both eyes followed by a slower return, a sign of severe pontine or other neurologic dysfunction.

Bocavirus

Bocavirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Bocavirus Infection

Infection caused by human bocavirus, a respiratory virus that most often affects young children. It may cause cough, wheezing, fever, runny nose, bronchiolitis, or pneumonia and can occur with other respiratory infections. Treatment is generally supportive; breathing difficulty, dehydration, or marked lethargy requires prompt assessment.

Bochdalek Hernia

A congenital diaphragmatic hernia occurring through the posterolateral foramen of Bochdalek, resulting from failure of the pleuroperitoneal membrane to close by the 8th week of gestation. The left side is affected in 85% of cases (the right pleuroperitoneal canal closes earlier).

Bock's Disease

Bock's disease is an older name for sarcoidosis, a granulomatous inflammatory disorder that can affect the lungs, lymph nodes, skin, eyes, and other organs.

BODE Index

A validated multidimensional scoring system and prognostic index for patients with chronic obstructive pulmonary disease. It evaluates four independent factors: Body mass index (B; 1 point if $\leq 21 \text{ kg/m}^2$), airflow Obstruction (O; FEV₁ percentage of predicted, 0–3 points), Dyspnea (D; modified Medical Research Council [mMRC] dyspnea scale, 0–3 points), and Exercise capacity (E; 6-minute walk distance in meters, 0–3 points). The composite score ranges from 0 to 10 points, where higher scores reflect greater severity, disease progression, and significantly increased risk of hospitalization and all-cause mortality.

Alternate Names:

BODE Score; BODE Index for COPD

Bodily Secretion

A substance produced and released by cells or glands, such as sweat, saliva, mucus, tears, digestive juices, or hormones. Secretions may protect surfaces, aid digestion, regulate body functions, or remove waste.

Body

The body is the physical organism and its integrated systems, tissues, organs, and functions; medical assessment considers structure, physiology, and health together.

Body Aches

Generalized muscle or body pain that may occur with viral infections, fever, exertion, inflammation, medication effects, or other systemic illness. Severe pain with weakness, dark urine, breathing difficulty, chest pain, or high fever requires prompt medical assessment.

Body Cavity

A fluid-containing space within the body that houses and protects organs, including the cranial, vertebral, thoracic, abdominal, and pelvic cavities.

Body Cell Mass

The metabolically active component of the body, including the cells of muscle, organs, blood, and other living tissues, excluding extracellular water and stored fat.

Body Composition

The proportions of fat, lean tissue, bone mineral, and body water in the body; it is assessed in nutrition, obesity, sports medicine, and some chronic diseases.

Body Dysmorphic Disorder

Preoccupation with one or more perceived defects or flaws in physical appearance that are not observable or appear slight to others. The individual performs repetitive behaviors in response to appearance concerns such as mirror checking, excessive grooming, skin picking, reassurance seeking, and comparing.

Alternate Names:

BDD

Body Fat

Body fat is adipose tissue that stores energy, cushions organs, insulates the body, and produces endocrine signals; excess or insufficient amounts affect health.

Body Fluid

Water and dissolved substances in intracellular or extracellular compartments, including plasma, interstitial fluid, cerebrospinal fluid, and lymph; abnormal volume or composition causes fluid and electrolyte disorders.

Body Fluids

Body fluids include blood, lymph, cerebrospinal fluid, urine, synovial fluid, and other water-based compartments that transport substances and support physiology.

Body Habitus

A person's general physical build or body composition, including features such as size, shape, muscle, and fat distribution.

Body Height

The vertical measurement from the soles of the feet to the top of the head; it is used to assess growth, nutrition, body-mass index, and medication or physiological calculations.

Body Identification

The process of establishing the identity of a deceased person, essential in forensic and mass-disaster contexts.

Body Image

A person's perception, thoughts, and feelings about their own body and appearance; severe disturbance may occur in body dysmorphic disorder or eating disorders.

Body Lice

Infestation of clothing and skin by the human body

louse, causing intense itching, excoriations, and sometimes secondary bacterial infection. Treatment requires hygiene, laundering or heat treatment of clothing and bedding, and appropriate antiparasitic measures.

Alternate Names:

Lice, Body

Body Mass Index

A screening measure calculated by dividing weight in kilograms by height in meters squared. It helps assess weight in relation to height, but it does not directly measure body fat or provide a complete assessment of health. Results are interpreted with age and other clinical factors; children and teens are assessed using age- and sex-specific percentiles.

Alternate Names:

BMI; BMI Calculation; Body Mass Index; Quetelet Index

Body Odor

Odour produced when skin secretions, especially sweat, are metabolized by microorganisms; a new or unusual odour may occasionally reflect infection, metabolic disease, or medication.

Body Odor And Sweating

Alternate index label; see *Sweating and body odor*.

Body Odour

Body odour is smell produced when skin secretions are metabolized by microorganisms; changes may reflect hygiene, diet, medicines, infection, or metabolic disease.

Body Piercing

The intentional creation of a skin or mucosal opening to insert jewellery; complications include bleeding, infection, scarring, allergic reaction, and damage to teeth or cartilage.

Body Plethysmography

Technique for measuring lung volumes using a sealed chamber. Gold standard for TLC measurement. Useful for diagnosing restriction and hyperinflation.

Body Position

The arrangement and orientation of the body or its parts, such as standing, supine, prone, or lateral positions; position affects examination findings, ventilation, circulation, and procedures.

Body Region

A named anatomical area used to locate structures, symptoms, injuries, or examination findings, such as the chest, abdomen, pelvis, or upper limb.

Body Ringworm

Dermatophyte infection of the glabrous skin, also called tinea corporis. It typically produces an expanding annular scaly lesion with a more active border and central clearing.

Alternate Names:

tinea corporis.

Body Space

An anatomical compartment or potential space within the body, such as the pleural, peritoneal, or retroperitoneal space, where fluid, air, blood, or infection may accumulate.

Body Substance

Any material originating from or present in the body, including blood, saliva, urine, faeces, respiratory secretions, and tissue; handling may require infection-control precautions.

Body Surface

The external area of the skin and accessible mucous membranes; body-surface area is also estimated clinically for burns, drug dosing, and physiologic calculations.

Body Surface Area

An estimate of the total external surface area of the body, calculated from height and weight and used in some medication dosing and physiologic calculations.

Body Temperature

The balance between heat production (metabolism,

muscle activity) and heat loss (radiation, conduction, convection, evaporation). See Temperature (Body Temperature).

Body Temperature Norms

Normal body temperature varies with age, time of day, activity, measurement site, and health. Fever and hypothermia are defined using clinical context and measurement method.

Body Type Ectomorph, Endomorph Or Mesomorph

Ectomorph, endomorph, and mesomorph are informal body-type categories that do not reliably predict metabolism, health, or exercise response. Fitness and nutrition decisions are better based on measured health, goals, activity, and individual response.

Body Wall

The layers of muscle, fascia, connective tissue, bone, and skin that enclose a body compartment, such as the thoracic or abdominal wall.

Body Weight

The measured mass of a person, influenced by fat, muscle, bone, water, and organs; it is used with height and clinical context to assess growth, nutrition, fluid balance, and medication dosing.

Boeck'S Sarcoid

An older eponym for sarcoidosis, a multisystem granulomatous disease characterized by noncaseating granulomas and commonly involving the lungs, intrathoracic lymph nodes, skin, or eyes.

Boerhaave Syndrome

Transmural spontaneous rupture of the esophagus typically occurring after forceful vomiting in the setting of a full stomach. Most commonly involves the left posterolateral wall of the distal esophagus. Presents with severe chest pain, subcutaneous emphysema, mediastinal emphysema, and rapidly progressive sepsis.

Boerhaave'S Syndrome

Boerhaave's syndrome is spontaneous full-thickness rupture of the esophagus, usually after forceful vomiting, causing severe chest pain, infection, and mediastinal contamination.

Bogros' Space

Bogros' space is a preperitoneal anatomical plane behind the inguinal region used as a surgical dissection plane in hernia repair.

Bohn Nodules

Small, benign keratin-filled cysts on the gums or palate of a newborn that usually resolve without treatment.

Bohr Effect

The decrease in hemoglobin's oxygen affinity (right shift of the dissociation curve) caused by decreased pH (increased H⁺) and increased PCO₂. In metabolically active tissues, CO₂ production and lactic acid accumulation lower the local pH, causing hemoglobin to release more oxygen precisely where it is needed most.

Bohring Syndrome

A clinical disorder or finding related to bohring syndrome; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Boil

Furuncle: an acute, deep infection of a hair follicle (folliculitis that extends to the dermis) by *Staphylococcus aureus*, forming a painful, red, swollen nodule with central purulent core that may spontaneously drain.

Boil (Furuncle)

A painful, deep-seated infection of a hair follicle caused most commonly by *Staphylococcus aureus*, extending into the surrounding subcutaneous tissue. Presents as a tender, erythematous, fluctuant nodule that eventually discharges pus. When multiple adjacent follicles coalesce, it forms a carbuncle, often with systemic symptoms.

Alternate Names:

Furuncle

Boils

Boils are localized infections of hair follicles, usually caused by *Staphylococcus aureus*, that form painful pus-filled nodules.

Alternate Names:

Abscesses Skin (Boils)

Boils And Carbuncles

Painful skin infections caused by pus-filled inflammation around hair follicles; a carbuncle is a connected group of boils.

Alternate Names:

Carbuncles And Boils

Bojlen'S Syndrome

Bojlen's syndrome is an uncommon historical eponym whose usage varies; the original clinical definition should be verified from the source.

Bolt Device

Bolt device is a procedure, device, treatment, or care intervention; indications, benefits, risks, and follow-up depend on the patient and clinical goal.

Bolus

A discrete amount of medication or fluid given over a short period, usually by injection or infusion. In digestion, a bolus is chewed food ready to be swallowed.

Bolus Infusion

A medical procedure, treatment, device, or care topic related to bolus infusion; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Bombay Blood Group

The Bombay blood group is a rare phenotype lacking the H antigen, causing apparent type O blood that can receive red cells only from compatible Bombay-type donors.

Bombe

Bombe is a historical term used in medical and biological naming; its precise meaning depends on the organism, apparatus, or source in which it appears.

Bombesin

Bombesin is a peptide signaling molecule related to gastrointestinal and neurologic hormones and used in research on digestion, appetite, and tumors.

Bombesin Receptor

Bombesin receptor is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Bonding With Your Newborn

Bonding with a newborn is the development of an emotional relationship through responsive caregiving, touch, feeding, and interaction.

Bone

A rigid, mineralised connective tissue that forms the skeleton, providing structural support, protection of vital organs, hematopoiesis (bone marrow), and mineral storage (calcium, phosphate). Bone composition: organic matrix (90% type I collagen) and inorganic mineral (hydroxyapatite, calcium phosphate).

Bone And Mineral Disorders

Conditions affecting bone strength, mineralization, calcium-phosphate balance, or skeletal remodeling, including osteoporosis, osteomalacia, hyperparathyroidism, and inherited metabolic bone disease.

Bone Callus

New tissue that forms around and between the ends of a broken bone during healing. It begins as soft tissue and cartilage, becomes hard immature bone, and is gradually reshaped during remodeling.

Bone Cancer

Cancer that begins in bone, such as osteosarcoma

or chondrosarcoma, or cancer that has spread to bone from another organ. Possible features include persistent localized bone pain, swelling, or a fracture after minor trauma. Diagnosis requires imaging and tissue evaluation; treatment depends on the type and extent of disease.

Alternate Names:

Sarcoma, Bone Cancer

Bone Cancer Overview

Primary bone cancer is a malignant tumor arising in bone, while cancer more commonly spreads to bone from another organ. Persistent focal bone pain, swelling, or an unexplained fracture may prompt imaging and biopsy; treatment depends on tumor type and stage.

Alternate Names:

Bone Tumor (Bone Cancer Overview)

Bone Cement

A material, usually polymethylmethacrylate, used to secure artificial joints or fill spaces in bone. It hardens chemically and may be used in joint replacement, vertebroplasty, kyphoplasty, or bone reconstruction. Possible complications include cement leakage, infection, loosening, and bone cement implantation syndrome.

**Bone Cement
(Polymethylmethacrylate—PMMA)**

Acrylic polymer used to fix prosthetic joints to bone. Mixes as dough, cures exothermically. Used in hip/knee arthroplasty, vertebroplasty, cranial reconstruction. Complications: cement implantation syndrome (hypotension, hypoxia), infection, loosening.

Alternate Names:

Polymethylmethacrylate—PMMA

Bone Conduction

Bone conduction transmits sound through skull vibration to the inner ear, bypassing the outer and middle ear; testing helps distinguish conductive from sensorineural hearing loss.

Bone Cyst

A fluid-filled or blood-filled space within bone. Many bone cysts are benign, but they can weaken the bone and cause pain, swelling, or fracture. Common types include simple bone cysts and aneurysmal bone cysts.

Bone Densitometry

See *Bone Mineral Density Test*.

Bone Density

The amount of mineral in bone, commonly estimated by dual-energy X-ray absorptiometry. Lower density is associated with increased fracture risk, and interpretation depends on the person's age, sex, and clinical context.

Bone Density Screening

See *Bone Mineral Density Test*.

Bone Development

The formation and growth of bone through intramembranous or endochondral ossification, followed by continued modeling and remodeling throughout life.

Bone Disease

A clinical disorder or finding related to bone disease; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Bone Fracture

A partial or complete break in a bone caused by injury, repeated stress, or weakened bone. Pain, swelling, bruising, deformity, and inability to use the limb may occur. Open fractures, loss of circulation or sensation, or major deformity require emergency care.

Bone Fractures (Broken Bone (Types Of Bone Fractures))

Alternate index label for Broken Bone (Types of Bone Fractures).

Bone Graft

Bone tissue or a bone-substitute material used to fill a defect, promote fusion, support reconstruction, or enhance fracture healing. Autograft provides osteogenic cells and biologic signals; allograft and synthetic substitutes provide variable osteoconductive and osteoinductive properties.

Bone Graft (Dental)

Material placed to augment alveolar bone. Types: autograft, allograft, xenograft, alloplast. Used for implants, ridge preservation.

Alternate Names:

Dental

Bone Health

The strength and condition of the skeleton, influenced by bone density, structure, nutrition, hormones, activity, medicines, and disease.

Bone Infection

Infection of bone, usually caused by bacteria and called osteomyelitis. It may arise from bloodstream spread, nearby tissue, surgery, or an open injury.

Bone Lesion Biopsy

Image-guided needle or open sampling of a bone lesion for histopathology and, when indicated, culture or molecular testing. Biopsy planning should account for the suspected tumor and any later surgical approach.

Bone Marrow

Vascular tissue within the medullary cavities and cancellous bone. Red marrow produces blood cells, whereas yellow marrow primarily stores fat and can revert to red marrow when demand increases.

Bone Marrow (Stem Cell) Donation

Bone marrow or stem-cell donation is collection of blood-forming stem cells from marrow or circulating blood for transplantation into another person.

Bone Marrow Aspiration

A procedure that withdraws liquid marrow, usually

from the posterior iliac crest, for cytologic, flow-cytometric, cytogenetic, molecular, and microbiologic evaluation. It complements the core biopsy, which assesses marrow architecture.

Bone Marrow Aspiration And Biopsy

Sampling of bone marrow, typically from the posterior iliac crest. Aspiration: liquid marrow for cytology (differential, morphology), flow cytometry, cytogenetics, and MRD. Biopsy: core of bone for cellularity, architecture, fibrosis, and infiltrative disease (lymphoma, metastatic cancer, granulomas).

Bone Marrow Biopsy

A procedure sampling bone marrow (typically posterior iliac crest) via aspiration (cytology, flow cytometry, cultures) and trephine core biopsy (histology, architecture, fibrosis, cellularity).

Bone Marrow Culture

A laboratory culture of bone-marrow material used to identify bacteria, fungi, or other organisms in suspected infection.

Bone Marrow Depression

Reduced production of blood cells by the bone marrow, causing anemia, low white-cell counts, low platelet counts, or combinations of these findings.

Bone Marrow Failure

A syndrome in which the bone marrow cannot produce enough blood cells, causing one or more cytopenias. Causes include aplastic anemia, inherited marrow-failure disorders, myelodysplasia, toxins, medicines, infection, and marrow infiltration.

Alternate Names:

Marrow Failure Bone Marrow Failure Syndrome

Bone Marrow Disorder

A clinical disorder or finding related to bone marrow disorder; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Bone Marrow Hyperplasia

Increased cellularity of the bone marrow resulting from accelerated hematopoiesis, typically compensating for chronic hemolytic anemia, hemorrhage, or systemic hypoxia.

Bone Marrow Necrosis

Death of hematopoietic and stromal cells within the bone marrow, often associated with sickle cell crisis, malignancy, or infections, presenting with severe bone pain and cytopenias.

Bone Marrow Transplantation

A treatment in which healthy blood-forming stem cells are infused to replace or restore bone-marrow cells damaged by disease or treatment. The cells may come from the patient or a donor. It is used for selected blood cancers, marrow failure, and some inherited disorders; risks include infection, graft failure, organ toxicity, and graft-versus-host disease after a donor transplant.

Alternate Names:

BMT; Bone Marrow Transplant

Bone Mass

The amount of mineralized bone tissue in the skeleton or in a specified region.

Alternate Names:

Bone mineral content; skeletal bone mass.

Bone Matrix

The material surrounding bone cells that gives bone its structure and strength. It contains a flexible protein framework, mainly type I collagen, reinforced with mineral crystals, mainly hydroxyapatite.

Bone Metastases

Cancer deposits that have spread to bone from another primary site. They may cause pain, fracture, high blood calcium, or spinal-cord compression.

Bone Metastasis

A deposit of cancer that has spread to bone from another primary site. It may cause pain, fracture, high blood calcium, or spinal-cord compression, and

treatment depends on the cancer and clinical context.

Bone Mineral Density

The concentration of mineral in bone, usually measured by dual-energy X-ray absorptiometry to assess fracture risk and osteoporosis.

Alternate Names:

BMD; bone density.

Bone Mineral Density Test

A low-dose dual-energy X-ray absorptiometry (DXA or DEXA) scan that measures calcium and other minerals in bone, usually at the hip and lumbar spine. It helps identify low bone density, diagnose osteoporosis, estimate fracture risk, and monitor treatment. T-scores are generally used for postmenopausal women and men aged 50 or older; Z-scores are used in younger adults and children, with interpretation based on clinical context.

Bone Necrosis

The death of bone tissue, usually caused by loss of blood supply, severe injury, infection, radiation, or long-term corticosteroid use. It may cause pain, bone collapse, joint damage, or fracture and is also called osteonecrosis or avascular necrosis.

Bone Neoplasm

A neoplastic process affecting bone neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Bone Pain

Aching or other discomfort arising from one or more bones. Causes include fractures, injury, infection, reduced blood supply, metabolic bone disease, overuse, and cancer involving bone. Persistent or unexplained bone pain requires medical evaluation.

Bone Pain Or Tenderness

Bone pain or tenderness may result from injury, infection, metabolic disease, inflammation, or cancer. Persistent or unexplained pain requires assessment.

Bone Plate

A metal or other biocompatible plate fixed to a bone with screws to stabilize a fracture, correct a deformity, or support bone healing.

Bone Regeneration

The formation of new bone to repair or replace damaged, missing, or diseased bone. It occurs naturally during healing and may be supported by bone grafts or other treatments.

Bone Remodeling

The continuous process in which osteoclasts remove old bone and osteoblasts form new bone, helping maintain bone strength, shape, and mineral balance.

Bone Resorption

The breakdown of bone tissue by osteoclasts, releasing minerals into the bloodstream. Excessive resorption can contribute to bone loss and osteoporosis.

Bone Saw

A surgical saw used to cut bone during procedures such as amputation or orthopedic reconstruction.

Bone Scan

Skeletal imaging using Tc-99m MDP. Detects osteoblastic activity. Sensitivity: high; specificity: low. Used for metastasis detection, fracture, infection, metabolic bone disease. "Hot spots" indicate increased uptake; "cold spots" indicate decreased uptake.

Bone Scan (Nuclear Scintigraphy)

Nuclear medicine imaging using Tc-99m MDP (methylene diphosphonate) that accumulates in areas of increased bone turnover. Three phases: perfusion (immediate), blood pool (5 min), and delayed (2-3 hours).

Alternate Names:

Nuclear Scintigraphy

Bone Screws

Surgical screws used to hold bone fragments to-

gether or attach plates, joint components, or other orthopedic implants to bone.

Bone Spurs

Bony projections that form along the edges of bones, usually near joints, often because of long-term wear or arthritis.

Alternate Names:

Osteophytes

Bone Transplantation

The transfer of bone tissue from the same person, a donor, or a processed graft to repair a bone defect, support fracture healing, or assist spinal or joint reconstruction.

Bone Tumor

A mass arising in bone that may be benign or malignant. Evaluation uses imaging and biopsy when indicated.

Bone Tumor (Bone Cancer Overview)

Alternate index label for Bone Cancer Overview.

Bone Turnover Markers

Biochemical markers reflecting the rate of bone remodeling. Formation markers include bone-specific alkaline phosphatase, osteocalcin, and N-terminal and C-terminal propeptides of type I procollagen. Resorption markers include C-terminal and N-terminal telopeptides of type I collagen and deoxypyridinoline.

Bone Type

A classification of bones by shape or structure, including long, short, flat, irregular, and sesamoid bones.

Bone X-Ray

An X-ray examination used to assess the specified body region for fracture, abnormal structure, or disease.

Bone-Marrow Transplantation

Bone-marrow transplantation replaces diseased or

damaged marrow with hematopoietic stem cells, either from the patient or a donor.

Bones And Joints

Bones and joints form the skeletal system, providing structural support, protection of organs, and enabling movement through articulations between bones. Joints are classified as fibrous, cartilaginous, or synovial, with synovial joints offering the greatest range of motion.

Bones Of The Arm, Wrist And Hand

The upper-limb bones: humerus; radius and ulna; eight carpal bones; five metacarpals; and the phalanges of the fingers and thumb.

Bones Of The Head

The skull bones, including the cranial bones that enclose the brain and the facial bones that form the face, nasal cavity, orbit, and jaws.

Bones Of The Leg, Ankle And Foot

The lower-limb bones from the knee downward: tibia and fibula; the ankle's talus and other tarsals; five metatarsals; and the toe phalanges.

Bones Of The Skeleton

The adult human skeleton normally contains 206 named bones, forming the axial skeleton and appendicular skeleton and providing support, protection, movement, mineral storage, and marrow for blood formation.

Bones Of The Trunk

The bones forming the central body framework, chiefly the vertebral column, ribs, and sternum, which support the trunk and protect the spinal cord and thoracic organs.

Bongkrek Poisoning

Bongkrek poisoning is severe food poisoning from bongkrelic acid produced by contaminated fermented foods, causing mitochondrial energy failure and potentially fatal liver or neurologic injury.

Bonnevie-Ullrich Syndrome

Bonnevie-Ullrich syndrome is a historical name for features associated with Turner syndrome, including short stature, webbed neck, gonadal dysfunction, and characteristic physical findings.

Bony Exostosis

A benign localized bony protuberance, including torus palatinus of the palate and torus mandibularis of the mandible. Treatment is usually unnecessary unless the lesion causes trauma, interferes with speech or prosthesis, or requires diagnostic clarification.

Bony Labyrinth

The system of fluid-filled spaces within the petrous temporal bone that encloses the membranous labyrinth of the inner ear and contains the cochlea, vestibule, and semicircular canals.

Boranes

Boranes are compounds of boron and hydrogen used in chemical synthesis and research; some are flammable or toxic.

Borborygmi

Rumbling or gurgling sounds produced by the movement of gas and fluid through the intestines. They may be normal or altered in gastrointestinal obstruction and other disorders.

Borborygmus

A rumbling or gurgling sound produced by movement of gas and fluid through the intestines.

Border Definition

In dermatology and pathology, the clarity or sharpness of a lesion's margin, which helps distinguish benign from malignant growths. Well-defined borders suggest benign lesions, while irregular or ill-defined borders raise concern for malignancy.

Borderline

A descriptor indicating a result or condition that falls at the boundary between normal and abnor-

mal, or between diagnostic categories. In laboratory medicine, borderline results require repeat testing or clinical correlation; in psychiatry, borderline describes a pattern of instability.

Borderline Personality Disorder

A pattern of instability in interpersonal relationships, self-image, and affects, with marked impulsivity.

Alternate Names:

BPD (Borderline Personality Disorder) Emotional Dysregulation Disorder (Borderline Personality Disorder) Personality Disorder, Borderline

Borderline Personality Disorder

Alternate index label for Borderline Personality Disorder.

Bordetella

A genus of small gram-negative bacteria that includes *B. pertussis*, the cause of whooping cough, and other respiratory or animal-associated species.

Bordetella Avium

A *Bordetella* species that mainly causes respiratory disease in turkeys and other birds; human infection is rare.

Bordetella Bronchiseptica

A respiratory bacterium of dogs and other animals that can rarely cause pneumonia or other opportunistic infection in humans, especially those with impaired immunity.

Bordetella Hinzii

A *Bordetella* species associated mainly with poultry that rarely causes opportunistic respiratory or bloodstream infection in humans.

Bordetella Parapertussis

A *Bordetella* species that causes a pertussis-like respiratory illness, usually milder than classic whooping cough caused by *B. pertussis*.

Bordetella Pertussis

A Gram-negative coccobacillus and the causative agent of whooping cough, a highly contagious respiratory infection. It produces pertussis toxin, which inhibits G-protein signaling in respiratory epithelial cells, and tracheal cytotoxin, which damages ciliated epithelium.

Bordetella Petrii

An environmental *Bordetella* species rarely associated with human infection; significance depends on recovery from a compatible clinical site.

Boric Acid

A weak boron-containing compound used in some antiseptic, insecticidal, and industrial products; ingestion can be toxic.

Boric Acid Poisoning

Toxic effects from ingesting or absorbing excessive boric acid. Severe poisoning may cause vomiting, diarrhea, skin changes, kidney injury, seizures, shock, or death.

Borna Disease

Borna disease is an uncommon neurologic infection caused by Borna disease virus, primarily recognized in animals and rarely considered in humans.

Bornholm Disease

An acute epidemic pleurodynia caused most often by coxsackieviruses. It produces sudden attacks of severe chest or upper-abdominal muscle pain, fever, and sometimes headache or vomiting.

Boron

A chemical element present in food, water, and industrial materials; it has no established essential human biological role, and excessive exposure can cause gastrointestinal, skin, reproductive, or developmental toxicity.

Borrelia

A genus of spiral-shaped bacteria (spirochetes) transmitted by arthropod vectors, most notably ticks

and lice. The major human pathogen is *Borrelia burgdorferi* sensu lato, the causative agent of Lyme disease, transmitted by Ixodes ticks.

Borrelia Burgdorferi

A spirochete bacterium transmitted by Ixodes ticks and the principal cause of Lyme disease in North America and Europe.

Borrelia Infection

Infection with a *Borrelia* spirochete, including Lyme disease, relapsing fever, or hard-tick relapsing fever; symptoms and treatment depend on the species and exposure.

Borrelia Miyamotoi

A tick-borne spirochete that causes hard-tick relapsing fever, usually with recurrent fever, headache, fatigue, and muscle aches.

Borreliosis

Infection caused by bacteria of the genus *Borrelia*. Human disease includes Lyme borreliosis and relapsing fever, with manifestations varying according to species, stage, and organ involvement.

Bortezomib

A proteasome inhibitor used mainly to treat multiple myeloma and some lymphomas; important adverse effects include peripheral neuropathy, cytopenias, gastrointestinal symptoms, and herpes-zoster reactivation.

Bosentan

Bosentan is an endothelin-receptor antagonist used for pulmonary arterial hypertension and requiring liver-function and pregnancy precautions.

Bosniak Classification

System classifying renal cysts by CT features. Categories I-II benign, IIF follow-up, III-IV likely malignant.

Bot Fly

A parasitic fly of the family Oestridae whose larvae

cause myiasis in humans and animals. *Dermatobia hominis*, the human bot fly, deposits eggs on blood-sucking insects; larvae penetrate the skin and develop subcutaneously, causing a painful boil-like lesion with a central breathing pore. Treatment is manual extraction of the larva.

Botany

Botany is the scientific study of plants, including their structure, growth, physiology, classification, and uses.

Botfly

A botfly is a fly whose larvae can infest animal or human tissue, causing furuncular or wound myiasis.

Bothriocephalus

Bothriocephalus is a historical or taxonomic tapeworm genus; related fish tapeworm infections may cause abdominal symptoms or vitamin-B12 deficiency.

Botryomycosis

A chronic bacterial infection characterized by granule-forming lesions that can resemble a fungal infection. It most often affects skin and subcutaneous tissue but may involve viscera in susceptible patients.

Bottle Feeding

Feeding an infant with expressed breast milk or infant formula using a bottle and teat, either as exclusive feeding or combined with breastfeeding. Proper preparation, sterilization, and feeding technique are important to prevent infection and ensure adequate nutrition.

Bottom Number Of Your Blood Pressure Is Over 100

A diastolic blood-pressure reading over 100 mm Hg is markedly elevated when confirmed with correct repeated measurements. Persistent elevation increases cardiovascular risk, while a very high reading with chest pain, breathlessness, neurologic symptoms, or confusion needs urgent care.

Botulinum Toxin

A neurotoxin produced by *Clostridium botulinum* that blocks acetylcholine release at neuromuscular junctions; purified preparations are used for selected muscle spasticity, dystonias, migraine, sweating, and cosmetic indications, but excess exposure can cause paralysis and respiratory failure.

Botulinum Toxin Injection - Larynx

A medical procedure involving botulinum toxin injection - larynx. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Botulism

A rare, potentially fatal neuroparalytic disease caused by botulinum toxin, a potent presynaptic neurotoxin produced by *Clostridium botulinum*, an anaerobic, spore-forming, gram-positive bacillus.

Bouchard Nodes

Bony enlargements of the proximal interphalangeal joints, usually associated with osteoarthritis.

Bouchard'S Node

A hard, palpable enlargement of the proximal interphalangeal joint caused by osteoarthritis. It reflects osteophyte formation and may be associated with pain, stiffness, and reduced finger motion.

Bouchard'S Nodes

Firm bony enlargements of the proximal interphalangeal joints, most commonly caused by hand osteoarthritis.

Alternate Names:

Proximal interphalangeal osteophytes; PIP-joint nodes.

Bougie

A slender, flexible instrument used to examine or gradually dilate a narrowed tubular structure, such as the esophagus or urethra.

Bougienage

The use of a bougie to dilate or explore a narrowed

tubular structure, such as the esophagus or urethra. It is a procedure that requires assessment of the cause and risks of the stricture.

Bouin'S Solution

A laboratory fixative containing picric acid, formaldehyde, and acetic acid. It preserves tissue structure for microscopic examination, but its use is limited because the solution is hazardous and can interfere with some molecular tests.

Bourbon Virus

A rare tick-borne viral infection associated with fever, fatigue, anorexia, nausea, and a low white-cell or platelet count. There is no established specific antiviral treatment or vaccine; management is supportive and requires exclusion of other serious tick-borne infections.

Boutonneuse Fever

A tick-borne rickettsial infection, usually caused by *Rickettsia conorii*, characterized by fever, headache, rash, and often a dark inoculation eschar called a tache noire.

Bouveret Syndrome

Rare variant of gallstone ileus where a large gallstone erodes through a cholecystoduodenal fistula and impacts in the duodenum or pylorus causing gastric outlet obstruction. Represents approximately two to three percent of gallstone ileus cases. Presents with nausea, vomiting, epigastric pain, and anorexia.

Bovine

Relating to cattle, including substances, diseases, or biological products derived from cattle.

Bovine Anaplasmosis

Bovine anaplasmosis is a tick- or blood-transmitted infection of cattle caused by *Anaplasma marginale*, producing red-cell destruction and anemia.

Bovine Papillomavirus

Bovine papillomavirus is a virus or viral group classified by its genetic and biologic features. Clinical

importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Bovine Spongiform Encephalopathy

A fatal transmissible prion disease of cattle causing progressive neurologic degeneration. It is related to variant Creutzfeldt–Jakob disease in humans through exposure to contaminated tissue.

Bow Legs (Genu Varum)

Outward curvature of the legs so that the knees remain apart when the ankles are together. It may be a normal developmental stage in young children or result from bone disease.

Alternate Names:

Genu Varum

Bowel

The small and large intestines, which receive digested food, absorb nutrients and water, and form and transport stool.

Alternate Names:

Intestine; intestinal tract.

Bowel Burn

Bowel burn is a procedure, device, treatment, or care intervention; indications, benefits, risks, and follow-up depend on the patient and clinical goal.

Bowel Cancer

Bowel cancer is malignant growth in the colon or rectum, commonly causing altered bowel habit, bleeding, anemia, weight loss, or abdominal symptoms.

Bowel Incontinence

The involuntary loss of solid or liquid stool through the anus, also known as fecal incontinence, affecting approximately 2-15 percent of the general population and increasing in prevalence with age.

Alternate Names:

fecal.

Bowel Infarction

Ischemic necrosis of bowel tissue from occlusive or

non-occlusive mesenteric ischemia, presenting with severe abdominal pain out of proportion to examination, bloody stools, and peritoneal signs. Causes include mesenteric arterial embolism or thrombosis, venous thrombosis, and low-flow states.

Bowel Ischemia

Reduced blood flow to the intestines causing mucosal injury, most commonly from mesenteric atherosclerosis, low-flow states, or mesenteric venous thrombosis. Presents with abdominal pain, bloody stools, and elevated lactate. Severe cases progress to transmural necrosis.

Bowel Obstruction

Blockage of intestinal lumen. X-ray, CT diagnosis. Differentiate small vs. large bowel obstruction. Mechanical vs.

Bowel Program

A planned routine using diet, fluids, medicines, and scheduled toileting to promote regular bowel movements and manage constipation or bowel dysfunction.

Bowel Retraining

Bowel retraining is a structured program using regular timing, adequate fluid and fiber, toileting posture, and other measures to establish predictable bowel movements.

Bowel Sounds

Sounds produced by intestinal contractions and movement of gas and fluid, heard with a stethoscope; their frequency and character may change with gastrointestinal disorders.

Bowel Transit Time

Bowel transit time is the time required for material to pass through the gastrointestinal tract, measured to evaluate selected motility disorders.

Bowel Urgency

A sudden, difficult-to-defer need to defecate, commonly associated with diarrhoea, inflammatory

bowel disease, irritable bowel syndrome, infection, or rectal disorders.

Bowel, Diseases Of

Diseases of the bowel include infection, inflammation, obstruction, malabsorption, functional disorders, bleeding, and tumors affecting the small or large intestine.

Bowen'S Disease

A clinical disorder or finding related to bowen's disease; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Bowenoid Papulosis

A low-grade squamous intraepithelial neoplasia of the anogenital skin caused by human papillomavirus infection, typically types 16 and 18, presenting as multiple pigmented papules on the genital mucosa. Histologically it resembles squamous cell carcinoma in situ but follows a benign clinical course.

Bowlegs

Outward bowing of the legs causing the knees to remain apart when the feet and ankles are together, also called genu varum. It is common in early childhood but requires assessment when severe, asymmetric, painful, or persistent beyond the expected age.

Alternate Names:

genu varum.

Bowman Capsule

Double-walled epithelial cup surrounding the glomerular capillaries forming the beginning of the nephron. The visceral layer is composed of podocytes whose foot processes interdigitate to form filtration slits. The parietal layer is composed of simple squamous epithelium.

Bowman Membrane

A tough, thin layer of connective tissue beneath the corneal epithelium that helps protect and support the cornea. It does not regenerate normally after significant injury.

Bowman'S Capsule

Bowman's capsule is the cup-shaped beginning of a nephron that surrounds the glomerulus and collects the filtrate produced from blood.

Bowman'S Gland

A serous gland in the olfactory mucosa whose secretion helps dissolve odorant molecules and cleanse the olfactory surface. It is located in the lamina propria beneath the olfactory epithelium.

Boxed Warning

The most prominent safety warning required in prescribing information for a medicine when there is a serious or life-threatening risk. It communicates major hazards and does not necessarily mean that the medication is never appropriate.

Boxer'S Ear

Boxer's ear is deformity of the outer ear caused by repeated trauma, usually involving cartilage thickening, scarring, or a cauliflower-ear appearance.

Boxer'S Fracture

A boxer's fracture is a fracture of the neck of the fifth metacarpal, commonly caused by punching, producing hand pain, swelling, and altered knuckle contour.

Braces And Retainers (Dental Braces)

Alternate index label for Dental Braces.

Brachial

Brachial means relating to the upper arm, especially its bones, vessels, or nerves.

Brachial Artery

The continuation of the axillary artery, running from the lower border of teres major to the cubital fossa, where it divides into radial and ulnar arteries. Lies medial to the biceps; palpated for blood pressure and brachial pulse. Branches: profunda brachii (deep brachial), superior/inferior ulnar collaterals.

Brachial Plexopathy

Dysfunction of the brachial plexus causing pain, weakness, sensory loss, or paralysis in the shoulder and upper limb. Causes include trauma, compression, tumors, inflammation, and radiation injury.

Brachial Plexus

Nerve network (C5-T1) formed by roots, trunks, divisions, and cords, giving rise to upper limb nerves. Roots unite into superior/middle/inferior trunks, dividing into anterior/posterior divisions, recombining into lateral/posterior/medial cords. Major branches: musculocutaneous, axillary, radial, median, ulnar.

Brachial Plexus Cords

Formed by divisions recombining behind the pectoralis minor. Lateral cord (anterior divisions of upper/middle trunks), posterior cord (all three posterior divisions), medial cord (anterior division of lower trunk). Named for relationship to the axillary artery. Lateral cord: lateral pectoral nerve, contributes to median nerve.

Brachial Plexus Injuries

Injuries to the network of nerves supplying the shoulder and upper limb, caused by traction, compression, or penetrating trauma. They may produce pain, weakness, sensory loss, or paralysis; the pattern and severity depend on which roots, trunks, or cords are affected.

Brachial Plexus Injury

Damage to the brachial plexus nerves during birth, most commonly from shoulder dystocia or excessive traction. Erb's palsy (C5-C6) presents with waiter's tip position (arm adducted, elbow extended, forearm pronated). Klumpke's palsy (C8-T1) affects the hand and may cause Horner syndrome. Most cases resolve spontaneously.

Brachial Plexus Injury In Newborns

A brachial plexus injury in a newborn affects the network of nerves supplying the shoulder, arm, and hand, usually after difficult delivery. Weakness may improve or require specialist care.

Brachial Plexus Neuropathies

Disorders caused by injury or dysfunction of the network of nerves supplying the shoulder and arm.

Brachial Plexus Roots

Ventral rami of C5-T1 emerging from intervertebral foramina, passing between anterior and middle scalene muscles (interscalene triangle). C5-C6 form the upper trunk, C7 continues as middle trunk, C8-T1 form the lower trunk. Root injuries cause specific dermatome/myotome deficits. C5 avulsion causes loss of deltoid/biceps function.

Brachial Plexus Trunks

Formed in the posterior neck triangle. Upper (C5-C6), middle (C7), and lower (C8-T1) trunks each divide into anterior/posterior divisions behind the clavicle. Upper trunk is most commonly injured in birth injuries (Erb-Duchenne palsy), causing waiter's tip position.

Brachialgia

Brachialgia is pain radiating along the arm, often from cervical nerve-root irritation or brachial plexus disease.

Brachialis Muscle

Flat strong muscle deep to the biceps, from distal anterior humerus to ulnar tuberosity and coronoid process. Innervated by musculocutaneous nerve (C5-C6). Primary elbow flexor regardless of forearm position ("workhorse" of elbow flexion). Less visible than biceps but significantly contributes to arm strength.

Brachiocephalic Trunk

The first and largest branch of the aortic arch. It ascends to the right sternoclavicular joint, where it divides into the right common carotid and right subclavian arteries.

Brachiocephalic Veins

Paired veins formed by the union of the internal jugular and subclavian veins. They join to form the superior vena cava and drain the head, neck, upper limbs, and upper chest.

Brachioplasty

An operation, commonly called an arm lift, that removes or reshapes excess skin and tissue of the upper arm; risks include scarring, sensory change, fluid collection, and wound complications.

Brachium

The brachium is the anatomical upper arm between the shoulder and elbow.

Brachycephalic

Brachycephalic means having a relatively short, broad skull; it may describe normal variation or a craniofacial pattern in a genetic condition.

Brachycephaly

Brachycephaly is a broad, shortened skull shape that may be positional or caused by premature fusion of coronal sutures.

Brachydactyly

Brachydactyly is congenital shortening of fingers or toes caused by abnormal development of the bones or surrounding tissues.

Brachytherapy

A type of internal radiation therapy in which a radioactive source is placed inside or near a tumor to deliver a concentrated dose to a limited area. The source may be temporary or permanent and is delivered through an applicator, catheter, needle, seed, or similar device.

Alternate Names:

Brachy Therapy Internal Radiation Therapy

Brachytherapy Seed

Radioactive source implanted directly into tumor. I-125 or Pd-103 seeds for prostate cancer. Permanent or temporary implantation.

Bradycardia

A heart rate <60 beats per minute. May be physiologic (athletes) or pathologic due to sinus node dysfunction, AV block, or medication effects. Symptomatic bradycardia causes fatigue, dizziness, syn-

cope, or heart failure. Evaluation includes ECG, Holter monitoring, and assessment of reversible causes.

Alternate Names:

Slow Heartbeat

Bradycardia Of Prematurity

Heart rate less than 100 beats per minute in a premature infant, often associated with apnea and oxygen desaturation. May be caused by vagal stimulation, reflux, or immaturity of the cardiac conduction system. Treatment: gentle stimulation, caffeine, and addressing underlying causes.

Bradycardia-Tachycardia Syndrome

Alternate index label; see *Sick sinus syndrome*.

Bradykinesia

Slowness and reduced amplitude of voluntary movement, often accompanied by difficulty initiating or sustaining repetitive actions. It is a cardinal feature of parkinsonism and may be associated with rigidity, resting tremor, postural instability, reduced arm swing, masked facial expression, and a shuffling gait.

Bradykinetic

Characterized by abnormally slow movement, often used for the movement slowness seen in Parkinson disease and related disorders.

Bradykinin

Bradykinin is a peptide mediator that increases blood-vessel permeability, dilates vessels, and contributes to pain and inflammation.

Bradyphrenia

Slowness of thought and mental processing, which may occur with neurologic disease, psychiatric illness, medication effects, or other conditions.

Bradypnea

Abnormally slow respiratory rate (adult <12 breaths/min). Causes: opioid/benzodiazepine/sedative overdose, central

nervous system depression (stroke, trauma, increased ICP), hypothyroidism, hypothermia, and metabolic (e.g., hypokalemia). May lead to hypoventilation, hypercapnia, and respiratory failure.

Bradypnoea

Bradypnoea is abnormally slow breathing, which may result from medicines, neurologic disease, metabolic disturbance, or severe illness.

Braille System

A tactile writing and reading system using patterns of raised dots to represent letters, numbers, punctuation, and symbols for people who are blind or have low vision.

Brain

The central organ of the nervous system, enclosed in the cranial cavity and composed of the cerebrum, cerebellum, and brainstem. The brain receives, processes, and integrates sensory information; initiates and coordinates motor responses; and serves as the seat of cognition, emotion, memory, and consciousness.

Brain Abscess

A focal, purulent infection within the brain parenchyma, most commonly occurring in the frontal or temporal lobes.

Brain Aneurysm

A brain aneurysm is a weakened area of a cerebral artery that can enlarge and rupture. Rupture causes sudden severe headache, vomiting, neck stiffness, loss of consciousness, or neurologic deficits and is a medical emergency.

Brain Aneurysm Repair

A medical procedure involving brain aneurysm repair. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Brain Arteriovenous Malformation

Alternate index label; see *Brain AVM (arteriovenous malformation)*.

Brain AVM (Arteriovenous Malformation)

An abnormal tangle of brain arteries and veins that can disrupt blood flow and may bleed or cause seizures, headaches, or neurologic deficits.

Alternate Names:

Brain Arteriovenous Malformation

Brain Cancer

Brain cancer consists of malignant tumors arising in or affecting brain tissue and its coverings. Symptoms depend on location and may include seizures, headache, weakness, speech or vision changes, personality change, or raised intracranial pressure; diagnosis uses imaging and tissue characterization.

Alternate Names:

Tumor Brain Cancer (Brain Cancer)

Brain Concussion

A mild traumatic brain injury caused by a blow, jolt, or force that temporarily changes brain function. Symptoms may include headache, dizziness, confusion, memory difficulty, nausea, and sensitivity to light or noise, sometimes without loss of consciousness. Worsening headache, repeated vomiting, seizure, weakness, or increasing drowsiness requires urgent assessment.

Brain Death

The irreversible cessation of all brain and brainstem function, including consciousness and brainstem reflexes, with absent respiratory drive—a legal definition of death.

Brain Disorder

A clinical disorder or finding related to brain disorder; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Brain Edema

Swelling caused by excess fluid in brain tissue, which

can raise pressure inside the skull. Causes include head injury, stroke, infection, tumor, bleeding, and severe metabolic disturbances. Symptoms may include headache, vomiting, confusion, seizures, weakness, or reduced consciousness; severe cases are medical emergencies.

Brain Embolism

Alternate index label; see *Embolic stroke (brain embolism)*.

Brain Fornix

A C-shaped fibre tract of the limbic system connecting the hippocampus with the mammillary bodies and septal region; it contributes to memory circuitry.

Brain Haemorrhage

Brain haemorrhage is bleeding within or around the brain, caused by trauma, hypertension, vascular lesions, clotting problems, or other disease.

Brain Health

Brain health encompasses the ability to perform cognitive functions such as memory, learning, decision-making, and emotional regulation across the lifespan. Key factors include cardiovascular health, nutrition, sleep, social engagement, cognitive stimulation, and management of chronic conditions like hypertension and diabetes.

Brain Hemorrhage

A brain hemorrhage is bleeding within or around the brain, often from hypertension, aneurysm, vascular malformation, trauma, or blood-thinning treatment. Sudden headache, weakness, speech difficulty, vomiting, seizure, or reduced consciousness requires emergency evaluation.

Brain Herniation

Displacement of brain tissue across rigid intracranial compartments or through the foramen magnum due to mass effect or raised ICP. It can compress the brainstem and cranial nerves and is a life-threatening neurosurgical emergency.

Brain Imaging

Medical imaging used to examine brain structure or function, including CT, MRI, vascular imaging, and selected nuclear medicine techniques.

Alternate Names:

Neuroimaging; cerebral imaging.

Brain Injuries

Brain injuries range from mild concussions to severe traumatic brain injury caused by falls, motor vehicle accidents, sports impacts, or penetrating trauma. Symptoms depend on injury location and severity, including headache, confusion, memory loss, seizures, and altered consciousness.

Brain Injury

Damage to the brain caused by trauma, lack of oxygen, stroke, infection, toxins, or other disease. Effects depend on the location and severity of injury.

Brain Injury (Head Injury)

Alternate index label for Head Injury.

Brain Lesions

Areas of abnormal brain tissue caused by tumors, stroke, infection, inflammation, trauma, demyelination, or vascular disease; symptoms depend on the lesion's location and size.

Brain Lesions (Lesions On The Brain)

Alternate index label for Lesions on the Brain.

Brain Malformations

Brain malformations are structural abnormalities of the brain that develop before birth. They may result from genetic factors, disrupted fetal development, infection, or other prenatal influences and can affect neurologic function.

Brain Mapping

Methods used to identify the anatomical or functional organization of the brain, including neuroimaging, electrophysiology, and cortical stimulation during selected neurosurgical procedures.

Brain Metastases

Cancer deposits that have spread to the brain from another primary site. They may cause headache, seizures, or focal neurologic deficits.

Alternate Names:

Secondary Brain Cancer

Brain Metastasis

A deposit of cancer that has spread to the brain from another primary site. It may cause headache, seizures, or focal neurologic deficits; treatment depends on the number, location, and biology of the lesions.

Brain Monitoring

Brain monitoring during anesthesia assesses the depth of hypnosis and neurological function. Bispectral index (BIS) processes the frontal EEG into a number from 0 (isoelectric) to 100 (fully awake), with 40 to 60 recommended for general anesthesia. BIS monitoring reduces awareness risk and may reduce anesthetic requirements.

B-Type Natriuretic Peptide

A hormone made mainly by the heart's ventricles and released when the heart wall is under increased stress. A blood BNP test can help assess suspected heart failure, but results are affected by age, kidney function, body size, heart rhythm, and other illnesses and must be interpreted with other clinical findings. NT-proBNP is a related but separate peptide measured by a different test.

Alternate Names:

BNP; B-Type Natriuretic Peptide; Brain Natriuretic Peptide

Brain Natriuretic Peptide Test

A laboratory test that measures BNP or NT-proBNP, peptides released by ventricular cardiomyocytes in response to wall stress. Elevated levels support the diagnosis of heart failure and help guide treatment; results are interpreted with kidney function, age, and clinical context.

Brain Neoplasm

A neoplastic process affecting brain neoplasm; di-

agnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Brain PET Scan

A positron-emission tomography study that shows metabolic or molecular activity in the brain using a radioactive tracer.

Brain Sand

Brain sand is an older name for corpora arenacea, tiny calcified deposits commonly found in the pineal gland and sometimes other brain regions.

Brain SPECT

Cerebral perfusion imaging using Tc-99m HMPAO or ECD. Used for epilepsy localization, dementia differentiation, cerebrovascular disease.

Brain Stem

The lower part of the brain connecting the cerebrum with the spinal cord and controlling essential functions such as breathing and consciousness.

Brain Stem Glioma

A tumor arising from glial cells in the brainstem.

Brain Surgery

A medical procedure involving brain surgery. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Brain Tumor

An abnormal growth of cells within the brain or its surrounding structures, classified as primary (originating from brain tissue or its coverings) or secondary/metastatic (spread from a cancer elsewhere in the body, which are more common, accounting for approximately 50 percent of all brain tumors).

Brain Tumor - Children

A primary brain tumor is an abnormal growth arising in brain tissue or nearby structures. It may be benign or malignant and can cause neurologic symptoms through invasion or pressure.

Brain Tumor - Primary - Adults

A primary brain tumor is an abnormal growth arising in brain tissue or nearby structures. It may be benign or malignant and can cause neurologic symptoms through invasion or pressure.

Brain Tumour

A brain tumour is an abnormal growth in or near the brain, which may be benign or malignant and can cause seizures, headache, cognitive change, or focal deficits.

Brain Ventricles

The four interconnected fluid-filled cavities within the brain that produce and contain cerebrospinal fluid (CSF).

Brain Wave

An oscillating pattern of electrical activity generated by populations of neurons and recorded by electroencephalography; frequency and amplitude vary with sleep, alertness, and neurological disease.

Brain Waves

Electrical activity generated by the coordinated firing of neurons and recorded by electroencephalography.

Alternate Names:

Cerebral electrical activity; EEG rhythms.

Brainstem

Connects cerebrum to spinal cord, with midbrain (superior), pons (middle), and medulla oblongata (inferior). Contains cranial nerve nuclei (III-XII), ascending/descending tracts, and vital autonomic centers for cardiac, respiratory, and vasomotor function. Reticular formation regulates consciousness and sleep-wake cycles.

Brainstem Tumor

A tumor arising in the brainstem, the part of the brain that connects the cerebrum with the spinal cord.

Bramwell'S Disease

Bramwell's disease is an older eponym with variable usage, historically associated with cardiac or circulatory disease; the source context is required for precision.

Branchial

Branchial means relating to embryonic gill-arch structures or their derivatives in the head and neck.

Branchial Arch

One of the paired embryonic arches that contribute to development of facial, pharyngeal, laryngeal, muscular, skeletal, vascular, and neural structures of the head and neck.

Branchial Cleft

An embryonic groove between adjacent branchial arches. Persistence of cleft remnants can produce congenital branchial cysts, sinuses, or fistulas, most commonly along the lateral neck.

Branchial Cleft Anomaly

A congenital remnant of the branchial apparatus presenting as a cyst, sinus, or fistula in the head and neck. Recurrent infection or drainage from a persistent opening warrants imaging and specialist management.

Branchial Cleft Cyst

Congenital cystic mass arising from incomplete obliteration of branchial cleft apparatus during embryogenesis.

Branchial Cleft Cyst (Branchial Cyst)

Alternate index label for Branchial Cyst.

Branchial Cyst

A branchial cyst is a congenital, usually benign fluid-filled lesion along an embryonic branchial-cleft remnant in the neck. It may present as a painless mass or become infected; examination and imaging guide management, and new adult neck masses require evaluation.

Alternate Names:

Branchial Cleft Cyst (Branchial Cyst)

Branchio-Oto-Renal Syndrome

A clinical disorder or finding related to branchio-oto-renal syndrome; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Branchioma

A branchioma is a tumor or cyst arising from tissues related to embryonic branchial structures, often presenting in the neck.

Brass-Founder'S Ague

Brass-founder's ague is metal-fume fever, an influenza-like illness after inhalation of fumes from heated metals such as zinc or brass.

Brassicasterol

Brassicasterol is a plant sterol found especially in rapeseed and related oils and used as a marker of plant-oil composition.

Brassidic Acid

Brassidic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Braxton Hicks Contractions

Irregular, painless uterine contractions occurring throughout pregnancy, becoming more frequent in the third trimester. Usually last 30-60 seconds and are felt as a tightening of the abdomen. Distinguished from true labour contractions by their irregularity, lack of cervical change, and resolution with rest or hydration.

Alternate Names:

Contractions Braxton-Hicks (Braxton Hicks Contractions) Labor False (Braxton Hicks Contractions)

BRCA

BRCA1 and BRCA2 are tumor-suppressor genes involved in repair of DNA double-strand breaks. Pathogenic variants increase susceptibility to breast, ovarian, fallopian-tube, primary peritoneal,

prostate, and pancreatic cancers.

BRCA1 And BRCA2

Tumor-suppressor genes involved in homologous-recombination DNA repair. Germline pathogenic variants increase susceptibility to breast, ovarian, fallopian-tube, primary peritoneal, prostate, and pancreatic cancers and may guide surveillance and treatment.

BRCA1 And BRCA2 Gene Testing

Genetic testing for pathogenic variants in the BRCA1 and BRCA2 tumor-suppressor genes, which increase risk of breast, ovarian, fallopian tube, prostate, and pancreatic cancers. Positive results guide surveillance, risk-reducing surgery, and targeted therapy decisions.

Break The Bonds Of Emotional Eating

Emotional eating is eating in response to stress or other emotions rather than physical hunger. Management may include behavioral strategies and treatment of underlying conditions.

Breakbone Fever

A traditional name for dengue fever, reflecting the severe myalgia and arthralgia that may accompany the acute viral illness.

Breakthrough Infection

An infection that occurs after a person has received a vaccine or other preventive intervention intended to reduce that infection; severity and protection vary by pathogen, product, and host factors.

Breakthrough Pain

A temporary flare of severe pain that occurs despite otherwise controlled persistent pain.

Breast

A paired mammary organ composed of glandular tissue, ducts, fat, connective tissue, and skin; it produces milk after pregnancy and is a site of benign and malignant disease.

Breast Abscess

A localized collection of pus within the breast tissue, most commonly a complication of lactational mastitis. Presents as a fluctuant, painful mass with overlying erythema and systemic symptoms. Diagnosis by ultrasound. Treatment requires incision and drainage plus antibiotics.

Breast Anatomy

The breast contains skin, fat, lobules that produce milk, ducts, connective tissue, blood vessels, nerves, and lymphatic drainage, extending toward the axilla. Normal structure varies with age, hormones, pregnancy, and lactation; a new persistent lump or skin change needs assessment.

Breast Augmentation

Surgery or another procedure intended to increase breast volume or alter breast shape, commonly using an implant or transferred tissue.

Alternate Names:

Breast enlargement; augmentation mammoplasty.

Breast Augmentation Surgery

A medical procedure involving breast augmentation surgery. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Breast Biopsy - Stereotactic

Image-guided core or vacuum-assisted sampling of a breast abnormality, usually mammographic calcifications or a mass not well seen on ultrasound. X-ray coordinates guide the needle to the target.

Breast Biopsy - Ultrasound

Real-time ultrasound-guided core-needle or vacuum-assisted sampling of a breast lesion visible on ultrasound. It provides tissue for diagnosis while allowing needle placement to be monitored.

Breast Bone

The sternum, a flat bone in the center of the chest that joins the ribs through costal cartilage and helps protect the heart and lungs.

Breast Calcifications

Small calcium deposits in breast tissue seen on mammography; they are often benign, but some patterns require additional imaging or biopsy.

Breast Cancer

A malignant tumor arising in breast tissue. Risk, screening, and treatment depend on age, inherited and acquired biology, stage, and the person's circumstances.

Breast Cancer And Lymphedema

Breast-cancer treatment can impair lymph drainage and cause swelling, heaviness, tightness, or skin changes in the arm, breast, or chest. Early assessment, compression when appropriate, exercise, skin protection, and specialized therapy can limit progression and complications.

Breast Cancer Genetics

The study of inherited variants that increase breast-cancer risk. Important genes and syndromes include BRCA1, BRCA2, PALB2, TP53, PTEN, and CHEK2. Genetic counseling can help determine whether testing is appropriate.

Breast Cancer In Men

Breast cancer in men is a malignant tumor of male breast tissue, usually presenting as a painless lump or nipple change and requiring diagnostic imaging and biopsy.

Breast Cancer Prevention

Breast-cancer risk can be lowered but not eliminated through physical activity, healthy weight, limiting alcohol, avoiding tobacco, appropriate screening, and discussing inherited risk. Preventive medicines or surgery may be considered for people at substantially elevated risk.

Breast Cancer Recurrence

Breast-cancer recurrence is return of cancer after treatment, either near the original site, in regional nodes, or at distant sites. New lumps, bone pain, persistent cough, neurologic symptoms, or other concerning changes should be evaluated, but many

symptoms have noncancer causes.

Breast Cancer Screening

Testing used to detect breast cancer before symptoms develop, usually with mammography and, for selected patients, additional imaging.

Breast Cancer Staging

Breast cancer staging describes the extent of disease using tumor features, lymph-node involvement, and distant spread to guide prognosis and treatment.

Breast Cancer, Inflammatory

Alternate index label; see *Inflammatory breast cancer*.

Breast Cancer, Male

Alternate index label; see *Male breast cancer*.

Breast Carcinoma

A neoplastic process affecting breast carcinoma; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Breast Cyst

A fluid-filled sac within breast tissue, usually benign and sometimes tender or larger before menstruation. Examination and imaging distinguish a cyst from a solid lesion; persistent, growing, bloody, or newly discovered breast lumps should be assessed by a clinician.

Breast Cysts

Fluid-filled sacs within breast tissue that may be tender or palpable and are usually benign.

Breast Cytology

Cytologic evaluation of breast lesions using fine-needle aspiration, nipple discharge, or other samples. It can help distinguish benign, suspicious, and malignant processes but may require core biopsy for definitive classification.

Breast Density

The relative proportion of fibroglandular tissue com-

pared with fat in a breast on mammography. Higher density can make small cancers harder to see and is associated with an increased risk of breast cancer.

Breast Development

Breast development is growth and maturation of mammary tissue during puberty, pregnancy, lactation, and hormonal life stages. or cancer. Bloody, clear, spontaneous, one-sided, or persistent discharge, especially with a lump or skin change, requires prompt breast evaluation.

Breast Duct

A channel that carries milk from mammary lobules toward the nipple; ductal obstruction, inflammation, atypia, or carcinoma can produce a breast lump or nipple discharge.

Breast Engorgement

Overfilling of the breasts with milk, causing bilateral pain, firmness, warmth, and nipple flattening. Occurs 3-5 days postpartum. Pathophysiology: vascular congestion and edema before milk secretion is established.

Breast Enlargement In Males

Enlargement of male breast tissue, called gynecomastia, caused by glandular proliferation. Causes include hormonal changes, medicines, obesity, and systemic disease.

Breast Examination

Systematic inspection and palpation of the breasts and regional lymph nodes to assess masses, skin or nipple changes, pain, or other abnormalities.

Breast Feed

To feed an infant with human milk directly from the breast; breastfeeding provides nutrition and immune factors, and may be affected by maternal or infant medical conditions and medicines.

Breast Finding

Any abnormal or notable breast observation, such as a lump, pain, skin change, nipple inversion, or

discharge; assessment may require examination and imaging or biopsy.

Breast Health

Breast health involves routine self-awareness, clinical examinations, and mammographic screening to detect abnormalities such as lumps, pain, or skin changes that may indicate benign conditions or breast cancer. Risk factors include genetics, age, hormone exposure, and lifestyle factors.

Breast Infection

Infection and inflammation of breast tissue, most commonly mastitis during breastfeeding. It causes localized pain, warmth, redness, swelling, and sometimes fever; an untreated infection may progress to a breast abscess.

Breast Lift

A breast lift, or mastopexy, surgically reshapes and raises sagging breast tissue by removing excess skin and repositioning the nipple when necessary.

Breast Lobe

A major subdivision of the mammary gland containing groups of lobules and ducts that converge toward the nipple.

Breast Lobule

A functional unit of the mammary gland composed of milk-producing acini and their small ducts; lobules enlarge and differentiate during pregnancy and lactation.

Breast Lump

A localized thickening, mass, or swelling in breast tissue caused by a cyst, fibroadenoma, infection, trauma, or malignancy. New, persistent, hard, fixed, or changing lumps require clinical and often imaging assessment.

Breast Lump Removal

A medical procedure involving breast lump removal. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the

specific procedure.

Breast Lumps In Women

A breast lump may be a cyst, fibroadenoma, infection, hormonal change, trauma-related finding, or cancer. Clinical examination and age-appropriate imaging, with biopsy when indicated, determine the cause; a new hard, fixed, enlarging, or skin-associated lump needs prompt review.

Alternate Names:

Breast Lumps

Breast Milk

Milk produced by the mammary glands after childbirth, providing nutrients, immune factors, and other substances important for infant growth and development.

Breast Milk - Pumping And Storing

Guidance on expressing, storing, labeling, and handling breast milk to preserve safety and nutritional quality.

Breast Milk Jaundice

A benign, prolonged unconjugated hyperbilirubinaemia occurring in thriving, breastfed term infants, typically developing after the first week of life (peaking around 2 weeks and lasting up to 3–12 weeks). Believed to be caused by substances in human breast milk (including beta-glucuronidase, epidermal growth factor, and non-esterified fatty acids) that inhibit hepatic glucuronyl transferase and increase enterohepatic bilirubin circulation. Distinguished from breastfeeding jaundice (lactation failure jaundice) by adequate weight gain, normal hydration, and vigorous feeding.

Alternate Names:

Human Milk Jaundice; Late-Onset Neonatal Jaundice

Breast MRI

Magnetic resonance imaging used to evaluate breast tissue. Contrast may be administered when clinically indicated.

Breast MRI Scan

A magnetic resonance imaging study that produces detailed images of the specified body region using magnetic fields and radiofrequency energy.

Breast Neoplasm

A neoplastic process affecting breast neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Breast Pain

Pain or tenderness in one or both breasts, caused by hormonal changes, cysts, infection, trauma, medications, or musculoskeletal disease. Persistent focal pain or an associated lump requires assessment.

Alternate Names:

Pain, Breast

Breast Rash

A rash affecting breast skin; causes include dermatitis, infection, irritation, or, less commonly, inflammatory breast cancer. Persistent one-sided changes require evaluation.

Breast Reconstruction

Surgical restoration of breast shape after mastectomy or trauma, using an implant, the patient's own tissue, or a combination of techniques.

Breast Reconstruction - Implants

A medical procedure involving breast reconstruction - implants. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Breast Reconstruction - Natural Tissue

A medical procedure involving breast reconstruction - natural tissue. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Breast Reduction

Breast reduction, or reduction mammoplasty, removes breast tissue, fat, and skin to reduce breast size and related symptoms.

Breast Screening

Examination of people without breast symptoms to detect breast cancer at an earlier stage, most commonly using mammography according to age and risk-based programs.

Breast Self-Examination

Inspection and palpation of one's own breasts to notice new lumps or changes; it may improve awareness but is not a substitute for recommended population screening or clinical assessment.

Breast Skin And Nipple Changes

Changes in breast skin or the nipple may result from benign conditions, infection, hormonal change, or cancer. New, persistent, or one-sided changes require assessment.

Breast Surgeon

A surgeon who evaluates and treats breast conditions, including benign disease, breast cancer, and reconstructive or risk-reducing surgical needs.

Breast Surgery

Breast cancer surgery: breast-conserving therapy (BCT: lumpectomy + radiation) equivalent survival to mastectomy for early stage. Margins: "no ink on tumor" for invasive cancer; 2mm for DCIS.

Breast And Thoracic Cancer Surgery

Surgical treatment of cancers of the breast and chest, including breast tumors and selected cancers of the lungs, pleura, esophagus, mediastinum, or chest wall. Procedures may remove the primary tumor and involved lymph nodes, with reconstruction or additional treatment planned according to the cancer type and stage.

Alternate Names:

Breast And Thoracic Oncologic Surgery

Breast Ultrasound

Ultrasound imaging of the breast. Indications: evaluating palpable masses (distinguishing solid vs. cystic), characterizing mammographic abnormalities, guiding biopsy (core needle, fine needle, or vacuum-

assisted), evaluating implant integrity, and imaging young/dense breast tissue.

Breastfeeding

The feeding of an infant with breast milk, recommended exclusively for the first 6 months and continued with complementary foods to 2 years and beyond. Benefits: infant (nutrition, immune protection—IgA, reduced infection/allergy/obesity, bonding) and maternal (uterine involution, contraception, reduced breast/ovarian cancer).

Breastfeeding - Skin And Nipple Changes

Breastfeeding-related skin or nipple changes may result from friction, dermatitis, infection, or other causes and require assessment when persistent or painful.

Breastfeeding With Rheumatoid Arthritis

People with rheumatoid arthritis may be able to breastfeed, but disease activity, fatigue, hand function, and medicine safety must be considered. Some medicines are compatible and others require avoidance or timing changes; treatment should be planned with rheumatology and obstetric or pediatric clinicians.

Breath

Breath is air moved into and out of the lungs; its odor, volume, pattern, and gas composition can provide clinical information.

Breath Alcohol Test

A measurement of ethanol concentration in exhaled breath, used to estimate blood alcohol content for clinical assessment of intoxication or in medicolegal contexts. Results correlate with blood alcohol levels but may be affected by temperature, lung disease, and device calibration.

Breath Bad (Bad Breath)

Alternate index label for Bad Breath.

Breath Focus

A mindfulness or relaxation practice that uses attention to breathing to stabilize awareness and reduce physiological or emotional arousal.

Alternate Names:

Breath-focused meditation; breathing-focused mindfulness.

Breath Odor

Unpleasant breath odor, or halitosis, commonly results from oral disease, dry mouth, tobacco, diet, or infection.

Breath Sounds

The sounds produced by airflow through the respiratory tract during breathing, assessed by auscultation with a stethoscope over the chest wall.

Breath Stacking

A technique using a manual resuscitation bag or glosopharyngeal breathing to deliver successive breaths without exhaling, stacking tidal volumes to achieve higher lung volumes. It improves cough effectiveness and secretion clearance in patients with weak respiratory muscles.

Breath Test

Analysis of exhaled air to detect or measure a substance produced by the body or by intestinal microorganisms; examples include the urea breath test for *Helicobacter pylori* and hydrogen or methane testing for carbohydrate malabsorption and intestinal overgrowth.

Breath-Holding Spell

A brief involuntary episode in a young child, usually after crying or upset, involving apnea, color change, and rapid recovery.

Breath-Holding Spells

A benign paroxysmal phenomenon in young children (6 months to 6 years), provoked by frustration, pain, or fear. Cyanotic type (more common): child cries, forcefully exhales, becomes cyanotic, and may lose consciousness briefly.

Breathalyzer

A breathalyzer estimates blood alcohol concentration by measuring alcohol in exhaled air; results can be affected by device, timing, and sampling conditions.

Breathing

Movement of air into and out of the lungs, enabling oxygen uptake and carbon-dioxide removal through ventilation.

Breathing - Slowed Or Stopped

Slowed or absent breathing, called respiratory depression or apnea, can result from airway obstruction, medicines, neurologic disease, or severe illness and requires immediate assessment.

Breathing Difficulties - First Aid

Breathing difficulty, or dyspnea, is a subjective sensation of difficult or uncomfortable breathing that may result from lung, heart, blood, metabolic, or other disorders.

Breathing Difficulty

Breathing difficulty, or dyspnea, is a subjective sensation of difficult or uncomfortable breathing that may result from lung, heart, blood, metabolic, or other disorders.

Breathing Difficulty - Lying Down

Breathing difficulty, or dyspnea, is a subjective sensation of difficult or uncomfortable breathing that may result from lung, heart, blood, metabolic, or other disorders.

Breathing Exercises

Breathing exercises deliberately change respiratory pattern to improve ventilation, relaxation, airway clearance, breath control, or rehabilitation.

Breathlessness

Breathlessness is the subjective sensation of difficult or uncomfortable breathing, caused by pulmonary, cardiac, metabolic, neuromuscular, or psychological conditions.

Breathlessness (Dyspnoea)

The uncomfortable awareness of difficult, labored, or insufficient breathing. Causes include respiratory, cardiac, metabolic, and psychological disorders.

Alternate Names:

Dyspnoea

Breech Birth

A breech birth occurs when the fetus presents buttocks or feet first rather than the head. Delivery planning depends on fetal position, gestational age, and maternal and fetal factors.

Breech Presentation

A fetal presentation where the buttocks or feet are positioned to deliver first. Types include frank breech (hips flexed, knees extended, most common), complete breech (hips and knees flexed), and footling breech (one or both feet presenting). Associated with increased risk of cord prolapse, head entrapment, and birth trauma.

Brefeldin A

Brefeldin A is a fungal compound used experimentally to disrupt Golgi trafficking and intracellular protein transport.

Bregma

Bregma is the skull landmark where the coronal and sagittal sutures meet; in infants it corresponds to the anterior fontanelle.

Bremsstrahlung

Braking radiation produced when beta particles decelerate in matter. Continuous spectrum. Detected in I-131 imaging (poor quality). Used for Y-90 bremsstrahlung SPECT.

Brenner Tumor

Rare ovarian epithelial neoplasm (1-2% of ovarian tumors) composed of nests of transitional (urothelial-like) epithelial cells embedded in a dense fibromatous stroma. Most are unilateral, small (< 2 cm), benign (90%) and incidental; borderline (proliferative) and malignant variants are uncommon.

Brenner Tumour

A Brenner tumour is usually a benign ovarian epithelial tumor with transitional-cell-like nests; rare borderline and malignant forms occur.

Breslau Bacillus

Breslau bacillus is an older name for *Salmonella enterica* serovar Typhimurium, a bacterium associated with gastroenteritis and foodborne disease.

Breslow Thickness

Microstaging measurement of melanoma invasion depth from the top of the granular layer of the epidermis to the deepest point of tumor invasion, in millimeters. Most important prognostic factor in primary cutaneous melanoma, used in the AJCC TNM staging system (T category).

Bretylum Tosylate

Bretylum tosylate is an older antiarrhythmic drug that suppresses ventricular fibrillation and is now rarely used because of alternatives and adverse effects.

Brevibacillus

A genus of Gram-positive, spore-forming rod-shaped bacteria previously classified within *Bacillus*. Some species are found in soil and food; opportunistic infections in humans are rare but have been reported.

Brevibacterium

A genus of Gram-positive, rod-shaped bacteria found on skin, in soil, and in some food products. Most species are harmless, but rare opportunistic infections can occur.

Brevibacterium Casei

A Gram-positive bacterial species found in environmental and human-associated settings that rarely causes opportunistic bloodstream or device-related infection.

Brevundimonas

A biomedical term referring to *brevundimonas*; its classification, distinguishing features, and medical

significance depend on the clinical or biologic context.

Brevundimonas Diminuta

A biomedical term referring to *brevundimonas diminuta*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Brevundimonas Vesicularis

A biomedical term referring to *brevundimonas vesicularis*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Bricklayer'S Disease

Bricklayer's disease is an occupational lung condition historically associated with inhalation of silica-containing dust, potentially causing silicosis.

Bridge (Dental)

Fixed prosthetic replacing missing teeth. Retained by adjacent teeth (abutments). Requires preparation of abutments.

Alternate Names:

Dental

Bridge To Transplant

Temporary mechanical circulatory support used to maintain a patient with advanced heart failure until heart transplantation can be performed. The device and duration depend on ventricular function, organ status, complications, and transplant eligibility.

Bridge Work

Bridge work is a fixed dental prosthesis that replaces one or more missing teeth by attaching artificial teeth to adjacent teeth or implants.

Brief Psychotic Disorder

A disturbance characterized by the sudden onset of at least one psychotic symptom: delusions, hallucinations, disorganized speech, or grossly disorganized or catatonic behavior. The disturbance lasts at least one day but less than one month, after which the

individual fully returns to premorbid functioning.

Alternate Names:

Psychotic Disorder Brief (Brief Psychotic Disorder)

Brief Resolved Unexplained Event

A sudden, brief, and now-resolved episode in an infant involving abnormal breathing, skin color, muscle tone, or responsiveness, with no explanation after the initial evaluation. Risk assessment is based on age, history, examination findings, and recurrence; higher-risk events require targeted investigation.

Brief Resolved Unexplained Event - BRUE

A brief resolved unexplained event is a sudden, brief, and resolved episode in an infant involving altered breathing, color, muscle tone, or responsiveness without an identifiable cause after evaluation.

Bright'S Disease

A historical term encompassing several forms of kidney disease, particularly glomerular disorders associated with proteinuria, edema, and altered urinary findings. Modern diagnosis specifies the underlying renal lesion.

Brill'S Disease

A recrudescence form of epidemic typhus caused by reactivation of latent *Rickettsia prowazekii* infection years after the original illness. It is usually milder than the initial infection but can transmit to body lice.

Brill-Zinsser Disease

Brill-Zinsser disease is reactivation of latent epidemic typhus years after the original infection, causing a usually milder typhus-like illness.

Brimonidine Tartrate

Brimonidine tartrate is an alpha-2 adrenergic agonist used in eye drops to lower intraocular pressure and topically for some facial redness.

Brine Bath

A brine bath is immersion in salty water, used recre-

ationally or therapeutically; benefits and risks depend on skin condition, temperature, and patient health.

Brittle Asthma

An uncommon term for severe asthma with unusually unpredictable and difficult-to-control airflow or attacks despite appropriate treatment. The term is not a universally standardized diagnostic category; specialist assessment is required to identify triggers, alternative diagnoses, and effective management.

Brittle Bone Disease

Brittle bone disease is osteogenesis imperfecta, an inherited connective-tissue disorder causing fragile bones, recurrent fractures, and variable hearing, dental, or skeletal findings.

Brittle Bone Disease (Osteogenesis Imperfecta Brittle Bone Disease)

Alternate index label for Osteogenesis Imperfecta Brittle Bone Disease.

Brittle Diabetes

Labile, unstable type 1 diabetes mellitus characterized by unpredictable, large, recurrent swings in blood glucose between severe hyperglycemia and recurrent hypoglycemia (often with ketoacidosis) despite optimal adherence, significantly disrupting quality of life.

Brivanib

An investigational oral anticancer drug that inhibits vascular endothelial growth factor and fibroblast growth factor receptors. It has been studied in advanced cancers but is not broadly approved for routine clinical use.

Brivanib Alaninate

An oral prodrug of brivanib, an investigational inhibitor of vascular endothelial growth factor and fibroblast growth factor receptors studied in advanced cancers.

Brivudine

Brivudine is an antiviral drug used in some countries for herpes zoster; it has a dangerous interaction with fluoropyrimidine chemotherapy.

Broad Ligament

A double layer of peritoneum extending from the sides of the uterus to the lateral pelvic walls. It helps support the uterus, uterine tubes, and ovaries and contains important vessels, nerves, and connective tissue.

Broad Nasal Bridge

A broad nasal bridge is increased width of the upper nose. It may be a normal familial feature or part of a congenital or genetic syndrome.

Broca Aphasia

Acquired language disorder of expressive speech (motor aphasia) caused by lesion of the posterior inferior frontal gyrus (Broca area, Brodmann area 44/45) of the dominant (usually left) hemisphere, most commonly from middle cerebral artery (MCA) superior division infarction.

Broca'S Aphasia

A nonfluent language disorder caused by dysfunction of the dominant inferior frontal language region. Speech is effortful and reduced, while comprehension is relatively preserved compared with repetition and fluent language production.

Broca'S Area

A language-related region of the dominant inferior frontal gyrus involved in speech production, language formulation, and aspects of syntax. Lesions may cause expressive aphasia.

Brodie'S Abscess

Brodie's abscess is a localized, usually subacute form of bone infection, often in the metaphysis of a long bone, causing persistent pain and a cavity in bone.

Brody Disease

A rare inherited skeletal muscle channelopathy inherited in an autosomal recessive fashion (Brody disease) or autosomal dominant fashion (Brody syndrome), caused by loss-of-function mutations in the *ATP2A1* gene on chromosome 16p12.1, which encodes the fast-twitch skeletal muscle sarcoplasmic/endoplasmic reticulum calcium ATPase 1 (SERCA1). Impaired ATP-dependent reuptake of cytosolic calcium into the sarcoplasmic reticulum leads to delayed myofibrillar relaxation. Manifests with childhood-onset, painless exercise-induced muscle stiffness and delayed relaxation predominantly affecting the face, eyelids, and hands after rapid strenuous exertion, notably in the complete absence of electrical myotonic discharges on needle electromyography (pseudomyotonia).

Alternate Names:

Brody Myopathy; SERCA1-Related Muscle Disease

Broken Ankle

A fracture of one or more bones forming the ankle joint, usually causing pain, swelling, bruising, and difficulty bearing weight.

Broken Arm

A fracture of one or more bones of the arm, commonly causing pain, swelling, bruising, deformity, or loss of function.

Broken Blood Vessel In Eye

Alternate index label; see *Subconjunctival hemorrhage (broken blood vessel in eye)*.

Broken Bone

A broken bone, or fracture, is a partial or complete disruption of bone continuity caused by injury, repetitive stress, or weakened bone.

Broken Bone (Types Of Bone Fractures)

Alternate index label for Types of Bone Fractures.

Broken Clavicle

Alternate index label; see *Broken collarbone*.

Broken Collarbone

A fracture of the clavicle, usually caused by a fall, direct impact, or contact injury and associated with shoulder pain and reduced arm movement.

Alternate Names:

Broken Clavicle

Broken Finger

A fracture of one of the finger bones caused by trauma, crushing, or twisting. Pain, swelling, bruising, deformity, and reduced movement may occur. Assessment should check alignment, tendon function, sensation, and circulation; some fractures require reduction, splinting, or surgery.

Alternate Names:

Pinky Broken Finger (Broken Finger)

Broken Foot

A fracture of one or more bones of the foot, commonly caused by a fall, direct impact, twisting injury, or repetitive stress. Pain, swelling, bruising, and difficulty bearing weight are typical. Deformity, an open wound, numbness, or inability to walk requires urgent evaluation and imaging.

Broken Hand

A fracture of one or more bones of the hand, causing pain, swelling, bruising, deformity, or impaired movement.

Broken Heart Syndrome

A temporary weakening of the heart muscle, often triggered by severe emotional or physical stress. It can resemble a heart attack but usually occurs without an obstructing coronary clot.

Broken Hip

Alternate index label; see *Hip fracture*.

Broken Leg

A fracture of one or more bones in the leg, commonly causing pain, swelling, bruising, deformity, or inability to bear weight.

Alternate Names:

Leg Fracture

Broken Nose

A fracture of one or more nasal bones that may cause pain, swelling, bleeding, deformity, or difficulty breathing through the nose.

Broken Or Dislocated Jaw

A fracture or dislocation of the jaw can cause pain, swelling, malocclusion, limited movement, and difficulty speaking or swallowing.

Broken Or Knocked Out Tooth

A broken or avulsed tooth is a dental injury caused by trauma. An avulsed permanent tooth requires urgent dental care.

Broken Ribs

Fractures of one or more ribs, usually causing localized chest pain that worsens with breathing, coughing, or movement.

Broken Toe

A fracture of a toe bone, usually after direct impact or crushing. Pain, swelling, bruising, and tenderness are common; deformity, an open injury, numbness, or impaired circulation requires prompt assessment. Treatment may include protection and immobilization, with reduction or surgery for displaced or unstable fractures.

Alternate Names:

Toe Broken (Broken Toe)

Broken Wrist

A fracture involving one or more bones near the wrist, often causing pain, swelling, bruising, deformity, or reduced hand function.

Bromelain

A mixture of proteolytic enzymes derived from pineapple; it is marketed as a supplement and has been studied for inflammation and pain, but evidence and product standardization vary and interactions or allergy are possible.

Bromhexine

A mucolytic medicine that reduces the viscosity

of bronchial mucus and may help expectoration; gastrointestinal upset, rash, or rare serious skin reactions can occur.

Bromhidrosis

Excessive or unpleasant body odor produced when microbial metabolism acts on sweat or skin secretions. It is often prominent in the axillae or feet and may be associated with hyperhidrosis or local skin disease.

Bromides

Bromides are salts or compounds containing bromide; prolonged or excessive exposure can cause bromism with neurologic, psychiatric, gastrointestinal, or skin symptoms.

Bromine

A reactive halogen used in industrial chemicals and flame retardants; elemental bromine and many bromine compounds can irritate or burn tissues, and excessive exposure may cause systemic toxicity.

Bromism

Toxicity from prolonged or excessive exposure to bromide compounds, which may cause neurologic, psychiatric, gastrointestinal, or skin symptoms.

Bromobenzenes

A group of aromatic compounds in which one or more bromine atoms are attached to benzene; they are mainly industrial chemicals and may cause irritation or toxic effects depending on the compound and exposure.

Bromocriptine

An ergot-derived dopamine D2-receptor agonist used for hyperprolactinaemia and selected Parkinson disease or endocrine disorders; nausea, postural hypotension, hallucinations, and rare fibrotic complications may occur.

Bromodeoxyuridine

A thymidine analogue incorporated into newly synthesized DNA during the S phase; it is used ex-

perimentally to measure cell proliferation and can interfere with DNA replication at high exposure.

Bromoderma

A skin eruption or dermatitis attributed to exposure to bromine or bromide compounds. It may present with acneiform, pustular, or other inflammatory lesions.

Bromodosis

Bromodosis is unusually offensive foot odor caused by sweat, moisture, and bacterial or fungal breakdown of skin debris.

Bromophos

An organophosphorus insecticide; exposure can inhibit acetylcholinesterase and produce cholinergic poisoning, including sweating, salivation, bronchospasm, vomiting, weakness, and seizures.

Bromouracil

A thymine analogue used mainly in experimental mutagenesis research; its altered base pairing can introduce DNA replication errors.

Bromovirus

Bromovirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Brompheniramine Overdose

Toxic effects from excessive brompheniramine, an antihistamine, including drowsiness or agitation, confusion, dry mouth, rapid heart rate, seizures, or abnormal heart rhythm.

Bromthymol Blue

A pH indicator that changes colour near neutrality and is used in laboratory demonstrations and some analytical tests; it is not a therapeutic medicine.

Bronchi

The major airways conducting air from the trachea into the lungs. The trachea bifurcates at the carina

(T4-T5) into right and left main bronchi. The right main bronchus is wider, shorter, and more vertical—aspirated foreign bodies preferentially lodge here.

Bronchial

Relating to the bronchi, the large airways branching from the trachea; bronchial disease may cause cough, wheeze, mucus production, or airflow obstruction.

Bronchial Artery

An artery arising from the systemic circulation that supplies the walls of the bronchi, connective tissue of the lungs, and visceral pleura. It differs from the pulmonary arteries, which carry blood for gas exchange.

Bronchial Artery Embolization

A minimally invasive endovascular catheter-directed procedure used as the primary, life-saving intervention for massive or recurrent hemoptysis. Under fluoroscopic guidance via femoral or radial arterial access, diagnostic bronchial arteriography identifies bleeding, hypertrophied, tortuous bronchial or non-bronchial systemic arteries; embolization is performed using particulate agents (polyvinyl alcohol particles or microspheres) or metallic microcoils while meticulously avoiding non-target embolization of the anterior spinal artery (artery of Adamkiewicz).

Alternate Names:

BAE; Bronchial Embolization; Embolization of Bronchial Arteries

Bronchial Asthma

Bronchial asthma is a chronic inflammatory airway disorder with variable, usually reversible bronchoconstriction, wheeze, cough, chest tightness, and breathlessness.

Bronchial Asthma (Asthma Overview)

Alternate index label for Asthma Overview.

Bronchial Biopsy

Tissue sampling of the bronchial wall during bronchoscopy. It is used to diagnose inflammatory con-

ditions (sarcoidosis, eosinophilic bronchitis), infections, and malignancy. Endobronchial biopsy of visible lesions and transbronchial biopsy of parenchymal lesions provide complementary diagnostic information.

Bronchial Carcinoid

Pulmonary neuroendocrine tumor arising from Kulchitsky cells of the bronchial mucosa; subclassified as typical carcinoid (TC, generally lower grade, with fewer mitoses and no necrosis) and atypical carcinoid (AC, generally higher grade, with more mitoses or necrosis and greater risk of recurrence or metastasis).

Bronchial Fistula

An abnormal connection involving a bronchus and another structure, such as the pleural space, esophagus, or skin; it can cause persistent air leak, infection, coughing, or respiratory compromise.

Bronchial Hemorrhage

Bleeding from bronchial vessels or airways, usually presenting as haemoptysis; severe bleeding can obstruct the airway or cause respiratory failure.

Bronchial Obstruction

Partial or complete blockage of a bronchus by mucus, a foreign body, tumour, swelling, or external compression, causing impaired ventilation and sometimes distal collapse or infection.

Bronchial Stenosis

Narrowing of a bronchus from trauma, infection, inflammation, or prior intervention. It causes localized wheezing, recurrent infections, and impaired secretion clearance. Diagnosis is by bronchoscopy. Treatment includes balloon dilation, stenting, or surgical resection.

Bronchial Stricture

Abnormal narrowing of a bronchus, caused by scarring, inflammation, tumour, infection, or prior instrumentation, which may produce localized wheeze, cough, recurrent infection, or atelectasis.

Bronchial Tree

The branching airway system formed by the trachea, main bronchi, lobar and segmental bronchi, bronchioles, and terminal airways leading toward the alveoli.

Bronchial Tubes

The right and left main bronchi and their branches that conduct air from the trachea into the lungs.

Alternate Names:

Bronchi; bronchial tree when referring to the branching airway system.

Bronchial Vein

A vein draining portions of the bronchial walls and larger airways. It empties partly into the azygos system and partly into pulmonary veins, creating a small physiologic venous admixture.

Bronchiectasis

Permanent, abnormal dilation of bronchi caused by destruction of the bronchial wall from chronic airway inflammation and infection. Causes include CF, immune deficiency, primary ciliary dyskinesia, and post-infectious damage. Features include chronic cough, copious purulent sputum, and recurrent hemoptysis.

Bronchiectasis (Acquired, Congenital)

Alternate index label for Acquired, Congenital.

Bronchiole

Small airways (less than 2 mm diameter) without cartilage, branching from segmental bronchi to terminal bronchioles (conducting zone) to respiratory bronchioles (transitional, alveoli bud from walls) to alveolar ducts to alveolar sacs to alveoli.

Bronchiolectasis

Permanent abnormal dilation of bronchioles, usually caused by chronic inflammation, infection, or obstruction. It may accompany bronchiectasis and contribute to impaired mucus clearance and recurrent infection.

Bronchioles

Bronchioles are small branching airways that conduct air from the bronchi toward the alveoli and regulate airflow through smooth muscle.

Bronchiolitis

Lower respiratory tract infection causing inflammation and obstruction of small airways. Most common cause is respiratory syncytial virus. Affects infants under 12 months. Presents with wheezing, tachypnea, nasal flaring, and retractions. Diagnosis is clinical.

Bronchiolitis Obliterans

An uncommon, often irreversible obstructive small-airway disease caused by inflammation and fibrotic narrowing or obliteration of the bronchioles. Causes include severe infection, inhalational injury, autoimmune disease, and complications after lung or bone-marrow transplantation.

Bronchitis

Inflammation of the bronchial mucosa. Acute bronchitis is typically viral, presents with cough (often productive), and resolves spontaneously. Cough with sputum production for at least 3 months in 2 consecutive years, often due to smoking.

Bronchitis (Acute)

Alternate index label for Acute.

Bronchitis Chronic (Chronic Bronchitis)

Alternate index label for Chronic Bronchitis.

Bronchoalveolar Lavage

Diagnostic procedure during bronchoscopy where saline is instilled and aspirated. Analyzed for cell counts, microbiology, and cytology. Useful for *Pneumocystis pneumonia*, pulmonary alveolar proteinosis, and sarcoidosis diagnosis.

Bronchocele

Bronchocele is an abnormal dilation or mucus-filled expansion of a bronchus, often related to obstruction or congenital airway abnormality.

Bronchoconstriction

Narrowing of the bronchi due to contraction of airway smooth muscle, mucosal swelling, or mucus; it increases airway resistance and occurs in asthma, allergic reactions, and some irritant exposures.

Bronchoconstrictor Agents

Substances that narrow the airways by contracting bronchial smooth muscle or causing airway inflammation, including histamine, leukotrienes, irritant gases, and some medicines.

Bronchodilator

A medicine that relaxes airway muscle and widens the airways to improve breathing.

Bronchodilator Response

The change in spirometric airflow after administration of a short-acting bronchodilator. A significant response can support variable airflow limitation, but a negative response does not exclude asthma and a positive response is not specific to one disease.

Bronchodilator Reversibility

Improvement in airflow measurements after an inhaled bronchodilator. A measurable increase may support variable airway obstruction, but no single response proves or excludes asthma or chronic obstructive pulmonary disease.

Bronchoesophageal Fistula

Abnormal communication between the bronchial tree and the esophagus, most commonly acquired from malignancy, mediastinal infection, or tuberculous lymphadenitis. Congenital forms are associated with esophageal atresia.

Bronchogenic Carcinoma

A neoplastic process affecting bronchogenic carcinoma; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Bronchogenic Cyst

A congenital cyst derived from abnormal budding of the ventral foregut. It is typically found in the

middle mediastinum or near the carina. It is usually asymptomatic but may cause compression of adjacent structures. Treatment is surgical resection if symptomatic.

Bronchomalacia

Weakness and excessive collapsibility of the bronchial walls due to deficient cartilage or structural support; it may cause noisy breathing, cough, recurrent infections, or expiratory airway obstruction.

Bronchophony

Increased transmission of spoken sounds to the stethoscope over consolidated lung tissue. It is normally heard over the central airways. Excessive bronchophony over peripheral lung fields indicates consolidation. It is part of the constellation of physical examination findings in pneumonia.

Bronchopleural Fistula

An abnormal communication between the bronchial tree and pleural space. It is a complication of pneumonectomy, lung surgery, or empyema. It causes persistent air leak, subcutaneous emphysema, and respiratory compromise. Treatment includes chest tube drainage, bronchoscopic intervention, and surgical repair.

Bronchopneumonia

Patchy consolidation centered on bronchioles across multiple lobes. Caused by *S. aureus*, gram-negative bacteria, and *Pseudomonas*.

Bronchopulmonary Dysplasia

A chronic lung disease of premature infants, resulting from lung injury from mechanical ventilation, oxygen toxicity, and inflammation. Most commonly occurs in infants born at less than 28 weeks gestation with birth weight less than 1500 grams.

Bronchopulmonary Segments

Anatomically distinct subdivisions of a lung lobe, each supplied by a segmental bronchus and pulmonary artery branch and separated from neighboring segments by connective-tissue planes.

Bronchoscope

A thin, lighted instrument used to examine the trachea and bronchi and, when needed, obtain tissue or fluid samples.

Bronchoscopic Culture

Bronchoscopic culture examines respiratory specimens collected through bronchoscopy for bacteria, fungi, or mycobacteria when ordinary samples are inadequate.

Bronchoscopy

Endoscopic examination of the tracheobronchial tree. Flexible bronchoscopy: most common, for diagnosis (lung mass, persistent cough, hemoptysis, suspected infection, ILD) and therapy (foreign body removal, bronchial washing, biopsy, BAL). Rigid bronchoscopy: for massive hemoptysis, large foreign bodies, and stent placement.

Bronchospasm

Involuntary constriction of bronchial smooth muscle causing airway obstruction, wheezing, and increased airway pressures. Triggers include airway manipulation, volatile anesthetics (sevoflurane is bronchodilating; desflurane can cause bronchospasm), histamine-releasing drugs (morphine, atracurium), and reactive airway disease.

Bronchostenosis

Narrowing of a bronchus that restricts airflow. Causes include scarring, tumors, foreign bodies, inflammation, external compression, or postinfectious and postprocedural changes.

Bronchus

One of the two main branches of the trachea that conduct air into the lungs. The right main bronchus is wider, shorter, and more vertical than the left, making it more susceptible to aspiration. Each main bronchus divides into lobar bronchi (secondary), segmental bronchi (tertiary), and progressively smaller branches (bronchioles).

Bronchus Disease

A clinical disorder or finding related to bronchus dis-

ease; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Bronchus Neoplasm

A neoplastic process affecting bronchus neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Bronze Diabetes

Historical eponym for hereditary hemochromatosis (HFE-related HH, C282Y homozygous mutation), characterized by the classic triad of skin pigmentation ("bronze"), cirrhosis, and diabetes mellitus from pancreatic iron deposition. Also known as "Troisier syndrome" or pigment cirrhosis.

Alternate Names:

"Troisier syndrome" or pigment cirrhosis.

Brooke Disease

Brooke disease is a rare inherited skin disorder historically associated with widespread keratinization or follicular abnormalities; the precise eponym requires source context.

Brooke-Spiegler Syndrome

An inherited skin-tumor syndrome, usually caused by pathogenic variants in *CYLD*, characterized by multiple benign adnexal tumors including trichoepitheliomas, cylindromas, and spiradenomas.

Broselow Tape

A standardized, color-coded, length-based pediatric emergency tape used worldwide to rapidly calculate estimated weight, appropriate resuscitation drug dosages, defibrillation joules, fluid bolus volumes, and equipment sizes (such as endotracheal tube internal diameter and blade size) in critically ill or injured children without requiring calculations.

Alternate Names:

Broselow Pediatric Emergency Tape; Pediatric Length-Based Resuscitation Tape

Brow Presentation

A fetal presentation where the fetal head is in a military position between full flexion (vertex) and

hyperextension (face). The presenting diameter is the largest (occipitofrontal, 13.5 cm). Incidence: 1:1000-1:2000. Vaginal delivery is impossible with a persistent brow in a term fetus unless it converts to vertex or face.

Brown Fat

Adipose tissue rich in mitochondria that produces heat through uncoupled oxidative metabolism; it is prominent in infants and can be activated by cold exposure in adults.

Brown Lung Disease

An occupational airway disease associated with inhalation of cotton, flax, or hemp dust, classically causing byssinosis. Symptoms may include chest tightness, cough, wheezing, and work-related airflow limitation.

Brown Recluse (Spider Bites (Black Widow And Brown Recluse))

Alternate index label for Spider Bites (Black Widow and Brown Recluse).

Brown Recluse Spider

The brown recluse spider is a venomous spider whose bite may cause local tissue injury and, rarely, systemic illness.

Brown Sequard Syndrome

Brown-Sequard syndrome (hemisection of the spinal cord) results from a lesion affecting one half of the spinal cord, producing a characteristic pattern of ipsilateral and contralateral neurological deficits.

Brown Stain

Extrinsic staining from chromogenic bacteria, diet, tobacco. Removed by prophylaxis.

Brown Teeth

Brown teeth are teeth with brown discoloration from staining, decay, developmental enamel defects, fluorosis, medicines, trauma, or other causes.

Brown-SéQuard Syndrome

Brown-Séquard syndrome follows hemisection or asymmetric injury of the spinal cord, causing ipsilateral weakness and loss of vibration with contralateral pain and temperature loss below the lesion.

Bruce Protocol

A standardized graded treadmill exercise test in which speed and incline increase at set intervals to assess exercise capacity and cardiovascular response.

Brucella

A genus of bacteria that causes brucellosis, usually acquired from infected animals, animal tissues, or unpasteurized dairy products.

Brucella Abortus

A Brucella species mainly associated with cattle that can cause brucellosis in people exposed to infected animals or unpasteurized dairy products.

Brucella Canis

A Brucella species associated with dogs that can rarely cause human brucellosis after contact with infected animals or reproductive material.

Brucella Melitensis

A highly pathogenic Brucella species associated with goats and sheep and a common cause of severe human brucellosis.

Brucella Suis

A Brucella species associated with pigs and some wildlife that can cause human brucellosis, sometimes with focal bone or organ infection.

Brucellaceae

A family of gram-negative bacteria that includes Brucella and related animal-associated organisms; Brucella species cause brucellosis.

Brucellosis

A zoonotic infection caused by Brucella bacteria, transmitted through contact with infected animals or consumption of unpasteurized animal products.

It commonly causes fever, sweats, fatigue, and joint or muscle pain.

Alternate Names:

Bruch's Membrane

Bruch's membrane is a thin extracellular matrix layer between the retinal pigment epithelium and choroid that supports retinal metabolism and transport.

Bruck's Syndrome

Bruck syndrome is a rare inherited disorder combining bone fragility resembling osteogenesis imperfecta with congenital joint contractures.

Brudzinski Sign

Flexion of the hips and knees that occurs involuntarily when the neck is passively flexed. It is a meningeal irritation sign that may occur in meningitis but lacks sufficient sensitivity or specificity to establish the diagnosis alone.

Brugada Phenocopy

A reversible clinical and electrocardiographic entity characterized by an ECG pattern identical to that of type 1 or type 2 Brugada syndrome (coved or saddleback ST-segment elevation ≥ 2 mm in right precordial leads V_1 – V_3), but provoked by identifiable, transient, underlying mechanical, metabolic, toxic, or ischemic conditions rather than true intrinsic genetic cardiac channelopathies. Common etiologies include severe hyperkalemia, acute anterior/inferior myocardial ischemia or infarction, pulmonary embolism, hypothermia, acute pericarditis, tricyclic antidepressant toxicity, and pectus excavatum. Differentiated from congenital Brugada syndrome by the complete normalization of the ECG upon resolution of the underlying condition, absence of personal or family history of syncope or sudden death, and negative sodium channel blocker (ajmaline/procainamide) provocation challenge.

Alternate Names:

Acquired Brugada Pattern; BrP

Brugada Syndrome

A genetic channelopathy causing abnormal sodium channel function (SCN5A mutation). ECG pattern: coved ST elevation >2 mm in V_1 – V_3 (type 1) followed by negative T wave. Patients are at risk for ventricular fibrillation and sudden cardiac death, often at rest or during sleep. More common in young Southeast Asian men.

Brugia

Brugia is a filarial nematode genus that includes parasites causing lymphatic filariasis, transmitted by mosquitoes.

Bruise

A contusion: localized hemorrhage into tissues due to blunt force trauma rupturing small blood vessels, with intact skin.

Bruised Rib Care

Care for a bruised rib includes pain control, normal gentle breathing, activity modification, and assessment for shortness of breath or other chest injury.

Bruises

A bruise is bleeding beneath the skin after small blood vessels are damaged. Frequent, large, spontaneous, painful, or unusually persistent bruises may reflect medicines, platelet or clotting disorders, liver disease, nutritional deficiency, or injury and should be assessed.

Bruising

Ecchymosis/contusion: a purplish-blue discoloration from blood extravasation into skin/subcutaneous tissue following trauma; color changes with age (red-blue purple green/yellow—bilirubin breakdown).

Bruit

An abnormal sound heard over a blood vessel, usually with a stethoscope, caused by turbulent blood flow.

Brunhilde Virus

A strain of poliovirus type 1 used in research and vaccine studies; poliovirus can invade the nervous system and cause paralytic disease.

Brunner Gland

A mucus-secreting gland in the duodenum that releases alkaline secretions to help protect the intestinal lining and neutralize stomach acid.

Brunner'S Gland

A mucus-secreting gland in the submucosa of the duodenum. Its alkaline secretion helps protect the mucosa from gastric acid and provides conditions favorable for intestinal enzymes.

Brunner'S Glands

Brunner's glands are mucus- and bicarbonate-secreting glands in the duodenum that help protect the intestinal lining from gastric acid.

Brush Burns

Brush burns are superficial friction burns caused by scraping skin against a rough surface, producing pain, redness, and abrasions.

Brushfield'S Spots

Small gray or white spots arranged around the iris, commonly associated with Down syndrome but also found in some people without it.

Brushing Your Child'S Teeth

Brushing a child's teeth removes plaque and helps prevent dental caries and gum disease. Fluoride toothpaste and age-appropriate supervision are recommended.

Bruxism

Grinding or clenching teeth, often during sleep. Causes: stress, occlusion, sleep disorders. Consequences: tooth wear, fractures, muscle pain, TMJ disorders. Treatment: occlusal guard, stress management, behavior modification.

Alternate Names:

Teeth Grinding (Bruxism)

Bruxism (Teeth Grinding (Bruxism))

Alternate index label for Teeth Grinding (Bruxism).

Bryant'S Traction

Bryant's traction is a historical traction technique for treating selected pediatric femoral fractures or hip conditions by elevating the child's legs; it is used cautiously.

Bryostatin

A group of marine-derived macrocyclic compounds that modulate protein kinase C. Bryostatin compounds have been investigated as experimental treatments for cancer and neurologic disease.

Biosafety Level

Biosafety level is a classification of laboratory containment practices and facilities required for work with biological agents of different risks.

Bubble Bath Soap Poisoning

Illness after swallowing or inhaling bubble-bath products. Small ingestions usually cause mild mouth or stomach irritation, but vomiting or aspiration can require medical assessment.

Bubo

A tender, enlarged lymph node, usually in the groin, armpit, or neck, caused by infection. Buboes are classically associated with plague but occur in other infections.

Bubonic Plague

Bubonic plague is infection with *Yersinia pestis* causing painful swollen lymph nodes, fever, and systemic illness, usually after a flea bite.

Buccal

Buccal means relating to the cheek or the inner cheek surface; buccal medicines are placed between cheek and gum for absorption.

Buccal Gland

A minor salivary gland located in the cheek. It secretes mucus into the oral cavity to lubricate the

buccal mucosa and assist formation and handling of the food bolus.

Buccal Mucosa

The moist mucous membrane lining the inside of the cheeks; it is examined for ulcers, infection, pigmentation, trauma, and oral cancer and can absorb some medicines.

Buccal Smear

A laboratory examination of cells collected from the specified site to identify abnormal, infectious, or malignant changes.

Buccalmucosa

A spelling variant of buccal mucosa, the moist lining of the inner cheeks.

Buccinator

The buccinator is a cheek muscle that compresses the cheek and assists chewing, sucking, blowing, and speech.

Buchanan'S Disease

Buchanan's disease is a historical eponym with inconsistent usage; its intended diagnosis must be established from the original clinical source.

Buckle Fracture

Alternate name; see *Torus Fracture*.

Budd-Chiari Syndrome

Obstruction of hepatic venous outflow at any level from small hepatic venules to the inferior vena cava junction with the right atrium. The most common predisposing condition is myeloproliferative disorders accounting for forty to fifty percent of cases. Other risk factors include thrombophilia and oral contraceptive use.

Budesonide

A corticosteroid used by inhalation for asthma or chronic obstructive pulmonary disease and by oral or rectal preparations for selected inflammatory bowel disorders; local and systemic corticosteroid adverse

effects are possible.

Budvicia

A genus of Gram-negative rod-shaped bacteria in the family Yersiniaceae, originally isolated from water sources. *Budvicia aquatica* has been occasionally associated with urinary tract infections and other clinical isolates.

Budvicia Aquatica

A Gram-negative rod bacterium in the family Yersiniaceae, isolated from aquatic environments. It has been reported as an occasional cause of urinary tract infection, particularly in catheterized patients.

Buerger Disease

An inflammatory disorder of small and medium arteries and veins, strongly associated with tobacco use, that can cause limb ischemia and tissue loss.

Buerger'S Disease

Thromboangiitis obliterans, a nonatherosclerotic inflammatory disease of small and medium arteries and veins, strongly associated with tobacco exposure. It causes distal ischemia, pain, ulcers, and sometimes gangrene.

Buffalo Hump

A buffalo hump is a pad of fat at the back of the neck and upper back, associated with obesity, corticosteroid excess, HIV therapy, or Cushing syndrome.

Buffer

A solution that resists pH change upon addition of acid or base, composed of a weak acid and its conjugate base (or weak base and conjugate acid). Buffer capacity is maximal when $\text{pH} = \text{pKa} \pm 1$.

Buffy Coat

The thin layer of white blood cells and platelets that forms between plasma and red blood cells after centrifugation of whole blood.

Bufonidae

The true toads, a family of anuran amphibians.

Some species produce bufotoxins, potent cardiotoxic substances secreted from parotoid glands, which can cause poisoning if ingested by predators or humans.

Buformin

Buformin is a former biguanide diabetes drug withdrawn or restricted because of lactic-acidosis risk.

Bug Spray Poisoning

Toxic effects from exposure to an insecticide product. Symptoms depend on the active ingredient and may include skin or eye irritation, nausea, breathing problems, abnormal sweating, confusion, seizures, or other neurologic effects.

Bulbar

Relating to the medulla oblongata or to functions controlled by its cranial-nerve nuclei, including swallowing, speech, and movements of the tongue and palate.

Bulbar Conjunctiva

The bulbar conjunctiva is the portion of the conjunctival membrane that covers the anterior surface of the sclera, extending from the limbus (the junction of the cornea and sclera) to the fornices where it reflects onto the inner surface of the eyelids.

Bulbar Palsy

Bulbar palsy is weakness or paralysis of muscles supplied by lower cranial nerves, causing dysarthria, dysphagia, impaired gag, and difficulty handling secretions.

Bulbar Paralysis

Weakness or paralysis of muscles supplied by the lower cranial nerves, causing dysarthria, dysphagia, impaired gag or cough, and sometimes respiratory compromise. Causes include brainstem disease, motor-neuron disease, and neuromuscular disorders.

Bulbocavernosus

A paired perineal muscle surrounding the bulb of the penis in males and the vestibular bulbs in females. It assists in emptying the urethra after urination,

contributes to erectile rigidity, and contracts during orgasm.

Bulbospongiosus

Bulbospongiosus is the paired perineal muscle that helps empty the urethra and contributes to erection and expulsion of genital secretions.

Bulbourethral Gland

One of two small male reproductive glands below the prostate that secrete lubricating, alkaline pre-ejaculate into the urethra.

Bulbourethral Glands

Paired male accessory glands, also called Cowper glands, that release a mucus-like fluid into the urethra during sexual arousal. The fluid lubricates the urethra and may help neutralize acidity.

Bulging

Protruding or swelling outward beyond the usual contour, as in a bulging intervertebral disk or bulging membrane.

Bulimia

An eating-disorder symptom characterized by recurrent episodes of binge eating with a sense of loss of control; in bulimia nervosa, episodes are followed by recurrent compensatory behaviours such as vomiting, fasting, or excessive exercise.

Bulimia Nervosa

An eating disorder characterized by recurrent episodes of binge eating followed by compensatory behaviors (purging, laxatives, excessive exercise). Self-esteem overly influenced by body shape and weight.

Alternate Names:

Eating Disorders, Bulimia

Bulla

A bulla is a large fluid-filled blister or a hollow air-filled structure, depending on the clinical context.

Bullae

Large, fluid-filled blisters that form within or beneath the skin or mucous membrane. Causes include friction, burns, infection, allergy, and autoimmune disease.

Bullectomy

A surgical procedure involving the excision of one or more large emphysematous bullae (subpleural air-filled spaces > 1 cm in diameter within the lung parenchyma), performed via video-assisted thoracoscopic surgery or thoracotomy. The procedure is indicated for giant bullae (occupying more than one-third of the hemithorax) causing severe compressive atelectasis of adjacent functional lung tissue, refractory dyspnea, or recurrent secondary spontaneous pneumothorax; removal decompresses the underlying normal lung and substantially improves ventilation mechanics.

Alternate Names:

Excision of Pulmonary Bullae; Resection of Giant Bulla

Bullet Track

The path a projectile traverses through the body, defined by the entry wound, internal course (through organs, vessels, bones), and exit wound (or retained bullet).

Bullet Wounds

Bullet wounds are penetrating or perforating injuries caused by firearm projectiles, potentially damaging skin, organs, vessels, nerves, and bone.

Bullous

Characterized by bullae, which are large, raised, fluid-filled blisters; bullous disorders may result from infection, burns, drugs, or autoimmune disease.

Bullous Impetigo

A superficial staphylococcal skin infection caused by exfoliative toxins that produce fragile, flaccid bullae and honey-coloured or varnish-like crusts, especially in infants and young children.

Bullous Pemphigoid

An autoimmune subepidermal blistering disease affecting elderly patients. It presents with tense blisters on erythematous or normal-appearing skin, often preceded by pruritic urticarial plaques. Autoantibodies target BP180 and BP230 (hemidesmosome proteins).

Alternate Names:

Pemphigoid Bullous (Bullous Pemphigoid)

Bully

Bullying is repeated harmful aggression or intimidation involving a power imbalance and can cause injury, anxiety, depression, and other health effects.

Bullying

Bullying is repeated aggression or intimidation involving a power imbalance and can be physical, verbal, social, or digital. It is associated with anxiety, depression, sleep problems, injury, school avoidance, and self-harm risk; safety planning and adult intervention are important.

Bumetanide

A potent loop diuretic inhibiting the Na-K-2Cl co-transporter in the thick ascending limb of the loop of Henle. Used for edema in heart failure, cirrhosis, and renal disease, and for hypertension in renal impairment. Approximately 40 times more potent than furosemide; more bioavailable orally, so equivalent IV-to-oral dosing is possible.

Bumps

Raised or palpable areas on the skin or under the skin; the cause may range from folliculitis and cysts to trauma, allergy, infection, or tumors.

BUN - Blood Test

A laboratory test that measures blood urea nitrogen concentration, a breakdown product of protein metabolism cleared by the kidneys. Elevated BUN indicates impaired renal function, dehydration, high protein intake, or gastrointestinal bleeding; it is interpreted alongside creatinine.

Bundle Branch Block

Delay or block in the electrical conduction through the bundle branches. Left Bundle Branch Block (LBBB): QRS >120 ms, broad notched R waves in lateral leads (I, V5-V6), deep S waves in V1-V2. Right Bundle Branch Block (RBBB): QRS >120 ms, rSR' pattern in V1-V2, wide S waves in I, V5-V6.

Bundle Of His

A narrow band of specialized cardiac tissue that conducts electrical impulses from the AV node through the fibrous cardiac skeleton to the interventricular septum. It divides into the right and left bundle branches. The bundle of His is the only electrical connection between atria and ventricles through the fibrous skeleton.

Bundle-Branch Block

Delayed or interrupted electrical conduction through one of the heart's ventricular bundle branches.

Bungarotoxins

Neurotoxins from kraits (*Bungarus* species) that bind or interfere with nicotinic acetylcholine receptors at the neuromuscular junction; envenomation can cause progressive paralysis and respiratory failure.

Bunion

A bony bump at the joint at the base of the big toe, also called hallux valgus. The big toe angles toward the second toe, causing the bump to form on the inner side of the foot. It may cause pain, redness, swelling, or stiffness.

Bunion Removal

A medical procedure involving bunion removal. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Bunionectomy

Surgical correction of hallux valgus (bunion) deformity, indicated for symptomatic bunions refractory to conservative measures (proper footwear, orthoses, toe spacers, NSAIDs). Procedure chosen based on

severity (hallux valgus angle and intermetatarsal angle) and patient factors (age, activity, comorbidities).

Bunionette

A painful prominence at the outer side of the fifth metatarsophalangeal joint, often associated with footwear pressure or altered foot mechanics.

Alternate Names:

Tailor's bunion; bunion of the fifth toe.

Bunions

A bony prominence and deformity at the base of the big toe caused by shifting of the joint, often producing pain and footwear difficulty.

Bunionoplasty

Bunionoplasty is surgery to correct a bunion or hallux valgus deformity, often by realigning bone and balancing surrounding soft tissues.

Bunyaviridae

Bunyaviridae is a virus or viral group classified by its genetic and biologic features; clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Bunyaviruses

Alternate index label; see *Viral hemorrhagic fevers*.

Buphthalmos

Enlargement of the eye caused by elevated intraocular pressure during early childhood, when the ocular coats are still distensible. It is classically associated with congenital or infantile glaucoma.

Bupivacaine

A long-acting amide local anesthetic used for regional, nerve-block, epidural, and local anesthesia. Excessive blood levels can cause seizures, severe cardiac toxicity, or cardiac arrest.

Bupivacaine Hydrochloride

The hydrochloride salt formulation of bupivacaine, a long-acting amide local anesthetic used for regional

and local anesthesia. Accidental intravascular exposure can cause serious neurologic or cardiac toxicity.

Buprenorphine

Buprenorphine is a partial opioid agonist used for opioid-use disorder and pain; it can cause dependence but has a ceiling effect on respiratory depression.

Bupropion

An antidepressant also used to help people stop smoking. It affects norepinephrine and dopamine signaling and can cause insomnia, dry mouth, or anxiety; it increases seizure risk and is contraindicated in people with certain seizure or eating disorders.

Burkholderia

A genus of environmental gram-negative bacteria that includes organisms causing melioidosis, glanders, and opportunistic infections in people with chronic lung disease or impaired immunity.

Burkholderia Cenocepacia

A member of the *B. cepacia* complex that can cause severe, difficult-to-treat respiratory infection in people with cystic fibrosis and other chronic lung disease.

Burkholderia Cepacia

A gram-negative environmental bacterium and member of the *B. cepacia* complex that can cause opportunistic lung, bloodstream, or device-associated infection.

Burkholderia Gladioli

An environmental bacterium that can cause opportunistic respiratory, bloodstream, or wound infection, especially in people with cystic fibrosis or impaired immunity.

Burkholderia Mallei

The bacterium that causes glanders, a zoonotic infection mainly affecting horses and related animals and occasionally transmitted to humans.

Burkholderia Pseudomallei

The bacterium that causes melioidosis, an infection acquired from contaminated soil or water and capable of causing pneumonia, abscesses, bloodstream infection, or chronic disease.

Burkholderiaceae

A family of gram-negative bacteria that includes *Burkholderia* and related environmental organisms, some of which cause serious human or animal infections.

Burkitt Lymphoma

A highly aggressive B-cell lymphoma characterized by MYC translocation (t(8;14) IGH-MYC, t(8;22) IGL-MYC, t(2;8) IGK-MYC) and a "starry sky" morphology (tingible body macrophages). Three variants: endemic (African, EBV+, jaw/facial bones), sporadic (abdominal, ileocecal), immunodeficiency-associated (HIV).

Alternate Names:

Burkitt's Tumor

Burkitt'S Lymphoma

A rapidly growing type of non-Hodgkin lymphoma arising from B lymphocytes.

Burkitt'S Tumor

Alternate index label; see *Burkitt lymphoma*.

Burn

An injury to skin or deeper tissue caused by heat, chemicals, electricity, radiation, or friction. Severity depends on depth, extent, location, and associated injury.

Burn (Oral)

Thermal or chemical injury to the lining of the mouth, often causing pain, redness, or ulceration. Severity depends on the substance, temperature, depth, and extent of injury.

Alternate Names:

Oral

Burn Injury

Alternate index label; see *Burns*.

Burn Surgery

Burn assessment: rule of 9s (adult), Lund-Browder (children). Depth: superficial (epidermis), superficial partial (dermis, blanch, blister, painful), deep partial (deeper dermis, white, sluggish capillary refill, less painful), full thickness (all dermis, white/charred, anesthetic).

Burning Feet

Burning, tingling, or painful sensations in the feet, commonly related to peripheral neuropathy, diabetes, nutritional deficiency, alcohol, medicines, or nerve compression.

Burning Mouth Syndrome

A chronic oral pain disorder causing daily burning, scalding, or tingling discomfort in the mouth, often involving the tongue, lips, or palate, usually with a normal-appearing mucosa. Altered taste or dry mouth may occur. It may be primary, with no identifiable cause, or secondary to conditions or exposures such as nutritional deficiency, oral infection, medication effects, allergy, diabetes, or hormonal change; evaluation should exclude treatable causes.

Alternate Names:

Scalded Mouth Syndrome Stomatodynia

Burning Thigh Pain

Alternate index label; see *Meralgia paresthetica*.

Burnout

Burnout is occupational exhaustion, mental distancing or cynicism, and reduced professional efficacy caused by chronic workplace stress; it is distinct from a medical diagnosis of depression.

Burns

Tissue injuries caused by heat, chemicals, electricity, radiation, or friction, with severity determined by depth, size, and location.

Alternate Names:

Burn Injury Burns First Aid (Burns)

Burns First Aid (Burns)

Alternate index label for Burns.

Burnt-Out Joint

A burnt-out joint is an older description of a joint in which active inflammatory disease has subsided but permanent structural damage and restricted movement remain.

Burroughs

Burroughs is a surname or eponym used inconsistently in medical sources; the associated condition or person must be identified from context.

Bursa

A small fluid-filled sac that reduces friction between moving tissues, such as a tendon and bone or skin and bone, especially near a joint. The plural is *bursae*; inflammation of a bursa is called bursitis.

Alternate Names:

Bursae Synovial bursae

Bursa Inflammation (Acute And Chronic Bursitis)

Alternate index label for Acute and Chronic Bursitis.

Bursa Of Fabricius

The bursa of Fabricius is a lymphoid organ in birds where B lymphocytes mature; it has no direct human counterpart.

Bursectomy

Surgical excision of a bursa, usually performed for persistent painful bursitis, recurrent infection, or another structural problem that has not responded to appropriate conservative management.

Bursitis

Inflammation of a bursa, a fluid-filled sac that reduces friction between tendons, muscles, bones, and skin. Bursae are lined by synovial cells and normally contain a thin layer of lubricating synovial fluid.

Bursitis Calcific (Calcific Bursitis)

Alternate index label for Calcific Bursitis.

Bursitis Of The Heel

Heel bursitis is inflammation of a fluid-filled bursa near the heel, producing localized tenderness and pain with pressure or walking.

Bursitis Of The Knee

Alternate index label; see *Knee bursitis*.

Bursitis Shoulder (Shoulder Bursitis)

Alternate index label for Shoulder Bursitis.

Bursopathy

Disease of a bursa, a fluid-filled sac that reduces friction around joints; bursopathy includes inflammation, infection, bleeding, or degeneration.

Buruli Ulcer

A skin infection caused by the bacterium *Mycobacterium ulcerans*. It mainly affects the skin, soft tissue, and sometimes bone, usually on the arms or legs. It often starts as a painless nodule and may develop into a large ulcer. Early diagnosis and treatment with antibiotics can cure it, but poorly managed disease may cause severe scars, deformities, and disability.

Burundanga (Hyoscine Hydrobromide)

A name associated with scopolamine, or hyoscine, an anticholinergic drug that can cause dry mouth, blurred vision, confusion, drowsiness, and memory impairment; exposure can be dangerous.

Buserelin

Buserelin is a gonadotropin-releasing hormone analogue that suppresses reproductive hormones after an initial stimulation phase and is used in selected fertility or hormone-sensitive conditions.

Buspirone

Buspirone is a nonbenzodiazepine anxiolytic used mainly for generalized anxiety and does not provide immediate relief like sedative medicines.

Busulfan

An alkylating chemotherapy medicine used chiefly as conditioning before haematopoietic stem-cell transplantation and, less commonly, for chronic myeloid leukaemia; it can cause prolonged myelosuppression, seizures, infertility, pulmonary injury, and hepatic veno-occlusive disease.

Butanol

Butanol is a flammable alcohol solvent; ingestion or substantial exposure can irritate tissues and depress the central nervous system.

Butazolidin Overdose

Toxic effects from excessive phenylbutazone, an older nonsteroidal anti-inflammatory drug. Severe exposure may cause gastrointestinal bleeding, kidney injury, bone-marrow suppression, metabolic disturbances, or seizures.

Buthionine Sulfoximine

An experimental compound that inhibits glutathione synthesis by blocking gamma-glutamylcysteine synthetase. It is used mainly in laboratory research to study oxidative stress and how glutathione affects drug or cancer-cell responses.

Butocarboxim

Butocarboxim is a carbamate insecticide that can inhibit acetylcholinesterase and cause cholinergic toxicity after exposure.

Butter, Margarine, And Cooking Oils

Butter, margarine, and cooking oils are dietary fat sources. Their nutritional effects depend on portion size, fatty-acid composition, and the overall eating pattern.

Buttiauxella

A genus of environmental gram-negative bacteria in the Enterobacterales; human infection is uncommon and usually opportunistic.

Buttiauxella Agrestis

A biomedical term referring to *buttiauxella agrestis*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Buttiauxella Ferragutiae

A biomedical term referring to *buttiauxella ferragutiae*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Buttiauxella Gaviniae

A biomedical term referring to *buttiauxella gaviniae*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Buttiauxella Noackiae

A biomedical term referring to *buttiauxella noackiae*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Buttock

The buttock is the rounded posterior region overlying the gluteal muscles, used for movement, sitting, examination, and selected injections.

Buttocks

The paired fleshy regions overlying the gluteal muscles at the back of the pelvis, formed mainly by muscle and subcutaneous fat.

Button Batteries

Button batteries are small batteries that can cause severe tissue injury if swallowed or inserted into the nose or ear. Suspected ingestion requires emergency evaluation.

Button Magnets In The Nasal Cavity

Ingestion or insertion of small magnets into the nose, posing a risk of nasal septal perforation or necrosis if multiple magnets are involved, requiring urgent removal.

Buttonhole Deformity

Buttonhole deformity is a finger deformity with flexion of the proximal interphalangeal joint and hyperextension of the distal interphalangeal joint, often due to central slip injury or rheumatoid disease.

Buttressing

Buttressing is reinforcement or thickening that supports a weakened structure; in bone it may describe cortical stress adaptation or surgical structural support.

Butylamine

Butylamine is an irritating industrial amine whose vapor or liquid can injure the eyes, skin, and respiratory tract after exposure.

Butyric Acid

Butyric acid is a short-chain fatty acid produced by gut bacteria during fiber fermentation and used by colon cells as an energy source; concentrated exposure irritates tissue.

Buying And Caring For Baby Bottles And Nipples

Safe selection and cleaning of baby bottles and nipples reduce contamination, feeding injury, and improper preparation risks.

Bx

A common medical abbreviation for biopsy, the removal of cells or tissue for microscopic or laboratory examination.

By-Pass

A by-pass is a route created around an obstruction, such as a vascular, gastrointestinal, or urinary surgical bypass, or a cardiac bypass circuit.

Bypass

Surgical creation of alternative route around blocked vessel. Uses vein or prosthetic graft.

Bypass (Vascular)

Surgical procedure creating detour around blocked vessel using autologous vein (saphenous) or prosthetic graft (PTFE, Dacron). See Bypass Graft.

Alternate Names:

Vascular

Bypass Graft

Surgical procedure creating detour around blocked vessel using autologous vein (saphenous) or prosthetic graft (PTFE, Dacron). Indications: peripheral arterial disease, coronary artery disease, aortic aneurysm. Long-term patency depends on graft type, inflow/outflow, patient factors.

Bypass Surgery

A surgical conduit that reroutes flow around an obstructed or diseased segment, applied to coronary arteries (CABG), peripheral arteries (autologous vein or prosthetic bypass), and the gastrointestinal tract (gastric bypass). Principal long-term concerns are conduit patency, graft infection, and anastomotic stenosis.

Byssinosis

An occupational obstructive airway disease caused by inhalation of textile fibers, especially cotton, flax, or hemp dust. It may cause chest tightness and cough that worsen during or after work exposure.

C

C And T Cells

B cells make antibodies, while T cells help coordinate the body's defenses or destroy infected and abnormal cells. Together, they are major types of white blood cells.

C Fiber

A slow nerve fiber without an insulating covering that carries dull pain, warmth, and itch.

C-Peptide

A small protein fragment released when the pancreas makes insulin. Measuring it helps show how much insulin a person's body is making, rather than how much injected insulin is present. Kidney function and blood-glucose levels can affect the result.

C-Reactive Protein

A protein made by the liver that rises when there is inflammation or infection. It can help monitor inflammatory diseases and possible infection, but it does not identify the cause by itself. A high-sensitivity CRP test may be used with other information to assess heart-disease risk.

C-Reactive Protein

See *C-Reactive Protein*.

C-Reactive Protein Test

A blood test that measures C-reactive protein. The level may rise with inflammation or infection, but the test cannot identify the cause by itself and must be interpreted with symptoms and other findings.

C-Section

Cesarean section, surgical delivery of a fetus through incisions in the abdominal wall and uterus.

C. Diff

Alternate index label; see *C. difficile infection*.

C. Difficile

Clostridioides difficile, a bacterium that can survive as resistant spores and cause diarrhea and inflammation of the colon, often after antibiotic use. It produces toxins that injure the colon. Risk is also increased by some acid-reducing medicines and older age. Treatment may include vancomycin or fidaxomicin.

C. Difficile Colitis

Alternate index label; see *Pseudomembranous colitis*.

C. Difficile Infection

Infection with *Clostridioides difficile* bacteria, often causing antibiotic-associated diarrhea and colitis.

Alternate Names:

C. Diff

C1Q Nephropathy

A kidney disorder in which a protein called C1q builds up in the kidney's filtering units. It may cause protein in the urine and reduced kidney function.

C282Y Mutation

A change in the HFE gene that is the most common genetic cause of hereditary hemochromatosis,

a condition in which the body stores too much iron. Having two copies of this mutation increases the risk, although not everyone with two copies develops the disease.

CA 19-9

A blood marker that may be high in pancreatic and other digestive-system cancers, but also in non-cancerous bile-duct or inflammatory conditions. It cannot screen for or diagnose cancer by itself.

CA 27-29

A blood tumor-marker test used mainly to monitor some people with breast cancer. It is not used by itself to screen for or diagnose cancer.

CA-125

A blood test marker used mainly to monitor some ovarian cancers or evaluate a known abnormality. It can also rise with noncancerous conditions, inflammation, liver disease, and other cancers, so it cannot by itself screen for or diagnose ovarian cancer.

Alternate Names:

CA125

CA-125 Blood Test

See *CA-125*.

Cabazitaxel

A medicine or chemical compound identified by cabazitaxel. Its uses, effects, and risks depend on the specific substance and clinical context.

Cabergoline

A dopamine D2-receptor agonist used to suppress prolactin secretion and treat hyperprolactinemia; it is also used in selected Parkinson disease settings and requires monitoring for adverse effects.

CABG

Coronary artery bypass grafting, surgery that creates a new path around a narrowed or blocked heart artery to improve blood flow to the heart muscle.

Alternate Names:

Coronary artery bypass grafting; coronary bypass

surgery.

Cabot Ring

A threadlike or ring-shaped mark inside a red blood cell. It may appear in severe anemia or when the bone marrow is not making red blood cells normally.

Cachectic

Affected by cachexia, a severe wasting syndrome involving involuntary weight loss, muscle loss, weakness, and often reduced appetite in chronic disease.

Cachetic

An alternate spelling of cachectic, meaning characterized by severe disease-related wasting and loss of muscle and body mass.

Cachexia

Severe, unintentional weight loss and loss of muscle caused by long-term illness. It may occur with cancer, heart failure, chronic lung disease, or advanced kidney disease and often involves inflammation and poor appetite.

Cacosmia

An unpleasant or distorted sense of smell in which odors may seem foul. It can occur with nose or sinus disease, infection, seizures, injury, or other nervous-system disorders.

CAD

Coronary artery disease, narrowing or blockage of the heart's arteries, usually from atherosclerotic plaque, that can cause angina or myocardial infarction.

CADASIL Syndrome

An inherited disorder of small brain arteries that can cause migraine with aura, strokes, mood symptoms, and cognitive decline.

Cadaver

A dead human body used for medical education, research, or examination. It may be preserved by embalming to slow decomposition.

Cadaveric Spasm

Rare form of instantaneous post-mortem stiffening of a specific group of voluntary muscles at the exact moment of death, without an intervening period of muscle relaxation.

Cadherin

A protein that helps neighboring cells stick together and keeps tissues organized. Changes in cadherins can help some cancers invade nearby tissue and spread.

Cadmium

A toxic heavy metal found in tobacco smoke, batteries, pigments, and contaminated food; chronic exposure can injure the kidneys, lungs, and bones and increases cancer risk.

Cadmium Chloride

A medicine or chemical compound identified by cadmium chloride. Its uses, effects, and risks depend on the specific substance and clinical context.

Caduceus

A staff with two entwined snakes and wings, traditionally associated with Hermes; it is often used as a medical symbol but is distinct from the single-snake Rod of Asclepius.

Cadusafos

An organophosphate pesticide that inhibits acetylcholinesterase; exposure can cause cholinergic poisoning with sweating, salivation, vomiting, weakness, breathing difficulty, or seizures.

Caecal

Relating to the cecum, the pouch at the beginning of the large intestine where the small intestine joins the colon.

Caecitis

Inflammation of the cecum. Causes include infection, inflammatory bowel disease, ischemia, neutropenic enterocolitis, and extension of adjacent inflammatory disease.

Caecopexy

Surgery that attaches the cecum, the first pouch of the large intestine, to nearby tissue to prevent or treat abnormal movement or twisting of the cecum.

Caecostomy

Surgery that creates an opening from the cecum, the first part of the large intestine, to the abdominal wall to release pressure or divert bowel contents. It may be temporary.

Caecotomy

Surgery to make an opening in the cecum, the first part of the large intestine, usually to remove an object, reach the bowel, or assist another abdominal operation.

Caecum

The first pouch-like part of the large intestine, located where the ileum joins the colon and attached to the appendix.

Caenorhabditis Elegans

A biomedical term referring to *caenorhabditis elegans*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Caesarean Birth

An operative procedure in which a baby is delivered through an incision in the abdominal wall and uterus. Indications: fetal distress, failure to progress in labor, breech presentation, placenta praevia, multiple pregnancy, previous cesarean, and maternal request.

Caesarean Section

A caesarean section is surgical delivery of a baby through incisions in the abdomen and uterus when vaginal birth is unsafe or not feasible.

Cafe-Au-Lait Macules

Flat, uniformly hyperpigmented, light-brown to dark-brown (cafe-au-lait, "coffee with milk") macules, > 5 mm in prepubertal children or > 15 mm

in postpubertal individuals.

Cafe-Au-Lait Spot

A flat, evenly pigmented light-brown skin macule; several lesions or characteristic size and distribution may suggest neurofibromatosis type 1 or another genetic disorder.

Café Au Lait Spot

A flat, sharply demarcated, light-brown skin macule caused by increased melanin. Multiple lesions may be a feature of neurofibromatosis type 1 or other genetic disorders.

Café Coronary

Sudden airway obstruction by food, usually occurring while eating and sometimes causing collapse or death if the airway is not rapidly cleared.

Café-Au-Lait Macules

Flat, light-brown skin patches with relatively uniform pigmentation. Multiple lesions may be associated with genetic disorders such as neurofibromatosis type 1.

Caffeic Acid

A plant-derived phenolic compound found in coffee, fruit, and vegetables; it participates in plant metabolism and is studied experimentally for antioxidant and other biologic effects.

Caffeine

A stimulant found in coffee, tea, chocolate, soft drinks, and energy drinks. It increases alertness and can raise heart rate, cause shakiness, or disturb sleep.

Caffeine Intoxication

A toxic state caused by excessive caffeine exposure, characterized by symptoms such as restlessness, nervousness, insomnia, rapid heartbeat, tremor, gastrointestinal upset, or agitation.

Caffeine Overdose

Caffeine overdose can cause agitation, tremor, vom-

iting, rapid heartbeat, abnormal rhythm, seizures, and dangerous metabolic disturbances.

Caffeine Withdrawal

A syndrome occurring after abrupt cessation or reduction of caffeine use. Symptoms include headache, fatigue, depressed mood, difficulty concentrating, and flu-like symptoms. Onset is typically 12–24 hours after last caffeine intake, peak at 20–51 hours, and duration of 2–9 days.

Caisson Disease

Decompression sickness caused by gas bubbles forming in the body after a rapid drop in surrounding pressure, such as during diving. It can cause joint pain, breathing problems, nervous-system injury, or collapse.

Cal (Calorie)

A unit of energy; in nutrition, a Calorie with a capital C is a kilocalorie equal to 1,000 small calories and is used to describe food energy.

Caladium Plant Poisoning

Poisoning caused by calcium oxalate crystals in the plant. Chewing may cause immediate burning and swelling of the mouth and throat; airway symptoms require emergency care.

Calamine

Calamine is a topical preparation containing zinc compounds used to soothe itching and irritation from minor rashes, insect bites, and sunburn.

Calbindin

A calcium-binding protein that buffers intracellular calcium and supports signaling, especially in neurons and intestinal epithelial cells; it is also used as a tissue marker.

Calcaneal Spur

A bony outgrowth from the heel bone, often at the plantar fascia attachment or Achilles tendon attachment; it may be asymptomatic or associated with heel pain.

Calcaneum

The calcaneum, or calcaneus, is the large heel bone that transfers load from the ankle and forms part of the hindfoot.

Calcaneus

Largest tarsal bone forming the heel. Articulates with talus proximally and cuboid anteriorly. Lever for the calf muscles via the Achilles tendon on its posterior tuberosity. Predominantly trabecular bone, the most commonly fractured tarsal bone from falls from height. Bohler's angle assesses fractures on lateral radiographs.

Calcifediol

25-hydroxyvitamin D, the major circulating storage form of vitamin D measured to assess vitamin-D status before activation to calcitriol.

Calciferol

Vitamin D is a fat-soluble vitamin (actually a pro-hormone) essential for calcium and phosphorus homeostasis and bone health. See Vitamin D: Calcitriol.

Calcific Bursitis

Calcific bursitis occurs when calcium deposits form in or around a bursa, often near the shoulder, causing pain and restricted movement. Treatment may include activity modification, anti-inflammatory care, physical therapy, aspiration, injection, or removal of deposits.

Alternate Names:

Bursitis Calcific (Calcific Bursitis)

Calcific Tendinitis

Deposition of calcium hydroxyapatite crystals within a tendon, most often the rotator cuff, causing acute pain and restricted movement.

Calcific Uremic Arteriopathy

A rare disorder, usually linked to advanced kidney disease, in which calcium builds up in small blood vessels and causes painful injury or tissue death in the skin and tissues beneath it.

Calcification

Calcification is abnormal deposition of calcium salts in tissue; its significance depends on the site, pattern, underlying injury, and associated disease. calcium or phosphate metabolism is abnormal.

Calcified Granuloma

A healed granulomatous inflammatory lesion in which calcium has been deposited, commonly seen as a small calcified focus in the lung or another organ.

Calcineurin

An enzyme that helps immune cells respond to signals. Medicines such as cyclosporine block this enzyme and reduce T-cell activity.

Calcinosis

Cutaneous calcinosis, the deposition of insoluble calcium phosphate salts within the skin and subcutaneous tissues, occurring in connective tissue diseases such as scleroderma and dermatomyositis, or as a localized phenomenon. Lesions present as firm, white, subcutaneous nodules that may ulcerate and extrude chalky material.

Calcinosis

The pathological deposition of calcium salts in cutaneous and subcutaneous tissues, presenting as firm, white to yellowish nodules or plaques that may ulcerate and extrude chalky material.

Calciophylaxis

A rare, serious condition in which calcium builds up in small blood vessels and causes painful skin wounds and tissue death, usually in people with advanced kidney disease. Risk is increased by obesity and some medicines, including warfarin. Treatment requires specialist care, wound care, and control of calcium and phosphate levels.

Calcitonin

A 32-amino-acid peptide hormone secreted by the parafollicular C-cells of the thyroid gland in response to elevated serum calcium. It lowers blood calcium by inhibiting osteoclast-mediated bone resorption

and increasing renal calcium excretion.

Calcitonin Blood Test

A blood test that measures calcitonin, a hormone produced by thyroid C cells. It is used mainly to evaluate or monitor medullary thyroid carcinoma and may be affected by other thyroid, kidney, gastrointestinal, or medicine-related factors.

Calcitriol

The active form of vitamin D. It helps the intestines absorb calcium and phosphate and helps regulate parathyroid hormone and bone health.

Calcitriol And Analogues

Active vitamin D forms suppressing PTH. Calcitriol, paricalcitol, doxercalciferol. Monitor calcium and phosphorus to avoid hypercalcemia and hyperphosphatemia. Part of CKD-MBD management.

Calcium

A mineral needed for strong bones and teeth, muscle movement, nerve signals, blood clotting, and normal cell function. Blood calcium exists in an active free form and forms attached to proteins or other substances.

Calcium - Ionized

A blood test measuring ionized calcium, the biologically active fraction of circulating calcium. It is useful when total calcium may be misleading, such as with abnormal albumin, critical illness, or acid-base disturbances.

Calcium - Urine

A urine test measuring calcium excretion over a specified collection period or in a spot specimen. It helps evaluate kidney stones, parathyroid disorders, bone disease, and abnormal calcium handling, and must be interpreted with kidney function and diet.

Calcium Blood Test

A blood test that measures calcium, usually as total calcium. The result may be interpreted with albumin, a blood protein, or measured as ionized

calcium when the active form is needed. It helps assess parathyroid, bone, and kidney disorders.

Calcium Carbonate

A medicine or chemical compound identified by calcium carbonate. Its uses, effects, and risks depend on the specific substance and clinical context.

Calcium Carbonate Overdose

Overdose of calcium carbonate that may cause gastrointestinal symptoms, hypercalcemia, alkalosis, kidney injury, or kidney stones.

Calcium Carbonate With Magnesium Overdose

Overdose of calcium and magnesium salts that may cause nausea, constipation, weakness, altered mental status, hypercalcemia, or kidney injury.

Calcium Channel

A protein in the cell membrane that controls calcium entry into cells. Calcium channels help regulate muscle contraction, the release of substances, nerve signals, and heart rhythm.

Calcium Chloride

A medicine or chemical compound identified by calcium chloride. Its uses, effects, and risks depend on the specific substance and clinical context.

Calcium Citrate

A medicine or chemical compound identified by calcium citrate. Its uses, effects, and risks depend on the specific substance and clinical context.

Calcium Deficiency

Inadequate calcium intake or availability; severe deficiency can contribute to hypocalcemia, muscle spasms, impaired bone mineralization, and increased fracture risk.

Calcium Elevated (Hypercalcemia)

Alternate index label for Hypercalcemia.

Calcium Gluconate

A medicine or chemical compound identified by calcium gluconate. Its uses, effects, and risks depend on the specific substance and clinical context.

Calcium Hydroxide Poisoning

Exposure to a caustic alkali that can burn the mouth, esophagus, skin, or eyes. Airway and gastrointestinal injury require urgent assessment.

Calcium Phosphate

The principal mineral constituent of bone and teeth, existing as hydroxyapatite crystals. In pathological states, calcium phosphate deposits form in soft tissues through dystrophic calcification in damaged tissue or metastatic calcification in hypercalcemic conditions.

Calcium Pyrophosphate Arthritis

Acute or chronic inflammatory arthritis caused by deposition of calcium pyrophosphate crystals in cartilage and joints. Acute attacks, often called pseudogout, commonly affect the knee or wrist and are diagnosed by identifying weakly positively birefringent crystals in synovial fluid.

Calcium Pyrophosphate Deposition

Alternate index label; see *Calcium Pyrophosphate Deposition Disease*.

Calcium Pyrophosphate Deposition Disease

A crystal arthropathy caused by deposition of calcium pyrophosphate dihydrate crystals in articular cartilage (chondrocalcinosis). Presents with acute pseudogout (most common): acute monoarticular inflammation of the knee or wrist, mimicking gout but less painful.

Alternate Names:

Calcium Pyrophosphate Deposition

Calcium Sensing Receptor

G-protein coupled receptor expressed primarily on parathyroid glands and renal tubular cells that senses extracellular ionized calcium levels. On

parathyroid glands activation suppresses PTH secretion. On renal tubular cells activation increases calcium excretion and decreases phosphate reabsorption.

Calcium Signaling

Cell communication in which changes in calcium concentration activate proteins controlling contraction, secretion, metabolism, and gene expression.

Calcium Stones

Most common kidney stone type composed of calcium oxalate or calcium phosphate. Risk factors include hypercalciuria, hyperoxaluria, hypocitraturia, and low urine volume. Treat with thiazides for hypercalciuria, dietary modification, and hydration. Potassium citrate for hypocitraturia.

Calculi

Hardened mineral concretions, such as gallstones, kidney stones, or salivary stones, formed when dissolved substances precipitate and aggregate within a body organ or duct.

Calculus

A calculus is a hard deposit formed from mineral salts, such as a kidney stone, gallstone, or dental tartar.

Calculus (Tartar)

Mineralized dental plaque. Cannot be removed by brushing. Requires professional scaling. Retains plaque, promotes inflammation.

Alternate Names:

Tartar

Calculus Removal

Professional cleaning to remove hardened plaque. Scaling and root planing. Ultrasonic and hand instruments.

Calculus, Renal

A kidney stone, a hard mass of mineral crystals formed in the urinary tract that may cause colicky pain, hematuria, obstruction, or infection.

Caldwell-Luc Operation

A surgical approach to the maxillary sinus through the canine fossa of the anterior maxilla. It is used selectively for access to maxillary sinus disease, lesions, or foreign bodies.

Calf

The posterior portion of the lower leg between the knee and ankle, containing the gastrocnemius, soleus, and other muscles.

Calf Pseudohypertrophy

A classic clinical physical examination sign seen in muscular dystrophies, most notably Duchenne and Becker muscular dystrophies, as well as some limb-girdle muscular dystrophies and sarcoglycanopathies. Refers to the paradoxically prominent, firm, enlarged, and bulky appearance of the calf muscles (gastrocnemius and soleus), which is caused not by true muscular hypertrophy, but by progressive chronic necrosis, degeneration of skeletal myofibers, and compensatory replacement by fibrous connective tissue and extensive adipocyte infiltration.

Alternate Names:

Pseudohypertrophy of the Calves; Muscular Pseudohypertrophy

Calibration

The process of adjusting and verifying the accuracy of medical instruments, diagnostic devices, or laboratory equipment against known standards to ensure reliable measurements and patient safety.

Caliciviridae

Caliciviridae is a virus or viral group classified by its genetic and biologic features; clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Calipers

An instrument with opposing arms used to measure distance, thickness, or body dimensions, including skinfold thickness in nutritional assessment.

Calla Lily Poisoning

Poisoning caused by calcium oxalate crystals in the plant. Chewing may cause immediate burning and swelling of the mouth and throat; airway symptoms require emergency care.

Callosal

Callosal means relating to the corpus callosum, the major bundle of fibers connecting the cerebral hemispheres.

Callosity

A callosity is a localized area of thickened skin caused by repeated pressure or friction.

Callosotomy

Surgical division of the corpus callosum to reduce the spread of certain severe, treatment-resistant seizures between the cerebral hemispheres.

Callus

A localized area of thickened, hardened skin caused by repeated pressure or friction.

Calmodulin

An intracellular calcium-binding messenger protein that changes shape when it binds calcium and regulates enzymes, ion channels, muscle contraction, and gene expression.

Calnexin

An endoplasmic-reticulum chaperone that binds newly synthesized glycoproteins and helps them fold correctly before they are transported or degraded.

Calor

Heat or increased warmth of a body part, one of the classic local signs of inflammation.

Calor, Dolor, Rubor, And Tumor

The classic signs of inflammation: heat, pain, redness, and swelling; inflammation may also impair function.

Caloric Restriction

Reduced calorie intake without malnutrition. Associated with longevity, reduced chronic disease risk. Requires careful planning.

Caloric Stimulation

A vestibular test that introduces warm or cool air or water into the ear canal to create convection in the semicircular canal and assess each vestibular nerve.

Calorie

A unit of energy measurement used in nutrition to quantify the energy content of foods and energy expenditure of the body. One kilocalorie (kcal, commonly called a Calorie with capital C) equals the energy required to raise 1 kilogram of water by 1 degree Celsius.

Alternate Names:

a Calorie.

Calories

Units used to express the energy supplied by food or expended by the body; food labels generally report kilocalories, commonly called calories.

Calorimetry

A technique for measuring heat production or exchange in the body, used to assess metabolic rate, energy expenditure, and nutritional requirements. Indirect calorimetry estimates energy expenditure from oxygen consumption and carbon dioxide production.

Calpain

A family of calcium-dependent cysteine proteases found in cytosol that cleave cytoskeletal and signaling proteins. Calpains are involved in cell migration, apoptosis, and platelet activation, and their dysregulation is implicated in neurodegenerative diseases and muscular dystrophy.

Calpainopathy

An autosomal recessive muscular dystrophy (Limb-Girdle Muscular Dystrophy R1 [LGMD R1], his-

torically LGMD 2A) caused by pathogenic biallelic loss-of-function mutations in the *CAPN3* gene on chromosome 15q15.1, encoding calpain-3, a skeletal muscle-specific intracellular calcium-activated neutral cysteine protease bound to titin. Represents the most common form of recessive limb-girdle muscular dystrophy worldwide. Manifests between ages 8 and 15 with slowly progressive, symmetrical weakness and wasting of the pelvic, hip girdle, and shoulder muscles, prominent early winging of the scapulae, lumbar lordosis, tiptoe walking due to Achilles tendon contractures, and laxity of abdominal wall muscles, generally sparing facial, extraocular, and cardiac muscles.

Alternate Names:

Limb-Girdle Muscular Dystrophy R1 (LGMD R1); LGMD 2A; Calpain-3 Deficiency

Calponin

An actin-associated protein in smooth muscle and other cells that modulates contraction and cytoskeletal organization; immunostaining can help identify myoid differentiation.

Calprotectin

Calprotectin is a protein released by activated neutrophils; fecal calprotectin helps detect intestinal inflammation, especially in inflammatory bowel disease.

Calreticulin

An endoplasmic-reticulum chaperone and calcium-binding protein involved in protein folding and immune signaling; CALR mutations are associated with some myeloproliferative neoplasms.

Calretinin

A calcium-binding protein expressed in selected neurons, mesothelial cells, and other tissues; immunohistochemical staining helps distinguish mesothelioma from some carcinomas.

Calvaria

The dome-shaped upper portion of the skull that protects the brain.

Calvarium

The dome-like portion of the skull formed mainly by the frontal, parietal, and occipital bones that encloses and protects the cerebral hemispheres.

Calyx

A cup-shaped structure that collects fluid or supports an organ. In the kidney, renal calyces collect urine from the renal pyramids and convey it to the renal pelvis.

Cambridge Classification

ERCP-based grading system for chronic pancreatitis severity evaluating the main pancreatic duct, branches, and common bile duct. Normal pancreas shows smooth duct without abnormalities. Equivocal shows fewer than three abnormal side branches. Mild shows three or more abnormal side branches. Moderate shows abnormal main duct.

Cameron Lesions

Linear gastric erosions or ulcers found on the mucosal folds at the diaphragmatic hiatus in patients with large hiatal hernias. They result from mechanical trauma as the gastric mucosa prolapses through the diaphragmatic crura during respiration and movement.

Campesterol

A plant sterol structurally related to cholesterol that is absorbed in small amounts from food; elevated absorption can contribute to rare sitosterolemia.

Campho-Phenique Overdose

Overdose of a topical product containing camphor and other ingredients. Toxicity may include gastrointestinal or neurologic symptoms, including seizures.

Camphor

A fragrant bicyclic compound used in topical products and some traditional remedies; ingestion can cause nausea, agitation, seizures, and potentially fatal toxicity.

Camphor Overdose

Overdose of camphor that can cause nausea, agitation, seizures, coma, and respiratory depression. Seizures may occur rapidly and require emergency treatment.

Campimetry

Measurement and mapping of the visual field, usually by presenting stimuli at different locations while the patient maintains fixation. It helps detect and characterize defects from retinal, optic-nerve, or retrochiasmal disease.

Campomelic Dysplasia

A rare SOX9-related skeletal disorder with bowed long bones, underdeveloped shoulder blades, and characteristic facial and airway abnormalities.

Camptocormia

A marked forward flexion of the trunk that increases while standing or walking and improves when lying down. It can occur with axial myopathy, Parkinsonian disorders, dystonia, or other neuromuscular disease.

Camptodactyly

A congenital or acquired flexion deformity of one or more fingers, usually caused by shortening or abnormality of tendons, muscles, fascia, or joints. It often affects the proximal interphalangeal joints.

Camptothecin

A plant-derived compound that inhibits topoisomerase I by stabilizing the enzyme–DNA cleavage complex; related drugs include irinotecan and topotecan.

Campylobacter

A genus of curved, motile Gram-negative bacteria; several species cause intestinal infection and some cause invasive disease.

Campylobacter Coli

A *Campylobacter* species that commonly causes foodborne gastroenteritis with diarrhea, abdominal

pain, and fever.

Campylobacter Concisus

A microbial taxon named campylobacter concisus that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Campylobacter Curvus

A microbial taxon named campylobacter curvus that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Campylobacter Fetus

A Campylobacter species that can cause bacteremia and infections of blood vessels, joints, or other tissues, especially in people with impaired immunity.

Campylobacter Gracilis

A microbial taxon named campylobacter gracilis that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Campylobacter Infection

Bacterial gastroenteritis caused most often by Campylobacter jejuni or C. coli, usually acquired from contaminated poultry, food, water, or unpasteurized milk. It causes fever, abdominal cramps, and diarrhea that may be bloody and can rarely precede Guillain-Barre syndrome.

Campylobacter Jejuni

A leading cause of foodborne gastroenteritis, commonly causing fever, abdominal cramps, and sometimes bloody diarrhea.

Campylobacter Lari

A microbial taxon named campylobacter lari that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Campylobacter Rectus

A microbial taxon named campylobacter rectus that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Campylobacter Serology Test

A blood test that looks for antibodies to Campylobacter species; it may support evaluation of selected postinfectious complications, but stool molecular testing or culture is generally used to diagnose active intestinal infection.

Campylobacter Upsaliensis

A microbial taxon named campylobacter upsaliensis that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Campylobacteraceae

A family of curved or spiral Gram-negative bacteria that includes Campylobacter and related genera.

Campylobacteriosis

Infection caused by Campylobacter species, commonly producing fever, abdominal pain, and inflammatory diarrhea that may be bloody. Complications can include dehydration, bacteremia, reactive arthritis, or Guillain-Barré syndrome.

Can You Boost Your Metabolism?

A health question about increasing energy expenditure; activity, muscle mass, sleep, diet, medicines, and endocrine disease influence metabolism, while quick “boosts” are usually modest.

Canagliflozin

An SGLT2 inhibitor that lowers blood glucose by increasing urinary glucose excretion and also reduces selected heart-failure and kidney risks; adverse effects include genital infections and ketoacidosis.

Canal Of Schlemm

A circular endothelial-lined channel at the corneoscleral junction that drains aqueous humor from

the anterior chamber into the episcleral venous circulation. Impaired outflow contributes to glaucoma.

Canaliculus

A canaliculus is a small anatomical channel, such as a lacrimal canaliculus that drains tears from the eyelid into the lacrimal sac.

Canalith Repositioning Maneuver

A series of head movements used to move displaced inner-ear crystals out of a semicircular canal, commonly to treat posterior-canal benign positional vertigo.

Canalith Repositioning Maneuvers

Canalith repositioning maneuvers move displaced inner-ear crystals from a semicircular canal back to the utricle. They are used for benign paroxysmal positional vertigo, with the maneuver selected according to the affected canal and side.

Canalization

Canalization is formation or restoration of a channel through tissue, a vessel, or an embryonic structure; in thrombosis it may describe reopening of an occluded vessel.

Canarypox Virus

Canarypox virus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Canavan Disease

An autosomal-recessive leukodystrophy caused by aspartoacylase deficiency, leading to accumulation of N-acetylaspartate and progressive spongy degeneration of cerebral white matter. It typically presents in infancy with hypotonia followed by developmental impairment and macrocephaly.

Cancell

A shortened or misspelled form of cancellous, referring to the porous, trabecular interior of bone that

contains marrow and helps distribute mechanical load.

Cancellous Bone

Porous, trabecular bone found inside many bones, especially near joints and in the vertebrae.

Cancer

Cancer is uncontrolled growth of abnormal cells that can invade nearby tissue and spread to distant organs; behavior and treatment depend on its tissue of origin and molecular features. or spread to distant sites. Cancer may arise in any tissue and is classified by its site and cell type. Early detection and treatment improve outcomes.

Cancer And Lymph Nodes

Cancer cells can spread to nearby or distant lymph nodes; lymph-node examination, imaging, or biopsy helps determine cancer stage and guide treatment.

Cancer Care Team

The multidisciplinary group of health professionals involved in cancer care, which may include oncologists, surgeons, radiologists, pathologists, nurses, pharmacists, rehabilitation specialists, psychologists, and palliative-care professionals.

Cancer Causes

Cancer develops when genetic and epigenetic changes allow cells to grow, survive, and invade abnormally. Causes and risks include inherited susceptibility, tobacco, alcohol, infections, radiation, ultraviolet exposure, carcinogens, chronic inflammation, and chance changes during cell division.

Alternate Names:

Cancer Risk (Cancer Causes)

Cancer Cell

A cell that has acquired genetic and epigenetic abnormalities allowing uncontrolled growth, impaired normal differentiation, resistance to cell death, invasion of surrounding tissues, or spread to distant sites.

Cancer Cluster

A greater-than-expected number of cancer cases in a defined group, place, or period; investigation compares the observed cases with expected rates and looks for shared exposures or chance variation.

Cancer Genetics

Inherited cancer syndromes. Examples: BRCA1/2 (breast, ovarian), Lynch syndrome (colon), Li-Fraumeni (various).

Cancer Grading

Assessment of how abnormal cancer cells appear under a microscope and how aggressively the tumor is likely to behave. It differs from cancer staging, which describes the extent of spread.

Cancer Immunotherapy

Treatment of cancer using agents that modify the immune response to recognize and destroy tumor cells, including checkpoint inhibitors, CAR T-cell therapy, cytokines, and cancer vaccines. It may be used alone or combined with chemotherapy or radiation.

Cancer Mutations

Genetic changes in cancer cells that can affect cell growth, survival, or spread. Most are acquired during a person's lifetime and are not inherited. Some are driver mutations that help cancer develop or progress, while others are passenger mutations with little effect. Testing for certain mutations may help guide diagnosis, prognosis, or targeted treatment.

Cancer In Lung (Lung Cancer)

Alternate index label for Lung Cancer.

Cancer Nutrition

Malnutrition common in cancer. Hypermetabolism, cachexia, treatment side effects. Nutritional support improves outcomes. High protein, high calorie.

Cancer Of Unknown Origin

Alternate index label; see *Carcinoma of unknown*

primary.

Cancer Of Unknown Primary

Metastatic cancer in which the primary site cannot be identified despite investigation. Workup includes histology, immunohistochemistry, and molecular profiling to guide treatment.

Cancer Pain

Pain caused by a tumor, its spread, or cancer treatment. It may be continuous or episodic and can involve bone, nerves, or internal organs. Management may include treating the cancer, non-opioid medicines, opioids, nerve-directed treatments, and supportive care.

Alternate Names:

Pain Cancer (Cancer Pain)

Cancer Prevention

Measures that lower cancer risk or detect precancerous change early, including tobacco avoidance, vaccination, healthy behaviors, reducing harmful exposures, and recommended screening.

Cancer Registry

System for collecting cancer data. Used for epidemiology, quality improvement, research.

Cancer Risk

The likelihood that a person will develop cancer, influenced by age, inherited variants, family history, tobacco, alcohol, infections, radiation, environmental exposures, and chance.

Cancer Risk (Cancer Causes)

Alternate index label for Cancer Causes.

Cancer Screening

Testing people without symptoms to detect certain cancers or precancerous changes early, when treatment may be more effective.

Cancer Staging

Classification of cancer extent. TNM system: T (tumor size/extent), N (nodal involvement), M (metas-

tasis). Stages: 0 (in situ), I (early), II-III (advanced), IV (metastatic). Guides treatment and prognosis.

Cancer Surveillance

The systematic, ongoing collection and analysis of information about cancer incidence, stage, treatment, survival, recurrence, and mortality for patient follow-up, research, and public-health planning.

Cancer Survivor

Person living with cancer from diagnosis through end of life and beyond.

Cancer Susceptibility Genes

Genes in which inherited pathogenic variants increase a person's risk of developing one or more cancers. Examples include BRCA1, BRCA2, APC, and mismatch-repair genes; a susceptibility variant does not guarantee that cancer will develop.

Cancer Treatment - Dealing With Pain

Pain during or after cancer treatment may arise from the tumor, surgery, radiation, chemotherapy, or nerve injury; management can combine treatment of the cause, analgesics, rehabilitation, and supportive care.

Cancer Treatment - Early Menopause

Menstrual periods may stop early after chemotherapy, pelvic radiation, ovarian surgery, or hormone-blocking treatment; symptoms and fertility effects require individualized oncology and gynecologic care.

Cancer Treatment: Dealing With Hot Flashes And Night Sweats

Hot flashes and night sweats may result from cancer treatment or hormone changes; options include cooling measures, behavioral strategies, and selected nonhormonal or hormonal medicines when appropriate.

Cancer Treatments

Treatments for cancer include surgery, radiation,

chemotherapy, immunotherapy, targeted therapy, hormone therapy, stem-cell transplantation, and supportive or palliative care, selected according to tumor type, stage, biology, and patient goals.

Cancer, Carcinoid Tumors

Alternate index label; see *Carcinoid tumors*.

Cancer, Chronic Lymphocytic Leukemia

Alternate index label; see *Chronic lymphocytic leukemia*.

Cancer, Chronic Myelogenous Leukemia

Alternate index label; see *Chronic myelogenous leukemia*.

Cancer, Colon

Alternate index label; see *Colon cancer*.

Cancer, Endometrial

Alternate index label; see *Endometrial cancer*.

Cancer, Esophageal

Alternate index label; see *Esophageal cancer*.

Cancer, Eye Melanoma

Alternate index label; see *Eye melanoma*.

Cancer, Floor Of The Mouth

Alternate index label; see *Floor of the mouth cancer*.

Cancer, Gallbladder

Alternate index label; see *Gallbladder cancer*.

Cancer, Gastric

Alternate index label; see *Stomach cancer*.

Cancer, Hairy Cell Leukemia

Alternate index label; see *Hairy cell leukemia*.

Cancer, Hodgkin Lymphoma

Alternate index label; see *Hodgkin lymphoma (Hodgkin disease)*.

Cancer, Hurthle Cell

Alternate index label; see *Hurthle cell cancer*.

Cancer, Inflammatory Breast

Alternate index label; see *Inflammatory breast cancer*.

Cancer, Kidney

Alternate index label; see *Kidney cancer*.

Cancer, Liver

Alternate index label; see *Liver cancer*.

Cancer, Lung

Alternate index label; see *Lung cancer*.

Cancer, Merkel Cell

Alternate index label; see *Merkel cell carcinoma*.

Cancer, Mouth

Alternate index label; see *Mouth cancer*.

Cancer, Multiple Myeloma

Alternate index label; see *Multiple myeloma*.

Cancer, Nasopharyngeal

Alternate index label; see *Nasopharyngeal carcinoma*.

Cancer, Neuroblastoma

Alternate index label; see *Neuroblastoma*.

Cancer, Neuroendocrine

Alternate index label; see *Neuroendocrine tumors*.

Cancer, Non-Hodgkin Lymphoma

Alternate index label; see *Non-Hodgkin lymphoma*.

Cancer, Ovarian

Alternate index label; see *Ovarian cancer*.

Cancer, Pancreatic

Alternate index label; see *Pancreatic cancer*.

Cancer, Paraneoplastic Syndromes

Alternate index label; see *Paraneoplastic syndromes of the nervous system*.

Cancer, Prostate

Alternate index label; see *Prostate cancer*.

Cancer, Rectal

Alternate index label; see *Rectal cancer*.

Cancer, Retinoblastoma

Alternate index label; see *Retinoblastoma*.

Cancer, Skin

Alternate index label; see *Skin cancer*.

Cancer, Soft Tissue Sarcoma

Alternate index label; see *Soft tissue sarcoma*.

Cancer, Squamous Cell

Alternate index label; see *Squamous cell carcinoma of the skin*.

Cancer, Stomach

Alternate index label; see *Stomach cancer*.

Cancer, Testicular

Alternate index label; see *Testicular cancer*.

Cancer, Throat

Alternate index label; see *Throat cancer*.

Cancer, Thyroid

Alternate index label; see *Thyroid cancer*.

Cancer, Uterine

Alternate index label; see *Endometrial cancer*.

Cancer, Vagina

Alternate index label; see *Vaginal cancer*.

Cancer, Vulvar

Alternate index label; see *Vulvar cancer*.

Cancer, Wilms' Tumor

Alternate index label; see *Wilms tumor*.

Cancer-Associated Thrombosis

Venous or arterial thrombosis occurring in the setting of active malignancy or its treatment. Tumor biology, immobilization, surgery, central venous devices, and systemic therapy contribute to risk; anticoagulation is individualized according to bleeding risk and cancer factors.

Cancrum Oris

Noma, a rapidly destructive gangrenous infection of the mouth and face that occurs mainly in severely malnourished or immunocompromised children. It begins with gingival or mucosal necrosis and can extend through facial tissues.

Candela

The SI unit of luminous intensity in a specified direction; it is used in photometry and is distinct from medical measures of light dose or illuminance.

Candida

A genus of yeast that normally colonizes skin and mucous membranes but can cause mucosal or invasive candidiasis.

Candida Albicans

The most common *Candida* species causing human disease, including thrush, genital infection, skin-fold infection, and invasive candidiasis.

Candida Albicans Infection

Infection caused by the yeast *Candida albicans*, ranging from mucosal disease to invasive bloodstream or organ infection in vulnerable patients.

Candida Auris

A multidrug-resistant yeast that can colonize skin and cause invasive bloodstream, wound, or ear infections, especially in hospitalized or immunocompromised patients. It spreads readily in healthcare settings and requires laboratory identification and strict infection-control measures.

Candida Auris Infection

Candida auris infection is an invasive or colonizing yeast infection notable for healthcare transmission and frequent resistance to antifungal medicines.

Candida Boidinii

A microbial taxon named *candida boidinii* that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Candida Dubliniensis

A microbial taxon named *candida dubliniensis* that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Candida Esophagitis

Candida infection of the esophagus, usually causing painful or difficult swallowing in people with impaired immunity.

Candida Famata

A microbial taxon named *candida famata* that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Candida Glabrata

A *Candida* species that can cause urinary, genital, bloodstream, or invasive infection and may resist some azole medicines.

Candida Inconspicua

A microbial taxon named *candida inconspicua* that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Candida Infection Of The Skin

A superficial fungal infection of the skin caused by *Candida* species, often affecting moist skin folds and causing an erythematous, itchy rash with possible satellite lesions.

Candida Intermedia

A microbial taxon named candida intermedia that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Candida Orthopsilosis

A microbial taxon named candida orthopsilosis that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Candida Parapsilosis

A Candida species associated with bloodstream and catheter-related infection, particularly in newborns and hospitalized patients.

Candida Sake

A microbial taxon named candida sake that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Candida Tropicalis

A Candida species that can cause mucosal or invasive infection, especially in people with neutropenia or other immune impairment.

Candida Zeylanoides

A microbial taxon named candida zeylanoides that is usually environmental or part of normal flora; disease, when it occurs, depends on the site and host defenses.

Candidal Intertrigo

A Candida infection of skin folds (axillae, inframammary, inguinal, intergluteal, pannus) characterized by erythematous, macerated plaques with satellite pustules at the periphery. Most common in obese, diabetic, and immunocompromised patients, and with prolonged antibiotic use. Presents with itching, burning, and soreness.

Candidal Vaginitis

Inflammation of the vagina caused by overgrowth of

Candida yeast, usually producing itching, soreness, and a thick white discharge.

Candidal Vulvovaginitis

Alternate index label; see *Vulvovaginal Candidiasis*.

Candidemia

The presence of Candida species in the bloodstream. It is an invasive fungal infection associated with intravascular devices, surgery, intensive care, immunosuppression, and substantial risk of organ dissemination.

Candidiasis

Fungal infection caused by Candida species, most commonly *C. albicans*. Mucocutaneous: oral thrush (white plaques on oral mucosa), esophagitis, vulvovaginitis (itching, discharge), and intertrigo. Invasive candidiasis: candidemia and deep organ infection, typically in immunocompromised or ICU patients. Diagnosis: culture, histopathology.

Candidiasis, Oral

Alternate index label; see *Oral thrush*.

Candidiasis, Vaginal

Alternate index label; see *Yeast infection (vaginal)*.

Candles Poisoning

Poisoning after ingestion of candle material or exposure to candle products. Effects are usually gastrointestinal, but product ingredients and aspiration risk determine severity.

Cane

Walking aid for balance. Types: quad (four-base), single-point. Held on opposite side of injury.

Canertinib Dihydrochloride

A medicine or chemical compound identified by canertinib dihydrochloride. Its uses, effects, and risks depend on the specific substance and clinical context.

Canfosfamide Hcl

An investigational glutathione-S-transferase P1-1–activated anticancer prodrug studied for tumor-selective cytotoxicity; it is not a routine approved treatment.

Canicola Fever

A form of leptospirosis associated particularly with *Leptospira interrogans* serovar canicola. It may cause fever, headache, myalgia, conjunctival suffusion, and renal or hepatic involvement.

Canine Teeth

Canine teeth are pointed teeth between the incisors and premolars that tear food and help guide jaw movement.

Canker

Canker is a nontechnical term often referring to an aphthous mouth ulcer, a painful shallow ulcer of the oral mucosa.

Canker Sore

A common aphthous ulcer of the oral mucosa, appearing as a painful, shallow, round or oval ulcer with an erythematous border. Most heal spontaneously, but recurrent, severe, or atypical ulcers require assessment.

Alternate Names:

Recurrent Oral Ulcers Ulcer, Aphthous

Canker Sores

Canker sores are painful, shallow ulcers on the moist lining of the mouth and are not contagious. Minor trauma, stress, nutritional deficiency, immune disease, medicines, or other triggers may contribute; persistent, severe, or recurrent ulcers need evaluation.

Alternate Names:

Sores Canker (Canker Sores)

Cannabidiol

A nonintoxicating cannabinoid from *Cannabis sativa* that influences several signaling systems; purified cannabidiol is used for selected seizure disorders,

while products vary in quality and interactions.

Cannabinoid

A compound that acts on cannabinoid receptors or related targets, including plant cannabinoids, endogenous endocannabinoids, and synthetic agents; effects vary from analgesia and appetite change to intoxication.

Cannabinoid Hyperemesis Syndrome

A syndrome of recurrent severe nausea, vomiting, and abdominal pain associated with prolonged frequent cannabis use; symptoms often improve after sustained cannabis cessation.

Cannabinoid Receptor

A G-protein-coupled receptor, chiefly CB1 or CB2, activated by endocannabinoids and some cannabis compounds to modulate pain, appetite, mood, memory, immunity, and inflammation.

Cannabinol

A mildly psychoactive cannabinoid produced as THC oxidizes; its clinical effects and therapeutic usefulness remain less established than those of several other cannabinoids.

Cannabis

Cannabis is a plant containing cannabinoids such as THC and CBD; effects include altered perception, impaired coordination, therapeutic actions, and possible dependence or adverse effects.

Cannabis Use Disorder

A problematic pattern of cannabis use leading to significant impairment. Criteria include tolerance and withdrawal (irritability, anger, decreased appetite, restlessness, anxiety, sleep disturbance). Cannabis intoxication includes euphoria, impaired judgment, and time distortion.

Cannot Intubate Cannot Oxygenate

The most critical airway emergency — inability to intubate the trachea and inability to oxygenate the patient by face mask or supraglottic airway despite

optimal efforts. CICO is a life-threatening situation requiring immediate surgical airway access. Signs: SpO2 decreasing despite 100% O2, absent end-tidal CO2, no chest rise.

Cannula

A hollow tube inserted into a blood vessel, body cavity, or tissue to deliver or remove fluids or to provide access for instruments.

Cannula Hub

The proximal connector of a cannula that joins it to tubing, a syringe, or another device and may contain a port or valve for access and fluid delivery.

Cannulation

The insertion of a cannula into a blood vessel, body cavity, duct, or other anatomical passage to permit drainage, infusion, monitoring, sampling, or access for a procedure.

Canola Oil

A vegetable oil low in saturated fat and relatively rich in monounsaturated and polyunsaturated fatty acids, commonly used as a cooking oil.

Canrenone

An active metabolite and potassium-sparing antimineralocorticoid related to spironolactone that antagonizes aldosterone at the mineralocorticoid receptor.

Cantharides

Cantharides are preparations made from blister beetles containing cantharidin, a potent irritant that can cause severe poisoning and urinary or gastrointestinal injury.

Canthaxanthin

A red-orange carotenoid pigment used as a color additive and formerly in tanning products; excessive ingestion can cause retinal crystal deposits and liver injury.

Canthotomy

A surgical incision through the outer canthus of the eyelids, usually performed urgently to relieve pressure in the orbit.

Canthus

The angle where the upper and lower eyelids meet.

Cantlie Line

Functional liver division extending from the gallbladder fossa to the IVC, corresponding to the middle hepatic vein plane. Unlike the falciform ligament (anatomical division), it represents the true functional blood supply/biliary drainage division. Right hepatic artery and portal vein supply the right half; left supply the left.

Cantuzumab Ravtansine

A medicine or chemical compound identified by cantuzumab ravtansine. Its uses, effects, and risks depend on the specific substance and clinical context.

Capacity

Autonomous voluntary authorization given by a competent patient to undergo a specific medical intervention, procedure, or clinical trial. See informed consent.

Capacity And Competence

Decision-making capacity is a clinical evaluation performed by a physician to determine whether a patient understands the nature, consequences, risks, and alternatives of a specific proposed medical decision at a given time.

Capecitabine

An oral prodrug converted in tissues to 5-fluorouracil, used for selected solid tumors; important toxicities include diarrhea, mucositis, hand-foot syndrome, and cardiotoxicity.

Capgras Syndrome

A delusional misidentification syndrome in which a person believes that someone familiar has been

replaced by an identical impostor. It can occur in neurodegenerative disease, psychosis, brain injury, or other neurologic conditions.

Capillaria

A genus of parasitic nematodes; human disease includes intestinal capillariasis and, rarely, infection of the liver or other tissues depending on the species.

Capillaria Philippinensis

A parasitic nematode acquired from raw or undercooked freshwater fish that causes intestinal capillariasis, with diarrhea, abdominal pain, malabsorption, and protein loss.

Capillaries

The smallest blood vessels, where oxygen, nutrients, and waste products move between blood and tissues.

Capillaroscopy

Examination of the small vessels in the nail-fold or another accessible surface using magnification. It can reveal capillary dilation, hemorrhage, avascular areas, and abnormal architecture in connective-tissue and microvascular disease.

Capillary

The smallest blood vessel in the body, forming a microscopic network (capillary bed) that connects arterioles to venules.

Capillary Bed

A network of interconnected capillaries that links arterioles to venules within a tissue. It provides a large surface area for exchange of oxygen, carbon dioxide, nutrients, hormones, and metabolic waste between blood and interstitial fluid.

Capillary Electrophoresis

A diagnostic test or assessment related to capillary electrophoresis. Results are interpreted with the clinical findings, test method, and reference range.

Capillary Leak Syndrome

A disorder of increased capillary permeability in

which plasma and proteins move from the intravascular space into surrounding tissues. It may cause hypotension, haemoconcentration, hypoalbuminaemia, oedema, and shock; causes include severe infection, immune-mediated disease, toxins, and rare idiopathic episodes.

Capillary Microscope

A capillary microscope magnifies small blood vessels, especially nail-fold capillaries, to assess structure and abnormalities in microcirculation.

Capillary Nail Refill Test

A bedside test of peripheral perfusion performed by pressing a nail bed until it blanches and observing the time for color to return. Results can be affected by temperature, lighting, age, shock, and peripheral vascular disease and should not be used alone.

Capillary Plasma

The liquid component of blood obtained from small vessels in skin or tissue, containing water, electrolytes, proteins, hormones, nutrients, and waste products.

Capillary Refill

A bedside assessment of peripheral perfusion performed by briefly pressing a nail bed or skin and measuring the time for normal color to return. Delayed refill may indicate poor circulation, shock, hypothermia, or vascular disease.

Capillary Refill Time

The time required for color to return after pressure briefly blanches a nail bed or area of skin. A delayed result may suggest impaired peripheral perfusion, but interpretation depends on age, temperature, technique, and the clinical setting.

Capillary Sample

A small blood specimen obtained by puncturing the skin, usually a finger or heel, for bedside glucose, hemoglobin, newborn screening, or other point-of-care tests.

Capillary Vessel

A microscopic vessel with a thin endothelial wall where blood exchanges oxygen, carbon dioxide, nutrients, fluid, and waste with surrounding tissue.

Capitate

Largest carpal bone, centrally located in the distal row as the carpal keystone. Rounded head articulates with scaphoid and lunate; distal surface with the third metacarpal. Last carpal bone to ossify. Wrist joint axis, bearing significant axial load. Avascular necrosis may occur with fractures similar to scaphoid injuries.

Capitation

Capitation is a healthcare payment arrangement in which a provider receives a fixed amount per enrolled person for defined services over a period.

Caplan'S Syndrome

Caplan's syndrome is pneumoconiosis with rheumatoid arthritis, producing lung nodules in people exposed to mineral or coal dust.

Capnocytophaga

A biomedical term referring to capnocytophaga; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Capnocytophaga Canimorsus

A Gram-negative, capnophilic bacillus found in the oral flora of dogs and cats that causes severe sepsis, meningitis, and cellulitis in humans, particularly the immunocompromised, asplenic, or elderly. Infection typically follows dog or cat bites. It is a fastidious organism requiring prolonged incubation for culture.

Capnocytophaga Gingivalis

A biomedical term referring to capnocytophaga gingivalis; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Capnocytophaga Infection

An infection caused by Capnocytophaga bacteria, sometimes transmitted through dog or cat bites or scratches.

Capnocytophaga Ochracea

A biomedical term referring to capnocytophaga ochracea; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Capnocytophaga Sputigena

A biomedical term referring to capnocytophaga sputigena; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Capnography

Continuous measurement and graphical display of exhaled carbon dioxide during anesthesia and sedation. The waveform and end-tidal value help assess ventilation, airway patency, circulation, and confirmation of tracheal tube placement.

Capreomycin

A medicine or chemical compound identified by capreomycin. Its uses, effects, and risks depend on the specific substance and clinical context.

Capromab Pendetide

A medicine or chemical compound identified by capromab pendetide. Its uses, effects, and risks depend on the specific substance and clinical context.

Caps

A prescription abbreviation for capsules, oral dosage forms in which medicine is enclosed in a soluble shell.

Capsaicin

A compound from chili peppers used in some topical medicines to reduce certain types of pain.

Capsid

The protein shell surrounding the genome of a virus, protecting nucleic acid and helping determine at-

tachment, entry, host range, and immune recognition.

Capsid Proteins

Viral structural proteins that assemble around the genome to form the capsid and often participate in host-cell attachment, entry, and immune recognition.

Capsular Contracture

Abnormal tightening and hardening of the fibrous capsule around an implanted device, especially a breast implant. It may cause pain, distortion, firmness, or displacement.

Capsule

A membrane, fibrous covering, or enclosed dosage form; in anatomy it may surround an organ or joint, while in pharmacology it contains medicine within a soluble shell.

Capsule Endoscope

A swallowable camera-containing capsule that transmits images while passing through the gastrointestinal tract, especially useful for examining the small intestine.

Capsule Endoscopy

Non-invasive diagnostic technique using a swallowable wireless endoscopic device to capture images of the small intestinal mucosa. The capsule contains a miniature video camera and transmitter that sends images to a receiver worn on the patient's belt.

Capsule Of Glisson

The connective-tissue capsule surrounding the liver, which sends fibrous septa inward along portal tracts and supports the organ's vessels, ducts, and nerves.

Capsule Of Tenon

A fascial sheath surrounding the globe of the eye and separating it from orbital fat. It contributes to eye movement and forms a potential space used in some ophthalmic procedures.

Capsulectomy

Surgical removal of part or all of a fibrous capsule surrounding an organ, implant, or other structure. The indication depends on contracture, infection, pathology, or the need to revise a reconstruction.

Captopril

A medicine or chemical compound identified by captopril. Its uses, effects, and risks depend on the specific substance and clinical context.

Captopril Renal Scan

A captopril renal scan uses a radiotracer before and after captopril to assess renovascular hypertension by detecting impaired renal perfusion or function.

Caput Medusae

Dilated periumbilical veins radiating outward from the umbilicus, seen in portal hypertension with recanalization of the umbilical vein. Not strictly a cardiac finding but relevant in right heart failure causing hepatic congestion and portal hypertension. It must be distinguished from caval collateral circulation.

Caput Succedaneum

Edema of the scalp presenting as a soft, boggy swelling that crosses suture lines. Caused by pressure of the vaginal walls on the fetal head during labor. Resolves within hours to days. Differentiated from cephalohematoma (subperiosteal, does not cross suture lines) and subgaleal hemorrhage (life-threatening, crosses suture lines).

CAR T-Cell

Chimeric antigen receptor T-cells. Engineered immune cells targeting cancer.

Chimeric Antigen Receptor T-Cell Therapy

A cancer treatment that collects a person's T cells, changes them in a laboratory so they can recognize cancer cells, and returns them to the body to attack the cancer.

Carbachol

A cholinergic agonist that stimulates muscarinic receptors; ophthalmic carbachol can constrict the pupil and lower intraocular pressure during selected procedures.

Carbadox

A quinoxaline antimicrobial growth-promoting drug formerly used in food-producing animals; safety concerns and carcinogenic potential have led to restrictions or bans.

Carbamate

A chemical group found in pesticides and medicines; carbamate insecticides reversibly inhibit acetylcholinesterase and can cause a cholinergic toxidrome.

Carbamazepine

An anticonvulsant and mood-stabilizing medicine used for focal seizures, trigeminal neuralgia, and selected bipolar disorder; serious risks include severe skin reactions, blood dyscrasias, liver injury, and hyponatremia.

Carbamide

Carbamide is urea, a nitrogen-containing compound produced by protein metabolism and excreted by the kidneys; it is also used in some topical preparations.

Carbapenem

A broad-spectrum beta-lactam antibiotic class that inhibits bacterial cell-wall synthesis and is reserved for serious infections, with resistance increasingly limiting use.

Carbapenem Resistant Enterobacteriaceae (CRE Infection)

Alternate index label for CRE Infection.

Carbapenem-Resistant Enterobacteriaceae Infection

Infection caused by Enterobacteriaceae bacteria resistant to carbapenem antibiotics, often involving the urinary tract, lungs, bloodstream, or wounds.

These infections are difficult to treat and may spread in healthcare settings. Culture and susceptibility testing guide therapy, with strict infection-control measures needed to prevent transmission.

Carbapenem-Resistant Organisms

Bacteria that are resistant to carbapenem antibiotics, typically through production of carbapenemases. The most important carbapenemases include KPC (*Klebsiella pneumoniae* carbapenemase), NDM (New Delhi metallo-beta-lactamase), and OXA-48-like enzymes. CRO are a major public health threat due to limited treatment options.

Carbaryl

A carbamate insecticide that reversibly inhibits acetylcholinesterase; substantial exposure can cause sweating, salivation, vomiting, muscle weakness, and breathing problems.

Carbazilquinone

An investigational quinone-containing anticancer compound studied for cytotoxic effects and bioreductive activity; it is not a routine clinical medicine.

Carbenicillin

A medicine or chemical compound identified by carbenicillin. Its uses, effects, and risks depend on the specific substance and clinical context.

Carbidopa

A peripheral dopa decarboxylase inhibitor that does not cross the blood-brain barrier; combined with levodopa to prevent its peripheral conversion to dopamine, reducing side effects (nausea, hypotension) and increasing CNS levodopa delivery.

Carbimazole

Carbimazole is an antithyroid drug converted to methimazole that reduces thyroid-hormone synthesis; rare serious adverse effects include agranulocytosis and liver injury.

Carbofuran

A highly toxic carbamate pesticide that inhibits

acetylcholinesterase and can cause rapid cholinergic poisoning after ingestion, inhalation, or skin exposure.

Carbogen

A breathing mixture containing oxygen and carbon dioxide, used experimentally or in selected treatments to stimulate ventilation or modify tissue oxygenation.

Carbohydrate

A class of organic compounds composed of carbon, hydrogen, and oxygen (general formula $C_m(H_2O)_n$), serving as the primary energy source for the body.

Carbohydrate Counting

Method for diabetes management. Matching insulin to carbohydrate intake. Education essential.

Carbohydrate Metabolism

The biochemical processes by which cells break down and synthesize carbohydrates, including glycolysis, glycogenesis, glycogenolysis, and gluconeogenesis. It maintains blood glucose levels and provides energy; disorders include diabetes mellitus and glycogen storage diseases.

Carbohydrates

Sugars, starches, and fiber that form one of the main types of nutrients in food. Digestible carbohydrates are broken down into glucose for energy, while fiber is not fully digested and helps support bowel function and blood-sugar control.

Carbolic Acid (Phenol)

An aromatic organic compound (C_6H_5OH) with antiseptic, disinfectant, and caustic properties. Historically significant as the first surgical antiseptic introduced by Joseph Lister in 1867, dramatically reducing post-operative infection and mortality. Mechanism: denatures bacterial proteins and disrupts cell membranes.

Alternate Names:

Phenol

Carbolic Acid Poisoning

Exposure to phenol, a corrosive and systemically toxic chemical that can cause burns, altered consciousness, metabolic acidosis, and organ injury.

Carbon

A chemical element forming the backbone of organic molecules such as carbohydrates, fats, proteins, and nucleic acids.

Carbon Dioxide

A colorless gas produced by cellular metabolism and removed mainly by the lungs during exhalation.

Carbon Dioxide Absorber

A safety device used in anesthesia breathing circuits that filters out carbon dioxide from a patient's exhaled breath, allowing the remaining anesthetic gases to be safely recycled and rebreathed.

Carbon Dioxide Transport

The movement of CO_2 from tissues to lungs. CO_2 is carried in blood in three forms: dissolved (approximately 10 percent), bound to hemoglobin as carbamino compounds (approximately 20 percent), and as bicarbonate (approximately 70 percent).

Carbon Ion Therapy

Radiation using carbon ions. Higher biological effectiveness. Limited availability.

Carbon Monoxide

A colorless, odorless gas that can prevent blood from carrying oxygen and cause life-threatening poisoning.

Carbon Monoxide Poisoning

Inhibits hemoglobin oxygen binding (affinity 200x oxygen). Causes headache, dizziness, nausea, cherry-red skin, syncope, death. Diagnosis: carboxyhemoglobin level (not routine). Treatment: 100% oxygen, hyperbaric oxygen if severe (neurological symptoms, $COHb >25\%$, pregnancy).

Carbon Tetrachloride

A medicine or chemical compound identified by carbon tetrachloride. Its uses, effects, and risks depend on the specific substance and clinical context.

Carbonate

A medicine or chemical compound identified by carbonate. Its uses, effects, and risks depend on the specific substance and clinical context.

Carbonic Anhydrase Inhibitors

Medicines that inhibit carbonic anhydrase and reduce formation of aqueous humor. They are used mainly to lower intraocular pressure and, in some circumstances, to treat edema or metabolic disorders.

Carbophenothion

An organophosphate insecticide that inhibits acetylcholinesterase; poisoning produces excessive cholinergic activity and may cause respiratory failure.

Carboplatin

A platinum chemotherapy drug that cross-links DNA and prevents replication; common dose-limiting toxicity is thrombocytopenia, with nausea, anemia, and hypersensitivity also possible.

Carboprost

A prostaglandin F₂-alpha analogue that contracts uterine muscle and is used for selected cases of postpartum hemorrhage or pregnancy termination; asthma is an important precaution.

Carbosulfan

A carbamate insecticide that inhibits acetylcholinesterase and can cause cholinergic poisoning after significant exposure.

Carboxy-Lyases

Enzymes that break a carbon-carbon bond and release carbon dioxide, participating in metabolic and biochemical reactions.

Carboxyhaemoglobin

Carboxyhaemoglobin is hemoglobin bound to carbon monoxide, reducing oxygen carriage and release; elevated levels indicate carbon-monoxide exposure.

Carboxyhemoglobin

Toxicity from inhalation of carbon monoxide, a colorless, odorless gas binding hemoglobin 200-300 times more avidly than oxygen, forming carboxyhemoglobin (COHb) and reducing oxygen delivery. See carbon monoxide poisoning.

Carboxyhemoglobinemia

An abnormally high level of carboxyhemoglobin formed when carbon monoxide binds hemoglobin, reducing oxygen delivery and impairing oxygen release to tissues.

Carboxyl Group

A functional chemical group containing carbon, oxygen, and hydroxyl; it gives many molecules acidic properties.

Carboxylesterase

An enzyme that hydrolyzes carboxylic esters, helping metabolize medicines, toxins, and endogenous lipids.

Carboxylic Acid

An organic compound containing a carboxyl group, which can donate a proton and form salts or esters; many biologically important fatty acids and metabolic intermediates are carboxylic acids.

Carboxypeptidase H

An enzyme involved in processing peptide hormones and neuropeptides by removing terminal amino acids from precursor molecules.

Carbuncle

A cluster of interconnected furuncles caused most often by *Staphylococcus aureus*, producing a painful inflamed swelling with multiple draining points and sometimes fever; diabetes increases risk.

Carbuncles And Boils

Alternate index label; see *Boils and carbuncles*.

Carcinoembryonal Antigen

The tumor marker carcinoembryonic antigen, or CEA, a glycoprotein that may be elevated in colorectal and other cancers but is not specific enough for general screening.

Carcinoembryonic Antigen

An oncofetal cell-surface glycoprotein measured in serum primarily as a tumor marker for colorectal adenocarcinoma, as well as gastric, pancreatic, and medullary thyroid carcinomas. Because CEA lacks sufficient sensitivity and specificity for primary cancer screening and can be elevated in benign conditions (cigarette smoking, cirrhosis, peptic ulcer disease, inflammatory bowel disease), its primary clinical utility lies in serial postoperative surveillance to detect early tumor recurrence and evaluate therapeutic response.

Carcinogen

Substance causing cancer. Types: chemical (benzene, asbestos, aflatoxin), physical (radiation), biological (HPV, HBV, HCV). Mechanisms: DNA damage, mutations, epigenetic changes. IARC classifications: Group 1 (carcinogenic), Group 2A (probably carcinogenic), Group 2B (possibly carcinogenic), Group 3 (not classifiable).

Carcinogen Testing

Testing used to assess whether a chemical, exposure, or treatment can cause cancer, using laboratory, animal, or population studies.

Carcinogenesis

The process by which normal cells acquire changes that lead to cancer.

Carcinogenic

Capable of causing or promoting cancer through genetic damage, altered cell regulation, or other mechanisms.

Carcinoid

A carcinoid is a usually slow-growing neuroendocrine tumor that may release hormones such as serotonin and occur in the gastrointestinal tract, appendix, or lungs.

Carcinoid Heart Disease

Cardiac complication of carcinoid syndrome occurring in approximately fifty percent of patients with hepatic metastases. Serotonin and other vasoactive substances bypass hepatic metabolism and cause endocardial fibrosis affecting the right-sided heart valves particularly the tricuspid and pulmonary valves.

Carcinoid Syndrome

A group of symptoms caused by substances such as serotonin released by a neuroendocrine tumor. It may cause facial flushing, diarrhea, wheezing, fast heartbeat, and, over time, right-sided heart disease. A twenty-four-hour urine test for 5-hydroxyindoleacetic acid may help confirm the diagnosis.

Carcinoid Tumor

Carcinoid tumors are well-differentiated neuroendocrine neoplasms that arise from enterochromaffin cells throughout the gastrointestinal tract and bronchial system. The most common primary sites are the appendix, small intestine, and rectum.

Carcinoid Tumors

Slow-growing neuroendocrine tumors that can release hormones and arise in organs such as the digestive tract or lungs.

Alternate Names:

Cancer, Carcinoid Tumors

Carcinoma

Malignant tumor of epithelial origin. Types: squamous cell carcinoma, adenocarcinoma, basal cell carcinoma, transitional cell carcinoma. Most common cancer type (80-90% of malignancies). Examples: lung carcinoma, breast carcinoma, colon carcinoma.

Carcinoma En Cuirasse

An unusual form of breast cancer characterized by diffuse, firm, indurated skin that resembles armor. It results from extensive dermal lymphatic invasion by tumor cells.

Carcinoma Erysipeloides

Cutaneous lymphatic spread of an internal carcinoma producing a sharply demarcated, erythematous, indurated plaque that may resemble erysipelas.

Carcinoma In Situ

Full-thickness dysplasia without invasion through basement membrane. Stage 0 cancer.

Carcinoma Of Unknown Primary

Cancer that has spread but whose original site cannot be identified despite evaluation.

Alternate Names:

Cancer Of Unknown Origin Occult Primary Cancer

Carcinomatosis

Widespread dissemination of carcinoma to multiple surfaces or organs.

Carcinosarcoma

A malignant tumor containing both carcinomatous and sarcomatous elements, most commonly of the uterus (malignant mixed Müllerian tumor), with an aggressive course.

Cardia

The part of the stomach surrounding the opening where the esophagus joins the stomach.

Cardiac

1. Pertaining to the heart (*cor*), including its muscular wall layers (endocardium, myocardium, epicardium), four chambers (atria and ventricles), atrioventricular and semilunar valves, intrinsic electrical conduction system (sinoatrial node, atrioventricular node, bundle of His, Purkinje network), and coronary arterial circulation. Pathology includes ischemic heart disease, valvular stenosis and regurgitation, cardiomyopathies, heart failure, and cardiac

arrhythmias. 2. In gastroenterology, pertaining to the cardia (cardiac orifice), the anatomical transition zone of the stomach immediately distal to the gastroesophageal junction where the tubular esophagus joins the gastric pouch, an important site for gastric adenocarcinoma (cardia cancer), hiatal herniation, and gastroesophageal reflux pathology.

Alternate Names:

Heart-related; cardiac-related.

Cardiac Ablation

A procedure that destroys a small area of heart tissue causing an abnormal rhythm.

Cardiac Ablation Procedures

A medical procedure involving cardiac ablation procedures. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Cardiac Allograft

A donor heart transplanted into a recipient with end-stage heart failure. The procedure requires lifelong immunosuppression to prevent rejection. Complications include rejection, cardiac allograft vasculopathy, infection, and malignancy; monitored by endomyocardial biopsy and echocardiography.

Cardiac Allograft Vasculopathy

A diffuse, progressive form of coronary arterial intimal thickening after heart transplantation that can cause silent ischemia, graft dysfunction, arrhythmia, or sudden death.

Cardiac Amyloidosis

Deposition of amyloid protein in the heart, causing stiff ventricular walls, impaired filling, arrhythmias, or heart failure. Important types include immunoglobulin light-chain (AL) and transthyretin (ATTR) amyloidosis; treatment depends on the type and organ involvement.

Cardiac Amyloidosis Diagnosis

Diagnosis involves a combination of clinical suspicion, imaging, and tissue biopsy. Non-invasive diagnosis using Tc-99m pyrophosphate (PYP) or DPD

scintigraphy for ATTR cardiac amyloidosis (with negative serum/urine free light chains to exclude AL amyloidosis).

Cardiac Amyloidosis Staging

Mayo Clinic staging system for ATTR cardiac amyloidosis: Stage I (NT-proBNP <3000 pg/mL), Stage II (NT-proBNP 3000-8499), Stage III (NT-proBNP ≥8500). Median survival: Stage I: 69 months, Stage II: 47 months, Stage III: 14 months. Treatment with tafamidis (TTR stabilizer) improves survival and functional capacity.

Cardiac Arrest

Sudden loss of effective heart pumping, causing unresponsiveness, absent normal breathing, and no detectable pulse. Initial rhythms may be shockable or nonshockable; treatment requires immediate cardiopulmonary resuscitation, defibrillation when indicated, and advanced resuscitation care.

Cardiac Arrest Survival

Outcome after cardiac arrest depends on the cause, location, initial rhythm, time to CPR and defibrillation, and post-arrest care. Early recognition, high-quality CPR, rapid defibrillation when indicated, advanced life support, and appropriate post-arrest treatment improve the chance of survival.

Cardiac Arrest, Sudden

Alternate index label; see *Sudden cardiac arrest*.

Cardiac Arrhythmias

Abnormalities of the heart rate or rhythm caused by disturbances in impulse formation or conduction. They include tachycardia, bradycardia, and irregular rhythms.

Cardiac Axis

The mean direction of ventricular depolarization in the frontal plane, estimated from the QRS complex on a 12-lead ECG. Left or right axis deviation may reflect conduction disease, ventricular hypertrophy, infarction, or altered cardiac position.

Cardiac Biomarkers

Blood measurements used to assess myocardial injury. Cardiac troponins I and T are the preferred specific biomarkers; CK-MB may assist in selected situations, while myoglobin is sensitive but nonspecific. Results must be interpreted with symptoms, electrocardiography, and serial testing.

Cardiac Cachexia

A severe wasting syndrome associated with advanced heart failure or other serious cardiac disease, characterized by involuntary weight loss, skeletal-muscle loss, weakness, and reduced functional capacity. It reflects neurohormonal, inflammatory, metabolic, and nutritional abnormalities and is associated with poor prognosis.

Cardiac Care

Medical care that prevents, diagnoses, and treats heart and circulation disorders through monitoring, medicines, procedures, and rehabilitation.

Cardiac Catheter

A thin tube inserted into a blood vessel and advanced to the heart for measurement, imaging, sampling, or treatment.

Cardiac Catheterization

Invasive procedure involving insertion of catheters into the heart via peripheral arteries or veins. Diagnostic catheterization measures intracardiac pressures, saturations, and performs coronary angiography. Indications include evaluation of CAD, valvular disease, cardiomyopathy, and congenital heart disease.

Cardiac Conduction System

The specialized tissue responsible for generating and conducting electrical impulses through the heart. Components include the sinoatrial (SA) node, internodal pathways, atrioventricular (AV) node, bundle of His, right and left bundle branches, and Purkinje fibers.

Cardiac CT

CT imaging of heart and coronary arteries. Coro-

nary calcium scoring, coronary CT angiography, cardiac function assessment.

Cardiac CT Angiography

Noninvasive imaging using CT to visualize coronary arteries and cardiac structures. It has high negative predictive value for excluding coronary artery disease, making it useful for low-to-intermediate risk patients with chest pain. Calcium scoring quantifies coronary artery calcification as a marker of atherosclerosis.

Cardiac Cycle

The sequence of events occurring during one complete heartbeat, consisting of systole (contraction) and diastole (relaxation). During diastole, the ventricles fill with blood; during systole, they contract and eject blood.

Cardiac Defibrillation

Delivery of an unsynchronized electrical shock for ventricular fibrillation or pulseless ventricular tachycardia. It is combined with high-quality CPR and advanced life support according to the cardiac-arrest rhythm and protocol.

Cardiac Device Infections

Infection involving a pacemaker, implantable defibrillator, generator pocket, or lead. Findings may include local inflammation, drainage, bacteremia, or endocarditis. Evaluation uses clinical examination, blood cultures, and imaging when indicated; treatment may require antimicrobial therapy and removal of infected hardware.

Cardiac Diet

A dietary pattern emphasizing vegetables, fruits, whole grains, legumes, fish, unsaturated fats, and limited sodium, added sugars, and saturated or trans fats. It supports cardiovascular risk reduction but should be individualized for heart failure, kidney disease, diabetes, and nutritional adequacy.

Cardiac Enzymes

Historically used blood markers of myocardial injury, especially creatine kinase-MB; cardiac troponins are

now the preferred biomarkers for detecting myocardial injury and must be interpreted with symptoms, ECG findings, and serial changes.

Cardiac Event Monitors

Portable devices that record or transmit an electrocardiogram when symptoms occur or at programmed intervals, helping detect intermittent arrhythmias.

Cardiac Failure

Alternate index label; see *Heart failure*.

Cardiac Fibroma

A rare benign fibrous tumor of the heart that can obstruct blood flow or disturb the heart rhythm.

Cardiac Gated SPECT

SPECT imaging synchronized with ECG. Assess ventricular function, wall motion, viability.

Cardiac Glycoside

A drug or plant compound that increases heart contraction and affects cardiac conduction; digoxin is the main clinical example.

Cardiac Glycoside Overdose

Overdose of a cardiac glycoside, such as digoxin, that may cause nausea, vomiting, visual changes, confusion, bradyarrhythmias, tachyarrhythmias, and hyperkalemia.

Cardiac Hemangioma

A rare benign tumor made of blood vessels in the heart; symptoms depend on its size and location.

Cardiac Imaging Tests

Noninvasive or minimally invasive studies used to evaluate cardiac structure, function, blood flow, or tissue characteristics. They include echocardiography, cardiac CT, cardiac MRI, nuclear perfusion imaging, and other modality-specific tests selected according to the clinical question.

Cardiac Index

Cardiac output normalized to body surface area, calculated as CO / BSA . Normal cardiac index is 2.5-4.2 L/min/m². It provides a more accurate assessment of cardiac function than cardiac output alone, accounting for body size. A cardiac index below 2.2 L/min/m² indicates cardiogenic shock.

Cardiac Intravascular Ultrasound

A catheter-based ultrasound technique that produces cross-sectional images from inside a coronary artery. It assesses plaque, vessel dimensions, calcification, dissections, and stent expansion during selected coronary procedures.

Cardiac Ischemia

Alternate index label; see *Myocardial ischemia*.

Cardiac Massage

Cardiac massage is manual compression of the heart, historically performed through the chest or during open surgery; modern cardiac arrest care uses chest compressions and advanced resuscitation.

Cardiac Metastases

More common than primary cardiac tumors. Most frequent sources: lung cancer, breast cancer, melanoma, lymphoma, and renal cell carcinoma. Most commonly involve the pericardium (effusion, tamponade), but can affect myocardium and endocardium.

Cardiac MRI

Magnetic-resonance imaging of the heart that characterizes ventricular function, myocardial tissue, perfusion, scar, edema, infiltration, and congenital anatomy without ionizing radiation. Gadolinium contrast is avoided or used cautiously in selected patients with severe renal dysfunction.

Cardiac Muscle

Involuntary, striated muscle forming the myocardium. Cardiac muscle cells are electrically connected by intercalated discs, allowing coordinated contraction of the heart.

Cardiac Myocyte

A specialized heart-muscle cell that contracts to pump blood and uses electrical signals and calcium cycling.

Cardiac Myocytes

Muscle cells of the heart that generate force for cardiac contraction.

Cardiac Myxoma

The most common primary benign cardiac tumor, usually arising in the left atrium from the interatrial septum. It may cause positional obstruction, emboli, or constitutional symptoms and is generally treated by surgical excision.

Cardiac Output

The volume of blood pumped by the heart per minute, calculated as the heart rate multiplied by the stroke volume.

Cardiac Output Measurement

Methods include: thermodilution via pulmonary artery catheter (bolus of cold saline and temperature change measurement), Fick principle (O_2 consumption / arteriovenous O_2 difference), echocardiographic Doppler (flow velocity time integral x valve area x heart rate), and non-invasive cardiac output monitors (bioimpedance, bioreactance).

Cardiac Pacemaker

A cardiac pacemaker is an implanted or external device that delivers electrical impulses to maintain an adequate heart rate when intrinsic conduction is too slow or unreliable.

Cardiac Pacemaker Types

Single-chamber: pacing one chamber (atrium or ventricle). Dual-chamber: paces and senses both atrium and ventricle, maintaining AV synchrony. Biventricular (CRT): paces both ventricles for resynchronization in heart failure.

Cardiac Perforation

A hole or tear through the heart wall that can

cause bleeding into the pericardial sac and cardiac tamponade.

Cardiac PET

Nuclear-medicine imaging of the heart using positron-emitting radiotracers. Depending on the tracer and protocol, it can assess myocardial perfusion and blood flow, identify viable or scarred myocardium, and evaluate selected inflammatory or infiltrative cardiac diseases.

Cardiac PET Imaging

Positron emission tomography using radiotracers (Rb-82, N-13 ammonia, O-15 water) for quantitative myocardial perfusion assessment. Advantages over SPECT: superior resolution, shorter scan time, attenuation correction, and absolute myocardial blood flow quantification (mL/g/min).

Cardiac Power Output

A calculated, comprehensive hemodynamic metric of true central cardiac performance, defined as the product of simultaneously measured mean arterial pressure (MAP, in mmHg) and cardiac output (CO, in L/min) divided by the conversion constant 451, expressed in Watts: $CPO = (MAP \times CO)/451$. Established by the SHOCK trial registry as the single strongest independent hemodynamic predictor of in-hospital mortality in patients presenting with cardiogenic shock, with a resting $CPO \leq 0.53\text{--}0.6\text{ W}$ identifying severe pump failure and guiding the timely escalation to percutaneous or surgical mechanical circulatory support.

Alternate Names:

CPO; Cardiac Power Index (CPI)

Cardiac Radiation Injury

Late complication of mediastinal radiation therapy (Hodgkin lymphoma, breast cancer, lung cancer). Manifestations include pericardial disease (constrictive pericarditis, effusion), coronary artery disease (accelerated atherosclerosis), valvular disease (fibrosis), myocardial fibrosis, and conduction abnormalities.

Cardiac Radionuclide Imaging

Cardiac imaging performed after administration of a radioactive tracer, usually with SPECT or PET, to assess myocardial perfusion, viability, metabolism, or inflammation. ECG-gated studies can also estimate ventricular volumes and ejection fraction; results must be interpreted with symptoms and other cardiac tests.

Cardiac Rehab Outcomes

Cardiac rehabilitation can improve exercise capacity, cardiovascular risk-factor control, quality of life, and psychological well-being after selected cardiac conditions. The benefits and participation rates vary among populations and programs.

Cardiac Rehabilitation

Supervised program for cardiac patients post-MI, PCI, CABG, heart failure. Phases: inpatient (early mobilization), outpatient (supervised exercise, education), maintenance. Components: exercise training, risk factor modification, psychological support, return to work counseling. Reduces mortality, hospitalizations.

Cardiac Resynchronization Therapy

Pacing therapy that coordinates ventricular contraction, usually with biventricular or conduction-system pacing. It is used in selected patients with symptomatic heart failure, reduced ejection fraction, and electrical dyssynchrony such as a prolonged QRS, especially left bundle-branch block, despite appropriate medical treatment. In suitable patients it can improve symptoms and function and reduce hospitalization and mortality.

Cardiac Rhabdomyoma

A benign striated-muscle tumor, usually in children and often associated with tuberous sclerosis.

Cardiac Sarcoidosis

Infiltrative disease characterized by non-caseating granulomas in the myocardium, affecting 25-30% of patients with systemic sarcoidosis. Presentation includes conduction abnormalities (AV block), ventricular arrhythmias, and heart failure.

Cardiac Sarcoma

A rare malignant tumor arising from connective or supporting tissue of the heart; symptoms may include obstruction, arrhythmia, heart failure, or emboli.

Cardiac Skeleton

A dense connective tissue framework composed of four fibrous rings that surround the heart valves and provide structural support. The cardiac skeleton includes the annuli fibrosi (valve rings), the membranous septum, and the left and right fibrous trigones.

Cardiac SPECT

A nuclear-imaging method that evaluates myocardial perfusion and can help identify ischemia, infarction, or viable myocardium. When synchronized with the heartbeat, gated images can also assess ventricular motion and function.

Cardiac Stress Test

Evaluation of cardiac function and ischemia under increased cardiac workload. Exercise stress test (treadmill, Bruce protocol): ECG monitoring for ST-segment changes, symptoms, BP response, and arrhythmias. Sensitivity 65-75% for CAD.

Cardiac Stress Testing

Assessment of cardiac function under physiological or pharmacological stress to detect ischemia. Exercise stress testing involves treadmill or bicycle exercise with continuous ECG monitoring. Pharmacological stress (dobutamine, adenosine, regadenoson) is used when patients cannot exercise.

Cardiac Syncope

Transient loss of consciousness caused by a cardiac disorder that reduces cerebral perfusion, usually through bradyarrhythmia, tachyarrhythmia, structural obstruction, or severe cardiac dysfunction. Syncope during exertion, while supine, or with palpitations or known heart disease requires urgent cardiovascular evaluation because it may indicate a risk of sudden cardiac death.

Cardiac Tamponade

Cardiac tamponade occurs when fluid or blood under pressure around the heart restricts filling, reducing cardiac output and potentially causing obstructive shock, and causing reduced circulation. It is a medical emergency.

Cardiac Tamponade Physiology

Accumulation of pericardial fluid causing compression of cardiac chambers, impairing diastolic filling and reducing cardiac output. Hemodynamic consequences include equalization of diastolic pressures ($RA = RV \text{ diastolic} = PCWP$), pulsus paradoxus, and decreased cardiac output.

Cardiac Transplantation

Replacement of a failing heart with a donor heart for carefully selected patients with end-stage heart failure. Long-term care requires immunosuppression and surveillance for rejection, infection, and graft vasculopathy.

Cardiac Tumors

Tumors involving the heart may be primary or metastatic. Primary tumors are rare; myxoma is a common benign primary tumor and angiosarcoma is a recognized malignant type. Symptoms and treatment depend on the tumor's location, tissue type, obstruction, embolic risk, and spread.

Cardiac Tumors Management

Myxoma: surgical excision is curative, with low recurrence rate. Benign tumors (rhabdomyoma, fibroma) may be observed if asymptomatic but require surgery if causing obstruction or arrhythmia. Malignant tumors: prognosis is poor; treatment may include surgery, chemotherapy, and radiation depending on tumor type.

Cardiac Ventriculography

Imaging of the heart ventricles to assess their size, wall motion, and pumping function.

Cardiac Volume

The amount of blood in a heart chamber at a given time; end-diastolic and end-systolic volumes help

assess ventricular function.

Cardiac Wall

The heart wall, consisting of endocardium, myocardium, and epicardium, which supports electrical conduction and contraction.

Cardinal Body Part

A major body region or structure identified for anatomical description or clinical examination.

Cardinal Cell Part

A principal cell component, such as the nucleus, membrane, cytoplasm, or an organelle, with a role in cell function.

Cardinal Ligament

A dense condensation of endopelvic connective tissue extending from the supravaginal cervix and lateral vaginal fornices to the pelvic sidewalls, located at the base of the broad ligament. Serves as the primary lateral fascial support maintaining uterine and vaginal apical suspension, and transmits the uterine artery and vein across the ureter (the “water under the bridge” anatomical relationship), making it a critical pedicle isolated and clamped during radical hysterectomy.

Alternate Names:

Mackenrodt’s Ligament; Lateral Cervical Ligament; Transverse Cervical Ligament

Cardio-Oncology

The subspecialty focusing on cardiovascular care of cancer patients, addressing cancer therapy-related cardiac dysfunction, prevention, and management. Key concerns include anthracycline cardiotoxicity, trastuzumab-related cardiomyopathy, immune checkpoint inhibitor myocarditis, and radiation-induced heart disease.

Cardiobacteriaceae

A family of Gram-negative HACEK bacteria that includes *Cardiobacterium* and can cause infective endocarditis.

Cardiobacterium

A Gram-negative HACEK bacterium that can cause slow-growing bloodstream infection and infective endocarditis.

Cardiobacterium Hominis

The *Cardiobacterium* species most often associated with human disease, especially subacute infective endocarditis.

Cardiofaciocutaneous Syndrome

A rare, clinically variable genetic disorder belonging to the RASopathy family, caused by heterozygous germline gain-of-function mutations in components of the MAPK/ERK signaling cascade, most commonly *BRAF* (approximately 75%), *MAP2K1*, or *MAP2K2*. Characterized by distinctive craniofacial dysmorphism (macrocephaly, prominent forehead, ocular hypertelorism, bitemporal constriction), congenital cardiac anomalies (pulmonic stenosis, hypertrophic cardiomyopathy, atrial septal defect), ectodermal abnormalities (sparse, curly, friable scalp hair, dystrophic nails, severe keratosis pilaris, progressive ichthyosis, eczema), and profound global developmental delay with hypotonia.

Alternate Names:

CFC Syndrome

Cardiogenic Shock

A state of inadequate tissue perfusion caused by severe cardiac dysfunction. It may produce low blood pressure, altered mental status, cool extremities, reduced urine output, metabolic acidosis, or other signs of organ hypoperfusion.

Cardiogenic Shock Management

Emergency management of shock caused by severe cardiac dysfunction. It includes support of oxygenation and circulation, treatment of the underlying cause, and selected use of vasopressors, inotropes, mechanical support, or revascularization.

Cardiolipin

A distinctive phospholipid concentrated in the inner mitochondrial membrane; antibodies to cardiolipin are among the laboratory markers of antiphospho-

lipid syndrome.

Cardiologist

Physician specializing in diagnosis and treatment of heart and cardiovascular diseases; trained in internal medicine and cardiology.

Cardiology

The branch of medicine devoted to the study, diagnosis, and treatment of disorders of the heart and circulatory system. Cardiologists manage conditions including coronary artery disease, heart failure, valvular heart disease, arrhythmias, cardiomyopathy, pericardial disease, congenital heart disease, and hypertension.

Cardiomegaly

Abnormal enlargement of the heart, identified clinically, on chest radiography, or with cardiac imaging. It may reflect chamber dilatation, myocardial hypertrophy, pericardial effusion, or a combination of these; causes include hypertension, valvular disease, cardiomyopathy, congenital heart disease, and high-output states.

Cardiomyopathy

Cardiomyopathy is disease of the heart muscle that can impair contraction, relaxation, rhythm, or valve function and may lead to heart failure or sudden death. disease, valvular disease, and congenital heart disease as primary causes. The three main types are dilated, hypertrophic, and restrictive cardiomyopathy.

Cardiomyopathy (Restrictive)

Alternate index label for Restrictive.

Cardiomyopathy Classification

The MOGE(S) classification system provides phenotypic and genotypic description: M (morphofunctional phenotype), O (organ/system involvement), G (genetic inheritance), E (etiology), S (functional status/NYHA class).

Cardiomyopathy, Dilated

Alternate index label; see *Dilated cardiomyopathy*.

Cardiomyopathy, Hypertrophic

Alternate index label; see *Hypertrophic cardiomyopathy*.

Cardiopathy

Any disease or disorder of the heart, including diseases of the heart muscle, valves, coronary arteries, rhythm, or congenital structure.

Cardioplegia

Deliberate temporary arrest and protection of the heart during cardiac surgery, usually achieved with a cold or blood-based solution that alters myocardial electrical and metabolic activity.

Cardiopulmonary

Relating to the heart and lungs or to their combined function.

Alternate Names:

Cardiorespiratory.

Cardiopulmonary Bypass

A technique that temporarily circulates and oxygenates blood during some heart operations.

Cardiopulmonary Bypass Machine

A machine that temporarily circulates and oxygenates blood during selected heart or major-vessel operations while the heart and lungs are not performing their usual functions.

Alternate Names:

Heart-lung machine; cardiopulmonary bypass pump.

Cardiopulmonary Exercise Testing

Comprehensive assessment of cardiovascular and pulmonary function during graded exercise on a treadmill or cycle ergometer. Measures: VO₂ max, anaerobic threshold, ventilatory efficiency (VE/VCO₂ slope), and hemodynamic responses.

Cardiopulmonary Resuscitation

A sequence of chest compressions, ventilation, defib-

rillation when indicated, and advanced life-support interventions for cardiac arrest. Its purpose is to maintain circulation and oxygen delivery until spontaneous circulation is restored or definitive treatment is provided.

Cardiopulmonary Resuscitation

CPR combines chest compressions and rescue breaths or assisted ventilation to maintain circulation and oxygenation during cardiac arrest.

Cardiorenal Syndrome

Dysfunction of the heart and kidneys where acute or chronic failure of one organ contributes to dysfunction of the other. Type 1: acute cardiac dysfunction causing acute kidney injury (e.g., cardiogenic shock). Type 2: chronic cardiac dysfunction causing progressive CKD. Type 3: acute kidney injury causing cardiac dysfunction.

Cardiorespiratory

Cardiorespiratory means relating to the heart, blood vessels, and respiratory system considered together, as in cardiorespiratory fitness or failure.

Cardiorespiratory Endurance

The ability of the heart, lungs, and circulation to deliver oxygen during sustained physical activity.

Alternate Names:

Cardiorespiratory fitness; aerobic endurance.

Cardioselective Beta-Blockers

Cardioselective beta-blockers selectively block beta-1 adrenergic receptors, which are predominant in the heart, while having minimal effects on beta-2 receptors in the bronchial smooth muscle and peripheral vasculature at therapeutic doses. Common cardioselective beta-blockers include metoprolol, atenolol, bisoprolol, and nebivolol.

Cardiospasm

Spasm or impaired relaxation of the lower esophageal sphincter, historically used as a term for achalasia. It causes dysphagia, regurgitation, and sometimes chest discomfort.

Cardiospasm (Achalasia)

Alternate index label for Achalasia.

Cardiotocography

A continuous electronic recording of the fetal heart rate and uterine contractions used to assess fetal well-being during pregnancy and labor. The CTG machine uses two transducers placed on the maternal abdomen: an ultrasound transducer records the fetal heart rate, and a tocodynamometer records uterine contractions.

Cardiotoxicity

Injury or functional impairment of the heart caused by a toxic exposure, systemic disease, or treatment. Manifestations include arrhythmia, myocardial dysfunction, ischemia, or heart failure.

Cardiotoxicity From Chemotherapy

Chemotherapy agents that damage the myocardium, most commonly anthracyclines (doxorubicin, daunorubicin). Cardiotoxicity presents as dilated cardiomyopathy with reduced ejection fraction. Risk is dose-dependent (doxorubicin cumulative dose >550 mg/m²).

Cardiotoxin

A substance that directly damages cardiac muscle or disrupts cardiac electrical or vascular function, potentially causing arrhythmia, heart failure, or myocardial injury.

Cardiovascular

Relating to the heart and blood vessels, including circulation, blood pressure, and diseases of the cardiac or vascular system.

Cardiovascular Agents

Medicines acting on the heart or blood vessels, including drugs for blood pressure, rhythm, angina, heart failure, thrombosis, and lipid disorders.

Cardiovascular Changes In Pregnancy

Plasma volume increases 40-50% (total volume 5 L), red cell mass increases 20-30% (physiologic anemia

of pregnancy due to dilution). Cardiac output increases 30-50% by 32 weeks (stroke volume increases initially, then heart rate).

Cardiovascular Disease

Disease affecting the heart or blood vessels, including coronary artery disease, heart failure, arrhythmias, valvular disease, and vascular disorders.

Cardiovascular Health

Cardio is shorthand for cardiovascular health or exercise that elevates heart rate and respiratory effort to strengthen the heart, lungs, and circulatory system. Aerobic activities such as walking, running, cycling, and swimming improve cardiac output, lower blood pressure, and enhance metabolic efficiency.

Cardiovascular Infection

An infection of the heart, blood vessels, or bloodstream, including endocarditis, myocarditis, and infected vascular devices.

Cardiovascular Injury

Damage to the heart or blood vessels caused by trauma, ischemia, inflammation, toxins, surgery, or other disease.

Cardiovascular Neoplasm

A benign or malignant tumor arising in the heart, blood vessels, or related tissues; evaluation depends on its location, tissue type, and effect on circulation.

Cardiovascular Pharmacology

The study of how medicines affect the heart, blood vessels, blood pressure, cardiac rhythm, coagulation, and lipid metabolism.

Cardiovascular Risk Reduction

Measures that lower the risk of myocardial infarction, stroke, heart failure, and vascular death through smoking cessation, blood-pressure and lipid control, diabetes management, physical activity, healthy diet, weight management, and indicated medicines.

Cardiovascular Screening

Evaluation of cardiovascular risk or disease in people with or without symptoms. It may include blood-pressure measurement, lipid testing, diabetes assessment, and targeted testing based on age and risk.

Cardiovascular Surgery

Surgical procedures performed on the heart and great vessels, including coronary artery bypass grafting, valve repair or replacement, aortic surgery, and heart transplantation. It requires cardiopulmonary bypass for many open procedures and specialized perioperative care.

Cardiovascular System

The organ system comprising the heart and blood vessels (arteries, arterioles, capillaries, venules, veins) responsible for circulating blood throughout the body.

Cardioversion

Restoration of a regular heart rhythm from an arrhythmia, using synchronized electrical shock or medicines; anticoagulation and assessment for atrial thrombus may be needed in atrial fibrillation.

Cardioversion (Synchronized)

Delivery of an electrical shock synchronized with the QRS complex to restore a more organized cardiac rhythm. It is used for selected tachyarrhythmias, with energy, sedation, anticoagulation, and monitoring determined by the rhythm and the patient's clinical condition.

Alternate Names:

Synchronized

Cardiovirus

Cardiovirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Carditis

Inflammation of the heart or one of its layers, includ-

ing the myocardium, endocardium, or pericardium. It may result from infection, immune-mediated disease, or systemic inflammation.

Care Coordination

Deliberate organization and communication of healthcare activities among professionals, services, patients, and caregivers to support safe, continuous, and appropriate care across settings.

Caregiver

A person who provides assistance with another person's health care, daily activities, or support. A caregiver may be a family member, friend, or paid professional.

Caregiver Burden

The emotional, physical, social, or financial strain associated with providing ongoing care for another person.

Caregiver Stress

Emotional, physical, social, or financial strain associated with providing ongoing care for another person.

Caries

A biofilm-mediated, multifactorial disease in which acids from bacterial metabolism of fermentable carbohydrates cause net demineralization of enamel, dentin, or cementum. Early lesions may be arrested or remineralized, but progressive disease forms cavities and can reach the pulp, causing pain or infection.

Caries (Dental)

A biofilm-mediated, sugar-driven, multifactorial disease causing net demineralization of enamel, dentin, or cementum. Risk is influenced by fermentable-carbohydrate frequency, fluoride exposure, saliva, oral hygiene, tooth anatomy, and previous disease.

Alternate Names:

Dental

Caries Risk Assessment

Systematic evaluation of an individual's likelihood of developing dental caries. Low risk: no active caries, good oral hygiene, regular fluoride exposure. Moderate risk: one or more active lesions in past 3 years, fair oral hygiene.

Carina

The ridge at the base of the trachea where it bifurcates into the right and left main bronchi. It is a landmark for endotracheal tube placement and bronchoscopy. The right main bronchus is wider, shorter, and more vertical than the left, making it the preferred path for aspirated foreign bodies.

Cariogenic

Capable of promoting dental caries, especially by producing acid from fermentable carbohydrates or supporting a plaque environment that demineralizes tooth structure.

Cariogram

A computer-based program that illustrates the interaction of caries-related factors and provides an overall caries risk assessment. Factors: diet (content, frequency), bacteria (mutans streptococci, lactobacilli), fluoride exposure, saliva (flow rate, buffering capacity), plaque amount, and clinical/social history.

Carisoprodol

A centrally acting skeletal-muscle relaxant metabolized partly to meprobamate and used short term for acute musculoskeletal pain; sedation, misuse, and dependence are important risks.

Carminative

A substance traditionally used to reduce intestinal gas or discomfort by promoting passage of gas through the digestive tract.

Alternate Names:

Antiflatulent; gas-relieving agent.

Carmine

A red pigment derived from cochineal insects and

used in foods, cosmetics, and medicines; it can rarely cause allergic reactions, including asthma or anaphylaxis.

Carmustine

A medicine or chemical compound identified by carmustine. Its uses, effects, and risks depend on the specific substance and clinical context.

Carney Complex

Autosomal dominant multiple endocrine neoplasia syndrome (NAME syndrome / LAMB syndrome), caused by PRKAR1A (17q22-24, protein kinase A regulatory subunit type 1A, 70% of cases) or PRKACA locus.

Carney Triad

A rare, usually nonhereditary association of gastric epithelioid leiomyosarcoma or gastrointestinal stromal tumor, pulmonary chondroma, and extra-adrenal paraganglioma.

Carnitine

Transports long-chain fatty acids into mitochondria for beta-oxidation. Deficiency causes hypoketotic hypoglycemia.

Carnitine Palmitoyltransferase II Deficiency

An autosomal recessive metabolic disorder of mitochondrial long-chain fatty acid oxidation caused by pathogenic variants in the *CPT2* gene. Impairs the conversion of acylcarnitines back into free fatty acyl-CoAs inside the mitochondrial matrix for β -oxidation. The adult myopathic form, which is the most common presentation, is characterized by recurrent episodes of severe exercise-, fasting-, cold-, or infection-induced myalgias, muscle stiffness, rhabdomyolysis, and marked myoglobinuria carrying a significant risk of acute kidney injury, typically while maintaining normal muscle histology and normal serum creatine kinase levels between acute crises.

Alternate Names:

CPT II Deficiency; CPT2 Deficiency

Carnobacteriaceae

A family of Gram-positive, lactic-acid-producing bacteria found in food, animals, and the environment.

Carnosine

A dipeptide of beta-alanine and histidine found in muscle and nervous tissue that buffers acidity and may limit protein glycation; many proposed supplement benefits remain unproven.

Carnosinemia

A rare inherited disorder of carnosinase deficiency causing elevated carnosine in blood and urine, usually with developmental or neurologic abnormalities of variable severity.

Caroli Disease

Congenital, autosomal recessive fibropolycystic liver disease (PKHD1 gene, encodes fibrocystin/polyductin; same gene as infantile polycystic kidney disease) characterized by non-obstructive, segmental, saccular dilation of the larger intrahepatic bile ducts.

Carotene

An orange-yellow plant pigment and provitamin-A carotenoid; beta-carotene can be converted to retinol, while excess dietary carotene may cause harmless yellow skin discoloration.

Carotenoid

A family of lipid-soluble yellow, orange, or red pigments, including beta-carotene and lutein, that act in photosynthesis, antioxidant pathways, and sometimes vitamin-A production.

Carotenoids

Plant pigments, including beta-carotene, that contribute yellow, orange, or red color and may have antioxidant or provitamin-A activity.

Alternate Names:

Carotenoid pigments; provitamin-A pigments.

Carotid

Relating to either of the major arteries in the neck that carry blood from the heart toward the brain and head.

Carotid Artery

The paired arterial system supplying the head and neck. Each common carotid artery divides into internal and external carotid arteries; the internal carotid supplies the brain and orbit, while the external carotid supplies most external head and neck structures.

Carotid Artery Disease

Narrowing or blockage of the carotid arteries, most often caused by atherosclerotic plaque. It can reduce cerebral blood flow or release emboli and is an important cause of transient ischemic attack and ischemic stroke.

Carotid Artery Stenting

Endovascular placement of a stent across a carotid stenosis, usually with embolic-protection strategies, to improve cerebral blood flow and reduce stroke risk in selected patients. Antiplatelet therapy and peri-procedural neurologic monitoring are required.

Carotid Body

A small cluster of chemoreceptor cells near the fork of the common carotid artery. It detects changes in blood oxygen, carbon dioxide, and acidity and helps regulate breathing through reflex signaling.

Carotid Bruit

An audible sound over a carotid artery caused by turbulent blood flow. It may occur with carotid atherosclerosis but does not reliably indicate the severity of stenosis; suspected disease is assessed with appropriate vascular imaging and risk evaluation.

Carotid Cavernous Fistula

An abnormal communication between the carotid artery or its branches and the cavernous sinus, most commonly traumatic but also spontaneous. Clinical features include pulsatile proptosis, chemosis, bruit

over the orbit, and cranial nerve palsies. Diagnosis is by CT or MR angiography.

Carotid Duplex Ultrasound

Ultrasound evaluation of the carotid arteries for atherosclerotic plaque and stenosis. Measures peak systolic velocity (PSV) and end-diastolic velocity (EDV) in the internal carotid artery (ICA).

Carotid Endarterectomy

Surgical removal of atherosclerotic plaque from a carotid artery to reduce stroke risk in selected patients. Suitability depends on symptoms, degree of narrowing, operative risk, life expectancy, and current clinical guidance.

Carotid Stenosis

Narrowing of a carotid artery, usually caused by atherosclerosis. It can reduce blood flow to the brain or serve as a source of emboli. Evaluation and treatment depend on symptoms, degree of narrowing, imaging, and overall stroke risk.

Carpal Bones

The eight wrist bones: scaphoid, lunate, triquetrum, pisiform, trapezium, trapezoid, capitate, and hamate.

Alternate Names:

Carpals

Carpal Joints

The articulations among the eight carpal bones and between the carpus, radius, ulna, and metacarpals that permit coordinated wrist movement.

Carpal Tunnel

Narrow passageway on the palmar wrist, bounded by carpal bones dorsally and the flexor retinaculum as the roof. Transmits the median nerve and flexor tendons. Carpal tunnel syndrome causes numbness in the lateral 3.5 digits and thenar weakness.

Carpal Tunnel Biopsy

Sampling of tissue from the carpal tunnel or its contents, performed selectively when an infiltrative,

inflammatory, infectious, or neoplastic process is suspected. It is not a routine test for carpal tunnel syndrome.

Carpal Tunnel Release

Surgery that cuts the transverse carpal ligament to relieve pressure on the median nerve.

Carpal Tunnel Syndrome

The most common entrapment neuropathy, caused by compression of the median nerve as it passes through the carpal tunnel, a narrow fibro-osseous canal bounded by the carpal bones and the flexor retinaculum (transverse carpal ligament) at the wrist.

Alternate Names:

Median Nerve Compression (Carpal Tunnel Syndrome)

Carpal Tunnel Syndrome Tests

Provocative tests for diagnosing carpal tunnel syndrome. Tinel sign: percuss over median nerve at the carpal tunnel tingling in median distribution. Phalen sign: hold wrists in full flexion for 60 seconds reproduces symptoms. Durkan test (carpal compression): direct compression over carpal tunnel for 30 seconds paresthesias.

Carpometacarpal Joint

A joint between a carpal bone and the base of a metacarpal; the thumb carpometacarpal joint is a highly mobile saddle joint commonly affected by osteoarthritis.

Carpus

The carpus is the group of eight wrist bones connecting the forearm to the metacarpals.

Carrageenan

A sulfated polysaccharide extracted from red seaweed and used as a thickener or stabilizer in foods and pharmaceutical preparations; its inflammatory effects depend on form and exposure.

Carrier

Individual heterozygous for a recessive genetic disorder. Typically asymptomatic but can pass mutant allele to offspring. Carrier screening recommended for certain populations (e.g., Ashkenazi Jewish for Tay-Sachs).

Carrier-Mediated Transport

Movement of a substance across a cell membrane with help from a specific carrier protein. It may be passive, moving down a concentration gradient without direct energy use, or active, using energy to move the substance against its gradient.

Carrier Excipient

An inactive substance that carries or supports an active medicine in a formulation, helping control dose, stability, absorption, appearance, or delivery.

Carrier Screening

Genetic testing offered to people without symptoms to determine whether they carry a pathogenic variant for a recessive or X-linked condition that could be passed to a child.

Carrier State

Asymptomatic infection with pathogen capable of transmission. Examples: typhoid carrier, hepatitis B carrier, meningococcal carrier. Importance: public health control, screening, treatment.

Carry-Over Effect

Persistence of an intervention's effect into a later treatment period or measurement, potentially confounding a crossover study or subsequent clinical assessment.

Carteolol

A nonselective beta blocker used mainly as eye drops to lower increased pressure inside the eye in open-angle glaucoma or ocular hypertension. It may be used alone or with other pressure-lowering medicines.

Cartilage

A tough, flexible avascular connective tissue found in joints, the ear, nose, rib cage, and intervertebral discs. Composed of chondrocytes embedded in an extracellular matrix of collagen fibers, proteoglycans, and water, providing resilience and compressive strength.

Cartilage Disease

A disorder damaging cartilage, the resilient connective tissue covering joints or supporting structures, causing pain, stiffness, deformity, impaired growth, or loss of smooth movement.

Cartilaginous Joint

A joint in which bones are connected by cartilage and allow limited movement.

Caruncles

Caruncles are small fleshy anatomical projections, such as the lacrimal caruncle at the inner corner of the eye or urethral caruncle.

Carvedilol

A nonselective beta-blocker with additional alpha-1 adrenergic blockade and antioxidant properties, producing vasodilation with balanced reduction of heart rate, contractility, and afterload.

CAS

Alternate index label; see *Childhood apraxia of speech*.

Casanthranol

A stimulant laxative that increases intestinal movement and water secretion; prolonged or excessive use can cause diarrhea, electrolyte abnormalities, and dependence on laxatives.

Cascade

A sequence in which one activated reaction or event triggers the next, such as the coagulation, complement, or inflammatory cascade.

Cascara

Cascara is a plant-derived stimulant laxative containing anthraquinones; prolonged or excessive use can cause diarrhea, electrolyte imbalance, and dependence.

Case Definition

Standardized clinical, laboratory, and epidemiologic criteria used to classify a person as a suspected, probable, or confirmed case for surveillance or outbreak investigation. It may differ from the criteria used for individual clinical diagnosis.

Case Fatality Rate

The proportion of people diagnosed with a disease who die from it during a defined period. It is used to describe severity, but the estimate depends on accurate case detection, follow-up, and the population studied.

Case Management

Coordinated planning and follow-up of health and social services to help a person obtain appropriate care, avoid duplication, and meet clinical and functional goals.

Case Study

An in-depth analysis of a single patient, disease episode, or clinical scenario, used in medical education, research, and quality improvement to illustrate diagnostic reasoning, treatment decisions, or unusual presentations.

Case-Control Studies

Observational studies that compare people with an outcome or disease to similar people without it and examine prior exposures or risk factors.

Case-Control Study

Observational study comparing cases with disease to controls without. Looks back at exposures.

Case-Fatality Rate

Proportion of cases dying from disease. Formula: deaths / cases. Measures severity.

Casein

A chemical, nutrient, pigment, or material whose physiologic or toxicologic effects depend on dose, route of exposure, metabolism, and tissue distribution.

Casein Kinase

A family of protein kinases that regulate cell growth, DNA repair, circadian rhythms, and other cellular processes.

Caseous

Having a cheese-like appearance or consistency, especially the necrotic material typical of some tuberculosis granulomas.

Caseous Necrosis

A distinctive form of cell death that characteristically occurs in granulomatous infections, most notably tuberculosis. The term derives from the cheese-like, friable, whitish-yellow appearance of the necrotic tissue on gross examination.

Cashew Allergy

An immune reaction to cashew proteins that can cause hives, swelling, vomiting, wheezing, or life-threatening anaphylaxis.

Caspase

A family of cysteine proteases involved in programmed cell death and inflammation.

Caspase Activation

Conversion of inactive procaspases into active caspases, initiating apoptosis or inflammatory pathways.

Caspofungin

An intravenous echinocandin antifungal that inhibits fungal cell-wall glucan synthesis and treats serious *Candida* infections.

Caspofungin Acetate

The acetate salt formulation of caspofungin, an intravenous echinocandin antifungal.

Cassette

A holder or container for recording or imaging material; in radiography, a cassette traditionally houses the film or image receptor during exposure.

Cast

A rigid or semi-rigid device applied around an injured limb or joint to immobilize it while bone or soft tissue heals.

Castleman Disease

Heterogeneous group of lymphoproliferative disorders (angiofollicular lymph node hyperplasia, giant lymph node hyperplasia) sharing lymph node histopathology.

Castor Oil

A vegetable oil extracted from the seeds of *Ricinus communis*. It is used mainly as an oral stimulant laxative and may also be used as an industrial lubricant or in topical preparations.

Castor Oil Overdose

Excessive ingestion of castor oil, which may cause gastrointestinal irritation, abdominal cramping, nausea, vomiting, and diarrhea. Severe exposure can lead to dehydration and fluid or electrolyte imbalance.

Castration

Removal or functional suppression of the gonads, reducing production of sex hormones; it may be surgical or achieved medically.

CAT Scan

Computed axial tomography, commonly called CT, an imaging method that uses multiple X-ray measurements and computer processing to create cross-sectional body images.

Cat'S Eye Reflex

The cat's-eye reflex is a bright whitish reflection from the pupil, called leukocoria in humans, which can signal serious eye disease and requires urgent assessment.

Cat-Nap

A cat-nap is a short period of sleep, usually during the day; frequent unintended sleep may warrant assessment for sleep deprivation or a sleep disorder.

Cat-Scratch Disease

An infection caused by *Bartonella henselae*, usually transmitted by a scratch or bite from an infected cat. It commonly causes a papule followed by regional lymph-node enlargement, fever, and fatigue; atypical disease may affect the eye, liver, spleen, or nervous system.

Alternate Names:

Cat-Scratch Fever

Catabolism

The set of metabolic pathways that break down complex molecules into simpler units, releasing energy captured as ATP, NADH, FADH₂, NADPH. Examples: glycolysis, beta-oxidation, proteolysis, glycogenolysis, nucleotide catabolism.

Catagen

The transitional phase of the hair-growth cycle between active growth and resting.

Alternate Names:

Catagen phase; regression phase of hair growth.

Catalase

An enzyme that breaks down hydrogen peroxide into water and oxygen, protecting cells from oxidative damage.

Catalepsy

A state of markedly reduced movement and responsiveness in which a person may maintain an imposed posture. It can occur in catatonia and other neurologic or psychiatric conditions.

Catalyst

A substance that increases the rate of a chemical reaction without being consumed, by lowering the activation energy (providing an alternative reaction pathway). Enzymes are biological catalysts (proteins or ribozymes). Catalysts do not alter the

reaction equilibrium (ΔG), only the kinetics.

Catamenial Pneumothorax

A manifestation of thoracic endometriosis syndrome characterized by recurrent spontaneous pneumothorax in ovulating women occurring temporally within 48 to 72 hours of the onset of menses. It predominantly affects the right hemithorax (in over 90 percent of cases) and arises from endometrial implants on the diaphragm or visceral pleura that undergo cyclic hormonal shedding, diaphragmatic fenestrations, or spontaneous alveolar rupture. Management combines video-assisted thoracoscopic surgery with diaphragmatic resection or mesh repair, pleurodesis, and prolonged ovarian suppression using GnRH agonists or continuous progestins.

Alternate Names:

Endometriosis-Related Pneumothorax; Catamenial Thorax

Cataplasma

A poultice: a soft, moist substance spread on cloth and applied to the body, historically for soothing, warming, or treating a local condition.

Cataplexy

Sudden temporary loss of muscle tone triggered by strong emotion, commonly associated with narcolepsy.

Cataract

A cataract is an opacity or clouding of the crystalline lens that impairs the transmission of light to the retina, resulting in blurred vision, glare, decreased contrast sensitivity, and progressive vision loss, and it is the leading cause of preventable blindness worldwide.

Cataract - Adult

An adult cataract is clouding of the eye lens that progressively reduces visual acuity, contrast, or glare tolerance; definitive treatment is lens-replacement surgery when function is impaired.

Cataract Removal

A surgical procedure that removes a cloudy natural

lens from the eye and usually replaces it with an artificial intraocular lens to improve vision.

Cataract Surgery

Surgical removal of the clouded lens and replacement with an intraocular lens implant to restore vision. It is the most commonly performed ophthalmic procedure and is indicated when cataract impairs daily function.

Cataracts

Clouding of the eye's natural lens that causes blurred vision, glare, faded colors, or difficulty seeing at night.

Catastrophic Aging

Rapid functional decline after stressor (illness, surgery, fall). Elderly have decreased reserve, less ability to recover. Prevention: comprehensive assessment, optimization, early intervention.

Catastrophic Reaction

A disproportionately intense emotional or behavioral response to a situation that may appear minor to others, often occurring with cognitive impairment or neurologic disease.

Alternate Names:

Catastrophic response; catastrophic emotional reaction.

Catatonია

A neuropsychiatric syndrome characterized by motor abnormalities including immobility, mutism, posturing, waxy flexibility, stereotypy, mannerisms, grimacing, echolalia, and echopraxia. It can occur in mood disorders, psychotic disorders, medical conditions, or as catatonia due to another medical condition.

Catatonია, Fatal

Fatal catatonia is a severe, life-threatening catatonic state with marked motor and behavioral disturbance, fever, autonomic instability, and risk of organ failure.

Catbite

A catbite is a puncture wound caused by a cat, with a high risk of bacterial infection, including *Pasteurella*, and possible tendon, joint, or bone involvement.

Catch-Up Immunization

Vaccination of a person who is behind the recommended schedule using age- and risk-appropriate intervals. Doses generally do not need to be restarted because of delay, but minimum ages and intervals must be respected.

Catechin

A chemical, nutrient, pigment, or material whose physiologic or toxicologic effects depend on dose, route of exposure, metabolism, and tissue distribution.

Catechol

A benzene compound with two adjacent hydroxyl groups and a chemical precursor or component of some biologically active compounds; exposure may be irritant or toxic.

Catecholamine

A group of hormones and neurotransmitters, including epinephrine, norepinephrine, and dopamine, involved in stress and cardiovascular responses.

Catecholamine Blood Test

A blood test that measures catecholamines such as epinephrine, norepinephrine, and dopamine; it may help evaluate suspected pheochromocytoma or related disorders, although plasma metanephrines are often preferred for screening.

Catecholamine Crisis

Acute life-threatening elevation of catecholamines causing severe hypertension, tachycardia, and multi-organ dysfunction. Occurs in pheochromocytoma and paraganglioma precipitated by anesthesia, surgery, trauma, or medications that stimulate catecholamine release.

Catecholaminergic Polymorphic Ventricular Tachycardia

A severe, genetically determined cardiac channelopathy characterized by exercise- or emotional stress-induced polymorphic or bidirectional ventricular arrhythmias and sudden cardiac death in children and young adults with structurally normal hearts and normal resting electrocardiograms. Caused by gain-of-function mutations in the cardiac ryanodine receptor gene (*RYR2*) or loss-of-function mutations in calsequestrin (*CASQ2*), leading to abnormal sarcoplasmic reticulum calcium release during sympathetic stimulation; management includes high-dose non-selective beta-blockers (nadolol), flecainide, left cardiac sympathetic denervation, and implantable cardioverter-defibrillators.

Alternate Names:

CPVT; Bidirectional Ventricular Tachycardia; Stress-Induced Polymorphic VT

Catecholamines

Catecholamines include adrenaline, noradrenaline, and dopamine, hormones and neurotransmitters involved in stress responses, blood pressure, heart function, and metabolism.

Catecholamines (Urine/Plasma)

Measurement of catecholamines or their metabolites in blood or urine. These tests help evaluate suspected pheochromocytoma or paraganglioma, but results must be interpreted with medications, stress, sampling conditions, and clinical findings.

Alternate Names:

Urine/Plasma

Catecholamines - Urine

A urine test measuring catecholamines or metabolites to help evaluate pheochromocytoma and related disorders.

Category Five Hurricane

A Saffir–Simpson hurricane with sustained winds of at least 157 miles per hour (252 km/h), capable of catastrophic damage and prolonged loss of habitability.

Category I Fetal Heart Tracing

A normal fetal heart rate tracing by NICHD 2008 criteria. All three characteristics present: baseline 110-160 bpm, moderate variability (6-25 bpm), and no late or variable decelerations. May have accelerations. Category I tracings are associated with normal fetal acid-base status.

Category II Fetal Heart Tracing

An indeterminate fetal heart rate tracing that falls between Category I and III.

Category III Fetal Heart Tracing

An abnormal fetal heart rate tracing indicating potential fetal hypoxia or acidosis. Two specific patterns: (1) absent baseline variability with recurrent late decelerations, (2) absent baseline variability with recurrent variable decelerations, and (3) sinusoidal pattern.

Category Three Hurricane

A Saffir–Simpson hurricane with sustained winds of 111–129 miles per hour (178–208 km/h), capable of devastating damage.

Category Two Hurricane

A Saffir–Simpson hurricane with sustained winds of 96–110 miles per hour (154–177 km/h), capable of extensive damage to structures and vegetation.

Catenin

A family of proteins involved in cell adhesion and Wnt signaling; abnormal beta-catenin activation can contribute to cancer.

Caterpillar Dermatitis

Caterpillar dermatitis is an itchy inflammatory skin reaction caused by contact with irritating or venomous caterpillar hairs or spines.

Caterpillars

The larval stage of butterflies and moths; contact with some species' hairs or spines can cause painful dermatitis, eye injury, or allergic reactions.

Catharsis

Catharsis is emotional release or expression; it is a historical psychological concept and not a specific medical treatment or diagnosis.

Cathartic

An agent that promotes bowel evacuation, usually by increasing stool water, intestinal motility, or both.

Alternate Names:

Purgative; laxative.

Cathepsin

A family of proteases involved in lysosomal protein breakdown, tissue remodeling, immunity, and bone resorption.

Cathepsin D

An aspartic protease found mainly in lysosomes that digests proteins and participates in apoptosis and tissue disease.

Cathepsin H

A lysosomal cysteine protease involved in protein breakdown and activation of some biologically active peptides.

Cathepsin K

A cysteine protease produced by osteoclasts that degrades bone matrix.

Cathepsin L

A lysosomal cysteine protease involved in protein degradation, antigen processing, tissue remodeling, and some infections.

Cathepsin Z

A lysosomal cysteine protease involved in protein processing and immune-cell function.

Cathepsins B

Cysteine proteases, especially cathepsin B, that participate in lysosomal protein breakdown, inflammation, tissue remodeling, and tumor invasion.

Catheter

Flexible tube for drainage or infusion. Types: urinary (Foley), intravenous, central venous, arterial. Materials: silicone, latex, PVC.

Catheter, Foley

An indwelling urinary catheter with an inflatable balloon that holds it in the bladder, allowing continuous drainage of urine.

Catheter-Associated UTI

Urinary tract infection related to indwelling urinary catheter. Most common healthcare-associated infection. Risk: prolonged catheterization, female sex, diabetes, immunosuppression. Prevention: avoid unnecessary catheters, aseptic insertion, closed drainage, daily hygiene, early removal.

Catheter-Related Infections

Infections caused by microorganisms entering along or contaminating an intravascular or other indwelling catheter. They may cause local inflammation or bloodstream infection; removal or replacement and antimicrobial treatment depend on the catheter and severity.

Catheter-Related UTI

A urinary tract infection associated with an indwelling urinary catheter, usually caused by bacteria entering or ascending along the catheter; diagnosis requires compatible symptoms and appropriately collected urine testing.

Catheterization

A medical procedure involving catheterization. Indications, risks, preparation, and follow-up depend on the patient and the specific procedure.

Catheterization (Urinary)

Insertion of a flexible tube (catheter) through the urethra or suprapubically into the urinary bladder to drain urine. Types: Foley catheter (indwelling balloon catheter), intermittent (in-out) catheter, and suprapubic catheter.

Alternate Names:

Urinary

Cathexis

The investment of emotional or psychological energy in a person, idea, object, or activity, a term used mainly in psychology and psychoanalytic theory.

Cation

An ion with a positive electrical charge, such as sodium, potassium, calcium, or hydrogen.

Alternate Names:

Positively charged ion.

Cauda Equina

The bundle of lumbar, sacral, and coccygeal nerve roots below the end of the spinal cord.

Alternate Names:

Cauda equina nerve roots; horse-tail nerve bundle.

Cauda Equina Syndrome

A neurosurgical emergency caused by compression of the cauda equina nerve roots below the level of the conus medullaris (typically at L1-L2 and below), resulting in bilateral lower extremity weakness, sensory loss, and bladder and bowel dysfunction.

Alternate Names:

Syndrome Cauda Equina (Cauda Equina Syndrome)

Caudad

Toward the tail or lower end of the body; in humans, it generally means toward the feet or away from the head.

Caudal

Positioned toward the tail or lower part of the body.

Caudal Dysplasia

A congenital disorder in which lower-spine and related body structures develop incompletely, sometimes causing limb, bowel, or bladder abnormalities.

Caudal Vein

An anatomic structure or region related to caudal Vein; its location and function determine its normal relationships and clinical significance.

Caudate Nucleus

An anatomic structure or region related to caudate Nucleus; its location and function determine its normal relationships and clinical significance.

Cauliflower Ear

A permanent deformity of the outer ear caused by blood or fluid collecting between the cartilage and its covering, usually after blunt trauma or repeated friction. Untreated separation can deprive cartilage of blood supply and produce a thickened, irregular ear. Prompt drainage and compression reduce the risk of deformity.

Alternate Names:

Perichondrial Hematoma (Cauliflower Ear)

Caulking Compound Poisoning

Exposure to caulking compounds that may irritate the skin, eyes, or airways; ingestion can cause gastrointestinal symptoms and aspiration risk depending on the product.

Caulobacter

A genus of stalked Gram-negative bacteria found mainly in freshwater and soil and rarely causing human disease.

Caulobacteraceae

A family of mostly aquatic Gram-negative bacteria that includes Caulobacter.

Causalgia

Severe persistent burning pain and sensory disturbance caused by injury to a peripheral nerve, now commonly classified within complex regional pain syndrome type II.

Alternate Names:

Complex regional pain syndrome type II; causalgic pain.

Cause Of Death

The specific disease, injury, or toxic agent responsible for initiating the lethal physiological sequence leading directly to death (e.g., Gunshot wound to the head, Acute myocardial infarction).

Causes Symptoms Of Hormonal Imbalances Women

Hormonal imbalance in women can reflect thyroid disease, polycystic ovary syndrome, menopause, pregnancy, adrenal or pituitary disorders, medicines, or changes in body weight. Symptoms may include irregular bleeding, acne, hair changes, infertility, hot flashes, fatigue, or mood changes.

Caustic

Caustic describes a substance that chemically burns or destroys tissue, such as a strong acid or alkali; ingestion or contact can be a medical emergency.

Cauterization

The deliberate destruction or sealing of tissue using heat, electricity, chemicals, or another energy source to control bleeding or remove abnormal tissue.

Cautery

Cautery destroys or seals tissue using heat, electricity, chemicals, or another energy source to control bleeding or remove abnormal tissue.

Cautery Device

An anatomic structure or region related to cautery Device; its location and function determine its normal relationships and clinical significance.

Caveola

An anatomic structure or region related to caveola; its location and function determine its normal relationships and clinical significance.

Caveolins

Membrane proteins that form or regulate caveolae involved in cell signaling, lipid handling, and membrane trafficking.

Cavernous Hemangioma

A cluster of enlarged, thin-walled blood vessels that may occur in the skin, brain, or other tissues.

Cavernous Malformations

Abnormally enlarged, thin-walled blood-vessel clus-

ters that can bleed or cause neurological symptoms.

Cavernous Sinus

The cavernous sinus is a venous channel beside the pituitary and sphenoid bone containing important cranial nerves and the internal carotid artery; thrombosis or infection can be life-threatening.

Cavernous Sinus Thrombosis

Life-threatening septic thrombosis of the cavernous sinus, usually a complication of mid-face, nasal, paranasal sinus (sphenoid, ethmoid), dental, or orbital infection spreading through the valveless ophthalmic veins.

Cavities

A cavity is tooth decay caused by acids produced when oral bacteria metabolize dietary carbohydrates. The process can progress from enamel demineralization to dentin and pulp infection; fluoride, oral hygiene, diet, dental sealants, fillings, or root-canal treatment may be needed.

Alternate Names:

Dental Caries (Cavities)

Cavities And Tooth Decay

Progressive destruction of tooth enamel and deeper tooth tissues by acids produced by oral bacteria.

Cavity

A localized hole or structural defect in a tooth caused by dental caries.

Alternate Names:

Dental cavity; carious lesion.

Cavum

Cavum means a cavity or hollow space, such as the cavum septi pellucidi in the brain; the specific anatomical meaning depends on the term.

CBC

See *Complete Blood Count*.

Alternate Names:

Complete blood count; full blood count.

CBC Blood Test

See *Complete Blood Count*.

CCP

Cyclic citrullinated peptide; anti-CCP antibody testing supports diagnosis of rheumatoid arthritis and may help assess the likelihood of erosive disease.

CCTA

Alternate name; see *Coronary Computed Tomography Angiography*.

CCU

A hospital unit providing continuous monitoring and specialized treatment for people with serious acute cardiac disease.

Alternate Names:

Coronary care unit; cardiac care unit.

Cd Antigens

Cell-surface markers identified by cluster-of-differentiation antibodies and used to classify immune cells.

CD4 Count

The number of CD4-positive T-helper lymphocytes in a blood sample. It helps assess immune function and risk of opportunistic infection in people with HIV, but interpretation depends on trends, viral load, clinical findings, and current treatment guidance.

CD4-Positive T Cell

A T lymphocyte that recognizes peptide presented by major histocompatibility complex class II molecules. Its differentiated subsets coordinate immune responses, including macrophage activation, B-cell help, mucosal defense, and immune regulation.

CD8-Positive T Cell

A T lymphocyte that recognizes peptide presented by major histocompatibility complex class I molecules. Cytotoxic CD8-positive cells can induce apoptosis in infected, malignant, or otherwise

altered target cells.

Cea

CEA is an abbreviation for carcinoembryonic antigen, a tumor marker used mainly to monitor selected cancers.

CEA Assay

A laboratory assay measuring carcinoembryonic antigen, mainly to monitor selected cancers rather than screen the general population.

CEA Blood Test

A blood test measuring carcinoembryonic antigen, a tumor marker used mainly to monitor selected cancers, especially colorectal cancer, and assess treatment response or recurrence. It is not sufficiently specific for general cancer screening.

Cecal Hemorrhage

Bleeding from the cecum, which may result from inflammation, infection, ischemia, diverticular disease, polyps, or cancer.

Cecal Infection

Infection involving the cecum, caused by organisms, inflammation, ischemia, or spread from nearby tissue.

Cecal Neoplasm

A benign or malignant growth arising in the cecum; symptoms may include bleeding, anemia, abdominal pain, or bowel changes, and diagnosis usually requires colonoscopy and biopsy.

Cecal Volvulus

Axial twisting of the cecum and ascending colon around its mesentery, accounting for approximately one-third of colonic volvulus cases. It presents with acute or subacute large bowel obstruction, abdominal pain, and distension. Diagnosis is by CT scan showing the characteristic whirl sign and a massively dilated cecum.

Cecostomy

A surgically created opening from the cecum to the abdominal wall to allow decompression or stool removal.

Cecum

First and widest large intestinal part in the right iliac fossa below the ileocecal valve, a blind-ended pouch approximately 6 cm long. Appendix arises from its posteromedial wall approximately 2 cm below the valve, containing abundant lymphoid tissue. Cecal volvulus causes large bowel obstruction.

Cedar Leaf Oil Poisoning

Ingestion or inhalation of cedar leaf oil, which may irritate tissues and cause gastrointestinal or neurologic toxicity depending on the dose.

Cedecea

A genus of gram-negative bacteria in the Enterobacterales that rarely causes opportunistic infection, usually in people with serious underlying illness.

Cedecea Davisae

A rare opportunistic gram-negative bacterium reported in bloodstream, respiratory, wound, and urinary infections; clinical significance depends on the specimen and symptoms.

Cedecea Lapagei

A gram-negative environmental bacterium rarely associated with human opportunistic infection, including bloodstream or respiratory infection.

Cedecea Neteri

An uncommon gram-negative bacterium that may cause opportunistic human infection; interpretation requires correlation with the clinical site and culture findings.

Cediranib Maleate

The maleate salt of cediranib, an investigational vascular endothelial growth factor receptor inhibitor studied as an anticancer medicine.

Cefaclor

A second-generation cephalosporin antibiotic used for selected susceptible bacterial infections; allergy, diarrhea, and antibiotic-associated colitis are important risks.

Cefadroxil

An oral first-generation cephalosporin antibiotic used for selected susceptible bacterial infections; allergy, gastrointestinal effects, and antibiotic-associated colitis are possible.

Cefamandole

A second-generation cephalosporin antibiotic used for selected susceptible bacterial infections; bleeding risk, allergy, diarrhea, and antibiotic-associated colitis are possible.

Cefatrizine

An older oral cephalosporin antibiotic used in some countries for susceptible bacterial infections; current use depends on local availability, resistance, and clinical guidance.

Cefazolin

A first-generation cephalosporin antibiotic given by injection for susceptible infections and commonly used for surgical antimicrobial prophylaxis.

Cefdinir

An oral third-generation cephalosporin antibiotic used for selected respiratory, skin, and other susceptible bacterial infections.

Cefepime

A fourth-generation cephalosporin antibiotic used for serious susceptible bacterial infections, including some hospital-acquired infections; renal dose adjustment is important.

Cefixime

An oral third-generation cephalosporin antibiotic used for selected susceptible infections, including some urinary and respiratory infections.

Cefmenoxime

A third-generation cephalosporin antibiotic used in some countries for susceptible bacterial infections; allergy, diarrhea, and antibiotic-associated colitis are possible.

Cefmetazole

A medicine or chemical compound identified by cefmetazole. Its uses, effects, and risks depend on the specific substance and clinical context.

Cefonicid

A long-acting second-generation cephalosporin antibiotic used for selected susceptible infections; dosing and use depend on kidney function and local resistance.

Cefoperazone

A third-generation cephalosporin with activity against many gram-negative bacteria, often combined with sulbactam; bleeding, allergy, and biliary effects may occur.

Cefotaxime

A third-generation cephalosporin antibiotic administered parenterally for serious susceptible infections, including meningitis, pneumonia, and sepsis.

Cefotetan

A second-generation cephamycin antibiotic used for selected serious infections and surgical prophylaxis, with activity against some anaerobic bacteria.

Cefotiam

A second-generation cephalosporin antibiotic used in some countries for susceptible respiratory, urinary, skin, and other bacterial infections.

Cefoxitin

A cephamycin antibiotic used for selected abdominal, pelvic, skin, and other infections, including some caused by anaerobic bacteria.

Cefsulodin

An older antipseudomonal cephalosporin used

mainly in laboratory or historical contexts because resistance and newer treatments limit routine use.

Ceftazidime

A third-generation cephalosporin antibiotic with activity against *Pseudomonas aeruginosa*, used for selected serious Gram-negative infections.

Ceftazidime-Avibactam

A combination of the third-generation cephalosporin ceftazidime with the non-beta-lactam beta-lactamase inhibitor avibactam. Avibactam inhibits class A, class C, and some class D beta-lactamases, including KPC carbapenemases. This combination has activity against many carbapenem-resistant Enterobacteriaceae and *Pseudomonas aeruginosa*.

Ceftibuten

An oral third-generation cephalosporin used for selected susceptible respiratory or urinary bacterial infections.

Ceftizoxime

A third-generation cephalosporin antibiotic used for selected serious infections caused by susceptible bacteria, including some anaerobes.

Ceftriaxone

A third-generation cephalosporin antibiotic administered by injection for many serious bacterial infections, including meningitis, pneumonia, and gonorrhea.

Cefuroxime

A second-generation cephalosporin available orally or by injection for selected respiratory, urinary, skin, and other bacterial infections.

Celiac Artery

The first major ventral branch of the abdominal aorta at T12, arising from the aorta just below the diaphragm and supplying the foregut (stomach, proximal duodenum, liver, gallbladder, spleen, pancreas). It trifurcates into the left gastric, splenic, and common hepatic arteries, with rich collateral

circulation between these branches.

Celiac Crisis

Acute life-threatening presentation of celiac disease characterized by severe dehydration, electrolyte imbalances, metabolic acidosis, hypocalcemic tetany, and hypovolemic shock. Most commonly occurs in infants and young children after a viral illness or introduction of gluten-containing foods.

Celiac Disease

An immune-mediated enteropathy triggered by gluten ingestion in genetically predisposed individuals (HLA-DQ2/DQ8). Gliadin peptides from wheat, barley, and rye activate an autoimmune response against tissue transglutaminase (tTG), causing villous atrophy, crypt hyperplasia, and intraepithelial lymphocytosis in the small intestine.

Alternate Names:

Nontropical Sprue

Celiac Disease (Gluten Enteropathy)

Alternate index label for Gluten Enteropathy.

Celiac Disease - Nutritional Considerations

Dietary management of celiac disease requiring strict lifelong avoidance of gluten, which is found in wheat, barley, rye, and contaminated products. Nutritional counseling addresses deficiencies of iron, B12, calcium, and vitamin D that result from intestinal damage.

Celiac Disease - Sprue

Celiac disease is an immune-mediated reaction to gluten that damages small-intestinal villi and can cause malabsorption, anemia, diarrhea, weight loss, or extraintestinal disease.

Celiac Plexus

A network of nerves in the upper abdomen that supplies abdominal organs and may be targeted for pain treatment.

Celiac Trunk

First major abdominal aortic branch at T12-L1, the foregut artery. Divides into left gastric (lesser curvature), splenic (spleen, pancreas, greater curvature), and common hepatic (proper hepatic/gastroduodenal) arteries. Surrounded by the celiac plexus.

Celiac Trunk Stenosis

Narrowing of the celiac trunk typically caused by extrinsic compression from the median arcuate ligament of the diaphragm, known as median arcuate ligament syndrome or celiac artery compression syndrome. Causes chronic abdominal pain particularly after eating and weight loss due to postprandial mesenteric ischemia.

Cell

The fundamental structural and functional unit of all living organisms. Human cells are eukaryotic, characterized by a nucleus containing DNA and membrane-bound organelles.

Cell Adhesion

The binding of cells to one another or to the extracellular matrix, mediated by adhesion molecules and important for tissue structure, signaling, and cell migration.

Cell Aggregation

Clumping of cells through adhesion molecules or extracellular proteins, important in development, immunity, wound healing, and clotting.

Cell Body

The part of a cell containing the nucleus and most organelles, excluding specialized extensions such as axons.

Cell Count

The number of cells in a sample, tissue, or fluid, measured to assess blood, inflammation, infection, or cell growth.

Cell Culture

Growth of cells outside the body in a controlled nutrient medium for research, testing, diagnosis, or production of biologic products.

Cell Cycle

The ordered series of cellular phases through which a cell grows, duplicates its DNA, and divides.

Cell Death

The regulated elimination of cells through programmed cellular pathways that help maintain tissue balance and remove damaged cells.

Cell Dedifferentiation

Loss of some specialized features by a mature cell, allowing it to become more developmentally flexible.

Cell Density

The number of cells within a defined volume, area, or tissue region.

Cell Differentiation

The process by which an unspecialized cell develops the structure and function of a specialized cell type.

Cell Division

The process by which one cell produces daughter cells; mitosis makes most body cells and meiosis makes reproductive cells.

Cell Fusion

Joining of two cells or their membranes to form one larger or multinucleated cell.

Cell Growth

An increase in cell size and biomass through synthesis of proteins, membranes, and other components.

Cell Maturation

Development of a cell from an immature precursor into a functionally specialized cell.

Cell Migration

Movement of cells through tissue or across a surface in response to chemical or mechanical signals.

Cell Motility

The ability of cells to move through or across a surface using the cytoskeleton, cell-adhesion structures, or specialized structures such as cilia and flagella. It is important in development, immune responses, wound healing, and cancer spread.

Alternate Names:

Cell Locomotion

Cell Movement

Directed or undirected movement of a cell, important in development, immunity, wound repair, and cancer spread.

Cell Nucleus

The membrane-bound compartment containing most cellular DNA and regulating gene expression and division.

Cell Pellet

The concentrated layer of cells formed at the bottom of a tube after centrifugation, usually collected for testing.

Cell Physiology

The study of how cells maintain energy, transport substances, communicate, respond to signals, and perform specialized functions.

Cell Polarity

Unequal distribution of structures and activities within a cell, enabling directional transport, signaling, movement, and tissue organization.

Cell Proliferation

An increase in cell number through division; excessive proliferation can contribute to cancer.

Cell Secretion

Release of substances such as hormones, enzymes, neurotransmitters, mucus, or cytokines from cells.

Cell Senescence

A state in which a cell remains metabolically active but permanently stops dividing, often in response to aging, stress, or cellular damage.

Alternate Names:

Cellular senescence; replicative senescence.

Cell Sorting

Separation of cells into groups based on properties such as size, density, surface markers, or fluorescence.

Cell Structure

The organization of a cell into membrane, cytoplasm, nucleus, and organelles that support its functions.

Cell Surface

The plasma membrane and associated molecules that control transport, adhesion, recognition, and communication.

Cell Survival

The ability of a cell to remain alive and functional despite stress.

Cell Therapy

A medical procedure, treatment, device, or care topic related to cell therapy; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Cell Transplantation

A medical procedure, treatment, device, or care topic related to cell transplantation; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Cell Wall

A rigid outer layer surrounding plant, fungal, and bacterial cells; human cells do not have cell walls.

Cell-Cell Junction

A specialized connection between adjacent cells that supports adhesion, communication, or controlled passage of substances.

Cell-Cell Signaling

Communication between cells through contact, electrical signals, hormones, neurotransmitters, or cytokines.

Cell-Free Fetal DNA Screening

A prenatal screening test that analyzes placental cell-free DNA in maternal blood to estimate the chance of selected chromosomal conditions. It is usually offered from early pregnancy onward. It is a screening test, not a diagnostic test, and a positive or inconclusive result requires appropriate counseling and follow-up.

Cell-Free System

A laboratory system containing selected cell components but no intact living cells, used to study biochemical reactions under controlled conditions.

Cell-Matrix Junction

A connection between a cell and extracellular matrix that anchors the cell and transmits signals.

Cell-Mediated Immune Response

Alternate index label; see *Cell-mediated immunity*.

Cell-Mediated Immunity

An immune response mediated mainly by T lymphocytes and other cells rather than by antibodies. It is important in defense against intracellular pathogens and in immune surveillance.

Alternate Names:

Cell-Mediated Immune Response

Cellophane

Cellophane is a thin transparent cellulose film; in medicine it may be used in dressings, packaging, or specialized laboratory techniques.

Cells Stem (Stem Cells)

Alternate index label; see *Stem Cell*.

Cellular

Relating to cells or processes occurring within or between cells.

Cellular Adaptation

Cellular adaptations are reversible changes in cell morphology and function that occur in response to altered physiological demands or mild pathological conditions.

Cellular Immunity

Immune defense mediated mainly by T cells, natural killer cells, and macrophages rather than antibodies alone.

Cellular Metabolic Process

A chemical reaction or pathway that produces energy, builds cell components, or breaks down nutrients and waste.

Cellular Respiration

The process by which cells extract energy from nutrients, usually producing ATP.

Cellular Swelling

An early manifestation of reversible injury caused by failure of ATP-dependent ion pumps and intracellular accumulation of sodium and water. It is also called hydropic change and is often most apparent in cells exposed to ischemia or toxins.

Alternate Names:
hydropic change.

Cellular Vesicle

A small membrane-bound compartment that transports, stores, or releases substances.

Cellulite

Cellulite refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Cellulitis

Cellulitis is a bacterial infection of the deeper skin and subcutaneous tissue, causing expanding redness, warmth, swelling, and tenderness with possible systemic illness. commonly caused by beta-hemolytic streptococci (particularly *Streptococcus pyogenes*)

or *Staphylococcus aureus*.

Cellulomonas

A genus of Gram-positive environmental bacteria found mainly in soil and plant material.

Cellulomonas Hominis

A rare Gram-positive bacterial species that can cause opportunistic human infection.

Cellulose

A chemical, nutrient, pigment, or material whose physiologic or toxicologic effects depend on dose, route of exposure, metabolism, and tissue distribution.

Cellulosimicrobium

A genus of Gram-positive environmental bacteria that rarely causes opportunistic invasive infection.

Cellulosome

A multi-enzyme complex produced by some microorganisms to break down cellulose.

Celosomy

A congenital body-wall closure defect in which abdominal or thoracic organs may protrude outside the body; the term is often used in descriptions of severe omphalocele-like malformations.

Cement

Medical cement, especially bone cement, is a material used to fix prostheses or fill spaces; other cements may be dental or industrial and require context.

Cemento-Enamel Junction

Junction between enamel and cementum. Three relationships: butt junction, enamel overlapping cementum, cementum overlapping enamel.

Cementoma

A benign fibro-osseous lesion of the jaw or tooth-supporting tissues; modern classification distinguishes several cementum-related lesions with dif-

ferent behavior and treatment.

Cementum

A mineralized connective tissue covering the tooth root. It provides attachment for periodontal-ligament fibers and is thinner near the cemento-enamel junction and thicker toward the root apex.

Cenesthesia

Cenesthesia is the general sense of the condition and position of one's body; disturbance may occur in neurologic or psychiatric illness.

Centenarian

Person aged 100 or older. 100,000 worldwide. More women than men (4:1). Often maintain functional independence, cognitive function. Models for healthy aging research.

Center

In anatomy or clinical measurement, a center is a central point, region, or facility organized around a function; the specific meaning depends on context.

Centigram

A metric unit of mass equal to one hundredth of a gram, or 10 milligrams, used for small quantities of medicines or specimens.

Centimeter

A metric unit of length equal to one hundredth of a meter, commonly used in clinical measurements such as wound size, tumor dimensions, and anatomical distances.

Centimeter (Cm)

A unit of length equal to one hundredth of a meter, or 10 millimeters.

Centipede

A many-legged arthropod whose bite can inject venom and cause local pain, swelling, and redness; severe allergic reactions are uncommon but possible.

Centipoise

A unit of dynamic viscosity equal to one millipascal-second, used to describe the flow resistance of blood, fluids, medicines, and other materials.

Central

Located near or toward the middle of the body or an organ.

Central (Brain) Fatigue

A sense of reduced mental energy, concentration, alertness, or motivation arising primarily from central nervous-system processes.

Alternate Names:

Central fatigue; mental fatigue.

Central Apnea

A pause in breathing caused by absent or reduced respiratory effort from impaired brainstem signaling, rather than upper-airway obstruction.

Central Auditory Processing Disorder

Impaired processing of auditory information by the central nervous system despite adequate peripheral hearing. It may cause difficulty understanding speech, especially in noise, and can affect learning and communication.

Central Core Disease

An autosomal dominant (or rarely recessive) congenital myopathy caused by heterozygous missense mutations in the ryanodine receptor 1 (*RYR1*) gene on chromosome 19q13.2, which encodes the skeletal muscle sarcoplasmic reticulum calcium release channel. Typically presents in infancy or early childhood with hypotonia, delayed motor milestones, and mild, non-progressive or slowly progressive proximal muscle weakness (affecting the hip and pelvic girdle) with skeletal deformities such as congenital hip dislocation, scoliosis, and pes cavus. Muscle biopsy stained with oxidative enzymes (NADH-TR and SDH) demonstrates characteristic well-circumscribed, pale, round, amorphous central areas ("cores") running the entire length of type 1 muscle fibers devoid of mitochondria and enzymatic activity. Affected patients carry an exceptionally high suscep-

tibility to malignant hyperthermia during volatile general anesthesia.

Alternate Names:

CCD; Core Myopathy; RYR1-Related Central Core Disease

Central Deafness

Loss or severe impairment of hearing caused by disease of the auditory pathways or auditory cortex in the brain, rather than by a disorder of the external, middle, or inner ear.

Central Diabetes Insipidus

Excessive urination and thirst caused by deficient secretion of antidiuretic hormone from the hypothalamus or posterior pituitary. The kidneys cannot appropriately concentrate urine, producing hypernatremia if water intake is insufficient.

Central Hypothyroidism

An inadequate thyroid-hormone state caused by insufficient stimulation from the pituitary gland or hypothalamus.

Central Line

A catheter inserted into a large vein with its tip in or near the heart, used to give medications or nutrition, draw blood, or monitor central venous pressures.

Central Line Infections - Hospitals

An infection related to a central venous catheter, including a central-line-associated bloodstream infection, in which microorganisms enter the bloodstream from or through the catheter.

Central Line Placement

Insertion of a catheter into a central vein (internal jugular, subclavian, or femoral), with the tip in the superior (or inferior) vena cava. Ultrasound guidance is standard. Indications: administration of vasopressors, TPN, chemotherapy, long-term antibiotics, central venous pressure monitoring, hemodialysis, and plasmapheresis.

Central Line-Associated Bloodstream Infection

Bloodstream infection associated with central venous catheter. Prevention: maximal sterile barriers, chlorhexidine skin preparation, optimal site selection (subclavian preferred), daily review of line necessity. Diagnosis: paired blood cultures (peripheral and line). Treatment: antibiotics, catheter removal if possible.

Central Nervous System

The part of the nervous system comprising the brain and spinal cord, enclosed within the skull and vertebral column.

Central Nervous System

The fundamental structural division of the nervous system comprising the brain and spinal cord, enclosed within the cranium and vertebral canal and cushioned by cerebrospinal fluid and three meningeal layers (dura, arachnoid, and pia mater). Integrates and coordinates sensory input, higher cognitive reasoning, voluntary motor execution, autonomic homeostatic regulation, and neuroendocrine signaling.

Central Nervous System Lymphoma

A lymphoma involving the brain, spinal cord, cerebrospinal fluid, or eye that can cause neurologic or visual symptoms.

Central Nervous System Lymphoma

A lymphoma that begins in the brain, spinal cord, eye, or cerebrospinal fluid.

Central Nervous System Vascular Malformations

Abnormal blood-vessel formations in or around the brain or spinal cord.

Central Neurofibromatosis

Alternate index label; see *NF2-related schwannomatosis (neurofibromatosis type 2)*.

Central Pontine Myelinolysis

A form of osmotic demyelination syndrome caused by overly rapid correction of hyponatremia, leading to demyelination of the central pons. Clinical features include flaccid quadriparesis, pseudobulbar palsy, dysarthria, dysphagia, and in severe cases locked-in syndrome.

Central Precocious Puberty

Central precocious puberty begins when the hypothalamus and pituitary activate the normal puberty pathway unusually early. It causes early secondary sexual characteristics and accelerated bone maturation; evaluation identifies causes and treatment may delay progression when appropriate.

Central Retinal Artery Occlusion

Acute obstruction of the central retinal artery causing sudden, usually painless, severe monocular visual loss. It is an ocular stroke and requires urgent systemic and vascular evaluation because it may signal a high risk of cerebral ischemia.

Central Serous Chorioretinopathy

Accumulation of serous fluid beneath the neurosensory retina, usually in the macular region, due to altered choroidal vascular permeability and retinal pigment-epithelium dysfunction. It often causes acute or subacute central visual distortion and may be associated with corticosteroid exposure.

Central Serous Choroidopathy

A disorder in which fluid accumulates beneath the neurosensory retina, usually near the macula, causing blurred or distorted central vision. It is associated with corticosteroid exposure and stress-related factors; persistent or recurrent disease requires ophthalmic evaluation.

Central Sleep Apnea

Apneas from variability in respiratory effort without airway obstruction. Associated with heart failure (Cheyne-Stokes respiration), high altitude, and opioid use.

Alternate Names:

Sleep Apnea, Central

Central Venous Catheter

A catheter whose tip terminates in a central vein, usually the lower superior vena cava or cavoatrial junction. It provides reliable access for vasoactive drugs, irritating infusions, hemodynamic monitoring, and prolonged therapy, but may cause bloodstream infection, thrombosis, or mechanical complications.

Central Venous Catheter - Dressing Change

Replacement of the sterile dressing covering a central venous catheter using aseptic technique to keep the insertion site clean, dry, and protected from contamination and infection.

Central Venous Catheter - Flushing

Instillation of a prescribed sterile solution through a central venous catheter to maintain patency and reduce blockage, using the device-specific technique and volume.

Central Venous Catheters - Ports

Implantable venous access ports are central venous catheters placed beneath the skin, accessed with a special needle to deliver medication, fluids, blood products, or obtain blood samples over prolonged treatment.

Central Venous Line

A vascular catheter whose tip lies in a major central vein, commonly the superior vena cava, for reliable long-term or high-concentration intravenous access.

Central Venous Line - Infants

A catheter whose tip lies in a large central vein of an infant, used for irritant medicines, nutrition, fluids, blood sampling, or monitoring; infection and thrombosis are key risks.

Central Venous Pressure

Pressure measured in or near the right atrium, usually through a central venous catheter. It reflects right-sided filling conditions but is affected by ventilation, heart function, vascular tone, and measurement technique; it should not be used alone to guide

fluid treatment.

Central Vision

Detailed vision directed at the point of gaze, provided primarily by the macula and fovea; it is needed for reading and recognizing faces.

Centriacinar Emphysema

A subtype of emphysema in which destruction begins in the respiratory bronchioles and extends peripherally. It predominantly affects the upper lobes and is strongly associated with cigarette smoking. It is the most common form of emphysema and often coexists with chronic bronchitis.

Centric-Fusion Translocation

A Robertsonian translocation in which the long arms of two acrocentric chromosomes fuse near their centromeres, leaving a combined chromosome and usually losing the short arms.

Centrifugation

Spinning a specimen at high speed to separate components by density, such as cells from plasma or precipitated material from a liquid.

Centrifuge

Spins samples to separate components by density. Used for blood processing, urine analysis, tissue preparation. Speed: RPM or RCF.

Centrilobular Emphysema

The most common morphological and pathological subtype of pulmonary emphysema, characterized by selective destruction and enlargement of the respiratory bronchioles and alveolar ducts in the central portion of the pulmonary acinus, while the distal alveolar sacs and alveoli are relatively preserved. It exhibits a marked predilection for the upper lobes and apical segments of the lungs and is overwhelmingly caused by long-term cigarette smoking and chronic inhalation of toxic combustion products.

Alternate Names:

Centriacinar Emphysema; Proximal Acinar Emphysema

Centriole

A cylindrical microtubule structure in the centrosome that helps organize the mitotic spindle and, in some cells, forms the basal body of a cilium or flagellum.

Centromere

The constricted chromosome region where sister chromatids remain attached and the kinetochore forms for accurate chromosome segregation during cell division.

Centronuclear Myopathy

A genetically and clinically heterogeneous congenital myopathy characterized histopathologically by the central, internal positioning of nuclei in a large majority of muscle fibers, resembling embryonic myotubes, accompanied by a surrounding peripheral halo devoid of myofibrils. Inherited in autosomal dominant patterns (mutations in dynamin 2 [*DNM2*]), autosomal recessive patterns (amphiphysin 2 [*BIN1*] or *RYR1*), or X-linked recessive form (myotubularin [*MTM1*]). Clinical presentation ranges from severe neonatal weakness to mild adult-onset limb-girdle weakness, classically featuring bilateral ptosis, external ophthalmoparesis, facial diplegia, neck flexor weakness, and foot drop.

Alternate Names:

CNM; Myotubular Myopathy (General)

Centrosome

The main microtubule-organizing center of an animal cell, containing a pair of centrioles and coordinating the spindle during mitosis.

Centrum

Centrum means a central part, especially the main body of a vertebra around which the neural arch develops.

Cephacetrile

An older first-generation cephalosporin antibiotic that inhibits bacterial cell-wall synthesis; it is not commonly used in current practice.

Cephalalgia

Headache or pain in the head, including pain arising from cranial or related structures. It is a symptom with many possible primary and secondary causes.

Cephalexin

An oral first-generation cephalosporin that inhibits bacterial cell-wall synthesis and treats selected skin, respiratory, urinary, and other susceptible infections.

Cephalgia

Headache or pain in the head. It may result from primary headache disorders or from an underlying illness or injury.

Cephalhaematoma

A collection of blood between the periosteum and the skull bone, caused by rupture of blood vessels during childbirth, most commonly associated with ventouse (vacuum) delivery. Unlike caput succedaneum, a cephalhaematoma does not cross suture lines and is limited by the periosteal attachments.

Cephalic

Relating to the head or toward the head; in obstetrics, a cephalic presentation means the fetus presents head first.

Cephalic Vein

A superficial vein on the lateral side of the upper limb that drains the hand and forearm and joins the axillary vein near the shoulder.

Cephalohematoma

A collection of blood beneath the periosteum of a newborn's skull, usually caused by birth trauma. It is confined by suture lines and generally resolves gradually.

Cephalometry

Measurement of the skull and facial structures, usually on standardized radiographs, used in orthodontic planning and assessment of craniofacial growth.

Cephalopelvic Disproportion

A mismatch between the size or shape of the maternal pelvis and the fetal head that prevents or impedes vaginal delivery. It may be suspected from clinical assessment but cannot always be diagnosed reliably before labor.

Cephaloridine

An older injectable first-generation cephalosporin antibiotic whose use declined because of nephrotoxicity and the availability of safer alternatives.

Cephalosporin

A class of beta-lactam antibiotics derived from the fungus *Cephalosporium acremonium*, with a mechanism identical to penicillins — inhibition of bacterial cell wall synthesis by binding to penicillin-binding proteins and inhibiting transpeptidase. Cephalosporins are classified into five generations based on their antimicrobial spectrum.

Cephalosporins

Cephalosporins are beta-lactam antibiotics that inhibit bacterial cell-wall synthesis; generations differ in spectrum, and allergy, diarrhea, and resistance are important considerations.

Cephalothin

An older injectable first-generation cephalosporin that inhibits bacterial cell-wall synthesis and was used for susceptible serious infections.

Cephapirin

A first-generation cephalosporin antibiotic formerly used for susceptible bacterial infections; current availability and indications vary by country.

Cephradine

A first-generation cephalosporin antibiotic related to cephalexin that inhibits bacterial cell-wall synthesis and treats selected susceptible infections.

Ceramide

A sphingolipid composed of a sphingoid base linked to a fatty acid. Ceramides are structural compo-

nents of cell membranes and the epidermal permeability barrier and also participate in cell signaling.

Cercaria

A cercaria is a free-swimming larval stage of a trematode parasite; contact with some species can cause swimmer's itch or other infection.

Cercarial Dermatitis

Alternate index label; see *Swimmer's itch*.

Cerclage

A surgical placement of a suture around the cervix to prevent preterm delivery in patients with cervical insufficiency. Types include transvaginal (McDonald or Shirodkar) and transabdominal. Usually placed at 12-14 weeks gestation and removed at 36-37 weeks.

Cerebellar

Cerebellar means relating to the cerebellum, which coordinates balance, posture, eye movements, and skilled movement.

Cerebellar Ataxia

A neurological sign characterized by impaired coordination of voluntary movements, resulting from dysfunction of the cerebellum or its connections. The cerebellum coordinates the timing, force, and accuracy of movements, and cerebellar dysfunction produces a characteristic constellation of findings.

Cerebellar Cortex

The folded outer layer of the cerebellum containing Purkinje and granular neurons that coordinates balance, posture, timing, and precision of movement.

Cerebellar Disease

A disorder of the cerebellum causing impaired coordination, gait ataxia, dysmetria, tremor, abnormal eye movements, or speech changes.

Cerebellar Dysmetria

Inability to control the range and force of a voluntary movement accurately, producing overshooting

or undershooting from cerebellar dysfunction.

Cerebellar Hemangioblastoma

A usually benign, highly vascular tumor of the cerebellum that may cause headache, imbalance, vomiting, or hydrocephalus and can occur with von Hippel-Lindau disease.

Cerebellar Hypoplasia

Underdevelopment of the cerebellum, usually beginning before birth, that may cause poor coordination, abnormal eye movements, seizures, and developmental impairment.

Cerebellar Neoplasm

A benign or malignant tumor involving the cerebellum; symptoms may include poor coordination, imbalance, abnormal eye movements, headache, vomiting, or hydrocephalus.

Cerebellar Nuclei

Deep clusters of cerebellar neurons that receive processed signals from the cerebellar cortex and send coordinated motor output to the brainstem and thalamus.

Cerebellar Tumor

A neoplasm involving the cerebellum that may cause ataxia, dysmetria, nystagmus, vomiting, or hydrocephalus.

Cerebellar Vermis

The midline portion of the cerebellum that helps control trunk posture, gait, balance, and coordination of eye and head movements.

Cerebellitis

Inflammation of the cerebellum, often after infection and sometimes associated with an autoimmune response. It may produce ataxia, dysarthria, nystagmus, headache, and altered consciousness.

Cerebellopontine Angle

The angle-shaped region at the junction of the cerebellum, pons, and petrous temporal bone. Lesions

in this space may affect cranial nerves and commonly include vestibular schwannoma.

Cerebellopontine Angle Tumor

Lesion at the cerebellopontine angle, an anatomic space between the cerebellum and pons, medial to the temporal lobe and lateral to the brainstem, containing CN V (trigeminal), VII (facial), VIII (vestibulocochlear), and the anterior inferior cerebellar artery (AICA).

Cerebellum

Largest brain portion (approximately 10 percent of volume but over 50 percent of neurons) in the posterior cranial fossa behind the brainstem. Two hemispheres and a vermis, divided into vestibulocerebellum (balance, eye movements), spinocerebellum (posture, gait, coordination), and cerebrocerebellum (motor planning, cognition).

Cerebral

Relating to the cerebrum or, more broadly, to the brain.

Cerebral Amyloid Angiopathy

Deposition of amyloid beta in the walls of small and medium cerebral and leptomeningeal arteries. It commonly causes recurrent lobar intracerebral hemorrhage in older adults and may also cause transient focal neurological episodes or cognitive decline.

Cerebral Aneurysm

An abnormal focal enlargement of an artery in or around the brain. It may remain asymptomatic or rupture and cause subarachnoid hemorrhage. Risk factors include hypertension, smoking, certain inherited disorders, and family history.

Cerebral Angiography

Contrast imaging of cerebral vessels. Evaluates aneurysms, AVMs, stenosis. Invasive procedure.

Cerebral Aqueduct

A narrow midbrain channel connecting the third

and fourth ventricles for cerebrospinal-fluid flow.

Cerebral Arteriovenous Malformation

An abnormal direct connection between cerebral arteries and veins without a normal capillary bed. It may cause intracranial hemorrhage, seizures, headache, or focal neurological deficits and is evaluated with brain imaging and angiography.

Cerebral Artery

An artery supplying the brain; blockage or rupture can cause stroke.

Cerebral Atrophy

Loss or shrinkage of brain tissue caused by aging, neurodegeneration, injury, infection, or vascular disease.

Cerebral Autoregulation

The intrinsic ability of cerebral arterioles to maintain relatively stable cerebral blood flow across a range of perfusion pressures. It may be impaired by severe brain injury, ischemia, hypercapnia, or anesthetic and vasoactive drugs.

Cerebral Calcification

Abnormal calcium deposition in brain tissue, with causes including inherited, metabolic, infectious, vascular, and age-related disorders.

Cerebral Cavernous Malformation

A low-flow vascular malformation composed of dilated thin-walled vascular spaces in the brain or spinal cord. It may cause seizures, headaches, or hemorrhage, although some lesions remain asymptomatic.

Cerebral Cavernous Malformations

Low-flow vascular malformations composed of dilated, thin-walled capillaries without intervening brain parenchyma, often multiple and familial (autosomal dominant with mutations in CCM1, CCM2, or CCM3 genes). They may present with seizures, focal neurological deficits, or intracerebral hemorrhage.

Cerebral Cortex

The outermost layer of the cerebrum, composed of gray matter (neuronal cell bodies) folded into gyri (ridges) and sulci (grooves), greatly increasing surface area.

Cerebral Crus

A midbrain fiber bundle carrying major descending motor pathways from the cortex.

Cerebral Diplegia

Cerebral diplegia is a form of cerebral palsy with predominant weakness and spasticity of both legs, often greater than in the arms.

Cerebral Edema

The accumulation of excess fluid within the brain parenchyma, leading to increased intracranial pressure and potentially fatal brain herniation. There are three main types with different pathophysiological mechanisms.

Cerebral Embolism

Alternate index label; see *Embolic stroke (brain embolism)*.

Cerebral Haemorrhage

Cerebral haemorrhage is bleeding within brain tissue or an adjacent intracranial space, causing pressure, tissue injury, and neurologic deficits.

Cerebral Hemisphere

One of the two large halves of the cerebrum, each containing specialized but interconnected cortical regions.

Cerebral Hemorrhage

Bleeding within the brain parenchyma, a type of hemorrhagic stroke causing mass effect, edema, and neurological deficits. Causes: hypertension (most common—basal ganglia, thalamus), cerebral amyloid angiopathy (lobar, elderly), arteriovenous malformation, aneurysm, anticoagulants, trauma, tumors.

Cerebral Hypoxia

Insufficient oxygen delivery to the brain, caused by cardiac arrest, respiratory failure, severe hypotension, or another interruption of oxygen supply. Prolonged hypoxia can produce permanent neuronal injury and hypoxic-ischemic encephalopathy.

Cerebral Infarction

Cerebral infarction, or ischemic stroke, results from occlusion of a cerebral artery causing irreversible neuronal damage. The most common cause is thromboembolism from atherosclerotic carotid or intracranial arteries, or cardioembolism from atrial fibrillation or cardiac thrombus.

Cerebral Ischemia

Inadequate blood flow and oxygen delivery to brain tissue, causing transient dysfunction or ischemic stroke.

Cerebral Lobes

Four lobes divided by sulci. Frontal (anterior to central sulcus): executive function, motor control (precentral gyrus), Broca's area. Parietal (between central and parieto-occipital sulci): somatosensory (postcentral gyrus), spatial awareness. Temporal (below lateral sulcus): auditory processing, memory (hippocampus), Wernicke's area.

Cerebral Malaria

Severe malaria with impaired consciousness, seizures, or other brain dysfunction, usually caused by *Plasmodium falciparum*.

Cerebral Palsy

A group of permanent disorders of movement and posture development causing activity limitation, attributed to non-progressive disturbances occurring in the developing fetal or infant brain. It is the most common motor disability in childhood, affecting approximately 1 in 500 live births.

Alternate Names:

Palsy, Cerebral Palsy Cerebral (Cerebral Palsy)

Cerebral Peduncle

Paired structures of the ventral midbrain carrying descending motor tracts (corticospinal, corticobulbar, corticopontine) connecting the cerebrum to the pons and below. Lesions produce contralateral hemiparesis; Weber syndrome (peduncle + CN III) causes ipsilateral CN III palsy with contralateral hemiparesis.

Cerebral Perfusion Pressure

The pressure gradient driving blood flow through the brain, approximated as mean arterial pressure minus intracranial pressure. CPP falls when systemic pressure decreases or ICP rises and must be interpreted with the clinical context.

Cerebral Salt Wasting

Renal sodium wasting causing hyponatremia and volume depletion occurring after intracranial surgery or in setting of intracranial disease. Distinguished from SIADH by the presence of volume depletion with negative sodium balance and compensatory elevation of renin and aldosterone.

Cerebral Thrombosis

Formation of a blood clot within a cerebral artery, reducing blood flow and potentially causing an ischemic stroke.

Cerebral Tumor

A neoplasm arising within or involving the brain, producing symptoms according to its location, growth rate, and effect on intracranial pressure.

Cerebral Tumours

Cerebral tumours are abnormal growths in the cerebrum, which may be benign or malignant and cause seizures, headache, cognitive change, or focal deficits.

Cerebral Vasospasm

Abnormal narrowing of cerebral arteries that can reduce brain perfusion and cause delayed cerebral ischemia, especially after aneurysmal subarachnoid hemorrhage.

Cerebral Venous Sinus Thrombosis

Thrombosis of the cerebral venous sinuses (most commonly the superior sagittal sinus; also transverse, sigmoid, straight; or cortical veins), with resultant venous hypertension, intracranial hypertension, hemorrhagic cortical infarcts, and venous congestion.

Cerebral Ventricle

One of the interconnected brain cavities containing and circulating cerebrospinal fluid.

Cerebral Ventriculography

Imaging of the cerebral ventricles to assess their size, shape, and possible obstruction.

Cerebral Ventriculomegaly

Enlargement of one or more cerebral ventricles, caused by hydrocephalus, brain-tissue loss, or developmental variation.

Cerebral Visual Impairment

Visual impairment caused by damage to the brain's visual pathways or occipital cortex rather than the eye itself; the leading cause of childhood visual impairment in developed countries. Features include poor visual attention, difficulty recognizing faces or objects, and variable vision; management is educational and rehabilitative.

Cerebriiform Nevus Sebaceous

A variant of nevus sebaceous presenting as a soft, cerebriiform (brain-like), pedunculated or sessile nodule, often found on the scalp or face. It is a benign adnexal hamartoma composed of sebaceous glands, but like other nevus sebaceous lesions, it carries a low risk of secondary neoplasms and may warrant monitoring or excision.

Cerebritis

Inflammatory process in brain parenchyma without abscess formation. May precede abscess. Causes: bacterial, fungal, parasitic infections. Diagnosis: MRI with contrast showing enhancement. Treatment: antibiotics/antifungals, surgical drainage if abscess forms.

Cerebroplacental Ratio

The ratio of the middle cerebral artery pulsatility index to the umbilical artery (UA) pulsatility index. CPR <5th percentile (or <1.0-1.1) suggests fetal brain-sparing effect (vasodilation of cerebral vessels in response to hypoxemia) due to placental insufficiency.

Cerebrosides

Glycosphingolipids composed of ceramide plus a single sugar, abundant in myelin and nervous tissue and involved in membrane structure.

Cerebrospinal Fluid

A clear, colorless fluid that fills the ventricles of the brain, the central canal of the spinal cord, and the subarachnoid space around the brain and spinal cord. Produced primarily by the choroid plexus of the lateral, third, and fourth ventricles (500 mL/day, total volume 150 mL in adults).

Alternate Names:

Spinal Fluid

Cerebrospinal Fluid

A clear fluid produced within the brain that surrounds and cushions the brain and spinal cord, helps regulate their chemical environment, and can be examined for disease.

Cerebrospinal Fluid Collection

A procedure, usually lumbar puncture, that collects cerebrospinal fluid for laboratory analysis; it can help evaluate infection, bleeding, inflammation, malignancy, and other disorders of the central nervous system.

Cerebrospinal Fluid Culture

A laboratory test that attempts to grow bacteria, fungi, or other microorganisms from cerebrospinal fluid to investigate meningitis or another central nervous system infection. Results depend on prior antimicrobial treatment, specimen volume, transport, and the clinical context.

Cerebrospinal Fluid Analysis

Laboratory examination of cerebrospinal fluid, usually obtained by lumbar puncture. It may assess infection, bleeding, inflammation, demyelinating disease, or malignant cells. Analysis can include opening pressure, appearance, cell counts, protein, glucose, microbiology, and other tests selected for the clinical question.

Cerebrospinal Fluid Cytology

Microscopic examination of cells in cerebrospinal fluid to evaluate suspected malignancy, infection, inflammation, or hemorrhage. It is especially useful for detecting leptomeningeal tumor involvement when interpreted with clinical and imaging findings.

Cerebrovascular

Relating to the blood vessels and circulation of the brain.

Alternate Names:

Brain-vascular; cerebral vascular.

Cerebrovascular Accident

An acute neurological deficit caused by interruption of blood flow to the brain or bleeding within the brain. See Stroke.

Cerebrovascular Accident

A cerebrovascular accident is a stroke caused by interrupted brain blood flow from ischemia or bleeding, producing sudden neurologic dysfunction.

Cerebrovascular Accident Prevention

Measures to reduce the risk of stroke, including controlling blood pressure, diabetes, and lipids, avoiding tobacco, treating atrial fibrillation when indicated, and using appropriate antithrombotic therapy.

Cerebrovascular Accident (Stroke)

Sudden brain dysfunction caused by interrupted blood flow or bleeding in the brain. See Stroke.

Alternate Names:

Stroke

Cerebrovascular Disorder

A disorder of brain blood vessels or cerebral blood flow, including ischemia, hemorrhage, aneurysm, malformation, and venous thrombosis.

Cerebrum

The largest part of the brain, comprising two cerebral hemispheres connected by the corpus callosum. Each hemisphere has four lobes: frontal (executive function, motor, Broca's area), parietal (somatosensory, spatial awareness), temporal (auditory, memory, Wernicke's area), occipital (visual processing).

Cerium

A rare-earth metal whose compounds have industrial and experimental medical uses; exposure to some forms can irritate tissues or cause toxicity.

Certification

A formal process confirming that a person, service, laboratory, or facility meets specified training, quality, safety, or professional standards.

Certified Nurse-Midwife

An advanced-practice nurse trained and credentialed to provide reproductive, pregnancy, birth, postpartum, and newborn care, including management of normal pregnancy and referral of complications.

Certolizumab Pegol

A medicine or chemical compound identified by certolizumab pegol. Its uses, effects, and risks depend on the specific substance and clinical context.

Ceruloplasmin

The principal copper-carrying protein in blood, synthesized mainly by the liver. Its concentration and copper-binding function are clinically relevant in disorders such as Wilson disease.

Alternate Names:

Caeruloplasmin

Ceruloplasmin Blood Test

A blood test measuring ceruloplasmin, the main copper-carrying protein in plasma; it is used with

other copper studies to evaluate Wilson disease and disorders of copper metabolism.

Cerumen

A substance that helps keep dirt out of the ear and lubricates the skin in the ear. More commonly known as earwax.

Cerumen Impaction

Accumulation of earwax in the external auditory canal that causes symptoms or prevents visualization of the tympanic membrane. It may cause aural fullness, conductive hearing loss, tinnitus, itching, or pain.

Ceruminous Adenocarcinoma

A malignant tumor of the ceruminous glands in the external ear canal that may cause pain, discharge, hearing loss, or a persistent ear-canal mass.

Cervical

Relating either to the neck portion of the spine or to the cervix of the uterus; the intended meaning is determined by the surrounding anatomic context.

Cervical Adenocarcinoma

A malignant gland-forming tumor arising in the uterine cervix; it may be HPV-associated or HPV-independent and commonly presents with abnormal bleeding or discharge.

Cervical Arthritis

Degenerative or inflammatory disease of the neck joints that can cause neck pain, stiffness, reduced motion, and sometimes nerve or spinal-cord compression.

Cervical Atlas

The first cervical vertebra, C1, a ring-shaped bone that supports the skull and rotates around the second cervical vertebra.

Cervical Biopsy

A cervical biopsy removes a small sample of cervix for microscopic examination, commonly to investi-

gate abnormal screening or suspected precancer or cancer.

Cervical Cancer

A malignant tumour arising from the cervix, most commonly caused by persistent infection with high-risk human papillomavirus. It may be asymptomatic early or cause abnormal vaginal bleeding, discharge, or pelvic pain; prevention and early detection rely on HPV vaccination, screening, and assessment of abnormal results.

Cervical Cancer - Screening And Prevention

Clinical measures used to prevent or detect cervical cancer early, including human papillomavirus vaccination and screening with HPV testing, cervical cytology, or both according to age and risk.

Cervical Cap

A cervical cap is a flexible barrier placed over the cervix to reduce sperm entry; it is used with spermicide as a contraceptive method.

Cervical Carcinoma

A malignant tumor of the uterine cervix, most commonly associated with persistent high-risk human papillomavirus infection.

Cervical Cerclage

Surgical placement of a non-absorbable suture around the cervix to provide mechanical support and prevent preterm delivery in patients with cervical insufficiency.

Cervical Change

A change in the uterine cervix or cervical spine, depending on context; clinical significance is determined by the specific structural, inflammatory, or pregnancy-related finding.

Cervical Dilatation

Progressive opening of the cervical os during labor, measured in centimeters from 0 (closed) to 10 cm (fully dilated). The cervix must also efface (thin out)

before significant dilatation occurs. Cervical status assessed by digital vaginal examination during labor.

Cervical Dilation

The opening of the cervical canal during labor, measured in centimeters from closed (0 cm) to fully dilated (10 cm); required for vaginal delivery. Occurs with effacement (thinning/shortening of the cervix) during the first stage of labor, driven by effective uterine contractions.

Cervical Disc Herniation

A common cause of neck pain and radiculopathy, occurring when the nucleus pulposus of an intervertebral disc in the cervical spine herniates through the annulus fibrosus, compressing adjacent nerve roots or the spinal cord.

Cervical Dislocation

Abnormal displacement of one cervical vertebra relative to another, usually from trauma, that can destabilize the neck and injure the spinal cord.

Cervical Dysplasia

Abnormal growth of cervical epithelial cells, representing a premalignant condition that may progress to cervical cancer if untreated. Also known as cervical intraepithelial neoplasia (CIN). CIN is caused by persistent infection with high-risk human papillomavirus (HPV) types, most commonly HPV 16 and 18.

Alternate Names:

cervical intraepithelial. Cervical Intraepithelial Neoplasia Cervical (Cervical Dysplasia) Dysplasia Cervical (Cervical Dysplasia) Squamous Intraepithelial Lesion Cervical (Cervical Dysplasia)

Cervical Dystocia

Failure of the uterine cervix to dilate normally during labor, which can delay or prevent vaginal delivery.

Cervical Dystonia

A focal movement disorder causing involuntary contractions of neck muscles and abnormal positioning of the head. It may produce pain, tremor, and

sustained or intermittent twisting of the neck.

Alternate Names:

Dystonia, Cervical Torticollis, Spasmodic

Cervical Effacement

Thinning and shortening of the cervix during labor, measured as a percentage from 0% (thick, normal length) to 100% (paper thin). Effacement precedes and accompanies cervical dilatation. Nulliparous women often efface before dilating; multiparous women may efface and dilate simultaneously.

Cervical Endometriosis

Endometrial-like tissue in or on the cervix that may cause bleeding after sex, pelvic pain, or an abnormal cervical lesion.

Cervical Exam In Labor

A digital vaginal examination used to assess cervical dilation, effacement, consistency, and position, together with the station and position of the fetal head. Serial examinations help assess labor progress.

Cervical Ganglion

A cluster of nerve-cell bodies in the sympathetic chain of the neck that relays autonomic signals to the head, neck, and upper chest.

Cervical Incompetence

The inability of the cervix to maintain pregnancy in the second trimester due to painless cervical dilatation. Results in recurrent second trimester pregnancy loss or preterm delivery. Risk factors include prior cervical trauma, cone biopsy, and congenital anomalies.

Cervical Insufficiency

Painless shortening or opening of the cervix during pregnancy before term, increasing risk of second-trimester loss or spontaneous preterm birth.

Cervical Intraepithelial Neoplasia

Abnormal precancerous changes in the squamous cells of the cervix, usually associated with persistent high-risk human papillomavirus infection. CIN is

graded by the severity of dysplasia.

Alternate Names:

CIN

Cervical Intraepithelial Neoplasia

Cervical (Cervical Dysplasia)

Alternate index label for Cervical Dysplasia.

Cervical Length Screening

Transvaginal ultrasound measurement of the cervix between 18-24 weeks to assess risk of spontaneous preterm birth. Normal cervix measures greater than 25 mm. Short cervix (less than 25 mm) increases preterm delivery risk. Cervical funneling and dilatation are additional concerning findings.

Cervical Lymphadenopathy

Enlarged lymph nodes in the neck, commonly caused by infection but sometimes reflecting malignancy, autoimmune disease, or other systemic illness.

Cervical MRI Scan

Magnetic resonance imaging of the cervical spine that uses a magnetic field and radio waves to produce detailed images of the neck vertebrae, discs, spinal cord, nerves, and surrounding soft tissues.

Cervical Mucus

Secretions from the cervical glands that change throughout the menstrual cycle under hormonal influence. Estrogen produces thin, clear, stretchy mucus favorable for sperm transport around ovulation (spinnbarkeit). Progesterone produces thick, sticky, opaque mucus that forms a plug in the cervical canal during pregnancy.

Cervical Myelopathy

Dysfunction of the spinal cord caused by compression in the cervical spine, often producing hand clumsiness, gait disturbance, weakness, sensory changes, or abnormal reflexes.

Cervical Osteoarthritis

Alternate index label; see *Cervical spondylosis*.

Cervical Pain

Alternate index label; see *Neck pain*.

Cervical Pain (Neck Pain)

Alternate index label for Neck Pain.

Cervical Polyp

A usually benign polypoid overgrowth of endocervical or ectocervical tissue that may cause intermenstrual or postcoital bleeding or may be found incidentally during examination.

Cervical Polyps

Usually benign, finger-like growths arising from the endocervical canal or cervix. They may cause irregular bleeding, postcoital bleeding, or vaginal discharge and are often found during a pelvic examination. Symptomatic or atypical polyps are removed and sent for pathology.

Cervical Pregnancy

A rare ectopic pregnancy (0.1-1%) implanted in the cervical canal. Presents with painless vaginal bleeding in the first trimester. Diagnosis: ultrasound (hourglass-shaped uterus, gestational sac in the cervix with absent sliding sign, empty uterine cavity). High risk of severe hemorrhage.

Cervical Radiculopathy

A condition caused by compression, irritation, or inflammation of a spinal nerve root in the cervical spine, resulting in pain, numbness, tingling, and weakness in the distribution of the affected nerve root.

Cervical Rib

An extra rib arising from a cervical vertebra, usually the seventh. It may be asymptomatic or contribute to thoracic outlet syndrome by compressing the brachial plexus or subclavian vessels.

Cervical Ripening Agents

Medicines or mechanical devices used to soften and prepare the cervix before labor induction. Options include prostaglandins and cervical balloons; selec-

tion and dosing depend on the cervix, pregnancy, contraindications, and local obstetric protocol.

Cervical Screening (Pap Smear)

Screening for cervical cancer by collecting cells from the transformation zone of the cervix. Liquid-based cytology (ThinPrep) or conventional Pap smear. HPV co-testing (high-risk HPV 16, 18, others) improves sensitivity.

Alternate Names:

Pap Smear

Cervical Spine

The upper segment of the vertebral column consisting of seven cervical vertebrae, C1 through C7, supporting the head and protecting the cervical spinal cord. It permits substantial flexion, extension, rotation, and lateral flexion while transmitting the vertebral arteries through the transverse foramina of C1-C6.

Cervical Spine CT Scan

Computed tomography of the cervical spine that uses x-rays and computer reconstruction to show the bones and alignment of the neck; it is particularly useful for detecting fractures and other bony abnormalities.

Cervical Spine Trauma

Injury to the cervical spine that may affect the vertebrae, discs, ligaments, spinal cord, or nerve roots. Evaluation includes the mechanism, examination, and appropriate imaging; classification systems can help guide treatment, which depends on stability and neurologic findings.

Cervical Spondyloarthropathy

A degenerative condition of the cervical spine involving disc degeneration, osteophyte formation, and facet joint arthropathy that may compress nerve roots or the spinal cord. It can cause neck pain, radiculopathy, or myelopathy, and is a common cause of cervical spinal stenosis in older adults.

Cervical Spondylosis

Age-related degenerative changes of the cervi-

cal spine—disc degeneration, osteophyte formation, facet joint arthritis, and ligament hypertrophy—affecting the intervertebral discs, vertebrae, and joints.

Alternate Names:

Cervical Osteoarthritis Osteoarthritis, Cervical Spondylosis, Cervical

Cervical Squamous Cell Carcinoma

A malignant squamous tumor of the cervix, most often caused by persistent high-risk human papillomavirus infection; it may present with abnormal bleeding, discharge, or an abnormal screening result.

Cervical Stenosis

Pathological narrowing or complete obliteration of the endocervical canal, commonly acquired secondary to cervical surgical procedures (e.g., cold knife conization, loop electrosurgical excision procedure [LEEP], or cryotherapy), radiation therapy, chronic infection, or postmenopausal hypoestrogenic atrophy. May result in dysmenorrhea, abnormal bleeding, hematometra, retrograde menstruation with secondary endometriosis, or mechanical infertility.

Cervical Vertebrae

The seven vertebrae of the neck. Typical cervical vertebrae have small bodies, bifid spinous processes, and transverse foramina transmitting the vertebral vessels; C1, C2, and C7 are specialized.

Cervical Vertebrae (C1-C7)

The seven vertebrae of the neck, numbered C1 through C7; they support the head, protect the cervical spinal cord, and allow neck movement.

Cervicitis

Cervicitis is inflammation of the uterine cervix, commonly from sexually transmitted infection, bacterial imbalance, irritation, or allergy.

Cervicogenic Headache

A headache caused by a disorder of the cervical spine or its soft tissues, with pain referred from the neck to the head.

Alternate Names:

Cervical headache; neck-related headache.

Cervix

The cervix is the lower part of the uterus projecting into the vagina and producing mucus, allowing menstrual flow out and sperm passage in.

Cervix Carcinoma

Cancer arising from the cervix, most often related to persistent high-risk human papillomavirus infection; screening and HPV vaccination reduce risk, and treatment depends on stage.

Cervix Cryosurgery

A medical procedure involving cervix cryosurgery. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Cervix Disease

A disorder of the lower uterine cervix, ranging from infection and benign polyps to precancer, cancer, scarring, or pregnancy-related change.

Cervix Neoplasm

A benign, precancerous, or malignant growth of the cervix; evaluation may include pelvic examination, HPV testing, cytology, colposcopy, and biopsy.

Cervix Uteri

The cervix uteri is the lower portion of the uterus that projects into the vagina and forms the passage between the uterus and birth canal.

Cesarean Delivery

Surgical delivery of the fetus through abdominal and uterine incisions. Performed when vaginal delivery is contraindicated or fails. Indications include malpresentation, placenta previa, previous cesarean, fetal distress, and failed induction. Most common uterine incision is low transverse (classical for very preterm or transverse lie).

Cesarean Section

C-section: surgical delivery of the infant through incisions in the abdominal wall and uterus. Indications: fetal distress, failure to progress, malpresentation (breech, transverse), placenta previa, abruption, previous cesarean (some), multiple pregnancy, preeclampsia, cord prolapse, and maternal conditions.

Cesarean Section (C-Section)

Surgical delivery of a fetus through incisions in the abdominal wall (laparotomy) and uterus (hysterotomy).

Alternate Names:

C-section

Cesium

A chemical element whose radioactive isotope cesium-137 can cause serious external or internal radiation injury after contamination or exposure.

Cestoda

The tapeworm class of flatworms; human infection follows ingestion of eggs or larval tissue and can cause intestinal disease or tissue cysts.

Cestode

A cestode is a tapeworm, a segmented parasitic flatworm that can infect humans or animals through contaminated food, water, or intermediate hosts.

Cestodiasis

Infection or infestation caused by cestodes, commonly called tapeworms. Disease may involve the intestine or tissues, depending on the species and life-cycle stage.

Alternate Names:

tapeworms.

Cetirizine

A second-generation H1 antihistamine used for allergic rhinitis and urticaria; it is less sedating than older antihistamines but can still cause drowsiness.

Cetoleic Acid

A long-chain monounsaturated fatty acid found mainly in marine oils and studied for effects on lipid metabolism and cellular membranes.

Cetrimide

Cetrimide is a quaternary ammonium antiseptic and detergent used in some topical or laboratory preparations; concentrated exposure irritates tissue.

Cetrorelix

A gonadotropin-releasing hormone antagonist used in controlled ovarian stimulation to prevent a premature luteinizing-hormone surge.

Cetuximab

A medicine or chemical compound identified by cetuximab. Its uses, effects, and risks depend on the specific substance and clinical context.

Cevimeline

A muscarinic receptor agonist used to stimulate salivary secretion and relieve dry mouth, especially in selected patients with Sjögren syndrome.

Cevimeline Hydrochloride

A medicine or chemical compound identified by cevimeline hydrochloride. Its uses, effects, and risks depend on the specific substance and clinical context.

CF

Alternate index label; see *Cystic fibrosis*.

CFS

Chronic fatigue syndrome, a condition involving substantial persistent fatigue with post-exertional worsening and associated symptoms that are not explained by another disorder.

Alternate Names:

Chronic fatigue syndrome; myalgic encephalomyelitis/chronic fatigue syndrome.

CFTR Modulator Therapy

Targeted therapies correcting the defective CFTR

protein. Elexacaftor-tezacaftor-ivacaftor (Trikafta) is the most effective combination for patients with at least one F508del mutation. These therapies have dramatically changed CF prognosis.

CHA2DS2-VASc Score

A clinical prediction score used universally to stratify the annual risk of ischemic stroke and systemic thromboembolism in adult patients with non-valvular atrial fibrillation. Points are assigned for Congestive heart failure (1), Hypertension (1), Age ≥ 75 years (2), Diabetes mellitus (1), prior Stroke or transient ischemic attack or thromboembolism (2), Vascular disease (1), Age 65–74 years (1), and Sex category female (1). A score of ≥ 2 in men or ≥ 3 in women indicates a high thromboembolic risk and strongly warrants oral anticoagulation with direct oral anticoagulants (DOACs) or warfarin.

Alternate Names:

CHA2DS2-VASc; Atrial Fibrillation Stroke Risk Score

Chaetomium

A genus of environmental molds that rarely causes opportunistic infection, including skin, eye, sinus, or disseminated disease in people with impaired immunity.

Chafing

Skin irritation caused by repeated friction, moisture, heat, or tight clothing, producing redness, burning, tenderness, and sometimes erosions in opposing or rubbing areas.

Chagas Cardiomyopathy

Heart damage caused by long-term Chagas disease. It can cause an irregular heartbeat, an enlarged heart, heart failure, or blood clots that may lead to a stroke.

Chagas Disease

An infection caused by the parasite *Trypanosoma cruzi*. It is usually spread when the faeces of an infected triatomine bug (kissing bug) enter a bite wound, the eyes, or the mouth; it can also spread from mother to baby, through contaminated blood

or transplanted organs, or through contaminated food. Early symptoms are often mild or absent, but may include fever, tiredness, or swelling around one eye. Years later, some people develop serious heart disease or enlargement of the oesophagus or colon, causing trouble swallowing or constipation. The infection can be treated with antiparasitic medicines, especially when treatment starts early.

Alternate Names:

Trypanosomiasis, American

Chagoma

A red, swollen, and sometimes painful area where the *Trypanosoma cruzi* parasite enters the body. It is an early sign of Chagas disease and may occur with swollen nearby lymph nodes.

Chain Of Custody

Meticulous paper trail and chronological documentation tracking the seizure, custody, control, transfer, analysis, and disposition of physical and electronic evidence in medico-legal investigations (e.g., blood samples for toxicology, bullets, sexual assault kits, biological fluids).

Chain-Termination Codon

A codon that signals the end of protein translation; the three standard stop codons are UAA, UAG, and UGA in messenger RNA.

Chalazion

A usually painless, firm eyelid lump caused by blockage and inflammation of a meibomian oil gland; persistent or atypical lesions require ophthalmic assessment.

Challenge Agent

A substance, organism, allergen, or physiologic stimulus deliberately administered in a controlled test to provoke and measure a response.

Challenge Testing

A controlled exposure to a suspected trigger, such as a food or medication, to assess whether it produces a reproducible reaction.

Alternate Names:

Provocation testing; challenge test.

Chamomile Extract

An extract of chamomile flowers used in traditional and complementary products; allergic reactions and medicine interactions are possible, while evidence varies by indication.

Chancre

A usually painless ulcer at the site of inoculation of an infection, especially the primary lesion of syphilis. It may be accompanied by regional lymph-node enlargement and heals without treatment, although infection persists.

Chancroid

A sexually transmitted infection caused by *Haemophilus ducreyi*, characterized by painful ragged genital ulcers and tender suppurative inguinal lymph nodes.

Changes In The Newborn At Birth

The newborn transition from placental support to independent breathing and circulation, including lung aeration, closure of fetal shunts, temperature control, and adaptation of glucose and feeding.

Changing Your Ostomy Pouch

Replacing the pouching system attached around a colostomy or ileostomy, including gentle removal, skin cleansing and measurement, barrier fitting, leak prevention, and disposal.

Changing Your Urostomy Pouch

Replacing the pouching system that collects urine from a urostomy, with attention to skin protection, a secure seal, drainage, hydration, and warning signs of infection or obstruction.

Channelopathy

A disorder caused by abnormal structure or function of an ion channel or its regulatory proteins. Channelopathies can affect excitable tissues and cause neurologic, muscular, or cardiac disease.

Chaplain

A trained spiritual-care professional who supports patients and families with religious, existential, emotional, or ethical concerns in healthcare settings.

Chapped Hands

Dry, cracked, inflamed skin of the hands caused by irritants, cold, low humidity, frequent washing, eczema, or allergy; fissures can become painful or infected.

Chapped Lips

Dry, scaling, fissured lips caused by weather, licking, dehydration, irritants, medicines, or cheilitis; persistent lesions need assessment for infection or another disorder.

Charcoal

Fine black powder produced by controlled pyrolysis of organic materials, processed to yield a massive micro-porous surface area (500-1500 m²/g). See activated charcoal.

Charcot Arthropathy

Progressive destruction and deformity of a joint caused by loss of protective pain or proprioceptive sensation. It is most often associated with peripheral neuropathy, particularly diabetic neuropathy.

Charcot Foot

A progressive destructive arthropathy of the foot and ankle, usually occurring in people with peripheral neuropathy, especially diabetes. Repeated unnoticed trauma causes warmth, swelling, deformity, instability, and an increased risk of ulceration and amputation.

Charcot Joint

Progressive, neurotrophic arthropathy caused by loss of sensation (proprioception and pain) leading to joint destruction with bone fragmentation, joint dislocation, deformity, and loss of normal pain and proprioceptive sensation in the affected limb.

Charcot Triad

Clinical constellation of fever, right upper quadrant pain, and jaundice representing the classic presentation of ascending cholangitis, infection of the biliary tree from common bile duct obstruction by a gallstone. The full triad is found in approximately fifty to seventy percent of patients.

Charcot Triad (Cholangitis)

Classic diagnostic triad of acute ascending cholangitis: (1) Fever, (2) Right Upper Quadrant abdominal pain, and (3) Jaundice. Reynolds Pentad adds Hypotension and Altered Mental Status. See Acute Cholangitis.

Alternate Names:

Cholangitis

Charcot's Joints

Charcot's joints are severely damaged, unstable joints caused by loss of protective sensation, classically in diabetic neuropathy or other neurologic disease.

Charcot's Triad

Charcot's triad is fever, jaundice, and right-upper-quadrant pain, a classic but incomplete clinical pattern of acute ascending cholangitis.

Charcot-Marie-Tooth Disease

Group of inherited peripheral neuropathies (hereditary motor and sensory neuropathy, HMSN), the most common inherited neuromuscular disorder (prevalence 1 in 2,500). More than 100 causative genes (PMP22, MPZ, GJB1, MFN2, NEFL).

Alternate Names:

Neuropathy, Hereditary Motor And Sensory Peroneal Muscular Atrophy

Charcot-Wilbrand Syndrome

A rare neuropsychiatric syndrome after brain injury characterized by loss of visual imagery and visual dreaming, sometimes with difficulty recognizing or interpreting visual information.

Chargaff Rule

The observation that double-stranded DNA contains approximately equal amounts of adenine and thymine and of guanine and cytosine, reflecting complementary base pairing.

CHARGE Syndrome

A genetic disorder caused by CHD7 gene mutations (autosomal dominant, most cases de novo) affecting multiple organ systems.

Charley Horse

A sudden painful involuntary contraction or spasm of a muscle, commonly in the calf or thigh, often related to exertion, dehydration, nerve disease, or medicines.

Chasing The Dragon

Inhalation of vapor produced by heating heroin, a method of drug use that can cause dependence, overdose, toxic exposures, and neurological injury.

Chaulmoogra Oil

Chaulmoogra oil is a historical plant oil once used to treat leprosy and some skin diseases; it has largely been replaced by effective antimicrobial therapy.

CHD

Alternate index label; see *Hip dysplasia*.

Check-Up

A check-up is a routine health evaluation used to review symptoms, risk factors, preventive care, examination findings, and appropriate screening.

Checkpoints

Immune regulatory molecules. Cancer exploits checkpoints. Inhibitors block checkpoints.

Chediak-Higashi Syndrome

Autosomal recessive (LYST mutation, lysosomal trafficking regulator). Giant lysosomal granules in neutrophils, melanocytes, neurons. Partial oculocutaneous albinism, silvery hair, photophobia, nystagmus.

Cheek

The facial region forming the side wall of the mouth, containing skin, muscle, fat, salivary structures, nerves, and blood vessels.

Cheek Pouch

A distensible pocket inside the cheek found in some rodents and other animals, used to temporarily store and transport food.

Cheilectropion

An eversion deformity of the lip, usually involving the lower lip. It may result from scarring, congenital malformation, or loss of tissue support.

Cheilitis

Inflammation of one or both lips, causing dryness, scaling, fissuring, erythema, swelling, pain, or ulceration. Causes include irritant or allergic contact exposure, infection, nutritional deficiency, chronic sun exposure, and inflammatory skin disease; persistent lesions require evaluation for premalignant or malignant change.

Cheilodysnia

Pain of the lips, which may result from inflammation, infection, trauma, allergy, dryness, nerve pain, or another oral or dermatologic disorder.

Cheilognathopalatoschisis

A congenital cleft involving the lip, jaw, and palate, caused by incomplete fusion of facial processes during embryonic development.

Cheilognathoschisis

A congenital cleft of the lip and alveolar part of the upper jaw, with or without associated palatal involvement.

Cheiloplasty

Plastic or reconstructive surgery of the lip, performed to repair congenital clefting, trauma, tumors, scars, or other deformity.

Cheiloschisis

Cleft lip, a congenital fissure of the upper lip caused by incomplete fusion of facial processes during embryonic development. It may occur alone or with cleft palate.

Cheilosis

A disorder of the lips characterized by dryness, scaling, fissuring, or inflammation. Causes include irritation, nutritional deficiency, infection, and systemic disease.

Cheiromegaly

Abnormal enlargement of one or both hands, classically associated with excess growth hormone but also possible with local overgrowth or other disorders.

Cheiopompholyx

A form of dyshidrotic eczema affecting the hands, with intensely itchy small vesicles on the palms or sides of the fingers.

Chelating Agent

A compound that binds a metal ion through multiple chemical sites, allowing the metal to be measured, mobilized, or removed in selected toxic exposures.

Chelating Agents

Chelating agents bind specific metal ions to form complexes that can facilitate their removal or reduce toxicity; examples include agents for lead, iron, or copper poisoning.

Chelation

The formation of a stable complex between a metal ion and a molecule with multiple binding sites. In medicine, chelation can be used under supervision to bind and promote elimination of selected toxic metals.

Chelation Study

Nuclear medicine study using chelating agents. Evaluates renal function, obstruction.

Chelation Therapy

Use of chelating agents to bind metals for excretion. Agents: EDTA (lead), deferoxamine (iron), dimercaprol (arsenic, mercury, gold), DMSA (lead, mercury), penicillamine (copper, lead). Complications: nephrotoxicity, electrolyte disturbances, redistribution of metals.

Chemexfoliation

Application of a chemical solution to remove controlled layers of skin, improving selected scars, pigmentation, wrinkles, or superficial lesions.

Chemical

A substance defined by its molecular composition and properties; medical effects depend on dose, route, exposure duration, metabolism, and biological target.

Chemical Ablation

Destruction of unwanted tissue by injecting or applying a chemical irritant or sclerosant, used in selected lesions, vessels, cysts, or other conditions.

Chemical Burn

Tissue injury caused by a corrosive acid, alkali, solvent, or other chemical, which may continue to penetrate until the substance is removed and neutralized appropriately.

Chemical Burn Or Reaction

Tissue injury caused by contact with a corrosive or irritating chemical, commonly affecting the skin or eyes; severity depends on the substance, concentration, contact time, and tissue involved.

Chemical Carcinogenesis

Development of cancer after exposure to chemicals that cause DNA damage, abnormal cell signaling, chronic tissue injury, or altered gene regulation.

Chemical Cleavage

Breaking a chemical bond with a reagent, used in laboratory analysis, molecular biology, synthesis, or controlled degradation of a larger molecule.

Chemical Cofactor

A nonprotein chemical helper required for an enzyme or biologic reaction, such as a metal ion, vitamin-derived coenzyme, or other small molecule.

Chemical Dependency

Legacy, nonpreferred label; see *Substance Use Disorder*.

Chemical Eye Injury

Ocular damage caused by exposure to an acidic or alkaline substance. Immediate copious irrigation, removal of particulate material, measurement of ocular-surface pH, and urgent ophthalmic assessment are required; alkali injuries may penetrate deeply and continue to worsen after exposure.

Chemical Fractionation

Separation of a mixture into fractions according to chemical or physical properties such as solubility, charge, size, or binding affinity.

Chemical Impurity

An unintended chemical present in a medicine, specimen, or material, potentially affecting safety, stability, measurement accuracy, or biologic activity.

Chemical Injury

Tissue damage from corrosive substances. Acid burns: coagulation necrosis, eschar formation. Alkali burns: liquefaction necrosis, deeper penetration. Management: copious irrigation, remove contaminated clothing, assess for airway involvement.

Chemical Peel

Application of a chemical solution to the skin to produce controlled exfoliation and regeneration of superficial or deeper skin layers.

Alternate Names:

Chemexfoliation; chemical resurfacing.

Chemical Peels

Chemical peels use acids to exfoliate skin layers. Superficial peels (glycolic acid 20 to 70 percent, salicylic acid 20 to 30 percent) affect the epidermis.

Medium peels (trichloroacetic acid 35 to 50 percent) reach the papillary dermis. Deep peels (phenol 88 percent) reach the reticular dermis.

Chemical Pneumonitis

Inflammation and lung injury caused by inhalation or aspiration of irritating chemicals, gastric contents, hydrocarbons, or other toxic substances. It may cause cough, dyspnea, hypoxemia, wheezing, and diffuse pulmonary infiltrates.

Chemistry

The study of substances, their composition, properties, reactions, and transformations, forming the basis of biochemistry, pharmacology, toxicology, and laboratory medicine.

Chemo

Informal shorthand for chemotherapy, treatment using drugs that destroy cancer cells, inhibit their growth, or suppress abnormal immune cells.

Chemo Brain

Subjective difficulties with memory, attention, or thinking that occur during or after cancer treatment.

Alternate Names:

Post-Chemotherapy Cognitive Impairment

Chemoattractant

A chemoattractant is a chemical signal that guides cells, especially immune cells, toward a source of the signal during inflammation or development.

Chemodectoma

A tumor arising from paraganglionic tissue, especially a carotid-body tumor. It is usually slow-growing and may compress nearby cranial nerves or vessels.

Chemoembolization

A procedure that combines delivery of an antineoplastic agent with embolization of the blood supply to a tumor, most commonly for selected liver cancers.

Chemokine

A family of small secreted cytokines that direct leukocyte migration (chemotaxis) to sites of infection, inflammation, and tissue repair. Classified by cysteine motif: CXC (e.g., IL-8/CXCL8, recruit neutrophils), CC (e.g., MCP-1/CCL2, attract monocytes), C, and CX3C.

Chemokine Receptor

A cell-surface receptor that binds chemokines and directs immune-cell movement; some also serve as entry cofactors for viruses such as HIV.

Chemokinesis

Nonspecific stimulation of cell movement by a chemical substance, without directional movement toward or away from a concentration gradient.

Chemonucleolysis

An intradiscal treatment intended to dissolve part of a herniated intervertebral disk, historically using an enzyme preparation.

Alternate Names:

Enzymatic disk dissolution; chemonucleolysis procedure.

Chemoprevention

The use of drugs or natural agents to prevent, delay, or reverse cancer development, e.g., tamoxifen or raloxifene in high-risk breast cancer, HPV and hepatitis B vaccination, and aspirin in colorectal cancer.

Chemoradiation

The concurrent combination of chemotherapy and radiation therapy, used to treat locoregionally advanced cancers such as anal, cervical, head and neck, esophageal, and limited-stage small cell lung cancer.

Chemoradiotherapy

Treatment that combines chemotherapy with radiation therapy, often to improve cancer control.

Chemoreceptor

A sensory receptor that responds to changes in blood

or CSF chemistry, involved in respiratory and cardiovascular regulation.

Chemoreceptor Cell

A sensory cell that detects chemical changes such as oxygen, carbon dioxide, pH, taste molecules, or odorants and converts them into neural or physiologic signals.

Chemoreceptor Trigger Zone

A region near the floor of the fourth ventricle that detects emetic stimuli in blood and cerebrospinal fluid and helps initiate vomiting.

Chemoreceptors

Chemoreceptors detect chemical changes such as oxygen, carbon dioxide, pH, or odor molecules and trigger neural or physiologic responses.

Chemoresistance

Reduced or absent responsiveness of cells, especially tumor cells, to a chemical treatment that would ordinarily inhibit or destroy them.

Chemosensitivity

The degree to which cells or a tumor respond to a chemical treatment. Testing may be used experimentally or clinically to help characterize treatment response.

Chemosensitivity Assay

A laboratory test exposing cells to medicines to assess response; results are mainly investigational and may not predict treatment benefit.

Chemosensitization

Increasing a tumor's or pathogen's susceptibility to a treatment by combining it with another agent that improves drug delivery, blocks resistance, or enhances cellular damage.

Chemosis

Chemosis is swelling of the conjunctiva, the membrane over the eye surface, caused by allergy, infection, irritation, trauma, or systemic fluid imbalance.

Chemosurgery

Chemosurgery uses a chemical agent to destroy selected abnormal tissue, historically including chemical treatment of skin lesions; modern use is specialized.

Chemotactic Factors

Chemical signals that attract or repel cells, especially leukocytes, by creating directional concentration gradients during inflammation, infection, development, or repair.

Chemotaxis

Directed movement of a cell or organism toward or away from a chemical stimulus. It is important in immune-cell recruitment and inflammatory responses.

Chemotherapy

Treatment using medicines that kill cancer cells, inhibit their growth, or prevent replication. It may be given alone or with surgery, radiation, immunotherapy, or targeted therapy; the drugs, schedule, benefits, and adverse effects depend on the cancer and treatment goal.

Chemotherapy Extravasation

Leakage of an intravenous anticancer drug into surrounding tissue. Vesicant extravasation can cause severe local injury, so infusion must stop immediately while the catheter is managed and drug-specific antidote and surgical guidance are obtained.

Chemotherapy Regimen

A specific combination of anticancer drugs, dosing schedule, and treatment cycles designed for a particular cancer type and stage. Regimens are standardized by evidence-based protocols and may be modified based on response, toxicity, and patient factors.

Chemotherapy Treatment

Chemotherapy uses medicines that damage or inhibit rapidly dividing cancer cells. It may be given before or after local treatment, for advanced disease, or for symptom control; effects and monitoring

depend on the drugs, cancer, treatment goal, and organ function.

Cherry Angioma

A common benign proliferation of small dermal blood vessels that appears as a bright-red to purple papule, usually on the trunk or limbs. Lesions may bleed after trauma but generally require no treatment unless diagnosis is uncertain or removal is desired.

Cherry Eye

Prolapse of the gland of the third eyelid, producing a rounded red mass at the inner corner of the eye. The term is used especially in veterinary ophthalmology.

Cherry-Red Spot

A striking fundoscopic finding in which the macula appears bright red against a pale, ischemic, or lipid-laden surrounding retina. Mechanism: the fovea centralis lacks inner retinal layers and receives blood supply from the underlying choroid, preserving its natural red color while adjacent edematous retina turns opaque white.

Cherubism

A rare inherited disorder characterized by bilateral enlargement of the jaws from expansile giant-cell lesions, usually beginning in childhood and often regressing after puberty.

Chest

The thorax, the region between the neck and abdomen containing the heart, lungs, major vessels, and other structures protected by the ribs and sternum.

Chest CT

Computed tomography of the chest using x-rays and computer reconstruction to produce cross-sectional images of the lungs, heart, blood vessels, mediastinum, pleura, and chest wall.

Chest MRI

Magnetic resonance imaging of the chest using a

magnetic field and radio waves to evaluate soft tissues, blood vessels, heart structures, mediastinum, chest wall, and selected lung or pleural abnormalities.

Chest Pain

A cardinal symptom with cardiac and non-cardiac causes requiring urgent distinction. Cardiac: acute coronary syndrome (angina, MI), pericarditis, aortic dissection, myocarditis. Pulmonary: pulmonary embolism, pneumothorax, pneumonia, pleurisy. GI: GERD, esophageal spasm, peptic ulcer. Musculoskeletal and psychogenic causes also common.

Alternate Names:

Pain, Chest

Chest Pain On The Left Side Above A Female Breast

Left-sided chest or breast-area pain can arise from muscle or rib injury, breast tissue, reflux, lung disease, anxiety, or heart disease. Pressure, exertional pain, shortness of breath, sweating, faintness, or pain spreading to the arm or jaw requires emergency assessment.

Chest Physiotherapy

Techniques to mobilize and clear secretions from the lungs, including postural drainage, percussion, vibration, and huffing. It is used in bronchiectasis, CF, and post-operative patients. Positive expiratory pressure devices and oscillating PEP devices are also used.

Chest Tightness

A sensation of pressure, constriction, or heaviness in the chest that may arise from heart, lung, gastrointestinal, musculoskeletal, or anxiety-related causes.

Chest Tube

A flexible tube inserted into the pleural space to remove air, blood, or fluid and help the lung re-expand.

Chest Tube Insertion

Placement of a thoracostomy tube in the safe triangle (latissimus dorsi, pectoralis major, nipple level)

to drain air, fluid, or blood from the pleural space. Connected to underwater seal drainage.

Chest Tube Thoracostomy

Insertion of a tube into the pleural space to drain air, fluid, or blood. Indications: pneumothorax (large, tension, or on mechanical ventilation), hemothorax, pleural effusion (empyema, malignant, transudative), and postoperative (cardiac, thoracic surgery).

Chest Wall

The musculoskeletal boundary of the thoracic cavity, formed by the skin, fascia, muscles, ribs, costal cartilages, sternum, thoracic vertebrae, and associated neurovascular structures. It protects the thoracic organs and participates in breathing through coordinated movement of the ribs and respiratory muscles.

Chest Wall Deformity

Abnormalities of the chest wall that can affect lung function. Pectus excavatum (sunken sternum) may compress the heart and lungs. Pectus carinatum (protruding sternum) is usually cosmetic. Scoliosis can cause restrictive lung disease. These may require surgical correction if severe.

Chest Wall Pain

Alternate index label; see *Costochondritis*.

Chest X-Ray

Initial imaging for respiratory symptoms. PA and lateral views. Evaluates pneumonia, pneumothorax, pleural effusion, heart size, mediastinal masses. Limited sensitivity for early disease.

Chewing

Mastication of food. Teeth, muscles, joints coordinated. Difficulties: dental pain, missing teeth, TMJ disorder.

Cheyne-Stokes Respiration

Crescendo-decrescendo breathing pattern with periods of apnea, seen in heart failure and stroke. It is a form of central sleep apnea.

CHF

Congestive heart failure, a syndrome in which the heart cannot pump or fill adequately, causing symptoms such as breathlessness, fatigue, and fluid retention.

Chiari I Malformation

Downward displacement of the cerebellar tonsils through the foramen magnum, which may cause headache, neck pain, balance problems, or no symptoms.

Chiari Malformation

Chiari malformations are structural abnormalities of the posterior fossa and cerebellum characterized by downward displacement of the cerebellar tonsils through the foramen magnum. Type I is the most common variant, defined as cerebellar tonsillar ectopia of five millimeters or more below the foramen magnum.

Chiari-Frommel Syndrome

Persistent postpartum secretion of milk with absence of menstruation, associated with hyperprolactinemia after pregnancy. It is a form of galactorrhea-amenorrhea syndrome.

Chick Embryo

The developing chicken embryo, widely used as an experimental model for embryology, organ development, virology, toxicology, and tissue growth.

Chicken Soup And Sickness

A supportive-care topic concerning the use of warm fluids and nutrition during illness; chicken soup may ease hydration or congestion but does not replace diagnosis or treatment.

Chickenpox

Chickenpox is a contagious varicella-zoster virus infection characterized by an itchy vesicular rash; complications are more likely in adults, pregnancy, immunosuppression, and some infants. In different stages (macules, papules, vesicles, crusts). Highly contagious via respiratory droplets and contact with lesions. Incubation 10-21 days. Complications in-

clude secondary bacterial infection, pneumonia, encephalitis, and Reye syndrome.

Chickenpox (Varicella)

Alternate index label for Varicella.

Chigger Bites

Itchy inflammatory skin reactions caused by the larvae of trombiculid mites. Small red bumps or blisters often occur around areas where clothing fits tightly and may itch intensely for several days. Washing the skin and clothing, topical antipruritics, and avoiding scratching can help; persistent or infected lesions require assessment.

Chiggers

Larval harvest mites whose bite causes intensely itchy grouped red papules or welts; symptoms result from skin irritation, not from the mite burrowing into human tissue.

Chiggers (Bites)

Alternate index label for Bites.

Chignon

A temporary swelling (edema) on the baby's scalp caused by the vacuum cap used during ventouse (vacuum) assisted delivery. The chignon forms as the vacuum creates negative pressure pulling the scalp tissue into the cup. It settles within 24-48 hours after birth without any treatment.

Chikungunya

A mosquito-borne viral disease caused by chikungunya virus, usually producing abrupt fever and severe, often symmetric joint pain. Headache, muscle pain, joint swelling, nausea, fatigue, and rash may occur; joint symptoms can persist for months or years.

Alternate Names:

Chikungunya fever; chikungunya virus disease; CHIKV infection

Chikungunya Virus

An alphavirus transmitted primarily by infected

female *Aedes aegypti* and *Aedes albopictus* mosquitoes. Infection can cause abrupt fever, severe polyarthralgia, joint swelling, rash, and prolonged inflammatory joint symptoms.

Chilblain

An inflammatory skin lesion caused by an abnormal response to cold and damp conditions. It produces itchy, red or violaceous swollen areas, usually on fingers, toes, ears, or nose.

Chilblains

Painful, itchy, swollen skin lesions caused by an abnormal response to cold and damp conditions.

Alternate Names:

Pernio

Chilblains (Pernio)

Itchy, painful, red or purplish inflammatory lesions caused by an abnormal response of small blood vessels to cold and subsequent rewarming.

Alternate Names:

Pernio

Child Abuse

Physical, sexual, or emotional harm or neglect of a child by a caregiver. Recognized by inconsistent histories, injury patterns inconsistent with mechanism, and behavioral indicators. Types: physical abuse, sexual abuse, neglect, and emotional abuse. Mandatory reporting requirements for healthcare providers.

Alternate Names:

Child Maltreatment

Alternate Names:

Child Adoption

Child adoption is the legal process by which parental rights and caregiving responsibility are transferred to adoptive parents; medical care includes records, consent, and family history.

Child Behavior

The observable actions, responses, and conduct of

children, influenced by developmental stage, temperament, environment, and neurobiological factors. Behavioral assessment helps identify developmental disorders, emotional difficulties, and parenting needs.

Child Development

Progressive physical, cognitive, language, emotional, and social changes from infancy through adolescence, assessed against developmental milestones and individual variation.

Child Health

The physical, developmental, mental, and social well-being of children, including prevention, nutrition, immunization, safety, and treatment of illness.

Child Health Services

Child health services provide preventive care, immunization, growth and developmental monitoring, screening, treatment, and support for children and families.

Child Neglect

The failure of a caregiver to provide adequate food, shelter, clothing, supervision, education, or medical care. The most common form of child maltreatment. Types: physical neglect, educational neglect, emotional neglect, and medical neglect. May result in failure to thrive, developmental delay, and chronic illness.

Child Neglect And Emotional Abuse

Failure to provide a child's basic needs or repeated behavior that harms emotional development, safety, or wellbeing; suspected abuse requires safeguarding assessment and appropriate reporting under local law.

Child Physical Abuse

Intentional physical injury or harm inflicted on a child by a caregiver or another person, including hitting, burning, shaking, or fractures; suspected abuse requires immediate safeguarding assessment.

Child Poisoning

Exposure of a child to a medicine, chemical, or other substance that may cause harm. Suspected poisoning requires prompt expert assessment.

Child Psychiatry

The medical specialty concerned with assessment and treatment of emotional, behavioral, developmental, and psychiatric disorders in children and adolescents.

Child Psychology

The study of children's cognitive, emotional, social, and behavioral development and the effects of relationships, learning, biology, and environment.

Child-Pugh Classification

Clinical scoring system assessing prognosis of chronic liver disease by evaluating five parameters: total bilirubin, serum albumin, international normalized ratio, ascites, and hepatic encephalopathy. Each parameter is scored one to three points.

Childbed Fever

Puerperal fever, usually infection of the reproductive tract after childbirth or miscarriage, often involving postpartum endometritis and requiring prompt evaluation and treatment.

Childbirth

The process of giving birth, encompassing three stages: (1) First stage: from the onset of regular uterine contractions (cervical effacement and dilation) to full dilation (10 cm). See Parturition (Childbirth).

Childhood

The developmental period between infancy and adolescence, during which physical growth, brain development, learning, and social skills progress.

Childhood Absence Epilepsy

An epilepsy syndrome beginning in childhood with frequent brief absence seizures, characterized by sudden impairment of awareness and a typical gen-

eralized spike-and-wave EEG pattern.

Childhood Angiosarcoma

A rare malignant tumor arising from blood-vessel or lymphatic-endothelial cells in a child; diagnosis and treatment require specialist pediatric oncology care.

Childhood Apraxia Of Speech

A speech-sound disorder in which a child has difficulty planning and coordinating the movements needed for clear speech.

Alternate Names:

CAS

Childhood Asthma

Asthma beginning during childhood, involving variable airway narrowing and inflammation that cause wheezing, cough, chest tightness, or shortness of breath.

Childhood Ependymoma

A tumor arising from ependymal cells lining the brain ventricles or spinal canal in children; symptoms depend on location and obstruction of cerebrospinal-fluid flow.

Childhood Fibrosarcoma

A malignant tumor of fibroblastic connective tissue in children, usually presenting as a growing mass; treatment depends on location, extent, and tumor biology.

Childhood Ganglioglioma

A usually slow-growing mixed neuronal and glial brain tumor in children or young adults, often presenting with long-standing focal seizures.

Childhood Leiomyosarcoma

A rare malignant tumor of smooth muscle in a child; it may arise in soft tissue or an organ and requires biopsy, staging, and specialist treatment.

Childhood Leukemia

Cancer of blood-forming cells in a child, most of-

ten acute lymphoblastic or acute myeloid leukemia; symptoms may include pallor, bruising, fever, bone pain, or recurrent infection.

Childhood Liposarcoma

A rare malignant tumor derived from fat-forming tissue in a child; diagnosis requires imaging and biopsy, and treatment is individualized by a sarcoma team.

Childhood Obesity

Alternate index label; see *Pediatric Obesity*.

Childhood Osteosarcoma

A malignant bone tumor that commonly develops near the ends of long bones during adolescence; pain, swelling, or a fracture may occur and treatment combines chemotherapy and surgery.

Childhood Rhabdomyosarcoma

A malignant tumor showing skeletal-muscle differentiation, occurring mainly in children and adolescents; it may arise in the head and neck, urinary or reproductive tract, or limbs.

Childhood Schizophrenia

A rare psychotic disorder beginning in childhood, involving disturbances in thinking, perception, behavior, and emotional functioning.

Alternate Names:

Juvenile Schizophrenia Schizophrenia, Adolescent Schizophrenia, Childhood Schizophrenia, Juvenile

Children

Humans in the developmental period between infancy and adolescence; medical assessment must account for age-specific anatomy, physiology, growth, communication, and consent.

Children And Grief

The emotional and behavioral response of children to death or major loss, which varies by developmental stage and may include sadness, regression, anger, worry, or physical symptoms.

Children'S Cancer Centers

Specialized services that coordinate diagnosis, treatment, rehabilitation, psychosocial support, survivorship care, and research for children and adolescents with cancer.

Alternate Names:

Childrens Health (Children's Health)

Chill

A chill is a sensation of cold, often with shivering, that can accompany fever, infection, exposure, anxiety, or other illness.

Chills

A sensation of cold with involuntary shivering, commonly accompanying fever or infection; chills without measured fever may also occur with anxiety, hypoglycaemia, or environmental cold.

Chilomastix Mesnili

A nonpathogenic intestinal flagellate protozoan that may be found in stool and indicates fecal exposure but usually does not require treatment.

Chimera

An organism or tissue containing genetically distinct cell populations derived from different zygotes or embryos, or a laboratory construct combining genetic material from different sources.

Chimerism

The presence in one individual of cell populations derived from genetically different zygotes or sources. It may occur naturally, after transplantation, or through exchange of cells between a pregnant person and fetus.

Chin

The lower facial prominence formed mainly by the mandible and overlying soft tissue, contributing to facial contour, speech, chewing, and airway anatomy.

Chin Augmentation

A cosmetic procedure that increases chin projection

with an implant, injectable filler, or bone surgery to change facial contour; risks include infection, asymmetry, nerve injury, and implant problems.

Chinese Restaurant Syndrome

An older, controversial term for short-lived symptoms such as headache, flushing, or palpitations attributed to eating restaurant food; evidence does not show that monosodium glutamate causes these symptoms in most people at usual dietary exposures.

Chionablepsia

Temporary visual impairment caused by exposure to intense reflected light from snow or ice, commonly called snow blindness. It is usually related to ultraviolet injury of the cornea.

Alternate Names:

snow blindness.

Chironomidae

The nonbiting midge family of insects; their larvae and body parts can act as environmental allergens and rarely trigger respiratory or skin reactions.

Chiropody

Chiropody is care of the feet, including assessment and treatment of skin, nail, pressure, and biomechanical problems; it is also called podiatry.

Chiropractic

A healthcare practice centered mainly on manual treatment of musculoskeletal disorders, especially spinal manipulation; benefits and risks vary by condition and technique.

Chiropractic Care For Back Pain

Manual therapy, primarily spinal manipulation and mobilization, provided for acute and chronic low back pain when indicated. Evidence supports its use for acute low back pain in particular, though it is not appropriate for all causes of back pain.

Chiropractor

A healthcare practitioner who provides manual or

manipulative treatment, especially for musculoskeletal conditions, according to the practitioner's training and local regulation.

Alternate Names:

Chiropractic practitioner; chiropractic clinician.

Chitin

A structural polysaccharide in fungal cell walls and arthropod exoskeletons.

Chitosan

A biodegradable polysaccharide derived from chitin, studied and used in wound dressings, drug delivery, hemostatic materials, and some nutritional products.

Chlamydia

A genus of obligate intracellular bacteria; *Chlamydia trachomatis* commonly causes sexually transmitted urethritis, cervicitis, pelvic inflammatory disease, and neonatal conjunctivitis or pneumonia.

Chlamydia Infections In Women

Infections in women caused by *Chlamydia trachomatis*, commonly involving the cervix or urethra and often producing no symptoms; untreated infection can cause pelvic inflammatory disease, infertility, or ectopic pregnancy.

Chlamydia Pneumoniae Infection

A respiratory infection caused by *Chlamydia pneumoniae*, formerly called *Chlamydophila pneumoniae*. It commonly causes gradual-onset atypical pneumonia, bronchitis, or upper-respiratory symptoms.

Chlamydia Psittaci

A bacterium that causes psittacosis, a zoonotic infection usually acquired by inhaling organisms from infected birds.

Chlamydia Trachomatis

A bacterium that commonly causes sexually transmitted infection of the urogenital tract and can also cause conjunctivitis or pneumonia.

Chlamydia Trachomatis Infection

A bacterial infection transmitted mainly through sexual contact. It may cause urethritis, cervicitis, pelvic inflammatory disease, proctitis, or no symptoms.

Chlamydiaceae

Chlamydiaceae is a bacterial taxon; disease risk and clinical significance depend on the strain, specimen source, host factors, and evidence of infection rather than colonization.

Chlamydial Infections - Male

Infections in men caused by *Chlamydia trachomatis*, most commonly urethritis and sometimes epididymitis or proctitis; many infections are asymptomatic and diagnosis is usually by nucleic acid amplification testing.

Chlamydial Pneumonia

Pneumonia caused by *Chlamydia* species, usually producing a gradual cough, fever, and malaise.

Chlamydiosis

Infection caused by bacteria of the genus *Chlamydia*. Depending on the species and site, it may cause sexually transmitted infection, conjunctivitis, pneumonia, or other disease.

Chloasma

Patchy brown hyperpigmentation, especially of the face, associated with pregnancy, hormones, sun exposure, or other causes. It is commonly called melasma.

Alternate Names:

melasma.

Chloasma (Melasma)

Alternate index label for Melasma.

Chloracne

An acne-like eruption caused by exposure to certain halogenated aromatic compounds, such as dioxins and related industrial chemicals.

Chloral Hydrate

A sedative-hypnotic compound with limited modern use because of toxicity, dependence, respiratory depression, and cardiac or gastrointestinal adverse effects.

Chlorambucil

A medicine or chemical compound identified by chlorambucil. Its uses, effects, and risks depend on the specific substance and clinical context.

Chloramphenicol

A broad-spectrum antibiotic with activity against gram-positive and gram-negative organisms, as well as Rickettsia. Chloramphenicol inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. It is indicated for serious infections when other agents are not appropriate, including bacterial meningitis and typhoid fever.

Chloranemia

Anemia in which the red blood cells are abnormally pale, usually because of reduced hemoglobin content, as in iron deficiency.

Chlordane

A persistent organochlorine insecticide formerly used for termite control; exposure can affect the nervous system and liver and has environmental persistence.

Chlordecone

A persistent organochlorine pesticide that can cause neurologic, reproductive, and liver effects after significant exposure and remains in the environment for long periods.

Chlordiazepoxide

Chlordiazepoxide is a defined chemical or biologic substance involved in chlordiazepoxide and studied in physiology, pharmacology, laboratory medicine, or treatment.

Chlordiazepoxide Overdose

Overdose of chlordiazepoxide, a benzodiazepine,

that can cause drowsiness, impaired coordination, respiratory depression, and coma, especially with other sedatives.

Chlorethoxyfos

An organophosphate insecticide that can inhibit acetylcholinesterase and cause cholinergic poisoning after substantial exposure.

Chlorfenvinphos

An organophosphate insecticide that can cause cholinergic toxicity by inhibiting acetylcholinesterase; use is restricted or prohibited in many places.

Chlorhexidine

A broad-spectrum antiseptic used for skin, mucosal, and oral preparation; it can cause irritation, allergy, or rare severe anaphylaxis.

Chloride

The principal negatively charged electrolyte in extracellular fluid, important for fluid balance, electrical neutrality, and gastric acid formation.

Chloride - Urine Test

A test that measures chloride excretion in urine; interpreted with serum electrolytes and clinical findings to evaluate fluid balance, acid-base disorders, and causes of abnormal chloride loss or retention.

Chloride Channel

A medicine or chemical compound identified by chloride channel. Its uses, effects, and risks depend on the specific substance and clinical context.

Chloride In Diet

A medicine or chemical compound identified by chloride in diet. Its uses, effects, and risks depend on the specific substance and clinical context.

Chloride Ion

The negatively charged form of chlorine and a major extracellular electrolyte that helps maintain fluid balance, osmotic pressure, acid-base status, and

electrical neutrality.

Chloride Test - Blood

A blood test measuring serum chloride, an electrolyte that helps regulate extracellular fluid volume and acid-base balance; results are interpreted with sodium, bicarbonate, kidney function, and the clinical context.

Chlorinated Lime

Chlorinated lime, or bleaching powder, releases chlorine and is used as a disinfectant; concentrated exposure can burn skin, eyes, and airways.

Chlorinated Lime Poisoning

Exposure to chlorinated lime, a hypochlorite-containing corrosive that can injure the skin, eyes, airways, or gastrointestinal tract.

Chlorine

A reactive halogen element and disinfectant; inhalation of chlorine gas can injure the eyes and airways and cause chemical pneumonitis or pulmonary edema.

Chlorine Poisoning

Inhalation or contact exposure to chlorine gas, which irritates the eyes and airways and can cause cough, bronchospasm, chemical pneumonitis, or pulmonary edema.

Chlormadinone Acetate

A synthetic progestogen used in some hormonal medicines; effects and risks include changes in bleeding, mood, metabolism, and thrombotic risk depending on the formulation.

Chlormerodrin

A historical radioactive mercury compound once used in diagnostic or research settings; it is not a routine modern medicine and mercury exposure is toxic.

Chlormethine

Chlormethine, or mechlorethamine, is an alkylating

anticancer drug used systemically or topically for selected cancers and cutaneous T-cell lymphoma.

Chlorodeoxyadenosine

Another name for cladribine, a purine-analog medicine used for selected blood cancers and multiple sclerosis; prolonged lymphocyte suppression and infection risk are important considerations.

Chloroform

A volatile solvent and environmental contaminant that can irritate the lungs and nervous system and damage the liver or kidneys after significant exposure.

Chlorogenic Acid

Chlorogenic acid is a defined chemical or biologic substance involved in chlorogenic acid and studied in physiology, pharmacology, laboratory medicine, or treatment.

Chloroleukemia

A rare form of myeloid leukemia associated with greenish tumor masses composed of myeloid cells, also called granulocytic sarcoma or chloroma.

Alternate Names:

granulocytic sarcoma or chloroma.

Chloroma

A greenish extramedullary mass of immature myeloid cells, usually associated with acute myeloid leukemia. It is also called granulocytic sarcoma or myeloid sarcoma.

Alternate Names:

granulocytic sarcoma or myeloid sarcoma.

Chlorophyll

The green pigment in plants and algae that absorbs light for photosynthesis.

Chlorophyll Poisoning

Ingestion of chlorophyll-containing products, which usually causes mild gastrointestinal effects; product ingredients and dose determine clinical risk.

Chlorophyllin

A water-soluble, copper-containing derivative of chlorophyll used in some products and studied for effects such as odor control; clinical uses depend on evidence and context.

Alternate Names:

Chlorophyllin copper complex; sodium copper chlorophyllin.

Chlorophyta

The group of green algae; some species produce toxins or contribute to harmful algal blooms, while others are used as food or in research.

Chloropicrin

A pungent soil fumigant and irritant that can injure the eyes, skin, and respiratory tract after exposure.

Chloroplast

A plant and algal cell organelle that performs photosynthesis and contains its own DNA; it is not found in human cells.

Chloroplast DNA

The genetic material inside chloroplasts, inherited mainly through the maternal line in many plants and used in studies of plant evolution and disease.

Chloroprene

An industrial chemical used to make chloroprene rubber; repeated or high exposure can irritate tissues and has carcinogenic concern.

Chloropsia

A visual disturbance in which objects appear green or have a greenish tint. It can occur with retinal disease, medication or toxin exposure, or other visual-pathway disorders.

Chloroquine

An antimalarial medicine used for susceptible Plasmodium species and selected autoimmune conditions; retinal toxicity, cardiotoxicity, and drug interactions require attention.

Chlorothiazide

A medicine or chemical compound identified by chlorothiazide. Its uses, effects, and risks depend on the specific substance and clinical context.

Chlorotoxin

A peptide from scorpion venom that binds some tumor and ion-channel targets; it is mainly studied experimentally for brain-tumor imaging or treatment.

Chlorotrianisene

A synthetic estrogen formerly used for menopausal symptoms; it is rarely used now because of estrogen-related risks and safer alternatives.

Chlorpheniramine

A first-generation antihistamine used for allergy symptoms; drowsiness, dry mouth, blurred vision, urinary retention, and other anticholinergic effects may occur.

Chlorphentermine

An older stimulant appetite suppressant related to amphetamines; it is not routinely used because of cardiovascular, psychiatric, and dependence risks.

Chlorpromazine

A first-generation antipsychotic and phenothiazine used for schizophrenia, severe nausea, and selected other conditions; adverse effects include sedation, hypotension, movement disorders, and QT prolongation.

Chlorpromazine Overdose

Overdose of chlorpromazine that may cause sedation, hypotension, anticholinergic effects, seizures, cardiac conduction abnormalities, or neuroleptic malignant syndrome.

Chlorpropamide

A first-generation sulfonylurea that lowers blood glucose by stimulating pancreatic insulin release; prolonged hypoglycemia and hyponatremia are important risks.

Chlorpyrifos

An organophosphate insecticide that inhibits acetylcholinesterase; significant exposure can cause cholinergic toxicity with secretions, bronchospasm, bradycardia, weakness, or seizures.

Chlortalidone

Chlortalidone, also called chlorthalidone, is a long-acting thiazide-like diuretic used mainly for hypertension and fluid retention.

Chlorthalidone

A long-acting thiazide-like diuretic used for hypertension and edema; it can cause hyponatremia, hypokalemia, hyperuricemia, and kidney-related electrolyte changes.

Chlorzoxazone

A centrally acting muscle relaxant used for painful muscle spasm in some countries; drowsiness and rare serious liver injury are important risks.

CHO Cell

A Chinese hamster ovary cell line widely used to produce recombinant proteins, antibodies, vaccines, and other biologic medicines.

Choanal Atresia

A congenital obstruction of the posterior nasal passage (choana) due to failure of the bucconasal membrane to rupture during embryogenesis. Bilateral choanal atresia presents as a respiratory emergency in the newborn, who is an obligate nasal breather — cyclic cyanosis that improves with crying.

Chocolate Cyst

A chocolate cyst is an ovarian endometrioma filled with old blood from endometriosis, which can cause pelvic pain, infertility, or an adnexal mass.

Choice Behavior

The process of selecting among alternatives, studied in psychology and neuroscience to understand decision-making, impulsivity, and reward processing. Abnormal choice behavior may be relevant to addic-

tion, eating disorders, and psychiatric conditions.

Choke

Airway blockage by food or another object that prevents normal airflow and can rapidly cause inability to speak, cyanosis, loss of consciousness, or death.

Choking

Airway obstruction caused by an object lodged in the pharynx or larynx, producing impaired ventilation. Complete obstruction can cause hypoxic brain injury and death within minutes; treatment depends on the patient's age, consciousness, and effectiveness of coughing.

Choking - Adult Or Child Over 1 Year

First aid for severe airway obstruction in a person older than one year, using back blows and abdominal thrusts when appropriate and CPR if the person becomes unresponsive.

Choking - Infant Under 1 Year

First aid for severe airway obstruction in an infant, using alternating back slaps and chest thrusts and avoiding blind finger sweeps; unresponsiveness requires infant CPR.

Choking - Unconscious Adult Or Child Over 1 Year

Resuscitation for an unresponsive person with suspected airway obstruction, combining emergency activation, airway inspection, CPR, and removal of a visible object without blind sweeping.

Cholagogue

A substance that promotes emptying of the gallbladder into the intestine.

Alternate Names:

Gallbladder stimulant; biliary evacuant.

Cholangitis

Inflammation of the bile ducts. It may be caused by infection, obstruction, autoimmune disease, or other biliary disorders.

Cholangiocarcinoma

Cholangiocarcinoma is cancer arising from bile-duct lining cells and may cause jaundice, itching, abdominal symptoms, or bile-flow obstruction.

Alternate Names:

Bile Duct Cancer (Cholangiocarcinoma)

Cholangiocarcinoma (Bile Duct Cancer (Cholangiocarcinoma))

Alternate index label for Bile Duct Cancer (Cholangiocarcinoma).

Cholangiocarcinoma (Bile Duct Cancer)

Cancer arising in the bile ducts, which may cause jaundice, itching, abdominal discomfort, weight loss, or impaired bile flow.

Cholangiography

Contrast imaging of bile ducts. Intraoperative, percutaneous, or ERCP approach. Evaluates biliary obstruction, stones, strictures.

Cholangiohepatitis

Inflammation involving both the bile ducts and liver, often associated with ascending bacterial infection, biliary obstruction, parasites, or recurrent cholangitis. It may cause fever, right-upper-quadrant pain, jaundice, and cholestatic liver-test abnormalities.

Cholangitis

Cholangitis is inflammation or infection of the bile ducts, often from obstruction by a stone or stricture, and can rapidly cause sepsis.

Cholangitis, Primary Sclerosing

Alternate index label; see *Primary sclerosing cholangitis*.

Cholate

The ionized form of cholic acid, a primary bile acid that helps emulsify dietary fat and is conjugated and secreted by the liver.

Cholecalciferol

Vitamin D3, produced in skin by ultraviolet light

and obtained from food or supplements, then converted to active metabolites that regulate calcium and phosphate.

Cholecyst

The gallbladder, a small sac beneath the liver that stores and concentrates bile.

Cholecystectomy

Surgical removal of the gallbladder, usually by a laparoscopic approach when appropriate. It may be performed for symptomatic gallstones, acute cholecystitis, selected gallstone pancreatitis, or other gallbladder disease.

Cholecystitis

Inflammation of the gallbladder, most commonly caused by obstruction of the cystic duct by a gallstone, which occurs in approximately 95 percent of cases. The obstruction leads to bile stasis, increased intraluminal pressure, and ischemia of the gallbladder wall, which may progress to bacterial infection.

Alternate Names:

Inflamed Gallbladder

Cholecystography

Radiographic examination of the gallbladder, usually after administration of a contrast substance, to assess its filling, concentration, and emptying.

Cholecystokinin

CCK is a peptide hormone produced by I-cells in the duodenal and jejunal mucosa in response to fatty acids (long-chain fatty acids are the most potent stimuli) and amino acids in the chyme.

Cholecystoses

Non-inflammatory disorders causing deposits, wall thickening, or other structural changes in the gallbladder, including cholesterosis and adenomyomatosis. Many are incidental, but symptoms and imaging findings determine whether further evaluation is needed.

Choledochal Cyst

Congenital dilatation of the biliary tree classified into five types according to the Todani classification. Type I involving dilation of the common bile duct is the most common. Presents with the classic triad of abdominal pain, jaundice, and palpable mass though all three are present in less than twenty percent of cases.

Choledochal Cysts

Choledochal cysts are congenital or acquired dilations of the bile ducts that can cause abdominal pain, jaundice, pancreatitis, cholangitis, or stones. Imaging establishes the anatomy, and surgical removal is often recommended because of recurrent complications and cancer risk.

Alternate Names:

Cysts Choledochal (Choledochal Cysts)

Choledocholithiasis

A gallstone within the common bile duct, which may cause biliary pain, jaundice, cholangitis, or pancreatitis and is usually evaluated with liver tests and imaging.

Cholelithiasis

The presence of stones in the gallbladder or biliary tract. Stones may contain cholesterol, pigment, or mixed material and may cause no symptoms or lead to biliary pain and complications.

Cholelithiasis (Gallstones)

Alternate index label for Gallstones.

Cholera

An acute intestinal infection caused by toxigenic *Vibrio cholerae*, producing profuse watery diarrhoea and potentially life-threatening dehydration; prompt oral or intravenous rehydration is central treatment.

Choleresis

Choleresis is production and secretion of bile by liver cells.

Choleretic

A substance that promotes production or secretion of bile by the liver.

Alternate Names:

Bile-secretory agent; bile-flow stimulant.

Cholestasis

Reduced formation or flow of bile, causing conjugated hyperbilirubinaemia, itching, dark urine, pale stools, and raised alkaline phosphatase; obstruction and intrahepatic causes must be distinguished.

Cholestasis Of Pregnancy

A liver disorder of pregnancy in which impaired bile flow causes itching and elevated bile acids.

Alternate Names:

Obstetric Cholestasis

Cholesteatoma

A keratin-filled sac of abnormal squamous epithelium in the middle ear or mastoid that can erode bone and cause chronic discharge, hearing loss, or intracranial complications.

Cholesterol

A lipid required for cell membranes and synthesis of steroid hormones and bile acids. Excess atherogenic cholesterol can contribute to arterial plaque.

Alternate Names:

Serum cholesterol; blood cholesterol.

Blood Cholesterol

Cholesterol carried in the blood. The body needs it to build cell membranes and make hormones and bile acids. Too much LDL cholesterol can form fatty buildup in the arteries and increase the risk of heart disease and stroke. Blood cholesterol is usually checked as part of a lipid profile.

Cholesterol And Lifestyle

A medicine or chemical compound identified by cholesterol and lifestyle. Its uses, effects, and risks depend on the specific substance and clinical context.

Cholesterol And Lipids

Fatty substances in the blood, including LDL cholesterol, HDL cholesterol, and triglycerides, that play critical roles in cellular function and energy metabolism. Abnormal lipid levels, particularly high LDL and triglycerides with low HDL, promote atherosclerosis and increase the risk of heart attack and stroke.

Cholesterol Crystal Embolization

Atheroembolic disease caused by cholesterol crystal fragments dislodging from proximal atherosclerotic plaques into distal small arteries, often after vascular procedures. Presents with livedo reticularis, blue-toe syndrome, renal failure, and eosinophilia. Diagnosis by skin, kidney, or muscle biopsy.

Cholesterol Crystals

Needle-shaped, cleft-like spaces seen in histological sections representing dissolved cholesterol esters, commonly found in atherosclerotic plaques, atheroembolic disease, and cholesterol-rich effusions. In atheroembolic disease, cholesterol crystals embolize from ruptured plaques to distal organs causing tissue ischemia.

Cholesterol Emboli

Cholesterol crystals released from an atherosclerotic plaque that lodge in small arteries and cause ischemia or inflammation in downstream tissues.

Cholesterol Embolism

Blockage of small arteries by cholesterol crystals, often after vascular instrumentation or plaque disruption.

Cholesterol Embolization

Release and downstream lodging of cholesterol crystals from an atherosclerotic arterial plaque.

Cholesterol Lowering (Cholesterol Management)

Alternate index label for Cholesterol Management.

Cholesterol Management

Cholesterol management combines cardiovascular-risk assessment with dietary pattern, physical activity, tobacco avoidance, weight care, and medicines when indicated. Statins and other agents lower atherogenic lipoproteins, with intensity chosen according to prior cardiovascular disease and overall risk.

Alternate Names:

Cholesterol Lowering (Cholesterol Management)

Cholesterol Synthesis

Occurs in all nucleated cells (primarily liver and intestine, approximately 1 g/day production), using acetyl-CoA as the precursor through approximately 30 enzymatic steps.

Cholesterol Testing And Results

Laboratory assessment of blood lipids, usually including total cholesterol, LDL cholesterol, HDL cholesterol, and triglycerides, used to estimate cardiovascular risk and guide prevention or treatment.

Cholesterolemia

The presence and concentration of cholesterol in the blood. Abnormal elevation is a risk factor for atherosclerotic cardiovascular disease.

Cholesterolosis

Deposition of cholesterol esters in the gallbladder mucosa, producing yellow speckling or a "strawberry gallbladder." It is often incidental but may coexist with gallstones or chronic gallbladder inflammation.

Cholestyramine

Cholestyramine is a bile-acid sequestrant used to lower LDL cholesterol and sometimes to treat bile-acid diarrhea or itching from cholestasis.

Choline

Choline is an essential nutrient used to make cell membranes and acetylcholine and to support liver and nervous-system function.

Cholinergic

Cholinergic means related to acetylcholine or receptors and pathways activated by it, including muscarinic and nicotinic signaling.

Cholinergic Crisis

Excess acetylcholine from organophosphate or carbamate poisoning. SLUDGE syndrome: salivation, lacrimation, urination, defecation, GI upset, emesis. Miosis, bradycardia, bronchospasm, muscle fasciculations. Treatment: atropine, pralidoxime.

Cholinergic Neuron

A nerve cell that releases acetylcholine as its principal neurotransmitter.

Alternate Names:

Acetylcholine-releasing neuron; cholinergic nerve cell.

Cholinergic Toxidrome

A poisoning pattern caused by excessive acetylcholine activity, often producing salivation, tearing, urination, diarrhea, vomiting, bronchial secretions, slow heart rate, muscle fasciculations, weakness, or respiratory failure.

Cholinesterase

Cholinesterase enzymes break down acetylcholine; inhibition increases cholinergic activity and is used therapeutically but can also cause poisoning.

Cholinesterase - Blood

A blood test measuring acetylcholinesterase or butyrylcholinesterase activity, useful in selected pesticide exposures, inherited enzyme variants, and some liver-function assessments.

Chondritis

Inflammation of cartilage, such as the cartilage of the ear, nose, ribs, or larynx. It may result from infection, trauma, or immune-mediated disease.

Chondroblastic Osteosarcoma

A variant of osteosarcoma in which the malignant cells produce a cartilaginous matrix. It typically af-

fects the long bones of adolescents and young adults, with a predilection for the metaphyseal regions.

Chondroblastoma

A chondroblastoma is an uncommon usually benign cartilage-forming bone tumor, often arising near the end of a long bone in adolescents or young adults.

Chondrocalcinosis

Calcification of cartilage visible on X-ray. Associated with CPPD disease. Found in knees, wrists, pubic symphysis. May be incidental or symptomatic.

Chondrocyte

A mature cartilage cell that produces and maintains the cartilage extracellular matrix.

Alternate Names:

Cartilage cell.

Chondrodysplasia

A disorder of cartilage and bone development that causes abnormal skeletal growth and, in some forms, disproportionate short stature.

Chondroitin Sulfate

A sulfated glycosaminoglycan found in cartilage and other connective tissues; it is also marketed in supplements for osteoarthritis, with variable evidence of benefit.

Chondroma

Benign cartilage-forming tumor composed of mature hyaline cartilage, most often intramedullary (enchondroma) or subperiosteal (juxtacortical / periosteal chondroma).

Chondromalacia

Softening and degeneration of cartilage, especially articular cartilage. The term is often used for changes affecting the patella and causing anterior knee pain.

Chondromyxoid Fibroma

A rare benign cartilage-forming bone tumor, usually arising in the metaphysis of a long bone and present-

ing with localized pain or an incidental radiographic lesion.

Chondrosarcoma

The second most common primary malignant bone tumor (after osteosarcoma) and the most common primary malignant bone tumor in adults (peak age 40-70 years).

Chondrosarcoma Of The Breast

An exceptionally rare malignant mesenchymal breast tumor producing cartilage. It is distinguished from metaplastic carcinoma and other sarcomas by histology and is managed with sarcoma-oriented complete excision.

Chopart'S Joint

The transverse tarsal joint, consisting of the talonavicular and calcaneocuboid joints, between the hindfoot and the midfoot.

Chorangioma

A benign vascular tumor of the placenta, usually small and incidental but occasionally associated with fetal anemia, hydrops, growth restriction, or other complications when large.

Chordae Tendineae

Thin tendinous cords connecting the papillary muscles to the free edges of the atrioventricular valve leaflets (mitral and tricuspid). They tether the leaflets and, with papillary muscle contraction during ventricular systole, prevent valve prolapse and regurgitation.

Chordee

Chordee is abnormal downward or other curvature of the penis, commonly associated with hypospadias or scar tissue and sometimes requiring surgical correction.

Chordoma

Slow-growing, locally aggressive malignant bone tumor originating from embryonic remnants of the notochord along the axial skeleton. Three anatomic

locations: (1) sacrococcygeal (50% of cases, most common), (2) sphenoccipital/clival (35%, base of skull), (3) vertebral (cervical thoracolumbar, 15%).

Chorea

Involuntary, irregular, brief, non-repetitive jerky movements flowing randomly from one body part to another ('dance-like'), due to basal ganglia (striatal) dysfunction.

Chorea, Huntington'S

Alternate index label; see *Huntington's disease*.

Choreoathetosis

Choreoathetosis combines irregular, dance-like chorea with slow twisting athetotic movements and may result from basal-ganglia injury or genetic disease.

Chorioamnionitis

Infection of the chorion and amnion, typically due to ascending bacterial infection from the genital tract. Risk factors include prolonged rupture of membranes, multiple vaginal exams, and preterm labor. Presents with maternal fever, uterine tenderness, and fetal tachycardia. Diagnosis is clinical; amniocentesis for confirmation.

Chorioamnionitis (Intraamniotic Infection)

Infection or inflammation of the amniotic fluid, membranes, placenta, or decidua, usually due to ascending polymicrobial infection. Maternal fever, uterine tenderness, fetal tachycardia, and maternal leukocytosis may occur; treatment requires broad-spectrum antibiotics and delivery planning.

Alternate Names:

Intraamniotic Infection

Choriocarcinoma

Choriocarcinoma is an aggressive trophoblastic cancer producing human chorionic gonadotropin, usually arising after pregnancy or a molar pregnancy.

Chorion

The outermost fetal membrane, derived from the trophoblast. The chorion frondosum (villous chorion) develops into the fetal portion of the placenta. The chorion laeve (smooth chorion) is the non-villous portion. Chorionic villi contain fetal blood vessels and are bathed in maternal blood.

Chorion Villus Sampling

Chorion villus sampling removes placental tissue for genetic or chromosomal testing, usually early in pregnancy; it carries a small procedure-related miscarriage risk.

Chorionic Gonadotrophin

Chorionic gonadotrophin, especially hCG, is a placental hormone that maintains early pregnancy and is measured in pregnancy tests and selected tumor evaluations.

Chorionic Villus Sampling

Prenatal diagnostic test at 10-13 weeks gestation, earlier than amniocentesis. Transabdominal or transcervical biopsy of chorionic villi (placental tissue). Indications: advanced maternal age, abnormal first-trimester screen, carrier screening, or family history of genetic disorders. Analyzed for karyotype, FISH, and microarray.

Chorioretinitis

Inflammation involving the choroid and retina, often caused by infection, autoimmune disease, or systemic inflammatory illness. It can produce floaters and impaired vision.

Chorioretinitis (Uveitis)

Alternate index label for Uveitis.

Chorioretinitis Toxoplasma (Toxoplasmosis)

Alternate index label for Toxoplasmosis.

Choroid

The vascular layer between retina and sclera that supplies the outer retina and absorbs stray light.

Choroid Plexus Carcinoma

A malignant tumor arising from the choroid plexus, the tissue that produces cerebrospinal fluid.

Choroid Plexus Cyst

A fluid-filled cyst within the choroid plexus of a fetal cerebral ventricle. An isolated cyst with otherwise reassuring prenatal assessment is usually benign, but its significance depends on other ultrasound findings and screening results.

Choroid Plexus Tumor

A papillary neoplasm of the choroid plexus, occurring most often in children, and frequently presenting with hydrocephalus from excessive cerebrospinal fluid production.

Choroidal Dystrophies

Inherited disorders causing progressive degeneration of the choroid, retinal pigment epithelium, and retina, leading to night blindness, visual-field loss, and eventual vision impairment.

Choroidal Neovascularization

The pathological growth of new, fragile blood vessels from the choroid through Bruch membrane into the subretinal or sub-retinal pigment epithelium space, and it is the hallmark of wet age-related macular degeneration and the primary cause of severe central vision loss in this condition.

Choroideremia

Choroideremia is an X-linked inherited retinal degeneration causing progressive night blindness and loss of peripheral then central vision, mainly in males.

Choroiditis

Choroiditis is inflammation of the choroid, the vascular layer behind the retina, often causing blurred vision, floaters, or visual-field disturbance.

Christmas Disease

Hemophilia B, an inherited bleeding disorder caused by deficiency or dysfunction of coagulation factor

IX.

Chromaffin

Chromaffin describes cells that stain with chromium salts because they contain catecholamine-rich granules, especially adrenal medullary cells and related neuroendocrine cells.

Chromaffinoma

A tumor of chromaffin cells, usually a pheochromocytoma of the adrenal medulla or a related extra-adrenal paraganglioma. It may secrete catecholamines and cause episodic hypertension.

Chromate

Chromate is a hexavalent chromium compound that can cause irritant or allergic dermatitis, nasal injury, respiratory disease, and cancer after occupational exposure.

Chromatic

Chromatic means relating to color or wavelengths of light; chromatic vision depends on cone photoreceptors and neural processing.

Chromatids

Chromatids are the two copies of a replicated chromosome joined at the centromere before separating during cell division.

Chromatin

The complex of DNA, histone proteins, and associated molecules that packages the genome in the nucleus. Its organization and chemical modification regulate DNA accessibility, replication, repair, and transcription.

Chromatography

A laboratory separation method in which substances partition between a stationary phase and a moving phase, allowing identification or measurement of components in a mixture.

Chromatopsia

A visual disturbance in which objects appear abnor-

mally colored. It may result from retinal disease, toxic exposure, medication effects, or other disorders of the visual system.

Chromhidrosis

A rare condition in which sweat is abnormally colored, often yellow, green, blue, or black, because of pigment in the sweat glands or on the skin surface.

Chromium

Chromium is a metallic element; trivalent chromium is a trace nutrient, while hexavalent compounds can be toxic, corrosive, mutagenic, and carcinogenic.

Chromium - Blood Test

A blood test measuring chromium exposure or body burden, generally used when occupational or medical exposure to chromium is suspected; results must be interpreted with the exposure history and clinical findings.

Chromium In Diet

Reference information on chromium as a trace element involved in carbohydrate and fat metabolism. Chromium deficiency is rare; most people obtain adequate chromium from normal dietary intake, and routine chromium supplementation is generally not recommended.

Chromoblastomycosis

A chronic fungal infection of skin and subcutaneous tissue that produces slowly enlarging warty or scaly lesions, usually after traumatic implantation of the fungus.

Chromopertubation

A diagnostic and operative laparoscopic procedure in which a dilute colored dye (such as methylene blue or indigo carmine) is introduced via a transcervical cannula or intrauterine catheter to evaluate fallopian tube patency under direct visual inspection. Immediate egress and spill of dye from the fimbriated ends confirms tubal patency, whereas lack of flow or loculated pooling identifies proximal or distal tubal occlusion, hydrosalpinx, or peritubal adhesions.

Chromosomal Abnormality

Change in chromosome number or structure. Numerical: aneuploidy (abnormal number, e.g., trisomy 21), polyploidy (extra sets). Structural: deletion, duplication, inversion, translocation.

Chromosomal Microarray

A genetic test that detects gains and losses of chromosome material, including copy-number variants and some regions of homozygosity; it is commonly used for developmental delay, intellectual disability, congenital anomalies, or autism-spectrum presentations.

Chromosome

A chromosome is a DNA-protein structure carrying genetic information; humans normally have 46 chromosomes in most body cells.

Chromosome Complement

The complete set of chromosomes in a cell, including their number and sex-chromosome composition, usually described by a karyotype.

Chromosome Duplication

A structural chromosome abnormality in which a segment of genetic material is copied one or more additional times, potentially altering gene dosage and development.

Chronic

Persisting or recurring over a long period, in contrast to acute; the appropriate duration depends on the disease or clinical context.

Chronic Adrenal Insufficiency

Alternate index label; see *Addison's disease*.

Chronic Allograft Nephropathy

Progressive transplant kidney function decline over months to years. Interstitial fibrosis and tubular atrophy on histology. Can be immunologic or non-immunologic. Optimize immunosuppression and manage risk factors including hypertension and recurrent disease.

Chronic Atrophic Gastritis

Long-term inflammation of the stomach lining that causes loss of gastric glands and may impair vitamin B12 absorption.

Chronic Autoimmune Hashimoto Thyroiditis

An autoimmune inflammatory disease of the thyroid that progressively destroys thyroid tissue and commonly causes hypothyroidism.

Chronic Autoimmune Thyroiditis

Hashimoto thyroiditis, an autoimmune destruction of the thyroid that commonly causes hypothyroidism and may be associated with anti-thyroid antibodies.

Chronic Bacterial Prostatitis

A chronic bacterial infection of the prostate gland classified as NIH Category II prostatitis, characterized by recurrent episodes of bacteriuria and urinary tract infection caused by persistence of the same uropathogen (most commonly *Escherichia coli*, *Klebsiella*, or *Proteus*) in prostatic tissue and fluid between clinical flares. Patients present with dysuria, urinary frequency and urgency, pelvic or perineal discomfort, and painful ejaculation. Diagnosis requires the Meares-Stamey four-glass test or two-glass pre- and post-prostatic massage urine culture demonstrating bacterial growth in expressed prostatic secretions or post-massage urine (VB3); treatment necessitates prolonged antimicrobial therapy (typically 4 to 6 weeks of a fluoroquinolone or trimethoprim-sulfamethoxazole) capable of penetrating the prostatic lipid-epithelial barrier.

Alternate Names:

NIH Category II Prostatitis; Recurrent Bacterial Prostatitis

Chronic Bronchitis

Defined clinically as a productive cough for at least three months in each of two consecutive years, after excluding other causes. It is characterized by hypertrophy of mucus-secreting glands in the bronchi (Reid index > 0.5), goblet cell hyperplasia, and chronic airway inflammation.

Alternate Names:

Bronchitis Chronic (Chronic Bronchitis)

Chronic Caries

Tooth decay that progresses slowly and may produce a firm, darkened lesion.

Chronic Cholecystitis

Repeated or persistent inflammation of the gallbladder, usually related to gallstones and impaired emptying. It may cause episodic right-upper-abdominal pain, nausea, or food intolerance and can lead to scarring or a contracted gallbladder.

Chronic Compartment Syndrome

Alternate index label; see *Chronic exertional compartment syndrome*.

Chronic Cough

Cough lasting > 8 weeks. Top 3 causes in non-smokers with normal CXR: upper airway cough syndrome, asthma, GERD. ACE inhibitor use is another common cause.

Alternate Names:

Cough Chronic (Chronic Cough)

Chronic Daily Headaches

Headaches occurring on at least 15 days each month for more than three months.

Chronic Disease

A disease or condition that persists for a prolonged period, often months or years, and may require continuing monitoring or management.

Chronic Endometritis

Persistent low-grade inflammation of the endometrium, often associated with infection, retained products, intrauterine devices, or reproductive failure; plasma cells support the diagnosis histologically. Risk factors include prolonged labor, multiple vaginal examinations, premature rupture of membranes, and cesarean section. Presents with fever, uterine tenderness, foul-smelling lochia, and leukocytosis.

Chronic Eosinophilic Leukaemia

A clonal myeloid neoplasm characterized by persistent eosinophilia, increased or dysplastic eosinophils, and evidence of an underlying clonal process or organ injury. Symptoms may include fatigue, fever, weight loss, rash, respiratory symptoms, splenomegaly, or cardiac and neurological complications.

Chronic Eosinophilic Pneumonia

An idiopathic inflammatory lung disease characterized by eosinophilic infiltration of the pulmonary parenchyma, presenting with fever, cough, dyspnea, and peripheral airspace opacities on imaging.

Chronic Exertional Compartment Syndrome

Exercise-related increased pressure within a muscle compartment that causes predictable pain, tightness, or neurologic symptoms during activity.

Alternate Names:

Chronic Compartment Syndrome

Chronic Fatigue Syndrome

A complex, debilitating disorder characterized by profound fatigue lasting >6 months that is not relieved by rest and is worsened by physical or mental exertion (post-exertional malaise). Associated symptoms: unrefreshing sleep, cognitive impairment (brain fog), orthostatic intolerance, joint pain, headaches, and sore throat.

Alternate Names:

CFS (Chronic Fatigue Syndrome) SEID (Chronic Fatigue Syndrome)

Chronic Fatigue Syndrome

Chronic fatigue syndrome, also called myalgic encephalomyelitis, involves substantial persistent fatigue with post-exertional worsening, unrefreshing sleep, and cognitive or orthostatic symptoms.

Chronic Gastritis

Chronic, progressive inflammation of the gastric mucosa leading over time to mucosal atrophy, loss of specialized parietal and chief cells, and intestinal metaplasia. Major causes include persistent *Helicobacter pylori* infection.

cobacter pylori infection and autoimmune gastritis targeting parietal cells and intrinsic factor.

Chronic Glomerulonephritis

Long-standing inflammation and scarring of the kidney glomeruli that can progressively reduce filtration and lead to chronic kidney disease or kidney failure.

Chronic Granulomatous Disease

A primary immunodeficiency caused by defects in the NADPH oxidase complex in phagocytes (neutrophils, monocytes, macrophages), impairing the respiratory burst and killing of catalase-positive organisms.

Chronic Hives

Recurrent itchy welts that occur on most days for more than six weeks.

Chronic Hypertension In Pregnancy

Hypertension that predates pregnancy or is diagnosed early in pregnancy. It increases the risk of preeclampsia, fetal growth problems, preterm birth, and placental complications and requires ongoing maternal and fetal monitoring.

Chronic Inflammation

A prolonged inflammatory response characterized by simultaneous tissue destruction and attempts at repair, lasting weeks to years. Unlike acute inflammation, it is predominantly mononuclear, involving macrophages, lymphocytes, and plasma cells rather than neutrophils.

Chronic Inflammatory Demyelinating Polyneuropathy

An acquired, immune-mediated peripheral neuropathy with a relapsing-remitting or progressive course. Presents with symmetric proximal and distal weakness, sensory loss, areflexia, and elevated CSF protein. Distinguishing from Guillain-Barre: progressive course >8 weeks (vs. rapid onset in GBS).

Chronic Inflammatory Response

Syndrome

A proposed multisystem illness involving persistent inflammatory symptoms after an exposure such as certain water-damaged building environments. Reported symptoms are nonspecific and may include fatigue, headache, cognitive difficulty, and respiratory complaints. Symptoms require a careful medical evaluation for established infectious, allergic, toxic, autoimmune, and other causes.

Chronic Inflammatory Response Syndrome

Alternate index label for CIRS.

Chronic Intestinal Pseudo-Obstruction

Chronic condition of recurrent episodes of intestinal obstruction without mechanical blockage due to neuromuscular dysfunction of the gastrointestinal tract. May involve the small bowel, colon, or both. Causes include systemic sclerosis, diabetes, Parkinson disease, amyloidosis, and paraneoplastic syndromes.

Chronic Kidney Disease

Chronic kidney disease is the progressive loss of kidney function over months to years, typically from diabetes, hypertension, or glomerular diseases. It is staged by estimated glomerular filtration rate and albuminuria, with end-stage renal disease requiring dialysis or transplantation.

Alternate Names:

Chronic Kidney Failure Chronic Renal Failure Kidney Disease, Chronic Kidney Failure, Chronic

Chronic Kidney Disease Staging

Classification of kidney disease by cause, eGFR category, and albuminuria category. G1 through G5 represent eGFR from normal or high to kidney failure; staging is applied when abnormalities persist for at least 3 months and helps estimate progression and complications.

Chronic Kidney Failure

Alternate index label; see *Chronic kidney disease*.

Chronic Lead Poisoning

Health effects caused by prolonged or repeated lead exposure. Possible findings include abdominal pain or constipation, fatigue, anemia, cognitive or behavioral changes, peripheral neuropathy, hypertension, and kidney injury; children may develop impaired learning, attention, growth, or development. Management centers on identifying and stopping the exposure, evaluating affected individuals, and considering specialist-directed chelation only when the blood lead level, symptoms, and clinical context warrant it.

Chronic Leukemia

A leukemia characterized by an insidious onset, clonal proliferation of relatively mature blood cells, and a more indolent clinical course.

Chronic Limb-Threatening Ischemia

The most severe clinical form of peripheral arterial disease, characterized by chronic ischemic rest pain, nonhealing ulceration, or gangrene attributable to arterial insufficiency. It carries high risks of limb loss and cardiovascular mortality and requires vascular evaluation for revascularization and wound management.

Chronic Liver Disease

Persistent hepatic injury and structural change lasting at least 6 months, caused by viral, metabolic, alcohol-related, autoimmune, cholestatic, vascular, or drug-related disease. Progressive fibrosis can lead to cirrhosis, portal hypertension, liver failure, and hepatocellular carcinoma.

Chronic Lymphocytic Leukemia

The most common leukemia in Western countries, characterized by the clonal proliferation of mature-appearing but functionally incompetent B lymphocytes that accumulate in blood, bone marrow, and lymphoid tissues.

Alternate Names:

Cancer, Chronic Lymphocytic Leukemia CLL Leukemia, Chronic Lymphocytic

Chronic Lymphocytic Leukemia

A usually indolent blood cancer characterized by clonal accumulation of mature-appearing but abnormal B lymphocytes in blood, marrow, lymph nodes, or spleen. It may cause lymphocytosis, enlarged nodes, infection risk, anemia, or thrombocytopenia.

Chronic Lymphocytic Thyroiditis

Alternate index label; see *Hashimoto's disease*.

Chronic Mesenteric Ischemia

Recurrent abdominal pain occurring after eating resulting from insufficient blood flow to meet the metabolic demands of the gastrointestinal tract. Most commonly caused by atherosclerotic stenosis of at least two of the three major mesenteric arteries.

Chronic Migraine

A migraine disorder with headache on at least 15 days per month for more than three months, with migraine features on at least eight days per month. Symptoms may include throbbing pain, nausea, vomiting, and light or sound sensitivity; frequent or disabling headaches require clinical evaluation.

Chronic Motor Or Vocal Tic Disorder

A neurodevelopmental disorder with persistent motor or vocal tics lasting more than one year; symptoms often fluctuate and may be treated with education, behavioral therapy, or medicines when impairment is significant.

Chronic Mucocutaneous Candidiasis

Persistent/recurrent *Candida* infections of skin, nails, mucosa (oral, esophageal, vaginal) without systemic dissemination. Autosomal dominant (STAT1 gain-of-function) or recessive (IL17F, IL17RA, ACT1, RORC). Associated with autoimmunity (hypothyroidism, AIHA, ITP, alopecia), endocrinopathies, Staph infections.

Chronic Myelogenous Leukemia

A myeloproliferative neoplasm associated with the BCR-ABL1 fusion gene, usually producing marked leukocytosis and splenomegaly.

Alternate Names:

Cancer, Chronic Myelogenous Leukemia Leukemia, Chronic Myelogenous

Chronic Myelogenous Leukemia

A myeloproliferative blood cancer driven in most cases by the BCR::ABL1 fusion gene from the Philadelphia chromosome. It causes excess myeloid cells and may progress from a chronic phase to accelerated or blast phase; tyrosine-kinase inhibitors are standard targeted treatment.

Chronic Myeloid Leukemia

A myeloproliferative neoplasm characterized by the Philadelphia chromosome [t(9;22)(q34;q11.2)], which creates the BCR-ABL1 fusion gene encoding a constitutively active tyrosine kinase.

Chronic Myelomonocytic Leukemia

A rare myelodysplastic/myeloproliferative neoplasm characterized by persistent monocytosis (absolute monocyte count greater than 1,000 per microliter and greater than 10 percent of white blood cells) in the peripheral blood and bone marrow, with dysplastic features and variable blast count.

Chronic Neutrophilic Leukaemia

A rare clonal myeloproliferative neoplasm characterized by sustained mature neutrophilic leukocytosis without a primary infectious or inflammatory cause. It may cause fatigue, fever, weight loss, splenomegaly, and progressive constitutional symptoms.

Chronic Obstructive Lung Disease

An older abbreviation for chronic obstructive lung disease, now usually called chronic obstructive pulmonary disease, characterized by persistent airflow limitation.

Chronic Obstructive Pulmonary Disease

COPD is a preventable lung disease characterized by persistent respiratory symptoms and airflow limitation due to airway and/or alveolar abnormalities, usually caused by significant exposure to noxious particles or gases (primarily tobacco smoke). In-

cludes emphysema and chronic bronchitis.

Alternate Names:

COPD (Chronic Obstructive Pulmonary Disease)

Chronic Obstructive Pulmonary Disease

Chronic obstructive pulmonary disease is persistent airflow limitation from airway and alveolar injury, commonly causing exertional breathlessness, cough, sputum, and exacerbations.

Chronic Pain

Pain that persists or recurs for longer than three months. It may have an identifiable ongoing cause or persist after healing, and is influenced by complex interactions among biological, psychological, and social factors.

Chronic Pain Management

Chronic pain management extends beyond the peri-operative period. Opioid therapy for chronic pain requires careful patient selection, informed consent, and regular reassessment.

Chronic Pain Syndrome

Persistent pain accompanied by significant physical, emotional, or social effects that interfere with daily life.

Chronic Pancreatitis

Long-lasting inflammation of the pancreas that causes permanent damage and may reduce the production of digestive enzymes and hormones such as insulin. Common causes include heavy alcohol use, smoking, genetic disorders, autoimmune disease, and blocked pancreatic ducts; sometimes no cause is found.

Chronic Paroxysmal Hemicrania

A primary headache disorder causing frequent, short-lasting, severe one-sided headaches with autonomic symptoms and a characteristic response to indomethacin.

Alternate Names:

CPH; chronic paroxysmal hemicrania syndrome.

Chronic Pelvic Pain

Persistent or recurrent pain in the lower abdomen or pelvis lasting at least six months.

Alternate Names:

Pelvic Pain, Chronic

Chronic Pelvic Pain Syndrome

A multifactorial chronic pain disorder classified as NIH Category III prostatitis, accounting for approximately 90–95% of all clinical prostatitis presentations. Defined as persistent or recurrent pelvic, perineal, scrotal, suprapubic, or lower back pain lasting at least 3 months within the preceding 6 months, commonly accompanied by voiding dysfunction (frequency, urgency, hesitance), pain during or after ejaculation, and erectile difficulties, in the complete absence of documented bacterial infection. Subclassified into Category IIIa (inflammatory, with leukocytes in expressed prostatic secretions) and Category IIIb (non-inflammatory). Pathophysiology involves neurogenic inflammation, pelvic floor muscular hypertonicity, central pain sensitization, and psychosocial distress, managed through multimodal UPOINT-guided therapy (alpha-blockers, anti-inflammatories, pelvic floor physical therapy, neuromodulators, and cognitive behavioral therapy).

Alternate Names:

CPPS; NIH Category III Prostatitis; Chronic Non-bacterial Prostatitis

Chronic Progressive External Ophthalmoplegia

A classic adult-onset mitochondrial myopathy caused by single large-scale deletions of mitochondrial DNA (mtDNA) or by nuclear gene mutations that destabilize mtDNA replication and maintenance, including *POLG*, *TWINK*, *SLC25A4*, and *RRM2B*. Typically manifests between the second and fourth decades of life with insidious, slowly progressive, symmetrical, bilateral ptosis and restriction of all extraocular eye movements (external ophthalmoplegia) without diplopia (because movement impairment is symmetrical). Muscle biopsy reveals ragged red fibers and cytochrome c oxidase-negative myofibers; can present as an isolated ocular

myopathy or as part of multisystem syndromes such as Kearns-Sayre syndrome.

Alternate Names:

CPEO; Progressive External Ophthalmoplegia (PEO)

Chronic Recurrent Multifocal Osteomyelitis

A rare autoinflammatory bone disorder, often beginning in children or adolescents, causing recurrent sterile inflammation and pain at multiple bone sites; it is also called CRMO.

Chronic Pyelonephritis

Chronic tubulointerstitial renal inflammation and scarring resulting from recurrent or persistent kidney infections, often in the setting of structural urinary tract obstruction or chronic vesicoureteral reflux (reflux nephropathy). Leads to progressive renal parenchymal thinning, calyceal blunting, and chronic kidney disease.

Chronic Renal Failure

Alternate index label; see *Chronic kidney disease*.

Chronic Rhinitis

Persistent inflammation or irritation of the nasal lining causing congestion, runny nose, sneezing, itching, or postnasal drainage. Allergic, nonallergic, medication-related, and structural causes are possible. Management depends on the cause and may include trigger avoidance, saline irrigation, and clinician-directed nasal treatment.

Alternate Names:

Post-Nasal Drip (Chronic Rhinitis)

Chronic Rhinosinusitis

Persistent inflammation of the nose and paranasal sinuses lasting at least 12 weeks, usually with nasal obstruction or discharge and sometimes facial pressure or reduced smell. It may occur with or without nasal polyps.

Chronic Sinusitis

Persistent inflammation of the paranasal sinuses

for at least 12 weeks, causing nasal obstruction, discharge, facial pressure, and reduced smell.

Alternate Names:

Sinusitis, Chronic

Chronic Subdural Hematoma

A collection of blood and breakdown products between the dura and arachnoid that develops over weeks, often after minor trauma in older adults, causing headache, confusion, weakness, or gait change.

Chronic Suppurative Otitis Media

Persistent inflammation and infection of the middle ear with recurrent or continuous ear discharge, usually through a perforated tympanic membrane. It can cause conductive hearing loss and may lead to cholesteatoma or other complications.

Alternate Names:

CSOM; chronic otitis media with discharge; chronic suppurative middle-ear infection.

Chronic Tension-Type Headache

A recurring headache occurring on at least 15 days per month for more than three months, usually causing bilateral pressing or tightening pain rather than throbbing. It may be associated with scalp or neck tenderness but typically lacks prominent nausea; a new or changing pattern requires clinical assessment.

Chronic Thromboembolic Pulmonary Hypertension

Pulmonary hypertension caused by organized, persistent thromboembolic obstruction and secondary pulmonary-vessel remodeling after pulmonary embolism. It is potentially curable in selected patients by pulmonary endarterectomy or balloon pulmonary angioplasty.

Chronic Thyroiditis (Hashimoto Disease)

An autoimmune disease in which immune attack gradually damages the thyroid, often causing hypothyroidism, fatigue, cold intolerance, constipation, dry skin, or goiter.

Chronic Toxicity

Adverse effects from repeated or prolonged exposure to low doses. Cumulative damage may be irreversible. Examples: heavy metals (lead, mercury), asbestos, alcohol, solvent exposure. Monitoring: biomarkers, organ function tests.

Chronic Traumatic Encephalopathy

A progressive brain disorder associated with repeated head impacts and characterized by cognitive, mood, or behavioral changes.

Chronic Venous Disease

A spectrum of venous outflow disorders caused by reflux, obstruction, or calf-muscle-pump dysfunction. Findings range from edema and varicose veins to skin changes, lipodermatosclerosis, and venous ulceration.

Chronic Venous Insufficiency

Impaired venous return from the lower extremities due to venous valve incompetence or obstruction. Leads to venous hypertension, edema, skin changes (lipodermatosclerosis, hyperpigmentation), and venous stasis ulcers (medial malleolus). Risk factors: varicose veins, DVT, obesity, prolonged standing.

Chronic Venous Stasis

Persistent pooling of blood in the lower limbs caused by impaired venous return. It can produce edema, skin discoloration, dermatitis, and venous ulcers.

Chronic Vulvar Pain

Alternate index label; see *Vulvodynia*.

Chronicity

The duration or long-standing nature of a disease, symptom, or process.

Chronobiology

Chronobiology studies biological timing and rhythms, including circadian sleep-wake cycles, seasonal rhythms, and their effects on health.

Chronotropic

Chronotropic describes an influence on heart rate; a positive chronotropic effect increases rate, while a negative effect decreases it.

Chrysarobin

Chrysarobin is a plant-derived anthracene compound formerly used topically for psoriasis and other skin disorders; it can irritate and stain skin.

Churg Strauss Syndrome (Churg-Strauss Syndrome)

Alternate index label for Churg-Strauss Syndrome.

Churg-Strauss Syndrome

Eosinophilic granulomatosis with polyangiitis, a necrotizing inflammation of small and medium blood vessels associated with asthma, eosinophilia, and possible injury to the lungs, nerves, skin, heart, or kidneys.

Alternate Names:

Churg Strauss Syndrome (Churg-Strauss Syndrome)

Chvostek Sign

A clinical sign of latent tetany caused by hypocalcemia (ionized $\text{Ca}^{2+} < 4.4 \text{ mg/dL}$). Elicited by tapping the facial nerve over the parotid gland just anterior to the ear canal. Positive response: involuntary twitching or contraction of ipsilateral facial muscles (lip twitching or facial spasm).

Chvostek's Sign

Chvostek's sign is twitching of facial muscles after tapping near the facial nerve, sometimes associated with hypocalcemia but neither sensitive nor specific.

Chylangioma

A benign malformation or tumor-like lesion of lymphatic vessels containing chyle or lymph. It may occur in the neck, mediastinum, or other tissues.

Chyle

Milky lymph from intestinal lymphatic vessels that is rich in absorbed dietary fat and transported through the thoracic duct.

Chylomicron

The largest and least-dense lipoprotein, made in the intestine after a fatty meal. It carries dietary triglycerides and cholesterol through the lymph and blood; its remnants are cleared by the liver.

Chylomicronemia Syndrome

Severe elevation of triglyceride-rich chylomicrons in blood, usually from genetic or acquired impairment of triglyceride clearance; it can cause abdominal pain, eruptive xanthomas, and recurrent acute pancreatitis.

Chylothorax

Pleural effusion with chyle from thoracic duct disruption. Triglycerides $> 110 \text{ mg/dL}$. Treatment includes dietary modification, octreotide, and surgical ligation if refractory.

Chyluria

Passage of lymphatic fluid into the urine, giving it a milky appearance. It usually results from an abnormal communication between lymphatic vessels and the urinary tract.

Chyme

A semifluid mixture of partially digested food and gastric or intestinal secretions that passes from the stomach into the small intestine.

Alternate Names:

Digestive chyme; gastric chyme.

Chymotrypsin

Chymotrypsin is a pancreatic digestive enzyme that cleaves proteins in the small intestine; abnormal pancreatic release can contribute to tissue injury.

Ci (Curie)

A unit of radioactivity equal to 3.7 billion nuclear disintegrations per second; the SI unit is the becquerel.

CIC

Commonly, clean intermittent catheterization: periodic insertion of a catheter to empty the bladder,

followed by removal; other expansions are possible by context.

Cicatricial Alopecia

Hair loss caused by destruction and scarring of hair follicles, usually leading to permanent loss in affected areas.

Alternate Names:

Scarring alopecia; cicatricial hair loss.

Cicatrization

Cicatrization is healing by formation of a scar after tissue injury, with replacement of damaged tissue by fibrous connective tissue.

Ciclopia

A rare severe congenital malformation in which the eyes are fused or a single median eye develops, usually as part of holoprosencephaly.

Cidofovir

Cidofovir is an antiviral nucleotide analogue used for selected serious DNA-virus infections; kidney toxicity requires careful dosing and monitoring.

CIDP

Chronic inflammatory demyelinating polyneuropathy is a chronic, immune-mediated, peripheral neuropathy characterized by progressive or relapsing symmetrical weakness and sensory loss in all four limbs, with areflexia or hyporeflexia, lasting for more than eight weeks.

Cigarette

A small cylinder of tobacco or other material wrapped in paper for smoking; smoke exposure delivers nicotine and many toxic and carcinogenic substances.

Ciguatera

Food poisoning caused by eating reef fish containing ciguatoxins. It may cause gastrointestinal illness followed by neurologic symptoms such as paresthesias and temperature-sensation disturbance.

Ciguatera Poisoning

Ciguatera poisoning follows eating reef fish containing ciguatoxins. Gastrointestinal symptoms may be followed by neurologic sensations such as temperature reversal, tingling, itching, weakness, or abnormal heart rate; treatment is supportive and prevention depends on avoiding high-risk fish.

Alternate Names:

Poisoning Ciguatera (Ciguatera Poisoning) Toxin Ciguatera (Ciguatera Poisoning)

Ciguatoxin

A heat-stable toxin produced by marine microorganisms and accumulated in some reef fish, causing ciguatera poisoning with gastrointestinal, neurologic, and temperature-sensation symptoms.

Cilia

Small hair-like structures on cells that move fluid, particles, or mucus across a tissue surface.

Ciliary Body

Circumferential structure between iris and choroid, with ciliary muscle (accommodation control via zonular fibers) and ciliary processes (aqueous humor production). Part of the uveal tract (iris, ciliary body, choroid).

Ciliary Dyskinesia

Abnormal movement of cilia that impairs clearance of mucus from the airways and can cause recurrent respiratory infections.

Ciliary Muscle

A ring of smooth muscle in the eye that changes the lens shape for focusing at different distances and helps regulate aqueous humor outflow through ciliary-body structures.

Ciliary Neuralgia

Pain arising from or felt along the ciliary nerves around the eye, often described as deep orbital or eye pain; the cause requires clinical evaluation.

Ciliopathy

A disorder caused by abnormal structure or function of cellular cilia. Ciliopathies may affect development, vision, kidney function, respiration, or multiple organ systems.

Cimetidine

Cimetidine is an H₂-receptor antagonist that reduces gastric acid secretion; it has numerous drug interactions and is used less often than newer agents.

Cinacalcet

A calcimimetic drug that activates calcium-sensing receptors and lowers parathyroid hormone, used for selected forms of hyperparathyroidism and hypercalcemia.

Cineangiocardiology

Radiographic recording of the heart and blood vessels during passage of contrast material, used to visualize cardiac chambers, valves, and coronary or great vessels.

Cinefluorography

Motion-picture recording of fluoroscopic images, used to document movement of organs or contrast material during a diagnostic examination.

Cineplasty

Surgical creation of a muscle-controlled stump or related reconstruction, historically used to improve control of a prosthetic limb.

Cingulotomy

Surgical interruption of selected fibers of the cingulum, sometimes performed as a psychosurgical treatment for severe, treatment-resistant psychiatric illness or chronic pain.

Cinnarizine

Cinnarizine is an antihistamine with calcium-channel-blocking properties used in some countries for motion sickness and vestibular symptoms.

Ciprofloxacin

Ciprofloxacin is a fluoroquinolone antibiotic used for selected bacterial infections; tendon injury, nerve effects, QT effects, and resistance limit indiscriminate use.

Circadian Rhythm

The approximately 24-hour endogenous biological cycle regulating sleep-wake behavior, hormone secretion, body temperature, metabolism, and other physiological processes. Generated by the suprachiasmatic nucleus of the hypothalamus, which is entrained to the environment primarily by light via the retinohypothalamic tract.

Circadian Rhythms

Approximately 24-hour biological cycles that regulate sleep and wakefulness, hormone release, body temperature, metabolism, and other physiological functions in response chiefly to light and darkness.

Circle Of Willis

A ring-like arterial anastomosis at the base of the brain that provides collateral blood flow between the anterior (internal carotid) and posterior (vertebrobasilar) circulations. It is formed by the anterior cerebral, anterior communicating, internal carotid, posterior cerebral, and posterior communicating arteries.

Circulating Tumor DNA

Fragments of tumor-derived DNA detectable in blood, enabling molecular profiling, treatment monitoring, and detection of minimal residual disease without tissue biopsy (liquid biopsy).

Circulation

Continuous movement of blood through the heart and blood vessels, or of lymph through lymphatic vessels, to transport oxygen, nutrients, hormones, and waste.

Circulatory

Relating to the movement of blood or lymph and to the organs and vessels that carry them.

Circulatory System

The heart and blood-vessel network that pumps and distributes blood through pulmonary and systemic circuits; broadly, the term may include the lymphatic system.

Circumcision

A medical procedure involving circumcision. Indications, risks, preparation, and follow-up depend on the patient and the specific procedure. medical reasons such as severe phimosis, recurrent balanoposthitis, paraphimosis, or lichen sclerosus (balanitis xerotica obliterans); elective infant circumcision for religious, cultural, or preventive health reasons.

Circumflex

Curving around; the circumflex coronary artery is a branch of the left coronary artery that travels around the heart's lateral and posterior surface.

Circumflex Artery

A branch of the left coronary artery that travels in the left atrioventricular groove. It supplies the lateral wall of the left ventricle, the left atrium, and in left-dominant systems, the inferior wall and posteromedial papillary muscle.

Circumvallate Papillae

Circumvallate papillae are large taste papillae arranged near the back of the tongue and surrounded by a trench containing taste buds and serous glands.

Circumvallate Placenta

An extrachorial placental configuration in which the chorionic plate is smaller than the basal plate, producing a raised circumferential edge; it may be associated with bleeding, abruption, or growth restriction.

Cirrhosis

Cirrhosis is advanced liver scarring with regenerative nodules that distorts blood flow and liver function, potentially causing portal hypertension, jaundice, ascites, and liver failure. distortion of the hepatic architecture and formation of regenerative nodules. The most common causes include chronic hepatitis

C infection, alcoholic liver disease, non-alcoholic steatohepatitis, chronic hepatitis B infection, and autoimmune hepatitis.

Cirrhosis Primary Biliary (Primary Biliary Cirrhosis)

Alternate index label for Primary Biliary Cirrhosis.

Cisplatin

A platinum-based chemotherapy drug that forms DNA cross-links and is used to treat several cancers; important toxicities include kidney injury, hearing loss, neuropathy, and nausea.

Citalopram

Citalopram is a selective serotonin-reuptake inhibitor used to treat depression and some anxiety disorders; nausea, sexual effects, and QT prolongation are important considerations.

Citrate

Citrate is a salt or ion of citric acid that participates in metabolism and binds calcium; it is used in anticoagulant blood collection and some stone-prevention treatments.

Citrate Synthase

First enzyme of TCA cycle. Condenses oxaloacetate and acetyl-CoA to citrate.

Citric Acid Cycle

The citric acid cycle is a mitochondrial pathway that oxidizes acetyl-CoA and generates reducing equivalents for ATP production.

Citric Acid Cycle (Krebs Cycle)

A central metabolic pathway occurring in the mitochondrial matrix that oxidizes acetyl-CoA (derived from carbohydrates, lipids, and proteins) into CO₂, generating 3 NADH, 1 FADH₂, and 1 GTP (ATP) per turn. Rate-limiting enzyme: isocitrate dehydrogenase (inhibited by ATP and NADH; activated by ADP).

Alternate Names:

Krebs Cycle

Citric Acid Urine Test

A urine test measuring citrate excretion, used mainly in the evaluation of recurrent kidney stones because low urinary citrate increases the risk of calcium stone formation.

Citrullinaemia

Citrullinaemia is an inherited urea-cycle disorder causing accumulation of citrulline and potentially life-threatening hyperammonemia.

Citrullinemia

An inherited metabolic disorder in which citrulline and ammonia accumulate because of impaired urea-cycle function.

CJD

Alternate index label; see *Creutzfeldt-Jakob disease*.

CK

Creatine kinase, an enzyme found in skeletal muscle, heart muscle, and brain whose blood level may rise after tissue injury.

Alternate Names:

Creatine phosphokinase; creatine kinase enzyme.

CKD-Mineral Bone Disorder

Systemic disorder of mineral and bone metabolism in CKD. Includes calcium, phosphorus, PTH, and vitamin D abnormalities with vascular calcification and bone disease. Renal osteodystrophy encompasses osteitis fibrosa, osteomalacia, adynamic bone disease, and mixed uremic osteodystrophy.

Cl (Chloride)

The chemical symbol for chloride, the negatively charged ion of chlorine and a major extracellular electrolyte.

Cladribine

Cladribine is a purine analogue used for selected blood cancers and multiple sclerosis; lymphocyte depletion and infection risk require monitoring.

Clairvoyance

Clairvoyance is an alleged ability to perceive information beyond ordinary sensory means; it is not an established medical diagnostic finding.

Clamp

A surgical instrument used to grasp, compress, occlude, or hold tissue, blood vessels, tubing, or other structures.

Clang Association

Clang association is speech in which word choice is driven by sound, rhyme, or pun rather than meaning, seen particularly in mania or psychosis.

Clarithromycin

Clarithromycin is a macrolide antibiotic used for selected bacterial and mycobacterial infections; interactions and QT prolongation are important risks.

Clark Level Of Invasion

A historical five-level system describing how deeply a melanoma has invaded skin layers; modern melanoma staging relies primarily on tumor thickness, ulceration, and other prognostic factors.

Class IA Antiarrhythmics

Sodium channel blockers that slow phase 0 depolarization and prolong the action potential duration and refractory period. Examples include quinidine, procainamide, and disopyramide. Used for atrial and ventricular arrhythmias. Quinidine can cause QT prolongation and torsades de pointes.

Class IB Antiarrhythmics

Sodium channel blockers that shorten the action potential duration and are preferentially taken up by ischemic tissue. Examples include lidocaine and mexiletine. Lidocaine is used for ventricular tachycardia and ventricular fibrillation, particularly in the setting of acute myocardial infarction.

Class IC Antiarrhythmics

Sodium channel blockers that significantly slow phase 0 depolarization without affecting action po-

tential duration. Examples include flecainide and propafenone. Used for supraventricular arrhythmias in patients without structural heart disease.

Class III Antiarrhythmics

Potassium channel blockers that prolong the action potential duration and refractory period. Examples include amiodarone, sotalol, dofetilide, and ibutilide. Amiodarone is the most commonly used but has significant toxicity. Sotalol has additional beta-blocking properties and can cause torsades de pointes.

Classic Migraine

A migraine associated with an aura, such as temporary visual or sensory symptoms.

Claudication

Muscle pain or cramping brought on by exertion and relieved by rest, usually because of reduced arterial blood flow to a limb.

Alternate Names:

Vascular Claudication

Claudication Distance

The distance a person can walk before exertional limb pain begins. It is used to describe symptoms and monitor change.

Claustrophobia

An intense fear of enclosed or confined spaces that may cause anxiety, panic, or avoidance. Symptoms can occur in places such as elevators, tunnels, or MRI scanners.

Clavicle

Only horizontally oriented long bone, serving as a shoulder-trunk strut. Medial end articulates with the manubrium (sternoclavicular joint); lateral end with the acromion (acromioclavicular joint). Most common fracture at the middle-lateral third junction from falls on an outstretched hand.

Clavicle Fracture

A break in the collarbone that may occur during

birth or after trauma. A newborn may show asymmetric arm movement or tenderness; older children and adults may have pain, swelling, or deformity. Diagnosis is usually made by examination and imaging.

Clavicle Fracture (Intentional)

Intentional fracture of the fetal clavicle as a last resort to reduce the bisacromial diameter in shoulder dystocia. The operator applies upward pressure on the midportion of the anterior fetal clavicle with a finger, fracturing it. The clavicle heals without long-term sequelae.

Alternate Names:

Intentional

Clavus

A clavus is a localized painful thickening of skin, or corn, caused by repeated pressure or friction.

Claw Foot

A deformity in which the toes are hyperextended at the metatarsophalangeal joints and flexed at the interphalangeal joints. It may result from muscle imbalance or neurologic disease.

Claw Hand

A hand deformity with hyperextension at the metacarpophalangeal joints and flexion at the interphalangeal joints, classically caused by ulnar-nerve palsy or lower brachial-plexus injury.

Claw Toe

A toe deformity characterized by extension at the metatarsophalangeal joint and flexion at the proximal and distal interphalangeal joints.

Clean Catch Urine Sample

A midstream urine specimen collected after cleaning the genital area to reduce contamination and improve interpretation of urinalysis or culture.

Clean-Catch Specimen

A urine sample collected after cleansing the surrounding area and beginning urination, with the

middle portion collected in a sterile container to reduce contamination.

Cleaning Supplies And Equipment

Materials and devices used to remove soil and microorganisms from surfaces or equipment; selection and disinfection level depend on the item and infection risk.

Cleaning To Prevent The Spread Of Germs

Cleaning and disinfection practices that remove or inactivate infectious organisms on hands, surfaces, equipment, and shared objects to reduce transmission.

Clear Cell Carcinoma

Clear cell carcinoma is a cancer whose cells appear clear because of glycogen or lipid; it can arise in organs such as kidney, ovary, or endometrium.

Clear Cell Carcinoma Of The Ovary

A distinct histological subtype of epithelial ovarian cancer characterized microscopically by hobnail and clear cells rich in glycogen arranged in tubulocystic, papillary, or solid architectural patterns. Strongly associated with preexisting pelvic endometriosis and *ARID1A* or *PIK3CA* mutations, it frequently presents at an early stage, carries an elevated risk of thromboembolic events, and demonstrates relative resistance to standard platinum- and taxane-based chemotherapy regimens compared to high-grade serous carcinoma.

Clear Cell Renal Cell Carcinoma

The predominant histologic subtype of renal cell carcinoma, accounting for approximately 75–80% of all adult primary kidney cancers. Arises from the proximal convoluted tubule and is histologically characterized by malignant cells with clear lipid- and glycogen-rich cytoplasm; strongly associated with loss or inactivation of the *VHL* tumor suppressor gene.

Alternate Names:

ccRCC; Conventional renal cell carcinoma

Clear Liquid Diet

A limited diet consisting of transparent liquids such as water, clear broth, apple juice, and black coffee that leave minimal residue. It is used for short periods before procedures, during acute illness, or after surgery to maintain hydration while resting the gastrointestinal tract.

Clear Margins

A histological finding indicating that no abnormal cells (e.g., cervical intraepithelial neoplasia or cancer cells) are present at the edge of a tissue specimen removed during a procedure such as LLETZ, cone biopsy, or surgical excision.

Clearance

The volume of plasma from which a drug is completely removed per unit time, typically expressed in milliliters per minute or liters per hour. Total clearance is the sum of all clearance mechanisms, primarily hepatic and renal clearance.

Cleft Lip

A birth defect in which the upper lip does not join together fully before birth. It may occur alone or with a cleft palate.

Cleft Lip And Cleft Palate

Birth defects in which the upper lip, the roof of the mouth, or both do not join together fully before birth. They can cause problems with feeding, speech, hearing, or teeth and may need treatment by several specialists.

Cleft Lip And Palate Repair

A medical procedure involving cleft lip and palate repair. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Cleft Palate

A birth defect in which there is an opening in the roof of the mouth. It may occur alone or with a cleft lip and can cause problems with feeding, speech, hearing, or teeth.

Cleidocranial Dysostosis

A genetic skeletal disorder, usually caused by RUNX2 variants, characterized by underdeveloped or absent clavicles, delayed closure of skull sutures, abnormal dentition, and short stature. Dental and orthopedic complications are common.

Cleidocranial Dysostosis

A usually inherited skeletal disorder, often caused by RUNX2 variants, characterized by hypoplastic or absent clavicles, delayed closure of skull sutures, short stature, and dental abnormalities.

Cleidocranial Dysplasia

An inherited skeletal disorder characterized by defective development or ossification of the clavicles and skull, delayed closure of cranial sutures, dental abnormalities, and short stature.

Clemastine

Clemastine is a first-generation antihistamine used for allergic symptoms; sedation, anticholinergic effects, and impaired driving are concerns.

Clendamyacin

Clendamyacin is a misspelling of clindamycin, a lincosamide antibiotic that inhibits bacterial protein synthesis and can cause severe antibiotic-associated colitis.

Click-Murmur Syndrome

Alternate index label; see *Mitral valve prolapse*.

Climacteric

The period of physiologic change around the end of reproductive capacity, including the menopausal transition. It may involve changes in menstruation, vasomotor symptoms, and other endocrine effects.

Climate

Climate is the long-term pattern of weather and environmental conditions; climate affects infectious disease, heat illness, allergies, and population health.

Clindamycin

Clindamycin is a lincosamide antibiotic used for selected infections; diarrhea and *Clostridioides difficile* colitis are important adverse effects.

Clinic Visit

Scheduled outpatient appointment for medical consultation, follow-up, or monitoring of a chronic condition.

Clinical

Relating to direct observation, examination, diagnosis, or treatment of a patient, rather than solely to laboratory or basic research findings.

Clinical Communication

Exchange of information between healthcare professionals, patients, and caregivers to support diagnosis, treatment, safety, and informed decisions.

Clinical Counseling

Guidance provided in healthcare to help patients make informed decisions about health, prevention, treatment, and self-management.

Clinical Depression

A depressive disorder marked by persistent depressed mood or loss of interest with associated cognitive, emotional, or physical symptoms that cause clinically significant distress or impairment.

Clinical Depression (Depression)

Alternate index label for Depression.

Clinical Deterioration

Worsening of physiologic function or symptoms that increases risk of organ failure, disability, or death. Changes in mental status, respiratory effort, perfusion, urine output, and vital signs may precede overt collapse.

Clinical Disease

Disease that produces recognizable signs or symptoms or other findings detectable in a patient, as opposed to a silent or subclinical state.

Clinical Documentation

Recording of patient history, examination, assessment, treatment, and follow-up in the medical record.

Clinical Frailty Scale

A clinician-rated scale that summarizes frailty from very fit to severely frail or terminally ill using activity, function, and disease burden. It supports communication and prognostic assessment but should be interpreted in the patient's clinical context.

Clinical Guideline

Systematically developed statements to assist practice. Examples: CDC guidelines, WHO guidelines, specialty society guidelines.

Clinical Handover

Transfer of responsibility and relevant patient information from one healthcare professional or team to another. Effective handover communicates diagnosis, current status, risks, pending tasks, contingency plans, and accountability.

Clinical Practice Guideline

Evidence-informed recommendations intended to help healthcare professionals and patients make decisions about a specific clinical condition or situation.

Clinical Presentation

Signs and symptoms of disease. Varies by disease, severity, patient factors.

Clinical Reasoning

The cognitive process of collecting, weighting, and integrating clinical information to formulate diagnostic and management decisions. It includes problem representation, differential diagnosis, probability assessment, and recognition of cognitive bias.

Clinical Research

Research involving people or human health data to study disease, diagnosis, prevention, treatment, or healthcare delivery.

Clinical Research Bias

Systematic error in clinical study design or analysis, including selection bias, information bias, and confounding.

Clinical Trial

A planned study in which people receive or are observed for an intervention to evaluate its safety, effects, or outcomes. Trials may be conducted in phases and for many types of health interventions.

Clinical Trial Phases

Stages used to study a medical intervention. Phase I primarily evaluates initial safety and dosing, Phase II evaluates preliminary efficacy and safety, Phase III compares treatment with standard care in larger groups, and Phase IV monitors use after approval.

Clinical Trials

Research studies involving human participants that evaluate the safety, effectiveness, risks, or best use of medical interventions or health strategies.

Clinically Proven

Clinically proven means supported by clinical evidence, but the strength, quality, population, and outcome of that evidence must be specified.

Clinicopathologic Correlation

Integration of clinical history, examination, imaging, laboratory data, and pathologic findings to reach a diagnosis. It is essential when tissue findings are limited, nonspecific, or potentially affected by sampling or artifact.

Clinistix

Clinistix is a reagent-strip method historically used to detect glucose in urine through an enzymatic color reaction.

Clinitest

Clinitest is a tablet-based copper-reduction test historically used to detect reducing substances in urine, including glucose; it is less specific than modern methods.

Clinodactyly

A congenital or developmental lateral curvature of a finger or toe, most often affecting the fifth finger and resulting from abnormal shape of a phalanx.

Clinoid Process

One of several bony projections of the sphenoid bone that form part of the cranial base and support structures around the optic canal and cavernous sinus.

CLIP

A device used to hold tissue or structures together; surgical clips can close blood vessels, mark tissue, or occlude structures such as the vas deferens.

Clip (Vascular)

A small device used to close, occlude, or secure a blood vessel during a procedure.

Alternate Names:

Vascular

Clitoral

Relating to the clitoris, an erectile organ with a dense concentration of sensory nerves at the front of the vulva.

Clitorectomy

Surgical removal of all or part of the clitoris, sometimes with removal of adjacent vulvar tissue; nonconsensual forms constitute female genital mutilation and can cause serious harm.

Clitoridectomy

Surgical removal of part or all of the clitoris. In medicine it may be performed for selected tumors or severe disease; nonmedical removal is a harmful practice with no health benefit.

Clitoris

A small, sensitive erectile organ located at the anterior junction of the labia minora, beneath the clitoral hood (prepuce). The clitoris is the primary source of female sexual pleasure and is homologous to the male penis.

Clitoromegaly

Clitoromegaly is enlargement of the clitoris, which may be congenital or acquired from androgen excess, endocrine disease, medication, or other causes.

CLL

Alternate index label; see *Chronic lymphocytic leukemia*.

Cloacal Exstrophy

The most severe congenital anomaly of the ventral abdominal wall, also called the OEIS complex (Omphalocele, Exstrophy of the bladder, Imperforate anus, Spinal defects). Incidence 1 in 200,000-400,000 births.

Alternate Names:

the OEIS.

Cloacitis

Inflammation of a cloaca or cloacal region, particularly in disorders affecting the common terminal outlet of the urinary, genital, and intestinal tracts.

Clodronate

Clodronate is a bisphosphonate that reduces bone resorption and is used in selected bone and cancer-related conditions.

Clofazimine

Clofazimine is a riminophenazine antimicrobial used mainly in multidrug treatment of leprosy and selected mycobacterial infections; skin discoloration is common.

Clomifene

Clomifene, or clomiphene, is a selective estrogen-receptor modulator used to induce ovulation in selected infertility settings.

Clomipramine

Clomipramine is a tricyclic antidepressant particularly effective for obsessive-compulsive disorder; anticholinergic, cardiac, and overdose toxicity are important.

Clonal Evolution

The selection and expansion of tumor-cell subclones with biologic or treatment-related advantages over time. It contributes to progression, metastasis, and acquired resistance to anticancer therapy.

Clonal Expansion

Rapid multiplication of genetically or immunologically related cells derived from a single precursor after activation.

Alternate Names:

Clonal proliferation; clone expansion.

Clonazepam

Clonazepam is a benzodiazepine anticonvulsant and anxiolytic that enhances GABA activity; sedation, dependence, and respiratory depression can occur.

Clonic Seizure

A seizure characterized by rhythmic, repetitive jerking caused by recurrent involuntary muscle contractions. It may be focal or generalized, with or without impaired awareness, and is classified using the clinical pattern, electroencephalography, and the underlying cause.

Clonidine

Clonidine stimulates central alpha-2 receptors to reduce sympathetic outflow and is used for hypertension and selected other conditions; abrupt withdrawal can cause rebound hypertension.

Clonorchiasis

Infection with the Chinese liver fluke, *Clonorchis sinensis*. It affects the biliary tract and may cause inflammation, obstruction, and long-term hepatobiliary complications.

Clonus

A rhythmic, involuntary oscillation of a muscle (usually ankle or patellar) produced by sustained stretch, indicating upper motor neuron (UMN) disease—hyperreflexia. Elicited by rapid dorsiflexion of the foot (sustained clonus—5 or more beats) or tapping. Associated with spasticity, hyperreflexia,

and UMN signs (Babinski).

Clopidogrel

Clopidogrel is an antiplatelet drug that irreversibly blocks the platelet P2Y₁₂ receptor, reducing arterial thrombosis risk after selected cardiovascular events or procedures.

Closed Comedo

A small skin lesion formed when a follicular opening is blocked by sebum and keratin but remains covered by the epidermis. It is commonly called a whitehead.

Alternate Names:

a whitehead.

Closed Fracture

A fracture in which the broken bone does not communicate through a wound with the external environment.

Closed Reduction

Nonoperative realignment of a fractured or dislocated bone or joint by manipulation without surgically opening the site.

Closed Reduction Of A Fractured Bone

A nonsurgical procedure that realigns a broken bone by manipulating it through the skin without opening the fracture site, usually followed by immobilization and imaging to confirm alignment.

Closed Suction Drain With Bulb

A flexible drain connected to a compressible bulb that creates gentle negative pressure to remove blood or fluid from a surgical or injured area; the bulb is emptied and output recorded as directed.

Closed-Angle Glaucoma

Glaucoma caused by obstruction of the drainage angle of the anterior chamber, preventing aqueous humor outflow and producing potentially rapid elevation of intraocular pressure.

Closed-Loop Insulin Delivery

Automated insulin delivery systems that combine

continuous glucose monitoring with an insulin pump and control algorithm. The algorithm adjusts basal insulin delivery based on sensor glucose values in real time. Hybrid closed-loop systems require user-initiated meal boluses but automatically manage basal rates.

Clostridioides Difficile

An anaerobic, spore-forming, toxin-producing gram-positive bacillus causing antibiotic-associated diarrhea and pseudomembranous colitis. Risk factors: antibiotics (especially clindamycin, fluoroquinolones, cephalosporins), hospitalization, age >65, PPIs, and immunosuppression. Toxins A and B damage colonic epithelium.

Clostridioides Difficile Infection

An infectious diarrhea caused by the bacterium *Clostridioides difficile*, typically occurring after disruption of normal gastrointestinal flora by antibiotics. *C. difficile* produces toxins A and B that cause colonic inflammation and diarrhea. Risk factors include antibiotic exposure, proton pump inhibitor use, and advanced age.

Clostridioides Difficile Pathogenesis

The mechanisms by which *C. difficile* causes disease. Antibiotic disruption of the normal gut flora allows the organism to colonize; its toxins injure the colonic mucosa and cause inflammation and diarrhea. Spores persist in the environment and facilitate transmission.

Clostridium

A genus of anaerobic, usually spore-forming bacteria that includes species capable of causing diseases such as tetanus, botulism, gas gangrene, and antibiotic-associated colitis.

Clostridium Botulinum

A gram-positive, spore-forming, anaerobic rod that produces botulinum toxin, the most potent toxin known. The toxin causes flaccid paralysis by blocking acetylcholine release at neuromuscular junctions. Infant botulism occurs from ingestion of spores in honey or soil. Wound botulism occurs from injection

of contaminated heroin.

Clostridium Difficile

Clostridioides difficile infection: a toxin-producing anaerobic gram-positive rod causing antibiotic-associated diarrhea and pseudomembranous colitis, most commonly after broad-spectrum antibiotics (clindamycin, cephalosporins, fluoroquinolones).

Clostridium Difficile Colitis

Clostridioides difficile colitis is antibiotic-associated inflammation of the colon caused by toxin-producing bacteria. It causes watery diarrhea, abdominal pain, fever, and sometimes severe colitis; diagnosis and treatment depend on symptoms, testing, recurrence, and severity.

Alternate Names:

Difficile Clostridium (*Clostridium Difficile* Colitis)

Clostridium Difficile Infection

An anaerobic, spore-forming, toxin-producing gram-positive bacillus causing antibiotic-associated diarrhea and pseudomembranous colitis. See *Clostridioides difficile*.

Clostridium Perfringens

A gram-positive, spore-forming, anaerobic rod that causes gas gangrene (clostridial myonecrosis) and food poisoning. The organism produces alpha-toxin, which causes tissue necrosis. Gas gangrene presents with crepitus, tissue necrosis, and systemic toxicity. Food poisoning presents with acute onset of diarrhea and abdominal cramps.

Clostridium Tetani

A gram-positive, spore-forming, anaerobic rod that causes tetanus. The organism produces tetanospasmin, a neurotoxin that causes spastic paralysis. Spores are found in soil and enter through wound contamination. The toxin blocks inhibitory neurotransmitters causing uncontrolled muscle contractions.

Closure

Closure is the joining or sealing of a wound, opening, vessel, duct, or anatomical structure, either

naturally or by treatment.

Clot

A clot is a semisolid mass of coagulated blood; clots may stop bleeding or obstruct a vessel and cause thrombosis or embolism.

Clot Blood (Blood Clots)

Alternate index label for Blood Clots.

Clot Buster

A medicine that dissolves a blood clot; the medical term is thrombolytic.

Cloth Dye Poisoning

Exposure to cloth dyes that may cause irritation, gastrointestinal symptoms, allergic reactions, or systemic toxicity depending on the chemical composition.

Clotrimazole

Clotrimazole is an azole antifungal used topically or orally in selected formulations for fungal infections of skin, mouth, or vagina.

Clotting

The cascade of enzymatic reactions converting circulating fibrinogen into an insoluble fibrin clot. See Blood Coagulation.

Cloudy Cornea

Loss of corneal transparency from edema, scarring, infection, dystrophy, trauma, or deposits, potentially causing blurred vision, pain, photophobia, or glare.

Clove Oil

Clove oil contains eugenol and is used in some dental and topical products; concentrated ingestion or exposure can cause irritation, liver injury, or seizures.

Clozapine

Clozapine is an atypical antipsychotic for treatment-resistant schizophrenia and selected suicidality risk; neutropenia, myocarditis, seizures, constipation,

and metabolic effects require monitoring.

Club Drugs List And Side Effects

Club drugs include substances such as MDMA, ketamine, GHB, and others used recreationally in social settings. Effects may include agitation, hallucinations, seizures, dangerous temperature elevation, respiratory depression, cardiovascular toxicity, impaired consent, and overdose.

Clubbing

Bulbous enlargement of the distal fingers or toes with increased nail curvature and loss of the normal nail-fold angle. It reflects proliferation of connective tissue in the distal digits and may be associated with chronic pulmonary disease, cyanotic heart disease, inflammatory bowel disease, or malignancy.

Clubbing Of The Fingers Or Toes

Bulbous enlargement of fingertip or toe soft tissue with increased nail curvature caused by chronic changes in the nail bed; it can accompany lung, heart, gastrointestinal, or other disease.

Clubfoot

A birth defect in which a baby's foot is turned inward and downward. The foot may be stiff and difficult to move into a normal position.

Clubfoot (Talipes Equinovarus)

Alternate name; see *Clubfoot*.

Clubfoot Repair

A medical procedure involving clubfoot repair. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Clubhand

A congenital deformity of the hand in which the wrist and hand deviate toward the radial or ulnar side, often associated with deficiency or abnormal development of forearm bones.

Clumping

Clumping is aggregation of cells, particles, or proteins into visible groups, as in red-cell agglutination or platelet aggregation.

Cluster

A group of similar events, findings, cells, or cases occurring together in space, time, or clinical pattern.

Cluster Headache

A primary headache disorder causing severe one-sided orbital or periorbital pain in attacks lasting about 15–180 minutes, often with tearing, conjunctival redness, nasal symptoms, eyelid swelling, or facial sweating on the same side.

Cluster Headaches

Cluster headache causes repeated attacks of severe one-sided orbital or temporal pain with tearing, nasal symptoms, eyelid drooping, or restlessness. Attacks occur in bouts and require targeted acute and preventive treatment; a first sudden severe headache needs urgent evaluation.

Clutton's Joints

Painless bilateral swelling of the knees or other large joints associated with congenital syphilis, caused by chronic synovitis.

Clysis

Clysis is administration of fluid into the rectum, historically used for hydration, medication, or bowel treatment; modern use is limited.

Cm (Centimeter)

The abbreviation for centimeter, a unit of length equal to one hundredth of a meter or 10 millimeters.

CMV

Cytomegalovirus, a common herpesvirus that may cause mild or asymptomatic infection but can cause serious disease in fetuses and people with weakened immunity.

Alternate Names:

Cytomegalovirus (CMV)

CMV - Gastroenteritis/Colitis

Inflammation and ulceration of the stomach or intestine caused by cytomegalovirus, usually in people with substantial immune suppression. It may cause abdominal pain, fever, diarrhea, bleeding, or perforation and requires tissue or other targeted evaluation.

CMV Blood Test

A blood test that detects cytomegalovirus antibodies, antigen, or viral DNA, depending on the method. Interpretation differs between antibody testing for past exposure and nucleic-acid or antigen testing for active replication, especially in immunocompromised patients.

CMV Colitis

Colonic inflammation caused by cytomegalovirus reactivation in immunocompromised patients including those with HIV and organ transplant recipients. Presents with bloody diarrhea and abdominal pain. Diagnosed by colonoscopy with biopsy showing characteristic owl-eye intranuclear inclusions in enlarged cells.

CMV Pneumonia

Pneumonia caused by cytomegalovirus, usually in people with substantial immune suppression; fever, cough, shortness of breath, and low oxygen may occur and require targeted evaluation and antiviral treatment.

CMV Retinitis

A retinal infection caused by cytomegalovirus, usually in people with severe cellular immunosuppression. It can cause floaters, blurred vision, field loss, or blindness and requires urgent ophthalmic and systemic antiviral management.

CNS Infection

Infection of the brain, spinal cord, meninges, or cerebrospinal-fluid spaces caused by bacteria, viruses, fungi, parasites, or other organisms; presentation and treatment depend on the pathogen and anatomic site.

CNS Lymphoma

A lymphoma involving the central nervous system, most often diffuse large B-cell lymphoma. It may involve brain parenchyma, meninges, eyes, or cerebrospinal fluid and requires urgent hematologic and neurologic evaluation.

CO Poisoning

Toxicity from inhalation of carbon monoxide, a colorless, odorless gas binding hemoglobin 200-300 times more avidly than oxygen, forming carboxyhemoglobin (COHb) and reducing oxygen delivery. See carbon monoxide poisoning.

CO2 Blood Test

A blood chemistry measurement that usually estimates serum total carbon dioxide, which largely reflects bicarbonate. It helps assess acid-base and electrolyte disorders but must be interpreted with sodium, chloride, kidney function, and, when needed, blood-gas results.

Coagulate

To change from a liquid to a semisolid or solid state, especially the formation of a blood clot through coagulation.

Alternate Names:

Clot; clot formation.

Coagulation

The physiologic enzymatic cascade (secondary hemostasis) through which circulating plasma proenzymes are sequentially activated to convert soluble fibrinogen into an insoluble fibrin polymer mesh that stabilizes the primary platelet plug. Initiated primarily in vivo via the tissue factor (extrinsic) pathway upon vascular injury and amplified by the contact (intrinsic) pathway, converging on factor Xa activation and prothrombinase assembly to generate a burst of thrombin.

Coagulation Cascade

The coagulation cascade is a series of enzymatic reactions leading to fibrin clot formation, involving serine protease activation cascades. The intrinsic pathway is initiated by contact activation when

factor XII binds exposed collagen, sequentially activating factors XI, IX, and the cofactor VIII.

Coagulation Defects

Coagulation defects are abnormalities of clotting proteins, platelets, vessels, or regulation that cause excessive bleeding or inappropriate thrombosis.

Coagulation Factor

A blood protein required for the coagulation cascade and formation of a stable fibrin clot; deficiencies or inhibitors can cause bleeding, while excessive activation can contribute to thrombosis.

Coagulation Factor Deficiencies

Inherited or acquired disorders in which one or more clotting factors are reduced or function abnormally, causing excessive bleeding. The clinical pattern and laboratory findings depend on the affected factor.

Coagulation Profile

Panel of tests assessing the coagulation cascade. PT (prothrombin time, 11-14 sec): measures extrinsic and common pathways; elevated in vitamin K deficiency, warfarin, liver disease, DIC. INR standardizes PT for warfarin monitoring.

Coagulation Time

Coagulation time is the time required for a blood sample to form a clot under specified test conditions; modern tests usually measure particular pathways more precisely.

Coagulative Necrosis

The most common form of necrosis, typically resulting from ischemic injury or infarction in most solid organs except the brain. The fundamental process involves denaturation of both structural proteins and enzymatic proteins, which prevents the immediate liquefaction of the dead tissue.

Coagulopathy

A disorder of hemostasis that impairs normal clot formation or regulation and predisposes to abnormal bleeding or thrombosis. It may result from

coagulation-factor deficiency, platelet dysfunction, anticoagulant exposure, liver disease, consumptive coagulopathy, or systemic illness.

Coal Worker'S Pneumoconiosis

Pneumoconiosis from coal dust. Simple CWP shows upper lobe macules. Complicated CWP (progressive massive fibrosis) involves coalescent macules. May be associated with Caplan syndrome.

Coarctation

A localized narrowing of a tubular anatomical structure, especially coarctation of the aorta, which obstructs blood flow and produces characteristic pressure differences.

Coarctation Of The Aorta

A congenital narrowing of the aorta, typically near the ductus arteriosus. Presents with upper extremity hypertension, diminished femoral pulses, and a blood pressure gradient between arms and legs. Associated with bicuspid aortic valve and other congenital heart defects.

Coarse Crackles

Lower-pitched, longer-duration crackles heard throughout inspiration, caused by fluid or secretions in larger airways. They are heard in COPD, bronchiectasis, pneumonia, and pulmonary edema. They are less specific than fine crackles.

Coats Disease

An uncommon retinal disorder characterized by abnormal retinal blood vessels, leakage, exudation, and sometimes retinal detachment, usually affecting one eye in childhood.

Coats Disease (Coats' Disease)

Alternate index label for Coats' Disease.

Cobalamin

Vitamin B12, a vitamin required for DNA synthesis, red-blood-cell formation, and normal neurologic function.

Cobalamin Deficiency

Deficiency of vitamin B12 (cobalamin), a cofactor for methionine synthase (methylcobalamin) and methylmalonyl-CoA mutase (adenosylcobalamin).

Cobalt

Cobalt is a metal present in vitamin B12 and some industrial materials; excess exposure can injure the heart, thyroid, lungs, or skin.

Cobalt Poisoning

Toxic exposure to cobalt that may cause gastrointestinal, neurologic, thyroid, cardiac, or lung effects after significant or prolonged exposure.

Cobalt-57

Calibration source for gamma cameras. Half-life 271 days. Used for quality control.

Cobalt-Chromium

Base metal alloy for cast restorations. Strong, corrosion-resistant. Used for partial dentures, crowns.

Cocaine

A stimulant drug that blocks monoamine reuptake and can also act as a local anesthetic; use can cause dependence, cardiovascular complications, seizures, stroke, and overdose.

Cocaine Use Disorder

A problematic pattern of cocaine use leading to clinically significant impairment or distress. It may involve craving, impaired control, continued use despite harm, tolerance, or withdrawal.

Alternate Names:

Cocaine And Crack Abuse

Cocaine Cardiotoxicity

Cocaine blocks norepinephrine and dopamine reuptake, causing sympathomimetic effects: tachycardia, hypertension, and coronary vasospasm. Acute myocardial infarction can occur even with normal coronary arteries due to vasospasm, accelerated atherosclerosis, and prothrombotic effects. Treat-

ment: benzodiazepines for sympatholysis.

Cocaine Intoxication

Cocaine intoxication causing sympathetic stimulation, agitation, hyperthermia, seizures, hypertension, arrhythmias, myocardial ischemia, or stroke.

Cocaine Toxicity

A toxic state caused by cocaine, often producing agitation, high blood pressure, rapid heart rate, hyperthermia, seizures, coronary vasospasm, myocardial infarction, or stroke. Treatment is supportive and may include sedation, temperature control, and management of cardiovascular or neurologic complications.

Cocaine Withdrawal

A medicine or chemical compound identified by cocaine withdrawal. Its uses, effects, and risks depend on the specific substance and clinical context.

Cocarboxylase

Cocarboxylase is thiamine pyrophosphate, the active coenzyme form of vitamin B1 required for several oxidative carbohydrate and amino-acid reactions.

Coccidioides Complement Fixation

A blood test that detects complement-fixing antibodies to *Coccidioides*; antibody levels can support diagnosis and help assess the severity or progression of coccidioidomycosis.

Coccidioides Immitis

A dimorphic soil fungus and a principal cause of coccidioidomycosis, or Valley fever, acquired by inhaling airborne arthroconidia. Infection is usually pulmonary but can disseminate, especially in people with impaired cellular immunity.

Coccidioides Precipitin Test

A serologic test that detects antibodies to *Coccidioides*, typically early immunoglobulin M antibodies, and supports evaluation of suspected coccidioidomycosis when interpreted with symptoms and other

testing.

Coccidioidomycosis

A fungal infection caused by *Coccidioides* species, usually acquired by inhaling spores.

Common name: Valley fever

Coccidiosis

Infection by coccidian protozoa. In humans it may cause intestinal disease or, in immunocompromised patients, infection of other organs.

Coccus

A spherical or approximately round bacterium; cocci may occur singly, in pairs, chains, or clusters.

Coccydynia

Pain in the region of the coccyx, often worsened by sitting. Causes include trauma, childbirth, degenerative change, and less commonly infection or tumor.

Alternate Names:

Pain Tailbone (Coccydynia)

Coccygectomy

Surgical removal of the coccyx, reserved for selected cases of severe persistent coccygeal pain or a lesion that has not responded to conservative treatment.

Coccyx

Terminal vertebral column part with 3-5 fused vertebrae. Articulates with the sacrum, allowing limited flexion important during childbirth. Provides attachment for gluteus maximus, coccygeus, and levator ani (pelvic floor). A common site of pain (coccydynia) following falls, direct trauma, and prolonged sitting.

Cochlea

Snail-shaped hearing organ in the inner ear, coiled approximately 2.5 turns around the modiolus. Three fluid-filled chambers: scala vestibuli (perilymph), scala media (endolymph, organ of Corti), scala tympani (perilymph).

Cochlear Implant

A cochlear implant is an electronic medical device that bypasses damaged or absent inner ear hair cells to provide direct electrical stimulation of the auditory nerve, restoring a perception of sound in individuals with severe to profound sensorineural hearing loss who derive limited benefit from hearing aids.

Cockayne Syndrome

A rare inherited disorder of DNA repair characterized by progressive neurologic dysfunction, growth failure, developmental impairment, photosensitivity, and a prematurely aged appearance.

Code Black

A hospital emergency alert whose meaning varies by institution, often indicating a severe security threat, bomb threat, or other critical incident; local policy determines the response.

Code Blue

A hospital emergency signal indicating that a person may be in cardiopulmonary arrest and requires immediate resuscitation response.

Code, Hospital

A standardized hospital announcement used to summon a specialized response to an emergency; the meaning of each code depends on the hospital's policy.

Codeine Overdose

Overdose of codeine, an opioid, that can cause respiratory depression, impaired consciousness, pinpoint pupils, and coma. Naloxone may be required in emergency care.

Codman Triangle

A characteristic periosteal reaction seen on plain radiographs, formed when an aggressive bone lesion elevates the periosteum away from the bone cortex, creating a triangular area of new reactive bone at the angle between the raised periosteum and underlying cortex.

Codon

Sequence of three nucleotides coding for one amino acid or stop signal. 64 possible codons: 61 code for amino acids, 3 are stop codons (UAA, UAG, UGA). Degeneracy: multiple codons code for same amino acid.

Coenzyme

A non-protein organic molecule that binds to an enzyme (apoenzyme) and is required for its catalytic activity. Coenzymes often act as transient carriers of specific chemical groups or electrons.

Coenzyme Q

A lipid-soluble molecule in mitochondrial membranes that transfers electrons during cellular energy production.

Coenzyme Q10

CoQ10, a lipid-soluble compound in mitochondrial electron transport and an antioxidant; it is also sold as a dietary supplement, although benefits vary by condition.

Cofactor

A non-protein chemical compound or metallic ion required for an enzyme's catalytic activity. Cofactors include: (1) Metal ions (Mg^{2+} , Zn^{2+} , Fe^{2+}/Fe^{3+} , Cu^{2+} , Mn^{2+} , Ca^{2+} , Mo, Se) that stabilize structure, participate in redox, or act as Lewis acids; (2) Coenzymes (organic cofactors, often vitamin-derived).

Coffin-Lowry Syndrome

An X-linked semi-dominant syndromic neurodevelopmental disorder caused by loss-of-function mutations in the *RPS6KA3* gene on Xp22.12, which encodes ribosomal protein S6 kinase alpha-3. Severely affected hemizygous males present with profound intellectual disability, characteristic craniofacial findings (prominent forehead, hypertelorism, downward-slanting palpebral fissures, thick prominent lips, wide nose with anteverted nares), tapered fingers, kyphoscoliosis, and pathognomonic stimulus-induced drop attacks (SIDAs) triggered by unexpected tactile or auditory stimuli.

Alternate Names:
CLS

Cogan'S Syndrome

An autoimmune vasculitis characterized by interstitial keratitis and vestibular dysfunction including vertigo, hearing loss, and tinnitus. May involve large vessels including the aorta, causing aortic aneurysm or regurgitation. Treatment includes corticosteroids and immunosuppressive agents.

Cognition

The mental processes involved in acquiring, storing, retrieving, and using information, including attention, memory, language, reasoning, and judgment.

Cognitive

Relating to mental processes such as attention, memory, learning, language, reasoning, perception, and decision-making.

Cognitive Behavior Therapy

A structured psychotherapy that identifies and changes unhelpful thoughts and behaviors.

Cognitive Behavioral Therapy

A structured, time-limited psychotherapy that identifies and modifies maladaptive thought patterns (cognitive distortions) and behaviors. It is based on the cognitive model that thoughts, feelings, and behaviors are interconnected. CBT uses techniques including cognitive restructuring, behavioral activation, exposure, and skills training.

Cognitive Behavioral Therapy For Back Pain

A structured psychological treatment that addresses pain-related thoughts, behaviors, sleep, activity, and coping to reduce disability and improve function in chronic back pain.

Cognitive Behavioral Therapy For Insomnia

Cognitive behavioral therapy for insomnia is a structured, evidence-based program that targets the

thoughts and behaviors perpetuating poor sleep. Components include sleep restriction, stimulus control, cognitive restructuring, and sleep hygiene education.

Cognitive Decline

A reduction over time in one or more cognitive abilities, such as memory, attention, language, reasoning, or visuospatial function.

Cognitive Distortions

Systematic errors in thinking that perpetuate negative emotional states and maladaptive behaviors. Common distortions include all-or-nothing thinking, catastrophizing, mind reading, overgeneralization, emotional reasoning, and should statements. CBT focuses on identifying and modifying these patterns.

Cognitive Fitness

The active maintenance of mental sharpness and brain health through stimulating activities, physical exercise, social engagement, and proper nutrition. Regular challenges such as learning new skills, puzzles, reading, and social interaction support neuroplasticity and cognitive reserve.

Cognitive Function

The mental processes involved in attention, memory, language, reasoning, learning, judgment, and problem solving.

Alternate Names:

Cognition; cognitive ability.

Cognitive Health And Memory

Cognitive health encompasses the ability to think clearly, learn, remember, and make decisions, with memory being a core component that involves encoding, storing, and retrieving information. Age-related changes in processing speed and recall are normal, but persistent decline may signal mild cognitive impairment or dementia.

Cognitive Impairment

Decline in one or more cognitive domains: memory, attention, language, visuospatial, executive function. Types: subjective cognitive decline, mild cognitive

impairment (MCI), dementia. MCI: functional independence preserved, impairment on testing. Dementia: functional impairment.

Cognitive Rehabilitation

Structured therapy to improve cognitive function or help a person compensate for difficulties with memory, attention, language, or executive skills after illness or injury.

Cognitive Reserve

The brain's ability to use alternative strategies or networks to maintain function despite injury or disease.

Cognitive Screening

A brief assessment used to identify possible problems with memory, attention, language, or other cognitive functions. It does not by itself diagnose dementia or another disorder.

Cognitive Therapy

A psychological treatment that helps a person identify and change unhelpful thoughts and beliefs and develop more adaptive behaviors; cognitive behavioral therapy combines it with behavioral methods.

Cohort

A group of people who share a defined characteristic or experience during a specified period and are studied over time or compared with another group.

Cohort Study

Follows exposed and unexposed groups over time. Prospective or retrospective. Measures incidence, relative risk.

Coil (Vascular)

A catheter-delivered wire device used to block selected blood vessels, often to treat aneurysms or bleeding.

Alternate Names:

Vascular

Coincidence Detection

Principle of PET imaging. Two 511 keV photons detected simultaneously (within nanoseconds) by opposing detectors. Line of response defined. Time-of-flight PET improves localization.

Coitus

Sexual intercourse, traditionally referring to penile-vaginal penetration; sexual activity is broader than this term.

Coitus Interruptus

The withdrawal method of contraception, in which the penis is removed from the vagina before ejaculation; it is less reliable than many other contraceptive methods.

Coke

Slang for cocaine, a stimulant drug; the word can also mean a carbon fuel product, so context is essential.

Cold (Common Cold)

Alternate index label for Common Cold.

Cold Abscess

An abscess that develops slowly with little heat, redness, or acute inflammation, classically in tuberculosis or other chronic infections.

Cold Agglutinin

An antibody that causes red blood cells to agglutinate at relatively low temperatures. Pathologic cold agglutinins can cause autoimmune hemolytic anemia.

Cold Agglutinin Disease

Autoimmune hemolytic anemia caused by IgM autoantibodies directed against the I/i antigens on RBC membranes (maximum reactivity at $< 30^{\circ}\text{C}$, optimal 4°C).

Cold Agglutinin Syndrome

An immune-mediated disorder in which antibodies react with red blood cells at low temperatures,

potentially causing hemolytic anemia or circulation-related symptoms.

Cold Agglutinins

IgM autoantibodies that bind red blood cell antigens at temperatures below 37 degrees Celsius, causing agglutination and potentially complement-mediated hemolysis in cooler peripheral circulation.

Cold Allergy

Alternate index label; see *Cold urticaria*.

Cold Exposure

Alternate index label; see *Hypothermia*.

Cold Flu Allergy Treatments (Cold, Flu, Allergy)

Alternate index label for Cold, Flu, Allergy.

Cold Hands

Hands that feel unusually cold, which may result from environmental exposure, reduced blood flow, Raynaud phenomenon, anemia, thyroid disease, or other medical conditions.

Cold Injury

Tissue or systemic injury caused by exposure to low temperature, including hypothermia, frostbite, chilblains, and non-freezing cold injury. Severity depends on temperature, exposure duration, moisture, wind, and perfusion; affected patients require assessment for core-temperature disturbance and local tissue damage.

Cold Intolerance

Unusual sensitivity or inability to tolerate cold temperatures, which may occur with hypothyroidism, anemia, low body weight, vascular disease, or other conditions.

Cold Medicines And Children

Safety guidance for giving medicines to children with cold symptoms; many combination products provide little benefit in young children and can cause dosing errors or serious adverse effects.

Cold Sore

A contagious infection of the lips or skin around the mouth, usually caused by herpes simplex virus type 1 and occasionally by HSV-2. It often begins with tingling or burning, followed by grouped fluid-filled blisters that crust and heal in about 1–2 weeks. The virus remains in the body and can reactivate, sometimes after illness, stress, fever, or sunlight exposure.

Cold Urticaria

Hives or swelling triggered by exposure of the skin to cold.

Alternate Names:

Cold Allergy

Cold Wave Lotion Poisoning

Exposure to cold-wave lotion, which may contain thioglycolates or other irritants and can cause burns, gastrointestinal symptoms, or respiratory irritation.

Cold, Common

Alternate index label; see *Common cold*.

Cold, Flu, Allergy

Colds, influenza, and allergies can all cause nasal symptoms and cough but differ in cause, fever pattern, duration, and contagiousness. Viral illness usually improves with supportive care, while allergies respond to trigger reduction and appropriate antihistamine or nasal treatment.

Alternate Names:

Cold Flu Allergy Treatments (Cold, Flu, Allergy)

Colds And Flu - Antibiotics

An explanation that antibiotics treat susceptible bacterial infections, not uncomplicated viral colds or influenza; unnecessary use causes adverse effects and antimicrobial resistance.

Colectomy

Surgical resection of part or all of the colon. Right hemicolectomy: for cecal/ascending colon cancer. Left hemicolectomy: for descending/sigmoid cancer. Sigmoidectomy: for diverticulitis or sigmoid cancer.

Total colectomy: for FAP, UC, or Lynch syndrome.

Colibacillosis

Disease caused by pathogenic strains of *Escherichia coli*, which may produce intestinal infection, urinary-tract infection, sepsis, or other illness.

Colic

Sudden, severe, often cramping pain caused by contraction or obstruction of a hollow organ. The term is also used for recurrent unexplained crying in an otherwise healthy infant.

Colicky Pain

Intermittent, wave-like cramping pain caused by contraction, distension, or obstruction of a hollow organ; common examples include biliary, renal, and intestinal colic.

Coliform

A group of generally gram-negative bacteria that ferment lactose with acid and gas production under defined laboratory conditions; their presence in water can indicate fecal contamination, although not all coliforms are fecal pathogens.

Colitis

Colitis is inflammation of the colon caused by infection, inflammatory bowel disease, reduced blood flow, medicines, radiation, or other injury.

Colitis Collagenous (Lymphocytic Colitis)

Alternate index label for Lymphocytic Colitis.

Colitis Lymphocytic (Lymphocytic Colitis)

Alternate index label for Lymphocytic Colitis.

Colitis Microscopic (Lymphocytic Colitis)

Alternate index label for Lymphocytic Colitis.

Colitis Ulcerative (Ulcerative Colitis)

Alternate index label for Ulcerative Colitis.

Colitis, Ischemic

Alternate index label; see *Ischemic colitis*.

Colitis, Microscopic

Alternate index label; see *Microscopic colitis*.

Colitis, Pseudomembranous

Alternate index label; see *Pseudomembranous colitis*.

Colitis, Ulcerative

Alternate index label; see *Ulcerative colitis*.

Collagen

The major structural protein of connective tissue, composed of three polypeptide chains arranged in a triple helix. Different collagen types provide tensile strength to skin, bone, cartilage, tendons, blood vessels, and basement membranes.

Collagen Disease

A systemic disorder involving connective tissue and often immune-mediated injury to collagen-rich structures. The term has historically included diseases such as systemic lupus erythematosus and systemic sclerosis.

Collagen Implant

Biocompatible scaffold for tissue engineering. Types: bovine, porcine, synthetic. Used for hernia repair, soft tissue augmentation.

Collagen Vascular Disease

An older term for systemic connective-tissue or autoimmune diseases such as lupus, systemic sclerosis, and inflammatory myopathies; modern diagnosis uses the specific disease name.

Collagenase

An enzyme that breaks down collagen in connective tissue.

Alternate Names:

Collagen-digesting enzyme; collagen-degrading enzyme.

Collagenous Colitis

Chronic inflammatory bowel disease characterized by thickening of the subepithelial collagen band in the colonic mucosa to greater than ten micrometers normally less than five micrometers with preservation of the epithelial surface. Presents with chronic watery non-bloody diarrhea primarily in middle-aged women.

Collapsed Arch

Alternate index label; see *Flatfeet*.

Collapsed Lung

A common description of pneumothorax, in which air enters the pleural space and partially or completely collapses the lung. It may cause sudden chest pain and shortness of breath and can be life-threatening when tension develops. See also Pneumothorax.

Collapsed Lung (Pneumothorax)

Air in the pleural space causing partial or complete lung collapse; symptoms may include sudden chest pain and shortness of breath, and tension pneumothorax is an emergency.

Collateral

Providing an alternative or parallel route, such as collateral blood vessels that supply tissue around an obstruction; in other contexts it means accompanying or secondary.

Collateral Circulation

Alternative vascular pathways that develop when coronary arteries become chronically narrowed. Collaterals connect different coronary artery territories and can partially compensate for stenosis. They develop gradually in response to ischemia and are more prominent in patients with chronic coronary artery disease.

Collateral Ligament Injury

MCL (medial) or LCL (lateral) knee ligament injury. MCL more common, often from valgus stress. Treatment: braces, physical therapy, surgical repair if severe.

Collecting Duct

Final segment of the nephron where fine-tuning of urine concentration and composition occurs. Principal cells respond to antidiuretic hormone by inserting aquaporin-2 water channels into the luminal membrane, enabling water reabsorption and urine concentration.

College Students And The Flu

Influenza prevention and care for college students, emphasizing vaccination, hand and respiratory hygiene, staying home when ill, and prompt assessment for severe symptoms or high-risk conditions.

Colles Fracture

A fracture of the distal radius with dorsal displacement and angulation of the distal fragment, usually caused by falling on an outstretched hand.

Collimator

Lead device with holes to define gamma ray direction. Types: parallel-hole, pinhole, converging, diverging. Resolution vs. sensitivity trade-off.

Colloids

Solutions containing large molecules that remain in the intravascular space, providing greater volume expansion per milliliter than crystalloids. Albumin (5 percent, 25 percent) is a natural plasma protein used for volume expansion and specific indications (cirrhosis with paracentesis, nephrotic syndrome).

Coloboma

A congenital defect caused by failure of the optic fissure to close during embryonic development, resulting in a keyhole-shaped gap in any ocular structure including the iris, retina, choroid, or optic nerve. Iris colobomas are the most common and appear as a comma-shaped defect in the inferior iris.

Coloboma Of The Iris

A congenital keyhole- or notch-shaped defect in the iris caused by incomplete embryonic closure, sometimes associated with colobomas in the retina, optic nerve, or other organs.

Colocolonic Intussusception

telescoping of a segment of colon into an adjacent segment, less common in adults than the ileocolic type and usually associated with a pathological lead point such as a polyp, tumor, or lipoma. Adults typically present with intermittent abdominal pain, bloating, and change in bowel habits.

Cologne Poisoning

Ingestion of cologne containing alcohol and fragrances that may cause intoxication, hypoglycemia, respiratory depression, or gastrointestinal irritation.

Cologuard

A stool-based colorectal-cancer screening test combining DNA-marker analysis with a fecal blood test; a positive result requires diagnostic colonoscopy.

Colon

The largest part of the large intestine, extending from the cecum to the rectum. Anatomically divided into ascending colon (right side), transverse colon (across the abdomen), descending colon (left side), and sigmoid colon (S-shaped, leading to the rectum).

Colon Cancer

Malignant neoplasm arising from the colonic epithelium, the third most common cancer worldwide and a leading cause of cancer-related mortality. Most arise from adenomatous polyps through the adenoma-carcinoma sequence over ten to fifteen years.

Alternate Names:

Cancer, Colon Colorectal Cancer (Colon Cancer)

Colon Cancer Prevention

Colorectal-cancer risk can be reduced through recommended screening, physical activity, healthy weight, limiting alcohol, avoiding tobacco, and a diet rich in fiber-containing plant foods. Screening can detect precancerous polyps and early cancer before symptoms develop.

Colon Cancer Screening

Testing intended to detect colorectal cancer or pre-

cancerous polyps before symptoms develop. Options include stool-based tests and direct visualization such as colonoscopy; the appropriate method and interval depend on age, risk, prior findings, and local guidelines.

Colon Polyps

Growths arising from the inner lining of the colon; some types can develop into colorectal cancer.

Alternate Names:

Polyps, Colon Polyps Colon (Colon Polyps) Polyps Rectal (Colon Polyps)

Colonic Diverticulitis (Diverticulosis)

Alternate index label for Diverticulosis.

Colonic Inertia

Markedly slow movement of contents through the colon, causing severe or persistent constipation.

Colonic Ischemia

Decreased blood flow to the colon causing mucosal or transmural injury. Most commonly affects watershed areas particularly the splenic flexure and rectosigmoid junction. Causes include non-occlusive hypoperfusion from cardiogenic shock, vasoconstrictive medications, and vasculitis.

Colonic Stenosis

Narrowing of the colon that may obstruct passage of intestinal contents.

Colonization

Presence and multiplication of microorganisms on or in a body site without tissue invasion or clinical disease. Colonization may precede infection but does not by itself establish that symptoms are caused by the organism.

Colonoscopy

Colonoscopy uses a flexible camera to examine the rectum and entire colon, allowing biopsy and removal of polyps or other lesions.

Colony-Forming Unit

A unit used to estimate the number of viable microorganisms in a sample, derived from counting colonies grown on agar after plating a diluted sample. Each colony is assumed to arise from a single viable microorganism (or a cluster); results are reported as CFU/mL or CFU/g.

Color Blindness

Reduced or absent ability to distinguish certain colors under normal lighting. It is often inherited, especially through X-linked variants, but may also result from eye disease, neurologic disease, aging, or medicines.

Alternate Names:

Deficient Color Vision Poor Color Vision

Color Doppler

An ultrasound method that displays the direction and relative speed of blood flow in color over a grayscale image.

Color Vision Deficiency

Color vision deficiency, commonly known as color blindness, is a condition in which the ability to perceive and discriminate certain colors is reduced or absent, most commonly affecting the perception of red and green hues, and it is caused by abnormalities in the cone photoreceptors responsible for color vision.

Color Vision Test

An eye test that assesses the ability to distinguish colors or identify color patterns. It helps detect inherited color-vision deficiency and acquired abnormalities caused by eye, optic-nerve, brain, medicine, or toxic disorders.

Colorado Tick Fever

An acute viral infection transmitted by ticks and characterized by fever, headache, myalgia, and sometimes leukopenia or rash.

Colorectal Adenocarcinoma

The third most common cancer worldwide and

the second leading cause of cancer death, arising from the colonic epithelium through the adenoma-carcinoma sequence over 10 to 15 years. The pathogenesis involves sequential accumulation of genetic mutations beginning with APC loss, followed by KRAS activation, TP53 loss, and other alterations.

Colorectal Adenoma

A gland-forming growth in the colon or rectum that may develop into colorectal cancer.

Colorectal Cancer

A malignant tumor of the colon or rectum. It may cause bleeding, altered bowel habits, anemia, abdominal pain, or no early symptoms; evaluation and treatment depend on the tumor's location and stage.

Colorectal Cancer (Colon Cancer)

Alternate index label for Colon Cancer.

Colorectal Cancer Genetics

The study of inherited variants that increase the risk of colorectal cancer. Important inherited syndromes include Lynch syndrome, involving DNA mismatch-repair genes, and familial adenomatous polyposis, involving the APC gene.

Colorectal Cancer Screening

Public health strategy to detect colorectal cancer and precancerous polyps in asymptomatic individuals. Current guidelines recommend screening beginning at age forty-five.

Colorectal Cancer Screening Tests

Tests used to detect colorectal cancer or precancerous polyps in people without symptoms. They include stool-based tests and examinations that visualize the colon, such as colonoscopy, CT colonography, and flexible sigmoidoscopy. The appropriate test and interval depend on age, risk, and local guidance.

Colorectal Polyps

Growths arising from the lining of the colon or rec-

tum. They may be non-neoplastic, adenomatous, or serrated; some can progress to colorectal cancer, so removal and microscopic examination guide surveillance.

Colorectal Surgery

Surgical treatment of diseases of the colon, rectum, and anus. Procedures may include segmental bowel resection, restorative reconstruction, stoma formation, or local procedures, depending on the disease and its extent.

Colostomy

A surgically created opening that brings part of the colon through the abdominal wall to divert stool into an external appliance.

Colostrum

The thick, yellowish fluid produced by the mammary glands in late pregnancy and the first few days postpartum. Rich in immunoglobulins (especially IgA), proteins, leukocytes, and growth factors.

Colpitis

Inflammation of the vagina, often due to infection, altered vaginal flora, irritation, or hormonal change.

Colporrhaphy

A surgical repair of vaginal wall defects performed to correct pelvic organ prolapse. Anterior colporrhaphy plicates and reinforces the attenuated pubocervical fascia to treat a cystocele (bladder prolapse), while posterior colporrhaphy plicates the rectovaginal fascia and perineal body to repair a rectocele (rectal prolapse) and restore normal vaginal caliber.

Colposcope

A lighted, magnifying instrument used to examine the cervix, vagina, or vulva in detail.

Colposcopy

Magnified visualization of the cervix, vagina, and vulva after application of acetic acid. Indication: abnormal Pap smear (ASCUS with high-risk HPV, LSIL, HSIL, AGC). Acetic acid causes acetowhite le-

sions in dysplastic epithelium. Lugol iodine (Schiller test): normal glycogenated epithelium stains brown; abnormal epithelium does not.

Colposcopy - Directed Biopsy

Removal of a small tissue sample from an abnormal area identified during colposcopy of the cervix, vagina, or vulva for histopathologic evaluation of dysplasia, precancer, or cancer.

Columnar Epithelium Lined Lower Esophagus

Alternate index label; see *Barrett's esophagus*.

Coma

A state of profound unresponsiveness in which a person cannot be awakened and does not show purposeful responses. It is a medical emergency with many possible causes.

Alternate Names:

Severe Brain Injury, Coma

Coma Myxedema (Myxedema Coma)

Alternate index label for Myxedema Coma.

Coma Scale

A standardized tool for describing level of consciousness using eye opening, verbal response, and motor response.

Combination Inhalers

Inhalers containing both an inhaled corticosteroid and a long-acting beta-2 agonist. Examples include fluticasone/salmeterol (Advair), budesonide/formoterol (Symbicort), and fluticasone furoate/vilanterol (Breo Ellipta). They simplify asthma and COPD maintenance therapy and improve adherence.

Combination Therapy

Use of two or more treatments together to address a disease or mechanism. The benefits and risks depend on the condition and the specific combination.

Combined Hormone Therapy

Hormone therapy containing both estrogen and a progestogen, generally used when endometrial protection is needed in a person with a uterus.

Alternate Names:

Estrogen-progestogen therapy; combined menopausal hormone therapy.

Combined Hyperlipidemia

A lipid disorder characterized by elevated cholesterol and triglycerides, often with increased atherogenic lipoproteins.

Alternate Names:

Mixed hyperlipidemia; combined hyperlipoproteinemia.

Combined Spinal-Epidural Analgesia

A neuraxial analgesia technique for labor that combines a spinal injection (short-acting opioid or opioid + low-dose local anesthetic) with an epidural catheter. Advantages: rapid onset of pain relief (3-5 minutes from the spinal component), minimal motor block, and the ability to dose through the epidural catheter for prolonged labor.

Comedocarcinoma

A form of ductal carcinoma in situ of the breast characterized by central necrosis within malignant ducts, which may produce a comedo-like appearance.

Comedones

Plugs of sebum and keratin within hair follicles, appearing as open blackheads or closed whiteheads and representing a primary lesion of acne.

Comfort Measures Only

A care plan focused on relief of pain, breathlessness, distress, and other symptoms rather than treatments intended to prolong life.

Commando Procedure

A high-complexity, heroic cardiac surgical operation performed for extensive, invasive infective endocarditis involving the aortic and mitral valves with destructive extension into the aortomitral con-

tinuity (the intervalvular fibrosa) and subaortic curtain. Involves radical en-bloc surgical debridement of infected, necrotic valvular and annular tissues, complete surgical reconstruction of the intervalvular fibrous skeleton using a folded bovine pericardial patch, and simultaneous double valve replacement (aortic and mitral valve replacements).

Alternate Names:

Aortomitral Fibrous Body Reconstruction; Intervalvular Fibrosa Reconstruction

Commensal

Describing an organism that lives with another organism and benefits from the relationship without normally harming or helping the host; normal skin and intestinal flora may be commensal.

Comminuted

Describing a fracture in which a bone is broken into multiple fragments.

Comminuted Fracture

A fracture in which the bone is broken into three or more fragments. It may result from high-energy trauma and can be associated with soft-tissue injury, instability, and impaired healing.

Commissurotomy

Surgical incision or separation of a commissure, especially a fused cardiac valve commissure, to relieve obstruction.

Commode

A portable toilet or chair placed near a bed for a person who cannot easily reach a bathroom.

Common Bile Duct

The duct formed by union of the common hepatic and cystic ducts. It conveys bile to the duodenum, usually joining the pancreatic duct at the hepatopancreatic ampulla.

Common Cold

An acute, self-limited viral infection of the upper respiratory tract, one of the most frequent illnesses

in humans, with adults experiencing 2-4 episodes annually and children 6-8 episodes.

Alternate Names:

Cold, Common Cold (Common Cold)

Common Cold In Babies

A viral infection of an infant's upper respiratory tract that commonly causes a runny nose, congestion, and cough.

Common Cold Stages And Timeline Of Symptoms

The common cold often begins with sore throat, sneezing, or fatigue, followed by nasal congestion, runny nose, and cough. Symptoms usually peak within several days and improve over one to two weeks, although cough and fatigue may last longer.

Common Early Symptoms Of Myasthenia Gravis

Early myasthenia gravis commonly causes fluctuating, fatigable weakness of the eyelids, eye movements, facial muscles, chewing, swallowing, speech, neck, or limbs. Symptoms worsen with use and improve with rest; breathing or swallowing difficulty is an emergency.

Common Iliac Arteries

Terminal abdominal aortic branches at L4, each approximately 4 cm long, dividing into internal and external iliac arteries. They supply the pelvis and lower extremities. The right common iliac artery crosses anterior to the right common iliac vein.

Common Medical Abbreviations And Terms

Medical abbreviations and terms summarize diagnoses, tests, medicines, anatomy, and treatment instructions. Because abbreviations can be ambiguous or unsafe, patients should ask clinicians or pharmacists to explain unfamiliar wording rather than infer its meaning.

Alternate Names:

Terms Medical Abbreviations (Common Medical Abbreviations And Terms)

Common Migraine

A recurrent, episodic neurovascular primary headache disorder occurring without a preceding neurological aura (migraine without aura, accounting for approximately 70–80% of migraine sufferers). Diagnostic criteria under ICHD-3 require recurrent attacks lasting 4 to 72 hours characterized by at least two pain features (unilateral location, pulsating quality, moderate-to-severe pain intensity, and aggravation by routine physical activity) accompanied by at least one associated symptom (nausea and/or vomiting, photophobia, and phonophobia). Pathophysiology involves trigeminovascular activation, cortical spreading depression, and release of calcitonin gene-related peptide; acutely treated with triptans, NSAIDs, or gepants, with CGRP antagonists or beta-blockers utilized for prevention.

Alternate Names:

Migraine without aura; migraine without neurologic aura.

Common Peroneal Nerve

Sciatic division (L4-S2) winding around the fibular neck, the body's most vulnerable major nerve to external compression. Divides into superficial peroneal (peroneus longus/brevis, dorsal foot sensation) and deep peroneal (tibialis anterior, extensors, first web space sensation).

Common Peroneal Nerve Dysfunction

Impairment of the common peroneal nerve, often from compression near the fibular head, trauma, surgery, or systemic nerve disease. It may cause foot drop, weakness of ankle eversion or dorsiflexion, and sensory loss over the lateral leg or dorsum of the foot.

Common Peroneal Neuropathy

A peripheral nerve disorder causing weakness of ankle dorsiflexion and eversion, sensory loss over the anterolateral leg and dorsum of the foot, and possible foot drop.

Common Symptoms During Pregnancy

Frequent pregnancy-related symptoms such as nausea, fatigue, breast tenderness, heartburn, consti-

pation, and urinary frequency; severe, sudden, or unusual symptoms may signal complications.

Common Variable Immunodeficiency

The most common symptomatic primary immunodeficiency in adults, affecting approximately 1 in 25,000 individuals. It is characterized by hypogammaglobulinemia (low IgG with low IgA and/or IgM), impaired antibody response to vaccines, and increased susceptibility to infections.

Common Warts (Verruca Vulgaris)

Benign, rough-surfaced skin growths caused by infection with human papillomavirus, commonly affecting the hands and fingers.

Alternate Names:

Verruca Vulgaris

Commotio Cordis

Sudden cardiac arrest caused by a blunt, nonpenetrating impact to the chest at a vulnerable moment in the cardiac cycle, without structural heart injury.

Communicable Disease

Any disease caused by bacteria, viruses, or other pathogens that is spread from person to person.

Communicating Hydrocephalus

A type of hydrocephalus in which there is free flow of cerebrospinal fluid between the ventricles, but impaired absorption of CSF at the arachnoid granulations or increased CSF production. The ventricles are uniformly dilated, including the fourth ventricle.

Communicating With Patients

The clinical exchange of information, concerns, preferences, and decisions between healthcare professionals and patients, using clear language, listening, empathy, and confirmation of understanding.

Communicating With Someone With Aphasia

Communication strategies for a person with impaired language from brain injury or disease, including short sentences, extra time, visual cues, and

supported conversation rather than assuming impaired intelligence.

Communicating With Someone With Dysarthria

Communication strategies for a person whose speech muscles are weak or poorly coordinated, including reducing background noise, confirming messages, and using writing or augmentative aids when needed.

Community Acquired Infection

An infection present or incubating at hospital admission, or developing within 48 hours of admission without recent healthcare exposure. Common examples include pneumonia, urinary tract infection, and meningitis caused by organisms such as *Streptococcus pneumoniae* and *Escherichia coli*, generally susceptible to narrow-spectrum antibiotics.

Community Health Worker

A trained member of a community who helps connect people with health services, education, prevention, and social support.

Community Rehabilitation

Rehabilitation delivered in community settings to improve a person's participation, independence, and daily functioning.

Community Reintegration

Return to community roles: work, leisure, social. Vocational rehabilitation, accessibility modifications.

Community-Acquired

Acquired outside a healthcare facility or before a person has been exposed to healthcare-associated conditions.

Community-Acquired Pneumonia

CAP: acquired outside hospital. CURB-65 (confusion, uremia >7 , respiratory rate ≥ 30 , BP $<90/60$, age ≥ 65) and PSI/PORT score for severity/site of care. Outpatient: amoxicillin 1g TID, doxycycline 100 mg BID, or macrolide (if resistance <25

percent); respiratory fluoroquinolone (levofloxacin, moxifloxacin) as alternative.

Community-Acquired Pneumonia In Adults

Pneumonia acquired outside a hospital or healthcare facility; assessment considers severity, oxygenation, comorbidities, likely pathogens, and the need for outpatient or inpatient treatment.

Comorbidity

Presence of two or more chronic conditions in same individual. Common in elderly: hypertension, diabetes, arthritis, COPD, depression. Increases complexity of care, polypharmacy risk, functional decline. Comprehensive assessment needed for each condition and interactions.

Computational Biomedicine

An interdisciplinary field that uses computers, mathematical models, simulations, and biomedical data to study health and disease. It may support research in areas such as genomics, drug development, medical imaging, disease prediction, diagnosis, and personalized treatment.

Alternate Names:

Computational Biomedical Science

Compact Bone

Dense outer bone tissue organized into osteons that provides strength and structural support.

Alternate Names:

Cortical bone; dense bone.

Comparison

A reference in an imaging report to earlier studies used to assess whether a finding has changed.

Compartment Pressure

The pressure within a closed osteofascial compartment. Markedly elevated pressure can impair capillary perfusion and cause acute compartment syndrome; pressure measurements support but do not replace clinical assessment.

Compartment Syndrome

A surgical emergency caused by increased pressure within a closed osteofascial compartment, leading to tissue ischemia, necrosis, and permanent functional loss. The lower leg (anterior, lateral, deep posterior, and superficial posterior compartments) and forearm (flexor and extensor compartments) are the most commonly affected sites.

Alternate Names:

Syndrome Compartment (Compartment Syndrome)

Competence

Autonomous voluntary authorization given by a competent patient to undergo a specific medical intervention, procedure, or clinical trial. See informed consent.

Competitive Antagonist

A competitive antagonist is a drug that binds to the same receptor site as an agonist and competes for binding, reducing the effect of the agonist without producing an effect of its own.

Competitive Inhibition

A type of enzyme inhibition where the inhibitor (I) competes with the substrate (S) for binding to the enzyme's active site. The inhibitor structurally resembles the substrate and binds reversibly to the free enzyme (E), forming an EI complex that cannot proceed to product.

Complement

Complement is a system of blood proteins that enhances antibody and immune-cell defense by marking, inflaming, or directly damaging targets.

Complement C1Q Deficiency

Autosomal recessive (C1QA, C1QB, C1QC). C1q essential for classical pathway activation and immune complex clearance. 85-90% develop SLE-like disease (malar rash, photosensitivity, nephritis, anti-nuclear antibodies). Recurrent infections with encapsulated bacteria. Low CH50, low C1q, normal C3/C4.

Complement C2/C4 Deficiency

C2 deficiency: most common early complement deficiency (1:20,000 Caucasians). SLE-like disease (10-30%), recurrent sinopulmonary infections. C4 deficiency: similar, strong SLE association (C4A/C4B null alleles). Low CH50, low C2 or C4, normal C3. Treatment: SLE management, prophylactic antibiotics, IVIG for recurrent infections.

Complement C3

The central complement component cleaved into C3a and C3b during activation of the classical, lectin, or alternative pathway. C3b promotes opsonization and formation of downstream convertases; severe C3 deficiency causes recurrent severe infections, especially with encapsulated bacteria.

Complement Cascade

The ordered activation of 30 serum and membrane proteins (complement) that opsonize pathogens, lyse microbes (membrane attack complex), and promote inflammation.

Complement Component 3 (C3)

A blood test measuring complement component C3, a major protein of the complement immune system; abnormal levels can support evaluation of immune-complex disease, infection, or complement deficiency.

Complement Component 4

A blood test measuring complement component C4, part of the classical complement pathway; low levels may occur with immune-complex disease, hereditary angioedema, or complement consumption.

Complement Deficiencies

Rare primary immunodeficiencies caused by genetic defects in complement pathway proteins, increasing susceptibility to specific infections and autoimmune diseases.

Complement Deficiency

Reduced or absent activity of one or more complement proteins, increasing susceptibility to some infections or autoimmune disease.

Complement Fixation Test For *C. burnetii*

A serologic test detecting complement-fixing antibodies to *Coxiella burnetii*, the bacterium that causes Q fever; antibody phase patterns and titers help distinguish recent from persistent infection.

Complement System

A group of blood proteins that supports immune defense by destroying microbes and promoting inflammation and phagocytosis.

Complementary Sequence

A nucleic-acid sequence whose bases pair with another sequence according to pairing rules, allowing hybridization between DNA or RNA strands.

Complete Blood Count

A blood test that measures red blood cells, white blood cells, platelets, hemoglobin, and hematocrit. It may also measure red-cell size and hemoglobin content and list the different types of white blood cells. It helps assess anemia, infection, inflammation, bleeding, and some bone-marrow disorders.

Complete Breech

A variation of breech fetal malpresentation in which both hips and both knees are flexed, causing the baby's feet and buttocks to present simultaneously at the pelvic inlet (the fetus sitting cross-legged). It accounts for approximately 5–10% of term breech presentations and carries an intermediate risk of umbilical cord prolapse compared with frank and footling breeches.

Alternate Names:

Flexed Breech; Complete Breech Presentation

Complete Cystectomy

Surgery to remove the entire urinary bladder, usually for invasive bladder cancer. Because urine can no longer leave through the bladder, the surgeon creates a urinary diversion such as an ileal conduit, a continent reservoir, or a new bladder from bowel.

Complete Denture

Removable prosthesis replacing all teeth in arch. Maxillary and mandibular. Stability challenging in edentulous patients.

Complete Miscarriage

The spontaneous expulsion of all products of conception (embryo/fetus and placenta) from the uterine cavity prior to 20 to 24 weeks of gestation. Clinically characterized by a history of vaginal bleeding and pelvic cramping that has substantially subsided, an internal cervical os that has closed again, and transvaginal ultrasonography demonstrating an empty endometrial cavity with a thin, regular lining and no retained gestational sac or heterogeneous tissue.

Alternate Names:

Complete Spontaneous Abortion

Complete Response

Disappearance of all detectable signs of a cancer after treatment. It does not necessarily mean that every cancer cell has been eliminated.

Complex Carbohydrates

Carbohydrates made up of longer chains of sugars, chiefly starches and dietary fiber, found in foods such as whole grains, legumes, vegetables, and potatoes. Fiber is not fully digested and supports bowel function, satiety, and metabolic health, while starch is digested into glucose at a rate that depends on food structure and processing.

Complex Febrile Seizure

A febrile seizure that is focal in onset or manifestation, prolonged (duration greater than 15 minutes), or recurrent (more than one episode within a 24-hour period or during the same febrile illness). Compared with simple febrile seizures, complex febrile seizures carry a higher risk of recurrent febrile seizures and a slightly increased long-term risk of developing epilepsy; they warrant thorough diagnostic evaluation to exclude central nervous system infection, structural lesion, or status epilepticus.

Alternate Names:

Atypical Febrile Seizure; Complicated Febrile

Seizure

Complex Regional Pain Syndrome

A chronic pain condition characterized by regional pain and sensory changes disproportionate to the inciting event. Type I (reflex sympathetic dystrophy): no identifiable nerve injury. Type II (causalgia): follows a confirmed nerve injury.

Compliance

The extent to which a person follows a treatment recommendation; adherence is generally preferred because it emphasizes an agreed plan.

Complicated Bereavement

Alternate index label; see *Complicated grief*.

Complicated Grief

Persistent, intense grief that substantially interferes with daily functioning and lasts longer than expected for the person's cultural and social context.

Alternate Names:

Complicated Bereavement Traumatic Grief

Complicated Migraine

An older, nonspecific term for migraine accompanied by prolonged or prominent neurologic symptoms; the specific syndrome should be identified when possible.

Alternate Names:

Migraine with prolonged neurologic symptoms; complex migraine, historical usage.

Complicated Urinary Tract Infection

A urinary infection associated with functional or structural abnormality, obstruction, catheterization, renal impairment, male anatomy, pregnancy, or another factor increasing treatment failure or complication risk. Management requires culture-directed therapy and correction of the underlying problem when possible.

Complication

Any adverse event arising during or after a surgical procedure, graded by the Clavien-Dindo classifica-

tion from minor deviation (Grade I) through organ failure (Grade IV) to death (Grade V). Common types include surgical site infection, hemorrhage, wound dehiscence, anastomotic leak, ileus, and venous thromboembolism.

Complications (Vascular)

Problems involving blood vessels after disease, injury, or a procedure, such as bleeding, hematoma, thrombosis, embolism, vessel injury, or reduced blood flow to a limb. Symptoms and urgency depend on the vessel and tissue affected.

Alternate Names:

Vascular

Composite Resin

Tooth-colored restorative material. Bis-GMA or UDMA resin matrix with silica filler. Adhesive bonding. Esthetic but less durable than ceramic.

Compound Fracture (Open Fracture)

A fracture associated with a wound that communicates between the fracture site and the external environment. It carries increased risks of infection and tissue damage.

Alternate Names:

Open Fracture

Compound Heterozygote

A person who has two different pathogenic variants in the two copies of the same gene, one inherited on each homologous chromosome.

Compound Presentation

A fetal presentation where an extremity (hand, arm, foot) presents alongside the vertex or breech. Most common: hand alongside the head. Incidence: 1:300-1:1000. Fetal prognosis is generally good; the prolapsed extremity often retracts as labor progresses. Management: expectant management in most cases.

Compounding

Preparation, mixing, or altering of a medicine to meet an individual patient's needs when an appropriate approved product is unavailable.

Compounding Pharmacy

A pharmacy that prepares customized formulations for an individual patient when a suitable commercially manufactured product is unavailable or unsuitable.

Alternate Names:

Compounding pharmacy practice; extemporaneous pharmacy.

Comprehensive Geriatric Assessment

A multidimensional assessment of an older person's medical conditions, medicines, function, cognition, mood, and social support, used to develop an individualized care plan.

Comprehensive Metabolic Panel

Extended version of BMP including albumin, total protein, alkaline phosphatase, ALT, AST, and bilirubin (total and direct). Evaluates liver function, kidney function, electrolytes, glucose, calcium, and protein status. Uses: preoperative assessment, medication monitoring, and liver/kidney disease evaluation.

Compress

To press together or apply pressure, as with a bandage or device, or to reduce the volume of something.

Compressed Nerve

Alternate index label; see *Pinched nerve*.

Compression

Pressure applied to tissue or a structure, which may reduce swelling or bleeding but can also impair nerves, blood vessels, or organs when excessive.

Compression Fracture

Collapse of a vertebra or other bone because of compression, often related to osteoporosis or trauma.

Compression Fractures Of The Back

Collapse of one or more vertebral bodies caused by weakened bone, often from osteoporosis, or by substantial trauma. It can cause sudden back pain,

height loss, kyphosis, and rarely nerve or spinal-cord compression.

Compression Stockings

Elastic garments that apply graduated pressure to the legs, supporting venous return and reducing swelling or selected risks of venous pooling; correct fit and arterial circulation matter.

Compression Therapy

Use of external pressure, usually from garments or bandages, to support venous or lymphatic flow and reduce swelling in selected conditions.

Compression Ultrasound

An ultrasound technique in which gentle pressure is applied to assess whether a vein collapses. Failure of a vein to compress may suggest thrombosis.

Compressive Atelectasis

Atelectasis caused by external compression of the lung by pleural fluid, air, or a mass. The lung volume is reduced by the external force. It is reversible when the compressing force is relieved (e.g., drainage of pleural effusion).

Compton Scattering

An interaction in which a photon transfers part of its energy to an electron and changes direction. It contributes to scattered radiation in imaging.

Compulsions

Repetitive behaviors or mental acts that the individual feels driven to perform in response to an obsession or according to rigid rules. They are aimed at preventing or reducing anxiety or preventing some dreaded event, but are not connected in a realistic way with what they are designed to neutralize.

Compulsive Gambling

Alternate, nonpreferred label; see *Gambling Disorder*.

Compulsive Obsessive Disorder

(Obsessive Compulsive Disorder)

Alternate index label for Obsessive Compulsive Disorder.

Compulsive Overeating

Alternate index label; see *Binge-eating disorder*.

Compulsive Sexual Behavior

Recurrent sexual thoughts or behaviors that are difficult to control and cause significant distress or impairment.

Alternate Names:

Sexual Obsession

Compulsive Stealing

Alternate index label; see *Kleptomania*.

Computed Tomography

Cross-sectional imaging using rotating X-ray tube and detectors. Produces axial images, reconstructable into sagittal, coronal, 3D. CT number measured in Hounsfield units (HU): water = 0, air = -1000, bone = +1000. Contrast: iodine-based, enhances vascular structures and perfused organs. Speed: seconds.

Computed Tomography Pulmonary Angiography

A specialized, rapid contrast-enhanced multidetector computed tomography examination of the pulmonary circulation that represents the reference-standard diagnostic test for acute pulmonary embolism. Following rapid intravenous iodinated contrast bolus timed to pulmonary arterial opacification, high-resolution thin-collimation images directly visualize intraluminal filling defects (thrombi) within the main, lobar, segmental, and subsegmental pulmonary arteries, while simultaneously providing assessment of right ventricular strain and alternative parenchymal or pleural diagnoses.

Alternate Names:

CTPA; CT Pulmonary Angiogram; Pulmonary CT Angiography

Computed Tomography Scan

Cross-sectional imaging using X-rays and computer processing. Emergency uses: head trauma (hemorrhage, fracture), chest (pulmonary embolism, aortic dissection), abdomen (appendicitis, bowel obstruction), CT pulmonary angiography (CTPA), CT angiography (CTA). Fast, widely available, high sensitivity for acute conditions.

Computerized Axial Tomography

Computed tomography, an imaging method that uses X-rays and computer processing to produce cross-sectional images of the body; it is also called a CT or CAT scan.

Concato'S Disease

A historical term for primary polycythemia, now generally called polycythemia vera, a myeloproliferative neoplasm causing increased red-cell mass and often elevated blood viscosity.

Concealed Conduction

An electrophysiological phenomenon wherein an electrical cardiac impulse partially penetrates into specialized conduction tissue (most frequently the atrioventricular [AV] node or bundle branches) without fully traversing it to emerge and depolarize adjacent myocardium. Although the propagating wavefront dies out and cannot be directly visualized on a surface electrocardiogram, it renders the traversed tissue refractory, thereby altering or delaying the conduction, velocity, or formation of the subsequent electrical impulse. Classically explains the irregular ventricular response and variable PR intervals observed during atrial fibrillation and atrial flutter with variable AV block.

Concealed Pregnancy

Pregnancy that is hidden or not recognized until delivery, often by denial (psychogenic) or deliberate concealment; a medicolegal concept relevant to infanticide.

Concentration

Amount of substance per unit volume. Units: mg/dL, mmol/L, mEq/L, ppm. Conversion: molec-

ular weight needed. Critical for toxicity assessment.

Conception

The beginning of pregnancy, usually referring to fertilization and the subsequent establishment of pregnancy.

Conception (Fertilization)

The fusion of a sperm and an ovum in the ampulla of the fallopian tube, forming a zygote. The sperm penetrates the corona radiata and zona pellucida (acrosome reaction). After fertilization, the zygote undergoes cleavage (mitotic division) as it travels toward the uterus. Implantation occurs approximately 6-10 days after ovulation.

Alternate Names:

Fertilization

Conchae

Curved bony structures covered by mucosa inside the nasal cavity that warm, humidify, and filter inhaled air.

Conclusion

The part of a report that summarizes the main findings and their clinical significance or recommended next steps.

Concomitant

Existing or occurring at the same time as another condition, treatment, or event.

Concordance

Probability both twins have same trait. Monozygotic (identical) twins > dizygotic (fraternal) for genetic traits. High concordance suggests genetic influence.

Concrescence

Union of adjacent teeth by cementum only. Usually posterior teeth.

Concretio Cordis

Abnormal adhesion or fusion of structures within or around the heart, including adhesion of the peri-

cardial surfaces or fusion of cardiac components.

Concussion

A form of mild traumatic brain injury caused by a biomechanical force (direct blow to the head, face, neck, or elsewhere on the body with impulsive force transmitted to the head), resulting in a transient disturbance of brain function without structural damage visible on conventional neuroimaging.

Condition

A general term for a state of bodily health or, in medical usage, a specific deviation from normal in structure or function that may or may not constitute a disease, including complaints lacking a discrete pathologic basis and overt diagnoses.

Conditioning

Learning in which repeated association or reinforcement changes a behavior or response; in medicine, conditioning can also mean preparing a patient or tissue for a treatment.

Conditions Caused By Deletion Mutations

Deletion mutations remove DNA segments and can disrupt one gene or several neighboring genes. The clinical effect depends on size, location, inheritance, and whether the remaining copy compensates; genetic counseling and testing may clarify diagnosis and recurrence risk.

Condom

A thin barrier sheath worn on the penis or inserted into the vagina that helps prevent pregnancy and reduces transmission of sexually transmitted infections when used correctly.

Condoms - Male

A sheath placed over the penis during sexual activity to reduce exchange of semen and genital secretions and lower pregnancy and sexually transmitted infection risk when used correctly.

Conduct Disorder

A childhood or adolescent disorder involving a persistent pattern of behavior that violates the rights of others or major age-appropriate rules. Examples include aggression, property destruction, deceitfulness, theft, and serious rule violations.

Conduction Deafness

Hearing loss caused by impaired transmission of sound through the external or middle ear, with the inner ear and neural auditory pathways relatively preserved.

Conductive Deafness

Hearing loss resulting from a problem in the external auditory canal, tympanic membrane, or middle ear that reduces conduction of sound to the inner ear.

Conductive Hearing Loss

Hearing loss caused by impaired transmission of sound through the outer or middle ear. Audiometry commonly shows an air-bone gap, and causes include earwax, middle-ear fluid, perforation, otosclerosis, and ossicular disease.

Condylar Process

The rounded articular projection of a bone, especially the mandibular process that articulates with the temporal bone to form the temporomandibular joint.

Condyle

A rounded projection at the end of a bone that contributes to a joint.

Condyloma

A wart-like growth, often in the genital or anal area and commonly caused by human papillomavirus; condyloma lata are lesions of secondary syphilis.

Condyloma Acuminata

Soft, wart-like genital or perianal growths caused by certain human papillomavirus types.

Condyloma Acuminatum

A benign, exophytic, papillomatous growth of the anogenital skin caused by human papillomavirus, most commonly types 6 and 11. They present as soft, flesh-colored, verrucous papules or plaques in the genital and perianal region. Condylomata are sexually transmitted and may be associated with cervical dysplasia.

Condyloma Latum

Mucocutaneous lesion of secondary syphilis caused by *Treponema pallidum*, distinct from condyloma acuminata (genital wart).

Condylomata Acuminata

Alternate index label; see *Genital warts*.

Condylomata Lata

Broad, flat, moist, grayish-white papules occurring in secondary syphilis, typically found in the anogenital region and other intertriginous areas. They are highly infectious and contain abundant *Treponema pallidum* organisms. Diagnosis is by darkfield microscopy or serological testing for syphilis.

Cone Biopsy

A surgical procedure that removes a cone-shaped portion of the cervix for diagnosis or treatment of high-grade cervical precancer and to determine whether invasive cancer is present.

Cone-Beam Computed Tomography

Three-dimensional dental imaging using a cone-shaped X-ray beam and a rotating detector. It provides high-resolution views of teeth and maxillofacial bone with generally less radiation than conventional medical CT, but should be used only when the additional information affects care.

Cones

Retinal photoreceptor cells responsible for color vision and detailed vision in relatively bright light.

Alternate Names:

Cone photoreceptors; cone cells.

Confabulation

Production of false or inaccurate statements or memories without conscious intent to deceive, often associated with memory impairment or frontal-lobe dysfunction.

Confidentiality

The ethical and legal duty of healthcare professionals to protect patient information from unauthorized disclosure. Founded on the principle of respect for patient autonomy and trust, confidentiality is a core requirement of medical ethics (GMC guidelines) and law (GDPR, Data Protection Act, Caldicott principles).

Confirmatory Test

A test used to verify a preliminary or screening result, often with a different method or greater specificity. Confirmation should be selected according to the condition, specimen, timing, and accepted clinical guidelines.

Conformal Radiation Therapy

Radiation treatment planned so that the high-dose region conforms closely to the tumor's three-dimensional shape while limiting exposure of nearby normal tissue.

Confounding

Distortion of association by third variable. Examples: age confounding relationship between exposure and disease. Controlled by matching, stratification, regression.

Confusion

Impaired ability to think clearly, orient, attend, remember, or respond appropriately; sudden confusion is delirium and may indicate infection, medication effect, metabolic illness, stroke, or another emergency.

Confusion (Delirium)

Acute onset fluctuating disturbance in attention and awareness. Usually reversible. Precipitants: infection, medications, metabolic derangement, pain, constipation, urinary retention. Differentiate from

dementia (chronic, progressive). Treatment: identify and treat underlying cause, environmental modification, avoid anticholinergics.

Alternate Names:

Delirium

Confusion Assessment Method

A standardized bedside algorithm for detecting delirium. It identifies acute or fluctuating change, inattention, disorganized thinking, and altered level of consciousness; delirium requires the first two features plus either of the latter two.

Congenital

Present at birth, whether inherited, caused by a developmental problem, or acquired before birth.

Congenital Adrenal Hyperplasia

Group of autosomal recessive disorders caused by enzyme deficiencies in the adrenal steroidogenesis pathway. Twenty-one hydroxylase deficiency is the most common accounting for approximately ninety-five percent of cases.

Congenital Antithrombin III Deficiency

An inherited deficiency or dysfunction of antithrombin, a natural anticoagulant, causing increased risk of venous thromboembolism.

Congenital Biliary Atresia

Alternate index label; see *Biliary atresia*.

Congenital Cataract

Opacification of the crystalline lens present at birth, a leading cause of treatable childhood blindness worldwide.

Congenital CMV

Cytomegalovirus infection acquired in utero, the most common congenital viral infection. Maternal primary infection poses highest risk of fetal transmission and disease. Can cause microcephaly, sensorineural hearing loss, chorioretinitis, hepatosplenomegaly, and petechiae. Diagnosis by CMV PCR in urine or saliva.

Congenital CNS Malformation

A structural abnormality of the brain, spinal cord, or related developmental structures present from birth, caused by disturbed neural development or other prenatal factors; severity ranges from incidental to life-threatening.

Congenital Contractural Arachnodactyly

An inherited connective-tissue disorder characterized by long slender fingers and toes, congenital flexion contractures, muscular hypoplasia, and sometimes kyphoscoliosis.

Congenital Cytomegalovirus

A medicine or chemical compound identified by congenital cytomegalovirus. Its uses, effects, and risks depend on the specific substance and clinical context.

Congenital Cytomegalovirus Infection

Cytomegalovirus infection acquired before birth. It may cause hearing loss, developmental problems, microcephaly, or no symptoms.

Congenital Diaphragmatic Hernia

A defect in the diaphragm allowing abdominal organs to herniate into the thoracic cavity, causing pulmonary hypoplasia and respiratory distress at birth. Most commonly a posterolateral (Bochdalek) defect on the left side. Presents with cyanosis, scaphoid abdomen, and bowel sounds in the chest.

Congenital Diaphragmatic Hernia

A birth defect in which an opening in the diaphragm allows abdominal organs to move into the chest and interfere with lung development.

Congenital Diaphragmatic Hernia Repair

A medical procedure involving congenital diaphragmatic hernia repair. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Congenital Disorder

A medical condition or structural abnormality

present in an individual from birth, which may be caused by genetic factors or environmental influences in the womb.

Congenital Fibrinogen Deficiency

An inherited absence or abnormality of fibrinogen that can cause bleeding, thrombosis, or both depending on the specific molecular defect.

Congenital Glaucoma

A rare form of glaucoma (1 in 10,000-20,000 births) caused by developmental anomalies of the trabecular meshwork and anterior chamber angle, impairing aqueous humor outflow. Most cases are sporadic; familial forms are autosomal recessive (CYP1B1 gene mutations).

Congenital Heart Defect - Corrective Surgery

Surgery or catheter-based intervention to repair a structural heart abnormality present at birth, with the specific procedure depending on the defect, its severity, and the patient's anatomy and condition.

Congenital Heart Defects

Problems in the heart or major blood vessels that are present at birth. They can be mild or serious and may affect how blood flows through the heart and body. Examples include holes in the heart, abnormal valves, and problems with the major blood vessels.

Congenital Heart Defects In Children

Alternate index label; see *Congenital Heart Defects*.

Congenital Heart Disease

A diverse spectrum of structural cardiac and great-vessel malformations present at birth, resulting from abnormal embryonic cardiovascular development during gestational weeks 3–8. Hemodynamically categorized into acyanotic lesions (left-to-right shunts: ventricular septal defect, atrial septal defect, patent ductus arteriosus; and obstructive lesions: coarctation of the aorta, congenital aortic stenosis, pulmonary stenosis) and cyanotic lesions characterized by right-to-left shunting or mixing (classically the

5 Ts: Tetralogy of Fallot, Transposition of the great arteries, Tricuspid atresia, Truncus arteriosus, and Total anomalous pulmonary venous return, as well as hypoplastic left heart syndrome). Clinical manifestations range from asymptomatic heart murmurs to neonatal cardiogenic shock, congestive heart failure, failure to thrive, and central cyanosis unresponsive to hyperoxia challenge; diagnosed by prenatal or neonatal transthoracic echocardiography and managed via surgical repair or transcatheter interventions.

Congenital Heart Disease In Adults

A heart defect present from birth that continues into adulthood; see *Congenital Heart Defects*.

Congenital Heart Murmur

A heart murmur heard in a newborn or infant. It may be innocent or may indicate a structural heart defect; assessment considers the murmur's characteristics and associated symptoms or examination findings.

Compensated Heart Failure

Heart failure in which symptoms and fluid buildup are stable or controlled with treatment. The heart failure is still present and can worsen, especially if a new illness or other problem develops.

Congenital Hip Dislocation

Abnormal development or displacement of the hip joint present at birth or developing in infancy, now generally called developmental dysplasia of the hip.

Congenital Hyperinsulinism

Condition of inappropriate insulin secretion in neonates and infants causing severe hypoglycemia. Most commonly caused by mutations in the KATP channel genes ABCC8 and KCNJ11. Histologically classified into focal and diffuse forms. Presents with severe symptomatic hypoglycemia in the first days of life.

Congenital Hyperthyroidism

Thyrotoxicosis present at birth most commonly caused by maternal Graves disease with transplacental

tal passage of thyroid-stimulating immunoglobulins. Presents with tachycardia, irritability, poor feeding, weight loss, and goiter. Transient neonatal hyperthyroidism resolves as maternal antibodies are cleared over weeks to months.

Congenital Hypothyroidism

Thyroid hormone deficiency present at birth occurring in approximately one in two thousand to four thousand live births. Most common cause is thyroid dysgenesis including aplasia, ectopia, and hypoplasia. Universal newborn screening with TSH or T4 allows early detection and treatment preventing intellectual disability.

Congenital Malformation

A structural abnormality that develops before birth and is present at birth, caused by genetic, environmental, developmental, or unknown factors.

Congenital Megacolon

Hirschsprung disease, a congenital absence of enteric ganglion cells in a segment of the bowel that causes functional obstruction and enlargement of the proximal colon.

Congenital Megacolon (Hirschsprung Disease)

Alternate index label for Hirschsprung Disease.

Congenital Melanocytic Nevus

A melanocytic nevus present at birth or appearing within the first few weeks of life, ranging from small (< 1.5 cm) to giant lesions covering large body surface areas. It results from proliferation of melanocytes during embryonic development and carries a small but important risk of melanoma, which is higher in larger lesions.

Congenital Mitral Valve Anomalies

Structural abnormalities of the mitral valve that are present at birth and may impair blood flow.

Congenital Muscular Dystrophy

A genetically and clinically heterogeneous group of

inherited primary muscle disorders characterized by onset of muscle weakness and hypotonia at birth or during the first six months of life, with muscle biopsy demonstrating marked dystrophic changes, including prominent myofiber necrosis, regeneration, and progressive endomysial and perimysial fibrosis. Subclassified into collagen VI-deficient dystrophies (Ullrich CMD), laminin- α 2 (merosin)-deficient CMD (MDC1A), and α -dystroglycanopathies (Walker-Warburg, Fukuyama, muscle-eye-brain disease) that frequently feature associated structural central nervous system malformations and eye anomalies.

Alternate Names:

CMD; Congenital Dystrophy

Congenital Myasthenic Syndromes

A group of inherited disorders that impair communication between nerves and muscles, causing fatigable weakness.

Congenital Nephrotic Syndrome

Severe protein-losing kidney disease beginning in fetal life or during the first three months after birth, causing edema, hypoalbuminemia, and increased infection and thrombosis risk.

Congenital Nevus

A melanocytic mole present at birth or developing soon afterward. It is composed of melanocytes in the epidermis and dermis. Size, location, and changes over time determine the need for clinical assessment and surveillance.

Congenital Penile Curvature

Curvature of the penis present from birth, usually caused by unequal development of the erectile tissues. Mild curvature may need no treatment; significant curvature causing functional difficulty may require specialist assessment.

Congenital Plagiocephaly

Alternate index label; see *Craniosynostosis*.

Congenital Platelet Function Defects

Inherited disorders in which platelets do not adhere, activate, or aggregate normally despite a normal

or near-normal platelet count. They cause mucocutaneous bleeding, easy bruising, heavy menstrual bleeding, or excessive bleeding after procedures.

Congenital Pneumonia

Lung infection acquired in utero, often from ascending chorioamnionitis. Presents with respiratory distress, fever or hypothermia, and sepsis. Common organisms include Group B Streptococcus, *E. coli*, and Ureaplasma. Diagnosis by chest X-ray showing infiltrates and blood culture.

Congenital Protein C Or S Deficiency

Inherited deficiency of natural anticoagulant protein C or S, increasing the risk of venous thromboembolism.

Congenital Rubella

Fetal infection with rubella virus acquired during pregnancy, potentially causing hearing loss, cataracts, congenital heart disease, developmental impairment, and other findings.

Congenital Rubella Syndrome

Fetal infection with rubella virus, particularly dangerous in the first trimester. Classic triad: congenital heart disease (patent ductus arteriosus, pulmonary stenosis), cataracts, and sensorineural hearing loss. Also causes hepatosplenomegaly, thrombocytopenia, and blueberry muffin rash.

Congenital Scoliosis

A lateral curvature of the spine caused by congenital vertebral anomalies that arise during embryogenesis (failure of formation or segmentation). Failure of formation: hemivertebra (wedge-shaped vertebra causing progressive curvature). Failure of segmentation: unilateral unsegmented bar (block of bone on one side).

Congenital Syphilis

Treponema pallidum infection acquired transplacentally. Can cause stillbirth, prematurity, hepatosplenomegaly, rash, snuffles, osteochondritis, and Hutchinson teeth. Diagnosed by darkfield microscopy, VDRL/RPR in neonatal serum, or PCR.

Congenital Toxoplasmosis

Toxoplasma gondii infection acquired in utero, potentially causing chorioretinitis, intracranial calcifications, hydrocephalus, seizures, or developmental impairment.

Congenital Zika Syndrome

Fetal brain infection with Zika virus, causing microcephaly, brain abnormalities, eye defects, hearing loss, and limb contractures. Transmission occurs via mosquito bite or sexual contact. Diagnosis by Zika virus PCR in maternal serum or amniotic fluid. No specific treatment. Prevention by avoiding endemic areas during pregnancy.

Congestion

A passive vascular phenomenon of engorgement of a tissue or organ with blood from impaired venous or lymphatic drainage. Chronic venous congestion produces stasis, hypoxia, dilated vessels, edema, hemorrhage, and hemosiderin-laden macrophages, exemplified by the lungs (heart failure cells) and liver (nutmeg liver).

Congestive Heart Failure

A historical term for heart failure with clinical fluid congestion. Current terminology is heart failure, which may occur with or without congestion and is classified by left-ventricular ejection fraction, symptoms, and disease stage.

Congestive Heart Failure Overview

Alternate index label; see *Heart failure*.

Conization

Surgical removal of a cone-shaped portion of the cervix for diagnosis or treatment of cervical dysplasia or selected early cervical cancers.

Conjoined Twins

Monozygotic, monochorionic, monoamniotic twins who are physically connected at birth, resulting from incomplete fission of the fertilized ovum after day 13 of fertilization. Incidence: 1 in 50,000-100,000 births.

Conjugated Equine Estrogens

A mixture of estrogenic compounds historically derived from the urine of pregnant mares and used in some hormone-therapy formulations.

Alternate Names:

CEE; conjugated estrogens.

Conjunctiva

A thin mucous membrane that lines the inner eyelids (palpebral conjunctiva) and covers the front surface of the sclera (bulbar conjunctiva). It protects and lubricates the eye and permits smooth movement of the eyelids.

Conjunctival

Relating to the conjunctiva, the thin mucous membrane covering the front of the eye and lining the inner eyelids.

Conjunctival Squamous-Cell Carcinoma

A malignant tumor of the conjunctival epithelium, typically arising from the limbal area. It is associated with UV radiation exposure and can invade locally if not treated promptly.

Conjunctivitis

Inflammation of the conjunctiva, the thin, transparent mucous membrane lining the inner surface of the eyelids and the anterior surface of the sclera. Conjunctivitis is the most common cause of red eye and is classified by etiology.

Conjunctivitis (Pink Eye)

Alternate index label for Pink Eye.

Conjunctivitis Or Pink Eye

Inflammation of the conjunctiva causing redness, discharge, irritation, itching, or tearing. Causes include viral or bacterial infection, allergy, and chemical irritation; severe pain, photophobia, vision loss, or contact-lens use requires prompt eye assessment.

Conjunctivoplasty

Plastic or reconstructive surgery of the conjunctiva, performed to repair defects, scarring, adhesions, or

other abnormalities of the ocular surface.

Conn'S Syndrome

Primary hyperaldosteronism caused by autonomous secretion of aldosterone, leading to hypertension and often low serum potassium.

Connective Tissue

A diverse type of tissue that supports, binds, protects, and insulates organs and structures throughout the body.

Connective Tissue Disease

Autoimmune diseases affecting connective tissue. Examples: SLE, scleroderma, RA, Sjogren syndrome, mixed connective tissue disease. Overlap syndromes common.

Connective Tissue Disease-Related Interstitial Lung Disease

Interstitial lung disease associated with autoimmune connective-tissue disease such as systemic sclerosis, rheumatoid arthritis, polymyositis, dermatomyositis, or Sjogren syndrome; inflammation and fibrosis can impair gas exchange.

Consanguinity

A biological relationship between people who share a recent common ancestor; it can increase the chance that both parents carry the same recessive disease variant.

Conscious Sedation For Surgical Procedures

A medical procedure involving conscious sedation for surgical procedures. Indications, risks, preparation, and follow-up depend on the patient and the specific procedure.

Consolidation

An area of lung in which air spaces are filled with fluid, cells, or other material, increasing its density on imaging. Causes include pneumonia, edema, hemorrhage, and tumor.

Consolidation Therapy

Additional treatment given after remission to eliminate residual disease and reduce the risk of relapse.

Constipation

A condition involving fewer than three bowel movements per week, hard or lumpy stools, difficult or painful passage, straining, or a feeling that the bowel has not emptied completely. Causes include diet, medicines, dehydration, functional bowel disorders, and structural or systemic disease. Rectal bleeding, vomiting, constant abdominal pain, inability to pass gas, fever, or unexplained weight loss requires medical evaluation.

Constipation (Laxatives For Constipation)

Alternate index label for Laxatives For Constipation.

Constipation In Infants And Children

Infrequent, hard, painful, or difficult passage of stool in a child, caused by stool withholding, diet, dehydration, illness, medicines, or structural and neurologic disorders.

Constitutional Delay In Puberty

A normal variant in which pubertal development is delayed but eventually progresses normally; growth and bone age are typically delayed.

Constraint-Induced Movement Therapy

Rehabilitation that limits use of a less-affected limb while providing structured practice with an affected limb to improve functional use in selected people.

Constrictive Pericarditis

Thickening and fibrosis of the pericardium restricting diastolic ventricular filling. Presents with jugular venous distension, peripheral edema, hepatomegaly, and Kussmaul sign. Causes include idiopathic, post-viral, radiation, and tuberculosis. Pericardiectomy is the definitive treatment.

Consultation-Liaison Psychiatry

The subspecialty of psychiatry focused on the interface between psychiatry and general medical illness. Psychiatrists provide consultation on medical-surgical wards for patients with psychiatric symptoms arising from or coexisting with medical conditions.

Consumption

An older term for tuberculosis, especially wasting pulmonary tuberculosis; in general usage it also means the act or amount of using something.

Contac Overdose

Overdose of a combination cold medicine that may cause acetaminophen toxicity, antihistamine or decongestant effects, sedation, seizures, or cardiac complications depending on ingredients.

Contact Allergen Testing

Patch testing identifies allergens causing allergic contact dermatitis. Standard panel includes 35 common allergens (nickel, fragrance mix, balsam of Peru, formaldehyde, rubber accelerators, topical antibiotics, preservatives). Patches are applied to the back for 48 hours and read at 48, 72, and 96 hours.

Contact Dermatitis

An inflammatory skin reaction caused by contact with an external substance. It may be allergic, involving delayed immune sensitization, or irritant, caused by direct chemical or physical injury.

Alternate Names:

Dermatitis, Contact

Contact Precautions

Infection control measures used for patients colonised or infected with multidrug-resistant organisms. Precautions include gown and glove use, patient placement in single rooms, and dedicated equipment. Contact precautions are used for MRSA, VRE, CRE, and other resistant organisms.

Contact Tracing

Identifying and monitoring persons exposed to in-

fectious disease. Critical for outbreak control. Examples: COVID-19, Ebola, TB.

Contagious Disease

An infectious disease that can be transmitted directly or indirectly from one person, animal, or source to another.

Contamination

Accidental presence of microorganisms in a specimen, on a surface, or in a normally sterile material without true infection of the host. Contamination can produce misleading culture results and must be distinguished from colonization and invasive disease.

Continuing Medical Education

Postgraduate education that maintains and updates a clinician's knowledge, skills, and professional competence.

Continuity Of Care

Consistent and coordinated care over time and across clinicians or settings, integrating the patient's history, current needs, treatment plan, and preferences. It supports safety, adherence, prevention, and chronic-disease outcomes.

Continuous Bladder Irrigation

Continuous instillation of sterile fluid through a urinary catheter to keep the bladder catheter patent and remove blood clots or debris.

Continuous Combined Hormone Therapy

A hormone regimen in which estrogen and progesterone are taken continuously rather than sequentially.

Alternate Names:

Continuous combined HRT; continuous estrogen-progesterone therapy.

Continuous Electronic Fetal Monitoring

The continuous recording of fetal heart rate and uterine contractions using external transducers (Doppler ultrasound and tocodynamometer) or internal devices (fetal scalp electrode and intrauterine pressure catheter). Used intrapartum to detect fetal hypoxia.

Continuous Glucose Monitoring

Technology measuring interstitial glucose levels continuously providing real-time glucose data, trends, and alerts. Sensors are inserted subcutaneously typically in the abdomen or arm and measure glucose every five minutes. Devices include Dexcom, FreeStyle Libre, and Medtronic Guardian systems.

Continuous Kidney Replacement Therapy

A prolonged extracorporeal blood-purification therapy, usually delivered continuously in an intensive-care setting. It provides gradual fluid and solute removal and is useful for hemodynamically unstable patients with severe AKI.

Continuous Murmurs

Murmurs that persist throughout systole and diastole without interruption, produced by blood flowing continuously through an abnormal connection. The most common cause is patent ductus arteriosus (PDA), where blood flows from the aorta to pulmonary artery throughout the cardiac cycle.

Continuous Positive Airway Pressure

Noninvasive respiratory support that delivers a constant positive pressure throughout the breathing cycle to keep the upper airway or alveoli open.

Continuous Subcutaneous Insulin Infusion

Delivery of rapid-acting insulin through a pump and a subcutaneous infusion set to provide basal insulin and mealtime doses.

Contraception

The deliberate prevention of conception or pregnancy through behavioral, mechanical, chemical, hormonal, or surgical methods. Tiered clinically by real-world efficacy (Pearl Index): Tier 1 (most effective, < 1% failure rate) includes long-acting reversible contraception (LARC: levonorgestrel-releasing and copper intrauterine devices [IUDs], etonogestrel subdermal implants) and permanent surgical sterilization (tubal ligation, bilateral salpingectomy, vasectomy); Tier 2 (moderate efficacy, 6–9% typical failure rate) includes combined oral

contraceptive pills, transdermal patches, vaginal rings, and depot medroxyprogesterone acetate injections; and Tier 3 (least effective, 12–28% typical failure) includes male and female condoms, diaphragms, spermicides, withdrawal, and fertility awareness-based methods.

Contraception And Birth Control

Contraception encompasses methods used to prevent pregnancy, including hormonal options like oral contraceptives, patches, rings, injections, and implants, as well as non-hormonal methods like copper IUDs, barrier methods, diaphragms, and permanent sterilization.

Contraception For Adolescents

Methods to prevent unintended pregnancy. Options: combined oral contraceptives, progestin-only pills, depot medroxyprogesterone acetate, contraceptive implant (Nexplanon), intrauterine devices (IUDs), condoms, and abstinence.

Contraceptive Device, Intrauterine

A small device placed inside the uterus to prevent pregnancy; copper IUDs impair sperm function, while hormonal IUDs release levonorgestrel and thicken cervical mucus.

Contraceptive, Pill

An oral hormonal contraceptive, usually a combined estrogen–progestin pill or a progestin-only pill, that prevents ovulation and/or changes cervical mucus and the uterine lining.

Contracted Pelvis

A pelvis whose size or shape is reduced or abnormal enough to interfere with passage of the fetus during labor.

Contractile Proteins

Proteins, especially actin and myosin, that interact to generate force and shorten muscle cells.

Alternate Names:

Muscle contractile proteins; actomyosin proteins.

Contraction

Shortening or tightening of a muscle; uterine contractions during labor thin and dilate the cervix and help deliver the fetus.

Contraction Stress Test

An antepartum test that evaluates the fetal heart rate response to uterine contractions. Contractions are induced with oxytocin or nipple stimulation (until 3 contractions in 10 minutes). Negative (normal): no late or significant variable decelerations. Positive (abnormal): late decelerations with $\geq 50\%$ of contractions.

Contractions

Rhythmic uterine muscular contractions of labor that produce cervical dilation and effacement, driving descent of the fetus. Characterized by frequency, duration, and intensity; progress is monitored by partogram and tocography (CTG). Braxton Hicks contractions: irregular, non-progressive, painless prelabor contractions.

Contractions Braxton-Hicks (Braxton Hicks Contractions)

Alternate index label for Braxton Hicks Contractions.

Contracture

Permanent or semi-permanent limitation of passive joint range of motion caused by shortening or structural change in muscle, tendon, capsule, fascia, skin, or other periarticular tissue. Prevention and treatment depend on cause and stage.

Contracture Deformity

A fixed limitation of movement caused by shortening or tightening of muscle, tendon, ligament, joint capsule, skin, or other soft tissue. It may follow neurologic disease, prolonged immobility, injury, burns, or inflammation.

Contracture Prevention

Measures such as positioning, stretching, exercise, splinting, orthoses, and serial casting that help preserve joint movement in people at risk of fixed short-

ening.

Contraindication

A condition or circumstance in which a treatment, procedure, or medication should not be used because the potential harm outweighs the expected benefit.

Alternate Names:

Contraindications

Contralateral

Pertaining to, positioned on, or affecting the opposite side of the body relative to a reference structure, lesion, or point of origin. In clinical neurology, motor and primary somatosensory pathways exhibit contralateral representation: an ischemic stroke or cortical lesion in the precentral gyrus of the left cerebral hemisphere results in contralateral (right-sided) hemiparesis due to decussation of the lateral corticospinal tract in the caudal medulla; similarly, pain and temperature signals ascend the anterolateral spinothalamic tract and project to the contralateral sensory cortex.

Contrast

A difference that makes structures distinguishable; in imaging, contrast material is administered to improve visualization of organs, vessels, or tissues.

Contrast Agent

Substance enhancing image visibility. X-ray/CT: iodine-based (positive contrast). MRI: gadolinium-based (shortens T1, bright on T1). Ultrasound: microbubble agents. Barium sulfate for GI studies.

Contrast Echocardiography

Use of microbubble contrast agents to enhance endocardial border definition and assess myocardial perfusion. Indicated for suboptimal non-contrast echo images and assessment of myocardial viability. Microbubbles remain entirely within the intravascular space, making them ideal for perfusion assessment.

Contrast Medium

A substance administered or placed in the body to make tissues, organs, or body spaces more visible during imaging.

Alternate Names:

Contrast agent; radiographic contrast material.

Contrast Radiology

Imaging of the gastrointestinal tract using contrast material, such as barium, to outline the esophagus, stomach, intestine, or colon on radiographs or fluoroscopy.

Contrast Reaction

Adverse reaction to contrast agent. Mild: nausea, urticaria. Severe: anaphylaxis, bronchospasm. Pre-medication for prior reactions.

Contrast-Associated Acute Kidney Injury

Acute worsening of kidney function occurring after administration of contrast, whether or not contrast is causal. Risk assessment includes baseline kidney function, hemodynamic status, diabetes, nephrotoxins, urgency, and the need to avoid unnecessary contrast exposure.

Contrast-Enhanced Ultrasound

Ultrasound using microbubble contrast agents. Improves lesion characterization, perfusion assessment.

Contrast-Induced Nephropathy

Acute kidney injury after iodinated contrast administration. Risk factors: pre-existing renal impairment, diabetes, dehydration. Prevention: hydration, lowest contrast volume, N-acetylcysteine controversial.

Contrecoup

An injury occurring on the side of an organ opposite the site of impact, especially a brain injury produced when the brain moves within the skull after trauma.

Control (Research)

A comparison participant or group in a study that does not receive the experimental intervention, or receives standard care or a placebo, allowing the intervention's effects to be estimated.

Control Group

The comparison group in a study, which may receive usual care, a placebo, or no intervention according to the study design.

Controlled Cord Traction

A technique for active management of third-stage labor: gentle, steady traction on the umbilical cord with one hand while the other hand applies suprapubic counter-pressure (to prevent uterine inversion). Performed after signs of placental separation. The Brandt-Andrews maneuver and the Finocchietti technique are CCT methods.

Controlled Ovarian Hyperstimulation

The pharmacological stimulation of the ovaries using exogenous gonadotropins (follicle-stimulating hormone [FSH] and luteinizing hormone [LH] preparations) alongside gonadotropin-releasing hormone (GnRH) agonists or antagonists to induce the simultaneous development and maturation of multiple ovarian follicles. Primarily utilized in in vitro fertilization and intrauterine insemination protocols, requiring close ultrasound follicular tracking and serum estradiol monitoring to optimize oocyte yield while preventing ovarian hyperstimulation syndrome.

Alternate Names:

COH; Ovarian Stimulation

Controllers

Long-term asthma medicines taken regularly to reduce airway inflammation and prevent symptoms or exacerbations.

Alternate Names:

Controller medicines; maintenance asthma therapy.

Controlling Your High Blood Pressure

Long-term blood-pressure management using accurate measurement, lifestyle measures, prescribed medicines, adherence, and monitoring for complications and treatment adverse effects.

Contusion

A closed soft-tissue injury caused by blunt force, producing bleeding into tissues without an open

wound. It commonly causes pain, swelling, and ecchymosis; severe contusions may involve muscle necrosis, compartment syndrome, internal bleeding, or associated bone and organ injury.

Conus Medullaris Syndrome

A clinical condition resulting from damage to the conus medullaris, the tapered, terminal end of the spinal cord located at approximately the L1-L2 vertebral level.

Convergence

The inward movement of both eyes so that they are directed toward the same nearby object.

Convergence Insufficiency

Inability to maintain binocular convergence on near objects, causing eyestrain, headaches, blurred near vision, and transient diplopia during reading. Common in children and young adults; treated with convergence exercises and base-in prism, with surgery rarely needed.

Conversion Disorder

Alternate index label; see *Functional neurologic disorder/conversion disorder*.

Conversion Disorder (Functional Neurological Symptom Disorder)

A condition in which neurological symptoms, such as weakness, abnormal movements, sensory loss, or seizures, are not explained by a recognized structural neurological disease and cause significant impairment.

Alternate Names:

Functional Neurological Symptom Disorder

Convulsion

A sudden episode of involuntary, often rhythmic muscle contractions that may occur with a seizure or other disturbance of brain function.

Alternate Names:

Convulsive episode; seizure-related motor activity.

Cooking Without Salt

Preparing food with little or no added sodium to reduce dietary sodium intake, often useful in blood-pressure or fluid-retention management when combined with an individualized eating plan.

Cooley'S Anemia

Alternate index label; see *Thalassemia*.

Coombs Test

A laboratory test that detects antibodies or complement attached to red blood cells, or antibodies in serum that can react with red cells. It is used in evaluating immune hemolysis and blood compatibility.

Cooper'S Ligament

Fibrous connective-tissue septa extending from the deep surface of the breast toward the skin and pectoral fascia. Tumor invasion or fibrosis may cause skin retraction.

Coordination

The ability to control movements smoothly, accurately, and in the correct order. It depends on the brain, nerves, muscles, vision, and balance and may be tested with finger-to-nose, heel-to-shin, or rapid alternating movements. movements.

COPD

Chronic obstructive pulmonary disease is a progressive, largely irreversible airflow limitation caused by chronic bronchitis, emphysema, or both, most commonly due to cigarette smoking. It presents with chronic cough, sputum production, and exertional dyspnea. Spirometry shows a reduced FEV1/FVC ratio.

COPD - Control Drugs

Long-term medicines used to reduce COPD symptoms and exacerbations, such as long-acting bronchodilators and, for selected patients, inhaled corticosteroids.

COPD - Managing Stress And Your

Mood

Support for anxiety, depression, and stress in chronic obstructive pulmonary disease through breathing strategies, activity, pulmonary rehabilitation, counseling, and treatment when indicated.

COPD - Quick-Relief Drugs

Fast-acting inhaled bronchodilators used for short-term relief of COPD breathlessness or bronchospasm; frequent need may indicate inadequate control or an exacerbation.

COPD And Other Health Problems

The interaction between chronic obstructive pulmonary disease and conditions such as cardiovascular disease, osteoporosis, depression, diabetes, and lung cancer, which can complicate symptoms and care.

COPD Exacerbation

Acute worsening of respiratory symptoms beyond normal day-to-day variation in a patient with COPD that leads to additional therapy. Common triggers include viral or bacterial infection, air pollution, and cardiopulmonary comorbidity.

COPD Flare-Ups

Acute worsening of breathlessness, cough, or sputum beyond usual daily variation in chronic obstructive pulmonary disease, often triggered by infection, pollution, or other irritants.

COPD Rehabilitation

Comprehensive intervention for COPD patients including exercise training, education, behavior change. Components: aerobic exercise, strength training, breathing techniques, nutrition, psychosocial support. Improves exercise capacity, quality of life, reduces hospitalizations.

Copper

An essential trace element required for iron transport, connective-tissue cross-linking, energy metabolism, neurotransmitter synthesis, and antioxidant enzymes. Deficiency can cause anemia, neutropenia, impaired neurologic function, and bone

abnormalities; excess is toxic and accumulates in Wilson disease.

Copper In Diet

A patient-care topic concerning copper in diet. It addresses the relevant purpose, risks, management, and follow-up.

Copper Poisoning

Toxic exposure to copper or copper salts that may cause vomiting, abdominal pain, hemolysis, liver injury, kidney injury, or shock.

Coprolalia

Involuntary utterance of obscene, taboo, or socially inappropriate words, most often described as a complex vocal tic in Tourette syndrome.

Copy Number Variant

A deletion or duplication that changes the number of copies of a DNA segment; its clinical significance depends on the region, size, genes involved, inheritance, and available evidence.

Copy Number Variation

Deletions or duplications of DNA segments >1 kb. Contribute to genetic diversity and disease. Examples: 22q11.2 deletion, Williams syndrome (7q11.23 deletion).

CoQ10

Coenzyme Q10, a lipid-soluble mitochondrial electron-transport compound and antioxidant also sold as a dietary supplement.

Cor

The Latin anatomical term for heart, used in medical names such as cor pulmonale.

Cor Pulmonale

Right ventricular hypertrophy and dilation from pulmonary hypertension due to lung disease. Features include peripheral edema, JVD, and hepatomegaly.

Cor Triatriatum

A rare congenital heart defect where the left atrium is divided into two chambers by a fibromuscular membrane, separating the pulmonary venous chamber (posterior) from the true left atrium (anterior) containing the left atrial appendage and mitral valve. The membrane obstructs pulmonary venous return, mimicking mitral stenosis.

Coracoid Process

Hook-like scapular projection extending anteriorly and laterally. Attachment for 3 muscles (pectoralis minor, coracobrachialis, short head of biceps) and 3 ligaments. Important surgical landmark. Fractures uncommon but may occur with direct trauma or shoulder dislocations.

Cord

A long, narrow structure such as the spinal cord, which contains the central nervous tissue, or the umbilical cord, which connects a fetus to the placenta.

Cord Blood Testing

Laboratory testing of blood collected from the umbilical cord at birth, which may include blood type, complete blood count, blood gases, glucose, or tests for infection or inherited disease when clinically indicated.

Cord Blood Stem Cells

Blood-forming stem cells collected from the umbilical cord and placenta after birth. They can make different types of blood and immune cells and may be used for stem-cell transplantation.

Cord Presentation

The presence of the umbilical cord between the presenting fetal part and the cervix, with intact membranes. Risk factors: malpresentation, unstable lie, low-lying placenta, and long cord. If the membranes rupture, the cord may prolapse (umbilical cord prolapse).

Cordocentesis

A prenatal diagnostic procedure in which fetal blood

is sampled from the umbilical cord, usually under ultrasound guidance.

Cordotomy

Surgical division of selected nerve fibers, especially a spinal cord tract, to interrupt transmission of severe chronic pain.

Core Biopsy

Removal of one or more cylindrical tissue samples with a hollow cutting needle for microscopic examination. It preserves more tissue architecture than fine-needle aspiration and is used for breast, liver, prostate, and other lesions.

Core Needle Biopsy

Biopsy using a hollow, relatively wide cutting needle to obtain tissue cylinders that preserve architecture; ultrasound, CT, MRI, or stereotactic guidance may be used.

Cori Cycle

Lactate produced by muscles transported to liver, converted to glucose via gluconeogenesis, returned to muscles.

Cori Disease

An autosomal recessive inborn error of glycogen degradation (Glycogen Storage Disease Type III [GSD III]) caused by mutations in the *AGL* gene, resulting in a deficiency of the bifunctional glycogen debranching enzyme (amylo-1,6-glucosidase and 4-alpha-glucanotransferase). Leads to incomplete glycogen breakdown and abnormal accumulation of limit dextrin in liver and muscle tissues. Clinically presents in infancy with hepatomegaly, hypoglycemia, hyperlipidemia, and growth impairment; GSD IIIa affects both liver and cardiac/skeletal muscle, resulting in late-onset progressive myopathy and cardiomyopathy, whereas GSD IIIb affects liver tissue exclusively.

Alternate Names:

Glycogen Storage Disease Type III (GSD III); Forbes Disease; Debrancher Deficiency

Corn

A localized thickening of skin caused by repeated pressure or friction, often occurring on the toes or feet.

Alternate Names:

Clavus; heloma.

Cornea

Transparent, avascular, dome-shaped anterior eye surface providing approximately two-thirds of total refractive power. Five layers: epithelium (stratified squamous, rapidly regenerating), Bowman's membrane, stroma (thickest, regular collagen), Descemet's membrane, endothelium (single cell layer maintaining transparency).

Corneal Abrasion

Loss of corneal epithelium caused by mechanical trauma — contact with fingernail, makeup brush, foreign body, contact lens overwear, dry eye, sports injury, paper edge, or surgical instrumentation. Most common eye injury presenting to emergency departments.

Corneal Cross-Linking

Corneal collagen cross-linking is a minimally invasive therapeutic procedure that strengthens the corneal stroma by creating additional chemical bonds between collagen fibrils, thereby stiffening the cornea and halting or slowing the progression of corneal ectatic disorders, particularly keratoconus and post-refractive surgery ectasia.

Corneal Disease

Corneal disease affects the transparent front surface of the eye through infection, inflammation, trauma, dystrophy, degeneration, dryness, or abnormal shape. Pain, redness, light sensitivity, discharge, or reduced vision can signal urgent disease because scarring may impair sight.

Corneal Edema

Excess fluid within the corneal stroma or epithelium causing corneal thickening, reduced transparency, glare, and blurred vision. Causes include endothelial dysfunction, acute glaucoma, inflammation, trauma,

and postoperative injury.

Corneal Graft (Corneal Transplant)

Surgical replacement of part or all of a diseased or damaged cornea with donor corneal tissue to restore vision or relieve pain.

Alternate Names:

Corneal Transplant

Corneal Implant

An implanted device or tissue substitute placed in or on the cornea to restore vision, replace diseased tissue, or alter corneal shape.

Corneal Inflammation (Keratitis)

Alternate index label for Keratitis.

Corneal Injury

Damage to the cornea, such as abrasion, foreign body, chemical burn, infection, or laceration.

Corneal Transplant

A medical procedure involving corneal transplant. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Corneal Ulcer

A corneal ulcer is a defect in the corneal epithelium with associated stromal inflammation and necrosis, typically caused by infectious or non-infectious mechanisms, and it is a sight-threatening condition that requires prompt diagnosis and treatment.

Corneal Ulcers And Infections

Infection or inflammation of the cornea that may produce an epithelial defect or ulcer, pain, redness, tearing, discharge, and light sensitivity. Bacterial, fungal, protozoal, and viral causes can threaten vision and require urgent ophthalmic assessment.

Corneitis

Inflammation of the cornea, often producing pain, photophobia, tearing, and impaired vision. It may be infectious, immune-mediated, traumatic, or re-

lated to exposure.

Cornelia De Lange Syndrome

A genetic disorder caused by mutations in genes of the cohesin complex (NIPBL 60%, SMC1A, SMC3, RAD21, HDAC8).

Corns

A corn is a localized thickening of skin caused by repeated pressure or friction, commonly on toes or other weight-bearing areas. Proper footwear, pressure relief, and treatment of the underlying deformity help; diabetes or poor circulation makes self-removal risky.

Corns And Calluses

Corns are small, localized areas of thickened skin caused by repeated pressure or friction, usually on the toes. Calluses are broader areas of thickened, hardened skin, usually on the soles of the feet or palms of the hands. Both are usually harmless but may become painful.

Coronal

Relating to a plane that divides the body into front and back portions, or to the crown or top of a structure.

Coronal Caries

Caries on clinical crown. Smooth surface or pits/fissures.

Coronal Plane

A vertical plane that divides the body or an organ into front (anterior) and back (posterior) portions.

Coronary

Relating to the arteries or circulation that supply the heart muscle.

Alternate Names:

Cardiac blood-vessel-related; heart-coronary.

Coronary Aneurysm

An abnormal widening or ballooning of a coronary artery. It may result from atherosclerosis, inflam-

mation, infection, injury, or a congenital condition and can cause clotting, blockage, or rarely rupture.

Coronary Angiography

Invasive imaging of the coronary arteries using catheter-delivered contrast and X-ray fluoroscopy. It shows coronary anatomy and stenosis and can support treatment planning; noninvasive tests may be appropriate in other clinical settings.

Coronary Arteries

The right and left arteries arising from the aortic sinuses that supply the myocardium. Their branches include the right coronary, anterior interventricular, and circumflex arteries.

Coronary Artery

One of the blood vessels branching from the aorta that supplies oxygen-rich blood directly to the heart muscle.

Coronary Artery Aneurysm

A localized dilatation of a coronary artery exceeding the normal vessel diameter by approximately 1.5 times. It may be caused by atherosclerosis, Kawasaki disease, connective-tissue disorders, infection, trauma, or an iatrogenic procedure, and can lead to thrombosis, distal embolization, myocardial ischemia, or rupture.

Coronary Artery Anomaly

A congenital abnormality in the origin, course, structure, or termination of a coronary artery. Some anomalies are incidental, whereas others can impair myocardial perfusion or increase the risk of ischemia, arrhythmia, or sudden cardiac death.

Coronary Artery Bypass

A surgical procedure that creates a new route for blood flow around a narrowed or blocked coronary artery using a vascular graft.

Coronary Artery Bypass Grafting

A surgical revascularization procedure for severe coronary artery disease. Grafts, most commonly

the left internal mammary artery (LIMA) and saphenous vein grafts (SVG), are used to bypass stenotic coronary arterial segments. Indicated for multi-vessel CAD, left main coronary artery stenosis, or severe CAD in diabetic patients.

Coronary Artery Bypass Grafting

Open heart surgery to bypass stenotic coronary arteries using autologous grafts. On-pump: cardiopulmonary bypass, cardioplegic arrest. Off-pump: beating heart, no bypass machine.

Alternate Names:

Coronary Artery Bypass Grafting

Coronary Artery Bypass Surgery

Surgery that uses a graft to route blood around a narrowed or blocked coronary artery.

Alternate Names:

CABG; coronary bypass surgery.

Coronary Artery Disease

Narrowing or occlusion of coronary arteries, usually due to atherosclerosis, reducing blood flow to the myocardium. Risk factors include hypertension, diabetes, hyperlipidemia, smoking, family history, and obesity. CAD can present as stable angina, acute coronary syndromes (unstable angina, NSTEMI, STEMI), or sudden cardiac death.

Coronary Artery Fistula

An abnormal connection between a coronary artery and a heart chamber or vessel that may cause ischemia, failure, arrhythmia, or endocarditis.

Coronary Artery Spasm

Transient intense constriction of an epicardial coronary artery that reduces myocardial blood flow. It may cause rest angina and transient ST-segment elevation and can be triggered by smoking, stimulants, or endothelial dysfunction; calcium-channel blockers and nitrates are commonly used for prevention and relief.

Coronary Calcium Score

A CT-based measurement of coronary artery cal-

cification, quantified as the Agatston score. A score of 0 indicates very low risk of cardiovascular events; >400 indicates high burden and significant atherosclerosis. It is used for cardiovascular risk stratification in asymptomatic individuals and may guide statin therapy decisions.

Coronary Calcium Scoring

Non-contrast CT measuring calcified plaque in coronary arteries. Agatston score predicts cardiovascular risk.

Coronary Care Unit

A hospital unit providing continuous monitoring and specialized care for acute cardiac illness.

Alternate Names:

CCU; cardiac care unit.

Coronary Care Unit

A specialized hospital unit providing continuous monitoring and treatment for patients with serious acute cardiac conditions.

Alternate Names:

CCU

Coronary Circulation

The vascular system supplying blood to the myocardium. The left coronary artery divides into the left anterior descending (LAD) and circumflex (LCx) arteries. The right coronary artery (RCA) supplies the right ventricle, SA node (in 60% of people), and inferior left ventricular wall.

Coronary Computed Tomography Angiography

A non-invasive imaging examination of the coronary circulation performed using ECG-gated multidetector computed tomography following intravenous administration of iodinated contrast. It provides high-resolution, three-dimensional visualization of the coronary arterial lumen and vessel wall, allowing precise detection and quantification of both calcified and non-calcified atherosclerotic plaques, assessment of luminal stenosis, and identification of high-risk plaque features (positive remodeling, low attenuation). Recommended as the first-line anatomical

test for evaluating suspected stable coronary artery disease in patients with low-to-intermediate pretest probability.

Alternate Names:

CCTA; Coronary CT Angiography; Cardiac CTA

Coronary Dominance

Determined by which artery gives rise to the posterior descending artery. Right coronary dominance (85% of people): RCA gives rise to PDA. Left coronary dominance (8%): LCx gives rise to PDA. Codominance (7%): both RCA and LCx contribute.

Coronary Endarterectomy

Surgical removal of atherosclerotic plaque from the inner lining of a coronary artery.

Alternate Names:

Coronary plaque endarterectomy; coronary endarterectomy procedure.

Coronary Heart Disease

Disease caused by reduced blood flow through the coronary arteries, usually due to atherosclerotic plaque. It may cause angina, myocardial infarction, heart failure, or sudden cardiac death. Also called coronary artery disease.

Alternate Names:

coronary artery disease.

Coronary Insufficiency

Inadequate blood flow through the coronary circulation to meet myocardial oxygen demand, usually because of coronary artery disease.

Coronary Microvascular Angina

Angina caused by impaired regulation or constriction of the small coronary vessels, often producing ischemic symptoms despite no flow-limiting epicardial coronary obstruction. Assessment may include evaluation for coronary microvascular dysfunction; management is individualized according to symptoms, cardiovascular risk, and the underlying mechanism.

Coronary Microvascular Disease

Alternate index label; see *Small vessel disease*.

Coronary Microvascular Dysfunction

Impairment of the small coronary arteries and arterioles causing impaired coronary flow reserve despite normal epicardial coronary arteries. Common in women, patients with diabetes, and hypertension. Causes chest pain (angina equivalent) and may contribute to heart failure with preserved ejection fraction.

Coronary Occlusion

Obstruction of a coronary artery, commonly by atherosclerotic plaque and thrombus, that can cause myocardial ischemia or infarction.

Coronary Sinus

A large vein on the back of the heart that collects de-oxygenated blood from the heart muscle and drains it into the right atrium.

Coronary Slow Flow Phenomenon

An angiographic microvascular disorder characterized by delayed opacification and clearance of radiopaque contrast in normal or near-normal epicardial coronary arteries (quantified by a corrected TIMI frame count > 27 frames), in the absence of obstructive epicardial stenosis, spasm, dissection, or severe left ventricular dysfunction. Patients typically present with recurrent episodes of acute rest or exertional angina; the pathophysiology involves coronary microvascular endothelial dysfunction, microvascular tone abnormalities, and platelet hyperactivity, managed with calcium channel blockers, dipyridamole, or statins.

Alternate Names:

CSFP; Slow Coronary Flow; Primary Coronary Slow Flow

Coronary Spasm

Temporary contraction of a coronary artery that reduces blood flow to the heart muscle and may cause angina or infarction.

Alternate Names:

Vasospastic angina mechanism; coronary artery

spasm.

Coronary Steal Phenomenon

A paradoxical hemodynamic reduction in coronary blood flow supplying an ischemic, maximally vasodilated myocardial territory dependent on a severely stenotic epicardial coronary artery or collateral network, provoked by pharmacological vasodilation of adjacent normal coronary vascular beds. When a potent, non-selective coronary arteriolar vasodilator (such as dipyridamole, adenosine, or regadenoson) is administered, arterioles in non-ischemic territories with preserved autoregulation dilate vigorously, reducing vascular resistance and shunting blood flow away from the diseased, fixed-resistance stenotic artery, precipitating or aggravating subendocardial ischemia. Forms the diagnostic basis of pharmacological myocardial perfusion stress testing.

Alternate Names:

Coronary Steal Syndrome; Myocardial Steal

Coronary Stent

Mesh tube keeping coronary artery open after angioplasty. Types: bare-metal (higher restenosis), drug-eluting (releases antiproliferative drug, lower restenosis but prolonged DAPT). Complications: stent thrombosis, restenosis, bleeding (from DAPT).

Coronary Thrombosis

Formation of a blood clot in a coronary artery, potentially obstructing blood flow and causing myocardial infarction.

Coronavirus

A medicine or chemical compound identified by coronavirus. Its uses, effects, and risks depend on the specific substance and clinical context.

Coronavirus Disease 2019

A respiratory and systemic infectious disease caused by SARS-CoV-2. Illness ranges from asymptomatic infection to pneumonia, respiratory failure, and multisystem complications.

Corporeal

Relating to the physical body rather than the mind

or spirit.

Corpse

The dead body of a human being; the subject of medicolegal examination. Forensic assessment of a corpse includes identification, determination of death and time of death (rigor mortis, livor mortis, algor mortis, vitreous potassium, gastric emptying), external and internal examination, and collection of evidence.

Corpus

Latin for body; in anatomy it refers to the main body of an organ or structure, such as the corpus callosum or corpus uteri.

Corpus Callosum

Brain's largest white matter tract with approximately 200 million axons connecting left and right hemispheres. Four parts: rostrum, genu, body, splenium. Facilitates interhemispheric communication for sensory, motor, and cognitive integration.

Corpus Cavernosum

One of two columns of erectile tissue in the penis that fill with blood during erection.

Alternate Names:

Cavernous body; penile corpus cavernosum.

Corpus Luteum

The temporary endocrine structure formed from an ovarian follicle after ovulation; it produces progesterone during the luteal phase and early pregnancy.

Alternate Names:

Yellow body; luteal body.

Corpus Luteum Cyst

A physiological functional ovarian cyst that develops when the post-ovulatory corpus luteum fails to regress, seals over, and fills with serous fluid or blood, typically exceeding 3 cm in diameter. Often asymptomatic and self-limiting over 1–3 menstrual cycles, but can cause localized pelvic pain, delayed menses, or acute hemoperitoneum following spontaneous rupture, particularly during early pregnancy

or anticoagulant therapy.

Corpus Spongiosum

A column of erectile tissue surrounding the penile urethra and expanding distally to form the glans.

Alternate Names:

Spongy body; corpus spongiosum penis.

Correct Code For Diagnostic Dilation And Curettage

Dilation and curettage is a procedure that opens the cervix and removes tissue from the uterus for diagnosis or treatment. Coding depends on the indication, setting, technique, pregnancy status, and tissue findings and should be assigned from the documented procedure.

Corrected Age

The age of a preterm infant calculated by subtracting the number of weeks born before 40 weeks of gestation from chronological age. Corrected age helps interpret growth and developmental milestones during early childhood, commonly until about 2 years of age.

Corrigan Pulse

A forceful, rapidly rising and collapsing arterial pulse, classically associated with aortic regurgitation.

Corrosive

Substance causing tissue destruction by chemical action. Acids: coagulation necrosis, eschar. Alkalis: liquefaction necrosis, deep penetration. Ingestion: avoid emesis, endoscopy, surgical consultation.

Corrugator Muscle

A small facial muscle that draws the eyebrows downward and inward, contributing to frowning.

Alternate Names:

Corrugator supercilii; glabellar muscle.

Cortex

The outer layer of an organ or structure, such as the cerebral cortex, adrenal cortex, or outer layer

of a hair shaft.

Alternate Names:

Cortical layer; outer cortex.

Corti'S Ganglion

The spiral ganglion of the cochlea, containing the cell bodies of sensory neurons that transmit auditory information from the organ of Corti.

Cortical

Relating to the cortex, the outer layer of an organ such as the brain, kidney, or adrenal gland.

Cortical Bone

Dense bone forming the outer shell of most bones and providing much of the skeleton's mechanical strength.

Alternate Names:

Compact bone; cortical tissue.

Cortical Cataract

A type of age-related lens opacity that begins in the peripheral lens cortex as wedge-shaped or spoke-like opacities that extend centripetally toward the visual axis, caused by disruption of the normally orderly arrangement of lens fibers and the accumulation of fluid between them.

Corticobasal Degeneration

A rare, progressive brain disorder affecting regions involved in movement and higher brain functions. It usually affects one side more than the other and may cause stiffness and slow movement, difficulty carrying out learned movements (apraxia), involuntary jerks, abnormal muscle contractions (dystonia), reduced sensation, or an "alien limb" that feels or acts outside the person's control.

Corticosteroid

A steroid hormone from the adrenal cortex or a synthetic analogue; glucocorticoids reduce inflammation and immune activity, while mineralocorticoids regulate salt and water balance.

Corticosteroids

Synthetic glucocorticoids with potent anti-inflammatory and immunosuppressive effects. Examples include prednisone, prednisolone, methylprednisolone, dexamethasone, and hydrocortisone. They suppress inflammatory gene expression and inhibit immune cell function.

Corticosteroids Mechanism

Corticosteroids exert their anti-inflammatory and immunosuppressive effects primarily by binding to intracellular glucocorticoid receptors, which then translocate to the nucleus and modulate gene transcription.

Corticosteroids Overdose

Overdose of corticosteroids that may cause gastrointestinal symptoms, mood changes, hyperglycemia, fluid retention, infection risk, or adrenal suppression.

Corticotropin-Releasing Factor

A hypothalamic hormone that stimulates the pituitary gland to release adrenocorticotrophic hormone and helps coordinate the stress response.

Alternate Names:

Corticotropin-releasing hormone; CRH.

Cortisol

The primary glucocorticoid produced by the adrenal cortex (zona fasciculata) under ACTH control; essential for stress response, metabolism, and homeostasis. Functions: gluconeogenesis, glycogenolysis, protein catabolism, lipolysis, anti-inflammatory/immunosuppressive effects, and maintenance of vascular tone.

Cortisol Blood Test

A blood test measuring cortisol, a hormone produced by the adrenal cortex. Results help evaluate adrenal insufficiency or cortisol excess, but vary substantially with time of day, stress, illness, medicines, and the assay used.

Cortisol Urine Test

A test measuring cortisol excreted in a timed urine collection, commonly 24-hour urine free cortisol, used to evaluate suspected excess cortisol production such as Cushing syndrome.

Cortisone

A corticosteroid that is converted in the body to cortisol and can be used to reduce inflammation or replace glucocorticoid activity.

Coryza

Acute inflammation of the nasal mucous membranes with nasal discharge and obstruction, commonly occurring in the common cold or other upper-respiratory infection.

Cosmeceutical

A cosmetic product marketed as having biologically active, drug-like benefits; the term has no uniform formal regulatory category in many jurisdictions.

Cosmetic Allergies

Cosmetic products can cause irritant dermatitis or allergic contact dermatitis, producing itching, redness, swelling, scaling, or blistering where applied. Stopping the suspected product and medical assessment, sometimes with patch testing, help identify the responsible ingredient.

Cosmetic Dermatology

Cosmetic dermatology encompasses aesthetic procedures. Botulinum toxin injections (Botox, Dysport, Xeomin) relax facial muscles reducing wrinkles. Dermal fillers (hyaluronic acid, calcium hydroxylapatite, poly-L-lactic acid, polymethylmethacrylate) restore volume.

Cosmetic Ear Surgery

A medical procedure involving cosmetic ear surgery. Indications, risks, preparation, and follow-up depend on the patient's clinical condition and the specific procedure.

Cost-Effectiveness

Comparison of costs and outcomes of interventions. Used for resource allocation.

Costello Syndrome

A rare multisystem disorder classified as a RASopathy, caused by heterozygous de novo germline missense mutations in the proto-oncogene *HRAS* on chromosome 11p15.5. Classically manifests with severe infantile failure to thrive, persistent feeding difficulties, short stature, characteristic coarse facial appearance, curly sparse hair, loose skin with deep creases on the palms and soles, periorificial papillomata, hypertrophic cardiomyopathy, and arrhythmias (multifocal atrial tachycardia). Patients carry a markedly elevated lifetime risk of benign and malignant embryonal neoplasms, particularly rhabdomyosarcoma and neuroblastoma.

Costochondritis

Costochondritis is inflammation or irritation where ribs join the breastbone, causing localized chest pain often worse with movement or pressing the area.

Alternate Names:

Chest Wall Pain Costosternal Chondrodynia Costosternal Syndrome

Costochondritis (Costochondritis And Tietze Syndrome)

Alternate index label for Costochondritis and Tietze Syndrome.

Costochondritis And Tietze Syndrome

Costochondritis is inflammation or irritation at the junction of ribs and the breastbone, causing reproducible chest-wall pain. Tietze syndrome is a less common related condition with visible swelling at one or a few joints; heart and lung emergencies must still be excluded.

Alternate Names:

Costochondritis (Costochondritis And Tietze Syndrome) Tietze Syndrome (Costochondritis And Tietze Syndrome)

Costosternal Chondrodynia

Alternate index label; see *Costochondritis*.

Costosternal Syndrome

Alternate index label; see *Costochondritis*.

Cot Death

Sudden infant death syndrome, the sudden unexplained death of an apparently healthy infant, usually during sleep, that remains unexplained after investigation.

Cough

Cough is a protective reflex that clears the airways; its duration, triggers, sputum, associated symptoms, and examination findings help distinguish infectious, inflammatory, cardiac, and other causes. foreign material from the tracheobronchial tree.

Cough Augmentation

Techniques to improve cough effectiveness in patients with neuromuscular disease or chest wall weakness. Methods include manually assisted cough (abdominal thrust timed with cough), mechanical insufflation-exsufflation (CoughAssist device), and breath stacking with a resuscitation bag.

Cough Chronic (Chronic Cough)

Alternate index label for Chronic Cough.

Cough Headaches

Headaches triggered by coughing, sneezing, straining, or other sudden increases in pressure in the head.

Cough In Adults

A cough in an adult may be acute, subacute, or chronic and may result from infection, asthma, chronic obstructive pulmonary disease, reflux, upper-airway cough syndrome, medication, or other disease. Coughing blood, breathing difficulty, chest pain, low oxygen, or persistent unexplained cough requires medical assessment.

Cough Suppressant

An antitussive medicine or measure that reduces the urge or reflex to cough; treatment should address the underlying cause and avoid suppressing a productive

cough when clearance is needed.

Coughing Up Blood

Expectoration of blood or blood-streaked sputum from the lower respiratory tract, called hemoptysis; heavy bleeding or breathing difficulty is an emergency.

Could A Stiff Neck Be A Sign Of Something Serious

Neck stiffness commonly follows muscle strain, but stiffness with fever, severe headache, light sensitivity, confusion, rash, weakness, trauma, or neurologic change can signal meningitis, bleeding, spinal injury, or another emergency.

Count Rate

Number of detected events per second. Dead time losses at high count rates.

Countercurrent Multiplication

The renal medullary process in which opposing tubular fluid flow and selective permeability along the loop of Henle generate a corticomedullary osmotic gradient. This gradient permits collecting-duct water reabsorption when antidiuretic hormone is present.

Counterpulsation

Timed inflation and deflation of a balloon or external cuffs in relation to the heartbeat to improve coronary perfusion or reduce cardiac workload, as in intra-aortic balloon counterpulsation.

Countertransference

The therapist's emotional reactions to the patient, which may be influenced by the therapist's own unresolved conflicts or may be a response to the patient's transference. Awareness of countertransference is essential for effective therapy and can provide valuable clinical information about the patient's interpersonal style.

Counting Carbohydrates

Estimating the grams of carbohydrate in foods and

drinks to help plan meals, match insulin, and manage blood glucose, especially in diabetes.

Coup-Contrecoup Injury

Brain injury from acceleration-deceleration trauma. Coup: injury at site of impact. Contrecoup: injury on opposite side. Common in falls, motor vehicle accidents.

Courvoisier Sign

Clinical finding of a palpable non-tender gallbladder in obstructive jaundice suggesting malignant biliary obstruction. In gallstone disease, the gallbladder is typically fibrotic and not palpable, whereas in malignant obstruction it is distended because the wall is initially normal and compliant.

Covered Stent (Vascular)

A vascular stent lined with a graft material that supports the vessel and excludes a leak or aneurysm from the bloodstream.

Alternate Names:

Vascular

COVID Toe

A chilblain-like skin lesion reported in association with COVID-19, usually affecting the toes or fingers and appearing red, purple, or swollen.

COVID-19

Coronavirus disease caused by infection with SARS-CoV-2. Illness ranges from asymptomatic or mild disease to pneumonia, respiratory failure, thrombosis, or death; common symptoms include fever, cough, shortness of breath, fatigue, muscle aches, and altered taste or smell. Some people develop persistent symptoms after the acute infection.

COVID-19 Antibody Test

A blood test that detects antibodies produced in response to SARS-CoV-2 infection or vaccination; it does not reliably diagnose current infection or establish complete immunity.

COVID-19 In Pregnancy

Pregnant women with COVID-19 are at increased risk of severe illness (ICU admission, mechanical ventilation, death) compared to non-pregnant women. Risks to the fetus: preterm birth, cesarean delivery, and possibly stillbirth. Vertical transmission is rare but possible.

COVID-19 Symptoms

Symptoms of SARS-CoV-2 infection, which may include fever, cough, sore throat, congestion, fatigue, muscle aches, headache, loss of smell or taste, vomiting, or diarrhea.

COVID-19 Test

A test that detects SARS-CoV-2 infection. Nucleic acid amplification tests, including RT-PCR, detect viral genetic material, while rapid antigen tests detect viral proteins. Test performance depends on timing, specimen quality, and the amount of virus present.

COVID-19 Vaccines

Vaccines that prime immune responses against SARS-CoV-2 and reduce severe disease, hospitalization, and death; products, schedules, effectiveness, and recommendations vary by age and risk.

COVID-19 Vaccines For Children Ages 6 Months And Older

COVID-19 vaccination guidance for children at least six months old, with product and dose schedules determined by age, prior doses, immune status, and current public-health recommendations.

COVID-19 Virus Test

A test for current SARS-CoV-2 infection, usually using a nucleic acid amplification test or antigen test on a respiratory specimen; results must be interpreted with symptoms, timing, and test performance.

Cow's Milk - Infants

Guidance on cow's milk in infancy: it should not replace breast milk or iron-fortified formula before 12 months because of low iron content and risks of

renal solute load and intestinal blood loss.

Cow'S Milk And Children

Nutritional guidance on cow's milk for children, balancing protein, calcium, vitamin D, fat, iron, allergy, and total intake so milk does not displace a varied diet.

Cowden Syndrome

Autosomal dominant PTEN hamartoma tumor syndrome, caused by loss-of-function mutations in PTEN (10q23) — a negative regulator of the PI3K/AKT/mTOR pathway. Multiple hamartomas and increased risk of benign and malignant tumors.

Cowper'S Gland

One of a pair of small male accessory sex glands below the prostate. Its secretion contributes mucus and fluid to the urethra before ejaculation.

Cowpox

A zoonotic viral infection caused by cowpox virus or related orthopoxviruses, producing localized pustular or ulcerative skin lesions and sometimes systemic illness.

Cox-2

Cyclooxygenase-2, an inducible enzyme that produces inflammatory prostaglandins; selective COX-2 inhibitors reduce pain and inflammation but may have cardiovascular, renal, and gastrointestinal risks.

COX-2 Inhibitors

Medicines that inhibit cyclooxygenase-2, an enzyme involved in prostaglandin production and inflammation.

Alternate Names:

Cyclooxygenase-2 inhibitors; selective COX-2 inhibitors.

Coxa Vara

A deformity in which the angle between the femoral neck and shaft is abnormally reduced, potentially causing a limp, shortened limb, or altered hip me-

chanics.

Coxalgia

Pain localized to the hip joint or periarticular structures (also termed coxodynia). Because the hip capsule receives sensory innervation from the femoral, obturator, and sciatic nerves, intra-articular hip pathology classically presents as groin pain that radiates along the anterior thigh to the knee, often misidentified as primary knee pathology. The differential diagnosis is age-dependent: in children, acute coxalgia mandates immediate exclusion of septic arthritis (using the Kocher criteria) versus transient synovitis, Legg-Calvé-Perthes disease, or slipped capital femoral epiphysis; in active young adults, common etiologies include femoroacetabular impingement, acetabular labral tears, and femoral neck stress fractures; in older adults, dominant causes include advanced osteoarthritis, femoral neck fractures, avascular necrosis (osteonecrosis), and greater trochanteric pain syndrome.

Coxiella Burnetii

An obligate intracellular bacterium that causes Q fever, a zoonotic infection transmitted from animals to humans. The organism is transmitted through inhalation of contaminated aerosols or ingestion of contaminated dairy products. Acute Q fever presents with fever, headache, and pneumonia. Chronic Q fever can cause endocarditis.

Coxsackie (Coxsackie Virus)

Alternate index label for Coxsackie Virus.

Coxsackie Virus Type A16

A member of the Enterovirus genus and the most common cause of hand, foot, and mouth disease, a mild, self-limited viral illness affecting children. It presents with fever, oral vesicles and ulcers, and a maculopapular or vesicular rash on the hands and feet. Complications are rare but may include aseptic meningitis.

Coxsackievirus

A group of enteroviruses that cause a range of human diseases. Group A viruses commonly cause hand-

foot-and-mouth disease and herpangina, whereas group B viruses are associated with myocarditis, pericarditis, pleurodynia, and aseptic meningitis.

CPAP

Continuous positive airway pressure delivered through a mask to splint open the upper airway during sleep. It is commonly used for obstructive sleep apnea and can reduce apneas and related symptoms when used consistently.

CPAP Machine

A device that sends a steady stream of air through a mask to keep the upper airway open during sleep. It is commonly used for obstructive sleep apnea; benefit depends on correct mask fit and regular use.

CpG

A DNA sequence in which cytosine is followed by guanine on the same strand, often occurring in regions whose methylation helps regulate gene expression.

CPK Isoenzymes Test

A blood test that separates creatine kinase isoenzymes, such as CK-MM, CK-MB, and CK-BB, to help identify the tissue source of elevated creatine kinase; cardiac troponin is preferred for suspected myocardial infarction.

CPR

Cardiopulmonary resuscitation is an emergency procedure using chest compressions and, when appropriate, rescue breaths or an automated external defibrillator to support circulation and breathing during cardiac arrest.

CPR - Infant

Cardiopulmonary resuscitation for an unresponsive infant who is not breathing normally, using emergency activation, infant chest compressions, rescue breaths when trained, and an AED according to local guidance.

CPR - Young Child (Age 1 Year To

Onset Of Puberty)

Cardiopulmonary resuscitation for a child from age one year until puberty who is unresponsive and not breathing normally, using emergency activation, compressions, breaths, and an AED as available.

CPT Code For Foreign Body Removal From The Ear

Removal of an ear foreign body is coded according to the documented method, location, anesthesia, complications, and whether a separate evaluation was performed. Objects causing pain, bleeding, hearing loss, or suspected eardrum injury should be removed by a trained clinician.

CPVT

Alternate name; see *Catecholaminergic Polymorphic Ventricular Tachycardia*.

Crabs

Alternate index label; see *Pubic lice (crabs)*.

Cracked Tooth Syndrome

Incomplete fracture of tooth causing pain on biting. Diagnosis difficult. Treatment: crown, root canal if pulpal involvement.

Crackles (Rales)

Discontinuous, non-musical adventitious lung sounds produced by the sudden explosive opening of collapsed small airways and alveoli during inspiration or the bubbling of air through fluid in the bronchioles.

Alternate Names:

Rales

Cradle Cap

A common form of seborrheic dermatitis in infants, causing greasy or flaky scales on the scalp with mild redness. It is usually harmless and improves with gentle scalp care; persistent inflammation or signs of infection require assessment.

Cramp, Muscle

Alternate index label; see *Muscle cramp*.

Cramping Leg Pain

Pain caused by involuntary, often painful muscle contraction in the leg; contributors include exertion, dehydration, electrolyte abnormalities, medications, nerve disease, and impaired circulation.

Cramps But No Period

Pelvic cramps without menstruation can result from ovulation, pregnancy, implantation, constipation, urinary disease, infection, endometriosis, cysts, or other pelvic conditions. Severe one-sided pain, faintness, fever, vomiting, or possible pregnancy requires urgent assessment.

Cramps Heat (Heat Cramps)

Alternate index label for Heat Cramps.

Cramps Menstrual (Menstrual Cramps)

Alternate index label for Menstrual Cramps.

Cramps, Menstrual

Alternate index label; see *Menstrual cramps*.

Cranial

Relating to the skull or toward the head, as opposed to caudal or toward the lower end of the body.

Cranial Arteritis

Giant-cell arteritis, an inflammatory disease of medium and large arteries, especially branches of the external carotid artery, that can cause headache, visual loss, and systemic symptoms.

Cranial Cavity

The hollow space inside the skull that contains and protects the brain.

Cranial Fasciitis

A benign, rapidly growing fibroblastic proliferation of the scalp or cranial soft tissues, usually occurring in infants and young children and potentially mimicking a malignant tumor.

Cranial Implant

An artificial device placed in or on the skull or cra-

nial cavity for reconstruction, protection, drainage, stimulation, or treatment.

Cranial Mononeuropathy III

Third-nerve damage causing some combination of double vision, eyelid drooping, and abnormal eye movement.

Cranial Mononeuropathy III - Diabetic Type

Diabetes-associated third-nerve palsy, typically causing painful double vision and eyelid drooping while sparing the pupil.

Cranial Mononeuropathy VI

Sixth-nerve damage causing weakness of outward eye movement and horizontal double vision.

Cranial Nerve

Twelve paired nerves (CN I-XII) emerging directly from the brain/brainstem, primarily serving head and neck.

Cranial Nerve Palsy

Weakness or paralysis of one or more cranial nerves, causing deficits such as impaired eye movement, facial weakness, hearing loss, swallowing difficulty, or tongue weakness. The cause may be vascular, traumatic, compressive, inflammatory, infectious, or metabolic.

Cranial Nerve X

The vagus nerve, the tenth cranial nerve, which carries parasympathetic and sensory fibers and contributes to swallowing, voice, and regulation of thoracic and abdominal organs.

Cranial Nerves

The 12 pairs of nerves arising from the brain or brainstem that provide sensory, motor, and autonomic functions to the head, neck, and parts of the trunk.

Cranial Sutures

Fibrous joints between skull bones that permit mold-

ing in infancy and accommodate brain growth before progressively fusing; premature fusion causes craniosynostosis.

Craniectomy

Surgical removal of part of the skull without immediate replacement, performed to relieve intracranial pressure or provide access to the brain.

Craniocerebral Injury

An injury involving the scalp, skull, brain, or more than one of these structures.

Craniofacial Abnormalities

Abnormalities of the skull, face, or related tissues that may be present at birth or develop later. Causes include genetic conditions, disrupted development, trauma, infection, and tumors.

Craniofacial Dysostosis

A congenital disorder involving premature fusion of cranial sutures and characteristic abnormalities of the face and skull, such as those seen in Crouzon syndrome.

Craniopharyngioma

A usually slow-growing epithelial tumor near the pituitary and hypothalamus that can cause visual loss, hormonal deficiencies, headache, or increased intracranial pressure.

Cranioplasty

Surgical repair or reconstruction of a skull defect using bone, an implant, or another suitable material.

Craniosynostosis

Craniosynostosis is premature fusion of one or more skull sutures, altering head shape and sometimes increasing pressure on the developing brain.

Alternate Names:

Congenital Plagiocephaly Craniostenosis

Craniosynostosis Repair

A medical procedure involving craniosynostosis repair. Indications, risks, preparation, and follow-up

depend on the patient's clinical condition and the specific procedure.

Craniotabes

Soft, thin, or pliable areas of an infant's skull that indent under gentle pressure, commonly associated with normal newborn molding but also with rickets or other bone disease.

Craniotomy

Surgical opening of the skull (bone flap) for access to intracranial contents. Indications: brain tumor resection (glioblastoma, meningioma, mets), hematoma evacuation (SDH, EDH, ICH), aneurysm clipping, AVM resection, and epilepsy surgery (lobectomy).

Cranium

The part of the skull that encloses and protects the brain, consisting of the cranial bones and their joints.

Crapulence

An unpleasant state of sickness or discomfort caused by excessive eating or drinking, especially excessive alcohol use.

Craving

A strong, often difficult-to-control desire for a substance, behavior, or experience, commonly seen in substance-use disorders.

Alternate Names:

Strong urge; compulsive desire.

Crazy Paving Pattern

A HRCT finding characterized by ground-glass opacification with superimposed interlobular septal thickening and intralobular lines. It is seen in pulmonary alveolar proteinosis, lipoid pneumonia, ARDS, pulmonary hemorrhage, and Pneumocystis pneumonia. Despite its name, it is not specific to any single diagnosis.

CRC

Colorectal cancer, malignant disease arising in the

colon or rectum; screening and evaluation depend on age, risk, symptoms, and test results.

CRE

Carbapenem-resistant Enterobacteriaceae, resistant to carbapenem antibiotics through production of carbapenemases including KPC, NDM, and OXA-48-like enzymes. CRE are a major public health threat due to limited treatment options. Infections are associated with high mortality rates.

CRE Infection

Carbapenem-resistant Enterobacterales are bacteria resistant to important antibiotics and can cause urinary, bloodstream, abdominal, wound, or lung infections. Infection-control precautions, culture-based treatment, and specialist advice are important because therapeutic options may be limited.

Alternate Names:

Carbapenem Resistant Enterobacteriaceae (CRE Infection)

Cream

A semisolid emulsion (oil in water or water in oil) for topical application, containing a drug in an emulsion base. Creams are lighter, less greasy, and easier to spread than ointments, and are preferred for moist or oozing skin lesions. They penetrate less deeply than ointments and are cosmetically acceptable.

Creatine

A nitrogen-containing compound stored mainly in skeletal muscle that helps supply energy for short, high-intensity activity.

Creatine Kinase

An enzyme found in muscle, brain, and heart. CK-MB: cardiac isoenzyme (used historically for MI diagnosis, largely replaced by troponin). CK-MM: skeletal muscle (elevated in rhabdomyolysis, trauma, statins, myopathy). Total CK: elevated in muscle injury, myocardial infarction, seizures, hypothyroidism, and statin myopathy.

Creatine Phosphokinase Test

A blood test measuring creatine kinase, an enzyme

released with muscle injury; elevated levels may occur in skeletal-muscle disease, strenuous exercise, rhabdomyolysis, or myocardial injury.

Creating A Family Health History

Collecting a record of relatives' diseases, ages at diagnosis, ancestry, and causes of death to identify inherited risk and guide screening, prevention, and genetic counseling.

Creatinine

A breakdown product of muscle creatine phosphate produced at a constant rate, filtered by the glomerulus and minimally reabsorbed; serum creatinine is the most used marker of kidney function.

Creatinine Blood Test

A blood test measuring serum creatinine, a waste product generated largely by skeletal muscle. It is used with age, body size, and other factors to estimate kidney filtration, but interpretation is affected by muscle mass, diet, medicines, and acute illness.

Creatinine Clearance Test

A test estimating kidney filtration from creatinine in a timed urine collection and blood sample. It can be affected by incomplete urine collection and creatinine secretion, so estimated glomerular filtration rate is often preferred for routine assessment.

Creatinine Test

A blood or urine test measuring creatinine to help assess kidney filtration and muscle-related creatinine production.

Alternate Names:

Creatinine measurement; renal function creatinine test.

Creatinine Urine Test

A urine test measuring creatinine concentration or excretion, used to assess kidney handling, interpret other urine measurements, or judge the completeness of a timed collection. Results depend on muscle mass, diet, collection quality, and kidney function.

Creeping Eruption

An intensely itchy winding skin track caused by larvae, usually hookworms from contaminated soil or sand, migrating within the superficial skin; it is also called cutaneous larva migrans.

Creptitation (Crepitus)

A crackling, grating, or popping sensation or sound produced by movement of tissues, air, or bone. Causes include fractures, joint disease, subcutaneous air, and lung disease.

Alternate Names:

Crepitus

Crepitus

A crackling, grating, or popping sensation or sound caused by movement of tissue, bone, joint surfaces, or air in soft tissue.

Alternate Names:

Crepitation; crackling sensation.

Crescentic Glomerulonephritis

A rapidly progressive kidney inflammation in which crescent-shaped lesions form in the glomeruli.

CREST Syndrome

A limited cutaneous form of systemic sclerosis characterized by calcinosis, Raynaud phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia. It may involve pulmonary vascular disease and other systemic complications.

Creutzfeldt-Jakob Disease

A rapidly progressive neurodegenerative disease caused by abnormal prion proteins, leading to cognitive decline, ataxia, myoclonus, and death.

Creutzfeldt Jakob Disease

Creutzfeldt-Jakob disease is a rare, fatal, transmissible spongiform encephalopathy caused by the accumulation of misfolded prion protein (PrP^{Sc}) in the brain, leading to rapid, progressive dementia and myoclonus.

Creutzfeldt-Jakob Disease

A rapidly progressive fatal neurodegenerative disease caused by abnormal prion protein, characterized by dementia, myoclonus, ataxia, and other neurologic signs.

Alternate Names:

CJD Mad Cow Disease Variant CJD VCJD

Cri Du Chat Syndrome

A congenital genetic disorder caused by deletion of part of the short arm of chromosome 5, characterized in infancy by a high-pitched cry, developmental delay, and characteristic facial features.

Cri-Du-Chat Syndrome

A genetic disorder caused by a deletion of the short arm of chromosome 5 (5p-), with the size of the deletion correlating with severity. Incidence 1 in 15,000-50,000.

Crib Death

Alternate index label; see *Sudden infant death syndrome*.

Cricoid Cartilage

The cricoid cartilage is a complete, ring-shaped cartilage located at the base of the larynx, directly above the trachea and below the thyroid cartilage, and it is the only complete cartilaginous ring in the airway, providing structural support and maintaining airway patency.

Cricoid Pressure

Cricoid pressure (Sellick maneuver) applies posterior displacement of the cricoid cartilage against the cervical vertebrae, occluding the esophagus and reducing aspiration risk during rapid sequence intubation.

Cricothyrotomy

An emergency airway procedure that creates an opening through the cricothyroid membrane to provide oxygenation and ventilation when a patient cannot be intubated and cannot be oxygenated by less invasive methods.

Crigler-Najjar Syndrome

An inherited disorder of bilirubin conjugation caused by reduced or absent UGT1A1 activity, producing unconjugated hyperbilirubinemia of variable severity.

Crime Scene Investigation

The systematic documentation, collection, preservation, and analysis of physical evidence from a suspected crime scene, often supporting forensic medical and legal conclusions.

Criminal Abortion

Abortion performed illegally, outside permitted legal indications and procedures (e.g., unsafe methods, unqualified practitioners, or non-medical settings), carrying high maternal risk.

CRISPR-Cas9

Genome editing tool derived from bacterial immune system. Cas9 enzyme guided by RNA to specific DNA sequence, creating double-strand break. Repaired by non-homologous end joining (NHEJ) or homology-directed repair (HDR). Applications: research, therapy.

CRISPR Therapy

A treatment that uses CRISPR-based genome-editing tools to make a targeted change to DNA in a patient's cells. Cells may be edited outside the body and returned to the patient, or edited directly inside the body. It is used or studied for selected genetic and other diseases; possible concerns include unintended genetic changes, immune reactions, and effects that require long-term monitoring.

Critical Care

Specialized care for people with life-threatening illness or injury who need close monitoring and support for failing organs.

Critical Care Nursing

Specialized nursing care for critically ill patients requiring close monitoring, organ support, rapid assessment, complex therapies, and coordination with multidisciplinary intensive-care teams.

Critical Findings

Findings requiring immediate communication. Examples: pneumothorax, brain herniation, aortic dissection.

Critical Illness Myopathy

An acute, acquired, primary skeletal muscle disorder that develops in critically ill patients treated in intensive care units, frequently triggered by systemic inflammatory response syndrome, sepsis, multiorgan failure, and prolonged administration of high-dose systemic corticosteroids or neuromuscular blocking agents. Characterized by profound, generalized, flaccid, symmetrical weakness affecting both proximal and distal limb musculature as well as diaphragmatic and respiratory muscles, resulting in prolonged failure to wean from mechanical ventilation. Serum creatine kinase is normal or only mildly elevated; electrophysiology reveals reduced muscle membrane excitability with normal sensory nerve action potentials, and muscle biopsy demonstrates pathognomonic selective loss of thick myosin filaments (myosin-deficient myopathy).

Alternate Names:

CIM; Acute Quadriplegic Myopathy; Critical Care Myopathy

Critical Limb Ischemia

Severe chronic reduction in limb blood flow causing ischemic rest pain, nonhealing wounds, or gangrene. It requires urgent vascular assessment.

Critical Results

Test results indicating a potentially life-threatening condition that require prompt communication and clinical action.

Critical Value

A laboratory result indicating a potentially life-threatening condition that requires prompt communication to a responsible clinician or care team. The threshold is test- and laboratory-specific and should be defined by institutional policy.

Crocodile Tears Syndrome

Gustatory lacrimation, in which eating or smelling

food triggers excessive tearing, usually after facial-nerve injury or aberrant regeneration following Bell palsy.

Crohn Disease

Chronic transmural inflammatory condition that can affect any segment of the gastrointestinal tract from the oral cavity to the perianal region, though the terminal ileum and ileocolonic region are most commonly involved.

Crohn Disease (Crohn's Disease)

Alternate index label for Crohn's Disease.

Crohn Disease Cobblestoning

Endoscopic appearance of the colonic or small bowel mucosa in Crohn disease characterized by a bumpy, irregular surface resembling cobblestones. This distinctive appearance results from intersecting lines of deep linear and serpiginous ulceration that surround areas of relatively preserved or edematous but intact mucosa.

Crohn's Disease

A chronic inflammatory bowel disease that can cause patchy, transmural inflammation anywhere from the mouth to the anus, most often involving the terminal ileum or colon. Common features include abdominal cramping or pain, persistent diarrhoea, weight loss, fatigue, fever, anaemia, and extraintestinal inflammation of the joints, eyes, or skin.

Alternate Names:

Crohn Disease (Crohn's Disease) Ileitis (Crohn's Disease)

Crohn's Disease Cause Dark Circles Around Eyes

Crohn disease does not directly cause dark circles in every patient, but anemia, fatigue, dehydration, poor sleep, nutritional deficiency, steroid effects, or atopic skin may contribute. Persistent pallor, fatigue, weight loss, bleeding, or active bowel symptoms deserve clinical review.

Cross-Matching

A laboratory test that checks donor and recipient

blood for compatibility before transfusion.

Cross-Section

A slice or plane made through a structure, or an image showing that slice, such as a cross-sectional CT image.

Cross-Sectional Study

Measures exposure and outcome at same time. Prevalence study. Cannot establish temporality.

Cross-Tolerance

Tolerance to one substance that extends to another substance with a similar mechanism of action. For example, tolerance to one benzodiazepine extends to all benzodiazepines, and tolerance to alcohol extends partially to benzodiazepines due to shared GABA-A receptor mechanisms.

Crossmatch

Laboratory test mixing donor and recipient blood to detect preformed antibodies that could cause transfusion or transplant reactions.

Crotch

The region where the inner thighs meet the trunk, including the groin and nearby perineal structures.

Croup

An acute viral infection causing inflammation and narrowing of the upper airway, typically producing a barking cough, hoarseness, and inspiratory stridor in young children.

Alternate Names:

Laryngotracheobronchitis (Croup)

Crouzon Syndrome

An autosomal dominant craniosynostosis syndrome caused by mutations in the FGFR2 or FGFR3 genes, leading to premature fusion of multiple cranial sutures. Features: brachycephaly (short, broad head from bilateral coronal synostosis), proptosis (shallow orbits), hypertelorism, midface hypoplasia, beaked nose, and Class III malocclusion.

Crown

The visible part of a tooth above the gum line, or a dental restoration that covers and protects a damaged tooth.

Alternate Names:

Dental crown; anatomic tooth crown.

Crown (Dental)

Cap covering damaged tooth. Materials: metal, porcelain-fused-to-metal, all-ceramic, zirconia. Indications: large caries, fractured tooth, root canal treated tooth.

Alternate Names:

Dental

Crown Lengthening

Surgical procedure exposing more tooth structure. Involves gingivectomy and bone removal. For restorative or aesthetic purposes.

Crown-Rump Length

The measurement from the top of the fetal head (crown) to the bottom of the buttocks (rump), the most accurate method for establishing gestational age in the first trimester. Measured at 7-13 weeks. Accuracy: ± 3 -5 days. The EDD is set using CRL if the difference from LMP dating exceeds 7 days. CRL growth: approximately 1 mm per day.

Crowning

The stage of the second stage of labor when the fetal head (widest diameter) distends the perineum and becomes visible at the introitus and is no longer receding between contractions.

CRP

C-reactive protein, a substance made by the liver that rises during inflammation or infection. A CRP test can help assess inflammation and response to treatment, but it does not identify the cause by itself.

Crush Injury

Tissue damage caused by prolonged compression, potentially producing muscle necrosis, compartment

syndrome, rhabdomyolysis, hyperkalemia, shock, and acute kidney injury.

Crush Syndrome

Systemic manifestations of prolonged muscle crush injury (rhabdomyolysis) following trauma — typically from collapsed buildings, landslides, entrapment under debris, or excessive compression of limbs. Bywaters syndrome.

Crushing Injury

Damage caused when a body part is compressed between heavy objects. It can produce fractures, nerve and blood-vessel injury, tissue death, and rhabdomyolysis. Severe limb compression or sudden release after prolonged entrapment requires emergency treatment.

Crust

Dried serum, blood, pus, or other exudate on the skin surface, often covering an erosion, ulcer, vesicle, pustule, or inflamed lesion.

Crusted Scabies

A severe, highly contagious form of scabies (Norwegian scabies) characterised by thick, hyperkeratotic crusts teeming with enormous numbers of *Sarcoptes scabiei* mites, typically occurring in immunocompromised, elderly, or neurologically impaired individuals.

Crutches

Walking aids. Types: axillary, forearm (Loftstrand). Axillary: under arm. Forearm: through loop.

Crutches And Children - Sitting And Getting Up From A Chair

Instruction for a child using crutches to approach a chair, keep the injured limb protected as prescribed, reach for the armrests, and sit or stand safely without losing balance.

Crutches And Children - Stairs

Instruction for safe stair use by a child with crutches, including handrail use, crutch placement, step se-

quencing, supervision, and adherence to weight-bearing restrictions.

Crutches And Children - Standing And Walking

Instruction for fitting and using crutches in children, including posture, handgrip support, gait pattern, weight-bearing limits, turning, and fall prevention.

Cry For Help

A verbal or behavioral expression of distress or need for assistance; in a mental-health setting it should be taken seriously and assessed for immediate safety concerns.

Crying In Childhood

Vocal expression of distress or emotion in a child; persistent, inconsolable, unusual, or pain-associated crying may indicate illness, injury, abuse, or unmet developmental needs.

Crying In Infancy

Crying is an infant's normal communication of hunger, discomfort, fatigue, or need for contact, but sudden inconsolable crying with lethargy, fever, vomiting, or injury needs urgent assessment.

Cryoablation

A procedure that freezes and destroys selected tissue, often using imaging to guide the treatment.

Cryobiopsy

A bronchoscopic technique using rapid freezing to obtain larger lung tissue samples than conventional forceps transbronchial biopsy. It can improve histopathologic sampling in selected interstitial lung diseases, but requires specialized expertise and has risks including bleeding and pneumothorax; anesthesia and procedural setting vary.

Cryoglobulinemia

Cryoglobulinemia occurs when abnormal proteins precipitate in the cold and can cause blood-vessel inflammation, rash, nerve disease, kidney injury, or hyperviscosity.

Cryoglobulinemic Vasculitis

Systemic vasculitis caused by cryoglobulin-containing immune complexes that deposit in small and medium vessels, classically presenting with purpura, arthralgia, and glomerulonephritis. Most cases are associated with hepatitis C virus infection.

Cryoglobulins

Immunoglobulins that precipitate at low temperatures and dissolve on warming; circulating cryoglobulins can cause vasculitis, purpura, neuropathy, kidney disease, or hyperviscosity.

Cryopexy

A procedure that uses controlled freezing to destroy or seal abnormal tissue, especially retinal breaks or small retinal detachments.

Cryoprecipitate

A plasma fraction rich in fibrinogen, factor VIII, von Willebrand factor, and factor XIII, used selectively to treat certain bleeding disorders.

Cryopreservation

Preservation of cells, tissues, or embryos by cooling them to very low temperatures, usually with protective substances that limit ice-related injury.

Cryopyrin-Associated Autoinflammatory Syndromes

Alternate index label; see *Cryopyrin-Associated Periodic Syndromes*; the group is also called CAPS.

Cryopyrin-Associated Periodic Syndromes

Autosomal dominant autoinflammatory disorders (NLRP3 mutations, gain-of-function) causing constitutive inflammasome activation and IL-1beta overproduction.

Alternate Names:

Cryopyrin-Associated Autoinflammatory Syndromes

Cryostat

An instrument that maintains very low tempera-

tures and cuts thin frozen sections of tissue for rapid microscopic examination.

Cryosurgery

Surgery that destroys abnormal tissue by controlled freezing, used in selected skin lesions, tumors, and other conditions.

Cryotherapy

The therapeutic application of extreme cold (typically using liquid nitrogen at -196°C) to destroy abnormal or diseased tissue.

Cryotherapy For Prostate Cancer

A treatment that destroys prostate tissue by repeated freezing and thawing, used selectively for localized or recurrent prostate cancer.

Cryotherapy For The Skin

Destruction of selected skin lesions by controlled freezing, usually with liquid nitrogen, for conditions such as warts, actinic keratoses, or some superficial tumors.

Crypt

A small cavity, gland, or recess, such as an intestinal crypt or tonsillar crypt.

Cryptitis

Inflammation of a crypt or small anatomical recess, especially inflammation of the anal glands and crypts.

Crypto

Clinical shorthand that may refer to cryptococcosis or *Cryptosporidium* infection; the intended meaning depends on the clinical context.

Cryptococcal Meningitis

Infection of the meninges by *Cryptococcus*, usually in people with impaired immunity, causing headache, fever, altered mental status, or other neurologic symptoms.

Cryptococcosis

Cryptococcosis is fungal infection caused by *Cryptococcus*, often affecting the lungs or brain in people with immune suppression.

Alternate Names:

Infection *Cryptococcus* (Cryptococcosis)

Cryptococcus Neoformans

An encapsulated yeast that can cause pulmonary or central nervous system infection, especially in people with impaired immunity.

Cryptogenic Organizing Pneumonia

An inflammatory lung disease with granulation tissue within alveolar ducts. May be idiopathic or secondary to infection, drugs, or connective tissue disease. Responds well to corticosteroids.

Cryptorchidism

Failure of one or both testes to descend into the scrotum, the most common congenital genitourinary anomaly in boys (3% at birth, 1% at 1 year). 70% of undescended testes are palpable (inguinal), 30% non-palpable (intra-abdominal, absent, or atrophic). Spontaneous descent by 6 months occurs in about one-third.

Cryptosporidiosis

A diarrhoeal disease caused by the protozoan parasite *Cryptosporidium parvum* (zoonotic) or *C. hominis* (anthroponotic), transmitted via contaminated water (chlorine-resistant oocysts), food, or direct contact. Presents with watery diarrhea, abdominal cramps, nausea, and low-grade fever, lasting 1-2 weeks.

Cryptosporidium Enteritis

Diarrheal intestinal infection caused by *Cryptosporidium* parasites, transmitted through contaminated water, food, or contact; illness can be prolonged or severe in immunocompromised people.

Crystal Arthropathy

Joint disease caused by crystal deposition. Gout (monosodium urate) and pseudogout (calcium py-

rophosphate). Diagnosis by joint aspiration and polarized microscopy.

Crystal-Induced Synovitis

Acute or chronic joint inflammation caused by deposition of crystals such as monosodium urate, calcium pyrophosphate, or basic calcium phosphate in or around the joint.

Crystalline Lens

The transparent flexible structure inside the eye that focuses light onto the retina.

Crystalloids

Aqueous solutions of mineral salts that distribute between intravascular and interstitial compartments. Normal saline (0.9 percent NaCl) has 154 mEq/L each of sodium and chloride, pH 5.4, and is associated with hyperchloremic metabolic acidosis at large volumes.

Crystalluria

The presence of crystals in urine, which may reflect precipitation of salts, medication effects, metabolic disease, or a tendency to form urinary stones.

CSF Analysis

Alternate index label; see *Cerebrospinal fluid analysis*.

CSF Cell Count

Laboratory measurement of red and white blood cells in cerebrospinal fluid, used to evaluate meningitis, encephalitis, subarachnoid hemorrhage, inflammation, malignancy, and other central nervous system disorders.

CSF Coccidioides Complement Fixation Test

A cerebrospinal-fluid test for complement-fixing antibodies to *Coccidioides*, used with other findings to evaluate suspected coccidioidal meningitis.

CSF Glucose Test

A laboratory measurement of glucose in cere-

brospinal fluid, usually interpreted with a simultaneous blood-glucose value. Low cerebrospinal-fluid glucose may occur in bacterial, fungal, or tuberculous meningitis and other disorders, but results require the complete clinical and cerebrospinal-fluid profile.

CSF Leak

Escape of cerebrospinal fluid through a defect in the skull base or spinal dura, causing positional headache, clear nasal or ear drainage, wound leakage, meningitis risk, or intracranial hypotension.

CSF Leak (Cerebrospinal Fluid Leak)

Escape of cerebrospinal fluid through a tear in the membranes surrounding the brain or spinal cord, often causing positional headache.

Alternate Names:

Spinal Fluid Leak

CSF Myelin Basic Protein

CSF myelin basic protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

CSF Oligoclonal Banding

An electrophoretic test for restricted immunoglobulin bands in cerebrospinal fluid, interpreted with serum bands to detect intrathecal antibody production such as in multiple sclerosis.

CSF Rhinorrhea

Cerebrospinal fluid rhinorrhea is the leakage of cerebrospinal fluid from the subarachnoid space through a defect in the skull base into the nasal cavity or paranasal sinuses, resulting in a clear, watery nasal discharge that may be mistaken for allergic rhinitis or sinusitis.

CSF Smear

Microscopic examination of cerebrospinal fluid for cells, organisms, or abnormal material, supporting evaluation of meningitis, hemorrhage, malignancy, or inflammation.

CSF Total Protein

CSF total protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

CSF-VDRL Test

A cerebrospinal-fluid test for reagin antibodies associated with *Treponema pallidum* infection; a reactive result is highly specific for neurosyphilis, but a nonreactive result does not exclude it.

CT

Computed tomography, an imaging method that uses multiple X-ray measurements and computer reconstruction to produce cross-sectional images.

Alternate Names:

Computed tomography; CAT scan.

CT Angiography

CT imaging with IV contrast timed to visualize arteries. Coronary CTA: assesses coronary artery stenosis and plaque, calcium scoring (Agatston). CTA chest: pulmonary embolism (PE protocol), aortic dissection. CTA head/neck: stroke evaluation, aneurysm, dissection, and vasculitis.

CT Angiography - Abdomen And Pelvis

Computed tomography angiography of the abdominal and pelvic blood vessels after intravenous contrast, used to assess aneurysm, stenosis, dissection, occlusion, bleeding, and vascular anatomy.

CT Angiography - Arms And Legs

Computed tomography angiography of the peripheral arteries after intravenous contrast, used to evaluate arterial narrowing, blockage, aneurysm, injury, or other vascular abnormalities in the limbs.

CT Angiography - Head And Neck

Computed tomography angiography of the blood vessels supplying the brain and neck after intravenous contrast, used to evaluate aneurysm, stenosis, dissection, occlusion, and vascular malformation.

CT Angiography – Chest

Computed tomography angiography of the chest uses intravenous contrast and computed tomography to visualize the thoracic aorta, pulmonary arteries, or other chest vessels. It can evaluate embolism, aneurysm, dissection, bleeding, and vascular malformation.

CT Chest

Detailed lung and mediastinal imaging. High-resolution CT for interstitial lung disease. CT pulmonary angiography (CTPA) for pulmonary embolism. Lung cancer screening with low-dose CT.

CT Colonography

CT imaging of the colon after bowel preparation and insufflation with air or carbon dioxide. Used for colorectal cancer screening in patients unable to undergo optical colonoscopy and for evaluating complete colonoscopy when the cecum was not reached. Detects polyps greater than six millimeters with high sensitivity.

CT Colonography (Virtual Colonoscopy)

CT imaging of the colon after bowel preparation and air insufflation. Indications: colorectal cancer screening (alternative to colonoscopy), incomplete colonoscopy due to obstructing mass, and patients unable to undergo colonoscopy. Sensitivity >90% for polyps >10mm. Requires same bowel prep as colonoscopy.

Alternate Names:

Virtual Colonoscopy

CT Contrast

A substance administered during computed tomography to make selected tissues, vessels, or body spaces more visible.

CT Coronary Angiography

CT imaging of coronary arteries after contrast. Detects stenosis, plaque. Alternative to invasive angiography in low-intermediate risk.

CT Dose-Length Product

A CT dose metric calculated from the volume CT dose index and scan length. It estimates the total scanner output for an examination and is used for protocol optimization and dose monitoring, not as a direct patient-specific risk estimate.

CT Enterography

CT imaging of the small bowel after administration of large volumes of neutral oral contrast distending the small bowel with IV contrast. Excellent for evaluating Crohn disease, small bowel tumors, and obscure gastrointestinal bleeding.

CT Perfusion

A CT technique that estimates blood flow and related perfusion parameters in tissue, commonly to assess selected cases of acute stroke or tumors.

CT Pulmonary Angiography

CT imaging performed after intravenous contrast to show the pulmonary arteries and evaluate suspected pulmonary embolism or other vascular abnormalities.

CT Scan

Computed tomography, an imaging method that uses X-rays and computer processing to produce cross-sectional images of the body.

CT Scan (Computed Tomography)

Cross-sectional imaging using X-rays and computer reconstruction. Non-contrast: evaluate hemorrhage, stones, and bone. IV contrast (iodinated): evaluate inflammation, infection, tumor, and vascular structures. Oral contrast: opacify bowel.

Alternate Names:

Computed Tomography

Ctenocephalides FelFlea Bite (Flea Bites In Humans)

Alternate index label for Flea Bites in Humans.

CTL

Cytotoxic T lymphocyte, an immune cell that rec-

ognizes and kills infected, abnormal, or target cells through antigen-specific mechanisms.

CTPA

Alternate name; see *Computed Tomography Pulmonary Angiography*.

CTS

Commonly, carpal tunnel syndrome: compression of the median nerve at the wrist causing numbness, tingling, pain, or weakness in the hand.

Cubital

Relating to the elbow or the forearm, especially the area around the elbow.

Cubital Tunnel Syndrome

Cubital tunnel syndrome: ulnar nerve compression at elbow (cubital tunnel: Osborne's ligament, FCU aponeurosis, medial epicondyle, olecranon). Clinical: paresthesias ulnar 1.5 digits, nocturnal, Tinel at elbow, Froment sign (IP flexion of thumb with key pinch), Wartenberg sign (abduction of small finger), thenar/hypothenar wasting late.

Cubitus

The elbow or the forearm; the term is also used in names of structures and conditions around the elbow.

Cubitus Valgus

An abnormal outward angulation of the forearm relative to the upper arm when the elbow is extended.

Cubitus Varus

An abnormal inward angulation of the forearm relative to the upper arm when the elbow is extended, often developing after a childhood elbow fracture.

Cuboid

Cube-shaped lateral foot tarsal bone articulating with calcaneus proximally and 4th/5th metatarsals distally. Attachment for the long plantar ligament and peroneus longus tendon. Keystone of the lateral longitudinal arch. Cuboid syndrome (subluxation)

is a common lateral foot pain cause in athletes from excessive pronation.

Cul-De-Sac

A blind-ending pouch or recess; in pelvic anatomy, the rectouterine pouch and rectovesical pouch are examples.

Culdocentesis

Surgical puncture of the posterior vaginal fornix to sample or drain fluid from the rectouterine pouch.

Cullen'S Sign

Bluish or violaceous periumbilical ecchymosis caused by blood tracking through the fascial planes to the umbilicus. It is a sign of intra-abdominal or retroperitoneal haemorrhage, classically severe haemorrhagic pancreatitis, and is associated with serious underlying disease.

Cultural Psychiatry

The study and clinical practice of how cultural factors influence mental health, illness expression, diagnosis, and treatment. Cultural concepts of distress, idioms of distress, and cultural syndromes must be considered. The DSM-5 includes a Cultural Formulation Interview.

Culture

A laboratory test in which a clinical specimen is placed in conditions that allow microorganisms to grow and be identified; susceptibility testing may help guide treatment.

Culture - Colonic Tissue

A laboratory culture of a colon tissue specimen to detect bacteria, fungi, or other microorganisms in suspected infection; results are interpreted with histopathology and clinical findings.

Culture - Duodenal Tissue

A laboratory culture of a duodenal tissue specimen to identify microorganisms causing infection or inflammation, interpreted with endoscopic findings and histopathology.

Culture And Sensitivity

Microbiological culture of a clinical specimen (blood, urine, sputum, wound swab, CSF) to identify pathogens and determine antimicrobial susceptibility. Culture media: blood agar (most bacteria), MacConkey (Gram-negative), chocolate agar (fastidious: *N. gonorrhoeae*, *H. influenzae*), and selective media.

Culture Media

Nutrient preparations used to grow microorganisms in the laboratory, classified by composition and purpose.

Culture Swab

A sterile swab used to collect material from skin, wounds, the throat, genital tract, or another site for laboratory culture. The test may identify bacteria or fungi and guide antibiotic choice; viral testing usually uses separate molecular or antigen tests.

Culture-Negative Endocarditis

Infective endocarditis in which routine blood cultures do not identify an organism, often because of prior antibiotics, fastidious bacteria, fungi, or noninfectious mimics; diagnosis requires integrated clinical, imaging, and laboratory assessment.

Cumulative Dose

Total amount of substance received over time. Important for carcinogens, teratogens, cumulative toxins.

Cumulative Trauma Disorder

A disorder caused by repeated forceful or awkward movements that progressively injure muscles, tendons, nerves, or other soft tissues.

Cup-To-Disc Ratio

The cup-to-disc ratio is a clinical measurement used to assess the severity of glaucomatous optic nerve damage, comparing the diameter of the optic cup (the central depression within the optic nerve head where neural tissue is absent) to the diameter of the optic disc (the entire optic nerve head).

Cupping

An abnormal enlargement or depression of the optic-disc cup, often assessed in evaluating glaucoma.

Alternate Names:

Optic-disc cupping; glaucomatous cupping.

CURB-65 Score

One point each for: Confusion, Urea > 7 mmol/L, RR > 30, BP < 90/60, age ≥ 65. 0–1: outpatient. 2: consider hospitalization. 3–5: severe.

Cure

Elimination or durable control of a disease so that it no longer causes active illness; the meaning and evidence for cure vary by condition.

Curettage

Removal of tissue from a body cavity or surface with a curette, used for diagnosis, treatment, or removal of abnormal tissue.

Curettage Specimen

Tissue obtained by scraping or scooping a surface or cavity, such as the endometrium or a bone lesion, for histologic examination. The limited and fragmented nature of the specimen may restrict assessment of invasion or architecture.

Curette

A surgical instrument featuring a spoon-shaped, ring-shaped, or scooped tip with sharp or blunt edges, designed for debriding, scraping, or excavating biological tissue from a cavity, bone, or mucosal surface. Widely used across surgical subspecialties: in obstetrics and gynecology for endometrial biopsy or dilation and curettage (D&C); in orthopedic surgery for excising benign bone cysts or necrotic bone; in dermatology for scoop excision of seborrheic keratoses and basal cell carcinomas (curettage and electrodesiccation); and in periodontics for subgingival scaling and root planing.

Curling Ulcer

Stress ulceration of the stomach or duodenum associated with severe burns. Part of the Curling

triad with hypovolemia and sepsis. The pathogenesis involves reduced mucosal blood flow during the systemic inflammatory response to burn injury.

Curling'S Ulcer

An acute peptic ulcer associated with severe burns, often occurring in the duodenum or stomach during physiologic stress.

Curvature Of The Penis

A bend of the erect or flaccid penis caused by congenital tissue asymmetry, scar tissue such as Peyronie disease, trauma, or other conditions; it may impair intercourse or cause pain.

Alternate Names:

Peyronie's Disease (Curvature Of The Penis)

Curvularia Alcornii

A dematiaceous fungal species causing phaeohyphomycosis, an opportunistic infection of the skin, sinuses, and occasionally the central nervous system in immunocompromised patients. It produces dark, septate hyphae in tissue. Diagnosis is by culture and histopathology showing pigmented fungal elements.

Cushing Disease

Cushing syndrome caused by an adrenocorticotrophic hormone-secreting pituitary tumor. Excess cortisol can cause central weight gain, rounded facial features, skin changes, muscle weakness, high blood pressure, high blood glucose, and menstrual or mood changes.

Cushing Syndrome

Clinical state resulting from chronic exposure to excess glucocorticoids regardless of cause. Most common endogenous cause is Cushing disease from pituitary ACTH-secreting adenoma accounting for approximately seventy percent.

Cushing Syndrome Complications

Untreated Cushing syndrome carries significant morbidity and mortality from cardiovascular disease, infections, thromboembolic events, and metabolic derangements. Cardiovascular complications include accelerated atherosclerosis, hypertension, and dys-

lipidemia. Infection risk is increased from immunosuppression.

Cushing Syndrome Due To Adrenal Tumor

Excess cortisol caused by an adrenal tumor, producing Cushing syndrome with features such as central weight gain, hypertension, diabetes, muscle weakness, and skin changes.

Cushing Ulcer

Stress ulceration associated with central nervous system disease, trauma, or surgery. Caused by vagally mediated gastric acid hypersecretion, distinguishing it from Curling ulcer which results from mucosal ischemia. The ulcers are typically solitary, located in the stomach or duodenum, and present with gastrointestinal bleeding.

Cushing'S Disease

A pituitary tumor that makes too much adrenocorticotrophic hormone (ACTH), causing the adrenal glands to produce too much cortisol. Features may include weight gain around the trunk, a round face, easy bruising, purple stretch marks, muscle weakness, high blood pressure, high blood glucose, and menstrual or mood changes.

Cushing'S Syndrome

An endocrine disorder caused by prolonged exposure to too much cortisol or glucocorticoid medicine. It may result from steroid treatment or from a pituitary, adrenal, or other tumor. Common features include weight gain mainly around the trunk, a round face, fat at the back of the neck, thin easily bruised skin with purple stretch marks, muscle weakness, high blood pressure, and high blood glucose. Untreated disease can cause infections, blood clots, bone loss, and cardiovascular complications.

Alternate Names:

Hypercortisolism; cortisol excess syndrome.

Cushingoid

Resembling the physical features of excess glucocorticoid exposure, such as a round face, central weight gain, thin skin, easy bruising, and muscle weakness.

Cusp

A pointed projection, such as a projection of a tooth or one of the leaflets of a heart valve.

Cusp Fracture

Breakage of tooth cusp. Often due to large restoration weakening tooth. Treatment: crown or onlay.

Cut

A break or incision made in tissue, usually by a sharp object; the depth, contamination, and location determine the required care.

Cutaneous

Pertaining to the skin (*cutis*), the largest organ of the human body, composed of the stratified keratinized squamous epithelium of the epidermis, the dense fibrovascular collagenous dermis, and the deeper subcutaneous adipose tissue (hypodermis). Functions include physical barrier defense, thermoregulation through dermal microvascular plexuses and eccrine sweating, tactile sensory transduction, and cutaneous vitamin D synthesis. Clinically, cutaneous manifestations frequently provide pathognomonic diagnostic clues to occult systemic illness—including the malar rash of systemic lupus erythematosus, heliotrope rash of dermatomyositis, necrobiosis lipidica in diabetes mellitus, erythema nodosum in sarcoidosis or inflammatory bowel disease, and palmar erythema in cirrhosis—as well as primary skin malignancies (melanoma, basal cell carcinoma, squamous cell carcinoma, and cutaneous T-cell lymphoma).

Cutaneous Angiosarcoma

An aggressive malignant vascular tumor of skin, typically affecting the scalp and face of older adults as a bruise-like or violaceous plaque that may ulcerate or metastasize.

Cutaneous Anthrax

The most frequent naturally occurring clinical presentation of infection with *Bacillus anthracis* (95% of human cases), typically contracted when bacterial spores enter broken skin after contact with infected animals or animal products. Characterized

by a pruritic macule or papule that rapidly ulcerates and develops into a pathognomonic, painless central black eschar surrounded by substantial non-pitting edema.

Cutaneous B-Cell Lymphoma

A type of non-Hodgkin lymphoma that primarily affects B lymphocytes in the skin.

Cutaneous Diphtheria

A skin infection caused by toxigenic or nontoxigenic *Corynebacterium diphtheriae*, producing chronic ulcers or a gray membrane.

Cutaneous Langerhans Cell Histiocytosis

A clonal proliferation of Langerhans cells in the skin that may cause scaly papules, seborrheic dermatitis-like plaques, or purpura. Evaluation determines whether disease is skin-limited or involves other organs.

Cutaneous Larva Migrans

A creeping eruption caused by hookworm larvae (*Ancylostoma braziliense*, *A. caninum*) from dog/cat feces penetrating the skin, typically in tropical/subtropical regions (walking barefoot on contaminated sand). Larvae migrate erratically under the skin causing intensely pruritic, serpiginous, erythematous tracks.

Cutaneous Leiomyoma

A benign smooth-muscle tumor of the skin arising from arrector pili muscles, vascular smooth muscle, or the dartos. Multiple painful lesions may indicate hereditary leiomyomatosis and renal-cell cancer syndrome.

Cutaneous Leishmaniasis

A skin infection caused by *Leishmania* parasites transmitted by sandflies, usually producing one or more chronic ulcers.

Cutaneous Lupus

A form of lupus erythematosus limited to the skin, manifesting as discoid lesions, subacute cutaneous

plaques, or acute malar erythema. It may occur in isolation or as a feature of systemic lupus erythematosus.

Cutaneous Lupus Erythematosus

Autoimmune skin disease associated with lupus-spectrum illness and characterized by inflammatory lesions that may scar, alter pigmentation, or cause photosensitivity. Subacute and chronic forms have different morphologies and systemic associations; evaluation may include biopsy, direct immunofluorescence, and autoimmune testing.

Cutaneous Lymphoma

Cutaneous lymphomas are heterogeneous group of T-cell and B-cell malignancies primarily involving the skin. Primary cutaneous B-cell lymphomas include marginal zone lymphoma, follicle center lymphoma, and diffuse large B-cell lymphoma leg type. Primary cutaneous T-cell lymphomas include mycosis fungoides and Sézary syndrome.

Cutaneous Metastases

Secondary deposits of cancer in the skin, originating from a primary tumor elsewhere in the body. They may present as nodules, plaques, or ulcers, and often indicate advanced or recurrent disease.

Cutaneous Metastasis

A deposit of malignant cells in the skin from a primary tumor elsewhere or, less often, another cutaneous site; it may present as a firm nodule, plaque, or inflammatory-appearing lesion.

Cutaneous Polyarteritis Nodosa

A skin-limited necrotizing vasculitis of small- and medium-sized arteries, typically causing tender subcutaneous nodules, livedo reticularis, and ulcers on the lower legs without systemic organ involvement.

Cutaneous Skin Tag

A small, soft, benign skin growth, usually on a narrow stalk in a skin fold.

Cutaneous Small-Vessel Vasculitis

Inflammation and destruction of small dermal blood vessels, commonly presenting as palpable purpura on dependent areas. Causes include infections, medications, autoimmune disease, and systemic vasculitis; skin biopsy with appropriate specimen handling is central to diagnosis.

Cutaneous T-Cell Lymphoma

A malignancy of skin-homing T lymphocytes. Mycosis fungoides is the most common subtype, presenting as patches, plaques, and tumors in a clothing distribution. Sézary syndrome is the leukemic variant with erythroderma, lymphadenopathy, and circulating malignant cells.

Cuticle

The thin outermost layer of a hair shaft formed by overlapping cells that protect the inner cortex.

Alternate Names:

Hair cuticle; cuticular layer.

Cuticle Remover Poisoning

Exposure to cuticle-remover products, which may contain alkaline or acidic chemicals that can injure the skin, eyes, mouth, or gastrointestinal tract.

Cutis Anserina

Gooseflesh or piloerection, in which contraction of tiny muscles around hair follicles makes the skin appear rough or dimpled.

Cutis Laxa

A group of inherited or acquired connective-tissue disorders characterized by loose, inelastic skin caused by abnormal elastic fibers. Some forms also affect the lungs, blood vessels, heart, gastrointestinal tract, or skeleton.

Cuts And Puncture Wounds

Breaks in the skin caused by sharp, blunt, or penetrating trauma; evaluation considers depth, contamination, retained foreign bodies, tendon or nerve injury, vascular damage, and tetanus protection.

Cuts Wounds (Cuts, Scrapes And Puncture Wounds)

Alternate index label for Cuts, Scrapes and Puncture Wounds.

Cuts, Scrapes And Puncture Wounds

Cuts, scrapes, and puncture wounds damage the skin to varying depths and can introduce bacteria or foreign material. Clean minor wounds with running water, control bleeding, cover them, and seek care for deep or contaminated wounds, bites, impaired circulation, infection, or uncertain tetanus protection.

Alternate Names:

Cuts Wounds (Cuts, Scrapes And Puncture Wounds)

Cutting Balloon (Vascular)

A balloon catheter with small blades or scoring elements that create controlled incisions in plaque during vessel dilation.

Alternate Names:

Vascular

Cutting For The Stone

An older term for surgically opening the urinary tract to remove a stone, historically called lithotomy.

CVC

Central venous catheter, a catheter whose tip lies in a large central vein and provides intravenous access for medications, fluids, nutrition, or monitoring.

Cyanide

A highly toxic chemical ion or compound that inhibits cellular cytochrome oxidase, preventing tissues from using oxygen and potentially causing rapid fatal poisoning.

Cyanide Poisoning

Toxicity from cyanide compounds (hydrogen cyanide, potassium/sodium cyanide) that inhibit cytochrome oxidase (complex IV), blocking oxidative phosphorylation and cellular respiration despite adequate oxygen—histotoxic hypoxia. Sources: smoke

inhalation (fires—plastic/acrylic), ingestion (industrial, suicidal), occupational exposure.

Cyanoacrylates Poisoning

Exposure to cyanoacrylate adhesives, commonly known as instant glue. Contact may bond tissues or irritate the skin, eyes, or airways; ingestion usually causes local adhesion or irritation.

Cyanosis

A bluish discoloration of the skin and mucous membranes resulting from excessive amounts of deoxygenated hemoglobin in the capillary blood. It becomes clinically apparent when the concentration of deoxygenated hemoglobin exceeds 5 grams per deciliter, which corresponds to an oxygen saturation of approximately 85 percent.

Cyanotic

Having a bluish or slate-gray discoloration of skin or mucous membranes, usually reflecting increased deoxygenated hemoglobin or abnormal blood flow.

Cyanotic Heart Disease

A congenital or acquired heart disorder that reduces systemic oxygen saturation and causes cyanosis, often through right-to-left shunting or inadequate pulmonary blood flow.

Cycle Time

Time for complete image acquisition. PET: minutes. SPECT: 20-30 minutes.

Cyclic Citrullinated Peptide

A laboratory antigen used to detect anti-CCP antibodies, which are highly specific for rheumatoid arthritis but are interpreted with clinical findings.

Cyclic Guanosine Monophosphate

A cellular signaling molecule involved in smooth-muscle relaxation, phototransduction, and other physiologic processes.

Alternate Names:

cGMP; cyclic GMP.

Cyclic Hormone Therapy

Hormone therapy in which estrogen is given continuously or for part of a cycle and progestogen is added for a defined number of days.

Alternate Names:

Sequential hormone therapy; cyclic estrogen-progestogen therapy.

Cyclic Neutropenia

Recurrent episodes of abnormally low neutrophil counts occurring at regular intervals.

Cyclic Vomiting Syndrome

Recurrent episodes of severe vomiting separated by symptom-free intervals.

Cyclic Vomiting Syndrome

Alternate index label for CVS.

Cyclooxygenase

An enzyme family that helps convert arachidonic acid into prostaglandins and related mediators involved in pain, inflammation, and platelet function.

Alternate Names:

COX enzyme; prostaglandin-endoperoxide synthase.

Cyclooxygenase-2

An inducible cyclooxygenase enzyme that generates prostaglandins involved in inflammation, pain, and fever and is inhibited by COX-2-selective medicines.

Cyclops

A term that may refer to cyclopia, a rare severe congenital malformation with a single central eye, or to a fibrous nodule causing loss of extension after knee surgery; context determines the meaning.

Cyclospora Infection

An intestinal infection caused by the parasite *Cyclospora cayentanensis*, usually acquired from contaminated food or water. It commonly causes prolonged watery diarrhea, abdominal cramps, nausea, fatigue, and weight loss. Diagnosis requires specialized stool testing, and antimicrobial treatment is often effective.

Cyclospora Infection (Cyclosporiasis)

Alternate index label for Cyclosporiasis.

Cyclosporiasis

An intestinal infection caused by the parasite *Cyclospora cayetanensis*, usually acquired by consuming contaminated food or water. It commonly causes prolonged or relapsing watery diarrhea, abdominal cramps, nausea, fatigue, loss of appetite, and weight loss; diagnosis requires specialized stool testing, and trimethoprim-sulfamethoxazole is the usual effective treatment when appropriate.

Alternate Names:

Cyclospora Infection (Cyclosporiasis)

Cyclosporiasis (Cyclospora)

An intestinal infection caused by *Cyclospora* parasites, typically producing prolonged watery diarrhea, cramps, nausea, and fatigue.

Cyclothymia

A long-term mood disorder with repeated periods of hypomanic and depressive symptoms that do not meet criteria for full episodes.

Cyclothymia (Cyclothymic Disorder)

A chronic mood disorder involving repeated periods of hypomanic and depressive symptoms that do not meet criteria for full episodes.

Cyclothymic Disorder

A chronic mood disorder characterized by numerous periods of hypomanic symptoms and periods of depressive symptoms over at least two years (one year in children and adolescents). The hypomanic and depressive symptoms do not meet criteria for a hypomanic or major depressive episode.

Cylindroma

A usually benign adnexal tumor composed of basaloid cells, often presenting as a firm pink or skin-coloured nodule on the scalp or face; multiple lesions may occur in Brooke-Spiegler syndrome.

CYP2C19 Polymorphism

CYP2C19 is a cytochrome P450 enzyme involved in the metabolism of approximately 10 percent of drugs, including the antiplatelet agent clopidogrel, proton pump inhibitors, some antidepressants, and the antiepileptic drug phenytoin.

CYP2D6 Polymorphism

CYP2D6 is a cytochrome P450 enzyme responsible for the metabolism of approximately 25 percent of commonly prescribed drugs, including codeine, tamoxifen, many antidepressants, and antipsychotics.

CYP450 System

A family of enzymes, found mainly in the liver, that metabolize many medications and other foreign substances. Differences in these enzymes and their inhibition or induction can alter drug effects and toxicity.

Cyproheptadine Overdose

Overdose of cyproheptadine, an antihistamine, that may cause sedation, agitation, anticholinergic effects, seizures, respiratory depression, or cardiac effects.

Cyst

A cyst is a closed sac containing fluid, air, cells, or semisolid material and may occur in skin, organs, glands, or other tissues.

Cyst Eyelid (Giardia Lamblia)

Alternate index label for *Giardia Lamblia*.

Cyst, Bartholin'S

Alternate index label; see *Bartholin's cyst*.

Cyst, Ganglion

Alternate index label; see *Ganglion cyst*.

Cyst, Kidney

Alternate index label; see *Kidney cysts*.

Cyst, Ovarian

Alternate index label; see *Ovarian cysts*.

Cyst, Pancreatic

Alternate index label; see *Pancreatic cysts*.

Cyst, Pilonidal

Alternate index label; see *Pilonidal cyst*.

Cyst, Spermatic

Alternate index label; see *Spermatocele*.

Cystadenoma

A usually benign glandular epithelial tumor that forms cystic spaces or cyst-like structures.

Cystic Acne

Cystic acne is severe inflammatory acne with deep painful nodules or cyst-like lesions that can scar. Treatment may require prescription retinoids, antimicrobial or hormonal therapy, or isotretinoin under specialist monitoring; squeezing lesions increases scarring.

Cystic Echinococcosis

Hydatid disease: infection with larval cysts of *Echinococcus granulosus* (dog tapeworm), transmitted by ingestion of eggs from dog feces. Cysts grow slowly in liver (most common, 60-70 percent), lungs (20-30 percent), and other organs. Hepatic: RUQ pain, hepatomegaly, biliary obstruction. Pulmonary: cough, chest pain, hemoptysis.

Cystic Fibrosis

An inherited disease caused by disease-causing variants in both copies of the CFTR gene. CFTR dysfunction disrupts salt and water movement, producing thick, sticky mucus and unusually salty sweat. It mainly affects the lungs, sinuses, pancreas, intestines, liver, and reproductive system. Common problems include recurrent lung infections, chronic cough, sinus disease, poor digestion and growth, intestinal blockage in newborns, and male infertility due to absent or blocked sperm ducts. Diagnosis uses newborn screening, sweat-chloride testing, genetic testing, and clinical assessment. Severity varies; treatment may include airway clearance, inhaled medicines, pancreatic enzymes, nutritional support, antibiotics, and CFTR modulators.

F508del is the most common CFTR variant.

Alternate Names:

CF

Cystic Fibrosis - Nutrition

A patient-care topic concerning cystic fibrosis - nutrition. It addresses the relevant purpose, risks, management, and follow-up.

Cystic Hygroma

Congenital lymphatic malformation composed of thin-walled lymphatic cysts, manifesting as a soft, fluid-filled, painless, translucent (transilluminates brilliantly) mass, typically in the posterior triangle of the neck (75%, left-sided), axilla (20%), mediastinum, retroperitoneum, or groin.

Cystic Medial Necrosis

Degeneration and loss of elastic fibers and smooth-muscle cells in the aortic media, creating areas of mucoid material and weakening the aortic wall. It may contribute to aortic aneurysm or dissection and is also called cystic medial degeneration.

Cysticercosis

Cysticercosis is tissue infection by larval *Taenia solium* after swallowing eggs, potentially causing cysts in muscle, skin, eyes, or brain.

Cystine And Triple Phosphate (Struvite) Crystals

Crystalline substances that may be found in urine sediment. Hexagonal cystine crystals support cystinuria in the appropriate setting, while struvite crystals may occur in infection-related alkaline urine; microscopy is interpreted with clinical and laboratory findings.

Alternate Names:

struvite

Cystine Stones

Rare stones from inherited cystinuria with defective cystine transport. Hexagonal crystals on urine microscopy. Difficult to dissolve. Treat with aggressive hydration, urine alkalinization to pH greater

than 7.5, and thiol agents including tiopronin and D-penicillamine that form soluble complexes with cystine.

Cystinuria

Cystinuria is an inherited kidney-transport disorder causing excess urinary cystine and recurrent cystine kidney stones.

Cystitis

Inflammation of the urinary bladder, typically from bacterial infection, one of the most common bacterial infections in women.

Alternate Names:

Bladder Infection

Cystitis (Bladder Infection)

Cystitis is inflammation of the bladder, most often from a bacterial urinary-tract infection. It causes burning urination, frequency, urgency, and lower-abdominal discomfort; fever, flank pain, vomiting, pregnancy, or recurrent infection requires prompt assessment.

Cystitis - Acute

Acute inflammation of the urinary bladder, most often caused by bacterial infection and producing dysuria, urinary frequency, urgency, and suprapubic discomfort.

Cystitis - Noninfectious

Inflammation of the urinary bladder without a proven bacterial infection. Causes include interstitial cystitis or bladder pain syndrome, medicines, radiation, chemical irritation, autoimmune disease, and injury; symptoms may include frequency, urgency, and painful urination.

Cystitis, Interstitial

Alternate index label; see *Interstitial cystitis*.

Cystocele

Anterior vaginal wall prolapse: herniation of the bladder into the anterior vaginal wall due to weakening of the pubocervical fascia, most commonly

after childbirth and menopause.

Cystography

Contrast imaging of bladder. Evaluates trauma, fistulas, vesicoureteral reflux.

Cystoid Macular Edema

Swelling of the macula caused by fluid collecting in small cyst-like spaces within the retina.

Cystometric Study

A diagnostic test or assessment related to cystometric study. Results are interpreted with the clinical findings, test method, and reference range.

Cystometrogram

Suite of diagnostic procedures designed to evaluate the physiological function of the lower urinary tract (bladder and urethral sphincter) during filling, storage, and emptying phases. See urodynamic testing.

Cystometry

A urodynamic test that measures bladder pressure and function during filling and emptying.

Cystoscope

A thin, lighted instrument used to examine the urethra and bladder and, when needed, perform procedures.

Cystoscopy

Cystoscopy uses a camera passed through the urethra to inspect the bladder and urethra and may allow biopsy or treatment.

Cystostomy

Surgical creation of an opening from the urinary bladder to the skin or another structure to permit drainage of urine.

Cystotome

A surgical instrument used to make an incision in the capsule of the lens during selected eye procedures.

Cystourethrography

Radiographic imaging of the urinary bladder and urethra, usually during filling and voiding after contrast has been introduced.

Cysts

Closed sac-like structures containing fluid, air, semisolid material, or other contents; their causes range from benign developmental or obstructive processes to infection or neoplasia.

Cysts Choledochal (Choledochal Cysts)

Alternate index label for Choledochal Cysts.

Cysts Ovary (Ovarian Cysts)

Alternate index label for Ovarian Cysts.

Cysts Pancreatic Inflammatory (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Cysts True Pancreas (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Cysts, Epidermoid

Alternate index label; see *Epidermoid cysts*.

Cysts, Sebaceous

Alternate index label; see *Epidermoid cysts*.

Cytochrome B5 Reductase

An enzyme that maintains hemoglobin iron in the ferrous state needed for oxygen binding; inherited deficiency can cause congenital methemoglobinemia, usually presenting with cyanosis and impaired oxygen delivery.

Cytochrome C

Mobile electron carrier between Complex III and IV. Also involved in apoptosis.

Cytochrome P450

The cytochrome P450 system is a superfamily of heme-containing monooxygenase enzymes primarily

located in the endoplasmic reticulum of hepatocytes, responsible for Phase I drug metabolism through oxidation, reduction, and hydrolysis reactions.

Cytogenetics

The study of chromosomes and their number, structure, and arrangement. Tests such as karyotyping, fluorescence in situ hybridization, and chromosome microarray can help identify inherited conditions, leukemia, other cancers, and some fetal abnormalities.

Cytogenetics (Karyotype)

Visualization of chromosomes under light microscopy from dividing cells (bone marrow, blood, amniotic fluid, tissue). Staining: G-banding (Giemsa, light and dark bands). Resolution: 400-550 bands per haploid genome.

Alternate Names:

Karyotype

Cytokine

Small signaling proteins (5-25 kDa) mediating cell-to-cell communication in immunity, inflammation, and hematopoiesis.

Cytokine Release Syndrome

CRS: systemic inflammatory response from massive cytokine release (IL-6, IFN-gamma, TNF, IL-10) after T cell engaging therapies (CAR-T, bispecific antibodies, TGN1412).

Cytokine Storm

A severe, dysregulated systemic inflammatory response involving excessive cytokine release that can cause tissue injury, organ dysfunction, and shock.

Cytokines

Small signaling proteins released by immune and other cells that regulate inflammation, immunity, blood-cell production, and tissue responses.

Alternate Names:

Immune signaling proteins; cytokine mediators.

Cytologic Evaluation

Microscopic examination of individual cells or small cell groups from a fluid, brushing, scraping, or fine-needle sample to detect infection, inflammation, or malignancy.

Cytology

The microscopic study of individual cells, distinct from histology in that tissue architecture is not examined, used for screening and diagnosis of neoplastic and infectious disease. Specimens are exfoliative (cervical smears, sputum) or aspirated (fine needle aspiration, body fluids), interpreted with Papanicolaou or special stains.

Cytology Exam Of Pleural Fluid

Microscopic examination of fluid around the lung for malignant cells and other cellular changes, performed with chemical and microbiologic testing when evaluating an effusion.

Cytology Exam Of Urine

Microscopic examination of shed urinary-tract cells, used mainly to detect high-grade urothelial malignancy and selected abnormal cells; a negative result does not exclude cancer.

Cytomegalovirus

A herpesvirus that causes significant disease in immunocompromised patients including transplant recipients and HIV-infected individuals. CMV infects a wide range of cell types and establishes latent infections in myeloid progenitor cells. Primary infection is usually asymptomatic in immunocompetent individuals.

Cytomegalovirus

Alternate index label for CMV.

Cytomegalovirus

Alternate index label for Cytomegalovirus.

Cytomegalovirus Infection

Cytomegalovirus infection is caused by a herpesvirus that is often mild in healthy people but can cause

congenital disease or severe organ infection in immunocompromised patients.

Cytomegalovirus Infection

A widespread herpesvirus infection. Primary infection in immunocompetent individuals is usually asymptomatic or causes mononucleosis-like syndrome. Congenital CMV: most common congenital infection, causes sensorineural hearing loss, microcephaly, and developmental delay.

Cytopathology

The branch of pathology using cytologic examination of cells to diagnose disease, particularly neoplastic and infectious processes, built on standardized reporting systems that guide management.

Cytopenia

A reduction in the number of one or more circulating blood-cell lineages: red cells, white cells, or platelets.

Cytoplasm

The material inside a cell between the cell membrane and nucleus, consisting of cytosol, organelles, and other cellular components.

Cytoreductive Surgery

Surgical removal of as much visible tumor as possible to reduce disease burden, a cornerstone of ovarian cancer management and combined with hyperthermic intraperitoneal chemotherapy for peritoneal carcinomatosis.

Cytoskeleton

Dynamic protein filament network providing structural support, cell shape, intracellular transport, and cell division. Three main components: (1) Microfilaments (actin, 7 nm): cell cortex, stress fibers, contractile ring (cytokinesis), lamellipodia/filopodia (motility), myosin interaction (muscle contraction).

Cytotoxic

Toxic to cells or capable of killing cells, as with some chemotherapy drugs, toxins, or cytotoxic immune cells.

Cytotoxic Alopecia

Hair loss caused by exposure to a cytotoxic treatment or other toxic injury to actively growing hair follicles.

Alternate Names:

Drug-induced alopecia; chemotherapy-related hair loss.

Cytotoxic T Cell

The effector cells of cell-mediated immunity that kill virus-infected cells and tumor cells. Recognize antigenic peptides presented on MHC class I molecules via the T-cell receptor; co-stimulation by CD4+ helper cells and cytokines (IL-2) is needed for activation.

Cytotoxicity

The capacity of an immune effector cell or immune mediator to injure or kill a target cell. T lymphocytes and natural killer cells commonly use perforin-granzyme pathways or death-receptor signaling to induce target-cell apoptosis.

Common Carotid Artery

One of the paired arteries supplying the head and neck. The right common carotid arises from the brachiocephalic trunk and the left from the aortic arch; each divides into the internal and external carotid arteries.

D

1p36 Deletion Syndrome

A contiguous gene microdeletion syndrome resulting from a terminal deletion of the subtelomeric region of chromosome 1p36, representing the most prevalent terminal deletion in humans. Phenotypic features include severe intellectual disability, microcephaly, developmental delay, infantile hypotonia, sensorineural hearing loss, structural congenital heart disease (specifically dilated cardiomyopathy and patent ductus arteriosus), seizures, and characteristic craniofacial dysmorphisms featuring deep-set eyes, straight horizontal eyebrows, midface retrusion, a broad flat nasal bridge, and a pointed chin.

Alternate Names:

Monosomy 1p36; Chromosome 1p36 Deletion Syndrome

22q11.2 Deletion Syndrome

The most common human chromosomal microdeletion syndrome, caused by non-allelic homologous recombination resulting in a heterozygous 1.5-to-3 Mb deletion in the 22q11.2 locus that encompasses the *TBX1* transcription factor gene. Unifies a diverse phenotypic spectrum historically split into DiGeorge syndrome, Velocardiofacial (Shprintzen) syndrome, and conotruncal anomaly face syndrome. Major clinical manifestations encompass conotruncal cardiac malformations (tetralogy of Fallot, interrupted aortic arch type B, truncus arteriosus), palatal anomalies (cleft palate, velopharyngeal insufficiency), thymic aplasia or hypoplasia causing T-cell immunodeficiency, parathyroid hypoplasia causing neonatal hypocalcemia, distinct facial features (tubular nose, bulbous nasal tip), developmental delays, and a markedly increased lifetime predisposition to schizophrenia.

Alternate Names:

DiGeorge Syndrome Spectrum; Velocardiofacial Syndrome (VCFS); 22q11.2DS

D & C

Dilation and curettage, a procedure in which the cervix is widened and tissue is removed from the uterine cavity for treatment or diagnosis.

D And C

Dilation and curettage, a procedure that opens the cervix and removes tissue from the uterine lining for treatment or diagnosis.

D And E

Dilation and evacuation, a surgical procedure in which the cervix is dilated and uterine contents are removed by suction and instruments.

D-Dimer

A substance released into the blood when a blood clot breaks down. A high level can occur with a blood clot, but also with infection, pregnancy, recent surgery, cancer, or other conditions. A normal result can help rule out a clot in carefully selected patients whose initial risk is low or moderate.

D-Dimer Test

D-dimer testing measures a fibrin-degradation product; an elevated result can support evaluation for thrombosis but is nonspecific and must be interpreted with pretest probability.

D-Limonene

A citrus-derived compound found in the oils of or-

ange, lemon, and other citrus fruits.

Alternate Names:

Limonene

D-Xylose Absorption

A test of small-intestinal absorption in which orally administered D-xylose is measured in blood or urine. Reduced absorption can support small-intestinal mucosal disease, although the test is now used less often and results are affected by kidney function and collection quality.

D.O.

Doctor of Osteopathic Medicine, a physician degree whose graduates are licensed to practice medicine and surgery like MD graduates in the United States.

D2 Receptor

A dopamine receptor involved in movement, motivation, reward, and some psychiatric disorders.

D&C Instruments

Instruments used to dilate the cervix and remove tissue from the uterus during dilation and curettage.

Da Costa's Syndrome

Da Costa's Syndrome is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Dacarbazine

An alkylating chemotherapy medicine used for Hodgkin lymphoma, melanoma, and selected other cancers; important toxicities include severe nausea, bone-marrow suppression, and liver injury.

Dacetuzumab

A medicine called dacetuzumab. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Daclizumab

A medicine called daclizumab. Its uses, dosing, adverse effects, and contraindications depend on

the specific indication and patient.

Dacomitinib

A medicine called dacomitinib. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Dacrocytes (Teardrop Cells)

Teardrop-shaped red blood cells seen on a blood smear. They may occur when red blood cells pass through scarred or crowded bone marrow, and can be seen in myelofibrosis, bone-marrow cancers, and severe thalassemia.

Alternate Names:

Teardrop Cells

Dacryoadenitis

Inflammation of a tear-producing gland, causing painful swelling near the outer upper eyelid. It may be caused by infection, inflammation, or a condition affecting the whole body.

Dacryocystitis

Inflammation of the lacrimal sac, most commonly caused by obstruction of the nasolacrimal duct leading to stagnation of tears and secondary bacterial infection (*Staphylococcus aureus*, *Streptococcus pneumoniae*, *Haemophilus influenzae*).

Dacryocystocele

A cystic swelling caused by blockage and dilation of the tear-duct system, usually near the inner corner of the eye.

Dacryocystography

An imaging study in which contrast material is introduced into the lacrimal drainage system to identify narrowing or obstruction.

Dacryocystorhinostomy

A surgical procedure creating a passage between the lacrimal sac and the nasal cavity to bypass an obstructed nasolacrimal duct.

Dacryocyte Analysis

Examination or counting of teardrop-shaped red blood cells on a blood smear; numerous dacryocytes may occur with marrow fibrosis, infiltration, severe anemia, or splenic disease.

Dacryolithiasis

Formation of a stone or concretion within the lacrimal drainage system, which may cause excessive tearing, discharge, recurrent dacryocystitis, or obstruction.

Dacryostenosis

Narrowing or obstruction of the lacrimal drainage passages, which can cause excessive tearing and recurrent infection of the lacrimal sac.

Dactinomycin

A medicine called dactinomycin. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Dactylitis

Inflammation of an entire digit (finger or toe), producing a diffuse, fusiform swelling often described as sausage-shaped.

Dactylomegaly

Abnormal enlargement of a finger or toe, which may occur as an isolated finding or as part of an endocrine, skeletal, vascular, or inflammatory disorder.

Dactylosymphysis

Abnormal fusion or webbing of adjacent digits, usually present from birth.

Daidzein

A naturally occurring isoflavone found in soy and other legumes that can be converted by intestinal bacteria into related compounds.

Daily Bowel Care Program

A planned routine for preventing and treating constipation or fecal incontinence, often including timing,

fluids, diet, activity, medication, and rectal measures when prescribed.

Daily Value

A reference amount used on U.S. Nutrition Facts labels to show how much a nutrient in one serving contributes to a typical daily diet.

Dakin Solution

A dilute sodium hypochlorite solution used as a topical antiseptic for selected contaminated or infected wounds. Concentration and exposure time must be controlled because stronger solutions can injure viable tissue.

Dalotuzumab

A medicine called dalotuzumab. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Dalteparin

A low-molecular-weight heparin that enhances antithrombin activity against factor Xa and is used to prevent or treat venous thromboembolism; bleeding and dose adjustment in severe kidney disease are important considerations.

Dalteparin Sodium

The sodium salt of dalteparin, a low-molecular-weight heparin used to prevent or treat venous thromboembolism; bleeding is the principal serious adverse effect.

Daltonism

Color-vision deficiency, especially difficulty distinguishing certain colors because of abnormal or absent retinal cone function.

DALY

Disability-adjusted life year, a measure combining years of healthy life lost because of premature death and disability.

Damage To The Amygdala

Damage to the amygdala can affect fear processing,

emotional learning, memory, threat recognition, and social behavior. Causes include trauma, stroke, infection, tumor, epilepsy, or degenerative disease, and effects depend on the side and extent of injury.

Damage-Control Resuscitation

A trauma strategy prioritizing rapid hemorrhage control, permissive approaches to blood pressure when appropriate, prevention of hypothermia, correction of coagulopathy, and balanced blood-product resuscitation rather than large-volume crystalloid administration.

Damage-Control Surgery

A staged operative strategy for critically ill or severely injured patients in whom definitive repair is unsafe because of hypothermia, acidosis, coagulopathy, or physiologic instability. Initial surgery controls hemorrhage and contamination, followed by resuscitation and later definitive reconstruction.

Damus-Kaye-Stansel Procedure

A palliative cardiac surgical operation utilized in single-ventricle congenital cardiac anomalies associated with systemic ventricular outflow tract obstruction or subaortic stenosis (e.g., double inlet left ventricle with transposition of the great arteries). Involves transecting the main pulmonary artery just proximal to its bifurcation, beveling the proximal end, and anastomosing it end-to-side or side-to-side to the ascending aorta, effectively unifying the aortic and pulmonary valves into a single, large, double-barrel systemic ventricular outflow conduit to prevent subaortic obstruction, while pulmonary blood flow is subsequently supplied via a systemic-to-pulmonary shunt or bidirectional Glenn anastomosis.

Alternate Names:

DKS Procedure; Damus-Stansel-Kaye Operation

Danazol

A synthetic androgen with antigonadotropic effects, used selectively for endometriosis, hereditary angioedema, or other conditions; androgenic, lipid, liver, and thrombotic adverse effects limit use.

Dandruff

A common scalp condition causing small white or yellow flakes and sometimes itching. It is often related to oily skin, seborrheic dermatitis, or the scalp's reaction to the normal yeast *Malassezia*; it is not usually contagious. Persistent severe flaking, redness, sores, or hair loss may indicate another scalp disorder.

Dandruff And Scalp Conditions

Dandruff is a common scalp condition presenting as flaking, itching, and mild inflammation caused by overgrowth of the yeast *Malassezia*, dry skin, or seborrheic dermatitis. Other scalp conditions include psoriasis, contact dermatitis, ringworm tinea capitis, and head lice.

Dandy Fever

Dengue fever, an acute mosquito-borne viral illness causing fever, headache, severe muscle or joint pain, and sometimes hemorrhage or shock.

Dandy-Walker Malformation

A condition present at birth in which part of the cerebellum does not develop normally and one of the brain's fluid spaces becomes enlarged. It may cause fluid buildup in the brain, delayed development, balance problems, or seizures.

Dandy-Walker Syndrome

A congenital abnormality involving the cerebellum and the fluid spaces around it, often causing hydrocephalus or developmental problems.

Dane Particle

The complete infectious particle of hepatitis B virus, consisting of the viral core and envelope.

Dangerous Abbreviations

Abbreviations that can be misread and cause medication or clinical errors; safer practice is to write the complete term when an abbreviation is ambiguous.

Danon Disease

A rare X-linked lysosomal storage disorder caused

by LAMP2 mutations, characterized by cardiomyopathy, skeletal-muscle weakness, and variable intellectual or retinal involvement.

Dantrolene

A skeletal-muscle relaxant that reduces calcium release from the sarcoplasmic reticulum; it is the specific treatment for malignant hyperthermia and may cause weakness or liver injury.

Danusertib

An investigational aurora-kinase inhibitor studied as an anticancer drug; it is not an established routine treatment.

Dapsone

A sulfone antimicrobial used for leprosy, selected dermatitis, and other infections; haemolysis, especially with G6PD deficiency, methaemoglobinaemia, agranulocytosis, and severe skin reactions are important risks.

Daptomycin

A medicine called daptomycin. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Daptomycin Resistance

Resistance to daptomycin can develop through mutations in genes encoding cell membrane phospholipid synthesis. The mechanism involves changes in the bacterial cell membrane that reduce daptomycin binding. Resistance is more common in VRE than MRSA. Resistance can emerge during therapy, particularly in endocarditis and bacteremia.

Darbepoetin Alfa

A long-acting erythropoiesis-stimulating agent used for selected anemias, including those associated with chronic kidney disease or chemotherapy; hemoglobin, blood pressure, and thrombotic risk require monitoring.

Darier'S Disease

An inherited skin disorder characterized by abnor-

mal keratinization, greasy crusted papules, and nail changes, often worsened by heat and sweating.

Dark Adaptation

The process by which the eyes become increasingly sensitive to dim light after moving from bright illumination into darkness.

Dartos

A layer of smooth muscle and connective tissue beneath the skin of the scrotum that wrinkles the scrotal skin in response to temperature and other stimuli.

Darunavir

A medicine called darunavir. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Darwin'S Tubercle

A variable small thickening or projection on the rim of the external ear, regarded as a remnant of an ancestral mammalian ear form.

Dasatinib

A medicine called dasatinib. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

DASH Diet

Dietary Approaches to Stop Hypertension. Emphasizes: fruits, vegetables, whole grains, low-fat dairy, reduced saturated fat, sodium <2300 mg/day. Lowers blood pressure effectively.

Dating Ultrasound

An ultrasound examination, usually performed in early pregnancy, used to estimate gestational age and calculate the expected date of delivery. First-trimester crown-rump length is generally the most accurate ultrasound measurement for dating.

Alternate Names:

Pregnancy dating scan; gestational-age ultrasound.

Daubenton'S Plane

An anatomical reference plane extending from the lower margin of the orbit to the upper margin of the external auditory opening, used in describing skull orientation.

Daunorubicin

An anthracycline chemotherapy medicine that intercalates DNA and inhibits topoisomerase II; important risks include myelosuppression and cumulative cardiotoxicity.

Daunorubicin Hydrochloride

The hydrochloride salt of daunorubicin, an anthracycline chemotherapy drug that interferes with DNA and is used mainly for acute leukemias. Important risks include bone-marrow suppression and heart damage.

Dav Regimen

The term is ambiguous without its source context; it may refer to a named treatment regimen, so the full regimen and indication must be specified before clinical use.

David Procedure

A valve-sparing aortic root replacement surgical operation (aortic valve reimplantation) developed by Tirone David for patients with ascending aortic and aortic root aneurysms with intact or repairable native aortic valve leaflets. The diseased aortic root and ascending aorta are excised while preserving the aortic valve apparatus, which is subsequently reimplanted and stabilized within a cylindrical Dacron vascular graft, and the coronary arteries are reimplanted as buttons. It eliminates the aneurysm and corrects aortic regurgitation without requiring a prosthetic valve, avoiding lifelong anticoagulation.

Alternate Names:

David Operation; Valve-Sparing Root Replacement; Reimplantation of Aortic Valve

Daw

An abbreviation meaning dispense as written, indicating that the pharmacist should dispense the prescribed product rather than automatically sub-

stitute a generic equivalent when permitted by law.

Dawn Phenomenon

Physiological increase in blood glucose occurring in the early morning hours between approximately three and eight AM driven by counter-regulatory hormone surges including growth hormone, cortisol, and catecholamines. More pronounced in patients with diabetes who lack the compensatory insulin response.

Day Hospital

A clinical care, prevention, behavioral, or public-health concept concerning day hospital; its use depends on the patient, setting, and clinical goal.

Day-Blindness

Poor vision in bright light, also called hemeralopia. It can result from retinal disease, cone dysfunction, or other eye disorders.

Day-Sickness

A term for symptoms or illness that occur predominantly during the daytime; the cause depends on the specific symptoms and underlying disorder.

Dcc Gene

A gene encoding the netrin-1 receptor DCC, which guides developing axons and may act as a tumor suppressor; pathogenic variants can cause developmental or cancer-related phenotypes.

DCIS

Alternate index label; see *Ductal carcinoma in situ*.

Dcmp Deaminase

An abbreviation or enzyme name requiring source confirmation; the full molecular name and clinical context should be verified before interpretation.

DDH

Alternate index label; see *Hip dysplasia*.

De Lange Syndrome

A congenital developmental disorder, also called

Cornelia de Lange syndrome, involving characteristic facial features, growth and developmental delay, limb abnormalities of variable severity, and intellectual disability.

Alternate Names:

Cornelia de Lange syndrome, involving characteristic facial features, growth and developmental delay, limb abnormalities of variable severity, and intellectual disability.

De Musset Sign

Rhythmic head bobbing with the heartbeat, classically associated with severe aortic regurgitation.

De Novo

Newly arising in an individual rather than inherited from either parent; the term is commonly used for genetic variants or disease presentations.

De Novo Variant

A genetic variant present in an affected individual that was not detected in either biological parent, usually arising in a parental gamete or early embryogenesis. It may cause a dominant disorder and can have implications for recurrence risk in future pregnancies.

De Quervain Tendinitis

Painful stenosing inflammation of the tendons of the abductor pollicis longus and extensor pollicis brevis in the first dorsal wrist compartment. It causes radial wrist pain worsened by thumb movement or grasping and is assessed clinically.

De Quervain Tenosynovitis

De Quervain tenosynovitis: first dorsal compartment stenosis. Clinical: radial wrist pain, Finkelstein test (thumb in palm, ulnar deviation), swelling over radial styloid.

Alternate Names:

Tenosynovitis, De Quervain's

De Quervain's Tendonitis

Painful inflammation or narrowing of the tendons on the thumb side of the wrist.

De Quervain's Tenosynovitis

De Quervain tenosynovitis is painful inflammation or thickening of the thumb-side wrist tendons, often aggravated by gripping, lifting, or repetitive wrist motion. Rest, splinting, activity changes, injections, or surgery may be used according to severity.

Alternate Names:

DeQuervains Tenosynovitis (De Quervain's Tenosynovitis)

De Winter ECG Pattern

A distinctive, hyperacute electrocardiographic pattern recognized as a high-acuity STEMI equivalent, diagnostic of acute, subtotal or total thrombotic occlusion of the proximal left anterior descending coronary artery, occurring in approximately 2% of acute LAD occlusions. Defined by: (1) upsloping ST-segment depression measuring 1 to 3 mm at the J-point in precordial leads V_1 through V_6 , which continues into (2) tall, prominent, symmetrical, hyperacute T waves; (3) 0.5 to 1 mm ST-segment elevation in lead aVR; and (4) absence of classic convex ST-segment elevation. Demands immediate activation of the cardiac catheterization laboratory for emergency primary percutaneous coronary intervention without waiting for classic ST elevation or biomarker elevation.

Alternate Names:

de Winter Pattern; de Winter T Waves; de Winter Sign

De-Escalation

A set of verbal and nonverbal techniques used to reduce agitation and prevent violence while preserving dignity and safety. It includes calm communication, appropriate personal space, clear limits, environmental modification, and offering choices.

Dead Space

The volume of ventilated air that does not participate in gas exchange. Anatomic dead space: the conducting airways (nose, pharynx, trachea, bronchi, terminal bronchioles) that warm, humidify, and filter inspired air but contain no alveoli for gas exchange. Approximately 150 mL in a 70 kg adult.

Dead Space Ventilation

Ventilation of alveoli without perfusion ($V/Q = \text{infinity}$). Anatomic dead space is the conducting airways (150 mL). Physiologic dead space includes alveoli without perfusion. Increased in PE, hypotension, and emphysema. Measured by the Bohr equation.

Dead Time

The brief period after an event during which a detector or recording system cannot register another event, limiting the measured count rate.

Deafferentation

Loss or interruption of nerve signals from a body area to the brain or spinal cord. It can cause unusual sensations or nerve pain.

Deafness

Complete or severe loss of hearing. Hearing loss may be conductive, sensorineural, or mixed, depending on whether the problem affects sound transmission, the inner ear or auditory nerve, or both.

Deaner

Not a standard medical term in common current usage; its intended meaning should be confirmed from the source or specialty context before clinical interpretation.

Death

The permanent end of the functions that maintain a living organism.

Death Certificate

A legal medical document recording a person's death, including the date, place, and certified causes of death.

Death Rate

The number of deaths in a defined population during a specified period, usually expressed per population size and used in public-health comparisons.

Death Rattle

Noisy, rattling breathing caused by secretions accumulating in the throat or upper airways, often occurring near the end of life when swallowing and coughing weaken.

Death Sudden Cardiac (Sudden Cardiac Death)

Alternate index label for Sudden Cardiac Death.

Deauville Criteria

A five-point scale used with a PET scan to compare activity in a lymphoma with normal reference tissues and judge how well treatment is working.

Debility

A state of weakness or loss of strength, often associated with illness, aging, or prolonged inactivity.

Alternate Names:

Weakness

Debride

To remove dead, damaged, infected, or contaminated tissue from a wound or body site to support healing and control infection.

Debridement

The surgical removal of dead, damaged, or infected tissue from a wound to promote healing and prevent infection.

Debrisoquin

An antihypertensive drug formerly used to reduce sympathetic nerve activity; it is now mainly relevant as a probe of CYP2D6 drug metabolism.

Debulk

To reduce the size or amount of a tumor or other abnormal tissue, usually by surgery or another treatment, without necessarily removing every abnormal cell.

Debulking

Surgical removal of as much tumor as possible when complete resection is not feasible, aiming to reduce

tumor burden (cytoreduction) and improve response to subsequent therapy; central to ovarian cancer management.

Decalcification

Loss of calcium salts from bone, teeth, or other mineralized tissue. In teeth it may weaken enamel; in bone it can contribute to reduced strength.

Decannulation

Removal of a cannula or tube, especially a tracheostomy tube, after the patient can maintain a safe airway and adequate breathing, secretion clearance, and oxygenation.

Decanoic Acid

A ten-carbon saturated fatty acid, also called capric acid, found in some fats and studied for antimicrobial, metabolic, and antiseizure effects.

Decapitation

Separation of the head from the body, seen in severe trauma (vehicle accidents, train injuries, explosions), industrial accidents, hanging with judicial execution, or homicide. Complete decapitation is almost always rapidly fatal due to exsanguination and brain ischemia.

Decay Correction

Mathematical adjustment for radioactive decay during imaging. Essential for quantitative comparison between studies.

Deceased Donor

Individual who has died (usually from brain death or cardiac death) and donated organs, tissues, or both for transplantation.

Decenoic Acid

An unsaturated ten-carbon fatty acid; its biologic effects depend on the position and geometry of its double bond and on tissue and dietary context.

Decerebrate Posture

Abnormal straightening of the arms and legs caused

by severe injury to the brainstem or nearby brain structures. It is a sign of serious neurologic injury and may occur after head injury, bleeding, or increased pressure in the skull.

Decerebrate Posturing

Abnormal extension of the arms and legs, often with inward rotation of the arms, caused by severe dysfunction below the cerebral hemispheres. It indicates serious neurologic injury, but posture alone does not determine outcome.

Decerebrate Rigidity

Abnormal extensor posturing caused by severe dysfunction below the red nucleus, often from brainstem injury or increased intracranial pressure, and indicating serious neurologic damage.

Decerebrate State

A severe neurologic posture marked by extension and inward rotation of the arms and extension of the legs, usually indicating serious injury to the brainstem or structures above it.

Decerebration

Severe loss of cerebral function, often with extension and rigid posturing of the limbs from brainstem dysfunction; it may result from major injury or increased intracranial pressure.

Decibel

A unit used to describe how loud a sound is. A higher decibel level means a louder sound; normal conversation is about 60 decibels.

Deciding To Have Knee Or Hip Replacement

A decision based on pain, functional limitation, imaging, nonoperative treatment response, surgical risk, expectations, and shared discussion with an orthopedic clinician.

Deciding To Quit Drinking Alcohol

The process of planning to stop or reduce alcohol use. Assessment should consider dependence, withdrawal

risk, support, and treatment options; people with heavy use may need medical supervision.

Decidua

The specialized lining of the uterus during pregnancy. It is shed after delivery and helps support the pregnancy and placenta.

Deciduitis

Inflammation of the decidua, the specialized uterine lining of pregnancy.

Deciduoma

Deciduoma is an old term for a decidual reaction or mass-like overgrowth of pregnancy-related uterine lining, not a true tumor.

Deciduosis

The presence of extensive or ectopic decidual tissue, usually associated with pregnancy and sometimes involving the cervix, peritoneum, or other sites.

Deciduous Teeth

Deciduous Teeth is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Deciduous Tooth

A primary or baby tooth that is eventually shed and replaced by a permanent tooth.

Decision Making

The process of comparing information, options, benefits, risks, and preferences to choose an action or plan, including shared decision-making in health-care.

Decision-Making Capacity

The ability to understand information about a medical decision, consider the possible results, compare choices, and communicate a choice. It applies to a specific decision, may change over time, and is different from legal competence.

Decitabine

A cytidine-analog antineoplastic agent that inhibits DNA methyltransferase and is used for selected myelodysplastic syndromes and acute myeloid leukemia.

Declarative Memory

Memory for facts and events that can be consciously recalled.

Declopramide

Not a standard generic drug name in common medical use; it may be a misspelling of metoclopramide and should be verified against the original source.

Decoction

A preparation made by simmering plant material in water to extract soluble substances.

Decompensated Cirrhosis

Cirrhosis complicated by clinically significant liver dysfunction or portal hypertension, commonly manifested by ascites, variceal hemorrhage, hepatic encephalopathy, or jaundice. It carries substantially higher mortality than compensated cirrhosis and warrants assessment for transplant and complication prevention.

Decomposition

Postmortem breakdown of the body through autolysis and putrefaction. It begins within 24 to 72 hours at ambient temperature, producing green ceal discoloration, venous marbling, gas distension, skin slippage, and purge fluid.

Decompression

Reduction of pressure on a tissue or body compartment, such as removing pressure from a spinal nerve, draining an obstructed organ, or treating pressure changes after diving.

Decompression Sickness

A systemic illness caused by nitrogen bubble formation in tissues during rapid decompression, presenting with joint pain, skin mottling, and neurological

deficits, treated with hyperbaric oxygen therapy.

Deconditioning

Physiological decline from reduced activity or immobilization. Effects: muscle atrophy, weakness, decreased cardiorespiratory fitness, balance impairment, increased fall risk. Preventable with early mobilization, exercise. Reversible with rehabilitation.

Decongestant

A medicine that reduces swelling of the nasal lining and relieves nasal congestion.

Decontamination

Measures that reduce continued absorption of a toxic substance from the skin, eyes, gastrointestinal tract, or clothing. Interventions are exposure-specific and may include removal from the source, irrigation, washing, activated charcoal, or specialist procedures.

Decorin

A small proteoglycan of the extracellular matrix that binds collagen and influences tissue structure, growth-factor signaling, and fibrosis.

Decorticate Posture

Abnormal bending of the arms with straightening of the legs, caused by severe injury to the brain. It is a sign of serious brain damage and reduced consciousness.

Decorticate Posturing

Abnormal flexion of the arms with extension of the legs, caused by severe dysfunction of the brain. It indicates serious neurologic injury, but posture alone does not determine outcome.

Decreased Alertness

Decreased alertness is reduced awareness or responsiveness that may result from metabolic illness, intoxication, infection, seizure, stroke, trauma, or sleep-related causes.

Decreased Breath Sounds

Diminished air movement heard on auscultation, indicating decreased ventilation. Causes include pneumothorax, pleural effusion, atelectasis, and mucus plugging. Unilateral decreased breath sounds may indicate tension pneumothorax or large effusion.

Decreased Concentration

Reduced ability to sustain attention or mental focus; causes include sleep loss, stress, medications, mood disorders, substance use, and neurologic or medical illness.

Decreased Dentin

Reduced thickness or formation of dentin, the mineralized tissue beneath tooth enamel; causes include developmental defects, wear, erosion, caries, or dental treatment.

Decreased Libido

Reduced sexual desire that may be related to relationships, mood, stress, medications, hormones, chronic illness, pain, or other health factors.

Decreased Tear Production

Alternate index label; see *Dry eyes*.

Decubitus

A lying or reclining position; the term also appears in decubitus ulcer, a pressure injury caused by prolonged pressure and shear.

Decubitus Ulcer

Localized damage to the skin and/or underlying soft tissue caused by prolonged pressure, shear, or friction, most commonly over bony prominences. See pressure injury.

Deep

Located further inward from the surface of the body.

Deep Femoral Artery

The profunda femoris artery, a major branch of the femoral artery that supplies the deep muscles of the

thigh and gives rise to perforating branches.

Deep Brain Stimulation

A treatment in which implanted electrodes deliver controlled electrical impulses to selected brain regions. It is used for conditions such as Parkinson disease, essential tremor, and some forms of dystonia.

Deep Infiltrating Endometriosis

A severe, aggressive form of endometriosis defined by the presence of ectopic endometrial glands and stroma penetrating ≥ 5 mm beneath the peritoneal surface into pelvic retroperitoneal structures. Commonly involves the uterosacral ligaments, rectovaginal septum, pouch of Douglas, rectosigmoid bowel, and bladder wall, presenting with chronic pelvic pain, severe dysmenorrhea, dyschezia, and deep dyspareunia, often necessitating complex multidisciplinary surgical excision.

Alternate Names:

DIE; Deep Endometriosis

Deep Sleep

The deepest stage of non-rapid-eye-movement sleep, during which arousal is difficult and physical restoration occurs.

Deep Tendon Reflex

An automatic muscle response caused by tapping a tendon with a reflex hammer. It helps assess nerves and the spinal cord. Common examples are the knee, ankle, biceps, and triceps reflexes.

Deep Vein Thrombosis

A clot in a deep vein, usually of the lower limb, caused by a combination of reduced blood flow, increased clotting tendency, or vessel-wall injury. Anticoagulant treatment is used when indicated.

Alternate Names:

DVT (Deep Vein Thrombosis)

Deep Vein Thrombosis

A blood clot in a deep vein, usually in the leg, that can cause swelling and pain and may travel to the

lungs as a pulmonary embolism.

Deep Vein Thrombosis Ultrasound

Venous duplex ultrasound used to evaluate suspected deep-vein thrombosis. The test assesses vein compressibility and blood flow; an abnormal result is interpreted with the clinical probability and, when needed, repeat or additional testing.

Deep Venous Reflux

Retrograde flow through incompetent venous valves in the deep venous system. It contributes to chronic venous hypertension, edema, skin changes, and venous ulcer formation.

Deep Venous Thrombosis

Formation of a blood clot in a deep vein, most often in the leg.

Defaecation

Defaecation is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Defecation

The process of eliminating feces from the body. Filling of the rectum triggers stretch receptors that initiate the defecation reflex via the sacral spinal cord (S2-S4), causing contraction of the rectal wall and relaxation of the internal anal sphincter.

Defecation Syncope

Fainting during or immediately after defecation, usually a situational reflex involving straining, reduced venous return, and a temporary fall in blood pressure.

Defecography

Dynamic imaging of the rectum and pelvic floor during defecation using fluoroscopy or MRI. The rectum is filled with contrast paste and the patient performs defecation while images are captured. Evaluates rectal prolapse, intussusception, rectocele, enterocele, and pelvic floor dyssynergia.

Defective Virus

Defective virus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Defence Mechanisms

A clinical care, prevention, behavioral, or public-health concept concerning defence mechanisms; its use depends on the patient, setting, and clinical goal.

Defense Mechanism

A psychological process, often unconscious, that helps a person manage conflict, anxiety, or distress; its usefulness depends on context and flexibility.

Defense Mechanisms

Psychological processes, often outside conscious awareness, that help a person manage distress or conflict. Examples include denial, projection, repression, humor, and sublimation.

Defense Mechanisms (Forensic)

Physical evidence patterns on a body demonstrating active resistance against an assailant prior to incapacitation.

Alternate Names:

Forensic

Defense Wounds

Injuries sustained by a victim attempting to ward off an attack, typically found on the flexor surfaces of hands, palms, wrists, and forearms (incised/stab wounds from knives or contusions/abrasions from blunt force).

Defense Wounds (Inscribed)

Cuts or other injuries on the hands or forearms that may occur when a person attempts to protect against an attack. The distribution and morphology of defensive injuries may help reconstruct an assault but are not independently proof of a particular event.

Alternate Names:

Inscribed

Defenses

The body's protective mechanisms against infection, injury, and other threats, including physical barriers and immune responses.

Alternate Names:

Body defenses

Defensin

An antimicrobial peptide produced by epithelial cells and immune cells that disrupts microbial membranes and contributes to innate immunity.

Deferasirox

An oral iron-chelating medicine used to reduce chronic iron overload when clinically indicated; kidney, liver, hearing, and gastrointestinal toxicity require monitoring.

Deferiprone

An oral iron chelator used for transfusional iron overload; neutropenia or agranulocytosis is an important adverse effect requiring blood-count monitoring.

Deferoxamine

A parenteral iron chelator used for severe or chronic iron overload and sometimes aluminum toxicity; treatment requires monitoring for ocular, auditory, renal, and infectious complications.

Defibrillation

Delivery of an unsynchronized electrical shock to terminate ventricular fibrillation or pulseless ventricular tachycardia by depolarizing a critical mass of myocardium. It is distinct from synchronized cardioversion, which times the shock to the QRS complex.

Defibrillator

Device delivering electric shock to heart. Types: external (AED, manual), internal (implantable). Used for ventricular fibrillation, pulseless ventricular tachycardia. Energy: biphasic 120-200 J.

Defibrillator Implant

An implanted cardioverter-defibrillator that detects and treats dangerous ventricular rhythms with electrical pacing or shock.

Deficiency Diseases

Deficiency Diseases is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Deficiency Primary Immunodeficiency (Primary Immunodeficiency Disease PIDD)

Alternate index label for Primary Immunodeficiency Disease PIDD.

Deficiency, FAO

Fatty-acid oxidation deficiency, an inherited metabolic disorder in which cells cannot properly use fatty acids for energy, potentially causing fasting- or illness-triggered hypoglycemia, muscle injury, or cardiomyopathy.

Deficiency, Iron

Inadequate iron stores or availability, which can impair hemoglobin production and cause iron-deficiency anemia, fatigue, pallor, and reduced exercise tolerance.

Deficiency, UDP-Glucuronosyltransferase

Reduced activity of the enzyme that conjugates bilirubin in the liver; inherited forms include Gilbert syndrome and the more severe Crigler-Najjar syndromes.

Deficient Color Vision

Alternate index label; see *Color blindness*.

Defined Observation

An observation recorded using a specified definition, measurement method, time point, or assessment rule so that it can be interpreted consistently.

Defining Overweight And Obesity In

Children

Classification of excess body mass in children using age- and sex-specific body-mass-index percentiles, growth patterns, health effects, and clinical context rather than adult cutoffs.

Deformation

An alteration in the shape or structure of a body part, tissue, or material caused by developmental forces, injury, disease, pressure, or mechanical stress.

Deformity

A deformity is an abnormal shape, alignment, or structure of a body part caused by development, injury, disease, or growth.

Degarelix

A gonadotropin-releasing hormone antagonist that rapidly suppresses testosterone production and is used for advanced prostate cancer.

Degeneration

Progressive structural or functional deterioration of cells, tissues, or organs caused by aging, disease, injury, inherited factors, or other processes.

Degenerative Arthritis

Alternate index label; see *Osteoarthritis*.

Degenerative Disc

Degenerative disc disease describes age- or wear-related changes in spinal discs that may cause neck or back pain, stiffness, or nerve compression. Imaging changes can occur without symptoms; treatment usually begins with activity, exercise-based rehabilitation, and pain management.

Alternate Names:

DDD (Degenerative Disc)

Degenerative Disc

Alternate index label for Degenerative Disc.

Degenerative Disease

A disorder in which the structure or function of tissues progressively deteriorates.

Alternate Names:

Degenerative disorder

Degenerative Disk Disease

Age-related or injury-related deterioration of an intervertebral disk that may cause back or neck pain.

Alternate Names:

Degenerative disc disease

Degenerative Disorder

A disease in which progressive cellular or tissue deterioration causes loss of structure or function, as in neurodegenerative, joint, retinal, or muscle disorders.

Degenerative Joint Disease

Progressive loss of cartilage and changes in a joint, commonly referring to osteoarthritis.

Alternate Names:

Osteoarthritis

Degenerative Spine Disease

Lumbar spinal stenosis: central (neurogenic claudication: bilateral leg pain with walking, relieved by sitting/flexion), lateral recess, foraminal. Imaging: MRI (central canal AP diameter <10mm, cross-sectional area <70mm²).

Degenerative Spondylolisthesis

Forward displacement of one vertebra over another caused by degenerative changes in the spine.

Deglutition

Deglutition is swallowing, the coordinated movement of food or liquid from the mouth through the pharynx and esophagus into the stomach.

Deglutition (Swallowing)

The coordinated process of moving food, liquid, or saliva from the mouth through the pharynx and esophagus into the stomach.

Alternate Names:

Swallowing

Degranulation

Release of stored granule contents from cells such as mast cells, basophils, neutrophils, and cytotoxic lymphocytes during immune or inflammatory responses.

Degrees Fahrenheit

A temperature measurement expressed on the Fahrenheit scale; 32 degrees Fahrenheit is the freezing point of water and 212 degrees is its boiling point at standard pressure.

Dehiscence

Separation or reopening of the edges of a wound or surgical closure. Wound dehiscence may be superficial or involve deeper tissues and requires clinical assessment.

DEHP

Di(2-ethylhexyl) phthalate, a plasticizer used in some medical and consumer products; exposure is monitored because of potential reproductive, developmental, and endocrine effects.

Dehydration

A condition in which the body loses more water than it takes in. It can cause thirst, less urination, dizziness, fast heartbeat, and changes in body salts; severe dehydration can lead to shock. Common causes include vomiting, diarrhea, fever, sweating, and not drinking enough.

Dehydration In Children

A deficit of total body water, usually caused by gastrointestinal losses, inadequate intake, fever, or renal losses. Clinical assessment includes mental status, heart rate, mucous membranes, tears, capillary refill, urine output, weight change, and perfusion rather than a single sign.

Dehydroascorbic Acid

The oxidized form of vitamin C that can be transported into cells and reduced back to ascorbate, contributing to redox and antioxidant chemistry.

Dehydroepiandrosterone

An adrenal steroid precursor that can be converted to androgens and estrogens; levels vary with age, adrenal disorders, medicines, and supplements.

Deinstitutionalize

To move a person from long-term institutional care toward community-based living and support, when safe and appropriate.

Deiodinase

An enzyme that removes iodine atoms from thyroid hormones, thereby activating or inactivating thyroxine and triiodothyronine and regulating local thyroid-hormone action.

Deja Vu

A feeling that a current situation has been experienced before, despite recognizing that it is probably new; frequent or distressing episodes may warrant evaluation.

Dejerine-Roussy Syndrome

Central post-stroke pain caused by a lesion of the thalamus or related sensory pathways. It may produce persistent contralateral pain, burning, abnormal sensation, or pain from normally nonpainful stimuli.

Delavirdine

A non-nucleoside reverse-transcriptase inhibitor formerly used in combination antiretroviral therapy for HIV; interactions and resistance limit its use.

Delayed Cord Clamping

The practice of delaying umbilical cord clamping for ≥ 30 –60 seconds after delivery, allowing placental transfusion of blood to the newborn. Benefits: increases neonatal blood volume by 30%, improves iron stores (ferritin, hemoglobin) for the first 6 months, reduces anemia, and improves transitional circulation.

Delayed Ejaculation

A sexual dysfunction characterized by a persistent

or recurrent delay in, infrequency of, or absence of ejaculation during partnered sexual activity, lasting ≥ 6 months and causing marked distress. Prevalence: 1-5% of men. Lifelong vs. acquired.

Delayed Emergence

Prolonged time to regain consciousness after discontinuation of anesthetic agents. Failure to regain consciousness within 30-60 minutes of completing surgery.

Delayed Gastric Emptying

Alternate index label; see *Gastroparesis*.

Delayed Gastric Emptying (Gastroparesis)

Alternate index label for Gastroparesis.

Delayed Growth

Slower-than-expected increase in height, weight, or head size compared with standardized growth patterns, caused by nutrition, chronic disease, endocrine disorders, genetics, or normal variation.

Delayed Hypersensitivity

A delayed immune reaction, usually mediated by T cells rather than antibodies, that develops hours to days after exposure. Examples include contact dermatitis, the tuberculin reaction, and some drug eruptions.

Delayed Orgasm

Persistent difficulty reaching orgasm despite adequate sexual stimulation, potentially related to medications, health conditions, psychological factors, or relationship context.

Delayed Puberty

Failure to develop secondary sexual characteristics by age thirteen in girls and age fourteen in boys. Causes include constitutional delay which is the most common, chronic illness, malnutrition, hypogonadotropic hypogonadism from Kallmann syndrome or pituitary disease, and hypergonadotropic hypogonadism from gonadal failure.

Delayed Puberty In Boys

Failure of testicular enlargement and other pubertal changes to begin by the expected age in a boy, caused by constitutional delay, chronic illness, hypogonadism, or hypothalamic-pituitary disease.

Delayed Puberty In Girls

Absence or inadequate progression of breast development or menstruation beyond expected age, caused by constitutional delay, low energy availability, chronic illness, ovarian failure, or hypothalamic-pituitary disease.

Delayed Sleep Phase

A circadian-rhythm disorder in which a person falls asleep and wakes later than desired; also called delayed sleep-wake phase disorder.

Alternate Names:

delayed sleep-wake phase disorder. Delayed Sleep-Wake Phase Sleep Disorder DSPS DSWPD

Delayed Sleep Phase Syndrome

A circadian-rhythm disorder in which sleep and waking occur substantially later than desired.

Delayed Sleep-Wake Phase Sleep Disorder

Alternate index label; see *Delayed sleep phase*.

Deletion

Loss of chromosomal segment. Terminal deletion: loss from end. Interstitial deletion: loss from middle. Microdeletion syndromes: 22q11.2 deletion (DiGeorge), 5p deletion (Cri-du-chat).

Deletion Mutation

A genetic variant in which one or more DNA bases are absent; its effect depends on the location, size, and whether it alters a gene or regulatory region.

Delftia

A genus of aerobic Gram-negative bacteria found in soil and water; human infection is uncommon but has been reported, especially in people with significant underlying illness.

Delftia Acidovorans

An environmental gram-negative bacterium that rarely causes bloodstream or other opportunistic infection, particularly in people with devices or impaired immunity.

Delirium

Delirium is an acute fluctuating disturbance of attention and awareness caused by an underlying medical, medication-related, toxic, or environmental factor.

Delirium In Anesthesia

Delirium is an acute confusional state characterized by fluctuating consciousness, inattention, disorganized thinking, and altered level of awareness. Postoperative delirium occurs in 15 to 50 percent of elderly surgical patients, with higher rates in hip fracture and cardiac surgery.

Delirium Tremens

A severe, potentially life-threatening form of alcohol withdrawal characterized by fluctuating confusion, agitation, hallucinations, and marked autonomic overactivity. It usually occurs after cessation or major reduction of prolonged heavy alcohol use and requires urgent medical treatment.

Delivery

The process of childbirth: vaginal delivery (spontaneous vertex most common; assisted—vacuum, forceps) or cesarean delivery (abdominal). Stages of labor culminate in delivery of the infant and placenta.

Delivery Of Posterior Arm

A shoulder dystocia maneuver: the operator's hand enters the posterior vagina along the fetal chest, finds the fetal hand, grasps the forearm, sweeps it across the chest and face, and delivers the posterior arm and hand over the perineum. This reduces the bisacromial diameter by 20% and is often highly effective.

Delivery Presentations

The part of the fetus that is positioned to enter the birth canal first, such as vertex, breech, or shoulder

presentation; presentation affects obstetric planning.

Delphi Technique

A structured research method that uses repeated anonymous surveys and feedback to develop expert consensus.

Delta Agent

An older name for hepatitis D virus, which requires hepatitis B virus for replication and can cause severe acute or chronic hepatitis.

Delta Waves

Large, slow brain waves associated with deep non-REM sleep.

Alternate Names:

Delta rhythm

Delta-ALA Urine Test

A urine test measuring delta-aminolevulinic acid, an intermediate in heme synthesis. Increased levels may occur in acute hepatic porphyrias or lead toxicity and must be interpreted with symptoms and other porphyrin studies.

Delta-Tocopherol

One naturally occurring form of vitamin E; its biological and antioxidant effects differ from those of alpha-tocopherol, the principal form used to assess human vitamin E status.

Deltacoronavirus

Deltacoronavirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Deltaretrovirus

Deltaretrovirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Deltoid

The deltoid is the large shoulder muscle that lifts the arm and helps stabilize the shoulder joint.

Deltoid Muscle

Large triangular muscle forming the shoulder contour, from lateral clavicle, acromion, and scapular spine to the deltoid tuberosity. Innervated by the axillary nerve (C5-C6). Anterior fibers flex/medially rotate, middle fibers abduct (primary abductor beyond 15 degrees), posterior fibers extend/laterally rotate.

Delusion

A fixed belief that is not consistent with available evidence or the person's sociocultural context and is not readily changed by reasoned discussion. Delusions may be persecutory, grandiose, referential, somatic, nihilistic, or erotomanic.

Delusional Disorder

A disorder characterized by the presence of one or more delusions lasting one month or more in the context of relatively preserved functioning. The delusions are non-bizarre (involving real-life situations that could conceivably occur, such as being followed, infected, loved from afar, or conspired against).

Demarcation

A clear boundary separating one area, tissue, process, or condition from another, such as the line between viable and necrotic tissue.

Demeclocycline

A medicine called demeclocycline. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Demecolcine

A colchicine-related antimitotic compound that disrupts microtubules; it has limited medical use and can be toxic.

Dementia

Dementia is acquired decline in one or more cognitive domains that interferes with independent function; causes include neurodegenerative, vascular, metabolic, infectious, and other disorders. Alzheimer disease, vascular disease, Lewy body disease, frontotemporal disorders, and other conditions. Evaluation is important because some causes of cognitive decline may be treatable.

Alternate Names:

Pugilistica Dementia (Dementia)

Dementia (Major Neurocognitive Disorder)

Dementia is the common clinical term for an acquired syndrome of decline in one or more cognitive domains that interferes with independence in daily activities. Major neurocognitive disorder is the preferred diagnostic classification term.

Alternate Names:

Major Neurocognitive Disorder

Dementia - Behavior And Sleep Problems

Behavioral and sleep changes that can occur in dementia, including agitation, wandering, apathy, hallucinations, or altered sleep-wake timing; causes and reversible contributors should be assessed.

Dementia And Driving

The effect of cognitive impairment on driving safety. Assessment considers memory, judgment, visuospatial skills, reaction time, medications, and local licensing requirements.

Dementia Due To Metabolic Causes

Cognitive impairment caused or worsened by metabolic problems such as thyroid disease, vitamin deficiency, organ failure, electrolyte disturbance, or medication toxicity; some causes are treatable.

Dementia Pugilistica

A progressive brain disorder associated with repeated head trauma, historically described in some professional boxers.

Dementia, Frontotemporal

Alternate index label; see *Frontotemporal dementia*.

Dementia, Lewy Body

Alternate index label; see *Lewy body dementia*.

Dementia, Vascular

Alternate index label; see *Vascular dementia*.

Demethylation

Removal of a methyl group from a molecule, including DNA or proteins; DNA demethylation can alter gene expression and cellular development.

Demineralization

Loss of mineral from a tissue, such as tooth enamel or bone, which may weaken the tissue.

Demographics Domain

The demographic characteristics of a population or study group, such as age, sex, ethnicity, location, education, or socioeconomic measures.

Demulcent

A substance that soothes irritated or inflamed mucous membranes by forming a protective coating.

Demyelinating Disease

A disease in which the myelin sheath surrounding nerve fibres is damaged, slowing or blocking neural conduction.

Demyelinating Disease Of The Central Nervous System

A disorder in which myelin is damaged within the brain, spinal cord, or optic pathways, impairing nerve conduction. Causes include autoimmune, infectious, genetic, toxic, and metabolic processes.

Demyelination

Loss or damage of the myelin sheath that normally insulates nerve fibers and speeds nerve signaling.

Denaturant

Denaturant is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Dendrimer

A highly branched, tree-like synthetic molecule studied for drug delivery, imaging, and gene-transfer applications; clinical use depends on safety and formulation.

Dendrite

The branched, tapering neuronal processes that receive synaptic input from other neurons and conduct graded potentials toward the cell body. Dendrites contain receptors, ion channels, and postsynaptic densities.

Dendrites

Branching extensions of nerve cells that receive signals from other neurons or sensory receptors.

Dendritic Cell

The most potent professional antigen-presenting cell, capturing antigens in tissues and migrating to lymph nodes to present peptides on MHC molecules to T cells, initiating adaptive immunity.

Dendritic Cells

Immune cells that capture foreign material and present antigens to T cells, helping initiate an adaptive immune response.

Dendritic Spine

A small protrusion on a neuronal dendrite that receives many excitatory synaptic inputs and can change with learning and disease.

Dendritic Ulcer

A branching epithelial ulcer of the cornea, classically caused by herpes simplex virus. It may cause a red eye, pain, light sensitivity, tearing, or reduced vision and requires ophthalmic assessment.

Denervate

Loss of nerve supply to a tissue or organ; may occur surgically, traumatically, or through disease processes.

Denervation

Denervation is loss of nerve supply to a muscle or tissue, causing weakness, atrophy, altered sensation, or autonomic dysfunction.

Dengue

A mosquito-borne viral infection that commonly causes fever, headache, muscle pain, joint pain, and rash. Severe disease can cause bleeding or shock.

Dengue Fever

Dengue fever is a mosquito-borne viral illness causing fever, headache, muscle and joint pain, rash, and sometimes plasma leakage, bleeding, or shock. flavivirus transmitted primarily by *Aedes aegypti* mosquitoes.

Alternate Names:

Dengue Hemorrhagic Fever

Dengue Hemorrhagic Fever

Alternate index label; see *Dengue fever*.

Dengue Hemorrhagic Fever

A severe, life-threatening manifestation of secondary dengue virus infection (flavivirus transmitted by *Aedes* mosquitoes), driven by antibody-dependent enhancement. Characterized by abrupt high fever, marked thrombocytopenia ($\leq 100,000/\mu\text{L}$), hemorrhagic diathesis (petechiae, mucosal bleeding), and acute systemic capillary hyperpermeability leading to plasma leakage into pleural and peritoneal cavities, hemoconcentration, and progression to dengue shock syndrome. Managed with meticulous fluid resuscitation and strict avoidance of NSAIDs.

Dengue Shock Syndrome

Life-threatening circulatory shock caused by severe dengue with plasma leakage and reduced effective blood volume, often accompanied by bleeding or organ dysfunction.

Dengue Virus

A positive-sense RNA virus belonging to the Flaviviridae family that causes dengue fever. The virus has four serotypes, and infection with one serotype provides lifelong immunity to that serotype but only temporary cross-protection against others.

Denial

A defense mechanism in which a person refuses to accept an unpleasant reality. It is a common initial response to traumatic events and terminal illness diagnoses. While adaptive in the short term, chronic denial can prevent appropriate medical treatment and psychological processing.

Denileukin Diftitox

An engineered fusion protein that combines interleukin-2 with diphtheria toxin and was used to treat selected cutaneous T-cell lymphomas. Its use is limited by toxicity and current product availability.

Denominator

The number below the line in a fraction; in a rate or proportion it identifies the population or total against which the numerator is compared.

Denosumab

A medicine called denosumab. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Denosumab For Osteoporosis

Fully human monoclonal antibody targeting RANK ligand preventing osteoclast formation activation and survival. Reduces fracture risk at the spine hip and non-vertebral sites. Administered as subcutaneous injection every six months. Advantages include no renal dose adjustment and effectiveness even with oral bisphosphonate intolerance.

Dens

The tooth-like projection of the second cervical vertebra that forms a pivot for rotation of the first vertebra.

Dens Evaginatus

An extra cusp or tubercle on a tooth, most often a premolar. It may contain an extension of the dental pulp and may fracture or become affected by caries.

Dens Invaginatus

An inward folding of enamel and dentin that may extend toward or into the dental pulp, allowing debris to collect and increasing the risk of caries and pulpal infection.

Densitometry

Measurement of density, most often the mineral content of bones. A bone-density scan helps assess osteoporosis and the risk of fractures.

Density

The amount of mass in a given volume; in medicine, density measurements help describe bone strength, tissue composition, or findings on imaging.

Dent Disease

An inherited X-linked proximal renal-tubule disorder causing low-molecular-weight proteinuria, hypercalciuria, kidney stones or nephrocalcinosis, and progressive kidney impairment.

Dental Abscess

A localized collection of pus arising from infection of a tooth or the surrounding periodontal tissues. It may cause toothache, percussion tenderness, gingival swelling, facial swelling, fever, trismus, or regional lymphadenopathy; spread into deep facial spaces can threaten the airway and requires urgent assessment.

Dental Age

An estimate of developmental age based on the formation, mineralization, and eruption of the teeth. It is useful in children but is not an exact measure of chronological age.

Dental Alloy

Dental alloy refers to a dental structure, material, procedure, or oral-health finding involving the teeth

or supporting tissues.

Dental Alveolus

Dental alveolus refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Amalgam

Dental amalgam refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Anesthesia

The use of local anesthetic, sedation, or, when necessary, general anesthesia to control pain and anxiety during dental care.

Dental Anxiety

Dental anxiety refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Arch

Dental arch refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Articulator

Dental articulator refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Braces

Dental braces apply controlled force to move teeth and align the bite over time. Treatment requires oral hygiene, dietary care, periodic adjustments, and retention afterward; risks include temporary soreness, enamel decalcification, gum problems, and root changes.

Alternate Names:

Braces And Retainers (Dental Braces)

Dental Braces (Orthodontics)

Fixed or removable orthodontic appliances that apply controlled forces to move teeth and correct mal-

occlusion or jaw-related alignment problems.

Dental Calculus

Hardened dental plaque formed when mineral salts accumulate on teeth.

Dental Caries

A disease in which bacterial acids progressively demineralize and damage the teeth. See Dental Cavities.

Dental Caries (Cavities)

Alternate index label for Cavities.

Dental Cavities

Permanent areas of tooth decay caused by acid-producing bacteria acting on dietary carbohydrates. Cavities progress from enamel demineralization to dentin and pulp involvement, causing sensitivity, pain, or infection.

Dental Cement

Dental cement refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Ceramic

A ceramic material used to make dental restorations such as crowns, veneers, and bridges. Examples include feldspathic ceramic, lithium disilicate, and zirconia.

Dental Clasp

Dental clasp refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Crowns

Custom-made restorations that cover most or all of a damaged tooth to restore its shape, strength, function, and appearance.

Dental Deposit

Dental deposit refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Disease

Disorders affecting the teeth, gums, or supporting tissues, including tooth decay, gum disease, oral infections, and tooth loss. Prevention includes fluoride, regular brushing and flossing, limiting frequent sugar exposure, and routine dental examinations.

Dental Dysplasia

Dental dysplasia refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Extraction

Removal of a tooth from its alveolar socket. Indications include unrestorable disease, advanced periodontal destruction, impaction, orthodontic treatment, or selected prosthodontic needs; risks include bleeding, infection, nerve injury, fracture, and alveolar osteitis.

Dental Fissure

Dental fissure refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Floss

Dental Floss is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Dental Hygiene

Dental hygiene refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Implant

A biocompatible fixture placed in the jawbone to support a replacement tooth. The fixture may become integrated with surrounding bone and is restored with a crown, bridge, or denture.

Dental Impression Material

Dental impression material refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Injuries

Dental injuries include fractures, displacement, loosening, or loss of teeth and damage to gums or jaw after trauma. Prompt dental assessment can preserve injured teeth; uncontrolled bleeding, jaw deformity, airway problems, or head injury requires emergency care.

Alternate Names:

Dental Trauma (Dental Injuries)

Dental Inlay

Dental inlay refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Material

Dental material refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Occlusion

Dental occlusion refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Overjet

Dental overjet refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Pain (Toothache)

Pain arising from a tooth or nearby tissues, commonly due to dental caries, pulp inflammation, infection, trauma, cracked teeth, or gum disease.

Dental Papilla

Dental papilla refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Pellicle

Dental pellicle refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Plaque

Biofilm of bacteria on tooth surfaces. Initiated by acquired pellicle. Maturation leads to disease if not removed. Disrupted by mechanical cleaning.

Dental Porcelain

Dental porcelain refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Procedure

A medical procedure, treatment, device, or care topic related to dental procedure; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Dental Prosthesis

Dental prosthesis refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Pulp

Dental pulp refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Pulp Necrosis

Death of the dental pulp, usually following untreated inflammation, trauma, or loss of blood supply. The tooth generally fails sensibility testing and may develop apical periodontitis or a periapical abscess.

Dental Pulpitis

Inflammation of the dental pulp caused by caries, trauma, restorative procedures, cracks, or other irritation. Reversible pulpitis produces brief provoked pain, whereas irreversible pulpitis causes spontaneous or lingering pain and may require pulpal treatment or extraction.

Dental Radiographs

Imaging for dental diagnosis. Intraoral: periapical (tooth root and surrounding bone), bitewing (interproximal caries detection, crestal bone level), and occlusal (larger area, localization).

Dental Radiography

Imaging techniques for dental diagnosis: bitewing (caries, bone levels), periapical (root, periapical tissues), panoramic (full mouth overview), CBCT (3D imaging). Radiation dose: dental radiography low exposure. ALARA principle applies.

Dental Restorative Materials

Materials used to restore teeth: amalgam, composite resin, glass ionomer, ceramic, gold alloy. Selection based on location, aesthetics, strength, cost, patient factors. Composite: tooth-colored, adhesive, conservative preparation. Ceramic: excellent aesthetics, brittle, expensive. Gold: durable, biocompatible, aesthetic concern.

Dental Sac

Dental sac refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dental Screening

Examination of the teeth, gums, and mouth to identify dental disease or risk before serious symptoms develop.

Dental Sealant

Thin protective coating applied to chewing surfaces of molars. Prevents caries by sealing pits and fissures.

Dental Sealants

Thin resin or glass-ionomer coatings applied to the pits and fissures of teeth, especially molars, to reduce retention of plaque and risk of dental caries.

Dental Trauma (Dental Injuries)

Alternate index label for Dental Injuries.

Dental Traumatology

Management of dental injuries. Types: enamel fractures, enamel-dentin fractures, pulp exposure, root fractures, avulsion. Immediate care critical for avulsed teeth.

Dental X-Rays

Radiographic images of the teeth and jaws used to detect caries, bone loss, impacted teeth, infection, developmental abnormalities, and other dental disease.

Dentate Gyrus

Dentate gyrus refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dentatorubral-Pallidoluysian Atrophy

An autosomal dominant neurodegenerative repeat expansion disorder caused by an unstable CAG trinucleotide expansion encoding a polyglutamine tract in the *ATN1* gene on chromosome 12p13.31. Exhibits striking genetic anticipation, especially with paternal transmission. Juvenile-onset cases (expansions > 65 repeats) present with progressive myoclonic epilepsy, cognitive decline, and ataxia; adult-onset presentations manifest with cerebellar ataxia, choreoathetosis, behavioral changes, and dementia, histopathologically marked by selective degeneration of the dentatorubral and pallidoluysian systems of the central nervous system.

Alternate Names:

DRPLA; Haw River Syndrome; Naito-Oyanagi Disease

Dentifrice

Dentifrice refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dentigerous Cyst

Dentigerous cyst refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dentin

Calcified tissue beneath enamel forming bulk of tooth. Contains dentinal tubules with fluid and processes. Sensitive to stimuli when exposed. Reparative dentin formed in response to irritation.

Dentin Sensitivity

Dentin sensitivity refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dentinogenesis

The formation of dentin by odontoblasts during tooth development and repair; inherited defects can cause abnormal tooth color, weakness, and rapid wear.

Dentinogenesis Imperfecta

An inherited disorder of dentin formation that can cause translucent blue-gray or amber teeth that fracture, wear, or become discolored easily. Some forms occur with osteogenesis imperfecta.

Dentist

Dentist refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dentistry

Dentistry refers to a dental structure, material, procedure, or oral-health finding involving the teeth or supporting tissues.

Dentistry (Geriatric)

Oral health challenges in elderly: xerostomia, root caries, periodontal disease, denture problems, oral cancer, medication side effects. Assessment: complete oral exam, cancer screening, denture fit, caries risk. Prevention: fluoride, oral hygiene, regular dental visits. Adaptations: handholds, magnification, positioning.

Alternate Names:

Geriatric

Dentition

The teeth present in the mouth, including their number, arrangement, development, and eruption.

Denture

A removable dental prosthesis that replaces some or all missing teeth and associated oral tissues. Partial

dentures replace selected teeth while complete dentures replace an entire dental arch; fit, oral hygiene, tissue health, and regular review are important for function and prevention of mucosal injury.

Denture Base

The part of a denture that rests on the oral tissues and supports the artificial teeth.

Denture Problems

Problems involving removable dental prostheses, such as poor fit, discomfort, sores, difficulty chewing or speaking, breakage, or oral infection.

Denturist

A dental professional authorized in some jurisdictions to provide, fit, and maintain removable dentures within the local scope of practice.

Denver Developmental Screening Test

A developmental screening tool for young children that assesses personal-social, fine-motor, language, and gross-motor skills; it screens for possible delay and is not a diagnostic test.

Denys-Drash Syndrome

A rare genetic disorder associated with abnormal kidney development, nephropathy, and an increased risk of Wilms tumor.

Deodorant Poisoning

Exposure to deodorant, which may irritate the skin, eyes, airways, or gastrointestinal tract; product ingredients determine severity.

Deoxycoformycin

A medicine called deoxycoformycin. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Deoxycytidine

A deoxyribonucleoside composed of cytosine attached to deoxyribose, used as a building block for DNA.

Deoxycytidine Kinase

An enzyme that phosphorylates deoxycytidine and related nucleosides; it activates several nucleoside-analog medicines used in cancer and antiviral treatment.

Deoxyglucose

A glucose analog used in metabolic research and, when labeled with fluorine-18, in FDG-PET imaging to identify tissues with increased glucose use.

Deoxyguanosine

A deoxyribonucleoside composed of guanine attached to deoxyribose, used as a building block for DNA.

Deoxyhemoglobin

Hemoglobin that is not bound to oxygen; increased levels in superficial blood can contribute to bluish cyanotic coloration.

Deoxyribonuclease

An enzyme that breaks down DNA; recombinant dornase alfa, a deoxyribonuclease, helps loosen thick mucus in cystic fibrosis.

Deoxyribonucleic Acid

The molecule that stores hereditary genetic information in cells.

Alternate Names:

DNA

Deoxyribonucleic Acid

The double-stranded helical macromolecule that stores hereditary genetic instructions in all cellular organisms and many viruses. Composed of two antiparallel polynucleotide strands with a deoxyribose-phosphate backbone linked by phosphodiester bonds, connected centrally through complementary hydrogen-bonded purine-pyrimidine base pairs (adenine with thymine, guanine with cytosine). Undergoes semiconservative replication catalyzed by DNA polymerases, packages into chromatin via histone octamers, and directs cellular protein synthesis through transcription into messenger RNA

and subsequent ribosomal translation.

Deoxyribonucleosides

Molecules made of a DNA base linked to deoxyribose; adenine, guanine, cytosine, and thymine deoxyribonucleosides supply DNA synthesis.

Deoxyribonucleotides

Nucleotides containing deoxyribose that serve as the building blocks of DNA, including dATP, dGTP, dCTP, and dTTP.

Deoxyuridine

A deoxyribonucleoside containing uracil; it can arise from DNA repair or abnormal nucleotide metabolism and is used in laboratory studies.

Department Of Health And Human Services

The U.S. federal department responsible for major public-health, human-services, biomedical research, and health-insurance programs.

Dependence

A state in which repeated exposure to a substance produces physiologic adaptation, so that stopping or reducing it may cause withdrawal. In substance-use disorders, dependence may occur with or without addiction.

Dependence, Nicotine

Alternate index label; see *Nicotine dependence*.

Dependent Personality Disorder

A pattern of submissive and clinging behavior related to an excessive need to be taken care of.

Dependovirus

Dependovirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Depersonalization

Depersonalization is a feeling of detachment or un-

reality regarding one's own body, thoughts, or self, often occurring with anxiety, trauma, or dissociation.

Depersonalization-Derealization Disorder

A disorder involving persistent or recurrent feelings of detachment from one's thoughts, feelings, body, or surroundings, while reality testing remains intact. Symptoms cause clinically significant distress or impairment.

Dephosphorylation

Removal of a phosphate group from a molecule. This can change how a protein works or how cells send signals.

Depigmentation

Loss or reduction of normal pigment, especially melanin in the skin, hair, or mucosa, caused by conditions such as vitiligo, inflammation, injury, or treatment.

Depilación

Removal of hair from the skin by shaving, cutting, or chemical treatment without removing the entire hair root.

Depilatory Poisoning

Exposure to depilatory, which may irritate the skin, eyes, airways, or gastrointestinal tract; product ingredients determine severity.

Depolarization

A change in a cell's electrical charge that makes the inside of the cell less negative. In nerve and muscle cells, this change can start an electrical signal.

Depot Implant

An implant designed to release a medicine gradually over an extended period after placement in the body.

Deprescribing

A planned, supervised process of reducing the dose or stopping a medication for which harms, burden, or lack of benefit outweigh expected benefit. It re-

quires review of indication, withdrawal risk, disease recurrence, and patient goals.

Depressed Fracture

A fracture in which broken bone is pushed inward, commonly involving the skull. It may injure underlying brain tissue and requires urgent assessment when associated with neurologic symptoms.

Depression

Depression is a mood disorder involving persistent depressed mood or loss of interest with cognitive, physical, and behavioral symptoms that impair functioning. persistent sadness and loss of interest or pleasure in activities, affecting approximately 7 percent of adults worldwide annually.

Alternate Names:

Clinical Depression (Depression)

Depression (Geriatric)

Mood disorder common in elderly, often underdiagnosed. Symptoms: sad mood, anhedonia, sleep/appetite changes, fatigue, guilt, concentration difficulty, suicidal ideation. May present as cognitive impairment ("pseudodementia"). Risk factors: chronic illness, social isolation, bereavement, medications.

Alternate Names:

Geriatric

Depression (Major Depressive Disorder)

A mood disorder causing persistent depressed mood or loss of interest along with cognitive, physical, and functional symptoms.

Depression - Stopping Your Medicines

Antidepressants should generally be tapered with a clinician when appropriate; abrupt stopping can cause discontinuation symptoms and relapse, while suicidal thoughts require urgent help.

Depression In Children

Depression in children may present as persistent sadness, irritability, loss of interest, withdrawal, school decline, sleep or appetite change, physical

complaints, guilt, or thoughts of death. Assessment considers development, family context, medical causes, safety, and treatment response.

Alternate Names:

Depression In Older Adults

Depression in later life may include sadness, loss of interest, sleep or appetite change, slowed thinking, physical complaints, or cognitive symptoms and should be assessed for medical and medication contributors.

Depression Screening

Major depressive disorder is a common, disabling mood disorder characterized by persistent sadness and loss of interest or pleasure in activities, affecting approximately 7 percent of adults worldwide annually. See depression.

Depression, Postpartum

Alternate index label; see *Postpartum depression*.

Depression, Teen

Alternate index label; see *Teen depression*.

Depressive Disorder

A mood disorder with persistent depressed mood or loss of interest plus cognitive, physical, and functional symptoms; severity, duration, and suicide risk guide care.

Depressive Illness

A disorder characterized by persistent depressed mood or loss of interest with associated cognitive, physical, or functional symptoms; severity and cause determine diagnosis and treatment.

Depth Perception

The visual ability to judge the relative distance and three-dimensional position of objects, produced largely by binocular disparity and supported by motion, perspective, and other visual cues.

Depurination

Loss of a purine base from DNA, leaving an abasic site that can contribute to mutation if not repaired.

DeQuervains Tenosynovitis (De Quervain'S Tenosynovitis)

Alternate index label for De Quervain's Tenosynovitis.

Derbyshire Neck

Derbyshire Neck is an anatomical or physiologic term describing a body structure or function; its clinical meaning depends on its location, role, and associated findings.

Dercum'S Disease

Dercum's Disease is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Derealization

Derealization is a feeling that the external world is unreal, dreamlike, distant, or visually altered, often associated with anxiety, trauma, dissociation, or substances.

Dermabacter

A genus of Gram-positive bacteria commonly found on skin and in the environment. Rare species can cause opportunistic infection, especially in people with underlying illness.

Dermabacter Hominis

A Gram-positive skin-associated bacterial species that rarely causes opportunistic infection, including bloodstream or wound infection.

Dermabrasion

A procedure that mechanically removes superficial layers of skin to improve selected scars, wrinkles, pigment changes, or lesions.

Dermal

Relating to the dermis, the connective-tissue layer

of skin beneath the epidermis.

Dermal Melanocytosis

A usually benign blue-gray patch caused by melanocytes located deeper than usual in the skin, often present at birth and commonly seen over the lower back or buttocks.

Dermal Papilla

A structure at the base of a hair follicle that supplies signals and nutrients important for hair growth.

Dermal Structure

The dermis has two layers: papillary dermis (superficial, thin, containing capillaries, lymphatics, and Meissner corpuscles) and reticular dermis (deep, thick, containing dense irregular connective tissue, collagen, elastin, hair follicles, sweat glands, and sebaceous glands).

Dermatan Sulfate

A sulfated glycosaminoglycan found in connective tissue and involved in extracellular-matrix structure and coagulation-related biology.

Dermatitis

Inflammation of the skin, often used interchangeably with eczema. Atopic dermatitis: chronic pruritic inflammation in flexural areas, associated with asthma and allergies, driven by skin barrier dysfunction and type 2 inflammation. Contact dermatitis: allergic (type IV hypersensitivity) or irritant (direct toxicity).

Dermatitis Herpetiformis

A long-lasting, very itchy skin condition that causes groups of small bumps or blisters, usually on the elbows, knees, buttocks, or scalp. It is linked to celiac disease and sensitivity to gluten.

Dermatitis, Atopic

Alternate index label; see *Atopic dermatitis* (eczema).

Dermatitis, Cercarial

Alternate index label; see *Swimmer's itch*.

Dermatitis, Contact

Alternate index label; see *Contact dermatitis*.

Dermatitis, Scratch

Alternate index label; see *Neurodermatitis*.

Dermatitis, Seborrheic

Alternate index label; see *Seborrheic dermatitis*.

Dermatofibroma

A common benign dermal nodule, typically on the lower extremities of young adults. It presents as a firm, hyperpigmented papule that dimples when pinched (Fitzpatrick sign). It is thought to result from reactive fibrosis following minor trauma or insect bites. Dermatofibromas are asymptomatic and require no treatment.

Dermatofibrosarcoma

Dermatofibrosarcoma protuberans is a locally aggressive fibrohistiocytic tumor of the skin. It typically presents as a slowly enlarging, indurated plaque on the trunk, which may develop nodules over time. It has a high local recurrence rate (up to 50 percent with simple excision) but low metastatic potential.

Dermatofibrosarcoma Protuberans

A rare locally aggressive cutaneous sarcoma arising from dermal fibroblasts. It presents as a slow-growing, firm, reddish-purple plaque or nodule, typically on the trunk. It has a high local recurrence rate but low metastatic potential. Treatment is wide local excision with Mohs micrographic surgery or imatinib for unresectable disease.

Alternate Names:

DFSP

Dermatofibrosarcoma Protuberans Of The Breast

A rare, usually slow-growing cutaneous sarcoma involving breast skin or superficial tissue. It is

locally infiltrative, has a high risk of local recurrence if incompletely removed, and typically shows COL1A1-PDGFB fusion.

Dermatoglyphics

Dermatoglyphics is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Dermatographia

A form of physical urticaria in which scratching or pressure causes raised, red, itchy lines on the skin.

Dermatographia (Dermatographism)

A condition in which light scratching or pressure causes temporary raised red lines or welts on the skin.

Alternate Names:

Skin Writing

Dermatographic Urticaria

A form of hives in which scratching or pressure causes raised, itchy lines on the skin.

Dermatographism

Urticaria factitia: the development of linear wheals (hives) within minutes of stroking or scratching the skin, the most common form of physical urticaria. Mast cell hyper-reactivity releases histamine after minor mechanical pressure. Associated with chronic spontaneous urticaria and atopy.

Dermatologic

Relating to the skin, hair, nails, or the medical specialty of dermatology.

Dermatologic Disorder

A disease affecting skin, hair, nails, or cutaneous glands, producing findings such as rash, scale, pigment change, ulcers, pain, or itching.

Dermatologist

A physician specializing in diagnosis and treatment of disorders of the skin, hair, nails, and related mucous membranes.

Dermatology

The medical specialty concerned with the diagnosis and treatment of diseases of the skin, hair, nails, and mucous membranes.

Dermatome

A surgical instrument used to remove a thin layer of skin for grafting. The term also describes a band of skin supplied by one spinal nerve.

Dermatomycosis

A fungal infection of the skin, hair, or nails.

Dermatomyositis

An autoimmune disease that causes muscle weakness and a characteristic skin rash. It can also affect swallowing or the lungs, and in some adults it is associated with an increased risk of cancer.

Dermatopathology

The study of skin disease through examination of tissue specimens, usually with microscopy and related laboratory methods. It helps identify inflammatory, infectious, benign, and malignant skin disorders.

Dermatophyte

Dermatophyte is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Dermatophytosis

A superficial infection of keratinized skin, hair, or nails caused by dermatophyte fungi, producing conditions such as tinea corporis, tinea pedis, and onychomycosis.

Dermatoplasty

Surgical repair or reconstruction of the skin, often using tissue rearrangement or a skin graft.

Dermatoses - Systemic

Skin manifestations of a disease affecting multiple organs, such as autoimmune, endocrine, infectious, nutritional, or malignant disease; the skin finding may provide a diagnostic clue.

Dermatosis

A disease or disorder of the skin, including inflammatory, infectious, allergic, genetic, and neoplastic conditions.

Dermis

Thick connective tissue layer beneath the epidermis providing structural support, elasticity, and nourishment. Two layers: papillary (superficial, loose connective tissue with dermal papillae interdigitating with epidermal ridges) and reticular (deep, dense irregular connective tissue).

Dermoid

A benign cystic teratoma containing elements derived from ectoderm, including skin, hair follicles, sebaceous glands, and sometimes teeth. Ovarian dermoid cysts are the most common germ cell tumor of the ovary, typically occurring in young women. They are usually asymptomatic but may cause torsion, rupture, or malignant transformation.

Dermoid Cyst

A usually benign cyst lined by tissue resembling skin and sometimes containing hair, glands, or other skin structures.

Dermoid Cyst Of The Ovary

A usually benign mature cystic teratoma arising from germ cells in the ovary and containing tissues such as skin, hair, fat, or teeth.

Dermoscopy

Noninvasive examination of skin lesions with magnification and polarized or nonpolarized illumination that reveals structures not visible to the unaided eye. It improves assessment of pigmented lesions and selected inflammatory or infectious conditions but does not replace biopsy when malignancy is suspected.

Descending Colon

From the splenic flexure to the left iliac fossa, approximately 25-30 cm, and mostly retroperitoneal and fixed to the posterior abdominal wall. The phrenicocolic ligament supports the splenic flexure.

Anteriorly it is related to small intestine; posteriorly to the left kidney. It absorbs water and electrolytes and compacts fecal matter.

Descending Thoracic Aorta

The portion of the aorta extending from the aortic arch to the diaphragm. It lies in the posterior mediastinum and gives branches to the thoracic wall and organs.

Desensitization

A treatment that gradually reduces an abnormal immune response by giving increasing or controlled exposure to a specific allergen.

Desferrioxamine

An iron-chelating medicine used for serious iron overload and, in selected cases, aluminum toxicity. It binds excess metal for removal from the body; hearing, vision, kidney, and infusion-related adverse effects are possible.

Desflurane

An inhaled anesthetic used to maintain general anesthesia. It acts rapidly but can irritate the airway and may cause low blood pressure, nausea, or rare malignant hyperthermia.

Desiccate

To dry out or remove moisture from a tissue, substance, or environment.

Desiccated Thyroid

A thyroid-hormone preparation made from dried animal thyroid tissue and containing thyroid hormones.

Alternate Names:

Desiccated thyroid extract; natural desiccated thyroid

Desipramine

A tricyclic antidepressant that increases norepinephrine signaling and is used for selected depressive disorders or pain conditions. It can cause dry mouth, constipation, sedation, abnormal heart

rhythms, and toxicity in overdose.

Desipramine Hydrochloride Overdose

Overdose of desipramine hydrochloride, a tricyclic antidepressant that may cause coma, seizures, hypotension, and life-threatening cardiac conduction abnormalities.

Deslorelin

A gonadotropin-releasing hormone agonist used mainly in veterinary medicine to suppress reproductive hormone activity. Continuous exposure initially raises and then suppresses gonadotropins and sex hormones.

Desmin

An intermediate-filament protein in muscle cells that helps maintain structural alignment; abnormal desmin or DES variants can cause myopathy and cardiomyopathy.

Desmoid Tumor

A locally aggressive but non-metastasizing fibrous tumor (fibromatosis), associated with familial adenomatous polyposis. Treatment includes surgery, radiation, and systemic therapy; recurrence after resection is common.

Desmoid Tumors

Locally aggressive fibrous tumors that do not usually metastasize but can invade nearby structures and recur; treatment may include observation, medicines, or surgery depending on site and symptoms.

Desmoid Tumour

A locally aggressive fibroblastic tumor that does not usually metastasize but can infiltrate adjacent tissues and recur.

Desmoplasia

Formation of dense fibrous connective tissue in response to invasive neoplasia, chronic inflammation, or tissue injury. The reaction is produced by activated stromal cells and extracellular-matrix deposition surrounding or infiltrating a lesion. Desmopla-

sia is especially common around invasive carcinomas and may make a tumor firm, tethered, or difficult to distinguish from scar tissue. Its presence is interpreted with the cellular morphology, location, and clinical findings to determine whether it represents a tumor-associated stromal response.

Desmoplastic

Describing dense fibrous connective-tissue formation, often as a reaction around a tumor or during chronic inflammation.

Desmoplastic Fibroblastoma

A rare benign collagen-rich fibroblastic tumor, also called collagenous fibroma, presenting as a slow-growing painless mass in superficial or deep soft tissue.

Desmoplastic Fibroma

A rare benign but locally aggressive fibrous tumor of bone.

Desmoplastic Infantile Ganglioglioma

A rare low-grade tumor of infancy containing astrocytic and neuronal components in a desmoplastic stroma, usually presenting as a large superficial cerebral mass with seizures or increased intracranial pressure.

Desmoplastic Melanoma

A rare spindle-cell melanoma with prominent fibrous stroma, usually arising on chronically sun-damaged skin; it may be amelanotic and can be difficult to recognize clinically and histologically.

Desmoplastic Reaction

Formation of dense fibrous connective tissue in response to tissue injury or, commonly, invasion by a carcinoma.

Desmoplastic Small Round Cell Tumors

Rare aggressive tumors that usually arise in the abdomen or pelvis and primarily affect children and young adults.

Alternate Names:

DSRCT

Desmoplastic Trichoepithelioma

A rare benign follicular adnexal tumor, usually a small solitary facial papule or annular lesion, that can resemble morpheaform basal-cell carcinoma and is distinguished by histopathologic examination.

Desmopressin

A synthetic analogue of vasopressin used for central diabetes insipidus, nighttime urination, and some bleeding disorders. Excessive fluid retention can cause dangerous low blood sodium.

Desmosine

A cross-linking amino acid formed from elastin; it is used as a marker of elastin breakdown in research and selected clinical studies.

Desmosome

A strong cell-to-cell junction that connects intermediate filaments between neighboring epithelial or cardiac muscle cells.

Desogestrel

A progestin used in some hormonal contraceptives. It prevents pregnancy mainly by suppressing ovulation and thickening cervical mucus; irregular bleeding and other hormonal adverse effects may occur.

Desonide

A low-potency topical corticosteroid used for short-term treatment of inflammatory skin conditions such as eczema and dermatitis. Prolonged or excessive use can thin the skin.

Desoximetasone

A topical corticosteroid used to reduce inflammation and itching in steroid-responsive dermatitis; potency, site, duration, and skin absorption determine risk.

Desoxycorticosterone

A mineralocorticoid steroid that promotes renal sodium retention and potassium loss, influencing

extracellular fluid volume and blood pressure.

Desquamation

Shedding or peeling of the outer layers of skin or another epithelial surface.

Desquamative Interstitial Pneumonia

A rare ILD characterized by uniform filling of alveoli with macrophages containing brown pigment (hemosiderin-laden). It is strongly associated with smoking. HRCT shows bilateral ground-glass opacities with lower lobe predominance. Smoking cessation and corticosteroids are the main treatments.

Desulfovibrio

A genus of anaerobic sulfate-reducing bacteria found in the environment and gut; invasive infection is uncommon and usually requires compatible clinical findings.

Desulfovibrio Desulfuricans

A sulfate-reducing anaerobic bacterium found mainly in environmental and gastrointestinal habitats; human disease is rare.

Desulfovibrio Piger

A sulfate-reducing anaerobic bacterium associated with the human intestinal microbiota; its clinical significance depends on the specimen and patient context.

Desulfovibrionaceae

A family of anaerobic sulfate-reducing bacteria that includes *Desulfovibrio* species.

Desvenlafaxine Succinate

The succinate salt of desvenlafaxine, a serotonin-norepinephrine reuptake inhibitor used for major depressive disorder. It may cause nausea, insomnia, increased blood pressure, or withdrawal symptoms if stopped suddenly.

Desynchronosis (Jet Lag)

Alternate index label for Jet Lag.

Detached

Detached means separated or no longer connected, as in a detached retina, tissue, device, or emotional state.

Detached Retina

Separation of the neurosensory retina from the underlying retinal pigment epithelium (RPE), a medical emergency requiring prompt surgical repair to prevent permanent vision loss.

Detergent

Detergent is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Detergent Poisoning

Exposure to detergent, which may irritate the skin, eyes, airways, or gastrointestinal tract; product ingredients determine severity.

Deterministic Effects

Tissue reactions caused by radiation that generally occur only above a threshold dose, with severity increasing as the dose rises. Examples include skin injury, cataracts, and tissue damage after substantial exposure.

Detoxification

Removal of toxic substance from body. Methods: activated charcoal, gastric lavage (rarely used), whole bowel irrigation, hemodialysis, hemoperfusion, chelation, supportive care.

Detritic Synovitis

A synovial inflammatory reaction to particulate debris, such as cartilage, bone, blood, crystals, or implant wear material, producing joint swelling and pain.

Detrusor Instability

Involuntary bladder-muscle contractions that can cause urgency and urge incontinence.

Detrusor Muscle

The smooth muscle in the bladder wall that contracts to expel urine.

Detumescence

The return of an erectile tissue to its non-erect state.

Deubiquitinating Enzymes

Enzymes that remove ubiquitin from proteins, thereby regulating protein stability, trafficking, signaling, DNA repair, and immune responses.

Deubiquitination

Removal of ubiquitin from a protein by deubiquitinating enzymes, which can prevent degradation or change the protein's location, activity, or signaling.

Deuteranopia

Red-green color blindness caused by absence or severe dysfunction of green-sensitive cone photopigments.

Deuterium

Deuterium is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Deuterium Oxide

Water in which some or all hydrogen atoms are the heavy isotope deuterium; it is also called heavy water and is used in research.

Development

The progressive process by which an organism, organ, or ability grows, differentiates, and acquires structure or function.

Developmental Coordination Disorder

A childhood neurodevelopmental disorder causing motor skills that are substantially below expectations for age and that interfere with daily activities, school, or coordination.

Developmental Differences Of The

Female Genital Tract

Congenital or developmental abnormalities of the female genital tract that may affect anatomy, menstruation, fertility, pregnancy, or urinary function.

Developmental Disabilities

A group of lifelong conditions beginning during development that affect physical, intellectual, language, learning, or adaptive functioning.

Developmental Disorder

A condition beginning during the developmental period that affects physical, intellectual, language, learning, behavioral, or social functioning.

Developmental Disorders

Developmental Disorders is a developmental, psychological, or behavioral concept; assessment considers age, duration, functional impact, medical causes, and social context.

Developmental Dysplasia Of The Hip

A spectrum of hip abnormalities in infants and children, ranging from mild acetabular dysplasia to complete hip dislocation. The femoral head is partially or completely displaced from the acetabulum due to capsular laxity and acetabular underdevelopment.

Developmental Expressive Language Disorder

A developmental disorder in which a child has difficulty expressing thoughts with spoken or signed language despite otherwise adequate opportunity to learn language.

Developmental Milestone

A skill or behavior generally acquired within an expected age range, such as motor, language, cognitive, or social development; delayed or lost milestones require assessment.

Developmental Milestones

Skills and abilities that children typically acquire at predictable ages. Examples: social smile at 2

months, rolling over at 4-6 months, sitting independently at 6-8 months, walking at 12 months, and speaking two-word phrases at 2 years. Delays in reaching milestones may indicate developmental delay, autism, or neurologic impairment.

Developmental Milestones Record

A record of age-related motor, language, cognitive, social, and self-care skills used to follow a child's development and identify possible delay.

Developmental Milestones Record - 12 Months

Expected developmental skills around one year, including pulling to stand, cruising, pointing, simple gestures, and interactive communication; individual variation exists.

Developmental Milestones Record - 18 Months

Expected developmental skills around 18 months, including walking, using several words, following simple directions, and beginning pretend or imitative play.

Developmental Milestones Record - 2 Months

Expected early-infant skills around two months, including social smiling, visual tracking, responding to sound, and improved head control.

Developmental Milestones Record - 2 Years

Expected skills around two years, including running, using two-word phrases, following two-step directions, and beginning parallel play.

Developmental Milestones Record - 3 Years

Expected skills around three years, including three-word speech, climbing stairs with support, drawing simple shapes, and cooperative social interaction.

Developmental Milestones Record - 4

Months

Expected skills around four months, including steady head control, rolling, reaching for objects, laughing, and responding to caregivers.

Developmental Milestones Record - 4 Years

Expected skills around four years, including understandable sentences, hopping, drawing a person with several parts, and imaginative cooperative play.

Developmental Milestones Record - 5 Years

Expected skills around five years, including running and jumping well, telling a simple story, counting, dressing with limited help, and following game rules.

Developmental Milestones Record - 6 Months

Expected skills around six months, including sitting with support, rolling, reaching and transferring objects, babbling, and recognizing familiar people.

Developmental Milestones Record - 9 Months

Expected skills around nine months, including sitting independently, crawling or moving purposefully, babbling, responding to name, and using a pincer-like grasp.

Developmental Process

The biological and psychological changes through which an organism, organ, or ability grows, differentiates, and acquires function.

Developmental Psychology

The study of cognitive, emotional, social, and behavioral change across the lifespan.

Developmental Reading Disorder

A learning disorder involving persistent difficulty with accurate or fluent reading, often including problems with word recognition, spelling, or decoding.

Developmental Regression

Loss of previously acquired developmental skills. Regression is a neurologic red flag requiring prompt evaluation for genetic, metabolic, neurodegenerative, epileptic, infectious, or structural causes.

Developmental Stage

A period of characteristic physical, cognitive, emotional, or social development used to describe progression over time.

Developmental Surveillance

The ongoing process of eliciting caregiver concerns, observing the child, documenting developmental history, and monitoring milestones during routine care. It differs from formal developmental screening, which uses standardized instruments at specified ages.

Deviated Septum

Displacement of the nasal septum from the midline, congenital or acquired from trauma, and present to some degree in most adults. Significant deviation narrows one nasal cavity, impairing airflow and sinus drainage and producing obstruction, congestion, recurrent sinusitis, epistaxis, and sleep disturbance.

Alternate Names:

Septum Deviated (Deviated Septum)

Deviation

A departure from an expected position, direction, value, or pattern; its meaning depends on the clinical or scientific context.

Device Embolisation

Migration of a medical device or device fragment through the bloodstream, potentially obstructing a vessel and requiring urgent assessment.

Devices For Hearing Loss

Assistive technologies such as hearing aids, cochlear implants, bone-conduction devices, alerting systems, and communication accessories that improve access to sound.

Dexamethasone

A potent, long-acting glucocorticoid used to suppress inflammation, cerebral oedema, allergic reactions, and selected malignancy-related or endocrine conditions; infection, hyperglycaemia, and adrenal suppression may occur.

Dexamethasone Suppression Test

A dynamic test used to assess cortisol regulation. In normal physiology, dexamethasone suppresses ACTH and cortisol; failure of suppression supports endogenous hypercortisolism but requires interpretation with medication, absorption, and comorbidity considerations.

Dexamphetamine

A central nervous system stimulant used in selected attention and sleep disorders. It increases norepinephrine and dopamine signaling and may cause appetite loss, insomnia, increased heart rate, or dependence.

Dexibuprofen

A medicine called dexibuprofen. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Dexketoprofen

The active enantiomer of ketoprofen, a nonsteroidal anti-inflammatory analgesic used for short-term pain; gastrointestinal, kidney, cardiovascular, and hypersensitivity risks apply.

Dexlansoprazole

A medicine called dexlansoprazole. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Dexmedetomidine

An alpha-2 adrenergic agonist used for monitored sedation. It usually causes less respiratory depression than many sedatives but may produce slow heart rate, low blood pressure, or withdrawal after prolonged use.

Dexmethylphenidate

A stimulant used to treat attention-deficit/hyperactivity disorder. It increases central dopamine and norepinephrine signaling and may cause appetite loss, insomnia, increased blood pressure, or misuse.

Dexrazoxane

A medicine used to reduce anthracycline-related heart injury and to treat tissue damage after accidental intravenous anthracycline leakage in selected settings. It can suppress bone marrow and increase infection risk.

Dexrazoxane Hydrochloride

The hydrochloride salt formulation of dexrazoxane, used to reduce selected anthracycline-related heart or tissue injury.

Dextrin

A group of shorter carbohydrate chains formed when starch breaks down, used in food, laboratory, and pharmaceutical products.

Dextroamphetamine

A central nervous system stimulant used for selected attention and sleep disorders. It increases dopamine and norepinephrine signaling and can cause insomnia, appetite loss, cardiovascular effects, or dependence.

Dextrocardia

A congenital position in which the heart lies predominantly on the right side of the chest, sometimes with reversed internal organs or associated heart defects.

Dextromethorphan

A cough suppressant that reduces activity in brain pathways involved in the cough reflex. Excessive doses can cause agitation, hallucinations, seizures, respiratory depression, or serotonin toxicity.

Dextromethorphan Overdose

Overdose of dextromethorphan that may cause

drowsiness, agitation, hallucinations, seizures, respiratory depression, or serotonin toxicity.

Dextroposition Of The Heart

Displacement of the heart toward the right side of the chest, usually because of an external shift such as lung disease or abnormal anatomy; it is distinct from dextrocardia.

Dextropropoxyphene

An opioid analgesic formerly used for pain but withdrawn or restricted in many countries because of cardiotoxicity, respiratory depression, overdose risk, and limited benefit.

Dextrose

D-glucose, a simple sugar used by cells for energy and given medically in oral or intravenous solutions to treat or prevent low blood glucose.

Dextrothyroxine

A stereoisomer of thyroxine formerly studied or used to lower cholesterol. It can affect thyroid function and is not a routine modern treatment for hypercholesterolemia.

Déjà Vu

The feeling that a current situation has been experienced before. It can occur normally but may also occur with temporal-lobe seizures or other neurologic conditions when frequent or associated with other symptoms.

DFSP

Alternate index label; see *Dermatofibrosarcoma protuberans*.

DHA

Docosahexaenoic acid, an omega-3 polyunsaturated fatty acid important in neural and retinal membranes and obtained from seafood or supplements.

DHEA

Dehydroepiandrosterone, an adrenal steroid precursor that can be converted into androgens and

estrogens; its levels vary with age and disease.

DHEA-Sulfate Test

A blood test measuring dehydroepiandrosterone sulfate, an adrenal androgen. It helps evaluate androgen excess, adrenal disorders, congenital adrenal hyperplasia, or an adrenal tumor, but interpretation depends on age, sex, medicines, and the laboratory range.

Diabetes

Diabetes is chronic hyperglycemia caused by inadequate insulin secretion, insulin resistance, or both, with long-term risks to the eyes, kidneys, nerves, heart, and blood vessels. inadequate insulin secretion, impaired insulin action, or both. Chronic hyperglycemia can damage the eyes, kidneys, nerves, heart, and blood vessels. Major forms include type 1, type 2, and gestational diabetes.

Diabetes (Diabetes Mellitus)

Alternate index label for Diabetes Mellitus.

Diabetes - Foot Ulcers

Open sores of the foot associated with diabetes, often involving neuropathy, pressure, deformity, or poor circulation; prompt wound and infection assessment helps prevent amputation.

Diabetes - Insulin Therapy

Treatment with injected or infused insulin to replace deficient insulin or improve glucose control; dosing must be individualized and balanced against hypoglycemia risk.

Diabetes - Preventing Heart Attack And Stroke

Cardiovascular risk reduction in diabetes includes glucose management, blood-pressure and lipid control, smoking cessation, physical activity, nutrition, and indicated medicines.

Diabetes - Taking Care Of Your Feet

A patient-care topic concerning diabetes - taking care of your feet. It addresses the relevant purpose,

risks, management, and follow-up.

Diabetes - When You Are Sick

Sick-day diabetes care includes more frequent glucose checks, hydration, continuation or adjustment of medicines as instructed, and ketone monitoring when indicated; severe symptoms need urgent care.

Diabetes And Alcohol

Alcohol can cause delayed hypoglycemia, interact with medicines, and complicate glucose control; risks depend on the amount, timing, food intake, diabetes type, and treatment.

Diabetes And Exercise

The relationship between physical activity and diabetes management. Exercise can improve glucose control and cardiovascular health, but insulin or medicine adjustments and glucose monitoring may be needed.

Diabetes And Eye Disease

Eye complications of diabetes, especially diabetic retinopathy and macular edema, caused by damage to small retinal blood vessels and potentially leading to vision loss.

Diabetes And Eye Problems

Diabetes can damage retinal blood vessels, causing diabetic retinopathy, macular edema, cataracts, glaucoma, and vision loss. Blood-glucose and blood-pressure control, regular dilated eye examinations, and timely laser, injection, or surgical treatment reduce risk.

Diabetes And Kidney Disease

Kidney damage caused by diabetes, also called diabetic kidney disease. It may begin with albumin in the urine and progress to reduced kidney function.

Diabetes And Metabolic Health

Closely linked through insulin resistance, obesity, dyslipidemia, and hypertension, collectively known as the metabolic syndrome. Poor metabolic health increases the risk of developing type 2 diabetes and

worsens glycemic control in existing diabetes.

Diabetes And Nerve Damage

Diabetic neuropathy is nerve injury associated with chronic metabolic disease and may cause pain, numbness, weakness, autonomic symptoms, or loss of protective sensation.

Diabetes Complications

Diabetes complications arise from chronic hyperglycemia damaging blood vessels and nerves over time. Microvascular complications include diabetic retinopathy leading to vision loss, nephropathy progressing to kidney failure, and neuropathy causing pain and loss of sensation.

Diabetes During Pregnancy (Gestational Diabetes)

Alternate index label for Gestational Diabetes.

Diabetes Eye Care

A patient-care topic concerning diabetes eye care. It addresses the relevant purpose, risks, management, and follow-up.

Diabetes Eye Exams

Eye examinations for diabetes screen for retinopathy, macular edema, cataracts, glaucoma, and other vision-threatening complications.

Diabetes Insipidus

A disorder that causes large amounts of very dilute urine and intense thirst. It happens when the body does not make enough vasopressin, a hormone that helps the kidneys retain water, or when the kidneys do not respond to it.

Diabetes Management

The coordinated use of nutrition, physical activity, medicines when indicated, glucose monitoring, and screening for complications to maintain health and reduce diabetes-related risk.

Diabetes Medications

Diabetes medications lower blood glucose through

various mechanisms: metformin reduces hepatic glucose production; sulfonylureas stimulate insulin secretion; SGLT2 inhibitors increase urinary glucose excretion; GLP-1 receptor agonists enhance incretin effects; and insulin replaces deficient hormone.

Diabetes Mellitus

A metabolic disorder characterized by chronic hyperglycemia resulting from insufficient insulin production, insulin resistance, or both. Type 1 diabetes results from autoimmune destruction of pancreatic beta cells, while type 2 is driven by genetic and lifestyle factors leading to insulin resistance.

Alternate Names:

Diabetes (Diabetes Mellitus)

Diabetes Mellitus Type 1

An autoimmune disease characterized by destruction of pancreatic beta cells leading to absolute insulin deficiency. Usually presents in childhood or young adulthood with acute onset of polyuria, polydipsia, weight loss, and diabetic ketoacidosis (DKA) at diagnosis in many.

Diabetes Mellitus Type 2

A progressive metabolic disorder characterized by insulin resistance and relative insulin deficiency, strongly associated with obesity and physical inactivity. Typically presents in adults but increasingly seen in younger populations. Diagnosis: fasting glucose ≥ 126 mg/dL, HbA1c $\geq 6.5\%$, or OGTT ≥ 200 mg/dL.

Diabetes Pre (Prediabetes)

Alternate index label for Prediabetes.

Diabetes Prevention

Preventing diabetes focuses on reducing the risk of developing type 2 diabetes through lifestyle interventions and risk factor management. Maintaining a healthy body weight, engaging in regular physical activity, and following a balanced diet low in refined sugars and saturated fats are cornerstone strategies.

Diabetes Symptoms In Men

Diabetes in men may cause thirst, frequent urina-

tion, fatigue, blurred vision, slow wound healing, recurrent infections, erectile dysfunction, or loss of sensation. Some people have no symptoms, so risk-based blood testing is important.

Diabetes Symptoms In Women

Diabetes in women may cause thirst, frequent urination, fatigue, blurred vision, recurrent vaginal or urinary infections, slow wound healing, or pregnancy complications. Symptoms can be absent, and screening depends on age, risk factors, and pregnancy status.

Diabetes Treatment

Diabetes treatment combines nutrition, physical activity, glucose monitoring, education, and medicines chosen for the type of diabetes and individual risks. Care also addresses blood pressure, lipids, kidneys, eyes, nerves, feet, vaccines, and cardiovascular health.

Diabetes Type 1 (Type 1 Diabetes)

Alternate index label for Type 1 Diabetes.

Diabetes Type 2 (Type 2 Diabetes)

Alternate index label for Type 2 Diabetes.

Diabetes Type 2 - Meal Planning

Meal planning for type 2 diabetes emphasizes appropriate portions, fiber-rich foods, carbohydrate awareness, adequate protein, and a pattern compatible with medicines, health goals, and preferences.

Diabetes, Gestational

Alternate index label; see *Gestational diabetes*.

Diabetes, Type 1

Alternate index label; see *Type 1 diabetes*.

Diabetes, Type 2

Alternate index label; see *Type 2 diabetes*.

Diabetic Angiopathy

Blood-vessel damage caused by diabetes, affecting small or large vessels.

Diabetic Autonomic Neuropathy

Dysfunction of autonomic nerves caused by diabetes, affecting cardiovascular, gastrointestinal, genitourinary, sudomotor, and pupillary systems. Manifestations include orthostatic hypotension, resting tachycardia, gastroparesis, bladder dysfunction, erectile dysfunction, and impaired hypoglycemia awareness.

Diabetic Cardiomyopathies

Structural or functional heart-muscle disease associated with diabetes, after considering other causes such as coronary artery disease and hypertension.

Diabetic Cardiomyopathy

Damage to the heart muscle associated with diabetes that occurs independently of coronary artery disease or hypertension. It may cause impaired relaxation, reduced pumping function, heart failure, and rhythm problems; glucose and cardiovascular risk control can reduce risk.

Diabetic Coma

A state of profound unconsciousness caused by a severe diabetes-related disturbance in blood glucose or metabolism.

Diabetic Complications

The chronic and acute consequences of diabetes mellitus. Microvascular: retinopathy (leading cause of blindness), nephropathy (leading cause of ESRD—albuminuria, declining GFR), neuropathy (peripheral—stocking-glove, autonomic—gastroparesis, erectile dysfunction, postural hypotension, cardiac autonomic).

Diabetic Diet

An individualized eating plan that considers carbohydrate amount and timing, overall nutrition, cultural preferences, and treatment goals to support blood-glucose control.

Alternate Names:

Diet Diabetes (Diabetic Diet)

Diabetic Diet For Type 2 Diabetes

A type 2 diabetes eating plan emphasizes consistent

carbohydrate awareness, fiber-rich foods, adequate protein, unsaturated fats, and portions that support glucose and weight goals. There is no single diabetic diet; medicines, kidney function, culture, and preferences guide the plan.

Diabetic Foot

Foot problems associated with diabetes, including neuropathy, deformity, ulcers, infection, and impaired healing.

Diabetic Foot Disease

Complex multifactorial condition resulting from the combination of peripheral neuropathy, peripheral arterial disease, and impaired immune function in diabetes. Neuropathy leads to loss of protective sensation and foot deformities. Peripheral arterial disease impairs healing. Minor trauma progresses to ulceration and infection.

Diabetic Foot Ulcer

A chronic wound of the foot in a patient with diabetes, resulting from peripheral neuropathy, peripheral artery disease, and impaired wound healing. Neuropathic ulcers: painless, at pressure points (plantar metatarsal heads, heel), punched-out appearance, surrounded by callus.

Diabetic Home Care And Monitoring

Home diabetes care may include checking glucose, taking medicines, inspecting feet, recognizing low or high glucose, recording results, and attending eye, kidney, dental, and cardiovascular follow-up. Sick-day plans and ketone guidance are important for people at risk of ketoacidosis.

Diabetic Hyperglycemic Hyperosmolar Syndrome

A life-threatening complication of diabetes with very high blood glucose, severe dehydration, and increased blood osmolality, usually without major ketoacidosis.

Diabetic Hyperosmolar Hyperglycemic State

A severe complication of diabetes marked by ex-

treme hyperglycemia, increased serum osmolality, and dehydration with little or no ketoacidosis. It can cause altered consciousness and requires urgent hospital treatment.

Diabetic Hypoglycemia

An abnormally low blood glucose level in a person with diabetes, often caused by insulin or other glucose-lowering medicines.

Diabetic Ketoacidosis

An acute, life-threatening complication of diabetes characterized by excessive ketone production and metabolic acidosis, usually with elevated blood glucose but sometimes with only modest glucose elevation. Triggers include infection, insufficient insulin, myocardial infarction, pancreatitis, certain medicines, fasting, or other physiologic stress.

Alternate Names:

DKA

Diabetic Ketoacidosis Monitoring

Monitoring during DKA management includes hourly blood glucose measurement, serum electrolytes every two to four hours, venous blood gas assessment, beta-hydroxybutyrate levels, and fluid balance.

Diabetic Macular Edema

The accumulation of fluid within the macula due to breakdown of the blood-retinal barrier in diabetic retinopathy, resulting in thickening of the macula and central vision loss, and it can occur at any stage of diabetic retinopathy, from mild non-proliferative to proliferative disease.

Diabetic Microvascular Complications

Small-vessel complications of diabetes, principally retinopathy, nephropathy, and neuropathy, caused by chronic metabolic and vascular injury; risk is reduced by glycemic, blood-pressure, and lipid control.

Diabetic Nephropathy

Kidney disease caused by diabetes mellitus, the leading cause of ESRD worldwide.

Pathophysiology: hyperglycemia-induced glomerular hyperfiltration, basement membrane thickening, mesangial expansion, and glomerulosclerosis. Presents with progressive albuminuria (microalbuminuria macroalbuminuria) and declining GFR.

Alternate Names:

DKD

Diabetic Nephropathy (Kidney Disease)

Kidney damage caused by diabetes, often beginning with protein in the urine and potentially progressing to chronic kidney disease.

Diabetic Neuropathy

Nerve dysfunction caused by diabetes. It may cause distal sensory loss and pain or affect autonomic functions such as digestion, blood pressure, sweating, bladder function, or sexual function.

Alternate Names:

Autonomic Neuropathy Diabetic (Diabetic Neuropathy) Neuropathy, Diabetic Neuropathy Autonomic Diabetic (Diabetic Neuropathy) Neuropathy Diabetic (Diabetic Neuropathy) Neuropathy Diabetic Peripheral (Diabetic Neuropathy) Neuropathy Focal Diabetic (Diabetic Neuropathy) Neuropathy Proximal Diabetic (Diabetic Neuropathy) Peripheral Neuropathy Diabetic (Diabetic Neuropathy) Proximal Neuropathy Diabetic (Diabetic Neuropathy)

Diabetic Peripheral Neuropathy

The most common complication of diabetes mellitus, affecting approximately fifty percent of patients with long-standing disease. It is a length-dependent, sensorimotor polyneuropathy that affects the distal lower extremities first, in a glove-and-stocking distribution, and progresses proximally over time.

Diabetic Retinopathy

A microvascular complication of diabetes mellitus and a leading cause of blindness in working-age adults. Non-proliferative DR (NPDR): microaneurysms, dot-blot hemorrhages, exudates (hard and cotton-wool spots), and venous beading. Proliferative DR (PDR): retinal neovascularization, vitreous hemorrhage, and tractional retinal detachment.

Alternate Names:

Retinopathy, Diabetic

Diabetogenic

Able to cause or promote diabetes or elevated blood glucose, as with some diseases, hormones, medicines, or environmental factors.

Diabetologist

A physician who specializes in the prevention, diagnosis, and treatment of diabetes and its complications, including glucose management, cardiovascular risk, kidney disease, neuropathy, and retinopathy.

Diabrosis

An obsolete or uncommon term whose meaning depends on its source; it may refer to tissue breakdown or ulcerative change and should be interpreted in context.

Diacetoxyscirpenol

A toxic trichothecene mycotoxin produced by some *Fusarium* molds. Contaminated food or feed can cause gastrointestinal irritation, reduced blood-cell production, and immune toxicity.

Diacritics

Marks added to letters to indicate pronunciation or distinguish words, such as accents or umlauts; in medical records they can affect names and data matching.

Diacylglycerol

A lipid second messenger produced from membrane phospholipids that activates protein kinase C and participates in cell signaling.

Diagnosis

The identification of a disease or other health condition through clinical history, examination, and appropriate investigations. A differential diagnosis considers alternative explanations for the findings.

Diagnostic Imaging

A diagnostic test or examination used to assess diagnostic imaging. Results are interpreted with

the clinical findings, method, and reference range.

Diagnostic Laparoscopy

A minimally invasive procedure using a camera inserted through abdominal incisions to inspect pelvic or abdominal organs and sometimes obtain biopsies or perform treatment.

Diagnostic Microbiology

The laboratory identification of microorganisms causing disease. Methods include microscopy, culture, biochemical testing, serology, and molecular methods. Rapid diagnostic tests provide same-day results. MALDI-TOF mass spectrometry revolutionized bacterial identification.

Diagnostic Procedure

A medical procedure, treatment, device, or care topic related to diagnostic procedure; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Diagnostic Radiology

The medical specialty that uses imaging methods to diagnose and characterize disease, including X-ray, ultrasound, computed tomography, magnetic resonance imaging, and nuclear medicine.

Diagnostic Service

A healthcare service that obtains or interprets information to detect, characterize, or monitor a disease or health condition.

Diagnostic Test

A test used to evaluate a person with symptoms, signs, or an abnormal screening result to establish, exclude, or characterize a disease. Its usefulness depends on analytic validity, clinical performance, and the consequences of the result.

Diagnostic Trial

A clinical study evaluating how well a test identifies or excludes a disease or predicts a clinically relevant outcome.

Diagnostic Uncertainty

The state in which available information does not establish a single diagnosis with sufficient confidence. It should be communicated transparently and managed with safety-netting, planned reassessment, and targeted additional evaluation.

Diakinesis

The final stage of prophase I of meiosis, when homologous chromosomes are highly condensed and attached by chiasmata before separating.

Dialectical Behavior Therapy

A comprehensive cognitive-behavioral treatment originally developed for borderline personality disorder. It combines cognitive-behavioral techniques with mindfulness and acceptance strategies. Core modules include mindfulness, distress tolerance, emotion regulation, and interpersonal effectiveness.

Dialister Micraerophilus

A microaerophilic anaerobic bacterium associated with the oral and genital microbiota; clinical significance depends on isolation from a compatible infection.

Dialister Pneumosintes

An anaerobic bacterium found in the oral cavity and respiratory tract that may contribute to polymicrobial infection in appropriate clinical settings.

Dialysis

Kidney replacement therapy that removes excess fluid and dissolved waste products from the body when kidney function is inadequate. The main forms are hemodialysis and peritoneal dialysis.

Dialysis - Hemodialysis

A form of dialysis in which blood is circulated through a dialyzer containing a semipermeable membrane to remove selected solutes and excess fluid; treatment requires vascular access and monitoring.

Dialysis - Peritoneal

A form of dialysis in which the peritoneal membrane

acts as the exchange surface while dialysis solution is instilled into and drained from the abdominal cavity through a catheter; peritonitis, catheter infection, and fluid-management problems require prompt attention.

Dialysis Access

A vascular route used for hemodialysis, such as an arteriovenous fistula, arteriovenous graft, or central venous catheter. Each option has different benefits and risks.

Dialysis Adequacy

The degree to which a dialysis prescription removes uremic solutes and controls fluid, electrolyte, and acid–base balance while meeting patient-centered treatment goals. Kt/V and urea reduction ratio are commonly used for hemodialysis, but interpretation and target values depend on the treatment schedule, modality, and applicable clinical guidance; clinical status remains essential.

Dialysis Disequilibrium Syndrome

A neurologic syndrome that can occur during or soon after hemodialysis, especially when treatment is first started in a severely uremic patient. Headache, nausea, confusion, seizures, and coma may result from rapid solute removal and cerebral edema. Risk is reduced by gradual initiation and shorter initial sessions.

Dialysis Machine

Equipment that circulates blood or dialysis fluid across a semipermeable membrane to remove waste, excess fluid, and electrolytes when kidney function is inadequate.

Dialysis Solutions

Dialysate: specially formulated fluids used in hemodialysis or peritoneal dialysis to exchange solutes and remove excess water from the body. Composition varies by modality and clinical need.

Dialysis Therapy

The use of hemodialysis or peritoneal dialysis to replace some kidney functions by removing waste

and excess fluid and helping manage electrolyte and acid–base disturbances. The prescription is individualized according to kidney function, symptoms, comorbidities, modality, and treatment goals.

Diamine

A molecule containing two amino groups; diamines occur in metabolism and are used in biochemical and industrial chemistry.

Diamorphine

A medicine called diamorphine. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Diamond-Blackfan Anemia

A rare congenital erythroid aplasia characterized by profound hypoplastic, macrocytic anemia, severe reticulocytopenia, and selective absence of erythroid precursors in the bone marrow, typically presenting in the first year of life. Inherited primarily in an autosomal dominant pattern caused by pathogenic variants in ribosomal protein genes (most commonly *RPS19*); roughly half of affected children have associated physical anomalies, such as craniofacial dysmorphism, thumb malformations (triphalaengeal or bifid thumbs), short stature, or congenital heart defects.

Alternate Names:

DBA; Diamond-Blackfan syndrome; Congenital hypoplastic anemia

Dianhydrogalactitol

An investigational alkylating compound studied as an anticancer drug, including for selected brain tumors and hematologic cancers.

Dianisidine

An aromatic chemical used in dyes and laboratory reagents; occupational or significant exposure can cause toxicity and has carcinogenic concern.

Diapause

Diapause is a period of suspended or slowed development in some animals, allowing survival during unfavorable environmental conditions.

Diapedesis

Movement of blood cells, especially leukocytes, through intact vessel walls into surrounding tissue.

Diaper Rash

Irritant inflammation of the skin covered by a diaper, commonly caused by prolonged contact with urine or stool, friction, moisture, or *Candida* infection. It causes redness and soreness; persistent, severe, or spreading rash requires evaluation for infection or another skin condition.

Diaphoresis

Excessive sweating, which may occur with heat, exertion, fever, pain, low blood sugar, or other medical conditions.

Diaphoretic

1. Pertaining to, producing, or characterized by profuse perspiration (diaphoresis), typically mediated by sympathetic nervous system activation of postganglionic muscarinic acetylcholine receptors on eccrine sweat glands. Clinically, acute cold diaphoresis serves as a cardinal autonomic warning sign of acute hemodynamic or metabolic compromise, including acute myocardial infarction, septic or cardiogenic shock, severe hypoglycemia, pheochromocytoma, and acute pulmonary embolism. 2. An agent, medication, or physical modality that induces or stimulates perspiration.

Diaphragm

Dome-shaped musculotendinous partition separating thoracic/abdominal cavities, primary inspiration muscle. Central tendon with muscular periphery from xiphoid, lower six costal cartilages, and lumbar vertebrae via crura and arcuate ligaments. Innervated by phrenic nerves (C3-C5). Contraction flattens the dome increasing thoracic volume.

Diaphragm (Muscle)

A dome-shaped skeletal muscle separating the chest from the abdomen; contraction enlarges the thoracic cavity and drives inhalation.

Diaphragmatic Eventration

Abnormal thinning and elevation of part or all of the diaphragm due to muscle weakness or defective development.

Diaphragmatic Hernia

Protrusion of abdominal contents through a defect or opening in the diaphragm into the thoracic cavity. It may be congenital or acquired after trauma and can cause respiratory compromise, bowel obstruction, or strangulation; diagnosis is based on imaging and management depends on symptoms, contents, and the risk of organ compromise.

Diaphragmatic Rupture

A tear in the diaphragm most commonly caused by blunt or penetrating thoracoabdominal trauma, with the left side affected in approximately 75 percent of cases due to the protective effect of the liver on the right. Herniation of abdominal viscera into the thorax may cause respiratory distress, bowel obstruction, or strangulation.

Diaphysis

The shaft of a long bone, comprising thick cortical (compact) bone surrounding the medullary cavity, which contains bone marrow in children and is largely yellow (fatty) marrow in adults. Enclosed by periosteum; the nutrient artery enters here. Fractures of the diaphysis heal by callus formation (enchondral and intramembranous).

Diarrhea

Diarrhea is passage of loose or watery stools more frequently than usual; duration, blood, fever, dehydration, exposures, and associated symptoms guide evaluation. Medicines, dietary factors, inflammation, malabsorption, or other conditions. Associated symptoms and the need for testing or treatment depend on severity, duration, and the person's clinical features.

Diarrhea IBS (IBS-D Irritable Bowel Syndrome With Diarrhea)

Alternate index label for IBS-D Irritable Bowel Syndrome with Diarrhea.

Diarrhea In Infants

Frequent loose or watery stools in a baby; dehydration, blood, fever, poor feeding, lethargy, or reduced urination require prompt assessment.

Diarrhea Travelers (Traveler'S Diarrhea)

Alternate index label for Traveler's Diarrhea.

Diarrhea, Constipation, Gas, And Bloating

These common gastrointestinal symptoms arise from disruptions in normal digestive function caused by diet, infections, medications, or underlying conditions like irritable bowel syndrome. Diarrhea involves increased stool frequency and water content, while constipation involves infrequent or difficult passage of stool.

Diarrhea, Traveler'S

Alternate index label; see *Traveler's diarrhea*.

Diarrhoea

Passing loose or watery stools more frequently than normal; acute diarrhea is often infectious, while persistent or chronic diarrhea may result from inflammatory, malabsorptive, functional, endocrine, or neoplastic disease.

Diarthrosis

A freely movable joint, usually a synovial joint, in which the bones are separated by a joint cavity.

Diastase

An older term for enzymes that break down starch into smaller sugars; it is now usually discussed as amylase activity.

Diastasis

Diastasis is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Diastasis Recti

Separation and thinning of the linea alba between the two rectus abdominis muscles, commonly oc-

curing during or after pregnancy. It produces a midline abdominal bulge that becomes prominent with raised intra-abdominal pressure.

Diastematomyelia

Diastematomyelia is a congenital spinal-cord malformation in which the cord is split into two portions, often with vertebral or skin abnormalities.

Diastole

The phase of the cardiac cycle when the heart muscle relaxes, allowing the chambers to fill with blood.

Diastolic

Relating to diastole, the phase when the heart muscle relaxes and the chambers fill; diastolic blood pressure is the lower number in a blood-pressure reading.

Diastolic Blood Pressure

The pressure in the arteries while the heart relaxes between contractions.

Diastolic Dysfunction

Impaired relaxation, compliance, or filling of the left or right ventricle during diastole. It can raise filling pressures and cause exertional dyspnoea, pulmonary congestion, or heart failure with preserved ejection fraction; echocardiography assesses relaxation, chamber structure, and estimated filling pressure.

Diastolic Heart Failure

An older or less-preferred term generally referring to heart failure associated with impaired ventricular relaxation, increased stiffness, or elevated filling pressures despite a relatively preserved ejection fraction. The term is often used for heart failure with preserved ejection fraction, but ejection fraction alone does not establish the diagnosis.

Diastolic Murmur

A heart murmur heard while the heart relaxes between contractions.

Diastolic Murmurs

Heart sounds heard during diastole, often caused by abnormal blood flow through a valve or great vessel and requiring clinical evaluation.

Diathermy

The use of high-frequency electrical current to generate heat for surgical purposes, including cutting tissue, coagulating blood vessels to stop bleeding (hemostasis), and destroying abnormal tissue (electrocautery, electrocoagulation).

Diathesis

A predisposition or susceptibility to developing a particular disease, symptom, or physiological response, such as a bleeding diathesis.

Diatom Test

Forensic technique used to support the diagnosis of drowning; microscopic silica-shelled diatoms present in drowning water enter lungs, penetrate alveolar capillaries, and circulate to distant closed organs (femur bone marrow, brain) only if active circulation was present at time of submersion.

Diatomaceous Earth

Diatomaceous earth is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Diatrizoate Meglumine

An iodinated radiographic contrast agent used in selected imaging studies. It can cause allergic-like reactions, kidney injury, or thyroid effects in susceptible people.

Diazepam

A benzodiazepine that enhances GABA-A receptor activity and is used for anxiety, muscle spasm, seizures, and alcohol withdrawal; sedation, respiratory depression, dependence, and withdrawal are important risks.

Diazepam Overdose

Overdose of diazepam, a benzodiazepine, that may cause sedation, impaired coordination, respiratory depression, or coma, especially with other depressants.

Diazinon

An organophosphate insecticide that inhibits acetylcholinesterase. Poisoning can cause sweating, salivation, vomiting, diarrhea, breathing difficulty, muscle weakness, or seizures.

Diazinon Poisoning

Organophosphate poisoning that can cause salivation, sweating, vomiting, diarrhea, bronchospasm, bradycardia, seizures, and respiratory failure.

Diaziquone

An investigational antineoplastic compound that damages DNA and was studied in experimental cancer treatment.

Diazonium Compounds

Chemical compounds containing a diazonium group, used in synthesis, dyes, and some laboratory reactions; certain members may be toxic or unstable.

Diazooxonorleucine

An experimental glutamine antagonist that interferes with nucleotide and amino-acid metabolism and has been studied as an antineoplastic compound.

Diazoxide

A potassium-channel opener used to lower blood pressure in selected settings and to suppress insulin release in some cases of hyperinsulinism; it can cause fluid retention and high blood glucose.

Dibekacin

An aminoglycoside antibiotic related to gentamicin, used in some regions for serious bacterial infections. Kidney and hearing toxicity are important risks.

Dibenzepin

A tricyclic antidepressant used in some countries for depressive disorders; sedation, anticholinergic effects, and cardiac toxicity may occur.

Dibucaine

A medicine called dibucaine. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Dicarboxylic Acid

An organic molecule containing two carboxyl groups; dicarboxylic acids participate in metabolism and can accumulate in selected inherited or acquired metabolic disorders.

Dichloralphenazone

A sedative component formerly used in combination headache medicines; excessive use can cause drowsiness and dependence.

Dichlorophen

Dichlorophen is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Dichlorvos

Dichlorvos is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Dichorionic Diamniotic Twin Placenta

A twin-placentation pattern in which each fetus has its own chorion and amniotic sac. It is usually associated with the lowest placental complication risk among twin pregnancies, although maternal and fetal monitoring remains necessary.

Dichorionic Twins

Twins that each have a separate chorion and amnion. All dizygotic twins are dichorionic, and some monozygotic twins are as well. Each fetus usually has a separate placental circulation, although the placentas may appear fused; complications include preterm birth and growth disturbance.

Dickeya Dadantii

A plant-pathogenic bacterium that causes soft rot and wilt in plants; it is not a usual human pathogen.

Diclofenac

A nonsteroidal anti-inflammatory drug used for pain and inflammation. It can cause gastrointestinal bleeding, kidney injury, cardiovascular events, or liver injury.

Diclofenac Sodium

The sodium salt of diclofenac, a nonsteroidal anti-inflammatory drug used for pain and inflammation.

Diclofenac Sodium Overdose

Overdose of diclofenac, a nonsteroidal anti-inflammatory drug, that may cause gastrointestinal bleeding, kidney injury, seizures, or metabolic abnormalities.

Dicloxacillin

A medicine called dicloxacillin. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

DICOM

Digital Imaging and Communications in Medicine. Standard for medical image exchange.

Dicoumarol

An oral anticoagulant and vitamin K antagonist used historically; it has largely been replaced by safer or more predictable anticoagulants.

Dicrotic

Relating to a second small wave or notch in an arterial pulse, often associated with closure of the aortic valve.

Dicrotophos

Dicrotophos is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Dictation Machine

A device or software used to record spoken information for later transcription into medical, legal, or administrative documents.

Dictyostelium

A genus of social amoebae used in cell and developmental biology research; it is not a usual human pathogen.

Dicumarol

Another name for dicoumarol, an older vitamin K antagonist anticoagulant.

Dicyclomine

An anticholinergic medicine used to relieve intestinal smooth-muscle spasm in selected functional bowel disorders.

Didanosine

A nucleoside reverse-transcriptase inhibitor formerly used in combination treatment for HIV infection; it can cause pancreatitis, neuropathy, and mitochondrial toxicity.

Dieffenbachia Poisoning

Poisoning caused by calcium oxalate crystals in *Dieffenbachia*, producing burning and swelling of the mouth and throat after chewing.

Diencephalon

The part of the brain between the cerebrum and brainstem that includes the thalamus and hypothalamus.

Dienoestrol

A synthetic nonsteroidal estrogen formerly used in selected hormone-related conditions; current use is limited because of estrogen-related risks.

Dientamoeba

Dientamoeba is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Dientamoeba Fragilis

Dientamoeba fragilis is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

DIEP Flap

A deep inferior epigastric perforator flap used for breast or other reconstruction, consisting of skin and fat supplied by perforating vessels from the lower abdomen while preserving most of the rectus muscle.

Diesel Oil

Toxic effects caused by excessive exposure to diesel oil. Severity depends on the dose, route, timing, and clinical findings; suspected poisoning requires prompt assessment.

Diestrus

The phase of the reproductive cycle in many mammals between estrus periods, characterized by activity of the corpus luteum or its regression.

Diet Diabetes (Diabetic Diet)

Alternate index label for Diabetic Diet.

Diet Therapy

Use of an individualized eating plan to prevent, manage, or treat disease, such as diabetes, kidney disease, food allergy, malnutrition, or gastrointestinal disorders.

Diet Ulcerative Colitis (Ulcerative Colitis Diet)

Alternate index label for Ulcerative Colitis Diet.

Dietary Assessment

The systematic evaluation of a person's food and nutrient intake, using methods such as a food diary, 24-hour recall, or food-frequency questionnaire.

Dietary Calcium

Dietary calcium is a healthcare or clinical-care term; its appropriate use depends on the patient's con-

dition, goals, risks, available evidence, and professional assessment.

Dietary Carbohydrate

Carbohydrate consumed in foods and drinks, including sugars, starches, and fiber; digestible carbohydrate affects blood glucose and provides energy.

Dietary Fat

A macronutrient composed mainly of triglycerides and fatty acids that provides approximately 9 kcal per gram. Fat supports cell membranes, energy storage, absorption of vitamins A, D, E, and K, and synthesis of signaling molecules; the type of fat affects cardiometabolic risk.

Dietary Fats

Lipids obtained from food, including saturated, monounsaturated, polyunsaturated, and trans fats; they provide energy and support cell membranes and absorption of fat-soluble vitamins.

Dietary Fiber

Plant material that is not fully digested and supports bowel function and other aspects of health.

Dietary Fibre

A clinical care, prevention, behavioral, or public-health concept concerning dietary fibre; its use depends on the patient, setting, and clinical goal.

Dietary Iron

Iron obtained from food or supplements and required for hemoglobin, myoglobin, and many enzymes; deficiency can cause iron-deficiency anemia, while excess can be toxic.

Dietary Phospholipid

A food-derived amphipathic lipid containing phosphate, used in cell membranes and lipoproteins and metabolized according to diet, absorption, and liver function.

Dietary Phosphorus

Phosphorus obtained from food, needed for bones,

cell membranes, and energy metabolism; people with kidney disease may need to limit absorbable phosphorus.

Dietary Protein

Protein consumed in food and digested into amino acids and peptides needed for tissue maintenance, enzymes, hormones, immunity, and energy when required.

Dietary Reference Intakes

Reference values for nutrient intake: RDA (Recommended Dietary Allowance), AI (Adequate Intake), UL (Tolerable Upper Intake Level), EAR (Estimated Average Requirement).

Dietary Sterol

A sterol present in foods, including cholesterol and plant sterols; intestinal absorption and metabolism influence blood lipid concentrations.

Dietary Supplement

A product taken by mouth that provides dietary ingredients such as vitamins, minerals, herbs, amino acids, or other substances and is intended to supplement the diet.

Dietary Supplementation

Dietary supplementation is deliberate addition of nutrients or other substances to food intake to address deficiency, support a goal, or alter health.

Dietary Supplements

Products such as vitamins, minerals, herbs, or other substances taken to supplement the diet.

Alternate Names:

Nutritional supplements

Dietetics

Dietetics is the applied science and professional practice of using nutrition and diet to promote health and manage disease.

Diethylcarbamazine

An antiparasitic medicine used mainly for lymphatic

filariasis and some other filarial infections. Reactions may result from dying parasites and can include fever, rash, swelling, or inflammation.

Diethylhexylphthalate

Diethylhexyl phthalate is a plasticizer used in some products; exposure can affect endocrine and reproductive systems and is regulated in many settings.

Diethylnitrosamine

Diethylnitrosamine is a chemical carcinogen used in research models and capable of causing liver and other organ injury.

Diethylpropion

A sympathomimetic appetite suppressant used short-term in selected obesity treatment. It may cause insomnia, increased heart rate or blood pressure, and dependence.

Diethylstilbestrol

Diethylstilbestrol is a synthetic estrogen formerly given during pregnancy; prenatal exposure is associated with reproductive-tract abnormalities and certain cancers.

Diethylstilbestrol Dipropionate

An ester formulation of the synthetic estrogen diethylstilbestrol, now largely obsolete because exposure can cause important reproductive and cancer risks.

Dietitian

A dietitian is a nutrition professional who assesses dietary needs and provides individualized nutrition counseling within the scope of local licensing.

Dieulafoy Lesion

Large caliber persistent submucosal artery in the stomach wall that can cause massive gastrointestinal bleeding. The artery is abnormally large and superficial, lacking the normal tapering into smaller arterioles. Presents with hematemesis or melena without a visible ulcer.

Diff Dx

Abbreviation for differential diagnosis, the reasoned comparison of possible causes of a patient's signs, symptoms, or test findings.

Difference Threshold

The smallest difference between two stimuli that a person can reliably detect, also called the just-noticeable difference.

Differences Of Sex Development

Congenital variations in chromosomal, gonadal, hormonal, or internal or external genital development that do not fit typical male or female patterns.

Differential Blood Count

A blood test measuring the proportions and often absolute numbers of different white-cell types, including neutrophils, lymphocytes, monocytes, eosinophils, and basophils.

Differential Diagnosis

The process of distinguishing a particular disease or condition from others that present with similar symptoms. Systematic approach: consider life-threatening conditions first, use pretest probability, apply diagnostic criteria, and order targeted tests.

Differentiated Cancer

A cancer whose cells retain recognizable structural or functional features of the tissue from which they arose; the degree of differentiation helps describe tumor grade.

Differentiation

The process by which cells develop specialized structures and functions. In pathology, the term also describes how closely tumor cells resemble the normal tissue from which they arose.

Differentiation Antigens

Cell-surface molecules whose expression identifies a cell's developmental or tissue lineage; they are used in immunophenotyping and cancer diagnosis.

Difficile Clostridium (Clostridium Difficile Colitis)

Alternate index label for Clostridium Difficile Colitis.

Difficult Airway

A clinical situation in which a trained clinician has difficulty with oxygenation, ventilation, supraglottic airway placement, laryngoscopy, or tracheal intubation. It may result from anatomical features, illness, trauma, or limited equipment and requires an appropriate airway plan.

Difficult Intubation

Difficulty placing an endotracheal tube despite appropriate technique and equipment. Risk may be increased by limited mouth opening or neck movement, abnormal airway anatomy, facial trauma, obesity, or a previous difficult airway. Preparation includes assessment, preoxygenation, and backup airway plans.

Difficult Urination

Trouble starting or maintaining urine flow, painful urination, weak stream, or incomplete emptying; causes include infection, obstruction, neurologic disease, and pelvic or prostate disorders.

Difficulty Speaking

Alternate index label; see *Voice disorders*.

Difficulty Swallowing

Alternate index label; see *Dysphagia*.

Difficulty Swallowing (Swallowing)

Alternate index label for Swallowing.

Difficulty Walking

Impaired gait or mobility that may result from pain, weakness, balance problems, sensory loss, joint disease, neurologic illness, or cardiopulmonary limitation.

Diffuse

Diffuse means spread broadly or throughout an area rather than confined to a small, well-defined focus.

Diffuse Astrocytoma

An infiltrating astrocytic glioma that was formerly classified as a diffuse astrocytoma. Modern diagnosis depends on molecular features such as IDH status and other genetic findings, which guide grading, prognosis, and treatment.

Diffuse Axonal Injury

Widespread injury to nerve fibers caused by acceleration or deceleration of the brain. It may produce prolonged impaired consciousness, and MRI can show characteristic lesions.

Diffuse Esophageal Spasm

A motility disorder characterised by premature, uncoordinated esophageal contractions, causing intermittent chest pain and dysphagia for liquids and solids. Endoscopy excludes structural disease; high-resolution manometry confirms premature contractions.

Diffuse Idiopathic Skeletal Hyperostosis

Diffuse idiopathic skeletal hyperostosis causes flowing calcification and ossification along spinal ligaments and may affect peripheral entheses. It can produce stiffness, pain, swallowing difficulty, or fractures after minor trauma; management is symptom-focused and addresses associated metabolic risks.

Alternate Names:

DISH (Diffuse Idiopathic Skeletal Hyperostosis)

Diffuse Idiopathic Skeletal Hyperostosis

Alternate index label for Diffuse Idiopathic Skeletal Hyperostosis.

Diffuse Large B-Cell Lymphoma

A fast-growing cancer of B lymphocytes and the most common type of non-Hodgkin lymphoma.

Diffuse Midline Glioma

An aggressive brain or spinal-cord tumor arising in midline structures, often in children.

Diffuse Midline Glioma

An aggressive central nervous system tumor arising

in midline brain structures, most often affecting children.

Alternate Names:

DMG

Diffuse Parenchymal Lung Diseases

Alternate index label; see *Interstitial Lung Disease*.

Diffuse Scleroderma

A form of systemic sclerosis with skin thickening extending proximally to the trunk or limbs and a higher risk of internal-organ involvement than limited cutaneous disease. Possible complications include interstitial lung disease, pulmonary hypertension, kidney crisis, and gastrointestinal dysfunction.

Diffusing Capacity

Ability of lungs to transfer gas from alveoli to blood. Reduced in emphysema, ILD, and anemia. Increased in pulmonary hemorrhage and left-to-right shunts.

Diffusing Capacity Of The Lung

The capacity of the respiratory membrane to transfer gas from alveolar air into pulmonary capillary blood. It depends on surface area, membrane thickness, gas properties, hemoglobin, and pulmonary capillary blood volume and is commonly measured with carbon monoxide as DLCO.

Diffusion

The net movement of molecules from an area of higher concentration to lower concentration due to random thermal motion (Brownian motion).

Diffusion Anisotropy

Direction-dependent movement of water molecules in tissue, especially along white-matter fibers; it is measured with diffusion MRI.

Diffusion Capacity

A measure of how efficiently oxygen and other gases pass from the air sacs into the pulmonary capillary blood; commonly reported as DLCO and reduced

in many interstitial and vascular lung disorders.

Diffusion Restriction

Reduced movement of water molecules detected by diffusion-sensitive MRI sequences, usually appearing bright on DWI with corresponding low ADC. It may indicate acute ischemia but can also occur with abscess, hypercellular tumor, seizure, and other processes.

Diffusion-Weighted Imaging

An MRI technique that detects differences in the movement of water molecules within tissue. Restricted diffusion may occur in acute stroke, abscess, hypercellular tumors, and other conditions.

Diflunisal

A salicylate nonsteroidal anti-inflammatory drug used for pain and arthritis; it can cause gastrointestinal, kidney, and cardiovascular adverse effects.

Digastric Muscle

Digastric muscle refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

DiGeorge Syndrome

A disorder usually caused by a deletion of part of chromosome 22 that can affect development of the thymus, parathyroid glands, heart, palate, and immune system. It is also called 22q11.2 deletion syndrome.

Alternate Names:

22q11.

DiGeorge Syndrome (22Q11.2 Deletion Syndrome)

A genetic disorder caused by deletion of part of chromosome 22, potentially affecting the heart, immune system, calcium regulation, palate, and development.

Alternate Names:

Velocardiofacial Syndrome

Digestion

The process by which the body breaks down ingested food into smaller molecules that can be absorbed and utilized. Digestion begins in the mouth (mechanical—mastication; enzymatic—salivary amylase for starch, lingual lipase).

Digestive Diseases

Disorders affecting the gastrointestinal tract, liver, gallbladder, or pancreas, including reflux, ulcers, inflammatory bowel disease, infections, obstruction, and cancers.

Digestive Disorder

A disease affecting the gastrointestinal tract or digestion, causing symptoms such as pain, nausea, reflux, diarrhea, constipation, bleeding, or malabsorption.

Digestive Health

The efficient functioning of the gastrointestinal tract in breaking down food, absorbing nutrients, and eliminating waste. Digestive health depends on factors such as diet, hydration, gut microbiome balance, and the integrity of the digestive mucosa.

Alternate Names:

Gut Health

Digestive System

The organs that digest food, absorb nutrients and water, and eliminate solid waste, including the gastrointestinal tract and accessory organs.

Digestive Tract

The continuous hollow passage from the mouth through the throat, esophagus, stomach, intestines, and anus.

Alternate Names:

Gastrointestinal tract; alimentary canal

Digit

An anatomical terminal extremity of the hand or foot (a finger or toe), numbered 1 through 5 from lateral to medial in the hand (thumb is digit 1) and medial to lateral in the foot (great toe or hallux is digit 1). Skeletal architecture consists of proximal,

middle, and distal phalanges articulated by interphalangeal joints, with the thumb and hallux possessing only two phalanges. Controlled by extrinsic flexor and extensor tendons and intrinsic hand or foot musculature. High-yield clinical pathology includes congenital anomalies (polydactyly, syndactyly), neurovascular lacerations requiring digital nerve blocks, tendon avulsions (mallet finger, jersey finger), closed compartment pulp space infections (felon), and surgical emergencies such as pyogenic flexor tenosynovitis diagnosed by Kanavel four cardinal signs (flexed posture, uniform symmetrical enlargement, tenderness over flexor sheath, and severe pain on passive extension).

Digital Anorectal Exam - DARE

A digital rectal examination in which a gloved, lubricated finger is inserted into the anus to assess the rectum, anal canal, sphincter tone, tenderness, masses, blood, and, when relevant, the prostate.

Digital Clubbing

Bulbous enlargement of distal phalanges with loss of nail-bed angle (> 180 degrees). Associated with lung cancer, ILD, bronchiectasis, and cyanotic heart disease.

Digital Ischemia

Reduced arterial blood flow to the fingers causing pain, pallor, coolness, and potential tissue necrosis. Causes include Raynaud phenomenon, thromboangiitis obliterans, cholesterol emboli, and arterial thrombosis. Severe cases may require amputation of affected digits.

Digital Mammography

Mammography that records x-ray images electronically, allowing image processing and evaluation for masses, asymmetry, architectural distortion, or calcifications.

Digital Mucous Cyst

A benign ganglion cyst arising from the distal interphalangeal joint, presenting as a translucent, dome-shaped nodule on the dorsal aspect of the finger near the nail fold. It is associated with osteoarthritis of

the DIP joint.

Digital Papillary Adenocarcinoma

A rare malignant sweat-gland tumor, usually arising on a finger, hand, or foot as a slow-growing nodule; it can recur locally and metastasize.

Digital Rectal Exam

A physical examination in which a gloved, lubricated finger assesses the anal canal, rectum, and, when relevant, the prostate or nearby pelvic structures. It may detect tenderness, masses, bleeding, or enlargement but does not replace indicated screening or imaging.

Digital Rectal Exam

An examination in which a gloved, lubricated finger is inserted into the rectum to assess the rectal wall, stool, prostate, tenderness, masses, or bleeding.

Digital Rectal Examination

Examination of the rectum and nearby structures using a lubricated, gloved finger.

Digital Subtraction Angiography

An imaging technique that subtracts precontrast images from contrast-enhanced images to show blood vessels more clearly. It can also guide selected vascular procedures.

Digitalis (Digoxin) Toxicity

Nausea, vomiting, visual changes (yellow halos), arrhythmias. Risk: narrow therapeutic index, renal impairment, hypokalemia. Treatment: digoxin immune Fab, cardiac monitoring, correct electrolytes.

Alternate Names:

Digoxin

Digitalis Preparation

A preparation containing cardiac glycosides such as digoxin or digitoxin that can increase cardiac contractility and slow atrioventricular conduction; toxicity can cause arrhythmias and visual or gastrointestinal symptoms.

Digitalis Toxicity

Toxicity from digitalis or related cardiac glycosides, causing gastrointestinal symptoms, visual changes, confusion, bradyarrhythmias, tachyarrhythmias, or hyperkalemia. digitoxin) characterized by a narrow therapeutic index (0.5-2.0 ng/mL for digoxin).

Digitate Warts (Verruca Digitata)

Filiform or finger-like warts, usually arising on the face or neck and caused by HPV infection.

Alternate Names:

Verruca Digitata

Digitonin

A steroidal saponin used mainly in laboratory work to permeabilize cell membranes by binding membrane cholesterol.

Digitoxin

A cardiac glycoside that inhibits sodium-potassium ATPase and increases cardiac contractility; its narrow therapeutic index makes toxicity and drug interactions clinically important.

Digoxin

A cardiac glycoside used in selected heart-failure and atrial-fibrillation patients; it slows atrioventricular conduction and has a narrow therapeutic range with potential for serious toxicity.

Digoxin Immune Fab

Antibody fragments that bind free digoxin and related cardiac glycosides, reducing their tissue activity. Indications include life-threatening dysrhythmia, severe hyperkalemia from acute toxicity, or a potentially massive ingestion.

Digoxin Test

A blood test measuring digoxin concentration to support dosing and evaluation of possible toxicity. Timing in relation to the dose, kidney function, electrolytes, interacting medicines, and symptoms are essential for interpretation; the level alone does not establish toxicity.

Dihydrocodeine

An opioid analgesic used for selected pain or cough indications; sedation, constipation, respiratory depression, dependence, and overdose are important risks.

Dihydroergotamine

An ergot medicine used to treat acute migraine attacks. It constricts certain blood vessels and is contraindicated in several vascular and pregnancy-related conditions.

Dihydrotestosterone

A potent androgen formed from testosterone that contributes to prostate growth and male sexual development.

Dihydroxyacetone

A three-carbon sugar used in metabolism; its derivative dihydroxyacetone phosphate is an intermediate in glycolysis and gluconeogenesis.

Dilantin Overdose

Overdose of phenytoin, also called Dilantin, that may cause nystagmus, ataxia, slurred speech, altered consciousness, hypotension, or cardiac toxicity.

Dilatation

Dilatation is widening or expansion of a hollow organ, vessel, duct, or other structure.

Dilatation And Curettage (D & C)

Dilatation and Curettage (D & C) is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Dilate

To become wider or cause a passage, vessel, or opening to become wider.

Alternate Names:

Expand; enlarge

Dilated Cardiomyopathy

A disease of the heart muscle in which one or more

chambers enlarge and contract poorly, usually causing reduced ejection fraction. Causes include inherited disease, infection, toxins, medicines, pregnancy, and ischemic or inflammatory injury.

Alternate Names:

Cardiomyopathy, Dilated

Dilated Cardiomyopathy Prognosis

Prognosis varies widely and depends on ventricular function, symptoms, cause, arrhythmias, comorbidities, and response to treatment. Guideline-directed therapy can improve cardiac function in some people, while severe disease may lead to heart failure, arrhythmia, or the need for advanced therapies.

Dilation

Dilation is widening or expansion of a hollow organ, vessel, duct, pupil, or other structure, whether normal, induced, or disease-related.

Dilation And Curettage

Surgical dilation of the cervix and curettage (scraping) of the uterine lining. Indications: incomplete or missed abortion (miscarriage management), therapeutic abortion, endometrial sampling (abnormal uterine bleeding, postmenopausal bleeding, endometrial hyperplasia/cancer), and removal of retained products of conception (RPOC).

Dilation And Evacuation

A procedure in which the cervix is dilated and pregnancy tissue is removed from the uterus using suction and surgical instruments.

Dilator

An instrument or device used to gradually widen a body opening or passage.

Diltiazem

A nondihydropyridine calcium-channel blocker used for hypertension, angina, and rate control in some supraventricular arrhythmias; it can cause bradycardia, hypotension, or conduction block.

Dilute Russell Viper Venom Time

A phospholipid-dependent clotting assay used mainly to detect lupus anticoagulant. It is commonly performed as part of laboratory evaluation for antiphospholipid syndrome, with confirmatory testing required because anticoagulants and other factor abnormalities can affect the result.

Dilution

The process of lowering the concentration of a substance by adding a solvent or another fluid. In laboratory work, accurate dilution is essential for preparing standards, reagents, and specimens.

Dimenhydrinate

An antihistamine used to prevent or treat motion sickness and nausea; drowsiness and anticholinergic effects are common concerns.

Dimenhydrinate Overdose

Overdose of dimenhydrinate, an antihistamine that may cause sedation, agitation, anticholinergic effects, seizures, arrhythmias, or respiratory depression.

Dimercaprol

Chelating agent for arsenic, mercury, gold. Given IM. Side effects: hypertension, tachycardia, nausea, pain at injection site.

Dimesna

A thiol-containing drug studied as a protective or detoxifying agent, including in some chemotherapy-related applications.

Dimethoate

An organophosphate insecticide that inhibits acetylcholinesterase and can cause cholinergic poisoning after significant exposure.

Dimethoxon

An organophosphate pesticide metabolite that inhibits acetylcholinesterase and may cause cholinergic poisoning.

Dimethyl Fumarate

An oral medicine used for multiple sclerosis and some skin disease. It can cause flushing, gastrointestinal symptoms, low lymphocyte counts, and rare serious infections.

Dimethyl Sulfoxide

Dimethyl sulfoxide is a defined chemical or biologic substance involved in dimethyl sulfoxide and studied in physiology, pharmacology, laboratory medicine, or treatment.

Dimethyl Sulphoxide

An organic solvent used in selected medical or investigational applications; topical or systemic exposure can cause irritation, odor, and other adverse effects.

Dimethylbenzanthracene

A laboratory chemical used to induce tumors in experimental animals and study carcinogenesis.

Dimethylhydrazines

A group of highly toxic hydrazine compounds used as rocket propellants; exposure can damage the nervous system, liver, and other organs.

Dimethylnitrosamine

A chemical carcinogen that can cause severe liver injury and tumors in experimental models.

Dimetilan

An organophosphate insecticide that inhibits acetylcholinesterase and can cause cholinergic poisoning.

Diminished Ovarian Reserve

A reduction in the quantity and quality of a woman's remaining oocyte pool compared with age-matched peers, resulting in reduced fecundity and suboptimal response to controlled ovarian hyperstimulation during assisted reproductive technology. It is clinically evaluated through serum anti-Müllerian hormone levels, day-3 follicle-stimulating hormone and estradiol, and transvaginal ultrasound assessment of the antral follicle count.

Alternate Names:

Decreased Ovarian Reserve; DOR

Dimocillin

A penicillin-type antibacterial drug used historically or in selected settings against susceptible bacterial infections.

Dimorphic

Existing in two distinct forms, such as a microorganism that has different forms depending on its environment or growth stage.

Dinaciclib

An investigational cyclin-dependent kinase inhibitor studied as an anticancer drug; it can suppress bone marrow and cause other treatment-related toxicities.

Dinitrochlorobenzene

A chemical sensitizer used in immunology research and historically in selected dermatologic treatments; it can cause severe skin irritation and allergic reactions.

Dinitrofluorobenzene

A reactive chemical used to label amino groups in biochemical research; it is irritating and toxic with significant exposure.

Dinophyceae

The class containing dinoflagellates, aquatic microorganisms that can produce toxins or harmful algal blooms affecting human health.

Dinoprost

A prostaglandin F2-alpha medicine used in veterinary medicine and selected reproductive-health applications to stimulate uterine contraction.

Dinoprostone

A prostaglandin E2 medicine used to soften the cervix and induce labor or manage selected pregnancy-related conditions.

Dinoprostone (Cervidil, Prepidil)

A prostaglandin E2 medicine used to soften the

cervix and help initiate labor. It is available in several formulations, and administration requires obstetric monitoring because it can cause excessive uterine activity or fetal effects.

Alternate Names:

Cervidil, Prepidil

Dinucleoside Phosphates

Molecules containing two nucleosides joined by phosphate groups; they occur in nucleotide metabolism and can participate in cellular signaling or energy transfer.

Dinutuximab

A medicine called dinutuximab. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Diocetyl Sodium Sulphosuccinate

A surfactant and stool-softening medicine, also called docusate sodium, used to help soften stool; evidence for constipation benefit is limited.

Diodone

An older iodinated contrast agent formerly used in radiographic imaging.

Dioecious

Having separate male and female individuals in a species, a term used mainly in botany and biology.

Dionin

An older name for ethylmorphine, an opioid formerly used as a cough suppressant or analgesic; it can cause sedation and respiratory depression.

Diopter

A unit of optical power equal to the reciprocal of the focal length in meters.

Dioscorides

A first-century Greek physician and pharmacologist whose work *De materia medica* described medicinal plants and substances and influenced pharmacology for centuries.

Dioxin

Dioxin is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Dipeptide

Dipeptide is a defined chemical or biologic substance involved in dipeptide and studied in physiology, pharmacology, laboratory medicine, or treatment.

Dipeptidyl Peptidase-4 Inhibitors

Oral glucose-lowering medications including sitagliptin, saxagliptin, linagliptin, and alogliptin that inhibit the enzyme degrading incretin hormones GLP-1 and GIP. Resulting increase in active incretin levels enhances glucose-dependent insulin secretion and suppresses glucagon. Neutral effect on weight and low hypoglycemia risk.

Dipetalonema

A genus of filarial nematodes; some species infect animals and rarely humans, with significance depending on species and exposure.

Dipetalonema Infection

Infection with a filarial nematode of the genus *Dipetalonema*, usually transmitted by arthropods and potentially causing skin or subcutaneous disease.

Diphallia

A rare congenital condition in which a person has duplication of the penis, sometimes accompanied by abnormalities of the urinary or digestive tract.

Diphenhydramine Overdose

Overdose of diphenhydramine, an antihistamine that may cause sedation, agitation, anticholinergic effects, seizures, arrhythmias, or respiratory depression.

Diphenoxylate

An opioid-related antidiarrheal medicine, often combined with atropine, that slows intestinal movement; excessive use can cause sedation or respiratory de-

pression.

Diphenylhydramine

An antihistamine with sedating and anticholinergic effects used for allergy symptoms or sleep in some products; it can impair alertness and cause toxicity in overdose.

Diphosphate

The anion or functional group containing two linked phosphate groups; diphosphate is released in reactions such as ATP use and nucleotide synthesis.

Diphosphonate

A chemical group or compound related to phosphonates; some diphosphonates bind bone mineral and are used or studied in bone and calcium disorders.

Diphtheria

Diphtheria is a toxin-producing *Corynebacterium* infection that can form a pharyngeal pseudomembrane and cause airway obstruction, myocarditis, and neuropathy. gram-positive aerobic bacillus that produces a potent exotoxin that inhibits protein synthesis via ADP-ribosylation of elongation factor 2, causing local tissue necrosis and systemic toxicity.

Diphtheria Toxin

Diphtheria toxin is a diagnostic or laboratory investigation; interpretation depends on the specimen, method, reference interval, timing, and clinical question.

Diphyllobothriasis

Intestinal infection caused by fish tapeworms, especially *Diphyllobothrium* or related *Dibothriocephalus* species; it may cause abdominal symptoms or vitamin B12 deficiency.

Diphyllobothrium Latum

The fish tapeworm, a large cestode infecting humans through consumption of raw or undercooked freshwater fish. It can grow up to 10 meters in length and absorbs vitamin B12 in the intestine, potentially causing megaloblastic anemia. Diagnosis

is by identification of eggs or proglottids in stool. Treatment is with praziquantel.

Diplacusis

The perception of a single sound as having two different pitches or qualities, usually reflecting asymmetric or abnormal auditory function.

Diplegia

Paralysis or marked weakness affecting corresponding parts on both sides of the body, classically the legs more than the arms in spastic diplegia.

Diplococcus

A bacterium that characteristically appears as pairs of spherical cells. The term describes shape rather than a single species; some diplococci cause human infection.

Diploid Cell

A cell containing two complete sets of chromosomes, one inherited from each parent; most human somatic cells are diploid.

Diploidy

Having two complete sets of chromosomes, one inherited from each parent; most human body cells are diploid.

Diplopia

Double vision, in which one object is perceived as two. It may result from ocular misalignment, nerve disease, muscle disease, or a brain disorder.

Diplopia (Double Vision)

The perception of two images of a single object, classified by onset, duration, and whether it resolves with covering one eye. Monocular diplopia (persists with one eye covered): caused by ocular surface abnormalities (dry eye, irregular astigmatism, cataract, corneal scar, lens dislocation)—usually benign.

Alternate Names:

Double Vision

Dipole

A pair of equal and opposite electrical or magnetic poles or charges separated by a distance; the term is used in physiology, imaging, and physics.

Dipper

A person whose blood pressure falls normally during sleep, usually by about 10

Diprosopus

A rare congenital craniofacial duplication in which parts or all of the face are duplicated, usually associated with severe developmental abnormalities.

Dipsomania

An outdated term for episodic compulsive alcohol drinking. Modern assessment uses alcohol use disorder and describes the person's pattern and consequences.

Dipsosis

Excessive thirst, which may result from dehydration, increased blood osmolality, diabetes, medication effects, or disorders of thirst regulation.

Dipstick Device

A narrow strip or small device carrying reagent pads that change color when dipped into a specimen, commonly urine, to screen for substances or abnormalities.

Diptera

The insect order containing flies and mosquitoes; some members transmit infections or cause myiasis.

Diphtheroid Bacillus

A term for coryneform, often Gram-positive rod-shaped bacteria resembling *Corynebacterium*. Many are normal skin flora, but some can cause opportunistic infection.

Dipylidium Caninum

A dog and cat tapeworm that can occasionally infect people, especially children, after swallowing an infected flea.

Dipyridamole

An antiplatelet and vasodilator medicine used in selected stroke-prevention regimens and cardiac testing; headache, dizziness, and bleeding may occur.

Direct Action

An effect produced by a substance or intervention through its immediate target rather than through an intermediate mediator or indirect pathway.

Direct Antiglobulin Test

An immunohematologic test that detects immunoglobulin and/or complement bound to the surface of a patient's erythrocytes. A positive result supports in-vivo red-cell coating, as in autoimmune hemolytic anemia, hemolytic transfusion reactions, and hemolytic disease of the fetus and newborn.

Direct Coombs Test

Direct antiglobulin test: detects antibodies or complement (C3) bound to the surface of RBCs in vivo. Positive in: autoimmune hemolytic anemia (warm: IgG, cold: C3), drug-induced hemolytic anemia, hemolytic transfusion reaction, and hemolytic disease of the newborn (HDN). Grading: 1+ to 4+.

Direct Hernia

An inguinal hernia protruding through a weakness in the posterior wall of the inguinal canal, medial to the inferior epigastric vessels.

Direct Oral Anticoagulants

DOACs directly inhibit specific coagulation factors. Direct thrombin inhibitors: dabigatran. Factor Xa inhibitors: rivaroxaban, apixaban, edoxaban, betrixaban. They have predictable pharmacokinetics, fixed dosing, and routine monitoring is not required. They have fewer drug interactions than warfarin.

Direct Transmission

Spread of an infectious organism through immediate contact or short-range respiratory droplets from an infected source.

Direct Veneer

Composite restoration placed chairside for aesthetic improvement. Conservative alternative to laboratory veneer.

Alternate Names:

Indirect Vs. Direct Hernia (Direct Vs Indirect Hernia)

Directly Observed Therapy

Supervision of medication ingestion to ensure adherence. Standard of care for TB treatment. Ensures completion of the full treatment course and reduces development of drug resistance.

Dirofilaria Repens

A filarial nematode transmitted by mosquitoes that causes subcutaneous or ocular dirofilariasis in humans, an accidental dead-end host. It presents as a solitary, migratory subcutaneous nodule, often on the trunk or extremities, or as a conjunctival or periorbital mass.

Dirt - Swallowing

A clinical term concerning dirt - swallowing. Its definition and clinical management depend on the context.

Dirt-Eating

Persistent eating of soil or other nonfood substances, a form of pica that may be associated with iron deficiency, pregnancy, nutritional problems, or mental-health conditions.

Dirty Necrosis

Tumor necrosis containing abundant nuclear debris and neutrophils, classically seen in some high-grade carcinomas such as colorectal adenocarcinoma.

Disability

A limitation in body function or structure, activity, or participation that results from a health condition and interaction with the person's environment.

Disability Evaluation

Assessment of functional limitations for legal, in-

surance, or benefits purposes. Objective measures, standardized tests.

Disability-Adjusted Life Year

A population-health measure combining years of life lost because of premature mortality with years lived with disability. One DALY represents one lost year of healthy life under the chosen methods and disability weights.

Disaccharidase

An intestinal enzyme that hydrolyzes a disaccharide into monosaccharides, such as lactase breaking down lactose.

Disaccharide

A carbohydrate made of two linked simple sugars, such as lactose, sucrose, or maltose, that must be digested before absorption.

Disarticulation

Amputation performed through a joint rather than by cutting through the shaft of a bone.

Disaster Preparedness

Planning and organizing resources, training, and procedures to reduce harm and support effective response and recovery during disasters.

Disc

A thin, round structure; in the spine, an intervertebral disc is a fibrocartilaginous cushion between adjacent vertebral bodies.

Discectomy

Surgical removal of a herniated intervertebral disc fragment compressing a nerve root. Microscopic discectomy (most common): small incision, microscope, minimal muscle dissection.

facility care to the next setting. It includes medication review, follow-up, education, equipment, self-management support, and communication of pending results.

Discitis

Inflammation or infection of an intervertebral disc, often involving adjacent vertebral endplates. It may cause severe localized back or neck pain, stiffness, fever, and reduced mobility; causes include haematogenous infection, postoperative infection, and inflammatory disease, with MRI being the preferred imaging study.

Discography

An imaging procedure in which contrast material is injected into an intervertebral disc to assess disc structure and reproduce pain. Its diagnostic value is limited and its use is selective.

Discoid

Coin-shaped or round, describing the morphology of a lesion or rash; the term may also refer to discoid lupus erythematosus in the appropriate clinical context.

Discoid Lupus

A chronic form of cutaneous lupus causing round, inflamed, scaly plaques that may scar, alter pigmentation, and cause permanent hair loss on the scalp.

Discoid Lupus Erythematosus

A chronic form of cutaneous lupus that causes persistent, inflamed skin plaques that may scar.

Discoloration

Discoloration is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Discolysis

Discolysis is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Discomfort

Discomfort is a medical or biological term describing

a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Discordance

Lack of agreement or correspondence between findings, measurements, genetic results, or the conditions of two related people.

Discordant

Not agreeing, matching, or corresponding; in medicine it may describe conflicting test results or differing conditions in related individuals.

Discordant Couple

A couple in which partners differ in a relevant medical status, such as one partner having HIV infection and the other not having it.

Discriminant Analysis

A statistical method that classifies observations into groups using measured characteristics; it is used in medical research and diagnostic modeling.

Discrimination Learning

Learning to respond differently to two or more stimuli based on their consequences or relevant features.

Disease

Abnormal condition affecting body. Can be infectious, degenerative, neoplastic, etc.

Disease Nomenclature

The system of names and classification terms used to identify, describe, and organize diseases and related clinical conditions.

Disease Progression

The change and usually worsening of a disease over time, assessed through symptoms, examination, biomarkers, imaging, function, complications, or treatment response.

Disease Reporting

Mandatory notification of specified diseases or

health events to public-health authorities.

Disease Reservoir

A person, animal, environmental site, or material in which an infectious agent normally lives and from which it can spread to a susceptible host.

Disease Response

The biologic or clinical reaction to a disease or its treatment, measured by symptoms, signs, laboratory results, imaging, function, or survival.

Disease Surveillance

Ongoing, systematic collection, analysis, interpretation, and reporting of health data to detect, monitor, and respond to disease patterns.

Disease Susceptibility

Increased likelihood of developing a disease because of genetic traits, immune status, age, exposures, behavior, comorbidities, or other risk factors.

Disease Vector

A living carrier, commonly a mosquito, tick, flea, or other arthropod, that transmits an infectious agent between hosts without necessarily becoming ill.

Disease, Osgood-Schlatter

Overuse-related traction injury of the tibial tubercle growth area in growing children or adolescents, causing activity-related pain and swelling below the knee.

Disease, Subclinical

Disease that produces no obvious symptoms or signs but can be detected by laboratory tests, imaging, or other evaluation.

Disease-Free Survival

The length of time after treatment during which a patient remains alive without detectable evidence of the disease. The precise definition depends on the cancer and the study protocol.

Disfigurement

Disfigurement is a visible change that substantially alters appearance, caused by injury, disease, surgery, congenital difference, or scarring.

Disinfectant

A chemical or physical agent used on inanimate surfaces to destroy or inactivate microorganisms; it is not necessarily safe for use on living tissue.

Disinfection

The elimination of most pathogenic microorganisms on inanimate surfaces. Disinfection does not eliminate all forms of microbial life, particularly spores. High-level disinfection eliminates all microorganisms except spores. Intermediate-level disinfection eliminates mycobacteria, most viruses, and bacteria.

Disinfestation

Disinfestation is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Disk

A flat, round structure; in the spine, an intervertebral disk separates adjacent vertebrae and cushions movement.

Alternate Names:

Disc

Disk Diffusion Susceptibility Test

A standardized method in which antimicrobial-impregnated disks are placed on an inoculated agar surface and zones of growth inhibition are measured. Zone diameters are interpreted using validated organism- and drug-specific breakpoints.

Disk Replacement - Lumbar Spine

Surgery replacing a damaged lumbar intervertebral disc with an artificial disc in selected patients; benefits, risks, and suitability require specialist assessment.

Diskectomy

A surgical procedure that removes part or all of a spinal disc to relieve pressure on a spinal nerve or the spinal cord, causing radiculopathy or myelopathy. Indicated in severe disk herniation causing unremitting pain, progressive motor weakness, or cauda equina syndrome.

Diskitis

Inflammation or infection of an intervertebral disc and adjacent vertebral endplates, often causing severe localized back pain and sometimes fever. usually caused by bacterial infection. It causes localized back pain, stiffness, fever, and reluctance to move, particularly in children.

Dislocated Ankle (Ankle Dislocation)

Alternate index label for Ankle Dislocation.

Dislocated Elbow

An elbow dislocation occurs when the joint surfaces lose alignment, usually after a fall or collision. It causes severe pain, deformity, and limited movement and requires urgent reduction with assessment of nerves, blood vessels, fractures, and later stiffness.

Dislocated Hip

A hip dislocation is displacement of the femoral head from the acetabulum, usually after major trauma or around a prosthetic joint. It is an emergency because blood supply to the femoral head and nearby nerves can be compromised.

Dislocated Knee

A knee dislocation is loss of alignment between the femur and tibia, often after high-energy trauma. Artery and nerve injury may be occult, so urgent reduction and vascular assessment are required even when the joint spontaneously relocates.

Dislocated Shoulder

Displacement of the upper arm bone from the shoulder joint.

Dislocation

Complete loss of normal contact between the surfaces of a joint, usually caused by trauma or structural disease and potentially accompanied by fracture, nerve, or blood-vessel injury.

Dislocation Of The Lens

Partial or complete displacement of the crystalline lens from its normal position within the pupillary axis due to disruption or weakness of the zonular fibers. Causes include Marfan syndrome, homocystinuria, trauma, and prior intraocular surgery; it may cause acute angle-closure glaucoma or blurred vision.

Disofenin Scan

Hepatobiliary imaging using Tc-99m disofenin. Similar to mebrofenin scan.

Disopyramide

A class IA antiarrhythmic drug used selectively for certain rhythm disorders; anticholinergic effects, negative inotropy, and proarrhythmia limit its use.

Disorder

A disturbance of normal structure or function that causes symptoms, impairment, or clinically important risk. The term does not necessarily indicate a particular level of severity.

Disorder Antisocial Personality (Antisocial Personality Disorder)

Alternate index label for Antisocial Personality Disorder.

Disorder Asperger (Asperger Syndrome)

Alternate index label for Asperger Syndrome.

Disordered Thinking

A disturbance in the organization, coherence, logic, or flow of thought that may occur with psychiatric illness, neurologic disease, substances, delirium, or other medical conditions.

Disorders Sleep (Sleep)

Alternate index label for Sleep.

Disorganized Schizophrenia

Disorganized Schizophrenia is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Disorientation

Difficulty knowing one's time, place, situation, or identity. It may occur with delirium, infection, metabolic disturbances, intoxication or withdrawal, brain injury, or dementia. Sudden disorientation is an urgent medical symptom.

Dispensary

A clinical care, prevention, behavioral, or public-health concept concerning dispensary; its use depends on the patient, setting, and clinical goal.

Dispensed Amount

The quantity of a medicine supplied to a patient, determined by the prescription, dose, duration, formulation, and applicable dispensing rules.

Dispensing

A clinical care, prevention, behavioral, or public-health concept concerning dispensing; its use depends on the patient, setting, and clinical goal.

Dispersal

Dispersal is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Displacement

Displacement is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Disruption

Disruption is interruption or breakdown of normal structure, function, continuity, or process.

Dissect

To dissect is to separate tissue layers or structures by cutting or careful blunt separation for examination, surgery, or anatomical study.

Dissecting Aneurysm

Alternate index label; see *Aortic dissection*.

Dissection

A tear in the inner layer of a blood vessel that allows blood to track between the vessel-wall layers and create a false channel. In the aorta, this is called aortic dissection.

Dissection Aorta (Aortic Dissection)

Alternate index label for Aortic Dissection.

Disseminate

To disseminate means to spread widely, as an infection, cancer, inflammation, or information.

Disseminated Coccidioidomycosis

Coccidioides infection that spreads beyond the lungs to skin, bone, meninges, or other organs.

Disseminated Cutaneous Blastomycosis

Blastomyces infection involving multiple areas of skin, usually after spread from a pulmonary infection.

Disseminated Intravascular Coagulation

A pathological state of simultaneous systemic activation of coagulation and fibrinolysis, leading to widespread microvascular thrombosis that causes organ ischemia and dysfunction, paradoxically accompanied by consumptive coagulopathy and increased bleeding risk.

Disseminated Intravascular Coagulation

Disseminated intravascular coagulation is uncontrolled systemic coagulation activation with consumption of platelets and clotting factors, producing simultaneous thrombosis and bleeding.

Disseminated Mycobacterium

Avium–Intracellulare Infection

Widespread infection with *Mycobacterium avium* complex, usually occurring in people with severe immune suppression.

Disseminated Neuroblastoma

Neuroblastoma that has spread from its primary site to distant organs, bone marrow, bone, liver, or lymph nodes; it requires risk-based specialist pediatric oncology treatment.

Disseminated Sclerosis

Disseminated Sclerosis is an anatomical structure, body position, physical finding, or physiological process; its clinical meaning depends on the examination and surrounding findings.

Disseminated Tuberculosis

Tuberculosis that has spread through the bloodstream or lymphatic system to multiple organs, potentially causing fever, weight loss, organ dysfunction, meningitis, or widespread small lesions.

Disseminated Varicella Infection

Varicella-zoster virus infection that spreads beyond the skin and may involve the lungs, brain, liver, or other organs.

Dissociation

A disruption in the normal integration of consciousness, memory, identity, emotion, perception, behavior, and sense of self. It exists on a spectrum from normal (daydreaming) to pathological (DID, depersonalization-derealization).

Dissociative Amnesia

Inability to recall important autobiographical information, usually of a traumatic or stressful nature, that is inconsistent with ordinary forgetting. The amnesia may be localized, selective, generalized, or continuous.

Dissociative Disorder

DSM-5 category comprising disruption of the normally integrated functions of consciousness, memory,

identity, perception, and behaviour, including dissociative amnesia, dissociative identity disorder, and depersonalization-derealization disorder. Strongly associated with severe early-life trauma.

Dissociative Disorders

Mental health disorders involving disruption in the normal integration of identity, memory, consciousness, emotion, perception, or behavior.

Dissociative Identity Disorder

The presence of two or more distinct personality states or an experience of possession involving marked discontinuity in sense of self, agency, affect, behavior, memory, perception, cognition, and/or sensory-motor functioning.

Distal

Farther from the point of attachment or origin of a structure, or farther from the centre of the body; the opposite directional term is proximal.

Distal Median Nerve Dysfunction

Impaired function of the median nerve near the forearm, wrist, or hand, causing sensory changes in its distribution and weakness of supplied muscles; carpal tunnel syndrome is a common compressive cause.

Distal Pancreatectomy

Surgical removal of the body or tail of the pancreas, usually with preservation of the pancreatic head.

Distal Radius Fractures

The most common upper extremity fracture. Classification: AO/OTA: A (extra-articular), B (partial articular), C (complete articular); Frykman. Colles fracture: dorsal angulation/displacement (fall on outstretched hand). Smith fracture: volar angulation (fall on flexed wrist).

Distal Renal Tubular Acidosis

A disorder in which the distal nephron cannot excrete acid effectively, causing non-anion-gap metabolic acidosis and sometimes low potassium,

kidney stones, or bone disease.

Distal Splenorenal Shunt

A surgical connection between the splenic vein and left renal vein that selectively decompresses esophageal varices while preserving some portal flow.

Distal Tubule

Segment of the nephron extending from the thick ascending limb of the loop of Henle to the collecting duct. The early distal tubule reabsorbs sodium and chloride through the NCC cotransporter, which is the target of thiazide diuretics.

Distamycins

A medicine called distamycins. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Distant

Distant means far from a reference point or primary site; in cancer, distant disease indicates spread to remote organs or lymph nodes.

Distemper

A serious viral disease of animals, especially dogs, caused by canine distemper virus; it can affect respiratory, gastrointestinal, and nervous systems.

Distended

Distended means abnormally stretched, swollen, or enlarged by fluid, gas, blood, tissue, or pressure.

Distension

Distension is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Distention

Enlargement or stretching of an organ, cavity, or body part, as in abdominal distention from gas, fluid, obstruction, or swelling.

Distichia

Distichia is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Distilling

Distilling is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Distortion

Distortion is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Distraction Osteogenesis

Surgical bone lengthening by gradual distraction. Used for jaw deformities, trauma.

Distress

Distress is significant physical or emotional suffering, discomfort, or strain that may accompany illness, injury, treatment, or difficult circumstances.

Distribution

The anatomical pattern and sites affected by a skin lesion or rash, including whether involvement is localized, generalized, symmetrical, flexural, extensor, dermatomal, or otherwise characteristic.

Distributive Shock

Shock caused by pathologic vasodilation and maldistribution of blood flow, producing inadequate effective circulating volume despite variable cardiac output. Causes include sepsis, anaphylaxis, neurogenic shock, and some toxic syndromes.

District Nurse

A clinical care, prevention, behavioral, or public-health concept concerning district nurse; its use depends on the patient, setting, and clinical goal.

Disulfaram-Like Reaction

An unpleasant reaction after alcohol is consumed with disulfiram or another interacting substance, potentially causing flushing, headache, nausea, vomiting, low blood pressure, or palpitations.

Disulfoton

Disulfoton is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Diterpenes

A class of compounds built from four isoprene units; they occur in plants, fungi, and some animal products and have diverse biological effects.

Dithiolethione

A sulfur-containing compound studied for its ability to activate cellular protective and detoxification pathways.

Ditiocarb

A chemical term for dithiocarbamate compounds, which can act as chelators or pesticides and may have toxic effects depending on the agent and exposure.

Diuresis

Increased urine production. Can be physiological (fluid overload) or pharmacological (diuretics). Osmotic diuresis (hyperglycemia), water diuresis (DI), saline diuresis (hypertonic saline).

Diurnal

Relating to daytime or occurring within a 24-hour daily cycle.

Diutina Rugosa

A yeast species that is an uncommon opportunistic human pathogen, mainly reported in people with underlying illness or immunosuppression.

Diverticula (Diverticulosis)

Alternate index label; see *Diverticulosis*.

Diverticular Disease

A condition in which colon diverticula cause symptoms, bleeding, inflammation, or complications. It includes symptomatic diverticulosis and diverticulitis.

Diverticulitis

Inflammation of one or more diverticula, usually in the colon. It may cause abdominal pain, fever, changes in bowel habits, or complications such as an abscess.

Diverticulosis

The presence of small pouches, called diverticula, in the wall of the colon. It often causes no symptoms but may lead to bleeding or diverticular disease.

Alternate Names:

Colonic Diverticulitis (Diverticulosis) Diverticula (Diverticulosis)

Diverticulosis And Diverticulitis

Diverticulosis is a condition in which small pouches (diverticula) form in the wall of the digestive tract, most commonly in the colon. Diverticulitis occurs when these pouches become inflamed, causing abdominal pain, fever, and changes in bowel habits.

Diverticulum

A pouch-like projection from the wall of a hollow organ, especially the intestine.

Divisum Pancreas (Pancreas Divisum)

Alternate index label for Pancreas Divisum.

Divorce

Divorce is legal dissolution of a marriage and can affect health through stress, finances, housing, caregiving, social support, and access to care.

Dix-Hallpike Maneuver

The Dix-Hallpike maneuver is a diagnostic test used to identify benign paroxysmal positional vertigo by detecting the characteristic nystagmus and vertigo that occur when displaced otoconia within the posterior semicircular canal move in response to gravity

during specific head positioning.

Dizygotic Twin

One of a pair of fraternal twins formed when two ova are fertilized by two sperm; the twins share about half their segregating genes on average.

Dizygotic Twins

Dizygotic Twins is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Dizziness

Dizziness is a subjective sensation that may represent vertigo, lightheadedness, imbalance, or nonspecific disequilibrium; history and examination distinguish vestibular, cardiovascular, neurologic, and other causes. include disorders of the vestibular system, cardiovascular system, nervous system, vision, medications, or metabolism.

Dizziness (Dizzy)

Alternate index label for Dizzy.

Dizziness: Causes, Symptoms, And Treatment

Dizziness may mean vertigo, lightheadedness, imbalance, or a floating sensation. Causes include inner-ear disease, dehydration, medication, anemia, heart rhythm problems, migraine, anxiety, or neurologic disease; treatment depends on the pattern and examination.

Dizziness: When To See A Doctor?

Dizziness needs urgent assessment when it begins suddenly or occurs with weakness, numbness, trouble speaking, severe headache, chest pain, fainting, inability to walk, persistent vomiting, or new hearing loss. Recurrent or unexplained dizziness also merits evaluation.

DKD

Alternate index label; see *Diabetic Nephropathy*.

DLI

Donor lymphocyte infusion, administration of lymphocytes from a stem-cell donor after transplantation to enhance graft-versus-leukemia effects or treat relapse; it can cause graft-versus-host disease.

DM

Most commonly diabetes mellitus, a group of disorders with chronically elevated blood glucose; in other contexts it may have another expansion.

DM2

Type 2 diabetes mellitus, a disorder characterized by insulin resistance and progressive impairment of insulin secretion.

DMFT Index

An epidemiologic index recording the number of permanent teeth that are decayed, missing because of caries, or filled. It estimates cumulative caries experience but does not measure current lesion activity or distinguish all causes of tooth loss.

DMG

Alternate index label; see *Diffuse midline glioma*.

DMSA Scan

A nuclear-medicine study using technetium-labeled dimercaptosuccinic acid to show renal cortical function and areas of cortical injury or scarring.

DNA

Deoxyribonucleic acid, the hereditary material in humans and most organisms. Double helix of two antiparallel strands (sugar-phosphate backbone, nitrogenous bases: A-T, G-C). Human genome: 3 billion base pairs, 20,000 protein-coding genes.

DNA Adduct

A chemical group covalently attached to DNA after exposure to a reactive compound; unrepaired adducts can distort replication and create mutations or cancer risk.

DNA Alkylation

Addition of an alkyl group to DNA bases or the phosphate backbone, which can alter base pairing, block replication, and cause mutations or cell death.

DNA Amplification

Production of many copies of a DNA sequence, naturally or through laboratory methods such as polymerase chain reaction, for detection, analysis, or research.

DNA Binding

The attachment of a protein or other molecule to a DNA sequence, influencing gene expression, replication, repair, or chromosome organization.

DNA Damage

A chemical or structural change in DNA that can interfere with gene function or replication.

DNA Demethylation

Removal or loss of methyl groups from DNA, changing chromatin accessibility and gene expression during development, reprogramming, or disease.

DNA Double Helix

The molecular structure of DNA described by Watson and Crick in 1953, consisting of two antiparallel polynucleotide chains wound around each other in a right-handed helix. The sugar-phosphate backbone is on the exterior; nitrogenous bases project inward.

DNA Fingerprinting

Analysis of variable DNA regions to distinguish individuals or establish biological relationships, used in forensic identification, paternity testing, and some research applications.

DNA Footprinting

A molecular method that identifies the DNA sequence protected by a bound protein because that region is less susceptible to enzymatic or chemical cleavage.

DNA Fragmentation

Breaking of DNA strands into smaller pieces, occurring during apoptosis, injury, radiation, or laboratory analysis; excessive or abnormal fragmentation can impair genome integrity.

DNA Gyrase

A bacterial type II topoisomerase that introduces negative supercoils and relieves DNA torsional stress; it is a target of fluoroquinolone antibiotics.

DNA Integration

Insertion of foreign or copied DNA into a host genome, occurring in evolution, gene therapy, or viral infection and potentially altering nearby gene activity.

DNA Ligases

Enzymes that join breaks in the sugar-phosphate backbone of DNA during replication and repair.

DNA Ligation

Joining of DNA fragments by forming a phosphodiester bond, catalyzed by DNA ligase during replication, repair, recombination, and molecular cloning.

DNA Methylation

Addition of methyl groups to DNA, usually at cytosine bases, that can alter chromatin and gene expression without changing the DNA sequence.

DNA Packaging

The organization of DNA around histone proteins and into higher-order chromatin so that it fits inside the nucleus and can be regulated.

DNA Primase

An enzyme that makes short RNA primers needed to start DNA synthesis during replication.

DNA Primers

Short nucleic-acid sequences that provide a starting point for DNA synthesis in cells or laboratory amplification.

DNA Probe

A labeled single-stranded nucleic-acid sequence that binds a complementary target and allows a specific gene, organism, mutation, or chromosome region to be detected.

DNA Profiling

Analysis of polymorphic DNA markers, commonly short tandem repeats, to compare biological samples for human identification or kinship assessment. Results are statistical associations and require validated laboratory procedures, appropriate controls, and documented sample integrity.

DNA Repair

The cellular processes that detect and correct damage or copying errors in DNA. Major pathways include base-excision, nucleotide-excision, mismatch, and double-strand-break repair.

DNA Repair Mechanisms

Cellular processes that correct DNA damage caused by replication errors, chemicals, radiation, and reactive oxygen species. Mismatch repair (MMR): corrects base-base mismatches and insertion-deletion loops that escape proofreading during replication.

DNA Replication

The semi-conservative process by which DNA duplicates before cell division, producing two identical daughter molecules each containing one original and one new strand.

DNA Sequencing

Determination of the nucleotide order in DNA. Sanger sequencing uses chain termination and is useful for selected small targets. Next-generation sequencing (NGS) uses massively parallel sequencing, whole genome, whole exome, targeted panels.

DNA Virus

DNA virus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Do Not Resuscitate

DNR: order not to perform CPR. Can be combined with DNI (do not intubate).

Do You Have A Drinking Problem?

A screening question about alcohol use, control, cravings, consequences, and withdrawal that can identify possible alcohol use disorder and prompt confidential assessment.

Do-Not-Resuscitate Order

A medical order instructing clinicians not to perform CPR if a person's breathing or heartbeat stops.

Alternate Names:

DNR order

DOACs

Direct oral anticoagulants are a class of anticoagulant drugs that directly inhibit specific coagulation factors, providing an alternative to warfarin with more predictable pharmacokinetics and no routine monitoring.

DOB

Date of birth, the calendar date on which a person was born.

Dobutamine

An intravenous beta-1-adrenergic agonist that increases cardiac contractility and is used short term for selected cases of acute low-output heart failure or shock.

Dobutamine Stress Echocardiography

Pharmacological stress test using dobutamine (beta-1 agonist) to increase heart rate, contractility, and myocardial oxygen demand. New or worsening wall motion abnormalities indicate ischemia. Preferred when patients cannot exercise. Beta-blockers should be held before testing.

Doc

Informal shorthand for doctor, a person with a professional qualification to practice medicine or another health profession.

Docetaxel

A medicine or chemical compound called docetaxel. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Doconexent

A pharmaceutical name for docosahexaenoic acid, an omega-3 fatty acid found in fish oils and cellular membranes.

Docosadienoic Acid

A 22-carbon fatty acid with two double bonds, present in some lipids and studied for effects on membranes and metabolism.

Docosahexaenoic Acid

The omega-3 fatty acid DHA, a major structural lipid in brain and retina obtained from marine foods or synthesized from precursors.

Docosapentaenoic Acid

A 22-carbon polyunsaturated fatty acid with five double bonds, occurring as omega-3 or omega-6 forms in tissues and diet.

Docosatrienoic Acid

A 22-carbon fatty acid containing three double bonds, present in small amounts in lipids and involved in experimental studies of metabolism.

Doctor

A person who has completed an appropriate professional program and is licensed or authorized to practice medicine; the title may also be used for other doctoral-level professionals.

Doctor-Patient Relationship

A clinical care, prevention, behavioral, or public-health concept concerning doctor-patient relationship; its use depends on the patient, setting, and clinical goal.

Doctors' Symbol

The Rod of Asclepius, a single snake around a staff, is the traditional symbol of medicine; it is often

confused with the two-snake caduceus.

Docusate

A stool softener (surfactant laxative) that increases water and fat incorporation into stool by lowering surface tension, facilitating defecation in patients who should avoid straining (post-MI, post-surgery, hemorrhoids) and for opioid-related constipation. Onset 12-72 hours.

Dodecenoic Acid

An unsaturated 12-carbon fatty acid whose biologic properties depend on the position and configuration of its double bond.

Dog Bite

An injury caused by a dog's teeth. Care includes prompt wound cleansing, assessment for tissue damage and infection, tetanus review, and evaluation of rabies exposure when relevant.

Alternate Names:

Dog bite injury

Dog Bite Treatment

Dog-bite care includes thorough wound cleaning, assessment of tissue and tendon injury, tetanus review, rabies-risk evaluation, and antibiotics when indicated. Deep punctures, bites to the hand or face, uncontrolled bleeding, infection, or immune compromise require prompt medical care.

Alternate Names:

Bite Dog (Dog Bite Treatment)

Dol

A historical unit intended to quantify subjective pain intensity; modern clinical practice uses validated pain scales rather than the dol.

Dolasetron

A medicine or chemical compound called dolasetron. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Dolasetron Mesylate

The mesylate salt of dolasetron, a serotonin 5-HT₃

receptor antagonist used to prevent nausea and vomiting in selected settings; it can affect cardiac rhythm.

Dolichocephalic

Dolichocephalic is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Dolichol

A long-chain lipid alcohol involved in assembling sugar chains for protein glycosylation in the endoplasmic reticulum.

Dolor

Latin for pain; it is one of the classic signs of inflammation.

Dolosigranulum Pigrum

A Gram-positive bacterium commonly found in the upper respiratory tract; it is usually a commensal, though rare infections have been reported.

Domain

A defined region or area of function; in biology, a protein domain is a structural or functional part of a protein, while in genetics it may mean a genomic or regulatory region.

Domestic Violence

A pattern of coercive control and physical, sexual, psychological, or economic abuse by one intimate partner or household member against another. Forensic evaluation documents patterned injuries, defensive wounds, strangulation signs, burns, and injuries of differing ages.

Dominant

Allele whose phenotype is expressed in heterozygotes. Autosomal dominant disorders: Huntington disease, marfan syndrome, achondroplasia. Requires only one copy for expression.

Domperidone

A peripheral dopamine-receptor antagonist used in some countries for nausea or gastrointestinal motility disorders; cardiac rhythm risks limit its use.

Donath-Landsteiner Test

A laboratory test that detects the biphasic hemolysin antibody associated with paroxysmal cold hemoglobinuria. The antibody binds red cells in the cold and causes complement-mediated hemolysis on warming.

Donepezil

A reversible acetylcholinesterase inhibitor used to treat symptoms of Alzheimer disease; it may cause gastrointestinal effects, bradycardia, syncope, or sleep disturbance.

Donnan Equilibrium

Donnan Equilibrium is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Donohue Syndrome

A severe inherited insulin-resistance disorder caused by pathogenic INSR variants, characterized by poor growth, loss of body fat, abnormal facial features, and early metabolic complications.

Donor

A clinical care, prevention, behavioral, or public-health concept concerning donor; its use depends on the patient, setting, and clinical goal.

Donor Insemination

A fertility treatment in which sperm from a donor (anonymous or known) is placed into a woman's vagina, cervix, or uterus to achieve pregnancy. Indications: male factor infertility (azoospermia, severe oligospermia), genetic disorders in the male partner, single women, and same-sex female couples.

Donor Suitability Assessment

Comprehensive evaluation of a potential organ

donor including medical history, laboratory testing, and organ function assessment to determine transplant eligibility.

Donor Testing

Laboratory and clinical evaluation of potential organ donors for infectious diseases, malignancy, and organ function.

Donovanosis

A chronic, progressive, ulcerative genital infection caused by the intracellular gram-negative bacterium *Klebsiella granulomatis* (formerly *Calymmatobacterium granulomatis*). Also known as granuloma inguinale.

Alternate Names:

granuloma inguinale.

Donovanosis (Granuloma Inguinale)

A sexually transmitted infection caused by *Klebsiella granulomatis* that produces slowly progressive, usually painless genital ulcers; diagnosis and antibiotic treatment require clinical evaluation.

Dopa

A precursor of dopamine; levodopa crosses the blood-brain barrier and is used with an aromatic L-amino-acid decarboxylase inhibitor to treat Parkinson disease.

Dopa-Responsive Dystonia

An inherited movement disorder with dystonia and sometimes parkinsonism that improves markedly with low-dose levodopa.

Dopamine

A neurotransmitter involved in movement, motivation, reward, attention, and regulation of several hormonal functions.

Dopamine Agonist

A class of medications that stimulate dopamine receptors (D1-D5), mimicking the effects of endogenous dopamine.

Dopamine Agonists

Medicines that stimulate dopamine receptors, used for conditions such as Parkinson disease, restless legs syndrome, or selected endocrine disorders; adverse effects include sleepiness, nausea, and impulse-control problems.

Dopamine Antagonist

A medicine or compound that blocks dopamine receptors, used in selected psychiatric, gastrointestinal, or antiemetic treatments; effects depend on receptor selectivity and dose.

Dopamine Receptor

Dopamine receptor is a defined chemical or biologic substance involved in dopamine receptor and studied in physiology, pharmacology, laboratory medicine, or treatment.

Dopamine Transporter

A membrane protein that recaptures dopamine from the synaptic cleft into presynaptic neurons and is a target of some stimulants and imaging tracers.

Dopaminergic Neuron

A neuron that synthesizes and releases dopamine as its principal neurotransmitter, contributing to movement, motivation, reward, and endocrine regulation.

Doppler Echocardiography

Echocardiographic imaging that uses Doppler ultrasound to assess the direction and speed of blood flow through the heart and great vessels.

Doppler Ultrasound

Ultrasound technique using the Doppler effect to evaluate blood flow. Color Doppler: direction and velocity of flow (red = toward transducer, blue = away). Spectral Doppler: waveform analysis (arterial: pulsatile; venous: continuous). Pulsed-wave: measures flow at a specific depth.

Doppler Ultrasound Exam Of An Arm

Or Leg

An ultrasound examination that uses the Doppler effect to assess blood flow through arteries and veins of an arm or leg. It can help evaluate obstruction, narrowing, thrombosis, aneurysm, or abnormal circulation without ionizing radiation.

Doppler Velocimetry

Ultrasound measuring blood flow velocity in umbilical and fetal vessels. Assesses fetal well-being.

Dor Procedure

A surgical ventricular restoration procedure designed to reconstruct and remodel an akinetic or dyskinetic left ventricular anterior and apical aneurysm following extensive transmural anterior myocardial infarction. Involves ventriculotomy through the scarred myocardium, placement of an endoventricular circumferential Dacron or bovine pericardial patch exclusion stitch (Manouguian/Dor purse-string stitch) to exclude the non-contractile, fibrotic ventricular wall, and patch closure of the defect. Restores a more physiological elliptical left ventricular geometry, significantly reduces end-diastolic and end-systolic wall tension, and improves left ventricular ejection fraction in selected patients with ischemic cardiomyopathy.

Alternate Names:

Endoventricular Circular Patch Plasty (EVCPP); Surgical Ventricular Restoration

Doravirine

A non-nucleoside reverse transcriptase inhibitor used in combination antiretroviral therapy for HIV-1 infection. Doravirine binds to the HIV reverse transcriptase enzyme and inhibits its activity through a unique mechanism that maintains activity against strains resistant to first-generation NNRTIs.

Dorea

A genus of anaerobic bacteria commonly found in the human intestine; detection generally reflects microbiota unless recovered from a compatible sterile-site infection.

Doripenem

A carbapenem antibiotic used intravenously for serious infections caused by susceptible bacteria.

Dormant

Dormant is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Dorsal

Relating to the back side of the body; synonym for posterior in human anatomy.

Dorsal Blocking Splint

A hand splint that holds the wrist and fingers in a protective position, often used after flexor-tendon repair to limit extension and protect healing tendons.

Dorsal Funiculus

A posterior white-matter column of the spinal cord containing ascending fibers that convey vibration, discriminative touch, and conscious proprioception to the brain.

Dorsal Kyphosis

Excessive forward curvature of the upper spine, often associated with vertebral fractures or degenerative change.

Dorsal Root Ganglion

A swelling on the dorsal root of a spinal nerve containing the cell bodies of primary sensory neurons. Its pseudounipolar neurons convey peripheral sensory information to the spinal cord.

Dorsiflexion

Upward movement of the foot at the ankle, bringing the toes toward the shin. It is produced mainly by the tibialis anterior and helps clear the foot during walking.

Dorsalis Pedis Artery

The continuation of the anterior tibial artery on the

dorsum of the foot. Its pulse helps assess arterial blood flow to the foot.

Dorsolateral

Dorsolateral means located toward the back and side of a body structure or region.

Dorsum

The back or upper surface of a body part, such as the dorsum of the hand, foot, or tongue.

Dorzolamide

A topical carbonic-anhydrase inhibitor that lowers intraocular pressure in glaucoma by reducing aqueous-humor production.

Dos And Donts During First Trimester Of Pregnancy

First-trimester care includes prenatal visits, recommended folic acid, avoidance of alcohol and tobacco, medication review, food-safety precautions, and appropriate activity. Severe pain, heavy bleeding, fainting, fever, or persistent vomiting needs urgent obstetric evaluation.

Dosage

The amount and frequency of a drug administered to achieve a therapeutic effect while minimizing toxicity. Components: dose (amount, e.g., mg), dosing interval (frequency), route, and duration.

Dosage Forms

The physical form in which a medicine is supplied, such as a tablet, capsule, liquid, cream, inhaler, injection, or patch.

Dose

Amount of substance administered. Factors: weight, age, renal/hepatic function, drug interactions. Toxic dose varies by substance and individual.

Dose Rate

The amount of a medicine, radiation, or other exposure delivered per unit of time, such as milligrams per hour or grays per minute.

Dose-Limiting Toxicity

An adverse effect of an anticancer treatment severe enough to prevent further dose escalation or require dose reduction according to a predefined protocol. DLTs are used in early-phase clinical trials to estimate a tolerable dose.

Dose-Response Curve

A graphical representation of the relationship between the dose of a drug and the magnitude of its biological effect. The x-axis represents dose (typically on a logarithmic scale), and the y-axis represents the response (as percentage of maximum effect).

Dose-Response Relationship

The dose-response relationship describes the quantitative relationship between the dose of a drug and the magnitude of the pharmacological response it produces. This relationship is typically represented as a sigmoidal curve when response is plotted against the log of the dose.

Dosimeter

An instrument that measures or records a person's or object's exposure to ionizing radiation.

Dosimetry

Measurement and calculation of radiation dose. Patient dosimetry for therapy.

Dotage

Dotage is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

DOTATATE

Somatostatin analog labeled with Ga-68 for PET imaging of neuroendocrine tumors.

DOTS

Directly observed treatment, short-course, a tuberculosis-control strategy in which health workers observe medication doses and monitor treatment response and adherence.

Double Aortic Arch

A congenital vascular ring in which two aortic arches surround the trachea and esophagus, potentially causing noisy breathing, feeding difficulty, cough, or swallowing problems.

Double Blind

A study design in which neither participants nor the researchers assessing outcomes know who receives each intervention during the blinded phase.

Alternate Names:

Double-masked

Double Blind Trial

A clinical trial in which participants and the investigators assessing outcomes do not know which intervention each participant received, reducing expectation bias.

Double Inlet Left Ventricle

A rare congenital heart defect in which both atria connect predominantly to one morphologic left ventricle, producing single-ventricle circulation and mixing of oxygenated blood.

Double Outlet Right Ventricle

A congenital heart defect where more than 50% of both great arteries arise from the right ventricle, with a VSD as the only outlet from the left ventricle.

Double Uterus

A congenital condition in which the uterus develops as two separate cavities or two uteruses.

Double Vision

Double vision may arise from an eye alignment problem, corneal or lens disease, muscle or nerve dysfunction, medication, migraine, or brain disease. Sudden double vision, especially with weakness, speech change, severe headache, or eye pain, is an emergency.

Double-Balloon Enteroscopy

Advanced endoscopic technique using an enteroscope equipped with balloons at the tip of both

the endoscope and an overtube allowing systematic advancement and visualization of the entire small bowel. The balloons are alternately inflated and deflated to anchor the bowel wall and advance the endoscope.

Double-Blind Method

A research method that conceals treatment assignment from participants and relevant study personnel to reduce bias.

Double-Blind Study

A clinical study in which neither participants nor the investigators assessing outcomes know the assigned treatment group.

Double-Contrast Barium Enema

An imaging examination of the colon using barium and air to outline the bowel wall on X-rays.

Alternate Names:

Air-contrast barium enema

Double-Outlet Right Ventricle

A congenital heart defect in which both major arteries arise mainly from the right ventricle.

Douche

A stream of water or solution directed into a body cavity, most commonly the vagina, for cleansing, therapeutic, or contraceptive purposes.

Douglas, Pouch Of

The rectouterine pouch, a peritoneal recess between the uterus and rectum; in people without a uterus, the related rectovesical pouch lies between the bladder and rectum.

Dowager'S Hump

An exaggerated upper-back curvature, usually caused by vertebral compression fractures or osteoporosis.

Alternate Names:

Kyphosis

Down Syndrome

A genetic condition caused by an extra copy of chromosome 21. It can affect how the brain and body develop. People with Down syndrome may have learning difficulties, low muscle tone, characteristic facial features, or heart, hearing, vision, thyroid, or digestive problems.

Alternate Names:

Trisomy 21 Syndrome Down Syndrome Overview Trisomy 21 (Down Syndrome Overview) Down's Syndrome

Downregulation

A decrease in the number, sensitivity, or activity of receptors, genes, or signaling pathways in response to sustained stimulation or other regulation.

Doxapram

A respiratory stimulant used rarely for selected breathing disorders; it can cause agitation, hypertension, seizures, or abnormal heart rhythms.

Doxazosin

An alpha-1 blocker used to relax smooth muscle and treat high blood pressure or urinary symptoms from an enlarged prostate.

Doxazosin Mesylate

The mesylate salt of doxazosin, an alpha-1 blocker used for hypertension and urinary symptoms from benign prostate enlargement.

Doxepin

A tricyclic antidepressant also used in low doses for insomnia or skin itching; sedation and anticholinergic effects are common concerns.

Doxepin Overdose

Overdose of doxepin, a tricyclic antidepressant that may cause coma, seizures, hypotension, and life-threatening cardiac conduction abnormalities.

Doxercalciferol

A vitamin D analog used to help control secondary hyperparathyroidism in chronic kidney disease.

Doxorubicin

An anthracycline chemotherapy medicine that damages DNA through intercalation, topoisomerase-II inhibition, and free-radical effects; cumulative cardiomyopathy and myelosuppression are major risks.

Doxorubicin Hydrochloride

The hydrochloride salt of doxorubicin, an anthracycline chemotherapy drug used for many cancers; major risks include bone-marrow suppression and dose-related heart damage.

Doxycycline

A medicine called doxycycline. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Doxycycline Hyclate

A medicine called doxycycline hyclate. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Doxylamine

A sedating antihistamine used in some sleep aids and pregnancy-related nausea products; it can cause drowsiness and anticholinergic effects.

DPP-4 Inhibitor

An oral glucose-lowering drug that inhibits dipeptidyl peptidase-4, prolonging the action of incretin hormones and increasing glucose-dependent insulin release while reducing glucagon secretion. It is used mainly in type 2 diabetes.

DPP-4 Inhibitors

Dipeptidyl peptidase-4 inhibitors that prevent degradation of endogenous GLP-1 and GIP, enhancing incretin effects on insulin secretion. Examples include sitagliptin, saxagliptin, linagliptin, and alogliptin. They are well-tolerated with low hypoglycemia risk and weight neutrality. They have modest glucose-lowering effects.

DPT Immunization

Immunization against diphtheria, pertussis, and

tetanus, now generally given in age-appropriate DTaP or Tdap formulations.

DR Syndrome

An abbreviation that may refer to diabetic retinopathy syndrome or another context-specific disorder; the full clinical term should be specified.

Dracunculiasis

Guinea-worm disease, a parasitic infection caused by *Dracunculus medinensis* and acquired by drinking water containing infected copepods.

Dracunculosis

Infection caused by the parasitic nematode *Dracunculus medinensis*, acquired by drinking water containing infected copepods. It classically produces a painful blister and emergence of a long adult worm through the skin.

Dracunculus Medinensis

The Guinea-worm parasite that causes dracunculiasis after ingestion of infected copepods in unsafe drinking water.

Drain

A tube or other device used to remove blood, pus, fluid, or air from a wound, body cavity, or organ; drain can also mean the act of removing it.

Drain (Surgical)

Tube for fluid evacuation. Types: closed suction (Jackson-Pratt, Hemovac), open (Penrose), pigtail. Indications: abscess, hematoma, seroma, biliary, urinary. Complications: infection, blockage, migration.

Alternate Names:

Surgical

Drain Cleaner Poisoning

Exposure to a caustic drain-cleaning product that can cause severe burns of the skin, eyes, airway, or gastrointestinal tract.

Drain Opener Poisoning

Exposure to a caustic drain-cleaning product that can cause severe burns of the skin, eyes, airway, or gastrointestinal tract.

Drainage

Removal of fluid, pus, blood, or air from a body site or wound, either naturally or using a procedure or drain.

Drainage Procedures

Image-guided drainage of fluid collections. Abscess, hematoma, biloma, urinoma. CT or ultrasound guidance.

Drainpipe Cleaners

A clinical term concerning drainpipe cleaners. Its definition and clinical management depend on the context.

Drastic

Drastic is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Draw-Sheet

A clinical care, prevention, behavioral, or public-health concept concerning draw-sheet; its use depends on the patient, setting, and clinical goal.

Dream

A dream is a sequence of perceptions, thoughts, or emotions during sleep, often most vivid during REM sleep.

Dreams

Subjective experiences, images, thoughts, or emotions occurring during sleep, especially during rapid-eye-movement sleep.

Dreams (Geriatric)

Sleep disturbances common. Nightmares, insomnia. Evaluate for delirium, depression, medications.

Alternate Names:

Geriatric

Dreamy State

Dreamy State is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Drepanocytosis

Drepanocytosis is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

DRESS Syndrome

Drug Reaction with Eosinophilia and Systemic Symptoms, a severe, potentially life-threatening adverse drug reaction. Typically occurs 2-8 weeks after drug initiation (latency longer than SJS/TEN). Causative drugs: anticonvulsants (phenytoin, carbamazepine, lamotrigine), allopurinol, sulfonamides, dapsone, minocycline.

Dressing

Material applied to a wound to protect it, absorb exudate, maintain a suitable healing environment, or deliver local treatment.

Dressler Syndrome

Inflammation of the sac around the heart that develops weeks after a heart attack, heart surgery, or other heart injury. It can cause fever, chest pain, and fluid around the heart or lungs.

Alternate Names:

Post-Cardiac Injury Syndrome Postmyocardial Infarction Syndrome

Dressler'S Syndrome

Alternate spelling; see *Dressler Syndrome*.

Dressler'S Syndrome

Alternate spelling; see *Dressler Syndrome*.

DRI

A set of nutrient reference values used to plan and

assess dietary intake for healthy people.

Alternate Names:

Dietary Reference Intakes

Dried Specimen

A biological sample that has been dried for storage, transport, or testing; the collection method and analyte determine whether drying is appropriate.

Drift

Drift is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Drift (Sit-To-Stand)

A functional assessment of lower-limb strength, balance, and mobility. Timed sit-to-stand, chair-rise, and timed-up-and-go tests may be used; results are interpreted according to the person's age, baseline function, assistive devices, and clinical purpose.

Alternate Names:

Sit-to-Stand

Drinking Water

Water intended for human consumption; it should meet local safety standards and be treated when contamination with pathogens or chemicals is possible.

Drinking Water Safely During Cancer Treatment

Food and water precautions during cancer treatment aim to reduce infection risk, especially when immunity is suppressed; advice should follow the oncology team and local water guidance.

Drip

A drop-by-drop flow of liquid or an intravenous infusion delivered at a controlled rate through tubing and a catheter.

Drip (Intravenous Infusion)

Controlled administration of fluid, blood, nutrients, or other substances directly into a vein through a

cannula or catheter.

Alternate Names:

Intravenous Infusion

Driving And Older Adults

Assessment of driving safety in older adults, considering vision, cognition, hearing, mobility, medications, and functional performance rather than age alone.

Driving Evaluation

Assess driving fitness after neurological, musculoskeletal conditions. On-road testing, vehicle modifications.

Driving Rehabilitation

Training for adaptive driving: hand controls, left-foot accelerator, other modifications.

Dronabinol

A synthetic form of delta-9-tetrahydrocannabinol used for selected cases of chemotherapy-related nausea or appetite loss; central nervous-system effects and interactions are important.

Drool

Excess saliva escaping from the mouth, which may result from increased production, impaired swallowing, poor oral closure, or neurologic disease.

Drooling

Unintentional escape of saliva from the mouth caused by excess production, impaired swallowing, poor lip closure, or neurologic and developmental disorders.

Drop Foot

Alternate index label; see *Foot drop*.

Drop Hand Syndrome

Weakness or paralysis of wrist and finger extension, usually from radial-nerve injury or compression, causing the hand to hang in flexion.

Droperidol

A butyrophenone dopamine antagonist used in selected settings for nausea or agitation; QT prolongation and arrhythmia risk require appropriate precautions.

Droplet Precautions

Infection control measures used for pathogens transmitted through respiratory droplets. Precautions include surgical masks, patient placement in single rooms, and spatial separation. Droplet precautions are used for influenza, pertussis, and meningococcal infections. The distance for spatial separation is typically 3-6 feet.

Dropper Bottle

A container fitted with a dropper for delivering a liquid in drops; the prescribed product, concentration, and number of drops must be followed carefully.

Drops

Drops is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Dropsy

An older term for generalized swelling caused by fluid accumulation, now usually called edema or anasarca.

Drosophila Proteins

Proteins produced by fruit flies and used as experimental models of development, signaling, genetics, immunity, and human disease mechanisms.

Drosophilidae

The family of insects commonly called fruit flies or vinegar flies; they are important in genetics research and are not typical human parasites.

Drowning

Respiratory impairment from submersion/immersion in liquid. Drowning can be fatal or non-fatal. Pathophysiology: laryngospasm

followed by aspiration of fluid, surfactant washout, pulmonary edema, and hypoxemia. Fresh water: washes out surfactant, causes alveolar collapse. Salt water: draws fluid into alveoli (pulmonary edema).

Alternate Names:

Wet Dry Drowning (Drowning)

Drowning (Dry/Wet Signs)

Respiratory impairment caused by submersion or immersion in liquid. So-called dry and wet signs are unreliable for determining the cause of death.

Alternate Names:

Dry/Wet Signs

Drowsiness

An abnormally strong tendency to fall asleep or reduced alertness, caused by insufficient sleep, medicines, substances, metabolic illness, sleep disorders, or neurologic disease.

Drug

A substance used to prevent, diagnose, treat, or otherwise affect a disease or body function; medicines may be prescribed or available without prescription.

Drug Addiction

Legacy, nonpreferred label; see *Substance Use Disorder*.

Drug Caution Code

A medication caution code indicating that a product contains acetylsalicylic acid, or aspirin.

Drug Implant

A solid or device-based dosage form placed in tissue to release a medicine locally or systemically over an extended period.

Drug Hypersensitivity

An immune-mediated reaction to a medicine that may cause skin, respiratory, gastrointestinal, blood, or systemic symptoms. Severe reactions can include anaphylaxis, Stevens–Johnson syndrome, toxic epidermal necrolysis, or drug hypersensitivity syndrome.

Drug-Induced Lupus

An autoimmune condition clinically and serologically resembling systemic lupus erythematosus, triggered by continuous exposure to specific medications (most classically procainamide, hydralazine, isoniazid, and minocycline). Characterized by arthralgias, serositis, constitutional symptoms, and high titers of anti-histone antibodies, with resolution usually following drug cessation.

Alternate Names:

Drug-Induced Lupus Erythematosus; DILE

Drug Safety

The identification, prevention, and monitoring of medication-related errors, interactions, adverse effects, and misuse.

Drug Screening

Urine, blood, hair testing for substances of abuse. Panels: opiates, cocaine, amphetamines, cannabinoids, PCP, benzodiazepines, barbiturates. False positives possible. Confirmation by GC/MS or LC-MS/MS.

Drug Testing

Laboratory analysis of biological specimens to detect drugs, metabolites, or poisons, used in clinical, workplace, sports, and forensic contexts. Specimens include blood, urine, oral fluid, hair, and postmortem tissues.

Drugs That May Cause Erection Problems

Medicines that may impair erection include some antihypertensives, antidepressants, antipsychotics, antiandrogens, and other drugs; review should be clinician-guided.

Drugs, Statin

Medicines that inhibit HMG-CoA reductase and lower LDL cholesterol, reducing the risk of atherosclerotic cardiovascular events.

Drusen

Yellowish-white deposits of extracellular material

that accumulate beneath the retinal pigment epithelium between the retinal pigment epithelium basement membrane and Bruch membrane, and they are the hallmark clinical finding in dry age-related macular degeneration.

Dry Age-Related Macular Degeneration

Alternate index label; see *Dry macular degeneration*.

Dry AMD

Alternate index label; see *Dry macular degeneration*.

Dry Cell Battery Poisoning

Ingestion or exposure to a dry-cell battery. Leakage or retention can cause corrosive tissue injury and requires urgent assessment.

Dry Cough

A cough that produces little or no sputum; causes include viral illness, asthma, reflux, medications such as ACE inhibitors, and other airway or lung disease.

Dry Cupping

Dry Cupping is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Dry Eye

A condition in which the eyes do not produce enough tears or tears evaporate too quickly, causing irritation and visual discomfort.

Alternate Names:

Dry-eye disease; keratoconjunctivitis sicca

Dry Eye Disease

Dry eye disease, also known as keratoconjunctivitis sicca, is a multifactorial disorder of the ocular surface characterized by loss of homeostasis of the tear film, resulting in symptoms of ocular discomfort, visual disturbance, and tear film instability, with potential damage to the ocular surface.

Alternate Names:

keratoconjunctivitis sicca, is a multifactorial disorder

der.

Dry Eye Syndrome

A multifactorial disease of the ocular surface characterized by tear film instability, hyperosmolarity, and inflammation. Aqueous-deficient dry eye: Sjogren syndrome (autoimmune), non-Sjogren (lacrimal gland dysfunction, age). Evaporative dry eye: meibomian gland dysfunction (most common), incomplete blink, environmental factors.

Dry Eyes

Irritation or discomfort caused by inadequate tear production or excessive tear evaporation.

Alternate Names:

Decreased Tear Production Keratoconjunctivitis Sicca (Dry Eyes) Tears, Decreased Production

Dry Gangrene

Tissue death caused mainly by loss of arterial blood supply, producing dry, shrunken, dark tissue with a relatively clear boundary from viable tissue.

Dry Hair

Hair that lacks moisture or appears rough, brittle, dull, or frizzy because of weather, heat or chemical damage, infrequent conditioning, nutritional disease, or an endocrine disorder.

Dry Ice

Dry ice is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Dry Macular Degeneration

The more common form of age-related macular degeneration, involving gradual damage to the macula without abnormal blood-vessel growth.

Alternate Names:

Dry Age-Related Macular Degeneration Dry AMD Macular Degeneration, Dry

Dry Mouth

A condition in which the mouth does not produce enough saliva; also called xerostomia.

Alternate Names:

xerostomia.

Dry Mouth During Cancer Treatment

Reduced saliva during or after cancer treatment, often from head-and-neck radiation or medicines, increasing discomfort, dental caries, swallowing difficulty, and oral infection risk.

Dry Powder Inhaler

A breath-actuated inhaler delivering powdered medication. Does not require hand-breath coordination. Requires adequate inspiratory flow. Examples include Advair Diskus and Spiriva HandiHaler.

Dry Skin

Abnormally dry, rough, or scaly skin caused by reduced moisture and disruption of the skin barrier. It is common with aging, low humidity, frequent washing, and eczema and may cause itching or fissuring.

Dry Socket

Painful inflammation of a tooth socket caused by loss or failure of the blood clot after tooth extraction. Also called alveolar osteitis.

Alternate Names:

alveolar osteitis.

Dry Socket (Alveolar Osteitis)

A painful postoperative complication of tooth extraction, most commonly occurring after the removal of mandibular third molars (wisdom teeth). It is caused by the premature dissolution or loss of the blood clot in the extraction socket, exposing the underlying alveolar bone and nerve endings to the oral cavity.

Alternate Names:

Alveolar Osteitis

Dry Socket Overview

Dry socket is painful inflammation after a tooth extraction when the protective blood clot is lost or breaks down before healing. Pain often worsens several days after extraction and may radiate to

the ear; dental cleaning, medicated dressing, and analgesia provide relief.

Alternate Names:

Alveolar Osteitis (Dry Socket Overview)

Dry-Powder Inhaler

A device that delivers powdered medicine to the lungs when the user inhales through it.

Dryness Vaginal (Vaginal Dryness And Vaginal Atrophy)

Alternate index label for Vaginal Dryness and Vaginal Atrophy.

DSM

The Diagnostic and Statistical Manual of Mental Disorders, a classification and diagnostic reference published by the American Psychiatric Association.

DSPS

Alternate index label; see *Delayed sleep phase*.

DSRCT

Alternate index label; see *Desmoplastic small round cell tumors*.

DSWPD

Alternate index label; see *Delayed sleep phase*.

DTaP

A childhood vaccine protecting against diphtheria, tetanus, and pertussis, containing diphtheria and tetanus toxoids and acellular pertussis components.

Dual Antiplatelet Therapy

The combination of aspirin and a P2Y₁₂ inhibitor (clopidogrel, prasugrel, or ticagrelor) for prevention of thrombotic events after acute coronary syndrome and percutaneous coronary intervention. Duration varies: 1 month after bare-metal stent, 6-12 months after drug-eluting stent, and 12 months after ACS.

Dual Energy X-Ray Absorptiometry

See *Bone Mineral Density Test*.

Alternate Names:

DXA; DEXA

Dual-Energy CT

CT using two different energy spectra. Material differentiation, virtual non-contrast images, gout detection.

Dual-Energy X-Ray Absorptiometry

See *Bone Mineral Density Test*.

Dual-Energy X-Ray Absorptiometry

See *Bone Mineral Density Test*.

Alternate Names:

Dual-Energy X-ray Absorptiometry

Dual-Photon Absorptiometry

An older bone-density measurement technique that uses two photon energies to estimate bone mineral content.

Duane Retraction Syndrome

A congenital eye-movement disorder in which abnormal eye-muscle innervation limits horizontal movement and may cause the eye to retract.

Dubin-Johnson Syndrome

An inherited disorder of bilirubin excretion causing predominantly conjugated hyperbilirubinemia and intermittent mild jaundice, usually without major liver dysfunction. mutations in the ABCC2 gene encoding the multidrug resistance-associated protein 2 (MRP2), which transports conjugated bilirubin from hepatocytes into bile.

Duboisin

An older anticholinergic alkaloid related to scopolamine, formerly used for selected eye or neurologic purposes and capable of causing anticholinergic toxicity.

Dubowitz Syndrome

A rare developmental disorder associated with growth restriction, characteristic facial features, microcephaly, intellectual or behavioral differences, and variable immune or skin findings.

Duchenne Muscular Dystrophy

An X-linked recessive neuromuscular disorder caused by mutations in the dystrophin gene (DMD gene) on chromosome Xp21.2, resulting in absent or severely reduced dystrophin protein, which is essential for maintaining the structural integrity of the sarcolemma during muscle contraction.

Duck Embryo Vaccine

Duck Embryo Vaccine is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Duck Syndrome

Duck syndrome is an informal term for appearing successful while privately experiencing severe stress, anxiety, exhaustion, or depression. It is not a formal diagnosis; persistent distress, impaired functioning, or self-harm thoughts warrant mental-health evaluation and support.

Duct

A tube or channel that carries a fluid, such as bile, pancreatic juice, or secretions from a gland.

Ductal Carcinoma

A carcinoma arising from the epithelial cells lining a duct, most commonly referring to ductal carcinoma of the breast or pancreas.

Ductal Carcinoma In Situ

A noninvasive breast cancer in which abnormal cells remain within the milk ducts; commonly abbreviated DCIS.

Ductal Carcinoma In Situ

A noninvasive breast cancer in which abnormal cells are confined to the milk ducts and have not spread into surrounding breast tissue.

Alternate Names:

DCIS Intraductal Carcinoma

Ductal-Dependent Lesions

Congenital heart defects that require the ductus

arteriosus to remain patent for adequate systemic or pulmonary blood flow. Includes pulmonary atresia, critical aortic stenosis, coarctation, hypoplastic left heart syndrome, and transposition of the great arteries.

Ductless Glands

Ductless Glands is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Ductus Arteriosus

A fetal blood vessel connecting the pulmonary artery to the descending aorta, allowing blood to bypass the non-functioning fetal lungs. In utero, the ductus arteriosus is kept patent by prostaglandin E2 and low oxygen tension.

Ductus Venosus

Ductus Venosus is an anatomical structure, physiologic process, or body function; its location or mechanism explains its clinical relevance.

Duffy-Null Associated Neutrophil Count

A naturally lower neutrophil count seen in some people with the Duffy-null blood-group type, especially people of African or Middle Eastern ancestry. It usually does not increase infection risk and should not automatically be diagnosed as neutropenia.

Duke Criteria

A standardized diagnostic framework for evaluating suspected infective endocarditis, combining microbiological, imaging, and clinical findings. Major criteria comprise persistently positive blood cultures with typical endocarditis organisms and echocardiographic evidence of endocardial involvement (vegetation, abscess, new partial dehiscence of a prosthetic valve, or new valvular regurgitation). Minor criteria include predisposing heart conditions or injection drug use, fever $\geq 38^{\circ}\text{C}$, vascular phenomena (arterial emboli, septic pulmonary infarcts, mycotic aneurysm, Janeway lesions), and immunologic phenomena (glomerulonephritis, Osler nodes, Roth spots, rheumatoid factor). Definite endocarditis is confirmed by 2 major, 1 major plus 3 minor,

or 5 minor criteria.

Alternate Names:

Modified Duke Criteria; Duke Endocarditis Criteria

Dukes Classification

An older staging system for colorectal cancer based mainly on how far the tumor has extended through the bowel wall and into lymph nodes or distant sites.

Dull Pain

A low-intensity, aching, or pressure-like pain that is not sharp or stabbing; its cause depends on the affected tissue and clinical context.

Duloxetine

A serotonin–norepinephrine reuptake inhibitor used for depression, anxiety, diabetic peripheral neuropathic pain, fibromyalgia, and some chronic musculoskeletal pain.

Duloxetine Hydrochloride

The hydrochloride salt of duloxetine, a serotonin–norepinephrine reuptake inhibitor used for depression, anxiety, neuropathic pain, fibromyalgia, and some chronic musculoskeletal pain.

Dumbbell Ossification

Bone formation or a bony lesion with a dumbbell shape, often describing extension through a neural foramen or across adjacent anatomical compartments.

Dumdum Fever

An older name for visceral leishmaniasis, a parasitic disease that can cause prolonged fever, enlarged spleen and liver, low blood counts, and weight loss.

Dumping Syndrome

Rapid gastric emptying of hypertonic contents into the small bowel after gastrectomy or gastric bypass surgery. Early dumping occurs within thirty minutes with vasomotor symptoms including flushing, palpitations, diarrhea, and dizziness from fluid shifts into the intestinal lumen.

Dunbar Syndrome

Alternate index label; see *Median arcuate ligament syndrome*.

Duocarmycins

A medicine called duocarmycins. Its uses, dosing, adverse effects, and contraindications depend on the specific indication and patient.

Duodenal

Relating to the duodenum, the first portion of the small intestine immediately beyond the stomach.

Duodenal Adenocarcinoma

A malignant gland-forming tumor of the duodenum that may cause abdominal pain, vomiting, bleeding, anemia, jaundice, or obstruction; diagnosis requires endoscopy and biopsy.

Duodenal Atresia

A congenital obstruction of the duodenum caused by failure of recanalization during the 8th–10th week of gestation. Associated with Down syndrome (30%), polyhydramnios (due to impaired fetal swallowing of amniotic fluid), and other anomalies (VACTERL, congenital heart disease, annular pancreas, malrotation).

Duodenal Bulb

The first, expanded portion of the duodenum immediately beyond the pylorus.

Duodenal Disease

A disorder of the first part of the small intestine, including inflammation, ulceration, obstruction, infection, bleeding, malabsorption, or neoplasia.

Duodenal Diverticulum

A duodenal diverticulum is an outpouching of the duodenal wall, often found incidentally near the ampulla. It can rarely cause pain, bleeding, inflammation, obstruction, pancreatitis, or bile-duct infection and is treated according to symptoms and complications.

Duodenal Fistula

An abnormal connection between the duodenum and another organ, bowel segment, or skin surface that can cause leakage, infection, malnutrition, or electrolyte loss.

Duodenal Hemorrhage

Bleeding from the duodenum, which may result from peptic ulcer disease, inflammation, vascular lesions, trauma, or cancer.

Duodenal Infection

Infection involving the duodenum, caused by organisms or spread from nearby gastrointestinal disease; symptoms may include abdominal pain, vomiting, diarrhea, or bleeding.

Duodenal Neoplasm

A benign or malignant growth arising in the duodenum; symptoms and treatment depend on the tumor type, location, size, obstruction, bleeding, and spread.

Duodenal Obstruction

Partial or complete blockage that prevents contents from passing through the duodenum.

Duodenal Papilla

A mucosal projection in the duodenum where the bile duct and pancreatic duct, usually through the major papilla, empty into the intestine.

Duodenal Perforation

Full-thickness breach of the duodenal wall, most commonly caused by penetrating peptic ulcer disease, trauma, or iatrogenic injury during endoscopy. Presents with sudden severe epigastric pain, board-like abdominal rigidity, and peritonitis. Free air under the diaphragm on upright chest radiograph is diagnostic.

Duodenal Stenosis

Narrowing of the duodenum that may obstruct passage of stomach contents and cause vomiting, abdominal distention, pain, or poor feeding.

Duodenal Ulcer

Peptic ulcer in the duodenal mucosa strongly associated with *H. pylori* infection. Presents with epigastric pain relieved by food typically occurring several hours after meals and at night. Complications include hemorrhage, perforation, and gastric outlet obstruction. Diagnosis is by upper endoscopy.

Duodenitis

Inflammation of the duodenal mucosa, caused by *Helicobacter pylori* infection, acid-related disease, non-steroidal anti-inflammatory drugs, infection, coeliac disease, or inflammatory bowel disease. Symptoms may include epigastric pain, nausea, vomiting, and gastrointestinal bleeding.

Duodenoscope

A flexible endoscope designed to examine the duodenum and access the bile and pancreatic ducts, commonly during ERCP.

Duodenostomy

A surgically created opening from the duodenum to the skin or another bowel segment, usually for drainage or feeding in selected circumstances.

Duodenum

The first part of the small intestine, extending from the pylorus to the jejunum; it receives bile and pancreatic secretions and is a common site of peptic ulcer disease.

Duplex Doppler Ultrasound Scanning

An ultrasound test that combines structural imaging with Doppler measurement of blood flow through vessels.

Alternate Names:

Duplex ultrasound

Duplex Surveillance

Scheduled clinical and ultrasound monitoring of an artery, vein, graft, stent, aneurysm, or dialysis access to detect restenosis, thrombosis, endoleak, or other failure before symptoms or limb threat develop.

Duplex Ultrasound

An ultrasound examination that combines grayscale imaging of anatomy with Doppler assessment of blood flow. It can evaluate vessels, organs, and selected soft tissues.

Duplication

Presence of extra copy of chromosomal segment. Can cause gene dosage effects. Examples: Charcot-Marie-Tooth disease type 1A (PMP22 duplication).

Dupuyten'S Contracture

A fibrosing disorder of the palm that causes progressive flexion contractures of one or more fingers.

Dupuytren Contracture

Dupuytren contracture is progressive shortening of palmar fascia that pulls one or more fingers into flexion and can limit hand function. cords, and contractures of the fingers, most commonly affecting the ring and small fingers.

Alternate Names:

Dupuytren's Contracture (Dupuytren Contracture)

Dupuytren'S Contracture

Dupuytren's Contracture is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Dupuytren'S Disease

Thickening and shortening of tissue in the palm that progressively bends one or more fingers toward the palm.

Dupuytren, Guillaume

A French surgeon whose name is associated with Dupuytren contracture, progressive thickening and shortening of palmar fascia that bends the fingers toward the palm.

Dupuytren's Contracture (Dupuytren Contracture)

Alternate index label for Dupuytren Contracture.

Dura

The dura mater, the tough outermost membrane surrounding the brain and spinal cord.

Dura Mater

The outermost, thickest meningeal layer, a tough fibrous membrane. In the cranium it is two layers: periosteal (outer, adherent to skull) and meningeal (inner), which separate to form dural venous sinuses (e.g., superior sagittal, cavernous). Dural folds: falx cerebri, tentorium cerebelli, falx cerebelli.

Durable Power Of Attorney

A legal authorization allowing a designated person to act or make decisions for another person when specified conditions are met. In healthcare, it may appoint a healthcare proxy.

Dural Arteriovenous Fistulas

Abnormal connections between arteries and veins in the dura mater surrounding the brain or spinal cord.

Dural Sac

The tubular sleeve of dura mater surrounding the spinal cord, cerebrospinal fluid, and spinal nerve roots within the vertebral canal.

Dural Venous Sinuses

Venous channels between dural layers, draining brain blood and CSF from the subarachnoid space into the internal jugular veins. Major sinuses: superior sagittal, transverse, sigmoid, cavernous, straight. Valveless with bidirectional flow possible.

Duramorph

A brand name for preservative-free morphine used by injection, including in epidural or intrathecal analgesia; respiratory depression is a serious risk.

Durapatite

Hydroxyapatite, the calcium-phosphate mineral that gives bone and teeth much of their hardness.

Durules

Durules is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on the surrounding context.

Dust

Dust is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Dust Disease

Dust Disease is a chemical, environmental, or toxicology term; health effects depend on the substance, dose, route, duration, and exposure circumstances.

Dust Mite Allergy

An allergic reaction to proteins from household dust mites, commonly causing sneezing, congestion, itchy eyes, or asthma symptoms.

Dutasteride

A 5-alpha-reductase inhibitor that lowers dihydrotestosterone and is used for benign prostatic hyperplasia; it reduces prostate volume and can affect sexual function and PSA interpretation.

DVT

Deep venous thrombosis, the formation of a blood clot in a deep vein, most commonly in the lower extremities.

DVT Prevention

Measures used to reduce the risk of deep venous thrombosis, such as early movement, mechanical compression, and anticoagulant medicine when clinically appropriate.

Dwarfism

Short stature resulting from a medical condition, most commonly achondroplasia (the most common form of skeletal dysplasia, caused by gain-of-function mutation in FGFR3, presenting with rhizomelic limb shortening, macrocephaly, frontal bossing, and trident hands).

Dwarfism (Achondroplasia)

Alternate index label for Achondroplasia.

DWI FLAIR Mismatch

Diffusion-weighted imaging and fluid-attenuated inversion recovery MRI sequences are used together to estimate the age of an ischemic stroke lesion and guide treatment decisions.

Dx

A common medical abbreviation for diagnosis or diagnostic assessment. It should be expanded in patient instructions when the meaning could be unclear.

DXA Scan (Dual-Energy X-Ray Absorptiometry)

See *Bone Mineral Density Test*.

Alternate Names:

Dual-Energy X-ray Absorptiometry

Dyad

A pair of related people, structures, cells, or variables considered together.

Dye

A colored substance used to stain tissue, label cells, visualize structures, or provide contrast in diagnostic or laboratory procedures.

Dye Lasers

Lasers that use an organic dye as the light-producing medium; they can generate tunable wavelengths for medical treatment and laboratory applications.

Dye Remover Poisoning

Exposure to dye remover, which may irritate the skin, eyes, airways, or gastrointestinal tract; product ingredients determine severity.

Dying

The process of progressive physical decline that occurs as a person approaches death, often accompanied by psychological, social, and spiritual changes.

Dynamin Gtpase

A family of GTP-binding proteins that constrict membranes during vesicle budding, endocytosis, and other cell-membrane processes.

Dynamometer

An instrument used to measure force, commonly hand-grip strength or muscle strength.

Dynein Atpase

The ATP-powered motor activity of dynein, which moves cargo toward the minus ends of microtubules and drives cilia and flagella movement.

Dynorphin

An endogenous opioid peptide that primarily activates kappa-opioid receptors and influences pain, stress, mood, and neuroendocrine signaling.

Dyphylline

A xanthine-derived bronchodilator used in some countries for asthma or chronic obstructive lung disease; nausea, tremor, and rapid heartbeat may occur.

Dysaesthesia

Dysaesthesia is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Dysarthria

Impaired clarity, strength, or coordination of speech caused by dysfunction of the muscles or nervous pathways used for speaking. components of speech production, including the lips, tongue, palate, vocal cords, and respiratory muscles.

Dysautonomia

Dysfunction of the autonomic nervous system that can affect blood pressure, heart rate, sweating, temperature regulation, digestion, bladder function, or sexual function. It may be primary or secondary to another disease, medication, or injury.

Dysbarism

Illness caused by changes in ambient pressure, including decompression sickness and barotrauma during diving, flying, or hyperbaric exposure.

Dysbiosis

An alteration in microbiome composition, diversity, or function relative to a healthy state, permitting overgrowth of pathobionts or loss of beneficial microbes. Causes include antibiotics, dietary change, stress, and infection. It is associated with inflammatory bowel disease, obesity, allergy, and *Clostridioides difficile* colitis.

Dyscalculia

A persistent difficulty with understanding or manipulating numerical quantities and performing arithmetic that is not explained by inadequate instruction alone. It may occur as a developmental learning disorder or result from acquired brain injury, particularly involving dominant parietal networks.

Dyschezia

Dyschezia is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Dyscrasia

An historic term broadly referring to any abnormal condition of the blood or bone marrow, now used primarily in the context of plasma cell dyscrasias (also called monoclonal gammopathies).

Alternate Names:

monoclonal gammopathies).

Dyscrasias

Dyscrasias is an older broad term for abnormal blood or body-fluid composition, often referring to hematologic disorders or abnormal blood cells.

Dysembryoplastic Neuroepithelial Tumor

A benign or very low-grade glioneuronal tumor, usually in the cerebral cortex of children or young adults, classically causing long-standing drug-

resistant focal seizures.

Dysentery

Severe diarrheal illness with blood and mucus in stool, typically caused by *Shigella*, *Salmonella*, *Campylobacter*, or *Entamoeba histolytica*. Presents with frequent bloody stools, abdominal cramps, tenesmus, and fever. Diagnosis: stool culture, microscopy, PCR.

Dysesthesia

An unpleasant, abnormal sensation produced by a normally non-painful stimulus, or an unpleasant, exaggerated response to a normal stimulus. It is distinguished from paresthesia (which is simply an abnormal sensation that may or may not be unpleasant) by its consistently unpleasant quality.

Dysferlinopathy

A spectrum of autosomal recessive muscular dystrophies caused by biallelic loss-of-function mutations in the *DYSF* gene on chromosome 2p13.2, which encodes dysferlin, a 230-kDa transmembrane protein essential for calcium-dependent sarcolemmal vesicle fusion and membrane patch repair following mechanical stress. Phenotypically manifests in late adolescence or early adulthood (ages 15–25) as either Limb-Girdle Muscular Dystrophy R2 (LGMD R2 / LGMD 2B) with predominantly pelvic- and shoulder-girdle weakness, or Miyoshi myopathy with selective, early wasting and weakness of distal calf muscles (gastrocnemius and soleus), both characterized by massive serum creatine kinase elevations ($> 20\text{--}50\times$ normal) and total absence of dysferlin on Western blot or immunohistochemistry.

Alternate Names:

DYSF-Related Muscular Dystrophy; Dysferlin Deficiency

Dysfibrinogenemia

An inherited or acquired disorder in which fibrinogen is abnormal and blood clotting may be impaired or excessive.

Dysfunction

Abnormal, impaired, or inadequate function of an

organ, tissue, system, or process.

Dysfunctional Uterine Bleeding

Dysfunctional Uterine Bleeding is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Dysgammaglobulinemia

Dysgammaglobulinemia concerns dysgammaglobulinemia, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Dysgerminoma

Dysgerminoma is a malignant germ-cell tumor of the ovary, analogous to seminoma of the testis, and often occurs in adolescents or young adults.

Dysgeusia

Distortion, reduction, or loss of the normal sense of taste. It may result from oral or nasal disease, viral infection, neurological injury, nutritional deficiency, systemic illness, or medication exposure and may occur with altered smell, dry mouth, or mucosal inflammation.

Dysgraphia

A learning disorder involving persistent difficulty with handwriting, spelling, written expression, or the motor and language skills required for writing. spelling, written composition, or the motor and language processes required for writing. It may occur alone or with other learning, language, attention, or neurological disorders.

Dyshidrosis

A skin disorder causing small, intensely itchy blisters on the palms, sides of the fingers, or soles.

Alternate Names:

Eczema, Dyshidrotic

Dyskeratosis Congenita

An inherited telomere-biology disorder causing variable bone-marrow failure, abnormal skin pigmenta-

tion, nail dystrophy, and oral leukoplakia. Patients may also develop pulmonary, liver, dental, skeletal, developmental, and malignancy complications.

Dyskinesia

Abnormal, often involuntary movement that may be excessive, repetitive, irregular, or purposeless. Causes include neurological disease and medication effects.

Dyslexia

A specific learning disorder with neurobiological origin, characterized by difficulties with accurate and fluent word recognition, spelling, and decoding ability, despite adequate intelligence, motivation, and educational opportunity. Affects approximately 5–10% of the population.

Alternate Names:

Reading Disability, Specific Reading Disability

Dyslipidemia

An abnormal concentration or composition of circulating lipoproteins, including elevated low-density lipoprotein cholesterol, elevated triglycerides, reduced high-density lipoprotein cholesterol, or mixed abnormalities.

Dyslipidemias

Abnormal blood-lipid concentrations, such as high LDL cholesterol, high triglycerides, or low HDL cholesterol, that can increase atherosclerotic and pancreatitis risk.

Dysmenorrhea

Painful menstruation, often causing cramping pelvic pain. See Menstrual Cramps.

Dysmenorrhoea

Painful menstruation, classified as primary (no identifiable pelvic pathology) or secondary (underlying structural or inflammatory condition).

Dysmetria

Inability to control the range, force, or speed of a

voluntary movement, commonly caused by cerebellar dysfunction.

Dysorexia

Dysorexia is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Dysostoses

Inherited or developmental disorders in which one or more bones form abnormally.

Dyspareunia

Recurrent or persistent genital or pelvic pain associated with sexual intercourse, categorized clinically by anatomical depth as superficial (introital) or deep. Superficial dyspareunia involves pain localized to the vestibule or introitus upon initial contact or attempted penetration, commonly caused by provoked vestibulodynia, vulvovaginal atrophy (genitourinary syndrome of menopause), local dermatoses (lichen sclerosus), infections (vulvovaginal candidiasis), or vaginismus. Deep dyspareunia involves aching or sharp visceral pain provoked by deep penile thrusting, classic for pelvic endometriosis, adenomyosis, pelvic inflammatory disease, interstitial cystitis, or pelvic adhesions. Requires systematic examination of the vulva, pelvic floor musculature, and bimanual palpation.

Dyspepsia

Chronic or recurrent pain or discomfort centered in the upper abdomen. Organic causes: peptic ulcer disease, GERD, gastritis, gastric cancer. Functional dyspepsia is diagnosed after exclusion of organic pathology with normal upper endoscopy. Rome IV criteria define subtypes: epigastric pain syndrome and postprandial distress syndrome.

Dyspeptic

Relating to dyspepsia, symptoms such as upper-abdominal discomfort, fullness, early satiety, belching, or nausea.

Dysphagia

Difficulty swallowing, classified as oropharyngeal (difficulty initiating a swallow, choking, coughing, nasal regurgitation) or esophageal (sensation of food sticking retrosternally).

Alternate Names:

Difficulty Swallowing Trouble Swallowing

Dysphagia (Geriatric)

Difficulty swallowing, common in elderly. Causes: neurological (stroke, Parkinson's), structural (cancer, stricture), muscular (presbyesophagus), medications. Symptoms: coughing with eating, choking, pocketing food, weight loss, aspiration. Assessment: clinical exam, videofluoroscopy, endoscopy.

Alternate Names:

Geriatric

Dysphagia Diet

A texture- and consistency-modified diet prescribed after assessment of swallowing safety. Liquids and foods may be thickened or altered to reduce aspiration risk while maintaining adequate energy, protein, hydration, and patient acceptability.

Dysphagia Rehabilitation

Therapy to improve safe swallowing and nutrition in a person with dysphagia. It may include swallow exercises, compensatory techniques, diet modification, and education based on the cause and swallowing assessment.

Dysphasia

A language disorder caused by damage to the language-dominant hemisphere of the brain (typically left), involving difficulty with expression, comprehension, reading, or writing. Dysphasia (often used interchangeably with aphasia): partial loss of language function.

Dysphonia

An abnormal voice quality encompassing hoarseness, breathiness, roughness, vocal fatigue, pitch breaks, and strained or whispered speech. It is organic when due to structural laryngeal pathology (nodules, polyps, laryngitis, laryngeal cancer) and functional

when due to voice misuse, muscle tension dysphonia, or psychogenic causes.

Dysphoria

A state of marked unease, dissatisfaction, or emotional distress; gender dysphoria specifically refers to distress related to incongruence between experienced gender and assigned sex.

Dysplasia

Abnormal cell growth, maturation, or organization within a tissue. In some epithelial tissues it is a precancerous change that may regress, persist, or progress to invasive cancer, depending on its grade and cause; developmental dysplasia is a different use of the term. Dysplasia is not itself invasive cancer.

Dysplasia Cervical (Cervical Dysplasia)

Alternate index label for Cervical Dysplasia.

Dysplastic Nevus

Atypical nevus: a melanocytic lesion with clinical and histologic atypia, larger (>5-6 mm), with irregular borders, variable color (pink-tan-brown), and a macular/papular component; often multiple in the dysplastic nevus syndrome (familial atypical mole-melanoma, FAMMM).

Dyspnea

The subjective sensation of difficult or uncomfortable breathing. See Shortness of Breath. It may result from cardiopulmonary, metabolic, neuromuscular, or psychological causes.

Dyspnea On Exertion

Shortness of breath precipitated by physical activity, a cardinal symptom of cardiac and pulmonary disease. Cardiac causes: heart failure, coronary artery disease (angina equivalent), valvular disease; pulmonary: COPD, asthma, ILD, anemia, deconditioning. Quantified by NYHA functional class (heart failure) and mMRC scale (dyspnea).

Dyspnoea

The subjective sensation of difficult or labored

breathing. See Shortness of Breath.

Dyspraxia

Dyspraxia is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Dyspraxia (Developmental Coordination Disorder)

Difficulty planning and coordinating voluntary movements that interferes with daily activities, learning, or motor skills and is not explained by a primary muscle or sensory disorder.

Alternate Names:

Developmental Coordination Disorder

Dysraphism

Dysraphism is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Dysreflexia (Autonomic Dysreflexia)

A potentially life-threatening episode of uncontrolled autonomic activity, usually in people with spinal-cord injury above T6, triggered by stimuli such as bladder or bowel distension.

Alternate Names:

Autonomic Dysreflexia

Dysrhythmia

Dysrhythmia is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Dyssomnias

Sleep disorders involving difficulty initiating or maintaining sleep, excessive sleepiness, or abnormal regulation of the sleep–wake cycle.

Dysthymia

Dysthymia is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend

on the history, examination, and appropriate investigations.

Dysthymia (Persistent Depressive Disorder)

A chronic form of depression with depressed mood for most of the day, more days than not, for ≥ 2 years (≥ 1 year in children/adolescents). Symptoms: poor appetite or overeating, insomnia or hypersomnia, low energy, low self-esteem, poor concentration, and hopelessness.

Alternate Names:

Persistent Depressive Disorder

Dystocia

Difficult or abnormal labor caused by problems with uterine contractions, the fetus, or the maternal pelvis and soft tissues.

Dystociais

Dystociais is a disease, syndrome, symptom, or clinical finding; diagnosis and management depend on the history, examination, and appropriate investigations.

Dystonia

A movement disorder characterized by sustained or intermittent muscle contractions causing abnormal, often repetitive, movements and postures.

Dystonia Disorder

A movement disorder causing sustained or intermittent involuntary muscle contractions that produce twisting movements, repetitive actions, or abnormal postures.

Dystonia, Cervical

Alternate index label; see *Cervical dystonia*.

Dystonin

A protein encoded by the DST gene that helps connect the cytoskeleton to cellular structures; harmful variants can cause neuropathy and skin blistering disorders.

Dystrophia Unguium

Abnormal nail growth or structure, which may result from trauma, infection, inflammatory disease, nutritional deficiency, medication, or an inherited disorder.

Dystrophic Calcification

The deposition of calcium phosphate salts in damaged, necrotic, or degenerating tissues despite normal serum calcium levels. It occurs in areas of tissue injury, inflammation, or chronic degeneration where the local tissue environment favors calcium precipitation.

Dystrophin

A muscle-cell protein that stabilizes the muscle-fiber membrane. Dystrophin deficiency causes Duchenne or Becker muscular dystrophy.

Dystrophin-Glycoprotein Complex

A large, multi-protein oligomeric complex spanning the sarcolemma of skeletal and cardiac muscle fibers, providing a vital mechanical and signaling bridge that links the intracellular actin cytoskeleton to the extracellular matrix. Consists of dystrophin, the dystroglycan subcomplex (α - and β -dystroglycan), the sarcoglycan subcomplex (α -, β -, γ -, and δ -sarcoglycan), sarcospan, syntrophins, α -dystrobrevin, and nNOS. Provides mechanical stabilization to the sarcolemma during the stress of muscle contraction and protects myofibers from contraction-induced micro-tears; mutations in constituent or modifying proteins cause Duchenne and Becker muscular dystrophies, sarcoglycanopathies, and dystroglycanopathies.

Alternate Names:

DGC; Dystrophin-Associated Glycoprotein Complex (DAGC)

Dystrophy

Abnormal development, growth, maintenance, or degeneration of a tissue or organ, often reflecting an inherited or metabolic disorder. The term is used in conditions such as muscular dystrophy, corneal dystrophy, retinal dystrophy, and nail dystrophy. Clinical effects depend on the affected tissue and

may include progressive weakness, visual impairment, structural fragility, or loss of function. Diagnosis generally combines clinical examination with laboratory, imaging, electrophysiologic, genetic, or histopathologic studies as appropriate.

Dysuria

Painful, burning, or uncomfortable sensation experienced during urination. Most commonly indicates inflammation or infection of the lower urinary tract. Causes in females: acute cystitis, urethritis (*Chlamydia trachomatis*, *Neisseria gonorrhoeae*), vaginitis (*Candida*, *Trichomonas*), and vulvar atrophy.

Alternate Names:

Painful Urination

E

E Coli Enteritis

Intestinal infection caused by pathogenic strains of *Escherichia coli*. Depending on the strain, it may cause watery or bloody diarrhea, abdominal cramps, vomiting, or systemic complications such as hemolytic uremic syndrome.

E Coli Infection

An infection caused by *E. coli* bacteria. It may cause urinary infections, diarrhea, wound or abdominal infections, meningitis in newborns, or bloodstream infection. Some strains cause bloody diarrhea and can damage the kidneys.

E-Cigarettes And E-Hookahs

Electronic nicotine-delivery devices that heat a liquid aerosol; they can cause nicotine dependence and expose users and bystanders to irritants and toxic chemicals.

E. Coli

A group of bacteria that normally inhabit the intestines, some strains of which cause diarrheal illness, urinary infections, or other infections.

E. Coli (E. Coli 0157:H7)

Alternate index label for *E. Coli* 0157:H7.

E. Coli 0157:H7

E. coli O157:H7 is a Shiga-toxin-producing strain that can cause severe diarrhea, abdominal cramps, and bloody stool. It may lead to hemolytic uremic syndrome, especially in children; antibiotics and antidiarrheals require careful medical guidance.

E. Coli Infection

An infection caused by *Escherichia coli* bacteria, which may cause diarrhea, urinary infection, or more serious illness.

Eales Disease

An uncommon eye condition in which small veins at the edge of the retina become inflamed or blocked. New fragile blood vessels may form and bleed inside the eye, causing vision problems.

Ear

The organ used for hearing and balance. It has three parts: the outer ear, the middle ear behind the eardrum, and the inner ear, which sends hearing and balance signals to the brain.

Ear - Blocked At High Altitudes

Ear - blocked at high altitudes concerns ear - blocked at high altitudes, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Ear Barotrauma

Ear injury caused by a pressure difference between the middle ear and the outside air, often during air travel or diving. It may cause pain, fullness, hearing loss, ringing, dizziness, or a torn eardrum.

Ear Canal

A tube leading from the eardrum to the outer ear.

Ear Cartilage

Flexible connective tissue forming much of the external ear and helping maintain its shape.

Ear Cleaning (By A Doctor)

Professional removal of excess or impacted earwax by a clinician using instruments, suction, irrigation when appropriate, or other safe methods after examining the ear.

Ear Disease

A disorder of the outer, middle, or inner ear causing pain, drainage, hearing loss, tinnitus, dizziness, infection, or balance disturbance.

Ear Diseases

Any disorder affecting the structures of the ear—external ear (pinna, ear canal), middle ear (tympanic membrane, ossicles, Eustachian tube), or inner ear (cochlea, vestibular apparatus).

Ear Drainage Culture

A laboratory culture of fluid draining from the ear to identify bacteria or fungi that may cause otitis externa, chronic middle-ear infection, or another ear infection. Results must be interpreted with examination findings because colonization and contamination are possible.

Ear Emergencies

Ear emergencies concerns ear emergencies, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Ear Examination

Assessment of the external ear, ear canal, and tympanic membrane using inspection and otoscopy, often with hearing and vestibular evaluation when indicated.

Ear Foreign Body

An object lodged in the external auditory canal, commonly occurring in children. It may cause pain, hearing reduction, discharge, or canal injury and should be removed with appropriate visualization and technique.

Alternate Names:

External auditory-canal foreign body; ear-canal for-

eign body.

Ear Infection

Infection or inflammation of the ear, commonly involving the external canal or middle ear. It may cause ear pain, fever, reduced hearing, discharge, or irritability in children. Also called otitis.

Alternate Names:

otitis. Otitis (Ear Infection)

Ear Infection (Middle Ear)

Infection or inflammation in the space behind the eardrum, often causing ear pain, fever, hearing changes, or irritability in children.

Ear Infection - Acute

Acute otitis media is a rapid-onset infection and inflammation of the middle ear, often following an upper-respiratory infection. It can cause ear pain, fever, reduced hearing, or a bulging tympanic membrane; treatment depends on age, severity, and cause.

Ear Infection - Chronic

Chronic ear infection refers to persistent or recurrent inflammation or infection of the middle ear or external ear. It may cause drainage, hearing loss, perforation, or structural damage and requires examination to identify the involved compartment and cause.

Ear Microbiome

The community of microorganisms living in and on the ear, especially the external auditory canal. These microorganisms interact with earwax and the skin and may help prevent infection. The composition varies between people and can be affected by ear cleaning, moisture, hearing aids, inflammation, and infection; research on the healthy middle-ear microbiome remains limited.

Ear Infection, Outer Ear

Alternate index label; see *Swimmer's ear*.

Ear Neoplasm

A benign or malignant growth arising in the outer, middle, or inner ear; symptoms may include a persistent mass, discharge, hearing loss, pain, or facial weakness.

Ear Object In (Objects Or Insects In Ear)

Alternate index label for Objects Or Insects In Ear.

Ear Tag

Ear tag concerns ear tag, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Ear Thermometer

A thermometer that estimates body temperature by measuring heat inside the ear canal.

Ear Tube Insertion

A medical procedure, treatment, device, or care topic related to ear tube insertion; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Ear Wax

Cerumen, a mixture of secretions and shed skin that protects and cleans the ear canal. Impacted wax may cause hearing loss, fullness, tinnitus, itching, or discomfort. Cotton swabs and inserted objects can worsen impaction or injure the ear; removal should use safe clinician-recommended methods, particularly when an eardrum perforation is possible.

Alternate Names:

Earwax (Ear Wax)

Ear, Diseases Of

Ear, Diseases of is an ear-related disease or condition; symptoms, examination findings, causes, and treatment depend on the affected structure.

Ear, Low-Set

An ear positioned lower on the head than usual; it may be a normal familial feature or one finding among others in a congenital syndrome.

Ear, Nose, And Throat

A medical specialty (otolaryngology) focused on the diagnosis and treatment of disorders affecting the ears, nose, throat, head, and neck. Conditions managed include hearing loss, sinusitis, tonsillitis, voice disorders, and head and neck cancers.

Ear-Trumpet

Ear-Trumpet is an ear-related disease or condition; symptoms, examination findings, causes, and treatment depend on the affected structure.

Earache

Earache (otalgia) is pain perceived in or around the ear. Causes include ear infection, ear-canal blockage or injury, dental or jaw disorders, and referred throat disease; persistent, severe, or discharge-associated pain requires evaluation.

Earache (Otalgia)

Pain originating from the ear, a common symptom with diverse causes. Otalgia is classified as primary (originating within the ear itself) or secondary/referred (pain perceived in the ear but arising from another site).

Alternate Names:

Otalgia

Eardrum

A thin membrane separating the ear canal and middle ear.

Eardrum Repair

A medical procedure, treatment, device, or care topic related to eardrum repair; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Eardrum, Ruptured

Alternate index label; see *Ruptured eardrum (perforated eardrum)*.

Earliest Signs Of Increased Intracranial Pressure

Early increased intracranial pressure may cause

worsening headache, vomiting, visual changes, drowsiness, slowed thinking, or a change in behavior. It can result from bleeding, tumor, swelling, infection, or hydrocephalus and requires prompt evaluation when progressive.

Earlobe Creases

Earlobe creases concerns earlobe creases, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Early Decelerations

A fetal heart-rate pattern with a gradual decrease and recovery that mirrors the uterine contraction, usually caused by fetal head compression. It is generally benign when isolated, but the complete tracing and clinical situation determine whether any action is needed.

Early Detection

Identification of a disease at an early stage, when treatment may be more effective. It may result from screening or prompt evaluation of symptoms.

Early Diagnosis

Identification of a disease at an early stage, before it has progressed or caused major complications; earlier diagnosis may expand treatment options.

Early Endosome

A membrane-bound compartment that receives material brought into a cell by endocytosis and sorts it for recycling or delivery to lysosomes.

Early Intervention

Services provided to infants and young children with developmental delays or disabilities to support development, function, and participation.

Early Puberty

Alternate index label; see *Precocious puberty*.

Early Repolarization Syndrome

A primary arrhythmogenic cardiac channelopathy

characterized by a malignant early repolarization pattern on resting 12-lead electrocardiography: J-point elevation ≥ 1 mm (0.1 mV) above the isoelectric baseline in ≥ 2 contiguous inferior (II, III, aVF) and/or lateral (I, aVL, V_4 – V_6) leads, manifested as QRS slurring or notching, in patients who have survived unexplained, idiopathic ventricular fibrillation or polymorphic ventricular tachycardia. Differentiated from the common, benign early repolarization pattern of young athletes by horizontal or downsloping ST segments following the J point and family or personal history of syncope, polymorphic ventricular tachycardia, or aborted sudden cardiac arrest.

Alternate Names:

ERS; Malignant Early Repolarization

Earwax

Cerumen, a waxy secretion produced by the ceruminous glands in the outer third of the external auditory canal. Earwax lubricates and protects the skin of the ear canal, trapping dust, debris, and microorganisms to prevent them from reaching the tympanic membrane.

Earwax (Ear Wax)

Alternate index label for Ear Wax.

Earwax Blockage

A buildup of earwax that blocks the ear canal and may cause hearing loss, pain, or ringing.

Eat

To consume food; nutritional needs and safe intake depend on age, health conditions, swallowing ability, and dietary requirements.

Eating Disorder

A group of conditions characterized by abnormal eating habits and distorted body image, including anorexia nervosa, bulimia nervosa, binge eating disorder, and avoidant/restrictive food intake disorder (ARFID). Anorexia: severe calorie restriction, intense fear of weight gain, and body image disturbance with dangerously low weight.

Eating Disorders

Serious mental and physical health conditions involving disturbed eating behaviors, body image, and relationships with food and weight.

Eating Disorders, Anorexia

Alternate index label; see *Anorexia nervosa*.

Eating Disorders, Binge Eating

Alternate index label; see *Binge-eating disorder*.

Eating Disorders, Bulimia

Alternate index label; see *Bulimia nervosa*.

Eating Extra Calories When Sick - Adults

A nutrition topic concerning increased calorie intake during adult illness. Energy needs depend on the illness, appetite, swallowing ability, and medical advice; forcing food is not appropriate when it is unsafe.

Eating Habits And Behaviors

Patterns of food choice, meal timing, appetite, restriction, overeating, and other behaviors that influence nutrition and health.

Eating Out

Eating meals prepared outside the home, which may require attention to portion size, nutrition, food allergies, and food safety.

EBCT

Electron-beam computed tomography, a rapid imaging method used especially to assess the heart and calcium in coronary arteries.

Alternate Names:

Electron-beam computed tomography

Ebola

Ebola virus disease: a severe, often fatal viral hemorrhagic fever caused by Ebola virus (Filoviridae, RNA), transmitted by direct contact with blood/body fluids of infected humans/animals. Incubation 2-21 days.

Ebola Hemorrhagic Fever (Ebola Hemorrhagic Fever (Ebola HF))

Alternate index label for Ebola Hemorrhagic Fever (Ebola HF).

Ebola Hemorrhagic Fever (Ebola HF)

Alternate index label for Ebola HF.

Ebola Virus

A filovirus that causes Ebola virus disease, a severe hemorrhagic fever. The virus is transmitted through contact with infected body fluids. The incubation period is 2-21 days. Symptoms include fever, headache, myalgia, and hemorrhage. Diagnosis is made by PCR and antigen detection.

Ebola Virus Disease

A severe viral infection caused by Ebola viruses, characterized by fever and systemic illness that may progress to bleeding, shock, or organ failure.

Ebolavirus

A genus of viruses that includes the viruses causing Ebola disease, a severe infection that can cause fever, bleeding, and organ damage.

Ebstein Anomaly

A rare congenital heart defect accounting for less than 1 percent of all congenital heart disease, characterized by apical displacement of the septal and posterior leaflets of the tricuspid valve into the right ventricle, resulting in atrialization of the proximal right ventricle and a small functional right ventricular chamber.

Ebstein'S Anomaly

A congenital heart defect in which the tricuspid valve is displaced toward the right ventricle and functions abnormally; it can cause tricuspid regurgitation, right-heart enlargement, cyanosis, or arrhythmias.

EBUS-TBNA

Endobronchial ultrasound-guided transbronchial needle aspiration, a minimally invasive broncho-

scopic procedure that combines flexible fiberoptic bronchoscopy with a dedicated convex curved-array ultrasound probe on the bronchoscope tip. It enables real-time ultrasound imaging and transbronchial needle aspiration of mediastinal and hilar lymph nodes (stations 2, 4, 7, 10, 11, and 12) and central peribronchial lesions for lung cancer staging, evaluation of sarcoidosis, lymphoma, or tuberculosis, largely replacing surgical mediastinoscopy.

Alternate Names:

Endobronchial Ultrasound Transbronchial Needle Aspiration; Convex Probe EBUS

Ecbolics

Medicines or substances that stimulate uterine contractions, including oxytocin and some prostaglandins, used in selected obstetric situations.

Eccentric

Eccentric is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Eccentric Action

Muscle action in which a muscle produces force while lengthening.

Alternate Names:

Eccentric contraction

Eccentrocyte Count

The number or proportion of red blood cells with hemoglobin displaced toward one side of the cell, a smear finding that can occur with oxidative injury and some hemolytic disorders.

Ecchymosis

A flat purple, blue, or brown area caused by blood leaking into skin or subcutaneous tissue, commonly after trauma but also with platelet disorders, anticoagulants, clotting disorders, or fragile blood vessels. Size alone does not determine the cause.

Eccrine Gland

A coiled sweat gland distributed throughout the skin that secretes a watery fluid directly onto the skin

surface through a duct. Eccrine sweating supports thermoregulation and is stimulated by sympathetic cholinergic fibres; the glands are especially numerous on the palms, soles, and forehead.

Eccrine Glands

Eccrine Glands is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Eccrine Porocarcinoma

A rare malignant skin adnexal tumor arising from eccrine sweat-gland ducts; it usually presents as a slowly enlarging nodule or plaque and may recur or metastasize, so biopsy and complete excision are important.

Eccrine Poroma

A usually benign tumor arising from the eccrine sweat-gland duct, commonly presenting as a flesh-colored or reddish nodule on the hands or feet.

Ecdysone

A steroid hormone in insects and other arthropods that triggers moulting and metamorphosis; it is relevant mainly to entomology and experimental biology, not routine human therapy.

ECG

A recording of the heart's electrical activity used to evaluate rhythm and other cardiac problems.

Alternate Names:

Electrocardiogram; EKG

ECG/EKG Machine

Records electrical activity of heart. 12-lead ECG standard. Components: electrodes, leads, amplifier, recorder. Essential for cardiac diagnosis.

Echinacea

A group of flowering plants used in some herbal products marketed for colds or immune support; evidence of benefit is inconsistent and allergic reactions or drug interactions may occur.

Echinocandins

A group of antifungal medicines that weaken the fungal cell wall. They are used mainly for serious *Candida* infections and some *Aspergillus* infections.

Echinococcal Cysts

Fluid-filled parasitic cysts caused by larval *Echinococcus* tapeworms, usually occurring in the liver or lungs.

Echinococcosis

A parasitic infection caused by *Echinococcus* tapeworms. Cystic disease commonly affects the liver or lungs, while alveolar disease can invade the liver and spread to other organs; diagnosis and treatment require specialist assessment.

Echinococcus

A genus of tapeworms that can cause echinococcosis when people ingest eggs from contaminated food, water, or contact with infected animals.

Echinococcus Granulosus

A tapeworm that causes hydatid disease (cystic echinococcosis). The adult worm infects dogs, and humans are accidental intermediate hosts. Infection occurs through ingestion of eggs from dog feces. Larval cysts form in the liver and lungs, causing space-occupying lesions.

Echinococcus Multilocularis

A tapeworm species that causes alveolar echinococcosis, a slowly progressive parasitic disease that mainly affects the liver.

Echinococcus Vogeli

A tapeworm species that causes polycystic echinococcosis, mainly affecting the liver and sometimes other abdominal organs.

Echinocytes (Burr Cells)

Red blood cells with many small, evenly spaced spikes around their edges. They may occur with kidney disease, some inherited blood disorders, or as a change caused by storing or preparing the blood

sample.

Alternate Names:

Burr Cells

Echinodermata

The animal phylum that includes starfish, sea urchins, and sea cucumbers; some species produce toxins or cause injury, but the group is mainly relevant to biology and marine research.

Echinomycin

A medicine or chemical compound called echinomycin. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Echo

A common abbreviation for echocardiography, an ultrasound examination used to assess the heart's structure and function.

Echocardiogram

An echocardiogram uses ultrasound to assess cardiac chambers, valves, wall motion, blood flow, and ventricular function without ionizing radiation. It helps evaluate heart failure, valve disease, congenital defects, and other cardiac disorders.

Echocardiogram - Children

An ultrasound examination of a child's heart that assesses chambers, valves, blood flow, wall motion, and congenital or acquired structural abnormalities without ionizing radiation.

Echocardiography

Ultrasound imaging of the heart. It shows the heart chambers, valves, pumping action, and blood flow. The most common type scans through the chest; another type uses a small probe passed into the esophagus for closer views.

Echoencephalography

Ultrasound imaging of the brain, historically used especially in infants, to detect midline shift or enlarged ventricles; CT or MRI is now usually preferred.

Echogenic Bowel

An ultrasound finding in which the fetal bowel appears unusually bright, sometimes as bright as nearby bone. It may be a normal variant or may be associated with infection, chromosomal conditions, cystic fibrosis, or other fetal abnormalities.

Echogenic Intracardiac Focus

A small bright area, usually in a papillary muscle of the fetal heart, seen on ultrasound. It is commonly an incidental variant when isolated and the remainder of the examination is normal. Its significance depends on prior screening, other ultrasound findings, and the clinical context.

Echography

Medical ultrasonography, which uses high-frequency sound waves to create images of tissues, organs, blood flow, or a developing fetus.

Echolalia

The involuntary, automatic repetition of words or phrases spoken by another person, a speech phenomenon observed in various neurologic and psychiatric conditions. Immediate echolalia involves repeating sounds immediately after hearing them; delayed echolalia involves repeating phrases heard hours, days, or weeks earlier.

Echolocation

Detection of objects by emitting sound and interpreting returning echoes; it is used by some animals and inspires medical ultrasound and assistive technologies.

Echopraxia

Involuntary imitation or repetition of another person's movements, seen in some neurologic and psychiatric disorders.

Echothiophate Iodide

An irreversible cholinesterase inhibitor formerly used topically to reduce intraocular pressure in glaucoma or produce prolonged miosis; systemic cholinergic toxicity is a major risk.

Echovirus

A group of enteroviruses that commonly cause asymptomatic or mild infection but may also cause aseptic meningitis, encephalitis, rash, respiratory illness, myocarditis, or neonatal sepsis. Transmission is mainly faecal-oral or respiratory, and diagnosis is usually by nucleic-acid testing or viral culture in selected cases.

Eclampsia

Eclampsia is a new-onset seizure during pregnancy or the postpartum period associated with preeclampsia or pregnancy-related hypertension. It is an emergency requiring airway and blood-pressure management, magnesium sulfate for seizure prevention or treatment, and urgent obstetric planning.

Eclampsia Pregnancy (Pregnancy: Preeclampsia And Eclampsia)

Alternate index label for Pregnancy: Preeclampsia and Eclampsia.

Eclosion

Emergence from a protective covering or developmental stage, such as hatching from an egg or emerging from a pupa.

ECOG Performance Status

A five-point scale describing functional impairment, from 0 for fully active to 5 for death. It is widely used in clinical trials, prognostic assessment, and decisions about systemic anticancer treatment.

Econazole

A topical imidazole antifungal that inhibits fungal ergosterol synthesis and treats dermatophyte or Candida skin infections; local irritation or contact dermatitis may occur.

Economy Class Syndrome

A colloquial term for travel-associated deep-vein thrombosis or pulmonary embolism after prolonged immobility, not limited to economy-class travel.

Ecotoxicology

The study of how chemicals and other pollutants affect organisms, populations, ecosystems, and environmental health.

Ecstasy

A common street name for MDMA, a psychoactive stimulant and empathogen that can cause dangerous overheating, dehydration or hyponatremia, cardiovascular stress, and neuropsychiatric effects.

Ectasia

Abnormal widening or dilation of a tubular structure, vessel, duct, or organ. The term is used in conditions such as vascular ectasia and duct ectasia.

Ecthyma

A bacterial skin infection that extends into the dermis and produces painful punched-out ulcers covered by thick crusts. It is commonly caused by *Streptococcus* or *Staphylococcus* and may require wound care and antimicrobial treatment.

Ecthyma Gangrenosum

A serious skin infection, usually caused by *Pseudomonas* bacteria, that produces dark, dying areas of skin and ulcers. It occurs most often in people with severely weakened immune systems or very low white-blood-cell counts and requires urgent treatment.

Ectoderm

The outer layer of cells in an early embryo. It develops into the skin, nervous system, and sense organs.

Ectodermal Dysplasia

A group of inherited disorders affecting tissues derived from the embryonic ectoderm, including hair, teeth, nails, skin, and sweat glands. Findings vary but may include sparse hair, abnormal teeth, reduced sweating, heat intolerance, and recurrent infections.

Ectodermal Dysplasias

A group of inherited disorders affecting tissues derived partly from the embryonic ectoderm, commonly the hair, teeth, nails, skin, and sweat glands. Features vary and may include abnormal temperature regulation, recurrent infections, and dental problems.

Ectomesenchymoma

A rare malignant mixed tumor containing neural and mesenchymal components, usually occurring in children and requiring specialist pathology and oncology evaluation.

Ectop Parasite

A clinical disorder or finding related to ectop parasite; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Ectoparasite

A parasite that lives on the external surface of a host, such as a louse, flea, tick, or mite.

Ectopia Cordis

A rare congenital defect in which the heart is located partly or completely outside the chest.

Ectopia Lentis

A displaced eye lens. It may be caused by an injury or by inherited conditions such as Marfan syndrome, and can cause blurred vision or other vision problems.

Ectopic

Located or occurring in an abnormal position; an ectopic pregnancy implants outside the uterine cavity, most often in a fallopian tube.

Ectopic ACTH Syndrome

A condition in which a tumor outside the pituitary gland makes ACTH, a hormone that causes the body to produce too much cortisol. It is most often linked to certain lung tumors and can cause Cushing syndrome.

Ectopic Beat

An extra or premature heartbeat arising outside the normal sinoatrial-node rhythm, such as a premature atrial contraction or premature ventricular contraction.

Ectopic Cushing Syndrome

Hypercortisolism caused by an extrapituitary tumor producing adrenocorticotrophic hormone or, rarely, corticotropin-releasing hormone. It may cause hypertension, diabetes, hypokalemia, muscle weakness, infection risk, and rapid features of cortisol excess.

Ectopic Heartbeat

A heartbeat originating outside the normal sinoatrial-node rhythm. Premature atrial or ventricular beats may feel like a skipped beat, flutter, or sudden thump and can occur in healthy people or with structural heart disease.

Ectopic Pregnancy

A pregnancy that grows outside the uterus, most often in a fallopian tube. It may cause missed periods, lower-abdominal pain, or vaginal bleeding. If it ruptures, it can cause severe internal bleeding and is an emergency.

Alternate Names:

Pregnancy Ectopic (Ectopic Pregnancy) Pregnancy Tubal (Ectopic Pregnancy)

Ectopic Thyroid

Thyroid tissue located outside its normal pretracheal position, most commonly along the path of thyroid descent from the foramen cecum to the normal cervical location. The most common site is the base of the tongue (lingual thyroid).

Ectopic Thyroid Tissue

Thyroid tissue located outside the normal pretracheal position resulting from failure of normal thyroid descent during embryological development. Most commonly found at the base of the tongue termed lingual thyroid. May be the only functioning thyroid tissue. Symptoms include dysphagia, dysphonia, and upper airway obstruction.

Ectopic Tissue

Tissue located in an abnormal anatomical position, usually because of developmental displacement or implantation.

Ectopion

An outward turning or eversion of a body structure, commonly used for the eyelid or cervix.

Ectopistes

A genus of birds that includes the extinct passenger pigeon, a term mainly used in zoology and historical biology.

Ectoplasm

The cytoplasm at the outer edge of a cell, especially in descriptions of amoeboid cells; the term is also used in nonmedical spiritual claims that lack scientific support.

Ectrodactyly

A condition present at birth in which one or more middle fingers or toes are missing, leaving a deep gap in the hand or foot. It is also called split hand/foot malformation.

Ectromelia

Congenital absence or severe underdevelopment of one or more limbs; the term may also refer to a viral disease of laboratory mice.

Ectropion

Ectropion is outward turning of an eyelid, usually the lower lid, causing tearing, irritation, dryness, and incomplete eyelid closure. It may be involutional, paralytic, cicatricial, or mechanical; persistent corneal exposure can damage vision.

Eculizumab

A medicine or chemical compound called eculizumab. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Eczema

A group of inflammatory skin conditions charac-

terized by pruritic, erythematous, scaly patches. Atopic dermatitis is the most common form. Nummular eczema presents as coin-shaped lesions. Dyshidrotic eczema affects palms and soles with vesicles. Eczema herpeticum is a disseminated HSV infection in eczema patients, a medical emergency.

Eczema (Atopic Dermatitis)

A chronic inflammatory skin condition characterized by dry, itchy, and inflamed patches of skin. It is driven by a combination of genetic predisposition, immune system dysfunction, and environmental triggers.

Alternate Names:

Atopic Dermatitis

Eczema Herpeticum

A widespread viral skin infection caused by herpes simplex virus occurring in patients with pre-existing skin barrier defects, most commonly atopic dermatitis. It presents with monomorphic, umbilicated vesicles, pustules, and hemorrhagic crusted erosions, often with fever and lymphadenopathy.

Dyshidrotic Eczema

A type of eczema that causes small, intensely itchy, fluid-filled blisters on the palms, sides of the fingers, or soles of the feet. The blisters may dry, peel, and leave cracked or thickened skin. It often comes and goes and may be triggered by irritation, allergies, sweating, or stress.

Alternate Names:

Pompholyx Dyshidrosis Dyshidrotic Dermatitis

Eczema, Dyshidrotic

Alternate index label for Dyshidrotic Eczema.

Eczematous Dermatitis

A pattern of inflammatory skin disease characterized by itching, redness, scaling, dryness, oozing, or vesicles. Causes include atopic, allergic contact, irritant contact, and other forms of dermatitis.

ED

Erectile dysfunction: inability to obtain or maintain

an erection adequate for sexual activity.

Alternate Names:

Erectile dysfunction

Edatrexate

An investigational antifolate related to methotrexate, studied as an anticancer drug by disrupting folate-dependent DNA synthesis.

Edema

Swelling caused by extra fluid in body tissues. It may leave a dent when pressed or feel firm. Common causes include heart, kidney, or liver disease, poor vein drainage, blocked lymph flow, and some medicines.

Edema (Historically Dropsy)

An historic term for edema, referring to the abnormal accumulation of serous fluid in the tissues or body cavities.

Alternate Names:

historically Dropsy

Edema Pulmonary (Pulmonary Edema)

Alternate index label; see *Pulmonary Edema*.

Edema Trunk

Abnormal accumulation of fluid in the soft tissues of the chest or abdomen, producing swelling from heart, kidney, liver, lymphatic, venous, inflammatory, or medication-related causes.

Edema, Pulmonary

Alternate index label; see *Pulmonary Edema*.

Edentulism

Complete or partial absence of natural teeth. It can impair mastication, speech, nutrition, facial support, and quality of life and may be managed with removable, fixed, or implant-supported prostheses.

Edentulous

The clinical state of lacking natural dentition, classified as completely edentulous (absence of all permanent teeth) or partially edentulous. Most commonly

secondary to severe periodontal disease, extensive dental caries, trauma, or congenital anodontia. The loss of functional occlusal load triggers progressive, irreversible alveolar bone resorption and loss of the vertical dimension of occlusion, leading to facial soft-tissue collapse, pseudoprognathism, and predispositions to angular cheilitis. Systemic consequences include impaired mastication, involuntary dietary shift toward soft carbohydrates, protein-calorie malnutrition, micronutrient deficiencies, and impaired phonetic articulation. Rehabilitative strategies encompass conventional complete or removable partial dentures, implant-supported overdentures, and fixed full-arch endosseous dental implants.

Edentulousness

Complete or partial absence of natural teeth. It may result from dental disease, trauma, developmental abnormalities, or tooth extraction and can impair chewing, speech, and oral function.

Edetic Acid

Another name for EDTA, a chemical that binds certain metals. It is used in blood-collection tubes, laboratories, and selected treatments for heavy-metal poisoning.

Edodekin Alfa

An investigational biologic drug name for an immune-signaling treatment; its use depends on the specific formulation and clinical indication.

Edotecarin

An investigational anticancer drug that inhibits topoisomerase I, an enzyme needed for DNA replication.

Edrecolomab

A medicine or chemical compound called edrecolomab. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Edrophonium

A short-acting acetylcholinesterase inhibitor formerly used to help diagnose myasthenia gravis or

reverse some nondepolarizing neuromuscular blockade; it can cause dangerous bradycardia or bronchospasm.

EDTA

Ethylenediaminetetraacetic acid, a chelating agent used in laboratory blood collection, research, and selected treatments for heavy-metal poisoning.

Edward Syndrome

Trisomy 18, a chromosomal condition associated with severe developmental impairment, characteristic hand and foot findings, congenital heart defects, growth restriction, and feeding difficulties. Survival and severity vary, but mortality in infancy is high.

Edwards Syndrome

Trisomy 18, caused by an extra chromosome 18 in all or some cells or by an unbalanced rearrangement. It is associated with multiple congenital anomalies, severe developmental impairment, and high infant mortality.

Alternate Names:

Trisomy 18 (Edwards Syndrome)

Edwardsiella

A genus of gram-negative bacteria associated with aquatic environments and animals; *E. tarda* can cause gastroenteritis, wound infection, or invasive disease in humans.

Edwardsiella Hoshinae

A bacterial species mainly associated with fish and the environment that rarely causes opportunistic human infection.

Edwardsiella Ictaluri

A fish pathogen that causes enteric septicemia in catfish and is not a common human pathogen.

Edwardsiella Tarda

A Gram-negative bacterium associated with fish, reptiles, and water that can rarely cause gastroenteritis, wound infection, or bloodstream infection in people.

EEG

An EEG records electrical activity generated by the brain through scalp electrodes and helps evaluate seizures, encephalopathy, and selected sleep or neurologic disorders.

Alternate Names:

Electroencephalogram

Efalizumab

A medicine or chemical compound called efalizumab. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Efaproxiral

An investigational modifier of hemoglobin oxygen affinity studied to increase oxygen release to tissues, including in cancer research.

EFAST Examination

A rapid bedside ultrasound examination used after serious injury to look for internal bleeding, air around the lung, or fluid around the heart. A normal EFAST scan does not rule out every internal injury.

Efavirenz

A medicine or chemical compound called efavirenz. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Effacement

Thinning (shortening) of the cervix during labor, expressed as a percentage (0-100 percent), as the cervix is taken up into the lower uterine segment, enabling dilation. Occurs with cervical dilation in the first stage of labor, driven by contractions and the presenting part.

Effective Dose

A radiation-safety estimate that combines the exposure to different organs into one number. It helps compare radiation risks between tests but does not predict an individual patient's exact cancer risk.

Effective Half-Life

The time required for the activity of a radionuclide in a tissue or body to fall by one half through both physical radioactive decay and biologic elimination. It determines the duration of radiation exposure in that compartment.

Effective Reproduction Number (Rt)

The average number of secondary infections caused by an infectious person at a particular time, accounting for existing immunity, behavior, interventions, and population structure. Sustained transmission generally requires Rt above 1.

Alternate Names:

Rt

Effectiveness

The degree to which an intervention produces its intended outcome in routine practice, as distinct from efficacy under ideal study conditions.

Effector

A cell, molecule, organ, or mechanism that carries out a response after receiving a signal.

Efferent

Carrying signals away from a central structure, especially motor or autonomic signals from the central nervous system to muscles or glands.

Efferent Nerve

Efferent nerve is a nerve pathway or nerve-related structure that carries signals between the nervous system and its target; symptoms depend on the site of involvement.

Efferent Neuron

A nerve cell that carries motor or autonomic signals away from the central nervous system to muscles, glands, or other target tissues.

Efferent Pathway

Efferent pathway is a nerve pathway or nerve-related structure that carries signals between the nervous system and its target; symptoms depend on the site

of involvement.

Effervescent Tablets

Tablets designed to dissolve in water and release carbon dioxide, producing a drinkable solution; they may contain substantial sodium.

Efficacy

How well a treatment works under ideal conditions, such as in a carefully controlled clinical trial. It is different from effectiveness, which describes how well it works in everyday care.

Effusion

Abnormal accumulation of fluid in a body space, such as a joint, pleural cavity, or pericardial sac.

Effusion Pleural (Pleural Effusion (Fluid In The Chest Or On Lung))

Alternate index label for Pleural Effusion (Fluid In the Chest or On Lung).

EGD

Esophagogastroduodenoscopy is an examination of the esophagus, stomach, and duodenum with a flexible endoscope. It can evaluate symptoms or bleeding, obtain biopsies, and provide selected treatments such as hemostasis or removal of a lesion.

EGD - Esophagogastroduodenoscopy

An upper endoscopy using a flexible camera to inspect the esophagus, stomach, and duodenum and to obtain biopsies or treat selected lesions.

EGFR

Epidermal growth factor receptor, a cell-surface signaling protein; activating EGFR variants can drive some cancers and may guide targeted treatment.

EGFR Inhibition

A targeted cancer therapy using monoclonal antibodies or small-molecule tyrosine kinase inhibitors that block the epidermal growth factor receptor signaling pathway, inhibiting tumor cell proliferation, survival, and metastasis. Used in colorectal can-

cer, non-small cell lung cancer, and head and neck squamous cell carcinoma.

Egg

Egg is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Egg Allergy

An immune reaction to proteins in eggs that may cause skin, digestive, or breathing symptoms.

Egg Shell

The hard calcium-containing outer covering of an egg; in medical descriptions, an “eggshell” calcification pattern may refer to a thin rim of calcification around a lesion or lymph node.

Egg-Cell

Egg-Cell is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Eggerthella

A genus of anaerobic bacteria found in the human intestine that can rarely cause bacteremia, abscesses, or other opportunistic infection.

Eggerthella Lenta

A Gram-positive anaerobic bacterium found in the human intestine that can rarely cause bloodstream or other opportunistic infection.

Eggshell Nail

A nail that becomes unusually thin, soft, flexible, and sometimes whitish or concave. It may be associated with nutritional deficiency, chronic illness, aging, or repeated trauma; persistent nail changes should be evaluated with the skin and other symptoms in mind.

Ego

Ego is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Egophony

A change in voice sounds heard through a stethoscope in which a spoken “E” sounds like “A.” It may occur when lung tissue is filled or hardened, such as with pneumonia.

Ehlers-Danlos Syndrome

A group of hereditary connective tissue disorders caused by defects in collagen synthesis or processing.

Alternate Names:

EDS (Ehlers-Danlos Syndrome) Elastic Skin (Ehlers-Danlos Syndrome)

Ehlers-Danlos Syndrome

Alternate index label for Ehlers-Danlos Syndrome.

EHR

Electronic health record, a digital record of a person’s health information that can include data from multiple clinicians and care settings.

Ehrlichia

A genus of bacteria transmitted mainly by ticks that infect white blood cells and cause ehrlichiosis, a febrile illness.

Ehrlichia And Anaplasma

Bacteria spread by ticks that can cause fever, headache, muscle aches, and low blood-cell counts. Ehrlichia mainly infects certain white blood cells, while Anaplasma mainly infects another type of white blood cell.

Ehrlichiosis

Ehrlichiosis is a tick-borne infection caused by Ehrlichia bacteria, commonly producing fever, headache, malaise, low blood counts, and elevated liver enzymes. ehrlichiosis) and E. ewingii (granulocytic). Endemic in the southeastern and south-central US. Presents with fever, headache, myalgia, malaise, and thrombocytopenia/leukopenia.

Ehrlichiosis And Anaplasmosis

Tick-borne bacterial infections that commonly cause fever, headache, muscle aches, and low blood-cell

counts.

Eicosadienoic Acid

A 20-carbon polyunsaturated fatty acid with two double bonds, present in membrane lipids and studied as a precursor in inflammatory and metabolic pathways.

Eicosanoid

A substance made from certain fats that helps control inflammation, blood-vessel widening or narrowing, platelets, and smooth-muscle activity. Prostaglandins, leukotrienes, and thromboxanes are examples.

Eicosanoid Degradation

Enzymatic inactivation and breakdown of eicosanoids, which limits the duration and strength of their inflammatory, vascular, and platelet effects.

Eicosanoid Metabolism

The synthesis and breakdown of prostaglandins, leukotrienes, thromboxanes, and related signaling lipids from polyunsaturated fatty acids.

Eicosanoid Receptor

A cell-surface or nuclear receptor that responds to prostaglandins, leukotrienes, thromboxanes, or related lipid mediators and regulates inflammation, vascular tone, or smooth muscle.

Eicosapentaenoic Acid

The omega-3 fatty acid EPA, found in oily fish and supplements, incorporated into membranes and used to produce signaling mediators with generally less inflammatory activity than arachidonic-acid products.

Eicosatetraenoic Acid

A 20-carbon fatty acid with four double bonds; the best-known form is arachidonic acid, a membrane precursor of prostaglandins, leukotrienes, and other eicosanoids.

Eicosatrienoic Acid

A 20-carbon fatty acid with three double bonds that can serve as a membrane lipid or precursor for biologically active lipid mediators.

Eicosenoic Acid

A 20-carbon monounsaturated fatty acid found in some plant and marine oils and incorporated into cellular and dietary lipids.

Eighth Cranial Nerve

The vestibulocochlear nerve, which carries hearing and balance signals from the inner ear to the brain.

Alternate Names:

Vestibulocochlear nerve; auditory nerve

Eikenella

A genus of gram-negative bacteria in the mouth; *Eikenella corrodens* can cause wound and soft-tissue infection, especially after human bites or clenched-fist injuries.

Eikenella Corrodens

A slow-growing Gram-negative bacterium found in the mouth that can cause human bite or clenched-fist wounds, abscesses, and endocarditis.

Eisenmenger Complex

Severe pulmonary hypertension caused by a long-standing congenital heart defect with abnormal blood flow between the heart chambers or vessels.

Eisenmenger Syndrome

Reversal of a left-to-right cardiac shunt due to irreversible pulmonary hypertension, resulting in right-to-left shunting and cyanosis. It occurs in untreated congenital heart defects with large left-to-right shunts (VSD, ASD, PDA) that cause progressive pulmonary vascular disease.

Eisenmenger's Syndrome

Pulmonary vascular disease caused by a long-standing congenital heart shunt, eventually reversing the shunt and causing cyanosis.

Ejaculation

The expulsion of seminal fluid from the male urethra during sexual climax, a reflex coordinated by the sympathetic nervous system (T10-L2) and involving contraction of the epididymis, vas deferens, seminal vesicles, prostate, and bulbospongiosus muscle.

Ejaculatory Duct

A duct formed by the union of the vas deferens and the duct of a seminal vesicle. It passes through the prostate and opens into the urethra.

Ejaculatory Ducts

Ejaculatory Ducts is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Ejection Click

A high-pitched sound occurring at the beginning of systole, caused by opening of a stenotic semilunar valve. Aortic ejection click is heard in bicuspid aortic valve and aortic stenosis with pliable leaflets. Pulmonary ejection click is heard in pulmonary stenosis and disappears as the valve becomes more calcified and immobile.

Ejection Fraction

The percentage of blood in a ventricle ejected during contraction, calculated as stroke volume divided by end-diastolic volume. It helps describe ventricular systolic function and classify some forms of heart failure, but it does not measure filling pressures, congestion, symptoms, or overall heart-failure severity by itself. Interpretation depends on the ventricle studied, imaging method, clinical findings, and applicable definitions.

EKG

An electrocardiogram, a recording of the heart's electrical activity through electrodes placed on the skin. It can show heart rate and rhythm and may provide clues to ischemia, heart attack, conduction problems, or heart enlargement.

Alternate Names:

Electrocardiogram; ECG

Elastic Cartilage

Flexible cartilage containing abundant elastic fibers in its matrix. It supports structures that require shape retention with flexibility, including the external ear, epiglottis, and auditory tube.

Elastic Fiber

Elastic fiber is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Elastic Skin (Ehlers-Danlos Syndrome)

Alternate index label for Ehlers-Danlos Syndrome.

Elastic Tissue

Connective tissue containing elastic fibers that allow structures such as arteries, lungs, skin, and ligaments to stretch and recoil.

Elasticity

The ability of a tissue or material to return toward its original shape after being stretched or compressed; it contributes to lung, vessel, skin, and muscle function.

Elastics

Elastic bands used in medical or dental treatment, such as orthodontic bands, compression supports, or surgical ligatures; their purpose depends on the application.

Elastin

A flexible connective-tissue protein that allows skin, blood vessels, and other tissues to stretch and recoil.

Elastofibroma

A benign, slowly growing fibroelastic tumor-like lesion, most often beneath the lower scapula in older adults.

Elastography

Imaging tissue stiffness. Used for liver fibrosis assessment, breast lesion characterization, thyroid nodule evaluation.

Elastomer

A flexible polymer that can stretch and return to its original shape, used in some medical devices, tubing, seals, and implants.

Elastosis

Abnormal accumulation or degeneration of elastic fibers in tissue, commonly associated with chronic sun damage and some degenerative or genetic disorders.

Elbow

The joint connecting the upper arm to the forearm, formed mainly by the humerus, radius, and ulna; it permits bending, straightening, and forearm rotation.

Elbow Extensor

A muscle that straightens the elbow, chiefly the triceps brachii and anconeus.

Elbow Flexors

Muscles that bend the elbow, chiefly the biceps brachii, brachialis, and brachioradialis.

Elbow Joint

A synovial hinge complex comprising the humeroulnar and humeroradial joints. It primarily permits flexion and extension, while the proximal radioulnar component permits forearm rotation.

Elbow Pain

Pain around the elbow caused by tendon disorders, arthritis, bursitis, nerve compression, fracture, instability, or referred pain from the neck or shoulder.

Alternate Names:

Pain Elbow (Elbow Pain)

Elbow Replacement

A medical procedure, treatment, device, or care topic related to elbow replacement; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Elbow, Cellulitis Of The

A bacterial infection of the skin and soft tissues around the elbow, causing redness, warmth, swelling, and tenderness and sometimes spreading rapidly.

Elbow, Tip Of The

The olecranon, the bony prominence at the back of the elbow formed by the proximal ulna.

Elder Abuse

Intentional or neglectful act causing harm to older person. Types: physical, emotional, sexual, financial, neglect. Risk factors: social isolation, caregiver stress, cognitive impairment, dependency. Mandatory reporting required. Assessment: direct question, observation, collateral history.

Elderly

Person aged 65 or older. Divisions: young-old (65-74), middle-old (75-84), old-old (85+). Heterogeneous population with varying health status, needs, preferences.

Elective

Planned in advance rather than required urgently; an elective procedure may be scheduled after considering its benefits, risks, and alternatives.

Elective Abortion

A planned termination of pregnancy, performed with medication or a procedure according to gestational age, patient preference, and local clinical requirements.

Elective Surgery

Surgery planned in advance because it is not required immediately to prevent serious harm; it may still be medically necessary.

Electric Impedance

Opposition to the flow of alternating electrical current, measured in ohms; changes in impedance are used in physiology, bioimpedance, and some medical devices.

Electric Organ

An organ that produces electric discharges, found in some fish and used for defense, hunting, navigation, or communication.

Electric Shock

Injury caused when electric current passes through the body. It may cause burns, muscle injury, abnormal heart rhythms, breathing failure, or neurologic effects.

Electrical Alternans

Beat-to-beat variation in ECG QRS amplitude or axis, classically caused by a swinging heart in a large pericardial effusion and potentially indicating tamponade.

Electrical Cardioversion

Delivery of a synchronized electrical shock timed to the R wave to terminate selected tachyarrhythmias, such as atrial fibrillation, atrial flutter, or certain supraventricular tachycardias. Sedation, thromboembolic risk, and the patient's hemodynamic status must be assessed.

Electrical Conductance

The ability of a material or body tissue to carry electrical current; it depends largely on the movement of charged particles.

Electrical Injury

Tissue damage caused by passage of electrical current through the body. Severity depends on voltage, current (alternating vs. direct), pathway, duration, and resistance. High-voltage (>1000V): extensive internal tissue damage (muscle necrosis, compartment syndrome, rhabdomyolysis) with relatively small skin burns (entry/exit wounds).

Electrical Stimulation

The therapeutic application of controlled electrical currents to nerves or muscles. Examples include transcutaneous electrical nerve stimulation for pain and neuromuscular stimulation for selected muscle or movement problems.

Electrical Synapse

A direct connection between cells through gap junctions that allows ions and electrical signals to pass rapidly from one cell to another.

Electrocardiogram

An electrocardiogram records the heart's electrical signals and helps identify rhythm disturbances, conduction disease, ischemia, infarction, and chamber strain.

Alternate Names:

ECG or EKG.

Electrocardiogram

A recording of the heart's electrical activity through skin electrodes, used to assess rhythm, conduction, ischemia, prior heart injury, and other cardiac abnormalities.

Electrocardiography

The process of recording the heart's electrical activity from skin electrodes, usually as a 12-lead ECG or EKG. It evaluates heart rate and rhythm, conduction, chamber enlargement, ischemia, infarction, and some electrolyte or drug effects.

Electrocauterization

A procedure that uses controlled electrical energy to heat and destroy tissue or seal blood vessels. It may control bleeding or remove selected lesions, but can cause burns, scarring, smoke exposure, or injury to adjacent tissue.

Electrocautery

Use of heat produced by an electrical current to cut tissue or control bleeding.

Electrocoagulation

A surgical technique using high-frequency electrical current to destroy tissue or stop bleeding by coagulating proteins, also known as diathermy or cauterization.

Alternate Names:

diathermy or cauterization.

Electroconvulsive Therapy

A neuromodulation treatment involving the induction of a generalized seizure under general anesthesia using electrical current applied to the scalp. It is the most effective treatment for severe depression, especially with psychotic features, catatonia, or suicidal ideation.

Electroconvulsive Therapy

A medical treatment that induces a brief controlled seizure under anesthesia and muscle relaxation, used mainly for severe depression, catatonia, or selected acute psychiatric illness.

Electrocution

Injury from electrical current. Types: low voltage (<1000V), high voltage (>1000V). Effects: cardiac arrhythmias, muscle contractions, burns, neurological damage. Management: ABCs, cardiac monitoring, tetanus prophylaxis, wound care, check for compartment syndrome.

Electrode

A conductive contact that transfers electrical current to or from tissue, a device, or a laboratory sample.

Electroencephalogram

EEG: a recording of electrical activity of the brain via scalp electrodes, reflecting cortical neuronal activity. Uses: diagnosis and classification of seizures/epilepsy (interictal epileptiform discharges, spikes/waves), status epilepticus monitoring, assessment of encephalopathy, sleep disorders, and determination of brain death.

Electroencephalogram

A recording of electrical activity from the brain through scalp electrodes, used to evaluate seizures, encephalopathy, sleep disorders, and selected disorders of consciousness.

Electroencephalography

A non-invasive neurophysiological test that records the electrical activity of the brain using electrodes placed on the scalp according to the international 10-20 system.

Electrogastrogram

A recording of the stomach's electrical activity, usually obtained from electrodes on the abdominal skin to assess gastric slow waves and rhythm.

Electrolysis

Permanent hair removal using electrical energy delivered through a fine probe into a hair follicle.

Electrolyte

A mineral that carries an electrical charge in body fluids, such as sodium, potassium, or calcium.

Electrolyte Balance

Homeostasis of sodium, potassium, calcium, magnesium, chloride, phosphate. Regulated by kidneys, hormones (aldosterone, ADH, PTH). Imbalance causes neuromuscular, cardiac, metabolic dysfunction.

Electrolyte Imbalance

Abnormal serum concentration of electrolytes (sodium, potassium, calcium, magnesium, phosphate, chloride). Hyponatremia (<135 mEq/L): most common electrolyte disorder; causes include SIADH, heart failure, hypovolemia, and polydipsia. Hypernatremia (>145 mEq/L): free water deficit from poor intake, diabetes insipidus, or diuretics.

Electrolytes

Electrically charged minerals in body fluids, including sodium, potassium, chloride, bicarbonate, calcium, magnesium, and phosphate. They support nerve conduction, muscle contraction, fluid balance, and acid-base regulation.

Electrolytes - Urine

A urine test measuring electrolytes such as sodium, potassium, chloride, and sometimes calcium or phosphate. It helps assess kidney handling of electrolytes and investigate volume status, acid-base disorders, hypertension, or abnormal blood electrolyte levels.

Electromagnetic Navigation

Bronchoscopy

A guidance system for bronchoscopy that uses electromagnetic fields to navigate to peripheral lung lesions. It allows biopsy of lesions not visible on standard bronchoscopy. It is more accurate than conventional transbronchial biopsy for peripheral lesions. It may reduce the need for CT-guided biopsy.

Electromyography

A test that records the electrical activity of skeletal muscles, often together with nerve-conduction studies, to evaluate peripheral nerve, nerve-root, neuromuscular-junction, and muscle disorders.

Electromyography

A test recording electrical activity in skeletal muscle, often combined with nerve-conduction studies, to evaluate neuropathy, radiculopathy, neuromuscular-junction disease, or myopathy.

Electron Beam Computerized Tomography

A rapid CT imaging technique using an electron beam to produce images, historically used especially to detect and quantify coronary-artery calcium.

Electron Microscope

A microscope that uses electrons rather than visible light to produce highly magnified images of cells, tissues, microorganisms, or materials.

Electron Transport Chain

Series of protein complexes (I-IV) and mobile carriers (ubiquinone, cytochrome c) in the inner mitochondrial membrane that transfer electrons from NADH and FADH₂ to oxygen, pumping protons to create electrochemical gradient used by ATP synthase. Complex I (NADH dehydrogenase): NADH $\rightarrow$ ubiquinone.

Electron-Beam Computed Tomography

A rapid computed-tomography technique used to image the heart and detect calcium deposits in arteries.

Alternate Names:

EBCT

Electronic Fetal Monitoring

Continuous or intermittent monitoring of fetal heart rate and uterine contractions to assess fetal response during pregnancy or labor.

Electronystagmography

A test that records eye movements using electrodes around the eyes, usually during vestibular stimulation. It helps assess nystagmus and disorders of the inner ear or vestibular pathways, although video-based testing is now often preferred.

Electrophoresis

A laboratory method that separates charged molecules according to their movement through a medium in an electric field.

Electrophysiologic Testing

A procedure that records or provokes electrical activity in the heart to investigate suspected arrhythmias.

Electrophysiology

The study of electrical activity in living cells and tissues; in clinical medicine it includes tests of heart rhythm and nerve or muscle function.

Electrophysiology Study

Invasive cardiac catheterization with multielectrode catheters placed in the heart to map the cardiac conduction system. Measures: sinus node function (SNRT, SACT), AV node function (AH interval, HV interval), and inducibility of arrhythmias (programmed electrical stimulation).

Electroretinography

A test that records the electrical responses of the retina to flashes or patterned light. It helps evaluate inherited or acquired retinal dysfunction when examination or imaging alone is insufficient.

Electrosurgery

A medical procedure involving electrosurgery. In-

dications, risks, preparation, and follow-up depend on the patient and the specific procedure.

Electrosurgical Unit

A generator that delivers high-frequency electrical energy through an active electrode to cut tissue or achieve hemostasis. Monopolar and bipolar configurations differ in current pathway and safety considerations; burns, unintended tissue injury, and interference with implanted devices are possible complications.

Element

Element is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Elephant Ear Poisoning

Toxic irritation after eating or handling plants such as taro or elephant ear that contain calcium oxalate crystals, causing burning and swelling of the mouth and throat.

Elephantiasis

Massive enlargement of a body part, typically a limb or the scrotum, caused by chronic lymphedema from obstruction of lymphatic vessels. The most common cause worldwide is lymphatic filariasis, a mosquito-borne parasitic infection by *Wuchereria bancrofti*, *Brugia malayi*, or *Brugia timori*.

Elesclomol Sodium

An investigational anticancer drug that binds copper and can increase oxidative stress inside cancer cells.

Elevated Blood Pressure

Blood pressure consistently above the healthy range, increasing long-term risk of heart, brain, kidney, and vascular disease.

Elevated Hemidiaphragm

An abnormally high position of one side of the diaphragm on imaging, which can result from volume loss, nerve or muscle dysfunction, abdominal disease,

or other causes.

Elevated Liver Enzymes

Higher-than-expected blood levels of enzymes released by injured or stressed liver cells, commonly including ALT and AST. Causes range from fatty liver disease and medicines to viral hepatitis, alcohol-related injury, and biliary obstruction; interpretation depends on the pattern and clinical context.

Alternate Names:

Abnormal liver enzymes; elevated hepatic enzymes.

Elevation

Superior movement of a body part, such as raising the mandible or shoulder girdle.

Elevator

A surgical instrument used to lift, separate, or strip tissue, bone, or a tooth from surrounding structures.

Eligible

Meeting the specified requirements for a treatment, clinical trial, screening program, benefit, or other healthcare service.

Elimination

Elimination is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Elimination Diets

Diets in which suspected foods are removed and later reintroduced to help identify food-related symptoms.

Alternate Names:

Elimination diet

Elimination Half-Life

Time for plasma concentration to decrease by 50%. Determines dosing interval, accumulation, withdrawal from exposure. Affected by renal/hepatic function, age, drug interactions.

Elimination Patterns

The frequency, timing, amount, and characteristics

of urination and bowel movements, which help assess hydration, gastrointestinal function, continence, and disease.

Elinafide

An investigational anticancer drug designed to bind DNA and interfere with the growth of cancer cells.

ELISA Blood Test

A blood test using an enzyme-linked immunosorbent assay to detect or measure a specific antibody, antigen, hormone, protein, or other substance. Results depend on the assay, timing, specimen, and pretest probability.

Elixir

Elixir is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Elizabethkingia

A genus of environmental Gram-negative bacteria that can cause severe opportunistic bloodstream, respiratory, or central nervous system infection.

Ellagic Acid

A plant-derived polyphenol found in foods such as berries and nuts.

Elliptocyte Count

The number or proportion of oval or elliptical red blood cells on a blood smear; increased elliptocytes may occur in hereditary elliptocytosis and some anemias.

Elliptocytosis

A condition in which many red blood cells have an oval or elliptical shape.

Ellis-Van Creveld Syndrome

An inherited skeletal dysplasia characterized by short limbs, extra fingers or toes, narrow thorax, abnormal teeth or nails, and frequent congenital heart disease.

Elongin

A protein complex that promotes transcriptional elongation by RNA polymerase II and participates in regulation of gene expression.

Elotuzumab

A medicine or chemical compound called elotuzumab. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Eltrombopag

An oral thrombopoietin-receptor agonist that stimulates platelet production and is used for selected immune thrombocytopenia, aplastic anemia, and other approved indications.

Eltrombopag Olamine

A medicine that stimulates the thrombopoietin receptor and increases platelet production in selected blood disorders.

Emaciation

Extreme wasting and loss of body mass, especially muscle and fat, caused by severe illness, malnutrition, or other chronic conditions.

Embolectomy

Surgical removal of embolus from blood vessel. Indications: acute arterial occlusion, pulmonary embolism. Techniques: Fogarty catheter, open surgical extraction. Often combined with thrombolysis.

Embolectomy (Vascular)

Removal of an embolus from a blood vessel by surgery or a catheter-based procedure.

Alternate Names:

Vascular

Emboli

Plural of embolus: particles or material carried through the bloodstream that can lodge in a smaller vessel and obstruct blood flow, such as a clot, fat, air, or infected material.

Embolic Stroke

An ischemic stroke caused by a clot or other material that travels from elsewhere and blocks a brain artery.

Embolic Stroke (Brain Embolism)

A stroke caused by a clot or other material traveling to and blocking an artery supplying the brain.

Alternate Names:

Brain Embolism Cerebral Embolism

Embolism

An embolism is the occlusion of a blood vessel by a detached mass that has traveled through the bloodstream from its site of origin. The most common form is thromboembolism, where a fragment of a thrombus dislodges and travels to a distant site through the venous or arterial circulation.

Embolism Pulmonary (Pulmonary Embolism)

Alternate index label for Pulmonary Embolism.

Embolization

Deliberate occlusion of a blood vessel or vascular malformation using materials such as coils, particles, glue, or plugs. It is used to control hemorrhage, treat aneurysms or malformations, reduce tumor vascularity, or induce organ infarction in selected therapies.

Embolization (Vascular)

Intentional blockage of a blood vessel using a catheter-delivered material to control bleeding, reduce blood flow, or treat a vascular lesion.

Alternate Names:

Vascular

Embolization Therapy

Image-guided occlusion of vessels. Used for hemorrhage control, tumor devascularization, aneurysm treatment, varices.

Embolus

A detached intravascular mass, such as a blood

clot, fat droplet, air bubble, or other material, that travels through the circulation and may obstruct a vessel.

Embryo

The developing human organism from fertilization through the end of the eighth week after fertilization. From the ninth week after fertilization, it is termed a fetus.

Embryo Neoplasm

A tumor arising from embryonal or primitive cells, usually in a child; diagnosis depends on the tissue of origin, microscopic features, molecular findings, and stage.

Embryo Transfer

The critical concluding step of an in vitro fertilization or intracytoplasmic sperm injection cycle, wherein one or more laboratory-fertilized cleavage-stage embryos or blastocysts are loaded into a sterile, flexible catheter and gently deposited into the mid-portion of the uterine endometrial cavity under transabdominal ultrasound guidance. May be performed as a fresh transfer or following cryopreservation and programmed endometrial preparation (frozen embryo transfer [FET]).

Alternate Names:

ET; Intrauterine Embryo Transfer

Embryogenesis

The sequence of fertilization, cell division, differentiation, migration, and organ formation through which an embryo develops.

Embryology

The study of development from fertilization through the embryonic and fetal stages, including the formation of organs and body structures.

Embryoma

Embryoma is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Embryonal Carcinoma

An aggressive germ-cell tumor composed of primitive malignant cells, occurring in the testis, ovary, or other sites and often treated with chemotherapy and surgery.

Embryonal Rhabdomyosarcoma

A malignant skeletal-muscle-lineage tumor that commonly affects young children and may arise in the head and neck, urinary tract, or reproductive tract.

Embryonal Tumor Of The Central Nervous System

A highly cellular malignant pediatric CNS tumor with primitive or embryonal morphology. Diagnosis depends on integrated histologic and molecular classification, and treatment usually combines surgery, radiation when appropriate, and chemotherapy.

Embryonal Tumors

Cancers that arise from cells resembling developing embryonic tissue, often occurring in children and commonly affecting the nervous system.

Embryonic

Embryonic is a developmental, genetic, reproductive, or family-related term describing a stage, trait, relationship, or pregnancy history; its meaning depends on the clinical context.

Embryonic Cell

A cell of an early embryo capable of division and differentiation into specialized tissues; embryonic stem cells have broad developmental potential.

Embryonic Induction

A developmental process in which one group of embryonic cells signals nearby cells to develop into a particular tissue or structure.

Embryonic Mosaic

Embryonic mosaic is a developmental, genetic, reproductive, or family-related term describing a stage, trait, relationship, or pregnancy history; its meaning depends on the clinical context.

Embryonic Stage

Embryonic stage is a developmental, genetic, reproductive, or family-related term describing a stage, trait, relationship, or pregnancy history; its meaning depends on the clinical context.

Emergence Agitation

A state of confusion, restlessness, and inconsolable crying during emergence from anesthesia, occurring in 10 to 80 percent of pediatric patients and 5 to 10 percent of adults. Risk factors include sevoflurane induction, short procedures, parental separation, and preexisting behavioral disorders.

Emergency Airway Puncture

An emergency procedure that creates an airway through the cricothyroid membrane or trachea when severe upper-airway obstruction prevents adequate ventilation and intubation is not possible. It is a temporary, high-risk lifesaving intervention requiring trained personnel.

Emergency Care

Immediate assessment and treatment for a potentially serious or time-sensitive illness or injury.

Emergency Code

A hospital alert used to summon a designated response to an urgent event; the meaning of a specific code varies by institution.

Emergency Contraception

Contraception used after unprotected intercourse or contraceptive failure to reduce the chance of pregnancy. Options include oral medicines and a copper intrauterine device; effectiveness depends on the method, timing, medical history, and access to care.

Emergency Department

Hospital department providing immediate medical care for acute illnesses and injuries. Staffed by emergency physicians and supported by specialists. Key principles: triage (prioritize by severity), stabilization, diagnosis, treatment, disposition.

Emergency Medicine

Emergency Medicine is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Emergency Procedures

Planned actions for responding to urgent events such as spills, contamination, equipment failure, or radiation exposure while protecting patients and staff.

Emerging Infectious Disease

New or reemerging infectious diseases. Examples: COVID-19, Zika, Ebola, MERS.

Emery-Dreifuss Muscular Dystrophy

A rare inherited muscular dystrophy characterized by early contractures, especially of the elbows, Achilles tendons, and neck, followed by slowly progressive humeroperoneal muscle weakness and wasting. Cardiac involvement, including conduction abnormalities, arrhythmias, and cardiomyopathy, is common and may occur before major muscle weakness; ongoing cardiac assessment is essential.

Emesis

The forceful expulsion of gastric contents through the mouth, a complex reflex coordinated by the vomiting center in the medulla oblongata.

Emetic

A pharmacologic agent or substance that stimulates or precipitates emesis (vomiting). Acts either peripherally by irritating gastric mucosal afferent sensory pathways (projecting via vagal fibers to the solitary nucleus) or centrally by activating the chemoreceptor trigger zone located in the area postrema of the fourth ventricle outside the blood-brain barrier. Historically utilized in acute toxicology (e.g., syrup of ipecac, apomorphine); however, routine use in toxic ingestions is no longer recommended due to risks of chemical aspiration pneumonitis, esophageal rupture (Boerhaave syndrome), and interference with activated charcoal administration.

Emetics

Substances that induce vomiting. They are generally not used to treat poisoning because vomiting may cause aspiration or additional injury.

Emetine

A plant-derived alkaloid formerly used to induce vomiting and treat some protozoal infections; it is toxic at high doses.

EMG

A test that evaluates muscle activity and the nerves controlling muscles.

Alternate Names:

Electromyography

Eminentia

An anatomical term for a raised or prominent area of a body structure.

Emission

The discharge or release of a substance, usually a fluid. In male reproduction, emission refers to the movement of semen into the posterior urethra in preparation for ejaculation.

Emitefur

An investigational fluoropyrimidine-related anticancer drug that interferes with DNA and RNA production.

Emmenagogue

A substance traditionally claimed to stimulate menstrual bleeding. Evidence and safety vary, and some herbal products can cause poisoning, miscarriage, or drug interactions; missed or abnormal periods need proper evaluation rather than self-treatment.

Emmetropia

The normal refractive state of the eye, in which parallel light focuses on the retina without accommodation.

Emodin

A naturally occurring anthraquinone compound

found in some plants and studied for laxative, anti-inflammatory, and anticancer effects; clinical safety and benefit are not established for routine treatment.

Emollient

A substance applied to the skin to soften and moisturize it by reducing water loss and improving the skin barrier. Emollients are commonly used in eczema and dry skin.

Emotion

A coordinated feeling state involving subjective experience, brain and body responses, behavior, and interpretation of events.

Emotional Adjustment

The process of adapting thoughts, feelings, and behavior to illness, loss, stress, or major life change; difficulty adjusting may cause distress or impairment.

Emotional Dependency

Reliance on another person for reassurance, self-worth, or emotional stability to a degree that may limit independence or contribute to unhealthy relationship patterns.

Emotional Disorders

A clinical disorder or finding related to emotional disorders; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Emotional Dysregulation Disorder (Borderline Personality Disorder)

Alternate index label for Borderline Personality Disorder.

Emotional Eating

Eating in response to emotions such as stress, boredom, or sadness rather than primarily in response to physical hunger. Cognitive behavioral therapy may help.

Empathy

The ability to understand or appreciate another person's feelings and perspective; it supports therapeutic communication and compassionate care.

Empedobacter

A genus of Gram-negative environmental bacteria that rarely causes opportunistic human infection.

Empedobacter Brevis

A Gram-negative environmental bacterial species that rarely causes opportunistic infection in people.

Empedobacter Falsenii

A Gram-negative bacterial species found in the environment and rarely associated with human opportunistic infection.

Emphysema

A form of COPD characterized by permanent, abnormal enlargement of airspaces distal to the terminal bronchioles, accompanied by destruction of their walls. Centriacinar emphysema affects the respiratory bronchioles and is associated with smoking.

Emphysema (Lung Condition)

Alternate index label for Lung Condition.

Emphysematous Aortitis Due To Clostridium Septicum

A rare, life-threatening infection of the aortic wall by *Clostridium septicum*, characterized by gas formation within the aortic wall visible on imaging. It typically occurs in immunocompromised patients and may present with sepsis, back pain, or aortoenteric fistula.

Emphysematous Cholecystitis

A severe, gas-forming infection of the gallbladder wall caused by anaerobic organisms, most commonly *Clostridium* species and *Escherichia coli*, occurring predominantly in diabetic men. Gas within the gallbladder wall or lumen is visible on imaging. It is a surgical emergency requiring urgent cholecystectomy and broad-spectrum antibiotics.

Emphysematous Hepatitis

A rare, life-threatening gas-forming infection of the liver parenchyma, most commonly caused by *Clostridium* species or mixed enteric organisms in immunocompromised or diabetic patients. Imaging reveals gas within the liver parenchyma.

Emphysematous Prostatitis

A rare, severe gas-forming infection of the prostate gland, most commonly caused by *Escherichia coli* in diabetic or immunocompromised patients. It presents with fever, perineal pain, urinary retention, and sepsis. Imaging shows gas within the prostatic parenchyma.

Emphysematous Pyelonephritis

Severe necrotizing infection of renal parenchyma with gas formation. Almost exclusively in diabetic patients. Presents with high fever, flank pain, and sepsis. CT shows gas within renal parenchyma. A urological emergency requiring aggressive antibiotics and usually nephrectomy.

Empiric Risk

A risk estimate based on observed population or clinical data rather than solely on a theoretical model; the estimate depends on the population and outcome studied.

Empirical

Based on observation, measurement, or experience rather than only on theory; empirical treatment may begin before a specific cause is confirmed.

Empirical Treatment

A medical procedure, treatment, device, or care topic related to empirical treatment; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Employee

A person who performs work for an employer under an employment agreement; occupational health considers the person's work exposures, safety, and ability to perform duties.

Employment Status

A person's current work situation, such as employed, unemployed, retired, on leave, or unable to work; it may affect health, insurance, income, and rehabilitation planning.

Empty Calories

A nontechnical term for foods or drinks that provide substantial energy but relatively few vitamins, minerals, fiber, or other beneficial nutrients.

Empty Sella Syndrome

Herniation of subarachnoid space into the sella turcica, compressing the pituitary gland and flattening it against the sellar floor. Primary: due to congenital weakness of the diaphragma sellae. Secondary: post-surgery, post-radiation, or pituitary apoplexy (Sheehan syndrome). Most are asymptomatic.

Empyema

A collection of pus in a normally sterile body cavity, most often infected pleural fluid around the lung, causing fever, chest pain, cough, and breathlessness.

Empyema Necessitans

Extension of pleural empyema through the chest wall. Presents as a painful, fluctuant mass. Requires drainage and antibiotics.

EMR

Electronic medical record, a digital record of health information maintained within a particular health-care organization or practice.

Emtricitabine

A nucleoside reverse-transcriptase inhibitor used in combination antiretroviral therapy for HIV and in selected preexposure prophylaxis regimens; kidney function and hepatitis B status are important considerations.

Emulsion

Emulsion is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and ex-

posure circumstances.

Enalapril

A medicine or chemical compound called enalapril. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Enalaprilat

A medicine or chemical compound called enalaprilat. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Enamel

Hard, highly mineralized outer layer of tooth crown. Avascular, aneural, cannot regenerate. Enamel hypoplasia causes defects. Dental caries begins in enamel. Fluoride strengthens enamel.

Enamel Hypoplasia

Defective enamel quantity due to systemic disturbance during development. Pits, grooves, or thin enamel.

Enamel Organ

The developing tooth structure that gives rise to enamel-producing ameloblasts and helps shape the crown of a tooth.

Enamel Pearl

Small mound of enamel on root surface. Usually bifurcation area of molars. May interfere with periodontal attachment.

Enanthem

A rash or lesion on a mucous membrane, especially inside the mouth; it may occur with infections, inflammatory disorders, or medication reactions.

Enanthema

A rash or lesion on a mucous membrane, such as the mouth, throat, or genital tract, often associated with infection, inflammation, or a systemic disease.

Enarthrosis

A ball-and-socket joint that allows movement in

multiple axes, such as the hip or shoulder.

Encapsulated

Enclosed by a capsule or well-defined layer of tissue; the term may describe an organ, cyst, infection, or tumor and does not by itself establish whether a lesion is benign or malignant.

Encapsulating Peritoneal Sclerosis

Rare condition characterized by a thick fibro-collagenous membrane encasing the small intestine, leading to recurrent bowel obstruction. Most commonly associated with long-term peritoneal dialysis but also seen after abdominal surgery, radiation, and beta-blocker use.

Encephalitis

Encephalitis is inflammation of brain tissue, often infectious or autoimmune, causing altered mental status, seizures, fever, focal deficits, or movement abnormalities. fever, headache, altered mental status, seizures, or focal neurologic deficits and requires urgent medical evaluation.

Encephalitis Virus

A virus that can infect the brain or its surrounding tissues and cause encephalitis, or inflammation of the brain.

Encephalitozoon

A genus of microsporidian parasites that can cause intestinal, eye, kidney, or disseminated infection, especially in people with impaired immunity.

Encephalocele

A neural tube defect where brain tissue and meninges herniate through a skull defect, most commonly in the occipital region. The herniated sac contains cerebrospinal fluid, meninges, and often brain tissue. Associated with other CNS anomalies including hydrocephalus and Chiari malformation. Presents as a visible, pulsatile midline mass.

Encephalomalacia

Softening or loss of brain tissue, usually after an

injury, stroke, or infection.

Encephalomyelitis

An acute or chronic inflammatory disease simultaneously involving both the brain (encephalitis) and the spinal cord (myelitis). Most frequently mediated by autoimmune or post-infectious demyelinating mechanisms, prominently exemplified by acute disseminated encephalomyelitis, an immune-mediated monophasic demyelinating encephalopathy triggered by viral exanthems or vaccinations in children. Can also arise from direct neurotropic viral invasions (enteroviruses, flaviviruses, rabies) or autoimmune neuromyelitis optica spectrum disorder. Presents with altered mental status, seizures, focal neurological deficits, and sensory levels; diagnosed by neuroimaging (MRI revealing widespread multifocal white-matter FLAIR hyperintensities) and CSF analysis, managed with high-dose intravenous methylprednisolone, plasma exchange, or IVIG.

Encephalopathy

Any diffuse disease of the brain that alters its structure or function, presenting with a global change in mental status (confusion, lethargy, personality change, coma) rather than focal neurologic deficits.

Enchondroma

A usually benign cartilage tumor within bone, often found incidentally in the hands or long bones; pain, growth, or atypical imaging may require evaluation for malignancy.

Enchondromatosis

A disorder with multiple enchondromas caused by abnormal cartilage growth in bones; extensive disease can cause deformity, fractures, and a small risk of malignant transformation.

Enclomiphene

The purified trans-isomer of clomiphene, studied for stimulating the body's own testosterone production.

Encoding

The process by which sensory or cognitive information is converted into a form that can be stored in

memory.

Encopresis

Repeated passage of stool into inappropriate places by a child beyond the expected toilet-training age, usually associated with chronic constipation and fecal retention.

Alternate Names:

Stool Holding

End Point

A predefined outcome or measurement used to assess the result of a clinical study, treatment, diagnostic test, or disease process.

End Stage

The advanced phase of a disease in which normal function is severely impaired; the exact criteria and treatment options depend on the disease, such as end-stage kidney or liver disease.

End-Of-Life Care

Care focused on comfort, symptom control, communication, and support for a person with advanced illness and for their family or caregivers.

End-Stage COPD: Signs, Symptoms, And Prognosis

Advanced COPD may cause severe persistent breathlessness, low exercise tolerance, frequent exacerbations, weight loss, oxygen dependence, and respiratory failure. Prognosis varies; pulmonary rehabilitation, inhaled treatment, oxygen when indicated, advance care planning, and palliative support are important.

End-Stage Kidney Disease

The advanced stage of chronic kidney disease in which the kidneys can no longer adequately maintain fluid, electrolyte, and waste balance. Treatment requires dialysis or kidney transplantation, together with management of complications.

End-Stage Renal Disease

The final stage of chronic kidney disease (CKD stage

5), defined by GFR <15 mL/min/1.73

m² or the need for renal replacement therapy. Uremia develops with accumulation of toxins causing nausea, anorexia, pruritus, pericarditis, encephalopathy, and bleeding diathesis.

Alternate Names:

ESRD

Endamoeba

A clinical disorder or finding related to endamoeba; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Endangiitis

A clinical disorder or finding related to endangiitis; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Endarterectomy

Surgical removal of atherosclerotic plaque and diseased inner lining from an artery to improve blood flow, commonly performed in the carotid arteries.

Endarteritis

Inflammation of the inner lining of an artery, which can narrow the vessel and reduce blood flow; causes include infection, autoimmune disease, and other vascular disorders.

Endemic

Regular presence of a disease or infectious agent in a particular population or geographic area at an expected baseline level.

Endemic Diseases

Diseases that persist at a predictable baseline in a defined population or region, including infections shaped by local ecology, immunity, behavior, and healthcare access.

Endemic Fungi

Fungi that occur naturally in particular geographic regions and can cause infection in people who inhale their spores.

Endemic Goiter

Enlargement of the thyroid occurring commonly in a population, classically from insufficient iodine intake and sometimes associated with thyroid dysfunction.

Endergonic

Describing a chemical reaction that requires an input of energy to occur.

Ending Pregnancy With Medicines

A medical procedure, treatment, device, or care topic related to ending pregnancy with medicines; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Endobronchial Foreign Body

A foreign object lodged in a bronchus, causing partial or complete airway obstruction. It may cause localized wheezing, atelectasis, and post-obstructive pneumonia. It is most common in children and may be initially misdiagnosed as asthma. Bronchoscopy is both diagnostic and therapeutic.

Endobronchial Tumor

A tumor growing within the bronchial lumen, causing obstruction, atelectasis, and post-obstructive pneumonia. It may be malignant (lung cancer, carcinoid tumor) or benign (hamartoma, papilloma). Symptoms include cough, hemoptysis, and wheezing. Treatment is bronchoscopic removal or debulking plus definitive therapy.

Endobronchial Ultrasound

Bronchoscopy with ultrasound guidance for trans-bronchial needle aspiration of mediastinal and hilar lymph nodes. Essential for lung cancer staging. Less invasive than mediastinoscopy.

Endobronchial Valve

A one-way valve placed bronchoscopically in a target airway to achieve lobar atelectasis in emphysema. It allows air and secretions to exit but prevents air from entering. It is used for bronchoscopic lung volume reduction in selected patients with heterogeneous emphysema and intact fissures.

Endocannabinoid

A naturally produced signaling molecule that activates cannabinoid receptors and helps regulate pain, appetite, mood, memory, and immune activity.

Endocardial Cushion Defect

A congenital heart defect caused by abnormal development of the atrioventricular cushions, producing an atrioventricular septal defect with abnormal atrioventricular valves. It is strongly associated with Down syndrome and may cause heart failure and pulmonary hypertension.

Endocardial Fibroelastosis

Abnormal thickening of the inner lining of the heart caused by excess fibrous and elastic tissue, usually in infants or children.

Endocarditis

Endocarditis is infection or inflammation of the endocardium, usually involving heart valves, and may cause fever, emboli, valve destruction, and heart failure. Acute bacterial endocarditis: *Staph aureus* (most common in IV drug users and prosthetic valves), rapid destruction. Subacute: viridans group streptococci (pre-existing valve disease).

Alternate Names:

Inflammation Of The Heart Valve (Endocarditis)

Endocarditis Prophylaxis

Infective endocarditis prophylaxis with antibiotics is recommended for patients at highest risk of adverse outcomes from endocarditis who are undergoing dental procedures that involve manipulation of gingival tissue, the periapical region of teeth, or perforation of the oral mucosa.

Endocarditis Surgical Indications

Surgery indicated for: heart failure from valve dysfunction, uncontrolled infection (abscess, fistula, vegetation >10 mm with persistent emboli), fungal endocarditis, prosthetic valve dehiscence, and large vegetations (>15 mm) with embolic events.

Endocardium

The innermost layer of the heart wall, a thin membrane of endothelial cells and underlying connective tissue that lines the cardiac chambers and covers the heart valves. The endocardium is continuous with the endothelium of blood vessels and provides a smooth, non-thrombogenic surface.

Endocervical

Relating to the inner canal or lining of the cervix, which connects the uterine cavity with the vagina.

Endocervical Culture

A laboratory test that grows microorganisms from a specimen collected from the endocervix. It may help investigate cervicitis or selected sexually transmitted infections, but nucleic-acid amplification tests are preferred for many pathogens.

Endocervical Gram Stain

Microscopic staining of cells and organisms collected from the endocervix; it may show inflammation or bacteria but is not by itself a definitive test for all sexually transmitted infections.

Endocrine

Relating to hormones released into the bloodstream by glands or specialized cells, such as the thyroid, adrenal, pituitary, pancreas, and gonads.

Endocrine Disease

A disorder of hormone-producing glands or hormone action, causing abnormal growth, metabolism, reproduction, stress responses, or fluid and electrolyte regulation.

Endocrine Disrupting Chemicals

Environmental chemicals that interfere with endocrine function including bisphenol A, phthalates, pesticides, and polychlorinated biphenyls. Potential associations with diabetes, obesity, thyroid dysfunction, reproductive disorders, and hormone-sensitive cancers. Evidence is strongest for associations with type two diabetes and obesity.

Endocrine Disruptor

Chemical interfering with hormone systems. Examples: BPA, phthalates, DDT, PCBs. Effects: reproductive disorders, developmental problems, cancer.

Endocrine Finding

An examination, laboratory, imaging, or symptom finding that may indicate abnormal hormone production, hormone action, or an endocrine-gland disorder.

Endocrine Function

Production, release, transport, and action of hormones that regulate metabolism, growth, reproduction, stress responses, and fluid or electrolyte balance.

Endocrine Gland

A gland that releases hormones directly into the bloodstream rather than through a duct.

Endocrine Glands

Ductless glands that secrete hormones directly into the bloodstream to regulate distant target organs.

Endocrine Neoplasm

A benign or malignant tumor arising in an endocrine gland or hormone-producing cell; it may produce excess hormones or cause symptoms by mass effect.

Endocrine System

The network of glands and hormone-producing tissues that regulate body functions by releasing hormones into the blood.

Endocrine Therapy

Systemic treatment that manipulates hormones or their receptors in hormone-sensitive cancers, including tamoxifen, aromatase inhibitors, luteinizing hormone-releasing hormone agonists, and antiandrogens.

Endocrine Tissues

Cells or groups of cells that produce and release hormones into the blood or nearby tissues.

Endocrinologist

A physician who specializes in diagnosing and treating disorders of hormones, endocrine glands, metabolism, and related conditions such as diabetes and thyroid disease.

Endocrinologists

Physicians who diagnose and treat hormone and endocrine-gland disorders, including diabetes, thyroid disease, adrenal disease, and reproductive hormone problems.

Endocrinology

The branch of medicine concerned with the study, diagnosis, and treatment of disorders of the endocrine system—the network of glands that secrete hormones regulating metabolism, growth and development, tissue function, sexual function, reproduction, sleep, and mood.

Endocrinopathy

A disorder of an endocrine gland or hormone system, causing excessive, deficient, or otherwise abnormal hormone action.

Endocytosis

Process by which cells internalize extracellular material by engulfing it with the cell membrane, forming intracellular vesicles. Phagocytosis ("cell eating") involves engulfing large particles (>0.5 micrometers) such as bacteria or debris, forming a phagosome that fuses with lysosomes for digestion.

Endoderm

The innermost germ layer in an embryo, which develops into the linings of the digestive and respiratory tracts, as well as the liver and pancreas.

Endodontic Therapy (Root Canal)

Treatment of dental pulp disease or necrosis. Indications: irreversible pulpitis, pulp necrosis, periapical abscess. Procedure: access cavity, locate canals, clean and shape, obturate with gutta-percha, seal. Success rate: >90%. Alternative: extraction.

Alternate Names:

Root Canal

Endodontics

The dental specialty concerned with the dental pulp, root canals, and tissues around tooth roots; common treatment includes root-canal therapy.

Endodontist

A dentist who specializes in diagnosing and treating diseases or injuries of the dental pulp and tissues around tooth roots.

Endogenous

Originating within the body or produced by its cells or tissues, rather than introduced from outside.

Endogenous Opiates

Naturally produced opioid peptides, such as endorphins, enkephalins, and dynorphins, that influence pain, reward, stress, and other nervous-system functions.

Endogenous Opioids

Opioid-like peptides made naturally by the body, including endorphins, enkephalins, and dynorphins. They act on opioid receptors to influence pain, stress, mood, reward, and movement.

Alternate Names:

Endogenous opioid peptides

Endogenous Retrovirus

A viral DNA sequence inherited in the genome after an ancient retrovirus infected a germ cell.

Endoleak

Persistent blood flow into an aneurysm sac after endovascular aneurysm repair. Type I and III endoleaks reflect inadequate sealing or device failure and generally require urgent correction; other types are managed according to sac expansion and risk.

Endolimax

A genus of intestinal amoebae; *Endolimax nana* is usually considered a nonpathogenic colonizing organism.

Endolimax Nana

A usually nonpathogenic intestinal amoeba that may be reported in stool testing.

Endolymph

The potassium-rich fluid within the membranous labyrinth of the inner ear that participates in hearing and balance.

Endometrial

Relating to the endometrium, the lining of the uterus that changes during the menstrual cycle and supports implantation of a pregnancy.

Endometrial Ablation

A procedure that destroys the uterine lining to reduce heavy menstrual bleeding. It is not a method of contraception and is not intended for people planning pregnancy.

Endometrial Adenocarcinoma

A gland-forming cancer of the uterine lining, often presenting with abnormal uterine bleeding; treatment depends on grade, stage, and molecular features.

Endometrial Biopsy

Office or procedural sampling of the uterine lining, commonly to evaluate abnormal uterine bleeding or suspected endometrial hyperplasia or cancer. Cramping and light bleeding are common afterward; the sample may be inadequate if tissue is scant.

Endometrial Cancer

The most common gynecologic malignancy in developed countries, arising from the lining of the uterine cavity, typically in postmenopausal women.

Alternate Names:

Cancer, Endometrial Cancer, Uterine

Endometrial Carcinoma

Cancer arising from the endometrium, usually presenting with abnormal bleeding, especially after menopause; diagnosis requires endometrial tissue evaluation.

Endometrial Disorder

A disorder of the uterine lining, including abnormal growth, inflammation, polyps, endometriosis-related change, hyperplasia, or cancer, often causing abnormal bleeding or infertility.

Endometrial Hyperplasia

Abnormal overgrowth of the lining of the uterus, often related to prolonged estrogen exposure without enough progesterone. It may cause heavy, irregular, or postmenopausal bleeding; atypical forms can precede endometrial cancer and require gynecologic evaluation, often including endometrial sampling.

Endometrial Intraepithelial Neoplasia

A premalignant clonal proliferation of architecturally crowded, genetically altered endometrial glands that serves as the direct precursor to type I (endometrioid) endometrial adenocarcinoma. Characterized histologically by crowded glands exceeding stroma (volume percentage of stroma <45%), cytologic demarcation from background endometrium, and PTEN/PAX2 alterations. EIN carries an estimated 40% risk of coexisting or impending invasive carcinoma, guiding recommendation for total hysterectomy or progestin therapy when fertility preservation is desired.

Alternate Names:

EIN; Atypical Endometrial Hyperplasia

Endometrial Neoplasm

A benign, precancerous, or malignant growth of the uterine lining; symptoms, pathology, and staging determine management.

Endometrial Polyp

A localized overgrowth of the uterine lining that may cause irregular bleeding, heavy periods, infertility, or no symptoms; some polyps contain precancerous or cancerous changes.

Endometrial Polyps

Localized overgrowths of endometrial tissue projecting into the uterine cavity that may cause irregular bleeding, infertility, or no symptoms.

Endometrial Receptivity Analysis

A molecular diagnostic test utilizing transcriptomic profiling of over 200 genes expressed in an endometrial biopsy taken during the receptive phase of the menstrual cycle. Aims to identify the personalized “window of implantation” and determine whether the endometrium is receptive, pre-receptive, or post-receptive, guiding personalized embryo transfer (pET) in patients with recurrent implantation failure.

Alternate Names:

ERA; Endometrial Receptivity Assay

Endometrial Stromal Nodule

A benign, well-circumscribed proliferation of endometrial stromal cells within the uterus, distinguished from endometrial stromal sarcoma by its lack of infiltrative margins or vascular invasion.

Endometrial Stromal Sarcoma

A rare malignant mesodermal uterine neoplasm composed of cells that recapitulate normal proliferative endometrial stroma. Categorized into low-grade endometrial stromal sarcoma, which exhibits indolent growth, estrogen- and progesterone-receptor positivity, characteristic *JAZF1-SUZ12* gene fusions, and prominent worm-like vascular permeation, and high-grade endometrial stromal sarcoma, which displays marked cytologic atypia, aggressive clinical behavior, and distinct gene alterations (such as *YWHAE-NUTM2*).

Alternate Names:

ESS

Endometrioid Carcinoma Of The Vulva

A rare vulvar adenocarcinoma showing endometrioid differentiation, sometimes arising in ectopic or specialized Müllerian-type tissue; diagnosis requires exclusion of a metastasis from the endometrium or ovary.

Endometrioid Ovarian Carcinoma

A malignant ovarian epithelial tumor resembling endometrial glands, sometimes associated with endometriosis and often showing alterations in the PI3K or mismatch-repair pathways.

Endometrioma

A chocolate cyst of the ovary formed by ectopic endometrial tissue that hemorrhages cyclically, appearing as a homogenous low-level echo mass on ultrasound and associated with chronic pelvic pain and infertility.

Endometriosis

A chronic inflammatory condition in which tissue resembling the uterine lining is found outside the uterus, often in the pelvis. It may cause pelvic pain, painful periods, pain during sex, infertility, or bowel and bladder symptoms, although some people have no symptoms. The condition can also cause scar tissue and ovarian cysts.

Alternate Names:

Implants Endometrial (Endometriosis)

Endometritis

Inflammation or infection of the uterine lining, often associated with childbirth, miscarriage, abortion, or uterine instrumentation.

Endometrium

The mucosal epithelial and stromal inner lining of the uterine cavity, composed of a superficial stratum functionale (which undergoes dynamic cyclic hormonal changes and sheds during menstruation) and a permanent deep stratum basale (which contains stem cells that regenerate the functional layer). Under the influence of ovarian steroid hormones, the endometrium transitions sequentially through the estrogen-driven proliferative phase, the post-ovulatory progesterone-driven secretory phase, and the ischemic menstrual phase. Pathologic proliferation occurs in endometrial hyperplasia and endometrioid adenocarcinoma, particularly driven by unopposed estrogen stimulation; evaluated via transvaginal ultrasonography (endometrial thickness) and definitive endometrial tissue biopsy.

Endomorph

A body-build classification traditionally describing a relatively round shape and a tendency to store body fat; it is not a medical diagnosis.

Endomyocardial Fibrosis

Scarring and thickening of the inner heart lining and underlying heart muscle that can impair filling.

Endoparasite

A clinical disorder or finding related to endoparasite; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Endopeptidase

An enzyme that cuts peptide bonds within a protein or peptide chain rather than removing amino acids only from an end.

Endophthalmitis

A severe inflammation, usually infectious, inside the eye that can rapidly threaten vision and requires urgent ophthalmic treatment. aqueous humor, and adjacent structures, representing an ophthalmologic emergency.

Endoplasmic Reticulum

Interconnected network of membranous tubules and sacs continuous with the nuclear envelope. Rough ER: studded with ribosomes, site of synthesis of secretory, membrane, and lysosomal proteins; co-translational insertion into ER lumen.

Endoprosthesis

An artificial device placed inside the body to replace, support, or restore a body structure. Examples include joint replacements, vascular grafts or stents, and some devices used to keep an airway open.

Endoribonuclease

An enzyme that cleaves RNA at internal phosphodiester bonds, helping process or degrade RNA molecules.

Endorphin

Endogenous opioid peptides (beta-endorphin, enkephalins, dynorphins) produced by the pituitary, hypothalamus, and immune cells that bind opioid receptors (mu, delta, kappa) to produce analgesia, euphoria, and modulation of pain perception and

stress responses.

Endorphins

Natural opioid peptides that reduce pain and can contribute to feelings of well-being.

Alternate Names:

Endogenous opioids

Endoscope

A flexible or rigid instrument with a light and camera used to inspect an internal body passage or cavity and sometimes perform procedures. bronchoscope, colonoscope, gastroscope, arthroscope, cystoscope. Essential for diagnosis and intervention.

Endoscopic Mucosal Resection

Advanced endoscopic technique for removal of superficial gastrointestinal neoplasms by lifting the lesion through submucosal injection and resecting with an electrosurgical snare.

Endoscopic Retrograde Cholangiopancreatography

Combined endoscopic and fluoroscopic procedure used to visualize and treat disorders of the biliary and pancreatic ducts. A duodenoscope is advanced to the second portion of the duodenum and the ampulla of Vater is cannulated. Contrast dye is injected under fluoroscopic guidance creating cholangiograms and pancreatograms.

Endoscopic Submucosal Dissection

Advanced endoscopic technique for en bloc resection of large gastrointestinal neoplasms using specialized electrosurgical knives to dissect through the submucosa. The technique allows complete histological assessment and is particularly valuable for early gastric and colorectal cancers.

Endoscopic Thoracic Sympathectomy

A minimally invasive operation that interrupts selected thoracic sympathetic nerves, most often to treat severe focal palmar hyperhidrosis. Possible adverse effects include compensatory sweating, Horner syndrome, pneumothorax, and cardiovascular changes.

Endoscopic Ultrasound

Combined endoscopy and ultrasound for high-resolution imaging of the GI wall and adjacent structures. Linear echoendoscope: allows FNA of mediastinal, pancreatic, and peripancreatic lesions. Radial echoendoscope: 360-degree view for staging GI cancers (TNM staging).

Endoscopy

Examination of an internal body passage or cavity with an endoscope; tissue sampling and treatment may be performed during the procedure. endoscopes. Upper endoscopy examines the esophagus, stomach, and duodenum. Colonoscopy examines the colon and terminal ileum. Both allow direct visualization, biopsy, and therapeutic interventions including polypectomy, hemostasis, and dilation.

Endosome

Endosome is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Endosteum

The thin connective tissue membrane lining the inner surface of bone, including the medullary cavity, Haversian canals, and trabeculae. Contains osteoprogenitor cells and osteoblasts, contributing to bone formation, remodeling, and repair (fracture healing).

Endostosis

Endostosis is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Endothelial

Relating to the endothelium, the thin layer of cells lining blood and lymphatic vessels and forming part of other internal surfaces.

Endothelial Cell

A cell lining blood and lymphatic vessels that regulates permeability, clotting, inflammation, vascular tone, and new-vessel formation.

Endothelin

Endothelin concerns endothelin, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Endothelin Receptor Antagonists

Medications that block endothelin-1, a potent vasoconstrictor implicated in pulmonary arterial hypertension. Bosentan (dual ERA), ambrisentan (selective ETA), and macitentan are used for PAH. Side effects include hepatotoxicity (bosentan), anemia, and teratogenicity.

Endothelins

Peptides made by vascular endothelium that constrict blood vessels and can raise blood pressure.

Endothelium

The thin layer of specialized cells lining blood and lymphatic vessels. It regulates exchange between blood and tissues and contributes to vascular tone, inflammation, clotting, and new-vessel formation.

Endothelium-Derived Relaxing Factor

A substance released by the endothelium that relaxes blood vessels; it is largely identified with nitric oxide.

Alternate Names:

EDRF; Nitric oxide

Endotoxin

Endotoxin concerns endotoxin, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Endotoxins

Lipopolysaccharide components of the outer membrane of Gram-negative bacteria that can trigger strong inflammatory responses, fever, hypotension, and septic shock when released systemically.

Endotracheal Intubation

Placement of a cuffed tube through the mouth or nose into the trachea to maintain airway patency, deliver positive-pressure ventilation, administer in-

haled anesthetics, and protect the airway in selected patients. Correct position must be confirmed clinically and with exhaled carbon dioxide when circulation is present.

Endotracheal Tube

A tube placed through the mouth or nose into the trachea to maintain an airway and support ventilation. A cuff may seal the airway around the tube.

Endourologist

A urologist who specializes in minimally invasive procedures performed through the urinary tract or small access sites, such as stone removal or ureteroscopy.

Endovaginal Ultrasound

Ultrasound imaging performed with a probe placed in the vagina to examine pelvic organs, including the uterus, endometrium, ovaries, and early pregnancy.

Endovascular

Located within or performed through the interior of a blood vessel, usually using a catheter-based technique.

Endovascular Aneurysm Repair

Minimally invasive technique for abdominal aortic aneurysm repair using stent graft. Accessed via femoral arteries. Advantages: lower perioperative mortality, faster recovery. Disadvantages: need for surveillance, endoleak risk, may require reintervention.

Endovascular Embolization

A medical procedure, treatment, device, or care topic related to endovascular embolization; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Endovascular Graft

A fabric-covered vascular prosthesis placed through a catheter to reinforce a diseased vessel or exclude an aneurysm. It is used in selected aortic and pe-

ripheral vascular conditions; anatomy, symptoms, operative risk, and imaging surveillance determine suitability and follow-up.

Endovascular Repair

A minimally invasive procedure that uses a catheter-delivered graft or other device to repair a blood vessel from inside its lumen.

Endurance

The ability to continue physical or mental activity without becoming exhausted quickly. It depends on heart and lung fitness, muscle conditioning, energy supply, sleep, and health; it can be assessed with exercise or functional tests.

Enema

The introduction of fluid into the rectum and colon through the anus for diagnostic or therapeutic purposes.

Energy

Energy is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Energy Conservation

Strategies that reduce unnecessary energy expenditure and fatigue during daily activities, such as pacing, prioritizing, planning, positioning, and using adaptive equipment.

Energy Increased

A subjective or behavioral increase in energy that may be normal or may occur with reduced need for sleep, stimulant use, endocrine disease, or a manic or hypomanic episode.

Energy Intake

Energy intake is a nutrition or dietary term; appropriate use depends on age, health status, allergies, intake, and medical treatment.

Energy Metabolism

The chemical processes that convert nutrients into

usable cellular energy, mainly through glycolysis, the citric-acid cycle, and oxidative phosphorylation.

Energy Resolution

The ability of a radiation detector to distinguish photons with different energies, commonly expressed as the width of a photopeak relative to its energy.

Energy Window

The selected range of photon energies accepted by a gamma camera or other radiation detector for image formation. Appropriate selection helps limit scattered radiation.

Enervate

To weaken, exhaust, or reduce strength and energy; it is a descriptive verb rather than a specific diagnosis.

Enflurane

An inhaled anesthetic formerly used to maintain general anesthesia; it is now uncommon in many settings.

Enfuvirtide

A medicine or chemical compound called enfuvirtide. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Engagement

The descent of the fetal presenting part into the pelvis, typically occurring 2-3 weeks before labor in nulliparas or at the onset of labor in multiparas.

Enhanced Elimination

Methods used to increase removal of a toxin from the body after absorption, including urinary alkalization, hemodialysis, hemoperfusion, or other extracorporeal treatments when indicated.

Enhanced Pause

An increased pause or interruption in breathing or another physiologic signal; the meaning depends on the specific test or clinical setting.

Enhanced Recovery After Surgery

A multimodal, evidence-based perioperative pathway that reduces surgical stress and supports early recovery through preoperative counseling, appropriate fasting, opioid-sparing analgesia, early mobilization, early nutrition, and prevention of complications.

Enhancement Artifact

Ultrasound artifact from fluid-filled structures. Enhanced by harmonic imaging.

Enhancer

DNA sequence increasing transcription of target gene, can be distant from gene, orientation-independent. Binds activator proteins. Examples: LCR (locus control region) for beta-globin.

Eniluracil

An investigational drug that blocks the breakdown of uracil and can alter the activity of fluoropyrimidine medicines.

Enkephalin

A naturally produced opioid peptide that binds mainly to delta and mu opioid receptors. It helps regulate pain, stress, movement, and other brain and nerve functions.

Alternate Names:

Endogenous opioid

Enlarged Adenoids

Overgrowth of lymphoid tissue behind the nose that can obstruct nasal breathing, cause mouth breathing, snoring, ear-fluid problems, or sleep-disordered breathing.

Enlarged Breasts In Men (Gynecomastia)

Enlargement of male breast tissue caused by hormonal imbalance, medicines, substance use, or an underlying illness.

Enlarged Heart

An increase in the size of the heart, which may result from high blood pressure, valve disease, cardiomy-

opathy, anemia, or other conditions and requires evaluation of the underlying cause.

Alternate Names:

Cardiomegaly

Enlarged Liver

Enlarged liver concerns enlarged liver, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Alternate Names:

Liver, Enlarged

Enlarged Prostate

Increase in prostate size, usually benign prostatic hyperplasia, that can obstruct urine flow and cause hesitancy, weak stream, frequency, urgency, or retention.

Enlarged Spleen

An enlarged spleen, or splenomegaly, may result from infection, liver or blood disease, cancer, inflammation, or congestion. It can cause fullness or pain and increases vulnerability to rupture; the cause and blood counts guide evaluation.

Alternate Names:

Spleen Enlarged (Enlarged Spleen)

Enlarged Spleen (Splenomegaly)

An increase in spleen size associated with infection, liver disease, blood disorders, inflammation, or malignancy.

Enophthalmos

Abnormal backward displacement of an eyeball within the eye socket.

Enoxaparin

A medicine or chemical compound called enoxaparin. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Enrichment

Restoration of nutrients lost during processing. Examples: enriched flour, restored B vitamins.

Enrollment

The process of registering a person in a healthcare program, insurance plan, or clinical study after eligibility and consent requirements are met.

ENT Physician

An otolaryngologist, a physician who diagnoses and treats disorders of the ear, nose, throat, and related head and neck structures.

Entamoeba

Entamoeba is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Entamoeba Coli

Entamoeba coli is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Entamoeba Hartmanni

Entamoeba hartmanni is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Entamoeba Histolytica

A protozoan parasite that causes amoebiasis, a gastrointestinal and extra-intestinal infection transmitted via the fecal-oral route. The organism exists in two forms: the infectious cyst (transmitted in contaminated food or water) and the invasive trophozoite.

Entamoeba Polecki

Entamoeba polecki is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Enteral

Relating to or delivered through the gastrointestinal tract; enteral nutrition provides food through

the mouth or a feeding tube into the stomach or intestine.

Enteral Formula

A nutritionally complete or modular liquid preparation delivered into the gastrointestinal tract. Formulas may be polymeric, semi-elemental, or elemental and are selected according to digestion, absorption, fluid, renal, hepatic, and metabolic requirements.

Enteral Nutrition

Delivery of nutrients via gastrointestinal tract. Routes: nasogastric, nasoduodenal, gastrostomy (PEG), jejunostomy. Indications: dysphagia, neurological disease, head/neck cancer, prolonged inability to eat. Complications: aspiration, diarrhea, tube displacement, sinusitis.

Enteric

Relating to the intestine or gastrointestinal tract; the term can also describe organisms or infections associated with the intestines.

Enteric Bacteria

Bacteria that live in or affect the intestinal tract, including normal flora and pathogens that cause gastrointestinal or extraintestinal infection.

Enteric Duplication Cyst

A congenital, smooth-muscle-lined cyst arising from the gastrointestinal tract, most commonly in the ileum, containing ectopic gastric or intestinal mucosa. It may present in childhood with abdominal pain, obstruction, or gastrointestinal bleeding from ectopic gastric acid secretion.

Enteric Fever (Typhoid Fever)

Alternate index label for Typhoid Fever.

Enteric Nervous System

The network of nerve cells in the gastrointestinal tract that regulates movement, secretion, and local blood flow.

Enteric-Coated

Covered with a coating designed to resist breakdown in the stomach and dissolve in the less-acidic small intestine.

Enteritis

Inflammation of the small intestine caused by infection, inflammatory disease, radiation, reduced blood flow, or other injury. bacterial, or parasitic). Viral enteritis (norovirus, rotavirus, adenovirus, astrovirus) presents with acute onset of watery diarrhea, nausea, vomiting, and abdominal cramps, typically self-limited over 2–7 days.

Enterobacter

A genus of Gram-negative bacteria in the Enterobacteriaceae family that can cause healthcare-associated urinary, bloodstream, wound, or respiratory infection.

Enterobacter Aerogenes

A former name for *Klebsiella aerogenes*, a Gram-negative bacterium that can cause healthcare-associated infection and may carry antibiotic resistance.

Enterobacter Asburiae

A Gram-negative *Enterobacter* species found in the environment and occasionally causing opportunistic human infection.

Enterobacter Cancerogenus

A Gram-negative environmental bacterium that can rarely cause wound or bloodstream infection.

Enterobacter Cloacae

A Gram-negative bacterium that can cause healthcare-associated urinary, bloodstream, respiratory, or wound infection and may develop antibiotic resistance.

Enterobacter Hormaechei

A member of the *Enterobacter cloacae* complex that can cause opportunistic healthcare-associated infection.

Enterobacteriaceae

A family of Gram-negative bacteria that includes *Escherichia*, *Klebsiella*, *Enterobacter*, *Salmonella*, and *Shigella*; members may be normal flora or pathogens.

Enterobacteriaceae Carbapenem Resistance

Resistance to carbapenem antibiotics in Enterobacteriaceae, primarily through production of carbapenemases. The most important carbapenemases include KPC, NDM, and OXA-48-like enzymes. Carbapenem-resistant Enterobacteriaceae (CRE) are a major public health threat due to limited treatment options.

Enterobiasis

Intestinal infection with the pinworm *Enterobius vermicularis*, commonly causing nighttime itching around the anus.

Enterobius

A genus of pinworms; *Enterobius vermicularis* causes the common intestinal infection known as pinworm infection.

Enterobius Vermicularis

A small intestinal nematode commonly called the pinworm or threadworm.

Alternate Names:

the pinworm or threadworm.

Enterocoele

A form of pelvic organ prolapse characterized by the herniation of the peritoneal pouch of Douglas (rectouterine or rectovaginal space) and loops of small intestine between the anterior rectal wall and posterior vaginal wall. Most commonly encountered following hysterectomy (apical vault defect) or in multiparous women with weakened endopelvic fascia and levator ani musculature, presenting with pelvic heaviness, a palpable vaginal bulge, and lower back or rectal discomfort.

Alternate Names:

Small Bowel Prolapse; Douglas Pouch Hernia

Enteroclysis

A contrast-imaging study in which material is delivered directly into the small intestine through a tube to evaluate mucosal disease, narrowing, or obstruction.

Enterococcaceae

A family of bacteria that includes *Enterococcus* species, which normally inhabit the intestine but can cause opportunistic infections.

Enterococcus

A genus of Gram-positive bacteria normally found in the intestine that can cause urinary, wound, bloodstream, and heart-valve infections.

Enterococcus Avium

A species of *Enterococcus* bacteria that can rarely cause urinary, bloodstream, wound, or other infections, especially in vulnerable patients.

Enterococcus Dispar

A species of *Enterococcus* bacteria identified mainly in microbiology studies and occasionally in clinical specimens.

Enterococcus Durans

A species of *Enterococcus* bacteria that can occasionally cause opportunistic infections.

Enterococcus Faecalis

A common *Enterococcus* species causing urinary, bloodstream, wound, and infective endocarditis infections.

Enterococcus Faecium

An *Enterococcus* species that causes healthcare-associated infection and may be resistant to vancomycin and other antibiotics.

Enterococcus Gallinarum

An *Enterococcus* species that may carry intrinsic low-level resistance to vancomycin and can rarely cause human infection.

Enterococcus Hirae

A species of *Enterococcus* bacteria that rarely causes opportunistic infections in humans.

Enterococcus Malodoratus

A rarely encountered *Enterococcus* species that may be isolated from clinical specimens.

Enterococcus Mundtii

A species of *Enterococcus* bacteria that is uncommon in human disease but can occasionally be clinically significant.

Enterococcus Pseudoavium

A rarely encountered *Enterococcus* species that may be isolated from human specimens.

Enterococcus Raffinosus

A species of *Enterococcus* bacteria that can cause occasional opportunistic infections.

Enterococcus Thailandicus

A recently recognized *Enterococcus* species that is uncommon in human clinical disease.

Enterocolitis

Inflammation of both the small intestine and colon, caused by infection, immune-mediated disease, reduced blood flow, toxins, or other injury. It may produce abdominal pain, diarrhoea, fever, vomiting, bleeding, dehydration, or systemic illness; diagnosis and management depend on the cause and severity.

Enterocyte

Enterocyte is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Enterocytozoon

A genus of microsporidia, including *Enterocytozoon bieneusi*, which can cause persistent diarrhea in people with weakened immunity.

Enteroendocrine Cell

A specialized intestinal epithelial cell that senses

nutrients and releases hormones such as GLP-1, GIP, secretin, or CCK to regulate digestion, appetite, and glucose metabolism.

Enterohepatic Circulation

Recycling of toxins between liver and intestine via bile. Prolonged exposure, increased toxicity. Interrupted by activated charcoal, cholestyramine. Examples: morphine, digoxin, estrogen, thyroxine.

Enteromonas Hominis

A usually nonpathogenic intestinal flagellate that may be found in stool specimens.

Enteropathic

Pertaining to, originating from, or associated with pathology of the intestines. Clinically prominent in rheumatology as enteropathic arthropathy, a seronegative, HLA-B27-associated spondyloarthritis that develops in patients with inflammatory bowel disease (Crohn disease or ulcerative colitis), presenting as peripheral asymmetric pauciarticular arthritis or axial ankylosing spondylitis. Also used to describe secondary extra-intestinal manifestations of gut barrier compromise, including enteropathic acrodermatitis (zinc deficiency) and enteropathic protein loss (protein-losing enteropathy in intestinal lymphangiectasia or severe celiac disease).

Enteropathic Arthritis

Inflammatory arthritis associated with inflammatory bowel disease, including Crohn disease and ulcerative colitis. It may affect the spine, sacroiliac joints, or peripheral joints.

Enteropathy

A disease or disorder affecting the intestine. Causes include infection, inflammation, gluten sensitivity, impaired blood flow, toxins, and inherited conditions.

Enteroscopy

Endoscopic examination of the small intestine, allowing visualization, biopsy, bleeding treatment, or removal of selected lesions beyond the reach of standard upper endoscopy.

Enterospasm

A spasm or involuntary contraction of the intestine, which may cause cramping, pain, or altered movement of intestinal contents.

Enterostomy

A surgically created opening from the intestine to the abdominal surface, such as an ileostomy or colostomy, to divert intestinal contents.

Enterostomy Procedure

A medical procedure, treatment, device, or care topic related to enterostomy procedure; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Enterotoxin

A toxin produced by some microorganisms that acts on the intestinal epithelium and can cause secretion, vomiting, abdominal cramps, or diarrhea.

Enterovesical Fistula

An abnormal connection between the intestine and urinary bladder.

Enterovirulent E. Coli

Alternate index label for EEC.

Enterovirulent E. Coli

Alternate index label for Enterovirulent E. Coli.

Enterovirus

A group of viruses that can infect the digestive and respiratory tracts. Many infections cause no symptoms or only mild illness, but enteroviruses can also cause hand-foot-and-mouth disease, rash, meningitis, myocarditis, or other disease. They commonly spread through respiratory secretions or contact with infected stool.

Enterovirus (Non-Polio Enterovirus Infection)

Alternate index label for Non-Polio Enterovirus Infection.

Enterovirus D68

An enterovirus that usually causes respiratory illness but can rarely cause severe pneumonia or acute flaccid myelitis, especially in children; spread occurs through respiratory secretions.

Enthesis

The site at which a tendon, ligament, or joint capsule attaches to bone.

Enthesitand Enthesopathy (Enthesitis And Enthesopathy)

Alternate index label for Enthesitis and Enthesopathy.

Enthesitis

Inflammation at the site where a tendon, ligament, or joint capsule inserts into bone (enthesis), a hallmark of spondyloarthritis (ankylosing spondylitis, psoriatic arthritis, reactive arthritis, IBD-associated). Affects Achilles tendon, plantar fascia, patellar, and costochondral junctions.

Enthesitis And Enthesopathy

Enthesitis is inflammation where a tendon, ligament, or fascia attaches to bone; enthesopathy includes degenerative and other disorders at that site. It is associated with overuse, arthritis, and spondyloarthritis and causes localized pain and stiffness.

Alternate Names:

Enthesitand Enthesopathy (Enthesitis And Enthesopathy)

Entinostat

An investigational histone-deacetylase inhibitor studied in cancer treatment, including combinations with immunotherapy or hormone therapy.

Entorhinal Cortex

Entorhinal cortex is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Entrance Wound

A skin defect created where a projectile enters the

body, distinguished from an exit wound by smaller size, rounded shape, and abraded inward margin. Contact wounds show a muzzle imprint and soot; close-range wounds show soot and powder tattooing; intermediate-range wounds show stippling; distant wounds show a clean abrasion ring.

Entropion

Inward turning of an eyelid margin, causing eyelashes to rub the eye and produce irritation, tearing, corneal injury, or infection.

Enucleation

Surgical removal of the eyeball while leaving the eyelid, eye muscles, and most orbital tissues in place.

Enucleation Procedure

A medical procedure, treatment, device, or care topic related to enucleation procedure; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Enuresis

Repeated involuntary urination, usually during sleep in a child who is developmentally expected to remain dry. It may be primary or secondary and requires assessment for constipation, urinary infection, diabetes, sleep or developmental disorders, psychosocial stressors, and lower-urinary-tract dysfunction when clinically indicated.

Enuresis (Bedwetting)

Repeated involuntary urination during sleep in a child who is developmentally expected to remain dry. It may be primary or secondary; assessment considers constipation, urinary infection, diabetes, sleep disorders, and developmental or psychosocial factors.

Alternate Names:

Bedwetting

Environment

Environment is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Environmental Exposure

Environmental exposure is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Environmental Health

Study of environmental factors affecting health. Examples: air pollution, water quality, occupational hazards, climate change.

Environmental Illness

Illness caused or worsened by environmental exposures; evaluation requires attention to timing, dose, setting, and alternative explanations.

Enzootic

Regularly present or persistently maintained in an animal population or geographic area; it is the animal-population counterpart of endemic disease.

Enzyme

A protein, or occasionally an RNA molecule, that catalyses a specific biochemical reaction without being consumed; enzyme activity measurements help assess tissue injury, metabolic disease, and drug effects. activation energy. Specificity: lock-and-key or induced fit models. Kinetics: Michaelis-Menten equation (V_{max} , K_m). Regulation: allosteric control, covalent modification (phosphorylation), proteolytic activation, feedback inhibition.

Enzyme Activity

The rate at which an enzyme catalyzes a reaction under defined conditions, influenced by substrate, temperature, pH, inhibitors, cofactors, and enzyme concentration.

Enzyme Deficiency

Reduced or absent activity of an enzyme, often inherited or acquired, causing accumulation of substrates, lack of products, and characteristic metabolic or cellular disease.

Enzyme Induction

A process by which a drug or xenobiotic increases the synthesis or activity of drug-metabolizing enzymes, primarily cytochrome P450 enzymes in the liver, accelerating the metabolism of itself (autoinduction) or other drugs. Enzyme induction typically occurs over days to weeks as it requires new protein synthesis.

Enzyme Inhibition

A process by which a drug inhibits the activity of drug-metabolizing enzymes, leading to increased plasma concentrations of co-administered drugs that are metabolized by the same enzyme. Enzyme inhibition occurs rapidly (hours to days) as it does not require new protein synthesis.

Enzyme Kinetics

The study of enzyme-catalyzed reaction rates and the factors that affect them. Important concepts include substrate concentration, maximum reaction rate, binding affinity, inhibitors, and the relationship between enzyme activity and temperature or pH.

Enzyme Markers

Enzyme markers concerns enzyme markers, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Enzyme Precursors

Inactive or less-active precursor proteins, also called zymogens, that are converted by cleavage or other modification into functional enzymes.

Enzyme Therapy

A medical procedure, treatment, device, or care topic related to enzyme therapy; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Enzyme-Linked Immunosorbent Assay

An immunoassay that uses an enzyme-labeled antibody or antigen and a substrate-generated signal to detect or quantify a target analyte. It is used for measurement of antibodies, antigens, hormones,

cytokines, and infectious disease markers.

Eosine Yellowish

Eosine yellowish is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Eosinophil

A granulocyte (1-4 percent of leukocytes) with bilobed nucleus and eosinophilic granules containing major basic protein, eosinophil cationic protein, and peroxidase. Roles: defense against parasitic (helminth) infections and modulation of allergic inflammation; contain histaminase and aryl sulfatase.

Eosinophil Count - Absolute

The absolute eosinophil count is the number of eosinophils in a volume of blood, calculated from the white-cell count and eosinophil percentage. Elevation may occur with allergy, parasitic infection, medicines, autoimmune disease, or hematologic disorders.

Eosinophilia

An absolute eosinophil count exceeding 500 cells per microliter in peripheral blood, most commonly caused by parasitic infections, allergic disorders, drug reactions, and autoimmune diseases. Hypereosinophilia, with counts above 1500 per microliter, may cause end-organ damage to the heart, lungs, skin, and nervous system.

Eosinophilic Bronchitis

A cause of chronic cough in which eosinophilic inflammation affects the airways without the variable airflow obstruction typical of asthma. Cough-variant asthma and reflux may mimic it; diagnosis relies on clinical assessment and airway inflammation testing.

Eosinophilic Esophagitis

Chronic immune-mediated condition characterized by dense eosinophilic infiltration of the esophageal mucosa defined as fifteen or more eosinophils per high-power field, in the absence of other causes of

esophageal eosinophilia.

Eosinophilic Fasciitis

A rare inflammatory disorder causing painful swelling and progressive thickening of fascia and skin, often with increased blood eosinophils. thickening of the deep fascia and adjacent muscle, presenting with painful swelling and induration of the extremities and a peau d'orange skin appearance. Peripheral eosinophilia and elevated inflammatory markers are typical.

Alternate Names:

Shulmans Syndrome (Eosinophilic Fasciitis)

Eosinophilic Gastroenteritis

Rare chronic inflammatory condition characterized by eosinophilic infiltration of the gastrointestinal tract beyond the esophagus involving any segment from stomach to rectum and extending beyond the mucosa to involve the muscularis propria or serosa.

Eosinophilic Granuloma

A localized form of Langerhans cell histiocytosis that commonly affects bone or skin.

Eosinophilic Granulomatosis With Polyangiitis

EGPA (formerly Churg-Strauss syndrome): an ANCA-associated vasculitis characterized by asthma, peripheral and tissue eosinophilia, and necrotizing vasculitis.

Eosinophils

White blood cells involved in allergic inflammation and defense against some parasites.

Alternate Names:

Eosinophil granulocytes

Ependyma

A specialized epithelial lining of the brain ventricles and spinal-cord central canal that is involved in cerebrospinal-fluid circulation and, in some regions, production.

Ependymal Cell

A glial cell lining brain ventricles and the spinal central canal that contributes to cerebrospinal-fluid circulation and the ventricular barrier.

Ependymal Tumor

A tumor arising from cells lining the brain ventricles or central spinal canal; ependymomas may obstruct cerebrospinal-fluid flow and require imaging, surgery, and sometimes radiation.

Ependyoblastoma

A rare, highly malignant embryonal brain tumor, now classified within specific molecular categories of pediatric central nervous-system tumors.

Ependymoma

A glioma arising from the ependymal cells lining the ventricles and spinal canal; more common in children and frequently associated with hydrocephalus.

Ephedrine

A sympathomimetic amine that increases noradrenergic activity and can raise blood pressure and heart rate; it has limited clinical uses and may cause palpitations, insomnia, hypertension, or misuse.

Ephedrinum

A name referring to ephedrine, a sympathomimetic medicine that can increase heart rate, blood pressure, and airway diameter.

Ephelides

Small, flat freckles caused by increased melanin production, often becoming darker after sun exposure.

Ephrins

Cell-surface signaling proteins that bind Eph receptors and guide cell positioning, axon growth, blood-vessel development, and tissue organization.

Epiblast

Epiblast is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Epicanthal Folds

Skin folds of the upper eyelid that partly cover the inner canthus, common in some populations and sometimes associated with chromosomal or developmental syndromes.

Epicardium

The outermost layer of the heart wall, which is the visceral layer of the serous pericardium. It consists of a single layer of mesothelial cells and underlying connective tissue containing blood vessels, nerves, and adipose tissue. The epicardium produces pericardial fluid that lubricates the heart surface.

Epichlorohydrin

Epichlorohydrin is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Epicondyle

A bony projection near a condyle that provides attachment for muscles and ligaments.

Epicondylitis

Painful inflammation or degeneration of a tendon attachment near a joint epicondyle, commonly at the elbow.

Alternate Names:

Tennis elbow; golfer's elbow

Epicondylitis, Lateral

Alternate index label; see *Tennis elbow*.

Epidemic

Sudden increase in disease cases above expected level. Examples: cholera outbreak, measles outbreak.

Epidemiologic Method

A method for measuring disease occurrence or investigating associations between exposures and health outcomes in populations.

Epidemiologic Studies

Research that examines the distribution, causes, risk factors, prevention, or outcomes of health events in populations.

Epidemiological Study

A population-based investigation of disease occurrence, causes, risks, or health outcomes.

Alternate Names:

Epidemiologic study

Epidemiologist

A scientist or public-health professional who studies the distribution, causes, risk factors, and prevention of health conditions in populations.

Epidemiology

Study of distribution and determinants of health-related states in populations. Branches: descriptive (who, when, where), analytic (why, how), experimental (intervention trials). Measures: incidence, prevalence, mortality rates, case-fatality rate.

Epidermal

Relating to the epidermis, the outermost layer of the skin.

Epidermal Acanthosis

Thickening of the epidermis caused by increased growth of its cells, seen in conditions such as psoriasis, chronic rubbing, and some inflammatory or genetic skin disorders.

Epidermal Cells

Cells forming the outer layer of skin, chiefly keratinocytes, which create a protective barrier; melanocytes, immune cells, and touch-associated cells are also present.

Epidermal Growth Factor

A signaling protein that binds the epidermal-growth-factor receptor and promotes cell growth, survival, and tissue repair; abnormal signaling can contribute to cancer.

Epidermal Inclusion Cyst

A common benign cyst lined by stratified squamous epithelium with keratin debris. It presents as a firm, mobile, subcutaneous nodule, typically on the face, neck, and back. A central punctum is characteristic. It may become inflamed or infected. Treatment includes complete excision (including the cyst wall) to prevent recurrence.

Epidermal Layers

From deep to superficial, the epidermis consists of the basal, spinous, granular, lucid, and cornified layers. The lucid layer is present mainly in thick skin such as the palms and soles.

Epidermis

Outermost skin layer, stratified squamous keratinized epithelium approximately 0.1 mm thick, with five layers (superficial to deep): stratum corneum (dead keratinized cells), lucidum (clear layer, thick skin only), granulosum (keratinization begins), spinosum (spiny layer, Langerhans cells), basale (stem cells, melanocytes, Merkel cells).

Epidermodysplasia Verruciformis

A rare inherited disorder of skin immunity associated with susceptibility to certain human papillomavirus infections, widespread wart-like lesions, and increased risk of some skin cancers.

Epidermoid Cyst

A benign, slow-growing cyst lined by squamous epithelium and filled with keratin, usually presenting as a firm subcutaneous nodule. Rupture can cause inflammation, tenderness, and a foul-smelling discharge.

Epidermoid Cysts

Slow-growing benign cysts beneath the skin filled with keratin, commonly presenting as a movable lump.

Alternate Names:

Cysts, Epidermoid Cysts, Sebaceous Sebaceous Cysts

Epidermolysis Bullosa

A group of genetic disorders causing skin fragility and blistering with minimal trauma. Simplex (intraepidermal blistering), junctional (blistering at lamina lucida), dystrophic (sublamina densa blistering), and Kindler syndrome (mixed). Dystrophic EB can cause scarring, contractures, and squamous cell carcinoma.

Epidermolysis Bullosa

A group of inherited disorders in which skin and sometimes mucous membranes are unusually fragile and blister or erode after minor friction or trauma.

Epidermolysis Bullosa Acquisita

An autoimmune subepidermal blistering disease caused by antibodies against type VII collagen in anchoring fibrils; it commonly causes skin fragility and trauma-induced blisters and may scar.

Epidermophyton

A genus of dermatophyte fungi that infect keratinized skin, hair, and nails.

Epidermophyton Floccosum

A dermatophyte fungus that commonly causes tinea of the body, groin, and feet and can infect nails.

Epididymal Cyst

A usually benign, fluid-filled cyst arising in the epididymis. It may be painless and may require treatment only if it causes significant symptoms or complications.

Epididymal Neoplasm

A benign or malignant growth arising in the epididymis; most palpable epididymal masses are benign, but persistent or enlarging lesions require evaluation.

Epididymides

The paired coiled ducts on the posterior surface of the testes where sperm mature and are stored before entering the vasa deferentia.

Alternate Names:

epididymis

Epididymis

Coiled duct approximately 6 meters long on the posterolateral aspect of each testis, divided into head (caput, receiving efferent tubules from the testis), body (corpus), and tail (cauda, continuing as the vas deferens). Functions include sperm maturation (gaining motility over approximately 12 days), storage (in the tail), and transport.

Epididymitis

Inflammation of the epididymis, commonly caused by infection and presenting with gradual unilateral scrotal pain and swelling. typically resulting from infection. In sexually active men under 35 years, the most common pathogens are sexually transmitted: *Chlamydia trachomatis* (most common) and *Neisseria gonorrhoeae*, with epididymitis often accompanied by urethritis.

Epidural

Regional analgesia (or anesthesia) delivering local anesthetic/opioid via a catheter in the epidural space (lumbar), used for pain relief during labor and for cesarean (higher dose) anesthesia. Provides excellent analgesia while allowing maternal mobility and pushing; reduces catecholamine stress.

Epidural Abscess

A collection of infected material between the dura mater and the skull or vertebral canal, capable of compressing the brain, spinal cord, or nerve roots. Spinal epidural abscess may cause severe focal back pain, fever, neurological deficit, or sphincter dysfunction and requires urgent imaging and specialist assessment.

Epidural Anaesthesia

Epidural Anaesthesia is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Epidural Analgesia

Segmental pain relief produced by delivering a local anesthetic, sometimes with an opioid, through a

catheter into the epidural space. It may be used during labor or selected operations; the medicines and concentrations depend on the patient and clinical setting.

Epidural Analgesia For Labor

A neuraxial technique for labor analgesia: a catheter is placed in the epidural space (L3-L4 or L2-L3 interspace) and local anesthetic (bupivacaine 0.0625-0.125%) with opioid (fentanyl 2 $\mu\text{g}/\text{mL}$) is infused continuously or in patient-controlled epidural analgesia (PCEA) boluses.

Epidural Anesthesia

A regional anesthetic technique involving injection of local anesthetic and/or opioid into the epidural space (the potential space between the dura mater and the ligamentum flavum within the spinal canal), producing segmental analgesia and sympathetic blockade.

Epidural Block - Pregnancy

A medical procedure, treatment, device, or care topic related to epidural block - pregnancy; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Epidural Blood Patch

A minimally invasive procedure used to treat post-dural puncture headache resulting from accidental dural puncture during epidural placement or spinal anesthesia. It involves injecting a small volume of the patient's autologous blood (typically 15-20 mL) into the epidural space at or near the site of the previous puncture.

Epidural Hematoma

Accumulation of blood between the skull and the dura mater, typically from arterial bleeding (middle meningeal artery) following temporal bone fracture. Presents with loss of consciousness, then a lucid interval (hours), followed by rapid neurological deterioration due to mass effect.

Epidural Injections For Back Pain

Injection of corticosteroid, local anesthetic, or both

into or near the epidural space to reduce inflammation and pain from selected spinal nerve or disc disorders.

Epidural Neoplasm

A tumor in the space outside the dura surrounding the spinal cord or brain; spinal epidural tumors can compress nerves or the spinal cord and cause pain, weakness, or sensory loss.

Epidural Space

The space outside the dura mater around the spinal cord, where local anesthetic may be injected.

Epigallocatechin Gallate

A catechin found in green tea and dietary supplements, studied for antioxidant and other effects; high-dose extracts can cause liver injury and clinical benefits are not established for most conditions.

Epigastric

Relating to the upper middle region of the abdomen between the ribs, where pain may arise from the stomach, pancreas, gallbladder, or other organs.

Epigastric Pain

Pain or discomfort in the upper middle part of the abdomen, below the ribs and above the navel. It may result from problems involving the stomach, esophagus, pancreas, gallbladder, or small intestine.

Epigastric Artery

An artery supplying the abdominal wall; the superior and inferior epigastric arteries contribute to its blood supply.

Epigastric Region

Central superior region between the hypochondriac regions and above the umbilical region. Overlies the stomach, duodenum, pancreas, and portions of the liver. The aorta and inferior vena cava pass through it. Epigastric pain associates with gastric ulcers, pancreatitis, GERD, and aortic aneurysms.

Epigastrium

The upper middle region of the abdomen, located between the right and left hypochondriac regions.

Epigenetics

Heritable changes in gene expression without DNA sequence alteration. Mechanisms: DNA methylation, histone modification, non-coding RNA. Influenced by environment, diet, stress. Reversible.

Epigenome

The collection of chemical modifications and chromatin features that regulate gene activity without changing the DNA sequence itself.

Epigenomics

The study of epigenetic modifications across the genome and how they influence development, disease, environmental responses, and treatment.

Epiglottis

A leaf-shaped elastic cartilage at the base of the tongue that covers the laryngeal inlet during swallowing, directing food and liquid into the esophagus and preventing aspiration into the airway. Elevates during swallowing (with the larynx) under glossopharyngeal/vagal innervation.

Epiglottitis

Epiglottitis is potentially life-threatening inflammation and swelling around the epiglottis that can rapidly obstruct the airway, especially in children. *Haemophilus influenzae* type b (now rare due to vaccination). Presents with high fever, drooling, dysphagia, tripod position, and rapid progression to airway obstruction. Do not examine the throat.

Epilepsy

A disorder of the brain that causes a lasting tendency to have seizures. It may be diagnosed after two unprovoked seizures occurring more than 24 hours apart, after one unprovoked seizure when the risk of another seizure is high, or when a recognized epilepsy syndrome is present. Seizures may begin in one brain region or involve both sides of the brain. Causes include genetic, structural, metabolic,

infectious, immune, and unknown factors.

Alternate Names:

Seizure Disorder

Epilepsy In Children

A neurologic disorder in which a child has recurrent unprovoked seizures or a seizure syndrome, with causes, seizure types, development, and treatment varying widely.

Epilepsy Surgery

Indicated for patients with drug-resistant epilepsy who have a concordant, resectable seizure focus identified on comprehensive presurgical evaluation. Failure of two or more appropriately chosen and adequately dosed antiepileptic drugs to achieve sustained seizure freedom.

Epilepsy, Frontal Lobe

Alternate index label; see *Frontal lobe seizures*.

Epilepticus, Status

A clinical disorder or finding related to epilepticus, status; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Epileptiform

Resembling an epileptic seizure or the electrical pattern associated with seizure activity.

Epinephrin (Adrenaline)

An alternate spelling of epinephrine, a catecholamine hormone and medicine that increases heart rate and airway diameter and constricts blood vessels.

Alternate Names:

Adrenaline

Epinephrine

A hormone and neurotransmitter that increases heart activity and narrows many blood vessels during stress.

Alternate Names:

Adrenaline

Epinephrine (Adrenaline)

A catecholamine hormone and neurotransmitter produced by the adrenal medulla and sympathetic nerve terminals, acting on alpha-1, alpha-2, beta-1, beta-2, and beta-3 adrenergic receptors.

Alternate Names:

Adrenaline

Epiphora

Excessive tearing caused by impaired tear drainage or reflex hypersecretion. Causes: nasolacrimal duct obstruction (most common in infants and elderly), punctal stenosis, eyelid malposition (ectropion, entropion), and reflex tearing from dry eye or ocular surface irritation.

Epiphyseal Plate

A layer of hyaline cartilage, also called the growth plate or physis, located between the epiphysis and metaphysis of a growing long bone. It permits longitudinal bone growth through endochondral ossification, progressing from resting cartilage through proliferation and hypertrophy to calcification and replacement by bone.

Alternate Names:

the growth plate or physis, located between.

Epiphysis

The expanded end of a long bone, usually covered by articular cartilage where it forms a synovial joint. It contains spongy (trabecular) bone with a thin outer shell of compact bone and contributes to the shape and function of the joint.

Epiphysitis

Epiphysitis is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Epipodophyllotoxin

Epipodophyllotoxin concerns epipodophyllotoxin, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Epiretinal Membrane

A thin layer of scar-like tissue that forms on the surface of the retina and may distort vision.

Epirubicin

An anthracycline chemotherapy medicine that intercalates DNA and inhibits topoisomerase II; cumulative cardiotoxicity and myelosuppression are important risks.

Epirubicin Hydrochloride

An anthracycline chemotherapy medicine that damages cancer-cell DNA and is used for selected cancers.

Episcleritis

Usually self-limited inflammation of the thin tissue overlying the white of the eye, causing localized redness and mild discomfort. connective tissue between the sclera and conjunctiva.

Episiotomy

A surgical incision in the perineum made during vaginal birth to enlarge the vaginal opening. Routine use is not recommended; selected indications may include urgent delivery or instrumental birth, and complications include pain, bleeding, infection, and extension of the tear.

Episiotomy Repair

Surgical repair of an episiotomy incision after delivery. Technique: continuous suturing of the vaginal mucosa (2-0 chromic), interrupted sutures for the perineal muscles, and subcuticular closure of the perineal skin. The rectum should be examined after repair (suture inadvertently through the rectal mucosa).

Episodic Ataxia

An inherited disorder causing recurrent brief attacks of impaired coordination and balance.

Epispadias

A congenital defect in which the urethral opening is on the upper surface of the penis or in an abnormal

position in females, often associated with bladder-exstrophy spectrum.

Epistasis

Interaction between genes where one gene masks or modifies expression of another. Examples: coat color in mice, ABO blood group modifications.

Epistaxis

Bleeding from the nasal mucosa that may range from minor bleeding to a medical emergency. See Nosebleed.

Epithalamic

Relating to the epithalamus, a region of the brain that includes the pineal gland and habenular nuclei and participates in circadian and emotional functions.

Epithelial

Relating to the sheets of cells that cover body surfaces, line cavities and organs, and form many glands.

Epithelial Cells

Cells that cover body surfaces and line organs, cavities, and glands.

Alternate Names:

Epithelial tissue cells

Epithelial Cyst

A fluid- or keratin-filled cyst lined by epithelial cells; most are benign, but symptoms, infection, growth, or an unusual location may require evaluation.

Epithelial Tissue

The tissue that covers external body surfaces, lines internal cavities and pathways, and forms glands.

Epithelioid Cell

A rounded or polygonal cell resembling epithelial cells, often describing activated macrophages in granulomas or the appearance of selected tumors.

Epithelioid Sarcoma

A rare soft-tissue cancer that often begins in the skin or tissues of the hands, forearms, or lower limbs.

Epithelioid Trophoblastic Tumor

A rare gestational trophoblastic neoplasm composed of chorionic-type intermediate trophoblast, often arising months or years after a pregnancy and requiring specialist oncologic management.

Epithelioma

An older or broad term for a tumor arising from epithelial cells; the specific diagnosis should identify whether it is benign, precancerous, or malignant.

Epithelium

Sheets of closely packed cells covering body surfaces, lining cavities and ducts, and forming glands. Epithelial structure varies with its function, such as protection, absorption, secretion, or exchange.

Epitope

Epitope concerns epitope, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Epitope Mapping

Epitope mapping concerns epitope mapping, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Epizootic

An outbreak of disease affecting many animals in a population or area at the same time; it is analogous to an epidemic in humans.

Eplerenone

A selective aldosterone-receptor antagonist used for hypertension and selected heart-failure or post-myocardial-infarction patients; hyperkalemia and kidney impairment require monitoring.

Epley Maneuver

A sequence of head and body movements used to move displaced otoconia from the posterior semicir-

cular canal into the utricle. It is commonly used to treat posterior-canal benign paroxysmal positional vertigo.

EPO Test

A blood test that measures erythropoietin, a hormone that stimulates red-blood-cell production; it may help evaluate anemia or erythrocytosis.

Epoetin Alfa

A recombinant erythropoiesis-stimulating agent used for selected anemias, including those related to chronic kidney disease or chemotherapy, with hemoglobin and thrombotic risk monitoring.

Epoetin Beta

A recombinant erythropoiesis-stimulating agent that stimulates red-cell production in selected anemias; treatment requires individualized hemoglobin targets and monitoring for hypertension and thrombosis.

Eponym

A name derived from a person, place, or other proper name and used for a disease, sign, procedure, or anatomical structure.

Epoprostenol

A short-acting prostacyclin administered by continuous intravenous infusion for severe pulmonary arterial hypertension; interruption can cause rapid clinical deterioration.

Epothilon

A family of microtubule-stabilizing compounds studied as anticancer drugs; some derivatives have entered clinical development.

Epothilone B

A compound that stabilizes microtubules and has been studied as an anticancer drug.

Epothilone D

A microtubule-stabilizing compound studied for its potential anticancer activity.

Epoxide Hydrolase

An enzyme that converts epoxide compounds into less reactive diols by adding water.

Epoxy Compounds

Epoxy compounds concerns epoxy compounds, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Epsom Salts

Magnesium sulfate crystals, sometimes used in baths or as a laxative; swallowing them can be unsafe in some conditions.

Epstein Pearls

Small, white keratin-filled cysts on the palate of a newborn that usually disappear without treatment.

Epstein-Barr Virus

EBV (human herpesvirus 4): ubiquitous virus causing infectious mononucleosis, transmitted via saliva ('kissing disease').

Epstein-Barr Virus

Alternate index label for EBV.

Epstein-Barr Virus

Alternate index label for Epstein-Barr Virus.

Epstein-Barr Virus Antibody Test

A blood test measuring antibodies to Epstein-Barr virus antigens to help determine whether infection is recent, past, or absent. Interpretation requires the antibody pattern, timing, symptoms, and sometimes molecular testing.

Eptifibatide

An injectable antiplatelet medicine that blocks platelet glycoprotein IIb/IIIa and is used in selected acute coronary syndromes.

Epulis

Epulis is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Equilibrium Receptors

Sensory receptors in the vestibular system that detect head position and movement and help maintain balance.

Equilin

A naturally occurring estrogen found mainly in pregnant mares and used in some conjugated-estrogen medicines.

Equine Morbillivirus

A virus in the morbillivirus group that has been associated with severe disease in horses and can infect other mammals.

Equinovarus Deformity

A foot deformity combining downward pointing and inward turning, often congenital or neuromuscular and commonly called clubfoot when present at birth.

Equipment Safety

Practices that reduce injury, infection, malfunction, and misuse of medical equipment through selection, inspection, cleaning, maintenance, and training.

Equivalence Trial

A clinical trial designed to determine whether a new treatment's effects are neither meaningfully better nor worse than those of an established treatment within predefined limits.

Eradication

Eradication is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Erb'S Palsy

Upper brachial plexus injury (C5-C6) causing weakness or paralysis of the shoulder and arm muscles. Presents with the characteristic waiter's tip position: arm adducted, elbow extended, forearm pronated, wrist flexed. Most commonly caused by shoulder dystocia during delivery. See also Brachial Plexus Injury.

ErbB Receptors

A family of receptor tyrosine kinases, including EGFR/HER1 and HER2, that respond to growth factors and regulate proliferation, survival, and differentiation; abnormal signaling drives some cancers.

Erbium

A chemical element used in specialized lasers and materials; erbium lasers can remove or remodel selected skin and dental tissues.

ERCP

Endoscopic retrograde cholangiopancreatography, a procedure combining endoscopy and contrast imaging to diagnose or treat bile-duct and pancreatic-duct obstruction, stones, leaks, or strictures.

Erdheim-Chester Disease

A rare non-Langerhans-cell histiocytosis caused by clonal histiocyte proliferation, commonly producing symmetrical long-bone pain or sclerosis and possible cardiovascular, renal, neurologic, or retroperitoneal involvement.

Erdheim–Chester Disease

A rare, non-Langerhans cell histiocytosis characterized by xanthogranulomatous infiltration of multiple organ systems, most commonly the long bones, retroperitoneum, and cardiovascular system. It causes symmetric osteosclerosis of the diaphyses and metaphyses of long bones, exophthalmos, and xanthelasma.

Erectile Disorder

A sexual dysfunction characterized by persistent or recurrent difficulty achieving or maintaining an erection sufficient for sexual activity, or a marked decrease in erectile rigidity, lasting ≥ 6 months and causing marked distress. Prevalence: 10-20% of men under 60, rising to 40-70% over age 70.

Erectile Dysfunction

Erectile dysfunction is recurrent or persistent difficulty achieving or maintaining a penile erection firm enough for satisfactory sexual activity. It may

result from vascular, neurological, hormonal, structural, medication-related, psychological, or mixed causes. Associated conditions include cardiovascular disease, diabetes, hypogonadism, neurological disorders, Peyronie disease, anxiety, depression, and relationship or other psychosocial factors.

Erectile Dysfunction (ED, Impotence)

Alternate index label for ED, Impotence.

Erectile Dysfunction (ED, Impotence)

Alternate index label for Erectile Dysfunction (ED, Impotence).

Erection

Enlargement and stiffening of the penis caused mainly by increased blood flow into erectile tissue and reduced venous outflow during sexual arousal or other stimulation.

Erection Problems

Persistent difficulty achieving or maintaining an erection sufficient for satisfactory sexual activity, also called erectile dysfunction. Causes may include vascular, neurologic, hormonal, psychological, medication-related, or relationship factors and may signal broader cardiovascular risk.

Erector Spinae

A group of muscles running along the back that extend and stabilize the spine.

Alternate Names:

Erector spinae muscles

Ergocalciferol

Vitamin D2, a form of vitamin D used to prevent or treat vitamin-D deficiency; excessive doses can cause high blood calcium and kidney injury.

Ergolines

A group of chemical compounds related to ergot alkaloids, including substances that act on serotonin, dopamine, or adrenergic receptors.

Ergometrine

An ergot alkaloid that causes uterine contraction and vasoconstriction, used selectively to control postpartum hemorrhage; it is contraindicated in some cardiovascular and hypertensive conditions.

Ergonomics

The design of tasks, tools, and environments to fit human capabilities and reduce discomfort, error, and injury.

Ergosterol

The principal sterol in fungal cell membranes; it is targeted by antifungal medicines such as amphotericin B and azole drugs.

Ergot

A fungus, especially *Claviceps purpurea*, that infects grains such as rye and produces alkaloids; ingestion can cause ergotism, with vascular, neurologic, or uterine effects.

Ergotamine

An ergot alkaloid formerly used to abort migraine attacks by constricting cranial vessels; excessive or interacting use can cause ergotism and ischemia.

Ergotism

A toxidrome caused by prolonged exposure to ergot alkaloids from the fungus *Claviceps purpurea* contaminating rye and other grains, or from ergot-derived medications.

Ergots

Compounds derived from a fungus and used in some treatments, including certain headache disorders.

Eribulin Mesylate

The mesylate salt of eribulin, a microtubule-inhibiting chemotherapy medicine used for selected advanced breast cancer and liposarcoma.

Erlotinib Hydrochloride

The hydrochloride salt of erlotinib, an epidermal growth factor receptor tyrosine-kinase inhibitor used

for selected cancers with appropriate molecular features.

Erosion

A superficial loss of skin or mucosal epithelium that does not extend through the full thickness of the tissue, caused by friction, inflammation, infection, trauma, or chemical injury.

Erosion (Tooth)

Chemical wear without bacterial involvement. Causes: acidic foods/drinks, gastric acid (GERD, bulimia), environmental acids. Management: dietary modification, fluoride, restoration.

Alternate Names:

Tooth

Erosion Lesion

A superficial epithelial defect caused by loss of the surface layer of skin or mucosa, without the deep tissue loss of an ulcer.

Erosive Esophagitis

Inflammation of the esophagus with visible breaks or erosions in its lining, most commonly caused by gastroesophageal reflux.

Erosive Gastritis

Inflammation of the stomach lining with superficial erosions but no full-thickness ulceration. Commonly caused by NSAIDs, alcohol, or critical illness. May present with epigastric pain or gastrointestinal bleeding. Diagnosis is by upper endoscopy showing erosions and erythema.

Erosive Lichen Planus

A severe, chronic T-cell-mediated autoimmune inflammatory mucocutaneous disease that targets mucosal surfaces including the vulva, vagina, and oral cavity. It presents with painful, glazed, erythematous mucosal erosions bordered by Wickham striae, profuse desquamative yellow discharge, introital burning, and dyspareunia. Chronic inflammation frequently leads to architectural resorption, loss of the labia minora, narrowing of the vaginal introitus, and complete vaginal synechiae.

Alternate Names:

Vulvovaginal Gingival Syndrome; Erosive Vulvovaginal Lichen Planus

Erratic Results

Test or measurement results that vary unpredictably or are inconsistent with repeated observations; causes may include specimen problems, device error, biological variation, or an unsuitable test.

Ertapenem

A once-daily carbapenem antibiotic used for selected serious infections caused by susceptible bacteria; it does not cover *Pseudomonas* or enterococci reliably.

Erucic Acid

A 22-carbon monounsaturated fatty acid found in mustard and rapeseed oils; high exposure has raised cardiac safety concerns in some animal studies.

Erucic Acids

Long-chain monounsaturated fatty acids of the erucic-acid family, found mainly in certain plant oils and metabolized according to chain length and dietary exposure.

Eructation

The release of swallowed or stomach gas through the mouth.

Alternate Names:

Belching; burping

Eruption Cyst

A fluid-filled swelling over a tooth that is about to erupt, often appearing blue or purple and usually resolving when the tooth emerges.

Eruptive Fever

A clinical disorder or finding related to eruptive fever; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Eruptive Xanthomas

Sudden eruption of numerous small, yellow-orange

papules with an erythematous halo, typically on the buttocks, elbows, knees, and trunk, caused by severe hypertriglyceridemia. They represent lipid-laden macrophages in the dermis and are a marker of significantly elevated serum triglycerides, often exceeding 1000 mg/dL.

Eruptive Xanthomatosis

A sudden outbreak of small yellow-red papules caused by lipid-laden macrophages, usually associated with severe hypertriglyceridemia or uncontrolled diabetes.

Eryptosis

A programmed-cell-death process in red blood cells that promotes their removal from circulation without the usual nuclear changes of apoptosis.

Erysipelas

An acute superficial bacterial skin infection, usually caused by beta-haemolytic streptococci, producing a sharply demarcated, raised, tender, bright-red plaque with fever.

Erysipeloid

A localized occupational skin infection caused by *Erysipelothrix rhusiopathiae*, usually producing a painful, well-demarcated violaceous lesion on a hand or finger after animal or fish exposure.

Erysipelothrix

A genus of bacteria that can cause erysipeloid, a skin infection usually acquired from handling infected animals or animal products.

Erythema

Redness of the skin or mucous membranes caused by increased blood flow to dilated superficial capillaries, often associated with inflammation, infection, or another underlying condition. It commonly blanches with pressure, unlike purpura.

Erythema Ab Igne

A reticulated, hyperpigmented erythematous patch caused by chronic, repeated exposure to moderate

heat or infrared radiation, such as from heating pads, laptops, or fireplaces. Histologically it shows increased melanin and dermal amyloid deposition.

Erythema Chronicum Migrans

The expanding skin lesion of early Lyme disease, usually developing days to weeks after infection with *Borrelia* following a tick bite; it may have central clearing but is not always a classic bull's-eye.

Erythema Dose

The amount of ultraviolet or other radiation that produces visible redness of the skin; the minimal erythema dose is the lowest amount causing a defined erythematous response.

Erythema Gydatum Repens

A distinctive, rapidly moving, concentric erythematous eruption with a wood-grain or serpiginous pattern. It is often associated with an underlying malignancy, especially lung cancer, and should prompt an appropriate evaluation while other causes are considered.

Erythema Induratum

A nodular lobular panniculitis, classically associated with tuberculosis but also caused by other conditions, producing recurrent tender nodules and ulcers on the backs of the lower legs.

Erythema Infectiosum

A common childhood exanthem caused by parvovirus B19, also called fifth disease. A mild prodrome is followed by a slapped-cheek facial eruption and a lacy rash on the trunk and limbs.

Alternate Names:

fifth disease.

Erythema Infectiosum (Fifth Disease)

Alternate index label for Fifth Disease.

Erythema Marginatum

A migratory, non-pruritic, erythematous rash with a serpiginous or annular pattern that blanches centrally, occurring as a major Jones criterion in acute

rheumatic fever. It typically appears on the trunk and proximal extremities and may come and go over days or weeks.

Erythema Migrans

The expanding annular skin lesion of early Lyme disease, appearing days to weeks after infection with *Borrelia* transmitted by Ixodes ticks. It may be uniformly erythematous rather than a classic bull's-eye lesion and is usually diagnosed clinically.

Erythema Multiforme

An acute, self-limited immune-mediated reaction, most commonly triggered by herpes simplex virus (90 percent of cases) and *Mycoplasma pneumoniae*. It presents as target lesions (iris lesions with three zones: central duskiness, middle pale edematous ring, outer erythematous halo) on extremities.

Erythema Nodosum

The most common form of panniculitis, characterized by painful, erythematous, tender nodules and plaques on the bilateral shins, representing an inflammatory reaction of the subcutaneous fat.

Erythema Toxicum

A common benign transient newborn rash consisting of erythematous macules, papules, or pustules that usually resolves without treatment.

Erythema Toxicum Neonatorum

A benign, self-limited cutaneous eruption occurring in up to 50 percent of healthy term newborns, typically presenting in the first 24 to 72 hours of life. Characterized by blotchy erythematous macules, papules, and small pustules on an erythematous base distributed over the face, trunk, and proximal extremities while sparing the palms and soles. Microscopic examination of pustule fluid demonstrates a predominance of eosinophils; no treatment is required as lesions resolve spontaneously within 1 to 2 weeks.

Alternate Names:

Erythema Neonatorum; Toxic Erythema of the Newborn

Erythrasma

A superficial skin infection caused by *Corynebacterium minutissimum*, producing sharply demarcated reddish-brown patches in skin folds that fluoresce coral-red under Wood lamp.

Erythroblast

Erythroblast is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Erythroblastosis Fetalis

Fetal or newborn red-cell destruction caused by maternal antibodies against fetal blood-cell antigens.

Erythroblastosis Foetalis

A clinical disorder or finding related to erythroblastosis foetalis; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Erythrocyte

A mature red blood cell specialized to transport oxygen and carbon dioxide through the circulation. Its biconcave, anucleate form maximizes surface area and flexibility; haemoglobin provides the principal oxygen-binding capacity.

Erythrocyte Enzyme Deficiency

An inherited or acquired deficiency of an enzyme needed for red-cell metabolism or protection. It can shorten erythrocyte survival and cause hemolytic anemia; examples include glucose-6-phosphate dehydrogenase and pyruvate-kinase deficiency.

Erythrocyte Membrane

The flexible membrane surrounding a red blood cell; it helps maintain cell shape and controls exchange with the blood.

Erythrocyte Sedimentation Rate

A non-specific inflammatory marker measuring the rate at which erythrocytes settle in 1 hour. Normals: males $\text{age}/2$, females $(\text{age}+10)/2$. Markedly elevated >100 mm/hr: suggests infection (endo-

carditis), autoimmune (giant cell arteritis, polymyalgia rheumatica, SLE, RA), malignancy (multiple myeloma), and chronic kidney disease.

Erythrocytosis

An increased red-cell mass, usually reflected by elevated haemoglobin and haematocrit. It may be relative because of reduced plasma volume or absolute because of increased erythropoiesis, which can be primary from a marrow disorder or secondary to hypoxia, erythropoietin production, or other causes.

Erythroderma

Generalized redness and scaling involving most of the skin surface, caused by psoriasis, eczema, drug reactions, lymphoma, or other disease and potentially causing fluid and temperature loss.

Erythrodermic Psoriasis

A severe, life-threatening variant of psoriasis characterised by widespread erythema and scaling involving more than 90% of the body surface area, often with systemic symptoms including fever, chills, and hemodynamic instability.

Erythrodysplasia

Abnormal development or maturation of red blood cells, usually referring to ineffective erythropoiesis in marrow or a dysplastic blood disorder.

Erythroid Hyperplasia

Increased production of red-cell precursors in bone marrow, often as a response to anemia, blood loss, hemolysis, or reduced oxygen availability.

Erythroleukoplakia

A mixed red-and-white oral mucosal lesion with a high risk of severe dysplasia or cancer, requiring specialist examination and often biopsy.

Erythromelalgia

Episodic burning pain, redness, and warmth of the extremities triggered by warmth or exercise and relieved by cooling, caused by arteriolar inflammation and platelet-mediated microvascular occlusion.

Erythromycin

A medicine or chemical compound called erythromycin. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Erythromycin Estolate

A medicine or chemical compound called erythromycin estolate. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Erythromycin Ophthalmic Prophylaxis

Application of ophthalmic erythromycin to a newborn soon after birth to reduce the risk of gonococcal ophthalmia neonatorum. Use, timing, and alternatives depend on local policy, availability, and the mother's infection-risk assessment.

Erythroplakia

A persistent red patch of mucosa, especially in the mouth, that may represent a premalignant or malignant lesion and requires clinical evaluation, often including biopsy.

Erythroplasia

A sharply demarcated red lesion of skin or mucosa; persistent oral or genital erythroplasia can represent severe dysplasia or carcinoma in situ.

Erythroplasia Of Queyrat

Squamous-cell carcinoma in situ of the glans penis or prepuce, appearing as a persistent well-demarcated red velvety plaque.

Erythropoiesis

The process by which haematopoietic stem and progenitor cells develop into mature red blood cells. It occurs primarily in adult bone marrow and is regulated chiefly by erythropoietin in response to tissue oxygenation; maturation requires iron, vitamin B12, folate, and appropriate marrow function.

Erythropoiesis-Stimulating Agents

Medicines such as epoetin alfa and darbepoetin alfa

that stimulate red-cell production, often used for selected forms of anemia in chronic kidney disease or cancer treatment. Use requires appropriate assessment of iron status, risks, and treatment targets.

Erythropoietin

A glycoprotein hormone that regulates erythropoiesis—the production of red blood cells in the bone marrow. Produced primarily by interstitial peritubular cells in the kidney (90%) and hepatocytes (10%) in response to tissue hypoxia.

Erythropoietin

A kidney-produced hormone that stimulates red-cell production in bone marrow in response to reduced tissue oxygenation.

Erythropoietin Deficiency

Insufficient erythropoietin production or an inadequate erythropoietin response, leading to reduced red-cell production and usually a hypoproliferative, often normocytic anemia. It is particularly important in chronic kidney disease.

Erythropoietin Receptor

A cell-surface receptor on erythroid precursor cells that responds to erythropoietin and promotes red-cell survival, proliferation, and maturation.

Erythropoietin Test

A blood test measuring erythropoietin, a hormone produced mainly by the kidneys that stimulates red-blood-cell production. It helps evaluate selected anemias or erythrocytosis and must be interpreted with the blood count and kidney function.

Erythrosine

A red food and pharmaceutical dye whose use is limited in some settings because of safety concerns.

ESBL

Extended-spectrum beta-lactamase, enzymes that hydrolyse penicillins, cephalosporins, and monobactams. ESBLs are common in Enterobacteriaceae,

particularly *Klebsiella pneumoniae* and *E. coli*. ESBL production confers resistance to most beta-lactam antibiotics except carbapenems.

ESBL-Producing Organisms

Bacteria that produce extended-spectrum beta-lactamases (ESBLs), enzymes that hydrolyze penicillins, cephalosporins, and monobactams. ESBLs are common in Enterobacteriaceae, particularly *Klebsiella pneumoniae* and *E. coli*. ESBL production confers resistance to most beta-lactam antibiotics except carbapenems.

Escape Reaction

A physiologic response that restores an organ's function after it has been suppressed by a drug, hormone, nerve influence, or feedback mechanism.

Eschar

A firm, usually black or brown layer of dead tissue over a wound, burn, pressure injury, infection, or vascular lesion that may conceal deeper damage.

Escharotomy

An incision through rigid eschar from a deep circumferential burn to relieve constriction and restore chest expansion or blood flow.

Escherichia

A genus of gram-negative bacteria that includes *E. coli* and related intestinal or environmental species; some strains cause intestinal or extraintestinal infection.

Escherichia Albertii

A diarrheagenic *Escherichia* species that can cause gastroenteritis and may be mistaken for certain *E. coli* strains in routine testing.

Escherichia Blattae

An *Escherichia* species originally associated with cockroaches and rarely reported in human specimens; pathogenic significance is uncommon and requires clinical correlation.

Escherichia Coli

A gram-negative bacterium that normally lives in the intestine but can cause urinary, intestinal, bloodstream, and other infections. Different strains produce different clinical syndromes, including traveler's diarrhea, bloody diarrhea, and hemolytic uremic syndrome.

Escherichia Coli Infection

Infection caused by *E. coli*, ranging from urinary-tract and bloodstream infection to diarrheal disease; symptoms and treatment depend on the strain and site.

Escherichia Coli O157:H7 Infection

Infection with a Shiga-toxin-producing strain of *E. coli*, usually acquired from contaminated food, water, or contact with infected animals or people. It commonly causes severe abdominal cramps and bloody diarrhea and can lead to hemolytic uremic syndrome, especially in children. Medical advice is important; antibiotics and antidiarrheal medicines may be inappropriate in suspected cases.

Escherichia Fergusoni

An uncommon *Escherichia* species that can cause opportunistic urinary, wound, bloodstream, or gastrointestinal infection.

Escherichia Hermannii

An uncommon gram-negative *Escherichia* species occasionally associated with opportunistic human infection, including urinary or bloodstream infection.

Escherichia Vulneris

An uncommon *Escherichia* species rarely associated with wound, urinary, or bloodstream infection, usually in people with underlying illness.

Escitalopram

A selective serotonin-reuptake inhibitor used mainly for depression and generalized anxiety disorder; common effects include nausea, sleep changes, sexual dysfunction, and increased bleeding risk.

ESKD

End-stage kidney disease defined as GFR less than 15 or need for dialysis. Medicare eligibility at GFR less than 15 or on dialysis. Palliative care option for those not wanting renal replacement therapy. Comprehensive advance care planning essential.

Eskimo Lung

Eskimo Lung is an anatomical or physiologic term describing a body structure or function; its clinical meaning depends on its location, role, and associated findings.

Esomeprazole

A medicine or chemical compound called esomeprazole. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Esomeprazole Magnesium

A medicine or chemical compound called esomeprazole magnesium. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Esophageal

Relating to the esophagus, the muscular tube that transports swallowed food and liquid from the pharynx to the stomach.

Esophageal Achalasia

Primary esophageal motility disorder characterized by failure of the lower esophageal sphincter to relax during swallowing and absent peristalsis in the distal esophagus, resulting from selective loss of inhibitory neurons in the myenteric plexus.

Esophageal Adenocarcinoma

A malignant gland-forming tumor of the esophagus, most often arising in the distal esophagus in the setting of Barrett Esophagus. It is associated with chronic gastroesophageal reflux, obesity, smoking, and other risk factors. Progressive dysphagia, weight loss, bleeding, or chest discomfort may occur, although early disease can be asymptomatic. Diagnosis requires endoscopic biopsy, followed by staging to guide multimodal oncologic treatment.

Esophageal Atresia

A congenital anomaly where the esophagus ends in a blind pouch, often with a tracheoesophageal fistula connecting the distal esophagus to the trachea. Presents with drooling, inability to pass a nasogastric tube, and choking with feedings. The 3 Es association: Esophageal atresia, Esophageal anomalies, and Extremity anomalies.

Esophageal Bleeding

Hemorrhage arising from the esophagus, commonly from varices, erosive inflammation, ulceration, tears, or cancer, and potentially presenting as vomiting blood or black stool.

Esophageal Cancer

Malignant tumor of the esophagus. Squamous cell carcinoma predominates in the upper and middle third (smoking, alcohol); adenocarcinoma of the lower third arises from Barrett esophagus. Presents with progressive dysphagia and weight loss; treatment depends on stage and includes surgery, chemoradiation, and systemic therapy.

Alternate Names:

Cancer, Esophageal

Esophageal Candidiasis

Fungal infection of the esophagus caused by *Candida* species common in immunocompromised patients including those with HIV, organ transplant recipients, and patients on inhaled corticosteroids. Presents with odynophagia and dysphagia. Diagnosed by endoscopy showing white adherent plaques and biopsy confirming fungal hyphae.

Esophageal Carcinoma

Cancer arising in the esophagus, most often squamous-cell carcinoma or adenocarcinoma; progressive swallowing difficulty, weight loss, pain, or bleeding may occur.

Esophageal Culture

A laboratory culture of a specimen from the esophagus to identify bacteria, fungi, or other microorganisms when infection is suspected. Sampling method and prior treatment affect results, which must be

interpreted with endoscopic and clinical findings.

Esophageal Dilation

Endoscopic passage of bougies or balloons to widen esophageal strictures. Used for symptomatic strictures from peptic disease, radiation, caustic injury, or eosinophilic esophagitis. Savary bougies are passed over a guidewire with progressive dilation. Through-the-scope balloons allow controlled radial dilation under direct visualization.

Esophageal Diverticulum

Outpouching of the esophageal wall classified by location. Zenker diverticulum occurs in the hypopharynx and is the most common esophageal diverticulum. Mid-esophageal diverticula are traction diverticula caused by extrinsic inflammatory disease including tuberculosis and histoplasmosis.

Esophageal Duplication

A rare congenital cyst or duplicated segment alongside the esophagus that may compress the airway or esophagus and cause respiratory or swallowing symptoms.

Esophageal Fistula

An abnormal passage between the esophagus and another organ, most often the airway, causing coughing with swallowing, aspiration, infection, or severe illness.

Esophageal Foreign Body

An object or food bolus lodged in the esophagus, causing difficulty swallowing, pain, drooling, or breathing symptoms. Batteries, sharp objects, complete obstruction, and perforation require urgent assessment and removal.

Esophageal Infection

Infection of the esophagus, commonly caused by *Candida*, viruses, or bacteria, producing painful swallowing, difficulty swallowing, chest pain, or ulcers.

Esophageal Intramural

Pseudodiverticulosis

Condition characterized by multiple small outpouchings of the esophageal submucosal glands seen on barium swallow as flask-shaped contrast collections oriented longitudinally along the esophagus. The pseudodiverticula are not true diverticula but rather dilated excretory ducts of submucosal glands.

Esophageal Leukoplakia

White plaques on the esophageal mucosa representing squamous cell hyperplasia with hyperkeratosis. Similar to oral leukoplakia it is considered a potentially premalignant condition. Associated with chronic irritation from smoking, alcohol, and gastroesophageal reflux.

Esophageal Manometry

Diagnostic test measuring esophageal motor function by recording intraluminal pressures during swallowing. A catheter with pressure sensors is passed through the nose into the stomach and withdrawn to record pressures at multiple levels. High-resolution manometry using closely spaced sensors provides a spatiotemporal color plot.

Esophageal Necrosis

Death of esophageal lining, classically producing a black-appearing esophagus in severe ischemia, shock, reflux, or caustic injury and carrying a risk of perforation.

Esophageal Perforation

Full-thickness rupture of the esophageal wall, most commonly Boerhaave syndrome from forceful vomiting occurring on the left posterolateral distal esophagus. Iatrogenic perforation during endoscopy is another common cause. Presents with chest pain, subcutaneous emphysema, and mediastinitis.

Esophageal PH Monitoring

24-hour ambulatory monitoring of esophageal acid exposure. Catheter-based (transnasal pH probe placed 5 cm above LES) or wireless (Bravo capsule endoscopically attached to esophageal mucosa). Measures: % time pH <4 (total, upright, supine), DeMeester score (>14.7 abnormal), and symptom-

reflux correlation.

Esophageal Spasm

Esophageal motility disorder characterized by uncoordinated or excessive contractions impairing bolus transport. Diffuse esophageal spasm shows simultaneous contractions on manometry. Jackhammer esophagus features extremely high-amplitude peristaltic contractions.

Esophageal Spasms

Uncoordinated or forceful contractions of the esophagus that can cause chest pain and difficulty swallowing.

Esophageal Squamous Cell Carcinoma

A malignant epithelial tumor of the esophagus arising from the squamous lining, most commonly localized in the upper and middle thirds of the thoracic esophagus. Globally the predominant histological type of esophageal cancer; strongly linked to tobacco use, heavy alcohol consumption, hot beverage ingestion, and nutritional deficiencies.

Esophageal Stenosis

Narrowing of the esophageal lumen from scar, inflammation, tumor, radiation, or congenital disease, causing progressive difficulty swallowing.

Esophageal Stent

Self-expanding metal stent for esophageal obstruction. Palliative for malignancy. Migration, tumor ingrowth common complications.

Esophageal Stricture

Narrowing of the esophageal lumen from chronic gastroesophageal reflux disease, caustic ingestion, or eosinophilic esophagitis. Presents with progressive dysphagia initially to solids then liquids. Diagnosed by upper endoscopy and barium swallow.

Esophageal Stricture - Benign

A noncancerous narrowing of the esophagus that can cause difficulty swallowing, food impaction, or regurgitation. Common causes include reflux-related

scarring, caustic injury, eosinophilic esophagitis, radiation, and procedures; endoscopy guides diagnosis and treatment.

Esophageal Surgery

Operations on the esophagus or its junction with the stomach, including procedures for reflux, stricture, cancer, perforation, or motility disorders. The operation is selected according to the disease, anatomy, symptoms, and surgical risk.

Esophageal Ulceration

A break in the esophageal lining caused by reflux, infection, medicines, caustic injury, inflammation, or cancer and capable of causing pain or bleeding.

Esophageal Varices

Dilated, tortuous submucosal veins in the lower esophagus, most commonly caused by portal hypertension (cirrhosis—hepatic fibrosis with increased portal venous pressure >10 mmHg). They develop as portosystemic collaterals connecting the portal venous system (left gastric vein) to the systemic venous system (azygos vein).

Alternate Names:

Varices, Esophageal

Esophageal Web

Thin mucosal fold protruding into the esophageal lumen most commonly in the postcricoid region or upper esophagus. Webs may be asymptomatic or cause dysphagia particularly to solids. Multiple webs can occur in association with iron deficiency anemia as part of Plummer-Vinson syndrome or Paterson-Kelly syndrome.

Esophagectomy

Surgical removal of part or all of the esophagus.

Esophagectomy - Minimally Invasive

Surgical removal of part or all of the esophagus using thoracoscopic or laparoscopic approaches, sometimes with robotic assistance, and reconstruction of the digestive tract. It is used for selected cancer or severe esophageal disease and carries substantial perioperative risks.

Esophagectomy - Open

Surgical removal of part or all of the esophagus through open chest and abdominal or neck incisions, followed by reconstruction of the digestive tract. It is used for selected esophageal disease and carries risks including leakage, infection, respiratory complications, and nutritional problems.

Esophagitis

Esophagitis is inflammation or injury of the esophageal lining caused by reflux, infection, medication, allergy, radiation, or other mechanisms.

Esophagogastric Bypass

Surgical creation of a conduit between the esophagus and stomach to restore gastrointestinal continuity, most commonly performed as part of esophagectomy or to palliate unresectable esophageal malignancy. Techniques include gastric pull-up with esophagogastrostomy and jejunal interposition.

Esophagogastric Junction

The transition where the esophagus meets the stomach, including the lower esophageal sphincter and anatomic structures that help prevent reflux.

Esophagogastric Tamponade

Temporary control of bleeding from esophageal or gastric varices by inflating a balloon device to apply pressure; it is a rescue measure with serious airway and aspiration risks.

Esophagogastroduodenoscopy

Upper endoscopy: flexible endoscopic examination of the esophagus, stomach, and duodenum.

Esophagoscope

A flexible or rigid instrument with a light and viewing system used to inspect the esophageal lining and obtain samples or perform treatment.

Esophagoscopy

Endoscopic examination of the esophagus, usually to investigate dysphagia, bleeding, reflux complications, strictures, foreign bodies, or suspected tu-

mors.

Esophagram

An imaging study of the esophagus performed after the patient swallows a contrast material.

Esophagus

A muscular tube that transports food and liquids from the pharynx (throat) to the stomach.

Esophagus (Esophagus)

A muscular tube approximately 25 cm long connecting the pharynx to the stomach, extending from the cricopharyngeus muscle (at the level of C6) to the gastroesophageal junction (T11).

Alternate Names:

esophagus

Esophagus Disease

A disorder of the esophagus causing swallowing difficulty, pain, reflux, regurgitation, bleeding, obstruction, inflammation, motility disturbance, or cancer.

Esophagus Neoplasm

A benign, precancerous, or malignant growth in the esophagus; evaluation may require endoscopy, biopsy, imaging, and staging.

Esotropia

A type of strabismus in which one eye deviates inward (toward the nose). Congenital/infantile esotropia: large-angle, constant, onset before 6 months; requires early surgical correction. Accommodative esotropia: associated with hyperopia, onset age 2-3 years; treatment with corrective glasses improves alignment.

ESR

The erythrocyte sedimentation rate, a nonspecific blood test measuring how quickly red cells settle in a tube; it can reflect inflammation but cannot identify its cause.

Alternate Names:

SED Rate

ESRD

Alternate index label; see *End-Stage Renal Disease*.

Essential

Necessary or fundamental; in a diagnosis such as essential hypertension, it indicates that no single secondary cause has been identified.

Essential Fats

Fatty acids that the body cannot make in sufficient amounts and must obtain from food.

Alternate Names:

Essential fatty acids

Essential Fatty Acid

A fatty acid the body cannot make in sufficient amounts and therefore must obtain from food; the main human examples are linoleic acid and alpha-linolenic acid.

Essential Hypertension

Persistent elevation of systemic arterial blood pressure without an identifiable single secondary cause. It results from interacting genetic, vascular, renal, metabolic, and environmental factors and increases the risk of stroke, coronary disease, heart failure, kidney disease, and retinopathy.

Essential Medicines

Drugs that satisfy the priority healthcare needs of a population, as defined by the WHO Model List of Essential Medicines. Selection criteria include efficacy, safety, comparative cost-effectiveness, and suitability for use in resource-limited settings.

Essential Mixed Cryoglobulinemia

A disorder in which circulating immunoglobulins precipitate in cooler temperatures and deposit in small blood vessels, causing vasculitis. Symptoms may include purpura, joint pain, weakness, nerve problems, kidney disease, and cold sensitivity. Evaluation includes cryoglobulin testing and assessment for infections, autoimmune disease, and lymphoproliferative disorders.

Essential Nutrient

Nutrient that cannot be synthesized by body, must be obtained from diet. Essential amino acids, fatty acids, vitamins, minerals.

Essential Thrombocythemia

A chronic myeloproliferative neoplasm characterized by sustained platelet count elevation (greater than 450,000 per microliter) due to clonal megakaryocytic proliferation, with a risk of both thrombotic and hemorrhagic complications.

Essential Tremor

The most common movement disorder, characterized by a bilateral, symmetrical, postural and kinetic tremor predominantly affecting the upper extremities, with a frequency of four to twelve hertz. It typically begins insidiously in the second or third decade of life and slowly progresses over years.

Essiac

Essiac is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Estazolam

A benzodiazepine sedative-hypnotic used short-term for insomnia; it can cause sedation, impaired coordination, dependence, respiratory depression, and withdrawal.

Ester

Ester is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Esterase

Esterase is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Esterification

Esterification is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Esterified Estrogens

A prescription mixture of estrogen compounds used for selected menopausal symptoms and other estrogen-deficiency conditions. Risks include blood clots, stroke, and endometrial cancer when used without appropriate progestogen in a person with a uterus; treatment should be individualized.

Esthesia

Sensation or perception, especially the ability to feel touch, pain, temperature, or position.

Esthesiometer

An instrument used to measure sensory perception, such as touch, pressure, or corneal sensitivity.

Esthesioneuroblastoma

A rare cancer arising from specialized nerve tissue in the upper nasal cavity, potentially causing nasal obstruction, bleeding, or loss of smell.

Esthetician

A person trained in nonmedical skin-care and cosmetic treatments.

Alternate Names:

Aesthetician

Estimated Average Glucose

An estimate of mean blood glucose calculated from the hemoglobin A1c result and reported in glucose units that correspond roughly to daily readings.

Estimated Date Of Confinement

The estimated date when a pregnancy is expected to reach delivery, also called the estimated due date; it is commonly calculated from the last menstrual period or early ultrasound.

Estimated Due Date

The expected date of delivery, calculated as 40 weeks (280 days) from the first day of the last menstrual period. Confirmed or adjusted by first-trimester ultrasound crown-rump length measurement (most accurate ± 5 -7 days). Second-trimester ultrasound is less accurate (± 10 -14 days). Third-trimester is least accurate (± 21 -30 days).

Estimated Fetal Weight

An ultrasound-derived estimate of fetal weight, calculated using formulas that combine BPD, HC, AC, and FL. The Hadlock formula (using HC, AC, FL) is the most commonly used. Accuracy: ± 10 -15%. Indications: assess fetal growth, determine size for gestational age, and plan delivery. EFW <10th percentile: SGA/IUGR.

Estimated Glomerular Filtration Rate

An estimate of glomerular filtration derived from serum creatinine and, depending on the validated equation, age and sex. It is used to stage chronic kidney disease and guide medication dosing, but is less reliable when creatinine is changing rapidly or in unusual body composition.

Estivation

A state of dormancy or reduced activity during hot or dry conditions, used mainly in zoology and biology.

Estradiol

The principal biologically active estrogen hormone during the reproductive years.

Alternate Names:

17 β -estradiol

Estradiol Blood Test

A blood test measuring estradiol, the principal biologically active estrogen during much of the reproductive years. It may help evaluate ovarian or testicular function, puberty, fertility treatment, menopause, or selected endocrine disorders; levels vary by age, sex, cycle stage, and medicines.

Estramustine

A medicine or chemical compound called estramustine. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Estriol

An estrogen produced substantially by the fetoplacental unit during pregnancy; maternal serum estriol may be included in prenatal screening.

Estrogen

A group of steroid hormones involved in reproductive development, menstrual regulation, bone health, and other physiologic functions.

Estrogen Test

A test that measures estrogen levels in a blood, urine, or saliva sample. It may help evaluate menstrual problems, infertility, menopause, puberty, pregnancy, or hormone-related conditions. Results depend on the type of estrogen measured, the person's age and sex, the menstrual-cycle stage, and whether the person is pregnant.

Alternate Names:

Estrogen Levels Test Estrogen Blood Test

Estrogen Overdose

Toxic or excessive exposure to estrogen, which may cause nausea, breast tenderness, headache, abnormal bleeding, thromboembolic risk, or other hormone-related effects.

Estrogen Receptor

A cellular protein that binds estrogen and mediates its effects on target tissues.

Estrogen Replacement Therapy

Treatment with estrogen to relieve menopausal symptoms or replace deficient estrogen; when the uterus is present, a progestogen is generally added to reduce endometrial cancer risk.

Estrogen-Replacement Therapy

Treatment using estrogen to replace hormone lost during menopause or because of ovarian failure.

Alternate Names:

Estrogen therapy

Estrogens

A group of steroid hormones, including estradiol, that regulate reproductive development, the menstrual cycle, bone, and other tissues; they are produced mainly by the ovaries before menopause and by other tissues at other times.

Estrone

A naturally occurring estrogen produced mainly in peripheral tissues from adrenal and ovarian precursors, especially after menopause.

Estrous Cycle

The reproductive cycle of many nonhuman female mammals, involving recurring hormonal and behavioral changes around ovulation.

Estrus

The period in an animal's reproductive cycle when the female is sexually receptive and ovulation may occur.

Eszopiclone

A nonbenzodiazepine hypnotic used for insomnia; it can cause sedation, impaired performance, dependence, and an unpleasant taste.

Etacrynic Acid

A loop diuretic that increases urine production by blocking salt reabsorption in the kidney.

Etanercept

A biologic medicine that binds tumor necrosis factor and is used for rheumatoid arthritis, psoriasis, spondyloarthritis, and other inflammatory diseases; infection and tuberculosis screening are important.

Etanidazole

A medicine or chemical compound called etanidazole. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Etaracizumab

A medicine or chemical compound called etaracizumab. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Ethacrynic Acid

A loop diuretic that blocks sodium and chloride reabsorption in the kidney; it can cause substantial fluid loss, low electrolytes, and dose-related ototoxicity.

Ethambutol

An antimycobacterial medicine used in combination treatment for tuberculosis; dose adjustment and monitoring for optic toxicity are essential.

Ethane

Ethane is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethanol

Ethanol is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethanol Metabolism

Ethanol metabolism is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethanol Poisoning

Toxic effects from excessive alcohol ingestion, ranging from impaired coordination and vomiting to respiratory depression, hypoglycemia, coma, and death.

Ethanolamine

Ethanolamine is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and ex-

posure circumstances.

Ethchlorvynol

An older sedative-hypnotic medicine formerly used for insomnia; dependence, overdose, and central-nervous-system depression limit its use.

Ether

A volatile organic solvent; diethyl ether was formerly used as a general anesthetic but is highly flammable.

Ethers

Ethers is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethics

Principles used to evaluate what is right, fair, respectful, and responsible in healthcare, research, and professional practice.

Ethidium

Ethidium is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethinyl Estradiol

A synthetic estrogen used in many combined hormonal contraceptives and other hormone treatments; risks include blood clots and cardiovascular complications in susceptible people.

Ethiodized Oil

An iodinated oil used as a contrast agent in selected lymphatic or liver procedures; allergy, oil embolization, and thyroid effects are possible.

Ethion

Ethion is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethionamide

An antibiotic used in some tuberculosis treatment regimens that interferes with production of the bacterial cell wall.

Ethmocephaly

A severe facial developmental anomaly with closely spaced eyes and a proboscis-like nasal structure, usually associated with holoprosencephaly.

Ethmoid Bone

Delicate bone between the orbits forming parts of the nasal septum and orbital/nasal roofs. Has the cribriform plate (olfactory nerve filaments), perpendicular plate (superior nasal septum), and labyrinths (nasal conchae, ethmoid air cells). The crista galli provides falx cerebri attachment.

Ethmoid Sinus

One of the air-filled ethmoid cells between the nose and the eye sockets; inflammation can cause ethmoid sinusitis and may threaten nearby orbital structures.

Ethmoid Sinusitis

Inflammation or infection of the ethmoid sinuses, causing nasal congestion, facial or eye pain, fever, or headache; orbital complications require urgent assessment.

Ethmoidal Air Cells

Air-filled spaces in the ethmoid bone that form part of the paranasal sinus system.

Ethmoidal Sinus

The ethmoidal sinuses, also known as ethmoidal air cells, are a collection of multiple small, thin-walled, air-containing cavities within the ethmoid bone between the orbits and above the nasal cavity, and they are divided into anterior and posterior groups based on their drainage sites.

Alternate Names:

ethmoidal air cells, are a collection of multiple.

Ethmoiditis

Inflammation of the ethmoid air cells, usually from

infection or chronic sinus disease.

Ethnicity

A social and cultural identity shaped by shared ancestry, history, language, traditions, or community; it is not a biological diagnosis and should not be used as a substitute for individual risk assessment.

Ethoglucid

An older sclerosing substance used historically to treat selected vascular lesions or hemorrhoids; modern use is limited by tissue-injury risks and alternatives.

Ethosuximide

An anticonvulsant used mainly for absence seizures; common adverse effects include gastrointestinal symptoms, fatigue, and headache, with rare blood or skin reactions.

Ethoxzolamide

A carbonic anhydrase inhibitor that can reduce fluid formation in the eye and other tissues.

Ethyl Chloride

A volatile liquid sprayed on the skin to produce brief cooling and local numbness.

Ethyl Ether

Ethyl ether is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethyl Methanesulfonate

Ethyl methanesulfonate is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethylene

Ethylene is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethylene Chlorohydrin

Ethylene chlorohydrin is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethylene Dibromide

Ethylene dibromide is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethylene Glycol

A toxic alcohol found in some antifreeze and industrial products. Ingestion can cause severe metabolic acidosis, neurologic impairment, kidney injury, and death.

Ethylene Glycol Blood Test

A blood test measuring ethylene glycol, a toxic alcohol found in some antifreeze products. Detection supports diagnosis of poisoning, but treatment must not wait for the result when the exposure and metabolic findings are strongly suggestive.

Ethylene Glycol Poisoning

Toxicity from ingestion of ethylene glycol, often found in antifreeze. Its metabolites can cause severe metabolic acidosis, neurologic impairment, kidney injury, and calcium-oxalate deposition. It requires urgent toxicology assessment and antidotal or extracorporeal treatment when indicated.

Ethylene Oxide

Ethylene oxide is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethylenethiourea

Ethylenethiourea is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Ethylnitrosourea

A laboratory alkylating chemical that damages DNA and induces mutations; it is used in research and is not a therapeutic medicine.

Ethinodiol Diacetate

A synthetic progestin formerly used in some oral contraceptives.

Etidronic Acid

A bisphosphonate that binds bone mineral and inhibits osteoclast-mediated bone resorption; it is also used as a chelating or scale-control agent in other settings.

Etilefrine

A sympathomimetic medicine that can raise blood pressure by increasing vascular tone.

Etiocholanolone

A metabolite of steroid hormones, especially androgens, measured in some endocrine or research studies.

Etiology

The cause or set of causes of a disease or health condition, including genetic, infectious, environmental, immune, traumatic, and behavioural factors.

Etioplast

A plant cell organelle that develops from a proplastid in darkness and can convert into a chloroplast when exposed to light.

Etodolac

A nonsteroidal anti-inflammatory drug used for pain and inflammation; gastrointestinal bleeding, kidney injury, cardiovascular events, and hypersensitivity are important risks.

Etomidate

A short-acting intravenous anesthetic used for induction, particularly when cardiovascular stability is desired; it can cause myoclonus, nausea, and transient adrenal suppression.

Etonogestrel

A progestin used in some hormonal contraceptive implants to prevent ovulation.

Etoposide

A chemotherapy medicine that blocks topoisomerase II and is used for several cancers.

Etoposide Phosphate

A water-soluble form of etoposide that is converted to etoposide in the body.

Etoricoxib

A selective cyclooxygenase-2 inhibitor used to relieve pain and inflammation in selected conditions.

Etretinate

An oral retinoid formerly used for severe psoriasis; it remains in the body for a long time and can cause birth defects.

Eubacterium

A genus of anaerobic bacteria found mainly in the intestine and oral cavity; some species are associated with opportunistic infection.

Eubacterium Limosum

An anaerobic intestinal bacterial species involved in fermentation and rarely associated with human infection.

Eubacterium Rectale

A common anaerobic intestinal bacterium involved in fiber fermentation and short-chain fatty-acid production.

Eubacterium Ventriosum

An anaerobic bacterium found in the human intestine that rarely causes opportunistic infection.

Eucalyptus Oil

An essential oil containing compounds such as eucalyptol; ingestion can be toxic, and concentrated oil should not be used internally without appropriate medical guidance.

Eucalyptus Oil Overdose

Toxic effects after ingesting a large amount of eucalyptus oil, which can cause burning, vomiting, drowsiness, seizures, and breathing problems.

Euchromatin

Euchromatin concerns euchromatin, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Eugenecist

Eugenecist is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Eugenics

Eugenics is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Eugenol Oil Overdose

Toxic effects after excessive exposure to eugenol oil, which may cause mouth or stomach irritation, liver injury, bleeding, seizures, or respiratory depression.

Euglena

A group of single-celled organisms with features of both plant-like and animal-like cells; they are mainly studied in biology.

Eukaryote

An organism whose cells contain a nucleus and other membrane-bound structures.

Eukaryotic Cell

A cell with a membrane-bound nucleus and specialized organelles, including the cells of animals, plants, fungi, and protists.

Eunuch

A historical term for a castrated male; it is not a current diagnosis and should be replaced with a specific description of anatomy or hormone status.

Euphoria

An intense feeling of well-being, elation, or happiness, which can be a normal emotional response or a pathologic symptom.

Europium

Europium is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Eurotium

A group of molds in the *Aspergillus* lineage that can grow in dry, salty, or sugar-rich environments.

Eustachian Tube

Auditory/pharyngotympanic tube connecting the middle ear to the nasopharynx, approximately 36 mm long in adults. Cartilaginous medial two-thirds and bony lateral third. Functions: equalizing middle ear/atmospheric pressure, draining middle ear secretions, and protecting from nasopharyngeal pathogens.

Eustachian Tube Dysfunction

Failure of the eustachian tube to open or close normally, causing ear pressure, muffled hearing, pain, or recurrent middle-ear fluid. It may follow an upper-respiratory infection, allergy, or rapid pressure changes.

Eustachian Tube Patency

The ability of the auditory tube to open and equalize pressure between the middle ear and nasopharynx; blockage causes pressure, muffled hearing, or effusion.

Eustachian Tube Problems

Eustachian-tube dysfunction prevents normal pressure equalization and middle-ear drainage, causing fullness, popping, muffled hearing, pain, or recurrent infection. Allergies, infection, inflammation, and structural factors contribute; persistent hearing loss or severe pain needs examination.

Alternate Names:

Tubes Ear Problems (Eustachian Tube Problems)

Eustachian-Tube Dysfunction

Failure of the auditory tube to equalize middle-ear pressure or clear secretions. It can cause aural fullness, popping, conductive hearing loss, and retracted tympanic membrane and is common with upper-respiratory infection, allergic rhinitis, or craniofacial disorders.

Euthanasia

The deliberate act of ending a person's life to relieve suffering, typically performed by a physician. Active euthanasia involves directly administering a lethal substance (e.g., barbiturate overdose followed by neuromuscular blockade) at the patient's explicit request.

Euthanasia And Assisted Dying

Ethical and legal concepts regarding intentional termination of life to relieve intractable pain and suffering. Active Euthanasia: deliberate administration of a lethal agent (e.g., lethal dose of barbiturates) by a physician to terminate a patient's life upon their explicit request.

Euthyroid

The clinical and biochemical state of possessing normal thyroid gland function and physiologic circulating concentrations of thyroid hormones. Characterized biochemically by serum thyroid-stimulating hormone, free thyroxine (free T_4), and free triiodothyronine (free T_3) within their respective laboratory reference ranges. Must be clinically distinguished from euthyroid sick syndrome (non-thyroidal illness syndrome), in which severe non-thyroidal systemic illness causes abnormal thyroid lab values (typically low T_3 with normal or low TSH and normal free T_4) in the absence of intrinsic thyroid gland pathology.

Euthyroid Goiter

An enlarged thyroid gland with thyroid-hormone levels that remain within the normal range.

Euthyroid Sick Syndrome

Abnormal thyroid function tests in patients with

non-thyroidal illness (critical illness, starvation, major surgery) without intrinsic thyroid disease. Pattern: low T3 (most common), low T4, reverse T3 elevated, normal or low TSH. The severity correlates with illness severity.

Eutocia

Normal labor and childbirth without major complications or abnormal obstruction.

Evacuation

Evacuation is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Evacuation Of Uterus

Evacuation of Uterus is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Evans Syndrome

An autoimmune disorder in which antibodies destroy red blood cells and platelets, separately or together.

Event

An occurrence relevant to health or care, such as a symptom, diagnosis, treatment, complication, or measurable outcome; its precise meaning depends on the clinical context.

Event Monitor

A portable device that records heart rhythm intermittently or when activated during symptoms over an extended period.

Event-Free Survival

The time from randomization or treatment start to a specified event such as disease progression, relapse, or death from any cause.

Everolimus

A medicine that inhibits the mTOR pathway and is used in selected cancers, transplant regimens, and other conditions.

Eversion

Turning the sole of the foot laterally, so that it faces away from the median plane.

Evidence-Based Practice

Integration of best research evidence, clinical expertise, patient values. Rehabilitation guidelines based on systematic reviews.

Evisceration

Protrusion of viscera through a dehiscd abdominal wound, typically occurring 5-8 days postoperatively and constituting a surgical emergency. Risk factors include obesity, wound infection, and inadequate fascial closure. Management requires sterile coverage, broad-spectrum antibiotics, and emergent operative reclosure.

Evoked Potential

An electrical response recorded from the nervous system after a specific sensory stimulus.

Evoked Potentials

Electrophysiologic tests measuring the CNS electrical response to specific sensory stimuli. Visual evoked potentials (VEP): checkboard pattern reversal, measures P100 latency; delayed in optic neuritis/MS. Brainstem auditory evoked responses (BAER): click stimuli, waves I-V; delayed in acoustic neuroma, MS, brainstem stroke.

Alternate Names:

VEP, BAER, SSEP

Evolution

Change in inherited characteristics of populations across generations; in clinical writing, it may also describe how a disease or symptom changes over time.

Evolutionarily Conserved Gene

A gene whose sequence or function has remained similar across species over evolutionary time, suggesting that it performs an important biological function.

Ewart Sign

Dullness to percussion, decreased breath sounds, and bronchial breathing at the left infrascapular region caused by compressive atelectasis from a large pericardial effusion. It is a physical examination finding of pericardial effusion. The effusion compresses the adjacent left lower lobe, creating a zone of consolidation.

Ewing Sarcoma

A highly malignant small round blue cell tumor of bone and soft tissue, representing the second most common primary bone malignancy in children and adolescents (peak age 10-20 years), with a strong predilection for males and Caucasian populations.

Ewing'S Tumour

A clinical disorder or finding related to ewing's tumour; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Ewingella

A genus of gram-negative bacteria that rarely causes opportunistic infection in humans.

Ewingella Americana

A rare Gram-negative environmental bacterium that can occasionally cause opportunistic infection, mainly in people with underlying illness.

Ewings Sarcoma

A malignant small-cell tumor of bone or soft tissue, usually affecting children or young adults and often associated with an EWSR1 gene fusion.

Exacerbate

To make a disease, symptom, or other medical problem more severe or troublesome.

Exacerbation

A worsening or flare of a disease or its symptoms, sometimes triggered by infection, exposure, treatment interruption, or another stressor.

Exacerbation Of COPD

An acute worsening of respiratory symptoms beyond normal day-to-day variations, requiring change in medication. It is commonly triggered by respiratory infections (viral, bacterial), air pollution, or non-compliance. Treatment includes increased bronchodilators, systemic corticosteroids, antibiotics if indicated, and oxygen therapy.

Exaggerated Placental Site

An exuberant but benign proliferation of implantation-site intermediate trophoblast in the uterus or placental bed; it can mimic gestational trophoblastic neoplasia and requires pathologic correlation.

Examination

Systematic assessment of a patient's health status. Components: inspection, palpation, percussion, auscultation (IPPA). General survey: appearance, posture, gait, vital signs. Focused examination based on chief complaint.

Examination Table

A table designed to support a patient during a physical examination or clinical procedure.

Exanthem

A widespread skin eruption, often caused by an infection or medication; measles and drug eruptions are examples.

Exanthema

A widespread skin eruption, usually macular or papular, associated with an acute systemic illness. Viral infections are common causes, but drug reactions, bacterial disease, immune disorders, and other systemic conditions may also produce exanthema; fever, mucosal involvement, blistering, or skin pain may indicate a severe process.

Exanthematous Pustulosis

An acute drug-related skin eruption with numerous small, sterile, non-follicular pustules on reddened skin. Fever and neutrophilia may occur. It often improves after the causative medicine is withdrawn,

but severe cases require medical assessment.

Exatecan Mesylate

An investigational anticancer drug related to camptothecin that inhibits topoisomerase I.

Excavation

A hollowing, depression, or removal of tissue or material; in medicine it may describe an anatomical depression or a surgical removal.

Exceptional Release

Regulatory process allowing use of an organ that does not meet standard criteria but is deemed acceptable for a specific recipient after informed consent.

Excess Facial Hair In Women

Alternate index label; see *Hirsutism*.

Excessive Bruising

Bruising that is unusually frequent, large, spontaneous, or prolonged and may reflect trauma, medications, low platelets, clotting disorders, or another illness.

Excessive Carrying Angle Of The Elbow

Excessive carrying angle of the elbow concerns excessive carrying angle of the elbow, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Excessive Crying In Infants

Crying that is unusually prolonged, frequent, or difficult to soothe; it may reflect normal colic but can also signal illness, injury, feeding problems, or unsafe caregiving stress.

Excessive Daytime Sleepiness

Alternate index label; see *Narcolepsy*.

Excessive Menstrual Bleeding

Alternate index label; see *Heavy menstrual bleeding*.

Excessive Or Unwanted Hair In Women

Increased hair growth that is cosmetically unwanted or hormonally patterned, including hirsutism from androgen excess and generalized hypertrichosis from other causes.

Excessive Sweating

Sweating more than needed for temperature control, either generally or in specific areas; causes include hyperhidrosis, medicines, infection, hormonal changes, or other medical conditions.

Excessive Tearing

Abnormally increased production or inadequate drainage of tears, causing watery eyes.

Exchange Transfusion

Emergent procedure to rapidly lower severe hyperbilirubinemia by removing small volumes of the infant's blood and replacing with compatible donor blood. Used when phototherapy fails or bilirubin approaches toxic levels (greater than 25 mg/dL in term infants or above exchange threshold on Bhutani curve).

Excise

To remove tissue or a lesion surgically by cutting it out.

Excision

The surgical removal of a piece of tissue, an organ, or a lesion. Excisional biopsy removes the entire lesion with a margin of normal tissue, used for definitive diagnosis and treatment of small tumors (suspected melanoma, breast lump, lymph node).

Excision Biopsy

Alternate index label; see *Excisional biopsy*.

Excisional Biopsy

Surgical removal of an entire lesion, often with a small margin of surrounding tissue, for pathologic diagnosis and possible definitive treatment when complete removal is appropriate.

Excitation

Activation or increased responsiveness of a cell, nerve, muscle, molecule, or physical system in response to a stimulus.

Excitation-Contraction Coupling

The sequence of events linking an action potential to muscle contraction. In skeletal muscle: the action potential travels down the T-tubule, triggering calcium release from the sarcoplasmic reticulum via ryanodine receptors (calcium-induced calcium release mediated by voltage-gated dihydropyridine receptors).

Excitatory Neurotransmitter

A chemical messenger that increases the likelihood that a receiving neuron will generate a signal.

Excitatory Synapse

A synapse in which neurotransmitter release increases the likelihood that the receiving neuron will generate an action potential.

Excitotoxin

A substance that overstimulates neurons and can injure or kill them.

Exclamation Point Hair

A short, broken hair that is narrower at the end away from the scalp and wider near the scalp, classically seen in alopecia areata.

Exclusion Diet

Exclusion Diet is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Excoriate

To scratch, abrade, or strip away the surface of the skin; an excoriation is the resulting superficial injury.

Excoriation

A superficial loss of skin caused by scratching, rubbing, or another form of mechanical trauma.

Excoriation Disorder

Recurrent skin picking resulting in skin lesions. The individual has made repeated attempts to decrease or stop the skin picking. The skin picking causes clinically significant distress or impairment. It is not attributable to a medical condition or substance. Common sites include face, arms, and hands.

Excreta

Material eliminated from the body, including feces, urine, sweat, and respiratory secretions.

Excretion

Removal of substance from body. Routes: renal (urine), hepatic (bile), pulmonary (gas), sweat, breast milk. Important for toxicity and drug clearance.

Executive Function

Mental processes used to plan, focus attention, remember instructions, control impulses, solve problems, and shift between tasks.

Executive Functions

Mental abilities used to plan, organize, make decisions, control impulses, and adapt behavior.

Alternate Names:

Executive control

Exemestane

An aromatase-inhibitor medicine that lowers estrogen production and is used mainly for hormone-sensitive breast cancer.

Exenatide

A glucagon-like peptide-1 receptor agonist used with diet and exercise to improve blood glucose control in type 2 diabetes.

Exencephaly

A severe neural-tube defect in which much of the developing brain lies outside the skull; it is usually incompatible with prolonged survival and is related to anencephaly.

Exenteration

Extensive surgical removal of the contents of a body cavity, such as pelvic, orbital, or ocular exenteration, usually for advanced cancer or destructive disease.

Exercise

Planned physical activity performed to improve or maintain fitness and health.

Alternate Names:

Physical exercise

Exercise And Activity For Weight Loss

Physical activity used with dietary changes to increase energy expenditure and support gradual weight loss.

Exercise And Age

The way physical activity needs and responses may change with age; safe exercise should match fitness, health, and ability.

Exercise And Immunity

The relationship between physical activity and immune function; regular moderate activity generally supports health, while excessive exertion may increase short-term illness risk.

Exercise Clothing And Shoes

Clothing and footwear chosen to allow comfortable movement, manage heat and moisture, and reduce the risk of foot injury.

Exercise Headaches

Headaches occurring during or after strenuous physical activity. Primary exercise headaches are recurrent and usually not caused by an underlying disorder, but a first, abrupt, severe, or atypical exertional headache requires prompt medical assessment to exclude secondary causes such as subarachnoid hemorrhage, arterial dissection, or reversible cerebral vasoconstriction syndrome.

Exercise Intolerance

Reduced ability to perform physical activity at the expected level because of exertional fatigue, breath-

lessness, chest discomfort, leg pain, or other symptoms. Causes include cardiac, pulmonary, vascular, metabolic, neuromuscular, haematological, and deconditioning states.

Exercise Stress Test

A noninvasive cardiac diagnostic test that evaluates the cardiovascular response to graded physical exertion, primarily used to diagnose coronary artery disease, assess functional capacity, and guide prognosis.

Exercise Tolerance

The ability to perform physical activity before symptoms such as breathlessness, fatigue, chest discomfort, or pain limit activity.

Exercise, Lifestyle, And Your Bones

Physical activity and lifestyle habits that help maintain bone strength, including weight-bearing exercise, adequate nutrition, and avoiding tobacco.

Exercise-Induced Arrhythmias

Catecholamine-sensitive ventricular tachycardia and right ventricular outflow tract VT can be triggered by exercise. HCM-related sudden death risk increases during vigorous exercise. Exercise-induced myocardial ischemia may trigger ventricular arrhythmias in CAD.

Exercise-Induced Asthma

Bronchoconstriction occurring during or shortly after physical exertion, causing cough, wheezing, chest tightness, or shortness of breath. It is associated with airway hyperresponsiveness and may occur in people with or without persistent asthma.

Exercise-Induced Bronchoconstriction

Airway narrowing during or after exercise, presenting with cough, wheeze, and dyspnea. It occurs in up to 90% of asthmatics and is triggered by heat and water loss from the airways. Prevention includes pre-exercise SABA.

Exercises To Help Prevent Falls

Balance, strength, flexibility, and walking exercises used to improve stability and reduce fall risk.

Exercising And Asthma At School

School-based asthma planning for physical activity, including access to a reliever inhaler, warm-up, recognition of symptoms, an action plan, and emergency response.

Exercising On A Budget

Planning safe physical activity with low-cost options such as walking, home exercises, stairs, or community facilities, adapted to health status and fitness goals.

Exfoliation Syndrome

An age-related disorder in which abnormal fibrillar material accumulates in the anterior eye, increasing risk of glaucoma, zonular weakness, and surgical complications.

Exfoliative Cytology

Examination of cells that naturally shed from a body surface or are gently collected, used to screen for or investigate infection, inflammation, or cancer.

Exfoliative Dermatitis

Severe widespread redness, scaling, and shedding of the skin, often involving most of the body. Causes include medicines, psoriasis, eczema, lymphoma, and other illness; it can cause fluid loss, temperature problems, infection, and requires urgent medical care.

Exhalation

The process of breathing air out of the lungs. It is normally passive at rest but becomes active during exercise or respiratory distress.

Exhaled Nitric Oxide

A non-invasive measure of airway eosinophilic inflammation. FeNO > 25 ppb in adults suggests eosinophilic airway inflammation and predicts response to inhaled corticosteroids. It is useful for

diagnosing eosinophilic asthma, monitoring treatment response, and differentiating asthma from other causes of cough.

Exhaustion

Exhaustion is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Exhaustion Heat (Heat Exhaustion)

Alternate index label for Heat Exhaustion.

Exhibitionism

Exposing one's genitals to an unsuspecting person for sexual arousal; it becomes exhibitionistic disorder when recurrent behavior causes distress or impairment or involves nonconsent.

Exhibitionistic Disorder

A paraphilic disorder involving recurrent, intense sexual arousal from exposing one's genitals to an unsuspecting person. The individual has acted on these urges or they cause marked distress or interpersonal difficulty. Duration $\geq$ 6 months. Onset typically before age 18. Treatment: CBT and SSRIs.

Exhumation

The lawful disinterment and removal of a buried body or remains for re-examination, re-testing, or identification. Indications: new evidence, suspicion of poisoning or foul play, DNA testing, paternity/maternity testing, exhumation of a wrongly buried body, or repatriation.

Exiguobacterium

A genus of Gram-positive bacteria found in soil, water, and extreme environments; human infection is rare.

Existential Distress

Emotional suffering related to questions of meaning, identity, freedom, mortality, or purpose, often occurring during serious illness.

Exit Wound

A skin defect produced where a projectile leaves the body, typically larger, irregular, and stellate, with everted edges, no abrasion ring or soot. Size depends on deformation, yaw, and fragmentation; small non-deformed bullets may produce shored exits, and an exit may be absent if the projectile is retained.

Exocervix

The portion of the cervix that projects into the vagina and is covered mainly by squamous epithelium.

Exocrine

Relating to glands that release secretions through ducts onto an epithelial surface or into a body cavity, such as sweat glands or the exocrine pancreas.

Exocrine Gland

A gland that releases its secretions through a duct onto a body surface or into a cavity, such as a sweat, salivary, or pancreatic gland.

Exocrine Glands

Exocrine Glands is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Exocrine Pancreas

Exocrine pancreas is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Exocrine Pancreatic Insufficiency

Inadequate secretion of pancreatic digestive enzymes, causing maldigestion and malabsorption, often with weight loss, bloating, and greasy or difficult-to-flush stools. Common causes include chronic pancreatitis, cystic fibrosis, and pancreatic surgery.

Exocrine Pancreatic Insufficiency

Inadequate pancreatic enzyme secretion causing impaired digestion and absorption, often with greasy

stools, weight loss, bloating, and fat-soluble vitamin deficiency.

Alternate Names:

EPI

Exocrine Pancreatic Insufficiency (Exocrine Pancreatic Insufficiency EPI)

Alternate index label for Exocrine Pancreatic Insufficiency EPI.

Exocrine Pancreatic Insufficiency EPI

Exocrine pancreatic insufficiency occurs when the pancreas does not release enough digestive enzymes, causing greasy stools, bloating, weight loss, and deficiencies of fat-soluble vitamins. Pancreatic enzyme replacement and nutritional monitoring are central to treatment.

Alternate Names:

Exocrine Pancreatic Insufficiency (Exocrine Pancreatic Insufficiency EPI)

Exocytosis

Process by which intracellular vesicles fuse with the plasma membrane, releasing their contents into the extracellular space and incorporating membrane proteins/lipids into the cell membrane. The SNARE complex (v-SNARE on vesicles and t-SNARE on target membrane) mediates vesicle docking and fusion.

Exodeoxyribonuclease

An enzyme that removes nucleotides from the ends of DNA strands and participates in DNA repair, replication, or degradation.

Exogenous

Originating outside the body or introduced from an external source, such as an exogenous medication or hormone.

Exogenous Cushing Syndrome

Cushing syndrome caused by prolonged exposure to corticosteroid medicines rather than excess cortisol production by the body.

Exomphalos

A congenital abdominal-wall defect in which abdominal organs protrude into the base of the umbilical cord and are covered by a membrane.

Exon

A segment of a DNA or RNA molecule containing information coding for a protein or peptide sequence.

Exonuclease

An enzyme that removes nucleotides one at a time from the end of a DNA or RNA strand.

Exopeptidase

An enzyme that removes amino acids from the end of a peptide or protein chain.

Exophiala

A genus of darkly pigmented fungi that can cause skin, lung, eye, or disseminated infections, especially in people with weakened immunity.

Exophiala Dermatitis

A darkly pigmented yeast that can cause skin, lung, or brain infection, mainly in people with weakened immunity.

Exophthalmic Goitre

A clinical disorder or finding related to exophthalmic goitre; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Exophthalmos

Abnormal protrusion of the eyeball (globe) from the orbit, most commonly caused by thyroid eye disease (Graves ophthalmopathy—an autoimmune condition associated with hyperthyroidism). The proptosis results from inflammation, edema, and fibrosis of the extraocular muscles and proliferation of orbital fat within the fixed bony orbit.

Exostosis

A benign bony growth; in the ear canal, repeated cold-water exposure can cause surfer's ear.

Alternate Names:

Surfer's ear

Exotoxins

A clinical disorder or finding related to exotoxins; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Exotropia

A type of strabismus in which one eye deviates outward (toward the temple). Intermittent exotropia: most common type, often noticed when tired or daydreaming, can be controlled with fusion. Constant exotropia: requires treatment to prevent amblyopia. Congenital exotropia: rare, requires early evaluation.

Expectant Treatment

A medical procedure, treatment, device, or care topic related to expectant treatment; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Expectorants

Medications that promote the clearance of mucus from the respiratory tract by increasing sputum volume and reducing its viscosity, facilitating productive cough. The most common expectorant is guaifenesin (glyceryl guaiacolate), which increases respiratory tract fluid secretion and reduces sputum adhesiveness.

Expectoration

The act of coughing up and spitting out mucus, sputum, or phlegm from the respiratory tract; sputum production.

Experience

A person's direct perception, knowledge, or emotional response to an event; in healthcare, patient experience includes how care is accessed and received.

Experience Sampling

A research method that collects brief reports about

a person's activities or feelings at randomly selected times.

Alternate Names:

Ecological momentary assessment

Expiration

The movement of air out of the lungs, usually caused by relaxation of the diaphragm and elastic recoil of the lungs.

Expiratory Grunting

A forced exhalation against a partially closed glottis, creating an audible sound. It is an end-expiratory maneuver that generates intrinsic PEEP, helping to maintain alveolar recruitment and improve oxygenation. It is commonly seen in children with respiratory distress.

Explant

To remove an implanted device or tissue; an explant can also be a piece of tissue removed from an organism and maintained in laboratory culture.

Explantation

Surgical removal of an implanted device or prosthesis. It may be required for infection, malfunction, intolerance, migration, fracture, tissue injury, or completion of a temporary therapy.

Exploratory Laparotomy

Open surgical entry into the abdominal cavity to diagnose or treat suspected intra-abdominal pathology, such as hemorrhage, perforation, ischemia, or obstruction, when less invasive evaluation is inadequate or inappropriate.

Exploratory Operation

Surgery performed to inspect an area directly and identify the cause or extent of disease when noninvasive tests are insufficient.

Exposure

Contact with a physical, chemical, biological, or psychological agent at a particular intensity and duration; risk depends on the agent and circum-

stances.

Exposure Route

Path by which substance enters body. Routes: inhalation, ingestion, dermal, injection, ocular. Determines absorption rate and toxicity.

Expressed Sequence Tag

A short DNA sequence obtained from a transcribed messenger-RNA copy and used as a marker for identifying or locating an expressed gene.

Expression

Expression is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Expression Hopping

A shift from one manifestation of an addiction to another, such as from substance use to compulsive gambling.

Expressivity

Degree to which a genotype is expressed in phenotype. Variable expressivity: same genotype produces different phenotypic severity in different individuals. Examples: neurofibromatosis type 1, Marfan syndrome.

Expulsion, Stage Of

The second stage of labor, beginning with complete cervical dilation and ending with delivery of the baby.

Exsanguination

Life-threatening loss of blood from severe internal or external bleeding. It can rapidly cause shock, organ failure, and death and requires immediate control of bleeding and emergency medical care.

Exserohilum

A genus of darkly pigmented molds that can cause sinus, eye, skin, or invasive infections.

Extend

To straighten a joint or increase the angle between bones.

Alternate Names:

Extension

Extension

Movement that increases the angle between two bones or straightens a joint.

Extensively Drug-Resistant Tuberculosis

A severe form of multidrug-resistant tuberculosis caused by *Mycobacterium tuberculosis* strains resistant to isoniazid, rifampin, any fluoroquinolone (such as levofloxacin or moxifloxacin), and at least one additional Group A priority agent (such as bedaquiline or linezolid).

Alternate Names:

XDR-TB; XDR tuberculosis

Extensor

Extensor is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Extensor Digitorum

Posterior forearm muscle from lateral epicondyle via the common extensor tendon, dividing into 4 tendons inserting on extensor expansions of digits 2-5. Innervated by posterior interosseous nerve (C7-C8). Extends MCP joints, assists PIP/DIP extension. Lateral epicondylitis (tennis elbow) is an overuse injury of its common origin.

Extent

The size, range, severity, or spread of a disease, injury, symptom, or treatment effect.

External Auditory Canal

Tube from the concha to the tympanic membrane, approximately 2.5 cm long in adults. Lateral third is cartilaginous; medial two-thirds are bony (temporal bone). Skin-lined with ceruminous glands (producing cerumen/earwax) and hair follicles. Cerumen has antimicrobial and insect-repellent properties.

External Beam Radiation Therapy

Radiation delivered from a machine outside the body to a planned treatment area. Techniques include three-dimensional conformal therapy, intensity-modulated therapy, and stereotactic therapy.

External Capsule

A thin white-matter tract lateral to the putamen that carries connections among the cerebral cortex, thalamus, basal ganglia, and other regions.

External Cephalic Version

A procedure that uses external pressure to turn a breech or transverse fetus into a head-down position. Timing, eligibility, monitoring, and contraindications depend on gestational age, placental and fetal factors, and the clinical setting.

External Ear

The visible auricle and external auditory canal that collect and conduct sound toward the tympanic membrane and help protect the middle ear.

External Fixation

Stabilization of a fracture or limb using pins or wires attached to an external frame. It may provide temporary damage-control stabilization or definitive treatment, particularly when soft-tissue injury or contamination limits internal fixation.

External Iliac Artery

Larger common iliac terminal branch, continuing as the femoral artery below the inguinal ligament. Gives off inferior epigastric and deep circumflex iliac arteries. Related to external iliac vein medially and ureter anteriorly. Commonly used for cardiac catheterization and endovascular access.

External Incontinence Devices

Devices worn or applied outside the body to collect urine or stool or help manage leakage, such as urinary sheaths, absorbent products, or fecal collection systems.

External Jugular Vein

Formed by posterior auricular and retromandibular vein divisions, crossing the sternocleidomastoid superficially to drain into the subclavian vein. Visible through the skin; used for peripheral IV access when other sites are unavailable. Distended in right heart failure and SVC obstruction. A useful physical examination landmark.

External Nose

The projecting nasal structure formed by bone, cartilage, skin, and soft tissue that shapes the nasal airway and supports smell and facial appearance.

External Oblique Muscle

Most superficial flat abdominal muscle from outer lower eight ribs to linea alba, pubic tubercle, and iliac crest. Innervated by T7-T12. Contralateral trunk rotation and flexion; compresses abdomen. Forms the inguinal ligament from its free inferior border. Superficial inguinal ring is an opening in its aponeurosis.

External Otitis

Inflammation or infection of the outer ear canal, commonly called swimmer's ear.

Alternate Names:

swimmer's ear.

External Resorption

Loss of tooth structure from outside. Inflammatory, replacement, surface, cervical resorption. Treatment depends on type and extent.

Extinction

Extinction is a physiologic, behavioral, environmental, or healthcare concept used in medical assessment, prevention, treatment, or research.

Extirpation

Complete surgical removal of a tissue, organ, or foreign body.

Extra-Dural

Located outside the dura mater, the tough outer

membrane surrounding the brain and spinal cord.

Extra-Gastrointestinal Stromal Tumor

A gastrointestinal-stromal-tumor-type neoplasm arising outside the gastrointestinal tract, most often in the mesentery, retroperitoneum, or omentum. Diagnosis uses morphology, KIT or DOG1 expression, and molecular testing.

Extraamniotic

Located outside the amniotic sac; the term may describe a procedure, device, or fluid placed between the amnion and surrounding tissues.

Extracapsular Cataract Surgery

Cataract surgery that removes the cloudy lens while leaving the lens capsule largely in place.

Extracellular

Extracellular is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Extracellular Exosome

A small membrane vesicle released by cells that carries proteins, lipids, and nucleic acids to other cells and participates in intercellular signaling.

Extracellular Fluid

The fluid outside cells, including plasma and interstitial fluid. Sodium, chloride, and bicarbonate are its principal effective solutes; ECF volume and composition are regulated by the kidneys, cardiovascular system, and endocrine mechanisms.

Extracellular Matrix

A complex network of proteins and carbohydrates secreted by cells that provides structural and chemical support to surrounding cells.

Extracellular Pathogens

Microorganisms that survive and replicate outside host cells. Extracellular bacteria include Streptococcus, Staphylococcus, and Escherichia coli. These pathogens are typically killed by antibodies and

complement. They produce toxins and enzymes to damage host tissues.

Extracellular Region

The space outside cells containing extracellular matrix, signaling molecules, ions, fluid, and other substances that influence tissue structure and cell behavior.

Extracellular Space

The fluid-filled and matrix-containing environment outside cells through which nutrients, waste, hormones, and signals move.

Extracellular Vesicle

A membrane-bound particle released by a cell that transfers molecular cargo between cells and is studied in physiology, disease, biomarkers, and drug delivery.

Extracorporeal

Occurring or performed outside the body; extracorporeal circulation and hemodialysis route blood through an external machine.

Extracorporeal Circulation

A medical procedure, treatment, device, or care topic related to extracorporeal circulation; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Extracorporeal Membrane Oxygenation

A temporary mechanical circulatory support system that provides cardiac and/or respiratory support. VA-ECMO (veno-arterial) provides both cardiac and respiratory support; VV-ECMO (veno-venous) provides respiratory support only.

Extracorporeal Shock Wave Lithotripsy

Non-invasive treatment method for urinary calculi that uses focused acoustic shock waves generated outside the body to fragment kidney or ureteral stones into smaller gravel-like pieces (dust/sand) that pass spontaneously in urine.

Extracranial

Located or occurring outside the skull or cranial cavity.

Extracranial Hematoma

A collection of blood outside the skull or cranial cavity, usually caused by bleeding into soft tissues after injury.

Extract

A concentrated preparation made by drawing active substances from a plant, herb, or other source.

Alternate Names:

Extractive

Extraction

Removal of a substance, tissue, tooth, foreign body, or other structure from the body, usually by a defined procedure.

Extrahepatic Biliary Atresia

Alternate index label; see *Biliary atresia*.

Extrahepatic Cholestasis

Impaired bile flow caused by blockage or disease outside the liver.

Extramammary Paget Disease

An intraepithelial adenocarcinoma usually affecting apocrine-rich skin such as the vulva, perianal region, or scrotum; it often presents as a persistent eczematous or erythematous plaque and requires biopsy.

Extranodal

Outside the lymph nodes; an extranodal lymphoma or lesion involves organs or tissues beyond the lymphatic nodes.

Extranodal Marginal Zone Lymphoma

Alternate index label; see *Extranodal marginal zone lymphoma of mucosa-associated lymphoid tissue*.

Extranodal Marginal Zone Lymphoma Of

Mucosa-Associated Lymphoid Tissue

A usually slow-growing B-cell lymphoma arising in mucosal tissues such as the stomach, salivary glands, thyroid, or lungs.

Alternate Names:

Extranodal Marginal Zone Lymphoma MALT Lymphoma Mucosa-Assisted Lymphoid Tissue Lymphoma

Extranodal Natural Killer T-Cell Lymphoma

An aggressive lymphoma arising from natural killer cells, most commonly involving the nasal cavity and upper aerodigestive tract. It is strongly associated with Epstein-Barr virus infection.

Extraocular

Outside the eye or involving the muscles, nerves, tissues, or structures surrounding the eyeball.

Extraocular Muscle

Extraocular muscle is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Extraocular Muscle Function Testing

Assessment of the six extraocular muscles innervated by CN III (medial rectus: adduction; superior rectus: elevation; inferior rectus: depression; inferior oblique: extorsion), CN IV (superior oblique: intorsion and depression), and CN VI (lateral rectus: abduction).

Extraocular Muscles

Six muscles controlling eye movement: four rectus (superior, inferior, medial, lateral) and two oblique (superior, inferior).

Extrapulmonary Manifestation Of Tuberculosis

Tuberculosis occurring outside the lungs, such as in lymph nodes, bone, kidneys, meninges, or the abdomen.

Extrapyramidal Disorder

A movement disorder caused by dysfunction of brain circuits that regulate posture and involuntary movement.

Extrapyramidal Symptoms

Drug-induced movement disorders caused by D2 receptor blockade in the nigrostriatal pathway.

Extrapyramidal System

Neural pathways and brain structures, including the basal ganglia, that help regulate movement, muscle tone, posture, and automatic motor activity.

Extrapyramidal Tracts

Extrapyramidal Tracts is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Extraskkeletal Chondroma

A benign hyaline-cartilage tumor occurring in soft tissue without attachment to bone, commonly in the hands or feet; it may calcify and resemble other cartilaginous lesions on imaging.

Extraskkeletal Mesenchymal Chondrosarcoma

A malignant small-cell sarcoma with islands of malignant cartilage arising in soft tissue or other extraskkeletal sites. It requires specialist pathologic and oncologic evaluation.

Extraskkeletal Osteosarcoma

A malignant tumor that produces bone or osteoid in soft tissue rather than arising primarily in bone, usually affecting adults.

Extrasystole

Alternate index label; see *Premature ventricular contractions (PVCs)*.

Extrasystole (Premature Ventricular Contractions)

Alternate index label for Premature Ventricular

Contractions.

Extrauterine Pregnancy

A pregnancy in which the fertilized ovum implants outside the uterine cavity, an alternative term for ectopic pregnancy. By far the most common site is the fallopian tube (ampulla 70%, isthmus 12%, fimbria 11%, cornual/interstitial 2–4%).

Extravasation

Leakage of urine outside urinary tract. Causes: trauma, surgery, obstruction. Pseudovesical extravasation (rectus sheath), extraperitoneal, intraperitoneal. Treatment: catheter drainage, surgical repair.

Extremities

Extremities is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Extremity

A limb of the body, including the upper extremity (shoulder, arm, forearm, wrist, and hand) or lower extremity (hip, thigh, leg, ankle, and foot).

Extremity Angiography

Imaging of arteries or veins in an arm or leg, usually with x-ray fluoroscopy after contrast injection, to identify narrowing, blockage, aneurysm, malformation, or vascular injury.

Extremity X-Ray

Plain radiography of an arm or leg used to evaluate fractures, alignment, arthritis, bone infection, tumors, or other abnormalities.

Extrinsic

Originating outside a structure, organ, or person; an extrinsic cause or factor acts from an external source.

Extrinsic Allergic Alveolitis (Hypersensitivity Pneumonitis)

An immune-mediated inflammation of the lung

caused by repeated inhalation of sensitizing organic particles, such as moulds, animal proteins, or contaminated agricultural dust.

Alternate Names:

Hypersensitivity Pneumonitis

Extrinsic Factor

An outside influence that affects a person, tissue, or biological process.

Alternate Names:

External factor

Extroversion

The outward turning or eversion of a structure; the term may also describe an outgoing personality trait depending on context.

Extubate

Removal of an endotracheal tube after restoration of adequate spontaneous breathing and airway protection.

Extubation

Removal of an endotracheal tube after the patient can maintain an open airway and adequate spontaneous breathing.

Extubation Criteria

Clinical findings used to determine whether a patient can safely have an endotracheal tube removed. They include adequate spontaneous breathing, oxygenation, airway protection, cough, neuromuscular recovery, and hemodynamic stability.

Exudate

Protein-rich fluid that escapes from inflamed or injured blood vessels into tissues or body cavities; unlike a transudate, it usually reflects increased vascular permeability.

Exudative Pleural Effusion

From local pleural disease. Causes include pneumonia, malignancy, TB, and PE. Elevated protein, LDH, and cell count.

Eye

The organ of vision, a spherical structure (approximately 24 mm diameter) in the bony orbit. Three layers: fibrous (sclera, cornea), vascular/uveal (choroid, ciliary body, iris), neural (retina). Anterior segment: cornea, anterior chamber (aqueous humor), iris/pupil, lens.

Eye - Foreign Object In

A condition in which material enters or becomes lodged on the ocular surface or within the eye; evaluation must exclude corneal penetration and intraocular injury.

Eye Allergy

An eye allergy is an immune reaction to airborne or contact allergens that causes itching, redness, tearing, and swollen eyelids. Avoidance, cool compresses, lubricants, and appropriate antiallergic drops help; pain or vision loss suggests a different urgent condition.

Alternate Names:

Allergy Eye (Eye Allergy)

Eye And Orbit Ultrasound

Ultrasound imaging of the eye and orbit used to evaluate structures that cannot be seen through an opaque lens or vitreous, including retinal detachment, masses, and hemorrhage.

Eye Bank

Eye Bank is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Eye Black (Black Eye)

Alternate index label for Black Eye.

Eye Bleed

Bleeding in or around the eye; a subconjunctival hemorrhage is blood beneath the clear surface membrane and often appears as a painless bright-red patch, while bleeding inside the eye can impair vision and requires medical assessment.

Eye Care

Eye care includes protective eyewear, regular examination based on age and risk, control of diabetes and blood pressure, safe contact-lens hygiene, and prompt attention to vision changes. Sudden vision loss, severe pain, flashes, or a curtain-like shadow is urgent.

Alternate Names:

Eye Care Eye Disorder (Eye Care)

Eye Care Eye Disorder (Eye Care)

Alternate index label for Eye Care.

Eye Color

The apparent color of the iris, determined mainly by the amount and distribution of melanin and the way light is scattered; a new or uneven color change can indicate disease or injury.

Eye Development

Formation and maturation of the eye and its structures during embryonic and postnatal development.

Eye Disorder

A disease of the eye or visual pathways causing altered vision, pain, redness, discharge, double vision, field loss, or other ocular findings.

Eye Disorders

The sensory ability to perceive light and interpret visual information processed by the eyes and brain, enabling recognition of shapes, colors, depth, and motion. See Vision.

Eye Drops

Liquid preparations instilled into the conjunctival sac for diagnosis, lubrication, or treatment of eye conditions. They should be used according to clinical or product instructions.

Eye Emergencies

Eye emergencies include acute vision loss, chemical injury, penetrating trauma, severe pain, and selected infections or pressure disorders requiring immediate assessment.

Eye Floaters

Spots, threads, or cobweb-like shapes that move across the visual field, usually caused by opacities in the vitreous. Sudden new floaters, flashes, or a curtain-like shadow require urgent evaluation for retinal tear or detachment. See also Floaters.

Eye Health

The maintenance of optimal vision and prevention of ocular disease through regular eye examinations, protective measures, and timely treatment. Common conditions affecting eye health include refractive errors, cataracts, glaucoma, diabetic retinopathy, and age-related macular degeneration.

Eye Hemorrhage

Bleeding in or around the eye, such as subconjunctival, vitreous, retinal, or orbital hemorrhage; symptoms depend on the location and amount of blood.

Eye Infection

Infection of the eyelid, conjunctiva, cornea, internal eye, or surrounding tissues caused by bacteria, viruses, fungi, or parasites.

Eye Injury

Damage to the eye or surrounding tissues from blunt or penetrating trauma, chemicals, foreign bodies, radiation, or other causes.

Eye Irritation

Chemical injury to eye. Immediate irrigation with water or saline for 15-20 minutes. Alkali burns more dangerous than acid. Ophthalmology consultation.

Eye Melanoma

A cancer that develops in the pigment-producing cells of the eye.

Alternate Names:

Cancer, Eye Melanoma

Eye Movement

Movement of the eyes that aligns the visual axes and directs gaze, controlled by the extraocular muscles and cranial nerves.

Eye Muscle Repair

A medical procedure, treatment, device, or care topic related to eye muscle repair; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Eye Neoplasm

A benign or malignant tumor arising in the eye or its surrounding tissues; symptoms and treatment depend on location, vision impact, and tumor type.

Eye Pain

Pain in or around the eye caused by surface irritation, infection, trauma, inflammation, glaucoma, or orbital disease. Severe pain with vision loss, photophobia, trauma, or nausea requires urgent assessment.

Eye Prosthesis

An artificial eye or ocular implant used to restore appearance after loss or severe damage to the eye; it does not restore vision.

Eye Redness

Red discoloration of the eye caused by dilation or inflammation of conjunctival or deeper ocular blood vessels. Causes range from irritation and conjunctivitis to keratitis, uveitis, and acute glaucoma.

Eye Trauma

Ocular injuries ranging from corneal abrasions to globe rupture. Chemical burns: irrigate immediately with saline or water for 30 minutes. Penetrating injuries: shield eye, no pressure, emergent ophthalmology consultation. Foreign bodies: remove superficial items; embedded objects require surgical removal.

Eye Twitching

Repeated involuntary spasms of an eyelid or muscles around the eye, commonly associated with fatigue, stress, caffeine, irritation, or eye strain.

Eye, Diseases Of

Eye, Diseases of is an anatomical structure, tissue,

cell, or body process; its location and function explain its role in normal physiology and disease.

Eyeball (Globe)

The roughly spherical organ of vision, consisting of the cornea, sclera, uveal tract, lens, retina, and internal chambers and fluids.

Alternate Names:

Globe

Eyebrow

A strip of hair above the orbit that helps protect the eye from sweat, moisture, and particles.

Eyeglass

A lens or pair of lenses mounted in a frame to correct refractive errors such as nearsightedness, farsightedness, astigmatism, or presbyopia.

Eyeglasses, Sunglasses, And Magnifiers

Eyeglasses correct refractive errors, sunglasses reduce glare and ultraviolet exposure, and magnifiers enlarge near targets for selected visual impairment. Prescription, fit, lens protection, and the underlying eye condition determine the appropriate aid.

Alternate Names:

Eyeglasses, Sunglasses, And Magnifying Glasses
(Eyeglasses, Sunglasses, And Magnifiers)

Eyeglasses, Sunglasses, And Magnifying Glasses (Eyeglasses, Sunglasses, And Magnifiers)

Alternate index label for Eyeglasses, Sunglasses, and Magnifiers.

Eyelash

A row of hairs along the eyelid margin that helps protect the eye from dust and triggers a protective blink reflex.

Eyelashes

Short, curved hairs arising from the eyelid margins that help protect the ocular surface from foreign particles and excessive light.

Eyelid

A movable fold of skin and muscle that protects the eye, spreads the tear film, and helps control light entry.

Eyelid Bump

Eyelid bump concerns eyelid bump, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Eyelid Coloboma

A congenital notch or defect in the eyelid caused by incomplete development; it may expose the eye and occur with other eye or facial abnormalities.

Eyelid Disorder

A disease or structural problem of the eyelid, such as inflammation, infection, malposition, tumor, or gland blockage, that can affect comfort or protect the eye.

Eyelid Drooping

Eyelid drooping concerns eyelid drooping, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Eyelid Inflammation

Alternate index label; see *Blepharitis*.

Eyelid Inflammation (Blepharitis)

Alternate index label for Blepharitis.

Eyelid Lift

Eyelid lift concerns eyelid lift, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Eyelid Neoplasm

A benign, precancerous, or malignant growth of the eyelid; persistent, changing, ulcerated, or destructive lesions require eye or skin evaluation.

Eyelid Swelling

Swelling of one or both eyelids caused by allergy,

infection, inflammation, trauma, a blocked gland, or angioedema. Pain, fever, impaired eye movement, proptosis, or vision change may indicate an urgent orbital or systemic condition.

Eyelid Twitch

Eyelid twitch concerns eyelid twitch, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Eyelids

Mobile folds of skin and muscle that protect the eyes, spread the tear film, and help regulate the amount of light entering the eye.

Eyes - Bulging

Eyes - bulging concerns eyes - bulging, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Eyes, Flashing Lights In The

Brief flashes or streaks of light, called photopsias, often caused by traction on the vitreous or retina; sudden new flashes, especially with floaters or a curtain-like loss of vision, require urgent eye evaluation for a retinal tear or detachment.

Eyesight

Eyesight is an anatomical structure, tissue, cell, or body process; its location and function explain its role in normal physiology and disease.

Eyesight (Vision)

The ability to perceive and interpret light, form, color, movement, and detail through the eyes and visual pathways of the nervous system.

Alternate Names:

Vision

Eyestrain

Eye discomfort, fatigue, or blurred vision caused by prolonged visual effort.

Ezetimibe

An oral cholesterol-lowering medicine that blocks

intestinal absorption of cholesterol and is used alone or with a statin.

External Carotid Artery

A branch of the common carotid artery that supplies much of the face, scalp, and neck through several branches.

F

Fab Immunoglobulins

The part of an antibody that recognizes and attaches to a specific target. Fab fragments are used in laboratory tests and some treatments.

Fabry Disease

An inherited disorder in which a missing or reduced enzyme causes a fatty substance to build up in cells. It can cause burning pain, skin changes, kidney disease, heart problems, and stroke.

Fabry'S Disease

An inherited disorder in which a lipid accumulates in cells, causing pain, skin findings, kidney disease, and heart or nervous-system complications.

Face

The face is the front part of the head, containing the eyes, nose, mouth, cheeks, and jaw; its movement is mainly supplied by the facial nerve and sensation by the trigeminal nerve.

Face Pain

Pain in the facial region caused by dental, sinus, nerve, eye, joint, skin, vascular, or other disorders; pattern and associated signs guide evaluation.

Face Powder Poisoning

Face powder poisoning involves exposure or injury from face powder poisoning; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Face Presentation

A fetal presentation in which the neck is hyperex-

tended and the face is the presenting part. The position of the fetal chin influences whether vaginal birth is possible and how labor is managed.

Facelift

Rhytidectomy: a surgical procedure to improve the appearance of the face and neck by removing excess skin and tightening underlying tissues. It can address sagging skin, deep folds, and jowls. Recovery includes swelling and bruising for several weeks.

Alternate Names:

Rhytidectomy

Facet Joint

A small synovial joint between neighboring vertebral arches that guides spinal movement and can cause localized neck or back pain when arthritic.

Facet Joints

Small joints at the back of the spine that connect neighboring vertebrae and guide movement. Arthritis or injury in these joints can cause neck or back pain.

Facet Rhizotomy

A procedure that interrupts selected sensory nerves supplying spinal facet joints to reduce chronic pain.

Facial Asymmetry

Unequal size, position, movement, or contour between the two sides of the face, which may be congenital or result from trauma, nerve dysfunction, muscle disease, or a mass.

Facial Cleft

A congenital gap or malformation caused by incomplete fusion of facial tissues during embryonic development, such as cleft lip or atypical facial cleft.

Facial Edema

Swelling caused by excess fluid in the tissues of the face. Causes include allergy, infection, trauma, kidney or heart disease, and venous obstruction. Sudden swelling of the lips, tongue, or throat with breathing difficulty is an emergency.

Facial Expression

Facial expression is visible movement or configuration of facial muscles conveying emotion, communication, pain, or neurologic information.

Facial Hair, Excess In Women

Hirsutism: excessive androgen-dependent hair growth in male-pattern distribution (face, chest, back) in women. Most common cause is polycystic ovary syndrome. Other causes include adrenal hyperplasia, tumors, and medications. Evaluation includes hormone levels and imaging.

Alternate Names:

Hirsutism

Facial Muscle

facial muscle refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Facial Neoplasm

A benign or malignant tumor arising in facial skin, soft tissue, bone, salivary glands, nerves, or other structures; symptoms and treatment depend on site and spread.

Facial Nerve

The seventh cranial nerve, which controls facial expression and carries taste sensation from part of the tongue.

Facial Nerve Palsy

Weakness or paralysis of muscles supplied by the seventh cranial nerve, causing facial asymmetry and difficulty closing the eye or moving the mouth. It may be idiopathic, infectious, inflammatory, traumatic, neoplastic, or vascular in origin. Sudden onset requires assessment for stroke and other neurologic emergencies, especially when limb weakness, speech disturbance, or sensory loss is present. Evaluation and treatment depend on the pattern, severity, cause, and risk of ocular or neurologic complications.

Facial Nerve Palsy Due To Birth Trauma

Facial nerve palsy due to birth trauma concerns facial nerve palsy due to birth trauma, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Facial Nerve Problems

Disorders affecting cranial nerve VII, which controls facial expression, taste from part of the tongue, and some tear and salivary function. Weakness may result from Bell palsy, stroke, infection, trauma, or a tumor. Sudden facial weakness, especially with arm weakness or speech difficulty, requires emergency assessment.

Alternate Names:

Bells Palsy (Facial Nerve Problems) Idiopathic Facial Paralysis (Facial Nerve Problems)

Facial Nucleus

A motor nucleus in the pons containing neurons whose axons form the facial nerve and control muscles of facial expression.

Facial Pain

Pain felt in the face that may arise from dental disease, sinus disease, temporomandibular disorders, nerve disorders, infection, or other causes.

Facial Palsy

Weakness or paralysis of the facial muscles on one side, most commonly caused by Bell palsy (idiopathic cranial nerve VII dysfunction). Other causes include stroke, infection (herpes zoster ophthalmicus), trauma, tumor, or Lyme disease. Sudden facial

weakness with other neurological signs requires urgent evaluation.

Alternate Names:

Bell's palsy; facial paralysis

Facial Paralysis

Weakness or loss of movement of the facial muscles caused by dysfunction of the facial nerve or its pathways; sudden onset requires urgent evaluation for stroke.

Facial Paresis

Weakness of the muscles supplied by the facial nerve, causing reduced facial movement and sometimes impaired eye closure, taste, or tearing.

Facial Spasm

Facial Spasm is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Facial Swelling

Enlargement of part or all of the face caused by infection, allergy, trauma, dental disease, salivary or sinus disease, fluid retention, or a mass; rapid swelling with breathing difficulty is an emergency.

Facial Tics

Sudden, repetitive, involuntary facial movements or sounds, such as blinking, grimacing, or nose wrinkling, commonly seen in tic disorders.

Facial Trauma

Injury to the face involving skin, soft tissue, teeth, nerves, vessels, orbit, nose, or facial bones, requiring assessment of airway, vision, and structural damage.

Facial Vein

A vein that drains blood from the superficial face and communicates with deeper venous channels before joining the internal jugular venous system.

Facies

A characteristic facial appearance that may provide clues to an underlying disease or physiologic

condition.

Facilitated Diffusion

Movement of a substance across a cell membrane through a protein channel or carrier, from an area of higher concentration to lower concentration. The cell does not need to use energy for this process.

Facioscapulohumeral Dystrophy

Facioscapulohumeral Dystrophy is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Facioscapulohumeral Muscular Dystrophy

An inherited muscular dystrophy causing progressive weakness of facial, shoulder-blade, and upper-arm muscles, often with asymmetric onset and variable severity.

Facklamia

A genus of gram-positive bacteria that can rarely cause bloodstream, urinary, wound, or other opportunistic infections.

Facklamia Hominis

A species of Facklamia bacteria that is rarely isolated from human clinical specimens and may cause opportunistic infection.

Factitious Disorder

Presentation of physical or psychological symptoms or imposing injury or disease on another, with the deceptive intention of assuming the sick role. In factitious disorder imposed on self, the individual falsifies signs or symptoms, induces injury or disease, and presents to medical providers.

Alternate Names:

Munchausen Syndrome

Factitious Disorders

Factitious disorder involves deliberately producing or falsifying illness behavior to assume the sick role, without an obvious external reward. It is a mental-health disorder requiring nonjudgmental assessment,

protection from unnecessary procedures, and coordinated psychiatric and medical care.

Factitious Hyperthyroidism

A state of excessive thyroid hormone in the body caused by ingestion of thyroid hormone or another external source rather than increased thyroid-gland production.

Factor H Deficiency

A rare inherited or acquired condition in which a protein that helps control the immune system is missing or does not work properly. It can cause repeated infections or uncontrolled inflammation and injury to small blood vessels, especially in the kidneys.

Factor

A blood protein needed to form a clot. Different clotting factors work together to stop bleeding; too little or abnormal activity can cause a bleeding disorder, while excessive clotting can contribute to thrombosis.

Factor I Deficiency

A rare condition in which a protein that helps control immune-system activity is missing or does not work properly. It may cause repeated bacterial infections or inflammation and injury to the kidneys.

Factor II (Prothrombin) Assay

A blood test measuring factor II activity, which helps assess the clotting pathway and investigate abnormal bleeding, prolonged prothrombin time, or suspected prothrombin deficiency or inhibition.

Factor IX

A vitamin K-dependent blood-clotting protein in the intrinsic coagulation pathway. Inherited deficiency causes hemophilia B.

Factor IX Assay

A blood test measuring factor IX activity, used to diagnose or monitor factor IX deficiency, the cause of hemophilia B, and to guide replacement treatment.

Factor V

A blood-clotting protein required for formation of thrombin and fibrin.

Factor V Assay

A blood test measuring factor V activity, used to investigate abnormal coagulation and diagnose inherited factor V deficiency or acquired reduction in factor V.

Factor V Deficiency

An inherited or acquired bleeding disorder caused by reduced factor V activity, impairing thrombin generation and fibrin formation. It may cause easy bruising, nosebleeds, heavy menstrual bleeding, postoperative bleeding, or severe bleeding in rare cases.

Factor V Leiden

An inherited change in a blood-clotting protein that makes clots less easily controlled. It increases the risk of blood clots in veins, although the risk varies with family history and other factors.

Factor V Leiden Mutation

A factor V variant resistant to activated protein C, causing inherited thrombophilia and increased venous thrombosis risk.

Factor VII

A blood-clotting protein made mainly in the liver that begins the tissue-factor pathway of coagulation. Inherited deficiency can cause abnormal bleeding, ranging from easy bruising to serious bleeding after injury or procedures.

Factor VII Assay

A blood test measuring factor VII activity, used to evaluate an isolated prolonged prothrombin time and investigate inherited or acquired factor VII deficiency.

Factor VII Deficiency

A bleeding disorder caused by inherited or acquired deficiency of coagulation factor VII, impairing the extrinsic and common coagulation pathways. It may

cause easy bruising, mucosal bleeding, postoperative bleeding, or severe hemorrhage; prothrombin time is usually prolonged.

Factor VIII

A clotting protein that works with factor IX to help form a stable blood clot. Inherited deficiency causes hemophilia A, characterized by prolonged or spontaneous bleeding into joints, muscles, or other tissues.

Factor VIII Assay

A blood test measuring factor VIII activity, used to diagnose and monitor hemophilia A and assess other causes of reduced factor VIII activity.

Factor X Assay

A blood test measuring factor X activity, used to investigate bleeding and diagnose or monitor inherited or acquired factor X deficiency.

Factor X Deficiency

A rare inherited or acquired bleeding disorder caused by insufficient coagulation factor X, which participates in the common pathway. Bleeding severity varies, and both prothrombin time and activated partial thromboplastin time may be prolonged.

Factor XI Deficiency

An inherited or acquired bleeding disorder caused by inadequate factor XI activity, producing variable bleeding that is often most apparent after surgery, dental procedures, or trauma.

Factor Xii

factor xii is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Factor XII (Hageman Factor) Deficiency

An inherited or acquired deficiency of coagulation factor XII that markedly prolongs the activated partial thromboplastin time but usually does not cause abnormal bleeding.

Factor XII Assay

A blood test measuring factor XII activity, used to investigate an isolated prolonged activated partial thromboplastin time and diagnose factor XII deficiency.

Factor Xiii

factor xiii is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Factorial Study

factorial study is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Facultative Anaerobe

A microorganism that can grow with or without oxygen.

Faculty

The teaching and research staff of an academic institution; in medicine, faculty may educate learners, conduct research, and supervise clinical training.

Faecal Fistula

Faecal Fistula is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Faecalibacterium

A genus of anaerobic bacteria commonly found in the intestine and associated with production of the short-chain fatty acid butyrate.

Faecalibacterium Prausnitzii

A common intestinal bacterium that produces butyrate and is being studied for its relationship to gut inflammation and health.

Faeces

Faeces is an anatomical, developmental, reproduc-

tive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Fahr Disease

A rare disorder in which calcium builds up in parts of the brain. It may cause movement problems, seizures, changes in thinking, or psychiatric symptoms. Inherited cases are often called primary familial brain calcification.

Failed Induction Of Labor

The inability to achieve active labor and progressive cervical change after an adequate trial of induction. Definition: no active labor (cervical change) after at least 12–18 hours of oxytocin with ruptured membranes and adequate contractions. Management: the latent phase may be prolonged (up to 24 hours or more).

Failed Intubation

Inability to successfully place an endotracheal tube after multiple attempts. In the operating room, a failed intubation algorithm is followed: call for help, maintain oxygenation (face mask, supraglottic airway), and decide to wake the patient (if non-emergent) or proceed with a supraglottic airway (if surgery is urgent).

Failure To Thrive

Inadequate growth in children, defined as weight below the 5th percentile or crossing two major percentile lines on growth charts. Causes include caloric insufficiency (neglect, poor feeding), organic disease (congenital heart disease, renal disease), and psychosocial factors.

Failure To Thrive

Inadequate physical growth or unexpected weight loss, most often used for children with poor weight gain and sometimes for adults with functional decline; evaluation seeks nutritional, medical, developmental, and social causes.

Fainting

Fainting is a brief loss of consciousness from transient inadequate brain perfusion, often caused by

reflex, orthostatic, or cardiac mechanisms; cardiac warning features require urgent evaluation.

Alternate Names:

syncope.

Fainting (Syncope)

A brief loss of consciousness and postural tone caused by temporary inadequate blood flow to the brain, followed by spontaneous recovery; cardiac causes require urgent assessment.

Fair Hair

Hair with relatively low melanin content, appearing blond, light brown, or red; color reflects inherited pigment production and can change with age or exposure.

Falciform Ligament

A fold of tissue that attaches the liver to the front wall of the abdomen and the diaphragm. It separates the right and left lobes of the liver and contains a remnant of a blood vessel from before birth.

Falciparum Malaria

Severe malaria caused by the parasite *Plasmodium falciparum* and transmitted by infected *Anopheles* mosquitoes. It can rapidly cause cerebral malaria, severe anemia, kidney injury, respiratory distress, or death and requires urgent treatment.

Falimarev

An investigational antiviral monoclonal antibody studied for prevention or treatment of selected viral infections; its use depends on trial evidence and authorization.

Fall

An unplanned movement to the ground or a lower level. Falls are more likely with poor balance, muscle weakness, vision problems, certain medicines, unsafe surroundings, or a drop in blood pressure when standing.

Fall Prevention

Measures that reduce the risk of falling, including

strength and balance training, medication review, vision assessment, safer footwear, and environmental modification.

Fall Risk

The likelihood that a person will fall, based on factors such as gait, balance, strength, vision, medicines, orthostatic blood-pressure changes, and the environment.

Fall Screening

Assessment of a person's history and risk factors for falling to guide prevention and further evaluation.

Fallen Arches

Pes planus: flattening of the medial longitudinal arch of the foot. May be flexible (appearing only when standing) or rigid. Can cause foot, ankle, or knee pain, or be asymptomatic. Treatment includes supportive footwear, orthotics, and stretching exercises.

Alternate Names:

Pes planus; flat feet

Fallopian Tube

One of two tubes connecting an ovary to the uterus. An egg usually travels through the tube, where fertilization may occur, before reaching the uterus.

Fallopian Tube Cancer

A rare tumor histologically similar to high-grade serous ovarian carcinoma. Presents with a pelvic mass, pain, or watery discharge; staging and treatment follow the principles of ovarian cancer.

Fallopian Tube Torsion

Twisting of the fallopian tube around its pedicle, usually around an underlying mass such as a hydrosalpinx or paraovarian cyst, causing acute pelvic pain and requiring urgent surgical intervention.

Fallopian Tubes

Fallopian Tubes is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Fallopian-Tube Endometriosis

Endometrial-type glands and stroma within or around the fallopian tube, which may cause pelvic pain, infertility, adhesions, or an incidental pathologic finding.

Fallot'S Tetralogy

Fallot's Tetralogy is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fallot, Tetralogy Of

A congenital heart defect consisting of four abnormalities: ventricular septal defect, right ventricular outflow tract obstruction, overriding aorta, and right ventricular hypertrophy. Causes cyanosis due to right-to-left shunting. Treatment includes surgical repair.

Alternate Names:

Tetralogy of Fallot

Falls

Falls is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

False Labor

Uterine contractions that may be uncomfortable but do not produce progressive cervical dilation and effacement; they are often irregular and may lessen with rest or hydration.

False Negative

A test result that indicates a condition is absent when it is actually present.

False Passages

Abnormal channels created through tissue, often during instrumentation, surgery, or trauma, that do not follow the normal anatomical pathway and can cause bleeding, perforation, or infection.

False Positive

A test result that indicates a disease or condition is present when it is not.

False Ribs

Ribs 8-10 not articulating directly with the sternum; their cartilages join rib 7's cartilage forming the costal margin. Greater mobility and flexibility of the lower thoracic wall. More susceptible to fractures at their angles due to greater length and less rigid attachment.

False-Negative Result

A negative test result in a person who does have the target condition. Causes include inadequate specimen, early or late testing, low organism or biomarker concentration, prior treatment, technical error, and limited test sensitivity.

False-Positive Result

A positive test result in a person who does not have the target condition. Causes include limited specificity, cross-reactivity, technical error, contamination, and testing in a low-prevalence population.

Faltering Weight

Inadequate weight gain or downward crossing of a child's growth trajectory, caused by insufficient intake, feeding difficulty, malabsorption, chronic disease, or psychosocial factors.

Famciclovir

A medicine or chemical compound called famciclovir. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Familial

Occurring in a family more often than expected, because of shared genes, environment, or both; familial does not necessarily mean a single-gene inherited disorder.

Familial Adenomatous Polyposis

An inherited condition that causes hundreds or thousands of growths in the colon. These growths can

become colon cancer if they are not treated, so regular screening and specialist care are needed.

Familial Combined Hyperlipidemia

An inherited disorder causing elevated cholesterol, triglycerides, or both.

Familial Dysautonomia

An inherited disorder affecting nerves that control sensation and automatic body functions such as blood pressure, sweating, digestion, and swallowing.

Familial Dysbetalipoproteinemia

An inherited condition in which the body does not clear certain blood fats properly. It can raise cholesterol and triglycerides and increase the risk of early artery disease.

Familial Hypercholesterolaemia

Familial Hypercholesterolaemia is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Familial Hypercholesterolemia

An inherited disorder causing persistently high low-density lipoprotein cholesterol, usually because of impaired LDL-receptor pathway function. It increases the risk of premature atherosclerotic disease; diagnosis uses lipid levels, family history, physical findings, and genetic testing when available.

Familial Hypertriglyceridemia

An inherited tendency toward elevated blood triglycerides, often worsened by obesity, diabetes, alcohol, or medicines and sometimes increasing pancreatitis risk.

Familial Hypocalciuric Hypercalcemia

Autosomal dominant condition caused by inactivating mutations in the calcium-sensing receptor gene leading to a higher set point for calcium-mediated PTH suppression. Presents with mild hypercalcemia, inappropriately normal or mildly elevated PTH, and low urinary calcium excretion

with calcium-to-creatinine clearance ratio less than 0.01.

Familial Lipoprotein Lipase Deficiency

A rare autosomal-recessive disorder in which impaired triglyceride breakdown causes severe chylomicronemia, recurrent pancreatitis, eruptive xanthomas, and abdominal pain.

Familial Mediterranean Fever

An autosomal recessive autoinflammatory disorder (MEFV gene mutations, pyrin/inflammasome dysregulation) causing recurrent self-limited febrile attacks with serositis (peritonitis, pleuritis, synovitis), erysipelas-like erythema, lasting 12-72 hours.

Familial Mediterranean Fever

An inherited autoinflammatory disorder, usually related to variants in *MEFV*, causing recurrent attacks of fever and inflammation of the abdominal, chest, or joint linings; amyloidosis is a possible complication.

Familial Multiple Lipomatosis

An autosomal dominant condition characterized by the development of numerous, symmetrically distributed subcutaneous lipomas, typically appearing in early adulthood. The lipomas are encapsulated, painless, and slow-growing, affecting the trunk and extremities. Unlike Dercum disease, they are not painful.

Familial Multiple Trichoepithelioma

An inherited disorder characterized by multiple benign hair-follicle tumors, usually skin-coloured papules on the face; it may be associated with pathogenic variants in *CYLD* and Brooke-Spiegler syndrome.

Familial Paroxysmal Peritonitis

Familial Mediterranean fever: an autosomal recessive autoinflammatory disorder causing recurrent fevers and serositis (peritonitis, pleuritis, synovitis). Caused by MEFV gene mutations affecting pyrin regulation.

Alternate Names:

Familial Mediterranean fever

Familial Testotoxicosis

Familial testotoxicosis is an activating mutation of the luteinizing-hormone receptor causing gonadotropin-independent precocious puberty in boys, with high testosterone but low prepubertal LH and FSH.

Family

A group of biologically or socially related people whose shared history, genes, environment, and support relationships can affect health risk and care.

Family Involvement

The participation of family members or other support persons in care planning, treatment, rehabilitation, and support, with attention to the patient's preferences and caregiver burden.

Family Planning

Healthcare and personal decision-making about whether and when to have children, including contraception, fertility care, preconception health, and pregnancy planning.

Family Study

family study is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Family Therapy

Family therapy is a medical test, procedure, treatment, or clinical measure used in patient care; its purpose and interpretation depend on the clinical question.

Famophos

Famophos is an organophosphate insecticide; exposure can inhibit acetylcholinesterase and cause cholinergic toxicity, including salivation, sweating, vomiting, bronchospasm, bradycardia, and muscle weakness.

Famotidine

Famotidine is a histamine H₂-receptor antagonist that reduces gastric-acid secretion and is used for peptic ulcer disease, gastro-oesophageal reflux, and hypersecretory states; dose adjustment may be needed in renal impairment.

Fanconi Anemia

An inherited disorder in which the bone marrow cannot make enough blood cells and the body has difficulty repairing damaged DNA. It may cause anemia, infections, bleeding, birth differences, and an increased risk of cancer.

Fanconi Syndrome

A kidney disorder in which the kidney tubules fail to return several useful substances to the blood. It can cause loss of sugar, minerals, and other substances in the urine, leading to weak bones, poor growth, or an abnormal blood-acid level.

FAP

Familial adenomatous polyposis: an inherited condition causing hundreds to thousands of colon polyps, with near-certain colorectal cancer if untreated. Caused by APC gene mutations with autosomal dominant inheritance. Managed with prophylactic colectomy.

Alternate Names:

Familial adenomatous polyposis

Farad

The SI unit of electrical capacitance, equal to one coulomb per volt, relevant to biomedical instrumentation and electrophysiology.

Farber Lipogranulomatosis

A rare autosomal-recessive lysosomal storage disorder caused by acid ceramidase deficiency, producing painful subcutaneous nodules, joint contractures, and hoarseness.

Farmer'S Lung

The most common HP, caused by thermophilic actinomycetes in moldy hay. Presents acutely 4–8 hours

after exposure with fever, cough, and dyspnea.

Farmer'S Lung

Hypersensitivity pneumonitis caused by inhalation of antigens from moldy hay, grain, or other agricultural materials. Repeated exposure produces immune-mediated inflammation of the distal airways and lung interstitium. Acute disease may cause fever, cough, chills, and dyspnea several hours after exposure, whereas chronic disease may cause progressive fibrosis. Diagnosis combines exposure history, imaging, pulmonary-function testing, and specialist assessment; avoidance of the antigen is central to management.

Farsightedness

Hyperopia, a refractive error in which light focuses behind the retina, causing near blur or eyestrain; glasses or contact lenses can correct it.

Alternate Names:

hyperopia, in which light focuses behind the retina when.

Fart

A colloquial term for passing intestinal gas through the anus, produced from swallowed air and bacterial fermentation in the digestive tract.

Fascia

A layer or band of connective tissue that surrounds, separates, and supports muscles, organs, nerves, and blood vessels.

Fascia Adherens

A connection between neighboring cells that helps hold them together, especially in heart muscle. It attaches parts of the cell's internal support system to the cell membrane.

Fascial Loop

A loop or sling made from a person's tissue or synthetic material and used to support or reinforce a body structure, such as the urethra.

Fasciculation

An involuntary, brief twitch of a small group of muscle fibers, often caused by irritation or disease of a motor nerve.

Fasciitis

Inflammation of fascia, the connective tissue surrounding muscles and other structures. Causes and severity vary widely, from localized plantar fasciitis to serious infectious fasciitis.

Fasciola

fasciola is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Fasciola Hepatica

fasciola hepatica is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Fascioliasis

A parasitic infection caused by liver flukes of the genus *Fasciola*, acquired by ingesting metacercariae on contaminated aquatic plants or in water. Acute hepatic migration may cause fever, right-upper-quadrant pain, hepatomegaly, and eosinophilia; chronic biliary infection may cause colic, cholangitis, or biliary obstruction.

Fasciolopsis

An intestinal fluke infection caused by *Fasciolopsis buski*, acquired by eating contaminated aquatic plants and potentially causing abdominal pain, diarrhea, or malnutrition.

Fasciolopsis Buski

Fasciolopsis buski is the large intestinal fluke that causes fasciolopsiasis after ingestion of metacercariae on contaminated aquatic plants; heavy infection can cause abdominal pain, diarrhoea, malnutrition, or oedema.

Fasciotomy

Surgical incision through the fascia to relieve compartment syndrome. Indications: acute compartment syndrome (trauma, reperfusion, burns, crush injury). Classic signs: pain out of proportion, pain with passive stretch, paresthesias, pallor, pulselessness (late).

Fast Heart Rate

A heart rate faster than expected for age or activity, usually called tachycardia. It may be a normal response to exercise, fever, pain, or stress, or a sign of arrhythmia or another medical disorder.

Fast Heart Rate - Tachycardia

Fast heart rate - tachycardia concerns fast heart rate - tachycardia, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Fast-Twitch Fiber

A muscle fiber that contracts quickly and powerfully but tires more rapidly than a slow-twitch fiber. It is important for sprinting, jumping, lifting, and other brief high-intensity activities.

Alternate Names:

Type II muscle fiber

Fasting

Fasting is abstaining from food, and sometimes drink, for a defined period before testing, treatment, surgery, or for personal or religious reasons.

Fasting Blood Glucose

The blood glucose concentration measured after a period without caloric intake, usually at least 8 hours; it is used to screen for and diagnose disorders such as diabetes.

Fasting Guidelines

Preoperative fasting guidelines minimize aspiration risk while reducing patient discomfort and catabolism. Clear liquids (water, clear juice without pulp, black coffee, sports drinks) are allowed up to 2 hours before surgery. Breast milk up to 4 hours.

Infant formula up to 6 hours. Light meal (toast and clear liquid) up to 6 hours.

Fasting Lipid Profile

A blood test measuring cholesterol and triglyceride levels, usually after fasting.

Alternate Names:

Lipid panel

Fasting Plasma Glucose Test

A blood test that measures glucose after fasting and helps diagnose diabetes or impaired glucose regulation.

Fat

A nutrient and energy source composed mainly of lipids.

Alternate Names:

Lipid

Fat Body

The fat body is an organ-like tissue in insects that stores lipid, produces metabolic proteins, and performs functions analogous to liver and adipose tissue.

Fat Embolism

A condition in which drops of fat enter the bloodstream, usually after a broken thigh or pelvic bone, major trauma, or orthopedic surgery. In serious cases, fat can reach the lungs or brain and cause breathing or neurologic problems.

Fat Measurement

Assessment of body fat or fat content using methods such as skinfolds, bioimpedance, DEXA, imaging, or laboratory analysis; results vary by method and are interpreted with overall health measures.

Fat Necrosis

A specialized form of necrosis that occurs in adipose tissue, most commonly following acute pancreatitis when activated pancreatic lipases escape into the surrounding tissues and digest triglycerides in mesenteric or subcutaneous fat cells. In the breast,

it may follow trauma or surgery and can produce a firm lump that mimics malignancy on examination or imaging.

Fat Pad

A fat pad is a localized collection of adipose tissue that cushions, supports, insulates, or permits movement around an organ or joint.

Fat-Soluble Toxins

Lipophilic toxins stored in fat, prolonged elimination. Examples: organochlorine pesticides, PCBs, dioxins, tetrachlorodiphenylmethane.

Fat-Soluble Vitamin

A vitamin that dissolves in fat and can be stored in the body. Vitamins A, D, E, and K are fat-soluble; taking too much can cause harmful buildup.

Fat-Soluble Vitamins

Fat-Soluble Vitamins is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fatal Familial Insomnia

Fatal Familial Insomnia is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fatigue

Fatigue is a persistent sense of low energy or reduced capacity that may result from sleep problems, anemia, infection, endocrine disease, medication, mood disorder, or systemic illness. relieved by rest. It may result from sleep disturbance, anemia, infection, endocrine disease, chronic organ disease, medication exposure, or psychological and neurological conditions.

Fatigue (Cancer)

Persistent tiredness or reduced energy associated with cancer or its treatment. It may result from the disease, anemia, sleep problems, medicines, mood,

or other factors.

Alternate Names:

Cancer

Fatty Acid

A carboxylic acid with a hydrocarbon chain that may be saturated or contain one or more double bonds. Fatty acids provide energy, form membrane lipids, and serve as precursors for signaling molecules.

Fatty Acid Synthase

A group of enzymes that helps the body make fatty acids, which are used for energy storage and cell structures.

Fatty Acids

Organic components of fats used for energy and for building cell membranes and signaling molecules.

Fatty Degeneration

Accumulation of abnormal lipid within cells, especially liver cells, caused by metabolic stress, alcohol, toxins, malnutrition, obesity, or other injury.

Fatty Liver (Steatosis)

Abnormal intrahepatic accumulation of triglycerides within hepatocytes exceeding 5% of total liver weight or visible in > 5% of hepatocytes on histological evaluation. Categorized etiologically into metabolic dysfunction-associated steatotic liver disease (MASLD, formerly designated NAFLD), which is tightly linked to obesity, type 2 diabetes mellitus, and dyslipidemia, and alcohol-associated liver disease. Can progress along an inflammatory continuum from isolated hepatic steatosis to steatohepatitis, characterized by hepatocyte ballooning, lobular inflammation, Mallory-Denk bodies, and progressive pericellular fibrosis that can culminate in cirrhosis, end-stage liver failure, and hepatocellular carcinoma. Evaluated with transabdominal ultrasonography, transient elastography (FibroScan), and biomarker panels; managed with lifestyle weight loss, cardiometabolic risk control, and GLP-1 receptor agonists.

Alternate Names:

NAFLD (Fatty Liver)

Fatty Liver

Alternate index label for Fatty Liver.

Fatty Liver Disease

A condition characterized by the accumulation of excess fat in liver cells, most commonly associated with obesity, insulin resistance, and metabolic syndrome.

Fatty Liver Disease

A condition in which excess fat accumulates in the liver, often associated with metabolic risk factors and potentially progressing to inflammation or scarring.

Alternate Names:

MASH MASLD Metabolic Dysfunction-Associated Steatohepatitis Metabolic Dysfunction-Associated Steatotic Liver Disease NAFLD

Fatty Streak

An early atherosclerotic lesion formed by lipid accumulation in an artery wall.

Favism

Acute hemolysis after eating fava beans or inhaling pollen in a person with glucose-6-phosphate dehydrogenase deficiency, triggered by oxidative stress.

Favus

Favus is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fazarabine

Fazarabine is a cytidine analogue antimetabolite investigated as an anticancer drug; it interferes with nucleic-acid synthesis and may cause myelosuppression and other chemotherapy toxicities.

Fc Receptor

A protein on the surface of some immune cells that attaches to antibodies and helps the cells identify,

remove, or destroy targets.

Fear

An emotional and physiologic response to perceived threat, involving vigilance, autonomic arousal, avoidance, and protective behavior; excessive fear can be pathological.

Fear Of Age

An excessive or persistent fear of growing old or of older people, sometimes called gerontophobia, that causes distress or avoidance.

Fear Of Children

An excessive or persistent fear of children, sometimes called pedophobia, that causes significant distress or avoidance.

Fear Of Crossing Streets

An excessive fear of crossing streets or roads, sometimes called agyrophobia, that may lead to marked avoidance and interfere with daily activities.

Fear Of Demons

An excessive or persistent fear of demons or evil supernatural beings, sometimes called demonophobia.

Fear Of Drinking Alcohol

An excessive or persistent fear of drinking alcoholic beverages, sometimes called methyphobia, that causes distress or avoidance.

Fear Of Dust

An excessive or persistent fear of dust, sometimes called amatophobia, that causes distress or avoidance.

Fear Of Feces

An excessive or persistent fear of feces or defecation, sometimes called coprophobia, that causes distress or avoidance.

Fear Of Fur

An excessive or persistent fear of fur, sometimes called doraphobia, that causes distress or avoidance.

Fear Of Heat

An excessive or persistent fear of heat, sometimes called thermophobia, that causes distress or avoidance.

Fear Of Nudity

An excessive or persistent fear of being naked or seeing others naked, sometimes called gymnophobia, that causes distress or avoidance.

Fear Of Oneself

An excessive or persistent fear of oneself or one's own thoughts or actions, sometimes called autophobia in some usage, that causes distress or avoidance.

Fear Of Worms (Parasitic)

An excessive or persistent fear of parasitic worms, sometimes called helminthophobia, that causes distress or avoidance.

Fears

Fears is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Febrile

Pertaining to, characterized by, or exhibiting pyrexia (fever), defined clinically as an abnormal elevation of core body temperature above the normal physiological circadian range (typically $\geq 38.0^{\circ}\text{C}$ [$\geq 100.4^{\circ}\text{F}$]). Regulated centrally by the preoptic area of the anterior hypothalamus in response to endogenous pyrogens (proinflammatory cytokines IL-1, IL-6, TNF- α) which stimulate endothelial production of prostaglandin E₂ (PGE₂) to elevate the hypothalamic thermoregulatory set point. Differentiated from hyperthermia (where the hypothalamic set point remains normal but heat dissipation fails, as in heat stroke). High-yield clinical presentations include febrile seizures in pediatric populations, fever of unknown origin, and neutropenic fever, which represents an oncologic emergency.

Febrile (Warm) And Cold Agglutinins

Antibodies that cause red cells to clump at warmer or colder temperatures; clinically important forms

can cause autoimmune hemolytic anemia.

Febrile Convulsion

A seizure occurring with fever in a young child without evidence of central nervous system infection or another acute cause; most are brief and generalized.

Febrile Convulsions

Febrile Convulsions is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Febrile Neutropenia

A fever in a person with a very low number of neutrophils, a type of white blood cell that fights infection. It may be the only sign of a serious infection and requires urgent medical evaluation, especially during cancer treatment.

Febrile Seizure

A seizure occurring in a child usually between 6 months and 5 years of age with fever but without evidence of central nervous system infection or a previous afebrile seizure. Simple febrile seizures are generalized, brief, and nonrecurrent within the same illness; prolonged or focal events require further evaluation.

Alternate Names:

Seizure, Febrile

Febrile Seizures

Febrile seizures are seizures associated with fever in young children without evidence of central nervous system infection or prior unprovoked seizures.

Alternate Names:

Seizure Febrile (Febrile Seizures) Seizure Fever Induced (Febrile Seizures)

Febuxostat

Febuxostat is a selective xanthine-oxidase inhibitor used to lower urate in chronic hyperuricaemia and gout; it is not a treatment for immediate pain and requires attention to cardiovascular risk and liver function.

FEC

Fuchs endothelial corneal dystrophy: a progressive condition affecting the corneal endothelium, leading to corneal edema and vision impairment. More common in women and older adults. Advanced cases may require corneal transplantation.

Alternate Names:

Fuchs dystrophy

Fec Protocol

A chemotherapy regimen combining fluorouracil, epirubicin, and cyclophosphamide, used for selected cancers with dosing and monitoring determined by oncology protocols.

Fecal Calprotectin

A stool test that measures a substance released by certain white blood cells. A high result suggests inflammation in the intestines and may help distinguish inflammatory bowel disease from disorders without inflammation.

Fecal Culture

A laboratory test that grows organisms from a stool specimen to identify bacterial causes of infectious diarrhea and guide antimicrobial treatment.

Fecal Elastase

A stool test that helps show whether the pancreas is making enough digestive enzymes. A low result may indicate pancreatic insufficiency, in which food is not digested properly.

Fecal Fat

Measurement or observation of excess fat in stool, suggesting impaired digestion or absorption from pancreatic, bile-acid, intestinal, or other disease.

Fecal Immunochemical Test

A stool-based colorectal-cancer screening test that uses antibodies to detect human hemoglobin, primarily from lower gastrointestinal bleeding. It is not diagnostic by itself; a positive result generally requires evaluation of the colon, commonly with colonoscopy. Test intervals and follow-up depend

on screening guidance and clinical risk.

Fecal Immunochemical Test

Alternate index label; see *Fecal immunochemical test*.

Fecal Impaction

A large, firm mass of stool retained in the rectum or colon that cannot be evacuated normally. It may cause abdominal or rectal pain, distention, overflow diarrhoea, urinary symptoms, nausea, or bowel obstruction and is associated with constipation, immobility, dehydration, neurological disease, and medications.

Fecal Incontinence

Inability to control bowel movements ranging from minor soiling to complete loss of continence. Causes include sphincter injury from obstetric trauma or surgery, neuropathy from diabetes or spinal cord injury, diarrhea, rectal prolapse, and cognitive impairment.

Alternate Names:

Incontinence, Bowel Incontinence, Fecal

Fecal Microbiota Transplant

Fecal microbiota transplantation transfers processed stool from a screened donor to restore intestinal microbial communities, particularly for recurrent *Clostridioides difficile* infection.

Fecal Microbiota Transplantation

Transfer of carefully screened and prepared stool from a donor into a patient's intestines to restore helpful bacteria. It may be used for selected repeated *C. difficile* infections when standard treatment has not worked.

Fecal Occult Blood Test

A stool test that detects small amounts of blood that are not visible to the naked eye. Methods include chemical guaiac testing and immunochemical testing such as FIT. It may be used for colorectal-cancer screening or evaluation of suspected gastrointestinal blood loss, but it is not diagnostic by itself; a positive result requires clinical follow-up and often

colon evaluation.

Fecal Smear

Fecal smear concerns fecal smear, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Fecalith

A hard mass of inspissated fecal material, commonly found in the appendix or colon, that can contribute to appendicitis or intestinal obstruction.

Fecaluria

Passage of fecal material or fecal odor in the urine, usually indicating an abnormal connection such as an enterovesical fistula.

FECD

Fuchs endothelial corneal dystrophy: a bilateral, progressive corneal disease characterized by loss of endothelial cells, resulting in corneal thickening and decreased vision, especially upon waking. Early stages may be asymptomatic.

Alternate Names:

Fuchs dystrophy

Feces

The solid or semisolid waste expelled from the bowel, containing undigested material, water, bacteria, shed cells, and metabolic waste.

Fecolith

A hard mass of inspissated fecal material, which may obstruct the bowel or appendix and contribute to inflammation or perforation.

Fecundity

The biological ability to conceive and produce a live birth. Fecundity is influenced by age (declines significantly after 35), ovarian reserve, tubal patency, uterine factors, and male fertility (sperm count, motility, morphology). Fecundability: the probability of achieving pregnancy in a single menstrual cycle.

Feedback

Information about a system's output that influences subsequent activity; negative feedback helps stabilize physiology, while positive feedback amplifies a process.

Feedback Inhibition

Regulatory mechanism where the end product of a metabolic pathway inhibits an earlier enzyme (usually the first committed step). Examples: isoleucine inhibiting threonine deaminase, CTP inhibiting aspartate transcarbamoylase, ATP inhibiting phosphofructokinase. Allows efficient resource utilization and prevents overproduction.

Feedback Mechanism

A regulatory process in which a system's output influences its future activity; negative feedback stabilizes functions such as temperature, glucose, and hormone levels.

Feeding Jejunostomy

Surgical or endoscopic placement of a feeding tube into the jejunum for enteral nutrition when upper gastrointestinal access is not feasible. Used in patients with head and neck cancers, esophageal obstruction, or severe dysphagia. Provides long-term nutritional support and avoids the complications of parenteral nutrition.

Feeding Methods

feeding methods is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Feeding Patterns And Diet - Babies And Infants

A patient-care topic concerning feeding patterns and diet - babies and infants. It addresses the relevant purpose, risks, management, and follow-up.

Feeding Patterns And Diet - Children 6 Months To 2 Years

A patient-care topic concerning feeding patterns and diet - children 6 months to 2 years. It addresses the

relevant purpose, risks, management, and follow-up.

Feeding Poor

Poor feeding means an infant or patient takes inadequate food or fluid and may reflect illness, swallowing difficulty, breathing problems, pain, or developmental issues.

Feeding Tube

A flexible tube used to deliver nutrition, fluids, or medicines to the stomach or intestine when oral intake is inadequate or unsafe.

Feeding Tube - Infants

A tube used to deliver breast milk, formula, fluids, or medicines to an infant who cannot safely or adequately feed by mouth; route and duration depend on the clinical need.

Feeding Tube Insertion - Gastrostomy

Feeding tube insertion - gastrostomy is a nutrition or dietary term; appropriate use depends on age, health status, allergies, intake, and medical treatment.

Feingold Diet

Feingold Diet is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Felbamate

Felbamate is an antiseizure medicine used selectively for refractory epilepsy, including Lennox-Gastaut syndrome; potentially fatal aplastic anaemia and liver failure require specialist use and monitoring.

Felodipine

Felodipine is a dihydropyridine calcium-channel blocker used mainly for hypertension; arteriolar vasodilation can cause ankle oedema, flushing, headache, palpitations, or gingival enlargement.

Felon

Felon is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and

severity.

Felty Syndrome

Triad of rheumatoid arthritis, splenomegaly, and neutropenia (ANC <2000). Complication of long-standing, severe, seropositive RA. Increased infection risk. Treatment: DMARDs (methotrexate, TNF inhibitors), G-CSF for neutropenia, splenectomy in refractory cases.

Felty'S Syndrome

Felty syndrome is a rare complication of long-standing seropositive rheumatoid arthritis characterized by neutropenia and often an enlarged spleen. It increases infection risk and may require disease-modifying therapy, growth factors, antibiotics, or specialist treatment.

Female

Female describes the sex category generally associated with ova production and female reproductive anatomy, while recognizing biological variation and gender identity distinctions.

Female Athlete Triad

A syndrome involving low energy availability, menstrual dysfunction, and decreased bone mineral density in physically active females. It is related to inadequate nutrition, excessive exercise, or both, and increases the risk of stress fractures and other health problems.

Female Breast

The mammary gland and associated skin and connective tissue on the anterior chest; it contains lobules and ducts and varies with age, hormones, pregnancy, and lactation.

Female Circumcision

Female Circumcision is an anatomical, developmental, reproductive, nutritional, behavioral, or health-care concept used in medical assessment or care.

Female Condom

A female condom is a sheath placed in the vagina

that provides a barrier against sperm and can reduce transmission of some sexually transmitted infections when used correctly.

Female Condoms

A flexible internal barrier placed in the vagina before sex that helps prevent pregnancy and reduces exposure to semen and some sexually transmitted infections.

Female Genital Mutilation

FGM: the partial or total removal of female external genitalia for non-medical reasons, a human rights violation practiced in parts of Africa, the Middle East, and Asia (types I-IV). Immediate complications: severe pain, hemorrhage, infection, urinary retention.

Female Genitalia

The external and internal reproductive organs of a female, including the vulva, vagina, cervix, uterus, fallopian tubes, and ovaries.

Female Gonad

An ovary, the female reproductive gland that produces oocytes and hormones such as estrogen and progesterone.

Female Infertility

Failure to achieve pregnancy after a defined period of regular unprotected intercourse, due to factors involving ovulation, the reproductive tract, ovarian reserve, or other causes.

Alternate Names:

Infertility, Female

Female Orgasmic Disorder (Anorgasmia)

Persistent or recurrent delay in, infrequency of, or absence of orgasm that causes personal distress. It may be lifelong or acquired and can have medical, psychological, relationship, medication-related, or other causes.

Alternate Names:

Anorgasmia

Female Pattern Baldness

Progressive scalp hair thinning in a typically female distribution, influenced by genetics and androgens; evaluation considers iron deficiency, thyroid disease, medicines, and other causes.

Female Pelvis

The bony pelvis and associated soft tissues in a person with typical female pelvic anatomy; it is generally broader and has a wider pelvic outlet than the typical male pelvis, with substantial individual variation.

Female Phenotype

A female phenotype is the observable combination of anatomical, physiologic, and developmental traits associated with female sex, which can vary with genes and hormones.

Female Reproductive System

The female reproductive system includes the ovaries, fallopian tubes, uterus, cervix, vagina, vulva, and related hormonal pathways. It supports ovulation, menstruation, pregnancy, birth, and sexual function and is affected by developmental, endocrine, infectious, and structural disorders.

Female Sexual Dysfunction

Persistent sexual problems in a woman that cause distress, such as low desire, arousal difficulty, or pain.

Alternate Names:

Sexual Dysfunction, Female

Female Sexual Interest/Arousal Disorder

Persistent or recurrent reduction in sexual interest, thoughts, responsiveness, or arousal that causes personal distress. Causes may include medical conditions, medicines, mood, relationship factors, pain, or other psychological and social factors.

Female Urethral Opening

The external urethral meatus, the opening through which urine leaves the female urethra, located in the vulvar vestibule anterior to the vaginal opening.

Feminization

Development of physical or hormonal traits typically associated with female sexual development, occurring normally, therapeutically, or through endocrine disorders.

Femoral

Relating to the femur or thigh, including the femoral artery, vein, nerve, and neck of the femur.

Femoral Artery

Continues from external iliac through the femoral triangle and adductor canal to become the popliteal artery. Branches: superficial/deep circumflex femoral, deep femoral (profunda femoris), muscular. Palpable in the femoral triangle; most common site for arterial catheterization. Related to femoral vein medially and nerve laterally.

Femoral Hernia

A hernia in which abdominal contents pass through the femoral ring into the femoral canal, below the inguinal ligament and medial to the femoral vessels. It may become incarcerated or strangulated and often requires surgical assessment.

Femoral Hernia Repair

Surgery to return herniated tissue through the femoral canal and close or reinforce the defect; repair is often recommended because femoral hernias have a relatively high risk of strangulation.

Femoral Nerve

Largest lumbar plexus nerve (L2-L4), passing through psoas major and entering the thigh under the inguinal ligament lateral to the femoral artery. Innervates quadriceps, sartorius, pectineus; sensation to anterior thigh and medial leg (saphenous nerve). Used for knee surgery nerve block anesthesia.

Femoral Nerve Dysfunction

Impaired function of the femoral nerve, causing weakness of hip flexion or knee extension and altered sensation over the anterior thigh and medial leg; causes include diabetes, compression, trauma, and

pelvic disease.

Femoral Neuropathy

Damage to the femoral nerve, causing weakness or sensory loss in the thigh and leg.

Femoral Pulse

Femoral Pulse is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Femoral Shaft Fractures

Femoral shaft fractures: high energy (MVC, fall from height) or low energy (elderly, pathologic). Winquist-Hansen classification of comminution: I (none), II (butterfly <50% width), III (butterfly >50%), IV (segmental).

Femoral Triangle

Upper anterior thigh space bounded superiorly by the inguinal ligament, laterally by sartorius, and medially by adductor longus. Contents from lateral to medial: femoral nerve, artery, vein, and deep inguinal lymph nodes. The femoral artery is used for arterial catheterization. The femoral vein is accessed for central venous lines.

Femoral Vein

The major deep vein of the thigh, continuation of the popliteal vein (through the adductor hiatus), running with the femoral artery in the femoral sheath (medial to artery) through the femoral triangle, becoming the external iliac vein at the inguinal ligament. Receives the great saphenous vein.

Femtogram

A femtogram is a unit of mass equal to one quadrillionth of a gram, or 10^{-15} grams.

Femtoliter

A femtoliter is a unit of volume equal to one quadrillionth of a liter, or 10^{-15} liters.

Femur

Longest, heaviest, strongest body bone. Proximal:

head, neck, trochanters. Shaft: linea aspera. Distal: condyles, epicondyles. The neck connects head to shaft at approximately 125 degrees.

Femur Fracture

A break in the thigh bone, caused by major trauma or weakened bone; location and displacement determine blood loss, mobility, nerve and vessel risk, and treatment.

Femur Length

The length of the fetal femur (diaphysis), measured from the greater trochanter to the distal femoral metaphysis. Used for dating (14-26 weeks) and growth assessment. Normal growth: 2 mm per week. Short femur: can be constitutional (especially with short parents), associated with aneuploidy (trisomy 21, skeletal dysplasias), or IUGR.

Fenamiphos

An organophosphate pesticide used mainly against nematodes; it can cause poisoning by inhibiting acetylcholinesterase.

Fencer'S Pose

The posture produced by the asymmetric tonic neck reflex, with extension of the arm and leg on the side toward which the head is turned and flexion on the opposite side.

Fenclonine

An experimental compound that reduces serotonin production by inhibiting tryptophan hydroxylase.

Fenfluramine

Fenfluramine is a serotonergic medicine used in some jurisdictions for selected severe epileptic encephalopathies; historical appetite-suppressant use was limited by valvular heart disease and pulmonary hypertension, so cardiac monitoring is required.

Fenitrothion

An organophosphate insecticide that can cause cholinergic poisoning by inhibiting acetylcholinesterase.

FeNO (Fractional Exhaled Nitric Oxide)

A breath test that estimates type 2, often eosinophilic, airway inflammation. The result is interpreted with symptoms, age, treatment, and other clinical findings.

Alternate Names:

Fractional Exhaled Nitric Oxide

Fenobucarb

A carbamate insecticide that can cause cholinergic poisoning if absorbed in a sufficient amount.

Fenofibrate

A fibric acid derivative (fibrate) activating PPAR-alpha, increasing HDL and reducing triglycerides, with modest LDL effects. Used for severe hypertriglyceridemia (pancreatitis risk), mixed dyslipidemia, and as an adjunct in residual hypertriglyceridemia.

Fenoldopam

A short-acting dopamine D1-receptor agonist used intravenously for severe hypertension in selected settings; it can cause hypotension, flushing, headache, and increased intraocular pressure.

Fenoprofen

Fenoprofen is a nonsteroidal anti-inflammatory drug used for pain and inflammatory arthritis; gastrointestinal bleeding, renal injury, fluid retention, and cardiovascular events are important risks.

Fenoprofen Calcium Overdose

Fenoprofen calcium overdose involves exposure or injury from fenoprofen calcium overdose; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Fenoterol

A medicine or chemical compound called fenoterol. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Fenoxycarb

An insect-growth regulator that mimics juvenile

hormone and disrupts development of immature insects.

Fenretinide

A synthetic retinoid studied for preventing or treating some cancers and other diseases.

Fentanyl Citrate

The citrate salt of fentanyl, a potent opioid analgesic used for severe pain and anesthesia; respiratory depression, sedation, dependence, and overdose are major risks.

Fentanyl Derivatives

Potent synthetic opioids related to fentanyl that may be used for analgesia or anesthesia. Examples include sufentanil, remifentanil, and alfentanil; their potency, duration, dosing, and risks differ, and they require careful monitoring because they can depress breathing.

Fenthion

An organophosphate insecticide that can cause cholinergic poisoning by inhibiting acetylcholinesterase.

Fermentation

A metabolic process in which microorganisms or cells convert organic substrates without using oxidative respiration, producing products such as acids, alcohols, or gases.

Ferredoxin

An iron-sulfur electron-transfer protein that participates in photosynthesis, steroid synthesis, and mitochondrial or bacterial redox reactions.

Ferritin

Intracellular iron storage protein reflecting total body iron stores. Low ferritin: iron deficiency anemia (<30 ng/mL, absolute; <100 can be consistent with iron deficiency in inflammation).

Ferritin Blood Test

A blood test measuring ferritin, a protein that stores

iron; low levels strongly support iron deficiency, while high levels may reflect iron excess, inflammation, infection, liver disease, or malignancy.

Ferroptosis

A regulated form of cell death caused by iron-dependent lipid-peroxide accumulation and loss of antioxidant protection, distinct from apoptosis and necrosis.

Ferrous Sulfate

Ferrous sulfate is an oral iron salt used to treat or prevent iron-deficiency anaemia; nausea, constipation, dark stools, and accidental overdose toxicity can occur.

Fertile Period

Fertile Period is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Fertilisation

The union of a sperm and an egg (oocyte) to form a zygote, the first cell of a new individual.

Fertility

The biological capacity to conceive and produce offspring. Fertility depends on the proper functioning of the reproductive system in both males and females, including ovulation, sperm production, and hormonal regulation.

Fertility Domain

The reproductive dimension of health involving ability to conceive, reproductive function, fertility goals, infertility, and related medical or psychosocial care.

Fertility Drugs

Fertility Drugs is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fertility Preservation

Measures taken before gonadotoxic cancer therapy

to preserve reproductive potential, including embryo or oocyte cryopreservation, sperm banking, and ovarian or testicular tissue cryopreservation.

Fertilization

Fusion of sperm and egg genetic material to form a zygote, usually in the fallopian tube, initiating embryonic development.

Ferumoxytol

An intravenous iron-carbohydrate complex used to treat iron-deficiency anemia in selected adults; hypersensitivity and hypotension require monitoring during infusion.

FESS

Functional endoscopic sinus surgery is a minimally invasive surgical technique that uses nasal endoscopes and specialized instruments to restore normal sinus drainage and ventilation by removing obstructive tissue and opening blocked sinus ostia, while preserving as much normal sinus mucosa and anatomy as possible.

Festinating Gait

A gait with progressively shortening, rapid steps and a forward-leaning posture, commonly associated with parkinsonism.

Fetal Acid-Base Status

Assessment of fetal oxygenation and acid-base balance, usually using umbilical cord blood gases at birth or other intrapartum information. Interpretation depends on the complete gas panel, sampling site, timing, clinical condition, and laboratory reference ranges; marked acidemia may indicate impaired oxygenation.

Fetal Alcohol Spectrum Disorder

A broad spectrum of physical, behavioral, and cognitive impairments caused by prenatal alcohol exposure.

Alternate Names:

Fetal Alcohol Spectrum Disorders

Fetal Alcohol Syndrome

A collection of birth defects resulting from exposure of the fetus to alcohol during pregnancy, representing the most severe form of fetal alcohol spectrum disorder (FASD).

Fetal Attitude

The relationship of the fetal body parts to one another, especially the degree of flexion or extension of the head and limbs. The usual attitude is flexion, but deflexed, neutral, or extended attitudes can affect the presenting part and labor.

Fetal Bradyarrhythmia

An abnormally slow or irregular fetal heart rhythm, which may result from conduction block, ectopic beats, maternal or fetal disease, or medication exposure.

Fetal Bradycardia

A fetal heart-rate baseline below 110 beats per minute for at least 10 minutes. Causes include medicines, maternal conditions, fetal conduction disease, hypoxia, cord problems, or placental abruption; persistent or severe bradycardia requires urgent obstetric assessment.

Fetal Circulation

The circulation before birth in which the placenta supplies oxygen and nutrients and fetal shunts direct much of the blood away from the lungs.

Fetal Death

Death of a fetus before delivery; medical and legal definitions vary according to gestational age and local regulations.

Fetal Development

The growth and maturation of the fetus from the end of the embryonic period until birth, including development of organs, body systems, movement, and physiologic function.

Fetal Disease

A disease, structural abnormality, or developmental

problem affecting a fetus before birth.

Fetal Distress

Abnormal fetal status during labor, indicated by abnormal fetal heart rate patterns (late or prolonged decelerations, minimal variability, absent accelerations). May indicate fetal hypoxia, cord compression, or placental insufficiency.

Fetal Echocardiography

A specialized ultrasound examination of the fetal heart used to assess cardiac structure, rhythm, and function, including possible congenital heart disease.

Fetal Growth Restriction

Failure of a fetus to reach its biologic growth potential, often associated with placental insufficiency. It is assessed using serial growth measurements, amniotic fluid, Doppler studies, and fetal surveillance rather than a single estimated-weight threshold alone.

Fetal Heart

The developing heart of a fetus, whose rate and rhythm can be assessed by ultrasound or electronic monitoring to evaluate fetal well-being.

Fetal Heart Rate

Normal fetal heart rate ranges from 110-160 beats per minute. Monitored via auscultation with a fetoscope, Doppler ultrasound, or continuous electronic fetal monitoring. Baseline FHR is determined over a 10-minute window and is modulated by the autonomic nervous system.

Fetal Hemoglobin

The predominant hemoglobin during fetal life, composed of two alpha and two gamma chains and having higher oxygen affinity than adult hemoglobin.

Fetal Hydrops

Abnormal accumulation of fluid in at least two fetal compartments, such as the skin, chest, abdomen, or around the heart. Causes include fetal anemia, heart disease, infection, chromosome conditions, and im-

mune incompatibility; it requires specialist obstetric evaluation.

Fetal Hypoxia

Inadequate oxygen available to the fetus before or during birth. It may result from placental problems, reduced maternal oxygenation, cord compression, or prolonged labor. Abnormal fetal heart-rate patterns can indicate risk and require prompt obstetric assessment.

Fetal Kick Counts (Fetal Movement Counting)

Observation and recording of fetal movements by a pregnant person. A noticeable reduction from the usual pattern should prompt timely obstetric advice.

Alternate Names:

Fetal Movement Counting

Fetal Lie

The relationship between the long axis of the fetus and that of the pregnant person. It may be longitudinal, transverse, or oblique. A persistent abnormal lie near delivery can affect the mode of birth.

Fetal Macrosomia

Excessive fetal size, often defined as estimated or birth weight of at least 4,000 or 4,500 g depending on the clinical context. Diabetes, excessive maternal weight gain, and prior macrosomia are risk factors; complications include shoulder dystocia, birth trauma, and cesarean delivery.

Fetal Malpresentation

A fetal presentation other than the usual vertex presentation, including breech, transverse, or oblique lie, which may affect labor and delivery planning.

Fetal Monitoring

Assessment of fetal heart rate and, sometimes, uterine contractions to evaluate fetal wellbeing during pregnancy or labor.

Fetal Monitoring, Electronic

Continuous or intermittent recording of fetal heart rate and uterine activity using external sensors or an internal electrode when indicated; patterns are interpreted with gestational age and labor context.

Fetal Presentation

The fetal part that enters the maternal pelvis first. Presentations include cephalic, breech, shoulder, face, brow, and compound forms; the presentation influences labor and delivery planning.

Fetal Pyelectasis

Dilation of the fetal renal pelvis seen on prenatal imaging; it may be transient or associated with urinary tract obstruction or aneuploidy and requires context-specific follow-up.

Fetal Scalp Electrode

A spiral wire electrode placed on the fetal scalp through the cervix to directly record the fetal heart rate. Provides a more accurate FHR signal than external Doppler, particularly in obese patients or when external monitoring is inadequate. Indications: poor-quality external tracing, need for internal monitoring.

Fetal Scalp PH Testing

Fetal scalp pH testing involves exposure or injury from fetal scalp pH testing; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Fetal Scalp Stimulation

A bedside test during labor in which the fetal scalp is gently stimulated during a vaginal examination. A resulting fetal heart-rate acceleration may be reassuring, but the result must be interpreted with the complete fetal monitoring pattern and clinical context.

Fetal Stage

The fetal stage begins after the embryonic period and continues until birth, characterized mainly by growth and maturation of established organs.

Fetal Tachycardia

A sustained fetal heart-rate baseline above 160 beats per minute for at least 10 minutes. Causes may include maternal fever, infection, medicines, fetal hypoxia, or fetal arrhythmia.

Fetal Tissue

Fetal tissue is tissue from a developing fetus and may be relevant to pregnancy care, pathology, research, transplantation, or genetic testing under applicable ethical and legal rules.

Fetal Ultrasound

Ultrasound imaging during pregnancy. First trimester (6-12 weeks): confirm pregnancy, gestational age (crown-rump length), number of fetuses, viability (heart activity), nuchal translucency (screening for aneuploidy at 11-13 weeks).

Fetal Viability

The potential for a fetus to survive outside the uterus with medical support. It depends on gestational age, fetal condition, available neonatal care, and other clinical factors; it is not a single fixed point.

Fetal Weight

Fetal weight is an estimate of a fetus's mass, usually from ultrasound measurements, used to assess growth and identify possible growth restriction or macrosomia.

Fetal-Maternal Erythrocyte Distribution Blood Test

A test that estimates the amount of fetal red blood cells in maternal circulation, helping guide Rh immune globulin dosing after fetomaternal hemorrhage.

Fetishism

A recurrent sexual interest in a nonliving object or a specific non-genital body part; it is a disorder only when it causes clinically significant distress or impairment or involves harm or lack of consent.

Fetishistic Disorder

A paraphilic disorder characterized by recurrent, intense sexual arousal from either the use of nonliving objects (fetish) or a highly specific focus on nongenital body parts (partialism), lasting ≥ 6 months and causing distress or impairment or involving acting on these urges.

Fetomaternal Hemorrhage

Transplacental passage of fetal red blood cells into the maternal circulation. Occurs in small amounts during normal pregnancy but can be massive during trauma, placental abruption, or cesarean delivery.

Fetomaternal Transfusion

A medical procedure, treatment, device, or care topic related to fetomaternal transfusion; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Fetoscope

An instrument used to hear or assess the fetal heart-beat through the pregnant person's abdominal wall, especially later in pregnancy.

Fetus

An unborn offspring from the 9th week of gestation (after the embryonic period) until birth. The fetal period is characterized by rapid growth, maturation of organ systems, and increasing complexity of function.

FEV1/FVC Ratio

Fraction of air exhaled in first second relative to total. < 0.70 post-bronchodilator indicates obstruction. Normal or elevated in restrictive disease.

Fever

Fever is a regulated rise in body temperature caused by altered hypothalamic set point, commonly in response to infection or inflammation; age, duration, and associated signs determine concern.

Fever And Headache

Fever with headache may occur with viral illness,

influenza, sinus disease, dehydration, meningitis, encephalitis, or other infection. Severe sudden headache, neck stiffness, confusion, rash, seizure, weakness, or reduced consciousness requires emergency assessment.

Fever Blister

Recurrent herpes simplex labialis: small, painful vesicles on the lips or around the mouth caused by HSV-1. The virus remains latent in trigeminal ganglia and reactivates with stress, illness, or sun exposure. Heals without scarring in 7-10 days.

Alternate Names:

Cold sore; herpes labialis

Fever Of Unknown Origin

Persistent or recurrent fever for which an appropriate initial evaluation has not established a diagnosis.

Fever, Acute, In Adults

A fever in an adult with recent onset, commonly caused by infection, inflammation, medication, heat illness, or other conditions. Severity, vital signs, immune status, source, and associated symptoms guide evaluation; confusion, breathing difficulty, stiff neck, shock, or severe localized pain requires urgent care.

Fever, Chronic

A prolonged or recurrent fever that remains unexplained after initial evaluation; fever of unknown origin is traditionally defined as a temperature of at least 101 degrees Fahrenheit (38.3 degrees Celsius) on several occasions lasting more than three weeks without a diagnosis after appropriate assessment. Causes include infection, malignancy, inflammatory disease, and medications. duration, height, and patient factors. Common causes: infections (bacterial, viral), autoimmune disorders, malignancies, drug reactions. Management: antipyretics (acetaminophen, ibuprofen), treat underlying cause.

Fever, Shin Bone

The shin bone is the tibia, the larger weight-bearing bone of the lower leg; fever is a separate term meaning an elevated body temperature, so this dictionary

label appears to combine two concepts.

Fever, Valley

Coccidioidomycosis: a fungal infection caused by *Coccidioides* species, endemic to arid regions of the southwestern US and parts of Mexico and Central/South America. Inhalation of spores causes respiratory illness ranging from asymptomatic to severe pneumonia. Dissemination can affect skin, bones, and meninges.

Alternate Names:

Valley fever; coccidioidomycosis

Fevers, Viral Hemorrhagic

Severe febrile illnesses caused by various RNA viruses (Ebola, Marburg, dengue, Lassa, Rift Valley fever) transmitted by arthropods or rodents. Characterized by fever, hemorrhage, and multi-organ failure. High mortality rates for some agents. Prevention focuses on vector control and infection prevention.

Alternate Names:

Viral hemorrhagic fevers

Fexofenadine

A second-generation (non-sedating) H1 antihistamine used for allergic rhinitis and chronic urticaria. Peripherally selective with minimal sedation and anticholinergic effects; no significant CNS penetration. Adverse effects: headache, dizziness, and dry mouth.

Fiber

Dietary fiber is nondigestible carbohydrate and plant material that reaches the colon; soluble and insoluble fiber affect stool bulk, bowel transit, glycaemic control, and blood lipids. cholesterol, blood glucose. Insoluble (whole grains, vegetables) adds bulk, prevents constipation. Recommended: 25-30 g/day.

Fiber Optic

A technology that transmits light through flexible glass or plastic fibers. It is used in medical endoscopes and other imaging or examination devices.

Fibrates

A class of lipid-lowering medicines that activate PPAR-alpha and reduce triglycerides while often increasing HDL cholesterol; adverse effects include liver, muscle, and gallbladder problems.

Fibre

A threadlike structure or material; in nutrition, dietary fiber is nondigestible plant carbohydrate that supports bowel function and can influence glucose and cholesterol.

Fibrillation

Rapid, irregular, and uncoordinated contraction of muscle fibers. Atrial fibrillation causes an irregular atrial rhythm; ventricular fibrillation is a cardiac arrest rhythm requiring immediate resuscitation.

Fibrin

A threadlike protein that forms the structural mesh of a blood clot.

Fibrin Degradation Products Blood Test

A blood test measuring fragments released when fibrin is broken down; increased levels may occur with widespread coagulation and fibrinolysis, including disseminated intravascular coagulation, but are nonspecific.

Fibrin Sealant

A biological adhesive containing fibrinogen and thrombin that forms a fibrin clot when applied to tissue. It may help control bleeding or seal selected surgical surfaces.

Fibrinogen

A soluble plasma protein, also called coagulation factor I, that is converted to fibrin during blood clot formation. Its concentration may increase with inflammation and decrease with severe liver disease or consumption coagulopathy.

Alternate Names:

coagulation factor I, that is converted to fibrin during blood clot formation.

Fibrinogen Assay

A fibrinogen assay measures plasma fibrinogen concentration or function, commonly by the Clauss method; low results may occur in consumption, severe liver disease, or inherited deficiency, while inflammation can increase the level.

Fibrinogen Blood Test

A blood test measuring fibrinogen, a liver-produced protein converted to fibrin during clot formation; it helps evaluate bleeding, excessive clotting, liver disease, and disseminated intravascular coagulation.

Fibrinogen Decreased

Decreased fibrinogen is a low plasma fibrinogen level that can impair clot formation and cause bleeding; causes include disseminated intravascular coagulation, severe hepatic failure, massive transfusion, and congenital deficiency.

Fibrinoid Necrosis

A specialized pattern of vascular necrosis characterized by the deposition of immune complexes, fibrin, and plasma proteins within the walls of blood vessels, creating an amorphous, bright pink, smudgy eosinophilic appearance on hematoxylin and eosin staining.

Fibrinolysis

The enzymatic breakdown of fibrin within a blood clot, principally mediated by plasmin generated from plasminogen. It limits clot extension and supports restoration of vascular patency; excessive fibrinolysis can cause bleeding, whereas impaired fibrinolysis can promote thrombosis.

Fibrinolysis - Primary Or Secondary

Breakdown of fibrin clots by plasmin; primary fibrinolysis occurs without preceding widespread coagulation activation, whereas secondary fibrinolysis follows clot formation and coagulation activation.

Fibrinolytic Agent

A fibrinolytic agent is a thrombolytic drug, such as alteplase or tenecteplase, that converts plasminogen to plasmin and dissolves fibrin clots; major bleeding,

including intracranial haemorrhage, is the principal danger.

Fibrinolytic Therapy

Treatment with medications that dissolve blood clots by activating plasminogen to plasmin, which breaks down fibrin. Common agents include tissue plasminogen activator (tPA), streptokinase, and urokinase. Used for acute ischemic stroke, pulmonary embolism, and myocardial infarction. Risks include bleeding complications.

Alternate Names:

Thrombolytic therapy

Fibrinopeptide A Blood Test

A blood test measuring fibrinopeptide A released when thrombin converts fibrinogen to fibrin; it may indicate thrombin generation but is not routinely used alone for diagnosis.

Fibrinous Pericarditis

Inflammation of the pericardium characterized by deposition of fibrin on the pericardial surfaces, giving them a rough, shaggy appearance described as bread-and-butter pericarditis.

Fibroadenoma

The most common benign tumor of the female breast, a proliferation of stromal and epithelial elements presenting as a firm, rubbery, smooth, mobile, painless mass, most often in women aged fifteen to thirty. The mass slips easily under the fingers (breast mouse), and ultrasonography shows a well-defined oval hypoechoic lesion.

Fibroadenoma Of The Breast

A common benign breast tumor composed of glandular and fibrous tissue, usually smooth, mobile, and painless, especially in adolescents and young adults.

Fibroblast

A connective-tissue cell that produces collagen and other extracellular-matrix components.

Fibrocartilage

Strong cartilage containing dense bundles of type I collagen, adapted to resist compression and tension. It forms intervertebral discs, the pubic symphysis, and menisci.

Fibrocystic Breast Changes

A benign condition characterized by lumpy, painful breast tissue, most common in women aged 20-50, driven by hormonal fluctuations during the menstrual cycle (estrogen and progesterone). Microscopic features include stromal fibrosis, cyst formation (macrocyts and microcyts), and adenosis (increased glandular tissue).

Fibrocystic Breast Condition

Fibrocystic breast change causes lumpiness, tenderness, or cysts that often vary with the menstrual cycle. It is usually benign, but a new persistent or changing mass, skin or nipple change, or bloody discharge needs clinical and imaging assessment.

Fibrocystic Breast Disease

Fibrocystic Breast Disease is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fibrocystic Breasts

Benign breast changes involving fibrosis and cyst formation, often producing lumpy or tender breasts that fluctuate with the menstrual cycle. They are common and are not themselves a premalignant condition.

Fibrodysplasia Ossificans Progressiva

An extremely rare, catastrophically disabling autosomal dominant genetic disorder caused in over 95% of patients by a recurrent, identical heterozygous activating missense mutation (c.617G>A, p.Arg206His) in the *ACVR1* gene on chromosome 2q24.1, which encodes the activin A receptor type I/ALK2 (a bone morphogenetic protein [BMP] type I receptor). Characterized by congenital malformation of the great toes (bilateral hallux valgus and digital monophalangism) evident at birth, followed in early

childhood by recurrent, painful inflammatory soft tissue swellings (flare-ups) that undergo progressive heterotopic endochondral ossification, irreversibly converting skeletal muscles, tendons, ligaments, and fascia into ribbons and plates of mature heterotopic bone, leading to complete ankylosis of the spine and peripheral joints (“stone man syndrome”).

Alternate Names:

FOP; Myositis Ossificans Progressiva; Stone Man Syndrome

Fibroepithelial Polyp Of The Vulva

A benign polypoid lesion composed of stellate stromal cells beneath squamous epithelium, often arising in the vulvovaginal region during reproductive years or pregnancy.

Fibrofolliculoma

A benign perifollicular skin tumor, often multiple on the face, that is characteristic of Birt-Hogg-Dubé syndrome when associated with other findings.

Fibroid

A benign smooth-muscle tumor, most often arising in the uterus.

Alternate Names:

Uterine leiomyoma

Fibroid Tumor

A benign smooth-muscle tumor of the uterus, also called a leiomyoma, which may cause heavy bleeding, pelvic pressure, pain, infertility, or no symptoms.

Fibroids

Fibroids is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fibroids, Cysts, And Polyps

Benign growths that can develop in the reproductive organs. Fibroids are smooth muscle tumors of the uterus; cysts are fluid-filled sacs that commonly form on the ovaries; and polyps are tissue overgrowths in the endometrium or cervix.

Fibroids, Uterine

Benign smooth muscle tumors of the uterus, also called leiomyomas. Common cause of abnormal uterine bleeding, pelvic pain, and bulk symptoms. May affect fertility. Treatment ranges from observation to medical management to surgical removal depending on symptoms and reproductive goals.

Alternate Names:

Uterine fibroids; leiomyomas

Fibrolipoma

A benign tumor composed of mature fat and abundant fibrous tissue, usually presenting as a slow-growing painless soft-tissue mass.

Fibroma

A benign cardiac tumour composed of fibroblasts and collagen, usually arising within the ventricular myocardium or interventricular septum in children. It may cause ventricular obstruction or clinically important arrhythmias; echocardiography and cardiac MRI help define its location and extent.

Fibroma Of Tendon Sheath

A benign, slow-growing fibroblastic tumor attached to a tendon sheath, most often affecting the fingers, hand, or wrist. It presents as a firm painless mass and may recur after excision.

Fibromatoses

A group of locally infiltrative fibroblastic proliferations that lack metastatic potential but may recur and damage adjacent structures; superficial and deep types are distinguished clinically and pathologically.

Fibromatosis

A proliferation of fibroblasts and myofibroblasts that is usually nonmetastatic but can infiltrate locally and recur, as in desmoid-type fibromatosis.

Fibromuscular Dysplasia

A non-atherosclerotic, non-inflammatory disease of medium-sized arteries, often involving the renal or carotid arteries. Imaging may show alternating

narrowing and dilation. Symptoms, vascular complications, and treatment depend on the arteries involved and the individual clinical situation.

Fibromyalgi

Fibromyalgi is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fibromyalgia

Fibromyalgia is a chronic pain-amplification disorder with widespread pain, fatigue, unrefreshing sleep, and cognitive symptoms without a single defining tissue lesion. months), fatigue, sleep disturbance, cognitive dysfunction (fibro fog), and somatic symptoms, affecting 2-4% of population (female:male 9:1).

Fibromyalgia Syndrome

A chronic disorder characterized by widespread musculoskeletal pain, fatigue, unrefreshing sleep, and cognitive symptoms. Diagnosis is clinical based on widespread pain index and symptom severity scale. Treatment includes exercise, cognitive behavioral therapy, and medications.

Alternate Names:

Fibromyalgia

Fibromyxosarcoma

A rare malignant soft-tissue tumor containing fibrous and myxoid tissue; diagnosis requires pathology and treatment is guided by location, grade, and stage.

Fibronectin

An extracellular-matrix glycoprotein that supports cell adhesion, migration, wound healing, and tissue organization and circulates in plasma.

Fibroproliferation

Abnormal proliferation of fibroblasts and deposition of connective tissue that contributes to scarring and fibrosis, including in the lungs.

Fibrosarcoma

A malignant tumor of fibroblasts, a soft tissue sarcoma of adulthood or infancy, arising in deep soft tissues, bones, or paranasal sinuses.

FibroScan

Non-invasive transient elastography technique measuring liver stiffness to assess hepatic fibrosis. Values correlate with fibrosis stage with higher values indicating more advanced fibrosis. The technique avoids the need for liver biopsy in many patients. Results may be unreliable in obesity, ascites, and narrow intercostal spaces.

Fibrosing Alopecia

A group of primary cicatricial (scarring) alopecias in which chronic inflammation leads to permanent destruction of hair follicles and replacement by fibrous tract remnants. It includes conditions such as lichen planopilaris, frontal fibrosing alopecia, and central centrifugal cicatricial alopecia.

Fibrosis

The excessive deposition of extracellular matrix components, primarily collagen types I and III, in response to chronic tissue injury. It represents a pathological wound healing response that leads to organ scarring and dysfunction.

Fibrosis, Interstitial Pulmonary

Chronic progressive scarring of lung interstitium, leading to impaired gas exchange. Causes include connective tissue diseases, medications, radiation, and idiopathic sources. Presents with dyspnea, dry cough, and restrictive physiology on pulmonary function tests.

Alternate Names:

Interstitial lung disease

Fibrosis, Pulmonary

Irreversible scarring of lung tissue with thickening of alveolar walls, reducing lung compliance and gas exchange. Idiopathic pulmonary fibrosis is the most common form. Presents with progressive dyspnea, dry cough, and characteristic honeycomb pattern on CT.

Alternate Names:

Pulmonary fibrosis

Fibrositis

Fibrositis is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fibrous Dysplasia

A benign skeletal disorder in which normal bone is replaced by fibro-osseous tissue, producing weakened or misshapen bone. It may affect one bone or multiple bones and can cause pain, deformity, pathological fracture, or nerve compression; imaging often shows a characteristic ground-glass appearance.

Fibrous Plaque

A raised arterial lesion containing fibrous tissue and lipid that can narrow the vessel.

Alternate Names:

Atherosclerotic plaque

Fibula

Slender lateral leg bone, not weight-bearing, serving for muscle attachment and forming the lateral malleolus. Head provides biceps femoris and lateral collateral ligament attachment; the common peroneal nerve wraps around its neck, vulnerable to injury. Lateral malleolus extends more distally and posteriorly than the medial malleolus.

Fick Principle

A fundamental physiological principle governing cardiovascular and pulmonary hemodynamics, stating that the rate of uptake or release of a substance by an organ is equal to the product of blood flow through that organ and the arteriovenous concentration difference of the substance across the organ. Applied during right heart catheterization as the gold standard method to calculate cardiac output (CO): $CO = VO_2 / (C_aO_2 - C_vO_2)$, where VO_2 represents whole-body oxygen consumption (measured directly by spirometric gas analysis or estimated from body surface area), and C_aO_2 and C_vO_2 represent arterial and mixed venous oxygen contents,

respectively.

Alternate Names:

Fick Method; Fick Cardiac Output Determination

Ficusin

A naturally occurring plant compound studied in chemistry and pharmacology; it is not a standard treatment for a specific disease.

Fiducial Markers

fiducial markers is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Field Fever

Field Fever is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Field Of View

The extent of anatomy or space captured in one image or measurement.

Fields Disease

An extremely rare, apparently congenital neuromuscular disorder reported in identical twins, characterized by progressive muscle degeneration and involuntary movements; its cause and natural history remain uncertain.

Fifth Disease

Erythema infectiosum caused by Parvovirus B19. Characterized by bright red cheeks (slapped cheek appearance) followed by a lacy, reticulated rash on the trunk and extremities. Mild febrile illness in children. Concern in pregnancy (hydrops fetalis risk) and immunocompromised patients (pure red cell aplasia). Diagnosis by serology.

Alternate Names:

Erythema Infectiosum (Fifth Disease)

FIGO Leiomyoma Classification

A universally adopted 9-category system established

by the International Federation of Gynecology and Obstetrics to standardize the anatomical subtyping of uterine fibroids based on their transmural location and relation to the endometrium and serosa: Sub-mucosal (Type 0: pedunculated intracavitary; Type 1: < 50% intramural; Type 2: $\geq$ 50% intramural); Other/Intramural (Type 3: 100% intramural contact with endometrium; Type 4: intramural entirely; Type 5: subserosal $\geq$ 50% intramural; Type 6: subserosal < 50% intramural; Type 7: pedunculated subserosal); and Hybrid/Cervical (Type 8).

Filamentous Fungus

A fungus that grows as thread-like structures called hyphae, forming a network known as mycelium.

Filaria Bancrofti

Filaria bancrofti is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Filaria Demarquayi

filaria demarquayi is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Filarial Elephantiasis

filarial elephantiasis is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Filarial Elephantiasis

Severe swelling and thickening of tissues caused by lymphatic obstruction from filarial infection.

Filariasis

A group of tissue-dwelling nematode infections transmitted by mosquitoes, blackflies, deerflies, or other vectors. Lymphatic filariasis is caused mainly by *Wuchereria bancrofti*, *Brugia malayi*, and *Brugia timori*; adult worms in lymphatic vessels can cause recurrent inflammation, lymphedema, hydrocele, and elephantiasis. Other filarial species cause skin,

eye, or subcutaneous disease, so diagnosis and treatment depend on the species and anatomic site.

Filial

Relating to a son or daughter, or to the relationship between offspring and a parent; in genetics, a filial generation is a generation descended from a parental generation.

Filler

A material injected or placed beneath the skin or in tissue to restore volume, smooth lines, alter contours, or replace lost structure; effects and risks depend on the material and site.

Filler Excipient

filler excipient is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Filopodium

A slender, actin-rich projection from a cell surface that senses the environment and helps cells migrate, adhere, or establish neuronal connections.

Filoviridae

A family of enveloped RNA viruses that includes the genera *Ebolavirus* and *Marburgvirus*.

Filoviruses

A family of negative-sense RNA viruses that includes Ebola and Marburg viruses. These viruses cause severe hemorrhagic fever with high mortality rates. The virus is transmitted through contact with infected body fluids. The incubation period is 2-21 days. Symptoms include fever, headache, myalgia, and hemorrhage.

Filter (Vascular)

A device placed in a large vein, usually the inferior vena cava, to trap clots and reduce the risk of pulmonary embolism in selected patients.

Alternate Names:

Vascular

Filterable Virus

An obsolete historical term for an infectious agent small enough to pass through filters that retain bacteria; it reflects the development of early virology rather than a modern virus classification.

Filtered Back Projection

An image-reconstruction method that mathematically back-projects measured data after applying a filter. It is fast but can be more sensitive to noise and artifacts than iterative methods.

Filtering Operations

Processes that separate particles, cells, molecules, or fluids by size or other properties using a filter, membrane, or computational rule.

Filtering Procedure

An operation that creates an alternate drainage pathway to lower pressure inside the eye, usually in glaucoma.

Filtration

Separation of particles or solutes from a fluid by passage through a porous material; kidney glomerular filtration removes water and small solutes from blood.

Filtration Fraction

The fraction of renal plasma flow that is filtered at the glomerulus, calculated as GFR divided by renal plasma flow (RPF). Normal GFR is approximately 125 mL/min and RPF is approximately 625 mL/min, yielding a filtration fraction of approximately 0.20 (20 percent).

Filum Terminale

Filum Terminale is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Fimbria

Fimbria is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Fimbriae

The finger-like projections at the distal end of each fallopian tube that sweep over the ovary during ovulation to capture the released egg and guide it into the fallopian tube. The fimbriae are lined with ciliated epithelium that creates a current to draw the egg into the tube.

Fimbriae Proteins

Bacterial surface proteins that form fimbriae or pili and mediate attachment to host cells, surfaces, or biofilms.

Fine Crackles

High-pitched, brief, discontinuous sounds heard at the end of inspiration, caused by opening of small airways. They are characteristic of ILD, pulmonary fibrosis, and heart failure. They are also called Velcro crackles due to their resemblance to the sound of Velcro being pulled apart.

Alternate Names:

Velcro crackles due to their resemblance to the sound of.

Fine Motor Control

The coordinated use of small muscles, especially in the hands and fingers, for precise tasks such as writing, buttoning, and manipulating objects.

Fine Motor Skills

The ability to use small muscles (hands/fingers) for precise movements, developing with age: reaching (3-4 months), grasping (5-6 months), pincer grasp (9-12 months), self-feeding (12-18 months), building towers/scribbling (18-24 months), drawing shapes and using utensils (3-5 years), and handwriting.

Fine Needle Aspiration

A percutaneous sampling technique in which a 22 to 25-gauge needle, often ultrasound- or palpation-guided, aspirates cellular material from a palpable or imaging-detected lesion for cytologic examination, with minimal trauma and lower morbidity than core or open biopsy.

Fine Needle Aspiration Of The Thyroid

A procedure using a thin needle to remove thyroid cells for cytologic examination, usually guided by ultrasound to evaluate a nodule.

Finegoldia

A genus of anaerobic gram-positive bacteria that can cause skin, soft-tissue, bone, joint, or bloodstream infections.

Finegoldia Magna

An anaerobic gram-positive bacterium found on skin and mucosal surfaces that can cause wound and other opportunistic infections.

Finger

One of the digits of the hand, consisting of phalangeal bones, joints, tendons, nerves, blood vessels, and soft tissue that support grasping and fine movement.

Finger Dislocation

Displacement of the bones forming a finger joint, usually caused by a direct blow, fall, or forced bending. Pain, swelling, deformity, and loss of movement are typical. The joint should be assessed for fracture, tendon injury, and neurovascular compromise before reduction and splinting.

Finger Flexors

Muscles and tendons that bend the fingers at the metacarpophalangeal and interphalangeal joints, primarily supplied by the median and ulnar nerves.

Finger Injury

Damage to a finger's bone, joint, tendon, ligament, nerve, vessel, nail, or skin caused by trauma and assessed for deformity, circulation, sensation, and function.

Finger Joint

A joint between bones of a finger, including the metacarpophalangeal and interphalangeal joints, permitting flexion, extension, and limited side-to-side movement.

Finger Pain

Pain in one or more fingers caused by injury, arthritis, tendon disease, nerve compression, infection, circulation problems, or inflammatory conditions.

Finger Stick

A small puncture of a fingertip used to obtain capillary blood, commonly for point-of-care glucose or other blood testing.

Fingernail

The hard keratin plate covering the dorsal end of a finger, produced by the nail matrix and protecting the fingertip.

Fingers That Change Color

Fingers that change color concerns fingers that change color, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Finkelstein Test

A wrist examination that stresses the tendons on the thumb side of the wrist. Pain near the radial styloid may support a diagnosis of de Quervain tenosynovitis but is not specific by itself.

Fire Ants

Stinging ants whose venom causes immediate burning pain and itchy sterile pustules; numerous stings can produce systemic allergic or toxic reactions.

Firesetting Behavior

Deliberate setting of fires, ranging from isolated behavior to recurrent fire-setting associated with impulse-control, psychiatric, developmental, substance-use, or social problems.

Firmicutes

A large bacterial group that includes many low-GC gram-positive bacteria, although modern genetic classification has changed its boundaries.

First Aid

Immediate basic care given to a person who is in-

jured or suddenly ill before definitive medical treatment is available. Priorities include scene safety, emergency activation, airway, breathing, circulation, and control of serious bleeding.

First Aid Kit

A collection of supplies for treating minor injuries and providing initial care until professional medical help is available.

First Stage Of Labor

The period from the onset of regular labor to complete cervical dilation. It includes a latent phase and an active phase, with progress varying widely among individuals. Assessment considers cervical change, contraction pattern, maternal and fetal condition, and the overall clinical course.

First Trimester

The period from conception to 12 weeks gestational age. Characterized by rapid embryonic organogenesis, rising hCG levels, and establishment of the placenta. Common symptoms include nausea, vomiting, fatigue, and breast tenderness. Dating ultrasound performed to confirm gestational age. Risk of miscarriage highest during this trimester.

First-Degree AV Block

A conduction delay in which every atrial impulse reaches the ventricles but with a prolonged PR interval (>0.20 seconds). It is caused by delayed conduction through the AV node or His-Purkinje system. First-degree block is usually benign and asymptomatic.

First-Degree Burn

A superficial burn affecting only the epidermis: erythema, pain, and dryness without blistering (e.g., mild sunburn); heals in 3-7 days without scarring. Assessment: size (percent BSA—rule of nines), depth, and cause (thermal, chemical, electrical).

First-Degree Heart Block

Delay in conduction from the atria to the ventricles, shown on ECG by a prolonged but constant PR interval greater than 200 milliseconds. It is often

asymptomatic, but the cause, medications, associated conduction disease, and progression risk should be assessed.

First-Episode Psychosis

The first clinically recognized episode of psychosis, regardless of eventual diagnosis. It requires urgent assessment for substance exposure, medication effects, mood symptoms, neurologic disease, systemic illness, safety risk, and functional decline.

First-Pass Effect

The metabolism of a drug by the liver and intestinal wall before it reaches the systemic circulation after oral administration.

First-Pass Metabolism

The metabolism of a drug by the liver before it reaches systemic circulation after oral administration. Drugs with high first-pass metabolism (e.g., morphine, nitroglycerin, lidocaine) have low oral bioavailability and may require alternative routes (sublingual, transdermal, IV).

Fish Oil

fish oil is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Fish Poisoning

Fish Poisoning is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fish Tapeworm Infection

Intestinal infection with a *Diphyllobothrium* or related fish tapeworm acquired from raw or undercooked fish; it can cause abdominal symptoms and vitamin B12 deficiency.

Fish Venom

fish venom is a diagnostic, research, imaging, or measurement term used to assess a condition or

question; interpretation depends on the indication, method, and result.

Fish-Handler'S Disease

Fish-Handler's Disease is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fisher'S Syndrome

Fisher's Syndrome is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fishhook Removal

Fishhook removal is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Fissure

A narrow crack, groove, or cleft in a body structure; an anal fissure is a painful tear in the lining of the anal canal.

Fissured Tongue

A usually harmless condition in which grooves or cracks develop on the surface of the tongue. Food and debris may collect in the fissures, causing irritation or odor. Gentle brushing of the tongue and routine oral care help prevent symptoms.

Fistula

A fistula is an abnormal connection between two epithelialized surfaces, such as bowel and skin or two organs, caused by inflammation, infection, surgery, trauma, or congenital development. congenital development.

Fistula (Pathology)

An abnormal epithelial-lined connection between two body surfaces or organs that normally do not communicate. Fistulas may be congenital, such as tracheoesophageal fistula, or acquired, resulting

from inflammation, infection, surgery, trauma, or malignancy.

Alternate Names:

Pathology

Fistula, Anal

An abnormal tunnel connecting the anal canal or rectum with the nearby skin, often developing after an anorectal abscess and causing drainage or recurrent infection.

Fistula, Arteriovenous

Alternate word order; see *Arteriovenous Fistula*.

Fistula-In-Ano

Abnormal epithelialized tract connecting the anal canal to the perianal skin, representing a common and distressing complication of Crohn disease and also occurring following perianal abscess drainage or from cryptoglandular infection.

Fit

Fit is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Fits (Seizures)

A common term for episodes caused by abnormal electrical activity in the brain, which may produce altered awareness, abnormal movements, sensory symptoms, or loss of consciousness.

Alternate Names:

Seizures

Fitz-Hugh-Curtis Syndrome

Perihepatitis associated with pelvic inflammatory disease caused by ascension of *Neisseria gonorrhoeae* or *Chlamydia trachomatis* from the pelvis to the liver capsule along the right paracolic gutter. Presents with right upper quadrant pain, fever, and tender hepatomegaly often in young sexually active women.

Five-Year Survival Rate

The proportion of patients alive five years after

diagnosis or treatment; a standard benchmark for comparing cancer outcomes.

Fixation

Stabilization of tissue or bone in a desired position, using methods such as sutures, wires, screws, plates, or external devices.

Fixation, Internal

Surgical stabilization of a fractured or unstable bone with implanted devices such as plates, screws, rods, or pins.

Fixative

fixative is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Fixed Joint

A joint with little or no movement, often because bones are joined by dense tissue.

Alternate Names:

Synarthrosis

Fixed Macrophage

A resident tissue macrophage that remains in a particular organ rather than circulating, where it removes debris and contributes to immune surveillance and repair.

Flaccid

Relaxed, soft, or lacking normal muscle tone or erection.

Flagellin

The structural protein forming bacterial flagella and an activator of innate immunity through Toll-like receptor 5.

Flagellum

A long whip-like cellular or bacterial appendage that produces movement or helps a microorganism swim.

Flail Chest

A clinical condition caused by three or more consecutive ribs fractured in two or more places, creating a free-floating segment of chest wall. The segment moves paradoxically inward during inspiration and outward during expiration. It causes severe pain, impaired ventilation, and may require mechanical ventilation.

Flakka

A street name for synthetic cathinone stimulants, particularly alpha-PVP. Intoxication may cause extreme agitation, paranoia, hallucinations, seizures, dangerous hyperthermia, muscle breakdown, and cardiovascular instability. Severe agitation, altered consciousness, high temperature, or chest pain is a medical emergency.

Flanged Needle

A needle with a widened rim or flange that helps position, stabilize, or connect it during a procedure.

Flank

The side of the body between the lower ribs and the hip, overlying part of the abdomen and back; flank pain may arise from the kidney, muscles, spine, or other structures.

Flank Pain

Pain arising from the lateral back between the ribs and iliac crest, often reflecting disease of the kidney, ureter, musculoskeletal structures, or adjacent abdominal organs. Associated fever, haematuria, vomiting, urinary symptoms, trauma, or hypotension helps distinguish infection, obstruction, stones, injury, and other causes.

Flap Operation

Flap Operation is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Flare

A temporary worsening or return of symptoms or inflammation in a disease that may otherwise be stable or controlled. Flares can have many triggers

and their severity varies; sudden severe symptoms may need urgent assessment.

Alternate Names:

Exacerbation

Flare-Up

A sudden or gradual increase in symptoms or inflammatory activity of a chronic disease after a period of stability or lower activity.

Flashes

Perception of flickering light or lightning-like streaks from mechanoelectrical stimulation of the retina rather than an external source. The leading cause is posterior vitreous detachment, often with floaters; retinal tears and detachment, migraine aura, and optic neuritis also produce flashes.

Flashing

The perception of brief flashes or streaks of light without an external light source, often called photopsia; new flashes with floaters or visual-field loss require urgent eye assessment.

Flat Affect

Markedly reduced outward expression of emotion, such as minimal facial expression, gestures, or vocal variation. It is an examination finding rather than a diagnosis and does not necessarily mean that the person lacks emotional experience; it may occur in schizophrenia, depression, neurologic disease, medication effects, or other conditions.

Flat Bones

Flat bones concerns flat bones, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Flat Feet

Reduction or absence of the medial longitudinal arch, causing the sole to approach or contact the ground during standing. Flexible flat feet are common and often asymptomatic; rigid or painful flat feet may result from structural deformity, tendon dysfunction, or arthritis.

Flatfoot

A reduction or collapse of the medial longitudinal arch of the foot, with increased contact of the sole with the ground. It may be flexible and asymptomatic or rigid and painful; causes include normal development, ligamentous laxity, tendon dysfunction, trauma, inflammatory disease, and structural deformity.

Flatulence

Passage of intestinal gas through the anus. It reflects swallowed air and bacterial fermentation of undigested carbohydrates and may be increased by dietary factors, constipation, malabsorption, altered gut motility, or functional gastrointestinal disorders; persistent symptoms require assessment for associated disease.

Flatus

Gas expelled from the gastrointestinal tract through the anus. Biochemically composed of greater than 99% odorless gases (N₂, H₂, CO₂, CH₄, and O₂), generated via aerophagia (swallowed air) and anaerobic bacterial fermentation of unabsorbed dietary carbohydrates (such as FODMAPs and resistant starches) in the colon. Odor is attributable to trace concentrations (< 1%) of volatile sulfur compounds, predominantly hydrogen sulfide (H₂S) and methanethiol. In postoperative surgical monitoring, the spontaneous passage of flatus is the cardinal clinical marker denoting the resolution of postoperative paralytic ileus and return of coordinated gastrointestinal motility. Conversely, obstipation (the complete cessation of both flatus and stool) is a pathognomonic finding of complete mechanical bowel obstruction.

Alternate Names:

Intestinal gas

Flavin

flavin is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Flavin Adenine Dinucleotide

A flavin-derived redox coenzyme, usually bound tightly to enzymes as a prosthetic group. FAD accepts two hydrogen atoms to form FADH₂, which donates electrons to the respiratory chain through selected dehydrogenases.

Alternate Names:

Flavin Adenine Dinucleotide

Flaviviridae

A family of mainly enveloped, single-stranded RNA viruses that includes hepatitis C virus, dengue virus, yellow fever virus, and Zika virus.

Flavivirus

A genus of RNA viruses that includes dengue, yellow fever, West Nile, Japanese encephalitis, and Zika viruses.

Flaviviruses

A genus of RNA viruses in the Flaviviridae family, transmitted by mosquitoes or ticks. Includes dengue, Zika, yellow fever, and West Nile virus. Many cause significant morbidity and mortality worldwide. Prevention relies on vector control and vaccination where available.

Flavobacteriaceae

A family of mainly gram-negative bacteria found in soil, water, and other environments; some species cause opportunistic infections.

Flavobacterium

A genus of gram-negative environmental bacteria; a few species can cause opportunistic infections in humans or animals.

Flavodoxin

Flavodoxin is a small flavin-containing electron-transfer protein, usually containing FMN, that participates in redox reactions in some bacteria, algae, and other organisms.

Flavonifractor

A genus of anaerobic intestinal bacteria, includ-

ing *Flavonifractor plautii*, whose effects on human health are still being studied.

Flavonoid

flavonoid is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Flavonoids

Plant polyphenols found in many fruits, vegetables, teas, and other foods.

Flavonol

flavonol is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Flavopiridol

An investigational cyclin-dependent kinase inhibitor studied as an anticancer medicine; it is not a routine standard treatment for most cancers.

Flavor Enhancer

A flavour enhancer is a food additive that intensifies existing taste, often savoury or umami flavour, without serving mainly as a sweetener or preservative; sensitivity and exposure depend on the substance and dose.

Flavoxate

A urinary antispasmodic used for symptomatic relief of bladder spasm or urgency; it does not treat the underlying cause of infection or obstruction.

Flaxseed Extract

A preparation derived from flaxseed containing variable amounts of alpha-linolenic acid, lignans, and soluble fiber; effects and interactions depend on formulation and dose.

Flea

A small wingless blood-feeding insect that can irritate skin and transmit infections between animals

or people.

Flea Bites

Small itchy or painful skin reactions caused by flea feeding, sometimes accompanied by papules, hypersensitivity, or secondary infection.

Flea Bites In Humans

Flea bites cause small itchy papules, often clustered around the ankles or lower legs, and may trigger allergic reactions. Fleas can transmit infections in some settings; control requires treating pets and the environment as well as symptoms.

Alternate Names:

Bites Flea (Flea Bites In Humans) Ctenocephalides FelFlea Bite (Flea Bites In Humans)

Fleas

Small wingless blood-feeding insects that can cause bites, allergic reactions, anemia in heavy infestations, and disease transmission.

Flecainide

A class IC sodium channel blocker that slows conduction velocity with little effect on refractoriness. Used for suppression of supraventricular arrhythmias (atrial fibrillation in patients without structural heart disease—'pill-in-the-pocket', paroxysmal SVT) and ventricular arrhythmias.

Flesh-Eating Bug

Flesh-Eating Bug is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fleshy Tab

A small projection of soft tissue, often a skin tag or mucosal appendage, whose cause and significance depend on its site and appearance.

Flex

To bend a joint and decrease the angle between its bones.

Alternate Names:

Flexion

Flexibility

The range through which a joint or group of joints can move, influenced by muscles, tendons, ligaments, joint structure, and pain.

Flexible Nasolaryngoscopy

Office examination of the nasal cavity, nasopharynx, pharynx, and larynx using a flexible fiberoptic or digital endoscope passed through the nose. It evaluates voice change, swallowing symptoms, nasal obstruction, lesions, vocal-fold movement, and airway anatomy.

Flexible Sigmoidoscopy

Endoscopic examination of the rectum and sigmoid colon (up to 60 cm). Indications: colorectal cancer screening (q5-10 years, combined with FIT), evaluation of rectal bleeding, chronic diarrhea, and surveillance of known distal colonic disease (ulcerative colitis).

Flexion

Movement that decreases the angle between two bones or bends a joint, generally in the sagittal plane.

Flexor

Flexor is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Flexor Digitorum Profundus

Deep anterior forearm muscle with ulnar and humeroulnar heads, inserting on distal phalanges bases of digits 2-5. Medial half (digits 4-5) innervated by ulnar nerve; lateral half (digits 2-3) by anterior interosseous nerve (C8-T1). Flexes DIP, PIP, MCP joints, and wrist.

Flexor Digitorum Superficialis

Large anterior forearm muscle with humeroulnar, radial, and ulnar heads, inserting on middle phalanges of digits 2-5. Innervated by median nerve

(C8-T1). Flexes PIP joints, MCP joints, and wrist. Important for median nerve function assessment. Tendons pass through carpal tunnel, susceptible to stenosing tenosynovitis (trigger finger).

Flexural

Relating to a skin fold or the inner surface of a joint, such as the antecubital, popliteal, axillary, groin, or inframammary fold.

Flibanserin

A centrally acting medicine used for acquired, generalized hypoactive sexual desire disorder in selected premenopausal women; it can cause hypotension, syncope, and sedation, especially with alcohol or interacting drugs.

Flight-Or-Fight Response

The coordinated stress response that prepares the body to confront or escape a perceived threat.

Alternate Names:

Fight-or-flight response; stress response

Floater

A spot, strand, or cobweb-like shape drifting across the field of vision, usually caused by opacities in the eye's vitreous; sudden numerous floaters may signal retinal disease.

Floaters

Mobile spots, threads, cobwebs, or rings in the visual field caused by vitreous opacities casting shadows on the retina. Most arise from posterior vitreous detachment, an age-related separation of the vitreous gel from the retina; hemorrhage, vitritis, and asteroid hyalosis are other causes.

Floating Kidney

Excessive downward movement of a kidney when standing, also called nephroptosis; it may be asymptomatic or cause positional pain, urinary symptoms, or rarely impaired drainage.

Floating Rib

One of the eleventh or twelfth pairs of ribs, attached

posteriorly to the spine but lacking an attachment to the sternum through cartilage.

Floating Ribs

Ribs 11-12 with short costal cartilages terminating in abdominal wall musculature, not articulating with the sternum or other ribs. Most mobile and least protected, susceptible to fracture from direct trauma. Fractures may injure underlying kidneys, requiring hematuria and flank tenderness assessment in trauma.

Flocculation

The aggregation of suspended particles into larger loose clumps, used in laboratory separation, water treatment, and some biologic or pharmaceutical processes.

Floor Of The Mouth Cancer

Cancer that develops in the tissues beneath the tongue.

Alternate Names:

Cancer, Floor Of The Mouth

Floppy Baby Syndrome

A descriptive term for infant hypotonia, characterized by reduced muscle tone and a limp posture; causes range from central nervous-system disorders to neuromuscular and genetic diseases and require evaluation.

Floppy Iris Syndrome

Floppy Iris Syndrome is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Floppy Valve Syndrome

Mitral valve prolapse: a condition in which the mitral valve leaflets bulge back into the left atrium during systole. Often asymptomatic but may cause palpitations, chest pain, or mitral regurgitation. Diagnosis by echocardiography.

Alternate Names:

Mitral valve prolapse

Flora

The microorganisms normally or currently living in a body site or environment; normal flora can protect against pathogens but may cause infection if displaced or if defenses are impaired.

Florida Flu

Florida Flu is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Flossing

Cleaning between teeth with dental floss or another interdental device to remove plaque and food debris from areas a toothbrush may not reach.

Flotation

A laboratory separation method that uses differences in density to concentrate particles, such as parasite eggs in a stool sample.

Flow

A state of highly focused, effortless involvement in an activity, often with reduced awareness of time.

Flow Cytometry

A laboratory technique that measures physical and molecular characteristics of cells or particles as they pass through a laser beam. It is used for cell counting, immunophenotyping, diagnosis of some leukemias and lymphomas, and assessment of selected immune disorders.

Flow-Volume Loop

Graphical representation of airflow during forced breathing. Obstructive: scooped expiratory limb. Restrictive: narrowed loop with preserved shape. Variable extrathoracic obstruction: flattening of inspiratory limb.

Flowmeter

flowmeter is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method,

and result.

Floxacin

A medicine or chemical compound called floxacin. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Floxuridine

A fluorinated pyrimidine antimetabolite that inhibits DNA synthesis and is used in selected regional or systemic chemotherapy protocols.

Flu

Influenza is an acute respiratory infection caused by influenza viruses, producing fever, cough, myalgia, headache, and fatigue with potential pulmonary or systemic complications. fever, cough, sore throat, headache, muscle aches, and fatigue. Complications may include pneumonia and worsening of underlying medical conditions.

Flu, Avian

Influenza A virus infection in birds, caused by H5N1, H7N9, and other subtypes. Primarily affects poultry but can occasionally infect humans through close contact with infected birds. High mortality rate in human cases. Pandemic potential if sustained human-to-human transmission develops.

Alternate Names:

Avian influenza; bird flu

Flu, Bird

Avian influenza: viral infection of birds that can occasionally spread to humans. H5N1 and H7N9 are the most concerning subtypes for human health. Characterized by severe respiratory illness with high mortality. Prevention includes poultry vaccination and avoiding contact with infected birds.

Alternate Names:

Avian influenza; bird flu

Flu, Swine

Influenza A virus infection in pigs that can occasionally spread to humans. The 2009 H1N1 pandemic strain originated from swine influenza viruses.

Symptoms resemble seasonal flu but can be severe in some individuals. Vaccination includes protection against swine flu strains.

Alternate Names:

H1N1 flu; swine influenza

Flucloxacillin

Flucloxacillin is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fluconazole

A medicine or chemical compound called fluconazole. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Flucytosine

An antifungal medicine converted within susceptible fungi to metabolites that inhibit nucleic-acid synthesis; it is used mainly with amphotericin for selected serious *Candida* or *Cryptococcus* infections.

Fludarabine

A purine-analogue chemotherapy medicine that inhibits DNA synthesis and is used for selected hematologic malignancies; prolonged immunosuppression and infection risk are important.

Fludarabine Phosphate

A purine-analog chemotherapy medicine used mainly for selected blood cancers; it can suppress bone-marrow and immune function.

Fludeoxyglucose F-18

Radiotracer used in PET imaging, analog of glucose labeled with fluorine-18. Taken up by cells via glucose transporters and phosphorylated by hexokinase, trapped intracellularly. Uptake reflects metabolic activity. Used for oncology (tumor detection, staging, response assessment), neurology (epilepsy, dementia), cardiology (viability).

Fludrocortisone

A synthetic mineralocorticoid used to replace al-

dosterone activity in adrenal insufficiency and selected orthostatic disorders; hypertension, edema, hypokalemia, and fluid retention may occur.

Fluid Around The Heart

Pericardial effusion: abnormal accumulation of fluid in the pericardial space. Can be caused by infection, malignancy, autoimmune disease, trauma, or metabolic conditions. Small effusions may be asymptomatic; large or rapid accumulations can cause cardiac tamponade.

Alternate Names:

Pericardial effusion

Fluid Balance

The relationship between fluid intake and output, regulated by the kidneys, hormones, thirst, and other mechanisms. Imbalance may cause dehydration, fluid overload, or electrolyte disturbances.

Fluid Imbalance

An abnormal excess or deficiency of body water or electrolytes such as sodium, potassium, calcium, or bicarbonate. It may result from dehydration, kidney or heart disease, endocrine disorders, gastrointestinal losses, or inappropriate fluid administration.

Fluid Management

A medical procedure, treatment, device, or care topic related to fluid management; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Fluid Overload

Excess total body fluid due to impaired renal sodium/water excretion or excessive intake, leading to edema, weight gain, hypertension, and pulmonary congestion. Causes: heart failure, renal failure (acute/chronic), nephrotic syndrome, cirrhosis, iatrogenic (IV fluids), and hypoalbuminemia.

Fluid Therapy

The intravenous administration of fluids to restore or maintain hydration, electrolyte balance, and circulatory volume. Types include isotonic (normal saline, lactated Ringer's), hypotonic, and hypertonic

solutions. Used for dehydration, shock, surgery, and electrolyte imbalances. Monitoring includes fluid balance, vital signs, and laboratory values.

Alternate Names:

Intravenous fluid therapy

Fluke

Fluke is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Flumazenil

A benzodiazepine-receptor antagonist that can reverse benzodiazepine sedation or respiratory depression in selected cases; it may precipitate seizures or withdrawal and is not appropriate for every mixed overdose.

Flunarizine

Flunarizine is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Flunisolide

Flunisolide is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fluocinolone

Fluocinolone is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fluocinolone Acetonide

A topical corticosteroid used to reduce inflammation and itching in selected dermatologic disorders; prolonged or extensive use can cause skin atrophy and systemic steroid effects.

Fluocinonide

A potent topical corticosteroid used for inflamma-

tory skin disease; application site, duration, and amount should be limited to reduce local and systemic adverse effects.

Fluocortolone

Fluocortolone is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fluorescein

fluorescein is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Fluorescein Angiography

An eye test in which fluorescein dye is injected into a vein and sequential photographs are taken of the retinal circulation. It helps identify leakage, blocked vessels, abnormal new vessels, and other retinal or choroidal abnormalities.

Fluorescein Eye Stain

An eye examination in which fluorescein dye highlights defects in the corneal or conjunctival surface under blue light, helping detect abrasions, ulcers, foreign-body injury, and some dry-eye changes.

Fluorescence

fluorescence is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Fluorescence In Situ Hybridization

A molecular cytogenetic technique using fluorescent DNA probes that hybridize to specific chromosomal regions. Faster than karyotype (24-48 hours, vs. 7-14 days).

Alternate Names:

Fluorescence In Situ Hybridization

Fluorescence In-Situ Hybridization

A targeted cytogenetic technique using fluorescent

DNA probes to detect or localize specific chromosome sequences. It can identify selected aneuploidies, microdeletions, gene rearrangements, or amplifications but does not survey the entire genome.

Fluorescence Microscopy

fluorescence microscopy is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Fluorescence Polarization

fluorescence polarization is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Fluorescent Dye

A dye that absorbs light at one wavelength and emits light at a longer wavelength, allowing cells, tissues, or molecules to be detected.

Fluoridation

Fluoridation is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fluoride

Mineral that strengthens tooth enamel and inhibits bacterial acid production. Sources: water fluoridation (0.7 ppm optimal), toothpaste, professional applications. Mechanism: incorporates into hydroxyapatite forming fluorapatite (more acid-resistant), promotes remineralization, inhibits bacterial enzymes.

Fluoride In Diet

A patient-care topic concerning fluoride in diet. It addresses the relevant purpose, risks, management, and follow-up.

Fluoride Overdose

Fluoride overdose involves exposure or injury from fluoride overdose; dose, route, timing, symptoms,

and tissue involvement determine the urgency and treatment approach.

Fluoride Poisoning

Illness caused by excessive fluoride exposure, which can produce nausea, abdominal symptoms, low calcium, heart-rhythm problems, or skeletal effects.

Fluoride Varnish

A concentrated fluoride preparation applied to teeth by a healthcare professional to help prevent or arrest dental caries. The frequency of application depends on age and caries risk.

Fluorimetry

fluorimetry is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Fluorine

fluorine is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Fluorine-18

PET radionuclide, half-life 110 minutes. Produced in cyclotron via O-18(p,n)F-18 reaction. Used in FDG and other tracers. Requires on-site or nearby cyclotron due to short half-life.

Fluorometholone

A corticosteroid ophthalmic medicine used for selected inflammatory eye conditions; prolonged use can raise intraocular pressure and increase infection or cataract risk.

Fluorophotometry

fluorophotometry is a diagnostic, research, imaging, or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Fluoropyrimidine

A fluoropyrimidine is an antimetabolite resembling pyrimidine bases; drugs such as fluorouracil and capecitabine disrupt thymidylate and nucleic-acid synthesis in cancer cells.

Fluoroquinolone

A fluoroquinolone is a bactericidal antibiotic that inhibits bacterial DNA gyrase and topoisomerase IV; class risks include tendinopathy or tendon rupture, neuropathy, QT prolongation, dysglycaemia, and central-nervous-system effects.

Fluoroquinolones

Broad-spectrum bactericidal antibiotics that inhibit bacterial DNA gyrase and topoisomerase IV, enzymes essential for DNA replication, transcription, and repair. They are synthetic derivatives of nalidixic acid and include ciprofloxacin, levofloxacin, moxifloxacin, and others.

Fluoroscopy

A type of X-ray that shows moving pictures of the inside of the body. It helps doctors guide procedures and examine swallowing, blood vessels, joints, and the digestive or urinary tract. Contrast material may be used, and the test involves radiation.

Fluorosis

Enamel hypomineralization from excess fluoride during development. Mild: white spots. Severe: brown stains, pitting. Prevention: monitor fluoride intake in children.

Fluorouracil

Fluorouracil is a fluoropyrimidine antimetabolite that inhibits thymidylate synthase; topical treatment is used for selected actinic or superficial lesions and systemic treatment for cancers, with mucositis, diarrhoea, myelosuppression, and DPYD-related severe toxicity possible.

Fluorouracil Side Effect

Toxicity from the antimetabolite 5-fluorouracil causing mucositis, diarrhea, hand-foot syndrome, myelosuppression, and potential cardiotoxicity, requiring

dose adjustment or discontinuation.

Fluoxetine

Fluoxetine is a selective serotonin-reuptake inhibitor used for depression, obsessive-compulsive disorder, panic disorder, bulimia nervosa, and premenstrual dysphoric disorder; its long half-life and serotonergic interactions are clinically important.

Fluoxymesterone

A synthetic androgen medicine used in selected hormone-related conditions; it can affect the liver, lipids, and reproductive function.

Flupentixol

Flupentixol is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fluphenazine

A medicine or chemical compound called fluphenazine. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Fluphenazine Hydrochloride

A medicine or chemical compound called fluphenazine hydrochloride. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Flurandrenolide

A topical corticosteroid used to reduce inflammation and itching in some skin disorders.

Flurazepam

A medicine or chemical compound called flurazepam. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Flurbiprofen

Flurbiprofen is an NSAID used for pain and inflammation and, in some formulations, ocular inflammation; gastrointestinal, renal, cardiovascular, and

hypersensitivity adverse effects are possible.

Flush

A sudden reddening or warm sensation of the skin from increased blood flow, or the act of washing a body passage or intravenous line with fluid; context determines the meaning.

Flushable Reagent Stool Blood Test

Alternate index label; see *Fecal occult blood test*.

Flushing

Transient reddening and warmth of the skin caused by increased cutaneous blood flow. It may result from heat, emotion, exertion, fever, alcohol, endocrine or neuroendocrine disease, mast-cell activation, medication exposure, or autonomic dysfunction; associated diarrhoea, wheeze, hypotension, or palpitations may indicate systemic disease.

Flutamide

A nonsteroidal androgen-receptor antagonist used with androgen-deprivation therapy for prostate cancer; hepatotoxicity and other adverse effects require monitoring.

Flutter

A rapid, regular, repetitive movement or contraction; atrial flutter is a cardiac arrhythmia caused by a rapid organized atrial circuit.

Flutter Valve

A handheld device combining PEP therapy with oscillations during exhalation. The oscillations create shear forces that loosen mucus from airway walls, while the PEP keeps airways open. It is used for airway clearance in bronchiectasis, CF, and COPD.

Fluvastatin

A medicine or chemical compound called fluvastatin. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Fluvoxamine

A selective serotonin-reuptake inhibitor used mainly

for obsessive-compulsive disorder and selected anxiety or depressive disorders; interactions and serotonin toxicity are important considerations.

Flux

Flux is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

FM

Fibromyalgia: a chronic disorder characterized by widespread pain, fatigue, and cognitive symptoms. Diagnosis is based on clinical criteria. Treatment includes exercise, cognitive behavioral therapy, and medications.

Alternate Names:

Fibromyalgia

FMRI

Functional magnetic resonance imaging (fMRI) uses magnetic resonance imaging to measure small changes in brain blood flow and oxygenation associated with brain activity. It can help map functions such as language and movement and guide planning for selected brain treatments. Unlike CT, it does not use ionizing radiation.

FMS

Fibromyalgia syndrome: a chronic condition causing widespread pain, tenderness, fatigue, and sleep disturbances. Affects 2-4% of the population, more common in women. Treatment is multimodal including exercise, therapy, and medications.

Alternate Names:

Fibromyalgia

FNA

A percutaneous sampling technique in which a 22 to 25-gauge needle, often ultrasound- or palpation-guided, aspirates cellular material from a palpable or imaging-detected lesion for cytologic examination, with minimal trauma and lower morbidity than core or open biopsy. See Fine Needle Aspiration (FNA).

FND

Functional neurological disorder: a condition where neurological symptoms (weakness, movement abnormalities, seizures, sensory loss) occur without structural disease. Symptoms are real and disabling. Diagnosis requires positive clinical signs. Treatment includes physical therapy, psychotherapy, and education.

Alternate Names:

Conversion disorder; functional neurologic disorder

Foam Cell

A lipid-filled macrophage or other cell with a foamy appearance, commonly found in atherosclerotic plaques and some inflammatory or storage disorders.

Foam Cells

Lipid-filled cells, usually macrophages, that contribute to atherosclerotic plaque.

Focal

Limited to a specific site or region rather than distributed diffusely; imaging, pathology, seizures, and neurologic deficits may be described as focal.

Focal Adhesion

A cell-matrix attachment complex linking integrins to the actin cytoskeleton and regulating adhesion, force sensing, migration, and signaling.

Focal Glomerulosclerosis

Scarring affecting some glomeruli or only parts of affected glomeruli, which can impair kidney filtration.

Focal Neurologic Deficits

Loss of a specific neurologic function, such as strength, sensation, language, vision, or coordination, caused by a localized brain, spinal-cord, or nerve lesion.

Focal Neuropathy

Damage to a single peripheral nerve or a small group of nerves.

Alternate Names:

Mononeuropathy

Focal Nodular Hyperplasia

Benign liver lesion with a characteristic central stellate scar visible on imaging. The lesion represents a hyperplastic response to a pre-existing vascular malformation rather than a true neoplasm. Associated with oral contraceptive use though it also occurs in men and children. No risk of hemorrhage or malignant transformation.

Focal Sclerosis

Localized hardening or scarring of tissue; in the kidney, focal segmental glomerulosclerosis is a pattern of scarring affecting some glomeruli and parts of affected glomeruli.

Focal Segmental Glomerulosclerosis

The most common cause of nephrotic syndrome in African American adults and Hispanic populations. Associated with HIV infection (HIV-associated nephropathy), heroin use, sickle cell disease, massive obesity, interferon therapy, and renal hyperfiltration (solitary kidney).

Focal Segmental Glomerulosclerosis

Kidney disease in which scar tissue affects some portions of some glomeruli, potentially causing protein in urine and progressive kidney failure.

Focal Seizure

A seizure arising from networks limited to one cerebral hemisphere at onset. Awareness may be preserved or impaired, and motor, sensory, autonomic, cognitive, or emotional features depend on the involved region; a focal seizure may spread to become bilateral tonic-clonic.

Focal-Onset Seizure

A seizure that begins within networks limited to one cerebral hemisphere. Awareness may be preserved or impaired, and motor, sensory, autonomic, cognitive, or emotional features reflect the involved cortical region; it may evolve to bilateral tonic-clonic activity.

Focus Words

Calming words or phrases used to anchor attention during relaxation or stress-management exercises.

FODMAP Diet

Restriction of Fermentable Oligo-, Di-, Mono-saccharides And Polyols. Used for IBS management. Low FODMAP diet followed by gradual reintroduction.

FODMAPs

Fermentable oligosaccharides, disaccharides, monosaccharides, and polyols: short-chain carbohydrates that may be poorly absorbed and produce gas, bloating, pain, or diarrhea in susceptible people.

Foetal Acidosis

Abnormally low fetal blood pH from inadequate oxygenation or impaired gas exchange, assessed through fetal monitoring and, when available, fetal blood sampling.

Foetal Alcohol Syndrome

Foetal Alcohol Syndrome is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Foetus

Foetus is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Folate

Folate is vitamin B9, required for one-carbon transfer, DNA synthesis, and normal red-cell formation; deficiency causes megaloblastic anaemia and in pregnancy increases neural-tube-defect risk, while supplementation can mask vitamin-B12 deficiency.

Folate Antagonist

A folate antagonist blocks folate-dependent enzymes or transport and thereby impairs nucleotide synthesis; examples include methotrexate and trimetho-

prim, used as anticancer, immunosuppressive, or antimicrobial drugs.

Folate Deficiency

Folate (vitamin B9) deficiency causes megaloblastic anemia identical to B12 deficiency but without neurologic manifestations (except possible cognitive changes). Dietary sources: leafy greens, legumes, liver; daily requirement 400 mcg. Absorption in jejunum.

Folate Deficiency Anemia

Megaloblastic anemia caused by insufficient folate (vitamin B9), characterized by large, immature red blood cells. Causes include poor diet, malabsorption, alcoholism, and medications. Symptoms include fatigue, weakness, and glossitis. Treatment with folic acid supplementation.

Alternate Names:

Vitamin B9 deficiency anemia

Folate-Deficiency Anemia

A megaloblastic anemia caused by inadequate folate, impaired absorption, increased need, or certain medicines; blood testing identifies the cause and folate replacement treats deficiency.

Foley Catheter

An indwelling urinary catheter held in the bladder by an inflatable balloon and connected to a drainage system.

Folic Acid

A synthetic form of folate, a B vitamin required for nucleotide synthesis and red-cell production; supplementation prevents neural-tube defects and treats deficiency.

Folic Acid - Test

A blood test measuring folate concentration, used to investigate macrocytic anemia and suspected folate deficiency; results are interpreted with vitamin B12 status, blood counts, diet, and clinical findings.

Folic Acid And Birth Defect Prevention

A patient-care topic concerning folic acid and birth defect prevention. It addresses the relevant purpose, risks, management, and follow-up.

Folic Acid In Diet

A patient-care topic concerning folic acid in diet. It addresses the relevant purpose, risks, management, and follow-up.

Folic Acid Supplementation

Use of folic acid before conception and during early pregnancy to reduce the risk of fetal neural-tube defects. The dose depends on individual risk and local clinical guidance; folate is also needed for DNA synthesis and red-cell production.

Folinic Acid

Folinic Acid is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Follicle

A small sac or tissue structure, such as a hair follicle or ovarian follicle.

Follicle Stimulating Hormone

FSH: a gonadotropin secreted by the anterior pituitary, regulating gonadal function. In females: stimulates ovarian follicle growth and estradiol production; LH surge triggers ovulation. In males: stimulates Sertoli cells and spermatogenesis. Controlled by GnRH and feedback (estrogen, inhibin).

Follicle-Stimulating Hormone

A pituitary hormone that supports ovarian follicle development and sperm production.

Alternate Names:

FSH

Follicle-Stimulating Hormone

A pituitary hormone that supports ovarian follicle development and estrogen production in females and sperm production in males.

Follicle-Stimulating Hormone Blood Test

A blood test measuring follicle-stimulating hormone, which regulates ovarian follicle development and sperm production; it helps evaluate infertility, menstrual disorders, gonadal failure, and pituitary disease.

Follicular

Relating to or centered on a hair follicle; follicular lesions may appear as papules, pustules, keratotic plugs, or inflammation around hairs.

Follicular Cyst

The most common type of physiological functional ovarian cyst, arising when a dominant mature ovarian follicle fails to rupture and release an ovum during the ovulatory cycle, or when an immature follicle fails to undergo atresia. Characteristically unilocular, thin-walled, filled with clear estrogen-rich fluid, and typically resolving spontaneously within two to three menstrual cycles without surgical intervention.

Follicular Fluid

Follicular fluid is the fluid within an ovarian follicle surrounding the developing oocyte and containing hormones and signaling molecules.

Follicular Hyperkeratosis

Follicular hyperkeratosis is excess keratin blocking hair follicles and producing rough small bumps, commonly seen in keratosis pilaris or deficiency states.

Follicular Lymphoma

An indolent B-cell non-Hodgkin lymphoma with a follicular growth pattern. Many cases have changes involving BCL2, but classification depends on tissue and molecular findings. It commonly causes painless lymph-node enlargement and may involve bone marrow or other sites.

Follicular Thyroid Carcinoma

Second most common thyroid carcinoma accounting for approximately ten percent of cases. Distinguished from follicular adenoma by capsular or vascular invasion on histological examination which cannot be determined by fine-needle aspiration. Metas-

tasizes hematogenously to bone, lung, and liver.

Follicular Unit

A natural hair-growth unit containing one or more follicles and associated structures.

Follicular Unit Transplantation

Hair-restoration surgery that transplants individual follicular units to thinning areas.

Alternate Names:

FUT

Folliculitis

Inflammation of a hair follicle, often caused by bacteria, yeast, friction, occlusion, or ingrown hairs, producing small itchy or tender bumps.

Alternate Names:

Barber's Itch

Follistatin

A medicine or chemical compound called follistatin. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Follow-Up Recommendations

Recommendations for further imaging or evaluation. Based on guidelines (Fleischner Society for lung nodules).

Fomentation

The application of warm, moist material to a body part, traditionally to soothe pain, relax muscles, or promote local blood flow.

Fomepizole

An alcohol dehydrogenase inhibitor used as an antidote for toxic alcohol exposure, particularly methanol and ethylene glycol. It prevents formation of toxic metabolites and is used with supportive care and hemodialysis when indicated.

Fomite

An inanimate object or surface capable of transmitting infectious agents (viruses, bacteria, fungi) between people (e.g., doorknobs, towels, medical

equipment, toys, needles). Transmission occurs when contaminated fomites contact mucous membranes or skin, or are involved in procedures.

Fomites

fomites is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Fondaparinux

A medicine or chemical compound called fondaparinux. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Fondaparinux Sodium

A medicine or chemical compound called fondaparinux sodium. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Fonofos

An organophosphate insecticide that can cause cholinergic poisoning by inhibiting acetylcholinesterase.

Fonsecaea

fonsecaea is a fungal taxon or fungus-related term; clinical interpretation depends on the species, site, host immune status, and evidence of invasive or superficial infection.

Fontan Procedure

The definitive, third-stage palliative surgical operation for patients with complex congenital single-ventricle physiology (e.g., tricuspid atresia, hypoplastic left heart syndrome, double inlet left ventricle), typically performed between 2 and 4 years of age following a prior bidirectional Glenn shunt. Involves directing all systemic venous blood return from the inferior vena cava directly into the pulmonary artery branches without the interposition of a subpulmonary pumping ventricle, modernly executed via an extracardiac PTFE conduit (or lateral tunnel) with or without a temporary fenestrated

tration. Creates the completed “Fontan circulation,” wherein pulmonary blood flow is entirely passive, dependent on low pulmonary vascular resistance, preserved single-ventricle diastolic function, and elevated central venous pressure.

Alternate Names:

Fontan Operation; Fontan Circulation; Total Cavopulmonary Connection (TCPC)

Fontanel

A soft membranous gap between the bones of an infant’s skull.

Fontanel (Fontanelle)

A soft membranous gap between the skull bones of an infant, allowing skull growth and molding during birth; a bulging or markedly sunken fontanel can signal illness.

Fontanelle

Soft membranous spaces between skull bones that allow brain growth and skull molding during delivery. Anterior fontanelle diamond-shaped, closes by 12-18 months. Posterior fontanelle triangular, closes by 2-3 months. Bulging fontanelle suggests increased intracranial pressure; sunken fontanelle indicates dehydration.

Fontanelles - Bulging

A tense or outwardly bulging soft spot on an infant’s skull when calm and upright, potentially indicating raised intracranial pressure or meningitis.

Fontanelles - Enlarged

Soft spots on an infant’s skull that are larger than expected, associated with normal variation, prematurity, hypothyroidism, rickets, or increased intracranial pressure.

Fontanelles - Sunken

Depression of an infant’s soft spot, commonly suggesting dehydration or reduced intracranial volume when assessed with the child calm and upright.

Food

Food is material eaten or drunk to provide energy, nutrients, hydration, and building components for the body.

Food Additives

Substances intentionally added to food for preservation, flavor, color, texture, or stability; reactions and safety depend on the additive and exposure.

Food Allergies

Immune reactions to particular food proteins that can cause symptoms ranging from hives to anaphylaxis.

Alternate Names:

Food allergy

Food Allergy

Immune-mediated reaction to food protein. IgE-mediated: anaphylaxis. Non-IgE: EoE, food protein-induced enterocolitis. Common allergens: milk, egg, peanut, tree nut, fish, shellfish, soy, wheat.

Alternate Names:

Allergy Food (Food Allergy)

Food Allergy, Egg

Immune-mediated hypersensitivity reaction to egg proteins, most common in children. Reactions range from mild urticaria to severe anaphylaxis. Diagnosis involves skin prick testing and specific IgE levels. Management includes strict avoidance and carrying epinephrine auto-injectors.

Alternate Names:

Egg allergy

Food Allergy, Milk

Hypersensitivity to milk proteins (casein, whey), the most common food allergy in infants and young children. Symptoms include eczema, gastrointestinal upset, and respiratory symptoms. Most children outgrow it by school age. Breastfeeding and hydrolyzed formulas are management options.

Alternate Names:

Milk allergy

Food Consumption

Food consumption is the amount and pattern of eating and drinking, influenced by appetite, culture, access, health, habits, and metabolic needs.

Food Hygiene

Food Hygiene is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Food Intolerance

Adverse reaction not immune-mediated. Lactose intolerance, gluten sensitivity (non-celiac), histamine intolerance, FODMAP sensitivity. Diagnosis: elimination diet, challenge.

Food Jags

A child's persistent preference for a very small number of foods or refusal of previously accepted foods, common in development but concerning when nutrition or growth suffers.

Food Labeling

Information on packaged food describing ingredients, allergens, serving size, nutrients, calories, and claims to support informed dietary choices.

Food Poisoning

Illness caused by consuming food contaminated with bacteria, viruses, parasites, or their toxins. Common pathogens include Salmonella, Escherichia coli, Listeria, and norovirus, with symptoms such as nausea, vomiting, diarrhea, abdominal cramps, and fever.

Food Poisoning (Foodborne Illness)

Illness caused by eating food or drinking water contaminated with infectious organisms, toxins, or harmful chemicals.

Food Poisoning Prevention

Food poisoning prevention involves exposure or injury from food poisoning prevention; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Food Poisoning, Campylobacter

Foodborne infection caused by *Campylobacter* bacteria, usually producing diarrhea, abdominal cramps, fever, and sometimes bloody stool; severe or persistent illness may require medical care.

Food Safety

Food safety involves preventing contamination, improper storage, undercooking, and allergen exposure that can cause foodborne illness.

Food Supplement

Food Supplement is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Food-Borne Disease

Illness caused by eating food or drinking water contaminated with infectious organisms, toxins, or harmful chemicals.

Food-Borne Illness

Illness caused by consuming food contaminated with bacteria, viruses, parasites, or toxins. Common pathogens include Salmonella, E. coli, Listeria, and Norovirus. Symptoms include nausea, vomiting, diarrhea, and fever. Prevention involves proper food handling and cooking.

Alternate Names:

Food poisoning; foodborne illness

Foodborne Disease

Illness caused by eating food or drinking water contaminated with infectious organisms, toxins, or harmful chemicals.

Foodborne Illness

Any illness resulting from the ingestion of contaminated food, encompassing infections and intoxications caused by pathogens, toxins, or chemical contaminants. It affects millions annually and poses particular risks to pregnant women, young children, older adults, and immunocompromised individuals.

Foot

The foot is the distal part of the lower limb below the ankle, formed by 26 bones, joints, muscles, tendons, and ligaments; its arches distribute load and support balance and walking.

Foot And Mouth Disease

Foot and Mouth Disease is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Foot Drop

Weakness or paralysis of ankle dorsiflexion causing the forefoot to drag or the patient to lift the knee abnormally while walking. Common causes include common peroneal nerve injury, L5 radiculopathy, motor-neuron disease, and peripheral neuropathy.

Alternate Names:

Drop Foot

Foot Fracture

A break in one or more of the 26 bones in the foot (tarsals, metatarsals, phalanges). Can result from trauma, stress, or underlying conditions like osteoporosis. Symptoms include pain, swelling, and inability to bear weight. Treatment ranges from immobilization to surgery.

Alternate Names:

Broken foot

Footling Breech

A variety of breech presentation in which one or both hips and knees are extended, so that one or both feet prolapse through the cervix below the fetal buttocks into the lower uterine segment or vagina. It carries the highest risk among all breech variations for umbilical cord prolapse and fetal entrapment, routinely mandating cesarean delivery.

Alternate Names:

Incomplete Breech; Footling Breech Presentation

Foot Pain

Pain in the foot caused by injury, overuse, arthritis, nerve disease, infection, or circulation problems. Persistent pain, inability to bear weight, numbness,

discoloration, fever, or an open wound requires medical assessment.

Foot, Leg, And Ankle Swelling

Accumulation of fluid or tissue enlargement in the lower limb caused by venous disease, heart or kidney failure, medicines, injury, infection, or inflammation; sudden one-sided swelling needs urgent assessment.

Foramen

An opening or passage through a bone or other anatomical structure, usually for nerves or blood vessels.

Foramen Ovale

An opening between the right and left atria of the fetal heart that allows blood to bypass the lungs. It normally closes functionally after birth; persistence is called a patent foramen ovale.

Foraminotomy

A medical procedure, treatment, device, or care topic related to foraminotomy; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Force Plates

Instruments that measure forces exerted by the feet on the ground during standing or movement, helping assess gait, balance, symmetry, and loading.

Forced Expiratory Volume

The volume of air forcibly exhaled in a specified time during a maximal expiratory effort. FEV1 is the volume exhaled in the first second; normally at least 70-80 percent of the forced vital capacity (FEV1/FVC ratio).

Forced Expiratory Volume

The volume of air forcefully exhaled during a specified time after a full breath in, measured by spirometry to assess airflow obstruction and lung function.

Forced Oscillation Technique

Similar to impulse oscillometry, applying small pres-

sure oscillations at various frequencies during tidal breathing to measure respiratory resistance and reactance. It provides information about airway mechanics without requiring forced expiratory maneuvers.

Forced Vital Capacity

Total air exhaled after maximal inspiration. Reduced in both obstructive and restrictive disease. In restrictive disease, FVC reduced with preserved ratio.

Forceps

Grasping instrument. Tissue forceps: with teeth. Dressing forceps: without teeth.

Forceps (Obstetric)

A paired instrument used in selected operative vaginal deliveries to assist delivery of the fetal head.

Alternate Names:

Obstetric

Forceps (Surgical)

Handheld instruments used to grasp, hold, move, or manipulate tissue or medical materials during a procedure.

Alternate Names:

Surgical

Forceps (Vascular)

A surgical instrument used to grasp, hold, or manipulate tissue, vessels, or medical materials.

Alternate Names:

Vascular

Forceps Delivery

Assisted vaginal delivery using forceps (curved blades applied to the fetal head) to expedite delivery for maternal or fetal indications (prolonged second stage, fetal distress, maternal exhaustion, medical conditions limiting pushing).

Forceps-Assisted Delivery

Operative vaginal delivery using forceps (two curved blades applied to the fetal head). Types: outlet

(forceps applied at the introitus, fetal scalp visible), low (station $\geq +2$, not at outlet), midpelvic (station $\geq +2$ but head above ischial spines), and rotational (forceps used to rotate from OP to OA).

Forcible Intercourse

Sexual intercourse imposed without freely given consent through force, threats, coercion, or inability to consent; it is a form of sexual violence requiring safety and medical support.

Fordyce'S Angiokeratomas

Benign vascular ectasias of the superficial dermal capillaries covered by an epidermal reaction, appearing as small, dark red to purple, asymptomatic papules on the scrotum, vulva, or oral mucosa. They are common, increase with age, and are clinically insignificant. They may bleed if traumatized.

Forearm

The portion of the upper limb between the elbow and wrist, containing the radius, ulna, muscles, nerves, and blood vessels.

Forearm Circumference

The distance around the forearm at a specified location, measured for anthropometry, nutritional assessment, growth monitoring, or evaluation of swelling or muscle loss.

Forehead

The part of the face above the eyebrows and in front of the frontal bone, covered by skin and soft tissue and contributing to facial expression and protection of the frontal skull.

Forehead Lift

Cosmetic surgery that reduces forehead lines and raises drooping eyebrows.

Alternate Names:

Brow lift

Foreign Body

Any object lodged in body. Types: organic (wood, thorn), inorganic (metal, glass, plastic). Ra-

diopaque vs. radiolucent. Removal: surgical, endoscopic, spontaneous passage.

Foreign Body Airway Obstruction

Partial or complete blockage of the nose, throat, or lower airway by an inhaled object or material; complete obstruction is a life-threatening choking emergency.

Foreign Body Aspiration

Inhalation of object into airway. Common in children (peanuts, coins, small toys). Signs: choking, coughing, stridor, wheezing, decreased breath sounds. Diagnosis: chest X-ray (may be negative), bronchoscopy.

Foreign Body In The Nose

An object lodged in a nasal passage, often in a child, causing unilateral blockage, discharge, bleeding, odor, or tissue injury and requiring safe removal.

Foreign Body Reaction

Inflammation and immune response to material that remains in tissue, such as a splinter, suture, implant, or other medical device. It may cause a lump, redness, drainage, pain, or scarring; treatment depends on the material and whether infection is present.

Foreign Object - Inhaled

Foreign object - inhaled involves exposure or injury from foreign object - inhaled; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Foreign Object - Swallowed

Foreign object - swallowed involves exposure or injury from foreign object - swallowed; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Foreign-Body Aspiration

Entry of an object into the larynx, trachea, or bronchi, most often in toddlers. Sudden cough, choking, unilateral wheeze, or focal air trapping may occur; complete airway obstruction is an im-

mediate emergency and requires age-appropriate resuscitation.

Forelimb

A forelimb is an anterior limb of an animal, corresponding broadly to an arm or wing in humans and other vertebrates.

Forensic

Relating to the application of scientific or medical knowledge to questions of law, investigation, or court proceedings.

Forensic Anthropology

The application of physical-anthropology methods to legal investigations involving human remains, especially skeletal material. It can help establish whether remains are human, estimate biological characteristics, assess trauma, and support identification.

Forensic Entomology

The study of insects and other arthropods in legal investigations, principally to estimate the postmortem interval (PMI) and to link suspects to scenes.

Forensic Evidence

Physical, biological, documentary, or testimonial material gathered, preserved, and analyzed to establish facts in a legal investigation. Categories include biological evidence (blood, semen, hair), trace evidence (fibers, glass, gunshot residue), pattern evidence (fingerprints, toolmarks, bloodstain patterns), and digital evidence.

Forensic Examination

Structured medicolegal evaluation of a living or deceased person, scene, or evidence to document findings relevant to a legal matter. Components include sexual assault examination with evidence collection, child abuse assessment, injury documentation, driving-impairment evaluation, and autopsy.

Forensic Genetics

The use of genetic testing and DNA analysis to iden-

tify individuals, establish biological relationships, or investigate evidence in legal and criminal cases.

Forensic Odontology

The application of dental science to legal investigations. It may support human identification by comparing dental records, radiographs, and distinctive dental anatomy and may assist in the assessment of dental injuries.

Forensic Pathology

A branch of pathology that applies medical knowledge to legal questions, especially the cause and manner of death and the documentation of injuries. Forensic pathologists examine deaths that are sudden, unexpected, violent, or otherwise subject to legal investigation.

Forensic Psychiatry

The medical specialty at the interface of psychiatry and the law. Core activities: competency to stand trial, criminal responsibility/insanity defense (mental state at time of offense), dangerousness/risk assessment, civil commitment, testamentary capacity, and personal injury claims.

Forensic Radiography

The use of imaging (X-ray, CT, MRI, ultrasound) in forensic and medico-legal investigation.

Forensic Toxicology

The analysis of drugs, poisons, and their metabolites in biological specimens (blood, urine, vitreous humor, hair, tissues) to determine their role in intoxication, poisoning, or death.

Forensic Toxicology Screen

Laboratory analysis of biological specimens for drugs, alcohols, poisons, and their metabolites. A screening result is presumptive until confirmed and quantified by a validated method such as mass spectrometry, with interpretation based on specimen type, timing, therapeutic use, and postmortem changes.

Foreskin

The retractable fold of skin covering the glans penis; it protects the glans and contains sensory nerve endings.

Foreskin (Prepuce)

A fold of skin covering and protecting the glans penis. It is normally retractable after separation from the glans, although the timing varies.

Alternate Names:

Prepuce

Foretinib

An investigational multikinase inhibitor targeting MET and other receptors, studied for selected advanced cancers and not broadly established as routine therapy.

Formaldehyde

Preservative, industrial chemical. Irritant, carcinogen (nasopharyngeal cancer). Sensitizer. Use in fume hood, PPE required.

Formalin

An aqueous solution of formaldehyde, commonly used to preserve biological tissue; it is toxic and irritating and must be handled with appropriate safety controls.

Formalin-Fixed Paraffin-Embedded Tissue

Tissue preserved in formalin, processed, and embedded in paraffin wax for sectioning and microscopic examination. FFPE blocks support histology, immunohistochemistry, and many molecular tests but may contain fixation-related nucleic-acid or antigen alterations.

Formate

formate is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Formed Elements Of Blood

The cellular components of blood: red blood cells, white blood cells, and platelets.

Formerly Churg-Strauss Syndrome

Eosinophilic granulomatosis with polyangiitis, an autoimmune vasculitis associated with asthma, eosinophilia, and inflammation of small and medium blood vessels.

Formetanate

formetanate is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Formication

A sensation that insects are crawling on or under the skin when none are present. It may occur with nerve disorders, menopause, substance withdrawal, medication effects, or psychiatric illness. Persistent or new symptoms should be medically evaluated.

Formoterol Fumarate

A medicine or chemical compound called formoterol fumarate. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Formothion

formothion is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Formparanate

formparanate is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Formula Feed

Prepared infant nutrition given by bottle, cup, or feeding tube when breast milk is unavailable, insufficient, or not selected; formula choice should match

the infant's age and medical needs.

Formula Feeding

Alternative to breastfeeding using commercially prepared infant formula. Types include cow's milk-based, soy-based, hydrolyzed protein, and amino acid-based formulas. Iron-fortified formulas standard to prevent iron deficiency anemia. Preparation requires clean water and proper sterilization of bottles.

Formula, Infant

Commercially prepared milk-based, soy-based, hydrolyzed, or specialized nutrition for infants; safe preparation and age-appropriate feeding are essential.

Formulae

Plural of formula; in healthcare it may refer to mathematical equations, medication formulations, or prepared nutritional products.

Formulary

A list of medicines approved or preferred for use by a health system, insurer, or institution, often including prescribing restrictions and coverage information.

Fornix

The conjunctival fornix is the redundant, dome-shaped fold of conjunctiva where the palpebral conjunctiva transitions to the bulbar conjunctiva, forming a deep recess superiorly (superior fornix), inferiorly (inferior fornix), and medially and laterally (medial and lateral fornices).

Forrester Classification

A hemodynamic classification system developed by James Forrester to categorize patients with acute myocardial infarction and cardiogenic shock based on invasive parameters obtained via pulmonary artery catheterization: cardiac index (CI; cut-off 2.2 L/min/m²) and pulmonary capillary wedge pressure (PCWP; cut-off 18 mmHg). It divides patients into four clinical subsets: Subset I (warm and dry: normal CI, normal PCWP; lowest mortal-

ity); Subset II (warm and wet: normal CI, elevated PCWP; pulmonary congestion treated with diuretics/vasodilators); Subset III (cold and dry: low CI, normal PCWP; hypovolemic shock treated with volume expansion); and Subset IV (cold and wet: low CI, elevated PCWP; cardiogenic shock with highest mortality, requiring inotropes, vasopressors, or mechanical circulatory support).

Alternate Names:

Forrester Hemodynamic Subsets; Forrester Score

Forssman Antigen

A glycolipid antigen found in some animal tissues and microorganisms; antibodies against it are used mainly in immunology and laboratory research.

Fortification

Addition of nutrients to food. Examples: iodized salt, vitamin D milk, folic acid grains. Public health strategy for deficiency prevention.

Fortified Wine

Fortified Wine is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fosaprepitant Dimeglumine

An intravenous prodrug converted to aprepitant that blocks neurokinin-1 receptors and prevents chemotherapy-induced nausea and vomiting.

Fosbretabulin

An investigational anticancer medicine that damages the blood vessels supplying tumors.

Foscarnet Sodium

The sodium salt of foscarnet, an antiviral that directly inhibits viral DNA polymerase and is used for selected resistant cytomegalovirus or herpesvirus infections; kidney and electrolyte toxicity are important.

Fosfomycin

A medicine or chemical compound called fosfomycin.

Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Fosinopril

A medicine or chemical compound called fosinopril. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Fossa

A shallow depression or hollow in a bone or other body structure.

Fossula

Fossula is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Fossus

Fossus is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Fotemustine

A nitrosourea chemotherapy drug that crosses the blood–brain barrier and is used in some settings for malignant melanoma or brain tumors; marrow toxicity is important.

Found Dead

A circumstance in which a person is discovered deceased outside medical observation; forensic evaluation may establish identity, timing, cause, and manner of death.

Four-Dimensional Ultrasound

Real-time 3D ultrasound. Used in obstetrics for fetal imaging.

Fourchette

Fourchette is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Fourier Analysis

fourier analysis is a diagnostic, research, imaging,

or measurement term used to assess a condition or question; interpretation depends on the indication, method, and result.

Fournier Gangrene

A rapidly progressive necrotizing infection of the perineum, external genitalia, or perianal region. It causes severe pain, swelling, fever, systemic toxicity, and tissue necrosis and requires immediate surgical debridement and broad-spectrum antimicrobial treatment.

Fourth Disease

An older name for Dukes' or Filatov–Dukes disease, a rash illness attributed to toxin-producing *Staphylococcus aureus*; the historical label has been debated and is not commonly used in modern diagnosis.

Fourth Finger

The fourth finger is the ring finger, located between the middle finger and little finger.

Fourth Stage Of Labor

The first one to two hours after delivery of the placenta, when the mother is observed for uterine contraction, bleeding, vital-sign changes, and early postpartum complications.

Fourth Venereal Disease

Fourth Venereal Disease is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fourth-Degree Tear

A severe perineal tear during childbirth that extends through the anal sphincter and into the rectal mucosa. It requires careful examination, repair, follow-up, and assessment for complications.

Fovea

A tiny depression in the center of the retina that provides the sharpest and most detailed visual acuity.

Fovea Centralis

Central macular pit (approximately 1.5 mm) with the highest cone density (approximately 200,000 per square millimeter) and no rods, providing the sharpest visual acuity and color vision. Avascular with overlying retinal layers displaced laterally for direct light access to cones.

Fowler Position

A semiupright position in which the head and upper body are elevated, usually about 45 to 60 degrees.

Foxglove Poisoning

Foxglove poisoning involves exposure or injury from foxglove poisoning; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Fractional Excretion Of Sodium

The percentage of filtered sodium excreted in urine, calculated from serum and urine sodium and creatinine. It may help distinguish prerenal physiology from tubular injury, although diuretics, CKD, sepsis, and other conditions limit interpretation.

Fractional Excretion Of Urea (FEUrea)

The percentage of filtered urea excreted in urine. It may be used as an adjunct when diuretics make FENa less interpretable, but it is also affected by catabolism, sepsis, gastrointestinal bleeding, and chronic kidney disease.

Alternate Names:

FEUrea

Fractional Flow Reserve

An invasive physiological measurement during coronary catheterization using a pressure wire to assess the hemodynamic significance of a coronary stenosis. $FFR = Pa/Pd$ (aortic pressure / distal coronary pressure) during maximal hyperemia (adenosine).

Fractional IPV

A reduced-dose schedule of inactivated polio vaccine used in the WHO and India Universal Immunization Schedule, where one-fifth of the standard intramus-

cular IPV dose (0.1 mL instead of 0.5 mL) is given by intradermal injection at each of two deltoid sites (total 0.2 mL per fractional dose), rather than the full 0.5 mL intramuscular dose.

Fracture

A break in the continuity of bone, classified by location, pattern (transverse, oblique, spiral, comminuted), displacement, and whether the fracture communicates with the external environment (open vs. closed). Fracture healing occurs through inflammatory, reparative, and remodeling phases.

Fracture Displacement

Loss of normal anatomic alignment between fracture fragments, described by translation, angulation, shortening, or rotation. The acceptable degree depends on the bone, joint involvement, patient age, and functional demands.

Fracture Fixation

Stabilization of a broken bone with casts, plates, screws, nails, wires, or external frames to restore alignment and permit healing or movement.

Fracture Liaison Service

A coordinated program that identifies patients with fragility fractures, evaluates bone health and secondary causes, initiates fracture prevention, and ensures follow-up to reduce recurrent fractures.

Fracture Of The Temporal Bone

A break in the thin bones forming the skull base and ear structures, potentially injuring the facial nerve, carotid artery, or middle ear and presenting with hearing loss, vertigo, or cerebrospinal fluid leak.

Fracture Prevention

Measures that reduce the risk of bone fracture, including assessment and treatment of osteoporosis, strength and balance exercise, fall prevention, and correction of relevant nutritional or medication-related risks.

Fracture, Arm

A break in any of the three arm bones: humerus (upper arm), radius, or ulna (forearm). Caused by trauma, falls, or sports injuries. Symptoms include pain, deformity, swelling, and loss of function. Treatment depends on fracture type and may require casting or surgical fixation.

Alternate Names:

Broken arm

Fracture, Growth Plate

A break through the physis (growth plate) in children and adolescents. Can affect future bone growth if not properly treated. Classified by Salter-Harris system. Requires prompt evaluation and appropriate management to prevent growth disturbances.

Alternate Names:

Growth plate fractures; physeal fractures

Fracture, Hip

A break in the proximal femur, common in elderly patients with osteoporosis and in young patients after high-energy trauma. Types include femoral neck, intertrochanteric, and subtrochanteric fractures. Treatment usually requires surgical fixation or joint replacement.

Alternate Names:

Hip fracture

Fracture, Leg

A break in the tibia (shinbone), fibula, or both bones of the lower leg. Caused by direct trauma, sports injuries, or falls. Symptoms include pain, swelling, deformity, and inability to bear weight. Treatment depends on fracture pattern and stability.

Alternate Names:

Broken leg

Fracture, Stress

A partial or complete bone fracture caused by repetitive force or overuse, commonly affecting weight-bearing bones (metatarsals, tibia, femur). Athletes and military recruits are at increased risk. Presents with activity-related pain that worsens with continued activity.

Alternate Names:

Stress fractures; fatigue fractures

Fracture-Fixation Device

An implant such as a plate, screw, intramedullary nail, or external-fixation component used to stabilize a broken bone and maintain alignment during healing. Selection depends on fracture pattern, bone quality, and soft-tissue condition.

Fractured Clavicle In The Newborn

A break in a newborn's collarbone, most often related to difficult delivery; it may cause reduced arm movement and usually heals with supportive care.

Fractured Elbow

A break involving the distal humerus, radial head, olecranon, or another elbow structure that can impair motion and injure nearby nerves or vessels.

Fractured Ilium

A break in the broad upper portion of the pelvic bone, usually from substantial trauma or weakened bone and potentially associated with pelvic-organ or blood-vessel injury.

Fractured Nose

A break in the nasal bones or cartilage, usually caused by direct trauma. Symptoms include pain, swelling, epistaxis, and deformity. May be associated with septal hematoma requiring drainage. Treatment ranges from closed reduction to surgical repair.

Alternate Names:

Broken nose; nasal fracture

Fractured Rib

A break in one or more of the 12 pairs of ribs. Usually caused by trauma. Symptoms include sharp chest pain that worsens with breathing, coughing, or movement. May be associated with pneumothorax, hemothorax, or organ injury. Treatment is mainly supportive.

Alternate Names:

Broken ribs; rib fracture

Fragile X Syndrome

An inherited neurodevelopmental disorder caused by expansion of a CGG repeat in the FMR1 gene, leading to reduced gene expression. It may cause intellectual disability, characteristic physical features, and behavioral or autism-spectrum features.

FRAIL Scale

A brief frailty screen assessing Fatigue, Resistance, Ambulation, Illnesses, and Loss of weight. It identifies risk of frailty but is not a substitute for a comprehensive geriatric assessment.

Frailty

Clinical syndrome of decreased reserve and resilience to stressors. Components: unintentional weight loss, exhaustion, low physical activity, slow gait speed, weak grip strength. Identification: Fried phenotype, frailty index. Consequences: adverse outcomes (falls, hospitalization, mortality), polypharmacy, functional decline.

Framboesia

Yaws, a chronic treponemal infection caused by *Treponema pallidum pertenue* and transmitted mainly through skin contact in endemic tropical regions.

Frameshift Mutation

Insertion or deletion of nucleotides not in multiples of three, shifting reading frame. Usually produces nonfunctional protein. Examples: some cystic fibrosis mutations, Tay-Sachs disease.

Framycetin

An aminoglycoside antibiotic used topically in selected bacterial infections; systemic absorption and allergy can cause toxicity or sensitization.

Francisella

A genus of small gram-negative bacteria that includes *F. tularensis*, the cause of tularemia, a zoonotic infection acquired from animals, ticks, contaminated water, or aerosols.

Francisella Tularensis

A gram-negative coccobacillus that causes tularemia, a zoonotic infection transmitted by ticks, deer flies, or direct contact with infected animals. The organism can be acquired through the skin, respiratory tract, or gastrointestinal tract. Ulceroglandular tularemia presents with skin ulcer and regional lymphadenopathy.

Francisellaceae

A family of small gram-negative bacteria that includes Francisella species, some of which cause tularemia or related animal infections.

Frank Breech

The most common type of breech presentation, accounting for 65–70% of all breech gestations at term. The fetus lies longitudinally with both hips flexed against the abdomen and both knees extended with the feet positioned up near the shoulders and head, allowing the buttocks to act as the primary presenting part. It has the lowest risk of umbilical cord prolapse among breech positions and is the most suitable variant for external cephalic version or trial of vaginal breech delivery.

Alternate Names:

Extended Breech; Frank Breech Presentation

Frank-Starling Mechanism

The physiological principle stating that stroke volume increases as end-diastolic volume (preload) increases, up to an optimal point. Within physiological limits, greater ventricular stretch leads to stronger contraction due to increased calcium sensitivity of myofilaments and optimal overlap of actin and myosin.

Frasier Syndrome

A rare genetic disorder associated with abnormal sexual development, kidney disease, and increased risk of gonadal tumors.

Fraternal Twins

Twins formed when two separate eggs are fertilized by two separate sperm, producing genetically distinct siblings who may be the same or different

sex.

FRAX Tool

A calculator that estimates a person's ten-year probability of major osteoporotic or hip fracture.

Alternate Names:

FRAX fracture-risk calculator

Freckle

A small, flat, light-brown or tan area of increased skin pigmentation, often becoming more apparent with ultraviolet exposure; a changing pigmented lesion should be assessed.

Freckles

Freckles are small flat areas of increased melanin that darken with ultraviolet exposure and often reflect genetic tendency. They are usually harmless, but a changing, bleeding, irregular, or distinctly different pigmented lesion should be examined.

Freckles (Ephelides)

Small, flat, light-brown macules caused by increased melanin production, usually becoming more noticeable with ultraviolet exposure.

Alternate Names:

Ephelides

Free Fragment

A piece of an intervertebral disk that has separated from the main disk.

Alternate Names:

Sequestered disk fragment

Free Radical

A molecule or atom containing an unpaired electron, making it chemically reactive. Excessive free-radical activity can contribute to oxidative injury.

Free Radical Injury

Free radicals are highly reactive molecules with unpaired electrons that can damage cellular components through oxidative stress. Reactive oxygen species include the superoxide anion, hydrogen peroxide, and the hydroxyl radical generated through

the Fenton reaction.

Free T4 Test

A blood test measuring unbound thyroxine, the biologically available thyroid hormone; it is interpreted with thyroid-stimulating hormone to evaluate hyperthyroidism, hypothyroidism, and treatment response.

Free Thyroxine Index

A calculated estimate of circulating free thyroxine based on the total thyroxine concentration and a measure of thyroid-hormone binding.

Free Water Clearance

The rate at which the kidneys excrete water in excess of or less than the amount needed to produce urine isotonic to plasma. Positive clearance indicates excretion of dilute urine; negative clearance indicates retention of water through concentration of urine.

Freedman Test

A historical pregnancy test based on biological activity in animals; it has been replaced by accurate immunologic tests for human chorionic gonadotropin.

Freeze Drying

Lyophilization, a process that freezes a material and removes ice by sublimation under low pressure to preserve medicines, biologic samples, or foods.

Freeze Etching

A preparation method in which frozen material is fractured and the exposed surface is etched by sublimating ice before microscopic examination.

Freeze Substitution

A microscopy preparation method replacing ice in a frozen specimen with an organic solvent at low temperature before fixation and embedding.

Freezing

Exposure to very low temperature that can cause tissue freezing, frostbite, or dangerous loss of body

heat; the term can also describe preservation by freezing.

Frei'S Test

A historical skin test for lymphogranuloma venereum using chlamydial antigen; it is no longer a routine diagnostic test.

Fremitus

A vibration felt or heard through body tissues, often used to assess transmitted voice sounds in the chest.

French Disease

A historical name for syphilis, a sexually transmitted infection caused by the bacterium *Treponema pallidum*.

Frenectomy

Surgical removal or repositioning of frenulum. Indications: diastema, gingival recession, tongue-tie, interference with denture.

Frenulum

Frenulum is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Frenulum Breve

An abnormally short penile frenulum that may cause pain, tearing, or bleeding during erection or sexual activity. Treatment depends on symptoms and may include observation or a minor surgical procedure.

Frenum

Frenum is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Frenum (Frenulum)

A fold of mucous membrane or skin that anchors or limits movement of a body part, such as the lingual frenulum beneath the tongue or the labial frenula.

Alternate Names:

Frenulum

Frequency

Increased daytime micturition, typically more than seven episodes per day, distinguished from polyuria (high urine volume) and nocturia (nighttime voiding). Common causes include urinary tract infection, benign prostatic hyperplasia, overactive bladder, and diabetes mellitus.

Frequency Of Urination

Urinating more often than usual, which may result from increased fluid intake, infection, pregnancy, diabetes, diuretics, bladder irritation, or other urinary disorders.

Frequency Range

The span from the lowest to highest sound frequencies amplified by a hearing aid.

Frequency Response

The amount of amplification a hearing aid provides across different sound frequencies.

Frequency, Urinary

The need to urinate more often than usual, sometimes with only small amounts passed; causes include infection, irritation, pregnancy, diuretics, and disorders of the bladder or prostate.

Frequent Bowel Movements

Passing stool more often than usual; causes include diet, infection, medicines, inflammation, malabsorption, or functional bowel disorders.

Frequent Or Urgent Urination

Increased need to urinate or sudden difficult-to-defer urges, caused by infection, bladder irritation, diabetes, medicines, pregnancy, obstruction, or overactive bladder.

Frequent Urination

The need to urinate more often than usual, which may result from increased fluid intake, urinary infection, diabetes, pregnancy, medicines, or bladder disorders.

Fresh Specimen

A newly collected biological sample examined promptly or preserved appropriately before degradation alters its cells, organisms, chemistry, or diagnostic features.

Fresolimumab

A medicine or chemical compound called fresolimumab. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Freund'S Adjuvant

A research adjuvant made from mineral oil; complete Freund's adjuvant also contains killed mycobacteria and is not used as a routine human vaccine additive.

Frey'S Syndrome

Aberrant regeneration of parasympathetic nerve fibers to sweat glands after injury to the auriculotemporal nerve, most commonly following parotid surgery or trauma. It results in gustatory sweating and facial flushing while eating. Diagnosis is confirmed by the starch-iodine test.

Friable

Easily crumbled, broken, or torn; tissue described as friable may bleed readily when touched.

Friction

Friction is an anatomical, developmental, reproductive, nutritional, behavioral, or healthcare concept used in medical assessment or care.

Friedman Curve

The classic labor curve described by Emanuel Friedman in the 1950s, plotting cervical dilatation against time. Defined the latent and active phases, with active phase beginning at 4 cm. The Friedman curve was the standard for labor management for decades.

Friedreich Ataxia

An inherited neurodegenerative disorder causing progressive gait and limb ataxia, sensory loss, weakness, and sometimes cardiomyopathy or diabetes.

Friedreich'S Ataxia

An inherited degenerative disorder causing progressive problems with coordination, sensation, and heart function.

Friedreichs Ataxia Syndrome

Friedreich ataxia is an inherited neurodegenerative disorder, usually autosomal recessive, caused by reduced frataxin and mitochondrial dysfunction. It causes progressive gait and limb ataxia, neuropathy, scoliosis, cardiomyopathy, and sometimes diabetes.

Friend Virus

A mouse retrovirus used in research that can cause erythroleukemia and help study blood-cell cancers.

Frigidity

Frigidity is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Frog Test

A historical pregnancy test in which urine was injected into a frog to look for an ovulation response; it has been replaced by laboratory and home pregnancy tests.

Frontal

Relating to the forehead or the frontal bone and frontal lobe of the brain.

Frontal Bone

Forms the forehead and superior orbits, with the squama frontalis, orbital plates, and nasal part. Contains frontal sinuses draining into the nasal cavity via the frontonasal duct. Articulates with parietal bones at the coronal suture. The metopic suture fuses by age six.

Frontal Bossing

Prominent forward bulging of the frontal bones and forehead, which may occur as a normal familial feature or with conditions such as rickets, hydrocephalus, or skeletal dysplasia.

Frontal Lobe

The largest lobe of the cerebral cortex, lying anterior to the central sulcus, comprising the primary motor cortex (precentral gyrus, Brodmann 4), premotor and supplementary motor areas, prefrontal cortex, and Broca's area (dominant hemisphere, expressive language).

Frontal Lobe Epilepsy

Seizures originating in the frontal lobe, characterized by motor symptoms, behavioral changes, or nocturnal events. Often occurs during sleep. Semiology depends on the specific frontal lobe region involved. May be refractory to medication.

Alternate Names:

Frontal lobe seizures

Frontal Lobe Seizures

Seizures beginning in the frontal lobes of the brain, often causing brief unusual movements, behaviors, sensations, or impaired awareness.

Alternate Names:

Epilepsy, Frontal Lobe

Frontal Sinus

The frontal sinuses are paired, air-filled cavities located within the frontal bone above the orbits and the root of the nose, separated by a bony septum that may be deviated to one side, and they are the last of the paranasal sinuses to develop, typically appearing around age seven and reaching full size by late adolescence.

Frontal Sinusitis

Inflammation or infection of the frontal sinuses, causing forehead pain or pressure, nasal congestion, and drainage. Severe headache, swelling around the eye, high fever, confusion, or neurologic symptoms may indicate a complication requiring urgent care.

Frontotemporal Dementia

Dementia affecting frontal and temporal lobes. Behavioral variant (personality changes) or language variant (aphasia). Younger onset (50-60).

Alternate Names:

Dementia, Frontotemporal

Frontotemporal Lobar Degeneration

A group of progressive disorders affecting the frontal and temporal brain lobes, causing changes in behavior, language, or executive function.

Frostbite

Frostbite is freezing injury to tissue, causing numbness, pallor, waxy skin, blistering, and possible tissue necrosis; rewarming and assessment are urgent. degree (superficial, numbness, erythema), second degree (blisters, clear fluid, pain), third degree (hemorrhagic blisters, necrosis, full-thickness skin loss), fourth degree (deep tissue necrosis, mummification, autoamputation).

Frotteuristic Disorder

A paraphilic disorder involving recurrent, intense sexual arousal from touching or rubbing against a nonconsenting person, lasting ≥ 6 months and causing distress or impairment or involving acting on these urges. Typically occurs in crowded public places (subways, buses, concerts).

Frozen Section

An intraoperative pathology consultation in which a fresh tissue specimen is rapidly frozen, sectioned on a cryostat, and stained with hematoxylin and eosin within 10 to 20 minutes to guide immediate surgical decision-making. It is used to establish benign versus malignant, assess margins, and identify lymph node metastases.

Frozen Shoulder

Adhesive capsulitis: progressive shoulder pain and stiffness. Stages: freezing (pain), frozen (stiffness), thawing (recovery). Associated with diabetes, thyroid disease, immobilization. Treatment: physical therapy, corticosteroid injection, manipulation under anesthesia, arthroscopic capsular release.

Alternate Names:

Adhesive Capsulitis Of Shoulder (Frozen Shoulder)

Fructosamine

A measurement of glycated serum proteins, pri-

marily albumin, that reflects average glycemia over approximately two to three weeks. It is useful when HbA1c is unreliable because of altered red-cell survival, but it is affected by changes in protein concentration and turnover.

Fructose

A simple sugar found naturally in fruit, honey, and some syrups.

Alternate Names:

Fruit sugar

Fructose Intolerance

Fructose Intolerance is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fructose Malabsorption

Poor absorption of fructose leading to GI symptoms. Fermentation by bacteria produces gas, osmotic diarrhea. Low-fructose diet beneficial.

Fructose Metabolism

The enzymatic conversion of fructose primarily in the liver through fructokinase, aldolase B, and downstream glycolytic intermediates. Essential fructosuria is usually benign, whereas aldolase B deficiency causes hereditary fructose intolerance and hypoglycemia after fructose ingestion.

Fruity Breath

Fruity Breath is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Frustration

An emotional response to blocked goals or unmet needs, potentially producing irritability, anger, distress, avoidance, or behavioral change.

Fryns Syndrome

A rare usually autosomal-recessive congenital syndrome with diaphragmatic hernia, distinctive facial

features, distal limb abnormalities, and variable brain, heart, kidney, or genital anomalies.

FSH

Follicle-stimulating hormone, a pituitary hormone involved in ovarian follicle development and sperm production.

Alternate Names:

Follicle-stimulating hormone

Ft.

Abbreviation for foot or feet, a unit of length equal to 12 inches or approximately 30.48 centimeters; in clinical notes, context should distinguish it from other abbreviations.

FTA-ABS Blood Test

A treponemal antibody blood test that detects antibodies to *Treponema pallidum*; it supports syphilis diagnosis but usually remains positive after treated infection and cannot determine disease activity alone.

FTA-ABS Test

Fluorescent treponemal antibody absorption test, a blood test that detects antibodies against *Treponema pallidum* and supports evaluation of syphilis; antibodies may remain detectable after treatment.

Fuchs Dystrophy

A progressive corneal endothelial disorder in which endothelial cell loss impairs fluid regulation and causes corneal oedema. Early disease may cause morning blur and glare; advanced disease produces persistent oedema, bullae, pain, and reduced vision. Diagnosis is clinical and supported by slit-lamp examination and corneal imaging.

Fuchs Endothelial Corneal Dystrophy

A bilateral, progressive disorder in which corneal endothelial cells are lost and abnormal Descemet-membrane excrescences called guttae develop. Reduced endothelial fluid-pump function causes morning-predominant blurred vision, corneal edema, and, in advanced disease, painful epithelial bullae. It commonly becomes clinically significant in middle

or older age, although early disease may be asymptomatic. Treatment ranges from monitoring and hypertonic measures to endothelial keratoplasty for visually disabling or painful disease.

Alternate Names:

Fuchs dystrophy

Fuchs' Dystrophy

A progressive disorder in which corneal endothelial cells deteriorate, causing corneal swelling and impaired vision.

Fucose

Fucose is a deoxy sugar found in cell-surface glycoproteins and glycolipids and involved in recognition, adhesion, and immune signaling.

Fucosidosis

A rare inherited metabolic disorder caused by deficiency of the enzyme alpha-L-fucosidase.

Fucosyltransferase

An enzyme that transfers fucose to glycans on proteins or lipids, affecting cell recognition, blood-group antigens, development, and host-pathogen interactions.

Fuel Oil Poisoning

Fuel oil poisoning involves exposure or injury from fuel oil poisoning; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Fugue

Fugue is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fukuyama Congenital Muscular Dystrophy

An autosomal recessive congenital muscular dystrophy-dystroglycanopathy characterized by severe muscular dystrophy combined with structural brain and ocular malformations. Caused by an

ancestral 3-kb retrotransposal insertion into the 3' untranslated region of the *FKTN* gene on chromosome 9q31, which encodes fukutin, an enzyme required for functional *O*-mannosyl glycosylation of α -dystroglycan; highly prevalent in the Japanese population. Manifests in the neonatal period with severe hypotonia, generalized muscle weakness, joint contractures, cobblestone (type II) lissencephaly with polymicrogyria, severe intellectual disability, microphthalmia, and epilepsy, typically progressing to loss of ambulation and childhood cardiomyopathy.

Alternate Names:

FCMD; Fukuyama Type Congenital Muscular Dystrophy

Fulguration

Destruction of tissue by applying electrical energy, commonly during a surgical procedure.

Full Agonist

A full agonist is a drug that binds to a receptor and produces the maximum possible biological response that the receptor can generate, achieving 100 percent of the maximal effect. Full agonists have both high affinity for the receptor and high intrinsic activity, meaning they are very effective at activating the receptor once bound.

Full Blood Count

See *Complete Blood Count*.

Alternate Names:

FBC

Full Liquid Diet

A patient-care topic concerning full liquid diet. It addresses the relevant purpose, risks, management, and follow-up.

Full Sibling

A person who shares both biological parents with another person, relevant to pedigree analysis, genetic risk assessment, and tissue-donor matching.

Full-Thickness Skin Graft

A skin graft containing the entire epidermis and dermis, transferred to cover a wound or reconstruct a defect.

Fullerene

fullerene is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Fulminant

Fulminant is a disease, syndrome, symptom, or clinical finding; its causes, distinguishing features, investigation, and treatment depend on the patient and severity.

Fulminant Hepatitis

Rapid, severe loss of liver function caused by extensive hepatic injury, often accompanied by coagulopathy and hepatic encephalopathy. It is a medical emergency that may require intensive care and liver transplantation.

Fulminant Myocarditis

A rapidly progressive form of myocarditis causing severe ventricular dysfunction, cardiogenic shock, or life-threatening arrhythmias. It requires urgent intensive-care assessment and treatment of shock and arrhythmias; temporary mechanical circulatory support may be needed while myocardial function recovers or definitive therapy is arranged.

Fulvestrant

An estrogen-receptor antagonist and degrader given by intramuscular injection for selected hormone-receptor-positive advanced breast cancer.

Fumigation

Treatment of an enclosed area with a gas, vapor, or smoke to kill pests or microorganisms; people must follow safety procedures before re-entry.

Functional Asplenia

Markedly reduced or absent splenic function despite

splenic tissue being present. It may result from repeated infarction in sickle cell disease or other disorders and increases the risk of severe bloodstream infection, especially from encapsulated bacteria.

Functional Assessment

Evaluation of ability to perform daily activities. Tools: Barthel Index, Lawton IADL Scale, Katz Index. Identifies needs for assistance, predicts outcomes, guides care planning.

Functional Capacity

The maximum level of physical, cognitive, communication, or self-care performance a person can safely achieve under specified conditions. It is assessed in relation to activities, participation, environment, and personal goals.

Functional Disorder

A condition in which symptoms or impaired function occur without a sufficient structural abnormality to explain them. Functional disorders are real and may involve altered nervous-system or physiological processing.

Functional Dyspepsia

Chronic epigastric pain or discomfort without identifiable structural cause. The Rome IV criteria include postprandial distress syndrome with early satiety and postprandial fullness, and epigastric pain syndrome with epigastric pain or burning unrelated to meals.

Alternate Names:

Nonulcer Dyspepsia Nonulcer Stomach Pain

Functional Electrical Stimulation

Application of electrical impulses to peripheral nerves or muscles to produce purposeful contraction during a functional task, such as foot clearance during gait or cycling after neurologic injury.

Functional Gastrointestinal Disorders

Digestive symptoms caused by altered gut function without an evident structural disease or infection.

Alternate Names:

Disorders of gut-brain interaction

Functional Hypothalamic Amenorrhea

A reversible form of secondary chronic anovulation and amenorrhea caused by suppression of the hypothalamic gonadotropin-releasing hormone (GnRH) pulse generator in the absence of an organic pituitary or intracranial lesion. Triggered by persistent negative energy balance, severe psychological stress, rigorous athletic training, or disordered eating, leading to hypoestrogenism, low luteinizing hormone and FSH levels, bone mineral density loss (osteopenia/osteoporosis), and subfertility.

Alternate Names:

Hypothalamic Amenorrhea; FHA

Functional Incontinence

Involuntary urine leakage because a person cannot reach or use the toilet in time despite recognizing the need to void.

Functional Independence Measure

FIM: standardized assessment of disability. Scores 18 (total assistance) to 7 (independent). 18 items: self-care, sphincter control, transfers, locomotion, communication, social cognition.

Functional MRI

An MRI method that estimates brain activity by detecting task-related changes in blood oxygenation and blood flow. It is used in research and in selected preoperative assessments.

Functional Murmur

A heart murmur caused by increased blood flow without a structural abnormality of the heart or valves.

Functional Neurologic Disorder

A condition in which changes in nervous-system function cause symptoms that are not explained by structural damage.

Functional Neurologic

Disorder/Conversion Disorder

Neurologic symptoms such as weakness, abnormal movement, seizures, or sensory changes that are not explained by structural neurologic disease.

Alternate Names:

Conversion Disorder

Functional Neurological Disorder

Neurologic symptoms such as weakness, abnormal movements, sensory loss, or seizures that are incompatible with recognized patterns of structural neurologic disease and arise from altered nervous-system function. Diagnosis is based on positive clinical signs, not merely on normal investigations.

Functional Neurological Symptom Disorder

A condition in which neurological symptoms such as weakness, abnormal movement, sensory loss, or episodes resembling seizures are incompatible with recognized neurological disease patterns and cause significant distress or impairment.

Functional Neurological Symptom Disorder (Conversion Disorder)

A condition in which patients present with neurological symptoms (weakness, paralysis, non-epileptic seizures, sensory loss, gait abnormalities, dystonia, speech difficulties) that cannot be explained by structural neurological disease. The symptoms are not intentionally produced (unlike malingering or factitious disorder).

Alternate Names:

Conversion Disorder

Functional Residual Capacity

The volume of air remaining in the lungs at the end of a normal expiration, equal to the sum of expiratory reserve volume plus residual volume (approximately 2.2-3 L in adults). FRC represents the lung volume at which the outward recoil of the chest wall balances the inward recoil of the lungs, the resting equilibrium position.

Functional Screening

A brief assessment of a person's ability to perform

daily activities, move safely, communicate, and manage basic self-care.

Functional Status

Ability to perform activities of daily living (ADLs) and instrumental ADLs (IADLs). ADLs: bathing, dressing, toileting, transferring, continence, feeding. IADLs: shopping, cooking, managing medications, finances, transportation, housekeeping. Decline indicates need for assistance.

Functional Urinary Incontinence

Urinary incontinence due to physical or cognitive impairment preventing toilet access.

Fundal Height

The distance from the symphysis pubis to the top of the uterus measured in centimeters. Used to estimate gestational age and fetal growth. Generally correlates with weeks of gestation between 20-36 weeks (e.g., 24 cm at 24 weeks). Discrepancies may suggest fetal growth restriction, macrosomia, polyhydramnios, or multiple gestation.

Fundoplication

Surgery that wraps the upper stomach around the lower esophagus to reduce gastroesophageal reflux.

Fundoscopy

Examination of the interior back of the eye, including the retina, optic disc, macula, and blood vessels, using an ophthalmoscope or retinal camera.

Fundus

The rounded or farthest part of a hollow organ; examples include the upper stomach and the posterior interior surface of the eye.

Fundus Oculi

The interior posterior surface of the eye visible through the pupil, including the retina, optic disc, macula, and retinal blood vessels.

Fundus Photography

The documentation of the appearance of the retina,

optic nerve, and retinal vasculature using a specialized camera that illuminates and photographs the posterior segment of the eye, providing a permanent, objective record of retinal findings that can be used for comparison over time, patient education, and telemedicine consultations.

Fundus Uteri

The rounded upper portion of the uterus above the openings of the fallopian tubes.

Fundus, Retinal

The inside back surface of the eye seen during ophthalmoscopy, including the retina, optic disc, macula, and retinal blood vessels.

Fungal

Relating to fungi or caused by a fungus; fungal infections may affect the skin, nails, mucous membranes, lungs, or other organs.

Fungal Arthritis

Infection and inflammation of a joint caused by a fungus, usually after surgery, trauma, injection, bloodstream spread, or in a person with impaired immunity. It often develops gradually with persistent pain, swelling, and reduced movement. Joint-fluid culture and tissue testing guide prolonged antifungal and sometimes surgical treatment.

Fungal Endophthalmitis

Fungal infection of the interior of the eye, often associated with bloodstream infection, surgery, trauma, or immunosuppression and capable of causing severe visual loss.

Fungal Infection

An infection caused by a fungus. It may be superficial, affecting the skin, nails, or mucous membranes, or invasive, affecting the lungs, bloodstream, or other organs. The presentation and risk depend on the organism and the person's immune status.

Fungal Infection

(Tinea/Dermatophytosis)

Superficial fungal infections of skin, hair, or nails caused by dermatophytes (*Trichophyton*, *Microsporum*, *Epidermophyton*). Tinea corporis (ringworm): annular scaling plaque with central clearing. Tinea pedis (athlete's foot): interdigital scaling, maceration. Tinea cruris (jock itch): groin rash sparing scrotum.

Alternate Names:

Tinea/Dermatophytosis

Fungal Infection, Nail

Onychomycosis: a fungal infection of the fingernails or toenails, causing discoloration, thickening, and brittleness. Most commonly caused by dermatophytes. Risk factors include age, diabetes, peripheral vascular disease, and immunosuppression. Treatment includes topical or oral antifungals.

Alternate Names:

Onychomycosis; nail fungus

Fungal Nail Infection

Infection of a fingernail or toenail by dermatophytes or other fungi, causing thickening, discoloration, brittleness, separation from the nail bed, or subungual debris.

Fungal Nails

A fungal infection of the nail, usually caused by dermatophytes, producing thickened, discolored, brittle, or crumbly nails. The infection may spread to surrounding skin and recur. Diagnosis may require microscopy or culture, and treatment depends on the organism, number of nails, severity, and the person's medical history.

Alternate Names:

Onychomycosis (Fungal Nails)

Fungal Pathogenesis

The mechanisms by which fungi cause disease. Fungal pathogenesis involves adhesion, invasion, and evasion of host defences. Adhesins mediate attachment to host cells. Invasive hyphae penetrate tissues. Capsules and melanin protect against immune responses.

Fungal Resistance Mechanisms

Mechanisms by which fungi develop resistance to antifungal agents. Azole resistance occurs through up-regulation of efflux pumps (CDR1, CDR2, MDR1), mutations in the target enzyme ERG11, and over-expression of ERG11. Echinocandin resistance is primarily due to mutations in the FKS1 and FKS2 genes encoding beta-(1,3)-D-glucan synthase.

Fungal Skin Infections

Skin infections caused by dermatophytes or yeasts, including ringworm (tinea), athlete's foot, and candidiasis. These infections thrive in warm, moist environments and present with red, itchy, scaly, or ring-shaped lesions. Treatment typically involves topical or oral antifungal medications and measures to keep the skin clean and dry.

Fungal Vaccines

Vaccines designed to prevent fungal infections; most remain experimental, and routine human use is limited.

Fungating

Describing a mass or lesion that grows outward with an irregular, ulcerated, necrotic, or malodorous surface, commonly in advanced malignancy or severe chronic wounds.

Fungemia

The presence of viable fungi in the bloodstream. *Candida* species are the most common cause in hospitalized patients; fungemia requires prompt evaluation for a source, metastatic infection, and appropriate antifungal treatment.

Fungi

Organisms such as yeasts, molds, and mushrooms whose cells usually have chitin-containing walls; some cause human infection.

Fungicide

Fungicide is a medicine, drug, vitamin, or chemical product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent.

Fungoid

Resembling a fungus or mushroom in appearance; in medicine it may describe a mushroom-like lesion or growth.

Fungus

A eukaryotic microorganism with a chitin cell wall and ergosterol-containing membrane, classified as yeasts, moulds, or dimorphic fungi. Reproducing by spores, fungi cause superficial, mucosal, sub-cutaneous, and systemic infections, especially in immunocompromised hosts.

Funiculus

Funiculus is a muscle, nerve, vessel, reproductive structure, or other body part; its location and function determine its clinical relevance.

Funisitis

Inflammation of the umbilical cord, usually associated with infection or an inflammatory response.

Funnel Chest

Pectus excavatum: a congenital chest wall deformity characterized by depression of the sternum and adjacent costal cartilages. Severity ranges from mild to severe. May cause cardiac or pulmonary compression in severe cases. Surgical correction is an option for symptomatic or severe deformity.

Alternate Names:

Pectus excavatum

Funnel-Web Spider Bite

Funnel-web spider bite involves exposure or injury from funnel-web spider bite; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Furan

furan is a chemical, toxicology, laboratory, or environmental term; significance depends on the substance, concentration, route, duration, and exposure circumstances.

Furazolidone

A nitrofuran antimicrobial formerly used for selected bacterial or protozoal infections; availability is restricted in many settings because of adverse effects and interactions.

Furniture Polish Poisoning

Furniture polish poisoning involves exposure or injury from furniture polish poisoning; dose, route, timing, symptoms, and tissue involvement determine the urgency and treatment approach.

Furocoumarin

A plant compound that can increase sensitivity to ultraviolet light and cause skin reactions when combined with sunlight.

Fursultiamin

A fat-soluble derivative of thiamine, or vitamin B1, used in some countries as a vitamin preparation.

Furuncle

A painful, pus-filled infection of a hair follicle, usually caused by *Staphylococcus aureus*. It may form an erythematous nodule with a central necrotic core or progress to a skin abscess. Treatment depends on size, location, recurrence, and systemic illness and may include drainage or antimicrobial therapy.

Furuncle And Carbuncle

Furuncle (boil) is a deep infection of the hair follicle caused by *Staphylococcus aureus*, presenting as a painful, erythematous nodule that may drain pus. Carbuncle is a cluster of interconnected furuncles causing a larger, more deeply invasive infection.

Furuncular Myiasis

A cutaneous infestation caused by larvae of certain fly species (such as *Dermatobia hominis* or *Cuterebra*) that develop within boil-like furuncular lesions in the skin, presenting with a painful, erythematous nodule with a central punctum through which the larva breathes.

Furunculosis

Recurrent or multiple furuncles, which are deep infections of hair follicles and surrounding tissue, usually caused by *Staphylococcus aureus*. Lesions are painful, erythematous, and purulent and may occur with colonization, close-contact transmission, skin disease, diabetes, immune dysfunction, or other risk factors.

Fusariosis

An opportunistic invasive fungal infection caused by *Fusarium* species, affecting predominantly immunocompromised patients, particularly those with prolonged neutropenia or hematological malignancies. It causes disseminated infection with skin lesions, bloodstream infection, and involvement of multiple organs.

Fusarium

fusarium is a fungal taxon or fungus-related term; clinical interpretation depends on the species, site, host immune status, and evidence of invasive or superficial infection.

Fusarium Chlamydosporum

fusarium chlamydosporum is a fungal taxon or fungus-related term; clinical interpretation depends on the species, site, host immune status, and evidence of invasive or superficial infection.

Fusarium Incarnatum

fusarium incarnatum is a fungal taxon or fungus-related term; clinical interpretation depends on the species, site, host immune status, and evidence of invasive or superficial infection.

Fusarium Infection

An opportunistic infection caused by *Fusarium* species, environmental molds that can produce keratitis, onychomycosis, or invasive disease. Invasive infection occurs mainly in people with severe neutropenia, hematologic malignancy, transplantation, or other major immunosuppression. Skin lesions, sinus or lung disease, and bloodstream infection may occur, with dissemination to multiple organs in severe cases. Diagnosis requires culture or

histopathology with species-level evaluation when possible; treatment requires urgent specialist antifungal management and correction of immunosuppression when feasible.

Fusarium Solani

fusarium solani is a fungal taxon or fungus-related term; clinical interpretation depends on the species, site, host immune status, and evidence of invasive or superficial infection.

Fused Placenta

A placenta in which normally separate lobes or placental regions are joined, sometimes affecting how the placenta is examined after delivery.

Fused Teeth

Alternate index label for dental fusion.

Fusidic Acid

An antibiotic that inhibits bacterial elongation factor G and is used mainly for selected staphylococcal infections; resistance and formulation vary by region.

Fusiform

Shaped like a spindle, wider in the middle and tapering toward both ends; the term can describe a swelling, muscle, cell, or blood-vessel abnormality.

Fusion

In dentistry, developmental union of two separate tooth germs, producing a joined or enlarged tooth and often a reduced apparent tooth count. Fusion with a supernumerary tooth may preserve the normal tooth count. It should be distinguished from gemination, which results from attempted division of a single tooth germ.

Alternate Names:

Dental fusion Tooth fusion

Fusion Gene

A hybrid gene formed when parts of two separate genes join through chromosomal rearrangement or other mechanism, sometimes producing an onco-

genic fusion protein.

Fusion Of The Ear Bones

Alternate descriptive label; see *Ossicular fusion*.

Fusion Protein

A protein encoded by a fusion gene or engineered by joining domains from different proteins, potentially changing signaling, localization, or therapeutic activity.

Fusobacterium

A genus of anaerobic gram-negative bacteria found especially in the mouth and intestine; some species cause abscesses and invasive infection.

Fusobacterium Infection

An infection caused by *Fusobacterium* bacteria, which may involve the mouth, throat, lungs, abdomen, skin, or bloodstream.

Fusobacterium Mortiferum

A *Fusobacterium* species that can rarely be involved in anaerobic infections, including infections of wounds or the bloodstream.

Fusobacterium Naviforme

A *Fusobacterium* species rarely identified in human specimens and occasionally associated with anaerobic infection.

Fusobacterium Necrogenes

A *Fusobacterium* species that may be found in mixed anaerobic infections and is uncommon in human disease.

Fusobacterium Necrophorum

A *Fusobacterium* species that can cause severe throat infection and Lemierre syndrome, in which infection spreads to the neck veins.

Fusobacterium Nucleatum

A common oral bacterium associated with periodontal disease and, occasionally, bloodstream or other invasive infections.

Fusobacterium Russii

A *Fusobacterium* species rarely associated with human anaerobic infection.

Fusobacterium Varium

A *Fusobacterium* species that can occasionally be isolated from mixed anaerobic infections.

Fusospirochetal Gingivitis

A severe form of gingival inflammation, traditionally associated with fusobacteria and spirochetes, causing painful swollen or bleeding gums, ulceration, and foul breath; it is also called acute necrotizing ulcerative gingivitis.

Fibular Artery

Also called the peroneal artery; a branch of the posterior tibial artery that supplies the lateral and posterior compartments of the leg and contributes to blood flow around the ankle.

G

G Spot

A proposed area of sexually sensitive tissue on the front wall of the vagina.

Alternate Names:

Grafenberg spot

G-Protein Coupled Receptors

Proteins on the surface of cells that receive signals from hormones, medicines, and other substances. They pass these signals into the cell and help control processes such as heart rate, mood, vision, and inflammation.

G-Quadruplexes

Four-stranded nucleic-acid structures formed by guanine-rich DNA or RNA sequences; they influence gene regulation and genome stability and are studied as possible therapeutic targets.

G6PD Deficiency

An inherited condition in which red blood cells are more easily damaged by oxidative stress. Certain medicines, infections, or fava beans can trigger their breakdown, causing anemia, jaundice, or dark urine.

Gaba Receptor

A protein that responds to GABA, a chemical messenger that usually slows activity in the brain. GABA receptors help regulate sleep, anxiety, muscle activity, and seizures.

Gabapentin

An anticonvulsant also used for neuropathic pain and selected other pain syndromes; dizziness, sedation, and dose adjustment in kidney disease are

important considerations.

GAD

Generalized Anxiety Disorder: a condition characterized by excessive, difficult-to-control anxiety and worry about multiple events or activities occurring more days than not for at least six months. Associated symptoms include restlessness, fatigue, difficulty concentrating, irritability, muscle tension, and sleep disturbance.

Alternate Names:

Generalized Anxiety Disorder

Gadoleic Acid

A 20-carbon monounsaturated fatty acid found in some plant and marine oils and incorporated into dietary and cellular lipids.

Gadolinium

Gadolinium is a lanthanide metal used in chelated form as a contrast agent for MRI; severe reactions are uncommon, but tissue retention and nephrogenic systemic fibrosis matter in advanced kidney disease.

Gadolinium-Based Contrast Agent

A substance injected for some MRI scans to make organs, blood vessels, or abnormal areas easier to see. The choice and dose depend on kidney function, pregnancy, and prior reactions.

Gadolinium-Induced Nephrogenic Systemic Fibrosis

A rare, serious condition that can occur mainly in people with severe kidney disease after some gadolinium MRI contrast exposure. It causes hardening

and thickening of the skin and may affect internal organs.

Gadopentetate Dimeglumine

Gadopentetate dimeglumine is a gadolinium-based extracellular MRI contrast agent that shortens T1 relaxation and increases lesion conspicuity; renal function and prior contrast reactions should be considered.

Gaffkya Homari

gaffkya homari is a microorganism or biological taxon; its clinical importance depends on the species, specimen, host, and evidence of infection or colonization.

Gag Gene

The retroviral gene encoding structural proteins that form the viral core, matrix, and capsid and help assemble new virus particles.

Gag Protein

A retroviral structural polyprotein processed into matrix, capsid, nucleocapsid, and related components required for virion assembly.

Gag Reflex (Pharyngeal Reflex)

A protective reflex in which stimulation of the back of the throat produces contraction of the pharyngeal muscles and may trigger retching. It helps protect the airway from foreign material.

Alternate Names:

Pharyngeal Reflex

Gagging

A protective retching response triggered by stimulation of the pharynx, posterior tongue, or nearby structures. It may occur during swallowing, examination, aspiration, reflux, infection, or emotional distress; persistent gagging with dysphagia, vomiting, respiratory symptoms, or weight loss requires assessment.

Gain

The amount by which a hearing aid increases sound

intensity, measured in decibels.

Gain-Of-Function Mutation

A genetic change that makes a gene or its protein more active, gives it a new function, or causes it to work in the wrong place or at the wrong time.

Gait

The pattern or style of walking. Gait is a complex motor function involving coordination of the central and peripheral nervous systems, muscles, and joints. Gait assessment evaluates stance, stride length, cadence, arm swing, truncal stability, and turning.

Gait Abnormality

A deviation from normal walking caused by musculoskeletal, neurological, vestibular, visual, cardiovascular, or systemic disease. Assessment considers speed, base, symmetry, balance, stride, turning, assistive-device use, and associated weakness, sensory loss, pain, or cognitive impairment.

Gait Apraxia

Difficulty initiating or correctly executing walking despite adequate strength, often associated with frontal-lobe, vascular, degenerative, or hydrocephalus-related disease.

Gait Assessment

An examination of how a person walks. It can provide clues about problems with strength, sensation, balance, coordination, joints, or thinking.

Gait Ataxia

An unsteady, poorly coordinated gait caused by dysfunction of the cerebellum, sensory pathways, vestibular system, or other neurologic structures.

Gait Cycle

The sequence of leg and foot movements from one heel strike to the next heel strike of the same foot.

Gait Disturbance

An abnormality of walking that may involve speed,

balance, coordination, strength, or the pattern of movement. Causes include neurologic, musculoskeletal, vestibular, and systemic disorders.

Gait Speed

Walking velocity measured over a standardized distance. It is a simple marker of mobility, frailty, disability, and mortality risk and can track functional change over time.

Gait Training

Teaching or retraining walking pattern. Techniques: parallel bars, walker, cane, stair training. Addresses abnormalities: limping, antalgic gait, hemiplegic gait, parkinsonian gait. Goals: safety, efficiency, independence.

Gaits

Gaits is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Galactans

Galactans are polysaccharides made mainly of repeating galactose units.

Galactoceles

A milk-filled retention cyst of the breast, occurring during lactation or after weaning. Results from obstruction of a milk duct. Presents as a painless, fluctuant breast mass. Diagnosis by ultrasound showing a well-defined, hypoechoic or anechoic mass. Treatment includes aspiration.

Galactocerebrosidase Deficiency

An inherited lack of an enzyme needed to maintain the protective covering around nerve cells. Severe deficiency causes Krabbe disease, which damages the nervous system.

Alternate Names:

Galactocerebrosidase deficiency; galactosylceramidase deficiency.

Galactagogue

A medicine, herb, or other substance intended to in-

crease breast-milk production. Evidence and safety vary, so feeding technique, milk removal, medical causes, and possible drug effects should be assessed before use.

Galactolipids

Lipids containing galactose, important in plant membranes and nervous-system myelin and involved in membrane structure and recognition.

Galactorrhoea

Milk-like nipple discharge unrelated to breastfeeding or the period after childbirth. It may result from increased prolactin, medicines, pregnancy, endocrine disease, breast stimulation, or other conditions and requires assessment when persistent or unexplained.

Galactorrhoea

The spontaneous flow of milk from the breast not associated with childbirth or breastfeeding. Galactorrhoea can be unilateral or bilateral and is caused by hyperprolactinaemia (elevated prolactin).

Galactosaemia

A rare inherited condition in which the body cannot properly break down galactose, a sugar found in milk. It can cause feeding problems, jaundice, liver injury, cataracts, serious infection, or developmental problems in infants.

Galactose

A simple sugar, commonly present in lactose, that is metabolized through the Leloir pathway; inherited galactosemia can make exposure dangerous in infancy.

Galactose Metabolism

The way the body breaks down galactose, a sugar found in milk, so it can be used for energy. Problems with this process cause galactosemia.

Galactose-1-Phosphate Uridyltransferase Blood Test

A blood test measuring galactose-1-phosphate uridyltransferase activity, used to screen for or di-

agnose classic galactosemia, an inherited disorder of galactose metabolism.

Galactosemia

Galactosemia is an inherited inability to metabolize galactose, causing neonatal feeding problems, jaundice, liver injury, infection risk, and developmental complications without dietary treatment. galactose-1-phosphate uridyltransferase deficiency. Untreated classic disease can cause jaundice, liver dysfunction, vomiting, cataracts, sepsis, and neurodevelopmental impairment after milk feeding.

Galactosialidosis

A rare lysosomal-storage disorder caused by deficiency of protective protein/cathepsin A, leading to accumulation of oligosaccharides and variable neurologic, skeletal, renal, and eye findings.

Galactoside

A galactoside is a compound in which galactose is linked to another molecule, such as a sugar, lipid, or protein.

Galactosyltransferase

Galactosyltransferase is an enzyme that transfers galactose to another molecule during the production of glycoconjugates.

Galanin

A neuropeptide distributed in the brain and peripheral nervous system that influences pain, appetite, mood, hormone release, and gastrointestinal function.

Galant Reflex

A primitive reflex present from birth to 4 months, elicited by stroking the skin along the side of the infant's back. The infant curves toward the stimulated side. Assessed by running a finger from the lateral back toward the hip while the infant is in ventral suspension. Absence may indicate neurologic abnormality.

Galeazzi Sign

A physical examination finding used to screen for unilateral developmental dysplasia of the hip or femoral shortening in infants. With the infant lying supine, hips and knees flexed to 90 degrees, and feet placed flat together on the examination table, an unequal knee height indicates a positive sign, caused by the posterior displacement or dislocation of the femoral head on the affected side.

Alternate Names:

Allis Sign; Galeazzi Test

Galiximab

A medicine or chemical compound called galiximab. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Gall Bladder Disease

A disorder of the gallbladder, including stones, inflammation, infection, polyps, dyskinesia, or cancer, often causing right-upper-abdominal pain or nausea.

Gall-Bladder

A small, pear-shaped sac beneath the liver that stores bile, a fluid that helps digest fat.

Gall-Bladder, Diseases Of

Conditions affecting the gallbladder, including gallstones, inflammation, infection, polyps, and cancer. Symptoms and treatment depend on the specific disorder.

Gallamine Triethiodide

An older muscle-paralyzing medicine that blocks nerve signals to muscles and was used during anesthesia. It is not commonly used today.

Gallavardin Phenomenon

The dissociation of the murmur of aortic stenosis into two components: a musical quality heard at the apex and a harsh, rough quality heard at the right upper sternal border. This occurs because high-frequency components are transmitted to the apex while low-frequency components are heard at the base.

Gallbladder

A small, pear-shaped sac under the liver that stores and concentrates bile before releasing it into the small intestine to aid fat digestion.

Gallbladder Absence

Congenital absence or surgical removal of the gallbladder; without a gallbladder, bile flows directly from the liver into the intestine.

Gallbladder Cancer

A malignant tumor arising in the gallbladder, often associated with chronic inflammation or gallstones. It may be discovered incidentally or present with abdominal pain, jaundice, or other biliary symptoms.

Alternate Names:

Cancer, Gallbladder

Gallbladder Carcinoma

A malignant tumour arising from the gallbladder, most often an adenocarcinoma. It may be discovered incidentally after cholecystectomy or present with right-upper-quadrant pain, jaundice, weight loss, or biliary obstruction; staging determines surgical and systemic treatment.

Gallbladder Duplication

A rare congenital anomaly in which two gallbladder structures develop, with either separate or shared ducts. It may be incidental or associated with gallstones, inflammation, or bile-duct injury risk during surgery; preoperative imaging helps define the anatomy.

Gallbladder Infection

Inflammation or infection of the gallbladder, often associated with blockage of bile flow.

Gallbladder Inflammation

Cholecystitis: inflammation of the gallbladder, most commonly caused by gallstones blocking the cystic duct. Presents with right upper quadrant pain, fever, nausea, and vomiting. Acute cholecystitis may require antibiotics and cholecystectomy.

Alternate Names:

Cholecystitis

Gallbladder Mucocele

Another name for hydrops of the gallbladder: distension caused by cystic-duct obstruction and accumulation of mucus or watery secretions, commonly from an impacted gallstone.

Gallbladder Necrosis

Death of gallbladder tissue, usually from severe inflammation or loss of blood supply, and a potential precursor to perforation and sepsis requiring urgent care.

Gallbladder Pain (Gall Bladder Pain)

Biliary colic: pain caused by gallstones temporarily blocking the cystic duct. Characterized by severe, steady right upper quadrant pain lasting 30 minutes to several hours, often after fatty meals. May be accompanied by nausea and vomiting.

Alternate Names:

Biliary colic; gallstone pain Pain Gallbladder (Gallbladder Pain (Gall Bladder Pain))

Gallbladder Perforation

A hole in the gallbladder wall that can release bile or infected contents into the abdomen, causing peritonitis, abscess, or sepsis.

Gallbladder Radionuclide Scan

A nuclear medicine study using a radiotracer to assess bile flow through the liver, gallbladder, and bile ducts, particularly for suspected acute cholecystitis or obstruction.

Alternate Names:

Laparoscopic cholecystectomy discharge

Alternate Names:

Open cholecystectomy discharge

Gallbladder Volvulus

Axial torsion of the gallbladder around its mesentery and cystic duct, most commonly affecting elderly women, leading to acute gallbladder ischemia. Presents with sudden onset severe right upper quadrant pain, nausea, vomiting, and signs of peritoneal

irritation.

Gallic Acid

A plant-derived phenolic acid found in tea, fruit, and other foods and studied for antioxidant, antimicrobial, and experimental anticancer effects.

Gallium

Gallium is a metallic element used in some radio-tracers and medical imaging and in semiconductor technology; its clinical interpretation depends on the isotope, compound, and distribution.

Gallium Nitrate

A gallium-containing compound formerly used in selected oncology settings and studied for antimicrobial or bone effects; use is limited by toxicity and clinical indication.

Gallium Scan

A nuclear medicine scan that uses a small amount of radioactive gallium to look for some infections, areas of inflammation, or tumors. Other scans are now often preferred.

Gallium-68

PET radionuclide, half-life 68 minutes. Produced from Ge-68/Ga-68 generator. Used in DOTATATE, PSMA, and other theranostic applications.

Gallop Rhythm

An extra heart sound producing a three- or four-beat cadence, usually from an S3 or S4; it can reflect altered ventricular filling or cardiac disease.

Gallstone

A solid concretion formed from bile components within the gallbladder or biliary tract, usually composed mainly of cholesterol, pigment, or a mixture. It may be asymptomatic or obstruct the cystic or common bile duct, causing biliary colic, cholecystitis, pancreatitis, cholangitis, or jaundice.

Gallstone Ileus

A bowel blockage caused by a large gallstone that

has passed from the gallbladder into the intestine. It can cause severe abdominal pain, vomiting, and swelling and usually needs urgent treatment.

Gallstone Pancreatitis

Acute pancreatitis caused by a gallstone transiently obstructing the ampulla, where the bile duct and pancreatic duct empty into the duodenum. It typically causes sudden persistent upper-abdominal pain with raised pancreatic enzymes; severity assessment and evaluation for retained common-bile-duct stones are required.

Gallstones

Gallstones are solid deposits in the gallbladder that may be asymptomatic or obstruct bile flow, causing biliary colic, cholecystitis, pancreatitis, or jaundice. obstruct the biliary tract, producing biliary pain, inflammation, jaundice, or pancreatitis.

Alternate Names:

Cholelithiasis (Gallstones)

Gallstones (Cholelithiasis)

Solid deposits that form in the gallbladder, usually from cholesterol or bile pigments. They may cause no symptoms or obstruct the biliary tract, producing biliary pain, inflammation, jaundice, or pancreatitis.

Alternate Names:

Cholelithiasis

Alternate Names:

Biliary colic discharge; cholelithiasis discharge

Galvanism

The application of a direct electric current for therapeutic or diagnostic purposes. Named after Luigi Galvani (1737-1798). Historical uses: galvanic stimulation of muscles, iontophoresis (introduction of ionised drugs through skin), electroconvulsive therapy (modified).

Galvanocautery

Galvanocautery uses heat produced by an electric current to destroy tissue or control bleeding.

Gambling Disorder

A persistent pattern of gambling with impaired control, increasing priority given to gambling, and continuation despite negative consequences. It must cause significant impairment or distress; the term includes the former labels gambling addiction, compulsive gambling, and pathological gambling.

Alternate Names:

Gambling Addiction; Compulsive Gambling; Gambling, Compulsive; Pathological Gambling

Gamete

A mature reproductive cell, such as a sperm or ovum, containing one set of chromosomes and combining with another gamete at fertilization.

Gamete Intrafallopian Transfer

An assisted-reproduction procedure in which eggs and sperm are placed into a fallopian tube so fertilization can occur in the body; it is now used less often than in-vitro fertilization.

Gametogenesis

Formation and maturation of sperm or eggs through meiosis and cellular differentiation in the testes or ovaries.

Gamma Camera

A machine used for nuclear medicine scans. It detects radiation from a small amount of injected radioactive medicine and creates pictures of organs or body functions.

Gamma Globulin

The fraction of blood plasma containing most immunoglobulins, or antibodies.

Gamma Hydroxybutyrate

A central-nervous-system depressant that can cause sedation, confusion, respiratory depression, coma, or death in overdose; it is also a drug of misuse.

Gamma Knife

Gamma Knife is a focused radiation treatment that uses many precisely aimed beams to treat selected

brain tumors and other brain lesions without an open operation.

Gamma Rays

Gamma rays are high-energy electromagnetic radiation produced by radioactive decay or other nuclear processes; they can damage cells and are used in some cancer treatments.

Gamma Synuclein

A neuronal and peripheral-tissue protein in the synuclein family that has been investigated in neurodegeneration and cancer; its clinical role remains context-dependent.

Gamma-Aminobutyric Acid

A chemical messenger that usually slows activity in the brain and spinal cord. It helps regulate sleep, anxiety, muscle activity, and seizure risk by acting through GABA-A and GABA-B receptors.

Alternate Names:

GABA

Gamma-Benzene Hexachloride

An insecticide also known as lindane that can cause neurotoxicity and seizures with excessive exposure; medical use is restricted because safer treatments are available.

Gamma-Chemokines

Chemokines that attract or activate immune cells through chemokine receptors and help coordinate inflammation and immune surveillance.

Gamma-Globulin

A fraction of plasma proteins containing antibodies (immunoglobulins). Historically the term referred to a group of serum proteins with the slowest electrophoretic mobility (gamma region on serum protein electrophoresis). Currently synonymous with immunoglobulin preparations used for passive immunisation and immunotherapy.

Gamma-Glutamyl Transferase Blood Test

A blood test measuring gamma-glutamyl transferase,

an enzyme associated with the liver and bile ducts; it helps evaluate cholestatic liver disease and support interpretation of an elevated alkaline phosphatase.

Gamma-Interferon

Interferon-gamma, a cytokine produced mainly by activated T lymphocytes and natural killer cells that activates macrophages and promotes cell-mediated immunity.

Gamma-Ray Therapy

Gamma-Ray Therapy is a medicine or pharmacologic product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent and patient.

Gammacoronavirus

gammacoronavirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Gammaherpesvirina

gammaherpesvirina is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Gammaretrovirus

gammaretrovirus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Ganciclovir

Ganciclovir is a guanosine analogue antiviral that inhibits viral DNA polymerase after phosphorylation and is used mainly against cytomegalovirus; neutropenia, thrombocytopenia, renal toxicity, and teratogenicity are important risks.

Ganglia

Ganglia are clusters of nerve-cell bodies outside the brain and spinal cord that help relay and organize nerve signals.

Ganglioglioma

A usually slow-growing brain or spinal-cord tumor made of nerve cells and supporting glial cells.

Ganglion

A collection of neuronal cell bodies located outside the central nervous system.

Alternate Names:

Bible Cyst (Ganglion)

Ganglion Cell

A neuron whose cell body lies in a peripheral or retinal ganglion; retinal ganglion cells transmit visual signals through the optic nerve.

Ganglion Cyst

A benign, fluid-filled cyst arising near a joint or tendon sheath, most often at the wrist or hand. It may be painless or cause pressure, discomfort, or limitation of movement.

Alternate Names:

Cyst, Ganglion

Ganglionectomy

Ganglionectomy is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Ganglioneuroblastoma

A neuroblastic tumor composed of mature ganglion cells and immature neuroblastic elements. Most often arises in the adrenal or paraspinal region and may present with an abdominal mass, pain, hypertension, or systemic symptoms. Intermediate between ganglioneuroma and neuroblastoma.

Ganglioneuroma

A ganglioneuroma is a usually benign tumor arising from nerve tissue of the autonomic nervous system.

Ganglioneuromatosis

Ganglioneuromatosis is a condition in which ganglion cells and related nerve tissue grow diffusely or in multiple areas.

Ganglionic Blockers

Ganglionic Blockers is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Ganglioside

A ganglioside is a fat-containing molecule in cell membranes, especially nerve-cell membranes, that helps with cell signaling and recognition.

Gangliosidosis

Gangliosidosis is a group of inherited lysosomal storage disorders in which gangliosides accumulate in cells, often damaging the nervous system.

Gangrene

Gangrene is tissue death caused by loss of blood supply, infection, or both; it can progress to systemic toxicity and requires urgent assessment of circulation and infection. bowel, and represents a form of coagulative necrosis. Dry gangrene occurs in tissues with gradual arterial occlusion, where the dead tissue becomes dry, shrunken, and dark brown or black.

Gap Junction

gap junction is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Gardner Syndrome

An inherited APC-related polyposis syndrome with numerous colorectal adenomas and extraintestinal findings such as osteomas, dental abnormalities, and desmoid tumors.

Gardnerella

Gardnerella is a group of bacteria commonly found in the genital tract; overgrowth, especially of *G.*

vaginalis, may contribute to bacterial vaginosis.

Gardnerella Vaginalis

Gardnerella vaginalis is a bacterium associated with bacterial vaginosis, an imbalance of vaginal bacteria that can cause discharge and odor.

Gargalesthesia

The sensation of tickling. A complex sensory experience mediated by light tactile stimulation of sensitive skin areas (axillae, soles, flanks, ribcage). Involves both spinal reflexes (goosebumps, withdrawal) and emotional processing in the anterior cingulate cortex and somatosensory cortex.

Gargatism

A clinical disorder or finding related to gargatism; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Gargle

The act of rinsing the mouth and throat with a liquid, holding it at the back of the throat while exhaling to create a bubbling action. Used for relief of sore throat or as a delivery method for antiseptics, anaesthetics, and anti-inflammatory agents.

Gargoylism

A clinical disorder or finding related to gargoylism; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Garlic

Garlic is an edible plant used as food and sometimes as a supplement; concentrated products may interact with medicines, including blood thinners.

Garré'S Osteomyelitis

A clinical disorder or finding related to garré's osteomyelitis; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Gartner Duct

A persistent remnant of the mesonephric duct in

the female genital tract that can form a vaginal or broad-ligament cyst.

Gartner Duct Cyst

A benign cyst arising from remnants of the mesonephric (Wolffian) duct along the lateral wall of the vagina or uterus. They are typically asymptomatic and discovered incidentally on pelvic examination or imaging. Most require no treatment unless symptomatic, in which case surgical excision is performed.

Gas

Gas is a state of matter without a fixed shape or volume; in the body, gas may refer to air or other gases in tissues, blood, or the digestive tract.

Alternate Names:

Strep Throat (GAS)

Gas And Gas Pains

Gas-related fullness, bloating, or abdominal discomfort caused by swallowed or intestinal gas.

Gas Chromatography

Gas chromatography is a laboratory method that separates and identifies substances that can be vaporized.

Gas Chromatography

An analytical method that vaporizes and separates volatile compounds in a gas stream, allowing their identification or measurement, often with mass spectrometry.

Gas Exchange

The movement of oxygen and carbon dioxide between air and blood in the lungs and between blood and tissues, essential for cellular respiration.

Gas Gangrene

A life-threatening infection in which bacteria produce toxins and gas within damaged tissue, causing rapid muscle destruction. It is commonly associated with *Clostridium* species and requires urgent surgical assessment, source control, and antimicrobial

treatment.

Gas Poisoning

An infectious process or toxic exposure involving gas poisoning; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Gas Scavenger

A gas scavenger is a substance or device that removes unwanted gases from a fluid, container, or medical breathing system.

Gas Transport

Movement of oxygen from the lungs to tissues and carbon dioxide from tissues to the lungs through the blood.

Gas – Flatulence

Flatulence: the passage of gas from the digestive system through the rectum, commonly known as passing gas or farting. Normal physiological process caused by swallowed air and bacterial fermentation of undigested carbohydrates in the colon. Excessive gas may cause bloating, cramps, or social embarrassment.

Alternate Names:

Flatulence; passing gas

Gaskin Maneuver (All-Fours Position)

A maneuver for shoulder dystocia resolution: the mother is rolled onto her hands and knees (all-fours position). Gravity and the change in pelvic dimensions (increased anteroposterior diameter of the pelvic outlet) may dislodge the impacted shoulder. The posterior arm may also be delivered more easily.

Alternate Names:

All-Fours Position

Gasoline Poisoning

Toxic exposure to gasoline (petrol) through ingestion, inhalation, or skin contact. Symptoms include nausea, vomiting, headache, dizziness, and central nervous system depression. Severe cases may cause chemical pneumonitis, seizures, or cardiac arrhythmia.

mias. Requires immediate medical attention.

Alternate Names:

Petrol poisoning

Gasserian Ganglion

Gasserian Ganglion is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gastralgia

Pain localized to the epigastric region of the abdomen or anatomical stomach (gastric pain / gastrodynia). The clinical differential diagnosis is guided by onset, relationship to food intake, and associated red-flag symptoms: burning postprandial epigastric discomfort relieved by antacids suggests peptic ulcer disease (gastric vs. duodenal ulcer) or gastroesophageal reflux; sudden, severe epigastric pain radiating to the back suggests acute pancreatitis; postprandial fullness and bloating without structural disease defines functional dyspepsia; and persistent epigastric pain accompanied by anorexia, early satiety, and unintentional weight loss requires upper gastrointestinal endoscopy to exclude gastric adenocarcinoma.

Gastrectomy

Surgical removal of all or part of the stomach. Indications: gastric cancer (most common), severe peptic ulcer disease, gastric lymphoma, gastrointestinal stromal tumor, perforation, bleeding. Types include total, partial, and laparoscopic gastrectomy.

Gastric Acid

Hydrochloric acid secreted by gastric parietal cells, producing an intragastric pH of 1-3. Secretion is stimulated by histamine (H₂ receptors), gastrin (CCK-B/gastrin receptors), and acetylcholine (M₃ muscarinic receptors), and inhibited by somatostatin, prostaglandins, and luminal acid (negative feedback).

Gastric Adenocarcinoma

The most common malignant epithelial tumor of the stomach, arising from gastric mucosa and associated

with factors such as *Helicobacter pylori* infection, chronic atrophic gastritis, intestinal metaplasia, and hereditary syndromes.

Gastric Analysis

Gastric analysis examines stomach contents, including their volume, acidity, and sometimes enzyme or hormone levels, to help evaluate selected gastric disorders.

Gastric Antral Vascular Ectasia

An uncommon cause of upper gastrointestinal bleeding and iron-deficiency anemia, characterized endoscopically by parallel longitudinal columns of dilated, tortuous mucosal vessels converging on the pylorus (classically termed “watermelon stomach”) or diffuse punctate lesions. Strongly associated with cirrhosis with portal hypertension and autoimmune connective-tissue diseases, particularly systemic sclerosis.

Alternate Names:

GAVE; Watermelon stomach

Gastric Balloon

A temporary endoscopically placed balloon that occupies space in the stomach to support weight loss; complications include nausea, vomiting, obstruction, and perforation.

Gastric Bezoar

Accumulation of indigestible material in the stomach. Phytobezoars consist of vegetable fiber, trichobezoars of hair, and pharmacobezoars of medications. Associated with gastroparesis, prior surgery, and impaired gastric motility. Diagnosed by endoscopy.

Gastric Bypass

Roux-en-Y gastric bypass creating a small proximal pouch drained by a Roux limb, combining restrictive and mildly malabsorptive mechanisms. Indicated for BMI over 40, or over 35 with obesity-related comorbidity. Mean excess weight loss approaches 60-70 percent.

Gastric Bypass Surgery

Bariatric surgical procedure that creates a small stomach pouch and reroutes the small intestine, combining restrictive and malabsorptive mechanisms. Indicated for severe obesity (BMI ≥ 40 or ≥ 35 with comorbidities). Results in significant weight loss and improvement in obesity-related conditions.

Alternate Names:

Roux-en-Y gastric bypass; bariatric surgery

Alternate Names:

Bariatric surgery discharge

Gastric Cancer

A malignant tumor arising from the stomach lining or other stomach tissue.

Gastric Carcinoid Tumor

Neuroendocrine tumor arising from enterochromaffin-like cells in the gastric fundus. Type I gastric carcinoids are associated with autoimmune atrophic gastritis and hypergastrinemia, are typically small and multifocal, and carry a good prognosis.

Gastric Culture

Laboratory test to identify bacteria in the stomach, most commonly used to detect *Helicobacter pylori* infection. Specimens are obtained via endoscopic biopsy. Results guide antibiotic treatment for *H. pylori* eradication.

Alternate Names:

H. pylori culture

Gastric Dysplasia

Abnormal precancerous-appearing growth of gastric lining cells, graded by pathology and associated with increased risk of gastric adenocarcinoma.

Gastric Emptying

The movement of stomach contents into the duodenum through coordinated muscular contractions and pyloric regulation. Delayed emptying is called gastroparesis.

Gastric Emptying Study

Nuclear medicine diagnostic test measuring the rate of gastric emptying. The protocol involves ingestion of a radiolabeled meal followed by serial gamma camera images obtained at one, two, and four hours. Gastric emptying is considered delayed if more than ten percent of the meal remains at four hours.

Gastric Emptying, Delayed

Gastroparesis: delayed gastric emptying without mechanical obstruction, causing early satiety, nausea, vomiting, bloating, and poor glucose control. Common causes include diabetes, post-surgical changes, and idiopathic causes. Diagnosed by gastric emptying scintigraphy.

Alternate Names:

Gastroparesis

Gastric Emptying, Rapid

Dumping Syndrome: rapid gastric emptying where food moves too quickly from the stomach into the small intestine, causing nausea, vomiting, cramping, diarrhea, dizziness, and sweating after eating. Common after gastric surgery. Managed with dietary modifications.

Alternate Names:

Dumping Syndrome

Gastric Fistula

An abnormal connection involving the stomach and another organ or the skin, which may cause leakage, infection, malnutrition, or drainage.

Gastric Flu

An informal name, usually for viral gastroenteritis—an infection of the stomach and intestines that causes nausea, vomiting, diarrhea, and abdominal cramps. It is not caused by influenza and can lead to dehydration.

Gastric Fundus

gastric fundus is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Gastric Hemorrhage

Bleeding from the stomach caused by ulcer, erosive gastritis, varices, tumor, vascular lesion, or other injury, potentially producing hematemesis or melena.

Gastric Juice

The acidic digestive secretion of the stomach containing hydrochloric acid, pepsinogen, mucus, intrinsic factor, water, and electrolytes.

Gastric Lavage

Gastric lavage is the flushing of the stomach through a tube, a rarely used procedure that may be considered in selected poisonings or to remove stomach contents.

Gastric Leiomyosarcoma

A neoplastic process affecting gastric leiomyosarcoma; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Gastric Lymphoma

Malignant proliferation of lymphoid tissue in the stomach most commonly mucosa-associated lymphoid tissue lymphoma. Strongly associated with *H. pylori* infection. Early disease may regress completely with *H. pylori* eradication therapy alone.

Gastric Malignancy

Any malignant neoplasm originating in the anatomical stomach, predominantly gastric adenocarcinoma (> 90% of cases), with gastric lymphoma (extranodal marginal zone B-cell lymphoma / MALToma, diffuse large B-cell lymphoma) and gastrointestinal stromal tumors (GIST, harboring activating mutations in KIT or PDGFRA) comprising the remainder. Gastric adenocarcinoma is categorized by the Lauren classification into intestinal type (gland-forming, linked to chronic *Helicobacter pylori* gastritis, smoking, and dietary nitrates) and diffuse type (infiltrative, containing signet ring cells, often driven by loss-of-function *CDH1* mutations, causing linitis plastica / "leather-bottle stomach"). Diagnosed via esophagogastroduodenoscopy with tissue biopsy; staged with CT and endoscopic ultrasound.

Gastric Mucosa

gastric mucosa is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Gastric Necrosis

Death of stomach tissue from ischemia, caustic injury, severe distension, infection, or other insult, with risk of perforation and systemic illness.

Gastric Obstruction

Impaired passage of stomach contents caused by narrowing or blockage at the stomach or gastric outlet, producing vomiting, pain, distension, and dehydration.

Gastric Outlet Obstruction

Blockage of gastric outflow at the pylorus or duodenum. Causes include peptic ulcer scarring, gastric cancer, and pyloric stenosis in infants. Presents with projectile vomiting, dehydration, and metabolic alkalosis. Diagnosis is by upper endoscopy and CT scan.

Gastric Pneumatosis

Presence of gas within the gastric wall, also called gastric emphysema, which may be benign (gastric emphysema) or life-threatening (emphysematous gastritis). Benign forms result from barotrauma, chronic obstructive pulmonary disease, or mechanical causes.

Alternate Names:

gastric emphysema, which may be.

Gastric Polyps

Protruding lesions of the gastric mucosa classified by histological type. Fundic gland polyps are the most common type associated with long-term proton pump inhibitor use and familial adenomatous polyposis. Hyperplastic polyps are associated with chronic gastritis and *H. pylori* infection and carry a small risk of dysplasia.

Gastric Stenosis

Abnormal narrowing of the stomach or gastric outlet,

from congenital disease, ulcer scarring, inflammation, tumor, or hypertrophy, that impedes emptying.

Gastric Suction

Nasogastric tube placement for removal of gastric contents, used for decompression in bowel obstruction, preparation for surgery, or treatment of gastric dilatation. Also called nasogastric aspiration or gastric lavage.

Alternate Names:

Nasogastric aspiration; gastric lavage

Gastric Surgery

Gastric cancer: D2 lymphadenectomy standard (stations 1-12a, 12b). Gastrectomy types: total, distal subtotal (Billroth I gastroduodenostomy, Billroth II gastrojejunostomy, Roux-en-Y), proximal, pylorus-preserving. Reconstruction: Roux-en-Y preferred (prevents bile reflux).

Gastric Tissue Biopsy And Culture

Endoscopic sampling of gastric mucosa for histology and, when indicated, culture or other testing. It may evaluate gastritis, *Helicobacter pylori* infection, unusual infection, dysplasia, or gastric neoplasia.

Alternate Names:

Endoscopic gastric biopsy

Gastric Ulcer

Peptic ulcer in the gastric mucosa associated with *H. pylori*, NSAIDs, and smoking. Pain is typically worsened by food. There is a higher risk of malignancy than duodenal ulcers. Diagnosis is by upper endoscopy with biopsy to exclude malignancy.

Gastric Varices

Enlarged veins in the stomach wall caused by portal hypertension, usually related to liver disease or obstruction of the splenic or portal venous circulation. Rupture can cause life-threatening gastrointestinal bleeding.

Gastric Volvulus

Rotation of the stomach along its axis causing obstruction. It may present with Borchardt triad of

epigastric pain, retching, and inability to pass a nasogastric tube. It is a surgical emergency requiring decompression and gastropexy. Organoaxial volvulus is the most common type where the stomach rotates along its longitudinal axis.

Gastrin

A peptide hormone produced by G cells in the gastric antrum and duodenum. Secreted in response to: (1) gastric distension, (2) amino acids and peptides (especially from protein-rich meals), (3) vagal stimulation (via gastrin-releasing peptide, GRP), (4) hypercalcaemia.

Gastrin Blood Test

Laboratory measurement of gastrin levels in the blood, used to evaluate suspected Zollinger-Ellison syndrome (gastrinoma), atrophic gastritis, or other causes of gastric acid hypersecretion. Levels are affected by proton pump inhibitors and fasting state.

Alternate Names:

Serum gastrin level

Gastrin Secretion

Gastrin is a peptide hormone produced by G-cells in the gastric antrum and duodenum, stimulating hydrochloric acid secretion by parietal cells, pepsinogen secretion by chief cells, and growth of the gastric mucosa.

Gastrinoma

A tumor that makes too much gastrin, a hormone that causes the stomach to produce acid. It usually starts in the pancreas or duodenum and may cause Zollinger–Ellison syndrome, repeated peptic ulcers, diarrhea, and acid reflux. Some cases are linked to MEN1 and may spread to the liver or nearby lymph nodes.

Gastritis

Gastritis is inflammation of the stomach lining. Common causes include *H. pylori* infection, certain medicines such as NSAIDs, alcohol, autoimmune disease, and severe physical illness.

Gastrocnemius

The large superficial muscle of the posterior calf, forming the bulk of the calf contour. Origin: medial and lateral femoral condyles. Insertion: calcaneus via the Achilles tendon. Nerve: tibial nerve (S1-S2). Action: plantar flexion of the ankle (standing, walking, running, jumping); also flexes the knee.

Gastrocnemius Muscle

Most superficial posterior leg muscle forming the calf prominence, with medial/lateral femoral condyle heads merging with soleus to form the Achilles tendon on the calcaneus. Innervated by tibial nerve (S1-S2). Plantarflexes ankle and flexes knee (two-joint muscle). Active in walking/running/jumping.

Gastroduodenal Crohn Disease

Involvement of the stomach and duodenum by Crohn disease occurring in approximately five percent of patients with Crohn. Gastric Crohn typically involves the antrum and pylorus and can cause gastric outlet obstruction from edema, fibrosis, or stricture. Duodenal Crohn presents with epigastric pain, nausea, vomiting, and weight loss.

Gastroduodenostomy

Surgical creation of an opening between the stomach and duodenum.

Gastroenteritis

Inflammation of the stomach and intestinal mucosa, most commonly caused by viral infections (norovirus, rotavirus, adenovirus), bacterial pathogens (*E. coli*, *Salmonella*, *Campylobacter*), or parasites. Presents with diarrhea, vomiting, abdominal cramps, and dehydration. Self-limited in most cases.

Gastroenteritis (Stomach Flu)

Viral gastroenteritis: inflammation of the stomach and intestines caused by viruses such as norovirus, rotavirus, or adenovirus. Presents with diarrhea, vomiting, abdominal cramps, and dehydration. Usually self-limited.

Alternate Names:

Stomach flu; viral gastroenteritis

Gastroenteritis Bacterial

Bacterial gastroenteritis is inflammation of the stomach or intestines caused by bacteria, often producing diarrhea, abdominal cramps, fever, and sometimes vomiting.

Gastroenteritis, Viral

Viral gastroenteritis (stomach flu): inflammation of the stomach and intestines caused by viruses such as norovirus, rotavirus, or adenovirus. Presents with diarrhea, vomiting, abdominal cramps, and dehydration. Usually self-limited.

Alternate Names:

Viral gastroenteritis; stomach flu

Gastroenterologist

A physician specializing in disorders of the digestive tract, liver, gallbladder, and pancreas.

Gastroenterology

The medical specialty focused on the diagnosis and management of diseases of the gastrointestinal tract (esophagus, stomach, small intestine, colon, rectum, pancreas, gallbladder, bile ducts, liver). Subspecialties: hepatology (liver), pancreatology, inflammatory bowel disease, gastrointestinal oncology, neurogastroenterology.

Gastroenterostomy

Surgical creation of an opening between the stomach and a segment of the small intestine, usually the jejunum, to bypass or replace a blocked or diseased gastric outlet.

Gastroepiploic Artery

The gastroepiploic arteries run along the greater curve of the stomach and supply blood to the stomach and nearby omentum.

Gastroesophageal

Relating to the stomach and esophagus, such as the gastroesophageal junction or gastroesophageal reflux.

Gastroesophageal Junction

Adenocarcinoma

A gland-forming cancer arising where the esophagus meets the stomach. Symptoms may include difficulty swallowing, reflux, weight loss, anemia, or bleeding. Diagnosis requires endoscopy and biopsy; treatment depends on stage and may include surgery, chemotherapy, radiation, immunotherapy, or targeted therapy.

Gastroesophageal Reflux

Backward flow of gastric contents into the oesophagus, caused by transient or persistent relaxation of the lower oesophageal sphincter or other mechanical factors.

Alternate Names:

GERD discharge

Gastroesophageal Reflux Disease

Chronic condition where stomach contents repeatedly flow back into the esophagus, causing symptoms such as heartburn, regurgitation, chest pain, and esophageal damage. Managed with lifestyle changes, medications, and possibly surgery.

Alternate Names:

GERD; acid reflux disease

Gastroesophageal Reflux Disease - Children

GERD in pediatric patients: chronic reflux of stomach contents into the esophagus causing symptoms such as regurgitation, vomiting, feeding difficulties, irritability, and poor weight gain. May require positional management, dietary changes, and acid-suppressing medications.

Alternate Names:

Pediatric GERD; infant reflux

Gastroesophageal Reflux In Infants

Gastroesophageal reflux in infants concerns gastroesophageal reflux in infants, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Gastrogastic Intussusception

Rare form of intussusception where one segment of

the stomach telescopes into an adjacent gastric segment, typically associated with a lead point such as a gastric polyp, leiomyoma, or carcinoma. Presents with abdominal pain, nausea, vomiting, and occasionally upper gastrointestinal bleeding.

Gastrografen Study

X-ray imaging using water-soluble contrast. Used when barium contraindicated (perforation risk). Less detailed than barium.

Gastrointestinal

Pertaining collectively to the stomach and intestines, or to the entire gastrointestinal tract (alimentary canal) extending from the oral cavity to the anus, along with its accessory digestive glands (liver, gallbladder, pancreas). The GI tract carries out ingestion, mechanical and chemical digestion, peristaltic propulsion, mucosal nutrient absorption, and waste elimination. The tract wall exhibits a conserved four-layer architecture: mucosa (epithelium, lamina propria, muscularis mucosae), submucosa (housing the Meissner submucosal nerve plexus), muscularis externa (inner circular and outer longitudinal smooth muscle sandwiching the Auerbach myenteric plexus), and outer serosa/adventitia.

Alternate Names:

GI

Gastrointestinal Agents

Gastrointestinal agents are medicines or other substances used to treat digestive symptoms or disorders, such as nausea, diarrhea, constipation, or excess stomach acid.

Gastrointestinal Amyloidosis

Deposition of misfolded amyloid fibrils in the gastrointestinal tract, either as part of systemic amyloidosis (AL, AA, or ATTR types) or rarely as primary localized disease.

Gastrointestinal Bleeding

Bleeding anywhere in the digestive tract. It may be acute or chronic and may come from the upper digestive tract, small intestine, or lower digestive tract. It can cause vomiting blood, black stools, or

red blood in the stool.

Gastrointestinal Bleeding Scan

Detection of GI bleeding using Tc-99m labeled RBCs. Sensitive for active bleeding (>0.1 mL/min). Localizes bleeding site. Alternative: tagged RBC scan, CT angiography.

Gastrointestinal Decontamination

Methods: activated charcoal, gastric lavage (rare), whole bowel irrigation (sorbitol, PEG). Timing critical for charcoal (1-2 hours post-ingestion).

Gastrointestinal Disorder

A gastrointestinal disorder is a disease or problem affecting the digestive tract, which may cause symptoms such as pain, nausea, vomiting, diarrhea, constipation, or bleeding.

Gastrointestinal Fistula

An abnormal epithelial-lined connection between the gastrointestinal tract and another organ, body surface, or adjacent bowel segment; it can cause leakage, infection, malnutrition, and fluid or electrolyte loss.

Gastrointestinal Gas

Gastrointestinal gas is air or other gas in the stomach or intestines; excess gas may cause belching, bloating, abdominal discomfort, or passing gas.

Gastrointestinal Hemorrhage

Gastrointestinal hemorrhage is bleeding anywhere in the digestive tract, which may appear as vomiting blood, black stools, or red blood in the stool.

Gastrointestinal Hormone

A gastrointestinal hormone is a chemical messenger made by digestive-tract cells that helps regulate digestion, movement of the gut, appetite, or blood sugar.

Gastrointestinal Neoplasm

A gastrointestinal neoplasm is an abnormal growth arising in the digestive tract; it may be benign or

cancerous.

Gastrointestinal Obstruction

Partial or complete blockage of the passage of intestinal contents through the gastrointestinal tract. It may be mechanical or functional and can cause colicky pain, vomiting, distention, constipation, obstipation, dehydration, bowel ischaemia, perforation, or sepsis.

Gastrointestinal Pain

Pain arising from the digestive tract or related organs. Causes include indigestion, infection, ulcer disease, inflammation, obstruction, gallstones, and disorders of the liver or pancreas. Severe, persistent, or worsening pain requires medical assessment.

Gastrointestinal Perforation

A full-thickness defect in the gastrointestinal wall allowing luminal contents to enter the peritoneal or adjacent space. It can cause peritonitis, sepsis, and shock and usually requires urgent imaging, antimicrobial therapy, resuscitation, and source control.

Gastrointestinal Stromal Tumor

A mesenchymal tumor arising from specialized cells in the gastrointestinal wall, most often in the stomach or small intestine. Many tumors contain activating changes in KIT or PDGFRA; their malignant potential varies.

Gastrointestinal Stromal Tumor

A tumor arising from specialized cells in the digestive tract, most often in the stomach or small intestine.

Gastrointestinal System

The gastrointestinal system is the group of organs that digests food, absorbs nutrients and water, and removes solid waste.

Gastrointestinal Tract

The continuous digestive passage from the mouth through the esophagus, stomach, intestines, rectum, and anus.

Gastrointestinal Villi

Finger-like projections of the small-intestinal mucosa that greatly increase absorptive surface area. Each villus contains blood capillaries and a central lacteal for transport of absorbed nutrients and lipids.

Gastrojejunostomy

Surgical creation of an opening between the stomach and jejunum.

Gastromalacia

Gastromalacia is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gastroparesis

Gastroparesis is delayed stomach emptying without mechanical obstruction, causing early satiety, nausea, vomiting, bloating, and poor glucose or nutritional control. mellitus, post-surgical changes, or idiopathic causes. Presents with nausea, vomiting of undigested food, early satiety, bloating, and abdominal pain. Diagnosed by gastric emptying scintigraphy showing more than ten percent retention at four hours.

Alternate Names:

Delayed Gastric Emptying Delayed Gastric Emptying (Gastroparesis)

Diabetic Gastroparesis

A complication of diabetes in which damage to the stomach's nerves slows the movement of food from the stomach into the small intestine. It may cause early fullness, nausea, vomiting, bloating, abdominal pain, and difficult-to-control blood sugar levels.

Gastropathy

Gastropathy is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gastroplasty

Gastroplasty is an anatomical structure or physiologic function; its location, mechanism, and associ-

ated findings determine its clinical significance.

Gastroschisis

Gastroschisis is a congenital full-thickness defect of the abdominal wall, usually beside the umbilicus, through which bowel or other viscera herniate without a protective sac. which bowel herniates without a covering membrane. Not associated with chromosomal abnormalities. Presents at birth with exposed bowel loops that may be edematous, thickened, and discolored from amniotic fluid exposure.

Gastroschisis Repair

Surgical repair of gastroschisis, a congenital abdominal wall defect where intestines protrude through an opening near the umbilicus. Surgery involves returning the bowel to the abdomen and closing the defect. May require staged closure with silo placement.

Alternate Names:

Abdominal wall defect repair

Gastroscope

A gastroscope is a thin, flexible viewing tube used to examine the inside of the esophagus, stomach, and first part of the small intestine.

Gastroscopy

Endoscopic examination of the esophagus, stomach, and duodenum, allowing direct visualization, biopsy, and selected treatments.

Gastrospenic Fistulization

Abnormal communication between the stomach and spleen, most commonly arising from gastric ulcer erosion into the splenic hilum or as a complication of splenic abscess. Presents with abdominal pain, gastrointestinal hemorrhage, and sepsis.

Gastrostomy

Surgical creation of an opening (stoma) through the abdominal wall into the stomach for enteral feeding or decompression.

Gastrostomy Feeding Tube - Bolus

Technique for administering formula or nutrition directly into the stomach through a gastrostomy tube using a syringe. Includes hand-washing, flushing the tube, administering formula in small boluses, and flushing after feeding. Provides supplemental or complete nutrition.

Alternate Names:

G-tube bolus feeding

Gastrostomy Feeding Tube - Pump - Child

Technique for continuous or intermittent tube feeding in pediatric patients using an enteral feeding pump. Provides controlled delivery of formula through a gastrostomy tube, allowing for adjustable rates and volume. Includes equipment setup, monitoring, and troubleshooting.

Alternate Names:

Pediatric G-tube pump feeding

Gastrostomy Tube

Tube through abdominal wall into stomach for feeding. Percutaneous endoscopic gastrostomy. Indications: prolonged enteral feeding, dysphagia, neurological disorders. Complications: infection, leakage, granulation tissue, burial bumper.

Gastrula

An early embryo stage formed after gastrulation, when cell layers reorganize into ectoderm, mesoderm, and endoderm that give rise to body tissues.

Gastrulation

A crucial early phase in embryonic development where the single-layered blastula reorganizes into a multi-layered structure with three germ layers.

Gated Blood Pool Scan

A radionuclide cardiac study, also called radionuclide ventriculography or MUGA, that uses ECG-gated images of labeled blood to measure ventricular ejection fraction and assess ventricular wall motion. It is used selectively when quantitative ventricular function is needed.

Gatifloxacin

Gatifloxacin is a fluoroquinolone antibiotic that inhibits bacterial DNA gyrase and topoisomerase IV; systemic use is restricted in many settings because of dysglycaemia and other class toxicities, while ophthalmic use remains available in some regions.

Gaucher Disease

An inherited lysosomal storage disorder caused by deficient glucocerebrosidase, leading to lipid accumulation in cells and possible enlargement of the spleen and liver, anemia, bone disease, or neurologic disease.

Alternate Names:

Splnomegaly Gaucher (Gaucher Disease)

Gaucher'S Disease

Another name for Gaucher disease, an inherited lysosomal storage disorder caused by deficient glucocerebrosidase.

Gaultheria Oil

Gaultheria Oil is a medicine or pharmacologic product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent and patient.

Gauss

A unit of magnetic flux density equal to 10^{-4} tesla, used in physics, imaging, and descriptions of magnetic fields.

Gauze

A woven or nonwoven material used to clean, dress, pack, or absorb fluid from a wound. It may be sterile or nonsterile and comes in several forms.

Gavage

Feeding through a tube passed into the stomach (nasogastric, orogastric, or gastrostomy tube). Used when oral feeding is impossible or inadequate. In neonates: gavage feeding is used for premature infants unable to coordinate suck-swallow-breathe.

GBM

Glioblastoma: the most aggressive primary brain tumor, classified as WHO grade IV astrocytoma. Characterized by rapid growth, necrosis, and microvascular proliferation. Presents with headaches, seizures, focal neurological deficits, and cognitive changes. Treatment includes surgery, radiation, and chemotherapy.

Alternate Names:

Glioblastoma multiforme

Gefitinib

An oral epidermal growth factor receptor tyrosine-kinase inhibitor used for selected advanced non-small-cell lung cancers with activating EGFR mutations.

Geiger Counter

A Geiger counter is an instrument that detects and measures ionizing radiation.

Gel

A semisolid pharmaceutical formulation in which a liquid (usually water) is immobilized within a three-dimensional network of a gelling agent (e.g., carbomer, cellulose, alginate).

Gel Electrophoresis

Gel electrophoresis separates DNA, RNA, or proteins by moving them through a gel in an electric field, usually according to size and charge.

Gelatin

Gelatin is a protein made from collagen and used in foods, medicines, capsules, and some medical materials.

Geldanamycin Analogue

A medicine or chemical compound called geldanamycin analogue. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Gelfoam

An absorbable hemostatic agent made from purified

porcine gelatin. Used in surgical settings to control bleeding where conventional hemostasis is difficult (liver, bone, neurosurgery). Mechanism: provides a physical matrix that promotes platelet aggregation and clot formation. Absorbed within 4-6 weeks.

Gelineau Syndrome

Another name for narcolepsy, a sleep disorder characterized by excessive daytime sleepiness and, in many patients, cataplexy, sleep paralysis, or vivid hallucinations around sleep.

Gelsenium

Gelsenium is a medicine or pharmacologic product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent and patient.

Gelsolin

Gelsolin is a protein that regulates actin filaments and helps control the shape and movement of cells.

Gemcitabine

Gemcitabine is a pyrimidine antimetabolite that inhibits DNA synthesis after intracellular activation and is used for several solid tumors; myelosuppression, flu-like symptoms, rash, and pulmonary or hepatic toxicity may occur.

Gemcitabine Hydrochloride

The hydrochloride salt of gemcitabine, a nucleoside-analogue antimetabolite that inhibits DNA synthesis and is used in several solid tumors.

Gemella

Gemella is a group of bacteria that normally may inhabit the mouth and upper airway but can rarely cause serious infections, including bloodstream infection.

Gemella Bergeri

Gemella bergeri is a bacterium occasionally recovered from humans; its clinical importance depends on the specimen and evidence of true infection.

Gemella Haemolysans

Gemella haemolysans is a bacterium found mainly in the mouth and upper airway that can rarely cause invasive infection, including endocarditis.

Gemella Morbillorum

Gemella morbillorum is a bacterium that may live in the mouth and can rarely cause infections such as endocarditis or abscesses.

Gemella Sanguinis

Gemella sanguinis is a bacterium associated with the mouth that can rarely enter the bloodstream and cause invasive infection.

Gemfibrozil

A fibrate (PPAR-alpha agonist) used primarily for severe hypertriglyceridemia (reducing pancreatitis risk) and mixed hyperlipidemia. Raises HDL and lowers triglycerides markedly; modest LDL lowering.

Gemifloxacin

A medicine or chemical compound called gemifloxacin. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Gemination

Attempted division of single tooth bud. Results in large tooth with bifid crown, normal tooth count.

Gemistocytic Astrocytoma

A former histologic variant of diffuse astrocytoma composed partly of gemistocytic astrocytes. The term is now used mainly descriptively because integrated molecular classification determines the modern tumor diagnosis and grade.

Gemtuzumab Ozogamicin

A medicine or chemical compound called gemtuzumab ozogamicin. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Gender

A social and personal construct involving roles, identity, expression, and expectations associated with sex; it is distinct from biological sex and sexual orientation.

Gender Dysphoria

Psychological distress resulting from incongruence between one's experienced gender and assigned gender at birth. Persists for ≥ 6 months and causes significant impairment. Onset in childhood (may resolve) or adolescence/adulthood (persistent).

Gender Identity

A person's internal sense of being a woman, man, both, neither, or another gender, which may or may not correspond to sex assigned at birth.

Gender Identity Disorder

Gender Dysphoria: psychological distress resulting from incongruence between one's experienced gender and assigned gender at birth. Persists for ≥ 6 months and causes significant impairment. Onset in childhood (may resolve) or adolescence/adulthood (persistent).

Alternate Names:

Gender Dysphoria

Gene

Basic unit of heredity, segment of DNA encoding a functional product (protein or RNA). Structure: promoter, coding sequence (exons), introns, terminator. Genes are located at specific positions (loci) on chromosomes.

Gene Amplification

An increase in the number of copies of a gene or genomic region, which can raise gene expression and drive some inherited conditions or cancers.

Gene Conversion

A nonreciprocal transfer of sequence information from one DNA region to a homologous region, potentially changing an allele during recombination or repair.

Gene Deletion

Loss of part or all of a gene from DNA, potentially abolishing its product or altering dosage and causing disease when clinically significant.

Gene Dosage

The number of functional copies of a gene and the amount of product they produce, influenced by deletions, duplications, chromosome number, and regulation.

Gene Duplication

An extra copy of a gene created by replication or rearrangement, which may have no effect, alter dosage, or provide material for new gene function.

Gene Expression

Process by which information from a gene is used to synthesize functional product (protein or RNA). Steps: transcription, RNA processing, translation, post-translational modification. Regulated at multiple levels.

Gene Frequency

The proportion of a particular allele among all copies of a gene in a population.

Gene Fusion

A hybrid gene formed when segments of two separate genes join, sometimes producing an abnormal protein that drives cancer or other disease.

Gene Library

A collection of cloned DNA fragments representing genomic DNA, complementary DNA, or selected sequences for laboratory screening and analysis.

Gene Modification

An intentional or naturally occurring alteration of DNA sequence or gene regulation that changes cellular function or inherited traits.

Gene Mutation

A permanent change in the DNA sequence of a gene, which may be harmless, beneficial, or pathogenic

depending on its effect and inheritance.

Gene Panel

A laboratory test that sequences a selected group of genes associated with a clinical phenotype. Panels can provide efficient, focused analysis but may miss variants outside the selected genes or changes poorly detected by the platform.

Gene Rearrangement

Reorganization of DNA segments within or between chromosomes that can alter gene structure, expression, immune-receptor diversity, or cancer biology.

Gene Silencing

Suppression of gene expression through DNA methylation, chromatin changes, RNA interference, or other mechanisms.

Gene Targeting

A molecular method that changes a chosen genomic locus through homologous recombination or engineered editing to study or alter gene function.

Gene Therapy

Treatment by introducing, removing, or changing genetic material. Approaches: gene addition (functional copy), gene silencing (siRNA, antisense), gene editing (CRISPR-Cas9). Examples: Luxturna (RPE65 mutation), Zolgensma (SMN1 in spinal muscular atrophy).

Gene Transfer

Movement of genetic material into a cell or organism, naturally or through laboratory vectors, potentially changing gene expression or correcting a defect.

Gene-Environment Interaction

A gene-environment interaction occurs when an environmental exposure changes disease risk differently according to a person's genotype, so inherited susceptibility and exposure jointly influence phenotype.

Gene-Modified

Describing cells or organisms whose genetic material

has been deliberately altered using recombinant DNA or genome-editing methods.

General Anesthesia

A controlled, reversible state of unconsciousness, analgesia, amnesia, and muscle relaxation induced by pharmacologic agents, enabling surgical procedures without patient awareness or pain.

General Anesthetic

An agent producing reversible loss of consciousness, amnesia, analgesia, and muscle relaxation sufficient for surgery.

General Paresis

Neurosyphilis: a chronic meningoencephalitis caused by *Treponema pallidum* infection, presenting with progressive cognitive decline, personality changes, psychiatric symptoms, and neurological deficits. Diagnosis is by CSF analysis and serology. Treatment with IV penicillin.

Alternate Names:

General paresis of the insane; neurosyphilis

Generalised

Widespread or involving multiple regions of the body rather than being confined to a single localized site.

Generalized

Distributed widely or involving most or all of a body region, system, or population rather than being confined to one focus.

Generalized Anxiety Disorder

Characterized by excessive, difficult-to-control anxiety and worry about multiple events or activities occurring more days than not for at least six months. Associated symptoms include restlessness, being easily fatigued, difficulty concentrating, irritability, muscle tension, and sleep disturbance.

Alternate Names:

GAD self-management

Generalized Anxiety Disorder In

Children

Excessive, difficult-to-control worry in children about multiple topics (school performance, health, family safety) occurring more days than not for at least 6 months. Associated with restlessness, fatigue, difficulty concentrating, irritability, muscle tension, and sleep disturbance. Treatment includes cognitive behavioral therapy and possibly medication.

Alternate Names:

Pediatric GAD

Generalized Granuloma Annulare

A widespread variant of granuloma annulare presenting with numerous small, flesh-coloured to erythematous papules distributed over the trunk and extremities, lacking the classic annular arrangement of localized disease. It is often associated with diabetes mellitus and may be more persistent than localized forms.

Generalized Lipodystrophy

A disorder with near-total loss of adipose tissue, severe insulin resistance, diabetes, high triglycerides, fatty liver, and abnormal fat deposition in other tissues.

Generalized Seizure

A seizure arising from or rapidly involving both cerebral hemispheres, commonly producing impaired awareness, bilateral motor activity, or brief absence episodes.

Generalized Tonic-Clonic Seizure

A generalized tonic-clonic seizure (formerly grand mal seizure) is a seizure involving bilateral tonic stiffening followed by rhythmic clonic jerking of the extremities, with loss of consciousness.

Generalized-Onset Seizure

A seizure that begins in bilaterally distributed networks and rapidly engages both cerebral hemispheres. Generalized absence, myoclonic, tonic, clonic, atonic, and tonic-clonic seizures are recognized clinical types.

Generator

System for producing short-lived radionuclide from longer-lived parent. Tc-99m from Mo-99. Elution with saline. Quality control: Mo-99 breakthrough, Al-3+ breakthrough, sterility.

Generic

Describing a medicine by its active ingredient rather than a brand name; an approved generic drug must meet applicable standards for quality and therapeutic equivalence to its reference product.

Genes

Basic units of heredity composed of DNA sequences that encode functional products (proteins or RNA). Located at specific positions (loci) on chromosomes. Genes determine traits and can carry mutations that cause disease.

Genesis

Origin, formation, or development, as in the genesis of a disease or organ.

Genetic Anticipation

Earlier onset or increasing severity of a hereditary disorder in successive generations, often caused by expansion of unstable repeated DNA sequences.

Genetic Carrier

A person with one pathogenic variant for a recessive disorder who is usually unaffected but can transmit the variant to children.

Genetic Code

The set of rules by which the nucleotide sequence of messenger RNA is read in groups of three bases (codons) to specify amino acids or stop signals during protein synthesis. The standard code is nearly universal and is redundant because several codons can specify the same amino acid.

Genetic Counseling

Process of helping individuals understand and adapt to genetic aspects of disease. Includes risk assessment, education, psychosocial support, informed

decision-making.

Genetic Counselling

Genetic Counselling is a biological, genetic, biochemical, or chemical term used in medicine; its significance depends on its structure, function, concentration, pathway, or clinical context.

Genetic Counselor

A health professional who evaluates inherited cancer risk, arranges and interprets genetic testing, and provides counseling to patients and families, including cascade testing of relatives.

Genetic Disease

A disorder caused in whole or in part by changes in genetic material; it may be inherited, arise anew, or result from chromosome or mitochondrial abnormalities.

Genetic Disorder

A disease caused wholly or partly by a pathogenic change in DNA, chromosome structure or number, or gene regulation.

Genetic Drift

Random changes in allele frequencies from one generation to the next, especially in small populations, independent of natural selection.

Genetic Engineering

Deliberate alteration, insertion, deletion, or transfer of genetic material using molecular biology techniques.

Genetic Heterogeneity

The occurrence of similar or related phenotypes caused by different genetic mechanisms. Locus heterogeneity refers to variants in different genes causing the same phenotype; allelic heterogeneity refers to different variants in the same gene.

Genetic Linkage

The tendency of nearby genes or DNA markers on the same chromosome to be inherited together

because crossing over is less likely between them.

Genetic Marker

A detectable DNA variation or feature used to track inheritance, identify a chromosome region, assess disease risk, or distinguish samples.

Genetic Polymorphism

A common DNA variation with at least two alleles in a population, which may influence traits, drug response, or disease susceptibility.

Genetic Profile

A set of genetic variants or DNA markers describing an individual, tumor, organism, or sample for diagnosis, risk assessment, identification, or research.

Genetic Recombination

Exchange or rearrangement of DNA segments that creates new allele combinations during meiosis, repair, immune-cell development, or laboratory manipulation.

Genetic Screening

Testing of asymptomatic individuals for genetic disorders. Newborn screening: metabolic disorders, hemoglobinopathies. Carrier screening: before pregnancy or during pregnancy. Prenatal screening: non-invasive prenatal testing (NIPT), maternal serum screening.

Genetic Sonogram

A targeted ultrasound performed at 18-22 weeks that combines standard anatomic survey with specific soft markers for aneuploidy.

Genetic Susceptibility

Inherited tendency to develop a disease or respond to an exposure, which usually modifies risk rather than determining outcome alone.

Genetic Technique

A laboratory method used to detect, amplify, sequence, edit, compare, or otherwise analyze DNA, RNA, chromosomes, or gene expression.

Genetic Testing

Analysis of DNA for disease risk or diagnosis. Examples: BRCA testing, carrier screening.

Genetic Testing (Sequencing)

Analysis of the order of DNA bases to identify variants associated with genetic disorders. The method is selected according to the gene, variant, and clinical question.

Alternate Names:
Sequencing

Genetic Testing And Your Cancer Risk

Genetic testing and your cancer risk involves toxic or environmental exposure to genetic testing and your cancer risk; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Genetic Transcription

Synthesis of an RNA copy from a DNA template by RNA polymerase, initiating expression of a gene.

Genetic Variant Of Uncertain Significance

A DNA variant for which available evidence is insufficient to classify it as benign or pathogenic. It should not generally be used alone to make predictive or treatment decisions and may be reclassified as evidence evolves.

Genetically Modified Organisms - GMOs

Genetically modified organisms - GMOs concerns genetically modified organisms - gmos, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Genetics

Genetics is the study of genes, heredity, genetic variation, and how DNA and gene regulation influence traits, disease risk, diagnosis, and treatment response.

Geniculate Body

geniculate body is a tissue, cell, or body structure whose location and function help explain its role in

normal physiology and disease.

Genistein

A plant compound found mainly in soy that can weakly interact with estrogen receptors and other cell-signaling pathways. It is studied for possible effects on bone, menopause, and cancer, but supplements are not a substitute for proven treatment.

Genital

Relating to the reproductive organs or external sexual structures.

Genital Edema

Swelling of genital tissues from fluid accumulation caused by infection, inflammation, allergy, trauma, venous or lymphatic obstruction, or systemic disease.

Genital Herpes

Herpes simplex virus infection of the genital region (HSV-2 most commonly; HSV-1 increasingly), transmitted sexually, causing painful grouped vesicles/ulcers, urethritis, dysuria, and lymphadenopathy; primary infection may cause fever and systemic symptoms. Recurrent episodes are shorter and milder.

Genital Herpes (Genital Herpes In Women Overview)

Herpes simplex virus infection of the female genitalia, causing painful blisters or ulcers. May be asymptomatic. Antiviral treatment reduces symptoms and transmission. Pregnancy requires obstetric planning due to risk of neonatal infection.

Alternate Names:

Genital herpes in women

Genital Injury

Trauma to the genital or anogenital tissues caused by sexual contact, assault, childbirth, accident, or disease. Examination should document location, morphology, healing stage, pain, bleeding, and associated injuries without assuming mechanism from appearance alone.

Genital Sores - Female

Ulcers, blisters, or lesions on the female genitalia (vulva, vagina, cervix). Causes include sexually transmitted infections (herpes, syphilis, chancroid), dermatological conditions, trauma, or malignancy. Requires medical evaluation for diagnosis and treatment.

Alternate Names:

Female genital ulcers

Genital Sores - Male

Ulcers, blisters, or lesions on the male genitalia (penis, scrotum, perianal area). Causes include sexually transmitted infections (herpes, syphilis, chancroid), dermatological conditions, trauma, or malignancy. Requires medical evaluation for diagnosis and treatment.

Alternate Names:

Male genital ulcers

Genital System

genital system is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Genital Warts (Condylomata Acuminata)

Warts of the genital or anal region caused most often by low-risk types of human papillomavirus. They spread through direct skin-to-skin sexual contact and may be visible or asymptomatic.

Alternate Names:

Condylomata Acuminata

Genitalia

The external and internal reproductive organs. The term includes the vulva and vagina, or the penis, scrotum, and associated structures, together with the gonads and reproductive ducts when used broadly; anatomy varies with sex, age, and developmental stage.

Genitals

The external and internal reproductive organs. Female genitals: external (vulva — mons pubis, labia majora, labia minora, clitoris, vaginal open-

ing, Bartholin glands) and internal (vagina, cervix, uterus, fallopian tubes, ovaries).

Genito-Pelvic Pain/Penetration Disorder

A sexual dysfunction characterized by persistent or recurrent difficulties with vaginal penetration, marked vulvovaginal or pelvic pain during intercourse, fear or anxiety about penetration, and/or tensing of the pelvic floor muscles, lasting ≥ 6 months and causing marked distress.

Genito-Urinary Medicine

A medicine or chemical compound called genitourinary medicine. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Genitourinary

Relating to the urinary system and the reproductive organs, which share developmental and anatomical relationships.

Genitourinary Neoplasm

A neoplastic process affecting genitourinary neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Genitourinary Tract

The combined urinary and reproductive organs and their associated ducts.

Genodermatoses

Genetic skin disorders. Inherited ichthyoses (lamellar, X-linked, congenital ichthyosiform erythroderma). Epidermolysis bullosa (simplex, junctional, dystrophic). Neurocutaneous syndromes (neurofibromatosis, tuberous sclerosis, Sturge-Weber). Ectodermal dysplasias (hypohidrotic, hypotrichosis, nail dystrophy).

Genome

The complete genetic material of an organism, including its genes and noncoding DNA.

Genomic Imprinting

Parent-of-origin-specific gene expression. Only one allele (maternal or paternal) is expressed. Examples: Prader-Willi (paternal deletion 15q11-q13), Angelman (maternal deletion 15q11-q13).

Genomic Instability

An increased tendency for genetic alterations to accumulate, contributing to cancer development and progression.

Genomic Library

A genomic library is a collection of cloned DNA fragments that together represent the genetic material of an organism.

Genomics

Genomics is the study of an organism complete genetic material, including its sequence, variation, and function.

Genotoxicity

The ability of a substance or physical agent to damage genetic material. Genotoxicity may cause mutations or chromosome damage, but a positive test does not by itself establish that an agent causes cancer in humans.

Genotype

Genetic makeup of an individual, the specific alleles present at one or more loci. Determines phenotype along with environmental factors.

Genotype Determination

Genotype determination is the use of laboratory testing to identify an individual's genetic variants or alleles at one or more locations.

Genotype-Phenotype Correlation

The relationship between a person's genetic makeup and the observable features, severity, age of onset, or clinical course of a condition; correlations may be incomplete because other genetic and environmental factors contribute.

Gentamicin

Gentamicin is an aminoglycoside antibiotic that binds the bacterial 30S ribosomal subunit and causes misreading of mRNA; nephrotoxicity and ototoxicity require dosing and level monitoring.

Gentamicin Sulfate

The sulfate salt of gentamicin, an aminoglycoside antibiotic that inhibits bacterial protein synthesis; nephrotoxicity and ototoxicity require careful dosing and monitoring.

Gentian

A plant of the *Gentiana* genus whose roots or other parts have been used in herbal preparations, traditionally as a bitter digestive; clinical evidence and product quality vary.

Gentian Violet

A purple antiseptic and dye with antifungal and antibacterial activity, used in selected topical applications but capable of irritating tissue and staining skin.

Genu

The knee or a knee-shaped part of an anatomical structure; the term also describes a bend, such as the genu of the corpus callosum.

Genu Recurvatum

Hyperextension of the knee beyond the usual straight position, caused by laxity, deformity, muscle imbalance, neurologic disease, or injury.

Genu Valgum (Knock-Knee)

An alignment abnormality in which the knees angle inward and may touch while the ankles remain apart. It may be developmental or caused by bone, joint, or metabolic disease.

Alternate Names:

Knock-Knee

Genu Varum

Outward bowing of the legs, commonly called bowlegs. It is often normal in young children and

improves with growth, but persistent, worsening, unequal, painful, or severe bowing may result from rickets, growth-plate disease, or another condition.

Alternate Names:

bowlegs.

Genus Dermatophilus

genus dermatophilus is a microorganism or biological taxon; its clinical importance depends on the species, specimen, host, and evidence of infection or colonization.

Genus Dickeya

genus dickeya is a microorganism or biological taxon; its clinical importance depends on the species, specimen, host, and evidence of infection or colonization.

Genus Leminorella

genus leminorella is a microorganism or biological taxon; its clinical importance depends on the species, specimen, host, and evidence of infection or colonization.

Genus Saimiri

genus saimiri is a microorganism or biological taxon; its clinical importance depends on the species, specimen, host, and evidence of infection or colonization.

Geographic Atrophy

The advanced form of dry age-related macular degeneration, characterized by well-demarcated, progressively enlarging areas of complete loss of the retinal pigment epithelium, choriocapillaris, and overlying photoreceptors in the macula, resulting in irreversible central vision loss.

Geographic Tongue

Benign inflammatory condition of the tongue characterized by irregular red patches with white or yellow borders, alternating with areas of depapillation. Often asymptomatic and of unknown cause. May cause burning or irritation with certain foods.

Alternate Names:

Benign migratory glossitis Benign Migratory Glossitis

Geophagia

Persistent eating of soil or other nonfood substances; it may cause poisoning, infection, intestinal obstruction, or iron-deficiency anemia.

Geophagy

The deliberate ingestion of soil or clay; it may cause lead or parasite exposure, gastrointestinal obstruction, or iron-deficiency anemia and can occur with pica.

Geotrichum

Geotrichum is a genus of fungi that may colonize humans and can rarely cause superficial or invasive infection, especially in people with weakened immunity.

Geotrichum Klebahnii

Geotrichum klebahnii is a species of fungus; recovery from a clinical specimen requires correlation with the site and evidence of infection.

Geriatric Assessment

Comprehensive multidimensional evaluation of elderly person. Domains: medical, functional, cognitive, psychological, social, environmental, pharmacological. Performed by interdisciplinary team. Guides care planning, identifies problems, predicts outcomes.

Geriatric Care Manager

A professional who helps coordinate health and support services for an older adult.

Alternate Names:

Aging-life care manager

Geriatric Dentistry

Dental care for elderly. Management of age-related changes, polypharmacy effects, xerostomia, root caries, periodontal disease, oral cancer screening.

Geriatric Depression Scale

A screening questionnaire designed to detect depressive symptoms in older adults while minimizing emphasis on somatic symptoms that may overlap

with medical illness. A positive screen requires clinical diagnostic assessment.

Geriatric Medical Ethics

Ethical principles applied to care of older adults, including autonomy, capacity, beneficence, non-maleficence, and justice.

Geriatric Nursing

Nursing care focused on the health, function, medication safety, cognition, mobility, and goals of older adults.

Geriatric Nutrition

Nutrition care addressing the effects of aging, chronic disease, medications, frailty, oral and swallowing problems, sensory changes, and social factors on intake and nutritional status. Goals include preserving function, muscle mass, hydration, and quality of life.

Geriatric Rehabilitation

Rehabilitation designed to improve function, mobility, independence, and safety in older adults after illness, injury, or functional decline.

Geriatric Syndrome

Multifactorial condition common in elderly, not attributable to single disease. Examples: falls, incontinence, frailty, delirium, immobility, pressure ulcers. Require comprehensive assessment and multidisciplinary management.

Geriatrician

A physician specializing in the health care of older adults, including complex illness, multiple conditions, medications, function, and goals of care.

Geriatrics

The medical specialty concerned with the health, function, and care of older adults.

Germ

A colloquial term for a microorganism that causes disease.

Germ Cell

A reproductive cell or precursor that gives rise to sperm or eggs and carries genetic material to the next generation.

Germ Cell Tumor

A tumor arising from germ cells (precursors of sperm and eggs), including seminoma, dysgerminoma, teratoma, yolk sac tumor, embryonal carcinoma, and choriocarcinoma. Occurs in the testis, ovary, and extragonadal sites (mediastinum, retroperitoneum, pineal); many are highly curable with platinum-based chemotherapy.

Germ Cell Tumors

Tumors arising from cells that normally develop into eggs or sperm, occurring in the gonads or other parts of the body.

Germ Layer

germ layer is a microorganism or biological taxon; its clinical importance depends on the species, specimen, host, and evidence of infection or colonization.

Germ Theory Of Disease

The scientific principle that specific microorganisms cause specific infectious diseases; established in the late 19th century by Koch (postulates—organism present in all cases, isolated in pure culture, reproduces disease when inoculated, re-isolated) and promoted by Pasteur and Lister.

German Measles

Rubella: a contagious viral infection caused by rubella virus, characterized by fever, lymph-node enlargement, and a fine rash. Infection during pregnancy can harm the fetus, causing congenital rubella syndrome with cardiac, auditory, and developmental defects.

Alternate Names:

Rubella

German Measles (Rubella)

A contagious viral infection caused by rubella virus, characterized by fever, lymph-node enlargement,

and a fine rash. Infection during pregnancy can harm the fetus.

Alternate Names:

Rubella

Germanium

germanium is a microorganism or biological taxon; its clinical importance depends on the species, specimen, host, and evidence of infection or colonization.

Germinoma

A germ-cell tumor most commonly arising in the pineal or suprasellar region, highly sensitive to radiation and chemotherapy. Ovarian (dysgerminoma) and testicular (seminoma) counterparts exist.

Germline Mutation

A genetic variant present in the egg or sperm and therefore present in essentially every cell of the individual. Pathogenic germline variants may confer inherited cancer susceptibility and can have implications for relatives and treatment.

Germline Testing

Genetic testing performed to identify inherited or potentially heritable variants, usually from blood or saliva. Results may affect the patient, biological relatives, cancer surveillance, reproductive planning, and genetic counseling.

Germline Variant

A DNA variant present in the cells that give rise to eggs or sperm and therefore potentially transmissible to offspring. It may also be present in most cells of the individual, unlike a variant restricted to a somatic tissue.

Gerontologist

A specialist who studies aging and its biological, psychological, social, or economic effects.

Gerontology

The multidisciplinary study of aging and the biological, psychological, social, and environmental factors that influence it.

Gesell Developmental Schedules

Standardized developmental assessment schedules that describe expected motor, language, adaptive, and social milestones in children; they support screening but do not by themselves diagnose a disorder.

Gestalt Psychology

A clinical condition or finding involving gestalt psychology. Its defining features, causes, evaluation, and management depend on the clinical context.

Gestation

The period of development of a fetus from fertilization to birth. Human gestation averages 280 days (40 weeks) from the first day of the last menstrual period (LMP) or 266 days (38 weeks) from conception.

Gestation Period

The time from conception to birth or, clinically, from the last menstrual period to delivery; human pregnancy is usually about 40 weeks from the last menstrual period.

Gestational Age

The age of pregnancy measured in weeks from the first day of the last menstrual period. Also called menstrual age. Typically 2 weeks longer than fertilization age. Calculated using LMP or first-trimester ultrasound. Used to estimate due date (40 weeks from LMP).

Alternate Names:
menstrual age.

Gestational Diabetes

Glucose intolerance first diagnosed during pregnancy. It may increase the risk of excessive fetal growth, birth complications, neonatal hypoglycemia, and hypertensive pregnancy disorders.

Alternate Names:
Diabetes, Gestational Diabetes During Pregnancy (Gestational Diabetes)

Gestational Diabetes Diet

Dietary management of gestational diabetes, focusing on controlled carbohydrate intake, balanced meals, regular meal timing, and appropriate portions to maintain blood glucose levels within target ranges. May require consultation with a dietitian.

Alternate Names:
GDM diet

Gestational Diabetes Mellitus

Glucose intolerance first recognized during pregnancy affecting approximately six to nine percent of pregnancies. Results from insulin resistance induced by placental hormones including human placental lactogen, cortisol, and progesterone combined with insufficient beta cell compensation.

Gestational Hypertension

New-onset hypertension after 20 weeks of pregnancy without proteinuria or other features of preeclampsia. It requires monitoring for progression and assessment of maternal and fetal well-being.

Gestational Pyelonephritis

A kidney infection occurring during pregnancy, usually causing fever, flank pain, nausea, or urinary symptoms. It can lead to sepsis, preterm labor, or respiratory complications and requires prompt clinical assessment.

Alternate Names:
Pyelonephritis in pregnancy; pregnancy-associated kidney infection.

Gestational Sac

The first sonographic sign of an intrauterine pregnancy, visible on transvaginal ultrasound at approximately 4.5-5 weeks gestational age. A fluid-filled structure within the decidua, surrounded by the chorionic ring. Mean sac diameter (MSD) correlates with gestational age.

Gestational Trophoblastic Disease

A spectrum of conditions arising from abnormal trophoblast proliferation, including complete and partial hydatidiform moles and gestational choriocarcinoma. Complete mole has 46,XX karyotype

(all paternal origin); partial mole is triploid (69 chromosomes). Presents with elevated hCG, uterus large for dates, and vaginal bleeding.

Gestational Trophoblastic Neoplasia

Persistent or malignant trophoblastic disease after a pregnancy event, identified by abnormal or rising beta-hCG, metastases, or histologic criteria. It includes invasive mole and choriocarcinoma and is highly responsive to risk-adapted chemotherapy.

Getting A Prescription Filled

The process of obtaining prescribed medications from a pharmacy, including presenting the prescription, insurance verification, payment, and receiving medication counseling. May involve prior authorization for certain medications.

Alternate Names:

Prescription fulfillment

Getting Support When Your Child Has Cancer

Resources and strategies for families coping with a child's cancer diagnosis, including emotional support, financial assistance, respite care, support groups, and connecting with other families facing similar challenges.

Alternate Names:

Pediatric cancer family support

Getting Your Home Ready - Knee Or Hip Surgery

Pre-surgical home preparation for knee or hip replacement, including removing tripping hazards, installing grab bars, arranging furniture for mobility, and preparing a recovery area on the main floor.

Alternate Names:

Pre-surgical home preparation

Getting Yourself Healthy Before Surgery

Pre-operative optimization strategies including smoking cessation, nutritional improvement, blood glucose control, medication management, and exercise to reduce surgical risks and improve recovery outcomes.

Alternate Names:

Pre-operative optimization

Gettler Test

Historical quantitative chemical test for drowning, comparing chloride concentration in blood harvested from the left and right heart ventricles.

GFR

Glomerular filtration rate, an estimate of how efficiently the kidneys filter blood.

Alternate Names:

Glomerular filtration rate

Ghrelin

A hormone produced mainly in the stomach that influences appetite, energy balance, and growth-hormone release.

Gianotti-Crosti Syndrome

A childhood skin condition characterized by a papular or papulovesicular rash on the face, buttocks, and extremities, often following viral infections (EBV, hepatitis B, CMV). Self-limited, lasting 2-8 weeks. Also called papular acrodermatitis of childhood.

Alternate Names:

Papular acrodermatitis of childhood

Giant Cell Arterit (Giant Cell Arteritis (GCA Or Temporal Arteritis))

Large-vessel vasculitis affecting adults >50, involving cranial branches of the carotid arteries. Presents with new-onset headache, scalp tenderness, jaw claudication, and visual disturbances. Risk of vision loss without treatment.

Alternate Names:

Giant cell arteritis; temporal arteritis

Giant Cell Arteritis

A large-vessel vasculitis of adults >50 (peak 70-80) affecting cranial branches of carotid arteries, with risk of permanent vision loss from anterior ischemic optic neuropathy. Clinical: new-onset headache (temporal), scalp tenderness, jaw claudication, visual disturbances (amaurosis fugax,

diplopia), polymyalgia rheumatica symptoms.

Giant Cell Arteritis (GCA Or Temporal Arteritis)

Inflammation of large and medium-sized arteries, predominantly affecting the temporal and ophthalmic arteries. Characterized by headache, visual symptoms, jaw claudication, and elevated ESR. Diagnosis by temporal artery biopsy. Treatment with high-dose corticosteroids.

Alternate Names:

Temporal arteritis; Horton disease

Giant Cell Arteritis Emergency

Suspected giant cell arteritis with visual symptoms, jaw or tongue claudication, or neurologic ischemia requires immediate glucocorticoid treatment to reduce risk of irreversible visual loss while diagnostic evaluation proceeds.

Giant Cell Tumor

A usually benign but locally aggressive bone tumor containing numerous giant cells.

Giant Cell Tumor Of Bone

A locally aggressive osteoclast-rich bone tumor, usually arising near the end of a long bone in skeletally mature adults. It may cause pain, swelling, cortical destruction, and occasional pulmonary metastases.

Giant Cell Tumor Of Soft Tissue

A usually low-grade soft-tissue neoplasm containing numerous osteoclast-like giant cells, most often arising in the extremities of adults and generally having a favorable outcome after excision.

Giant Cell Tumor Of Tendon Sheath

A common benign localized tenosynovial tumor, usually presenting as a slow-growing firm nodule on a finger or hand; recurrence can occur after excision.

Giant Cells

Large cells formed by fusion of macrophages or abnormal cellular growth, seen in granulomatous inflammation, foreign-body reactions, and some tu-

mors.

Giant Congenital Nevus

A large pigmented skin lesion present at birth, typically exceeding 20 cm in diameter. Composed of melanocytes and may involve dermis, subcutaneous tissue, and underlying structures. Risk of melanoma transformation. May require surgical excision or monitoring.

Alternate Names:

Giant pigmented nevus; garment nevus

Giant Coronary Aneurysm

An unusually large dilation of a coronary artery, with risk of thrombosis, rupture, or impaired coronary blood flow.

Giant Hypertrophic Gastritis

Severe enlargement and thickening of the stomach's mucosal folds, classically associated with Ménétrier disease, which can cause protein loss, abdominal symptoms, and increased gastric-cancer risk.

Giant Lymph Node Hyperplasia

Castleman Disease: a rare lymphoproliferative disorder characterized by non-cancerous overgrowth of lymph node tissue. May be unicentric (localized) or multicentric (systemic symptoms). Associated with HHV-8 and IL-6 overproduction. Treatment depends on subtype.

Alternate Names:

Castleman Disease

Giant Platelet Syndrome

A disorder characterized by unusually large platelets, often with a low platelet count and impaired clotting; inherited examples include Bernard-Soulier syndrome and MYH9-related disease.

Giant Platelet Syndrome (Bernard-Soulier Disease)

An inherited platelet disorder caused by deficiency of glycoprotein Ib/IX/V complex, resulting in large platelets and impaired platelet adhesion. Presents with mucocutaneous bleeding, thrombocytopenia,

and giant platelets on blood smear.

Alternate Names:

Bernard-Soulier syndrome

Giant Right Atrium

Marked dilatation of the right atrium, most commonly caused by chronic severe tricuspid regurgitation, Ebstein anomaly, or long-standing atrial fibrillation with right heart failure. The enlarged atrium may cause compression of adjacent structures, arrhythmias, and thrombus formation. Diagnosis is by echocardiography or cardiac imaging.

Giardia

giardia is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Giardia Infection

Intestinal infection caused by the protozoan *Giardia duodenalis*, usually acquired from contaminated water, food, or person-to-person spread. It commonly causes foul-smelling diarrhea, bloating, abdominal cramps, and malabsorption.

Giardia Infection (Giardiasis)

An intestinal infection caused by *Giardia* parasites, commonly producing diarrhea, cramps, bloating, gas, and malabsorption.

Giardia Lamblia

A protozoan parasite that causes giardiasis, a diarrhoeal illness transmitted through the fecal-oral route. The organism has a trophozoite and cyst stage. Infection occurs through ingestion of cysts in contaminated water. Symptoms include foul-smelling diarrhea, abdominal cramps, and bloating.

Alternate Names:

Cyst Eyelid (*Giardia Lamblia*)

Giardiasis

Infection with *Giardia lamblia* (*G. intestinalis*), a protozoan parasite causing diarrheal illness. Transmission via contaminated water (fecal-oral).

Presents with foul-smelling, greasy diarrhea, bloating, abdominal cramps, weight loss, and steatorrhea. Diagnosis: stool antigen testing or microscopy.

Gibraltar Fever

A clinical disorder or finding related to gibraltar fever; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Giemsa Stain

A Romanowsky-type stain of methylene blue and eosin dyes used to visualise blood cells and microorganisms in smears and tissue sections. In microbiology it stains *Plasmodium*, *Babesia*, *Trypanosoma*, *Leishmania*, *Borrelia*, and intracellular cytomegalovirus inclusions, supporting rapid diagnosis of blood-borne parasitic and other infections.

Gigabecquerel

A gigabecquerel is a unit of radioactivity equal to one billion nuclear disintegrations per second.

Gigantism

Excessive growth hormone hypersecretion that begins before epiphyseal plate closure in childhood, before the onset of puberty. Results in excessive linear growth and tall stature. Usually caused by a pituitary adenoma. Treatment includes surgery, radiation, and medical therapy.

Gigli Saw

A flexible wire saw used to cut bone, especially during selected orthopedic or amputation procedures.

Gilbert Syndrome

A common, benign autosomal recessive hereditary hyperbilirubinemia caused by a promoter mutation in the *UGT1A1* gene leading to reduced activity (30% of normal) of UDP-glucuronosyltransferase.

Gilbert'S Syndrome

A clinical disorder or finding related to gilbert's syndrome; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

tions.

Gilles De La Tourette Syndrome

A clinical disorder or finding related to gilles de la tourette syndrome; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Gimatecan

An investigational camptothecin-derived topoisomerase-I inhibitor studied as an anti-cancer agent; availability and clinical use depend on regulatory approval and trial evidence.

Gingiva

The oral tissue surrounding and protecting the roots of the teeth.

Alternate Names:

Gums; gum tissue

Gingiva Disease

A disorder of the gums, including plaque-induced gingivitis, periodontitis, infection, immune disease, recession, or tumor.

Gingival Hemorrhage

Bleeding from gum tissue, commonly due to inflammation and plaque but also caused by trauma, periodontal disease, medicines, or bleeding disorders.

Gingival Hyperplasia

An abnormal enlargement or overgrowth of gingival tissue around the teeth that can impair cleaning, cause bleeding, and increase periodontal risk. Causes include plaque-related inflammation and medicines such as phenytoin, cyclosporine, and some calcium-channel blockers; hereditary and systemic conditions can also contribute.

Gingival Infections

An infectious process or toxic exposure involving gingival infections; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Gingival Margin

gingival margin is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Gingival Melanoma

A rare and aggressive mucosal melanoma arising from melanocytes in the gingival mucosa, presenting as a pigmented or amelanotic nodule or ulcer on the gums. Mucosal melanomas of the head and neck carry a poor prognosis compared to cutaneous melanoma, often presenting at an advanced stage.

Gingival Neoplasm

A neoplastic process affecting gingival neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Gingival Overgrowth

Gingival overgrowth is an abnormal increase in gum tissue, which may be caused by medicines, inflammation, or systemic disease.

Gingival Pocket

gingival pocket is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Gingival Recession

Apical displacement of the gingival margin from the cemento-enamel junction, exposing the root surface. Causes include periodontal inflammation, traumatic toothbrushing, thin periodontal tissue, tooth position, and iatrogenic factors; it may produce sensitivity and root-caries risk.

Gingivectomy

Surgical removal of gingival tissue. Indications: gingival hyperplasia, cosmetic recontouring, access to caries.

Gingivitis

Inflammation of the gingiva characterized by redness, swelling, and bleeding with brushing or probing, usually due to dental plaque. It is generally

reversible with effective oral hygiene and professional cleaning; it can precede periodontitis, but does not inevitably progress to it.

Alternate Names:

Periodontal Disease, Gingivitis

Gingivitis Contagious (Is Gingivitis Contagious)

Gingivitis itself is not directly contagious, but the bacteria that cause it (particularly *Porphyromonas gingivalis*) can be transmitted through saliva. Poor oral hygiene is the primary risk factor. Prevention focuses on personal oral care.

Alternate Names:

Gingivitis transmission

Gingivoplasty

Surgical reshaping or contouring of the gums to improve their form, remove diseased tissue, or make cleaning easier. It may be combined with gingivectomy or other periodontal treatment.

Gingivostomatitis

Common viral infection of the mouth and gums, most often caused by primary herpes simplex virus infection but also associated with other infections, trauma, or systemic disease. It may cause painful ulcers, swollen bleeding gums, fever, and difficulty eating or drinking.

Ginkgo Biloba

An herbal product derived from *Ginkgo biloba* leaves and marketed for cognitive or circulatory benefits; evidence is variable and it may increase bleeding risk or interact with medicines.

Ginseng

A group of plants whose roots are used in traditional medicine and dietary supplements; products vary in composition and may interact with medicines or affect blood glucose and clotting.

Ginseng Preparation

ginseng preparation is a biological, genetic, biochemical, or chemical term used in medicine; its

significance depends on its structure, function, concentration, pathway, or clinical context.

Girdle Pain

Girdle Pain is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gitelman Syndrome

Inherited autosomal recessive disorder of the distal convoluted tubule caused by mutations in the NCC sodium-chloride cotransporter gene. Milder than Bartter syndrome typically presenting in adolescence or adulthood.

Githism

A clinical disorder or finding related to githism; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Give Your Heart A Workout

Cardiovascular exercise recommendations and guidelines for improving heart health through regular physical activity, including aerobic exercises, intensity targets, duration, and frequency to strengthen the heart and improve circulation.

Alternate Names:

Cardiovascular exercise; heart-healthy exercise

Giving An IM (Intramuscular) Injection

Technique for administering medication into muscle tissue using a needle and syringe. Common sites include deltoid, vastus lateralis, and ventrogluteal muscles. Proper technique includes site selection, needle insertion at 90 degrees, aspiration, and slow injection.

Alternate Names:

Intramuscular injection technique

Giving An Insulin Injection

Technique for administering insulin subcutaneously using a syringe or insulin pen. Includes site selection (abdomen, thigh, upper arm), rotation of injection sites, proper needle insertion, and blood glucose monitoring.

Alternate Names:

Insulin administration technique

Gizzard

gizzard is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Glabella

The smooth area of the forehead between the eyebrows and above the bridge of the nose.

Gland

An organized group of epithelial cells specialized for secretion.

Glanders

A biomedical term referring to glanders; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Glands Swollen Lymph (Swollen Lymph Nodes)

Lymphadenopathy: enlargement of lymph nodes, usually in response to infection, inflammation, or malignancy. Common locations include neck, armpits, and groin. Most cases are benign and self-limited. Persistent or unexplained enlargement requires evaluation.

Alternate Names:

Lymphadenopathy; swollen lymph nodes

Glands Swollen Nodes (Swollen Lymph Nodes)

Enlarged lymph nodes: palpable increase in lymph node size, typically >1 cm, caused by immune activation, infection, autoimmune disease, or malignancy. Usually tender and mobile in benign causes. Fixed or hard nodes raise concern for malignancy.

Alternate Names:

Lymphadenopathy; enlarged lymph nodes

Glandular Fever

Glandular Fever is an anatomical structure or physiologic function; its location, mechanism, and asso-

ciated findings determine its clinical significance.

Glandular Fever (Infectious Mononucleosis)

An acute illness usually caused by Epstein-Barr virus, characterized by fever, sore throat, swollen lymph nodes, and fatigue.

Alternate Names:

Infectious Mononucleosis

Glandular Stomach

glandular stomach is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Glans Penis

The expanded, sensitive distal end of the penis containing the external urethral opening and covered by the foreskin when present.

Glanzmann Thrombasthenia

An inherited platelet function disorder caused by deficiency or dysfunction of platelet glycoprotein IIb/IIIa, producing mucocutaneous bleeding despite a normal platelet count.

Glasgow Coma Scale

A standardized assessment of consciousness based on eye opening, verbal response, and motor response, with a total score from 3 to 15. It supports serial neurologic assessment but should be recorded with the individual component scores and confounders.

Glass Eye

A glass eye is an ocular prosthesis made to replace the appearance of an eye that is missing or severely damaged.

Glass Ionomer

Restorative material releasing fluoride. Bonds to tooth structure. Used for pediatric dentistry, cervical lesions, bases.

Glasses

Optical devices containing corrective lenses posi-

tioned in a frame in front of the eyes to compensate for refractive errors or provide protection.

Glassy Cell Carcinoma

A rare poorly differentiated adenosquamous carcinoma of the cervix with distinctive glassy cytoplasm, prominent nucleoli, and aggressive clinical behavior.

Glaucoma

Glaucoma is progressive optic-nerve damage often associated with elevated or dysregulated intraocular pressure, leading to irreversible visual-field loss if untreated. ganglion cells and their axons, resulting in characteristic optic nerve head changes and visual field loss, with elevated intraocular pressure being the most important modifiable risk factor.

Glaucoma Drainage Implant

A surgically implanted tube-and-plate device that diverts aqueous humor from the anterior chamber to a subconjunctival reservoir, lowering intraocular pressure in selected glaucomas.

Glaze Poisoning

Toxic exposure to ceramic or pottery glaze, which may contain lead, cadmium, or other heavy metals. Symptoms depend on the metal involved and may include abdominal pain, neurological symptoms, and organ damage. Requires medical evaluation and chelation therapy in severe cases.

Alternate Names:

Ceramic glaze poisoning

Gleason Grading System

A histological grading system for prostate cancer based on glandular pattern differentiation. Score ranges from 2-10, with higher grades indicating more aggressive cancer. Used to guide treatment decisions and predict prognosis.

Alternate Names:

Gleason score; Gleason grade

Gleason Score

A grading system that estimates the aggressiveness of prostate cancer from its microscopic appearance.

Alternate Names:

Gleason grade group

Gleason Tumor Grade

A histopathologic grade of prostate adenocarcinoma based on the architectural pattern of the tumor glands; it contributes to the Gleason score and Grade Group.

Glenn Procedure

An intermediate, second-stage palliative surgical operation for children with functional single-ventricle congenital heart disease, performed between 4 and 6 months of age to transition the infant toward single-ventricle staging. Modernly performed as the Bidirectional Glenn shunt: involves dividing the superior vena cava from the right atrium and anastomosing its cardiac end-to-side to the undivided right pulmonary artery. Directs passive venous blood from the head and upper extremities directly into both the right and left pulmonary capillary beds for oxygenation, unloading the single ventricle from systemic-to-pulmonary volume overload while preserving pulmonary arterial architecture prior to the completion Fontan operation.

Alternate Names:

Bidirectional Glenn (BDG) Procedure; Bidirectional Cavopulmonary Anastomosis

Glenohumeral Joint

Ball-and-socket synovial joint between humeral head and glenoid cavity. Most mobile body joint allowing flexion, extension, abduction, adduction, medial/lateral rotation, circumduction. Mobility trades for stability, making it the most commonly dislocated joint. Stabilized by glenoid labrum, glenohumeral ligaments, and rotator cuff.

Glenoid

The shallow socket of the scapula that forms the shoulder joint with the humeral head.

Alternate Names:

Glenoid cavity

Glenoid Cavity

Shallow pear-shaped depression on the lateral scapu-

lar angle articulating with the humeral head. The glenoid labrum deepens the socket by approximately 50 percent. Due to its shallowness, the glenohumeral joint is the most commonly dislocated joint.

Gliadin

A group of wheat storage proteins that, together with glutenin, form gluten; gliadin peptides trigger the immune response in celiac disease.

Glial Cell

A non-neuronal cell that supports, nourishes, insulates, and regulates neurons; examples include astrocytes, oligodendrocytes, microglia, and Schwann cells.

Glial Cells

Glial Cells is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Glibenclamide

Glibenclamide is a medicine or pharmacologic product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent and patient.

Gliclazide

Gliclazide is a sulfonylurea for type 2 diabetes that stimulates pancreatic beta-cell insulin release; hypoglycaemia, weight gain, and dose accumulation in organ impairment are important considerations.

GLIM Criteria

A consensus framework for diagnosing malnutrition using at least one phenotypic criterion, such as weight loss, low body-mass index, or reduced muscle mass, and at least one etiologic criterion, such as reduced intake or assimilation or disease-related inflammation. Severity is graded using phenotypic thresholds.

Glimepiride

Glimepiride is a medicine or pharmacologic product; its indication, dose, interactions, contraindications,

and adverse effects depend on the specific agent and patient.

Glioblastoma

Grade IV astrocytoma, the most common and aggressive primary malignant brain tumor in adults. Arises de novo (primary GBM, IDH-wildtype) or from low-grade astrocytoma (secondary, IDH-mutant—better prognosis).

Glioblastoma Multiforme

A neoplastic process affecting glioblastoma multiforme; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Gliocladium

A biomedical term referring to gliocladium; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Glioma

A tumor arising from glial cells (astrocytes, oligodendrocytes, ependymal cells) in the brain or spinal cord. Includes astrocytomas, oligodendrogliomas, and ependymomas; behavior ranges from low-grade to highly aggressive (glioblastoma).

Gliomatosis Cerebri

A diffuse infiltrative growth pattern of glioma involving multiple brain regions, rather than a separate tumor type in current classifications; it may cause progressive cognitive, motor, or seizure-related symptoms.

Gliosarcoma

A neoplastic process affecting gliosarcoma; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Gliososis

A reactive increase in glial cells after injury, infection, ischemia, or other damage to the central nervous system.

Glipizide

Glipizide is a short-acting sulfonylurea that lowers blood glucose by stimulating insulin secretion from functioning pancreatic beta cells; hypoglycaemia is the principal adverse effect.

Glisson'S Capsule

Thin fibrous connective tissue envelope surrounding the liver, sending septa inward dividing it into lobules. Richly innervated by phrenic and intercostal nerve pain fibers, sensitive to distension and inflammation. Covered by visceral peritoneum except at the bare area.

Global Aphasia

A severe language disorder caused by extensive damage to the dominant cerebral hemisphere, impairing speech production, comprehension, reading, and writing.

Global Developmental Delay

A significant performance delay (defined as performance scores at least two standard deviations below the age-appropriate mean on standardized developmental testing) across two or more developmental domains: gross or fine motor skills, speech and language, cognitive function, social and personal interaction, and activities of daily living. The diagnosis is reserved for children younger than five years of age whose developmental levels cannot be reliably assessed with formal intellectual testing.

Alternate Names:

GDD (Global Developmental Delay); Generalized Developmental Delay

Global Health

Health issues transcending national boundaries. Examples: pandemics, health disparities, climate change.

Global Longitudinal Strain

An advanced quantitative echocardiographic parameter derived from two-dimensional speckle-tracking imaging that measures the percentage of myocardial shortening in the longitudinal plane from the base to the apex of the left ventricle during systole.

Expressed as a negative percentage (normal value typically more negative than -18 to -20 percent), it is significantly more sensitive than conventional left ventricular ejection fraction for detecting early, sub-clinical myocardial contractile impairment, particularly in cancer chemotherapy-induced cardiotoxicity (e.g., anthracyclines, trastuzumab), amyloidosis, aortic stenosis, and hypertrophic cardiomyopathy.

Alternate Names:

GLS; 2D Speckle-Tracking GLS; Left Ventricular Longitudinal Strain

Globicatella

Globicatella is a rare group of bacteria that can cause opportunistic infections, particularly in people with serious underlying illness.

Globicatella Sanguinis

A biomedical term referring to globicatella sanguinis; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Globin

A family of heme-binding proteins involved in oxygen transport and storage. Structure: globin fold (8 alpha-helices arranged in a globular structure), binds one heme molecule (iron-protoporphyrin IX) per subunit.

Globozoospermia

A rare sperm disorder in which sperm heads are round and lack a normal acrosome, causing impaired fertilization and male infertility.

Globule Unit

A small rounded particle or droplet used as a descriptive or measurement unit in laboratory, pharmaceutical, or biologic contexts; exact meaning depends on the source.

Globulin

A group of serum proteins (globular proteins) that are less soluble in water but soluble in dilute salt solutions. On serum protein electrophoresis, globulins migrate in the alpha-1, alpha-2, beta, and gamma

regions. Alpha-1 globulins: alpha-1-antitrypsin, alpha-fetoprotein.

Globus

A persistent or intermittent sensation of a lump or tightness in the throat without a true obstructing mass, often called globus pharyngeus.

Globus Hystericus

A sensation of a lump or tightness in the throat without a structural obstruction, now called globus sensation; reflux, muscle tension, and anxiety may contribute.

Globus Pallidus

The globus pallidus is a group of deep brain structures involved in regulating movement as part of the basal ganglia.

Globus Sensation

The persistent or intermittent sensation of a lump or foreign body in the throat without an anatomic lesion or dysphagia, also termed globus pharyngeus, typically noticed between meals. There is no impedance to swallowing and no weight loss; the feeling worsens with stress, anxiety, or laryngopharyngeal reflux.

Glomangiomyoma

A benign glomus tumor showing both vascular and smooth-muscle differentiation, usually presenting as a small painful subcutaneous nodule.

Glomangiosarcoma

A rare malignant glomus tumor showing cytologic atypia, infiltrative growth, or other malignant features; complete excision and specialist evaluation are required.

Glomerular Capsule

The cup-shaped structure surrounding a kidney glomerulus that receives filtered fluid.

Glomerular Filtrate

Fluid filtered from blood into the kidney tubules at

the glomerulus.

Glomerular Filtration

Glomerular filtration is the process by which the kidney filters water and small dissolved substances from blood into the kidney tubules to form the starting fluid for urine.

Glomerular Filtration Rate

Volume of plasma filtered by the glomeruli per unit time normally ninety to one hundred twenty milliliters per minute per one point seven three square meters body surface area. Estimated by the CKD-EPI equation using serum creatinine, age, sex, and race.

Glomerular Filtration Rate

The glomerular filtration rate is an estimate of how much blood the kidneys filter each minute and is a key measure of kidney function.

Glomeruli

Microscopic capillary networks in the kidneys where blood filtration begins.

Alternate Names:

Renal glomeruli

Glomerulomegaly

Abnormal enlargement of kidney glomeruli, seen with diabetes, obesity-related hyperfiltration, congenital syndromes, or other causes of glomerular stress.

Glomerulonephritis

Glomerulonephritis is inflammatory injury of the kidney glomeruli, producing hematuria, proteinuria, edema, hypertension, and variable decline in filtration.

nephrotic syndrome) and histopathology (e.g., IgA nephropathy, post-streptococcal, membranoproliferative, crescentic). Presents with hematuria, proteinuria, hypertension, and reduced GFR.

Alternate Names:

GN

Glomerulonephritis Acute

Sudden inflammation of the kidney glomeruli that can cause blood or protein in urine, edema, hypertension, and reduced kidney function.

Glomerulopathy

Any disease affecting the renal glomeruli, the capillary structures responsible for blood filtration. It may produce albuminuria or proteinuria, haematuria, reduced filtration, oedema, hypertension, or kidney failure and may be primary or secondary to systemic, metabolic, infectious, or immune disease.

Glomerulosclerosis

Scarring of the glomeruli, the kidney structures that filter blood.

Glomerulus

The tuft of capillaries in the renal corpuscle (Bowman's capsule) where blood is filtered to form urine.

Glomus Jugulare Tumor

A paraganglioma arising from the jugular bulb in the temporal bone. Benign but locally aggressive. Presents with pulsatile tinnitus, hearing loss, and cranial nerve palsies. Treatment includes surgery, radiation, and embolization.

Alternate Names:

Jugular paraganglioma

Glomus Tumor

A rare benign neoplasm arising from glomus bodies (thermoregulatory shunts), presenting as a small, blue-red, exquisitely tender subungual or digital nodule. The classic triad is severe pain, point tenderness, and cold sensitivity. Diagnosis is clinical with positive Love pin test and Hildreth sign (tumor disappears with ischemia).

Glomus Tympanicum

A benign vascular tumor arising from the glomus bodies on the promontory of the middle ear, presenting with pulsatile tinnitus, hearing loss, and a reddish mass behind the tympanic membrane.

Glomus Tympanicum Tumor

A paraganglioma arising from the glomus bodies on the promontory of the middle ear. Benign tumor causing pulsatile tinnitus and conductive hearing loss. Diagnosis by otoscopy and imaging. Treatment includes surgery or observation.

Alternate Names:

Middle ear paraganglioma

Glossal

Pertaining to the tongue (extitglossa), the muscular organ in the oral cavity essential for mastication, deglutition, taste sensation (gustation), and speech articulation. Composed of eight paired skeletal muscles: four intrinsic muscles (superior longitudinal, inferior longitudinal, transverse, vertical) that alter tongue shape, and four extrinsic muscles (genioglossus, hyoglossus, styloglossus, palatoglossus) that alter position, all innervated by the hypoglossal nerve (CN XII), except the palatoglossus (CN X). Somatosensory innervation involves CN V3 (anterior two-thirds) and CN IX (posterior third); taste involves CN VII (chorda tympani, anterior two-thirds) and CN IX (posterior third). Clinical conditions include glossitis, atrophic tongue (vitamin B₁₂/iron deficiency), and lingual motor weakness.

Glossalgia

Pain or burning of the tongue, with causes including trauma, infection, nutritional deficiency, neuropathy, medication, or burning-mouth syndrome.

Glossectomy

Surgical removal of all or part of the tongue, performed for malignancies (most common), severe trauma, or macroglossia. Procedures range from partial glossectomy to total glossectomy. Speech and swallowing rehabilitation are often required.

Glossinidae

The family of flies belonging to the genus *Glossina*, commonly known as tsetse flies. These insects are vectors of trypanosomes that cause African trypanosomiasis (sleeping sickness) in humans and nagana in animals.

Alternate Names:

Tsetse flies

Glossitis

Inflammation of the tongue, which can cause pain, swelling, and changes in tongue appearance. Causes include nutritional deficiencies (iron, B12, folate), infections (bacterial, viral, fungal), allergic reactions, and autoimmune conditions. Treatment depends on the underlying cause.

Glossodynia

Burning Mouth Syndrome: a chronic pain condition characterized by a burning sensation in the mouth, often affecting the tongue, lips, and palate, without visible abnormalities. May be associated with dry mouth, altered taste, or tingling. Diagnosis is clinical after excluding other causes.

Alternate Names:

Burning Mouth Syndrome

Glossopharyngeal Nerve

glossopharyngeal nerve is a nerve pathway or nerve-related structure that carries signals between the nervous system and its target; symptoms depend on the site of involvement.

Glossopharyngeal Neuralgia

Recurrent, brief attacks of severe stabbing or electric-shock-like pain in the throat, tonsillar region, base of the tongue, or ear, caused by irritation of the glossopharyngeal nerve. Swallowing, coughing, or talking may trigger attacks.

Glossoptosis

Backward displacement of the tongue that can narrow the upper airway, particularly in infants with craniofacial disorders.

Glottis

The opening between the vocal folds (true vocal cords) in the larynx, through which air passes during respiration and phonation. Boundaries: anterior (anterior commissure, attachment of vocal folds to thyroid cartilage), posterior (arytenoid cartilages and posterior commissure).

Gloves

Hand coverings used to protect the wearer, the patient, or both from contamination, chemicals, or injury. Medical gloves may be examination or surgical gloves and may be sterile or nonsterile.

Glucagon-Like Peptide-1 Receptor Agonists (GLP-1 Receptor Agonists)

A class of medicines that activate the glucagon-like peptide-1 receptor. They increase glucose-dependent insulin release, reduce inappropriate glucagon release, slow gastric emptying, and increase fullness. They are used for selected people with type 2 diabetes or chronic weight management. Examples include semaglutide, liraglutide, and dulaglutide.

Alternate Names:

GLP-1 Agonists; GLP-1 receptor agonists; GLP-1 Receptor Agonist; incretin mimetics

GLS

Alternate name; see *Global Longitudinal Strain*.

Glu

Commonly an abbreviation for glucose in laboratory or clinical notes; in biochemical notation, Glu can also denote the amino acid glutamic acid, so context matters.

Glucagon

A peptide hormone produced by the alpha cells of the pancreatic islets (Langerhans).

Glucagon Blood Test

Laboratory measurement of glucagon levels in the blood, used to evaluate suspected glucagonoma, hypoglycemia, or other pancreatic endocrine disorders. Levels are affected by fasting, meals, and medications.

Alternate Names:

Serum glucagon level

Glucagon Signaling

Glucagon binds GPCR, activates adenylate cyclase, increases cAMP, activates PKA. Promotes

glycogenolysis, gluconeogenesis.

Glucagonoma

A rare pancreatic neuroendocrine tumor that secretes excess glucagon, causing diabetes mellitus (mild, new onset), dermatitis (necrolytic migratory erythema, characteristic painful, erythematous, blistering rash on groin, perineum, extremities), diarrhea, and deep vein thrombosis.

Glucan

A glucan is a polysaccharide made of linked glucose units; different glucans occur in foods, fungi, plants, and bacteria.

Glucarpidase

An enzyme that rapidly breaks down methotrexate in the bloodstream and is used for toxic methotrexate concentrations when renal clearance is inadequate.

Glucocerebrosidase Deficiency

Gaucher Disease: an inherited lysosomal storage disorder caused by deficient glucocerebrosidase, leading to lipid accumulation in cells and possible enlargement of the spleen and liver, anemia, bone disease, or neurologic disease.

Alternate Names:

Gaucher Disease

Glucocorticoid

A corticosteroid hormone or medicine that regulates inflammation, immunity, metabolism, and stress responses; prolonged treatment has systemic adverse effects.

Glucocorticoid Replacement Therapy

Replacement of cortisol deficiency in adrenal insufficiency using hydrocortisone in divided doses. Typical regimen is ten to fifteen milligrams in the morning and five to ten milligrams in the early afternoon mimicking the physiological cortisol diurnal rhythm.

Glucocorticoid-Induced Osteoporosis

Bone loss and increased fracture risk caused or accelerated by prolonged or high-dose glucocorticoid treatment; risk depends on dose, duration, baseline bone health, and other factors.

Glucocorticoid-Sparing Therapy

Use of a non-glucocorticoid treatment strategy to control inflammation while reducing cumulative corticosteroid exposure. It aims to limit osteoporosis, infection, diabetes, hypertension, adrenal suppression, and other glucocorticoid complications.

Glucocorticoids

A class of steroid hormones produced by the zona fasciculata of the adrenal cortex, regulated by the hypothalamic-pituitary-adrenal (HPA) axis.

Glucometer

A handheld device that measures blood glucose, usually from a small capillary-blood sample obtained by finger prick.

Gluconacetobacter Liquefaciens

An acetic-acid bacterium found mainly in environmental and plant-associated settings; it is not a common human pathogen.

Gluconates

Gluconates are salts or esters of gluconic acid, used in some medicines and supplements such as calcium gluconate.

Gluconeogenesis

Metabolic pathway generating glucose from non-carbohydrate precursors (lactate, glycerol, glucogenic amino acids). Occurs primarily in liver (90%) and kidney (10%).

Glucosamine

A naturally occurring compound used in cartilage formation and sometimes taken as a supplement for joint symptoms.

Glucose

A monosaccharide (C₆H₁₂O₆) that is the primary energy source for cells. Types: D-glucose (dextrose, naturally occurring enantiomer) and L-glucose (synthetic). Routes of entry: dietary carbohydrates (digested to glucose), gluconeogenesis (synthesis from non-carbohydrate precursors), glycogenolysis (breakdown of stored glycogen).

Glucose Challenge Test

A screening test for gestational diabetes mellitus, commonly using a nonfasting 50-g glucose drink followed by a one-hour plasma-glucose measurement. An abnormal screening result is followed by a diagnostic oral glucose tolerance test; the threshold depends on the protocol used.

Alternate Names:

GCT; one-hour glucose challenge test; 50-g glucose screening test.

Glucose Intolerance

Impaired handling of glucose resulting in blood-glucose concentrations above normal but below the diagnostic threshold for diabetes in a defined testing context. It often reflects insulin resistance or inadequate insulin secretion and is associated with increased risk of type 2 diabetes and cardiovascular disease.

Glucose Management Indicator

Derived metric from continuous glucose monitoring that estimates the hemoglobin A1c equivalent based on mean sensor glucose values. Formula converts mean glucose to an A1c-equivalent value allowing comparison between CGM data and traditional A1c.

Glucose Screening Tests During Pregnancy

Laboratory tests to detect gestational diabetes mellitus, including the oral glucose tolerance test and glucose challenge test. Usually performed between 24-28 weeks of gestation. Abnormal results require further evaluation and management.

Alternate Names:

Gestational diabetes screening

Glucose Sensor

Continuous glucose monitor for diabetes. Subcutaneous sensor measures interstitial glucose. Transmitter sends data to receiver/phone. Examples: Dexcom, Freestyle Libre, Medtronic. Reduces hypoglycemia, improves A1c.

Glucose Tolerance Test

Testing for diabetes mellitus and gestational diabetes. 75g OGTT (oral glucose tolerance test): measure plasma glucose at 0, 60, and 120 minutes. Diagnostic: fasting ≥ 126 mg/dL, 2-hour ≥ 200 mg/dL. Gestational diabetes screening: 50g glucose challenge test (GCT) at 24-28 weeks; if >140 mg/dL at 1 hour, proceed to 100g 3-hour OGTT.

Alternate Names:

Oral glucose tolerance test (OGTT); gestational diabetes glucose tolerance test.

Glucose Tolerance Test - Non-Pregnant

Laboratory test measuring blood glucose levels after oral glucose administration to evaluate glucose metabolism and diagnose diabetes or prediabetes. Involves fasting, blood glucose measurement, ingestion of glucose solution, and serial blood draws over 2-3 hours.

Alternate Names:

OGTT; oral glucose tolerance test

Glucose Transport

glucose transport is a nutrition or dietary term; appropriate use depends on age, health status, allergies, intake, and medical treatment.

Glucose Transporter

A membrane protein that moves glucose across a cell membrane; transporter defects or altered regulation can affect glucose metabolism.

Glucose Urine Test

Laboratory test to detect glucose in the urine (glycosuria). Normally absent; presence indicates blood glucose exceeding renal threshold, suggesting diabetes mellitus or renal tubular dysfunction. May be detected by dipstick or laboratory analysis.

Alternate Names:

Urinary glucose; glycosuria test

Glucose-6-Phosphatase

Enzyme converting glucose-6-phosphate to glucose. Located in liver and kidney. Deficiency causes Von Gierke disease.

Glucose-6-Phosphate Dehydrogenase Deficiency

Glucose-6-phosphate dehydrogenase deficiency is an inherited red-cell enzyme disorder in which oxidative stress can trigger hemolytic anemia after selected infections, foods, or medicines. episodes of hemolytic anemia triggered by oxidative stress from infections, certain drugs including primaquine and sulfonamides, and fava beans. G6PD is essential for generating NADPH to protect red blood cells from oxidative damage.

Glucose-6-Phosphate Dehydrogenase Test

Laboratory test to measure G6PD enzyme activity in red blood cells, used to diagnose G6PD deficiency. Important before prescribing certain medications (primaquine, dapsone) that can cause hemolytic anemia in deficient individuals.

Alternate Names:

G6PD assay

Glucosinolate

A sulfur- and nitrogen-containing plant compound found especially in cruciferous vegetables; its breakdown products influence flavor and are being studied for possible biologic effects.

Glucosuria

The presence of glucose in urine, most commonly from blood glucose exceeding the renal reabsorption threshold, diabetes, pregnancy, or a proximal tubular reabsorption defect.

Glucosylceramidase

Glucosylceramidase is a lysosomal enzyme that hydrolyses glucosylceramide; inherited deficiency causes Gaucher disease with lipid-laden macrophages in organs such as spleen, liver, bone

marrow, and bone.

Glucuronate

The anion or salt of glucuronic acid, a sugar-acid component of connective-tissue substances and a molecule used in hepatic conjugation reactions.

Glucuronic Acid

A sugar acid used by the liver to conjugate bilirubin, drugs, hormones, and toxins, increasing their water solubility for excretion.

Glucuronide

A compound formed when glucuronic acid is conjugated to a drug, hormone, bilirubin, or other molecule, usually increasing water solubility and facilitating excretion.

Glucuronyl Transferase

UDP-glucuronosyltransferase: a liver enzyme involved in phase II drug metabolism, catalyzing glucuronidation of bilirubin, drugs, and toxins to make them water-soluble for excretion. Deficiency causes Gilbert syndrome or Crigler-Najjar syndrome.

Alternate Names:

UDP-glucuronosyltransferase; UGT

Glue (Vascular)

A vascular embolization material used to occlude selected vessels under image guidance.

Alternate Names:

Vascular

Glue Ear

Glue ear is persistent fluid behind the eardrum without the acute signs of a middle-ear infection and may cause reduced hearing.

Glue Ear (Secretory Otitis Media)

Persistent fluid in the middle ear without acute infection, often causing conductive hearing loss and a feeling of blockage. It is common in children.

Alternate Names:

Secretory Otitis Media

Glutamate

The principal excitatory neurotransmitter in the central nervous system. Synthesized from glutamine or alpha-ketoglutarate and acting on ionotropic receptors (AMPA, kainate, NMDA) for fast synaptic transmission and metabotropic receptors (mGluR1-8) for modulation.

Glutamate Decarboxylase

Glutamate decarboxylase is an enzyme that converts glutamate into the neurotransmitter GABA; antibodies against it are associated with some autoimmune disorders.

Glutamate Receptor

A neuronal receptor activated by glutamate, the major excitatory neurotransmitter, including NMDA, AMPA, kainate, and metabotropic receptor classes.

Glutamic Acid

An amino acid that acts as a metabolic intermediate and neurotransmitter precursor; its charged form, glutamate, mediates most fast excitatory signaling in the brain.

Glutamic Acid Decarboxylase

An enzyme that converts glutamate to GABA; antibodies to it can occur in type 1 diabetes.

Alternate Names:

GAD

Glutamine

Glutamine is an amino acid used in protein production and nitrogen transport and serves as an energy source for some cells.

Glutaral

Glutaral is a chemical disinfectant and tissue fixative that can irritate the skin, eyes, and airways.

Glutaric Acidemia Type 1

An autosomal recessive cerebral organic aciduria caused by deficiency of glutaryl-CoA dehydrogenase (*GCDH*), a key mitochondrial flavoprotein enzyme in lysine, hydroxylysine, and tryptophan

catabolism. Leads to accumulation of glutaric acid and 3-hydroxyglutaric acid in tissues and urine. Classically presents with macrocephaly at birth followed by an acute, irreversible striatal encephalopathic crisis typically triggered by a routine febrile infant illness between 6 and 18 months of age, resulting in bilateral striatal necrosis and severe dystonia. Brain MRI shows frontotemporal atrophy and pathognomonic enlarged sylvian fissures (“bat-wing” appearance) accompanied by subdural hematomas.

Alternate Names:

Glutaric Aciduria Type 1 (GA-1); Glutaryl-CoA Dehydrogenase Deficiency

Glutaric Aciduria

A clinical disorder or finding related to glutaric aciduria; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Glutathione

Antioxidant tripeptide (Glu-Cys-Gly). Detoxifies reactive metabolites (e.g., NAPQI from acetaminophen). Depletion causes toxicity. N-acetylcysteine replenishes glutathione.

Glutathione Disulfide

Glutathione disulfide is the oxidized form of glutathione, produced when glutathione helps neutralize oxidants in cells.

Glutathione Transferase

Glutathione transferase is an enzyme family that attaches glutathione to certain chemicals, helping detoxify them and process medicines.

Gluteal

Relating to the buttock or the gluteal muscles, which extend and rotate the hip and help stabilize the pelvis.

Gluteal Injury

Damage to the muscles, tendons, nerves, or soft tissues of the buttock caused by trauma, overuse, injection, or falls. Pain, bruising, weakness, or difficulty walking may occur. Severe swelling, numbness,

inability to bear weight, or bowel or bladder changes requires urgent evaluation.

Gluteal Muscle

gluteal muscle is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Gluten

A complex of storage proteins (prolamins and glutelins) found in wheat (gliadin), barley (hordein), and rye (secalin). Gluten provides elasticity to dough. In susceptible individuals, gluten triggers an autoimmune response (celiac disease) or non-autoimmune sensitivity (non-celiac gluten sensitivity, NCGS).

Gluten Enteropathy

A clinical disorder or finding related to gluten enteropathy; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Alternate Names:

Celiac Disease (Gluten Enteropathy)

Gluten Sensitivity (Nonceliac Gluten Sensitivity (Intolerance))

Adverse symptoms triggered by gluten ingestion in individuals without celiac disease or wheat allergy. Symptoms include bloating, abdominal pain, diarrhea, and fatigue. Diagnosis is by exclusion. Management involves gluten avoidance.

Alternate Names:

Nonceliac gluten sensitivity; NCGS

Gluten-Free Diet

Eliminates gluten (wheat, barley, rye). Essential for celiac disease, non-celiac gluten sensitivity. Risk of nutrient deficiency if not supplemented. May be unnecessarily restrictive for others.

Gluten-Sensitive Enteropathy

Celiac Disease: an autoimmune disorder triggered by gluten ingestion in genetically susceptible individuals, causing damage to the small intestinal mucosa.

Presents with diarrhea, bloating, weight loss, and malabsorption. Diagnosis by serology and intestinal biopsy.

Alternate Names:

Celiac Disease; celiac sprue

Gluten-Sensitive Enteropathy (Celiac Disease)

An immune-mediated disorder in which ingestion of gluten damages the lining of the small intestine in genetically susceptible people. It may cause diarrhea, malabsorption, anemia, or other systemic effects.

Alternate Names:

Celiac Disease

Gluteus

Gluteus is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gluteus Maximus Muscle

Largest, most superficial gluteal muscle forming the buttock bulk, from posterior iliac gluteal line, sacrum, coccyx, and sacrotuberous ligament to the iliotibial tract and gluteal tuberosity. Innervated by inferior gluteal nerve (L5-S2). Primary hip extensor and lateral rotator.

Gluteus Medius Muscle

Thick fan-shaped muscle between maximus and minimus, from external ilium to the greater trochanter. Innervated by superior gluteal nerve (L4-S1). Primary hip abductor essential for pelvic stability during single-leg stance. Superior gluteal nerve injury causes Trendelenburg sign with contralateral pelvic drop.

Gluteus Minimus Muscle

The smallest and deepest gluteal muscle, extending from the outer ilium to the greater trochanter. It abducts and medially rotates the hip and helps stabilize the pelvis during walking.

Glyburide

An oral sulfonylurea that stimulates pancreatic insulin release for type 2 diabetes; prolonged or severe hypoglycemia is an important risk.

Glycaemia

The concentration of glucose in the blood, normally maintained within a narrow range (fasting: 3.9-5.5 mmol/L or 70-100 mg/dL) through the coordinated action of insulin (lowers glucose) and counter-regulatory hormones (glucagon, cortisol, catecholamines, growth hormone).

Glycated Hemoglobin (A1C)

Glycated hemoglobin reflecting average blood glucose concentration over the preceding two to three months corresponding to the red blood cell lifespan. Formed by non-enzymatic glycation of hemoglobin at the N-terminal valine of the beta chain. The percent of glycated hemoglobin is directly proportional to mean glucose levels.

Alternate Names:

A1C A1C test Hemoglobin A1C HbA1C

Glycemic Control

Management of blood glucose to individualized targets that reduce symptoms and long-term complications while avoiding hypoglycemia. Goals and monitoring methods depend on age, health, treatment, pregnancy, and patient circumstances.

Glycemic Index

Ranking of carbohydrates by effect on blood glucose. High GI foods spike glucose. Low GI foods provide sustained energy. Useful for diabetes management.

Glycemic Index And Diabetes

The glycemic index ranks carbohydrate-containing foods by their effect on blood glucose levels. Low GI foods (≤ 55) cause gradual glucose rise; high GI foods (≥ 70) cause rapid spikes. Important for diabetes management and dietary planning.

Alternate Names:

GI; glycemic index

Glycemic Load

A measure that combines a food's glycemic index with the amount of available carbohydrate in a serving to estimate its effect on blood glucose.

Glyceraldehyde

Glyceraldehyde is a three-carbon sugar and an intermediate in carbohydrate metabolism.

Glyceride

A glyceride is a compound formed when glycerol is linked to one or more fatty acids, including many dietary fats.

Glycerin

A colorless, odorless, sweet-tasting trihydroxy alcohol (glycerol) with humectant, osmotic, and solvent properties.

Glycerol

Glycerol is a three-carbon alcohol found in fats and used as a solvent, humectant, and osmotic medicine in selected settings.

Glycerol (Glycerine)

A triol compound ($\text{CH}_2\text{OH}-\text{CHOH}-\text{CH}_2\text{OH}$), a colourless, odourless, viscous liquid with a sweet taste.

Alternate Names:

Glycerine

Glycerol Kinase

Glycerol kinase is an enzyme that adds phosphate to glycerol, allowing it to enter pathways for energy production and fat metabolism.

Glycerophosphates

Glycerophosphates are compounds containing glycerol and phosphate that participate in cell membranes and metabolism.

Glycerophospholipids

The major membrane phospholipids built from glycerol, fatty acids, phosphate, and a polar head group, including phosphatidylcholine and phos-

phatidylethanolamine.

Glyceryl Trinitrate

Glyceryl Trinitrate is a medicine or pharmacologic product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent and patient.

Glycerylphosphorylcholine

Glycerylphosphorylcholine is a choline-containing phospholipid metabolite found in cell membranes and body fluids.

Glycine

Glycine is a small amino acid used to make proteins and heme and also acts as an inhibitory neurotransmitter in the nervous system.

Glycine Receptor

A ligand-gated chloride channel activated by glycine that mediates inhibitory neurotransmission, especially in the spinal cord and brainstem.

Glycocalyx

The glycocalyx is a sugar-rich layer on the surface of many cells that helps with protection, adhesion, and cell recognition.

Glycogen

A branched polysaccharide composed of glucose units linked by alpha-1,4 glycosidic bonds (linear) and alpha-1,6 bonds (branching points, every 8-10 residues). Glycogen serves as the main storage form of glucose in animals.

Glycogen Storage Disease

A group of inherited metabolic disorders caused by defects in glycogen synthesis, breakdown, or regulation, resulting in abnormal glycogen accumulation in tissues. Major types: Type I (von Gierke, G6PC deficiency) — fasting hypoglycemia, lactic acidosis, hepatomegaly, hyperuricemia, and hypertriglyceridemia.

Glycogen Storage Diseases

A clinical disorder or finding related to glycogen storage diseases; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Glycogenesis

The synthesis of glycogen from glucose, primarily in liver and skeletal muscle. Glucose-1-phosphate is converted to UDP-glucose, and glycogen synthase and the branching enzyme extend and branch the polymer; insulin promotes the pathway.

Glycogenolysis

Breakdown of glycogen to glucose-1-phosphate (converted to glucose-6-phosphate). Liver glycogenolysis maintains blood glucose during fasting; muscle glycogenolysis provides energy for contraction. Regulated by hormonal signals (glucagon, epinephrine) and allosteric effectors (AMP activates phosphorylase).

Glycohemoglobin

Hemoglobin with glucose chemically attached to it; glycated hemoglobin A1c reflects average blood glucose over roughly the preceding two to three months.

Glycolates

Glycolates are salts or esters of glycolic acid and may occur as metabolic products or in chemical and medical preparations.

Glycolipid

A lipid containing a carbohydrate group that contributes to cell membranes, recognition, signaling, and myelin structure.

Glycolysis

Metabolic pathway breaking down glucose (6C) to pyruvate (3C), generating 2 ATP (net) and 2 NADH. Occurs in cytoplasm. Ten enzymatic steps: investment phase (uses 2 ATP) and payoff phase (produces 4 ATP, 2 NADH). Key regulatory enzymes: hexokinase, phosphofructokinase-1 (PFK-1, rate-limiting), pyruvate kinase.

Glycomics

Glycomics is the study of all or many of the sugar chains attached to proteins, lipids, and cells in a biological system.

Glycopeptide

glycopeptide is a defined chemical or biologic substance involved in glycopeptide and studied in physiology, pharmacology, laboratory medicine, or treatment.

Glycophorin

Glycophorin is a glycoprotein in the red-cell membrane that contributes to the cell surface and carries blood-group-related markers.

Glycoprotein

A protein with covalently attached carbohydrate chains, important in receptors, antibodies, enzymes, mucus, cell adhesion, and extracellular matrix.

Glycopyrrolate

Glycopyrrolate is an anticholinergic medicine used to reduce secretions and, in some inhaled forms, to help control chronic obstructive lung disease; it may cause dry mouth, constipation, or urinary retention.

Glycosaminoglycan

A long, unbranched carbohydrate chain found in the extracellular matrix and connective tissues. Abnormal breakdown or accumulation of GAGs occurs in mucopolysaccharidoses.

Alternate Names:

Glycosaminoglycan

Glycosaminoglycans

Glycosaminoglycans are long chains of sugars found in connective tissue and involved in its structure, hydration, and signaling.

Glycoside

A glycoside is a compound in which a sugar is linked to a nonsugar molecule; some glycosides are medicines or biologically active substances.

Glycosphingolipid

A membrane lipid containing ceramide and one or more sugars, involved in cell recognition, signaling, and nervous-system structure.

Glycosuria

The presence of glucose in the urine. Normally, the renal tubules reabsorb filtered glucose via sodium-glucose co-transporters (SGLT2 in proximal tubule). Glycosuria occurs when blood glucose exceeds the renal threshold (approximately 10 mmol/L or 180 mg/dL), overwhelming reabsorptive capacity.

Glycosylated Hemoglobin

Hemoglobin with glucose attached, reflecting average blood glucose over roughly two to three months.

Alternate Names:

Hemoglobin A1c; HbA1c

Glycosylation

Glycosylation is the attachment of sugar chains to proteins or lipids, a process that affects their function, stability, and location.

Glycosylphosphatidylinositol

Glycosylphosphatidylinositol is a lipid-based anchor that attaches certain proteins to the outer surface of a cell membrane.

Glycosyltransferase

Glycosyltransferase is an enzyme that transfers a sugar to a protein, lipid, or other molecule during glycan synthesis.

Glycyrrhetic Acid

An active metabolite of licorice glycosides that inhibits 11-beta-hydroxysteroid dehydrogenase; excess exposure can cause hypertension, low potassium, and fluid retention.

Glycyrrhizic Acid

The major saponin in licorice root that is converted to glycyrrhetic acid and can produce mineralocorticoid-like toxicity when consumed excessively.

Glyoxal

A reactive dialdehyde formed in metabolism and food processing that can modify proteins and DNA and contribute to advanced glycation products.

Glyoxysome

A specialized type of peroxisome in plant cells that converts stored fatty acids into sugars during seed germination; it has no direct human organ equivalent.

Glyphosate

A broad-spectrum herbicide that inhibits plant EPSP synthase; concentrated exposure can irritate tissues and cause systemic poisoning, especially after ingestion.

Glyphosine

A chemical compound used experimentally or in agriculture; its medical significance depends on the specific formulation and exposure context.

Gm (Gram)

An abbreviation for gram, a metric unit of mass equal to one-thousandth of a kilogram.

GN

Alternate index label; see *Glomerulonephritis*.

Gnathic

Gnathic is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gnathion

Gnathion is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gnathitis

Gnathitis is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gnathoplasty

Gnathoplasty is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gnathostoma

gnathostoma is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Gnathostoma Spinigerum

gnathostoma spinigerum is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Gnathostomiasis

A parasitic infection caused by *Gnathostoma* larvae acquired from undercooked freshwater fish or other hosts, producing migratory skin swellings or deeper neurologic or ocular disease.

Gnawing Pain

A biomedical term referring to gnawing pain; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

GNE Myopathy

An autosomal recessive adult-onset distal myopathy caused by biallelic mutations in the *GNE* gene on chromosome 9p13.3, which encodes the bifunctional rate-limiting enzyme in sialic acid biosynthesis, UDP-*N*-acetylglucosamine 2-epimerase/*N*-acetylmannosamine kinase. Typically manifests in the third decade of life with slowly progressive bilateral foot drop due to selective weakness of the tibialis anterior muscle, followed by distal lower and upper extremity involvement, but with pathognomonic, striking, and permanent sparing of the quadriceps femoris muscles even in late, wheelchair-dependent stages. Muscle biopsy shows rimmed vacuoles and filamentous inclusions under electron microscopy without inflammatory infiltrates.

Alternate Names:

Nonaka Myopathy; Distal Myopathy with Rimmed Vacuoles (DMRV); Hereditary Inclusion Body Myopathy (HIBM)

GnRH

Gonadotropin-releasing hormone, which stimulates pituitary release of LH and FSH.

Alternate Names:

Gonadotropin-releasing hormone

GnRH Agonist

Gonadotropin-releasing hormone agonists are a class of medications that initially stimulate (flare effect, 1-3 weeks) and then suppress pituitary gonadotropin (LH and FSH) secretion through receptor down-regulation, leading to profound suppression of sex steroid production (testosterone in men, estradiol in women).

GnRH Antagonist

A class of medications that competitively block GnRH receptors on pituitary gonadotrophs, producing immediate, reversible suppression of LH and FSH secretion without the initial flare effect seen with GnRH agonists.

GnRH Antagonists

Medicines that block GnRH signaling and reduce pituitary release of reproductive hormones.

Goal Attainment Scaling

An individualized outcome method in which patient-specific goals are defined with graded levels of expected attainment and scored after intervention. It captures meaningful changes that standardized measures may miss.

Goal Setting

SMART goals: Specific, Measurable, Achievable, Relevant, Time-bound. Patient-centered goals improve adherence.

Goals-Of-Care Discussion

A clinical conversation that aligns treatment recommendations with a patient's values, priorities, prog-

nosis, acceptable burdens, and preferred outcomes. It may include decisions about hospitalization, resuscitation, artificial nutrition, and comfort-focused care.

Goblet Cell

A mucus-secreting epithelial cell, abundant in the respiratory and intestinal tracts, that protects surfaces with a mucin-rich layer.

Goblet Cell Adenocarcinoma

A distinctive appendiceal adenocarcinoma composed of goblet-like mucinous cells, formerly called goblet cell carcinoid. It can spread within the peritoneum or to the ovaries and is managed according to grade, stage, and dissemination.

Goblet Cells

Specialized epithelial cells that secrete mucus, especially in the intestine and airways.

Goiter

Enlargement of the thyroid gland, which can be diffuse or nodular. Causes: iodine deficiency (most common worldwide), Hashimoto thyroiditis, Graves disease, multinodular goiter, and thyroid cancer. May be associated with normal, hyperthyroid, or hypothyroid function.

Alternate Names:

Thyroid Gland Enlargement

Goitre

A clinical disorder or finding related to goitre; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Goitrogenic Foods

Foods or food components that can interfere with thyroid hormone production or iodine use under some conditions; risk depends on intake, preparation, iodine status, and thyroid health.

Gold

A metal used in selected dental restorations, implants, laboratory applications, and historical medic-

inal preparations; exposure can cause contact allergy or toxicity depending on the compound and dose.

Gold Alloy

Noble metal alloy for dental restorations. Excellent biocompatibility, durability, marginal adaptation. Expensive.

Gold Alloys

Metal mixtures containing gold and other elements, used in selected dental restorations or medical devices; properties and biocompatibility depend on composition.

Gold Therapy

Gold Therapy is a medicine or pharmacologic product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent and patient.

Golden Hour

The golden hour is the early period after a serious injury or medical emergency when rapid assessment and treatment may improve the chance of survival and recovery.

Goldenhar Syndrome

A variant of hemifacial microsomia characterized by the triad: epibulbar dermoids (benign conjunctival tumors), preauricular skin tags or sinuses, and vertebral anomalies (hemivertebrae, fused vertebrae, scoliosis). Also involves facial asymmetry, microtia, conductive hearing loss, and macrostomia.

Goldmann Tonometry

A slit-lamp examination that estimates intraocular pressure by measuring the force needed to flatten a defined area of the cornea.

Golfer Elbow

Medial epicondylitis, a tendinopathy of the common flexor/pronator tendon at the medial epicondyle. Less common than tennis elbow. Caused by repetitive wrist flexion and forearm pronation. Presents with medial elbow pain worsened by wrist flex-

ion and gripping. Tenderness over the medial epicondyle.

Golfer'S Elbow

Painful inflammation or degeneration of tendons on the inner side of the elbow.

Golfer'S Elbow (Medial Epicondylitis)

Overuse injury involving tendons on the inside of the elbow, causing pain and tenderness with gripping or wrist movement.

Alternate Names:

Medial Epicondylitis

Golgi Apparatus

Membrane-bound organelle (Golgi complex) consisting of flattened cisternae (cis, medial, trans) that modifies, sorts, and packages proteins and lipids from the ER for secretion or delivery to other organelles.

Golgi Cisternae

Flattened membrane-bound sacs that form the Golgi apparatus and modify, sort, and package proteins and lipids for transport within or outside the cell.

Golgi Tendon Organ

A mechanoreceptor located at the junction between a skeletal muscle and its tendon, innervated by Ib afferent fibers. The Golgi tendon organ detects tension generated by muscle contraction or passive stretch.

Gomori'S Stain

Gomori's stain is a group of laboratory stains used to highlight structures such as connective tissue, carbohydrates, fungi, or muscle components in tissue samples.

Gonad

A testis or ovary, an organ that produces reproductive cells and sex hormones.

Gonad Disorder

A disease affecting testes or ovaries and causing

abnormal reproductive-cell production, hormone secretion, development, fertility, or sexual function.

Gonadal Agenesis

Congenital absence or severe underdevelopment of a gonad, resulting in impaired sex-hormone production and infertility or differences in sexual development.

Gonadal Dysgenesis

Incomplete or abnormal development of the ovaries or testes, often resulting from chromosomal abnormalities, gene variants, or disordered sex development. It may cause atypical genital development, delayed or absent puberty, infertility, and abnormal gonadal hormone production.

Gonadal Vein

A vein draining a testis or ovary. The right gonadal vein usually drains directly into the inferior vena cava, while the left usually drains into the left renal vein.

Gonadoblastoma

A rare tumor of an abnormal or developmentally disordered gonad that may contain germ cells and sex-cord cells.

Gonadorelin

Synthetic gonadotropin-releasing hormone that stimulates pituitary release of luteinizing hormone and follicle-stimulating hormone when given intermittently.

Gonadotrophin

Gonadotrophin is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gonadotrophin-Releasing Factors

Gonadotrophin-Releasing Factors is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gonadotrophin-Releasing Hormone (GnRH)

Gonadotrophin-Releasing Hormone (GnRH) is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gonadotrophins

Hormones that stimulate the gonads. The anterior pituitary secretes follicle-stimulating hormone (FSH) and luteinising hormone (LH), both glycoprotein hormones sharing a common alpha subunit (identical to TSH and hCG alpha) with a unique beta subunit.

Gonadotrophs

Anterior-pituitary cells that secrete luteinizing hormone and follicle-stimulating hormone in response to gonadotropin-releasing hormone.

Gonadotropin

A hormone, chiefly luteinizing hormone or follicle-stimulating hormone, that acts on the ovaries or testes to regulate gamete production and sex-hormone secretion.

Gonadotropin-Releasing Hormone

A hypothalamic hormone that controls pituitary secretion of FSH and LH.

Alternate Names:

GnRH

Gonadotropins

Hormones that act on the gonads. In humans, the pituitary gonadotropins are follicle-stimulating hormone and luteinizing hormone, which regulate gamete production and sex-hormone secretion.

Gonads

The primary reproductive organs: testes in males and ovaries in females. Functions: (1) gamete production (spermatogenesis in testes, oogenesis in ovaries); (2) sex steroid hormone secretion (testosterone from testicular Leydig cells; estrogen and progesterone from ovarian follicular cells and corpus

luteum).

Gondoic Acid

A 20-carbon monounsaturated fatty acid found in some seed oils and incorporated into dietary and tissue lipids.

Goniometer

An instrument used to measure the angle or range of motion of a joint.

Gonioscopy

An ophthalmological examination technique that allows direct visualization of the anterior chamber angle, the anatomical region where the cornea and iris meet and where the trabecular meshwork is located, and it is essential for distinguishing between open-angle and angle-closure forms of glaucoma.

Gonococcal

A clinical disorder or finding related to gonococcal; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Gonococcal Arthritis

A disseminated infection of joints caused by *Neisseria gonorrhoeae*, often with fever, migratory joint or tendon pain, skin pustules, and tenosynovitis. It requires prompt antimicrobial treatment and evaluation for concomitant sexually transmitted infection.

Gonococcal Pharyngitis

Infection of the throat by *Neisseria gonorrhoeae*, usually acquired through sexual contact. It may cause sore throat or no symptoms.

Gonococcal Urethritis

Inflammation of the urethra caused by infection with *Neisseria gonorrhoeae*.

Gonococcus

A clinical disorder or finding related to gonococcus; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Gonorrhea

Gonorrhea is a sexually transmitted infection caused by *Neisseria gonorrhoeae*, producing urethritis, cervicitis, pelvic inflammatory disease, rectal or pharyngeal infection, and neonatal disease. *gonorrhoeae*. Epidemiology: second most common bacterial STI (after Chlamydia), highest incidence in 15-24 year olds.

Gonorrhoea

A clinical disorder or finding related to gonorrhoea; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Good Samaritan

A person who voluntarily helps someone in an emergency, especially by providing immediate aid before professional responders arrive.

Good Samaritan Law

A law that may protect a person who provides reasonable emergency assistance from certain civil claims, subject to jurisdiction-specific conditions and exceptions such as gross negligence.

Goodpasture Syndrome

Anti-GBM disease affecting both kidneys and lungs. Pulmonary hemorrhage with hemoptysis and crescentic GN. Linear IgG deposits on renal biopsy. Emergent plasmapheresis, cyclophosphamide, and steroids. Early treatment essential to prevent irreversible renal and pulmonary damage.

Goodpasture's Syndrome

A clinical disorder or finding related to goodpasture's syndrome; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Goose Bump

A small raised area around a hair follicle caused by contraction of tiny arrector pili muscles, commonly in response to cold, fear, or strong emotion.

Goose-Skin

Goose skin is the temporary raising of hair follicles, usually caused by cold, fear, or strong emotion.

Gooseflesh

Another term for goose bumps, the temporary raised appearance of skin caused by piloerection.

Gordonia Aichiensis

A biomedical term referring to *gordonia aichiensis*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Gordonia Bacterium

A *Gordonia* bacterium is an environmental bacterium that rarely causes opportunistic infection, especially in people with weakened immunity or indwelling devices.

Gordonia Bronchialis

A biomedical term referring to *gordonia bronchialis*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Gordonia Rubripertincta

A biomedical term referring to *gordonia rubripertincta*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Gordonia Sputi

A biomedical term referring to *gordonia sputi*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Gordonia Terrae

A biomedical term referring to *gordonia terrae*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Gorlin'S Syndrome

A clinical disorder or finding related to gorlin's syndrome; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Gornall'S Test

Gornall's test is a color-based laboratory method used to measure the amount of protein in a sample.

Goserelin

A gonadotropin-releasing hormone agonist that initially stimulates and then suppresses pituitary gonadotropins; it is used in hormone-sensitive prostate or breast cancer and selected gynecologic conditions.

Gossypium

A clinical condition or finding involving gossypium. Its defining features, causes, evaluation, and management depend on the clinical context.

Gossypol

A polyphenolic compound from cottonseed that can be toxic and has been investigated as an antifertility agent; it is not a standard treatment.

Gott'S Shunt

Gott's shunt is a historical surgical connection used to reroute blood around a blocked section of a major artery.

Gotttron Papules

A characteristic skin finding of dermatomyositis, consisting of erythematous to violaceous lichenoid papules located over the dorsal aspects of the interphalangeal (PIP/DIP) and metacarpophalangeal (MCP) joints. May undergo necrosis or ulceration.

Gougerot-Carteaud Syndrome

A clinical disorder or finding related to gougerot-carteaud syndrome; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Gout

An inflammatory arthritis caused by deposition of monosodium urate crystals in and around joints, usually due to hyperuricemia. Attacks cause sudden severe pain, swelling, warmth, and redness—often in the first metatarsophalangeal joint; chronic disease can produce tophi and kidney stones. Treatment of an acute attack may include an NSAID, colchicine, or a corticosteroid; urate-lowering therapy helps prevent recurrent attacks.

Alternate Names:

Gouty arthritis; Gout (Gouty Arthritis); crystal arthropathy

Gout Flare

An acute inflammatory response to monosodium urate crystal deposition in or around a joint. It commonly causes abrupt severe pain, swelling, warmth, and erythema, often involving the first metatarsophalangeal joint; aspiration can confirm crystals and exclude infection.

Gout Nodulosis

The formation of tophi—large, firm, subcutaneous nodules composed of monosodium urate crystals—resulting from chronic, poorly controlled hyperuricemia. Tophi most commonly occur at the ears, fingers, toes, olecranon bursae, and Achilles tendon, and may ulcerate and discharge chalky white material.

Gout Suppressants

Medicines that lower uric-acid production or increase its excretion to prevent recurrent gout attacks and urate crystal deposition.

Gower Syndrome (Situational Syncope)

A form of reflex syncope, also called vasovagal or vasodepressor syncope, in which a trigger causes a fall in blood pressure and sometimes heart rate, briefly reducing brain perfusion.

Gowers Sign

A maneuver in which a person uses the hands to push up from the floor or thighs because of proximal muscle weakness, classically seen in muscular

dystrophy.

Graafian Follicle

A mature ovarian follicle that contains an egg ready for ovulation.

GRACE Score

A validated multivariable risk assessment model developed from the Global Registry of Acute Coronary Events used to predict in-hospital and 6-month all-cause mortality and recurrent myocardial infarction across the full spectrum of acute coronary syndromes (STEMI, NSTEMI, and unstable angina). It incorporates eight independent clinical prognostic factors: age, heart rate, systolic blood pressure, serum creatinine, Killip classification of heart failure, presence of cardiac arrest at presentation, ST-segment deviation on ECG, and elevated cardiac biomarkers (troponins). High-risk scores (GRACE > 140) guide rapid, early invasive coronary angiography within 24 hours.

Alternate Names:

Global Registry of Acute Coronary Events Score; GRACE Risk Model

Gracilibacillus

Gracilibacillus is a group of environmental bacteria; recovery from a clinical sample must be interpreted with the specimen source and evidence of infection.

Grade

Degree of cellular differentiation. Grade 1: well-differentiated. Grade 2: moderately differentiated. Grade 3: poorly differentiated. Grade 4: undifferentiated.

Grade IV Astrocytoma

Glioblastoma: the most malignant primary brain tumor, characterized by rapid growth, necrosis, and microvascular proliferation. Presents with headaches, seizures, focal neurological deficits, and cognitive changes. Treatment includes surgery, radiation, and chemotherapy.

Alternate Names:

Glioblastoma

Graded Potentials

Local changes in membrane potential that vary in magnitude proportional to stimulus strength, unlike all-or-none action potentials. They include receptor potentials (in sensory receptors), postsynaptic potentials (EPSPs and IPSPs at synapses), and pacemaker potentials (in cardiac/specialized cells).

Graefe'S Sign

A delayed downward movement of the upper eyelid during downward gaze, classically associated with thyroid eye disease but not specific to it.

Graft

Tissue or a synthetic material transferred to another site to replace, repair, reinforce, or bypass a damaged body structure.

Graft (Periodontal)

Tissue used to cover exposed root or augment gingiva. Connective tissue graft (gold standard), free gingival graft, allograft, xenograft.

Alternate Names:

Periodontal

Graft Versus Host

Graft-versus-host disease occurs when immunocompetent donor T cells in an allogeneic transplant attack recipient tissues, especially skin, liver, and gastrointestinal tract; it may be acute or chronic.

Graft Versus Host Disease

A condition occurring after allogeneic stem cell or bone marrow transplantation where donor immune cells attack the recipient's tissues. Can be acute (within 100 days) or chronic (after 100 days). Affects skin, liver, and gastrointestinal tract.

Alternate Names:

GVHD

Graft Versus Host Disease

A condition occurring after allogeneic stem cell or bone marrow transplantation where donor immune cells attack the recipient's tissues. Can be acute (within 100 days) or chronic (after 100 days). Affects

skin, liver, and gastrointestinal tract. Prevention includes immunosuppressive prophylaxis.

Alternate Names:

Graft Versus Host Disease

Graft-Versus-Host Disease

An immune-mediated complication of allogeneic hematopoietic transplantation in which donor-derived immunocompetent T cells attack recipient tissues. Acute or chronic GVHD commonly affects skin, liver, and gastrointestinal tract and requires prevention and immunosuppressive treatment.

Graft-Versus-Host Reaction

An immune reaction in which transplanted donor immune cells attack the recipient's tissues.

Grafting

The surgical transfer of tissue or a synthetic substitute to repair, replace, or support a body structure. An autograft comes from the same person, whereas an allograft comes from another person of the same species.

Gram

A metric unit of mass equal to one-thousandth of a kilogram.

Alternate Names:

g

Gram (G)

A unit of mass equal to one-thousandth of a kilogram; medication doses and laboratory measurements are often expressed in grams or smaller metric units.

Gram (Measure)

A metric unit of mass equal to one-thousandth of a kilogram or approximately 15.4 grains.

Gram Calorie (Calorie)

A small calorie is the energy needed to raise 1 gram of water by 1 degree Celsius; the dietary Calorie, written with a capital C, is a kilocalorie equal to 1,000 small calories.

Gram Negative Bacilli

Rod-shaped gram-negative bacteria appearing pink on Gram stain owing to a thin peptidoglycan layer and lipopolysaccharide outer membrane.

Gram Positive Cocci

Spherical gram-positive bacteria staining purple owing to a thick peptidoglycan layer and absent outer membrane. Clinically important genera include *Staphylococcus*, *Streptococcus*, and *Enterococcus*, identifiable by arrangement (clusters, chains, pairs) and hemolysis, and causing skin, pharyngeal, respiratory, and bloodstream infections.

Gram Stain

A laboratory staining method that classifies bacteria as Gram-positive or Gram-negative according to cell-wall properties and helps identify organisms in clinical specimens before or alongside culture. on cell wall structure. The procedure involves crystal violet staining, iodine mordant, alcohol decolorisation, and safranin counterstain.

Gram Stain Of Skin Lesion

Microscopic examination of a skin lesion specimen stained with Gram stain to identify bacteria. Useful for diagnosing bacterial skin infections and guiding antibiotic therapy. Results available within hours.

Alternate Names:

Gram stain culture

Gram Stain Of Tissue Biopsy

Microscopic examination of tissue biopsy specimen stained with Gram stain to identify bacteria. Used to diagnose tissue infections and guide antimicrobial therapy. May be performed alongside culture and histopathology.

Alternate Names:

Tissue Gram stain

Alternate Names:

Urethral Gram stain

Gram'S Stain

Gram's stain is a laboratory method that classifies bacteria by how their cell walls retain the stain,

helping guide identification and initial antibiotic choices.

Gram-Negative

Describing bacteria with a thin peptidoglycan cell wall and an outer membrane containing lipopolysaccharide; they stain pink or red with the Gram stain.

Gram-Negative Bacteria

Bacteria that do not retain crystal violet during Gram staining and therefore appear pink or red. They have a thin peptidoglycan layer and an outer membrane that may contain lipopolysaccharide. Examples include *Escherichia coli*, *Klebsiella*, *Pseudomonas*, and *Neisseria*.

Gram-Negative Meningitis

Meningitis caused by gram-negative bacteria, including *Escherichia coli*, *Klebsiella*, and *Neisseria* species. It can progress rapidly to sepsis, neurologic injury, and death and requires urgent cerebrospinal fluid evaluation and antimicrobial treatment.

Gram-Positive

Describing bacteria with a thick peptidoglycan cell wall that retains crystal violet during Gram staining and therefore appears purple.

Gram-Positive Bacteria

Bacteria that retain crystal violet in the Gram stain (appear purple/blue), characterized by a thick peptidoglycan cell wall and no outer membrane.

Gram-Positive Rods

Gram-positive rods are rod-shaped bacteria that retain the purple part of the Gram stain; some are harmless colonizers, while others cause infection.

Gramicidin

Gramicidin is a topical antibiotic that forms channels in bacterial membranes; it is not generally used systemically because it can be toxic.

Grand Mal

A type of generalised seizure characterized by bi-

lateral synchronous epileptic activity involving the entire brain. Formerly called grand mal, now classified as bilateral tonic-clonic seizure according to the ILAE 2017 classification.

Grand Multipara

A person who has given birth five or more times beyond the gestational threshold used to count parity; definitions can vary by obstetric system.

Grand Rounds

An educational medical conference in which clinicians present a case, diagnosis, management, or clinical topic to a professional audience.

Granisetron

Granisetron is a selective 5-HT₃-receptor antagonist used to prevent chemotherapy-, radiotherapy-, or surgery-related nausea and vomiting; headache, constipation, and QT prolongation can occur.

Granisetron Hydrochloride

A medicine or chemical compound called granisetron hydrochloride. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Granular Casts

Degenerated cellular debris within tubular lumen appearing as coarse or fine-grained structures. Coarse casts represent more acute process, fine casts indicate chronic degenerative process. Common in acute tubular necrosis reflecting tubular epithelial cell damage. Also seen in chronic kidney disease, nephrotic syndrome, and hypertension.

Granular Cell Tumor Of The Breast

An uncommon tumor showing Schwann-cell differentiation that can occur in the breast and mimic carcinoma clinically or on imaging. Histologic examination distinguishes it from malignancy; most are benign.

Granular Leukocyte

A granulocyte, a white blood cell containing cyto-

plasmic granules; the main types are neutrophils, eosinophils, and basophils.

Granulation

The formation of new vascular connective tissue during wound healing, producing moist red or pink granulation tissue made of capillaries, fibroblasts, and extracellular matrix.

Granulation Tissue

granulation tissue is a biological, genetic, biochemical, or chemical term used in medicine; its significance depends on its structure, function, concentration, pathway, or clinical context.

Granulicatella

Granulicatella is a group of bacteria that may live in the mouth and can rarely cause bloodstream infection or endocarditis.

Granulicatella Adiacens

A biomedical term referring to granulicatella adiacens; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Granulicatella Elegans

A biomedical term referring to granulicatella elegans; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Granulocyte

A granulocyte is a white blood cell containing cytoplasmic granules; neutrophils, eosinophils, and basophils are granulocytes involved in innate immunity, inflammation, and host defense. on light microscopy. Granulocytes are produced in the bone marrow (myelopoiesis, 3-4 days maturation) and are part of the innate immune system.

Granulocyte Count

A granulocyte count measures the number of neutrophils, eosinophils, and basophils in the blood and helps assess infection, inflammation, and bone-

marrow function.

Granulocytic Sarcoma

A neoplastic process affecting granulocytic sarcoma; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Granulocytopenia

An abnormally low number of granulocytes, especially neutrophils, which can increase susceptibility to bacterial and fungal infection.

Granuloma

A distinctive pattern of chronic inflammation characterized by the aggregation of modified macrophages (epithelioid histiocytes), often with multinucleated giant cells (Langhans or foreign-body type), surrounded by lymphocytes. Granulomas form when macrophages cannot eliminate a persistent antigen or pathogen.

Granuloma Annulare

A benign, noninfectious inflammatory skin condition that causes raised red, pink, or skin-colored bumps arranged in rings, often with clearer skin in the center. It commonly affects the hands, feet, or elbows and may cause little or no discomfort. Widespread disease may be associated with diabetes or thyroid disease, but the cause is often unknown.

Granuloma Inguinale

A chronic sexually transmitted infection that causes ulcerative lesions of the genital region.

Granulomatosis

A disorder characterized by formation of granulomas, which are organized collections of immune cells caused by chronic inflammation.

Granulomatosis With Polyangiitis

GPA (formerly Wegener granulomatosis): an ANCA-associated small-vessel necrotizing vasculitis affecting the upper and lower respiratory tract and kidneys.

Alternate Names:

Wegeners Granulomatosis (Granulomatosis With Polyangiitis)

Granulomatosis With Polyangiitis

A systemic vasculitis affecting small to medium vessels, characterized by necrotizing granulomatous inflammation of the respiratory tract and glomerulonephritis. Presents with sinusitis, pulmonary nodules, and renal involvement. Associated with c-ANCA/PR3 antibodies.

Alternate Names:

Granulomatosis with polyangiitis; Wegener granulomatosis

Granulomatous Amebic Encephalitis

A rare, usually fatal central nervous system infection caused by free-living amoebae such as *Acanthamoeba* species, typically affecting immunocompromised individuals. It presents with a subacute to chronic meningoencephalitis and forms granulomatous lesions in the brain.

Granulomatous Colitis

Inflammation of the colon in which granulomas are found on tissue examination, often associated with Crohn disease but also possible with infections or other inflammatory conditions.

Granulomatous Inflammation

A distinctive pattern of chronic inflammation characterized by organized collections of activated macrophages that have undergone morphological transformation into epithelioid cells, often surrounded by a rim of lymphocytes and occasionally fibroblasts.

Granulomatous Mastitis

A chronic inflammatory breast disease characterized by noncaseating granulomas after other causes, especially infection and sarcoidosis, have been excluded. It may present as a painful mass, abscesses, or draining sinuses.

Granulosa Cell

A granulosa cell is a cell surrounding an ovarian egg that supports its development and produces

hormones such as estrogen and inhibin.

Granulosa Cell Tumor

An estrogen-secreting ovarian sex cord-stromal tumor that can cause precocious puberty, endometrial hyperplasia, or carcinoma. Generally low-grade with a tendency to late recurrence.

Granzyme

A protease released by cytotoxic T lymphocytes and natural killer cells that enters target cells through perforin-associated pores and promotes programmed cell death.

Grasp Reflex

Grasp Reflex is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Grass Allergy

Hypersensitivity reaction to grass pollen, causing allergic rhinitis (hay fever) with symptoms including sneezing, nasal congestion, itchy eyes, and throat irritation. Seasonal, typically spring through fall. Treatment includes antihistamines, nasal corticosteroids, and allergen avoidance.

Alternate Names:

Hay fever; seasonal allergic rhinitis

Grass And Weed Killer Poisoning

Toxic exposure to herbicides containing glyphosate, 2,4-D, or other chemicals. Symptoms range from mild skin and eye irritation to severe gastrointestinal, respiratory, and neurological effects depending on the product and exposure route. Requires immediate medical attention.

Alternate Names:

Herbicide poisoning; pesticide poisoning

Graves Disease

An autoimmune disease in which antibodies stimulate the thyroid gland to make too much thyroid hormone, causing hyperthyroidism. It is the most common cause of hyperthyroidism and may cause an enlarged thyroid (goiter) and eye problems.

Gravida

The number of times a person has been pregnant, regardless of pregnancy outcome; the current pregnancy is included in the count.

Gravidity

Gravidity is the number of times a person has been pregnant, including the current pregnancy and pregnancies that did not result in a live birth.

Gravitropism

Gravitropism is the growth of a plant in response to gravity, such as roots growing downward and shoots growing upward.

Gray (Gy)

The SI unit of absorbed radiation dose, equal to one joule of ionizing radiation energy deposited per kilogram of matter. It describes physical energy absorption and differs from the sievert, which incorporates biologic weighting.

Alternate Names:

Gy

Gray Baby Syndrome

A serious form of chloramphenicol toxicity in newborns, caused by immature drug metabolism and characterized by abdominal distension, vomiting, cyanosis or gray skin color, cardiovascular collapse, and potentially death.

Gray Matter

Regions of the brain and spinal cord composed primarily of nerve cell bodies, dendrites, and synapses, responsible for processing information.

Gray'S Anatomy

A major illustrated human-anatomy reference book, first published in 1858 by Henry Gray and Henry Vandyke Carter and subsequently revised in many editions.

Greater Curvature

Convex lateral stomach border from gastroesophageal junction to pylorus, approximately four

times longer than the lesser curvature. Attachment site for the greater omentum. Blood supply from short gastric and left gastroepiploic vessels. Common site for gastric ulcers and cancer.

Greater Omentum

A fold of fatty peritoneal tissue that hangs from the stomach and partly covers the abdominal organs.

Greater Trochanter

Large quadrilateral lateral proximal femoral prominence. Insertion for gluteus medius/minimus and piriformis. Palpable landmark and site of trochanteric bursa. Intertrochanteric fractures between trochanters are extracapsular with lower avascular necrosis risk because periosteal blood supply is preserved.

Greater Trochanteric Pain Syndrome

Pain over the lateral hip centered on the greater trochanter, commonly caused by tendinopathy or bursitis of the gluteal tendons. Presents with lateral hip pain worse with walking, climbing stairs, or lying on the affected side. Treatment includes physical therapy and activity modification.

Alternate Names:

Trochanteric bursitis; gluteal tendinopathy

Greater Vestibular Glands

Paired glands near the vaginal opening that secrete mucus into the vestibule, contributing to lubrication. They are also called Bartholin glands.

Alternate Names:

Bartholin glands; greater vestibular glands

Green Blindness

A clinical disorder or finding related to green blindness; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Green Nail Syndrome

Green discoloration of one or more nails, usually caused by colonization or infection with *Pseudomonas aeruginosa*, often in an onycholytic or chronically moist nail.

Green Sickness

A clinical disorder or finding related to green sickness; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Green Stool

Stool with a green color that may result from diet, food coloring, rapid intestinal transit, medicines, or infection.

Greenstick Fracture

An incomplete fracture in which a young bone bends and cracks rather than breaking all the way through.

Grey Ankle Phenomenon

A gray or dusky discoloration around the ankle associated with chronic venous insufficiency and hemosiderin deposition; it should be interpreted with other signs of venous disease.

Grey Matter

Grey Matter is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Grey Turner'S Sign

Bluish or violaceous ecchymosis of the flanks caused by retroperitoneal bleeding tracking through the abdominal wall. It is classically associated with severe haemorrhagic pancreatitis but may occur with other retroperitoneal haemorrhage.

Grid Implant

A flexible electrode array placed on or near the brain or another tissue surface to record electrical activity or deliver stimulation.

Grief

The natural emotional response to loss, particularly the death of a loved one, but also following other significant losses such as divorce, job loss, or declining health. It encompasses a range of feelings including sadness, anger, guilt, and numbness, often occurring in waves rather than linear stages.

Grief (Loss, Grief, And Bereavement)

The natural emotional response to loss, encompassing the process of coping with the death of a loved one or other significant losses. Includes grief (emotional response), loss (the event), and bereavement (the period of mourning). May involve physical, emotional, social, and spiritual dimensions.

Alternate Names:

Bereavement; mourning

Grifulvin

Grifulvin is a medicine or pharmacologic product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent and patient.

Grimontia Hollisae

A biomedical term referring to *grimontia hollisae*; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Grip Strength

The force generated when the hand closes around an object, commonly measured with a dynamometer to assess hand function, muscle strength, nutrition, or rehabilitation progress.

Gripping

Cramping or spasmodic abdominal pain, often caused by contractions or obstruction of a hollow organ.

Grippe

An older name for influenza, a viral respiratory illness that commonly causes fever, cough, body aches, and fatigue.

Griscelli Syndrome

Three types: Type 1 (MYO5A mutation): primary neurologic impairment, no immunodeficiency; Type 2 (RAB27A mutation): immunodeficiency + partial albinism, HLH risk; Type 3 (MLPH mutation): only partial albinism. Type 2: silvery hair, large melanosomes in melanocytes, defective cytotoxic

granule release -> HLH.

Griseofulvin

Griseofulvin is an oral antifungal that deposits in newly formed keratin and disrupts dermatophyte mitosis, treating infections of skin, hair, and nails; treatment is prolonged and hepatic or drug-interaction risks apply.

Groin

The region where the lower abdomen meets the upper thigh, containing the inguinal canal and related muscles, vessels, nerves, and lymph nodes.

Groin Lump

A palpable mass in the inguinal region, which may represent lymphadenopathy, hernia, cyst, tumor, or other pathology. Requires medical evaluation to determine the cause and appropriate treatment.

Alternate Names:

Inguinal mass

Groin Pain

Pain in the inguinal region, which may originate from musculoskeletal, genitourinary, gastrointestinal, or vascular causes. Common causes include muscle strain, hernia, lymphadenopathy, and hip pathology. Requires clinical evaluation for diagnosis.

Alternate Names:

Inguinal pain

Groin Pain (Male)

Pain in the groin or adjacent lower abdomen, hip, or testicle in a male; causes include muscle strain, hernia, infection, urinary stones, or referred pain.

Groove

Groove is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gross Examination

The macroscopic, naked-eye inspection of a specimen in the pathology laboratory, performed before

microscopic sectioning and the essential first step in pathologic diagnosis of resected tissues and organs.

Gross Hematuria

Visible blood in the urine, appearing pink, red, or brown. Causes include urinary tract infections, kidney stones, bladder or kidney cancer, trauma, and glomerular disease. Requires medical evaluation to determine the underlying cause.

Alternate Names:

Visible hematuria

Gross Motor Control

The ability to use large muscle groups for coordinated movements such as walking, running, jumping, and maintaining posture. Develops progressively from infancy through childhood and is assessed as part of developmental milestones.

Alternate Names:

Large motor skills

Gross Motor Skills

The ability to use large muscle groups for movement and posture, developing in a predictable sequence in infancy/childhood: head control (3 months), sitting (6 months), crawling (9 months), standing (10-12 months), walking (12-18 months), running/jumping (2-3 years), hopping and balance (3-5 years).

Gross Virus

gross virus is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Ground Itch

A clinical disorder or finding related to ground itch; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Ground Substance

ground substance is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Ground-Glass Opacity

A hazy area on a chest CT scan in which underlying lung vessels remain partly visible. It is a radiologic finding, not a diagnosis, and may result from infection, inflammation, fluid, bleeding, scarring, or sometimes a tumor; its meaning depends on the pattern and clinical context.

Group A Strep

Group A Streptococcus (*Streptococcus pyogenes*), a gram-positive coccus in chains, beta-hemolytic. Infections: pharyngitis/tonsillitis, impetigo, erysipelas, cellulitis, scarlet fever, necrotizing fasciitis, streptococcal toxic shock syndrome, pneumonia, and puerperal sepsis.

Group A Streptococcus

Streptococcus pyogenes, a bacterium that can cause pharyngitis, skin infections, invasive disease, and immune-mediated complications.

Group B Strep

Group B Streptococcus (GBS, *Streptococcus agalactiae*): a gram-positive coccus colonizing the GI and genitourinary tract in 10-30 percent of women; major cause of neonatal sepsis and meningitis.

Group B Strep

Group B Streptococcus: bacteria that can colonize the gastrointestinal and genital tracts, potentially causing serious infections in newborns during delivery. Routine screening in pregnancy guides intrapartum antibiotic prophylaxis.

Alternate Names:

Group B Streptococcus; GBS colonization

Group B Strep Disease

Illness caused by group B Streptococcus bacteria, which can cause serious infections in newborns and infections in pregnant people or adults with medical risks.

Group B Streptococcal Disease

Infection caused by group B Streptococcus bacteria, which can be serious in newborns and vulnerable

adults.

Group B Streptococcal Infection In Pregnancy

Maternal colonization or infection with *Streptococcus agalactiae*, which can be transmitted to the newborn during labor and cause serious early-onset infection. Screening and intrapartum management reduce neonatal risk.

Group B Streptococcal Septicemia Of The Newborn

Serious bloodstream infection caused by Group B *Streptococcus* in neonates, typically acquired during birth. Presents with fever, lethargy, poor feeding, and respiratory distress. Requires prompt antibiotic treatment and supportive care.

Alternate Names:

Neonatal GBS sepsis

Group B Streptococcus

A gram-positive bacterium (*Streptococcus agalactiae*) colonizing the genitourinary and gastrointestinal tracts. Asymptomatic vaginal colonization occurs in 20-30% of pregnant women. Screening is performed at 36-37 weeks gestation via rectovaginal swab culture.

Group B Streptococcus - Pregnancy

Colonization with Group B *Streptococcus* during pregnancy, which can cause neonatal sepsis, pneumonia, and meningitis during delivery. Routine screening at 36-37 weeks gestation identifies colonized women who receive intrapartum antibiotic prophylaxis.

Alternate Names:

GBS screening; intrapartum GBS prophylaxis

Group Practice

A clinical condition or finding involving group practice. Its defining features, causes, evaluation, and management depend on the clinical context.

Group Therapy

A form of psychotherapy where a small group of indi-

viduals with similar issues meet to discuss, interact, and develop coping strategies under the guidance of a trained therapist. Used for depression, anxiety, substance use disorders, grief, and chronic illness. Benefits include peer support and shared experiences.

Grover Disease

A transient or recurrent acantholytic dermatosis causing itchy red papules or papulovesicles, usually on the trunk of middle-aged or older adults; heat, sweating, and occlusion may aggravate it.

Growing Pains

Recurrent, bilateral leg pain in children ages 3-12, typically occurring in the late afternoon or evening and resolving by morning. Pain is deep, aching in the calves, shins, and thighs (not joints). No limping, fever, swelling, or functional impairment.

Growth

The process of increasing in physical size and development, including physical, cognitive, and emotional maturation. In medicine, growth is monitored using growth charts and anthropometric measurements to assess health and development in children.

Growth Chart

A graphical representation of growth (weight, height/length, head circumference, BMI) plotted against age, standardized by population (WHO growth standards 0-5 years; CDC growth charts 2-20 years; percentiles/z-scores).

Growth Cone

growth cone is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Growth Disorder

A clinical disorder or finding related to growth disorder; evaluation identifies the underlying mechanism, affected site, severity, and relevant complications.

Growth Factor

Proteins that stimulate cell growth, proliferation, differentiation, and survival. Major families: EGF (epidermal growth factor), FGF (fibroblast growth factor), PDGF (platelet-derived), VEGF (vascular endothelial), TGF-beta (transforming), NGF (nerve), IGF (insulin-like), HGF (hepatocyte).

Growth Failure

Abnormally slow physical growth or inadequate weight gain compared with expected development.

Growth Hormone

A hormone released by the anterior pituitary that promotes growth partly through insulin-like growth factor 1 and also affects protein, fat, and carbohydrate metabolism.

Growth Hormone Deficiency

Insufficient secretion of growth hormone from the anterior pituitary, causing short stature in children and metabolic effects in adults. In children: delayed growth, short stature, delayed bone age, and increased fat mass. In adults: decreased muscle mass, increased fat mass, reduced bone density, fatigue, and dyslipidemia.

Growth Hormone Deficiency - Children

Insufficient growth hormone production in children, resulting in short stature, delayed bone age, and reduced growth velocity. Causes include congenital anomalies, tumors, radiation, or idiopathic. Diagnosis involves GH stimulation tests. Treatment with recombinant GH.

Alternate Names:

Pediatric GH deficiency

Growth Hormone Physiology

Growth hormone is secreted in a pulsatile pattern by pituitary somatotroph cells regulated by hypothalamic GHRH which stimulates release and somatostatin which inhibits release. GH acts directly on tissues and indirectly through IGF-1 produced primarily by the liver.

Growth Hormone Stimulation Test

Laboratory tests to evaluate growth hormone secretion by stimulating the pituitary gland with insulin, arginine, glucagon, or clonidine. Used to diagnose GH deficiency when random GH levels are inconclusive. Peak GH <10 ng/mL suggests deficiency.

Alternate Names:

GH stimulation test

Growth Hormone Suppression Test

Laboratory test to evaluate excessive growth hormone secretion, typically using oral glucose tolerance. In normal individuals, glucose suppresses GH to <1 ng/mL. Failure to suppress suggests acromegaly or gigantism.

Alternate Names:

GH suppression test; oral glucose tolerance test for acromegaly

Growth Hormone Test

Laboratory measurement of growth hormone levels in the blood, used to evaluate GH deficiency or excess. Random GH levels have limited utility; stimulation or suppression tests are more informative.

Alternate Names:

GH level; serum growth hormone

Growth Medium

A nutrient preparation supporting microbial multiplication in vitro, supplying carbon, nitrogen, minerals, and growth factors in liquid or solid form.

Growth Plate Fractures

Injuries involving the cartilage growth plates at the ends of growing children's bones, with possible effects on future bone growth.

Growth Retardation

Abnormally slow physical growth or failure to reach expected height or weight for age and developmental stage. Causes include inadequate nutrition, chronic systemic disease, endocrine or genetic disorders, malabsorption, psychosocial deprivation, and constitutional variation; assessment uses growth charts

and clinical context.

Growth Scan

An ultrasound examination used to assess fetal size, growth pattern, amniotic fluid, and sometimes blood flow. Measurements are compared with gestational-age standards to identify possible growth restriction or excessive growth.

Gtp Binding

GTP binding is the attachment of guanosine triphosphate to a protein, often switching that protein into an active state for cell signaling.

Gtt.

An abbreviation from the Latin *guttae*, meaning drops; because abbreviations can be misread, prescriptions should state liquid doses clearly.

Guaifenesin

An expectorant used to loosen respiratory mucus and make coughing more productive. It provides symptomatic relief in some respiratory illnesses; dosing depends on the formulation and patient.

Guanazole

A medicine or chemical compound called guanazole. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Guanethidine

Guanethidine is a medicine or pharmacologic product; its indication, dose, interactions, contraindications, and adverse effects depend on the specific agent and patient.

Guanfacine

Guanfacine is a centrally acting alpha-2A adrenergic agonist used for hypertension and, in extended-release form, ADHD; sedation, hypotension, bradycardia, and rebound hypertension after abrupt withdrawal are important risks.

Guanine

Guanine is a purine nitrogenous base in DNA and

RNA; it pairs with cytosine and is present in nucleotides such as GMP and GTP that support energy transfer and signaling.

Guanosine

A purine nucleoside composed of guanine linked to ribose and used as a building block of RNA and guanine nucleotides.

Guanosine Diphosphate

A guanine nucleotide containing two phosphate groups that participates in cellular energy transfer and G-protein signaling.

Guanosine Monophosphate

A guanine nucleotide containing one phosphate group and serving as a component of RNA and a precursor of other guanine nucleotides.

Guanosine Triphosphate

A guanine nucleotide that provides energy and acts as a molecular switch in protein synthesis, cytoskeletal regulation, and G-protein signaling.

Guanosinetriphosphatase

An enzyme that hydrolyzes guanosine triphosphate to guanosine diphosphate and phosphate, regulating signaling and other cellular processes.

Guanylate Cyclase

Guanylate cyclase is an enzyme that converts GTP to cyclic GMP, a second messenger involved in nitric oxide signaling, smooth-muscle relaxation, phototransduction, and intestinal fluid secretion.

Guardianship

A legal arrangement in which an appointed person makes decisions for someone unable to make them independently.

Alternate Names:

Legal guardianship

Guevedoces

A term used in the Dominican Republic for individuals with 46,XY differences in sex development

caused by 5-alpha-reductase type 2 deficiency; they may have female-appearing or ambiguous external genitalia at birth and increased virilization at puberty.

Guide Device

A guide device is a tool used to direct, position, or stabilize another medical instrument during a procedure.

Guided Imagery

Guided imagery is a relaxation technique that uses imagined sights, sounds, or other sensations to promote calm and sometimes help manage pain or stress.

Guided Tissue Regeneration

Periodontal procedure using barrier membrane to promote regeneration of bone, cementum, periodontal ligament.

Guillain-Barre Syndrome

An acute, immune-mediated polyneuropathy typically presenting with ascending paralysis following an antecedent infection (*Campylobacter jejuni*, CMV, EBV, Zika). Variants: acute inflammatory demyelinating polyneuropathy (AIDP, most common in North America), acute motor axonal neuropathy (AMAN, more common in Asia).

Alternate Names:

AIDP

Guinea Worm

A parasitic infection caused by *Dracunculus medienensis*, acquired by drinking water containing infected copepods and characterized by emergence of a long worm through the skin.

Gullet

The esophagus, the tube carrying food from the throat to the stomach.

Alternate Names:

Esophagus

Gum

Gum is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gum Biopsy

Histological examination of gingival tissue obtained by biopsy, used to diagnose gingival diseases, infections, or malignancies. May involve excisional or incisional biopsy techniques.

Alternate Names:

Gingival biopsy

Gum Disease

Inflammation or infection affecting the gums and supporting tissues of the teeth.

Alternate Names:

Periodontal disease Periodontitis (Gum Disease)

Gum Disease, Gingivitis

Gingivitis: inflammation of the gingiva (gums) without loss of supporting bone. Characterized by red, swollen, bleeding gums. Commonly caused by plaque accumulation. Reversible with good oral hygiene and professional cleaning.

Alternate Names:

Gingivitis

Gum Disease, Periodontitis

Periodontitis: progressive inflammation and destruction of the supporting structures of the teeth, including the periodontal ligament and alveolar bone. Results from untreated gingivitis. Can lead to tooth loss. Requires professional treatment including scaling and root planing.

Alternate Names:

Periodontitis

Gum Problems

Gum problems include inflammation, bleeding, recession, ulcers, infection, trauma, and growths. Causes include plaque, tobacco, dry mouth, medicines, nutritional or immune disease, and poorly fitting appliances; persistent lesions or bleeding need dental assessment.

Gumma

A soft, non-cancerous granulomatous lesion occurring in tertiary syphilis, characterized by central necrosis surrounded by inflammatory cells. Can affect any organ but commonly occurs in skin, bone, and liver. Treatment with antibiotics.

Alternate Names:

Syphilitic gumma

Gummata

Gummata is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gummy Smile

Excessive gingival display when smiling. Caused by hypermobile lip, vertical maxillary excess, altered passive eruption. Treatment: surgery, Botox, orthodontics.

Gums

The soft oral tissue surrounding and supporting the teeth.

Alternate Names:

Gingiva

Gums - Swollen

Gingival swelling: enlargement of the gum tissue, which may be caused by inflammation (gingivitis, periodontitis), infection, medications (phenytoin, cyclosporine), systemic diseases, or allergic reactions. May cause pain, bleeding, and difficulty eating.

Alternate Names:

Swollen gums; gingival enlargement

Gunning'S Splint

Gunning's splint is a historical dental device used to stabilize broken jaws and keep the teeth aligned while the bones heal.

Gunshot Wound

An injury produced by a projectile discharged from a firearm; forensic classification by range and direction. Entrance wound: rounded/ovoid defect with an abraded ring and often a zone of soot (close

range) or stippling/tattooing (intermediate range); contact wounds show muzzle imprint.

Alternate Names:

Ballistic trauma aftercare

Gustation

The physiological chemosensory special sense of taste perception, mediated by specialized taste receptor cells organized within taste buds located predominantly on the fungiform, foliate, and circumvallate papillae of the dorsal tongue, as well as the soft palate, pharynx, and epiglottis. Discriminates five primary taste modalities: sweet, umami (savory), and bitter (transduced via G-protein-coupled receptors and second messenger cascades involving gustducin), and salty and sour (transduced via direct ion flux through epithelial sodium channels [ENaC] and proton-permeable OTO1 channels). Chemosensory signals travel to the solitary tract nucleus in the medulla via cranial nerves VII (facial), IX (glossopharyngeal), and X (vagus), projecting to the thalamus and primary gustatory cortex in the anterior insula and frontal operculum.

Gustatory

Gustatory is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gustatory Sweating

Sweating triggered by eating or thinking about food, usually over the cheek or temple after injury or surgery involving the parotid region; it is also called Frey syndrome.

Gut

Gut is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gut Epithelium

gut epithelium is a tissue, cell, or body structure whose location and function help explain its role in normal physiology and disease.

Gut Microbiome

The complex community of trillions of microorganisms, including bacteria, viruses, and fungi, residing in the gastrointestinal tract. The gut microbiome plays a crucial role in digestion, immune function, vitamin synthesis, and protection against pathogens.

Guthrie Test

Guthrie test is a medical test, procedure, treatment, or clinical measure used in patient care; its purpose and interpretation depend on the clinical question.

Guttae

Guttae is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Guttate Psoriasis

An acute eruption of numerous small, scaly, drop-like psoriatic papules and plaques, often following streptococcal pharyngitis in children or young adults. Diagnosis is clinical; the eruption may resolve spontaneously or evolve into chronic plaque psoriasis.

Guttural

Guttural is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gynaecologist

A medical doctor who specializes in the health of the female reproductive system (vagina, cervix, uterus, fallopian tubes, ovaries) and breasts.

Gynaecology

Gynaecology is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gynaecomastia

The benign enlargement of breast tissue in males. Pathophysiology: an imbalance between estrogen (stimulates breast tissue growth) and androgen (suppresses) activity, resulting in increased estrogen-to-

testosterone ratio.

Gynatresia

Congenital or acquired obstruction or absence of an opening in the female genital tract, potentially causing retained menstrual blood and outflow obstruction.

Gynecic

Gynecic is an anatomical structure or physiologic function; its location, mechanism, and associated findings determine its clinical significance.

Gynecoid Pelvis

A commonly described pelvic shape with a rounded inlet, relatively broad dimensions, and a spacious outlet; it is associated with typical female pelvic anatomy but individual variation is substantial.

Gynecological Examination

A gynecological examination assesses the female reproductive organs and may include inspection, a pelvic examination, and screening tests when appropriate.

Gynecologist

A gynecologist is a physician who provides care for the female reproductive system, including preventive care, diagnosis, and treatment.

Gynecology

The medical specialty concerned with the health of the female reproductive system (uterus, ovaries, fallopian tubes, cervix, vagina, and vulva).

Gynecomastia

Enlargement of breast tissue in males, often bilateral, occurring physiologically in neonates, during puberty (40-60% of boys), and in aging men. Pathologic causes include Klinefelter syndrome, testicular tumors, medications (spironolactone, cannabis), and liver disease. Persistent or asymmetric gynecomastia warrants evaluation.

Alternate Names:

Man Boobs (Gynecomastia)

Gyrate Atrophy

An inherited retinal disorder causing progressive loss of peripheral vision and night vision.

Gyrus

A ridge or convolution on the surface of the cerebral cortex, separated by sulci (grooves). The folded structure triples the surface area of the brain (approximately 2500 cm²) while fitting within the cranial vault.

H

21-Hydroxylase Deficiency

An inherited condition in which the adrenal glands cannot make enough of certain hormones. It can cause salt loss, low blood pressure, and excess male-type hormones, and is the most common cause of congenital adrenal hyperplasia.

H And E Staining

H and E Staining is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

H And H

Abbreviation for hemoglobin and hematocrit, two blood measurements commonly used to assess red-blood-cell levels and anemia.

H And P

Abbreviation for history and physical examination, the clinical history and examination performed to assess a patient.

HER2-Positive Breast Cancer

A distinct molecular subtype of breast cancer defined by overexpression of the human epidermal growth factor receptor 2 (HER2/neu) protein or amplification of the *ERBB2* gene, occurring in approximately 15–20% of invasive breast carcinomas. Historically aggressive, but highly responsive to HER2-targeted therapies such as trastuzumab and pertuzumab.

H Influenzae Meningitis

A serious infection of the coverings of the brain

and spinal cord caused by *Haemophilus influenzae* bacteria. It may cause fever, headache, stiff neck, confusion, seizures, or brain injury, especially in unvaccinated children.

H1N1 Flu (Swine Flu)

Influenza caused by the H1N1 virus strain, spread through respiratory droplets and capable of causing ordinary or severe flu illness.

Alternate Names:

Influenza, H1N1 Influenza, Swine Flu

H1N1 Influenza

Influenza A infection caused by an H1N1 virus strain. It usually produces fever, cough, sore throat, myalgia, and fatigue but can cause viral pneumonia and severe respiratory failure in high-risk patients.

H1N1 Influenza (Swine Flu)

H1N1 influenza is a strain of influenza A virus that causes an acute respiratory illness with fever, cough, myalgia, and variable pulmonary complications.

H2 Blockers

Medicines that reduce stomach acid. They are used for heartburn, indigestion, stomach ulcers, and some conditions that cause too much acid.

H2 Receptor Antagonists Overdose

H2 receptor antagonists overdose involves toxic or environmental exposure to h2 receptor antagonists overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Habenula

A small brain structure involved in processing reward, unpleasant experiences, stress, and motivation.

Habit

A repeated behavior or pattern of action that may be ordinary, helpful, or harmful; in clinical assessment, its frequency, control, and effect on functioning determine whether it is a concern.

Habituation

Habituation is a developmental, immunologic, reproductive, behavioral, environmental, or healthcare concept used in medicine or biomedical research.

Habitus

A person's general body build, physical appearance, or constitutional type as observed clinically.

Hache'S Sign

Hache's sign is a clinical sign historically described in assessment of certain vascular disorders; its value depends on the examination findings and clinical setting.

Haem

The iron-containing part of hemoglobin that allows red blood cells to carry oxygen.

Haemangioma

A usually harmless growth made of extra blood vessels. It is common in babies and may appear as a red or purple mark or lump.

Haemarthrosis

Bleeding into a joint, causing swelling, pain, and reduced movement. It may follow an injury or occur with hemophilia, blood-thinning medicines, surgery, or certain joint disorders.

Haematemesis

Haematemesis is vomiting blood, usually from bleeding in the esophagus, stomach, or first part of the small intestine.

Haematin

Haematin is an oxidized form of haem produced when hemoglobin is broken down.

Haematinics

Haematinics are nutrients or medicines, such as iron, vitamin B12, and folate, needed to make healthy red blood cells.

Haematite

Haematite is an iron-containing mineral; it is not a blood disorder, although its name derives from its blood-red appearance in powdered form.

Haematocele

A haematocele is a collection of blood within a body cavity or tissue, such as the scrotum after injury or surgery.

Haematocolpos

Haematocolpos is accumulation of menstrual blood in the vagina, usually because the vaginal outlet is blocked.

Haematocrit

The percentage of blood made up of red blood cells. The expected range varies with age, sex, pregnancy, altitude, and the laboratory.

Haematogenous

Spread through the bloodstream, as when an infection or cancer moves from one body area to another.

Haematologist

A medical doctor who specializes in the diagnosis, treatment, and prevention of diseases of the blood and blood-forming organs.

Haematoma

A localised collection of blood outside blood vessels, usually clotted, within an organ or tissue. Haematomas form when trauma or disease causes a blood vessel to rupture.

Haematometra

Haematometra is accumulation of menstrual blood in the uterus, usually because the cervical or vaginal outflow tract is blocked.

Haematomyelia

Haematomyelia is bleeding into the spinal cord, which can cause weakness, sensory loss, or paralysis.

Haematopericardium

Haematopericardium is blood within the sac around the heart and may cause cardiac tamponade if it compresses the heart.

Haematopoiesis

Haematopoiesis is the production of blood cells, mainly in the bone marrow.

Haematoporphyria

Haematoporphyria is an older term related to disorders of porphyrin metabolism that can cause photosensitivity and other symptoms.

Haematosalpinx

Haematosalpinx is blood collected in a fallopian tube, often related to ectopic pregnancy, endometriosis, or pelvic injury.

Haematuria

Blood in the urine. It may be visible or found only with a urine test and can have many causes, including infection, stones, medicines, or kidney disease.

Haemochromatosis

Haemochromatosis is a disorder in which the body absorbs and stores too much iron, potentially damaging the liver, heart, pancreas, and joints.

Haemodialysis

A treatment for severe kidney failure in which a machine filters waste and extra fluid from the blood.

Haemoflagellate

A haemoflagellate is a flagellated parasite that lives

in blood or tissues, such as the organisms that cause sleeping sickness and Chagas disease.

Haemoglobin

Haemoglobin is the iron-containing protein in red blood cells that carries oxygen from the lungs and helps transport carbon dioxide back to the lungs.

Haemoglobinopathy

A haemoglobinopathy is an inherited disorder affecting the structure or production of hemoglobin, such as sickle-cell disease or thalassemia.

Haemoglobinuria

Haemoglobinuria is hemoglobin in the urine, often caused by destruction of red blood cells within the bloodstream.

Haemoglobinuria (Hemoglobinuria)

The presence of free hemoglobin in the urine, often causing red or dark-brown discoloration. It may result from intravascular hemolysis or other blood disorders.

Alternate Names:

Hemoglobinuria

Haemolysis

The breakdown of red blood cells, which releases hemoglobin into the blood. It can happen normally as old cells are removed or too quickly because of disease.

Haemolytic Anaemia

Haemolytic anaemia is anemia caused by red blood cells being destroyed faster than the bone marrow can replace them.

Haemolytic Disease Of The Newborn

Destruction of a newborn's red blood cells by maternal antibodies that crossed the placenta, causing anemia and jaundice.

Haemoperitoneum

Haemoperitoneum is bleeding into the abdominal cavity, commonly from injury, a ruptured ectopic

pregnancy, or a bleeding organ.

Haemophilia

An inherited bleeding disorder in which the blood does not clot properly. Hemophilia A is caused by low factor VIII, and hemophilia B by low factor IX. It can cause prolonged bleeding or bleeding into joints and muscles.

Haemophilus

Haemophilus is a genus of bacteria that can colonize the respiratory tract and can cause illnesses such as ear infection, pneumonia, or meningitis.

Haemophilus Aegyptius

Haemophilus aegyptius is a bacterium associated with conjunctivitis and, rarely, severe bloodstream infection.

Haemophilus Haemolyticus

Haemophilus haemolyticus commonly colonizes the upper airway and can occasionally cause respiratory or invasive infection.

Haemophilus Infections

Haemophilus infections are bacterial infections that may affect the ears, sinuses, lungs, bloodstream, or meninges; risk varies by species and vaccination status.

Haemophilus Influenzae

A type of bacteria that can cause ear infections, pneumonia, meningitis, and other infections. Type b, or Hib, can cause severe disease in children, but vaccination greatly reduces this risk.

Haemophilus Paragallinarum

Haemophilus paragallinarum is a bacterium that causes infectious coryza, a respiratory disease of chickens; it is not a usual human pathogen.

Haemophilus Parainfluenzae

A small gram-negative bacterium that normally colonizes the upper respiratory tract but can rarely cause respiratory, bloodstream, or other opportunis-

tic infections.

Haemophilus Somnus

Haemophilus somnus, now commonly called Histophilus somni, is a bacterium that causes respiratory and other infections mainly in cattle.

Haemoptysis

Coughing up blood or blood-stained sputum from the lower respiratory tract.

Haemorrhage

British spelling of Hemorrhage; see Hemorrhage.

Haemorrhagic Disease Of The Newborn (Vitamin K Deficiency Bleeding)

Bleeding in a newborn or young infant caused by inadequate vitamin K-dependent clotting activity. It may occur early, classically, or late and can involve the brain or gastrointestinal tract.

Alternate Names:

Vitamin K Deficiency Bleeding

Haemorrhagic Fever

Haemorrhagic fever is a severe infection or illness that can damage blood vessels and cause bleeding, shock, and organ failure.

Haemorrhagic Fever With Renal Syndrome

Haemorrhagic fever with renal syndrome is a hantavirus infection that can cause fever, bleeding, low blood pressure, and sudden kidney dysfunction.

Haemorrhagic Stroke

A haemorrhagic stroke occurs when a blood vessel in or around the brain ruptures and causes bleeding that injures brain tissue.

Haemorrhoids

Enlarged symptomatic vascular cushions of the anal canal that may cause painless bright-red rectal bleeding, prolapse, itching, or discomfort.

Haemosiderin

Haemosiderin is an iron-storage pigment formed when iron accumulates in cells, often after repeated bleeding or excess iron.

Haemosiderosis

Haemosiderosis is excess deposition of iron-containing haemosiderin in tissues, which may follow transfusions, bleeding, or increased iron absorption.

Haemostasis

Haemostasis is the normal process that stops bleeding through blood-vessel constriction, platelet plugs, and clot formation.

Haemostat

A haemostat is a substance, device, or instrument used to control bleeding.

Haemothorax

Haemothorax is blood in the space between the lung and chest wall, usually caused by injury or a medical procedure.

Haff Disease

A clinical disorder related to haff disease; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hafnia

A genus of gram-negative intestinal bacteria; *Hafnia alvei* is the species most often identified in human specimens.

Hafnia Alvei

A gram-negative bacterium found in the intestine and environment that rarely causes opportunistic infection.

Hageman Factor

Hageman factor, also called factor XII, is a blood-clotting protein that helps initiate the contact pathway of coagulation.

Hailey-Hailey Disease

An inherited acantholytic disorder caused by pathogenic variants in *ATP2C1*, producing recurrent erosions and fragile vesicles in flexures such as the neck, axillae, and groins.

Hair

Keratinized filaments produced by follicles in the skin. Hair protects the scalp, contributes to temperature regulation and sensation, and grows through anagen, catagen, and telogen phases.

Hair Bleach Poisoning

Hair bleach poisoning involves toxic or environmental exposure to hair bleach poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hair Bulb

The enlarged base of a hair follicle where the hair shaft is produced.

Hair Cells

Special cells in the inner ear that change sound vibrations and head movement into nerve signals for hearing and balance.

Hair Disease

A disorder affecting hair growth, the hair shaft, or the scalp; causes include genetics, inflammation, infection, medicines, and nutritional problems.

Hair Dye

Hair dye is a cosmetic product used to change hair color; some ingredients can cause skin irritation or allergic contact dermatitis.

Hair Dye Poisoning

Hair dye poisoning involves toxic or environmental exposure to hair dye poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hair Follicle

A small tube in the skin that produces a hair. Each

follicle has a growth phase, a brief transition phase, and a resting phase.

Hair Loss

Loss or thinning of hair on the scalp or body. It may be temporary or permanent, patchy or widespread, and caused by genetics, illness, medicines, autoimmune disease, infection, or scarring of the scalp.

Alternate Names:

Alopecia (Hair Loss)

Hair Loss (Alopecia)

The partial or complete loss of hair from areas where it normally grows, most commonly the scalp. Causes include genetic predisposition (androgenetic alopecia), autoimmune conditions (alopecia areata), hormonal changes, medications, stress, and nutritional deficiencies.

Alternate Names:

Alopecia

Hair Loss Classification

Hair loss (alopecia) is classified as scarring (cicatricial) or non-scarring. Non-scarring alopecias are potentially reversible: androgenetic alopecia (male and female pattern), alopecia areata, telogen effluvium, anagen effluvium, and tinea capitis.

Hair Shaft

The portion of a hair that projects above the skin surface.

Hair Spray Poisoning

Hair spray poisoning involves toxic or environmental exposure to hair spray poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hair Straightener Poisoning

Hair straightener poisoning involves toxic or environmental exposure to hair straightener poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hair Tonic Poisoning

Hair tonic poisoning involves toxic or environmental exposure to hair tonic poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hair Transplant

A procedure that moves hair follicles, usually from a donor area of the scalp, to areas affected by permanent hair loss; results depend on diagnosis, donor supply, technique, and postoperative care.

Hair, Lanugal

Lanugo: the fine, soft, usually unpigmented hair covering a fetus and sometimes present on a premature newborn; it is normally shed before or shortly after birth.

Hair-Ball

A hair-ball, or trichobezoar, is a mass of swallowed hair in the stomach or intestine, usually associated with repeated hair eating.

Hairless

Hairless means lacking hair; localized or generalized hair loss may result from genetics, autoimmune disease, medicines, or other conditions.

Hairy Cell Leukaemia

Hairy cell leukaemia is a rare slow-growing blood cancer in which abnormal B lymphocytes accumulate in the bone marrow, spleen, and blood.

Hairy Cell Leukemia

A rare, indolent B-cell chronic lymphoproliferative disorder characterized by the accumulation of mature B lymphocytes with characteristic cytoplasmic projections ("hairy cells") in the bone marrow, peripheral blood, and spleen.

Alternate Names:

Cancer, Hairy Cell Leukemia Leukemia, Hairy Cell

Hairy Tongue

A usually harmless overgrowth and discoloration of the tongue's papillae, often appearing black,

brown, or white. It may be associated with smoking, poor oral hygiene, dry mouth, or antibiotics; gentle tongue cleaning and removal of contributing factors usually help.

Hajdu-Cheney Syndrome

A clinical disorder related to hajdu-cheney syndrome; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Halcinonide

A potent topical corticosteroid used to reduce inflammation and itching in steroid-responsive dermatitis and other inflammatory skin disorders; prolonged or extensive use can cause skin atrophy and systemic effects.

Haldane Effect

The way oxygen binding to hemoglobin in the lungs helps release carbon dioxide for exhalation. Hemoglobin without oxygen carries more carbon dioxide in the body's tissues.

Half-Life

The time it takes for the amount of a medicine in the body, or its blood level, to decrease by half. It helps determine how often a medicine should be taken and how long it remains in the body.

Half-Life (Physical)

The time it takes for half of a radioactive substance to lose its radioactivity. It helps determine the timing of scans and the amount of radiation exposure.

Alternate Names:

Physical

Half-Life (T_{1/2})

The time required for plasma drug concentration to decrease by 50%. It determines dosing frequency and time to steady state (4-5 half-lives). Drugs with long half-lives (e.g., amiodarone: 40-55 days, warfarin: 40 hours) can be given once daily but take longer to reach steady state and to be eliminated.

Alternate Names:

t_{1/2}

Halitosis

An unpleasant odour of the breath that may be socially distressing. The most common cause (85-90%) is poor oral hygiene—accumulation of bacteria on the tongue, dental plaque, periodontitis, and gingivitis.

Hallermann'S Syndrome

A rare congenital disorder characterized by distinctive craniofacial development, sparse hair, dental abnormalities, and sometimes eye or skeletal findings; it is also called Hallermann–Streiff syndrome.

Hallervorden-Spatz Disease

A clinical disorder related to hallervorden-spatz disease; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hallucination

A perception in the absence of an external stimulus, experienced as real and occurring in objective space.

Hallucinations

Hallucinations are perceptions without an external sensory stimulus and can arise from psychiatric illness, neurologic disease, delirium, substances, medications, sensory loss, or sleep transitions.

Hallucinogen

A psychoactive substance that can alter perception, thought, and mood and may produce hallucinations; effects and toxicity vary by substance, dose, co-ingestants, and mental state.

Hallux

The first and most medial digit of the human foot (the great toe or big toe), designated digit 1. Unlike digits 2 through 5 (which contain three phalanges), the hallux contains only two phalanges (proximal and distal) articulated by a single interphalangeal joint, and articulates with the first metatarsal at the first metatarsophalangeal joint, flanked by two plantar sesamoid bones. Crucial for biomechanical stability and propulsion during the push-off phase of the gait cycle, bearing up to twice the body weight. High-yield clinical pathology includes hallux valgus

(bunion deformity), hallux rigidus (osteoarthritis of the first MTP joint), turf toe (sprain of the plantar capsuloligamentous complex), and podagra (acute gouty arthritis classic to the first MTP joint).

Hallux Rigidus

Stiffness and painful loss of motion at the joint where the big toe meets the foot.

Hallux Valgus

Lateral deviation of the great toe (hallux) with medial prominence of the first metatarsal head (bunion). More common in women, associated with narrow-toed shoes, high heels, and genetic predisposition. Presents with pain over the medial eminence, difficulty with shoe wear, and progressive deformity.

Hallux Valgus (Bunion)

A progressive deformity of the first metatarsophalangeal joint, characterized by lateral deviation of the great toe (hallux valgus) and medial deviation of the first metatarsal head (metatarsus primus varus), resulting in a bony prominence on the medial side of the forefoot (the bunion).

Alternate Names:

Bunion

Hallux Varus

Abnormal medial deviation of the great toe, often following surgery, trauma, neuromuscular imbalance, or congenital deformity; it can cause shoe difficulty and pressure-related pain.

Halo Nevus

A benign melanocytic nevus surrounded by a rim of depigmentation (Sutton nevus), caused by an immune-mediated T-cell response directed against melanocytes in the nevus. It is most common on the trunk of children and young adults and typically resolves by central depigmentation, leaving a white macule.

Halobacillus

A genus of salt-tolerant, spore-forming bacteria found mainly in saline environments and rarely associated with human disease.

Halobacteriaceae

A family of salt-loving archaea found in very salty environments; they are not usual causes of human infection.

Halofenate

An investigational lipid-lowering drug related to fenofibrate that was studied for effects on triglycerides and cholesterol but is not a routine current treatment.

Halofuginone Hydrobromide

The hydrobromide salt of halofuginone, a compound with antiparasitic and antifibrotic activity that has been studied experimentally; clinical use depends on the specific approved indication.

Halogen

An element in group 17 of the periodic table, including fluorine, chlorine, bromine, iodine, and astatine; halogens and their compounds have medical, biologic, and toxicologic importance.

Halogenation

A chemical reaction that adds a halogen such as chlorine, bromine, or iodine to a molecule; in biology it can alter drug activity, metabolism, or biomolecular function.

Haloperidol

Haloperidol is a first-generation antipsychotic that blocks dopamine D2 receptors and is used for psychosis, agitation, delirium, and tics; extrapyramidal reactions, tardive dyskinesia, hyperprolactinaemia, and QT prolongation are important risks.

Halothane

A volatile inhaled anesthetic formerly used to induce or maintain general anesthesia; it can cause hypotension, arrhythmias, and rare severe immune-mediated liver injury.

Hamamelis

A botanical preparation from witch hazel used topically as an astringent for minor skin or mucosal

irritation; evidence and product composition vary.

Hamartoma

A benign growth composed of a disorganized mixture of tissue elements normally present at that site, e.g., pulmonary hamartoma, Peutz-Jeghers polyps, and PTEN-related (Cowden) lesions.

Hamartomatous Polyposis

A condition in which multiple hamartomatous gastrointestinal polyps develop from disorganized but benign tissue overgrowth, sometimes as part of an inherited cancer-predisposition syndrome.

Hamate

Wedge-shaped distal row carpal bone on the ulnar side with the hook of the hamate. The hook forms Guyon's canal medial border and serves as flexor retinaculum attachment. Articulates with 4th/5th metacarpals. Hook fractures associated with racquet sports and ulnar nerve compression.

Hamate Bone

The hamate is a wedge-shaped wrist bone on the little-finger side that has a hook serving as an attachment and passage point for tendons and nerves.

Hamman'S Sign

A crunching or crackling sound heard on auscultation synchronized with the heartbeat, caused by air in the mediastinum (pneumomediastinum). It is a rare but specific finding. It is also described as a mediastinal crunch heard in spontaneous pneumomediastinum.

Hammer

The common name for the malleus, one of the three small bones of the middle ear.

Hammer Toe

A deformity of the second, third, or fourth toe characterized by flexion at the PIP joint and extension at the MTP and DIP joints. Caused by muscle imbalance (intrinsic foot muscle weakness) and narrow-toed shoes. Presents with pain over the

dorsal PIP joint (corn/callus) and the tip of the toe. Flexible hammer toe: correctable passively.

Hammer Toe Repair

Surgery to correct a fixed or painful flexion deformity of a lesser toe, using soft-tissue release, tendon balancing, bone correction, or temporary fixation as indicated.

Hammertoe

A deformity in which a toe bends downward at its middle joint, giving it a claw-like appearance. It may cause pain, stiffness, corns, or calluses, especially when wearing shoes.

Hampton Hump

Wedge-shaped opacification with its base abutting the pleura on chest radiography, representing pulmonary infarction secondary to Pulmonary Embolism. See Pulmonary Embolism (PE).

Hampton'S Hump

Wedge-shaped, pleural-based opacity on CXR representing pulmonary infarction distal to PE.

Hamstring

One of three muscles forming the posterior compartment of the thigh: semitendinosus, semimembranosus, and biceps femoris (long and short heads). Common origin: ischial tuberosity (except short head of biceps femoris which originates from the linea aspera of the femur).

Hamstring Injury

A strain or tear of one or more muscles or tendons at the back of the thigh.

Alternate Names:

Pulled Hamstring (Hamstring Injury)

Hamstring Muscles

Three posterior thigh muscles: biceps femoris (long/short heads), semitendinosus, semimembranosus, mostly from the ischial tuberosity (short head of biceps from linea aspera). Long head/semimembranosus/semitendinosus innervated

by tibial sciatic nerve; short head by common peroneal division. Flex knee, extend hip.

Hand

The part of the upper limb beyond the wrist, containing the palm, fingers, bones, joints, muscles, nerves, and blood vessels used for grasping and sensation.

Hand Deformity

An abnormal shape, alignment, or loss of motion of the hand or fingers caused by congenital difference, trauma, arthritis, nerve dysfunction, tendon disease, or other conditions.

Hand Foot And Mouth Disease

A common viral illness (mostly children <10 years) caused by coxsackievirus A16 (and enterovirus 71), transmitted by fecal-oral/respiratory.

Hand Hygiene

The most effective measure for preventing healthcare-associated infections. Hand hygiene includes handwashing with soap and water and hand rubbing with alcohol-based rubs. Alcohol-based rubs are preferred for routine hand hygiene as they are faster and more effective than soap and water.

Hand Lotion Poisoning

Hand lotion poisoning involves toxic or environmental exposure to hand lotion poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hand Microsurgery

Specialized hand surgery performed with magnification and fine instruments to repair or reconstruct small blood vessels, nerves, and other delicate tissues. It may be used for complex injuries, replantation, and restoration of blood flow to a threatened digit or hand.

Hand Or Foot Spasms

Hand or foot spasms are involuntary muscle contractions that may result from electrolyte abnormalities,

nerve disorders, overuse, or reduced blood flow.

Hand Prolapse

Prolapse of a fetal hand or arm alongside or below the presenting part. May occur spontaneously or after manual rotation. The prolapsed arm may be mistaken for a cord or foot. Management: if the hand is above the presenting part (compound presentation), it may be gently pushed back during contractions.

Hand X-Ray

Radiographic imaging of the hand used to evaluate fractures, dislocations, arthritis, infection, tumors, bone age, and other abnormalities of the bones and joints.

Hand, Foot, And Mouth Disease

A contagious viral illness, usually in children, causing fever and vesicles or ulcers on the hands, feet, and mouth. Enteroviruses are common causes.

Hand-Arm Vibration Syndrome

An occupational disorder from prolonged exposure to vibrating tools, causing numbness, tingling, reduced grip, and episodic whitening of the fingers from vascular spasm.

Hand-Foot-And-Mouth Disease

A contagious viral illness causing sores in the mouth and a rash on the hands and feet, usually in children.

Hand-Foot-And-Mouth Syndrome

A usually mild, contagious viral illness, commonly caused by enteroviruses, producing fever, mouth sores, and small spots or blisters on the hands and feet. It mainly affects young children but can occur in adults. Hydration, pain relief, and hygiene are important; severe dehydration, neurologic symptoms, or persistent fever requires care.

Hand-Foot-Mouth Disease

A common enteroviral infection, usually in young children, causing fever, painful oral ulcers, and a vesicular rash on the hands, feet, and sometimes

buttocks. It is highly contagious and usually self-limited.

Hand-Schüller-Christian Disease

Hand-Schüller-Christian disease is an older name for a multisystem form of Langerhans cell histiocytosis, often involving bone, skin, and the pituitary gland.

Handling Sharps And Needles

Handling sharps and needles means using, passing, disposing of, and storing sharp medical instruments safely to prevent needlestick injuries and infection transmission.

Handwashing

Handwashing is cleaning the hands with soap and water or an alcohol-based product to remove germs and reduce transmission of infection.

Hang-Nail

A small torn flap of skin beside a fingernail or toenail, often caused by dryness or trauma and vulnerable to local infection if torn or contaminated.

Hanging And Strangulation Signs

Medico-legal findings resulting from mechanical constriction of the neck structures. Hanging: neck constriction by a ligature tightened by the weight of the victim's body. Ligature mark: pale, hard, parchment-like groove running obliquely upward toward the knot, usually situated above the thyroid cartilage.

Hangover

A group of symptoms after alcohol use, such as headache, nausea, thirst, fatigue, dizziness, and sensitivity to light; symptoms reflect effects of alcohol and its metabolites, dehydration, sleep disruption, and inflammation.

Hangover Treatment

Supportive care for alcohol-related next-day symptoms such as headache, nausea, thirst, fatigue, and poor concentration; hydration, rest, and time help, while more alcohol or unsafe medication combina-

tions can worsen harm.

Hannaella Luteola

A yeast species that is usually environmental but can rarely cause opportunistic infection, especially in people with weakened immunity.

Hansen Disease

A long-lasting bacterial infection that mainly affects the skin and nerves. It can cause pale or reddish skin patches, loss of feeling, weakness, and eye problems. It spreads mainly through prolonged close contact with an untreated person. Antibiotic treatment can cure it and help prevent lasting disability.

Alternate Names:

Leprosy; Hansen's Disease; Hansen's Disease (Leprosy)

Hanseniaspora

A genus of yeasts found on fruits and in fermented foods; some species are used in fermentation and rarely cause infection.

Hanseniaspora Uvarum

A yeast commonly associated with fruit and wine fermentation; human infection is rare.

Hanseniaspora Valbyensis

A yeast found in fruit and fermented products; it is rarely reported as an opportunistic human pathogen.

Hantavirus

Hantavirus infection is acquired from exposure to infected rodent excreta and can cause hemorrhagic fever with renal syndrome or severe cardiopulmonary syndrome. to humans primarily through inhalation of aerosolized rodent excreta (urine, feces, saliva).

Hantavirus Infection

A zoonotic viral infection acquired mainly through inhalation of contaminated rodent excreta. It may cause hemorrhagic fever with renal syndrome or hantavirus pulmonary syndrome, which can progress

rapidly to respiratory failure.

Hantavirus Pulmonary Syndrome

A severe respiratory illness caused by hantaviruses, usually acquired by inhaling aerosolized particles from infected rodent urine, droppings, or nesting material. Fever, muscle aches, headache, and gastrointestinal symptoms may rapidly progress to cough, pulmonary edema, and respiratory failure. Immediate hospital care is essential.

Alternate Names:

Sin Nombre Virus Infection

Haploid

Haploid is a developmental, immunologic, reproductive, behavioral, environmental, or healthcare concept used in medicine or biomedical research.

Haploidentical Donor

A donor who shares one human leukocyte antigen haplotype with the recipient, commonly used for selected hematopoietic stem-cell or organ transplantation when a fully matched donor is unavailable.

Haploidy

The state of having one complete set of chromosomes, as in human gametes; haploid cells contain 23 chromosomes.

Haploinsufficiency

A genetic state in which one functional copy of a gene does not produce enough gene product for normal function, so loss or inactivation of the other copy can cause disease.

Haplotype

Set of alleles inherited together on same chromosome. Used in linkage analysis, population genetics, disease association studies.

Hapten

A hapten is a small molecule that is not immunogenic by itself but can trigger an immune response after binding to a larger carrier protein, as in some drug allergies.

Haptoglobin

An alpha-2 globulin that binds free hemoglobin, preventing oxidative damage and renal excretion. Low/undetectable: acute hemolysis (intravascular and extravascular), liver disease (decreased production), and hereditary deficiency. Elevated: acute phase reaction (inflammation, infection, malignancy).

Haptoglobin Analyses

Laboratory tests measuring haptoglobin, a plasma protein that binds free hemoglobin; low values can support evaluation of intravascular hemolysis but are also affected by liver disease and inflammation.

Haptoglobin Blood Test

A blood test measuring haptoglobin, a protein that binds free hemoglobin; low levels may support intravascular hemolysis but can also reflect liver disease or inflammation.

Haptoglobin Decreased

A low blood haptoglobin result, commonly caused by intravascular red-cell destruction, reduced hepatic production, or severe liver disease; it is interpreted with hemoglobin, bilirubin, reticulocytes, and a blood smear.

Hard Palate

The bony, mucosa-covered front portion of the roof of the mouth that separates the oral and nasal cavities and provides a surface for chewing and speech.

Harderian Gland

A gland associated with the eye in many vertebrates that produces lipid-rich secretions; it is not a normal distinct gland in the same form in humans.

Hardness

A measurable resistance to deformation, used in medicine and materials science; in clinical contexts it may describe tissue induration, a hard mass, or hard stool depending on the setting.

Hardware Removal - Extremity

Surgical removal of plates, screws, rods, or other orthopedic hardware from an arm or leg when it causes pain, infection, prominence, nonunion-related problems, or other selected complications.

Harlequin Color Change

A transient, benign skin color change in newborns where one half of the body becomes flushed while the other half remains pale, due to asymmetric vasomotor tone. It occurs in up to 10 percent of healthy neonates, typically when the infant is placed on their side.

Harlequin Foetus

A severe congenital ichthyosis, now called harlequin ichthyosis, characterized at birth by thick armor-like plates of keratin separated by deep fissures.

Harm Reduction

Evidence-based strategies that decrease health, social, and legal harms associated with substance use without requiring immediate abstinence. Examples include naloxone access, sterile injecting equipment, drug-checking where available, safer-use education, and linkage to treatment.

Harm Reduction Therapy

Treatment that aims to reduce health and social harms from addictive behavior, even when abstinence is not immediate.

Alternate Names:

Harm reduction

Harmless Murmur

An innocent or physiologic heart murmur produced by normal blood flow without structural heart disease; it is diagnosed only after appropriate clinical assessment.

Harringtonines

A group of plant-derived polyol esters and related alkaloid compounds investigated for anticancer activity; their pharmacology and toxicity depend on the individual compound.

Harris-Imvoden Equation

The Harris-Imvoden equation is a clinical calculation used to estimate a physiologic or nutritional measure; its result is an approximation and must be interpreted with the patient's clinical context.

Harrison's Sulcus

Harrison's sulcus is a horizontal groove along the lower chest wall caused by inward pulling of the ribs, classically in long-standing childhood breathing difficulty.

Hartmann Procedure

Surgical resection of the rectosigmoid colon with creation of an end colostomy and rectal stump. Used for emergency resection of complicated diverticulitis with peritonitis and colorectal cancer with obstruction or perforation. The procedure avoids primary anastomosis in a contaminated field.

Hartnup Disease

An autosomal-recessive disorder of neutral-amino-acid transport, especially in the intestine and renal tubules. Defective tryptophan handling can cause pellagra-like photosensitive dermatitis, cerebellar ataxia, and neuropsychiatric symptoms, often intermittently.

Hartnup Disorder

An inherited disorder of neutral amino-acid transport that can cause pellagra-like skin changes, neurologic symptoms, or no symptoms.

HAS-BLED Score

A validated clinical scoring system used to quantify the one-year risk of major bleeding in patients with atrial fibrillation receiving oral anticoagulation (vitamin K antagonists or direct oral anticoagulants). Points are assigned for Hypertension (systolic BP > 160 mmHg, 1), Abnormal renal or liver function (1 or 2), Stroke history (1), Bleeding history or predisposition (1), Labile international normalized ratio (1), Elderly age > 65 years (1), and Drugs or alcohol excess (1 or 2). A score of ≥ 3 indicates high bleeding risk and prompts closer clinical monitoring and aggressive correction of modifiable bleeding risk.

factors rather than withholding anticoagulation.

Alternate Names:

HAS-BLED; Anticoagulation Bleeding Risk Score

Hashimoto Disease

An autoimmune disorder in which antibodies and lymphocytes progressively damage the thyroid, commonly causing hypothyroidism, goiter, fatigue, cold intolerance, constipation, and weight change.

Hashimoto Thyroiditis

An autoimmune disorder in which immune-mediated inflammation gradually damages the thyroid gland and may cause hypothyroidism. The gland may be enlarged or atrophic, and thyroid autoantibodies are often present.

Hashimoto'S Disease

An autoimmune disorder in which the immune system attacks the thyroid, often leading to hypothyroidism.

Alternate Names:

Chronic Lymphocytic Thyroiditis Thyroiditis, Chronic Lymphocytic

Hashimoto'S Thyroiditis

A chronic autoimmune inflammation of the thyroid that commonly causes progressive hypothyroidism and may produce a firm, enlarged thyroid gland.

Hashish

Hashish is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hashitoxicosis

A temporary period of excess thyroid hormone caused by release of stored hormone during Hashimoto thyroiditis.

Hassall'S Corpuscles

Hassall's corpuscles are rounded structures in the thymus made of specialized epithelial cells and involved in the development of immune tolerance.

Haustra

Pouch-like sacculations of the colon produced by the arrangement of the circular and longitudinal smooth muscle layers.

Haverhill Fever

A clinical disorder related to haverhill fever; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hay Fever

Seasonal or persistent allergic rhinitis caused by inhaled allergens.

Alternate Names:

Allergic rhinitis Rhinitis, Allergic Seasonal Allergy Seasonal Allergies (Hay Fever)

Hay Fever (Allergic Rhinitis)

Inflammation of the nasal lining caused by an allergic response to airborne substances, including pollens, dust mites, animal dander, or mold. It causes sneezing, congestion, nasal itching, and watery discharge.

Alternate Names:

Allergic Rhinitis

Hazardous Materials

Substances that can cause injury, poisoning, fire, radiation exposure, or environmental harm; safe handling requires labeling, protective equipment, exposure control, and emergency procedures.

Hazardous Substance

Hazardous Substance is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hb (Hemoglobin)

Hemoglobin, the iron-containing protein in red blood cells that carries oxygen and contributes to carbon-dioxide transport and blood's acid-base buffering.

HbA1C

Hemoglobin with glucose attached; its measurement estimates average blood glucose over the previous two to three months.

Alternate Names:

Glycosylated hemoglobin; hemoglobin A1c

HbA1C (Glycated Hemoglobin)

Hemoglobin containing glucose-derived adducts formed by nonenzymatic glycation. It reflects average glycemic exposure over approximately two to three months, with interpretation affected by altered red-cell survival and some hemoglobin variants.

Alternate Names:

Glycated Hemoglobin

HCG

Human chorionic gonadotropin is a hormone produced mainly during pregnancy; its measurement is used in pregnancy testing and evaluation of selected tumors.

HCG Blood Test - Qualitative

A blood test that detects whether human chorionic gonadotropin is present, most commonly to identify pregnancy.

HCG Blood Test - Quantitative

A blood test measuring the concentration of human chorionic gonadotropin, used to confirm and monitor pregnancy and to evaluate selected gestational or germ-cell tumors.

HCG In Urine

A urine test detecting human chorionic gonadotropin, commonly used as a pregnancy test; accuracy depends on timing, urine concentration, and test performance.

Hct (Hematocrit)

The percentage of blood volume occupied by red blood cells, measured as part of a complete blood count.

HCV PCR

A polymerase chain reaction test that detects or measures hepatitis C virus RNA in blood, helping identify active infection and monitor treatment response.

HDL

High-density lipoprotein, a class of particles that transports cholesterol and other lipids in the blood. HDL concentration is one component of cardiovascular risk assessment; it should not be interpreted as a protective or harmful measure by itself.

HDL Cholesterol

Cholesterol carried in high-density lipoprotein particles; HDL participates in transporting cholesterol from tissues toward the liver, but its concentration alone does not determine cardiovascular risk.

HDL Test

A blood test measuring high-density lipoprotein cholesterol, which is used with the rest of the lipid profile to assess cardiovascular risk.

HDR Syndrome

A genetic disorder characterized by hypoparathyroidism, sensorineural deafness, and renal dysplasia or other kidney abnormalities, usually caused by a pathogenic variant in *GATA3*.

Head

The body region containing the brain, eyes, ears, nose, mouth, skull, and face, connected to the trunk by the neck and supplied by major cranial nerves and vessels.

Head And Face Reconstruction

Reconstructive surgery to restore the form or function of the skull, face, or associated soft tissues after trauma, cancer treatment, congenital difference, or other disease.

Head And Neck Cancer

Cancers arising in the oral cavity, pharynx, larynx, sinonasal tract, and salivary glands, most commonly

squamous cell carcinoma. Risk factors include tobacco, alcohol, and HPV (oropharyngeal). Treatment involves surgery, radiation, and chemotherapy, often combined to preserve function.

Head And Neck Oncologic Surgery

Surgical treatment of cancers of the oral cavity, throat, larynx, nose, sinuses, salivary glands, or neck lymph nodes. Procedures may remove the tumor, treat involved lymph nodes, and reconstruct affected tissues while preserving breathing, swallowing, speech, and appearance when possible.

Alternate Names:

Head And Neck Onco Surgery

Head Circumference

Measured at the largest circumference above the ears and eyebrows, reflecting brain growth. Plotted on growth charts to detect microcephaly (below 3rd percentile) or macrocephaly (above 98th percentile). Rapid increase may indicate hydrocephalus or intracranial hemorrhage; failure to grow may indicate microcephaly or brain atrophy.

Head CT Scan

Computed tomography of the head using x-rays and computer reconstruction to evaluate the brain, skull, bleeding, stroke, tumors, trauma, and other abnormalities.

Head Injury

Trauma to scalp, skull, or brain. Scalp lacerations: bleed profusely, require suturing. Skull fracture: basilar (Battle sign, raccoon eyes) or depressed (surgical emergency). Concussion: transient neurological dysfunction, manage with rest, gradual return to activity.

Alternate Names:

Brain Injury (Head Injury)

Head Injury - First Aid

Initial care after a head injury includes protecting the neck, controlling external bleeding gently, monitoring breathing and alertness, and seeking urgent help for loss of consciousness, repeated vomiting, seizure, confusion, or worsening headache.

Head Movement

Motion of the head produced mainly by the cervical spine and neck muscles, including flexion, extension, rotation, and side bending; pain or restricted movement may indicate injury or disease.

Head MRI

Magnetic resonance imaging of the head using a magnetic field and radio waves to produce detailed images of the brain, cranial nerves, blood vessels, and surrounding structures.

Head-Lice

Head lice are small insects that live on the scalp and cause itching; they spread mainly through close head-to-head contact.

Headache

Headache is pain in the head or upper neck; onset, pattern, associated neurologic findings, fever, trauma, and systemic symptoms distinguish primary from secondary causes. tension-type, cluster, and trigeminal autonomic cephalalgias) or secondary (attributed to an underlying structural, infectious, vascular, or metabolic cause).

Headache And Migraine

Headaches are pain in the head or upper neck region, classified as primary (tension-type, migraine, cluster) or secondary to underlying conditions. Migraines are a neurological disorder characterized by recurrent episodes of moderate to severe throbbing pain, often accompanied by nausea, photophobia, and aura.

Headaches - Danger Signs

Headache danger signs include sudden severe onset, new neurologic symptoms, fever with neck stiffness, confusion, head injury, or a new headache during pregnancy; urgent assessment may be needed.

Healing

Healing is a developmental, immunologic, reproductive, behavioral, environmental, or healthcare concept used in medicine or biomedical research.

Health Behavior

An action or pattern of action that affects health, such as eating, physical activity, tobacco use, alcohol use, sexual practices, or taking medicines.

Health Behavior Change

Modification of behaviors that affect health, such as tobacco use, diet, physical activity, and adherence to treatment.

Health Behavior Motivation

Factors that influence initiation and maintenance of health-related behaviors, including adherence, physical activity, and smoking cessation.

Health Care

The prevention, diagnosis, treatment, and ongoing support provided to maintain or improve health, delivered by professionals, institutions, public-health systems, and informal caregivers.

Health Care Agents

A patient-care topic concerning health care agents. It addresses the relevant purpose, risks, management, and follow-up.

Health Care Facility

A place where health services are delivered, such as a clinic, hospital, laboratory, nursing facility, or rehabilitation center; its services and level of care determine appropriate referrals.

Health Care Personnel

Professionals and support workers involved in patient care, including clinicians, nurses, therapists, technicians, aides, and public-health staff.

Health Care Proxy

A legal appointment authorizing another person to make healthcare decisions when a patient cannot make them.

Health Disparities

Differences in health outcomes or access to care among groups, often associated with social, eco-

nomic, environmental, geographic, or structural factors.

Health Disparity

Difference in health outcomes between populations. Examples: racial disparities, socioeconomic disparities, geographic disparities.

Health Education

The process of giving people understandable information and skills to prevent illness, recognize symptoms, use treatment safely, and make informed health decisions.

Health Equity

The fair opportunity for all people to attain their highest possible level of health, without avoidable differences caused by social or economic disadvantage.

Health For All

A public-health goal emphasizing that all people should have access to needed health services and conditions that support health, without financial hardship.

Health Inequity

A systematic, avoidable, and unjust difference in health or access to health resources between social groups. It differs from health variation that is not preventable or not linked to unfair social conditions.

Health Information Technology

Use of technology for health information. Examples: electronic health records, telemedicine, mobile health, decision support.

Health Literacy

Ability to obtain, process, understand health information. Examples: reading labels, following instructions, communicating with providers.

Health Maintenance Organization

Health Maintenance Organization is a developmental, immunologic, reproductive, behavioral, environ-

mental, or healthcare concept used in medicine or biomedical research.

Health Promotion

Activities maintaining/improving health in elderly. Components: vaccination, screening, exercise, nutrition, cognitive stimulation, social engagement, fall prevention. Tailored to individual capacity and preferences.

Health Risk

The likelihood that a behavior, exposure, condition, or other factor will cause harm to health, considered together with the severity and timing of possible harm.

Health Risks Of Alcohol Use

Alcohol use can increase the risk of liver disease, several cancers, high blood pressure, injuries, dependence, and problems affecting mental health and relationships.

Health Risks Of Obesity

Obesity increases the risk of type 2 diabetes, hypertension, cardiovascular disease, sleep apnea, fatty liver disease, osteoarthritis, some cancers, and pregnancy complications.

Health Services Research

Research that evaluates how healthcare is organized, financed, delivered, accessed, and experienced, including quality, safety, outcomes, cost, and inequity.

Health Surveillance

The ongoing collection and analysis of health data to identify patterns, monitor disease, and guide public-health action.

Health Survey

A structured set of questions used to collect information about health, symptoms, behaviors, healthcare use, or population needs.

Health Visitor

Health Visitor is a developmental, immunologic,

reproductive, behavioral, environmental, or health-care concept used in medicine or biomedical research.

Health Workforce

The people and organizations that provide, support, manage, or study health services, including clinicians, public-health workers, and support staff.

Health-Related QoL

A person's perceived physical, psychological, and social well-being as affected by health and illness.

Healthcare Associated Infection

An infection developing as a consequence of health-care delivery, typically beyond 48 hours after admission or within 30 days of a procedure.

Healthcare Provider

A healthcare professional or organization that provides medical care, such as a physician, nurse, therapist, or clinic.

Healthcare Proxy

A person designated to make healthcare decisions for someone who cannot make or communicate those decisions. The appointment may be recorded in an advance directive or other legal document.

Healthcare-Associated Pneumonia

HCAP concept retired (2016 IDSA/ATS); risk factors for MDR now assessed individually. Risk factors for MDR pathogens (prior antibiotics, structural lung disease, frequent hospitalization) should be assessed individually rather than using the HCAP category.

Healthy Aging

The process of developing and maintaining the functional ability that supports well-being in older age.

Healthy Eating For Diabetes

A dietary approach designed to maintain stable blood glucose levels in individuals with diabetes mellitus. It emphasizes carbohydrate consistency,

high fiber intake, lean proteins, and healthy fats while minimizing refined sugars and simple carbohydrates.

Healthy Years

Years of life lived in good health, without substantial disease, disability, or loss of function.

Healthy Years Lost

Years of life lost due to premature death or disability. DALYs combine YLL and YLD.

Hearing

The sensory perception of sound produced when vibrations stimulate the cochlea and auditory nerve and are interpreted by the brain; hearing loss may be conductive, sensorineural, or mixed.

Hearing Aid

Amplification device for hearing loss. Types: behind-the-ear, in-the-ear, receiver-in-canal (RIC). Components: microphone, amplifier, speaker, battery. Fitting: audiogram-based. Adjustments: volume, programs.

Hearing Aids And Devices

Electronic devices worn in or behind the ear that amplify sound for individuals with hearing loss. Modern hearing aids include digital signal processing, directional microphones, and connectivity to smartphones and other audio sources.

Hearing Disorder

A condition that reduces or distorts hearing through disease of the outer or middle ear, cochlea, auditory nerve, or central pathways; audiometry and examination help distinguish conductive from sensorineural loss.

Hearing Impairment

Partial or complete reduction in the ability to detect or understand sound. It may be conductive, sensorineural, or mixed and can result from congenital disorders, infection, noise, ototoxic injury, trauma, ageing, or disease of the ear or auditory pathways.

Hearing Loss

Reduced ability to hear, classified as conductive (outer/middle ear pathology: cerumen impaction, otitis media, otosclerosis, tympanic membrane perforation), sensorineural (inner ear or auditory nerve: presbycusis, noise exposure, ototoxic drugs, Meniere disease, acoustic neuroma), or mixed.

Alternate Names:

Noise-Related Hearing Loss

Hearing Loss - Infants

Hearing loss in infants is reduced ability to hear that may be present at birth or develop later; screening and early intervention support language development.

Hearing Loss And Music

Hearing loss and music concerns reduced ability to hear musical pitch, loudness, or timbre; hearing protection and individualized amplification may help.

Hearing Loss: Sudden Deafness

A rapid sensorineural hearing loss, usually developing over hours to a few days, that may affect one or both ears. It is an otologic emergency; urgent hearing testing and evaluation are needed because early treatment may improve recovery.

Hearing Screening

Newborn hearing screening (universal): otoacoustic emissions and/or automated auditory brainstem response (AABR) before discharge; referral for comprehensive audiology if failed.

Hearing Test

An audiologic assessment that measures hearing thresholds and speech perception across frequencies to identify conductive, sensorineural, or mixed hearing loss.

Heart

A four-chambered muscular pump (two atria, two ventricles) located in the mediastinum, responsible for circulating blood throughout the body.

Heart And Vascular Services

Heart and vascular services include prevention, diagnosis, and treatment of diseases affecting the heart and blood vessels.

Heart Aneurysm

A localized bulging or outpouching of a weakened heart wall, often after myocardial infarction or from cardiomyopathy, that can impair contraction or form clots and arrhythmias.

Heart Arrhythmia

An abnormal heart rate or rhythm caused by altered electrical activity in the heart; it may be too fast, too slow, or irregular.

Alternate Names:

Dysrhythmias

Heart Attack

Acute myocardial injury caused most often by blockage of a coronary artery, usually from rupture of an atherosclerotic plaque and thrombosis. It may cause chest pressure, shortness of breath, sweating, nausea, or atypical symptoms and requires emergency treatment.

Alternate Names:

Myocardial Infarction (Heart Attack)

Heart Attack (Myocardial Infarction)

The death of heart muscle tissue due to sudden blockage of a coronary artery, typically by a blood clot. It presents with chest pain or pressure, shortness of breath, nausea, and diaphoresis, though symptoms may be atypical in women and diabetics.

Alternate Names:

Myocardial Infarction

Heart Attack First Aid

Heart attack first aid means calling emergency services immediately, helping the person rest, and following dispatcher instructions; aspirin may be appropriate for some adults who can safely take it.

Heart Attack Prevention

Heart-attack prevention targets blood pressure,

atherogenic cholesterol, diabetes, tobacco exposure, diet, physical activity, weight, sleep, and adherence to prescribed medicines. Risk-based preventive treatment is more effective than relying on one supplement or symptom.

Heart Block

A delay or interruption in the conduction of cardiac impulses from the atria to the ventricles through the atrioventricular (AV) node and His-Purkinje system. Classification: first-degree AV block: PR interval >200 ms, each atrial impulse conducts to the ventricles, typically asymptomatic.

Heart Bypass Surgery

Coronary artery bypass grafting creates new routes around blocked coronary arteries using graft vessels to improve myocardial blood flow and relieve ischemic symptoms or reduce selected risks.

Heart Bypass Surgery - Minimally Invasive

Coronary bypass performed through smaller chest incisions or with limited access rather than a full sternotomy; suitability depends on coronary anatomy, surgeon expertise, and patient factors.

Heart Catheterization

A procedure in which a catheter is passed through a blood vessel into the heart to measure pressures, assess coronary arteries, inject contrast, obtain samples, or perform treatment.

Heart Contraction

The coordinated systolic shortening of cardiac muscle that ejects blood from the ventricles into the pulmonary artery and aorta; electrical activation and calcium handling control it.

Heart CT Scan

Computed tomography of the heart and nearby structures, sometimes with ECG synchronization or contrast, used to evaluate coronary arteries, cardiac anatomy, calcium, or selected masses.

Heart Disease

A broad term encompassing various conditions affecting the heart and blood vessels, most commonly coronary artery disease, heart failure, arrhythmias, and valvular heart disease. Atherosclerosis, the buildup of plaque in the arteries, is the underlying cause of most cardiovascular events.

Heart Disease And Depression

Depression can occur with heart disease and may worsen quality of life, treatment adherence, and recovery; both conditions should be assessed and treated.

Heart Disease And Intimacy

Heart disease or its treatment may affect sexual activity, desire, and confidence; individualized medical advice can help address symptoms and safety.

Heart Disease And Women

Heart disease in women may involve different risk factors, symptoms, and timing than in men; chest pressure, shortness of breath, nausea, fatigue, or jaw or back pain can be important symptoms.

Heart Failure

A condition in which the heart cannot pump or fill with enough blood to meet the body's needs. It does not mean that the heart has stopped. In systolic heart failure, the heart does not squeeze strongly enough; in diastolic heart failure, the heart is stiff and does not relax or fill properly. Symptoms may include shortness of breath, tiredness, reduced ability to exercise, and swelling caused by fluid buildup.

Heart Failure In Children

Heart failure in children is inability of the heart to meet body demands, caused by congenital defects, cardiomyopathy, rhythm disease, infection, or other illness.

Heart Failure In Children - Home Care

Home care for a child with heart failure includes following the prescribed medicines and nutrition plan, monitoring breathing, feeding, activity, swelling,

and weight when advised, attending cardiology follow-up, and keeping recommended vaccines and illness plans current. Fluid or salt limits should be individualized. Worsening breathlessness, blue or gray color, fainting, marked lethargy, poor feeding, rapidly increasing swelling, or reduced urine requires urgent medical assessment.

Heart Failure Stages

The ACC/AHA framework describes Stage A as people at risk for heart failure without symptoms or current structural heart disease; Stage B, or pre-heart failure, as structural heart disease, abnormal cardiac function, or relevant biomarker evidence without symptoms; Stage C as current or previous symptoms or signs of heart failure with underlying cardiac disease; and Stage D as advanced heart failure with persistent symptoms that may require specialized therapies. Stages describe disease progression and are distinct from NYHA symptom classes.

Heart Failure With Preserved Ejection Fraction

A clinical syndrome of heart failure with a relatively preserved left-ventricular ejection fraction, often associated with impaired relaxation, increased ventricular stiffness, or elevated filling pressures. Diagnosis requires compatible symptoms or signs and objective evidence of cardiac dysfunction or elevated filling pressures; a preserved ejection fraction alone is not diagnostic.

Heart Failure With Mildly Reduced Ejection Fraction

A clinical syndrome of heart failure with mildly reduced left-ventricular ejection fraction, often defined by applicable guidelines as an ejection fraction of about 41–49%. Diagnosis requires compatible symptoms or signs and evidence of cardiac dysfunction or elevated filling pressures; the ejection-fraction range and treatment recommendations may vary by guideline.

Heart Failure With Reduced Ejection

Fraction

A clinical syndrome of heart failure with reduced left-ventricular systolic function, commonly associated with ischemic heart disease, cardiomyopathy, hypertension, or valvular disease. The ejection-fraction threshold is defined by the applicable guideline and must be interpreted with symptoms, signs, imaging, and other evidence of heart failure.

Heart Health

The state of cardiovascular well-being, characterized by efficient pumping function, clear arteries, normal blood pressure, and regular heart rhythm. Maintaining heart health involves a heart-healthy diet, regular aerobic and resistance exercise, weight management, stress reduction, and avoidance of tobacco.

Heart Injury

Damage to heart muscle or surrounding cardiac structures caused by trauma, ischemia, inflammation, toxins, or procedures; symptoms and tests depend on the injured tissue and severity.

Heart MRI

Magnetic resonance imaging of the heart that evaluates cardiac structure, function, tissue characteristics, blood flow, and selected congenital, ischemic, inflammatory, or infiltrative disorders.

Heart Murmur

An audible extra sound produced by turbulent blood flow within the heart or great vessels, heard on auscultation.

Alternate Names:

Murmur Congenital (Heart Murmur) Murmur Heart (Heart Murmur)

Heart Murmurs

Heart murmurs are extra or unusual sounds caused by turbulent blood flow through the heart or nearby vessels; some are harmless and others reflect heart disease.

Heart Muscle

The myocardium, the specialized cardiac muscle that contracts rhythmically to pump blood through the heart and circulation.

Heart Neoplasm

A neoplastic process affecting heart neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Heart Pacemaker

A heart pacemaker is a device that sends electrical impulses to maintain an appropriate heartbeat when the heart's natural rhythm is too slow or unreliable.

Heart Palpitations

An awareness of the heartbeat as pounding, racing, fluttering, or irregular. Palpitations may occur with normal rhythm or arrhythmia and can be triggered by stress, exertion, fever, stimulants, endocrine disease, or structural heart disease.

Alternate Names:

Skipped Heartbeats

Heart PET Scan

Positron-emission tomography of the heart using a radiotracer to assess myocardial perfusion, metabolism, viability, inflammation, or selected cardiac masses.

Heart Rate

The number of cardiac contractions (heartbeats) per minute. Normal resting heart rate (sinus rhythm): 60-100 bpm in adults; higher in children (70-120 at age 1-2 years, 80-130 in neonates).

Heart Rhythm

The rate and pattern of cardiac electrical activation and contraction, normally initiated by the sinoatrial node; abnormal rhythms include bradycardia, tachycardia, atrial fibrillation, and ventricular arrhythmias.

Heart Rhythm Disorders

Arrhythmias, in which the heart beats too fast, too

slowly, or irregularly because of abnormal electrical impulse formation or conduction.

Alternate Names:

Abnormal Heart Rhythms (Heart Rhythm Disorders)

Heart Septum

The muscular or membranous wall separating the right and left sides of the heart, including the atrial and ventricular septa; defects can permit abnormal blood flow between chambers.

Heart Sounds

Audible sounds produced by the heart during the cardiac cycle, normally heard as two sounds (lub-dub). S1 (lub) is caused by closure of the mitral and tricuspid valves at the beginning of systole. S2 (dub) is caused by closure of the aortic and pulmonary valves at the end of systole. Additional sounds (S3, S4) may indicate pathology.

Heart Surgery

Surgical procedures performed on the heart or great vessels to treat structural or functional cardiac conditions. Common operations include coronary artery bypass grafting (CABG), valve repair or replacement, and repair of congenital heart defects.

Heart Tamponade

Life-threatening compression of the heart by fluid, blood, or gas in the pericardial space, restricting filling and causing hypotension, raised venous pressure, tachycardia, and shock.

Heart Test, EPS

An electrophysiology study of the heart, using catheters to record and stimulate cardiac electrical activity and evaluate arrhythmias, conduction disorders, or unexplained fainting.

Heart Transplant

Replacement of a failing heart with a donor heart, followed by lifelong immunosuppression and surveillance for rejection, infection, coronary allograft disease, kidney injury, and medication complications.

Heart Transplantation

Surgical replacement of a failing heart with a donor heart, usually for selected patients with advanced heart failure who meet specialist criteria.

Heart Valve

A flap-like structure that directs one-way blood flow through the heart. The four valves are the tricuspid, pulmonary, mitral, and aortic valves.

Heart Valve Disease

Structural or functional abnormality of one or more cardiac valves causing stenosis, regurgitation, or both. Common causes include degeneration, congenital malformation, rheumatic disease, and infective endocarditis.

Heart Valve Disorder

Abnormal narrowing, leakage, or structural deformation of a cardiac valve that can impair forward blood flow or enlarge the heart and cause murmur, breathlessness, fatigue, or edema.

Heart Valve Disorders

Disorders affecting any of the four heart valves, including stenosis (narrowing) and regurgitation (leaking), which impair blood flow through the heart. Causes include congenital malformations, age-related degeneration, rheumatic fever, and endocarditis.

Heart Valve Prosthesis

Replacement heart valve. Types: mechanical (titanium, carbon, durable, requires lifelong anticoagulation), bioprosthetic (cow/pig tissue, no anticoagulation, degenerates). Mechanical: more durable, higher thrombotic risk. Bioprosthetic: better quality of life, reoperation risk.

Heart Valve Surgery

Repair or replacement of a diseased cardiac valve to correct severe narrowing or leakage and improve blood flow; choice of repair, mechanical valve, or tissue valve depends on anatomy and patient factors.

Heart Ventricle

One of the two lower heart chambers that pumps blood to the lungs or systemic circulation; the right ventricle pumps to the pulmonary artery and the left to the aorta.

Heart Ventricles

The two lower chambers of the heart. The right ventricle pumps blood to the lungs, and the left ventricle pumps blood to the body.

Heart-Healthy Eating

A dietary approach emphasizing foods that support cardiovascular function and reduce the risk of heart disease. Key principles include limiting saturated fats, trans fats, sodium, and added sugars while prioritizing fruits, vegetables, whole grains, lean proteins, and healthy fats like those from olive oil and fatty fish.

Heart-Lung Machine

A heart-lung machine temporarily circulates and oxygenates blood during some heart or major blood-vessel operations while the heart is stopped.

Heart-Lung Machine (Cardiopulmonary Bypass Machine)

A device that temporarily takes over the pumping and gas-exchange functions of the heart and lungs during selected cardiac operations.

Alternate Names:

Cardiopulmonary Bypass Machine

Heartburn

Heartburn is a burning retrosternal sensation caused by reflux of gastric contents; frequent or complicated symptoms may indicate gastroesophageal reflux disease. the throat and is often associated with regurgitation of sour or bitter-tasting fluid into the mouth. Despite its name, heartburn is a symptom of gastro-oesophageal reflux disease (GORD), not a cardiac symptom.

Heat

Thermal energy that raises tissue or body tempera-

ture; excessive heat exposure can cause heat cramps, heat exhaustion, heatstroke, or thermal injury.

Heat Cramps

Painful, involuntary muscle spasms occurring during or after strenuous activity in heat, often related to fluid and electrolyte loss. Resting in a cool place and replacing fluids and electrolytes may help. Confusion, fainting, or ongoing symptoms may signal heat illness requiring urgent medical attention.

Alternate Names:

Cramps Heat (Heat Cramps)

Heat Emergencies

Heat emergencies range from heat cramps and exhaustion to heat stroke, in which dangerous core-temperature elevation causes central nervous system dysfunction and organ injury.

Heat Exhaustion

A heat-related illness caused by excessive loss of water and salt, producing weakness, dizziness, nausea, headache, sweating, and sometimes fainting without severe central nervous system dysfunction.

Alternate Names:

Exhaustion Heat (Heat Exhaustion)

Heat Intolerance

An abnormal inability to tolerate warm environments or increased body heat, often causing excessive sweating, flushing, fatigue, or dizziness. Common causes include hyperthyroidism, menopause, fever, medications, autonomic disorders, and dehydration.

Heat Pack

A device that applies warmth to the body surface to relieve some muscle pain or stiffness. Excessive heat or prolonged contact can cause burns.

Heat Prostration

An older term for severe heat exhaustion, marked by weakness, fatigue, dizziness, headache, nausea, and sweating after heat exposure; confusion or loss of consciousness suggests heat stroke.

Heat Rash

An itchy or prickly rash caused by blocked sweat ducts during hot, humid conditions. Small red bumps or clear vesicles often appear in skin folds or areas covered by clothing. Cooling, keeping the skin dry, and avoiding occlusive products usually help; infection or persistent symptoms warrants assessment.

Alternate Names:

Profunda Miliaria (Heat Rash) Rash Heat (Heat Rash) Ruba Miliaria (Heat Rash)

Heat Rash (Pictures)

Miliaria, an itchy or prickly rash caused by blocked sweat ducts and trapped sweat, usually in hot humid conditions.

Heat Shock Protein

A stress-response protein that helps other proteins fold, avoid aggregation, and recover after heat, oxidative, infectious, or other cellular stress; some also influence immunity and cancer biology.

Heat Stroke

A life-threatening heat-related illness with markedly elevated body temperature and altered mental status. It requires immediate cooling and emergency medical treatment.

Alternate Names:

Stroke Heat (Heat Stroke)

Heatstroke

A life-threatening rise in body temperature caused by failure of the body's heat regulation.

Heavy Menstrual Bleeding

Menstrual bleeding that is unusually heavy or prolonged and interferes with daily activities.

Alternate Names:

Excessive Menstrual Bleeding

Heavy Metal Poisoning

Lead, mercury, arsenic, cadmium, thallium. Chelation therapy, supportive care, prevention. Sources: paint, water pipes, fish, occupational exposure.

Hebefrenia

An outdated term for a disorganized form of schizophrenia, now generally classified under schizophrenia-spectrum disorders with prominent disorganization.

Heberden Node

A firm bony enlargement of the distal interphalangeal joint, usually caused by hand osteoarthritis. It may be tender during progression and can produce deformity, reduced fine-motor function, or restricted finger movement.

Heberden Nodes

Bony enlargements at the end joints of fingers, commonly associated with osteoarthritis.

Alternate Names:

Heberden's nodes

Heberden'S Node

A hard, bony enlargement of the distal interphalangeal joint caused by osteoarthritis, usually affecting the fingers and sometimes producing pain, stiffness, or reduced dexterity.

Heberden'S Nodes

Firm bony enlargements of the distal interphalangeal finger joints, usually caused by osteoarthritis.

Hecht'S Pneumonia

A rare, often fatal giant-cell pneumonia associated with measles, especially in immunocompromised children, characterized by multinucleated giant cells in the lung alveoli.

Hectocotylyzation

A reproductive process in some cephalopods in which a modified arm transfers sperm to the female.

Hedgehog Proteins

A family of secreted developmental signaling proteins, including Sonic hedgehog, that regulate embryonic patterning, tissue differentiation, and se-

lected stem-cell and cancer pathways.

Heel

The heel is the back and underside of the foot, formed mainly by the calcaneus and a cushioning fat pad.

Heel Pain

Pain beneath or around the heel, commonly caused by plantar fasciitis, Achilles tendinopathy, bursitis, stress fracture, nerve compression, or inflammatory arthritis. Persistent pain, swelling, inability to bear weight, or trauma requires assessment.

Alternate Names:

Plantar Heel Pain

Heel Spur

A bony growth arising from the heel bone, often associated with plantar-fascia strain. The spur itself may cause no symptoms; heel pain is more often related to surrounding soft-tissue inflammation.

Hegar'S Dilators

Hegar's dilators are graduated metal instruments used to gently widen the cervix or other body openings during selected procedures.

Hegar'S Sign

Hegar's sign is softening and compressibility of the lower uterus, traditionally described as an early sign of pregnancy.

Height

The vertical distance from the base of a person to the top of the head, measured for growth assessment, nutritional evaluation, drug dosing, and calculation of body-mass index.

Heimlich Maneuver On Self

A self-administered abdominal-thrust technique sometimes used for severe choking when a person cannot breathe, speak, or cough effectively; emergency services should be activated immediately.

Heimlich Manoeuvre

An emergency procedure for relieving upper airway obstruction caused by a foreign body (choking). The maneuver is performed on a conscious, choking adult who cannot speak, cough effectively, or breathe.

Heine-Medin Disease

A clinical disorder related to heine-medin disease; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Heinz Bodies

Clumps of denatured hemoglobin inside red blood cells caused by oxidative injury. They may occur in glucose-6-phosphate dehydrogenase deficiency after exposure to oxidizing substances and are identified with special stains.

Hejama

An Arabic term for cupping, a traditional practice in which suction is applied to the skin; wet cupping also involves superficial skin incisions and carries bleeding and infection risks.

Hela Cell

A continuously dividing human cell line derived from cervical cancer cells of Henrietta Lacks and widely used in biomedical research, virology, and drug testing.

Helcococcus

A genus of gram-positive cocci that can rarely cause opportunistic skin, wound, bone, or bloodstream infections.

Helcococcus Kunzii

A gram-positive coccus usually found on skin that can rarely cause opportunistic infection.

Helcococcus Pyogenes

A rarely encountered gram-positive coccus that may be associated with opportunistic human infection.

Helcococcus Sueciensis

A rarely encountered gram-positive coccus isolated

from human clinical specimens and occasionally associated with infection.

Helicase

An enzyme that uses energy from nucleoside triphosphates to separate paired nucleic-acid strands during DNA replication, repair, recombination, or RNA processing.

Helicobacter

A genus of spiral-shaped bacteria that colonize the stomach or intestine; some species cause ulcers, gastritis, or other disease.

Helicobacter Cinaedi

A *Helicobacter* species that can cause bloodstream, skin, bone, joint, or other invasive infections, especially in people with weakened immunity.

Helicobacter Felis

A gastric *Helicobacter* species found mainly in cats and dogs that can rarely infect humans.

Helicobacter Fennelliae

An intestinal *Helicobacter* species that can rarely cause bacteremia or other opportunistic infection.

Helicobacter Mustelae

A gastric *Helicobacter* species associated mainly with ferrets and studied as a cause of gastritis in animal models.

Helicobacter Pylori

A gram-negative, spiral-shaped bacterium adapted to colonize the stomach. It can cause chronic gastritis, peptic ulcer disease, and increase the risk of gastric cancer and gastric MALT lymphoma.

Helicobacter Pylori Infection

Infection of the stomach by *H. pylori*, a spiral-shaped bacterium associated with chronic gastritis, peptic ulcer disease, gastric adenocarcinoma, and gastric MALT lymphoma. It is diagnosed with noninvasive or endoscopic testing and treated with combination eradication therapy.

Helicobacter Rappini

A name used for a group of gastric *Helicobacter* organisms found mainly in animals and rarely reported in human specimens.

Helicobacteraceae

A family of spiral-shaped bacteria that includes *Helicobacter* and related genera, many of which inhabit the digestive tract.

Heliotherapy

Heliotherapy is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Helium

A chemically inert gas used in respiratory mixtures, cryogenics, imaging, and laboratory applications; inhalation can displace oxygen and cause hypoxia despite its lack of direct toxicity.

Helix

The helix is the curved outer rim of the external ear.

Heller Myotomy

A surgical procedure that cuts the muscles of the lower esophageal sphincter to relieve obstruction in achalasia, usually performed laparoscopically and often combined with an antireflux procedure.

Heller'S Operation

A surgical procedure that cuts the lower esophageal muscle to relieve achalasia, usually with an antireflux procedure.

HELLP Syndrome

A serious pregnancy-associated disorder characterized by hemolysis, elevated liver enzymes, and low platelet count. It may occur with or without classic hypertension or proteinuria and requires urgent assessment.

Helminth

A parasitic worm, including roundworms, tapeworms, and flukes, that may live in or on a host.

Helminthiasis

Infection or infestation caused by parasitic worms such as roundworms, tapeworms, or flukes.

Helminths

Parasitic worms that cause disease in humans. The major groups are nematodes (roundworms), cestodes (tapeworms), and trematodes (flukes). Helminths have complex life cycles often involving intermediate hosts. Infection occurs through ingestion of eggs or larvae, penetration of skin, or insect vectors.

Help Prevent Hospital Errors

Actions that reduce hospital errors include confirming identity, checking medicines, communicating clearly, asking questions, and reporting changes or concerns promptly.

Helper T Cell

A type of CD4-positive T lymphocyte that coordinates immune responses by releasing cytokines and activating other immune cells.

Helper T Cells

CD4-positive immune cells that coordinate responses by activating other immune cells.

Alternate Names:

CD4 T cells

Helper Virus

A virus that supplies functions needed for another virus or defective viral genome to replicate.

HELPER Mnemonic

A mnemonic for the sequential management of shoulder dystocia: H — Help (call for additional personnel, anesthesia, and pediatrics). E — Evaluate for episiotomy (generous episiotomy to provide more room). L — Legs (McRoberts maneuver: hyperflex maternal legs onto the abdomen).

Helping A Loved One With A Drinking Problem

Support for someone with alcohol use disorder includes calm nonjudgmental communication, boundaries, encouragement toward medical treatment, attention to withdrawal risk, and emergency action for overdose or dangerous withdrawal.

Helping Your Teen With Depression

Support for an adolescent with depression includes listening without judgment, arranging professional assessment, supporting treatment and routines, reducing access to lethal means, and acting urgently for self-harm risk.

Hemagglutination Test

A test in which antibodies or other agents cause red blood cells to clump; the reaction can help detect antibodies, antigens, or selected infectious agents.

Hemagglutinin

A surface glycoprotein, especially on influenza viruses, that binds sialic-acid receptors and helps the virus attach to and enter host cells; it is a major target of protective antibodies.

Hemangioblastoma

A benign, well-circumscribed vascular tumor of the central nervous system, most commonly the cerebellum, often cystic with a mural nodule. Associated with von Hippel-Lindau disease; treated by surgical resection.

Hemangioendothelioma

A vascular neoplasm arising from endothelial cells with malignant potential between benign hemangioma and high-grade angiosarcoma; behavior depends on subtype and site.

Hemangioma

A benign tumor of blood vessel endothelial cells, common in infants. Most involute spontaneously; treatment includes laser therapy or surgery for problematic lesions.

Alternate Names:

Infantile Hemangioma Strawberry Hemangioma

Hemangiopericytoma

A rare tumor formerly defined by branching “staghorn” vessels and perivascular cells; many lesions classified this way are now recognized as solitary fibrous tumors.

Hemangiosarcoma

A malignant tumor of vascular endothelial cells, an aggressive sarcoma that may arise in the skin, liver, or bone.

Hemarthrosis

The accumulation of blood within a synovial joint cavity, most commonly involving the knee. Major etiologies include trauma (acute anterior cruciate ligament [ACL] rupture, patellar dislocation, intra-articular osteochondral fractures) and coagulopathies, classically representing the hallmark manifestation of severe congenital hemophilia A (factor VIII deficiency) or hemophilia B (factor IX deficiency). Presents with rapid, tense joint swelling, severe pain, warmth, and restricted range of motion developing within hours of injury. Evaluated via diagnostic arthrocentesis yielding gross bloody synovial fluid; in chronic hemophilia, recurrent hemarthroses induce synovial hemosiderin deposition, chronic inflammatory synovitis, and progressive cartilage destruction known as hemophilic arthropathy.

Hematemesis

Vomiting of blood, indicating upper gastrointestinal bleeding proximal to the ligament of Treitz. Blood may be bright red (acute, brisk bleeding) or resemble coffee grounds (altered blood that has been in contact with gastric acid).

Hematocele

An abnormal accumulation of blood within the potential space between the visceral and parietal layers of the tunica vaginalis surrounding the testis. Typically caused by blunt or penetrating scrotal trauma, testicular rupture, recent scrotal surgery, testicular biopsy, or underlying testicular neoplasm. Clinically

presents as an acute or enlarging, heavy, firm scrotal mass that does not transilluminate on light examination. Prompt scrotal ultrasonography with color Doppler is essential to assess testicular parenchymal integrity and blood flow; large or traumatic hematoceles often warrant surgical exploration, evacuation of clot, and testicular repair to prevent testicular necrosis, infection, or chronic fibrosis.

Alternate Names:

Scrotal Hematocele; Hematocele of the Tunica Vaginalis

Hematochezia

The passage of bright red or maroon blood per rectum, usually indicating active lower gastrointestinal (LGI) bleeding distal to the ligament of Treitz (e.g., diverticulosis, arteriovenous malformations/angiodysplasia, colorectal polyps/cancer, inflammatory bowel disease, hemorrhoids).

Hematocolpos

Accumulation of menstrual blood in the vagina, usually because an imperforate hymen or another obstructing anomaly prevents outflow; it may cause cyclic pelvic pain and a bulging vaginal membrane.

Hematocrit

The proportion of blood volume occupied by packed red blood cells, expressed as a percentage. One of the most commonly ordered laboratory tests in clinical medicine.

Hematogenous

Spread through the bloodstream, as when infection, tumor cells, or embolic material travels from one body site to another.

Hematologic Agents

Medicines or biologic therapies that act on blood cells, coagulation, marrow, or immune pathways, including agents used for anemia, thrombosis, leukemia, and lymphoma.

Hematologic Changes In Pregnancy

Plasma volume increases 40-50% while red cell mass increases 20-30%, resulting in physiologic anemia

(Hb 10-11 g/dL). Leukocytosis (WBC up to 15,000-16,000/ μ L, predominantly neutrophils) is normal.

Hematologic Malignancy

Alternate index label; see *Blood Cancer* and relevant leukemia, lymphoma, or myeloma entries.

Hematologic Response

A measurable improvement or change in blood counts or marrow findings after treatment, such as recovery of neutrophils, platelets, or hemoglobin or reduction of malignant cells.

Hematological Disease

A disorder of blood cells, bone marrow, lymphatic tissue, or coagulation, including anemia, leukemia, lymphoma, bleeding disorders, and clotting disorders.

Hematologist

A physician who diagnoses and treats disorders of blood, bone marrow, lymph nodes, clotting, and hematologic cancers.

Hematology

The branch of medicine concerned with the study, diagnosis, treatment, and prevention of disorders of the blood and blood-forming organs (bone marrow, lymph nodes, spleen).

Hematology-Oncology

The medical specialty concerned with blood disorders and cancers, including anemia, leukemia, lymphoma, myeloma, and clotting disorders.

Hematoma

A localized collection of blood outside blood vessels, usually caused by injury or a bleeding disorder.

Hematometra

A pathological distension and accumulation of blood within the uterine cavity resulting from structural obstruction of the lower outflow tract, such as congenital cervical agenesis, transverse vaginal septum, imperforate hymen, or acquired cervical stenosis fol-

lowing surgical conization, loop excision, or pelvic radiation. Clinically manifests with severe amenorrhea or cryptomenorrhea, cyclic lower midline abdominal/pelvic pain, and an enlarged, tender, globular uterus on bimanual examination.

Hematometrocolpos

Accumulation of blood in the uterine cavity and vagina due to outflow obstruction such as an imperforate hymen or transverse vaginal septum, presenting with cyclic pelvic pain and primary amenorrhea in adolescents.

Hematomyelia

Hemorrhage occurring directly within the parenchyma of the spinal cord, most often localized centrally within the gray matter. Primarily caused by acute mechanical spinal cord trauma (such as fracture-dislocations or severe hyperflexion/hyperextension injuries), followed by rupture of spinal vascular malformations (arteriovenous malformations [AVMs], cavernous angiomas), bleeding diatheses, or therapeutic anticoagulation. Manifests acutely with sudden, severe back or neck pain followed by rapid-onset flaccid paraparesis or quadriparesis, sensory loss below the lesion level, loss of deep tendon reflexes (spinal shock), and bowel/bladder dysfunction; urgently diagnosed via emergent spinal MRI showing intramedullary hematoma with surrounding edema.

Hematophobia

An excessive or persistent fear of blood, blood-related sights, or medical procedures involving blood that causes distress or avoidance; some people have a vasovagal fainting response.

Hematopoiesis

The production and maturation of blood cells from multipotent hematopoietic stem cells, principally in the bone marrow after birth. It generates erythrocytes, leukocytes, and platelets through regulated progenitor-cell differentiation.

Hematopoietic Growth Factor

A cytokine or medicine that stimulates production,

maturation, or survival of blood-forming cells in bone marrow, such as erythropoietin or granulocyte colony-stimulating factor.

Hematopoietic Neoplasm

A neoplastic process affecting hematopoietic neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Hematopoietic Stem Cell Transplantation

Infusion of hematopoietic stem cells after a conditioning regimen to restore marrow function or replace diseased hematopoiesis. Autologous HSCT uses the patient's own cells; allogeneic HSCT uses cells from a donor and carries risks including graft-versus-host disease and graft failure.

Hematopoietic System

The bone marrow and related organs and cells that produce, mature, circulate, and remove blood cells, including red cells, white cells, and platelets.

Hematoporphyrin Derivative

A porphyrin-based photosensitizer that accumulates in some tumors and can be activated by light in photodynamic treatment; it may also cause prolonged photosensitivity.

Hematosalpinx

Accumulation of blood within a fallopian tube, often caused by ectopic pregnancy, endometriosis, pelvic surgery, or obstruction, and potentially producing pelvic pain or infertility.

Hematosis

The exchange of oxygen and carbon dioxide between alveolar air and pulmonary capillary blood, resulting in oxygenation of the blood.

Hemospermia

Blood in the semen, often temporary and associated with infection, inflammation, trauma, or a recent urologic procedure. Persistent or recurrent bleeding, pain, fever, urinary symptoms, or bleeding in an older adult requires medical evaluation.

Hematoxylin

A dye used in histology, usually with eosin, to stain cell nuclei blue-purple and help show tissue structure.

Hematuria

Presence of red blood cells in the urine, categorized as microscopic (>3 RBCs per high-power field on urine microscopy) or gross (macroscopic, visible color change from pink to dark red). Microscopic hematuria is often asymptomatic and detected on routine urinalysis.

Heme

Heme is an iron-containing porphyrin group that binds oxygen in haemoglobin and myoglobin and forms the functional centre of several enzymes; abnormal heme metabolism contributes to porphyrias and jaundice.

Hemeralopia

Hemeralopia is reduced vision in bright light, sometimes associated with retinal disorders or abnormal cone function.

Hemerythrin

An oxygen-binding protein found in some marine invertebrates; it is not the principal oxygen-carrying protein in humans but is relevant to comparative physiology.

Hemianopia

Loss of vision in half the visual field of one or both eyes. Homonymous hemianopia affects the same side in both eyes and localizes to a contralateral lesion posterior to the optic chiasm, most often stroke or tumor.

Hemianopsia

Loss of vision in half of the visual field of one or both eyes, usually from a lesion of the optic tract, radiations, or occipital cortex; sudden onset may indicate stroke.

Hemiballism

A movement disorder causing sudden, large-amplitude, flinging movements on one side of the body, usually from a lesion involving the subthalamic region.

Hemiballismus

Hemiballismus is involuntary, rapid, flinging movement of one side of the body, usually caused by a lesion of the opposite subthalamic region.

Hemicolectomy

Surgical removal of the right or left portion of the colon, sometimes with nearby lymph nodes and reconnection of the bowel.

Hemidesmosome

A specialized junction that anchors basal epithelial cells to the underlying basement membrane by linking intracellular keratin filaments with extracellular matrix proteins.

Hemidiaphragm

One of the two muscular halves of the diaphragm separating the chest from the abdomen and assisting breathing.

Hemifacial Atrophy

Progressive wasting of skin, subcutaneous tissue, muscle, and sometimes bone on one side of the face, also called Parry–Romberg syndrome.

Hemifacial Microsomia

A congenital craniofacial disorder involving asymmetric hypoplasia of the structures derived from the first and second branchial arches, most commonly affecting the mandible, ear, and soft tissues on one side. Also known as craniofacial microsomia or Goldenhar syndrome when epibulbar dermoids and vertebral anomalies are present.

Alternate Names:

craniofacial microsomia or.

Hemifacial Spasm

Involuntary, unilateral, intermittent twitching of the

facial muscles innervated by the facial nerve (CN VII). Typically begins around the eye (orbicularis oculi) and spreads to the lower face.

Hemihypertrophy

Unequal overgrowth of one side or region of the body, producing asymmetry of limbs, trunk, or face; some childhood cases carry increased risk of embryonal tumors.

Hemilaryngectomy

Hemilaryngectomy is surgery to remove part of the larynx, usually to treat a selected laryngeal cancer while preserving as much speech and swallowing function as possible.

Hemimandibulectomy

Hemimandibulectomy is surgery to remove one side of the lower jaw, usually for a tumor or severe bone disease, sometimes followed by reconstruction.

Hemimegalencephaly

A rare developmental brain malformation in which one cerebral hemisphere is abnormally enlarged and dysplastic, often causing infantile seizures, developmental impairment, and contralateral neurologic signs.

Hemimelia

A congenital limb-reduction defect in which part or all of a limb segment is absent, such as fibular or radial hemimelia.

Hemiparesis

Weakness affecting one side of the body, less severe than hemiplegia (paralysis). It results from a lesion of the corticospinal tract anywhere from the motor cortex to the spinal cord, most commonly from stroke, but also from tumors, trauma, or demyelination. The pattern of weakness helps localize the lesion.

Hemipelvectomy

Hemipelvectomy is major surgery that removes part or all of one side of the pelvis, with or without

removal of the leg, usually for extensive cancer or severe trauma.

Hemiplegia

Paralysis of one side of the body (the face, arm, and leg on the same side), caused by damage to the corticospinal tract (pyramidal tract) at any level from the cerebral cortex to the spinal cord.

Hemispatial Neglect

Hemispatial neglect is reduced awareness of one side of the body or space, usually after injury to the opposite cerebral hemisphere.

Hemispherectomy

Hemispherectomy is surgery that removes or disconnects much of one cerebral hemisphere to treat severe, otherwise uncontrolled epilepsy in selected patients.

Hemisternebra

A congenital or developmental partial segment of the sternum, caused by incomplete formation or fusion of sternal elements and sometimes associated with chest-wall anomalies.

Hemithorax

One side of the thorax, including the lung, pleural space, ribs, and other structures on that side of the chest.

Hemivertebrae

Congenital vertebrae formed from only part of the normal vertebral body, producing wedge-shaped bone and potentially causing congenital scoliosis or spinal-cord abnormalities.

Hemizygous

Having only one copy of a gene or chromosome region rather than the usual pair, as occurs for many X-linked genes in people with one X chromosome.

Hemochromatosis

Hemochromatosis is excessive iron accumulation that can injure the liver, pancreas, heart, joints,

skin, and endocrine organs; inherited and acquired forms exist. progressive iron deposition in parenchymal organs. Hereditary hemochromatosis is most commonly caused by homozygous C282Y mutations in the HFE gene affecting approximately one in two hundred individuals of Northern European descent.

Hemochromatosis Cardiomyopathy

Iron overload cardiomyopathy from hereditary (HFE gene mutations, most common) or secondary causes. Iron deposition in the myocardium causes restrictive then dilated cardiomyopathy. Diagnosis: elevated transferrin saturation (>45%), ferritin, and genetic testing. Cardiac MRI T2* <20 ms indicates myocardial iron overload.

Hemocyanin

A copper-containing oxygen-transport protein in many mollusks and arthropods; unlike human hemoglobin, it circulates dissolved in their hemolymph.

Hemocytometer

A ruled counting chamber used with a microscope to determine the concentration of cells or other particles in a measured volume of fluid.

Hemodialysis

Extracorporeal blood purification in which blood passes through a dialyzer containing a semipermeable membrane to remove selected solutes and excess fluid. Conventional outpatient treatment is often scheduled about three times weekly for several hours, but frequency and duration vary with the clinical situation and modality. Vascular access is required; an arteriovenous fistula is generally preferred when feasible. Adequacy is assessed with measures such as Kt/V or urea reduction ratio, together with clinical status. Complications can include hypotension, access infection, thrombosis, and dialyzer reactions.

Hemodialysis Access - Self Care

Daily care of an arteriovenous fistula, graft, or catheter preserves blood flow and prevents infection; check for a thrill, avoid compression, and report redness, drainage, swelling, or loss of flow.

Hemodialysis Access Procedures

Procedures that create, maintain, repair, or revise vascular access for hemodialysis, including fistula or graft creation, catheter placement, angioplasty, thrombectomy, and treatment of stenosis or infection.

Hemodialysis Clearance

The volume of plasma from which hemodialysis removes a solute per unit time; it depends on blood and dialysate flow, membrane properties, treatment time, and the substance's distribution.

Hemodialysis Solution

The dialysate fluid circulated across a semipermeable membrane during hemodialysis to remove waste and help correct electrolyte, acid-base, and fluid abnormalities.

Hemodynamics

Study of blood flow dynamics within the cardiovascular system; includes assessment of blood pressure, cardiac output, and vascular resistance. Its significance is established from its characteristic features, causes, and clinical context.

Hemofiltration

An extracorporeal process that removes plasma water and dissolved solutes by convection across a semipermeable membrane, with replacement fluid used to maintain volume. It is used in continuous and intermittent kidney-replacement modalities.

Hemoglobin

Hemoglobin is the iron-containing protein in red blood cells that carries oxygen and carbon dioxide; its concentration and structure help evaluate anemia and hemoglobin disorders. heme groups that bind oxygen. Each heme contains iron (Fe^{2+}) that reversibly binds O_2 . Cooperative binding produces sigmoidal oxygen dissociation curve. P_{50} (pO_2 at 50% saturation) 27 mmHg.

Hemoglobin A

Haemoglobin A is the predominant adult haemoglobin, composed of two alpha and

two beta chains ($\alpha_2\beta_2$), and transports oxygen in normal red blood cells.

Hemoglobin A1C Variability

Factors that can make hemoglobin A1c less reliable include altered red-cell survival, hemoglobin variants, anemia, pregnancy, kidney disease, transfusion, and some treatments. An alternative measure of glucose may be needed when A1c does not reflect recent glycemia.

Hemoglobin C

A beta-globin variant caused by a hemoglobin gene mutation; homozygosity can cause mild hemolytic anemia and splenomegaly, while inheritance with hemoglobin S causes a clinically significant sickling disorder.

Hemoglobin C Disease

An inherited hemoglobinopathy caused by a beta-globin variant that produces hemoglobin C. It may cause mild hemolytic anemia, jaundice, splenomegaly, or, when combined with hemoglobin S, clinically significant sickling disease.

Hemoglobin Electrophoresis

Separation of hemoglobin variants by electrophoresis or HPLC. Normal adult: HbA (95-98%), HbA2 (2-3%), HbF (<1%). Sick cell disease (HbSS): HbS only, no HbA. Sick cell trait (HbAS): HbA > HbS. Thalassemia: elevated HbA2 or HbF.

Hemoglobin H

A beta-chain tetramer formed in alpha-thalassemia, which is unstable and causes chronic hemolytic anemia of variable severity.

Hemoglobin Increased

An increased hemoglobin concentration, which may reflect dehydration, chronic hypoxemia, erythrocytosis, or a myeloproliferative neoplasm and must be interpreted with hematocrit and clinical context.

Hemoglobin M

A group of abnormal hemoglobins that stabilize

iron in the ferric state and can cause congenital methemoglobinemia and cyanosis.

Hemoglobin Normal Values

Reference ranges for blood hemoglobin concentration, which vary with age, sex, pregnancy, altitude, laboratory method, and other factors; results should be interpreted using the reporting laboratory's range.

Hemoglobin S

Haemoglobin S is a beta-globin variant caused by the HbS mutation; when deoxygenated it can polymerize, producing sickling, haemolysis, and vaso-occlusion in sickle-cell disease.

Hemoglobin SC Disease

A major symptomatic sickling hemoglobinopathy resulting from compound heterozygosity for the hemoglobin S ($\beta^6 \text{Glu} \rightarrow \text{Val}$) and hemoglobin C ($\beta^6 \text{Glu} \rightarrow \text{Lys}$) mutations. Patients produce roughly equal amounts of HbS and HbC without normal adult hemoglobin A. Although hemolytic anemia and painful vaso-occlusive crises are generally milder than in homozygous sickle cell anemia (HbSS), patients carry a markedly higher risk of proliferative retinopathy, avascular necrosis of the femoral head, and thromboembolic events.

Alternate Names:

HbSC; HbSC disease; Sickle-hemoglobin C disease

Hemoglobin Trait

The inherited presence of a hemoglobin variant in a heterozygous state, usually causing no symptoms but potentially affecting screening results or reproductive risk.

Hemoglobinopathies: Thalassemias

Thalassemias are inherited disorders of globin chain synthesis imbalance: alpha-thalassemia (HBA1/HBA2 deletions) and beta-thalassemia (HBB mutations).

Hemoglobinopathy

A group of inherited disorders of hemoglobin structure or synthesis. Major types include sickle-cell

disease and thalassemia; manifestations range from asymptomatic microcytosis to haemolytic anaemia, vaso-occlusion, and organ injury.

Hemoglobinuria

Haemoglobinuria is free haemoglobin in urine, usually from intravascular haemolysis; urine may be dipstick-positive for blood with few or no red cells, and causes include transfusion reactions, malaria, and paroxysmal nocturnal haemoglobinuria.

Hemoglobinuria Test

A laboratory test evaluating urine for hemoglobin or related findings, used to investigate intravascular hemolysis, hemoglobinuria, transfusion reactions, or other causes of dark urine.

Hemolymph

The circulating fluid of arthropods and many other invertebrates, serving functions analogous to blood and lymph; it is not the normal circulatory fluid of humans.

Hemolysin

A toxin or antibody that damages red-cell membranes and causes hemolysis.

Hemolysis

The premature destruction of red blood cells, releasing hemoglobin into the plasma. Extravascular hemolysis (most common): RBCs are destroyed by macrophages in the spleen and liver — causes unconjugated hyperbilirubinemia, increased LDH, and decreased haptoglobin.

Hemolytic

Relating to hemolysis, the premature destruction of red blood cells, which can release hemoglobin and produce anemia or jaundice.

Hemolytic Anemia

Anemia due to accelerated destruction of red blood cells (RBCs), classified as intrinsic (RBC defect) or extrinsic (external factors). Intrinsic: hereditary spherocytosis, G6PD deficiency, sickle cell disease,

thalassemia.

Hemolytic Anemia Caused By Chemicals And Toxins

Anemia caused by accelerated destruction of red blood cells after exposure to a chemical or toxin; evaluation identifies hemolysis and removes or treats the responsible exposure.

Hemolytic Anemias: Autoimmune

Autoimmune hemolytic anemia occurs when antibodies or complement cause premature destruction of red blood cells. Warm-antibody and cold-antibody forms differ in their triggers, laboratory findings, and sites of red-cell destruction.

Hemolytic Anemias: Hereditary

Hereditary spherocytosis: cytoskeletal protein defects (ankyrin, band 3, protein 4.2, alpha/beta spectrin) -> spherical RBCs, splenic trapping, hemolysis. AD (75%) or AR. Clinical: anemia, jaundice, splenomegaly, gallstones.

Hemolytic Crisis

A sudden increase in red-cell destruction causing worsening anemia, jaundice, dark urine, pain, or other manifestations of hemolysis.

Hemolytic Disease Of The Newborn

A condition in which maternal antibodies destroy fetal/neonatal red blood cells (RBCs), causing hemolytic anemia, hyperbilirubinaemia, and in severe cases hydrops fetalis.

Hemolytic Index

A laboratory-quality indicator estimating hemolysis in a blood specimen; substantial in-vitro hemolysis can interfere with measurement of several analytes.

Hemolytic Jaundice

Jaundice resulting from excessive hemolysis (breakdown of red blood cells) causing overproduction of unconjugated bilirubin that exceeds the liver's capacity for conjugation and excretion. See Jaundice for complete overview.

Hemolytic Jaundice, Congenital

Jaundice present from or caused by an inherited condition that increases red-blood-cell destruction, producing excess unconjugated bilirubin; examples include hereditary spherocytosis and some enzymatic defects.

Hemolytic Transfusion Reaction

An immune or nonimmune complication in which transfused red blood cells are destroyed, potentially causing fever, pain, hemoglobinuria, kidney injury, shock, or disseminated intravascular coagulation.

Hemolytic Uremic Syndrome

A thrombotic microangiopathy characterized by red-cell destruction, low platelets, and acute kidney injury. It may follow Shiga-toxin-producing gastrointestinal infection or result from complement dysregulation and other causes.

Hemolytic-Uremic Syndrome

A potentially life-threatening disorder characterized by microangiopathic hemolytic anemia, thrombocytopenia, and acute kidney injury. It often follows Shiga-toxin-producing bacterial gastroenteritis but may also result from complement dysregulation or other triggers.

Hemoperfusion

Blood passed through adsorbent column (charcoal, resin) to remove toxins. Used for drug overdoses with high protein binding, large volume of distribution.

Hemopericardium

Accumulation of blood in the sac surrounding the heart.

Hemoperitoneum

Blood collecting inside the abdominal cavity, usually from trauma, a ruptured blood vessel or organ, surgery, or a ruptured ectopic pregnancy. It can cause abdominal pain, distention, dizziness, or shock and may require emergency imaging and surgery or another procedure.

Hemopexin

A plasma glycoprotein that binds free heme and transports it to the liver, helping limit oxidative injury during hemolysis.

Hemophagocytic Lymphohistiocytosis

HLH: uncontrolled immune activation with cytokine storm (IFN-gamma, IL-6, IL-10, TNF). Familial HLH (FHL): PRF1 (FHL2), UNC13D (FHL3), STX11 (FHL4), STXBP2 (FHL5), RAB27A (FHL6/GrisCELLI type 2). Secondary HLH: infections (EBV, CMV), malignancy (lymphoma), autoimmune (SJIA/MAS, SLE), immunodeficiency.

Hemophagocytosis Of Sick Cells

Activation of macrophages that engulf erythrocytes, leukocytes, and platelets within the bone marrow, occurring as a complication of sickle cell disease during severe hemolytic crises or infection.

Hemophilia

Hemophilia is an inherited coagulation-factor deficiency, usually factor VIII or IX, causing prolonged bleeding into joints, muscles, and tissues after injury or spontaneously. VIII, IX, or another coagulation component. It causes prolonged bleeding, particularly into joints and muscles, after injury or procedures.

Alternate Names:

A Hemophilia (Hemophilia) B Hemophilia (Hemophilia)

Hemophilia A

An X-linked recessive deficiency of coagulation factor VIII, causing prolonged bleeding into joints, muscles, and soft tissues. Severity correlates with factor VIII activity levels.

Hemophilia B

An X-linked recessive deficiency of coagulation factor IX (Christmas disease), causing prolonged bleeding similar to hemophilia A. Severity correlates with factor IX activity levels.

Hemophilus

An older spelling or variant of Haemophilus, a group of small gram-negative bacteria that includes respiratory and genital pathogens.

Hemophilus Ducreyi

A gram-negative bacterium that causes chancroid, a sexually transmitted infection with painful genital ulcers and tender lymph nodes.

Hemophobia

An excessive or persistent fear of blood, also called hematophobia, that causes significant distress, avoidance, or sometimes a vasovagal faint.

Hemopoietic Stem Cell

A blood-forming stem cell found mainly in the bone marrow. It can make copies of itself and develop into red blood cells, white blood cells, or platelets.

Alternate Names:

Hematopoietic Stem Cell

Hemopoietic Tissue

Tissue that produces blood cells, chiefly bone marrow, containing stem cells and supporting cells that generate erythrocytes, leukocytes, and platelets.

Hemoprotein

A protein containing a heme group with an iron atom, such as hemoglobin, myoglobin, cytochromes, and many oxidoreductive enzymes.

Hemoptysis

Expectoration of blood from the lower respiratory tract. Must be differentiated from hematemesis and epistaxis. Causes: bronchitis, pneumonia, TB, bronchiectasis, lung cancer, PE.

Hemorrhage

Abnormal escape of blood from a damaged blood vessel or, less commonly, from the heart. It may be external or internal, acute or chronic, and range from minor bleeding to life-threatening blood loss causing anemia, shock, or organ compression; the cause and site determine management.

Hemorrhagic

Pertaining to or characterized by hemorrhage (extravasation of blood from vessels). Used in contexts such as hemorrhagic stroke, hemorrhagic shock, and hemorrhagic fever.

Hemorrhagic Anemia

A decrease in total circulating red blood cell mass, hemoglobin concentration, and oxygen-carrying capacity resulting directly from acute or chronic hemorrhage. Acute hemorrhagic anemia (from major trauma, ruptured aortic aneurysm, ectopic pregnancy, or massive gastrointestinal bleeding) presents with hypovolemia, postural hypotension, tachycardia, and potentially hemorrhagic shock; immediately following acute blood loss, hemoglobin and hematocrit appear deceptively normal until extravascular fluid shifts or IV resuscitation causes hemodilution. Chronic hemorrhagic anemia (from occult gastrointestinal bleeding, heavy menstrual bleeding, or angiodysplasia) presents indolently with fatigue, pallor, and exertional dyspnea, classically resulting in secondary iron deficiency anemia with microcytic, hypochromic red blood cell indices.

Hemorrhagic Disorder

A condition that causes abnormal bleeding because of platelet number or function, clotting-factor deficiency, excessive fibrinolysis, or blood-vessel fragility.

Hemorrhagic Shock

Circulatory failure caused by severe blood loss. It may produce rapid pulse, low blood pressure, cool skin, reduced urine output, confusion, or collapse. Treatment requires urgent control of bleeding, restoration of circulating volume, blood products when indicated, and correction of associated injuries or clotting abnormalities.

Hemorrhagic Stroke

A neurological deficit caused by rupture of a cerebral blood vessel, accounting for 13% of all strokes but 40% of stroke mortality. Intracerebral hemorrhage: hypertensive (basal ganglia, thalamus, pons, cerebellum), amyloid angiopathy (lobar, elderly),

vascular malformations, anticoagulation. Subarachnoid hemorrhage: aneurysm rupture.

Hemorrhoid

Pathologically engorged, dilated vascular cushions composed of submucosal arteriovenous plexuses and connective tissue located within the lower rectum and anal canal. Categorized anatomically by their position relative to the dentate (pectinate) line: internal hemorrhoids originate above the dentate line (covered by insensitive columnar/transitional epithelium, innervated by visceral nerves, typically presenting with painless bright red rectal bleeding during defecation or prolapse, graded from I to IV based on degree of protrusion), whereas external hemorrhoids originate below the dentate line (covered by sensitive squamous epithelium, innervated by somatic pudendal nerves, causing severe, acute localized anal pain upon thrombosis). Managed conservatively with dietary fiber, hydration, sitz baths, rubber band ligation, or surgical hemorrhoidectomy.

Hemorrhoid Surgery

An operation to remove or reduce symptomatic internal or external hemorrhoids when bleeding, prolapse, thrombosis, or pain persists despite appropriate conservative treatment.

Hemorrhoidal Hemorrhage

Rectal bleeding from enlarged or injured hemorrhoidal vessels, usually bright red and associated with defecation; heavy, persistent, painful, or unexplained bleeding requires evaluation for other causes.

Hemorrhoidectomy

Surgical excision of symptomatic hemorrhoids. Indications: Grade III-IV internal hemorrhoids (prolapsed, irreducible), external hemorrhoids with uncontrolled bleeding or recurrent thrombosis, mixed hemorrhoids.

Hemorrhoids

Enlarged vascular cushions of the anal canal that may cause painless rectal bleeding, prolapse, itching, or discomfort; they may be internal or external.

Hemorrhoids (Piles)

Vascular cushions in the anal canal composed of arteriovenous anastomoses (normal structures present from birth), which become symptomatic when they bleed, prolapse, or thrombose.

Alternate Names:

Piles Piles (Hemorrhoids (Piles))

Hemosiderin

A golden-brown, granular, iron-containing pigment derived from the degradation of hemoglobin and stored within macrophages as an insoluble complex. It forms when red blood cells are phagocytosed and degraded, typically in conditions of hemorrhage, hemolysis, or excessive red blood cell destruction.

Hemosiderosis

Excessive deposition of the iron-storage pigment hemosiderin in tissues.

Hemosiderotic Synovitis

Synovial inflammation with hemosiderin deposition caused by recurrent bleeding into a joint, commonly associated with hemophilia, anticoagulation, trauma, or a vascular synovial lesion.

Hemospermia

The presence of blood in semen, usually caused by inflammation or infection of the genitourinary tract and often self-limited, but persistent or recurrent cases require evaluation.

Hemostasis

The coordinated physiological process that stops bleeding after vascular injury through vessel constriction, platelet adhesion and aggregation, coagulation-factor activation, fibrin formation, clot stabilization, and regulated fibrinolysis. Failure can cause bleeding or thrombosis.

Hemostasis And Coagulation

Hemostasis is the coordinated response that limits bleeding through vessel constriction, platelet adhesion and aggregation, coagulation-factor activation, fibrin formation, clot retraction, and later fibrinoly-

sis; imbalance can cause bleeding or thrombosis.

Hemostasis Valve

A valve or sealing device used with an intravascular catheter or sheath to limit blood loss while allowing passage of wires or instruments.

Hemostat

A clamp used to compress a blood vessel or tissue and control bleeding.

Hemostat (Perineal)

A clamp used during procedures to compress a blood vessel or hold tissue and control bleeding.

Alternate Names:

Perineal

Hemostat (Surgical)

A clamping surgical instrument used to control bleeding by compressing a blood vessel or tissue.

Alternate Names:

Surgical

Hemostat (Vascular)

A surgical clamp used to temporarily occlude a blood vessel and control bleeding.

Alternate Names:

Vascular

Hemostatic Agent

A haemostatic agent promotes control of bleeding by supporting clot formation, providing a physical barrier, or inhibiting fibrinolysis; examples include topical agents, tranexamic acid, and factor replacement.

Hemothorax

Accumulation of blood in the pleural space, most commonly from trauma. Presents with decreased breath sounds, dullness to percussion. Diagnosis: chest X-ray, ultrasound, CT. Treatment: tube thoracostomy, blood transfusion, surgery if ongoing bleeding.

Hemovac Drain

A closed-suction surgical drain with a compressible reservoir used to remove blood or other fluid from a wound or operative site.

Hempa

A historical or context-dependent biomedical abbreviation or term; its meaning must be confirmed from the source because it does not identify one universally accepted diagnosis or treatment.

Henipavirus

A genus of paramyxoviruses that includes Nipah and Hendra viruses, which can cause severe encephalitis and respiratory disease.

Henoch-Schonlein Purpura

IgA vasculitis: palpable purpura on buttocks/legs, abdominal pain, arthralgia, renal involvement (IgA nephropathy), most common in children. Self-limited in most cases; corticosteroids for severe GI/renal involvement.

Alternate Names:

Spring Fever (Henoch-Schonlein Purpura)

Henoch-SchöNlein Purpura

An older name for IgA vasculitis, an inflammation of small blood vessels that can cause palpable purpura, usually on the lower limbs and buttocks, with joint pain, abdominal symptoms, and kidney involvement.

HEPA Filter

A high-efficiency particulate air filter that removes a large proportion of airborne particles.

Alternate Names:

High-efficiency particulate air filter

Hepadnaviridae

A family of small DNA viruses that includes hepatitis B virus and related viruses infecting mammals and birds.

Hepar

Latin for liver; it is used in medical terminology and

in names of substances or preparations relating to the liver.

Heparan Sulfate

A sulfated glycosaminoglycan in cell surfaces and extracellular matrix that binds growth factors, regulates signaling, and contributes to filtration-barrier and connective-tissue structure.

Heparin

An anticoagulant medicine that reduces blood clotting by enhancing antithrombin activity. It may be given by injection to prevent or treat blood clots.

Heparin Binding

A medicine or chemical compound called heparin binding. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Heparin Lyase

A medicine or chemical compound called heparin lyase. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Heparin-Induced Thrombocytopenia

An immune-mediated adverse drug reaction caused by IgG antibodies against platelet factor 4 (PF4)-heparin complexes, leading to platelet activation, thrombocytopenia, and a prothrombotic state. HIT typically occurs 5-14 days after heparin exposure (unfractionated heparin > low molecular weight heparin).

Hepatectomy

Surgical removal of part or all of the liver for primary or metastatic tumors, requiring an adequate future liver remnant; portal vein embolization can hypertrophy the remnant before resection.

Hepatic

Pertaining to the liver; hepatic function includes metabolism, detoxification, bile production, and protein synthesis. Its significance is established from its characteristic features, causes, and clinical

context.

Hepatic Artery

The artery that supplies oxygenated blood to the liver, arising from the celiac circulation and dividing into right and left branches within the porta hepatis.

Hepatic Biopsy

Alternate index label; see *Liver biopsy*. It may evaluate inflammation, fibrosis, fatty liver disease, infection, infiltrative disease, or a mass when noninvasive tests are insufficient.

Hepatic Coma

A clinical disorder related to hepatic coma; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hepatic Cyst

Fluid-filled cavity in the liver most commonly congenital and asymptomatic. Simple hepatic cysts are lined by biliary epithelium and contain clear fluid. May cause pain or compressive symptoms if large or complicated by hemorrhage, infection, or rupture. Simple cysts require no treatment.

Hepatic Dose Adjustment

Modification of medication selection or dosing because impaired liver function may reduce metabolism, transport, protein synthesis, or biliary excretion. Standardized liver tests do not always predict drug clearance, so drug-specific guidance is required.

Hepatic Duct

The duct formed by the union of the right and left liver ducts, carrying bile from the liver toward the common bile duct and intestine.

Hepatic Encephalopathy

Neuropsychiatric syndrome occurring in advanced liver disease resulting from accumulation of neurotoxic substances, particularly ammonia. Ammonia crosses the blood-brain barrier and is metabolized by astrocytes into glutamine causing osmotic swelling

and impaired neurotransmission.

Hepatic Fissure

A hepatic fissure is a groove or cleft on the liver surface that separates anatomical regions and provides passage for vessels, ducts, or ligaments.

Hepatic Flexure

The bend where the ascending colon becomes the transverse colon near the liver.

Hepatic Hemangioma

Most common benign liver tumor composed of dilated blood-filled vascular spaces lined by endothelial cells. Usually asymptomatic and found incidentally on imaging. Most are small and require no treatment.

Hepatic Hydrothorax

Large pleural effusion occurring in patients with cirrhosis and portal hypertension in the absence of primary cardiac or pulmonary disease. Most commonly right-sided resulting from transit of ascitic fluid through small defects in the diaphragm into the pleural space driven by the pressure gradient between the abdomen and thorax.

Hepatic Ischemia

Liver injury caused by inadequate hepatic blood flow or oxygen delivery, often during shock, severe hypoxemia, cardiac failure, or hepatic vascular occlusion. It can produce a rapid, marked rise in aminotransferases and may progress to acute liver failure.

Hepatic Jaundice (Hepatocellular Jaundice)

Jaundice caused by impaired processing or excretion of bilirubin by damaged liver cells.

Alternate Names:

Hepatocellular Jaundice

Hepatic Lamina

The hepatic lamina is an embryologic or microscopic layer involved in formation and organization of liver

tissue and bile ducts.

Hepatic Necrosis

Death of liver cells caused by severe ischemia, toxins, viral infection, autoimmune injury, or other insults; extensive necrosis can produce acute liver failure.

Hepatic Portal System

The venous network draining blood from the gastrointestinal tract, spleen, and pancreas to the liver before it reaches the systemic circulation. Formed by the union of the superior mesenteric and splenic veins behind the pancreatic neck; receives inferior mesenteric, left gastric, and other tributaries.

Hepatic Steatosis

Excess accumulation of fat in liver cells, commonly called fatty liver; causes include metabolic disease, alcohol, medicines, and other conditions, and inflammation or fibrosis may develop in some people.

Hepatic Surgery

Surgical treatment of liver disease, including partial liver resection, transplantation, and procedures for tumors or biliary disorders. The operation depends on the diagnosis, tumor extent, liver function, and overall health.

Hepatic Synthetic Dysfunction

Impaired hepatic production of proteins and other molecules, including albumin and coagulation factors. It may cause hypoalbuminemia, edema, prolonged prothrombin time, and increased INR, although INR can also be altered by vitamin K deficiency, anticoagulants, and other conditions.

Hepatic Trauma

Liver injury from trauma. CT staging guides management. Grades I-V based on injury severity.

Hepatic Vein

Veins draining the liver into the inferior vena cava. The right, middle, and left hepatic veins emerge from the posterior surface of the liver at the bare area and drain into the IVC just below the di-

aphragm. They carry deoxygenated blood leaving the liver (after the hepatic portal system delivers blood for processing).

Hepatic Vein Obstruction (Budd-Chiari)

Obstruction of hepatic venous outflow, usually from thrombosis or compression, causing abdominal pain, hepatomegaly, ascites, and potentially liver failure.

Hepatitis

Hepatitis is inflammation of the liver caused by viruses, alcohol, medications, toxins, metabolic disease, or autoimmunity and may be acute or chronic.

Hepatitis (Viral Hepatitis)

Inflammation of the liver caused by hepatotropic viruses. Hepatitis A: fecal-oral, acute self-limited, no chronic carrier state; vaccine-preventable. Hepatitis B (HBV): blood/sexual/perinatal transmission; acute and chronic infection; leading cause of cirrhosis and HCC.

Alternate Names:

Viral Hepatitis

Hepatitis A

Hepatitis A is an acute liver infection caused by hepatitis A virus, spread mainly by the fecal-oral route through close contact or contaminated food and drink. The incubation period is about 15–50 days. It may cause fever, fatigue, nausea, abdominal discomfort, and jaundice, although young children may have few or no symptoms. Most people recover completely; it does not cause chronic infection, but rarely causes acute liver failure. Vaccination provides effective prevention.

Hepatitis A - Children

Hepatitis A in children is acute liver infection spread mainly by fecal contamination and usually resolves, though dehydration or severe liver injury can occur.

Hepatitis A Virus

Picornavirus transmitted through the fecal-oral route most commonly via contaminated food or water. The incubation period is fifteen to fifty days. Clinical presentation ranges from asymptomatic in-

fection in young children to symptomatic acute hepatitis with jaundice, fatigue, nausea, elevated transaminases, dark urine, and pale stools.

Hepatitis A, B, C, D, E

Five viruses that can cause viral hepatitis. They differ in transmission, risk of chronic infection, prevention, and treatment.

Hepatitis B

Hepatitis B is a liver infection caused by hepatitis B virus, spread through infected blood and certain body fluids, including semen and vaginal fluids, and from an infected parent to a newborn during childbirth. It may cause acute illness or chronic infection; chronic infection is more likely when acquired during infancy or childhood and can lead to cirrhosis, liver failure, or hepatocellular carcinoma. Vaccination is the main method of prevention.

Hepatitis B - Children

Hepatitis B in children is infection with hepatitis B virus, which may be acute or chronic and can cause liver inflammation, cirrhosis, or cancer.

Hepatitis B Virus

Hepadnavirus transmitted through percutaneous exposure, sexual contact, and perinatally causing acute and potentially chronic hepatitis. The virus has a partially double-stranded circular DNA genome.

Hepatitis C

Hepatitis C is a liver infection caused by hepatitis C virus, spread mainly through exposure to infected blood, especially through shared injection equipment; transmission during childbirth or sexual contact is less common. Acute infection is often asymptomatic, and more than half of infected people develop chronic infection. Untreated chronic infection can cause liver fibrosis, cirrhosis, liver failure, or hepatocellular carcinoma. Direct-acting antiviral medicines cure more than 95% of cases, but no vaccine is available.

Hepatitis C - Children

Hepatitis C in children is infection with hepatitis C virus, often acquired around birth or through blood exposure and now frequently curable with direct-acting antivirals.

Hepatitis C Virus

Flavivirus transmitted primarily through percutaneous exposure with injection drug use being the most common route. Approximately seventy to eighty percent of acute infections progress to chronic hepatitis C, the leading cause of cirrhosis, hepatocellular carcinoma, and liver transplantation in many Western countries.

Hepatitis D

Liver inflammation caused by hepatitis D virus, which occurs only with hepatitis B infection.

Hepatitis D (Delta Agent)

Hepatitis caused by hepatitis D virus, which requires hepatitis B virus for replication; coinfection or superinfection can produce severe acute or chronic liver disease.

Hepatitis D Virus

Defective RNA virus requiring co-infection with hepatitis B to replicate, using hepatitis B surface antigen for its envelope. Hepatitis D causes more severe liver disease than hepatitis B alone with accelerated progression to cirrhosis and higher risk of hepatocellular carcinoma.

Hepatitis E

Liver inflammation caused by hepatitis E virus, usually transmitted through contaminated food or water.

Hepatitis E Virus

A non-enveloped RNA virus transmitted mainly by the fecal-oral route, often through contaminated water or food. It causes acute hepatitis and, in some people, chronic infection or severe disease. Clinical presentation may resemble hepatitis A, with jaundice and elevated transaminases.

Hepatitis F

A historical or proposed label for an additional hepatitis virus; no distinct, confirmed human hepatitis F virus is currently recognized as an established cause of hepatitis.

Hepatitis G

A clinical disorder related to hepatitis g; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hepatitis G Virus

A name formerly used for GB virus C, an RNA virus found in blood; its role in causing clinically important hepatitis remains uncertain.

Hepatitis Serology

Panel of serologic markers for hepatitis A, B, C, and E. HAV: anti-HAV IgM (acute), anti-HAV IgG (past infection/vaccination). HBV: HBsAg (active infection), anti-HBs (immunity/vaccination), anti-HBc IgM (acute), anti-HBc IgG (past), HBeAg (high replication), anti-HBe (low replication). HBV DNA: quantifies viral load.

Hepatitis Virus

A virus that can inflame the liver; hepatitis viruses include A, B, C, D, and E, which differ in transmission, chronicity, and prevention.

Hepatitis Virus Panel

A group of blood tests detecting antigens or antibodies associated with hepatitis A, B, C, and sometimes other hepatitis viruses to determine infection, immunity, or prior exposure.

Hepatitis Viruses

Hepatitis A: fecal-oral, acute self-limited, vaccine (inactivated). Hepatitis B: blood/sexual/perinatal. Acute: HBsAg, IgM anti-HBc. Chronic: HBsAg >6 months, HBV DNA, HBeAg. Treatment: entecavir, tenofovir (TAF/TDF) - suppress DNA, prevent cirrhosis/HCC.

Hepatobiliary

Relating to the liver, gallbladder, and bile ducts. Hepatobiliary disease can affect bile flow, digestion, or liver function.

Hepatobiliary Scintigraphy

Liver/gallbladder imaging using Tc-99m mebrofenin or disofenin. Assess biliary patency, gallbladder function, bile leak, acute cholecystitis (non-visualization of gallbladder).

Hepatobiliary System

The connected liver, gallbladder, and bile ducts that produce, store, transport, and secrete bile and process many nutrients, drugs, and toxins.

Hepatoblastoma

A malignant embryonal liver tumor occurring almost exclusively in infants and young children, associated with elevated AFP and prematurity.

Hepatocellular Adenoma

A benign liver-cell tumor, seen more often in women using estrogen-containing medicines and in some metabolic disorders, that can bleed or rarely undergo malignant transformation.

Hepatocellular Carcinoma

Primary liver cancer arising from hepatocytes. Risk factors: cirrhosis, chronic hepatitis B/C, alcohol, and metabolic dysfunction-associated steatotic liver disease. Surveillance with ultrasound and AFP; treatment includes resection, transplantation, ablation, TACE, and systemic therapy.

Hepatocellular Jaundice

Jaundice caused by dysfunction of hepatocytes, impairing the liver's ability to take up, conjugate, or excrete bilirubin. See Jaundice for complete overview. Causes: viral hepatitis (A, B, C, D, E), alcoholic hepatitis, drug-induced liver injury (paracetamol, isoniazid), cirrhosis, autoimmune hepatitis, Wilson disease, haemochromatosis.

Hepatocerebral Degeneration

Hepatocerebral degeneration is a brain disorder associated with chronic liver disease or abnormal manganese accumulation and may cause movement, speech, or cognitive problems.

Hepatocyte

The principal functional cell of the liver, responsible for metabolism, detoxification, bile production, plasma-protein synthesis, and many other hepatic functions.

Hepatodiaphragmatic Nodule

A nodule located at the interface of the liver and diaphragm; it may represent an accessory liver tissue, benign lesion, inflammatory focus, or metastatic deposit, so imaging and context matter.

Hepatojugular Reflux

Sustained pressure applied over the liver for 15-30 seconds causes a sustained rise in JVP of more than 3 cm above baseline. It indicates right heart failure or fluid overload, reflecting the inability of the right ventricle to accommodate increased venous return.

Hepatolenticular Degeneration

Wilson disease, an inherited ATP7B disorder causing copper accumulation in the liver, brain, cornea, and other tissues, with hepatic, neurologic, psychiatric, or Kayser-Fleischer-ring findings.

Hepatologist

Physician specializing in disorders of the liver, biliary system, and pancreas.

Hepatology

The medical specialty concerned with the liver, gallbladder, bile ducts, and related diseases.

Hepatoma

A tumor arising from liver cells, usually referring to hepatocellular carcinoma, the most common primary malignant liver tumor.

Hepatomegaly

Enlarged liver detected on imaging. Causes: fatty liver, hepatitis, congestion, infiltration, malignancy.

Hepatopulmonary Syndrome

Triad of liver disease, intrapulmonary vascular dilatation, and impaired oxygenation causing hypoxemia. Presents with platypnea which is dyspnea worsening in the upright position and orthodeoxia which is oxygen desaturation in the upright position.

Hepatorenal Syndrome

A form of acute or chronic kidney dysfunction associated with advanced liver disease and portal hypertension, without another sufficient cause of renal failure. It reflects altered circulation and renal blood flow and requires prompt assessment and treatment of both the liver disease and kidney dysfunction.

Hepatosplenomegaly

Enlargement of both the liver and spleen. It may result from infection, portal hypertension, storage disease, malignancy, haemolysis, or systemic inflammatory and infiltrative disorders; evaluation depends on associated symptoms and laboratory and imaging findings.

Hepatotoxicity

Liver injury caused by a drug, toxin, chemical, or other exposure. Acetaminophen is an important cause of acute liver failure in some regions. Evaluation may include liver tests, clotting studies, medication review, and clinical assessment.

Hepatovirus

A genus of picornaviruses that includes hepatitis A virus, which usually causes an acute fecal-oral liver infection.

Hepcidin

A liver-produced peptide hormone that controls systemic iron balance by binding ferroportin and reducing intestinal iron absorption and iron release from macrophages.

Heptachlor

Heptachlor is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

HER2

Human epidermal growth factor receptor 2, also called ERBB2, a cell-surface receptor whose increased expression or gene amplification drives some cancers and may guide targeted treatment.

HER2-Positive

A cancer whose cells have increased expression or amplification of human epidermal growth factor receptor 2 (HER2). Some HER2-targeted medicines may be used according to the cancer type and clinical context.

Herbal Remedies And Supplements For Weight Loss

Plant-based weight-loss products have variable evidence, uncertain composition, and potential stimulant, liver, cardiovascular, or drug-interaction risks; they should not replace evidence-based obesity treatment.

Herbal Supplement

A concentrated plant-derived product marketed for health or symptom relief; composition and evidence vary, and contamination, drug interactions, liver injury, and pregnancy risks must be considered.

Herbaspirillum

A genus of gram-negative bacteria found in soil and plants; human infection is rare but can occur opportunistically.

Herbaspirillum Huttiense

A species of Herbaspirillum rarely isolated from human specimens and occasionally associated with opportunistic infection.

Herbicide

A chemical used to kill or control unwanted plants;

exposure can irritate skin, eyes, lungs, or the gastrointestinal tract, with toxicity determined by the specific compound and dose.

Herd Immunity

Resistance of population to invasion by infectious agent when sufficient proportion immune. Threshold varies by disease (measles 95%, polio 80%).

Hereditary

Transmitted through genetic material from a parent or arising from an inherited variant; a hereditary trait may be dominant, recessive, X-linked, mitochondrial, or multifactorial.

Hereditary Amyloidosis

A group of inherited disorders in which mutations cause misfolded proteins to form amyloid deposits in organs such as the nerves, heart, kidneys, or gastrointestinal tract.

Hereditary Angioedema

Recurrent episodes of nonpitting swelling of the skin, gastrointestinal tract, or upper airway caused most often by deficiency or dysfunction of C1 inhibitor. Attacks can cause abdominal pain, vomiting, airway obstruction, or life-threatening laryngeal edema without typical urticaria.

Hereditary Cancer

Cancer caused by inherited genetic mutations. Genetic counseling recommended.

Hereditary Coproporphyrria

An inherited hepatic porphyria, usually caused by reduced coproporphyrinogen oxidase activity, that can cause episodic neurovisceral attacks and, in some people, photosensitive skin symptoms.

Hereditary Diffuse Gastric Cancer

An autosomal dominant cancer susceptibility syndrome caused by heterozygous germline inactivating mutations or deletions in the *CDH1* gene on chromosome 16q22.1, which encodes the cell adhesion glycoprotein E-cadherin. Confers a lifetime cumulative

risk of 70% in men and 56% in women for developing diffuse, signet-ring cell adenocarcinoma of the stomach, which infiltrates submucosally without forming an exophytic mass (linitis plastica) and is frequently undetected on surveillance endoscopy. Women carry an additional 40–50% lifetime risk of invasive lobular breast carcinoma. Asymptomatic pathogenic variant carriers are strongly recommended to undergo prophylactic total gastrectomy.

Alternate Names:

HDGC; CDH1-Related Gastric Cancer Syndrome

Hereditary Elliptocytosis

An inherited red-cell membrane disorder in which erythrocytes are elliptical on the blood smear; most people are asymptomatic, but hemolytic anemia can occur in severe forms.

Hereditary Fructose Intolerance

An autosomal-recessive disorder caused by aldolase B deficiency, leading to accumulation of fructose metabolites and hypoglycemia, vomiting, abdominal symptoms, or liver injury after fructose or sucrose ingestion.

Hereditary Hemorrhagic Telangiectasia

An autosomal dominant vascular disorder (also called Osler-Weber-Rendu syndrome) characterized by telangiectasias on skin and mucous membranes, pulmonary AVMs, hepatic AVMs, and cerebral AVMs. Pulmonary AVMs can cause hypoxemia and paradoxical emboli. Treatment is embolization of pulmonary AVMs.

Alternate Names:

Osler-Weber-Rendu syndrome).

Hereditary Hemorrhagic Telangiectasia (Osler-Weber-Rendu Disease)

An autosomal dominant vascular dysplasia characterized by malformed blood vessels lacking smooth muscle layers. Mutations in *ENG* (endoglin) or *ACVRL1* (*ALK1*) genes.

Alternate Names:

Osler-Weber-Rendu Disease

Hereditary Hyperbilirubinemia

An inherited disorder of bilirubin production, conjugation, or excretion that causes persistent or intermittent jaundice, such as Gilbert, Crigler–Najjar, Dubin–Johnson, or Rotor syndrome.

Hereditary Leiomyomatosis And Renal Cell Cancer

An autosomal dominant tumor predisposition syndrome caused by heterozygous germline mutations in the fumarate hydratase (*FH*) gene, a Krebs cycle tumor suppressor. Clinically characterized by multiple painful cutaneous piloleiomyomas (smooth muscle tumors of the arrector pili muscles), multiple large and early-onset uterine leiomyomas that typically present in the second or third decade of life and necessitate early myomectomy or hysterectomy, and an aggressive, solitary, early-metastasizing subtype of type 2 papillary renal cell carcinoma that can arise even in early adulthood.

Alternate Names:

HLRCC; Reed Syndrome

Hereditary Ovalocytosis

An inherited red-cell membrane disorder in which many erythrocytes are oval or elliptical; it is often mild but can cause hemolytic anemia in some individuals.

Hereditary Paraganglioma-Pheochromocytoma Syndrome

An autosomal dominant familial neuroendocrine tumor syndrome caused by germline mutations in succinate dehydrogenase subunit genes, including *SDHD* (maternally imprinted), *SDHB*, *SDHC*, and *SDHA*. Predisposes to the development of extra-adrenal sympathetic and parasympathetic paragangliomas (including carotid body, jugulotympanic, vagal, and retroperitoneal tumors) and adrenal pheochromocytomas. Mutations in *SDHB* are particularly notorious for conferring a high rate of malignancy (up to 30–50%) and early metastatic dissemination, as well as predisposing to renal cell carcinoma and gastrointestinal stromal tumors.

Alternate Names:

Hereditary Pheochromocytoma-Paraganglioma;
SDH-Related Paraganglioma

Hereditary Spherocytic Anemia

An inherited red-cell membrane disorder, usually hereditary spherocytosis, in which fragile spherical erythrocytes are destroyed prematurely, causing hemolytic anemia, jaundice, and splenomegaly.

Hereditary Spherocytosis

The most common inherited hemolytic anemia in Northern Europeans (autosomal dominant 75%, recessive 25%). Caused by defects in RBC membrane proteins (spectrin, ankyrin, band 3, protein 4.2), leading to loss of membrane surface area, reduced deformability, and premature destruction in the spleen.

Hereditary Urea Cycle Abnormality

An inherited defect in ammonia disposal that can cause neonatal or episodic hyperammonemia, vomiting, lethargy, seizures, coma, and neurologic injury; acute decompensation requires urgent metabolic treatment.

Heredity

The transmission of genetic characteristics from parents to offspring through genes and chromosomes.

Heredo-Ataxia

A clinical disorder related to heredo-ataxia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Heredopolyneuritis

A clinical disorder related to heredopolyneuritis; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hering-Breuer Reflex

The Hering-Breuer reflex helps limit overinflation of the lungs by reducing inspiration when lung stretch receptors are strongly activated.

Hermanski-Pudlak Syndrome

A group of inherited disorders involving abnormal lysosome-related organelles, typically causing oculocutaneous albinism, platelet-storage-pool deficiency with bleeding, and sometimes lung or bowel disease.

Hermansky-Pudlak Syndrome

Autosomal recessive (10+ genes: HPS1-10, BLOC/AP-3 complexes). Oculocutaneous albinism, bleeding disorder (absent dense bodies in platelets -> prolonged bleeding time), pulmonary fibrosis (HPS-1, HPS-4), granulomatous colitis, neutropenia. Diagnosis: electron microscopy of platelets (absent dense bodies), genetic testing.

Hernia

A hernia is protrusion of tissue through a weakness or opening in the surrounding muscle or connective tissue and may become incarcerated or strangulated. cavity that normally contains it. Most hernias occur in the abdominal wall.

Hernia Mesh

Synthetic or biologic material for hernia repair. Polypropylene most common. Complications: infection, recurrence, chronic pain, mesh migration.

Hernia Repair (Herniorrhaphy)

Surgical repair of a hernia (inguinal, femoral, umbilical, incisional, hiatal). Inguinal hernia repair: open (Lichtenstein tension-free mesh, Shouldice, Bassini) or laparoscopic (TEP: totally extraperitoneal, TAPP: transabdominal preperitoneal). Mesh reinforcement reduces recurrence (1-3% for mesh vs. 5-15% for primary tissue repair).

Alternate Names:

Herniorrhaphy

Herniated Disc

Protrusion of the nucleus pulposus through a tear in the annulus fibrosus of the intervertebral disc, causing compression or inflammation of adjacent nerve roots. Most common at L4-L5 and L5-S1 (lumbar) and C5-C6, C6-C7 (cervical). Lumbar: sciatica (radiating leg pain), paresthesias, sensory loss, and motor weakness (foot drop).

Herniated Disk

A herniated disk occurs when spinal disk material extends through a fissure and may compress a nerve root, producing back pain, radiating pain, numbness, or weakness. Intervertebral disc, most commonly occurring in the lumbar spine (L4-L5 and L5-S1 account for 95 percent of lumbar herniations) and cervical spine (C5-C6 and C6-C7), potentially compressing spinal nerves.

Herniation

Protrusion of an organ, tissue, or structure through an abnormal opening or weak area, as in a herniated disk or abdominal-wall hernia.

Herniorrhaphy

The surgical operation to repair a hernia defect by anatomical reconstruction, suture approximation, and reinforcement of the weakened abdominal wall or groin musculature. Distinct from herniotomy (simple excision of the hernia sac) and classic non-mesh suture repairs (such as the Bassini, Shouldice, or McVay techniques, which close tissues under tension), modern hernia surgery predominantly utilizes tension-free mesh hernioplasty (e.g., the Lichtenstein open repair or minimally invasive laparoscopic/robotic approaches including transabdominal preperitoneal [TAPP] and totally extraperitoneal [TEP] repairs). Indicated to relieve symptoms and to prevent acute surgical emergencies such as hernia incarceration and ischemic strangulation.

Heroin

Heroin is a highly addictive opioid converted to morphine in the body; overdose causes respiratory depression, coma, and death, while repeated use causes dependence and withdrawal.

Heroin Dependence

Legacy, drug-specific label; see *Opioid Use Disorder*.

Heroin Overdose

Heroin overdose involves toxic or environmental exposure to heroin overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Herpangina

A common childhood viral illness caused by Coxsackievirus A (enterovirus), presenting with fever, sore throat, and small vesicles or ulcers on the soft palate and posterior pharynx. It is most common in children and spreads through respiratory and fecal-oral secretions.

Herpes

Infection caused by a herpesvirus. The term commonly refers to herpes simplex infection, which causes oral or genital lesions, but other herpesviruses cause distinct diseases.

Herpes - Oral

Oral herpes is usually caused by herpes simplex virus type 1 and produces recurrent grouped blisters or ulcers around the lips or mouth, with asymptomatic viral shedding.

Herpes Encephalitis

Inflammation of the brain caused by herpes simplex virus infection.

Herpes Gestationis

An autoimmune blistering skin disease of pregnancy mediated by anti-basement membrane zone antibodies, presenting with pruritic urticarial plaques and vesicles, typically in the second or third trimester.

Herpes Labialis

Recurrent blisters or erosions on or around the lips caused most often by herpes simplex virus type 1; tingling is commonly followed by clustered vesicles that crust and heal.

Herpes Simplex

Infection with herpes simplex virus type 1 or type 2, commonly causing recurrent oral or genital vesicles and ulcers.

Herpes Simplex (Cold Sores / Genital Herpes)

A viral infection caused by herpes simplex virus type 1 (HSV-1, typically orofacial, though increas-

ingly anogenital) and type 2 (HSV-2, predominantly anogenital). HSV is a double-stranded DNA virus of the Herpesviridae family with latency in sensory ganglia and periodic reactivation.

Alternate Names:

Cold Sores / Genital Herpes

Herpes Simplex Encephalitis

The most common cause of sporadic fatal encephalitis, caused by herpes simplex virus type 1 (HSV-1) in over ninety percent of cases. The virus typically infects the temporal lobes, orbitofrontal cortex, and insular cortex, causing hemorrhagic necrosis.

Herpes Simplex Skin Infection

Infection of skin or mucosa by herpes simplex virus, producing grouped vesicles or erosions on an erythematous base. Recurrent disease commonly affects the oral or genital region; widespread infection in patients with eczema is eczema herpeticum and requires urgent medical assessment.

Herpes Simplex Type 1

HSV-1, a DNA virus commonly causing oral or perioral blisters; it can also cause genital infection and remains latent with possible recurrence.

Herpes Simplex Virus

A double-stranded DNA virus belonging to the Herpesviridae family that causes oral and genital herpes. HSV has two serotypes: HSV-1, which typically causes oral herpes, and HSV-2, which typically causes genital herpes. The virus establishes latent infections in sensory ganglia and can reactivate periodically.

Herpes Simplex Virus Infection

Infection with herpes simplex virus type 1 or type 2, commonly causing recurrent oral, genital, skin, eye, or central nervous system disease.

Herpes Vegetans

A rare, chronic, verrucous or vegetative form of herpes simplex virus infection occurring in immunocompromised patients, particularly those with HIV/AIDS. It presents as exophytic, proliferative,

ulcerative plaques on the genital or perianal skin, often refractory to standard antiviral therapy.

Herpes Viral Culture Of Lesion

A laboratory culture of fluid or tissue from a lesion to detect herpes simplex virus or varicella-zoster virus; molecular testing is generally more sensitive and is often preferred.

Herpes Zoster

A painful, usually one-sided blistering rash caused by reactivation of latent varicella-zoster virus in a sensory nerve ganglion. It typically follows a band-like distribution and may cause persistent nerve pain after the rash resolves.

Alternate Names:

Shingles (Herpes Zoster)

Herpes Zoster (Shingles)

A painful, vesicular rash caused by reactivation of latent varicella-zoster virus (VZV, human herpesvirus 3) in the dorsal root ganglia.

Alternate Names:

Shingles

Herpes Zoster Ophthalmicus

Reactivation of varicella-zoster virus latency in the trigeminal ganglion, with viral spread along the ophthalmic division (V1) causing painful vesicular skin eruption on the forehead, eyelid, and nose, with potential ocular involvement including keratitis, uveitis, and secondary glaucoma.

Herpesviridae

A family of DNA viruses that establish lifelong latent infection; human members include herpes simplex, varicella-zoster, cytomegalovirus, and Epstein-Barr virus.

Herpesviridae Infection

An infection caused by a herpesvirus; symptoms range from localized lesions to serious disease in newborns or people with weakened immunity.

Herpesvirus Saimiri

A herpesvirus that naturally infects squirrel monkeys and can cause lymphoid tumors in some other primates used in research.

Herpetic Glossitis

Inflammation of the tongue caused by herpes simplex virus infection, presenting with painful vesicles and ulceration on the tongue surface, often recurring in immunocompromised patients.

Herpetic Stomatitis

Inflammation and ulceration of the mouth caused by herpes simplex virus, most commonly primary HSV-1 infection, with painful vesicles or ulcers and sometimes fever or gingivitis.

Herpetic Whitlow

A painful viral infection of the finger caused by the herpes simplex virus (typically HSV-1), characterized by swelling, redness, and painful vesicular lesions on the finger.

Herpetiform

Resembling herpes lesions, typically describing grouped vesicles or an arrangement of lesions; the term does not by itself establish herpesvirus infection.

Hertz

The SI unit of frequency, equal to one cycle per second.

Herxheimer Reaction

A clinical disorder related to herxheimer reaction; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hesitation Wounds

Multiple superficial, tentative parallel incised cuts surrounding a deeper primary fatal cut wound, classically observed in suicides involving sharp instruments (wrists, neck).

Hesperidin

A flavanone glycoside found mainly in citrus fruits and studied for antioxidant, vascular, and anti-inflammatory effects; clinical benefits and supplement dosing are not established by the term alone.

Hesselbach Triangle

A weak area of the anterior abdominal wall bounded by the lateral border of the rectus abdominis, the inferior epigastric vessels, and the inguinal ligament; direct inguinal hernias pass through it.

Heterochromatin

A densely packed form of chromatin that is relatively transcriptionally inactive and enriched in repetitive DNA, commonly located near centromeres and telomeres.

Heterochromia

A difference in iris color between the eyes or within one iris, which may be congenital or acquired from disease, trauma, medication, or inflammation.

Heterocyclic Amine

A nitrogen-containing chemical compound formed mainly when meat or fish is cooked at high temperatures; several heterocyclic amines are mutagens and potential dietary carcinogens.

Heteroduplex Analysis

A molecular method that compares paired DNA strands containing a sequence mismatch, allowing some mutations or genetic variants to be detected by altered migration or enzyme sensitivity.

Heterogeneity

Variation in composition, structure, or behavior within a tissue, sample, lesion, or population; in tumors, heterogeneity can affect biopsy interpretation and treatment response.

Heterogeneous Echotexture

A mixed ultrasound appearance with regions of differing echogenicity, suggesting nonuniform tissue composition from inflammation, fibrosis, fat, hem-

orrhage, tumor, or other structural change.

Heterologous

Heterologous is a developmental, immunologic, reproductive, behavioral, environmental, or health-care concept used in medicine or biomedical research.

Heterophil Antibody

A heterophil antibody reacts with antigens from unrelated animal species; testing for these antibodies can help diagnose some infections, including infectious mononucleosis.

Heterophile Antibody

A heterophile antibody is an antibody that reacts with antigens from unrelated species and can interfere with some laboratory immunoassays.

Heterophyes Heterophyes

A small intestinal fluke acquired by eating raw or undercooked fish; it can cause abdominal symptoms and occasionally tissue complications.

Heteroplasia

Heteroplasia is a developmental, immunologic, reproductive, behavioral, environmental, or healthcare concept used in medicine or biomedical research.

Heteroplasmy

The presence of more than one mitochondrial DNA sequence in a cell or person, usually because mutant and normal mitochondrial genomes coexist.

Heterosexual

Heterosexual is a developmental, immunologic, reproductive, behavioral, environmental, or health-care concept used in medicine or biomedical research.

Heterotaxy Syndrome

A congenital disorder in which internal organs are arranged abnormally and may have associated heart defects.

Heterotopic

Located in an abnormal or unusual place; a heterotopic pregnancy is simultaneous implantation outside the uterus and, usually, inside the uterus.

Heterotopic Breast Tissue Of The Vulva

Ectopic mammary-like tissue in the vulva that may undergo benign, hormonal, inflammatory, or malignant changes similar to breast tissue; diagnosis is histologic.

Heterotopic Pregnancy

Simultaneous intrauterine and ectopic pregnancy. Rare in spontaneous conception (1:30,000) but more common with ART (1:100-1:500). Risk factors: ovulation induction, IVF with multiple embryo transfer. The presence of an intrauterine pregnancy does not exclude an ectopic pregnancy.

Heterotransplant

Heterotransplant is a developmental, immunologic, reproductive, behavioral, environmental, or health-care concept used in medicine or biomedical research.

Heterozygote

A heterozygote has two different alleles at a genetic locus, one inherited from each parent; phenotype depends on dominance, the variant's effect, and the surrounding genetic and environmental context.

Heterozygous

Having two different alleles of a particular gene at a genetic locus. The observable effect depends on the gene and the relationship between the alleles.

Hexachlorobenzene

A persistent organochlorine compound that can cause porphyria, liver injury, neurologic effects, and other toxicity after significant exposure.

Hexachlorophane

Hexachlorophane is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific

product or circumstance.

Hexachlorophene

A topical antiseptic phenolic compound whose excessive or prolonged use can cause neurotoxicity and skin injury; many uses are restricted.

Hexadactyly

The presence of six fingers or six toes on a hand or foot, also called polydactyly.

Hexadecadienoic Acid

A long-chain fatty acid containing 16 carbon atoms and two double bonds, occurring in lipid metabolism and studied as a biochemical or signaling molecule.

Hexadecatetraenoic Acid

A 16-carbon polyunsaturated fatty acid containing four double bonds, present in lipid pathways and investigated for effects on membranes and signaling.

Hexadecatrienoic Acid

A 16-carbon fatty acid with three double bonds, found in lipid metabolism and plant or animal tissues depending on its isomer.

Hexamethonium

Hexamethonium is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hexamethonium Compounds

Hexamethonium compounds are ganglion-blocking drugs that were formerly used to lower blood pressure but are now rarely used because of extensive side effects.

Hexamine

Hexamine is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hexane

Hexane is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hexanoic Acid

A six-carbon saturated fatty acid, also called caproic acid, found in some fats and odors and used in chemical synthesis; concentrated exposure irritates skin and mucosa.

Hexestrol

A synthetic nonsteroidal estrogen formerly used therapeutically; it is no longer routinely used because of adverse effects and carcinogenic concerns.

Hexoestrol

Hexoestrol is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hexokinase

Phosphorylates glucose to glucose-6-phosphate. Low K_m , inhibited by product. Glucokinase (liver) has high K_m , not inhibited.

Hexose

A six-carbon monosaccharide, such as glucose, fructose, or galactose, that serves as an energy substrate or metabolic intermediate.

Hexylresorcinol

A phenolic antiseptic and local anesthetic used in some throat lozenges and topical products; excessive use or allergy can cause adverse effects.

Hg

The chemical symbol for mercury; in measurements such as mmHg, it refers to millimeters of mercury, a unit of pressure.

Hiatal

Relating to a hiatus, an opening or passage in an

anatomical structure; a hiatal hernia occurs when abdominal tissue passes through the diaphragm's esophageal hiatus.

Hiatal Hernia

Protrusion of part of the stomach through the esophageal opening in the diaphragm into the chest. It may be sliding or paraesophageal and can be associated with reflux or other symptoms.

Hiatal Hernia Overview

A hiatal hernia occurs when part of the stomach moves through the diaphragm into the chest. It may cause reflux, chest or upper-abdominal discomfort, swallowing problems, or no symptoms; treatment ranges from lifestyle and acid control to surgery.

Hiatus

An opening, gap, or passage in an anatomical structure, such as the esophageal hiatus in the diaphragm.

Hiatus Hernia

Another term for hiatal hernia: protrusion of part of the stomach through the esophageal hiatus of the diaphragm into the chest. It may be sliding or paraesophageal.

Hib

Haemophilus influenzae type b, an encapsulated bacterium that can cause meningitis, epiglottitis, pneumonia, septic arthritis, cellulitis, or sepsis. Routine conjugate vaccination substantially reduces invasive Hib disease.

Hibernation

A prolonged state of markedly reduced metabolism, body temperature, heart rate, and activity used by some animals to survive cold or food scarcity; medically, the term may describe induced metabolic suppression.

Hibernation, Artificial

An experimental state of reduced body temperature and metabolism induced for medical or research

purposes; it is not routine human care.

Hibernoma

A rare benign tumor of brown fat, usually a slow-growing painless soft-tissue mass that may feel warm because of its high metabolic and vascular activity.

Hiccough

Another term for hiccup, a sudden involuntary contraction of the diaphragm followed by rapid closure of the vocal cords that produces a characteristic sound.

Hiccough (Hiccup)

An involuntary, repetitive, spasmodic contraction of the diaphragm followed by sudden closure of the glottis, producing the characteristic 'hic' sound. Hiccups are caused by irritation or stimulation of the phrenic nerve (C3-C5, innervates the diaphragm) or the vagus nerve (CN X).

Alternate Names:

Hiccup

Hiccup

An involuntary, repeated contraction of the diaphragm followed by sudden closure of the glottis, producing the characteristic sound; persistent hiccups may signal irritation, metabolic disease, or neurologic illness.

Hiccups

Involuntary, spasmodic contractions of the diaphragm followed by sudden closure of the glottis, producing a characteristic sound. Persistent or intractable hiccups may reflect irritation of the diaphragm or phrenic and vagal pathways, metabolic disease, medication exposure, or central nervous system disease.

Hiccups (Singultus)

Involuntary, spasmodic contractions of the diaphragm and intercostal muscles followed by sudden glottic closure, producing the characteristic "hic" sound. The reflex arc involves the phrenic nerve (C3-C5), vagus nerve, and brainstem centers.

Alternate Names:

Singultus

Hidradenitis

Hidradenitis suppurativa is a chronic inflammatory follicular occlusion disease affecting apocrine gland-bearing areas (axillae, groin, inframammary). It presents with recurrent painful nodules, abscesses, sinus tracts, and scarring.

Hidradenitis Suppurativa

A chronic inflammatory skin disorder involving recurrent painful nodules, abscesses, draining tunnels, and scars, usually in skin folds such as the axillae, groin, buttocks, or breasts. Severity varies and is influenced by genetic, hormonal, and environmental factors.

Hidradenocarcinoma

A rare malignant sweat-gland tumor that may arise de novo or from a hidradenoma, presenting as a firm nodule and capable of local recurrence and metastasis.

Hidradenoma

A usually benign adnexal neoplasm showing differentiation toward sweat-gland epithelium and presenting as a solitary dermal or subcutaneous nodule. It is commonly slow growing and may be flesh colored, erythematous, cystic, or ulcerated, which can make clinical distinction from other skin tumors difficult. Histology typically demonstrates a well-circumscribed proliferation with polygonal or clear cells, ductal structures, and variable cystic change. Excision is generally diagnostic and therapeutic; atypia, infiltrative growth, recurrence, or metastasis raises concern for hidradenocarcinoma or another malignant adnexal tumor.

Hidradenoma Papilliferum

A benign papillary tumor of apocrine or mammary-like glands, usually presenting as a solitary flesh-colored or red nodule in the vulvar or anogenital region.

Hidrocystoma

A benign cyst of a sweat gland near the eyelid or skin, appearing as a small translucent or bluish papule that may enlarge with heat or sweating.

Hidrosis

Sweating or perspiration; abnormal hidrosis may mean excessive sweating, reduced sweating, or sweating in an unusual pattern.

High Altitude

Elevation high enough that reduced atmospheric pressure lowers available oxygen, commonly beginning around 2,500 meters and increasing the risk of altitude illness.

High Altitude Illness

A clinical disorder related to high altitude illness; diagnosis and treatment depend on its cause, characteristic findings, and severity.

High Arch

An abnormally elevated medial longitudinal arch of the foot, also called pes cavus. It may be idiopathic or associated with neuromuscular disease and can cause overload of the heel and forefoot, calluses, pain, instability, or difficulty fitting footwear.

Alternate Names:

pes cavus.

High Blood Cholesterol Levels

Excess circulating cholesterol, especially LDL cholesterol, promotes atherosclerotic plaque and increases risk of coronary disease and stroke; assessment includes overall cardiovascular risk and treatment response.

High Blood Pressure

Blood pressure that stays above the healthy range, also called hypertension. It often has no symptoms but can damage the heart, brain, kidneys, and eyes over time.

Alternate Names:

Hypertension

High Blood Pressure And Eye Disease

Long-standing high blood pressure can damage retinal blood vessels and the optic nerve, causing hypertensive retinopathy or vision loss.

High Blood Pressure Hypertension

Hypertension is persistently elevated arterial pressure that increases the risk of stroke, heart disease, kidney disease, eye damage, and dementia. Diagnosis uses repeated accurate measurements, and management combines lifestyle measures with medicines when indicated.

High Blood Pressure In Adults - Hypertension

Persistent elevation of arterial blood pressure in an adult, diagnosed using properly measured readings over time and associated with increased cardiovascular, cerebrovascular, and kidney risk.

High Blood Pressure In Children - Hypertension

Blood pressure in a child or adolescent that is persistently above the age-, sex-, and height-specific reference range; evaluation considers secondary causes and target-organ effects.

High Blood Pressure In Infants - Hypertension

Infant hypertension is elevated blood pressure in a baby and may result from kidney, vascular, heart, lung, endocrine, medication, or intensive-care conditions.

High Blood Pressure Medications

Medicines used to lower arterial blood pressure, including diuretics, ACE inhibitors, ARBs, calcium-channel blockers, beta blockers, and other agents selected according to comorbidities and risk.

High Blood Protein

An abnormally increased concentration of proteins in the blood, which may result from dehydration, inflammation, infection, or a plasma-cell disorder.

High Blood Sugar

Hyperglycemia, an abnormally elevated concentration of glucose in the blood; persistent hyperglycemia is commonly associated with diabetes but can have other causes.

High Cholesterol - Children

Elevated cholesterol in a child may reflect inherited familial hypercholesterolemia, diet, obesity, endocrine disease, or medication; confirmation and family assessment guide lifestyle and specialist treatment.

High Hemoglobin Count

A hemoglobin level above the expected range, which may reflect dehydration, chronic low oxygen, smoking, medication effects, or a blood disorder.

High Output Failure

High Output Failure is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

High Potassium (Hyperkalemia)

An abnormally high potassium level in the blood that can affect muscle and heart function and may result from kidney disease, medicines, tissue injury, or other conditions.

High Potassium Level

Hyperkalemia is an elevated blood potassium concentration that can cause dangerous cardiac conduction abnormalities; confirmation, medication review, kidney assessment, ECG, and urgent treatment depend on level and symptoms.

High Protein Diet

An eating pattern providing more protein than a person's usual or estimated requirement; suitability depends on nutritional needs, kidney and liver function, and the overall diet.

High Red Blood Cell Count

An increased number or concentration of red blood

cells in the blood, which may thicken the blood and result from dehydration, low oxygen, smoking, medications, or a bone marrow disorder.

Alternate Names:

Polycythemia (High Red Blood Cell Count)

High Uric Acid Level

An increased concentration of uric acid in the blood, which may contribute to gout or kidney stones and can result from reduced kidney clearance, diet, medicines, or metabolic disease.

High White Blood Cell Count

An increased number of white blood cells, often associated with infection, inflammation, stress, medicines, or a blood or bone marrow disorder.

High White Blood Cell Count In Pregnancy

A mild rise in white blood cells can be normal during pregnancy, labor, or stress. Marked or persistent elevation with fever, pain, urinary or respiratory symptoms, or abnormal differential counts may indicate infection, inflammation, medication effect, or another disorder.

High-Altitude Pulmonary Edema

A potentially fatal accumulation of fluid in the lungs after rapid ascent to high altitude, characterized by breathlessness, cough, reduced exercise tolerance, and sometimes frothy sputum. Immediate descent and oxygen are essential.

High-Density Lipoprotein

A lipoprotein that transports cholesterol in the blood and is commonly called HDL.

Alternate Names:

HDL.

High-Density Lipoprotein

A lipoprotein that transports cholesterol and other fats in the blood. HDL level is one part of cardiovascular risk assessment and should not be interpreted alone.

High-Dependency Unit

A ward or area within a hospital that provides a level of care between a general ward and an intensive care unit, for patients who require intensive observation or treatment but not mechanical ventilation.

High-Fiber Foods

Foods rich in soluble or insoluble dietary fiber, including legumes, whole grains, vegetables, fruit, nuts, and seeds, support bowel regularity and may improve cholesterol and glucose control when increased with adequate fluid.

High-Flow Nasal Cannula

A device delivering heated and humidified oxygen at flow rates up to 60 L/min through a nasal cannula. It provides a modest amount of PEEP, reduces dead space ventilation, and improves patient comfort compared to non-invasive ventilation. It is used for acute hypoxemic respiratory failure and post-extubation support.

High-Fructose Corn Syrup

High-Fructose Corn Syrup is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

High-Grade Endometrial Stromal Sarcoma

An aggressive malignant uterine stromal tumor with high-grade cytology and characteristic molecular alterations in some cases; it may invade the myometrium, vessels, or distant sites.

High-Grade Serous Ovarian Carcinoma

A high-grade tubo-ovarian epithelial carcinoma characterized by marked nuclear atypia and nearly universal abnormality of the TP53 pathway; many arise from the fimbrial fallopian tube.

High-Grade Surface Osteosarcoma

A high-grade osteogenic sarcoma arising on the surface of bone, including periosteal or parosteal regions, with aggressive local and metastatic potential.

High-Intensity Focused Ultrasound

A treatment that concentrates ultrasound energy to heat and destroy a targeted area of tissue. Its uses depend on the device, disease, and clinical setting.

High-Level Disinfection

Disinfection that inactivates all microorganisms except high numbers of bacterial spores. It is commonly used for heat-sensitive semicritical equipment such as some flexible endoscopes after meticulous cleaning.

High-Pitched Cry

An unusually shrill infant cry that can occur with pain, neurologic dysfunction, drug withdrawal, or serious illness and warrants assessment when persistent or accompanied by abnormal behavior or feeding.

Highly Pathogenic Avian Influenza

Highly pathogenic influenza A infection in birds that can cause severe disease and mortality in poultry; selected strains can rarely infect humans after animal exposure.

High-Resolution Computed Tomography

An advanced computed tomography technique optimized for imaging the lung parenchyma by utilizing thin slice collimation (0.625 to 1.5 mm), high-spatial-frequency reconstruction algorithms, and optional inspiratory, expiratory, and prone acquisitions. It allows exquisite visualization of the secondary pulmonary lobule and is the gold-standard non-invasive diagnostic modality for diffuse parenchymal lung diseases, interstitial lung diseases (differentiating usual interstitial pneumonia from nonspecific interstitial pneumonia), bronchiectasis, and small airway disease.

Alternate Names:

HRCT; HRCT Chest; High-Resolution Chest CT

High-Sensitivity Cardiac Troponin

An analytically refined immunoassay that quantifies circulating cardiac troponin I (hs-cTnI) or T (hs-cTnT) at concentrations down to single-digit nanograms per liter, with an analytical coefficient

of variation ≤ 10 percent at the 99th percentile upper reference limit of a healthy population. It enables early detection of subclinical myocardial injury and forms the core of validated European and American rapid diagnostic pathways (0/1-hour and 0/2-hour algorithms) to rapidly rule out or confirm acute myocardial infarction in emergency departments.

Alternate Names:

hs-cTn; High-Sensitivity Troponin; hs-Troponin

Hilar

Relating to a hilum, the gateway where vessels, nerves, ducts, or bronchi enter or leave an organ, especially the lung or kidney.

Hilar Cholangiocarcinoma

A cancer of the bile ducts where the ducts leave the liver.

Hill'S Sign

Hill's sign is an older physical finding described in some cases of aortic regurgitation; it is not sufficient by itself to diagnose the condition.

Hill-Sachs Lesion

Impaction fracture of posterolateral humeral head from anterior shoulder dislocation. Associated with Bankart lesion.

Hilum

A depression or opening where blood vessels, nerves, ducts, or other structures enter or leave an organ.

Hilus

The hilus is an indentation or entry point on an organ where vessels, nerves, ducts, or airways enter or leave.

Hind

Hind means toward the back or rear of an organism or structure, especially in comparative anatomy.

Hindbrain

The hindbrain is the lower part of the developing

brain, giving rise to the medulla, pons, and cerebellum and helping control vital functions and coordination.

Hindgut

The hindgut is the embryonic part of the digestive tract that forms much of the distal colon, rectum, and upper anal canal.

Hip

The joint and surrounding region where the thigh bone meets the pelvis, allowing movement and bearing body weight.

Hip Arthroscopy

A minimally invasive procedure that uses a camera and instruments inserted through small incisions to diagnose and treat selected disorders inside or around the hip joint.

Hip Bursitis

Hip bursitis is inflammation of a bursa near the hip, commonly causing pain over the outer hip that worsens with lying on that side, stairs, or prolonged walking. Activity changes, physical therapy, medicines, or injection may help after other causes are considered.

Alternate Names:

Trochanteric Bursitis (Hip Bursitis)

Hip Circumference

The measurement around the widest part of the hips or buttocks, used in anthropometry and sometimes combined with waist circumference to assess body-fat distribution.

Hip Dislocation

Loss of normal congruence between the femoral head and acetabulum. It is usually traumatic in adults and may damage the sciatic nerve or femoral-head blood supply; congenital or developmental dislocation occurs in developmental dysplasia of the hip.

Hip Disorders

The hip joints are ball-and-socket joints connecting the femur to the pelvis, providing stability and a wide range of motion for walking, running, and weight-bearing activities. Common hip conditions include osteoarthritis, fractures, bursitis, and labral tears.

Hip Dysplasia

Developmental dysplasia of the hip: abnormal development of the acetabulum/femoral head causing instability, subluxation, or dislocation. Risk factors: female, breech presentation, family history, firstborn, and tight swaddling.

Alternate Names:

CHD DDH

Hip Fracture

Fracture of proximal femur, usually femoral neck or intertrochanteric region. Mechanism: low-energy fall in osteoporotic bone. Symptoms: pain, inability to bear weight, shortened/externally rotated leg. Diagnosis: X-ray, MRI if X-ray negative. Treatment: surgical fixation (internal fixation, arthroplasty).

Alternate Names:

Broken Hip

Hip Fracture Surgery

An operation to stabilize or replace a fractured proximal femur, using internal fixation, hemiarthroplasty, or total hip arthroplasty according to fracture pattern and patient factors.

Hip Injury

Damage involving the hip joint, surrounding muscles, tendons, ligaments, or bones; causes range from strain and bursitis to fracture and dislocation, with inability to bear weight requiring prompt assessment.

Hip Joint

Multiaxial ball-and-socket synovial joint between femoral head and acetabulum. Designed for stability and mobility, bearing entire upper body weight. Acetabular labrum deepens the socket; strong ligaments (iliofemoral, pubofemoral, ischiofemoral) re-

inforce it. Ligamentum teres contains the femoral head artery.

Hip Joint Injection

Injection of local anesthetic, corticosteroid, or another prescribed agent into or around the hip joint for diagnosis or temporary relief of pain and inflammation.

Hip Joint Replacement

Arthroplasty replaces damaged parts of the hip with prosthetic components to relieve pain and improve mobility, commonly for advanced osteoarthritis, fracture, or other destructive joint disease.

Hip Labral Tear

A tear in the ring of cartilage that surrounds the hip socket.

Hip Pain

Pain in or around the hip caused by joint disease, tendon or muscle injury, bursitis, fracture, nerve problems, or pain referred from the back.

Hip Prosthesis

Total hip replacement components: femoral stem (metal), head (metal/ceramic), acetabular cup (polyethylene/metal/ceramic). Fixation: cemented (older patients), cementless (younger, porous coating for bone ingrowth). Materials: cobalt-chrome, titanium, ceramics.

Hip Replacement

A surgical procedure in which the diseased hip joint is replaced with a prosthetic implant (arthroplasty). Total hip replacement (THR) replaces both the femoral head and the acetabulum.

Hip-Fracture Rehabilitation

Multidisciplinary postoperative management aimed at restoring mobility, preventing complications, and returning the older person to the highest feasible level of independence. It includes early mobilization, pain control, osteoporosis treatment, delirium prevention, nutrition, and discharge planning.

HIPAA

The United States Health Insurance Portability and Accountability Act, including federal rules governing protected health information, privacy, security, and certain electronic health-care transactions.

Hippocampus

Curved medial temporal lobe structure, part of the limbic system, critical for new declarative memory formation (episodic and spatial) and short-to-long-term memory conversion. Has dentate gyrus, hippocampus proper (CA1-CA4), subiculum, and entorhinal cortex. One of few adult neurogenesis sites (dentate gyrus).

Hippocrates

An ancient Greek physician traditionally regarded as a major founder of Western clinical medicine; the Hippocratic corpus emphasizes observation, prognosis, and professional conduct.

Hippocratic Oath

A traditional promise that doctors make to practice medicine ethically, help patients, protect their privacy, and avoid harm. Modern versions differ by country and institution.

Hippocratic Succussion

Hippocratic succussion is an old examination method in which the body is shaken to detect a splashing sound from fluid and air in a body cavity.

Hippotherapy

Therapeutic horseback riding for neurological, musculoskeletal conditions. Improves balance, posture, coordination.

Hippus

Hippus is rhythmic, involuntary widening and narrowing of the pupil, which may occur normally or with neurologic or eye disease.

Hirayama Disease

A benign, non-progressive cervical myelopathy predominantly affecting adolescent males (ages 15–25),

caused by dynamic displacement of the lower cervical posterior dural sac forward against the spinal cord during neck flexion, resulting in chronic microcirculatory ischemia of the anterior horns (C7–T1). Clinically characterized by insidious, slowly progressive, asymmetrical or unilateral weakness and muscular atrophy restricted to the distal upper limb, selectively involving the intrinsic hand muscles, thenar and hypothenar eminences, and the medial forearm flexors and extensors while characteristically sparing the brachioradialis (giving rise to a distinctive “oblique amyotrophy” appearance). Typically arrests spontaneously within 3 to 5 years; diagnosed by dynamic flexion MRI of the cervical spine.

Alternate Names:

Monomelic Amyotrophy; Juvenile Asymmetric Segmental Spinal Muscular Atrophy

Hirschsprung Disease

Congenital condition characterized by the absence of enteric ganglion cells in a segment of the distal colon, resulting from failure of neural crest cell migration during embryonic development.

Alternate Names:

Congenital Megacolon (Hirschsprung Disease)

Hirschsprung'S Disease

A congenital disorder characterized by the absence of parasympathetic ganglion cells (aganglionosis) in the distal colon, resulting in failure of relaxation and functional obstruction. Incidence: 1 in 5,000 live births, male predominance (4:1).

Hirsutism

Excess terminal hair growth in women in a male-pattern distribution (face—upper lip, chin, sideburns; chest, back, lower abdomen, inner thighs). Hirsutism affects 5-10% of women of reproductive age. Pathophysiology: increased androgen production or increased end-organ sensitivity.

Alternate Names:

Excess Facial Hair In Women

Hirudin

A potent anticoagulant peptide originally identi-

fied in medicinal leech saliva that directly inhibits thrombin and prevents conversion of fibrinogen to fibrin.

Hirudins

A family of thrombin-inhibiting anticoagulant peptides, including natural and recombinant agents, used or studied to prevent or treat clotting in selected clinical situations.

Hirudo Medicinalis

The medicinal leech, used in selected medical procedures to relieve venous congestion after reconstructive surgery.

His

Usually referring to the bundle of His, the atrioventricular conduction bundle that carries electrical impulses from the AV node into the right and left bundle branches.

His Bundle Electrophory

An invasive electrophysiologic recording of electrical activity in the atrioventricular His bundle, used to assess conduction-system disease and localize atrioventricular block.

His Disease

Trench fever, a louse-borne infection caused by *Bartonella quintana*, classically producing recurrent or five-day fever, headache, and severe shin or leg pain.

Histamine

A biogenic amine (imidazolyethylamine) synthesised from L-histidine by histidine decarboxylase, stored in basophils, mast cells, and platelets, and released during allergic and inflammatory responses.

Histamine Degradation

The enzymatic breakdown of histamine, mainly through diamine oxidase and histamine N-methyltransferase, limiting its effects in the gut, airway, skin, and nervous system.

Histamine Receptor

A cell-surface receptor activated by histamine; H1, H2, H3, and H4 receptors mediate different effects involving allergy, gastric acid, neurotransmission, and immune-cell activity.

Histamine Release

Discharge of histamine from mast cells or basophils during allergic, immune, or nonimmune activation, causing effects such as itching, vasodilation, swelling, mucus production, and bronchoconstriction.

Histidine

Histidine is an amino acid used in protein synthesis and a precursor of histamine; it is essential during growth and can be metabolized to urocanic acid and other products.

Histidine Decarboxylase

A medicine or chemical compound called histidine decarboxylase. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Histidinemia

An inherited metabolic disorder caused by reduced histidase activity, leading to elevated histidine in blood and urine; many affected people have few or no clinically important symptoms.

Histiocyte

A tissue-resident immune cell of the mononuclear phagocyte system, generally derived from monocytes and capable of engulfing foreign material and presenting antigen.

Histiocytic Sarcoma

A neoplastic process affecting histiocytic sarcoma; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Histiocytoma

A tumor or tumor-like proliferation of histiocytes; the term may refer to a benign fibrous skin lesion or, less commonly, a malignant histiocytic neoplasm

depending on classification.

Histiocytosis

A group of disorders characterized by abnormal proliferation of histiocytes (tissue macrophages/dendritic cells), including Langerhans cell histiocytosis, Erdheim-Chester disease, and hemophagocytic lymphohistiocytosis.

Histochemical

Histochemical means using chemical stains or reactions in tissue sections to identify particular substances or cell components.

Histocompatibility

Degree of immunological similarity between donor and recipient tissues; determined primarily by HLA antigen matching, critical for transplant success. Its significance is established from its characteristic features, causes, and clinical context.

Histocompatibility Antigen Test

A test identifying human leukocyte antigen types, which influence tissue compatibility for transplantation and are relevant to some immune-mediated and genetic associations.

Histocompatibility Testing

Laboratory typing of HLA and other immune markers to assess donor–recipient compatibility before transplantation and to investigate selected immune or transfusion-related questions.

Histogenesis

The developmental process by which unspecialized cells differentiate and organize into mature tissues with characteristic structures and functions.

Histological Techniques

Laboratory methods for preparing, staining, sectioning, and examining tissue under a microscope to reveal cellular architecture, deposits, organisms, or disease-related changes.

Histology

The microscopic study of the structure of tissues and cells. Histology involves preparing thin tissue sections (3-10 micrometres) by fixation (formalin, paraffin embedding), sectioning (microtome), staining (haematoxylin and eosin, H&E), and examination under a light or electron microscope.

Histology Cassette

A labeled perforated container that holds a tissue specimen during processing, allowing fixation, dehydration, paraffin infiltration, and later embedding for microscopic sectioning.

Histone

A histone is a basic DNA-binding protein around which DNA winds to form nucleosomes; histone modifications regulate chromatin accessibility and gene expression.

Histone Acetylation

Addition of acetyl groups to histone proteins that usually loosens DNA packaging and can increase access of transcriptional machinery to genes.

Histone Acetyltransferase

An enzyme that transfers acetyl groups to histone proteins, generally promoting a more open chromatin state and regulating gene transcription.

Histone Deacetylases

Enzymes that remove acetyl groups from histones and other proteins, influencing chromatin structure, gene expression, cell differentiation, and disease biology.

Histone Deacetylation

Removal of acetyl groups from histones, often tightening chromatin and reducing transcription; histone deacetylases are targets of some anticancer drugs.

Histone Modification

A chemical change to histone proteins, such as acetylation, methylation, phosphorylation, or ubiquitination, that alters chromatin structure and gene

expression.

Histopathologic Grade

A microscopic estimate of how abnormal and aggressive tumor cells appear compared with their tissue of origin; grade contributes to prognosis and treatment planning alongside stage.

Histopathology

The laboratory-based diagnosis of disease through microscopic examination of tissue sections from biopsy, excision, or autopsy, built on fixation, paraffin embedding, sectioning, and hematoxylin and eosin staining.

Histophilus

A genus of gram-negative bacteria, including *Histophilus somni*, that can cause respiratory or systemic disease mainly in cattle.

Histoplasma

A genus of fungi that includes *Histoplasma capsulatum*, which can cause histoplasmosis after inhalation of environmental spores.

Histoplasma Capsulatum

A fungus found in soil contaminated with bird or bat droppings that can cause histoplasmosis when its spores are inhaled.

Histoplasma Complement Fixation

A blood test that detects complement-fixing antibodies to *Histoplasma* and can support evaluation of histoplasmosis, although sensitivity, timing, cross-reactions, and immune status affect interpretation.

Histoplasma Skin Test

An older delayed-hypersensitivity skin test using *Histoplasma* antigen; it indicates prior exposure but does not reliably distinguish active infection and is rarely used.

Histoplasmosis

A fungal infection caused by *Histoplasma capsulatum*, acquired by inhaling spores found in nitro-

genrich soil enriched with bat or bird guano, most commonly in the Ohio and Mississippi River valleys of the United States, Central and South America, and parts of Africa and Asia.

Histoplasmosis - Acute (Primary) Pulmonary

A lung infection caused by recently inhaled Histoplasma spores; it may cause fever, cough, chest discomfort, and fatigue, or no symptoms.

History, Developmental

A record of a child's developmental milestones and abilities, including motor, language, cognitive, social, and adaptive skills, used to identify possible delays or regression.

History, Social

Information about a person's living situation, relationships, occupation, education, tobacco or substance use, diet, safety, and other social factors relevant to health and care.

Histrionic Personality Disorder

A long-term pattern of intense emotional expression, strong need for attention, and behavior that may seem dramatic, suggestible, or uncomfortable when not the focus. Diagnosis requires a persistent pattern that causes significant problems and is made by a qualified mental-health professional.

HIV

Human immunodeficiency virus: a retrovirus causing progressive CD4+ T-cell depletion and AIDS if untreated. Transmitted via sexual contact, blood/blood products, mother-to-child.

Hiv Antibodies

Antibodies produced after exposure to HIV and detected by screening or confirmatory tests; results are interpreted with antigen testing, nucleic-acid testing, timing, and clinical context.

HIV Infection

Infection with human immunodeficiency virus,

which progressively damages immune function if untreated. Antiretroviral therapy suppresses viral replication and reduces transmission and complications.

Hiv Seropositivity

The presence of detectable antibodies or other serologic evidence of HIV infection; a reactive screening result requires the recommended confirmatory testing algorithm.

HIV Test

Screening and confirmatory testing for HIV. Fourth-generation HIV antigen/antibody combo test (p24 antigen + HIV-1/2 antibodies): window period 2-4 weeks. Positive screening requires confirmatory HIV-1/2 antibody differentiation assay. Rapid point-of-care tests (fingerstick, oral swab): results in 20 minutes.

HIV Transmission And Prevention

The spread of HIV through infected blood, semen, vaginal fluids, rectal fluids, or breast milk, prevented by testing, treatment that suppresses viral load, condoms, sterile equipment, and pre- or post-exposure prophylaxis.

HIV/AIDS

HIV-1/2: retrovirus, CD4+ T cell depletion. Transmission: sexual, blood, perinatal, breast milk. Acute retroviral syndrome: fever, lymphadenopathy, pharyngitis, rash, high viral load. Clinical latency: CD4 decline over years. AIDS: CD4 <200 or AIDS-defining illness (PJP, toxo, CMV, MAC, crypto, KS, lymphoma, TB).

HIV/AIDS Immunopathogenesis

HIV-1 infects CD4+ T cells, macrophages, dendritic cells via gp120 binding to CD4 + CCR5 (R5-tropic, early) or CXCR4 (X4-tropic, late). Reverse transcription -> integration -> transcription/translation -> budding. Acute retroviral syndrome (2-4 weeks): fever, lymphadenopathy, pharyngitis, rash, high viral load.

HIV/AIDS In Pregnant Women And

Infants

HIV care in pregnancy and infancy uses maternal antiretroviral therapy, viral-load monitoring, delivery and feeding planning, infant prophylaxis, and early infant diagnostic testing to reduce vertical transmission.

HIV/AIDS Nutrition

Malnutrition due to increased need requirement, malabsorption, side effects. Supplementation, opportunistic infection management.

Hives

Hives, or urticaria, are transient raised itchy wheals caused by dermal edema, commonly from allergy, infection, medications, physical triggers, or spontaneous mast-cell activation. or chronic and can result from infection, foods, medications, physical triggers, or autoimmune mechanisms; swelling of the airway is an emergency.

Alternate Names:

Angioedema (Hives)

Hives (Urticaria)

A common skin condition characterized by transient, well-circumscribed, intensely pruritic wheals that blanch with pressure and resolve within 24 hours without scarring, caused by mast-cell mediator release in the dermis. Acute urticaria lasts less than 6 weeks; chronic urticaria recurs for 6 weeks or longer and is often spontaneous.

Alternate Names:

Urticaria

Hives And Angioedema

Hives are raised itchy welts; angioedema is deeper swelling of the skin or mucous membranes, often caused by allergy, medicines, or other immune reactions.

HLA

Human leukocyte antigen, a cell-surface protein system involved in immune recognition and tissue compatibility.

Alternate Names:

Human leukocyte antigen

Hla Antigen

A cell-surface human leukocyte antigen encoded by the major histocompatibility complex that presents peptides to T cells and influences transplant compatibility and immune disease risk.

HLA System

The HLA system is a group of cell-surface proteins that present antigens to immune cells and influence transplant matching and autoimmune risk.

HLA-B27

A class I human-leukocyte-antigen molecule associated with several spondyloarthropathies, including ankylosing spondylitis and reactive arthritis. A positive result supports but does not establish a diagnosis.

HLA-B27 Antigen

A cell-surface immune marker associated with increased risk of ankylosing spondylitis and related spondyloarthritis, reactive arthritis, and uveitis; a positive result is not diagnostic by itself.

HMG-CoA Reductase

Rate-limiting enzyme of cholesterol synthesis. Target of statin drugs.

Hmgb Proteins

High-mobility group box proteins that bind and bend DNA in the nucleus and can act as inflammatory danger signals when released from damaged or activated cells.

Hmgn Proteins

High-mobility-group nucleosome-binding proteins that regulate chromatin accessibility and influence transcription, DNA repair, and cell differentiation.

Hoarding Disorder

Persistent difficulty discarding or parting with possessions, regardless of their actual value. The individual perceives a need to save items and experiences

distress at the thought of discarding them. This results in accumulation of possessions that congest and clutter active living areas.

Hoarseness

A change in vocal quality due to abnormal vibration of the vocal folds, resulting in altered pitch; a subset of dysphonia rather than a diagnosis. Common causes include acute viral laryngitis, voice overuse, smoking, reflux, and vocal cord lesions (nodules, polyps, papillomas, laryngeal carcinoma).

Hodgkin Lymphoma

A B-cell lymphoma characterized by the presence of Reed-Sternberg cells (large, binucleate or multinucleate cells with owl-eye nucleoli) in a background of reactive inflammatory cells.

Hodgkin Lymphoma (Hodgkin Disease)

A cancer of the lymphatic system characterized by abnormal lymphocytes and often presenting with painless lymph-node enlargement, fever, sweats, or weight loss.

Alternate Names:

Cancer, Hodgkin Lymphoma Lymphoma, Hodgkin

Hodgkin Lymphoma In Children

A malignant lymphoma arising from lymphoid tissue in children, characterized by Hodgkin and Reed-Sternberg cells and treated according to subtype, stage, and risk.

Hodgkin'S And Non-Hodgkin'S Lymphoma Comparison

Hodgkin lymphoma is defined by characteristic Reed-Sternberg cells and commonly spreads in an orderly pattern, while non-Hodgkin lymphomas comprise many B-cell, T-cell, and other lymphoid cancers. Biopsy, immunophenotyping, and staging determine treatment and prognosis.

Hodgkin'S Disease

A cancer of the lymphatic system characterized by abnormal lymphocytes, including Reed-Sternberg cells.

Hodgkin'S Lymphoma

A B-cell lymphoma characterized by the presence of Reed-Sternberg cells (large, binucleate or multinucleate cells with owl-eye nucleoli) in a background of reactive inflammatory cells. See Hodgkin Lymphoma.

Hoffmann Syndrome

A rare adult manifestation of severe, chronic, untreated hypothyroid myopathy characterized by the clinical triad of proximal skeletal muscle weakness, painful muscle stiffness and cramping, and pseudohypertrophy of skeletal muscles, most conspicuously involving the gastrocnemius, deltoid, and forearm musculature, accompanied by marked delay in the relaxation phase of deep tendon reflexes ("hung-up" reflexes) and significantly elevated serum creatine kinase levels ($> 10\text{--}50\times$ normal). All symptoms, muscular pseudohypertrophy, and enzymatic elevations resolve completely following thyroid hormone replacement therapy with levothyroxine.

Alternate Names:

Hypothyroid Myopathy with Pseudohypertrophy (Adult Form)

Hofmann'S Syndrome

A clinical disorder related to hofmann's syndrome; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Holdemania

A genus of anaerobic bacteria found mainly in the intestinal microbiome and rarely associated with human infection.

Holdemania Filiformis

An anaerobic intestinal bacterium that is rarely isolated from human clinical specimens.

Holistic Medicine

Holistic Medicine is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

Holmium

Holmium is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Holmium Laser Enucleation Of The Prostate

A modern, minimally invasive transurethral endoscopic surgical procedure for the definitive treatment of bladder outlet obstruction secondary to benign prostatic hyperplasia. Utilizing a high-powered holmium:YAG laser fiber, the surgeon dissects and enucleates intact entire transition-zone adenomatous lobes along the surgical capsule of the prostate into the bladder lumen, followed by endoscopic mechanical morcellation and tissue evacuation for histopathologic evaluation. HoLEP provides size-independent adenoma clearance comparable to open prostatectomy with reduced intraoperative blood loss, minimal risk of transurethral resection syndrome, shorter catheterization times, and durable relief of lower urinary tract symptoms.

Alternate Names:

HoLEP; Holmium Laser Prostate Enucleation

Holocaine

Holocaine is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Holography

A method of recording and reconstructing the wavefront of light to produce a three-dimensional image; medical applications include experimental imaging, microscopy, and visualization.

Holoprosencephaly

A structural brain malformation resulting from incomplete or failed division of the forebrain (prosencephalon) into the cerebral hemispheres during the fifth week of gestation.

Holorachischisis

A severe neural-tube defect in which the spinal cord

and vertebral canal fail to close along an extensive length of the spine, usually causing major neurologic impairment.

Holosystolic Murmur Differential

Pansystolic (holosystolic) murmurs are plateau-shaped and extend from S1 to S2.

Holt-Oram Syndrome

An autosomal dominant disorder caused by TBX5 gene mutations, characterized by congenital heart disease and upper limb malformations. Cardiac anomalies: atrial septal defect (most common), VSD, and conduction abnormalities (heart block, atrial fibrillation).

Holter Electrocardiography

Continuous ambulatory ECG recording, usually for 24 hours or longer, used to detect intermittent arrhythmias, correlate symptoms with rhythm, and assess heart-rate patterns.

Holter Monitor

Continuous ECG recording for 24-48 hours to detect arrhythmias, quantify burden (atrial fibrillation, PVCs, PACs), and correlate symptoms with rhythm. Patient diary logs symptoms. Indications: palpitations, syncope, presyncope, and assessment of arrhythmia control. Reveals paroxysmal arrhythmias not captured on standard 12-lead ECG.

Holter Monitor (24H)

A portable ambulatory electrocardiographic recorder that continuously monitors heart rhythm, usually for 24 hours or longer, to detect intermittent arrhythmias and correlate them with symptoms.

Holter Monitoring

Portable, continuous electrocardiographic recording over an extended period, commonly 24 hours or longer, to detect intermittent rhythm abnormalities and correlate them with symptoms.

Homans Sign

Calf pain elicited by forced ankle dorsiflexion, his-

torically associated with deep-vein thrombosis but insufficiently accurate for diagnosis and not recommended as a standalone test.

Homeobox Gene

A gene containing a homeobox sequence that encodes a DNA-binding homeodomain and helps control embryonic patterning, organ development, and cell identity.

Homeodomain Proteins

Transcription factors containing a homeodomain that bind regulatory DNA sequences and direct developmental patterning and cell differentiation.

Homeopathy

An alternative-health system that uses highly diluted preparations selected according to “like cures like”; reliable evidence has not shown homeopathic products to treat serious disease beyond placebo effects.

Homeostasis

The maintenance of stable internal conditions despite external changes. Key regulated variables: temperature, pH, glucose, electrolytes, blood pressure, fluid balance. Mechanisms: negative feedback loops (most common), positive feedback (rare, e.g., labor, blood clotting).

Homeostasis, Psychological

Homeostasis, Psychological is a developmental, immunologic, reproductive, behavioral, environmental, or healthcare concept used in medicine or biomedical research.

Homeotherm

An organism that maintains a relatively stable internal body temperature despite changes in environmental temperature, largely through physiological regulation.

Homicide

The killing of one person by another, classified legally according to intent and circumstances; med-

ical assessment may address injury patterns, cause of death, toxicology, and safeguarding.

Homo Sapiens

The scientific name for modern humans, the species to which all living people belong.

Homocysteine

An amino acid formed during methionine metabolism. Persistently elevated blood levels may occur with genetic, nutritional, renal, or other disorders.

Homocystinuria

An autosomal recessive metabolic disorder caused by deficiency of cystathionine beta-synthase (CBS), leading to accumulation of homocysteine and methionine.

Homograft

Homograft is a developmental, immunologic, reproductive, behavioral, environmental, or healthcare concept used in medicine or biomedical research.

Homologous

Homologous is a developmental, immunologic, reproductive, behavioral, environmental, or healthcare concept used in medicine or biomedical research.

Homologous Recombination

A DNA-repair and genetic-exchange process that uses a matching or nearly matching sequence as a template to repair breaks or reshuffle genetic material.

Homologous Transplantation

Transplantation of cells, tissue, or an organ between genetically different people of the same species, requiring attention to compatibility, rejection, infection, and immunosuppression.

Homophile

Homophile is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

Homosexual

Describing a person who experiences enduring romantic or sexual attraction to people of the same sex; sexual orientation is not a mental disorder.

Homosexuality

Sexual or romantic attraction to people of the same sex or gender; it is not a mental disorder or pathology.

Homovanillic Acid

A breakdown product of dopamine found in cerebrospinal fluid, blood, and urine; levels may reflect dopamine metabolism but are interpreted cautiously because many factors affect them.

Homozygote

An individual or cell carrying two identical alleles at a particular gene locus, whether the alleles are normal, pathogenic, or another variant.

Homozygous

Having two identical alleles of a particular gene (AA or aa). Can be homozygous dominant or homozygous recessive.

Honey-Bee Stings

Honey-Bee Stings is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hong Kong Flu

Hong Kong Flu is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

Honolulu Fever

A clinical disorder related to honolulu fever; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hoof Wall

The hard keratinized outer covering of a hoof that

bears weight and protects the deeper tissues; disease or cracking can impair movement and cause pain in hoofed animals.

Hookworm

Intestinal nematode infection by *Ancylostoma duodenale* or *Necator americanus*, acquired by skin penetration of filariform larvae (walking barefoot on contaminated soil).

Hookworm Infection

Intestinal infection caused by hookworms acquired when larvae in contaminated soil penetrate the skin or are ingested. Chronic infection can cause abdominal symptoms, iron-deficiency anemia, protein loss, and impaired growth in children.

Hookworm Infestation

Intestinal infection by hookworm nematodes, which attach to the intestinal wall and may cause iron-deficiency anemia, abdominal symptoms, or poor growth.

Hordeolum

A tender, localized infection or inflammation of an eyelid gland, commonly called a sty, producing a painful red lump near the eyelid margin.

Hordeolum (Stye)

A tender, localized infection or inflammation of an eyelid gland that produces a small red lump near the eyelid margin.

Alternate Names:

Stye

Horizontal Gene Transfer

The transfer of genetic material between bacteria through mechanisms other than parent-to-offspring transmission. The three mechanisms are transformation (uptake of free DNA), transduction (transfer via bacteriophages), and conjugation (direct cell-to-cell transfer).

Hormonal Effects In Newborns

Transient newborn findings caused by maternal hor-

mones or normal neonatal endocrine transitions, such as breast enlargement or vaginal bleeding; persistent or severe findings require evaluation.

Hormone

Chemical messenger secreted by endocrine glands into blood, acting on distant target organs with specific receptors. Classes: peptides/proteins (insulin, GH), steroids (cortisol, estrogen), amines (thyroid hormone, catecholamines). Mechanisms: cell surface receptors (GPCRs, RTKs) or intracellular receptors (nuclear receptors).

Hormone Levels

Measurements of hormones in blood, urine, or other specimens used to evaluate endocrine function; interpretation depends on the specific hormone, timing, age, sex, medications, and clinical context.

Hormone Receptor

A protein that binds a specific hormone and converts the signal into cellular responses, either at the cell surface or inside the cell nucleus.

Hormone Replacement Therapy

HRT (menopause hormone therapy): estrogen ± progestin to relieve menopausal symptoms (vasomotor, vaginal atrophy, sleep, mood) and prevent osteoporosis. Regimens: estrogen-only (post-hysterectomy) or combined (with progestin for endometrial protection in uterus-intact women); routes: oral, transdermal, vaginal.

Hormone Replacement Therapy

Pharmacologic administration of exogenous hormones to supplement deficient endogenous levels. Most commonly refers to menopausal hormone therapy utilizing estrogens (with a progestogen in women with an intact uterus to prevent endometrial hyperplasia and adenocarcinoma) to manage moderate-to-severe vasomotor symptoms (hot flashes, night sweats) and genitourinary syndrome of menopause, as well as to prevent postmenopausal osteoporosis. Also encompasses hormone replacement in primary hypogonadism, hypopituitarism, hypothyroidism, and adrenal insufficiency.

Hormone Therapy

Treatment blocking hormone effects in hormone-sensitive cancers. Examples: tamoxifen (breast cancer), leuprolide (prostate cancer), aromatase inhibitors.

Hormone Therapy For Breast Cancer

Treatment that blocks or lowers estrogen or progesterone signaling in hormone-receptor-positive breast cancer.

Hormone Therapy For Prostate Cancer

Treatment that lowers androgen production or blocks androgen action to slow growth of hormone-sensitive prostate cancer.

Hormone-Receptor Positive

A cancer whose cells express receptors for hormones such as estrogen or progesterone. Hormone-blocking treatment may be useful for selected cancers with these receptors.

Hormone-Sensitive Lipase

Breaks down triglycerides in adipose tissue. Activated by glucagon, epinephrine via cAMP.

Hormones

Chemical messengers released by glands or tissues that regulate processes throughout the body.

Horn

A horn is a pointed, hard growth of keratinized tissue, usually on the skin or as a normal anatomical projection such as a uterine or spinal horn.

Horner Syndrome

A clinical triad caused by disruption of the sympathetic pathway to the eye and face, resulting in ipsilateral miosis (constricted pupil), ptosis (drooping of the upper eyelid due to paralysis of the superior tarsal muscle), and anhidrosis (loss of sweating on the affected side of the face).

Alternate Names:

Oculosympathetic Palsy

Horner'S Syndrome

A group of findings caused by disruption of sympathetic nerves, including a drooping eyelid, a small pupil, and reduced sweating on one side of the face.

Hornet Stings

Hornet stings inject venom and usually cause local pain and swelling; multiple stings or allergy can cause a serious systemic reaction.

Horripilation

Piloerection, or the temporary raising of hair and development of gooseflesh due to cold, fear, emotion, or autonomic stimulation.

Horror Autotoxicus

A historical immunology concept describing the supposed inability of the immune system to attack the body's own tissues; autoimmune disease showed that this idea was not universally correct.

Horseradish Peroxidase

An enzyme widely used as a laboratory reporter because it catalyzes color-producing or light-producing reactions when supplied with an appropriate substrate.

Horseshoe Kidney

The most common renal fusion anomaly (1 in 400-500), where the lower poles of both kidneys are fused across the midline by an isthmus of renal parenchyma or fibrous tissue.

Hospice

Palliative care for terminal illness with prognosis <6 months. Focus on comfort, quality of life, support for patient and family. Provided at home, hospice facility, or hospital. Interdisciplinary team: physician, nurse, social worker, chaplain, volunteer.

Hospice Care

Team-based care for a person nearing the end of life that prioritizes comfort, symptom control, communication, and family support rather than disease-directed cure.

Hospice Rehabilitation

Rehabilitation in hospice setting. Symptom management, positioning, caregiver education.

Hospital

A healthcare facility that provides inpatient and often emergency, surgical, diagnostic, and specialist care.

Hospital Medicine

The medical specialty focused on comprehensive care of hospitalized patients, including diagnosis, treatment, coordination across teams, transitions of care, and prevention of inpatient complications.

Hospital-Acquired Infection

Infection acquired during hospitalization. Also: healthcare-associated infection. Examples: CLABSI, CAUTI, surgical site infection.

Hospital-Acquired Infections

Infections that develop during or after hospitalisation. Major types include central line-associated bloodstream infections, catheter-associated urinary tract infections, ventilator-associated pneumonia, and surgical site infections. Risk factors include invasive devices, prolonged hospitalisation, and antimicrobial use.

Hospital-Acquired Pneumonia

Pneumonia developing > 48 hours after hospital admission. Common pathogens include *Pseudomonas* and MRSA. Empiric treatment covers resistant organisms.

Hospitalism

Hospitalism is a developmental, immunologic, reproductive, behavioral, environmental, or healthcare concept used in medicine or biomedical research.

Hospitalization

Admission to a hospital for monitoring, diagnosis, treatment, surgery, rehabilitation, or intensive support that cannot safely be delivered in an outpatient setting.

Hospitalization (Geriatric)

Inpatient stay for elderly patients. High risk for adverse outcomes: delirium, functional decline, institutionalization. Precautions: fall prevention, early mobilization, medication reconciliation, delirium prevention, nutrition support. Geriatrician consultation beneficial.

Alternate Names:

Geriatric

Hospitals As Health Educators

Hospitals educate patients, families, staff, and communities about prevention, treatment, medicines, recovery, and when to seek care.

Host

A person or other living organism that can be infected by a virus or other pathogen under natural conditions. The host may be a human, animal, or plant. In medical contexts, the host is the patient in whom an infection or disease develops.

Host Cell

A cell in which an infectious agent, parasite, or symbiont lives, replicates, or uses cellular machinery; the host-cell response influences disease and clearance.

Host Response

Immunological reaction to implant. Foreign body response, encapsulation, rejection.

Hostility

An attitude or behavior marked by antagonism, anger, distrust, or intent to oppose another person; persistent severe hostility may occur with psychiatric, neurologic, substance-related, or situational problems.

Hot Flash

A sudden episode of heat, flushing, sweating, and sometimes palpitations, commonly occurring during menopause.

Alternate Names:

Hot flush

Hot Flashes

Vasomotor symptoms: sudden sensations of intense heat, sweating, flushing, and palpitations, most commonly from estrogen withdrawal around menopause; also in hypogonadism, chemotherapy-induced ovarian failure, and some medications (SERMs, GnRH agonists). Mechanism: hypothalamic thermoregulatory dysfunction.

Alternate Names:

Menopause Hot Flashes (Hot Flashes)

Hot Flushes

Hot Flushes is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

Hot Tub Folliculitis

A temporary bacterial infection of hair follicles, commonly caused by *Pseudomonas aeruginosa* after exposure to contaminated hot tubs or pools, producing itchy follicular papules or pustules.

Hounsfield Unit

A CT attenuation value that expresses tissue density relative to water, conventionally assigning air about -1000 HU, water 0 HU, and dense bone positive values.

Hounsfield Units

CT number scale referencing water (0 HU) and air (-1000 HU). Fat: -50 to -100 HU. Water: 0 HU. Soft tissue: +20 to +50 HU. Bone: +400 to +3000 HU.

Hour

A unit of time equal to 60 minutes, commonly used in medicine to describe symptom duration, dosing intervals, monitoring periods, and treatment timing.

Hour-Glass Contraction

Hour-Glass Contraction is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

House Officer

A physician receiving postgraduate clinical training in a hospital, such as an intern, resident, or fellow; usage varies by country and institution.

House Staff

The physicians and other trainees who provide patient care within a hospital under the supervision of attending clinicians, commonly including interns, residents, and fellows.

House Surgeon

A hospital-based physician, historically a junior surgical trainee or resident responsible for inpatient surgical care; usage varies by institution.

Household Glue Poisoning

Household glue poisoning involves toxic or environmental exposure to household glue poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Housemaid's Knee

Housemaid's Knee is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

Howell-Jolly Bodies

Basophilic nuclear remnants (clusters of DNA) normally removed from circulating red blood cells by splenic macrophages (pitting). Presence on peripheral blood smear indicates functional asplenia or anatomical splenectomy (e.g., trauma, sickle cell auto-splenectomy). Also seen in severe megaloblastic anemia.

HPA

The hypothalamic-pituitary-adrenal axis, a hormone system central to the stress response.

Alternate Names:

Hypothalamic-pituitary-adrenal axis

HPLC

High-performance liquid chromatography, a labora-

tory method that pushes a liquid sample through a special column to separate, identify, and measure its components. It is used for medicines, metabolites, chemicals, and purity testing, including some radiopharmaceuticals.

HPV

Human papillomavirus is a group of DNA viruses that infect skin and mucous membranes. Some types cause warts, whereas persistent infection with high-risk types can cause precancerous lesions and cancers of the cervix, anus, penis, vulva, vagina, or oropharynx.

HPV Test

A test that detects high-risk human papillomavirus genetic material or related markers in cervical samples to screen for cervical cancer risk or evaluate abnormal cervical cytology.

HPylori

HPylori is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

HRCT

Alternate name; see *High-Resolution Computed Tomography*.

HRT

Hormone replacement therapy uses estrogen, with a progestogen when needed, to treat selected menopausal symptoms caused by declining ovarian hormone production. It can relieve vasomotor and genitourinary symptoms and help prevent bone loss, but benefits and risks depend on the person's age, medical history, regimen, and duration of use.

Hs-cTn

Alternate name; see *High-Sensitivity Cardiac Troponin*.

Ht

Common abbreviation for hematocrit, the percentage of blood volume occupied by red blood cells; in

other contexts it may mean height.

Huber Needle

Huber Needle is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Human Calcitonin

A peptide hormone produced by thyroid parafollicular cells that can lower blood calcium by inhibiting osteoclast activity; synthetic calcitonin is used selectively for bone and calcium disorders.

Human Chorionic Gonadotropin

A glycoprotein hormone produced by trophoblast cells after implantation. Detectable in serum 8-11 days post-conception and in urine 12-14 days. Levels double approximately every 48-72 hours in early pregnancy, peaking at 8-11 weeks then plateauing.

Human Chorionic Gonadotropin

A hormone produced mainly by placental tissue and measured in urine or blood to help detect and monitor pregnancy. Concentrations normally rise early in pregnancy, but the rate and interpretation vary; results must be considered with timing and clinical findings.

Alternate Names:

Human Chorionic Gonadotropin

Human Chromosome Count

Most human body cells contain 46 chromosomes arranged in 23 pairs, including one pair of sex chromosomes; mature egg and sperm cells normally contain 23.

Human Embryo

The developing human organism from fertilization through approximately the eighth week, during which the major body plan and organ systems are established.

Human Gastrin

A peptide hormone produced mainly by gastric G

cells that stimulates gastric acid secretion and supports growth of the gastric mucosa after food enters the stomach.

Human Genome

The complete DNA sequence and genetic content of a human, organized into nuclear chromosomes and mitochondrial DNA; variants influence traits, disease risk, and treatment response.

Human Herpesvirus 6

A herpesvirus that commonly infects children and can cause roseola; it may reactivate and cause serious disease in people with weakened immunity.

Human Herpesvirus 8

A herpesvirus associated with Kaposi sarcoma, primary effusion lymphoma, and multicentric Castleman disease, especially in people with weakened immunity.

Human Immunodeficiency Virus

A retrovirus belonging to the Lentivirus genus that causes acquired immunodeficiency syndrome (AIDS). HIV has two species: HIV-1 and HIV-2. The virus targets CD4+ T cells, macrophages, and dendritic cells. HIV has a complex genome encoding structural proteins, enzymes (reverse transcriptase, integrase, protease), and regulatory proteins.

Human Immunodeficiency Virus

A retrovirus (family Retroviridae, genus Lentivirus) that causes acquired immunodeficiency syndrome by infecting and destroying CD4+ T lymphocytes (helper T cells). Two types: HIV-1 (pandemic, most common worldwide) and HIV-2 (less virulent, West Africa).

Alternate Names:

Human Immunodeficiency Virus

Human Leukocyte Antigen

A protein on cell surfaces that helps the immune system distinguish self from foreign material.

Alternate Names:

HLA

Human Leukocyte Antigen

A group of cell-surface proteins encoded by genes in the major histocompatibility complex. HLA molecules present antigens to T cells and are important in immune responses and tissue matching for transplantation.

Alternate Names:

HLA

Human Metapneumovirus

A respiratory virus that can cause cough, fever, bronchiolitis, or pneumonia, especially in infants, older adults, and people with chronic disease.

Human Metapneumovirus

A medicine or chemical compound called human metapneumovirus (hmpv). Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Human Microbiome

The collection of microorganisms that live on and in the human body. The microbiome includes bacteria, viruses, fungi, and archaea. The gut microbiome is the most studied and plays a role in digestion, immune function, and disease.

Human Nociceptin

A neuropeptide that binds the nociceptin/orphanin FQ receptor and modulates pain, stress, reward, autonomic function, and other brain and spinal-cord processes.

Human Papillomavirus

The most common STI, with over 100 types. High-risk types (16, 18) cause cervical, oropharyngeal, anal, and genital cancers. Low-risk types (6, 11) cause genital warts. Prevention: HPV vaccine (Gardasil 9) protects against 9 HPV types. Screening: Pap smear and HPV testing for cervical cancer prevention.

Human Placental Lactogen

A peptide hormone produced by the placenta that contributes to mammary-gland development and

shifts maternal metabolism toward increased lipolysis and reduced insulin sensitivity during pregnancy. It supports nutrient availability for the fetus and contributes to physiologic gestational insulin resistance.

Humectant

A substance that attracts and retains water, commonly used to hydrate skin.

Humeral Epicondyle

A bony prominence at the distal humerus that provides attachment for forearm muscles and ligaments; the medial and lateral epicondyles are located on opposite sides of the elbow.

Humeral Fracture

A break in the upper-arm bone, involving the proximal, shaft, or distal humerus; pain, swelling, deformity, and possible nerve or blood-vessel injury guide urgent assessment and treatment.

Humeroradial Joint

Between capitulum and radial head, allowing flexion-extension and contributing to pronation-supination via radial head rotation. Radial collateral ligament provides lateral stability. Capitulum has articular cartilage only anteriorly.

Humeroulnar Joint

Uniaxial hinge between trochlea and trochlear notch of the ulna, allowing only flexion/extension (0 to approximately 145 degrees). Ulnar collateral ligament is the primary valgus restraint. Trochlear notch provides inherent bony stability, making it the most congruent elbow joint.

Humerus

Single upper arm bone, the longest of the upper limb. Proximal end: head, anatomical/surgical necks, tubercles. Shaft: deltoid tuberosity, radial groove, epicondyles. Distal end: trochlea, capitulum, olecranon fossa. The surgical neck is the common fracture site potentially injuring the axillary nerve.

Humidifier

A device that adds moisture to air; medical humidifiers require appropriate cleaning and water handling because contaminated equipment can disperse infectious or inflammatory material.

Humidifier Fever

A flu-like illness caused by inhalation of microorganisms or biologic contaminants from a contaminated humidifier or ventilation system, often involving fever, chills, cough, and malaise.

Humidifiers And Health

Humidifiers may ease dryness when used correctly, but dirty devices can spread microbes and excess humidity can promote mold; regular cleaning and moderate humidity are important.

Humidity

The amount of water vapor in air, influencing heat loss, respiratory comfort, skin hydration, and survival or transmission of some microorganisms.

Humor

In anatomy, a fluid of the body, especially the aqueous or vitreous humor of the eye; in ordinary language, it means a sense of amusement or comedy.

Humoral Immunity

The component of adaptive immunity mediated by antibodies produced by B lymphocytes (plasma cells) that neutralize extracellular pathogens and toxins. Humoral responses defend against bacteria, viruses outside cells, and parasites.

Humoralism

The historical medical theory that health depended on balance among four bodily fluids, or humors: blood, phlegm, yellow bile, and black bile.

Humorism

Another name for the ancient theory of humoral balance, which attributed disease to excess or deficiency of blood, phlegm, yellow bile, or black bile.

Hump On Upper Back (Dorsocervical Fat Pad)

A visible accumulation of fat at the back of the neck and upper shoulders, associated with obesity, corticosteroid excess, some medicines, or lipodystrophy; it differs from spinal kyphosis.

Hunchback

Hunchback is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

Hunger

Physiological drive to eat. Regulated by ghrelin, blood glucose, stomach distension. Psychological factors also influence.

Hungry Bone Syndrome

Severe hypocalcemia occurring after parathyroidectomy particularly in patients with severe preoperative hyperparathyroidism and high bone turnover. Rapid bone remineralization after removal of excess PTH sequesters calcium and phosphorus from the serum causing symptomatic hypocalcemia.

Hunter Syndrome

Mucopolysaccharidosis type II (MPS II), an X-linked recessive lysosomal storage disorder caused by iduronate-2-sulfatase deficiency (IDS gene mutations). Accumulation of dermatan sulfate and heparan sulfate (same as Hurler).

Hunter'S Syndrome

A clinical disorder related to hunter's syndrome; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Huntington'S Disease

An inherited, progressive neurodegenerative disease caused by an expanded CAG repeat in the HTT (huntingtin) gene on chromosome 4p16.3. It usually begins in adulthood and affects movement, thinking, mood, and behavior; involuntary movements called chorea are common. Symptoms may include cognitive decline and psychiatric changes. Earlier onset can occur in successive generations, especially with

paternal transmission. Because it is autosomal dominant, a person with the disease has a 50% chance of passing the altered gene to each child. There is no cure, but treatment can address symptoms and support function.

Alternate Names:

Huntington Disease Huntington's Chorea Chorea, Huntington's Huntington's Disease (Chorea) Chorea

Hurler Syndrome

Mucopolysaccharidosis type I (MPS I), the severe form, an autosomal recessive lysosomal storage disorder caused by alpha-L-iduronidase deficiency (IDUA gene mutations). Accumulation of dermatan sulfate and heparan sulfate leads to progressive multisystem involvement.

Hurthle Cell

A thyroid follicular cell with abundant granular mitochondria-rich cytoplasm, seen in Hashimoto thyroiditis and Hürthle-cell neoplasms.

Hurthle Cell Cancer

A rare thyroid cancer arising from Hurthle cells, with behavior ranging from indolent to aggressive.

Alternate Names:

Cancer, Hurthle Cell Oxyphil Cell Carcinoma

Hutchinson'S Teeth

Hutchinson's teeth are permanently notched, widely spaced upper front teeth classically associated with congenital syphilis.

Hürthle Cell Tumour

A clinical disorder related to hürthle cell tumour; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyaline

A glassy, translucent material or appearance, commonly describing proteinaceous deposits or homogeneous extracellular material seen microscopically in tissues.

Hyaline Cartilage

The most common type of cartilage, containing fine type II collagen fibers in a firm, glassy matrix. It supports airways, covers articular surfaces, and forms much of the fetal skeleton and growth plates.

Hyaline Casts

Transparent colorless cylindrical structures composed primarily of Tamm-Horsfall protein secreted by the thick ascending limb. Most common type of urine cast. Normal in concentrated urine and after exercise. Increased in dehydration, prerenal states, and volume depletion.

Hyaline Membrane

A layer of protein-rich material lining damaged alveoli, classically seen in respiratory distress syndrome. It impairs gas exchange.

Hyaline Membrane Disease

An older name for neonatal respiratory distress syndrome, caused mainly by surfactant deficiency in premature infants. See Respiratory Distress Syndrome.

Hyalitis

Hyalitis is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

Hyaloid Canal

A narrow channel through the vitreous body of the eye that marks the former path of the hyaloid artery from the optic disc toward the lens during fetal development.

Hyaluronic Acid

A naturally occurring carbohydrate polymer that binds water in connective tissue, skin, joints, and the eye.

Alternate Names:

Hyaluronan

Hyaluronidase

Hyaluronidase is a medicine, chemical, treatment,

device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hyaluronoglucosaminidase

An enzyme that degrades hyaluronan and related glycosaminoglycans; hyaluronidase preparations can increase tissue permeability and assist dispersion of injected fluids or medicines.

Hybridoma

A cell line formed by fusing an antibody-producing B lymphocyte with an immortal myeloma cell, allowing continuous production of a monoclonal antibody.

Hycanthone

An older antiparasitic drug formerly used against schistosomiasis; its use was abandoned or restricted because of serious toxicity, including liver injury and carcinogenic concern.

Hydatid Cyst

A cyst caused by the larval stage of *Echinococcus granulosus*, a tapeworm transmitted from dogs to humans through ingestion of contaminated food. The liver is the most commonly affected organ, followed by the lungs. Cysts grow slowly and may become symptomatic from mass effect or rupture, causing anaphylaxis.

Hydatid Disease

Cystic echinococcosis caused by the larval stage of *Echinococcus granulosus* and related species. Humans acquire infection by ingesting eggs shed in dog feces; hydatid cysts develop most often in the liver or lungs and may remain silent or cause pain, obstruction, infection, rupture, or allergic reaction. Diagnosis relies on characteristic imaging with supportive serology, and treatment must be individualized by specialists because unsafe puncture or rupture can cause anaphylaxis and secondary spread.

Hydatidiform Mole

A form of gestational trophoblastic disease in which abnormal trophoblastic tissue grows in the uterus. A complete mole has no viable fetus; a partial mole

may contain abnormal fetal tissue. Both require follow-up because persistent disease can occur.

Hydralazine

Hydralazine is a direct arteriolar vasodilator used for selected hypertension, including severe hypertension in pregnancy; reflex tachycardia, fluid retention, headache, and drug-induced lupus are recognized adverse effects.

Hydramnios

Excess amniotic fluid during pregnancy, also called polyhydramnios, caused by fetal swallowing impairment, maternal diabetes, multiple gestation, fetal anomalies, infection, or sometimes no identifiable cause.

Hydranencephaly

A rare congenital condition where the cerebral hemispheres are nearly completely absent, replaced by thin-walled, fluid-filled sacs, while the brainstem, basal ganglia, and cerebellum are relatively preserved. Caused by bilateral internal carotid artery occlusion in utero (vascular insult, infection, or twin-twin transfusion).

Hydrarthrosis

Abnormal accumulation of clear synovial fluid within a joint, producing swelling and restricted movement; causes include osteoarthritis, inflammatory arthritis, trauma, and infection.

Hydration

The body's water balance and the process of replacing water and electrolytes lost through urine, stool, sweat, breathing, vomiting, or bleeding.

Hydrazine

A highly reactive chemical used in fuels and industry that can cause irritant, neurologic, hepatic, and systemic toxicity after significant exposure.

Hydrazine Sulfate

A hydrazine-containing chemical once promoted experimentally for cancer-related symptoms; evidence

of benefit is inadequate and toxicity can include liver injury, neuropathy, and seizures.

Hydrazines

A family of nitrogen-containing chemical compounds used in industry, agriculture, and pharmaceuticals; some are irritants, toxic, mutagenic, or hepatotoxic.

Hydrazones

Chemical compounds formed by reaction of hydrazine derivatives with aldehydes or ketones, important in medicinal chemistry, analytical chemistry, and some drug structures.

Hydroa Vacciniiforme

A rare photodermatosis causing recurrent vesicles or papules on sun-exposed skin that heal with variciform scars; some cases are associated with Epstein-Barr virus.

Hydrocarbon

An organic compound composed only of carbon and hydrogen; ingestion or inhalation of petroleum hydrocarbons can cause chemical pneumonitis and systemic toxicity.

Hydrocarbon Pneumonia

Lung inflammation caused by inhaling or aspirating petroleum products or other hydrocarbons, which can produce cough, breathing difficulty, and chemical pneumonitis.

Hydrocele

A collection of serous fluid within the tunica vaginalis surrounding the testis. Congenital hydrocele results from patent processus vaginalis; acquired hydrocele may follow infection, trauma, or tumor. Transilluminates on examination. Ultrasound confirms fluid collection.

Hydrocele Repair

Surgery to close or remove the fluid-containing sac around a testicle when a hydrocele is persistent, large, painful, associated with a hernia, or obscures examination.

Hydrocephalus

Accumulation of cerebrospinal fluid within the ventricular system, leading to increased intracranial pressure. Communicating hydrocephalus: impaired CSF absorption (subarachnoid hemorrhage, meningitis, leptomeningeal carcinomatosis).

Hydrocephalus Acquired

Acquired hydrocephalus is abnormal accumulation of cerebrospinal fluid after birth due to obstruction, impaired absorption, or rarely excess production; raised intracranial pressure may cause headache, vomiting, gait or cognitive change, and visual symptoms.

Hydrocephalus Ex-Vacuo

Enlargement of the brain's ventricles secondary to loss or shrinkage of surrounding brain tissue, rather than primarily from obstructed cerebrospinal-fluid flow.

Hydrocephaly

Hydrocephalus: abnormal accumulation or circulation of cerebrospinal fluid that enlarges the ventricles and may increase pressure on the brain.

Hydrochloric Acid

The acid in gastric juice that helps digest food and kill microorganisms.

Hydrochloric Acid Poisoning

Hydrochloric acid poisoning involves toxic or environmental exposure to hydrochloric acid poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hydrochlorothiazide

Hydrochlorothiazide is a thiazide diuretic that inhibits sodium-chloride reabsorption in the distal convoluted tubule and treats hypertension and mild oedema; hypokalaemia, hyponatraemia, hyperuricaemia, and photosensitivity may occur.

Hydrocodone

An opioid analgesic used for moderate to severe pain

and, in some formulations, cough; it can cause respiratory depression, sedation, constipation, tolerance, dependence, and overdose.

Hydrocodone And Acetaminophen Overdose

Hydrocodone and acetaminophen overdose involves toxic or environmental exposure to hydrocodone and acetaminophen overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hydrocodone Bitartrate

The bitartrate salt of hydrocodone, an opioid analgesic that can cause sedation, constipation, respiratory depression, tolerance, dependence, and overdose.

Hydrocodone/Oxycodone Overdose

Hydrocodone/oxycodone overdose involves toxic or environmental exposure to hydrocodone/oxycodone overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hydrocolpos

Distension of the vagina with fluid due to obstruction of the vaginal outlet, often from an imperforate hymen or transverse septum, presenting as a pelvic mass in neonates or adolescents.

Hydrocortisone

Hydrocortisone is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hydrocyanic Acid Poisoning

Hydrocyanic Acid Poisoning is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hydroflumethiazide

A medicine or chemical compound called hydroflumethiazide. Its uses, effects, interactions, and

risks depend on the specific substance and clinical indication.

Hydrofluoric Acid Exposure

Contact with hydrofluoric acid, which can cause deep tissue injury and systemic fluoride toxicity, including dangerous low calcium and cardiac arrhythmias.

Hydrofluoric Acid Poisoning

Hydrofluoric acid poisoning involves toxic or environmental exposure to hydrofluoric acid poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hydrogen Bond

Weak electrostatic interaction between a hydrogen atom covalently bonded to an electronegative atom (N, O, F) and another electronegative atom. Critical for protein structure (alpha helix, beta sheet), DNA double helix (base pairing), enzyme-substrate binding, and water properties (high boiling point, solvent properties).

Hydrogen Breath Test

Non-invasive test for carbohydrate malabsorption (lactose, fructose) and small intestinal bacterial overgrowth (SIBO). Patient ingests a substrate (lactose, glucose, or lactulose). Breath hydrogen is measured at baseline and 15-30 minutes for 2-3 hours. Rise in H₂ >20 ppm from baseline: positive (bacterial fermentation).

Hydrogen Cyanide

A rapidly acting cellular toxin that inhibits cytochrome oxidase and prevents oxidative phosphorylation; significant exposure can cause severe lactic acidosis, seizures, cardiovascular collapse, and death.

Hydrogen Peroxide

An oxidizing chemical used in some antiseptic and industrial products; ingestion or concentrated exposure can injure tissue and release gas, causing embolic or respiratory complications.

Hydrogen Peroxide Poisoning

Hydrogen peroxide poisoning involves toxic or environmental exposure to hydrogen peroxide poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hydrogen Sulfide

A toxic, flammable gas with a rotten-egg odor at low concentrations that inhibits cellular respiration and can rapidly cause respiratory failure, neurologic effects, or death.

Hydrogen-Ion Concentration

The concentration of hydrogen ions in a solution, expressed through pH and central to acid-base physiology, enzyme activity, and interpretation of blood-gas results.

Hydrogenation

A chemical process that adds hydrogen to a compound; in foods it can convert liquid oils into more solid fats.

Hydromorphone

Hydromorphone is a potent opioid analgesic for severe pain; sedation, constipation, respiratory depression, tolerance, and dependence are important risks, especially with alcohol or other sedatives.

Hydromorphone Overdose

Hydromorphone overdose involves toxic or environmental exposure to hydromorphone overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hydromyelia

Abnormal widening of the central canal of the spinal cord, which may be congenital or associated with obstruction of cerebrospinal-fluid flow and neurologic symptoms.

Hydronephrosis

Dilation of renal collecting system proximal to obstruction. Causes include stones, strictures, malignancy, and BPH. Graded on ultrasound from mild

to severe. Prolonged obstruction leads to cortical thinning and renal failure. Bilateral more clinically significant.

Hydronephrosis Of One Kidney

Dilation of one kidney's renal pelvis and calyces caused by impaired urine drainage, often from a stone, stricture, reflux, or obstruction of the ureter or ureteropelvic junction.

Hydrophobia

Painful spasms or fear triggered by attempts to drink, classically occurring in advanced rabies because of inflammation of the brain and nerves involved in swallowing.

Hydropneumothorax

The simultaneous presence of both fluid and air in the pleural space, causing respiratory distress and dullness to percussion with absent breath sounds, typically requiring chest tube drainage.

Hydrops

Abnormal accumulation of fluid in tissues or body cavities. The term may refer to generalized fetal edema, a localized collection, or organ-specific fluid distension; the cause depends on the clinical setting.

Hydrops Fetalis

Abnormal accumulation of fluid in two or more fetal compartments, including pleural effusion, pericardial effusion, ascites, and subcutaneous edema. Causes include Rh isoimmunization, chromosomal abnormalities, cardiac defects, infections (parvovirus B19), and twin-to-twin transfusion syndrome.

Hydrops Of The Gallbladder

Distension of the gallbladder by clear mucus or watery fluid, usually caused by prolonged obstruction of the cystic duct by a gallstone. It may cause right-upper-quadrant pain and can progress to inflammation or infection.

Hydrosalpinx

Dilatation of a fallopian tube caused by accumulation of clear fluid, usually after inflammation, infection, endometriosis, or previous pelvic surgery. It may affect fertility.

Hydrotherapy

Exercise in water using buoyancy, resistance, warmth. Benefits: reduced weight-bearing, improved circulation, pain relief, increased range of motion. Used for arthritis, spinal cord injury, amputation, neurological conditions.

Hydrothorax

An abnormal collection of clear fluid in the pleural space around a lung.

Hydroureter

Abnormal enlargement of a ureter caused by impaired urine drainage.

Hydroxamic Acid

An organic functional group containing hydroxylamine attached to a carbonyl, found in chelators and some medicines, including histone-deacetylase inhibitors.

Hydroxocobalamin

A cobalt-containing form of vitamin B12 used as an antidote for cyanide poisoning. It binds cyanide to form cyanocobalamin, which is excreted in urine; it can cause red discoloration of skin and body fluids and interfere with some laboratory tests.

Hydroxyapatite

A calcium-phosphate mineral forming much of the hardness of bones and teeth.

Hydroxychloroquine

Hydroxychloroquine is an antimalarial and immunomodulatory drug used for malaria prevention or treatment and selected autoimmune diseases; cumulative retinal toxicity requires ophthalmic monitoring.

Hydroxycholesterols

A medicine or chemical compound called hydroxycholesterols. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Hydroxylation

An enzymatic or chemical reaction that adds a hydroxyl group to a molecule, often altering the activity, solubility, metabolism, or excretion of a drug or biomolecule.

Hydroxyproline

An amino acid formed mainly by post-translational modification of collagen; its measurement has been used as an indicator of collagen turnover but is not a routine diagnostic test.

Hydroxyprolinemia

A rare inherited metabolic finding caused by impaired hydroxyproline breakdown, leading to elevated hydroxyproline in blood or urine and usually little clinical illness.

Hydroxysteroid

A steroid hormone or steroid-related compound containing a hydroxyl group; hydroxysteroids include intermediates and products of adrenal and gonadal steroid metabolism.

Hydroxyurea

Hydroxyurea inhibits ribonucleotide reductase and reduces DNA synthesis; it is used for myeloproliferative disorders and to increase fetal haemoglobin in sickle-cell disease, with myelosuppression and other toxicities requiring monitoring.

Hydroxyzine

Hydroxyzine is a sedating H1-antihistamine used for itching, anxiety, and preoperative sedation; drowsiness, anticholinergic effects, and QT prolongation can occur.

Hydroxyzine Overdose

Hydroxyzine overdose involves toxic or environmen-

tal exposure to hydroxyzine overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Hydrozoa

Hydrozoa is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Hygiene

Practices that reduce exposure to pathogens and support health, including hand hygiene, oral care, skin care, food safety, and appropriate environmental cleanliness.

Hygiene Hypothesis

The theory that limited early-life microbial exposure may increase susceptibility to allergies and immune disorders.

Hygroma

Hygroma is a developmental, immunologic, reproductive, behavioral, environmental, or healthcare concept used in medicine or biomedical research.

Hymen

A thin fold of mucous membrane that partially covers the external opening of the vagina. The hymen is a remnant of fetal development, with no known physiological function in postnatal life.

Hymenectomy

Hymenectomy is a procedure to remove or open part of the hymen, usually to treat an obstructed hymenal opening or another symptomatic abnormality.

Hymenolepiasis

Intestinal infection with dwarf tapeworms, especially *Hymenolepis nana*, acquired by ingesting eggs; it may cause abdominal symptoms, diarrhea, or be asymptomatic.

Hymenolepis Diminuta

hymenolepis diminuta is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Hymenolepis Nana

hymenolepis nana is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Hymenoptera

An insect order that includes bees, wasps, hornets, and ants; stings or bites can cause pain, allergy, or life-threatening anaphylaxis.

Hyoid Bone

U-shaped bone in the anterior neck at C3, unique for not directly articulating with any other bone. Suspended by muscles and the stylohyoid ligament. Provides attachment for suprahyoid (digastric, stylohyoid, mylohyoid, geniohyoid) and infrahyoid (sternohyoid, sternothyroid, thyrohyoid, omohyoid) muscles.

Hyoscine

Hyoscine is a medicine, chemical, treatment, device, or exposure; its indication, mechanism, dose, toxicity, and precautions depend on the specific product or circumstance.

Hyoscyamine

An antimuscarinic medicine that reduces smooth-muscle spasm and secretions; it may be used for selected gastrointestinal or urinary symptoms and can cause dry mouth, blurred vision, urinary retention, or confusion.

Hyper-IgM Syndromes

A group of disorders characterized by normal or elevated IgM with low IgG, IgA, and IgE, due to defective class switch recombination (CSR) and somatic hypermutation (SHM).

Hyper-IgM Syndrome

See *Hyper-IgM Syndromes*.

Hyperactivity

Excessive or developmentally inappropriate motor activity, restlessness, or difficulty remaining still; persistent impairment may occur in ADHD, mania, substance effects, or other conditions.

Hyperacusis

Reduced tolerance to ordinary environmental sounds, which may be perceived as excessively loud, painful, or distressing. It can occur with hearing disorders, migraine, head injury, neurologic conditions, or without an identifiable cause.

Hyperadrenocorticism

Excess production or exposure to adrenal glucocorticoids, especially cortisol; in humans it causes Cushing syndrome, which may result from adrenal, pituitary, ectopic, or medication-related causes.

Hyperaemia

A clinical disorder related to hyperaemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyperaesthesia

A clinical disorder related to hyperaesthesia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyperaldosteronism

Hyperaldosteronism, also known as Conn syndrome when caused by an adrenal adenoma, is characterized by excessive aldosterone production causing hypertension, hypokalemia, and metabolic alkalosis.

Alternate Names:

Conn syndrome when caused by an adrenal adenoma, is.

Hyperaldosteronism Complications

Beyond hypertension and hypokalemia, primary hyperaldosteronism causes target organ damage in-

cluding left ventricular hypertrophy, fibrosis, stroke, coronary artery disease, atrial fibrillation, and chronic kidney disease. These complications occur at higher rates than in essential hypertension of similar severity.

Hyperalgesia

Feeling more pain than expected from something that normally hurts. It can happen with inflammation, nerve injury, changes in nervous-system pain processing, or opioid use; it differs from allodynia, which is pain from a normally painless touch.

Hyperammonemia

An abnormally elevated concentration of ammonia in the blood. It may result from an inherited urea-cycle or organic-acid disorder, severe liver dysfunction, or other acquired causes and can produce encephalopathy, cerebral oedema, seizures, and coma.

Hyperamylasemia

An elevated blood amylase concentration caused by pancreatic or salivary-gland disease, reduced clearance, macroamylasemia, or other conditions; it is not diagnostic of pancreatitis by itself.

Hyperargininemia

An inherited urea-cycle disorder caused by arginase deficiency, leading to high arginine and progressive spasticity, developmental impairment, seizures, or liver-related findings.

Hyperbaric

Having a pressure greater than the surrounding atmospheric pressure; hyperbaric oxygen therapy delivers oxygen at increased pressure for selected medical indications.

Hyperbaric Chamber

A pressurized chamber in which a person breathes oxygen or another gas at pressure above sea level, used for conditions such as decompression sickness, air embolism, and selected wounds.

Hyperbaric Oxygen Therapy

Administration of 100% oxygen at increased atmospheric pressure. Indications: carbon monoxide poisoning, decompression sickness, gas gangrene, necrotizing soft tissue infections, delayed radiation injury, crush injury, compromised grafts. Mechanism: increases dissolved oxygen, vasoconstriction, antimicrobial effect.

Hyperbilirubinaemia

Hyperbilirubinaemia is an abnormally high level of bilirubin in the blood, which can cause jaundice and may result from increased breakdown of red cells, liver disease, or blocked bile flow.

Hyperbilirubinemia

An abnormally high concentration of bilirubin in blood that may cause jaundice; causes include hemolysis, impaired liver processing, hepatocellular injury, and bile-duct obstruction.

Hypercalcaemia

A clinical disorder related to hypercalcaemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypercalcemia

Elevated serum calcium level (total calcium >10.5 mg/dL or ionized calcium >5.3 mg/dL). mnemonic MALIGNANT: Myeloma, Metastases, Metabolic bone disease, Hyperparathyroidism, Adrenal insufficiency, Paget disease, Immobilization, Thiazide diuretics, Addison disease, Neonatal, Tumor-derived PTHrP.

Alternate Names:

Calcium Elevated (Hypercalcemia)

Hypercalcemia Of Malignancy

Elevated serum calcium caused by cancer, a common paraneoplastic and oncologic emergency. Mechanisms: PTHrP secretion (most common—squamous cell lung, breast, renal, ovarian), osteolytic metastases (breast, myeloma), 1,25-dihydroxyvitamin D (lymphoma), and ectopic PTH (rare).

Hypercalciuria

Excessive urinary calcium excretion, which can promote calcium kidney stones and, in some settings, bone mineral loss; evaluation considers diet, medications, kidney function, and endocrine causes.

Hypercapnia

An abnormally elevated arterial partial pressure of carbon dioxide, usually caused by alveolar hypoventilation. It may produce headache, somnolence, confusion, asterixis, respiratory acidosis, and, when severe or rapidly developing, coma.

Hypercementosis

Excessive deposition of cementum around a tooth root, producing a bulbous root contour that can complicate extraction but is often asymptomatic.

Hypercholesterolaemia

A clinical disorder related to hypercholesterolaemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypercholesterolemia

Elevated serum total cholesterol and LDL cholesterol levels, a major modifiable risk factor for atherosclerotic cardiovascular disease. It may be primary, due to familial hypercholesterolemia from LDL receptor mutations, or secondary to diet, obesity, hypothyroidism, or nephrotic syndrome.

Hypercoagulable State

Condition increasing thrombosis risk. Inherited: Factor V Leiden, prothrombin mutation, protein C/S deficiency, antithrombin deficiency. Acquired: cancer, pregnancy, oral contraceptives, antiphospholipid syndrome, myeloproliferative disorders.

Hypercyanotic Tetralogy

A complex congenital heart defect with four features: ventricular septal defect, overriding aorta, right ventricular outflow tract obstruction (pulmonary stenosis), and right ventricular hypertrophy. It causes right-to-left shunting and cyanosis. Treatment is surgical repair, typically in infancy.

Hyperelastic Skin

Skin that stretches farther than usual and recoils abnormally; generalized hyperextensibility may occur in connective-tissue disorders such as Ehlers-Danlos syndromes.

Hyperemesis

Excessive or persistent vomiting that may cause dehydration, electrolyte abnormalities, and nutritional deficiency.

Hyperemesis Gravidarum

Severe, persistent nausea and vomiting during pregnancy resulting in weight loss greater than 5%, dehydration, electrolyte imbalances, and ketosis. Occurs in 0.3-2% of pregnancies, typically presenting in the first trimester. Differentiated from normal morning sickness by severity and metabolic derangement.

Hyperemia

Increased blood flow to a tissue, causing redness and warmth; it may be an active physiologic response to exercise or inflammation or a reactive response after temporary ischemia.

Hyperemia Grade

A graded clinical or endoscopic estimate of tissue redness and vascular congestion, used to describe the severity of inflammation or mucosal irritation.

Hyperesthesia

Increased sensitivity to sensory stimulation—touch (tactile), pain (hyperalgesia), or other stimuli—so that normally non-painful or mildly painful stimuli produce exaggerated discomfort.

Hyperextension

Movement that extends a joint beyond its usual neutral or anatomical range, potentially stretching or tearing ligaments, capsules, muscles, or nerves.

Hyperfibrinogenemia

An abnormally high blood fibrinogen concentration, commonly reflecting inflammation, infection, pregnancy, obesity, or estrogen exposure and contribut-

ing to a prothrombotic state.

Hyperflexion

Excessive bending of a joint beyond its normal range, which can injure ligaments, discs, muscles, or other supporting structures.

Hypergammaglobulinemia

An excess of gamma globulins, usually immunoglobulins, in blood caused by polyclonal immune activation or a monoclonal plasma-cell or B-cell disorder.

Hyperglycaemia

A clinical disorder related to hyperglycaemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyperglycemia

Blood glucose above the expected range for the testing conditions. Acute hyperglycemia can cause osmotic diuresis, dehydration, and electrolyte disturbances. Chronic hyperglycemia leads to microvascular (retinopathy, nephropathy, neuropathy) and macrovascular complications.

Hyperglycemia - Infants

An abnormally high blood glucose concentration in an infant, which may result from prematurity, stress, infection, medications, excessive glucose delivery, or endocrine or metabolic disease.

Hyperglycinemia

An abnormally high glycine concentration in blood, most notably in nonketotic hyperglycinemia, a disorder of glycine cleavage that can cause neonatal hypotonia, seizures, and developmental impairment.

Hypergonadotropic Hypogonadism

Primary testicular failure with elevated LH/FSH. Causes: Klinefelter syndrome, cryptorchidism, orchitis, chemotherapy, radiation.

Hypergraphia

Excessive, compulsive, or unusually prolonged writing or drawing, sometimes associated with temporal-

lobe epilepsy, mania, or other neurologic or psychiatric conditions.

Hyperhidrosis

Excessive sweating beyond what is needed for thermoregulation. palms, soles, axillae, and face without underlying cause. Secondary generalized hyperhidrosis is caused by medications, endocrine disorders, infections, or malignancies.

Alternate Names:

Palm Sweating Excessive (Hyperhidrosis) Sweating, Abnormally Excessive Sole Sweating Excessive (Hyperhidrosis)

Hyperhomocysteinemia

An elevated blood homocysteine concentration associated with genetic enzyme defects, folate or vitamin B12 deficiency, kidney disease, medicines, and increased vascular or pregnancy-related risk in selected settings.

Hypericum Perforatum

St. John's wort, a plant used in some complementary products for depressive symptoms; it can cause drug interactions by inducing hepatic enzymes and transporters and should not be combined casually with medicines.

Hyperimmunization

Hyperimmunization is an unusually strong or repeated immune exposure, sometimes referring to an over-vaccinated state or production of high-titer antibodies.

Hyperimmunoglobulin D Syndrome

A hereditary autoinflammatory disorder, usually caused by mevalonate kinase deficiency, characterized by recurrent fever, enlarged lymph nodes, abdominal symptoms, rash, and joint or muscle pain; it is also called HIDS or MKD.

Hyperimmunoglobulin E Syndrome

A rare primary immunodeficiency characterized by very high IgE, eczema, recurrent skin and lung infections, and sometimes skeletal or dental abnormalities.

Hyper-IgE Syndrome

See *Hyperimmunoglobulin E Syndrome*.

Hyperinsulinemia

An abnormally high concentration of insulin in the blood.

Hyperinsulinism

Excessive insulin secretion or action causing recurrent or persistent hypoglycemia; causes include congenital disorders, tumors, medications, and critical illness.

Hyperkalaemia

British spelling of Hyperkalemia; see Hyperkalemia.

Hyperkalemia

Serum potassium >5.0 mEq/L. ECG changes: peaked T waves, widened QRS, sine wave. Causes: renal failure, medications (ACE inhibitors, potassium-sparing diuretics), tissue breakdown, acidosis.

Hyperkalemic Periodic Paralysis

An inherited skeletal-muscle channelopathy causing episodic weakness or paralysis, often triggered by rest after exercise or potassium exposure, with attacks associated with elevated or rapidly changing serum potassium.

Hyperkeratosis

Thickening of the outer keratin layer of skin or mucosa, producing callus, scale, or rough plaques in response to friction, inflammation, infection, inherited disorders, or other causes.

Hyperkinesia

Excessive or involuntary movement, such as chorea, dystonia, tics, or tremor, resulting from abnormal activity in motor-control circuits or from drugs and other diseases.

Hyperlipasaemia

An abnormally high blood lipase concentration, often associated with pancreatic inflammation but

also seen with kidney failure, gastrointestinal disease, medications, and other conditions.

Hyperlipidaemia

A clinical disorder related to hyperlipidaemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyperlipidemia

Elevated levels of lipids in the blood—total cholesterol, LDL, triglycerides, and/or reduced HDL—a major modifiable risk factor for atherosclerosis and cardiovascular disease. Primary (genetic: familial hypercholesterolemia) or secondary (hypothyroidism, diabetes, nephrotic syndrome, obesity, alcohol).

Hyperlipoproteinemia

An excess of one or more lipoproteins in blood, producing high cholesterol or triglycerides and increasing risk of atherosclerosis, pancreatitis, or inherited lipid disease.

Hyperlordosis

Exaggerated inward curvature of the lumbar spine, often producing a swayback appearance and sometimes associated with pain, muscular imbalance, obesity, pregnancy, or structural spinal disease.

Hyperlordotic Posture

An exaggerated inward curvature of the lumbar spine, often producing increased anterior pelvic tilt and a prominent lower-back arch.

Hyperlysinemia

An inherited metabolic abnormality with elevated lysine in blood caused by altered lysine degradation; many people have few symptoms, but some have developmental or neurologic findings.

Hypermagnesemia

An abnormally high concentration of magnesium in the blood, often related to kidney failure or excessive magnesium intake; severe cases can cause weakness, low blood pressure, depressed reflexes, or

arrhythmias.

Hypermetropia

A clinical disorder related to hypermetropia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypermobile Ehlers-Danlos Syndrome

A heritable connective-tissue disorder characterized mainly by generalized joint hypermobility, instability, chronic musculoskeletal pain, and other variable features involving skin, autonomic function, or digestion. Diagnosis is clinical and requires assessment for other connective-tissue disorders and related conditions.

Hypermobile Joints

A clinical disorder related to hypermobile joints; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypermobility Syndrome

Excessive joint movement associated with pain, instability, fatigue, or other symptoms; evaluation distinguishes generalized joint hypermobility from a defined hypermobility spectrum disorder or hypermobile Ehlers-Danlos syndrome.

Hypermotility

A clinical disorder related to hypermotility; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypernatraemia

A clinical disorder related to hypernatraemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypernatremia

A serum sodium concentration above the reference range, usually reflecting too little free water relative to body sodium or, less often, excess sodium. Causes include impaired water intake, diabetes insipidus, osmotic diuresis, gastrointestinal losses, and sodium administration.

Hypernatremia Result

A high blood sodium result, usually reflecting too little free water relative to body sodium; severity and rate of onset determine neurologic risk and the urgency and speed of correction.

Hyperopia

A refractive error in which light focuses behind the retina when the eye is relaxed. See Farsightedness.

Hyperornithinemia

An elevated blood ornithine concentration caused by defects in ornithine metabolism or transport, including disorders that can produce retinal degeneration or neurologic disease.

Hyperosmolar

Having a higher concentration of osmotically active particles than a reference fluid; increased plasma osmolality draws water out of cells and can impair neurologic function.

Hyperosmolar Hyperglycemic State

Hyperglycemic emergency characterized by severe hyperglycemia greater than 600 milligrams per deciliter, hyperosmolality greater than 320 milliosmoles per kilogram, and profound dehydration without significant ketosis.

Hyperosmolar Non-Ketotic Coma

A clinical disorder related to hyperosmolar non-ketotic coma; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyperostosis

Abnormal excessive formation or thickening of bone, which may be localized or widespread and may occur in skeletal dysplasias, inflammation, endocrine disease, or medication effects.

Hyperoxaluria

Excess oxalate production or absorption leading to high urinary oxalate, calcium oxalate stones, nephrocalcinosis, and in severe inherited forms progressive kidney injury.

Hyperoxia

Exposure to an abnormally high oxygen concentration or tissue oxygen level, which can cause oxygen toxicity, absorption atelectasis, retinal injury in premature infants, or lung injury depending on dose and duration.

Hyperparakeratosis

Excessive retention of nuclei in the outer keratin layer of skin or mucosa, producing thickened scale and commonly reflecting inflammation, irritation, or a keratinization disorder.

Hyperparathyroidism

Excessive secretion of parathyroid hormone leading to hypercalcemia. Classified as primary (parathyroid adenoma or hyperplasia), secondary (chronic kidney disease, vitamin D deficiency), or tertiary (autonomous PTH secretion after prolonged secondary hyperparathyroidism).

Hyperpathia

A clinical disorder related to hyperpathia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyperphagia

An abnormally increased, voracious appetite and pathological compulsion to consume excessive quantities of food (polyphagia), resulting in caloric intake far exceeding physiological requirements. Primarily mediated by dysregulation of hypothalamic satiety circuits: damage to the ventromedial hypothalamus (the satiety center) from craniopharyngiomas, trauma, or radiation causes intractable hyperphagia and rapid-onset hypothalamic obesity. Also classic for genetic obesity syndromes—most notably Prader-Willi syndrome (caused by loss of paternal expression on chromosome 15q11–q13, leading to lack of satiety and severe early-childhood hyperphagia)—as well as monogenic leptin/leptin-receptor deficiencies, uncontrolled diabetes mellitus, and atypical depression.

Hyperphosphatemia

A phosphate concentration above the laboratory ref-

erence range, often caused by impaired renal excretion, hypoparathyroidism, tumor lysis, rhabdomyolysis, or excess intake. Consequences may include secondary hyperparathyroidism, vascular/metastatic calcification, and when combined with hypocalcemia, tetany.

Hyperpigmentation

Darkening of the skin caused by increased melanin production, deposition of pigment in the dermis, or accumulation of other pigments.

Hyperpituitarism

Excess production of one or more pituitary hormones, most often from a pituitary adenoma, causing disorders such as acromegaly, Cushing disease, hyperprolactinemia, or precocious puberty.

Hyperplasia

An increase in the number of cells in a tissue or organ, resulting in enlargement while cellular architecture may remain relatively preserved. It may be physiologic, such as hormonal endometrial proliferation, or pathologic, such as benign prostatic hyperplasia or endocrine gland hyperplasia. Hyperplasia differs from hypertrophy, which is enlargement caused primarily by increased cell size rather than increased cell number. The clinical significance depends on the cause, distribution, degree of proliferation, cellular atypia, and whether the process can regress after the stimulus is removed.

Hyperplastic Polyp

A usually benign overgrowth of mucosal cells, often found in the colon or rectum.

Hyperpnea

Breathing that is deeper than normal and may also be faster, increasing overall ventilation.

Hyperpnoea

A clinical disorder related to hyperpnoea; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyperpolarization

A change in membrane potential to a more negative value, moving it further from threshold and making the cell less excitable. Occurs when potassium leaves the cell (opening of voltage-gated potassium channels, as in the afterhyperpolarization after an action potential) or when chloride enters (GABA-A receptor activation).

Hyperprolactinaemia

An abnormally elevated concentration of prolactin in the blood. Causes include medicines, pregnancy, hypothyroidism, pituitary tumors, and other disorders.

Hyperprolactinemia

Elevated serum prolactin levels most commonly caused by prolactin-secreting pituitary adenomas termed prolactinomas. Other causes include medications particularly antipsychotics and metoclopramide, hypothyroidism, stalk effect from non-functioning pituitary adenomas, renal failure, and chest wall lesions.

Hyperproteinemia

An abnormally high total blood-protein concentration, usually from dehydration, chronic inflammation, or a monoclonal immunoglobulin disorder.

Hyperpyrexia

An extreme elevation of body temperature, generally above 41.5 °C, that can result from severe infection, heatstroke, malignant hyperthermia, drug toxicity, or other life-threatening conditions.

Hyperreactio Luteinalis

Bilateral ovarian enlargement with multiple theca-lutein cysts caused by an exaggerated response to human chorionic gonadotropin, usually during pregnancy or with trophoblastic disease.

Hyperreflexia

Exaggerated or overactive deep-tendon (stretch) reflexes. It is a neurologic sign, not a diagnosis, and often reflects impaired descending upper-motor-neuron control; it may occur with spasticity, clonus,

or an extensor plantar response. Distribution and associated findings help distinguish central nervous-system disease from systemic or toxic causes.

Hypersensitivity

An exaggerated or inappropriate immune response to an antigen, ranging from a mild reaction to anaphylaxis.

Hypersensitivity Pneumonitis

An immune-mediated lung disease from environmental antigen inhalation (molds, bird proteins). Acute, subacute, and chronic forms exist. HRCT shows ground-glass opacities, air trapping, and centrilobular nodules. Chronic HP progresses to fibrosis.

Hypersensitivity Reactions

Inappropriate or excessive immune responses causing tissue damage or disease.

Hypersensitivity Type I

Type I hypersensitivity, also known as immediate hypersensitivity or allergic reactions, is mediated by IgE antibodies and mast cells, producing symptoms within minutes of allergen re-exposure.

Alternate Names:

immediate hypersensitivity or allergic reactions,.

Hypersensitivity Type II

Type II hypersensitivity, also known as cytotoxic hypersensitivity, is mediated by IgG or IgM antibodies directed against antigens on cell surfaces or the extracellular matrix, causing tissue damage through complement activation, antibody-dependent cellular cytotoxicity, and phagocytosis.

Alternate Names:

cytotoxic hypersensitivity, is mediated by IgG.

Hypersensitivity Type III

Type III hypersensitivity is caused by deposition of antigen-antibody immune complexes in tissues and blood vessel walls, activating complement and recruiting neutrophils that release destructive enzymes and reactive oxygen species.

Hypersensitivity Type IV

Type IV hypersensitivity, also known as delayed-type hypersensitivity, is mediated by T lymphocytes rather than antibodies and takes 48 to 72 hours to develop after antigen exposure.

Alternate Names:

delayed-type hypersensitivity, is mediated by T.

Hypersensitivity Vasculitis

A clinical disorder related to hypersensitivity vasculitis; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypersomnia

Being very sleepy during the day or sleeping much longer than usual even after enough sleep. It can be caused by sleep disorders, medicines, depression, substance use, or other illness; it is different from feeling tired without wanting to sleep.

Hypersplenism

Cytopenia caused by splenomegaly-related sequestration and increased destruction of circulating blood cells. It may affect red cells, white cells, platelets, or all three, with compensatory bone-marrow activity and an underlying liver, blood, infectious, or infiltrative disorder often present.

Hypertelorism

Increased distance between the orbits (the bony eye sockets), measured as an increased interorbital distance on imaging.

Hypertension

Persistently elevated arterial blood pressure. The diagnostic threshold depends on the guideline and the measurement method; diagnosis generally requires properly obtained readings on more than one occasion or other appropriate confirmation. It increases the risk of cardiovascular, cerebrovascular, and kidney disease.

Hypertension (High Blood Pressure)

Persistently elevated blood pressure, defined as systolic BP >130 mmHg or diastolic BP >80 mmHg

(ACC/AHA guidelines). See Hypertension.

Alternate Names:

High Blood Pressure

Hypertension In Anesthesia

Hypertension during anesthesia causes increased bleeding, myocardial ischemia, and stroke risk. Common causes include light anesthesia, pain, hypercarbia, hypoxia, bladder distension, shivering, and drug withdrawal (beta-blockers, clonidine).

Hypertensive

Having blood pressure that is persistently above the accepted range for the clinical context.

Hypertensive Cerebellar Hemorrhage

Bleeding into the cerebellum caused by rupture of a vessel damaged by high blood pressure.

Hypertensive Crisis

Severe elevation in blood pressure (SBP >180 or DBP >120) requiring urgent medical attention. Two types: hypertensive emergency (with end-organ damage) and hypertensive urgency (no acute end-organ damage). Hypertensive urgency: can be managed with oral antihypertensives (captopril, clonidine, labetalol) and careful outpatient follow-up.

Hypertensive Emergency

Severe hypertension (SBP >180 or DBP >120) with acute target organ damage. Organs involved: brain (hypertensive encephalopathy, stroke), heart (acute coronary syndrome, acute pulmonary edema, aortic dissection), kidneys (acute kidney injury), and eyes (papilledema).

Hypertensive Encephalopathy

A clinical disorder related to hypertensive encephalopathy; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypertensive Heart Disease

Left ventricular hypertrophy and diastolic/systolic dysfunction resulting from chronic hypertension. LVH is an independent cardiovascular risk factor

and precursor to heart failure. Diagnosis by ECG (voltage criteria) or echocardiography (wall thickness >1.1 cm). Treatment of hypertension can regress LVH.

Hypertensive Intracerebral Hemorrhage

Bleeding into brain tissue caused by rupture of a blood vessel associated with hypertension.

Hypertensive Nephropathy

Kidney damage caused by chronic hypertension, leading to nephrosclerosis (arteriolar hyalinosis and glomerulosclerosis). Presents with slowly progressive proteinuria (usually subnephrotic), hypertension, and gradual decline in kidney function. More common in African American patients. Diagnosis of exclusion.

Hypertensive Nephrosclerosis

Kidney damage from chronic hypertension. Benign form with hyaline arteriolosclerosis. Malignant form with hyperplastic arteriolosclerosis and onion-skinning. Second leading cause of ESRD after diabetes. Tight BP control with ACE inhibitors or ARBs essential to slow progression.

Hypertensive Retinopathy

Retinal changes caused by chronic or acute hypertension. Grading (Keith-Wagener-Barker): Grade I: generalized arteriolar narrowing; Grade II: arteriovenous nicking; Grade III: cotton-wool spots, flame hemorrhages, hard exudates; Grade IV: papilledema (malignant hypertension).

Hypertensive Urgency

A severe elevation in systemic blood pressure (typically systolic ≥ 180 mmHg or diastolic ≥ 120 mmHg) occurring in the absence of progressive acute target-organ damage (such as encephalopathy, myocardial infarction, aortic dissection, or acute pulmonary edema). Managed with oral antihypertensive medications over 24–48 hours rather than intravenous infusions.

Hyperthermia

Hyperthermia is an abnormally high body tempera-

ture caused by excessive heat production or impaired heat loss, without a change in the hypothalamic temperature set point. Causes include heat exposure, exertion, medicines, endocrine disorders, infection-related illness, and impaired sweating. Severe hyperthermia can cause confusion, seizures, organ injury, or death and requires urgent treatment.

Hyperthermia For Treating Cancer

A clinical disorder related to hyperthermia for treating cancer; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyperthermia In Anesthesia

Perioperative hyperthermia (temperature greater than 38 degrees Celsius) has multiple causes. Malignant hyperthermia (discussed separately) is the most urgent. Sepsis, thyroid storm, serotonin syndrome, neuroleptic malignant syndrome, and central fever are other causes.

Hyperthermic Intraperitoneal Chemotherapy

Heated chemotherapy circulated through the peritoneal cavity during surgery, following cytoreductive surgery, for peritoneal carcinomatosis from appendiceal, colorectal, and ovarian cancers and peritoneal mesothelioma.

Hyperthyroidism

Excessive thyroid hormone production causing thyrotoxicosis (elevated serum T4 and T3 concentrations). Causes include Graves disease (most common), toxic multinodular goiter, and toxic adenoma. Symptoms include weight loss, tremor, heat intolerance, palpitations, and anxiety.

Alternate Names:

Overactive Thyroid (Hyperthyroidism)

Hyperthyroidism (Thyrotoxicosis)

A state in which excess thyroid hormone effect causes symptoms such as heat intolerance, sweating, tremor, weight loss, or rapid heartbeat. It may result from increased thyroid production or other causes of thyrotoxicosis.

Alternate Names:

Thyrotoxicosis

Hyperthyroxinemia

An abnormally high concentration of circulating thyroxine, caused by hyperthyroidism, thyroiditis, medication, binding-protein changes, or assay interference; symptoms and TSH clarify the cause.

Hypertonia

Abnormally increased muscle tone or resistance to passive movement, occurring with upper motor-neuron disease, dystonia, cerebral palsy, spinal injury, or other neurologic disorders.

Hypertonic Saline

Intravenous saline with a sodium concentration greater than plasma, commonly 3% sodium chloride. It is used selectively for severe symptomatic hyponatremia or other urgent indications, with careful monitoring to avoid overly rapid correction and osmotic demyelination.

Hypertonic Saline Inhalation

Nebulized 3–7% sodium chloride solution that draws water into the airway surface liquid by osmosis, improving mucociliary clearance. It is used in CF and bronchiectasis. It may cause bronchospasm and should be preceded by a bronchodilator. It reduces exacerbations in CF.

Hypertonic Solution

A solution with greater effective osmotic concentration than the fluid with which it is compared, causing water to move out of cells across a semipermeable membrane.

Hypertrichosis

Excessive hair growth in androgen-independent areas, affecting both sexes, either congenital or acquired. Generalized: genetic syndromes, medications (minoxidil, cyclosporine, phenytoin, glucocorticoids), malnutrition, porphyria, hypothyroidism, and malignancy (paraneoplastic).

Hypertriglyceridemia

Elevated serum triglyceride levels, classified as mild at 150 to 499 mg per dL, moderate at 500 to 999, and severe above 1000. Severe hypertriglyceridemia increases the risk of acute pancreatitis. Causes include obesity, diabetes, alcohol use, and genetic disorders of lipoprotein metabolism.

Hypertrophic

Relating to hypertrophy, an enlargement of cells or tissue that increases the size of an organ or structure.

Hypertrophic Cardiomyopathy

A disease in which the heart muscle, especially the left ventricle, becomes abnormally thick and stiff. It may obstruct blood flow, impair filling, cause chest pain or breathlessness, and increase the risk of arrhythmia and sudden death.

Alternate Names:

Cardiomyopathy, Hypertrophic

Hypertrophic Cardiomyopathy Genetics

HCM is caused by mutations in genes encoding sarcomeric proteins, most commonly beta-myosin heavy chain (MYH7) and myosin-binding protein C (MYBPC3). Inheritance is autosomal dominant with variable penetrance. Genetic testing identifies causative mutations in 40-60% of cases.

Hypertrophic Cardiomyopathy Treatment

Management may include medicines to relieve symptoms, treatment of arrhythmias or obstruction, an implantable defibrillator for selected patients, and surgery or ablation when indicated. Treatment is guided by symptoms, anatomy, risk, and specialist assessment.

Hypertrophic Gastritis

Thickening of the gastric mucosal folds caused by glandular overgrowth or inflammation, as in Ménétrier disease; protein loss, pain, nausea, and increased cancer risk may occur.

Hypertrophic Myocardopathy

Abnormal thickening of the heart muscle, usually the

left ventricle, that can impair relaxation, obstruct outflow, cause arrhythmias, and increase sudden-death risk; it is often inherited.

Hypertrophic Pyloric Stenosis

Hypertrophy and hyperplasia of the pyloric muscle causing gastric outlet obstruction, typically in young infants. Progressive nonbilious projectile vomiting, hunger after vomiting, dehydration, and hypochloremic hypokalemic metabolic alkalosis may occur.

Hypertrophic Scar

A raised, thick scar that stays within the boundaries of the original wound. It forms when the body makes too much scar tissue during healing, often after burns, surgery, or injury. Unlike a keloid, it does not grow beyond the wound edges and may become flatter over time.

Hypertrophy

Enlargement of a tissue or organ caused primarily by an increase in the size of its constituent cells. It may be physiologic, as in skeletal-muscle adaptation, or pathologic, as in pressure-induced left ventricular hypertrophy. Hypertrophy can preserve function initially but may eventually cause stiffness, impaired relaxation, oxygen-demand mismatch, or mechanical compression. Evaluation considers the affected organ, underlying stimulus, functional consequences, and whether enlargement is accompanied by fibrosis, hyperplasia, or cellular injury.

Hypertropia

Vertical misalignment of the eyes in which one eye deviates upward relative to the other, causing double vision, eye strain, or abnormal head posture.

Hyperuricemia

An abnormally high concentration of uric acid in the blood, which may be asymptomatic or contribute to gout and uric-acid kidney stones.

Hyperventilating

A clinical disorder related to hyperventilating; diagnosis and treatment depend on its cause, character-

istic findings, and severity.

Hyperventilation

Excessive alveolar ventilation that reduces arterial carbon dioxide (PaCO₂) below 35 mmHg, causing respiratory alkalosis. Symptoms include lightheadedness, tingling, chest tightness, and carpopedal spasm. It may result from anxiety, pain, metabolic acidosis, or pulmonary embolism.

Hyperventilation Syndrome

Recurrent or persistent overbreathing that exceeds metabolic requirements, usually producing reduced arterial carbon dioxide and respiratory alkalosis in the absence of primary pulmonary disease. Symptoms may include breathlessness, chest tightness, dizziness, palpitations, paraesthesia, and carpopedal spasm.

Hypervigilance

A state of excessive alertness to possible threat, often accompanied by scanning, startle, irritability, and sleep difficulty in anxiety, trauma-related disorders, stimulant use, or medical illness.

Hypervitaminosis A

A clinical disorder related to hypervitaminosis a; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypervitaminosis D

A clinical disorder related to hypervitaminosis d; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypervolaemia

A clinical disorder related to hypervolaemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypervolemia

Excess circulating blood or extracellular fluid volume, producing edema, weight gain, breathlessness, elevated venous pressure, or hypertension in settings such as heart, kidney, or liver failure.

Hypha

A branching tubular filament that forms the basic structural unit of most molds and many fungi; networks of hyphae form a mycelium and may invade tissue in fungal disease.

Hyphema

Blood in the anterior chamber of the eye, typically resulting from blunt ocular trauma that tears the blood vessels of the iris or ciliary body. It presents as a layered collection of red blood cells in the lower portion of the anterior chamber and can cause elevated intraocular pressure and corneal blood staining.

Hypnagogic

Hypnagogic is a medical or biological term describing a defined process, structure, substance, or clinical finding; its precise meaning depends on context.

Hypnagogic Hallucinations

Vivid sensory experiences occurring while a person is falling asleep.

Hypnic Headache

A rare primary headache that repeatedly awakens a person during sleep, usually in later life and often at a similar time each night.

Hypnogram

A chart showing the timing and stages of sleep during a recording.

Alternate Names:

Sleep-stage diagram

Hypnosis

A state of focused attention, reduced peripheral awareness, and heightened suggestibility, resembling a trance-like state but distinct from sleep. Hypnosis is induced by a procedure (hypnotic induction) involving focused concentration, relaxation, and guided imagery.

Hypnotherapy

Hypnotherapy uses focused attention and guided

suggestion as an adjunct to care for selected problems such as pain, anxiety, or unwanted habits.

Hypnotic

A medication that induces, maintains, or improves sleep, used for insomnia.

Hypoactive Sexual Desire Disorder

A sexual dysfunction characterized by persistently or recurrently deficient or absent sexual thoughts, fantasies, and desire for sexual activity, causing marked distress or interpersonal difficulty. In DSM-5, HSDD in women is merged with female sexual arousal disorder into female sexual interest/arousal disorder.

Hypoadrenalism

Insufficient adrenal-cortex hormone production, especially cortisol and sometimes aldosterone; primary disease can cause fatigue, weight loss, low blood pressure, hyperpigmentation, and adrenal crisis.

Hypoalbuminemia

An abnormally low blood albumin concentration caused by reduced production, inflammation, kidney or gastrointestinal loss, or severe malnutrition; it can contribute to edema and alter drug binding.

Hypoalphalipoproteinemias

Disorders with low HDL-related apolipoprotein levels, which may impair reverse cholesterol transport and increase cardiovascular risk depending on the cause and overall lipid profile.

Hypoblast

The inner cell layer of the early embryonic disc that contributes to the yolk-sac lining and helps establish the embryonic body axes during gastrulation.

Hypocalcaemia

British spelling of Hypocalcemia; see Hypocalcemia.

Hypocalcemia

Serum calcium less than 8.5 milligrams per deciliter or ionized calcium less than 4.5 milligrams per

deciliter. Causes include hypoparathyroidism, vitamin D deficiency, hypomagnesemia, hyperphosphatemia, and critical illness.

Hypocalcemia, Hypophosphatemia, And Hyperkalemia

The classic electrolyte triad of acute hypoparathyroidism, most commonly occurring after thyroid or parathyroid surgery from inadvertent removal or devascularization of the parathyroid glands. Hypocalcemia causes tetany, paresthesias, and Chvostek and Trousseau signs.

Hypocapnia

An abnormally low arterial carbon-dioxide tension, usually from excessive ventilation due to anxiety, pain, hypoxia, lung disease, pregnancy, or mechanical ventilation.

Hypochlorite

A clinical disorder related to hypochlorite; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypocholesterolemia

An unusually low blood cholesterol concentration caused by malnutrition, malabsorption, hyperthyroidism, chronic illness, liver disease, or inherited lipid disorders.

Hypochondriasis

A clinical disorder related to hypochondriasis; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypochondroplasia

An autosomal dominant skeletal dysplasia caused by heterozygous gain-of-function mutations in the fibroblast growth factor receptor 3 (*FGFR3*) gene on chromosome 4p16.3, most frequently the p.Asn540Lys missense substitution. Represents a milder allelic variant of achondroplasia, characterized by disproportionate short stature, rhizomelic micromelia (shortening of proximal limbs), broad hands, lumbar lordosis, and limited elbow extension, but typically features a normal craniofacial

appearance and normal head circumference without the prominent frontal bossing or midface retrusion characteristic of classic achondroplasia.

Alternate Names:

HCH

Hypochromia

Reduced hemoglobin color in red blood cells, seen as increased central pallor on a blood smear and commonly associated with iron deficiency or thalassemia.

Hypochromic Anaemia

Hypochromic anaemia is anemia in which red blood cells contain too little hemoglobin and appear unusually pale, commonly because of iron deficiency or thalassemia.

Hypochromic Anemia

Anemia in which red cells contain too little hemoglobin and appear pale, most often because of iron deficiency, thalassemia, chronic disease, or sideroblastic disorders.

Hypocitraturia

Low urinary citrate excretion, which reduces inhibition of calcium-crystal formation and increases risk of calcium kidney stones; potassium citrate may be used when clinically appropriate.

Hypocupremia

An abnormally low copper concentration in blood or tissues, caused by inadequate intake, malabsorption, excess zinc, or inherited disorders, and potentially producing anemia, neutropenia, or neurologic disease.

Hypodermic Needle

A hollow needle designed to inject medicines or withdraw fluid through the skin, with gauge and length selected according to the substance, route, body site, and patient.

Hypodermis

Subcutaneous tissue/superficial fascia, the deepest

skin layer with loose connective tissue and adipose tissue. Attaches skin to underlying muscles/bones via fibrous septa. Provides thermal insulation, energy storage (fat), mechanical cushioning, and contains blood vessels, lymphatics, and nerves.

Hypodontia

Congenital absence of one or more teeth. Common: third molars, maxillary lateral incisors, second premolars.

Hypoesthesia

Reduced sensitivity to touch, temperature, pain, or other sensations. It may result from nerve compression, neuropathy, spinal disease, stroke, injury, or other neurologic conditions; sudden or spreading numbness requires prompt assessment.

Hypogalactia

Insufficient breast-milk production for an infant's needs, caused by ineffective milk removal, hormonal or breast disorders, medications, illness, or other feeding difficulties.

Hypogammaglobulinemia

A reduction in circulating immunoglobulins that predisposes to recurrent infections, especially with encapsulated bacteria; it may be inherited or acquired.

Hypogastric Region

Central inferior region below the umbilical region, containing the urinary bladder, sigmoid colon, rectum, and in females the uterus and ovaries. Assessed for bladder distension, uterine enlargement, and pelvic masses. Suprapubic pain may indicate cystitis, calculi, or pelvic pathology.

Hypoglossal Nerve

Cranial nerve XII, which supplies the muscles that move the tongue; a lesion causes tongue weakness, wasting, and deviation toward the affected side on protrusion.

Hypoglossal Nerve Injury

Damage to the twelfth cranial nerve (CN XII), which innervates the ipsilateral intrinsic and extrinsic tongue muscles. Causes include skull base fractures, surgical trauma, tumors, or infection.

Hypoglossal Nerve Palsy

Weakness or paralysis of the hypoglossal nerve (CN XII), leading to ipsilateral tongue weakness, atrophy, and fasciculations. Causes include tumors at the skull base or in the hypoglossal canal, surgical injury, or central lesions. Unilateral palsy causes the tongue to deviate toward the affected side on protrusion.

Hypoglycaemia

A clinical disorder related to hypoglycaemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypoglycemia

An abnormally low blood-glucose concentration that may cause sweating, tremor, palpitations, hunger, confusion, seizures, or loss of consciousness. Causes include glucose-lowering medicines, missed meals, alcohol, and critical illness. Treatment depends on the person's ability to swallow and the cause; severe or recurrent episodes require urgent assessment.

Hypoglycemia In Diabetes

Blood glucose less than 70 milligrams per deciliter in patients with diabetes. Classification by level: level one less than 70, level two less than 54 which is clinically significant hypoglycemia, and level three severe hypoglycemia requiring assistance.

Hypoglycemia Unawareness

Loss of the warning symptoms of hypoglycemia including tremor, sweating, and palpitations occurring in patients with recurrent hypoglycemia particularly type one diabetes. Results from attenuated counter-regulatory hormone responses particularly impaired glucagon and epinephrine responses.

Hypoglycemic

Relating to an abnormally low blood glucose level, or describing a medicine that lowers blood glucose.

Hypogonadism

Reduced or absent function of the gonads, leading to inadequate production of sex hormones (testosterone in males, estrogen in females) and impaired gamete production. Male hypogonadism: low testosterone causing decreased libido, erectile dysfunction, infertility, reduced muscle mass, and osteoporosis. Female: estrogen deficiency causing amenorrhea, hot flashes, and osteoporosis.

Hypogonadism In Men

Testosterone deficiency resulting from inadequate testicular function classified as primary testicular failure or secondary hypothalamic-pituitary dysfunction. Primary causes include Klinefelter syndrome, cryptorchidism, orchitis, trauma, and chemotherapy.

Hypogonadism In Women

Testosterone deficiency in women causing decreased libido, fatigue, depressed mood, and reduced sense of well-being. Causes include oophorectomy, premature ovarian insufficiency, adrenal insufficiency, hypopituitarism, and medications.

Hypogonadotropic Hypogonadism

Secondary testicular failure with low/normal LH/FSH. Causes: Kallmann syndrome, pituitary tumors, obesity, medications, systemic illness.

Hypohidrosis

Reduced or absent sweating, leading to heat intolerance and risk of hyperthermia. Causes include autonomic neuropathy (diabetes, Parkinson disease), skin diseases (scleroderma, burns), medications (anticholinergics), and genetic disorders (hypohidrotic ectodermal dysplasia).

Hypokalaemia

British spelling of Hypokalemia; see Hypokalemia.

Hypokalemia

Serum potassium less than 3.5 mEq/L. Causes include gastrointestinal losses from vomiting and diarrhea, renal losses from diuretics, RTA, and hyperaldosteronism, intracellular shift from alkalosis and

insulin, and inadequate intake.

Hypokalemic Periodic Paralysis

A clinical disorder related to hypokalemic periodic paralysis; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypokinesia

Reduced amplitude or amount of voluntary movement, commonly seen in Parkinson disease and other basal-ganglia disorders; it may appear as a short-stepped gait, reduced facial expression, or small handwriting.

Hypolipidemic Agents

Medicines that lower cholesterol or triglycerides, including statins, ezetimibe, fibrates, bile-acid sequestrants, niacin, and selected injectable therapies.

Hypolipoproteinemia

An abnormally low concentration of one or more blood lipoproteins, caused by inherited disorders, malabsorption, malnutrition, liver disease, or other systemic illness.

Hypomagnesaemia

A clinical disorder related to hypomagnesaemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypomagnesemia

Serum magnesium less than 1.7 milligrams per deciliter. Causes include gastrointestinal losses from diarrhea, renal losses from diuretics and alcohol, malnutrition, and medications including proton pump inhibitors.

Hypomania

Distinct period of abnormally elevated, expansive, or irritable mood with increased energy lasting at least four consecutive days, observable by others but without marked functional impairment, hospitalisation, or psychotic features.

Hypomanic Episode

A distinct period of abnormally elevated, expansive, or irritable mood with increased energy lasting at least four consecutive days. The episode is not severe enough to cause marked impairment in functioning or necessitate hospitalization, and there are no psychotic features. The change in functioning is observable by others.

Hypomelanosis Of Ito

A clinical disorder related to hypomelanosis of ito; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypomethylation

Reduced addition of methyl groups to DNA or other molecules, which can alter gene expression, genomic stability, imprinting, and cancer biology.

Hypomnesia

Reduced ability to remember or recall information, which may occur with aging, neurologic disease, psychiatric conditions, medications, or other causes.

Hyponatraemia

A clinical disorder related to hyponatraemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hyponatremia

Serum sodium less than 135 mEq/L. Most common electrolyte disorder in hospitalized patients. Causes are classified by volume status: hypovolemic from GI losses and diuretics, euvolemic from SIADH and hypothyroidism, and hypervolemic from heart failure, cirrhosis, and nephrotic syndrome.

Alternate Names:

Sodium Low Levels In The Blood (Hyponatremia)

Hypoparathyroidism

Reduced or absent parathyroid hormone secretion, resulting in hypocalcemia and hyperphosphatemia. Most commonly occurs after thyroid or parathyroid surgery from inadvertent removal of parathyroid glands. Autoimmune hypoparathyroidism may oc-

cur alone or as part of autoimmune polyendocrine syndrome type one.

Hypoparathyroidism Management

Management aims to correct low calcium and inadequate parathyroid-hormone activity while avoiding complications. It may include calcium and active vitamin D, monitoring of calcium and kidney function, and treatment of the underlying cause; specialist guidance is often required.

Hypopharyngeal Neoplasm

A neoplastic process affecting hypopharyngeal neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Hypopharynx

The lower part of the pharynx behind and beside the larynx, extending from the oropharynx to the esophagus. It directs food toward the esophagus during swallowing.

Hypophosphataemia

A clinical disorder related to hypophosphataemia; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypophosphatasia

A rare inherited disorder of bone and tooth mineralization caused by deficient tissue-nonspecific alkaline phosphatase.

Hypophosphatemia

A serum phosphate concentration below the laboratory reference range, typically caused by increased renal phosphate excretion, intracellular shifting, or renal phosphate wasting. Severe deficiency can cause muscle weakness, respiratory failure, hemolysis, impaired myocardial function, and neurologic dysfunction.

Hypophosphatemic Ricket

Rickets caused by low phosphate availability, often from renal phosphate wasting, leading to defective mineralization, bowed legs, bone pain, growth im-

pairment, and dental abnormalities.

Hypophysectomy

Hypophysectomy is surgical removal of all or part of the pituitary gland, usually to treat a pituitary tumor or another serious pituitary disorder.

Hypophysis

A clinical disorder related to hypophysis; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypophysis (Pituitary Gland)

An endocrine gland at the base of the brain that regulates growth, thyroid and adrenal function, reproduction, lactation, and water balance through its hormones.

Alternate Names:

Pituitary Gland

Hypophysitis

Inflammation of the pituitary gland that can cause headache, visual symptoms, and deficiency or excess of pituitary hormones; it may be autoimmune, related to pregnancy, or triggered by immune-checkpoint therapy.

Hypopigmentation

Reduced skin pigmentation producing paler areas than the surrounding skin, caused by decreased melanin, reduced melanocyte function or number, inflammation, injury, infection, or inherited disorders.

Hypopituitarism

Deficiency of one or more pituitary hormones due to disorders of the pituitary gland or hypothalamus. Causes include craniopharyngiomas, pituitary surgery or radiation, traumatic brain injury, or infiltrative diseases. Growth hormone deficiency is typically the first hormone lost followed by gonadotropins then TSH and ACTH.

Alternate Names:

Pituitary Insufficiency

Hypoplasia

Incomplete or underdevelopment of a tissue, organ, or body structure, resulting in reduced size or functional capacity. It may be congenital, developmental, genetic, or acquired and ranges from a mild anatomical variant to severe organ dysfunction. Assessment considers the timing of onset, associated malformations, tissue function, and whether the process is isolated or syndromic.

Hypoplastic Left Heart Syndrome

A severe congenital heart defect where the left heart structures (mitral valve, left ventricle, aortic valve, ascending aorta) are underdeveloped and unable to support systemic circulation. The right ventricle pumps blood to both the pulmonary and systemic circulations via a patent ductus arteriosus.

Hypoploidy

A chromosome number below the normal diploid complement, caused by loss of one or more chromosomes; in tumor cytogenetics it may have prognostic significance.

Hypopnea

Abnormally shallow or reduced breathing, especially when it occurs repeatedly during sleep.

Hypoproteinaemia

Hypoproteinaemia is an abnormally low level of proteins in the blood, which may result from malnutrition, liver disease, kidney protein loss, or intestinal loss.

Hypoproteinemia

An abnormally low concentration of proteins in the blood, often involving low albumin levels.

Hypopyon

A layering of inflammatory cells (purulent exudate/leukocytes) in the anterior chamber of the eye, forming a visible white-to-yellow fluid level in the lower portion of the cornea. Caused by severe anterior uveitis (HLA-B27 spondyloarthropathies, Behçet disease), infectious keratitis (bacterial/fungal corneal ulcers), or endophthalmitis.

Hyposmia

Decreased sensitivity to odors in which olfaction is reduced but not absent, distinct from anosmia and parosmia, and the most common olfactory complaint.

Hypospadias

A congenital anomaly in which the urethral meatus opens on the ventral surface of the penis (or in females, below the normal location) rather than at the tip of the glans. It is frequently associated with a curved penis or an incomplete foreskin, and severity depends on the location of the opening and associated abnormalities.

Hypospadias Repair

Surgery reconstructs the urethral opening and, when needed, straightens the penis in a child born with hypospadias; timing and technique depend on anatomy and surgeon assessment.

Hypotension

Abnormally low blood pressure, generally systolic <90 mmHg or mean arterial pressure <65 mmHg. Causes include hypovolemia, cardiac failure, sepsis, anaphylaxis, and medications. Orthostatic hypotension is a drop upon standing. Severe hypotension (shock) requires urgent fluid resuscitation and vasopressors.

Hypotension In Anesthesia

Abnormally low blood pressure occurring during anesthesia. The clinically important threshold depends on the patient, baseline pressure, duration, and surgical context.

Hypotensive

Having or causing abnormally low blood pressure; symptoms may include dizziness, fainting, weakness, confusion, or shock depending on the cause and severity.

Hypothalamic Dysfunction

Abnormal function of the hypothalamus, which regulates endocrine, autonomic, temperature, appetite, thirst, sleep, and reproductive functions; causes in-

clude tumors, trauma, inflammation, and genetic disease.

Hypothalamic Hamartoma

A congenital, noncancerous mass of disorganized hypothalamic tissue that can cause gelastic seizures, precocious puberty, behavioral symptoms, or cognitive decline.

Hypothalamic Hormone

A releasing or inhibiting hormone produced by the hypothalamus that regulates pituitary secretion, including signals controlling thyroid, adrenal, reproductive, growth, and lactation axes.

Hypothalamic Neoplasm

A neoplastic process affecting hypothalamic neoplasm; diagnosis establishes the tumor type, extent, biologic behavior, and appropriate management.

Hypothalamic Tumor

A clinical disorder related to hypothalamic tumor; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypothalamic-Pituitary-Adrenal Axis

A network linking the hypothalamus, pituitary gland, and adrenal glands to regulate stress hormones and other functions.

Alternate Names:

HPA axis

Hypothalamic-Pituitary-Adrenal Axis (HPA Axis)

An endocrine regulatory pathway in which hypothalamic corticotropin-releasing hormone stimulates pituitary ACTH secretion, which stimulates adrenal cortisol production. Cortisol exerts negative feedback and coordinates metabolic, immune, and stress responses.

Alternate Names:

HPA Axis

Hypothalamic-Pituitary-Adrenal Axis

Physiology

The hypothalamus releases CRH in a diurnal rhythm peaking in the early morning stimulating pituitary corticotrophs to secrete ACTH which stimulates the adrenal cortex to produce cortisol. Cortisol exerts negative feedback on both the hypothalamus and pituitary suppressing CRH and ACTH release.

Hypothalamic-Pituitary-Gonadal Axis Physiology

The hypothalamus releases GnRH in a pulsatile fashion stimulating pituitary gonadotrophs to secrete LH and FSH. In males LH stimulates Leydig cells to produce testosterone while FSH stimulates Sertoli cells to support spermatogenesis.

Hypothalamic-Pituitary-Thyroid Axis Physiology

The hypothalamus secretes TRH stimulating pituitary thyrotrophs to release TSH which stimulates the thyroid gland to produce T4 and T3. T3 the more biologically active form is produced by peripheral conversion of T4 by deiodinase enzymes.

Hypothalamus

A small region of the brain at the base of the third ventricle, bounded by the optic chiasm, mammillary bodies, and infundibulum. It weighs approximately 4 grams and is the master homeostasis regulator. Controls autonomic and endocrine systems (via the pituitary through the hypophyseal portal system), temperature, hunger, thirst, sleep-wake cycles, and emotion.

Hypothermia

A core body temperature below 35 degrees Celsius, classified as mild (32 to 35), moderate (28 to 32), or severe (<28). J-wave (Osborn wave) on ECG. Management: passive rewarming (mild), active external rewarming (moderate), active internal rewarming (severe).

Alternate Names:

Cold Exposure Low Body Temperature

Hypothermia In Anesthesia

Hypothermia (core temperature less than 36 degrees

Celsius) occurs in 50 to 80 percent of surgical patients due to anesthetic-induced thermoregulatory dysfunction and heat redistribution. Initial phase (first 30 to 60 minutes) involves core-to-peripheral heat redistribution causing 0.5 to 1.5 degree Celsius drop.

Hypothesis

A clinical disorder related to hypothesis; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hypothyroidism

Insufficient thyroid hormone production causing decreased metabolic rate. Most common endocrine disorder, affecting up to 5 percent in the general population and up to 20 percent in women over 60 years.

Hypotonia

Decreased muscle tone presenting as a floppy or poorly supported posture. It may result from muscle, peripheral nerve, neuromuscular-junction, spinal-cord, brain, genetic, or metabolic disorders and requires evaluation when persistent or associated with weakness or developmental delay.

Hypotonic Solution

A solution with lower effective osmotic concentration than the fluid with which it is compared, causing water to move into cells across a semipermeable membrane.

Hypotrichosis

Abnormally sparse or absent hair caused by inherited follicle disorders, autoimmune disease, endocrine or nutritional problems, medications, or scarring alopecia.

Hypotrophy

Reduction in the volume or development of a tissue or organ without complete absence of the structure. It may reflect inadequate growth, reduced trophic stimulation, chronic underuse, nutritional deficiency, vascular insufficiency, or a systemic disorder. Hypotrophy is distinguished from aplasia, in which

development fails almost completely, and from atrophy, which usually describes acquired wasting after prior development. Clinical significance depends on the affected tissue, age of onset, degree of functional impairment, and whether the cause is congenital, neurologic, vascular, or metabolic.

Hypotropia

Vertical eye misalignment in which one eye deviates downward relative to the other, potentially causing diplopia, abnormal head posture, or suppression of vision.

Hypoventilation

Reduced alveolar ventilation leading to elevated carbon dioxide levels in the blood, potentially causing respiratory acidosis and hypoxemia.

Hypovolaemia

Hypovolaemia is an abnormally low volume of circulating blood or body fluid, commonly caused by bleeding, dehydration, or major fluid loss.

Hypovolemia

Hypovolemia is reduced effective circulating blood volume, usually from bleeding, fluid loss, or inadequate intake; it can cause tachycardia, hypotension, poor perfusion, kidney injury, and hypovolaemic shock.

Hypovolemic Shock

Circulatory failure caused by insufficient circulating blood or fluid volume, leading to reduced tissue perfusion. Causes include hemorrhage, severe dehydration, and fluid loss.

Hypoxaemia

Hypoxaemia is an abnormally low level of oxygen in the arterial blood.

Hypoxanthine

A naturally occurring purine base and intermediate in purine metabolism that is converted to xanthine by xanthine oxidase and can be salvaged to form inosine monophosphate.

Hypoxemia

Abnormally low oxygen concentration in arterial blood. Causes include hypoventilation, ventilation-perfusion mismatch, shunting, impaired diffusion, and low inspired oxygen.

Hypoxia

Inadequate oxygen availability at the tissue level. It may result from low arterial oxygen, anemia, reduced blood flow, or impaired cellular oxygen use.

Hypoxia And Hypoxemia

Hypoxemia is low oxygen in arterial blood, while hypoxia is inadequate oxygen at the tissue level. Causes include lung disease, airway obstruction, heart disease, anemia, shock, or high altitude; severe breathlessness, cyanosis, confusion, or collapse is urgent.

Hypoxic Cell Injury

Cellular injury caused by inadequate oxygen availability, reducing oxidative phosphorylation and ATP production. Causes include ischemia, hypoxemia, anemia, and impaired oxygen utilization; severity and duration determine reversibility.

Hypoxic Pulmonary Vasoconstriction

The unique response of pulmonary arterioles to low alveolar oxygen tension (PO₂ less than approximately 60 mmHg), causing vasoconstriction.

Hypoxic-Ischemic Brain Injury

Brain injury caused by inadequate oxygen delivery or cerebral blood flow. It may follow cardiac arrest, asphyxia, shock, respiratory failure, or severe hypoglycemia and can produce coma, seizures, cognitive impairment, or death.

Hypsarrhythmia

A pathognomonic, high-amplitude, chaotic, and disorganized interictal electroencephalographic pattern characterized by continuous, high-voltage slow waves interspersed with multi-focal spikes and sharp waves without consistent rhythmicity or hemispheric synchronization. It is the hallmark diagnostic finding of infantile spasms (West syndrome) and rep-

resents severe diffuse cerebral cortical dysfunction; treatment consists of adrenocorticotrophic hormone, high-dose oral corticosteroids, or vigabatrin.

Alternate Names:

Hypsarrhythmic Pattern; Chaotic Interictal EEG

Hysterectomy

Surgical removal of the uterus. The cervix, fallopian tubes, and ovaries may or may not also be removed, depending on the procedure and clinical indication. Hysterectomy permanently ends menstruation and the ability to carry a pregnancy, but it does not cause immediate menopause when functioning ovaries remain. Indications include cancer, fibroids, severe bleeding, adenomyosis, endometriosis, chronic pelvic pain, and uterine prolapse.

Hysterectomy And Surgical Menopause

Hysterectomy removes the uterus and ends menstruation and the ability to carry a pregnancy. Surgical menopause occurs when both ovaries are removed or ovarian function otherwise stops abruptly; hysterectomy alone does not necessarily cause menopause if the ovaries remain.

Hysteria

A clinical disorder related to hysteria; diagnosis and treatment depend on its cause, characteristic findings, and severity.

Hysterosalpingo-Contrast-Sonography

An ultrasound-based technique for evaluating the uterine cavity and fallopian tube patency. Sterile saline or contrast medium is injected through the cervix while transvaginal ultrasound is performed.

Hysterosalpingo-Contrast-Sonography (HyCoSy)

Ultrasound evaluation of tubal patency using contrast. Alternative to hysterosalpingography.

Alternate Names:

HyCoSy

Hysterosalpingogram

An X-ray procedure used to evaluate the uterine

cavity and fallopian tube patency. A radiopaque contrast medium is injected through the cervix, and serial X-ray images are taken as the contrast fills the uterine cavity and spills from the fallopian tubes.

Hysterosalpingogram

An X-ray study in which contrast material is placed through the cervix to outline the uterine cavity and fallopian tubes, commonly to assess uterine structure and tubal patency during infertility evaluation.

Hysterosalpingography

An X-ray imaging procedure in which radiopaque contrast is injected through a cervical cannula. Used for infertility evaluation: assesses tubal patency (normal: contrast spills freely into peritoneal cavity), uterine cavity shape (fibroids, polyps, synechiae, Mullerian anomalies), and proximal/distal tubal occlusion.

Hysteroscope

A thin camera-equipped instrument passed through the cervix to inspect the uterine cavity and, when designed for operative use, remove polyps, treat bleeding, or perform other procedures.

Hysteroscopic Metroplasty

A minimally invasive transcervical surgical procedure in which a congenital uterine septum (septate uterus) is divided or resected under direct hysteroscopic visualization using cold endoscopic scissors, electrosurgical needle/knife, or laser. Restores normal anatomical contours to the endometrial cavity, significantly reducing spontaneous pregnancy loss and improving live birth rates in women with recurrent pregnancy loss.

Hysteroscopic Myomectomy

An operative transcervical endoscopic procedure performed to excise intracavitary and submucosal uterine leiomyomas (FIGO Types 0, 1, and selected Type 2 fibroids) using a resectoscope loop with radiofrequency energy or a mechanical hysteroscopic tissue morcellator. Relieves abnormal uterine bleeding, prevents myoma-related recurrent pregnancy loss, and enhances fertility while preserving the

uterus and avoiding abdominal incisions.

Hysteroscopy

A procedure that passes a small camera through the cervix to inspect the inside of the uterus and, when needed, obtain tissue or treat conditions such as polyps, fibroids, or intrauterine adhesions.

Hysterotomy

A surgical incision into the uterus, performed in selected obstetric or gynecologic procedures.

I

Iatrogenic

An illness or condition caused by medical examination or treatment. Examples: hospital-acquired infections, drug side effects, surgical complications, radiation-induced damage. Prevention: proper technique, informed consent, monitoring, reporting systems.

Iatrogenic Disease

A disease or injury caused unintentionally by a medical examination, treatment, medication, procedure, or other healthcare intervention.

Iatrogenic Injury

An injury caused unintentionally by a medical examination, treatment, procedure, or medicine.

Ibandronic Acid

ibandronic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

IBD

Inflammatory bowel disease, including Crohn's disease and ulcerative colitis, causing chronic intestinal inflammation.

Alternate Names:

Inflammatory bowel disease

Ibogaine

Ibogaine is a psychoactive plant alkaloid studied for addiction treatment but associated with dangerous cardiac arrhythmias, seizures, and death.

Ibritumomab Tiuxetan

A medicine or chemical compound called ibritumomab tiuxetan. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

IBS

Alternate index label; see *Irritable bowel syndrome*.

IBS-D Irritable Bowel Syndrome With Diarrhea

IBS with diarrhea causes recurrent abdominal pain associated with loose stools without structural disease explaining the symptoms. Management may include diet changes, soluble fiber, gut-directed medicines, behavioral therapy, and evaluation for alarm features or alternative diagnoses.

Alternate Names:

Diarrhea IBS (IBS-D Irritable Bowel Syndrome With Diarrhea)

Ibuprofen

A medicine that reduces pain, fever, and inflammation. It can irritate or bleed the stomach, harm the kidneys, cause fluid retention, or increase heart risks in some people.

Ibuprofen Dosing For Children

Children's ibuprofen dosing is based mainly on weight, age, product concentration, and medical advice; the label dose should not be exceeded and it should be avoided in dehydration or certain kidney conditions.

Ibuprofen Overdose

Toxic exposure to more ibuprofen than recommended, which can cause nausea, vomiting, abdominal pain, gastrointestinal bleeding, kidney injury, metabolic acidosis, drowsiness, seizures, or coma.

ICD

Implantable cardioverter-defibrillator, a device that detects dangerous heart rhythms and delivers a corrective shock.

Alternate Names:

Implantable cardioverter defibrillator

ICD Shock Therapy

Treatment delivered by an implanted defibrillator to stop dangerous heart rhythms. The device may use painless pacing or an electrical shock, depending on the rhythm.

ICD-O Neoplasm Morphology Classification

A coding system used by cancer registries and pathology services to record what type of tumor is present and whether it is benign or cancerous. It does not replace the diagnosis, location, or stage of the cancer.

Ice Pack

A device that applies cold to a body area to help relieve pain or swelling. It should be wrapped to protect the skin and used for limited periods.

Ice Pick Headache

A sudden, severe, stabbing headache that usually lasts only seconds.

Alternate Names:

Primary stabbing headache

ICF Framework

A World Health Organization framework for describing how a health condition affects the body, daily activities, participation in life, and the person's environment.

Ichthyosis

A group of conditions that cause very dry, rough, scaly skin. Some forms are inherited and may be present from birth; others are linked to allergies or another illness.

Ichthyosis Vulgaris

A common inherited disorder of skin keratinization characterized by dry, scaling skin, especially on the extensor limbs. Symptoms often improve with age and with regular emollient and keratolytic skin care.

Icteric

Describing yellow discoloration of the skin, sclerae, or other tissues caused by increased bilirubin, commonly called jaundiced.

Icteric Index

The icteric index is a laboratory estimate of serum yellow discoloration related mainly to bilirubin and can signal jaundice or interfere with testing.

Icterus

Yellow discoloration of the skin or eyes caused by excess bilirubin; also called jaundice.

Alternate Names:

jaundice.

Ictus

A sudden neurological event, such as a stroke or seizure.

ICU Psychosis

A colloquial term for delirium occurring in an intensive-care setting, with acute changes in attention, awareness, and cognition. Causes include severe illness, infection, medicines, sleep disruption, pain, and metabolic abnormalities. It requires prompt evaluation and treatment of reversible causes, with supportive measures to orient and calm the patient.

Alternate Names:

Psychosis ICU (ICU Psychosis)

Idaiovirus

Likely a misspelling of *Idaeovirus*, a genus of plant-infecting RNA viruses that includes viruses associated with raspberry and privet; it is not a recognized human pathogen.

Idarubicin

Idarubicin is an anthracycline chemotherapy drug that intercalates DNA and inhibits topoisomerase II, used mainly in acute myeloid leukaemia; myelosuppression and cumulative cardiotoxicity are important risks.

IDC

Alternate index label; see *Invasive ductal carcinoma*.

Idea

A thought, belief, or mental representation; in clinical writing, its significance depends on whether it is ordinary, intrusive, fixed, or associated with impaired reality testing.

Idea Of Reference

A belief that ordinary events, comments, or media messages have a special personal meaning, often seen in psychotic or mood disorders when held with strong conviction.

Ideas Of Reference

The false belief that unrelated events, objects, or people have a particular and unusual significance specifically for the patient. For example, believing that TV newscasters are sending personal messages or that strangers are conspiring against them.

Ideation

A chemical substance or compound called ideation. Its properties, uses, and risks depend on its composition and clinical context.

Identical Twins

Identical Twins is an immunologic, genetic, infectious, developmental, or cellular concept used to explain disease mechanisms, diagnosis, or treatment.

Ideopathic

A misspelling of idiopathic, meaning that no specific cause has been identified.

Idiopathic

Of unknown cause. Examples: idiopathic pulmonary fibrosis, idiopathic thrombocytopenia.

Idiopathic Facial Paralysis (Facial Nerve Problems)

Alternate index label for Facial Nerve Problems.

Idiopathic Hypercalciuria

Excess calcium in the urine without an identified cause. It can increase the risk of kidney stones and, less often, bone loss.

Idiopathic Hypersomnia

A sleep disorder that causes extreme daytime sleepiness, long or unrefreshing sleep, and difficulty waking up, even when a person has enough time to sleep. Other causes must be ruled out.

Idiopathic Interstitial Pneumonias

A group of interstitial lung diseases with no identified cause, characterized by varying degrees of lung inflammation and fibrosis; diagnosis requires clinical, radiologic, and sometimes pathologic assessment.

Idiopathic Intracranial Hypertension

Increased pressure around the brain without a tumor or other clear cause. It can cause headaches, a pulsing sound in the ears, and vision problems and may damage sight if untreated.

Idiopathic Myelofibrosis

Alternate index label; see *Myelofibrosis*.

Idiopathic Pulmonary Fibrosis

A long-term lung disease of unknown cause in which scar tissue gradually builds up in the lungs. It causes increasing shortness of breath and a dry cough and requires specialist care.

Idiopathic Pulmonary Fibrosis

Alternate index label for IPF.

Idiopathic Thrombocytopenic Purpura

Alternate index label; see *Immune thrombocytopenia*.

Idiopathic Thrombocytopenic Purpura

Alternate index label for ITP.

Idiopathic Thrombocytopenic Purpura

Alternate index label for Idiopathic Thrombocytopenic Purpura.

Idiopathic Toe Walking

Alternate index label; see *Toe walking in children*.

Idiosyncrasy

An unusual individual reaction to a medicine or exposure that is not explained by the usual pharmacologic effect.

Idiosyncratic Reaction

An unusual and unexpected reaction to a medicine that is not explained by its usual effects and may occur even at a normal dose.

Idiotypic

Idiotypic is an immunologic, genetic, infectious, developmental, or cellular concept used to explain disease mechanisms, diagnosis, or treatment.

Idioventricular Rhythm

A slow heart rhythm that starts in the lower chambers of the heart instead of the normal pacemaker. It may occur when the usual electrical signal is blocked or when the heart is severely ill.

Idoxuridine

Idoxuridine is a thymidine analogue antiviral formerly used mainly topically for herpes simplex keratitis; it inhibits viral DNA synthesis but can irritate or damage the corneal epithelium.

Ifosfamide

Ifosfamide is an alkylating chemotherapy drug that forms DNA cross-links and is used for selected sarcomas and other cancers; haemorrhagic cystitis, encephalopathy, infertility, and myelosuppression require monitoring and mesna protection.

iFR

Alternate name; see *Instantaneous Wave-Free Ratio*.

Ig

Abbreviation for immunoglobulin, an antibody protein produced by B cells and plasma cells; major classes include IgG, IgA, IgM, IgE, and IgD.

IgA

A type of antibody found mainly in body fluids and linings, including saliva, tears, breast milk, and the digestive and breathing passages. It helps protect these surfaces from infection.

IgA Nephropathy

A kidney disease in which IgA antibodies build up in the kidney's filtering units. It can cause blood or protein in the urine, high blood pressure, and sometimes kidney failure.

IgA Nephropathy (Berger Disease)

A kidney disease caused by deposits of immunoglobulin A in the kidney filters, potentially leading to blood in urine, protein loss, high blood pressure, or kidney failure.

Alternate Names:

Berger's Disease

IgA Vasculitis

An illness in which inflammation damages small blood vessels, often causing a raised purple rash, joint pain, stomach pain, and kidney problems. It is most common in children and may follow an infection.

IgA Vasculitis (Henoch-Schonlein Purpura)

The most common systemic vasculitis in chil-

dren (peak age 2-8 years), characterized by IgA1-containing immune complex deposition in small vessels. Often preceded by an upper respiratory tract infection (e.g., group A *Streptococcus*).

Alternate Names:

Henoch-Schonlein Purpura

IgD

Immunoglobulin D, a membrane-bound antibody expressed on the surface of mature naive B cells (with IgM) serving as a B-cell receptor. Serum IgD is present in trace amounts and its secreted function is unclear. Its role in B-cell activation and signaling is an area of research.

IgE

A type of antibody involved in allergies and defense against some parasites. When it reacts to an allergen, it can trigger symptoms such as sneezing, hives, asthma, or anaphylaxis.

IgG

The most common type of antibody in the blood. It helps neutralize germs and toxins, supports immune protection after infection or vaccination, and can cross the placenta to protect a developing baby.

IgG And Complement Deposition At The Dermis-Epidermis Junction

An immunofluorescence finding that may occur in the lupus band test, in which immune complexes containing immunoglobulin and complement deposit along the dermoepidermal junction. The pattern and clinical context are needed for interpretation.

IgG4-Related Disease

A fibro-inflammatory disorder that can produce tumefactive lesions and organ enlargement in multiple sites. Histology may show IgG4-positive plasma cells, storiform fibrosis, and obliterative phlebitis; diagnosis requires clinicopathologic correlation because no single test is sufficient.

IgG4-Related Disease Arteritis

A form of IgG4-related disease affecting large and medium-sized arteries, characterized by transmural

inflammation with abundant IgG4-positive plasma cells, storiform fibrosis, and obliterative phlebitis. It may involve the aorta and its branches, causing stenosis or aneurysm formation.

IgG4-Related Ophthalmic Disease

An immune-mediated fibroinflammatory disorder that can affect the lacrimal glands, orbit, eyelids, or extraocular muscles. It may cause swelling, mass lesions, double vision, or visual symptoms and can occur with IgG4-related disease in other organs.

Igh Gene

igh gene is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

IgM

Immunoglobulin M, the largest antibody (pentamer), produced during the primary immune response (first antibody to appear). Functions: potent complement activation (classical pathway) and agglutination; expressed as a membrane-bound B-cell receptor on naive B cells.

IIH

Alternate index label; see *Pseudotumor cerebri (idiopathic intracranial hypertension)*.

Ikappab Kinase

A group of enzymes that helps turn on genes involved in inflammation and immune responses.

Ileal Atresia

A condition present at birth in which part of the ileum, the last section of the small intestine, is closed or missing. It causes bowel blockage in a newborn, with a swollen abdomen, green vomiting, and failure to pass the first stool.

Ileal Conduit

A surgically created passage that diverts urine through a segment of ileum to an opening in the abdominal wall.

Ileal Fistula

An abnormal connection involving the ileum and another organ, bowel segment, or skin surface, which may cause intestinal contents to leak and lead to infection, malnutrition, or fluid loss.

Ileal Hemorrhage

Bleeding from the ileum, the distal portion of the small intestine, caused by ulceration, inflammation, vascular lesions, tumors, trauma, or other disease.

Ileal Neoplasm

An ileal neoplasm is an abnormal growth in the ileum, the last part of the small intestine; it may be benign or cancerous.

Ileal Obstruction

Partial or complete blockage of the ileum, the final portion of the small intestine. Causes include adhesions, hernia, inflammatory bowel disease, tumors, and twisting of the bowel. Symptoms may include cramping pain, vomiting, abdominal distension, and inability to pass stool or gas.

Ileal Pouch-Anal Anastomosis

An operation that removes the colon and rectum and creates a pouch from the end of the small intestine. The pouch is connected to the anus so bowel movements can continue through the usual route.

Ileal Stenosis

Narrowing of the ileum that may impair passage of intestinal contents and result from Crohn disease, inflammation, scarring, ischemia, tumor, or surgery.

Ileal Ulcer

An ileal ulcer is a break in the lining of the ileum caused by inflammation, infection, medicines, ischemia, trauma, or another intestinal disorder.

Ileitis

Inflammation of the ileum, the distal segment of the small intestine. Causes include Crohn disease, infection, ischemia, medication-related injury, and other inflammatory disorders; symptoms may in-

clude right-lower-quadrant pain, diarrhea, fever, or bleeding.

Ileitis (Crohn'S Disease)

Alternate index label for Crohn's Disease.

Ileitis, Crohn

Crohn disease involving the ileum, the final part of the small intestine, causing chronic or relapsing inflammation that may produce abdominal pain, diarrhea, weight loss, strictures, or fistulas.

Ileocaecal

Ileocaecal means relating to the junction between the ileum and the caecum, where the small intestine joins the large intestine.

Ileocecal Valve

A sphincter-like fold at the junction of the terminal ileum and caecum. It regulates passage of intestinal contents into the colon and helps limit reflux of colonic contents into the small intestine.

Ileocolitis, Crohn

Crohn disease affecting both the ileum and colon, causing patchy chronic inflammation that may lead to abdominal pain, diarrhea, weight loss, strictures, fistulas, or perianal disease.

Ileostomy

An operation that brings the end of the small intestine through an opening in the abdomen so stool can drain into an external pouch. The output is often liquid, so fluids and skin care are important. after proctocolectomy for ulcerative colitis or familial adenomatous polyposis, or to protect a distal anastomosis. Output is liquid and enzyme-rich, requiring careful fluid and skin management.

Ileostomy And Your Child

Care for a child with an ileostomy includes protecting the skin, measuring output, maintaining hydration, and watching for blockage, leakage, dehydration, or infection.

Ileum

The final and longest part of the small intestine. It absorbs vitamin B12, bile salts, and remaining nutrients before food passes into the large intestine.

Ileus

Failure of coordinated intestinal propulsion without a mechanical blockage, usually from postoperative bowel inhibition, inflammation, metabolic disturbance, or medication effect. It causes distension, nausea, vomiting, and reduced passage of flatus or stool.

Iliac

Relating to the ilium or the iliac region, including the iliac bones and iliac blood vessels.

Iliac Artery

An artery that supplies blood to the pelvis and lower limb.

Iliac Crest

The iliac crest is the curved upper border of the ilium and a landmark and attachment site for abdominal, back, and pelvic muscles.

Iliac Vein

An iliac vein drains blood from the pelvis and lower limb toward the inferior vena cava; thrombosis can cause swelling and pulmonary embolism.

Common Iliac Vein

A vein formed by the union of the internal and external iliac veins. The right and left common iliac veins unite to form the inferior vena cava.

External Iliac Vein

A continuation of the femoral vein above the inguinal ligament that joins the internal iliac vein to form the common iliac vein.

Internal Iliac Vein

A vein draining the pelvic organs, pelvic wall, and gluteal region that joins the external iliac vein to form the common iliac vein.

Iliopsoas Muscles

The psoas major and iliacus muscles, which work together to flex the hip.

Alternate Names:

Iliopsoas

Iliotibial Band

A strong band of tissue running along the outer side of the thigh from the hip to the knee. It helps stabilize the hip and knee during movement.

Iliotibial Band Syndrome

Pain on the outer side of the knee caused by irritation of the iliotibial band, especially during repeated activities such as running or cycling.

Alternate Names:

ITBS (Iliotibial Band Syndrome)

Iliotibial Band Syndrome

Alternate index label for Iliotibial Band Syndrome.

Ilium

Largest hip bone forming the superior portion. Body contributes to acetabulum; wing has a concave iliac fossa. Iliac crest runs ASIS to PSIS, a palpable landmark for lumbar puncture level. Sacroiliac joint articulates with sacrum. Gluteal muscles originate from its external surface; iliacus fills its fossa.

Illness Anxiety Disorder

Preoccupation with having or acquiring a serious illness, with minimal or no somatic symptoms present. The individual has a high level of anxiety about health, performs excessive health-related behaviors, and experiences maladaptive avoidance. Symptoms persist for at least six months. The individual is not delusional and insight may vary.

Illusion

A chemical substance or compound called illusion. Its properties, uses, and risks depend on its composition and clinical context.

Iloprost

A prostacyclin analogue that causes vasodilation

and inhibits platelet aggregation; inhaled or infused formulations are used for selected forms of pulmonary hypertension and severe peripheral ischemia.

Image View

An image view is a particular angle or projection used when producing a medical image, such as a radiograph.

Image-Guided Ablation

Destruction of abnormal tissue using an energy source such as heat, cold, or chemicals while imaging guides placement and monitors the procedure.

Image-Guided Biopsy

Removal of a tissue sample while ultrasound, CT, MRI, mammography, or fluoroscopy guides the needle or instrument to the target and helps avoid critical structures.

Image-Guided Drainage

Placement of a catheter or needle under imaging guidance to evacuate fluid, pus, blood, or bile from a pathologic collection. Indications include abscess, biloma, urinoma, pleural collection, and symptomatic cysts.

Image-Guided Therapy

A minimally invasive procedure performed with imaging to guide instruments or treatment to a target tissue.

IGRT

Image-guided radiation therapy uses medical imaging, such as CT, MRI, ultrasound, or X-ray, to position the patient and target radiation accurately before or during treatment. It helps deliver radiation to a tumor while reducing exposure to nearby healthy tissue.

Imaging

The production of pictures of internal body structures using methods such as radiography, ultrasound, computed tomography, magnetic resonance

imaging, or nuclear medicine.

Imaging And Radiology

Imaging and radiology use methods such as X-ray, ultrasound, CT, and MRI to create pictures that help diagnose or guide treatment of disease.

Imaging Technique

An imaging technique is a method for producing pictures of body structures or processes, such as ultrasound, CT, MRI, or nuclear imaging.

Imaging-Guided Biopsy

Alternate index label; see *Image-guided biopsy*. Appropriate specimen handling, bleeding-risk assessment, informed consent, and post-procedure monitoring remain essential.

Imatinib Mesylate

The mesylate salt of imatinib, a tyrosine-kinase inhibitor used to treat BCR-ABL-positive chronic myeloid leukemia, gastrointestinal stromal tumors with KIT alterations, and other selected cancers.

Imerslund-Gräsbeck Syndrome

An inherited disorder of selective vitamin B12 malabsorption caused by defects in cubilin or amnionless, often producing megaloblastic anemia and low B12 levels despite adequate intake.

Imidazoles

A group of antifungal medicines, including clotrimazole, miconazole, and ketoconazole, used to treat fungal infections of the skin, mouth, or other body areas.

Imide

An imide is a chemical compound containing nitrogen bonded to two carbonyl groups and may have pharmaceutical, metabolic, or industrial significance.

Imines

Imines are compounds containing a carbon-nitrogen double bond and occur in chemical synthesis,

metabolism, and drug chemistry.

Imino Acid

imino acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Iminoacetic Acid Scan

A nuclear medicine scan that uses a small amount of radioactive medicine to show how bile moves through the liver, gallbladder, and bile ducts. It is also called a HIDA scan.

Imipenem

A strong antibiotic, usually given with cilastatin, used for serious bacterial infections. It can cause allergic reactions, stomach problems, seizures, or antibiotic resistance.

Imipramine

A medicine used for depression and some bladder problems. It can cause dry mouth, constipation, dizziness, abnormal heart rhythms, and dangerous effects in overdose.

Imipramine Overdose

Toxic ingestion of imipramine, a tricyclic antidepressant, which can cause anticholinergic effects, seizures, hypotension, ventricular arrhythmias, coma, and life-threatening cardiac toxicity.

Imiquimod

Imiquimod is a topical immune-response modifier that activates Toll-like receptor 7 and induces cytokines; it treats selected actinic keratoses, superficial basal-cell carcinomas, and external genital warts, often causing local inflammation.

Immature

Immature means not fully developed or differentiated, such as an immature cell, organ, tissue, or developmental stage.

Immature Ovarian Teratoma

A malignant germ-cell tumor containing immature

embryonal tissues, usually diagnosed in adolescents or young adults and graded by the amount of immature neuroepithelium.

Immaturity

Incomplete physical, neurologic, emotional, or developmental maturity relative to the expected stage of life.

Immediate Hypersensitivity

A rapid, typically immunoglobulin-E-mediated hypersensitivity reaction in which allergen cross-linking of mast-cell or basophil receptors releases inflammatory mediators. It can cause urticaria, bronchospasm, gastrointestinal symptoms, or anaphylaxis.

Immersion Foot (Trench Foot)

A painful cold injury caused by prolonged exposure of the feet to cold, wet conditions without freezing, leading to numbness, swelling, and skin changes.

Alternate Names:

Trench Foot

Immobilization

A chemical substance or compound called immobilization. Its properties, uses, and risks depend on its composition and clinical context.

Immobilize

To restrict movement of a body part, often with a splint or cast, to support healing.

Alternate Names:

Immobilization

Immune

Having an immune response that recognizes and helps protect against a particular antigen, infection, or toxin; immunity may be innate or acquired.

Immune And Infectious Diseases

A medical field encompassing disorders of the immune system and illnesses caused by pathogenic microorganisms. Immune system disorders include autoimmune diseases (e.g., lupus, rheumatoid arthritis).

tis), immunodeficiencies, and hypersensitivity reactions.

Immune Checkpoint

An inhibitory or stimulatory receptor-ligand pathway that regulates lymphocyte activation and limits tissue injury. Therapeutic blockade of inhibitory checkpoints such as CTLA-4 or PD-1 can enhance antitumor immunity but may cause immune-related adverse effects.

Immune Complex

A soluble or tissue-associated aggregate formed by antigen bound to antibody. Persistent or inadequately cleared immune complexes may activate complement and deposit in tissues, producing type III hypersensitivity and vasculitic or renal injury.

Immune Evasion

The ability of a pathogen, tumor, or other abnormal cell to avoid recognition or destruction by the immune system.

Immune Hemolytic Anemia

Anemia caused by immune-mediated destruction of red blood cells, either by autoantibodies or antibodies directed against foreign red-cell antigens. It may cause fatigue, jaundice, dark urine, splenomegaly, and laboratory evidence of hemolysis.

Immunoheolysis

Destruction of red blood cells by an immune reaction involving antibodies, complement, or both. It may occur in autoimmune hemolytic anemia, incompatible transfusion reactions, or other antibody-mediated disorders.

Immune Response

The coordinated reaction of the immune system to foreign substances (antigens), pathogens, or abnormal cells. The innate immune response acts rapidly and non-specifically (physical barriers, phagocytes, complement, NK cells, inflammation).

Immune Status

Immune status describes the functional state of a person's immune defenses, including immunocompetence, prior infection or vaccination, immunosuppression, and susceptibility to infection.

Immune System

Complex network of cells, tissues, and organs that defends the body against pathogens; includes innate and adaptive immunity. Its significance is established from its characteristic features, causes, and clinical context.

Immune Thrombocytopenia

An autoimmune disorder characterized by isolated thrombocytopenia (platelet count $<100 \times 10^9/L$) due to autoantibody-mediated platelet destruction and impaired platelet production. Primary ITP (no identifiable cause) vs secondary (SLE, CLL, HIV, HCV, drugs).

Immune Thrombocytopenic Purpura

An autoimmune bleeding disorder characterized by isolated thrombocytopenia (platelet count $<100,000/L$), usually with otherwise preserved white blood cell and hemoglobin levels. Caused by autoantibodies (primarily IgG) directed against platelet surface glycoproteins (IIb/IIIa or Ib/IX), leading to splenic macrophage phagocytosis of platelets.

Immune Thrombocytopenic Purpura (Idiopathic Thrombocytopenic Purpura)

Alternate index label for Idiopathic Thrombocytopenic Purpura.

Immune Thrombocytopenic Purpura

Immune thrombocytopenic purpura is immune-mediated platelet destruction causing isolated thrombocytopenia with bruising, petechiae, mucosal bleeding, or no symptoms.

Immune Tolerance

The state of unresponsiveness of the immune system to self-antigens and to specific foreign antigens, preventing autoimmunity while preserving responses to pathogens. Central tolerance: deletion of self-

reactive lymphocytes in the thymus (T cells) and bone marrow (B cells).

Immune-Mediated Necrotizing Myopathy

A subacute severe proximal myopathy with very high CK (often >10,000), minimal inflammation on biopsy, necrotic fibers. Two major autoantibodies: anti-HMGCR (3-hydroxy-3-methylglutaryl-CoA reductase, statin-associated or spontaneous) and anti-SRP (signal recognition particle).

Immune-Related Adverse Event

An inflammatory or autoimmune toxicity caused by immune checkpoint blockade or another immune-activating cancer treatment. It may affect skin, bowel, liver, endocrine organs, lungs, heart, kidneys, or nervous system and often requires treatment-specific immunosuppression.

Immunity

The ability of the body to recognize and resist infectious agents or other foreign substances through innate or adaptive defenses.

Immunization

The process of inducing protection against an infection, usually through vaccination. Recommended vaccines and schedules depend on age, previous doses, pregnancy, immune status, medical conditions, travel, exposure risk, and current public-health guidance.

Immunizations For People With Diabetes

Vaccination is especially important in diabetes because infections may be more severe; recommendations depend on age, health status, vaccines received, and local guidance.

Immunoassay

An immunoassay is a laboratory test that uses antibodies to detect or measure a substance such as a hormone, drug, antigen, or antibody.

Immunoblast Count

An immunoblast count measures activated lympho-

cytes, usually in a tissue or blood sample, and may help evaluate immune responses or lymphoid disease.

Immunoblotting

Immunoblotting, or Western blotting, separates proteins by electrophoresis and detects a target with specific antibodies; band pattern and intensity are interpreted with controls and clinical context.

Immunochemistry

The study or laboratory measurement of biological molecules using specific antigen-antibody reactions, including immunoassays and immunostaining.

Immunocompetence

Immunocompetence is the capacity of the immune system to recognize antigens and mount an effective, appropriately regulated response; deficiency increases infection risk and may impair vaccine responses.

Immunocompetent

Having an immune system capable of mounting an appropriate response to common infectious or foreign antigens.

Immunocompromised

Having weakened or impaired immune defenses because of disease, medication, cancer treatment, transplantation, or another cause, increasing susceptibility to some infections and complications.

Immunocompromised Host

A person whose immune defenses are impaired by congenital disease, infection, malignancy, immunosuppressive treatment, organ failure, or another condition. Such patients have increased susceptibility to opportunistic infection, severe disease, and atypical presentations.

Immunoconjugate

A therapeutic or diagnostic molecule in which an antibody or other targeting protein is chemically linked to a drug, toxin, enzyme, or imaging agent.

Immunocytochemistry

A laboratory method using labeled antibodies to identify specific proteins or antigens within cells or tissue sections.

Immunodeficiency

Impaired development or function of one or more components of the immune system, resulting in increased susceptibility to infection, malignancy, or autoimmunity. It may be primary and genetic or secondary to disease, medication, malnutrition, or another acquired cause.

Immunodeficiency And Infections

Immunocompromised individuals are at increased risk for opportunistic infections. Primary immunodeficiencies are genetic disorders affecting immune function. Secondary immunodeficiencies include HIV/AIDS, organ transplantation, chemotherapy, and corticosteroid use.

Immunodeficiency Diseases

Disorders in which one or more components of the immune system are absent, deficient, or dysfunctional. They may be inherited or acquired and can cause recurrent, severe, unusual, or persistent infections, autoimmune disease, and increased malignancy risk.

Immunodeficiency Disorders

Disorders caused by impaired development or function of the immune system, leading to increased susceptibility to infections and sometimes autoimmunity or malignancy. See *Immunodeficiency Diseases*.

Immuno-electrophoresis

A laboratory method that separates proteins by electrophoresis and identifies them with specific antibodies, often to study immunoglobulins.

Immuno-electrophoresis - Urine

A laboratory technique that separates and identifies urinary proteins with specific antibodies, used to investigate abnormal immunoglobulin excretion and monoclonal light chains.

Immuno-electrophoresis – Blood

A laboratory technique that separates blood proteins by electrophoresis and identifies them with specific antibodies, used historically to evaluate abnormal immunoglobulins and monoclonal proteins.

Immunofixation - Urine

A urine test that uses electrophoresis and antibodies to identify and type monoclonal immunoglobulins or light chains, including Bence Jones proteins.

Immunofixation Blood Test

A blood test that identifies and types monoclonal immunoglobulins, helping evaluate monoclonal gammopathy, multiple myeloma, and related plasma-cell disorders.

Immunofluorescence

A technique using fluorescently labeled antibodies to detect specific antigens in tissues, cells, or microorganisms; visualized by fluorescence microscopy.

Immunofluorescence Technique

An immunofluorescence technique uses fluorescently labelled antibodies to locate specific proteins, organisms, or immune deposits in a sample.

Immunogenetics

immunogenetics is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

Immunogenicity

The ability of an antigen, vaccine, or other substance to provoke an immune response, including antibody or cellular responses.

Immunoglobulin

An antibody protein produced by immune cells to recognize and help neutralize foreign substances.

Immunoglobulin (Antibody)

Y-shaped glycoprotein produced by plasma cells that specifically binds antigens. Composed of two

identical heavy chains and two identical light chains linked by disulfide bonds. Variable regions (Fab) bind antigen; constant region (Fc) mediates effector functions (complement activation, Fc receptor binding).

Alternate Names:

Antibody

Immunoglobulin A

An antibody found mainly in mucosal secretions, including saliva, tears, and intestinal fluid.

Immunoglobulin D

immunoglobulin d concerns immunoglobulin d, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Immunoglobulin E

An antibody class that plays a central role in many immediate allergic reactions.

Alternate Names:

IgE

Immunoglobulin G

The most abundant antibody in blood, providing long-lasting protection against many infections.

Immunoglobulin Idiotyp

immunoglobulin idiotype concerns immunoglobulin idiotype, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Immunoglobulin Isotype

immunoglobulin isotype concerns immunoglobulin isotype, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Immunoglobulin M

The first major antibody produced during an initial immune response.

Immunoglobulin Replacement Therapy

Indications: primary antibody deficiencies (XLA,

CVID, hyper-IgM, specific antibody deficiency), secondary hypogammaglobulinemia (CLL, myeloma, post-rituximab, post-transplant) with recurrent infections.

Immunohistochemistry

A technique that uses labeled antibodies to detect specific antigens in tissue sections. It supports tumor classification, identification of lineage, assessment of predictive biomarkers, and detection of infectious or immune-mediated processes.

Immunologic Adjuvant

A substance added to an antigen or vaccine to strengthen or shape the immune response.

Immunologic Factor

A substance, cell, gene, or process that contributes to immune-system function or regulation.

Immunologic Memory

The capacity of the adaptive immune system to respond more rapidly and strongly after re-exposure to an antigen because of long-lived memory lymphocytes.

Immunologic Skin Test

An immunologic skin test places a small amount of an antigen or allergen in or on the skin to assess immune reactivity, such as allergy or tuberculosis exposure.

Immunologic Surveillance

The immune system's recognition and removal of abnormal, infected, or damaged cells.

Immunologic Technique

A laboratory or clinical method that uses antibodies, antigens, or immune reactions to detect or measure a substance.

Immunologic Test

An immunologic test examines immune cells, antibodies, antigens, or immune-related proteins to help evaluate infection, allergy, autoimmune disease, or

immune deficiency.

Immunological Memory

The ability of the adaptive immune system to mount a faster, larger, and more effective response upon re-exposure to an antigen, conferred by long-lived memory B cells and memory T cells generated during the primary response.

Immunological Synapse

The organized contact interface between a T lymphocyte or other immune cell and its target or antigen-presenting cell, where receptor and signaling molecules coordinate immune activation.

Immunological Tolerance

The reduced or absent immune response to a specific antigen, including the normal lack of attack on the body's own tissues.

Immunologically Privileged Site

A body site where immune responses are limited to reduce risk of tissue damage.

Immunology

The medical and biological study of the immune system, immune responses, and immune disorders.

Immunology

Antigen-presenting cell, an immune cell that processes antigens and displays peptide fragments on major histocompatibility complex molecules to activate T cells.

Immunomodulation

A chemical substance or compound called immunomodulation. Its properties, uses, and risks depend on its composition and clinical context.

Immunomodulators In Dermatology

Topical immunomodulators treat inflammatory skin diseases without corticosteroid side effects. Pimecrolimus 1 percent cream and tacrolimus 0.03 and 0.1 percent ointments are calcineurin inhibitors used for atopic dermatitis, facial eczema, and intertrigi-

nous areas. They are steroid-sparing agents.

Immunophenotyping

Identification and characterization of cells according to their surface, cytoplasmic, or nuclear antigens, usually by flow cytometry or immunohistochemistry. It is used to classify leukemias, lymphomas, immune-cell subsets, and some immunodeficiency disorders.

Immunoprecipitation

A chemical substance or compound called immunoprecipitation. Its properties, uses, and risks depend on its composition and clinical context.

Immunoprotein

immunoprotein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Immunoradiometric Assay

An immunoradiometric assay uses radiolabelled antibodies to measure a specific substance in a sample.

Immunoregulation

A chemical substance or compound called immunoregulation. Its properties, uses, and risks depend on its composition and clinical context.

Immunosenescence

Age-related changes in innate and adaptive immunity that alter infection susceptibility, vaccine responses, inflammation, and cancer surveillance. It is influenced by chronic disease, nutrition, medications, and functional status.

Immunosuppression

Reduction of immune-system activity, caused by disease or intentionally induced with medications or other treatment.

Immunosuppression In Transplant

Medications to prevent rejection. Induction with thymoglobulin or basiliximab. Maintenance with

tacrolimus, mycophenolate, and prednisone. Monitor drug levels and renal function. Calcineurin inhibitor nephrotoxicity is major concern requiring dose optimization.

Immunosuppression Screening

Pre-treatment evaluation for latent or active infection, vaccination status, malignancy risk, and organ dysfunction before immunosuppressive therapy. Testing commonly includes tuberculosis, hepatitis B and C, HIV, blood counts, liver and kidney function, and pregnancy when relevant.

Immunosuppressive

Agent or therapy that suppresses immune system activity; used to prevent organ transplant rejection and treat autoimmune diseases. Its significance is established from its characteristic features, causes, and clinical context.

Immunotherapy

Treatment using immune system to fight cancer. Types: checkpoint inhibitors (PD-1, PD-L1, CTLA-4), CAR T-cell therapy, cytokines, cancer vaccines. Examples: pembrolizumab, nivolumab, ipilimumab, axicabtagene ciloleucel.

Immunotherapy And Biologic Treatments

Advanced therapies that harness or modify the immune system to treat diseases, particularly cancer, autoimmune disorders, and inflammatory conditions. Biologics are large-molecule drugs derived from living organisms that target specific immune pathways, such as TNF inhibitors, monoclonal antibodies, and checkpoint inhibitors.

Immunotherapy For Cancer

Cancer treatment that stimulates, restores, redirects, or supplies immune activity to recognize and attack malignant cells, including immune-checkpoint inhibitors, cellular therapies, and antibodies.

Immunotherapy In Dermatology

Cancer immunotherapy uses medicines that alter immune activity against skin cancers. Checkpoint inhibitors target pathways such as PD-1, PD-L1,

or CTLA-4. Their use depends on the tumor type, stage, prior treatment, and the patient's health.

Immunotoxicity

Immune system damage from chemicals. Effects: immunosuppression, hypersensitivity, autoimmunity. Examples: benzene, chemotherapy, biologics.

Immunotoxin

A targeted toxin made by linking a toxic protein or drug to an antibody or binding molecule that directs it toward selected cells.

Immunoturbidimetry

A laboratory method that measures cloudiness produced when antibodies react with a target antigen, allowing estimation of the target's concentration.

Imp Dehydrogenase

IMP dehydrogenase converts inosine monophosphate to xanthosine monophosphate in guanine-nucleotide synthesis and is a target of some immunosuppressive drugs.

Impact

The forceful contact or collision of one object or body part with another, or the effect a disease, treatment, or event has on health; context determines the meaning.

Impacted

Firmly wedged or unable to move from its normal position, as with an impacted tooth.

Impacted Fracture

A fracture in which one broken bone fragment is driven firmly into another, often producing relative stability but sometimes causing shortening or deformity.

Impacted Tooth

Tooth unable to erupt into functional position. Most common: mandibular third molars, maxillary canines. May cause pathology: cyst, root resorption, caries. Treatment: monitoring or extraction.

Impacted Wisdom Teeth

Wisdom teeth that cannot emerge normally because they are blocked by bone, gum, or another tooth.

Alternate Names:

Wisdom Teeth, Impacted

Impaction

Obstruction caused by material that is firmly lodged, such as hardened stool in the bowel.

Impaired Fasting Glucose

Blood glucose above the normal range after fasting but below the diagnostic threshold for diabetes.

Impaired Glucose Tolerance

Blood glucose above the normal range after a glucose challenge but below the diagnostic threshold for diabetes.

Impedance

Impedance is the opposition to the flow of an alternating electrical current; in medicine it can help assess tissue properties or body fluids.

Impedance Hearing Testing

A group of tests, including tympanometry, that measure the movement of the eardrum and middle-ear function.

Impedance Plethysmography

A noninvasive method that detects changes in limb electrical impedance caused by changes in blood volume. It has been used to assess venous blood flow and suspected deep-vein thrombosis, although duplex ultrasonography is generally preferred for current diagnostic evaluation.

Impella Device

Percutaneous microaxial left ventricular assist device that actively pumps blood from the LV into the aorta, reducing LV workload and increasing cardiac output. Types: Impella 2.5 (up to 2.5 L/min), Impella CP (up to 3.7 L/min), Impella 5.0 (up to 5 L/min, surgically placed).

Imperforate Anus

A congenital anomaly where the anus is completely absent or fails to open at the perineum. Classified as low (distal to the levator ani complex) or high (proximal to the levator ani). Often associated with other anomalies including vertebral, anal, cardiac, tracheoesophageal, renal, and limb defects.

Imperforate Anus Repair

A medical procedure involving imperforate anus repair. Indications, risks, preparation, and follow-up depend on the patient and the specific procedure.

Imperforate Hymen

A congenital obstruction of the vaginal opening by an intact hymen without a normal central opening. After puberty it may cause absent menstruation, cyclic pelvic pain, urinary retention, or a pelvic mass from retained menstrual blood.

Imperforate Vagina

Imperforate vagina is a rare congenital absence or obstruction of the vaginal canal, often associated with genital outflow obstruction, cyclical pelvic pain, haematocolpos, and urinary or reproductive complications.

Impetigo

Impetigo is a superficial contagious bacterial skin infection producing fragile blisters or honey-colored crusts, commonly caused by *Staphylococcus aureus* or *Streptococcus pyogenes*, affecting children.

Alternate Names:

School Sores (Impetigo)

Impingement Syndrome

Pain and restricted movement caused when a tendon or other soft tissue is compressed between bones, commonly in the shoulder during elevation of the arm.

Implant

A medical device, tissue, or material placed inside the body to replace, support, monitor, or deliver treatment.

Implant (Dental)

A biocompatible fixture placed in the jawbone to support a dental prosthesis. It may be used for one missing tooth or to support a partial or complete denture; complications include infection, bone loss, and mechanical failure.

Alternate Names:

Dental

Implant Failure

Loss of an implant's intended structural, biologic, or functional performance. Causes include infection, fracture, loosening, wear, malposition, thrombosis, material degradation, and patient or tissue factors.

Implant Sterilization

Validated processing that removes or inactivates all viable microorganisms, including bacterial spores, from an implant. Methods include moist heat, ethylene oxide, radiation, and low-temperature plasma according to material and device requirements.

Implant-Associated Infection

Infection involving an implanted medical device or the tissue surrounding it. Microbial adherence and biofilm formation can impair antimicrobial penetration and often require device-directed therapy, source control, or removal.

Implant-Supported Prosthesis

A fixed or removable dental prosthesis that is supported or retained by one or more dental implants.

Implantable Cardioverter Defibrillator

An implanted device that detects dangerous ventricular rhythms and treats them with antitachycardia pacing or an electrical shock.

Implantable Cardioverter-Defibrillator

An implanted device that monitors heart rhythm and delivers electrical therapy (antitachycardia pacing or defibrillation) for life-threatening ventricular arrhythmias.

Implantable Defibrillator

A surgically implanted device that detects dangerous ventricular arrhythmias and delivers pacing or an electrical shock to restore an effective rhythm.

Implantable Loop Recorder

Subcutaneous device providing continuous ECG monitoring for up to 3 years. Used for evaluation of unexplained syncope, recurrent palpitations, cryptogenic stroke, and occult atrial fibrillation detection. Small size (coin-sized), inserted under local anesthesia in the chest.

Implantable Port

A subcutaneously placed vascular-access device consisting of a reservoir connected to a catheter, used for repeated administration of medication, fluids, blood products, or parenteral nutrition. It requires sterile access and may be complicated by infection, thrombosis, occlusion, or mechanical failure.

Implantation

The embedding of the blastocyst into the endometrium, occurring approximately 6-10 days after ovulation. The blastocyst hatches from the zona pellucida, and the trophoblast invades the decidua. Implantation triggers hCG production, which maintains the corpus luteum. Failure of implantation is a common cause of early pregnancy loss.

Implantation Procedure

A medical procedure involving implantation procedure. Indications, risks, preparation, and follow-up depend on the patient and the specific procedure.

Implants Endometrial (Endometriosis)

Alternate index label for Endometriosis.

Importance Weight

An importance weight is a statistical weight used to adjust observations so a sample better represents a target population or distribution.

Impotence (Erectile Dysfunction [ED])

The persistent inability to attain or maintain a pe-

nile erection of sufficient rigidity for satisfactory sexual intercourse. Pathophysiologically categorized into vasculogenic (arteriogenic insufficiency from atherosclerosis, veno-occlusive dysfunction), neurogenic (diabetic autonomic neuropathy, radical prostatectomy, spinal cord injury), endocrinologic (hypogonadism, hyperprolactinemia), drug-induced (antihypertensives, SSRIs, antiandrogens), and psychogenic etiologies. Often serves as an early harbinger of occult systemic coronary artery disease due to endothelial dysfunction in smaller penile arteries; evaluated with fasting glucose/HbA1c, lipid panels, and morning serum testosterone, and treated first-line with oral phosphodiesterase-5 (PDE5) inhibitors (e.g., sildenafil, tadalafil).

Alternate Names:

Erectile dysfunction

Impotence (Erectile Dysfunction (ED, Impotence))

Alternate index label for Erectile Dysfunction (ED, Impotence).

Impression

Summary of key findings and recommendations in radiology report.

Imprinting

A pattern of gene expression determined by whether a gene copy was inherited from the mother or father.

Impulse

An impulse is a brief signal or force; in physiology it commonly refers to an electrical signal travelling along a nerve or through the heart.

Impulse Oscillometry

A pulmonary function testing technique that measures respiratory impedance by applying small pressure oscillations during tidal breathing. It assesses large airway resistance (R5), small airway resistance (R5-R20), and reactance.

Impulsive Behavior

Impulsive behavior is action with limited forethought or inhibition despite possible harm and

may occur with psychiatric, neurologic, substance-related, or developmental conditions.

Impulsivity

A tendency to act quickly without adequate consideration of consequences or difficulty delaying a response; it can occur in ADHD, mania, substance use, and other conditions.

In

In clinical measurements, an abbreviation for inch or inches; the same letters may have other meanings in different contexts.

In Situ

Confined to the exact site of anatomical origin without invasion into adjacent tissue planes or across anatomical barriers. In surgical pathology and oncology, carcinoma in situ (CIS; e.g., ductal carcinoma in situ [DCIS] of the breast, high-grade squamous intraepithelial lesion [HSIL] of the cervix) denotes a malignant epithelial proliferation that has not breached the basement membrane (basal lamina); because lymphovascular structures reside exclusively in the underlying stroma, lesions strictly in situ possess zero potential for regional lymphatic or distant hematogenous metastasis.

Alternate Names:

In place

In Situ Carcinoma

Abnormal malignant cells confined to the tissue layer in which they arose and not yet invading through the basement membrane.

In Situ Hybridization

A chemical substance or compound called in situ hybridization. Its properties, uses, and risks depend on its composition and clinical context.

In Vitro

In vitro means performed outside a living organism, such as in a test tube, culture dish, or laboratory system.

In Vitro Fertilisation

An assisted reproductive technology in which eggs are retrieved from the ovaries, fertilized with sperm in a laboratory, and resulting embryos are transferred to the uterus.

In Vitro Fertilization

A fertility treatment in which an egg and sperm are combined in a laboratory and a resulting embryo is transferred to the uterus.

In Vitro Fertilization

In vitro fertilization fertilizes retrieved eggs in a laboratory and transfers resulting embryos to the uterus, forming part of assisted reproductive treatment.

Intracytoplasmic Sperm Injection

A fertility treatment in which a single sperm is injected directly into an egg in a laboratory to help achieve fertilisation. It is also called ICSI.

In Vivo

In vivo means occurring or being studied within a living organism.

In-Vivo

In-vivo means occurring or being studied within a living organism.

Inactivated Vaccines

Vaccines made from microorganisms that have been killed and cannot reproduce.

Inactivator

Inactivator is an immunologic, genetic, infectious, developmental, or cellular concept used to explain disease mechanisms, diagnosis, or treatment.

Inalimarev

Inalimarev is an investigational antiviral agent studied for activity against selected viral infections.

Inamrinone

Inamrinone is an intravenous medicine that in-

creases heart contraction and widens blood vessels; it was used short term in severe heart failure but can cause low blood pressure and abnormal rhythms.

Inanition

A chemical substance or compound called inanition. Its properties, uses, and risks depend on its composition and clinical context.

Inborn Error Of Metabolism

An inherited disorder caused by a defect in a metabolic pathway, often resulting in accumulation of harmful substances or deficiency of essential products.

Inborn Errors Of Metabolism

Inherited disorders caused by defects in enzymes, transporters, or other metabolic proteins, leading to abnormal buildup or deficiency of substances.

Inbreed

To reproduce between genetically related individuals; inbreeding can increase the chance that recessive inherited disorders will occur.

Inbreeding

Inbreeding is an immunologic, genetic, infectious, developmental, or cellular concept used to explain disease mechanisms, diagnosis, or treatment.

Inbreeding, Coefficient Of

The probability that the two copies of a gene at a locus in an individual are identical by descent from a common ancestor; it is used in population and medical genetics.

Incarcerated Hernia

A hernia whose protruded tissue cannot be returned to the abdominal cavity. It may progress to strangulation and requires urgent clinical assessment.

Incentive Spirometry

A breathing device that provides visual feedback while a person inhales slowly and deeply. It may be used with coughing, mobilization, and other mea-

tures to encourage lung expansion after illness or surgery; it is a lung-expansion exercise and is not the same as diagnostic spirometry.

Incest

Incest is an immunologic, genetic, infectious, developmental, or cellular concept used to explain disease mechanisms, diagnosis, or treatment.

Incidence

Incidence is the number of new cases of a disease or event arising in a population during a specified period.

Incidental Finding

An unexpected observation discovered during an examination performed for another reason. Its importance depends on the finding and the clinical context.

Incidentaloma

Incidental imaging finding of uncertain significance. May require further evaluation.

Incised Wound

Clean-cut wound produced by a sharp-edged instrument (scalpel, knife, glass) that is longer on the skin surface than it is deep, featuring clean edges without tissue bridging or marginal abrasion.

Incision

A deliberate surgical cut made into the skin or other tissue.

Incisional Biopsy

Surgical removal of only part of a lesion for diagnosis, generally used when the lesion is too large or located where complete removal would be inappropriate as an initial procedure.

Incisional Hernia

A hernia through a weakness in a previous surgical incision. It may appear as a bulge that becomes more prominent with standing or straining. Risk factors include wound infection, obesity, smoking,

poor nutrition, chronic cough, and repeated surgery.

Incisor

One of the front teeth, adapted mainly for cutting or biting food.

Incisors

Incisors are the flat front teeth used mainly for biting and cutting food.

Inclusion Body

A circumscribed mass of foreign, metabolic, or abnormal material within a cell's cytoplasm or nucleus. Inclusion bodies may contain aggregated or misfolded proteins, viral components, glycogen, lipids, or other substances, and their appearance and location can help identify cellular injury or disease.

Inclusion Body Myositis

A slowly progressive inflammatory and degenerative muscle disease, usually causing asymmetric weakness of the finger flexors and quadriceps. Muscle biopsy may show endomysial inflammation and rimmed vacuoles. Response to immunotherapy is often limited.

Inclusion Conjunctivitis

A chemical substance or compound called inclusion conjunctivitis. Its properties, uses, and risks depend on its composition and clinical context.

Inclusion-Cell (I-Cell) Disease

A rare inherited lysosomal-storage disorder caused by impaired tagging and trafficking of lysosomal enzymes, leading to their secretion outside cells. It causes severe developmental delay, skeletal abnormalities, coarse facial features, and progressive multisystem disease. Management is supportive and coordinated by specialists.

Alternate Names:

I-Cell

Incompetence

Failure of a structure to close or function adequately; examples include valvular incompetence of the heart

and cervical incompetence during pregnancy.

Incompetent Cervix

Early painless opening of the cervix during pregnancy, which may lead to pregnancy loss or premature birth.

Incomplete Miscarriage

Partial expulsion of products of conception, with some pregnancy tissue remaining in the uterus.

Incomplete Ossification

A chemical substance or compound called incomplete ossification. Its properties, uses, and risks depend on its composition and clinical context.

Incontinence

The involuntary loss of control over urination (urinary incontinence) or defecation (fecal incontinence).

Incontinence (Fecal)

Involuntary fecal leakage. Causes: diarrhea, constipation with overflow, sphincter damage, neurological disease, medications. Assessment: history, examination, anorectal manometry. Treatment: bowel regimen, sphincter exercises, biofeedback, surgery if indicated.

Alternate Names:

Fecal

Incontinence (Urinary)

Involuntary urine leakage. Types: stress (cough/sneeze), urge (overactive bladder), overflow (retention), functional (immobility/cognition), mixed. Assessment: history, bladder diary, urinalysis, post-void residual.

Alternate Names:

Urinary

Incontinence, Bowel

Alternate index label; see *Fecal incontinence*.

Incontinence, Fecal

Alternate index label; see *Fecal incontinence*.

Incontinence, Urinary

Alternate index label; see *Urinary incontinence*.

Incontinentia Pigmenti

An X-linked dominant disorder affecting skin, teeth, eyes, and the nervous system, caused by pathogenic variants in IKBKG. Skin lesions evolve through vesicular, verrucous, hyperpigmented, and atrophic stages and may be associated with retinal disease or seizures.

Incoordination

Lack of coordination between muscles or body movements; the term is often used in older medical language.

Incoordination

A chemical substance or compound called incoordination. Its properties, uses, and risks depend on its composition and clinical context.

Incoordination (Ataxia)

Impaired coordination of voluntary movement, often caused by disease affecting the cerebellum or other parts of the nervous system.

Alternate Names:

Ataxia

Increased Head Circumference

Increased head circumference means head size above the expected range for age and sex and may reflect familial size, hydrocephalus, swelling, or a genetic condition.

Increased Intracranial Pressure

Abnormally elevated pressure within the skull caused by brain swelling, hemorrhage, mass lesion, impaired cerebrospinal-fluid circulation, or venous obstruction. It can cause headache, vomiting, visual disturbance, reduced consciousness, and brain herniation.

Increased Libido

Increased libido is heightened sexual desire that may be normal or associated with hormones, medicines,

mood elevation, neurologic disease, or substance use.

Increased Osteoid

Increased osteoid is excess unmineralized bone matrix, seen when bone formation outpaces mineralization as in osteomalacia or some metabolic disorders.

Increased Vascular Permeability

A fundamental feature of acute inflammation that allows plasma proteins, fluid, and leukocytes to exit the vascular space and enter the extravascular tissue.

Increment

An increase or stepwise change in an amount, measurement, dose, or treatment intensity.

Incretins

Incretins are gut hormones, mainly GLP-1 and GIP, released after meals that enhance glucose-dependent insulin secretion; GLP-1 also suppresses glucagon, slows gastric emptying, and reduces appetite.

Incubation

Incubation is the time between exposure to an infectious agent and the appearance of symptoms.

Incubation Period

The time between when a person is exposed to an infection and when symptoms appear.

Incubator

A controlled-environment chamber culturing microorganisms, cells, or tissues at defined temperature, humidity, and atmospheric composition.

Incus

The incus, commonly known as the anvil, is the middle ossicle of the three bones in the middle ear, articulating with the malleus at the incudomalleolar joint and with the stapes at the incudostapedial joint.

Incyclinide

Incyclinide is an investigational tetracycline-derived matrix-metalloproteinase inhibitor studied for anti-inflammatory or anticancer effects.

Indapamide

Indapamide is a thiazide-like diuretic used mainly for hypertension; it promotes renal sodium excretion and may cause hyponatraemia, hypokalaemia, dehydration, hyperuricaemia, or photosensitivity.

Indel Mutation

indel mutation is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

Independence

The ability to perform activities and make decisions without assistance, although a person may still use devices, adaptations, or occasional support.

Indeterminate

Indeterminate means not definitively classified or interpreted because available findings are insufficient, conflicting, or nonspecific.

India Immunization Schedule

The Universal Immunization Schedule of the Government of India, providing free vaccines to all children, adolescents, and pregnant women through the public health system.

Indican

Indican is a compound formed from tryptophan metabolism and excreted in urine; abnormal levels may occur with intestinal bacterial overgrowth or metabolic disease.

Indicate

To show, suggest, or point to a finding or conclusion; an indication is a clinical reason for using a test, treatment, or procedure.

Indication

A clinical condition, symptom, or question that provides the reason for performing a test, procedure, or treatment.

Indigestion

A chemical substance or compound called indigestion. Its properties, uses, and risks depend on its composition and clinical context.

Indigo Carmine

Indigo carmine is a blue dye used in selected medical procedures to visualize urinary or other fluid flow and in laboratory staining.

Indinavir

A medicine or chemical compound called indinavir. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Indirect Antiglobulin Test

A test that detects free, clinically significant antibodies in a patient's serum or plasma by incubating it with reagent erythrocytes. It is used for antibody screening, compatibility testing, and prenatal red-cell antibody evaluation.

Indirect Calorimetry

Measurement of oxygen consumption and carbon-dioxide production to estimate resting energy expenditure. It is the preferred method for determining energy needs when prediction equations are unreliable, particularly in critically ill or metabolically complex patients.

Indirect Veneer

Ceramic restoration fabricated in lab. Porcelain or composite. Aesthetic enhancement.

Indium

Indium is a metal used in industry and medical radioisotopes; exposure effects depend on the compound, dose, and route.

Indium-111

SPECT radionuclide, half-life 2.8 days. Used in WBC labeling, octreotide scanning. Emits gamma rays at 171 and 245 keV.

Indium-111 Scan

Nuclear medicine imaging using In-111. Used for WBC labeling, octreotide scanning.

Indium-Labelled WBC Scan

A nuclear medicine study in which a patient's white blood cells are labeled with indium-111 and reinjected to localize selected sites of inflammation or infection.

Indocyanine Green

Indocyanine green is a fluorescent dye injected intravenously to assess liver function, blood flow, and retinal or choroidal circulation.

Indole

A nitrogen-containing chemical ring found in tryptophan and many biologically active compounds.

Indole-3-Carbinol

A compound formed from glucosinolates in cruciferous vegetables and studied for effects on estrogen metabolism and cancer biology.

Indolent

Slow to develop, progress, or cause symptoms; it often describes a relatively slow-growing disease or tumor.

Indoles

A group of compounds containing an indole ring, including substances involved in biological signaling, metabolism, and medicines.

Indometacin

Indometacin is a nonsteroidal anti-inflammatory medicine used for pain and inflammation; it can cause stomach bleeding, kidney problems, and cardiovascular complications.

Indomethacin

Indomethacin is a potent NSAID used for inflammatory pain and selected acute conditions such as gout; gastrointestinal ulceration or bleeding, renal injury, cardiovascular events, and headache are important adverse effects.

Indomethacin Overdose

Toxic exposure to excessive indomethacin, which can cause nausea, vomiting, stomach bleeding, kidney injury, seizures, or other serious effects.

Indoor Allergens

Substances inside buildings that can trigger allergic or asthma symptoms, including dust mites, mold, cockroach particles, and animal dander. Exposure may cause rhinitis, conjunctivitis, wheezing, or eczema. Reduction of exposure and appropriate allergy or asthma treatment depend on the identified trigger.

Indoximod

Indoximod is an investigational immunotherapy that modulates tryptophan metabolism and has been studied in combination cancer treatment.

Induced Abortion

A chemical substance or compound called induced abortion. Its properties, uses, and risks depend on its composition and clinical context.

Induced Hyperthermia

Induced hyperthermia is deliberate elevation of body or tissue temperature for a therapeutic purpose, with careful monitoring to prevent heat injury.

Induced Labor

Medical initiation of uterine contractions before labor begins naturally to achieve delivery when continuing pregnancy may pose greater risk.

Induced Sputum Analysis

A method of assessing airway inflammation by examining sputum produced after inhaled saline stimulation. The types and proportions of inflammatory

cells may support assessment of asthma, infection, COPD, or bronchiectasis.

Inducing Labor

The process of starting labor artificially with medicines, membrane rupture, or other obstetric methods.

Induction

A chemical substance or compound called induction. Its properties, uses, and risks depend on its composition and clinical context.

Induction Of Labor

The artificial initiation of uterine contractions before the onset of spontaneous labor.

Induction Therapy

Intensive initial treatment aimed at achieving remission, especially in acute leukemias.

Indurated

Abnormally firm or hardened when palpated, usually because of inflammation, edema, fibrosis, infiltration, or a tumor within the skin or underlying tissue.

Induration

Abnormal hardening of tissue, often detected by palpation and caused by inflammation, infection, edema, fibrosis, or a mass.

Industrial Bronchitis

Industrial bronchitis is chronic airway irritation and inflammation related to workplace dusts, fumes, or chemicals, causing cough, phlegm, and breathing limitation.

Industrial Disease

A disease caused or worsened by workplace exposures or conditions, such as chemicals, dust, noise, radiation, infection, or repetitive strain.

Industrial Health

The branch of occupational health concerned with

preventing and managing illness and injury related to industrial work, workplace exposures, and working conditions.

Indwelling Catheter

A catheter left in place for ongoing drainage, infusion, monitoring, or access; urinary indwelling catheters require appropriate indications and care because they increase infection and trauma risk.

Indwelling Catheter Care

A patient-care topic concerning indwelling catheter care. It addresses the relevant purpose, risks, management, and follow-up.

Indwelling Pleural Catheter

A small-bore, flexible silicone tube with a polyester cuff and a one-way valve mechanism (such as the PleurX catheter) placed percutaneously into the pleural space and tunneled subcutaneously through the chest wall. It allows safe, intermittent outpatient drainage of recurrent symptomatic pleural effusions (most commonly malignant pleural effusion or trapped lung) by the patient or a caregiver, relieving dyspnea without requiring repeated thoracentesis or prolonged hospital stays, and often inducing spontaneous pleurodesis.

Alternate Names:

IPC; Tunneled Pleural Catheter; PleurX Catheter

Inertia

Resistance to a change in motion; in obstetrics, uterine inertia means weak or ineffective contractions that slow labor.

Infant

An infant is a child in the first year of life, a period of rapid growth and changing feeding, immune, neurologic, and developmental needs.

Infant - Newborn Development

The early growth of a newborn's body, brain, senses, movement, feeding, communication, and social responses during the first weeks and months.

Infant Botulism

A neuromuscular illness caused by intestinal colonization with *Clostridium botulinum* and production of botulinum neurotoxin in infants. It causes constipation, poor feeding, weak cry, hypotonia, facial weakness, and descending paralysis.

Infant Feeding

Provision of breast milk, appropriately prepared formula, or both to meet an infant's energy, protein, fluid, and micronutrient requirements. Feeding recommendations depend on age, growth, medical conditions, developmental readiness, and safe preparation practices.

Infant Formula

Infant formula is manufactured nutrition designed to provide adequate nutrients when breast milk is unavailable, insufficient, or not chosen; preparation and storage must be safe.

Infant Formulas

Commercial preparations designed to provide nutrition for infants when breast milk is unavailable, insufficient, or not chosen.

Infant Gynecomastia

Infant gynecomastia is temporary breast enlargement in a newborn or young infant caused by maternal or neonatal hormones and usually resolves spontaneously.

Infant Harlequin Syndrome

A transient, benign cutaneous vascular phenomenon in neonates characterized by asymmetric flushing of one half of the body with sharp midline demarcation, caused by transient asymmetric sympathetic tone. The affected side is red and warm while the contralateral side is pale.

Infant Incubators

Enclosed devices that provide controlled warmth, humidity, and monitoring for premature or ill newborns.

Infant Jaundice

Yellowing of a newborn's skin or eyes caused by excess bilirubin in the blood.

Infant Leukemia

Infant leukemia is leukemia diagnosed during infancy, in which abnormal blood-forming cells multiply and interfere with normal blood-cell production.

Infant Mortality

The death of an infant before the first birthday; the infant mortality rate is usually expressed per 1,000 live births.

Infant Nutrition

A chemical substance or compound called infant nutrition. Its properties, uses, and risks depend on its composition and clinical context.

Infant Of A Substance-Using Mother

A newborn exposed to alcohol or other substances during pregnancy who may need assessment for withdrawal, toxicity, feeding problems, or developmental effects.

Infant Of Diabetic Mother

A newborn born to a mother with diabetes who may be at increased risk of large size, low blood sugar, breathing problems, jaundice, or congenital abnormalities.

Infant Reflex

An infant reflex is an automatic response present during early development, such as rooting, sucking, grasping, or startle, whose timing and symmetry help assess neurologic function.

Infant Reflexes

Automatic newborn responses such as rooting, sucking, Moro, grasp, and stepping that help assess neurologic development.

Infant Reflux

Backflow of stomach contents into an infant's esophagus, often causing spitting up.

Infant Sleep Position

Placing an infant on the back for every sleep is recommended to reduce the risk of sleep-related death. Infants should sleep on a firm, separate surface without pillows, loose bedding, bumpers, or stuffed toys.

Infant Test Procedure Preparation

Infant test procedure preparation includes age-appropriate instructions, feeding or fasting guidance, comfort measures, and safety steps before a test.

Infant, Post-Term

An infant born at or after 42 weeks of gestation, beyond the usual term-pregnancy period; risks depend on placental function, fetal condition, and circumstances of delivery.

Infanticide

The killing of a child, in law typically defined as the killing of a newborn or infant (usually within the first year, or specifically within 24 hours of birth in some jurisdictions) by its mother while mentally disturbed by the effects of childbirth or lactation.

Infantile Colic

A behavioral syndrome in healthy infants characterized by episodes of excessive, inconsolable crying for >3 hours per day, >3 days per week, for >3 weeks (Wessel criteria). Onset typically at 2-3 weeks, peaks at 6-8 weeks, and resolves by 3-4 months of age.

Infantile Fibrosarcoma

Infantile fibrosarcoma is a rare soft-tissue cancer that usually develops in infants or young children and requires specialist treatment.

Infantile Hemangioma

Alternate index label; see *Hemangioma*.

Infantile Hepatic Hemangiomas

Benign vascular tumors of the liver occurring in infants, typically presenting within the first six months of life with hepatomegaly, high-output cardiac fail-

ure, consumptive hypothyroidism, or cutaneous hemangiomas. Large tumors may cause abdominal compartment syndrome and thrombocytopenia from Kasabach-Merritt phenomenon.

Infantile Paralysis (Polio)

Paralytic poliomyelitis caused by poliovirus infection involving motor neurons, which can produce acute flaccid paralysis and respiratory failure; vaccination prevents most cases.

Infantile Spasm

Infantile spasms are brief epileptic seizures in infancy, often in clusters with flexion or extension movements, and require urgent diagnosis and treatment to protect development.

Infantile Spasms

A severe seizure disorder of infancy characterized by brief, sudden flexion or extension spasms, often occurring in clusters and associated with developmental impairment.

Infarct

An area of dead tissue caused by inadequate blood supply.

Infarction

Tissue death from prolonged ischemia. Examples: myocardial infarction, cerebral infarction.

Infection

Measures to prevent healthcare-associated infections: hand hygiene, personal protective equipment (PPE), isolation precautions (contact, droplet, airborne), sterilization, disinfection. See infection control.

Infection (Geriatric)

Infectious diseases in elderly often present atypically: fever may be absent, confusion may be sole symptom. Common infections: UTI, pneumonia, skin/soft tissue, bloodstream. Prevention: vaccination, hand hygiene, aspiration precautions. Treatment: earlier presentation, broader differential, drug interactions,

renal dosing.

Alternate Names:

Geriatric

Infection Control

Measures to prevent healthcare-associated infections: hand hygiene, personal protective equipment (PPE), isolation precautions (contact, droplet, airborne), sterilization, disinfection. Multidrug-resistant organisms: MRSA, VRE, C. difficile, carbapenem-resistant Enterobacteriaceae.

Infection Cryptococcus (Cryptococcosis)

Alternate index label for Cryptococcosis.

Infection Cyclospora (Cyclospora Infection (Cyclosporiasis))

Alternate index label for Cyclospora Infection (Cyclosporiasis).

Infection Enterovirus (Enterovirus (Non-Polio Enterovirus Infection))

Alternate index label for Enterovirus (Non-Polio Enterovirus Infection).

Infection HPV (HPV Infection Human Papillomavirus)

Alternate index label for HPV Infection Human Papillomavirus.

Infection Human Papillomavirus (HPV Infection Human Papillomavirus)

Alternate index label for HPV Infection Human Papillomavirus.

Infection Imaging

WBC scanning, gallium scanning, FDG PET. Detection of occult infection.

Infection Listeria (Listeria)

Alternate index label for Listeria.

Infection Non-Polio Enterovirus (Enterovirus (Non-Polio Enterovirus Infection))

Infection))

Alternate index label for Enterovirus (Non-Polio Enterovirus Infection).

Infection Pinworms (Pinworm Infection)

Alternate index label for Pinworm Infection.

Infection Prevention

Measures that reduce the risk of infection, including hand hygiene, personal protective equipment, environmental cleaning, sterilization, disinfection, vaccination, and appropriate isolation precautions.

Infection Testicle (Testicular Disorders)

Alternate index label for Testicular Disorders.

Infection, Hospital-Acquired

An infection acquired during or as a result of health-care that was not present or incubating at the time of admission; it may be caused by bacteria, viruses, fungi, or other organisms.

Infection, Urinary Tract

Infection of part of the urinary system, commonly the bladder or urethra and sometimes the kidneys, usually caused by bacteria and producing symptoms such as burning urination, urgency, or flank pain.

Infections In Pregnancy

Listeriosis: *Listeria monocytogenes* (deli meats, soft cheese), meningoen­cephalitis, fetal loss, neonatal sepsis. Ampicillin + gentamicin. GBS: vaginal colonization 20-30 percent of pregnant women; universal screening at 35-37 weeks; intrapartum prophylaxis (penicillin G) for GBS-positive women to prevent early-onset neonatal disease.

Infectious Arteritis

Inflammation and damage of an artery caused by infection, which can weaken the vessel, reduce blood flow, or produce aneurysm.

Infectious Arthritis

Inflammation of a joint caused by infection, usually bacterial and sometimes viral or fungal. Acute pain,

swelling, warmth, fever, and restricted movement require urgent evaluation because joint damage can develop quickly.

Infectious Conjunctivitis

Inflammation of the conjunctiva caused by a virus, bacterium, or less commonly another infectious organism, often causing redness and discharge.

Infectious Disease

Disease caused by pathogen. Examples: bacterial, viral, fungal, parasitic.

Infectious Dose

The number of viable organisms required to establish infection in a host under specified conditions. The dose depends on the pathogen, route of exposure, host defenses, and organism viability.

Infectious Encephalitis

Inflammation of the brain caused by an infectious organism, most often a virus.

Infectious Encephalomyelitis

Infection-related inflammation of the brain and spinal cord that may cause fever, headache, confusion, weakness, seizures, or other neurologic signs.

Infectious Enterocolitis

Infection-related inflammation of the small intestine and colon, commonly causing diarrhea, abdominal pain, fever, or vomiting.

Infectious Esophagitis

Inflammation of the esophagus caused by an infectious organism, most often *Candida*, herpes simplex virus, or cytomegalovirus in immunocompromised patients. It commonly causes painful swallowing, difficulty swallowing, and retrosternal pain.

Infectious Hepatitis

Alternate index label; see *Viral hepatitis*.

Infectious Mediastinitis

Infection and inflammation of the tissues in the

middle of the chest, often after esophageal injury, surgery, or spread from a nearby infection.

Infectious Meningitis

Infection and inflammation of the membranes surrounding the brain and spinal cord. Bacterial meningitis is a medical emergency; headache, fever, neck stiffness, vomiting, confusion, or a new rash requires immediate medical attention.

Infectious Mononucleosis

A viral illness, usually caused by Epstein-Barr virus, characterized by fatigue, sore throat, and swollen lymph nodes.

Alternate Names:

Mono (Infectious Mononucleosis)

Infectious Myringitis

Inflammation and infection of the tympanic membrane, often associated with viral or bacterial respiratory infection. It can cause severe ear pain, fever, hearing reduction, and small fluid-filled blisters on the eardrum.

Infective Cervicitis

Infective cervicitis is inflammation of the cervix caused by organisms such as chlamydia, gonorrhea, trichomonas, or herpes, producing discharge, bleeding, or pelvic discomfort.

Infective Cystitis

Infective cystitis is bacterial or other microbial inflammation of the bladder, causing urinary frequency, urgency, burning, and sometimes blood or fever.

Infective Endocarditis

Infection of the endocardial surface of the heart, most commonly affecting the heart valves. It is diagnosed using the Duke criteria, which include major criteria (positive blood cultures, echocardiographic evidence of vegetations) and minor criteria (predisposing heart condition, fever, vascular phenomena, immunologic phenomena).

Infective Phlebitis

Infective phlebitis is infection and inflammation of a vein, often around an intravenous catheter, causing pain, redness, swelling, and possible bloodstream infection.

Infective Vaginitis

Infective vaginitis is infection or inflammation of the vagina caused by yeast, bacteria, parasites, or viruses and may cause discharge, itching, odor, or pain.

Inferior

In standard anatomical directional terminology, situated below, downward, or directed toward the feet or caudal extremity of the body relative to another anatomical landmark (the opposite of superior). For example, the diaphragm lies inferior to the thoracic lungs and heart, and the inferior vena cava drains venous blood from structures located inferior to the anatomical respiratory diaphragm.

Inferior Alveolar Nerve Block

A regional anesthetic injection near the mandibular foramen to anesthetize the inferior alveolar nerve before it enters the mandibular canal. It usually anesthetizes mandibular teeth on that side; the lingual nerve is often anesthetized incidentally.

Inferior Colliculus

The inferior colliculus is a midbrain auditory relay that integrates sound information and helps orient reflexively to auditory stimuli.

Inferior Hemifield Loss

Loss of vision in the lower half of the visual field in one or both eyes, which localises to lesions of the superior retina or superior optic radiations passing through the parietal lobe. Causes include superior retinal detachment, branch retinal artery occlusion, and parietal lobe stroke.

Inferior Mesenteric Artery

Third major abdominal aortic branch at L3, the hindgut artery. Supplies distal transverse colon through rectum. Branches: left colic, sigmoid, su-

perior rectal. The marginal artery of Drummond provides SMA-IMA collateral circulation. Occlusion may cause ischemic colitis at the splenic flexure watershed (Griffiths' point).

Inferior Myocardial Infarction

A heart attack affecting the lower or inferior portion of the heart muscle.

Inferior Vena Cava

Body's largest vein, formed by common iliac vein union at L5, returning lower body deoxygenated blood to the right atrium. Ascends right of the aorta, pierces the diaphragm at T8, enters the pericardium. Receives hepatic, renal, gonadal, and lumbar veins. IVC thrombosis/compression causes bilateral lower extremity edema and DVT.

Inferolateral

Located below and toward the side of a reference structure.

Infertility

Infertility is failure to achieve pregnancy after a defined period of regular unprotected intercourse, reflecting factors involving ovulation, sperm, reproductive anatomy, age, or unexplained causes. to conceive or carry a pregnancy. Causes may involve ovulation, reproductive anatomy, sperm, age-related fertility, medical conditions, or unexplained factors.

Alternate Names:

Subfertility

Infertility, Female

Alternate index label; see *Female infertility*.

Infertility, Male

Alternate index label; see *Male infertility*.

Infestation

The presence or invasion of parasitic organisms, such as lice, mites, or worms, on or in the body.

Infiltrate

Patchy lung opacity detected on imaging. Non-

specific, may represent infection, edema, hemorrhage, inflammation.

Infiltrating Cancer

Cancer that has grown beyond its site of origin into surrounding tissue.

Alternate Names:

Invasive cancer

Infiltrating Ductal Carcinoma

Alternate index label; see *Invasive ductal carcinoma*.

Infiltration

A chemical substance or compound called infiltration. Its properties, uses, and risks depend on its composition and clinical context.

Inflamed Gallbladder

Alternate index label; see *Cholecystitis*.

Inflamed Pancreas

Alternate index label; see *Pancreatitis*.

Inflamed Pericardium

Alternate index label; see *Pericarditis*.

Inflammaging

Persistent, low-grade, age-associated inflammatory activity linked to cellular senescence, immune dysregulation, chronic infection, tissue injury, and comorbidity. It contributes to frailty, atherosclerosis, sarcopenia, and neurodegenerative disease.

Inflammasomes

Multiprotein complexes activating caspase-1 -> cleavage of pro-IL-1 β /pro-IL-18 -> mature IL-1 β /IL-18 + pyroptosis (gasdermin D pores). Canonical: NLRP3 (crystals, ATP, K⁺ efflux, ROS, lysosomal rupture), NLRC4 (flagellin, T3SS), AIM2 (cytosolic DNA), Pyrin (RhoA inactivation).

Inflammation

The body's protective response to injury or infection. Cardinal signs: redness (rubor), heat (calor), swelling (tumor), pain (dolor), loss of function (func-

tio laesa). Mediators: histamine, prostaglandins, cytokines, complement. Acute: neutrophil-predominant. Chronic: lymphocyte/macrophage-predominant.

Inflammation (Dietary)

Chronic low-grade inflammation linked to diet. Western diet pro-inflammatory (saturated fat, sugar, refined grains). Mediterranean diet anti-inflammatory.

Alternate Names:

Dietary

Inflammation Imaging

FDG PET sensitive for inflammatory processes. Differentiate infection from malignancy challenging.

Inflammation Of The Heart Valve (Endocarditis)

Alternate index label for Endocarditis.

Inflammation Sclera (Scleritis)

Alternate index label for Scleritis.

Inflammation Vaginal (Vaginitis Overview)

Alternate index label for Vaginitis Overview.

Inflammatory Back Pain

A pattern of chronic back pain characterized by insidious onset, improvement with exercise, lack of improvement with rest, nocturnal pain, and prolonged morning stiffness. It suggests axial spondyloarthritis but is not diagnostic without supportive clinical, laboratory, and imaging findings.

Inflammatory Bowel Disease

A group of chronic conditions that cause inflammation in the digestive tract, mainly Crohn disease and ulcerative colitis. Symptoms may come and go and can include abdominal pain, diarrhea, rectal bleeding, and weight loss. The exact cause is unknown, but genes, the immune system, the environment, and gut bacteria may all play a role.

Inflammatory Bowel Disease Diet

Crohn disease, ulcerative colitis. Nutritional support during flares. Exclusive enteral nutrition for Crohn induction. Individual triggers vary.

Inflammatory Bowel Disease: Intestinal Problems

Inflammatory bowel disease, including Crohn disease and ulcerative colitis, causes chronic immune-mediated intestinal inflammation with diarrhea, pain, bleeding, weight loss, and extraintestinal effects. Treatment aims for sustained remission and prevention of structural and nutritional complications.

Alternate Names:

IBD (Inflammatory Bowel Disease: Intestinal Problems)

Inflammatory Bowel Disease: Intestinal Problems

Alternate index label for Inflammatory Bowel Disease: Intestinal Problems.

Inflammatory Breast Cancer

A rare, rapidly progressing breast cancer that causes redness, swelling, and skin thickening.

Alternate Names:

Breast Cancer, Inflammatory Cancer, Inflammatory Breast

Inflammatory Myopathies

A group of autoimmune disorders characterized by chronic muscle inflammation and progressive proximal muscle weakness. The three main types are dermatomyositis, polymyositis, and inclusion body myositis.

Inflammatory Response

The coordinated vascular and immune reaction to infection, tissue injury, or other danger signals, producing mediator release, leukocyte recruitment, and changes in blood flow and permeability.

Infliximab

A medicine or chemical compound called infliximab.

Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Influenza

A contagious respiratory illness caused by influenza A or B viruses, characterized by sudden onset of fever, myalgia, headache, malaise, nonproductive cough, sore throat, and nasal congestion. Seasonal epidemics occur annually. Complications: pneumonia (primary viral or secondary bacterial), exacerbation of underlying conditions.

Influenza (Flu)

A contagious respiratory infection caused by influenza viruses, commonly producing fever, cough, body aches, fatigue, and respiratory symptoms.

Influenza A Virus

An influenza virus with segmented RNA that infects people and animals and can cause seasonal epidemics or, after major genetic changes, pandemics.

Influenza Test

A test for influenza A or B using a respiratory specimen. Rapid antigen tests provide quick results but may miss infection. Molecular tests are generally more sensitive and may identify or subtype influenza viruses; results must be interpreted with symptoms, timing, and local laboratory methods.

Influenza Vaccination In Pregnancy

Annual inactivated influenza vaccine recommended for all pregnant women. Pregnancy increases the risk of severe influenza (pneumonia, ICU admission, death) due to changes in immune function and cardiopulmonary physiology.

Influenza Virus

A segmented, negative-sense RNA virus belonging to the Orthomyxoviridae family. Influenza causes seasonal epidemics and occasional pandemics of respiratory illness. The virus has two surface glycoproteins: hemagglutinin (HA), which mediates attachment, and neuraminidase (NA), which facilitates release.

Influenza, Avian

Alternate index label; see *Bird flu (avian influenza)*.

Influenza, H1N1

Alternate index label; see *H1N1 flu (swine flu)*.

Influenza, Swine Flu

Alternate index label; see *H1N1 flu (swine flu)*.

Influenzavirus A

The virus genus containing influenza A viruses, which are classified into subtypes by their hemagglutinin and neuraminidase proteins.

Influenzavirus B

The virus genus containing influenza B viruses, which mainly infect humans and cause seasonal respiratory illness.

Informatics

Informatics applies information science, data, computing, and human factors to organize and improve healthcare, biology, research, or another field.

Informative

Informative means providing useful information or evidence that improves understanding, decision-making, diagnosis, or evaluation.

Informed Consent

A patient's voluntary agreement to a proposed intervention after receiving understandable information about its purpose, benefits, risks, alternatives, and the option of declining, with decision-making capacity or appropriate authorization.

Informed Consent (Geriatric)

Process of obtaining voluntary authorization from elderly patient with decision-making capacity. Capacity assessment: understanding, appreciation, reasoning, communication of choice. Impaired capacity may require surrogate decision-maker.

Alternate Names:

Geriatric

Informed Consent - Adults

Informed consent requires an adult patient to receive understandable information about a proposed intervention, benefits, risks, alternatives, and refusal before deciding voluntarily.

Infrared Rays

Infrared rays are electromagnetic radiation with wavelengths longer than visible red light; excessive exposure can heat tissue and cause burns.

Infrared Spectrometry

Infrared spectrometry identifies or measures substances by how they absorb infrared light.

Infrarenal Aorta

The infrarenal aorta is the portion of the abdominal aorta below the renal arteries and above its division into the iliac arteries.

Infratentorial Neoplasm

An infratentorial neoplasm is an abnormal growth in the posterior fossa, the part of the skull containing the cerebellum and brainstem.

Infrequent

Infrequent means occurring less often than usual or expected, such as infrequent bowel movements, symptoms, or events.

Infundibulopelvic Ligament

A prominent peritoneal fold extending from the tubal pole of each ovary to the lateral pelvic sidewall. Serves as the vital anatomical conduit transmitting the ovarian artery, ovarian venous plexus, lymphatic channels, and autonomic nerve fibers to and from the ovary; represents the critical vascular pedicle isolated, doubly ligated, and divided during oophorectomy, with surgical care required to avoid injuring the underlying ureter crossing into the pelvis.

Alternate Names:

Suspensory Ligament of the Ovary; IP Ligament

Infuse

To introduce a fluid, medicine, or other substance gradually into the body, commonly through an intravenous line; it can also mean to soak a substance in a liquid.

Infusion

Slow administration of a fluid or medicine into a vein or tissue.

Alternate Names:

Intravenous infusion

Infusion Pump

Device controlling IV fluid delivery rate. Types: syringe pump, volumetric pump, elastomer. Precision: mL/hour. Alarms: occlusion, air embolism, completion.

Ingestion

The act of taking a substance into the digestive tract by swallowing. Ingestion of a toxic substance can cause poisoning.

Ingredient

An ingredient is a substance included in a medicine, food, cosmetic, or other preparation, whether active or inactive.

Ingrown Hair

A hair that curls back into the skin or grows sideways, causing an inflamed bump.

Ingrown Nail

An ingrown nail occurs when the nail edge grows into surrounding skin, causing pain, redness, swelling, and sometimes infection or granulation tissue.

Ingrown Toenail

Ingrown toenail (onychocryptosis) occurs when the nail edge grows into the surrounding skin, causing pain, redness, and swelling. It is common in adolescents and adults, exacerbated by improper nail trimming, tight shoes, and trauma. Treatment includes partial nail avulsion with matrixectomy (phenol application) for recurrent cases.

Alternate Names:

Onychocryptosis (Ingrown Toenail)

Inguinal

Relating to the groin, the region where the lower abdomen meets the upper thigh.

Inguinal Canal

An oblique passage (4 cm) through the anterior abdominal wall, from the deep (internal) inguinal ring to the superficial (external) inguinal ring, transmitting the spermatic cord in males and round ligament in females, plus ilioinguinal and genital branch of genitofemoral nerves.

Inguinal Hernia

Protrusion of abdominal contents through the inguinal canal. Indirect (lateral to inferior epigastric vessels, through deep inguinal ring): congenital, patent processus vaginalis; most common overall (60-70%), male predominance.

Inguinal Hernia Repair

A medical procedure involving inguinal hernia repair. Indications, risks, preparation, and follow-up depend on the patient and the specific procedure.

Inguinal Orchiectomy

Inguinal orchiectomy is removal of a testicle through an incision in the groin, commonly used when treating suspected testicular cancer.

Inguinal Region

Area at the thigh-trunk junction traversed by the inguinal ligament (ASIS to pubic tubercle). Clinically significant for inguinal hernias: indirect (through deep inguinal ring, most common) and direct (through Hesselbach's triangle).

Inhalant

A substance breathed into the lungs as a gas, vapor, aerosol, or smoke; inhalants may be medicines, workplace substances, or abused chemicals.

Inhalation

A chemical substance or compound called inhalation. Its properties, uses, and risks depend on its composition and clinical context.

Inhalation Injury

Thermal or chemical injury to the airway or lungs caused by smoke, heated gases, or toxic combustion products. It can produce upper-airway edema, bronchospasm, impaired mucociliary clearance, and carbon-monoxide or cyanide toxicity and may worsen after the initial burn.

Inhalational Anesthesia

Anesthetic maintenance or induction using volatile liquids or gases delivered through the respiratory tract. Depth is influenced by inspired concentration, alveolar ventilation, uptake, distribution, and the agent's blood-gas solubility.

Inhaled Corticosteroids

Anti-inflammatory medications delivered directly to the airways, reducing airway inflammation, mucus production, and bronchial hyperresponsiveness. Examples include fluticasone, budesonide, beclomethasone, and mometasone. They are the cornerstone of asthma maintenance therapy. Side effects include oral candidiasis and dysphonia.

Inhaled Nitric Oxide

A selective pulmonary vasodilator used in acute settings (acute respiratory failure, right heart failure, post-cardiac surgery). It reduces pulmonary vascular resistance and improves oxygenation by dilating well-ventilated pulmonary vessels (improving V/Q matching). It has a short half-life and requires continuous delivery.

Inhaler

Device for administering aerosolized medication. Types: MDI (metered-dose), DPI (dry powder).

Inherited Disease

A disease caused wholly or partly by genetic changes passed from parents or arising in a reproductive cell or early embryo.

Inhibin

A glycoprotein hormone produced mainly by the gonads that suppresses pituitary follicle-stimulating hormone secretion and contributes to regulation of reproductive function.

Inhibition

Inhibition is a reduction or suppression of an activity, such as nerve signalling, enzyme action, muscle contraction, or cell growth.

Inhibitory Neurochemical

A chemical messenger that reduces signaling between nerve cells.

Alternate Names:

Inhibitory neurotransmitter

Inhibitory Synapse

A connection between nerve cells that reduces the chance that the receiving cell will generate an electrical signal.

Iniiencephaly

A rare, usually lethal neural-tube defect characterized by occipital bone defect, severe retroflexion of the head, and marked shortening or distortion of the cervical and thoracic spine. It is often identified prenatally and may occur with other congenital anomalies.

Iniparib

An investigational anticancer drug studied for effects on DNA repair and tumor-cell survival; it is not a standard treatment for most cancers.

Initial Insomnia

Initial insomnia is difficulty falling asleep at the beginning of the sleep period, often related to stress, habits, medicines, mood, or another sleep disorder.

Injectable Fillers

Materials injected beneath or into skin to restore volume or soften wrinkles.

Alternate Names:

Dermal fillers

Injection

Introduction of a medication, fluid, or other substance into the body with a needle or other injection device.

Injury

Damage to the body's tissues or function caused by trauma, force, heat, chemicals, radiation, or another harmful exposure.

Injury - Kidney And Ureter

Damage to the kidney or ureter from trauma, obstruction, surgery, infection, or other causes; severe injury can impair urine flow or kidney function.

Injury Classification (Abrasion, Laceration, Contusion, Incised Wounds)

Forensic categorization of mechanical wounds. Abrasion: superficial friction injury affecting only the epidermis (e.g., scrape, graze, pattern abrasion).

Alternate Names:

Abrasion, Laceration, Contusion, Incised Wounds

Injury Dating

Estimation of the age of an injury using clinical appearance, histopathology, imaging, and the history of symptoms. Exact dating is often uncertain, particularly for bruises and soft-tissue injuries, and findings should be reported with appropriate limitations.

Injury Site

The body location where an injury occurred or where its effects are being evaluated.

Ink Poisoning

Illness after swallowing or significant exposure to ink; most small pen-ink exposures cause little toxicity, but symptoms depend on the product and amount.

Ink Remover Poisoning

Poisoning caused by chemicals used to remove ink, which may irritate the skin, eyes, lungs, or digestive tract and may cause systemic toxicity.

Innate Immune Evasion

Mechanisms by which pathogens evade the innate immune system. Strategies include capsules that prevent phagocytosis, biofilms that protect against immune cells, and intracellular survival within macrophages. Some bacteria produce enzymes that degrade complement components. Others modify their surface molecules to avoid recognition.

Innate Immunity

First line of defense against pathogens, present from birth. Rapid (minutes to hours), germline-encoded, lacks immunological memory.

Innate Lymphoid Cells

ILCs: lymphoid lineage, no rearranged antigen receptors, mirror Th subsets. ILC1 (T-bet, IFN-gamma): intracellular pathogens, IBD. ILC2 (GATA3, IL-5/13): helminths, allergy, asthma, tissue repair. ILC3 (ROR-gammat, IL-17/22): extracellular bacteria/fungi, lymphoid organogenesis.

Inner Ear

The deepest part of the ear, consisting of the cochlea and the labyrinth.

Inner Ear Infection

Inner-ear inflammation can affect hearing and balance, producing vertigo, nausea, tinnitus, or hearing loss. Causes include viral infection, bacterial disease, autoimmune inflammation, or other disorders; sudden hearing loss, severe imbalance, or neurologic signs requires urgent assessment.

Innervation

The nerve supply to an organ, muscle, or area of the body, including the functional control provided by those nerves.

Innominate

Innominate means unnamed; in anatomy it may refer to structures such as the innominate artery or innominate bone, depending on context.

INOCA

Ischemia with non-obstructive coronary arteries: objective myocardial ischemia or ischemic symptoms in a patient whose coronary arteries do not have an obstructive epicardial stenosis. Mechanisms include coronary microvascular dysfunction and vasospastic angina; evaluation should identify the mechanism so treatment can be tailored.

Inoculation

The introduction of microorganisms or clinical material into a culture medium, host, or sterile site to initiate growth, infection, or immune response.

Inoperable

Not suitable for surgical treatment, often because surgery cannot safely remove or correct the condition.

Inorganic

Inorganic is an immunologic, genetic, infectious, developmental, or cellular concept used to explain disease mechanisms, diagnosis, or treatment.

Inorganic Chemical

inorganic chemical is a biochemical reaction, pathway, or chemical process in which molecules are transformed; its rate and effects depend on substrates, enzymes, and cellular conditions.

Inosculation

A chemical substance or compound called inosculation. Its properties, uses, and risks depend on its composition and clinical context.

Inosine

A nucleoside made of hypoxanthine linked to ribose that participates in purine metabolism and cellular energy pathways.

Inosine Monophosphate

A nucleotide containing inosine and one phosphate group; it is an intermediate in purine metabolism and a precursor of other nucleotides.

Inosine Pranobex

A medicine combination marketed in some countries as an immunomodulator or antiviral; evidence and approved uses vary by jurisdiction.

Inositol

A sugar-like compound involved in cell membranes and signaling; some forms are studied or used for selected metabolic, reproductive, or mental-health conditions.

Inositol Phosphates

Phosphorylated forms of inositol that act in cell signaling, calcium regulation, and other intracellular processes.

Inotropes In Anesthesia

Inotropes increase myocardial contractility. Dobutamine is a beta-1 and beta-2 agonist increasing cardiac output with minimal effect on heart rate. Dose: 2 to 20 mcg/kg/min infusion. Milrinone is a phosphodiesterase-3 inhibitor increasing contractility and causing vasodilation.

Inotropic

Affecting the force of cardiac muscle contraction; a positive inotropic drug increases contractility, while a negative inotropic effect decreases it.

Inpatient

An inpatient is a person admitted to a healthcare facility for monitoring, diagnosis, treatment, surgery, or care requiring an overnight or formal stay.

Inpatient Rehabilitation

Intensive inpatient rehabilitation (3+ hours therapy/day) for acute conditions: stroke, spinal cord injury, traumatic brain injury, major trauma. See acute rehabilitation.

Inquilinus

A genus of gram-negative bacteria associated mainly with water and industrial environments; human infection is rare.

Inquilinus Limosus

A species of Inquilinus bacteria rarely isolated from human clinical specimens and occasionally associated with opportunistic infection.

INR

International normalized ratio: a standardized measure of the prothrombin time, calculated as $(\text{patient PT}/\text{mean normal PT})^{\text{ISI}}$, to ensure consistent monitoring of vitamin K antagonist (warfarin) therapy across laboratories. Target INR: 2-3 for most indications (AF, VTE); 2.5-3.5 for mechanical heart valves.

Insect Bite

A cutaneous reaction resulting from the injection of salivary contents from an insect during feeding, causing a localized or systemic hypersensitivity response. Common presentations include a pruritic papule or wheal with surrounding erythema. Severe reactions may include anaphylaxis, particularly with Hymenoptera stings.

Insect Bites

Skin injuries caused by an insect feeding or injecting saliva or venom, producing local inflammation and sometimes allergic, infectious, or systemic reactions.

Insect Bites And Stings

Insect bites and stings can cause local pain, itching, swelling, allergic reactions, infection, or toxin effects; breathing difficulty or widespread symptoms is an emergency.

Insect Sting Allergies

Immune reactions to venom from insects such as bees, wasps, hornets, and fire ants. Reactions may be local or systemic; generalized hives, breathing difficulty, dizziness, vomiting, or throat swelling may indicate anaphylaxis and require emergency treatment.

Alternate Names:

Allergy Insect (Insect Sting Allergies)

Insect Virus

A virus that infects insects or other arthropods; some are studied for ecology or biological pest control rather than human disease.

Insecta

insecta is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Insecticide

A chemical substance or compound called insecticide. Its properties, uses, and risks depend on its composition and clinical context.

Insecticide Poisoning

Toxic effects caused by exposure to an insect-killing chemical; symptoms vary by agent and may include gastrointestinal, neurologic, respiratory, or cholinergic manifestations.

Insemination

A chemical substance or compound called insemination. Its properties, uses, and risks depend on its composition and clinical context.

Insertion

Movement of a muscle attachment toward its fixed attachment, or placement of one structure into another.

Insidious

Developing gradually and subtly, often without obvious early symptoms.

Insinuate

To introduce or insert gradually into a space, or to suggest something indirectly; in anatomy, a structure may insinuate between or around other structures.

Insomnia

Insomnia is persistent difficulty initiating or maintaining sleep, or early awakening, with daytime

impairment despite adequate opportunity and circumstances for sleep. too early despite adequate opportunity for sleep, with resulting daytime impairment. It may be acute or chronic and is often triggered by stress, anxiety, depression, or poor sleep habits.

Alternate Names:

Sleeplessness

Insomnia Disorder

Difficulty initiating or maintaining sleep, or early-morning awakening with inability to return to sleep, at least three nights per week for at least three months. The individual experiences clinically significant distress or impairment in functioning.

Inspection

Systematic visual examination of the body or a specimen for abnormalities such as color change, swelling, deformity, lesions, movement, or discharge. the patient's overall appearance, posture, body habitus, gait, skin color, nutritional status, respiratory effort, and specific anatomical regions. Requires adequate lighting and proper patient exposure.

Inspiration (Inhalation)

The active physiological phase of the respiratory ventilatory cycle during which ambient air is drawn into the lungs through the tracheobronchial tree. Initiated by contraction of the diaphragm (innervated by phrenic nerves C3–C5), which flattens downward, along with external intercostal muscle contraction, expanding thoracic cavity dimensions. This generates negative intrapleural and intrapulmonary (subatmospheric) pressure relative to atmospheric pressure, overcoming airway resistance and lung elastic recoil to inflate the pulmonary alveoli until alveolar pressure equilibrates with atmospheric pressure.

Inspiratory Capacity

Inspiratory capacity is the maximum volume inhaled after a normal exhalation and equals tidal volume plus inspiratory reserve volume.

Inspiratory Crackles

Fine, discontinuous sounds heard during inspiration, caused by the sudden opening of collapsed airways. They are a hallmark of interstitial lung disease (bibasilar, Velcro-like crackles) and pulmonary fibrosis. They are also heard in pneumonia, atelectasis, and heart failure.

Inspiratory Time

Inspiratory time is the duration of inhalation during a breath or mechanical-ventilation cycle.

Instantaneous Wave-Free Ratio

A resting invasive physiological index of coronary artery stenosis severity measured during catheterization using a specialized intracoronary pressure wire. Unlike fractional flow reserve, which requires maximal pharmacological hyperemia induced by adenosine, iFR calculates the ratio of distal coronary pressure to aortic pressure (P_d/P_a) specifically during a designated wave-free period in diastole when intracoronary microvascular resistance is naturally minimal and constant. Large randomized trials (DEFINE-FLAIR and iFR-SWEDEHEART) confirmed that an iFR threshold of ≤ 0.89 is non-inferior to FFR (≤ 0.80) for guiding revascularization.

Alternate Names:

iFR; Wave-Free Ratio; Resting Coronary Physiology

Instillation

Instillation is the gradual introduction of a liquid medicine or solution into a body cavity, organ, or tissue.

Instinct

An innate or largely automatic pattern of behavior or response, though human behavior is also shaped by learning and context.

Instrument

A tool or device used for examination, measurement, diagnosis, surgery, treatment, or laboratory work.

Instrument Sterilization

A validated process that destroys all forms of microbial life, including bacterial spores, on or within a medical instrument. Common methods include steam under pressure, ethylene oxide, hydrogen-peroxide plasma, and dry heat, selected according to device material and design.

Instrumental Activities Of Daily Living

More complex activities required for independent community living, including medication management, shopping, cooking, finances, communication, transportation, and household tasks.

Instrumental ADL

A complex activity needed for independent community living, such as shopping, cooking, managing medicines or finances, and using transportation.

Insufficient Cervix

Painless cervical shortening and dilation during the second trimester, also called cervical insufficiency. It can cause recurrent mid-trimester pregnancy loss or spontaneous preterm birth without regular uterine contractions.

Alternate Names:

cervical insufficiency.

Insula

The insula is a cerebral cortical region involved in sensation, taste, emotion, interoception, pain, and integration of internal and external information.

Insulin

A pancreatic hormone that helps cells take up glucose and lowers blood glucose.

Insulin And Syringes - Storage And Safety

A patient-care topic concerning insulin and syringes - storage and safety. It addresses the relevant purpose, risks, management, and follow-up.

Insulin Antagonist

A hormone, medicine, or other factor that opposes

insulin's effects and can raise blood glucose.

Insulin Aspart

A rapid-acting insulin analog (modified amino acid sequence) with onset 10-20 minutes, peak 1-3 hours, duration 3-5 hours. Used to cover prandial glucose (injected at or just after meals) in type 1 and type 2 diabetes; also used in continuous subcutaneous insulin infusion (pump) and IV in acute settings.

Insulin Autoantibodies

Autoantibodies directed against insulin or other pancreatic beta-cell antigens, including GAD, IA-2, and ZnT8 antibodies. Their presence supports autoimmune type 1 diabetes but a negative panel does not exclude it.

Insulin C-Peptide Test

A blood or urine test measuring C-peptide released in equal amounts with endogenous insulin; it helps distinguish native insulin production from injected insulin and evaluate hypoglycemia or diabetes classification.

Insulin Coma

A state of severe hypoglycemia and unconsciousness caused by excessive insulin; it is a medical emergency and is not a current routine treatment.

Insulin Detemir

A long-acting insulin analogue used to provide basal insulin replacement in diabetes mellitus; dosing is individualized and hypoglycemia is the principal serious adverse effect.

Insulin Glargine

A long-acting basal insulin analog with a flat, peakless profile providing near-24-hour coverage, administered once daily (or twice in some patients). Formed by modification of human insulin producing slow, steady absorption from the injection depot. Used for type 1 and type 2 diabetes (basal component).

Insulin Injection Technique

Insulin is injected into subcutaneous tissue using the prescribed device, dose, and site. Common sites include the abdomen, thighs, buttocks, and upper arms; rotating sites helps reduce lipohypertrophy and variable absorption. Technique should be individualized to the person and device.

Insulin Isophane

Intermediate-acting human insulin, also called neutral protamine Hagedorn insulin, used for basal glucose control in diabetes mellitus.

Insulin Lipohypertrophy

Thickened fatty tissue at repeated insulin-injection sites. Injecting into these areas can make insulin absorption unpredictable.

Insulin Lispro

A rapid-acting insulin analogue used around meals or for correction of high blood glucose in diabetes mellitus.

Insulin Pen Devices

Pre-filled or reusable insulin pen devices providing convenient and accurate insulin delivery. Benefits include portability precision dosing discrete injection and reduced needle phobia.

Insulin Preparations

Hormone synthesized by pancreatic beta cells that regulates glucose homeostasis by promoting glucose uptake in skeletal muscle and adipose tissue, suppressing hepatic glucose production, and inhibiting lipolysis.

Insulin Pump Therapy

Continuous subcutaneous insulin infusion delivering rapid-acting insulin analog through a catheter placed subcutaneously typically in the abdomen. Provides basal insulin at programmed rates and bolus doses at mealtimes.

Insulin Pumps

Insulin pumps deliver rapid-acting insulin contin-

uously and through programmed meal or correction doses, allowing flexible management of diabetes while requiring monitoring and device safety.

Insulin Receptor

A transmembrane receptor tyrosine kinase that binds insulin and initiates signaling controlling glucose uptake, metabolism, growth, and anabolic processes.

Insulin Resistance

Diminished biological response to physiological insulin concentrations resulting in impaired glucose disposal in peripheral tissues and failure to suppress hepatic glucose production. Pathophysiological hallmark of type two diabetes and metabolic syndrome.

Alternate Names:

IR (Insulin Resistance)

Insulin Resistance

Alternate index label for Insulin Resistance.

Insulin Resistance Syndrome

Alternate index label; see *Metabolic syndrome*.

Insulin Resistance Syndromes

Rare inherited severe insulin resistance syndromes include Donohue syndrome also known as leprechaunism, Rabson-Mendenhall syndrome, and type A insulin resistance syndrome caused by mutations in the insulin receptor gene. Donohue syndrome is the most severe with death in infancy.

Insulin Secretion

Release of insulin from pancreatic beta cells in response primarily to increased plasma glucose, with amplification by incretin hormones and nutrients. Insulin promotes glucose uptake, glycogen synthesis, lipogenesis, and protein synthesis while suppressing hepatic glucose production.

Insulin Shock

Severe hypoglycemia, historically associated with excessive insulin and characterized by confusion, seizures, loss of consciousness, or other neurologic

dysfunction.

Insulin Signaling

The cellular process initiated when insulin binds its receptor and activates pathways that regulate glucose uptake, glycogen formation, protein synthesis, and fat metabolism.

Insulin-Dependent Diabetes

Type 1 diabetes, in which the pancreas produces little or no insulin and replacement insulin is required.

Alternate Names:

Type 1 diabetes

Insulin-Induced Lipohypertrophy

A common complication of subcutaneous insulin therapy characterized by localized accumulation of adipose tissue at injection sites, caused by the lipogenic effect of insulin on subcutaneous fat. It presents as firm, rubbery, non-tender subcutaneous nodules that can cause unpredictable insulin absorption and glycemic variability.

Insulin-Like Growth Factor

IGF-1 promotes growth. Produced in liver in response to GH.

Insulinase

An enzyme that breaks down insulin, helping regulate how long insulin remains active.

Insulinoma

An insulinoma is a usually benign pancreatic beta-cell neuroendocrine tumour that secretes excessive insulin, causing recurrent fasting or exertional hypoglycaemia relieved by glucose.

Integrase

Integrase is an enzyme that inserts viral DNA into a host genome, a process targeted by integrase inhibitors used in HIV treatment.

Integrative Medicine

Patient-centered care that combines evidence-based conventional treatment with appropriate comple-

mentary approaches, emphasizing whole-person health, prevention, lifestyle, and shared decision-making.

Integrative Rehabilitation

Combination of conventional and complementary therapies: acupuncture, massage, yoga, meditation.

Integrin

A cell-surface receptor that links cells to the extracellular matrix and helps control adhesion, movement, signaling, and immune-cell activity.

Integrin Binding

The attachment of a molecule or cell to an integrin receptor, which can influence adhesion, migration, signaling, or clot formation.

Integron

A genetic element containing an integrase and an attachment site that can capture and express gene cassettes; integrons commonly contribute to bacterial antimicrobial resistance.

Integrons

Genetic elements that capture and express gene cassettes through site-specific recombination. Integrons contain an integrase gene and a recombination site for gene cassette insertion. They are important vehicles for the spread of antibiotic resistance genes, particularly in gram-negative bacteria.

Integumentary System

The body system comprising the skin and its appendages: hair, nails, sweat glands, and sebaceous glands. It forms a protective barrier, limits fluid loss, contributes to thermoregulation and sensation, supports immune defence, and participates in vitamin D synthesis.

Intein

An intein is a protein segment that excises itself from a precursor protein while joining the surrounding extein sequences.

Intellectual Disability

Neurodevelopmental disorder with onset in the developmental period, characterised by deficits in intellectual and adaptive functioning across conceptual, social, and practical domains. DSM-5 grades severity by adaptive functioning: mild, moderate, severe, profound.

Intelligence

Intelligence is the capacity to learn, reason, solve problems, understand information, and adapt; tests measure selected abilities rather than a person's entire worth or potential.

Intelligence Test

An intelligence test uses standardized tasks to estimate cognitive abilities such as reasoning, memory, language, and problem-solving.

Intense Pulsed Light

Broadband light treatment used for selected cosmetic skin conditions and hair reduction.

Alternate Names:

IPL

Intensity

The degree of effort or physiological demand during physical activity.

Intensity-Modulated Radiation Therapy

A computer-optimized form of three-dimensional radiation therapy that shapes and modulates the radiation dose to conform tightly to the target while sparing adjacent normal tissue.

Intensive Care

A patient-care topic concerning intensive care. It addresses the relevant purpose, risks, management, and follow-up.

Intensive Care Rehabilitation

Early rehabilitation in ICU setting. Early mobilization, positioning, passive range of motion. Improves outcomes, reduces complications.

Intensive Care Unit

A specialist hospital unit providing continuous monitoring and life support for critically ill patients. In obstetrics, women may require ICU admission for: severe pre-eclampsia or eclampsia, massive postpartum hemorrhage, amniotic fluid embolism, sepsis, cardiac disease in pregnancy, or complications of anesthesia.

Intention Tremor

A chemical substance or compound called intention tremor. Its properties, uses, and risks depend on its composition and clinical context.

Interatrial Septum

The wall separating the right and left atria. It contains the fossa ovalis, a remnant of the foramen ovale from fetal circulation. The interatrial septum has a thinner area where transseptal puncture can be performed during cardiac catheterization.

Intercalated Disc

An intercalated disc is a specialized junction connecting cardiac muscle cells and transmitting electrical and mechanical force through the heart.

Intercalating Agent

An intercalating agent inserts between DNA base pairs and can disrupt replication or transcription; some are laboratory reagents or anticancer drugs.

Intercellular

Intercellular means located between cells or occurring in the space between cells.

Intercostal

Intercostal means located between the ribs, especially the muscles, nerves, and blood vessels in those spaces.

Intercostal Muscles

Muscles between the ribs that help expand and compress the chest during breathing and also assist movement of the trunk.

Intercostal Retraction

Inward movement of the skin and soft tissues between the ribs during inspiration, indicating increased work of breathing. It is common in infants and children with respiratory distress and may accompany airway obstruction, bronchiolitis, pneumonia, asthma, or other pulmonary disease.

Intercostal Retractions

Inward pulling of the skin and soft tissues between the ribs during inspiration, indicating increased work of breathing and potentially significant respiratory distress.

Intercurrent Disease

An illness or complication that develops during the course of another disease or treatment and may affect its management or outcome.

Interdental Brush

Small brush for cleaning between teeth. Used where spaces exist.

Interdigital Neuroma

Alternate index label; see *Morton neuroma*.

Interference

A factor that alters the accuracy or interpretation of a laboratory test, measurement, or drug effect.

Interferometry

A measurement method that uses interference between waves to determine distance, motion, thickness, optical properties, or other physical quantities.

Interferon

Interferons are cytokines that induce antiviral defenses, alter immune-cell activity, and regulate cell growth; type I, II, and III interferons have different receptors and clinical uses.

Interferon Inducer

A substance that stimulates cells to produce interferons, proteins involved in antiviral and immune responses.

Interferon Signaling

Type I IFN (IFN-alpha/beta): all nucleated cells produce; receptor IFNAR (ubiquitous); JAK1/TYK2 -> STAT1/2/IRF9 (ISGF3) -> ISGs (hundreds: antiviral, antiproliferative, immunomodulatory). Type II IFN (IFN-gamma): T/NK cells; receptor IFNGR; JAK1/JAK2 -> STAT1 -> GAS elements. Type III IFN (IFN-lambda): mucosal epithelium; receptor IFNLR.

Interferon-Alpha

A type I interferon cytokine with antiviral, antiproliferative, and immune-modulating effects; recombinant forms have been used for selected viral infections and cancers.

Interferon-Beta

A type I interferon cytokine used mainly as an immunomodulatory treatment for relapsing forms of multiple sclerosis.

Interferon-Gamma Release Assay

Blood tests measuring IFN-gamma release in response to *M. tuberculosis* antigens (ESAT-6, CFP-10). More specific than PPD and unaffected by BCG. Cannot distinguish latent from active TB.

Interferon-Gamma Release Assay

Blood test for *Mycobacterium tuberculosis* infection (latent or active). T-SPOT.TB(ELISPOT) or QuantiFERON-TB Gold Plus. Measures IFN-gamma released by T cells after exposure to MTB-specific antigens (ESAT-6, CFP-10, TB7.7).

Alternate Names:

Interferon-Gamma Release Assay

Interferonopathies

A group of rare disorders caused by abnormal activation of type I interferon pathways. They can produce inflammation, neurologic disease, skin lesions, blood-vessel injury, or lung disease, often beginning in childhood.

Interferons

Signaling proteins that help coordinate immune

responses, especially against viral infections.

Interleukin

A family of cytokines (interleukins) produced mainly by leukocytes that regulate immune cell growth, differentiation, activation, and communication. Examples: IL-1 and IL-6 (inflammation, fever), IL-2 (T-cell growth), IL-4/IL-13 (Th2, IgE), IL-5 (eosinophils), IL-10 (anti-inflammatory), IL-12 (Th1), IL-17 (Th17, autoimmunity), IL-23.

Interleukin 1

A group of proinflammatory cytokines, especially IL-1 alpha and IL-1 beta, that promote fever, leukocyte activation, and inflammatory responses.

Interleukin 13

A cytokine produced by type 2 helper T cells and other immune cells that contributes to allergic inflammation, mucus production, and tissue fibrosis.

Interleukin 2

Interleukin-2 is a T-cell growth factor that promotes proliferation and survival of activated T cells and natural-killer cells; high-dose therapy can cause capillary-leak syndrome and systemic toxicity.

Interleukin 7

A cytokine important for development, survival, and homeostasis of T lymphocytes and some B-cell precursors.

Interleukin-6

A cytokine involved in immune regulation and inflammation that stimulates production of acute-phase proteins by the liver.

Interleukins

Cytokines that carry signals between immune cells and regulate inflammation.

Alternate Names:

ILs

Interlocutory

Interlocutory means provisional or occurring during

an ongoing process, such as an interim clinical, legal, or research decision.

INTERMACS Profile

A standardized, internationally recognized 7-tier clinical staging classification established by the Interagency Registry for Mechanically Assisted Circulatory Support to stratify the severity of illness and hemodynamic compromise in patients with advanced, end-stage (Stage D) heart failure. Classifies patients from Profile 1 (“crash and burn”: critical cardiogenic shock with persistent severe hypoperfusion despite escalating inotropic support) through Profile 2 (progressive decline on inotropes), Profile 3 (inotrope-dependent stability), Profile 4 (recurrent frequent hospitalizations), Profile 5 (housebound/exertion intolerant), Profile 6 (exertion limited), to Profile 7 (advanced NYHA Class III), guiding clinical urgency, patient selection, and timing for mechanical circulatory support implantation or emergency heart transplantation.

Alternate Names:

INTERMACS Staging; INTERMACS Scale

Intermediate

Intermediate means between two stages, positions, or levels, such as an intermediate metabolite, cell, or developmental phase.

Intermediate Cuneiform

Smallest cuneiform, centrally located between medial and lateral cuneiforms. Articulates with navicular, medial/lateral cuneiforms, and second metatarsal. Forms the transverse arch apex. Relatively fixed, contributing to midfoot stability. Injuries are typically associated with high-energy midfoot trauma.

Intermediate Filament

An intermediate filament is a strong cytoskeletal fiber that supports cell shape, mechanical stability, and tissue integrity.

Intermediate Uveitis

Inflammation primarily affecting the vitreous body, peripheral retina, and ciliary body, with the classic

presentation being pars planitis, which is characterized by the presence of snowbanking (a white, inflammatory exudate on the inferior pars plana) and snowballs (vitreous inflammatory aggregates).

Intermediate-Density Lipoprotein

A lipoprotein remnant formed as very-low-density lipoprotein is metabolized toward LDL.

Alternate Names:

IDL

Intermittent Auscultation

Periodic assessment of the fetal heart rate with a fetoscope or Doppler device during labor rather than continuous electronic monitoring. The frequency of assessment depends on the stage of labor, risk factors, and local clinical protocol.

Intermittent Claudication

Reproducible muscle discomfort, usually in the calf, thigh, or buttock, induced by walking and relieved by rest, caused by exertional arterial insufficiency. It is distinguished from rest pain and nonvascular musculoskeletal pain by history and vascular testing.

Intermittent Explosive Disorder

A disorder characterized by recurrent, impulsive, problematic outbursts of verbal or physical aggression that are grossly disproportionate to the provocation. The outbursts are not premeditated and cause marked distress or functional impairment. Age >6 years.

Alternate Names:

Road Rage

Intermittent Fasting

Cycling between fasting and eating periods. Methods: alternate-day fasting, 5:2 diet, time-restricted eating. Potential benefits: weight loss, metabolic health. Evidence emerging.

Intermittent Fever

A fever that rises and falls with periods when temperature returns to normal or near normal.

Intern

A physician in the early stage of postgraduate medical training, traditionally the first year after medical school; the exact training structure varies by country.

Internal

Internal means located within the body, an organ, a structure, or a system rather than on its outer surface.

Internal Bleeding

Internal bleeding is blood loss into a body cavity, tissue, or organ rather than through an obvious external wound. Trauma, ulcers, aneurysm, pregnancy complications, medicines, and tumors can cause it; faintness, shock, abdominal swelling, black stool, or vomiting blood is an emergency.

Internal Capsule

A compact band of white-matter nerve fibres in the brain carrying important motor and sensory pathways between the cerebral cortex and lower centres.

Internal Environment

The fluid and physiologic conditions inside the body that cells require, including temperature, pH, oxygen, electrolytes, and nutrients.

Internal Fixation

Surgical stabilization of a fracture or unstable bone using implanted hardware such as plates, screws, rods, wires, or pins.

Internal Fixator

An internal fixator is a surgically implanted plate, screw, rod, nail, or frame used to stabilize a fracture or correct skeletal alignment.

Internal Hernia

Herniation of bowel through a mesenteric defect common after Roux-en-Y gastric bypass creating a defect at the jejunojejunostomy. May cause intermittent or acute small bowel obstruction. Presents

with episodic abdominal pain, nausea, and vomiting.

Internal Iliac Artery

Smaller common iliac terminal branch (hypogastric), supplying the pelvis, perineum, gluteal region, and medial thigh. Posterior division: iliolumbar, lateral sacral, superior gluteal. Anterior division: inferior gluteal, internal pudendal, obturator, vesical, uterine, vaginal, middle rectal arteries.

Internal Jugular Vein

Descends in the carotid sheath with the internal carotid artery and vagus nerve, draining brain, face, and neck. Begins at the jugular foramen as a sigmoid sinus continuation, joining the subclavian vein to form the brachiocephalic vein. Contains backflow-preventing valves. Used for central venous catheterization.

Internal Oblique Muscle

Middle flat abdominal layer deep to external oblique, from thoracolumbar fascia, iliac crest, and lateral inguinal ligament to lower ribs, linea alba, and pubic crest. Innervated by T7-T12 and L1. Ipsilateral rotation (opposite of external oblique). Contributes to inguinal canal and abdominal wall integrity.

Internal Podalic Version

A maneuver where the operator inserts a hand into the uterus, grasps one or both fetal feet, and converts a transverse lie or cephalic presentation to a breech, then delivers by total breech extraction. Indication: delivery of the second twin (transverse lie) or severe fetal distress in the second stage. Rare in modern obstetrics.

Internal Resorption

Loss of dentin from pulp cavity inward. Pink spot may be visible. Treatment: root canal therapy.

International Normalized Ratio

The INR, a standardized calculation from the prothrombin time used to assess blood clotting and guide treatment with vitamin K antagonists such as warfarin.

International Unit

A chemical substance or compound called international unit. Its properties, uses, and risks depend on its composition and clinical context.

International Unit

A standardized internationally accepted unit of measurement for quantifying the biological activity, potency, or pharmacological effect of a substance (such as fat-soluble vitamins, insulin, heparin, and vaccines), established by the World Health Organization. Unlike metric mass units (milligrams or micrograms), the exact mass equivalent of one IU varies depending on the specific biological substance and reference preparation.

Interneuron

An interneuron connects neurons within the central nervous system and helps integrate, inhibit, or coordinate neural signals.

Interneurons

Neurons within the central nervous system that connect other neurons and help process or coordinate neural signals.

Internist

An internist is a physician specializing in adult internal medicine, including diagnosis and management of complex medical conditions.

Internuclear Ophthalmoplegia

A conjugate horizontal gaze disorder caused by a lesion of the medial longitudinal fasciculus (MLF), a heavily myelinated fiber tract that connects the abducens nucleus (CN VI) on one side to the oculomotor nucleus (CN III) on the contralateral side, coordinating horizontal gaze.

Interosseous

Interosseous means located between bones, such as the membrane or muscles between the forearm or lower-leg bones.

Interpersonal Psychotherapy

A time-limited, evidence-based psychotherapy for depression that focuses on the interpersonal context of mood symptoms. It identifies four problem areas: grief, role disputes, role transitions, and interpersonal deficits. IPT helps patients improve interpersonal functioning and social support.

Interphase

Interphase is the portion of the cell cycle between divisions, including growth, DNA replication, and preparation for mitosis or meiosis.

Interphase Cell

interphase cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Interproximal Caries

Dental caries occurring on the contact surfaces between adjacent teeth. It may be difficult to detect clinically and is commonly assessed with bitewing radiographs when imaging is indicated.

Intersexual

Intersexual is an immunologic, genetic, infectious, developmental, or cellular concept used to explain disease mechanisms, diagnosis, or treatment.

Interstitial

Relating to the spaces between cells or tissues, such as interstitial fluid or interstitial lung disease.

Interstitial (Cornual) Pregnancy

An ectopic pregnancy implanted in the part of the fallopian tube that passes through the muscular wall of the uterus. It can enlarge before rupture and cause severe internal bleeding.

Alternate Names:

Cornual

Interstitial Cystitis

Chronic bladder pain syndrome. Symptoms: suprapubic pain, frequency, urgency. Diagnosis: exclu-

sion, cystoscopy with hydrodistension. Treatment: lifestyle modification, medications, bladder instillations, surgery.

Alternate Names:

Bladder Inflammation Cystitis, Interstitial IC (Interstitial Cystitis) Painful Bladder Syndrome

Interstitial Cystitis

A chronic bladder pain syndrome with urinary urgency, frequency, or pain without an identifiable infection or other cause sufficient to explain the symptoms.

Interstitial Fluid

Interstitial fluid is the fluid surrounding tissue cells, formed from blood plasma and returned to circulation through lymphatic vessels.

Interstitial Granulomatous Dermatitis

A rare inflammatory dermatosis with interstitial or palisading histiocytes around collagen, often presenting with symmetric erythematous or violaceous plaques and associated systemic disease or medication exposure.

Interstitial Keratitis

Inflammation of the corneal stroma without primary involvement of the corneal surface. It may follow infection, autoimmune disease, or trauma and can cause pain, photophobia, redness, and reduced vision.

Interstitial Lung Disease

A heterogeneous group of disorders with inflammation and fibrosis of lung parenchyma, causing restrictive physiology (reduced FVC and TLC with normal or elevated FEV1/FVC ratio).

Alternate Names:

Diffuse Parenchymal Lung Diseases Interstitial Lung Diseases Lung Disease, Interstitial

Interstitial Lung Disease (Interstitial Pneumonia)

Alternate index label for Interstitial Pneumonia.

Interstitial Nephritis

Inflammation of the renal interstitium and tubules, which can be acute or chronic. Acute interstitial nephritis is most commonly caused by medications, particularly NSAIDs, antibiotics including penicillins and cephalosporins, and proton pump inhibitors.

Interstitial Pneumonia

Inflammation centered on the interstitium. Caused by Mycoplasma, Chlamydophila, Legionella, and viruses. Presents with dry cough and diffuse crackles.

Alternate Names:

Interstitial Lung Disease (Interstitial Pneumonia)

Interstitialium

The supporting tissue between and around the lung air sacs, capillaries, and small airways; inflammation or fibrosis of this tissue can impair lung expansion and gas exchange.

Intertrigo

Intertrigo is inflammation in opposing skin folds caused by moisture, heat, friction, and maceration; Candida or bacterial infection may complicate the tender erythematous rash.

Alternate Names:

Allergy Diaper (Intertrigo)

Interval

An interval is the time, distance, or range between two points or events, such as a dosing or follow-up interval.

Intervening Sequence

A nucleotide sequence within a gene transcript that lies between coding sequences; in many genes it is an intron removed during RNA processing.

Intervention

An action intended to prevent, diagnose, treat, or change a health condition or outcome, such as a medicine, procedure, counseling, or public-health measure.

Intervention Study

A study in which researchers assign or provide an intervention and compare its outcomes with a control.

Alternate Names:

Clinical trial

Interventional

Relating to a procedure or active treatment used to diagnose or manage disease, often through minimally invasive access rather than open surgery.

Interventional Cardiology

A subspecialty using catheter-based procedures to diagnose and treat coronary, structural, and selected vascular heart conditions, including angioplasty, stenting, and transcatheter valve procedures.

Interventional Procedure

A medical procedure involving interventional procedure. Indications, risks, preparation, and follow-up depend on the patient and the specific procedure.

Interventional Procedures

Image-guided treatments: angioplasty, stenting, embolization, ablation, biopsy, drain placement. Advantages: minimally invasive, faster recovery, lower complication rates.

Interventional Radiologist

A physician trained to perform minimally invasive diagnostic and therapeutic procedures using medical imaging for guidance.

Interventional Radiology

Image-guided minimally invasive procedures. Examples: angioplasty, stenting, embolization, biopsy, ablation, drain placement. Reduces need for open surgery.

Interventional Radiology

A specialty using imaging guidance to perform minimally invasive diagnostic and therapeutic procedures, such as biopsies, drainages, embolization, and vascular treatments.

Interventricular Septum

The thick muscular wall separating the left and right ventricles. It has two portions: the muscular (inferior, thicker) and membranous (superior, thinner) portions. The septum contains parts of the cardiac conduction system, including the bundle of His.

Intervertebral Disc

A fibrocartilaginous cushion between adjacent vertebral bodies, composed of a gelatinous nucleus pulposus surrounded by a tough annulus fibrosus. It absorbs shock and permits limited movement of the vertebral column.

Intervertebral Disk

A fibrocartilaginous cushion between spinal vertebrae that absorbs shock and permits movement.

Alternate Names:

Intervertebral disc

Intervertebral Foramen

An opening between adjacent vertebrae through which a spinal nerve exits.

Alternate Names:

Neural foramen

Intestinal Absorption

Intestinal absorption is the movement of digested nutrients, water, and electrolytes from the intestine into the blood or lymph.

Intestinal Atresia

A congenital obstruction of the small intestine (jejunum or ileum) caused by an intrauterine vascular accident (mesenteric ischemia) leading to aseptic necrosis and resorption of a segment of bowel. In contrast to duodenal atresia, it is not associated with Down syndrome.

Intestinal Cancer

Intestinal cancer is a malignant growth arising in the small or large intestine; symptoms and treatment depend on its location and stage.

Intestinal Duplication

Intestinal duplication is a congenital extra segment or cyst of bowel that may cause obstruction, bleeding, pain, or infection.

Intestinal Fistula

An intestinal fistula is an abnormal passage connecting the intestine to another organ or the skin, often after inflammation, surgery, or injury.

Intestinal Ganglioneuromatosis

Benign proliferation of ganglion cells and nerve fibers in the gastrointestinal tract wall. May be diffuse involving the entire gastrointestinal tract or localized as ganglioneuromas. Associated with multiple endocrine neoplasia type 2B where intestinal ganglioneuromatosis may precede the development of medullary thyroid carcinoma.

Intestinal Gas

Gas in the digestive tract that may cause belching, bloating, abdominal discomfort, or flatulence.

Intestinal Gas (Belching, Bloating, Flatulence)

Alternate index label for Belching, Bloating, Flatulence.

Intestinal Gas (Intestinal Gas (Belching, Bloating, Flatulence))

Alternate index label for Intestinal Gas (Belching, Bloating, Flatulence).

Intestinal Ischemia

Inadequate blood flow to the intestines, which can damage intestinal tissue.

Alternate Names:

Ischemia, Intestinal

Intestinal Leiomyoma

A usually benign smooth-muscle tumor arising in the wall of the intestine; symptoms depend on size and location and may include bleeding, obstruction, or abdominal pain.

Intestinal Lipodystrophy

Alternate index label; see *Whipple's disease*.

Intestinal Lipomatosis

Benign overgrowth of adipose tissue within the intestinal wall, most commonly affecting the ileum and colon. Often asymptomatic and discovered incidentally, but large lipomas can cause intussusception, gastrointestinal bleeding, or bowel obstruction.

Intestinal Lymphoma

Malignant proliferation of lymphoid cells arising in the small or large intestine. Important types include enteropathy-associated T-cell lymphoma, monomorphic epitheliotropic intestinal T-cell lymphoma, and diffuse large B-cell lymphoma.

Intestinal Malrotation

Congenital failure of normal midgut rotation and fixation, predisposing to a narrow mesenteric base, volvulus, and duodenal obstruction. Infants may present with bilious vomiting, abdominal distension, or bowel ischemia; sudden symptoms require emergency assessment.

Intestinal Metaplasia

Replacement of the normal lining of an intestinal or gastric region by intestinal-type epithelium, usually in response to chronic injury or inflammation and sometimes associated with increased cancer risk.

Intestinal Mucosa

The intestinal mucosa is the inner lining of the intestine, where digestion, nutrient absorption, mucus production, and immune defence occur.

Intestinal Obstruction

A mechanical or functional blockage that prevents normal passage of intestinal contents. Causes include adhesions, hernias, tumors, inflammation, volvulus, and disorders of intestinal motility.

Intestinal Obstruction And Ileus

Impaired passage of intestinal contents caused by a mechanical blockage or by reduced bowel motility

without a physical blockage; symptoms commonly include abdominal pain, distension, vomiting, and inability to pass stool or gas.

Intestinal Obstruction Repair

Surgery or another intervention to relieve an intestinal blockage, remove a diseased segment when necessary, and restore bowel continuity or create a stoma when required.

Intestinal Parasite

A protozoan or worm that lives in the gastrointestinal tract and may cause diarrhea, abdominal pain, anemia, malnutrition, or no symptoms.

Intestinal Perforation

Full-thickness breach in the bowel wall allowing luminal contents to enter the peritoneal cavity. Causes include diverticulitis, appendicitis, peptic ulcer disease, and trauma. Presents with acute abdomen, peritonitis, and free air on imaging.

Intestinal Polyp

An intestinal polyp is a growth projecting from the lining of the intestine; some types can become cancerous.

Intestinal Polyposis

Intestinal polyposis is the presence of numerous polyps in the intestine, sometimes as part of an inherited cancer-predisposition syndrome.

Intestinal Pseudo-Obstruction

Acute or chronic dilation of the bowel without mechanical obstruction due to neuromuscular dysfunction. Acute colonic pseudo-obstruction also called Ogilvie syndrome occurs in hospitalized or post-surgical patients.

Alternate Names:

Ogilvie syndrome.

Intestinal Villi

Intestinal villi are tiny finger-like projections of the small-intestinal lining that increase surface area for nutrient absorption.

Intestine

The part of the digestive tract extending from the stomach to the anus, consisting of the small intestine and large intestine.

Intestine Disorder

An intestine disorder is a disease affecting the small or large intestine, such as infection, inflammation, obstruction, or cancer.

Intestine Neoplasm

An intestine neoplasm is an abnormal growth in the small or large intestine; it may be benign or malignant.

Intestine Obstruction

Intestinal obstruction is partial or complete blockage of the passage of intestinal contents, causing pain, swelling, vomiting, and failure to pass stool or gas.

Intestine, Diseases Of

Diseases of the intestine include infections, inflammatory disorders, malabsorption, obstruction, polyps, and cancer.

Intimal Dissection

A chemical substance or compound called intimal dissection. Its properties, uses, and risks depend on its composition and clinical context.

Intimal Hyperplasia

Proliferation and migration of vascular smooth-muscle cells with extracellular-matrix deposition within the intima after vascular injury or graft placement. It can produce restenosis at anastomoses, stents, dialysis access, or bypass grafts.

Intimate Partner Violence In Pregnancy

Physical, sexual, or psychological abuse by a current or former partner during pregnancy. It can harm the pregnant person and fetus and is associated with complications such as injury, preterm birth, and placental abruption.

Intolerance

An adverse response to a substance that does not involve the immune mechanism of an allergy.

Alternate Names:

Nonallergic sensitivity Nonceliac Gluten Sensitivity (Intolerance)

Intolerance, Gluten (Nonceliac Gluten Sensitivity (Intolerance))

Alternate index label for Nonceliac Gluten Sensitivity (Intolerance).

Intoplicine

Intoplicine is an investigational topoisomerase inhibitor studied for anticancer effects by disrupting DNA structure and replication.

Intoxication

A state of physical or mental impairment caused by recent exposure to a substance, such as alcohol, a medicine, or another drug.

Intra-Abdominal Abscess

Localized collection of pus within the abdomen commonly in the pelvis, between bowel loops, or beneath the diaphragm. Results from intra-abdominal infection or surgery. Presents with fever, abdominal pain, and leukocytosis. Diagnosed by CT with contrast showing rim-enhancing fluid collection.

Intra-Abdominal Hypertension

Sustained elevation of intra-abdominal pressure above twelve millimeters of mercury measured through the urinary bladder. Graded as grade one twelve to fifteen, grade two sixteen to twenty, grade three twenty-one to twenty-five, and grade four greater than twenty-five millimeters of mercury.

Intra-Aortic Balloon Pump

Mechanical circulatory support. Inflatable catheter in descending aorta. Inflates during diastole (increases coronary perfusion), deflates during systole (reduces afterload). Indications: cardiogenic shock, unstable angina, high-risk PCI. Complications: limb ischemia, bleeding, infection.

Intra-Arterial

Intra-arterial means within an artery, including a catheter, injection, infusion, or measurement performed directly in an arterial vessel.

Intraabdominal Cancer

A broad term for malignancies arising within the abdominal cavity, including cancers of the colon, stomach, pancreas, liver, ovaries, and peritoneum. Presentation varies widely depending on the specific organ involved.

Intraabdominal Hemorrhage

Bleeding inside the abdominal cavity, which may follow trauma, surgery, organ rupture, pregnancy complications, or internal disease.

Intraabdominal Hernia

Herniation of abdominal viscera through a congenital or acquired defect or abnormal mesenteric aperture within the peritoneal cavity, distinct from ventral or inguinal hernias. Types include paraduodenal, foramen of Winslow, transmesocolic, and transomental hernias.

Intraabdominal Infection

An infection within the abdominal cavity, such as peritonitis or an abscess, often requiring antibiotics and sometimes drainage or surgery.

Intraaortic Balloon Pump

A temporary circulatory-support device with a balloon in the descending aorta that inflates and deflates in coordination with the cardiac cycle.

Intraarticular

Intraarticular means within a joint, such as an injection, fracture, loose body, or inflammatory process inside the joint space.

Intrabronchial

Intrabronchial means within a bronchus, one of the air passages branching from the trachea into the lungs.

Intracardiac Electrophysiology Study

An invasive test using electrode catheters inside the heart to record electrical activity and provoke or map arrhythmias, helping guide diagnosis and treatment.

Intracartilaginous

Intracartilaginous means located within cartilage, such as a lesion, injection, or developmental structure.

Intracavernosal Injection

A second-line treatment for erectile dysfunction involving the direct self-injection of vasoactive medications into the corpus cavernosum of the penis using a fine-gauge needle. The injected agents (commonly alprostadil [prostaglandin E1] as monotherapy, or combinations with papaverine and phentolamine known as Bimix or Trimix) stimulate smooth-muscle relaxation and arterial dilation within the cavernosal sinusoids to induce an erection independent of sexual stimulation. It is effective in organic, vasculogenic, and neurogenic erectile dysfunction (such as following radical prostatectomy); potential complications include localized pain, hematoma, cavernosal fibrosis, and prolonged erection or ischemic priapism requiring urgent intervention.

Alternate Names:

ICI; Penile Injection Therapy; Intracavernous Injection

Intracavitary

Intracavitary means within a body cavity, such as a treatment, device, fluid, or lesion located in the thorax, abdomen, uterus, or another cavity.

Intracavity Radiotherapy

Intracavity radiotherapy places a radiation source inside a body cavity near a tumor, commonly to treat selected cancers of the cervix, uterus, or vagina.

Intracellular

Pertaining to, positioned inside, or occurring within the bounding plasma membrane of a biological cell. The intracellular fluid compartment comprises approximately two-thirds of total body water in adults,

characterized electrochemically by high concentrations of potassium (K^+), magnesium (Mg^{2+}), and organic phosphate/protein polyanions, and low concentrations of sodium (Na^+) and chloride (Cl^-), maintained continuously across the plasma membrane by ATP-dependent Na^+/K^+ -ATPase pump activity.

Intracellular Fluid

The fluid contained within cells, comprising approximately two thirds of total body water in a typical adult. Its major solutes include potassium, magnesium, phosphate, and intracellular proteins, and its composition is maintained by selective membrane transport.

Intracellular Membrane

intracellular membrane refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Intracellular Pathogens

Microorganisms that survive and replicate within host cells. Intracellular bacteria include Mycobacterium, Listeria, and Chlamydia. Viruses are obligate intracellular pathogens. Intracellular survival protects pathogens from antibiotics and immune responses.

Intracellular Transport

intracellular transport refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Intracerebral

Intracerebral means within the brain tissue, as in an intracerebral hemorrhage, injection, electrode, or lesion.

Intracerebral Hemorrhage

Bleeding directly into the brain parenchyma, accounting for approximately ten to fifteen percent of all strokes and carrying a high mortality rate of approximately forty percent at thirty days.

Intracisternal

Intracisternal means within a cistern, especially a cerebrospinal-fluid space at the base of the brain.

Intracoronaral

Intracoronaral means within the crown of a tooth or within a coronary structure, with the exact meaning determined by context.

Intracoronary

Intracoronary means within a coronary artery or the coronary circulation, such as a catheter, stent, imaging study, or medicine delivery.

Intracranial

Situated or occurring within the bony cranium (skull cavity). The rigid, non-distensible cranial vault contains three physiological volume components: brain parenchyma ($\approx 80\%$), cerebrospinal fluid (CSF; $\approx 10\%$), and intracranial blood ($\approx 10\%$). Governed by the Monro-Kellie doctrine, an increase in any one intracranial volume component (e.g., from cerebral edema, intracranial hemorrhage, or brain tumor) must be compensated by a reciprocal reduction in CSF or venous blood; when compensatory capacity is exhausted, intracranial pressure rises steeply (> 20 mmHg), risking cerebral hypoperfusion, ischemic brain injury, and fatal brain herniation syndromes.

Intracranial Aneurysm

A weakened, bulging area in a blood vessel within the skull.

Intracranial Hemorrhage

Bleeding within the cranial cavity. Epidural hematoma: arterial bleeding (middle meningeal artery), typically from temporal bone fracture; lucid interval then rapid deterioration. Subdural hematoma: venous bleeding from bridging veins; acute (<72 h), subacute, or chronic (elderly, alcoholics).

Intracranial Pressure

The pressure within the cranial vault generated by brain tissue, cerebrospinal fluid, and intracra-

nial blood. Sustained elevation can reduce cerebral perfusion and cause headache, vomiting, altered consciousness, pupillary abnormalities, and brain herniation.

Intracranial Pressure Monitoring

Continuous or intermittent measurement of pressure inside the skull, usually in critically ill patients with severe brain injury, hemorrhage, hydrocephalus, or other risk of intracranial hypertension.

Intracranial Venous Malformations

Abnormal collections or connections of veins within the skull that may bleed or affect nearby brain tissue.

Intractable

Difficult to control, relieve, or cure despite appropriate treatment, as in intractable pain or seizures.

Intradermal

Located within the dermis, the layer of skin beneath the epidermis; an intradermal injection places medication or test material into that layer.

Intraductal Carcinoma

Alternate index label; see *Ductal carcinoma in situ*.

Intraductal Papillary Mucinous Nneoplasm Pancreas (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Intraductal Papilloma

A benign papillary tumor arising within a breast duct, often presenting with spontaneous unilateral bloody or clear nipple discharge. It may be solitary near the nipple or multiple in the peripheral breast.

Intraduodenal

Intraduodenal means within the duodenum, the first part of the small intestine beyond the stomach.

Intraepidermal

Intraepidermal means within the epidermis, the outermost layer of skin.

Intraepithelial

Located within or occurring among the cells of an epithelial layer.

Intraesophageal

Intraesophageal means within the esophagus, such as a probe, pressure sensor, lesion, or medication delivery.

Intrahepatic

Situated, occurring, or originating within the anatomical parenchyma or capsule of the liver. Clinically relevant in hepatobiliary disease, including intrahepatic cholestasis (impaired bile formation or secretion by hepatocytes or small intrahepatic cholangioles, as in primary biliary cholangitis), intrahepatic bile duct dilatation on ultrasound (differentiating intrahepatic biliary obstruction from extrahepatic choledocholithiasis), and primary intrahepatic malignancies (hepatocellular carcinoma and intrahepatic cholangiocarcinoma).

Intrahepatic Cholestasis

Impaired bile flow within the liver, causing bile acids to accumulate in the blood.

Intrahepatic Cholestasis Of Pregnancy

A pregnancy-associated liver disorder characterized mainly by itching without a primary rash and increased bile acids in the blood. It may increase fetal risk and requires clinical evaluation and monitoring.

Intralesional

A chemical substance or compound called intralesional. Its properties, uses, and risks depend on its composition and clinical context.

Intralymphatic

Intralymphatic means within a lymphatic vessel or channel, such as a medicine, tumor spread, infection, or imaging contrast.

Intramedullary

Intramedullary means within the medulla of an organ or the central canal of a bone, especially

an implant placed inside a long bone.

Intramedullary Nail

A metal implant inserted into the medullary canal of a long bone to align and stabilize a fracture. It shares load along the bone axis and is commonly used for femoral and tibial shaft fractures.

Intramural Hematoma

Collection of blood within the wall of the gastrointestinal tract most commonly the duodenum. Caused by trauma, anticoagulation, or coagulopathy. Presents with abdominal pain, vomiting, and sometimes obstructive symptoms. Diagnosis is by CT showing circumferential wall thickening with high-attenuation hematoma.

Intramural Plaque

An intramural plaque is an atherosclerotic deposit within an artery wall that can narrow the lumen or destabilize and cause thrombosis.

Intramuscular

A route of drug administration in which the drug is injected into skeletal muscle (deltoid, vastus lateralis, gluteus maximus).

Intramuscular

Pertaining to the interior of a skeletal muscle, or describing a parenteral route of medication administration injected deep into muscular tissue (commonly the deltoid, vastus lateralis, or ventrogluteal muscles). Provides more rapid systemic vascular absorption than subcutaneous injection owing to greater muscle vascularity, suitable for vaccines, depot antipsychotics, and medications unsuited for oral or IV administration.

Intramuscular Injection

A medical procedure involving intramuscular injection. Indications, risks, preparation, and follow-up depend on the patient and the specific procedure.

Intramuscular Myxoma

An intramuscular myxoma is a rare benign tumor

of gelatinous myxoid tissue located within skeletal muscle.

Intraocular

Situated, occurring, or administered within the eyeball (globe). The intraocular compartment is divided into the anterior segment (anterior and posterior chambers filled with clear aqueous humor) and the posterior segment (vitreous chamber filled with gelatinous vitreous humor). Maintained at a normal intraocular pressure (IOP; 10–21 mmHg) by a delicate balance between aqueous humor secretion by the ciliary body epithelium and outflow through the trabecular meshwork; chronically elevated intraocular pressure is the primary modifiable risk factor for glaucomatous optic neuropathy and irreversible visual field loss.

Intraocular Infection

An infection inside the eye, such as endophthalmitis, that can rapidly threaten vision and requires urgent ophthalmic treatment.

Intraocular Lens

Artificial lens replacing natural lens after cataract extraction. Types: monofocal (distance vision), multifocal/bifocal (near and distance), toric (astigmatism), extended depth of focus. Implantation via small incision, folds for insertion. Complications: posterior capsule opacification, dysphotopsia, infection, retinal detachment.

Intraocular Lens Dislocation

Postoperative displacement of an implanted artificial intraocular lens from its intended position within the capsular bag or ciliary sulcus into the vitreous cavity or anterior chamber. It is most commonly caused by progressive zonular dehiscence and may require surgical repositioning or exchange.

Intraocular Lymphoma

Intraocular lymphoma is lymphoma involving the eye, often affecting the vitreous or retina and sometimes occurring with lymphoma in the brain or elsewhere.

Intraocular Pressure

The pressure generated by aqueous humor within the eye, normally maintained by a balance between aqueous production and outflow. Abnormal IOP is an important risk factor for glaucomatous optic neuropathy, but glaucoma can occur at statistically normal pressures.

Intraocular Pressure

The pressure of fluid within the eye, usually measured to assess risk for or management of glaucoma and other eye disorders.

Intraocular Tumor

Any neoplasm arising within the eye, with uveal melanoma being the most common primary malignancy in adults and retinoblastoma the most common in children. Diagnosis requires dilated fundus examination, ultrasound, and MRI; treatment depends on tumour type, size, and location.

Intraoperative

Occurring during a surgical operation, including monitoring, findings, complications, or treatment decisions made during the procedure.

Intraoperative Blood Transfusion

Intraoperative blood transfusion replaces lost blood volume and maintains oxygen-carrying capacity during major surgery. Indications include hemoglobin less than 7 g/dL in stable patients, less than 8 g/dL in cardiac disease, or active hemorrhage with hemodynamic instability.

Intraoperative Neuromonitoring

Intraoperative neuromonitoring uses electrophysiological techniques to monitor neural pathway integrity during surgery. Somatosensory evoked potentials (SSEP) monitor sensory pathways. Motor evoked potentials (MEP) monitor motor pathways. Electromyography (EMG) monitors cranial nerves and spinal roots.

Intraoral

Located within the mouth or performed through the mouth.

Intraosseous Access

Placement of a vascular-access needle into the medullary cavity of a suitable bone to permit rapid infusion of fluids, blood products, and medications when intravenous access is delayed or impossible. It is a temporary emergency route and can cause extravasation, infection, or compartment injury.

Intraosseous Infusion

Administration of fluids, blood products, or medications through an intraosseous access device. Infusion should be monitored for resistance, swelling, pain, and extravasation and replaced with definitive vascular access as soon as practical.

Intrapartum

The period during labor and childbirth, from the onset of regular uterine contractions leading to cervical dilation until delivery of the baby and placenta.

Intrapartum Fever

Fever during labor, which may result from infection, epidural analgesia, dehydration, or other causes and requires maternal and fetal assessment.

Intrapartum Stillbirth

Intrapartum stillbirth is fetal death occurring during labor after a live fetus was documented before labor, requiring obstetric investigation and support.

Intrapericardial

Located inside the pericardial sac, the thin membrane surrounding the heart. An intrapericardial mass or fluid collection may affect how the heart fills or beats.

Intraperitoneal

Intraperitoneal means within the peritoneal cavity or administered into it, including selected medicines, dialysis fluid, or procedures.

Intraperitonealroute

The intraperitoneal route delivers a medicine or fluid into the peritoneal cavity, commonly for peritoneal dialysis or selected treatments.

Intraprostatic

Intraprostatic means within the prostate gland, such as a lesion, injection, infection, or procedure.

Intrapulmonary

Located within the lung or occurring inside the pulmonary tissue or blood vessels.

Intrapulmonary Shunt

Perfusion of alveoli without ventilation ($V/Q = 0$). Blood passes through the pulmonary circulation without gas exchange, mixing with oxygenated blood and causing hypoxemia. It is seen in ARDS, pneumonia, and atelectasis. It is refractory to supplemental oxygen.

Intrasynovial

Intrasynovial means within the synovial lining or fluid-filled space of a joint or tendon sheath.

Intratendinous

Intratendinous means within a tendon, such as a tear, deposit, injection, or degenerative change.

Intratesticular

Located within testicular tissue. An intratesticular mass may be benign or malignant and is usually evaluated with physical examination and scrotal ultrasound, interpreted in the clinical context.

Intrathecal

Intrathecal means within the fluid-filled space surrounding the brain and spinal cord; medicines may be given there by lumbar puncture.

Intrathecal Chemotherapy

Administration of chemotherapy directly into the cerebrospinal fluid via lumbar puncture or an Ommaya reservoir, used for leptomeningeal metastasis and central nervous system prophylaxis in leukemias and lymphomas.

Intrathoracic

Intrathoracic means within the chest cavity, including the lungs, heart, mediastinum, vessels, and tho-

racic lymph nodes.

Intrathoracic Localized Obstruction

A focal blockage within the thoracic cavity affecting the airways, esophagus, or great vessels, leading to localized symptoms depending on the structure involved. Causes include endobronchial tumors, foreign bodies, esophageal strictures, or vascular anomalies. Diagnosis is by imaging and endoscopy.

Intratracheal

Intratracheal means within the trachea, such as a tube, medicine, foreign body, lesion, or route of administration.

Intratympanic

Intratympanic means delivered through the eardrum into the middle ear, often for selected inner-ear disorders or local treatment.

Intrauterine

Intrauterine means within the uterus, including a pregnancy, device, medicine, lesion, or procedure.

Intrauterine Contraceptive Device

An intrauterine contraceptive device is a small device placed in the uterus to prevent pregnancy; copper and hormone-releasing types are available.

Intrauterine Contraceptive Device

A small device placed in the uterus to prevent pregnancy; copper IUDs release copper, while hormonal IUDs release a progestin, and both are highly effective reversible contraception.

Intrauterine Device

IUD: a long-acting reversible contraceptive device placed in the uterus. Types: copper IUD (ParaGard—hormone-free, up to 10-12 years, also emergency contraception) and levonorgestrel-releasing IUS (Mirena/Skyla—progestin, up to 3-8 years, reduces bleeding).

Intrauterine System

A small T-shaped contraceptive device placed in-

side the uterus that slowly releases a progestogen hormone to prevent pregnancy. It is also called IUS.

Intrauterine Devices

Small devices placed in the uterus to prevent pregnancy; copper and hormonal IUDs are highly effective long-acting reversible contraceptives.

Intrauterine Fetal Demise / Stillbirth

Death of a fetus before complete delivery. The gestational-age definition varies by jurisdiction. Causes may include placental disease, umbilical-cord complications, infection, genetic abnormalities, maternal disease, or unexplained factors.

Intrauterine Growth Restriction

IUGR: a fetus with estimated weight below the 10th percentile (or growth velocity decline), usually with placental insufficiency or maternal/fetal factors.

Intrauterine Pressure Catheter

A fluid-filled catheter inserted into the uterine cavity alongside the fetus to measure the intensity and frequency of contractions in mm Hg (Montevideo units, MVU).

Intrauterine Transfusion

Ultrasonically guided intravascular or intraperitoneal transfusion of fetal red blood cells to treat severe fetal anemia, most commonly from Rh isoimmunization. Performed through the umbilical vein using a 22-25 gauge needle. Packed red blood cells, CMV-negative and crossmatched, are transfused slowly.

Intravascular

Intravascular means within a blood vessel, including a catheter, clot, injection, or process occurring in the circulation.

Intravascular Catheter Infections

CLABSI: central line-associated bloodstream infection. Diagnosis: quantitative blood cultures (catheter vs peripheral >5:1), differential time to positivity (>2h), catheter tip culture (>15 CFU).

Catheter-related BSIs: coagulase-negative staphylococci (most common), *S. aureus*, enterococci, gram-negatives, *Candida*.

Intravascular Filter

A device placed in a large vein to trap venous clots and reduce pulmonary embolism risk in selected patients who cannot receive or have not responded adequately to anticoagulation.

Intravascular Hemolysis

Destruction of red blood cells within the bloodstream, releasing hemoglobin into plasma; causes include immune hemolysis, mechanical trauma, infections, toxins, and transfusion reactions.

Intravascular Lymphoma

A rare aggressive B-cell non-Hodgkin lymphoma characterized by proliferation of neoplastic lymphocytes within the lumina of small blood vessels, predominantly affecting the skin and central nervous system. Presents with livedo reticularis, skin nodules, neurological deficits, and fever of unknown origin.

Intravascular Ultrasound

Catheter-based imaging using a miniature ultrasound probe inside coronary arteries to visualize vessel wall layers, plaque composition, and stent deployment. Provides more detailed information than angiography alone, including vessel remodeling, plaque burden, and true vessel diameter.

Intravenous

Administered into a vein, allowing a medicine or fluid to enter the bloodstream directly.

Intravenous

Into or within a vein; intravenous medicines, fluids, or blood products are delivered directly into the bloodstream.

Intravenous Anesthesia

Anesthetic induction or maintenance using drugs delivered into a vein. It permits rapid titration

but requires attention to dose, context-sensitive recovery, cardiovascular and respiratory effects, and the availability of a secure airway.

Intravenous Catheters

Flexible devices inserted into a vein to deliver fluids, medicines, blood products, nutrition, or contrast, or to obtain blood samples; complications include infection, thrombosis, infiltration, and extravasation.

Intravenous Immunoglobulin

A purified preparation of pooled antibodies given through a vein (IVIg) for selected immune deficiencies, autoimmune diseases, inflammatory disorders, and some neurologic conditions.

Intravenous Leiomyomatosis

Extension of benign uterine smooth-muscle tumor into venous channels, sometimes reaching the inferior vena cava or right heart. Imaging is important because extensive disease can cause serious cardiovascular complications.

Intravenous Lipid Emulsion Therapy

Administration of a lipid emulsion as a rescue treatment for selected severe lipophilic drug toxicities, especially local-anesthetic systemic toxicity. It may act as a lipid sink and through metabolic and cardiovascular effects and requires critical-care monitoring.

Intravenous Pyelogram

An x-ray examination of the urinary tract after intravenous iodinated contrast, historically used to visualize the kidneys, ureters, and bladder; CT urography or ultrasound often replaces it. of iodinated contrast material. The contrast is filtered by the glomeruli and excreted through the collecting system, allowing sequential imaging of the renal parenchyma, calyces, renal pelvis, ureters, and bladder.

Intravenous Pyelography

An X-ray examination of the urinary tract after intravenous contrast administration; it is now often replaced by CT urography or ultrasound.

Intravenous Sedation

Administration of sedative medicines through a vein to reduce anxiety, discomfort, or awareness during a procedure while maintaining appropriate monitoring and readiness to support breathing.

Intraventricular Hemorrhage

Bleeding into the germinal matrix and ventricular system, primarily in premature infants less than 32 weeks gestation. Graded I-IV by severity on cranial ultrasound. Grade I (germinal matrix only) and Grade II (intraventricular without dilation) have good prognosis.

Intraventricular Hemorrhage Of The Newborn

Intraventricular hemorrhage in a newborn is bleeding into the brain's ventricular system, most often in premature infants, and can cause hydrocephalus or neurologic injury.

Intravital Microscopy

Intravital microscopy uses a microscope to observe cells or tissues in a living organism, usually in research.

Intravitreal Injection

An ophthalmological procedure in which medication is injected directly into the vitreous cavity of the eye through the pars plana, a safe entry zone located approximately 3.5 to 4 millimeters posterior to the limbus, to deliver therapeutic agents directly to the retina and vitreous for the treatment of various retinal diseases.

Intrinsic

Belonging naturally to a structure or process and arising from within it, rather than being externally imposed.

Intrinsic Factor

A protein secreted by stomach parietal cells that is required for vitamin B12 absorption in the terminal ileum.

Intrinsic Renal AKI

Acute kidney injury caused by damage to renal tissue. Causes include acute tubular injury, acute interstitial nephritis, glomerulonephritis, and vascular disorders. Laboratory findings vary with the cause and should not be interpreted in isolation.

Intrinsic Sphincter Deficiency

Weakness of the urinary sphincter that causes stress urinary incontinence.

Alternate Names:

ISD

Introitus

An entrance or opening, especially the entrance to the vagina.

Intron

A non-coding segment of a gene that is transcribed into RNA but spliced out before the messenger RNA is translated into a protein.

Intrusive Thought

An unwanted, distressing thought, image, or impulse that enters awareness repeatedly; it does not necessarily reflect intent or desire.

Intubate

To place a tube into a body passage, especially an endotracheal tube through the mouth or nose into the trachea to maintain an airway or support ventilation.

Intubation

Placement of a tube into the airway or another body passage to maintain it or deliver treatment.

Intussusception

Condition in which a segment of intestine telescopes into the adjacent distal segment causing bowel obstruction and compromise of the mesenteric blood supply. It is the most common cause of intestinal obstruction in infants and children between six months and three years of age.

Intussusception – Children

Intussusception occurs when one segment of intestine telescopes into another, causing intermittent pain, vomiting, obstruction, and possible bowel ischemia in children.

Inulin

Inulin is a plant fructan used as a dietary fiber and in tests of glomerular filtration because it is freely filtered and minimally handled by the kidney.

Inuresis

A rare or obsolete spelling related to enuresis, meaning involuntary urination, especially during sleep.

Inutero

A variant or misspelling of in utero, meaning within the uterus before birth.

Invasive

Able to enter and spread through tissues; the term may describe an infection, tumor, procedure, or disease that penetrates nearby structures.

Invasive Candidiasis

A serious *Candida* infection that enters the bloodstream or deep organs, most often in people who are critically ill or immunocompromised. It may cause persistent fever or organ dysfunction and requires prompt medical diagnosis and treatment.

Invasive Ductal Carcinoma

The most common type of breast cancer, characterized by malignant cells that have broken through the duct wall and invaded surrounding breast tissue. It can metastasize via lymphatic or hematogenous spread.

Invasive Lobular Carcinoma

A breast cancer characterized by discohesive cells infiltrating in single-file lines. More frequently bilateral and multifocal than ductal carcinoma and harder to detect on imaging; usually hormone receptor positive.

Invasive Mole

A hydatidiform mole that invades the uterine myometrium or vascular spaces; it is a form of gestational trophoblastic neoplasia and may persist or metastasize.

Invasive Procedure

A medical procedure involving invasive procedure. Indications, risks, preparation, and follow-up depend on the patient and the specific procedure.

Inverse Agonist

A type of drug that binds to a cell receptor and actively produces the opposite biological effect of an agonist, effectively turning down the receptor's baseline activity.

Inversion

Chromosomal segment reversed end-to-end. Paracentric: does not include centromere. Pericentric: includes centromere. Usually benign unless breakpoint disrupts gene.

Invert

To turn upside down, reverse, or turn inward; in anatomy, inversion can describe movement of the sole of the foot toward the midline.

Invert Sugar

A chemical substance or compound called invert sugar. Its properties, uses, and risks depend on its composition and clinical context.

Invertase

An enzyme that breaks sucrose into glucose and fructose.

Invertebrata

Invertebrata is a broad nonformal grouping of animals without a vertebral column, including worms, mollusks, arthropods, and many other groups.

Invertebrate

An animal without a vertebral column, including insects, worms, mollusks, and many other groups.

Invertebrate Hormones

Chemical signals that regulate growth, development, reproduction, molting, or metabolism in invertebrate animals.

Inverted Papilloma

A usually benign but locally destructive tumor arising most often in the nasal cavity or sinuses; it can recur and rarely become cancerous.

Investigational

A chemical substance or compound called investigational. Its properties, uses, and risks depend on its composition and clinical context.

Inveterate

Long-standing or deeply established, often referring to a persistent habit, condition, or behavior.

Involucrum

An involucrum is a sheath of new bone that forms around dead bone, classically in chronic osteomyelitis.

Involuntary

Occurring without conscious control or deliberate choice, such as a reflex, autonomic function, or involuntary muscle movement.

Involuntary Movement

A movement that occurs without conscious control, such as a tremor, tic, chorea, dystonia, or myoclonus.

Involuntary Muscle

Muscle that contracts without conscious control, including smooth muscle in internal organs and cardiac muscle in the heart.

Involuntary Psychiatric Treatment

Assessment, hospitalization, or treatment without the patient's consent under applicable law when serious mental disorder creates a substantial risk of harm or severe inability to meet basic needs. The least restrictive alternative, due process, and

ongoing reassessment are required.

Involute

To shrink, return toward a former size, or undergo involution; in anatomy, an involute structure may be tightly coiled or rolled inward.

Involution

The process by which the uterus returns to its prepregnancy size after delivery. The uterus weighs approximately 1000 grams immediately postpartum and returns to approximately 60 grams by 6 weeks. Fundal height decreases approximately 1 cm per day.

Iodamoeba Butschlii

iodamoeba butschlii is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Iodide

The negatively charged form of iodine, used by the thyroid to make hormones and present in medicines, supplements, and contrast compounds.

Iodide-Associated Sialadenitis

Inflammation and swelling of the salivary glands caused by excessive iodide intake, typically from medications such as amiodarone or iodine-containing contrast, resolving with dose reduction.

Iodides

Salts containing iodine, sometimes used in carefully selected thyroid disorders.

Iodinated Contrast Media

Contrast agents containing iodine used primarily for radiography, CT, angiography, and some fluoroscopic procedures. Important risks include immediate hypersensitivity-like reactions, extravasation, thyroid effects, and kidney injury in susceptible patients.

Iodine

Required for thyroid hormones. Deficiency: goiter, hypothyroidism, cretinism. Prevention: iodized salt, seafood. Excess can cause thyroid dysfunction.

Iodine Deficiency

Inadequate iodine intake or availability, which can impair thyroid-hormone production and cause goiter, hypothyroidism, or developmental problems in a fetus or child.

Iodine Poisoning

Iodine poisoning follows excessive iodine exposure and can irritate the gastrointestinal tract, disrupt thyroid function, and cause systemic toxicity depending on dose and formulation.

Iodine-123

Pure gamma emitter, half-life 13 hours. Used for thyroid imaging. No beta emission, lower radiation dose than I-131.

Iodine-131

A radioactive iodine isotope used in selected diagnostic and therapeutic procedures, especially for thyroid disease.

Iodipamide

An iodinated contrast medium formerly used for imaging the biliary system and urinary tract; it can cause allergic or kidney-related reactions.

Iodoacetamide

A chemical reagent that irreversibly modifies sulfhydryl groups, especially cysteine residues, and is used in biochemical research.

Iododerma

A papulopustular or acneiform skin eruption caused by iodine excess, presenting on the face and extensor surfaces, resolving with iodine restriction and symptomatic treatment.

Iodoform

A chemical substance or compound called iodoform.

Its properties, uses, and risks depend on its composition and clinical context.

Iohexol

A nonionic, water-soluble iodinated contrast medium used for CT, angiography, and other imaging studies.

Ion

A chemical substance or compound called ion. Its properties, uses, and risks depend on its composition and clinical context.

Ion Transport

A chemical substance or compound called ion transport. Its properties, uses, and risks depend on its composition and clinical context.

Ionic Strength

A chemical substance or compound called ionic strength. Its properties, uses, and risks depend on its composition and clinical context.

Ionization

A chemical substance or compound called ionization. Its properties, uses, and risks depend on its composition and clinical context.

Ionizing Radiation

Radiation with enough energy to remove electrons from atoms and create ions, including X-rays, gamma rays, and certain particulate emissions. It can cause deterministic tissue effects at sufficient doses and stochastic cancer risk at lower doses.

Ionomycin

A medicine or chemical compound called ionomycin. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Ionophore

A chemical substance or compound called ionophore. Its properties, uses, and risks depend on its composition and clinical context.

Ions

A chemical substance or compound called ions. Its properties, uses, and risks depend on its composition and clinical context.

Iontophoresis

A chemical substance or compound called iontophoresis. Its properties, uses, and risks depend on its composition and clinical context.

Iopamidol

A nonionic iodinated contrast medium used in CT, angiography, and other radiologic examinations.

Iothalamate Clearance

Iothalamate clearance estimates glomerular filtration by measuring how quickly injected iothalamate is cleared from blood or urine.

Iothalamate Meglumine

Iothalamate meglumine is an iodinated contrast agent formerly used in radiographic imaging and associated with dose- and kidney-related risks.

Iothalamic Acid

A chemical substance or compound called iothalamic acid. Its properties, uses, and risks depend on its composition and clinical context.

Ioxaglic Acid

A chemical substance or compound called ioxaglic acid. Its properties, uses, and risks depend on its composition and clinical context.

IPC

Alternate name; see *Indwelling Pleural Catheter*.

Ipecac

Emetic no longer recommended for routine poisoning management. Risk of aspiration, delays charcoal administration.

Ipilimumab

A medicine or chemical compound called ipilimumab. Its uses, effects, interactions, and risks

depend on the specific substance and clinical indication.

IPMN Pancreas (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Ipratropium

Ipratropium is an inhaled short-acting muscarinic antagonist that reduces vagally mediated bronchoconstriction and nasal secretion; it is used in COPD and acute asthma with bronchodilators.

Ipratropium Bromide

An inhaled anticholinergic (antimuscarinic) bronchodilator that blocks acetylcholine at airway smooth muscle muscarinic receptors. It reduces bronchospasm and mucus secretion. It is used in COPD and acute asthma exacerbations. It is less effective than beta-2 agonists in asthma but more effective in COPD.

Iproniazid

Iproniazid is an older monoamine oxidase inhibitor antidepressant that is no longer commonly used because of serious liver toxicity and drug interactions.

Ipsilateral

Pertaining to, situated on, or affecting the same side of the body relative to a given point of reference or pathological lesion (the opposite of contralateral). In clinical neurology, unilateral cerebellar hemisphere lesions produce ipsilateral limb ataxia, dysmetria, and intention tremor, and a disruption of the cervical sympathetic chain causes ipsilateral Horner syndrome (ptosis, miosis, and anhidrosis on the same side as the lesion).

IQ Testing

IQ testing uses standardized tasks to estimate intellectual functioning; results should be interpreted with education, language, development, and clinical context.

Iratumumab

A medicine or chemical compound called iratu-

mumab. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Irbesartan

Irbesartan is an angiotensin-II receptor blocker used for hypertension and diabetic kidney disease; it can cause hyperkalaemia, hypotension, and renal-function changes and is contraindicated in pregnancy.

Iridium

Iridium is a dense metal whose radioactive isotopes, especially iridium-192, are used in industrial testing and some medical brachytherapy.

Iridocyclitis

Inflammation involving the iris and ciliary body, a form of anterior uveitis. It may cause eye pain, photophobia, redness, blurred vision, and cells or flare in the anterior chamber and may be associated with autoimmune disease, infection, trauma, or other causes.

Iridotomy

A procedure that creates a small opening in the iris to allow aqueous humor to pass between the posterior and anterior chambers. Laser peripheral iridotomy is used to treat or prevent selected forms of angle-closure glaucoma.

Irinotecan

A chemotherapy medicine that inhibits topoisomerase I and is used for colorectal and several other cancers.

Irinotecan Hydrochloride

The hydrochloride salt form of irinotecan, a topoisomerase I inhibitor used in cancer treatment.

Iris

The colored, muscular ring between the cornea and lens that controls the size of the pupil and therefore the amount of light entering the eye. Its muscles constrict the pupil in bright light and enlarge it in

dim light.

Iris Atrophy

Thinning or loss of iris tissue resulting from trauma, chronic inflammation, ischaemia, or neurodegenerative disease, leading to a translucent or patched iris appearance. It may be sectoral or diffuse and can be associated with decreased pupillary response, photophobia, or secondary glaucoma.

Iris Cyst

An iris cyst is a fluid-filled lesion on or within the iris, usually benign but sometimes affecting vision, pressure, or the appearance of the eye.

Iris Disease

A disorder of the colored part of the eye, including inflammation, injury, infection, developmental abnormalities, or tumors.

Iris Melanoma

Iris melanoma is a malignant tumor of pigment cells in the iris and may appear as a growing pigmented lesion, pupil distortion, or glaucoma.

Iris Neoplasm

An iris neoplasm is an abnormal growth arising in the colored part of the eye; it may be benign or malignant and can affect vision or eye pressure.

Iritis

Inflammation of the iris, the most common form of anterior uveitis. Presents with acute onset eye pain, photophobia, blurred vision, and redness (ciliary flush). Signs: cells and flare in the anterior chamber, keratic precipitates (KP), synechiae (adhesions between iris and lens).

Irofulven

Irofulven is an investigational anticancer compound derived from illudin that damages DNA and has been studied in solid tumors.

Iron

Essential for hemoglobin, myoglobin, and electron

transport. Heme iron from animal foods is generally more bioavailable than non-heme iron from plant foods. Deficiency is the most common nutritional deficiency worldwide and may result from blood loss, increased requirements, poor intake, or malabsorption.

Iron Deficiency

The most common nutritional deficiency worldwide, resulting from inadequate iron intake, impaired absorption, or chronic blood loss, leading to reduced hemoglobin synthesis and microcytic hypochromic anemia. Clinical features include fatigue, pallor, koilonychia, pica, and angular cheilitis.

Iron Deficiency Anemia

Anemia caused by insufficient iron for hemoglobin production. It is often microcytic and hypochromic; treatment requires identifying the cause and replacing iron when appropriate.

Iron Overdose

Iron overdose can cause gastrointestinal injury, metabolic acidosis, shock, liver failure, and multiorgan toxicity, particularly after a large ingestion by a child.

Iron Overload

Excess iron stored in the body, which can damage the liver, heart, pancreas, joints, or endocrine organs.

Alternate Names:

Overload Iron (Iron Overload)

Iron-Deficiency Anaemia

Anaemia caused by insufficient iron for haemoglobin production, commonly due to blood loss, inadequate intake, or impaired absorption.

Iron-Deficiency Anaemia In Adolescents

Iron-deficiency anaemia during adolescence, commonly related to rapid growth, inadequate dietary iron, or menstrual blood loss. Assessment should identify the cause and guide dietary advice and clinical treatment.

Iron-Deficiency Anemia

Anemia caused by insufficient iron for normal hemoglobin production. It may cause fatigue, weakness, shortness of breath, paleness, headache, or unusual cravings such as ice; the cause of iron loss or inadequate intake should be identified.

Irradiated Foods

Irradiated foods are exposed to controlled ionizing radiation to reduce pathogens, insects, or spoilage; the process does not make food radioactive.

Irradiation

Exposure to or treatment with radiation; irradiation may be used diagnostically or therapeutically, or may cause tissue injury depending on dose and type.

Irregular

Not following a regular pattern, rhythm, shape, timing, or expected sequence.

Irregular Menstruation

A chemical substance or compound called irregular menstruation. Its properties, uses, and risks depend on its composition and clinical context.

Irregular Respiration

A chemical substance or compound called irregular respiration. Its properties, uses, and risks depend on its composition and clinical context.

Irregular Sleep-Wake Syndrome

A circadian rhythm sleep-wake disorder with multiple irregular periods of sleep and wakefulness across the 24-hour day rather than one consolidated nighttime sleep period. It is associated with neurological disease, dementia, aging, and reduced exposure to environmental time cues.

Irreversible Cell Injury

Cell damage beyond the point of recovery, resulting in death by necrosis, apoptosis, or another regulated cell-death pathway. Key features include inability to reverse mitochondrial dysfunction and severe membrane or nuclear damage.

Irrigate

To wash or flush a body part, wound, cavity, or medical device with a fluid to remove debris, contaminants, or secretions.

Irrigation

Flushing a wound, body cavity, or organ with fluid to remove blood, debris, microorganisms, or unwanted material.

Irrigation Solution

A chemical substance or compound called irrigation solution. Its properties, uses, and risks depend on its composition and clinical context.

Irritable Bowel Syndrome

A disorder of gut–brain interaction that causes repeated abdominal pain and changes in bowel habits, such as diarrhea, constipation, or both. No visible structural damage is usually found in the digestive tract.

Alternate Names:

IBS

Irritable Bowel Syndrome

Alternate index label for IBS.

Irritable Bowel Syndrome

Alternate index label for Irritable Bowel Syndrome.

Irritable Bowel Syndrome Diet

An individualized eating plan used to reduce irritable bowel syndrome symptoms. A limited trial of a low-FODMAP diet may help some people and should be followed by structured food reintroduction.

Irritable Mood

A mood marked by increased sensitivity, impatience, anger, or a tendency to become upset easily.

Irritant Contact Dermatitis

Inflammation of the skin caused by direct chemical or physical injury rather than an immune-mediated allergy. It commonly produces redness, burning,

scaling, fissures, or pain at the area of exposure.

Irritants

Substances that provoke inflammation or allergy-like symptoms without necessarily causing an immune allergy.

Irritation

A chemical substance or compound called irritation. Its properties, uses, and risks depend on its composition and clinical context.

Isaacs Syndrome

A rare peripheral nerve hyperexcitability disorder causing continuous muscle-fibre activity, stiffness, cramps, fasciculations, myokymia, and delayed muscle relaxation. It may be autoimmune, paraneoplastic, or associated with inherited or acquired peripheral-nerve disorders.

Isavuconazole

A triazole antifungal agent used in the treatment of invasive aspergillosis and mucormycosis. Isavuconazole inhibits fungal lanosterol 14- α demethylase, disrupting ergosterol synthesis and fungal cell membrane integrity.

Isazophos

Isazophos is an organophosphate pesticide that can inhibit acetylcholinesterase and cause cholinergic toxicity after exposure.

Ischaemia

Inadequate blood flow and oxygen delivery to a tissue, usually because of narrowed or blocked blood vessels, causing dysfunction or injury.

Ischaemic Heart Disease

Heart disease caused by reduced blood supply to the heart muscle, usually as a result of coronary artery disease; it may cause angina or myocardial infarction.

Ischemia

Reduced blood flow causing oxygen deprivation.

Can lead to tissue damage or death.

Ischemia, Intestinal

Alternate index label; see *Intestinal ischemia*.

Ischemia-Reperfusion Injury

The paradoxical tissue damage that occurs when blood flow is restored to previously ischemic tissue, a phenomenon with major clinical significance in myocardial infarction treatment, stroke management, organ transplantation, and shock resuscitation.

Ischemic Bone Necrosis

Alternate index label; see *Avascular necrosis (osteonecrosis)*.

Ischemic Cardiomyopathy

Weakening of the heart muscle caused by coronary artery disease and previous or ongoing myocardial ischemia. Scarred or poorly perfused myocardium can reduce contractile function and increase the risk of heart failure and ventricular arrhythmias.

Ischemic Chest Pain (Angina Symptoms)

Alternate index label for Angina Symptoms.

Ischemic Colitis

Inflammation and injury of the colon caused by reduced blood flow. It often causes sudden abdominal pain, bloody diarrhea, or rectal bleeding and commonly affects areas such as the splenic flexure.

Alternate Names:

Colitis, Ischemic

Ischemic Fasciitis

A benign fibroblastic and myofibroblastic proliferation associated with pressure-related ischemic injury, usually arising in deep soft tissue over bony prominences in debilitated or immobile patients.

Ischemic Gangrene

Death of tissue due to prolonged severe arterial insufficiency, most commonly in the extremities. Presents as dry, black, mummified tissue with a clear line of demarcation from viable tissue. Re-

quires revascularization when possible and surgical debridement or amputation of necrotic tissue.

Ischemic Heart Disease

Heart disease caused by inadequate blood flow through narrowed or blocked coronary arteries.

Alternate Names:

Coronary artery disease; coronary heart disease

Ischemic Preconditioning

A chemical substance or compound called ischemic preconditioning. Its properties, uses, and risks depend on its composition and clinical context.

Ischemic Priapism

Alternate index label; see *Priapism*.

Ischemic Stroke

A neurologic deficit caused by inadequate blood flow to part of the brain, usually from thrombotic or embolic arterial occlusion. Causes include atherosclerotic disease, cardiac embolism, and small-vessel disease.

Ischium

Inferior-posterior hip bone. Ischial tuberosity bears seated weight and serves as hamstring attachment. Ischial spine is a pudendal nerve block landmark and determines pelvic outlet level. Lesser sciatic notch lies between spine and tuberosity.

Ishak Score

ishak score is a laboratory test, imaging method, instrument, score, or measurement used to characterize a specimen or clinical finding; interpretation depends on the method and clinical context.

Islet Cell

islet cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Islet Cell Cancer

Alternate index label; see *Pancreatic neuroendocrine*

tumors.

Islets Of Langerhans

Clusters of endocrine cells in the pancreas that produce hormones including insulin and glucagon.

Iso-Sulfan Blue

A blue dye used to map lymphatic drainage or identify sentinel lymph nodes during selected surgical procedures.

Isoaminile

Isoaminile is an older sympathomimetic appetite suppressant that is no longer widely used because of cardiovascular and other adverse effects.

Isoantibodies

Antibodies made against antigens from another individual of the same species, such as antibodies involved in blood-group incompatibility.

Isoantigen

An antigen that differs between individuals of the same species and can trigger an immune response after exposure.

Isocarboxazid

Isocarboxazid is a monoamine oxidase inhibitor used for depression; it has important interactions with other medicines and tyramine-rich foods.

Isochromosomes

Abnormal chromosomes with two identical arms caused by division across the wrong axis of the centromere.

Isocitrate Dehydrogenase

A regulated enzyme of the citric acid cycle that converts isocitrate to alpha-ketoglutarate while producing NADH and releasing carbon dioxide. Mitochondrial isoforms are stimulated by ADP and calcium and inhibited by ATP and NADH. Mutations in IDH1 or IDH2 can produce the oncometabolite 2-hydroxyglutarate.

Isocyanate

A reactive chemical group found in some industrial products; exposure can irritate the airways and cause occupational asthma or skin reactions.

Isoelectric Point

The pH at which a molecule, especially a protein, has no net electrical charge.

Isoenzyme

Isoenzyme is an immunologic, genetic, infectious, developmental, or cellular concept used to explain disease mechanisms, diagnosis, or treatment.

Isoetharine

A bronchodilator related to beta-adrenergic agonists that was formerly used to relieve airway narrowing.

Isofenphos

An organophosphate insecticide that can cause cholinergic poisoning by inhibiting acetylcholinesterase.

Isoflavone

A plant compound, especially from soy, with weak estrogen-like activity that is studied in relation to bone, menopause, and cancer risk.

Isoflurane

Isoflurane is an inhaled anesthetic used to maintain general anesthesia; it can lower blood pressure and depress breathing.

Isoflurophate

Isoflurophate is a long-acting organophosphate cholinesterase inhibitor formerly used in eye treatments; it can cause prolonged cholinergic effects.

Isogeneic Graft

A transplant between genetically identical individuals, such as identical twins, so immune rejection risk is relatively low.

Isokinetic Exercise

Muscle contraction at constant speed throughout

range of motion. Uses specialized equipment. Assesses and improves strength.

Isolate

To separate a person, organism, tissue, or substance from others; infection-control isolation separates people who may transmit a pathogen.

Isolate Compound

A chemical substance or compound called isolate compound. Its properties, uses, and risks depend on its composition and clinical context.

Isolated Systolic Hypertension

Elevated systolic blood pressure with normal diastolic pressure, commonly related to arterial stiffness in older adults and associated with increased cardiovascular risk.

Isolation

Separation of sick persons with contagious disease. Examples: tuberculosis isolation, airborne infection isolation rooms.

Isolation (Social)

Lack of social contacts and interactions. Common in elderly: bereavement, mobility limitation, sensory impairment, retirement. Consequences: depression, cognitive decline, functional impairment, mortality. Interventions: social programs, transportation, technology, volunteer visits.

Alternate Names:

Social

Isolation Precautions

Measures used with standard precautions to reduce transmission of infection. Additional precautions may be contact, droplet, or airborne, depending on the mode of spread.

Isoleucine

An essential branched-chain amino acid needed for protein synthesis, energy metabolism, and tissue repair.

Isomerase

A chemical substance or compound called isomerase. Its properties, uses, and risks depend on its composition and clinical context.

Isometheptene

Isometheptene is a vasoconstrictor formerly used in some headache medicines; it may raise blood pressure and is unsuitable for some cardiovascular conditions.

Isometric

Describing muscle contraction that produces force without a meaningful change in muscle length or joint movement.

Isometropia

The condition in which both eyes have approximately equal refractive power, so neither eye has substantially more near-sightedness, far-sightedness, or astigmatism than the other.

Isoniazid

Isoniazid is a first-line antituberculous drug that inhibits mycolic-acid synthesis in *Mycobacterium tuberculosis*; hepatotoxicity and peripheral neuropathy are important risks, and pyridoxine is often given.

Isoniazid Toxicity

Hepatotoxicity, peripheral neuropathy, seizures. Treatment: pyridoxine (vitamin B6) supplementation, activated charcoal, seizure control.

Isoprenaline

Isoprenaline, also called isoproterenol, is a beta-adrenergic medicine that can increase heart rate and is used in selected emergency cardiac situations.

Isoprenylation

A chemical substance or compound called isoprenylation. Its properties, uses, and risks depend on its composition and clinical context.

Isoprocarb

Isoprocarb is a carbamate insecticide that can cause cholinergic poisoning if swallowed or absorbed in a significant amount.

Isopropamide

Isopropamide is an anticholinergic medicine formerly used for gastrointestinal spasms; it may cause dry mouth, blurred vision, constipation, or urinary retention.

Isopropanol Alcohol Poisoning

Poisoning from isopropanol, often found in rubbing alcohol, causing intoxication, vomiting, abdominal pain, low blood pressure, and sometimes coma.

Isopropyl Alcohol

A chemical substance or compound called isopropyl alcohol. Its properties, uses, and risks depend on its composition and clinical context.

Isoprostane

A compound formed when free radicals oxidize membrane lipids; it is used as a marker of oxidative stress.

Isoproterenol

Isoproterenol, also called isoprenaline, is a beta-adrenergic medicine that increases heart rate and contractility and is used in selected emergency settings.

Isoquinolines

A group of nitrogen-containing ring compounds that includes chemicals studied in pharmacology, natural products, and organic chemistry.

Isosorbide

Isosorbide is a nitrate medicine that relaxes blood vessels and is used mainly to prevent or relieve angina; headache and low blood pressure are common adverse effects.

Isosorbide Dinitrate

An organic nitrate (and prodrug) that releases

nitric oxide, relaxing vascular smooth muscle—predominantly venous (preload reduction) with coronary dilation.

Isospora

isospora is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Isospora Belli

isospora belli is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Isosporiasis

Intestinal infection caused by the protozoan *Cystoisospora belli*. It typically causes watery diarrhea, abdominal cramps, nausea, fever, and weight loss and may be prolonged or severe in people with impaired cellular immunity.

Isothiocyanate

A reactive compound formed from glucosinolates in some plants, especially cruciferous vegetables, and studied for antimicrobial and cancer-related effects.

Isothiocyanates

A group of reactive plant compounds formed from glucosinolates, especially in cruciferous vegetables.

Isotonic

Describing muscle contraction in which muscle length changes to move a joint under a relatively constant external load.

Isotonic Exercise

Exercise in which muscles change length while producing force against a relatively constant load, as in lifting or lowering a weight.

Isotonic Solution

A chemical substance or compound called isotonic solution. Its properties, uses, and risks depend on

its composition and clinical context.

Isotope

One of two or more atoms of the same element with the same number of protons but different numbers of neutrons.

Isotretinoin

A systemic retinoid used for severe, recalcitrant nodular acne. It reduces sebum production, normalizes follicular keratinization, decreases *Propionibacterium acnes*, and has anti-inflammatory effects. Dose is 0.5 to 1 mg/kg/day for 15 to 20 weeks, targeting cumulative dose of 120 to 150 mg/kg.

Isovaleric Acidemia

An autosomal recessive inborn error of leucine catabolism caused by a deficiency of the mitochondrial enzyme isovaleryl-CoA dehydrogenase (IVD). Results in the pathological accumulation of isovaleryl-CoA and its derivatives, notably isovaleric acid, isovalerylglycine, and isovalerylcarnitine. Characterized by intermittent acute episodes of severe metabolic ketoacidosis, vomiting, lethargy, encephalopathy, cerebral edema, and a distinctive, pungent odor resembling “sweaty feet” or stale cheese, accompanied by hyperammonemia, pancytopenia, and hypocalcemia during acute decompensation.

Alternate Names:

IVA; Isovaleryl-CoA Dehydrogenase Deficiency

Isoxsuprine

Isoxsuprine is a vasodilating medicine formerly used for selected circulation disorders; it may cause dizziness, fast heartbeat, or low blood pressure.

Isozyme

One of several molecular forms of an enzyme that catalyze the same reaction but differ in structure, tissue distribution, or regulation.

Ispaghula Husk

Ispaghula husk, or psyllium, is a soluble fiber that absorbs water and helps regulate stool consistency; adequate fluid is important when using it.

Ispinesib

Ispinesib is an investigational kinesin-spindle protein inhibitor studied for anticancer effects by disrupting chromosome separation during cell division.

Isradipine

Isradipine is a calcium-channel blocker used to lower blood pressure; it may cause flushing, headache, ankle swelling, or dizziness.

ISUP Grade Group

The standardized five-tier pathological grading system established by the International Society of Urological Pathology and endorsed by the World Health Organization to stage prostate adenocarcinoma, addressing the clinical ambiguity of historical Gleason scoring. The system stratifies cancer prognosis into Grade Group 1 (Gleason score ≤ 6 , indolent with excellent prognosis), Grade Group 2 (Gleason score $3+4 = 7$, predominantly favorable prognosis), Grade Group 3 (Gleason score $4+3 = 7$, unfavorable prognosis with predominant pattern 4), Grade Group 4 (Gleason score 8, high risk), and Grade Group 5 (Gleason score 9–10, highest risk of systemic progression and mortality). It provides patients and clinicians with clearer risk stratification for active surveillance versus definitive local or systemic therapy.

Alternate Names:

Grade Group; ISUP Prostate Grading System; WHO/ISUP Grade Group

Itch

Itch, or pruritus, is an unpleasant skin sensation that provokes scratching and may arise from dermatitis, allergy, infection, medicines, kidney or liver disease, blood disorders, nerve disease, or systemic illness. Persistent generalized itch requires evaluation.

Alternate Names:

Pruritus (Itch)

Itching

Itching, or pruritus, is an unpleasant sensation that provokes scratching and may arise from skin disease, allergy, systemic illness, medications, neuropathy, or psychogenic factors. be acute or chronic and may

arise from skin disease, systemic illness, medication, or neuropathic mechanisms.

Itching Anal (Anal Itching)

Alternate index label for Anal Itching.

Itching, Anal

Itching around the anus, commonly caused by irritant dermatitis, hemorrhoids, pinworms, infection, inflammatory skin disease, or hygiene-related irritation. Persistent, severe, bleeding, painful, or unexplained symptoms require examination.

Itchy Ear

An itching sensation of the external ear or ear canal, which may result from dermatitis, allergy, infection, dryness, or irritation from objects placed in the ear.

Iterative Reconstruction

Modern reconstruction method. OSEM (Ordered Subsets Expectation Maximization). Better quality, longer computation.

ITP

Alternate index label; see *Immune thrombocytopenia*.

Itraconazole

Itraconazole is a triazole antifungal that inhibits fungal ergosterol synthesis and treats several superficial and systemic mycoses; absorption, extensive drug interactions, liver injury, and negative inotropic effects require attention.

IUCD

An intrauterine contraceptive device, usually referring to an intrauterine device placed in the uterus to prevent pregnancy.

IV Fluids

Intravenous fluid therapy for volume resuscitation and maintenance. Crystalloids: normal saline (0.9% NaCl, increases chloride, risk of hyperchloremic metabolic acidosis), lactated Ringer (balanced, contains K⁺, Ca²⁺, lactate buffer, safer in large vol-

umes), Plasma-Lyte (balanced, closest to plasma).

IV Treatment At Home

Intravenous treatment given outside a hospital, usually through a catheter with instructions for safe infusion, line care, monitoring, and follow-up.

Ivabradine

Ivabradine slows the heart rate by acting on the sinus node and is used for selected patients with chronic heart failure or stable angina.

IVC Filter

Device placed in inferior vena cava to prevent pulmonary embolism. Indications: DVT/PE with contraindication to anticoagulation, recurrent PE despite anticoagulation. Complications: filter thrombosis, migration, perforation, IVC occlusion.

Ivermectin

Ivermectin is an antiparasitic medicine used for infections such as strongyloidiasis, onchocerciasis, and scabies.

IVF

In vitro fertilization: the assisted reproductive technology in which oocytes are retrieved, fertilized by sperm outside the body (in vitro), and the resulting embryos transferred to the uterus. Indications: tubal factor, male factor (with ICSI), endometriosis, unexplained, ovulation disorders, genetic testing, and same-sex couples or individuals using donor sperm or eggs.

Ixabepilone

Ixabepilone is a chemotherapy medicine used for some advanced breast cancers; important risks include nerve damage, low blood counts, and allergic reactions.

Ixodes

Ixodes is an immunologic, genetic, infectious, developmental, or cellular concept used to explain disease mechanisms, diagnosis, or treatment.

Internal Carotid Artery

A branch of the common carotid artery that enters the skull and supplies the brain, eyes, and related structures. It contributes to the cerebral arterial circulation.

J

Jaccoud Arthropathy

A deforming but usually nonerosive arthropathy, classically associated with previous rheumatic fever or systemic lupus erythematosus, causing reducible ulnar deviation and other hand deformities.

Jaccoud'S Arthropathy

A deforming but usually non-damaging form of arthritis that can cause the fingers to drift toward the little-finger side. The deformity may improve when the hand is moved and is often linked to lupus or past rheumatic fever.

Jack-In-The-Pulpit Poisoning

Jack-in-the-pulpit poisoning follows ingestion of calcium oxalate crystals from the plant, causing intense burning and swelling of the mouth, throat, and gastrointestinal tract.

Jackknife Position

A prone position in which the operating table is flexed at the hips, raising the buttocks and lowering the head and legs.

Jacksonian Epilepsy

Jacksonian epilepsy is a focal motor seizure that spreads in an orderly pattern across adjacent body regions as abnormal activity propagates through the motor cortex.

Jacksonian Epilepsy (Focal Seizures)

A focal seizure that begins with twitching or stiffening in one body part and spreads in an orderly way to nearby body parts. Awareness may remain normal at first.

Alternate Names:

Focal Seizures

Jacksonian Seizure

A focal motor seizure that begins in one body part and may spread in an orderly pattern to adjacent body regions; awareness may remain intact at first.

Jadassohn'S Naevus

Jadassohn's naevus is a congenital hairless skin lesion, usually a sebaceous naevus, that may change during life and rarely develop secondary tumors.

JAK-STAT Pathway

A system cells use to respond to signals from hormones and immune chemicals. It helps control growth, blood-cell production, inflammation, and immune activity.

JAK2

A protein inside cells that passes signals from certain hormones and immune chemicals. Changes that keep JAK2 active can cause some blood cancers.

Jaktitation

Jaktitation is restless, repetitive, or jerky movement, especially during severe illness or delirium; it is a descriptive historical term rather than a specific diagnosis.

Jammed Finger

A finger injury caused by sudden force, often involving a sprain, joint dislocation, fracture, or tendon injury; persistent deformity, severe pain, or inability to move requires assessment.

Janeway Lesions

Small, usually painless red or purple spots on the palms or soles that can occur with infective endocarditis, an infection of the heart lining or valves. They result from infected particles blocking tiny blood vessels.

Janus Kinase

One of a family of proteins inside cells that pass signals from hormones and immune chemicals. Janus kinases help control inflammation, blood-cell production, and immune responses.

Janus Kinase Inhibitors

Medicines that block Janus kinase proteins and reduce certain immune signals. They are used for selected inflammatory and blood disorders, but can increase the risk of infections, blood clots, and other serious effects.

Japanese Encephalitis

A viral infection spread by mosquitoes that can inflame the brain. Most infections cause no symptoms, but severe illness can cause fever, confusion, seizures, movement problems, or lasting disability.

Jarisch-Herxheimer Reaction

A short-lived reaction that may occur soon after antibiotics are started for certain infections. It can cause fever, chills, headache, muscle aches, and temporarily worse symptoms.

Jaundice

Yellowing of the skin or eyes caused by too much bilirubin in the blood. It can result from breakdown of red blood cells, liver disease, or blockage of bile flow.

Jaundice (Icterus)

Yellow discoloration of the skin, sclerae, or mucous membranes caused by an increased bilirubin concentration in the blood.

Alternate Names:

Icterus

Jaundice And Breastfeeding

Jaundice associated with breastfeeding may reflect inadequate milk intake in the early days or persistent breast-milk jaundice after other serious causes have been excluded.

Jaundice Causes

Jaundice causes include hemolysis, inherited bilirubin disorders, hepatitis, medication or toxin injury, and obstruction of bile drainage; evaluation distinguishes unconjugated from conjugated hyperbilirubinemia.

Jaundice In Adults

Adult jaundice is yellowing of skin or eyes from elevated bilirubin due to red-cell breakdown, liver injury, impaired bile flow, or inherited metabolism. Dark urine, pale stool, abdominal pain, fever, confusion, or rapid onset requires prompt assessment.

Jaundice In Neonates

Yellowing of a newborn's skin or eyes from elevated bilirubin. It is often physiologic, but jaundice in the first 24 hours, rapidly increasing jaundice, poor feeding, lethargy, fever, pale stools, or dark urine requires urgent evaluation.

Jaundice In Newborns

Yellow discoloration of a newborn's skin or eyes caused by elevated bilirubin.

Jaw

The upper and lower bony structures of the mouth—the maxilla and mandible—which support the teeth and contribute to chewing, speech, facial shape, and the oral airway.

Jaw Fracture

A traumatic structural disruption or fracture involving either the mandible (lower jaw) or the maxilla (upper jaw). Mandibular fractures are far more common, frequently occurring at multiple sites simultaneously (e.g., parasymphysis and contralateral condylar neck) owing to the rigid ring-like architecture of the mandible; presents with post-traumatic malocclusion, step-off deformities, sublin-

gual hematoma (Coleman sign), and mental nerve paresthesia. Maxillary fractures typically follow severe blunt facial impact and are classified according to the Le Fort system (Types I, II, and III). Critical management priorities include immediate airway stabilization (especially with posterior tongue displacement in bilateral symphyseal fractures), reduction, and maxillomandibular fixation or open reduction internal fixation.

Jaw Jerk Reflex

A brief closing movement of the jaw when the chin is tapped with the mouth slightly open. An unusually strong response may suggest a problem in the brain's motor pathways.

Jaw Neoplasm

A neoplasm arising in the jawbone or adjacent oral and facial tissues. Diagnosis establishes the tumour type, extent, biological behaviour, and appropriate management.

Jaw Pain

Pain in the jaw, jaw joint, or chewing muscles caused by temporomandibular disorders, dental disease, infection, injury, or less commonly heart or nerve disease. Pain with chest discomfort, sweating, breathlessness, or exertion requires urgent assessment; persistent pain or restricted jaw movement requires dental or medical review.

Jaw Tumors And Cysts

Abnormal growths or fluid-filled sacs that develop in the jawbone or nearby tissues.

Alternate Names:

Odontogenic Tumors And Cysts

Jeb Protocol

An experimental oncology treatment protocol; its exact medicines, schedule, indication, and safety profile depend on the study or institution using the name.

Jehovah'S Witness

A member of a Christian religious group whose beliefs may lead them to decline transfusions of whole

blood and certain blood components. Individual choices about blood products, procedures, and treatment vary, so clinicians should discuss the patient's informed preferences and document them carefully.

Jejunal Atresia

A condition present at birth in which part of the jejunum, a section of the small intestine, is closed or missing. It causes bowel blockage in a newborn, with green vomiting, a swollen abdomen, and failure to pass the first stool.

Jejunal Fistula

An abnormal connection involving the jejunum and another organ, skin, or body surface. It may cause intestinal contents to leak, infection, malnutrition, dehydration, or sepsis and requires assessment of its cause and output.

Jejunal Hemorrhage

Bleeding from the jejunum, the middle segment of the small intestine, which may cause maroon or black stool, anemia, abdominal pain, or shock depending on its rate and cause.

Jejunal Neoplasm

A neoplasm arising in the jejunum, the middle part of the small intestine. Diagnosis establishes the tumour type, extent, biological behaviour, and appropriate management.

Jejunal Stenosis

Narrowing of the jejunal lumen that can impede intestinal contents and cause cramping, vomiting, distension, malabsorption, or obstruction; causes include Crohn disease, adhesions, tumors, and ischemic injury.

Jejunal Ulcer

An ulcerated break in the lining of the jejunum, caused by inflammation, infection, ischemia, medicines, tumor, or anastomotic disease and potentially producing pain, bleeding, or perforation.

Jejunitis

Jejunitis is inflammation of the jejunum, which may result from infection, inflammatory bowel disease, ischemia, medication, or other intestinal disorders.

Jejunioleal Atresia

Congenital complete intrinsic occlusion of the lumen of the jejunum or ileum, resulting from an intrauterine mesenteric vascular disruption (ischemic accident, volvulus, intussusception, or internal herniation) during late gestation. Neonates present within the first 24 to 48 hours of life with bilious vomiting, progressive abdominal distension, and failure to pass meconium. Abdominal radiography classically demonstrates multiple dilated bowel loops with air-fluid levels and an absence of distal gas; definitive management is surgical bowel resection with primary enterostomy or end-to-end anastomosis.

Alternate Names:

Jejunal Atresia; Ileal Atresia; Small Bowel Atresia

Jejunostomy

A surgically created opening into the jejunum, often used for placement of a feeding tube when nutrition cannot safely be given through the stomach.

Jejunostomy Feeding Tube

A jejunostomy feeding tube delivers nutrition directly into the jejunum when oral or gastric feeding is unsafe or inadequate, with attention to formula, rate, position, and complications.

Jejunum

The middle part of the small intestine. It absorbs much of the nutrients from food before the remaining contents pass into the ileum.

Jellyfish Stings

Skin injury caused by contact with jellyfish tentacles, which release venom and may cause pain or systemic symptoms.

Alternate Names:

Stings, Jellyfish

Jendrassik'S Maneuver

Jendrassik's maneuver reinforces deep-tendon reflexes by having the patient interlock and pull the hands or tense another muscle group.

Jeotgalibacillus

Jeotgalibacillus is a genus of Gram-positive, usually halotolerant bacteria reported mainly from salt-rich or fermented environments; isolation from a clinical specimen must be distinguished from contamination or colonization.

Jerusalem Cherry Poisoning

Jerusalem cherry poisoning follows ingestion of *Solanum pseudocapsicum* berries, which contain toxic compounds that may cause gastrointestinal symptoms, confusion, weakness, or cardiac effects.

Jervell And Lange-Nielsen Syndrome

Jervell and Lange-Nielsen syndrome is an inherited disorder combining congenital profound sensorineural deafness with prolonged QT and risk of life-threatening ventricular arrhythmia.

Jet Lag

Temporary circadian rhythm disruption after rapid travel across time zones, causing sleep disturbance, fatigue, impaired concentration, and sometimes gastrointestinal symptoms.

Alternate Names:

Desynchronization (Jet Lag)

Jet Lag Prevention

Jet lag prevention uses timed light exposure, sleep adjustment, hydration, and carefully planned activity or melatonin strategies to shift the circadian rhythm across time zones.

Jewelry Cleaner Poisoning

Jewelry cleaner poisoning can injure the mouth, esophagus, stomach, lungs, or eyes depending on the product and exposure route; immediate poison-control guidance is appropriate.

Jimsonweed Poisoning

Jimsonweed poisoning causes anticholinergic toxicity, including dilated pupils, dry mouth, rapid heart rate, urinary retention, agitation, delirium, seizures, or coma.

Job Syndrome

A primary immunodeficiency, usually caused by pathogenic variants affecting STAT3, characterized by eczema, recurrent staphylococcal skin and lung infections, and markedly elevated immunoglobulin E. Skeletal, dental, and characteristic facial findings may also occur.

Jock Itch

A fungal infection of the groin and inner thighs, also called tinea cruris. It causes an itchy, scaly, ring-shaped or spreading rash, often worsened by warmth and moisture. Pain, fever, extensive involvement, or failure to improve with appropriate care requires clinical evaluation.

Alternate Names:

tinea cruris. Tinea Cruris

Jod-Basedow Phenomenon

The Jod-Basedow phenomenon is iodine-induced hyperthyroidism, usually in people with autonomous thyroid tissue exposed to excess iodine.

Johnny

A colloquial term for a hospital gown, usually a short gown that ties in the back; in some regional slang it can also mean a condom.

Johnsonella

Johnsonella is a genus of anaerobic bacteria found mainly in the mouth; recovery from a clinical specimen should be interpreted with the site and evidence of infection.

Johnsonella Ignava

An anaerobic bacterium associated mainly with the human mouth and occasionally isolated from mixed infections.

Joint

The place where two or more bones meet. Joints may be fixed, slightly movable, or freely movable, such as the knee, hip, or shoulder.

Joint Capsule

A joint capsule is the fibrous envelope surrounding a freely movable joint; its inner lining produces synovial fluid that lubricates the joint.

Joint Contracture

Persistent limitation of passive joint movement caused by shortening or fibrosis of muscle, tendon, capsule, ligament, fascia, skin, or other periarticular tissue. It may follow immobilization, neurological disease, trauma, inflammation, burns, or congenital disorders.

Joint Diseases

A broad category of pathological conditions affecting the structure and function of joints. Classification by etiology: (1) Degenerative: osteoarthritis (most common, non-inflammatory, affects load-bearing joints—knee, hip, spine).

Joint Effusion

Excess fluid within a joint space. Common sites: knee (most accessible), ankle, wrist, shoulder. Causes: trauma, infection, inflammatory arthritis, crystal arthropathy. Detected by ballottement or bulge sign.

Joint Fluid Culture

Joint fluid culture tests synovial fluid for bacterial, fungal, or other organisms when septic arthritis is suspected; results are interpreted with cell count, crystals, and clinical findings.

Joint Fluid Gram Stain

Joint fluid Gram stain applies differential staining to synovial fluid to look for bacteria, helping evaluate septic arthritis alongside culture and fluid analysis.

Joint Inflammation (Arthritis)

Alternate index label for Arthritis.

Joint Mice

Joint mice are loose bodies within a joint, consisting of cartilage, bone, or other tissue that can cause catching, locking, pain, or swelling.

Joint Pain

Discomfort arising from articular structures including cartilage, synovium, ligaments, and periarticular soft tissues due to inflammation, degeneration, injury, or infection.

Joint Pain, Monoarticular

Pain affecting one joint, caused by trauma, infection, crystal arthritis, osteoarthritis, or inflammatory arthritis. Acute monoarticular pain with swelling, warmth, fever, or inability to bear weight requires urgent assessment for septic arthritis.

Joint Pain, Polyarticular

Pain affecting multiple joints, caused by systemic inflammatory disease, infection, osteoarthritis, crystal disease, medication, or other conditions. The pattern, duration, swelling, and associated symptoms guide evaluation.

Joint Prosthesis

An artificial joint or implant designed to replace a damaged or diseased biological joint, restoring mobility and relieving chronic pain.

Joint Replacement

Surgical replacement of damaged joint surfaces with prosthetic components to reduce pain and improve function.

Joint Replacement (Arthroplasty)

A surgical procedure in which a damaged or arthritic joint surface is removed and replaced with a prosthetic implant, most commonly performed for the hip (total hip arthroplasty) and knee (total knee arthroplasty).

Alternate Names:

Arthroplasty

Joint Replacement Rehabilitation

Exercise, education, and functional training after joint replacement to restore safe movement and independence.

Joint Stiffness

Reduced ease or range of joint movement, often worse after rest. Causes include arthritis, injury, inflammation, prolonged immobility, or muscle and tendon disorders. Sudden swelling, redness, fever, severe pain, or inability to use the joint requires assessment.

Joint Swelling

Enlargement of a joint caused by excess synovial fluid, blood, inflammation, infection, trauma, or structural disease. It may be accompanied by pain, warmth, stiffness, or restricted movement and can indicate inflammatory or septic arthritis.

Joint X-Ray

A joint x-ray uses ionizing radiation to assess alignment, fractures, degenerative change, erosions, calcification, and selected soft-tissue or bone abnormalities.

Jondice

Jondice is a misspelling or variant form of jaundice, the yellow discoloration caused by elevated bilirubin in blood and tissues.

Joubert Syndrome

An autosomal recessive and X-linked primary developmental ciliopathy caused by pathogenic variants in over 35 genes involved in primary ciliary structure and function (including *AHI1*, *CC2D2A*, and *CEP290*). Neuropathologically and radiographically characterized by the pathognomonic “molar tooth sign” on axial brain MRI, resulting from cerebellar vermis hypoplasia, deepened interpeduncular fossa, and elongated, horizontally oriented superior cerebellar peduncles. Clinically manifests in infancy with episodic hyperpnea/apnea, profound hypotonia, oculomotor apraxia, cerebellar ataxia, and variable retinal dystrophy, cystic kidney disease, and polydactyly.

Alternate Names:

Joubert Syndrome and Related Disorders (JSRD)

Jugular

Jugular means relating to the neck, especially the jugular veins that drain blood from the head and brain to the heart.

Jugular Foramen

An opening at the base of the skull transmitting the internal jugular vein and cranial nerves IX, X, and XI. Lesions in this region can cause swallowing difficulty, hoarseness, shoulder weakness, hearing symptoms, or other neurologic deficits.

Jugular Vein

One of the major veins draining blood from the head, face, and neck. There are two pairs: internal jugular vein (IJV) and external jugular vein (EJV).

Jugular Venous Pressure

The visible height of blood in the neck veins. It helps clinicians estimate pressure in the right side of the heart and assess fluid buildup.

Alternate Names:

JVP

Jumper'S Knee

Alternate index label; see *Patellar tendinitis*.

Junctional Nevus

A usually benign mole formed by melanocytes at the junction between the epidermis and dermis; it is often flat and should be assessed if it changes in size, shape, color, or symptoms.

Junctional Rhythm

A cardiac rhythm arising from the atrioventricular junction rather than the sinoatrial node, usually with a narrow QRS complex and a slow or absent preceding atrial impulse.

Jungian Analysis

Jungian analysis is a psychotherapy approach based on Carl Jung's theories, using exploration of uncon-

scious processes, symbols, dreams, and personality development.

Juvenile Angiofibroma

Juvenile angiofibroma is a benign but highly vascular tumor arising near the nasal cavity in adolescent boys, causing progressive nasal obstruction and recurrent nosebleeds. predominantly in adolescent boys. It may cause recurrent nosebleeds, nasal obstruction, or facial swelling.

Juvenile Arthritis

A general term for inflammatory arthritis beginning in childhood. Persistent joint swelling, pain, stiffness, or limited movement may occur, and causes and subtypes vary.

Alternate Names:

Arthritis Juvenile (Juvenile Arthritis)

Juvenile Astrocytoma

An astrocytic glioma arising in a child or adolescent. Many are low-grade and may cause seizures, headache, visual symptoms, endocrine disturbance, or focal neurological deficits depending on location; diagnosis requires neuroimaging and tumour evaluation.

Juvenile Dermatomyositis

An inflammatory muscle disease of childhood characterized by proximal weakness and distinctive skin findings such as heliotrope rash and Gottron papules. It may also affect blood vessels, muscles, or other organs.

Juvenile Hypermobility Syndrome

Excessive joint movement in a child or adolescent associated with pain, instability, fatigue, or recurrent sprains, without another defined connective-tissue disorder. Assessment considers the pattern of hypermobility, skin and systemic features, and functional impact; management may include physiotherapy and joint protection.

Juvenile Idiopathic Arthritis

A group of inflammatory arthritis disorders that begin before age 16 and persist for at least six weeks

without another identified cause. Subtypes differ in the number of joints affected, systemic symptoms, associated eye disease, and laboratory findings.

Juvenile Idiopathic Arthritis

A group of inflammatory arthritides beginning before age 16 and lasting at least six weeks, sometimes affecting the eyes or growth.

Juvenile Myoclonic Epilepsy

A generalized epilepsy syndrome that typically begins in adolescence and causes myoclonic jerks, often after awakening, with possible generalized tonic-clonic or absence seizures.

Juvenile Osteoporosis

Reduced bone mass or impaired bone strength occurring during childhood or adolescence, increasing fracture risk. It may be primary or secondary to chronic disease, nutritional deficiency, endocrine disorder, immobility, or medication exposure.

Juvenile Polyposis Syndrome

An inherited disorder characterized by multiple hamartomatous polyps in the gastrointestinal tract and an increased risk of colorectal and other gastrointestinal cancers.

Juvenile Schizophrenia

Alternate index label; see *Childhood schizophrenia*.

Juvenile Xanthogranuloma

A rare, usually harmless skin disorder in infants and young children that causes one or more firm red, yellow, or brown bumps, often on the head or neck. The bumps usually go away on their own, but rarely the eyes or internal organs can also be affected.

Juxta-Articular

Positioned immediately adjacent to or in the immediate anatomical vicinity of a synovial joint. Clinically prominent in musculoskeletal radiology to describe periarticular bone and soft-tissue alterations: juxta-articular osteopenia (periarticular demineralization characteristic of early inflammatory rheumatoid

arthritis), juxta-articular erosions and subchondral cysts in advanced arthropathies, and juxta-articular tophi in chronic tophaceous gout.

Juxtaglomerular Apparatus

Specialized structure located at the vascular pole of the glomerulus where the distal tubule comes into contact with the afferent arteriole. Composed of macula densa cells in the tubular wall, juxtaglomerular cells in the afferent arteriolar wall that produce renin, and extraglomerular mesangial cells.

Juxtavesical

Positioned anatomically adjacent to or in the immediate proximity of the urinary bladder (exit vesica urinaria). Most frequently used in urology to describe the distal terminal segment of the ureter (the intramural or juxtavesical ureter) as it approaches the ureterovesical junction, a common site of impaction for migrating distal ureteral calculi causing acute renal colic, urinary frequency, and dysuria mimicking cystitis.

K

Kabuki Syndrome

A rare, genetically heterogeneous congenital malformation and intellectual disability syndrome caused by pathogenic heterozygous mutations in the histone methyltransferase gene *KMT2D* (type 1, autosomal dominant, 70%) or deletions/variants in the histone demethylase gene *KDM6A* (type 2, X-linked dominant, 5%). Clinically characterized by a distinctive facial appearance reminiscent of Japanese Kabuki actors (elongated palpebral fissures with eversion of the lateral third of the lower eyelid, arched eyebrows with sparse lateral thirds, broad depressed nasal tip, prominent ears), persistent fetal fingertip pads, skeletal and spinal abnormalities, short stature, and congenital heart defects.

Alternate Names:

Niikawa-Kuroki Syndrome

Kahlers Disease (Multiple Myeloma)

Alternate index label for Multiple Myeloma.

Kala-Azar

Visceral leishmaniasis, a potentially fatal parasitic infection affecting internal organs such as the spleen, liver, and bone marrow, usually transmitted by sandflies.

Kala-Azar (Visceral Leishmaniasis)

A serious parasitic infection caused by *Leishmania* species and transmitted by sandflies. It may cause prolonged fever, weight loss, enlargement of the spleen and liver, and suppression of blood-cell production.

Alternate Names:

Visceral Leishmaniasis

Kallmann Syndrome

An inherited condition that causes delayed or absent puberty and a reduced or absent sense of smell. It occurs when the brain does not send the hormone signals needed to start puberty.

Kallmann'S Syndrome

Inherited hypogonadotropic hypogonadism caused by deficient gonadotropin-releasing hormone, typically accompanied by reduced or absent sense of smell.

Kanamycin

An antibiotic used for selected serious bacterial infections. It can permanently damage hearing and harm the kidneys, so careful monitoring is needed.

Kangaroo Care (Skin-To-Skin Contact)

Holding a newborn directly against a caregiver's bare chest while monitoring the infant's condition. It supports thermal regulation, physiologic stability, bonding, and feeding, and is particularly useful for many preterm or low-birth-weight infants.

Alternate Names:

Skin-to-Skin Contact

Kant'S Disease

An uncommon historical eponym whose precise clinical meaning varies between sources; the original reference should be checked before using it clinically.

Kaolin

A clay mineral containing hydrated aluminium silicate, used as an adsorbent or protective substance; inhalation of kaolin dust can cause occupational

lung disease.

Kaposi Sarcoma

A cancer linked to human herpesvirus 8 that causes purple, red, or brown spots, plaques, or lumps on the skin or inside the body. It is more likely when the immune system is weakened.

Kaposi'S Sarcoma

A cancer of blood-vessel or lymphatic-vessel lining cells, often associated with human herpesvirus 8.

Kaposi'S Sarcoma

A vascular tumor associated with human herpesvirus 8. It may cause purple or reddish skin lesions and can involve internal organs, especially in people with impaired immunity.

Karnofsky Performance Status

A scale used to describe how well a person can perform daily activities and care for themselves. It helps clinicians assess overall function and ability to tolerate treatment, especially in cancer care.

Kartagener Syndrome

An inherited condition in which tiny hair-like structures that clear mucus do not work properly. It causes repeated sinus and lung infections and may place the internal organs in a reversed position.

Kartagener Triad

The combination of situs inversus, chronic sinusitis, and bronchiectasis caused by primary ciliary dyskinesia. The underlying ciliary defect impairs mucociliary clearance and embryonic left-right development.

Kartagener'S Syndrome

A form of primary ciliary dyskinesia characterized by chronic sinusitis, bronchiectasis, and situs inversus.

Karyocyte

A cell that contains a nucleus; the term is used especially when distinguishing nucleated cells from cells without nuclei.

Karyokinesis

Division of a cell's nucleus during cell division, producing two new nuclei with matching sets of chromosomes.

Karyomegaly

Karyomegaly is abnormal enlargement of a cell nucleus, often reflecting cellular injury, regeneration, infection, toxic exposure, or neoplasia; its significance depends on the tissue and accompanying microscopic changes.

Karyometry

Karyometry is the measurement of cell-nucleus size, shape, and related features, usually in tissue samples examined under a microscope.

Karyotype

An organized picture of a person's chromosomes, arranged in pairs. It can identify some chromosome changes and conditions.

Karyotype Result

A karyotype result reports the number and visible structure of a person's chromosomes and can identify some chromosome abnormalities.

Karyotyping

A test that examines the number and structure of chromosomes in cells to identify some genetic conditions.

Kasai Procedure

A pediatric surgical operation (hepatoportoenterostomy) performed for biliary atresia in infants, typically before 60 to 90 days of age to optimize outcomes. The atretic extrahepatic biliary tree is completely resected to the fibrous cone at the porta hepatis, and an end-to-side Roux-en-Y jejunal loop is anastomosed to the transected liver hilum to permit drainage of microscopic bile ducts. It restores bile flow, reduces cholestatic jaundice, and delays or eliminates the need for liver transplantation in responders.

Alternate Names:

Hepatoportoenterostomy; Kasai Operation; Por-toenterostomy

Kashin-Beck Disease

A long-term disease of the bones and joints that can cause pain, stiffness, deformity, and poor growth. It is linked to low selenium and environmental factors in some regions.

Katayama Fever (Acute Schistosomiasis)

An acute systemic reaction to schistosome infection, usually occurring weeks after exposure and causing fever, rash, cough, abdominal symptoms, and eosinophilia.

Alternate Names:

Acute Schistosomiasis

Katz Index

A scale that assesses independence in six basic activities of daily living: bathing, dressing, toileting, transferring, continence, and feeding.

Kawasaki Disease

An acute inflammatory disease of childhood that can affect the coronary arteries. It commonly causes persistent fever, rash, conjunctival redness, mouth or extremity changes, and swollen lymph nodes; diagnosis is clinical and incomplete forms occur.

Alternate Names:

Mucocutaneous Lymph Node Syndrome Mucocuta-neous Lymph Node Syndrome (Kawasaki Disease)

Kawasaki'S Disease

An acute, self-limited systemic vasculitis of child-hood, also known as mucocutaneous lymph node syndrome, primarily affecting medium-sized arteries with a predilection for the coronary arteries.

Alternate Names:

mucocutaneous.

Kayser-Fleischer Rings

Golden-brown or greenish rings around the colored part of the eye caused by copper buildup. They strongly suggest Wilson disease when other findings fit and are seen with a special eye examination.

Kearns-Sayre Syndrome

A rare inherited disorder affecting the mitochondria, the parts of cells that make energy. It usually begins in childhood or young adulthood and can cause drooping or limited eye movement, vision problems, heart-rhythm problems, and muscle weakness.

Keel Breast

A protruding breastbone and anterior chest-wall deformity, generally referring to pectus carinatum.

Keeping Your Medicines Organized

Medicine organization uses an accurate list, labeled containers, schedules, pillboxes when appropriate, refill planning, and review for duplication or inter-actions.

Kegel

A pelvic-floor muscle contraction exercise used to improve pelvic-floor strength and control of selected types of urinary or fecal incontinence.

Kegel Exercise

Repeated contraction and relaxation of the pelvic-floor muscles to improve their strength and coordi-nation.

Keller'S Operation

An excision arthroplasty of the first metatarsopha-langeal joint, historically used for hallux rigidus or painful bunion deformity.

Kelly Clamp

A hinged surgical clamp used to grasp tissue or compress blood vessels and control bleeding.

Keloid

An overgrowth of scar tissue extending beyond the original wound margins, caused by excessive col-lagen deposition during wound healing. Keloids are more common in darker skin types (Fitzpatrick IV to VI) and occur at sites of trauma, surgery, piercings, and burns. They are pruritic and painful.

Alternate Names:

Scar Excessive (Keloid)

Keloid Scar

A raised scar caused by excessive growth of scar tissue after skin injury.

Keloids

Keloids are raised scars that extend beyond the original wound because of excessive collagen deposition and may cause itching, pain, or cosmetic concern.

Keratan Sulfate

Keratan sulfate is a sulfated glycosaminoglycan found especially in cornea, cartilage, and brain; impaired breakdown causes accumulation in some mucopolysaccharidoses and can produce skeletal or ocular disease.

Keratan Sulphate

A sulfated glycosaminoglycan found especially in the cornea and cartilage; abnormal accumulation occurs in some mucopolysaccharidoses.

Keratotomy

Surgical removal or reshaping of part of the cornea, often to correct vision.

Keratin

A tough structural protein found in skin, hair, nails, and other epithelial tissues.

Keratinocyte

The predominant cell of the epidermis, producing keratin and contributing to the skin's barrier and immune functions.

Keratinocytes

The predominant cells of the epidermis that produce keratin and form the skin's protective barrier.

Keratitis

Inflammation or infection of the clear front surface of the eye. It can cause eye pain, redness, tearing, sensitivity to light, blurred vision, or a cloudy spot and needs prompt assessment.

Alternate Names:

Corneal Inflammation (Keratitis)

Keratoacanthoma

A low-grade or well-differentiated variant of squamous cell carcinoma originating from hair follicles. Characterized by rapid growth over 4-6 weeks into a dome-shaped, flesh-colored or erythematous nodular lesion with a central keratin-filled crater. Often resolves spontaneously over months via involution, leaving a depressed scar.

Keratoconjunctivitis

Keratoconjunctivitis is inflammation of both the cornea and conjunctiva, caused by infection, allergy, dryness, immune disease, or chemical injury.

Keratoconjunctivitis Sicca

Chronic dryness and inflammation of the cornea and conjunctiva, commonly called dry-eye syndrome. Symptoms include burning, grittiness, redness, fluctuating vision, and light sensitivity; causes include reduced tear production, excessive evaporation, medicines, and autoimmune disease.

Alternate Names:

dry-eye.

Keratoconjunctivitis Sicca (Dry Eyes)

Alternate index label for Dry Eyes.

Keratoconus

A condition in which the clear front surface of the eye becomes thin and bulges into a cone shape. It can cause blurred or distorted vision, glare, and sensitivity to light. Glasses, special contact lenses, or treatment to strengthen the cornea may help.

Keratocyte Count

A keratocyte count estimates the number or density of stromal cells in the cornea, usually by microscopy or imaging; it may help assess corneal injury, inflammation, degeneration, or response to surgery.

Keratoderma

A group of disorders characterised by excessive thickening of the skin of the palms and soles, which may be hereditary or acquired. Acquired forms can be diffuse, focal, or punctate and may be associated

with systemic diseases, malignancy, or medications.

Keratoderma Blennorrhagica

Thickened, scaly, sometimes pustular skin lesions, usually on the soles or palms, associated with reactive arthritis. Treatment focuses on the underlying infection or inflammation and the skin symptoms.

Keratoderma Blennorrhagicum

A keratoderma associated with reactive arthritis (formerly Reiter syndrome), characterised by waxy, yellowish, vesiculopustular papules and plaques on the palms and soles that may coalesce and extend to the extremities. It mimics pustular psoriasis histologically and clinically.

Keratoderma Blennorrhagicum

Alternate spelling of keratoderma blennorrhagicum, a cutaneous manifestation of reactive arthritis with waxy, crusty, vesiculopustular lesions on the palms and soles. It is a reactive phenomenon typically occurring in the context of genitourinary or gastrointestinal infection.

Keratoderma Blennorrhagicum

Thickened, scaly, or wart-like skin lesions, often on the palms and soles, associated with reactive arthritis and sometimes infection-triggered inflammation.

Keratolysis

Breakdown or loss of keratinized tissue, especially the superficial layer of the skin; it may occur in some infections and inflammatory skin disorders.

Keratolytics

Substances such as salicylic acid or urea that soften and remove excess keratin from thickened skin, corns, or warts.

Keratoma

A localized thickening or tumor-like growth of keratinized skin, such as a callus or corn; the exact type and need for evaluation depend on its appearance and behavior.

Keratomalacia

Softening and ulceration of the cornea caused most often by severe vitamin A deficiency, potentially leading to blindness.

Keratometer

An instrument that measures the curvature of the cornea.

Keratoplasty

Surgical transplantation or reshaping of corneal tissue. Penetrating keratoplasty replaces the full corneal thickness, whereas lamellar techniques replace selected layers and may reduce rejection and recovery time.

Keratoprosthesis

Artificial cornea for severe corneal blindness. Boston KPro most common. Indications: chemical burns, limbal stem cell deficiency.

Keratosis

Keratosis is a thickened or abnormal growth of keratinized skin and may be benign, inflammatory, infectious, or premalignant.

Keratosis Follicularis

Keratosis follicularis is excessive keratinization around hair follicles, producing rough follicular papules; it may refer to keratosis pilaris or, in some contexts, inherited follicular keratosis disorders.

Keratosis Obturans

Keratosis obturans is accumulation of shed keratin blocking the external ear canal, causing pain, hearing loss, and canal widening.

Keratosis Pilaris

A common, harmless skin condition that causes small, rough bumps around hair follicles, usually on the upper arms, thighs, buttocks, or face.

Alternate Names:

KP

Keratosis, Actinic

Alternate index label; see *Actinic keratosis*.

Keratosis, Seborrheic

Alternate index label; see *Seborrheic keratosis*.

Kerion

An inflamed, boggy, pus-forming mass caused by a fungal infection of hair-bearing skin, usually a severe form of tinea capitis.

Kernicterus

Brain damage caused by very high levels of bilirubin in a newborn. It can cause poor feeding, unusual sleepiness, a high-pitched cry, seizures, movement problems, or permanent disability.

Kernicterus (Nuclear Jaundice)

Brain injury caused by deposition of excessive unconjugated bilirubin in the basal ganglia and other brain regions, usually in a severely jaundiced newborn.

Kernig Sign

Pain or resistance when the knee is straightened while the hip is bent. It may suggest irritation of the coverings of the brain and spinal cord, such as with meningitis, but it is not diagnostic by itself.

Kernig'S Sign

Alternate index label; see *Kernig sign*.

Kerosene Poisoning

Kerosene poisoning involves toxic exposure to kerosene poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Kerstesia Gyiorum

A nonfermenting gram-negative bacterium found in environmental and clinical samples that can rarely cause opportunistic infection.

Keshan Disease

Heart disease caused by a lack of selenium, an ele-

ment that the body needs to function properly.

Ketamine

Ketamine is a dissociative anaesthetic and NMDA-receptor antagonist used for anaesthesia, procedural sedation, and selected treatment-resistant depression; emergence reactions, increased blood pressure, nausea, misuse, and respiratory complications are possible.

Ketamine Hydrochloride

Ketamine hydrochloride is a dissociative anesthetic used for anesthesia, procedural sedation, and selected pain or depression treatments; it can cause increased blood pressure, hallucinations, and impaired coordination.

Ketanserin

A selective serotonin 5-HT₂-receptor antagonist formerly used in some countries for hypertension and studied for vascular and platelet effects; availability and clinical use are limited and adverse effects may include dizziness and hypotension.

Keto Acid

An organic acid containing a ketone group, often formed during amino-acid metabolism and used in energy production or biosynthesis.

Ketoacidosis

A metabolic emergency in which excessive ketone production causes acidosis, most commonly in diabetic ketoacidosis.

Ketoconazole

Ketoconazole is an imidazole antifungal that inhibits fungal ergosterol synthesis; topical forms treat superficial mycoses, while oral use is restricted because of potentially severe liver injury, adrenal suppression, and drug interactions.

Ketogenesis

The production of ketone bodies from fatty acids, mainly in the liver during fasting, prolonged exercise, or inadequate insulin activity.

Ketogenic Diet

High fat, adequate protein, very low carbohydrate diet (<50g/day). Induces ketosis. Medical uses: epilepsy, obesity, metabolic syndrome. Risks: nutrient deficiency, kidney stones, dyslipidemia.

Ketone

A ketone is an organic compound containing a carbonyl group between two carbon atoms; medically important ketone bodies include acetoacetate, beta-hydroxybutyrate, and acetone, which rise during fasting or diabetic ketoacidosis.

Ketone Bodies

Water-soluble molecules produced mainly by the liver from fatty-acid-derived acetyl-CoA during fasting, starvation, prolonged exercise, or insulin deficiency. The principal ketone bodies are acetoacetate, beta-hydroxybutyrate, and acetone.

Ketones

Water-soluble substances produced mainly by the liver when the body breaks down fat for energy, such as during fasting, a low-carbohydrate state, or insufficient insulin. The main ketone bodies are beta-hydroxybutyrate, acetoacetate, and acetone. Small amounts can be normal, but high levels may cause ketoacidosis, especially in diabetes; ketones can be measured in blood or urine.

Ketones Blood Test

A blood test measuring ketone bodies, such as beta-hydroxybutyrate, used to assess ketosis and help diagnose or monitor diabetic ketoacidosis and other causes of ketone production.

Ketones Urine Test

A urine test detecting ketone bodies produced when the body metabolizes fat; it can support evaluation of diabetic ketoacidosis, starvation, vomiting, or other metabolic states but may lag behind blood ketone levels.

Ketonuria

Ketonuria is the presence of ketone bodies in urine, occurring with fasting, vomiting, low-carbohydrate

intake, pregnancy, or uncontrolled diabetes; marked ketonuria with hyperglycaemia may indicate diabetic ketoacidosis.

Ketonuria (Acetonuria)

The presence of ketone bodies in the urine, occurring in conditions such as diabetic ketoacidosis, prolonged fasting, or severe vomiting.

Alternate Names:

Acetonuria

Ketoprofen

A non-steroidal anti-inflammatory drug that inhibits cyclo-oxygenase and reduces pain, inflammation, and fever; gastrointestinal bleeding, kidney injury, cardiovascular events, and hypersensitivity are important risks.

Ketoprofen Overdose

Ketoprofen overdose involves toxic exposure to ketoprofen overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Ketorolac

A potent non-steroidal anti-inflammatory analgesic used for short-term management of moderate-to-severe acute pain; treatment duration is restricted because gastrointestinal bleeding, kidney injury, and impaired healing can occur.

Ketorolac Tromethamine

The tromethamine salt of ketorolac, a potent non-steroidal anti-inflammatory analgesic used short term for moderate-to-severe acute pain; it is contraindicated in several bleeding, renal, pregnancy, and perioperative situations.

Ketosis

Metabolic state with elevated ketone bodies. Physiological: fasting, ketogenic diet. Pathological: diabetic ketoacidosis. Distinguish from ketonemia vs ketonuria.

Ketotic Hypoglycemia

Ketotic hypoglycemia is low blood glucose with ketones after fasting or illness, often occurring in young children and usually improving with age.

Ketotifen

An antihistamine with mast-cell-stabilizing activity used in some countries for allergic disorders and, in ophthalmic form, allergic conjunctivitis; drowsiness, increased appetite, and irritation may occur.

Kidney

One of two bean-shaped organs that filter waste from the blood into urine. The kidneys also help control body water, salts, blood pressure, and the balance of acids in the blood.

Kidney And Urinary Health

Kidney and urinary health encompasses the function of the kidneys, ureters, bladder, and urethra in filtering waste, regulating fluid/electrolyte balance, and eliminating urine.

Kidney Biopsy

A procedure in which a small sample of kidney tissue is removed, usually with a needle guided by ultrasound, to identify kidney disease. Bleeding is the main risk.

Kidney Calculi

Solid mineral deposits that form in the kidneys; also called kidney stones.

Alternate Names:

kidney stones.

Kidney Cancer

A malignant tumor arising in the kidney. Renal cell carcinoma is the most common type in adults, but other types occur, especially in children.

Alternate Names:

Cancer, Kidney Renal Cancer

Kidney Cysts

Fluid-filled sacs in or on the kidneys that are often harmless but may occasionally cause pain, infection,

bleeding, or impaired kidney function.

Alternate Names:

Cyst, Kidney

Kidney Disease, Chronic

Alternate index label; see *Chronic kidney disease*.

Kidney Failure

A condition in which the kidneys can no longer remove enough waste or control fluid and body salts. It may develop suddenly or gradually and can require medicines, dialysis, or a transplant.

Alternate Names:

Renal Failure (Kidney Failure)

Kidney Failure, Chronic

Alternate index label; see *Chronic kidney disease*.

Kidney Function Tests

Blood and urine tests used to assess how well the kidneys filter waste and regulate fluid and electrolytes. Common measures include creatinine, estimated glomerular filtration rate, urine albumin, and urinalysis; results are interpreted with the clinical context.

Kidney Infection

An infection of one or both kidneys, usually caused by bacteria that spread from the urinary tract.

Kidney Neoplasm

A kidney neoplasm is an abnormal growth in the kidney that may be benign or cancerous; symptoms and treatment depend on its type, size, and spread.

Kidney Pain

Pain felt in the side or back below the ribs that may be caused by kidney stones, infection, obstruction, inflammation, or referred pain.

Alternate Names:

Pain Kidney (Kidney Pain)

Kidney Position

A modified lateral position in which the operating table is flexed at the waist to elevate the flank and

improve access to the kidney or retroperitoneum.

Kidney Removal

A medical procedure, treatment, device, or care topic related to kidney removal; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Kidney Stone

A hard mineral deposit formed from substances in urine that can obstruct the urinary tract and cause colicky flank or abdominal pain, blood in the urine, vomiting, or infection. Urgent assessment is needed when obstruction may be infected or pain is severe or persistent.

Alternate Names:

Kidney Stones Kidney Stones (Nephrolithiasis / Urolithiasis) Nephrolithiasis (Kidney Stones) Nephrolithiasis / Urolithiasis Renal Lithiasis

Kidney Transplant

Surgical placement of a healthy donor kidney into a person with kidney failure to restore renal function. Lifelong immunosuppressive treatment is generally required to reduce rejection risk.

Kidney Transplant Rejection

Immune-mediated injury to a transplanted kidney caused by recipient antibodies, T cells, or both. Hyperacute, acute, and chronic rejection differ in timing and pathology and require correlation of graft function, imaging, serology, and often biopsy.

Kidney Transplantation

Best treatment for ESRD with improved survival over dialysis. Living donor superior to deceased donor. Lifelong immunosuppression required. Rejection types include hyperacute, acute, and chronic. Complications include rejection, infection, vascular complications, and recurrent disease.

Kidney, Artificial

A dialysis device that removes metabolic waste and excess fluid from blood and helps correct electrolyte disturbances when renal function is inadequate.

Kidney, Diseases Of

Disorders affecting the kidneys or urinary collecting system that can impair filtration, urine formation, fluid balance, blood pressure, or electrolyte regulation.

Kienbock Disease

Kienbock disease (lunatomalacia): AVN of lunate. Lichtman classification: I (normal XR, MRI edema), II (lunate sclerosis), IIIA (lunate collapse, no scaphoid rotation), IIIB (lunate collapse, fixed scaphoid rotation, DISI), IV (perilunate arthritis). Etiology: negative ulnar variance, trauma, vascular variation.

Kieselguhr

Diatomaceous earth containing substantial silica; inhalation of its dust can produce occupational pneumoconiosis and other lung injury.

Kiesselbach Plexus

A network of small arteries on the front part of the nasal septum and a common source of nosebleeds.

Killer Cold Virus (Adenovirus Infection, Ad14)

Alternate index label for Adenovirus Infection, Ad14.

Killian's Triangle

An area of potential weakness in the pharyngeal wall between the thyropharyngeus and cricopharyngeus muscles, predisposing to Zenker diverticulum formation.

Killip Classification

A clinical classification system for assessing the severity of myocardial infarction based on physical examination findings. Class I: no heart failure signs. Class II: mild heart failure (crackles, S3 gallop, elevated JVD). Class III: pulmonary edema. Class IV: cardiogenic shock.

Kilocalorie

A unit of energy equal to 1,000 small calories; the

dietary Calorie, written with a capital C, is one kilocalorie.

Kilojoule

The SI unit of energy, equal to 1,000 joules; approximately 4.184 kilojoules equal one kilocalorie.

Kimmelstiel-Wilson Nodules

Characteristic round eosinophilic nodular mesangial expansions in diabetic nephropathy. Represent advanced mesangial matrix accumulation. Found in peripheral mesangium associated with diffuse mesangial expansion, GBM thickening, and arteriolar hyalinosis.

Kimmelstiel-Wilson Syndrome

Nodular glomerulosclerosis caused by long-standing diabetic kidney disease, with thickened glomerular structures, proteinuria, declining kidney function, and possible kidney failure.

Kimura Disease

A chronic inflammatory disorder characterized by painless subcutaneous masses, commonly in the head and neck, regional lymphadenopathy, eosinophilia, and elevated immunoglobulin E. It is distinct from angiolymphoid hyperplasia with eosinophilia and may involve the kidneys.

Kinase

An enzyme that transfers a phosphate group, usually from ATP, to another molecule; kinases regulate many cellular processes.

Kinesin

A motor protein that moves along microtubules and transports cellular cargo using energy from ATP.

Kinesio Taping

Application of an elastic adhesive tape to the skin to provide support or sensory input while allowing movement. Its clinical benefit varies by condition.

Kinesiology

The study of human movement and the forces that

produce or affect it, with applications in health care, rehabilitation, sport, and ergonomics.

Kinesis

A change in the rate or direction of an organism's movement in response to a stimulus, without movement toward or away from its source.

Kinesthesia

The sense of movement and position of the body's parts.

Kinesthesia

The sense of body movement and position produced by signals from muscles, tendons, joints, and skin.

Kinetic

Relating to movement or motion; in clinical usage it may describe movement-related phenomena such as a kinetic tremor.

Kinetic Modeling

Mathematical modeling of how a tracer or drug moves through and is exchanged among tissues or body compartments.

Kinetics

The study of how a process changes over time. In pharmacology, it includes the absorption, distribution, metabolism, and elimination of a drug.

Kinetochore

A protein structure assembled on a chromosome's centromere that attaches to spindle fibers during cell division.

Kinetoplast

A DNA-containing region of the mitochondrion in certain protozoa, including *Trypanosoma* and *Leishmania*.

Kinetosome

A kinetosome is a basal body that organizes the microtubules of a cilium or flagellum.

King-Copeland-Coons Syndrome

A rare historical eponym whose definition is not consistent across medical references; its source-specific meaning should be verified.

Kingella

Kingella is a genus of bacteria that can live in the mouth and can occasionally cause bloodstream, bone, joint, or heart-valve infections.

Kingella Denitrificans

Kingella denitrificans is a rare bacterium found mainly in the upper airway; it can occasionally cause invasive infection.

Kingella Kingae

Kingella kingae is a bacterium that can cause bone or joint infection, especially in young children, and can rarely cause endocarditis.

Kingella Oralis

Kingella oralis is a mouth-associated bacterium that rarely causes invasive infection, including bloodstream infection or endocarditis.

Kingella Potus

Kingella potus is a rare bacterium that has been associated with occasional human infections; clinical importance depends on the specimen and evidence of true infection.

Kinin

A short peptide, such as bradykinin, that widens blood vessels, increases vascular permeability, and contributes to pain and inflammation.

Kininogenase

An enzyme that releases kinins from kininogen proteins, contributing to regulation of blood flow, inflammation, and pain.

Kinins

A group of vasoactive peptides, chiefly bradykinin and kallidin, generated from kininogens during inflammation; they promote vasodilation, increased

vascular permeability, smooth-muscle effects, and pain.

Kinocilium

A specialized cilium, especially the prominent cilium of an inner-ear sensory hair cell, that helps establish directional sensitivity.

Kiss

A touch or caress with the lips; kissing can transmit some infections through saliva or close contact, depending on the pathogen.

Kissing Disease

Alternate index label; see *Mononucleosis*.

Kit Device

A medical procedure, device, or treatment involving kit device. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Klatskin Tumor

A Klatskin tumor is a cancer arising at the junction where the right and left bile ducts meet near the liver.

Klebsiella

A genus of Gram-negative bacteria that can cause pneumonia, urinary-tract infection, bloodstream infection, and other healthcare-associated infections.

Klebsiella Oxytoca

Klebsiella oxytoca is a bacterium that can cause urinary, bloodstream, wound, and hospital-acquired infections and can sometimes cause antibiotic-associated hemorrhagic colitis.

Klebsiella Pneumonia

Severe, necrotizing pneumonia in alcoholic or debilitated patients. Upper lobe involvement with “bulging fissure sign” and “currant jelly” sputum.

Klebsiella Pneumoniae

Klebsiella pneumoniae is a bacterium that can cause pneumonia, urinary infection, bloodstream infection,

and liver abscess; some strains are highly antibiotic-resistant.

Klebsiella Rhinoscleromatis

A gram-negative bacillus causing chronic granulomatous infection of the nasal passages and upper airway, leading to rhinoscleroma with progressive nasal obstruction and tissue destruction.

Klebsiella Singaporensis

Klebsiella singaporensis is a rare environmental bacterium that has occasionally been recovered from human clinical specimens; its significance depends on the specimen and symptoms.

Klebsiella Variicola

Klebsiella variicola is a bacterium closely related to *K. pneumoniae* that can cause human infections, including bloodstream and urinary infections.

Kleptomania

A mental health disorder marked by repeated inability to resist urges to steal items that are not needed.

Alternate Names:

Compulsive Stealing Pathological Stealing

Klinefelter Syndrome

Most common sex chromosome aneuploidy in males with a karyotype of 47,XXY occurring in approximately one in six hundred live male births. Characterized by small firm testes, gynecomastia, tall stature with long extremities, and infertility from azoospermia.

Klinefelter's Syndrome

A chromosomal condition in which a male has one or more extra X chromosomes, often causing small testes and reduced fertility.

Klippel Trenaunay Weber Syndrome (Klippel-Trenaunay-Weber Syndrome)

Alternate index label for Klippel-Trenaunay-Weber Syndrome.

Klippel-Feil Syndrome

A congenital condition characterized by fusion of two or more cervical vertebrae, resulting from failure of normal segmentation of the cervical somites during weeks 3-8 of gestation. Classic triad: short neck, low posterior hairline, and limited cervical range of motion (present in <50% of cases).

Klippel-Trenaunay Syndrome

A congenital vascular disorder characterized by the triad: capillary malformations (port-wine stains), venous malformations (varicose veins, persistent embryonic veins), and soft tissue or bone hypertrophy of the affected limb (typically a lower extremity).

Klippel-Trenaunay-Weber Syndrome

An older, imprecise label for a congenital vascular and limb-overgrowth disorder; modern terminology distinguishes Klippel-Trenaunay syndrome from Parkes Weber syndrome, which includes arteriovenous malformations.

Klippel-Trenaunay-Weber Syndrome

A rare congenital overgrowth disorder involving capillary or venous malformations and enlargement of bone or soft tissue, usually affecting one limb. It may cause limb asymmetry, pain, bleeding, thrombosis, or lymphedema. Care is multidisciplinary and may include compression, vascular treatment, and orthopedic management.

Alternate Names:

Klippel Trenaunay Weber Syndrome (Klippel-Trenaunay-Weber Syndrome)

Klumpke Paralysis

Lower-trunk brachial-plexus injury affecting primarily the C8 and T1 nerve roots. It causes weakness or paralysis of the intrinsic hand muscles, sensory loss along the medial forearm and hand, and occasionally Horner syndrome when sympathetic fibres are involved.

Klumpke's Paralysis

Lower brachial-plexus palsy involving mainly the C8 and T1 nerve roots, causing hand weakness and sensory loss; Horner syndrome may accompany

severe cases.

Kluyvera

A genus of gram-negative bacteria in the Enterobacteriales group; species can rarely cause opportunistic human infections.

Kluyvera Ascorbata

A gram-negative Kluyvera species found in the environment and occasionally associated with urinary, bloodstream, or other opportunistic infections.

Kluyvera Cryocrescens

A gram-negative environmental bacterium that rarely causes opportunistic human infection.

Kluyvera Georgiana

A rarely encountered gram-negative Kluyvera species that may be isolated from clinical specimens.

Kluyvera Intermedia

A gram-negative Kluyvera species occasionally isolated from human specimens and rarely associated with infection.

Kluyveromyces

A genus of yeasts used in fermentation and biotechnology; some species can rarely cause opportunistic infection.

Kluyveromyces Lactis

A yeast used in dairy and biotechnology applications, including production of enzymes and recombinant proteins.

Kluyveromyces Marxianus

A yeast used in fermentation and biotechnology; it can grow on lactose and other sugars and rarely causes infection.

Knee

The synovial joint between the femur, tibia, and patella, stabilized by the cruciate and collateral ligaments and menisci; it permits flexion and extension and bears body weight.

Knee Arthroscopy

Knee arthroscopy is a minimally invasive procedure in which a camera and small instruments are inserted into the knee to diagnose or treat selected joint problems.

Knee Braces - Unloading

A medical procedure, device, or treatment involving knee braces - unloading. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Knee Bursitis

Knee bursitis is inflammation of a fluid-filled sac around the knee, causing localized swelling, tenderness, and pain with kneeling or movement. Repetitive pressure, trauma, arthritis, gout, or infection can contribute; fever or redness requires urgent review.

Alternate Names:

Bursitis Of The Knee

Knee CT Scan

Computed tomography of the knee using x-rays and computer reconstruction to evaluate fractures, bone alignment, arthritis, masses, infection, and other osseous or soft-tissue abnormalities.

Knee Disorders

The knee is a complex hinge-type synovial joint connecting the femur, tibia, and patella, stabilized by ligaments (ACL, PCL, MCL, LCL), menisci (medial and lateral), and surrounding muscles/tendons.

Knee Injury And Meniscus Tears

A meniscus tear is damage to the cartilage cushion between the thighbone and shinbone, often after twisting or with degeneration. Pain, swelling, catching, or locking may occur; treatment ranges from rehabilitation to surgery according to tear, symptoms, and stability.

Knee Joint

A synovial joint formed mainly by the femur, tibia, and patella. It permits flexion and extension with

limited rotation and is stabilized by ligaments, menisci, muscles, and the joint capsule.

Knee Joint Replacement

A medical procedure, treatment, device, or care topic related to knee joint replacement; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Knee Microfracture Surgery

A medical procedure, treatment, device, or care topic related to knee microfracture surgery; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Knee MRI Scan

Magnetic resonance imaging of the knee using a magnetic field and radio waves to evaluate cartilage, menisci, ligaments, tendons, bone marrow, muscles, and joint fluid.

Knee Pain

Pain felt in or around the knee, commonly caused by injury, osteoarthritis, bursitis, tendinitis, or inflammatory disease. Persistent swelling, instability, inability to bear weight, or fever requires medical assessment.

Alternate Names:

Pain, Knee Pain Knee Knee Pain: Symptom

Knee Prosthesis

An artificial joint component used to replace damaged surfaces of the knee, usually including femoral and tibial components and sometimes a patellar component.

Knee-Chest Position

A position in which the patient rests on the knees and chest, with the hips flexed; it may be used for spinal or anorectal procedures.

Knee-Jerk

The patellar tendon deep-tendon reflex, in which tapping the tendon produces extension of the knee, chiefly testing the L3-L4 spinal segments.

Kneecap

The largest sesamoid bone in the human body, embedded within the quadriceps tendon anterior to the knee joint. See Patella (Kneecap).

Kneecap Dislocation

A kneecap dislocation occurs when the patella moves out of its femoral groove, usually laterally, causing pain, deformity, swelling, and difficulty walking.

Kniest Dysplasia

A rare skeletal dysplasia caused most often by pathogenic variants in COL2A1. It produces disproportionate short stature, enlarged joints, progressive joint disease, distinctive facial features, and sometimes severe ocular, auditory, or respiratory complications.

Knife-Cut Ulcers From Crohn'S Disease

Deep, linear, serpiginous ulcerations of the intestinal mucosa in Crohn disease, resembling cuts made by a knife. These longitudinal ulcers with heaped-up margins are characteristic of Crohn disease and help distinguish it from ulcerative colitis on endoscopy. They may cause strictures, fistulae, and bleeding.

Knock Knees

Medial angulation of the knees causing the knees to touch while the ankles remain apart, also called genu valgum. It is common during childhood development but may be pathologic when severe, asymmetric, persistent, or associated with pain or growth disturbance.

Alternate Names:

genu valgum.

Knock-Knee

Genu valgum, a leg alignment in which the knees angle inward and may touch while the ankles remain apart.

Knoellia

A genus of gram-positive, aerobic bacteria found in the environment and rarely isolated from human clinical specimens.

Knotless Anchor

Suture anchor for soft tissue repair without tying knots. Used in orthopedic surgery.

Knuckle

A projecting finger joint, usually a metacarpophalangeal joint; it may be affected by trauma, inflammation, or arthritis.

Knuckle Pad

A knuckle pad is a firm thickening over a finger joint, usually benign and associated with friction, trauma, or fibromatosis.

Koch'S Bacillus

Mycobacterium tuberculosis, the bacterium that causes tuberculosis; the term is a historical eponym.

Koch'S Bacillus (*Mycobacterium Tuberculosis*)

The bacterium that causes tuberculosis, an infection most often affecting the lungs but capable of involving many organs.

Alternate Names:

Mycobacterium Tuberculosis

Koch'S Postulates

Historical criteria proposed to link a microorganism to a disease, including its presence in affected cases, isolation and culture, reproduction of disease, and re-isolation; molecular and modern infectious-disease methods have expanded beyond these criteria.

Kocher Clamp

A toothed surgical clamp used to hold firm tissue or clamp vessels.

Kocher-Debré-SéMéLaigne Syndrome

A rare pediatric syndrome of generalized skeletal muscle pseudohypertrophy occurring in infants and young children with severe, long-standing, untreated congenital hypothyroidism (cretinism). Characterized by generalized, prominent, diffuse muscle enlargement giving affected children a striking, athletic, or "Herculean" appearance, accompanied by

slow muscle contractions, proximal limb weakness, severe developmental delay, short stature, puffy coarse facial features, and myxedema. Like Hoffmann syndrome in adults, the apparent muscle hypertrophy and weakness resolve entirely upon prompt correction of the hypothyroid state with thyroid hormone replacement.

Alternate Names:

Cretinism with Muscular Hypertrophy; Infantile Hypothyroid Myopathy

Kocuria

A genus of gram-positive cocci commonly found on skin and in the environment that can rarely cause opportunistic infection.

Kocuria Rosea

A pink-pigmented gram-positive coccus usually found on skin or in the environment and rarely associated with opportunistic infection.

Kocuria Varians

A gram-positive coccus found on skin and in the environment that can rarely cause opportunistic infection.

Kodamaea Ohmeri

A yeast that can rarely cause bloodstream or other opportunistic infections, especially in people with weakened immunity.

Koebner Phenomenon

Development of lesions of a skin disease, classically psoriasis or vitiligo, at sites of injury or other trauma.

Koebnerisation

Development of lesions of an existing skin disease at sites of trauma or injury, classically seen in psoriasis, vitiligo, and lichen planus.

KOH Preparation

Microscopic examination of skin, hair, or nail scrapings treated with potassium hydroxide to dissolve keratin and reveal fungal hyphae or yeast. A neg-

ative preparation does not exclude infection when sampling is inadequate or prior treatment has been given.

Koilonychia

Spoon nails: nails with a concave (spoon-shaped) contour, most commonly associated with iron deficiency anemia; also seen in hemochromatosis, Plummer-Vinson syndrome, malnutrition, and trauma (nail dystrophy). The nail becomes thin, brittle, and turns up at the edges so a drop of water rests in the concavity.

Konno Procedure

A complex reconstructive cardiac surgical operation designed to treat severe, diffuse, tunnel-like subaortic left ventricular outflow tract obstruction, complex multi-level LVOT obstruction, or an excessively small/hypoplastic aortic annulus. Involves aortoventriculoplasty: a full-thickness longitudinal incision is made through the anterior aortic root and carried inferiorly across the aortic annulus deep into the muscular interventricular septum (avoiding the conduction bundle), followed by patch enlargement of the ventricular septum and outflow tract, and subsequent surgical aortic valve replacement with a prosthetic valve of appropriate adult caliber.

Alternate Names:

Konno-Rastan Procedure; Aortoventriculoplasty

Koplik Spots

A characteristic enanthem of measles. It consists of small, bluish-white spots on an erythematous bluish-white spots on an erythematous background located on the buccal mucosa opposite the lower molars ("grains of salt on a red background"). Appear 1-2 days prior to the onset of the maculopapular rash and disappear 1-2 days after rash onset.

Koplik'S Spots

Small white or bluish-gray spots on the inner cheek, often with a red base, that are characteristic of measles and may appear shortly before the skin rash.

Korotkoff Sounds

Sounds heard through a stethoscope over an artery as a blood-pressure cuff is deflated; their appearance and disappearance are used to estimate systolic and diastolic pressure.

Korsakoff Syndrome

A chronic amnesic disorder caused most often by severe thiamine deficiency associated with alcohol-use disorder. It causes impaired formation of new memories, loss of past memories, and sometimes confabulation.

Korsakoff'S Syndrome

A chronic amnesic disorder, commonly associated with severe thiamine deficiency and alcohol-use disorder, causing impaired new learning, memory loss, and sometimes confabulation.

Kounis Syndrome

An acute coronary syndrome caused by an allergic or hypersensitivity reaction, involving coronary vasospasm, plaque erosion or rupture, or stent thrombosis. It may occur with allergic symptoms and requires simultaneous assessment and treatment of the allergic reaction and myocardial ischemia.

KP

Alternate index label; see *Keratosis pilaris*.

Krabbe Disease

An autosomal-recessive lysosomal storage disorder caused by deficient galactocerebrosidase activity and accumulation of psychosine. Infantile disease causes irritability, stiffness, developmental regression, seizures, and progressive loss of neurological function; later-onset forms are variable.

Krabbe'S Disease

A rare inherited disorder in which loss of myelin progressively damages the nervous system.

KRAS Gene

KRAS encodes a small GTPase that relays signals from growth-factor receptors through the RAS-

RAF–MEK–ERK pathway; activating mutations drive cancers and can predict resistance to some targeted therapies.

Kraurosis

An older term for atrophic and sclerotic vulvar disease, often referring to lichen sclerosus and associated with itching, discomfort, and tissue fragility.

Krebs Cycle (Citric Acid Cycle)

A series of chemical reactions in mitochondria that oxidizes acetyl-CoA and generates energy-rich molecules used for cellular energy production.

Alternate Names:

Citric Acid Cycle

Krebs' Cycle

The Krebs cycle, or citric-acid cycle, is a mitochondrial pathway that oxidizes acetyl-CoA to carbon dioxide and generates NADH, FADH₂, and GTP for oxidative phosphorylation.

Kringles

Compact protein domains stabilized by disulfide bonds that mediate binding in proteins such as plasminogen and some growth factors.

Krukenberg Tumor

An ovarian tumor caused by metastatic cancer, commonly originating in the stomach or gastrointestinal tract.

Krukenberg'S Tumor

A metastatic ovarian tumour composed commonly of signet-ring cells, most often arising from gastric or other gastrointestinal adenocarcinoma.

Krunkenberg Tumour

A misspelled variant of Krukenberg tumour, meaning a metastatic ovarian signet-ring-cell adenocarcinoma, usually originating in the stomach or another gastrointestinal site.

Kt/V

Dialysis adequacy measure representing the frac-

tional clearance of urea. K is dialyzer clearance, t is treatment time, V is volume of distribution of urea. Target Kt/V greater than 1.4 for thrice-weekly hemodialysis. Urea reduction ratio greater than 70% is alternative adequacy measure.

Kugelberg-Welander Disease

The mild, juvenile-onset form of 5q-related spinal muscular atrophy (Spinal Muscular Atrophy Type 3 [SMA 3]), inherited in an autosomal recessive pattern and caused by homozygous deletions or loss-of-function variants in the *SMN1* gene on chromosome 5q13, with disease severity modulated by the number of copies of the homologous *SMN2* backup gene (typically 3 to 4 copies). Manifests after 18 months of age with slowly progressive, symmetrical proximal muscle weakness and wasting, predominantly affecting the lower extremities and pelvic girdle, waddling gait, difficulty climbing stairs, fine postural tremor of the fingers, and preserved deep tendon reflexes initially, with most patients achieving independent ambulation and normal life expectancy.

Alternate Names:

Spinal Muscular Atrophy Type 3 (SMA 3); Juvenile Spinal Muscular Atrophy

Kulchitsky Cells (Enterochromaffin Cells)

Specialized neuroendocrine cells, especially in the gastrointestinal and respiratory tracts, that release signalling substances such as serotonin.

Alternate Names:

Enterochromaffin Cells

Kupffer'S Cell

A Kupffer cell is a resident macrophage lining the liver sinusoids; it removes microbes, damaged blood cells, and particles from portal blood and contributes to hepatic inflammation and iron handling.

Kurthia

A genus of gram-positive, spore-forming bacteria found mainly in soil and food; human infection is rare.

Kuru

Kuru is a rare fatal prion disease historically asso-

ciated with ritual cannibalism, causing progressive ataxia, tremor, and neurologic decline.

Kussmaul Breathing

A deep, rapid breathing pattern seen in severe metabolic acidosis (diabetic ketoacidosis, lactic acidosis, uremia). It represents the body's attempt to compensate by blowing off CO₂. The increased respiratory rate and depth create the characteristic labored breathing.

Kussmaul Respiration

Deep, rapid breathing commonly associated with severe metabolic acidosis, especially diabetic ketoacidosis.

Kussmaul Sign

Paradoxical increase in jugular venous pressure with inspiration, normally the JVP decreases with inspiration. It indicates impaired right ventricular filling during inspiration. Seen in constrictive pericarditis, restrictive cardiomyopathy, severe right heart failure, cardiac tamponade, and massive pulmonary embolism.

Kwarshiorkor

A misspelling of kwashiorkor, a severe protein-energy malnutrition syndrome characterized by oedema, growth failure, muscle wasting, and often fatty liver.

Alternate Names:

Nuclear Jaundice

Kwashiorkor

Kwashiorkor is severe protein-energy malnutrition characterized by edema, growth failure, muscle wasting, skin or hair changes, and impaired immune function. deficiency relative to calorie intake (often occurring in children when weaned from breast milk to a high-carbohydrate starch diet).

Kynurenine

A metabolite formed during breakdown of tryptophan and a precursor of several neuroactive and immune-regulating compounds.

Kyphoplasty

A minimally invasive procedure that uses a balloon to create space in a collapsed vertebral body and then injects bone cement to stabilize a painful compression fracture.

Kyphoscoliosis

An abnormal curvature of the spine in both the coronal (scoliosis) and sagittal (kyphosis) planes. Severe forms cause restrictive lung disease by reducing chest wall compliance and lung volumes. It can lead to chronic respiratory failure and pulmonary hypertension.

Kyphosis

Kyphosis is excessive forward curvature of the thoracic spine, which may result from posture, vertebral compression, developmental conditions, degeneration, or inflammatory disease. angle on lateral spine x-ray (normal thoracic kyphosis is 20-40 degrees).

Alternate Names:

Round Back

Kytococcus

A genus of gram-positive cocci found on skin and in the environment that can rarely cause opportunistic infection.

Kytococcus Sedentarius

A skin-associated gram-positive coccus that produces enzymes linked to pitted keratolysis and can rarely cause invasive infection.

L

L-Dopa

A medicine that the brain converts into dopamine. It is usually combined with carbidopa or benserazide to treat Parkinson disease.

La Crosse Encephalitis

A mosquito-borne viral encephalitis caused by La Crosse virus, usually affecting children and causing fever, headache, seizures, or altered consciousness.

LAAO

Alternate name; see *Left Atrial Appendage Occlusion*.

Labetalol

A medicine that lowers blood pressure by slowing the heart and relaxing blood vessels. It is also used for severe high blood pressure during pregnancy and may cause a slow heartbeat, low blood pressure, or breathing problems in people with asthma.

Labia

The folds of skin surrounding the entrance to the vagina and urethra, consisting mainly of the labia majora and labia minora.

Labia Majora

The paired outer folds of skin of the vulva that help protect the urethral and vaginal openings and contain connective tissue, glands, and fat.

Labia Minora

The paired inner folds of tissue of the vulva, lying within the labia majora and surrounding the vestibule and its openings.

Labial Sounds

Speech sounds produced with one or both lips, such as /p/, /b/, and /m/; lip movement and coordination can be relevant in speech assessment.

Labiaplasty

Surgical reshaping or reduction of the labia, usually the labia minora, performed for functional symptoms, reconstruction, or selected cosmetic indications. Risks include pain, bleeding, infection, scarring, and altered sensation.

Labile

Easily or rapidly changing, as in labile blood pressure, blood glucose, or mood.

Labile Hypertension

Blood pressure that changes substantially over short periods. Stress, pain, medicines, illness, and measurement conditions can contribute, so repeated properly performed readings help confirm the pattern.

Lability

A tendency to change readily or unpredictably, such as emotional lability or labile blood pressure.

Labium

A lip-like anatomical structure; in obstetrics and gynecology, usually one of the labia of the vulva.

Labium Majus

One of the paired outer folds of the vulva containing skin, connective tissue, glands, vessels, and nerves and protecting the more internal genital structures.

Labium Minus

One of the paired inner folds of the vulva bordering the vestibule and helping protect the urethral and vaginal openings.

Labor

The process in which regular contractions open the cervix and lead to delivery of the baby and placenta. It includes the stages of cervical opening, birth of the baby, and delivery of the placenta.

Labor Complication

A maternal or fetal problem occurring during labor or delivery, such as abnormal labor progress, hemorrhage, infection, fetal distress, or malpresentation, requiring assessment and management based on the cause.

Labor Dystocia

Labor that is unusually slow or difficult because of weak contractions, the baby's position or size, or the shape of the mother's pelvis. Treatment depends on the cause and may include medicines or an operation.

Labor False (Braxton Hicks Contractions)

Alternate index label for Braxton Hicks Contractions.

Labor Pain

Pain caused by uterine contractions, cervical dilation, and stretching of pelvic tissues during labor; its intensity and management vary with labor stage and individual factors.

Laboratory

A facility or department where clinical specimens are examined using microbiologic, hematologic, chemical, molecular, or other diagnostic methods.

Laboratory Medicine

The medical specialty that uses laboratory tests on blood, tissue, and other specimens for diagnosis, screening, monitoring, and treatment.

Laboratory Procedure

A medical procedure, treatment, device, or care topic related to laboratory procedure; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Laboratory Test

An examination of a body fluid, tissue, or other specimen to obtain information for diagnosis, monitoring, screening, or treatment.

Labour

The process of childbirth, consisting of regular uterine contractions that cause progressive cervical effacement and dilation, leading to delivery of the baby and placenta.

Labra

Plural of labrum, referring to fibrocartilaginous rims that deepen certain joint sockets, such as the shoulder or hip labra.

Labradimil

An investigational pharmacologic compound; its mechanism, indication, safety, and clinical significance depend on the specific study or product context.

Labral Tear

A tear in the fibrocartilaginous rim of a joint socket, commonly the shoulder or hip, causing pain, clicking, instability, or reduced range of motion.

Labrum

A ring of tough, flexible tissue around the socket of a joint. It deepens and stabilizes the shoulder or hip socket.

Labyrinth

The inner ear. It contains the cochlea, which is responsible for hearing, as well as structures that are needed for balance.

Labyrinth Disorder

A disorder of the inner-ear balance structures that

may cause vertigo, nausea, or imbalance.

Labyrinthine Membrane

The membranous labyrinth of the inner ear, a fluid-filled system containing the cochlear duct and vestibular organs responsible for hearing and balance.

Labyrinthitis

Inflammation of the inner ear that can cause sudden spinning dizziness, imbalance, nausea, ringing in the ear, and sometimes hearing loss.

Labyrinthitis (Inner Ear Inflammation)

Alternate index label for Inner Ear Inflammation.

Lac Operon

A group of bacterial genes that are switched on or off to help the bacterium use lactose, a sugar found in milk.

Laceration

A wound caused by tearing of body tissue, typically caused by trauma from a sharp object (knife, glass, metal) or blunt force that splits the skin. Lacerations involve the skin and may extend into subcutaneous tissue, muscle, tendon, nerve, or blood vessels.

Laceration (Forensic)

Tear or disruption of tissue produced by blunt force trauma exceeding skin elasticity, featuring irregular bruised edges, tissue bridging (intact nerves, vessels across wound floor), and hair crushing.

Alternate Names:

Forensic

Laceration - Sutures Or Staples - At Home

Home care after sutures or staples includes keeping the wound clean and protected, following dressing instructions, avoiding strain, and watching for infection or wound separation.

Lacerations - Liquid Bandage

Liquid bandage is a film-forming adhesive applied over small, clean, shallow cuts to protect the wound; it should not be used on deep, infected, heavily bleeding, or gaping wounds.

Lachman Test

A knee examination in which the lower leg is gently pulled forward while the knee is slightly bent. Too much movement may suggest an injury to the anterior cruciate ligament.

Lachrymal

Relating to tears or the structures that produce and drain them; also spelled lacrimal.

Lacosamide

An antiseizure medicine used mainly for focal seizures. It may cause dizziness, double vision, poor coordination, or changes in the heart's electrical rhythm.

Lacquer Poisoning

Lacquer poisoning involves toxic or environmental exposure to lacquer poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Lacrimal

Relating to tears or the lacrimal glands and drainage passages.

Lacrimal Apparatus

The glands and passages that make, spread, and drain tears. It includes the tear gland, small openings near the inner eyelids, the tear sac, and the duct that drains tears into the nose.

Lacrimal Disease

A disorder of the tear-producing or tear-draining system, including dry eye, dacryocystitis, obstruction, inflammation, or abnormal tearing.

Lacrimal Duct

A drainage channel that carries tears from the

lacrimal sac into the nasal cavity. Blockage can cause watering of the eye, recurrent infection, swelling near the inner corner, or discharge.

Lacrimal Gland

The paired almond-shaped exocrine glands that secrete the aqueous layer of the tear film, located in the lacrimal fossa of the superior lateral orbit (orbital part) and the upper eyelid (palpebral part).

Lacrimal Gland Tumor

A benign or malignant growth arising in the tear-producing lacrimal gland, often causing swelling, pain, or displacement of the eye.

Lacrimation

Production and release of tears by the lacrimal glands; excessive tearing may result from irritation, allergy, infection, or blocked tear drainage.

Lactam

A type of chemical structure found in some medicines. Beta-lactam antibiotics, such as penicillins and cephalosporins, kill bacteria by weakening their cell walls.

Lactase

An enzyme that breaks down the milk sugar lactose in the small intestine.

Lactase Deficiency

Alternate index label; see *Lactose intolerance*.

Lactate

A product of cellular metabolism measured in blood. An increased level may occur with strenuous exercise, impaired oxygen delivery, infection, medicines, liver disease, or other conditions and must be interpreted clinically.

Lactate Dehydrogenase

An enzyme found in many body tissues. Its blood level may rise when cells are damaged, but the test does not identify the cause by itself.

Lactate Dehydrogenase Test

A blood test measuring lactate dehydrogenase, an enzyme released when cells are injured; increased activity is nonspecific and may occur with hemolysis, liver disease, muscle injury, infection, or malignancy.

Lactation

The physiological process of milk production and secretion from the mammary glands, occurring after childbirth (puerperium). See Lactation (Breastfeeding).

Lactation (Breastfeeding)

The production and release of breast milk after childbirth. Early milk, called colostrum, is followed by mature milk; hormones help make and release the milk.

Alternate Names:

Breastfeeding

Lactation Disorder

Abnormal milk production or milk transfer after childbirth, including low supply, oversupply, engorgement, mastitis, or impaired infant latch.

Lactation Mastitis

Alternate index label; see *Mastitis*.

Lactation Nutrition

Nutrition and hydration during breastfeeding sufficient to support maternal health and milk production. Requirements vary with maternal stores, dietary pattern, and the degree of breastfeeding; selected supplements may be required according to local guidance and clinical assessment.

Lactational Amenorrhea

The temporary absence of menstrual periods during breastfeeding because ovulation is suppressed. Breastfeeding prevents pregnancy reliably only when specific conditions are met, so other contraception may be needed.

Lacteals

Lymphatic vessels in intestinal villi that absorb

dietary fats.

Lactic Acid

An organic acid produced during normal metabolism, especially when pyruvate is converted to lactate. Blood lactate may rise for many reasons, including impaired oxygen delivery, strenuous exercise, or medicines.

Lactic Acid Test

A blood test measuring lactate, which accumulates when production exceeds clearance; it helps assess tissue hypoperfusion, sepsis, severe exercise, metabolic disease, or other causes of lactic acidosis.

Lactic Acidosis

Metabolic acidosis associated with increased lactate production or reduced lactate clearance. Causes include impaired tissue oxygen delivery, medicines, liver disease, seizures, and other disorders.

Lactobacillaceae

Lactobacillaceae is a family of bacteria that includes many lactic-acid-producing organisms found in food and normal human microbiota; most are harmless or beneficial.

Lactobacillus

A genus of bacteria that produces lactic acid and normally forms part of the healthy vaginal and intestinal microbiota; some species are used as probiotics.

Lactobacillus Acidophilus

A lactic-acid-producing bacterium found in some fermented foods and probiotic products; it normally inhabits parts of the human body but is rarely opportunistically invasive.

Lactobacillus Brevis

A lactic-acid-producing bacterium found in fermented foods and some body sites; it rarely causes opportunistic infection.

Lactobacillus Casei

A lactic-acid-producing bacterium found in fermented foods and the human intestine; it is used in some probiotic products and rarely causes infection.

Lactobacillus Fermentum

A lactic-acid-producing bacterium found in fermented foods and the human microbiome that can rarely cause opportunistic infection.

Lactobacillus Gallinarum

A lactic-acid-producing bacterial species found in animals and occasionally in human specimens; human infection is rare.

Lactobacillus Gasseri

A lactic-acid-producing bacterium found in the gastrointestinal and genital microbiota and used in some probiotic products.

Lactobacillus Jensenii

A lactic-acid-producing bacterium associated with the vaginal microbiota that may help maintain an acidic environment.

Lactobacillus Johnsonii

A lactic-acid-producing bacterium found mainly in the gastrointestinal tract and studied as a probiotic organism.

Lactobacillus Oris

A lactic-acid-producing bacterium found in the mouth and occasionally associated with opportunistic infection.

Lactobacillus Paracasei

A lactic-acid-producing bacterium found in fermented foods and the intestine and used in some probiotic preparations.

Lactobacillus Rhamnosus

A lactic-acid-producing bacterium used in some probiotic products; it is usually harmless but can rarely cause invasive infection.

Lactobacillus Salivarius

A lactic-acid-producing bacterium found in the mouth and intestine and studied for possible probiotic effects.

Lactobacillus Vaginalis

A lactic-acid-producing bacterium associated with the vaginal microbiota; it is usually a colonizer rather than a pathogen.

Lactococcus

A genus of lactic-acid-producing bacteria used in dairy fermentation; some species rarely cause opportunistic infection.

Lactococcus Garvieae

A lactic-acid-producing bacterium that mainly affects fish but can rarely cause infection in humans, including bloodstream or endocardial infection.

Lactococcus Lactis

A lactic-acid-producing bacterium widely used to make cheese and other fermented dairy foods; human infection is rare.

Lactoferrin

Lactoferrin is an iron-binding glycoprotein in milk and mucosal secretions that limits microbial iron access and contributes to innate immune defense.

Lactone

A cyclic ester formed when a hydroxy acid undergoes internal esterification; lactone structures occur in many natural products and medicines.

Lactose

The principal sugar in milk, composed of glucose and galactose. Lactase in the small intestine digests it.

Lactose Fermentation

The ability of bacteria to break down lactose and produce acid or gas. It is used in the laboratory to help identify different types of bacteria.

Lactose Intolerance

A condition in which the body cannot fully digest lactose, the sugar in milk and dairy products, because it does not make enough lactase. It may cause bloating, gas, abdominal pain, nausea, or diarrhea after consuming lactose.

Alternate Names:

Lactase Deficiency Milk Intolerance Milk Intolerance (Lactose Intolerance)

Lactose Tolerance Test

Non-invasive test for carbohydrate malabsorption (lactose, fructose) and small intestinal bacterial overgrowth (SIBO). See Hydrogen Breath Test.

Lactose Tolerance Tests

Tests that evaluate digestion and absorption of lactose by measuring blood glucose response after lactose ingestion or detecting hydrogen in exhaled breath; they may support evaluation of lactose malabsorption.

Lactulose

A medicine that softens stool by drawing water into the bowel. It is also used to lower harmful ammonia levels in some people with severe liver disease. Gas and diarrhea can occur.

Lacuna

A small space or cavity in tissue, such as a space containing an osteocyte in bone.

Lacunar Stroke

An ischemic stroke caused by blockage of a small artery deep within the brain.

Ladd Bands

Congenital, anomalous peritoneal fibrous bands running across the second and third portions of the duodenum from the abnormally placed cecum and ascending colon to the right posterior abdominal wall, associated with intestinal malrotation. They cause extrinsic duodenal compression, obstruction, and bilious emesis in infants; division of these bands is an integral step of the Ladd procedure.

Alternate Names:

Ladd's Peritoneal Bands; Ladd Fibrous Bands

Ladd Procedure

The classic emergency pediatric surgical operation for intestinal malrotation and midgut volvulus. The procedure consists of four sequential steps: counter-clockwise derotation of the midgut volvulus, surgical division of Ladd bands and relief of duodenal compression, widening of the narrow mesenteric base to prevent future volvulus, and incidental appendectomy (as the mobile cecum is relocated into the left lower quadrant).

Alternate Names:

Ladd Operation; Ladd's Procedure

Laennec's Cirrhosis

An older term for alcohol-associated cirrhosis, characterized by diffuse liver fibrosis and regenerative nodules.

Laetrile

A semisynthetic amygdalin-related compound promoted as an alternative cancer treatment; credible evidence has not established anticancer benefit, and cyanide poisoning is a serious risk.

Lafora Disease

A progressive autosomal-recessive myoclonic epilepsy caused by pathogenic variants affecting glycogen metabolism. It usually begins in adolescence with generalized seizures, visual symptoms, myoclonus, cognitive decline, and progressive neurological deterioration.

Lagophthalmos

Incomplete closure of the eyelids, which can expose and dry the cornea.

LAMB Syndrome

An older acronym for lentigines, atrial myxomas, mucocutaneous myxomas, and blue nevi, findings within the spectrum of Carney complex.

Lambda

The junction of the sagittal and lambdoid sutures of the skull; it is also a symbol used in medical notation.

Lambert-Eaton Myasthenic Syndrome

A rare condition in which the immune system interferes with nerve signals to muscles. It causes weakness, especially in the hips and shoulders, reduced reflexes, and sometimes dry mouth. It can be associated with small-cell lung cancer.

Lamellar

Arranged in, composed of, or characterized by thin plates, scales, or concentric parallel sheets (lamellae). In osteology, lamellar bone represents mature, mechanically strong, highly organized bone tissue composed of parallel or concentric collagen sheets with alternating fibril orientations, contrasting with disorganized immature woven bone. In histology, also describes lamellar granules (Odland bodies in keratinocytes of the stratum granulosum that discharge lipid barrier sheets) and lamellar corpuscles (Pacinian corpuscles, encapsulated mechanoreceptors sensitive to vibration and deep pressure).

Lamellar Body

A membrane-bound secretory organelle containing layered lipid material, especially in alveolar type II cells where it releases pulmonary surfactant.

Lamellar Bone

Mature bone made of organized layers of mineralized tissue. It forms most adult compact and spongy bone and provides strength while being continually remodeled.

Lamellar Ichthyosis

An inherited skin disorder present from birth that causes dry skin covered with large, dark, plate-like scales over much of the body, including the arms and legs. It may reduce sweating and cause overheating, and can affect the hair, nails, eyes, or palms and soles.

Lamellipodium

A broad, sheet-like extension at the edge of a moving cell that helps it spread and migrate across a surface.

Lamin

A structural protein forming part of the nuclear lamina, which supports the inner surface of the cell nucleus.

Lamina

A thin plate of bone forming part of a vertebral arch.

Lamina Propria

A layer of supporting tissue beneath the surface lining of organs. It contains small blood vessels, immune cells, and other structures.

Laminectomy

An operation that removes part of a spinal bone to make more room for the spinal cord or nerves. It may be used for spinal narrowing, a tumor, or selected disk problems.

Laminin

A large extracellular-matrix protein that helps cells attach to basement membranes and influences tissue development and repair.

Lamivudine

Lamivudine is a nucleoside reverse-transcriptase inhibitor used with other antiretrovirals for HIV and as an antiviral for chronic hepatitis B; resistance and rare lactic acidosis or hepatic complications require monitoring.

Lamotrigine

Lamotrigine is an antiseizure medicine that inhibits voltage-gated sodium channels and is also used for bipolar-maintenance treatment; rapid dose escalation increases the risk of serious rash including Stevens–Johnson syndrome.

Lancet

A small, sterile, usually single-use device with a

sharp point used to puncture the skin and obtain a capillary-blood sample. Lancets are commonly used for fingertip glucose testing and other point-of-care blood tests; used lancets must be discarded in an appropriate sharps container.

Alternate Names:

Blood lancet Fingerstick lancet

Landolt Ring

A visual-acuity test symbol shaped like a ring with a gap; the person identifies the gap's direction.

Landry Ascending Paralysis

An older name for Guillain–Barré syndrome, an acute immune-mediated polyradiculoneuropathy causing progressive weakness that often begins in the legs and ascends; respiratory failure can occur.

Landry'S Paralysis

An older name for the rapidly ascending paralysis of Guillain-Barré syndrome.

Langer Lines

Natural lines of skin tension that reflect the predominant orientation of collagen fibers in the dermis and may influence surgical scar direction.

Langerhans Cell Histiocytosis

A rare disorder in which abnormal immune cells build up in one or more organs, commonly the bones, skin, lungs, lymph nodes, or pituitary gland. Symptoms and treatment depend on which organs are affected.

Langerhans Cells

Immune cells in the epidermis that capture antigens and help initiate immune responses.

Langerhans' Cell

Langerhans' cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Langerhans-Cell Histiocytosis

A group of disorders characterized by clonal proliferation of Langerhans cells (antigen-presenting dendritic cells), which can infiltrate bone, skin, and central nervous system. CNS involvement may cause diabetes insipidus, hypothalamic-pituitary dysfunction, or cerebellar degeneration.

Language Barrier

Difficulty communicating because a patient and healthcare professional do not share a language or have limited proficiency in the language used for care.

Language Development

Language development is the progressive acquisition of understanding, sounds, words, grammar, and communication during childhood.

Language Disorder

A developmental or acquired impairment in understanding, producing, or using spoken, written, or signed language that affects communication and daily function.

Language Therapy

Treatment that improves understanding, expression, reading, writing, or other language skills after developmental or acquired impairment.

Laniquidar

Laniquidar is an investigational inhibitor of drug-efflux transporters studied to overcome multidrug resistance in cancer cells.

Lanolin

Lanolin is a waxy substance derived from sheep wool used as an emollient and in topical products; allergy or irritation can occur.

Lanolin Poisoning

Lanolin poisoning involves toxic or environmental exposure to lanolin poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Lanosterol

A sterol intermediate used by cells to synthesize cholesterol and other sterols.

Lansoprazole

A medicine that strongly reduces stomach acid. It is used for heartburn, ulcers, and some *H. pylori* infections; long-term use may affect mineral or vitamin levels and infection risk.

Lanthanum

A chemical element whose carbonate salt can bind dietary phosphate and lower phosphate absorption in people with advanced kidney disease.

Lanugo

Fine, soft hair that grows all over the body of a fetus and is typically shed before birth.

Laparatomy

A misspelling of laparotomy, an operation involving incision through the abdominal wall to enter the abdominal cavity.

Laparoscope

A slender instrument with a light and camera used to view the abdominal or pelvic cavity through a small incision during minimally invasive surgery.

Laparoscopic Cholecystectomy

Surgical removal of the gallbladder through small abdominal incisions using a laparoscope and instruments, commonly performed for symptomatic gallstones or gallbladder inflammation.

Laparoscopic Colectomy

Laparoscopic colectomy is removal of part or all of the colon through small incisions using a camera and specialized instruments.

Laparoscopic Gallbladder Removal

A medical procedure, treatment, device, or care topic related to laparoscopic gallbladder removal; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Laparoscopic Gastric Banding

A bariatric operation that places an adjustable band around the upper stomach to create a small pouch and restrict food passage; it is less commonly performed than other bariatric procedures.

Laparoscopic Hysterectomy

Removal of the uterus using laparoscopic instruments through small abdominal incisions. The cervix, fallopian tubes, and ovaries may or may not also be removed; the approach and extent depend on the indication, anatomy, and surgical plan.

Laparoscopy

A type of minimally invasive surgery in which a thin camera is placed through a small cut in the abdomen. Other small cuts may be used to insert surgical instruments.

Laparotomy

An operation that uses a larger cut in the abdomen to reach the abdominal or pelvic organs.

Lapatinib

Lapatinib is a targeted cancer medicine that blocks HER2 and epidermal-growth-factor receptors and is used for selected advanced breast cancers.

Lapatinib Ditosylate

Lapatinib ditosylate is the salt form of lapatinib, a targeted medicine used for selected HER2-positive breast cancers.

Large Airway Obstruction

A blockage in the windpipe or large breathing tubes. It can cause noisy breathing, shortness of breath, and difficulty clearing mucus and may be caused by narrowing, a tumor, or weakness of the airway wall.

Large Bowel Obstruction

Mechanical blockage of the colon most commonly caused by colorectal cancer, volvulus, or diverticular stricture. Clinical presentation includes progressive abdominal distension, constipation, and obstipation. The cecum is at highest risk of perforation with dis-

tal obstruction. Diagnosis is by CT scan identifying the site and cause.

Large Bowel Resection

Surgical removal of a segment of the colon or rectum, followed by reconnection of the bowel or creation of a stoma when necessary to treat cancer, obstruction, ischemia, inflammation, or other disease.

Large Colon

The large colon is the main portion of the large intestine that absorbs water, stores fecal material, and moves contents toward the rectum.

Large For Gestational Age

Birth weight or fetal size above the 90th percentile for gestational age. It is associated with maternal diabetes, excessive maternal weight gain, and genetic or constitutional factors and increases the risk of shoulder dystocia, birth trauma, and neonatal hypoglycemia.

Large For Gestational Age

A fetus or newborn whose estimated or measured weight is above the expected range for gestational age, commonly above the 90th percentile.

Large Intestine

The final part of the digestive tract, extending from the small intestine to the rectum. It absorbs water and helps form and move stool.

Large Solitary Luteinized Follicle Cyst Of Pregnancy

A benign ovarian cystic lesion with prominent luteinization, usually occurring during pregnancy and often discovered incidentally or as an adnexal mass.

Laromustine

Laromustine is an investigational alkylating antineoplastic agent studied for DNA damage and cancer-cell killing.

Laron Syndrome

An inherited disorder of growth-hormone signaling, usually caused by growth-hormone-receptor defects. It produces severe postnatal growth failure, characteristic facial features, and very low insulin-like growth factor 1 despite elevated circulating growth hormone.

Larotaxel

Larotaxel is an investigational taxane that disrupts microtubules and has been studied for anticancer activity.

Larva

An immature life stage of an animal that differs in form from the adult and later undergoes development or metamorphosis.

Larva Currens From Strongyloides

A rapidly moving, itchy skin eruption caused by larvae of *Strongyloides stercoralis*.

Larva Migrans

A condition caused by migrating larval parasites in human tissues, including cutaneous larva migrans in the skin and visceral larva migrans in internal organs.

Laryngeal

Relating to the larynx, the organ in the throat containing the vocal cords and helping protect the airway.

Laryngeal Cancer

Malignant tumor arising in the larynx, most commonly squamous cell carcinoma. Risk factors include tobacco, alcohol, and selected HPV-related disease; persistent hoarseness, dysphagia, odynophagia, stridor, or neck mass requires laryngoscopic evaluation and biopsy.

Laryngeal Candidiasis

Fungal infection of the larynx by *Candida* species, causing hoarseness, dysphagia, and throat pain, most commonly in immunocompromised patients

or those using inhaled corticosteroids.

Laryngeal Carcinoma

Laryngeal carcinoma is cancer arising in the voice box and may cause hoarseness, swallowing difficulty, pain, or breathing problems.

Laryngeal Cartilage

Laryngeal cartilage forms the structural framework of the voice box, maintains the airway, and supports vocal-fold movement.

Laryngeal Cleft

A laryngeal cleft is a congenital opening between the airway and esophagus that can allow aspiration and cause feeding or breathing problems.

Laryngeal Edema

Swelling of the tissues of the larynx that can narrow the upper airway and cause hoarseness, stridor, or respiratory distress. It may result from allergy, infection, trauma, or airway instrumentation.

Laryngeal Fistula

A laryngeal fistula is an abnormal passage connecting the larynx with nearby tissue or the digestive tract and may cause aspiration or infection.

Laryngeal Hemorrhage

Bleeding within or around the larynx, which may result from trauma, instrumentation, vocal strain, vascular lesions, infection, or tumor and can threaten the airway.

Laryngeal Masks

Laryngeal masks are supraglottic airway devices placed above the vocal cords to maintain ventilation during anesthesia or emergencies.

Laryngeal Mucositis

Laryngeal mucositis is inflammation and ulceration of the laryngeal lining, often after radiation, chemotherapy, infection, or irritation.

Laryngeal Nerve

laryngeal nerve refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Laryngeal Nerve Damage

Injury to the recurrent or superior laryngeal nerves causing hoarseness, vocal-cord weakness, swallowing difficulty, or airway compromise. Common causes include thyroid or neck surgery, malignancy, trauma, and compression along the nerve pathway.

Laryngeal Obstruction

Partial or complete blockage of the laryngeal airway caused by edema, foreign body, tumor, trauma, infection, vocal-cord dysfunction, or other structural disease.

Laryngeal Prosthesis

A device used to help restore speech after removal of the larynx, commonly through a one-way valve placed between the trachea and esophagus.

Laryngeal Stridor

A harsh, high-pitched noise caused by turbulent airflow through a narrowed upper airway, often most noticeable during inspiration.

Laryngeal Warts (Recurrent Respiratory Papillomatosis)

Benign papillomatous growths caused by HPV in the larynx or respiratory tract. They may cause hoarseness, stridor, or airway obstruction.

Alternate Names:

Recurrent Respiratory Papillomatosis

Laryngeal Web

A laryngeal web is a thin membrane across part of the airway, usually congenital or acquired after injury, causing hoarseness or airway narrowing.

Laryngectomy

A person who has undergone laryngectomy, surgical removal of all or part of the larynx; total laryn-

gectomy separates the airway from the mouth and nose.

Laryngectomy

Surgical removal of part or all of the larynx, performed for selected laryngeal cancers or severe structural disease; total laryngectomy separates the airway from the mouth and creates a permanent neck stoma. the larynx or hypopharynx and less often for severe trauma or refractory aspiration. Total laryngectomy requires a permanent tracheostomy, so the patient breathes through a neck stoma; partial laryngectomy preserves voice and swallowing.

Laryngectomy Tube

Tube maintaining airway after total laryngectomy. Types: silicone, metal. Indications: laryngeal cancer, trauma. Complications: obstruction, infection, granulation.

Laryngismus Stridulus

An older term for brief laryngeal spasm causing inspiratory stridor, historically associated with childhood spasmodic croup or tetany.

Laryngitis

Laryngitis is inflammation of the larynx causing hoarseness or voice loss, commonly from viral infection, vocal overuse, irritation, reflux, or less often structural disease. sore throat, dry cough, and a sensation of a lump in the throat (globus sensation), and it may be acute or chronic.

Laryngocele

Abnormal air-filled dilation of the laryngeal saccule that may be internal, external, or combined, presenting with hoarseness, neck swelling, or airway obstruction.

Laryngomalacia

Dynamic collapse of supraglottic laryngeal tissue during inspiration, causing congenital stridor in infants. Most cases improve with growth, but feeding difficulty, apnea, hypoxemia, or failure to thrive requires specialist assessment and possible intervention.

Laryngopharyngeal Reflux

Retrograde movement of gastric contents above the upper esophageal sphincter into the larynx and pharynx, producing chronic hoarseness, throat clearing, globus sensation, cough, excessive throat mucus, and laryngospasm, often without heartburn.

Laryngopharynx

The lower part of the pharynx that connects the throat with the larynx and esophagus.

Laryngoscope

Device for visualizing vocal cords during intubation. Blades: Mac (curved), Miller (straight).

Laryngoscopy

A medical procedure used to visualise the larynx (voice box) and adjacent structures of the throat. Types: (1) Indirect laryngoscopy: the simplest technique, using a laryngeal mirror angled at 120 degrees to reflect the laryngeal image, with a headlight or external light source.

Laryngoscopy And Nasolaryngoscopy

Endoscopic examination of the larynx and adjacent upper airway, using a rigid or flexible scope passed through the mouth or nose to assess voice, swallowing, airway, and structural abnormalities.

Laryngospasm

A protective reflex causing involuntary adduction of the vocal cords, producing partial or complete airway obstruction. It is common during light anesthesia, emergence, and in pediatric patients. Triggers include secretions, blood, airway manipulation, and suctioning.

Laryngostenosis

Narrowing of the voice box, or larynx, that can restrict breathing. Causes include scarring after intubation, injury, inflammation, tumors, and congenital narrowing; noisy breathing or breathlessness needs medical assessment.

Laryngotracheobronchitis (Croup)

Alternate index label for Croup.

Larynx

The larynx, commonly known as the voice box, is a complex organ located in the anterior neck between the pharynx and the trachea that serves three primary functions: phonation (voice production), protection of the lower airway from aspiration during swallowing, and regulation of airflow during respiration.

Larynx Cancer

Laryngeal cancer is a malignant tumor of the voice box and may cause persistent hoarseness, swallowing pain, neck mass, ear pain, cough, or breathing difficulty. Diagnosis requires examination and biopsy; treatment depends on site and stage.

Alternate Names:

Throat Cancer (Larynx Cancer)

Larynx Disease

Any disorder affecting the larynx, including inflammation, infection, structural lesions, vocal-cord dysfunction, neurologic disease, or cancer.

Larynx Muscle

larynx muscle refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Larynx Neoplasm

A larynx neoplasm is an abnormal growth in the voice box that may be benign or malignant.

Laser

A device that produces a concentrated beam of light used for diagnosis, treatment, or surgery.

Laser (Vascular)

A catheter-based laser system used to modify or remove selected vascular lesions or obstructive tissue.

Alternate Names:

Vascular

Laser Ablation

The use of focused laser energy to heat, destroy, or seal targeted tissue. The technique is used in selected vascular, ophthalmic, dermatologic, and other procedures.

Laser Assisted Uvula Palatoplasty

A procedure that reshapes or removes selected soft-palate and uvula tissue to reduce snoring.

Alternate Names:

LAUP

Laser Capture Microdissection

A laboratory method that uses a laser to isolate selected cells or tissue regions for microscopic, genetic, or molecular analysis.

Laser Dentistry

Use of lasers for various procedures: caries detection, soft tissue surgery, whitening, periodontal therapy.

Laser Fiber

Optical fiber for laser delivery in surgery. Used in lithotripsy, tumor ablation.

Laser Hair Removal

Hair reduction using laser energy absorbed by pigment in the hair follicle.

Laser Photocoagulation

A laser treatment that seals or destroys abnormal blood vessels, especially in the retina.

Laser Photocoagulation - Eye

A medical procedure, device, or treatment involving laser photocoagulation - eye. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Laser Prostatectomy

A procedure that uses laser energy to remove or vaporize enlarged prostate tissue.

Laser Surgery

Laser surgery uses a focused beam of light to cut, re-

move, seal, or reshape tissue in selected procedures.

Laser Surgery For The Skin

A medical procedure, treatment, device, or care topic related to laser surgery for the skin; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Laser Therapy

Laser therapy uses controlled light energy to treat selected skin, eye, vascular, dental, or other medical conditions.

Laser Therapy For Cancer

A medical procedure, treatment, device, or care topic related to laser therapy for cancer; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Laser Trabeculoplasty

A minimally invasive laser procedure used in the treatment of open-angle glaucoma that lowers intraocular pressure by improving the outflow of aqueous humor through the trabecular meshwork.

Laser-Assisted In Situ Keratomileusis

A refractive surgical procedure that reshapes the cornea using an excimer laser to correct myopia, hyperopia, and astigmatism, thereby reducing dependence on glasses or contact lenses.

Alternate Names:

Laser-Assisted In Situ Keratomileusis

LASIK

LASIK, or laser-assisted in situ keratomileusis, is a refractive surgical procedure that corrects myopia, hyperopia, and astigmatism by reshaping the corneal stroma using an excimer laser, after creating a thin corneal flap with either a microkeratome (a mechanical blade) or a femtosecond laser.

LASIK Eye Surgery

A medical procedure, treatment, device, or care topic related to lasik eye surgery; indications, benefits, risks, and follow-up depend on the specific

intervention and patient.

Lassa Fever

An acute viral hemorrhagic illness caused by Lassa virus, an arenavirus, endemic in West Africa (Nigeria, Sierra Leone, Guinea, Liberia). Transmission: exposure to excreta (urine, feces) of the multimammate rat (*Mastomys natalensis*), the natural reservoir. Incubation: 6-21 days.

Lassa Virus

An arenavirus that causes Lassa fever, a potentially severe hemorrhagic illness acquired mainly through contact with infected rodent excreta or body fluids.

Lassitude

A state of marked fatigue, weakness, or lack of energy that may occur with systemic illness, anemia, endocrine disease, infection, medication effects, or psychological conditions.

Last Menstrual Period

The first day of the last normal menstrual period, used to estimate gestational age and the due date. Requires regular menstrual cycles and reliable recall. Limitations: irregular cycles, recent oral contraceptive use, early pregnancy bleeding mistaken for menses, and poor recall.

Latanoprost

Latanoprost is a prostaglandin F2-alpha analogue eye drop that increases uveoscleral outflow and lowers intraocular pressure in glaucoma or ocular hypertension; iris pigmentation and eyelash growth may occur.

Latarjet Procedure

Surgical procedure for shoulder instability using coracoid process transfer. Provides bony block to prevent recurrent dislocation.

Late Decelerations

A fetal heart rate pattern with a gradual decrease (onset to nadir >30 seconds) that starts at or after the peak of the contraction and returns to baseline

after the contraction ends. Associated with uteroplacental insufficiency (reduced oxygen transfer across the placenta).

Late Effects

Health problems occurring months to years after cancer treatment. Cardiac, pulmonary, endocrine, secondary cancers.

Late Endosome

A late endosome is a membrane-bound cell compartment that receives cargo from early endosomes and matures toward lysosomal degradation.

Late-Onset Hypogonadism

A clinical and biochemical syndrome occurring in aging men characterized by low circulating serum total or free testosterone levels accompanied by typical systemic symptoms of androgen deficiency. Common clinical features include decreased sexual desire (libido), reduced spontaneous morning erections, erectile dysfunction, decreased lean muscle mass and physical strength, increased visceral adiposity, osteopenia or osteoporosis, fatigue, and depressive mood changes. Diagnosis requires persistent symptoms confirmed by morning fasting total testosterone levels consistently below clinical thresholds on repeated measurements; management involves lifestyle modification, management of comorbidities (such as metabolic syndrome and sleep apnea), and consideration of testosterone replacement therapy with routine hematologic and prostate monitoring.

Alternate Names:

LOH; Age-Related Testosterone Deficiency; Testosterone Deficiency Syndrome

Late Gadolinium Enhancement

A specialized tissue characterization technique in cardiac magnetic resonance imaging performed 10 to 20 minutes after intravenous administration of an extracellular gadolinium-based contrast agent. Gadolinium washes out slowly from expanded extracellular spaces (fibrosis, collagen deposition, necrosis, or amyloid), appearing hyperintense (bright white) against nulled, dark normal myocardium.

The distribution pattern reliably differentiates ischemic myocardial infarction (subendocardial or transmural enhancement corresponding to a coronary vascular territory) from non-ischemic cardiomyopathies (mid-wall, subepicardial, or patchy enhancement seen in myocarditis, sarcoidosis, or hypertrophic cardiomyopathy).

Alternate Names:

LGE; Delayed Contrast Enhancement; Myocardial Scar Imaging

Late Preterm (34-36 Weeks)

Births occurring from 34 weeks 0 days through 36 weeks 6 days of gestation. These infants have greater risks than term infants of respiratory, temperature, glucose, jaundice, feeding, and readmission problems.

Alternate Names:

34-36 weeks

Latent

Present but inactive, hidden, or not currently producing symptoms, as in latent tuberculosis infection or a latent virus.

Latent Autoimmune Diabetes In Adults

A form of autoimmune diabetes with adult onset, characterized by the presence of pancreatic islet autoantibodies (particularly anti-GAD65) and a progressive, gradual loss of beta-cell function. Patients initially do not require insulin at diagnosis and are often misdiagnosed with type 2 diabetes, but eventually progress to absolute insulin dependence within months to years.

Alternate Names:

Type 1.5 diabetes; Late-onset autoimmune diabetes of adulthood

Latent Infection

Persistent infection in which an infectious agent remains in a host in a nonreplicating or intermittently replicating state without active clinical disease. Reactivation can occur when immune control is impaired, as in latent tuberculosis.

Latent Phase Of Labor

The early part of the first stage of labor, beginning with regular contractions and gradual cervical change before active labor. The transition to active labor is assessed clinically, and the duration varies widely between individuals.

Latent Syphilis

Latent syphilis is syphilis infection without current symptoms, identified by testing and classified by duration to guide treatment and follow-up.

Latent Tuberculosis

Infection with *Mycobacterium tuberculosis* in which the bacteria remain inactive and cause no symptoms but may later become active.

Latent Tuberculosis Infection

An immune response to *Mycobacterium tuberculosis* without clinical or microbiologic evidence of active disease. It is usually asymptomatic and not contagious, but it can reactivate; treatment decisions depend on age, immune status, exposure risk, test results, and drug interactions.

Lateral

Lateral means toward or situated on the side away from the body's midline; it contrasts with medial and is used to describe anatomical position, movement, or imaging findings. lateral to the little finger in anatomical position.

Lateral Canthotomy And Cantholysis

An emergency procedure that cuts the outer corner of the eyelids and releases the lateral canthal tendon to reduce orbital pressure in suspected orbital compartment syndrome and protect vision.

Lateral Collateral Ligament Injury

Tear of the LCL, less common than MCL injury but more significant. Mechanism: varus stress to the knee (direct blow to medial knee). Presents with lateral knee pain, swelling, and varus laxity. Part of the posterolateral corner (PLC) — when injured, causes significant rotational instability.

Lateral Cuneiform

Located between intermediate cuneiform and cuboid, articulating with navicular, intermediate cuneiform, cuboid, and third metatarsal. Contributes to the transverse arch and lateral foot column. Injuries typically from high-energy midfoot trauma.

Lateral Decubitus Position

A side-lying position in which the patient rests on the nonoperative side to provide access to the chest, hip, or flank.

Lateral Elbow Tendinopathy

Alternate index label; see *Tennis elbow*.

Lateral Epicondylitis

Alternate index label; see *Tennis elbow*.

Lateral Epicondylitis (Tennis Elbow (Lateral Epicondylitis))

Alternate index label for Tennis Elbow (Lateral Epicondylitis).

Lateral Epicondylosis

Alternate index label; see *Tennis elbow*.

Lateral Meniscus

The lateral meniscus is a crescent-shaped cartilage pad on the outer side of the knee that distributes load and stabilizes movement.

Lateral Rotation

Rotation of a limb away from the median plane of the body.

Lateral Sclerosis

Lateral sclerosis describes hardening or degeneration involving a lateral structure, often referring to spinal motor pathways in motor-neuron disease.

Lateral Traction

Lateral traction is a pulling force applied sideways to a body part to improve alignment, reduce a dislocation, or relieve pressure.

Lateral Ventricle

A lateral ventricle is one of the paired cerebrospinal-fluid-filled cavities within a cerebral hemisphere.

Lateral X-Ray

An X-ray image taken from the side, providing a lateral view of the body part being examined.

Latex Agglutination Test

A laboratory test in which latex particles coated with antigen or antibody visibly clump when the matching target is present; it can detect selected microbial antigens, antibodies, or proteins.

Latex Allergies - For Hospital Patients

Latex allergy can cause contact dermatitis, hives, wheeze, or anaphylaxis; hospitals reduce risk with latex-free supplies and clear documentation of the allergy.

Latex Allergy

An immune-mediated hypersensitivity reaction to natural rubber latex. Contact may cause dermatitis, urticaria, rhinitis, asthma, or anaphylaxis, especially in healthcare and occupational settings.

Alternate Names:

Allergy Latex (Latex Allergy)

Lathyrism

A neurologic disorder caused by prolonged or excessive ingestion of certain Lathyrus legumes, especially grass pea, characterized primarily by progressive spastic weakness of the legs.

Latissimus Dorsi

A broad back muscle that extends and internally rotates the arm and assists movements such as climbing.

Latissimus Dorsi Muscle

Broadest back muscle forming the posterior axillary wall, from T7-L5 spinous processes, thoracolumbar fascia, iliac crest, and lower ribs to the intertubercular groove floor. Innervated by thoracodorsal nerve (C6-C8). Adducts, medially rotates, extends

the arm powerfully in swimming/rowing/climbing. Common reconstructive surgery flap.

Lats

Informal name for the latissimus dorsi muscles, broad muscles of the back that help pull the arm downward and backward and rotate it inward.

Lauric Acid

lauric acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lauroleic Acid

lauroleic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lavage

Washing or irrigating a body cavity, organ, or wound with fluid.

Lavage Fluid

lavage fluid is a diagnostic or laboratory investigation; interpretation depends on the specimen, method, reference interval, timing, and clinical question.

Lavender Oil

Lavender oil is an aromatic essential oil used in fragrances and complementary products; ingestion, skin reactions, and interactions are possible.

Lawton IADL Scale

Instrumental ADLs: phone use, shopping, food preparation, housekeeping, laundry, transportation, medication management, finance management.

Laxative Overdose

Laxative overdose involves toxic or environmental exposure to laxative overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Laxatives For Constipation

Laxatives treat constipation through fiber and water retention, stool softening, osmotic effects, or increased bowel contractions. Choice depends on cause, duration, age, medicines, kidney function, and obstruction risk; persistent bleeding, vomiting, or severe pain needs evaluation.

Alternate Names:

Constipation (Laxatives For Constipation)

Laying On Of Hands

A religious or spiritual practice in which a person places hands on another for blessing, prayer, or healing; it should not replace evidence-based medical evaluation or treatment.

Lazy Eye

Reduced vision caused by abnormal visual development during childhood, usually in one eye but sometimes in both. It may result from eye misalignment, unequal focusing, or an obstruction such as a cataract; the medical term is amblyopia.

Alternate Names:

amblyopia.

Lazy Eye (Amblyopia)

Reduced vision in one or both eyes caused by abnormal visual development during childhood, often related to misalignment, unequal focusing, or visual obstruction.

Lb.

Abbreviation for pound, a unit of mass equal to 16 ounces or approximately 0.454 kilograms; medication weights should use unambiguous units.

LCIS

Alternate index label; see *Lobular carcinoma in situ*.

LD50

Lethal dose for 50% of test population. Measure of acute toxicity. Lower LD50 = more toxic. Measured in mg/kg body weight. Ethical concerns limit animal testing.

LDH Isoenzyme Blood Test

A blood test separating lactate dehydrogenase isoenzymes to help suggest the tissue source of an elevated total LDH, although more specific tests are preferred for most conditions.

LDL

Low-density lipoprotein; the 'bad' cholesterol that transports cholesterol to peripheral tissues and is the principal driver of atherosclerosis. Elevated LDL is a major modifiable cardiovascular risk factor; guidelines target LDL levels based on global risk (statin intensity).

LDL Test

A blood test measuring low-density lipoprotein cholesterol, a major atherogenic lipoprotein used with the lipid profile to estimate cardiovascular risk and guide lipid-lowering treatment.

Lead

A toxic heavy metal that can harm the nervous system, blood-forming system, kidneys, cardiovascular system, and reproductive system. Exposure may occur through contaminated paint dust or soil, drinking water, workplaces, hobbies, imported products, or other sources; children and fetuses are especially vulnerable to neurodevelopmental effects.

Lead - Nutritional Considerations

Adequate iron, calcium, and vitamin C intake may help reduce lead absorption in people who are deficient, especially children, but nutrition does not remove lead from the body or replace exposure control. Suspected lead exposure requires evaluation of the source and appropriate blood lead testing.

Lead And Tap Water

Drinking water can become contaminated with lead when it contacts lead-containing service lines, plumbing, fixtures, or solder. Risk reduction may require testing, use of an appropriately certified filter, flushing or other interim measures, and repair or replacement of the source according to local public-health guidance; boiling water does not remove lead.

Lead Levels - Blood

A blood lead test measures the amount of lead in blood, usually reported in micrograms per deciliter (µg/dL). Results are interpreted with age, symptoms, exposure history, and testing method; an elevated capillary result may require confirmation with a venous sample. Management focuses on removing the exposure and arranging medical or public-health follow-up, with chelation reserved for selected cases.

Lead Poisoning

Toxic effects caused by lead exposure. Lead can affect the nervous system, cognition and behavior, child development, blood formation, gastrointestinal tract, kidneys, blood pressure, and reproductive health. Common sources include deteriorating lead paint and dust, contaminated soil or water, occupational or hobby exposure, and some imported products. Evaluation includes exposure assessment and blood lead testing, with venous confirmation when indicated. Treatment prioritizes removing the exposure and correcting nutritional deficiencies; specialist-directed chelation is used only for selected patients based on blood lead level, symptoms, age, and other clinical factors.

Lead-Time Bias

The appearance of improved survival after screening because diagnosis occurs earlier, even when the time of death is unchanged. It can make screening appear beneficial without reducing mortality.

Learn About Gluten-Free Diets

A gluten-free diet excludes wheat, barley, rye, and their derivatives and is essential for people with celiac disease; dietitian guidance helps prevent nutritional gaps.

Learn To Love Exercise

Building a regular exercise habit involves choosing enjoyable activities, starting gradually, setting realistic goals, and adjusting activity for health conditions.

Learn To Manage Stress

Stress management includes regular sleep, physical

activity, relaxation skills, social support, and professional help when stress causes persistent distress or impairment.

Learn To Manage Your Anger

Anger management uses techniques such as recognizing triggers, pausing before reacting, communicating clearly, and seeking counseling when anger becomes unsafe or disruptive.

Learned Insomnia

Persistent difficulty sleeping that is maintained by anxiety and conditioned behaviors associated with sleep.

Learning About Depression

Depression is a persistent disorder of mood and interest that can affect sleep, energy, thinking, appetite, and functioning; it is treatable with professional care.

Learning About Ventilators

A ventilator supports or replaces breathing by delivering controlled airflow and oxygen; understanding alarms, tubes, sedation, and weaning helps families participate in care.

Learning Difficulties

Persistent problems acquiring or using academic skills such as reading, writing, or mathematics, not explained solely by low intelligence or inadequate education.

Learning Disabilities

Persistent difficulties acquiring or using skills such as reading, writing, mathematics, language, attention, or coordination despite appropriate opportunity and instruction; assessment considers development, education, and other neurologic or sensory factors.

Learning Disorder

A neurodevelopmental disorder causing persistent difficulty acquiring or using academic skills such as reading, writing, or mathematics despite appropriate instruction.

Leber Congenital Amaurosis

The most severe inherited retinal dystrophy, presenting at birth or within the first few months of life with profound visual impairment, nystagmus, and pupillary light-near dissociation.

Leber Hereditary Optic Neuropathy

A maternally inherited mitochondrial genetic disorder caused by primary point mutations in the mitochondrial genome (mtDNA) encoding subunits of respiratory Complex I (NADH dehydrogenase), most commonly *m.11778G>A* (*MT-ND4*, 70%), *m.3460G>A* (*MT-ND1*), or *m.14484T>C* (*MT-ND6*). Classically presents in young adult males (male-to-female ratio 4:1) with rapid, painless, sequential bilateral central vision loss, profound central scotomas, impaired color vision, and telangiectatic microangiopathy around the optic disc, progressing to permanent bilateral optic atrophy.

Alternate Names:

LHON

Lecithin

A group of phospholipids found in cell membranes and foods; lecithin is also used as an emulsifier and studied in nutrition and medicine.

Leclercia

Leclercia is a genus of bacteria found in the environment and occasionally associated with opportunistic human infections.

Leclercia Adecarboxylata

Leclercia adecarboxylata is an uncommon bacterium that can cause opportunistic bloodstream, wound, urinary, or other infections.

Lectin

A protein or carbohydrate-binding molecule that recognizes specific sugar structures on cells or other molecules.

LED Photomodulation

Use of low-energy light-emitting diodes to influence skin cells and improve selected skin concerns.

Alternate Names:

LED light therapy

LED Therapy

LED therapy uses specific wavelengths of light for selected skin or wound-care applications; effectiveness depends on the condition and treatment protocol.

Leech

A blood-feeding annelid used in selected reconstructive procedures to relieve venous congestion; leech bites can cause bleeding and infection.

Leech Therapy

A medical procedure, treatment, device, or care topic related to leech therapy; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Leeche

An obsolete or variant spelling of leech, a segmented blood-feeding worm used in selected medical treatments.

Leflunomide

Leflunomide is a disease-modifying antirheumatic drug that inhibits dihydroorotate dehydrogenase and lymphocyte proliferation; it treats rheumatoid or psoriatic arthritis but can cause hepatotoxicity, cytopenias, hypertension, and fetal harm.

Left Anterior Descending Artery

A major branch of the left coronary artery that travels in the anterior interventricular groove and supplies the anterior left ventricle, much of the interventricular septum, and the apex. Its occlusion can cause extensive anterior myocardial infarction.

Left Anterior Descending Coronary Artery

A major coronary artery supplying the front wall and septum of the left ventricle.

Alternate Names:

LAD artery

Left Atrial Appendage Occlusion

A catheter-based percutaneous intervention used as a non-pharmacological alternative for stroke prevention in patients with non-valvular atrial fibrillation who have a high thromboembolic risk and contraindications to long-term oral anticoagulation. Under transesophageal echocardiographic and fluoroscopic guidance via transseptal puncture, a dedicated self-expanding nitinol occlusion device (such as the Watchman or Amulet device) is deployed into the orifice of the left atrial appendage, mechanically sealing the source of over 90 percent of cardiac thrombi in non-valvular atrial fibrillation.

Alternate Names:

LAAO; Left Atrial Appendage Closure; Watchman Procedure

Left Atrial Myxoma

A usually benign primary cardiac tumor, commonly arising from the interatrial septum in the left atrium. It may obstruct mitral inflow or cause embolic and constitutional symptoms such as fever, weight loss, or fatigue.

Left Atrium

Receives oxygenated blood from 4 pulmonary veins. Most posterior chamber forming the cardiac base on chest radiography. Smooth walls from absorbed pulmonary veins; small auricle with pectinate muscles. Mitral valve separates it from the left ventricle.

Left Bundle Branch

A continuation of the bundle of His that divides into anterior and posterior fascicles. The anterior fascicle supplies the anterior and superior left ventricular wall; the posterior fascicle supplies the inferior and posterior wall.

Left Circumflex Coronary Artery

A coronary artery that curves around the left side and back of the heart.

Alternate Names:

Circumflex artery

Left Heart Catheterization

A medical procedure, device, or treatment involving

left heart catheterization. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Left Heart Ventricular Angiography

An imaging study that injects contrast into the left ventricle to assess ventricular contraction, wall motion, ejection fraction, and selected structural abnormalities.

Left Hemiparesis, Gaze Deviated To The Right

A clinical finding indicating a right hemisphere lesion, likely a right middle cerebral artery stroke. Gaze deviation toward the right suggests the lesion is in the right frontal eye field, which normally drives gaze to the contralateral (left) side; when damaged, the intact left frontal eye field drives gaze ipsilaterally (to the right).

Left Hypochondriac Region

Upper left abdomen lateral to the epigastric region, containing the spleen, stomach fundus, splenic flexure, and pancreatic tail. Important for evaluating splenomegaly, gastric pathology, and splenic infarction. The spleen is normally not palpable below the costal margin; enlargement into this region is a significant finding.

Left Iliac Region

Lower left abdomen containing the descending colon, sigmoid colon, and left ureter. Important for evaluating diverticulosis and diverticulitis, most commonly affecting the sigmoid colon here. Palpation may reveal masses from inflammatory bowel disease or colorectal cancer.

Left Main Coronary Artery

The short vessel arising from the left aortic sinus and dividing principally into the left anterior descending and circumflex arteries. Significant left-main disease jeopardizes a large proportion of left-ventricular myocardium.

Left Testicular Torsion

Acute rotation of the left testis on its spermatic

cord, obstructing blood flow and causing ischemia. It is a urological emergency presenting with sudden severe left scrotal pain, nausea, vomiting, and a high-riding, horizontally oriented testis with absent cremasteric reflex.

Left Ventricle

Thickest cardiac chamber (13-15 mm), pumping oxygenated blood into the aorta for systemic distribution. Generates approximately 120 mmHg systolic pressure. Receives blood through the mitral valve; ejects through the aortic valve.

Left Ventricular Hypertrophy

Thickening of the muscular wall of the heart's left ventricle.

Left Ventricular Noncompaction

A rare cardiomyopathy characterized by prominent myocardial trabeculations with deep intertrabecular recesses, thought to result from arrested myocardial compaction during embryogenesis. It may present with heart failure, arrhythmias, or thromboembolism.

Left-Handed

Having a preference for using the left hand for skilled activities; handedness is a normal form of human variation.

Left-Sided Superior Vena Cava

A congenital venous anomaly where the left superior vena cava drains into the coronary sinus instead of the right atrium via the right SVC. It is the most common thoracic venous anomaly, present in 0.3–0.5% of the general population. Usually asymptomatic but relevant during central venous catheterization or cardiac surgery.

Left-Ventricular Assist Device

An implanted mechanical pump that helps the left ventricle circulate blood.

Leg

The lower limb, often meaning the region between

the knee and ankle, although it may also refer broadly to the entire lower extremity.

Leg CT Scan

Computed tomography of the leg using x-rays and computer reconstruction to evaluate bones, joints, soft tissues, blood vessels, tumors, infection, and trauma.

Leg Fracture

Alternate index label; see *Broken leg*.

Leg Lengthening And Shortening

Orthopedic procedures that gradually increase or decrease the length of a bone to correct limb-length discrepancy or deformity, usually using specialized fixation devices.

Leg MRI Scan

Magnetic resonance imaging of the leg using a magnetic field and radio waves to evaluate muscles, tendons, ligaments, nerves, marrow, joints, and soft-tissue masses.

Leg Or Foot Amputation

A medical procedure, device, or treatment involving leg or foot amputation. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Leg Or Foot Amputation - Dressing Change

A medical procedure, device, or treatment involving leg or foot amputation - dressing change. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Leg Pain

Pain in the lower limb caused by muscle or tendon injury, joint disease, nerve problems, vascular disease, infection, or other conditions.

Leg Pain With Cramping

Painful tightening or spasms in a leg muscle; causes include exertion, dehydration, medicines, nerve or

muscle disorders, and reduced arterial blood flow causing intermittent claudication.

Leg Swelling

Enlargement or puffiness of one or both legs caused by fluid retention, venous disease, injury, infection, inflammation, or a blood clot.

Leg Ulcer

An open sore on the leg that fails to heal normally, often caused by poor venous circulation, arterial disease, diabetes, pressure, or inflammation. Pain, spreading redness, fever, or black tissue requires prompt assessment.

Legg-Calvé-Perthes Disease

Idiopathic avascular necrosis of the femoral head in children, typically ages 4-8 years. More common in boys (4:1) and Caucasian children. Presents with limping (antalgic gait), hip/knee pain, and limited hip range of motion (especially abduction and internal rotation). Etiology: temporary disruption of blood supply to the femoral head.

Legg-Calvé-Perthes Disease

A childhood disorder caused by temporary loss of blood supply to the femoral head, resulting in avascular necrosis and hip pain or limp.

Legg-Calvé-Perthes Disease (Perthes Disease)

A childhood disorder in which the blood supply to the femoral head is temporarily impaired, causing avascular necrosis, hip pain, stiffness, and possible deformity.

Alternate Names:

Perthes Disease

Legg-Calvé-Perthes Disease

Idiopathic avascular necrosis of the femoral head in children, typically between ages two and twelve, resulting from disrupted blood supply to the capital femoral epiphysis. It presents with insidious onset of hip pain, limp, and restricted range of motion.

Legionella

Legionella is a group of water-associated bacteria that can cause Legionnaires' disease, a severe pneumonia, or Pontiac fever, a milder illness.

Legionella Longbeachae

Legionella longbeachae is a Legionella species associated particularly with exposure to contaminated potting mix and capable of causing pneumonia.

Legionella Pneumonia

From Legionella pneumophila in water systems. Presents with high fever, diarrhea, confusion, and hyponatremia. Diagnosis by urinary antigen test. Treatment is fluoroquinolones or macrolides.

Legionella Pneumonia (Legionnaire Disease And Pontiac Fever)

Alternate index label for Legionnaire Disease and Pontiac Fever.

Legionella Pneumophila

A gram-negative rod that causes Legionnaires' disease, a severe pneumonia, and Pontiac fever, a milder illness. The organism is found in aquatic environments and is transmitted through inhalation of contaminated water droplets. Risk factors include age, smoking, and immunosuppression.

Legionellosis

An infection caused by Legionella bacteria, ranging from a mild flu-like illness to Legionnaires' disease with pneumonia.

Legionnaire Disease And Pontiac Fever

Legionnaires disease is a severe pneumonia caused by Legionella, while Pontiac fever is a milder flu-like illness from the same bacteria. Exposure to contaminated water systems is typical; pneumonia, confusion, low oxygen, or high fever requires urgent treatment.

Alternate Names:

Legionella Pneumonia (Legionnaire Disease And Pontiac Fever) Legionnaires Disease (Legionnaire Disease And Pontiac Fever)

Legionnaires Disease

A severe pneumonia caused by Legionella bacteria, usually acquired by inhaling contaminated water aerosols. It may also cause gastrointestinal symptoms, confusion, and low serum sodium.

Legionnaires Disease (Legionnaire Disease And Pontiac Fever)

Alternate index label for Legionnaire Disease and Pontiac Fever.

Legionnaires' Disease

A severe form of pneumonia caused by Legionella pneumophila, a Gram-negative aerobic bacillus that is found naturally in freshwater environments (lakes, streams) and artificially in water systems (air conditioning cooling towers, hot tubs, decorative fountains, plumbing systems, showers, humidifiers).

Legs Restless (Restless Leg Syndrome)

Alternate index label for Restless Leg Syndrome.

Leifsonia

A genus of gram-positive bacteria found mainly in soil and plants; human infection is rare.

Leifsonia Aquatica

A water-associated gram-positive bacterium rarely isolated from human specimens and only occasionally linked to opportunistic infection.

Leigh Disease

A progressive mitochondrial disorder of early life characterized by symmetric lesions of the brainstem and basal ganglia, developmental regression, movement abnormalities, and lactic acidosis.

Leiomyoma

Uterine fibroid: a benign smooth muscle tumor of the uterus (estrogen/progesterone-sensitive), the most common pelvic tumor in women. Locations: subserosal, intramural, submucosal (most symptomatic).

Leiomyoma (Fibroid)

A benign tumor of uterine smooth muscle that may cause heavy menstrual bleeding, pelvic pressure, pain, or no symptoms.

Alternate Names:

Fibroid

Leiomyomatosis

Leiomyomatosis is development of multiple benign smooth-muscle tumors or proliferations, which may affect the uterus, skin, lungs, or other sites.

Leiomyomatosis Peritonealis Disseminata

A rare benign condition with multiple smooth-muscle nodules scattered over the peritoneum, often associated with pregnancy or hormonal stimulation and sometimes mimicking peritoneal carcinomatosis.

Leiomyosarcoma

A malignant tumor of smooth muscle, most often arising in the uterus, retroperitoneum, or gastrointestinal tract.

Leiomyosarcoma Of The Vulva

A rare malignant smooth-muscle tumor of the vulva, usually presenting as a solitary enlarging mass. Histologic atypia, mitotic activity, necrosis, and invasion help determine malignant behavior.

Leishmania

leishmania is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Leishmania Chagasi

leishmania chagasi is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Leishmania Donovanii

A protozoan parasite that causes visceral leishmaniasis (kala-azar), transmitted by the bite of infected

female phlebotomine sandflies. The organism exists as flagellated promastigotes in the sandfly gut and as non-flagellated amastigotes within human macrophages.

Leishmania Infection

A parasitic infection caused by *Leishmania* protozoa and transmitted by infected sandflies; it can cause cutaneous, mucocutaneous, or visceral leishmaniasis.

Leishmania Major

leishmania major is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Leishmania Mexicana

leishmania mexicana is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Leishmania Tropica

leishmania tropica is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Leishmaniasis

Leishmaniasis is infection by *Leishmania* parasites transmitted by sandflies, producing cutaneous, mucosal, or visceral disease depending on species and host response. of infected female phlebotomine sandflies. Visceral leishmaniasis (kala-azar): caused by *L. donovani*/*L. infantum* – fever, hepatosplenomegaly, pancytopenia, weight loss, hypergammaglobulinaemia; fatal if untreated.

Lelliottia Amnigena

A gram-negative environmental bacterium in the Enterobacterales group that can rarely cause opportunistic human infection.

Lemaitre'S Sign

Pain or discomfort in the urethra or penis during digital rectal examination, traditionally associated with posterior urethral or prostatic inflammation.

Lemierre Syndrome

A potentially severe oropharyngeal infection complicated by septic thrombophlebitis of the internal jugular vein and frequently metastatic infection, classically caused by *Fusobacterium necrophorum*.

Leminorella Grimontii

A rare gram-negative bacterium that may be isolated from clinical specimens and can occasionally cause opportunistic infection.

Leminorella Richardii

A rare gram-negative bacterium occasionally recovered from human clinical specimens.

Lenalidomide

Lenalidomide is an immunomodulatory anticancer drug used for multiple myeloma and selected myelodysplastic or lymphoma disorders; myelosuppression, thrombosis, infection, and severe teratogenicity are important risks.

Length-Time Bias

The tendency of screening to preferentially detect slower-growing or less aggressive disease because it remains detectable for a longer preclinical period. It can exaggerate the apparent benefit of screening.

Lengthening Of Limb

A surgical or orthopedic process that gradually increases the length of a bone, often using controlled distraction after osteotomy.

Lennox-Gastaut Syndrome

A severe childhood-onset developmental and epileptic encephalopathy with multiple seizure types, characteristic slow spike-and-wave EEG patterns, cognitive impairment, and often treatment-resistant seizures.

Lenograstim

A recombinant granulocyte colony-stimulating factor that increases neutrophil production and is used to prevent or treat some forms of neutropenia.

Lens

Transparent biconvex avascular structure behind the iris/pupil, suspended by zonular fibers from the ciliary body. Provides approximately one-third of refractive power and allows accommodation (focusing on near objects by changing shape). Concentric lens fiber layers from surface epithelium, surrounded by a capsule.

Lens Coloboma

A lens coloboma is a congenital segmental defect or notch in the eye lens caused by incomplete development.

Lens Disease

A disorder of the eye lens, including cataract, dislocation, inflammation-related opacity, or developmental abnormality, that may impair focusing or vision.

Lens Dislocation

Displacement of the crystalline lens from its normal position because of trauma, zonular weakness, surgery, or connective-tissue disease; it can cause blurred vision, glaucoma, or inflammation.

Lens Prescription

OD in an eyeglass prescription means the right eye, from the Latin *oculus dexter*.

Lens Prescription

OS in an eyeglass prescription means the left eye, from the Latin *oculus sinister*.

Lens Subluxation

Partial displacement of the crystalline lens while some zonular attachments remain, potentially causing blurred or fluctuating vision, refractive change, or glaucoma.

Lensometer

An instrument used to measure the prescription and optical properties of eyeglass lenses.

Alternate Names:

Lensmeter; focimeter

Lente Insulin

Lente insulin is an intermediate-acting insulin preparation formerly used to control blood glucose; availability varies and it can cause low blood sugar.

Lentiform Nucleus

The lentiform nucleus is a basal-ganglia structure composed of the putamen and globus pallidus and involved in motor control.

Lentigines, Solar

Alternate index label; see *Age spots (liver spots)*.

Lentigo

A lentigo is a sharply defined flat brown skin spot caused by increased melanocytes or pigment and may be benign or, rarely, atypical.

Lentigo Maligna

A melanoma in situ that occurs on chronically sun-damaged skin, most commonly on the face of elderly individuals. It presents as a large, irregularly pigmented, slow-growing macule with variegated shades of brown and black. The radial growth phase may persist for years before invasion occurs.

Lentigo Maligna Melanoma

An invasive melanoma arising in a lentigo maligna on chronically sun-damaged skin, usually of the head or neck; it presents as an enlarging irregular pigmented patch or plaque.

Lentinan

Lentinan is a beta-glucan from shiitake mushrooms studied as an immune modulator and adjunctive cancer treatment.

LEOPARD Syndrome

A rare genetic disorder, usually related to

pathogenic variants in PTPN11 or associated genes, characterized by lentigines, electrocardiographic conduction abnormalities, ocular hypertelorism, pulmonary-valve stenosis, abnormal genitalia, growth retardation, and deafness.

Leopold Maneuvers

A systematic abdominal palpation technique used to determine fetal position and presentation. Four maneuvers: (1) fundal grip identifies fetal lie (head vs breech), (2) umbilical grip locates the fetal back, (3) Pawlik grip identifies the presenting part and its position, (4) pelvic grip confirms presentation and attitude.

Lepidoptera

The insect order containing butterflies and moths; contact with some caterpillars can cause skin, eye, or systemic reactions.

Lepirudin

Lepirudin is a direct thrombin inhibitor formerly used as an anticoagulant in heparin-induced thrombocytopenia; bleeding and kidney clearance are important considerations.

Lepromatous Leprosy

The disseminated, multibacillary pole of the leprosy spectrum characterised by poor cell-mediated immunity to *Mycobacterium leprae*, resulting in widespread skin nodules, plaques, and diffuse infiltration with thickening of the skin and facial features (leonine facies).

Lepromin Skin Test

A delayed skin-reaction test using killed *Mycobacterium leprae* antigens; it helps classify leprosy and assess cell-mediated immunity but does not diagnose active infection.

Leprosy (Hansen Disease)

A chronic granulomatous infection caused by the obligate intracellular acid-fast bacilli *Mycobacterium leprae* and *Mycobacterium lepromatosis*, primarily targeting the skin, cooler peripheral nerves, upper respiratory mucosa, and anterior eye. Classi-

fied by the Ridley-Jopling system based on host cell-mediated Th1 immunity, ranging from tuberculoid leprosy (paucibacillary, robust Th1 immunity with solitary anesthetic hypopigmented plaques and palpable thickened nerves) to lepromatous leprosy (multibacillary, defective Th1 immunity with extensive foamy histiocytes, diffuse skin nodules, leonine facies, saddle-nose collapse, and symmetric peripheral sensorimotor glove-and-stocking neuropathy). Characterized by hypopigmented anesthetic skin macules and peripheral nerve enlargement. Diagnosed by slit-skin smear or full-thickness skin biopsy with Fite-Faraco staining; treated with WHO multidrug therapy (rifampicin, dapsone, and clofazimine) to prevent permanent deformities and blindness.

Leptin

A hormone produced mainly by adipose tissue that signals stored-energy status to the brain and influences appetite and energy balance.

Leptin Receptor

leptin receptor is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Leptomeningeal

Relating to the pia mater and arachnoid mater, the two thin inner layers covering the brain and spinal cord; cancer or infection can spread in these layers.

Leptomeningeal Metastasis

Spread of cancer to the leptomeninges and cerebrospinal fluid (carcinomatous meningitis), causing multifocal neurologic deficits; managed with intrathecal chemotherapy and involved-field radiation.

Leptomeningeal Sarcoma

Leptomeningeal sarcoma is a rare malignant tumor involving the membranes and fluid spaces around the brain or spinal cord.

Leptomeninges

The arachnoid mater and pia mater, the two delicate inner membranes covering the brain and spinal cord.

Leptomeningitis

Inflammation or infection of the leptomeninges, the arachnoid and pia mater covering the brain and spinal cord.

Leptospira

A genus of spiral-shaped bacteria that includes the causes of leptospirosis, usually acquired through water or soil contaminated with infected animal urine.

Leptospiraceae

A family of spiral-shaped bacteria that includes *Leptospira* and related organisms, many of which are associated with animals or the environment.

Leptospirosis

Leptospirosis is zoonotic infection with *Leptospira* bacteria acquired from water or soil contaminated by animal urine, ranging from fever and myalgia to kidney and liver failure. transmitted through contact with water or soil contaminated with the urine of infected animals (rodents, cattle, dogs). Incubation period 5-14 days. Mild form: fever, headache, myalgia (especially calves), conjunctival suffusion.

Alternate Names:

Weils Disease (Leptospirosis)

Leptotrichia

A genus of anaerobic gram-negative bacteria found in the mouth and genital tract that can rarely cause invasive infection.

Leriche Syndrome

Aortoiliac occlusive disease causing reduced blood flow to the pelvis and lower limbs. The classic presentation includes buttock or thigh claudication, diminished femoral pulses, and erectile dysfunction, although symptoms vary with the extent of collateral circulation.

Leridistim

Leridistim is an investigational hematopoietic growth factor designed to stimulate production of neutrophils and other blood cells.

Lerisetron

Lerisetron is an investigational serotonin 5-HT₃ receptor antagonist studied to prevent nausea and vomiting.

Lesch-Nyhan Syndrome

An inherited disorder of purine metabolism caused by severe HPRT enzyme deficiency, leading to high uric acid, neurologic problems, and characteristic self-injurious behavior.

Lesion

A general term for any abnormal change in tissue structure due to disease or injury.

Lesion Identification

A diagnostic or laboratory method related to lesion identification. Results are interpreted with the method, reference range, and clinical findings.

Lesions On The Brain (Brain Lesions (Lesions On The Brain))

Alternate index label for Brain Lesions (Lesions on the Brain).

Less Than

Less than indicates a value below a stated amount or threshold, such as a laboratory reference limit, dose, age, or measurement.

Lesser Curvature

Concave medial stomach border from gastroesophageal junction to pylorus. Attachment site for the lesser omentum connecting to the liver, containing right/left gastric arteries. Most common gastric ulcer site. The angular incisure marks the body-antrum junction. Cancer here may cause early obstruction from the narrow lumen.

Lesser Trochanter

Small conical posteromedial proximal femoral prominence. Insertion for iliopsoas (primary hip flexor). Isolated fractures in adults may indicate pathologic fracture from metastatic disease; in adolescents may result from iliopsoas avulsion. Visible on lateral hip radiographs.

Lestaurtinib

An investigational kinase-inhibitor medicine studied for leukemia and other cancers.

Let-Down Reflex

Release of milk from the breast when oxytocin causes contraction of myoepithelial cells around mammary alveoli.

Lethal

Sufficient to cause death; fatal or destructive to biological life. In pharmacology and toxicology, quantified by parameters such as the median lethal dose (LD₅₀), the dose of a substance that proves fatal to 50% of an animal test population. In medical genetics, lethal mutations denote genetic variations that result in non-viability of the embryo, intrauterine fetal demise, or death in early neonatal life (e.g., homozygous achondroplasia, thanatophoric dysplasia).

Lethal Injection

A medical procedure, treatment, device, or care topic related to lethal injection; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Lethal-Means Counseling

A preventive intervention that reduces access to firearms, medications, toxic substances, or other methods of self-harm during periods of elevated risk. It is conducted collaboratively and is an important component of suicide prevention.

Lethargy

Lethargy is reduced energy, alertness, or responsiveness; it may result from sleep loss, infection, metabolic disease, medicines, depression, or neuro-

logic illness, and sudden marked lethargy requires urgent assessment.

Letrozole

Letrozole is an aromatase inhibitor that lowers oestrogen production and is used mainly for hormone-receptor-positive breast cancer and ovulation induction; hot flushes, arthralgia, bone loss, and fetal harm are important considerations.

Leu

The standard amino-acid abbreviation for leucine, an essential branched-chain amino acid used in protein synthesis.

Leucine

An essential branched-chain amino acid needed for protein synthesis, muscle metabolism, and regulation of cellular growth pathways.

Leucine Aminopeptidase - Urine

A biochemical laboratory test related to leucine aminopeptidase - urine. Results are interpreted with the specimen, method, reference range, and clinical findings.

Leucine Zipper

A protein-structure motif containing leucines at regular intervals that helps two protein chains bind together.

Leucocyte

Another spelling of leukocyte, meaning a white blood cell. Leukocytes help defend against infection and other threats; major groups include neutrophils, lymphocytes, monocytes, eosinophils, and basophils.

Leucocyte (White Blood Cell)

A blood cell involved in immune defense against infection, foreign substances, and abnormal cells.

Alternate Names:

White Blood Cell

Leucocytosis

An abnormally increased number of white blood

cells in the blood.

Leucoderma

Loss or marked reduction of skin pigmentation, often used as a descriptive term for depigmented patches.

Leucoderma (Vitiligo)

Loss of skin pigment producing sharply defined pale or white patches, usually because melanocytes have been destroyed or have stopped functioning.

Alternate Names:

Vitiligo

Leucomalacia

Softening of tissue caused by injury, classically white-matter injury in periventricular leukomalacia.

Leuconostoc

A biomedical term referring to leuconostoc; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Leuconostoc Citreum

A biomedical term referring to leuconostoc citreum; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Leuconostoc Lactis

A biomedical term referring to leuconostoc lactis; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Leuconostoc Mesenteroides

A biomedical term referring to leuconostoc mesenteroides; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Leuconostocaceae

Leuconostocaceae is a family of lactic-acid-producing bacteria found in plants, fermented foods, and occasionally clinical specimens.

Leucopenia

A decreased total white blood cell count below the normal reference range ($< 4.0 \times 10^9/\text{L}$). Leucopenia is often due to a decrease in neutrophils (neutropenia) or lymphocytes (lymphopenia).

Leucoplakia

A persistent white mucosal patch that cannot be scraped off and may require assessment for dysplasia or cancer risk.

Leucopoiesis

The production and maturation of white blood cells, mainly in the bone marrow. It is regulated by growth factors and increases or changes during infection, inflammation, and some blood disorders.

Leucorrhoea (Leukorrhoea)

A whitish vaginal discharge that may be physiological or may result from infection, inflammation, or another genital disorder.

Alternate Names:

Leukorrhoea

Leucovorin

Leucovorin is a folate medicine used to reduce the toxicity of methotrexate, enhance some chemotherapy regimens, and treat certain folate-related problems.

Leucovorin Calcium

Leucovorin calcium is the calcium salt of leucovorin, used as rescue therapy after methotrexate and in selected colorectal-cancer treatments.

Leukaemia

Cancer of blood-forming tissues in which abnormal leukocytes proliferate and disrupt normal marrow function.

Leukapheresis

A medical procedure, treatment, device, or care topic related to leukapheresis; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Leukemia

Leukemia is cancer of blood-forming cells that disrupts normal marrow production and may cause anemia, infection, bleeding, abnormal blood counts, and organ infiltration. Leukemias are classified by the cell lineage and by clinical course, including acute and chronic forms.

Alternate Names:

Blood Cancer (Leukemia) Leukemia, General

Leukemia Cutis

Infiltration of the skin by leukemic cells, presenting as papules, nodules, or plaques. It is more common in acute myeloid leukemia and indicates a poorer prognosis.

Leukemia Genetics

Acute lymphoblastic leukemia: Philadelphia chromosome (t9;22), Hyperdiploidy, ETV6-RUNX1 fusion. Acute myeloid leukemia (AML): t8;21, t15;17 (APL), FLT3 mutations. Chronic myeloid leukemia (CML): Philadelphia chromosome (BCR-ABL).

Leukemia, Acute Lymphocytic

Alternate index label; see *Acute lymphocytic leukemia*.

Leukemia, Acute Myelogenous

Alternate index label; see *Acute myelogenous leukemia*.

Leukemia, Chronic Lymphocytic

Alternate index label; see *Chronic lymphocytic leukemia*.

Leukemia, Chronic Myelogenous

Alternate index label; see *Chronic myelogenous leukemia*.

Leukemia, General

Alternate index label; see *Leukemia*.

Leukemia, Hairy Cell

Alternate index label; see *Hairy cell leukemia*.

Leukemic Infiltration Of The Gingiva

Gingival enlargement caused by infiltration of leukemic cells, most commonly in acute myeloid leukemia, particularly the monocytic subtype. The gingiva becomes swollen, erythematous, friable, and may bleed spontaneously. It results from leukemic cell proliferation within the gingival connective tissue.

Leukemoid Reaction

A marked reactive increase in the white-cell count, usually with prominent neutrophilia and immature granulocytes, caused by severe infection, inflammation, malignancy, or tissue injury. It resembles leukemia but is not a clonal hematologic malignancy.

Leukoaraiosis

Leukoaraiosis describes white-matter changes seen on brain imaging, commonly associated with small-vessel disease, aging, hypertension, and vascular risk.

Leukocyte

White blood cell; component of the immune system responsible for fighting infection and disease. Types include neutrophils, lymphocytes, monocytes, eosinophils, and basophils. Its significance is established from its characteristic features, causes, and clinical context.

Leukocyte Adhesion Deficiency

LAD-I (ITGB2 mutation, CD18 deficiency): absent CD11/CD18 integrins -> impaired neutrophil adhesion/transmigration -> delayed umbilical cord separation (>30 days), recurrent soft tissue infections without pus formation (leukocytosis >20,000), poor wound healing, periodontitis. Severe: death by age 2 without HSCT.

Leukocyte Alkaline Phosphatase

An enzyme activity measured in mature neutrophils using a cytochemical stain, historically reported as a leukocyte alkaline phosphatase (LAP) score. The score is typically low in chronic myeloid leukaemia and increased in many leukemoid reactions, infections, pregnancy, and other reactive states.

Leukocyte Differentiation

Leukocyte differentiation is the process by which immature white blood cells develop into specialized immune-cell types.

Leukocyte Disorder

A disorder involving the number, function, or development of white blood cells, including leukopenia, leukocytosis, immune deficiency, and malignant leukocyte proliferation.

Leukocyte Elastase

Leukocyte elastase is a proteolytic enzyme released mainly by neutrophils that can destroy microbes but also injure tissue during excessive inflammation.

Leukocyte Esterase Urine Test

A urine dipstick test detecting leukocyte esterase released by white blood cells; a positive result supports pyuria but does not by itself diagnose urinary tract infection.

Leukocyte Trafficking

Leukocyte trafficking is movement of white blood cells through blood and tissues, directed by adhesion molecules and chemical signals during immune surveillance or inflammation.

Leukocyte Transfusion

A medical procedure, treatment, device, or care topic related to leukocyte transfusion; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Leukocytes

White blood cells that help defend the body against infection and other threats.

Leukocytoclastic Vasculitis

A histopathologic pattern of small-vessel vasculitis characterized by neutrophilic debris, fibrinoid change, and vessel-wall inflammation. Clinically it often presents with palpable purpura; causes may be cutaneous-limited or associated with systemic disease.

Leukocytosis

An increased total white blood cell count above the normal reference range ($> 11 \times 10^9/\text{L}$ in adults).

Leukodystrophy

A group of inherited disorders affecting the formation or maintenance of central-nervous-system myelin, causing progressive neurologic impairment.

Leukoencephalopathy

Leukoencephalopathy is disease of the brain's white matter caused by inherited, infectious, toxic, vascular, metabolic, inflammatory, or other processes.

Leukonychia

White discoloration of the nails. True leukonychia: involves the nail matrix and moves distally with nail growth (punctate, striate/Mees lines from arsenic poisoning, or total).

Leukonychia Striata

Longitudinal white bands running along the nail plate, usually caused by changes in the nail matrix. They may be inherited or follow repeated minor trauma, skin disease, or some medicines; persistent or widespread changes should be medically assessed.

Leukonychia Totalis

Uniform white discoloration involving most or all of one or more nail plates. It is often inherited but can occasionally be associated with illness or medication; persistent changes affecting many nails should be evaluated with the medical history and other findings.

Leukopenia

An abnormally low white-blood-cell count, which may increase susceptibility to infection depending on the cells affected and the severity of the reduction.

Leukoplakia

Leukoplakia is a persistent white patch on mucosa that cannot be scraped away and may reflect irritation, infection, or premalignant epithelial change requiring evaluation. and cannot be characterized clin-

ically as any other definable lesion (WHO). Leukoplakia is a potentially malignant disorder of the oral cavity, with a risk of malignant transformation to squamous cell carcinoma of approximately 1-5% per year.

Alternate Names:

Smokers Keratosis (Leukoplakia)

Leukopoiesis

The production and development of white blood cells (leukocytes) from hematopoietic stem cells in the bone marrow, regulated by colony-stimulating factors and interleukins.

Leukorrhea

A whitish vaginal discharge that may be normal or may indicate infection or another disorder.

Leukostasis

A medical emergency in which very high numbers of abnormal leukocytes impair microvascular blood flow, potentially causing respiratory distress, neurologic symptoms, organ ischemia, or hemorrhage.

Leukotomy

A neurosurgical procedure involving the cutting of white matter fibers in the prefrontal cortex, also known as frontal lobotomy or prefrontal leukotomy. Developed by Egas Moniz (Nobel Prize 1949) and Walter Freeman.

Alternate Names:

frontal lobotomy or prefrontal leukotomy.

Leukotriene

A leukotriene is an inflammatory lipid mediator made from arachidonic acid that contributes to bronchoconstriction, mucus, vascular leak, and immune-cell recruitment.

Leukotriene Blockers

Medicines that reduce leukotriene effects and help prevent asthma or allergy symptoms.

Alternate Names:

Leukotriene antagonists

Leukotriene Degradation

Leukotriene degradation is the breakdown and removal of leukotrienes, inflammatory lipid mediators involved in airway and immune responses.

Leukotriene Modifiers

Medicines that reduce asthma or allergy symptoms by blocking leukotriene production or action.

Leukotriene Receptor

leukotriene receptor is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Leukotrienes

Inflammatory signaling molecules that can narrow and swell airways during allergic reactions.

Leuprolide

Leuprolide is a gonadotropin-releasing hormone agonist used to suppress sex-hormone production in conditions such as prostate cancer, endometriosis, and fibroids.

Leuprolide Acetate

Leuprolide acetate is the salt form of leuprolide, used to suppress ovarian or testicular hormone production for selected medical conditions.

Levalbuterol

Levalbuterol is an inhaled short-acting beta-2 agonist that relaxes airway smooth muscle and relieves bronchospasm.

Levamisole

Levamisole is an antiparasitic and immune-modulating drug used selectively; serious low white-cell counts and vasculitic reactions can occur.

Levator Ani

The primary pelvic floor muscle complex supporting the pelvic organs, forming a hammock from the pubic symphysis to the coccyx. Composed of

the pubococcygeus, puborectalis, and iliococcygeus muscles. Innervated by the pudendal nerve (S2-S4).

Levetiracetam

Levetiracetam is a broad-spectrum antiseizure medicine that binds synaptic vesicle protein SV2A; it treats focal and generalized seizures and may cause somnolence, dizziness, or behavioral and mood changes.

Levobunolol

Levobunolol is a beta-blocker eye drop used to lower pressure inside the eye in glaucoma or ocular hypertension.

Levocardia

Levocardia is a normal left-sided position of the heart; it may occur with normal organ arrangement or with other congenital laterality abnormalities.

Levocarnitine

Levocarnitine is the biologically active form of carnitine used to treat selected carnitine deficiencies and help transport fatty acids into mitochondria.

Levodopa

Levodopa is a dopamine precursor that crosses the blood–brain barrier and is converted to dopamine, usually with carbidopa or benserazide, to treat Parkinson disease; dyskinesias, hallucinations, nausea, and motor fluctuations can develop.

Levofloxacin

Levofloxacin is a fluoroquinolone antibiotic that inhibits bacterial DNA gyrase and topoisomerase IV; tendon injury, peripheral neuropathy, QT prolongation, dysglycaemia, and central-nervous-system effects are important class risks.

Levoleucovorin

Levoleucovorin is the active isomer of leucovorin, used as methotrexate rescue and with selected chemotherapy regimens.

Levoleucovorin Calcium

Levoleucovorin calcium is the calcium salt of levoleucovorin, used to reduce methotrexate toxicity and support selected cancer treatments.

Levomepromazine

A phenothiazine medicine with antipsychotic, sedative, antiemetic, and analgesic properties, used in selected palliative-care and psychiatric settings.

Levonorgestrel

Levonorgestrel is a progestin used in contraceptive pills, hormonal intrauterine devices, and emergency contraception.

Levorphanol

Levorphanol is a potent opioid analgesic that can relieve severe pain but also cause sedation, dependence, and respiratory depression.

Levothyroxine

A synthetic form of thyroxine (T4) used as thyroid hormone replacement in hypothyroidism and in selected other thyroid conditions; dose is individualized and monitored with laboratory tests.

Levothyroxine Sodium

The sodium salt of synthetic thyroxine used to replace deficient thyroid hormone.

Alternate Names:

Levothyroxine

Levulose

An older name for fructose, a simple sugar found in fruit, honey, and many sweeteners.

Lewy Body

A Lewy body is an abnormal neuronal inclusion containing alpha-synuclein, characteristic of Parkinson disease and Lewy-body dementia.

Lewy Body Dementia

A progressive brain disorder caused by abnormal alpha-synuclein protein deposits called Lewy bodies. It causes dementia with changing attention or

alertness, recurrent visual hallucinations, Parkinson-like movement problems, sleep disorders such as acting out dreams, and sometimes automatic body-function problems such as constipation or dizziness on standing. Some people are very sensitive to antipsychotic medicines, so treatment must be individualized.

Alternate Names:

Dementia, Lewy Body

Lewy Body Disease

A neurodegenerative disorder associated with abnormal neuronal alpha-synuclein aggregates. Dementia with Lewy bodies commonly features fluctuating cognition, recurrent visual hallucinations, REM sleep behavior disorder, and spontaneous parkinsonism; sensitivity to antipsychotics may occur.

Lexatumumab

Lexatumumab is an investigational monoclonal antibody that targets TRAIL receptor 2 and has been studied as an anticancer treatment.

Leydig Cell

Leydig cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Leydig Cell Testicular Tumor

A usually rare sex-cord stromal tumor arising from testosterone-producing Leydig cells in the testis; it may cause a testicular mass or hormone-related signs.

Leydig Cell Tumor

A testicular sex cord-stromal tumor that secretes testosterone or estrogen, causing precocious puberty or gynecomastia; most are benign and treated by orchidectomy.

Leydig Cells

Testicular interstitial cells that produce testosterone in response to luteinizing hormone.

Lfgren Syndrome

Acute sarcoidosis with bilateral hilar lymphadenopathy, erythema nodosum, and ankle polyarthralgia. It often resolves spontaneously and generally has a favorable prognosis, although follow-up is needed. Associated with HLA-DR3.

LGE

Alternate name; see *Late Gadolinium Enhancement*.

LH

Luteinizing hormone, a pituitary hormone involved in ovulation and testosterone production.

Alternate Names:

Luteinizing hormone

LH Response To GnRH Blood Test

A dynamic endocrine test measuring luteinizing hormone before and after administration of gonadotropin-releasing hormone to assess hypothalamic-pituitary-gonadal function and some pubertal disorders.

Libman-Sacks Endocarditis

A form of nonbacterial verrucous endocarditis characterized by sterile, small platelet-fibrin vegetations deposited on cardiac valve leaflets and mural endocardium. It is strongly associated with systemic lupus erythematosus and antiphospholipid syndrome, frequently involving the ventricular surfaces of the mitral and aortic valves.

Alternate Names:

Verrucous endocarditis; Nonbacterial verrucous endocarditis

Li-Fraumeni Syndrome

An inherited cancer-predisposition syndrome most often caused by a pathogenic TP53 variant. It confers high lifetime risks of sarcomas, breast cancer, brain tumors, adrenocortical carcinoma, leukemia, and other early-onset malignancies.

LI-RADS

Liver Imaging-Reporting and Data System. Standardizes reporting of liver imaging in patients with

cirrhosis. Used for hepatocellular carcinoma diagnosis.

Liarozole

Liarozole is an investigational medicine that inhibits retinoic-acid metabolism and has been studied for disorders of keratinization and cancer.

Libido

Sexual desire or drive, influenced by biological, psychological, and social factors.

Libido And Sexual Performance

Libido (sexual desire) is modulated by a complex interplay of hormonal (testosterone, estrogen, prolactin), neurologic, psychologic, and relational factors.

Lice

Pediculosis: infestation by blood-sucking lice—head lice (*Pediculus humanus capitis*, common in children, scalp), body lice (*Pediculus humanus corporis*, clothing/seams, associated with poor hygiene and epidemic typhus transmission), and pubic/crab lice (*Phthirus pubis*, sexually transmitted).

Lice Infestation

A biomedical term referring to lice infestation; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Lice Infestations

Pediculosis is infestation by head, body, or pubic lice. Head lice cause scalp pruritus and nits attached to hair shafts; body lice live mainly in clothing; pubic lice affect coarse hair-bearing regions.

Lice, Body

Alternate index label; see *Body lice*.

Licensed Clinical Social Worker

A social worker licensed to provide clinical assessment and psychotherapy or other mental-health services within the scope of local law and training.

Lichen

A descriptive dermatological term referring to a flat-topped, firm, papular skin lesion resembling the lichen plant (a symbiotic organism of algae and fungi growing on rocks and trees). Lichenoid describes lesions that resemble lichen planus or lichenoid drug eruptions.

Lichen Planopilaris

A lymphocytic inflammatory scalp disorder that attacks hair follicles and can cause scarring alopecia, perifollicular scale, redness, and permanent hair loss.

Lichen Planus

A T-cell mediated inflammatory condition affecting skin, mucous membranes, nails, and hair. Classic presentation: the 6 Ps (pruritic, purple, polygonal, planar, papules, plaques), often with Wickham striae (white lacy network on surface). Oral lichen planus affects buccal mucosa and may be erosive.

Alternate Names:

Lichen Ruber Planus (Lichen Planus)

Lichen Planus, Oral

Alternate index label; see *Oral lichen planus*.

Lichen Ruber Planus (Lichen Planus)

Alternate index label for Lichen Planus.

Lichen Sclerosus

A chronic inflammatory skin disease most commonly affecting anogenital skin. It presents as ivory-white, atrophic plaques with wrinkled paper-like appearance. In women, it predominantly affects the vulva and perianal region (figure-eight distribution). In men, it affects the glans penis and foreskin (balanitis xerotica obliterans).

Alternate Names:

White Spot Disease

Lichen Simplex Chronicus

A thickened, leathery plaque caused by chronic scratching and rubbing (itch-scratch cycle). It commonly affects the neck, ankles, lower legs, and genital area. It is a form of neurodermatitis associated

with stress and anxiety.

Lichenification

Thickening and accentuation of skin markings caused by chronic scratching or rubbing.

Lichenified

Lichenified skin is thickened and leathery from chronic rubbing or scratching, commonly occurring in longstanding eczema or itch disorders.

Lichenoid Drug Eruption

A lichenoid inflammatory skin reaction caused by a medicine, often resembling lichen planus but with a broader distribution or different clinical and histologic features; improvement may follow withdrawal of the culprit drug.

Liddle Syndrome

Autosomal dominant condition caused by gain-of-function mutations in the ENaC sodium channel subunits leading to constitutive sodium reabsorption and potassium secretion in the collecting duct. Presents with severe hypertension, hypokalemia, and metabolic alkalosis occurring in childhood or young adulthood.

Liddle Syndrome Treatment

Liddle syndrome is an autosomal dominant cause of early-onset hypertension with hypokalemia and metabolic alkalosis caused by gain-of-function mutations in ENaC sodium channel subunits. Aldosterone levels are suppressed.

Lidocaine

Lidocaine is a local anaesthetic that blocks voltage-gated sodium channels and, intravenously, can treat selected ventricular arrhythmias; excessive exposure may cause seizures, hypotension, arrhythmia, or cardiac arrest.

Lieberkühn's Crypts

Tubular intestinal glands between the villi that contain absorptive, secretory, and stem cells.

Life Expectancy

The average number of additional years a person of a specified age is expected to live under current mortality conditions.

Lifestyle Disease

A noncommunicable disease influenced by behaviors and living conditions, such as tobacco use, diet, inactivity, or alcohol; genetics, environment, and access to care also contribute.

Lifetime Risk

The probability that a person will develop a specified disease or experience an outcome during the remainder of their life or over a defined lifetime period.

Lifting And Bending The Right Way

Safe lifting and bending use planning, a stable base, load close to the body, hip and knee movement, and assistance for heavy or awkward objects.

Ligament

A dense, fibrous connective tissue band that connects bone to bone, providing passive joint stability and guiding joint motion. Ligaments are composed predominantly of type I collagen fibers (parallel orientation, crimped pattern) and elastin fibers (10-20% in some ligaments), arranged in bundles.

Ligand

Any molecule that binds specifically to a biological receptor, macromolecule, or active site. In pharmacology and physiology, ligands include hormones, neurotransmitters, drugs, and toxins that bind receptors to initiate (agonist), block (antagonist), or modulate signaling.

Ligase

A ligase is an enzyme that joins two molecules, often using energy from ATP or another high-energy compound.

Ligation

The surgical process of tying off a blood vessel or

duct using a suture or clip to prevent bleeding or block fluid flow.

Ligation (Vascular)

Surgical tying, clipping, or sealing of a blood vessel to stop or redirect blood flow.

Alternate Names:

Vascular

Ligature

Any cord, wire, rope, fabric, or flexible object used to constrict the neck, bind limbs, suspend the body, or inflict injury. The resulting pale groove-shaped abrasion may be horizontal with petechiae in homicidal strangulation, or oblique and upward in hanging.

Light

Visible electromagnetic radiation; in medicine it is used for examination, phototherapy, imaging, and selected treatments.

Light Coagulation

Light coagulation uses intense focused light, usually a laser, to heat and seal tissue or blood vessels.

Light Cycle

A light cycle is the recurring pattern of light and darkness that helps regulate circadian rhythms and biological activity.

Light Microscope

light microscope is a diagnostic test, specimen, imaging method, score, instrument, or measurement used to answer a clinical or laboratory question; results must be interpreted with the method and reference range.

Light Microscopy

light microscopy is a diagnostic test, specimen, imaging method, score, instrument, or measurement used to answer a clinical or laboratory question; results must be interpreted with the method and reference range.

Light Reflex

Constriction of the pupil in response to light, mediated by the retinal, optic, midbrain, and oculomotor pathways.

Light Scattering

light scattering is a diagnostic test, specimen, imaging method, score, instrument, or measurement used to answer a clinical or laboratory question; results must be interpreted with the method and reference range.

Light Therapy

Treatment that uses selected wavelengths or intensity of light for conditions such as some skin disorders, seasonal depression, or neonatal jaundice.

Light'S Criteria

An effusion is exudative if any one is met: pleural protein/serum protein > 0.5 , pleural LDH/serum LDH > 0.6 , or pleural LDH $> 2/3$ upper limit of normal serum LDH.

Light-Near Dissociation

A pupillary abnormality in which the pupil does not constrict to light but does constrict during accommodation (near response). Classically seen in Argyll Robertson pupils (neurosyphilis) and also in midbrain lesions, diabetes mellitus, and optic nerve disease. Differentiates from tonic pupil (Adie pupil) by preserved near response.

Lightening

In late pregnancy, descent of the fetus into the pelvis before labor, which may reduce upper-abdominal pressure but increase pelvic pressure and urinary frequency.

Lighter Fluid Poisoning

Lighter fluid poisoning involves toxic or environmental exposure to lighter fluid poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Lightheadedness

A feeling of near-fainting, floating, or unsteadiness without necessarily perceiving that the surroundings are spinning; causes include dehydration, low blood pressure, anemia, medications, and heart or neurologic conditions.

Lightning Pains

Brief, stabbing, electric-shock-like pains, classically occurring in tabes dorsalis or other nerve disorders.

Lignan

A lignan is a plant compound found in foods such as seeds and whole grains; some lignans can be converted by gut bacteria into hormone-like compounds.

Lignans

Plant compounds found in seeds, grains, fruits, and vegetables that have antioxidant activity.

Lignin

Lignin is a tough plant polymer that strengthens cell walls and contributes to dietary fiber.

Lignocaine

The international nonproprietary name for lidocaine, a local anesthetic and antiarrhythmic drug.

Lignoceric Acid

lignoceric acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Likelihood Ratio

A ratio that describes how much a test result changes the odds of disease. The positive likelihood ratio is sensitivity divided by one minus specificity; the negative likelihood ratio is one minus sensitivity divided by specificity.

Lilliputian Hallucination

A visual hallucination of unusually small people, animals, or objects; it can occur with neurologic, psychiatric, toxic, infectious, or sensory conditions.

Lily Of The Valley Poisoning

Lily of the valley poisoning involves toxic or environmental exposure to lily of the valley poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Limb

A limb is an arm or leg and its associated bones, joints, muscles, nerves, vessels, and soft tissues.

Limb Bud

A limb bud is an embryonic outgrowth that develops into an arm or leg through coordinated signaling, growth, and tissue differentiation.

Limb Development

Limb development is embryonic formation and growth of the bones, joints, muscles, nerves, vessels, and skin of the arms and legs.

Limb Reduction Defects

Birth defects in which part or all of an arm or leg does not form normally. The missing or shortened part may involve a bone, joint, hand, foot, or the whole limb.

Limb Fracture

A break in a bone of an upper or lower limb, assessed according to location, displacement, open or closed status, joint involvement, and associated neurovascular injury.

Limb Girdle

A limb girdle is the skeletal connection of a limb to the trunk, such as the shoulder or pelvic girdle.

Limb Injury

Trauma affecting an arm or leg, including injury to bone, joint, muscle, tendon, ligament, nerve, blood vessel, or skin.

Limb Pain

Pain arising from an arm or leg, with causes including trauma, nerve disease, vascular insufficiency, infection, inflammation, and musculoskeletal disorders.

ders.

Limb Plethysmography

A noninvasive test measuring changes in limb volume or blood flow, used to evaluate arterial or venous obstruction and sometimes to assess vascular responses.

Limb Prosthesis

A limb prosthesis is an artificial device replacing part or all of an arm or leg to restore mobility, function, appearance, or independence.

Limb-Girdle Muscular Dystrophies

Limb-girdle muscular dystrophies are inherited muscle diseases causing progressive weakness predominantly around the hips and shoulders, with variable cardiac and respiratory involvement.

Limbic Encephalitis

Inflammation of the limbic system, often presenting subacutely with memory impairment, seizures, psychiatric or behavioral change, and sleep disturbance. It may be autoimmune, paraneoplastic, infectious, or of unknown cause.

Limbic Lobe

A curved group of cortical structures on the medial cerebral hemisphere involved in emotion, memory, and motivation.

Limbic System

A set of interconnected brain structures located in the medial temporal lobe and around the corpus callosum, involved in emotion, memory, motivation, behaviour, and autonomic regulation. The term was coined by Paul Broca (1878, le grand lobe limbique) and expanded by Papez (1937, Papez circuit) and MacLean (1952, 'visceral brain').

Limbs

The arms and legs, including their bones, joints, muscles, nerves, and blood vessels.

Limbus

The border where the cornea meets the white outer coat of the eye.

Alternate Names:

Corneoscleral limbus

Limbus Corneae

The corneal limbus is the circular border where the transparent cornea meets the white sclera and conjunctiva, containing stem cells important for corneal repair.

Limited Range Of Motion

Limited range of motion is reduced ability to move a joint through its normal extent because of pain, stiffness, weakness, swelling, or structural change.

Linagliptin

Linagliptin is a DPP-4 inhibitor used with diet and exercise to improve blood glucose control in type 2 diabetes.

Lincomycin

Lincomycin is a lincosamide antibiotic used for selected serious bacterial infections; it can cause antibiotic-associated diarrhea and colitis.

Lindane

Lindane is an organochlorine insecticide formerly used for lice or scabies; neurotoxicity and environmental persistence limit its use.

Line, Central

A catheter whose tip lies in a large central vein, used to deliver medicines or fluids, monitor central venous pressure, draw blood, or provide specialized nutrition.

Line, Central Venous

A central venous catheter inserted into a large vein such as the internal jugular, subclavian, or femoral vein and advanced toward the superior or inferior vena cava.

Linea

A line or linear anatomical marking, often used in names such as linea alba or linea nigra.

Linea Aspera

Prominent longitudinal posterior femoral shaft ridge from converging muscle attachment lines. Insertion for adductor magnus, vastus medialis/lateralis, and short head of biceps femoris. Thickest femoral cortex, resisting bending forces. Diverges distally into supracondylar lines.

Linea Nigra

A dark brown or black vertical line appearing on the abdomen during pregnancy, extending from the pubic symphysis to the umbilicus or beyond. Caused by hormonal stimulation of melanocytes, resulting in hyperpigmentation. More prominent in darker-skinned women.

Linear

Arranged in a straight or nearly straight line, describing the shape or distribution of a skin lesion or eruption.

Linear Accelerator

A device that accelerates electrons to produce high-energy X-rays or electron beams for radiotherapy.

Linear Energy Transfer

Energy deposited per unit path length. High LET (alpha, neutrons) more biologically effective.

Linear Fracture

A narrow crack in a bone without significant displacement or fragmentation.

Linear IgA Bullous Dermatitis

An autoimmune blistering disease caused by linear deposition of IgA at the basement membrane zone. It produces tense blisters, sometimes in annular or "string of pearls" patterns, and may be idiopathic or drug-induced.

Lined Canister

A medical procedure, device, or treatment involving lined canister. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Linezolid

Linezolid is an oxazolidinone antibiotic that blocks formation of the bacterial 70S initiation complex and treats resistant Gram-positive infections; thrombocytopenia, neuropathy, lactic acidosis, and serotonin interactions require caution.

Lingua

The anatomical Latin term for the tongue, the versatile muscular organ situated on the floor of the oral cavity. Essential for mastication, oral bolus formation, deglutition, phonetic sound articulation, and gustatory taste sensation. Divided into an anterior two-thirds (oral body) and posterior third (pharyngeal base) by the V-shaped sulcus terminalis. Clinical derivatives include sublingual administration (rapid systemic absorption through the rich sublingual venous plexus, bypassing hepatic first-pass metabolism, as with sublingual nitroglycerin) and lingual tonsils (lymphoid tissue aggregations at the tongue base contributing to Waldeyer's ring).

Lingua Villosa Nigra

A benign condition characterized by elongation and dark pigmentation of the filiform papillae of the tongue, often associated with tobacco use, poor oral hygiene, or antimicrobial therapy.

Lingual

Relating to the tongue, including its muscles, nerves, blood supply, or surface.

Lingual Artery

A branch of the external carotid artery supplying the tongue, floor of mouth, and sublingual structures. It arises anterior to the hypoglossal nerve and courses deep to the hyoglossus muscle. Lingual artery aneurysm or thrombosis is rare but can cause life-threatening hemorrhage or tongue ischemia.

Lingual Thyroid

A lingual thyroid is thyroid tissue at the base of the tongue caused by failure of normal thyroid descent and may obstruct swallowing or breathing or cause hypothyroidism.

Lingual Tonsil

The lingual tonsil is lymphoid tissue at the base of the tongue that contributes to immune surveillance of the throat.

Lingual Tonsils

Collections of lymphoid tissue at the base of the tongue.

Liniment

A medical procedure, device, or treatment involving liniment. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Linitis Plastica

A diffuse infiltrative form of gastric carcinoma in which tumour cells spread through the stomach wall, producing marked thickening and rigidity with loss of normal distensibility. It may cause early satiety, weight loss, nausea, vomiting, or vague upper-abdominal symptoms.

Linkage

Co-inheritance of nearby genetic markers and disease gene. Measured by recombination fraction (theta). LOD score >3 indicates linkage.

Linkage Analysis

A genetic method that studies whether a disease trait and a genetic marker are inherited together, helping locate disease-associated regions in a family or population.

Linkage Disequilibrium

Linkage disequilibrium is the nonrandom association of genetic variants at different locations in a population.

Linkage Map

A genetic map showing the relative positions of genes or markers based on recombination frequency, usually expressed in centimorgans rather than physical distance.

Linoleic Acid

linoleic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Linolenic Acid

linolenic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Linsitinib

Linsitinib is an investigational drug that blocks insulin-like growth factor receptors and has been studied in cancer.

Lintuzumab

Lintuzumab is an investigational monoclonal antibody targeting CD33 on myeloid cells, studied mainly for acute myeloid leukemia.

Liothyronine Sodium

A synthetic form of the thyroid hormone triiodothyronine used in selected thyroid disorders.

Lip

The lips are mobile folds of skin and mucosa containing muscle, blood vessels, and sensory nerves; they form the oral seal and assist speech, facial expression, feeding, and swallowing.

Lip Cancer

Cancer that develops in the tissues of the upper or lower lip.

Lip Disease

A disorder of the lips, such as infection, inflammation, ulceration, trauma, pigmentation change, benign tumor, or malignancy.

Lip Infection

An infectious process or toxic exposure involving lip infection; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Lip Moisturizer Poisoning

Lip moisturizer poisoning involves toxic or environmental exposure to lip moisturizer poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Lip Neoplasm

A lip neoplasm is an abnormal growth in the lip that may be benign or malignant; persistent sores or lumps require assessment.

Lip Swelling

Enlargement of one or both lips caused by local injury, infection, inflammation, dental disease, medication, or angioedema. Sudden swelling of the lips with tongue or throat swelling, breathing difficulty, wheezing, faintness, or widespread hives may indicate a severe allergic reaction and requires emergency care.

Alternate Names:

Fats

Lipase

An enzyme that breaks down fats into smaller molecules. Blood lipase testing is commonly used when pancreatitis is suspected, although an abnormal result is not specific in every case.

Lipase Test

A lipase test measures a pancreatic digestive enzyme in blood and supports evaluation of acute pancreatitis and selected pancreatic or gastrointestinal disorders.

Lipedema

A chronic disorder of abnormal, usually symmetrical deposition of painful subcutaneous adipose tissue, predominantly affecting the legs and sometimes the arms. The feet and hands are typically spared, and easy bruising, tenderness, and progressive functional

limitation may occur.

Lipemia Retinalis

A creamy, pinkish-white discolouration of the retinal vessels caused by severely elevated serum triglycerides, typically exceeding 2,000 mg/dL. It is a sign of extreme hypertriglyceridaemia and indicates a risk for acute pancreatitis, requiring urgent lipid-lowering therapy and dietary management.

Lipemic Index

lipemic index is a diagnostic test, specimen, imaging method, score, instrument, or measurement used to answer a clinical or laboratory question; results must be interpreted with the method and reference range.

Lipid

Diverse group of hydrophobic molecules including triglycerides (energy storage), phospholipids (membranes), cholesterol (membranes, steroid precursor), fat-soluble vitamins (A, D, E, K). Triglycerides: glycerol + 3 fatty acids. Fatty acids: saturated (no double bonds) vs. unsaturated (one or more double bonds).

Lipid Bilayer

lipid bilayer is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lipid Binding

lipid binding is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Lipid Disorder

lipid disorder is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lipid Droplet

lipid droplet is a fatty acid or other lipid-related molecule whose chain structure influences mem-

branes, energy storage, signaling, and metabolism.

Lipid Emulsion Therapy

The antidote for local anesthetic systemic toxicity. The mechanism involves the lipid sink theory (lipid phase sequesters lipophilic local anesthetics from neural and cardiac tissue) and metabolic theory (lipid provides fatty acid substrate to ischemic myocardium).

Lipid Metabolism

lipid metabolism is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lipid Panel

Measurement of total cholesterol, LDL-C, HDL-C, and triglycerides. Fasting (9-12 hours) or non-fasting acceptable for screening. Desirable levels: total cholesterol <200 mg/dL, LDL <100 mg/dL (<70 for very high risk), HDL >40 (men) or >50 (women) mg/dL, triglycerides <150 mg/dL.

Lipid Peroxidation

lipid peroxidation is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lipid Peroxide

lipid peroxide is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lipid Profile

A blood test panel measuring lipids and lipoproteins, commonly including total cholesterol, LDL cholesterol, HDL cholesterol, and triglycerides.

Lipid Profile Test

A blood test usually measuring total cholesterol, LDL cholesterol, HDL cholesterol, and triglycerides to assess atherosclerotic cardiovascular risk and monitor treatment.

Lipid Transport

lipid transport is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lipidomics

lipidomics is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lipidosis

A disorder involving abnormal accumulation or metabolism of lipids in tissues, often due to inherited enzyme defects.

Lipids

A group of mostly water-insoluble substances that includes fats, oils, fatty acids, phospholipids, and cholesterol. Lipids store energy, form cell membranes, help make hormones, and help the body absorb vitamins A, D, E, and K.

Lipoatrophy

Loss of subcutaneous adipose tissue at specific sites, most commonly caused by repeated subcutaneous injections of insulin or other medications, or associated with autoimmune conditions and HIV lipodystrophy.

Lipoblastoma

A usually benign tumor of immature adipose tissue occurring mainly in infants and young children; it may be localized or diffuse and can recur if incompletely removed.

Lipocaic

A historical term for a pancreatic substance once thought to regulate fat metabolism; it is not a current standard clinical diagnosis.

Lipochrome

A fat-soluble pigment, such as a carotenoid or lipofuscin-related pigment, that can impart yellow coloration to tissues.

Lipocyte

A lipid-storing parenchymal cell of adipose tissue; synonymous with adipocyte. In hepatic physiology, the term lipocyte historically also designates the hepatic stellate cell of Ito, situated within the subendothelial space of Disse; these quiescent perisinusoidal cells store approximately 80% of the body's total vitamin A in cytoplasmic lipid droplets and, upon chronic hepatic injury or inflammation, transdifferentiate into proliferative, contractile, α -smooth muscle actin (α -SMA)-positive myofibroblasts that synthesize type I collagen, serving as the primary drivers of hepatic fibrosis and cirrhosis.

Lipodystrophy

Abnormal distribution or loss of body fat, which may be inherited or caused by medicines, metabolic disease, or repeated injections.

Lipodystrophy (Acquired Generalized And Inherited Lipodystrophy)

Alternate index label for Acquired Generalized and Inherited Lipodystrophy.

Lipodystrophy Syndromes

Rare conditions characterized by selective loss of adipose tissue leading to severe insulin resistance, diabetes, and hypertriglyceridemia. Congenital generalized lipodystrophy involves loss of all body fat from birth. Familial partial lipodystrophy involves loss of limb and truncal fat with fat accumulation in face and neck.

Lipodystrophy, Intestinal

Alternate index label; see *Whipple's disease*.

Lipofuscin

Lipofuscin is a yellow-brown, non-degradable pigment made of oxidized lipid and protein residues that accumulates in long-lived cells with ageing and in some degenerative diseases.

Lipogenesis

The synthesis of fatty acids and triglycerides from acetyl-CoA and other carbon sources. It occurs mainly in liver and adipose tissue when energy and

carbohydrate are abundant and is stimulated by insulin.

Lipogranulomatosis

Lipogranulomatosis is inflammation caused by lipid accumulation and granuloma formation, which may occur in skin, soft tissue, or internal organs.

Lipohypertrophy

Localized thickening or accumulation of fatty tissue, commonly at repeated insulin-injection sites.

Lipoid Nephrosis

A historical term for minimal change disease, a glomerular disorder causing nephrotic syndrome through increased glomerular permeability to proteins. It is especially common in children and typically shows diffuse podocyte foot-process effacement on electron microscopy.

Lipoid Pneumonia

Lung inflammation caused by accumulation of lipid-laden macrophages in the alveoli. Exogenous lipoid pneumonia results from aspiration of mineral oil, petroleum jelly, or oil-based nose drops. Endogenous lipoid pneumonia occurs distal to bronchial obstruction. HRCT shows ground-glass opacities with low-attenuation areas (fat density).

Lipoidosis

Lipoidosis is a disorder involving abnormal accumulation of lipids in cells or tissues, often from inherited storage disease or another metabolic problem.

Lipolysis

Hydrolysis of stored triglycerides in adipocytes to release non-esterified fatty acids and glycerol. It is stimulated by catecholamines and fasting-related hormonal signals and inhibited by insulin.

Lipoma

The most common benign soft tissue tumor, composed of mature adipocytes. It presents as a soft, mobile, painless subcutaneous nodule, commonly on the trunk, neck, and proximal extremities. Lipomas

grow slowly and may be multiple (familial multiple lipomatosis). Diagnosis is clinical; ultrasound or MRI for deep or atypical lesions.

Lipomatosis

A rare condition characterized by the diffuse, symmetric, non-encapsulated proliferation of mature adipose tissue, most commonly affecting the pelvis and lower extremities. Unlike simple lipomas, lipomatosis involves infiltration of fat into surrounding structures without encapsulation, making surgical excision difficult.

Lipomatous Neoplasm

A lipomatous neoplasm is a growth made partly or mainly of fat cells and may be benign, such as a lipoma, or malignant, such as a liposarcoma.

Lipophagy

Lipophagy is the cellular process of breaking down lipid droplets through autophagy to release fatty acids for energy or metabolism.

Lipophilicity

Lipophilicity is the tendency of a substance to dissolve in fats or oils rather than water, influencing drug absorption and distribution.

Lipopolysaccharide

Lipopolysaccharide is the outer-membrane endotoxin of Gram-negative bacteria, consisting of lipid A, core polysaccharide, and O antigen; circulating lipid A can trigger cytokine release, fever, shock, and disseminated inflammation.

Lipoprotein

Complex of lipids and proteins transporting hydrophobic lipids in aqueous blood. Classes: chylomicrons (dietary lipids), VLDL (endogenous triglycerides), LDL (cholesterol delivery), HDL (reverse cholesterol transport). ApoB-100 in LDL; apoA-1 in HDL. Elevated LDL cholesterol is a major risk factor for atherosclerosis.

Lipoprotein Analysis

A blood test measuring cholesterol, triglycerides, and lipoprotein fractions such as LDL and HDL.

Alternate Names:

Lipid panel; lipid profile

Lipoprotein Lipase

Hydrolyzes triglycerides in chylomicrons and VLDL. Activated by insulin.

Lipoprotein Metabolism

The production, transport, modification, and clearance of particles that carry lipids through the blood.

Lipoprotein(A)

An LDL-like lipoprotein whose elevated level is an inherited cardiovascular risk factor.

Alternate Names:

Lp(a)

Liposarcoma

A malignant tumor of fat cells, usually arising in the deep soft tissues of the thigh or retroperitoneum; well-differentiated, myxoid, and pleomorphic subtypes are recognized.

Liposarcoma Of The Breast

A rare malignant mesenchymal tumor with adipocytic differentiation arising in breast soft tissue. Diagnosis requires histopathologic and molecular evaluation, and treatment is generally wide excision with sarcoma-focused care.

Liposomal Amphotericin B

A liposomal formulation of amphotericin B used in the treatment of severe systemic fungal infections, with reduced nephrotoxicity compared to conventional amphotericin B. The liposomal formulation encapsulates amphotericin B within lipid vesicles, which alters the drug's biodistribution and reduces renal tubular damage.

Liposome

A liposome is a tiny vesicle made of a lipid bilayer that can carry medicines or other substances into

or around the body.

Liposuction

A surgical procedure that uses a cannula and suction to remove localized deposits of subcutaneous fat for body contouring. It is not a treatment for obesity or a substitute for diet and exercise. Risks may include bleeding, infection, contour irregularity, fluid accumulation or shifts, sensory changes, and anesthesia-related complications.

Alternate Names:

Lipoplasty

Lips

The paired movable folds surrounding the opening of the mouth, involved in speech, facial expression, eating, and oral sensation.

Lipschütz Bodies

An older term for nonspecific acute genital ulcers, often occurring in adolescents or young women after an infection.

Liquefaction

Conversion of a solid or semisolid material into a liquid, as in liquefactive necrosis.

Liquefactive Necrosis

Characterized by the complete enzymatic digestion of dead cells, resulting in a liquid, viscous mass that eventually liquefies and forms a cystic cavity.

Liquid Biopsy

A test of blood, urine, or another body fluid for cancer cells or DNA, RNA, and other material released by a tumor. It may help identify genetic changes, guide treatment, or monitor response and recurrence, but it does not always replace a tissue biopsy.

Liquid Chromatography

liquid chromatography is a diagnostic test, specimen, imaging method, score, instrument, or measurement used to answer a clinical or laboratory question; results must be interpreted with the method and

reference range.

Liquid Diet

A diet consisting partly or entirely of liquids; clear-liquid and full-liquid diets differ in allowed foods and may be prescribed temporarily for procedures, illness, or swallowing or digestive problems.

Liquid Incense

Liquid Incense involves toxic or environmental exposure to liquid incense; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Liquid Nitrogen

Liquid nitrogen is extremely cold nitrogen used to freeze tissue in cryotherapy; contact can cause frostbite and its gas can displace oxygen.

Liquor

A liquid; in clinical usage it may refer to cerebrospinal fluid or a medicinal preparation, depending on context.

Liraglutide

Liraglutide is a GLP-1 receptor agonist used to improve glucose control in type 2 diabetes and, at some doses, support weight management.

Lisch Nodule

A Lisch nodule is a harmless pigmented hamartoma of the iris, commonly associated with neurofibromatosis type 1.

Lisch Nodules

Benign, dome-shaped melanocytic hamartomas on the iris, commonly seen in neurofibromatosis type 1. They usually do not affect vision and support the diagnosis when interpreted with other clinical findings.

Lisinopril

Lisinopril is an angiotensin-converting-enzyme inhibitor used for hypertension, heart failure, and kidney protection in some diabetes; cough, hyper-

kalaemia, renal-function changes, angioedema, and fetal toxicity are important risks.

Lissencephaly

A severe brain malformation characterized by a smooth cerebral surface due to absent (agyria) or decreased (pachygyria) sulcation and gyration, caused by impaired neuronal migration during 12-16 weeks of gestation.

Listeria

Listeria is a genus of bacteria; *L. monocytogenes* can cause foodborne illness, bloodstream infection, and meningitis, especially in pregnancy, older adults, and people with weakened immunity.

Alternate Names:

Infection Listeria (Listeria)

Listeria Grayi

Listeria grayi is a rare Listeria species mainly found in the environment; its significance in a clinical specimen depends on evidence of true infection.

Listeria Infection

An infection caused by Listeria bacteria, often acquired from contaminated food.

Listeria Innocua

Listeria innocua is usually considered nonpathogenic but rare invasive infections have been reported, particularly in people with weakened immunity.

Listeria Ivanovii

Listeria ivanovii is mainly an animal pathogen that rarely infects humans.

Listeria Monocytogenes

A gram-positive, facultatively anaerobic rod that causes listeriosis, a foodborne infection. The organism is transmitted through contaminated food, particularly deli meats, soft cheeses, and ready-to-eat foods. It can grow at refrigeration temperatures.

Listeria Seeligeri

Listeria seeligeri is an environmental Listeria species

that rarely causes human infection.

Listeria Welshimeri

Listeria welshimeri is an environmental and food-associated species that is not a usual human pathogen.

Listeriosis

Listeriosis is infection with *Listeria monocytogenes*, causing febrile gastroenteritis or invasive disease such as meningitis, especially in pregnancy, newborns, older adults, and immunocompromised people. food (deli meats, soft cheeses, unpasteurized dairy, produce). Risk groups: pregnant women (fever, flu-like illness), neonates (sepsis, meningitis), immunocompromised, and elderly.

Listeriosis In Pregnancy

Infection with *Listeria monocytogenes*, a foodborne Gram-positive rod. Pregnant women are 10-20 times more susceptible than the general population. Ingestion of contaminated food (deli meats, soft cheeses, unpasteurized dairy, raw sprouts). Maternal symptoms: fever, myalgias, GI symptoms (nausea, diarrhea).

Liter

A metric unit of volume equal to 1,000 milliliters or 1 cubic decimeter.

Lithiasis

The formation or presence of stones (calculi) in an organ or duct. The term is often named for the site, such as nephrolithiasis in the kidney, ureterolithiasis in the ureter, or cholelithiasis in the gallbladder.

Lithium

Lithium is a mood stabilizer used mainly for bipolar disorder and selected augmentation treatment; its narrow therapeutic range requires monitoring for tremor, thyroid or renal dysfunction, dehydration, drug interactions, and toxicity.

Lithium Carbonate

A mood-stabilizing medicine used mainly to treat

bipolar disorder and reduce the risk of manic or depressive episodes; serum levels, kidney function, thyroid function, and drug interactions require monitoring.

Lithium Toxicity

A potentially life-threatening condition caused by lithium levels exceeding the therapeutic range (0.6–1.2 mEq/L). Mild toxicity (1.5–2.0 mEq/L): nausea, vomiting, diarrhea, coarse tremor, muscle weakness. Moderate toxicity (2.0–2.5 mEq/L): confusion, ataxia, blurred vision, dysarthria.

Lithotomy

A surgical procedure for removing a stone, or the position with the hips and knees flexed and legs supported.

Lithotomy Position

A supine position in which the hips and knees are flexed and the legs are raised and supported to provide access to the pelvis and perineum.

Lithotripsy

Lithotripsy breaks urinary stones into smaller fragments using externally delivered shock waves or an endoscopic energy source so they can pass or be removed. smaller pieces that can pass through the urinary tract. See Shock Wave Lithotripsy.

Lithotripsy Probe

A device that fragments a stone using mechanical, ultrasonic, laser, or shock-wave energy during a stone-removal procedure.

Lithotripter

A device that fragments urinary or other calculi using shock waves or endoscopic energy.

Lithotripter Probe

A probe used to break urinary or biliary stones into smaller fragments for removal.

Litmus

A dye that changes color with acidity or alkalinity

and is used as a simple pH indicator.

Little Area

The highly vascular region of the anterior nasal septum where several arteries meet and where most anterior nosebleeds originate.

Little'S Disease

A clinical condition or finding involving little's disease. Its features, causes, evaluation, and management depend on the clinical context.

Live Attenuated Vaccines

Vaccines containing weakened organisms that stimulate immunity without usually causing disease.

Alternate Names:

Live vaccines

Live Birth

A live birth is delivery of an infant showing signs of life such as breathing, heartbeat, or movement, regardless of gestational age.

Livedo Racemosa

A persistent, irregular, branching, broken-net pattern of skin discoloration caused by disturbance of small or medium blood vessels and associated with vasculopathy, autoimmune disease, or other systemic disorders.

Livedo Reticularis

A net-like, violaceous discoloration of the skin caused by altered blood flow in small cutaneous vessels. It may be temporary and cold-induced or associated with vascular, autoimmune, infectious, or clotting disorders.

Livedoid Vasculopathy

A chronic occlusive disorder of small dermal vessels causing painful purpuric lesions, ulcers, and porcelain-white stellate scars, usually on the lower legs and ankles.

Liver

The largest internal organ, located in the right upper

quadrant of the abdomen beneath the diaphragm.

Liver (Anatomy And Function)

Alternate index label for Anatomy and Function.

Liver Abscess

Focal collection of pus within the liver parenchyma. Pyogenic abscesses are most commonly from biliary tract infection or hematogenous spread from portal or systemic infection. Amoebic abscesses are caused by *Entamoeba histolytica* predominantly in tropical regions. Presents with fever, right upper quadrant pain, and hepatomegaly.

Liver Biopsy

Percutaneous, transjugular, or laparoscopic sampling of liver tissue for histologic evaluation when noninvasive tests do not answer the clinical question. It may stage fibrosis or characterize inflammation, fatty liver disease, infiltrative disease, or a mass; complications include pain and bleeding.

Liver Cancer

A malignant tumor arising in the liver. Hepatocellular carcinoma is the most common primary type in adults; risk factors include chronic liver disease, viral hepatitis, alcohol-related liver injury, and some toxins.

Alternate Names:

Cancer, Liver

Liver Cancer - Hepatocellular Carcinoma

Liver cancer - hepatocellular carcinoma refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Liver Circulation

Liver circulation is the flow of blood through the hepatic artery and portal vein into the liver and out through the hepatic veins.

Liver Cirrhosis

Advanced hepatic fibrosis with regenerative nodules surrounded by dense fibrous septa representing the

end result of chronic liver injury. Common etiologies include chronic viral hepatitis, alcoholic liver disease, and nonalcoholic steatohepatitis.

Liver Damage

Liver damage is injury or loss of liver-cell function caused by infection, alcohol, medicines, toxins, fatty disease, autoimmune disease, or other processes.

Liver Disease

Any disorder that damages or impairs the structure or function of the liver.

Liver Disease Diet

An individualized eating plan for a person with liver disease. It may address adequate energy and protein, sodium or fluid intake, alcohol avoidance, and nutrient deficiencies according to the condition.

Liver Disorder

A clinical condition or finding involving liver disorder. Its features, causes, evaluation, and management depend on the clinical context.

Liver Dysfunction

Liver dysfunction is reduced ability of the liver to perform tasks such as processing nutrients, making proteins, clearing toxins, and producing bile.

Liver Enzymes

ALT, AST, ALP, GGT. Markers of liver injury. ALT more specific for liver.

Liver Failure

A life-threatening condition resulting from severe impairment of hepatic synthetic, metabolic, and detoxification functions.

Liver Failure, Acute

Alternate index label; see *Acute liver failure*.

Liver Fibrosis

Excessive formation of scar tissue in the liver after ongoing injury or inflammation.

Liver Fluke

A parasitic flatworm that infects the liver or bile ducts, including *Clonorchis*, *Opisthorchis*, and *Fasciola* species.

Liver Function (Liver (Anatomy And Function))

Alternate index label for Liver (Anatomy and Function).

Liver Function Tests

Blood tests used to assess liver injury, bile-flow abnormalities, and liver synthetic function. They commonly include aminotransferases, alkaline phosphatase, bilirubin, albumin, and clotting tests; results must be interpreted together with the clinical context.

Liver Function Tests (LFTs)

Blood tests assessing liver-cell injury, bile flow, and synthetic function, commonly including aminotransferases, alkaline phosphatase, bilirubin, albumin, and clotting measures.

Liver Health

Essential for metabolism, detoxification, protein synthesis, bile production, and immune function.

Liver Infection

An infectious process or toxic exposure involving liver infection; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Liver Lobule

The functional unit of the liver: a hexagonal structural unit centered on a central (hepatic) venule, with portal triads (portal vein, hepatic artery, bile ductule) at the corners.

Liver Mass

Abnormality in liver detected on imaging. Benign: hemangioma, focal nodular hyperplasia, adenoma. Malignant: hepatocellular carcinoma, metastasis, cholangiocarcinoma.

Liver Metastases

Liver metastases are cancer deposits that spread to the liver from another primary tumor and may cause no symptoms or produce pain, weight loss, or liver dysfunction.

Liver Neoplasm

A liver neoplasm is an abnormal growth in the liver that may be benign or malignant and may arise from liver cells, bile ducts, or metastasis from another cancer.

Liver Pain

A clinical condition or finding involving liver pain. Its features, causes, evaluation, and management depend on the clinical context.

Liver Regeneration

liver regeneration is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

Liver Scan

A nuclear medicine or imaging examination of the liver used to evaluate its size, shape, tissue distribution, focal lesions, blood flow, or selected functional abnormalities.

Liver Shunt

An abnormal or surgically created channel that diverts blood around the liver, such as a portosystemic shunt; it can reduce detoxification and contribute to hepatic encephalopathy.

Liver Spot

A common lay term for a solar lentigo, a flat brown skin patch caused by chronic ultraviolet exposure.

Liver Spots

Liver spots are flat brown areas of sun-related pigmentation, also called solar lentigines, and are not caused by liver disease.

Liver Transplant

Surgical replacement of a diseased liver with a healthy liver or part of a liver from a donor.

Liver Transplantation

Surgical replacement of a diseased liver with a healthy donor organ indicated for end-stage liver disease, acute liver failure, and select hepatocellular carcinomas within Milan criteria. Requires lifelong immunosuppression with calcineurin inhibitors, antimetabolites, and corticosteroids.

Liver, Diseases Of

Disorders of the liver, including hepatitis, fatty liver, cirrhosis, tumors, and biliary disease, that can impair metabolism and detoxification.

Liver, Enlarged

Alternate index label; see *Enlarged liver*.

Living Donor

Individual who donates an organ (kidney, liver segment, or other tissue) while alive for transplantation into a recipient.

Living Will

A type of advance directive that records a person's preferences for future medical treatment if the person cannot communicate or decide.

Living Will (Advance Directive)

A document stating a person's preferences for future medical treatment if they later lose the capacity to make or communicate decisions.

Alternate Names:

Advance Directive

Living Will (Advance Statements)

A legally binding written document that specifies a person's preferences for medical treatment if they become unable to make decisions for themselves in the future (loss of capacity). Living wills are a type of advance directive (or advance decision/advance statement in the UK).

Alternate Names:

Living With A Chronic Illness - Dealing With Feelings

Living with a chronic illness can bring grief, anxiety, anger, or low mood; support from healthcare professionals, counseling, and trusted people can help.

Living With A Chronic Illness - Reaching Out To Others

Reaching out to family, friends, support groups, or healthcare professionals can reduce isolation and help with practical and emotional effects of chronic illness.

Living With Endometriosis

Living with endometriosis may involve pain management, hormonal or surgical treatment, pacing activities, fertility counseling, and follow-up for changing symptoms.

Living With Hearing Loss

Living with hearing loss may include hearing aids or other devices, communication strategies, hearing protection, and treatment of an underlying cause when possible.

Living With Heart Disease And Angina

Living with heart disease and angina involves prescribed medicines, activity and risk-factor management, follow-up, and urgent care for new or worsening chest pain.

Living With Uterine Fibroids

Living with uterine fibroids may involve monitoring, medicines for bleeding or pain, procedures, or surgery depending on symptoms, size, location, and fertility plans.

Living With Vision Loss

Living with vision loss may involve treating its cause, low-vision rehabilitation, assistive devices, home safety changes, and support with daily activities.

Living With Your Ileostomy

Living with an ileostomy involves caring for the stoma and pouch, maintaining hydration and nutrition, and seeking help for blockage, severe skin problems, or unusual output.

Livor Mortis (Lividity)

Post-mortem purplish-red discoloration of dependent body parts resulting from gravitational settling of blood in uncompressed capillaries and small veins following cessation of circulation.

Alternate Names:

Lividity

Livor Mortis (Postmortem Lividity)

Purplish-red discoloration of dependent body parts following death, caused by gravitational pooling of blood in uncompressed capillaries and venules. Starts within 30 min–2 hours, becomes fixed after 6–12 hours.

Alternate Names:

Postmortem Lividity

Lizard Bite

A puncture or tissue injury caused by a lizard; care includes cleaning and assessing for infection, tetanus, venom exposure, and significant tissue damage according to the species and circumstances.

LLETZ

Large Loop Excision of the Transformation Zone, also called LEEP (Loop Electrosurgical Excision Procedure). A common outpatient procedure used to diagnose and treat cervical intraepithelial neoplasia (CIN).

Alternate Names:

LEEP (Loop Electrosurgical).

Lloyd-Davies Position

A modified lithotomy position in which the hips and knees are flexed and the legs are supported, commonly for pelvic or colorectal surgery.

Loa Loa

The African eye worm, a filarial nematode trans-

mitted by Chrysops deer flies, endemic to the rainforests of West and Central Africa. Adult worms migrate subcutaneously, causing Calabar swellings, and may cross the subconjunctiva of the eye, causing visible worm migration. Microfilariae are detected in daytime peripheral blood.

Loading Dose

A higher initial dose of a drug administered to rapidly achieve therapeutic plasma concentrations, followed by lower maintenance doses to sustain the steady state. The loading dose is calculated as: $\text{Loading dose} = (\text{Vd} \times \text{desired concentration}) / \text{bioavailability}$.

Lobaplatin

Lobaplatin is a platinum-based antineoplastic drug used in some countries and studied for its ability to damage cancer-cell DNA.

Lobar Hemorrhage

Bleeding into a lobe of the brain, beneath the cerebral cortex.

Lobar Pneumonia

Consolidation involving an entire lobe. Most commonly caused by *S. pneumoniae*. Classic progression: congestion, red hepatization, gray hepatization, resolution.

Lobe

A lobe is a distinct subdivision of an organ or structure, such as a lung, liver, brain, thyroid, or gland.

Lobectomy

Surgical removal of a lobe of an organ. Most commonly lung lobectomy (removal of one lobe of the lung, performed for lung cancer, bronchiectasis, abscess, or benign tumor) or temporal lobectomy (removal of the anterior temporal lobe for drug-resistant temporal lobe epilepsy).

Lobeline

Lobeline is a plant alkaloid that affects nicotinic receptors and has been studied for respiratory, neu-

rologic, and smoking-cessation effects.

Lobotomy

A historical brain operation that severed connections involving the frontal lobes.

Lobstein Disease

An older name for osteogenesis imperfecta, an inherited disorder of collagen or bone formation causing bone fragility, fractures, short stature, dentin abnormalities, or hearing loss.

Lobucavir

Lobucavir is an investigational antiviral medicine that has been studied mainly for herpesvirus infections.

Lobular Carcinoma

Lobular carcinoma is a cancer arising from cells of a breast lobule, the gland that produces milk; it may be invasive or confined to the lobule.

Lobular Carcinoma In Situ

Abnormal cells confined to the breast lobules that increase the risk of developing invasive breast cancer.

Lobule

A lobule is a small subdivision of an organ or tissue, such as a liver lobule, lung lobule, or breast lobule.

Lobules

Small subdivisions of an organ; in the breast, milk-producing glandular units.

Local

Local means limited to a particular site or region rather than involving the whole body.

Local Anaesthesia

Loss of sensation in a limited body area caused by a local anesthetic without loss of consciousness.

Local Anesthesia

The reversible abolition of sensation in a circum-

scribed area by infiltration or topical application of local anesthetic agents (lidocaine, bupivacaine) without loss of consciousness. Maximum safe dosing must be observed to avoid systemic toxicity. Epinephrine prolongs duration but is avoided in end-arterial territories.

Local Anesthetic Systemic Toxicity

A life-threatening complication caused by elevated plasma concentration of local anesthetic, most commonly from inadvertent intravascular injection or rapid absorption.

Local Anesthetic Toxicity

CNS toxicity: tinnitus, perioral numbness, seizures. Cardiovascular toxicity: arrhythmias, hypotension, cardiac arrest. Treatment: lipid emulsion therapy (Intralipid), seizure control, ACLS.

Local Invasion

Tumor growth into surrounding tissues. Distinguishes malignant from benign.

Local Therapy

Local therapy treats a disease at or near the affected area rather than throughout the body, for example with a cream, injection, surgery, or focused radiation.

Local Treatment

Treatment directed at a specific body site or lesion rather than the whole body, such as a topical medicine, local injection, radiation field, or localized procedure.

Localised

Confined to a limited area or a small number of adjacent body sites rather than widespread.

Localization

Determining the anatomical site responsible for a symptom, sign, lesion, or physiological function.

Localized

Limited to a particular area or organ, without evi-

dence of distant spread.

Alternate Names:

Localized disease

Localized Edema

Swelling confined to a particular body region because of fluid accumulation, commonly caused by venous or lymphatic obstruction, inflammation, trauma, allergy, or local infection.

Localized Scleroderma

An autoimmune skin disorder causing localized thickening and hardening of the skin, with or without involvement of deeper tissues; it differs from systemic sclerosis.

Lochia

Vaginal discharge following delivery that undergoes characteristic changes over the puerperium. Lochia rubra (days 1-4): red, bloody discharge containing blood, decidual tissue, and fetal membranes. Lochia serosa (days 5-10): pinkish-brown discharge with serous fluid, leukocytes, and organisms.

Locked In Syndrome (Locked-In Syndrome)

Alternate index label for Locked-in Syndrome.

Locked-In Syndrome

A neurological condition in which the patient is fully conscious and aware but cannot move or speak due to complete paralysis of all voluntary muscles except for vertical eye movements and blinking (total locked-in syndrome: loss of all eye movements).

Alternate Names:

Locked In Syndrome (Locked-In Syndrome)

Lockjaw

Alternate index label; see *Tetanus*.

Lockjaw (Tetanus)

A serious neurologic illness caused by tetanospasmin produced by *Clostridium tetani*. It causes painful muscle stiffness and spasms, often beginning in the jaw.

Alternate Names:

Tetanus

Locomotion

Movement of an organism or person from one place to another using coordinated muscles, bones, nerves, and balance systems.

Locomotive System

The musculoskeletal and nervous systems working together to produce posture, balance, and movement from place to place.

Locomotor Ataxia

An older term for tabes dorsalis, a neurologic complication of untreated syphilis causing sensory ataxia and lightning pains.

Locomotor Ataxia (Tabes Dorsalis)

A late manifestation of untreated tertiary syphilis (neurosyphilis) caused by infection with *Treponema pallidum*, involving progressive degeneration of the dorsal columns (fasciculus gracilis and cuneatus) and dorsal roots of the spinal cord.

Alternate Names:

Tabes Dorsalis

Locoregional

Affecting a defined local area and nearby tissues or lymph nodes, without necessarily involving the whole body.

Locus

Specific position of gene on chromosome. Designated by chromosome number, arm (p/q), band, subband. Example: 7q31.2 (CFTR gene).

Locus Ceruleus

A brainstem nucleus involved in alertness, attention, stress responses, and norepinephrine release.

Alternate Names:

Locus coeruleus

Locus Coeruleus

The locus coeruleus is a brainstem nucleus whose

noradrenaline-producing neurons influence arousal, attention, stress, sleep, and pain.

Locus Minoris Resistentiae

Latin for a site of lesser resistance, meaning a body area considered more vulnerable to injury, infection, or disease because of local weakness or prior damage.

Lod Score

lod score is a diagnostic test, specimen, imaging method, score, instrument, or measurement used to answer a clinical or laboratory question; results must be interpreted with the method and reference range.

Loeys-Dietz Syndrome

Loeys-Dietz syndrome is an inherited connective-tissue disorder associated with arterial aneurysm or dissection, distinctive skeletal and craniofacial features, allergic disease, and variable organ involvement. Lifelong vascular imaging and coordinated specialist care are important.

Alternate Names:

Syndrome Loeys-Dietz (Loeys-Dietz Syndrome)

Loiasis

Loiasis is infection with the filarial worm *Loa loa*, transmitted by deerflies and sometimes causing transient eye-worm migration and localized swelling.

Loin

The region of the lower back between the ribs and pelvis, including the flank.

Lometrexol

Lometrexol is an investigational antifolate that inhibits glycinamide ribonucleotide transformylase and has been studied as an anticancer agent.

Lomotil Overdose

Lomotil overdose involves toxic or environmental exposure to lomotil overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Lomustine

Lomustine is an oral chemotherapy medicine used for selected brain tumors and other cancers; delayed low blood counts are an important risk.

Lonafarnib

Lonafarnib is a farnesyltransferase inhibitor used for selected progeroid syndromes and studied in other diseases.

Loneliness

Subjective feeling of isolation. Associated with depression, cognitive decline, mortality.

Long Arm Of A Chromosome

The longer segment of a chromosome, designated q, extending from the centromere toward the chromosome's end.

Long Bones

Long bones concerns long bones, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Long COVID

Long COVID is a condition of persistent or newly developing symptoms after SARS-CoV-2 infection, potentially involving fatigue, exertional intolerance, cognition, breathing, autonomic function, or multiple organs. several weeks and sometimes months or years. Common manifestations include fatigue, post-exertional malaise, cognitive difficulty, breathlessness, dysautonomia, and altered smell or taste.

Long QT Syndrome

A disorder of cardiac repolarization that prolongs the QT interval and increases the risk of torsades de pointes and sudden cardiac death. It may be inherited or acquired from medicines, electrolyte abnormalities, or other conditions.

Long-Acting Beta-2 Agonists

Bronchodilators that provide sustained relaxation of airway smooth muscle for maintenance treatment of asthma or chronic obstructive pulmonary disease. In

asthma, they are used with an inhaled corticosteroid rather than alone.

Long-Sightedness

Hyperopia, a refractive error in which distant objects may be clearer than near objects and accommodation is required for focusing.

Long-Term Care

Assistance with ADLs/IADLs for extended period. Settings: home, assisted living, nursing home. Funding: Medicaid, long-term care insurance, out-of-pocket. Planning essential for elderly and families.

Long-Term Care Facility

A residential or care setting providing ongoing assistance, nursing, rehabilitation, or personal care to people who cannot live independently or need sustained support.

Long-Term Complications Of Diabetes

Chronic complications of diabetes caused mainly by vascular and metabolic injury, including retinopathy, kidney disease, neuropathy, cardiovascular disease, and foot disease.

Long-Term Effect

A health effect that develops or persists after prolonged exposure, treatment, or disease. It may be beneficial, harmful, or uncertain.

Long-Term Memory

Memory that can persist from minutes to many years.

Long-Term Potentiation

Long-term potentiation is a lasting increase in the strength of communication between nerve cells after repeated stimulation and is important in learning and memory.

Long-Term Rehabilitation

Chronic condition management. Focus on maintaining function, preventing complications.

Longevity

Longevity is length of life or survival and is influenced by biology, disease, behavior, environment, social conditions, and healthcare.

Longitudinal Section

A cut or image made along the long axis of an organ, limb, vessel, or other structure.

Longitudinal Studies

Longitudinal studies follow participants or measurements over time to examine change, incidence, exposures, outcomes, and temporal relationships.

Longitudinal Vaginal Septum

A congenital sagittal partition running lengthwise along the vagina, arising from incomplete fusion or failure of resorption of the fused paired Müllerian ducts during embryogenesis. Frequently associated with uterine anomalies such as uterus didelphys or septate uterus, it may be completely asymptomatic, cause obstructive hemivagina (in Herlyn-Werner-Wunderlich syndrome), or produce persistent dyspareunia and difficulty with tampon insertion, corrected surgically by longitudinal excision.

Lontucirev

Lontucirev is an investigational targeted anticancer antibody or drug studied for effects on tumor-associated targets.

Loop Diuretic

A diuretic that inhibits sodium and chloride reabsorption in the thick ascending loop of Henle, producing powerful natriuresis.

Loop Diuretics

A class of potent diuretics that act on the thick ascending limb of the loop of Henle by inhibiting the sodium-potassium-chloride cotransporter, blocking sodium and chloride reabsorption and producing a powerful natriuresis and diuresis. The prototypical loop diuretic is furosemide, with other agents including bumetanide and torsemide.

Loop Electrosurgical Excision Procedure

An outpatient gynecologic operative procedure that utilizes a thin, electrified wire loop powered by high-frequency alternating current to excise abnormal cervical tissue and the transformation zone under colposcopic guidance. Widely employed for both the diagnosis and definitive treatment of high-grade cervical intraepithelial neoplasia (CIN 2 and CIN 3) and persistent cervical dysplasia, providing an intact tissue specimen with surgical margins for histopathologic evaluation.

Alternate Names:

LEEP; Large Loop Excision of the Transformation Zone

Loop Of Henle

U-shaped tubular segment consisting of a thin descending limb, thin ascending limb, and thick ascending limb. The descending limb is permeable to water but not solutes, allowing water reabsorption into the hypertonic medullary interstitium.

Loose Bodies

Free fragments of cartilage, bone, or other tissue within a joint or body cavity that may cause pain, catching, or locking.

Loperamide

Loperamide is a peripherally acting opioid-receptor agonist that slows intestinal motility and reduces diarrhoea; it should be avoided in bloody or febrile diarrhoea and excessive doses can cause severe cardiac arrhythmias.

Loperamide Hydrochloride

Loperamide hydrochloride is an antidiarrheal medicine that slows movement through the intestine; excessive doses can cause dangerous heart rhythms.

Lopinavir

Lopinavir is an HIV protease inhibitor used with ritonavir in antiretroviral treatment; drug interactions are common.

Loratadine

A second-generation (non-sedating) H1 antihistamine used for allergic rhinitis and chronic urticaria. Peripherally selective—minimal CNS penetration, minimal anticholinergic effects. Once daily; prodrug (desloratadine is active). Adverse effects: headache, dry mouth, and rare sedation. Available OTC.

Lorazepam

Lorazepam is a benzodiazepine used short term for severe anxiety, seizures, or sedation; it can cause drowsiness, dependence, and breathing suppression.

Lordosis

The normal inward curvature of the cervical and lumbar spine. Excessive lumbar lordosis (hyperlordosis): increased pelvic tilt, commonly due to hip flexor tightness, weak abdominals, or pregnancy. Presents with low back pain and forward protrusion of the abdomen and buttocks.

Lordosis - Lumbar

Lumbar lordosis is the inward curve of the lower spine; excessive or reduced curvature may be postural, structural, degenerative, or related to pain.

Lorenzo'S Oil

Lorenzo's oil is a mixture of fatty acids used experimentally to reduce very-long-chain fatty acids in adrenoleukodystrophy, with benefit depending on disease stage and context.

Losartan

Losartan is an angiotensin-receptor blocker used for high blood pressure, some kidney disease, and heart failure; it can raise potassium levels.

Losartan Potassium

Losartan potassium is the salt form of losartan, an angiotensin-receptor blocker used to lower blood pressure and protect selected patients with kidney or heart disease.

Losoxantrone

Losoxantrone is an investigational anthracenedione-

like antineoplastic compound that interferes with DNA and cancer-cell growth.

Loss Of Appetite

Reduced desire to eat. It may result from illness, medicines, mood disorders, pain, or other causes.

Loss Of Bladder Control

Alternate index label; see *Urinary incontinence*.

Loss Of Brain Function - Liver Disease

Hepatic encephalopathy, a brain dysfunction caused by severe liver disease or portosystemic shunting and characterized by changes in attention, behavior, consciousness, or neuromuscular function.

Loss Of Heterozygosity

Loss of heterozygosity is loss of one of two different versions of a gene in a cell, a change that can contribute to cancer when a tumor-suppressor gene is affected.

Loss Of Sense Of Smell And Taste With COVID-19

COVID-19 can temporarily impair smell and taste through infection or inflammation of nasal and olfactory pathways. Most people improve, but persistent loss, distortion, poor nutrition, or neurologic symptoms warrants follow-up and smell-retraining advice.

Loss Of Smell (Anosmia)

Partial or complete inability to detect odors, commonly caused by nasal obstruction, infection, inflammation, head injury, neurologic disease, or certain medicines.

Loss-Of-Function Mutation

A genetic change that reduces or abolishes the normal activity or production of a gene product; its clinical effect depends on the gene, variant, and inheritance pattern.

Loss-Of-Function Variant

A genetic variant that reduces or abolishes the nor-

mal activity of the encoded gene product. Disease risk depends on the gene, inheritance pattern, residual function, and whether one or both copies are affected.

Loteprednol Etabonate

Loteprednol etabonate is a corticosteroid eye medicine used for eye inflammation; prolonged use can raise eye pressure and increase infection risk.

Lou Gehrig Disease

Amyotrophic lateral sclerosis, a progressive neurodegenerative disease affecting upper and lower motor neurons and causing weakness, muscle wasting, and eventual respiratory impairment.

Lou Gehrig'S Disease

Alternate index label; see *Amyotrophic lateral sclerosis*.

Louse

A biomedical term referring to louse; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Lovastatin

Lovastatin is a statin medicine that lowers LDL cholesterol and reduces cardiovascular risk; muscle injury and liver effects are uncommon but important risks.

Low Back Pain

Pain or discomfort in the lower back, commonly caused by muscle or ligament strain, disc disease, arthritis, or spinal nerve irritation. Weakness, numbness, new bladder or bowel problems, fever, major trauma, or unexplained weight loss are warning signs requiring prompt medical assessment.

Alternate Names:

Lumbago (Low Back Pain)

Low Back Pain - Acute

A clinical condition involving low Back Pain - Acute; its causes, manifestations, severity, and complications guide diagnosis and management.

Low Back Pain - Chronic

A clinical condition involving low Back Pain - Chronic; its causes, manifestations, severity, and complications guide diagnosis and management.

Low Birth Weight

Birth weight <2500 g, often due to prematurity and/or intrauterine growth restriction; further classified: very low (<1500 g), extremely low (<1000 g), and small for gestational age.

Low Blood Oxygen (Hypoxemia)

An abnormally low oxygen level in arterial blood, usually related to lung disease, impaired breathing, circulation problems, or reduced oxygen availability.

Low Blood Potassium

Low blood potassium, or hypokalemia, can cause weakness, cramps, constipation, abnormal heart rhythms, or paralysis and may result from losses, medicines, or hormonal disease.

Low Blood Pressure

Low blood pressure can cause dizziness, fainting, weakness, or shock and may result from dehydration, medicines, bleeding, infection, heart disease, or autonomic dysfunction.

Low Blood Pressure (Hypotension) Causes

Alternate index label for Hypotension) Cause.

Low Blood Sodium

Low blood sodium, or hyponatremia, reflects excess water relative to sodium and can cause headache, confusion, seizures, or coma when severe or rapidly developing.

Low Blood Sugar

Low blood sugar, or hypoglycemia, occurs when circulating glucose is insufficient for normal brain and body function, causing sweating, tremor, confusion, seizures, or loss of consciousness.

Low Blood Sugar - Newborns

Low blood sugar in a newborn can cause jitteriness, poor feeding, lethargy, apnea, or seizures and is more likely with prematurity, growth problems, maternal diabetes, or illness.

Low Body Temperature

Alternate index label; see *Hypothermia*.

Low Calcium Level - Infants

Hypocalcemia in an infant, defined using age-appropriate values, which may cause jitteriness, poor feeding, lethargy, apnea, seizures, or cardiac rhythm abnormalities.

Low FODMAP Diet

Dietary intervention for irritable bowel syndrome that restricts fermentable oligosaccharides, disaccharides, monosaccharides, and polyols, which are short-chain carbohydrates poorly absorbed in the small intestine and rapidly fermented by colonic bacteria producing gas and drawing water into the intestinal lumen.

Low Grade

Low grade describes a lesion or disease with relatively less aggressive microscopic features or behavior, though prognosis depends on the specific condition.

Low Hemoglobin Count

A hemoglobin level below the expected range, often indicating anemia from blood loss, reduced production, nutritional deficiency, inflammation, or increased destruction of red blood cells.

Low Molecular Weight Heparin

A shorter heparin preparation that predominantly inhibits factor Xa through antithrombin and is used to prevent or treat thrombosis.

Low Nasal Bridge

A low nasal bridge is reduced height of the nasal root and may be a normal familial feature or part of a congenital, genetic, or developmental syndrome.

Low Neutrophil Count (Neutropenia)

Alternate index label for Neutropenia.

Low Platelet Count

Alternate index label; see *Thrombocytopenia*.

Low Potassium (Hypokalemia)

An abnormally low potassium level in the blood, which can cause weakness, cramps, abnormal heart rhythms, or paralysis and may result from losses, medicines, or hormonal disorders.

Low Sex Drive In Women

Persistently reduced sexual desire in a woman that causes personal distress.

Low Sperm Count

A semen condition in which the sperm concentration is below the expected reference range.

Low Testosterone

Alternate index label; see *Male hypogonadism*.

Low Testosterone (Hypogonadism)

Low testosterone (male hypogonadism) is a clinical syndrome of deficient testosterone production by the testes (primary) or inadequate stimulation by the pituitary/hypothalamus (secondary), defined as total testosterone <300 ng/dL on two morning measurements with consistent symptoms.

Alternate Names:

Hypogonadism

Low Testosterone (Low T)

Alternate index label for Low T.

Low Vision

Visual impairment that cannot be fully corrected with spectacles, contact lenses, medication, or surgery, impairing daily activities. Managed with low-vision aids (magnifiers, telescopes, adaptive devices), lighting, and occupational therapy.

Low White Blood Cell Count

A reduced number of white blood cells, which can

increase susceptibility to infection and may result from medicines, infection, autoimmune disease, nutritional deficiency, or bone marrow disorders.

Low White Blood Cell Count And Cancer

Leukopenia or neutropenia occurring with cancer or its treatment, increasing susceptibility to infection and requiring assessment of the cell line affected and the cause.

Low-Calorie Diet

A diet that restricts energy intake to support weight loss or another therapeutic goal.

Alternate Names:

Calorie-restricted diet

Low-Density Lipoprotein

A lipoprotein that transports cholesterol and can contribute to artery-wall plaque when elevated.

Alternate Names:

LDL; bad cholesterol

Low-Density Lipoprotein

The primary circulating cholesterol-carrying lipoprotein in human plasma, generated by the catabolism of very-low-density lipoproteins via intermediate-density lipoproteins. Characterized by a core of cholesteryl esters and triglycerides surrounded by a phospholipid monolayer containing apolipoprotein B-100 (ApoB-100). Elevated circulating LDL particles penetrate the arterial intima, undergo oxidative modification, and are engulfed by macrophages to form foam cells, driving atherogenesis. Target of primary and secondary cardiovascular prevention using HMG-CoA reductase inhibitors (statins), ezetimibe, and PCSK9 inhibitors.

Low-Dose CT Scan

CT protocol using reduced radiation for lung cancer screening. Detects small nodules at earlier stages. False positives require follow-up per Fleischner guidelines.

Low-Fiber Diet

A low-fiber diet limits foods that leave substantial

residue in the intestine and may be used temporarily for selected bowel conditions or before procedures.

Low-Flow Low-Gradient Aortic Stenosis

A challenging clinical and echocardiographic presentation of severe aortic stenosis defined by an indexed aortic valve area $\leq 1.0 \text{ cm}^2$ (or $\leq 0.6 \text{ cm}^2/\text{m}^2$) occurring despite a low mean transvalvular gradient ($< 40 \text{ mmHg}$) and a low stroke volume index ($\text{SVi} < 35 \text{ mL}/\text{m}^2$). Subdivided into: (1) classical low-flow, low-gradient aortic stenosis, in which left ventricular ejection fraction is reduced ($\text{LVEF} < 50\%$), requiring low-dose dobutamine stress echocardiography to differentiate true-severe aortic stenosis (valve area remains small with contractile reserve) from pseudo-severe aortic stenosis (valve area increases $> 1.0 \text{ cm}^2$ with increased flow); and (2) paradoxical low-flow, low-gradient aortic stenosis, in which LVEF is preserved ($\geq 50\%$), but a small, hypertrophied, non-compliant LV cavity generates low stroke volumes.

Alternate Names:

LF-LG Aortic Stenosis; Low-Gradient Severe Aortic Stenosis

Low-Flow Priapism

Alternate index label; see *Priapism*.

Low-Grade Appendiceal Mucinous Neoplasm

A mucin-producing appendiceal neoplasm with low-grade cytologic atypia that may dilate the appendix and rupture, disseminating mucin into the peritoneum as pseudomyxoma peritonei.

Low-Grade Central Osteosarcoma

A rare low-grade malignant osteogenic tumor arising within the medullary cavity of bone. It may resemble fibrous dysplasia or a benign fibro-osseous lesion and requires complete oncologic excision.

Low-Grade Endometrial Stromal Sarcoma

A malignant uterine stromal tumor resembling proliferative-phase endometrial stroma, characterized by infiltrative myometrial or vascular growth and often driven by characteristic gene fusions.

Low-Incision

A low or small surgical incision; the precise meaning depends on the procedure and anatomical site.

Low-Phosphorus Diet

A dietary pattern limiting phosphorus intake for selected patients with chronic kidney disease or persistent hyperphosphatemia. It emphasizes avoidance of phosphate additives while maintaining adequate protein and overall nutritional status.

Low-Potassium Diet

A dietary pattern limiting potassium intake when clinically indicated, usually for hyperkalemia or advanced kidney disease. It must be individualized because unnecessary restriction can reduce diet quality and cause inadequate intake of fruits and vegetables.

Low-Residue Diet

A short-term diet limiting dietary fiber and foods that leave substantial undigested material in the bowel. It may be used in selected gastrointestinal conditions or around certain procedures but is not a universal treatment for inflammatory bowel disease.

Low-Salt Diet

A low-salt diet limits sodium intake and may help control blood pressure or fluid retention in conditions such as heart failure and kidney disease.

Low-Set Ear

An ear positioned lower on the side of the head than expected relative to facial landmarks; it may be an isolated variation or a feature of a genetic or developmental condition.

Low-Set Ears And Pinna Abnormalities

Low-set ears and pinna abnormalities concerns low-set ears and pinna abnormalities, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Lower Airway Obstruction

Narrowing of airways below the carina, causing expiratory wheezing and airflow obstruction on spirome-

try. Causes include asthma, COPD, bronchiectasis, and foreign body. It is characterized by scooped-out expiratory limb on flow-volume loops and reduced FEV1/FVC ratio.

Lower Esophageal Ring

Lower esophageal ring concerns lower esophageal ring, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Lower Esophageal Sphincter

A ring of muscle between the esophagus and stomach that helps prevent acid reflux.

Alternate Names:

LES

Lower Extremity

The lower extremity is the hip, thigh, knee, leg, ankle, and foot, including their bones, joints, muscles, nerves, vessels, and skin.

Lower Extremity Venous Doppler

An ultrasound examination of the leg veins that assesses compressibility and blood flow, commonly to evaluate suspected deep venous thrombosis.

Lower Leg

The part of the limb between the knee and ankle, containing the tibia, fibula, muscles, nerves, and blood vessels.

Lower Motor Neuron Syndrome

A pattern caused by injury to an anterior horn cell, motor cranial-nerve nucleus, nerve root, plexus, or peripheral motor nerve. It typically causes weakness, reduced tone and reflexes, muscle atrophy, and fasciculations in affected muscles.

Lower Segment Cesarean Section

A cesarean delivery performed through an incision in the lower uterine segment, usually through a low transverse uterine incision.

Lower Urinary Tract Symptoms

Urinary symptoms arising from the bladder, bladder outlet, or urethra, classified as storage symptoms, voiding symptoms, and post-micturition symptoms. Causes include benign prostatic obstruction, overactive bladder, infection, neurologic disease, stricture, and malignancy.

Lower-Calorie Choices For Alcoholic Drinks

Lower-calorie alcoholic choices contain less sugar or alcohol energy, but alcohol still carries health risks and moderation or avoidance may be medically advised.

Lozenges

Solid medicinal preparations that dissolve slowly in the mouth, usually to provide local treatment for the throat or oral cavity.

LSD

Lysergic acid diethylamide is a potent hallucinogen that can cause altered perception, visual hallucinations, autonomic effects, and unpredictable changes in mood and behavior.

Lubiprostone

Lubiprostone activates intestinal chloride channels and is used for selected forms of chronic constipation and irritable bowel syndrome with constipation.

Lubricant

A substance that reduces friction between surfaces; medical lubricants are used on skin, mucosa, instruments, or devices and should be selected for tissue compatibility and intended use.

Lucanthone

Lucanthone is an older antiparasitic and investigational antineoplastic compound with effects on parasite or tumor-cell DNA metabolism.

Luciferase

Luciferase is an enzyme that produces light in a chemical reaction; it is widely used as a reporter in

biological research.

Lucilia

A genus of blowflies; some species can cause wound myiasis, while sterile larvae have occasionally been used in maggot debridement.

Ludwig Angina

A rapidly spreading cellulitis of the floor of the mouth and submandibular spaces, usually arising from an infected lower molar. Tongue elevation, dysphagia, drooling, and airway obstruction can develop rapidly and require emergency management.

Ludwig'S Angina

A rapidly progressive, life-threatening cellulitis of the submandibular, sublingual, and submental spaces, typically originating from an odontogenic infection of the lower second or third molars.

Lues

An older name for syphilis, an infection caused by the bacterium *Treponema pallidum*.

Luetic

Pertaining to, caused by, or characteristic of syphilis (lues venerea), the chronic multistage sexually transmitted and congenital systemic infection caused by the spirochete bacterium *Treponema pallidum*. Denotes syphilis-related manifestations across clinical stages: primary luetic chancre (painless, indurated genital ulcer), secondary luetic exanthem (diffuse copper-colored maculopapular rash on palms and soles, condylomata lata), tertiary luetic aortitis (obliterative endarteritis of aortic vasa vasorum causing ascending aortic aneurysm), and tertiary neurosyphilis, including tabes dorsalis (luetic posterior column spinal cord degeneration causing sensory ataxia and Argyll Robertson pupils).

Lugano Classification

The consensus staging and treatment response evaluation criteria for pediatric and adult non-Hodgkin lymphoma and Hodgkin lymphoma, adapted from the historical Ann Arbor staging system. Incorporates 18F-fluorodeoxyglucose positron emission

tomography combined with computed tomography (FDG-PET/CT) as the primary imaging standard for FDG-avid lymphomas. Staged from Stage I (single lymph node region or localized extralymphatic site) to Stage IV (diffuse or disseminated involvement of one or more extralymphatic organs), utilizing the 5-point Deauville score to assess metabolic response during interim and end-of-treatment evaluations.

Lugol'S Iodine

An aqueous solution of iodine and potassium iodide used as an antiseptic, in selected thyroid treatments, and in some laboratory procedures.

Lugol'S Solution

Lugol's solution is a mixture of iodine and potassium iodide used in selected laboratory, diagnostic, and medical applications; it can irritate tissue and affect thyroid function.

Lumbago

A non-specific term for low back pain, typically referring to acute or chronic pain in the lumbar region (between the lower ribs and the gluteal folds). Lumbago is a symptom, not a diagnosis.

Lumbago (Low Back Pain)

Alternate index label for Low Back Pain.

Lumbar

Relating to the lower back, especially the region or five vertebrae between the thoracic spine and sacrum.

Lumbar Disc Herniation

Protrusion or extrusion of lumbar disc material through the annulus fibrosus, which may irritate or compress a nerve root and cause back or leg pain. Symptoms, examination findings, and imaging determine its significance and treatment.

Lumbar Lordosis Lower Spine

Lumbar lordosis is the normal inward curve of the lower spine; excessive or reduced curvature may

be associated with posture, muscle imbalance, hip disease, vertebral deformity, or pain. Examination and imaging are guided by symptoms rather than appearance alone.

Lumbar MRI Scan

Magnetic resonance imaging of the lumbar spine used to evaluate vertebrae, discs, spinal canal, nerve roots, marrow, and soft tissues in conditions such as radiculopathy or spinal stenosis.

Lumbar Plexus

The lumbar plexus is a network of nerves from the upper lumbar spinal roots that supplies the lower abdomen, pelvis, and parts of the thigh and leg.

Lumbar Puncture

A sterile procedure in which a needle is inserted, usually between the L3–L4 or L4–L5 vertebrae below the conus medullaris, to enter the lumbar subarachnoid space. It can measure opening pressure, collect cerebrospinal fluid for cell count, protein, glucose, microbiology, cytology, or other tests, or deliver selected intrathecal medicines. Indications include suspected meningitis or encephalitis, subarachnoid hemorrhage, inflammatory or demyelinating disease, malignancy, and planned intrathecal therapy. Before the procedure, clinicians assess for features suggesting a mass lesion, obstructive hydrocephalus, or another risk of brain herniation, as well as bleeding risk, anticoagulant use, thrombocytopenia, and infection at the puncture site; neuroimaging is obtained before puncture when clinically indicated. Possible complications include post-dural-puncture headache, back pain, bleeding, infection, persistent CSF leak, and, rarely, neurologic deterioration.

Alternate Names:

LP Spinal Tap

Lumbar Radiculopathy

A condition caused by compression, irritation, or inflammation of a spinal nerve root in the lumbar spine, resulting in pain radiating into the lower extremity along the distribution of the affected nerve root (sciatica). The most commonly affected nerve roots are L4, L5, and S1.

Lumbar Region

The lumbar region is the lower part of the back between the ribs and pelvis, containing the five lumbar vertebrae.

Lumbar Spine

The inferior mobile segment of the human vertebral column, typically composed of five robust vertebrae (L1 through L5) situated between the thoracic spine and the sacrum. Anatomically characterized by a secondary lordotic curvature, broad kidney-shaped vertebral bodies, and short, sturdy spinous and transverse processes designed to bear heavy axial compression loads. Houses the conus medullaris (terminating at the L1–L2 interspace) and the cauda equina nerve roots bathed in CSF within the lumbar thecal sac. Common pathologies include lumbar intervertebral disc herniation with radiculopathy (sciatica), lumbar spinal canal stenosis with neurogenic claudication, spondylolysis, and degenerative spondylolisthesis.

Alternate Names:

Lumbar region

Lumbar Spine CT Scan

Computed tomography of the lumbar spine that provides detailed images of vertebral bone and alignment and is particularly useful for detecting fractures or other bony abnormalities.

Lumbar Stenosis

Narrowing of the spinal canal or nerve openings in the lower back.

Alternate Names:

Stenosis Lumbar (Lumbar Stenosis) Stenosis Spinal (Lumbar Stenosis)

Lumbar Vertebrae

Largest and strongest mobile vertebrae for weight-bearing. Large kidney-shaped bodies, short hatchet-shaped spinous processes, sagittally-oriented facets allowing primarily flexion/extension. Lordotic curvature. Most common disc herniation site, particularly L4-L5 and L5-S1 causing sciatica.

Lumbar Vertebrae (L1-L5)

The five lumbar vertebrae in the lower back, numbered L1 through L5, between the thoracic spine and sacrum and bearing much of the body's load.

Lumbosacral Region

The lumbosacral region is the junction between the lower lumbar spine and the sacrum at the back of the pelvis.

Lumbosacral Spine CT

Computed tomography of the lower spine and sacrum using x-rays and computer reconstruction to evaluate fractures, bone lesions, alignment, and selected causes of nerve compression.

Lumbosacral Spine X-Ray

Plain radiographic imaging of the lumbar spine and sacrum used to evaluate alignment, fractures, degenerative changes, deformity, and some bone lesions.

Lumbricals

Small hand or foot muscles arising from tendons of the flexor digitorum profundus or flexor digitorum longus and assisting flexion-extension coordination.

Lumen

The inner space of a hollow organ, vessel, or tube.

Luminescence

Luminescence is emission of light from a substance through a process other than high-temperature incandescence, such as fluorescence or chemiluminescence.

Luminescence Type

Luminescence type identifies the mechanism by which a material emits light, such as fluorescence, phosphorescence, bioluminescence, or chemiluminescence.

Luminescent

Emitting visible light without being incandescent, as in some diagnostic markers or biological phenomena.

Lump In The Abdomen

An abdominal lump may be a hernia, enlarged organ, cyst, stool, inflammation, or tumor; pain, vomiting, fever, growth, or a pulsatile mass requires prompt assessment.

Lumpectomy

Breast-conserving surgery excising a malignant lesion with a margin of normal tissue, combined with whole-breast radiation to yield survival equivalent to mastectomy for early-stage cancer. An acceptable margin is no ink on tumor for invasive cancer and 2 mm for ductal carcinoma in situ.

Lumpy Jaw

A chronic granulomatous infection of the jaw, usually cervicofacial actinomycosis, causing swelling and sinus formation.

Lunar Caustic

An old name for silver nitrate, a caustic agent formerly used to cauterize tissue and treat some lesions.

Lunate

Crescent-shaped proximal row carpal bone between scaphoid and triquetrum. Most commonly dislocated carpal bone from falls on extended wrists. Dislocation may compress the median nerve causing acute carpal tunnel syndrome. Vulnerable to avascular necrosis (Kienbock disease) from its precarious blood supply.

Lung

The paired, cone-shaped respiratory organs located in the thoracic cavity, responsible for gas exchange (oxygen uptake, carbon dioxide elimination). Each lung is divided into lobes: the right lung has three (superior, middle, inferior), the left lung has two (superior, inferior) with a cardiac notch accommodating the heart.

Lung Abscess

Localized necrosis with cavitation containing purulent material. Causes include aspiration and anaerobic bacteria. CXR shows thick-walled cavity with air-fluid level.

Lung Anatomy (Lungs Design And Purpose)

Alternate index label for Lungs Design And Purpose.

Lung Cancer

Leading cancer death worldwide. Types: small cell, non-small cell. Risk factors: smoking, radon, asbestos.

Alternate Names:

Cancer, Lung Cancer In Lung (Lung Cancer)

Lung Cancer - Small Cell

Lung cancer - small cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Lung Collapse

Alternate index label; see *Atelectasis*; see also *Pneumothorax*.

Lung Compliance

The change in lung volume produced by a given change in transpulmonary pressure. Reduced compliance increases the work of breathing, whereas excessive compliance may occur when elastic recoil is lost, as in emphysema.

Lung Cyst

A lung cyst is a round air- or fluid-containing space in lung tissue, caused by developmental disease, infection, emphysema, inflammation, or another disorder.

Lung Development

Lung development is embryonic and postnatal formation and maturation of airways, alveoli, blood vessels, and gas-exchange capacity.

Lung Diffusion Testing

Pulmonary-function testing that measures transfer of gas, usually carbon monoxide, from the air sacs into pulmonary capillary blood and helps assess emphysema, interstitial lung disease, or pulmonary

vascular disease.

Lung Disease

Lung disease encompasses disorders affecting the airways, parenchyma, pleura, pulmonary vasculature, and chest wall.

Lung Disease, Interstitial

Alternate index label; see *Interstitial lung disease*.

Lung Fibroma

A lung fibroma is a usually benign fibrous tumor in or around lung tissue, often found incidentally and assessed by imaging or biopsy when needed.

Lung Fluke

A parasitic trematode, especially *Paragonimus* species, whose infection can cause chronic cough, hemoptysis, and pulmonary lesions.

Lung Function Tests

Physiologic tests that measure aspects of breathing, including airflow, lung volumes, and gas transfer, to assess respiratory function.

Lung Gallium Scan

A nuclear medicine study using gallium-67 to identify areas of inflammation or infection in the lungs and other tissues; it has largely been replaced by more specific imaging.

Lung Herniation

Protrusion of lung tissue through a defect in the chest wall, most commonly occurring after thoracic trauma, thoracotomy, or in association with rib fractures and chronic obstructive pulmonary disease. Presents with a reducible chest wall mass that increases with coughing or straining and may cause pain and respiratory compromise.

Lung Injury

Damage to lung tissue or the pulmonary circulation caused by trauma, infection, inhaled substances, inflammation, aspiration, oxygen toxicity, or other disease.

Lung Lavage

Lung lavage is washing or flushing of the airways or lungs with fluid, used in selected conditions such as pulmonary alveolar proteinosis or sampling.

Lung Metastases

Lung metastases are cancer deposits that have spread to the lungs from another primary tumor and may cause cough, breathlessness, chest pain, or no symptoms.

Lung Needle Biopsy

CT- or ultrasound-guided percutaneous sampling of a lung or pleural lesion to evaluate selected masses, infection, or diffuse disease. Pneumothorax and pulmonary hemorrhage are important complications.

Lung Neoplasm

A lung neoplasm is an abnormal growth in the lung that may be benign or malignant; evaluation commonly uses imaging and tissue sampling.

Lung Nodule

A rounded, usually well-defined opacity in the lung measuring up to 3 cm on imaging. It may be benign, infectious, inflammatory, or an early manifestation of lung cancer and often requires follow-up assessment.

Lung PET Scan

Positron-emission tomography, often combined with computed tomography, using a radiotracer to assess metabolic activity in lung nodules, cancer, infection, or inflammation.

Lung Plethysmography

A pulmonary-function test performed in a sealed chamber to measure lung volumes, including trapped gas and functional residual capacity, and airway resistance.

Lung Problems And Volcanic Smog

Volcanic smog contains gases and fine particles that irritate eyes and airways and can worsen asthma or lung disease; limiting exposure and following

public-health advice reduce harm.

Lung Protective Ventilation

A ventilation strategy using low tidal volumes (6 mL/kg ideal body weight), plateau pressure < 30 cmH₂O, appropriate PEEP, and permissive hypercapnia to minimize ventilator-induced lung injury. It is the standard of care for ARDS and is used in all mechanically ventilated patients to reduce barotrauma, volutrauma, and biotrauma.

Lung Surgery

A medical procedure, treatment, device, or care topic related to lung surgery; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Lung Transplant

Surgical replacement of one or both lungs from a donor for end-stage lung disease: COPD (most common), idiopathic pulmonary fibrosis, cystic fibrosis, pulmonary hypertension, and other advanced conditions. Types: single, bilateral, heart-lung.

Lung Transplantation

Surgical replacement of a diseased lung with a healthy lung from a donor. Indications include IPF, COPD, CF, PAH, and bronchiectasis. Options include single lung transplant, double lung transplant, or heart-lung transplant. Post-transplant immunosuppression is required lifelong. Complications include rejection and infection.

Lung, Diseases Of

Conditions affecting the airways, alveoli, pleura, or pulmonary circulation, including infection, asthma, COPD, fibrosis, and cancer.

Lung Volume Reduction Surgery

A surgical palliative procedure for patients with severe, heterogeneous emphysema (predominantly upper-lobe emphysema with low baseline exercise capacity) refractory to medical therapy. Performed through video-assisted thoracoscopic surgery or median sternotomy, approximately 20 to 30 percent of the most severely diseased, hyperinflated, non-

functional lung tissue is excised, which decompresses adjacent healthier parenchyma, improves diaphragm curvature, enhances chest wall respiratory mechanics, and increases dynamic lung elastic recoil.

Alternate Names:

LVRs; Reduction Pneumoplasty; Bilateral Lung Volume Reduction

Lungs

The paired respiratory organs in the chest that exchange oxygen and carbon dioxide between air and blood; they also contribute to acid–base regulation.

Lungs Design And Purpose

The lungs exchange oxygen and carbon dioxide through branching airways and microscopic alveoli surrounded by capillaries. The airways filter, warm, and humidify air, while respiratory muscles drive ventilation; disease can impair airflow, diffusion, or blood flow.

Alternate Names:

Lung Anatomy (Lungs Design And Purpose)

Lungs Fluid (Pleural Effusion (Fluid In The Chest Or On Lung))

Alternate index label for Pleural Effusion (Fluid In the Chest or On Lung).

Lunula

The pale crescent-shaped area at the base of a fingernail or toenail.

Lupus

A group of autoimmune diseases in which the immune system attacks the body's own tissues. Systemic lupus erythematosus may affect the skin, joints, kidneys, blood cells, nervous system, and other organs. Also called SLE.

Alternate Names:

SLE.

Lupus (Systemic Lupus)

Alternate index label for Systemic Lupus.

Lupus Erythematosus

Systemic lupus erythematosus is a chronic autoimmune disease characterized by production of autoantibodies against nuclear antigens, leading to immune complex-mediated inflammation and tissue damage in multiple organ systems. The disease predominantly affects women of childbearing age, with a striking female predominance of 9 to 1.

Lupus Erythematosus (Systemic Lupus Erythematosus - SLE)

A chronic multisystem autoimmune disease predominantly affecting women (9:1 female:male) of childbearing age, characterized by autoantibody production (ANA 98%, anti-dsDNA 70%, anti-Sm 30%) and immune complex deposition causing tissue injury.

Alternate Names:

Systemic Lupus Erythematosus - SLE

Lupus Nephritis

Kidney inflammation caused by systemic lupus erythematosus. It can cause blood or protein in the urine, high blood pressure, and reduced kidney function. Classification is based largely on kidney-biopsy findings, and treatment depends on the class, severity, kidney function, and individual risks.

Lupus Panniculitis

A chronic inflammatory form of cutaneous lupus affecting subcutaneous fat, producing firm deep nodules that may heal with lipoatrophy, scarring, or overlying discoid changes.

Lupus Pernio

A chronic, violaceous skin lesion of the face and extremities associated with sarcoidosis, indicating a higher risk of systemic involvement and chronic disease.

Lupus Vulgaris

A chronic, progressive form of cutaneous tuberculosis (TB of the skin) caused by *Mycobacterium tuberculosis*. Once common in Europe, now rare in developed countries but still seen in developing regions.

Lurtotecan

Lurtotecan is an investigational topoisomerase-I inhibitor studied for anticancer activity by interfering with DNA replication and repair.

Luteal Cell

luteal cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Luteal Phase Defect

An endocrinopathy of the menstrual cycle characterized by insufficient progesterone production by the corpus luteum or inadequate endometrial secretory responsiveness, resulting in an abnormally shortened luteal phase (typically fewer than 10 to 11 days from ovulation to menses). Associated with abnormal endometrial maturation, failed embryonic implantation, and recurrent early pregnancy loss.

Alternate Names:

Luteal Phase Deficiency; LPD

Lutein

A carotenoid pigment found in leafy greens and concentrated in the retina.

Luteinization

Luteinization is the transformation of an ovarian follicle after ovulation into a corpus luteum that produces progesterone.

Luteinizing Hormone

LH: a gonadotropin secreted by the anterior pituitary, regulating gonadal steroidogenesis. In females: stimulates theca cells to produce androgens (aromatized to estradiol); mid-cycle LH surge triggers ovulation (used in ovulation prediction and timing of intercourse/IVF). In males: stimulates Leydig cells to produce testosterone.

Luteinizing Hormone

A pituitary gonadotropin that triggers ovulation and corpus-luteum formation and stimulates testosterone production in the testes.

Luteinizing Hormone Blood Test

A blood test measuring luteinizing hormone, which stimulates ovulation and ovarian hormone production in females and testosterone production in males; it helps evaluate puberty, fertility, and pituitary or gonadal disorders.

Luteinizing Hormone-Releasing Hormone

An older name for gonadotropin-releasing hormone (GnRH), a hypothalamic hormone released in pulses that stimulates the anterior pituitary to secrete luteinizing hormone and follicle-stimulating hormone.

Luteinizing Hormone-Releasing Hormone

Another name for gonadotropin-releasing hormone, which stimulates pituitary release of LH and FSH.

Luteolin

Luteolin is a plant flavonoid studied for antioxidant and anti-inflammatory effects; clinical benefits and safe therapeutic dosing are not established for routine treatment.

Luteolysis

Luteolysis is the breakdown of the corpus luteum, causing progesterone levels to fall when pregnancy has not occurred.

Luteoma

A luteoma is a usually benign ovarian mass of hormone-producing lutein cells, sometimes associated with pregnancy and androgen excess.

Luteoma Of Pregnancy

A benign, pregnancy-associated ovarian proliferation of luteinized stromal cells driven by hormonal stimulation. It may cause maternal or fetal androgen excess and usually regresses after delivery.

Luteotropic Hormone

A hormone that supports the corpus luteum; the term traditionally refers mainly to prolactin, although usage varies by species and source.

Lutetic

Relating to syphilis, an infection caused by the bacterium *Treponema pallidum*. This term is related to “luetetic,” from an older name for syphilis; “luteal” is the term for the corpus luteum.

Lutetium-177

Beta and gamma emitter, half-life 6.7 days. Used in peptide receptor radionuclide therapy (PRRT) and PSMA therapy. Both therapeutic and imaging capabilities.

Luxation

Complete dislocation of a joint, in which the articular surfaces lose their normal contact.

LVAD

Left-ventricular assist device, an implanted pump that helps the left ventricle circulate blood.

Alternate Names:

Left-ventricular assist device

LVNC

Alternate name; see *Left Ventricular Noncompaction*.

LVRS

Alternate name; see *Lung Volume Reduction Surgery*.

Lyase

A lyase is an enzyme that breaks or forms chemical bonds without using hydrolysis or oxidation as the main reaction.

Lycopene

Lycopene is a red plant pigment and antioxidant found in tomatoes and some other foods; supplements have not been proven to prevent disease reliably.

Lyme Disease

A tick-borne infection caused by *Borrelia burgdorferi* (*Ixodes scapularis*, deer tick), endemic to the northeastern and upper midwestern US. Early lo-

calized: erythema migrans (bullseye rash, >5cm) at tick bite site, fatigue, fever, myalgia.

Alternate Names:

Arthritis Lyme (Lyme Disease)

Lyme Disease Blood Test

Serologic testing for antibodies to *Borrelia burgdorferi* and related organisms, usually using a two-step algorithm; results must be interpreted with symptoms, exposure, and timing.

Lymph

A clear, watery fluid that circulates through the lymphatic system, derived from interstitial fluid that has entered the lymphatic capillaries.

Lymph Leakage

Lymph leakage is escape of lymphatic fluid after injury, surgery, or obstruction and may cause swelling, fluid collections, or chylous drainage.

Lymph Node

A lymph node is a small immune organ that filters lymph and helps activate immune responses to infection, inflammation, or abnormal cells.

Lymph Node / Gland

Small, bean-shaped organs of the lymphatic system that filter lymph and serve as sites for immune cell activation. Lymph nodes are distributed throughout the body (approximately 500-600 in total) and are concentrated in regions: cervical, axillary, inguinal, mediastinal, mesenteric, para-aortic, and iliac chains.

Lymph Node Biopsy

Removal of part or all of a lymph node for microscopic and sometimes microbiologic or molecular examination to evaluate malignancy, infection, or inflammatory disease.

Lymph Node Culture

Laboratory culture of lymph-node tissue or aspirate to identify bacteria, fungi, or mycobacteria causing lymphadenitis; culture is interpreted with histology

and other tests.

Lymph Node Dissection

Removal of lymph nodes for staging or treatment. Sentinel node biopsy less invasive alternative.

Lymph Node Structure

Small bean-shaped lymphoid organs (1-25 mm) distributed along lymphatic vessels, filtering lymph and mounting immune responses. Each has a cortex (outer, containing B-cell follicles with germinal centers), paracortex (deep cortex, containing T-cells and dendritic cells), and medulla (inner, containing medullary cords and sinuses).

Lymph Nodes

Small immune organs that filter lymph and help detect infection or abnormal cells.

Alternate Names:

Lymph glands

Lymph Nodes, Swollen

Alternate index label; see *Swollen lymph nodes*.

Lymph Swollen Glands (Swollen Lymph Nodes)

Alternate index label for Swollen Lymph Nodes.

Lymph Swollen Nodes (Swollen Lymph Nodes)

Alternate index label for Swollen Lymph Nodes.

Lymph System

The lymph system includes lymph, lymphatic vessels, lymph nodes, spleen, thymus, and related tissues that return fluid and support immune defense.

Lymphadenectomy

Surgical removal of lymph nodes, either sentinel or regional, for staging or therapeutic purposes; examples include axillary, inguinal, cervical, and retroperitoneal lymph node dissection.

Lymphadenitis

Lymphadenitis is inflammation or infection of lymph

nodes, causing localized tenderness and enlargement from bacterial, viral, mycobacterial, cat-scratch, or other disease. infection. Affected nodes may be tender, warm, or fluctuant, and the condition may be localized or generalized.

Lymphadenitis, Mesenteric

Alternate index label; see *Mesenteric lymphadenitis*.

Lymphadenopathy

Enlargement of lymph nodes (greater than 1 cm in diameter, or >1.5 cm for inguinal nodes). Lymphadenopathy can be localised (single or contiguous region) or generalised (three or more non-contiguous regions).

Lymphangiectasis

Abnormal dilatation of lymphatic vessels. It may be congenital or acquired and can cause lymph leakage, oedema, protein-losing enteropathy, chylous effusions, or pulmonary and pleural complications depending on the affected site.

Lymphangiogenesis

lymphangiogenesis is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

Lymphangiogram

A lymphangiogram uses contrast imaging to show lymphatic vessels and nodes, helping evaluate obstruction, leakage, malformation, or spread of disease.

Lymphangiography

Contrast imaging of lymphatic vessels. Evaluates lymphatic obstruction, fistulas. Largely replaced by MRI/CT.

Lymphangioleiomyomatosis

A rare, progressive cystic lung disease caused by proliferation of abnormal smooth muscle-like cells (LAM cells) in the lungs, lymphatics, and kidneys. It occurs almost exclusively in women of childbearing

age. HRCT shows thin-walled cysts throughout both lungs. It may be associated with tuberous sclerosis complex.

Lymphangioma

A benign congenital malformation of the lymphatic system, composed of dilated lymphatic channels. Lymphangiomas are typically present at birth (90% diagnosed by age 2) and result from failure of the lymphatic system to develop normal connections with the venous system.

Lymphangiomatosis

An uncommon disorder involving excessive or abnormal growth of lymphatic vessels in one or more tissues.

Lymphangiomyoma

Lymphangiomyoma is a benign proliferation of lymphatic-type smooth-muscle cells, usually discussed in relation to lymphangioleiomyomatosis.

Lymphangiosarcoma

Lymphangiosarcoma is a rare aggressive cancer arising from cells of lymphatic vessels, sometimes developing in chronically lymphedematous tissue.

Lymphangitis

Inflammation and infection of lymphatic vessels, usually spreading from an infected skin wound or cellulitis; tender red streaks, fever, and enlarged regional lymph nodes may occur.

Lymphatic Basin

A lymphatic basin is the group of lymph nodes and drainage pathways receiving lymph from a defined body region.

Lymphatic Disease

A clinical condition or finding involving lymphatic disease. Its features, causes, evaluation, and management depend on the clinical context.

Lymphatic Filariasis

A mosquito-borne parasitic infection caused by fi-

larial worms, most commonly *Wuchereria bancrofti*; adult worms inhabit lymphatic vessels and can cause chronic lymphatic obstruction, lymphedema, and elephantiasis.

Lymphatic Mapping

A clinical condition or finding involving lymphatic mapping. Its features, causes, evaluation, and management depend on the clinical context.

Lymphatic Obstruction

Impaired flow of lymph through lymphatic vessels or nodes, causing accumulation of protein-rich interstitial fluid and swelling known as lymphedema.

Lymphatic Obstruction (Lymphedema)

Alternate index label for Lymphedema.

Lymphatic System

A network of lymphatic vessels, lymph nodes, and lymphoid organs (tonsils, thymus, spleen, Peyer patches) that returns interstitial fluid (lymph) to the bloodstream, transports dietary fat (chyle), and serves as the conduit for immune surveillance.

Lymphatic Tissue

Lymphatic tissue contains immune cells and structures that filter lymph, trap antigens, and coordinate immune defense.

Lymphatic Vessel

A lymphatic vessel transports lymph from tissues through lymph nodes and eventually returns fluid to the bloodstream.

Lymphedema

Persistent swelling caused by accumulation of lymphatic fluid in tissues, usually in an arm or leg.

Alternate Names:

Lymphatic Obstruction (Lymphedema)

Lymphedema (Lymphoedema)

Chronic swelling of a body part due to impaired lymphatic drainage, leading to accumulation of protein-rich interstitial fluid. Primary lymphedema: caused

by congenital or hereditary lymphatic maldevelopment.

Alternate Names:

Lymphoedema

Lymphedema Management

Measures used to reduce swelling and preserve function, including compression, exercise, skin care, and selected forms of manual lymphatic therapy.

Lymphoedema

A misspelling of lymphoedema, swelling caused by accumulation of protein-rich lymph when lymphatic drainage is impaired.

Lymphoblast

A lymphoblast is an immature lymphoid precursor that normally develops into a B or T lymphocyte; uncontrolled accumulation occurs in acute lymphoblastic leukemia.

Lymphoblast Count

A lymphoblast count measures immature lymphoid cells in blood or bone marrow and may help evaluate or monitor acute lymphoblastic leukemia.

Lymphocele

A lymphocele is a collection of lymphatic fluid caused by disruption or blockage of lymph vessels, often after surgery or transplantation.

Lymphocyte

A type of white blood cell (leukocyte) that is the primary cellular component of the adaptive immune system. Lymphocytes account for 20-40% of circulating white blood cells ($1.0\text{--}4.8 \times 10^9/\text{L}$). Three main types: (1) T lymphocytes (T cells): develop in the thymus, responsible for cell-mediated immunity.

Lymphocyte Activation

A clinical condition or finding involving lymphocyte activation. Its features, causes, evaluation, and management depend on the clinical context.

Lymphocyte Count

A lymphocyte count measures lymphocytes in the blood and can help assess infection, immune deficiency, inflammation, or blood disorders.

Lymphocyte Proliferation

A clinical condition or finding involving lymphocyte proliferation. Its features, causes, evaluation, and management depend on the clinical context.

Lymphocytic

Relating to lymphocytes or characterized by an increased or predominant number of lymphocytes.

Lymphocytic Colitis

Chronic inflammatory bowel disease characterized by increased intraepithelial lymphocytes exceeding twenty per one hundred surface epithelial cells in the colonic mucosa without significant subepithelial collagen band thickening.

Alternate Names:

Colitis Collagenous (Lymphocytic Colitis) Colitis Lymphocytic (Lymphocytic Colitis) Colitis Microscopic (Lymphocytic Colitis)

Lymphocytic Interstitial Pneumonia

A rare ILD characterized by diffuse polyclonal lymphocytic infiltration of the lung parenchyma. It is strongly associated with Sjgren syndrome and HIV. HRCT shows ground-glass opacities, thin-walled cysts, and lymphadenopathy. Treatment is corticosteroids for symptomatic disease.

Lymphocytic Leukemia

Lymphocytic leukemia is a blood cancer in which abnormal lymphoid cells multiply in the bone marrow, blood, lymph nodes, or other tissues.

Lymphocytic Mastitis

A benign fibroinflammatory breast lesion with dense lymphocytic infiltration and stromal fibrosis, classically associated with diabetes or autoimmune disease and also called diabetic mastopathy. It can mimic carcinoma clinically and radiologically.

Lymphocytoma Cutis

Lymphocytoma cutis is a benign skin lymphoid proliferation presenting as red or violaceous plaques or nodules and sometimes associated with infection or immune stimulation.

Lymphocytopenia

A reduced lymphocyte count ($< 1.0 \times 10^9/L$). Causes: acute (stress, steroids, sepsis, trauma, burns), infections (HIV, viral, tuberculosis, typhoid), immunosuppressive drugs, radiation, marrow failure (aplasia, leukemia, lymphoma), autoimmune (SLE), and congenital immunodeficiencies.

Lymphocytosis

An abnormally increased number or proportion of lymphocytes in the blood, often associated with viral infection, some bacterial infections, or lymphoproliferative disorders.

Lymphocytosis (High Lymphocyte Count)

An increased number of lymphocytes in the blood, often associated with infection, inflammation, or a lymphoproliferative disorder.

Lymphoedema

Chronic swelling caused by impaired lymphatic drainage, most often affecting a limb but potentially involving other regions.

Lymphofollicular Hyperplasia

Lymphofollicular hyperplasia is enlargement or increased formation of lymphoid follicles in response to immune stimulation, infection, inflammation, or other triggers.

Lymphogranuloma Venereum

Lymphogranuloma venereum is a sexually transmitted infection caused by invasive Chlamydia trachomatis strains, often producing a small lesion followed by painful regional lymphadenopathy or proctitis. L1–L3, classically producing transient genital lesions and painful inguinal lymphadenopathy.

Lymphoid

Resembling or relating to lymphatic tissue, lymphocytes, or organs involved in lymphocyte development and immune function.

Lymphoid Hyperplasia

Lymphoid hyperplasia is benign increase in lymphoid tissue caused by immune stimulation, infection, inflammation, or sometimes a reactive response near a tumor.

Lymphoid Tissue

Tissue containing lymphocytes and supporting immune responses, found in lymph nodes, spleen, thymus, tonsils, and mucosal sites.

Lymphokine

A cytokine produced by lymphocytes (a subset of cytokines; the term lymphokine is largely historical, now 'cytokine'). Examples: IL-2, IL-4, IFN-gamma (from T cells), and B-cell growth factors. Mediate lymphocyte communication, activation, and regulation of immune responses.

Lymphoma

A malignancy of lymphocytes, divided into Hodgkin lymphoma (HL—Reed-Sternberg cells, cervical/supraclavicular nodes, B symptoms, younger adults) and non-Hodgkin lymphoma (NHL—heterogeneous, more common, aggressive or indolent).

Lymphoma (Hodgkin'S And Non-Hodgkin'S)

A group of cancers arising from lymphocytes, including Hodgkin lymphoma and the many types of non-Hodgkin lymphoma.

Alternate Names:

Hodgkin's and Non-Hodgkin's

Lymphoma, Hodgkin

Alternate index label; see *Hodgkin lymphoma* (*Hodgkin disease*).

Lymphoma, Non-Hodgkin

Alternate index label; see *Non-Hodgkin lymphoma*.

Lymphomatoid Papulosis

Lymphomatoid papulosis is a chronic recurrent skin disorder with cancer-like lymphoid cells; the bumps often heal spontaneously but require monitoring for lymphoma.

Lymphopenia

Lymphopenia is an abnormally low number of lymphocytes, caused by infection, medicines, immune disease, malnutrition, marrow disorders, or other conditions.

Lymphoplasmacytic Lymphoma

Alternate index label; see *Waldenstrom macroglobulinemia*.

Lymphopoiesis

The production and maturation of lymphocytes from hematopoietic stem and progenitor cells. It occurs in primary lymphoid organs such as bone marrow and thymus and continues in peripheral lymphoid tissues during immune responses.

Lymphoproliferative Disorder

A clinical condition or finding involving lymphoproliferative disorder. Its features, causes, evaluation, and management depend on the clinical context.

Lymphoscintigraphy

Lymphatic system imaging using Tc-99m nanocolloid injected near lesion. Used for sentinel lymph node biopsy (melanoma, breast cancer), lymphedema evaluation.

Lynch Syndrome

Hereditary nonpolyposis colorectal cancer syndrome caused by mismatch repair gene mutations including MLH1, MSH2, MSH6, and PMS2 with autosomal dominant inheritance. It increases the risk of colorectal, endometrial, ovarian, gastric, and urinary tract cancers.

Lyonnaise

A historical term associated with a Lyon-based medical classification or method; its precise meaning depends on the source and should be verified.

Lysergic Acid Diethylamide

A potent hallucinogenic drug (psychedelic) classified as a Schedule 1 controlled substance (high abuse potential, no accepted medical use). LSD acts as a partial agonist at serotonin 5-HT_{2A} receptors in the prefrontal cortex, leading to altered sensory perception, mood, and cognition.

Lysine

Lysine is an essential amino acid needed for protein production, growth, and tissue repair; it must be obtained from food.

Lysinibacillus

Lysinibacillus is a group of environmental, spore-forming bacteria; recovery from a clinical sample requires correlation with symptoms and specimen quality.

Lysis

The breakdown, dissolution, or destruction of cells or tissue; examples include hemolysis, the breakdown of red blood cells, and thrombolysis, the dissolution of a blood clot.

Lysogenic Conversion

The acquisition of new properties (genes) by a bacterium when a temperate bacteriophage integrates its genome as a prophage and expresses phage-encoded genes.

Lysogeny

Lysogeny is a state in which a bacteriophage's genetic material is maintained inside a bacterial cell and replicated with the host genome.

Lysol

A historical trade name for cresol-containing disinfectant preparations; products and formulations vary, and ingestion or concentrated exposure is haz-

ardous.

Lysophagy

A biochemical process or molecule involving lysophagy. Its role depends on its structure, pathway, and physiological context.

Lysophosphatidylcholine

A biochemical process or molecule involving lysophosphatidylcholine. Its role depends on its structure, pathway, and physiological context.

Lysophospholipids

lysophospholipids is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Lysosome

Membrane-bound organelles containing hydrolytic enzymes (acid hydrolases: proteases, lipases, nucleases, glycosidases) functioning at low pH (4.5-5.0) maintained by V-ATPase proton pump. Degrade macromolecules from endocytosis, phagocytosis, and autophagy.

Lysozyme

An antimicrobial enzyme that cleaves bacterial cell-wall peptidoglycan and is found in tears, saliva, breast milk, and some leukocytes.

Lysozyme Test

A lysozyme test measures the enzyme lysozyme in a sample and may support evaluation of selected blood, inflammatory, or infectious disorders.

Lyssa

Rabies, a fatal viral infection of the central nervous system transmitted most often through saliva from an infected animal.

Alternate Names:

Compact bone

Lyssavirus

lyssavirus is a virus or viral group classified by its

genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Lytic

Causing or involving lysis, or the breakdown of cells, tissue, or another biological structure; a lytic bone lesion is an area of bone destruction.

Lytic Lesion

A lytic lesion is an area of bone destruction caused by processes such as cancer, infection, or some metabolic bone disorders.

M

M Phase

The part of the cell cycle when a cell separates its copied genetic material and divides into two new cells.

MacConkey Agar

A laboratory growth medium used mainly to grow and distinguish certain intestinal bacteria. It helps show whether bacteria can break down lactose, a sugar.

Macerate

To soften tissue by soaking and enzymatic digestion, as may occur in a stillborn fetus after death.

Maceration

Softening and breakdown of tissue caused by prolonged exposure to moisture or fluid; it may occur in skin, wounds, or a fetus after death.

Macewen Sign

An unusually hollow sound heard when part of the skull is tapped. It may occur with fluid buildup in the brain or, rarely, a brain abscess, but it is not diagnostic by itself.

Machine Learning

A computational approach in which algorithms learn patterns from data and use them to make predictions or decisions without being explicitly programmed for every rule; medical uses include image analysis and risk prediction.

Machine, CPM

A continuous passive motion machine that repeat-

edly moves a limb, such as a surgically treated knee, through a prescribed range of motion while the patient relaxes.

Machupo Virus

An arenavirus that causes Bolivian hemorrhagic fever, a potentially fatal illness characterized by fever, pain, and bleeding; it is endemic in parts of Bolivia.

Macroamylasemia

Macroamylasemia is a benign condition in which amylase binds to immunoglobulin and remains in blood, causing high serum amylase with low urinary clearance.

Macroangiopathy

Disease of large blood vessels, often involving atherosclerotic buildup or thrombosis that narrows or blocks blood flow; examples include coronary, cerebrovascular, and peripheral vascular disease.

Macroautophagy

A process cells use to enclose and break down damaged cell parts or proteins so their materials can be reused.

Macrobiotic

Relating to the macrobiota, the living organisms in a region that are large enough to be seen with the naked eye.

Macrobiotics

Diet emphasizing whole grains, vegetables, beans. May be nutritionally inadequate if restrictive. Long-

term followers generally healthy but may have deficiencies.

Macrocephalus

Having an abnormally large head, usually defined by head circumference above the expected range for age and sex.

Macrocephaly

An unusually large head for a person's age and sex. It may run in a family or be linked to fluid buildup, increased brain growth, or another condition.

Macrococcus

A genus of gram-positive cocci related to staphylococci and found mainly on animals; human infection is rare.

Macrococcus Caseolyticus

A gram-positive bacterium found mainly on animals and dairy products that is rarely associated with human infection.

Macrocyte

An abnormally large red blood cell, commonly seen in macrocytosis and megaloblastic or other anemias.

Macrocytic

Describing a cell, especially a red blood cell, that is abnormally large; macrocytic anemia features enlarged red blood cells.

Macrocytic Anaemia

Anemia in which red blood cells are larger than normal. Common causes include low vitamin B12 or folate, other illnesses, and some medicines.

Macrocytic Anemia

Anemia in which red blood cells are larger than normal, commonly because of vitamin B12 or folate deficiency.

Macroductyly

Abnormally large fingers or toes, usually caused by overgrowth of bone and soft tissue.

Macrodonia

Teeth that are larger than expected. It may affect one tooth or many teeth and can cause crowding, bite problems, or difficulty cleaning; treatment depends on the cause and symptoms.

Macrogonitosomia

Premature or abnormal enlargement of the external genitalia, usually associated with a hormonal disorder that may also affect the internal sex organs.

Macroglobulin

A plasma globulin with a high molecular mass, generally formed by the association of several immunoglobulin or other protein subunits; immunoglobulin M is a principal example.

Macroglossia

Macroglossia is an abnormally enlarged tongue that can affect feeding, speech, airway, or dental development and may be congenital or acquired.

Macrognathia

Abnormal enlargement of the jaw, which may be associated with pituitary gigantism, tumors, or other disorders; also called a prognathic mandible.

Macrolide

A group of antibiotics, including erythromycin, azithromycin, and clarithromycin, that stop bacteria from making proteins.

Macrolides

A class of bacteriostatic antibiotics that bind to the 50S ribosomal subunit and inhibit translocation of peptidyl-tRNA during protein synthesis. The major macrolides include erythromycin, clarithromycin, and azithromycin.

Macronucleus

The larger, transcriptionally active nucleus of a ciliate protozoan that controls routine cellular functions; it differs from the smaller micronucleus involved in genetic reproduction.

Macronutrients

Nutrients needed in large amounts: carbohydrates, protein, fat. Provide energy and structural components.

Macrophage

A large immune cell that surrounds and removes germs, dead cells, and debris. It also helps coordinate inflammation and tissue repair.

Macrophage Activation

The process by which macrophages respond to signals such as microbial products or cytokines and increase phagocytosis, antigen presentation, and production of inflammatory or tissue-repair mediators.

Macrophage Activation Syndrome

A dangerous complication of some rheumatic diseases in which immune cells become overactive and cause severe inflammation, fever, low blood counts, and organ problems.

Macrophage Count

The measured number or proportion of macrophages in a tissue, fluid, or laboratory sample; interpretation depends on the specimen, method, and clinical context.

Macropsia

A visual disturbance in which objects appear larger than they actually are.

Macroscopic

Visible to the unaided eye, as opposed to microscopic.

Macroscopic Hematuria

Visible blood in the urine, also called gross hematuria; it may arise from the kidneys, ureters, bladder, prostate, or urethra and requires evaluation for causes such as infection, stones, trauma, or malignancy.

Macrosomia

Macrosomia means unusually large fetal or newborn size, often associated with diabetes, prolonged pregnancy, parental size, or other factors.

Macrostomia

An abnormally wide mouth or oral opening, usually a congenital facial cleft caused by incomplete fusion of the embryonic facial processes.

Macula

The central part of the retina that provides sharp, detailed, and color vision.

Macula Lutea

The small yellowish area in the center of the retina responsible for sharp vision. Its center, called the fovea, provides the clearest vision.

Macular

Relating to a macule, a flat, circumscribed change in skin color without a change in texture or elevation. Macules may be caused by pigment, inflammation, vascular change, or other processes and are distinguished from raised lesions such as papules.

Macular (Skin)

Relating to a macule, a small, flat, circumscribed change in skin color that is neither raised nor depressed.

Macular Cyst

A cystic defect in the macula, the central retinal area responsible for sharp vision, often associated with vitreous traction and potentially causing distortion or loss of central vision.

Macular Degeneration

A progressive degenerative disease of the central retina (macula) and the leading cause of irreversible vision loss in individuals over age 50 in developed countries, affecting approximately 10 percent of those over 65 and 30 percent over 80.

Alternate Names:

Age Related Macular Degeneration (Macular De-

generation)

Macular Degeneration, Dry

Alternate index label; see *Dry macular degeneration*.

Macular Degeneration, Wet

Alternate index label; see *Wet macular degeneration*.

Macular Drusen

Small yellow deposits beneath the retina. A few may occur with aging, while many or larger deposits may be linked to age-related macular degeneration.

Macular Edema

Swelling of the macula caused by fluid leaking from retinal blood vessels.

Macular Hole

A small opening in the center of the retina that can cause a missing or distorted area in central vision. Some cases need surgery.

Macular Pucker

A thin layer of scar-like tissue on the retina that can wrinkle the macula and cause distorted or blurred central vision.

Macule

A macule is a flat, circumscribed change in skin color without palpable elevation or depression.

Macules

Small, flat, circumscribed changes in skin color that are neither raised nor depressed; they are distinguished from raised papules and fluid-filled vesicles.

Maculopapular Lesion

A skin lesion combining flat areas of altered color (macules) with small raised solid lesions (papules); it can occur in infections, drug eruptions, inflammatory diseases, and other conditions.

Mad Cow Disease

Alternate index label; see *Creutzfeldt-Jakob disease*.

Madarosis

Loss or absence of eyelashes, eyebrows, or other hair, caused by local disease, inflammation, trauma, or systemic conditions.

Maddock'S Disease

A rare historical eponym whose meaning is not consistent across medical references; the original source should be checked before clinical use.

Madura Foot

A chronic localized infection of the foot or other tissue, also called mycetoma, producing swelling, draining sinuses, and grains.

Madura Foot (Mycetoma)

A chronic localized infection of the skin and deeper tissues of the foot, caused by fungi or certain bacteria and characterized by swelling, draining sinuses, and granules.

Alternate Names:

Mycetoma

Maduramycosis

Mycetoma, a chronic subcutaneous infection caused by true fungi or filamentous bacteria and characterized by swelling, sinuses, and granules.

Madurella

Madurella is a genus of fungi that can cause mycetoma, a chronic infection of skin and deeper tissue after implantation through the skin.

Madurella Mycetomatis

A fungus that commonly causes eumycetoma, a chronic infection producing nodules, draining sinuses, and grains, usually after traumatic implantation in the foot or limb.

Maf Protein

maf protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Maffucci Syndrome

A rare congenital disorder characterized by multiple enchondromas together with soft-tissue hemangiomas or lymphangiomas. Skeletal deformity, limb shortening, pathological fractures, and increased risk of cartilage or other malignancy may occur.

Mafosfamide

An activated oxazaphosphorine alkylating compound related to ifosfamide and cyclophosphamide, studied for antineoplastic activity and capable of cross-linking DNA.

MAG3 Renal Scan

A nuclear medicine scan that shows how well each kidney works and whether urine is draining properly. It can help identify a blockage.

Magenblase Syndrome

Stomach discomfort and bloating caused by swallowing excessive air while eating.

Magic Bullet

An idealized treatment that selectively cures a disease with little or no harm to the patient; the term was introduced by Paul Ehrlich in connection with targeted treatment of infection.

Magnesium

A mineral needed for healthy nerves, muscles, bones, heart rhythm, and many body processes. Too little can cause weakness, tremor, muscle cramps, or abnormal heart rhythms.

Magnesium Blood Test

A blood test measuring serum magnesium, an electrolyte involved in neuromuscular function, cardiac rhythm, and metabolism; it helps evaluate deficiency, excess, and related electrolyte disorders.

Magnesium Carbonate

A magnesium salt used in some antacid or laxative preparations and as a magnesium supplement; excessive intake may cause toxicity, especially in renal failure.

Magnesium Chloride

A magnesium salt used in some oral supplements and medical preparations to provide magnesium or, in concentrated solutions, to help correct magnesium deficiency; dosing depends on the elemental magnesium content.

Magnesium Deficiency

An abnormally low body magnesium level, often caused by poor intake, gastrointestinal loss, renal wasting, or medications. It may cause weakness, tremor, muscle cramps, seizures, cardiac arrhythmias, hypokalemia, and hypocalcemia.

Magnesium Hydroxide

An osmotic laxative and antacid that neutralizes gastric acid and draws water into the bowel.

Magnesium In Diet

Dietary magnesium comes from foods such as nuts, seeds, legumes, whole grains, and leafy vegetables and supports nerve, muscle, bone, and heart function.

Magnesium Oxide

An inorganic magnesium compound used in some supplements and antacids; it can also have a laxative effect, and excessive use may cause hypermagnesemia, especially with kidney impairment.

Magnesium Sulfate

A magnesium salt used intravenously or intramuscularly to treat selected magnesium deficiencies and conditions such as eclampsia, and used orally as an osmotic laxative; it is also known as Epsom salt in its hydrated form.

Magnesium Sulfate For Neuroprotection

Magnesium given through a vein before some very early preterm births to lower the baby's risk of brain injury. The timing and dose are guided by obstetric and newborn-care specialists.

Magnesium Sulphate

A magnesium salt used medically for severe magne-

sium deficiency, prevention or treatment of eclamptic seizures, and selected arrhythmias or asthma indications.

Magnet

An object that produces a magnetic field and attracts ferromagnetic materials; magnets are used in devices such as magnetic-resonance imaging systems, while unproven magnet therapies should not replace effective treatment.

Magnet Therapy

An alternative therapy that places magnets on or near the body to relieve pain or treat disease. Scientific evidence does not support the claimed therapeutic benefits.

Magnetic Field

The region around a magnet or electric current in which magnetic forces act; in medicine, controlled magnetic fields are used in MRI and some other technologies.

Magnetic Resonance Angiography

A special MRI that creates pictures of blood vessels. It can help find narrowing, bulging, tears, blockages, or abnormal vessels without using x-rays.

Magnetic Resonance Cholangiopancreatography

A special MRI that shows the bile ducts and pancreatic duct. It can help find gallstones, narrowing, inflammation, or other problems without placing a scope inside the ducts.

Magnetic Resonance Cholangiopancreatography

Non-invasive MRI of the biliary and pancreatic ducts using heavily T2-weighted sequences that highlight static fluid. Used for: choledocholithiasis (CBD stones), biliary strictures, primary sclerosing cholangitis, pancreatic duct anatomy, and congenital biliary anomalies.

Alternate Names:

Magnetic Resonance Cholangiopancreatography

Magnetic Resonance Imaging

A scan that uses magnets and radio waves to make detailed pictures of organs and tissues. It does not use x-rays and is especially useful for viewing soft tissues.

Magnetic Resonance Imaging

A non-invasive diagnostic imaging modality that utilizes strong static magnetic fields (typically 1.5 to 3.0 Tesla), magnetic field gradients, and radiofrequency pulses to generate detailed cross-sectional anatomical images without ionizing radiation. Operates by exciting the nuclear spin of abundant tissue hydrogen protons and detecting RF signals emitted during T1 (longitudinal) and T2 (transverse) relaxation. Provides soft-tissue contrast resolution superior to CT for imaging the central nervous system, musculoskeletal joints, spinal cord, and pelvic organs.

Magnetic Resonance Safety

The practices used to prevent injury from the static magnetic field, gradient fields, radiofrequency energy, and implanted or ferromagnetic objects in the MRI environment. Screening must address implants, foreign bodies, devices, pregnancy, burns, hearing protection, and projectile risk.

Magnetic Resonance Spectroscopy

MRI technique measuring metabolic compounds. Used for tumor characterization, metabolic disorders.

Magnetic Resonance Venography

MRI technique to visualize veins, using time-of-flight or contrast-enhanced sequences. Indications: cerebral venous sinus thrombosis, dural sinus stenosis, abdominal/pelvic venous thrombosis, portal vein thrombosis, and renal vein thrombosis. Differentiates acute vs. chronic thrombus.

Alternate Names:

Magnetic Resonance Venography

Magnetic Seed

Sentinel lymph node localization using magnetic seed. Alternative to radioactive tracer.

Magnetic Therapy

The clinical application of static or pulsed electromagnetic fields to biological tissues to stimulate healing, modulate pain, or alter neuronal activity. While consumer static magnet products lack robust empirical clinical evidence, pulsed electromagnetic field therapy is FDA-approved as an adjunctive bioelectric treatment for established orthopedic nonunions and failed joint fusions to stimulate osteoblast proliferation and extracellular matrix synthesis. In neuropsychiatry, high-intensity pulsed magnetic stimulation is utilized in repetitive transcranial magnetic stimulation (rTMS) to modulate focal cortical excitability in treatment-resistant major depressive disorder.

Magnetism

The physical property by which magnetic fields exert forces; it is important in MRI and requires screening for ferromagnetic hazards.

Magnetocardiography

Recording of the magnetic fields produced by electrical activity of the heart, usually with highly sensitive external sensors.

Magnetoencephalography

Recording of the magnetic fields generated by neuronal electrical activity, used to localize brain function or epileptic activity.

Magnetosomes

Intracellular, membrane-bound structures in certain bacteria that contain magnetic minerals, commonly magnetite or greigite, and help the bacteria orient along magnetic-field lines.

Magnusiomyces Capitatus

An opportunistic yeast formerly known as *Geotrichum capitatum* or *Saprochaete capitata*; it can cause invasive infections, especially in people with severe immunosuppression or hematologic malignancy.

Mahvash Disease

A rare inherited disorder of glucagon signaling

caused by inactivating variants in the glucagon receptor. It may cause pancreatic alpha-cell hyperplasia, enlarged pancreas, abnormal glucose regulation, and elevated circulating glucagon.

Main-En-Griffe

A claw-hand deformity caused by lower brachial-plexus or ulnar-nerve dysfunction, with hyperextension at the MCP joints and flexion at the IP joints.

Maintenance

Ongoing care or treatment intended to preserve health, prevent relapse or deterioration, or sustain a therapeutic result; maintenance therapy is continued after initial control of a condition.

Maintenance Dose

The regular amount of a medicine needed to keep its effect at the desired level. The dose depends on how quickly the body removes the medicine and how much the body absorbs.

Maintenance Of Wakefulness Test

A sleep test that measures how long a person can remain awake in a quiet setting during the day.

Maintenance Therapy

Long-term treatment to prevent recurrence. Examples: hormonal therapy, targeted therapy.

Maize

Corn, a cereal grain used as food and feed; contaminated maize can expose people to fungal toxins such as aflatoxin.

Majeed Syndrome

A rare autoinflammatory disorder caused by LPIN2 variants, characterized by chronic recurrent multifocal osteomyelitis, congenital dyserythropoietic anemia, and inflammatory skin disease.

Majewski Syndrome

A rare inherited skeletal disorder involving abnormal bone growth and distinctive craniofacial or limb

findings; the exact features depend on the genetic subtype.

Major

Larger, greater, or more significant than another related structure or condition; for example, the teres major muscle is larger than the teres minor muscle.

Major Depression

A depressive disorder causing persistent low mood or loss of interest with significant functional impairment.

Alternate Names:

Major depressive disorder

Major Depression With Psychotic Features

A major depressive episode accompanied by delusions or hallucinations, usually mood-congruent, requiring prompt psychiatric assessment and treatment.

Major Depressive Disorder

A mood disorder characterized by at least one major depressive episode consisting of five or more symptoms during the same two-week period. At least one symptom must be either depressed mood or loss of interest or pleasure (anhedonia).

Major Histocompatibility Complex

A group of genes and cell-surface proteins that display pieces of germs or abnormal cells to the immune system. In humans, these proteins are called HLA.

Major Histocompatibility Complex Molecule

A cell-surface molecule that presents protein fragments to immune cells.

Alternate Names:

MHC molecule

Major Tranquillizers

An older term for antipsychotic medicines, used to reduce psychosis, severe agitation, or mania.

Make Peak Flow A Habit

Regular peak-flow measurement can help some people with asthma recognize worsening airflow early and follow an agreed action plan.

Malabsorption

A condition in which the body cannot properly digest or absorb nutrients from food. It may cause chronic diarrhea, greasy stools, weight loss, anemia, or vitamin and mineral deficiencies.

Malabsorption Syndrome

Impaired absorption of nutrients from the small intestine, causing diarrhea, weight loss, anemia, vitamin deficiencies, or other nutritional complications.

Malacia

Abnormal softening of tissue, such as cartilage, bone, or brain tissue.

Malacoplakia

A rare long-term inflammation in which immune cells collect mineral deposits. It most often affects the bladder or digestive tract and can form a mass that looks like cancer.

Maladaptation

An adjustment to a situation or environment that is ineffective or harmful to health or functioning.

Maladaptive Stress Response

A persistent or disproportionate stress reaction that harms health or functioning.

Malaise

Malaise is a general feeling of illness, discomfort, or lack of well-being that commonly accompanies infection, inflammation, chronic disease, medication effects, or other systemic conditions.

Malakoplakia

Rare chronic inflammatory condition characterized by accumulation of macrophages containing calcified inclusion bodies called Michaelis-Gutmann bodies, most commonly affecting the urinary bladder but

also occurring in the gastrointestinal tract.

Malar

Pertaining anatomically to the zygomatic bone (cheekbone) or the prominence of the cheek. Clinically prominent in rheumatology for the pathognomonic malar rash (the classic "butterfly rash" of systemic lupus erythematosus, featuring an erythematous, fixed, flat or raised eruption over the malar prominences and nasal bridge while characteristically sparing the nasolabial folds) and in maxillofacial trauma for malar or zygomaticomaxillary complex (ZMC / tripod) fractures caused by direct lateral facial impact, presenting with cheek flattening, trismus, and infraorbital nerve hypoesthesia.

Malar Rash

A red or violaceous rash across the cheeks and bridge of the nose, often sparing the folds beside the nose. It is classically associated with systemic lupus erythematosus but can occur with other conditions; appearance alone does not establish a diagnosis.

Malaria

Malaria is a mosquito-borne infection caused by Plasmodium parasites. It commonly causes cyclical fever, chills, headache, and hemolytic anaemia; severe disease may cause cerebral malaria, respiratory distress, kidney injury, hypoglycaemia, shock, organ failure, or death. Diagnosis and treatment depend on the species, illness severity, and exposure or travel history.

Malassezia

Malassezia is a yeast normally found on skin that can contribute to dandruff, seborrheic dermatitis, pityriasis versicolor, and some folliculitis.

Malassezia Furfur

Malassezia furfur is a skin yeast associated with dandruff, seborrheic dermatitis, and pityriasis versicolor.

Malassezia Pachydermatis

Malassezia pachydermatis is a yeast common on

dogs that can rarely cause opportunistic infection in people with weakened immunity.

Malassimilation

Failure to properly digest or absorb nutrients, encompassing maldigestion and malabsorption.

Malate Dehydrogenase

An enzyme involved in the process cells use to make energy from nutrients.

Malathion

An organophosphate insecticide that inhibits acetylcholinesterase; topical formulations have also been used to treat head lice, but exposure or ingestion can cause cholinergic toxicity.

Malathion Poisoning

Malathion poisoning involves toxic or environmental exposure to malathion poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

MALDI-TOF Mass Spectrometry

A laboratory method that identifies bacteria or fungi by analyzing their protein pattern. It can provide a quick identification but does not show which antibiotics will work.

Male

Describing a sex associated with reproductive anatomy that produces or is organized around small gametes (sperm); chromosomes, hormones, anatomy, and gender identity do not always fit a simple binary pattern.

Male Breast Cancer

Cancer that develops in breast tissue in a man.

Alternate Names:

Breast Cancer, Male

Male Breast Pathology

Disorders affecting male breast tissue, including gynecomastia, benign epithelial lesions, inflammatory conditions, and carcinoma. Evaluation of a

persistent mass or nipple change requires clinical examination and appropriate imaging, with biopsy when indicated.

Male Condom

A thin sheath placed over the penis before sexual contact to reduce the risk of pregnancy and transmission of many sexually transmitted infections by acting as a physical barrier.

Male Genitalia

The external and internal reproductive organs of a male, including the penis, scrotum, testes, epididymides, vas deferens, seminal vesicles, prostate, and associated ducts and glands.

Male Hypoactive Sexual Desire Disorder

A persistent or recurrent lack of sexual or erotic thoughts and desire in a man that causes personal distress or relationship difficulty. It is distinct from erectile dysfunction, although the conditions may coexist.

Male Hypogonadism

Reduced production of testosterone, impaired sperm production, or both due to disease of the testes or the hypothalamic-pituitary system. Possible features include low libido, erectile difficulty, infertility, reduced muscle mass, fatigue, and loss of bone density.

Alternate Names:

Low Testosterone Testosterone Deficiency

Male Infertility

Reduced ability to achieve a pregnancy because of factors affecting sperm production, sperm transport, ejaculation, sexual function, or other reproductive processes.

Alternate Names:

Infertility, Male

Male Menopause

A nontechnical term sometimes used for age-related decline in testosterone, although it is not equivalent to female menopause. Low testosterone may

cause reduced libido, erectile difficulty, fatigue, depressed mood, or loss of muscle mass, but symptoms have many causes and require appropriate hormone testing before treatment.

Male Pattern Baldness

Androgenetic alopecia, a patterned progressive loss of scalp hair caused by genetic susceptibility and androgen effects on hair follicles.

Male Pelvis

The lower part of the abdomen between the hip bones in a male; compared with the female pelvis, it is generally narrower, taller, and more robust, with a narrower pubic arch and sacrum.

Male Phenotype

The observable reproductive anatomy, secondary sex characteristics, and other traits associated with male sex development, produced by interacting genetic, hormonal, and developmental factors and showing natural variation.

Male Reproductive System

The organs and tissues that produce sperm and male sex hormones and deliver sperm, including the testes, ducts, accessory glands, penis, and scrotum.

Malformation

An abnormal structural development present from birth, caused by disturbed formation of an organ or body part.

Malignancy

Malignancy is a cancerous neoplasm capable of invading nearby tissue or spreading to distant sites; diagnosis requires classification of tumor type, grade, stage, and molecular features.

Malignancy-Associated Hypercoagulability

A cancer-related tendency toward excessive clot formation caused by tumor-associated coagulation activation, inflammation, platelet interaction, treatment, immobility, or vascular injury. See *Cancer-*

Associated Thrombosis.

Malignant

Cancerous and capable of invading nearby tissue or spreading to distant sites.

Malignant Acanthosis Nigricans

A paraneoplastic syndrome in which velvety, hyperpigmented, verrucous plaques develop rapidly in flexural and mucosal areas as a marker of an underlying internal malignancy, most commonly gastric adenocarcinoma. Unlike benign acanthosis nigricans, the malignant form is extensive, abrupt in onset, and may involve the oral mucosa.

Malignant Ameloblastoma

A rare cancer arising from ameloblastoma-forming tissue, usually in the jaw, with potential to invade locally or spread to distant organs.

Malignant Ascites

Intraperitoneal fluid accumulation caused by peritoneal tumor spread, most often ovarian or gastrointestinal cancer. Management includes therapeutic paracentesis and, in selected cases, intraperitoneal chemotherapy or shunt placement.

Malignant Bone Neoplasm

A malignant bone neoplasm is a cancer arising in bone or nearby tissue; treatment depends on the tumor type, location, and spread.

Malignant Fibrous Histiocytoma

Alternate index label; see *Undifferentiated pleomorphic sarcoma*.

Malignant Giant Cell Tumor Of Tendon Sheath

A rare locally aggressive malignant tumor resembling a giant cell tumor of tendon sheath, characterized by destructive growth and potential for metastasis; diagnosis requires expert soft-tissue pathology review.

Malignant Glioma

An aggressive primary tumor arising from glial or glial-like cells of the central nervous system; high-grade examples include anaplastic astrocytoma and glioblastoma.

Malignant Hypertension

Severely elevated BP greater than 180/120 with end-organ damage. Renal findings include hyperplastic arteriolosclerosis with onion-skinning and fibrinoid necrosis. Treat with IV antihypertensives. Poor prognosis if untreated. May cause AKI, encephalopathy, and retinopathy.

Malignant Hyperthermia

A rare, life-threatening pharmacogenetic disorder of skeletal muscle triggered by volatile inhalation anesthetics and succinylcholine, with incidence of approximately 1 in 15,000 to 50,000 anesthetics.

Malignant Mastocytosis

An older term for an aggressive systemic mast-cell neoplasm characterized by abnormal mast-cell proliferation and organ infiltration; modern classification distinguishes aggressive systemic mastocytosis and mast-cell leukemia.

Malignant Melanoma

A highly aggressive malignancy arising from melanocytes, characterized by uncontrolled proliferation and potential for early metastasis via lymphatic and haematogenous spread.

Malignant Melanoma (Melanoma)

Alternate index label for Melanoma.

Malignant Mesothelioma

A cancer of the mesothelial lining of the pleura (most common), peritoneum, or pericardium. Strongly associated with asbestos exposure, presenting with effusion, chest pain, and dyspnea; prognosis is generally poor.

Malignant Myoepithelioma

A malignant tumor showing myoepithelial differenti-

ation, arising most often in a salivary gland or other glandular tissue and capable of local invasion and metastasis.

Malignant Neoplasm

Malignant neoplasms, commonly called cancers, are uncontrolled proliferations of cells that invade surrounding tissues and have the capacity to metastasize to distant sites through lymphatic, hematogenous, or transcoelomic routes.

Alternate Names:

cancers, are uncontrolled proliferations of cells.

Malignant Otitis Externa

An invasive infection of the external ear canal and skull base, usually caused by *Pseudomonas aeruginosa* in older adults with diabetes or immunocompromise; it can cause severe pain and cranial-nerve complications.

Malignant Peripheral Nerve Sheath Tumor

A cancer arising from cells associated with peripheral nerves.

Malignant Pleural Effusion

Pleural fluid accumulation from pleural or metastatic disease, causing dyspnea. Managed with thoracentesis, pleurodesis, or an indwelling pleural catheter.

Malignant Pustule

An older name for the skin form of anthrax, caused by *Bacillus anthracis* and beginning as an itchy papule that develops into a painless black eschar.

Malignant Syphilis

A severe, destructive form of secondary syphilis presenting with necrotic, gummatous, or ulcerative skin lesions, often in immunocompromised patients, particularly those with HIV co-infection. It represents an aberrant immune response to heavy *Treponema pallidum* infection and requires aggressive intravenous penicillin therapy.

Malignant Teratoma Of The Mediastinum

A malignant germ-cell tumor containing tissues from more than one embryonic layer and arising in the mediastinum; evaluation uses imaging, tumor markers, and tissue diagnosis.

Malignant Thymoma

An older term for an invasive thymic epithelial tumor; modern usage distinguishes invasive thymoma from thymic carcinoma, which is a separate, more aggressive thymic malignancy.

Malignant Tumor

A cancerous growth capable of invading nearby tissue or spreading to distant sites. Its behavior and treatment depend on its type, grade, stage, and molecular features.

Malingering

The intentional fabrication or exaggeration of physical or psychological symptoms for external incentives (avoiding work, obtaining financial compensation, evading criminal prosecution, obtaining drugs). Not a psychiatric illness (coded as a V-code/Z-code).

Mallampati Classification

A simple scoring system used to predict the ease of endotracheal intubation based on the anatomical structures visible in the oral cavity when the patient opens their mouth wide and protrudes their tongue without phonating. Class I: soft palate, fauces, uvula, and tonsillar pillars visible.

Malleolus

A bony prominence at the ankle formed by the lower end of the tibia medially or fibula laterally.

Mallet

A surgical hammer used with instruments such as osteotomes to cut or shape bone.

Mallet Finger

An injury to the extensor tendon at the distal finger joint, causing inability to actively straighten the fingertip. It may result from a forced flexion injury

and sometimes includes an avulsion fracture.

Mallet Toe And Hammertoe

Alternate index label; see *Hammertoe and mallet toe*.

Malleus

The largest of the three ossicles in the middle ear.

Common name: Hammer

Mallory Body

An eosinophilic cytoplasmic inclusion in a hepatocyte, composed largely of damaged intermediate filaments and associated proteins; it is classically seen in alcohol-associated liver disease but is not specific to it.

Mallory-Weiss Tear

A superficial tear in the lining of the lower esophagus or upper stomach, usually caused by forceful vomiting or retching. It commonly causes vomiting of blood. Many tears stop bleeding on their own within a few days, but upper endoscopy may be needed for diagnosis and treatment.

Malnutrition

Malnutrition includes inadequate, excessive, or imbalanced nutrient intake and can impair growth, immunity, wound healing, muscle function, and organ reserve. form, and function. Undernutrition: protein-energy malnutrition, micronutrient deficiencies. Overnutrition: overweight, obesity, related comorbidities. Screen with MNA (Mini Nutritional Assessment), MUST (Malnutrition Universal Screening Tool).

Malocclusion

Misalignment of teeth or incorrect relation between dental arches. Types: Class I (normal molar relationship, crowding), Class II (retrognathic mandible), Class III (prognathic mandible).

Malocclusion Of Teeth

Misalignment of the upper and lower teeth or jaws that affects the bite, chewing, speech, appearance,

or oral hygiene.

Malondialdehyde

A reactive aldehyde produced during oxidation of polyunsaturated lipids; it is commonly measured as an indirect marker of lipid peroxidation and oxidative stress.

Malpighian Bodies

Renal glomeruli within Bowman's capsules, where blood filtration begins; the term may also refer historically to lymphoid nodules in the spleen.

Malposition

An abnormal position of an organ, body part, or fetus.

Malpresentation

A fetal presentation other than the usual vertex presentation, including breech, face, brow, shoulder, or compound presentation. It may complicate labor and influence delivery planning.

Malrotation

Abnormal rotation or fixation of an organ during development, most often referring to abnormal rotation of the intestine and its risk of volvulus.

MALS

Alternate index label; see *Median arcuate ligament syndrome*.

MALT Lymphoma

Alternate index label; see *Extranodal marginal zone lymphoma of mucosa-associated lymphoid tissue*.

Malt Soup Extract

Malt extract is a concentrated food ingredient made from germinated grain and contains sugars and other grain components. Its nutritional value and suitability depend on portion, ingredients, gluten content, diabetes management, and the reason it is being used.

Malt-Worker'S Lung

An occupational hypersensitivity pneumonitis caused by inhalation of organic dusts associated with malting or grain processing.

Malta Fever

Brucellosis, a zoonotic infection caused by *Brucella* species and characterized by intermittent fever, sweats, fatigue, and joint or systemic symptoms.

Maltase

An intestinal enzyme that breaks maltose into two glucose molecules.

Maltodextrin

A mixture of short glucose polymers produced by partial hydrolysis of starch, used as a rapidly digestible carbohydrate and as a thickener, filler, or carrier in foods and medicines.

Maltol

A flavoring compound used in some foods and medicines to provide a sweet, caramel-like odor or taste.

Maltose

A reducing disaccharide composed of two glucose molecules joined by an alpha-1,4 glycosidic bond, produced during starch digestion and hydrolyzed by maltase.

Mamillary

A misspelling or variant of mamillary, meaning relating to the breast or to a rounded nipple-like structure.

Mammaglobin A

A secretory protein expressed predominantly in mammary tissue and used as an immunohistochemical marker that can help identify breast origin in some metastatic carcinomas.

Mammals

Warm-blooded vertebrate animals that have hair or fur, three middle-ear bones, and mammary glands

that produce milk; humans are mammals.

Mammaplasty

Mammaplasty is surgery to reshape, enlarge, reduce, or reconstruct the breast.

Mammary

Pertaining to the breast or the mammary gland, the specialized apocrine exocrine gland present in both sexes but functionally developed in females for lactation (milk synthesis and ejection). Composed of 15 to 20 glandular lobes embedded within adipose tissue, supported by fibrous Cooper suspensory ligaments and drained by lactiferous ducts converging at the nipple-areolar complex. Glandular growth and alveolar differentiation are regulated by estrogen, progesterone, and prolactin during puberty and pregnancy, with oxytocin mediating alveolar myoepithelial contraction during milk ejection. Major clinical conditions include fibrocystic changes, fibroadenomas, and invasive mammary (breast) carcinomas.

Mammary Disease

A disorder of breast tissue, including benign proliferative disease, infection, cysts, tumors, and malignant neoplasms.

Mammary Duct Ectasia

Widening and inflammation of a milk duct beneath the nipple.

Mammary Epithelium

The epithelial lining of the mammary ducts and secretory alveoli; these cells help produce, secrete, and transport milk and can give rise to many breast carcinomas.

Mammary Gland

Modified sweat (apocrine) gland in the breast, overlying pectoralis major, with 15-20 lactiferous ducts converging on the nipple. Composed of glandular lobules (alveoli), fibrous stroma (Cooper's ligaments), and fat. Development regulated by estrogen, progesterone, prolactin, and oxytocin (milk let-down).

Mammary Glands

Paired milk-producing glands in the breasts, composed of lobules and ducts that develop and function under hormonal control.

Mammary Glands (Breasts)

Paired glands of the anterior chest that produce milk after childbirth. Each breast contains lobules, ducts, connective tissue, and fat and is influenced by hormonal and reproductive changes.

Alternate Names:

breasts; mammary glands

Mammary Implant

Breast augmentation/reconstruction device. Types: saline (silicone shell, saline fill), silicone gel (cohesive gel). Incisions: inframammary, periareolar, axillary. Placement: subglandular or submuscular. Complications: capsular contracture, rupture, infection, BIA-ALCL (anaplastic large cell lymphoma).

Mammary Paget Disease

An intraepidermal spread of malignant glandular cells into the nipple epidermis, usually associated with underlying breast carcinoma and presenting as persistent eczematous or crusted nipple change.

Mammillary Body

One of a pair of rounded nuclei on the underside of the hypothalamus that participate in memory circuits connecting the hippocampus with the thalamus; injury can contribute to memory impairment.

Mammogram

A mammogram is a low-dose x-ray examination of breast tissue used for screening and evaluation of masses, asymmetry, calcifications, or other abnormalities.

Mammogram - Calcifications

Mammographic assessment of calcium deposits in breast tissue; their morphology and distribution help distinguish benign findings from changes requiring additional imaging or biopsy.

Mammography

X-ray imaging of the breast used to screen for or evaluate breast abnormalities. Screening intervals depend on age, risk, symptoms, and the guideline being followed. Standard views include the cranio-caudal and mediolateral-oblique projections.

Mammoplasty

Plastic or reconstructive surgery to alter the size, shape, or contour of the breast.

Man Boobs (Gynecomastia)

Alternate index label for Gynecomastia.

Managed Care

A system of organizing and financing health care through provider networks, utilization review, care coordination, and negotiated payment arrangements intended to control costs and maintain quality.

Managing Pain During Labor

Pain relief during labor may include breathing and movement, support, injected or intravenous medicines, nitrous oxide, or neuraxial anesthesia, chosen according to preference and clinical circumstances.

Mandatory Reporting

A legal or regulatory duty to report specified findings or circumstances—such as suspected child or elder abuse, certain communicable diseases, or some injuries—to designated authorities; requirements vary by jurisdiction.

Mandible

Largest facial bone forming the lower jaw, with horseshoe-shaped body and two rami ending in condylar (TMJ) and coronoid processes. Mental foramen transmits the mental nerve. Mandibular foramen is the entry point for inferior alveolar nerve blocks. Most commonly fractured facial bone, typically at the condyle, angle, or symphysis.

Mandible (Lower Jaw)

The large movable bone forming the lower jaw and

carrying the lower teeth.

Alternate Names:

Lower Jaw

Mandibular

Pertaining to the mandible, the horseshoe-shaped single bone forming the lower jaw and the only mobile osseous structure of the facial skeleton. Articulates bilaterally with the temporal bones at the synovial temporomandibular joints and houses the lower dentition within its alveolar process. Traversed internally by the mandibular canal transmitting the inferior alveolar neurovascular bundle (supplying lower teeth and sensory branches to the chin and lower lip via the mental nerve). Important in maxillofacial trauma, micrognathia (Pierre Robin sequence), and surgical mandibular osteotomy for severe malocclusion.

Mandibular Advancement Device

An oral appliance that holds the lower jaw or tongue forward during sleep to enlarge or stabilize the upper airway, usually for selected patients with obstructive sleep apnea or snoring.

Mandibular Condyle

The rounded superior end of each mandibular ramus that articulates with the temporal bone at the temporomandibular joint and permits jaw movement.

Mandibular Fracture

A break in the lower jaw bone typically caused by trauma, presenting with pain, swelling, malocclusion, and difficulty opening the mouth; may require surgical fixation.

Mandibular Prosthesis

An artificial implant or device used to replace, reconstruct, stabilize, or restore part of the mandible after trauma, tumor removal, congenital deformity, or other loss of jaw structure.

Mandibulofacial Dysostosis

A congenital craniofacial disorder, classically Treacher Collins syndrome, involving underdevelopment of the mandible and cheekbones with variable

abnormalities of the ears, eyes, and palate.

Manganese

An essential trace element required in small amounts. Excessive exposure, especially by inhalation, can injure the nervous system and cause manganism, a movement disorder resembling parkinsonism.

Mania

A distinct period of abnormally elevated, expansive, or irritable mood with increased goal-directed activity or energy, lasting ≥ 1 week (or any duration if hospitalization required).

Maniac-Depressive Psychosis

An obsolete term for bipolar disorder, characterized by episodes of mania or hypomania and episodes of depression.

Manic

Relating to mania, a pathological episode of abnormally elevated, expansive, or irritable mood with increased energy or activity and associated changes in thought, speech, sleep, or judgment.

Manic Disorder

A condition characterized by a distinct period of abnormally elevated, expansive, or irritable mood with increased energy and activity; persistent mania is a defining feature of bipolar I disorder.

Manic Episode

A distinct period lasting at least seven days (or any duration if hospitalization is required) of abnormally elevated, expansive, or irritable mood with increased energy or activity.

Manic Mood

An abnormally elevated, expansive, or persistently irritable mood accompanied by increased energy or activity, reduced need for sleep, rapid speech, racing thoughts, inflated self-esteem, or risky behavior; it is characteristic of a manic episode.

Manic-Depression

An older term for bipolar disorder, a mood disorder characterized by episodes of mania or hypomania and episodes of depression.

Manic-Depressive Illness

Alternate index label; see *Bipolar disorder*.

Manifest

Clearly apparent or expressed; in clinical usage, a disease may manifest through particular signs or symptoms.

Manifestation

A sign, symptom, or other observable expression of a disease or condition.

Manipulation

Deliberate physical movement of a body part or joint, often performed therapeutically or diagnostically by a trained clinician.

Manipulation Procedure

A medical procedure, treatment, device, or care topic related to manipulation procedure; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Mannan

A polysaccharide composed primarily of mannose units, found in the cell walls or capsules of many fungi and in some plants; fungal mannans can be recognized by the immune system and used in laboratory testing.

Mannheimia

A genus of gram-negative bacteria that mainly infects livestock, especially the respiratory tract of cattle and sheep.

Mannheimia Haemolytica

A gram-negative bacterium that causes bovine respiratory disease and pneumonia in other animals; human infection is rare.

Mannitol

An osmotic diuretic that is freely filtered at the glomerulus but not reabsorbed, increasing tubular osmotic pressure and water excretion. Used to reduce intracranial pressure (cerebral edema) and intraocular pressure (acute glaucoma), and to promote diuresis in rhabdomyolysis/contrast nephropathy prophylaxis.

Mannomustine

Mannomustine is an older nitrogen-mustard alkylating agent investigated or used historically in lymphoma and other malignancies; it cross-links DNA and can cause myelosuppression and other cytotoxic effects.

Mannose

A six-carbon monosaccharide related to glucose that forms part of glycoproteins and glycolipids and is important in protein glycosylation and cell recognition.

Mannose-Binding Lectin Deficiency

An inherited or acquired deficiency of mannose-binding lectin, a protein that activates the lectin pathway of complement. It may increase susceptibility to infection in selected people, especially when other immune defenses are impaired.

Mannosidosis

An inherited lysosomal storage disorder caused by deficient alpha-mannosidase activity, leading to accumulation of mannose-rich oligosaccharides and variable intellectual, skeletal, immune, hearing, and neurologic abnormalities.

Manometer

An instrument for measuring pressure in a fluid or gas, including pressure in blood vessels or body cavities.

Manometry

Measurement of pressure within a body organ or passage, such as the esophagus or rectum.

Manorexia

An informal, nonmedical term sometimes used for anorexia nervosa in males; it is not an officially recognized diagnostic term.

Mansonella Perstans

A filarial worm that causes mansonellosis, usually acquired through bites of infected midges and often producing mild or no symptoms.

Mansonella Streptocerca

A filarial worm transmitted by biting midges that lives mainly in the skin and can cause itchy skin disease or small nodules.

Mansonelliasis

Infection with filarial nematodes of the genus *Mansonella*, transmitted by biting midges or blackflies; many infections are asymptomatic, but some cause itching, skin lesions, fever, or eosinophilia.

Mantle Cell Lymphoma

An aggressive B-cell lymphoma characterized by t(11;14) translocation leading to cyclin D1 overexpression and CCND1-IGH fusion. Clinical: older males, widespread lymphadenopathy, splenomegaly, GI involvement (lymphomatous polyposis), bone marrow, peripheral blood (leukemic phase).

Mantle Cell Lymphoma

Alternate index label for MCL.

Mantle Field

A historical radiation-treatment field encompassing lymph-node regions in the neck, mediastinum, and upper chest, formerly used for Hodgkin lymphoma; modern treatment generally uses more limited involved-site fields.

Mantoux Test

An intradermal tuberculin skin test in which induration is measured after 48 to 72 hours; it indicates immune sensitization, not by itself active tuberculosis.

Manual

Performed by hand, or a written guide describing procedures, instructions, or reference information.

Manual Removal Of Placenta

A procedure in which a clinician removes a placenta that has not separated or cannot be delivered after birth. The timing and need depend on bleeding, maternal condition, and local clinical protocols.

Manual Vacuum Aspiration

A minor surgical method of uterine evacuation that uses a handheld, portable locking vacuum syringe attached to a semi-rigid plastic cannula to generate suction equivalent to an electric vacuum pump. Recommended by the World Health Organization and American College of Obstetricians and Gynecologists as a safe, effective, and cost-efficient technique for managing first-trimester incomplete miscarriage, missed abortion, retained products of conception, and legal pregnancy termination under local paracervical block.

Alternate Names:

MVA; Mini-Suction Curettage

Manus

The hand, including the wrist and fingers, in anatomical terminology.

MAOIs

Monoamine oxidase inhibitors are a class of antidepressant drugs that inhibit the enzyme monoamine oxidase, which breaks down serotonin, norepinephrine, and dopamine in the brain. By inhibiting this enzyme, MAOIs increase the levels of these neurotransmitters in the synaptic cleft.

Map-Dot-Fingerprint Type Corneal Dystrophy

A usually bilateral corneal epithelial basement membrane dystrophy producing map-like, dot-like, or fingerprint-like patterns and sometimes recurrent corneal erosions or visual blurring.

Mapatumumab

An investigational monoclonal antibody that targets a death receptor on cells and has been studied as an anticancer treatment.

Maple Syrup Urine Disease

An autosomal recessive inborn error of branched-chain amino acid metabolism caused by deficiency of the branched-chain alpha-ketoacid dehydrogenase complex (BCKDC) — mutations in BCKDHA, BCKDHB, DBT, or DLD.

Mapping

Charting the location of genes or other identifiable regions on chromosomes; the term may also describe charting functional or anatomical regions in other medical contexts.

Marantic

Relating to a sterile, noninfective process associated with wasting; marantic endocarditis refers to nonbacterial thrombotic endocarditis.

Marasmus

A severe form of protein-energy malnutrition resulting from chronic total calorie and protein deprivation (starvation). Clinical features include severe muscle wasting, loss of subcutaneous fat ("skin and bones" appearance), stunted growth, a "monkey-like" facial appearance, and alertness.

Maraviroc

A CCR5 antagonist (entry inhibitor) used in combination antiretroviral therapy for HIV-1 infection in patients with CCR5-tropic virus. Maraviroc binds to the human CCR5 coreceptor on CD4+ T cells, preventing HIV-1 gp120 binding and viral entry.

Marble Bone Disease

Osteopetrosis, an inherited disorder of defective bone resorption causing abnormally dense but fragile bones and possible marrow failure.

Marburg Virus History, Symptoms, And

Treatment

Marburg virus disease is a severe filovirus infection spread through contact with infected body fluids and contaminated materials. Fever, weakness, gastrointestinal symptoms, rash, bleeding, and shock may occur; isolation, supportive critical care, and public-health response are essential.

Marburgvirus

A genus of filoviruses that causes Marburg virus disease, a severe viral hemorrhagic fever transmitted through infected body fluids or animal exposure.

Marek'S Disease

A contagious herpesvirus disease of chickens that can cause nerve paralysis, tumors, weight loss, or death.

Marfan Syndrome

An autosomal dominant connective tissue disorder caused by mutations in the FBN1 gene (fibrillin-1). Cardiovascular manifestations include aortic root dilation and dissection, mitral valve prolapse, and aortic regurgitation. Skeletal features include tall stature, arachnodactyly, pectus deformity, and joint hypermobility.

Marfan'S Syndrome

An inherited connective-tissue disorder, usually caused by FBN1 variants, affecting the aorta, eyes, skeleton, and other organs.

Marfanoid Habitus

A body habitus characterized by tall stature, long limbs, long fingers (arachnodactyly), and joint laxity. Seen in Marfan syndrome, but also in other connective tissue disorders and homocystinuria. Differentiated from Marfan syndrome by absence of aortic root dilation and lens subluxation.

Marfan'S Syndrome

An autosomal dominant connective tissue disorder caused by mutations in the FBN1 gene encoding fibrillin-1. Neurological manifestations include dural ectasia (widening of the dural sac, causing back pain and headaches), spontaneous intracranial hy-

potension from CSF leaks, and increased risk of stroke from arterial dissection.

Margaric Acid

margaric acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Marginal Ulcer

An ulcer occurring at or near a gastrojejunal anastomosis, often after gastric bypass or other gastric surgery, where acid, ischemia, smoking, nonsteroidal anti-inflammatory drugs, or other factors may contribute.

Marginal Zone Lymphoma

A slow-growing non-Hodgkin lymphoma that begins in immune cells in the marginal zone of lymphoid tissue.

Alternate Names:

MZL

Maribavir

Maribavir is an antiviral medicine used to treat certain cytomegalovirus infections that are resistant or refractory to other treatment after transplantation.

Marie-Strumpell Disease

Ankylosing spondylitis, a chronic inflammatory disease primarily affecting the sacroiliac joints and spine that can cause back pain, stiffness, and progressive spinal fusion.

Marijuana

Cannabis preparations containing cannabinoids such as THC; medical effects and risks depend on dose, formulation, route, and patient factors.

Alternate Names:

Pot Marijuana (Marijuana)

Marijuana Intoxication

Marijuana intoxication involves toxic or environmental exposure to marijuana intoxication; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Marimastat

An investigational synthetic broad-spectrum inhibitor of matrix metalloproteinases, enzymes involved in extracellular-matrix breakdown, studied as a potential anticancer and anti-inflammatory drug.

Marine Animal Stings Or Bites

Injuries caused by marine animals through biting, stinging, spines, or venom; effects range from local pain and swelling to allergic, neurologic, or systemic toxicity.

Marine Toxin

A poisonous substance produced or accumulated by marine organisms, such as some fish, shellfish, algae, or bacteria, that can cause neurologic, gastrointestinal, or other poisoning.

Marital Disharmony

Persistent conflict or distress in a marriage or partnership that may contribute to psychological symptoms or require relationship-focused support.

Marital Status

A demographic or social-status variable describing whether a person is unmarried, married or partnered, separated, divorced, widowed, or in another legally or culturally recognized relationship; it may be relevant to social and health assessments.

Marital Therapy

Marital therapy is counseling for couples that addresses communication, conflict, emotional distress, and relationship problems.

Marked

Pronounced, conspicuous, or substantially greater than usual; in a clinical report, the exact meaning depends on the finding and the comparison standard.

Marker

An identifiable biological feature used to track a gene, cell, disease, or process; a genetic marker is a detectable DNA region inherited with a nearby

gene, while a blood or tumor marker is a measurable substance associated with a condition.

Marker, Blood

A measurable sign of a disease or condition that can be isolated or detected in a blood sample, such as a disease-associated antibody, hormone, protein, or other substance.

Maroteaux-Lamy Syndrome

An autosomal recessive lysosomal storage disease (Mucopolysaccharidosis Type VI [MPS VI]) caused by deficiency of the lysosomal enzyme *N*-acetylgalactosamine-4-sulfatase (arylsulfatase B, *ARSB*), which leads to the progressive accumulation of the glycosaminoglycan dermatan sulfate in tissues. Characterized by progressive coarse facial features, severe dysostosis multiplex (skeletal dysplasia), joint stiffness, corneal clouding, cardiac valvular thickening, hepatosplenomegaly, and restrictive pulmonary compromise, notably with normal cognitive development and preserved intellectual function.

Alternate Names:

Mucopolysaccharidosis Type VI (MPS VI); Arylsulfatase B Deficiency

Married

Describing a person who is legally or socially joined to a spouse or partner; marital status is a demographic characteristic and does not by itself indicate health or relationship quality.

Marrow

The soft tissue inside bones, including red marrow that produces blood cells and yellow marrow rich in fat.

Marrow Hyperplasia

Increased cellularity of the bone marrow due to accelerated production of blood cells, typically seen as a compensatory response to chronic hemolysis, hemorrhage, or hypoxia.

Marshall Syndrome

A rare periodic-fever syndrome, usually associated with PFAPA features, including recurrent fever, aph-

thous ulcers, pharyngitis, and cervical adenitis, often beginning in childhood.

Marsupialization

A surgical technique that opens a cyst or abscess and sutures its edges to the surrounding skin or mucosa to permit continuing drainage.

Martin-Bell Syndrome

An older name for fragile X syndrome, an inherited neurodevelopmental disorder caused by CGG-repeat expansion in the FMR1 gene.

MASA Syndrome

An X-linked neurodevelopmental disorder characterized by intellectual disability, absent or limited speech, shuffling gait, spasticity, and adducted or clasped thumbs; it may also include short stature and lumbar lordosis. The name is an older acronym and terminology should be used sensitively.

Masculinization

Development of male-typical physical characteristics, caused by normal puberty, androgen excess, androgen treatment, or certain disorders.

MASH

Alternate index label; see *Fatty liver disease*.

Mask

A device placed over the face to deliver oxygen, anesthesia, ventilation, or protection; the meaning depends on clinical context.

Masked Data

Data altered, obscured, or replaced so that identifying values or sensitive details are hidden while selected analytical properties may be retained; masking is used to protect privacy and can be reversible or irreversible.

Masklike Face

An expressionless face with little spontaneous animation, also called masklike facies; it may occur in Parkinson disease, myotonic dystrophy, and other

disorders.

MASLD

Alternate index label; see *Fatty liver disease*.

Masochism

Sexual or nonsexual arousal or gratification associated with receiving pain, humiliation, or degradation; distress or impairment determines clinical significance.

Masoprocol

A synthetic derivative of nordihydroguaiaretic acid that inhibits lipoxygenase pathways and has been investigated for topical treatment of actinic keratoses and for antineoplastic activity.

Mass

A localized area of tissue that is larger or more substantial than surrounding tissue and may be benign, malignant, inflammatory, or cystic.

Mass Casualty Incident

Event overwhelming available resources, requiring triage and resource allocation. Triage systems: START (Simple Triage and Rapid Treatment), SALT. Categories: immediate (red), delayed (yellow), minor (green), expectant (black). Incident command system (ICS) coordinates response.

Mass Screening

A diagnostic or laboratory method related to mass screening. Results are interpreted with the method, reference range, and clinical findings.

Mass Spectrometer

An instrument that ionizes chemical substances and separates or detects the resulting ions according to their mass-to-charge ratios, producing a mass spectrum.

Mass Spectrometry

An analytical method that identifies and measures molecules by their mass-to-charge ratios; coupled with chromatography, it is widely used in clinical

chemistry, toxicology, pharmacology, and proteomics.

Massage

Manual manipulation of soft tissues using techniques such as stroking, kneading, or pressure, used for relaxation or symptom relief; it is not a substitute for diagnosis or treatment of an underlying disease.

Massage Therapy

Massage therapy uses hands-on manipulation of soft tissues to promote relaxation or help manage selected pain and movement problems.

Masseter

A powerful muscle of chewing that elevates the mandible and lies over the lateral surface of the mandibular ramus.

Masseter Muscle

Thick rectangular muscle of mastication on the mandibular ramus, from the zygomatic arch to mandibular angle/ramus. Innervated by the masseteric nerve (V3). Most powerful masticatory muscle, elevating the mandible with up to 70 kg force. Involved in bruxism, TM disorders, and myofascial pain. Hypertrophy can widen facial appearance.

Massive Hemoptysis

Life-threatening coughing of blood that threatens the airway, breathing, or circulation; a fixed volume threshold is not required. It requires urgent airway assessment, stabilization, and definitive treatment of the bleeding source.

Massive Ovarian Edema

Marked stromal edema causing enlargement of one or both ovaries, usually due to partial torsion or hormonal stimulation. It can mimic an ovarian neoplasm and may cause pain or virilization.

Massive Transfusion Protocol

A coordinated protocol for life-threatening hemorrhage that provides rapid, balanced delivery of red cells, plasma, and platelets with monitoring and

correction of fibrinogen, calcium, temperature, and acid-base abnormalities.

Mast Cell

A tissue-resident immune cell with large granules containing histamine, heparin, tryptase, and cytokines; bears high-affinity IgE receptors (Fc-epsilon-RI).

Mast Cell Activation Syndrome

A condition involving recurrent symptoms from inappropriate release of mast-cell mediators, such as flushing, itching, hives, abdominal symptoms, wheezing, or low blood pressure. Diagnosis requires compatible episodes, objective evidence of mediator release, and response to appropriate treatment; many similar conditions must be excluded.

Mast Cell Sarcoma

An exceptionally rare, aggressive neoplasm composed of immature or atypical mast cells forming a destructive mass in tissue or bone. Unlike systemic mastocytosis, it is characterized by a localized sarcomatous tumor and may progress to disseminated mast-cell disease or acute leukemia. Clinical manifestations depend on the involved site and may include pain, swelling, organ dysfunction, or mediator-related symptoms such as flushing and hypotension. Diagnosis requires histopathology, mast-cell immunophenotyping, and systemic staging; management generally involves specialist-directed intensive antineoplastic therapy.

Mast-Cell Sarcoma

mast-cell sarcoma refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Mastalgia

Breast pain or tenderness, which may be cyclical with menstruation or noncyclical. Causes include hormonal changes, cysts, infection, trauma, and musculoskeletal pain.

Mastectomy

Surgical removal of all or part of a breast, performed to treat or prevent selected breast cancers and sometimes for other breast conditions. areola, and nipple. Modified radical mastectomy (MRM): total mastectomy + axillary lymph node dissection (levels I-III), for invasive breast cancer.

Masticate

To chew food, using the teeth, tongue, and muscles of mastication to mechanically break it down and mix it with saliva before swallowing.

Mastication

Chewing, the mechanical breakdown of food in the mouth before swallowing.

Masticatory Muscle

One of the muscles that moves the mandible during chewing, chiefly the masseter, temporalis, medial pterygoid, and lateral pterygoid muscles.

Mastitis

Inflammation of breast tissue, often occurring during lactation but also possible outside lactation. It may cause localized pain, warmth, swelling, and fever; infection and inflammatory causes must be distinguished clinically.

Alternate Names:

Lactation Mastitis

Mastocytoma

A localized collection or tumor of mast cells, usually presenting in the skin as a solitary cutaneous mastocytoma; rubbing the lesion can cause itching, redness, or swelling from mediator release.

Mastocytosis

A clonal disorder in which abnormal mast cells accumulate in the skin or internal organs. Cutaneous disease commonly causes urticaria pigmentosa: tan-brown macules or papules that swell and itch after rubbing (Darier sign).

Mastodynia

Pain or tenderness in one or both breasts, also called mastalgia. It may follow the menstrual cycle or result from injury, infection, medicines, cysts, or other breast conditions; a new persistent lump, skin change, or bloody discharge needs assessment.

Mastoid

Relating to or forming the mastoid portion of the temporal bone behind the ear.

Mastoid Antrum

An air-containing cavity in the mastoid portion of the temporal bone that communicates with the middle-ear attic.

Mastoid Bone

The portion of the temporal bone located behind the ear.

Mastoid Process

The rounded bony projection of the temporal bone behind the ear, providing attachment for neck muscles.

Mastoidectomy

Surgical removal of infected, diseased, or abnormal mastoid air cells behind the ear, commonly performed for chronic ear infection, cholesteatoma, or complications of otitis media. from the mastoid portion of the temporal bone, performed for chronic otitis media with cholesteatoma, mastoiditis, or as a component of surgery for acoustic neuroma and other skull base tumors.

Mastoiditis

Mastoiditis is infection and inflammation of the mastoid air cells behind the ear, usually following middle-ear infection and potentially causing serious spread.

Mastopexy

Breast-lift surgery that removes extra skin and reshapes breast tissue to raise the breasts and nipples. It can improve shape but does not necessarily reduce

breast size; scars, altered sensation, asymmetry, and changes in breastfeeding ability are possible risks.

Alternate Names:

Breast lift

Masturbation

Self-stimulation of the genitals for sexual pleasure. It is a common human behavior; it is generally not harmful unless it causes injury, distress, or interferes with daily life.

Alternate Names:

Self Gratification Sexual (Masturbation)

Match-Box Test

A historical clinical test whose exact purpose varies by source; it should not be interpreted without the original examination description.

Materia Medica

An older term for the body of medicinal substances and information about their properties and uses.

Material

Any substance or matter, including a tissue, specimen, biological product, implant, or foreign body, considered in a medical or scientific context.

Maternal

Maternal means relating to a mother or the pregnant person, especially in connection with pregnancy, childbirth, or postpartum health.

Maternal Age

Maternal age is the age of the pregnant person and can influence fertility, pregnancy risks, screening choices, and obstetric care.

Maternal Mortality

Maternal mortality is death during pregnancy or within the defined postpartum period from a pregnancy-related cause, measured as a population health indicator.

Maternal Mortality Rate

The number of maternal deaths related to pregnancy

per 100,000 live births in a specified population and period.

Maternal Obesity

Obesity during pregnancy, generally defined using pre-pregnancy or early-pregnancy body mass index, which can increase risks for diabetes, hypertension, cesarean birth, and complications for the fetus.

Maternal Physiologic Changes In Pregnancy

Extensive adaptive changes across all organ systems to support the growing fetus.

Maternal Pushing (Second Stage)

Maternal expulsive efforts during the second stage of labor to facilitate fetal descent. Directed (closed-glottis, Valsalva) pushing: sustained breath-hold for 10 seconds with three pushes per contraction. Spontaneous (open-glottis) pushing: shorter pushes with the mother pushing when she feels an urge.

Alternate Names:

Second Stage

Maternal Sepsis

Organ dysfunction caused by a dysregulated maternal response to infection during pregnancy, childbirth, or the postpartum period. It may arise from uterine, urinary, respiratory, wound, or other infection and requires prompt cultures, antimicrobial therapy, source control, and hemodynamic support.

Maternal Serum Alpha-Fetoprotein

A second-trimester maternal serum screening biomarker measured at 15–20 weeks gestation, reported as multiples of the median (MoM). Elevated levels (> 2.0 – 2.5 MoM) suggest open neural tube defects (spina bifida, anencephaly), ventral wall defects (gastroschisis, omphalocele), multiple gestation, or underestimation of gestational age. Decreased levels are associated with fetal chromosomal aneuploidies (trisomy 21 / Down syndrome, trisomy 18 / Edwards syndrome). Abnormal values warrant targeted diagnostic fetal ultrasonography and genetic counseling.

Maternal Vaccine

A vaccine given during pregnancy to protect the pregnant person and, through transferred antibodies, provide early protection to the fetus or newborn; recommended products and timing depend on current guidance.

Maternity

The state or service relating to pregnancy, childbirth, and the care of the mother and newborn.

Maternity Services

Healthcare services providing antenatal, intrapartum, postpartum, and newborn care.

Materteral

Relating to a maternal aunt or to the genetic relationship between an aunt and her niece or nephew; synonymous with materterine.

Mathematics

The study of numbers, quantities, patterns, structures, and logical relationships; it supports medical statistics, modeling, imaging, epidemiology, and quantitative research.

Mathematics Disorder

A learning disorder involving persistent difficulty understanding numbers, calculation, quantity, or mathematical reasoning despite appropriate instruction.

Mating

Sexual pairing or reproduction between organisms that permits exchange or transfer of gametes and may lead to fertilization.

Matrix Band

A thin dental strip placed around a prepared tooth to temporarily replace a missing wall and contain restorative material while a filling or crown buildup is formed.

Matrix Metalloproteinase

matrix metalloproteinase is a protein or protein-

related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Maturation

The process by which cells, tissues, organs, or an individual develop toward functional maturity.

Mature Onset Diabetes

An older term generally referring to type 2 diabetes mellitus, which often develops in adulthood, though diabetes classification depends on cause rather than age alone.

Mature Teratoma

A germ-cell tumor composed of well-differentiated tissues from one or more embryonic germ layers; many are benign, but behavior depends on the site, age, and tumor type, and some can undergo malignant transformation.

Maturity Onset Diabetes Of The Young

A group of monogenic forms of diabetes mellitus inherited in an autosomal dominant pattern, typically presenting before age 25 in non-obese individuals without autoantibodies. Caused by single-gene mutations affecting pancreatic beta-cell development and insulin secretion, most commonly involving *HNF1A* (MODY 3) and *GCK* (glucokinase, MODY 2).

Alternate Names:

MODY; Maturity-Onset Diabetes of the Young

Matuzumab

An investigational monoclonal antibody targeting the epidermal growth factor receptor and studied for cancer treatment.

Maxilla

Paired bone forming the upper jaw, central face, and parts of orbit, nasal cavity, and hard palate. Body contains the maxillary sinus (largest paranasal sinus). Four processes: frontal, zygomatic, alveolar (upper teeth), and palatine (anterior hard palate). Fractures classified using the Le Fort system.

Maxilla (Upper Jaw)

The paired bones forming the upper jaw, the front part of the hard palate, and parts of the walls of the nose and eye sockets.

Alternate Names:

Upper Jaw

Maxillary

Pertaining to the maxilla, the paired irregular facial bone that forms the upper jaw, the floor of the orbit, the lateral and inferior boundaries of the nasal cavity, and the anterior hard palate. Houses the maxillary teeth within the maxillary alveolar ridge and encloses the large air-filled maxillary sinus (antrum of Highmore), the largest paranasal sinus, which drains into the middle meatus via the ostiomeatal complex. Subject to odontogenic infections, acute and chronic maxillary sinusitis, and complex midface impact fractures categorized by the Le Fort classification.

Maxillary Sinus

The maxillary sinus is a paired, air-filled cavity in the maxilla beneath each orbit. It drains into the nasal cavity and may become inflamed or infected in sinusitis.

Maxillary Sinusitis

Inflammation or infection of the air-filled sinus cavities behind the cheekbones.

Maxillofacial Prosthesis

Artificial replacement of maxillofacial structures. Obturator for cleft palate or surgical defect. Ocular, nasal, auricular prostheses.

Maximal Breathing Capacity

The greatest volume of air that can be moved in and out of the lungs in a specified period during maximal effort; it is related to but distinct from standard spirometric measures.

Maximum

The maximum is the greatest value in a set of measurements or the highest level reached by a variable,

such as a recorded temperature, blood pressure, or drug concentration.

Maximum Allowable Concentration

The highest permitted concentration of a substance in a workplace or environment under a specified regulatory standard and exposure period.

Maximum Plasma Concentration (C_{max})

The highest measured plasma concentration after a dose. C_{max} and the time to reach it are used to describe absorption rate, peak-related toxicity, and comparative pharmacokinetics.

Alternate Names:

C_{max}

Maximum Thickness

Maximum thickness is the greatest measured thickness of a structure or specimen. It may be reported in imaging, pathology, or assessment of a lesion, organ, or tissue layer.

Maximum Tolerated Dose

The highest dose of a drug that produces acceptable, manageable toxicity without causing unacceptable adverse effects, determined through dose-escalation studies, most commonly in Phase I clinical trials. The MTD identifies the dose range for subsequent efficacy studies.

May-Hegglin Anomaly

An inherited MYH9-related disorder with macrothrombocytopenia, giant platelets, and characteristic leukocyte inclusions; hearing, kidney, or cataract abnormalities may occur.

May-Thurner Syndrome

Compression of left common iliac vein by right common iliac artery. Causes left lower extremity DVT. Diagnosis: CT/MR venography, phlebography. Treatment: angioplasty and stenting.

Mayer-Rokitansky-Küster-Hauser Syndrome

A congenital malformation syndrome characterized

by agenesis or severe hypoplasia of the Müllerian duct derivatives, resulting in the congenital absence of the uterus and the upper two-thirds of the vagina in phenotypic and genetic 46,XX females. Affected individuals have normal functional ovaries, normal androgen and estrogen levels, and normal female secondary sexual characteristics (breast development and pubic hair), classically presenting in adolescence with primary amenorrhea. Associated with renal anomalies (such as unilateral renal agenesis or pelvic kidney) and skeletal defects.

Alternate Names:

MRKH Syndrome; Müllerian Agenesis; Vaginal Agenesis

Mayo Scissors

Heavy surgical scissors used to cut fascia, muscle, sutures, and other relatively tough material.

Maytansine

Maytansine is a potent antimetabolic compound that binds tubulin and inhibits microtubule assembly. Its derivatives are used as cytotoxic payloads in some antibody–drug conjugates.

Maze Procedure

A cardiac surgical operation developed by James Cox (the Cox-Maze procedure) designed to cure atrial fibrillation. Through surgical incisions or lines of thermal ablation (bipolar radiofrequency or cryoablation), a precise maze-like pattern of transmural scar lesions is created across both atria, which prevents the propagation of macro-reentrant electrical wavelets and isolates pulmonary vein triggers while preserving an organized electrical conduit from the sinoatrial node to the atrioventricular node, restoring atrial transport function.

Alternate Names:

Cox-Maze Procedure; Surgical Ablation for Atrial Fibrillation; Maze Operation

Mazindol

Mazindol is a sympathomimetic appetite suppressant that affects central catecholamine signaling. It has been used for short-term obesity treatment in some countries and may cause stimulant-related

adverse effects.

MB ChB

Abbreviation for Bachelor of Medicine and Bachelor of Surgery, a medical degree awarded in many countries.

MC Lymphoma (Mantle Cell Lymphoma)

Alternate index label for Mantle Cell Lymphoma.

MCA Peak Systolic Velocity

Doppler measurement of the peak systolic velocity in the fetal middle cerebral artery. An indicator of fetal anemia: elevated MCA PSV (>1.5 MoM) correlates with moderate-to-severe anemia. The most important application is surveillance of Rh-sensitized pregnancies (detect anemia before hydrops develops).

MCAD Deficiency

An inherited disorder in which the body cannot properly break down certain medium-chain fats for energy.

McArdle Disease

An autosomal recessive metabolic myopathy and inborn error of glycogen metabolism (Glycogen Storage Disease Type V [GSD V]) caused by homozygous or compound heterozygous mutations in the *PYGM* gene on chromosome 11q13, resulting in deficiency of muscle glycogen phosphorylase (myophosphorylase). Prevents the breakdown of muscle glycogen into glucose-1-phosphate during anaerobic exercise. Characterized by exercise-induced muscle fatigue, stiffness, severe muscle cramps, and recurrent rhabdomyolysis with myoglobinuria, with characteristic relief and resurgence of muscle endurance after 8–10 minutes of moderate activity known as the “second-wind” phenomenon.

Alternate Names:

Glycogen Storage Disease Type V (GSD V);
Myophosphorylase Deficiency

McBurney’S Point

A surface point in the right lower abdomen, classically near the junction of the lateral and middle

thirds of a line from the anterior superior iliac spine to the umbilicus, where maximal tenderness may occur in acute appendicitis.

Mcc Gene

mcc gene is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

McCune-Albright Syndrome

Sporadic condition caused by activating mutations in the *GNAS1* gene encoding the Gs alpha subunit leading to constitutive activation of adenylyl cyclase.

McDonald Cerclage

A transvaginal cervical cerclage using a purse-string suture (Mersilene tape or Prolene) placed at the cervicovaginal junction, without bladder dissection. The most common cerclage technique. Indications: cervical insufficiency (history-indicated or ultrasound-indicated short cervix <25 mm before 24 weeks). Placed at 12–14 weeks.

MCF-7 Cell

MCF-7 cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

MCH

Mean corpuscular hemoglobin: the calculated average amount of hemoglobin in each red blood cell, reported as part of a complete blood count.

MCHC

Mean corpuscular hemoglobin concentration: the calculated average concentration of hemoglobin in a given volume of packed red blood cells, derived from hemoglobin and hematocrit values and reported in a complete blood count.

MCL Injury (Medial Collateral Ligament Injury)

Alternate index label for Medial Collateral Ligament

Injury.

McMurray Test

A knee examination in which the leg is flexed, rotated, and extended to place stress on the menisci. Pain or a palpable click may suggest a meniscal lesion but is not diagnostic alone.

McRoberts Maneuver

A positioning maneuver for managing shoulder dystocia where the mother's legs are hyperflexed onto her abdomen, with thighs against the abdomen. This flattens the sacrum, increases the anteroposterior diameter of the pelvis, and rotates the symphysis pubis superiorly. Often combined with suprapubic pressure.

MCRPC

Alternate index label; see *Metastatic castration-resistant prostate cancer (mCRPC)*.

MCV

Mean corpuscular volume: the calculated average volume of a red blood cell, derived from hematocrit and red-cell count and reported as part of a complete blood count.

MDMA

3,4-Methylenedioxymethamphetamine, a psychoactive stimulant and hallucinogen commonly known as ecstasy or molly; it can produce serious cardiovascular, neurologic, and temperature-regulation effects.

MDS

Alternate index label; see *Myelodysplastic syndromes*.

ME/CFS

Alternate index label; see *Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS)*.

Mead Acid

mead acid is a fatty acid or other lipid-related molecule whose chain structure influences mem-

branes, energy storage, signaling, and metabolism.

Meadow Saffron

Colchicum autumnale, a plant containing colchicine; ingestion can cause severe gastrointestinal illness, multiorgan toxicity, and death.

Mean

The mean is the arithmetic average, calculated by adding all values and dividing by the number of values. It is commonly used to summarize clinical measurements but can be distorted by extreme values.

Mean Arterial Pressure

The average pressure driving systemic blood flow during one cardiac cycle. It is influenced by cardiac output and systemic vascular resistance and is approximated clinically as diastolic pressure plus one-third of pulse pressure at normal heart rates.

Mean Cell Volume

The average volume of a red blood cell, calculated from the hematocrit and red-cell count and reported as MCV in a complete blood count.

Mean Corpuscular Hemoglobin Concentration

The mean concentration of hemoglobin within packed erythrocytes, calculated from hemoglobin and hematocrit. A reduced MCHC indicates hypochromia; an increased value may occur with spherocytosis or analytical interference.

Mean Corpuscular Volume

The average volume of an erythrocyte, reported in femtoliters as part of the CBC. A low MCV indicates microcytosis, whereas a high MCV indicates macrocytosis; interpretation requires correlation with hemoglobin, reticulocytes, and other indices.

Mean Diameter

Mean diameter is the average diameter of a structure or lesion, usually calculated from one or more

measured dimensions. The method used should be specified when measurements are clinically compared.

Mean Thickness

Mean thickness is the average thickness calculated from multiple measurements of a tissue, structure, or specimen. It is used in imaging, pathology, and physiologic measurements.

Measles

Measles is a highly contagious measles-virus infection causing fever, cough, conjunctivitis, and a spreading maculopapular rash, with possible pneumonia, encephalitis, or other complications. droplets. Prodrome of fever, coryza, cough, and conjunctivitis, followed by Koplik spots (white buccal mucosal lesions) and maculopapular rash spreading cephalocaudally. Complications: pneumonia, encephalitis, and subacute sclerosing panencephalitis.

Measles (Rubeola)

Alternate index label for Rubeola.

Measles Virus

A negative-sense RNA virus that causes measles, a highly contagious respiratory illness. The virus is transmitted by respiratory droplets and is one of the most infectious pathogens known. Symptoms include fever, cough, coryza, conjunctivitis, and rash.

Measles, German

Alternate index label; see *Rubella*.

Measurable Residual Disease

The burden of cancer remaining after therapy that can be quantified using a validated assay. The term is increasingly preferred to minimal residual disease because residual disease may be clinically important even when not minimal.

Measurement

The process of determining the size, amount, concentration, or value of a clinical or biological variable.

Meatal

Pertaining to a meatus, a natural anatomical canal, opening, or passageway in the body. Most commonly refers in urology to the external urethral meatus (with meatal stenosis describing pathologic narrowing of the urethral orifice, often secondary to instrumentation or circumcisions, causing a forceful, narrow urinary stream) and in otolaryngology to the external auditory meatus (the canal leading from the auricle to the tympanic membrane) or the nasal meatuses (superior, middle, inferior) underlying the nasal turbinates.

Meatal Stenosis

Narrowing of the urethral meatus, which may cause a deflected or narrow urinary stream, spraying, dysuria, or difficulty emptying the bladder.

Meatus

An opening or passage leading into or out of a body structure, such as the urethral or auditory meatus.

Meatus, Female Urethral

The external opening of the female urethra, through which urine leaves the body; it is located anterior and superior to the vaginal opening.

Meatuses

The nasal meatuses are the air passages located beneath each corresponding turbinate on the lateral wall of the nasal cavity, and they serve as the drainage pathways for the paranasal sinuses and the nasolacrimal duct.

Mebendazole

An anthelmintic medicine used to treat several intestinal worm infections by impairing parasite microtubules.

Mebrofenin Scan

Hepatobiliary imaging using Tc-99m mebrofenin. Evaluates liver function, biliary obstruction.

Mecamylamine

Mecamylamine is an orally active ganglionic

nicotinic-receptor blocker that reduces sympathetic nerve transmission and lowers blood pressure. It has been used for severe hypertension and may cause postural hypotension, dizziness, constipation, and dry mouth.

Mecarzole

Mecarzole is a synonym for carbendazim, a benzimidazole fungicide that interferes with fungal microtubule formation. It is an agricultural chemical rather than a routine human medicine and may be toxic if exposure occurs.

Mechanical Circulatory Support

Devices that provide temporary or permanent cardiac support. Options include intra-aortic balloon pump (IABP), Impella (percutaneous ventricular assist device), TandemHeart, extracorporeal membrane oxygenation (ECMO), and durable left ventricular assist devices (LVADs).

Mechanical Insufflation-Exsufflation

A device that delivers positive pressure (insufflation) followed by rapid negative pressure (exsufflation) to simulate a cough. It is used in neuromuscular disease (Duchenne muscular dystrophy, spinal cord injury, ALS) to augment cough effectiveness and clear secretions.

Mechanical Ventilation

Mechanical ventilation provides respiratory support by delivering controlled or spontaneous breaths through an endotracheal tube or tracheostomy.

Mechanical Ventilator

A mechanical ventilator is a machine that supports or replaces spontaneous breathing by delivering controlled or assisted breaths, usually through a mask or an endotracheal tube. Settings are adjusted to oxygenation, ventilation, mechanics, and patient needs.

Mechanical Ventilator - Infants

A machine that supports or replaces spontaneous breathing in an infant by delivering controlled breaths through an endotracheal tube or, in selected

cases, another airway interface.

Mechanism Of Death

The physiologic or biochemical derangement that causes death (the final failure) as opposed to the underlying disease or injury (cause of death) and the circumstances (manner). Examples: ventricular fibrillation, cerebral hemorrhage, sepsis, respiratory failure, exsanguination, cardiac tamponade.

Mechanism Of Toxicity

How substance causes harm. Examples: enzyme inhibition, membrane disruption, receptor binding, oxidative stress, DNA damage.

Mechanoreceptor

mechanoreceptor is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Mechlorethamine

Mechlorethamine is an alkylating antineoplastic agent that cross-links DNA and inhibits cell division. Topical gel is used for cutaneous T-cell lymphoma; systemic use is limited by toxicities such as myelosuppression and tissue injury.

Meckel Diverticulectomy

Surgical removal of a Meckel diverticulum, usually performed when it causes bleeding, inflammation, obstruction, perforation, or is found incidentally in a high-risk setting.

Meckel Diverticulum

Congenital remnant of the omphalomesenteric duct located on the antimesenteric border of the distal ileum approximately sixty centimeters from the ileocecal valve.

Meckel'S Diverticulum

A congenital pouch of the small intestine caused by persistence of the embryonic vitelline duct; it may bleed, obstruct, or become inflamed.

Meckel-Gruber Syndrome

A lethal autosomal recessive primary developmental ciliopathy caused by pathogenic biallelic variants in genes encoding components of the primary ciliary transition zone (including *TMEM216*, *RPGRIP1L*, *CC2D2A*, and *CEP290*). Classically diagnosed prenatally by the pathognomonic diagnostic triad of an occipital encephalocele, bilateral massive multicystic dysplastic kidneys, and postaxial polydactyly (affecting both hands and feet), often accompanied by hepatic ductal plate malformations (microscopic bile duct proliferation) and pulmonary hypoplasia secondary to severe oligohydramnios.

Alternate Names:

Meckel Syndrome; Dysencephalia Splanchnocystica

Meclizine

Meclizine is a first-generation antihistamine with anticholinergic and vestibular-suppressant effects. It is used to prevent or treat motion sickness and vertigo and may cause drowsiness, dry mouth, and blurred vision.

Meclofenamate Overdose

Meclofenamate overdose involves toxic or environmental exposure to meclufenamate overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Meconium

The thick, dark first stool of a newborn, composed of intestinal secretions, shed cells, mucus, and swallowed amniotic material.

Meconium Aspiration

Meconium aspiration syndrome: aspiration of meconium-stained amniotic fluid into the fetal/neonatal airway causing airway obstruction, chemical pneumonitis, and surfactant dysfunction—respiratory distress in term/post-term infants. Risk: fetal distress/hypoxia (passage of meconium in utero).

Meconium Aspiration Syndrome

Respiratory distress in newborns born through meconium-stained amniotic fluid, caused by aspira-

tion of meconium into the airways. Presents with respiratory distress, barrel-shaped chest, and coarse crackles. Meconium is thick, greenish-black fetal stool passed in utero. Risk factors include post-term pregnancy and fetal distress.

Meconium Ileus

Intestinal obstruction in neonates caused by inspissated meconium most commonly associated with cystic fibrosis. Presents with abdominal distension, bilious vomiting, and failure to pass meconium within forty-eight hours.

Meconium Peritonitis

Sterile chemical peritonitis resulting from in utero intestinal perforation allowing meconium to spill into the peritoneal cavity. Most commonly associated with cystic fibrosis causing inspissated meconium obstruction, intestinal atresia, and volvulus. May be detected prenatally as fetal ascites with echogenic peritoneal calcifications.

Meconium-Stained Amniotic Fluid

Greenish or yellowish discoloration of amniotic fluid due to fetal meconium passage in utero. Associated with fetal distress, post-term pregnancy, and fetal growth restriction. Thick meconium increases risk of meconium aspiration syndrome. Management of non-vigorous newborns: tracheal suctioning before positive pressure ventilation.

Medial

Medial means toward or situated nearer the body's midline; it contrasts with lateral and is used to describe anatomical position, movement, or imaging findings.

Medial Collateral Ligament Injury

Alternate index label for MCL) Injur.

Medial Collateral Ligament Injury

Tear of the MCL, the most common knee ligament injury. Mechanism: valgus stress to the knee (direct blow to lateral knee). Presents with medial knee pain, swelling, and valgus laxity. Grading: I (pain, no laxity), II (partial tear, some laxity), III

(complete tear, significant laxity).

Medial Cuneiform

Largest cuneiform, on the medial foot between navicular and first metatarsal. Insertion for tibialis anterior and posterior tendons supporting the medial longitudinal arch. The first tarsometatarsal joint (Lisfranc joint) is an important landmark; Lisfranc injuries involve fracture-dislocation at this level.

Medial Epicondylitis

Alternate index label; see *Golfer's elbow (Medial epicondylitis)*.

Medial Epicondylitis - Golfer'S Elbow

Overuse-related tendinopathy at the common flexor tendon origin on the medial elbow, causing pain with gripping or resisted wrist and forearm flexion.

Medial Meniscus

The medial meniscus is a crescent-shaped fibrocartilage pad between the medial femoral condyle and medial tibial plateau. It distributes load and contributes to knee stability; tears may cause joint-line pain, swelling, or locking.

Medial Rotation

Rotation of a limb toward the median plane of the body.

Medial Tibial Stress Syndrome

Alternate index label; see *Shin splints*.

Median

Located in or relating to the middle plane or midpoint of a structure.

Median Arcuate Ligament Syndrome

Compression of the celiac artery by a ligament near the diaphragm, sometimes causing abdominal pain.

Median Arcuate Ligament Syndrome

Compression of the celiac artery and nearby nerves by the median arcuate ligament, potentially causing

post-meal abdominal pain and weight loss.

Alternate Names:

Dunbar Syndrome MALS

Median Nerve

From lateral and medial cords (C5-T1), passing through the arm without branching, through the cubital fossa, into the forearm innervating most flexor muscles. In the hand innervates thenar muscles (opponens, abductor brevis, flexor brevis pollicis) and first two lumbricals. Sensation to lateral 3.5 digits.

Median Nerve Compression (Carpal Tunnel Syndrome)

Alternate index label for Carpal Tunnel Syndrome.

Median Plane

A sagittal plane that passes directly through the midline of the body, dividing it into equal right and left halves.

Median Rhomboid Glossitis

A smooth, red, diamond-shaped depapillated area on the dorsum of the tongue caused by chronic *Candida* infection, usually asymptomatic and managed with antifungal therapy.

Median Survival

The time by which half of the people in a study have experienced the specified event, such as death. It is a summary measure of survival in a defined population.

Mediastinal Cyst

A mediastinal cyst is a fluid- or tissue-filled cystic lesion in the mediastinum. Common types include bronchogenic, thymic, and pericardial cysts; symptoms depend on size and compression, and imaging guides evaluation.

Mediastinal Disease

A disorder affecting structures in the mediastinum, including the heart, great vessels, thymus, lymph nodes, nerves, and tumors or cysts within the central chest.

Mediastinal Emphysema

Mediastinal emphysema, or pneumomediastinum, is the presence of free air in the mediastinum. It can follow alveolar rupture, trauma, surgery, or esophageal injury and may cause chest pain, neck crepitus, or respiratory compromise.

Mediastinal Hemorrhage

Bleeding into the mediastinum, commonly caused by trauma, vascular rupture, surgery, or invasive procedures, and potentially causing airway or cardiovascular compression.

Mediastinal Mass

An abnormal growth or collection of tissue in the mediastinum, the central compartment of the chest. Causes include tumors, enlarged lymph nodes, cysts, and vascular abnormalities.

Mediastinal Neoplasm

A mediastinal neoplasm is an abnormal growth in the central space of the chest between the lungs; it may arise from lymph nodes, thymus, nerves, or other tissues.

Mediastinal Pleura

The mediastinal pleura is the portion of parietal pleura lining the lateral surfaces of the mediastinum. It separates the mediastinal contents from the pleural cavity and reflects around the lung root.

Mediastinal Shift

Mediastinal shift is displacement of the mediastinal structures away from their usual position. It may result from increased pressure on one side, such as a tension pneumothorax, or loss of volume on the other, such as lung collapse.

Mediastinal Tumor

A mediastinal tumor is a mass in the central chest compartment between the lungs and may be benign or malignant.

Mediastinitis

Mediastinitis is inflammation or infection of the

central chest compartment, often after esophageal rupture, surgery, trauma, or spread of infection.

Mediastinoscopes

Mediastinoscopes are slender endoscopic instruments used during mediastinoscopy to inspect the mediastinum and obtain tissue samples, especially from mediastinal lymph nodes. The procedure is performed through a small incision near the sternum.

Mediastinoscopy

A surgical procedure that accesses the mediastinum through a small incision near the sternum to inspect or biopsy mediastinal lymph nodes or masses. It is one option for staging lung cancer when less-invasive sampling is nondiagnostic or unsuitable.

Mediastinoscopy With Biopsy

A procedure using an instrument inserted through a small incision near the sternum to inspect the mediastinum and obtain lymph-node or mass samples for diagnosis and staging.

Mediastinum

The central compartment of the thoracic cavity between the lungs, containing the heart, major blood vessels, esophagus, and trachea.

Mediators Of Inflammation

Inflammatory mediators are soluble molecules that initiate, amplify, or regulate the inflammatory response.

Medical Cannabis And Cannabidiol

Cannabis (marijuana) and cannabidiol are cannabinoid compounds derived from *Cannabis sativa* that interact with the endocannabinoid system (CB1 and CB2 receptors) to modulate pain, appetite, mood, and inflammation.

Medical Certificate

A document completed by an authorized healthcare professional that records a medical finding, diagnosis, fitness status, or period of illness.

Medical Condition

A state affecting health or bodily function, whether acute, chronic, symptomatic, or asymptomatic.

Medical Device

A medical device is an instrument, apparatus, implant, software, or similar article intended for diagnosis, prevention, monitoring, treatment, or alleviation of disease or injury. Its primary action is not achieved by chemical or metabolic means.

Medical Economics

Medical economics is the study of how health-care resources are produced, distributed, financed, and used. It evaluates costs, outcomes, efficiency, access, and value in health-care decisions.

Medical Entomology

The study of insects and related arthropods that cause disease or transmit disease-causing organisms to humans, including their biology, ecology, physiology, and genetics.

Medical Ethics

Moral principles guiding healthcare practice, including autonomy, beneficence, non-maleficence, and justice.

Medical Etymology

The study of the origins and historical development of medical words and terms, including their roots, prefixes, suffixes, and changes in meaning.

Medical Examiner

An appointed physician, usually a board-certified forensic pathologist, charged with investigating sudden, unexpected, violent, or suspicious deaths, ordering autopsies, determining cause and manner of death, and certifying the death certificate. Unlike the elected coroner, a medical examiner must meet professional qualifications.

Medical Geography

The study of how diseases, health, health-care resources, and health-related environmental factors

are distributed and vary across geographic areas and populations.

Medical History

A medical history is a structured account of a person's current concerns, past illnesses and operations, medications, allergies, family history, social history, and relevant exposures. It helps guide examination, diagnosis, and treatment.

Medical Illness

The subjective experience of being unwell, which may or may not correspond to an identifiable disease.

Medical Indigency

Medical indigency is inability to obtain needed medical care because of insufficient financial resources or lack of adequate coverage. The term describes a social and economic circumstance, not a disease.

Medical Jargon

Specialized terminology used by healthcare professionals. Unexplained medical jargon can hinder communication with patients.

Medical Jurisprudence

The branch of law (and its study) that deals with the rights and duties of physicians and the legal aspects of medical practice, distinct from but overlapping with forensic medicine.

Medical Marijuana

Medical marijuana involves toxic or environmental exposure to medical marijuana; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Medical Marijuana (Medical Cannabis)

Alternate index label for Medical Cannabis.

Medical Marijuana (Medical Marijuana (Medical Cannabis))

Alternate index label for Medical Marijuana (Medical Cannabis).

Medical Negligence (Malpractice)

Legal tort occurring when a healthcare provider fails to exercise the standard of reasonable care, skill, or diligence expected of a prudent practitioner of similar standing, resulting in injury, harm, or death to a patient.

Alternate Names:

Malpractice

Medical Nutrition Therapy

Individualized nutrition intervention by registered dietitian. Conditions: diabetes, renal disease, GI disorders, cancer, obesity.

Medical Officer

A physician appointed to provide clinical care or medical oversight for an institution, service, or organization.

Medical Physicist

A board-certified healthcare professional with specialized advanced training in medical physics, applying the principles of physics to diagnostic imaging, nuclear medicine, and radiation oncology. Responsibilities include calculating and verifying individualized radiation therapy dose distributions (treatment planning for external beam radiotherapy, stereotactic radiosurgery, and brachytherapy), ensuring radiation dosimetry calibration accuracy, maintaining strict radiation safety protocols, and conducting quality assurance for linear accelerators, CT, MRI, and PET scanners to protect patients and clinical personnel.

Medical Physics

The application of physics to medicine, including diagnostic imaging, radiation therapy, radiation safety, and physiological measurement.

Medical Pin

A medical pin is a slender metal fixation device inserted into bone or across a joint to stabilize a fracture or support orthopedic treatment. Pins may be temporary or permanent and require monitoring for infection and displacement.

Medical Practitioner

A licensed physician qualified to diagnose, prevent, and treat illness.

Medical Privacy

Protection of identifiable patient information and control of its disclosure in healthcare.

Medical Record

A medical record is the documented account of a patient's health information, including history, findings, diagnoses, test results, treatments, and clinical decisions. It supports continuity, communication, quality improvement, and legal documentation.

Medical Records

A documented account of a patient's clinical history, examination, investigations, treatment, and outcome maintained by providers. In medicolegal practice, records are contemporaneous evidence in malpractice, injury, criminal, and death cases, so accuracy, timeliness, and unaltered amendments are essential.

Medical Repository

The first medical magazine and scientific journal published in the United States; founded in New York in 1797, it was published quarterly until 1824.

Medical School

An institution whose curriculum leads to a medical degree and includes foundational sciences, clinical training, patient care, and often medical research.

Medical Symbol

A commonly recognized sign representing medicine, especially the Rod of Asclepius: a staff with one serpent. The winged, two-serpent caduceus is often used incorrectly as a medical symbol.

Medical Tattooing

Medical tattooing (permanent makeup, micropigmentation) covers depigmented skin (vitiligo, vitiligo camouflage), restores areolas after mastectomy, and masks scars. Trichopigmentation simulates hair

on bald scalp. Medical tattooing uses specialized pigments and techniques.

Medical Tests

Diagnostic procedures used to screen for, confirm, or monitor disease through analysis of biologic specimens, imaging, physiologic measurements, or genetic material.

Medication

A combination of aspirin, phenacetin, and caffeine; phenacetin-containing APC products were associated with kidney injury and are no longer widely used.

Medications And Treatments

all pharmacologic and therapeutic interventions used to prevent, manage, or cure disease, classified by mechanism of action, therapeutic class, and route of administration.

Medications Type 2 Diabetes (Antidiabetics-Oral)

Alternate index label for antidiabetics-oral.

Medicinal

Used for the prevention, diagnosis, relief, or treatment of disease.

Medicines For Back Pain

Medicines for back pain may include selected anti-inflammatory drugs, acetaminophen, muscle relaxants, or other treatments depending on the cause and patient risks.

Medicines For Osteoporosis

Osteoporosis medicines reduce bone loss or increase bone formation and may include bisphosphonates, denosumab, or anabolic treatments chosen according to fracture risk.

Medicines For Sleep

Medicines for sleep may help short-term insomnia but can cause dependence, falls, confusion, or next-day impairment; treatment should also address the

cause of poor sleep.

Medico-Legal

Relating to the application of medical knowledge to legal questions or proceedings.

Medicolegal

Pertaining to the application of medical knowledge to legal questions and the intersection of medicine and the law. Covers forensic pathology, toxicology, psychiatry, and clinical issues: informed consent, malpractice, medical records, disability, personal injury, occupational disease, and public health law.

Meditation

Meditation is a practice of intentionally directing attention, often toward the breath, a sensation, or an object, while observing thoughts without judgment. It may promote relaxation and stress management but does not replace indicated medical treatment.

Mediterranean Anemia

Alternate index label; see *Thalassemia*.

Mediterranean Fever

An older name for brucellosis, a bacterial infection that can cause prolonged fever, sweats, fatigue, and joint or muscle pain.

Medium

A substance or condition through which something occurs; in microbiology, a culture medium supports the growth of microorganisms.

MEDLARS

Medical Literature Analysis and Retrieval System, an electronic bibliographic system developed by the U.S. National Library of Medicine to index and retrieve biomedical literature; it preceded MEDLINE.

Medroxyprogesterone

Medroxyprogesterone is a progestogen used for contraception, abnormal uterine bleeding, endometriosis, and hormone therapy; depot injections may cause irregular bleeding, weight change, delayed

fertility return, and reduced bone density.

Medroxyprogesterone Acetate

Medroxyprogesterone acetate is a synthetic progestin used in hormonal contraception, treatment of endometriosis or abnormal uterine bleeding, and hormone therapy. It suppresses ovulation and produces progestational effects; formulation-specific risks and contraindications apply.

Medulla

The inner or central part of an organ, such as the renal medulla or adrenal medulla.

Medulla Oblongata

Most inferior brainstem segment, continuous with the spinal cord at the foramen magnum. Contains vital autonomic centers: cardiovascular center (heart rate, blood pressure), respiratory center (dorsal/ventral respiratory groups), and area postrema (vomiting center).

Medullary

Relating to the medulla or inner portion of an organ.

Medullary Carcinoma

Medullary carcinoma is a cancer with sheets of malignant cells and a prominent immune-cell response; the exact meaning depends on the organ involved, especially the breast or thyroid.

Medullary Sponge Kidney

Congenital malformation characterized by cystic dilatation of the collecting ducts in the renal papillae predisposing to nephrolithiasis and nephrocalcinosis. Presents with recurrent kidney stones, hematuria, and urinary tract infections.

Medullary Thyroid Carcinoma

Neuroendocrine tumor arising from parafollicular C cells that produce calcitonin. Accounts for approximately five percent of thyroid carcinomas. May be sporadic or hereditary as part of multiple endocrine neoplasia type two A or two B. Serum calcitonin serves as a tumor marker. Diagnosis requires FNA

with calcitonin immunohistochemistry.

Medulloblastoma

A fast-growing embryonal tumor of the cerebellum and the most common malignant brain tumor of childhood. Treated with surgical resection, craniospinal radiation, and chemotherapy.

Medulloepithelioma

A medulloepithelioma is a rare embryonal tumor arising most often from the nonpigmented ciliary epithelium of the eye in children. It may cause visual symptoms, neovascular glaucoma, or a visible mass and is treated according to its extent and pathology.

Mefenamic Acid

mefenamic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Mefloquine

Mefloquine is an antimalarial drug used for prevention and treatment of susceptible malaria. It has a long half-life but can cause neuropsychiatric, vestibular, and cardiac adverse effects, so patient-specific contraindications and warnings are important.

Mega-Trial

A mega-trial is a very large clinical trial designed to enroll enough participants to detect clinically important differences, including relatively uncommon outcomes or modest treatment effects.

Megabecquerel

A megabecquerel (MBq) is a unit of radioactivity equal to one million nuclear disintegrations per second. It is used to state the activity of radiopharmaceuticals and other radioactive materials.

Megacalycosis

Megacalycosis is congenital, nonobstructive dilatation of the renal calyces, usually with otherwise preserved renal architecture and function. It is often incidental but must be distinguished from

obstructive hydronephrosis.

Megacolon

Abnormal enlargement and dilation of the colon, which may be congenital, acquired, toxic, or inflammatory.

Megakaryocyte

A large bone-marrow cell that extends cytoplasmic processes and releases platelets.

Megakaryocytopoiesis

Megakaryocytopoiesis is the formation and maturation of megakaryocytes in the bone marrow, followed by platelet production. It is regulated in large part by thrombopoietin and other hematopoietic signals.

Megaloblast

An abnormally large, immature red-cell precursor in bone marrow, typically reflecting impaired DNA synthesis.

Megaloblastic Anaemia

Macrocytic anemia caused by defective DNA synthesis, most often from vitamin B12 or folate deficiency.

Megaloblastic Anemia

Anemia caused by impaired DNA synthesis, usually related to vitamin B12 or folate deficiency, with abnormally large red blood cells.

Megalocephaly

Abnormally large head size, which may be familial, developmental, metabolic, or associated with brain disease.

Megamonas

Megamonas is a genus of anaerobic, gram-negative bacteria found mainly in the intestinal microbiota of humans and animals. Some species have been recovered from clinical specimens, although their disease-causing role depends on the clinical setting.

Megamonas Funiformis

Megamonas funiformis is an anaerobic bacterium of

the genus Megamonas, commonly associated with the intestinal microbiota. Its detection in a clinical specimen requires interpretation alongside symptoms, specimen quality, and other microbiologic findings.

Megamonas Hypermegale

Megamonas hypermegale is an anaerobic bacterium in the genus Megamonas, primarily associated with the gastrointestinal microbiota. Its presence in a specimen is interpreted in relation to the site, clinical symptoms, and other microbiologic results.

Megasphaera

Megasphaera is a genus of anaerobic bacteria found in the oral cavity and gastrointestinal and genital microbiota. Species identification may be relevant in microbiome studies or when recovered from a clinical specimen.

Megasphaera Elsdenii

Megasphaera elsdenii is an anaerobic bacterium commonly found in the rumen and other microbial communities. It ferments organic acids and is mainly of relevance in microbiology and animal or microbiome research.

Megestrol

Megestrol is a synthetic progestin used in some settings to treat hormone-responsive cancers and, in certain formulations, anorexia or severe weight loss. It can cause glucocorticoid-like effects, thromboembolic risk, and adrenal suppression.

Megestrol Acetate

Megestrol acetate is the acetate salt of the progestin megestrol. Depending on formulation and indication, it is used for advanced endometrial or breast cancer and for appetite stimulation in selected patients with severe weight loss.

Meglitinides

Short-acting medicines that stimulate pancreatic insulin release to lower blood glucose.

Megrim

An older term for migraine, a recurrent headache disorder often associated with nausea and sensitivity to light or sound.

Meibomian Glands

Sebaceous glands in the eyelids that produce the oily layer of the tear film and help prevent tear evaporation.

Meibomianitis

Meibomianitis is inflammation or blockage of oil-producing eyelid glands, causing irritated eyelid margins, dry eye, crusting, or recurrent styes.

Meige Syndrome

A segmental cranial dystonia characterized by involuntary bilateral blepharospasm with dystonic contractions of the lower face, jaw, or tongue. Symptoms fluctuate and may be worsened by stress, speech, chewing, or sensory stimuli.

Meigs Syndrome

The triad of benign ovarian fibroma, ascites, and pleural effusion. The effusions typically resolve after removal of the ovarian tumour and should be distinguished from malignant ovarian disease.

Meigs' Syndrome

The triad of an ovarian fibroma, ascites, and pleural effusion, with resolution of the fluid after tumor removal.

Meiosis

A specialized cell division that produces haploid eggs or sperm from a diploid precursor cell and creates genetic variation through recombination and chromosome assortment.

Meiotic Recombination

Meiotic recombination is exchange of DNA between homologous chromosomes during meiosis, usually through crossing over. It creates new combinations of alleles and contributes to genetic variation.

Meissner's Corpuscles

Encapsulated sensory receptors in hairless skin that detect light touch and low-frequency vibration.

Melancholia

An older term for severe depression, often describing pervasive sadness, loss of pleasure, psychomotor change, and biological symptoms.

Melancholic Features

A specifier for major depressive episodes characterized by a loss of interest or pleasure in all or almost all activities, a depressed mood that is qualitatively distinct from grief, prominent psychomotor retardation or agitation, significant anorexia or weight loss, excessive guilt, and worsening of mood in the morning.

Melanin

Melanin is a pigment produced by melanocytes that gives skin, hair, and eyes their colour and absorbs ultraviolet radiation; its amount and distribution influence pigmentation disorders and melanoma biology.

Melanoblast

A precursor cell that develops into a melanocyte and produces pigment after migrating into the skin and other tissues.

Melanocyte

A melanocyte is a pigment-producing cell derived from the neural crest. It makes melanin in specialized organelles called melanosomes and transfers the pigment to neighboring keratinocytes, helping protect skin from ultraviolet radiation.

Melanocytes

Dendritic pigment-producing neural crest-derived cells in the epidermal basal layer and hair follicles, comprising approximately 5-10 percent of basal keratinocytes. Produce melanin (eumelanin, brown-black; pheomelanin, yellow-red) in melanosomes, transferred to keratinocytes via dendritic processes for UV protection.

Melanocytic Nevus

A melanocytic nevus is a usually benign, localized proliferation of melanocytes that appears as a macule, papule, or nodule. Changes in size, color, border, or symptoms may require assessment for atypia or melanoma.

Melanocytic Tumor Of The Central Nervous System

A tumor showing melanocytic differentiation within the CNS or meninges, ranging from benign melanocytoma to malignant melanoma. Evaluation includes imaging, histology, and molecular assessment to determine behavior.

Melanoderma

Increased pigmentation or darkening of the skin caused by increased melanin production or an increased number of melanocytes.

Melanoma

Melanoma is a malignant tumor of melanocytes, most often arising in skin but also in eye or mucosal sites; early detection matters because it can invade and metastasize.

Alternate Names:

Malignant Melanoma (Melanoma) Skin Cancer, Melanoma

Melanoma (Skin Cancer Overview)

Alternate index label for Skin Cancer Overview.

Melanoma Of The Eye

Melanoma of the eye concerns melanoma of the eye, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Melanoma Surveillance

regular skin examination, patient education on self-examination, and dermoscopic monitoring of atypical nevi. High-risk patients (personal or family history, numerous atypical nevi, immunosuppression) require 6-month follow-up. Standard patients require annual examination. Total body photogra-

phy establishes baseline for change detection.

Melanoma Treatment

Stage-dependent. Stage 0 to IA: wide local excision. Stage IB to IIA: wide excision with sentinel lymph node biopsy. Stage IIB to IIC: wide excision, SLNB, and adjuvant therapy.

Melanoma, Benign

An older or imprecise term for a benign melanocytic growth, such as a melanocytic nevus or mole; unlike malignant melanoma, it is not cancerous.

Melanophore

A melanophore is a pigment-containing cell, especially one in lower vertebrates, whose melanin granules can disperse or aggregate to alter visible coloration. In mammals, melanocytes are the principal melanin-producing cells.

Melanosis

Abnormal accumulation or increased production of melanin in skin, mucosa, an organ, or other tissue.

Melanosis Coli

Melanosis coli is dark pigmentation of the colon lining caused by lipofuscin accumulation, classically associated with prolonged stimulant-laxative use. It is usually benign and may fade after stopping the trigger, but the underlying constipation and medication use should be reviewed.

Melanosome

A melanosome is a specialized organelle in a melanocyte in which melanin is synthesized, stored, and transported. Melanosomes are transferred to keratinocytes and influence skin, hair, and eye pigmentation.

MELAS Syndrome

A mitochondrial disorder involving encephalopathy, lactic acidosis, and stroke-like episodes, often beginning before adulthood and caused by pathogenic mitochondrial DNA variants.

Melasma

Melasma is acquired, usually symmetric brown or gray facial hyperpigmentation associated with ultra-violet exposure, hormones, pregnancy, and genetic susceptibility; it is benign but often recurrent.

Alternate Names:

Chloasma (Melasma)

Melasma (Chloasma)

An acquired, chronic pigmentary disorder characterized by symmetric, hyperpigmented, light-to-dark brown macules and patches on sun-exposed areas of the face (malar, forehead, upper lip, chin). It primarily affects women of childbearing age and individuals with darker skin types (Fitzpatrick types III-V).

Alternate Names:

Chloasma

Melatonin

A hormone produced mainly by the pineal gland that helps regulate sleep-wake timing.

Melatonin Receptor

melatonin receptor is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Melatonin Receptor Agonists

Medications that act on MT1 and MT2 receptors in the suprachiasmatic nucleus to regulate circadian rhythms. Ramelteon is approved for insomnia with sleep onset difficulty. It has no abuse potential and does not cause dependence. It is a useful alternative to benzodiazepines for sleep initiation problems.

MELD

Model for End-Stage Liver Disease, a laboratory-based score used to estimate severity of advanced liver disease and help prioritize adults for liver transplantation. Traditional versions use bilirubin, creatinine, and INR; current implementations may include additional variables.

MELD Score

Model for End-Stage Liver Disease predicting three-month mortality in advanced liver disease used for liver transplant allocation. The score is calculated using serum bilirubin, creatinine, and international normalized ratio generating a continuous score from approximately six to forty points.

Melena

Black, tarry, often foul-smelling stool caused usually by digested blood from upper gastrointestinal bleeding.

Melengestrol Acetate

Melengestrol acetate is a synthetic progestin used primarily in veterinary medicine to suppress estrus and improve feed efficiency in cattle. It is not approved for use in humans.

Melibiose

A disaccharide composed of galactose and glucose, produced by hydrolysis of raffinose and metabolized by some microorganisms.

Melio

An older shortened form related to melioidosis, the infection caused by *Burkholderia pseudomallei*.

Melioidosis

An infectious disease caused by *Burkholderia pseudomallei*, usually acquired from contaminated soil or water through inoculation or inhalation. It may cause pneumonia, skin infection, abscesses, sepsis, or chronic disease and is more severe in people with diabetes or other risk factors.

Melitensis

A species designation, especially *Brucella melitensis*, a bacterium that causes brucellosis and is associated with goats and sheep.

Melitten

Melitten, more commonly spelled melittin, is the principal peptide toxin in honeybee venom. It disrupts cell membranes and contributes to pain, in-

flammation, hemolysis, and other effects of envenomation.

Melorheostosis

A rare sclerosing bone dysplasia producing irregular, flowing cortical hyperostosis on imaging. It may cause pain, stiffness, limb deformity, restricted movement, contractures, or soft-tissue fibrosis and can affect one or several contiguous bones.

Meloxicam

Meloxicam is a nonsteroidal anti-inflammatory drug that inhibits cyclooxygenase enzymes and reduces prostaglandin synthesis. It is used for pain and inflammation, but can cause gastrointestinal, renal, cardiovascular, and hypersensitivity adverse effects.

Melphalan

An alkylating antineoplastic medicine used in selected plasma-cell disorders and other cancers.

Memantine

An NMDA receptor antagonist (uncompetitive, low-to-moderate affinity) used for moderate to severe Alzheimer disease, often added to an acetylcholinesterase inhibitor. Blocks excessive glutamate-mediated excitotoxicity while preserving normal synaptic transmission. Once or twice daily oral.

Memantine Hydrochloride

Memantine hydrochloride is an NMDA-receptor antagonist used to help manage moderate-to-severe Alzheimer disease. It reduces pathologic glutamate-mediated excitotoxicity; common adverse effects include dizziness, headache, constipation, and confusion.

Membrane

A thin tissue layer that surrounds a cell, organ, or body cavity.

Membrane Channel

A membrane channel is a protein-forming pore across a cell membrane that permits selected ions or molecules to pass down an electrochemical or

concentration gradient. Many channels are gated by voltage, ligands, or mechanical forces.

Membrane Fusion

Membrane fusion is merging of two lipid bilayers into one continuous membrane. It is required for processes such as synaptic vesicle release, enveloped-virus entry, fertilization, and intracellular vesicle trafficking.

Membrane Lipids

membrane lipids is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Membrane Microdomain

A membrane microdomain is a small, specialized region of a cell membrane with a distinct composition of lipids and proteins. It can organize receptors, signaling molecules, transporters, and cytoskeletal interactions.

Membrane Part

A membrane part is a structural or molecular component of a biological membrane, such as a lipid, protein, carbohydrate, channel, or receptor. These components determine membrane integrity, transport, signaling, and cell recognition.

Membrane Potential

The electrical potential difference across a cell membrane, approximately -70 mV in most neurons (resting membrane potential). Established by the unequal distribution of ions (primarily K⁺, Na⁺, Cl⁻) across the membrane and selective permeability.

Membrane Protein

membrane protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Membrane Raft

A membrane raft is a cholesterol- and sphingolipid-

enriched membrane microdomain that helps cluster selected proteins and signaling molecules. Raft organization is dynamic and may change during receptor signaling or membrane trafficking.

Membranoproliferative Glomerulonephritis

Pattern of glomerular injury with mesangial proliferation and capillary wall thickening creating tram-track appearance on silver stain. Hepatitis C is common cause of immune complex MPGN. C3 glomerulopathy represents complement-mediated disease. Associated with monoclonal gammopathies.

Membranous

Relating to, consisting of, or resembling a membrane. In medicine, the term may describe a thin tissue layer or a disease pattern involving such a layer.

Membranous Nephropathy

A glomerular disease caused by immune-complex deposition along the glomerular basement membrane. It commonly causes proteinuria and nephrotic syndrome and may be primary or secondary to another disease, medication, or infection.

Memory

The cognitive process of encoding, storing, and retrieving information, mediated by distributed neural networks involving the hippocampus, medial temporal lobe, prefrontal cortex, and limbic system.

Memory Cell

Long-lived lymphocytes (memory B cells and memory T cells) that survive after an immune response and mount rapid, enhanced responses to re-exposure to the same antigen (immunological memory).

Memory Disorder

Impairment of encoding, storage, retrieval, or recognition of information that exceeds normal variation and may result from neurodegeneration, injury, vascular disease, medication, or other causes.

Memory Impairment

Memory impairment is reduced ability to learn, store, retrieve, or recognize information. It may be transient or persistent and can result from normal aging, medications, sleep problems, neurologic disease, psychiatric illness, or other medical conditions.

Memory Loss

Reduced ability to recall or learn information. It may result from normal aging, medical illness, medicines, brain injury, or a neurocognitive disorder.

Memory Loss (Amnesia)

Reduced ability to recall past information, form new memories, or both. Causes include normal aging, medications, head injury, seizures, stroke, infection, substance use, neurocognitive disorders, and psychological conditions; sudden memory loss requires urgent assessment.

Memory Span

The number of items, such as words or digits, that a person can briefly retain and recall, usually in the original order; it is a measure of short-term or working memory.

Memory T Cells And B Cells

Long-lived immune cells that respond rapidly when they encounter a previously recognized antigen.

Alternate Names:

Memory lymphocytes

Memory Training

Memory training consists of structured exercises or strategies intended to improve encoding, storage, retrieval, or compensatory organization of information. Benefits vary by method and condition and do not establish treatment for an underlying disorder.

MEN 2

Alternate index label; see *Multiple endocrine neoplasia, type 2 (MEN 2)*.

Men Night Sweats (Night Sweats)

Alternate index label for Night Sweats.

Menarche

The first menstrual period, marking the onset of menstruation. Its timing varies with genetics, nutrition, health, and pubertal development; markedly early or delayed menarche may require assessment.

Mendelian Inheritance

Inheritance of traits determined by single genes and transmitted according to dominant, recessive, or sex-linked patterns.

Mendelism

The principles of inheritance based on Gregor Mendel's work, including segregation of alleles and independent assortment of genes; Mendelian traits may follow autosomal or X-linked dominant or recessive patterns.

Mendelson Syndrome

A chemical pneumonitis caused by inhalation of acidic stomach contents, classically during anesthesia or childbirth. It can cause cough, breathing difficulty, and impaired oxygenation.

Meniere Disease

An inner ear disorder characterized by recurrent episodes of vertigo lasting 20 minutes to 12 hours, with fluctuating sensorineural hearing loss, tinnitus, and aural fullness. Due to endolymphatic hydrops. Diagnosis: clinical criteria; audiometry shows low-frequency SNHL during attacks.

Alternate Names:

Menieres Disease (Meniere Disease)

Meniere'S Disease

An inner-ear disorder causing episodic vertigo, fluctuating sensorineural hearing loss, tinnitus, and aural pressure.

Menieres Disease (Meniere Disease)

Alternate index label for Meniere Disease.

Meningeal Carcinomatosis

Meningeal carcinomatosis is spread of cancer cells to the membranes and fluid surrounding the brain and spinal cord.

Meningeal Leukemia

Meningeal leukemia is leukemia involving the membranes or fluid around the brain and spinal cord, potentially causing headache, weakness, or cranial-nerve problems.

Meningeal Lymphoma

Meningeal lymphoma is lymphoma affecting the membranes or fluid spaces around the brain and spinal cord.

Meningeal Neoplasm

A meningeal neoplasm is an abnormal growth arising in or near the membranes surrounding the brain and spinal cord.

Meninges

The three membranes enveloping the brain and spinal cord: dura mater (outermost, tough), arachnoid mater (middle, avascular), and pia mater (innermost, adherent to the CNS surface). The subarachnoid space (between arachnoid and pia) contains cerebrospinal fluid. The epidural and subdural spaces are sites of hemorrhage.

Meningioma

A usually slow-growing tumor arising from cells associated with the meninges. Most are benign, but some are atypical or malignant and can cause symptoms by compressing the brain, spinal cord, or cranial nerves.

Meningism

Meningism is a syndrome of neck stiffness, headache, and sometimes photophobia or vomiting that resembles meningeal irritation without necessarily indicating meningitis. Serious causes, including infection or subarachnoid hemorrhage, require prompt evaluation.

Meningismus

Neck stiffness and other meningeal-irritation signs without proven inflammation of the meninges.

Meningitis

Meningitis is inflammation of the membranes around the brain and spinal cord, caused by infection or noninfectious inflammation and presenting with headache, fever, neck stiffness, or altered mental status. Infection but also by bacteria, fungi, parasites, medicines, or autoimmune disease. Symptoms may include fever, headache, neck stiffness, light sensitivity, or altered mental status and require urgent assessment.

Meningitis - Cryptococcal

Meningitis caused by *Cryptococcus* species, usually affecting people with impaired cellular immunity; symptoms may develop gradually and diagnosis uses cerebrospinal-fluid studies and fungal testing.

Meningitis - Tuberculous

Subacute or chronic meningitis caused by *Mycobacterium tuberculosis*, characterized by inflammation of the meninges and requiring urgent multidrug treatment.

Meningocele

A congenital defect in which the membranes surrounding the spinal cord or brain protrude through an opening in the skull or spine, forming a fluid-filled sac. It may occur with other neural-tube defects and can cause neurologic problems; evaluation commonly includes imaging and specialist assessment.

Meningocele Repair

A medical procedure, treatment, device, or care topic related to meningocele repair; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Meningococcal Disease

A bacterial infection caused by *Neisseria meningitidis* (serogroups A, B, C, W, Y), transmitted via respiratory droplets. Meningococcemia: sudden onset fever, petechial/purpuric rash, hypotension,

and DIC. Meningococcal meningitis: fever, nuchal rigidity, headache, photophobia.

Meningococcal Infection

Infection with *Neisseria meningitidis*, which may cause meningitis, bloodstream infection, or localized respiratory disease. Severe illness can progress rapidly.

Meningococcal Meningitis

Acute bacterial meningitis caused by *Neisseria meningitidis*, often accompanied by bloodstream infection; it is a medical emergency requiring immediate antibiotics and public-health measures.

Meningococcemia

Meningococcemia is bloodstream infection with *Neisseria meningitidis*, causing fever, rash, shock, clotting failure, and potentially fatal sepsis.

Meningococcus

Neisseria meningitidis, a bacterium that can cause meningitis and rapidly progressive bloodstream infection.

Meningoencephalitis

Inflammation of both the meninges surrounding the brain and the brain tissue itself, usually caused by infection or autoimmune disease. Fever, headache, neck stiffness, confusion, seizures, or reduced consciousness requires urgent medical assessment.

Meningoencephalitis Toxoplasma (Toxoplasmosis)

Alternate index label for Toxoplasmosis.

Meningomyelocele

A neural-tube defect in which meninges and spinal cord tissue protrude through a vertebral opening.

Meniscal Allograft Transplantation

A medical procedure, treatment, device, or care topic related to meniscal allograft transplantation; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Meniscal Injuries

Tears or other damage to the knee menisci, which are crescent-shaped pieces of cartilage that help distribute load and stabilize the knee. Symptoms may include joint-line pain, swelling, clicking, or restricted motion. Diagnosis is based on examination and, when needed, imaging.

Meniscal Tear

Tear of the C-shaped fibrocartilage (medial or lateral meniscus) that provides shock absorption and stability in the knee. Medial meniscus tears more common (attached to capsule). Mechanism: twisting injury with weight-bearing. Presents with joint line tenderness, pain with deep squatting, catching, locking, and giving way.

Meniscectomy

Surgical removal of part or all of a torn knee meniscus.

Meniscus

A crescent-shaped fibrocartilage structure in the knee, or a curved fluid surface in a container.

Meniscus Tear

Alternate index label; see *Torn meniscus*.

Menkes Disease

An X-linked recessive disorder of copper metabolism caused by ATP7A mutations, resulting in impaired intestinal copper absorption and defective copper transport across the blood-brain barrier. Copper accumulates in intestinal and renal cells but is deficient in the liver and brain.

Menkes Kinky Hair Syndrome

An X-linked disorder of copper transport caused by ATP7A dysfunction. It produces progressive neurodegeneration, seizures, developmental delay, hypopigmented or brittle kinky hair, connective-tissue abnormalities, and failure to thrive.

Menogaril

Menogaril is a semisynthetic anthracycline deriva-

tive of nogalamycin. It intercalates into DNA and inhibits topoisomerase II, thereby impairing DNA replication and repair; it was investigated as an antineoplastic agent.

Menometrorrhagia

Excessive uterine bleeding occurring both at expected menstrual times and at irregular intervals; the older term corresponds broadly to heavy and irregular menstrual bleeding and warrants evaluation for underlying causes.

Menopausal

Relating to menopause or the transition around the end of menstrual periods.

Menopausal Hormone Therapy

Menopausal hormone therapy, formerly called hormone replacement therapy, is the use of estrogen with or without progestogen to alleviate moderate-to-severe menopausal vasomotor symptoms (hot flashes, night sweats), genitourinary syndrome of menopause (vaginal dryness, dyspareunia, urinary urgency), and prevent osteoporosis.

Menopause

The permanent end of menstrual periods because of loss of ovarian activity, usually diagnosed after 12 consecutive months without menstruation. Hormonal changes may cause hot flashes, night sweats, vaginal symptoms, sleep problems, or other symptoms.

Menopause And Perimenopause

Menopause is the permanent cessation of menstruation for 12 consecutive months, reflecting ovarian follicle depletion and estrogen decline, occurring at a median age of 51–52 years in North America.

Menopause Hot Flashes (Hot Flashes)

Alternate index label for Hot Flashes.

Menopause Status

Menopause status describes whether a person is premenopausal, perimenopausal, or postmenopausal.

Natural menopause is clinically defined after 12 consecutive months without menstruation when no other cause explains the amenorrhea.

Menopause Syndrome

A nontechnical term for the group of symptoms that may occur during the menopausal transition, such as hot flashes, night sweats, sleep problems, mood changes, vaginal dryness, and menstrual irregularity. Menopause itself is a normal life stage, not a disease.

Menorrhagia

Heavy or prolonged menstrual bleeding that interferes with physical, social, emotional, or material quality of life. Causes include fibroids, adenomyosis, endometrial polyps, bleeding disorders, thyroid disease, and ovulatory dysfunction; it may cause iron-deficiency anemia.

Menotropin

Menotropin is a urinary-derived preparation containing follicle-stimulating and luteinizing hormone activity, used to stimulate follicular development or spermatogenesis; ovarian hyperstimulation and multiple pregnancy are important risks.

Mensa

A table or flat surface; in anatomy, the term may describe a flat surface of a structure and is context-dependent.

Menses Duration

Menses duration is the number of days of menstrual bleeding in a cycle, counted from the first day of bleeding until bleeding stops. Persistently prolonged, heavy, or irregular bleeding warrants clinical assessment.

Menstrual

Relating to menstruation or the menstrual cycle, including the monthly shedding of the uterine lining and the bleeding that follows. Cycle length and bleeding vary, but very heavy, painful, irregular, or unexpected bleeding may need evaluation.

Menstrual Cramps

Cramping pain in the lower abdomen during or around menstruation, medically called dysmenorrhea. Pain may radiate to the back or thighs and may occur with nausea, diarrhea, headache, or fatigue. Severe, worsening, or newly developed pain requires evaluation for causes such as endometriosis or fibroids.

Alternate Names:

Cramps, Menstrual Painful periods Cramps Menstrual (Menstrual Cramps) Painful Periods

Menstrual Cycle

The recurring hormonal and uterine changes that prepare the body for possible pregnancy. Cycle length and bleeding pattern vary among individuals and across life stages.

Menstruation

The periodic shedding of the endometrial lining as part of the menstrual cycle when pregnancy does not occur. The timing, duration, and amount of bleeding vary with age, hormonal state, medicines, and health conditions.

Menstruation Disturbance

A menstruation disturbance is an abnormality in the timing, regularity, duration, amount, or associated symptoms of menstrual bleeding. Causes include pregnancy, endocrine disorders, structural disease, medications, systemic illness, and perimenopause.

Menstruation, Cessation Of

Cessation of menstrual periods, called amenorrhea; it is a normal part of menopause but may also result from pregnancy, illness, endocrine disorders, medications, stress, excessive exercise, or inadequate nutrition.

Mental

Relating to thinking, memory, emotion, behavior, or other functions of the mind. It does not by itself mean that a condition is psychiatric.

Mental Age

An estimate of cognitive functioning expressed as the age level at which a standardized test score is typical; it is not the same as chronological age.

Mental Disorder

A clinically significant disturbance in cognition, emotion regulation, or behavior that causes distress, impairment, or both.

Mental Disorders

Health conditions involving clinically significant disturbances in cognition, emotion regulation, behavior, or functioning, including mood, anxiety, psychotic, neurodevelopmental, and substance-use disorders.

Mental Fatigue

Mental fatigue is a subjective feeling of reduced mental energy, concentration, or cognitive endurance after sustained effort or illness. Persistent or severe symptoms may accompany sleep disorders, depression, neurologic disease, medication effects, or other medical conditions.

Mental Health

Mental health encompasses emotional, psychologic, and social well-being, affecting how individuals think, feel, behave, and handle stress, relationships, and decision-making.

Mental Health (Psychology)

Alternate index label for Psychology.

Mental Illness

A broad term for mental disorders that affect mood, thinking, behavior, or functioning.

Mental Illness In Children

Mental illness in children may appear as persistent changes in mood, behavior, learning, sleep, appetite, relationships, or development. Evaluation considers age, medical causes, trauma, school context, family history, safety, and the child's strengths and supports.

Mental Process

A mental process is a cognitive or affective activity of the mind, such as perception, attention, memory, reasoning, language, emotion, or decision-making.

Mental Recall

Mental recall is retrieval of previously learned or stored information from memory. It may be assessed by asking a person to remember facts, events, words, or experiences after a delay.

Mental Status Examination

A structured assessment of the patient's current psychological functioning performed during the clinical interview. It evaluates appearance, behaviour, speech, mood, affect, thought process, thought content, perceptions, cognition, insight, and judgment.

Mental Status Testing

A structured assessment of appearance, behavior, alertness, orientation, attention, memory, language, thought, perception, mood, insight, and judgment.

Mentation

The process of thinking, perceiving, reasoning, and other mental activity.

Menthol

A mint-derived or synthetic compound producing a cooling sensation and used in some topical and respiratory preparations.

Menthol Poisoning

Menthol poisoning involves toxic or environmental exposure to menthol poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Mepacrine

An older antimalarial and antiprotozoal medicine, also called quinacrine, now used only in selected indications.

Meperidine

An opioid analgesic, also called pethidine, whose use

is limited by adverse effects and an active neurotoxic metabolite.

Meperidine Hydrochloride

Meperidine hydrochloride is the hydrochloride salt of the opioid analgesic meperidine. It produces analgesia through opioid-receptor activity and carries risks of sedation, respiratory depression, seizures, dependence, and serotonin toxicity with interacting drugs.

Meperidine Hydrochloride Overdose

Meperidine hydrochloride overdose involves toxic or environmental exposure to meperidine hydrochloride overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Mephenesin

An older centrally acting muscle relaxant with sedative properties.

Mephentermine

A sympathomimetic medicine formerly used to raise blood pressure in selected hypotensive states.

Mephénytoin

Mephénytoin is a hydantoin anticonvulsant formerly used to treat seizures. It is metabolized partly to ethotoin and has limited modern use because of potentially serious adverse effects, including hypersensitivity and blood dyscrasias.

Mepivacaine

An amide local anesthetic used for infiltration, nerve blocks, and regional anesthesia.

Meprobamate

An older anxiolytic and sedative medicine with dependence and overdose risks.

Meprobamate Overdose

Meprobamate overdose involves toxic or environmental exposure to meprobamate overdose; dose, route, timing, symptoms, and organ involvement

determine urgency and treatment.

Mepyramine

An antihistamine that blocks H1 receptors and may cause sedation; it is also called pyrilamine.

Meralgia Paresthetica

Burning, tingling, or numbness over the outer thigh caused by compression of the lateral femoral cutaneous nerve.

Alternate Names:

Burning Thigh Pain

Merbromin Poisoning

Merbromin poisoning involves toxic or environmental exposure to merbromin poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Mercaptopurine

A thiopurine antimetabolite used in selected leukemias and inflammatory disorders, requiring monitoring for marrow and liver toxicity.

Mercurialism

Chronic poisoning by mercury, potentially causing neurologic, psychiatric, renal, gastrointestinal, or oral effects.

Mercuric Chloride Poisoning

Mercuric chloride poisoning involves toxic or environmental exposure to mercuric chloride poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Mercuric Oxide Poisoning

Mercuric oxide poisoning involves toxic or environmental exposure to mercuric oxide poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Mercury

Liquid metal combining with other metals to form amalgam. Concerns about toxicity, proper handling essential.

Mercury Poisoning

Three forms: elemental (inhalation, neurological), inorganic (renal, GI), organic (methylmercury, neurological). Minamata disease from methylmercury. Treatment: chelation (dimercaprol, DMSA), remove exposure.

Meridian

A reference line or plane; in traditional acupuncture it denotes a proposed pathway for qi, while in anatomy it may describe a longitudinal line.

Merkel Cell Carcinoma

An aggressive neuroendocrine skin cancer arising from Merkel cells. It presents as a rapidly growing, painless, red-violet nodule on sun-exposed skin of elderly patients. Approximately 80 percent are associated with Merkel cell polyomavirus. Risk factors include immunosuppression, UV exposure, and fair skin.

Alternate Names:

Cancer, Merkel Cell Neuroendocrine Carcinoma Of The Skin

Merkel Cells

Specialized touch-sensing cells in the deepest layer of the skin that help detect light touch and sustained pressure.

Mermaid Syndrome

An extremely rare congenital anomaly in which the legs are fused to varying degrees, also called sirenomelia.

Meropenem

Meropenem is a broad-spectrum carbapenem antibiotic that inhibits bacterial cell-wall synthesis and treats severe susceptible infections, including meningitis; allergy, gastrointestinal effects, seizures, and resistance are possible.

Meropenem-Vaborbactam

A combination of the carbapenem meropenem with the cyclic boronate beta-lactamase inhibitor vaborbactam. Vaborbactam inhibits class A and

class C beta-lactamases, including KPC carbapenemases, but does not inhibit class D or metallo-beta-lactamases. This combination has activity against many carbapenem-resistant Enterobacteriaceae.

MERRF Syndrome

Myoclonic epilepsy with ragged-red fibres, a mitochondrial disorder usually caused by a pathogenic mitochondrial DNA variant. It features myoclonus, generalized epilepsy, ataxia, and skeletal-muscle mitochondrial abnormalities, with variable multisystem involvement.

Mersalyl

Mersalyl is an organomercury compound formerly used as a diuretic. Because mercury compounds can cause serious renal and neurologic toxicity, it has been replaced by safer diuretics.

Merthiolate Poisoning

Merthiolate poisoning involves toxic or environmental exposure to merthiolate poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Merycism

Rumination, the repeated regurgitation and rechewing of food after eating.

Mesalamine

Mesalamine, or 5-aminosalicylic acid, is an anti-inflammatory drug used to induce and maintain remission in ulcerative colitis and some other inflammatory bowel conditions. It acts mainly topically in the intestinal mucosa and can rarely affect renal function.

Mesangiolytic

Mesangiolytic is dissolution or disruption of the mesangial tissue within a renal glomerulus. It may occur in severe endothelial injury and can be associated with microvascular lesions and glomerular capillary aneurysmal changes.

Mesangium

The mesangium is the supportive tissue between glomerular capillary loops, composed of mesangial cells and extracellular matrix. Mesangial cells provide structural support, regulate capillary flow, and clear certain substances; abnormal expansion occurs in several glomerular diseases.

Mescaline

A naturally occurring psychedelic substance found in certain cacti that alters perception and may cause hallucinations and autonomic effects.

Mesencephalon

The mesencephalon, or midbrain, is the part of the brainstem between the pons and diencephalon. It contains pathways and nuclei involved in eye movements, motor control, arousal, and auditory and visual reflexes.

Mesenchymal Cell

mesenchymal cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Mesenchymal Chondrosarcoma

Mesenchymal chondrosarcoma is a rare aggressive cancer containing cartilage-forming tissue, often arising in bone or soft tissue.

Mesenchymal Tumor Of The Central Nervous System

A tumor arising from non-neural supporting tissues of the CNS, including meningeal, vascular, fibroblastic, or other mesenchymal lineages. Diagnosis requires correlation of morphology, immunophenotype, molecular findings, and site.

Mesenchyme

Embryonic connective tissue that gives rise to much of the skeleton, connective tissue, blood vessels, and smooth muscle.

Mesenchymoma

A mesenchymoma is a rare tumor containing more than one type of mesenchymal tissue, sometimes including cartilage, bone, fat, or vascular elements. Behavior depends on the site, age, histology, and presence of malignant components.

Mesenteric

Relating to the mesentery or its blood vessels, nerves, lymphatics, and fat.

Mesenteric Angiography

Contrast imaging of the arteries supplying the intestines, used to evaluate mesenteric ischemia, bleeding, aneurysm, stenosis, occlusion, or vascular anatomy.

Mesenteric Artery

A mesenteric artery supplies blood to portions of the intestine and associated abdominal organs. The superior and inferior mesenteric arteries are the major vessels; obstruction or embolism can cause acute mesenteric ischemia.

Mesenteric Artery Ischemia

Inadequate blood flow through the mesenteric arteries, usually caused by embolism, thrombosis, or severe low-flow states. Acute disease presents with severe abdominal pain out of proportion to examination and can progress to bowel infarction.

Mesenteric Cyst

A mesenteric cyst is a fluid-filled cystic lesion within the mesentery. It may be congenital, lymphatic, enteric, or traumatic; symptoms depend on size and complications such as infection, hemorrhage, torsion, or bowel obstruction.

Mesenteric Ischemia

Reduced blood flow to intestines. Acute: embolism, thrombosis. Chronic: atherosclerosis. CT angiography diagnosis.

Mesenteric Lymphadenitis

Inflammation of lymph nodes in the membrane that

attaches the intestines to the abdominal wall.

Alternate Names:

Lymphadenitis, Mesenteric

Mesenteric Veins

Mesenteric veins drain blood from the small and large intestines and carry it mainly through the superior and inferior mesenteric veins to the portal venous system. Thrombosis or impaired flow can cause intestinal congestion and ischemia.

Mesenteric Venous Thrombosis

A blood clot in a mesenteric vein that impairs intestinal venous drainage and can cause abdominal pain, bowel edema, ischemia, infarction, or gastrointestinal bleeding.

Mesentery

The fan-shaped double layer of peritoneum attaching the small intestine (jejunum and ileum) to the posterior abdominal wall, carrying the superior mesenteric artery, vein, lymphatics, and autonomic nerves between its layers. Root extends from duodenojejunal junction to ileocecal junction.

Mesh

A synthetic or biological implant used to provide structural support in pelvic organ prolapse and stress urinary incontinence surgery. Types: polypropylene (most common, mid-urethral slings for SUI, vaginal mesh for prolapse), biological (porcine, bovine, or cadaveric dermis).

Mesial

Situated toward the midline; in dentistry, toward the front or midline of a tooth.

Mesial Surface

The mesial surface is the surface of a tooth or other structure oriented toward the midline. In dentistry, mesial surfaces contact adjacent teeth and can be involved in caries, periodontal disease, or restoration margins.

Mesna

Mesna is a sulfhydryl-containing uroprotective drug that binds the toxic acrolein metabolite of ifosfamide and cyclophosphamide in the urine. It is used to reduce hemorrhagic cystitis during treatment with these oxazaphosphorine chemotherapy agents.

Mesoderm

The middle germ layer in an embryo, which develops into muscles, bones, cartilage, blood vessels, and the kidneys.

Mesomorph

A body-build classification characterized traditionally by relatively prominent muscle and bone; it is descriptive, not a diagnosis.

Mesonephric

Mesonephric means relating to the mesonephros, a temporary embryonic kidney, or to structures derived from its ducts and tubules. Mesonephric remnants can give rise to uncommon benign or malignant lesions of the genital tract.

Mesonephric Adenocarcinoma

A rare HPV-independent cervical adenocarcinoma arising from mesonephric remnants, usually presenting as a cervical mass or abnormal bleeding and requiring histopathologic diagnosis.

Mesonephric Tumor

A mesonephric tumor is a rare tumor arising from remnants of embryonic mesonephric ducts, usually in the female reproductive tract.

Mesonephroma

Mesonephroma is an older, context-dependent term for a tumor formerly thought to arise from mesonephric remnants, including some clear-cell or mesonephric tumors of the female genital tract and historically some yolk-sac tumors. Modern pathology uses more specific diagnostic names.

Mesothelial Neoplasm

A mesothelial neoplasm is a growth arising from

the lining of body cavities, such as the pleura, peritoneum, or pericardium.

Mesothelin

Mesothelin is a cell-surface glycoprotein expressed mainly by mesothelial cells and overexpressed in several cancers, including mesothelioma and some ovarian and pancreatic cancers. It is being studied as a biomarker and therapeutic target.

Mesothelioma

A malignant tumor of mesothelial surfaces, most often the pleura and commonly associated with asbestos exposure.

Mesothelium

The mesothelium is a single layer of specialized cells lining serous body cavities and covering their organs, including the pleura, peritoneum, and pericardium. It produces lubricating fluid and participates in inflammation and repair.

Messenger RNA

A single-stranded RNA molecule transcribed from a DNA template that carries coding information to ribosomes for protein synthesis. Mature eukaryotic mRNA typically contains 5' and 3' untranslated regions, a coding sequence, a 5' cap, and a poly(A) tail; it is usually produced by processing a precursor RNA, including splicing, capping, and polyadenylation. Also called mRNA.

Mestranol

A synthetic estrogen formerly used in some combined oral contraceptives.

Met

Abbreviation for methionine, an essential sulfur-containing amino acid; it is also represented by the one-letter code M and the codons AUG.

Meta-Analysis

A statistical method that combines quantitative results from multiple studies addressing a related question. Its validity depends on the quality and

comparability of the studies.

Metabolic

Relating to metabolism, the chemical processes by which cells produce and use energy and materials.

Metabolic Acidosis

A primary acid-base disorder characterized by decreased serum bicarbonate (<22 mEq/L) and pH <7.35 . Causes are classified by anion gap (AG). High AG: diabetic ketoacidosis, lactic acidosis, renal failure, starvation, toxins (methanol, ethylene glycol, salicylates).

Metabolic Alkalosis

A primary increase in plasma bicarbonate or loss of hydrogen ions. Causes include vomiting, diuretic use, mineralocorticoid excess, and chloride depletion; respiratory compensation increases carbon dioxide but is limited by hypoxemia and acidosis.

Metabolic Disease

A disorder of biochemical pathways or energy metabolism that disrupts normal processing of carbohydrates, fats, proteins, minerals, or hormones.

Metabolic Dysfunction-Associated Steatohepatitis

Alternate index label; see *Fatty liver disease*.

Metabolic Dysfunction-Associated Steatotic Liver Disease

Alternate index label; see *Fatty liver disease*.

Metabolic Equivalents

Units estimating the energy cost of physical activity relative to resting metabolism.

Alternate Names:

METs

Metabolic Myopathies

Muscle disorders caused by impaired energy metabolism, often producing exercise intolerance, cramps, weakness, muscle pain, or episodes of rhabdomyolysis.

Metabolic Neuropathies

Metabolic neuropathies concerns metabolic neuropathies, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Metabolic Rate

The rate at which the body uses energy to maintain life and perform activity.

Metabolic Response

A metabolic response is a change in energy use, substrate metabolism, or biochemical pathways in response to conditions such as fasting, exercise, infection, trauma, or medication. The response may be adaptive or contribute to illness when excessive or prolonged.

Metabolic Syndrome

Cluster of conditions: abdominal obesity, hypertension, hyperglycemia, high triglycerides, low HDL. Increases cardiovascular disease and diabetes risk.

Alternate Names:

Insulin Resistance Syndrome Syndrome X

Metabolic Tumor Volume

Total volume of metabolically active tumor on PET. Prognostic in lymphoma.

Metabolism

Metabolism comprises the chemical reactions that produce energy and synthesize or break down molecules in cells, including catabolism, anabolism, and regulated handling of nutrients and drugs. anabolism (synthesis, consumes energy). Central pathways: glycolysis, TCA cycle, oxidative phosphorylation, pentose phosphate pathway. Metabolic regulation ensures energy homeostasis.

Metabolite

A substance produced or consumed during metabolism of a drug, nutrient, or endogenous molecule.

Metabolomics

Metabolomics is the comprehensive study of small-molecule metabolites in cells, tissues, or biological fluids. It uses analytical methods such as mass spectrometry or nuclear magnetic resonance to investigate physiology, disease, and treatment effects.

Metacarpal

Relating to one of the five bones forming the palm of the hand.

Metacarpal Bone

A metacarpal bone is one of the five long bones of the hand between the wrist bones and the proximal phalanges. Each has a base, shaft, and head; fractures can affect alignment, grip, and joint function.

Metacarpals

Five long bones (I-V) forming the palm framework. Each has base, shaft, and head. The first is shortest and most mobile, articulating with the trapezium at a saddle joint for opposition. Second/third are firmly fixed; fourth/fifth more mobile. Bennett's fracture affects the first metacarpal base; boxer's fracture affects the fifth neck.

Metacarpophalangeal Joint

A condyloid synovial joint between a metacarpal head and the base of a proximal phalanx. It permits flexion, extension, abduction, adduction, and limited circumduction.

Metacarpus

The part of the hand between the wrist and fingers, containing the five metacarpal bones.

Metachromatic Leukodystrophy

An inherited lysosomal-storage disorder caused by deficient arylsulfatase A activity, leading to sulfatide accumulation and progressive demyelination.

Metadata

Metadata is information describing other data, such as its source, format, time, units, author, processing history, or quality. In health care it helps identify,

interpret, exchange, and govern clinical and research data.

Metagenome

A metagenome is the total genetic material recovered from a microbial community or other mixed biological sample, including genomes from organisms that may not be cultured.

Metagenomic Sequencing

Sequencing of nucleic acid recovered directly from a clinical or environmental specimen to identify microbial genetic material without requiring growth in culture. It can detect unusual or polymicrobial infections but is vulnerable to contamination and requires clinical interpretation.

Metagenomics

Metagenomics is the study of genetic material collected directly from an environmental or clinical sample containing multiple organisms. It can characterize microbial composition and functional genes without isolating each organism in culture.

Metagonimus Yokogawai

Metagonimus yokogawai is a small intestinal fluke acquired by eating raw or inadequately cooked freshwater fish. Infection can cause abdominal symptoms and, rarely, ectopic migration of eggs or adult worms.

Metal

A metal is an element or alloy with characteristic physical and chemical properties such as electrical conductivity and metallic bonding. In medicine, metals may be nutrients, toxic exposures, laboratory analytes, or components of implants and devices.

Metal Artifact

Image degradation from metal implants. Reduced by metal artifact reduction sequences (MRI), iterative reconstruction (CT).

Metal Cleaner Poisoning

Metal cleaner poisoning involves toxic or environ-

mental exposure to metal cleaner poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Metal Fume Fever

A short-lived influenza-like illness caused by inhalation of fumes from heated metals, especially zinc; symptoms include fever, chills, headache, muscle aches, and cough.

Metal Polish Poisoning

Metal polish poisoning involves toxic or environmental exposure to metal polish poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Metalloendopeptidase

A metalloendopeptidase is an endopeptidase whose catalytic activity depends on a bound metal ion, commonly zinc. It cleaves peptide bonds within proteins and includes enzymes involved in extracellular matrix remodeling and signaling.

Metalloporphyrins

Metalloporphyrins are porphyrin molecules containing a centrally coordinated metal ion. Heme, which contains iron, is a biologically important example and forms the functional group of hemoglobin, myoglobin, and many enzymes.

Metalloprotease

A metalloprotease is a proteolytic enzyme that requires a metal ion, commonly zinc, for catalytic activity. Metalloproteases regulate extracellular matrix turnover, cell signaling, inflammation, and tissue remodeling; dysregulation contributes to disease.

Metalloprotein

metalloprotein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Metallothionein

Metallothionein is a small, cysteine-rich protein that binds metals such as zinc, copper, and cadmium. It helps regulate essential metals and may protect cells from metal toxicity and oxidative stress.

Metamorphosis

Metamorphosis is a major change in body form during development, especially the transition from larva to adult in insects or other animals.

Metanephrine

Metanephrine is a metabolite of epinephrine formed mainly by catechol-O-methyltransferase. Plasma or urinary metanephrine measurements are used in evaluation for catecholamine-secreting tumors such as pheochromocytoma and paraganglioma.

Metaphase

The stage of cell division in which chromosomes align at the cell's equatorial plane.

Metaphase Plate

The metaphase plate is the plane at the center of a dividing cell where chromosomes align during metaphase. It is a geometric arrangement rather than a discrete anatomical structure.

Metaphyseal Corner Fractures

Bucket-handle or corner fractures at the metaphysis of long bones, characteristic of child abuse. Caused by twisting or pulling forces on extremities. Most common in the knee, ankle, and elbow. Seen on skeletal survey; may not be visible initially. Highly specific for non-accidental trauma in infants.

Metaphyseal Widening

Metaphyseal widening is enlargement of the growth-plate or metaphyseal region of a developing long bone. It may be seen in disorders such as rickets, skeletal dysplasia, infection, or healing injury and must be interpreted with imaging and clinical findings.

Metaphysis

The region of a long bone between the epiphysis (end) and diaphysis (shaft), containing the epiphyseal growth plate (physis) in children—the zone of active cartilage growth (endochondral ossification).

Metaplasia

A reversible adaptive change in which one differentiated adult cell type is replaced by another, representing the body's attempt to better withstand a chronic stress or irritation. The process involves reprogramming of tissue stem cells to differentiate along a different lineage that is more resilient to the new environment. Persistent metaplasia may impair normal tissue function and, in some settings, increase susceptibility to dysplasia and malignant transformation. Examples include intestinal metaplasia in the stomach and squamous metaplasia in chronically irritated respiratory epithelium.

Metaplastic Carcinoma

Metaplastic carcinoma is a cancer in which malignant cells have changed into a different cell type; the term is used especially for an aggressive form of breast cancer.

Metapneumovirus

A respiratory virus that can cause cough, fever, bronchiolitis, or pneumonia, especially in infants, older adults, and people with chronic disease.

Metastasectomy

Surgical removal of one or more metastases, which may be curative in selected oligometastatic disease, e.g., colorectal liver and lung metastases and sarcoma pulmonary metastases.

Metastases To The Breast

Secondary malignant deposits in breast tissue from a nonmammary primary tumor, most often melanoma, lymphoma, or carcinoma from another organ. Imaging and biopsy are needed because the treatment differs from primary breast carcinoma.

Metastasis

Metastasis is spread of malignant cells from a pri-

mary tumor to noncontiguous tissues through lymphatic, blood, or body-cavity routes, forming secondary tumors. hematogenous, direct extension, transcoelomic. Common sites: lungs, liver, bone, brain. Distant metastasis indicates Stage IV disease.

Metastasis Of Colorectal Cancer

Spread of colorectal malignancy to distant organs via lymphatic, hematogenous, or peritoneal routes, most commonly involving the liver, lungs, and peritoneum. The liver is the first site of metastasis due to portal venous drainage, and hepatic metastases may be resectable with curative intent.

Metastasis-Free Survival

The time from treatment until the development of distant metastasis or death; an important endpoint in adjuvant therapy trials.

Metastatic Adenocarcinoma

An adenocarcinoma that has spread from its primary site to distant organs, commonly the liver, lungs, bones, or brain. The prognosis and treatment depend on the primary tumor origin and the extent of metastatic disease.

Metastatic Brain Tumor

A metastatic brain tumor is cancer that has spread to the brain from a primary tumor elsewhere in the body.

Metastatic Breast Cancer

Breast cancer that has spread from the breast to distant parts of the body.

Metastatic Cancer

Cancer that has spread from its original site to distant organs or tissues. The spread may occur through the blood or lymphatic system and can form new tumors. Metastatic cancer is often called stage 4 cancer, although staging depends on the cancer type. The new tumors retain the characteristics of the original cancer; for example, breast cancer that spreads to the lung remains metastatic breast cancer, not lung cancer.

Alternate Names:

Stage 4 Cancer Advanced Metastatic Cancer Secondary Cancer

Metastatic Calcification

The deposition of calcium salts in otherwise normal tissues due to systemic hypercalcemia, where elevated serum calcium levels exceed the tissue ability to maintain calcium homeostasis.

Metastatic Cascade

Sequence of events in metastasis: local invasion, intravasation, circulation, extravasation, colonization. Many cells die during process.

Metastatic Castration-Resistant Prostate Cancer

Prostate cancer that has spread and continues to grow despite treatments that lower testosterone.

Metastatic Colon Cancer

Advanced colonic malignancy that has spread beyond the bowel to distant sites, most commonly the liver, lungs, and peritoneum, via lymphatic and hematogenous dissemination. Staging uses CT of chest, abdomen, and pelvis, and PET-CT may identify occult metastases.

Metastatic Colorectal Cancer

Alternate index label; see *Stage 4 (metastatic) colon cancer*.

Metastatic Inguinal Lymphadenopathy

Enlargement of inguinal lymph nodes due to metastatic cancer cells, often from primary tumors of the lower extremities, pelvis, or anorectal region. It may indicate advanced disease requiring systemic therapy.

Metastatic Lung Cancer

Lung cancer that has spread beyond the lungs to other organs, such as the brain, bones, liver, or adrenal glands. It represents advanced stage disease with limited treatment options.

Metastatic Pleural Tumor

A cancer deposit in the pleura that has spread from another primary tumor, commonly from the lung, breast, or other organs.

Metastin

Metastin is an older name for kisspeptin, a peptide derived from the KISS1 gene. It stimulates hypothalamic gonadotropin-releasing hormone secretion and helps regulate puberty, reproductive function, and tumor-cell signaling.

Metatarsal

Relating to one of the five long bones forming the forefoot.

Metatarsal Bone

A metatarsal bone is one of the five long bones of the forefoot between the tarsal bones and proximal phalanges. Metatarsal fractures and alignment abnormalities can affect weight bearing and gait.

Metatarsal Valgus

Metatarsal valgus is outward deviation of one or more metatarsal bones relative to the foot's normal axis. It may contribute to forefoot deformity and altered load distribution and is assessed clinically and radiographically.

Metatarsalgia

Pain in the front or ball of the foot, often arising from pressure or irritation around the metatarsal heads.

Metatarsals

Five long bones (I-V) forming the forefoot. Each has base, shaft, and head. First is shortest/thickest, bearing approximately one-third of forefoot load. Stress fractures commonly affect the second and third (march fractures). Morton's neuroma typically affects the nerve between 3rd/4th metatarsal heads.

Metatarsus

The part of the foot between the tarsal bones and toes, containing the five metatarsal bones.

Metatarsus Adductus

A foot deformity in which the forefoot is angled inward relative to the hindfoot, often recognized in infancy and ranging from flexible to rigid.

Metatarsus Varus

Metatarsus varus is inward deviation of the forefoot caused by medial angulation of the metatarsal bones. It is often recognized in infancy and may be flexible or rigid; management depends on severity and persistence.

Metathesis

Transposition or rearrangement of sounds in a word; in chemistry or biology, it can mean exchange of parts or groups.

Metencephalon

The metencephalon is the embryonic brain region that develops into the pons and cerebellum. These structures participate in motor coordination, balance, cranial-nerve functions, and relay of signals between the brain and spinal cord.

Meteorism

Abdominal distension caused by excessive gas in the gastrointestinal tract.

Meter

A metric unit of length equal to 100 centimeters; medical usage may also refer to a measuring device.

Metered-Dose Inhaler

A pressurized inhaler delivering a metered dose of medication. Requires proper technique for effective delivery. Spacer devices improve drug delivery and reduce oropharyngeal deposition. Most common delivery device for asthma and COPD maintenance.

Metestrus

A diagnostic or laboratory method related to metestrus. Results are interpreted with the method, reference range, and clinical findings.

Metformin

Metformin is a biguanide antihyperglycemic drug that lowers hepatic glucose production and improves insulin sensitivity. It is commonly used for type 2 diabetes; gastrointestinal effects are common, and rare lactic acidosis risk increases with certain severe conditions.

Metformin Hydrochloride

Metformin hydrochloride is the hydrochloride salt of metformin, an oral biguanide used chiefly for type 2 diabetes. It lowers blood glucose without directly stimulating pancreatic insulin release and is available in immediate- and extended-release formulations.

Metformin In Prediabetes

Metformin may reduce progression from prediabetes to type 2 diabetes, especially in people at high risk. Its use depends on age, body size, prior gestational diabetes, kidney function, other conditions, and patient preference; lifestyle intervention remains important.

Methacholine Challenge Test

A bronchoprovocation test used when asthma is suspected but routine spirometry is nondiagnostic. Inhaled methacholine may cause bronchoconstriction, and spirometry is repeated after increasing doses. The result must be interpreted with symptoms, treatment, and other test findings because the test is not specific for asthma.

Methacholine Chloride

Methacholine chloride is a muscarinic cholinergic agonist used in bronchoprovocation testing. In susceptible people it causes airway narrowing, allowing assessment for bronchial hyperresponsiveness under controlled medical supervision.

Methadone

A long-acting opioid used for pain in selected cases and for opioid-use-disorder treatment.

Methadone Hydrochloride

Methadone hydrochloride is the hydrochloride salt

of methadone. It provides long-lasting opioid-receptor activity for selected analgesic and opioid-use-disorder indications, with risks of respiratory depression, dependence, overdose, and cardiac arrhythmia.

Methadone Overdose

Methadone overdose involves toxic or environmental exposure to methadone overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Methaemoglobinaemia

A condition in which haemoglobin is oxidized to methaemoglobin and cannot release oxygen effectively, causing tissue hypoxia and sometimes cyanosis.

Methamidophos

Methamidophos is a highly toxic organophosphorus insecticide that inhibits acetylcholinesterase. Exposure can cause cholinergic poisoning with sweating, salivation, vomiting, bronchospasm, bradycardia, muscle weakness, and potentially respiratory failure.

Methamphetamine

Methamphetamine is a potent central nervous system stimulant that increases synaptic catecholamines. It has limited medical use in some settings but is commonly misused; toxicity and chronic use can cause cardiovascular complications, psychosis, dependence, and neurocognitive harm.

Methamphetamine Overdose

Methamphetamine overdose involves toxic or environmental exposure to methamphetamine overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Methamphetamines

Powerful synthetic stimulant drugs with substantial risks of dependence, cardiovascular toxicity, psychosis, and neurotoxicity.

Methane

Methane is a colorless, odorless, highly flammable gas produced by anaerobic decomposition and normal intestinal microbial activity. It can displace oxygen in enclosed spaces and cause asphyxiation; it is not itself a direct systemic poison at ordinary exposure levels.

Methane Sulfonate

Methane sulfonate, or mesylate, is the anion or ester derived from methanesulfonic acid. Mesylate salts are commonly used to improve the formulation of medicines; some methanesulfonate esters are potent alkylating agents and toxic impurities.

Methanobacteriaceae

A family of methane-producing archaea found in oxygen-free environments such as sediments, wetlands, and the digestive tracts of animals.

Methanobrevibacter Smithii

A methane-producing microorganism commonly found in the human intestine.

Methanol

A toxic alcohol whose metabolites can cause metabolic acidosis, visual injury, and death.

Methanol Poisoning

Toxic ingestion or exposure to methanol, whose metabolites cause high-anion-gap metabolic acidosis and visual or neurologic injury. Treatment requires urgent toxicology consultation, antidotal fomepizole or selected ethanol therapy, correction of acidosis, and hemodialysis in severe cases.

Methanol Test

Methanol test involves toxic or environmental exposure to methanol test; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Methantheline

An older anticholinergic medicine formerly used for gastrointestinal spasm and hyperhidrosis.

Methaqualone

Methaqualone is a sedative-hypnotic formerly used for insomnia and anxiety. It is no longer widely marketed because of misuse, dependence, dangerous respiratory and central nervous system depression, and overdose risk.

Methazolamide

Methazolamide is a carbonic anhydrase inhibitor used mainly to lower intraocular pressure in glaucoma. It reduces aqueous humor production and can cause metabolic acidosis, electrolyte abnormalities, kidney stones, or sulfonamide-related reactions.

Methemoglobin

Hemoglobin containing oxidized iron that cannot bind oxygen normally.

Methemoglobinemia

Methemoglobinemia occurs when hemoglobin iron is oxidized and cannot carry oxygen normally, causing cyanosis, headache, breathlessness, or severe tissue hypoxia.

Methemoglobinemia - Acquired

Abnormally high methemoglobin caused by oxidizing drugs, chemicals, or toxins, impairing hemoglobin oxygen delivery and potentially causing cyanosis, headache, confusion, or hypoxia.

Methenamine

Methenamine is a urinary antiseptic that decomposes in acidic urine to release formaldehyde, which inhibits bacterial growth. It is used for prevention of recurrent urinary tract infection in selected patients, not for treatment of active upper-tract infection.

Methicillin

A former penicillin antibiotic used historically to treat staphylococcal infections; resistance is mediated by altered penicillin-binding protein.

Methicillin-Resistant Staphylococcus Aureus

Staphylococcus aureus resistant to beta-lactam an-

tibiotics through acquisition of the *mecA* gene encoding penicillin-binding protein 2a (PBP2a). MRSA is a major cause of healthcare-associated and community-associated infections. Treatment options include vancomycin, daptomycin, linezolid, and trimethoprim-sulfamethoxazole.

Methicillin-Resistant Staphylococcus Aureus

Methicillin-resistant *Staphylococcus aureus* is a strain of *S. aureus* resistant to several beta-lactam antibiotics and capable of causing skin, wound, lung, bloodstream, or other infections.

Methicillin-Resistant Staphylococcus Aureus Infection

Alternate index label; see *MRSA infection*.

Methidathion

Methidathion is an organophosphorus insecticide and acaricide that inhibits acetylcholinesterase. Exposure can produce a cholinergic toxidrome with secretions, bronchospasm, bradycardia, gastrointestinal symptoms, weakness, and respiratory failure.

Methimazole

A medicine that reduces production of thyroid hormones and is used mainly for hyperthyroidism and Graves disease. It can rarely cause severe low white-cell counts or liver injury; fever or sore throat during treatment needs prompt medical advice.

Methiocarb

Methiocarb is a carbamate pesticide that inhibits acetylcholinesterase reversibly. Poisoning causes excessive muscarinic and nicotinic cholinergic activity and may require urgent supportive care and antidotal treatment.

Methionine

An essential sulfur-containing amino acid used in protein synthesis and methyl-group metabolism.

Methionine Enkephalin

Methionine enkephalin is an endogenous opioid pen-

tapeptide that binds opioid receptors and participates in modulation of pain, stress, reward, and neuroendocrine function. It is one of the products of proenkephalin processing.

Methisazone

An older antiviral compound investigated or used historically against poxvirus infections.

Methocarbamol

Methocarbamol is a centrally acting skeletal-muscle relaxant used with rest and physical measures for acute painful musculoskeletal conditions. It can cause drowsiness, dizziness, impaired coordination, and additive central nervous system depression.

Methodology

Methodology is the systematic set of principles, methods, procedures, and assumptions used to conduct and evaluate an investigation or clinical study. Sound methodology addresses design, measurement, bias, analysis, and reproducibility.

Methohexital

Methohexital is an ultra-short-acting barbiturate anesthetic used for induction of general anesthesia and, in selected settings, procedural sedation. It depresses central nervous system activity and may cause respiratory or cardiovascular depression.

Methomyl

Methomyl is a carbamate insecticide that reversibly inhibits acetylcholinesterase. Poisoning produces cholinergic symptoms such as salivation, sweating, vomiting, bronchospasm, bradycardia, weakness, and respiratory compromise.

Methotrexate

An antimetabolite that inhibits folate-dependent enzymes and is used in cancer and selected autoimmune or inflammatory disorders.

Methotrexate In Dermatology

An immunomodulatory antimetabolite used in selected inflammatory and autoimmune skin diseases,

including psoriasis, eczema, and some connective-tissue disorders. Treatment requires attention to liver disease, blood counts, infection risk, drug interactions, and strict pregnancy precautions.

Methotrimeprazine

Methotrimeprazine, also called levomepromazine, is a phenothiazine with antipsychotic, sedative, antiemetic, and analgesic properties. It is used in some countries for psychiatric illness and palliative-care symptoms and can cause sedation, hypotension, and anticholinergic effects.

Methoxamine

Methoxamine is a predominantly alpha-adrenergic agonist that increases vascular tone and blood pressure. It has been used as a vasopressor for hypotension but is not a routine first-line agent in many current practice settings.

Methoxsalen

Methoxsalen is a psoralen that intercalates into DNA and, after exposure to UVA, forms cross-links that suppress abnormal lymphocyte or skin-cell activity. It is used with UVA phototherapy for selected skin diseases; photosensitivity and skin-cancer risk require precautions.

Methoxyamine Hydrochloride

Methoxyamine hydrochloride is the hydrochloride salt of methoxyamine, a sympathomimetic compound with predominantly alpha-adrenergic activity. It has been used experimentally or historically as a vasopressor and pharmacologic probe.

Methoxyflurane

A volatile anesthetic formerly used for general anesthesia and, in some settings, low-dose analgesia.

Methoxyhydroxyphenylglycol

Methoxyhydroxyphenylglycol is a major metabolite of norepinephrine. Measurement in blood, urine, or cerebrospinal fluid has been used in research and selected neuroendocrine evaluations of sympathetic or central noradrenergic activity.

Methyclothiazide

Methyclothiazide is a thiazide diuretic that inhibits sodium and chloride reabsorption in the distal convoluted tubule. It is used for hypertension or edema and may cause hypokalemia, hyponatremia, hyperuricemia, and glucose intolerance.

Methyl Alcohol

Another name for methanol, a toxic alcohol that can cause severe acidosis and blindness.

Methyl Bromide

Methyl bromide, or bromomethane, is a volatile fumigant and alkylating chemical. Inhalation can cause severe neurologic, respiratory, renal, and cardiovascular toxicity, including delayed seizures and pulmonary edema.

Methyl Chloride

Methyl chloride, or chloromethane, is a volatile industrial gas formerly used in refrigerants and other applications. High-level inhalation exposure can depress the central nervous system and cause headache, ataxia, seizures, or cardiac effects.

Methyl Green

Methyl green is a cationic dye used in histology, often to stain DNA and nuclei. It is a laboratory reagent rather than a therapeutic drug, and handling should follow appropriate chemical-safety procedures.

Methyl Group

A methyl group is a chemical group consisting of one carbon atom bonded to three hydrogen atoms, written $-\text{CH}_3$. Methyl transfer reactions are important in metabolism, epigenetic regulation, and drug biotransformation.

Methyl Isocyanate

Methyl isocyanate is a highly volatile and reactive industrial chemical. Inhalation or eye and skin exposure can cause severe irritation, pulmonary edema, and delayed respiratory injury; it was the toxic agent in the Bhopal disaster.

Methyl Methanesulfonate

Methyl methanesulfonate is an alkylating agent that methylates DNA and other cellular molecules. It is used experimentally to induce mutagenesis and is hazardous because of its genotoxic and carcinogenic potential.

Methyl Parathion

Methyl parathion is an organophosphorus insecticide that inhibits acetylcholinesterase. Exposure can produce severe cholinergic poisoning with secretions, bronchospasm, bradycardia, weakness, seizures, and respiratory failure.

Methyl Salicylate Overdose

Methyl salicylate overdose involves toxic or environmental exposure to methyl salicylate overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Methylation

Addition of methyl group to DNA (cytosine) or histones. DNA methylation at CpG islands typically silences genes. Hypomethylation associated with cancer.

Methylcellulose

A cellulose derivative used as a bulk-forming laxative, thickener, and excipient.

Methylcholanthrene

Methylcholanthrene is a polycyclic aromatic hydrocarbon used experimentally as a carcinogen. It can form reactive metabolites that damage DNA and is used in research on tumor initiation and chemical carcinogenesis.

Methyldopa

A centrally acting alpha-2 agonist used mainly in selected cases of hypertension, including pregnancy.

Methylene Blue

An oxidizing-reducing agent used primarily to treat clinically significant methemoglobinemia by facilitating conversion of methemoglobin to functional

hemoglobin. It may be ineffective or hazardous in glucose-6-phosphate dehydrogenase deficiency and can interact with serotonergic drugs.

Methylene Blue Test

Methylene blue test involves toxic or environmental exposure to methylene blue test; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Methylene Chloride

Methylene chloride, or dichloromethane, is a volatile industrial solvent. Inhalation can cause central nervous system depression and metabolism to carbon monoxide can produce tissue hypoxia; high exposure may be fatal.

Methylenedioxymethamphetamine

3,4-Methylenedioxymethamphetamine, abbreviated MDMA and commonly called ecstasy or molly, is a psychoactive drug with stimulant and hallucinogenic effects that increases serotonergic signaling and can cause serious toxicity.

Methylergometrine

An ergot alkaloid that increases uterine contraction and may be used to control postpartum hemorrhage when appropriate.

Methylguanidine

Methylguanidine is a guanidine compound formed during protein and amino-acid metabolism. It can accumulate in advanced kidney failure and is studied as a uremic toxin that may contribute to oxidative and cellular injury.

Methylisazone

Another spelling of methisazone, an older antiviral compound investigated for poxvirus infection.

Methylmalonic Acid Blood Test

Methylmalonic acid blood test involves toxic or environmental exposure to methylmalonic acid blood test; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Methylmalonic Acid Level

A blood or urine measurement used to assess functional vitamin B12 status. Elevated methylmalonic acid supports vitamin B12 deficiency, particularly when the serum vitamin B12 concentration is borderline. Renal impairment can also increase the level and should be considered during interpretation.

Methylmalonic Acidemia

Methylmalonic acidemia involves toxic or environmental exposure to methylmalonic acidemia; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Methylmalonic Aciduria

methylmalonic aciduria is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Methylmercury Compounds

Methylmercury compounds are organic mercury compounds that can accumulate in aquatic food chains and cross the blood–brain and placental barriers. Exposure primarily injures the nervous system and can impair fetal neurodevelopment.

Methylmercury Poisoning

Methylmercury poisoning involves toxic or environmental exposure to methylmercury poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Methylmethacrylate

Methyl methacrylate is a volatile monomer used to make acrylic plastics and dental or orthopedic bone cement. Vapors may irritate the eyes and airways, and liquid exposure can cause dermatitis or sensitization.

Methylmorphine

Another name for codeine, an opioid analgesic and antitussive.

Methylnaltrexone

Methylnaltrexone is a peripherally acting mu-opioid-receptor antagonist used to treat opioid-induced constipation when laxatives are insufficient. It generally preserves central analgesia but can cause abdominal pain, diarrhea, and rarely gastrointestinal perforation.

Methylnaltrexone Bromide

Methylnaltrexone bromide is the bromide salt of methylnaltrexone, a peripherally restricted opioid antagonist. It reverses opioid effects in the gastrointestinal tract and is used for opioid-induced constipation under indication-specific precautions.

Methylnitrosourea

Methylnitrosourea is a potent alkylating and mutagenic compound used experimentally to induce DNA damage and tumors. It is not a therapeutic drug and requires strict laboratory handling because of its carcinogenicity.

Methylobacteriaceae

A family of bacteria that can use one-carbon compounds such as methanol and are common in soil, water, and plants.

Methylobacterium

A genus of pink-pigmented, methylotrophic bacteria found in water, soil, and plants that can rarely cause opportunistic infection.

Methylphenidate

A central nervous system stimulant used mainly for ADHD and narcolepsy.

Methylprednisolone

A synthetic glucocorticoid used to suppress inflammation and immune activity.

Methyltestosterone

An androgen medicine formerly used in selected hypogonadal or palliative indications; it can cause hepatic and other adverse effects.

Methylthioinosine

Methylthioinosine is a sulfur-containing purine nucleoside involved in cellular methylthio and purine metabolism. Its derivatives and related metabolites are studied in biochemical pathways and cancer biology rather than used routinely as medicines.

Methyltransferase

A methyltransferase is an enzyme that transfers a methyl group from a donor, commonly S-adenosylmethionine, to DNA, RNA, proteins, lipids, or small molecules. Methylation can alter gene expression, protein activity, or metabolism.

Metoclopramide

A dopamine (D2) receptor antagonist and 5-HT3 antagonist used as a prokinetic antiemetic for gastroparesis, chemotherapy-induced and postoperative nausea, and GERD (facilitates gastric emptying).

Metolazone

A thiazide-like diuretic used for hypertension and fluid retention, often with a loop diuretic.

Metolcarb

Metolcarb is a carbamate insecticide that can inhibit acetylcholinesterase. Exposure may cause cholinergic poisoning with excessive secretions, gastrointestinal symptoms, bronchospasm, bradycardia, weakness, or respiratory compromise.

Metopic Ridge

A metopic ridge is a palpable or visible line along the forehead from fusion of the metopic suture; it may be normal or associated with trigonocephaly.

Metoprolol

Metoprolol is a relatively beta-1-selective blocker used for hypertension, angina, heart failure, rate control, and some post-infarction settings; bradycardia, hypotension, fatigue, and bronchospasm can occur.

Metric System

The metric system is a decimal system of measure-

ment based on units such as the meter, kilogram, and liter. Modern medicine generally uses the International System of Units, with conventional units retained for some measurements.

Metritis

Inflammation of the uterus, often occurring after childbirth, miscarriage, or uterine infection.

Metronidazole

A nitroimidazole antibiotic and antiprotozoal agent that is selectively toxic to anaerobic organisms and certain parasites. The drug is activated by reduction of its nitro group within anaerobic cells, generating toxic free radicals that damage DNA.

Metronidazole-Associated Encephalopathy

A rare complication of prolonged or high-dose metronidazole therapy, causing cerebellar dysfunction (ataxia, dysarthria, nystagmus) and sometimes confusion or seizures. MRI may show T2-hyperintense lesions in the cerebellar dentate nuclei, corpus callosum splenium, and brainstem. Symptoms typically resolve with drug discontinuation.

Metronomic Chemotherapy

Metronomic chemotherapy gives chemotherapy in frequent low doses over an extended period, aiming to inhibit tumor growth and its blood supply with less intensive cycles.

Metrorrhagia

Uterine bleeding occurring at irregular times, including between expected menstrual periods.

METs

Metabolic equivalents, units used to express the intensity or energy cost of physical activity.

Alternate Names:

Metabolic equivalents

Metschnikowia

A genus of yeasts found on plants and in fermented foods; human infection is rare but can occur oppor-

tunistically.

Metyrapone

A medicine that inhibits 11-beta-hydroxylase and is used in selected endocrine tests or treatment.

Metzenbaum Scissors

Fine, lightweight surgical scissors used to dissect and cut delicate soft tissue.

Mevinphos

Mevinphos is a highly toxic organophosphorus insecticide that inhibits acetylcholinesterase. Exposure can rapidly cause cholinergic crisis, including bronchorrhea, bronchospasm, muscle weakness, seizures, and respiratory failure.

Mexacarbate

Mexacarbate is a carbamate insecticide that interferes with acetylcholinesterase. Toxic exposure may produce muscarinic and nicotinic cholinergic symptoms and requires prompt decontamination and medical evaluation.

Mexican Tea

A traditional name for medicinal plants or preparations used in Mexico; the exact species and safety depend on the source.

Mexiletine

Mexiletine is an oral class IB antiarrhythmic and sodium-channel blocker used for selected ventricular arrhythmias and sometimes neuropathic pain. It can cause gastrointestinal and neurologic effects and may worsen or provoke arrhythmias in some patients.

Meyerozyma Guilliermondii

Meyerozyma guilliermondii is a yeast formerly classified as *Candida guilliermondii*. It is usually an environmental or colonizing organism but can cause opportunistic bloodstream or other infections, particularly in immunocompromised patients.

Mezlocillin

A penicillin antibiotic of the acylureido group, formerly used against susceptible gram-negative and other bacteria.

MéNéTrier'S Disease

A rare acquired gastropathy characterized by massive hypertrophy of the gastric mucosal folds, particularly in the body and fundus, with foveolar hyperplasia, cystic dilation of gastric glands, and glandular atrophy. It presents with epigastric pain, nausea, vomiting, weight loss, and protein-losing enteropathy causing hypoalbuminemia.

MéNièRe'S Disease

An inner-ear disorder causing episodes of vertigo, fluctuating hearing loss, tinnitus, and ear pressure.

MéNièRe'S Disease

An inner-ear disorder associated with episodic vertigo, fluctuating hearing loss, tinnitus, and ear pressure.

Alternate Names:

Meniere disease

MF

Alternate index label; see *Mycosis fungoides*.

MFS

Alternate index label; see *Myxofibrosarcoma*.

MG

Alternate index label; see *Myasthenia gravis*.

MGUS

See *Monoclonal Gammopathy of Undetermined Significance*.

Monoclonal Gammopathy Of Undetermined Significance

A condition in which abnormal plasma cells make a small amount of one unusual antibody. It usually causes no symptoms or organ damage but can rarely progress to multiple myeloma or another blood cancer, so monitoring is recommended.

MHC Molecule

A major histocompatibility complex molecule that displays antigens to T cells.

Alternate Names:

Major histocompatibility complex molecule

Mianserin

A tetracyclic antidepressant with noradrenergic and antihistamine effects, used in some countries.

Miasma

A poisonous vapor or foul-smelling mist once believed to arise from decomposing matter and cause disease; the miasma theory is historically important but scientifically incorrect.

MIBG Scan

Nuclear medicine imaging using I-123 or In-111 MIBG. Detects pheochromocytoma, neuroblastoma.

MIBG Scintiscan

A nuclear medicine scan using radiolabeled metaiodobenzylguanidine to image tissues derived from the sympathetic nervous system, including selected pheochromocytomas, paragangliomas, and neuroblastomas.

Mica

A group of sheet silicate minerals; inhalation of mica dust can cause occupational lung disease.

Micafungin

An echinocandin antifungal agent used in the treatment and prophylaxis of invasive candidiasis and esophageal candidiasis. Micafungin inhibits beta-(1,3)-D-glucan synthase, disrupting fungal cell wall synthesis and leading to osmotic instability and cell death.

Mice

Plural of mouse; laboratory mice are widely used in biomedical research and can also transmit zoonotic infections.

Micelle

micelle refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Michaelis-Menten Kinetics

The mathematical model describing the rate of enzyme-catalyzed reactions as a function of substrate concentration.

Miconazole

An imidazole antifungal used topically or orally in selected fungal infections.

Micro-Surgery

Surgery performed with magnification and fine instruments to repair very small structures such as nerves and vessels.

Microabscess

A microabscess is a small, localized collection of pus consisting of neutrophils, necrotic material, and microorganisms or inflammatory debris. It may occur in skin, glands, organs, or tissue affected by infection or inflammatory disease.

Microalbuminuria

Excretion of 30 to 300 mg of albumin daily undetectable by standard dipstick. Earliest marker of diabetic nephropathy and cardiovascular risk predictor. Detected by urine albumin-to-creatinine ratio on spot specimen. Normal less than 30 mg/g creatinine. Signals early glomerular damage in diabetes warranting RAAS blockade initiation.

Microalbuminuria Test

A urine test measuring small amounts of albumin, usually by an albumin-to-creatinine ratio, to detect early kidney damage, especially in diabetes or hypertension.

Microaneurysm

A minute, localized, focal saccular outpouching of a capillary wall, typically measuring 10 to 100 mi-

chrometers in diameter. In ophthalmology, retinal capillary microaneurysms represent the earliest clinically detectable sign of non-proliferative diabetic retinopathy. Caused by selective loss of vascular pericytes and basement membrane thickening under chronic hyperglycemic stress, leading to weakened capillary walls that balloon under intravascular hydrostatic pressure; appear on fundoscopy as tiny, well-demarcated red dots and on fluorescein angiography as hyperfluorescent pinpoint foci that may leak fluid, causing macular edema.

Microangiopathic Hemolysis

Red-cell fragmentation and intravascular hemolysis caused by passage through abnormal small vessels, producing schistocytes, elevated LDH, and reduced haptoglobin.

Microangiopathic Hemolytic Anemia

An anemia caused by mechanical fragmentation of red blood cells as they pass through damaged or narrowed small blood vessels, producing schistocytes and often elevated hemolysis markers.

Microangiopathy

Disease of small blood vessels, especially capillaries, in which vessel walls become thickened or damaged, impairing flow and allowing leakage or bleeding; diabetes is a common cause.

Microarray

Technology measuring expression of thousands of genes simultaneously or detecting copy number variations. Gene expression microarrays: compare expression patterns. SNP arrays: detect genetic variations. Comparative genomic hybridization (CGH): detect deletions/duplications.

Microarray Analysis

Microarray analysis measures the abundance or hybridization of many nucleic-acid sequences simultaneously using an array of immobilized probes. It can assess gene expression, copy-number changes, or selected genetic variants.

Microbacterium

A genus of gram-positive bacteria found in soil and water that can rarely cause bloodstream or other opportunistic infections.

Microbe

A microorganism, such as a bacterium, fungus, protozoan, or virus.

Alternate Names:

Microorganism

Microbial Biofilms

Microbial biofilms are organized communities of microorganisms attached to a surface and enclosed in a self-produced extracellular matrix. Biofilms can form on tissues and medical devices and often show increased tolerance to antimicrobials and host defenses.

Microbicide

A microbicide is a chemical or biological agent that kills microorganisms or inactivates them. The term may refer to disinfectants, topical agents, or investigational products intended to reduce sexual transmission of infection.

Microbiology

The study of microorganisms, including bacteria, viruses, fungi, and parasites.

Microbiome

The collection of microorganisms (bacteria, archaea, fungi, viruses—the microbiota) and their genetic material inhabiting a host's body surfaces, particularly the gastrointestinal tract.

Microbody

A microbody is a small membrane-bound cytoplasmic organelle, a term that includes peroxisomes and related organelles. Microbodies contain enzymes involved in oxidation, detoxification, and lipid metabolism.

Microcalcification

A very small calcium deposit in tissue, sometimes

an early sign of breast cancer.

Microcephaly

Microcephaly is head size substantially smaller than expected for age and sex, reflecting genetic, developmental, infectious, toxic, or acquired brain causes.

Microchip Site

A microchip site is the anatomical location where an implanted identification or monitoring microchip is placed, commonly in subcutaneous tissue. The site should be recorded for identification and assessed for migration, irritation, or infection when relevant.

Microcirculation

Blood flow through arterioles, capillaries, and venules, where tissue exchange occurs.

Micrococcaceae

Micrococcaceae is a family of generally gram-positive, catalase-positive cocci that includes *Micrococcus* and related genera. These organisms commonly inhabit skin and the environment, although some can cause opportunistic infection.

Micrococcus

A genus of generally harmless gram-positive cocci that can occasionally cause opportunistic infection.

Micrococcus Luteus

Micrococcus luteus is a gram-positive, aerobic coccus commonly present on human skin and in the environment. It is usually a contaminant or colonizer but can rarely cause bacteremia or device-associated infection in susceptible patients.

Micrococcus Lylae

Micrococcus lylae is a gram-positive, aerobic coccus associated mainly with skin flora and environmental samples. Its recovery from a clinical specimen requires correlation with symptoms and the possibility of contamination or colonization.

Microcolon

Microcolon is an abnormally small-caliber colon,

often identified on contrast imaging. It may result from reduced fetal intestinal use or obstruction and is associated with conditions such as small-left-colon syndrome, meconium ileus, or distal intestinal obstruction.

Microcornea

Microcornea is an abnormally small cornea, usually defined in adults by a horizontal diameter below about 10 mm. It may be isolated or associated with microphthalmia, glaucoma, cataract, or genetic syndromes.

Microcurie

A microcurie (μCi) is a unit of radioactivity equal to one-millionth of a curie, or approximately 37 kilobecquerels. It may be used to express the activity of radioactive tracers or sources.

Microcystic Adnexal Carcinoma

A rare, slow-growing but locally aggressive malignant adnexal tumor, usually on the head or neck, with follicular and ductal differentiation; it may extend deeply and requires complete excision.

Microcytic Anemia

Anemia in which red blood cells are smaller than normal, commonly because of iron deficiency or a thalassemia.

Microdermabrasion

A cosmetic procedure that mechanically removes a superficial layer of skin.

Microdialysis

Microdialysis is a sampling technique in which a semipermeable probe perfused with fluid collects small molecules from extracellular tissue fluid. It can monitor local concentrations of metabolites or drugs, including in the brain or subcutaneous tissue.

Microdissection

Microdissection is separation or isolation of very small tissue regions or structures under microscopic or highly magnified visualization. It is used in

surgery, pathology, and molecular research to obtain selected cells or tissue.

Microdissection Testicular Sperm Extraction

A specialized advanced microsurgical sperm retrieval procedure performed under an operating microscope (15× to 25× magnification) to locate and extract focal areas of spermatogenesis in men with non-obstructive azoospermia. Through a wide testicular incision, the surgeon microsurgically surveys the testicular parenchyma to selectively identify and excise enlarged, opaque, fuller seminiferous tubules that are most likely to harbor active spermatogenesis while sparing thin, collapsed tubules and minimizing parenchymal devascularization. Retrieved spermatozoa are subsequently utilized for intracytoplasmic sperm injection.

Alternate Names:

Micro-TESE; Microsurgical TESE

Microdontia

Abnormally small teeth, which may affect one or several teeth and can be inherited or developmental.

Microelectrode

A microelectrode is a very small electrode used to record electrical activity from, or deliver stimulation to, individual cells, nerves, or small tissue regions. It is common in electrophysiology and neuroscience research.

Microfauna

Small or microscopic animals, including protozoa and tiny invertebrates, found in an environment or specimen.

Microfibril

A microfibril is a fine, threadlike structural unit composed of proteins such as fibrillin or cellulose, depending on the tissue. In connective tissue, microfibrils help organize elastic fibers and provide extracellular support.

Microfilament

A microfilament is a thin cytoskeletal filament com-

posed mainly of actin, about 7 nm in diameter. Microfilaments support cell shape, motility, contraction, cytokinesis, and intracellular transport.

Microfilament Proteins

microfilament proteins is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Microfilaria

The larval form of filarial worms, circulating in blood or skin and transmitted by insect vectors.

Microflora

An older term for the microorganisms inhabiting a body site or environment; microbiota is preferred.

Microfluidics

Microfluidics is the manipulation of very small volumes of fluid in channels or systems with dimensions typically in the micrometer range. It enables compact assays, cell sorting, diagnostics, and laboratory-on-a-chip devices.

Microglandular Adenosis

Microglandular adenosis is an uncommon benign proliferation of small glands in the breast, often associated with hormonal influences such as pregnancy or progestin use. It can mimic invasive carcinoma and is diagnosed histologically.

Microglia

Microglia are resident immune cells of the central nervous system. They survey neural tissue, remove debris, and respond to injury or infection; persistent activation can contribute to neuroinflammation.

Microgliosis

Microgliosis is proliferation or increased activation of microglia in the central nervous system, usually in response to injury, infection, degeneration, or other inflammation. It is a histopathologic finding rather than a specific disease.

Micrognathia

Micrognathia is an unusually small lower jaw that may affect feeding, teeth, facial development, or airway breathing.

Micrognathism

Micrognathism is abnormal smallness of the jaw, usually referring to the mandible. It may be congenital or acquired and can affect feeding, dentition, airway patency, speech, or facial development.

Micrograms

A unit of mass equal to one millionth of a gram, written as μg .

Micrographia

Abnormally small handwriting, commonly associated with Parkinson disease and other neurologic disorders.

Microinjection

Microinjection is delivery of a very small volume of fluid into a cell, tissue, or other microscopic target using a fine needle or micropipette. It is used in assisted reproduction, experimental biology, and some therapeutic procedures.

Micrometastasis

A small deposit of cancer cells that has spread from the primary tumor but is not detectable clinically or on standard imaging, e.g., isolated tumor cells in a sentinel lymph node; relevant to staging and prognosis.

Micrometer

An instrument or unit used to measure very small distances; one micrometer is one-millionth of a meter.

Micromolar

A concentration unit equal to one millionth of a mole per liter, written as $\mu\text{mol/L}$.

Micromole

A micromole (μmol) is an amount of substance

equal to one-millionth of a mole, or approximately 6.022×10^{17} entities. It is commonly used for small quantities and concentrations of biochemical substances.

Micron

A unit of length equal to one micrometer, or one-millionth of a meter.

Microneme

A microneme is a small secretory organelle in many apicomplexan parasites. It releases proteins that support host-cell attachment, motility, and invasion during the parasite life cycle.

Micronization

Micronization is a process that reduces particles to micrometer-scale dimensions, often to increase surface area and improve dissolution or absorption of a drug. The resulting particle size and formulation still determine clinical performance.

Micronized

Processed into very small particles to improve dispersion or absorption.

Micronutrient

A micronutrient is a vitamin or mineral required in small amounts for normal growth, metabolism, and physiologic function. Deficiency or excess can cause disease, and requirements vary with age, pregnancy, illness, and other factors.

Micronutrients

Vitamins and minerals required in small amounts for enzymatic, hormonal, structural, hematologic, neurologic, and immune functions. They include fat-soluble vitamins (A, D, E, and K), water-soluble vitamins, and essential minerals.

Microorchidism

Abnormally small testes, which may result from impaired development, genetic syndromes, or testicular atrophy after injury or infection.

Microorganism

A microscopic living organism, such as a bacterium, fungus, protozoan, or alga; viruses are acellular infectious agents.

Microorganism Genetics

The study of genetic material, inheritance, variation, and gene function in microorganisms.

Micropenis

An unusually small penis caused by inadequate androgen effect or other developmental conditions, defined clinically by a stretched penile length more than 2.5 standard deviations below the mean for age and population. Evaluation may identify endocrine or genetic causes.

Micropexophagy

Micropexophagy is a form of selective autophagy in which a cell degrades peroxisomes by enclosing them in an autophagic structure and delivering them to the vacuole or lysosome. It is studied mainly in fungal and cellular biology.

Microphallus

An abnormally small penis, also called micropenis, usually defined relative to age and developmental standards; it may reflect hormonal or developmental disorders.

Microphone

A device that converts sound into an electrical signal, used in hearing assessment, recording, or communication equipment.

Microphthalmia

A birth defect in which one or both eyes are smaller than usual. It may occur with other eye or genetic conditions and can reduce vision. Treatment depends on the cause and may include glasses, treatment for poor vision, or surgery.

Microphthalmos

A developmental ocular malformation characterized by an abnormally small eyeball with an axial

length at least two standard deviations below the age-adjusted mean. May be unilateral or bilateral and can occur as an isolated ocular defect or as a component of genetic syndromes (such as trisomy 13, CHARGE syndrome, or microphthalmia with linear skin defects) or intrauterine congenital infections (TORCH complex: cytomegalovirus, toxoplasmosis, rubella). Can be associated with coloboma, persistent fetal vasculature, cataracts, and orbital hypoplasia, which often requires sequential orbital expanders and conformers to promote symmetric craniofacial development and maximize visual potential.

Micropsia

A visual disturbance in which objects appear smaller than they are.

Microradiography

Microradiography is radiographic imaging of very small specimens or structures, often using high magnification and fine-detail techniques. It can demonstrate microarchitecture in bone, tissue, teeth, or experimental samples.

MicroRNA

A very small, non-coding RNA that helps control which genes are active, mainly by reducing the production of specific proteins. MicroRNAs are being studied as possible disease markers and treatment targets.

Microsatellite Instability

A molecular pattern caused by deficient DNA mismatch-repair function, resulting in length changes in short repetitive DNA sequences. MSI-high tumors may have increased mutation burden and can be eligible for particular immunotherapies.

Microscope

Magnifies small structures. Types: light, electron, fluorescent. Essential for histology, hematology, microbiology.

Microscopic

Visible only with a microscope or relating to micro-

scopic examination.

Microscopic Anatomy

The study of cells, tissues, and fine structural details that require magnification for examination; it includes cytology and histology.

Microscopic Colitis

Umbrella term encompassing both collagenous colitis and lymphocytic colitis which share the clinical presentation of chronic watery non-bloody diarrhea and the pathophysiological feature of colonic mucosal inflammation. Endoscopic appearance is typically normal necessitating random biopsies for diagnosis.

Alternate Names:

Colitis, Microscopic

Microscopic Examination

The pathologic evaluation of tissue or cells under the light or electron microscope, complementing the gross examination to form the complete morphologic diagnosis. Routine light microscopy uses hematoxylin and eosin sections to assess architecture, cellular morphology, mitotic activity, and relation to margins.

Microscopic Hematuria

Alternate index label; see *Blood in urine (hematuria)*.

Microscopic Polyangiitis

A necrotizing vasculitis affecting small blood vessels (capillaries, venules, arterioles) with few or no immune deposits (pauci-immune). Part of the ANCA-associated vasculitis group. Strongly associated with MPO-ANCA (p-ANCA, 80%).

Microscopy

Examination of cells, tissues, organisms, or materials using a microscope.

Microsecond

A unit of time equal to one-millionth of a second.

Microsome

A microsome is a small membrane vesicle formed from fragments of endoplasmic reticulum when cells are disrupted and centrifuged. Microsomes are laboratory preparations used to study membrane enzymes, including drug-metabolizing cytochrome P450 systems.

Microspectrophotometry

Microspectrophotometry measures the absorption or emission spectrum of a microscopic region or individual cell. It can quantify pigments, nucleic acids, proteins, or other chromophores in small biological samples.

Microsphere

A microsphere is a small spherical particle, often made of polymer, glass, or another material, used in drug delivery, imaging, embolization, or laboratory assays. Its size, composition, and surface properties determine its biologic behavior.

Microspherophakia

Microspherophakia is an abnormally small, spherical crystalline lens. It can occur in inherited disorders such as Weill–Marchesani syndrome and may cause lens displacement, pupillary block, angle-closure glaucoma, or reduced vision.

Microsporidia

A group of microscopic, spore-forming parasites now classified among fungi; several species can infect humans.

Microsporidiosis

An infection caused by microsporidia, often affecting the intestine, eyes, muscles, or other organs in people with weakened immunity.

Microsporum

Microsporum is a genus of dermatophyte fungi that can cause ringworm of the skin, scalp, or hair.

Microsporum Audouinii

Microsporum audouinii is a dermatophyte that

mainly infects the scalp and can cause tinea capitis, especially in children.

Microsporium Distortum

Microsporium distortum is a rare dermatophyte related to *M. canis* that can cause skin or scalp infection.

Microsporium Equinum

Microsporium equinum is an animal-associated dermatophyte that rarely causes infection in humans.

Microsporium Ferrugineum

Microsporium ferrugineum is a dermatophyte that can cause scalp ringworm, particularly in children.

Microsporium Gypseum

Microsporium gypseum is a soil-associated dermatophyte that can cause ringworm of the skin or scalp.

Microstomia

Microstomia is abnormal narrowing or smallness of the mouth opening. It may be congenital or result from scarring, burns, surgery, trauma, or connective-tissue disease and can impair eating, speech, oral hygiene, and dental treatment.

Microsurgery

Microsurgery uses magnification and very small instruments to repair tiny blood vessels, nerves, or other structures.

Microtechnology

Microtechnology is the design and manufacture of devices or structures with micrometer-scale features. Medical applications include microfluidic diagnostics, sensors, drug-delivery systems, and minimally invasive instruments.

Microtia

A birth defect in which the outer ear is smaller than usual or does not form fully. The ear canal may also be narrow or absent, which can reduce hearing.

Microtome

An instrument that cuts extremely thin sections of tissue for microscopic examination.

Microtubule

A microtubule is a hollow cytoskeletal polymer made of tubulin dimers. Microtubules maintain cell shape, organize intracellular transport, form the mitotic spindle, and support cilia and flagella.

Microtubule Bundle

A microtubule bundle is an organized group of parallel or otherwise coordinated microtubules within a cell. Bundles provide structural support and guide transport or movement, including in axons, cilia, and dividing cells.

Microvascular

Relating to the microvasculature, the network of small blood vessels—generally those with an internal diameter of about 100 micrometers or less—including arterioles, capillaries, and venules.

Microvascular Disease

Disease affecting small blood vessels, which may reduce blood flow to organs or tissues.

Microvasculature

The network of the body's smallest blood vessels, including arterioles, capillaries, and venules.

Microvillus

A microvillus is a microscopic, actin-supported projection of a cell membrane that increases surface area. Microvilli are abundant on intestinal epithelial cells, where they support nutrient absorption, and on renal tubular cells.

Microvolt

A microvolt (μV) is one-millionth of a volt. It is used to express small bioelectric signals, such as those recorded by electroencephalography, electromyography, electrocardiography, or evoked-potential testing.

Microwave Ablation

A procedure that uses microwave energy to heat and destroy selected tissue, usually under imaging guidance.

Microwave Therapy

Microwave therapy uses electromagnetic energy to heat tissue and is used in selected medical or surgical treatments.

Microwell Plate

A microwell plate is a laboratory plate containing an array of small wells used to hold separate samples or reactions. It supports high-throughput assays, cell culture, immunoassays, and biochemical testing.

Micturition

The process of urination, coordinated by the pontine micturition center, parasympathetic (pelvic nerve, S2-S4) contraction of the detrusor muscle, and relaxation of the internal (involuntary) and external (voluntary, pudendal nerve) urethral sphincters.

Mid-Brain

The midbrain, or mesencephalon, is the upper part of the brainstem involved in eye movements, motor pathways, and auditory and visual reflexes.

Mid-Day

The middle part of the day; in clinical instructions it may specify a medication or measurement time.

Mid-Dorsal

Situated in the middle region of the back or on the dorsal side of a structure.

Mid-Forearm

The middle part of the forearm, between the elbow and wrist. Measurements or injections described as mid-forearm use this region as a reference point.

Mid-Gut

The embryonic middle portion of the primitive gut, which develops into much of the small intestine and proximal colon.

Midgut Volvulus

A life-threatening pediatric surgical emergency caused by clockwise twisting of the small intestine and its mesenteric vascular pedicle around the superior mesenteric artery axis, occurring on the background of congenital intestinal malrotation with a narrow mesenteric base. Leads rapidly to midgut venous occlusion, arterial compromise, bowel ischemia, transmural infarction, and catastrophic short bowel syndrome or death. Classically presents in a previously thriving neonate or young infant with sudden bilious vomiting, abdominal tenderness, and hematochezia; diagnosis is confirmed urgently by upper GI contrast series showing a "corkscrew" duodenal sign, followed immediately by the Ladd procedure.

Alternate Names:

Volvulus Neonatorum; Midgut Torsion

Mid-Leg

The central region of the lower limb between the knee and ankle.

Mid-Line

The imaginary or anatomical line dividing a structure or body into right and left portions.

Mid-Sized

Of intermediate size; it is a descriptive term rather than a specific medical diagnosis.

Mid-Stream Urine Specimen

A urine sample collected after initial urine is voided, reducing contamination from the urethral opening and used commonly for culture.

Mid-Systolic Click

A sharp, high-pitched sound occurring in mid-to-late systole, associated with mitral valve prolapse. The click represents sudden tensing of the elongated chordae tendineae as the prolapsing leaflet billows into the left atrium. It may be followed by a late systolic murmur of mitral regurgitation.

Mid-Upper-Arm Circumference

The circumference of the upper arm measured at

the midpoint between the acromion and olecranon processes. It is a practical anthropometric measure of muscle and fat stores, particularly useful when body weight or height cannot be measured reliably.

Midaxillary Line

An imaginary vertical line on the side of the body passing through the apex of the axilla (armpit).

Midazolam

Midazolam is a short-acting benzodiazepine used for procedural sedation, anesthesia, and treatment of acute seizures in appropriate settings. It enhances GABA-A receptor activity and can cause profound sedation and respiratory depression, especially with opioids or other depressants.

Midazolam Hydrochloride

Midazolam hydrochloride is the hydrochloride salt of the short-acting benzodiazepine midazolam. It is used for procedural sedation, anesthesia, and selected seizure emergencies and can cause dose-related sedation, hypotension, and respiratory depression.

Midbrain

Most superior brainstem segment connecting pons to diencephalon. Contains cerebral peduncles (corticospinal/corticobulbar tracts), substantia nigra (dopaminergic neurons degenerated in Parkinson's), red nucleus (rubrospinal tract for coordination), and periaqueductal gray (pain modulation).

Midclavicular Line

An imaginary vertical line on the front of the body passing through the midpoint of the clavicle (collarbone).

Middle Ear

An air-filled space between the eardrum and inner ear that contains the malleus, incus, and stapes. These ossicles transmit sound vibrations to the inner ear, and the auditory tube helps equalize middle-ear pressure.

Middle East Respiratory Syndrome

A severe respiratory infection caused by Middle East respiratory syndrome coronavirus, usually acquired through close contact with infected camels or people. It can cause pneumonia, respiratory failure, and renal or multiorgan complications.

Alternate Names:

MERS (Middle East Respiratory Syndrome)

Middle East Respiratory Syndrome (MERS (Middle East Respiratory Syndrome))

Alternate index label for MERS (Middle East Respiratory Syndrome).

Middle East Respiratory Syndrome

A serious respiratory illness caused by MERS coronavirus, usually associated with fever, cough, breathing difficulty, and exposure to infected people or camels.

Middle Insomnia

Middle insomnia is repeated or prolonged awakening during the middle portion of the sleep period, often with difficulty returning to sleep. It may occur with stress, mood disorders, sleep apnea, pain, substances, medications, or other sleep disorders.

Middle Lobe Syndrome

Chronic or recurrent atelectasis and infection of the right middle lobe due to bronchial obstruction. The long, narrow right middle lobe bronchus is susceptible to compression by lymph nodes, tumors, or mucous plugging. It may be caused by bronchiectasis, malignancy, or granulomatous disease. Treatment depends on the underlying cause.

Midline Venous Catheters - Infants

A medical procedure, device, or treatment involving midline venous catheters - infants. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Midnight Disease

An older term for hypergraphia, an unusual or ex-

cessive drive to write, sometimes associated with temporal-lobe epilepsy or other neurologic or psychiatric conditions.

Midodrine

Midodrine is an oral prodrug whose active metabolite stimulates alpha-1 adrenergic receptors and increases vascular tone. It is used for symptomatic orthostatic hypotension and may cause supine hypertension, piloerection, urinary retention, or scalp tingling.

Midomafetamine

Midomafetamine is another name for 3,4-methylenedioxymethamphetamine, a psychoactive amphetamine derivative. It is being studied in controlled psychotherapy-assisted research but can cause hyperthermia, hyponatremia, cardiovascular toxicity, serotonin toxicity, and misuse.

Midostaurin

Midostaurin is a multikinase inhibitor used with chemotherapy for newly diagnosed acute myeloid leukemia with a susceptible FLT3 mutation and for advanced systemic mastocytosis. It can cause nausea, cytopenias, infection, and other treatment-related toxicities.

Midriff

The region of the trunk around the diaphragm and upper abdomen.

Midtrimester

The middle third of pregnancy, from 14 to 28 weeks gestation. The midtrimester is when many women feel their best: nausea and fatigue of the first trimester resolve, and the physical discomforts of the third trimester have not yet developed.

Midwife

A trained and regulated professional who provides care during pregnancy, labor, birth, and the postnatal period.

Midwifery

The professional practice of supporting and managing normal pregnancy, childbirth, postpartum recovery, and newborn care.

Mifepristone

Mifepristone is a progesterone-receptor antagonist and, at higher doses, a glucocorticoid-receptor antagonist. It is used with misoprostol for medication abortion under applicable protocols and for selected cases of hyperglycemia associated with Cushing syndrome.

Migraine

A recurrent neurologic disorder that causes attacks of moderate or severe headache and may also cause nausea, vomiting, sensitivity to light or sound, or reversible aura symptoms. Some attacks occur without headache.

Alternate Names:

Migraine Headache

Migraine And Stroke Symptoms

Migraine aura usually develops gradually and resolves, whereas stroke symptoms often begin suddenly and cause persistent focal deficits such as weakness, speech trouble, facial droop, or vision loss. New or atypical neurologic symptoms require emergency stroke assessment.

Migraine With Aura

A primary headache disorder characterized by recurrent attacks of fully reversible visual, sensory, or speech symptoms that develop gradually over at least five minutes and last no more than sixty minutes, followed by headache with the characteristics of migraine.

Migraine Without Aura

A primary headache disorder characterized by recurrent attacks of moderate to severe, pulsating, unilateral headache lasting four to seventy-two hours, associated with nausea, vomiting, photophobia, and phonophobia.

Migrainous Neuralgia

An older term for cluster headache, a disorder causing recurrent attacks of severe unilateral pain, usually around one eye, with ipsilateral autonomic symptoms such as tearing or nasal congestion.

Migrating Motor Complex

Cyclical patterns of intense gastrointestinal motility occurring during fasting (interdigestive period), sweeping residual food, bacteria, and debris from the stomach through the small intestine to the colon.

Migration

Movement of cells, organisms, substances, or a body structure from one location to another.

Migratory Thrombophlebitis

Recurrent or shifting venous thrombosis associated with an underlying malignancy; an alternate term for the thrombotic manifestation of Trousseau syndrome.

Mikulicz Disease

An older term for persistent, usually symmetric enlargement and inflammation of the lacrimal and salivary glands, now commonly associated with IgG4-related disease.

Mikulicz's Disease

An older term for persistent, usually symmetrical enlargement of the lacrimal and salivary glands, now often considered within IgG4-related disease or Sjögren syndrome.

Milan System For Reporting Salivary Gland Cytopathology

A standardized six-category framework for salivary-gland fine-needle aspiration, reporting nondiagnostic, non-neoplastic, benign neoplasm, uncertain, suspicious, and malignant findings with associated risk estimates.

Milatuzumab

An investigational monoclonal antibody directed at CD74 and studied for blood cancers and other

immune-related conditions.

Mild

Mild describes relatively low intensity, severity, or extent of a symptom, sign, disease, or abnormality. The meaning depends on the condition and the clinical scale used and does not by itself indicate that no treatment or evaluation is needed.

Mild Cognitive Impairment

A transitional cognitive state between normal aging and dementia, characterized by subjective cognitive complaints confirmed by objective cognitive impairment that is not severe enough to significantly interfere with daily activities.

Mildew Remover Poisoning

Mildew remover poisoning involves toxic or environmental exposure to mildew remover poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Mile

A mile is a unit of distance equal to 1,609.344 meters. It may be used in descriptions of walking or running distance, although scientific and clinical measurements generally use SI units.

Milia

Milia are tiny, firm, white or yellow bumps caused by small cysts containing keratin, a protein in the skin. They are common in newborns and may appear on the face in adults; they are harmless and usually go away on their own.

Miliaria

A sweat-retention disorder caused by blockage of eccrine ducts, producing small itchy or prickly papules.

Miliaria Crystallina

A benign, asymptomatic eruption of small, clear, superficial vesicles caused by obstruction of eccrine sweat ducts in the stratum corneum, occurring in hot, humid environments or during high fevers. The vesicles resemble dewdrops on the skin and rupture

easily, leaving fine scales.

Miliaria Rubra

An itchy prickly heat rash caused by sweat-duct blockage in the epidermis, producing small red papules or vesicles.

Miliary

Resembling millet seeds; in medicine, describing numerous tiny lesions, especially the pattern of disseminated tuberculosis.

Miliary TB

Hematogenous dissemination of *M. tuberculosis* causing innumerable 1–3 mm granulomas throughout lungs and other organs. CXR shows “millet seed” pattern. More common in immunocompromised patients.

Miliary Tuberculosis

Disseminated tuberculosis caused by hematogenous spread of *Mycobacterium tuberculosis*, producing numerous tiny granulomatous lesions in the lungs and potentially multiple organs. Symptoms may include prolonged fever, weight loss, night sweats, cough, and organ-specific manifestations.

Milieu Therapy

Milieu therapy uses the structured social environment of a treatment setting as part of care, especially in some psychiatric and rehabilitation programs.

Military Personnel

Military personnel are members of the armed forces, including active-duty, reserve, and related service members. Their health care may address occupational exposures, injuries, infectious risks, combat-related trauma, and deployment-related mental-health conditions.

Milium

A small, firm, white or yellow keratin-filled cyst, usually on the face; plural milia.

Milk

A nutrient-rich secretion of mammary glands used to feed an infant.

Milk Allergy

An immune reaction to proteins in cow’s milk that may cause skin, gastrointestinal, or respiratory symptoms and, rarely, anaphylaxis.

Milk Fever

Hypocalcemia around calving in dairy cattle; in human medicine, the term is sometimes used loosely for postpartum hypocalcemia but is not a standard diagnosis.

Milk Intolerance

Alternate index label; see *Lactose intolerance*.

Milk Intolerance (Lactose Intolerance)

Alternate index label for Lactose Intolerance.

Milk Of Magnesia

A suspension of magnesium hydroxide used as an antacid and osmotic laxative.

Milk-Alkali Syndrome

Hypercalcemia, metabolic alkalosis, and kidney dysfunction caused by excessive calcium and absorbable alkali intake.

Milk-Leg

An older term for postpartum swelling of a leg, historically associated with deep venous thrombosis or lymphatic obstruction.

Milk-Sickness

A historical poisoning caused by drinking milk from cattle that have eaten white snakeroot containing tremetol.

Milker’S Node

A localized nodular skin infection caused by parapoxvirus acquired from infected cattle, also called orf-like milker’s nodule.

Milky Urine

Urine with a cloudy or white appearance caused by chyle, pus, crystals, fat, or other substances; evaluation identifies the cause.

Miller'S Lung

Hypersensitivity pneumonitis caused by inhaling material associated with the granary weevil (wheat weevil, *Sitophilus granarius*), especially in people working with contaminated grain or flour.

Millicurie

A millicurie (mCi) is one-thousandth of a curie, equal to approximately 37 megabecquerels. It is used to express the activity of radioactive sources and some radiopharmaceutical doses.

Milliequivalent

A unit expressing the chemical combining power of an ion, equal to one-thousandth of an equivalent.

Milligram

A unit of mass equal to one-thousandth of a gram.

Milligram-Percent

An older concentration unit meaning milligrams per 100 milliliters, numerically equivalent to mg/dL.

Milligray

A unit of absorbed ionizing-radiation dose equal to one-thousandth of a gray, or 0.001 joule per kilogram; its abbreviation is mGy.

Milliliter

A unit of volume equal to one-thousandth of a liter and one cubic centimeter.

Millimeter

A unit of length equal to one-thousandth of a meter.

Millimolar

A concentration unit equal to one thousandth of a mole per liter, written as mmol/L.

Millimole

A unit of amount of substance equal to one-thousandth of a mole.

Million

A number equal to 1,000,000; laboratory counts may use it as a numerical multiplier.

Milliosmole

A milliosmole (mOsm) is one-thousandth of an osmole, a unit reflecting the number of osmotically active particles in a solution. Osmolality and osmolarity are commonly reported in mOsm/kg or mOsm/L.

Millipede Toxin

Millipede toxin involves toxic or environmental exposure to millipede toxin; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Millisecond

A millisecond (ms) is one-thousandth of a second. It is used to measure brief physiologic events, reaction times, nerve conduction intervals, and timing of electrical or imaging signals.

Milliunit

A milliunit (mU) is one-thousandth of a unit of biologic or enzymatic activity. Because a unit is defined according to the substance or assay, the clinical meaning of a milliunit depends on the specific test or medication.

Millivolt

A unit of electrical potential equal to one-thousandth of a volt.

Milrinone

A phosphodiesterase-3 inhibitor with positive inotropic and vasodilatory (inodilator) actions, increasing cAMP in cardiac and vascular smooth muscle.

Milroy Disease

An inherited primary lymphedema caused most often by pathogenic variants affecting VEGFR3 signaling. It typically presents with chronic swelling of the lower limbs from birth or early infancy and may be associated with papillomatosis, hydrocele, or recurrent skin infection.

Milroy'S Disease

An inherited form of primary lymphedema, usually present from birth or early life and caused commonly by lymphatic-development abnormalities.

Mima

A rare or historical term whose clinical meaning is not consistent across references; the source-specific definition should be verified.

Mimesis

Imitation or mimicry; in medicine, the term can refer to simulation of an organic disease by psychological symptoms or to one organic disease resembling another.

Mimicry

Resemblance of one organism, structure, or clinical pattern to another; in medicine, it may describe a disorder that imitates another disease.

Minamata Disease

A neurologic disorder caused by methylmercury poisoning, classically from contaminated seafood or industrial pollution.

Mind

The collection of mental processes and capacities involved in consciousness, thought, perception, memory, emotion, judgment, and behavior.

Mindful Eating

Paying attention to hunger cues, eating slowly, savoring food. May reduce overeating, improve digestion.

Miner'S Anaemia

Anemia associated historically with mining, com-

monly due to hookworm infection, chronic blood loss, nutritional deficiency, or occupational exposure.

Miner'S Asthma

Asthma or asthma-like airway disease associated with inhalation of dusts encountered in mining.

Miner'S Nystagmus

An older term for visual disturbance and nystagmus reported in underground miners, associated with poor lighting and occupational conditions.

Mineral

An inorganic element or compound required for body structure, fluid balance, enzyme activity, or other physiological functions.

Mineral Oil

Mineral oil is a purified mixture of liquid hydrocarbons derived from petroleum. It is used in some topical products, cosmetics, and laxatives; aspiration of liquid mineral oil can cause lipid pneumonia.

Mineral Oil Overdose

Mineral oil overdose involves toxic or environmental exposure to mineral oil overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Mineral Spirits Poisoning

Mineral spirits poisoning involves toxic or environmental exposure to mineral spirits poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Mineralocorticoid

A mineralocorticoid is a steroid hormone that regulates sodium, potassium, and extracellular-fluid balance, chiefly through receptors in the renal distal nephron. Aldosterone is the principal endogenous mineralocorticoid; excess causes hypertension and hypokalemia.

Mineralocorticoid Receptor

An intracellular steroid receptor activated primarily by aldosterone in the kidney, colon, sweat glands, and other tissues. Its activation increases sodium reabsorption and potassium and hydrogen secretion and contributes to blood-pressure regulation and tissue remodeling.

Mineralocorticoids

Steroid hormones, chiefly aldosterone, that promote sodium retention and potassium and hydrogen secretion in the kidney.

Mini Stroke (Transient Ischemic Attack (TIA, Mini-Stroke))

Alternate index label for Transient Ischemic Attack (TIA, Mini-Stroke).

Mini-Mental State Examination

A brief structured cognitive screening instrument assessing orientation, registration, attention, calculation, recall, language, and visuospatial ability. Scores are affected by education, language, sensory impairment, culture, and delirium and do not establish a dementia diagnosis alone.

Mini-Stroke

A lay term usually referring to a transient ischemic attack, in which focal neurologic symptoms resolve without persistent infarction.

Minim

A very small amount or unit; the term is historical and may refer to a small volume in old pharmaceutical measurement.

Minimal Brain Dysfunction

An obsolete term formerly used for children with attention, learning, or behavior problems; it has no precise modern diagnostic meaning.

Minimal Change Disease

A cause of nephrotic syndrome, especially in children, in which the glomeruli appear nearly normal by light microscopy but podocyte foot processes are

effaced on electron microscopy. It causes marked protein loss in the urine and edema.

Minimal Residual Disease

The small number of malignant cells that remain after treatment and are below detection by conventional clinical examination or routine microscopy. Sensitive flow-cytometric, molecular, or imaging methods may detect MRD and refine relapse-risk assessment.

Minimally Invasive Gastrointestinal Surgery

Gastrointestinal operations performed through small incisions or natural orifices using laparoscopic, robotic, or endoscopic techniques, often reducing recovery time while retaining procedure-specific risks.

Minimally Invasive Hip Replacement

A medical procedure, treatment, device, or care topic related to minimally invasive hip replacement; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Minimally Invasive Surgery

Surgery performed through small incisions or natural openings using specialized instruments, usually reducing tissue disruption and recovery time.

Minimization

A psychological or communication process of downplaying the importance or severity of symptoms, events, or problems.

Minimum Alveolar Concentration

The alveolar concentration of an inhaled anesthetic that prevents movement in half of people exposed to a standardized painful surgical stimulus. It is used to compare the potency of volatile anesthetics.

Minimum Bactericidal Concentration

The lowest concentration of an antimicrobial that kills a defined proportion of the original bacterial inoculum, commonly demonstrated by subculture without growth. MBC is distinct from MIC and is

not routinely required for every infection.

Minimum Effect Concentration

The lowest concentration of a drug in the blood or at its site of action that produces a defined measurable effect.

Minimum Inhibitory Concentration

The lowest concentration of an antimicrobial that prevents visible growth of a microorganism under standardized laboratory conditions. It helps characterize susceptibility, but results must be interpreted with the organism, specimen, assay, and clinical context.

Minimum Lethal Concentration

The lowest concentration of a substance associated with death in a defined test population or experimental setting.

Minimum Lethal Dose

The smallest dose of a substance reported to cause death under specified conditions; it varies by species, route, and individual.

Minisatellite Repeat

A minisatellite repeat is a tandemly repeated DNA sequence with repeat units commonly ranging from about 10 to 100 base pairs. Variation in repeat number can serve as a genetic marker and, in some loci, influence gene regulation or disease risk.

MINOCA

Myocardial infarction with non-obstructive coronary arteries: an acute myocardial infarction with no coronary stenosis of 50 percent or greater and no other immediately apparent cause for the presentation. It is a working diagnosis requiring evaluation for mechanisms such as plaque disruption, coronary spasm, coronary embolism, spontaneous coronary artery dissection, myocarditis, or Takotsubo syndrome.

Minocycline

A tetracycline antibiotic used for selected bacterial

infections and inflammatory acne.

Minor

Smaller, lesser, or less significant than a related structure or condition; for example, the teres minor muscle is smaller than the teres major muscle.

Minor Response

A minor response is a limited improvement after treatment that does not meet the predefined criteria for a partial or complete response. The exact threshold depends on the disease, study protocol, and outcome being measured.

Minority Health

Minority health concerns the health status, health determinants, access, quality of care, and outcomes of populations that experience social, economic, cultural, or structural marginalization. Differences in outcomes often reflect modifiable inequities rather than inherent biology.

Minoxidil

A vasodilator used topically for pattern hair loss and orally in selected cases of resistant hypertension.

Minretumomab

An investigational monoclonal antibody studied as a targeted treatment for cancers expressing its selected tumor-associated antigen.

Minute Ventilation

The total volume of air breathed in one minute (tidal volume x respiratory rate). Normal is 6–8 L/min. Elevated in increased metabolic demand, pain, anxiety, and sepsis. Reduced in neuromuscular disease and sedation.

Miosis

Constriction of the pupil, produced mainly by parasympathetic activity through the oculomotor nerve and sphincter pupillae muscle. It may be physiologic or caused by medicines, toxins, neurologic disease, or ocular conditions.

Miosis Disorder

Abnormally small pupils or an abnormal pupillary constriction response, caused by medications, neurologic disease, ocular disease, or exposure to certain toxins.

Miotic

A medicine or substance that makes the pupil smaller. Miotics may be used in glaucoma or during some eye procedures and can affect vision in dim light.

Alternate Names:

Miotic agent

Miracidium

The ciliated larval stage of a trematode that hatches from an egg and infects a suitable snail host.

Mirage Syndrome

A rare genetic disorder, usually related to SAMD9 variants, that can cause growth restriction, adrenal insufficiency, recurrent infections, and distinctive developmental findings.

MIRD Schema

Medical Internal Radiation Dose. Method for calculating radiation absorbed doses.

Mirex

Mirex is a persistent organochlorine insecticide and industrial chemical. It bioaccumulates and is environmentally hazardous; exposure has been associated with toxic effects in experimental studies and is controlled or prohibited in many jurisdictions.

Mirizzi Syndrome

Condition where a gallstone impacted in the cystic duct or Hartmann pouch extrinsic compression of the common hepatic duct causing obstructive jaundice. Classified into four types based on the degree of erosion and fistula formation. Type one is extrinsic compression without fistula.

Mirror Therapy

Mirror reflection of unaffected limb movement cre-

ates visual illusion of affected limb moving. Reduces phantom limb pain, improves stroke recovery.

Mirror-Writing

Writing in which letters or words are reversed as if seen in a mirror; it may occur transiently in childhood or with neurologic disease.

Mirtazapine

Mirtazapine is a noradrenergic and specific serotonergic antidepressant used for major depressive disorder. It commonly causes sedation and increased appetite or weight and can rarely cause neutropenia or other serious adverse effects.

Misalignment

Misalignment is abnormal positioning or orientation of two or more structures relative to their intended relationship. It may affect bones, joints, teeth, eyes, implants, or medical devices and can impair function or healing.

Miscarriage

Spontaneous loss of a pregnancy before fetal viability, commonly caused by chromosomal abnormalities but requiring clinical assessment.

Alternate Names:

Spontaneous Abortion

Miscarriage - Threatened

Vaginal bleeding during early pregnancy without cervical dilation or passage of pregnancy tissue; the pregnancy may continue or progress to pregnancy loss.

Miscarriage, Spontaneous

Pregnancy loss occurring without an intentional medical or surgical procedure.

Misdiagnosis

Misdiagnosis is assignment of an incorrect diagnosis or failure to recognize the correct condition. It can lead to inappropriate treatment, delayed care, avoidable harm, or unnecessary testing and may result from atypical presentation, incomplete information,

or system factors.

Miserere

An obsolete term for feculent vomiting associated with severe intestinal obstruction.

Mismatch Repair

Mismatch repair is a DNA-repair pathway that recognizes and corrects incorrectly paired bases or small insertion–deletion loops generated during replication. Defects in mismatch-repair genes cause microsatellite instability and increase risk of several cancers, including Lynch-syndrome cancers.

Mismatch Repair Deficiency

Loss of DNA mismatch-repair function that produces microsatellite instability and a high tumor mutational burden. Seen in Lynch syndrome and some sporadic colorectal, endometrial, and other cancers, and associated with responsiveness to immune checkpoint inhibitors.

MISME

Alternate index label; see *NF2-related schwannomatosis (neurofibromatosis type 2)*.

Misonidazole

Misonidazole is an older investigational radiosensitizer designed to make low-oxygen tumor cells more sensitive to radiation.

Misophonia

Misophonia is an intense emotional or physiologic reaction to specific repetitive sounds, such as chewing or tapping. It is not simple annoyance; behavioral strategies, sound management, and therapy can reduce impairment, while assessment considers anxiety, obsessive-compulsive symptoms, and sensory sensitivity.

Misoprostol

A prostaglandin E1 analogue used for ulcer prevention in selected patients, medical abortion, cervical preparation, and obstetric indications.

Misoprostol (Cytotec)

A synthetic prostaglandin analogue used for selected gynecologic and obstetric indications, including cervical ripening, labor induction, and medical management of some pregnancies. It can cause uterine contractions, bleeding, gastrointestinal effects, or uterine tachysystole; use depends on the clinical setting and gestational age.

Alternate Names:

Cytotec

Missed Dose

A missed dose is a scheduled dose of a medicine that was not taken or administered. The appropriate response depends on the drug and timing; doubling a dose can be dangerous, so the product instructions or a clinician or pharmacist should guide the next dose.

Missed Miscarriage

A nonviable intrauterine pregnancy that has not been expelled and may produce few or no symptoms of miscarriage. Diagnosis requires appropriate ultrasound or other clinical evaluation before management is selected.

Alternate Names:

Missed abortion; delayed miscarriage.

Missense Mutation

A missense mutation is a DNA sequence change that substitutes one amino acid for another in a protein. Its effect may be harmful, beneficial, or neutral depending on the affected residue, protein function, and genetic context.

Missense Variant

A single-nucleotide substitution that changes one amino acid in a protein. Its clinical effect ranges from benign to pathogenic and must be assessed using molecular, functional, population, and clinical evidence.

Mistletoe Poisoning

Mistletoe poisoning involves toxic or environmental exposure to mistletoe poisoning; dose, route, timing, symptoms, and organ involvement determine

urgency and treatment.

Mite

A tiny arachnid; some mites cause dermatitis or transmit disease, and house-dust mites can trigger allergy.

Mithramycin

An older antineoplastic antibiotic, also called plitamicin, whose use is limited by substantial toxicity.

Mitigate

To mitigate means to lessen the severity, impact, or likelihood of a symptom, risk, injury, or disease complication. Mitigation may involve prevention, treatment, environmental changes, or supportive care.

Mitobronitol

Mitobronitol is an alkylating antineoplastic compound formerly investigated or used in selected hematologic malignancies. Its cytotoxic effects can damage DNA and cause myelosuppression and other treatment-related toxicities.

Mitochondria

Double-membrane organelles known as the "powerhouse of the cell," generating ATP through oxidative phosphorylation. Outer membrane is permeable; inner membrane is highly folded (cristae) and contains the electron transport chain and ATP synthase.

Mitochondrial Cristae

Mitochondrial cristae are folds of the inner mitochondrial membrane that increase surface area for the electron-transport chain and ATP synthase. Their structure supports oxidative phosphorylation and changes with cellular energy demands.

Mitochondrial Diabetes

Diabetes caused by mutations in mitochondrial DNA particularly the m.3243A>G mutation in the MT-TL1 gene. Maternally inherited. Often associated with sensorineural hearing loss termed maternally inherited diabetes and deafness. Patients

are typically lean with impaired insulin secretion. Diagnosis requires genetic testing.

Mitochondrial Disease

A group of genetic disorders caused by impaired mitochondrial energy production. Symptoms vary and may affect the brain, muscles, nerves, heart, eyes, or multiple organs; onset and severity can differ widely even within one family.

Mitochondrial DNA

Small circular DNA molecules located primarily in mitochondria that encode some components of oxidative phosphorylation and are inherited predominantly through the maternal line.

Mitochondrial DNA Depletion Syndrome MDS MDDS

A group of inherited disorders in which affected tissues contain abnormally reduced amounts of mitochondrial DNA, impairing cellular energy production. Forms may predominantly affect muscle, liver, or brain and muscle; many are autosomal recessive.

Mitochondrial Fission

Mitochondrial fission is division of one mitochondrion into two or more mitochondria. It helps distribute mitochondria, remove damaged portions through mitophagy, and respond to cell-cycle or metabolic signals; excessive fission may contribute to disease.

Mitochondrial Fusion

Mitochondrial fusion is joining of mitochondria into a connected network. It permits exchange of mitochondrial contents and helps maintain bioenergetic function and genome integrity; fusion and fission are normally balanced dynamically.

Mitochondrial Inheritance

Inheritance of mitochondrial DNA, which is transmitted predominantly through the egg and therefore usually from the mother. Examples include Leber hereditary optic neuropathy and MELAS. Heteroplasmy means that mutant and normal mitochondrial DNA coexist in a cell.

Mitochondrial Matrix

The mitochondrial matrix is the protein-rich compartment enclosed by the inner mitochondrial membrane. It contains mitochondrial DNA, ribosomes, and enzymes for the citric-acid cycle, fatty-acid oxidation, and other metabolic processes.

Mitochondrial Membrane

The mitochondrial membrane system consists of an outer membrane and a highly folded inner membrane separated by an intermembrane space. It controls transport and houses the respiratory chain and ATP-producing machinery.

Mitochondrial Myopathy

Muscle weakness or dysfunction caused by abnormalities of mitochondria, the energy-producing structures of cells.

Mitochondrial Ribosome

A mitochondrial ribosome is a ribosome within mitochondria that translates proteins encoded by mitochondrial DNA. Its structure differs from cytosolic ribosomes and defects in mitochondrial translation can cause energy-production disorders.

Mitochondrial Swelling

Enlargement of mitochondria caused by disruption of ion and water balance, often indicating cellular injury, impaired oxidative phosphorylation, or permeability-transition changes.

Mitochondrion

A mitochondrion is a membrane-bound organelle that generates much of a cell's ATP through oxidative phosphorylation. It also participates in apoptosis, calcium handling, metabolism, and production or regulation of reactive oxygen species.

Mitogen

A mitogen is a substance that stimulates a cell, especially a lymphocyte, to enter the cell cycle and proliferate. Mitogens include growth factors, lectins, and other signals that activate proliferative pathways.

Mitoguazone

Mitoguazone, also called methylglyoxal bis(guanyldrazone), is an experimental antineoplastic agent that interferes with polyamine metabolism and cell proliferation. It has been investigated in hematologic and other cancers.

Mitolactol

Mitolactol is an alkylating antineoplastic compound formerly used or investigated in cancer treatment. It damages DNA and can cause cytotoxic effects such as myelosuppression and gastrointestinal toxicity.

Mitomycin

Mitomycin is a chemotherapy medicine that damages DNA and is used for selected cancers; it can suppress bone marrow and injure the kidneys or bladder.

Mitomycins

Mitomycins are antitumor antibiotics that become activated in cells and cross-link DNA, inhibiting replication and transcription. Mitomycin C is the best-known member and is used in selected cancers and ophthalmic or surgical applications.

Mitophagy

Mitophagy is selective autophagic removal of damaged, excess, or otherwise targeted mitochondria. It maintains mitochondrial quality and cellular homeostasis; impaired or excessive mitophagy is implicated in some neurodegenerative and metabolic disorders.

Mitosis

Cell division that produces two genetically similar daughter cells with the same chromosome number as the parent cell.

Mitosporic Fungi

Fungi that reproduce by asexual spores; the term is mainly used for fungi whose sexual stage is unknown or not observed.

Mitotane

Mitotane is an adrenolytic drug used for selected adrenocortical carcinomas and to suppress excess adrenal steroid production. It can cause gastrointestinal and neurologic effects, adrenal insufficiency, and numerous drug interactions.

Mitotic Anaphase

Mitotic anaphase is the stage of mitosis in which sister chromatids separate and move toward opposite poles of the cell. Accurate spindle attachment and chromosome segregation are required for formation of genetically similar daughter cells.

Mitotic Figure

A cell undergoing nuclear division that can be identified microscopically. Mitotic count contributes to assessment of proliferative activity and tumor grade, although counting method and area must be standardized.

Mitotic Metaphase

Mitotic metaphase is the stage of mitosis in which duplicated chromosomes align at the cell equator with sister kinetochores attached to opposite spindle poles. A checkpoint helps prevent premature chromosome separation.

Mitotic Prophase

Mitotic prophase is the early stage of mitosis in which chromatin condenses into visible chromosomes, the nucleolus disappears, and the mitotic spindle begins to form. In late prophase or prometaphase, the nuclear envelope breaks down.

Mitotic Recombination

Mitotic recombination is exchange or rearrangement of DNA between homologous chromosomes during or around mitotic cell division. It can repair DNA or generate loss of heterozygosity and is relevant to tumor evolution and genetic disease.

Mitotic Telophase

Mitotic telophase is the stage after chromosome separation when chromosomes reach opposite poles, decondense, and become enclosed in newly forming

nuclear envelopes. Cytokinesis usually follows or overlaps with telophase.

Mitoxantrone

A chemotherapy and immunosuppressive medicine that interferes with DNA and is used for selected cancers and some forms of multiple sclerosis.

MitraClip

A transcatheter device that creates a double orifice in the mitral valve by clipping the anterior and posterior leaflets together, reducing mitral regurgitation. It is indicated for patients with significant (3+ or 4+) mitral regurgitation who are at high risk for surgical mitral valve repair.

Mitral

Relating to the mitral or bicuspid valve between the left atrium and left ventricle.

Mitral Atresia

Congenital absence or complete closure of the mitral-valve opening, causing severe obstruction of blood flow into the left ventricle.

Mitral Insufficiency

Failure of the mitral valve to close completely, allowing blood to leak back into the left atrium during systole.

Mitral Regurgitation

Incomplete closure of the mitral valve allowing blood to flow backward from the left ventricle to the left atrium during systole. Causes include mitral valve prolapse, papillary muscle dysfunction, rheumatic disease, and ischemic heart disease. Murmur: pansystolic (holosystolic) blowing murmur at the apex radiating to the axilla.

Mitral Stenosis

Narrowing of the mitral valve opening, obstructing left atrial to left ventricular blood flow. Most commonly caused by rheumatic heart disease. Symptoms include dyspnea, fatigue, and atrial fibrillation. Murmur: low-pitched, rumbling mid-diastolic mur-

mur at the apex with opening snap.

Mitral Stenosis Murmur

Low-pitched, rumbling mid-diastolic murmur at the apex, best heard with the bell of the stethoscope in the left lateral decubitus position. Accompanied by an opening snap (early diastolic high-pitched click) and loud S1 in pliable valves. Severe MS may show a diastolic murmur without a preceding S2 gap.

Mitral Valve

Left atrioventricular (bicuspid) valve with anterior and posterior cusps supported by chordae tendineae and papillary muscles. Anterior cusp is larger and proximate to the aortic valve. Mitral valve prolapse is the most common valvular abnormality. Mitral stenosis (most commonly rheumatic) and regurgitation are significant.

Mitral Valve Insufficiency

Leakage of the mitral valve that allows blood to flow back into the left atrium.

Mitral Valve Prolapse

A valvular heart disease where the mitral valve leaflets prolapse into the left atrium during systole. Most patients are asymptomatic; some develop palpitations, atypical chest pain, or dyspnea. Diagnosis by echocardiography. Complications include mitral regurgitation, infective endocarditis, and arrhythmias.

Alternate Names:

Barlow's Syndrome Barlow Syndrome (Mitral Valve Prolapse) Click-Murmur Syndrome Prolapse, Mitral Valve

Mitral Valve Regurgitation

Backward leakage of blood through the mitral valve when the left ventricle contracts, which can cause breathlessness, fatigue, atrial enlargement, or heart failure.

Alternate Names:

Regurgitation, Mitral Valve

Mitral Valve Stenosis

Narrowing of the mitral valve that restricts blood flow from the left atrium to the left ventricle.

Alternate Names:

Stenosis, Mitral Valve

Mitral Valve Surgery - Minimally Invasive

A medical procedure, treatment, device, or care topic related to mitral valve surgery - minimally invasive; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Mitral Valve Surgery - Open

A medical procedure, treatment, device, or care topic related to mitral valve surgery - open; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Mitral Valvotomy

An operation or catheter procedure that splits or widens a narrowed mitral valve.

Mitral Valvuloplasty

Mitral valvuloplasty is a procedure that widens a narrowed mitral valve, often with a balloon catheter or surgical repair.

Mitral-Valve Repair

A surgical or transcatheter procedure to restore mitral valve competence without replacement, typically for regurgitation from prolapse, chordal rupture, or ischemic dysfunction. Repair preserves native valve tissue, maintains left ventricular function, and avoids lifelong anticoagulation required by mechanical prostheses.

Mitsuokella

Mitsuokella is a genus of anaerobic bacteria found mainly in the mouth and digestive tract; human infection is uncommon.

Mitsuokella Multacida

An anaerobic bacterium found mainly in the mouth and intestine that is rarely associated with human

infection.

Mittelschmerz

Recurrent one-sided lower-abdominal or pelvic pain occurring around ovulation. It is usually brief or lasts up to a day or two, but severe, persistent, or atypical pain requires assessment for other causes.

Mixed Connective Tissue Disease

An overlap syndrome with features of systemic lupus erythematosus, systemic sclerosis, and polymyositis/dermatomyositis. Highly associated with anti-U1 RNP antibodies. Presents with Raynaud phenomenon (nearly universal), puffy hands, arthritis, myositis, esophageal dysmotility, and serositis. Renal disease is less common than in SLE.

Alternate Names:

Arthritis MCTD (Mixed Connective Tissue Disease)

Mixed Glioma

Mixed glioma is an older term for a glial tumor showing features of more than one glial lineage, historically including oligoastrocytoma. Modern classification relies mainly on molecular findings and generally assigns such tumors to a more specific entity.

Mixed Hearing Loss

A combination of both conductive and sensorineural hearing loss occurring in the same ear, resulting in a hearing deficit that involves impaired sound transmission through the outer or middle ear in addition to cochlear or neural dysfunction.

Mixed Oedema

Edema caused by more than one mechanism or affecting more than one tissue compartment; interpretation depends on clinical context.

Mixed Urinary Incontinence

Involuntary leakage of urine, significantly affecting psychological, social, and physical well-being. See urinary incontinence.

Mixed-Lineage Leukemia

An acute leukemia in which the malignant blasts show defining features of more than one hematopoietic lineage, commonly both myeloid and lymphoid; it is also called mixed-phenotype acute leukemia.

Mixovirus

An older group name for certain myxovirus-like viruses, including viruses historically associated with influenza and other respiratory infections.

MME

Membrane metalloendopeptidase, also called neprilysin, CD10, neutral endopeptidase, or CALLA; it is a cell-surface enzyme and marker expressed in several tissues and in some acute lymphoblastic leukemias.

Miyoshi Myopathy

An autosomal recessive distal muscular dystrophy belonging to the dysferlinopathy spectrum, caused by homozygous or compound heterozygous mutations in the *DYSF* gene on chromosome 2p13.2. Characteristically presents in early adulthood (ages 15–25) with selective, progressive weakness and severe atrophy of the posterior calf muscles (gastrocnemius and soleus), rendering patients unable to stand on their tiptoes or perform heel rises, while the anterior compartment (tibialis anterior) is initially spared. Associated with massive serum creatine kinase elevations (often > 10,000 U/L), progressing slowly over decades to involve the proximal thigh and pelvic girdle musculature.

Alternate Names:

Miyoshi Distal Myopathy; Distal Dysferlinopathy

Mobile

Capable of moving or being moved; a mass or structure described as mobile is not fixed to adjacent tissue.

Mobile Genetic Elements

Genetic elements that can move within and between genomes. These include plasmids, transposons, and integrons. Mobile genetic elements can carry antibiotic resistance genes and virulence factors. They

facilitate horizontal gene transfer and contribute to the rapid evolution of bacteria.

Mobile Health (MHealth)

Use of mobile devices for health. Examples: health apps, text message reminders, wearable devices.

Alternate Names:

mHealth

Mobile Health Unit

A mobile health unit is a vehicle or transportable facility equipped to provide health services at changing locations. It may deliver screening, vaccination, primary care, diagnostic testing, emergency support, or outreach to underserved communities.

Mobility

The ability to move or be moved, including functional movement of a person or range of motion of a joint.

Mobilization

The process of restoring or encouraging movement, such as early patient activity, joint movement, or surgical release of tissue.

Mobiluncus

A genus of curved anaerobic bacteria associated with bacterial vaginosis and occasionally isolated from other clinical specimens.

Mobiluncus Curtisi

An anaerobic curved bacterium associated with bacterial vaginosis and changes in the vaginal microbiota.

Mobiluncus Mulieris

An anaerobic curved bacterium associated with bacterial vaginosis and vaginal microbial imbalance.

Mobius Syndrome

A rare congenital disorder causing facial weakness or paralysis and limited outward eye movement.

Mocetinostat

Mocetinostat is an investigational histone deacetylase inhibitor studied as an antineoplastic agent. By altering histone acetylation and gene expression, it may inhibit tumor-cell growth; clinical use depends on regulatory approval and indication.

Modafinil

A wakefulness-promoting agent (mechanism not fully understood, likely dopaminergic) used for narcolepsy, obstructive sleep apnea residual sleepiness, and shift work sleep disorder; used off-label for fatigue and ADHD. Less abuse potential and fewer sympathomimetic effects than amphetamines.

Modality

A method or form of treatment, diagnosis, imaging, or sensory stimulation.

Mode

The most frequent value in a dataset or a particular manner or method of operation.

Model

A simplified representation used to describe, explain, predict, or study a biological or clinical process.

Moderate To Severe Disabilities

Moderate to severe disability describes substantial limitations in physical, cognitive, sensory, communication, or adaptive functioning that require ongoing support. Needs vary widely and may include medical care, rehabilitation, assistive technology, education, and assistance with daily activities.

Moderation

Moderation is reduction or control of the intensity, amount, or severity of a process, symptom, behavior, or treatment effect. In clinical research, a moderator is a factor that changes the strength or direction of an association.

Moderator

A factor that changes the strength or direction of a relationship or treatment effect; it may also mean a

person who guides a discussion.

Modes Of Inheritance

Patterns by which genetic traits pass through families, including autosomal dominant, autosomal recessive, X-linked, mitochondrial, and other patterns.

Modification

A change made to a structure, process, treatment, or behavior.

Modified Biophysical Profile (Modified BPP)

A simplified antepartum test combining the non-stress test with the amniotic fluid index (AFI). The NST assesses acute fetal status (heart rate reactivity over 20-40 minutes) and the AFI assesses chronic uteroplacental function (oligohydramnios suggests chronic hypoxia or placental insufficiency).

Alternate Names:

Modified BPP

Modulator

A substance or factor that alters the activity of a biological pathway, receptor, gene, or physiological process.

Modus Operandi

The characteristic method by which a person performs an action; it is chiefly a forensic or legal term.

Modus Vivendi

A practical arrangement or way of living that allows people or groups with differences to coexist.

Moebius Syndrome

A congenital disorder characterized mainly by facial paralysis and inability to abduct the eyes, caused by developmental cranial-nerve abnormalities.

Moeller'S Glossitis

An older term for a smooth, painful, inflamed tongue associated especially with nutritional deficiency, including iron, folate, or vitamin B12 deficiency.

Moellerella

Moellerella is a rare gram-negative bacterium that can occasionally cause opportunistic human infection.

Moellerella Wisconsinensis

A rare gram-negative bacterium that can occasionally cause urinary, bloodstream, or other opportunistic infection.

MOG Antibody-Associated Disorder

Alternate index label; see *Myelin oligodendrocyte glycoprotein antibody-associated disease*.

MOGAD

Alternate index label; see *Myelin oligodendrocyte glycoprotein antibody-associated disease*.

Mogibacterium

A genus of anaerobic bacteria found in the mouth and intestine and occasionally associated with dental or other infection.

Mohs Micrographic Surgery

A medical procedure, treatment, device, or care topic related to mohs micrographic surgery; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Mohs Surgery

A micrographically controlled technique in which skin cancer is excised in thin layers and examined margin-by-margin, achieving high cure rates for basal cell and squamous cell carcinoma while sparing normal tissue.

Moisture

Water or other liquid present on a surface or in tissue; excessive moisture can cause maceration and skin breakdown.

Molar

Relating to a molar tooth or, in chemistry, to one mole per liter.

Molar Pregnancy

A type of gestational trophoblastic disease characterized by abnormal proliferation of trophoblastic tissue. See Hydatidiform Mole.

Molar Pregnancy (Hydatidiform Mole)

An abnormal pregnancy in which placental tissue proliferates excessively and the embryo is absent or abnormal; it is a form of gestational trophoblastic disease.

Alternate Names:

Hydatidiform Mole

Molasses

A viscous syrup produced during sugar refining; it is rich in sugar and may contain variable amounts of minerals.

Mold

A multicellular fungus that grows as branching filaments; molds can cause allergy, food spoilage, or opportunistic infection.

Mold Allergy

An allergic response to mold spores that may cause nasal, eye, skin, or lower-respiratory symptoms.

Mold Exposure

Mold exposure can irritate the eyes, skin, and airways and may worsen asthma or allergic rhinitis; invasive infection is uncommon and mainly affects severely immunocompromised people. Reducing moisture and remediating contaminated areas are more useful than unproven detoxification.

Mole

Nevus: a benign proliferation of melanocytes, the most common skin lesion; also used for other nevi (moles are melanocytic nevi). Types: junctional (flat, brown—childhood), compound (slightly raised), intradermal (dome-shaped, flesh/pigmented).

Molecular

Relating to molecules or processes occurring at the

molecular level.

Molecular Chaperone

A molecular chaperone is a protein that assists other proteins with folding, assembly, transport, or refolding without becoming part of the final structure. Chaperones can also help prevent or resolve protein aggregation.

Molecular Diagnosis

Molecular diagnosis uses analysis of DNA, RNA, proteins, or other molecular markers to identify a disease, pathogen, inherited variant, tumor subtype, or treatment-relevant characteristic. Results require appropriate assay validation and clinical interpretation.

Molecular Dynamic

Molecular dynamics is a computational method that simulates movement and interactions of atoms or molecules over time using physical force fields. It is used to study biomolecular structure, flexibility, binding, and drug interactions.

Molecular Imaging

A diagnostic or laboratory method related to molecular imaging. Results are interpreted with the method, reference range, and clinical findings.

Molecular Pathology

The investigation of disease using nucleic-acid, genomic, epigenomic, or molecular protein alterations in cells and tissues. It may establish diagnosis, estimate prognosis, identify inherited risk, or select targeted therapy.

Molecular Pathology Of Colorectal Cancer

The assessment of molecular alterations that classify colorectal tumors and guide prognosis or therapy, including mismatch-repair deficiency or microsatellite instability, RAS mutations, BRAF mutation, HER2 amplification, and selected gene fusions.

Molecular Probe

A molecular probe is a labeled or otherwise detectable molecule designed to bind a particular nucleic-acid sequence, protein, metabolite, or other target. Probes are used in imaging, assays, microscopy, and molecular diagnostics.

Molecular Response

A molecular response is a measurable change in a molecular marker, such as a pathogen load, oncogenic transcript, mutation burden, or biomarker, after disease progression or treatment. Its definition and clinical significance depend on the disease and assay.

Molecular Testing

Analysis of DNA, RNA, or other molecular features to identify variants, fusions, expression patterns, or infectious agents for diagnosis, prognosis, or treatment selection.

Molecular Weight

The mass of one molecule, usually expressed in atomic mass units; for a substance used in chemistry, its numerical value corresponds to molar mass in grams per mole. It helps calculate concentrations and doses.

Molecule

A group of chemically bonded atoms forming the smallest unit of a substance that retains its chemical properties.

Moles

Moles are common collections of pigment-producing melanocytes and are usually benign. A lesion that changes in size, shape, color, symmetry, or symptoms, or differs markedly from others, should be examined for melanoma.

Molimina

The symptoms and physical changes that occur before menstruation, including breast tenderness, bloating, or mood changes.

Mollusc

A soft-bodied invertebrate; the term is also used in names of mollusc-borne or mollusc-associated parasitic diseases.

Mollusca

mollusca is a parasitic or invertebrate taxon; its medical relevance depends on the species, life cycle, exposure, host, and site of infection or tissue involvement.

Molluscipoxvirus

A poxvirus genus containing molluscum contagiosum virus, which causes umbilicated skin papules in humans.

Molluscum Contagiosum

A viral skin infection caused by a poxvirus, presenting as firm, dome-shaped, umbilicated papules with central dimpling, 2 to 5 mm in diameter. It is transmitted by direct skin contact, autoinoculation, and sexual contact in adults. Immunocompetent hosts clear lesions spontaneously within 6 to 12 months.

Molluscum Sebaceum

An obsolete name historically applied to sebaceous-gland lesions; terminology is inconsistent and the original source should be checked.

Molt

A molt is shedding and replacement of an outer covering, such as the exoskeleton of an arthropod or the skin, feathers, or hair of an animal. The process is controlled by developmental and hormonal signals.

Molten Metal Fumes

Fumes produced during welding, smelting, or casting that can cause metal-fume fever, airway irritation, or metal toxicity.

Molybdenum

An essential trace element required in small amounts for enzymes; excessive exposure can affect copper balance.

Molybdenum-99

Parent isotope of Tc-99m, half-life 66 hours. Produced in nuclear reactors. Generator systems provide Tc-99m for 1 week.

Molybdenosis

Illness caused by excessive molybdenum exposure, which may disturb copper metabolism and cause systemic effects.

Moment

A brief interval or, in biomechanics, the turning effect of a force around a point or joint.

Mometasone

Mometasone is a potent synthetic glucocorticoid used topically or by inhalation for inflammatory skin or airway disease. It reduces local inflammation but may cause skin atrophy, oral candidiasis, adrenal effects, or other corticosteroid adverse effects.

Mometasone Furoate

Mometasone furoate is the furoate ester of the corticosteroid mometasone. Depending on formulation, it is used for inflammatory skin disease, allergic rhinitis, or asthma; local and systemic corticosteroid precautions apply.

Momordica

A genus of plants including bitter melon; preparations may affect blood glucose and can interact with medicines.

Monascus

Monascus is a genus of fungi used in some fermented foods; some species produce pigments, while contamination can produce harmful mycotoxins.

Monascus Ruber

Monascus ruber is a red-pigment-producing fungus found in environmental and food settings; it is not a usual human pathogen.

Mondor's Disease

Superficial thrombophlebitis of the anterior chest

wall veins, typically the lateral thoracic, thoracoepigastric, or superior epigastric veins. Presents as a tender, cord-like subcutaneous structure with skin retraction. Usually self-limited; associated with breast surgery, trauma, or hypercoagulable states.

Monensin

Monensin is a polyether ionophore antibiotic used mainly in veterinary and agricultural settings to alter rumen fermentation or prevent coccidiosis. It can be toxic to animals and humans, especially through effects on cardiac and skeletal muscle.

Mongol

An obsolete and offensive term formerly used for a person with Down syndrome; it should not be used clinically.

Mongolian Spot

A Mongolian spot, now preferably called congenital dermal melanocytosis, is a flat blue-gray patch caused by melanocytes in the dermis, usually present at birth. It is benign and often fades during childhood but should be documented to distinguish it from bruising.

Mongolism

An obsolete and offensive term for Down syndrome, a chromosomal condition usually caused by trisomy 21.

Monilethrix

Monilethrix is an inherited hair-shaft disorder in which hair has regularly spaced narrow segments and breaks easily, producing a beaded appearance. It commonly causes short, brittle scalp hair and may involve follicles elsewhere.

Monitored Anesthesia Care

A distinct service where an anesthesia professional provides premedication, administration of analgesics and/or sedatives, continuous monitoring of the patient's cardiorespiratory status, and intervention for any emergent or unavoidable complications that may arise during a diagnostic or therapeutic procedure.

Monitoring Device

A monitoring device measures and displays or records a physiologic variable, such as heart rhythm, oxygen saturation, blood pressure, glucose, temperature, or respiratory rate. Accuracy, calibration, and clinical context determine how results should be interpreted.

Monitoring Test

A test repeated over time to follow a disease, treatment response, medicine safety, or a possible complication. The result is interpreted with symptoms and earlier measurements.

Monkeypox

Monkeypox, now generally called mpox, is a zoonotic infection caused by monkeypox virus, an orthopoxvirus. It can cause fever, lymphadenopathy, and a characteristic rash or lesions and spreads through close contact, including skin-to-skin or respiratory contact.

Mono

Alternate index label; see *Mononucleosis*.

Mono (Infectious Mononucleosis)

Alternate index label for Infectious Mononucleosis.

Monoamine Oxidase Inhibitors

Antidepressants that inhibit monoamine oxidase, the enzyme responsible for breaking down serotonin, norepinephrine, and dopamine. MAOIs include phenelzine, tranylcypromine, isocarboxazid, and selegiline (transdermal). They are effective for atypical depression and treatment-resistant depression.

Monobactams

Monobactams are beta-lactam antibiotics containing a single beta-lactam ring, rather than the fused rings of penicillins or cephalosporins. Aztreonam is the principal clinical example and primarily targets aerobic gram-negative bacteria.

Monoblast

An immature precursor cell in the monocyte lineage found mainly in bone marrow.

Monoblast Count

A monoblast count is the number or proportion of monoblasts, immature precursors of monocytes, in a blood or bone-marrow sample. Increased blasts may support evaluation for acute myeloid leukemia or another marrow disorder and require expert morphologic or immunophenotypic interpretation.

Monoclonal

Derived from a single cell and therefore composed of cells or molecules that are genetically or immunologically identical to that original cell; monoclonal antibodies recognize one target or epitope.

Monoclonal Antibodies

Antibodies produced from a single B-cell clone and directed against one antigenic target; they are used in diagnosis and treatment.

Monoclonal Antibody

An antibody produced by a single cloned B-cell line, specific for one epitope, generated by hybridoma technology or recombinant methods (name suffix -mab).

Monocrotophos

Monocrotophos is a highly toxic organophosphorus insecticide that inhibits acetylcholinesterase. Exposure can cause acute cholinergic poisoning with secretions, bronchospasm, vomiting, bradycardia, weakness, seizures, and respiratory failure.

Monocular

Involving or affecting one eye, or occurring with vision from one eye.

Monocyte

A circulating leukocyte (2-8 percent) that differentiates into macrophages and dendritic cells in tissues. Monocytes phagocytose pathogens, present antigens, and secrete cytokines (TNF, IL-1, IL-6);

participate in innate immunity, inflammation, and tissue repair. Subsets differ in function (classical, nonclassical, intermediate).

Monocytes

Large circulating white blood cells that can enter tissues and become macrophages or dendritic cells.

Monocytopenia

An abnormally low number of monocytes in the blood.

Monocytosis

An abnormally increased number of monocytes in the blood.

Monogenic

Pertaining to or caused by a mutation or pathogenic variant in a single gene locus (Mendelian disorder). Contrasts with polygenic and multifactorial traits, which result from the cumulative interactions of multiple genes and environmental factors. Monogenic diseases follow classic Mendelian inheritance patterns (autosomal dominant, autosomal recessive, X-linked, or mitochondrial); classic examples include cystic fibrosis (*extitCFTR* gene), Huntington disease (*extitHTT* gene), sickle cell disease (*extitHBB* gene), and Duchenne muscular dystrophy (*extitDMD* gene). Diagnosed with targeted single-gene sequencing or next-generation sequencing gene panels.

Monogenic Obesity

Monogenic obesity is severe or early-onset obesity caused primarily by a pathogenic variant in a single gene involved in appetite, energy balance, or leptin–melanocortin signaling. Genetic evaluation may guide counseling and selected targeted treatment.

Monograph

A monograph is a detailed, authoritative document devoted to one subject. In health care it may describe a drug's identity, quality, pharmacology, indications, dosing, safety, or regulatory standards.

Monoiodotyrosine

Monoiodotyrosine is an iodinated tyrosine residue formed during thyroid-hormone synthesis. Coupling of iodinated tyrosine residues within thyroglobulin contributes to production of thyroxine and triiodothyronine.

Monokines

Monokines are cytokines produced chiefly by monocytes and macrophages, including mediators such as tumor necrosis factor and interleukin-1. They regulate inflammation, immune responses, fever, and communication among cells.

Monomer

A monomer is a relatively small molecule that can chemically bond with other monomers to form a polymer. Biological examples include amino acids and nucleotides; some industrial monomers can irritate tissues or be toxic.

Monomethylhydrazine

Monomethylhydrazine is a volatile, corrosive, highly toxic chemical used as a rocket propellant. Exposure can injure the eyes, skin, lungs, liver, and nervous system and may cause seizures or systemic poisoning.

Mononegavirales

An order of viruses with nonsegmented, negative-sense RNA genomes, including rabies, measles, Ebola, and respiratory syncytial viruses.

Mononeuropathy

Mononeuropathy is dysfunction or injury of a single peripheral nerve. Common causes include compression, trauma, ischemia, inflammation, and metabolic disease; symptoms follow the nerve's distribution and may include pain, numbness, weakness, or autonomic changes.

Mononuclear Leukocyte

A mononuclear leukocyte is a white blood cell with a single, non-lobed nucleus, chiefly a lymphocyte or monocyte. These cells participate in adaptive immunity, innate immune responses, antigen presentation,

and inflammation.

Mononucleosis

Mononucleosis is an infectious syndrome commonly caused by Epstein–Barr virus, with fever, sore throat, swollen lymph nodes, and fatigue.

Alternate Names:

Kissing Disease Mono

Mononucleosis Spot Test

A rapid blood test detecting heterophile antibodies associated with infectious mononucleosis; it may be negative early in illness or in young children, so other testing may be needed.

Monoplegia

Complete motor paralysis affecting a single anatomical limb (an arm or a leg); distinct from monoparesis, which describes partial weakness. Caused either centrally by a focal lesion in the contralateral primary motor cortex (e.g., a small cortical ischemic stroke, localized tumor, or cerebral abscess involving a discrete segment of the motor homunculus) or peripherally by a severe focal nerve trunk lesion, spinal nerve root avulsion, or isolated brachial or lumbosacral plexopathy. Evaluated through detailed neurological localization testing, electromyography/nerve conduction studies (EMG/NCS), and brain/spine MRI.

Monopodium

A single foot or one-sided foot structure; the term is uncommon and may be used descriptively in developmental anatomy.

Monorchidism

The presence of only one testis, either because the other failed to develop, descended, or was removed.

Monosaccharide

A simple sugar that cannot be hydrolyzed into a smaller carbohydrate, such as glucose or fructose.

Monosodium Glutamate

MSG, the sodium salt of glutamic acid, used as a

flavor enhancer because it contributes umami taste; it occurs naturally in foods such as tomatoes and aged cheese.

Monosodium Urate

The crystalline salt that precipitates in tissues when serum uric acid levels exceed the saturation point, forming needle-shaped, negatively birefringent crystals under polarized light microscopy. It is the pathological hallmark of gout, depositing in joints, soft tissues, and kidneys.

Monosomy

Presence of one copy of a chromosome instead of the usual two. Complete autosomal monosomies are generally not compatible with life; monosomy X and some mosaic or structural forms can survive and have variable features.

Monospot (EBV Heterophile Antibody)

A rapid blood test that detects heterophile antibodies associated with infectious mononucleosis. It may be negative early in illness or in young children and does not directly detect Epstein–Barr virus; specific antibody testing may be needed.

Alternate Names:

EBV Heterophile Antibody

Monoterpenes

Monoterpenes are terpenoid compounds built from two isoprene units and containing 10 carbon atoms. They occur widely in plant essential oils and may have aromatic, biologic, or toxic effects depending on the compound and exposure.

Monotest

A rapid heterophile-antibody test historically used to support diagnosis of infectious mononucleosis.

Monounsaturated Fat

A fat whose fatty acids contain one carbon-carbon double bond. It occurs in foods such as olive oil, avocados, nuts, and some seeds.

Monozygotic Twin

A monozygotic twin develops when one fertilized egg divides into two embryos. The twins share nearly identical nuclear genomes but can differ through developmental, epigenetic, mitochondrial, and environmental factors.

Mons Veneris

The rounded fatty prominence over the pubic bone in front of the external female genitalia.

Monster

An obsolete and stigmatizing term for a person or fetus with severe congenital anomalies; modern medical language describes the specific anomalies.

Montelukast

Montelukast is a leukotriene CysLT1-receptor antagonist used for asthma prevention and allergic rhinitis; it is not a rescue bronchodilator, and neuropsychiatric adverse effects require attention.

Montelukast Sodium

Montelukast sodium is the sodium salt of montelukast, a leukotriene-receptor antagonist used for asthma prevention and allergic rhinitis. It is not a rescue bronchodilator and carries an important warning for possible neuropsychiatric adverse effects.

Montenegro Test

A delayed-type hypersensitivity skin test using Leishmania antigen, historically used to indicate exposure to leishmaniasis.

Montevideo Units

A quantitative measure of uterine contraction intensity, calculated by summing the peak contraction pressures (in mm Hg) minus the resting tone for each contraction in a 10-minute window. Adequate labor: ≥ 200 MVU (spontaneous) or ≥ 250 MVU (augmented). Measured with an intrauterine pressure catheter (IUPC).

Montgomery Tubercles

Small, raised sebaceous glands visible on the areola of the breast during pregnancy and lactation. They appear as small bumps surrounding the nipple and secrete an oily substance that lubricates the nipple and areola, protecting against infection. Their prominence increases during pregnancy under hormonal influence.

Montreal Cognitive Assessment

A brief cognitive screening instrument with greater emphasis on executive function, visuospatial ability, attention, memory, language, and orientation than many shorter screens. Interpretation requires consideration of education, language, sensory deficits, and acute illness.

Mood

A sustained emotional state perceived by the person and observed by others.

Mood Disorder

A group of mental disorders characterized primarily by pathological depression, mania, hypomania, or their combination.

Mood Stabilizers

Medications used to treat mood episodes in bipolar disorder and prevent recurrence. The primary mood stabilizers are lithium, valproate, carbamazepine, and lamotrigine. Each has a different mechanism and side effect profile. Lithium is the only agent with demonstrated anti-suicide properties.

Mood Swing

A mood swing is a noticeable change in emotional state or affect over time. Occasional changes can be normal, while frequent, severe, or impairing swings may occur with mood disorders, substance use, medical illness, medications, or sleep disruption.

Mopidamol

Mopidamol is a phosphodiesterase inhibitor that inhibits platelet aggregation. It was formerly investigated or used as an antineoplastic and is not a routine contemporary treatment in most clinical

settings.

MOPP Protocol

The MOPP protocol is a chemotherapy regimen consisting of mechlorethamine, vincristine (Oncovin), procarbazine, and prednisone. It was historically used for Hodgkin lymphoma and has largely been replaced by regimens with different efficacy and toxicity profiles.

Moral

Moral refers to principles or judgments concerning right and wrong conduct. In health care, moral reasoning may inform ethical decisions but is distinct from clinical diagnosis and from legally enforceable rules.

Morale

Morale is the confidence, motivation, and sense of shared purpose within a person or group. It can affect teamwork and occupational well-being but is not itself a specific medical diagnosis.

Moraxella

Moraxella is a genus of bacteria that can colonize the respiratory tract and cause ear, sinus, eye, or respiratory infections.

Moraxella Atlantae

Moraxella atlantae is a rare Moraxella species; its recovery from a clinical sample should be interpreted with the site and evidence of infection.

Moraxella Bovis

Moraxella bovis is a bacterium that causes infectious bovine keratoconjunctivitis, commonly called pinkeye, in cattle.

Moraxella Canis

Moraxella canis is a bacterium associated with dogs and can rarely cause opportunistic infection in humans.

Moraxella Lacunata

Moraxella lacunata is a bacterium historically associ-

ated with conjunctivitis and occasionally recovered from the upper respiratory tract.

Moraxella Lincolnii

Moraxella lincolnii is a rare respiratory-tract bacterium; clinical importance depends on the specimen and evidence of infection.

Moraxella Nonliquefaciens

Moraxella nonliquefaciens commonly colonizes the upper airway and can occasionally cause respiratory or invasive infection.

Moraxella Oblonga

Moraxella oblonga is a rare Moraxella species found in respiratory or oral specimens; significance depends on clinical evidence of infection.

Moraxella Osloensis

Moraxella osloensis is an uncommon bacterium that can cause opportunistic bloodstream, respiratory, or other infections.

Moraxellaceae

Moraxellaceae is a family of gram-negative bacteria that includes Moraxella and related respiratory-tract organisms.

Morbid Obesity

Severe obesity defined clinically as BMI ≥ 40 kg/m² or BMI ≥ 35 kg/m² with obesity-related comorbidities (type 2 diabetes, hypertension, obstructive sleep apnoea, non-alcoholic fatty liver disease, osteoarthritis, cardiovascular disease).

Morbidity

Illness or disability in a person or group of people. It can also mean how many people have a disease or how much illness or disability a disease causes.

Morbili

The medical or Latin name for measles, an acute contagious viral infection with fever and a characteristic rash.

Morbilli Form

Resembling measles in appearance, especially a rash with widespread maculopapular lesions.

Morbillivirus

A genus of paramyxoviruses that includes measles virus and related animal viruses causing fever, respiratory illness, rash, or neurologic disease.

Morbus

A Latin term meaning disease, used in some historical disease names.

Morcellation

The division of tissue into smaller fragments, often during minimally invasive surgery; it may spread an unrecognized cancer and requires appropriate case selection.

Mordant

A substance that helps fix a dye to tissue or a material, used in some histological staining methods.

Morgagni Hernia

A congenital diaphragmatic hernia occurring through the anterior retrosternal foramen of Morgagni, resulting from incomplete fusion of parts of the diaphragm. Abdominal contents may move into the chest and cause respiratory or gastrointestinal symptoms.

Morgagni'S Syndrome

A historical eponym used inconsistently; in some sources it refers to a combination of obesity, virilism, and diabetes associated with adrenal or endocrine disease.

Morganella

Morganella is a genus of gram-negative bacteria that can cause urinary, wound, bloodstream, and other opportunistic infections.

Morganella Morganii

Morganella morganii is a gram-negative bacterium that can cause urinary, wound, bloodstream, and

postoperative infections.

Morgellons Disease

Morgellons is a controversial condition involving distressing skin sensations and the belief that fibers or materials emerge from the skin. Symptoms are real to the person and require respectful evaluation for dermatologic, neurologic, psychiatric, toxic, or infectious causes without reinforcing an unproven explanation.

Moribund

In a state of imminent dying; rapidly approaching physiological expiration. In clinical medicine, describes a patient in the final, irreversible agonal phase of a terminal illness or catastrophic physiologic collapse. Characterized by profound obtundation, mottled or cyanotic extremities with cold skin from extreme circulatory failure, Cheyne-Stokes respiration or agonal respiratory efforts with accumulation of pharyngeal secretions (terminal respiratory secretions / "death rattle"), progressive bradycardia, and profound hypotension; marks the clinical transition to comfort-focused palliative end-of-life care.

Morning Sickness

Nausea and vomiting of pregnancy: nausea (and sometimes vomiting) commonly in the first trimester (6-12 weeks), typically resolving by 14-16 weeks; multifactorial (hCG, estrogen, progesterone, odor sensitivity).

Moro Reflex

A primitive reflex present from birth to 3-6 months, elicited by suddenly startling the infant (loud noise, head drop). The infant extends both arms outward with fingers spread, then draws arms inward as if embracing. Absence at birth suggests central nervous system injury or brachial plexus palsy.

Morphea

A localized sclerosing skin disorder that produces areas of thickened, hardened, or discolored skin and differs from systemic sclerosis by usually lacking internal-organ involvement.

Morpheus

The Greek and Roman mythological god of dreams; the name is the source of *morphine*, an opioid named for its association with sleep and dreams.

Morphinans

Morphinans are a group of chemical compounds with a fused-ring structure related to morphine. The class includes opioid analgesics and other compounds with varying receptor activity, potency, and clinical effects.

Morphine

Morphine is a potent opioid analgesic used for selected moderate-to-severe pain and palliative-care indications. It activates opioid receptors and can cause respiratory depression, sedation, constipation, nausea, tolerance, dependence, and overdose.

Morphine Overdose

Morphine overdose involves toxic or environmental exposure to morphine overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Morphine Sulfate

Morphine sulfate is a salt formulation of the opioid analgesic morphine. It is available in immediate- and extended-release preparations for selected pain indications and carries risks of respiratory depression, misuse, dependence, and fatal overdose.

Morphogenesis

morphogenesis is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

Morpholino

A morpholino is a synthetic antisense oligomer with a morpholine-ring backbone that binds complementary RNA sequences and alters splicing or translation. Morpholinos are used mainly in research and selected experimental applications.

Morphologic Response

A morphologic response is a treatment-related change in the visible or microscopic structure of a lesion, tumor, tissue, or cell population. It may include shrinkage, necrosis, differentiation, or altered cellular features and is interpreted using disease-specific criteria.

Morphology

The study or description of the form, shape, and structural features of organisms, organs, tissues, cells, or other biological structures.

Morquio Syndrome

An autosomal recessive lysosomal storage disorder (Mucopolysaccharidosis Type IV [MPS IV]) characterized by the defective degradation of keratan sulfate and chondroitin-6-sulfate. Subtyped into MPS IVA (caused by deficiency of *N*-acetylgalactosamine-6-sulfate sulfatase, encoded by *GALNS*) and MPS IVB (caused by β -galactosidase deficiency, encoded by *GLB1*). Clinically distinguished from other mucopolysaccharidoses by severe spondyloepiphyseal dysplasia, marked short-trunk dwarfism, pectus carinatum, flail joints, corneal opacification, and severe odontoid hypoplasia predisposing to atlantoaxial subluxation and life-threatening cervical spinal cord compression, while preserving normal intellectual function.

Alternate Names:

Mucopolysaccharidosis Type IV (MPS IV)

Mortality

Death rate in population. Elderly mortality: higher rates, common causes: cardiovascular disease, cancer, dementia, respiratory disease, fall-related injuries. Life expectancy varies by sex, race, socioeconomic status.

Mortality Rate

Deaths from disease in population. Crude: total deaths / total population. Specific: disease-specific deaths / population at risk.

Morton Neuroma

A perineural fibrosis of the plantar digital nerve,

most commonly between the third and fourth metatarsal heads. It presents as burning pain, numbness, and a feeling of walking on a pebble. It is not a true neuroma but a mechanical entrapment neuropathy.

Alternate Names:

Interdigital Neuroma Neuroma, Plantar Plantar Neuroma

Morton'S Metatarsalgia

Pain in the forefoot beneath the metatarsal heads, caused by pressure or irritation and distinct from Morton neuroma.

Morton'S Neuroma

Thickening of tissue around a nerve in the forefoot, causing burning pain or numbness between the toes.

Alternate Names:

Mortons Neuroma (Morton's Neuroma)

Mortons Neuroma (Morton'S Neuroma)

Alternate index label for Morton's Neuroma.

Morton'S Neuroma

Thickening and irritation of a nerve, usually between the toes, causing burning forefoot pain.

Alternate Names:

Morton neuroma; interdigital neuroma

Morula

A morula is an early embryo consisting of a compact ball of blastomeres formed after several cleavage divisions. It develops into a blastocyst before implantation in the uterine lining.

Morvan'S Disease

A rare autoimmune neurologic syndrome with severe insomnia, hallucinations, confusion, pain, and peripheral nerve hyperexcitability.

Mosaic

Describing a person or tissue composed of two or more genetically different cell populations derived from one zygote, often because of a post-zygotic mutation or chromosomal event; mosaicism can alter

the severity and distribution of a genetic disorder.

Mosaicism

The presence of two or more genetically distinct cell populations arising from a single zygote, which can cause variable distribution and severity of clinical features.

Mosquito

A small blood-feeding insect whose bites may transmit malaria, dengue, West Nile virus, and other infections.

Mosquito Clamp

A small hemostatic clamp used to compress small blood vessels.

Mosquito-Borne

Transmitted by mosquitoes, usually referring to an infectious disease or parasite.

Motexafin Gadolinium

Motexafin gadolinium is an experimental texaphyrin compound designed to catalyze redox reactions and increase oxidative stress in tumor cells. It was investigated as a radiosensitizer and antineoplastic agent but is not a routine standard therapy.

Motexafin Lutetium

Motexafin lutetium is an investigational texaphyrin compound containing lutetium that can participate in redox chemistry and has been studied for photodynamic or other antineoplastic applications. It is not a routinely used approved treatment.

Motilin

A hormone that helps regulate contractions and movement in the upper gastrointestinal tract.

Motility

The ability of an organ, especially the intestine, to move its contents.

Motion Analysis

Assessment of movement patterns. 3D gait analy-

sis, video analysis. Identifies abnormalities, guides treatment.

Motion Artifact

Image distortion or blurring caused by movement of the patient, organs, or equipment during image acquisition.

Motion Correction

Correction for patient motion during acquisition. Respiratory gating, navigation.

Motion Perception

Motion perception is the ability of the visual, vestibular, and proprioceptive systems to detect and interpret movement of the body or surrounding objects. Disturbance may cause vertigo, oscillopsia, imbalance, or visual-motion sensitivity.

Motion Sickness

Nausea, dizziness, sweating, or vomiting caused by sensory mismatch during travel or other motion. It can occur in cars, boats, aircraft, amusement rides, or virtual-reality environments and usually improves when motion stops. Severe or persistent symptoms require assessment for other causes.

Motion Sickness (Sea Sickness, Car Sickness)

Alternate index label for Sea Sickness, Car Sickness.

Motivation

Motivation is the set of processes that initiate, direct, and sustain goal-directed behavior. Reduced or excessive motivation may occur with mood disorders, neurologic disease, substance use, medication effects, or other conditions.

Motivational Interviewing

A client-centered, directive counseling method for enhancing intrinsic motivation to change by exploring and resolving ambivalence. Developed by Miller and Rollnick, it uses the spirit of partnership, acceptance, compassion, and evocation. Core skills include open-ended questions, affirmations, reflec-

tions, and summaries.

Motor

Relating to movement or to nerves and pathways that control movement.

Motor Aphasia

Motor aphasia, often called expressive or Broca aphasia, is impaired ability to produce spoken or written language despite relatively preserved motivation to communicate. It commonly follows injury to dominant frontal language networks, especially from stroke.

Motor Cortex

The frontal-lobe regions that plan and initiate voluntary movement. The primary motor cortex lies in the precentral gyrus and controls the opposite side of the body in an orderly somatotopic map.

Motor Delay

Motor delay is slower-than-expected acquisition of gross- or fine-motor skills in a child. It may reflect neuromuscular, developmental, genetic, sensory, environmental, or systemic causes and warrants age-appropriate developmental assessment.

Motor End-Plate

The specialized postsynaptic region of a skeletal-muscle fiber at the neuromuscular junction.

Motor Fluctuations Causes In Parkinson'S Disease

Motor fluctuations in Parkinson disease are changes between good mobility and wearing-off or medication-related involuntary movements as dopamine treatment duration and disease progression change. Adjusting timing, formulations, adjunct medicines, infusion, or surgery may help.

Motor Learning

A relatively lasting change in the ability to perform a motor skill resulting from practice, experience, feedback, and adaptation. Rehabilitation uses task-specific repetition and meaningful practice to

promote motor learning.

Motor Neuron

A nerve cell that controls skeletal or smooth muscle activity.

Alternate Names:

Motor nerve cell

Motor Neurone Disease

A group of progressive disorders affecting upper and lower motor neurons, including amyotrophic lateral sclerosis.

Motor Neurones

Nerve cells that transmit signals to skeletal muscle or other effectors to produce movement.

Motor Stereotypy

A motor stereotypy is a repetitive, patterned, seemingly purposeless movement, such as hand flapping, rocking, or body waving. It may occur in typical development or with autism, intellectual disability, neurologic disease, or other conditions.

Motor Tic

A motor tic is a sudden, brief, repetitive movement such as blinking, facial grimacing, or shoulder shrugging. Tics are often temporarily suppressible and may occur in transient tic disorder, persistent tic disorder, or Tourette syndrome.

Motor Unit

A single alpha motor neuron and all the skeletal muscle fibers it innervates. The number of muscle fibers per motor unit determines precision of control: small motor units (few fibers, e.g., extraocular and hand muscles) provide fine movement; large motor units (hundreds to thousands of fibers, e.g., gastrocnemius) provide gross power.

Motor Vehicle Accident

A collision or other road-traffic event causing injury or property damage. Medical evaluation may be needed for visible or delayed injuries, including head, neck, chest, abdominal, or musculoskeletal trauma.

Mottled Enamel

Patchy, discolored, or structurally altered tooth enamel, classically caused by excessive fluoride during tooth development.

Mottled Teeth

Teeth with irregular white, brown, or pitted enamel, often due to dental fluorosis or developmental enamel defects.

Moulage

A model, cast, or realistic reproduction of a body part or lesion used for teaching, simulation, or forensic documentation.

Mould

British spelling of mold, a multicellular fungus that grows as branching filaments.

Mountain Sickness

Illness caused by rapid ascent to high altitude, ranging from acute mountain sickness to life-threatening pulmonary or cerebral edema.

Mouth

The oral cavity that serves as the entry point of the digestive tract, used for eating, breathing, and speaking.

Mouth (Oral Cavity)

The entry to the digestive tract, containing the teeth, tongue, and openings of salivary glands. It supports chewing, taste, swallowing, speech, and initial digestion.

Alternate Names:

oral cavity; mouth

Mouth Cancer

Cancer arising in the lips, gums, tongue, cheeks, floor of the mouth, palate, or other oral tissues.

Alternate Names:

Cancer, Mouth

Mouth Diseases

Disorders affecting the oral cavity, including infec-

tions, ulcers, inflammatory disease, dental disease, mucosal lesions, and tumors.

Mouth Neoplasm

A mouth neoplasm is an abnormal growth in the oral cavity that may be benign or malignant; persistent sores, lumps, or patches need assessment.

Mouth Sores

Mouth sores are painful ulcers, blisters, erosions, or inflamed areas caused by trauma, infection, immune disease, medicines, nutritional deficiency, or cancer.

Mouth Ulcer

Open sores affecting the oral mucosa, commonly caused by aphthous ulcers, trauma, infection, irritation, or systemic inflammatory disease. Most resolve within two weeks; persistent, recurrent, unusually large, or indurated ulcers require evaluation.

Alternate Names:

Oral Ulcer

Mouth-To-Mouth Resuscitation

Rescue breathing in which air is delivered from a rescuer to a person who is not breathing normally, usually as part of CPR.

Mouth-Wash

A liquid used to rinse the mouth, freshen breath, or deliver an antiseptic, fluoride, or other active ingredient.

Mouthwash

A liquid rinsed around the mouth and then expelled to freshen breath, deliver an active ingredient, or reduce selected oral microorganisms.

Mouthwash Overdose

Mouthwash overdose involves toxic or environmental exposure to mouthwash overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Movement

Movement is a change in position of the body or

a body part produced by coordinated activity of the nervous system, muscles, and joints. Abnormal movement may be excessive, reduced, involuntary, poorly coordinated, or restricted.

Movement - Uncontrollable

Involuntary movement that cannot be suppressed, such as tremor, chorea, dystonia, myoclonus, or tics; its pattern helps identify the cause.

Movement - Uncontrolled Or Slow

Abnormally slow or poorly controlled movement, which may reflect bradykinesia, rigidity, dystonia, medication effects, or basal-ganglia disease.

Movement - Uncoordinated

Ataxic or poorly coordinated movement caused by dysfunction of the cerebellum, sensory pathways, vestibular system, peripheral nerves, or their connections.

Movement - Unpredictable Or Jerky

Sudden, irregular, or jerky involuntary movements, including myoclonus, chorea, or dyskinesia, arising from neurologic disease, medications, metabolic disorders, or seizures.

Movement Disorder

A neurologic condition causing excessive, reduced, or otherwise abnormal movement. Hyperkinetic disorders include tremor, dystonia, chorea, myoclonus, tics, and dyskinesia; hypokinetic disorders include bradykinesia and parkinsonism.

Movement Disorders

Neurologic conditions characterized by excessive, reduced, or abnormal voluntary or involuntary movement, including tremor, dystonia, chorea, tics, ataxia, and parkinsonism.

Moxalactam

Moxalactam is a beta-lactam antibiotic of the oxacephem group with activity against many gram-negative bacteria and some gram-positive organisms. It is rarely used today and may cause allergic re-

actions, diarrhea, or bleeding related to impaired coagulation.

Moxibustion

A traditional therapy involving heat from burning mugwort near or on selected skin points; burns and other adverse effects can occur.

Moxifloxacin

Moxifloxacin is a fluoroquinolone antibiotic that inhibits bacterial DNA gyrase and topoisomerase IV; tendon injury, neuropathy, QT prolongation, dysglycaemia, and other class toxicities limit use.

Moyamoya Disease

A progressive occlusive arteriopathy of the terminal internal carotid arteries and their proximal branches, with development of fragile collateral vessels at the brain base. It can cause transient ischemic attacks, strokes, seizures, headaches, or cognitive impairment.

MPH

Master of Public Health, a graduate degree focused on population health, disease prevention, health policy, epidemiology, and related public-health practice.

MPO

Myeloperoxidase, an enzyme in neutrophil and monocyte granules that uses hydrogen peroxide and halide ions to generate antimicrobial oxidants; MPO antibodies are also used in evaluating some autoimmune vasculitides.

Mpox

Mpox is a zoonotic infection caused by monkeypox virus, producing fever, lymph-node enlargement, and a characteristic rash or lesions that can occur on skin, mouth, or genital areas. nodes, and a distinctive rash that may include deep-seated vesicles or pustules.

Mpox (Monkeypox)

A zoonotic viral disease caused by the monkeypox

virus, an orthopoxvirus in the same genus as smallpox (variola) and vaccinia. The virus has two clades: clade I (formerly Congo Basin, Central African) with higher virulence and mortality (up to 10 percent), and clade II (formerly West African) with lower mortality (less than 1 percent).

Alternate Names:

monkeypox

MPS I Mucopolysaccharidosis Type I Hurler Syndrome

Mucopolysaccharidosis type I is an inherited lysosomal storage disease caused by alpha-L-iduronidase deficiency. Hurler syndrome is the severe form, with developmental regression, skeletal and airway disease, organ enlargement, corneal clouding, and cardiac involvement; early specialist treatment matters.

Alternate Names:

MPS I (MPS I Mucopolysaccharidosis Type I Hurler Syndrome)

MPS I Mucopolysaccharidosis Type I Hurler Syndrome (MPS I)

Alternate index label for MPS I Mucopolysaccharidosis Type I Hurler Syndrome.

MPS II Mucopolysaccharidosis II Hunter Syndrome (Hunter Syndrome Mucopolysaccharidosis II MPS)

Alternate index label for Hunter Syndrome Mucopolysaccharidosis II MPS.

MPTP

A neurotoxin that can produce permanent parkinsonism by damaging dopaminergic neurons in the substantia nigra.

MR Angiography

Vascular imaging using MRI without ionizing radiation. Time-of-flight or contrast-enhanced (CE-MRA). Used for cerebral aneurysms, renal artery stenosis, peripheral arterial disease.

MR Enterography

MRI of the small bowel after oral contrast administration. Superior soft tissue contrast for evaluating Crohn disease activity, fistulas, and abscesses without ionizing radiation. Demonstrates mural thickening, edema, enhancement patterns, and extraintestinal complications. Diffusion-weighted imaging helps assess disease activity.

MR Neurography

Magnetic resonance imaging optimized to show peripheral nerves and abnormalities such as swelling, compression, or injury.

MRA

Magnetic resonance angiography, a noninvasive MRI-based technique that produces images of blood vessels and can help evaluate aneurysms, stenoses, malformations, and atherosclerosis.

MRCP

Magnetic resonance cholangiopancreatography, an MRI technique that noninvasively images the bile ducts, gallbladder, and pancreatic duct without requiring endoscopic catheterization.

MRD

Minimal residual disease: a small number of malignant cells remaining after treatment, detectable by sensitive methods such as flow cytometry or PCR even when routine tests show remission.

MRI

Magnetic resonance imaging uses a strong magnetic field, radiofrequency energy, and computer processing to produce detailed images of body tissues without ionizing radiation. Contrast material may be used when clinically appropriate.

MRI And Low Back Pain

MRI may be used for low back pain when serious disease is suspected, neurologic deficits are present, or symptoms persist despite appropriate care; many uncomplicated cases do not need early MRI.

MRI Brain

Excellent for brain pathology: tumors, stroke, demyelination, infection, degeneration. Sequences: T1, T2, FLAIR, DWI, SWI, contrast-enhanced. DWI detects acute ischemia within minutes.

MRI Contrast

A contrast agent administered during magnetic resonance imaging to improve visibility of selected tissues or abnormalities.

MRI Joint

Evaluation of musculoskeletal structures: cartilage, ligaments, tendons, menisci, labrum. Knee: ACL, meniscus tears. Shoulder: rotator cuff, labrum. Hip: labral tears, FAI.

MRI Sequences

T1-weighted: anatomy, fat bright. T2-weighted: pathology, fluid bright. FLAIR: suppresses CSF, lesions bright. DWI: restricted diffusion bright (acute stroke, abscess). STIR: fat suppression, fluid bright.

MRI-Targeted Fusion Prostate Biopsy

An advanced diagnostic biopsy technique for prostate cancer detection that digitally co-registers and overlays pre-acquired multiparametric magnetic resonance imaging (mpMRI) images with real-time transrectal ultrasound. Suspicious lesions identified on mpMRI (scored using the PI-RADS classification) are digitally mapped and tracked, enabling the urologist to accurately guide biopsy needles directly into the targeted index lesions under real-time ultrasound imaging. Compared with conventional random systematic 12-core TRUS biopsies, MRI-fusion biopsy significantly improves the detection rate of clinically significant prostate cancers while reducing the overdiagnosis of clinically insignificant indolent tumors.

Alternate Names:

Fusion Biopsy; MRI-Ultrasound Fusion Biopsy

Mrna Precursor

An initial RNA transcript that is processed by capping, splicing, and polyadenylation to form mature messenger RNA.

MRSA

Methicillin-resistant *Staphylococcus aureus*, resistant to beta-lactam antibiotics through acquisition of the *mecA* gene encoding PBP2a. MRSA is a major cause of healthcare-associated and community-associated infections. Treatment options include vancomycin, daptomycin, and linezolid.

MRSA Decolonization

Measures intended to reduce MRSA carriage on the skin or in the nose. Protocols may include topical agents and antiseptic bathing, but the indication, regimen, and duration depend on the person's risk, infection history, setting, and local guidance.

MRSA Infection

Methicillin-resistant *Staphylococcus aureus* is a strain of staphylococcus resistant to several beta-lactam antibiotics. It can colonize without illness or cause skin, wound, lung, bone, joint, or bloodstream infection; culture-guided treatment and hygiene reduce spread.

Alternate Names:

Methicillin-Resistant *Staphylococcus Aureus* Infection

MS Alternative And Complementary Therapies (Alternative Therapy For Multiple Sclerosis)

Alternate index label for Alternative Therapy for Multiple Sclerosis.

MS/MS

Tandem mass spectrometry, an analytical technique in which ions are selected, fragmented, and analyzed by a second stage of mass spectrometry; it is used to identify and quantify molecules in clinical and research samples.

MSG Symptom Complex

A group of symptoms such as headache, flushing, or palpitations attributed by some people to monosodium glutamate; evidence for a consistent syndrome is limited and symptoms may have other causes.

MST

Multiple subpial transection, a neurosurgical procedure in which shallow cuts are made in the cerebral cortex to interrupt seizure pathways when the epileptogenic region cannot safely be removed.

MTOR

Mammalian target of rapamycin. Controls cell growth, proliferation. Inhibited by rapamycin.

Mu-Chain Disease

A rare B-cell disorder in which abnormal immunoglobulin heavy chains are produced, sometimes causing anemia, enlarged lymph nodes, or other features of lymphoproliferative disease.

Mucilage

A viscous plant-derived polysaccharide or similar substance that swells in water and may act as a soothing or bulking agent.

Mucin

A glycoprotein forming the principal component of mucus and helping protect epithelial surfaces.

Mucinous

Mucinous means containing, producing, or resembling mucin. In pathology, the term describes lesions or tumors with extracellular or intracellular mucin and must be interpreted with the organ, architecture, and other histologic features.

Mucinous Adenocarcinoma

Mucinous adenocarcinoma is a gland-forming cancer that produces abundant mucus and can arise in organs such as the colon, pancreas, lung, or breast.

Mucinous Borderline Ovarian Tumor

An ovarian mucinous epithelial tumor with cellular proliferation and atypia but without destructive stromal invasion; it may be large and unilateral.

Mucinous Cyst Adenomas Pancreas (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Mucinous Cyst Of The Vulva

A benign mucus-filled cyst of the vulva, arising from obstruction or metaplastic mucinous epithelium in a vulvar gland or duct; persistent or atypical masses require examination and possible biopsy.

Mucinous Cystadenoma

A mucinous cystadenoma is a usually benign cystic epithelial tumor that produces mucin, commonly arising in the ovary or pancreas. Large lesions may cause pain or pressure, and pathology is needed to distinguish benign from borderline or malignant mucinous tumors.

Mucinous Ovarian Carcinoma

A malignant ovarian epithelial tumor composed of invasive mucin-producing cells; metastatic gastrointestinal or pancreatobiliary carcinoma must be excluded.

Mucocele

Mucus extravasation cyst of minor salivary gland. Lower lip most common. Treatment: excision.

Mucocutaneous Lymph Node Syndrome

Alternate index label; see *Kawasaki disease*.

Mucocutaneous Lymph Node Syndrome (Kawasaki Disease)

Alternate index label for Kawasaki Disease.

Mucoepidermoid Carcinoma

Mucoepidermoid carcinoma is a cancer containing mucus-producing and squamous-like cells, most often arising in salivary glands or the lungs.

Mucoid

Resembling mucus or having a thick, slippery, mucus-like appearance. The word describes a substance or finding and does not by itself identify its cause.

Mucolipidoses

mucolipidoses is a fatty acid or other lipid-related molecule whose chain structure influences mem-

branes, energy storage, signaling, and metabolism.

Mucolytic Therapy

Medications that reduce mucus viscosity and elasticity, facilitating clearance. N-acetylcysteine (NAC) breaks disulfide bonds in mucus glycoproteins. Dornase alfa (Pulmozyme) cleaves extracellular DNA in CF sputum. Hypertonic saline aerosol draws water into the airway surface liquid.

Mucolytics

Medicines that reduce the viscosity of mucus and facilitate its clearance from the airways.

Mucopolysaccharides

Mucopolysaccharides, or glycosaminoglycans, are long sugar chains in connective-tissue matrix; abnormal breakdown causes mucopolysaccharidosis storage disorders.

Mucopolysaccharidoses

Mucopolysaccharidoses are inherited lysosomal storage diseases in which glycosaminoglycans accumulate, causing variable skeletal, organ, airway, neurologic, and developmental problems.

Mucopolysaccharidoses II Hunter Syndrome (Hunter Syndrome Mucopolysaccharidosis II MPS)

Alternate index label for Hunter Syndrome Mucopolysaccharidosis II MPS.

Mucopolysaccharidosis

A group of inherited lysosomal-storage disorders in which glycosaminoglycans accumulate, causing variable skeletal, organ, neurologic, and developmental abnormalities.

Mucopolysaccharidosis Type I

Mucopolysaccharidosis type I is an inherited alpha-L-iduronidase deficiency ranging from severe Hurler syndrome to milder forms, with skeletal, airway, cardiac, eye, and neurologic disease.

Mucopolysaccharidosis Type II

Mucopolysaccharidosis type II, or Hunter syndrome, is an X-linked iduronate-2-sulfatase deficiency causing accumulation of glycosaminoglycans and multi-system disease.

Mucopolysaccharidosis Type III

Mucopolysaccharidosis type III, or Sanfilippo syndrome, is an inherited disorder of heparan-sulfate breakdown causing progressive neurodevelopmental and behavioral decline.

Mucopolysaccharidosis Type IV

Mucopolysaccharidosis type IV, or Morquio syndrome, is an inherited disorder of keratan-sulfate breakdown causing skeletal and connective-tissue abnormalities with variable neurologic involvement.

Mucopolysaccharidosis VII

Mucopolysaccharidosis VII, or Sly syndrome, is a lysosomal storage disorder caused by deficient beta-glucuronidase activity. Accumulation of glycosaminoglycans can cause skeletal, connective-tissue, visceral, developmental, and sometimes neurologic abnormalities.

Mucopus

Pus mixed with mucus, commonly seen in infected mucous membranes or sinuses.

Mucor

Mucor is a genus of molds; related Mucorales fungi can cause mucormycosis, a serious invasive infection mainly in people with severe immune or metabolic problems.

Mucor Ramosissimus

Mucor ramosissimus is an environmental mold that rarely causes opportunistic infection.

Mucorales

Mucorales is an order of molds that includes fungi capable of causing mucormycosis, a rapidly invasive infection of the sinuses, lungs, skin, or other tissues.

Mucormycosis

Mucormycosis is an invasive infection by Mucorales fungi, typically affecting immunocompromised people and causing rapidly destructive sinus, orbital, pulmonary, cutaneous, or disseminated disease. Mucor, and Rhizomucor species. The infection primarily affects immunocompromised patients, particularly those with uncontrolled diabetes mellitus and hematologic malignancies. The fungi invade blood vessels causing thrombosis and tissue infarction.

Mucosa

Mucosa is the moist epithelial lining of body passages such as the mouth, nose, airways, gastrointestinal tract, and reproductive or urinary tracts.

Mucosa-Assisted Lymphoid Tissue Lymphoma

Alternate index label; see *Extranodal marginal zone lymphoma of mucosa-associated lymphoid tissue*.

Mucosal

Pertaining to a mucous membrane (mucosa), the moist, epithelial lining that lines body cavities and hollow organs opening to the external environment (gastrointestinal, respiratory, urogenital tracts). Composed of an overlying epithelial layer, a vascular connective tissue lamina propria, and often a thin smooth-muscle muscularis mucosae. Functions in selective barrier defense, nutrient absorption, and mucosal immunity, supported by mucosal-associated lymphoid tissue and plasma cell secretion of dimeric secretory immunoglobulin A (sIgA) to neutralize luminal pathogens without inducing tissue-damaging inflammation.

Mucosal Immunity

Mucosal immunity is immune defense at surfaces lining the respiratory, gastrointestinal, genitourinary, and other tracts. It includes epithelial barriers, mucus, secretory IgA, antimicrobial molecules, innate cells, and locally regulated adaptive responses.

Mucosal Infection

An infection involving a moist epithelial lining such as the mouth, eye, nose, airway, gastrointestinal

tract, or genital tract.

Mucosal Irritation

Chemical injury to mucous membranes. Eyes, nose, throat, GI tract. Treatment: irrigation, supportive care.

Mucosal Melanoma

A melanoma arising from melanocytes in mucosal surfaces such as the sinonasal tract, oral cavity, anorectal region, or genital tract; it is uncommon and often diagnosed at an advanced stage.

Mucosal Warts

HPV-associated warts arising on mucous membranes, including the oral, genital, or anorectal mucosa.

Mucositis

Inflammation and ulceration of the oral or gastrointestinal mucosa caused by chemotherapy or radiation, causing pain, dysphagia, and poor intake and predisposing to local infection.

Mucous

Relating to mucus, the viscous fluid produced by mucous membranes and glands that lubricates surfaces and helps trap particles and microorganisms.

Mucous Cysts

Small fluid-filled cysts near the end joint of a finger, often associated with osteoarthritis.

Alternate Names:

Digital mucous cyst

Mucous Membrane

A moist tissue lining body cavities that open to the outside, including the nose, mouth, and lungs.

Alternate Names:

Mucosa

Mucous Membrane Pemphigoid

A chronic autoimmune subepidermal blistering disease affecting mucous membranes, especially the eyes and mouth; scarring can impair vision or func-

tion.

Mucus

A slippery secretion containing mucins, water, salts, and cells that protects and lubricates mucous membranes.

Mucus Plugging

Accumulation of thick, tenacious mucus within airways causing obstruction. It is common in asthma, CF, bronchiectasis, and chronic bronchitis. It impairs gas exchange, promotes infection, and causes atelectasis. Treatment includes airway clearance techniques, mucolytics, and bronchodilators.

Muehrcke'S Nails

Paired transverse white bands in the nail beds, usually associated with hypoalbuminemia or other systemic illness. The bands remain fixed as the nail grows because they arise from the nail bed rather than the nail plate.

Muenke Syndrome

A craniosynostosis syndrome caused by a pathogenic FGFR3 variant. Premature fusion of one or more skull sutures may produce an abnormal head shape and can be accompanied by developmental, hearing, limb, or neurological findings.

MUGA Scan (Multigated Acquisition Scan)

ECG-gated radionuclide imaging using technetium-labeled red blood cells to measure ventricular ejection fraction and wall motion. It can be used when reproducible ejection-fraction measurement is needed, although echocardiography and cardiac MRI are also important alternatives.

Alternate Names:

Multigated Acquisition Scan

Mulberry Molars

Permanent molars with irregular, multiple rounded cusps, classically associated with congenital syphilis.

Mule-Spinner'S Cancer

An obsolete occupational term for scrotal squamous-cell cancer associated historically with soot exposure in chimney sweeps and textile workers.

Mulibrey Nanism

Mulibrey nanism is a rare inherited multisystem disorder, usually caused by biallelic variants in *TRIM37*. It may cause severe growth restriction, characteristic facial features, pericardial or hepatic disease, insulin resistance, and other organ involvement.

Muller Cells

Supporting glial cells of the retina that help nourish retinal nerve cells and maintain the retina's structure and chemical environment.

Multi-Infarct Dementia

Cognitive decline caused by the cumulative effects of multiple small strokes.

Alternate Names:

Vascular dementia

Multi-Minicore Disease

An autosomal recessive congenital myopathy caused in most cases by mutations in the selenoprotein N (*SEPN1*) gene or the ryanodine receptor 1 (*RYR1*) gene. Characterized histopathologically by the presence of multiple, small, circumscribed, longitudinally restricted zones of sarcomeric disorganization devoid of oxidative enzyme activity (minicores) throughout muscle fibers. Clinically features neonatal hypotonia, early progressive axial and neck muscle weakness, rigid spine syndrome, progressive scoliosis, and early disproportionate diaphragmatic weakness leading to nocturnal hypoventilation and respiratory failure.

Alternate Names:

MmD; Minicore Myopathy; Multicore Disease

Multicenter Study

A multicenter study is a clinical or research investigation conducted at two or more sites under a coordinated protocol. It can improve enrollment and generalizability but requires harmonized meth-

ods, quality control, and site-specific oversight.

Multicentric

Having more than one center or focus; in oncology, describing separate primary tumor foci, often also called multifocal or polycentric.

Multicystic Dysplastic Kidney

A congenital renal anomaly where one or both kidneys are replaced by multiple non-communicating cysts of varying sizes, with no functioning renal parenchyma. The most common cause of an abdominal mass in a newborn (unilateral). The ureter is typically atretic. The contralateral kidney may have VUR (20%) or UPJ obstruction (10%).

Multidetector Computed Tomography

MDCT is CT imaging using a two-dimensional array of detector elements to acquire multiple slices simultaneously, increasing acquisition speed and enabling high-resolution applications such as CT angiography and virtual colonoscopy.

Multidisciplinary Disaster Victim Identification

The coordinated process of identifying victims of mass-fatality incidents (natural disasters, transport crashes, terrorist attacks) using multiple disciplines: fingerprinting, dental records, DNA profiling, forensic anthropology/radiology, and personal effects, with ante-mortem data collection and comparison.

Multidisciplinary Team

Rehabilitation team includes: physiatrist, physical therapist, occupational therapist, speech-language pathologist, psychologist, social worker, case manager, nursing, recreation therapist.

Multidrug-Resistant TB

TB resistant to at least isoniazid and rifampin. Requires second-line drugs for 18–24 months. Worse outcomes and higher mortality than drug-susceptible TB.

Multienzyme Complexes

Groups of enzymes arranged together so sequential reactions occur efficiently within a shared molecular complex.

Multifactorial

Caused or influenced by multiple genetic, environmental, behavioral, or other factors.

Multifocal

Arising from or involving two or more distinct foci or sites, such as separate tumor foci or vascular lesions.

Multifocal Atrial Tachycardia

An atrial tachyarrhythmia with at least three distinct P-wave morphologies, irregular rhythm, and a rate above 100 beats per minute. It is strongly associated with severe pulmonary disease, hypoxia, and metabolic stress.

Multigravida

A person who has been pregnant two or more times, regardless of outcome.

Multilocular

Having many compartments or locules, as in a cyst or lesion.

Multimodal Analgesia

Use of analgesic techniques with different mechanisms to improve pain control while reducing reliance on any single drug, especially opioids. Components may include acetaminophen, nonsteroidal anti-inflammatory drugs, regional anesthesia, local anesthetics, and selected adjuvants.

Multimodal Imaging

A diagnostic or laboratory method related to multimodal imaging. Results are interpreted with the method, reference range, and clinical findings.

Multimorbidity

The coexistence of two or more chronic health conditions in the same person, without requiring one

to be considered the primary disease. It complicates treatment, increases polypharmacy risk, and requires coordinated patient-centered care.

Multinodular Goiter

An enlarged thyroid containing multiple nodules. Thyroid function may be normal, underactive, or overactive; evaluation commonly includes thyroid-function tests and ultrasound, with further assessment of suspicious nodules or compressive symptoms.

Multinucleated Hepatocyte

A multinucleated hepatocyte is a liver cell containing two or more nuclei, usually from nuclear division without complete cytokinesis. It may be a normal or reactive finding, but its significance depends on the number, morphology, and associated liver abnormalities.

Multipara

A person who has had two or more pregnancies that reached the stage of potential fetal viability, regardless of whether the pregnancies resulted in live births.

Multiparametric MRI Of The Prostate

MRI protocol combining T2-weighted, diffusion-weighted, and dynamic contrast-enhanced sequences to detect and characterize clinically significant prostate cancer. Findings are commonly reported using the PI-RADS system.

Multiparous

Having given birth two or more times to a fetus or fetuses beyond the stage of viability.

Multiple Bacterial Abscesses

Numerous circumscribed collections of pus within an organ or tissue, caused by hematogenous seeding or direct extension of bacterial infection. They appear as multiple, low-attenuation lesions on imaging, often with peripheral enhancement. Common sites include the liver, brain, and kidneys.

Multiple Biliary Hamartomas

Benign, congenital malformations of the intrahepatic bile ducts appearing as multiple, small, well-defined cystic lesions scattered throughout the liver parenchyma on imaging. They are typically asymptomatic and discovered incidentally. They must be distinguished from metastatic disease and Caroli disease.

Multiple Endocrine Neoplasia

An inherited disorder causing tumors or overactivity in two or more endocrine glands.

Multiple Endocrine Neoplasia Type 2A

An autosomal dominant hereditary cancer syndrome caused by activating germline mutations in the *RET* proto-oncogene on chromosome 10. Characterized by the triad of medullary thyroid carcinoma (occurring in almost 100% of affected individuals and prompting prophylactic thyroidectomy in childhood), pheochromocytoma (typically bilateral, 50%), and primary hyperparathyroidism due to parathyroid hyperplasia (20–30%).

Alternate Names:

MEN 2A; MEN2A; Sipple syndrome

Multiple Endocrine Neoplasia I

MEN type 1 is an inherited tumor syndrome involving parathyroid, pituitary, and pancreatic or duodenal neuroendocrine tumors, often caused by MEN1 variants.

Multiple Endocrine Neoplasia II

MEN type 2 is an inherited syndrome with medullary thyroid cancer, pheochromocytoma, and sometimes parathyroid disease, commonly involving RET variants.

Multiple Endocrine Neoplasia Type 1

An autosomal dominant tumor syndrome caused by MEN1 gene mutation (menin).

Multiple Endocrine Neoplasia Type 2

An inherited tumor syndrome caused by pathogenic variants in the RET gene. Type 2A is associated

with medullary thyroid carcinoma, pheochromocytoma, and hyperparathyroidism. Type 2B is associated with medullary thyroid carcinoma, pheochromocytoma, mucosal neuromas, and a marfanoid habitus.

Multiple Endocrine Neoplasia Type 2B

An inherited RET-associated syndrome characterized by medullary thyroid carcinoma, pheochromocytoma, mucosal neuromas, and a marfanoid habitus. The thyroid-cancer risk is high and requires early specialist genetic and surgical assessment.

Multiple Exostoses

Multiple exostoses, or hereditary multiple osteochondromas, is an inherited disorder causing multiple cartilage-capped bony projections near growth plates. Lesions can deform bones, restrict motion, compress nerves or vessels, and rarely undergo malignant transformation.

Multiple Fibrofolliculomas

Multiple fibrofolliculomas are benign hair-follicle tumors appearing as numerous small skin-colored papules, especially on the face and trunk. They are characteristic of Birt–Hogg–Dubé syndrome, which also carries risks of renal tumors and spontaneous pneumothorax.

Multiple Inherited Schwannomas, Meningiomas, And Ependymomas

Alternate index label; see *NF2-related schwannomatosis (neurofibromatosis type 2)*.

Multiple Mononeuropathy

Multiple mononeuropathy is patchy dysfunction of separate individual nerves, often caused by vasculitis, diabetes, infection, inflammation, or infiltrative disease.

Multiple Myeloma

A cancer of plasma cells that usually develops in the bone marrow and often produces a monoclonal immunoglobulin. It can cause bone lesions, anemia, kidney dysfunction, high calcium, infection, or other organ damage.

Alternate Names:

Cancer, Multiple Myeloma Kahlers Disease (Multiple Myeloma)

Multiple Myeloma: Deadly To Chronic

Multiple myeloma is a plasma-cell cancer that produces a monoclonal protein and can cause bone lesions, anemia, kidney injury, high calcium, infections, and fatigue. It is often treatable as a chronic illness, though relapse and complications remain possible.

Multiple Organ Failure

Multiple organ failure, also called multiple organ dysfunction syndrome, is progressive dysfunction of two or more organ systems, usually during severe critical illness. Causes include sepsis, shock, trauma, and major inflammation; treatment is supportive and directed at the underlying cause.

Multiple Pregnancy

Gestation with more than one fetus: twins (most common), triplets, higher order. Types: dizygotic (two eggs, two placentas—fraternal, more common, familial and increased with IVF) vs. monozygotic (one egg split—identical, sharing determined by timing: dichorionic-diamniotic, monochorionic-diamniotic, monoamniotic, conjoined).

Multiple Sclerosis

A chronic disease of the central nervous system—the brain, spinal cord, and optic nerves. The immune system damages myelin, the protective covering of nerve fibers, which can disrupt messages between the brain and body. Symptoms may include vision problems, numbness, weakness, balance problems, fatigue, or bladder and bowel difficulties. The disease may occur in attacks with periods of improvement, progress gradually, or follow another pattern. Diagnosis is based on the medical history, examination, magnetic resonance imaging, and sometimes other tests.

Alternate Names:

Sclerosis, Multiple

Multiple Sclerosis

Alternate index label for MS.

Multiple Sclerosis

Alternate index label for Multiple Sclerosis.

Multiple Sclerosis MS And Pregnancy

Multiple sclerosis usually does not prevent pregnancy, but symptoms, disease-modifying medicines, relapse risk, delivery planning, and postpartum care require coordination. Some treatments must be stopped or changed, and relapse risk can shift after delivery.

Multiple Sclerosis MS Early Signs And Types

Multiple sclerosis is an immune-mediated disease of the central nervous system with episodes or progression of visual, sensory, motor, balance, bladder, cognitive, or fatigue symptoms. Relapsing, progressive, and other patterns guide testing and treatment.

Multiple Sleep Latency Test

A daytime sleep study that measures how quickly a person falls asleep during scheduled naps.

Multiple System Atrophy

A progressive, fatal neurodegenerative disorder characterized by varying combinations of autonomic failure, cerebellar ataxia, parkinsonism, and pyramidal signs.

Alternate Names:

Shy-Drager Syndrome

Multiple System Atrophy - Cerebellar Subtype

The cerebellar subtype of multiple-system atrophy is a progressive neurodegenerative disorder causing ataxia, coordination problems, autonomic dysfunction, and sometimes parkinsonism.

Multiple System Atrophy - Parkinsonian Type

The parkinsonian subtype of multiple-system atrophy causes poorly levodopa-responsive rigidity and

slowed movement together with autonomic failure and possible cerebellar signs.

Multiple System Failure

Failure of two or more organ systems, also called multiple-organ failure or multiple-organ dysfunction syndrome; it may occur during severe illness such as sepsis, shock, or major trauma.

Multiple Vitamin Overdose

Multiple vitamin overdose involves toxic or environmental exposure to multiple vitamin overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Multiplex PCR (BioFire, FilmArray)

A molecular test that detects genetic material from multiple organisms in one clinical sample. Different panels are designed for respiratory, gastrointestinal, bloodstream, or other specimens. Results require interpretation with the symptoms and may detect colonization or nonviable organisms as well as active infection.

Alternate Names:

BioFire, FilmArray

Multiplicity

Multiplicity is the state of having more than one occurrence, copy, lesion, tumor, or pregnancy component. Its interpretation depends on context, such as multiple gestation, multifocal disease, or copy number in genetics.

Multipolar Neuron

A multipolar neuron is a nerve cell with one axon and multiple dendrites. It is the most common neuronal form in the central nervous system and includes many motor neurons and interneurons.

Multisystem Inflammatory Syndrome In Children

A serious inflammatory condition occurring after SARS-CoV-2 infection, affecting multiple organs and sometimes causing shock or cardiac dysfunction.

Multivesicular Body

A multivesicular body is an endosomal compartment containing many small intraluminal vesicles. It can fuse with the plasma membrane to release exosomes or with a lysosome to degrade its cargo.

Mummification

Postmortem alteration occurring in hot, dry, arid, ventilated environments where rapid desiccation of tissues occurs before putrefaction can proceed, resulting in dry, leathery, dark brown skin clinging to underlying skeleton.

Mumps

Mumps is a contagious paramyxovirus infection causing painful salivary-gland swelling and fever, with possible meningitis, orchitis, pancreatitis, or hearing complications. include orchitis (rarely causing infertility), meningitis, pancreatitis, and sensorineural hearing loss. Prevention by MMR vaccination. Outbreaks occur in unvaccinated populations. Diagnosis by serology or RT-PCR from saliva.

Alternate Names:

Parotitis (Mumps)

Mumps Virus

A paramyxovirus that causes mumps, a contagious respiratory illness. The virus is transmitted by respiratory droplets. Symptoms include fever, headache, and parotitis. Complications include orchitis, oophoritis, meningitis, and deafness. Diagnosis involves serology and PCR.

Munchausen Syndrome

Alternate index label; see *Factitious disorder*.

Munchausen Syndrome By Proxy

An obsolete term for factitious disorder imposed on another, in which a caregiver falsifies, exaggerates, or causes illness in someone under their care.

Alternate Names:

MSBP (Munchausen Syndrome By Proxy)

Munchausen Syndrome By Proxy

Alternate index label for Munchausen Syndrome by Proxy.

Muparfostat

Muparfostat is a mixture of highly sulfated, phosphorylated mannose oligosaccharides investigated as a heparanase inhibitor and antiangiogenic agent. It has been studied experimentally in cancer and is not a routine approved therapy.

Mupirocin

Mupirocin is a topical antibiotic that inhibits bacterial isoleucyl-tRNA synthetase. It is used for selected skin infections and nasal eradication of certain staphylococci; resistance can develop with inappropriate or prolonged use.

Mural

Relating to a wall, especially the wall of a body cavity, organ, or blood vessel.

Mural Thrombosis

Formation of a blood clot along the wall of a heart chamber or blood vessel rather than freely within the lumen.

Mural Thrombus

A mural thrombus is a blood clot that forms on the wall of a heart chamber or large blood vessel, typically in areas of damaged endothelium, stasis, or turbulent flow.

Muramidase

Muramidase, or lysozyme, is an enzyme that cleaves the peptidoglycan backbone of bacterial cell walls. It is present in secretions and innate immune cells and contributes to defense against susceptible bacteria.

Murmur

An extra sound heard during a heartbeat, usually caused by fast or turbulent blood flow through the heart or nearby blood vessels. Some murmurs are harmless, while others may be caused by heart valve or structural problems.

Murmur Classification

Heart murmurs are classified by timing, grade, shape, location, radiation, and pitch. Timing: systolic (between S1 and S2), diastolic (between S2 and S1), or continuous. Grade: I (barely audible) to VI (heard without stethoscope). Shape: crescendo, decrescendo, or plateau.

Murmur Congenital (Heart Murmur)

Alternate index label for Heart Murmur.

Murmur Heart (Heart Murmur)

Alternate index label for Heart Murmur.

Murphy'S Sign

Inspiratory arrest caused by pain when the right upper quadrant is palpated, traditionally suggesting acute cholecystitis.

Musca

A genus of flies, including the housefly; some species can mechanically carry pathogens.

Muscae Volitantes

Moving spots, threads, or cobweb-like shapes seen in the visual field, usually caused by shadows from vitreous opacities.

Muscarine

A mushroom-derived parasympathomimetic toxin that stimulates muscarinic receptors and can cause salivation, sweating, bradycardia, and bronchospasm.

Muscarinic Antagonist

A muscarinic antagonist blocks acetylcholine at muscarinic receptors in smooth muscle, glands, the heart, and the central nervous system. Examples include atropine, ipratropium, and oxybutynin; effects can include dry mouth, blurred vision, urinary retention, tachycardia, and confusion.

Muscidae

Muscidae is a family of flies that includes the housefly. Some species can mechanically carry pathogens

to food or wounds, and their larvae may occasionally cause myiasis.

Muscimol

Muscimol is a psychoactive compound found in some *Amanita* mushrooms and an agonist at GABA-A receptors. Ingestion can cause altered perception, sedation, agitation, incoordination, or seizures and may require toxicologic evaluation.

Muscle

A specialized contractile tissue responsible for movement, posture, and heat production.

Muscle Aches

Pain or soreness in one or more muscles. Common causes include overuse, injury, infection, medicines, and conditions affecting the whole body.

Muscle Atrophy

Loss or wasting of muscle tissue caused by disuse, nerve injury, inflammation, malnutrition, or systemic disease. It may lead to weakness and reduced function; treatment focuses on the cause, nutrition, and appropriately supervised rehabilitation.

Muscle Biopsy

Removal of a small sample of skeletal muscle for microscopic, histochemical, immunologic, or molecular examination in suspected myopathy, inflammation, infection, or infiltrative disease.

Muscle Cell

muscle cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Muscle Contraction

The shortening or tension generation of muscle produced by the interaction of actin and myosin myofilaments. In skeletal and cardiac muscle, contraction follows excitation-contraction coupling: calcium binds troponin, tropomyosin moves aside, and myosin cross-bridges cycle against actin (sliding fil-

ament mechanism) using ATP.

Muscle Contracture

A sustained, involuntary, non-relaxing shortening and hardening of muscle tissue that resists passive stretch and limits joint range of motion. Occurs either as a mechanical, structural contracture caused by chronic fibrosis and shortening of connective tissue, muscle fibers, and ligaments following prolonged spasticity or immobilization, or as an electrically silent metabolic contracture, which occurs in inborn errors of glycogenolysis (such as McArdle disease) due to complete depletion of intracellular ATP, preventing the dissociation of actin-myosin cross-bridges and calcium reuptake into the sarcoplasmic reticulum.

Alternate Names:

Skeletal Muscle Contracture; Metabolic Contracture

Muscle Cramp

A muscle cramp is a sudden, involuntary, painful contraction of a muscle. It may follow exertion, dehydration, electrolyte disturbance, pregnancy, medication use, nerve disease, or vascular disease; recurrent cramps need assessment of the cause.

Alternate Names:

Cramp, Muscle Muscle Cramps

Muscle Development

Muscle development is the formation, growth, differentiation, and maturation of muscle tissue. It depends on genetic programs, neural input, hormones, nutrition, mechanical loading, and repair processes.

Muscle Disorder

A disease affecting muscle structure or function, causing weakness, pain, cramps, stiffness, or abnormal movement.

Muscle Fatigue

Reduced ability of a muscle to generate force after sustained or repeated activity.

Muscle Fiber

A muscle fiber is an individual muscle cell containing contractile myofibrils. Skeletal muscle fibers vary in size and metabolic properties, while cardiac and smooth muscle cells have distinct structures and functions.

Muscle Fibers

Individual contractile cells that combine to form skeletal muscle.

Alternate Names:

Muscle cells

Muscle Function Loss

Reduced ability of a muscle or muscle group to generate movement or force, caused by muscle disease, nerve injury, neuromuscular-junction disorders, pain, or disuse.

Muscle Hypertrophy

Muscle hypertrophy is enlargement of individual muscle fibers, usually in response to resistance, hormonal stimulation, or compensatory workload. It can be physiologic or associated with disease, depending on the muscle and cause.

Muscle Neoplasm

A muscle neoplasm is an abnormal growth in skeletal or smooth muscle or nearby soft tissue and may be benign or malignant.

Muscle Pain

Pain, soreness, or aching in one or more muscles caused by exertion, injury, infection, inflammation, medicines, or an underlying medical condition. Persistent, severe, or weakness-associated pain needs assessment.

Alternate Names:

Myalgia Muscle Pain (Myalgia)

Muscle Protein

muscle protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute

to disease.

Muscle Relaxant

A muscle relaxant is a medicine that reduces skeletal-muscle spasm or contraction. Centrally acting agents reduce muscle spasm through the nervous system, whereas neuromuscular blockers interrupt transmission at the neuromuscular junction and require respiratory support.

Muscle Relaxants

Medicines that reduce skeletal-muscle spasm or neuromuscular transmission; they include centrally acting drugs and peripheral neuromuscular blockers.

Muscle Rigidity

Muscle rigidity is abnormal resistance to passive movement caused by sustained muscle contraction or altered tone. It occurs in conditions such as Parkinson disease, dystonia, meningitis, neuroleptic malignant syndrome, and tetanus.

Muscle Spasm

Sudden involuntary contraction of a muscle that may cause pain, stiffness, or abnormal posture.

Muscle Spasms

Muscle spasms are sudden involuntary contractions caused by exertion, dehydration, electrolyte disturbance, nerve disease, medicines, pregnancy, or musculoskeletal problems. Recurrent, widespread, painful, or weakness-associated spasms need evaluation for an underlying cause.

Muscle Spindle

A muscle spindle is a sensory receptor within skeletal muscle that detects muscle length and the rate of length change. Its afferent signals contribute to the stretch reflex, muscle tone, posture, and coordinated movement.

Muscle Strain Treatment

A medical procedure, treatment, device, or care topic related to muscle strain treatment; indications, benefits, risks, and follow-up depend on the specific

intervention and patient.

Muscle Strength

Muscle strength is the maximal force a muscle or muscle group can generate under specified conditions. It is assessed by functional or resisted testing and may be reduced by muscle, nerve, neuromuscular-junction, central, or systemic disorders.

Muscle Tissue

A highly specialized tissue capable of contracting to produce body movement, posture, and heat.

Muscle Tonus

Muscle tonus, or muscle tone, is the continuous, low-level activity and resistance to passive stretch present in resting muscle. It depends on spinal and supraspinal reflexes and may be abnormally increased or decreased in neurologic disease.

Muscle Twitching

Brief involuntary contractions of a small group of muscle fibers, often called fasciculations; persistent or widespread twitching may require evaluation.

Muscle Weakness

Reduced ability to generate force with one or more muscles. Causes include nerve or muscle disease, pain, inactivity, medicines, metabolic disorders, and aging.

Muscle-Contraction Headache

A pressure-like headache related to sustained contraction of scalp, neck, or jaw muscles.

Alternate Names:

Tension-type headache

Muscular

Relating to muscle or made of muscle tissue. It may describe strength, body build, pain, weakness, or a structure containing muscle.

Muscular Dystrophy

A group of inherited disorders characterized by pro-

gressive skeletal-muscle weakness and degeneration. The pattern, age of onset, and complications vary by genetic type; some forms also affect the heart, respiratory muscles, or other organs.

Alternate Names:

MD

Muscular Endurance

The ability of a muscle or muscle group to sustain repeated or prolonged effort.

Muscularis

A smooth-muscle layer in the wall of a hollow organ, especially the intestine.

Alternate Names:

Muscular layer

Musculocutaneous Nerve

From lateral cord (C5-C7), piercing coracobrachialis, running between biceps and brachialis to supply these muscles and lateral forearm skin (lateral antebrachial cutaneous nerve). Sole motor supply to the anterior arm compartment. Injury causes weakness of elbow flexion/forearm supination with lateral forearm sensation loss.

Musculoskeletal

Relating to muscles, bones, joints, tendons, ligaments, and related tissues.

Musculoskeletal Deformity

A musculoskeletal deformity is an abnormal shape, alignment, or structure of bone, joint, muscle, or related tissue. It may be congenital, developmental, traumatic, degenerative, neuromuscular, or caused by disease.

Musculoskeletal Disease

A disorder affecting bones, joints, muscles, tendons, ligaments, cartilage, or related connective tissues.

Musculoskeletal Injury

Trauma to muscles, bones, joints, tendons, ligaments, cartilage, or related soft tissues, ranging from strains and sprains to fractures and disloca-

tions.

Musculoskeletal Pain

Pain arising from muscles, bones, joints, ligaments, tendons, or related soft tissues. Common causes include injury, overuse, arthritis, and inflammation; persistent pain, weakness, swelling, or neurologic symptoms should be assessed.

Musculoskeletal System

The musculoskeletal system consists of bones, joints, cartilage, ligaments, tendons, muscles, and associated connective tissues. It provides support, protects organs, enables movement, and contributes to mineral storage and blood-cell production.

Musculus

The formal anatomical Latin noun for muscle (abbreviated extitM. in the singular and extitMm. in the plural), utilized in international Nomina Anatomica terminology to designate individual skeletal muscles based on their geometry, attachments, fiber orientation, or physiological action (e.g., extitMusculus rectus abdominis, extitMusculus sternocleidomastoideus, extitMusculus deltoideus).

MUSE

A medicated urethral system that delivers a pellet into the urethra to treat erectile dysfunction.

Alternate Names:

Medicated urethral system for erection

Mushroom Poisoning

Illness caused by eating toxic fungi, with effects ranging from gastrointestinal upset to liver failure, neurologic toxicity, or death.

Music Therapy

Use of music for therapeutic purposes. Pain reduction, emotional expression, cognitive stimulation.

MuSK-Positive Myasthenia Gravis

A distinct, clinically aggressive subtype of autoimmune myasthenia gravis, accounting for approximately 5–10% of generalized myasthenia gravis

cases (and up to 40–50% of AChR-seronegative generalized cases), mediated by pathogenic IgG4 autoantibodies against muscle-specific tyrosine kinase (*MuSK*), an agrin-activated receptor essential for postsynaptic acetylcholine receptor clustering. Marked by a strong female preponderance, prominent early involvement of bulbar, facial, neck flexor, and respiratory muscles with frequent tongue and facial muscle atrophy, relative sparing of extraocular muscles, high risk of precipitous myasthenic crisis, and characteristically poor responsiveness (or clinical worsening) with standard acetylcholinesterase inhibitors (pyridostigmine).

Alternate Names:

MuSK Myasthenia Gravis; MuSK-Ab-Positive MG

Mussolini'S Disease

An obsolete eponym historically used for a form of peripheral neuropathy; its meaning is inconsistent and should be verified from the original source.

Mustard Compounds

Mustard compounds are chemical agents containing reactive sulfur or nitrogen mustard groups that can alkylate DNA and other molecules. Some are chemotherapy drugs; others are highly toxic chemical-warfare agents.

Mustard Gas

Mustard gas, or sulfur mustard, is a vesicant chemical-warfare agent that damages skin, eyes, and respiratory tissues and can cause blistering, airway injury, and delayed complications. It is not a true gas at ordinary temperatures and has no specific widely available antidote.

Mutagen

A chemical, physical agent, or biological factor capable of causing a genetic mutation. Mutagenicity does not by itself prove that an agent causes cancer.

Mutagenicity Test

A diagnostic or laboratory method related to mutagenicity test. Results are interpreted with the method, reference range, and clinical findings.

Mutant

A mutant is a cell, organism, or genetic sequence carrying a mutation relative to a reference or parental form. The change may have no effect, alter a trait, or cause disease, depending on its location and functional consequence.

Mutant Proteins

mutant proteins is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Mutate

To mutate means to undergo a change in genetic sequence. Mutations arise spontaneously or from replication errors, DNA damage, or mutagen exposure and may be inherited, acquired in somatic cells, harmful, neutral, or beneficial.

Mutation

A heritable or acquired alteration in the DNA sequence. Variants may be single-nucleotide substitutions, insertions, deletions, copy-number changes, or larger rearrangements and may be benign, pathogenic, or of uncertain significance.

Mutation Abnormality

A mutation abnormality is an atypical change in DNA sequence or chromosome structure identified relative to a reference. Its clinical significance must be classified using evidence, such as pathogenic, likely pathogenic, uncertain significance, likely benign, or benign.

Mutation Rate

Frequency of new mutations per generation or per base pair per replication. Varies by gene, genomic region, organism. High mutation rate in repetitive sequences.

Mute

Describing a person who does not speak, whether because of a physical, neurologic, developmental, or psychological reason or by choice; current clinical

language should describe the specific communication limitation or preference.

Mutism

Absence or refusal of speech despite adequate ability to speak, occurring in conditions such as catatonia, selective mutism, or neurologic disease.

Myalgia

Pain, aching, or tenderness originating within skeletal muscle tissue. May be localized (secondary to unaccustomed eccentric exercise causing delayed-onset muscle soreness [DOMS], muscle strain, or local trauma) or diffuse and systemic. Systemic myalgia is a cardinal symptom of acute viral infections (influenza, COVID-19, dengue), autoimmune rheumatologic disorders (polymyalgia rheumatica, systemic lupus erythematosus), drug-induced myopathies (HMG-CoA reductase inhibitors / statin-associated muscle symptoms [SAMS]), endocrine dysfunction (hypothyroidism), and chronic centralized pain syndromes (fibromyalgia). Evaluated with serum creatine kinase to rule out rhabdomyolysis and inflammatory markers.

Alternate Names:

Myodynia

Myalgic Encephalomyelitis

A complex illness characterized by substantial reduction in activity, post-exertional malaise, and unrefreshing sleep, often with cognitive or autonomic symptoms.

Myalgic Encephalomyelitis/Chronic Fatigue Syndrome

A complex illness characterized by substantial reduction in activity, post-exertional malaise, unrefreshing sleep, and cognitive or autonomic symptoms. Management is individualized and focuses on pacing, symptom relief, and treatment of coexisting conditions.

Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS)

A complex illness involving substantial activity-limiting fatigue, post-exertional worsening, unre-

freshening sleep, and often cognitive or orthostatic symptoms.

Alternate Names:

ME/CFS SEID Systemic Exertional Intolerance Disease

Myasthenia

Muscle weakness, often becoming worse with activity and improving with rest. It may refer generally to weakness or to conditions such as myasthenia gravis; drooping eyelids, trouble swallowing, or breathing difficulty require prompt assessment.

Myasthenia Gravis

A chronic autoimmune disease that disrupts communication between nerves and voluntary muscles. It causes weakness that worsens with activity and improves with rest. It commonly affects the eyelids, eye movements, facial muscles, speech, swallowing, neck, arms, or legs. Severe weakness can affect breathing and requires emergency care. Antibody and other tests may help confirm the diagnosis.

Alternate Names:

MG

Myasthenia Gravis

Alternate index label for Myasthenia Gravis.

Myasthenic (Lambert-Eaton) Syndrome

An autoimmune presynaptic neuromuscular-junction disorder caused by antibodies against voltage-gated calcium channels, producing proximal weakness and autonomic symptoms.

Alternate Names:

Lambert-Eaton

Myasthenic Crisis

A life-threatening, acute neurological emergency occurring in patients with myasthenia gravis, defined as exacerbation of myasthenic muscle weakness resulting in acute respiratory failure requiring endotracheal intubation and mechanical ventilation or non-invasive positive pressure ventilation. Triggered in up to 80% of cases by acute intercurrent respiratory tract infections, surgery, pregnancy, emotional stress, or administration of medications that

impair neuromuscular transmission (such as aminoglycosides, fluoroquinolones, macrolides, or beta-blockers). Differentiated from cholinergic crisis by lack of muscarinic toxidrome signs and managed with prompt ventilatory support, plasmapheresis or intravenous immunoglobulin, and aggressive treatment of the underlying trigger.

Alternate Names:

Myasthenia Gravis Crisis; Acute Myasthenic Exacerbation

Myasthenic Syndrome

A disorder of neuromuscular transmission causing fluctuating weakness; it may refer to myasthenia gravis or a congenital or Lambert-Eaton syndrome depending on context.

Mycelium

A mycelium is the network of branching hyphae that forms the vegetative body of a fungus. It grows through soil, tissue, or other substrates and produces or supports formation of reproductive structures.

Mycetoma

A chronic localized infection of skin and subcutaneous tissue caused by fungi or filamentous bacteria, producing swelling, sinuses, and grains.

Mycobacteriaceae

A family of slow-growing or rapidly growing, acid-fast bacteria that includes *Mycobacterium* species causing tuberculosis, leprosy, and nontuberculous infections.

Mycobacterial Culture

Laboratory culture of a clinical specimen for *Mycobacterium* species, including tuberculosis and nontuberculous mycobacteria; cultures may require prolonged incubation.

Mycobacterium

A genus of acid-fast bacteria that includes the causes of tuberculosis and leprosy and several nontuberculous infections.

Mycobacterium Abscessus

A rapidly growing nontuberculous mycobacterium that can cause pulmonary, skin, and soft-tissue infections and is often difficult to treat because of multidrug resistance.

Mycobacterium Avium

A member of the *Mycobacterium avium* complex that can cause lung disease, lymph-node infection, or disseminated infection, especially in people with weakened immunity.

Mycobacterium Avium Complex

A group of nontuberculous mycobacteria that can cause chronic pulmonary disease or disseminated infection, particularly in people with impaired immunity. Treatment depends on the species, site, severity, susceptibility, and patient factors.

Mycobacterium Bovis

A tuberculosis-complex bacterium that mainly infects cattle but can cause tuberculosis in humans through unpasteurized dairy or animal exposure.

Mycobacterium Chelonae

A rapidly growing nontuberculous mycobacterium that can cause skin, soft-tissue, bone, eye, or disseminated infection.

Mycobacterium Flavescens

A rapidly growing or environmental nontuberculous mycobacterium that rarely causes human infection.

Mycobacterium Fortuitum

A rapidly growing nontuberculous mycobacterium that can cause wound, catheter, bone, lung, or other opportunistic infections.

Mycobacterium Gordona

A usually nonpathogenic water-associated nontuberculous mycobacterium that can occasionally cause disease in people with weakened immunity.

Mycobacterium Haemophilum

A nontuberculous mycobacterium that can cause

skin, joint, and bone infections, especially in people with impaired immunity.

Mycobacterium Infection

Infection caused by a *Mycobacterium* species, including tuberculosis, leprosy, and nontuberculous mycobacterial disease.

Mycobacterium Kansaii

A nontuberculous mycobacterium that commonly causes chronic lung disease resembling tuberculosis and can occasionally cause disseminated infection.

Mycobacterium Leprae

An acid-fast bacillus that causes leprosy (Hansen's disease), a chronic infection affecting the skin and peripheral nerves. The organism is transmitted through respiratory droplets, though the exact mechanism is not fully understood. The incubation period is long (2-5 years).

Mycobacterium Marinum

A nontuberculous mycobacterium associated with fish tanks and aquatic exposure that can cause chronic skin and soft-tissue infection.

Mycobacterium Scrofulaceum

A nontuberculous mycobacterium associated especially with cervical lymph-node infection in children and occasional other disease.

Mycobacterium Smegmatis

A rapidly growing environmental mycobacterium that rarely causes opportunistic wound, catheter, or other infection.

Mycobacterium Szulgai

A rare nontuberculous mycobacterium that can cause chronic lung disease or other opportunistic infection.

Mycobacterium Terrae

A nontuberculous mycobacterium found in soil and water that rarely causes human disease.

Mycobacterium Tuberculosis

An acid-fast bacillus that causes tuberculosis, a chronic respiratory infection. The organism is transmitted by respiratory droplets and primarily affects the lungs but can disseminate to other organs. Latent TB infection is asymptomatic and non-transmissible. Active TB presents with cough, haemoptysis, fever, and weight loss.

Mycobacterium Ulcerans

A mycobacterium that causes Buruli ulcer, a chronic destructive skin infection associated with the toxin mycolactone.

Mycobiome

The mycobiome is the community of fungi living in a particular environment or body site, including the human microbiome. Its composition can be influenced by immunity, antibiotics, diet, and disease, and detection does not by itself prove infection.

Mycology

The branch of microbiology that studies fungi, including yeasts, moulds, and mushrooms. Medical mycology focuses on fungal pathogens that cause disease in humans. Fungal infections (mycoses) are classified as superficial, cutaneous, subcutaneous, or systemic.

Mycophenolate Mofetil

Mycophenolate mofetil is a prodrug of mycophenolic acid that inhibits inosine monophosphate dehydrogenase and lymphocyte proliferation. It is used as an immunosuppressant, especially after organ transplantation, and can cause infection, cytopenias, gastrointestinal effects, and serious fetal harm.

Mycophenolic Acid

mycophenolic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Mycoplasma

A genus of very small bacteria lacking a cell wall, including organisms that cause respiratory and gen-

itourinary infections.

Mycoplasma Genitalium

A sexually transmitted bacterium that can cause urethritis, cervicitis, pelvic inflammatory disease, and, less commonly, proctitis. Many infections are asymptomatic. Diagnosis is usually by a nucleic acid amplification test, and macrolide resistance is common, so treatment should follow current susceptibility-informed guidance.

Mycoplasma Hominis

A wall-less bacterium associated with the genital tract that can cause pelvic, postpartum, urinary, wound, or joint infection.

Mycoplasma Orale

A bacterium found mainly in the mouth and respiratory tract that rarely causes opportunistic infection.

Mycoplasma Pneumonia

Atypical pneumonia most common in ages 5–35. Dry cough, low-grade fever, headache. Patchy infiltrates on CXR disproportionate to symptoms (“walking pneumonia”).

Mycoplasma Pneumoniae

A wall-less bacterium that causes atypical respiratory infection, including tracheobronchitis and community-acquired pneumonia.

Mycoplasma Pneumoniae-Associated Mucositis

Inflammatory lesions of the oral, ocular, or genital mucosa occurring as an extrapulmonary manifestation of *Mycoplasma pneumoniae* infection, often presenting with target-like skin lesions.

Mycoplasmal Pneumonia

Pneumonia caused by *Mycoplasma* species, most often *Mycoplasma pneumoniae*, usually producing a gradual cough and fever.

Mycoplasmataceae

A family of very small bacteria that lack a cell wall

and includes *Mycoplasma* and related genera.

Mycosis

Any pathological infectious disease caused by a microscopic fungus (yeast, mold, or dimorphic fungus). Categorized by anatomical depth and invasiveness: superficial mycoses (tinea versicolor, pityriasis), cutaneous mycoses or dermatophytoses (tinea pedis, tinea capitis, tinea corporis caused by *Trichophyton*, *Microsporum*, or *Epidermophyton*), subcutaneous mycoses (sporotrichosis, mycetoma), and invasive systemic/opportunistic mycoses (candidiasis, aspergillosis, mucormycosis, cryptococcosis, and endemic dimorphic mycoses like histoplasmosis and coccidioidomycosis). Diagnosed via direct microscopy (10–20% KOH prep), fungal culture on Sabouraud dextrose agar, histopathology (GMS, PAS stains), and fungal antigen testing (galactomannan, beta-D-glucan).

Mycosis Fungoides

The most common cutaneous T-cell lymphoma, presenting with progressive patches, plaques, and tumors on the skin with an indolent course. Staging and therapy depend on skin involvement and extracutaneous spread.

Alternate Names:

MF

Mycosis Fungoides Lymphoma

Mycosis fungoides is a type of cutaneous T-cell lymphoma that usually begins with persistent patches or plaques on the skin.

Mycotic Aneurysm

An infected arterial wall or aneurysm, usually caused by bacteria rather than fungi. It may develop from bloodstream infection or an infected embolus and can rupture, causing life-threatening hemorrhage.

Mycotoxin

A toxic substance produced by certain fungi, including aflatoxins and ergot alkaloids.

Mydriasis

Dilation of the pupil, produced mainly by sympathetic activity through the dilator pupillae muscle or by reduced parasympathetic activity. Causes include darkness, medicines, toxins, neurologic disease, and ocular injury.

Mydriatic (Mydriatics)

A pharmacological agent or substance that induces pupillary dilation (mydriasis). Functions through one of two distinct autonomic mechanisms: parasympatholytic agents (anticholinergics/cycloplegics such as tropicamide, cyclopentolate, and atropine) that block muscarinic receptors on the circular pupillary sphincter muscle and ciliary body, paralyzing accommodation; or sympathomimetic agents (such as phenylephrine) that stimulate alpha-1 adrenergic receptors on the radially oriented pupillary dilator muscle without inducing cycloplegia. Indicated to facilitate comprehensive diagnostic fundoscopic examinations of the retina and vitreous, and therapeutically in anterior uveitis to alleviate ciliary muscle spasm and prevent posterior synechiae formation; can precipitate acute angle-closure glaucoma in patients with anatomically narrow anterior chamber angles.

Myelin

Myelin is an insulating sheath around nerve fibers that speeds electrical conduction; it is produced by oligodendrocytes in the CNS and Schwann cells in the PNS.

Myelin Sheath

The lipid-rich insulating layer around axons formed by Schwann cells (peripheral nervous system) and oligodendrocytes (central nervous system), which wrap the axon in concentric membrane layers.

Myelination

Myelination is formation of a myelin sheath around an axon by oligodendrocytes in the central nervous system or Schwann cells in the peripheral nervous system. It increases the speed and efficiency of nerve conduction.

Myelitis

Myelitis is inflammation of the spinal cord or, less commonly, other neural tissue described by the term. Transverse myelitis can cause weakness, sensory changes, and bladder or bowel dysfunction and requires urgent evaluation for infectious, immune, vascular, or compressive causes.

Myeloablative Agonist

A myeloablative agent, likely intended by this term, is a treatment that severely or completely destroys bone-marrow blood-forming cells. It is used before selected stem-cell transplants and requires stem-cell rescue because of profound, prolonged cytopenias.

Myeloablative Chemotherapy

Myeloablative chemotherapy destroys or severely suppresses bone-marrow blood-cell production and is often used before a stem-cell transplant.

Myeloblast

An immature precursor of granulocytes in the bone marrow; excess myeloblasts are characteristic of acute myeloid leukemia.

Myelocyte

A developing granulocyte precursor in bone marrow that is more mature than a myeloblast and normally absent or rare in peripheral blood.

Myelocytic Leukaemia

Leukemia arising from the myeloid cell line, including acute myeloid and chronic myeloid leukemia.

Myelodysplastic Neoplasms

Alternate index label; see *Myelodysplastic syndromes*.

Myelodysplastic Or Myeloproliferative Neoplasms

Myelodysplastic and myeloproliferative neoplasms are clonal bone-marrow disorders causing abnormal blood-cell production. They may produce anemia, infection, bleeding, high counts, enlarged spleen, thrombosis, or progression to acute leukemia; clas-

sification and mutation testing guide care.

Myelodysplastic Syndrome

MDS: a group of clonal stem cell disorders characterized by ineffective hematopoiesis, cytopenias, and risk of transformation to acute myeloid leukemia (AML). Features: anemia (most common), neutropenia, thrombocytopenia; dysplasia (abnormal morphology) in marrow and blood; cytogenetic abnormalities (del5q, trisomy 8, del7q).

Myelodysplastic Syndromes

Alternate index label for MDS.

Myeloencephalitis

Myeloencephalitis is inflammation involving the spinal cord and brain, caused by infection, immune disease, toxins, or other neurologic processes.

Myelofibrosis

Myelofibrosis is a myeloproliferative neoplasm in which abnormal marrow signaling causes fibrosis, disordered blood production, anemia, enlarged spleen, and variable constitutional symptoms. hematopoiesis, and JAK2, CALR, or MPL mutations. Presents with anemia, massive splenomegaly, constitutional symptoms (fever, night sweats, weight loss), and leukoerythroblastic blood smear.

Alternate Names:

Idiopathic Myelofibrosis

Myelogram

An imaging study of the spinal canal performed after contrast material is introduced around the spinal cord.

Myelography

Contrast injected into subarachnoid space for spinal imaging. Used when MRI contraindicated. CT myelography more common now.

Myelogenous

Relating to blood-forming cells that develop from the bone marrow's myeloid cell line.

Alternate Names:

Myeloid

Myeloid

Relating to bone marrow or to the non-lymphoid blood-cell lineages derived from hematopoietic stem cells.

Myeloid Cell

myeloid cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Myeloid Leukemia

Myeloid leukemia is a blood cancer arising from abnormal myeloid precursor cells in the bone marrow and blood.

Myeloid Sarcoma

Myeloid sarcoma is a mass of immature myeloid cells occurring outside the bone marrow, often associated with acute myeloid leukemia.

Myelolipoma

A usually benign tumor composed of mature adipose tissue and blood-forming tissue, most often arising in the adrenal gland and often found incidentally.

Myeloma

Multiple myeloma: plasma cell malignancy. Features: bone pain, anemia, renal failure, hypercalcemia.

Myeloma Cast Nephropathy

Light chain cast formation in tubules causing AKI in multiple myeloma. Friable waxy casts with giant cells. Bence Jones proteinuria. Treat myeloma and consider plasmapheresis to reduce light chain burden.

Myelomalacia

Softening of the spinal cord, usually due to ischemia, trauma, hemorrhage, or other cord injury.

Myelomeningocele

Myelomeningocele is an open neural-tube defect in which spinal cord and meninges protrude through a vertebral opening, causing variable paralysis, sensory loss, bladder dysfunction, and hydrocephalus. through a vertebral defect into a sac on the back. Presents with a visible, fluid-filled sac at birth. Associated with hydrocephalus, Arnold-Chiari malformation, and lower extremity paralysis.

Myelomonocyte

A bone-marrow precursor that can develop toward either the monocyte or granulocyte lineages.

Myelopathy

A general term for any neurological deficit resulting from spinal cord pathology, encompassing a wide range of causes and clinical presentations.

Myeloperoxidase

An enzyme in neutrophil and monocyte granules that uses hydrogen peroxide and chloride to generate antimicrobial oxidants; antibodies against it occur in some autoimmune diseases.

Myelopoiesis

The production and development of myeloid blood cells, including granulocytes, monocytes, and some precursor cells, in the bone marrow.

Myeloproliferative Disorders

Clonal bone-marrow disorders causing overproduction of one or more blood-cell lineages, including polycythemia vera, essential thrombocythemia, and myelofibrosis.

Myeloproliferative Neoplasm

A myeloproliferative neoplasm is a blood cancer in which the bone marrow produces too many mature or immature blood cells.

Myelosuppression

Reduced bone marrow production of red cells, white cells, and platelets, causing anemia, neutropenia, and thrombocytopenia; a common dose-limiting tox-

icity of chemotherapy and radiation that increases infection and bleeding risk.

Myelosuppressive Therapy

Myelosuppressive therapy reduces bone-marrow production of blood cells, sometimes intentionally during cancer treatment but with risk of anemia, infection, and bleeding.

Myenteric Plexus

A network of enteric nerves between the circular and longitudinal muscle layers of the gastrointestinal tract that coordinates intestinal motility.

Myhre Syndrome

A rare inherited connective-tissue disorder caused by variants in SMAD4 and characterized by short stature, stiff joints, distinctive facial features, hearing loss, and cardiovascular abnormalities.

Myiasis

Infestation of living tissue by larvae (maggots) of various fly species, which feed on host tissue and can cause pain, secondary infection, and tissue destruction. It may be cutaneous, involving the skin and subcutaneous tissue, or affect other body sites including the nose, ears, and wounds.

Myoblast

An embryonic precursor cell that fuses or differentiates to form skeletal muscle fibers.

Myoblastoma

A rare tumor composed of cells resembling developing muscle; the exact entity depends on the source and modern classification.

Alternate Names:

Aerophagia; stomach bubble syndrome

Myocardial

Relating to the myocardium, the muscular wall of the heart.

Myocardial Biopsy

Removal of a small sample of heart muscle, usually

by catheter, for microscopic or molecular examination of suspected myocarditis, rejection, infiltrative disease, or other cardiomyopathy.

Myocardial Bridging

Myocardial bridging occurs when a segment of a coronary artery tunnels through heart muscle and may be compressed during contraction.

Myocardial Contusion

A bruise of the heart muscle caused by blunt chest trauma, which may produce chest pain, arrhythmias, or impaired heart function.

Myocardial Infarction

Irreversible death of heart-muscle tissue caused by prolonged inadequate blood flow. See Heart Attack.

Myocardial Infarction (Heart Attack)

Alternate index label for Heart Attack.

Myocardial Ischemia

Reduced blood flow and oxygen supply to heart muscle, usually from narrowed coronary arteries, potentially causing chest pain or heart damage.

Alternate Names:

Cardiac Ischemia

Myocardial Perfusion Imaging

Nuclear cardiology technique using radiotracers (Tc-99m sestamibi, Tl-201) to assess blood flow to the myocardium. Performed at rest and during stress (exercise or pharmacological) to detect reversible ischemia (stress defect with rest redistribution) or fixed defects (scar from prior MI).

Myocardial Reperfusion

Restoration of blood flow to heart muscle after ischemia, which can limit injury but may also cause reperfusion-related tissue damage.

Myocardial Rupture

A tear in the heart wall or other heart structure, usually a severe complication of myocardial infarction.

Myocardial Viability

PET assessment of hibernating myocardium. FDG uptake indicates viable myocardium.

Myocarditis

Myocarditis is inflammation of heart muscle, often infectious or immune-mediated, that can cause chest pain, arrhythmias, heart failure, or sudden cardiac dysfunction. (coxsackievirus, adenovirus, SARS-CoV-2). Presents with chest pain, dyspnea, arrhythmias, and elevated troponin. Diagnosis: ECG, echocardiography, cardiac MRI with Lake Louise criteria, and endomyocardial biopsy.

Myocarditis - Pediatric

Pediatric myocarditis is inflammation of the heart muscle, often after infection or immune injury, causing chest symptoms, arrhythmia, heart failure, or shock.

Myocarditis Diagnosis

Cardiac MRI (Lake Louise criteria) shows myocardial edema (T2-weighted imaging), late gadolinium enhancement (typically epicardial/mid-wall in non-ischemic distribution), and increased myocardial T1/T2 mapping values.

Myocarditis Treatment

No specific antiviral therapy for most cases. Treatment is supportive: heart failure therapy (diuretics, ACE inhibitors, beta-blockers) for reduced EF, arrhythmia management, and avoidance of exercise during acute phase (risk of sudden death). Immunosuppression may be considered in autoimmune myocarditis or cardiac sarcoidosis.

Myocardium

The muscular middle layer of the heart wall that contracts to pump blood.

Myoclonic Epilepsy

An epilepsy syndrome in which seizures include brief, shock-like involuntary muscle jerks; the specific syndrome determines associated seizure types, age of onset, and prognosis.

Myoclonic Seizure

A generalized seizure characterized by brief, shock-like, involuntary single or multiple muscle contractions without loss of consciousness. Commonly affects the upper extremities, resulting in sudden dropping or throwing of objects, and is a hallmark feature of juvenile myoclonic epilepsy.

Myoclonus

A sudden, brief, involuntary muscle jerk or group of jerks.

Myocyte

A differentiated muscle cell specialized for contraction. Skeletal myocytes are long, multinucleated fibers; cardiac myocytes are branching cells joined by intercalated discs; and smooth-muscle myocytes are spindle-shaped cells without organized sarcomeres. Myocyte injury or death contributes to myopathy, rhabdomyolysis, and myocardial infarction.

Alternate Names:

Muscle cell

Myodegeneration

Degeneration or breakdown of muscle fibers, caused by inherited, inflammatory, ischemic, toxic, or other disorders.

Myoepithelial Cell

myoepithelial cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Myoepithelioma

A myoepithelioma is a tumor composed of myoepithelial cells and can be benign or malignant, commonly arising in salivary or other glands.

Myoepithelioma Of Soft Tissue

A rare soft-tissue neoplasm showing myoepithelial differentiation, ranging from benign to malignant. Diagnosis uses morphology and immunohistochemistry, with treatment guided by behavior and complete excision.

Myofascial Pain Syndrome

A regional pain disorder involving sensitive trigger points in muscle and fascia, with pain that may be referred to other areas and often worsens with pressure or movement.

Myofibril

A contractile thread-like structure inside a muscle fiber, composed of repeating sarcomeres containing actin and myosin.

Myofibrillar Myopathy

Myofibrillar myopathy is a group of inherited muscle disorders with abnormal protein aggregates and progressive skeletal-muscle weakness, sometimes involving heart or breathing muscles.

Myofibrils

Contractile structures inside muscle fibers made of repeating sarcomeres.

Myofibroblastoma

A benign mesenchymal tumor composed of myofibroblastic cells in collagenous or myxoid stroma, most often arising in the mammary or inguinal region but potentially occurring at other soft-tissue sites.

Myofibroblastoma Of The Breast

A rare benign stromal tumor composed of spindle-shaped myofibroblastic cells, usually occurring in older adults. It is typically a well-circumscribed painless mass and is diagnosed by histology with immunohistochemistry.

Myofibroblasts

Cells that combine features of fibroblasts and smooth muscle cells and help contract wounds and produce scar tissue.

Myofibromatosis

Myofibromatosis is a benign proliferation of myofibroblastic tumors, often in infants or children and involving skin, muscle, bone, or internal organs.

Myofilament

A contractile protein filament within a muscle fiber; actin and myosin myofilaments interact through cross-bridges to produce muscle contraction.

Myofilaments

Actin and myosin protein filaments that slide past each other to produce muscle contraction.

Alternate Names:

Contractile filaments

Myogenesis

myogenesis is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

Myogenin

A muscle-specific transcription factor required for skeletal-muscle differentiation and used as an immunohistochemical marker in diagnosing some soft-tissue tumors.

Myoglobin

An oxygen-binding protein in skeletal and cardiac muscle; it can enter blood and urine after muscle injury.

Myoglobin Blood Test

A blood test measuring myoglobin released from injured skeletal or cardiac muscle; it rises early after muscle injury but is nonspecific and is not preferred over troponin for myocardial infarction.

Myoglobin Urine Test

A urine test detecting myoglobin released during muscle breakdown, used in evaluating rhabdomyolysis; substantial myoglobinuria can contribute to acute kidney injury.

Myoglobinuria

Presence of myoglobin in urine, usually due to muscle breakdown such as rhabdomyolysis.

Myokymia

Involuntary, fine, continuous rippling contractions of muscle fibers, often seen around the eye but also associated with nerve hyperexcitability or neurologic disease.

Myoma

A benign tumor arising from muscle tissue, commonly referring to a uterine leiomyoma.

Myomalacia

Softening or degeneration of muscle tissue, including the heart muscle after severe injury.

Myomectomy

Surgical removal of uterine fibroids while preserving the uterus.

Myometrium

The thick smooth-muscle layer of the uterus responsible for contractions during menstruation and labor.

Myopathy

A disease of muscle causing weakness, fatigue, pain, or abnormal muscle enzymes.

Myopericytoma

A myopericytoma is a usually benign perivascular tumor composed of cells around small blood vessels, often occurring in skin or soft tissue.

Myopia

Short-sightedness, a refractive error in which distant objects focus in front of the retina and appear blurred.

Myosin

A motor protein that interacts with actin to generate muscle force.

Myosin Atpase

The enzyme activity of myosin that uses ATP to power interactions with actin and generate muscle contraction.

Myosis (Miosis)

Abnormal, excessive pupillary constriction, typically resulting in a pupil diameter < 2 mm under ambient lighting conditions. Results from either hyperactivation of the parasympathetic pupillary constrictor pathway (cranial nerve III via Edinger-Westphal nucleus) or loss of sympathetic pupillary dilator tone. Cardinal clinical causes include acute opioid intoxication (classic "pinpoint pupils"), organophosphate or nerve agent toxicity (cholinergic excess), pontine hemorrhage, Horner syndrome (unilateral miosis with ptosis and anhidrosis), and topical miotic eye drops (e.g., pilocarpine for angle-closure glaucoma).

Myositis

Inflammation of skeletal muscle, caused by autoimmune disease, infection, medicines, or other injury.

Myositis Ossifican

Abnormal bone formation within muscle or other soft tissue, often after injury; the usual term is myositis ossificans.

Myositis Ossificans

Formation of mature bone within muscle or other soft tissue, usually after trauma or repeated injury.

Myotatic (Stretch Reflex)

Pertaining to the myotatic reflex (deep tendon reflex / muscle stretch reflex), a fundamental monosynaptic spinal reflex that resists unexpected passive elongation of a skeletal muscle to maintain muscle tone and posture. Initiated when rapid stretch activates intrafusal primary (Ia) sensory afferents within neuromuscular spindles; these fibers enter the dorsal horn of the spinal cord and synapse directly on alpha motor neurons innervating the same (homonymous) muscle, eliciting rapid muscle contraction while simultaneously inhibiting the antagonist muscle via Ia inhibitory interneurons (reciprocal inhibition). Clinically assessed at the patellar, Achilles, biceps, and triceps tendons to evaluate lower and upper motor neuron integrity.

Myotonia

Delayed relaxation of a muscle after voluntary con-

traction or stimulation, often caused by an inherited or acquired muscle-channel disorder.

Myotonia Congenita

An inherited skeletal-muscle channelopathy causing delayed relaxation after voluntary contraction. Stiffness often improves with repeated movement and may be accompanied by muscle hypertrophy; onset and severity vary between genetic forms.

Myotonia Fluctuans

A mild, atypical form of non-dystrophic skeletal muscle sodium channelopathy belonging to the potassium-aggravated myotonias, caused by heterozygous missense mutations in the *SCN4A* gene on chromosome 17q23. Characterized by fluctuating, episodic muscle stiffness and delayed relaxation that is characteristically exacerbated or triggered 10 to 30 minutes following strenuous physical exercise (delayed post-exercise stiffness) or by oral potassium loading, but completely lacks the cold-sensitivity characteristic of paramyotonia congenita and is not associated with episodes of permanent or prolonged muscle weakness.

Alternate Names:

Potassium-Aggravated Myotonia Fluctuans

Myotonia Permanens

A severe, continuous form of non-dystrophic sodium channelopathy caused by specific heterozygous gain-of-function missense mutations in the *SCN4A* gene (such as p.Gly1306Glu). Clinically characterized by persistent, severe, generalized muscle stiffness that is continuous and present even at rest, profoundly exacerbated by potassium intake, and often presenting in infancy with severe generalized stiffness, facial and laryngeal myotonia, stridor, and life-threatening episodic respiratory insufficiency due to persistent rigidity of intercostal and laryngeal musculature.

Alternate Names:

Severe Sodium Channel Myotonia; Continuous Myotonia

Myotonic Disorder

A neuromuscular disorder characterized by delayed relaxation after muscle contraction, often with weak-

ness, stiffness, or progressive muscle wasting.

Myotonic Dystrophy

The most common adult-onset muscular dystrophy, an autosomal dominant multisystem disorder caused by CTG trinucleotide repeat expansions in the DMPK gene (type 1, DM1) or CCTG tetranucleotide repeat expansions in the CNBP gene (type 2, DM2) on chromosome 19q13.3. DM1 is more common and more severe than DM2.

Myotonic Dystrophy Type 2

An inherited multisystem disorder caused by CCTG repeat expansion in CNBP, producing myotonia, proximal or axial weakness, muscle pain, cataracts, and variable endocrine or cardiac features.

Myriad

Myriad means a very large or indefinite number; in healthcare writing, specific counts or ranges are preferable when they are known.

Myringitis

Inflammation of the tympanic membrane, sometimes producing painful blisters and associated hearing symptoms.

Myringotomy

A small incision in the eardrum, often used to drain middle-ear fluid and insert a ventilation tube.

Myringotomy With Tympanostomy Tube

Incision of the tympanic membrane with placement of a ventilation tube to equalize middle-ear pressure and permit drainage. It is used selectively for persistent middle-ear effusion, recurrent acute otitis media, or associated hearing and developmental problems.

Myristic Acid

A saturated 14-carbon fatty acid found in some animal fats, coconut oil, and palm oil; high dietary intake can raise LDL cholesterol.

Myristica Oil Poisoning

Myristica oil poisoning involves toxic or environmental exposure to myristica oil poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Myroides

A genus of gram-negative environmental bacteria that can rarely cause opportunistic infections, often with notable antibiotic resistance.

Myroides Odoratus

A gram-negative environmental bacterium that can rarely cause wound, bloodstream, urinary, or other opportunistic infections.

Myxedema

Severe hypothyroidism causing accumulation of mucin in tissues, with swelling, slowed body functions, and other symptoms.

Myxedema Coma

Severe decompensated hypothyroidism characterized by altered mental status, hypothermia, bradycardia, hypoventilation, hyponatremia, and hypoglycemia. Most commonly occurs in elderly patients with long-standing untreated hypothyroidism precipitated by infection, surgery, or sedative medications. Mortality rates exceed fifty percent.

Alternate Names:

Coma Myxedema (Myxedema Coma)

Myxococcales

An order of soil bacteria known for coordinated movement, predation on other microbes, and formation of multicellular fruiting structures.

Myxococcus

A genus of predatory soil bacteria that move in groups and form fruiting bodies when nutrients are scarce.

Myxofibroma

A myxofibroma is a benign tumor containing fibrous and gelatinous myxoid tissue, often arising in the

jaw or other connective tissue.

Myxoid Cyst

A myxoid cyst is a gelatinous fluid-filled lesion near a joint or tendon, commonly on a finger near the distal joint and nail.

Myxoma

A usually benign primary cardiac tumour composed of myxoid tissue, most often arising in the left atrium from the interatrial septum. It may cause positional obstruction of the mitral valve, constitutional symptoms, or systemic embolism; echocardiography is the usual initial diagnostic test.

Myxomycete

A slime mold, a fungus-like organism that forms mobile amoeboid stages and spore-producing structures; it is not a typical human pathogen.

Myxopapillary Ependymoma

Myxopapillary ependymoma is a usually slow-growing spinal tumor arising from ependymal cells, commonly in the conus or filum terminale.

Myxosarcoma

Myxosarcoma is a rare malignant soft-tissue tumor containing abundant gelatinous or myxoid material.

Alternate Names:

It may cause vaginal bleeding, cramping, and an open cervix and requires clinical assessment. Incomplete abortion; incomplete pregnancy loss.

MZL

Alternate index label; see *Marginal zone lymphoma*.

N

N-Acetylcysteine

An antidote used mainly for acetaminophen overdose. It helps protect the liver and may still help when treatment starts late or liver injury has already begun.

Na (Sodium)

A clinical or physiological concept involving na (sodium). Its meaning and clinical importance depend on the relevant body system and context.

Na⁺/K⁺ ATPase

A cell-membrane pump that moves sodium out of cells and potassium into cells. It helps maintain the electrical balance needed for nerves, muscles, and other cell functions.

Nabothian Cyst

A benign mucus-retention cyst of the uterine cervix caused by obstruction of an endocervical gland. It is common, usually asymptomatic, and generally requires no treatment.

Nabumetone

A nonsteroidal anti-inflammatory drug used to reduce pain and inflammation in selected musculoskeletal disorders.

NAD⁺

Nicotinamide adenine dinucleotide in its oxidized form, a coenzyme that carries electrons in cellular energy metabolism.

NAD⁺ (Nicotinamide Adenine

Dinucleotide)

A helper molecule that carries energy during the chemical reactions cells use to turn food into energy. It changes between NAD⁺ and NADH forms.

Alternate Names:

Nicotinamide Adenine Dinucleotide

Nadir

The lowest point of something. After chemotherapy, it often refers to the time when the white blood cell count is lowest and infection risk is highest.

Naegle Rule

A way to estimate a baby's due date by adding 7 days and about 9 months to the first day of the last menstrual period. It is most accurate when menstrual cycles are regular.

Naegleria Fowleri

A free-living amoeba that can cause primary amoebic meningoencephalitis after contaminated water enters the nose. The infection is rare but rapidly progressive, with fever, severe headache, meningism, altered consciousness, seizures, and high mortality.

Naegleria Infection

A rare infection caused by the free-living amoeba *Naegleria fowleri*, usually acquired when contaminated warm fresh water enters the nose. It can cause rapidly progressive primary amoebic meningoencephalitis with severe headache, fever, vomiting, confusion, seizures, and coma. It is a medical emergency.

Naevus

The British spelling of nevus, a usually benign localized skin or mucosal lesion such as a mole.

NAFLD

Alternate index label; see *Fatty liver disease*.

Nail

A protective plate of hardened keratin covering the dorsal surface of the distal phalanges of the fingers and toes.

Nail Abnormalities

Changes in nail color, shape, thickness, texture, or attachment caused by trauma, infection, inflammatory disease, systemic illness, medication, or congenital conditions.

Nail Care

A medical procedure, treatment, device, or care topic related to nail care; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Nail Care For Newborns

A medical procedure, treatment, device, or care topic related to nail care for newborns; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Nail Changes

Alterations in nail color, shape, texture, or attachment that may result from local injury, skin disease, infection, medicines, nutritional deficiency, or systemic illness. Their significance depends on the pattern and other clinical findings.

Nail Clubbing

Abnormal enlargement of the distal digit with increased nail curvature and loss of the normal nail-fold angle. It may be familial or associated with pulmonary, cardiovascular, gastrointestinal, or malignant disease and should be interpreted with the clinical context.

Nail Fungus

A fungal infection of a fingernail or toenail causing thickening, discoloration, brittleness, distortion, or separation from the nail bed.

Alternate Names:

Toenail Fungus

Nail Furrows

Nail furrows are grooves or depressions running across or along a nail; transverse Beau lines can follow systemic illness or trauma, while longitudinal ridges are often age-related but may reflect local or systemic disease.

Nail Infection

Infection of the nail or surrounding tissue, commonly caused by fungi or bacteria. It may cause thickening, discoloration, separation of the nail, pain, swelling, or drainage; diabetes and poor circulation increase the risk of complications.

Nail Injuries

Trauma to the nail plate, nail bed, nail folds, or fingertip that may cause bruising, laceration, avulsion, fracture, or later nail-growth abnormalities.

Nail Pits

Small, sharply circumscribed depressions in the nail plate caused by abnormal focal keratinization of the proximal nail matrix. They are classically associated with psoriasis but may also occur in alopecia areata, eczema, lichen planus, or other nail disorders.

Nail Polish Poisoning

Nail polish poisoning involves toxic exposure to nail polish poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Nail Psoriasis

Nail psoriasis affects 50 percent of patients with psoriasis and 80 percent with nail involvement having joint disease. Features include pitting (most common), onycholysis (nail separation), oil drop sign (salmon patch), subungual hyperkeratosis, and nail

plate crumbling. Nail psoriasis is associated with psoriatic arthritis.

Nail-Patella Syndrome

A rare inherited disorder, usually caused by a change in the *LMX1B* gene, affecting the fingernails, kneecaps, elbows, and pelvis. The kneecaps may be small, absent, or dislocated, and some people develop glaucoma or kidney disease. It is usually inherited in an autosomal-dominant pattern.

Nalbuphine

An opioid analgesic with mixed agonist-antagonist activity, used for moderate to severe pain.

Nalidixic Acid

An older quinolone antibacterial formerly used for some urinary-tract infections.

Naloxone

A medicine that blocks opioids and can quickly restore breathing during an opioid overdose. It comes as a nasal spray or injection. Emergency help is still needed because its effect may wear off before the opioid does.

Naltrexone

A longer-acting opioid antagonist used in selected treatment of opioid or alcohol use disorder.

Nanism

An older term for dwarfism or markedly short stature; modern usage specifies the underlying condition.

Nanocolloid

Tc-99m labeled colloidal particles for lymphatic imaging. Particle size 10-100 nm. Used for sentinel node detection.

Nanomedicine

The use of very small materials or devices in medicine to help diagnose, prevent, monitor, or treat disease. Examples include nanoparticle drug carriers, imaging agents, biosensors, and some vac-

cine or cancer treatments; their effects and safety depend on the material, size, dose, and how the body handles them.

Nanosomia

Abnormally small body size or stature; the term is uncommon and context-dependent.

Nantenine

An alkaloid found in *Nandina domestica*; it has pharmacologic activity but is not a routine clinical medicine.

Napelline

A toxic diterpenoid alkaloid related to aconite, capable of causing neurologic and cardiac poisoning.

Naphazoline

A topical alpha-adrenergic agonist used as a nasal decongestant or ocular vasoconstrictor; prolonged use can worsen congestion.

Naphthalene Poisoning

Naphthalene poisoning involves toxic exposure to naphthalene poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Naproxen

A nonsteroidal anti-inflammatory drug used for pain, fever, and inflammation.

Naproxen Sodium Overdose

Naproxen sodium overdose involves toxic exposure to naproxen sodium overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Narcissism

Excessive self-focus or need for admiration; clinically significant persistent patterns may occur in narcissistic personality disorder.

Narcissistic Personality Disorder

A pattern of grandiosity, need for admiration, and

lack of empathy.

Alternate Names:

NPD (Narcissistic Personality Disorder) Personality Disorder, Narcissistic

Narcissistic Personality Disorder

Alternate index label for Narcissistic Personality Disorder.

Narcolepsy

Narcolepsy is a chronic sleep-wake disorder with excessive daytime sleepiness and, in some people, cataplexy, sleep paralysis, and hypnagogic hallucinations. sleep attacks, with or without cataplexy (sudden bilateral loss of muscle tone triggered by strong emotions). Narcolepsy type 1: with cataplexy, caused by loss of hypocretin/orexin neurons in the hypothalamus.

Alternate Names:

Excessive Daytime Sleepiness Sleepiness, Daytime

Narcotic

A historical term for opioid drugs and, in legal usage, certain controlled substances; meaning varies by context.

NARP Syndrome

A maternally inherited mitochondrial disorder caused by a heteroplasmic point mutation at nucleotide position 8993 of mitochondrial DNA (most commonly *m.8993T>G* or *m.8993T>C*) in the *MT-ATP6* gene, which encodes subunit 6 of mitochondrial ATP synthase (Complex V). The clinical acronym reflects its core manifestations: Neurogenic muscle weakness, Ataxia, and Retinitis Pigmentosa. Typically manifests with proximal sensory neuropathy, progressive cerebellar ataxia, pigmentary retinal retinopathy, developmental delay, and dementia, corresponding to moderate mutant heteroplasmy levels (70–90%), whereas higher heteroplasmy (> 90%) results in the more severe, fatal maternally inherited Leigh syndrome.

Alternate Names:

Neurogenic Muscle Weakness, Ataxia, and Retinitis Pigmentosa Syndrome

Narrow Angle Glaucoma

A condition in which the drainage area inside the eye is unusually narrow, raising the risk of a sudden rise in eye pressure. Severe eye pain, blurred vision, halos, nausea, or vomiting requires emergency care.

Narrow Therapeutic Index

A medicine for which the helpful dose is close to the harmful dose. It requires careful dosing and sometimes blood tests to avoid toxicity.

Narrow-Complex Tachycardia

A fast heart rhythm that usually starts in the upper chambers of the heart. The electrical signals travel through the usual pathways, producing narrow beats on an ECG.

Narrowband UVB Phototherapy

A treatment that uses carefully measured ultraviolet light for conditions such as psoriasis, vitiligo, and eczema. Treatment requires eye protection and monitoring of the skin.

Nasal

Pertaining to the nose (*extitnasus*) or the internal nasal cavity, the primary upper respiratory airway responsible for warming, humidifying, and filtering inspired air, as well as providing olfaction and speech resonance. Divided by the midline osteo-cartilaginous nasal septum and flanked laterally by three scroll-like turbinates (conchae) that increase mucosal surface area. Houses the Kiesselbach plexus (Little area) on the anterior septum, the site of over 90% of anterior epistaxis episodes. Serves as a direct route for topical and systemic intranasal drug delivery (e.g., corticosteroids for allergic rhinitis, naloxone for acute opioid overdose).

Nasal Cannula

A device with two small prongs placed in the nostrils to deliver supplemental oxygen. The flow rate is adjusted to the patient's needs and the device used.

Nasal Cavity

The paired space inside the nose, extending from the nostrils to the nasopharynx. It warms, humidifies,

and filters inspired air and contains the receptors responsible for smell.

Nasal Congestion

A blocked or stuffy nose caused by swelling inside the nasal passages. It can occur with colds, allergies, sinus disease, irritants, or structural problems.

Nasal Congestion And Rhinorrhea

Nasal obstruction or stuffiness together with a runny nose, commonly caused by viral infection, allergic or nonallergic rhinitis, sinus disease, irritants, or structural blockage. Persistent unilateral symptoms, recurrent bleeding, fever, facial swelling, or breathing difficulty requires evaluation.

Nasal Corticosteroid Sprays

Nasal corticosteroid sprays reduce mucosal inflammation and congestion in allergic rhinitis, nasal polyps, and some chronic sinus disorders; correct direction away from the septum limits irritation and nosebleeds.

Nasal Endoscopy

An examination in which a thin camera is used to look inside the nose and upper throat. It can help find swelling, polyps, a foreign object, bleeding, or a growth.

Nasal Flaring

Widening and outward movement of the nostrils during breathing. It is a visible sign of increased work of breathing, particularly in infants and children, and may accompany airway obstruction, bronchiolitis, pneumonia, asthma, or severe systemic illness.

Nasal Implant

A device placed in or near the nose to support nasal structures or improve airflow. Its purpose depends on the device and condition.

Nasal Mucosal Biopsy

Removal of a small sample of nasal lining for microscopic, immunologic, or microbiologic examination of suspected inflammatory, infectious, neoplastic, or

infiltrative disease.

Nasal Obstruction

Blocked airflow through one or both nasal passages. Causes include congestion, allergy, infection, a deviated septum, polyps, trauma, or a mass. Persistent one-sided obstruction or bleeding requires medical evaluation.

Nasal Passage

The nasal passages are air channels inside the nose that warm, humidify, filter, and conduct air toward the throat and lungs.

Nasal Polyps

Soft, usually noncancerous growths in the lining of the nose or sinuses. They can cause a blocked nose, runny nose, and reduced sense of smell.

Alternate Names:

Polyps, Nasal

Nasal Septal Hematoma

A collection of blood inside the wall dividing the nostrils, usually after a nose injury. It can block breathing and damage the nose if it is not drained promptly.

Nasal Septum

The wall that divides the nasal cavity into right and left sides. It consists mainly of cartilage in front and bone behind and helps support the nose and direct airflow.

Nasal Septum Perforation

A hole through the cartilage or bone separating the nasal passages, which may cause crusting, nosebleeds, whistling, or obstruction and can result from trauma, surgery, intranasal drugs, or inflammatory disease.

Nasal Stent

Support for nasal septum/airway. Materials: silicone, metal. Indications: septal hematoma, post-septoplasty support, nasal valve collapse. Complications: crusting, infection, pressure necrosis.

Nasal Vestibulitis

Inflammation or infection of the nasal vestibule, commonly caused by *Staphylococcus* bacteria and producing tenderness, crusting, redness, or small pustules near the nostrils.

Nasogastric Feeding Tube

A flexible tube passed through the nose into the stomach to deliver enteral nutrition or medicines, or to decompress or aspirate the stomach.

Nasogastric Tube

A tube passed through the nose into the stomach to remove contents or deliver nutrition, fluids, or medicines when clinically indicated.

Nasolabial Folds

Skin creases running from the sides of the nose toward the corners of the mouth.

Nasopharyngeal Airway

A soft tube inserted through the nose to maintain a passage between the nares and pharynx. It may be useful in a spontaneously breathing patient with partial upper-airway obstruction but is relatively contraindicated in suspected basal skull fracture or significant nasal trauma.

Nasopharyngeal Carcinoma

A malignancy arising from the epithelial lining of the nasopharynx, and it has a unique geographical and ethnic distribution, with highest incidence in southern China, Southeast Asia, and North Africa.

Alternate Names:

Cancer, Nasopharyngeal

Nasopharyngeal Culture

A laboratory culture of a specimen from the nasopharynx used to identify selected respiratory pathogens; molecular tests are often more sensitive for many organisms.

Nasopharynx

The upper part of the pharynx behind the nose, above the soft palate and connected to the auditory

tubes.

Natal Cleft

The groove between the buttocks, also called the intergluteal or sacral cleft.

Natal Teeth

Teeth present at birth, most often involving the lower central incisors. They may be loose, interfere with feeding, or injure the infant's tongue and require dental assessment.

National Formulary

A clinical or physiological concept involving national formulary. Its meaning and clinical importance depend on the relevant body system and context.

Natriuresis

The physiological or pharmacological excretion of sodium in the urine. Driven by neurohormonal mediators—most notably atrial natriuretic peptide and B-type natriuretic peptide, released from cardiac atria and ventricles in response to myocardial stretch and intravascular volume expansion—which inhibit renal tubular sodium reabsorption in the collecting ducts and suppress renin and aldosterone secretion. Pathologic pressure natriuresis occurs with elevated renal perfusion pressure; pharmacologically induced by diuretic agents (loop, thiazide, and potassium-sparing diuretics) to treat fluid overload and hypertension.

Natriuretic

Pertaining to, stimulating, or promoting the excretion of sodium ions in the urine (natriuresis). Refers to endogenous peptide hormones (atrial natriuretic peptide [ANP], B-type natriuretic peptide [BNP], C-type natriuretic peptide [CNP], and urodilatin) that promote urinary sodium and water loss while reducing systemic vascular resistance, as well as pharmacological diuretic medications. In clinical diagnostics, elevated serum concentrations of BNP or N-terminal pro-BNP (NT-proBNP) serve as primary biomarkers for diagnosing heart failure and monitoring hemodynamic congestion.

Natural Immunity

A clinical or physiological concept involving natural immunity. Its meaning and clinical importance depend on the relevant body system and context.

Natural Killer Cell

An immune cell that can quickly destroy virus-infected or abnormal cells without needing prior exposure to a specific germ.

Natural Killer Cells

Immune cells that destroy infected or abnormal cells without prior exposure to a specific antigen.

Alternate Names:

NK cells

Natural Short Sleeper

A natural short sleeper is a person who functions well with substantially less sleep than usual without daytime sleepiness, often because of an inherited sleep pattern.

Naturopathy

Naturopathy is a system of complementary care emphasizing lifestyle and natural products; evidence and safety vary, and proven medical treatment should not be delayed.

Nausea

An unpleasant sensation of an impending need to vomit, often accompanied by autonomic symptoms such as salivation, sweating, pallor, or dizziness. Common causes include gastrointestinal illness, pregnancy, medication effects, migraine, vestibular disorders, metabolic disease, and serious abdominal or neurologic conditions.

Nausea And Acupressure

Acupressure may help some people with nausea by applying pressure to selected points, but evidence varies and serious or persistent nausea requires evaluation.

Nausea And Vomiting

Nausea is the urge to vomit; vomiting is forceful

emptying of stomach contents. Causes include infection, medicines, pregnancy, migraine, obstruction, and other illnesses.

Nausea And Vomiting During Early Pregnancy

Nausea and vomiting are common in early pregnancy, but persistent symptoms with inability to keep fluids down, weight loss, dehydration, abdominal pain, fever, blood, or altered mental status may indicate hyperemesis gravidarum or another condition and require medical evaluation.

Nausea And Vomiting – Adults

Nausea and vomiting in adults can result from gastrointestinal disease, infection, medication, pregnancy, metabolic illness, neurologic disease, or obstruction; dehydration and alarm features guide urgency.

Navel

The navel, or umbilicus, is the scar left after the umbilical cord separates and marks the former connection between fetus and placenta.

Navicular

A boat-shaped bone in the middle of the foot that helps support the arch. Injury to it can cause foot pain and may heal slowly because part of the bone has a limited blood supply.

NCS

NCS commonly means nerve-conduction study, which measures electrical signals in peripheral nerves to evaluate neuropathy, nerve compression, or other disorders.

NDM-1

An enzyme made by some bacteria that makes them resistant to many important antibiotics. Infections caused by these bacteria can be difficult to treat and require laboratory testing and careful infection control.

Nearsightedness

Nearsightedness, or myopia, occurs when light focuses in front of the retina, causing distant blur that is corrected with lenses, contact lenses, or selected procedures. light focuses in front of the retina. It is corrected with diverging lenses, contact lenses, or refractive surgery.

Alternate Names:

myopia, in which distant objects appear blurred because.

Nebulizer

A device that converts a liquid medicine into an inhalable aerosol, using a jet compressor, ultrasonic mechanism, or vibrating mesh. Medication is delivered through a mouthpiece or face mask and may include a bronchodilator, corticosteroid, antibiotic, or saline solution when prescribed for the relevant condition. Nebulizers can be useful when a person cannot coordinate an inhaler, is very young, or is severely breathless, but a properly used inhaler with a spacer may be equally effective for many medicines. Treatment choice depends on the medication, device, patient, and clinical situation. Follow the prescribed dose and treatment time, and clean and dry the device according to the manufacturer's and medication instructions to reduce contamination.

Alternate Names:

Nebuliser

NEC

Alternate index label; see *Neuroendocrine tumors*.

Necator

A genus of hookworms that inhabit the small intestine and can cause iron-deficiency anemia.

Necator Americanus

A hookworm species that infects humans through skin penetration by larvae and causes intestinal blood loss.

Neck

The neck connects the head to the trunk and contains the cervical spine, airway, esophagus, thyroid, vessels, nerves, muscles, and lymph nodes.

Neck And Back Pain

Pain in the cervical, thoracic, or lumbar spine and surrounding tissues, commonly caused by strain, degenerative disease, disc problems, or trauma. Weakness, numbness, gait difficulty, bowel or bladder changes, fever, cancer history, or major trauma requires urgent assessment.

Neck Cancer

Human papillomavirus, particularly HPV-16, is an increasingly important etiological factor in head and neck squamous cell carcinoma, particularly oropharyngeal cancer, and it is responsible for the rising incidence of oropharyngeal cancer in younger patients without traditional risk factors of tobacco and alcohol use.

Neck Dissection

Surgical removal of cervical lymph nodes and sometimes surrounding tissues to treat or stage head and neck cancer.

Neck Lump

A localized swelling or mass in the neck caused by enlarged lymph nodes, thyroid or salivary disease, cyst, infection, or tumor; persistent or concerning masses require evaluation.

Neck Mass

A lump or swelling in the neck caused by an enlarged lymph node, congenital cyst, thyroid or salivary-gland lesion, infection, or tumor. A persistent, enlarging, firm, fixed, or unexplained mass—especially in an adult—requires prompt medical evaluation.

Neck Pain

Pain or stiffness in the cervical region, commonly caused by muscle strain, degenerative disease, poor posture, or injury. Severe pain after trauma or pain with weakness, numbness, fever, or neurologic deficits requires prompt assessment.

Alternate Names:

Pain, neck

Alternate Names:

Cervical Pain Cervical Pain (Neck Pain) Pain, Neck

Neck Pain And Dizziness

Neck pain with dizziness may reflect muscle strain, migraine, inner-ear disease, medication, blood-pressure change, or less commonly vascular or neurologic disease. Sudden severe symptoms, weakness, speech or vision change, fainting, or inability to walk require emergency care.

Neck Pain Or Spasms - Self Care

Self-care for uncomplicated neck pain or muscle spasm may include gentle movement, heat or cold, posture changes, and appropriate over-the-counter pain relief; weakness, numbness, fever, trauma, or severe worsening requires medical assessment.

Neck Pain: Symptom

Pain or stiffness in the neck caused by muscle strain, injury, arthritis, disc disease, nerve compression, or other conditions.

Neck X-Ray

Plain radiographic imaging of the cervical spine or neck soft tissues used to evaluate alignment, fractures, degenerative change, foreign bodies, or selected airway abnormalities.

Neck, Chronic Stiff

Chronic neck stiffness may result from muscle strain, arthritis, disc disease, inflammatory illness, or neurologic problems; weakness, fever, trauma, or severe headache needs assessment.

Necrobiosis Lipoidica

A chronic inflammatory skin disorder characterized by yellow-brown, atrophic plaques, usually on the shins. It is associated with diabetes but can occur without diabetes.

Necrobiosis Lipoidica Diabeticorum

A chronic inflammatory skin disorder producing yellow-brown atrophic plaques with a raised border, most often on the shins and associated with diabetes, although it can occur without diabetes.

Necrolytic Migratory Erythema

A recurrent, erosive, erythematous eruption that often affects periorificial and intertriginous skin. It is classically associated with glucagonoma but may have other causes, so evaluation should consider nutritional, hepatic, pancreatic, and metabolic disease.

Necropsy

A post-mortem examination of an animal or human body to determine disease or cause of death.

Necroptosis

A regulated, programmed form of necrosis that shares features of both classical necrosis and apoptosis, serving as a backup cell death mechanism when apoptosis is inhibited. It is particularly important during viral infections, as many viruses encode proteins that block caspase-mediated apoptosis to ensure their own replication.

Necrosis

Necrosis is death of cells or tissue within a living body caused by ischemia, infection, toxins, trauma, inflammation, or other injury.

Necrotizing Anterior Scleritis

A severe, sight-threatening form of scleral inflammation characterised by ischaemic necrosis of the sclera with visible underlying uvea through thinned, grey-white scleral patches.

Necrotizing Enterocolitis

An acquired inflammatory disease that can cause intestinal necrosis, primarily affecting premature infants. Risk factors include prematurity, enteral feeding, and bacterial colonization. It may cause abdominal distension, bloody stools, bilious vomiting, and feeding intolerance. Pneumatosis intestinalis on imaging strongly supports the diagnosis but must be interpreted with the clinical findings.

Necrotizing Fasciitis

A life-threatening soft tissue infection characterized by rapid necrosis of the subcutaneous fascia and subcutaneous tissue, with relative sparing of

the overlying skin in early stages. It is a surgical emergency requiring immediate wide surgical debridement and broad-spectrum antibiotics.

Necrotizing Granuloma

A granulomatous inflammatory lesion containing central tissue necrosis, commonly associated with infections such as tuberculosis or fungal disease but requiring clinical and histologic correlation.

Necrotizing Infections

Severe infections that cause tissue death through microbial invasion, toxin production, thrombosis, or intense inflammation. Examples include necrotizing fasciitis, necrotizing pneumonia, and necrotizing enterocolitis; rapidly progressive pain, swelling, skin changes, systemic toxicity, or organ dysfunction requires emergency evaluation and source control.

Necrotizing Pneumonia

Severe pneumonia in which infection causes necrosis and cavitation of lung tissue, often accompanied by sepsis, respiratory failure, or abscess formation.

Necrotizing Soft Tissue Infection

A rapidly progressive infection of fascia, subcutaneous tissue, or muscle causing severe pain, swelling, systemic toxicity, and tissue necrosis. It is a surgical emergency requiring immediate assessment, broad antimicrobial therapy, and aggressive debridement.

Necrotizing Vasculitis

Inflammation of blood-vessel walls causing fibrinoid necrosis and potential vessel narrowing, thrombosis, hemorrhage, ischemia, or organ injury.

Needle

Hollow metal tube for fluid withdrawal or injection. Gauge: larger number = smaller diameter. Common gauges: 18G (blood draw), 22G (IV), 25-27G (injection).

Needle Biopsy

Removal of cells, fluid, or a tissue sample through a needle for cytologic or histopathologic examina-

tion. Fine-needle aspiration primarily collects cells, whereas core biopsy collects tissue architecture.

Needle Driver

A surgical instrument with gripping jaws and handles used to hold and guide a suture needle through tissue. Its design helps control the needle while protecting the operator's fingers.

Needle Driver (Perineal)

A hand instrument used to hold and guide a suture needle during a perineal repair.

Alternate Names:

Perineal

Needle Driver (Vascular)

A hand instrument used to hold and guide a suture needle during vascular repair.

Alternate Names:

Vascular

Needle Holder

A surgical instrument with jaws used to grasp and guide a suture needle during tissue repair.

Needlestick Injury

A puncture from a needle or sharp object that may expose someone to blood or body fluids; wash the area, report it promptly, and obtain urgent assessment for infection testing and possible HIV or hepatitis post-exposure treatment.

Negative Feedback

A regulatory mechanism in which the output of a system feeds back to decrease or reverse the initiating stimulus, maintaining homeostasis around a set point.

Negative Predictive Value

The chance that a person with a negative test result truly does not have the condition being tested for. This chance depends on how common the condition is and on the person's initial risk.

Negative Pressure Ventilation

Breathing support that uses pressure around the chest to help the lungs expand, instead of pushing air directly through the airway. It is used rarely, including for some neuromuscular conditions.

Negative Symptoms

Reduced normal abilities such as emotional expression, speech, motivation, pleasure, or social interest. They can occur in schizophrenia and other mental-health or neurologic conditions.

Negative-Strand RNA Virus

A virus whose genetic material must first be copied into a readable form before the virus can make its proteins inside a cell.

Neglect

A form of child maltreatment: the failure to provide for a child's basic needs—physical (food, clothing, shelter, medical care), educational, or emotional—causing harm or risk of harm. Features: failure to thrive, poor hygiene, untreated illness/injury, missed healthcare appointments, inadequate supervision, and developmental delay.

Neglect Syndrome

Hemispatial neglect (unilateral neglect, hemineglect) is a neurological condition characterized by failure to attend to, respond to, or orient toward stimuli on the side of space contralateral to the brain lesion, in the absence of primary sensory or motor deficits sufficient to explain the neglect.

Neisseria Gonorrhoeae

A gram-negative diplococcus that causes gonorrhea, a sexually transmitted infection. The organism has developed resistance to many antibiotics including fluoroquinolones and macrolides.

Neisseria Meningitidis

A gram-negative diplococcus that causes meningitis and meningococcemia. The organism has a polysaccharide capsule that is the target of conjugate vaccines. Serogroups A, B, C, W, X, and Y cause most disease. Meningitis presents with fever,

headache, neck stiffness, and altered mental status.

Nelson Syndrome

Enlargement and hypersecretion of an adrenocorticotrophic-hormone-producing pituitary tumour after bilateral adrenalectomy, usually for Cushing disease. It may cause rapid skin hyperpigmentation, headache, visual-field loss, and other effects of an expanding sellar mass.

Nemaline Myopathy

A genetically heterogeneous congenital myopathy characterized histopathologically by the presence of numerous abnormal, thread-like or rod-shaped structures (nemaline rods) composed of α -actinin and actin within muscle fibers on modified Gömöri trichrome staining. Caused by mutations in thin-filament and sarcomeric structural genes, most frequently nebulin (*NEB*) and skeletal muscle α -actin (*ACTA1*). Clinically ranges from severe neonatal forms with profound hypotonia and respiratory failure to mild adult-onset weakness, typically presenting with infantile hypotonia, symmetrical proximal muscle weakness, facial weakness with an elongated “tent-shaped” upper lip, high-arched palate, and early respiratory muscle compromise.

Alternate Names:

Rod Body Myopathy; Nemaline Rod Myopathy

Neoadjuvant Therapy

Treatment before primary treatment to shrink tumor. Examples: chemotherapy before surgery.

Neonatal

Relating to a newborn, especially during the first 28 days after birth.

Neonatal Abstinence Syndrome

A withdrawal syndrome in neonates following in utero exposure to opioids (most common) or other substances (benzodiazepines, SSRIs, alcohol). Onset within 24-72 hours of birth (varies by half-life of substance).

Neonatal Cephalohematoma

A collection of blood beneath the periosteum of a

newborn's skull, usually caused by birth trauma. It is limited by skull sutures and generally resolves over time, although it requires assessment for associated injury or jaundice.

Neonatal Conjunctivitis

Inflammation of the conjunctiva during the first month of life, caused by chemical irritation, bacteria, or viruses acquired during delivery. Purulent discharge, eyelid swelling, and corneal involvement require urgent evaluation.

Neonatal Cystic Fibrosis Screening Test

A newborn screening test, usually measuring immunoreactive trypsinogen with reflex DNA or sweat testing, used to identify infants at increased risk for cystic fibrosis.

Neonatal Diabetes Mellitus

Diabetes diagnosed within the first six months of life. Can be transient or permanent. Transient neonatal diabetes mellitus is caused by abnormalities at chromosome 6q24 and usually resolves but may relapse in adolescence.

Neonatal Herpes Simplex

Herpes simplex virus infection in the first 28 days of life, usually HSV-2 from maternal genital infection. Three forms: skin, eye, and mouth disease; CNS disease; and disseminated disease. Disseminated form mimics sepsis with high mortality. Diagnosis by viral PCR, culture, or Tzanck smear. Treatment: IV acyclovir for 14-21 days.

Neonatal Herpes Simplex Virus

A severe neonatal infection caused by herpes simplex virus, typically acquired during passage through the birth canal, presenting within the first month of life as skin, eye, and mouth disease, central nervous system disease, or disseminated disease with multiorgan involvement.

Neonatal Hyperbilirubinemia

Elevated unconjugated bilirubin levels in newborns exceeding 5 mg per deciliter, potentially causing jaundice and, if severe, kernicterus with neurological

damage. Phototherapy is the standard treatment.

Neonatal Hypocalcemia

Low serum calcium in a newborn, which may cause jitteriness, poor feeding, hypotonia, apnea, or seizures. Causes include prematurity, maternal diabetes, birth asphyxia, and disorders of parathyroid function.

Neonatal Hypoglycemia

Blood glucose less than 40-45 mg/dL in term infants or less than 25 mg/dL in preterm infants. Risk factors include prematurity, small for gestational age, large for gestational age, and maternal diabetes. Presents with jitteriness, lethargy, poor feeding, and seizures. Treatment: early feeding, IV dextrose infusion, and monitoring.

Neonatal Hypothermia

Core body temperature below 36.5 degrees Celsius in a newborn. Classified as mild (36-36.4), moderate (32-35.9), or severe (below 32). Occurs due to rapid heat loss through evaporation, convection, conduction, and radiation. Prevention: skin-to-skin contact, drying, warm environment, and delayed bathing.

Neonatal Hypothyroidism

Insufficient thyroid hormone production in a newborn, which can impair brain development and growth if untreated; newborn screening enables early diagnosis and treatment.

Neonatal Hypoxic-Ischemic Encephalopathy

Brain dysfunction in a newborn caused by inadequate oxygenation and/or cerebral blood flow around the time of birth. Moderate or severe HIE may produce altered consciousness, abnormal tone, seizures, and multiorgan injury; selected term or near-term infants may benefit from early therapeutic hypothermia.

Neonatal Infections

Early-onset sepsis (EOS, <72h): GBS, *E. coli*, *Listeria*. Risk factors: maternal GBS+, chorioamnionitis,

PROM >18h. Kaiser sepsis calculator for management. Late-onset sepsis (LOS, >72h): CoNS, S.

Neonatal Jaundice

Yellow discoloration of a newborn's skin or eyes caused by elevated bilirubin. It is common but may require treatment or investigation when severe, early, or prolonged.

Alternate Names:

Newborn Jaundice (Neonatal Jaundice)

Neonatal Jaundice (Jaundice Of The Newborn)

Yellow discoloration of a newborn's skin or eyes caused by elevated bilirubin. It is often physiologic but requires assessment when severe, early, rapidly increasing, or prolonged.

Alternate Names:

Jaundice of the Newborn

Neonatal Opioid Withdrawal Syndrome

A complex neurobehavioral and autonomic withdrawal syndrome affecting newborns following intrauterine exposure to opioids (including methadone, buprenorphine, heroin, or prescribed analgesics). Clinical manifestations include central nervous system hyperirritability (high-pitched crying, tremors, hypertonia, sleep fragmentation), autonomic dysregulation (sweating, sneezing, nasal congestion, tachypnea, temperature instability), and gastrointestinal dysfunction (poor feeding, uncoordinated sucking, vomiting, diarrhea). Managed using non-pharmacological care (Eat, Sleep, Console [ESC] model, low-stimulation environment, skin-to-skin contact, rooming-in) and pharmacological therapy (morphine or methadone) when supportive measures are insufficient.

Alternate Names:

NOWS; Neonatal Abstinence Syndrome (Opioid); Opioid-Induced NAS

Neonatal Polycythemia

An abnormally high red-cell concentration in a newborn, often defined by a venous hematocrit of at least 65 percent. It may increase blood viscosity and cause respiratory, neurologic, or metabolic symp-

toms.

Neonatal Respiratory Distress Syndrome

Breathing difficulty in a premature newborn caused mainly by insufficient pulmonary surfactant, leading to stiff lungs and impaired oxygen exchange.

Neonatal Resuscitation

The systematic approach to stabilizing a newborn after delivery, guided by the Neonatal Resuscitation Program (NRP).

Neonatal Sepsis

Systemic bacterial infection in the first 28 days of life, classified as early-onset (within 72 hours, usually from vertical transmission) or late-onset (after 72 hours, often nosocomial). Common organisms: Group B Streptococcus, E. coli, Listeria.

Neonatal Skin Conditions

Common and usually benign. Erythema toxicum neonatorum (red papules and pustules on erythematous base, appears day 2 to 3, resolves by 2 weeks). Milia (tiny white cysts on nose and cheeks). Miliaria crystallina (clear vesicles from sweat duct obstruction). Sebaceous hyperplasia (yellow papules on nose and cheeks).

Neonatal Thyrotoxicosis

Transient hyperthyroidism in newborns caused by transplacental transfer of maternal thyroid-stimulating immunoglobulins, presenting with tachycardia, irritability, and poor weight gain in the first weeks of life.

Neonatal Transition

The physiologic adaptation from placental gas exchange to pulmonary respiration after birth. It includes lung aeration, falling pulmonary vascular resistance, closure of fetal circulatory shunts, and changes in cardiac output and thermoregulation.

Neonatal Weight Gain And Nutrition

A clinical or physiological concept involving neonatal weight gain and nutrition. Its meaning and

clinical importance depend on the relevant body system and context.

Neonate

A neonate is a newborn infant during the first 28 completed days after birth, a period requiring attention to feeding, temperature, jaundice, infection, breathing, and transition to extrauterine life.

Neonatologist

A pediatrician who specializes in the medical care of newborn infants, particularly premature and critically ill neonates.

Neoplasia

Abnormal clonal proliferation of cells producing a new growth, called a neoplasm, that may be benign or malignant. Neoplastic growth persists beyond normal regulatory controls and may show genetic, epigenetic, morphologic, or functional abnormalities. Benign neoplasms are usually localized, whereas malignant neoplasms can invade adjacent tissues and metastasize to distant sites. Classification requires integration of morphology, immunophenotype, molecular findings, anatomic behavior, and clinical staging when applicable.

Neoplasm

Abnormal growth of tissue, may be benign or malignant. Benign neoplasms: localized, slow-growing, do not metastasize. Malignant neoplasms (cancer): invasive, metastasize, can be fatal. Classification: carcinoma (epithelial origin), sarcoma (mesenchymal origin), leukemia (hematopoietic), lymphoma (lymphoid).

Neoplasm Thymomic (Thymoma)

Alternate index label for Thymoma.

Neoplastic

Relating to a neoplasm or abnormal new growth of tissue.

Neostigmine

An acetylcholinesterase inhibitor used to reverse se-

lected nondepolarizing neuromuscular blockade and to treat some disorders of neuromuscular transmission or bowel and bladder motility.

Nephoptosis

Abnormal downward movement or mobility of a kidney, also called floating kidney.

Nephrectomy

Surgical removal of a kidney. Radical nephrectomy: kidney + Gerota fascia + adrenal gland (for renal cell carcinoma). Partial nephrectomy (nephron-sparing): for small renal tumors (<7 cm, renal function preservation). Simple nephrectomy: for benign disease (non-functioning kidney, chronic infection, massive hydronephrosis).

Nephric

Relating to the kidney or to embryonic kidney development.

Nephritic Syndrome

Characterized by hematuria with dysmorphic RBCs and RBC casts, oliguria, hypertension, and mild proteinuria less than 3.5 g/day. Results from glomerular inflammation increasing permeability to cells and reducing GFR. Hypertension from sodium and water retention.

Nephritis

Inflammation of kidney tissue, often affecting the glomeruli or interstitium and potentially impairing renal function.

Nephroblastoma

Wilms tumor, a malignant embryonal renal tumor of childhood that commonly presents as an asymptomatic abdominal mass.

Nephrocalcinosis

Nephrocalcinosis is deposition of calcium salts within kidney tissue, caused by metabolic, hormonal, genetic, medication-related, or renal disorders.

Nephrogenic Diabetes Insipidus

Inability of the renal collecting ducts to respond adequately to antidiuretic hormone, causing excessive dilute urine production and intense thirst. Causes include inherited defects, chronic kidney disease, electrolyte disorders, and medication exposure.

Nephrogenic Fibrosing Dermopathy

A rare fibrosing disorder of the skin causing thickening and hardening, primarily of the extremities and trunk, associated with kidney failure and gadolinium-based contrast exposure.

Nephrogenic Systemic Fibrosis

A rare fibrosing disorder associated mainly with exposure to certain gadolinium contrast agents in people with severe kidney impairment.

Nephrolithiasis

Formation of stones in the urinary tract. Calcium-containing stones are common, but stone composition varies and may include uric acid, struvite, or cystine. Risk factors include dehydration, hypercalciuria, hyperoxaluria, hypocitraturia, and low urine volume. Evaluation includes 24-hour urine analysis.

Nephrolithiasis (Kidney Stones)

Alternate index label for Kidney Stones.

Nephrologist

Physician specializing in kidney diseases, including dialysis management and transplant care.

Nephrology

The medical specialty concerned with kidney disease, fluid and electrolyte disorders, hypertension, dialysis, and kidney transplantation.

Nephrolysis

Surgical freeing of a kidney from surrounding adhesions or tissue.

Nephron

Functional unit of the kidney comprising the glomerulus, proximal tubule, loop of Henle, distal

tubule, and collecting duct. Each kidney contains approximately one million nephrons responsible for filtering blood, reabsorbing solutes and water, and excreting waste to form urine.

Nephron Function

The nephron is the functional unit of the kidney, with approximately 1 million per kidney. Each nephron consists of a renal corpuscle (glomerulus and Bowman's capsule) and a renal tubule (proximal convoluted tubule, loop of Henle, distal convoluted tubule, and collecting duct).

Nephronophthisis

An autosomal-recessive tubulointerstitial kidney disease causing progressive loss of concentrating ability, polyuria, polydipsia, anemia, and chronic kidney failure. It may be isolated or part of a ciliopathy with retinal, skeletal, hepatic, or neurological manifestations.

Nephropathy

Disease or damage of the kidneys, which may impair filtration, fluid balance, or other kidney functions.

Nephroptosis (Floating Kidney)

Abnormal downward movement of a kidney when a person stands, sometimes causing flank discomfort, urinary symptoms, or intermittent obstruction.

Alternate Names:

Floating Kidney

Nephrosclerosis

Hardening and scarring of kidney tissue, commonly caused by long-standing hypertension or vascular disease.

Nephrosis

An older broad term for noninflammatory kidney disease, historically associated particularly with nephrotic states.

Nephrostomy

An artificial opening created between the kidney and the skin to drain urine directly when the normal

pathway is blocked. See urology.

Nephrostomy Tube

A percutaneously placed catheter in the renal pelvis to drain urine, bypassing a ureteral obstruction. Indications: ureteral obstruction (calculus, tumor, stricture) causing hydronephrosis, pyonephrosis (infected, obstructed kidney — emergency drainage), and as a temporary measure before definitive surgery.

Nephrotic Range Proteinuria

Protein excretion greater than 3.5 g/day or urine protein-to-creatinine ratio greater than 3.5 g/g. Reflects severe glomerular damage with significant protein loss. Degree correlates with severity of injury and risk of complications including thromboembolism, infection, and progression to ESRD.

Nephrotic Syndrome

Clinical constellation of heavy proteinuria greater than 3.5 g/day, hypoalbuminemia less than 3.0 g/dL, generalized edema, hyperlipidemia, and lipiduria. Results from increased glomerular permeability to albumin. Reduced oncotic pressure causes fluid extravasation. Hepatic lipoprotein synthesis increases causing hyperlipidemia.

Nephrotoxic

Describing a substance or condition capable of damaging the kidneys or impairing their function.

Nephrotoxicity

Kidney damage from toxins, drugs, contrast. Aminoglycosides, cisplatin, contrast media, ethylene glycol. Monitoring: creatinine, urine output. Prevention: hydration, dose adjustment.

Nerve

A bundle of axons (nerve fibers) surrounded by connective tissue sheaths that transmits electrical impulses between the central nervous system and the periphery. Structure: epineurium (outer layer), perineurium (surrounds fascicles), endoneurium (surrounds individual axons).

Nerve Biopsy

Removal of a small segment of a peripheral nerve for microscopic examination, used selectively to evaluate vasculitis, infiltrative disease, leprosy, amyloidosis, or selected neuropathies.

Nerve Block

Injection of local anesthetic, sometimes with other agents, near a nerve or nerve plexus to interrupt pain signals.

Nerve Compression

Alternate index label; see *Pinched nerve*.

Nerve Conduction Studies

Measurement of the speed and amplitude of electrical impulses along peripheral nerves. Sensory NCS: stimulate at one point, record distally. Motor NCS: stimulate at two or more points along the nerve, record from muscle. Parameters: conduction velocity (m/s), amplitude (microvolts/millivolts), and distal latency.

Nerve Conduction Study

A diagnostic test that measures the speed and strength of electrical signals traveling through a nerve to detect nerve damage or dysfunction.

Nerve Conduction Velocity

Measurement of the speed and strength of electrical signals traveling along peripheral nerves, used to evaluate neuropathy, nerve compression, demyelination, and axonal injury.

Nerve Deafness (Sensorineural Deafness)

Hearing loss caused by damage to the cochlea, auditory nerve, or central auditory pathways rather than blockage of sound transmission through the outer or middle ear.

Alternate Names:

Sensorineural Deafness

Nerve Endings

The terminal branches of sensory or motor nerves that detect stimuli or communicate with muscles

and glands.

Nerve Growth Factor

A neurotrophin that supports the survival, growth, and maintenance of selected sensory and sympathetic neurons.

Nerve Injury

Damage to a peripheral or central nerve causing pain, numbness, weakness, altered reflexes, or autonomic dysfunction.

Nerve Pinched (Pinched Nerve Overview)

Alternate index label for Pinched Nerve Overview.

Nerve Sheath Myxoma

A benign myxoid peripheral-nerve-sheath tumor, usually presenting as a slow-growing papule or nodule on the fingers, hands, or upper extremities.

Nerve Sparing

Surgical preservation of nerves needed for erectile function, especially during prostate surgery.

Nerve Tract

A bundle of nerve fibers traveling together within the central nervous system.

Nerve, Trigeminal

The trigeminal nerve is the fifth cranial nerve and carries sensation from the face while also controlling the main muscles used for chewing.

Nerve-Muscle Preparation

An experimental isolated nerve and muscle arrangement used to study neuromuscular transmission and muscle contraction.

Nervine

A clinical or physiological concept involving nervine. Its meaning and clinical importance depend on the relevant body system and context.

Nervous Breakdown

A nonmedical lay term for severe emotional or psy-

chological distress; a clinician should assess for a specific mental or physical disorder.

Nervous Colon Syndrome

An older name for irritable bowel syndrome, a disorder causing recurrent abdominal pain with changes in bowel habits without an identifiable structural cause.

Nervous System

The brain, spinal cord, peripheral nerves, and related cells that coordinate sensation, movement, autonomic function, cognition, and behavior.

Nervous System, Peripheral

The peripheral nervous system includes nerves and ganglia outside the brain and spinal cord and carries sensory, motor, and autonomic signals.

Nervous Tissue

The tissue of the nervous system, composed of neurons and supporting cells, that detects stimuli and transmits electrical signals.

Nervus

The formal anatomical Latin noun for a nerve (abbreviated extitN. in the singular and extitNn. in the plural), utilized in Nomina Anatomica terminology to designate specific cranial nerves, spinal nerves, or peripheral nerve branches (e.g., extitNervus vagus for cranial nerve X, extitNervus ischiadicus for the sciatic nerve, extitNervus medianus for the median nerve).

NET

Alternate index label; see *Neuroendocrine tumors*.

Nettle Rash

Urticaria, an itchy eruption of transient raised wheals caused by release of inflammatory mediators in the skin.

Nettles

Plants of the genus *Urtica* whose hairs inject irritants and cause transient stinging, itching, and

wheals.

Neural

Pertaining to a nerve, neuron, neuroglia, or the nervous system in its entirety. Derived embryologically from the neuroectoderm, which folds during primary neurulation (gestational weeks 3–4) to form the neural tube (giving rise to the brain, spinal cord, and motor neurons) and the neural crest (giving rise to the peripheral nervous system, dorsal root sensory ganglia, autonomic ganglia, Schwann cells, and adrenal chromaffin cells). Failure of neural tube closure results in open neural tube defects, including spina bifida and anencephaly.

Neural Tube Defects

Congenital malformations resulting from failure of the neural tube to close during the third to fourth week of gestation. Include anencephaly (incomplete cranial closure), encephalocele (herniation through skull defect), and spina bifida (incomplete spinal closure).

Neuralgia

Neuralgia is sharp, burning, or electric pain along the distribution of a nerve, caused by irritation, compression, infection, injury, or other disease.

Neuralgia, Postherpetic

Alternate index label; see *Postherpetic neuralgia*.

Neuralgia, Trigeminal

Alternate index label; see *Trigeminal neuralgia*.

Neuraminidase Inhibitors

A class of antiviral drugs that inhibit the neuraminidase enzyme on the surface of influenza viruses, preventing the release of newly formed viral particles from infected cells and thereby limiting viral spread. The two available agents are oseltamivir, administered orally, and zanamivir, administered by inhalation.

Neurilemma

The outer nucleated layer of a Schwann cell sur-

rounding a peripheral nerve fiber.

Neurilemmoma

Alternate index label; see *Schwannoma*.

Neurinomatosis

Alternate index label; see *Schwannomatosis*.

Neuritic Plaques

Abnormal deposits of beta-amyloid surrounded by damaged nerve processes in Alzheimer disease.

Alternate Names:

Amyloid plaques

Neuritis

Inflammation or injury of a nerve, causing pain, sensory change, weakness, or altered nerve function.

Neuro-Ophthalmology

The subspecialty dealing with visual disorders caused by the nervous system, optic nerve, or eye-movement pathways.

Neuroanatomy

The study of the structure and organization of the nervous system.

Neurobiological Disorder

A disorder involving the structure, function, development, or chemistry of the nervous system and its biological pathways.

Neuroblastoma

Neuroblastoma is a childhood cancer of immature sympathetic nervous-system cells, commonly arising in the adrenal or paraspinal region and ranging from spontaneously resolving to aggressive disease. as abdominal mass, opsoclonus-myoelonus syndrome. Spontaneous regression possible in infants.

Alternate Names:

Cancer, Neuroblastoma

Neurocardiogenic Syncope

Alternate index label; see *Vasovagal syncope*.

Neurocognitive Disorder

A condition involving acquired decline in one or more cognitive abilities that affects daily function. Causes include neurodegenerative disease, vascular disease, injury, infection, and medicines.

Neurocysticercosis

The most common parasitic infection of the central nervous system worldwide and is caused by the larval stage (cysticercus) of the pork tapeworm *Taenia solium*. It is endemic in Latin America, Asia, and Africa, and is an important cause of epilepsy in endemic regions.

Neurocytoma

A neuronal tumor composed of relatively uniform neurocytes, most often arising in the lateral ventricles as a central neurocytoma. It may obstruct cerebrospinal-fluid flow and is graded by proliferative activity and other features.

Neurodegeneration

The progressive loss of structure and function of neurons, culminating in their death, and represents the underlying mechanism of many devastating neurological disorders.

Neurodegeneration With Brain Iron Accumulation

NBIA is a group of inherited disorders in which iron accumulates in specific brain regions and causes progressive movement, muscle, or cognitive problems.

Neurodermatitis

A chronic itchy skin condition in which repeated scratching or rubbing causes thickened, irritated patches.

Alternate Names:

Dermatitis, Scratch

Neuroendocrine

Relating to interaction between the nervous and endocrine systems or to cells with both neural and hormone-related features.

Neuroendocrine Carcinoma

A malignant epithelial neoplasm showing neuroendocrine differentiation, characterized by tumor cells that express neuroendocrine markers and may secrete bioactive substances. It can arise in the lung, gastrointestinal tract, pancreas, genitourinary tract, or other organs and may be classified as well differentiated or poorly differentiated. Clinical findings vary by the primary site and stage; advanced disease may cause obstruction, bleeding, metastases, or hormone-mediated syndromes. Diagnosis integrates morphology, immunohistochemistry, grading, and staging, while treatment depends on differentiation, anatomic site, molecular features, and disease extent.

Neuroendocrine Carcinoma Of The Skin

Alternate index label; see *Merkel cell carcinoma*.

Neuroendocrine Integration

Coordination between the nervous and endocrine systems, in which neural signals regulate hormone secretion and hormones influence neural activity, behavior, metabolism, and physiological responses.

Neuroendocrine Tumor

A tumor arising from cells with both neural and hormone-producing features. These tumors can occur throughout the body and may produce hormones or be nonfunctioning.

Neuroendocrine Tumors

Tumors arising from cells that receive signals from the nervous system and produce hormones; they may be benign or malignant.

Alternate Names:

Cancer, Neuroendocrine NEC NET

Neuroendocrine Tumors Grading

Classification of neuroendocrine tumors by proliferative rate using Ki-67 index and mitotic count. Well-differentiated low-grade tumors have Ki-67 less than three percent and fewer than two mitoses per ten high-power fields. Intermediate-grade tumors have Ki-67 of three to twenty percent.

Neuroendocrinology

The study of interactions between the nervous system and the endocrine system, including how the brain regulates hormone release.

Neurofibrillary Tangles

Twisted tau-protein fibers inside damaged neurons, characteristic of Alzheimer disease.

Alternate Names:

Tau tangles

Neurofibroma

A benign peripheral nerve sheath tumor. Solitary neurofibromas are common and present as soft, flesh-colored papules or nodules. Multiple neurofibromas are characteristic of neurofibromatosis type 1 (NF1).

Neurofibromatosis

A group of autosomal dominant genetic disorders causing tumor growth along nerves. Type 1 (NF1, chromosome 17) causes café-au-lait spots, neurofibromas, optic gliomas, and learning disabilities. Type 2 (NF2, chromosome 22) causes bilateral vestibular schwannomas, meningiomas, and ependymomas.

Neurofibromatosis 2

Neurofibromatosis type 2 is an inherited tumor-predisposition disorder characterized especially by bilateral vestibular schwannomas and possible meningiomas, ependymomas, cataracts, or neuropathy.

Neurofibromatosis Type 1

Autosomal dominant disorder caused by mutations in the NF1 gene encoding neurofibromin on chromosome seventeen.

Alternate Names:

Neurofibromatosis Type I NF1 Peripheral Neurofibromatosis Von Recklinghausen's Disease Von Recklinghausen's Neurofibromatosis Von Recklinghausen's Phakomatosis

Neurofibromatosis Type 2

An autosomal dominant disorder caused by NF2

gene mutation on chromosome 22. Characterized by bilateral vestibular schwannomas (acoustic neuromas), meningiomas, and ependymomas. Features: hearing loss, tinnitus, and balance problems. Diagnosis: MRI showing bilateral vestibular schwannomas.

Neurofibromatosis Type I

Alternate index label; see *Neurofibromatosis type 1*.

Neurofibromatosis Type II

Alternate index label; see *NF2-related schwannomatosis (neurofibromatosis type 2)*.

Neurofibromatosis-1

Neurofibromatosis type 1 is an inherited condition featuring café-au-lait macules, neurofibromas, axillary freckling, optic pathway gliomas, bone lesions, and variable learning or vascular complications.

Neurofibrosarcoma

Alternate index label; see *Malignant peripheral nerve sheath tumors*.

Neurogenic

A clinical or physiological concept involving neurogenic. Its meaning and clinical importance depend on the relevant body system and context.

Neurogenic Bladder

Lower urinary-tract dysfunction caused by a neurologic lesion affecting central or peripheral control of storage and voiding. It may produce urgency, retention, incontinence, high bladder pressures, recurrent infection, and upper-tract damage.

Neurogenic Megacolon

Massive dilation of the colon due to loss of autonomic innervation, most commonly from Chagas disease, Hirschsprung disease, or diabetic autonomic neuropathy. It presents with chronic constipation, abdominal distension, fecal impaction, and risk of toxic megacolon.

Neurogenic Orthostatic Hypotension

A fall in blood pressure on standing caused by failure of the autonomic nervous system to produce an adequate vasoconstrictor response. It may cause dizziness, fainting, or falls.

Neurogenic Shock

Distributive shock caused by loss of sympathetic vascular tone, usually after acute spinal-cord injury above T6. It is characterized by hypotension with relative bradycardia and warm skin, distinguishing it from hemorrhagic shock when the clinical findings are complete.

Neuroglia

Supporting cells of the nervous system that nourish, protect, and maintain neurons.

Neurography

MRI technique visualizing peripheral nerves. Used for nerve tumors, entrapment, trauma.

Neurohormone

A hormone released by a neuron or neurosecretory cell into the blood or surrounding tissue.

Neurohypophysis

The posterior lobe of the pituitary gland, which stores and releases oxytocin and vasopressin.

Neurolathyrism

A toxic neurologic disorder caused by excessive consumption of *Lathyrus sativus*, producing progressive spastic weakness of the legs.

Neuroleptic

An older term for antipsychotic medications, derived from their ability to produce a state of mental indifference and motor slowing. The term is now largely replaced by “antipsychotic,” though it is still used in specific contexts such as neuroleptic malignant syndrome. Typical neuroleptics are first-generation antipsychotics.

Neuroleptic Agents

Antipsychotic medicines used to treat disorders such as schizophrenia.

Alternate Names:

Antipsychotics

Neuroleptic Malignant Syndrome

A rare but life-threatening idiosyncratic reaction to dopamine receptor-blocking agents, typically antipsychotics, characterized by hyperthermia, severe muscle rigidity, altered mental status, and autonomic dysfunction.

Neurologic Deficit

Loss or impairment of a neurologic function, such as strength, sensation, coordination, vision, language, or consciousness, caused by dysfunction of the nervous system.

Neurologic Localization

The process of using the pattern of deficits in strength, sensation, reflexes, coordination, cranial-nerve function, and cognition to identify the anatomic site of nervous-system dysfunction. Localization guides differential diagnosis and selection of investigations.

Neurological

A clinical or physiological concept involving neurological. Its meaning and clinical importance depend on the relevant body system and context.

Neurological Examination

A clinical assessment of mental status, cranial nerves, motor function, reflexes, coordination, sensation, and gait. It helps localize nervous-system dysfunction and determine the need for further testing.

Neurological Rehabilitation

Rehabilitation for stroke, spinal cord injury, traumatic brain injury, Parkinson disease, multiple sclerosis. Goals: maximize function, independence, quality of life. Multidisciplinary team: physiatrist, PT, OT, speech therapy, psychology.

Neurologist

A physician specializing in disorders of the brain, spinal cord, nerves, and muscles.

Neurology

The medical specialty concerned with diseases of the brain, spinal cord, peripheral nerves, and muscles.

Neurology PET

PET imaging used in selected neurologic disorders to assess brain metabolism or receptor activity, such as in some dementias, epilepsy evaluations, and movement disorders.

Neurolysis

Surgical or chemical freeing or destruction of a nerve, depending on context.

Neuroma

A benign growth or reactive thickening involving nerve tissue, often producing pain or altered sensation.

Neuroma, Acoustic

Alternate index label; see *Acoustic neuroma* (*vestibular schwannoma*).

Neuroma, Plantar

Alternate index label; see *Morton neuroma*.

Neuromuscular

Relating to nerves and muscles or to their junction and communication.

Neuromuscular Blocking Agents

Medications administered intraoperatively to induce skeletal muscle relaxation and facilitate endotracheal intubation and surgical exposure.

Neuromuscular Hamartoma

A rare benign developmental lesion in which mature skeletal muscle and nerve fascicles are intermixed, typically involving a major peripheral nerve and causing enlargement or neuropathic symptoms.

Neuromuscular Junction

The neuromuscular junction is the synapse between motor neuron and skeletal muscle fiber, where acetylcholine released from presynaptic terminals binds to nicotinic acetylcholine receptors on the postsynaptic motor end plate, causing depolarization and muscle contraction.

Neuromuscular Junction (Myoneural Junction)

The specialized synapse where a motor nerve ending communicates with a skeletal-muscle fibre to initiate contraction.

Alternate Names:

Myoneural Junction

Neuromyelitis Optica Spectrum Disorder

An autoimmune disorder of the central nervous system that causes attacks affecting the optic nerves, spinal cord, or certain brain regions. Attacks may cause optic neuritis, extensive transverse myelitis, area postrema syndrome, or brainstem and diencephalic symptoms. Aquaporin-4 immunoglobulin G (AQP4-IgG) testing can support the diagnosis, but results must be interpreted with the clinical findings and imaging.

Alternate Names:

Neuromyelitis Optica NMO NMOSD Devic Disease Devic's Disease

Neuromyotonia

A rare disorder of peripheral nerve hyperexcitability characterized by continuous, spontaneous motor unit activity of peripheral nerve origin that drives involuntary skeletal muscle contraction. May be acquired (autoimmune, frequently associated with autoantibodies against voltage-gated potassium channel complex proteins, particularly *CASPR2*) or hereditary. Clinically manifested by undulating, visible, involuntary muscular rippling (myokymia), generalized muscle stiffness, delayed muscle relaxation (pseudomyotonia), muscle cramps, persistent fasciculations, muscle hypertrophy, and profuse hyperhidrosis, with electromyography demonstrating diagnostic continuous high-frequency doublet, triplet, or multiplet discharges (neuromyotonic discharges).

Alternate Names:

Isaacs Syndrome; Continuous Muscle Fiber Activity Syndrome

Neuron

The fundamental excitable cell of the nervous system specialized for receiving, integrating, and transmitting electrical and chemical signals. Structural components: cell body (soma), dendrites (receptive), axon (conducting), and synaptic terminals (releasing neurotransmitters).

Neuron-Specific Enolase

Neuron-specific enolase is an enzyme marker that can rise in neuroendocrine tumors or neuronal injury but is not diagnostic by itself.

Neuronal Ceroid Lipofuscinoses

A group of inherited lysosomal storage disorders characterized by accumulation of ceroid lipopigment and progressive loss of vision, cognition, motor function, and sometimes seizures.

Neurone

The British spelling of neuron, an excitable nerve cell that receives and transmits signals.

Neuropathic

Caused by disease or dysfunction of nerves; neuropathic skin symptoms may include burning, electric-shock pain, altered sensation, numbness, or itch.

Neuropathic Pain

Chronic pain caused by damage or disease affecting the somatosensory nervous system, including the peripheral nerves, dorsal root ganglia, spinal cord, or brain.

Neuropathic Pain (Nerve Pain)

Alternate index label for Nerve Pain.

Neuropathic Pain (Pain Management: Neuropathic Pain)

Alternate index label for Pain Management: Neuropathic Pain.

Neuropathic Ulcer

A chronic wound of the foot in a patient with diabetes, resulting from peripheral neuropathy, peripheral artery disease, and impaired wound healing. See diabetic foot ulcer.

Neuropathy

Disease or dysfunction of peripheral nerves causing pain, numbness, tingling, weakness, or altered reflexes. Common causes include diabetes, alcohol, nutritional deficiency, toxins, compression, autoimmune disease, and inherited disorders.

Neuropathy (Peripheral Neuropathy)

Damage to peripheral nerves causing sensory, motor, or autonomic dysfunction. Common causes: diabetes mellitus, alcohol use disorder, vitamin B12 deficiency, chemotherapy, autoimmune diseases, and hereditary conditions (Charcot-Marie-Tooth disease). Presents with numbness, tingling, burning pain, weakness, and gait instability.

Alternate Names:

Peripheral Neuropathy

Neuropathy Autonomic Diabetic (Diabetic Neuropathy)

Alternate index label for Diabetic Neuropathy.

Neuropathy Diabetic (Diabetic Neuropathy)

Alternate index label for Diabetic Neuropathy.

Neuropathy Diabetic Peripheral (Diabetic Neuropathy)

Alternate index label for Diabetic Neuropathy.

Neuropathy Focal Diabetic (Diabetic Neuropathy)

Alternate index label for Diabetic Neuropathy.

Neuropathy Proximal Diabetic (Diabetic Neuropathy)

Alternate index label for Diabetic Neuropathy.

Neuropathy, Autonomic

Alternate index label; see *Autonomic neuropathy*.

Neuropathy, Diabetic

Alternate index label; see *Diabetic neuropathy*.

Neuropathy, Hereditary Motor And Sensory

Alternate index label; see *Charcot-Marie-Tooth disease*.

Neuropathy, Peripheral

Alternate index label; see *Peripheral neuropathy*.

Neuropeptide

A peptide released by neurons that acts as a signaling molecule in the nervous or endocrine system.

Neuropeptides

Short chains of amino acids that act as signaling molecules in the nervous system.

Neuropharmacology

The study of how medicines and other substances affect the nervous system.

Neuroplasticity

The brain's ability to reorganize its structure, function, and connections in response to experience, learning, and injury. It is relevant to psychiatric treatment because antidepressants, psychotherapy, and ECT are thought to promote neuroplastic changes including neurogenesis in the hippocampus and synaptic remodeling.

Neuroplegic

Causing paralysis or profound reduction of nervous or muscular activity; the term is uncommon.

Neuropsychiatry

The clinical field linking neurologic and psychiatric disorders, including cognitive and behavioral consequences of brain disease.

Neuroradiology

The use of imaging to diagnose disorders of the brain, spine, head and neck, and peripheral nerves.

Neuroretinitis

Neuroretinitis is inflammation of the optic nerve and retina, often causing vision loss and a characteristic macular star, with infectious or inflammatory causes.

Neurosarcoidosis

Neurosarcoidosis is sarcoid inflammation affecting the nervous system, potentially involving cranial nerves, meninges, brain, spinal cord, or peripheral nerves.

Neurosciences

Neurosciences are the scientific disciplines studying the nervous system, including its structure, function, development, disease, and treatment.

Neurosurgeon

Surgeon specializing in surgical treatment of disorders of the brain, spine, and peripheral nerves.

Neurosurgery

Surgical treatment of disorders of the brain, spinal cord, peripheral nerves, and supporting structures.

Neurosyphilis

Neurosyphilis is infection of the nervous system by *Treponema pallidum* and can cause meningitis, stroke, sensory loss, cognitive change, or tabes dorsalis.

Neurotoxicity

Injury or dysfunction of the nervous system caused by a medicine, drug, toxin, heavy metal, infection, or other harmful exposure.

Neurotransmitter

Neurotransmitters are endogenous chemical messengers that transmit signals across synapses from one neuron to another or to target cells such as muscle or gland cells.

Neurotransmitter Receptors

Cell proteins that recognize neurotransmitters and alter cell activity when bound.

Neurotrophic

Relating to factors that support the growth, survival, or maintenance of nerve cells.

Neurotropic

Having an affinity for or acting primarily on nervous tissue.

Neurovascular Examination

Assessment of distal motor function, sensation, pulses, capillary refill, skin temperature, and color in an injured limb. Serial examination is essential for detecting nerve injury, vascular compromise, compartment syndrome, and evolving ischemia.

Neutral Alignment

A posture in which the body is arranged in a straight, balanced line while preserving normal spinal curves.

Neutralization

Removal or reduction of an effect by counteraction, such as an acid-base reaction or antibody-antigen binding.

Neutralizing Antibodies

Antibodies that bind a pathogen or its toxin and block its ability to enter cells, attach to tissues, or cause its usual biologic effects.

Neutropenia

A reduced neutrophil count ($< 1.5 \times 10^9/L$; severe < 0.5), increasing risk of bacterial and fungal infection, especially febrile neutropenia.

Alternate Names:

Low Neutrophil Count (Neutropenia)

Neutropenia - Infants

Infant neutropenia is an abnormally low absolute neutrophil count for age, increasing susceptibility to bacterial or fungal infection; causes include infection, immune destruction, marrow disease, and medicines.

Neutrophil

The most common circulating granulocyte, an innate immune cell important in early defense against bacterial and fungal infection.

Neutrophil Extracellular Traps

NETs: web-like structures of chromatin + granular proteins (MPO, NE, histone) released by neutrophils (NETosis: suicidal vs vital).

Neutrophil Recruitment

A multistep process essential for acute inflammation and host defense, involving sequential adhesive interactions between neutrophils and endothelial cells. The process begins with the rolling phase, where neutrophils loosely attach to endothelial cells through selectin-mediated interactions.

Neutrophilic Dermatoses

A group of inflammatory skin conditions characterized by dense neutrophilic infiltrate in the dermis without evidence of infection, which includes Sweet syndrome (acute febrile neutrophilic dermatosis), pyoderma gangrenosum, and Behçet disease.

Neutrophilic Eccrine Hidradenitis

A neutrophilic dermatosis characterized by neutrophilic infiltration of the eccrine glands and surrounding dermis, typically occurring in patients receiving chemotherapy, particularly cytarabine. It presents with painful, erythematous plaques or nodules, most commonly on the trunk and extremities.

Neutrophils

The most abundant type of white blood cell. Neutrophils rapidly migrate to sites of infection or injury and ingest microorganisms and cellular debris.

Nevoid Basal Cell Carcinoma Syndrome

Nevoid basal cell carcinoma syndrome refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Nevus

A benign hamartoma of the skin; most commonly melanocytic nevus (mole). Other nevi: congenital, dysplastic/atypical, Spitz, blue nevus, halo nevus; non-melanocytic: epidermal, sebaceous, vascular (strawberry hemangioma, port-wine stain).

Nevus Comedonicus

A congenital hamartoma of the pilosebaceous unit presenting as a localised cluster of dilated follicular ostia filled with keratinous plugs, resembling closely grouped open comedones. It typically appears at birth or in early childhood on the face, neck, or trunk, and may be associated with other developmental anomalies.

Nevus Sebaceus

A congenital or early-onset hamartoma of sebaceous and other adnexal structures, usually a yellow-orange hairless plaque on the scalp or face; secondary tumors may rarely arise within it.

New Daily Persistent Headache

A headache that begins on a clearly remembered day and becomes continuous within 24 hours, persisting for at least three months. It may resemble migraine or tension-type headache; a new persistent headache requires medical evaluation to exclude secondary causes such as infection, pressure disorders, or vascular disease.

Newborn

An infant in the neonatal period, conventionally the first 28 days after birth.

Newborn Circumcision

Surgical removal of the foreskin from the penis, typically performed within the first few days of life. Medical benefits include reduced risk of urinary tract infections, sexually transmitted infections, and penile cancer. Complications include bleeding, infection, and meatal stenosis.

Newborn Head Molding

Temporary shaping or overlapping of a newborn's skull bones during passage through the birth canal;

it usually resolves as the skull returns to its normal contour.

Newborn Hearing Screen

Universal screening for congenital hearing loss, performed before hospital discharge. Two-step protocol: otoacoustic emissions (OAE: measures cochlear outer hair cell response to click stimulus) and automated auditory brainstem response (AABR: measures neural response from the cochlea to the brainstem). Pass/refer.

Newborn Jaundice

Newborn jaundice is yellow discoloration from elevated bilirubin; physiologic patterns are common, but early, severe, rapidly rising, or conjugated jaundice requires evaluation for serious disease.

Newborn Jaundice (Neonatal Jaundice)

Alternate index label for Neonatal Jaundice.

Newborn Metabolic Screen

A dried blood spot test (heel stick) collected 24-48 hours after birth to screen for congenital disorders before symptoms develop.

Newborn Screening

A public health program testing newborns for treatable genetic, metabolic, and endocrine disorders 24-48 hours after birth. Blood spot (heel stick) on filter paper (Guthrie card). Conditions vary by state (Recommended Uniform Screening Panel: 35 core conditions).

Newborn Screening Tests

Tests performed shortly after birth to detect selected treatable genetic, endocrine, metabolic, hematologic, hearing, or critical congenital heart conditions before symptoms develop.

Next-Generation Sequencing

Massively parallel sequencing technology that analyzes many DNA molecules simultaneously. It supports gene panels, whole-exome sequencing, whole-genome sequencing, and tumor sequencing, with

limitations involving coverage, structural variants, mosaicism, and interpretation.

Nexus

A clinical or physiological concept involving nexus. Its meaning and clinical importance depend on the relevant body system and context.

NF1

Alternate index label; see *Neurofibromatosis type 1*.

NF2

Alternate index label; see *NF2-related schwannomatosis (neurofibromatosis type 2)*.

NF2-Related Schwannomatosis (Neurofibromatosis Type 2)

A genetic disorder characterized by tumors involving nerves, especially bilateral vestibular schwannomas, with possible hearing loss, balance problems, and other nerve tumors.

Alternate Names:

BANF Bilateral Acoustic Neurofibromatosis
Central Neurofibromatosis
MISME Multiple Inherited Schwannomas, Meningiomas, And Ependymomas
Neurofibromatosis Type II
NF2

Niacin

Vitamin B3, required to form NAD and NADP coenzymes; deficiency causes pellagra.

Niacin (Nicotinic Acid, Vitamin B3)

A water-soluble vitamin required for cellular energy metabolism; severe deficiency causes pellagra, with dermatitis, diarrhoea, and cognitive changes.

Alternate Names:

Nicotinic Acid, Vitamin B3

Niacin (Vitamin B3)

A water-soluble vitamin that forms NAD⁺ and NADP⁺, essential electron carriers in oxidation-reduction reactions and biosynthetic pathways. Severe deficiency causes pellagra, characterized by dermatitis, diarrhea, and dementia.

Alternate Names:

Vitamin B3

Niacin For Cholesterol

Niacin lowers triglycerides and can raise HDL cholesterol, but its routine cardiovascular benefit is limited when added to effective statin therapy; flushing, liver injury, hyperglycaemia, and hyperuricaemia limit high-dose use.

Niacinamide

The amide form of vitamin B3, used as a nutritional supplement and in selected dermatologic preparations.

Nialamide

An older monoamine-oxidase-inhibitor antidepressant that is no longer commonly used.

Nicardipine

A dihydropyridine calcium-channel blocker used to lower blood pressure and treat selected vascular conditions.

Niche

A specialized site or microenvironment supporting particular cells, organisms, or biological processes.

Nickel Allergy

An allergic skin reaction caused by contact with nickel-containing materials.

Nickel-Chromium

Base metal alloy for crowns. Less expensive than noble metals. Nickel allergy concern.

Niclosamide

An antiparasitic medicine used in some countries to treat selected intestinal tapeworm infections. It acts mainly in the gut and does not reliably treat larval tapeworm infections in tissues, such as neurocysticercosis.

Nicotinamide

Another name for niacinamide, a form of vitamin B3.

Nicotine

An addictive stimulant found in tobacco and some vaping products that acts on neuronal nicotinic receptors.

Nicotine Addiction

Alternate index label; see *Nicotine dependence*.

Nicotine And Tobacco

Nicotine and tobacco involves toxic exposure to nicotine and tobacco; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Nicotine Dependence

A pattern of compulsive nicotine use with cravings, loss of control, tolerance, and withdrawal symptoms when use is reduced or stopped.

Alternate Names:

Dependence, Nicotine Nicotine Addiction

Nicotine Poisoning

Nicotine poisoning causes early nausea, vomiting, salivation, dizziness, tachycardia, and agitation followed by weakness, bradycardia, seizures, or respiratory failure in severe exposure.

Nicotine Replacement Therapy

Nicotine replacement therapy involves toxic exposure to nicotine replacement therapy; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Nicotine Replacement Therapy

Use of nicotine patches, gum, lozenges, spray, or inhaler to reduce withdrawal and cravings while stopping tobacco use.

Nicotinic

Relating to nicotine or to nicotinic acetylcholine receptors. These receptors respond to acetylcholine and help control nerve signaling, skeletal-muscle contraction, and some automatic body functions.

Nictitating Membrane

A transparent or translucent third eyelid found in some animals; humans have only a small vestigial fold.

NICU Consultants And Support Staff

NICU consultants and support staff are specialists and professionals who help care for critically ill or premature newborns, including physicians, nurses, therapists, and social workers.

NICU Staff

NICU staff are the healthcare professionals who monitor and treat newborns in a neonatal intensive care unit and support their families.

Nidation

Implantation of a fertilized ovum into the uterine lining.

Nidus

A nidus is a focus or central site from which a process develops, such as a stone-forming core, infection focus, or vascular malformation.

Niemann-Pick Disease

A group of autosomal recessive lysosomal storage disorders caused by sphingomyelinase deficiency (ASM, types A and B, SMPD1 mutations) or NPC1/NPC2 cholesterol transport defects (type C).

Nifedipine

A dihydropyridine calcium-channel blocker used for hypertension, angina, and selected pregnancy-related hypertensive conditions.

Night Blindness

Impaired vision in dim light (nyctalopia). Causes: vitamin A deficiency (most common cause worldwide, leading to retinol deficiency and rod dysfunction), retinitis pigmentosa (progressive rod-cone degeneration, genetic), congenital stationary night blindness, advanced glaucoma, and cataract.

Night Eating Syndrome

Delayed circadian pattern of eating. Evening hyperphagia, morning anorexia, insomnia. Associated with obesity, depression.

Night Leg Cramps

Sudden, painful involuntary contractions of leg muscles occurring during sleep or rest, often affecting the calf or foot.

Night Sweats

Episodes of excessive sweating during sleep that soak clothing or bedding. They may occur with infection, hormonal changes, cancer, medication effects, or other systemic illness; persistent unexplained night sweats warrant assessment.

Alternate Names:

Men Night Sweats (Night Sweats)

Night Terrors

Episodes of abrupt partial awakening from deep non-REM sleep with intense fear, screaming, autonomic arousal, and limited recall afterward, occurring most often in children.

Night Terrors In Children

Night terrors are episodes of sudden fear, screaming, autonomic arousal, and apparent confusion during non-REM sleep; children usually remain difficult to awaken and have little memory afterward.

Night-Blindness

Difficulty seeing in dim light, called nyctalopia; causes include vitamin A deficiency, retinal disease, and inherited disorders.

Night-Sweats

Episodes of excessive sweating during sleep; causes include infection, malignancy, hormonal change, medicines, and environmental heat.

Nightmare Disorder

Repeated disturbing dreams that cause significant distress or interfere with sleep and daytime function.

Nightmares

Disturbing, vivid dreams that cause fear or distress and may awaken the sleeper.

NIH Stroke Scale

The National Institutes of Health Stroke Scale is a standardized, validated clinical assessment tool used to quantify the severity of acute ischemic stroke.

Nijmegen Breakage Syndrome

A rare autosomal recessive chromosomal instability and DNA double-strand break repair deficiency disorder caused by homozygous hypomorphic mutations in the *NBN* gene on chromosome 8q21, which encodes nibrin (p95), a critical component of the MRN complex (*MRE11-RAD50-NBN*). Characterized by progressive microcephaly, characteristic facial dysmorphism (bird-like facies with sloping forehead, prominent nose, and retrognathia), mild-to-moderate intellectual disability, severe combined immunodeficiency with hypogammaglobulinemia, extreme sensitivity to ionizing radiation, and a striking predisposition to early-onset lymphoid malignancies, particularly B-cell non-Hodgkin lymphomas.

Alternate Names:

NBS; Ataxia-Telangiectasia Variant 1

Nikethamide

An obsolete respiratory stimulant with limited historical use and potential toxicity.

Nikolsky Sign

A clinical dermatological sign in which light lateral pressure applied to normal-appearing skin causes epidermal detachment or sloughing. Positive sign indicates intraepidermal loss of cell-cell cohesion (acantholysis).

Nikolsky'S Sign

Shearing or peeling of the superficial epidermis when apparently normal skin beside a blister is rubbed; it may occur in pemphigus and other blistering disorders.

Nil Disease

A phrase meaning that no disease or abnormality was found; it is not a specific medical diagnosis.

NIncontinence

Likely a misspelling or truncated form of incontinence, meaning loss of voluntary control over urination, bowel movements, or another bodily function.

Nipah Virus Infection

A zoonotic viral infection that may cause fever, respiratory illness, encephalitis, seizures, and coma. It can spread from infected animals or people through close contact. There is no widely available licensed vaccine or specific antiviral treatment; suspected infection requires urgent isolation and specialist care.

Nipple

The projection at the centre of the breast through which milk is released from the mammary ducts; it contains smooth muscle and sensory nerve endings.

Nipple Inversion Or Retraction

Inward turning or pulling-in of the nipple. It may be longstanding and benign, but new unilateral inversion or retraction requires assessment for duct disease or breast cancer.

Niridazole

An older antischistosomal medicine with significant adverse effects and limited current use.

Nisbet'S Canker

A rare historical term whose clinical meaning is not consistent across references; the original source should be checked.

Nisin

An antimicrobial peptide produced by *Lactococcus lactis* and used mainly as a food preservative.

Nit

A clinical or physiological concept involving nit. Its meaning and clinical importance depend on the relevant body system and context.

Nitrates

Vasodilator drugs that release nitric oxide, causing relaxation of vascular smooth muscle through activation of guanylyl cyclase and increased cyclic GMP. They primarily cause venodilation at low doses, reducing preload and myocardial oxygen demand, and arterial dilation at higher doses, reducing afterload.

Nitrazepam

A benzodiazepine hypnotic with sedative and anticonvulsant effects; dependence and respiratory depression are risks.

Nitric Acid

A corrosive mineral acid that can cause severe chemical burns and toxic inhalational injury.

Nitric Acid Poisoning

Nitric acid poisoning involves toxic exposure to nitric acid poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Nitric Oxide

A short-lived signaling molecule that relaxes vascular smooth muscle and participates in immune and neural signaling.

Nitroblue Tetrazolium Blood Test

A functional test of neutrophil oxidative burst in which activated cells reduce nitroblue tetrazolium; it was historically used to evaluate chronic granulomatous disease and has largely been replaced by flow cytometry.

Nitrofurantoin

An antibacterial medicine used mainly for susceptible lower urinary-tract infections.

Nitrogen

An essential element in amino acids, proteins, nucleic acids, and other biological molecules; atmospheric nitrogen is largely inert to humans.

Nitrogen Balance

The difference between dietary nitrogen intake, estimated mainly from protein intake, and nitrogen loss, primarily in urine. Positive balance supports growth or tissue repair; negative balance occurs with inadequate intake, inflammation, trauma, or catabolic disease.

Nitrogenous Base

A nitrogen-containing component of nucleic acids; adenine, guanine, cytosine, thymine, and uracil pair in DNA or RNA to encode genetic information.

Nitroglycerin Overdose

Nitroglycerin overdose involves toxic exposure to nitroglycerin overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Nitromors

A trade name for paint or varnish stripper formulations; exposure can cause chemical burns or toxic inhalation.

Nitrosoureas

A group of chemotherapy medicines that can cross the blood-brain barrier and damage DNA; examples include carmustine and lomustine.

Nitrous Oxide

An inhaled anesthetic gas with analgesic and anesthetic-sparing properties. It has rapid onset and offset but relatively low potency, expands closed gas spaces, and may contribute to diffusion hypoxia or postoperative nausea.

Nitrous Oxide For Labor

A 50:50 mixture of nitrous oxide and oxygen (Entonox) inhaled by the mother during labor for analgesia. Self-administered via a mask or mouthpiece; the woman controls her own dosing, taking a breath 30 seconds before a contraction.

Nizatidine

An H₂-receptor antagonist that reduces gastric acid

secretion; availability varies by country.

NMR

NMR, or nuclear magnetic resonance, measures how atomic nuclei respond to a magnetic field and is used in chemical analysis and imaging.

No-Reflow Phenomenon

Failure of microvascular perfusion to resume despite reopening of a previously blocked larger vessel, especially after myocardial infarction or stroke.

Nocardia Asteroides Infection

Infection by *Nocardia asteroides*, an environmental bacterium that can cause lung, skin, or brain infection, especially in immunocompromised people.

Nocardia Infection

An infection caused by *Nocardia* bacteria, often affecting the lungs, skin, or brain.

Nocardiosis

A bacterial infection caused by *Nocardia* species (*N. asteroides*, *N. brasiliensis*), found in soil. Most common in immunocompromised patients (transplant, HIV, chronic steroid use). Pulmonary nocardiosis (most common): cough, fever, necrotizing pneumonia, cavitary lesions.

Nocebo

A harmful or unpleasant symptom or treatment effect caused or amplified by negative expectations rather than by the treatment's specific pharmacologic action.

Nociception

Neural processing of noxious thermal, mechanical, or chemical stimuli that may result in the perception of pain. It includes transduction, transmission, modulation, and projection; nociception and pain are related but not identical concepts.

Nociceptor

A sensory receptor that detects potentially tissue-damaging mechanical, thermal, or chemical stimuli.

Nociceptors

Sensory nerve endings that detect potentially tissue-damaging stimuli and transmit pain signals.

Nocturnal Enuresis

Alternate index label; see *Bed-wetting*.

Nocturia

Complaint of waking one or more times during the night to void, with each void preceded and followed by sleep. Highly prevalent in older adults and significantly impairs sleep quality and quality of life.

Nocturnal

Pertaining to, occurring in, or characteristic of the nighttime hours or periods of sleep. Clinically prominent in sleep-related disorders, chronobiology, and diagnostic symptom patterns: nocturnal enuresis (bedwetting), paroxysmal nocturnal dyspnea (PND; sudden severe breathlessness awakening a patient from sleep in congestive heart failure), nocturnal hypoxemia in obstructive sleep apnea, nocturnal cough in asthma or gastroesophageal reflux disease, and nocturnal hypertension on 24-hour ambulatory blood pressure monitoring (loss of the normal $\geq 10\%$ nighttime blood pressure "dip").

Nocturnal Emission

Involuntary ejaculation during sleep, sometimes called a wet dream. It is a common and normal event during male puberty and adolescence.

Nocturnal Enuresis

Involuntary urination during sleep in a child old enough for continence, without another defining lower urinary-tract disorder.

Nocturnal Enuresis (Bed-Wetting)

Repeated involuntary urination during sleep in a child old enough to have achieved bladder control.

Alternate Names:

Bed-Wetting

Nocturnal Lagophthalmos

Incomplete closure of the eyelids during sleep, which

may cause dryness, irritation, and exposure injury to the cornea.

Nodal Rhythm

A cardiac rhythm arising from the atrioventricular junction when the sinoatrial node does not control the heartbeat.

Node

A localized swelling, structure, or point of connection; examples include lymph nodes and electrical conduction nodes.

Nodular

Pertaining to, characterized by, or consisting of one or more nodules (circumscribed, solid, palpable anatomical or pathological lesions typically $> 0.5\text{--}1.0$ cm in diameter). In dermatology, describes firm, raised, dermal or subcutaneous lesions such as nodular melanoma or nodular basal cell carcinoma; in endocrinology, describes multi-nodular goiter of the thyroid; and in rheumatology, describes subcutaneous rheumatoid nodules developing over mechanical pressure points (e.g., the olecranon).

Nodular Fasciitis

A rapidly growing benign self-limited fibroblastic and myofibroblastic proliferation, usually presenting as a painful or tender superficial mass. It can mimic sarcoma clinically and microscopically.

Nodular Lymphoid Hyperplasia

Benign aggregates of lymphoid tissue occurring in the gastrointestinal tract, most commonly in the small intestine, presenting as multiple small nodules on endoscopy or imaging. It may be associated with immunodeficiency states, particularly common variable immunodeficiency, and *Giardia lamblia* infection.

Nodular Melanoma

An invasive melanoma characterized by early vertical growth and a rapidly enlarging papule or nodule that may be pigmented or amelanotic.

Nodule

Small, palpable mass <2 cm. Can be solid or cystic. Examples: thyroid nodule, rheumatoid nodule, pulmonary nodule. Evaluation depends on location.

Nodule Thyroid (Thyroid Nodules)

Alternate index label for Thyroid Nodules.

Noise

Unwanted sound or signal that can interfere with communication, measurement, imaging, or hearing.

Noise Induced Hearing Loss And Its Prevention

Noise-induced hearing loss is permanent injury to inner-ear hair cells from repeated or intense sound exposure, often with tinnitus or difficulty understanding speech. Limiting exposure, increasing distance, taking quiet breaks, and using well-fitted hearing protection reduce risk.

Noise-Related Hearing Loss

Alternate index label; see *Hearing loss*.

Noma

Noma is a rapidly progressive gangrenous infection of the mouth and face, usually in severely malnourished or immunocompromised children; it destroys oral and facial tissue and requires urgent antibiotics, nutrition, and surgery.

Nomenclature

Nomenclature is the organized system used to name and classify diseases, organisms, chemicals, anatomy, medicines, or other entities.

Nomogram

A graphical tool that estimates a quantity or risk from related measurements. The Rumack–Matthew nomogram is used in selected cases of acute acetaminophen ingestion.

Non Communicating Hydrocephalus

Non-communicating (obstructive) hydrocephalus is a type of hydrocephalus in which there is obstruction

to the flow of cerebrospinal fluid (CSF) within the ventricular system, preventing CSF from reaching the arachnoid granulations for absorption.

Non-Accidental Trauma

Injuries inflicted on a child, suspected when history and examination findings are inconsistent with the reported mechanism. Red flags: multiple fractures at different stages of healing, posterior rib fractures, metaphyseal corner fractures, and retinal hemorrhages.

Non-Adherence

Failure to follow an agreed treatment plan, which may result from access barriers, side effects, beliefs, misunderstanding, or other circumstances.

Non-Alcoholic Fatty Liver Disease

Hepatic steatosis in the absence of significant alcohol consumption, strongly associated with metabolic syndrome, obesity, and type 2 diabetes. NAFLD includes simple steatosis and non-alcoholic steatohepatitis (NASH) with inflammation and fibrosis. NASH can progress to cirrhosis and hepatocellular carcinoma.

Non-Allergic Rhinitis

Nasal congestion, discharge, or sneezing not caused by an allergic immune response. Triggers may include changes in temperature or humidity, smoke, odors, medicines, or hormonal changes.

Non-Articular Rheumatism

An older term for painful disorders of muscles, tendons, bursae, or other tissues outside the joints.

Non-Asthmatic Eosinophilic Bronchitis

A cause of chronic cough characterized by airway eosinophilia without the variable airflow obstruction or bronchial hyperresponsiveness typical of asthma. Symptoms often improve with inhaled corticosteroids, whereas bronchodilators alone may provide little benefit.

Non-Bacterial Thrombotic Endocarditis

A form of non-infectious endocarditis pathologically characterized by the deposition of sterile, small, friable vegetations composed almost exclusively of platelets and fibrin aggregates along the coaptation lines of heart valves (most frequently aortic and mitral valves), occurring in the absence of systemic bacteremia or destructive valvular inflammation. Pathophysiologically driven by endothelial injury combined with systemic hypercoagulable states, classically associated with advanced mucin-secreting adenocarcinomas of the pancreas, lung, or gastrointestinal tract (marantic endocarditis, Trousseau syndrome), systemic lupus erythematosus (Libman-Sacks endocarditis), and antiphospholipid syndrome; vegetations are loosely attached and carry an exceptionally high risk of recurrent systemic thromboembolism (ischemic stroke).

Alternate Names:

NBTE; Marantic Endocarditis; Libman-Sacks Endocarditis Spectrum

Non-Celiac Gluten Sensitivity

Symptom improvement with gluten-free diet without celiac disease or wheat allergy. Diagnosis of exclusion.

Non-Communicable Disease

Disease not transmitted between persons. Examples: cancer, heart disease, diabetes.

Non-Contact Tonometry

A method of estimating intraocular pressure with a brief puff of air directed at the cornea. It does not touch the eye and is useful for screening, although abnormal or uncertain results may require confirmation with another method.

Non-Distensible Balloon (Vascular)

A catheter balloon designed to maintain its shape during high-pressure dilation of a narrowed vessel.

Alternate Names:

Vascular

Non-Fatal Drowning

Survival after respiratory impairment caused by sub-

mersion or immersion in liquid; complications may include hypoxic brain injury, lung injury, aspiration, and secondary respiratory failure.

Non-Functioning Pituitary Adenoma

Pituitary adenomas that do not secrete clinically significant hormones. The most common pituitary adenomas to present with mass effects.

Nonbacterial Thrombotic Endocarditis

A form of non-infective endocarditis characterized by the deposition of sterile, friable, non-inflammatory platelet and fibrin vegetations along the closure lines of cardiac valves (most frequently aortic and mitral). Strongly associated with advanced malignancies (especially mucin-secreting adenocarcinomas of the lung, pancreas, or GI tract, termed marantic endocarditis), systemic lupus erythematosus, and hypercoagulable states; vegetations easily detach, causing recurrent systemic and cerebral embolic strokes.

Alternate Names:

NBTE; Marantic endocarditis; Libman-Sacks-like nonbacterial endocarditis

Non-Hodgkin Lymphoma

A group of malignant lymphoid tumors arising from B cells, T cells, or natural-killer cells. Subtypes differ in biology, clinical behavior, affected organs, and treatment.

Alternate Names:

Cancer, Non-Hodgkin Lymphoma Lymphoma, Non-Hodgkin

Non-Hodgkin Lymphoma In Children

A group of malignant lymphoid tumors in children, including B-cell, T-cell, and anaplastic large-cell types, classified and treated according to histology and stage.

Non-Hodgkin'S Lymphoma

A diverse group of malignant lymphoid tumors that may arise from B cells, T cells, or natural-killer cells.

Non-Hodgkins Lymphomas

Non-Hodgkin lymphomas are a diverse group of cancers of B, T, or other lymphocytes, ranging from slow-growing to aggressive. Painless lymph-node enlargement, fever, night sweats, weight loss, or extranodal symptoms may occur; biopsy and staging determine treatment.

Alternate Names:

NHL (Non-Hodgkins Lymphomas)

Non-Hodgkins Lymphomas

Alternate index label for Non-Hodgkins Lymphomas.

Non-Infectious Cystitis

Inflammation of the bladder that is not caused by an infectious organism.

Non-Invasive Prenatal Testing

Screening of cell-free placental DNA in maternal blood for selected fetal chromosomal abnormalities, especially trisomies 21, 18, and 13. It is highly sensitive screening, not a definitive diagnosis; positive results require confirmatory invasive testing.

Non-Invasive Procedure

A procedure that does not intentionally enter a body cavity, organ, or tissue with an instrument.

Non-Invasive Ventilation

Positive pressure ventilation delivered via a face mask or nasal mask without endotracheal intubation. BiPAP (bilevel positive airway pressure) is most commonly used for acute hypercapnic respiratory failure (COPD exacerbation) and cardiogenic pulmonary edema. It reduces the need for intubation and improves outcomes in selected patients.

Non-Obstructive Azoospermia

A severe form of male factor infertility characterized by the complete absence of spermatozoa in the ejaculate due to severe intrinsic testicular spermatogenic failure rather than physical blockage of the ductal tract. Underlying causes include genetic abnormalities (such as Klinefelter syndrome 47,XXY

and Y-chromosome microdeletions in the AZF regions), cryptorchidism, mumps orchitis, gonadotoxic chemotherapy or radiation, varicocele-induced atrophy, or idiopathic maturation arrest and Sertoli-cell-only syndrome. Clinically distinguished by elevated serum follicle-stimulating hormone, decreased testicular volume, and low inhibin B levels. Fertility management relies on microsurgical testicular sperm extraction (Micro-TESE) combined with in-vitro fertilization and intracytoplasmic sperm injection.

Alternate Names:

NOA; Secretory Azoospermia; Testicular Failure Azoospermia

Non-Proliferative Diabetic Retinopathy

The early stage of diabetic retinopathy, characterized by structural abnormalities of the retinal vasculature resulting from chronic hyperglycemia-induced damage to the endothelial cells and pericytes of retinal capillaries, without neovascularization.

Non-Radiographic Axial Spondyloarthritis

Inflammatory arthritis affecting the spine and sacroiliac region without definite structural changes on plain X-ray. It may cause chronic inflammatory back pain, morning stiffness, enthesitis, uveitis, psoriasis, or inflammatory bowel disease. MRI, laboratory findings, and clinical assessment help establish the diagnosis.

Non-Rapid Eye Movement Sleep

Sleep consisting of stages N1, N2, and N3, with progressively deeper sleep and reduced responsiveness to the environment.

Non-Rebreather Mask

A face mask with a reservoir bag and one-way valves that can deliver a high concentration of supplemental oxygen when fitted and used correctly.

Non-REM Sleep

Sleep stages that do not include rapid eye movements, including deep slow-wave sleep.

Alternate Names:

NREM sleep

Non-Responder

A person or disease that shows little or no expected response to a treatment, vaccine, or test stimulus.

Non-Small Cell Lung Cancer

Most common lung cancer type. Includes adenocarcinoma, squamous cell, large cell.

Non-Specific Urethritis

Urethral inflammation without identification of a specific cause, often associated with sexually transmitted organisms such as *Chlamydia trachomatis*.

Non-Steroidal Anti-Inflammatory Drugs (NSAIDs)

Medicines that inhibit cyclooxygenase and reduce prostaglandin production, providing analgesic, antipyretic, and anti-inflammatory effects.

Non-Stress Test

Antepartum fetal monitoring assessing fetal heart rate reactivity. Reactive NST: ≥ 2 accelerations of 15 bpm for 15 seconds (≥ 32 weeks) or 10 bpm for 10 seconds (<32 weeks) within 20 minutes. Non-reactive: insufficient accelerations after 40 minutes (may indicate fetal sleep, acidosis, or CNS depression). False positive rate high.

Non-Tuberculous Mycobacteria

NTM: environmental mycobacteria (*M. avium* complex - MAC, *M. kansasii*, *M. abscessus*). Pulmonary NTM: ATS/IDSA criteria (clinical + radiographic + microbiologic: 2+ sputum or 1 BAL + nodules/cavities/bronchiectasis on CT).

Nonagenarian

A nonagenarian is a person aged 90 through 99 years.

Nonalcoholic Fatty Liver Disease

NAFLD: hepatic steatosis (>5 percent of hepatocytes) in the absence of significant alcohol use, associated with the metabolic syndrome (obesity, type 2 diabetes, dyslipidemia, hypertension).

Nonalcoholic Steatohepatitis

Progressive form of NAFLD characterized by hepatic steatosis with lobular inflammation and hepatocyte ballooning with significant risk of progression to fibrosis, cirrhosis, and hepatocellular carcinoma. NASH affects approximately three to five percent of the general population.

Nonallergic Rhinitis

Nasal congestion, sneezing, or runny nose caused by irritants or other triggers rather than an allergy.

Alternate Names:

Rhinitis, Nonallergic Rhinitis, Vasomotor

Nonallergic Rhinopathy

Nonallergic rhinopathy causes congestion, runny nose, or sneezing without an allergic immune reaction, often triggered by odors, smoke, weather changes, or irritants.

Nonceliac Gluten Intolerance (Nonceliac Gluten Sensitivity (Intolerance))

Alternate index label for Nonceliac Gluten Sensitivity (Intolerance).

Nonceliac Gluten Sensitivity (Intolerance)

Alternate index label for Intolerance.

Noncompetitive Antagonist

A noncompetitive antagonist is a drug that reduces the effect of an agonist by a mechanism that cannot be overcome by increasing the agonist concentration.

Noncompetitive Inhibition

A type of enzyme inhibition where the inhibitor binds at a site distinct from the active site (allosteric site), and can bind to both the free enzyme (E) and the enzyme-substrate complex (ES) with equal affinity. The inhibitor does not compete with the substrate and cannot be overcome by increasing substrate concentration.

Nonconvulsive Status Epilepticus

Ongoing seizure activity without prominent rhyth-

mic limb jerking, often presenting as prolonged confusion, impaired awareness, or subtle twitching and requiring electroencephalographic confirmation and urgent treatment.

Nondisjunction

Failure of homologous chromosomes or sister chromatids to separate properly during cell division, producing cells with abnormal chromosome numbers.

Noninvasive

A medical device or care approach involving noninvasive. Its purpose, indications, risks, and follow-up depend on the patient and clinical setting.

Noninvasive Prenatal Testing

A maternal blood test that analyzes cell-free DNA to estimate the chance of selected fetal chromosomal conditions. It is a screening test, not a definitive diagnosis.

Noninvasive Ventilation

Positive-pressure ventilatory support delivered without an endotracheal tube, commonly by a mask or helmet. CPAP provides continuous positive pressure; bilevel support provides separate inspiratory and expiratory pressures. It is used selectively when the patient can protect the airway and cooperate.

Nonischemic Priapism

Alternate index label; see *Priapism*.

Nonketotic Hyperglycemic Hyperosmolar Syndrome

A severe complication of diabetes marked by very high blood glucose, dehydration, and increased blood osmolality, usually without substantial ketoacidosis.

Nonmelanoma Skin Cancer

A group of skin cancers that arise from the cells of the epidermis, excluding melanoma. The two most common types are basal cell carcinoma (BCC) and squamous cell carcinoma (SCC), which together

account for the vast majority of all skin cancers diagnosed globally.

Nonpathogenic

Nonpathogenic means not ordinarily capable of causing disease in a given host or setting; an organism can still cause opportunistic infection when immunity or barriers are impaired.

Nonproliferative Retinopathy

Retinal disease in which small vessels leak or become damaged without abnormal new-vessel growth.

Alternate Names:

Background retinopathy

Nonselective Beta-Blockers

Nonselective beta-adrenergic blocking agents block both beta-1 and beta-2 adrenergic receptors, reducing heart rate, myocardial contractility, and cardiac output through beta-1 blockade, while also blocking beta-2 receptors in the bronchial smooth muscle and peripheral vasculature.

Nonseminomatous Germ Cell Tumor

A major category of malignant testicular and extragonadal germ cell tumors, comprising embryonal carcinoma, yolk sac tumor, choriocarcinoma, teratoma, and mixed histologic subtypes. Characterized by a younger age of onset, aggressive early hematogenous metastasis, and frequent secretion of serum tumor markers (alpha-fetoprotein and beta-human chorionic gonadotropin), which are used for risk stratification, staging, and monitoring response to platinum-based chemotherapy.

Alternate Names:

NSGCT; Nonseminoma

Nonsense Variant

A sequence variant that creates a premature stop codon, potentially producing a truncated protein or triggering nonsense-mediated mRNA decay. The effect depends on the gene, location, transcript, and residual protein production.

Nonspecific Interstitial Pneumonia

A pattern with uniform, temporally homogeneous inflammation and/or fibrosis. Most common ILD associated with connective tissue disease. HRCT shows bilateral ground-glass opacities with subpleural sparing.

Nonspecific Interstitial Pneumonitis

An interstitial lung disease characterized by relatively uniform inflammation and fibrosis of the lung interstitium, often associated with connective-tissue disease, medications, or an unknown cause.

Nonsteroidal Anti-Inflammatory Drugs

Analgesic and anti-inflammatory drugs that inhibit cyclooxygenase enzymes and prostaglandin synthesis. Perioperative use requires assessment of renal function, gastrointestinal bleeding risk, platelet effects, cardiovascular risk, and surgical bleeding considerations.

Nonsteroidal Anti-Inflammatory Drugs And Ulcers

Nonsteroidal anti-inflammatory drugs can reduce stomach-protective prostaglandins and cause gastritis, ulcers, and gastrointestinal bleeding. Risk rises with age, prior ulcer, anticoagulants, steroids, kidney disease, and high doses; safer alternatives or acid protection may be needed.

Alternate Names:

Nonsteroidal Antiinflammatory Drugs (Nonsteroidal Anti-Inflammatory Drugs And Ulcers)

Nonsteroidal Antiinflammatory Drugs (Nonsteroidal Anti-Inflammatory Drugs And Ulcers)

Alternate index label for Nonsteroidal Anti-inflammatory Drugs and Ulcers.

Nonsyndromic

Nonsyndromic means occurring without the additional characteristic features of a recognized syndrome; it may still be genetic, congenital, or clinically significant.

Nontropical Sprue

Alternate index label; see *Celiac disease*.

Nontuberculous Mycobacterial Infection

Infection caused by *Mycobacterium* species other than *M. tuberculosis*, often affecting the lungs but sometimes the lymph nodes, skin, or other organs. Disease is more likely in people with structural lung disease or impaired immunity.

Nontuberculous Mycobacterial Lung Disease

Chronic lung infection caused by environmental *Mycobacterium* species other than *M. tuberculosis*. It may produce cough, fatigue, weight loss, breathlessness, or coughing blood, especially in people with structural lung disease. Diagnosis requires compatible imaging and repeated microbiologic testing; treatment is prolonged and individualized.

Nonulcer Dyspepsia

Alternate index label; see *Functional dyspepsia*.

Nonulcer Stomach Pain

Alternate index label; see *Functional dyspepsia*.

Nonunion And Malunion

Nonunion is failure of a fractured bone to heal after an adequate period (generally 6-9 months for long bones) without clinical or radiographic evidence of progression toward union for 3 months. Classification: hypertrophic (viable ends, inadequate stability), atrophic (devitalized ends, poor biology), oligotrophic (intermediate).

Nonunion Fracture

Failure of a broken bone to heal within the expected time and without progressive healing, often causing persistent pain, instability, or deformity.

Noonan Syndrome

An inherited disorder of the RAS-MAPK signaling pathway, characterized by distinctive facial features, short stature, congenital heart disease, and variable developmental or bleeding problems.

Noonan Syndrome

Noonan syndrome is a RASopathy causing characteristic facial features, short stature, congenital heart disease, developmental variation, and sometimes bleeding or lymphatic abnormalities; it is usually autosomal dominant and genetically heterogeneous.

Noonan Syndrome With Multiple Lentigines

A genetic disorder also called LEOPARD syndrome, characterized by multiple lentigines, distinctive facial features, heart disease, short stature, and sometimes deafness or genital abnormalities.

Noradrenaline

A hormone and neurotransmitter involved in alertness, stress responses, and blood-vessel constriction.

Alternate Names:

Norepinephrine

Norepinephrine

A hormone and neurotransmitter that supports alertness, heart function, and the stress response.

Alternate Names:

Noradrenaline

Norethisterone

A synthetic progestogen used for contraception and selected menstrual or endometriosis-related disorders.

Norfloxacin

A quinolone antibiotic formerly used for selected urinary or gastrointestinal bacterial infections.

Norgestrel

A synthetic progestogen used in some hormonal contraceptive preparations.

Norketamine

Norketamine is an active metabolite of ketamine with NMDA-receptor antagonist activity; it contributes to ketamine's analgesic and psychoactive effects and may persist after the parent drug is

cleared.

Normal

Within an expected range or consistent with usual anatomy, physiology, or function for a specified population.

Normal Flora

The resident microorganisms (commensals) colonizing body surfaces—skin, oropharynx, gastrointestinal tract, genitourinary tract—without causing disease under normal conditions. They provide protection (colonization resistance, immune priming), synthesis of vitamins, and nutrient metabolism.

Normal Growth And Development

Normal growth and development describes expected physical growth, movement, language, learning, and social progress for a person's age and stage of life.

Normal Labour

Spontaneous labour at term with regular uterine contractions, progressive cervical dilatation, and delivery through the birth canal without major complication.

Normal Pressure Hydrocephalus

A disorder of cerebrospinal-fluid circulation causing enlarged ventricles and a clinical pattern that may include gait impairment, cognitive decline, and urinary urgency or incontinence.

Normal Range

A normal range is the interval of results commonly found in a reference population; a result outside it does not by itself prove disease.

Normal Saline

A sterile approximately 0.9

Normal-Pressure Hydrocephalus

A disorder in which cerebrospinal fluid accumulates and causes gait difficulty, cognitive decline, and urinary problems despite near-normal measured pressure.

Normalization

Correction for detector efficiency variations. Performed daily with uniform source.

Normoblast

A nucleated immature red-cell precursor in bone marrow that normally loses its nucleus before entering circulation.

Normocalcemic Hyperparathyroidism

Condition of elevated parathyroid hormone with persistently normal serum calcium and normal vitamin D levels. Exclusion of secondary causes of elevated PTH including vitamin D deficiency, renal insufficiency, and medications is required. Represents an early or mild form of primary hyperparathyroidism. May progress to overt hypercalcemia.

Normocytic Anaemia

Anemia in which red blood cells are of approximately normal mean volume, despite a reduced total red-cell mass.

Normotensive

Having blood pressure within the expected range for the individual and measurement setting.

Norovirus

A single-stranded RNA virus that is the leading cause of gastroenteritis outbreaks worldwide. The virus is highly contagious and can cause large outbreaks in closed settings including hospitals, cruise ships, and nursing homes. Norovirus causes acute onset of vomiting and diarrhea. Diagnosis is made by PCR or antigen detection in stool.

Norovirus - Hospital

Hospital-associated norovirus is an acute, highly contagious gastroenteritis causing vomiting and watery diarrhoea; control depends on isolation, handwashing with soap and water, environmental disinfection, and rehydration.

Norovirus Infection

A highly contagious viral gastroenteritis causing sud-

den nausea, vomiting, watery diarrhea, abdominal cramps, and sometimes fever. Illness is usually short but can cause dangerous dehydration, especially in infants, older adults, or medically vulnerable people. Oral rehydration and careful hand hygiene are central to management and prevention.

Norrie Disease

An X-linked retinal disorder, usually caused by pathogenic variants in NDP, characterized by congenital or early-onset blindness from abnormal retinal development. Hearing loss, developmental differences, and peripheral vascular or vitreoretinal complications may occur.

Nortriptyline

A tricyclic antidepressant used for depression and selected neuropathic pain disorders.

Norwegian Scabies

A severe, highly contagious form of crusted scabies with large numbers of mites, often occurring in immunocompromised people.

Norwood Procedure

The first-stage, high-complexity neonatal reconstructive cardiac surgical operation for hypoplastic left heart syndrome and related single-right-ventricle anomalies, performed within the first few days of life. Consists of three core surgical maneuvers: (1) neo-aortic reconstruction, transecting the main pulmonary trunk and anastomosing it to the hypoplastic ascending aorta and aortic arch with extensive patch augmentation; (2) wide atrial septectomy to guarantee uninhibited pulmonary venous return to the single right ventricle; and (3) placement of a source of controlled pulmonary blood flow, classically utilizing a modified Blalock-Taussig shunt (from innominate artery to pulmonary artery) or a Sano shunt (valveless conduit from right ventricle to pulmonary artery confluence).

Alternate Names:

Stage 1 Norwood Reconstruction; Norwood Operation

Nose

The external nasal structure and nasal cavities, the primary organ of olfaction and the entry to the respiratory tract. External nose: bone (nasal bones, frontal process of maxilla) and cartilage (septal, upper/lower lateral cartilages).

Nose Fracture

A break in one or more nasal bones caused by trauma, potentially causing pain, swelling, deformity, bleeding, or nasal obstruction.

Nose, Disorders Of

Conditions affecting the nose or nasal passages, including infection, allergy, trauma, obstruction, polyps, and tumors.

Nosebleed

A nosebleed, or epistaxis, is bleeding from nasal blood vessels caused by dryness, trauma, inflammation, medications, hypertension, vascular disease, or less commonly a tumor or bleeding disorder. anterior nasal septum and result from local trauma, dryness, inflammation, or vascular disease; persistent or severe bleeding may require medical intervention.

Alternate Names:

epistaxis.

Nosebleeds

Bleeding from blood vessels in the nose, commonly caused by dryness, irritation, injury, infection, allergies, medicines, or high blood pressure.

Nosocomial

Acquired during healthcare, especially in a hospital, rather than being present or incubating when the person entered care. A nosocomial infection may be caused by bacteria, viruses, fungi, or other organisms and can involve drug-resistant organisms.

Nosocomial Infection

A hospital-acquired infection developing during or after hospitalization that was not present or incubating at admission, classically defined by onset beyond 48 hours. Common sites are bloodstream, urinary

tract, lungs, and surgical wounds; causative organisms are frequently multidrug-resistant, prevented by hand hygiene and stewardship.

Nostril

The anterior external opening or aperture of the nasal cavity; anatomically termed the anterior naris (plural: nares). Flanked laterally by the mobile fibrofatty ala of the nose and separated medially by the columella and caudal nasal septum. Leads directly into the nasal vestibule, which is lined by stratified squamous keratinized epithelium and coarse protective hairs (vibrissae) that trap large airborne particulates and foreign debris before inspired air enters the respiratory nasal cavity.

Notencephalocele

A rare congenital neural-tube defect involving protrusion of brain tissue or meninges through a skull defect; terminology varies by source.

Notifiable Disease

Disease required by law to be reported to health authorities. Examples: tuberculosis, measles, HIV, COVID-19.

Notochord

A flexible embryonic rod that guides axial development and largely disappears as the vertebral column forms.

Novobiocin

An older aminocoumarin antibiotic formerly used for selected staphylococcal infections and in laboratory differentiation of staphylococci.

NOWS

Alternate name; see *Neonatal Opioid Withdrawal Syndrome*.

Noxious

Harmful or capable of causing tissue injury or pain.

NREM Sleep

NREM sleep is non-rapid-eye-movement sleep, com-

prising stages from light sleep to deep slow-wave sleep and alternating with REM sleep.

NSAID

A nonsteroidal anti-inflammatory drug used to reduce pain, fever, and inflammation.

Alternate Names:

Nonsteroidal anti-inflammatory drug

NSAIDs (Nonsteroidal Anti-Inflammatory Drugs)

A class of analgesic, antipyretic, and anti-inflammatory drugs that inhibit cyclooxygenase (COX-1 and/or COX-2), reducing prostaglandin and thromboxane synthesis.

Alternate Names:

Nonsteroidal Anti-Inflammatory Drugs

NSTEMI

Non-ST-elevation myocardial infarction: acute myocardial necrosis (elevated cardiac troponin) without persistent ST-segment elevation on ECG, caused by subtotal occlusion or distal embolization of a coronary artery. Features: chest pain, ECG changes (ST depression, T-wave inversion), elevated troponin.

NTD

NTD commonly means neural-tube defect, a congenital brain or spinal-cord malformation associated with inadequate folate and other factors.

Nuchal

Pertaining anatomically to the back or nape of the neck (the nuchal region). Clinically prominent in infectious disease and neurology for nuchal rigidity (inability to passively flex the neck toward the chest due to severe spasm of extensor neck muscles, a cardinal physical finding of meningeal irritation in bacterial or viral meningitis and subarachnoid hemorrhage). Also high-yield in prenatal ultrasonography for the first-trimester nuchal translucency measurement at 11–14 weeks gestation, where increased subcutaneous fluid thickness at the fetal nape indicates elevated risk for fetal chromosomal aneuploidies (trisomy 21 / Down syndrome) and congenital cardiac malformations.

Nuchal Cord

Wrapping of the umbilical cord around the fetal neck, occurring once or more times. It is common and often harmless, but monitoring during labor is needed if fetal heart-rate abnormalities occur.

Nuchal Rigidity

Abnormal stiffness and resistance to passive flexion of the neck, classically associated with irritation or inflammation of the meninges, including meningitis or subarachnoid hemorrhage.

Nuchal Translucency Screening

An ultrasound measurement of the fluid-filled space at the posterior fetal nuchal region, performed at 11–14 weeks (crown-rump length 45–84 mm). Increased NT (>95th percentile or >3.0–3.5 mm) is associated with trisomies 21, 18, 13, Turner syndrome, and cardiac defects.

Nuchal Translucency Test

An ultrasound measurement of fluid at the back of the fetal neck during early pregnancy, used with other factors for screening chromosomal abnormalities and selected structural conditions.

Nuclear

Relating to a cell nucleus, atomic nucleus, or nuclear medicine depending on context.

Nuclear Magnetic Resonance

The absorption and emission of electromagnetic energy by atomic nuclei in a magnetic field; it is the physical basis of MRI and spectroscopy.

Nuclear Sclerosis

A common age-related cataract in which the central lens becomes increasingly yellow and hard. It can gradually reduce vision, alter color perception, and sometimes temporarily improve near vision.

Nuclear Stress Test

Myocardial perfusion imaging during stress. Detects coronary artery disease.

Nuclear Tests

Diagnostic tests using small amounts of radioactive material to assess organs or identify disease.

Alternate Names:

Nuclear medicine tests

Nuclear Ventriculography

A radionuclide imaging test that tracks labeled blood through the heart to measure ventricular ejection fraction and wall motion.

Nucleic Acid

DNA or RNA, polymers that store, transmit, or express genetic information.

Nucleic Acid Hybridization

A laboratory method in which complementary DNA or RNA strands bind to each other to detect, identify, or measure genetic sequences.

Nucleolus

A dense region within the cell nucleus where ribosomal RNA is produced and ribosome subunits begin assembly.

Nucleoside

A nitrogenous base linked to a sugar, without a phosphate group.

Nucleoside Analogs

A class of antiviral drugs that structurally resemble natural nucleosides and inhibit viral replication by interfering with DNA or RNA synthesis.

Nucleoside Bypass Therapy

Nucleoside bypass therapy is a medical test, procedure, treatment, or clinical measure used in patient care; its purpose and interpretation depend on the clinical question.

Nucleotide

A nucleoside linked to one or more phosphate groups; nucleotides form DNA and RNA and carry cellular energy.

Nucleus

The membrane-bound cell organelle containing most genetic material, or a defined cluster of neurons in the central nervous system.

Nucleus Accumbens

A brain region involved in reward, motivation, and reinforcement.

Nucleus Pulposus

The soft, gel-like central portion of an intervertebral disk, surrounded by the annulus fibrosus.

Null Mutation

A genetic variant that prevents a gene from producing a functional protein or RNA product.

Nulligravida

An obstetrical term designating a woman who is not currently pregnant and has never experienced a pregnancy (gravidity = 0). Differentiated from nullipara, which designates a woman who has never carried a pregnancy beyond 20 weeks of gestation (or delivered an infant weighing ≥ 500 g), even if she has experienced previous early miscarriages or elective terminations. Nulligravidity is an epidemiological risk factor for endometrial cancer and breast cancer (due to a greater cumulative number of lifetime ovulatory cycles and uninterrupted estrogen exposure).

Nullipara

A person who has never carried a pregnancy beyond the stage of fetal viability; the adjective is nulliparous.

Numb

Numb means lacking normal sensation or emotional responsiveness; physical numbness can result from nerve, brain, circulation, metabolic, or medication-related problems.

Number Needed To Harm

The number of people exposed to an intervention for one additional person to experience a specified

adverse outcome compared with the control over a defined time. It is the reciprocal of absolute risk increase.

Number Needed To Treat

The number of people who must receive an intervention for one additional person to experience the specified beneficial outcome compared with the control over a defined time. It is the reciprocal of absolute risk reduction.

Numbness

Reduced or absent sensation, often caused by nerve compression, injury, neuropathy, circulation problems, or brain or spinal disease.

Numbness (Numb)

Numbness is reduced or absent sensation, often from nerve compression, neuropathy, stroke, spinal disease, circulation problems, or anesthetic effects.

Numbness And Tingling

Abnormal sensations, also called paresthesias, that may result from nerve compression, peripheral neuropathy, spinal disease, vitamin deficiency, vascular problems, or neurologic disorders. Sudden one-sided symptoms require urgent evaluation for stroke.

Alternate Names:

paresthesias, that may result from nerve compression,.

Numbness In Hands

Reduced or absent sensation in one or both hands, commonly caused by nerve compression, neuropathy, injury, circulation problems, or systemic disease.

Nummular

Coin-shaped or round. The term is commonly used for nummular eczema, which causes itchy, coin-shaped patches of inflamed skin.

Nummular Eczema

Nummular eczema is an inflammatory skin disorder producing intensely itchy coin-shaped patches, often worsened by dry skin, irritation, or barrier disruption.

tion. erythematous, vesicular, or crusted plaques, typically on the extremities. The lesions are intensely pruritic and may be triggered by dry skin, skin trauma, or allergic contact dermatitis. It is more common in adults with atopic predisposition.

Nurse

A regulated healthcare professional who provides assessment, treatment, education, monitoring, and supportive care.

Nurse Assistant

A nurse assistant helps provide basic patient care such as bathing, mobility, feeding, measurements, and comfort under the direction of licensed staff.

Nurse Practitioner

A nurse practitioner is an advanced-practice nurse who can assess patients, diagnose and treat many conditions, prescribe medicines where authorized, and provide preventive care.

Nursemaid'S Elbow

Nursemaid's elbow is subluxation of the radial head after sudden traction on a young child's extended or pronated arm; the child avoids using the arm, and gentle clinician reduction usually restores movement.

Nursery

A place or service for the care of infants, including specialized newborn care in a hospital.

Nursing

The professional practice of caring for people, promoting health, preventing illness, and supporting recovery or comfort.

Nursing Home

A residential facility that provides ongoing nursing care, assistance with daily activities, and medical or rehabilitation services for people who cannot live independently.

Nut

A seed or fruit used as food; it may cause IgE-mediated food allergy in susceptible people.

Nut Allergies

Nut allergy is an immune-mediated reaction to tree nuts or peanuts that can cause hives, gastrointestinal or respiratory symptoms, or life-threatening anaphylaxis.

Nut-Allergy

An immune-mediated reaction to one or more nuts, potentially causing urticaria, anaphylaxis, or gastrointestinal and respiratory symptoms.

Nutation

A nodding or rocking movement; in anatomy it commonly describes movement of the sacrum relative to the pelvis.

Nutcracker Esophagus

Esophageal motility disorder characterized by high-amplitude peristaltic contractions exceeding 180 mmHg in the distal esophagus on manometry, with preserved peristaltic coordination. Presents with intermittent chest pain and dysphagia to solids and liquids.

Nutcracker Syndrome

Compression of left renal vein between aorta and SMA causing hematuria and flank pain. Managed conservatively. Stenting or surgery for severe cases.

Nutraceutical

A food-derived product marketed as providing health or medicinal benefits beyond basic nutrition; evidence, composition, and regulatory oversight vary by product.

Nutrient

A substance required by the body for energy, growth, repair, or physiological function.

Nutrient Density

Amount of nutrients per calorie. Whole foods more

nutrient-dense than processed foods. Importance for weight management, health.

Nutriment

Food or nourishment that supports bodily maintenance and growth.

Nutrition

The intake, digestion, absorption, metabolism, and use of food components by the body.

Nutrition (Geriatric)

Nutritional assessment and intervention in elderly. Risks: malnutrition (undernutrition, obesity), micronutrient deficiencies (vitamin D, B12, calcium), dehydration. Assessment: weight, BMI, dietary intake, lab values. Interventions: supplementation, meal delivery, appetite stimulants, feeding assistance.

Alternate Names:

Geriatric

Nutrition (Oncology)

Nutritional care for people with cancer, addressing intake, symptoms, treatment effects, weight, muscle mass, and personal goals.

Alternate Names:

Oncology

Nutrition And Athletic Performance

A clinical or physiological concept involving nutrition and athletic performance. Its meaning and clinical importance depend on the relevant body system and context.

Nutrition Education

Teaching patients about healthy eating. Cultural sensitivity, health literacy appropriate. Motivational interviewing effective.

Nutrition In Pregnancy

A balanced diet providing adequate calories, protein, vitamins, and minerals to support maternal and fetal health. Additional caloric needs: 300 kcal/day in the second and third trimesters. Key nutrients:

folic acid (400 mcg/day), iron (27 mg/day), calcium (1000 mg/day), vitamin D (600 IU/day), iodine (220 mcg/day), DHA (200-300 mg/day).

Nutrition Risk Screening 2002

A validated screening tool for hospitalized adults that considers nutritional status and disease severity to identify patients likely to benefit from nutrition support. Screening is not itself a complete malnutrition diagnosis.

Nutrition-Focused Physical Examination

A structured examination for physical findings associated with malnutrition, including loss of subcutaneous fat, muscle wasting, edema, ascites, reduced functional status, and micronutrient-related signs. It complements dietary history, anthropometry, laboratory data, and disease assessment.

Nutritional Anemia

Anemia caused by inadequate intake, absorption, or use of nutrients needed for red-cell production, especially iron, vitamin B12, or folate.

Nutritional Assessment

Comprehensive evaluation: dietary intake, anthropometrics, biochemical tests, clinical examination. Dietitian involvement essential.

Nutritional Dermatoses

Nutritional deficiencies cause distinctive skin findings. Vitamin C deficiency (scurvy): perifollicular hemorrhage, corkscrew hairs, gingival bleeding. Niacin deficiency (pellagra): photosensitive dermatitis in sun-exposed areas, diarrhea, dementia.

Nutritional Disorders

Conditions caused by deficiency, excess, imbalance, or impaired use of nutrients.

Nutritional Science

The study of nutrients, dietary patterns, metabolism, and their effects on health and disease.

Nutritional Screening

Rapid assessment for malnutrition risk. Tools: MST, MNA, MUST. Screening all elderly patients recommended. Identifies need for comprehensive assessment.

Nutritive

Pertaining to, promoting, or providing biological nutrition and sustenance. Refers to dietary constituents (macronutrients: carbohydrates, proteins, fats; and micronutrients: vitamins, essential minerals) capable of undergoing metabolic absorption to supply cellular energy, sustain cellular homeostasis, and support tissue growth and repair. Contrasts with non-nutritive substances (e.g., non-nutritive artificial sweeteners like aspartame and sucralose) that supply sweetness without contributing significant metabolic calories.

Nux Vomica

The seed of *Strychnos nux-vomica*, containing strychnine and brucine; ingestion can cause severe convulsant poisoning.

Nyctalga

Pain occurring at night; the term is uncommon and requires clinical context.

Nyctalopia

Impaired vision in dim light or darkness, also called night blindness.

Nycteralgia

Night pain, an uncommon term sometimes used interchangeably with nyctalga.

NYHA Functional Classification

A four-class system grading heart failure severity based on symptoms and exercise tolerance. Class I: no limitation of physical activity. Class II: slight limitation; comfortable at rest but ordinary activity causes symptoms. Class III: marked limitation; comfortable at rest but less than ordinary activity causes symptoms.

Nystagmus

Nystagmus is involuntary rhythmic eye movement that can arise from vestibular disease, neurologic disorders, impaired vision, medication, or normal infant development. rotational, or mixed in pattern, and it is classified based on the underlying cause as infantile (developing in the first six months of life), acquired (developing later in life), or periodic alternating.

O

OAB

Alternate index label; see *Overactive bladder*.

Oat Cell Carcinoma

An older name for small-cell carcinoma of the lung, an aggressive neuroendocrine cancer strongly associated with tobacco exposure.

Oath Of Maimonides

The Oath of Maimonides is a historical ethical prayer emphasizing compassion, dedication, humility, and responsibility in the practice of medicine.

OB/GYN

OB/GYN refers to obstetrics and gynecology, the medical specialties concerned with pregnancy and reproductive-system health.

Obatoclax

An investigational anticancer drug that blocks BCL-2 family proteins, which help cancer cells avoid programmed cell death.

Obatoclax Mesylate

The mesylate salt form of obatoclax, an investigational BCL-2 family inhibitor studied as an anti-cancer medicine.

Obese

Obese describes a body-mass index of 30 or higher in adults, though BMI is an imperfect measure and health assessment should include overall risk and function.

Obesity

A condition in which excess body fat affects health. It is influenced by genes, metabolism, environment, behavior, and social factors and can increase the risk of diabetes, high blood pressure, heart disease, sleep apnea, joint disease, and some cancers.

Obesity (Weight Loss)

Alternate index label for Weight Loss.

Obesity Hypoventilation Syndrome

A condition in which a person with obesity does not breathe deeply enough while awake, causing too much carbon dioxide in the blood. It often occurs with obstructive sleep apnea and may require weight treatment and breathing support.

Obesity Hypoventilation Syndrome

Alternate index label; see *Obesity Hypoventilation Syndrome*.

Obesity Rating

A classification or score describing excess body weight, commonly based on body-mass index and sometimes waist circumference or health risk.

Obesity, Exogenous

Obesity attributed mainly to excess energy intake relative to expenditure, rather than to a primary endocrine disorder; genetics, medicines, sleep, and environment can also contribute.

Obex Region

A small area at the bottom of one of the brain's fluid-filled spaces, where it narrows toward the spinal

cord.

Objective

Based on observable findings or measurements rather than personal feelings or impressions.

Objective Response Rate

The percentage of people in a cancer study whose tumors shrink by a defined amount during treatment.

Objects Or Insects In Ear

An object or insect in the ear canal can cause pain, hearing loss, bleeding, discharge, or a sensation of movement. Do not insert tools or liquids; removal by a clinician is safest when the object is deep, sharp, chemical, or near the eardrum.

Alternate Names:

Ear Object In (Objects Or Insects In Ear)

Obligate

Required for a process or able to live only under certain conditions, such as a microorganism that cannot live with oxygen.

Oblimersen Sodium

Oblimersen sodium is an investigational antisense drug designed to reduce BCL-2 protein expression and increase cancer-cell sensitivity to treatment.

Oblique Fracture

A break in a bone that runs diagonally across it.

Oblique Plane

A plane that passes through the body or an organ at an angle other than a right angle (diagonal plane).

Obliteration

Complete closure, destruction, or disappearance of a space, channel, or lumen.

Observation

Deliberate monitoring of a person, symptom, disease, or test result without immediate intervention.

Observational Study

observational study is a developmental, reproductive, family, or study-design term describing how a person, embryo, pregnancy, or research subject is classified or observed.

Observer Variation

Differences between observers when they assess, measure, or interpret the same finding, which can affect clinical reliability and research results.

Obsession

A recurrent, intrusive, unwanted thought, image, or urge that causes distress and is difficult to control.

Obsessions

Recurrent, persistent, intrusive, and unwanted thoughts, urges, or images that are experienced as intrusive and cause marked anxiety or distress. Common obsessions include contamination fears, fears of harming oneself or others, need for symmetry, taboo thoughts (sexual, religious, or aggressive), and doubt.

Obsessive Compulsive Disorder

Alternate index label for OCD.

Obsessive-Compulsive Disorder

Characterized by the presence of obsessions, compulsions, or both. Obsessions are recurrent, persistent, intrusive, and unwanted thoughts, urges, or images that cause marked anxiety or distress. The individual attempts to ignore or suppress them or neutralize them with some other thought or action.

Obsessive-Compulsive Disorder

A disorder involving recurrent obsessions, compulsions, or both, which are time-consuming or cause significant distress or impairment.

Obsessive-Compulsive Personality Disorder

A pattern of preoccupation with orderliness, perfectionism, and mental and interpersonal control, at the expense of flexibility and efficiency.

Obstetric Anal Sphincter Injury

A severe tear during childbirth that involves the muscles controlling bowel movements. It needs careful repair and follow-up because it can affect bowel control.

Obstetric Cholestasis

Alternate index label; see *Cholestasis of pregnancy*.

Obstetric Fistula

A hole that forms between the birth canal and the bladder or rectum after prolonged, obstructed labour without timely medical care. It causes continuous leakage of urine or stool and may lead to skin infections, kidney problems, and social isolation. Most cases can be prevented with timely maternity care, and many can be repaired with surgery.

Obstetric Nursing

Obstetric nursing provides care during pregnancy, labor, birth, and the postpartum period, including monitoring, education, support, and recognition of complications.

Obstetric Ultrasound

An ultrasound scan during pregnancy used to check the baby's age, growth, anatomy, position, heart-beat, and sometimes the placenta and fluid.

Obstetrical Forceps

A surgical instrument used in selected vaginal deliveries to guide the baby's head when delivery needs assistance; use requires appropriate position, skilled assessment, and informed consent.

Obstetrician

A medical doctor who specializes in the care of women during pregnancy, labor, and the postpartum period (obstetrics). Obstetricians manage normal and high-risk pregnancies, perform cesarean deliveries and assisted vaginal deliveries, and treat pregnancy complications (pre-eclampsia, gestational diabetes, hemorrhage, preterm labor).

Obstetrics

The branch of medicine and surgery concerned with the care of women during pregnancy, childbirth, and the postpartum period.

Obstipation

Severe constipation with inability to pass stool or gas, often suggesting intestinal obstruction.

Obstructed Infected Urinary System

A blocked urinary tract with an infection behind the blockage. It is a medical emergency because infection can spread quickly and the blockage often must be relieved urgently.

Obstructed Labour

Failure of the fetus to pass through the birth canal despite adequate uterine contractions, commonly because of disproportion, abnormal presentation, or an obstructing lesion.

Obstruction

A blockage that prevents urine from flowing normally. Stones, narrowing, tumors, or an enlarged prostate can cause it and may lead to infection or kidney damage.

Obstructive Azoospermia

A form of male infertility characterized by the complete absence of spermatozoa in the ejaculate despite normal spermatogenesis within the testicular parenchyma, caused by physical obstruction or absence of the excurrent reproductive duct system. Etiologies include congenital bilateral absence of the vas deferens (CBAVD, strongly associated with CFTR gene variants in cystic fibrosis), previous vasectomy, post-infectious scarring (epididymitis, gonococcal urethritis), ejaculatory duct obstruction, or iatrogenic surgical injury during hernia repair. It is characterized by normal testicular volume, normal serum FSH and testosterone levels, and dilated epididymides. Treatment options include microsurgical reconstructive surgery (vasovasostomy or vasoepididymostomy) or percutaneous/microsurgical epididymal or testicular sperm aspiration (PESA/MESA/TESE) for use in in-vitro

fertilization.

Alternate Names:

OA; Excretory Azoospermia

Obstructive Hydrocephalus

A buildup of cerebrospinal fluid caused by a blockage in its normal flow through the brain's ventricles.

Obstructive Jaundice

Jaundice caused by obstruction of bile flow from the liver to the duodenum, resulting in accumulation of conjugated bilirubin in the blood. See Jaundice for complete overview.

Obstructive Nephropathy

Kidney damage caused by blockage of urine flow anywhere from the kidney to the urethra. Causes include stones, tumors, strictures, enlarged prostate, and congenital abnormalities; untreated obstruction can lead to hydronephrosis, infection, and kidney failure.

Obstructive Parotitis

Inflammation of the parotid gland resulting from obstruction of the parotid duct, most commonly by sialolithiasis, presenting with painful glandular swelling that worsens with meals. The obstruction leads to stasis, secondary bacterial infection, and recurrent episodes of swelling and tenderness.

Obstructive Shock

A life-threatening drop in blood flow caused by a physical blockage, such as a large blood clot in the lungs, pressure around the heart, or a collapsed lung under pressure.

Obstructive Sleep Apnea

Repeated narrowing or closing of the upper airway during sleep, causing pauses in breathing, snoring, poor sleep, and daytime tiredness. Treatment may include a breathing device, weight management, an oral appliance, position changes, or surgery.

Alternate Names:

Sleep Apnea, Obstructive

Obstructive Sleep Apnea - Adults

Obstructive sleep apnea is repeated upper-airway collapse during sleep, causing pauses in breathing, oxygen fluctuation, snoring, fragmented sleep, and daytime impairment.

Obstructive Uropathy

Urinary tract obstruction at any level from the renal pelvis to the urethra, leading to impaired urine flow and potential renal damage. Causes include nephrolithiasis, benign prostatic hyperplasia, ureteral strictures, tumors of the bladder, prostate, or cervix, and external compression from lymphadenopathy or retroperitoneal fibrosis.

Obtunded

Having reduced alertness and slowed responses, but being more responsive than a person who is comatose.

Obturator

A structure that occludes an opening. In anatomy, the obturator foramen is a large opening in the pelvic bone, closed by the obturator membrane. The obturator nerve (L2–L4) innervates the adductor muscles of the thigh and provides sensory innervation to the medial thigh. The obturator artery supplies the adductor compartment.

Obturator Hernia

A rare hernia in which part of the intestine pushes through an opening in the pelvic bone. It can block the bowel and may cause pain in the inner thigh.

Obtuse Marginal Artery

A branch of the circumflex coronary artery that runs along the obtuse margin of the heart and supplies the lateral wall of the left ventricle. Its size and number vary with coronary anatomy.

Occam's Razor

Occam's razor is the reasoning principle favoring the simplest explanation that accounts for the findings, while recognizing that patients can have multiple conditions.

Occipital

Relating to the back of the head, the occipital bone, or the occipital lobe.

Occipital Bone

The bone at the back and base of the skull. It surrounds the opening where the spinal cord joins the brain and connects with the first neck vertebra.

Occipital Diastasis

Occipital diastasis is separation or widening of an occipital skull suture, usually related to birth trauma or developmental abnormality and assessed with imaging.

Occipital Hematoma

A collection of blood in the scalp or tissues over the back of the head, often after trauma; a rapidly enlarging swelling or associated neurologic symptoms requires urgent assessment.

Occipital Lobe

The part of the brain at the back of the head that processes vision.

Occipital Neuralgia

A paroxysmal, severe, stabbing or electric shock-like pain in the distribution of the greater, lesser, or third occipital nerves (posterior scalp, from the nuchal line to the vertex).

Occiput

The posterior part of the skull or the back of the head.

Occiput Posterior Position

A fetal head position in which the occiput faces the maternal sacrum. It may cause back pain or slow labor, although the fetus may rotate during labor.

Occiput Transverse Position

The fetal head presenting with the sagittal suture oriented transversely. Usually the normal descent position; most rotate to OA with contractions. When persistent, it can cause protracted descent

and prolonged second stage.

Occlude

To occlude means to block or close a passage, vessel, airway, duct, or opening.

Occlusal Adjustment

Selective reshaping of tooth surfaces to improve how the upper and lower teeth meet and reduce abnormal contact or bite interference.

Occlusal Guard

Appliance protecting teeth from attrition. Night guard for bruxism. Sports mouthguard.

Occlusal Splint

A removable dental appliance that changes or protects tooth contact and may reduce effects of grinding, jaw-joint strain, or selected bite problems.

Occlusion

Complete blockage of vessel. May be natural (thrombosis) or iatrogenic (embolization).

Occlusive Vascular Disease

Disease in which narrowing or blockage of a blood vessel reduces or stops blood flow to tissues.

Occult

Hidden or not visible without microscopy or laboratory testing.

Occult Blood

Blood that is not visible to the naked eye and is detected by laboratory testing. In stool, it may be detected by a chemical guaiac test or an immunochemical test; the method and specimen determine the collection instructions and possible dietary or medicine effects. A positive stool test indicates blood is present but does not establish its cause.

Occult Blood Screen

Alternate index label; see *Fecal occult blood test*.

Occult Gastrointestinal Bleeding

Chronic or intermittent gastrointestinal blood loss that is not visible to the patient and may not produce melaena or haematochezia. It commonly presents with iron-deficiency anaemia or a positive faecal blood test.

Occult Primary Cancer

Alternate index label; see *Carcinoma of unknown primary*.

Occupation

A person's usual work or activity; occupational history can identify exposures relevant to health.

Occupational Asthma

Asthma from workplace exposures to sensitizers (isocyanates, flour, latex) or irritants. Improves away from work and worsens at work. Serial peak flow monitoring supports diagnosis.

Occupational COPD

Chronic obstructive pulmonary disease associated with workplace exposure to dust, chemicals, fumes, or other inhaled hazards. Symptoms and risk depend on the exposure and other factors, including smoking.

Occupational Dermatoses

Work-related skin diseases. Occupational contact dermatitis accounts for 90 percent of occupational skin diseases. Irritant contact dermatitis is more common (hands, wet work) than allergic contact dermatitis. Occupations at risk include healthcare workers, hairdressers, construction workers, food handlers, and metalworkers.

Occupational Disease

An illness caused or substantially contributed to by workplace exposures or conditions.

Occupational Exposure

Radiation exposure to nuclear medicine staff. Monitored with dosimeters. Limits: 50 mSv/year whole body, 150 mSv/year lens, 500 mSv/year extremities.

Occupational Exposure Limit

A regulatory or advisory concentration or dose limit intended to reduce worker exposure to a hazardous substance over a defined averaging period. Limits vary by substance, jurisdiction, route, and whether the value is a ceiling, short-term limit, or time-weighted average.

Occupational Health

The field concerned with protecting and promoting workers' physical, mental, and social health.

Occupational Hearing Loss

Permanent sensorineural hearing impairment caused by repeated or prolonged exposure to workplace noise or ototoxic hazards. Prevention depends on reducing exposure and using effective hearing protection.

Occupational Rehabilitation

Services that help a person regain or adapt the skills needed for work after illness, injury, or disability.

Occupational Safety And Health

Policies and practices that prevent work-related injury, illness, hazardous exposure, and unsafe conditions.

Occupational Therapist

A healthcare professional who helps people perform daily activities through rehabilitation, adaptive techniques, environmental changes, and assistive equipment.

Occupational Therapy

OT helps patients perform daily activities (ADLs, IADLs) through therapeutic activities, adaptive equipment, environmental modification. Focus on upper extremity function, cognition, sensory integration.

Occupational Therapy Assessment

Evaluation of a person's ability to perform daily activities, work, and other meaningful occupations, including factors affecting performance and the need

for adaptive strategies or equipment.

OCD

Obsessive-compulsive disorder, a condition involving recurrent unwanted thoughts or urges and/or repetitive behaviors or mental acts that cause distress or interfere with daily life.

Alternate Names:

Obsessive Compulsive Disorder (OCD)

Ochre

A yellow-brown earth pigment; ochre discoloration may describe skin or tissue pigmentation.

Ochrobactrum

A genus of gram-negative environmental bacteria that can rarely cause opportunistic human infection.

Ochrobactrum Anthropi

A gram-negative environmental bacterium that can cause opportunistic bloodstream, wound, device-related, or other infections, especially in vulnerable patients.

Ochrobactrum Intermedium

A gram-negative *Ochrobactrum* species found in the environment that can rarely cause opportunistic infection.

Ochronosis

A blue-black or brownish pigmentation of connective tissues (cartilage, skin, sclerae, tendons) caused by accumulation of homogentisic acid and its oxidation products.

OCP

OCP commonly means oral contraceptive pill, a medicine containing estrogen, progestin, or both to prevent pregnancy and sometimes treat menstrual disorders.

Octadecadienoic Acid

octadecadienoic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and

metabolism.

Octanoic Acid

octanoic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Octopamine

A chemical messenger found mainly in invertebrates that regulates movement, metabolism, arousal, and other physiological functions.

Octreoscan

Nuclear medicine imaging using In-111 octreotide. Detects neuroendocrine tumors.

Octreotide

Octreotide is a somatostatin analogue that suppresses growth hormone, several gastrointestinal hormones, and intestinal secretion; it is used for acromegaly, certain neuroendocrine tumors, and selected secretory diarrhoeas.

Ocular

Relating to, affecting, or situated within the eye (*oculus*) or the visual apparatus. Clinically used in ophthalmology to describe structures (ocular adnexa, ocular media, ocular fundus), diagnostic examinations (optical coherence tomography [OCT], ocular ultrasonography), and systemic diseases with eye involvement, such as ocular toxoplasmosis, ocular manifestations of rheumatoid arthritis (scleritis, peripheral ulcerative keratitis), and ocular emergencies including acute angle-closure glaucoma and central retinal artery occlusion.

Ocular Albinism

Ocular albinism is an inherited disorder with reduced pigment primarily in the eyes, causing decreased visual acuity, nystagmus, photophobia, and abnormal retinal development.

Ocular Fixation

The ability to hold the eyes steadily on a target; abnormal fixation can impair visual acuity, tracking,

or alignment.

Ocular Histoplasmosis Syndrome

An eye condition linked to an earlier histoplasmosis infection. It can cause scars in the retina and, in some people, abnormal blood vessels that damage central vision.

Ocular Hypertension

Higher-than-normal pressure inside the eye without current signs of glaucoma damage. It increases the future risk of glaucoma and needs regular eye checks.

Ocular Melanoma

A cancer that begins in pigment-producing cells of the eye.

Ocular Migraine

A migraine-related visual disturbance, often involving flashing lights, zigzag lines, or a temporary blind spot. It may occur with or without headache.

Ocular Myasthenia Gravis

A localized, organ-specific clinical presentation of autoimmune myasthenia gravis in which muscle weakness is strictly confined to the extraocular muscles, levator palpebrae superioris, and orbicularis oculi, with no clinical involvement of bulbar, facial, or extremity musculature. Manifests with fluctuating, fatigable, asymmetric ptosis and binocular diplopia that characteristically worsens toward the end of the day or with prolonged gaze; serologically positive for acetylcholine receptor antibodies in approximately 50–70% of cases, with up to 50–80% of patients progressing to generalized myasthenia gravis within two years of symptom onset if untreated.

Alternate Names:

OMG; Ocular MG

Ocular Myiasis

Infestation of the eye or nearby tissues by fly larvae. It can cause pain, irritation, tearing, and vision problems and requires prompt eye care.

Ocular Orbit

The ocular orbit is the bony cavity containing the eye, extraocular muscles, nerves, vessels, glands, and protective connective tissue.

Ocular Pain

Pain in or around the eye caused by surface irritation, infection, inflammation, trauma, glaucoma, or orbital disease. Severe pain with vision loss, light sensitivity, nausea, or injury requires urgent ophthalmic assessment.

Ocular Rosacea

Inflammation affecting the eyes and eyelids in people with rosacea, causing irritation, redness, or dryness.

Ocular Surface Disease

A group of conditions affecting the eye's surface, eyelids, or tears, including dry eye and eyelid inflammation. Symptoms may include burning, redness, irritation, and blurred or changing vision.

Ocular Toxoplasmosis

A retinal infection or inflammation caused by *Toxoplasma gondii*, which can produce blurred vision, floaters, pain, and recurrent retinal scars.

Oculist

An older term for an eye specialist; modern usage distinguishes ophthalmologists and optometrists.

Oculocardiac Reflex

A slowing of the heartbeat or similar body response caused by pressure or pulling on the eye or tissues around it.

Oculocerebrorenal Syndrome

A rare inherited disorder, also called Zellweger spectrum disorder, that affects the brain, eyes, kidneys, liver, and other organs. Severe forms begin in newborns and can cause weak muscles, seizures, feeding problems, and major developmental problems.

Alternate Names:

Zellweger spectrum disorder, affecting.

Oculocutaneous Albinism

Alternate index label; see *Albinism*.

Oculogyric Crisis

A prolonged, involuntary upward turning of the eyes, often caused by a medicine or neurologic disorder.

Oculomotor Nerve

Cranial nerve III, controlling most eye movements, eyelid elevation, and pupil constriction.

Oculomotor Nerve Palsy

Weakness or paralysis of muscles supplied by the third cranial nerve, causing eyelid drooping, double vision, abnormal eye position, and sometimes a dilated pupil.

Oculomotor Nucleus

The oculomotor nucleus is a midbrain group of motor neurons controlling most eye movements, eyelid elevation, and pupil constriction through the third cranial nerve.

Oculopharyngeal Muscular Dystrophy

An adult-onset, autosomal dominant (or rarely autosomal recessive) myopathy caused by short expansions of a GCN trinucleotide repeat encoding an expanded polyalanine tract in the polyadenylate-binding protein nuclear 1 (*PABPN1*) gene on chromosome 14q11. Typically presents in the fifth to sixth decade of life with progressive bilateral ptosis, severe dysphagia due to pharyngeal muscle involvement, tongue atrophy, and proximal lower and upper limb girdle weakness. Characterized histopathologically by intranuclear filamentous tubulofilamentous inclusions in skeletal muscle fibers.

Alternate Names:

OPMD

Oculoplastic Procedures

A medical procedure, treatment, device, or care topic related to oculoplastic procedures; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Oculosympathetic Palsy

Alternate index label; see *Horner syndrome*.

ODD

Alternate index label; see *Oppositional defiant disorder*.

Oddi, Sphincter Of

A circular muscle regulating the flow of bile and pancreatic juice into the duodenum.

Odontalgia

Toothache or pain arising from a tooth or its supporting tissues.

Odontoblast

A cell that produces dentin during tooth development and repair.

Odontoclast

A cell that resorbs dental hard tissue, analogous to an osteoclast.

Odontodysplasia

Odontodysplasia is abnormal development of tooth enamel and dentin, producing thin, weak, discolored teeth that may erupt late or be easily damaged.

Odontogenesis

The complex, highly regulated developmental process by which teeth form from embryonic oral ectoderm (giving rise to enamel via ameloblasts) and neural crest-derived ectomesenchyme (giving rise to dentin, dental pulp, cementum, and periodontal ligament via odontoblasts and cementoblasts). Progresses through discrete morphological stages: dental lamina formation, bud stage, cap stage, bell stage (histodifferentiation and morphodifferentiation), and root development. Perturbations during odontogenesis produce developmental anomalies, including hypodontia, amelogenesis imperfecta, and dentinogenesis imperfecta.

Odontogenic Cutaneous Fistula

An abnormal tract connecting the skin surface to

a periapical dental infection, typically presenting as a draining sinus tract on the chin, cheek, or submandibular region. It results from chronic dental infection eroding through the bone and soft tissue to the skin surface.

Odontogenic Cyst

A fluid-filled sac that develops from tissues involved in tooth formation, usually in the jaw.

Odontogenic Fibroma

An odontogenic fibroma is a rare benign tumor arising from tooth-forming connective tissue, usually presenting as a jaw lesion.

Odontogenic Keratocyst

Aggressive cyst with high recurrence. Associated with nevoid basal cell carcinoma syndrome (Gorlin syndrome). Treatment: enucleation, resection, or decompression.

Odontogenic Neoplasm

An odontogenic neoplasm is a growth arising from tissues involved in tooth formation, usually in the jaw; it may be benign or malignant.

Odontogenic Tumors And Cysts

Alternate index label; see *Jaw tumors and cysts*.

Odontoid Process

A peg-like projection from the second neck vertebra that fits into the first neck vertebra and allows the head to turn from side to side.

Odontology

The study of teeth, their structure, development, diseases, and treatment.

Odontoma

Hamartomatous dental tissue mass. Compound (multiple tooth-like structures) or complex (amorphous calcified mass). Treatment: surgical removal.

Odontopathy

Any disease, structural anomaly, or pathological dis-

order affecting the teeth or their immediate supporting structures. Encompasses infectious processes (dental caries driven by *Streptococcus mutans*, pulpitis, periapical abscesses), non-carious hard tissue loss (attrition, abrasion, erosion, abfraction), developmental defects (microdontia, enamel hypoplasia), and traumatic dental injuries. Without timely operative or endodontic intervention, localized odontopathies can breach the alveolar bone to trigger deep neck space infections (e.g., Ludwig angina) or systemic odontogenic bacteremia.

Odor Threshold

Lowest concentration detectable by smell. Important for warning properties of toxins.

Odorant

An odorant is a chemical molecule detected by olfactory receptors and perceived as a smell.

Odoribacter Splanchnicus

An anaerobic bacterium found in the human intestine that produces short-chain fatty acids and is rarely associated with infection.

Odour

A smell detected by the olfactory system; unusual body or breath odor can be a clinical clue.

Odynophagia

Painful swallowing, characterized by a burning, squeezing, or sticking sensation behind the sternum during deglutition. Strongly suggests mucosal inflammation or ulceration of the esophagus.

Oedema

Abnormal accumulation of fluid in interstitial tissues, causing swelling.

Oedipus Complex

A historical psychoanalytic concept describing unconscious sexual attachment to one parent and rivalry with the other; it is not a modern medical diagnosis.

Oerskovia

A genus of gram-positive bacteria, now often classified within *Cellulosimicrobium*, found in soil and occasionally in clinical specimens.

Oerskovia Turbata

A gram-positive environmental bacterium now generally called *Cellulosimicrobium turbata*; it rarely causes opportunistic infection.

Oesophageal

Relating to the oesophagus, the muscular tube carrying swallowed material from the pharynx to the stomach.

Oesophageal Atresia

A congenital interruption or blind ending of the oesophagus, often associated with a tracheoesophageal fistula.

Oesophageal Ulcer

A mucosal ulceration of the oesophagus caused by gastro-oesophageal reflux, infection, medication or corrosive injury, inflammatory disease, or malignancy. It may cause odynophagia, dysphagia, chest pain, haematemesis, or melaena.

Oesophageal Varices

Dilated submucosal veins of the oesophagus, usually caused by portal hypertension and capable of life-threatening bleeding.

Oesophagitis

Inflammation of the oesophageal mucosa. Common causes include gastro-oesophageal reflux, infection in immunocompromised patients, medication-related injury, eosinophilic inflammation, radiation, and caustic ingestion. Symptoms include heartburn, odynophagia, dysphagia, and chest pain.

Oesophagoscopy

Direct endoscopic visualization of the lumen and mucosal lining of the esophagus using either a flexible fiberoptic/video endoscope (most common, typically as part of esophagogastroduodenoscopy [EGD]) or a

rigid metal esophagoscope (primarily used in ENT and thoracic surgery for foreign body removal or stent deployment). Indicated for diagnostic evaluation of dysphagia, odynophagia, gastroesophageal reflux disease refractory to medical therapy, suspected Barrett esophagus, and upper gastrointestinal hemorrhage (e.g., esophageal varices requiring endoscopic band ligation), as well as therapeutic dilation of benign esophageal strictures.

Oesophagus

The muscular tube that transports swallowed food and liquid from the pharynx to the stomach.

Oesophagus, Diseases Of

Disorders affecting the oesophagus, including reflux, inflammation, strictures, motility disorders, cancer, and congenital abnormalities.

Oestradiol

The principal naturally occurring estrogen during the reproductive years.

Oestriol

An estrogen produced in substantial amounts by the placenta during pregnancy.

Oestrogen

The primary female sex hormone, produced mainly by the ovaries (granulosa cells of developing follicles) and in smaller amounts by adipose tissue, the adrenal glands, and the placenta during pregnancy.

Oestrogen Replacement Therapy

The administration of exogenous estrogen (with or without progestogen) to relieve symptoms of estrogen deficiency and prevent long-term health consequences in postmenopausal women. See Hormone Replacement Therapy (HRT).

Oestrogens

A group of steroid hormones involved in reproductive development, menstrual cycling, pregnancy, bone, and other physiological functions.

Oestrone

An estrogen produced by the ovaries and peripheral conversion of androgens, especially after menopause.

Ofatumumab

A monoclonal antibody against CD20 used for selected blood cancers and relapsing multiple sclerosis; it can increase infection risk and reduce B cells.

Off-Label

Use of an approved medicine for an indication, dose, route, or population not specified in its regulatory approval.

Off-Label Use

The use of a licensed medication for an indication, dose, route, or patient population not included in its marketing authorisation (product licence).

Offspring

Offspring are the descendants or young produced by an organism or pair of organisms.

Ofloxacin

A fluoroquinolone antibiotic used for selected susceptible bacterial infections.

Ogilvie Syndrome

Acute colonic pseudo-obstruction characterized by massive colonic dilation in the absence of mechanical obstruction typically occurring in hospitalized patients with serious comorbidities including post-surgical states, severe infections, and medication effects.

Ogilvie'S Syndrome

Acute colonic pseudo-obstruction characterized by massive dilation of the colon, particularly the cecum and right colon, without mechanical obstruction. It occurs in hospitalized patients with serious medical or surgical conditions, often postoperatively, and results from autonomic nervous system imbalance.

Ohm

The SI unit of electrical resistance, equal to one volt

per ampere.

OHSS

Alternate index label; see *Ovarian hyperstimulation syndrome*.

Oidiodendron

Oidiodendron is a genus of environmental molds that rarely causes opportunistic infection.

Oidiomycosis

An older term for infection caused by yeast-like fungi, historically including candidiasis.

Oidium

A fragment or spore-like fungal element formed by fragmentation of a hypha; the term is also used in older fungal classifications.

Oil

A hydrophobic liquid used as a nutrient, solvent, emollient, or medicinal preparation; safety depends on the specific oil.

Oil-Based Paint Poisoning

Oil-based paint poisoning involves toxic or environmental exposure to oil-based paint poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Oily Hair

Oily hair involves toxic or environmental exposure to oily hair; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Oily Skin

Oily skin involves toxic or environmental exposure to oily skin; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Ointment

A semi-solid, greasy topical preparation for application to the skin, mucous membranes, or eyes. Ointments are typically composed of a hydrophobic base (petrolatum, lanolin, paraffin, beeswax, veg-

etable oils) and may contain dissolved or suspended active pharmaceutical ingredients.

Ointment Bases

Semisolid vehicles used to carry medicines applied to the skin or mucous membranes; they influence spread, absorption, moisture retention, and stability.

Okadaic Acid

okadaic acid is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Okhihiro Syndrome

A congenital disorder, usually caused by pathogenic variants in SALL4, combining radial-ray limb abnormalities with Duane retraction anomaly or other eye-movement defects. Hearing, cardiac, renal, and vertebral anomalies may also occur.

Olanzapine

Olanzapine is an atypical antipsychotic that blocks dopamine and serotonin receptors and treats schizophrenia and bipolar disorder; weight gain, diabetes, dyslipidaemia, sedation, and movement disorders are important risks.

Olaparib

Olaparib is a PARP inhibitor used for selected cancers with BRCA or other homologous-recombination repair defects; anaemia, nausea, fatigue, myelosuppression, and rare myelodysplastic syndrome or acute myeloid leukaemia may occur.

Olaratumab

A monoclonal antibody formerly studied for soft-tissue sarcoma; it was withdrawn in several jurisdictions after confirmatory benefit was not demonstrated.

Old Age

Later life; health assessment should consider function, comorbidity, cognition, and social circumstances rather than age alone.

Oleander Poisoning

Oleander poisoning involves toxic or environmental exposure to oleander poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Oleandomycin

Oleandomycin is an older macrolide antibiotic that is now rarely used because of limited availability, resistance, and adverse effects.

Olecranon

The prominent bony process at the proximal end of the ulna forming the point of the elbow.

Oleic Acid

A monounsaturated fatty acid abundant in olive oil and many other dietary fats.

Oleum

A concentrated preparation of sulfur trioxide in sulfuric acid, highly corrosive and hazardous.

Olfaction

The sense of smell. Odor molecules stimulate special cells in the upper part of the nose, which send signals to the brain to identify and perceive odors.

Olfactory

Relating to the sense of smell or the olfactory nerve and pathways.

Olfactory Bulb

The olfactory bulb is a forebrain structure that receives signals from nasal smell receptors and begins processing odor information.

Olfactory Cortex

The olfactory cortex is a group of brain regions that interprets smell signals and links them with memory, emotion, and behavior.

Olfactory Gyrus

The olfactory gyrus is a cortical region associated with processing olfactory information and includes

areas near the medial temporal and frontal lobes.

Olfactory Learning

Olfactory learning is acquisition or modification of behavior or memory through exposure to odors and their associations.

Olfactory Mucosa

The olfactory mucosa is specialized nasal lining containing receptor neurons, supporting cells, and glands that detect odor molecules.

Olfactory Nerve

Cranial nerve I (CN I), responsible for the sense of smell. Olfactory receptor neurons in the olfactory epithelium of the nasal roof project axons through the cribriform plate of the ethmoid bone to the olfactory bulb, then via the olfactory tract to the primary olfactory cortex (piriform cortex, amygdala, entorhinal cortex).

Olfactory Neuroblastoma

A rare malignant tumor arising in the upper nasal cavity near the olfactory nerves, often causing nasal blockage, nosebleeds, or reduced smell.

Olfactory Tract

The olfactory tract is a bundle of nerve fibers carrying smell information from the olfactory bulb to cerebral regions involved in odor perception.

Olfactory Tubercle

The olfactory tubercle is a ventral forebrain region involved in odor processing, reward, motivation, and integration with limbic circuits.

Oligella

Oligella is a genus of bacteria that can rarely cause opportunistic urinary, bloodstream, or other infections.

Oligella Ureolytica

A gram-negative bacterium that can rarely cause urinary, bloodstream, wound, or other opportunistic infections.

Oligella Urethralis

A gram-negative bacterium associated with the urinary tract that rarely causes opportunistic infection.

Oligoasthenoteratozoospermia

A combined semen abnormality characterized by concurrent deficits across all three primary quantitative and qualitative semen parameters: low sperm count or concentration (oligozoospermia, <16 million/mL), impaired progressive motility (asthenozoospermia, <32% progressively motile sperm), and an abnormally low percentage of morphologically normal forms (teratozoospermia, <4% normal morphology by strict Kruger criteria). OAT syndrome is a frequent presentation in idiopathic male subfertility and can result from clinical varicoceles, heat exposure, endocrine disruption, oxidative stress, tobacco or gonadotoxin exposure, or genetic factors. When severe, assisted reproduction via intracytoplasmic sperm injection is frequently indicated.

Alternate Names:

OAT Syndrome; OAT

Oligoclonal Bands

Immunoglobulin G (IgG) proteins produced by a limited number of B-cell clones in the cerebrospinal fluid (CSF), detected by isoelectric focusing of CSF and serum samples.

Oligocythaemia

An abnormally low number of cells in the blood, especially red blood cells; the term is uncommon.

Oligodendrocyte

A central-nervous-system glial cell that forms myelin around axons.

Oligodendroglia

Oligodendroglia are glial cells of the central nervous system that produce myelin around axons and support nerve conduction.

Oligodendroglial Neoplasm

An oligodendroglial neoplasm is a brain tumor with features of oligodendrocyte-lineage cells; diagnosis

depends on microscopic and molecular findings.

Oligodendroglioma

A slow-growing glioma arising from oligodendrocytes, most often in the cerebral hemispheres. Associated with IDH mutation and 1p/19q codeletion, which confer a better prognosis and chemosensitivity.

Oligodontia

Congenital absence of many teeth, usually more than six permanent teeth, excluding third molars.

Oligogenic Inheritance

Oligogenic inheritance occurs when variants in a small number of genes together influence a trait or disease, often with variable expression.

Oligohydramnios

Insufficient amniotic fluid volume, defined as amniotic fluid index less than 5 cm or single deepest pocket less than 2 cm. Causes include fetal renal anomalies, post-term pregnancy, premature rupture of membranes, maternal dehydration, and placental insufficiency.

Oligomenorrhea

Oligomenorrhea is infrequent or widely spaced menstrual bleeding, often from anovulation, polycystic ovary syndrome, endocrine disease, undernutrition, excessive exercise, stress, or perimenopause.

Oligomenorrhoea

Infrequent menstrual periods, usually with cycles longer than about 35 days.

Oligometastasis

A limited number of metastatic deposits, often defined as up to five, amenable to aggressive local treatment (surgery, stereotactic radiation, or ablation) with the potential for prolonged control.

Oligomycin

Oligomycin is a research compound that blocks ATP production in mitochondria; it is not a routine clinical

medicine.

Oligonucleotide

An oligonucleotide is a short single- or double-stranded DNA or RNA sequence used in molecular biology, diagnostics, antisense therapy, gene silencing, or sequencing.

Oligopeptide

Oligopeptide is a defined chemical or biologic substance involved in oligopeptide and studied in physiology, pharmacology, laboratory medicine, or treatment.

Oligosaccharide

An oligosaccharide is a short chain of monosaccharides found in glycoproteins, glycolipids, and foods; its structure influences cell recognition, digestion, and the intestinal microbiome.

Oligospermia

A low sperm concentration in semen, assessed using current laboratory reference standards.

Oliguria

Reduced urine output, defined as <400 mL/day or <0.5 mL/kg/hour in adults. Oliguria is an important sign of acute kidney injury (AKI) and prerenal azotaemia.

Olivary

Relating to the olive-shaped inferior olivary nucleus of the medulla or to an olive-like structure.

Olivary Nucleus

The olivary nucleus is a brainstem structure that sends timing and motor-learning signals to the cerebellum through climbing fibers.

Olive Oil

A plant oil rich in monounsaturated fatty acids, used as food and in some topical preparations.

Olive Pontocerebellar Atrophy

A group of neurodegenerative disorders character-

ized by progressive loss of neurons in the inferior olives, pons, and cerebellum. They cause worsening ataxia, dysarthria, incoordination, and sometimes pyramidal, autonomic, or cognitive features.

Olivomycin

Olivomycin is an older antitumor antibiotic studied experimentally; it is not a standard routine treatment.

Olivopontocerebellar Atrophy

Olivopontocerebellar atrophy is a degenerative condition affecting the inferior olives, pons, and cerebellum, causing progressive ataxia and other neurologic dysfunction.

Ollier'S Disease

A nonhereditary skeletal disorder with multiple enchondromas, causing deformity, limb-length discrepancy, and an increased risk of malignancy.

Olmesartan Medoxomil

An oral prodrug converted to olmesartan, an angiotensin II receptor blocker used to treat high blood pressure.

Olopatadine Hydrochloride

Olopatadine hydrochloride is an antihistamine used in eye drops or nasal spray to treat allergic conjunctivitis or rhinitis.

OLP

Alternate index label; see *Oral lichen planus*.

Olsenella

Olsenella is a genus of anaerobic bacteria found mainly in the mouth and digestive tract; human infection is uncommon.

Olsenella Uli

An anaerobic gram-positive bacterium found in the mouth and associated with periodontal disease; invasive infection is rare.

Oltipraz

Oltipraz is an investigational dithiole compound studied for activation of cellular antioxidant defenses and cancer prevention.

Omacetaxine Mepesuccinate

Omacetaxine mepesuccinate is a protein-synthesis inhibitor used for selected chronic myeloid leukemia resistant or intolerant to tyrosine-kinase inhibitors.

Omalizumab

Omalizumab is a monoclonal antibody that binds circulating IgE and reduces allergic inflammation; it is used for selected allergic asthma, chronic urticaria, nasal polyps, and food allergy, with rare anaphylaxis.

Omasum

The omasum is the third compartment of a ruminant stomach, absorbing water and small particles before contents reach the abomasum.

Omega-3

Polyunsaturated fatty acids. Anti-inflammatory, cardiovascular benefits. Sources: fatty fish, flaxseed, walnuts. Supplementation common.

Omega-3 Fatty Acids

Polyunsaturated fats found especially in fatty fish and some plant foods.

Omega-6

Polyunsaturated fatty acids (linoleic acid). Pro-inflammatory in excess. Balance with omega-3 important. Sources: vegetable oils, nuts, seeds.

Omega-6 Fatty Acids

Essential polyunsaturated fats found in many plant oils and foods.

Alternate Names:

n-6 fatty acids

Omega-N-Methylarginine

Omega-N-methylarginine is a methylated arginine compound that can inhibit nitric-oxide synthase

and affect vascular signaling.

Omenn Syndrome

A severe combined immunodeficiency caused by hypomorphic defects affecting lymphocyte-receptor gene rearrangement or related pathways. It presents in infancy with erythroderma, lymphadenopathy, hepatosplenomegaly, diarrhea, failure to thrive, and recurrent or opportunistic infections.

Omentectomy

Removal of part or all of the greater omentum, commonly performed during staging and cytoreduction for ovarian and other intra-abdominal cancers.

Omentum

A double layer of peritoneum connecting the stomach to other abdominal organs. The greater omentum (gastrocolic omentum) hangs from the greater curvature of the stomach, covers the transverse colon and small intestine, and is rich in fat and immune cells (milky spots—aggregates of macrophages and lymphocytes).

Omeprazole

Omeprazole is a proton-pump inhibitor that irreversibly suppresses gastric H⁺/K⁺-ATPase; it treats reflux and peptic-ulcer disease and supports *Helicobacter pylori* eradication, while prolonged use may affect magnesium, B12, bone, and infection risk.

Ommaya Reservoir

A surgically implanted reservoir connected to a catheter in a brain ventricle, used to sample cerebrospinal fluid or deliver selected medicines.

Omnivore

An organism that eats both plant and animal matter; it is a dietary term rather than a disease.

Omphalitis

Infection of the umbilical stump in neonates, typically occurring within the first 1–2 weeks of life. Omphalitis is a potentially serious condition requir-

ing prompt treatment due to the risk of spread to the liver (portal vein thrombosis, portal hypertension) and peritoneum via the umbilical vessels.

Omphalocele

Omphalocele is a congenital abdominal-wall defect in which abdominal organs protrude into a membrane-covered sac at the umbilicus and may coexist with chromosomal or structural anomalies. umbilical cord, covered by a peritoneal-amniotic membrane. Associated with chromosomal abnormalities (trisomies 13, 18) and other anomalies. Differentiated from gastroschisis by membrane coverage and midline location.

Omphalocele Repair

A medical procedure, treatment, device, or care topic related to omphalocele repair; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

On Call

On call means available and responsible to respond to clinical or operational needs during a designated period outside routine scheduled activity.

Onchocerca Volvulus

A filarial nematode that causes onchocerciasis (river blindness), transmitted by repeated bites of infected blackflies (*Simulium* species) breeding near fast-flowing rivers. Adult worms reside in subcutaneous nodules (onchocercomata), where they can survive for years, producing microfilariae that migrate through the skin and eyes.

Onchocerciasis

A parasitic disease caused by the filarial nematode *Onchocerca volvulus*, transmitted by repeated bites of infected blackflies (*Simulium* species) breeding near fast-flowing rivers. Adult worms form subcutaneous nodules (onchocercomata).

Oncocytic Neoplasm

An oncocytic neoplasm is a growth made of cells with abundant mitochondria and granular cytoplasm; it may be benign or malignant depending on

the organ and tumor type.

Oncocytoma

A usually benign epithelial tumor, commonly arising in the kidney and composed of cells with abundant eosinophilic mitochondria-rich cytoplasm.

Oncogene

A gene that, when activated, overexpressed, or altered, can promote uncontrolled cell proliferation or survival. Oncogenes commonly arise from normal proto-oncogenes through mutation, amplification, translocation, or abnormal regulation.

Oncogenic

Capable of causing or promoting cancer. An oncogenic change, virus, gene, or chemical can increase abnormal cell growth or reduce normal control of cell division.

Oncogenic Virus

A virus capable of contributing to cancer by altering cell growth, suppressing tumor defenses, or integrating or maintaining oncogenic genetic material.

Oncogenicity

The ability of a substance, infection, genetic change, or other factor to contribute to the development of cancer.

Oncologic Emergency

A cancer-related condition that threatens life, organ function, or neurologic integrity and requires urgent treatment. Examples include malignant spinal-cord compression, superior vena cava syndrome, tumor lysis syndrome, hypercalcemia, airway obstruction, and febrile neutropenia.

Oncologist

A physician who specializes in the diagnosis and treatment of cancer. Includes medical oncologists (systemic therapy), radiation oncologists (radiotherapy), and surgical oncologists (operative management).

Oncology

The medical specialty concerned with cancer prevention, diagnosis, treatment, and survivorship.

Oncology Nursing

Oncology nursing provides cancer-related assessment, treatment support, symptom control, education, coordination, survivorship, and end-of-life care.

Oncology PET

PET, often combined with CT, used in selected cancers to assess disease distribution, treatment response, or recurrence.

Oncology Rehabilitation

Cancer rehabilitation: fatigue, pain, neuropathy, lymphedema, functional impairment. Pre- and post-treatment support.

Oncolytic Virotherapy

Oncolytic virotherapy uses modified viruses that preferentially infect and destroy cancer cells and may stimulate an immune response against the tumor.

Oncolytic Virus

A genetically modified virus that selectively replicates in and destroys cancer cells while sparing normal tissue, e.g., talimogene laherparepvec (T-VEC) for melanoma.

Oncolytic Viruses

Viruses selected or engineered to infect and destroy cancer cells while stimulating an immune response against the tumor.

Oncoprotein

oncoprotein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Oncosis

A form of cell death characterized by cellular

swelling, derived from the Greek word *onkos* meaning swelling, representing a pre-lethal state that precedes necrosis. It is typically caused by ischemic injury leading to ATP depletion, which impairs the sodium-potassium pump and causes osmotic water influx.

Oncostatin M

Oncostatin M is a cytokine involved in inflammation, immune regulation, and tissue repair; it is mainly studied as a biological research target.

Oncotic Pressure

The osmotic pressure exerted by plasma proteins (predominantly albumin) that do not freely cross the capillary wall. Normal plasma oncotic pressure is approximately 25-28 mmHg. It opposes the hydrostatic pressure that drives fluid out of capillaries, favoring fluid reabsorption at the venular end.

Oncotoxic

Toxic to tumor cells or relating to substances that damage cancer cells.

Ondansetron

Ondansetron is a selective 5-HT₃-receptor antagonist used to prevent nausea and vomiting after chemotherapy, radiotherapy, surgery, or other triggers; constipation, headache, and QT prolongation can occur.

Ondansetron Hydrochloride

The hydrochloride salt of ondansetron, a serotonin 5-HT₃ receptor blocker used to prevent nausea and vomiting.

Ondine'S Curse

A rare disorder in which automatic breathing is impaired, especially during sleep; also called congenital central hypoventilation syndrome.

Alternate Names:

congenital central hypoventilation syndrome.

Ondine'S Curse

A rare disorder of automatic breathing in which

ventilation may fail during sleep.

Alternate Names:

Congenital central hypoventilation syndrome

One-Hour Glucose Challenge Test

A screening test for gestational diabetes mellitus performed at 24-28 weeks. The patient drinks 50 g of glucose (regardless of fasting state), and plasma glucose is measured at 1 hour. Thresholds vary by protocol; a result in the range of 130–140 mg/dL positive, warranting diagnostic 3-hour oral glucose tolerance test (OGTT).

Oneirophrenia

A dreamlike state with confusion or impaired reality testing; the term is uncommon and does not represent a specific modern psychiatric diagnosis.

Onset

Onset is the time or manner in which a symptom, disease, or physiologic change begins, such as sudden, gradual, acute, or insidious onset.

Onychia

Inflammation or infection of the nail matrix (the proliferative tissue beneath the proximal nail fold that generates the nail plate), frequently extending to the nail bed (onychia and paronychia complex). Etiologies include localized pyogenic bacterial infection (frequently *Staphylococcus aureus* following microtrauma or nail-biting), fungal invasion (tinea unguium / candidal onychia), or chronic dermatologic disorders such as nail psoriasis or lichen planus. Clinically manifests with erythema, throbbing pain, edema of the proximal nail fold, and potential nail dystrophy, onycholysis, or permanent subungual scarring if untreated.

Onychocryptosis

Onychocryptosis is an ingrown nail, usually at the great toe, in which the nail edge penetrates surrounding skin and causes pain or inflammation.

Onychocryptosis (Ingrowing Toenail)

A condition in which the edge of a nail grows into the surrounding skin, causing pain, inflammation,

and sometimes infection.

Alternate Names:

Ingrowing Toenail

Onychocryptosis (Ingrown Toenail)

Alternate index label for Ingrown Toenail.

Onychogryphosis

Marked thickening and curvature of a nail, often resembling a claw.

Onycholysis

Separation of the nail plate from the underlying nail bed, beginning distally or laterally. Causes include trauma, psoriasis, fungal infection, thyroid disease, medications, and chemical exposure; the underlying cause should be identified.

Onychomadesis

Onychomadesis is proximal separation and shedding of a nail caused by temporary arrest of nail-matrix growth, often after infection, systemic illness, trauma, or medication exposure.

Onychomycosis

Fungal nail infection, most commonly caused by dermatophytes (*Trichophyton rubrum*), affecting toenails more than fingernails. It presents as nail thickening, discoloration (yellow-brown), subungual debris, and nail separation. Diagnosis is confirmed by KOH preparation or fungal culture.

Onychomycosis (Fungal Nails)

Alternate index label for Fungal Nails.

Oocyte

The female gamete (egg cell) produced in the ovary through oogenesis. Oocytes develop from primordial germ cells, which undergo mitosis to form oogonia, then enter meiosis I before birth and arrest in prophase I until puberty.

Oocyte Cryopreservation

An assisted reproductive technology procedure involving the retrieval, rapid dehydration, and ultra-

fast cooling (vitrification) of mature unfertilized female gametes (metaphase II oocytes) in liquid nitrogen at -196°C for long-term storage. Indicated for medical fertility preservation prior to gonadotoxic cancer therapy, surgical oophorectomy, or genetic conditions causing premature ovarian insufficiency, as well as for elective (planned) age-related fertility preservation.

Alternate Names:

Egg Freezing; Oocyte Vitrification

Oocyte Retrieval

An operative assisted reproductive technology procedure in which mature oocytes are harvested from ovarian follicles following controlled ovarian stimulation and a trigger injection. Performed under transvaginal ultrasound guidance using a specialized needle puncturing the vaginal fornix into each ovarian follicle, with gentle negative-pressure aspiration of follicular fluid containing the cumulus-oocyte complexes into sterile tubes for embryological evaluation and fertilization.

Alternate Names:

Ovum Pick-Up (OPU); Egg Retrieval; Follicular Aspiration

Oogenesis

Formation and maturation of egg cells in the ovary.

Oogonium

An oogonium is an early female germ cell that proliferates during embryonic development and gives rise to primary oocytes.

Oophorectomy

Surgical removal of one (unilateral) or both (bilateral) ovaries.

Oophoritis

Inflammation of an ovary, usually associated with pelvic infection or, less commonly, autoimmune disease. It may cause lower abdominal or pelvic pain, fever, abnormal discharge, or menstrual changes; severe pain requires assessment to exclude ovarian torsion or abscess.

Ooplasm

Ooplasm is the cytoplasm of an egg cell, containing organelles, nutrients, and molecular factors that support fertilization and early embryo development.

Opacity

Lack of transparency of a normally transparent ocular structure (cornea, lens, vitreous, or aqueous humour). Corneal opacity results from scarring (trauma, infection—herpes simplex keratitis, trachoma), dystrophies (Fuchs endothelial dystrophy, granular/lattice/macular dystrophies), or edema.

Open Bite

An open bite is a malocclusion in which upper and lower teeth do not meet in part of the mouth when the jaws are closed.

Open Fractures

Fractures in which the broken bone or fracture hematoma communicates with an external wound. They have increased risks of infection and soft-tissue injury and require urgent assessment and specialized treatment.

Open Gallbladder Removal

A medical procedure, treatment, device, or care topic related to open gallbladder removal; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Open Lung Biopsy

An open lung biopsy removes a small piece of lung through a surgical chest incision for microscopic diagnosis when less invasive tests are insufficient.

Open Pleural Biopsy

An open pleural biopsy removes a sample of the lining around the lung through surgery to investigate cancer, infection, or unexplained pleural disease.

Open Prostatectomy

Surgical removal of enlarged prostate tissue through an abdominal incision.

Open Reduction And Internal Fixation

Operative exposure and realignment of a fracture followed by fixation with implants such as plates, screws, wires, or nails. It is used when closed treatment cannot provide acceptable alignment, stability, or joint restoration.

Open Surgery

Traditional surgical approach with incision. Used for bypass, aneurysm repair, thrombectomy.

Open-Angle Glaucoma

A chronic optic-nerve disease in which the drainage angle of the eye remains anatomically open but aqueous humour outflow is impaired, often causing progressive, initially symptomless visual-field loss.

Open-Book Fracture

A pelvic ring disruption with diastasis of the pubic symphysis or sacroiliac joints, often from high-energy trauma, carrying risk of massive hemorrhage and organ injury.

Open-Heart Surgery

Any surgical procedure involving opening the chest wall (sternotomy or thoracotomy) and operating on the heart, usually with cardiopulmonary bypass (the heart–lung machine).

Operable

Able to be treated or corrected by surgery, or suitable for surgical intervention.

Operant Conditioning

Learning in which the likelihood of a behavior changes because of its consequences, such as reinforcement or punishment.

Operating Room

An operating room is a controlled sterile or semi-sterile clinical area equipped for surgical procedures, anesthesia, monitoring, and recovery support.

Operating Tables

Adjustable tables designed to support and position

patients during surgery or other procedures while allowing access and maintaining safety.

Operation

A planned surgical procedure or, more generally, an action or process.

Operative Dentistry

The branch of dentistry concerned with restoring teeth affected by caries, fracture, or defects.

Operative Vaginal Delivery

Vaginal delivery assisted by vacuum or forceps. Indications: prolonged second stage, suspected fetal compromise, need to shorten the second stage for maternal benefit (exhaustion, cardiac disease, hypertensive crisis).

Operator

A person who performs a procedure or operates equipment; in healthcare, training and authorization are essential.

Operculum

A lid, covering, or flap-like structure; in anatomy it may cover an opening or cortical region.

Operon

An operon is a group of adjacent prokaryotic genes controlled by a shared promoter and regulatory elements and transcribed as one messenger-RNA unit.

Ophthalmia

Inflammation of the eye, especially the conjunctiva or other superficial ocular tissues.

Ophthalmia Neonatorum

Eye inflammation or infection in a newborn, often acquired during delivery.

Ophthalmic

Relating to the eye; an ophthalmic medicine is intended for ocular use.

Ophthalmic Emergency

An eye or visual condition requiring urgent evaluation to prevent irreversible vision loss or severe ocular injury. Examples include acute angle closure, retinal detachment, central retinal artery occlusion, endophthalmitis, chemical injury, orbital cellulitis, and open-globe trauma.

Ophthalmic Solution

A sterile liquid medicine or preparation intended for application to the eye.

Ophthalmitis

Ophthalmitis is inflammation of the eye and can result from infection, immune disease, trauma, or surgery.

Ophthalmologist

A medical doctor specializing in diseases and surgery of the eyes.

Ophthalmology

The medical specialty concerned with eye disease, vision, and ocular surgery.

Ophthalmomyiasis

Infestation of the eye by fly larvae (most commonly Oestridae), which may be superficial (conjunctival) or deep (intraocular), with the latter carrying a risk of retinal detachment, endophthalmitis, and permanent vision loss.

Ophthalmoplegia

Weakness or paralysis of one or more eye muscles, causing limited eye movement, double vision, or abnormal eye position. Causes include nerve disorders, stroke, diabetes, thyroid disease, trauma, and neuromuscular conditions.

Ophthalmoscope

Instrument for examining retina and optic nerve. Direct: larger, brighter image. Indirect: wider field. Essential for fundoscopic examination.

Ophthalmoscopy

Ophthalmoscopy examines the inside of the eye, including the retina, optic nerve, and blood vessels, using an ophthalmoscope.

Opiate

A natural alkaloid derived from the opium poppy (*Papaver somniferum*) or a semi-synthetic derivative that binds to opioid receptors.

Opiate Alkaloid

An opiate alkaloid is a naturally occurring alkaloid from the opium poppy, such as morphine or codeine, with opioid analgesic and dependence potential.

Opiate And Opioid Withdrawal

Opioid withdrawal is a predictable syndrome after reducing or stopping opioid exposure, producing anxiety, sweating, muscle aches, diarrhea, abdominal cramps, nausea, and autonomic symptoms.

Opiate Assay

A diagnostic or laboratory method related to opiate assay. Results are interpreted with the method, reference range, and clinical findings.

Opiate Overdose

opiate overdose refers to harmful exposure to the named substance or condition; severity depends on dose, route, timing, symptoms, and patient factors, and urgent professional assessment may be required.

Opioid

A natural or synthetic substance that acts on opioid receptors to relieve pain and may cause sedation or dependence.

Opioid Agonist Treatment

Treatment of opioid use disorder with a prescribed opioid agonist or partial agonist, most commonly methadone or buprenorphine, together with psychosocial and medical support. It reduces withdrawal, craving, illicit opioid use, overdose, and mortality.

Opioid Intoxication

Opioid intoxication can cause euphoria, drowsiness, slowed breathing, pinpoint pupils, impaired coordination, coma, and death; respiratory depression requires immediate emergency treatment.

Opioid Overdose

A medical emergency characterized by the classic triad of respiratory depression, miosis (pinpoint pupils), and decreased level of consciousness. It can progress to respiratory arrest, cardiovascular collapse, and death.

Opioid Receptor

opioid receptor is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Opioid Reversal

Emergency treatment of opioid toxicity with naloxone, a competitive opioid-receptor antagonist. It is titrated to restore adequate breathing, with repeated doses or monitoring sometimes needed because its effect may wear off before the opioid.

Opioid Toxicity

Toxic effects of opioid agonists, classically respiratory depression, central nervous system depression, and miosis. Management prioritizes ventilation and airway support; naloxone is titrated to restore adequate breathing while recognizing the risk of recurrent toxicity and withdrawal.

Opioid Toxicodrome

A poisoning pattern caused by opioid excess, characterized by respiratory depression, reduced consciousness, and small pupils; naloxone and ventilatory support may be required.

Opioid Use Disorder

A problematic pattern of opioid use leading to significant impairment or distress. Opioids include heroin, fentanyl, morphine, oxycodone, and hydrocodone. Criteria are the same eleven as for substance use

disorders generally. Overdose presents with respiratory depression, miosis, and decreased level of consciousness.

Opioid Withdrawal

A characteristic group of symptoms occurring after abrupt reduction or cessation of prolonged opioid use, including anxiety, muscle aches, sweating, abdominal cramps, diarrhea, nausea, and insomnia.

Opisthorchis Felineus

A liver fluke acquired by eating raw or undercooked freshwater fish; chronic infection can cause bile-duct inflammation and increase cholangiocarcinoma risk.

Opisthotonos

Opisthotonos is severe backward arching of the head, neck, and spine from sustained muscle spasm, classically associated with tetanus, meningitis, strychnine poisoning, or severe neurologic injury.

Opium

The dried latex of *Papaver somniferum* containing opioid alkaloids; it can cause dependence and respiratory depression.

Opponens

A muscle that produces opposition, especially the thumb's movement across the palm.

Opportunistic Infection

An infection caused by organisms that ordinarily do not cause disease in immunocompetent hosts but cause significant illness when host defences are impaired.

Opposition

Movement of the thumb across the palm to contact the fingertips. It combines abduction, flexion, and medial rotation at the first carpometacarpal joint.

Oppositional Defiant Disorder

A childhood/adolescent disorder characterized by a persistent pattern (≥ 6 months) of angry/irritable mood, argumentative/defiant behavior, and vindic-

tiveness.

Oprelvekin

A recombinant interleukin-11 medicine formerly used to reduce chemotherapy-related thrombocytopenia; fluid retention and cardiac effects limit its use.

Opsoclonus

Rapid, involuntary, multidirectional, conjugate eye movements without an intersaccadic interval. It may occur with myoclonus and ataxia as part of an autoimmune or paraneoplastic syndrome, particularly in children with neuroblastoma.

Opsonin

A molecule, such as IgG or complement C3b, that coats a microbe and enhances its uptake by phagocytes.

Opsonization

The process by which pathogens or particles are coated with opsonins (antibodies—IgG, complement—C3b, or acute-phase proteins) that enhance recognition and phagocytosis by phagocytes bearing corresponding receptors (Fc receptors, complement receptors).

Optic Atrophy

End-stage degeneration of the optic nerve fibers resulting from any optic neuropathy. Causes: glaucoma, ischemic optic neuropathy, multiple sclerosis, compressive lesions (pituitary tumor, aneurysm), toxins (methanol, ethambutol), and hereditary (Leber hereditary optic neuropathy).

Optic Chiasm

The X-shaped junction where nasal retinal fibers of the optic nerves cross.

Optic Chiasma

The optic chiasma is the brain structure where fibers from the nasal halves of the optic nerves cross, enabling coordinated visual-field processing.

Optic Disc

The point on the retina where the optic nerve exits the eye, lacking photoreceptors and creating the natural blind spot.

Optic Disc Edema

Swelling of the optic-nerve head caused by axoplasmic-flow disruption. Bilateral optic-disc edema from raised intracranial pressure is papilledema; unilateral or non-pressure causes require separate evaluation for optic-nerve, retinal, inflammatory, ischemic, or infiltrative disease.

Optic Disk

The point where the optic nerve leaves the retina; it has no photoreceptors and forms the blind spot.

Optic Glioma

A usually slow-growing glioma of the optic nerve or optic pathway, occurring most often in children. It may cause progressive visual loss, optic atrophy, proptosis, or visual-field defects and is associated with neurofibromatosis type 1.

Alternate Names:

Optic Nerve Glioma Optic Pathway Glioma

Optic Nerve

CN II exiting through the optic disc (blind spot), passing through the optic canal, and partially decussating at the optic chiasm. Technically a CNS tract with oligodendrocyte myelination and meningeal coverings, making it susceptible to demyelinating disease (optic neuritis in MS).

Optic Nerve Atrophy

Pallor and degeneration of the optic nerve caused by loss of retinal ganglion-cell axons. It produces permanent visual impairment and may result from ischemia, inflammation, compression, glaucoma, trauma, or raised intracranial pressure.

Optic Nerve Disorder

A disease affecting the optic nerve, which carries visual signals from the eye to the brain; it can cause reduced vision, color loss, or visual-field defects.

Optic Nerve Glioma

Alternate index label; see *Optic glioma*.

Optic Neuritis

Inflammation or demyelination of the optic nerve, often causing subacute vision loss, impaired color vision, and pain with eye movement. It may be associated with multiple sclerosis or other inflammatory, infectious, or immune-mediated disorders.

Optic Pathway Glioma

Alternate index label; see *Optic glioma*.

Optic Tract

The optic tract is a neural pathway carrying post-chiasmal visual signals to the lateral geniculate nucleus and other brain regions.

Optical Coherence Tomography

A non-invasive imaging technique that uses low-coherence interferometry to produce high-resolution, cross-sectional images of the retina and other ocular structures, providing quantitative measurements of retinal layer thickness and detecting structural abnormalities with micrometer-level resolution.

Optical Illusion

optical illusion is a pharmacologic, physiologic, or clinical-care concept used to describe a measurable effect, intervention, or patient-management issue.

Optical Rotation

optical rotation is a pharmacologic, physiologic, or clinical-care concept used to describe a measurable effect, intervention, or patient-management issue.

Optical Trap

optical trap is a pharmacologic, physiologic, or clinical-care concept used to describe a measurable effect, intervention, or patient-management issue.

Optician

A technician who fits and dispenses eyeglasses or contact lenses.

Optics

The study and application of light, lenses, and image formation, including visual optics.

Optometrist

A trained eye-care professional who examines vision and many ocular conditions and prescribes optical correction within local scope of practice.

Optometry

A healthcare profession concerned with examining the eyes and visual system for defects and abnormalities, prescribing corrective lenses (spectacles, contact lenses), diagnosing eye disease, and providing pre- and post-operative care for surgical eye treatments.

Oral

Relating to the mouth or administered through the mouth.

Oral Administration

Taking a medicine by mouth so it is absorbed through the digestive tract; absorption can be affected by food, vomiting, interactions, or gastrointestinal disease.

Oral Appliance Therapy

Treatment using a removable dental appliance during sleep, commonly to advance the mandible or retain the tongue and reduce upper-airway collapse in selected obstructive sleep apnea.

Oral Cancer

Malignancy of oral cavity, including lips, tongue, floor of mouth, buccal mucosa, palate, tonsils, oropharynx. Major risk factors: tobacco use, alcohol consumption, human papillomavirus (HPV, especially p16-positive oropharyngeal cancer). Presentation: persistent ulcer, leukoplakia, erythroplakia, mass, pain, difficulty swallowing.

Oral Candidiasis

Infection or overgrowth of *Candida* species on the oral mucosa, often presenting as removable white

plaques, erythematous patches, or angular cheilitis. Risk factors include antibiotics, inhaled corticosteroids, immunosuppression, diabetes, xerostomia, and dentures.

Oral Cavity

The space extending from the lips to the oropharynx, bounded by the cheeks, palate, and tongue. It receives food, begins mechanical and chemical digestion, and contributes to speech.

Oral Cavity Cancer

Squamous cell carcinoma of the lip, tongue, buccal mucosa, floor of mouth, or palate, associated with tobacco, alcohol, and betel quid use. Treatment is primarily surgical resection, with adjuvant radiation or chemoradiation for adverse features.

Oral Cholecystogram

Oral cholecystogram is a laboratory test, imaging method, instrument, score, or measurement used to characterize a specimen or clinical finding; interpretation depends on the method and clinical context.

Oral Contraceptive

Birth control pill: combined oral contraceptive (COC—ethinylestradiol + progestin) or progestin-only pill (POP/minipill). Mechanisms: ovulation inhibition, cervical mucus thickening, endometrial change.

Oral Glucose Tolerance Test

A diagnostic test measuring plasma glucose response after a standardized oral glucose load. It is used in selected evaluations of diabetes, impaired glucose tolerance, and gestational diabetes when fasting glucose or HbA1c is insufficient.

Oral Health

Oral health is the condition of the teeth, gums, mouth, jaw, and related functions such as eating, speaking, and swallowing.

Oral Hemorrhage

Bleeding from the mouth, gums, tongue, throat, or nearby tissues, caused by trauma, dental disease, infection, medicines, or a bleeding disorder.

Oral Human Papillomavirus Infection

Oral human papillomavirus infection affects the mouth or throat; many infections clear without symptoms, while persistent high-risk types can contribute to some cancers.

Oral Hygiene

Practices maintaining oral health: brushing twice daily with fluoride toothpaste, flossing daily, mouthwash (optional), regular dental visits. Proper brushing technique: soft brush, 45-degree angle to gum-line, gentle circular motions. Interdental cleaning essential for plaque removal.

Oral Hypoglycemics Overdose

Oral hypoglycemics overdose involves toxic or environmental exposure to oral hypoglycemics overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Oral Leukoplakia

A white patch or plaque on the oral mucosa that cannot be scraped off and cannot be characterized as any other diagnosable disease, considered a potentially malignant disorder. Risk factors: tobacco use (most common), alcohol, and betel nut chewing. Most common sites: buccal mucosa, alveolar ridge, and tongue.

Oral Lichen Planus

Chronic inflammatory condition. Reticular (lace-like white lines), erosive (painful ulcers). Associated with hepatitis C. Malignant transformation rare but possible.

Alternate Names:

Lichen Planus, Oral OLP

Oral Moniliasis

Oral moniliasis is oral candidiasis, a *Candida* yeast infection causing white plaques, soreness, burning,

or altered taste.

Oral Mucosa

The moist stratified squamous epithelial lining of the oral cavity. It includes masticatory, lining, and specialized mucosa and provides barrier protection, sensation, lubrication, and sites for minor salivary glands.

Oral Mucous Cyst

An oral mucous cyst is a mucus-filled lesion caused by salivary-gland duct injury or blockage, often appearing on the lower lip or floor of the mouth.

Oral Nutritional Supplement

A calorie-, protein-, or micronutrient-containing product consumed by mouth to supplement, not automatically replace, usual food intake. The product and dose should match the patient's needs, swallowing ability, disease restrictions, and tolerance.

Oral Pathology

Study of diseases affecting oral tissues. Includes neoplasms, cysts, inflammatory conditions, developmental anomalies.

Oral Radiology

Specialty of producing and interpreting images of oral and maxillofacial region.

Oral Rehydration Salts

A balanced mixture of glucose and electrolytes dissolved in clean water to replace fluid and salts lost through diarrhea or vomiting.

Oral Rehydration Therapy

Replacement of water and electrolytes by drinking a balanced glucose-electrolyte solution, especially during diarrheal illness.

Oral Rehydration Therapy

Replacement of water and electrolytes by mouth using a balanced glucose-and-salt solution, especially to prevent or treat dehydration from diarrhoea or vomiting.

Alternate Names:

ORT

Oral Sex (Fellatio, Cunnilingus)

Oral stimulation of the genitals for sexual pleasure. Fellatio: oral stimulation of the penis. Cunnilingus: oral stimulation of the vulva and clitoris. Analingus (rimming): oral stimulation of the anus.

Alternate Names:

Fellatio, Cunnilingus

Oral Spray

A liquid medicine or preparation sprayed into the mouth or throat for local treatment or absorption through oral tissues.

Oral Squamous Cell Carcinoma

A malignant neoplasm arising from oral stratified squamous epithelium. Persistent induration, ulceration, red or white patches, unexplained bleeding, pain, or neck mass requires evaluation and biopsy; staging incorporates tumor, nodal, and metastatic findings.

Oral Surgery

Surgical procedures in oral and maxillofacial region. Extractions, implant placement, biopsies, cyst removal, corrective jaw surgery.

Oral Thrush

A fungal infection of the mouth, usually caused by *Candida*, producing creamy white patches, soreness, or altered taste.

Alternate Names:

Candidiasis, Oral Thrush, Oral

Oral Toxicity

Harm caused by swallowing a toxic substance, medicine, or excessive dose.

Orange Peel Skin

Peau d'orange, a dimpled skin appearance caused by edema or lymphatic obstruction, classically considered in breast disease.

Orantinib

An investigational tyrosine-kinase inhibitor studied for cancer and abnormal blood-vessel growth.

Orbicularis

A circular muscle surrounding an opening, such as orbicularis oculi around the eye or orbicularis oris around the mouth.

Orbit

The bony cavity containing the eye, extraocular muscles, nerves, vessels, and supporting tissues.

Orbit CT Scan

An orbit CT scan uses X-rays and computer reconstruction to assess the eye sockets, eye muscles, bones, and surrounding tissues.

Orbital

Relating to the orbit, the bony cavity containing the eye, or to structures within it. Orbital disease may affect the eye, muscles, nerves, or surrounding tissues.

Orbital Cellulitis

A serious infection of the orbital tissues posterior to the orbital septum, potentially sight-threatening and life-threatening due to the risk of optic nerve compression, cavernous sinus thrombosis, and intracranial extension.

Orbital Contusion

A bruise of the tissues around the eye caused by blunt trauma, often producing swelling, bruising, and tenderness.

Orbital Fracture

A break in one or more bones forming the eye socket, usually caused by facial trauma and potentially associated with double vision, eye-movement restriction, or injury to the eye.

Orbital Fractures

Breaks in the bones surrounding the eye, which may cause swelling, double vision, numbness, restricted

eye movement, or injury to the eye itself.

Orbital Irradiation

Radiation treatment directed at tissues within the eye socket, sometimes used for severe thyroid eye disease.

Orbital Metastasis

Secondary cancer deposits in the orbit, most commonly from breast, lung, or prostate primaries. Symptoms include proptosis, diplopia, pain, and visual impairment.

Orbital Myositis

Orbital myositis is inflammation of extraocular muscles causing eye pain, swelling, double vision, and restricted movement.

Orbital Neoplasm

An orbital neoplasm is an abnormal growth in the eye socket that may affect the eye, optic nerve, muscles, or surrounding tissues.

Orbital Pseudotumor

Noninfectious, idiopathic inflammation of orbital tissues, also called idiopathic orbital inflammation. It can cause painful eye movement, proptosis, eyelid swelling, diplopia, and restricted ocular motility.

Alternate Names:

idiopathic.

Orbivirus

A genus of double-stranded RNA viruses transmitted mainly by arthropods and associated with diseases in animals; some cause human illness.

Orchialgia

Pain originating within one or both testicles, categorized chronologically as acute (a surgical emergency) or chronic (persisting > 3 months, chronic orchialgia / testicular pain syndrome). Acute orchialgia mandates immediate clinical triage to rule out testicular torsion (which requires emergent surgical detorsion within 6 hours to preserve testicular viability) versus acute epididymitis, orchitis, incarcerated

inguinal hernia, or torsion of testicular appendages. Chronic orchialgia may stem from varicocele, post-vasectomy pain syndrome, intermittent torsion, or pudendal nerve entrapment, often evaluated with scrotal Doppler ultrasonography.

Orchidectomy

Surgical removal of one or both testes, also called orchiectomy.

Orchiectomy

Surgical removal of one or both testicles. Radical inguinal orchiectomy is diagnostic and therapeutic for testicular cancer; bilateral orchiectomy is a hormonal therapy for prostate cancer.

Orchitis

Orchitis is inflammation of one or both testes, often from infection such as mumps or bacteria, causing pain, swelling, and sometimes impaired fertility.

Order Coleoptera

The insect order containing beetles, characterized by hardened forewings that protect the flying wings; some species affect health as pests or vectors.

Order Spirochaetales

An order of spiral-shaped bacteria that includes medically important families such as Spirochaetaceae and Leptospiraceae.

Oregovomab

An investigational monoclonal antibody targeting the tumor-associated CA-125 antigen, studied mainly in ovarian cancer.

Orexin

Orexin, or hypocretin, is a hypothalamic neuropeptide that promotes wakefulness, arousal, appetite, and energy regulation; loss of orexin signaling is strongly associated with narcolepsy type 1.

Orexin Receptor Antagonists

Medications that block orexin (hypocretin) receptors to promote sleep. Suvorexant and lemborexant

are approved for insomnia. They reduce wakefulness by blocking the arousal system. They have lower abuse potential than benzodiazepines and do not cause respiratory depression.

Orf

A zoonotic viral skin infection caused by the parapoxvirus, transmitted to humans from infected sheep and goats (direct contact with lesions, fomites—virus survives in scabs for months). Orf is an occupational disease of farmers, butchers, and veterinarians. Incubation: 3–7 days.

Organ

A structure composed of two or more different types of tissues that has a recognizable shape and performs specific functions.

Organ Culture

Organ culture maintains a piece of tissue or an organ outside the body under controlled conditions for research or selected testing.

Organ Failure

Organ failure refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Organ System

A group of related organs that work together to perform major body functions.

Organ Transplant

Surgical transfer of an organ from a donor to a recipient to replace failing organ function.

Organ Transplantation

A medical procedure, treatment, device, or care topic related to organ transplantation; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Organ Transplants

The surgical transfer of an organ from a donor to a recipient whose organ function is severely impaired.

Solid-organ transplantation may involve the kidney, liver, heart, lung, pancreas, or intestine and requires donor assessment, recipient selection, and ongoing follow-up.

Organelle

A specialized structure within a cell that performs a particular function.

Organic

Relating to living matter or carbon-containing chemistry; in psychiatry, an older term for symptoms caused by a physical brain disorder.

Organic Brain Syndrome

An obsolete term formerly used for cognitive or behavioral impairment attributed to a physical brain disorder. Current practice identifies the specific condition and symptoms, such as delirium, dementia, brain injury, or another neurological disease.

Organic Chemical

organic chemical is a biochemical reaction, pathway, or chemical process in which molecules are transformed; its rate and effects depend on substrates, enzymes, and cellular conditions.

Organic Disease

A disorder caused by demonstrable structural or physiological pathology, as opposed to symptoms without an identified organic basis.

Organic Impurity

An organic impurity is an unintended carbon-containing chemical in a medicine, material, or specimen that may affect safety, quality, or measurement.

Organic Oxide

An organic oxide is a carbon-containing compound bearing oxygen in an oxidized form; properties and toxicity depend on its molecular structure.

Organic Psychosis

An obsolete term for psychosis caused by a physical brain disease, substance, or medical condition.

Organism

An individual living thing, including a human, animal, plant, fungus, or microorganism.

Organization Name

The formal name of a group, institution, company, or agency; in healthcare records it should identify the responsible organization accurately.

Organogenesis

organogenesis is a gene, genetic trait, chromosome-related concept, or inherited process that affects biological function through DNA variation or gene regulation.

Organoid

An organoid is a three-dimensional cell culture that self-organizes to model selected structure or function of an organ for research or testing.

Organometallic Compound

An organometallic compound contains a chemical bond between carbon and a metal and may have industrial, laboratory, medicinal, or toxicologic significance.

Organophosphate

A class of chemicals that can inhibit acetylcholinesterase. Some are pesticides or nerve agents and can cause cholinergic poisoning.

Organophosphate Poisoning

Acetylcholinesterase inhibitors. SLUDGE syndrome: salivation, lacrimation, urination, defecation, GI upset, emesis. Miosis, bradycardia, bronchospasm, muscle fasciculations. Treatment: atropine, pralidoxime, supportive care.

Organosilicon Compounds

Organosilicon compounds contain silicon bonded to carbon and are used in materials, medicines, devices, and industry; biological effects depend on the specific compound.

Organotherapy

An obsolete treatment approach using extracts or tissues from organs to treat disease.

Orgasm

The peak of sexual pleasure accompanied by characteristic physiologic responses.

Orgasm Male (Sexual (Sex) Problems In Men)

Alternate index label for Sexual (Sex) Problems in Men.

Orgasmic Dysfunction

Persistent difficulty, delay, absence, or marked reduction of orgasm causing distress or impairment.

Orgasmic Dysfunction In Women

Orgasmic dysfunction in women is persistent difficulty reaching orgasm or reduced orgasmic intensity that causes distress and may have psychological, relationship, medication, hormonal, or medical causes.

Orientation

Awareness of personal identity, place, time, and situation. Clinicians assess orientation as part of mental-status and neurologic examinations.

Orifice

An opening into or out of a body structure, organ, or anatomical passage.

Orlistat

A gastrointestinal lipase inhibitor used with dietary measures for weight management.

Ornithine

A nonessential amino acid and intermediate in the urea cycle that helps convert toxic ammonia into urea.

Ornithine Decarboxylase

An enzyme that converts ornithine to putrescine, an early step in polyamine synthesis involved in cell growth.

Ornithine Transcarbamylase Deficiency

OTC deficiency is an inherited urea-cycle disorder causing ammonia accumulation, vomiting, confusion, seizures, or coma, especially during metabolic stress.

Orogastric Tube

Tube through mouth to stomach. For infants, unconscious patients.

Oropharyngeal

Oropharyngeal means relating to the mouth and the part of the throat behind it, including structures involved in breathing, swallowing, and speech.

Oropharyngeal Airway

A rigid airway adjunct inserted through the mouth to displace the tongue and help maintain upper-airway patency in an unconscious patient without an intact gag reflex. It does not protect against aspiration.

Oropharyngeal Carcinoma

Malignancy arising in the oropharynx, including the tonsil, base of tongue, soft palate, and posterior pharyngeal wall. HPV-associated tumors form an important subgroup; evaluation includes endoscopy, biopsy, imaging, and cervical-node assessment.

Oropharyngeal Dysphagia

Difficulty transferring food or liquid from the mouth through the throat, often causing coughing, choking, nasal regurgitation, or aspiration.

Oropharyngeal Neoplasm

An oropharyngeal neoplasm is an abnormal growth in the throat behind the mouth, including the tonsils and base of the tongue; it may be benign or malignant.

Oropharynx

The posterior portion of the mouth leading to the esophagus, containing the tonsils and base of tongue. Area for swallowing and speech. Examination with tongue depressor and light source. Pathol-

ogy: pharyngitis, tonsillitis, malignancy.

Oropharynx Lesion Biopsy

Sampling of an oropharyngeal lesion, usually through the mouth under local or procedural anesthesia, for histopathologic evaluation of infection, dysplasia, or cancer. Persistent or suspicious lesions require specialist assessment.

Orotic Aciduria

orotic aciduria is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Orphan Disease

A rare disease affecting relatively few people; definitions and eligibility for orphan-drug programs vary by jurisdiction.

Orphenadrine

A centrally acting medicine with anticholinergic properties used in some countries for painful muscle spasm; it can cause drowsiness, dry mouth, and confusion.

Ortataxel

Ortataxel is an investigational taxane-like antineoplastic drug studied for its ability to disrupt microtubules and inhibit cancer-cell division.

Orthobunyavirus

A genus of segmented negative-sense RNA viruses, many transmitted by mosquitoes or other arthropods; some cause fever, encephalitis, or fetal disease in humans or animals.

Orthodontic Appliance

A device used to move teeth, guide jaw growth, maintain alignment, or correct a bite problem.

Orthodontic Bracket

A bonded or banded attachment that transmits forces from an orthodontic archwire to a tooth. Bracket position and wire sequence guide controlled tooth movement.

Orthodontic Retainer

A removable or fixed appliance used after active orthodontic treatment to maintain tooth position while periodontal and alveolar tissues remodel. Retention is often needed long term because relapse can occur.

Orthodontic Treatment (Dental Braces)

Orthodontic treatment (dental braces) is a medical test, procedure, treatment, or clinical measure used in patient care; its purpose and interpretation depend on the clinical question.

Orthodontic Wires

Metal or other wires used with braces to apply controlled forces that move teeth or help maintain their position.

Orthodontics

Specialty of dental management of malocclusion and dental-facial discrepancies. Appliances: fixed braces, removable appliances, clear aligners. Treatment indications: crowding, spacing, overbite, underbite, crossbite, open bite, midline discrepancy. Treatment duration: 12-24 months typically. Retention phase essential for stability.

Orthodontist

A dentist specializing in diagnosis and treatment of malocclusion and tooth or jaw alignment.

Orthognathic Surgery

Corrective jaw surgery for skeletal malocclusion. Le Fort I, Bilateral Sagittal Split Osteotomy. Combined with orthodontics.

Orthohepadnavirus

A genus of hepadnaviruses that includes hepatitis B virus and related viruses infecting mammals.

Orthokeratosis

Orthokeratosis is thickening of the outer skin layer with mature keratin cells lacking nuclei, occurring normally or in several skin disorders.

Orthomyxoviridae

A family of segmented, negative-sense RNA viruses that includes influenza A, B, and C viruses.

Orthopaedic Surgery

Surgical treatment of disorders and injuries of bones, joints, muscles, ligaments, and related tissues.

Orthopaedics

The British spelling of orthopedics, the medical and surgical specialty concerned with the musculoskeletal system.

Orthopedic Biologics

Biological or biomaterial products used to support bone or soft-tissue healing. Examples include autograft, allograft, demineralized bone matrix, bone-marrow products, and synthetic graft substitutes; their uses and evidence vary by procedure and condition.

Orthopedic Implant

Devices for bone/joint reconstruction. Types: plates, screws, rods, hinges, prostheses. Materials: titanium, cobalt-chrome, stainless steel, polyethylene, ceramic. Used for fractures, arthritis, tumors, deformities. Complications: infection, loosening, fracture, wear.

Orthopedic Infections

Infections involving bones, joints, muscles, or orthopedic implants. Examples include osteomyelitis, septic arthritis, prosthetic-joint infection, and some diabetic foot infections.

Orthopedic Rehabilitation

Therapy that restores movement, strength, function, and independence after musculoskeletal injury or surgery.

Orthopedic Screws

Fixation devices for fractures. Materials: titanium, stainless steel. Types: cortical, cancellous, interfragmentary. Used with plates, rods, nails. Complications: loosening, breakage, infection.

Orthopedic Services

Orthopedic services diagnose and treat problems of bones, joints, muscles, ligaments, tendons, and related movement systems.

Orthopedics

The branch of surgery concerned with the musculoskeletal system, including bones, joints, ligaments, tendons, muscles, and nerves.

Orthopedist

A physician specializing in disorders and injuries of bones, joints, muscles, and related tissues.

Alternate Names:

Orthopedic surgeon

Orthopnea

Dyspnea that occurs when lying flat and is relieved by sitting upright. It is measured by the number of pillows the patient needs to sleep (two-pillow orthopnea). It is a hallmark of left heart failure and is caused by redistribution of fluid from the lower extremities to the lungs in the supine position.

Orthopneic Position

An upright sitting position that helps a person breathe more easily, often with the arms supported.

Orthopnoea

Shortness of breath that occurs when lying flat and is relieved by sitting or standing upright. Orthopnoea is a classic symptom of left-sided heart failure (elevated pulmonary venous pressure worsens pulmonary congestion in the supine position). Other causes: severe COPD, asthma exacerbation, diaphragmatic paralysis, ascites, and obesity.

Orthopod

Orthopod is an informal term for an orthopedic physician who diagnoses and treats disorders of bones, joints, muscles, and related tissues.

Orthopoxvirus

A genus of poxviruses that includes variola virus, monkeypox virus, vaccinia virus, and cowpox virus.

Orthoptera

The insect order containing grasshoppers, crickets, katydids, and locusts; some species cause bites, allergies, or crop-related health effects.

Orthoptic

Relating to assessment and treatment of binocular vision, eye alignment, and disorders such as strabismus.

Orthoreovirus

A genus of nonenveloped, segmented double-stranded RNA viruses; human infection is often mild or asymptomatic but can occasionally cause disease.

Orthorexia

A term used for an excessive or rigid preoccupation with eating foods considered healthy. It is not a universally recognized formal psychiatric diagnosis but may cause nutritional or social impairment.

Orthoscopic

Orthoscopic means providing a straight or undistorted view, as in an optical instrument or examination.

Orthosis

An externally applied device that supports, aligns, prevents deformity, protects, or assists movement of a body segment. It differs from a prosthesis, which replaces a missing body part.

Orthostatic Hypotension

Decrease in SBP ≥ 20 mmHg or DBP ≥ 10 mmHg within 3 minutes of standing. Causes: dehydration, medications (antihypertensives, diuretics, alpha-blockers, antidepressants), autonomic dysfunction (diabetes, Parkinson disease), and prolonged bed rest. Symptoms: dizziness, lightheadedness, syncope, falls.

Orthostatic Intolerance

Symptoms or signs that develop when standing because the cardiovascular and autonomic systems do

not maintain adequate cerebral perfusion. Manifestations include light-headedness, palpitations, fatigue, visual dimming, presyncope, or syncope and may occur with or without orthostatic tachycardia or hypotension.

Orthotic

Orthotic means relating to a brace, splint, insert, or device that supports, aligns, protects, or improves function of a body part.

Orthotic Devices

A medical procedure, device, or treatment involving orthotic devices. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Orthotic Training

Training for brace users: donning/doffing, skin care, gait training with brace.

Ortolani'S Sign

A palpable or audible clunk produced when a dislocated but reducible infant hip is gently abducted and reduced; it is a screening sign of developmental dysplasia of the hip.

Ortolani'S Test

A hip examination maneuver in infants that detects a reducible dislocation by producing a palpable clunk as the femoral head enters the acetabulum.

Os

A bone or bony structure; in Latin anatomical terminology it means bone, while in obstetrics it means an opening.

Os Odontoideum

Os odontoideum is a separate ossicle near the second cervical vertebra's odontoid process that can cause atlantoaxial instability or spinal-cord risk.

Osborn Wave

A positive deflection at the junction of the QRS complex and ST segment on electrocardiography, also known as a J wave. It is classically associated

with hypothermia, where its amplitude correlates with the degree of cooling.

Alternate Names:

a J wave.

Oschophoria

A rare or historical term whose precise clinical meaning varies across references; the source-specific definition should be verified.

Oscillation

Repeated movement or variation around a central position or value.

Oscillator

A device or system that produces repeated periodic movement or an electrical signal.

Oscillograph

An instrument that records changing electrical or mechanical signals over time.

Oscillometry

A method that measures pressure oscillations, commonly used in automated blood-pressure cuffs or vascular testing.

Oscillopsia

The disturbing illusion that the visual environment is moving or bouncing, often caused by vestibular or eye-movement dysfunction.

Oscitation

Yawning, a reflex involving opening the mouth and deep inhalation that is associated with sleepiness, fatigue, or other physiologic states.

Oseltamivir

Oseltamivir is an antiviral medicine used to treat or prevent influenza, with the greatest benefit when started early.

Oseltamivir Phosphate

Oseltamivir phosphate is the salt form of oseltamivir, an antiviral medicine used for treatment or preven-

tion of influenza.

OSFED

Other Specified Feeding or Eating Disorder, used when the clinical presentation causes significant distress or impairment but does not meet full criteria for any of the specific eating disorders.

Osgood-Schlatter Disease

A traction apophysitis of the tibial tuberosity at the insertion of the patellar tendon, caused by repetitive microtrauma during periods of rapid growth.

Osiander'S Sign

An older sign described in pregnancy assessment; its use and definition vary by source and should be verified.

Osler Nodes

Painful, erythematous nodules on the pads of fingers and toes, caused by immune complex deposition in infective endocarditis. They are a minor criterion in the Duke criteria. Osler nodes are tender and may last hours to days.

Osmidrosis

Offensive-smelling perspiration, usually from bacterial breakdown of sweat and skin material.

Osmium

A dense metallic element; some osmium compounds are highly toxic and can damage the eyes and respiratory tract.

Osmium Tetroxide

A highly toxic and volatile chemical used mainly in laboratory fixation and staining; exposure can severely injure the eyes, skin, and respiratory tract.

Osmolality

The concentration of osmotically active particles per kilogram of solvent. Plasma osmolality is determined mainly by sodium and its accompanying anions, glucose, and urea and governs water movement across cell membranes.

Osmolality Blood Test

A blood osmolality test measures the concentration of dissolved particles in blood and helps assess hydration, electrolyte disorders, and certain poisonings.

Osmolality Urine Test

A urine osmolality test measures how concentrated the urine is and helps evaluate the kidneys' ability to regulate water balance.

Osmolar

Relating to osmolarity, the concentration of osmotically active particles per liter of solution.

Osmolarity

The concentration of osmotically active particles per liter of solution.

Osmole

A unit representing the amount of a substance that contributes one mole of osmotically active particles.

Osmoreceptor

A sensory cell or neural receptor that detects changes in osmotic concentration, especially in the hypothalamus.

Osmoregulation

Control of water and dissolved-solute balance in the body or an organism, maintaining appropriate cell volume and internal concentration.

Osmosis

The net movement of water molecules across a semipermeable membrane from a region of lower solute concentration (higher water concentration) to a region of higher solute concentration (lower water concentration). Driven by the osmotic pressure difference, which is proportional to the number of solute particles.

Osmotic Demyelination Syndrome

Demyelinating process affecting the pons and extrapontine regions caused by overly rapid correc-

tion of chronic hyponatremia. The brain adapts to chronic hyponatremia by extruding osmoles and when sodium is corrected too quickly water shifts out of brain cells causing osmotic injury to oligodendrocytes.

Osmotic Diarrhea

Diarrhea caused by poorly absorbed solutes retaining water in the intestinal lumen. It typically improves with fasting and may result from carbohydrate malabsorption, laxatives, or other osmotically active substances.

Osmotic Diuresis

Increased urine production caused by poorly reabsorbed solutes in the kidney tubules, such as glucose during marked hyperglycemia.

Osmotic Fragility Test

A test that measures the susceptibility of red blood cells to haemolysis when placed in progressively more dilute saline solutions. Increased fragility supports hereditary spherocytosis, although the result may also be affected by other haemolytic disorders and recent transfusion.

Osmotic Pressure

The pressure needed to stop water from crossing a semipermeable membrane toward the side with the higher effective solute concentration. It depends mainly on the number of dissolved particles rather than their identity; for dilute solutions, it is described by the van't Hoff relation $\pi = iCRT$. In physiology, osmotic pressure helps determine how water moves between body compartments.

Osseointegration

Direct structural and functional connection between living bone and the surface of a load-bearing implant under appropriate conditions. It depends on implant stability, surface characteristics, bone quality, healing, and control of infection and occlusal forces.

Osseous

Relating to or composed of bone. For example, an

osseous lesion is a bone lesion.

Osseous Metaplasia

Osseous metaplasia is formation of mature bone in tissue where bone is not normally present, sometimes after injury, inflammation, or within a lesion.

Ossicle

A small bone, especially one of the three auditory bones in the middle ear.

Ossicles

Three smallest body bones forming the ossicular chain in the middle ear: malleus (hammer, attached to tympanic membrane), incus (anvil), stapes (stirrup, footplate in oval window).

Ossicular Discontinuity

Partial or complete separation of the middle-ear ossicles, usually caused by trauma, chronic ear disease, or surgery. It produces conductive hearing loss and may be evaluated with otoscopy, audiometry, and imaging.

Alternate Names:

Ossicular-chain discontinuity; ossicular disruption.

Ossicular Fusion

Abnormal joining or fixation of the middle-ear ossicles (malleus, incus, or stapes), which can restrict sound transmission and cause conductive hearing loss. Evaluation and treatment depend on the underlying congenital, inflammatory, traumatic, or postoperative cause.

Ossification

The physiological or pathological process of bone tissue formation, in which osteoblasts synthesize and secrete osteoid matrix that subsequently undergoes calcium hydroxyapatite mineralization. Occurs through two primary embryological pathways: intramembranous ossification (direct transformation of mesenchymal connective tissue into flat bones, such as the cranial vault, facial bones, and clavicles) and endochondral ossification (replacement of a hyaline cartilage intermediate scaffold, forming the long bones, axial skeleton, and base of the

skull at epiphyseal growth plates). Pathologic heterotopic ossification entails abnormal extraskkeletal bone formation within soft tissues or muscle (myositis ossificans) following severe trauma, burns, or central neurological injury.

Ossifying Fibroma

An ossifying fibroma is a usually benign fibroosseous tumor that replaces normal bone with fibrous tissue and mineralized material, often in the jaw.

Osteitis

Inflammation of bone, caused by infection, trauma, inflammatory disease, or other processes.

Osteitis Deformans

Osteitis deformans is Paget disease of bone, in which abnormal remodeling produces enlarged, weakened, or deformed bones.

Osteitis Deformans (Paget'S Disease)

Alternate index label for Paget's Disease.

Osteitis Fibrosa

Osteitis fibrosa is fibrous replacement and weakening of bone caused by prolonged excess parathyroid hormone, usually in severe hyperparathyroidism.

Osteoarthritis

A misspelling of osteoarthritis, a degenerative joint disorder involving cartilage and subchondral bone.

Osteoarthritis

Osteoarthritis is a degenerative joint disorder involving cartilage loss, subchondral bone change, pain, stiffness, and functional limitation, influenced by age, injury, and joint loading. bone sclerosis, osteophyte formation, and synovial inflammation, affecting knees, hips, hands (DIP/PIP Heberden/Bouchard nodes), spine. Risk factors: age, obesity, joint injury, malalignment, genetics, female sex.

Alternate Names:

Arthritis Degenerative (Osteoarthritis) Degenera-

tive Arthritis

Osteoarthritis (Geriatric)

Osteoarthritis occurring in later life, commonly affecting the knees, hips, hands, or spine and causing joint pain, stiffness, and reduced movement. Management may include exercise, weight management, medicines, assistive measures, or joint replacement when appropriate.

Alternate Names:

Geriatric

Osteoarthritis, Cervical

Alternate index label; see *Cervical spondylosis*.

Osteoarthropathy

Osteoarthropathy is disease affecting bones and joints, often describing degenerative, hypertrophic, metabolic, or systemic-associated changes.

Osteoarthrosis

Another name for osteoarthritis, a joint disorder in which cartilage and nearby bone gradually change, causing pain, stiffness, and reduced movement.

Osteoblast

A bone-forming cell derived from mesenchymal stem cells in the bone marrow. Osteoblasts synthesise and deposit the bone matrix (osteoid), composed primarily of type I collagen and non-collagenous proteins (osteocalcin, osteonectin, alkaline phosphatase).

Osteoblastoma

A rare benign bone-forming tumor, usually in the spine or long bones, that can cause persistent pain and sometimes neurologic symptoms.

Osteoblasts

Specialized, mononuclear, bone-forming cells derived from multipotent mesenchymal stem cells under the direction of master transcription factors Runx2 and Osterix. Located along bone surfaces, active osteoblasts synthesize and secrete the organic components of the bone matrix (osteoid), predominantly type I collagen, osteocalcin, and alkaline

phosphatase, and regulate its mineralization via calcium and phosphate deposition. Furthermore, osteoblasts orchestrate bone remodeling by expressing receptor activator of nuclear factor κ B ligand and its decoy receptor osteoprotegerin to regulate osteoclast differentiation and activity; upon entrapment within the mineralized matrix, they terminally differentiate into osteocytes.

Osteocalcin

A protein made mainly by osteoblasts that participates in bone mineralization and serves as a marker of bone formation.

Osteocalcin Assay

A diagnostic or laboratory method related to osteocalcin assay. Results are interpreted with the method, reference range, and clinical findings.

Osteochondritis

Inflammation or injury involving bone and its overlying cartilage.

Osteochondritis Dissecans

A condition characterized by separation of a fragment of articular cartilage and subchondral bone from the joint surface, most commonly in the knee (lateral aspect of the medial femoral condyle). Most common in adolescents and young adults. Etiology: repetitive microtrauma, ischemia, or genetic predisposition.

Osteochondroma

A usually benign cartilage-capped bony projection arising from the surface of a bone and continuous with its cortex and medullary cavity. It may be solitary or occur in hereditary multiple exostoses.

Osteochondromatosis

Osteochondromatosis is formation of multiple cartilage-capped bony outgrowths, usually around growth plates, that can cause deformity, pain, or nerve and vessel compression.

Osteochondrosis

A disorder of bone and cartilage development, often affecting a growth center or joint surface.

Osteoclasia

Osteoclasia is surgical or mechanical breaking of bone, historically used to correct deformity.

Osteoclasis

Osteoclasis is deliberate fracture or manipulation of bone, usually to correct deformity or restore alignment.

Osteoclast

A large, multinucleated bone-resorbing cell derived from the monocyte–macrophage lineage (hematopoietic stem cells).

Osteoclasts

Large, multinucleated giant cells derived from the monocyte/macrophage hematopoietic lineage, responsible for bone resorption and mineral mobilization during skeletal growth, repair, and calcium homeostasis. Activated by macrophage colony-stimulating factor (M-CSF) and RANKL binding to surface RANK receptors, osteoclasts seal against mineralized bone surfaces to create an isolated resorptive microenvironment (Howship lacunae). Resorption proceeds via proton pump (H^+ -ATPase) acidification of the sealed compartment to dissolve inorganic hydroxyapatite, coupled with secretion of cathepsin K and matrix metalloproteinases (MMPs) to degrade collagenous organic osteoid. Excess osteoclast activity drives bone loss in postmenopausal osteoporosis, Paget disease, and osteolytic bone metastases, and is targeted clinically by antiresorptive therapies including bisphosphonates and denosumab.

Osteocyte

The mature, terminally differentiated bone cell embedded within the bone matrix in lacunae, derived from osteoblasts. Osteocytes are connected to each other and to bone surface cells via canaliculi, forming a mechanosensory network that senses mechanical loading and fluid flow.

Osteocytes

Mature bone cells embedded within bone tissue that help maintain it.

Osteodystrophy

Abnormal bone growth or remodeling caused by a systemic or metabolic disorder.

Osteogenesis

The biological formation and structural development of bone during embryonic morphogenesis, postnatal skeletal maturation, and fracture repair. Encompasses both cellular differentiation (osteoblastogenesis from mesenchymal progenitors) and matrix deposition/mineralization (intramembranous and endochondral pathways). Derangements in osteogenesis underlie congenital skeletal dysplasias, most notably osteogenesis imperfecta (brittle bone disease), a heterogeneous genetic disorder caused by mutations in *COL1A1* or *COL1A2* encoding type I procollagen, characterized by bone fragility, recurrent fractures, blue sclerae, dentinogenesis imperfecta, and conductive hearing loss.

Osteogenesis Imperfecta

A connective tissue disorder caused by collagen type I mutations, ranging from mild to lethal. Features: fractures with minimal trauma, blue sclerae, dental abnormalities, and hearing loss. Severe forms (type II) cause multiple fractures in utero and perinatal death. Diagnosis: clinical features, genetic testing, and bone biopsy.

Osteogenesis Imperfecta Brittle Bone Disease

Osteogenesis imperfecta is an inherited collagen disorder causing fragile bones, recurrent fractures, short stature, dental abnormalities, hearing loss, and variable skeletal deformity. Care may include fracture prevention, rehabilitation, medicines, surgery, and monitoring of hearing and breathing.

Alternate Names:

Brittle Bone Disease (Osteogenesis Imperfecta Brittle Bone Disease)

Osteogenic Sarcoma

Osteosarcoma, a malignant bone tumor whose cells produce osteoid or immature bone.

Osteoid

The newly formed, unmineralized organic framework of bone, produced mainly by osteoblasts. Minerals are later deposited in it to make mature hard bone.

Osteoid Osteoma

A benign bone-forming tumor, usually in an adolescent or young adult, characterized by a small vascular nidus surrounded by reactive sclerosis. It commonly causes localized pain that is worse at night and may improve with nonsteroidal anti-inflammatory drugs.

Osteology

The scientific and medical branch of anatomy dedicated to the structure, morphology, developmental biology, biomechanics, and pathology of the skeletal system and individual bones. In clinical practice, osteological principles underpin orthopedic surgery, musculoskeletal radiology, physical anthropology, and forensic medicine, guiding the classification and biomechanical management of fractures, skeletal deformities, metabolic bone disorders (rickets, osteomalacia), and neoplastic bone processes.

Osteolysis

Osteolysis is progressive destruction or resorption of bone caused by tumors, inflammation, prosthesis reactions, infection, metabolic disease, or other processes.

Osteolytic

Osteolytic means causing or characterized by breakdown and loss of bone tissue.

Osteoma

A benign, slow-growing, circumscribed osteogenic neoplasm composed predominantly of dense, mature lamellar or cortical bone (compact osteoma) or trabecular bone (cancellous osteoma). Most commonly arises in the craniofacial skeleton, particularly the paranasal sinuses (especially frontal and ethmoid

sinuses) and the outer cortex of the mandible or calvarium. While frequently asymptomatic and discovered incidentally on facial CT scans, large lesions may cause mechanical sinus obstruction, recurrent sinusitis, headache, or facial asymmetry. Multiple osteomas of the skull, sinuses, and mandible are a cardinal diagnostic sign of Gardner syndrome (a phenotypic variant of familial adenomatous polyposis [FAP] carrying a nearly 100% risk of colorectal adenocarcinoma).

Osteoma Cutis

Osteoma cutis is formation of bone within the skin or subcutaneous tissue, occurring as a primary disorder or secondary to acne, trauma, or another condition.

Osteomalacia

Osteomalacia is inadequate mineralization of adult bone, commonly from vitamin-D deficiency or phosphate disorders, causing bone pain, weakness, and fractures.

Osteomalacia (Rickets)

Alternate index label for Rickets.

Osteomyelitis

Osteomyelitis is infection and inflammation of bone caused by bacteria, fungi, or other organisms, producing pain, fever, swelling, and possible bone destruction or chronic drainage. *Staphylococcus aureus* responsible for the majority of cases.

Osteomyelitis In Children

Childhood osteomyelitis is infection of bone, often caused by bacteria entering through blood or trauma, producing pain, fever, swelling, and impaired movement.

Osteon

An osteon, or Haversian system, is the cylindrical structural unit of compact bone surrounding a central blood-vessel and nerve canal.

Osteonecrosis

Osteonecrosis is death of bone tissue from inade-

quate blood supply, causing pain, collapse, and joint damage, commonly in the hip or knee.

Osteonecrosis (Avascular Necrosis)

Osteonecrosis is bone death from vascular compromise, affecting femoral head (most common), humeral head, femoral condyles, talus. Risk factors: glucocorticoids (high dose, cumulative), alcohol, trauma, sickle cell disease, SLE, antiphospholipid syndrome, pancreatitis, decompression sickness, radiation, organ transplantation.

Alternate Names:

Avascular Necrosis

Osteonecrosis Of The Jaw

Death of jawbone tissue, often associated with antiresorptive or antiangiogenic medications, radiation, infection, or trauma, and characterized by exposed bone or a nonhealing oral wound.

Osteonectin

An extracellular-matrix protein involved in collagen organization, mineral binding, and tissue remodeling; it is also called SPARC.

Osteons

Cylindrical structural units of compact bone arranged around central canals.

Alternate Names:

Haversian systems

Osteopath

A physician trained in medicine and surgery who earns a Doctor of Osteopathic Medicine degree.

Alternate Names:

DO

Osteopathy

A system of manual medicine and healthcare founded by Andrew Taylor Still (1828–1917) based on the principle that the body's structure (musculoskeletal system) and function are interrelated, and that the body has inherent self-regulatory and healing mechanisms.

Osteopenia

Bone mineral density that is lower than normal but not low enough to be classified as osteoporosis. T-score between -1.0 and -2.5 on DXA scan. A risk factor for osteoporosis and fractures.

Osteopenia - Premature Infants

Osteopenia of prematurity is reduced bone mineralization in premature infants due to immature mineral stores, limited intake, illness, or prolonged intensive care.

Osteopetrosis

A rare inherited bone disease characterized by defective osteoclast function, leading to impaired bone resorption and abnormally dense, sclerotic bones. The bone is structurally abnormal—brittle despite increased density—and prone to fracture.

Osteophyte

A bony projection arising at a joint margin, commonly associated with osteoarthritis.

Osteoplasty

Surgery to repair, reshape, or reconstruct bone to restore structure, movement, or function. The technique depends on the bone problem and may be combined with fixation, grafting, or joint surgery.

Osteopoikilosis

Osteopoikilosis is an inherited benign bone condition with multiple small dense spots on imaging, usually asymptomatic and discovered incidentally.

Osteopontin

A phosphorylated protein involved in bone remodeling, immune signaling, inflammation, and cell adhesion.

Osteoporosis

Osteoporosis is a systemic skeletal disorder characterized by reduced bone strength and increased susceptibility to fragility fractures, which may occur after minimal trauma or ordinary activity. It commonly affects the spine, hip, wrist, and other

bones and may remain asymptomatic until a fracture occurs. medicines, genetics, nutrition, smoking, alcohol, and other conditions; diagnosis and management use clinical assessment and bone-density testing when appropriate.

Osteoporosis Fracture Risk Assessment

Evaluation combining clinical risk factors, prior fragility fracture history, bone mineral density, and tools such as FRAX to estimate future fracture probability and guide treatment.

Osteoporotic Fracture

A fracture resulting from weakened, porous bone after a low-energy event that would not usually break healthy bone.

Osteoprotegerin Test

A diagnostic or laboratory method related to osteoprotegerin test. Results are interpreted with the method, reference range, and clinical findings.

Osteoradionecrosis

Death of irradiated bone with poor healing, most often affecting the jaw after head-and-neck radiotherapy.

Osteosarcoma

Osteosarcoma is an aggressive malignant bone tumor that produces osteoid, most often affecting the metaphysis of long bones in children and young adults. leukemia), with a peak incidence during the adolescent growth spurt (ages 10-20 years), and a second smaller peak in older adults (usually secondary to Paget disease or prior radiation).

Osteosarcoma, Secondary

An osteogenic sarcoma arising in previously abnormal or treated bone, such as bone affected by Paget disease, prior radiation, or chronic bone lesions; prognosis and treatment depend on stage and resectability.

Osteosclerosis

Osteosclerosis is abnormal increased bone density

caused by defective remodeling, inherited disease, chronic stress, inflammation, medicines, or malignancy.

Osteosynthesis

Surgical stabilization and repair of a broken bone using implants such as plates, screws, nails, or wires.

Osteotome

A sharp surgical instrument used to cut or shape bone.

Osteotomy

A surgical procedure involving cutting and repositioning of bone to correct deformity, realign joint surfaces, alter weight-bearing forces, or change limb length.

Osteotomy Of The Knee

Surgery that cuts and reshapes a knee-region bone to correct alignment and redistribute weight across the joint.

Ostium

A natural opening or mouth of a vessel, duct, cavity, or organ.

Ostomy

Surgical opening into bowel or urinary tract. Colostomy, ileostomy, urostomy. Appliances for collection.

Ostomy Appliance

Collection device for feces/urine after ostomy surgery. Types: urostomy (urine), colostomy (feces), ileostomy (ileum). One-piece or two-piece systems. Open or closed end. Skin care essential.

Ostraceous Psoriasis

A severe, chronic variant of plaque psoriasis in which the lesions become thick, hard, and shell-like (ostraceous), with laminated, greyish-brown crusts resembling oyster shells. It is often refractory to standard therapies and may require systemic agents, biologics, or combination treatment.

Otalgia

Pain originating in or referred to the ear, classified clinically as primary (otogenic) or secondary (referred). Primary otalgia stems from pathology within the external, middle, or inner ear, such as acute otitis media, otitis externa (swimmer's ear), mastoiditis, impacted cerumen, or trauma. Secondary (referred) otalgia accounts for up to 50% of adult ear pain complaints and arises because the external ear and auditory canal receive sensory innervation from cranial nerves V, VII, IX, and X, as well as cervical nerves C2–C3. Referred pain can originate from temporomandibular joint dysfunction, dental abscesses, tonsillitis, pharyngitis, cervical spine disorders, or base-of-tongue and laryngeal malignancies, requiring a systematic head and neck exam in adults with normal otoscopic findings.

Otalgia (Earache)

Pain in or around the ear, arising from the ear itself or referred from structures such as the teeth, jaw, throat, or cervical spine.

Alternate Names:

Earache

Othello Syndrome

A delusional belief that a partner is unfaithful despite insufficient evidence; it may occur with psychiatric, neurologic, or substance-related disorders.

Other

Other indicates a category or finding not fitting the listed alternatives; its medical meaning depends on the surrounding classification or clinical context.

Other Pain Conditions

Pain conditions that do not fit neatly into a single category, including nociplastic pain, fibromyalgia, chronic tension-type headache, temporomandibular pain, chronic pelvic pain, and complex regional pain syndrome. Diagnosis and treatment depend on the specific condition.

Other Race

Other race is a demographic category for a person who does not select or fit the available listed race

categories; classification varies by system and should not be treated as a biological diagnosis.

Other Specified Sexual Dysfunction

A category for sexual dysfunction symptoms that cause distress but do not meet full criteria for any specific sexual dysfunction.

Otic Barotrauma

Injury to the ear caused by failure to equalize pressure during flying, diving, or other pressure changes.

Otic Capsule

The otic capsule is the dense bony shell (otic capsule bone) that surrounds and protects the inner ear structures, including the cochlea, vestibule, and semicircular canals, forming the hardest bone in the human body.

Otic Ointment

A semisolid medicine formulated for application to the outer ear or ear canal; it should be used only as directed, especially when the eardrum may be perforated.

Otitis

Inflammation or infection involving any anatomical division of the ear, classified by location into otitis externa (infection of the external auditory canal, commonly bacterial [*Pseudomonas aeruginosa*, *Staphylococcus aureus*] or fungal), otitis media (inflammation of the middle ear cleft, subdivided into acute otitis media [AOM], otitis media with effusion [OME / serous otitis], and chronic suppurative otitis media [CSOM]), and otitis interna (labyrinthitis, involving the vestibulocochlear apparatus). Clinical presentation varies by subtype, ranging from canal erythema and tragus tenderness in otitis externa, to bulging erythematous tympanic membrane, fever, and otalgia in AOM, to vertigo and sensorineural hearing loss in labyrinthitis.

Otitis (Ear Infection)

Alternate index label for Ear Infection.

Otitis Externa

Inflammation or infection of the external ear canal, commonly causing ear pain and tenderness.

Alternate Names:

Swimmers Ear (Otitis Externa)

Otitis Media

Inflammation or infection of the middle ear, often following an upper-respiratory infection.

Otitis Media With Effusion

Nonpurulent fluid in the middle ear without acute signs of infection. It may cause conductive hearing loss, aural fullness, or speech and learning effects in children; many cases resolve spontaneously, but persistent or consequential effusion requires hearing assessment and ENT review.

Oto-Rhino-Laryngology

Another name for otolaryngology, the specialty of the ear, nose, throat, head, and neck.

Otoacoustic Emissions

Low-level sounds generated by the cochlea, usually by outer hair cells, that can be recorded with a sensitive probe in the ear canal. Testing is used for newborn hearing screening and for assessing cochlear function.

Alternate Names:

OAE; otoacoustic-emission testing; cochlear emissions.

Otocephaly

A severe congenital craniofacial malformation characterized by absent or markedly underdeveloped mandible, ventral displacement or fusion of the ears, and associated facial and airway abnormalities. It is usually lethal because of profound impairment of breathing and feeding.

Otolaryngologist

An otolaryngologist is a physician specializing in disorders of the ear, nose, throat, head, neck, voice, swallowing, and related surgery.

Otolaryngology

The medical specialty concerned with the ear, nose, throat, head, and neck.

Otolith

A small calcium-carbonate crystal in the vestibular organs of the inner ear that contributes to sensing gravity and linear acceleration; displaced otoliths can cause benign positional vertigo.

Otoliths

Calcium-carbonate crystals in the vestibular organs that sense gravity and linear acceleration.

Otology

The study and medical care of the ear and hearing.

Otoplasty

Surgery to alter the shape, position, or proportion of the external ear, commonly to correct prominent or congenital ear differences.

Otorhinolaryngology

The medical specialty concerned with diseases of the ear, nose, throat, head, and neck; also called otolaryngology.

Alternate Names:

otolaryngology.

Otorrhea

Discharge from the ear. Classification: acute otorrhea (<6 weeks) and chronic otorrhea (>6 weeks).

Otorrhoea

The drainage of fluid, discharge, or pus from the external auditory canal. Pathophysiologically categorized by fluid characteristics: purulent or mucopurulent discharge indicates active infection (acute otitis media with tympanic membrane perforation, otitis externa, or cholesteatoma); serosanguineous discharge may indicate ear canal trauma, acute myringitis, or barotrauma; clear watery discharge following head trauma or skull base fracture warns of cerebrospinal fluid otorrhoea (caused by temporal bone fracture tearing the dura and tympanic

membrane, confirmed by beta-2 transferrin testing and carrying high risk of bacterial meningitis).

Otosclerosis

Abnormal remodeling and fixation of middle-ear bone, commonly causing progressive conductive hearing loss.

Otoscope

An instrument used to examine the ear canal and eardrum.

Otoscopy

Examination of the ear canal and tympanic membrane with an otoscope.

Ototoxicity

Functional damage to the cochlea or vestibular system caused by exposure to a toxic substance, including some medicines or industrial chemicals. It may present with hearing loss, tinnitus, or imbalance and can be temporary or permanent.

Alternate Names:

Auditory toxicity; vestibular toxicity; ototoxic effect.

Ouabain

A cardiac glycoside that inhibits sodium-potassium ATPase; it is mainly of historical or research interest.

Ounce

An ounce is a unit of mass or fluid volume whose exact conversion depends on the measurement system; medication instructions should use unambiguous units.

Outcome

The result or endpoint of a disease, treatment, exposure, or healthcare intervention.

Outcome Measurement

The use of defined measures to assess a patient's symptoms, function, quality of life, or other results before and after care.

Outcome Research

Research that studies what happens to patients after a treatment, procedure, or health program. It may measure survival, symptoms, function, quality of life, safety, patient experience, or cost in real-world care.

Outcomes Research

Outcomes research studies how healthcare interventions, services, and policies affect survival, function, quality of life, safety, cost, and patient experience.

Outdoor Fitness Routine

A planned program of physical activity performed outdoors, including aerobic, strength, balance, or flexibility exercise adapted to weather, terrain, and fitness level.

Outer Ear

The visible auricle and the external auditory canal, which collect and conduct sound toward the eardrum.

Outer Ear Infection

Alternate index label; see *Swimmer's ear*.

Outpatient

A person who receives medical evaluation or treatment without being admitted overnight to a hospital or other inpatient facility.

Outpatient Rehabilitation

Rehabilitation provided without hospital admission to improve function after illness, injury, surgery, or disability. The setting, frequency, and duration are individualized.

Output

The amount of a substance produced or eliminated by the body or by an organ over a specified time, such as urine output or cardiac output.

Ova

Plural of ovum; eggs or egg cells, and in parasitology, eggs of worms or other parasites.

Ovalbumin

The main protein in egg white, used in nutrition and laboratory research and capable of triggering egg allergy.

Ovalocyte Count

An ovalocyte count estimates the number or proportion of oval-shaped red blood cells on a blood smear and may help evaluate anemia.

Ovarian

Pertaining to, originating from, or affecting the ovary, the primary female gonadal organ responsible for gametogenesis (oocyte maturation and ovulation) and endocrine secretion of sex steroid hormones (estrogens, progesterone, and androgens). Clinically relevant across gynecologic, reproductive, and oncologic conditions, including ovarian cysts, polycystic ovary syndrome, ovarian torsion (an acute gynecologic surgical emergency requiring urgent detorsion to salvage ovarian blood flow), and ovarian neoplasms (epithelial, germ cell, and sex cord-stromal tumors).

Ovarian Cancer

A malignant tumor arising in the ovary or related tissues. It may cause abdominal or pelvic symptoms, but early disease can be asymptomatic.

Alternate Names:

Cancer, Ovarian

Ovarian Cyst

A fluid-filled sac within or on the ovary, most commonly functional (follicular or corpus luteum cysts) related to the menstrual cycle. Usually asymptomatic; can present with pelvic pain, pressure, and bloating. Complications: rupture, hemorrhage, torsion (sudden severe pain with nausea/vomiting).

Ovarian Cysts

Fluid-filled or partly solid sacs in or on an ovary; many are benign and resolve spontaneously, but some cause pain or complications.

Alternate Names:

Cyst, Ovarian Cysts Ovary (Ovarian Cysts)

Ovarian Failure

Ovarian failure is inadequate ovarian hormone production or ovulation, causing irregular or absent periods, infertility, and low-estrogen symptoms.

Ovarian Fibromata

Ovarian fibromata are usually benign fibrous tumors of the ovary that may be incidental or cause pelvic pressure, pain, or fluid in the abdomen.

Ovarian Follicle

An ovarian follicle is a fluid-containing structure holding an immature egg and supporting cells; one usually matures and ovulates during each cycle.

Ovarian Gonadoblastoma

A rare ovarian tumor composed of germ-cell and sex-cord elements, often associated with abnormal gonadal development or a Y-chromosome material and potentially producing hormones.

Ovarian Hemorrhage

Bleeding into or around an ovary, which may result from a ruptured cyst, ovulation, trauma, pregnancy-related conditions, or a bleeding disorder.

Ovarian Hyperstimulation Syndrome

A potentially serious complication of ovarian stimulation in assisted reproduction caused by increased vascular permeability and fluid shift. It produces enlarged ovaries, abdominal distension, ascites, hemoconcentration, electrolyte abnormalities, thrombosis risk, and, in severe cases, organ dysfunction.

Alternate Names:

OHSS

Ovarian Infection

Infection involving an ovary, often as part of pelvic inflammatory disease or an abscess, causing pelvic pain, fever, and abnormal discharge.

Ovarian Metastasis

Secondary involvement of the ovary by a malignant tumor arising elsewhere, commonly from the gastrointestinal tract, breast, or gynecologic tract; the

distinction from a primary ovarian tumor guides treatment.

Ovarian Overproduction Of Androgens

Ovarian androgen overproduction is excessive production of male-pattern hormones by the ovaries, which may cause acne, excess facial hair, irregular periods, or infertility.

Ovarian Pregnancy

A rare ectopic pregnancy implanted in ovarian tissue rather than in the uterine cavity or fallopian tube. Diagnosis uses clinical findings, serial human chorionic gonadotropin measurements, ultrasound, and sometimes surgical or pathologic confirmation; treatment is individualized.

Ovarian Rupture

Ovarian rupture is tearing of an ovary or ovarian cyst, which may cause sudden pelvic pain and internal bleeding and sometimes requires urgent treatment.

Ovarian Teratoma

An ovarian teratoma is a germ-cell tumor containing tissues from one or more embryonic layers; mature teratomas are usually benign, while immature forms can be malignant.

Ovarian Thecoma

An ovarian thecoma is a usually benign sex-cord stromal tumor that may produce estrogen and cause abnormal uterine bleeding or other hormonal effects.

Ovarian Torsion

Twisting of the ovary (and often the fallopian tube) on its vascular pedicle, causing obstruction of blood flow—a gynecologic emergency. Risk: ovarian cysts/tumors (especially >5 cm), longer utero-ovarian ligament, younger women, pregnancy.

Ovariectomy (Oophorectomy)

The surgical excision of one ovary (unilateral oophorectomy) or both ovaries (bilateral oophorectomy), performed via laparoscopy or laparotomy.

Indicated therapeutically for benign or malignant ovarian neoplasms, severe ovarian torsion with tissue necrosis, tubo-ovarian abscess refractory to medical therapy, endometriosis, or as risk-reducing bilateral salpingo-oophorectomy in patients carrying high-penetrance genetic mutations such as *BRCA1* or *BRCA2*. In premenopausal women, bilateral oophorectomy induces immediate surgical menopause (abrupt loss of ovarian estradiol and progesterone), precipitating vasomotor symptoms, accelerated bone mineral density loss, and elevated cardiovascular risk, often warranting hormone replacement therapy when not contraindicated by hormone-receptor-positive malignancy.

Ovaries

The paired female gonads that produce oocytes and secrete sex hormones, mainly estrogen and progesterone. They are located on either side of the uterus and participate in the menstrual cycle and reproduction.

Alternate Names:

ovary

Ovariectomy

An older term for surgical removal or incision of an ovary.

Ovary

A female reproductive organ that releases ova and produces hormones including estrogen and progesterone.

Ovary Disease

Any disorder affecting an ovary, including cysts, infection, inflammation, tumors, hormonal dysfunction, torsion, or bleeding.

Ovary Neoplasm

An ovarian neoplasm is an abnormal growth in an ovary that may be benign, borderline, or malignant.

Oven Cleaner Poisoning

Oven cleaner poisoning involves toxic or environmental exposure to oven cleaner poisoning; dose, route, timing, symptoms, and organ involvement

determine urgency and treatment.

Over-Counter

An incomplete or nonstandard form usually intended to mean over-the-counter, describing medicines available without a prescription.

Over-Hydration

Excess accumulation of body water, which may cause edema, dilutional hyponatremia, or heart and lung complications.

Over-Nutrition

Excessive intake or accumulation of energy or nutrients that can contribute to obesity, metabolic disease, or toxicity.

Over-The-Counter

Medications available without a prescription, considered safe and effective for self-care with appropriate labeling and low abuse potential. Examples: acetaminophen, ibuprofen, antacids, antihistamines, decongestants, laxatives. Regulated in the US by the FDA (OTC monograph system) and Rx-to-OTC switches (e.g., omeprazole, loratadine).

Over-The-Counter

Over-the-counter means available for purchase without a prescription, although safe use still requires attention to directions, interactions, and warnings.

Over-The-Counter Medications

Nonprescription drugs approved as safe and effective for self-management of common symptoms, available at pharmacies and retail stores without a prescription.

Over-The-Counter Medicines

Over-the-counter medicines can be bought without a prescription for common problems, but they may still cause side effects or interact with prescribed medicines.

Over-The-Counter Pain Relievers

Over-the-counter pain relievers include medicines

such as acetaminophen and nonsteroidal anti-inflammatory drugs; choice depends on age, health conditions, and other medicines.

Overactive Bladder

A clinical syndrome of urinary urgency, usually with frequency and nocturia, with or without urge incontinence, due to detrusor overactivity during filling. Causes: idiopathic (most), neurogenic (stroke, spinal injury, MS, Parkinson), bladder outlet obstruction, and aging; worsened by caffeine and diuretics.

Alternate Names:

OAB OAB (Overactive Bladder)

Overactive Bladder

Alternate index label for Overactive Bladder.

Overactive Thyroid

Alternate index label; see *Hyperthyroidism* (*overactive thyroid*).

Overactive Thyroid (Hyperthyroidism)

Alternate index label for Hyperthyroidism.

Overall Survival

In clinical oncology and clinical trial methodology, the duration of time measured from the date of trial randomization (or initiation of treatment) until death from any cause. Regarded by regulatory authorities (such as the FDA and EMA) as the gold-standard, most objective and definitive endpoint for demonstrating clinical efficacy in Phase III cancer trials, as it directly reflects clinical benefit without subjective interpretation or measurement bias. Statistically analyzed using Kaplan-Meier survival curves and Cox proportional hazards regression models, and compared with surrogate survival endpoints such as progression-free survival and disease-free survival.

Overbite

Vertical overlap of the upper front teeth over the lower front teeth; excessive overbite can damage gums, teeth, or jaw function.

Overcoming Breastfeeding Problems

Breastfeeding problems such as pain, poor latch, low supply, engorgement, or blocked ducts can often improve with feeding assessment, positioning help, and timely lactation or medical support.

Overcoming Job Stress

Managing job stress may include setting boundaries, taking breaks, improving workload or support, practicing relaxation, and seeking professional help when distress persists.

Overdenture

Removable denture supported by implants or natural teeth. Preservation of alveolar bone, improved stability.

Overdose

Exposure to a dose of a medicine or substance that exceeds the person's safe or intended amount and may cause toxicity.

Overflow Urinary Incontinence

Leakage of urine caused by incomplete bladder emptying and excessive bladder filling. It may occur with weak urinary flow, dribbling, urgency, or urinary retention.

Overgrowth Syndrome

A group of congenital disorders characterized by excessive growth of the body, a region, or specific tissues, often with developmental or tumor-predisposition features. Causes include genomic imprinting abnormalities, mosaic variants, and alterations in growth-signaling pathways.

Overlap Myositis

Patients with features of inflammatory myopathy (DM/PM/IMNM) plus another systemic autoimmune disease (SLE, SSc, MCTD, Sjogren). Anti-PM-Scl (Scl/PM overlap, limited SSc + myositis, good prognosis); anti-Ku (SLE/PM overlap); anti-U1-RNP (MCTD with myositis).

Overlay Denture

A removable denture that fits over remaining teeth, roots, or implants to improve support, chewing, and retention.

Overload Iron (Iron Overload)

Alternate index label for Iron Overload.

Overriding Aorta

A congenital heart defect in which the aorta is positioned over a ventricular septal defect and receives blood from both ventricles, classically in tetralogy of Fallot.

Oversedate

To give excessive sedation, causing unusually deep sleep, slowed responses, impaired breathing, or difficulty maintaining the airway.

Overt Proteinuria

Abnormally high protein loss in the urine that is detectable by routine testing and may indicate kidney disease.

Overuse Syndrome (Repetitive Motion Disorders (RMDs))

Alternate index label for Repetitive Motion Disorders (RMDs).

Overweight

Body weight above the usual range for height, commonly defined in adults as BMI 25 to less than 30 kg/m², with context-dependent limitations.

Oviduct

A tube carrying an ovum from the ovary toward the uterus; in humans it is the uterine or fallopian tube.

Oviposition

The laying or depositing of eggs by a female animal or insect.

Ovomucin

A glycoprotein in egg white that contributes to its gel-like viscosity and helps stabilize the egg-white

structure.

Ovotestis

A diagnostic or laboratory method related to ovotestis. Results are interpreted with the method, reference range, and clinical findings.

Ovulation

Release of a mature egg from an ovarian follicle, usually around the middle of the menstrual cycle.

Ovulation Home Test

An ovulation home test detects the rise in urinary luteinizing hormone that usually occurs shortly before ovulation.

Ovule

An immature seed in plants or, historically, an ovum or small egg-like structure.

Ovum

The mature female haploid reproductive gamete (unfertilized secondary oocyte arrested at metaphase of meiosis II) released from the mature ovarian Graafian follicle during ovulation. Surrounded by the inner glycoprotein matrix of the zona pellucida and outer follicular support cells of the corona radiata. Upon penetration by a viable spermatozoon, the secondary oocyte undergoes cortical granule reaction (preventing polyspermy), completes meiosis II to form a mature ovum and a second polar body, and fuses paternal and maternal pronuclei to form a single diploid zygote (46 chromosomes).

Ownership

Ownership is legal or practical control of property, data, decisions, or responsibilities; healthcare decisions remain with the patient or authorized surrogate.

Oxacillin

A penicillinase-resistant penicillin used mainly to test for methicillin resistance in staphylococci; it is not commonly used as the preferred treatment for most infections.

Oxalate

A salt or ester of oxalic acid that can form kidney stones and binds calcium.

Oxalates

Oxalates are salts or esters of oxalic acid that can bind calcium and form calcium-oxalate kidney stones; excess dietary absorption or inherited hyperoxaluria can cause renal injury.

Oxalic Acid

A dicarboxylic acid whose oxalate salts can contribute to calcium-oxalate kidney stones; concentrated exposure is toxic.

Oxalic Acid Poisoning

Oxalic acid poisoning involves toxic or environmental exposure to oxalic acid poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Oxaliplatin

Oxaliplatin is a platinum chemotherapy drug that forms DNA cross-links and is used especially in colorectal cancer regimens; acute cold-triggered neuropathy and cumulative sensory neuropathy are characteristic toxicities.

Oxaloacetate

A four-carbon citric-acid-cycle intermediate that condenses with acetyl-CoA to form citrate. It is also a precursor for gluconeogenesis and participates in amino-acid transamination to aspartate.

Oxandrolone

An anabolic steroid used in selected conditions involving severe weight loss or catabolic stress; it can affect the liver, lipids, and hormone function.

Oxaprozin

A nonsteroidal anti-inflammatory drug used to relieve pain and inflammation in conditions such as osteoarthritis and rheumatoid arthritis.

Oxazepam

A benzodiazepine anxiolytic and sedative with risks of dependence, sedation, and respiratory depression.

Oxazepam Overdose

Oxazepam overdose involves toxic or environmental exposure to oxazepam overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Oxazole

Oxazole is a five-membered chemical ring containing oxygen and nitrogen that is used as a building block in medicinal and other chemistry.

Oxcarbazepine

Oxcarbazepine is an antiseizure medicine that blocks voltage-gated sodium channels and treats focal seizures; dizziness, ataxia, rash, and clinically important hyponatraemia may occur.

Oxidant

A reactive molecule that can damage cells and tissues by promoting oxidation.

Alternate Names:

Oxidizing agent

Oxidants

Substances or reactive molecules that accept electrons or promote oxidation; excessive oxidant activity can damage proteins, lipids, and DNA.

Oxidation

A chemical process that changes a substance through loss of electrons or reaction with oxygen.

Oxidation-Reduction

A chemical reaction involving transfer of electrons, in which one substance is oxidized and another is reduced.

Oxidative Phosphorylation

Process generating ATP using energy released by electron transfer through the electron transport chain. Protons pumped across inner mitochondrial

membrane create electrochemical gradient (proton motive force). ATP synthase (Complex V) uses proton flow to phosphorylate ADP to ATP. P/O ratio: 2.5 ATP per NADH, 1.5 ATP per FADH₂.

Oxidative Stress

Imbalance between reactive oxygen species and antioxidants. DNA damage, protein oxidation, lipid peroxidation.

Oxide

A compound formed when oxygen combines with another element.

Oxidopamine

A toxic catecholamine analogue used experimentally to damage sympathetic nerve terminals and study autonomic function.

Oxidoreductase

An oxidoreductase is an enzyme that catalyzes transfer of electrons or hydrogen between molecules during oxidation-reduction reactions.

Oxime

A compound containing a carbon-nitrogen double bond with a hydroxyl group; some oximes, such as pralidoxime, help reactivate acetylcholinesterase after organophosphate poisoning.

Oximeter

A device that measures oxygen saturation or related optical signals, commonly by pulse oximetry.

Oximetry

oximetry is a laboratory test, imaging method, instrument, score, or measurement used to characterize a specimen or clinical finding; interpretation depends on the method and clinical context.

Oxonic Acid

A uricase inhibitor used mainly in laboratory research to increase uric-acid levels in animal models.

Oxprenolol

Oxprenolol is a beta-blocker formerly used for high blood pressure and heart disease; it can slow the heart rate and worsen asthma in susceptible people.

Oxter

Oxter is an anatomical term for the armpit or axilla.

Oxybuprocaine

A topical ester local anesthetic used mainly for ocular surface anesthesia.

Oxycephaly

A cone-shaped skull caused by premature fusion of cranial sutures, also called acrocephaly.

Oxycodone

Oxycodone is an opioid analgesic used for moderate-to-severe pain; sedation, constipation, respiratory depression, tolerance, dependence, and overdose risk increase with alcohol or other sedatives.

Oxycodone Hydrochloride

The hydrochloride salt of oxycodone, an opioid analgesic used for moderate-to-severe pain; it can cause sedation, constipation, dependence, and fatal respiratory depression in overdose.

OxyContin

OxyContin is an extended-release oxycodone opioid formulation used for selected severe persistent pain and carrying risks of misuse, dependence, and respiratory depression.

Oxygen

A gas essential for aerobic metabolism and tissue energy production.

Oxygen Consumption

The volume of oxygen consumed by the body per unit time, reflecting overall metabolic rate. Resting oxygen consumption (VO₂) in a normal adult is approximately 250 mL/min (about 3.5 mL/kg/min, 1 MET). It is calculated as cardiac output x (arterial O₂ content - venous O₂ content) by the Fick

principle.

Oxygen Content

The amount of oxygen carried in blood, mainly bound to hemoglobin with a small dissolved component.

Oxygen Debt

An older term for the increased oxygen consumption after exercise needed to restore metabolic and physiologic conditions; excess post-exercise oxygen consumption is preferred.

Oxygen Delivery

The rate at which oxygen is transported from the lungs to the tissues, equal to cardiac output multiplied by arterial oxygen content: $DO_2 = CO \times CaO_2$. Arterial oxygen content = $(1.34 \times \text{hemoglobin} \times SaO_2) + (0.003 \times PaO_2)$. Normal oxygen delivery is approximately 1000 mL/min.

Oxygen Desaturation Index

The average number of oxygen desaturation events per hour of sleep, usually reported from a sleep study as the ODI; interpretation depends on the recording method and clinical context.

Oxygen Enhancement Ratio

Ratio of dose under hypoxic vs. aerobic conditions. Higher for low LET radiation.

Oxygen Mask

Delivers supplemental oxygen. Types: simple, nasal cannula, non-rebreather, Venturi.

Oxygen Pulse

The amount of oxygen used by the body with each heartbeat, estimated during exercise as oxygen uptake divided by heart rate.

Oxygen Safety

Practices that reduce fire, explosion, pressure, and treatment risks during oxygen use, including keeping oxygen away from flames, smoking, oils, and unsafe equipment.

Oxygen Saturation

The percentage of hemoglobin carrying oxygen in blood. It may be estimated by pulse oximetry (SpO_2) or measured in an arterial blood sample by co-oximetry (SaO_2). The appropriate range depends on the person's condition, baseline oxygenation, and treatment goals.

Oxygen Tent

A canopy or enclosure used to deliver oxygen around a patient's head; modern oxygen masks, nasal cannulas, or ventilatory devices are usually more precise.

Oxygen Therapy

Administration of supplemental oxygen to correct or prevent inadequate tissue oxygenation.

Oxygen Therapy In Infants

Oxygen therapy in infants supplements inspired oxygen when gas exchange is inadequate, while careful monitoring limits complications from both hypoxemia and excessive oxygen exposure.

Oxygen Toxicity

Tissue injury caused by exposure to oxygen at excessively high partial pressures or concentrations, potentially affecting the lungs, eyes, or central nervous system.

Oxygen Transport

The movement of oxygen from the lungs to tissues, mainly bound to hemoglobin and partly dissolved in plasma, with delivery affected by blood flow and hemoglobin concentration.

Oxygen-Hemoglobin Dissociation Curve

The sigmoidal relationship between partial pressure of oxygen (PO_2) and hemoglobin oxygen saturation (SO_2). The curve reflects hemoglobin's cooperative binding: oxygen binding to one subunit increases affinity for subsequent oxygen molecules. At PO_2 of 100 mmHg (alveolar), hemoglobin is approximately 98 percent saturated.

Oxygenases

Enzymes that incorporate oxygen from molecular oxygen into a substrate; monooxygenases add one oxygen atom and dioxygenases add both.

Oxygenated Blood

Blood that has acquired oxygen in the lungs and carries it to body tissues.

Oxygenation

The process of supplying oxygen to blood and tissues, or the degree to which hemoglobin and tissues are adequately oxygenated.

Oxygenator

A device that adds oxygen to blood and removes carbon dioxide, used in cardiopulmonary bypass and extracorporeal life support.

Oxyhaemoglobin

The oxygen-saturated physiological form of hemoglobin formed within pulmonary alveolar capillaries when molecular oxygen (O_2) reversibly binds to the ferrous iron (Fe^{2+}) atom within the protoporphyrin IX heme group of hemoglobin tetramers. Exhibits cooperative binding kinetics, giving rise to the characteristic sigmoidal oxyhemoglobin dissociation curve. Rightward shifts in the curve (facilitating oxygen unloading to metabolically active peripheral tissues) are driven by acidosis (low pH), hypercapnia (elevated pCO_2 ; the Bohr effect), elevated temperature, and increased red blood cell 2,3-bisphosphoglycerate (2,3-BPG); leftward shifts (increasing oxygen affinity) occur with alkalosis, hypothermia, decreased 2,3-BPG, and carboxyhemoglobinemia.

Oxyhemoglobin

Oxyhemoglobin is hemoglobin bound to oxygen, formed mainly in the lungs and carrying oxygen through arterial blood to tissues.

Oxylipins

Oxylipins are oxygenated fatty-acid derivatives that act as local mediators of inflammation, vascular tone, pain, and other physiologic responses.

Oxymetazoline

A topical alpha-adrenergic agonist used as a nasal decongestant or ocular vasoconstrictor; prolonged nasal use can cause rebound congestion.

Oxymetholone

Oxymetholone is an anabolic steroid used in selected severe anemias; it can cause liver injury, fluid retention, and hormonal side effects.

Oxymorphone

A potent opioid analgesic used for severe pain; it can cause sedation, constipation, dependence, and life-threatening respiratory depression.

Oxyphil Cell

oxyphil cell refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Oxyphil Cell Carcinoma

Alternate index label; see *Hurthle cell cancer*.

Oxyphilic Adenoma

An oxyphilic adenoma is a usually benign glandular tumor composed of cells rich in mitochondria, such as an oncocytic parathyroid or salivary-gland tumor.

Oxytocic

Causing or promoting uterine contraction, especially a medicine such as oxytocin.

Oxytocics

A class of medications that stimulate uterine contractions, used for induction and augmentation of labor and prevention/treatment of postpartum hemorrhage. Oxytocin (Syntocinon) — synthetic analog of endogenous oxytocin, administered IV infusion for labor induction or IM/IV for PPH prevention.

Oxytocin

A hormone that stimulates uterine contractions and milk ejection; it is also used therapeutically in obstetrics.

Oxytocin Infusion

Intravenous synthetic oxytocin used to induce or augment labor or, in selected settings, to manage postpartum bleeding. The dose is started and adjusted according to uterine activity, fetal status, maternal condition, and local protocol; excessive uterine stimulation can be harmful.

Oxyuriasis

Infection with pinworms, especially *Enterobius vermicularis*, causing perianal itching and disturbed sleep.

Alternate Names:

n-3 fatty acids

Oz.

Oz. is an abbreviation for ounce, a unit of mass or fluid volume whose exact conversion depends on the measurement system.

Ozaena

A chronic condition characterized by atrophic rhinitis with crusting, foul-smelling nasal discharge (fetor), and a sensation of nasal obstruction despite a patent airway. Ozaena is caused by progressive atrophy of the nasal mucosa and underlying bone, often associated with *Klebsiella ozaenae* infection.

Ozena

Ozena is a chronic form of atrophic rhinitis with widened nasal passages, crusting, foul odor, and impaired mucosal function.

Ozone

Ozone is a reactive form of oxygen that can irritate and injure the airways and lungs when inhaled, despite useful atmospheric and disinfecting roles.

P

Oral Route

The administration of a medicine through the mouth for swallowing and absorption through the gastrointestinal tract. Clear instructions should spell out “by mouth” to reduce medication-route errors.

Alternate Names:

P.O.; Per Os; By Mouth

P Arm Of A Chromosome

The short arm of a chromosome, designated p for petit, opposite the longer q arm; cytogenetic locations are written with p-band coordinates.

P Wave

The first deflection on the ECG, representing atrial depolarization. Normal P waves are upright in leads I and II, inverted in aVR, and less than 0.12 seconds (3 small boxes) in duration and 2.5 mm in amplitude.

P-Glycoprotein

An ATP-dependent membrane transporter encoded by ABCB1 that pumps many drugs and other compounds out of cells, affecting intestinal absorption, tissue penetration, and drug interactions.

P-Value

A p-value is the probability, under a specified null hypothesis, of observing results at least as extreme as those found; it does not by itself measure clinical importance.

P53 Tumor Suppressor

The p53 protein, encoded by the TP53 gene on chromosome 17p, is a transcription factor and the most commonly mutated gene in human cancer, with mu-

tations found in over 50 percent of all malignancies.

PA X-Ray

A PA X-ray is a posteroanterior radiograph, commonly of the chest, taken with the beam traveling from back to front to reduce heart magnification.

PA/IVS

Alternate index label; see *Pulmonary atresia with intact ventricular septum*.

Pacemaker

An implanted electronic device that delivers electrical impulses to the heart to regulate heart rate and rhythm. Indications include symptomatic bradycardia, high-degree AV block, and sick sinus syndrome. Single-chamber pacemakers pace the atrium or ventricle; dual-chamber pacemakers pace both.

Alternate Names:

Pacemaker, artificial

Pacemaker Complications

Early: pneumothorax, hematoma, cardiac perforation/tamponade, lead dislodgement. Late: lead fracture, battery depletion, infection, tricuspid regurgitation from lead interference, venous thrombosis, and pacemaker syndrome (AV dyssynchrony in VVI pacing).

Pachyderma

Abnormal thickening and coarsening of the skin, which may occur with chronic inflammation, lymphedema, endocrine disease, or pachydermoperiostosis.

Pachygyria

A cortical malformation in which the cerebral gyri are abnormally broad and reduced in number because of impaired neuronal migration. It may cause developmental delay, intellectual disability, seizures, and motor or neurological impairment of variable severity.

Pachymeningitis

Inflammation of the dura mater (pachymeninx), the outermost layer of the meninges. Pachymeningitis is less common than leptomeningitis (inflammation of pia and arachnoid mater).

Pachyonychia Congenita

An inherited keratin disorder, usually caused by KRT6A, KRT16, or KRT17 variants, producing thick dystrophic nails, painful palmoplantar keratoderma, and sometimes cysts or oral leukokeratosis.

Pacifier

A pacifier is a nipple-shaped device used to soothe an infant by sucking; safe use includes avoiding cords and sweet coatings and stopping according to dental and developmental guidance.

Pacing

Planned regulation of activity, rest, and intensity to avoid symptom exacerbation while gradually increasing tolerance. It is used in fatigue, chronic pain, cardiopulmonary disease, and post-viral conditions.

Pacinian Corpuscle

A pressure- and vibration-sensitive nerve ending located deep in the skin, connective tissue, and some joints.

Packaging Tube

A packaging tube is a squeezable container for medicines, creams, gels, pastes, or other products, designed to protect contents and allow controlled dispensing.

Paclitaxel

Paclitaxel is a taxane chemotherapy drug that sta-

bilizes microtubules and prevents cell division; it treats several cancers and can cause myelosuppression, neuropathy, alopecia, myalgia, and hypersensitivity reactions.

Paclitaxel Liposome

A liposomal formulation of paclitaxel designed to alter drug distribution and delivery; paclitaxel disrupts microtubules and is used as chemotherapy in selected cancers.

Paclitaxel Poliglumex

An investigational formulation in which paclitaxel is linked to a biodegradable polymer to alter drug delivery and reduce solvent-related toxicity.

PACS

Picture Archiving and Communication System. Digital image storage and retrieval.

Paecilomyces

A genus of environmental molds, some of which can cause opportunistic infection in people with weakened immunity.

Paenibacillus

A genus of spore-forming gram-positive bacteria found in soil, water, and plants; a few species rarely cause human infection.

Paenibacillus Alvei

A spore-forming environmental bacterium that rarely causes opportunistic human infection.

Paenibacillus Phoenicis

A soil-associated *Paenibacillus* species rarely recovered from human clinical specimens.

Paenibacillus Polymyxa

A soil bacterium used in agriculture and biotechnology that rarely causes opportunistic infection.

PAES

Alternate index label; see *Popliteal artery entrapment*.

Page Kidney

Renal compression from a perinephric hematoma or mass causing hypertension through activation of the renin-angiotensin system, typically following trauma or occurring in patients on anticoagulation.

Paget Disease Of Bone

A chronic bone disorder characterized by excessive, disorganized bone remodeling with increased osteoclastic bone resorption followed by compensatory osteoblastic bone formation, producing structurally abnormal bone that is enlarged, weak, and prone to fracture.

Alternate Names:

Paget's disease; Paget disease of the bone; Osteitis deformans

Paget Disease Of The Anus

An intraepithelial adenocarcinoma of perianal skin or anal mucosa presenting as a persistent eczematous, itchy, or eroded lesion. Biopsy is required, and associated underlying or distant malignancy must be evaluated.

Paget Disease Of The Nipple

A rare breast cancer involving the nipple and areola, often causing persistent eczema-like scaling, redness, itching, burning, or discharge. It may be associated with underlying ductal carcinoma in situ or invasive breast cancer.

Paget Disease Of The Vulva

An intraepithelial adenocarcinoma of vulvar skin, usually presenting as a persistent itchy, eczematous, or erythematous plaque. Biopsy and evaluation for an associated underlying or distant malignancy are required.

Pagetoid Reticulosis

A rare localized variant of cutaneous T-cell lymphoma with solitary slowly enlarging scaly plaques containing atypical pagetoid lymphocytes.

Pain

An unpleasant sensory and emotional experience

associated with actual or potential tissue damage. It is influenced by biological, psychological, and social factors and may persist or recur after the original injury has healed.

Pain (Geriatric)

Pain in an older adult. Assessment considers communication, cognition, function, medicines, and possible causes; treatment is individualized to the person and condition.

Pain (Vascular)

Pain caused by impaired blood flow, often appearing in a limb with exertion as claudication or, in more severe disease, at rest.

Pain And Your Emotions

Pain can affect mood, sleep, attention, and stress, while anxiety or depression can increase how pain is experienced; both the pain and emotional effects may need treatment.

Pain Assessment Scales

Pain assessment uses standardized scales to quantify pain intensity. Numeric rating scale (NRS) asks patients to rate pain from 0 (no pain) to 10 (worst pain imaginable). Visual analog scale (VAS) is a 10 cm line with anchors. The Wong-Baker FACES scale uses facial expressions, particularly when a person can understand the scale but has difficulty using numbers; observational tools may be needed when self-report is not possible.

Pain Cancer (Cancer Pain)

Alternate index label for Cancer Pain.

Pain Elbow (Elbow Pain)

Alternate index label for Elbow Pain.

Pain Gallbladder (Gallbladder Pain (Gall Bladder Pain))

Alternate index label for Gallbladder Pain (Gall Bladder Pain).

Pain Heel (Heel Spurs)

Alternate index label for Heel Spurs.

Pain Kidney (Kidney Pain)

Alternate index label for Kidney Pain.

Pain Lower Right Abdomen (Appendicitis)

Alternate index label for Appendicitis.

Pain Management

Assessment and treatment of pain using appropriate pharmacologic, physical, psychological, and interventional approaches. Treatment is individualized according to the cause, severity, duration, functional effect, and risks of each option.

Pain Management: Neuropathic Pain

Neuropathic pain arises from injury or dysfunction of nerves and may feel burning, electric, shooting, numb, or hypersensitive. Treatment can include the cause-specific therapy, physical and psychological rehabilitation, topical agents, anticonvulsants, antidepressants, or specialist procedures.

Alternate Names:

Neuropathic Pain (Pain Management: Neuropathic Pain)

Pain Medications - Opioid Analgesics

Opioid analgesics can relieve selected moderate-to-severe pain but may cause constipation, sedation, tolerance, dependence, overdose, and slowed breathing. They require individualized prescribing and monitoring.

Alternate Names:

Pain Medications - Narcotics

Pain Medicine

The medical specialty concerned with assessing and treating acute, chronic, and complex pain through medication, procedures, rehabilitation, and multidisciplinary care.

Alternate Names:

Pain clinic

Pain Nerve (Neuropathic Pain (Nerve Pain))

Alternate index label for Neuropathic Pain (Nerve Pain).

Pain Perception

The sensory and emotional process by which the nervous system detects, transmits, and interprets potentially tissue-damaging stimuli as pain.

Pain Relief (Analgesia)

Achieved through pharmacologic, interventional, physical, and psychologic modalities targeting nociceptive transmission and pain perception.

Alternate Names:

Analgesia

Pain Scrotum (Testicular Disorders)

Alternate index label for Testicular Disorders.

Pain Tailbone (Coccydynia)

Alternate index label for Coccydynia.

Pain Threshold

The minimum intensity of a stimulus that a person perceives as painful; it varies with tissue, context, attention, inflammation, medications, and neurologic state.

Pain Tooth (Toothache Overview)

Alternate index label for Toothache Overview.

Pain Unit

A clinical service specializing in assessment and treatment of acute, chronic, cancer, neuropathic, or procedural pain using medicines, procedures, rehabilitation, and psychological care.

Pain Vaginal (Vaginal Pain (Vulvodynia))

Alternate index label for Vaginal Pain (Vulvodynia).

Pain Whiplash (Whiplash)

Alternate index label for Whiplash.

Pain, Back

Alternate index label; see *Back pain*.

Pain, Breast

Alternate index label; see *Breast pain*.

Painful Bladder Syndrome

Alternate index label; see *Interstitial cystitis*.

Painful Intercourse (Dyspareunia)

Persistent or recurrent genital pain before, during, or after sexual intercourse, with physical, hormonal, emotional, or relationship-related causes.

Painful Menstrual Periods

Painful menstrual periods, or dysmenorrhea, are cramps during menstruation caused by uterine contractions or conditions such as endometriosis or fibroids.

Painful Swallowing

Painful swallowing, or odynophagia, is discomfort when swallowing and may result from infection, inflammation, injury, reflux, or a blockage.

Paint, Lacquer, And Varnish Remover Poisoning

Poisoning caused by ingestion, inhalation, or skin or eye exposure to solvents and other chemicals in paint, lacquer, or varnish removers. Effects may include irritation, central nervous system depression, respiratory injury, metabolic abnormalities, or organ damage; severity depends on the substance, dose, route, and timing.

Palatal Muscle

Muscle of the soft palate that elevates or tenses the palate and helps separate the nasal and oral cavities during swallowing and speech.

Palatal Myoclonus

Palatal myoclonus is rhythmic involuntary contraction of muscles of the soft palate, sometimes caused by lesions in brainstem motor pathways and producing clicking in the ear.

Palate

The roof of the mouth, consisting of a bony hard palate and muscular soft palate.

Palate Neoplasm

A palate neoplasm is an abnormal growth in the roof of the mouth that may be benign or malignant.

Palatine

Relating to the palate (the roof of the mouth separating the oral and nasal cavities) or the paired L-shaped palatine bones of the facial skeleton. Clinically relevant structures include the palatine tonsils (lymphoid tissue situated in the tonsillar fossa between the palatoglossal and palatopharyngeal arches, frequently affected by acute exudative tonsillitis and peritonsillar abscess), the greater and lesser palatine canals transmitting palatine nerves and vessels for regional dental anesthesia, and congenital cleft palate resulting from failure of fusion of the lateral palatine shelves.

Palatine Tonsil

One of two masses of lymphoid tissue at the sides of the throat that help detect inhaled or swallowed pathogens.

Palatorrhaphy

Palatorrhaphy is surgical repair of a cleft or defect of the palate to restore separation between the oral and nasal cavities and improve feeding and speech.

Palatoschisis

A congenital cleft or failure of fusion of the palate, commonly called cleft palate, which may affect feeding, speech, hearing, and dental development.

Paleness

Unusual lightening of the skin or mucous membranes, which may result from reduced blood flow, anemia, shock, cold, or normal variation.

Paleostriatum

The paleostriatum is a historical or comparative-anatomy term for a forebrain basal-ganglia region

involved in motor control, corresponding in part to mammalian pallidal structures.

Palindrome

A palindrome is a word or sequence that reads the same forward and backward; in genetics, palindromic DNA sequences can have specialized structural or recognition roles.

Palindromic Rheumatism

Characterized by recurrent, self-limited attacks of mono- or oligoarthritis (hours to days), periarticular soft tissue swelling, no radiographic erosions between attacks. 30-50% evolve to RA (anti-CCP positive predicts progression).

Paliperidone

An atypical antipsychotic medicine used to treat schizophrenia and schizoaffective disorder.

Palivizumab

A monoclonal antibody against respiratory syncytial virus fusion protein, used prophylactically in selected high-risk infants and young children during RSV season.

Palladium

A chemical element used in some dental materials, medical devices, and industrial products; exposure can cause allergy in susceptible people.

Pallesthesia

The ability to perceive vibration, tested with a tuning fork and mediated by large sensory fibers.

Palliative Care

Specialized medical care for serious illness focusing on symptom relief and quality of life. Not limited to end-of-life. Can be provided alongside curative treatment. Components: pain management, symptom control, psychosocial support, spiritual care, advance care planning. Referral early in disease course.

Palliative Rehabilitation

Rehabilitation for people with serious or life-limiting illness, focused on comfort, function, participation, symptom management, and quality of life.

Palliative Radiation

Radiation therapy given to relieve symptoms caused by cancer, such as pain, bleeding, obstruction, or pressure from a tumor. It is usually intended to improve comfort or function rather than cure the cancer and may be delivered in a short treatment course.

Palliative Surgery

Surgery intended to relieve symptoms or improve function without necessarily curing the underlying disease.

Palliative Treatment

Treatment intended to relieve symptoms and improve comfort or quality of life rather than cure the underlying disease.

Pallidotomy

A neurosurgical procedure creating a lesion in the globus pallidus to reduce selected symptoms of advanced Parkinson disease.

Pallister-Hall Syndrome

A congenital disorder usually caused by pathogenic variants in *GLI3*. It is characterized by hypothalamic hamartoma and mesoaxial polydactyly and may include imperforate anus, renal or airway anomalies, and pituitary hormone abnormalities.

Pallor

Unusual paleness of skin or mucous membranes, often due to reduced blood flow, anemia, or other illness.

Palm

The inner surface of the hand between the wrist and fingers.

PALM-COEIN Classification

The standardized classification system established by the International Federation of Gynecology and Obstetrics to categorize causes of abnormal uterine bleeding in non-pregnant women of reproductive age. The acronym stratifies etiologies into nine distinct categories, divided into structural abnormalities confirmed by imaging or pathology (Polyp, Adenomyosis, Leiomyoma, Malignancy and hyperplasia) and non-structural entities (Coagulopathy, Ovulatory dysfunction, Endometrial disorders, Iatrogenic, and Not otherwise classified).

Alternate Names:

FIGO AUB Classification; PALM-COEIN System

Palm Sweating Excessive (Hyperhidrosis)

Alternate index label for Hyperhidrosis.

Palmar

Pertaining to the anterior flexor surface of the human hand (the palm; *palma manus*). Anatomically characterized by thick, specialized glabrous skin rich in eccrine sweat glands, firm subcutaneous fibrous tethering via the dense palmar aponeurosis (which undergoes progressive fibromatous contracture in Dupuytren disease), and superficial and deep palmar arterial arches supplying digital circulation. Clinically significant in dermatology and neurology (e.g., palmar erythema associated with hyperestrogenism in cirrhosis, palmar xanthomas in dysbetalipoproteinemia, and the primitive grasp reflex).

Palmar Erythema Of Pregnancy

Symmetrical redness of the palms occurring during pregnancy, thought to be related to elevated circulating oestrogen levels causing vasodilatation. It is a benign, physiological finding that typically resolves after delivery and requires no specific treatment.

Palmar Grasp Reflex

A primitive reflex in which an infant flexes the fingers around an object placed in the palm. It is normally present at birth and gradually diminishes as voluntary grasp develops; asymmetry or persistence may warrant neurologic assessment.

Palmar Reflex

The palmar reflex is an infant response in which stroking the palm causes finger flexion; persistence or asymmetry beyond the expected age may indicate neurologic dysfunction.

Palmitate

Palmitate is the salt or ester form of palmitic acid and participates in fatty-acid metabolism, lipid storage, and protein modification.

Palmitic Acid

A 16-carbon saturated fatty acid common in dietary and stored fats; excess saturated-fat intake can contribute to adverse lipid profiles, although the body also synthesizes it normally.

Palmoplantar Hyperkeratosis

Excessive thickening of the skin on the palms and soles, inherited or acquired, causing calluses, fissures, pain, impaired walking, and sometimes infection.

Palonosetron Hydrochloride

A long-acting 5-HT₃ serotonin-receptor antagonist used to prevent nausea and vomiting caused by chemotherapy, surgery, or selected other treatments.

Palpable

Palpable means detectable by touch during physical examination, such as a mass, enlarged organ, lymph node, pulse, tenderness, or abnormal surface.

Palpate

To examine body tissues by carefully feeling them with the hands.

Palpation

Examination by touching the body to assess tenderness, temperature, texture, masses, pulses, organ size, and movement.

Palpebra

The eyelid; an anatomically stratified, mobile musculo-membranous tissue fold that protects the globe, distributes the tear film across the cornea

during blinking, and limits light entry. Composed of skin, orbicularis oculi muscle (innervated by CN VII, responsible for eye closure), dense fibrous tarsal plates containing meibomian (sebaceous) glands, and the inner palpebral conjunctiva. Pathologies include ptosis (drooping due to levator palpebrae superioris / CN III or sympathetic Müller muscle dysfunction), ectropion, entropion, blepharitis, chalazion, and hordeolum.

Palpebral Conjunctiva

The part of the conjunctiva that lines the inner surfaces of the eyelids. It joins the bulbar conjunctiva at the fornices and helps protect and lubricate the eye.

Palpebral Slant - Eye

The direction of the opening between the upper and lower eyelids. Its shape varies normally, but an unusual slant may be associated with a genetic or developmental condition.

Palpitation

The subjective sensation of a rapid, forceful, or irregular heartbeat. Palpitations are a common symptom with a broad differential.

Palpitations Overview

Palpitations are awareness of a racing, pounding, fluttering, or skipped heartbeat. They may result from normal stress, stimulants, anemia, thyroid disease, medicines, or arrhythmia; fainting, chest pain, breathlessness, or sustained rapid rhythm requires urgent assessment.

PALS

Alternate name; see *Pediatric Advanced Life Support*.

Palsy

Weakness or paralysis, often involving a specific nerve or muscle group.

Palsy Cerebral (Cerebral Palsy)

Alternate index label for Cerebral Palsy.

Palsy Of The Long Thoracic Nerve

Weakness or paralysis of the long thoracic nerve (C5-C7), which innervates the serratus anterior muscle. Causes include trauma, compression, viral infection, or idiopathic. Results in winging of the scapula, particularly when pushing against a wall, and impaired overhead arm elevation. Most cases resolve spontaneously.

Palsy, Bell'S

Alternate index label; see *Bell's palsy*.

Palsy, Cerebral

Alternate index label; see *Cerebral palsy*.

Palsy, Facial

Alternate index label; see *Bell's palsy*.

Pamidronic Acid

An intravenous bisphosphonate that inhibits osteoclast-mediated bone resorption and is used for selected hypercalcemia, bone complications of cancer, and Paget disease.

Pan Troglodyte

Pan troglodytes is the scientific name for the common chimpanzee, a great ape and close evolutionary relative of humans used in behavioral and biomedical research.

Panacea

A supposed remedy for all diseases; it is not a recognized medical treatment.

Panacinar Emphysema

A pathological subtype of pulmonary emphysema characterized by uniform, panlobular destruction and dilation of all air spaces within the acinus, from the respiratory bronchioles through the alveolar ducts down to the terminal alveoli. It demonstrates a marked predilection for the lower lung zones and bases and is classically associated with severe inherited alpha-1 antitrypsin deficiency (PiZZ phenotype), although it can also occur in older smokers.

Alternate Names:

Panlobular Emphysema; Lower Lobe Emphysema

Panax

Panax is a plant genus that includes ginseng species whose roots are used in traditional products; composition, evidence, interactions, and adverse effects vary by preparation.

Pancoast Syndrome

A clinical syndrome caused by an apical lung tumour invading the lower brachial plexus, sympathetic chain, or adjacent structures. It may produce severe shoulder and arm pain, hand weakness, sensory loss, and Horner syndrome.

Pancoast Tumor

Superior sulcus tumor invading brachial plexus, subclavian vessels, and sympathetic chain. Causes Horner syndrome, shoulder/arm pain, and hand muscle atrophy. Treated with chemoradiation followed by surgery.

Pancolitis

Inflammation affecting the entire colon, often as an extensive form of ulcerative colitis.

Pancreas

Retroperitoneal gland (15-20 cm) behind the stomach, spanning from duodenum to spleen. Exocrine function: acinar cells secrete digestive enzymes (amylase, lipase, proteases) and bicarbonate into pancreatic duct to duodenum.

Pancreas Divisum

A congenital anomaly in which the dorsal and ventral pancreatic ducts fail to fuse. Most people are asymptomatic, but impaired drainage through the minor papilla can predispose to recurrent pancreatitis.

Alternate Names:

Divisum Pancreas (Pancreas Divisum)

Pancreas Enzyme

An enzyme produced by the pancreas, such as amylase, lipase, or proteases, that helps digest carbohy-

drate, fat, or protein in the small intestine.

Pancreas Infection

Infection involving the pancreas or a pancreatic collection, usually occurring with pancreatitis, surgery, trauma, or spread from nearby infection.

Pancreas Inflammation

Alternate index label; see *Pancreatitis*.

Pancreas Inflammation (Pancreatitis)

Alternate index label for Pancreatitis.

Pancreas Inflammatory Cysts (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Pancreas Intraductal Papillary Mucinous Neoplasm (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Pancreas IPMN (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Pancreas Mucinous Cyst Adenomas (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Pancreas Neoplasm

A pancreatic neoplasm is an abnormal growth in the pancreas that may be benign, precancerous, or malignant.

Pancreas Non-Inflammatory Cysts (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Pancreas Pseudocysts (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Pancreas Serous Cyst Adenomas (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Pancreas Solid Pseudopapillary Tumor (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Pancreas Transplant

A procedure that replaces a diseased pancreas with a healthy donor pancreas, usually to restore insulin production in selected people with severe diabetes.

Pancreas Transplantation

Alternate index label; see *Pancreas Transplant*.

Pancreas, Diseases Of

Disorders affecting the pancreas, including pancreatitis, cysts, tumors, diabetes, and exocrine insufficiency.

Pancreatectomy

Removal of all (total) or part of the pancreas. Pancreaticoduodenectomy (Whipple) is used for pancreatic head and periampullary tumors; distal pancreatectomy for body and tail lesions.

Pancreatic Abscess

A localized collection of pus within or adjacent to the pancreas, usually developing as a complication of infected pancreatic necrosis or severe pancreatitis. It may cause persistent fever, abdominal pain, sepsis, and organ failure.

Pancreatic Adenocarcinoma

A malignant tumor arising from the exocrine ductal cells of the pancreas. It may cause jaundice, abdominal or back pain, weight loss, or no early symptoms; risk factors include smoking, chronic pancreatitis, diabetes, obesity, and inherited cancer syndromes.

Pancreatic Cancer

Malignancy arising in the pancreas, most often pancreatic ductal adenocarcinoma. Symptoms may include jaundice, abdominal or back pain, weight loss, or digestive problems, although early disease may be asymptomatic. Treatment depends on tumor type, extent, and patient factors.

Alternate Names:

Cancer, Pancreatic

Pancreatic Carcinoma

Pancreatic carcinoma is cancer arising in the pancreas, most often from duct cells, and may cause jaundice, pain, weight loss, or digestive problems.

Pancreatic Cysts

Fluid-filled sacs in or around the pancreas, which may be benign or precancerous.

Alternate Names:

Cyst, Pancreatic Cysts Pancreatic Inflammatory (Pancreatic Cysts) Cysts True Pancreas (Pancreatic Cysts) Intraductal Papillary Mucinous Neoplasm Pancreas (Pancreatic Cysts) IPMN Pancreas (Pancreatic Cysts) Mucinous Cyst Adenomas Pancreas (Pancreatic Cysts) Pancreas Inflammatory Cysts (Pancreatic Cysts) Pancreas Intraductal Papillary Mucinous Neoplasm (Pancreatic Cysts) Pancreas IPMN (Pancreatic Cysts) Pancreas Mucinous Cyst Adenomas (Pancreatic Cysts) Pancreas Non-Inflammatory Cysts (Pancreatic Cysts) Pancreas Pseudocysts (Pancreatic Cysts) Pancreas Serous Cyst Adenomas (Pancreatic Cysts) Pancreas Solid Pseudopapillary Tumor (Pancreatic Cysts) Serous Cyst Adenomas Pancreas (Pancreatic Cysts)

Pancreatic Diseases

Disorders of the exocrine or endocrine pancreas, including pancreatitis, cysts, tumors, duct obstruction, and diabetes-related dysfunction.

Pancreatic Duct

The main duct of the pancreas, conveying exocrine pancreatic secretions to the duodenum. It commonly joins the common bile duct before opening at the major duodenal papilla.

Pancreatic Ductal Adenocarcinoma

The most common exocrine pancreatic malignancy, arising from ductal epithelium and often presenting late with painless jaundice, weight loss, pain, or pancreatitis. Diagnosis and staging rely on imaging, tissue sampling when needed, and multidisciplinary assessment.

Pancreatic Elastase

A stool test that measures elastase-1 as an indirect estimate of pancreatic exocrine function. Low results support pancreatic exocrine insufficiency, but interpretation depends on stool consistency, the assay, and the clinical setting. Replacement pancreatic enzymes do not usually interfere with the test.

Pancreatic Enzyme

Digestive enzymes secreted by pancreatic acinar cells into the duodenum, including proteases (trypsinogen, chymotrypsinogen, proelastase, procarboxypeptidases), amylase (starch digestion), and lipase (triglyceride digestion), along with bicarbonate from ductal cells to neutralize gastric acid.

Pancreatic Exocrine Insufficiency

Inadequate secretion or delivery of pancreatic digestive enzymes and bicarbonate, causing maldigestion and malabsorption. Chronic pancreatitis, cystic fibrosis, pancreatic resection, and advanced pancreatic disease are important causes; steatorrhea and weight loss may respond to pancreatic enzyme replacement.

Pancreatic Fistula

A pancreatic fistula is an abnormal passage through which pancreatic juice leaks from the pancreas, often after trauma or surgery, causing fluid collections, ascites, or skin drainage.

Pancreatic Hormones

Hormones produced by pancreatic islet cells, including insulin, glucagon, somatostatin, and pancreatic polypeptide, which regulate glucose, digestion, and energy metabolism.

Pancreatic Insufficiency

Inadequate production or delivery of pancreatic digestive enzymes, causing impaired digestion and absorption, weight loss, diarrhea, or fatty stools. Treatment may include pancreatic enzyme replacement and nutritional support.

Pancreatic Islet

A pancreatic islet is a cluster of endocrine cells that releases hormones such as insulin, glucagon, and somatostatin to regulate blood glucose and digestion.

Pancreatic Juice

Pancreatic juice is an alkaline digestive secretion containing bicarbonate and enzymes such as amylase, lipase, and proteases, delivered into the duodenum.

Pancreatic Mass

Abnormality in pancreas detected on imaging. Pancreatic adenocarcinoma most common malignancy. Neuroendocrine tumors also important.

Pancreatic Neuroendocrine Tumor

A neoplasm arising from pancreatic neuroendocrine cells that may secrete hormones or be nonfunctioning. Behavior ranges from indolent to aggressive and is assessed by differentiation, grade, proliferative index, and stage.

Pancreatic Neuroendocrine Tumors

Tumors arising from neuroendocrine cells of the pancreas. They may produce hormones, such as insulin or gastrin, or be nonfunctioning. Classification and prognosis use differentiation, grade, proliferation measures, stage, and clinical behavior.

Alternate Names:

Islet Cell Cancer

Pancreatic Necrosis

Non-viable, devitalized areas of pancreatic parenchymal tissue and/or peripancreatic fatty tissue resulting from enzymatic autodigestion and ischemia during severe acute pancreatitis. Identified by non-enhancing areas on contrast-enhanced computed tomography, categorized as sterile or infected necrosis; infected necrosis carries a high risk of sepsis and mortality.

Alternate Names:

Necrotizing pancreatitis; Acute necrotizing pancreatitis

Pancreatic Pseudocyst

A pancreatic pseudocyst is a localized collection of pancreatic fluid surrounded by a wall of granulation tissue without an epithelial lining, typically occurring as a complication of acute or chronic pancreatitis. Unlike true cysts, pseudocysts lack an epithelial lining, which is why they are called pseudocysts.

Pancreatic Surgery

Pancreaticoduodenectomy (Whipple): pancreatic head, duodenum, gallbladder, distal stomach (pylorus-preserving standard), common bile duct; reconstruction: pancreaticojejunostomy (duct-to-mucosa or invagination), hepaticojejunostomy, gastrojejunostomy/duodenojejunostomy.

Pancreatic Trauma

Injury to the pancreas caused by blunt or penetrating trauma. Assessment may require CT, laboratory testing, and observation or intervention according to the injury.

Pancreaticobiliary Cytology

Cytologic evaluation of pancreatic and biliary lesions, usually from fine-needle aspiration, brushings, or bile-duct sampling. It is used to assess neoplasia, inflammation, and obstructive lesions alongside imaging and histology.

Pancreaticoduodenectomy

Pancreaticoduodenectomy, or the Whipple procedure, removes the head of the pancreas and nearby parts of the digestive tract to treat selected tumors and other serious disease.

Pancreatitis

Inflammation of the pancreas that may be acute or chronic. Causes include gallstones, alcohol, medicines, metabolic disorders, and other conditions; severity ranges from mild to necrotizing disease.

Alternate Names:

Inflamed Pancreas Pancreas Inflammation Pancreas Inflammation (Pancreatitis)

Pancreatitis - Children

Pancreatitis in children is inflammation of the pancreas causing abdominal pain, nausea, vomiting, and elevated pancreatic enzymes from structural, genetic, medication, infectious, or other causes.

Pancreatoblastoma

A rare malignant pancreatic epithelial tumour occurring mainly in children. It may present with an abdominal mass, pain, vomiting, weight loss, or hormonal symptoms and requires imaging, tissue diagnosis, staging, and multidisciplinary treatment.

Pancreatography

Contrast imaging of pancreatic duct. ERCP approach. Evaluates pancreatic duct abnormalities.

Pancreazym

An older name for cholecystokinin, a hormone that stimulates pancreatic enzyme secretion and gallbladder contraction.

Pancrelipase

Pancrelipase is a mixture of pancreatic digestive enzymes used as replacement therapy for exocrine pancreatic insufficiency, including that caused by cystic fibrosis or pancreatic disease.

Pancuronium

Pancuronium is a long-acting nondepolarizing neuromuscular blocker that prevents skeletal-muscle contraction during anesthesia or ventilation; it must be given with airway support.

Pancytopenia

Simultaneous reduction of red cells, white cells, and platelets, indicating bone marrow failure or infiltration.

PANDAS

Pediatric autoimmune neuropsychiatric disorder associated with streptococcal infection, a proposed syndrome involving abrupt onset or worsening of obsessive-compulsive symptoms or tics in a child after infection. Diagnosis is clinical and remains con-

troversial; evaluation should also consider common neurologic, psychiatric, and infectious causes.

Pandemic

Epidemic spread across multiple countries/continents. Examples: COVID-19, Spanish flu, HIV/AIDS.

Pandoraea

A genus of gram-negative environmental bacteria that can cause rare opportunistic respiratory or bloodstream infections.

Panendoscopy

Endoscopic examination of several connected parts of the upper aerodigestive tract, often the pharynx, larynx, and oesophagus.

Paneth Cell

Specialized secretory epithelial cell at the base of a small-intestinal crypt that produces antimicrobial substances.

Panhypopituitarism

Deficiency of most or all anterior pituitary hormones due to disease of the pituitary gland or hypothalamus.

Panic

Panic is sudden intense fear or distress with autonomic symptoms such as racing heart, sweating, trembling, breathlessness, or a sense of impending danger.

Panic Attack

An abrupt surge of intense fear or discomfort reaching a peak within minutes.

Panic Attacks And Panic Disorder

Panic attacks are sudden episodes of intense fear; panic disorder involves recurrent attacks and persistent concern about having more.

Panic Disorder

Recurrent unexpected panic attacks, which are

abrupt surges of intense fear or discomfort peaking within minutes.

Panlobular Emphysema

A subtype involving diffuse destruction of the entire acinus. It predominantly affects the lower lobes and is classically associated with alpha-1 antitrypsin deficiency. The destruction is more uniform compared to centriacinar emphysema.

Panmyelopathy

Disease affecting multiple cell lines or functions of the bone marrow and spinal cord; the term is uncommon and context-dependent.

Panniculectomy

Surgical removal of a hanging apron of excess lower-abdominal skin and fat, usually after substantial weight loss; it is distinct from a full abdominoplasty.

Panniculitis

Inflammation of the subcutaneous fat, presenting as tender, erythematous or violaceous nodules and plaques, most commonly on the lower extremities. The classification is based on whether the septa or lobules of the fat are primarily affected, and the aetiology includes infections, autoimmune diseases, medications, and pancreatic disease.

Panniculitis Idiopathic Nodular (Weber-Christian Disease)

Alternate index label for Weber-Christian Disease.

Panniculus

A panniculus is a fold or layer of subcutaneous fatty tissue, especially the lower abdominal apron that may overhang the pubic region.

Pannus

An abnormal layer of vascularized tissue that grows over a joint or the cornea.

Panophthalmitis

Severe inflammation involving all coats and internal contents of the eye, usually due to infection and

threatening vision.

Panoramic Radiograph

An extraoral radiograph displaying both jaws, teeth, temporomandibular regions, and surrounding structures in one image. It is useful for surveying dentition, development, impaction, and gross osseous pathology but has less fine detail than intraoral radiographs.

Panretinal Photocoagulation

A laser treatment that applies scattered burns to the peripheral retina to reduce abnormal blood-vessel growth. It is used for selected proliferative retinal diseases, including proliferative diabetic retinopathy, and can affect peripheral or night vision.

Pantothenic Acid

Vitamin B5, a component of coenzyme A required for fatty-acid and energy metabolism.

Pantothenic Acid And Biotin

Two B vitamins involved in energy metabolism, fatty-acid synthesis, and other enzyme reactions; deficiency is uncommon but can occur with severe malnutrition, malabsorption, or specific inherited disorders.

Panuveitis

Inflammation involving the anterior uvea, vitreous, and posterior uveal tissues. It can threaten vision and may be caused by infection, autoimmune disease, trauma, or other disorders; treatment depends on the cause and severity.

Pap Smear

Papanicolaou test (cervical cytology): a screening test collecting cervical cells (conventional or liquid-based) to detect precancerous cervical changes (CIN) and cancer, usually combined with HPV DNA testing (co-testing).

Pap Test

A cervical cytology test in which cells collected from the cervix are examined for precancerous changes,

cancer, or selected infections.

Papain

A proteolytic enzyme from papaya latex used in some food and historical medicinal preparations.

Papanicolaou (Pap) Test

A screening test in which cervical cells are examined for precancerous changes and cancer.

Papillary Fibroelastoma

A small benign cardiac valvular tumor composed of avascular frond-like elastic tissue. It may be incidental but can cause embolic events, particularly when mobile or located on the left-sided valves.

Papillary Muscle

Muscular projections of the ventricular myocardium attached by chordae tendineae to the atrioventricular valve leaflets (anterolateral and posteromedial in the left ventricle; anterior, septal, and posterior in the right ventricle).

Papillary Thyroid Carcinoma

Most common histological subtype of thyroid carcinoma accounting for approximately eighty percent of cases. Characterized by nuclear features including nuclear grooves, intranuclear cytoplasmic inclusions, and Orphan Annie eye nuclei. Often multifocal and bilateral with lymph node metastases.

Papillectomy

Surgical removal of a papilla or papillary lesion, such as an ampullary tumor.

Papilledema

Swelling of the optic disc caused by elevated intracranial pressure, and it is a critical clinical sign that requires urgent investigation to identify the underlying cause, which may be life-threatening.

Papilloedema

Swelling of the optic discs caused by raised intracranial pressure.

Papilloma

A benign epithelial neoplasm characterised by finger-like projections of fibrovascular tissue covered by hyperplastic epithelium, most commonly caused by human papillomavirus infection in cutaneous warts or mucosal sites. Cutaneous papillomas present as flesh-coloured, rough, or pedunculated growths.

Papillomatosis Cutis Lymphostatica

Diffuse, warty, papillomatous thickening of the skin resulting from chronic lymphoedema, most commonly affecting the lower extremities in the setting of long-standing venous or lymphatic insufficiency. The skin develops cobblestone-like hyperkeratotic papules and plaques due to lymph stasis and chronic inflammation.

Papillomavirus

A papillomavirus, especially human papillomavirus, which can cause warts and some anogenital, oropharyngeal, and other cancers.

Papular

A papule: a small, solid, elevated skin lesion <1 cm in diameter (palpable), of various colors and etiologies.

Papular Urticaria

A hypersensitivity reaction to insect bites, especially from fleas, bedbugs, or mosquitoes, causing recurrent intensely itchy papules or small vesicles, often clustered on exposed areas in children. Diagnosis is clinical.

Papule

A small, solid, raised skin lesion, usually less than 1 centimeter in diameter.

PAPVR

Alternate index label; see *Partial anomalous pulmonary venous return*.

Para-Aminobenzoic Acid

A folate-related compound used by bacteria in folate synthesis and formerly used in some medicines;

humans do not require it as an essential vitamin.

Paracentesis

Percutaneous needle aspiration of ascitic fluid from the peritoneal cavity. Used for diagnostic analysis to determine the cause of ascites and for therapeutic removal of large volumes for symptom relief. The serum-ascites albumin gradient is calculated to differentiate portal hypertensive from non-portal hypertensive causes.

Paracetamol Overdose

Toxic ingestion of paracetamol that can cause dose-dependent hepatic necrosis; risk assessment uses the timing and serum concentration of the ingestion.

Paracetamol Toxicity

Liver injury caused by excessive acetaminophen (paracetamol) exposure. Severe poisoning may cause nausea, abdominal pain, liver failure, or death and requires urgent assessment.

Paracusia

A disorder of hearing in which sounds are perceived abnormally or inaccurately.

Paradichlorobenzene Poisoning

Paradichlorobenzene poisoning involves toxic or environmental exposure to paradichlorobenzene poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Paraesophageal Hernia

A hiatal hernia in which part of the stomach moves through the diaphragm beside the esophagus. It may cause obstruction or strangulation even when reflux is absent.

Paraesthesia

An abnormal sensation such as tingling, pins and needles, burning, or numbness.

Paraffin

A mixture of hydrocarbons used in laboratory tissue processing, topical preparations, and other applica-

tions; some forms are flammable.

Paraffin Poisoning

Paraffin poisoning involves toxic or environmental exposure to paraffin poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Paraganglioma

A neuroendocrine tumor arising from extra-adrenal paraganglia, related to pheochromocytoma; may secrete catecholamines. Associations with hereditary syndromes (SDH mutations, VHL, NF1) warrant genetic counseling.

Parageusia (Dysgeusia)

A qualitative sensory distortion or alteration in the perception of taste, in which normal tastants evoke an unpleasant, metallic, bitter, salty, or rancid flavor. Etiologies include pharmacologic adverse effects (common with metronidazole, clarithromycin, ACE inhibitors, chemotherapy, and terbinafine), nutritional deficiencies (severe zinc or vitamin B12 deficiency), head trauma injuring the chorda tympani or glossopharyngeal nerve, upper respiratory viral infections (including SARS-CoV-2), and chronic periodontal or salivary gland disease.

Paragonimiasis

A parasitic infection caused by *Paragonimus* lung flukes, usually acquired by eating undercooked crustaceans. It commonly affects the lungs.

Paragraphia

A language disorder involving incorrect or unintended words or written symbols.

Parainfectious

Parainfectious describes a condition occurring during or around an infection, often from an immune response rather than direct invasion by the infectious organism.

Parainfluenza

Parainfluenza viruses are respiratory viruses that

commonly cause colds, croup, bronchitis, or pneumonia, especially in children; illness ranges from mild to severe airway disease.

Paraldehyde

An older sedative and anticonvulsant medicine with limited current use.

Paralysis

Loss of voluntary muscle movement in part or all of the body due to nervous-system or muscle disease.

Paralysis Agitans

An older name for Parkinson disease, characterized by tremor, rigidity, and bradykinesia.

Paralysis, Stomach

Stomach paralysis, or gastroparesis, is delayed gastric emptying without a mechanical blockage, causing early fullness, nausea, vomiting, bloating, and variable glucose control.

Paralytic Ileus

Failure of intestinal peristalsis without a mechanical blockage, causing abdominal distension, pain, vomiting, and absent or reduced bowel sounds.

Paramedical

Relating to healthcare professionals or services that support medical practice, such as physiotherapy, radiography, or emergency medical care.

Parametritis

Inflammation of the connective tissue beside the uterus, often following pelvic infection or childbirth.

Paramnesia

A memory disturbance involving false, distorted, or inaccurately familiar recollections.

Paramyotonia Congenita

An autosomal dominant skeletal muscle channelopathy caused by gain-of-function missense mutations in the *SCN4A* gene, which encodes the α -subunit of the skeletal muscle voltage-gated sodium channel

Nav1.4. Characterized by “paradoxical myotonia,” in which muscle stiffness and impaired relaxation are dramatically worsened rather than alleviated by repeated voluntary contractions (contrasting with the warm-up phenomenon of myotonia congenita) and exquisitely aggravated by exposure to cold environments. Severe cold-induced stiffness is typically followed by hours of flaccid muscle weakness and paralysis, predominantly affecting the muscles of the face, eyelids, neck, and hands.

Alternate Names:

Eulenburg Disease; Paramyotonia Congenita of Eulenburg

Paranasal Sinuses

Air-filled cavities in the bones around the nose that connect with the nasal cavity through small openings. They are lined with mucosa and contribute to mucus production and resonance of the voice.

Paraneoplastic Disorder

A systemic syndrome caused by a malignancy through mechanisms other than direct invasion, metastasis, or metabolic effects of the tumor. Paraneoplastic syndromes result from hormone production, immune-mediated mechanisms, or cytokine release by the tumor, affecting distant organs.

Paraneoplastic Neurologic Syndrome

Neurologic symptoms from immune response to cancer. Examples: Lambert-Eaton, cerebellar degeneration.

Paraneoplastic Neurological Disorder

Alternate index label; see *Paraneoplastic syndromes of the nervous system*.

Paraneoplastic Neurological Syndrome

Alternate index label; see *Paraneoplastic syndromes of the nervous system*.

Paraneoplastic Syndrome

Symptoms caused by cancer but not by tumor mass or metastasis. Mechanisms: hormone production, immune cross-reaction. Examples: SIADH, Cushing syndrome, Lambert-Eaton myasthenia, hyper-

calcemia.

Paraneoplastic Syndromes Of The Nervous System

Neurologic disorders caused by an immune response to cancer rather than by direct tumor invasion or metastasis.

Alternate Names:

Cancer, Paraneoplastic Syndromes Paraneoplastic Neurological Disorder Paraneoplastic Neurological Syndrome

Paranoia

A thought process characterized by excessive or irrational suspiciousness and distrust of others, often manifesting as delusions of persecution, reference, or grandeur.

Paranoid

Characterized by persistent or excessive suspicion and distrust, often involving beliefs that others intend harm. Severe or fixed beliefs may occur in a psychotic disorder and need professional assessment.

Paranoid Personality Disorder

A longstanding pattern of distrust and suspiciousness in which other people’s motives are interpreted as threatening or deceitful.

Paranoid Psychosis

An older term for psychosis dominated by delusions of persecution, reference, or grandeur.

Paranoid Schizophrenia

An obsolete subtype label for schizophrenia with prominent delusions and hallucinations; current classification uses symptom dimensions.

Paraparesis

Weakness affecting both legs, usually due to spinal-cord, brain, nerve, or muscle disease.

Paraphasia

Substitution, omission, or distortion of sounds or words during speech, usually due to aphasia.

Paraphilia

A condition characterized by intense and persistent sexual interest in atypical objects, situations, or individuals. Examples: exhibitionism, fetishism, frotteurism, pedophilic disorder, sexual masochism, sexual sadism, transvestic disorder, voyeurism, and paraphilia not otherwise specified.

Paraphimosis

Paraphimosis occurs when a retracted foreskin cannot be returned over the glans, causing painful swelling and possible loss of blood flow; it requires urgent reduction.

Paraplegia

Paralysis of both legs and often part of the trunk, usually from spinal-cord or neurologic disease.

Parapneumonic Effusion

Pleural effusion complicating pneumonia. Simple effusions resolve with antibiotics. Complicated effusions (pH < 7.2, glucose < 60 mg/dL) require chest tube drainage.

Parapneumonic Pleural Effusion

Pleural fluid accumulation associated with bacterial pneumonia. It may be uncomplicated or progress to a complicated effusion or empyema, requiring imaging, pleural-fluid analysis, antibiotics, and sometimes drainage.

Paraquat

A highly toxic herbicide. Ingestion can cause injury to the mouth and gastrointestinal tract, kidney failure, and delayed lung injury; severe poisoning has a high risk of death.

Paraquat Poisoning

Paraquat poisoning involves toxic or environmental exposure to paraquat poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Paraseptal Emphysema

A form of emphysema characterized by destruction

of alveoli adjacent to the pleura. It typically affects young adults and predisposes to spontaneous pneumothorax due to rupture of subpleural blebs. It may be associated with Marfan syndrome.

Parasite

An organism that lives on or within a host organism, deriving nutrients at the host's expense.

Parasitemia

The presence and measurable burden of parasites in the blood. It is used particularly for malaria and some blood-borne protozoal infections and may correlate with disease severity.

Parasitic

Parasitic means relating to an organism that lives in or on a host and obtains nutrients at the host's expense, sometimes causing disease.

Parasitic Infections

Malaria: *Plasmodium* (*falciparum*, *vivax*, *ovale*, *malariae*, *knowlesi*). *Falciparum*: cerebral malaria, severe anemia, ARDS, AKI, hypoglycemia. Diagnosis: thick/thin smear, RDT (HRP2 for *P. falciparum*, pLDH for all species, aldolase). Amoebiasis: *E.*

Parasitic Worm

A parasitic worm is a helminth that lives in a host; human infection may involve the intestine, blood, lymphatics, lungs, liver, or tissues.

Parasitology

The branch of microbiology that studies parasites, including protozoa, helminths, and ectoparasites. Parasitic infections are a major global health problem, particularly in tropical and subtropical regions. Protozoan parasites include *Plasmodium* (malaria), *Trypanosoma* (sleeping sickness, Chagas disease), and *Leishmania* (leishmaniasis).

Parasitosis

Infestation or infection by a parasite; the term may describe a confirmed parasitic disease or, in delusional parasitosis, a fixed false belief of infestation

without evidence.

Parasomnia

A parasomnia is an undesirable movement, behavior, emotion, or experience occurring during sleep or transitions between sleep and wakefulness.

Parasomnias

Unwanted movements, behaviors, emotions, or experiences that occur during sleep or at the transition between sleep and wakefulness.

Parasuicide

Parasuicide is an older term for a suicide attempt or intentional self-harm without fatal outcome; current assessment focuses on self-harm, intent, safety, and urgent support.

Parasympathetic Nervous System

The craniosacral division of the autonomic nervous system mediating 'rest-and-digest' functions. Pre-ganglionic fibers from cranial nerves III, VII, IX, X, and sacral nerves S2-S4 synapse on ganglia near or within target organs (short postganglionic fibers).

Parathyroid

One of the small endocrine glands behind the thyroid that secretes parathyroid hormone and regulates calcium and phosphate balance.

Parathyroid Adenoma

A parathyroid adenoma is a usually benign tumor of a parathyroid gland that can secrete excess parathyroid hormone and cause primary hyperparathyroidism with high calcium.

Parathyroid Cancer

Parathyroid cancer is a rare malignant tumor of the parathyroid gland that often causes excessive parathyroid hormone and high blood calcium.

Parathyroid Carcinoma

A rare malignant tumor of the parathyroid gland that commonly causes excessive parathyroid-hormone secretion, hypercalcemia, kidney stones,

bone disease, or related symptoms.

Parathyroid Crisis

Severe hypercalcemia with calcium greater than fourteen milligrams per deciliter causing altered mental status, cardiac arrhythmias, and acute kidney injury. Most commonly occurs in patients with primary hyperparathyroidism precipitated by dehydration, immobility, or thiazide diuretics.

Parathyroid Gland

Parathyroid gland refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Parathyroid Gland Removal

A medical procedure, treatment, device, or care topic related to parathyroid gland removal; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Parathyroid Glands

Usually four small endocrine glands behind the thyroid that secrete parathyroid hormone to regulate blood calcium and phosphate levels.

Parathyroid Hormone

A hormone that regulates calcium and phosphate levels in the blood and supports bone remodeling.

Parathyroid Hormone Blood Test

A blood test measuring parathyroid hormone, interpreted with calcium, phosphate, vitamin D, and kidney function to evaluate parathyroid and calcium disorders.

Parathyroid Hormone Physiology

Parathyroid hormone is the primary regulator of calcium homeostasis secreted by the four parathyroid glands in response to low ionized calcium levels detected by the calcium-sensing receptor.

Parathyroid Hormone-Related Protein Blood Test

A blood test measuring parathyroid hormone-

related protein, used mainly to evaluate humoral hypercalcemia of malignancy and selected disorders of calcium regulation.

Parathyroid Hyperplasia

Enlargement and increased cellularity of one or more parathyroid glands causing excessive parathyroid hormone secretion and primary hyperparathyroidism. It may occur sporadically or as part of multiple endocrine neoplasia.

Parathyroid Surgery

Operations on the parathyroid glands, usually performed for selected cases of primary or secondary hyperparathyroidism. The decision depends on symptoms, calcium levels, kidney involvement, bone disease, overall health, and current clinical guidance.

Parathyroidectomy

Surgical removal of parathyroid glands for refractory secondary hyperparathyroidism unresponsive to medical therapy. Indicated when calcium-phosphorus product persistently elevated despite maximal medical management, symptomatic bone disease, or calciphylaxis. Subtotal or total parathyroidectomy with autotransplantation.

Paratubal Cyst

A usually benign fluid-filled cyst adjacent to the fallopian tube, arising from paratubal or embryologic remnants; most are asymptomatic but large cysts may cause pain or torsion.

Paratyphoid Fever

An enteric fever caused by *Salmonella Paratyphi*, transmitted mainly through contaminated food or water.

Parenchyma

The functional tissue of an organ, distinguished from its supporting connective tissue or stroma.

Parenchymal

Parenchymal means relating to the functional cells of an organ, distinguished from its supporting con-

nective tissue, vessels, and framework.

Parenchymal Diffuse Lung Disease (Interstitial Lung Disease (Interstitial Pneumonia))

Alternate index label for Interstitial Lung Disease (Interstitial Pneumonia).

Parentectomy

Parentectomy is surgical removal of a parent or source structure, such as the parent vessel or lesion, and its precise meaning depends on the procedure.

Parenteral

Parenteral means delivered by a route other than the gastrointestinal tract, commonly by injection or infusion such as intravenous, intramuscular, or subcutaneous administration.

Parenteral Nutrition

Intravenous nutrition bypassing GI tract. Total parenteral nutrition provides complete nutrition. Indications: bowel failure, short bowel syndrome, prolonged NPO, malabsorption. Risks: infection, liver dysfunction, metabolic complications, line complications.

Paresis

A neurological state of partial motor weakness, reduced voluntary muscle strength, or incomplete paralysis, distinguished from plegia (which denotes complete, total loss of voluntary muscle motor function). Classified according to anatomical distribution: monoparesis (single limb), hemiparesis (unilateral arm and leg, classic for contralateral cortical or internal capsule ischemic stroke), paraparesis (both lower extremities, seen in thoracic spinal cord lesions or cauda equina syndrome), and tetraparesis/quadruparesis (all four limbs, resulting from cervical spinal cord compromise or generalized polyneuropathies like Guillain-Barré syndrome).

Paresthesia

An abnormal sensation such as tingling, prickling, burning, or pins and needles.

Pargyline

Pargyline is an irreversible monoamine-oxidase inhibitor formerly used for hypertension; it is now mainly of historical or research interest because of interactions and adverse effects.

Paricalcitol

Paricalcitol is a vitamin-D receptor activator used to reduce secondary hyperparathyroidism in selected chronic kidney disease, with monitoring for calcium and phosphate changes.

Parietal

Relating to a wall of a body cavity or to the parietal bone or lobe of the skull and brain.

Parietal Bone

One of two bones forming much of the sides and roof of the skull.

Parietal Lobe

The cerebral lobe posterior to the central sulcus and above the lateral sulcus, containing the primary somatosensory cortex (postcentral gyrus, Brodmann 3,1,2) processing touch, pain, temperature, and proprioception, plus the somatosensory association cortex.

Parietal Pericardium

The outer serous layer of the pericardium that lines the inner surface of the fibrous sac surrounding the heart.

Parietal Pleura

The pleural layer lining the chest wall, diaphragm, and mediastinum; it is sensitive to pain when inflamed.

Parinaric Acid

A conjugated polyunsaturated fatty acid used as a fluorescent probe of lipid membranes and studied in oxidation and lipid-chemistry research.

Parinaud Oculoglandular Syndrome

Conjunctivitis with nearby lymph-node swelling,

usually caused by infections such as Bartonella, tularemia, or certain viruses.

Parinaud Syndrome

A disorder caused by dysfunction of the dorsal midbrain, characterized by impaired upward gaze, abnormal pupil responses, and convergence-retraction eye movements. Causes include tumors, stroke, multiple sclerosis, and hydrocephalus.

Paris System For Reporting Urinary Cytology

A standardized framework for urinary cytology that emphasizes detection of high-grade urothelial carcinoma and uses defined categories to improve diagnostic consistency and clinical communication.

Parity

Parity is the number of pregnancies that reached a viable gestational age, regardless of whether the infants were born alive or stillborn; it differs from gravidity.

Park-Bench Position

A modified lateral position in which the upper body is partly rotated toward prone; it is used for selected posterior brain and neck procedures.

Parkinson Disease

A progressive neurodegenerative disorder characterized by the loss of dopaminergic neurons in the substantia nigra pars compacta of the midbrain, leading to reduced dopamine levels in the striatum and the cardinal motor features of bradykinesia, resting tremor, rigidity, and postural instability.

Parkinson'S Disease

A progressive neurodegenerative disorder caused by selective loss of dopaminergic neurons in the substantia nigra pars compacta, leading to dopamine depletion in the striatum. The pathological hallmark is the presence of Lewy bodies, intracellular inclusions composed of aggregated alpha-synuclein protein, within surviving neurons.

Alternate Names:

Parkinsons Disease (Parkinson's Disease)

Parkinsonism

A clinical syndrome characterized by bradykinesia, rigidity, tremor (resting, pill-rolling), and postural instability. Idiopathic Parkinson disease: most common cause, responds well to levodopa.

Parkinsons Disease (Parkinson'S Disease)

Alternate index label for Parkinson's Disease.

Paromomycin

Paromomycin is an aminoglycoside antibiotic used mainly in the intestine for selected parasitic infections and sometimes for other infections under specialist care.

Paronychia

Paronychia is infection or inflammation of tissue around a fingernail or toenail, producing pain, redness, swelling, and sometimes a localized abscess. (*Streptococcus*), presenting as painful, erythematous, swollen nail fold with possible abscess. Treatment includes warm compresses, antibiotics, and incision and drainage if abscessed.

Parosteal Osteosarcoma

Parosteal osteosarcoma is a usually lower-grade bone cancer that grows on the outer surface of a bone, often near the back of the thigh.

Parotid Gland

The largest of the three major salivary glands, located in the preauricular region, extending from the zygomatic arch superiorly to the angle of the mandible inferiorly. The parotid gland is encased in the parotid fascia (investing layer of deep cervical fascia).

Parotid Neoplasm

A parotid neoplasm is an abnormal growth in the parotid salivary gland and may be benign or malignant.

Parotid Tumors

Abnormal growths arising in the parotid salivary glands, located in front of and below the ears.

Parotidectomy

Parotidectomy is surgical removal of part or all of a parotid salivary gland, usually to treat a tumor.

Parotitis

Inflammation or infection of one or both parotid salivary glands, the largest of the major salivary glands located anterior and inferior to the external ear. Etiologies are categorized into acute viral parotitis (classically mumps caused by paramyxovirus, presenting with bilateral painful glandular swelling), acute suppurative bacterial parotitis (predominantly caused by *Staphylococcus aureus* in dehydrated, elderly, or postoperative patients with poor oral hygiene and ductal stasis), and chronic autoimmune parotitis (prominent in Sjögren syndrome and sarcoidosis). Bacterial parotitis manifests with exquisite preauricular erythema, induration, fever, and expression of gross pus from the orifice of Stensen duct.

Parotitis (Mumps)

Alternate index label for Mumps.

Paroxetine Hydrochloride

Paroxetine hydrochloride is an SSRI antidepressant used for depression and several anxiety-related disorders; it can cause sexual side effects and withdrawal symptoms if stopped abruptly.

Paroxysm

A sudden attack or burst of symptoms or physiologic activity.

Paroxysmal Atrial Tachycardia

Sudden episodes of rapid, regular atrial rhythm that start and stop abruptly, often producing palpitations, lightheadedness, chest discomfort, or shortness of breath.

Paroxysmal Cold Hemoglobinuria

An autoimmune hemolytic anemia caused by a biphasic Donath–Landsteiner antibody that binds red cells in the cold and triggers complement-mediated destruction on rewarming.

Paroxysmal Hemicrania

A rare headache disorder causing brief, repeated attacks of severe one-sided head or facial pain.

Paroxysmal Nocturnal Dyspnea

Episodes of breathlessness that awaken a person from sleep after a period of lying down. It can occur with heart failure or other cardiopulmonary conditions and differs from orthopnea, which begins soon after lying down.

Paroxysmal Nocturnal Haemoglobinuria

An acquired clonal stem-cell disorder causing complement-mediated hemolysis, thrombosis, and variable bone-marrow failure.

Paroxysmal Nocturnal Hemoglobinuria

A rare, acquired clonal disorder of hematopoietic stem cells characterized by complement-mediated intravascular hemolysis, thrombosis (particularly in unusual sites such as hepatic, portal, and cerebral veins), and bone marrow failure.

Paroxysmal Nocturnal Hemoglobinuria (Paroxysmal Nocturnal Hemoglobinuria PNH)

Alternate index label for Paroxysmal Nocturnal Hemoglobinuria PNH.

Paroxysmal Nocturnal Hemoglobinuria

An acquired clonal stem-cell disorder caused by deficient GPI-anchored proteins, leading to complement-mediated hemolysis, thrombosis, and bone-marrow failure.

Paroxysmal Supraventricular Tachycardia

Sudden-onset and sudden-termination episodes of rapid regular tachycardia arising above the ventricles, usually involving an atrioventricular re-entry or nodal re-entry circuit. Palpitations, dizziness, chest discomfort, or syncope may occur.

Paroxysmal Supraventricular Tachycardia

Paroxysmal supraventricular tachycardia is an abrupt-onset and abrupt-ending rapid rhythm arising

above the ventricles, causing palpitations, dizziness, chest discomfort, or syncope.

Alternate Names:

PSVT (Paroxysmal Supraventricular Tachycardia (PSVT)) Supraventricular Tachycardia Paroxysmal (Paroxysmal Supraventricular Tachycardia (PSVT)) Tachycardia Paroxysmal Atrial (Paroxysmal Supraventricular Tachycardia (PSVT)) Tachycardia Paroxysmal Supraventricular (Paroxysmal Supraventricular Tachycardia (PSVT))

Paroxysmal Supraventricular Tachycardia

Alternate index label for Paroxysmal Supraventricular Tachycardia.

Pars

A part or portion of an anatomical structure, used in names such as pars interarticularis.

Pars Compacta

The substantia nigra pars compacta is a midbrain region containing dopamine-producing neurons whose loss contributes to the motor features of Parkinson disease.

Pars Planitis

A form of intermediate uveitis involving inflammatory exudates near the pars plana, causing floaters, blurred vision, and sometimes macular edema or retinal complications.

Parthenogenesis

Development of an embryo from an unfertilized egg; it occurs naturally in some animals and plants and is not a normal mechanism of human reproduction.

Partial

Partial means incomplete or involving only a portion, such as partial loss, partial response, partial thickness, or partial seizure; the specific meaning depends on context.

Partial Agonist

A partial agonist is a drug that binds to a receptor and produces a biological response that is less

than the maximum possible response, even when all receptors are occupied. Partial agonists have affinity for the receptor but lower intrinsic activity compared to full agonists, producing a submaximal response regardless of dose.

Partial Anomalous Pulmonary Venous Connection

Alternate index label; see *Partial anomalous pulmonary venous return*.

Partial Breast Brachytherapy

A medical procedure, treatment, device, or care topic related to partial breast brachytherapy; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Partial Breast Radiation Therapy - External Beam

A medical procedure, treatment, device, or care topic related to partial breast radiation therapy - external beam; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Partial Cystectomy

Partial cystectomy removes part of the urinary bladder, usually to treat a selected bladder tumor while preserving the remaining bladder.

Partial Denture

Removable prosthesis replacing some teeth. Metal framework or acrylic. Requires remaining teeth for support.

Partial Hysterectomy

A common but imprecise term usually referring to supracervical or subtotal hysterectomy, in which the uterine body is removed while the cervix remains. The fallopian tubes and ovaries may or may not be removed separately.

Partial Knee Replacement

A medical procedure, treatment, device, or care topic related to partial knee replacement; indica-

tions, benefits, risks, and follow-up depend on the specific intervention and patient.

Partial Nephrectomy

Partial nephrectomy removes a diseased part of a kidney while preserving the remaining healthy tissue.

Partial Remission

A reduction in signs, symptoms, or measurable disease burden that does not meet criteria for complete remission; the definition depends on the disease and response criteria used.

Partial Response

In clinical oncology and therapeutic trial evaluation under the standardized Response Evaluation Criteria in Solid Tumors (RECIST v1.1), a documented reduction of at least 30% in the sum of the longest diameters of all measured target tumor lesions, compared to baseline, in the absence of unequivocal progression in non-target lesions or the emergence of new metastatic lesions. In hematologic malignancies, partial response is defined by disease-specific reduction thresholds in circulating malignant clone burden, marrow blast percentage, or paraprotein levels. Serves as an established surrogate marker of anti-tumor biological activity and therapeutic response.

Partial Seizure

A seizure beginning in a localized region of one cerebral hemisphere, now called a focal seizure, with symptoms determined by the involved network and awareness either preserved or impaired.

Partial Thromboplastin Time

A coagulation test measuring the intrinsic and common pathways (factors XII, XI, IX, VIII, X, V, II, fibrinogen), used to monitor unfractionated heparin and detect factor deficiencies.

Partial Thromboplastin Time

The partial thromboplastin time measures the intrinsic and common coagulation pathways and helps monitor unfractionated heparin or investigate bleed-

ing and clotting abnormalities.

Partial Volume Artifact

CT artifact from voxel containing multiple tissues. Reduced by thinner slices.

Partial-Volume Effect

Underestimation or distortion of measured activity in small or low-contrast structures because the object size approaches the scanner's spatial resolution. It affects quantitative PET and SPECT and may require recovery or correction methods.

Participation

A person's involvement in life situations and activities, a central concept in rehabilitation and the International Classification of Functioning that extends beyond impairment alone.

Particle (Vascular)

An embolic particle used to occlude selected vessels under image guidance.

Alternate Names:

Vascular

Particle Size

Particle size is the diameter or size distribution of particles in a powder, suspension, aerosol, or formulation, influencing dissolution, absorption, stability, and delivery.

Particulate Matter

Small solid or liquid particles suspended in air. Fine particles can penetrate the lungs and increase cardiovascular and respiratory risk.

Partograph (Partogram)

A graphic record of cervical change, fetal descent, contractions, and maternal or fetal observations during labor. It can support communication and recognition of abnormal progress, but findings must be interpreted with current labor guidance and the overall clinical situation.

Alternate Names:

Partogram

Parturition

The process of childbirth, including cervical ripening, labor contractions, delivery of the fetus, and expulsion of the placenta.

Parturition (Childbirth)

The process of giving birth, encompassing three stages: (1) First stage: from the onset of regular uterine contractions (cervical effacement and dilation) to full dilation (10 cm). Latent phase—slow dilation (0–4 cm) over hours, variable duration. Active phase—more rapid dilation (4–10 cm), typically 1–1.5 cm/hour in nulliparous women.

Alternate Names:

Childbirth

Parvimonas Micra

An anaerobic gram-positive bacterium found in the mouth and gastrointestinal tract that can cause dental, bone, joint, or bloodstream infections.

Parvoviridae

A family of very small DNA viruses; human parvovirus B19 can cause fifth disease, anemia, or fetal complications.

Parvovirus

A small DNA virus; parvovirus B19 is the main human pathogen and can cause rash, joint symptoms, or red-cell aplasia.

Parvovirus B19

Human parvovirus B19: a small DNA virus causing erythema infectiosum ('fifth disease') in children, transmitted by respiratory droplets.

Parvovirus B19 Infection

A viral infection that may cause slapped-cheek rash, joint symptoms, or temporary suppression of red-cell production.

Parvovirus Infection

An infection caused by parvovirus B19 that commonly causes mild fever and rash but may be serious in some people.

Alternate Names:
Slapped Cheek Disease

Passive Exercise

A medical procedure, treatment, device, or care topic related to passive exercise; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Passive Immunity

Protection produced by transferred antibodies rather than by the person's own immune response, such as maternal antibodies or immunoglobulin treatment.

Passive Immunization

Protection produced by giving preformed antibodies, such as immunoglobulin or antivenom, which acts quickly but is temporary because the recipient does not generate durable immune memory.

Paste

A paste is a semisolid preparation or material containing finely divided solids in a liquid or base, used in medicines, dentistry, skin care, and laboratory work.

Pasteurella

A genus of bacteria carried by animals that can cause wound infections after bites and other human disease.

Pasteurella Aerogenes

Pasteurella aerogenes is a rare *Pasteurella* species; its recovery from a clinical sample should be interpreted with the source and evidence of infection.

Pasteurella Bettyae

Pasteurella bettyae is a rare bacterium associated with animals and occasional human wound or soft-tissue infections.

Pasteurella Canis

Pasteurella canis is a bacterium found in dogs and cats that can cause wound or soft-tissue infection

after an animal bite.

Pasteurella Multocida

A bacterium commonly carried by animals that can cause wound infections after bites or scratches and, less often, respiratory or bloodstream infection.

Pasteurella Pneumotropica

Pasteurella pneumotropica is mainly an animal-associated bacterium that rarely causes opportunistic human infection.

Pasteurellaceae

Pasteurellaceae is a family of bacteria that includes *Pasteurella* and related animal-associated organisms capable of causing bite-wound and respiratory infections.

Pasteurellosis

Infection caused by *Pasteurella* bacteria, commonly introduced through animal bites or scratches and causing cellulitis, abscesses, or occasionally invasive disease.

Pasteurization

A heat treatment reducing microbial load in liquids by heating below boiling point to inactivate vegetative pathogens and spoilage organisms while preserving quality.

Patau Syndrome

Trisomy 13, affecting 1 in 10,000-20,000 live births. Features: holoprosencephaly (most characteristic, incomplete forebrain division), microcephaly, microphthalmia or anophthalmia, cleft lip and palate, polydactyly (postaxial), clenched fists with overlapping fingers, and congenital heart defects (VSD, PDA, dextrocardia, 80%).

Patch

A flat, circumscribed area of altered skin color larger than approximately 1 cm. It is the larger counterpart of a macule and may be caused by pigmentary, vascular, inflammatory, or neoplastic disease.

Patch (Vascular)

A piece of biological or synthetic material sewn onto a blood vessel to widen or repair it.

Alternate Names:

Vascular

Patch Graft

A piece of natural or synthetic material sewn onto a blood vessel to widen or repair it during patch angioplasty. Materials may include a patient's tissue, bovine pericardium, or synthetic PTFE.

Patch Test

A skin test used to identify delayed allergic contact reactions to substances applied under adhesive patches.

Patch Testing

Patch testing identifies allergens causing allergic contact dermatitis. The standard series includes 35 allergens. Additional custom trays include occupational and personal allergens. Patches are applied to the upper back for 48 hours and removed. Reading at 48 hours (initial) and 72 to 96 hours (delayed) identifies positive reactions.

Patches

Patches are flat, distinct areas of altered skin color larger than a macule; their appearance, distribution, and associated symptoms help identify the cause.

Patchy Hair Loss (Alopecia Areata)

Alternate index label for Alopecia Areata.

Patchy Skin Color

Patchy skin color concerns patchy skin color, a body structure, substance, or clinical concept whose meaning depends on location, function, and patient context.

Patella

The largest sesamoid bone, embedded in the quadriceps tendon. It articulates with the femoral trochlea to form the patellofemoral joint and improves the mechanical efficiency of knee extension. It is trian-

gular and covered by thick articular cartilage.

Patella (Kneecap)

The largest sesamoid bone in the human body, embedded within the quadriceps tendon at the front of the knee. It improves the mechanical efficiency of knee extension, protects the front of the joint, and distributes forces across the femoral trochlea.

Alternate Names:

Kneecap

Patellar Dislocation

Complete displacement of the kneecap from the femoral groove, usually laterally, causing acute pain, deformity, swelling, and difficulty extending the knee.

Patellofemoral Instability

A tendency of the patella to move abnormally or dislocate from the femoral trochlea. Risk factors include abnormal trochlear shape, a high-riding patella, ligament insufficiency, altered limb alignment, and previous dislocation.

Patellofemoral Pain Syndrome

Anterior knee pain related to the patellofemoral joint, common in adolescents and young adults (especially runners). Etiology: multifactorial (malalignment, quadriceps imbalance, overuse, tight lateral structures, weak VMO, and patellar instability).

Alternate Names:

Runner's Knee

Patellofemoral Syndrome

Pain around or behind the kneecap related to movement and loading of the patellofemoral joint, often without a major structural injury.

Patent (Adjective)

Patent means open, unobstructed, or functioning, as in a patent airway, vessel, duct, or congenital fetal channel.

Patent (Noun)

A patent is a legal grant giving an inventor defined rights over an invention for a limited period, subject to applicable law.

Patent Ductus Arteriosus

A congenital heart defect characterized by persistent patency of the ductus arteriosus, a vessel connecting the pulmonary artery to the proximal descending aorta, which normally closes within the first few days of life. The resulting left-to-right shunt leads to pulmonary overcirculation and left heart volume overload.

Patent Foramen Ovale

A persistent flap-like opening between the atria that normally permits fetal blood flow and usually closes after birth. It is often an incidental finding but may allow a clot to pass from the venous to the arterial circulation in selected circumstances.

Patent Urachus

A patent urachus is persistence of the fetal channel between the bladder and umbilicus after birth, allowing urine to drain from the umbilicus and predisposing to infection.

Patent Urachus Repair

A medical procedure, treatment, device, or care topic related to patent urachus repair; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Paternal Behavior

Paternal behavior includes caregiving, protection, bonding, and other behaviors by a father or male caregiver; it is shaped by biology, experience, culture, and circumstances.

Paternal Uniparental Disomy 14 (UDP 14 Pat) Thoracic Dysplasia

A rare genetic condition resulting from inheritance of both copies of chromosome 14 from the father, leading to dysregulated expression of imprinted genes on chromosome 14. It presents with severe thoracic hypoplasia, pulmonary hypoplasia, and

characteristic facial features. It is a lethal condition in most cases.

Alternate Names:

UDP 14 pat

Paternity Testing

The use of genetic markers to establish or exclude a biological parent-child relationship, the reference method being short tandem repeat DNA profiling of the child, alleged father, and mother. Modern STR analysis commonly yields paternity probability exceeding 99.99 percent for true fathers.

Pathogen

A tiny organism such as a virus, bacterium, or parasite that can invade the body and produce disease.

Pathogenesis

The biological process by which a disease begins and develops, including its causes, cellular changes, and progression.

Pathogenic

Pathogenic means capable of causing disease in a susceptible host; pathogenicity depends on the organism or factor, dose, route, and host defenses.

Pathogenic Variant

A genetic variant with sufficient validated evidence to be interpreted as causing or contributing to disease. Classification uses standardized evidence from population data, segregation, functional studies, computational analysis, and phenotype correlation.

Pathogenicity

The capacity of a microorganism to cause disease in a susceptible host. Pathogenicity is a qualitative property, whereas virulence describes the relative degree of disease-causing capacity among strains or organisms.

Pathognomonic

So characteristic of a disease that it strongly identifies that disease when present.

Pathologic

Pathologic means related to disease or an abnormal structural or functional change, rather than normal anatomy or physiology.

Pathologic Dilatation

Abnormal widening of a hollow organ, duct, blood vessel, or other structure beyond its expected size because of obstruction, weakness, pressure, or disease.

Pathologic Fracture

A fracture occurring through bone weakened by an underlying disease such as metastasis, myeloma, infection, benign tumor, or metabolic bone disorder. Evaluation includes characterization of the lesion and treatment of both the fracture and its cause.

Pathologic Hypertrophy

The enlargement of an organ resulting from an increase in the size of its constituent cells without an increase in cell number, typically occurring as a response to chronically increased workload or hormonal stimulation.

Pathologic Ossification

Abnormal formation of bone in soft tissue or in a location where bone should not develop, as in heterotopic ossification after trauma, neurologic injury, or surgery.

Pathologic Process

A pathologic process is an abnormal biological change that produces disease, such as infection, inflammation, ischemia, degeneration, abnormal growth, or immune injury.

Pathologic Response

A pathologic response is an abnormal reaction of cells, tissues, or organs to injury, infection, immune activation, toxins, or other stressors that contributes to disease.

Pathological Fracture

A fracture occurring in bone weakened by disease,

such as osteoporosis, tumor, infection, or metabolic bone disorder, rather than by normal force alone.

Pathological Gambling

Legacy, nonpreferred label; see *Gambling Disorder*.

Pathological Laughter And Crying

Alternate index label; see *Pseudobulbar affect*.

Pathological Stealing

Alternate index label; see *Kleptomania*.

Pathologist

A pathologist is a physician who diagnoses disease by examining tissues, cells, blood, or body fluids and may guide laboratory testing, cancer classification, and treatment decisions.

Pathology

Study of disease. Includes general pathology (cell injury, inflammation, neoplasia) and systemic pathology (organ-specific diseases).

Pathology Artifact

A distortion, alteration, or contaminant introduced during specimen collection, fixation, processing, sectioning, staining, or imaging rather than representing true disease. Artifacts must be recognized to avoid diagnostic error.

Pathophysiology

The study of the functional changes produced by disease or injury and the mechanisms through which they cause signs and symptoms.

Patient

Person receiving medical care. Also: patient zero (initial case), patient safety.

Patient Advocacy

Acting on behalf of a patient to ensure that the patient's needs, rights, and preferences are addressed in healthcare.

Patient Autonomy

A patient's right and capacity to make informed voluntary decisions about healthcare after receiving understandable information about options, benefits, risks, and alternatives.

Patient Communication

The exchange of information and understanding between a patient, caregivers, and healthcare professionals during care.

Patient Education

The process of explaining a health condition, treatment, medicines, exercises, equipment, warning signs, and prevention strategies in a way the patient can understand and use.

Patient Observation

Systematic monitoring of a patient's symptoms, vital signs, examination findings, behavior, and response to treatment to detect deterioration or guide management.

Patient Positioning

Placing and supporting a patient in a position that permits safe care, accurate examination or imaging, and adequate comfort.

Patient Preparation

The instructions and assessments completed before a test or procedure, such as fasting, medicine review, hydration, consent, or skin preparation when required.

Patient Rights

Ethical and legal protections for people receiving care, including informed consent, privacy, access to information, respectful treatment, participation in decisions, and refusal of treatment where lawful.

Patient Safety

Prevention of harm from healthcare. Examples: medication errors, healthcare-associated infections, surgical errors. Strategies: checklists, hand hygiene, barcoding.

Patient-Centered Care

Care respecting patient preferences, needs, values. Examples: shared decision-making, communication, coordination.

Patient-Centered Outcome

An outcome that reflects what matters to the patient, including symptoms, function, quality of life, treatment burden, and ability to achieve personal goals, rather than relying only on laboratory or imaging measures.

Patient-Controlled Analgesia

A method in which a patient self-administers prescribed doses of analgesic medication using a programmable pump with preset safety limits. The medication, dose, and lockout interval are individualized according to the patient's condition and monitoring needs.

Patient-Controlled Analgesia

A system allowing a patient to self-administer preset doses of analgesic through a programmed pump with safety limits.

Patient-Reported Outcome Measure

A validated questionnaire completed directly by a patient to assess symptoms, function, health status, or quality of life without clinician interpretation of the response.

Patlak Plot (Patlak-Rutland Analysis)

A graphical, kinetic modeling method utilized in dynamic positron emission tomography and nuclear medicine imaging to quantitatively determine the uptake and irreversible trapping rate (K_i) of radiotracers within biological tissues. Most prominently applied in clinical and research oncology using 2-[^{18}F]fluoro-2-deoxy-D-glucose (^{18}F -FDG), where FDG is transported into cells and phosphorylated by hexokinase into ^{18}F -FDG-6-phosphate, which cannot be metabolized further and remains trapped. By plotting tissue activity normalized to plasma concentration against normalized integral plasma time, the slope of the resulting linear phase directly yields the influx rate constant, providing

operator-independent quantification of metabolic tumor activity.

Pattern Recognition Receptors

PRRs recognize PAMPs (pathogen) and DAMPs (damage). TLRs: TLR1/2 (lipoproteins), TLR2/6 (lipoteichoic acid), TLR3 (dsRNA), TLR4 (LPS, MD-2, CD14), TLR5 (flagellin), TLR7/8 (ssRNA), TLR9 (CpG DNA). Cytosolic: RIG-I/MDA5 (RNA), cGAS (DNA → cGAMP → STING → IFN), NOD1/2 (peptidoglycan).

Pattern-Recognition Receptors

Immune receptors that detect common molecular features of pathogens or damaged cells.

Pavlov Pouch

A Pavlov pouch is a surgically created isolated stomach pouch used in experimental physiology to study gastric secretion and neural control.

Pavor Nocturnus (Night Terrors)

Episodes of sudden intense fear, screaming, and autonomic arousal during sleep, usually arising from deep non-REM sleep and often followed by little or no memory.

Alternate Names:

Night Terrors

Paxillin

Paxillin is a focal-adhesion adaptor protein that links integrin signaling to the actin cytoskeleton and helps regulate cell adhesion, migration, and mechanical sensing.

Pazopanib

An oral tyrosine-kinase inhibitor targeting vascular endothelial growth factor and other receptors, used for selected advanced renal-cell carcinoma and soft-tissue sarcoma.

PBA

Alternate index label; see *Pseudobulbar affect*.

PBG Urine Test

A urine test measuring porphobilinogen, used urgently to evaluate suspected acute hepatic porphyria during compatible neurovisceral or abdominal attacks.

PCBs

PCBs, or polychlorinated biphenyls, are persistent industrial chemicals that accumulate in tissues and can affect the nervous, endocrine, immune, and reproductive systems.

PCL Injuries

Posterior cruciate ligament tears: dashboard injury (posterior tibial force), hyperflexion. Clinical: posterior drawer (most sensitive), quadriceps active test, posterior sag sign. Grade I (1-5mm), II (6-10mm), III (>10mm). Isolated PCL: nonoperative (brace, rehab) for Grade I/II; surgical for Grade III or combined.

PCO

PCO may mean polycystic ovaries or posterior capsular opacification; the intended condition depends on the clinical context.

PCO Disease

Polycystic ovary syndrome, a hormonal and metabolic disorder involving irregular ovulation, androgen excess, and sometimes polycystic-appearing ovaries.

PCOS

Common endocrine disorder affecting six to twelve percent of reproductive-aged women characterized by hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology on ultrasound. See Polycystic Ovary Syndrome.

PCP

Pneumocystis jirovecii pneumonia (formerly Pneumocystis carinii pneumonia); opportunistic infection in immunocompromised patients, especially transplant recipients.

PDA

Alternate index label; see *Patent ductus arteriosus*.

PDE5 Inhibitors

Medicines that increase penile blood flow by inhibiting phosphodiesterase type 5; they are used mainly to treat erectile dysfunction.

PE Anticoagulation

Anticoagulant treatment for pulmonary embolism to prevent clot extension and recurrence. Options include direct oral anticoagulants, heparins, and vitamin K antagonists; the choice and duration depend on bleeding risk, pregnancy, kidney function, cancer, provoking factors, and other clinical features.

Peak Expiratory Flow Rate

The maximum flow rate achieved during a forced expiratory manoeuvre after maximal inspiration, measured in litres per minute (L/min) using a peak flow meter. PEFR reflects the degree of airflow obstruction and is effort-dependent.

Peak Flow Meter

A handheld device measuring peak expiratory flow, useful for monitoring variable airflow obstruction such as asthma.

Peanut Allergy

An immune reaction to peanut proteins that can cause symptoms ranging from hives to life-threatening anaphylaxis.

Peanut Oil

Peanut oil is oil extracted from peanuts; refined oil usually contains little allergenic protein, whereas unrefined products may trigger reactions in susceptible people.

Pearly Penile Papules

Small, benign, dome-shaped papules arranged circumferentially around the glans penis, representing normal anatomical variants requiring no treatment.

Peau D'Orange

Peau d'orange is dimpled skin resembling an orange peel, caused by edema tethered around hair follicles; it can occur with breast cancer, infection, or lymphatic obstruction.

Pectin

Pectin is a soluble plant fiber that forms gels and may alter stool consistency and absorption; it is used in foods, formulations, and some nutritional products.

Pectinate Line

The pectinate line is the junction in the anal canal between regions with different embryologic origin, blood supply, lymphatic drainage, sensation, and venous anatomy.

Pectoral

Relating to the anterior thoracic chest wall, breast, or shoulder girdle. Encompasses key musculoskeletal structures including the pectoralis major and pectoralis minor muscles, which act to adduct, medially rotate, and draw the humerus and scapula forward. Important in cardiology and trauma for localization of chest wall pain (e.g., musculoskeletal pectoralis strain vs. ischemic angina), in surgical oncology during modified radical mastectomy and axillary lymph node dissection, and in plastic surgery for subpectoral placement of breast implants.

Pectoral Girdle

The pectoral girdle consists of the clavicles and scapulae, attaching the upper limbs to the trunk and providing broad surfaces for shoulder and arm movement.

Pectoralis

A muscle of the chest, chiefly pectoralis major or pectoralis minor.

Pectoralis Major Muscle

Thick fan-shaped chest muscle with clavicular and sternocostal heads, inserting on the lateral intertubercular groove lip. Innervated by medial/lateral pectoral nerves (C5-T1). Clavicular head flexes;

sternocostal head extends from flexion. Adducts and medially rotates powerfully.

Pectoralis Minor Muscle

Thin triangular muscle deep to pectoralis major, from ribs 3-5 to the coracoid process. Innervated by medial pectoral nerve (C8-T1). Draws scapula anteriorly/inferiorly, stabilizing it against the thorax. Assists forced inspiration. Tightness contributes to rounded shoulder posture and thoracic outlet syndrome.

Pectoralis Muscles

The pectoralis major and minor muscles of the chest, which move and stabilize the shoulder and assist with movements such as pushing, adduction, and breathing.

Pectoriloquy

Abnormally clear transmission of spoken sounds through the chest wall, often suggesting lung consolidation or another change in lung tissue.

Pectus Carinatum

A congenital chest wall deformity characterized by anterior protrusion of the sternum (pigeon chest), caused by excessive growth of the costal cartilages pushing the sternum forward. Less common than pectus excavatum. Associated with congenital heart disease, Marfan syndrome, Noonan syndrome, and homocystinuria.

Pectus Excavatum

The most common congenital chest wall deformity, characterized by depression of the sternum (funnel chest), caused by excessive growth of the costal cartilages that push the sternum inward. Typically sporadic, but associated with Marfan syndrome, Ehlers-Danlos syndrome, Noonan syndrome, and Poland syndrome.

Alternate Names:

Sunken Chest

Pectus Excavatum Repair

Surgical correction of a sunken sternum and adjacent costal cartilage, performed when chest-wall

deformity causes significant symptoms, physiologic impairment, or psychosocial impact.

Pediatric Advanced Life Support

A comprehensive, standardized, systematic training curriculum and clinical protocol developed by the American Heart Association and American Academy of Pediatrics for the evaluation, resuscitation, and stabilization of critically ill or injured infants and children experiencing cardiopulmonary arrest, severe respiratory failure, or circulatory shock.

Alternate Names:

PALS (Pediatric Advanced Life Support); Pediatric Resuscitation Protocol

Pediatric AIDS

Pediatric AIDS is advanced HIV infection in a child, with severe immune suppression or AIDS-defining illness; antiretroviral therapy and prevention of opportunistic infection are essential.

Pediatric Anesthesia

Pediatric anesthesia requires age-appropriate dosing, equipment, and techniques. Children have higher metabolic rates, increased oxygen consumption, smaller functional residual capacity, and higher risk of desaturation. Airway anatomy differs (larger tongue, more anterior larynx, narrower subglottic area).

Pediatric Appendicitis

Acute inflammation of the vermiform appendix, usually caused by luminal obstruction. Children may have evolving periumbilical-to-right-lower-quadrant pain, anorexia, vomiting, fever, and guarding, but presentation can be atypical in young children; timely surgical and antimicrobial assessment reduces perforation risk.

Pediatric Assessment Triangle

A rapid, standardized observational assessment tool used by emergency and prehospital clinicians to evaluate the physiological acuity of an infant or child within seconds and without touching the patient. It evaluates three independent observational components: Appearance (tone, interactiveness, consola-

bility, look/gaze, speech/cry; TICLS mnemonic), Work of Breathing (abnormal airway sounds, retractions, flaring, positioning), and Circulation to Skin (pallor, mottling, cyanosis).

Alternate Names:

PAT (Pediatric Assessment Triangle); Pediatric Visual Triage Triangle

Pediatric Brain Tumors

Tumors that develop in the brain or nearby tissues of children.

Alternate Names:

Pediatric Cancer

Cancer occurring in infants, children, or adolescents. Common types include leukemia, brain tumors, lymphoma, neuroblastoma, and Wilms tumor; prognosis varies by type and stage.

Pediatric Dentistry

Specialty of dental care for children. Manages primary dentition, growth, behavior. Preventive focus. Special needs adaptation.

Pediatric Dermatology

The medical specialty concerned with skin, hair, and nail disorders in infants, children, and adolescents. It includes congenital lesions, inflammatory diseases, infections, vascular lesions, and tumors.

Pediatric Dermatology Conditions

Distinct from adult dermatology. Birthmarks include vascular (hemangiomas, port-wine stains) and pigmented (café-au-lait, congenital nevi, Mongolian spots). Infantile hemangiomas proliferate in first months then involute. Congenital melanocytic nevi carry melanoma risk proportional to size.

Pediatric Diabetic Ketoacidosis

A life-threatening complication of type 1 diabetes characterized by hyperglycemia, metabolic acidosis, and ketosis. Presents with polyuria, polydipsia, vomiting, abdominal pain, Kussmaul respirations, and fruity breath odor.

Pediatric Early Warning Score

An objective clinical bedside scoring system used in pediatric inpatient settings to detect early physiological deterioration in hospitalized children. It calculates a composite score by grading vital signs and bedside parameters across three domains: behavioral state/neurologic responsiveness, cardiovascular status (heart rate, capillary refill time, skin color), and respiratory effort (respiratory rate, retractions, oxygen requirement), prompting timely escalation of care or pediatric rapid response activation.

Alternate Names:

PEWS; Pediatric Early Warning System

Pediatric Eczema

Atopic dermatitis in children commonly begins in infancy with facial and extensor involvement; flexural disease predominates in older children. Pruritus, xerosis, sleep disturbance, and scratching are characteristic.

Pediatric Fever

An elevated body temperature in a child, usually caused by infection but sometimes by inflammation, medicines, or environmental heat. Evaluation depends on age, appearance, duration, and associated findings.

Pediatric Health

Pediatric health encompasses the physical, cognitive, emotional, and social development of children from infancy through adolescence, with emphasis on preventive care, developmental surveillance, and age-appropriate management of acute and chronic conditions.

Pediatric Heart Surgery

A medical procedure, treatment, device, or care topic related to pediatric heart surgery; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Pediatric Meningitis

Inflammation of the meninges in children, bacterial forms being the most dangerous. Presents with fever, headache, neck stiffness, photophobia, altered

mental status, and petechial rash (meningococcal). Infants may present with irritability, poor feeding, bulging fontanelle, and high-pitched cry.

Pediatric Nursing

Pediatric nursing provides age-appropriate care to infants, children, and adolescents, accounting for growth, development, family involvement, communication, safety, and medication differences.

Pediatric Obesity

Excess body fat in a child or adolescent defined using age- and sex-specific BMI percentiles, associated with metabolic, orthopedic, respiratory, psychosocial, and cardiovascular risks.

Pediatric Obstructive Sleep Apnea

Repeated blockage of a child's upper airway during sleep, causing pauses in breathing.

Alternate Names:

Pediatric Ophthalmology

The subspecialty caring for eye and vision disorders in infants, children, and adolescents, including strabismus, amblyopia, congenital abnormalities, refractive error, and childhood eye disease.

Pediatric Rehabilitation

Rehabilitation for children with congenital or acquired disabilities. Developmental considerations, family-centered care, school collaboration.

Pediatric Shock

Inadequate tissue oxygen delivery or utilization in a child, most often caused by hypovolemia, distributive disease, cardiogenic dysfunction, or obstruction. Children may maintain blood pressure until late; tachycardia, abnormal perfusion, altered mental status, and oliguria are important early warning signs.

Pediatric Skin Tumors

Pediatric skin tumors differ from adult tumors. Infantile hemangioma is the most common benign tumor of infancy. Congenital melanocytic nevus

carries risk of melanoma (larger nevi have higher risk). Juvenile xanthogranuloma is a benign histiocytic lesion. Pyogenic granuloma is a common vascular lesion.

Pediatric Sleep Apnea

Pediatric sleep apnea is repeated interruption or reduction of breathing during sleep, commonly from enlarged tonsils or adenoids, causing snoring, restless sleep, behavior changes, or impaired growth.

Pediatric Toxicology

Poisoning in children. Common: iron, medications, household cleaners, plants. Prevention: child-proof containers, education.

Pediatrician

A medical doctor who specializes in the health and medical care of children from birth through adolescence (typically up to age 16-18).

Alternate Names:

Paediatrician

Pediatrics

The branch of medicine concerned with the health and medical care of infants, children, and adolescents from birth to age 18 (or up to age 16 in the UK).

Alternate Names:

Paediatrics

Pedicle

A pedicle is a stalk or attachment containing vessels, nerves, or other supporting structures, such as the base of a polyp, flap, or vertebral arch.

Pediculosis

An ectoparasitic cutaneous infestation of humans by blood-feeding lice of the order Phthiraptera. Classified by anatomical site and species into pediculosis capitis (head lice, caused by *Pediculus humanus capitis*, common in school-aged children and characterized by pruritus, viable nits cemented to proximal hair shafts, and excoriations), pediculosis corporis (body lice, caused by *Pediculus humanus corporis*,

residing in clothing seams of unhoused individuals and serving as vectors for *Rickettsia prowazekii* [epidemic typhus], *Bartonella quintana* [trench fever], and *Borrelia recurrentis*), and pediculosis pubis (crab lice, caused by *Phthirus pubis*, transmitted sexually). First-line treatment utilizes topical pediculicides (e.g., permethrin 1% or 5%) and oral ivermectin.

Pediculosis Capitis

Infestation of the scalp by head lice, causing itching and attached eggs or lice in the hair.

Pediculus Humanus Capitis

A blood-sucking parasite commonly known as the louse (plural, lice) that can cause an itchy scalp; infestations are highly contagious and especially common in school-age children.

Pedigree

A diagram showing family relationships and inheritance of traits or disease.

Pediococcus

A genus of lactic-acid-producing bacteria used in food fermentation and rarely associated with opportunistic infection.

Pediococcus Pentosaceus

A lactic-acid-producing bacterium used in fermented foods that rarely causes opportunistic infection.

Pedodontics

The branch of dentistry concerned with infants, children, and adolescents.

Pedophilia

Pedophilia is a psychiatric term for a persistent sexual interest in prepubescent children; sexual contact with a child is abuse and illegal, and safety-focused professional help is required.

Pedophilic Disorder

A paraphilic disorder involving recurrent, intense sexual fantasies, urges, or behaviors toward pre-

pubescent children, generally beginning in adolescence or adulthood. Clinical diagnosis requires the criteria to cause significant distress or impairment, or to be acted upon, and requires the person to meet established age-difference criteria.

Pedunculate

Pedunculate describes a structure, lesion, or polyp attached by a narrow stalk rather than a broad base.

Pedunculated

Attached to the skin or another surface by a narrow stalk, rather than having a broad base.

Peeling Skin

Loss or shedding of the outer layer of skin, which may result from dryness, sunburn, irritation, allergy, infection, inflammation, or a systemic illness.

PEEP

PEEP, or positive end-expiratory pressure, is pressure maintained in the airways at the end of exhalation during mechanical ventilation to prevent alveolar collapse.

Peeping Tom

Peeping tom is a nonclinical term for a person who secretly observes others, often for sexual gratification; assessment focuses on consent, harm, and safety.

Peer Review

Peer review is evaluation of research, clinical work, or scholarly writing by qualified independent reviewers to assess methods, evidence, clarity, and significance.

PEF

PEF, or peak expiratory flow, is the fastest airflow produced during a forceful exhalation and is used chiefly to monitor variable airway obstruction such as asthma.

PEG Tube

A feeding tube placed through the abdominal wall into the stomach using endoscopic guidance.

Pegdinetanib

An investigational pegylated antibody or biologic studied for blocking platelet-derived growth factor receptor signaling in cancer or eye disease.

Peldesine

Peldesine is an investigational sphingosine-1-phosphate lyase inhibitor studied for effects on immune-cell trafficking and inflammatory or autoimmune disease.

Peliosis Hepatis

Blood-filled cyst-like spaces within the liver, associated with medicines, toxins, infection, malignancy, or systemic disease and occasionally complicated by liver failure or rupture.

Pelitrexol

Pelitrexol is an investigational antifolate drug that inhibits thymidylate synthase and has been studied as an antineoplastic agent.

Pelizaeus-Merzbacher Disease

An X-linked recessive hypomyelinating leukodystrophy caused by copy-number duplications (60–70%), point mutations, or null deletions of the proteolipid protein 1 (*PLP1*) gene on chromosome Xq22.2. Clinically characterized by the presentation of coarse pendular nystagmus and hypotonia in early infancy, followed by progressive spastic quadriplegia, choreoathetosis, cerebellar ataxia, and intellectual decline. Brain MRI pathognomically reveals diffusely abnormal, homogeneous high signal intensity on T2-weighted images throughout the cerebral white matter, reflecting near-total failure of central nervous system myelination.

Alternate Names:

PMD; PLP1-Related Hypomyelinating Leukodystrophy

Pellagra

A disease caused by niacin deficiency, classically producing dermatitis, diarrhea, and neuropsychiatric changes.

Pellet

A pellet is a small compacted mass of medication, material, or tissue; drug pellets may be formulated for implantation or controlled release.

Pelvic

Relating to the pelvis, the bony ring and associated organs, muscles, vessels, and nerves in the lower trunk.

Pelvic Binder

A circumferential device applied around the pelvis to reduce motion and hemorrhage from suspected unstable pelvic-ring injury. It should be centered over the greater trochanters and used with definitive imaging and hemorrhage-control management.

Pelvic Bone

The ring of bones formed by the paired hip bones, sacrum, and coccyx that supports the trunk, protects pelvic organs, and transmits weight to the lower limbs.

Alternate Names:

Hip Bone Hip Bones

Pelvic Cavity

The lower abdominal cavity enclosed by the pelvis, containing the urinary bladder, internal reproductive organs, and rectum.

Pelvic Congestion

Enlargement or pooling of blood in the veins of the pelvis, which may cause pelvic pressure, discomfort, or pain.

Pelvic Congestion Syndrome

Chronic pelvic pain associated with enlarged or inefficient pelvic veins, often worsening with prolonged standing, before menstruation, or after intercourse. Diagnosis requires evaluation for other causes of

pelvic pain and may involve pelvic imaging; treatment is individualized by gynecology or vascular specialists.

Pelvic CT Scan

Computed tomography of the pelvis using x-rays and computer reconstruction to evaluate bones, organs, soft tissues, masses, trauma, infection, and bleeding.

Pelvic Diaphragm

The muscular and fascial floor of the pelvis, mainly formed by the levator ani and coccygeus muscles, supporting pelvic organs.

Pelvic Examination

Clinical inspection and palpation of the external genitalia, vagina, cervix, uterus, and adnexa to evaluate bleeding, pain, discharge, lesions, pregnancy-related findings, or infection.

Pelvic Exenteration

Extensive surgery removing selected pelvic organs and surrounding tissues, reserved for carefully selected locally advanced or recurrent cancers when less radical treatment is inadequate.

Pelvic Floor

A group of muscles and connective tissues supporting the pelvic organs and controlling urinary and fecal continence.

Pelvic Floor Dysfunction

Weakness, overactivity, poor coordination, or injury of the muscles and connective tissues supporting the bladder, bowel, and reproductive organs. Symptoms may include urinary or fecal problems, pelvic pain, constipation, pain with sex, or pelvic organ prolapse; assessment may involve pelvic-health clinicians.

Pelvic Floor Health And Disorders

The pelvic floor is a musculofascial sling (levator ani, coccygeus, and surrounding fascia) spanning the pelvic outlet, supporting the pelvic organs (bladder, uterus, rectum), maintaining continence, and

facilitating urination, defecation, and sexual function.

Pelvic Floor Muscle Training Exercises

Pelvic floor muscle exercises strengthen the muscles supporting the bladder, bowel, and reproductive organs and may improve selected forms of urinary or fecal leakage.

Pelvic Girdle

The ring of bones connecting the lower limbs to the trunk, consisting of the hip bones, sacrum, and associated joints.

Pelvic Girdle Pain In Pregnancy

Pain arising from the pelvic joints and surrounding tissues during pregnancy, often affecting the pubic symphysis, sacroiliac joints, hips, or lower back. It may impair walking, turning in bed, or climbing stairs.

Pelvic Infection

An infection involving pelvic organs or tissues, including pelvic inflammatory disease, urinary infection, abscess, or postoperative infection.

Pelvic Inflammatory Disease

Infection and inflammation of the upper female genital tract (uterus, fallopian tubes, ovaries, and peritoneum), most commonly caused by sexually transmitted infections (*Chlamydia trachomatis*, *Neisseria gonorrhoeae*). Presents with lower abdominal pain, cervical motion tenderness, adnexal tenderness, and fever.

Pelvic Inflammatory Disease

Pelvic inflammatory disease is infection and inflammation of the upper female reproductive tract, often caused by sexually transmitted organisms and capable of causing infertility, ectopic pregnancy, or chronic pain.

Alternate Names:

PID

Pelvic Inlet

The superior opening of the true pelvis, bounded by the sacral promontory posteriorly, the arcuate lines of the ilia laterally, and the symphysis pubis anteriorly. Its anteroposterior diameter (true conjugate) normally measures at least 11 cm. The transverse diameter is typically 13 cm.

Pelvic Kidney

A kidney that remains in the pelvis rather than ascending to its usual lumbar position during development; it may be asymptomatic or associated with abnormal drainage, stones, infection, or vascular anatomy.

Pelvic Laparoscopy

A minimally invasive procedure using a camera inserted through small abdominal incisions to inspect and treat pelvic organs and structures.

Pelvic Lymphadenectomy

Pelvic lymphadenectomy is surgical removal of lymph nodes in the pelvis to check for or treat spread of cancer.

Pelvic Mass

A lump or abnormal enlargement in the pelvis arising from reproductive, urinary, gastrointestinal, lymphatic, or other tissues.

Pelvic Neoplasm

A pelvic neoplasm is an abnormal growth in the pelvis, arising from organs, bones, soft tissues, or lymph nodes; it may be benign or malignant.

Pelvic Organ Prolapse

POP: descent of the pelvic organs through the vagina due to pelvic floor weakness. Types: cystocele (anterior—bladder), rectocele (posterior—rectum), uterine prolapse (apical), enterocele, and vault prolapse (post-hysterectomy).

Pelvic Organ Prolapse Quantification System

An internationally validated, objective, repro-

ducible staging system established by the International Continence Society to quantify the site and severity of female pelvic organ prolapse. Utilizes nine discrete anatomical measuring points (A_a , B_a , C , D , a_p , b_p , gh , pb , tv) measured in centimeters during maximum Valsalva relative to the plane of the hymen (assigned a value of 0), classifying prolapse into five distinct clinical stages (Stage 0 to Stage IV).

Alternate Names:

POP-Q System; POP-Q Classification

Pelvic Outlet

The inferior opening of the true pelvis, bounded by the pubic symphysis, ischial tuberosities, sacrotuberous ligaments, and coccyx. Normal anteroposterior diameter is 11-13 cm (from the lower border of the pubic symphysis to the tip of the coccyx), and the transverse diameter (between the ischial tuberosities) is approximately 10-11 cm.

Pelvic Pain

Pain localized to the lower abdomen, pelvis, or perineum, lasting <3 months (acute) or >3-6 months (chronic), and can arise from gynecologic, urologic, gastrointestinal, musculoskeletal, or psychologic causes.

Pelvic Pain During Early Pregnancy

Pain in the pelvis or lower abdomen during early pregnancy, caused by normal uterine changes, miscarriage, ectopic pregnancy, ovarian disease, infection, or other conditions. Severe or one-sided pain, shoulder pain, fainting, fever, or vaginal bleeding requires emergency assessment.

Pelvic Pain, Chronic

Alternate index label; see *Chronic pelvic pain*.

Pelvic Spleen

An ectopic or wandering spleen that has descended into the pelvic cavity, resulting from laxity or absence of the normal ligamentous attachments that fix the spleen in the left upper quadrant. It may present as a palpable pelvic mass, with risk of torsion, infarction, or rupture. Diagnosis is by CT or

ultrasound.

Pelvic Support Problems, Uterine Prolapse

Alternate index label; see *Uterine prolapse*.

Pelvic Ultrasound - Abdominal

Ultrasound imaging of pelvic organs through the lower abdominal wall, usually with a full bladder, to evaluate the uterus, ovaries, bladder, and surrounding structures.

Pelvic Viscus

An organ located in the pelvic cavity, such as the bladder, rectum, uterus, ovaries, prostate, or related structures.

Pelvimeter

An instrument used to estimate the size and shape of the pelvis, especially during obstetric assessment. Pelvic measurements are only one part of judging whether vaginal birth is likely and are interpreted with the clinical situation.

Pelvimetry

Measurement of the dimensions of the maternal pelvis, usually to assess selected obstetric concerns.

Pelvis

The bony ring and associated soft tissues at the base of the trunk, connecting the spine with the lower limbs.

Pelvis MRI Scan

Magnetic resonance imaging of the pelvis used to evaluate organs, muscles, bones, joints, nerves, blood vessels, and soft-tissue disease without ionizing radiation.

Pelvis X-Ray

Plain radiographic imaging of the pelvic bones and hips used to evaluate fractures, alignment, arthritis, deformity, and selected bone lesions.

Pemberton Sign

Facial flushing, cyanosis, or breathing difficulty induced by raising both arms, suggesting obstruction at the thoracic inlet, often from a large retrosternal goiter.

Pemberton'S Sign

A clinical sign elicited by asking the patient to raise both arms above the head for one minute, causing facial plethora, cyanosis, and distended neck veins in patients with a substernal goiter or mediastinal mass compressing the thoracic inlet.

Pemetrexed

Pemetrexed is a folate-antimetabolite chemotherapy drug that inhibits several enzymes involved in nucleotide synthesis, used especially for mesothelioma and nonsquamous non-small-cell lung cancer; folic acid and vitamin B12 reduce toxicity.

Pemetrexed Disodium

The disodium salt of pemetrexed, an antifolate chemotherapy medicine used with other treatment for selected non-small-cell lung cancer and mesothelioma.

Pemoline

Pemoline is a stimulant formerly used for attention-deficit disorders; serious liver toxicity led to withdrawal or severe restriction in many countries.

Pemphigoid

A group of autoimmune subepidermal blistering disorders, including bullous pemphigoid.

Pemphigoid Bullous (Bullous Pemphigoid)

Alternate index label for Bullous Pemphigoid.

Pemphigoid Gestationis

An autoimmune blistering disease of pregnancy characterised by intensely pruritic urticarial plaques and tense vesicles or bullae, typically arising in the second or third trimester around the umbilicus and spreading to the trunk and extremities.

Pemphigus

A group of autoimmune intraepidermal blistering disorders caused by antibodies against desmosomal proteins.

Pemphigus Vulgaris

An autoimmune intraepidermal blistering disease caused by IgG antibodies against desmoglein 1 and 3. It presents with flaccid blisters that rupture easily, leaving painful erosions of skin and mucous membranes. Nikolsky sign is positive (lateral pressure causes epidermal sloughing).

Penbutolol

Penbutolol is a beta-blocker formerly used for high blood pressure; it can slow the heart rate and worsen asthma in susceptible people.

Pencil Eraser Swallowing

Pencil eraser swallowing involves toxic or environmental exposure to pencil eraser swallowing; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pencil Swallowing

Pencil swallowing involves toxic or environmental exposure to pencil swallowing; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Penclomedine

Penclomedine is an investigational antineoplastic compound studied for cytotoxic effects against cancer cells and is not a routine clinical medicine.

Pendular

Moving or swinging freely like a pendulum; pendular nystagmus has a similar speed in both directions.

Pendulous

Pendulous means hanging loosely from a supporting structure, as in a pedunculated lesion, drooping tissue, or an organ suspended by a stalk.

Penetrance

Proportion of individuals with a particular genotype who express the expected phenotype. Complete penetrance: all with genotype show phenotype. Incomplete penetrance: some with genotype do not show phenotype.

Penetrating Abdominal Trauma

Injury to the abdomen caused by a projectile, sharp object, or other penetrating force. It may damage abdominal organs or blood vessels and requires urgent assessment for internal bleeding or organ injury; management depends on the mechanism, examination, stability, and imaging findings.

Penetrating Wounds

Wounds caused by an object entering tissue and potentially damaging deeper structures.

Penicillin

A family of antibiotics that kill bacteria by stopping them from making cell walls. Different types treat different bacterial infections; they do not work against viruses. Allergy, resistance, and kidney problems can affect the choice of medicine.

Penicillin Allergy

An immune reaction to penicillin antibiotics that may cause rash, swelling, breathing difficulty, or anaphylaxis.

Penicillin G

Penicillin G is an injectable penicillin antibiotic used for infections such as syphilis and selected streptococcal or other susceptible bacterial infections.

Penicillin V

Penicillin V is an oral penicillin antibiotic used for selected susceptible infections, including some throat and dental infections.

Penicillium

A genus of molds that includes species used to make medicines and foods and species that can cause allergy or opportunistic infection.

Penile

Penile means relating to the penis, the external male organ involved in urination and sexual reproduction.

Penile Adhesions

Abnormal attachment of penile skin to the glans, often after circumcision or during childhood. They may cause irritation or conceal part of the glans and are assessed and treated according to severity.

Penile Cancer

Rare malignancy, mostly squamous cell carcinoma. Risk factors: HPV, phimosis, poor hygiene, smoking. Presentation: penile lesion, ulcer, mass. Treatment: surgery (partial/total penectomy), radiation, chemotherapy.

Penile Disease

A disorder affecting the penis, including infection, inflammation, erectile or structural disease, trauma, skin disease, or malignancy.

Penile Edema

Swelling of the penis caused by inflammation, infection, trauma, allergic reaction, impaired lymphatic drainage, or a constricting ring or retracted foreskin. Sudden painful swelling with an irreducible foreskin is an emergency.

Penile Erection

Rigidity of the penis caused by increased arterial inflow and reduced venous outflow after sexual or reflex stimulation; persistent difficulty obtaining or maintaining it is erectile dysfunction.

Penile Fibrosis (Peyronie's Disease (Curvature Of The Penis))

Alternate index label for Peyronie's Disease (Curvature of the Penis).

Penile Fracture

Rupture of tunica albuginea during erection. "Pop" sound, immediate detumescence, swelling, ecchymosis. Surgical emergency.

Penile Hygiene

Routine cleaning and drying of the penis and foreskin, with gentle retraction only when it moves freely and replacement afterward. Good hygiene helps reduce irritation and infection without harsh soaps or forceful retraction.

Penile Implant

A surgically implanted device used to produce penile rigidity in selected people with erectile dysfunction that has not responded to other treatments.

Penile Infection

Infection of the penis or foreskin caused by bacteria, fungi, viruses, or parasites, producing pain, redness, discharge, sores, or swelling.

Penile Intraepithelial Neoplasia

The standardized World Health Organization histopathologic designation for premalignant, non-invasive intraepithelial dysplastic lesions of the squamous epithelium of the penis. PeIN is classified into two major clinicopathologic categories: HPV-associated PeIN (undifferentiated type, including warty, basaloid, and warty-basaloid morphologies, frequently presenting on the glans or prepuce as red velvety plaques termed erythroplasia of Queyrat or crusted scaly plaques termed Bowen disease) and HPV-independent PeIN (differentiated type, strongly associated with chronic balanitis xerotica obliterans/lichen sclerosus). If untreated, PeIN carries a significant risk of progression to invasive penile squamous cell carcinoma, requiring treatment with topical immunomodulators (imiquimod, 5-fluorouracil), laser ablation, photodynamic therapy, or surgical glans resurfacing.

Alternate Names:

PeIN; Penile Precancerous Lesion

Penile Leukoplakia

A persistent white patch on penile skin or mucosa that may reflect irritation, inflammation, or a precancerous disorder and requires assessment.

Penile Prosthesis

Inflatable or malleative device for erectile dysfunction.

tion. Types: two-piece, three-piece (reservoir, pump, cylinders). Surgical implantation. Complications: infection, erosion, mechanical failure, dissatisfaction.

Penile Torsion

Rotation of the penis around its longitudinal axis, usually a congenital condition in which the glans or urethral opening is rotated. Symptomatic or marked torsion may require urologic evaluation and surgical correction.

Penis

The male external genital organ used for urination and sexual reproduction.

Penis Carcinoma

Penis carcinoma is cancer of the penis, most often arising from skin or squamous cells and sometimes linked to HPV infection.

Penis Care (Uncircumcised)

Care of an uncircumcised penis includes gently retracting the foreskin only as far as comfortable, cleaning underneath, drying, and returning the foreskin to its normal position.

Penis Disorder (Priapism (Penis Disorder))

Alternate index label for Priapism (Penis Disorder).

Penis Disorder Balantis (Balanitis (Penis Disorder))

Alternate index label for Balanitis (Penis Disorder).

Penis Pain

Penis pain may result from infection, injury, inflammation, urinary problems, or sexual conditions; severe pain, swelling, discharge, or inability to urinate requires prompt assessment.

Penis, Erection Of The

Penile rigidity produced by increased arterial inflow and reduced venous outflow after sexual or reflex stimulation; persistent difficulty is erectile dysfunction.

tion.

Penis, Small

A penis that is substantially below expected stretched length for age, usually assessed as micropenis in a newborn or child after excluding measurement and developmental variation.

Penoscrotal Transposition

A congenital difference in which the penis and scrotum are positioned abnormally relative to each other, often associated with hypospadias or other genital and urinary anomalies.

Pentad Of TTP

Classic clinical pentad of Thrombotic Thrombocytopenic Purpura: (1) Thrombocytopenia, (2) Microangiopathic Hemolytic Anemia (schistocytes), (3) Neurological symptoms, (4) Renal insufficiency, and (5) Fever. See Thrombotic Thrombocytopenic Purpura (TTP).

Pentadecanoic Acid

A 15-carbon saturated fatty acid found in small amounts in dairy and ruminant fats and used in nutritional and lipid-metabolism research.

Pentaerythritol Tetranitrate

An organic nitrate with vasodilator activity, mainly of historical or specialized pharmaceutical interest.

Pentamidine

An antiprotozoal and antimicrobial medicine used for selected leishmaniasis, trypanosomiasis, and Pneumocystis infections.

Pentamidine Isethionate

An antiprotozoal and antimicrobial drug used in selected cases of Pneumocystis pneumonia, leishmaniasis, trypanosomiasis, and other infections; toxicity includes hypoglycemia, kidney injury, and arrhythmia.

Pentatrichomonas Hominis

Pentatrichomonas hominis is a flagellated intesti-

nal protozoan generally regarded as nonpathogenic; finding it in stool usually indicates exposure to fecal contamination.

Pentazocine

An opioid pain medicine with mixed agonist and antagonist effects, used for some moderate or severe pain. It can cause drowsiness, dependence, and slowed breathing and may trigger withdrawal in a person dependent on other opioids.

Pentazocine Overdose

Pentazocine overdose involves toxic or environmental exposure to pentazocine overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pentetic Acid

A chelating compound that binds metal ions and is used in selected radiopharmaceuticals, contrast agents, or laboratory applications.

Pentobarbital Overdose

Pentobarbital overdose involves toxic or environmental exposure to pentobarbital overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pentosan Polysulfate

A semi-synthetic sulfated polysaccharide used for selected bladder pain syndrome and historically as an anticoagulant; prolonged use can cause pigmentary maculopathy and bleeding.

Pentose

A pentose is a five-carbon sugar, including ribose in RNA and deoxyribose in DNA, and participates in nucleotide and carbohydrate metabolism.

Pentose Phosphate Pathway

A cytoplasmic glucose oxidation pathway running parallel to glycolysis, primarily in the liver, adipose tissue, adrenal cortex, and red blood cells. It has two phases: the oxidative phase (producing NADPH and ribose-5-phosphate) and the non-oxidative phase

(interconverting sugar phosphates).

Pentoxifylline

A methylxanthine derivative that improves red-cell flexibility and is used in some countries for peripheral vascular disease; benefit varies by condition.

Pentylenetetrazole

Pentylenetetrazole is a convulsant research compound formerly used as a respiratory stimulant; it is not a routine modern medicine.

Penumbra

Brain tissue surrounding an area of infarction that is functionally impaired but potentially salvageable if blood flow is restored promptly.

Peplomycin

Peplomycin is a bleomycin-related chemotherapy medicine used in some countries for selected cancers; lung toxicity is an important risk.

Peppermint

Mentha piperita, used as a flavoring and in some preparations for dyspeptic symptoms; concentrated oil can be hazardous.

Peppermint Oil Overdose

Peppermint oil overdose involves toxic or environmental exposure to peppermint oil overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pepsin

A stomach enzyme that breaks proteins into smaller fragments. It is released as inactive pepsinogen and activated by stomach acid.

Pepsin A

Pepsin A is a stomach protease activated by acid that begins digestion of dietary proteins into smaller peptides.

Pepsinogen

An inactive precursor released by gastric chief cells

and converted to pepsin in acidic conditions.

Peptic Esophagitis

Inflammation and injury of the esophageal lining caused mainly by repeated gastroesophageal reflux of acid or bile, producing heartburn, chest discomfort, swallowing pain, or strictures.

Peptic Ulcer

An open sore in the lining of the stomach or duodenum, most commonly caused by *Helicobacter pylori* infection or NSAID use.

Alternate Names:

Stomach Ulcer (Peptic Ulcer) Ulcer, Peptic

Peptic Ulcer Bleeding

Haemorrhage caused by erosion of an artery or other blood vessel beneath a gastric or duodenal peptic ulcer. It is a common cause of upper gastrointestinal bleeding and may present with haematemesis, coffee-ground vomiting, melaena, or haemodynamic instability.

Peptic Ulcer Disease

Condition characterized by ulceration of the gastrointestinal mucosa extending through the muscularis mucosae into the submucosa or deeper layers, occurring most commonly in the duodenal bulb and gastric antrum.

Peptichemio

An investigational or historical anticancer drug name related to peptide-linked chemotherapy compounds.

Peptidase

An enzyme that breaks peptide bonds in proteins or peptides.

Peptide

A short chain of amino acids linked by peptide bonds.

Peptide Biosynthesis

peptide biosynthesis concerns peptide biosynthesis,

a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Peptide Bond

A covalent amide bond formed between the carboxyl group of one amino acid and the amino group of another, releasing water. Peptide bonds link amino acids into polypeptide chains and have partial double-bond character that restricts rotation.

Peptide Hormone

A hormone made of amino acids that binds a cell-surface receptor and regulates processes such as metabolism, growth, reproduction, stress, and fluid balance.

Peptide Library

peptide library concerns peptide library, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Peptide Metabolism

peptide metabolism concerns peptide metabolism, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Peptide Transport

peptide transport concerns peptide transport, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Peptide Yy

peptide yy concerns peptide yy, a biologic or physiologic concept used in research, laboratory medicine, and the explanation of disease.

Peptidoglycan

A mesh-like polymer forming the bacterial cell wall, made of sugars and short peptides and targeted by several antibiotics.

Peptococcaceae

A family of anaerobic gram-positive cocci found in the human microbiota and sometimes associated with infection.

Peptococcus

A genus of anaerobic gram-positive cocci that can contribute to mixed infections, especially in damaged or poorly oxygenated tissue.

Peptococcus Niger

An anaerobic gram-positive coccus found in the human microbiota and rarely associated with mixed infection.

Peptoniphilus

A genus of anaerobic gram-positive cocci found on skin and mucosal surfaces that can cause opportunistic infection.

Peptoniphilus Hareii

An anaerobic gram-positive coccus occasionally recovered from wounds, abscesses, or other clinical specimens.

Peptoniphilus Indolicus

An anaerobic gram-positive coccus found in the genital and other microbiota that can occasionally contribute to mixed infection.

Peptostreptococcus

A group of anaerobic gram-positive cocci associated with dental, skin, abdominal, pelvic, and other mixed infections.

Peracetic Acid

A strong oxidizing disinfectant and sterilant used for equipment and surfaces; concentrated exposure irritates or burns skin, eyes, and airways.

Peracute

Peracute describes disease or a clinical event that develops extremely rapidly, often over minutes to hours, with little time for adaptation or diagnosis.

PERC Rule

The Pulmonary Embolism Rule-out Criteria, a validated clinical decision rule designed to identify emergency department patients with suspected acute pulmonary embolism who have an exceptionally low

pretest probability (less than 1.5 to 2 percent), in whom diagnostic testing with D-dimer or CT pulmonary angiography can be safely omitted without increasing adverse thrombotic outcomes. It comprises eight clinical criteria: age under 50 years, heart rate under 100 bpm, pulse oximetry saturation ≥ 95 percent on room air, no unilateral leg swelling, no hemoptysis, no recent surgery or trauma within 4 weeks, no prior history of venous thromboembolism, and no estrogen use; all eight criteria must be satisfied to be PERC-negative.

Alternate Names:

Pulmonary Embolism Rule-out Criteria; PERC Score

Percentile

A percentile indicates the percentage of a reference population with a value below a measurement, such as height, weight, or test score.

Perception

The brain's process of organizing and interpreting sensory information into an experience of objects, events, body sensations, or the environment.

Perceptual

Relating to the process of detecting and interpreting sensory information.

Perceptual Disorder

A clinically significant disturbance in the way sensory information is interpreted, including altered visual, auditory, tactile, or body perception.

Perceptual Distortion

An inaccurate or altered interpretation of a real sensory stimulus, which may occur with neurologic disease, psychiatric illness, drugs, migraine, seizures, or sensory impairment.

Perceptual Masking

Reduction in the ability to detect or identify one sensory stimulus because another stimulus overlaps it in time, space, or processing.

Percussion

Tapping the body surface to assess the underlying tissue by its sound and transmitted vibration.

Percussion Note

The sound produced by percussing the chest wall. Normal lung produces a resonant note. Dullness suggests consolidation or fluid. Hyperresonance suggests pneumothorax or emphysema. Stony dullness suggests pleural effusion.

Percutaneous

Performed through the skin, usually with a needle, catheter, or other instrument.

Percutaneous Biopsy

Removal of tissue through the skin using a needle or other instrument, often guided by ultrasound, CT, or another imaging method.

Percutaneous Coronary Intervention

A catheter-based procedure to open blocked coronary arteries, typically involving balloon angioplasty and stent placement. Indications include acute STEMI (primary PCI), NSTEMI (early invasive strategy), and stable CAD with significant stenosis. Drug-eluting stents have reduced restenosis rates compared to bare-metal stents.

Percutaneous Discectomy

Percutaneous discectomy removes or decompresses part of a herniated spinal disc through a small needle or incision.

Percutaneous Diskectomy

Removal of part of a herniated spinal disk through a small skin incision.

Percutaneous Drainage

Placement of a needle or catheter through the skin to drain a fluid collection or obstructed body space, often with imaging guidance.

Percutaneous Electrical Nerve

Stimulation

Electrical stimulation delivered through the skin to reduce pain signals.

Alternate Names:

PENS

Percutaneous Endoscopic Gastrostomy

Endoscopic placement of a feeding tube through the abdominal wall directly into the stomach. Indications: long-term enteral feeding (>4 weeks) for dysphagia (stroke, ALS, dementia, head/neck cancer) and malnutrition.

Percutaneous Kidney Procedures

A medical procedure, treatment, device, or care topic related to percutaneous kidney procedures; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Percutaneous Nephrolithotomy

Minimally invasive surgical procedure for removing large, complex, or staghorn renal calculi. Access: a needle track is established under fluoroscopic or ultrasound guidance from the flank through the renal parenchyma into the target renal calyx, dilated to allow passage of a nephroscope and working sheath.

Percutaneous Nephrolithotripsy

Percutaneous nephrolithotripsy removes or fragments kidney stones through a tract made through the skin into the kidney, usually for larger or complex stones.

Percutaneous Transhepatic Cholangiogram

A percutaneous transhepatic cholangiogram uses a needle through the liver to inject contrast into bile ducts for imaging and may guide drainage or stent placement.

Percutaneous Transhepatic Cholangiography

PTC is imaging of the bile ducts after contrast is injected through a needle passed through the skin and liver; drainage or stenting may follow.

Percutaneous Transluminal Angioplasty

Balloon dilation of arterial stenosis. Often combined with stenting.

Percutaneous Transluminal Coronary Angioplasty

Balloon-based treatment that widens a narrowed coronary artery, often followed by stent placement.

Alternate Names:

PTCA; balloon angioplasty

Percutaneous Umbilical Blood Sampling

An invasive procedure for fetal blood sampling from the umbilical vein under ultrasound guidance, typically performed after 18-20 weeks.

Percutaneously Inserted Central Catheter - Infants

A medical procedure, device, or treatment involving percutaneously inserted central catheter - infants. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Perforant Pathway

A major hippocampal input pathway carrying signals from the entorhinal cortex to the dentate gyrus and related memory structures.

Perforated Bowel

A perforated bowel is a hole through the intestinal wall allowing contents to leak into the abdomen, causing peritonitis and potentially life-threatening sepsis.

Perforated Eardrum

Alternate index label; see *Ruptured eardrum (perforated eardrum)*.

Perforation

A hole or rupture through the wall of an organ or tissue.

Perforation Of A Viscus

Full-thickness breach of the wall of a hollow abdominal organ, most commonly the stomach, duodenum,

or colon, resulting in leakage of luminal contents into the peritoneal cavity. Causes include peptic ulcer disease, diverticulitis, appendicitis, inflammatory bowel disease, ischemia, and trauma.

Performance

Performance describes how effectively a person, organ, treatment, test, or system carries out a defined task; the relevant measure depends on the clinical context.

Performance Anxiety

Anxiety about sexual performance that can interfere with sexual function.

Performance Status

A standardized measure of a patient's functional ability and activity level, commonly recorded with ECOG or Karnofsky scales. It helps estimate prognosis, determine treatment eligibility, and quantify change during cancer care.

Performed Diagnosis

A diagnosis established after clinical assessment and appropriate testing identifies the condition explaining a patient's findings; it should be distinguished from a symptom or provisional impression.

Performed Histopathology

Microscopic examination of tissue that has been processed and stained to identify disease, tumor type, inflammation, or structural change.

Perfusion

Passage of blood or another fluid through a tissue or organ.

Perfusion Culture

A laboratory culture system in which cells or tissue receive continuously renewed nutrient medium, supporting controlled study of growth, metabolism, infection, or drug effects.

Perfusion Defect

An area receiving abnormally low blood flow, iden-

tified by an imaging or functional test.

Perfusion Imaging

Imaging tissue blood flow. CT perfusion, MR perfusion, PET perfusion. Used for stroke, tumor characterization.

Perfusion PET

Tracers: Rb-82, N-13 ammonia, O-15 water. Measure blood flow.

Perianal

Located around the anus; the term describes the skin, glands, pain, inflammation, abscesses, fistulas, and other findings in that region.

Perianal Abscess

A localized collection of pus near the anus caused by infected anal glands or surrounding tissue, producing severe throbbing pain, swelling, fever, and sometimes a fistula.

Perianal Glands

Small glands near the anus that contribute secretions to the local tissue; obstruction or infection can contribute to perianal abscesses and fistulas.

Perianal Streptococcal Cellulitis

A bacterial skin infection around the anus, usually caused by group A *Streptococcus*. It may cause bright-red skin, pain, itching, discharge, or pain during bowel movements.

Periapical Abscess

An acute or chronic localized collection of pus at the apex of a tooth's root, arising from pulpal infection and necrosis (usually due to untreated dental caries or trauma).

Periapical Granuloma

A chronic inflammatory lesion at the apex of a tooth, usually resulting from pulpal necrosis and persistent apical infection. It may be asymptomatic and is diagnosed through clinical, radiographic, and sometimes histopathologic assessment.

Periappendiceal

Periappendiceal means located around the appendix, as in inflammation, fluid, an abscess, or fat stranding near the appendiceal tissue.

Periarteritis Nodosa

An older name for polyarteritis nodosa, a necrotizing vasculitis of medium-sized arteries that can cause skin, nerve, muscle, kidney, and abdominal ischemic symptoms without typical glomerulonephritis.

Periarticular

Periarticular means located around a joint, including nearby tendons, bursae, ligaments, capsule, and soft tissues rather than the joint surface itself.

Pericardial

Relating to the pericardium, the sac and membranes surrounding the heart.

Pericardial Cavity

The narrow, fluid-filled space between the layers of the pericardium surrounding the heart.

Pericardial Disease (Pericarditis)

Alternate index label for Pericarditis.

Pericardial Effusion

Excess fluid in the pericardial sac surrounding the heart; rapid accumulation can cause tamponade.

Pericardial Effusion Etiology

Causes include infection, malignancy, kidney failure, hypothyroidism, autoimmune disease, inflammation after myocardial injury or cardiac surgery, radiation, trauma, and some medications. The likely cause varies with geography, age, medical history, and immune status.

Pericardial Fluid Culture

A microbiologic culture of fluid from the pericardial space to identify bacteria, fungi, or other organisms in suspected infectious pericarditis.

Pericardial Fluid Gram Stain

A microscopic stain of pericardial fluid that can rapidly suggest bacterial organisms, although a negative result does not exclude infection.

Pericardial Inflammation

Alternate index label; see *Pericarditis*.

Pericardial Knock

A high-pitched, early diastolic sound caused by sudden cessation of ventricular filling in constrictive pericarditis. It is heard at the left sternal border and may be confused with an S3 or opening snap. The knock is typically louder with inspiration and is a characteristic finding in constrictive pericarditis.

Pericardial Mesothelioma

A rare malignant tumor arising from the mesothelial lining around the heart, which can cause effusion, constriction, or impaired cardiac filling.

Pericardial Metastases

Cancer deposits in the pericardium, which may cause inflammation or accumulation of fluid around the heart.

Pericardial Stripping

Surgical removal of diseased pericardium, usually performed to relieve constrictive pericarditis.

Pericardial Tamponade

Fluid accumulation in pericardial space compromising cardiac filling. Beck triad: hypotension, jugular venous distension, muffled heart sounds. Electrical alternans on ECG. Diagnosis: echocardiography. Treatment: emergent pericardiocentesis, pericardial window.

Pericardial Wall

The layered wall of the pericardial sac that surrounds and protects the heart.

Pericardial Window

A surgical procedure in which a portion of the pericardium is surgically excised to create a permanent

or semi-permanent communication between the pericardial space and the pleural cavity or subcutaneous tissues. Performed via a subxiphoid approach or video-assisted thoracoscopic surgery, it allows continuous internal drainage and absorption of pericardial fluid, preventing recurrent cardiac tamponade and providing tissue biopsy in patients with large, recurrent, or malignant pericardial effusions.

Alternate Names:

Subxiphoid Pericardiostomy; Pleuropericardial Window; Pericardiostomy

Pericardiectomy

Pericardiectomy removes part or all of the sac around the heart, usually to treat severe constrictive pericarditis.

Pericardiocentesis

Pericardiocentesis is needle or catheter drainage of fluid from the pericardial sac, usually to relieve cardiac tamponade or diagnose an effusion; bleeding, arrhythmia, coronary injury, and pneumothorax are possible complications. Cardiac tamponade or diagnostically for pericardial effusion. The subxiphoid approach is most commonly used, with the needle directed toward the left shoulder under echocardiographic guidance.

Pericardiostomy

A surgical opening made in the pericardium to drain fluid or prevent recurrent accumulation around the heart.

Pericarditis

Inflammation of the sac around the heart. It often causes sharp chest pain that worsens with deep breathing or lying down and improves when sitting up and leaning forward. It may be caused by infections, autoimmune diseases, heart injury, kidney disease, cancer, or some medicines. Severe cases can cause fluid buildup around the heart.

Alternate Names:

Inflamed Pericardium Pericardial Inflammation Pericardial Disease (Pericarditis)

Post-Myocardial Infarction Pericarditis

Inflammation of the sac around the heart that occurs after a heart attack. It may begin soon after the heart attack or develop later as part of a delayed immune reaction.

Pericarditis - Constrictive

A chronic condition in which a thickened, scarred, or calcified pericardium restricts ventricular filling and causes systemic venous congestion and heart-failure symptoms.

Pericardium

The double-layered sac surrounding the heart and the roots of the great vessels.

Perichondrial Hematoma (Cauliflower Ear)

Alternate index label for Cauliflower Ear.

Perichondritis

Perichondritis is inflammation or infection of tissue surrounding cartilage, often affecting the outer ear after trauma or piercing and potentially damaging cartilage.

Perichondritis Of The Ear

Infection or inflammation of the auricular perichondrium, often after trauma, piercing, or surgery. *Pseudomonas aeruginosa* is an important cause; delayed treatment can produce cartilage necrosis and permanent deformity.

Perichondrium

A vascular connective-tissue sheath surrounding most cartilage, with an outer fibrous layer and an inner layer containing cartilage-forming cells. It supports cartilage nutrition, growth, and repair but is absent from articular cartilage and fibrocartilage.

Pericoronitis

Inflammation of the soft tissue around the crown of a partially erupted tooth, most often a mandibular third molar. Food retention and bacterial accumulation may cause pain, swelling, trismus, fever, or

spread into deeper spaces.

Pericytes

Pericytes are contractile support cells wrapped around small blood vessels that help regulate capillary stability, blood flow, repair, and the blood-brain barrier.

Periductal Mastitis

Inflammation around the major subareolar ducts, often associated with smoking, causing a painful periareolar mass, discharge, or recurrent breast abscess.

Perifosine

Perifosine is an investigational alkylphospholipid that affects cell-membrane signaling, including AKT pathways, and has been studied in cancer treatment.

Perillyl Alcohol

Perillyl alcohol is a monoterpene found in some plant oils and studied experimentally for effects on cell signaling and cancer growth.

Perilymph

Fluid in the bony labyrinth of the inner ear that surrounds the membranous labyrinth and helps transmit sound and balance signals.

Perilymphatic Fistula

Abnormal communication between the inner ear fluid spaces and the middle ear or mastoid, often after trauma, barotrauma, or surgery. It may cause episodic vertigo, hearing loss, tinnitus, or pressure-induced symptoms and requires specialist evaluation.

Perimenopausal

Perimenopausal describes the years of hormonal and menstrual transition before and around menopause, when cycles, bleeding, hot flashes, sleep, and fertility may change.

Perimenopause

The transition time in a woman's life that begins

when ovaries produce less estrogen and menstruation becomes less frequent, and ends when the ovaries no longer produce eggs and menstruation stops.

Perinatal

Perinatal refers to the period around birth, conventionally including late pregnancy and the first days or weeks after delivery depending on the definition used.

Perinatal Asphyxia

Impaired oxygenation or ventilation before, during, or shortly after birth that may cause neonatal encephalopathy, organ injury, or death. Immediate newborn assessment and resuscitation are essential.

Perinatal Death

Death of a fetus or newborn during the late pregnancy, birth, or early newborn period; the exact time range varies by definition.

Perinatal Psychiatry

The subspecialty focused on psychiatric disorders during pregnancy and the postpartum period. Common conditions include perinatal depression, perinatal anxiety, perinatal OCD, and postpartum psychosis. Medication selection must balance maternal mental health with fetal or neonatal exposure risks.

Perinatal Transmission

Transmission of an infection or condition from mother to infant during pregnancy, labor, delivery, or breastfeeding, as with HIV, hepatitis B, syphilis, or certain genetic and congenital disorders.

Perinatologist

A perinatologist is an obstetrician specializing in high-risk pregnancy, fetal medicine, prenatal diagnosis, and complications affecting the pregnant person or fetus.

Perindopril

Perindopril is an ACE inhibitor used for high blood pressure and cardiovascular risk reduction; it can cause cough, high potassium, or rarely angioedema.

Perineal

Relating to the perineum, the region between the external genitalia and the anus.

Perineal Laceration

Tearing of the perineal tissues during vaginal delivery. The injury is graded by depth: it may involve skin, perineal muscle, the anal sphincter, or the rectal mucosa.

Perineal Laceration Classification

Obstetric perineal tears are graded by depth: first degree involves skin, second degree involves perineal muscle, third degree involves the anal sphincter complex, and fourth degree extends through the rectal mucosa. Accurate examination and repair reduce long-term pelvic-floor and continence complications.

Perineal Protection

Techniques used during the second stage to minimize perineal trauma (lacerations, episiotomy). Manual protection: the attendant's hand applies gentle pressure to the fetal head to control the speed of delivery (not too fast), supporting the perineum and flexing the fetal head.

Perineal Tear

Laceration of the perineum during vaginal delivery, classified by degree: first degree (skin only), second degree (perineal muscles), third degree (sphincter complex involvement), and fourth degree (rectal mucosa). Most deliveries involve first or second degree tears. Third and fourth degree tears risk fecal incontinence.

Perineum

The area between the external genitalia and the anus. It contains skin, muscles, nerves, and connective tissue and may be affected by injury, infection, childbirth, or surgery.

Perineural

Perineural means around a nerve; the term may describe normal connective tissue, inflammation, tumor spread, or imaging findings surrounding a nerve.

Perineuronal Satellitosis

Clustering of glial cells around a neuron, seen microscopically in some brain tumors, inflammatory conditions, and reactive processes; its significance depends on the surrounding pathology.

Perinuclear Space

The narrow compartment between the inner and outer nuclear membranes, continuous with the endoplasmic-reticulum lumen and involved in nuclear-envelope structure.

Periodic Abstinence

Periodic abstinence is avoiding intercourse during estimated fertile days to reduce pregnancy likelihood; cycle timing is imperfect and does not prevent infections.

Periodic Acid

An oxidizing acid used in histochemistry, especially the periodic-acid-Schiff reaction, which highlights carbohydrates, mucosubstances, fungi, and basement membranes.

Periodic Breathing

Periodic breathing consists of repeated cycles of shallow or absent breathing followed by deeper breaths, occurring in some infants, during sleep, or with neurologic or cardiopulmonary disease.

Periodic Fever, Aphthous Stomatitis, Pharyngitis, Adenitis Syndrome

A recurrent fever syndrome, usually beginning in childhood, characterized by periodic fever with some combination of mouth ulcers, sore throat, and swollen neck lymph nodes; it is commonly abbreviated PFAPA.

Periodic Limb Movement Disorder

Repeated involuntary limb movements during sleep that may disrupt rest.

Alternate Names:

PLMD Sleep Disorder Periodic Limb Movement (Periodic Limb Movement Disorder)

Periodic Paralysis Syndrome

Periodic paralysis syndromes are rare inherited or acquired disorders causing episodic muscle weakness associated with abnormal potassium handling or thyroid disease. Attacks may follow rest after exercise, meals, or stress; ECG, potassium testing, and genetic or endocrine evaluation guide care.

Periodization

Planned variation of exercise volume and intensity over time.

Periodontal Abscess

A localized accumulation of pus within periodontal tissues, commonly associated with a periodontal pocket, foreign material, or impaired drainage. It may cause pain, swelling, suppuration, tooth mobility, and systemic symptoms and requires urgent dental assessment.

Periodontal Cyst

A periodontal cyst is a fluid-filled epithelial-lined lesion near the tooth-supporting tissues, with effects ranging from incidental discovery to local bone destruction.

Periodontal Disease

Inflammatory or infectious disease of the tissues supporting the teeth, including gingivitis and periodontitis.

Periodontal Disease, Gingivitis

Alternate index label; see *Gingivitis*.

Periodontal Disease, Periodontitis

Alternate index label; see *Periodontitis*.

Periodontal Dressing

A medical procedure, device, or treatment involving periodontal dressing. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Periodontal Flap Surgery

Surgical access for deep debridement. Flap reflected, calculus removed, bone contoured, flap repositioned.

Periodontal Ligament

Connective tissue attaching tooth to bone. Contains collagen fibers, cells, blood vessels, nerves. Provides proprioception, shock absorption. Expanded in bruxism.

Periodontal Pocket

A pathologically deepened gingival sulcus caused by apical migration of the junctional epithelium and loss of periodontal attachment. Pocket depth and clinical attachment level are measured with a periodontal probe and interpreted with bleeding, inflammation, and radiographic bone loss.

Periodontal Probing

Measurement of the depth of the gingival sulcus/pocket using a periodontal probe with millimeter markings. Normal: 1–3 mm. Periodontal pockets: ≥ 4 mm (indicates attachment loss and periodontitis). Full-mouth probing at 6 sites per tooth: mesio-buccal, mid-buccal, disto-buccal, mesio-lingual, mid-lingual, disto-lingual.

Periodontics

The dental specialty concerned with the gums and supporting tissues of the teeth.

Periodontist

A periodontist is a dentist specializing in prevention, diagnosis, and treatment of gum disease and supporting tissues, including periodontal surgery and implants.

Periodontitis

A chronic inflammatory disease caused by dysbiotic dental plaque that destroys the periodontal ligament and alveolar bone supporting the teeth. It can cause bleeding gums, periodontal pockets, tooth mobility, recession, bad breath, and tooth loss; severity and progression vary and require dental evaluation.

Alternate Names:

Periodontal Disease, Periodontitis

Periodontitis (Gum Disease)

Alternate index label for Gum Disease.

Periodontium

The tissues supporting a tooth: gingiva, periodontal ligament, cementum, and alveolar bone.

Periodontium Disease

Disease of the tissues supporting the teeth, including gingiva, periodontal ligament, cementum, and alveolar bone; periodontitis can cause attachment loss and tooth loss.

Perionychium

The skin and soft tissue surrounding a fingernail or toenail, including the nail folds and nearby cuticle; infection or inflammation causes paronychia.

Perioperative

Perioperative means relating to the period before, during, and after an operation, including preparation, anesthesia, recovery, and complication prevention.

Perioral Dermatitis

A papulopustular eruption around the mouth, sparing the vermilion border. It is more common in young women and associated with topical corticosteroid use. Other triggers include fluoridated toothpaste, cosmetics, and inhaler steroids.

Periorbital

Located around the eye socket or eyelids. Periorbital swelling may result from injury, allergy, infection, or fluid retention.

Periorbital Cellulitis

A bacterial infection of the eyelid and tissues around the eye that causes redness, swelling, and tenderness. It must be distinguished from orbital cellulitis, which can threaten vision and spread more deeply.

Periorbital Edema

Swelling of the tissues around the eye, caused by allergy, infection, trauma, venous or lymphatic congestion, kidney disease, thyroid eye disease, or other systemic conditions.

Periorbital Infection

Infection of the tissues around the eye; it can cause swelling and redness and must be distinguished from orbital cellulitis, which threatens vision.

Periosteal Osteosarcoma

A surface osteosarcoma arising beneath the periosteum, usually producing cartilage and affecting the shaft of a long bone. It is generally intermediate grade and requires specialist orthopedic-oncology management.

Periosteum

A vascular connective-tissue membrane covering the outer surface of bone except at articular cartilage.

Periostitis

Inflammation of the periosteum, the dense, richly vascularized and innervated fibrous connective tissue membrane covering the outer cortical surfaces of bones. Manifests as localized bone tenderness, swelling, and characteristic radiographic periosteal reaction (e.g., solid, laminated "onion-skin", or spicular "sunburst" patterns). Etiologies range from chronic mechanical overuse (medial tibial stress syndrome / "shin splints" in athletes), acute or chronic infectious osteomyelitis, vascular stasis in severe chronic venous insufficiency, and systemic diseases such as secondary syphilis, hypertrophic osteoarthropathy (associated with lung carcinoma), and osteogenic malignancies.

Peripancreatic Malignancy

Neoplastic process involving tissues surrounding the pancreas, including primary pancreatic adenocarcinoma with peripancreatic extension and metastatic disease from adjacent or distant organs. Often presents at an advanced stage with abdominal pain, weight loss, obstructive jaundice, and new-onset diabetes.

Peripartum

The period immediately before, during, and shortly after childbirth.

Peripartum Cardiomyopathy

A form of dilated cardiomyopathy occurring in the last month of pregnancy or within 5 months postpartum, with no other identifiable cause. Risk factors include advanced maternal age, multiple gestations, preeclampsia, and African American race. Presents with heart failure symptoms.

Peripartum Cardiomyopathy Prognosis

Recovery of ventricular function varies widely. Prognosis is influenced by the initial ejection fraction, ventricular size, symptoms, treatment, complications, and follow-up; persistent dysfunction requires ongoing cardiac care.

Peripheral

Peripheral means away from the center or central nervous system; in clinical use it may describe nerves, vessels, lesions, circulation, or symptoms outside the brain, spinal cord, or central structures.

Peripheral Arterial Line - Infants

A small catheter placed in an infant's artery to continuously measure blood pressure and obtain blood samples. It requires careful monitoring for bleeding, infection, or reduced blood flow to the limb.

Peripheral Artery Bypass - Leg

Surgery that creates a new route around a blocked leg artery to improve blood flow. A vein or artificial graft may be used as the bypass.

Peripheral Artery Disease

Narrowing of arteries supplying the limbs, usually caused by atherosclerosis. It may cause exertional leg pain relieved by rest, reduced pulses, nonhealing wounds, or critical limb ischemia. Also called peripheral arterial disease.

Alternate Names:

Peripheral Arterial Disease; PAD

Peripheral Artery Disease - Legs

Peripheral artery disease of the legs is atherosclerotic reduction in arterial blood flow causing exer-

tional claudication, diminished pulses, nonhealing wounds, or critical limb ischemia.

Peripheral Blood

Peripheral blood is blood obtained from a vein or capillary rather than directly from an organ or marrow, used for counts, smears, chemistry, and other testing.

Peripheral Blood Smear

Microscopic examination of a stained blood smear (Wright-Giemsa stain).

Peripheral Blood Stem Cells

Hematopoietic stem cells collected from circulating blood, usually after mobilization from the bone marrow; they can be used for autologous or allogeneic stem-cell transplantation.

Peripheral Edema

Swelling due to accumulation of fluid in the interstitial space of the limbs, most commonly the lower extremities.

Peripheral Intravenous Line - Infants

A small tube placed in an infant's vein, usually in the hand, foot, or scalp, to give medicines or fluids. The site should be checked for swelling, leakage, or infection.

Peripheral Ischemia

Inadequate arterial blood flow to an arm, leg, digit, or other peripheral tissue, causing pain, pallor, coolness, weak pulses, numbness, ulcers, or tissue loss.

Peripheral Nerve

A nerve outside the brain and spinal cord that carries motor, sensory, and autonomic signals between the central nervous system and the rest of the body.

Peripheral Nerve Blocks

Peripheral nerve blocks provide targeted anesthesia and analgesia for specific surgical sites. Ultrasound-guided blocks improve success rates and reduce complications.

Peripheral Nerve Injuries

Damage to nerves outside the brain and spinal cord caused by trauma, compression, stretching, disease, or toxins. It may cause pain, numbness, weakness, or loss of function; recovery depends on the severity and location of the injury.

Peripheral Nerve Tumors

Tumors that arise from or around nerves outside the brain and spinal cord.

Peripheral Nervous System

The portion of the nervous system consisting of nerves and ganglia outside the brain and spinal cord. The PNS connects the central nervous system (CNS) to peripheral tissues and organs. Components: (1) Cranial nerves (12 pairs)—emerging from the brain/brainstem, innervating head, neck, and viscera.

Peripheral Nervous System

The peripheral nervous system includes nerves and ganglia outside the brain and spinal cord and carries sensory, motor, and autonomic signals between the body and central nervous system.

Peripheral Neuritis (Peripheral Neuropathy)

Alternate index label for Peripheral Neuropathy.

Peripheral Neurofibromatosis

Alternate index label; see *Neurofibromatosis type 1*.

Peripheral Neuropathy

A disorder of peripheral nerves causing sensory, motor, and/or autonomic symptoms in a distal, symmetric 'stocking-glove' pattern.

Alternate Names:

Neuropathy, Peripheral Peripheral Neuritis (Peripheral Neuropathy)

Peripheral Neuropathy Diabetic (Diabetic Neuropathy)

Alternate index label for Diabetic Neuropathy.

Peripheral Oedema

Alternate spelling of Peripheral Edema; swelling caused by excess interstitial fluid, usually in the limbs.

Peripheral Vascular Disease

Disease involving narrowed or blocked blood vessels outside the heart and brain.

Alternate Names:

Peripheral arterial disease PVD (Peripheral Vascular Disease)

Peripheral Vascular Disease

Alternate index label for Peripheral Vascular Disease.

Peripheral Vision

The portion of the visual field outside the direct line of sight.

Alternate Names:

Side vision

Peripherally Inserted Central Catheter

A long, flexible catheter inserted into the basilic or cephalic vein and advanced so the tip sits in the superior vena cava. Inserted at the bedside, often by a specialized PICC nurse, with ultrasound and ECG or X-ray confirmation. Indications: long-term IV antibiotics (>2 weeks), TPN, chemotherapy, and frequent blood draws.

Peripherally Inserted Central Catheter - Dressing Change

A medical procedure, device, or treatment involving peripherally inserted central catheter - dressing change. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Peripherally Inserted Central Catheter - Flushing

A medical procedure, device, or treatment involving peripherally inserted central catheter - flushing. Its purpose, risks, and follow-up depend on the specific intervention and patient.

Peripherally Inserted Central Catheter - Insertion

A medical procedure, treatment, device, or care topic related to peripherally inserted central catheter - insertion; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Periplasm

The periplasm is the space between the inner and outer membranes of Gram-negative bacteria, containing enzymes, transport proteins, and structural components.

Periprosthetic Joint Infection

Infection involving a joint prosthesis and adjacent tissues. Diagnosis integrates clinical findings, inflammatory markers, joint-fluid analysis and culture, and imaging; treatment may require debridement, implant retention, staged exchange, or resection arthroplasty.

Perirenal Abscess

A perirenal abscess is a collection of pus around the kidney, usually from spread of a urinary or kidney infection, causing fever, flank pain, and possible sepsis.

Perirenal Extramedullary Hematopoiesis

Formation of blood cells in the tissue surrounding the kidney, occurring as a compensatory response to chronic severe anemia such as myelofibrosis or thalassemia major.

Perirenal Hematoma

A collection of blood around the kidney, usually caused by trauma, a bleeding lesion, a procedure, or anticoagulation.

Peristalsis

Peristalsis is coordinated, wave-like contraction of smooth muscle that propels food, fluid, or other contents through the gastrointestinal tract and some ducts.

Peritoneal

Peritoneal means relating to the peritoneum, the thin membrane lining the abdominal cavity and covering many abdominal organs.

Peritoneal Carcinomatosis

Diffuse metastatic seeding of the peritoneal cavity from gastrointestinal or ovarian malignancies. Presents with ascites, bowel obstruction, and cachexia. Diagnosed by CT and paracentesis showing malignant cells. Prognosis is generally poor.

Peritoneal Cavity

The potential space between the parietal and visceral peritoneum surrounding abdominal and pelvic organs; fluid, blood, pus, or tumor cells can accumulate there.

Peritoneal Dialysis

Dialysis using the peritoneal membrane as the exchange surface. Continuous ambulatory peritoneal dialysis uses manual exchanges, whereas automated peritoneal dialysis uses a cycler, often overnight. It can provide home-based therapy and may offer hemodynamic stability. Complications include peritonitis, catheter infection, inadequate ultrafiltration, and fluid imbalance. A Tenckhoff catheter is commonly used.

Peritoneal Dialysis Catheter

Flexible tube surgically placed into the peritoneal cavity for peritoneal dialysis. The Tenckhoff catheter is the most commonly used type with one or two cuffs that promote tissue ingrowth and anchor the catheter. Placement can be done surgically, laparoscopically, or percutaneously under radiological guidance.

Peritoneal Diseases

Disorders affecting the peritoneum, including infection, inflammation, malignancy, hemorrhage, adhesions, and fluid accumulation.

Peritoneal Fluid

Peritoneal fluid is the small amount of lubricating fluid in the abdominal cavity; excess or infected

fluid may cause ascites, peritonitis, or diagnostic findings.

Peritoneal Fluid Analysis

Laboratory examination of ascitic fluid, including cell count, albumin, protein, culture, and selected chemical tests, to determine the cause of ascites and detect infection or malignancy.

Peritoneal Fluid Culture

Culture of ascitic fluid to detect microorganisms, particularly in suspected spontaneous bacterial peritonitis or secondary peritoneal infection.

Peritoneal Infection

Infection of the lining of the abdominal cavity, often called peritonitis. Causes include a ruptured appendix or bowel, abdominal surgery, pelvic infection, or contamination during dialysis. Severe abdominal pain, fever, vomiting, or a rigid abdomen requires urgent assessment.

Peritoneal Lavage

Peritoneal lavage is washing or sampling the abdominal cavity with sterile fluid, historically used to detect internal bleeding or contamination and sometimes used therapeutically.

Peritoneal Necrosis

Death of peritoneal tissue caused by ischemia, severe infection, inflammation, trauma, or tumor, potentially leading to peritonitis, adhesions, sepsis, or organ failure.

Peritoneal Sac

The peritoneal sac is the potential space enclosed by the peritoneum, containing lubricating fluid and allowing abdominal organs to move with minimal friction.

Peritoneum

The serous membrane lining the abdominal cavity and covering the abdominal organs. Parietal peritoneum lines the abdominal wall (innervated by somatic nerves—pain is well-localized); visceral

peritoneum covers organs (innervated by visceral afferents—pain is dull, poorly localized).

Peritoneum Neoplasm

A peritoneal neoplasm is an abnormal growth in the lining of the abdominal cavity and may be primary or spread from another cancer.

Peritonitis

Peritonitis is inflammation or infection of the peritoneum, often from perforation, surgery, ascites infection, or abdominal disease, and may cause severe pain, ileus, sepsis, and shock. organ, appendicitis, abdominal trauma, or infection of ascitic fluid. Severe abdominal pain, guarding, fever, vomiting, or shock requires urgent medical evaluation.

Peritonitis - Secondary

Secondary peritonitis is infection of the abdominal lining caused by a perforated or injured gastrointestinal or other abdominal organ and may rapidly lead to sepsis.

Peritonitis - Spontaneous Bacterial

Infection of ascitic fluid without an evident surgically treatable intra-abdominal source, most often in advanced cirrhosis or other severe liver disease.

Peritonitis In PD

Infection or inflammation of the peritoneum in a person receiving peritoneal dialysis. Cloudy effluent, abdominal pain, or fever requires prompt evaluation of the effluent and other causes. Treatment usually includes antibiotics selected for the suspected organisms; recurrent or refractory infection may require catheter removal.

Peritonsillar Abscess

A collection of pus beside the tonsil, usually developing as a complication of tonsillar infection. See Quinsy.

Periungual Fibroma

Benign, flesh-colored or pink fibrous tumors arising from the nail fold or nail bed, presenting as firm, exo-

phytic nodules that may distort the nail plate. They occur sporadically or in association with tuberous sclerosis complex. When multiple periungual fibromas are present, they raise suspicion for tuberous sclerosis.

Periurethral

Situated, occurring, or distributed immediately surrounding the female or male urethra. Clinically significant in urology and gynecology, describing periurethral glands (such as the Skene glands in females, which can develop infection, abscess, or urethral diverticula), periurethral collagen or bulking agent injections for female stress urinary incontinence, and periurethral prostate zones (the transition zone surrounding the proximal urethra, which undergoes progressive hyperplasia in benign prostatic hyperplasia [BPH], producing lower urinary tract obstructive symptoms).

Periventricular Leukomalacia

Ischemic white matter injury in the premature infant brain, a major cause of cerebral palsy. Risk factors include prematurity, hypotension, and inflammation. Diagnosis by cranial ultrasound showing periventricular echodensities or cystic changes. Prevention: avoiding hypotension, maintaining normothermia, and managing infection.

Perivitelline Space

The perivitelline space is the narrow region between an egg's plasma membrane and surrounding vitelline or zona material, important in fertilization and early development.

Perleche

Angular cheilitis, inflammation and cracking at the corners of the mouth.

Permanent Teeth

The adult set of teeth that replaces the deciduous teeth.

Permease

A permease is a membrane transport protein that facilitates movement of a specific molecule or ion

across a cell membrane.

Permethrin

A topical pyrethroid medicine used mainly to treat scabies and head lice.

Permissive Hypercapnia

A ventilator strategy that allows PaCO₂ to rise above normal (and pH to fall) in exchange for lung-protective ventilation with lower tidal volumes and airway pressures. It is used in severe ARDS when achieving normal PaCO₂ would require harmful ventilation parameters.

Permissiveness

Permissiveness is the ability of a cell or host environment to allow a biological process, such as viral replication, gene expression, or tissue response, to occur.

Pernicious Anaemia

An autoimmune cause of vitamin B12 deficiency due to impaired intrinsic-factor-mediated absorption.

Pernicious Anemia

A type of vitamin B12 deficiency anemia caused by autoimmune destruction of gastric parietal cells, leading to lack of intrinsic factor and impaired B12 absorption in the terminal ileum.

Alternate Names:

Addisons Anemia (Pernicious Anemia)

Pernio

Alternate index label; see *Chilblains*.

Peroneal

Relating to the fibula or the muscles and nerve on the outer side of the lower leg.

Peroneal Muscular Atrophy

Alternate index label; see *Charcot-Marie-Tooth disease*.

Peroneal Nerve

The common fibular nerve, a branch of the sciatic

nerve that winds around the fibular neck and supplies sensation and muscles responsible for ankle dorsiflexion and eversion.

Peroxidase

An enzyme that uses hydrogen peroxide or another peroxide to oxidize a substrate; thyroid peroxidase is essential for thyroid-hormone synthesis.

Peroxisomal Disorder

An inherited disorder of peroxisome formation or enzyme function that disrupts lipid and plasmalogen metabolism and may cause neurologic, hepatic, skeletal, or sensory disease.

Peroxisome

Single-membrane organelles containing oxidative enzymes (catalase, urate oxidase, D-amino acid oxidase) that produce and degrade hydrogen peroxide (H₂O₂).

Peroxisome Proliferation

An increase in the number or size of peroxisomes, organelles involved in fatty-acid oxidation and reactive-oxygen metabolism, triggered by selected chemicals, hormones, or metabolic states.

Perphenazine

A first-generation (typical) phenothiazine antipsychotic used for schizophrenia and, in some settings, severe nausea or vomiting. Adverse effects can include sedation, orthostatic hypotension, extrapyramidal symptoms, tardive dyskinesia, and neuroleptic malignant syndrome.

Perseveration

Involuntary repetition of a word, idea, movement, or response after the appropriate stimulus has changed, associated with frontal or basal-ganglia disease, aphasia, delirium, or psychiatric illness.

Persistent Asthma

Asthma requiring daily controller therapy. Mild persistent: symptoms > 2 days/week but not daily. Moderate persistent: daily symptoms, nighttime

awakenings > 1/week. Severe persistent: symptoms throughout the day, frequent nighttime awakenings, SABA use several times/day, extremely limited activity.

Persistent Cloaca

Persistent cloaca is a congenital malformation in which the rectum, vagina, and urinary tract join into one channel, requiring specialist evaluation and reconstructive care.

Persistent Depressive Disorder

formerly called dysthymia, a chronic depressive mood lasting at least two years in adults (one year in children and adolescents).

Persistent Organic Pollutant

POCPs: organochlorines, PCBs, dioxins. Bioaccumulate, long-range transport, persistent. Stockholm Convention bans/restricts.

Persistent Pulmonary Hypertension Of The Newborn

Failure of the normal fall in pulmonary vascular resistance after birth, causing right-to-left shunting and hypoxaemia. Management requires urgent neonatal respiratory and circulatory support.

Persistent Pupillary Membranes

Remnants of the fetal tunica vasculosa lentis that fail to regress after birth, appearing as fine pigmented strands bridging across the pupil or attaching from the iris collarette to the lens capsule. They are usually benign and asymptomatic but may rarely obstruct the visual axis requiring surgical or laser treatment.

Person With Disability

A person with disability has a long-term physical, sensory, intellectual, or mental-health impairment that may interact with barriers to limit participation; care should be accessible and person-centered.

Personal Emergency Response System

A wearable or nearby device that lets a person

summon help during an emergency.

Alternate Names:

Medical alert system

Personal Protective Equipment

PPE: gloves, gowns, masks, eye protection. Used based on transmission-based precautions.

Personality

Personality is the relatively enduring pattern of thoughts, emotions, motivations, and behaviors through which a person experiences and responds to the world.

Personality Change

Personality change is a new or marked alteration in behavior, emotional responses, judgment, or interpersonal style, which may reflect neurologic disease, psychiatric illness, substances, or medical conditions.

Personality Disorder

A group of mental disorders characterized by enduring maladaptive patterns of thinking, feeling, and behaving that deviate from cultural expectations and cause functional impairment. Cluster A (odd/eccentric): paranoid, schizoid, schizotypal. Cluster B (dramatic/erratic): antisocial, borderline, histrionic, narcissistic.

Personality Disorder Antisocial (Antisocial Personality Disorder)

Alternate index label for Antisocial Personality Disorder.

Personality Disorder, Antisocial

Alternate index label; see *Antisocial personality disorder*.

Personality Disorder, Borderline

Alternate index label; see *Borderline personality disorder*.

Personality Disorder, Narcissistic

Alternate index label; see *Narcissistic personality*

disorder.

Personality Disorder, Schizoid

Alternate index label; see *Schizoid personality disorder*.

Personality Disorder, Schizotypal

Alternate index label; see *Schizotypal personality disorder*.

Personality Disorders

Personality disorders are enduring patterns of thought, emotion, and behavior that differ markedly from cultural expectations and cause distress or impaired functioning.

Personality Test

A psychological assessment using structured questions or stimuli to describe enduring traits, patterns, or possible personality pathology; results require validated instruments and clinical interpretation.

Perspiration

Sweat released by skin glands. Eccrine glands mainly help cool the body, while apocrine glands contribute to odor after their secretions are changed by skin bacteria; unusually heavy sweating is called hyperhidrosis.

Persuasive Communication

Communication intended to influence beliefs, decisions, or behavior through information, reasoning, emotion, or framing; healthcare use should preserve informed choice and avoid coercion.

Pertussis

Whooping cough, a contagious respiratory infection caused by *Bordetella pertussis*.

Pertussis Toxin

A toxin produced by *Bordetella pertussis* that disrupts cell signaling and contributes to the cough and systemic effects of whooping cough.

Pertuzumab

A monoclonal antibody targeting the HER2 receptor and blocking its dimerization, used with other medicines for selected HER2-positive cancers.

Pervasive Developmental Disorders

An obsolete group term for disorders involving social communication and restricted repetitive behavior, including autism-spectrum conditions.

Pes

Pes means foot; it is also used in terms describing foot shape or deformity, such as pes planus or pes cavus.

Pes Anserinus

The combined tendon insertion of the sartorius, gracilis, and semitendinosus muscles on the antero-medial proximal tibia; inflammation there is pes-anserine bursitis.

Pes Planus

Alternate index label; see *Flatfeet*.

Pes Planus (Flatfoot (Pes Planus))

Alternate index label for Flatfoot (Pes Planus).

Pessaries

Devices inserted into the vagina to support pelvic organs in cases of pelvic organ prolapse (cystocele, rectocele, uterine prolapse). Types: ring pessary (most common, with or without supportive membrane), shelf pessary (for severe prolapse), cube pessary (suction-based), and Gellhorn pessary (for advanced prolapse).

Pessary

A device placed in the vagina to support pelvic organs or deliver medication.

Pessimism

A tendency to expect negative outcomes or interpret situations unfavorably.

Pestalotiopsis

A genus of plant-associated molds that can rarely cause eye, skin, or subcutaneous infection in humans.

Pesticide

A substance used to control insects, weeds, fungi, or other pests; exposure can cause poisoning.

Pesticides

Pesticides involves toxic or environmental exposure to pesticides; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pesticides On Fruits And Vegetables

Pesticides on fruits and vegetables involves toxic or environmental exposure to pesticides on fruits and vegetables; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pestilence

Pestilence is an old term for a severe, widespread, often fatal infectious disease outbreak; modern usage usually specifies the pathogen and epidemiologic event.

Pet Allergy

An allergic reaction to proteins in an animal's skin flakes, saliva, or urine, commonly causing rhinitis, conjunctivitis, or asthma.

PET Scan

An imaging study using a positron-emitting tracer to measure metabolism, receptor activity, or tissue function.

PET Scan (Positron Emission Tomography)

Nuclear medicine imaging using radioactive tracers (most commonly FDG, a glucose analog) to detect metabolically active tissues.

Alternate Names:

Positron Emission Tomography

PET Scan For Breast Cancer

Positron-emission tomography, usually combined with computed tomography, used selectively to assess metabolically active breast cancer, staging, recurrence, or treatment response.

PET Scanner

Ring of detectors capturing coincidence events. BGO, LSO, or LYSO crystals. Time-of-flight PET improves signal-to-noise.

PET Tracers

18F-FDG: glucose metabolism (cancer, infection, inflammation). 18F-fluoride: bone turnover. 11C-metionine: protein synthesis. 68Ga-DOTATATE: somatostatin receptors. 68Ga-PSMA: prostate cancer.

PET/CT

Combined PET and CT imaging. CT provides anatomical localization and attenuation correction. Essential for accurate interpretation. Low-dose CT for attenuation correction; diagnostic CT for detailed anatomy.

PET/MRI

Combined PET and MRI. Superior soft tissue contrast, no ionizing radiation from CT. Useful for neurology, oncology, cardiology. Longer scan time, higher cost.

Petechia

A petechia is a tiny nonblanching red, purple, or brown spot caused by bleeding from small vessels into the skin or mucosa; numerous or rapidly spreading petechiae need assessment.

Petechiae

Tiny nonblanching red or purple spots caused by small areas of bleeding into the skin or mucosa.

Petit Mal

An older term for absence seizures, brief episodes of impaired awareness with characteristic generalized EEG activity.

Petit Mal Seizure

Alternate index label; see *Absence seizure*.

Petri Dish

A Petri dish is a shallow lidded laboratory container used to culture microorganisms or cells and to hold specimens or small samples.

Petrissage

A massage technique involving kneading or lifting soft tissues.

Petrolatum

A semisolid mixture of hydrocarbons used as a protective skin ointment and moisture barrier.

Petroleum Jelly Overdose

Petroleum jelly overdose involves toxic or environmental exposure to petroleum jelly overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Petrositis

Inflammation or infection of the petrous portion of the temporal bone, usually arising from complicated otitis media and potentially affecting nearby cranial nerves.

Petrous

Dense and rock-like; usually referring to the petrous part of the temporal bone.

Pets And The Immunocompromised Person

People with weakened immunity can often keep pets but should reduce exposure to animal waste, scratches, bites, and unsafe food and follow individualized medical advice.

Peutz-Jeghers Syndrome

An autosomal dominant neurocutaneous disorder caused by germline loss-of-function mutations in the STK11 (LKB1) tumor suppressor gene on chromosome 19p13.

Alternate Names:

PJS

Peutz-Jeghers'S Syndrome

An autosomal dominant condition caused by STK11 gene mutations, characterized by hamartomatous polyps throughout the gastrointestinal tract and mucocutaneous hyperpigmented macules on the lips, buccal mucosa, and digits. The polyps may cause gastrointestinal bleeding, intussusception, and obstruction.

PEWS

Alternate name; see *Pediatric Early Warning Score*.

Pexophagy

Pexophagy is selective autophagic degradation of peroxisomes, allowing cells to remove damaged or excess organelles and recycle their components.

Peyer Patches

Aggregated lymphoid follicles in the mucosa and submucosa of the ileum that monitor intestinal contents and support immune defense.

Peyer'S Patches

Organized lymphoid tissue in the mucosa and submucosa of the ileum. These aggregates sample intestinal contents and contribute to immune surveillance of the gastrointestinal tract.

Alternate Names:

ileal lymphoid nodules

Peyronie Disease

Fibrous plaque in penis causing curvature, pain, erectile dysfunction. Treatment: observation, medications, injection therapy, surgery.

Peyronie'S Disease

A condition in which scar tissue in the penis causes curvature or pain during erection.

Peyronie'S Disease (Curvature Of The Penis)

Alternate index label for Curvature of the Penis.

Peyronie'S Disease

Scar tissue within the penis that can cause curvature and pain during erection.

Alternate Names:

Peyronie disease

Pfeiffer Syndrome

An autosomal dominant craniosynostosis syndrome caused by FGFR1 or FGFR2 mutations. Features: craniosynostosis (bilateral coronal), midface hypoplasia, proptosis, hypertelorism, broad thumbs and great toes (medial deviation), and variable syndactyly.

Alternate Names:

Syndrome Pfeiffer (Pfeiffer Syndrome)

PH

Measure of hydrogen ion concentration: $\text{pH} = -\log[\text{H}^+]$. Normal blood pH: 7.35-7.45. Acidemia: $\text{pH} < 7.35$. Alkalemia: $\text{pH} > 7.45$. Buffers minimize pH changes.

PH Meter

An instrument that measures acidity or alkalinity as pH in a solution or specimen; clinical applications include gastric, urinary, laboratory, and environmental measurements.

PH Monitoring

Measurement of acidity, usually in the esophagus, over time with a small sensor. It can help detect and measure acid reflux when symptoms or endoscopy do not give a clear answer.

Ph.D.

A Ph.D. is a doctoral research degree awarded after advanced study and an original dissertation; it does not by itself confer a clinical license.

Phacoemulsification

A cataract-surgery technique that uses ultrasound to break up and remove the cloudy lens through a small incision. An intraocular lens is usually implanted to restore focusing ability.

Phagocyte

A cell that engulfs and digests microorganisms, dead cells, or foreign particles.

Phagocytes

Immune cells that engulf and digest microorganisms, dead cells, and other particles.

Phagocytic Vesicle

A phagocytic vesicle, or phagosome, is the membrane-bound compartment formed when a cell engulfs a particle before fusion with lysosomes for digestion.

Phagocytosis

The process by which a cell surrounds, ingests, and digests a particle or microorganism.

Phagolysosome

A phagolysosome forms when a phagosome fuses with a lysosome, exposing engulfed microbes or debris to enzymes and toxic substances for destruction.

Phagophores

Phagophores are isolation membranes that expand around cytoplasmic material during autophagy and close to form autophagosomes.

Phalanges

The 56 tubular small long bones constituting the skeletal architecture of the digits (fingers and toes) in humans. Each human hand and foot contains 14 phalanges: two in the first digit (pollex/thumb and hallux/great toe, consisting of proximal and distal phalanges) and three in each of the four lateral digits (digits II–V, consisting of proximal, middle, and distal phalanges). Connected by hinge-type interphalangeal joints (proximal PIP and distal DIP joints) reinforced by collateral ligaments and volar plates; commonly fractured in manual trauma, affected by inflammatory erosions in rheumatoid arthritis (PIP synovitis) and osteoarthritis (Bouchard and Heberden nodes).

Phalanx

A single individual tubular bone of any digit of the hand or foot. Anatomically consists of an expanded base (articulating proximally), a slender shaft (body), and a distal head (trochlea) articulating with the adjacent phalanx or terminating in a flattened, spatulate apical tuft on distal phalanges. Distal phalanges support the overlying nail bed and dense fibrofatty pulp space of the digital pad, where closed space infection can lead to a painful digital felon.

Phalanx Unspecified

A phalanx is one of the bones of a finger or toe; “unspecified” indicates that the particular digit or proximal, middle, or distal segment is not identified.

Phalen Test

Wrist flexion test for carpal tunnel syndrome. Symptoms reproduced within 60 seconds.

Phallic Stage

A historical psychoanalytic developmental stage centered on genital awareness; it is not a modern disease classification.

Phantom

Device simulating human anatomy for quality control and calibration. Types: line source, uniform, Jaszczak, NEMA IU body phantom.

Phantom Limb

The sensation that an amputated limb is still present, sometimes with pain.

Alternate Names:

Phantom-limb sensation

Phantom Limb Pain

Painful sensations perceived in a limb that has been amputated, caused by persistent or reorganized activity in peripheral nerves and central sensory pathways. It may feel burning, shooting, cramping, or electric.

Phantom Pain

Pain perceived in a body part that has been amputated, caused by changes in peripheral nerves and central pain-processing pathways.

Phantosmia

Perception of a smell when no corresponding odor is present. It may occur with sinonasal disease, migraine, seizures involving olfactory networks, head injury, neurodegenerative disease, infection, or other neurological conditions.

Pharmaceutical Adjuvant

A pharmaceutical adjuvant is an inactive or auxiliary ingredient added to improve a medicine’s stability, absorption, preservation, delivery, or therapeutic effect.

Pharmaceutical Excipient

A pharmaceutical excipient is a non-active ingredient in a dosage form, such as a filler, binder, coating, solvent, preservative, or disintegrant.

Pharmaceutical Solutions

Liquid preparations in which one or more active pharmaceutical ingredients are dissolved in a suitable vehicle for administration, with concentration, sterility, stability, and route determining safe use.

Pharmacist

A healthcare professional trained in medicines, including their preparation, dispensing, safety, and use.

Pharmacodynamics

The study of what a drug does to the body, including its molecular targets, mechanism of action, dose-response relationship, therapeutic effects, and adverse effects.

Pharmacogenetic Testing

Testing for genetic variants that influence a person’s response, metabolism, efficacy, or toxicity of particular medicines and may support individualized prescribing.

Pharmacogenetics

Pharmacogenetics studies how inherited genetic variation alters a person's response to medicines, including efficacy, metabolism, and adverse-effect risk, to support individualized prescribing. toxicity), CYP2C9/VKORC1 (warfarin dosing), HLA-B*1502 (carbamazepine SJS in Asians), HLA-B*5701 (abacavir hypersensitivity), UGT1A1 (irinotecan toxicity).

Pharmacogenomic Testing

Testing that examines genetic variants affecting drug metabolism, transport, targets, or adverse reactions to help select or dose certain medicines.

Pharmacogenomics

The study of how genetic variations influence individual responses to drugs, including drug efficacy, toxicity, and metabolism. Genetic polymorphisms in drug-metabolizing enzymes, transporters, and receptors can lead to significant inter-individual variability in drug response.

Pharmacokinetics

The study of what the body does to a drug, including absorption, distribution, metabolism, and excretion, and how drug concentrations change over time.

Pharmacologic Substance

A pharmacologic substance is a chemical that produces a biological effect by interacting with cells, receptors, enzymes, transporters, or other targets.

Pharmacological Stress Test

Stress testing using medications (dobutamine, adenosine, regadenoson). For patients unable to exercise.

Pharmacologist

A pharmacologist studies how medicines and chemicals act in living systems, including mechanisms, metabolism, toxicity, and therapeutic effects.

Pharmacology

The study of drugs, their actions, uses, adverse

effects, and interactions.

Pharmacopoeia

A pharmacopoeia is an official reference specifying standards for identity, purity, strength, testing, and preparation of medicines.

Pharmacotherapy

Pharmacotherapy is treatment with medicines chosen to prevent, relieve, or control a disease or symptom.

Pharmacovigilance

The science and activities relating to the detection, assessment, understanding, and prevention of adverse drug reactions and other drug-related problems.

Pharmacy

The science and practice of preparing, supplying, and advising on medicines.

Pharmacy Facility

A pharmacy facility is an authorized location where medicines may be stored, prepared, dispensed, monitored, and supplied under applicable professional and safety standards.

Pharyngeal

Relating to or located in the pharynx, the fibromuscular conduit extending from the base of the skull to the lower border of the cricoid cartilage (level of C6), connecting the oral and nasal cavities to the larynx and esophagus. Divided anatomically into three regions: nasopharynx, oropharynx, and laryngopharynx (hypopharynx). Pathologies include acute infectious pharyngitis (viral or group A beta-hemolytic *Streptococcus*), pharyngeal diverticulum (Zenker diverticulum through Killian triangle), obstructive sleep apnea due to retropalatal pharyngeal collapse, and squamous cell carcinomas (frequently HPV-positive in the oropharynx).

Pharyngeal Carcinoma

Pharyngeal carcinoma is cancer arising in the throat

and may cause persistent sore throat, swallowing difficulty, hoarseness, or a neck mass.

Pharyngeal Disease

A disorder affecting the pharynx, including infection, inflammation, obstruction, trauma, neuromuscular dysfunction, or tumor.

Pharyngeal Hemorrhage

Bleeding from the pharynx caused by trauma, surgery, infection, vascular lesions, tumor, or severe mucosal disease and potentially threatening the airway.

Pharyngeal Mucositis

Inflammation and ulceration of the throat lining, commonly caused by radiation, chemotherapy, infection, trauma, or medications and producing pain, difficulty swallowing, and reduced intake.

Pharyngeal Stenosis

Narrowing of the pharyngeal passage caused by scarring, congenital abnormality, tumor, inflammation, or surgery, which may impair swallowing or breathing.

Pharyngeal Tonsil

The adenoid tissue located in the roof of the nasopharynx, behind the nose.

Pharyngectomy

Pharyngectomy is surgical removal of part or all of the pharynx, usually to treat cancer.

Pharyngitis

Inflammation of the pharynx, commonly causing sore throat and pain on swallowing.

Alternate Names:

Sore Throat (Pharyngitis)

Pharyngitis (Sore Throat (Pharyngitis))

Alternate index label for Sore Throat (Pharyngitis).

Pharyngitis - Sore Throat

Pharyngitis is inflammation of the throat causing

pain with swallowing, commonly from viral infection and sometimes from group A streptococcal infection or other causes.

Pharyngitis - Viral

Viral pharyngitis is inflammation of the throat caused by a virus, commonly producing sore throat, cough, runny nose, or fever and usually resolving with supportive care.

Pharyngolaryngeal Pain

Pain involving the pharynx and larynx, commonly associated with infection, inflammation, irritation, trauma, reflux, or malignancy.

Pharyngomaxillary Space Abscess

A pharyngomaxillary space abscess is a deep infection containing pus near the pharynx and maxilla that can cause severe pain, swelling, difficulty swallowing, and airway risk.

Pharynx

The muscular throat passage behind the nose and mouth that leads to the larynx and oesophagus.

Pharynx Necrosis

Death of tissue in the throat caused by severe infection, ischemia, radiation, caustic injury, or tumor, with risk of deep infection, bleeding, airway compromise, and sepsis.

Pharynx Neoplasm

A pharynx neoplasm is an abnormal growth in the throat that may be benign or malignant.

Phase I Metabolism

Phase I metabolism involves chemical modification of drug molecules through oxidation, reduction, and hydrolysis reactions, primarily catalyzed by the cytochrome P450 enzyme system in the liver.

Phase II Metabolism

Phase II metabolism, also known as conjugation reactions, involves the attachment of endogenous hydrophilic molecules to drugs or their Phase I metabo-

lites, dramatically increasing water solubility and facilitating renal excretion.

Alternate Names:

conjugation reactions, involves the attachment of.

Phase-Contrast Microscopy

A diagnostic or laboratory method related to phase-contrast microscopy. Results are interpreted with the method, reference range, and clinical findings.

Phe

Phe is the standard abbreviation for phenylalanine, an essential amino acid incorporated into proteins and metabolized through a pathway disrupted in phenylketonuria.

Phelan-McDermid Syndrome

A contiguous gene microdeletion syndrome caused by terminal microdeletions of chromosome 22q13.3 or pathogenic loss-of-function variants in the *SHANK3* gene, which encodes a master scaffold protein in the postsynaptic density of excitatory glutamatergic synapses. Characterized by neonatal hypotonia, global developmental delay, severe intellectual disability, profound expressive speech delay or complete absence of speech, autistic spectrum behaviors, high pain tolerance, and minor dysmorphic findings such as long eyelashes, large fleshy hands, and hypoplastic dysplastic toenails.

Alternate Names:

22q13.3 Deletion Syndrome

Phenacetin

Phenacetin is a former analgesic and fever-reducing drug withdrawn or restricted because long-term use can cause kidney damage and increase cancer risk.

Phenazopyridine

Phenazopyridine is a urinary analgesic that temporarily relieves dysuria without treating its cause; it can turn urine orange and prolonged or excessive use may cause haemolysis or methemoglobinaemia.

Phencyclidine

A dissociative drug that can cause altered percep-

tion, agitation, hallucinations, impaired coordination, or dangerous behavior.

Phencyclidine Overdose

Phencyclidine overdose involves toxic or environmental exposure to phencyclidine overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Phenelzine

A monoamine oxidase inhibitor antidepressant requiring attention to interactions and dietary precautions.

Phenethyl Isothiocyanate

Phenethyl isothiocyanate is a sulfur-containing compound found in some plants and studied experimentally for antimicrobial and anticancer effects.

Phenethylamines

Phenethylamines are a chemical family including neurotransmitters, medicines, and psychoactive substances that can affect adrenergic, dopaminergic, or serotonergic signaling.

Phenformin

Phenformin is a former biguanide diabetes medicine withdrawn because it can cause life-threatening lactic acidosis.

Phenindione

An older oral anticoagulant that interferes with vitamin-K-dependent clotting factors.

Pheniramine Overdose

Pheniramine overdose involves toxic or environmental exposure to pheniramine overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Phenobarbital

Phenobarbital is a long-acting barbiturate anti-seizure medicine that enhances GABA-A receptor inhibition; sedation, respiratory depression, cognitive effects, dependence, and enzyme-inducing drug

interactions are important risks.

Phenobarbital Overdose

Phenobarbital overdose involves toxic or environmental exposure to phenobarbital overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Phenobarbitone

The British name for phenobarbital, a barbiturate anticonvulsant and sedative.

Phenocopy

A phenocopy is a trait resembling an inherited phenotype but produced by an environmental exposure, disease, or other non-genetic cause.

Phenol

A corrosive aromatic compound used in chemical and laboratory applications and in some medical preparations.

Phenolphthalein

A chemical indicator formerly used in some laxatives; long-term use is avoided because of safety concerns and more suitable alternatives.

Phenols

A group of aromatic chemicals that can burn skin and mucous membranes and cause systemic poisoning if absorbed or swallowed.

Phenomenon

An observable biological occurrence, clinical physical sign, physiological event, or characteristic disease manifestation recognized in clinical medicine. Prominent medical eponyms include the Raynaud phenomenon (episodic digital vasospasm triggered by cold or emotional stress, progressing through classic triphasic pallor, cyanosis, and reactive rubor), the Koebner phenomenon (isomorphic skin lesion development at sites of cutaneous trauma in psoriasis or lichen planus), the Dawn phenomenon (early-morning hyperglycemia secondary to physiological circadian cortisol and growth hormone surges

in diabetes mellitus), and the Bell phenomenon (normal reflex upward and outward deviation of the eye during forced eyelid closure).

Phenothiazine

Phenothiazine is a chemical class that includes several antipsychotic and antiemetic medicines, with possible sedation, movement, and cardiac adverse effects.

Phenothiazine Overdose

Phenothiazine overdose involves toxic or environmental exposure to phenothiazine overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Phenothiazines

A class of medicines including antipsychotic and antiemetic drugs.

Phenotype

Observable characteristics of an individual, resulting from genotype and environmental interactions. Includes morphology, development, biochemical properties, behavior.

Phenoxodiol

Phenoxodiol is an investigational anticancer compound studied for effects on apoptosis and cellular signaling.

Phenoxybenzamine

An irreversible alpha-adrenergic blocker used mainly before surgery for pheochromocytoma.

Phenoxymethylpenicillin

Penicillin V, an oral penicillin used for selected susceptible infections.

Phenprocoumon

Phenprocoumon is a long-acting vitamin-K antagonist anticoagulant that reduces clotting-factor production and requires coagulation monitoring.

Phentermine

A sympathomimetic appetite suppressant used short term in selected weight-management plans.

Phenthoate

Phenthoate is an organophosphate insecticide that can inhibit acetylcholinesterase and cause cholinergic poisoning after exposure.

Phentolamine

Phentolamine is a short-acting alpha-adrenergic blocker used for catecholamine-related hypertension and to limit tissue injury after certain vasoconstrictor leaks.

Phenylacetate

A salt or ester of phenylacetic acid used in some metabolic and experimental medical applications; it can bind nitrogen-containing metabolites.

Phenylalanine

An essential amino acid incorporated into proteins and converted to tyrosine; impaired metabolism causes phenylketonuria and requires dietary or medical management.

Phenylbutazone

An older NSAID with serious blood, kidney, gastrointestinal, and other toxicity risks.

Phenylbutyrate

A nitrogen-scavenging medicine converted to phenylacetate, which combines with glutamine and facilitates nitrogen excretion in selected urea-cycle disorders.

Phenylephrine

An alpha-adrenergic agonist used as a decongestant, vasopressor, or pupil dilator.

Phenylketonuria

An inherited metabolic disease, usually caused by reduced phenylalanine hydroxylase activity, that leads to phenylalanine accumulation and reduced tyrosine availability. Without early treatment, it

can impair brain development and cause intellectual disability, seizures, eczema, or a musty odor. Newborn screening and lifelong dietary or medical treatment can prevent or reduce complications.

Alternate Names:

Phenylketonuria (PKU) PKU

Phenylpropanolamine

A sympathomimetic decongestant formerly used in some products; it can raise blood pressure and has been restricted because of safety concerns.

Phenytoin

An anticonvulsant that blocks voltage-gated sodium channels and treats selected seizures.

Phenytoin Overdose

Phenytoin overdose involves toxic or environmental exposure to phenytoin overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Phenytoin Sodium

The sodium salt of phenytoin, an anticonvulsant used for selected focal and generalized tonic-clonic seizures; toxicity can cause nystagmus, ataxia, arrhythmia, or severe skin reactions.

Pheochromocytoma

A catecholamine-producing tumor of chromaffin cells, usually in the adrenal medulla, causing episodic or sustained hypertension.

Alternate Names:

Tumor Adrenal Gland (Pheochromocytoma)

Pheresis

Pheresis is removal of a selected component from blood by an apheresis machine while returning the remaining blood components to the donor or patient.

Pheromone

A pheromone is a chemical signal released by an organism that alters behavior or physiology in another member of the same species.

Phialemonium Atrogriseum

A darkly pigmented environmental mold that can rarely cause opportunistic eye, skin, bloodstream, or disseminated infection.

Phialophora

A genus of darkly pigmented fungi that can cause chromoblastomycosis or other subcutaneous and opportunistic infections.

Phialophora Verrucosa

A dematiaceous fungus that commonly causes chromoblastomycosis, a chronic warty skin and subcutaneous infection.

Philadelphia Chromosome

A chromosome rearrangement, usually t(9;22), that creates the BCR::ABL1 fusion gene and is characteristic of most cases of chronic myeloid leukemia.

Philodendron Poisoning

Philodendron poisoning involves toxic or environmental exposure to philodendron poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Phimosis

Inability to retract the foreskin behind the glans; pathologic cases may cause pain, infection, or urinary difficulty.

Phimosis And Paraphimosis (Penis Disorders)

Alternate index label for Penis Disorders.

Phlebitis

Inflammation of a vein, causing pain, warmth, redness, or tenderness along its course. It may occur with a blood clot, called thrombophlebitis, or after irritation from an intravenous line.

Phlebitis And Thrombophlebitis

Phlebitis is inflammation of a vein, while thrombophlebitis includes a clot within the inflamed vein. Superficial disease causes localized pain and redness;

deep-vein thrombosis can embolize to the lungs and is suspected with limb swelling, pain, or breathlessness.

Phlebography

Venography using contrast dye. Used for diagnosing venous thrombosis, malformations. Largely replaced by duplex ultrasound.

Phlebotomy

The collection of blood from a vein, usually with a needle.

Alternate Names:

Venipuncture

Phlebovirus

A genus of segmented RNA viruses, many transmitted by sandflies or mosquitoes; some cause fever, encephalitis, or hemorrhagic disease.

Phlegm

Thick mucus coughed up from the lower respiratory tract.

Phlegmasia Cerulea Dolens

Phlegmasia cerulea dolens is a severe, extensive deep-vein thrombosis that obstructs venous outflow, causing painful swollen blue discoloration and risking limb ischemia or pulmonary embolism.

Phobia

An excessive, persistent fear of a specific object or situation that leads to avoidance or distress.

Phobia - Simple/Specific

A specific phobia is intense, persistent fear of a particular object or situation that is excessive, leads to avoidance, and interferes with life.

Phobia, Social

Alternate index label; see *Social anxiety disorder (social phobia)*.

Phobias

Persistent, excessive fear of a specific object or situ-

ation that leads to avoidance or significant distress.

Phobic Disorder

A disorder characterized by clinically significant fear and avoidance of a particular object or situation.

Phocomelia

A rare congenital limb-development defect in which hands or feet are attached close to the trunk because of absent or severely shortened long bones.

Phonation

Production of voiced sound when exhaled air makes the vocal folds vibrate in the larynx; disorders cause hoarseness, weak voice, or loss of voice.

Phonocardiogram

A graphic recording of heart sounds and murmurs, traditionally made with a microphone over the chest. It can document timing and duration but is usually interpreted with examination and echocardiography.

Phonocardiography

Graphic recording of heart sounds and murmurs using a microphone and recording system.

Phonological Disorder

A communication disorder involving persistent difficulty producing or organizing speech sounds beyond what is expected for age and language development.

Phonophobia

Abnormal sensitivity to or fear of sound, often associated with migraine.

Phorate

Phorate is a highly toxic organophosphate pesticide whose exposure can produce excessive cholinergic activity and medical emergency.

Phorbol Ester

A phorbol ester is a plant-derived compound that activates protein kinase C and is used in research on inflammation, signaling, and tumor promotion.

Phorbols

Phorbols are diterpene compounds related to phorbol esters that can activate cellular protein kinase C signaling.

Phorometer

An instrument used to measure ocular alignment and the amount of prism correction needed for binocular vision disorders.

Phosgene

A highly toxic industrial gas that reacts with moist respiratory tissues and can cause delayed pulmonary edema, cough, chest tightness, and respiratory failure after exposure.

Phosgene Oxime

A highly corrosive chemical warfare agent that causes immediate severe pain, skin and eye injury, airway damage, and tissue necrosis on contact.

Phosmet

Phosmet is an organophosphate insecticide that can inhibit acetylcholinesterase and cause cholinergic toxicity after substantial exposure.

Phosphamidon

An organophosphate insecticide that can cause cholinergic poisoning by inhibiting acetylcholinesterase.

Phosphate

A salt or ester of phosphoric acid; phosphate is important in bone, energy metabolism, and cellular signaling.

Phosphate Binders

Calcium acetate, sevelamer, lanthanum. Bind dietary phosphate in GI tract. Target normal serum phosphorus in CKD-MBD. Avoid calcium-based binders in patients with vascular calcification.

Phosphate Homeostasis

Serum phosphate is regulated by the interplay of PTH vitamin D and fibroblast growth factor 23.

PTH promotes renal phosphate excretion. Active vitamin D stimulates intestinal phosphate absorption. FGF23 produced by osteocytes inhibits renal phosphate reabsorption and suppresses 1-alpha-hydroxylase reducing active vitamin D production.

Phosphatidic Acid

A membrane phospholipid and signaling intermediate that serves as a precursor for other glycerolipids and influences membrane curvature, trafficking, and cell growth.

Phosphatidylcholine

A major membrane phospholipid and component of pulmonary surfactant, lipoproteins, and bile; it also participates in cell signaling.

Phosphatidylethanolamines

Membrane phospholipids containing ethanolamine that help determine membrane structure, fusion, and cell signaling.

Phosphatidylinositol

A membrane phospholipid that serves as a precursor for signaling molecules and helps anchor proteins to cell membranes.

Phosphatidylserine

A membrane phospholipid normally concentrated on the inner cell surface that becomes exposed during apoptosis and helps signal cell removal.

Phosphatidylserines

Molecules of phosphatidylserine, a membrane phospholipid involved in apoptosis, coagulation, membrane structure, and cell signaling.

Phosphaturic Mesenchymal Tumor

A usually benign mesenchymal tumor that secretes fibroblast growth factor 23 and causes tumor-induced osteomalacia with phosphate wasting, bone pain, weakness, and fractures; removal is usually curative.

Phosphine

A highly toxic gas used in some industrial or pest-control settings that can cause severe lung, heart, liver, kidney, and neurologic injury.

Phosphodiesterase-5 Inhibitors

Medications that inhibit PDE-5, increasing cGMP and causing smooth muscle relaxation and vasodilation. Sildenafil and tadalafil are used for PAH and erectile dysfunction. In PAH, they reduce pulmonary vascular resistance and improve exercise capacity.

Phosphofructokinase-1

Rate-limiting enzyme of glycolysis. Activated by AMP, fructose-2,6-bisphosphate. Inhibited by ATP, citrate.

Phospholipase C Pathway

A GPCR signaling pathway activated primarily by Gq-protein coupled receptors (e.g., alpha-1 adrenergic, muscarinic M1/M3, vasopressin V1, angiotensin II AT1 receptors).

Phospholipid

A type of lipid with a water-attracting phosphate-containing head and water-repelling fatty-acid tails. Phospholipids form the basic double-layered structure of cell membranes and also help with cell signaling and transport.

Phosphorus

An essential element in bone, nucleic acids, phospholipids, and ATP.

Phosphorus Blood Test

A blood test measuring inorganic phosphate, interpreted with calcium, parathyroid hormone, vitamin D, and kidney function to evaluate mineral and renal disorders.

Phosphorus In Diet

Dietary phosphorus comes from protein-rich foods and many processed foods; people with kidney disease may need individualized limits.

Phosphorylation

Addition of a phosphate group to a molecule, often regulating protein activity or energy transfer.

Photoallergy

An immune-mediated skin reaction that occurs when ultraviolet light activates a sensitizing substance on or in the skin, causing an eczematous rash.

Photocoagulation

Coagulation of tissue by intense light, often laser light, to treat lesions such as abnormal retinal vessels.

Photodermatosis

A skin disorder caused or aggravated by exposure to ultraviolet or visible light, including polymorphous light eruption, phototoxic reactions, and photoallergic reactions.

Photodynamic Therapy

A treatment that combines a photosensitizing agent with light to generate reactive oxygen species and destroy cancer cells, used for actinic keratosis, superficial skin cancers, and selected visceral tumors.

Photodynamic Therapy For Cancer

A medical procedure, treatment, device, or care topic related to photodynamic therapy for cancer; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Photoelectric Effect

Photon absorption by inner shell electron. Major interaction at low energies, high Z materials.

Photographic Fixative Poisoning

Photographic fixative poisoning involves toxic or environmental exposure to photographic fixative poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Photomultiplier Tube

Converts light photons to electrical signals. Array of

PMTs positioned behind crystal. Positioning circuit determines interaction location.

Photopeak

Full-energy peak in gamma spectrum. Used for energy window setting.

Photopheresis

Extracorporeal treatment in which white blood cells are exposed to a photosensitizer and ultraviolet A light before being returned to the patient.

Photophobia

An abnormal, pathological ocular discomfort, intolerance, or painful avoidance of light exposure. Not a true phobia, but a sensory symptom mediated by trigeminal nerve innervation of the anterior eye and central thalamocortical visual pathways. Cardinal ocular causes include anterior uveitis (iritis/iridocyclitis, where ciliary muscle contraction provokes severe pain), corneal abrasions or infectious keratitis, and acute angle-closure glaucoma. Systemic and neurologic causes include acute bacterial or viral meningitis, subarachnoid hemorrhage (meningeal irritation), migraine headache attacks, and concussion.

Photoreceptors

Specialized retinal cells that detect light; rods and cones are the two main types.

Photorefractive Keratectomy

Laser eye surgery that reshapes the cornea to correct refractive errors.

Photosensitivity

An abnormal skin reaction to sunlight (UVA/UVB), manifesting as rash, erythema, or burning on sun-exposed areas.

Photosensitization

Increased sensitivity to ultraviolet or visible light caused by a medication, chemical, or disease, producing an exaggerated skin reaction.

Phototherapy

Treatment using controlled light, such as for neonatal jaundice, psoriasis, or seasonal affective disorder.

Phototoxic Dermatitis

An acute, dose-dependent inflammatory skin reaction occurring when certain ingested or topically applied chemicals (such as psoralens, tetracyclines, NSAIDs, or thiazides) absorb ultraviolet radiation, leading to direct cellular damage and a sunburn-like eruption in exposed areas.

Phototoxicity

Skin reaction from UV light + chemical. Sunburn-like. Examples: tetracyclines, thiazides, NSAIDs, psoralens.

Phrenic

Relating to the diaphragm or the phrenic nerves, which arise from the neck and carry the main motor supply to the diaphragm for breathing.

Phrenic Nerve

The nerve arising mainly from C3-C5 that supplies the diaphragm and carries sensory fibers from nearby structures.

PHT

Alternate index label; see *Portal hypertension*.

Phthiriasis

Infestation with lice, especially pubic lice, causing itching and sometimes visible lice or eggs attached to hair. Treatment usually includes an approved lice-killing product, washing or sealing recently used clothing and bedding, and evaluation or treatment of close or sexual contacts.

Phthisis

An obsolete term for wasting or shrinkage of an organ, especially a severely damaged eye or lung.

Phyllodes Tumor

A breast tumor composed of both stromal and epithelial elements, ranging from benign through bor-

derline to malignant. Typically a large, rapidly growing mass; treatment is surgical excision with clear margins.

Physiatrist

Physician specialty focusing on function, quality of life for patients with impairment, disability. See rehabilitation medicine.

Physical Activity

Any movement produced by skeletal muscles that increases energy expenditure, including occupational, household, transportation, and recreational movement. Regular activity supports cardiovascular, metabolic, musculoskeletal, and mental health.

Physical Child Abuse

Physical child abuse is intentional physical injury or harmful force inflicted on a child, requiring immediate safety assessment, medical care, and safeguarding action.

Physical Dependence

A state in which the body adapts to a substance or medication and develops withdrawal symptoms when it is reduced or stopped.

Physical Exam Frequency

Physical exam frequency is a laboratory test, imaging method, instrument, score, or measurement used to characterize a specimen or clinical finding; interpretation depends on the method and clinical context.

Physical Examination

A systematic assessment of the body using inspection, palpation, percussion, auscultation, and other examination methods.

Physical Half-Life

The time required for half of the radioactive nuclei in a sample to undergo nuclear decay. It is distinct from biologic half-life, which describes biological elimination, and effective half-life, which combines both processes.

Physical Medicine

The medical specialty focused on function, rehabilitation, pain, and restoration of independence after illness or injury.

Physical Therapy

Rehabilitation using exercise, movement training, manual techniques, and physical modalities to improve function, mobility, and reduce disability.

Physical Therapy Assessment

Comprehensive evaluation: history, observation, ROM, strength, sensation, coordination, balance, gait, functional tasks.

Physician

A licensed medical doctor who diagnoses, treats, and prevents disease.

Physician Assistant

A licensed clinician who evaluates patients and provides medical care under a physician's practice framework.

Alternate Names:

PA

Physician Assistant

A nationally certified and state-licensed medical clinician who practices medicine collaboratively with or under the supervision of physicians. PAs obtain medical histories, perform physical examinations, order and interpret diagnostic studies, formulate medical diagnoses, prescribe pharmacotherapy, assist in surgery, and perform clinical procedures across primary care and surgical specialties.

Physicians' Desk Reference

A reference source containing prescribing information, indications, contraindications, dosing, warnings, and adverse effects for medicines and selected products.

Physiologic

Physiologic means relating to normal body function or a process occurring within the expected biological

range.

Physiologic Jaundice Of Newborn

A common, benign condition characterized by unconjugated hyperbilirubinemia appearing after 24 hours of age (peak at 3-5 days) in healthy term newborns. Caused by increased bilirubin production (high RBC turnover, short RBC lifespan) and immature hepatic conjugation (low UDP-glucuronyl transferase activity).

Physiologic Jaundice Of The Newborn

Physiologic newborn jaundice is a temporary bilirubin rise after birth caused by immature bilirubin processing, but early or severe jaundice needs urgent evaluation.

Physiological Dead Space

Volume of ventilated respiratory tract that does not participate in gas exchange, including anatomic and alveolar dead space.

Physiology

The study of normal functions and processes of living organisms and their parts.

Physiotherapy

Assessment and treatment using movement, exercise, manual techniques, education, and physical modalities to improve function.

Physostigmine

A reversible acetylcholinesterase inhibitor that may reverse severe central anticholinergic delirium in carefully selected patients after assessment for contraindications and cardiac conduction abnormalities.

Phytomedicine

The evidence-based use of medicinal products derived from plants to prevent or treat disease.

Phytomenadione

Vitamin K is a fat-soluble vitamin essential for the post-translational gamma-carboxylation of gluta-

mate residues to gamma-carboxyglutamate (Gla) in specific clotting factors (II, VII, IX, X, protein C, protein S) and anticoagulant proteins (protein C, protein S, protein Z). See Vitamin K: Phylloquinone.

PI-RADS

Prostate Imaging-Reporting and Data System. Standardizes multiparametric MRI reporting for prostate cancer detection and staging.

Pia Mater

The delicate innermost membrane closely covering the brain and spinal cord.

Pica

Persistent eating of nonnutritive substances inappropriate for developmental level or culture.

Pick'S Disease

An older term for frontotemporal dementia associated with characteristic Pick bodies; current classification uses specific frontotemporal dementia syndromes.

Pickwickian Syndrome

An older name for obesity hypoventilation syndrome, involving obesity, daytime hypercapnia, and sleep-disordered breathing.

PID

Alternate index label; see *Pelvic inflammatory disease*.

Piebaldism

A rare congenital disorder of melanocyte development caused most often by pathogenic variants affecting the KIT signaling pathway. It produces stable, sharply demarcated depigmented patches of skin and a characteristic white forelock or localized poliosis from birth. The lesions usually remain non-progressive, although their distribution and degree of pigment loss vary among affected individuals. Diagnosis is generally clinical, with genetic assessment considered when features suggest a broader

pigmentary or syndromic disorder.

Pierre Robin Sequence

A congenital anomaly triad: micrognathia (small mandible), glossoptosis (posteriorly displaced tongue causing airway obstruction), and U-shaped cleft palate. The sequence arises from early mandibular hypoplasia, which prevents the tongue from descending, thereby impairing palatal shelf fusion.

Pierson Syndrome

A rare autosomal-recessive disorder caused by LAMB2 variants, classically combining congenital nephrotic syndrome with microcoria and structural abnormalities of the eyes. Kidney failure, developmental impairment, and severe visual disability may occur.

Pigment

A substance that gives tissue, hair, skin, or other material its color.

Pigment Cirrhosis

Liver cirrhosis associated with abnormal pigment or iron deposition, as may occur in hereditary hemochromatosis.

Pigmentary Mosaicism

A pattern of alternating hyperpigmented and hypopigmented skin patches arranged in a linear, segmental, or blaschkoid distribution, caused by genetic mosaicism from post-zygotic mutations affecting melanocyte development. It is a cosmetic condition without systemic involvement.

Pigmentation

Coloration of skin, hair, eyes, or other tissue produced mainly by melanin, carotenoids, blood, or exogenous material; abnormal changes may reflect inflammation, endocrine disease, drugs, or tumors.

Pigmentation Disorder

A condition causing abnormal increase, decrease, or uneven distribution of pigment in the skin, mucosa,

hair, or eye.

Pilar Cyst

A benign cyst arising from the outer root sheath of hair follicles, most commonly on the scalp. It presents as a firm, mobile, subcutaneous nodule without a central punctum. Unlike epidermoid cysts, pilar cysts have smooth walls and contain dense keratin. They may be multiple in familial cases. Treatment is simple excision.

Piles

Enlarged or swollen veins in the anus or lower rectum.

Alternate Names:

Hemorrhoids

Piles (Hemorrhoids (Piles))

Alternate index label for Hemorrhoids (Piles).

Piles (Hemorrhoids)

Swollen, inflamed vascular cushions in the anal canal that normally aid continence. Internal hemorrhoids (above the dentate line, painless) present with painless bright red rectal bleeding and prolapse; graded I-IV by degree of prolapse.

Alternate Names:

Hemorrhoids

Pill

A solid oral dosage form containing medicine or another active ingredient.

Alternate Names:

Tablet

Pilocarpine

A muscarinic agonist used to constrict the pupil, lower intraocular pressure, or stimulate salivation.

Pilocytic Astrocytoma

Pilocytic astrocytoma is usually a slow-growing low-grade astrocytic brain tumor, often occurring in children and commonly involving the cerebellum or optic pathway.

Piloerection

Elevation of hair shafts producing gooseflesh, caused by contraction of arrector pili muscles under sympathetic stimulation from cold, fear, pain, fever, or certain drugs.

Pilomatrix Carcinoma

A rare malignant tumor showing matrical hair-follicle differentiation, usually presenting as a firm dermal or subcutaneous nodule on the head, neck, or upper extremity; it may recur or metastasize.

Pilomatrixoma

Pilomatrixoma is a usually benign skin tumor derived from hair-matrix cells, presenting as a firm subcutaneous nodule, often on the head or neck.

Pilonidal Cyst

A cyst or sinus in the cleft between the buttocks, usually containing hair and skin debris. It may become infected, painful, swollen, and drain pus.

Alternate Names:

Cyst, Pilonidal Pilonidal Sinus (Pilonidal Cyst)

Pilonidal Dimple

Alternate index label; see *Sacral dimple*.

Pilonidal Disease

A chronic inflammatory condition of the sacrococcygeal region involving hair follicles. Pilonidal sinus: epithelialized tract containing hair and debris. Pilonidal abscess: acute infection of the sinus. Most common in young, hirsute males, exacerbated by prolonged sitting, obesity, and local trauma.

Pilonidal Fistula

An abnormal tunnel from a pilonidal sinus to the skin, usually in the natal cleft, that can drain pus or recur after abscess formation.

Pilonidal Sinus (Pilonidal Cyst)

Alternate index label for Pilonidal Cyst.

Pilonidal Sinus Disease

A chronic sinus or cyst in the natal cleft, usually

containing hair and debris. It may cause recurrent pain, swelling, drainage, or abscess formation.

Pilot Project

A pilot project is a small preliminary implementation used to test feasibility, identify problems, and refine methods before broader rollout.

Pilus

A pilus is a hair-like bacterial surface structure used for attachment, movement, or transfer of genetic material between cells.

Pimozide

An antipsychotic medicine used in selected disorders, with important cardiac and neurologic adverse effects.

Pimple

A common lay term for a small inflamed or blocked skin lesion, often an acne papule or pustule.

Pimples (Acne)

Alternate index label for Acne.

Pin Care

Pin care keeps the skin around external fixation or traction pins clean and monitored for redness, swelling, drainage, loosening, or infection.

Pinch Strength

Pinch strength is the force produced when the thumb presses against one or more fingers and can help assess hand, nerve, muscle, and functional impairment.

Pinched Nerve

Compression or irritation of a nerve that can cause pain, tingling, numbness, or weakness along the nerve's distribution.

Alternate Names:

Compressed Nerve Nerve Compression

Pinched Nerve Overview

A pinched nerve is compression or irritation of a

nerve root or peripheral nerve, causing pain, tingling, numbness, weakness, or radiating symptoms. Causes include disc disease, narrowing, injury, or entrapment; progressive weakness or bowel or bladder change is urgent.

Alternate Names:

Nerve Pinched (Pinched Nerve Overview)

Pindolol

Pindolol is a beta-blocker used for selected cardiovascular conditions; it can slow the heart rate and worsen asthma in susceptible people.

Pine Oil Poisoning

Pine oil poisoning involves toxic or environmental exposure to pine oil poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pineal Cyst

A pineal cyst is a fluid-filled lesion of the pineal gland, usually incidental and harmless but occasionally associated with headache, hydrocephalus, or visual symptoms.

Pineal Germinoma

A pineal germinoma is a malignant germ-cell tumor in the pineal region, often causing headache, eye-movement abnormalities, or hydrocephalus and usually responsive to radiotherapy and chemotherapy.

Pineal Gland

An endocrine gland in the brain that secretes melatonin and helps regulate circadian timing.

Pineal Gland Tumor

A tumor of the pineal region, including germinoma, pineocytoma, and pineoblastoma. Presents with Parinaud syndrome, hydrocephalus, and raised intracranial pressure; germinomas are highly radiosensitive.

Pineal Parenchymal Tumor

A tumor arising from pineal cells, ranging from pi-

neocytoma to pineal parenchymal tumor of intermediate differentiation and pineoblastoma. Symptoms may result from hydrocephalus or compression of nearby visual and ocular-motor pathways.

Pineoblastoma

A malignant embryonal tumor of the pineal region, occurring mainly in children and requiring neuroimaging, histologic diagnosis, staging, and multimodal treatment.

Pineocytoma

Pineocytoma is a usually slow-growing tumor of pineal cells that may obstruct cerebrospinal-fluid flow and cause raised intracranial pressure.

Pinguecula

A pinguecula is a benign yellowish raised growth on the conjunctiva near the cornea, often associated with ultraviolet light, wind, and dust; it may irritate the eye.

Pinhole Meatus

A pinhole meatus is an abnormally narrow external opening, commonly referring to the urethral meatus, that can obstruct urine flow or complicate catheterization.

Pink Eye

A lay term for conjunctivitis, inflammation of the conjunctiva causing redness and discharge.

Alternate Names:

Conjunctivitis (Pink Eye)

Pink Eye (Conjunctivitis)

Inflammation or infection of the conjunctiva causing eye redness, discharge, irritation, itching, or tearing.

Pink Puffer

A clinical phenotype of emphysema in COPD. Patients present with severe dyspnea, minimal cough, thin habitus, pursed-lip breathing, accessory muscle use, and relatively normal PaO₂ with late hypercapnia. They tend to be older and have more extensive emphysema.

Pinky Broken Finger (Broken Finger)

Alternate index label for Broken Finger.

Pinna

The visible external part of the ear that collects sound.

Pinocytosis

Cellular uptake of extracellular fluid and dissolved substances in small vesicles.

Pint

A unit of volume; its exact value varies by measurement system and country.

Pinta

A chronic skin infection caused by *Treponema carateum*, producing changing stages of pigmentary lesions.

Pinworm

The intestinal nematode *Enterobius vermicularis*, which commonly causes nocturnal perianal itching.

Pinworm Infection

Pinworm infection is an intestinal infestation by *Enterobius vermicularis*, causing nighttime anal itching, disturbed sleep, and sometimes abdominal or genital irritation. Treatment of the person and close household contacts, hygiene, and repeat dosing reduce reinfection.

Alternate Names:

Infection Pinworms (Pinworm Infection) Worms Pinworms (Pinworm Infection)

Pinworm Test

A test for *Enterobius vermicularis*, usually the early-morning adhesive-tape test applied to the perianal skin before bathing or defecation.

Pinworms

Pinworms are intestinal nematodes, *Enterobius vermicularis*, that cause nighttime anal itching; treatment, hand hygiene, and laundering help prevent household reinfection.

Pioglitazone

Pioglitazone is a PPAR-gamma agonist that improves insulin sensitivity in type 2 diabetes; weight gain, oedema, heart-failure worsening, fractures, and bladder-cancer concerns require patient-specific assessment.

Pipemidic Acid

An older quinolone antibacterial formerly used mainly for urinary-tract infections; resistance and safety limitations have greatly reduced its clinical use.

Pipendoxifene

Pipendoxifene is an investigational selective estrogen-receptor modulator studied for possible effects on breast, bone, and reproductive tissues.

Piperacillin

Piperacillin is an extended-spectrum penicillin antibiotic often combined with tazobactam to treat serious infections including *Pseudomonas*; allergy, diarrhoea, cytopenias, electrolyte effects, and renal dosing are important.

Piperazine

An older antiparasitic medicine that paralyzes certain intestinal worms, especially roundworms and pinworms. Its use depends on local guidance and the particular infection.

Piperonyl Butoxide

A pesticide synergist that inhibits insect detoxification enzymes and increases the activity of certain insecticides; it can irritate skin, eyes, or airways.

Piperonyl Butoxide With Pyrethrins Poisoning

Piperonyl butoxide with pyrethrins poisoning involves toxic or environmental exposure to piperonyl butoxide with pyrethrins poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pipette

A pipette is a laboratory instrument used to measure and transfer small volumes of liquid accurately.

Pipobroman

Pipobroman is an oral alkylating antineoplastic drug used historically for some blood disorders and associated with marrow suppression and later cancer risk.

Pipotiazine

A long-acting phenothiazine antipsychotic given by injection in selected settings. It can cause movement disorders, drowsiness, and other adverse effects and requires clinical monitoring.

Pipradrol

An older central nervous system stimulant with limited current use.

Piracetam

A nootropic compound used in some countries for selected neurologic indications; evidence and approval vary.

Pirfenidone

Pirfenidone is an antifibrotic medicine used to slow decline in idiopathic pulmonary fibrosis; nausea, photosensitivity, fatigue, and liver-enzyme elevation may occur and require monitoring.

Piriformis Syndrome

Sciatic nerve compression by piriformis muscle. Buttock pain, radiating down leg. Diagnosis: clinical, exclude disc herniation. Treatment: stretching, injection, surgical release.

Alternate Names:

PiriformSyndrome (Piriformis Syndrome)

PiriformSyndrome (Piriformis Syndrome)

Alternate index label for Piriformis Syndrome.

Pirimicarb

Pirimicarb is a selective carbamate insecticide targeting aphids; exposure can inhibit acetyl-

cholinesterase and cause cholinergic symptoms.

Pirinitramide

Pirinitramide is a synthetic opioid analgesic used historically in some countries; opioid effects include sedation, constipation, dependence, and respiratory depression.

Piritrexim

Piritrexim is an investigational antifolate that inhibits dihydrofolate reductase and has been studied as an anticancer agent.

Piromidic Acid

An older quinolone-like antibacterial investigated or used for urinary and gastrointestinal infections; current use depends on local availability, susceptibility, and safety guidance.

Piroplasmosis

An animal disease caused by *Babesia* or *Theileria* parasites; human babesiosis is a related infection.

Piroxicam

A long-acting nonsteroidal anti-inflammatory drug used for pain and inflammation; gastrointestinal bleeding, kidney injury, cardiovascular events, and hypersensitivity are important risks.

Piroxicam Overdose

Piroxicam overdose involves toxic or environmental exposure to piroxicam overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pisa Syndrome

A sustained involuntary sideways flexion of the trunk, often associated with medicines, Parkinson disease, or other neurologic disorders.

Pisiform

Small pea-shaped sesamoid bone in the flexor carpi ulnaris tendon, in the proximal row. Articulates only with the triquetrum. Forms the medial border of Guyon's canal (ulnar nerve/artery passage).

Only sesamoid carpal bone, subcutaneous and easily palpable on the ulnar palm.

Pisiform Bone

A small pea-shaped carpal bone embedded in the tendon of flexor carpi ulnaris on the ulnar side of the wrist; fracture or arthritis can cause localized wrist pain.

Pithomyces

A genus of environmental molds associated with plant disease and rarely with human opportunistic infection.

Pits

Small depressions or holes; in dermatology, pits may occur in nails or skin lesions.

Pitting Edema

Swelling in which pressure applied to the skin leaves a temporary indentation, usually because of excess interstitial fluid.

Pituicytoma

Pituicytoma is a rare typically slow-growing tumor arising from specialized glial cells of the pituitary stalk or posterior pituitary region.

Pituitary

The pituitary is an endocrine gland at the base of the brain that releases hormones controlling growth, thyroid and adrenal function, reproduction, lactation, and water balance.

Pituitary Adenoma

Pituitary adenomas are benign neoplasms arising from the adenohypophysis (anterior pituitary gland), accounting for approximately ten to fifteen percent of all intracranial neoplasms.

Pituitary Apoplexy

Acute hemorrhage or infarction within a pituitary adenoma causing sudden headache, visual field defects, ophthalmoplegia, and altered consciousness. Medical emergency requiring urgent glucocorticoid

administration to prevent adrenal crisis from acute ACTH deficiency.

Pituitary Carcinoma

Pituitary carcinoma is a rare malignant pituitary tumor defined by spread to distant or separate sites, often producing excess hormones.

Pituitary Disease

A disorder of the pituitary gland that alters hormone secretion or causes mass effects, including adenomas, inflammation, hemorrhage, genetic disease, or hypopituitarism.

Pituitary Dwarfism

Pituitary dwarfism is severe short stature caused by deficient growth hormone or broader pituitary dysfunction, with evaluation focused on endocrine and genetic causes.

Pituitary Gland

An endocrine gland at the base of the brain that regulates growth, thyroid, adrenal, reproductive, lactation, and water-balance hormones.

Pituitary Hormone

A hormone secreted by the anterior or posterior pituitary that regulates growth, thyroid and adrenal function, reproduction, lactation, water balance, or other endocrine organs.

Pituitary Insufficiency

Alternate index label; see *Hypopituitarism*.

Pituitary Neoplasm

A pituitary neoplasm is an abnormal growth in or near the pituitary gland and may affect hormones, vision, or nearby brain structures.

Pituitary Neuroendocrine Tumor

Alternate index label; see *Pituitary tumors and adenomas*.

Pituitary Tumor

A pituitary tumor is a growth in the pituitary gland

that may be noncancerous but can cause hormone excess, hormone deficiency, or visual problems.

Pituitary Tumor (Prolactinoma)

Alternate index label for Prolactinoma.

Pityriasis

A group of skin disorders characterized by fine scaling, including pityriasis rosea and versicolor.

Pityriasis Alba

Pityriasis alba is a common benign skin condition causing faint, dry, lighter patches, especially on children's faces, often more noticeable after tanning.

Pityriasis Lichenoides

An uncommon inflammatory skin disorder producing recurrent small scaly papules, with forms ranging from mild chronic lesions to an acute ulceronecrotic eruption.

Pityriasis Rosea

An acute, self-limited eruption often beginning with a herald patch followed by oval scaly lesions distributed along skin-cleavage lines. It usually resolves within several weeks; treatment is symptomatic, and atypical, severe, recurrent, or persistent disease should be reassessed.

Alternate Names:

Pityriasis Rosea Gibert (Pityriasis Rosea)

Pityriasis Rosea Gibert (Pityriasis Rosea)

Alternate index label for Pityriasis Rosea.

Pityriasis Rubra Pilaris

Pityriasis rubra pilaris is an uncommon inflammatory skin disorder producing reddish-orange scaly plaques, follicular bumps, and areas of relatively normal skin.

Pityriasis Versicolor

Alternate index label; see *Tinea versicolor*.

Pivampicillin

An orally absorbed prodrug of ampicillin used for

selected bacterial infections; its use is limited in some settings by adverse effects and availability.

Pixantrone

Pixantrone is an anthracenedione antineoplastic drug developed for selected aggressive lymphomas, with potential marrow, infection, and cardiac risks.

PJI

Alternate index label; see *Periprosthetic joint infection*.

PJS

Alternate index label; see *Peutz-Jeghers syndrome*.

PKD

Alternate index label; see *Polycystic kidney disease*.

Place

Place describes a location or position; in healthcare, the relevant place may be an anatomic site, care setting, or environmental context.

Placebo

An inactive substance, treatment, or procedure designed to resemble an active treatment. In a clinical trial, it is given to a control group so researchers can compare outcomes and estimate the specific effect of the treatment being studied.

Alternate Names:

Dummy Treatment Inactive Treatment

Placebo Control

A placebo control is an inactive treatment designed to resemble the intervention in a clinical trial, helping separate treatment effects from expectation and natural change.

Placebo Effect

A psychological and physiological phenomenon where a patient experiences a real improvement in health or symptoms after receiving an inactive treatment (like a sugar pill) due to their expectations of healing.

Placenta

The temporary organ connecting mother and fetus, allowing exchange of oxygen, nutrients, wastes, and hormones.

Placenta Abruptio

Placental abruption is premature separation of the placenta from the uterine wall, causing bleeding, abdominal pain, uterine tenderness, fetal compromise, or maternal shock.

Placenta Abruptio - Definition

Placental abruption is premature separation of the placenta from the uterine wall before birth, causing bleeding, abdominal pain, uterine tenderness, and possible fetal or maternal compromise.

Placenta Accreta

Placenta accreta spectrum occurs when placental tissue abnormally attaches to or invades the uterine wall, creating major risk of hemorrhage during attempted delivery or removal.

Placenta Accreta Spectrum

Abnormal adherence or invasion of placental villi into the uterine wall, ranging from accreta to increta and percreta; it can cause severe postpartum hemorrhage and requires planned obstetric management.

Placenta Disorder

A disorder of placental implantation, structure, perfusion, or separation, including previa, accreta spectrum, abruption, infarction, or placental insufficiency.

Placenta Increta

Abnormal placental invasion into the myometrium without reaching the serosa, increasing the risk of massive hemorrhage at delivery and often requiring cesarean hysterectomy.

Placenta Percreta

Placenta percreta is the most invasive form of placenta accreta spectrum, with villi growing through

the uterine wall and sometimes into nearby organs such as the bladder.

Placenta Praevia

Placental implantation over or near the cervical opening, causing risk of painless late-pregnancy bleeding.

Placenta Previa

Implantation of the placenta over or near the internal cervical opening, which can cause painless bleeding during the second half of pregnancy.

Placenta, Succenturiate

A succenturiate placenta has one or more accessory placental lobes separate from the main placenta, which can cause retained tissue or fetal-vessel complications after delivery.

Placental Abnormalities

Disorders of placental location, attachment, structure, or separation that may affect pregnancy or cause bleeding.

Placental Abruption

Premature separation of the placenta from the uterine wall before delivery. It may cause vaginal bleeding, abdominal pain, uterine tenderness, fetal distress, and maternal or fetal compromise.

Placental Accreta Spectrum

Abnormal placental attachment to the myometrium, ranging from accreta (adherent to decidua basalis) to increta (invading myometrium) to percreta (invading through serosa). Major risk factor is prior caesarean section with anterior placenta previa.

Placental Hormone

A hormone produced by placental tissue, such as hCG, progesterone, estrogens, or human placental lactogen, that supports pregnancy, fetal growth, and maternal metabolic adaptation.

Placental Infarction

A region of placental tissue damaged by inadequate

maternal blood flow and coagulation, which may be incidental or, when extensive, associated with fetal growth restriction or pregnancy complications.

Placental Insufficiency

Placental insufficiency is inadequate placental delivery of oxygen and nutrients, which can cause fetal growth restriction, abnormal Doppler findings, reduced fluid, or fetal distress.

Placental Lactogen

Placental lactogen is a hormone made by the placenta that alters maternal metabolism and supports breast development and fetal nutrient availability during pregnancy.

Placental Mesenchymal Dysplasia

A rare placental lesion with enlarged cystic stem villi and abnormal chorionic vessels that can mimic a partial mole and may be associated with fetal growth restriction or other fetal abnormalities.

Placental Separation Signs

Clinical signs indicating the placenta has separated from the uterine wall and is ready for delivery. Classic signs: lengthening of the umbilical cord (cord lengthens as the placenta descends), a sudden gush of blood, and uterine elevation and firming (the uterus rises and becomes globular as the placenta moves into the lower segment).

Placental Site Nodule

A benign, usually incidental remnant of intermediate trophoblast at a prior implantation site, composed of small nests of chorionic-type or extravillous trophoblast.

Placentation

The developmental process by which the placenta forms and attaches to the uterus, establishes maternal–fetal circulation, and develops the exchange and endocrine functions needed for pregnancy.

Plagiocephaly

Plagiocephaly is asymmetric flattening or shaping of an infant's skull, caused by position or premature suture fusion; its evaluation distinguishes benign molding from craniosynostosis.

Plague

A serious infection caused by *Yersinia pestis*, transmitted mainly by infected fleas or respiratory droplets in pneumonic disease.

Alternate Names:

Plan B

Plan B is a levonorgestrel emergency contraceptive intended to reduce pregnancy risk after unprotected intercourse or contraceptive failure; it does not end an established pregnancy.

Planar

Flat or lying in a single plane; planar imaging produces a two-dimensional projection.

Plane Warts (Verruca Plana)

Small, smooth, relatively flat warts caused by HPV, often occurring in groups on the face, hands, or legs.

Alternate Names:

Verruca Plana

Planococcaceae

A family of gram-positive bacteria found in soil, water, and other environments; human infection is uncommon.

Planococcus Bacteria

Bacteria in the genus *Planococcus*, generally environmental gram-positive cocci that rarely cause human infection.

Plant Development

Plant development is the progression from seed or meristem through growth, differentiation, flowering, reproduction, and senescence.

Plant Extract

A plant extract is a preparation made by removing selected constituents from plant material; composition, dose, contamination, interactions, and evidence vary widely.

Plant Fertilizer Poisoning

Plant fertilizer poisoning involves toxic or environmental exposure to plant fertilizer poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Plant Oils

Plant oils are lipid mixtures extracted from seeds, fruits, or other plant parts and used as foods, medicines, cosmetics, or industrial materials.

Plant Part

A plant part is a structural component such as a root, stem, leaf, flower, fruit, or seed that may contain distinct nutrients, chemicals, or medicinal constituents.

Plant Poisoning

Illness caused by eating, touching, or inhaling a toxic plant or plant product; effects range from local irritation to organ failure.

Plant Proteins

Proteins supplied by legumes, grains, nuts, seeds, vegetables, and other plants; combining varied sources can provide essential amino acids, with nutritional needs depending on age and health.

Plant Thorn Arthritis

Plant thorn arthritis is persistent joint inflammation after a thorn puncture introduces plant material, sometimes requiring removal of foreign material and treatment of infection or inflammation.

Plant Thorn Synovitis

Inflammation of a joint after penetration by a plant thorn or splinter, sometimes involving an infection introduced by the injury. Symptoms may develop gradually as persistent joint pain, swelling, and

limited movement; evaluation may require removal of retained material and targeted treatment.

Alternate Names:

Arthritis Plant Thorn (Plant Thorn Synovitis)

Plant Tumor

A plant tumor is an abnormal plant growth caused by uncontrolled cell multiplication, infection, insects, or other environmental factors.

Plant Virus

A virus that infects plants; plant viruses do not generally cause infection in humans, although contaminated plants may carry other hazards.

Plantar

Pertaining to the inferior weight-bearing surface or sole of the human foot (*planta pedis*). Structurally supported by thick keratinized epidermis, a dense shock-absorbing heel fat pad, and the longitudinally spanning plantar fascia (plantar aponeurosis). Pathologies include plantar fasciitis (degenerative enthesopathy at the medial calcaneal tuberosity causing sharp post-rest heel pain), plantar warts (verruca plantaris induced by human papillomavirus [HPV]), plantar ulcers (neuropathic pressure ulcers in diabetic feet), and the plantar reflex (Babinski sign in upper motor neuron lesions).

Plantar Flexion

Downward movement of the foot at the ankle, pointing the toes away from the shin. It is produced mainly by the calf muscles and is used in standing on tiptoe, walking, running, and jumping.

Plantar Fasciitis

The most common cause of heel pain, caused by degeneration and microtears of the plantar fascia at its calcaneal origin. Presents with sharp, stabbing heel pain with the first steps in the morning (post-static dyskinesia), improving after a few minutes but worsening with prolonged standing or activity.

Plantar Nerve

A nerve supplying sensation to the sole of the foot and motor branches to intrinsic foot muscles; com-

pression can cause burning pain, numbness, or altered toe movement.

Plantar Neuroma

Alternate index label; see *Morton neuroma*.

Plantar Reflex

The response produced by stroking the sole of the foot; an extensor response in an adult suggests corticospinal-tract dysfunction.

Plantar Response

Movement of the toes and sole after stimulation of the foot; an extensor response in an adult may indicate corticospinal-tract dysfunction.

Plantar Wart (Verruca Plantaris)

A wart (verruca) located on the sole of the foot, caused by human papillomavirus (HPV, typically types 1, 2, 4, 63). Plantar warts are endophytic (grow inward due to pressure) rather than exophytic like common warts, and are often painful with direct pressure (walking, standing).

Alternate Names:

Verruca Plantaris

Plantaris Muscle

A small posterior-leg muscle with a long tendon that assists weakly with knee flexion and plantar flexion; it can be strained during sudden athletic movement.

Plaque

A flat, raised, or deposited area of material or tissue; examples include dental plaque, arterial plaque, and skin plaques.

Plaque And Tartar On Teeth

Dental plaque is a bacterial biofilm that forms on teeth, while tartar is hardened mineralized plaque; both contribute to cavities, gingivitis, and periodontitis.

Plaque Assay

A method for quantifying viruses: serial dilutions of

a viral suspension are added to a monolayer of host cells, forming zones of cell destruction (plaques), each plaque representing one infectious unit; results expressed as plaque-forming units per mL (PFU/mL).

Plaque Psoriasis

The most common form of psoriasis, producing well-demarcated raised red plaques with silvery scale, commonly on elbows, knees, scalp, and lower back and sometimes associated with arthritis.

Plasma

The liquid portion of blood in which blood cells and dissolved substances circulate.

Plasma Amino Acids

A blood test that measures amino acids in plasma. It can help evaluate inherited metabolic disorders, poor nutrition, liver disease, or other conditions affecting amino-acid processing.

Plasma Cell

The terminally differentiated antibody-secreting cell derived from activated B lymphocytes (after antigen stimulation, class switching, and affinity maturation in germinal centers).

Plasma Cell Mastitis

A chronic benign inflammatory breast lesion with prominent plasma-cell infiltration, often involving ducts beneath the nipple and potentially causing a mass, nipple retraction, or discharge. It can clinically mimic carcinoma.

Plasma Cell Myeloma

A malignant neoplasm of clonal plasma cells, usually producing a monoclonal immunoglobulin or light chain. It may cause hypercalcemia, renal impairment, anemia, and destructive bone lesions, collectively known as CRAB features, together with recurrent infection and hyperviscosity in some patients.

Plasma Cell Neoplasm

A clonal disorder of antibody-producing plasma cells, ranging from localized plasmacytoma to multiple myeloma with marrow, bone, kidney, calcium, or blood abnormalities.

Plasma Cells

Fully differentiated, terminal effector B lymphocytes specialized for the high-volume synthesis and secretion of antigen-specific immunoglobulin antibodies. Characteristically demonstrate eccentric round nuclei with distinctive "clock-face" or "cartwheel" chromatin patterns, abundant deeply basophilic cytoplasm, and a prominent pale perinuclear halo (representing an extensive Golgi apparatus). While essential for humoral immunity against pathogens, clonal neoplastic proliferation of plasma cells drives plasma cell dyscrasias, prominently multiple myeloma (characterized by monoclonal protein spikes, osteolytic bone lesions, hypercalcemia, renal impairment, and anemia) and Waldenström macroglobulinemia.

Plasma Concentration

The amount of a substance present in a defined volume of plasma at a given time. It is used to interpret exposure, toxicity, therapeutic drug monitoring, and pharmacokinetic relationships.

Plasma Expanders

Intravenous fluids intended to increase circulating volume, including crystalloids and selected colloids.

Plasma Kallikrein

A serine protease in the contact and kallikrein-kinin systems that generates bradykinin and contributes to vascular permeability and inflammation.

Plasma Membrane

The selectively permeable lipid bilayer surrounding a cell, containing proteins that control transport, signaling, adhesion, recognition, and communication with the extracellular environment.

Plasma Osmolarity

A chemical or physiological term involving plasma osmolarity. Its effects and clinical importance de-

pend on its concentration, route of exposure, and patient context.

Plasma Substitutes

Intravenous fluids intended to expand plasma volume, including crystalloids and colloid solutions; choice depends on the clinical indication and risks.

Plasmablastic Lymphoma

Plasmablastic lymphoma is an aggressive B-cell lymphoma often associated with immune deficiency, HIV infection, or immunosuppression.

Plasmacytoma

A localized tumor of abnormal plasma cells, occurring in bone or soft tissue and sometimes progressing to multiple myeloma.

Plasmalogen

An ether phospholipid abundant in cell membranes, especially in heart, brain, and myelin; impaired synthesis occurs in some peroxisomal disorders.

Plasmapheresis

Removal of plasma from blood, with return of cellular components and replacement fluid, to remove pathogenic antibodies or substances.

Plasmid

A plasmid is a small usually circular DNA molecule separate from a bacterial chromosome that can replicate independently and sometimes carries antibiotic-resistance genes.

Plasmids

Small, circular, extrachromosomal DNA elements that replicate independently of the bacterial chromosome. Plasmids carry non-essential genes that provide selective advantages including antibiotic resistance, virulence factors, and metabolic capabilities.

Plasmin

A proteolytic enzyme that breaks down fibrin and helps dissolve blood clots.

Plasminogen

Plasminogen is the inactive circulating precursor of plasmin, the enzyme that degrades fibrin; tissue or urokinase-type activators convert it to plasmin during fibrinolysis.

Plasminogen Activator

A plasminogen activator is an enzyme, such as tissue plasminogen activator or urokinase, that converts plasminogen to plasmin and promotes fibrin clot breakdown; therapeutic use carries bleeding risk.

Plasminogen Inactivators

Substances or medicines that reduce plasminogen activation and thereby limit conversion of plasminogen to the clot-dissolving enzyme plasmin.

Plasmodesma

A plasmodesma is a microscopic channel through a plant cell wall connecting neighboring cells and allowing controlled movement of molecules and signals.

Plasmodium

A genus of parasites transmitted by *Anopheles* mosquitoes and responsible for malaria.

Plasmodium Vivax

A malaria parasite transmitted by *Anopheles* mosquitoes. It infects red blood cells and can persist in the liver as dormant hypnozoites that cause relapses.

Plasmodium Vivax Malaria

Malaria caused by the protozoan parasite *Plasmodium vivax*, the most geographically widespread form of human malaria. Characterized by recurring paroxysms of fever every 48 hours (benign tertian malaria) and the unique ability of dormant hypnozoite liver stages to cause late clinical relapses months to years after the initial mosquito bite, requiring radical cure with primaquine or tafenoquine.

Alternate Names:

Vivax malaria; Benign tertian malaria

Plaster Of Paris

Calcium sulfate hemihydrate that sets into a hard material, used for casts, splints, and models.

Plastic Bronchitis

A rare condition producing large, branching mucous casts that form in the airways, causing life-threatening airway obstruction, often associated with congenital heart disease or post-fontan circulation.

Plastic Casting Resin Poisoning

Plastic casting resin poisoning involves toxic or environmental exposure to plastic casting resin poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Plastic Resin Hardener Poisoning

Plastic resin hardener poisoning involves toxic or environmental exposure to plastic resin hardener poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Plastic Surgery

Surgery that reconstructs, restores, or alters body tissues for functional or cosmetic reasons.

Plate Method

Simple portion control: half plate vegetables, quarter plate protein, quarter plate grains. Promotes balanced eating.

Platelet

A small piece of a blood cell that helps stop bleeding. Platelets have no nucleus, gather at damaged blood vessels, and stick together to form an early blood clot. They are made in the bone marrow from larger cells called megakaryocytes.

Platelet Aggregation

The adhesion and clumping of platelets to form a hemostatic plug at sites of vascular injury.

Platelet Aggregation Test

A laboratory test measuring platelet clumping in

response to agonists such as ADP, collagen, or epinephrine, used to evaluate platelet function disorders and medication effects.

Platelet Antibodies Blood Test

A blood test detecting antibodies directed against platelet antigens, used selectively in suspected immune thrombocytopenia, neonatal alloimmune thrombocytopenia, or platelet-transfusion refractoriness.

Platelet Count

The number of platelets in a given volume of blood, used to assess bleeding or clotting risk.

Platelet Count Low (Thrombocytopenia (Low Platelet Count))

Alternate index label for Thrombocytopenia (Low Platelet Count).

Platelet Disorder - Immune Thrombocytopenia

Alternate index label for ITP.

Platelet Function Disorders

Conditions in which platelets do not adhere, activate, or aggregate normally, causing impaired primary hemostasis. They may be inherited or acquired and can cause easy bruising, nosebleeds, heavy menstrual bleeding, or bleeding after procedures.

Platelet Plug Formation

The initial hemostatic response to vascular injury, involving platelet adhesion, activation, and aggregation. When the endothelium is disrupted, subendothelial von Willebrand factor and collagen are exposed to flowing blood.

Alternate Names:

Thrombocytes

Platelets (Thrombocytes)

Small, anucleate, disc-shaped cell fragments (2–4 μm diameter) derived from megakaryocytes in the bone marrow, circulating at $150\text{--}400 \times 10^9/\text{L}$.

Alternate Names:

Thrombocytes

Platybasia

Flattening of the skull base, measured by an increased basal angle and sometimes associated with neurologic compression.

Platypnea

Dyspnea that occurs in the upright position and is relieved by lying down. It is the opposite of orthopnea. It is rare and may be associated with hepatopulmonary syndrome, intracardiac shunting, or pulmonary arteriovenous malformations.

Platypnea-Orthodeoxia Syndrome

A rare clinical syndrome characterized by dyspnea (platypnea) and arterial oxygen desaturation (orthodeoxia) that is provoked by assuming an upright seated or standing posture and completely relieved by lying flat in the recumbent position. It is caused by position-dependent right-to-left shunting of deoxygenated blood into the systemic arterial circulation, most commonly arising from an anatomical communication (such as a patent foramen ovale or atrial septal defect) combined with an upright functional alteration (e.g., aortic root dilatation, pneumonectomy, or kyphoscoliosis), or from intrapulmonary vascular shunts in hepatopulmonary syndrome.

Alternate Names:

Orthodeoxia-Platypnea Syndrome; Postural Dyspnea and Deoxygenation

Platysma

A thin superficial muscle of the neck that tenses the skin and assists lower-jaw and lower-lip movement.

Pleat

A fold of tissue or material; in anatomy it may describe a fold of mucosa or serosa.

Pleiotropic Cytokine

A cytokine that acts on multiple cell types or produces several biological effects, such as changes in inflammation, blood formation, immunity, or tissue

repair.

Pleiotropy

Single gene influencing multiple phenotypic traits. Examples: phenylketonuria affects multiple systems; sickle cell anemia causes anemia, pain crises, organ damage.

Pleomorphic

Pleomorphic means having multiple forms or marked variation in cell or tissue appearance; in pathology, it may be a feature of malignancy but is not diagnostic alone.

Pleomorphic Adenoma

A usually benign salivary-gland tumor composed of epithelial and connective-tissue elements. It most often occurs in the parotid gland and may slowly enlarge; long-standing lesions can rarely undergo malignant transformation.

Pleomorphic Adenoma Of The Vulva

An exceptionally rare benign mixed epithelial and mesenchymal tumor of the vulva, resembling pleomorphic adenoma of salivary tissue and requiring histopathologic confirmation.

Pleomorphic Dermal Sarcoma

An uncommon high-grade cutaneous sarcoma, usually on chronically sun-damaged skin of older adults, composed of markedly pleomorphic cells and capable of local recurrence or metastasis.

Pleomorphic Xanthoastrocytoma

A circumscribed astrocytic tumor, often arising superficially in the cerebral hemispheres of younger patients and associated with seizures. Most are low grade, but anaplastic transformation can occur.

Pleomorphism

Variation in the size and shape of cells or their nuclei within a tissue. Marked pleomorphism is a feature of many high-grade malignancies but can also occur in some reactive or degenerative conditions.

Plethoric

Plethoric means unusually full, congested, or ruddy, often describing the appearance associated with increased blood volume or red-cell mass.

Plethysmography

A method measuring changes in volume or blood flow in a body part, used to evaluate vascular obstruction, respiratory volumes, or other physiologic responses.

Pleura

The double-layered membrane surrounding each lung and lining the inner chest wall.

Pleural

Relating to the pleura, the membranes surrounding the lungs and lining the inside of the chest.

Pleural Cavity

The narrow, fluid-filled space between the two layers of the pleura surrounding each lung.

Pleural Decortication

A major thoracic surgical operation involving the painstaking surgical excision and stripping of a thick, fibrous, non-compliant inflammatory membrane or peel (fibrothorax) from the visceral and parietal pleural surfaces of the lung and chest wall. The procedure is indicated for organized, stage III empyema thoracis, chronic hemothorax, or tuberculous pleurisy where the fibrous cortex entraps the underlying lung parenchyma and prevents re-expansion; successful decortication eliminates the restrictive pleural cage, restores diaphragmatic excursion, and obliterates the pleural dead space.

Alternate Names:

Decortication of the Lung; Pulmonary Decortication; Thoracoscopic Decortication

Pleural Disorders

Conditions affecting the pleura, including pleural effusion, pneumothorax, pleurisy, infection, and tumors.

Pleural Effusion

An abnormal buildup of fluid in the space between a lung and the chest wall. It may result from heart failure, infection, inflammation, cancer, or injury, and can cause shortness of breath, cough, or chest pain.

Pleural Effusion (Fluid In The Chest Or On Lung)

Alternate index label for Fluid In the Chest or On Lung.

Pleural Fluid Analysis

Laboratory examination of pleural fluid, including cell count, protein, lactate dehydrogenase, glucose, pH, and microbiology, to classify an effusion and identify its cause.

Pleural Fluid Culture

Culture of pleural fluid to detect bacteria, fungi, or mycobacteria in suspected pleural-space infection; specimens should be collected appropriately, often in blood-culture bottles.

Pleural Fluid Glucose

Pleural-fluid glucose may be reduced in empyema, complicated parapneumonic effusion, rheumatoid pleuritis, tuberculosis, and malignancy. It is interpreted with pH, cell count, protein, LDH, microbiology, imaging, and the clinical setting.

Pleural Fluid Gram Stain

A pleural-fluid Gram stain looks for bacteria in fluid around the lung and may support evaluation of empyema or bacterial pleural infection.

Pleural Fluid PH

Pleural-fluid pH helps assess infection and other causes of effusion. A markedly low pH supports a complicated parapneumonic effusion in the appropriate clinical setting, but drainage decisions also depend on imaging, fluid appearance, microbiology, glucose, and the patient's condition.

Pleural Fluid Smear

Pleural fluid smear is a diagnostic or laboratory investigation; interpretation depends on the specimen, method, reference interval, timing, and clinical question.

Pleural Friction Rub

A grating, leathery sound heard on auscultation caused by inflamed pleural surfaces rubbing against each other during breathing. It is a sign of pleurisy and may be heard in pneumonia, PE, and pleural effusion. It disappears when fluid accumulates and separates the pleural surfaces.

Pleural Needle Biopsy

A pleural needle biopsy removes a small sample of the pleural lining through a needle for evaluation of cancer, infection, or inflammation.

Pleural Neoplasm

A pleural neoplasm is an abnormal growth in the lining around the lung, such as mesothelioma or metastatic cancer.

Pleural Perforation

A breach in the pleural lining allowing air or fluid to enter the pleural space, potentially causing pneumothorax or hydropneumothorax, often from trauma, procedures, or underlying lung disease.

Pleural Plaque

Circumscribed fibrous tissue on parietal pleura, hallmark of asbestos exposure. Bilateral, often calcified. Does not indicate asbestosis.

Pleural Sac

The potential space and enclosing pleural membranes around a lung, normally containing a small amount of lubricating fluid.

Pleural Space

The potential space between the visceral and parietal pleura surrounding each lung.

Pleural Tuberculosis

Tuberculous infection or inflammation of the pleura, often producing a lymphocyte-rich pleural effusion, chest pain, fever, and breathlessness and requiring microbiologic or tissue evaluation.

Pleurectomy

Surgical removal of part or all of the pleura, used in the treatment of malignant pleural mesothelioma and recurrent malignant pleural effusion.

Pleurisy

Inflammation of the pleura, often causing sharp pain worsened by breathing.

Alternate Names:

Pleuritis (Pleurisy)

Pleuritic Pain

Sharp chest pain that worsens with breathing or coughing, often caused by inflammation of the pleura or nearby tissues.

Pleuritis

Inflammation of the pleural membranes (visceral and parietal pleura). Presents with pleuritic chest pain: sharp, stabbing pain exacerbated by deep breathing, coughing, or sneezing.

Pleuritis (Pleurisy)

Alternate index label for Pleurisy.

Pleurodesis

Fusion of visceral and parietal pleura to prevent recurrence of effusions or pneumothorax. Chemical pleurodesis uses talc, doxycycline, or bleomycin.

Pleurodynia

Severe, often episodic chest or upper-abdominal muscle pain associated with viral infection, especially coxsackievirus.

Pleuropulmonary Blastoma

A rare aggressive childhood cancer arising in the lung or pleura, usually associated with DICER1-related tumor predisposition.

Plexiform Fibrohistiocytic Tumor

A rare low-grade malignant fibroblastic and histiocytic tumor, usually arising in the dermis or subcutis of the upper extremities of young people; local recurrence is common and metastasis is uncommon.

Plexiform Neurofibroma

A plexiform neurofibroma is a diffuse nerve-sheath tumor, strongly associated with neurofibromatosis type 1, that can grow along multiple nerve branches and sometimes become malignant.

Alternate Names:

PN PNF

Plexus

A network of intersecting nerves, vessels, or other structures.

Pliability

Pliability is the ability of a material or tissue to bend, stretch, or deform without breaking; it is relevant to skin, vessels, lungs, and medical devices.

Plica

A fold of tissue, especially a synovial fold within a joint.

Plitidepsin

Plitidepsin is an antineoplastic compound that targets eukaryotic elongation factor 1A and has been studied for selected cancers and viral infections.

Ploidies

Ploidy describes the number of complete chromosome sets in a cell, such as haploid, diploid, or polyploid; abnormal ploidy can characterize tumors or developmental disorders.

Plozalizumab

An investigational monoclonal antibody studied as a targeted treatment for inflammatory or malignant disease.

PLS

Alternate index label; see *Primary lateral sclerosis*.

Plug (Vascular)

A vascular occlusion device used to block a selected vessel under image guidance.

Alternate Names:

Vascular

Plummer-Vinson Syndrome

Iron-deficiency anemia with dysphagia and esophageal webs, associated with increased upper gastrointestinal cancer risk.

Pluralibacter Gergoviae

A gram-negative environmental bacterium that can contaminate medicines or cosmetics and rarely cause opportunistic infection.

Pluripotent Stem Cell

A stem cell capable of producing derivatives of all three embryonic germ layers but not, by itself, a complete organism; embryonic and induced pluripotent stem cells are major research sources.

Pluripotential

Capable of differentiating into many, but not all, cell types.

Plutonium

A radioactive heavy metal whose inhalation or ingestion can cause internal radiation exposure and toxicity.

Pm

An abbreviation that may mean afternoon, post-mortem, or picometer; meaning depends on clinical context.

PMBCL

Alternate index label; see *Primary mediastinal large B-cell lymphoma*.

PMBL

Alternate index label; see *Primary mediastinal large B-cell lymphoma*.

PMDD

Premenstrual dysphoric disorder is a severe cyclic mood disorder occurring before menstruation with irritability, depression, anxiety, and functional impairment that improve after menses.

PML

PML commonly means progressive multifocal leukoencephalopathy, a JC-virus demyelinating brain infection during severe immune suppression, but it may have other meanings.

PN

Alternate index label; see *Plexiform neurofibroma*.

Pneumatic

Pertaining to, containing, operated by, or inflated with air or gas under pressure. In clinical medicine, utilized in diagnostic instruments such as the pneumatic otoscope (delivering a controlled air puff against the tympanic membrane to assess drum mobility, essential for diagnosing otitis media with effusion), medical devices such as pneumatic compression stockings (intermittent pneumatic compression [IPC] for deep vein thrombosis prophylaxis), and pneumatic tourniquets utilized to maintain a bloodless operative field during orthopedic extremity surgery.

Pneumatic Otoscopy

A diagnostic technique that combines visualization of the tympanic membrane with assessment of its mobility in response to positive and negative air pressure changes, providing information about the presence of middle ear fluid and the condition of the Eustachian tube.

Pneumatic Retinopexy

A procedure for selected retinal detachments in which an intravitreal gas bubble is injected and laser or cryotherapy seals the retinal break while positioning helps the bubble support the repair.

Pneumatic Tourniquet

An inflatable cuff used to temporarily reduce blood flow to a limb during surgery or selected procedures.

Pneumatocele

A thin-walled, air-filled cavity in the lung, often developing after infection or trauma.

Pneumatosis

Presence of gas within a tissue or organ wall, such as pneumatosis intestinalis.

Pneumatosis Cystoides Coli

Condition characterized by multiple gas-filled cysts within the wall of the colon, which may be primary (idiopathic) or secondary to intestinal mucosal disruption from conditions such as chronic obstructive pulmonary disease, ischemic bowel disease, or intestinal obstruction. Often discovered incidentally on imaging or colonoscopy.

Pneumatosis Cystoides Intestinalis

Gas-filled cysts within the bowel wall that may be primary or secondary to bowel ischemia, infection, chronic lung disease, or medications. Often an incidental finding on imaging. Primary pneumatosis is benign and asymptomatic. Secondary pneumatosis may indicate serious underlying disease. Managed by treating the underlying condition.

Pneumatosis Intestinalis

Presence of gas within the bowel wall appearing as linear or cystic radiolucencies on imaging. Primary pneumatosis is benign and idiopathic accounting for approximately fifteen percent of cases.

Pneumocephalus

The presence of air or gas within the cranial cavity, most commonly following trauma, surgery, or sinus disease, and potentially causing increased intracranial pressure if tension develops.

Pneumococcal Infection

Infection caused by *Streptococcus pneumoniae*, ranging from ear or sinus infection to pneumonia, meningitis, or bloodstream infection.

Pneumococcal Meningitis

Meningitis caused by *Streptococcus pneumoniae*, a

medical emergency requiring immediate antibiotics and supportive care.

Pneumococcal Sepsis

Sepsis caused by *Streptococcus pneumoniae*, often arising from pneumonia, meningitis, or another invasive pneumococcal infection.

Pneumoconiosis

A group of occupational lung diseases caused by inhalation of mineral or industrial dusts.

Pneumocystis

A genus of fungi that includes *Pneumocystis jirovecii*, which can cause pneumonia in people with weakened cellular immunity.

Pneumocystis Carinii

An older name associated with *Pneumocystis*; in humans, the usual species causing pneumonia is *Pneumocystis jirovecii*.

Pneumocystis Carinii Pneumonia

The historical name for *Pneumocystis jirovecii* pneumonia, an opportunistic lung infection in immunocompromised people.

Pneumocystis Infection

Infection by *Pneumocystis jirovecii*, most often causing pneumonia with dry cough, fever, and progressive shortness of breath in immunocompromised people.

Pneumocystis Jiroveci

Pneumocystis jirovecii is a fungus that causes opportunistic pneumonia, especially in advanced HIV or other cellular immunodeficiency; symptoms include progressive dry cough, fever, and hypoxaemia, and diagnosis requires appropriate respiratory testing.

Pneumocystis Jirovecii Pneumonia

Opportunistic pneumonia caused by *Pneumocystis jirovecii*, especially in people with advanced HIV infection or other cellular immunodeficiency. It causes progressive exertional dyspnea, dry cough,

fever, and hypoxemia.

Pneumocystis Pneumonia

Opportunistic pneumonia caused by *Pneumocystis jirovecii*, usually in people with impaired cell-mediated immunity. It may cause progressive exertional dyspnea, dry cough, fever, and hypoxemia with diffuse ground-glass changes. Diagnosis and treatment require clinical, microbiologic, and radiologic assessment; trimethoprim-sulfamethoxazole is commonly used.

Pneumocyte

An epithelial cell lining the alveoli; type I cells exchange gases and type II cells produce surfactant.

Pneumoencephalography

An obsolete imaging procedure that replaced cerebrospinal fluid with air to outline the brain on X-ray.

Pneumography

Contrast imaging of body cavities. Pneumoencephalography obsolete. Pneumoperitoneum for laparoscopy.

Pneumolysis

Surgical separation or freeing of lung tissue or pleural adhesions.

Pneumomediastinum

Air within the mediastinum, caused by trauma, rupture, barotrauma, or air tracking from the lungs or neck.

Pneumonectomy

The major surgical resection and removal of an entire lung (either right or left), historically performed via posterolateral thoracotomy and increasingly via minimally invasive video-assisted or robotic techniques. Indicated primarily for centrally located non-small cell lung cancer involving mainstem bronchi or the pulmonary hilum that cannot be excised by anatomical lobectomy or sleeve resection, and rarely for extensive unilateral destruction from multidrug-resistant tuberculosis or bronchiectasis.

Carries significant perioperative morbidity and mortality (higher on the right side due to larger resected lung volume, higher risk of post-pneumonectomy pulmonary edema, and bronchopleural fistula formation).

Pneumonia

Infection or inflammation of lung tissue causing symptoms such as cough, fever, breathlessness, and abnormal imaging.

Pneumonia - Weakened Immune System

Pneumonia in a person with impaired immunity, which may be caused by bacteria, viruses, fungi, or opportunistic organisms and can progress rapidly.

Pneumonia, Bilateral

Pneumonia affecting both lungs, which may cause more extensive oxygen-exchange impairment than disease confined to one lung.

Pneumonitis

Inflammation of lung tissue, often from immune, drug, radiation, aspiration, or environmental causes rather than infection.

Pneumopericardium

Air in the pericardial space, usually from trauma (penetrating chest injury), esophageal perforation, or iatrogenic causes (central line placement, cardiac surgery). Can cause cardiac tamponade if under tension. Diagnosis by chest X-ray showing air surrounding the heart (air-fluid level if combined with effusion).

Pneumoperitoneum

Air or gas within the peritoneal cavity, commonly from perforated bowel or abdominal surgery.

Pneumoplasty

Surgical repair or reconstruction of lung tissue or an airway.

Pneumotachometer

An instrument that measures the rate of airflow

during breathing.

Pneumothorax

Air in the pleural space causing partial or complete lung collapse.

Pneumothorax - Infants

Pneumothorax in an infant is air outside the lung that can compress it and cause breathing difficulty, low oxygen, or cardiovascular instability.

Pneumovirinae

Pneumovirinae is a subfamily of viruses that includes respiratory pathogens such as respiratory syncytial virus and human metapneumovirus.

Pneumovirus

A group or genus of respiratory viruses; human pneumovirus terminology often refers to human metapneumovirus or related paramyxoviruses.

PNF

Alternate index label; see *Plexiform neurofibroma*.

Pocket Erosion

Loss of tissue at the margin or base of a periodontal or surgical pocket, potentially exposing deeper structures and promoting infection, bleeding, or impaired healing.

Podagra

Acute gout affecting the first metatarsophalangeal joint, causing severe pain, redness, and swelling.

Podiatrist

A healthcare professional specializing in disorders of the feet and lower limbs.

Podiatry

The healthcare field concerned with the feet, ankles, and related lower-limb disorders.

Podocyte

A podocyte is a specialized epithelial cell covering glomerular capillaries; injury to its filtration slits

contributes to proteinuria and kidney disease.

Podofilox

Podofilox is a topical antimitotic medicine used to treat selected external anogenital warts caused by human papillomavirus; irritation and correct application are important.

Podophyllin

Podophyllin is a plant-derived resin with antimitotic activity formerly used topically for warts; it can be severely toxic if absorbed and is generally replaced by safer products.

Podophyllum

A plant genus containing podophyllotoxin, used in some topical wart treatments but toxic if misused.

Podosome

A podosome is an actin-rich adhesion and matrix-degrading structure in cells such as macrophages and osteoclasts.

POEMS Syndrome

A rare plasma-cell disorder named for polyneuropathy, organomegaly, endocrinopathy, monoclonal plasma-cell disorder, and skin changes. Other manifestations include sclerotic bone lesions, papilledema, fluid overload, thrombocytosis, and elevated vascular endothelial growth factor.

Poikilocytosis

The presence of an abnormal, marked variation in the physical shape of mature circulating red blood cells (erythrocytes) on a peripheral blood smear. While normal erythrocytes are uniform biconcave discs, poikilocytosis reflects disrupted erythropoiesis, mechanical shear injury, or hemoglobin defects. Specific morphologic poikilocytes offer vital diagnostic clues: schistocytes (fragmented helmet cells in microangiopathic hemolytic anemias like TTP, HUS, and DIC), spherocytes (hereditary spherocytosis, autoimmune hemolytic anemia), dacryocytes (teardrop cells in primary myelofibrosis), drepanocytes (sickle cells in sickle cell anemia), and target cells (codocytes in thalassemia and

hemoglobin C disease).

Poikiloderma

A skin pattern combining mottled pigmentation, visible small vessels, and epidermal thinning, seen in connective-tissue disease, chronic sun damage, inflammation, and some cutaneous lymphomas.

Poikilotherm

An organism whose body temperature varies substantially with environmental temperature; humans are homeothermic.

Poinsettia Plant Exposure

Poinsettia plant exposure involves toxic or environmental exposure to poinsettia plant exposure; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Point

A specific location, spot, or value; in medicine it may refer to a tender point, pressure point, or measurement point.

Point Mutation

Change in single nucleotide base pair. Substitutions: transition (purine to purine, pyrimidine to pyrimidine) or transversion (purine to pyrimidine). Effects: silent (same amino acid), missense (different amino acid), nonsense (stop codon).

Point Tenderness - Abdomen

Abdominal point tenderness is sharply localized pain when a particular abdominal site is pressed, which may indicate inflammation, infection, obstruction, or another focal process.

Point-Of-Care Testing

Diagnostic testing performed near the patient rather than in a central laboratory. POCT can shorten turnaround time but requires validated devices, quality control, operator training, and appropriate interpretation.

Point-Of-Care Ultrasound

Bedside ultrasonography performed and interpreted by the treating clinician to answer focused diagnostic or procedural questions. Emergency applications include cardiac activity, volume status, pulmonary pathology, abdominal free fluid, vascular thrombosis, and procedural guidance.

Poison

A substance capable of causing injury, illness, or death when absorbed in sufficient amount.

Poison Control

A specialized service that provides immediate advice about exposure to medicines, chemicals, or other potential poisons.

Poison Control Center - Emergency Number

A poison-control center provides immediate expert advice after a possible poisoning or toxic exposure; the correct emergency number depends on the country and life-threatening symptoms require emergency services.

Poison Ivy - Oak - Sumac

Poison ivy, oak, and sumac cause allergic contact dermatitis through urushiol oil, producing intensely itchy red patches and blisters after skin exposure.

Poison Ivy - Oak - Sumac Rash

A poison-ivy, oak, or sumac rash is itchy allergic contact dermatitis caused by urushiol, often appearing in streaks and spreading as exposed oil remains on skin or clothing.

Poison Ivy, Oak, And Sumac

Poison ivy, oak, and sumac cause allergic contact dermatitis through urushiol oil, producing intensely itchy streaks or patches of redness, blisters, and swelling. Washing exposed skin and clothing promptly helps; breathing difficulty or facial swelling is an emergency.

Alternate Names:

Allergy Plant Contact (Poison Ivy, Oak, And

Sumac)

Poison-Center Consultation

Specialist toxicology advice used to identify hazards, interpret exposure and laboratory data, select antidotes, guide observation or disposition, and coordinate public-health response. Consultation is appropriate for significant, uncertain, mixed, pediatric, or unusual exposures.

Poisoning

Harmful effects caused by exposure to a toxic substance.

Poisoning - Fish And Shellfish

Poisoning - fish and shellfish involves toxic or environmental exposure to poisoning - fish and shellfish; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Poisoning Arsenic (Arsenic Poisoning)

Alternate index label for Arsenic Poisoning.

Poisoning Ciguatera (Ciguatera Poisoning)

Alternate index label for Ciguatera Poisoning.

Poisoning First Aid

Poisoning first aid involves toxic or environmental exposure to poisoning first aid; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Poisoning Thallium (Thallium)

Alternate index label for Thallium.

Poisson Distribution

A probability distribution modeling the number of events occurring in a fixed interval when events occur independently at a stable average rate.

Pokeweed Mitogen

Pokeweed mitogen is a plant lectin used experimentally to stimulate lymphocyte proliferation and investigate immune-cell responses.

Pokeweed Poisoning

Pokeweed poisoning involves toxic or environmental exposure to pokeweed poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Pol Gene

The HIV gene encoding enzymes required for viral replication, including protease, reverse transcriptase, and integrase; antiretroviral drugs target these products.

Poland Syndrome

A congenital condition characterized by absence or underdevelopment of part of the pectoral muscles on one side, sometimes accompanied by abnormalities of the hand on the same side.

Polar

Relating to a pole, direction, or molecule with separated electrical charges.

Polarity

The distribution or orientation of positive and negative electrical or biological properties.

Polarography

Polarography is an electrochemical analytical method that measures current while voltage changes at an electrode to identify or quantify reducible substances.

Polidocanol

Polidocanol is a detergent sclerosant injected to close selected varicose veins or vascular lesions by damaging the vessel lining and promoting fibrosis.

Polio

Polio is poliomyelitis, an infection caused by poliovirus, an enterovirus of the Picornaviridae family. It is usually asymptomatic or mild but can cause meningitis or irreversible asymmetric flaccid paralysis; transmission is primarily fecal-oral.

Poliomyelitis

An infection caused by poliovirus, usually spread by the fecal-oral route. Most infections are asymptomatic or cause a mild febrile illness, but some cause meningitis or damage to anterior-horn motor neurons, resulting in acute asymmetric flaccid paralysis and sometimes permanent weakness.

Poliosis

A localized patch of white or depigmented hair caused by reduced melanin in follicles, occurring in vitiligo, alopecia areata, inflammatory disease, genetic syndromes, or after injury.

Poliovirus

An enterovirus that causes poliomyelitis. It is transmitted mainly by the fecal-oral route; most infections are asymptomatic, but a small proportion cause meningitis or acute flaccid paralysis by damaging motor neurons in the spinal cord. Diagnosis uses molecular testing, usually reverse-transcription PCR of stool or other appropriate specimens.

Polish (Dental)

Smoothing restorative surfaces to reduce plaque accumulation. Essential for longevity.

Alternate Names:

Dental

Politzerization

A historical method of inflating the middle ear through the nose during swallowing to improve Eustachian-tube ventilation.

Pollakisuria

Pollakisuria is unusually frequent urination, often in small amounts, caused by irritation, infection, stones, bladder disorders, anxiety, or other urinary conditions.

Pollakiuria

Frequent urination, often in small amounts. It may occur with urinary infection, irritation, diabetes, pregnancy, medicines, or anxiety; in some children it is a temporary benign pattern, but pain, fever,

blood, thirst, or weight loss needs assessment.

Pollen

Fine grains released by plants that can trigger allergic rhinitis, asthma, or conjunctivitis.

Pollen Tube

A pollen tube is a tube grown by a germinating pollen grain that carries sperm cells through plant tissue to the ovule for fertilization.

Pollex

The thumb; the first, most radial, and opposable digit of the human hand (digit I). Uniquely comprised of only two phalanges (proximal and distal) and articulated with the carpal trapezium bone via a mobile saddle-shaped carpometacarpal joint. Endowed with specialized intrinsic and extrinsic musculature (the thenar muscles, innervated by the recurrent branch of the median nerve, and adductor pollicis, innervated by the ulnar nerve), enabling precise tip-to-tip and pad-to-pad opposition that is critical for fine motor dexterity and human grasp.

Pollination

Transfer of pollen from anthers to a receptive stigma or ovule, allowing fertilization in flowering plants; it is not a human reproductive process.

Pollution

Contamination of air, water, soil, or surroundings by substances that may harm health or ecosystems.

Poly A

A sequence of adenine nucleotides added to the end of many messenger RNA molecules, helping regulate stability, transport, and translation.

Poly I-C

A synthetic double-stranded RNA molecule used experimentally to stimulate antiviral innate immune responses through pattern-recognition receptors.

Poly ICLC

A stabilized form of poly I-C studied as an experi-

mental immune stimulant and antiviral or anticancer treatment.

Polyadenylation

Addition of a polyadenosine tail to the 3' end of many eukaryotic messenger RNAs, promoting stability, nuclear export, and efficient translation.

Polyamine

A small positively charged molecule, such as putrescine, spermidine, or spermine, required for cell growth and nucleic-acid function.

Polyamine Catabolism

Enzymatic breakdown of polyamines, producing metabolites that influence cell growth, oxidative stress, and signaling.

Polyarteritis

Inflammation of multiple arteries; polyarteritis nodosa is a systemic necrotizing vasculitis.

Polyarteritis Nodosa

Necrotizing inflammation of medium-sized muscular arteries, spares lungs. Associated with hepatitis B (10-30%). Clinical: systemic symptoms, mononeuritis multiplex, livedo reticularis, renal involvement (hypertension, renal insufficiency—no glomerulonephritis), GI ischemia.

Alternate Names:

PAN (Polyarteritis Nodosa)

Polyarteritis Nodosa

Alternate index label for Polyarteritis Nodosa.

Polyarthrititis

Inflammation involving multiple joints, commonly defined as five or more. It is a clinical pattern rather than a single diagnosis and may result from autoimmune, infectious, crystal-related, or other inflammatory disorders.

Polybrominated Biphenyl

A group of brominated industrial chemicals that can persist in the environment and cause toxic effects

after exposure.

Polychlorinated Biphenyl

A group of persistent industrial chemicals that can accumulate in body fat and are associated with cancer, endocrine, immune, and neurologic effects.

Polychondritis

Relapsing inflammatory disorder targeting cartilage throughout the body, causing pain, swelling, and destruction of the ears, nose, larynx, trachea, and joints. May lead to airway collapse and hearing loss.

Polychromasia

Presence of immature, bluish red blood cells in peripheral blood, reflecting increased marrow release.

Polychromatophilia

Polychromatophilia is bluish-gray staining of red blood cells on a blood smear, reflecting increased circulating immature reticulocytes from stimulated red-cell production.

Polycystic Kidney Disease

An inherited disorder in which numerous kidney cysts enlarge the kidneys and may cause hypertension and renal failure.

Alternate Names:

PKD

Polycystic Liver Disease

Dominant polycystic liver disease is an autosomal dominant condition characterized by multiple hepatic cysts that may cause massive hepatomegaly, abdominal pain, and compressive symptoms. Usually associated with autosomal dominant polycystic kidney disease. Liver function is preserved even with extensive cyst involvement.

Polycystic Ovarian Syndrome

A hormonal and metabolic disorder involving ovulatory dysfunction and androgen excess, often causing irregular periods, acne, excess hair, infertility, or metabolic risk.

Polycystic Ovary

Polycystic ovaries contain many small follicles and may occur as a normal variant or with polycystic ovary syndrome. PCOS additionally involves ovulatory dysfunction, androgen excess, irregular periods, infertility, acne, or metabolic risk and requires clinical—not ultrasound alone—diagnosis.

Alternate Names:

PCOS (Polycystic Ovary)

Polycystic Ovary

Alternate index label for Polycystic Ovary.

Polycystic Ovary Syndrome

Common endocrine disorder affecting six to twelve percent of reproductive-aged women characterized by hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology on ultrasound.

Polycystic Ovary Syndrome

A common endocrine disorder involving ovulatory dysfunction, androgen excess, and polycystic ovarian morphology in varying combinations.

Polycythaemia

An increased red-cell mass or concentration, which may be primary or secondary to another condition.

Polycythemia

Elevated red blood cell mass/hematocrit. Primary: polycythemia vera (JAK2 V617F mutation, tri-lineage proliferation, low EPO, risk of thrombosis and myelofibrosis/leukemia). Secondary: hypoxia (COPD, high altitude, congenital heart disease, sleep apnea) with high EPO, or EPO-secreting tumors (renal cell, hepatoma).

Polycythemia (High Red Blood Cell Count)

Alternate index label for High Red Blood Cell Count.

Polycythemia - Newborn

Neonatal polycythemia is an abnormally high red-cell concentration in a newborn that can increase

blood viscosity and cause poor feeding, lethargy, breathing problems, or low glucose.

Polycythemia Vera

A chronic myeloproliferative neoplasm characterized by increased red blood cell mass, often with increased white blood cell and platelet counts, driven by constitutive activation of the JAK-STAT signaling pathway.

Alternate Names:

Primary Polycythemia

Polycythemia Vera

Polycythemia vera is a myeloproliferative blood cancer, usually driven by a JAK2 mutation, causing excess red cells and increased risks of thrombosis, bleeding, and transformation.

Polydactylism (Polydactyly)

A common congenital limb malformation characterized by the presence of supernumerary (extra) fingers or toes. Classified anatomically into preaxial polydactyly (radial/thumb side in the hand, tibial/great toe side in the foot), postaxial polydactyly (ulnar/little finger side, fibular/little toe side; the most prevalent form, frequently inherited as an autosomal dominant trait in individuals of African ancestry), and central polydactyly (duplication of digits II, III, or IV). Can present as an isolated nonsyndromic trait or as part of complex genetic syndromes (e.g., Bardet-Biedl syndrome, Ellis-van Creveld syndrome, Patau syndrome [trisomy 13]); treated surgically by reconstructive excision and ligamentous balancing.

Polydactyly

The congenital presence of one or more additional digits (fingers or toes) beyond the normal pentadactyl complement of five per limb. Arises from aberrant signaling during embryonic limb bud patterning, particularly involving the zone of polarizing activity and the sonic hedgehog (*SHH*) signaling pathway. Categorized by location as preaxial (radial/tibial), postaxial (ulnar/fibular), or central. Supernumerary digits range from rudimentary, hypoplastic soft-tissue tags ("floating" digits) to fully

articulated, osseous-containing digits requiring complex orthopedic and reconstructive surgical excision to preserve normal hand or foot function.

Polydipsia

Excessive, abnormal, and unquenchable thirst leading to high fluid intake. Cardinal clinical symptom of osmotic diuresis driven by severe hyperglycemia in uncontrolled diabetes mellitus (type 1 or type 2), alongside polyuria and weight loss. Must be clinically differentiated from central diabetes insipidus (deficient antidiuretic hormone [ADH/vasopressin] secretion), nephrogenic diabetes insipidus (renal collecting duct resistance to ADH, frequently caused by lithium therapy), and psychogenic polydipsia (compulsive water drinking in psychiatric disorders, which can precipitate life-threatening dilutional hyponatremia and cerebral edema).

Polyendocrine Metabolic Ovarian Syndrome

An older or nonstandard term generally referring to polycystic ovary syndrome or a related endocrine-metabolic disorder affecting ovarian function.

Polyene

A molecule containing multiple carbon-carbon double bonds; polyene medicines such as amphotericin B act against fungal cell membranes.

Polyethylene

A plastic polymer used in packaging and medical devices; implanted polyethylene components may wear over time and produce foreign-body reactions.

Polyethylene Glycol

A polymer used as an osmotic laxative, excipient, and in various medical products.

Polygenic

Influenced by or caused by variants in multiple genes.

Polygenic Disease

A disease influenced by variants in many genes to

gether with environmental and lifestyle factors.

Polygenic Inheritance

Trait influenced by multiple genes, each with small effect. Examples: height, skin color, common diseases (diabetes, heart disease). Produces continuous variation.

Polygenic Obesity

Obesity arising from the combined effects of many genetic variants interacting with diet, activity, sleep, medications, and the wider environment rather than a single gene defect.

Polygenic Risk Score

A statistical estimate of disease susceptibility calculated from the combined effects of many genetic variants, each usually contributing a small amount of risk. It modifies but does not replace clinical risk assessment or environmental factors.

Polyglandular Autoimmune Syndrome

An autoimmune disorder in which two or more endocrine glands are damaged, causing combinations of thyroid, adrenal, parathyroid, pancreatic, or gonadal dysfunction.

Polyglutamate Camptothecin

An investigational polymer-linked form of camptothecin designed to alter delivery of a topoisomerase-I-inhibiting anticancer drug.

Polygraphy

Polygraphy is the simultaneous recording of several physiologic signals, such as breathing, heart rate, oxygen level, and movement, often during sleep testing.

Polyhydramnios

Polyhydramnios is excessive amniotic fluid during pregnancy and may increase the risk of maternal discomfort, preterm birth, or complications during delivery.

Polyketide Synthase

An enzyme system that builds complex natural products from repeated carbon units, producing many antibiotics, toxins, and other biologically active molecules.

Polylysine

A polymer made of lysine residues used in research, biomaterials, food preservation, and some medical applications.

Polymagia (Polyphagia)

An abnormally ravenous appetite and excessive consumption of food (hyperphagia). In clinical internal medicine, polyphagia forms one of the classic "3 Ps" of newly diagnosed or poorly controlled diabetes mellitus (along with polyuria and polydipsia), arising because cellular glucose starvation from absolute or relative insulin deficiency triggers hypothalamic appetite centers despite severe intravascular hyperglycemia. Also observed in thyrotoxicosis/hyperthyroidism (high metabolic rate), Prader-Willi syndrome (ghrelin dysregulation and loss of hypothalamic satiety), and atypical depressive or bulimic eating disorders.

Polymastia

The presence of extra breast tissue or additional nipples.

Polymenorrhea

Menstrual periods that occur unusually often, commonly at intervals shorter than about 21 days. Causes include hormonal changes, thyroid disease, fibroids, bleeding disorders, medicines, and the menopausal transition; persistent frequent bleeding needs evaluation.

Polymenorrhoea

British spelling of polymenorrhea: menstrual periods occurring unusually often, commonly at intervals shorter than about 21 days.

Polymer

A large molecule made of repeating smaller units called monomers; polymers are widely used in biol-

ogy, medicines, and medical materials.

Polymerase Chain Reaction

An in-vitro method that exponentially amplifies a defined DNA segment through repeated cycles of denaturation, primer annealing, and DNA synthesis by a thermostable polymerase. It is used in diagnosis, research, forensics, and pathogen detection.

Polymerase Chain Reaction

An enzymatic molecular biology technique for exponentially amplifying targeted DNA sequences in vitro. Driven by repeated thermal cycles consisting of three steps: thermal denaturation of double-stranded DNA (94–98°C), sequence-specific oligonucleotide primer annealing (50–65°C), and primer extension by a thermostable DNA polymerase (e.g., *Taq* polymerase) at 72°C. Used in infectious disease diagnostics, forensic identification, oncologic mutation testing, and quantitative real-time PCR (RT-qPCR) for viral load quantification.

Polymerization

Chemical joining of many small monomers into a larger polymer, a process important in biomolecules, plastics, dental materials, drug delivery, and laboratory techniques.

Polymicrobial Infections

Infections caused by multiple microbial species. Polymicrobial infections are common in wound infections, intra-abdominal infections, and respiratory infections. The interactions between organisms can be synergistic or antagonistic. Understanding polymicrobial infections is essential for selecting appropriate antimicrobial therapy.

Polymicrogyria

A brain malformation characterized by an excessive number of abnormally small, fused gyri, resulting in an irregular, bumpy cortical surface with shallow sulci. Caused by abnormal late neuronal migration or post-migrational cortical organization. May be focal or diffuse, unilateral or bilateral, and symmetric or asymmetric.

Polymorphic

A clinical or therapeutic concept involving polymorphic. Its features, uses, and clinical significance depend on the relevant medical context.

Polymorphism

The occurrence of two or more genetic forms or phenotypes in a population.

Polymorphonuclear Leukocyte

A white blood cell with a multilobed nucleus, especially a neutrophil, eosinophil, or basophil, involved in innate immune defense and inflammation.

Polymorphous

Having lesions or clinical features that differ in shape, size, or stage and therefore show multiple forms.

Polymorphous Light Eruption

Polymorphous light eruption is an itchy inflammatory rash triggered by sunlight, usually appearing hours to days after exposure and recurring seasonally in susceptible people.

Alternate Names:

Sun Poisoning

Polymyalgia Rheumatica

An inflammatory disorder causing aching and morning stiffness around the shoulders and hips in older adults.

Alternate Names:

PMR (Polymyalgia Rheumatica)

Polymyalgia Rheumatica

Polymyalgia rheumatica causes abrupt bilateral shoulder and hip-girdle aching and stiffness in older adults and can occur with giant-cell arteritis.

Polymyositis

An autoimmune inflammatory muscle disease causing symmetric proximal muscle weakness.

Alternate Names:

PM (Polymyositis)

Polymyositis

Alternate index label for Polymyositis.

Polymyositis - Adult

Adult polymyositis is an uncommon immune-mediated inflammatory muscle disease causing progressive symmetric proximal weakness and elevated muscle enzymes, after exclusion of mimics.

Polymyxin

A group of antibiotics that disrupt bacterial cell membranes, used mainly for selected gram-negative infections.

Polymyxin B

An antibiotic active against selected gram-negative bacteria; kidney and nerve toxicity limit its use to difficult infections.

Polyneuritis

Inflammation or injury affecting multiple peripheral nerves, causing symmetric pain, numbness, weakness, or autonomic dysfunction; causes include immune disease, infection, toxins, and deficiency.

Polyneuritis, Acute Idiopathic

A clinical or therapeutic concept involving polyneuritis, acute idiopathic. Its features, uses, and clinical significance depend on the relevant medical context.

Polyneuropathy

A generalized disorder affecting multiple peripheral nerves, often causing symmetric numbness, tingling, burning pain, weakness, or impaired reflexes in the limbs.

Polynucleotide

A chain of nucleotides forming DNA, RNA, or a related laboratory molecule.

Polyoma Virus

A polyomavirus; some human polyomaviruses are widespread and cause serious disease mainly in immunocompromised people.

Polyomavirus Hominis 2

A human polyomavirus designation associated with a virus found in people; clinically important disease is uncommon and mainly associated with immune suppression.

Polyp

A growth projecting from a mucous membrane; it may be benign, precancerous, or rarely cancerous.

Polyp Biopsy

A polyp biopsy removes part or all of a polyp for microscopic examination to determine whether it is benign, precancerous, or cancerous.

Polyp Surveillance Colonoscopy

Interval-based endoscopic follow-up of patients with previously identified colorectal polyps. Intervals are determined by polyp number, size, and histological features. One to two small adenomas require surveillance at seven to ten years. Three to four adenomas at three to five years.

Polypeptide

A chain of amino acids linked by peptide bonds.

Polyphagia

Excessive, uncontrollable hunger or increased appetite. Classic cardinal symptom of uncontrolled Diabetes Mellitus (cellular starvation due to lack of insulin or insulin resistance), Hyperthyroidism (elevated basal metabolic rate), Prader-Willi Syndrome, and atypical depression.

Polypharmacy

Use of multiple medications by one patient, often defined numerically as five or more concurrent medicines. It increases the risk of interactions, adverse effects, prescribing cascades, nonadherence, and medication errors, although some multidrug regimens are clinically appropriate.

Polyphenol

A plant-derived compound containing multiple phenolic groups, including flavonoids and tannins, with

antioxidant and cell-signaling effects.

Polyploid

Having more than two complete sets of chromosomes in a cell or organism.

Polyposis

A condition in which numerous polyps develop, especially in the gastrointestinal tract.

Polypropylene

A synthetic plastic polymer used in medical supplies, sutures, meshes, and containers; implanted materials may rarely cause infection or foreign-body reactions.

Polyps Colon (Colon Polyps)

Alternate index label for Colon Polyps.

Polyps Rectal (Colon Polyps)

Alternate index label for Colon Polyps.

Polyps Uterus (Uterine Fibroids)

Alternate index label for Uterine Fibroids.

Polyps, Colon

Alternate index label; see *Colon polyps*.

Polyps, Endometrial

Alternate index label; see *Uterine polyps*.

Polyps, Nasal

Alternate index label; see *Nasal polyps*.

Polyps, Stomach

Alternate index label; see *Stomach polyps*.

Polyps, Uterine

Alternate index label; see *Uterine polyps*.

Polyradiculitis

Inflammation or other disease affecting several spinal nerve roots. It may cause weakness, numbness, pain, and reduced reflexes; causes include in-

fections, autoimmune disorders, and toxins.

Polyradiculopathy

Disease affecting multiple spinal nerve roots, producing pain, weakness, sensory loss, or reflex changes in a distribution determined by the involved roots.

Polysaccharide

A carbohydrate made of many linked simple sugar units. Examples include starch, glycogen, cellulose, and some bacterial capsules.

Polysome

A group of ribosomes translating the same messenger RNA molecule at the same time.

Polysomia

The presence of extra copies of one or more chromosomes or, in developmental terminology, multiple body structures.

Polysomnography

An overnight sleep study that records brain activity, eye movements, muscle activity, heart rhythm, airflow, breathing effort, and blood oxygen. It helps evaluate sleep apnea, movement disorders, parasomnias, and other sleep conditions.

Polysomnography (Sleep Study)

Overnight monitoring for sleep-disordered breathing. Parameters: EEG (sleep staging: N1, N2, N3, REM), EOG (eye movements), EMG (chin and leg movements), airflow (nasal pressure, thermistor), respiratory effort (chest/abdomen belts), snore microphone, body position, oximetry (SpO2), and ECG.

Alternate Names:

Sleep Study

Polysplenia

A congenital condition with multiple small spleens and variable abnormalities of organ arrangement, heart, blood vessels, or intestinal rotation.

Polystyrene

A synthetic plastic polymer used in containers and laboratory materials; burning it can release hazardous fumes.

Polytene Chromosome

An enlarged chromosome formed by repeated DNA replication without cell division, displaying visible bands and commonly used to study gene activity in insect tissues.

Polythelia

The presence of one or more additional nipples, usually along the embryonic milk line. It is often harmless, although accessory tissue can rarely have the same disorders as normal breast tissue.

Polythiazide

A thiazide diuretic formerly used to treat high blood pressure and fluid retention by increasing renal salt and water excretion.

Polyunsaturated Fat

Liquid at room temperature. Sources: fish, vegetable oils, nuts. Essential fatty acids.

Polyurethane

A family of synthetic polymers used in medical tubing, implants, dressings, and other products; tissue response depends on formulation and site.

Polyuria

Excessive urine production, commonly more than about 3 liters per day in an adult.

Pompe Disease

An autosomal recessive lysosomal storage and glycogen storage disorder (GSD Type II) caused by deficiency of acid alpha-glucosidase (acid maltase), leading to progressive accumulation of glycogen in lysosomes, particularly in cardiac and skeletal muscle. The infantile-onset form presents with massive cardiomegaly, severe hypotonia (“floppy baby”), respiratory failure, and death in early infancy without treatment; late-onset forms present as progressive

limb-girdle and diaphragm muscle weakness, treatable with recombinant enzyme replacement therapy.

Alternate Names:

Glycogen storage disease type II; Acid maltase deficiency; GSD II

Polyvalent

Effective against or involving multiple types, strains, antigens, or targets.

Polyvinyl

A family of polymers used in plastics and medical materials; properties vary according to attached chemical groups.

Pomalidomide

An immunomodulatory medicine used with other treatments for selected multiple myeloma and related blood cancers; it can cause severe birth defects and blood abnormalities.

Pompholyx

A form of dyshidrotic eczema causing intensely itchy small blisters on the hands or feet.

Pompholyx Eczema

Pompholyx, or dyshidrotic eczema, causes intensely itchy small blisters on the palms, sides of fingers, or soles and may leave painful peeling or cracks.

Pons

The middle part of the brainstem, containing pathways and nuclei involved in movement, sensation, breathing, and cranial-nerve function.

Pontiac Fever

A mild, self-limited flu-like illness caused by Legionella bacteria without the pneumonia typical of Legionnaires’ disease.

Pontine

Pertaining to or located within the pons, the middle division of the brainstem situated between the midbrain superiorly and the medulla oblongata inferiorly, and anterior to the fourth ventricle and cere-

bellum. Contains essential descending motor tracts (corticospinal and corticobulbar tracts), ascending sensory pathways (medial lemniscus, spinothalamic tract), the locus coeruleus, the pontine micturition center, and the cranial nerve nuclei for CN V (trigeminal), VI (abducens), VII (facial), and VIII (vestibulocochlear). Pathologies include acute pontine ischemic stroke (presenting with crossed hemiparesis or catastrophic locked-in syndrome), pontine hemorrhage (pinpoint reactive pupils, coma, quadriparesis, and hyperpyrexia), and osmotic demyelination syndrome (central pontine myelinolysis from overly rapid sodium correction in severe hyponatremia).

Poor Circulation

Reduced or abnormal blood flow, usually referring to arterial insufficiency or venous disease, causing coolness, color change, pain, swelling, ulcers, or delayed wound healing.

Poor Color Vision

Alternate index label; see *Color blindness*.

Popcorn Lung Bronchiolitis Obliterans

Bronchiolitis obliterans is irreversible scarring and narrowing of small airways that can follow toxic inhalation, infection, transplantation, or autoimmune disease. It causes persistent cough and breathlessness with obstructive lung testing; exposure avoidance and specialist care are important.

Popliteal

Pertaining to the popliteal fossa, the diamond-shaped anatomical depression located on the posterior aspect of the knee joint. Bounded by the biceps femoris laterally and semimembranosus/semitendinosus medially superiorly, and the heads of the gastrocnemius inferiorly. Transmits critical neurovascular structures from superficial to deep: the tibial nerve, common fibular nerve, popliteal vein, and popliteal artery (the deepest structure, palpated for peripheral pulses and prone to popliteal artery aneurysms and entrapment syndrome). The fossa is also the classic site for popliteal synovial cysts (Baker cysts) associated with intra-

articular knee osteoarthritis or meniscal tears.

Popliteal Aneurysm

Aneurysm of popliteal artery behind knee. Associated with AAA. Risk: thrombosis, embolism, compression. Treatment: surgical repair or endovascular stenting.

Popliteal Artery

Continues from femoral artery at the adductor hiatus, dividing into anterior/posterior tibial arteries at the lower popliteus border. Deepest popliteal fossa structure, related anteriorly to vein and tibial nerve. Palpable at the fossa's inferior margin.

Popliteal Vein

A deep vein behind the knee formed by the union of the anterior and posterior tibial veins. It continues as the femoral vein and can be affected by deep-vein thrombosis.

Popliteal Artery Aneurysm

An abnormal enlargement of the artery behind the knee that may cause limb ischemia, clotting, or embolization.

Popliteal Cyst

A fluid-filled swelling behind the knee, also called a Baker cyst, caused by communication between the joint and a bursa and often associated with arthritis, meniscal injury, or inflammation.

Popliteal Cyst (Baker Cyst)

Alternate index label for Baker Cyst.

Popliteal Fossa

Diamond-shaped space on the posterior knee, bounded by hamstrings superiorly and gastrocnemius heads inferiorly. Contains popliteal artery/vein, tibial nerve, and common peroneal nerve. The popliteal pulse is palpated at the inferior margin.

Population Attributable Fraction

The proportion of cases or disease burden in the

total population that would theoretically be prevented if the exposure were eliminated, assuming the exposure is causal and other conditions remain unchanged.

Population Attributable Risk

Proportion of disease in population attributable to exposure. $PAR = (\text{incidence total} - \text{incidence unexposed}) / \text{incidence total}$.

Population Density

The number of people living in a defined area, a demographic measure relevant to crowding, resource needs, environmental exposure, and spread of communicable disease.

Population Genetics

The study of how gene frequencies and genetic variation change within and between populations through inheritance, mutation, selection, migration, and genetic drift.

Population Group

A defined set of people sharing a characteristic such as age, location, exposure, ancestry, occupation, or health condition for epidemiologic or clinical analysis.

Population Health

The health outcomes of a defined group and the distribution of those outcomes, considered in relation to social conditions, environment, behavior, healthcare, and policy.

Porcelain Gallbladder

Calcification of the gallbladder wall visible on imaging representing chronic inflammatory condition associated with cholelithiasis and chronic cholecystitis. Historically considered a strong risk factor for gallbladder carcinoma, recent studies suggest the cancer risk may be lower than previously estimated.

Porcine

Porcine means derived from or relating to pigs; porcine tissues, enzymes, and biologic products may

be used in medicine after appropriate processing and screening.

Pore

A small opening in the skin or another surface, often associated with a gland or follicle.

Porencephaly

A cavity or cyst within the brain, usually caused by developmental abnormality or prenatal or acquired injury.

Porfimer Sodium

A photosensitizing medicine used with light therapy to destroy selected abnormal tissues, including some tumors; it causes prolonged light sensitivity.

Porfiromycin

Porfiromycin is an older investigational chemotherapy compound related to mitomycin and is not a routine modern treatment.

Porifera

The animal phylum containing sponges, aquatic organisms studied in biology and as sources of bioactive compounds.

Porin

A porin is a channel-forming protein in an outer membrane, especially of Gram-negative bacteria, allowing passage of selected small molecules.

Pork Tapeworm

Taenia solium, a tapeworm acquired from undercooked pork; its larvae can cause cysticercosis.

Porokeratosis

A group of keratinization disorders producing characteristic ridge-like borders called cornoid lamellae around annular scaly lesions, with some forms carrying a small skin-cancer risk.

Poroma

A poroma is a usually benign sweat-gland tumor presenting as a flesh-colored or reddish papule or

nodule, often on palms or soles.

Porphria

A misspelling of porphyria, a group of disorders involving abnormal heme synthesis.

Porphyria

A group of inherited or acquired disorders caused by defects in heme biosynthesis, producing neurologic, abdominal, or skin symptoms.

Porphyria Cutanea Tarda

The most common porphyria, caused by a deficiency of hepatic uroporphyrinogen decarboxylase (UROD), leading to accumulation of uroporphyrinogens. Triggered by hepatitis C virus (HCV), alcohol abuse, iron overload (hemochromatosis), and estrogen therapy.

Porphyrias

Metabolic disorders of heme synthesis causing cutaneous and neurological symptoms. Cutaneous porphyrias include porphyria cutanea tarda (blisters on sun-exposed skin, hypertrichosis, sclerodermoid changes), erythropoietic protoporphyria (immediate pain and burning on sun exposure without visible changes), and variegate porphyria.

Porphyrin

A ring-shaped compound involved in heme and other biological molecules; abnormal accumulation occurs in porphyrias.

Porphyrins Blood Test

A blood porphyrin test measures porphyrins or related compounds and helps evaluate suspected porphyria and disorders of haem production.

Porphyrins Urine Test

A urine porphyrin test measures porphyrins or their precursors and helps investigate suspected porphyria or abnormal haem metabolism.

Porphyromonas

A genus of anaerobic gram-negative bacteria found

especially in the mouth and associated with periodontal and other infections.

Porphyromonas Bennonis

An anaerobic oral bacterium rarely isolated from human specimens and occasionally associated with mixed infection.

Porphyromonas Catoniae

An anaerobic gram-negative bacterium found in the mouth and occasionally associated with periodontal or mixed infection.

Porphyromonas Endodontalis

An anaerobic oral bacterium associated with infected dental pulp and root-canal disease.

Porphyromonas Levii

A *Porphyromonas* species found mainly in animals and rarely associated with human infection.

Porphyromonas Somerae

An anaerobic gram-negative bacterium occasionally isolated from human wounds, abscesses, or other clinical specimens.

Port-Wine Stain

A permanent capillary malformation producing a flat pink, red, or purple skin patch.

Portacaval Shunting

Surgical or radiologic diversion of portal blood into the vena cava to reduce portal hypertension; it can worsen hepatic encephalopathy.

Portal Hypertension

Elevated pressure in the portal venous system defined as hepatic venous pressure gradient exceeding five millimeters of mercury with clinically significant portal hypertension occurring when the gradient exceeds ten.

Alternate Names:

PHT

Portal Hypertensive Gastropathy

Diffuse mucosal changes in the stomach due to portal hypertension causing chronic blood loss. Characterized by mosaic pattern and cherry red spots on endoscopy representing dilated mucosal vessels. Most common in patients with advanced cirrhosis and portal hypertension.

Portal System

Venous network draining the GI tract, spleen, and pancreas to the liver via the portal vein (formed by superior mesenteric and splenic vein union behind the pancreatic neck). Carries nutrient-rich deoxygenated blood for hepatic processing before systemic entry. Receives left gastric, right gastric, cystic, and paraumbilical veins.

Portal Vein

The vein carrying nutrient-rich blood from the gastrointestinal tract, spleen, pancreas, and gallbladder to the liver, formed behind the pancreatic neck by the union of the superior mesenteric and splenic veins. It enters the porta hepatis with the hepatic artery and bile duct.

Portal Vein Thrombosis

Partial or complete thrombosis of the portal vein or its branches. It may result from cirrhosis, malignancy, inflammation, thrombophilia, or abdominal surgery and can cause portal hypertension, intestinal ischemia, or variceal formation.

Portion Control

Managing serving sizes. Larger portions available today contribute to obesity. Using smaller plates, reading labels helpful.

Portion Of Cytosol

A region of the cell's aqueous internal fluid outside membrane-bound organelles, containing ions, metabolites, proteins, and enzymes that support cellular reactions.

Portion Size

The amount of food or drink served or eaten at one time, distinct from a standardized serving size

and important for estimating energy, carbohydrate, sodium, and nutrient intake.

Portopulmonary Syndrome

Development of pulmonary arterial hypertension in the setting of portal hypertension or liver disease. Results from increased cardiac output and splanchnic vasodilation leading to pulmonary vascular remodeling.

Position

The location or posture of a body part, organ, fetus, or patient.

Positional Asphyxia

Fatal or potentially fatal impairment of breathing caused by body position that restricts chest movement, airway patency, or venous return. Risk is increased by restraint, intoxication, obesity, and inability to reposition; findings are nonspecific and require scene and medical correlation.

Positioning Your Baby For Breastfeeding

Placing an infant's head, neck, and body in alignment with the breast so the mouth opens widely, the latch is deep, and feeding is comfortable and effective.

Positive

Positive means that a test, finding, result, or response is present or above a defined threshold; interpretation depends on the test and clinical setting.

Positive Airway Pressure Treatment

A medical procedure, treatment, device, or care topic related to positive airway pressure treatment; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Positive End-Expiratory Pressure

Pressure maintained at end-expiration to prevent alveolar collapse. Improves oxygenation by recruiting alveoli and increasing FRC. Balances oxygenation with hemodynamic effects.

Positive Feedback

A regulatory process in which an output amplifies the process that produced it, as in labor contractions.

Positive Finding

A positive finding is an observed symptom, examination sign, image, laboratory result, or test response that supports the presence of a specified condition or feature.

Positive Predictive Value

The probability that a person with a positive test result truly has the target condition. PPV depends on test specificity and sensitivity and on disease prevalence or pretest probability in the tested population.

Positive Predictive Value In Surveillance

The proportion of reported signals or alerts that represent true events meeting the surveillance case definition. It is influenced by the specificity of the signal and the underlying prevalence of the condition.

Positive Pressure Ventilation

Mechanical ventilation that delivers air/oxygen under positive pressure to support or replace spontaneous breathing. Noninvasive (NIV): CPAP (continuous) and BiPAP (two levels) via mask for obstructive sleep apnea, cardiogenic pulmonary edema, and hypercapnic COPD.

Positive Psychology

The study of well-being, strengths, resilience, and factors that help people thrive.

Positive Symptoms

Symptoms reflecting an excess or distortion of normal mental functions, such as hallucinations, delusions, disorganized speech, or grossly disorganized behavior. They are contrasted with negative symptoms, which reflect loss or reduction of function.

Positive-Strand RNA Virus

A positive-strand RNA virus has a genome that can function directly as messenger RNA in host cells, allowing translation soon after infection.

Positron

Anti-particle of electron. Emitted by PET radionuclides. Annihilation with electron produces two 511 keV photons.

Positron Camera

A positron camera detects paired gamma rays from positron annihilation and is used in positron-emission tomography to map tracer distribution.

Positron Emission Mammography

PET imaging of breast. Higher sensitivity than mammography for dense breasts. Limited availability.

Positron Emission Tomography

Functional imaging modality using positron-emitting radionuclides. Common isotope: fluorine-18 (110 min half-life). Detection: annihilation photons (511 keV) detected in coincidence. Tracers: FDG (metabolism), florbetapir (amyloid), flortaucipir (tau), PSMA (prostate cancer).

Positron Emission Tomography

Imaging that detects paired photons from a positron-emitting tracer to map metabolism or molecular activity.

Post Cibum

A medication instruction meaning that a medicine should be taken after meals. Clear instructions should spell this out to prevent confusion with other uses of the abbreviation P.C.

Alternate Names:

P.C.; After Meals

Post-Acute Rehabilitation

Time-limited interdisciplinary care after an acute illness, injury, or hospitalization to restore mobility, self-care, cognition, communication, and safe com-

munity function. Intensity and setting depend on medical stability and rehabilitation potential.

Post-Cardiac Arrest Management

Care after return of spontaneous circulation that supports breathing and circulation, identifies and treats the cause of the arrest, manages temperature and seizures when indicated, and assesses neurologic recovery over time.

Post-Cardiac Injury Syndrome

Alternate index label; see *Dressler syndrome*.

Post-Chemotherapy Cognitive Impairment

Alternate index label; see *Chemo brain*.

Post-Concussion Syndrome

Persistent symptoms after concussion: headache, dizziness, cognitive difficulties, sleep disturbance, mood changes. Management: symptom-directed, gradual return to activity.

Post-Concussional Syndrome

Persistent physical, cognitive, emotional, or sleep symptoms after a concussion or mild traumatic brain injury.

Post-Dural Puncture Headache

A positional headache occurring after accidental dural puncture during epidural placement or lumbar puncture, caused by cerebrospinal fluid leakage and intracranial hypotension. Symptoms include frontal or occipital headache worse when upright and relieved when supine, neck stiffness, tinnitus, photophobia, and nausea.

Post-Exposure Prophylaxis

The administration of antimicrobial agents after exposure to a pathogen to prevent infection. PEP is used for HIV, hepatitis B, hepatitis C, rabies, and other infections. The timing of PEP is critical, with early administration being most effective. Understanding PEP is essential for managing occupational exposures and outbreaks.

Post-Hepatic Jaundice (Obstructive Jaundice)

Jaundice caused by obstruction of bile flow after bilirubin has been processed by the liver, commonly from gallstones, strictures, or tumours.

Alternate Names:

Obstructive Jaundice

Post-Hoc

Performed or analyzed after an event; in statistics, a post-hoc analysis follows the planned primary analysis.

Post-Infectious Cough

A chronic cough persisting 3–8 weeks after a URI, caused by transient bronchial hyperresponsiveness and airway inflammation. It is self-limited and usually resolves without treatment. Inhaled corticosteroids, antitussives, and time are the mainstays of management.

Post-Mortem

Occurring after death or relating to examination of a body after death.

Post-Nasal Drip (Chronic Rhinitis)

Alternate index label for Chronic Rhinitis.

Post-Natal Depression

Depression beginning during pregnancy or after childbirth, with persistent low mood or loss of interest and possible effects on sleep, bonding, and daily functioning.

Post-Obstructive Pneumonia

Pneumonia occurring distal to a bronchial obstruction caused by tumor, foreign body, mucus plug, or stenosis. The obstruction prevents clearance of secretions, creating a favorable environment for infection. It does not resolve with antibiotics alone until the obstruction is relieved.

Post-Operative Atelectasis

Collapse of some alveoli after surgery, promoted by anesthesia, shallow breathing, pain, immobil-

ity, mucus retention, or underlying lung disease. Management addresses the cause and may include mobilization, effective analgesia, deep breathing, and treatment of complications.

Post-Orgasmic Illness Syndrome

A rare, disabling chronic neuroendocrine and immunologic disorder in men characterized by the rapid onset of severe, flu-like symptoms and cognitive dysfunction following ejaculation (whether through sexual intercourse, masturbation, or spontaneous nocturnal emissions). Symptoms appear within seconds to hours after ejaculation and persist for two to seven days, encompassing extreme physical exhaustion, muscle weakness, feverish sensations, sweating, rhinitis, conjunctival injection, severe headache, irritability, and profound brain fog. The exact pathophysiology remains under investigation, with evidence pointing toward an autoimmune type I or type IV allergic response to autologous seminal fluid, central neurochemical dysregulation, or transient autonomic instability.

Alternate Names:

POIS

Post-Partum Haemorrhage

Excessive bleeding after childbirth, most commonly caused by failure of the uterus to contract, retained placental tissue, genital-tract trauma, or clotting problems.

Post-Pneumonectomy Syndrome

A rare complication following pneumonectomy where the remaining lung herniates across the mediastinum, causing compression of airways and vessels. It presents with dyspnea, stridor, and recurrent infections. Treatment is surgical correction with mediastinal repositioning and prosthetic implant placement.

Post-Prandial

Occurring after, or pertaining to the period following, the consumption of a meal. In metabolic and endocrinologic evaluation, postprandial glycemic excursions (typically measured 2 hours post-meal) provide critical data regarding glucose tolerance,

insulin secretion, and diabetes control, with 2-hour postprandial glucose levels ≥ 200 mg/dL during an oral glucose tolerance test establishing a diagnosis of diabetes mellitus. Postprandial physiology also encompasses postprandial hypotension (a sudden drop in blood pressure caused by splanchnic blood pooling in autonomic failure) and postprandial angina in severe coronary artery disease.

Post-Splenectomy Complications

Post-splenectomy complications include infection by encapsulated bacteria, blood clots, bleeding, and changes in blood-cell counts; vaccination and follow-up reduce risk.

Post-Streptococcal Glomerulonephritis

Immune-mediated glomerulonephritis following infection with nephritogenic strains of group A streptococcus typically 1 to 3 weeks after pharyngitis or 3 to 6 weeks after skin infection. Subepithelial humps on electron microscopy representing immune complex deposits.

Post-Surgical Pain

Acute pain following operative procedures, caused by tissue incision, retraction, dissection, and inflammation, mediated by nociceptors sensitized by prostaglandins, bradykinin, and cytokines. Intensity varies by procedure (thoracotomy, joint replacement, and laparotomy produce more severe pain).

Post-Surgical Rehabilitation

Recovery program after surgery. Phases: acute (pain control, wound care), subacute (range of motion, strengthening), functional (return to activities). Protocols vary by procedure.

Post-Term Pregnancy

Pregnancy continuing beyond 42 weeks gestation (294 days from LMP). Risk factors include prior post-term pregnancy, nulliparity, and genetic factors. Complications include macrosomia, oligohydramnios, meconium aspiration, placental insufficiency, and stillbirth.

Post-Test Probability

The probability of disease after incorporating a test result with the pretest probability. It can be estimated using Bayes' theorem or by converting pretest odds to post-test odds with the appropriate likelihood ratio.

Post-Thrombotic Syndrome

Chronic pain, heaviness, edema, skin change, and sometimes ulceration developing after deep-vein thrombosis because of persistent venous obstruction and valvular damage. Compression, exercise, skin care, and selected interventions may reduce disability.

Post-Transfusion Purpura

A rare immune-mediated thrombocytopenic disorder occurring several days after transfusion of blood products, usually in previously sensitized individuals. It causes abrupt, severe thrombocytopenia with petechiae, mucosal bleeding, or more serious haemorrhage.

Post-Traumatic Headache

A headache that begins or persists after a head or neck injury.

Post-Traumatic Stress Disorder

A disorder that may develop after exposure to actual or threatened death, serious injury, or sexual violence, causing persistent re-experiencing, avoidance, negative mood or cognition, and heightened arousal.

Post-Vasectomy Pain Syndrome

A chronic, disabling complication of vasectomy characterized by persistent or intermittent, dull, aching, or sharp scrotal pain lasting for at least three months post-procedure, of sufficient severity to interfere with daily activities or sexual function. Affecting approximately 1–2% of vasectomized men, proposed etiologies include epididymal congestion and back-pressure behind the blocked vas, chronic inflammation, sperm granuloma formation, perineural fibrosis, or direct nerve entrapment in scar tissue (such as the ilioinguinal or genital branch of the

genitofemoral nerve). Conservative management includes scrotal support, anti-inflammatory medications, and pelvic floor physical therapy; refractory cases may require spermatic cord nerve blocks, microsurgical denervation of the spermatic cord, epididymectomy, or vasectomy reversal (vasovasostomy).

Alternate Names:

PVPS; Chronic Post-Vasectomy Pain

Post-Void Residual

The amount of urine remaining in the bladder after urination, measured by ultrasound or catheterization. A persistently high residual may indicate incomplete emptying, but its significance depends on the volume, timing, symptoms, and clinical context.

Post-Void Residual Volume

The amount of urine remaining in the bladder immediately after urination, measured by ultrasound or catheterization. An elevated residual suggests incomplete emptying but must be interpreted with timing, bladder volume, symptoms, and repeated measurements.

Postal Service

A postal service transports letters, packages, specimens, or medicines; healthcare shipping requires appropriate packaging, labeling, temperature control, and privacy.

Postanesthesia Care Unit

The postanesthesia care unit is a monitored recovery area where patients are observed during the immediate postoperative period. Admission assessment includes vital signs, level of consciousness, pain assessment, and surgical site evaluation.

Postcholecystectomy Syndrome

Persistence or recurrence of symptoms after cholecystectomy. Causes include retained common bile duct stones, sphincter of Oddi dysfunction, bile salt malabsorption, irritable bowel syndrome, and other gastrointestinal conditions. Evaluation includes liver function tests, MRCP, and endoscopic ultrasound.

Postconcussion Syndrome

Persistent physical, cognitive, or emotional symptoms after a concussion.

Posterior

Situated behind or toward the back of the body or a structure.

Posterior Chamber

The part of the eye between the iris and lens, containing aqueous humor.

Posterior Cruciate Ligament Injury

Tear of the PCL, less common than ACL injury. Mechanism: dashboard injury (tibia striking dashboard), hyperflexion, or fall on a flexed knee with foot plantarflexed. Presents with posterior knee pain, mild swelling, and posterior sag of the tibia. Positive posterior drawer test (most sensitive), posterior sag sign (Godfrey test).

Posterior Cruciate Ligament Tear

PCL tear: less common, usually from direct blow to tibia (dashboard injury). Symptoms: knee pain, swelling, instability. Diagnosis: posterior drawer test, MRI. Treatment: usually conservative, surgical if combined injuries.

Posterior Descending Artery

An artery running in the posterior interventricular groove and supplying the inferior ventricular walls and posterior interventricular septum. It usually arises from the right coronary artery in right-dominant circulation but may arise from the circumflex artery.

Posterior Drawer Test

PCL integrity test. Knee flexed 90 degrees, posterior tibial translation.

Posterior Fossa Tumor

A posterior fossa tumor is a growth in the part of the skull containing the cerebellum and brainstem and may obstruct fluid flow or affect coordination and vital functions.

Posterior Heel Bursitis

Inflammation of a bursa near the back of the heel, causing pain and swelling.

Alternate Names:

Retrocalcaneal bursitis

Posterior Keyhole Foraminotomy

A minimally invasive operation that enlarges a spinal nerve opening to relieve compression.

Posterior Myocardial Infarction

A heart attack involving the back wall of the heart.

Posterior Pituitary

The neural lobe of the pituitary that stores and releases vasopressin and oxytocin synthesized in the hypothalamus. Vasopressin regulates renal water retention and vascular tone; oxytocin contributes to uterine contraction and milk ejection.

Posterior Prolapse

Alternate index label; see *Posterior vaginal prolapse (rectocele)*.

Posterior Reversible Encephalopathy Syndrome

A neurotoxic state caused by failed cerebrovascular autoregulation, typically from acute hypertension, eclampsia, immunosuppressive drugs (tacrolimus, cyclosporine), or sepsis. Presents with headache, seizures, altered consciousness, and visual disturbances.

Posterior Rhinorrhea

Posterior rhinorrhea is mucus draining from the nose into the throat, commonly causing throat clearing or cough from allergy, infection, irritation, or sinus disease.

Posterior Subcapsular Cataract

A type of lens opacity located at the posterior aspect of the lens, just anterior to the posterior lens capsule, in the visual axis, and it is particularly visually disabling because of its central location.

Posterior Tibial Artery

Arises from the popliteal artery, courses deep in the posterior leg, palpable behind the medial malleolus. Divides into medial/lateral plantar arteries for the sole. Supplies posterior leg muscles. Important for peripheral vascular disease assessment. The pedal pulse behind the medial malleolus is a key clinical examination component.

Posterior Urethral Valves

The most common cause of lower urinary tract obstruction in male infants, caused by congenital mucosal folds (valves) in the posterior urethra (prostatic urethra) that obstruct urine outflow.

Posterior Uveitis

Inflammation affecting the retina, choroid, or both. It may cause floaters, blurred vision, visual-field defects, or eye pain and can result from infection, autoimmune disease, or other inflammatory conditions.

Posterior Vitreous Detachment

Separation of the posterior vitreous cortex from the retinal surface, commonly related to age-related vitreous liquefaction. New flashes or floaters require dilated retinal examination because an associated retinal tear or detachment may be present.

Posteroanterior

Posteroanterior means directed from back to front; a PA chest radiograph uses this projection to reduce magnification of the heart.

Postganglionic Fiber

An autonomic nerve fiber extending from a ganglion to its target organ after the synapse, carrying sympathetic or parasympathetic signals to smooth muscle, glands, or the heart.

Postherpetic Neuralgia

Persistent nerve pain in an area affected by shingles after the rash has healed.

Alternate Names:

Neuralgia, Postherpetic

Posthitis

Inflammation of the foreskin, causing redness, swelling, itching, pain, discharge, or difficulty retracting it; infection, irritation, diabetes, and poor hygiene are possible contributors.

Posthumous

Occurring, discovered, or published after a person's death. A posthumous diagnosis may be based on records, stored samples, or autopsy findings.

Postinflammatory Hyperpigmentation

Darkening of skin after inflammation or injury, caused by increased melanin production or its deposition in the epidermis or dermis.

Postmenopausal

Postmenopausal describes the period after menopause, when ovarian estrogen production and menstrual periods have permanently ceased.

Postmenopausal Osteoporosis

Bone loss caused by lower estrogen levels associated with menopause. Sometimes called type I osteoporosis.

Postmenopause

The period in a woman's life lasting from the end of perimenopause until the end of life.

Postmortem Artifact

Any structural, chemical, or physical change introduced into a body after death (e.g., embalming artifacts, insect activity, postmortem animal scavenging, resuscitation trauma) that mimics antemortem disease or trauma.

Postmortem Interval

Estimated elapsed time between actual death and the time of autopsy or body discovery, calculated utilizing algor mortis, livor mortis, rigor mortis, vitreous potassium levels, and insect succession patterns.

Postmortem Toxicology

Toxicologic analysis of specimens collected after death to identify drugs, alcohols, poisons, and metabolites. Results must account for therapeutic administration, redistribution, decomposition, specimen site, contamination, and tolerance before conclusions about causation are made.

Postmyocardial Infarction Syndrome

Alternate index label; see *Dressler syndrome*.

Postnatal

Pertaining to or occurring during the period immediately following birth, specifically referring to the infant (contrasted with postpartum, which specifically refers to the mother). The postnatal stage encompasses the crucial neonatal period (the first 28 days of life) and infancy, characterized by dramatic physiological adaptations including pulmonary vascular resistance drop, closure of fetal circulatory shunts (ductus arteriosus, foramen ovale), metabolic shifts, routine newborn screening for metabolic and genetic disorders, and neurodevelopmental milestone tracking.

Postnatal Care

A medical procedure, treatment, device, or care topic related to postnatal care; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Postnatal Depression

A depressive episode occurring after childbirth, with persistent low mood, anhedonia, sleep or appetite disturbance, guilt, impaired bonding, or suicidal thoughts.

Postop

Postop is an abbreviation for postoperative, meaning relating to the period after a surgical procedure.

Postoperative

Postoperative means occurring or provided after surgery, including recovery, wound care, pain control, rehabilitation, and complication monitoring.

Postoperative Care

Structured post-surgical management encompassing pain control, fluid balance, wound care, early mobilization, and venous thromboembolism prophylaxis.

Postoperative Cognitive Dysfunction

A decline in cognitive abilities such as memory, attention, or executive function after surgery. It may be temporary or persistent and is more frequent in older adults and after major or complicated procedures. Assessment requires comparison with the person's baseline and consideration of delirium, medication effects, and other medical causes.

Postoperative Complications

Clavien-Dindo classification: Grade I (any deviation, no treatment), Grade II (pharmacologic), Grade III (surgical/endoscopic/radiologic intervention: IIIa no GA, IIIb GA), Grade IV (ICU: IVa single organ, IVb multi-organ), Grade V (death).

Postoperative Delirium

A common complication after major surgery, particularly in elderly patients. It is categorized as hyperactive (agitation, restlessness, hallucinations), hypoactive (lethargy, withdrawal, decreased responsiveness), or mixed. Hypoactive delirium is the most common and most missed form.

Postoperative Hemorrhage

Bleeding after surgery that may occur at the operative site or internally and can cause shock, anemia, or the need for urgent intervention.

Postoperative Ileus

Temporary impairment of gastrointestinal motility after surgery, most common after abdominal surgery. Pathophysiology: autonomic dysfunction (sympathetic overactivity), inflammatory mediators, opioids, and electrolyte abnormalities.

Postoperative Infection

An infection developing after surgery, involving the incision, an organ space, implanted material, or the bloodstream.

Postoperative Nausea And Vomiting

Nausea, retching, or vomiting occurring after surgery. Risk is influenced by factors such as previous motion sickness or postoperative nausea, anesthetic technique, opioid use, and patient characteristics. Prevention and treatment are tailored to the individual risk and procedure.

Postoperative Pain Management

Management of pain after surgery using a combination of medicines, regional techniques, physical measures, and rehabilitation. The plan is tailored to the procedure, pain severity, medical conditions, and risks of adverse effects.

Postoperative Period

The postoperative period is the time after an operation when recovery, pain control, wound care, mobility, nutrition, complications, and follow-up are monitored.

Postoperative Pulmonary Complication

Respiratory deterioration after surgery, including atelectasis, pneumonia, aspiration, respiratory failure, bronchospasm, and exacerbation of chronic lung disease. Risk reduction includes smoking cessation, lung-protective ventilation, analgesia, mobilization, and appropriate respiratory exercises.

Postpartum

Pertaining to or occurring during the time interval following childbirth, typically defined as the 6 to 12 weeks following delivery (the puerperium) during which maternal physiological, hormonal, and anatomical systems return toward their non-pregnant baseline. High-risk period for maternal complications including postpartum hemorrhage (PPH; predominantly caused by uterine atony, retained placenta, genital tract trauma, or coagulopathy), puerperal infections (endometritis, mastitis), postpartum preeclampsia, deep vein thrombosis, and psychiatric disorders ranging from transient postpartum blues to major postpartum depression and postpartum psychosis.

Alternate Names:

Postnatal

Postpartum Anxiety

Excessive, persistent, or distressing anxiety during pregnancy or after childbirth that interferes with sleep, daily function, bonding, or self-care. It may occur with panic, intrusive worries, or depression and deserves clinical assessment; thoughts of self-harm or harm to the baby require emergency help.

Postpartum Blues

A common, short-lived period of tearfulness, mood changes, irritability, or anxiety beginning soon after childbirth and usually improving within two weeks. Support, rest, and practical help are important; persistent, severe, psychotic, or suicidal symptoms require prompt assessment for a postpartum mental-health disorder.

Postpartum Cardiomyopathy (Peripartum Cardiomyopathy)

A dilated cardiomyopathy causing heart failure toward the end of pregnancy or during the months after delivery. Symptoms may include breathlessness, difficulty lying flat, swelling, fatigue, or chest discomfort and require prompt assessment.

Alternate Names:

Peripartum Cardiomyopathy

Postpartum Care

Clinical care during the weeks after childbirth that assesses bleeding, uterine involution, blood pressure, infection, wound healing, breastfeeding, mental health, contraception, and recovery.

Postpartum Contraception

Contraceptive counseling and care after childbirth, taking account of breastfeeding, venous-thromboembolism risk, medical conditions, fertility intentions, and the timing of method initiation.

Postpartum Depression

A depressive episode occurring during pregnancy or after childbirth, causing significant distress or impairment.

Alternate Names:

Depression, Postpartum PPD (Postpartum Depression)

Postpartum Depression

Alternate index label for Postpartum Depression.

Postpartum Endometritis

Infection of the decidua and endometrial cavity after delivery, occurring more often after cesarean birth. Fever, uterine tenderness, pelvic pain, and foul-smelling lochia are typical; treatment generally requires intravenous broad-spectrum antibiotics.

Postpartum Haemorrhage

Excessive bleeding after childbirth, commonly defined by blood loss of at least 1,000 mL or bleeding with hypovolemia after birth.

Postpartum Hemorrhage

Excessive bleeding after childbirth, usually caused by poor uterine contraction, retained placental tissue, genital-tract trauma, or a clotting disorder. It can cause shock and requires urgent obstetric assessment and treatment.

Postpartum Hemorrhage (Secondary)

Excessive bleeding occurring between 24 hours and 12 weeks postpartum. Causes: retained products of conception (most common), endometritis, subinvolution of the placental site, and coagulopathy. Presents with heavy or prolonged bleeding, cramping, and passing clots. Diagnosis: ultrasound (retained tissue, thickened endometrium).

Alternate Names:

Secondary

Postpartum Infection

Infections occurring after delivery, a leading cause of maternal morbidity. Types: endometritis (most common, especially after cesarean), wound infection (cesarean incision, episiotomy, perineal laceration), urinary tract infection, mastitis, septic pelvic thrombophlebitis, and pneumonia.

Postpartum Preeclampsia

New-onset hypertension with pre-eclampsia features after childbirth, sometimes occurring several days or weeks postpartum. Severe headache, visual symp-

toms, chest pain, shortness of breath, or seizures require urgent assessment.

Postpartum Psychosis

A rare but severe psychiatric emergency that can begin suddenly during the days or weeks after childbirth. Symptoms may include hallucinations, delusions, paranoia, severe confusion, mania, depression, disorganized behavior, or rapidly changing moods. It can worsen quickly and may threaten the safety of the mother or baby, so immediate medical assessment is required.

Alternate Names:

Postnatal Psychosis Puerperal Psychosis

Postpartum Thyroiditis

Autoimmune inflammation of the thyroid occurring during the first year after delivery. It may cause a temporary hyperthyroid phase followed by hypothyroidism, and symptoms and recovery vary; thyroid-function testing guides follow-up and treatment.

Postpartum Urinary Retention

Inability to empty the bladder adequately after childbirth, sometimes associated with epidural anesthesia, perineal trauma, swelling, or prolonged labor. Prompt assessment prevents bladder overdistension and injury.

Postprandial

Postprandial means occurring after a meal, especially the changes in glucose, insulin, digestion, and circulation after eating.

Postprandial Period

The postprandial period is the interval after eating when digestion and nutrient absorption occur and glucose, insulin, and other metabolic responses change.

Postrenal AKI

Acute kidney injury caused by obstruction of urine flow from the kidneys to the urethra. Causes include prostate enlargement, stones, tumors, fibrosis, and neurogenic bladder. The effect depends on the

location, duration, completeness, and whether one or both kidneys are affected.

Poststreptococcal Glomerulonephritis

Poststreptococcal glomerulonephritis is kidney inflammation occurring after certain streptococcal infections, causing blood or protein in urine, swelling, high blood pressure, and reduced kidney function.

Poststreptococcal Pustulosis

A sterile, non-follicular pustular eruption occurring as a rare sequela of streptococcal pharyngitis, presenting with scattered pustules on an erythematous base, typically on the trunk and extremities. It is thought to be an immune-mediated hypersensitivity reaction to streptococcal antigens.

Postsurgical Pain Treatment In Adults

Adult postsurgical pain treatment may combine acetaminophen, anti-inflammatory medicines, regional anesthesia, limited opioids, movement, and wound care according to the operation and patient risks.

Postsynaptic Membrane

The specialized receiving surface of a neuron or muscle cell at a synapse, containing receptors and signaling proteins that respond to neurotransmitter released by the presynaptic cell.

Postterm Pregnancy

Pregnancy that continues beyond 42 completed weeks of gestation. Risks and management depend on gestational dating, fetal condition, and maternal health.

Posttraumatic Stress Disorder

A psychiatric disorder that develops after exposure to a traumatic event (death, serious injury, sexual violence).

Alternate Names:

PTSD (Posttraumatic Stress Disorder)

Posttraumatic Stress Disorder

Alternate index label for Posttraumatic Stress Disorder.

Postural

Postural means relating to body position or posture, including symptoms or vital signs that change when a person sits, stands, or lies down.

Postural Drainage

Positioning a person so gravity assists drainage of secretions from selected parts of the lungs.

Postural Hypotension

A significant fall in blood pressure on standing, often causing dizziness or fainting.

Postural Orthostatic Tachycardia Syndrome

An autonomic disorder causing an excessive increase in heart rate after standing, with symptoms such as light-headedness, palpitations, fatigue, weakness, brain fog, nausea, or fainting. Diagnosis requires clinical assessment and exclusion of other causes of orthostatic symptoms.

Posture

The alignment and position of the body at rest or during activity.

Pot Marijuana (Marijuana)

Alternate index label for Marijuana.

Potassium

An essential intracellular electrolyte important for nerve conduction, muscle contraction, and cardiac rhythm.

Potassium (K)

The chemical symbol for potassium, a major intracellular electrolyte needed for nerve signaling, muscle contraction, and cardiac electrical activity.

Potassium Acetate

A potassium salt used to replace potassium or provide acetate in selected medical and laboratory applications; dosing requires monitoring of kidney function and serum potassium.

Potassium Binders

Sodium zirconium cyclosilicate and patiromer. Exchange potassium in GI tract for sodium or calcium. Treat chronic hyperkalemia allowing continued RAAS inhibitor use in CKD.

Potassium Carbonate Poisoning

Potassium carbonate poisoning involves toxic or environmental exposure to potassium carbonate poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Potassium Channel

A membrane protein that allows potassium ions to cross cells and helps control electrical excitability, heart rhythm, and hormone release.

Potassium Chloride

A chemical or physiological term involving potassium chloride. Its effects and clinical importance depend on its concentration, route of exposure, and patient context.

Potassium Citrate

A chemical or physiological term involving potassium citrate. Its effects and clinical importance depend on its concentration, route of exposure, and patient context.

Potassium Cyanide

A chemical or physiological term involving potassium cyanide. Its effects and clinical importance depend on its concentration, route of exposure, and patient context.

Potassium Deficiency

An abnormally low blood potassium level, which can cause weakness, cramps, constipation, abnormal heart rhythms, or paralysis.

Potassium Dichromate

A chemical or physiological term involving potassium dichromate. Its effects and clinical importance depend on its concentration, route of exposure, and patient context.

Potassium Hydroxide Poisoning

Potassium hydroxide poisoning involves toxic or environmental exposure to potassium hydroxide poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Potassium In Diet

Dietary potassium comes from fruits, vegetables, beans, dairy, and other foods; intake may need adjustment in kidney disease or with medicines that change potassium levels.

Potassium Iodide

Potassium iodide supplies iodide and is used in selected thyroid conditions, radiation emergencies, and some expectorant preparations; excess can disturb thyroid function.

Potassium Ion

The positively charged form of potassium, the principal intracellular cation that helps establish resting membrane potential, nerve transmission, muscle contraction, and cardiac rhythm.

Potassium Permanganate

An oxidizing antiseptic used in very dilute topical solutions; concentrated material can burn tissue.

Potassium Test

A potassium test measures potassium in blood, usually to assess heart, muscle, kidney, or medication-related electrolyte problems.

Potassium Urine Test

A urine potassium test measures potassium excretion and helps determine whether abnormal blood potassium is related to kidney losses or other causes.

Potassium-Sparing Diuretic

A diuretic that promotes sodium excretion while reducing potassium loss, such as spironolactone or amiloride.

Potassium-Sparing Diuretics

Potassium-sparing diuretics act on the distal

nephron to promote sodium excretion while minimizing potassium loss. They include epithelial sodium channel blockers such as amiloride and triamterene, and mineralocorticoid receptor antagonists such as spironolactone and eplerenone.

Potato Plant Poisoning - Green Tubers And Sprouts

Potato plant poisoning - green tubers and sprouts involves toxic or environmental exposure to potato plant poisoning - green tubers and sprouts; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Potbellies And Toddlers

A rounded abdomen in a toddler is often normal body shape or posture, but persistent distension with pain, vomiting, poor growth, constipation, or a mass needs medical assessment.

Potency

The amount of a drug required to produce a specified effect. A more potent drug produces that effect at a lower dose, but potency does not indicate the maximum effect.

Potential

Potential describes a capacity, possibility, or stored energy that may become clinically relevant depending on circumstances and additional findings.

Potential Acuity Meter

Potential acuity meter is a laboratory test, imaging method, instrument, score, or measurement used to characterize a specimen or clinical finding; interpretation depends on the method and clinical context.

Potentiometry

potentiometry is a laboratory test, imaging method, instrument, score, or measurement used to characterize a specimen or clinical finding; interpretation depends on the method and clinical context.

Potocki-Lupski Syndrome

A contiguous gene microduplication syndrome resulting from a reciprocal 3.7 Mb interstitial duplication of chromosome 17p11.2, encompassing the dosage-sensitive retinoic acid-induced 1 (*RAI1*) gene (the reciprocal duplication of the Smith-Magenis syndrome deletion locus). Manifests with neonatal hypotonia, failure to thrive with severe feeding difficulties, intellectual disability, verbal apraxia, high rates of autism spectrum disorder, hyperactive behaviors, and congenital cardiovascular malformations (most notably hypoplastic left heart syndrome, dilated aortic root, and ventricular septal defects).

Alternate Names:

PTLS; Duplication 17p11.2 Syndrome

Pott Disease

TB of the spine involving thoracic or lumbar vertebrae. Causes vertebral destruction, kyphosis (gibbus deformity), and paravertebral abscess. May cause spinal cord compression.

Pott'S Disease

Tuberculosis of the spine, causing vertebral destruction, back pain, deformity, or neurologic compromise.

Pott'S Fracture

An ankle fracture-dislocation involving the malleoli, historically defined variably.

Potter Syndrome

A sequence of anomalies resulting from severe oligohydramnios (low amniotic fluid volume), most commonly caused by bilateral renal agenesis, but also from autosomal recessive polycystic kidney disease, posterior urethral valves, or chronic amniotic fluid leak.

Pouch

A sac-like body structure or a surgically created reservoir. Examples include a cheek pouch, a pouch in the digestive tract, and an ileal pouch made from small intestine.

Pouch Of Douglas

The pouch of Douglas is the peritoneal recess between the uterus and rectum, also called the rectouterine pouch, where fluid may collect.

Poultice

A soft, moist substance applied to the body, historically to soothe or draw fluid from tissue.

Pound

A unit of weight or pressure; the exact meaning depends on context.

Poverty

Insufficient material resources to meet basic needs, a social determinant strongly affecting health and access to care.

Povidone

Povidone is a water-soluble polymer used in medicines, eye products, and other preparations; it is also part of povidone-iodine antiseptic.

Povidone-Iodine

Povidone-iodine is an iodine-containing antiseptic used on skin or tissue before selected procedures; it can irritate skin and affect the thyroid with extensive use.

Powassan Encephalitis

Inflammation of the brain caused by Powassan virus, a tick-borne flavivirus. It can cause fever, headache, vomiting, confusion, seizures, or weakness and may require hospitalization.

Powassan Virus Disease

Powassan virus disease is a rare tick-borne infection that can cause fever, headache, vomiting, confusion, seizures, meningitis, or encephalitis. Treatment is supportive, and prevention relies on avoiding ticks and removing them promptly.

Powder Diffraction

An analytical method that identifies crystalline substances by measuring how an X-ray or other beam

is diffracted by powdered material.

Power

The rate at which work is performed; in exercise, it combines force and movement speed.

Power Training

Exercise designed to improve the ability to produce force rapidly.

Alternate Names:

Explosive training

Pox

A disease producing vesicles, pustules, or pitted scars; the term is used in names such as smallpox and cowpox.

Poxviridae

A family of large DNA viruses that includes orthopoxviruses, molluscipoxvirus, and parapoxviruses; some cause significant human disease.

Poxviridae Infection

An infection caused by a poxvirus, ranging from localized skin lesions to severe systemic disease depending on the virus and host.

Poxvirus

A family of large DNA viruses that includes variola, vaccinia, mpox, and molluscum contagiosum viruses.

Ppar Alpha

PPAR-alpha is a nuclear receptor that regulates fatty-acid oxidation and lipid metabolism and is activated by fibrate medicines.

Ppar Delta

PPAR-delta is a nuclear receptor involved in fatty-acid use, energy balance, inflammation, and tissue differentiation.

PPD

PPD may mean purified protein derivative used in tuberculosis skin testing or post-partum depression;

context determines the intended meaning.

PPD Skin Test

A PPD skin test places a small amount of tuberculin under the skin to assess immune evidence of tuberculosis exposure; interpretation depends on risk and the size of the reaction.

PR Interval

The ECG interval from the start of atrial depolarization to the start of ventricular depolarization, reflecting atrioventricular conduction.

Pracinostat

Pracinostat is an investigational histone-deacetylase inhibitor studied in myeloid and other cancers by altering gene expression and cell survival.

Practitioner

A person qualified and authorized to practice a healthcare profession.

Prader-Labhart-Willi Syndrome

Alternate index label; see *Prader-Willi syndrome*.

Prader-Willi Syndrome

A genetic disorder caused by loss of function of genes on chromosome 15 (paternal deletion or maternal uniparental disomy). Features: neonatal hypotonia and failure to thrive, followed by hyperphagia and obesity in childhood. Intellectual disability, short stature, hypogonadism, and behavioral problems.

Alternate Names:

Prader-Labhart-Willi Syndrome

Pragmatic

Focused on what works in ordinary real-world conditions. A pragmatic clinical trial studies how an intervention performs in usual healthcare rather than under highly controlled conditions.

Pragmatic Trial

A pragmatic trial evaluates an intervention under usual real-world care conditions to estimate how well it works in routine practice.

Pralatrexate

Pralatrexate is an antifolate chemotherapy medicine used for relapsed or refractory peripheral T-cell lymphoma.

Pralidoxime

An oxime that can reactivate inhibited acetylcholinesterase when administered early after organophosphate poisoning, particularly for nicotinic weakness and respiratory-muscle dysfunction. It is used with atropine and supportive care according to exposure and local guidance.

Pramipexole

A dopamine agonist used mainly for Parkinson disease and restless legs syndrome.

Prandial

Prandial means relating to a meal, as in preprandial before eating or postprandial after eating.

Prasugrel

An antiplatelet P2Y₁₂ inhibitor used in selected patients with acute coronary syndrome undergoing PCI.

Pravastatin

A statin that lowers LDL cholesterol by inhibiting HMG-CoA reductase.

Pravastatin Sodium

Pravastatin sodium is a statin medicine that lowers LDL cholesterol and reduces cardiovascular risk.

Praziquantel

An anthelmintic used for schistosomiasis and several tapeworm and fluke infections.

Pre-Cancerous

Showing cellular changes that may progress to cancer but are not invasive cancer.

Pre-Eclampsia

New-onset hypertension after 20 weeks of pregnancy with proteinuria or other maternal organ or placen-

tal dysfunction.

Pre-Existing Diabetes And Pregnancy

Pre-existing type 1 or type 2 diabetes during pregnancy increases risks to parent and fetus, so glucose control, medicines, fetal growth, blood pressure, and eye and kidney health need coordinated care.

Pre-Hepatic Jaundice (Haemolytic Jaundice)

Jaundice caused by excessive breakdown of red blood cells before bilirubin reaches the liver, producing increased unconjugated bilirubin.

Alternate Names:

Haemolytic Jaundice

Pre-MiRNA

A pre-miRNA is an early hairpin-shaped microRNA transcript processed in the nucleus before becoming a mature regulator of gene expression.

Pre-Synaptic

Situated before a synapse, especially the nerve-terminal side that releases a neurotransmitter.

Prealbumin

Prealbumin, or transthyretin, is a liver-produced transport protein for thyroid hormone and retinol-binding protein; its level is influenced by inflammation and nutrition.

Prebiotics

Non-digestible fibers feeding beneficial gut bacteria. Sources: garlic, onions, leeks, asparagus, bananas, whole grains.

Prebiotics And Probiotics

Prebiotics are nondigestible food ingredients (fiber and oligosaccharides—inulin, fructooligosaccharides, galactooligosaccharides) that selectively stimulate the growth and activity of beneficial gut bacteria (*Bifidobacterium*, *Lactobacillus*), promoting host health.

Precancerous Condition

A precancerous condition contains abnormal changes that may progress to cancer but are not cancer themselves.

Precancerous Polyp

A precancerous polyp is a growth with cellular changes that can become cancerous over time if not removed or monitored appropriately.

Precaution

An action taken to reduce risk of harm, infection, or error.

Precipitate Labour

Unusually rapid labour, often defined as delivery within three hours of the onset of regular contractions.

Precipitating Factor

A precipitating factor is an event or exposure that triggers the onset or worsening of symptoms or disease in a susceptible person.

Preclinical

Before human clinical testing or relating to disease changes before symptoms become evident.

Preclinical Study

A preclinical study evaluates a medicine, device, or treatment in laboratory models or animals before testing in humans.

Precocious Puberty

Onset of secondary sexual characteristics earlier than the accepted normal age range.

Alternate Names:

Early Puberty Premature Puberty

Preconception Care

A medical procedure, treatment, device, or care topic related to preconception care; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Preconception Counseling

Counseling provided to women of reproductive age before pregnancy to optimize outcomes.

Precordial Catch Syndrome

Brief, sharp, localized chest pain near the heart that worsens with breathing and usually occurs in otherwise healthy children or adolescents.

Precordium

The anterior anatomical region of the chest wall overlying the heart, pericardium, and great vessels, extending from the second to the sixth intercostal spaces and from the right sternal border to the left midclavicular line. A central focus of cardiovascular physical examination, evaluated systematically by inspection, palpation (for the apical impulse / point of maximal impulse [PMI], right ventricular heaves, and thrills), and auscultation of heart sounds (S_1 , S_2 , S_3 , S_4) and murmurs across the four standard valve auscultatory areas.

Precursor

A substance that can be converted into another biologically active substance.

Prediabetes

Blood glucose higher than normal but below the diagnostic threshold for diabetes, usually identified by A1C, fasting glucose, or an oral glucose-tolerance test; it increases future diabetes and cardiovascular risk.

Alternate Names:

Diabetes Pre (Prediabetes)

Prediabetes Syndrome

Blood glucose levels above normal but below the diabetes threshold, associated with increased future risk of type 2 diabetes and cardiovascular disease.

Predictive Test

A test that estimates whether a person or disease is likely to respond to a particular treatment.

Predispose

To predispose is to increase a person's susceptibility or likelihood of developing a disease or response when additional triggers occur.

Prednimustine

Prednimustine is a former antineoplastic drug combining a corticosteroid-related group with an alkylating component and causing immunosuppression and marrow toxicity.

Prednisolone

Prednisolone is a corticosteroid used to reduce inflammation and immune activity; prolonged or high-dose use can cause infection, bone loss, and other effects.

Prednisone

Prednisone is a corticosteroid used for inflammatory, allergic, and autoimmune conditions; it should often be tapered after prolonged use.

Preeclampsia

Preeclampsia is pregnancy-related hypertension with proteinuria or other maternal organ dysfunction after midpregnancy and can progress to seizures, stroke, liver injury, or placental complications. pressure 140/90 mmHg or higher) after 20 weeks gestation with proteinuria or other end-organ damage.

Alternate Names:

Pregnancy-Related Hypertension

Preeclampsia With Severe Features

Preeclampsia accompanied by severe hypertension or signs of significant organ involvement, such as low platelets, liver or kidney dysfunction, pulmonary edema, severe headache, or visual symptoms. It requires urgent obstetric evaluation and management.

Preexcitation Syndrome

Early activation of the ventricles through an accessory electrical pathway, which can cause tachycardia and is classically associated with Wolff-Parkinson-White syndrome.

Prefrontal

Relating to the anterior part of the frontal lobe involved in planning, judgment, working memory, and behavior.

Prefrontal Cortex

The prefrontal cortex is the front part of the frontal lobe involved in planning, judgment, working memory, impulse control, and social behavior.

Pregabalin

Pregabalin is used for certain nerve pain and seizures and may cause dizziness, sleepiness, swelling, or dependence.

Preganglionic Fiber

An autonomic nerve fiber that runs from the central nervous system to an autonomic ganglion, where it synapses with a postganglionic neuron.

Pregnancy

The physiological state in which a developing embryo or fetus grows within the female uterus, typically lasting approximately 40 weeks from the last menstrual period to childbirth.

Pregnancy - Health Risks

Health risks in pregnancy vary with medical conditions, medicines, infections, age, exposures, and obstetric history; prenatal assessment identifies preventable complications and needed monitoring.

Pregnancy - Identifying Fertile Days

Fertile days are the interval around ovulation when intercourse can lead to pregnancy; cycle timing, cervical mucus, basal temperature, and ovulation tests can provide estimates but are imperfect.

Pregnancy And Herpes

Genital herpes during pregnancy can affect delivery planning and rarely infect the newborn; antiviral treatment and examination of lesions near labor help reduce transmission risk.

Pregnancy And The Flu

Influenza during pregnancy can be more severe and affect the fetus; vaccination is recommended in pregnancy, and suspected illness should receive prompt clinical advice about antiviral treatment.

Pregnancy And Travel

Travel during pregnancy depends on gestational age, health, destination infections, transport, and access to care; prevention includes vaccines as appropriate, food and water safety, movement, and medical planning.

Pregnancy And Work

Work during pregnancy is usually possible but may require adjustments for heavy lifting, prolonged standing, heat, chemicals, radiation, infection, night work, or other occupational hazards.

Pregnancy Care

Pregnancy care includes regular maternal and fetal checkups, screening, nutrition and medicine review, vaccination, and advice about warning symptoms.

Pregnancy Complication

A maternal, fetal, placental, or labor-related problem that disrupts an otherwise expected pregnancy, such as preeclampsia, hemorrhage, diabetes, growth restriction, or preterm labor.

Pregnancy Ectopic (Ectopic Pregnancy)

Alternate index label for Ectopic Pregnancy.

Pregnancy High Blood Pressure (Pregnancy: Preeclampsia And Eclampsia)

Alternate index label for Pregnancy: Preeclampsia and Eclampsia.

Pregnancy Nutrition

Nutrition care during pregnancy that supports maternal health, placental and fetal growth, and appropriate gestational weight gain. Energy, protein, folate, iron, iodine, choline, vitamin D, and other requirements vary by stage and individual clinical

factors.

Pregnancy Of Unknown Location

A positive pregnancy test without a confirmed intrauterine or ectopic pregnancy on ultrasound. Serial human chorionic gonadotropin measurements and repeat imaging help determine whether the pregnancy is early, failing, or ectopic.

Pregnancy Outcome

Pregnancy outcome describes what happens to a pregnancy, such as live birth, miscarriage, stillbirth, preterm birth, or other maternal or fetal result.

Pregnancy Planning (Preparing For Pregnancy)

Alternate index label for Preparing for Pregnancy.

Pregnancy Status

Pregnancy status indicates whether a person is pregnant, not pregnant, or of unknown status and can affect medication, imaging, and treatment decisions.

Pregnancy Test

A test detecting human chorionic gonadotropin in urine or blood to identify pregnancy.

Pregnancy Trimester

A pregnancy trimester is one of three approximate stages of gestation used to organize fetal development, screening, risks, and clinical care.

Pregnancy Tubal (Ectopic Pregnancy)

Alternate index label for Ectopic Pregnancy.

Pregnancy-Induced Hypertension

New-onset hypertension during pregnancy without the proteinuria or organ dysfunction that defines preeclampsia. Diagnosis and treatment depend on repeated blood-pressure measurements, gestational age, symptoms, and evidence of maternal or fetal complications.

Pregnancy-Related Hypertension

Alternate index label; see *Preeclampsia*.

Pregnancy: Bleeding During The First Trimester

First-trimester bleeding may occur with implantation, cervical changes, miscarriage, ectopic pregnancy, or other causes. Assessment uses symptoms, examination, serial pregnancy hormone testing, and ultrasound; heavy bleeding, severe pain, faintness, or shoulder pain is urgent.

Pregnancy: Pain Relief Options For Birth

Labor pain relief may include breathing and movement, water, massage, nitrous oxide, intravenous medicines, epidural analgesia, or spinal anesthesia. Choice depends on labor stage, maternal and fetal status, contraindications, availability, and informed preference.

Pregnancy: Placenta Previa

Placenta previa occurs when the placenta lies over or near the cervical opening and can cause painless bright-red bleeding, especially later in pregnancy. Digital vaginal examination is avoided until location is known; recurrent bleeding requires obstetric planning and sometimes cesarean delivery.

Pregnancy: Preeclampsia And Eclampsia

Preeclampsia is new hypertension during pregnancy with proteinuria or organ dysfunction; eclampsia is the occurrence of seizures. Severe headache, visual symptoms, upper-abdominal pain, shortness of breath, or seizure requires emergency obstetric treatment.

Alternate Names:

Eclampsia Pregnancy (Pregnancy: Preeclampsia And Eclampsia) Pregnancy High Blood Pressure (Pregnancy: Preeclampsia And Eclampsia)

Pregnane

Pregnane is a 21-carbon steroid hydrocarbon that forms the structural backbone of progesterone and several corticosteroid and progestin compounds.

Pregnanolone

Pregnanolone is a neuroactive steroid metabolite that can modulate GABA-A receptors and influence neuronal excitability, mood, and sedation.

Pregnant

Pregnant means carrying a developing embryo or fetus in the uterus after conception; pregnancy status affects medicines, imaging, and clinical decisions.

Pregnant (Pregnancy Planning (Preparing For Pregnancy))

Alternate index label for Pregnancy Planning (Preparing for Pregnancy).

Pregnenolone

Pregnenolone is a steroid hormone precursor made from cholesterol that can be converted into progesterone, corticosteroids, and sex hormones.

Prehn Sign

A classical physical examination finding used in the emergency clinical evaluation of acute scrotal pain to aid in differentiating epididymitis from testicular torsion. The sign is considered positive when manual elevation and support of the affected scrotum and testis relieves or significantly reduces the patient's pain, typically supporting a diagnosis of acute epididymitis or epididymo-orchitis due to relief of gravitational suspension strain. Conversely, a negative Prehn sign (pain unchanged or worsened by elevation) is classically associated with testicular torsion; however, because the sign is neither completely sensitive nor specific, emergency color Doppler ultrasonography or immediate surgical scrotal exploration remains mandatory whenever testicular torsion cannot be definitively excluded.

Alternate Names:

Prehn's Sign

Prehypertension

Blood pressure above normal but below the former threshold for hypertension.

Alternate Names:

Elevated blood pressure

Preimplantation Genetic Testing

Genetic testing of cells from an embryo created through in-vitro fertilization before possible transfer to the uterus, used in selected situations to identify specific inherited disorders or chromosome-number

abnormalities.

Preleukaemia

An obsolete term formerly used for myelodysplastic syndromes or other states preceding leukemia.

Preload

The end-diastolic volume or pressure in the ventricles, representing the degree of ventricular stretch before contraction. Preload is determined by venous return and is estimated by the pulmonary capillary wedge pressure (PCWP) or left ventricular end-diastolic pressure (LVEDP).

Preload Reduction

Lowering the amount of blood returning to the heart to reduce cardiac workload.

Premalignant

Having cellular or tissue changes that may progress to invasive cancer.

Premature

Occurring earlier than expected or before the usual developmental or gestational time.

Premature (Early) Ejaculation

A persistent or recurrent pattern of ejaculation that occurs sooner than desired, with limited control and associated personal distress. Assessment considers the person's typical experience and sexual context.

Alternate Names:

Early

Premature Adrenarche

Early appearance of pubic or axillary hair, body odor, or mild acne from adrenal androgen production without true central puberty; evaluation excludes virilizing disorders.

Premature Atrial Contraction

A premature beat arising from an ectopic focus in the atria, producing an early P wave (often abnormal morphology) followed by a normal QRS; common and usually benign, increasing with age

and caffeine/stress. ECG: early abnormal P wave, normal PR and QRS, followed by a compensatory pause.

Premature Baby

An infant born before 37 completed weeks of pregnancy, with risks related to immature lungs, brain, feeding, temperature control, immunity, and other organ systems.

Premature Birth

Birth before 37 completed weeks of pregnancy; the earlier the birth, the greater the risks of respiratory, neurological, and developmental complications.

Premature Closure

Closure of a body opening, developmental structure, or growth plate earlier than expected, potentially causing obstruction, abnormal shape, or impaired growth depending on the site.

Premature Ejaculation

Ejaculation that occurs sooner than desired with poor control and causes persistent personal or relationship distress.

Premature Infant

An infant born before 37 completed weeks of gestation.

Premature Labor

Regular uterine contractions with cervical change before 37 completed weeks of pregnancy, which can lead to preterm birth and requires assessment of membranes, infection, and fetal well-being.

Premature Labour (Preterm Labour)

Labour that begins before 37 completed weeks of pregnancy and may result in preterm birth.

Alternate Names:

Preterm Labour

Premature Menopause

Loss of ovarian function and menstruation before age 40, now often termed primary ovarian insufficiency.

ciency.

Premature Menopause (Medical Procedural Causes)

Alternate index label for Medical Procedural Causes.

Premature Ovarian Failure

Premature ovarian failure is loss of normal ovarian function before age 40, causing irregular or absent periods, infertility, and low-estrogen symptoms; causes may be genetic, autoimmune, toxic, or unknown.

Premature Ovarian Insufficiency

Loss of ovarian function before age forty characterized by amenorrhea, hypergonadotropic hypogonadism, and estrogen deficiency. Causes include genetic disorders particularly Turner syndrome and fragile X premutations, autoimmune oophoritis, iatrogenic from chemotherapy or radiation, and idiopathic.

Premature Puberty

Alternate index label; see *Precocious puberty*.

Premature Rupture Of Membranes

PROM: spontaneous rupture of the amniotic membranes before the onset of labor; preterm PROM (PPROM) occurs before 37 weeks. Features: gush or leakage of fluid per vaginam; confirmed by sterile speculum (pooling, ferning, nitrazine pH test) or amniotic fluid biomarkers (IGFBP-1, PAMG-1).

Premature Thelarche

Isolated breast development in a young child without other signs of puberty, often transient but assessed for growth acceleration, estrogen exposure, or true precocious puberty.

Premature Ventricular Contraction

An early heartbeat that starts in a ventricle instead of following the heart's usual rhythm. It may feel like a skipped or extra beat and is often harmless, but frequent PVCs or symptoms such as fainting,

chest pain, or breathlessness need assessment.

Alternate Names:

PVC

Premature Ventricular Contractions

Ectopic beats originating from the ventricles, appearing as wide, bizarre QRS complexes that occur earlier than expected. PVCs are common in the general population and are usually benign. Frequent PVCs (>10% of total beats) may cause left ventricular dysfunction.

Alternate Names:

Extrasystole (Premature Ventricular Contractions)
PVC (Premature Ventricular Contractions)

Premature Ventricular Contractions

Alternate index label for Premature Ventricular Contractions.

Premature Ventricular Contractions (PVCs)

Early extra heartbeats originating in the heart's ventricles, often felt as fluttering, skipped beats, or pounding and frequently benign.

Alternate Names:

Extrasystole PVCs Ventricular Premature Contraction VPCs

Premenstrual

Occurring during the days before menstruation; the term commonly refers to cyclical symptoms such as mood change, breast tenderness, bloating, headache, or fatigue.

Premenstrual Breast Changes

Premenstrual breast changes include cyclic fullness, tenderness, swelling, or lumpiness caused by hormonal variation; a persistent one-sided mass, skin change, or bloody discharge needs evaluation.

Premenstrual Dysphoric Disorder

A severe, debilitating form of premenstrual syndrome characterized by marked mood symptoms (depression, anxiety, irritability, mood swings), physical symptoms (breast tenderness, bloating, fatigue),

and functional impairment that occur during the luteal phase and remit within days of menses onset.

Premenstrual Dysphoric Disorder

A severe cyclical mood disorder with affective and physical symptoms in the luteal phase that remit after menstruation.

Alternate Names:

PMDD (Premenstrual Dysphoric Disorder (PMDD))

Premenstrual Dysphoric Disorder

Alternate index label for Premenstrual Dysphoric Disorder.

Premenstrual Syndrome

A recurrent pattern of physical, emotional, and behavioral symptoms during the luteal phase of the menstrual cycle that resolve with menses. Affects up to 50% of reproductive-age women. Symptoms: irritability, mood swings, depression, anxiety, bloating, breast tenderness, fatigue, and appetite changes.

Premenstrual Syndrome

Recurring physical, emotional, or behavioral symptoms before menstruation that resolve around or after its onset.

Premie

Premie is an informal term for a premature infant, born before 37 completed weeks of gestation and at risk for complications related to immature organs.

Premolar (Bicuspid)

Any of the eight transitional permanent teeth situated between the anterior canines and the posterior molars in the human dental arch (two in each quadrant of the maxilla and mandible). Typically feature two distinct cusps (buccal and lingual) on their occlusal crowns and are unique to the permanent dentition, where they replace deciduous (primary) molars. Function mechanically to shear and crush food boluses before fine grinding by permanent molars; frequently extracted selectively in orthodontic treatment to resolve severe dental crowding.

Prenatal

Prenatal means occurring during pregnancy before birth, including care, testing, development, counseling, and treatment of the pregnant person or fetus.

Prenatal Care

Regular medical care during pregnancy to monitor maternal and fetal health. Includes history and physical exam, weight and blood pressure monitoring, urine testing, blood work, fundal height measurements, fetal heart rate auscultation, and ultrasound.

Prenatal Care In Your First Trimester

First-trimester prenatal care confirms pregnancy and dating, reviews health and medicines, offers screening, and addresses nutrition, folic acid, and warning symptoms.

Prenatal Care In Your Second Trimester

Second-trimester prenatal care monitors parent and fetus, offers anatomy and other screening, and discusses movement, nutrition, vaccinations, and warning symptoms.

Prenatal Care In Your Third Trimester

Third-trimester prenatal care monitors growth, blood pressure, fetal wellbeing, and birth planning and prepares for labor and postpartum care.

Prenatal Development

Prenatal development is growth and differentiation of the embryo and fetus from conception to birth, influenced by genes, nutrition, maternal health, and exposures.

Prenatal Diagnosis

Detection of genetic or structural disorders in fetus before birth. Methods: chorionic villus sampling (CVS, 10-13 weeks), amniocentesis (15-20 weeks), percutaneous umbilical blood sampling (PUBS, after 18 weeks). Non-invasive: NIPT (cell-free fetal DNA).

Prenatal Exposure

Contact of a fetus with a medicine, substance, infection, or environmental factor during pregnancy. Effects depend on the agent, dose, timing, and fetal susceptibility.

Prenatal Genetic Counseling

A consultation that reviews the risk of inherited or chromosomal conditions, available screening and diagnostic tests, possible results, and reproductive options before or during pregnancy.

Prenatal Testing

Screening and diagnostic tests performed during pregnancy to assess fetal health and development or identify genetic, chromosomal, and structural conditions. Tests may include ultrasound, maternal blood tests, cell-free DNA screening, chorionic-villous sampling, and amniocentesis.

Prenatal Vitamins

Multivitamin supplements formulated for pregnancy, containing folic acid (400-800 mcg), iron (27-60 mg), calcium (200-300 mg), vitamin D (400-600 IU), DHA (200 mg), and iodine (150-220 mcg). Started preconceptionally or at the first prenatal visit. Evidence supports NTD reduction from folic acid. Iron prevents maternal anemia.

Prenylation

Attachment of a lipid group to a protein, helping determine its membrane localization, signaling, stability, and interactions; abnormal prenylation contributes to some diseases and is a drug target.

Preop

Preop is an abbreviation for preoperative, meaning relating to preparation and assessment before surgery.

Preoperative Assessment

Preoperative assessment evaluates surgical risk and optimizes medical conditions before anesthesia. It includes medical history, physical examination, airway assessment, cardiovascular risk stratification (Revised Cardiac Risk Index), pulmonary function

assessment, and laboratory testing based on clinical indication.

Preoperative Care

A medical procedure, treatment, device, or care topic related to preoperative care; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Preoperative Evaluation

Medical assessment before elective surgery to optimize outcomes and minimize perioperative risk.

Preoptic Area

The preoptic area is a hypothalamic region involved in temperature regulation, sleep, reproductive behavior, and hormone control.

Preparing Children For Pregnancy And A New Baby

Preparing a child for a new baby involves age-appropriate explanation, routine and attention planning, safe involvement, reassurance, and support for emotional or behavioral changes.

Preparing For Surgery When You Have Diabetes

A medical procedure, treatment, device, or care topic related to preparing for surgery when you have diabetes; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Prepubertal

Prepubertal means before the physical and hormonal changes of puberty have begun or progressed to reproductive maturity.

Prepuce

The fold of skin covering the glans penis or clitoris; penile prepuce is the foreskin.

Prerenal Azotemia

A rise in nitrogen-containing waste products caused by reduced effective blood flow to the kidneys without primary renal-tissue injury. Causes include

dehydration, hemorrhage, heart failure, sepsis, and hepatorenal syndrome; urine indices may support but do not by themselves establish the diagnosis.

Presbyacusis

Age-related progressive sensorineural hearing loss, usually affecting high frequencies.

Presbyacusis (Presbycusis)

Age-related, usually gradual sensorineural hearing loss, particularly affecting the ability to hear high-frequency sounds and understand speech in noise.

Alternate Names:

Presbycusis

Presbycusis

Bilateral, usually symmetric age-related sensorineural hearing loss, initially affecting high frequencies and speech discrimination in noise. Management includes hearing rehabilitation, hearing aids, communication strategies, and cochlear implantation in selected severe cases.

Presbyopia

Age-related loss of near focusing caused by reduced lens accommodation.

Preschooler Development

Preschooler development includes rapid gains in language, motor skills, thinking, self-care, play, and social-emotional regulation; milestones vary, but loss of skills warrants assessment.

Preschooler Test Or Procedure Preparation

Preparing a preschooler for a test or procedure uses simple explanations, play, comfort objects, and caregiver support while following any fasting or medicine instructions.

Prescription

A written or electronic instruction from an authorized prescriber for a medicine or treatment.

Prescription Medications

Pharmaceutical drugs that require authorization from a licensed healthcare provider before they can be dispensed to a patient. These medications are regulated because they carry risks that necessitate professional oversight, including potential side effects, drug interactions, and the need for precise dosing.

Presenile Dementia

An older term for dementia beginning before old age; modern diagnosis specifies the cause and syndrome.

Presenilin

Presenilin is a component of the gamma-secretase complex that processes membrane proteins; inherited variants can cause familial early-onset Alzheimer disease.

Presentation

The part of a fetus that is directed toward the birth canal, or the symptoms and signs with which a patient seeks care.

Preseptal Cellulitis

Preseptal cellulitis, also known as periorbital cellulitis, is an infection of the eyelid and periorbital tissues anterior to the orbital septum, without involvement of the orbital contents, and it is the most common orbital infection, particularly in children.

Alternate Names:

periorbital cellulitis, is an infection of the.

Preservation

Maintenance of tissue, cells, organs, or function by preventing deterioration.

Preservative

A substance added to prevent microbial growth or chemical deterioration of a product.

Pressor

Raising blood pressure, usually by constricting blood vessels or increasing cardiac output.

Pressoreceptor

A pressure-sensitive receptor, especially a baroreceptor in the carotid sinus or aortic arch, that detects arterial stretch and helps regulate heart rate and blood pressure.

Pressure

Force exerted over an area; clinically measured in units such as mmHg.

Pressure Atrophy

The gradual wasting of tissue resulting from sustained external compression that impairs blood supply and causes cellular shrinkage and eventual death. The mechanism involves compression of blood vessels, which reduces perfusion and causes ischemia of the dependent tissue.

Pressure Injury

Localized damage to the skin and/or underlying soft tissue caused by prolonged pressure, shear, or friction, most commonly over bony prominences.

Pressure Sore

A localized injury to skin and underlying tissue caused by prolonged pressure or shear, also called a pressure ulcer.

Pressure Support Ventilation

Spontaneous breathing mode with ventilator-provided pressure support above PEEP for each breath. Reduces work of breathing. Used for weaning from mechanical ventilation.

Pressure Ulcer

Localized damage to skin and underlying tissue caused by sustained pressure or pressure with shear, typically over a bony prominence. Staging includes non-blanchable erythema, partial-thickness loss, full-thickness loss, and full-thickness tissue loss with exposed or palpable deep structures; obscured wounds are unstageable.

Pressure Ulcer Prevention

Measures that reduce prolonged pressure and tissue

injury, including repositioning, support surfaces, skin inspection, moisture control, nutrition, and mobility.

Pressure Urticaria

A form of chronic inducible urticaria in which sustained pressure causes localized swelling, erythema, and pain, usually after a delay of several hours. Lesions may last 8 to 72 hours and commonly affect the hands, feet, buttocks, or areas compressed by clothing or sitting.

Pressure-Flow Study

Suite of diagnostic procedures designed to evaluate the physiological function of the lower urinary tract (bladder and urethral sphincter) during filling, storage, and emptying phases. See urodynamic testing.

Presynaptic Terminals

Presynaptic terminals are the ends of neurons that release neurotransmitters into synapses in response to electrical activity.

Presyncope

The sensation of imminent fainting (lightheadedness, weakness, visual blurring) without actual loss of consciousness, due to transient global cerebral hypoperfusion that resolves before LOC. Causes: orthostatic hypotension (volume depletion, autonomic failure, medications), vasovagal, cardiac (arrhythmias, aortic stenosis), and metabolic.

Preterm Birth

Birth before 37 completed weeks of gestation. It may follow spontaneous labor or rupture of the membranes, or may be medically indicated because of maternal or fetal complications.

Preterm Infant

An infant born before 37 completed weeks of gestation; classified by gestational age (late preterm 34-36, moderate 32-33, very preterm <32, extremely <28 weeks) and by birth weight. Causes: preterm labor, PPROM, preeclampsia, multiple pregnancy, infection, and iatrogenic.

Preterm Labor

Regular uterine contractions causing cervical change before 37 weeks gestation. Risk factors include prior preterm delivery, multiple gestation, infection, and cervical insufficiency. Diagnosis requires regular contractions with documented cervical change.

Preterm Premature Rupture Of Membranes

Rupture of the amniotic membranes before the onset of labor at <37 weeks gestation. Causes: infection (subclinical chorioamnionitis), cervical insufficiency, and prior PPROM. Presents with sudden gush or leakage of fluid per vagina.

Pretest Probability

The estimated probability that a patient has a disease before the result of a diagnostic test is known. It is based on prevalence, symptoms, examination findings, risk factors, and prior test results and should guide test selection and interpretation.

Pretibial Myxedema

Localized dermopathy occurring in approximately five percent of patients with Graves disease caused by glycosaminoglycan deposition in the dermis particularly over the pretibial area. Presents as non-pitting indurated plaques with a peau d'orange appearance.

Pretibial Myxoedema

Thickened, waxy swelling of the skin over the shins, most often associated with Graves disease.

Prevalence

The proportion of a population with a disease or characteristic at a specified time or period.

Preventing Falls

Fall prevention combines strength and balance activity, vision and medication review, safe footwear, adequate lighting, handrails, removal of hazards, and appropriate assistive devices.

Preventing Food Poisoning

Preventing food poisoning involves toxic or environmental exposure to preventing food poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Preventing Head Injuries In Children

Children's head injuries can be reduced with age-appropriate car seats, helmets, supervision, safe play areas, window and stair protection, and prevention of abusive shaking or impact.

Preventing Hepatitis A

Hepatitis A prevention relies on vaccination, hand-washing, safe food and water, sanitation, and post-exposure advice when close contact with an infected person occurs.

Preventing Hepatitis B Or C

Hepatitis B prevention includes vaccination and avoiding blood or sexual exposure; hepatitis C prevention centers on sterile injection equipment, screened blood, and testing and treatment of infection.

Preventing Pressure Injuries

Pressure-injury prevention uses regular repositioning, skin inspection, moisture control, nutrition, pressure-redistributing surfaces, and minimizing friction and shear in people with limited mobility.

Preventing Stroke

Stroke prevention includes controlling blood pressure, diabetes and cholesterol, avoiding tobacco, treating atrial fibrillation when indicated, exercising, and addressing other vascular risks.

Prevention

Measures intended to avoid disease, detect it early, limit complications, or reduce disability.

Preventive Dentistry

Practice of preventing dental disease. Methods: oral hygiene instruction, fluoride, sealants, diet counseling, regular check-ups.

Preventive Health Care

Preventive health care aims to prevent disease or detect it early through vaccination, screening, healthy behaviors, and routine medical care.

Preventive Medicine

The field concerned with preventing disease and promoting health in individuals and populations.

Prevertebral Air

Gas in the prevertebral soft tissues on lateral cervical spine radiograph, indicating a breach in the aerodigestive tract. It may result from esophageal perforation, tracheal injury, retropharyngeal abscess with gas-forming organisms, or trauma.

Priapism

A persistent, usually painful erection unrelated to sexual stimulation and lasting long enough to require urgent assessment.

Alternate Names:

Ischemic Priapism Low-Flow Priapism Nonischemic Priapism

Priapism (Penis Disorder)

Alternate index label for Penis Disorder.

Prick Test

A skin test in which small amounts of suspected allergens are introduced into the skin.

Alternate Names:

Skin-prick test

Prickle Cell

An obsolete or imprecise term sometimes used for a dendritic epidermal cell or for a sickle cell; the intended meaning requires context.

Prickly Heat

Miliaria, an itchy rash caused by blockage of sweat ducts.

Primary

First in time, origin, importance, or site; in disease classification it often means arising independently

rather than secondary to another cause.

Primary Adrenal Insufficiency

Alternate index label; see *Addison's disease*.

Primary Adrenal Insufficiency (Addison's Disease)

Destruction or dysfunction of the adrenal cortex causing deficiency of cortisol and aldosterone, presenting with fatigue, hyperpigmentation, weight loss, and salt craving. Lifelong hormone replacement is required.

Alternate Names:

Addison's disease

Primary Adrenal Lymphoma

A rare lymphoma arising primarily in the adrenal gland, most commonly diffuse large B-cell lymphoma. It often presents bilaterally and may cause adrenal insufficiency.

Primary Aldosteronism

Excess production of aldosterone by the adrenal glands, usually from an adrenal nodule or enlargement of both glands. It increases sodium retention and may cause hypertension, low potassium, or suppressed renin.

Primary Alveolar Hypoventilation

A rare disorder of inadequate automatic breathing, often related to PHOX2B or other autonomic-control abnormalities, causing elevated carbon dioxide especially during sleep.

Primary Amyloidosis

Primary amyloidosis, usually AL amyloidosis, results from abnormal plasma-cell proteins depositing as amyloid in organs such as the heart, kidneys, nerves, or gastrointestinal tract.

Primary And Secondary Hyperaldosteronism

Excess aldosterone production, from an adrenal source in primary disease or from increased renin stimulation in secondary disease, causing hyperten-

sion, sodium retention, and sometimes low potassium.

Primary Angioplasty

Angioplasty used as the initial treatment to reopen an artery during an acute heart attack.

Primary Biliary Cholangitis

A chronic autoimmune liver disease in which small intrahepatic bile ducts are progressively damaged. It can cause cholestasis, itching, fatigue, and eventually cirrhosis; antimitochondrial antibodies often support the diagnosis.

Primary Biliary Cirrhosis

Alternate index label; see *Primary biliary cholangitis*.

Primary Biliary Cirrhosis

Alternate index label for PBC.

Primary Biliary Cirrhosis

Alternate index label for Primary Biliary Cirrhosis.

Primary Biliary Cirrhosis Treatment

Alternate index label for PBC.

Primary Care

First-contact, continuing, comprehensive healthcare, including prevention and management of common conditions.

Primary Ciliary Dyskinesia

An inherited disorder of ciliary structure or function that impairs mucociliary clearance. It can cause chronic sinus and lung infections, bronchiectasis, infertility, and abnormal organ laterality.

Primary Cutaneous B-Cell Lymphoma

A B-cell lymphoma presenting primarily in the skin without evidence of extracutaneous disease at diagnosis; clinical behavior and treatment depend on the specific subtype.

Primary Endpoint

A primary endpoint is the main prespecified outcome used to judge the effect of an intervention in a clinical trial.

Primary Hyperaldosteronism

Excess autonomous production of aldosterone by the adrenal glands. It can cause hypertension, low potassium, metabolic alkalosis, and suppressed renin. Causes include an adrenal adenoma or bilateral adrenal hyperplasia.

Primary Hyperoxaluria

A group of autosomal recessive metabolic disorders characterized by overproduction of oxalate by the liver, leading to calcium oxalate deposition in the kidneys and other organs. Type 1 is caused by alanine-glyoxylate aminotransferase deficiency. It presents with recurrent nephrolithiasis, nephrocalcinosis, and progressive renal failure.

Primary Hyperparathyroidism

Autonomous overproduction of parathyroid hormone from one or more abnormal parathyroid glands causing hypercalcemia. Most commonly caused by a single parathyroid adenoma accounting for approximately eighty-five percent. Multigland hyperplasia accounts for approximately ten to fifteen percent and parathyroid carcinoma is rare.

Primary Hypersomnia

Excessive daytime sleepiness not better explained by insufficient sleep, another sleep disorder, substance, or medical condition; modern classification uses specific central-hypersomnolence diagnoses.

Primary Hypertension

High blood pressure without an identifiable single underlying cause.

Primary Hypertrophic Osteoarthropathy

A rare inherited syndrome characterized by digital clubbing, periostosis of long bones, and pachydermia, presenting with joint pain, skin thickening, and visible nail-bed enlargement.

Primary Hypoparathyroidism

Inadequate parathyroid hormone production causing low calcium and high phosphate, with possible tingling, cramps, tetany, seizures, cataracts, or chronic bone and dental effects.

Primary Hypothyroidism

Reduced thyroid-gland production of thyroxine and triiodothyronine, usually causing high TSH and symptoms such as fatigue, cold intolerance, constipation, dry skin, and slowed heart rate.

Primary Immune Response

The adaptive immune response to first exposure to an antigen, characterized by a slow onset (lag phase: days, during which antigen processing, clonal expansion, and differentiation occur), followed by rising antibody titers (predominantly IgM, then IgG), and generation of memory cells.

Primary Immunodeficiency

A group of genetically determined disorders of the immune system causing increased susceptibility to infection, autoimmunity, and malignancy.

Primary Immunodeficiency Disease

An inherited or genetic disorder of immune development or function that causes unusual, recurrent, severe, or persistent infections and sometimes autoimmunity.

Primary Immunodeficiency Disease PIDD

Primary immunodeficiency diseases are inherited disorders of immune development or function that cause unusual, severe, recurrent, or persistent infections and sometimes autoimmunity or inflammation. Evaluation may include blood counts, immunoglobulins, functional tests, and genetic testing.

Alternate Names:

Deficiency Primary Immunodeficiency (Primary Immunodeficiency Disease PIDD)

Primary Lactase Deficiency

Reduced lactase enzyme activity developing with age, causing lactose malabsorption and possible

bloating, abdominal pain, gas, or diarrhea after dairy consumption.

Primary Lateral Sclerosis

A rare neurodegenerative disorder confined to the upper motor neurons (corticospinal tract) of the motor cortex and brainstem. Presents with slowly progressive spasticity, hyperreflexia, Babinski sign, and pseudobulbar affect (emotional lability).

Primary Lymphoma Of The Brain

Primary central-nervous-system lymphoma is a lymphoma arising in the brain, spinal cord, eyes, or surrounding fluid without systemic disease at diagnosis.

Primary Metabolism

The essential biochemical processes by which cells produce energy, synthesize molecules, and maintain their functions.

Primary Microrna

A primary microRNA is the initial RNA transcript made from a microRNA gene before nuclear and cytoplasmic processing into mature microRNA.

Primary Myelofibrosis

A chronic myeloproliferative neoplasm characterized by bone marrow fibrosis (reticulin and/or collagen fibrosis), extramedullary hematopoiesis (primarily splenic and hepatic), and a leukoerythroblastic peripheral blood picture.

Primary Open-Angle Glaucoma

A chronic optic neuropathy with characteristic retinal ganglion-cell and optic-nerve loss, usually accompanied by progressive visual-field defects, despite an anatomically open anterior-chamber angle. Intraocular pressure is an important modifiable risk factor, and treatment lowers pressure to slow progression.

Primary Osteoporosis

Bone loss due to aging, menopause, or other normal physiological processes rather than another disease.

Primary Ovarian Insufficiency

Loss of normal ovarian activity before age 40, characterized by irregular or absent menses, reduced estrogen, and intermittently or persistently elevated gonadotropins. Causes include genetic, autoimmune, iatrogenic, toxic, infectious, and unknown factors.

Primary Polycythemia

Alternate index label; see *Polycythemia vera*.

Primary Prevention

Preventing cancer development. Strategies: smoking cessation, vaccination (HPV, hepatitis B), sun protection, healthy diet, exercise.

Primary Progressive Multiple Sclerosis

A clinical phenotype of multiple sclerosis accounting for approximately 10–15% of patients, characterized from onset by continuous, insidious neurologic decline without distinct relapses and remissions. Typically presents with progressive spastic paraparesis, gait ataxia, and neurodegenerative axonal loss on spinal cord magnetic resonance imaging, managed primarily with B-cell targeted therapies such as ocrelizumab.

Alternate Names:

PPMS; Primary-progressive MS

Primary Progressive Aphasia

A neurodegenerative disorder causing gradual worsening of language abilities, including speaking, naming, comprehension, reading, or writing.

Primary Sclerosing Cholangitis

Chronic cholestatic liver disease characterized by progressive fibrosis and strictures of intrahepatic and extrahepatic bile ducts. The disease has a strong association with inflammatory bowel disease, especially ulcerative colitis, occurring in a substantial proportion of patients.

Primary Sclerosing Cholangitis

Alternate index label for *PSC*.

Primary Sclerosing Cholangitis

Alternate index label for Primary Sclerosing Cholangitis.

Primary Tuberculosis

The first infection with *Mycobacterium tuberculosis*. It may produce a small pulmonary focus and regional lymph-node involvement; the infection may be contained as latent tuberculosis or progress to active disease.

Primary Tumor

The original tumor, or site where a cancer began, from which cancer cells may spread to form secondary or metastatic tumors. A cancer may be described as having an unknown primary when the original site cannot be identified.

Primidone

Primidone is an antiseizure medicine used for epilepsy and sometimes essential tremor; it can cause sleepiness, dizziness, and drug interactions.

Primigravida

A clinical obstetrical term denoting a female patient who is pregnant for the very first time, designated in clinical gravidity and parity notation as Gravidity 1 (G1). Primigravida patients exhibit distinct labor dynamics compared to multigravida women, characterized by a longer first stage of labor, slower cervical dilation, and an increased incidence of hypertensive disorders of pregnancy, notably preeclampsia, which develops with substantially higher frequency during a first gestation.

Primipara

A person who has given birth once after the stage of fetal viability.

Primitive

Relating to an early developmental stage or a basic, less differentiated form.

Primitive Node

The primitive node is an embryonic organizer at

the cranial end of the primitive streak that helps establish body axes and germ-layer formation.

Primitive Streak

The primitive streak is an embryonic structure through which cells migrate during gastrulation to form the three germ layers and establish body organization.

Primordial

Present at the earliest stage of development or existing in an original, basic form. Examples include primordial germ cells and primordial follicles.

Primordial Prevention

Preventing the emergence of disease risk factors by addressing social, economic, behavioural, and environmental conditions early. Examples: promoting healthy diets and physical activity in children, reducing tobacco exposure, and creating safe environments for exercise.

Prinomastat

Prinomastat is an investigational matrix-metalloproteinase inhibitor studied for blocking enzymes involved in tumor invasion and tissue remodeling.

Prion

An abnormally folded infectious protein that can cause progressive neurodegenerative disease.

Prion Disease

A neurodegenerative disease caused by abnormal folding and propagation of prion protein. These disorders are rare and usually progressive and fatal, although their course varies by the specific disease.

Prism

A transparent optical device that bends light; prisms may correct or measure ocular alignment.

Privacy

Privacy is a person's right and expectation that personal information, communications, and bodily

matters are protected from unauthorized access or disclosure.

Privacy Breach

A privacy breach is unauthorized access, use, disclosure, loss, or exposure of personal or health information.

Private Mutation

A genetic variant confined to a family, individual, tumor, or small population rather than commonly seen in the general population; its significance requires genetic and clinical interpretation.

PRK

Photorefractive keratectomy, commonly known as PRK, is a refractive surgical procedure that corrects myopia, hyperopia, and astigmatism by reshaping the corneal stroma using an excimer laser after mechanical removal of the corneal epithelium, without creating a corneal flap.

Pro Re Nata

A medication or treatment instruction meaning that it should be used only when a specified symptom or clinical circumstance occurs, within clearly prescribed dose and frequency limits.

Alternate Names:

P.R.N.; As Needed

Prob Problem

An incomplete or nonstandard label; its intended clinical meaning cannot be determined without source context.

Probability

Probability is a numerical expression of how likely an event is, ranging from impossible to certain and used in diagnosis, risk assessment, and research.

Probable Cause

A legally recognized reasonable basis for believing that a person committed an offense or that evidence of an offense will be found.

Probe

A slender, flexible or rigid handheld surgical, dental, or diagnostic instrument with a blunt, rounded tip, designed for exploring anatomical cavities, sinus tracts, fistulas, or deep wounds without creating false passages. Prominent clinical variants include periodontal probes (calibrated in millimeters to measure periodontal pocket depths and epithelial attachment loss), lacrimal probes (used to dilate and confirm patency of obstructed nasolacrimal ducts), and surgical fistula probes (used during anal fistulotomy to trace fistulous tracts between the anal canal and perianal skin).

Probenecid

A uricosuric medicine used to prevent gout attacks in selected patients and to alter renal excretion of some drugs.

Probiotic

Live microorganisms administered in adequate amounts to confer a health benefit by modulating the microbiome and immune responses.

Probiotics

Probiotics are live microorganisms that may provide a health benefit when consumed in adequate amounts, although effects are strain-specific and vary by condition.

Problem

A symptom, sign, diagnosis, or issue requiring assessment or management.

Problems Sleeping During Pregnancy

Sleep problems during pregnancy may result from discomfort, reflux, urination, anxiety, hormonal change, or sleep apnea; safe sleep habits and clinician-guided treatment are important.

Probocis

A misspelling of proboscis, a projecting snout or appendage.

Proboscis

An elongated nose, snout, or tubular feeding structure in an animal. In human description, it may refer to an unusually prominent or elongated nose.

Probucol

Probucol is a lipid-lowering drug that reduces cholesterol but may prolong the QT interval and is not widely used in current practice.

PROC

PROC is the gene symbol for protein C, an anticoagulant protein that limits clot formation; inherited deficiency increases thrombosis risk.

Procainamide

An antiarrhythmic medicine used for selected cardiac rhythm disorders.

Procaine

An ester local anesthetic, now used mainly in limited or historical settings.

Procalcitonin

A biomarker that may rise during some bacterial infections and systemic inflammation. It can support assessment of infection and antibiotic decisions, but it cannot by itself diagnose or exclude bacterial infection or determine sepsis severity.

Procarbazine

An alkylating-like chemotherapy medicine used in selected lymphomas and brain tumors.

Procarbazine Hydrochloride

Procarbazine hydrochloride is the hydrochloride salt formulation of the antineoplastic drug procarbazine.

Procedural Memory

Memory for how to perform learned skills, such as riding a bicycle, typing, or playing an instrument. It usually works without consciously recalling each step and differs from memory for facts or events.

Procedural Sedation

Administration of sedative and analgesic medicines to facilitate a painful or anxiety-provoking procedure while preserving spontaneous ventilation and protective reflexes to the intended depth. Continuous monitoring and immediate rescue capability are required.

Procedure

A planned medical or surgical action performed for diagnosis, treatment, prevention, or palliation.

Procentriole

A procentriole is an immature centriole that duplicates beside a parent centriole and later matures to organize microtubules.

Procerus Muscle

A small facial muscle over the bridge of the nose that draws the medial eyebrows downward and produces horizontal wrinkles between the eyebrows.

Process

A process is an ordered series of biological, physical, chemical, or operational changes that produces a result.

Processing Bodies

Processing bodies are cytoplasmic RNA-protein granules where messenger RNAs may be stored, decapped, degraded, or regulated.

Processor

A processor is a device or biological system that performs a defined sequence of operations; in computing it executes instructions, while in medicine the meaning is context-dependent.

Prochlorperazine

A phenothiazine medicine used as an antiemetic and antipsychotic.

Prochlorperazine Overdose

Prochlorperazine overdose involves toxic or environmental exposure to prochlorperazine overdose; dose,

route, timing, symptoms, and organ involvement determine urgency and treatment.

Procidentia

Complete eversion and severe uterine prolapse (Stage IV pelvic organ prolapse) in which the entire uterus, cervix, and surrounding anterior and posterior vaginal walls have descended completely beyond the introitus and outside the vulva. Typically accompanied by secondary decubitus ulceration, urinary tract obstruction, hydronephrosis, and recurrent cystitis due to kinking of the ureters.

Alternate Names:

Complete Uterovaginal Prolapse; Third-Degree Uterine Prolapse

Proclivity

Proclivity is a tendency or predisposition toward a particular behavior, response, disease, or outcome.

Procollagen

An immature precursor of collagen that is modified and cleaved outside cells to form mature collagen fibers.

Proctalgia

Pain originating within or referred to the rectum or anal canal. Categorized etiologically into organic conditions (such as acute anal fissures, thrombosed external hemorrhoids, perianal abscesses, proctitis, or rectal foreign bodies) and functional anorectal pain disorders. Among functional syndromes, proctalgia fugax presents as severe, sudden-onset, episodic cramping rectal spasms lasting from seconds to minutes, typically occurring at night and resolving spontaneously, whereas levator ani syndrome features chronic, vague aching or pressure in the rectum exacerbated by prolonged sitting.

Proctitis

Inflammation of the rectal mucosal lining (*rectum*), extending up to 15 cm from the anal verge. Presents clinically with tenesmus (a painful, persistent urge to defecate), rectal bleeding (hematochezia), passage of mucus or pus, and pelvic discomfort. Etiologies include inflammatory bowel disease (ulcerative

proctitis as the most distal manifestation of ulcerative colitis), sexually transmitted infections (STIs; *Neisseria gonorrhoeae*, *Chlamydia trachomatis* [including lymphogranuloma venereum], *Treponema pallidum*, and herpes simplex virus [HSV]), radiation proctitis (acute or chronic sequela of pelvic radiotherapy for prostate, cervical, or rectal cancer), and diversion proctitis following fecal diversion. Diagnosed by flexible sigmoidoscopy with mucosal biopsy and rectal swabs.

Alternate Names:

Rectal Inflammation

Proctoclysis

Administration of fluid into the rectum, an obsolete method of fluid or medicine delivery.

Proctocolitis

Inflammation of the rectum and the adjacent portion of the colon.

Proctology

The medical specialty concerned with diseases of the anus and rectum.

Proctoscope

An instrument used to inspect the anal canal and rectum.

Proctoscopy

Examination of the anal canal and rectum with a proctoscope.

Proctostomy

Surgical creation of an opening from the rectum to the skin.

Procurement

Process of recovering organs and tissues from donors for transplantation.

Prodromal

Describing early, nonspecific signs or symptoms that occur before the characteristic clinical features of a disease become apparent.

Prodrome

An early symptom or group of symptoms preceding a disease or major clinical event.

Prodrug

A prodrug is a pharmacologically inactive compound that is converted to an active drug through metabolic processes within the body. Prodrugs are designed to overcome limitations of the parent drug including poor bioavailability, rapid metabolism, lack of tissue selectivity, or unacceptable side effects.

Product Liability

Legal responsibility for injury or damage caused by a defective or unsafe product.

Product Recall

Removal or correction of a medicine, device, food, or other product because it may be defective, contaminated, mislabeled, or unsafe for use.

Productive Cough

A cough that successfully mobilizes and expectorates respiratory secretions, mucus, or purulent exudate (sputum) from the tracheobronchial tree. Distinguished from a dry (non-productive) irritative cough. Clinically evaluated by sputum color, consistency, volume, and odor: clear or white mucoid sputum suggests viral bronchitis or chronic obstructive pulmonary disease; yellow or green purulent sputum reflects neutrophil myeloperoxidase activity in acute bacterial bronchitis or pneumonia; foul-smelling sputum indicates anaerobic pulmonary abscess or bronchiectasis; and rust-colored sputum is classic for *Streptococcus pneumoniae* lobar pneumonia.

Proestrus

Proestrus is the phase of the reproductive cycle before estrus, characterized by follicular development and rising estrogen in many mammals.

Professional

Relating to a trained occupation or to the conduct expected of a qualified practitioner.

Profunda

A deep artery or vessel, commonly referring to the profunda femoris artery.

Profunda Miliaria (Heat Rash)

Alternate index label for Heat Rash.

Progenitor

A cell that has partially differentiated and can give rise to one or more mature cell types.

Progeny

The biological offspring, descendants, or successive generational cellular/organismal progeny derived from a parent or ancestral lineage. In clinical genetics and pedigree analysis, progeny are evaluated to trace Mendelian and non-Mendelian inheritance patterns, penetrance, variable expressivity, and carrier states across familial generations. In cellular biology and immunology, clonal progeny refers to identical daughter cells arising from the mitotic proliferation of a single committed parent cell, such as antigen-stimulated B or T lymphocyte clonal expansion.

Progeria

A rare genetic disorder causing features of accelerated aging beginning in childhood.

Progestational

Relating to progesterone-like effects or to preparation of the uterus for pregnancy.

Progesterone

A steroid hormone produced mainly by the corpus luteum and placenta that supports the endometrium and pregnancy.

Progesterone Test

A blood test that measures the level of progesterone, a hormone involved in ovulation and pregnancy. It may help assess ovulation, infertility, abnormal uterine bleeding, fertility treatment, or certain pregnancy problems. Results depend on the timing of the menstrual cycle and whether the person is

pregnant.

Alternate Names:

Serum Progesterone Progesterone Blood Test

Progesterone Antagonist

A progesterone antagonist blocks progesterone receptors and may be used for selected reproductive, gynecologic, or other medical purposes.

Progesterone Supplementation

Administration of progesterone to prevent preterm birth in high-risk women. Natural progesterone (17-alpha hydroxyprogesterone caproate) shown to reduce preterm birth rate in women with prior spontaneous preterm delivery. Also beneficial for short cervix detected on screening. Typically started in the second trimester.

Progestin

A progestin is a synthetic progesterone-like hormone used in contraception, hormone therapy, and treatment of selected gynecologic conditions.

Progestogen

A natural or synthetic hormone with effects like progesterone. Progestogens help regulate the menstrual cycle and pregnancy and are used in some contraceptives and hormone treatments.

Proglottid

One segment of a tapeworm. Each mature segment contains reproductive organs and may release eggs in stool.

Prognathism

Forward projection of the upper or lower jaw relative to the facial skeleton, affecting bite, appearance, chewing, speech, and sometimes airway function.

Prognosis

The expected course and outcome of a disease or condition.

Prognostic Factor

A prognostic factor is a clinical, pathologic, or bio-

logic feature associated with the likely course, outcome, or survival of a disease.

Prognostic Test

A test that provides information about the likely course or outcome of a disease.

Program

A planned set of activities or services, such as a screening or rehabilitation program.

Program Development

Program development is the planned design, testing, implementation, and evaluation of a service, intervention, or software system.

Programmed Instruction

A structured teaching method that presents small units of information with questions and immediate feedback, allowing learners to progress through material step by step.

Progression

Worsening or advancement of a disease, such as increasing tumor size, new lesions, or spread to another site.

Progression-Free Survival

The length of time from a defined starting point, such as diagnosis or treatment, until cancer progression or death from any cause, whichever occurs first.

Progressive

Increasing or worsening over time, or advancing in a specified direction.

Progressive Disease

Worsening of cancer during or after treatment, defined per RECIST as growth of target lesions, appearance of new lesions, or unequivocal progression of non-target lesions.

Progressive Lenticular Degeneration

Wilson disease, an inherited copper-accumulation

disorder affecting the liver, brain, and eyes.

Progressive Multifocal Leukoencephalopathy

Progressive multifocal leukoencephalopathy is a potentially fatal demyelinating brain infection caused by JC virus reactivation during significant immune suppression.

Progressive Muscular Atrophy

A motor-neuron disorder predominantly affecting lower motor neurons and causing progressive muscle wasting and weakness.

Progressive Supranuclear Palsy

A neurodegenerative tauopathy characterized by the accumulation of tau protein in the basal ganglia, brainstem, and cerebral cortex, belonging to the family of Parkinson plus syndromes.

Alternate Names:

Steele-Richardson-Olszewski Syndrome Supranuclear Palsy

Proinflammatory

Proinflammatory describes a substance, cell, or response that promotes inflammation through mediators such as cytokines, chemokines, or lipid signals.

Proinsulin

Proinsulin is the precursor molecule that pancreatic beta cells process into insulin and C-peptide.

Projectile Vomiting

Forceful vomiting expelled with unusual intensity, which may occur with gastric outlet obstruction, raised intracranial pressure, infection, migraine, or other causes; severity and associated signs guide evaluation.

Projection

The act of extending, displaying, or estimating something; in imaging, a view produced by directing rays through a body.

Prokaryote

A prokaryote is a cell or organism without a membrane-bound nucleus, including bacteria and archaea.

Prokinetic

A medicine that promotes gastrointestinal motility and movement of contents through the digestive tract; use depends on the underlying disorder and safety profile.

Prolactin

A pituitary hormone that stimulates breast development and milk production and influences reproductive function.

Prolactin Blood Test

A prolactin blood test measures the hormone prolactin and helps evaluate abnormal milk production, menstrual or fertility problems, and some pituitary disorders.

Prolactin Physiology

Prolactin is the only pituitary hormone primarily under tonic inhibitory control by the hypothalamus through dopamine acting on D2 receptors. Prolactin stimulates lactation suppresses GnRH and is important for reproductive function. Physiological elevations occur during pregnancy lactation breast stimulation sleep and stress.

Prolactin Receptor

A cell-surface receptor activated by prolactin to regulate mammary development, milk production, reproduction, immune effects, and other tissue responses.

Prolactinoma

A usually benign pituitary adenoma that secretes excess prolactin.

Alternate Names:

Pituitary Tumor (Prolactinoma)

Prolactinoma Treatment

Treatment may include a dopamine agonist, surgery,

radiation, or observation, depending on tumor size, symptoms, hormone levels, treatment response, and patient preferences. Dopamine agonists often lower prolactin levels and reduce tumor size.

Prolapse

Downward displacement or protrusion of an organ or tissue from its normal position, often through a supporting structure or body opening.

Prolapse Of Orbital Fat

Herniation of intraorbital fat through a dehiscence in the orbital septum, most commonly presenting as a painless, soft, reducible mass in the upper eyelid or lower eyelid. It is often age-related and may be confused with a lipoma or dermoid cyst.

Prolapse Of The Uterus (Uterine Prolapse)

Descent of the uterus into the vaginal canal due to weakening of pelvic floor supports, most commonly the uterosacral and cardinal ligaments.

Alternate Names:

Uterine Prolapse

Prolapse, Mitral Valve

Alternate index label; see *Mitral valve prolapse*.

Prolapsed Bladder

Alternate index label; see *Anterior vaginal prolapse (cystocele)*.

Prolapsed Cord

A prolapsed umbilical cord descends beside or ahead of the fetus after membrane rupture, risking cord compression and reduced fetal oxygen; it is an obstetric emergency.

Prolapsed Disk

A spinal disk that bulges or ruptures and may compress a nearby nerve; also called a herniated disk.

Alternate Names:

a herniated disk.

Prolapsed Uterus

Alternate index label; see *Uterine prolapse*.

Proliferating Trichilemmal Tumor

A usually low-grade follicular tumor arising from a trichilemmal cyst, most often on the scalp; large or recurrent lesions may show locally aggressive or malignant behavior.

Proliferation

Rapid increase in the number of cells or organisms.

Proliferative

Characterized by increased cell multiplication or new tissue growth. The meaning depends on context: some proliferative changes are normal or healing, while others are precancerous or cancerous.

Proliferative Diabetic Retinopathy

The advanced stage of diabetic retinopathy, characterized by the growth of new, fragile blood vessels (neovascularization) on the retinal surface, optic disc, or iris in response to severe retinal ischemia and the release of vascular endothelial growth factor.

Proliferative Fasciitis

A benign, reactive fibroblastic proliferation resembling nodular fasciitis but containing distinctive ganglion-like cells; it usually grows quickly and resolves or stabilizes after local treatment.

Proliferative Myositis

A benign reactive proliferation within skeletal muscle that may present as a rapidly enlarging mass and mimic sarcoma. Imaging and biopsy are used to establish the diagnosis.

Proliferative Retinopathy

Advanced retinal disease in which abnormal new blood vessels grow and may bleed or impair vision.

Prolindac

Prolindac is an investigational prodrug designed to release an active anticancer agent within tumors.

Proline

A protein-forming amino acid with a cyclic side chain.

Prolonged Deceleration

A fetal heart rate deceleration lasting ≥ 2 minutes but <10 minutes. If the deceleration persists ≥ 10 minutes, it is a baseline change. Causes: maternal hypotension (epidural-induced aortocaval compression), uterine hyperstimulation, umbilical cord prolapse, placental abruption, and maternal seizure.

Prolonged Second Stage

Second stage of labor exceeding the time limits: nulliparas — 3 hours with regional anesthesia (2 hours without); multiparas — 2 hours with regional (1 hour without). Management: reassess position, contractions, and fetal status. For inadequate contractions: oxytocin augmentation. For malposition (OP, OT): manual rotation.

Prolymphocyte Count

The number or proportion of prolymphocytes, immature-appearing lymphoid cells, in blood or marrow; an increased count may occur in selected lymphoid leukemias and requires morphology and immunophenotyping.

Prolymphocytic Leukemia

Prolymphocytic leukemia is a rare aggressive leukemia with large abnormal lymphocytes in the blood, bone marrow, spleen, or lymph nodes.

Promazine

A first-generation phenothiazine antipsychotic with sedative properties; it is now used infrequently in many settings and availability varies by country. It can cause drowsiness, orthostatic hypotension, anticholinergic effects, extrapyramidal symptoms, and other serious antipsychotic adverse effects.

Promecarb

Promecarb is a carbamate insecticide that inhibits acetylcholinesterase and can cause cholinergic poisoning after substantial exposure.

Promegapoietin

Promegapoietin is an investigational hematopoietic growth-factor molecule designed to stimulate production of blood cells or platelets.

Prometaphase

Prometaphase is the stage of mitosis after nuclear-envelope breakdown when spindle microtubules attach to chromosome kinetochores.

Promethazine

Promethazine is an antihistamine used for allergy symptoms, nausea, motion sickness, and sometimes sedation; it can cause marked sleepiness and breathing problems in young children.

Promethazine Overdose

Promethazine overdose involves toxic or environmental exposure to promethazine overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Prominent Iris Collarette

A visible thickening or ridge of the iris at the junction of the pupillary and ciliary zones, representing the site where the embryonic pupillary membrane was attached. It is typically a normal anatomical variant but can be a clinical sign in conditions such as Axenfeld-Rieger syndrome or iridocorneal endothelial syndrome.

Promonocyte Count

The number or proportion of promonocytes, immature monocyte precursors, in blood or marrow; increased cells may support evaluation of acute or chronic myeloid leukemia.

Promontory

A projecting anatomical prominence, such as the sacral or cochlear promontory.

Promoter

DNA sequence upstream of gene where RNA polymerase and transcription factors bind to initiate transcription. Core promoter: TATA box, tran-

scription start site. Regulatory elements: enhancers, silencers.

Promoters

DNA regions that initiate transcription of a gene when bound by regulatory proteins.

Promyelocytes

Promyelocytes are immature myeloid blood-cell precursors that develop from myeloblasts and mature into myelocytes; abnormal accumulation occurs in some leukemias.

Pronation

Rotation of the forearm so the palm faces posteriorly or downward, or a related foot movement.

Pronator

A muscle that turns a limb or part of it into a pronated position.

Prone

Prone means lying face downward with the front of the body against a surface; positioning affects breathing, pressure injury, and procedures.

Prone Position

A position in which the patient lies face down, commonly providing access to the back, spine, buttocks, or posterior legs.

Prone Positioning

Turning supine patients to prone for > 12 hours daily in severe ARDS. Improves oxygenation by recruiting dorsal lung segments. Reduces mortality when $\text{PaO}_2/\text{FiO}_2 < 150$.

Pronephric

Pronephric refers to the earliest embryonic kidney system, which is temporary and helps organize development of later renal structures.

Pronephros

The earliest embryonic kidney structure, which is nonfunctional in humans and largely regresses.

Prontosil

An early sulfonamide antibacterial drug, historically important in antimicrobial therapy.

Propafenone

A class IC antiarrhythmic with sodium channel blockade and weak beta-blocking activity. Used for paroxysmal atrial fibrillation (in patients without structural heart disease, including 'pill-in-the-pocket'), supraventricular tachycardias, and ventricular arrhythmias.

Propane

Propane is a flammable hydrocarbon gas used as fuel; inhalation or displacement of oxygen can cause suffocation, and liquid contact can cause cold burns.

Propane Poisoning

Propane poisoning involves toxic or environmental exposure to propane poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Propanol

Propanol is an alcohol solvent with two structural isomers; ingestion or substantial exposure can depress the nervous system and irritate tissues.

Propanolol

A nonselective beta-adrenergic blocker used for hypertension, arrhythmias, angina, tremor, migraine prevention, and other indications.

Propantheline

An antimuscarinic medicine formerly used for gastrointestinal spasm or excessive sweating.

Propellant

A propellant is a gas or volatile substance that expels a product from a container or delivers medicine in an inhaler.

Properdin

Properdin is a positive regulator of the alternative complement pathway that stabilizes complement

convertase and supports innate immune defense.

Properdin Deficiency

X-linked recessive (CFP gene). Properdin stabilizes alternative pathway C3 convertase. Recurrent Neisseria infections (meningococcemia, gonococcemia). Low AH50, normal CH50, low properdin. Treatment: meningococcal vaccination (ACWY + B), prophylactic penicillin, IVIG for severe infections.

Propetamphos

Propetamphos is an organophosphate insecticide that can inhibit acetylcholinesterase and cause cholinergic toxicity after exposure.

Prophage

A prophage is bacteriophage genetic material integrated into a bacterial chromosome or maintained in a dormant state inside a bacterium.

Prophase

Prophase is the early stage of cell division when chromosomes condense, the spindle begins forming, and the nucleolus disappears.

Prophenofos

Prophenofos is an organophosphate pesticide whose toxicity results mainly from acetylcholinesterase inhibition and excess cholinergic activity.

Prophylactic

Prophylactic means intended to prevent disease, infection, injury, or another unwanted outcome before it occurs.

Prophylactic Mastectomy

Prophylactic mastectomy removes one or both breasts to greatly reduce cancer risk in selected people at very high genetic or clinical risk.

Prophylaxis

Treatment or preventive measures intended to reduce the risk of disease or infection.

Propiolactone

Propiolactone is a reactive chemical formerly used as a disinfectant or sterilant; it is corrosive and potentially carcinogenic.

Propionates

Salts or esters of propionic acid used in food preservation, metabolism, and some medicines; clinical effects depend on the specific compound and exposure.

Propionibacteriaceae

A family of gram-positive, often anaerobic bacteria that includes Cutibacterium and related genera found on skin and in other environments.

Propionibacterium

A former bacterial genus name now largely replaced by Cutibacterium; its members are common skin organisms and can cause acne or device-related infection.

Propionibacterium Acnes

The former name for Cutibacterium acnes, a skin bacterium associated with acne and occasional implant or bloodstream infection.

Propionic Acidemia

An autosomal recessive inborn error of branched-chain amino acid and odd-chain fatty acid metabolism caused by deficiency of the biotin-dependent mitochondrial enzyme propionyl-CoA carboxylase, composed of alpha and beta subunits encoded by *PCCA* and *PCCB*. Prevents conversion of propionyl-CoA to methylmalonyl-CoA, causing accumulation of propionic acid, 3-hydroxypropionate, and methylcitrate. Manifests in the neonatal period with life-threatening ketoacidosis, lethargy, vomiting, hyperammonemia, pancytopenia, and metabolic stroke of the basal ganglia, predisposing to long-term intellectual disability and dilated cardiomyopathy.

Alternate Names:

Propionic Aciduria (PA); Propionyl-CoA Carboxylase Deficiency

Propofol

A short-acting intravenous anesthetic used for induction and maintenance of anesthesia and procedural sedation.

Propofol Infusion Syndrome

Rare but fatal complication of prolonged high-dose propofol infusion. Features: metabolic acidosis, rhabdomyolysis, cardiac failure, hyperlipidemia. Risk: critical illness, children, high doses ($>4\text{--}5\text{ mg/kg/hr}$), prolonged infusion (>48 hours). Prevention: limit dose and duration.

Propoxur

Propoxur is a carbamate insecticide that inhibits acetylcholinesterase and may cause sweating, salivation, vomiting, breathing difficulty, or weakness after exposure.

Propoxycaine

A local anesthetic formerly used in some topical or dental preparations; toxicity can affect the nervous system and heart.

Propoxyphene

Propoxyphene is a former opioid analgesic withdrawn in many countries because of overdose toxicity, respiratory depression, and dangerous cardiac arrhythmias.

Propranolol

A nonselective beta blocker used for cardiovascular conditions, tremor, migraine prevention, and other indications.

Proprioception

Awareness of body position and movement from receptors in muscles, tendons, joints, and skin.

Proprioceptor

A sensory receptor that detects position, movement, or muscle stretch.

Proptosis (Exophthalmos)

Abnormal anterior protrusion, forward displace-

ment, or bulging of one or both eyeballs (globes) from the bony orbital cavity. Exophthalmos is traditionally used when proptosis is caused by endocrine ophthalmopathy, specifically Graves disease (thyroid eye disease), in which autoantibodies directed against the TSH receptor stimulate orbital fibroblasts and adipocytes to produce glycosaminoglycans, causing severe retro-orbital edema, extraocular muscle enlargement, and bilateral proptosis. Unilateral proptosis mandates urgent imaging (CT/MRI of the orbit) to rule out orbital cellulitis, retrobulbar hematoma, cavernous sinus thrombosis, orbital pseudotumor, or primary orbital neoplasms. Quantified clinically using a Hertel exophthalmometer.

Propulsion

The act of driving or moving something forward; in Parkinson disease, propulsion may be impaired.

Propyl Alcohol

Propyl alcohols are flammable solvents; ingestion or substantial exposure can irritate tissues and depress the central nervous system, with toxicity depending on the isomer and dose.

Propylene Glycol

Propylene glycol is a solvent and humectant used in medicines, foods, and cosmetics; large exposures can cause metabolic or kidney problems.

Propylthiouracil

An antithyroid medicine that inhibits thyroid hormone synthesis and peripheral conversion of T4 to T3.

Propylthiouracil

Propylthiouracil is an antithyroid drug that reduces thyroid-hormone synthesis and conversion of T4 to T3; liver injury and fetal risks require careful use.

Prosencephalon

The prosencephalon is the embryonic forebrain, which develops into the telencephalon and diencephalon and their associated brain structures.

Prosopagnosia

Inability to recognize familiar faces despite otherwise adequate vision.

Prosoplasia

Development of a new or more specialized cellular function or phenotype within a tissue. The term describes progressive differentiation toward an advanced functional state rather than loss of differentiation or disordered maturation. It may be used in developmental biology and pathology to characterize cellular specialization in response to genetic, environmental, or tissue-specific signals. Interpretation depends on the tissue, developmental stage, molecular pathways, and whether the change is physiologic, adaptive, or associated with disease.

Prosoposchisis

A rare congenital facial cleft or fissure caused by abnormal fusion of facial processes during embryonic development, potentially affecting the nose, mouth, eyes, and feeding or breathing.

Prostacyclin Analogs

Medications that mimic prostacyclin (PGI₂), causing vasodilation and inhibiting platelet aggregation in the pulmonary vasculature. They include epoprostenol (continuous IV), treprostinil (subcutaneous, IV, inhaled, oral), and iloprost (inhaled). They are used for severe PAH.

Prostaglandin

A locally acting lipid mediator derived from arachidonic acid that influences pain, inflammation, fever, vascular tone, platelet function, gastric protection, kidney function, and uterine contraction.

Prostaglandin Analogs

A class of topical medications used in the treatment of open-angle glaucoma and ocular hypertension that lower intraocular pressure by increasing the uveoscleral outflow of aqueous humor, reducing intraocular pressure by approximately twenty-five to thirty-five percent.

Prostaglandin Degradation

Enzymatic inactivation and removal of prostaglandins, which limits the duration and intensity of their local effects on vessels, inflammation, smooth muscle, kidneys, stomach, and reproductive tissues.

Prostaglandin E₂

A prostaglandin that mediates inflammation, pain, fever, vasodilation, gastric protection, cervical ripening, and uterine contraction and is used in selected obstetric and cardiovascular treatments.

Prostaglandin F₂-Alpha

A prostaglandin that promotes smooth-muscle contraction and luteolysis; analogues are used in glaucoma, postpartum hemorrhage, labor-related care, and veterinary medicine.

Prostaglandin Receptor

A membrane receptor that binds a specific prostaglandin and activates intracellular signaling controlling vascular tone, inflammation, pain, smooth muscle, platelet activity, or reproduction.

Prostaglandins

Locally acting lipid mediators involved in pain, inflammation, vascular tone, gastric protection, and reproductive physiology.

Prostaglandins E

The E series of prostaglandins, lipid mediators that influence inflammation, vascular tone, gastric protection, kidney blood flow, cervical ripening, and uterine contraction.

Prostaglandins F

The F series of prostaglandins, lipid mediators involved in smooth-muscle contraction, luteal function, vascular responses, inflammation, and labor-related uterine activity.

Prostate

A gland below the bladder surrounding the proximal urethra that contributes fluid to semen.

Prostate Adenocarcinoma

The most common non-cutaneous malignancy in men and the second leading cause of cancer death in men after lung cancer. The disease typically arises in the peripheral zone of the prostate and is hormone-dependent, requiring androgens for growth.

Prostate Biopsy

Sampling of prostate tissue, commonly by transperineal or transrectal needle technique, for histologic diagnosis of prostate cancer. Targeted and systematic cores may be combined, with attention to infection prevention, bleeding, urinary retention, and sepsis risk.

Prostate Brachytherapy

A medical procedure, treatment, device, or care topic related to prostate brachytherapy; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Prostate Cancer

Cancer that forms in the prostate gland. It may grow slowly or behave aggressively and may cause no symptoms early on. Risk increases with age, family history, inherited gene changes, and certain racial or ethnic backgrounds. Prostate-specific antigen testing and sometimes a digital rectal examination may help evaluate risk, but diagnosis requires a prostate biopsy. Tumor grade and stage help guide treatment.

Alternate Names:

Cancer, Prostate Prostatic Carcinoma (Prostate Cancer)

Prostate Cancer Screening

Prostate cancer screening may use a PSA blood test with or without a digital rectal examination after discussing possible benefits, false positives, and harms.

Prostate Cancer Staging

Assessment of how far prostate cancer has spread using tumor features, PSA, grade, imaging, and whether disease is confined to the prostate or metastatic.

Prostate Cancer Treatment

A medical procedure, treatment, device, or care topic related to prostate cancer treatment; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Prostate Carcinoma

Prostate carcinoma is cancer arising from the prostate gland and may grow slowly or behave aggressively.

Prostate Disease

Any disorder affecting the prostate, including benign enlargement, infection, inflammation, or cancer.

Prostate Enlargement

An increase in prostate size, most often benign prostatic hyperplasia, which can obstruct urine flow and cause lower urinary-tract symptoms.

Prostate Gland

The prostate, a gland below the bladder that surrounds the urethra and contributes secretions to semen.

Prostate Gland Enlargement

Alternate index label; see *Benign prostatic hyperplasia*.

Prostate Infection

Infection or inflammation of the prostate, usually causing pelvic or urinary symptoms and sometimes fever or bloodstream infection.

Prostate Inflammation (Prostatitis (Inflammation Of The Prostate Gland))

Alternate index label for Prostatitis (Inflammation of the Prostate Gland).

Prostate Neoplasm

A prostate neoplasm is an abnormal growth in the prostate that may be benign, such as benign enlargement, or malignant.

Prostate Resection - Minimally Invasive

Minimally invasive prostate resection removes obstructing prostate tissue through the urethra or small access points to improve urine flow.

Prostate-Specific Antigen

A serine protease produced mainly by prostate epithelial cells and secreted into seminal fluid. PSA can be measured in blood, where levels may rise with benign enlargement, inflammation, infection, or prostate cancer. Interpretation depends on age, prostate size, medicines, symptoms, and changes over time.

Prostate-Specific Antigen

A kallikrein-related serine protease (human kallikrein-3 [hK3]) produced primarily by prostate glandular epithelial cells, functioning physiologically to liquefy seminal coagulum. Measured in serum as a biomarker for prostate cancer screening, risk stratification, and post-treatment recurrence monitoring. Elevated serum PSA (> 4.0 ng/mL) is organ-specific but not cancer-specific, also rising in benign prostatic hyperplasia, acute prostatitis, urinary retention, and recent instrumentation or ejaculation.

Prostate-Specific Antigen Blood Test

A PSA blood test measures prostate-specific antigen; an elevated result may reflect enlargement, inflammation, infection, or cancer and is not diagnostic by itself.

Prostatectomy

Surgical removal of part or all of the prostate.

Prostatic Carcinoma (Prostate Cancer)

Alternate index label for Prostate Cancer.

Prostatic Fluid

Prostatic fluid is a mildly alkaline secretion that forms part of semen and contains substances supporting sperm function and survival.

Prostatic Obstruction

Blockage of urine flow caused by enlargement, inflammation, scarring, stones, or a tumor of the prostate or nearby structures.

Prostatic Urethral Stent

A small device placed in the prostatic urethra to help maintain urine flow when the prostate obstructs it.

Prostatism

Lower urinary-tract symptoms caused by bladder-outlet obstruction or dysfunction, often associated with prostate enlargement.

Prostatitis

Inflammation or infection of the prostate, causing pelvic pain, urinary symptoms, or systemic illness.

Prostatitis (Inflammation Of The Prostate Gland)

Alternate index label for Inflammation of the Prostate Gland.

Prostatitis - Bacterial

Bacterial prostatitis is infection and inflammation of the prostate, causing pelvic or urinary pain, fever, urinary symptoms, or recurrent infection.

Prostatitis - Nonbacterial

Nonbacterial prostatitis refers to persistent pelvic or urinary symptoms without proven bacterial infection and is often managed with several approaches.

Prostatodynia

Prostatodynia is an older term for pelvic or prostate-region pain without proven bacterial infection, now often considered within chronic pelvic pain syndrome.

Prosthesis

An artificial device replacing or supporting a missing body part or function.

Prosthetic

Pertaining to, designed for, or functioning as a prosthesis—an artificial, biocompatible device fabricated to substitute for, augment, or replace a missing, amputated, or defective body part, joint, or organ. Encompasses extremity prostheses (e.g., transtibial, transfemoral, and myoelectric bionic limbs), orthopedic joint arthroplasties (total hip and total knee replacements), cardiovascular prostheses (mechanical and bioprosthetic prosthetic heart valves, endovascular stent grafts), and dental/maxillofacial prostheses. A major clinical challenge is managing prosthetic complications, including wear, aseptic mechanical loosening, and prosthetic joint or valve infections (frequently driven by biofilm-forming staphylococci).

Prosthetic Graft

A prosthetic graft is an artificial tube or patch used to replace or bypass a damaged blood vessel or other tubular structure; infection and thrombosis are important risks.

Prosthetic Heart Valve

An artificial valve replacing or supplementing a diseased cardiac valve; mechanical valves require anticoagulation, while tissue valves have different durability and anticoagulation considerations.

Prosthetic Limb

Artificial replacement for missing limb. Types: above/below knee, above/below elbow. Components: socket, suspension, joint, pylon, terminal device. Materials: carbon fiber, silicone, metal. Rehabilitation: gait training, strength, ADL.

Prosthetic Training

Training for amputees: stump care, donning/doffing, gait training, balance, stair negotiation, community ambulation.

Prosthetics

Artificial replacement of missing body part. Elderly use: lower limb (transtibial, transfemoral), upper limb. Considerations: mobility level, cognition, vision, comorbidities, support system. Rehabilitation:

physical therapy, strength training, balance.

Prosthodontics

Dental specialty concerned with replacement of missing teeth and restoration of oral function. Types: fixed (crowns, bridges, implants), removable (partial dentures, complete dentures), maxillofacial prosthetics (ocular, nasal, auricular). Goals: aesthetics, function, phonetics, preservation of remaining structures.

Prostrate

Lying flat or face downward; not to be confused with prostate.

Protamine

A strongly basic protein used to reverse the anticoagulant effect of heparin.

Protanopia

Protanopia is complete absence of red-cone function, causing a form of red-green color-vision deficiency and altered perception of red and related colors.

Protean

Highly variable in form, appearance, or behavior; in medicine it may describe a disease with changing or diverse manifestations.

Proteaphagy

Selective autophagic degradation of proteasomes, helping cells regulate protein-degradation capacity during stress or nutrient changes.

Protease

An enzyme that breaks down proteins by cleaving peptide bonds.

Protease Inhibitors

A class of antiviral drugs that block viral proteases, enzymes essential for processing viral polyproteins into functional components needed for viral assembly and maturation.

Proteasome

A large protein complex that breaks down proteins tagged with ubiquitin, helping regulate cell-cycle proteins, signaling, and damaged proteins.

Proteasome Binding

Attachment of a molecule to a proteasome or its regulatory particle, which can alter protein degradation or proteasome activity.

Protecting Yourself From Cancer Scams

Protection from cancer scams includes checking claims with qualified clinicians, avoiding guaranteed cures, reviewing evidence, and being cautious about costly unproven treatments.

Protective

Serving to prevent or reduce injury, infection, or other harm.

Protective Agent

A clinical or therapeutic concept involving protective agent. Its features, uses, and clinical significance depend on the relevant medical context.

Protective Factor

Attribute decreasing probability of disease. Examples: exercise (cardiovascular disease), vaccination (infectious disease).

Protegrin

A small antimicrobial peptide, usually derived from pig immune cells, that can disrupt microbial membranes.

Protein

A molecule made of amino acids that performs structural, enzymatic, transport, signaling, or immune functions.

Protein Acetylation

Addition of an acetyl group to a protein, often changing its activity, localization, stability, or interactions; histone acetylation is a major regulator of gene expression.

Protein Binding

The reversible association of drug molecules with plasma proteins, primarily albumin for acidic drugs and alpha-1 acid glycoprotein for basic drugs. Only unbound or free drug is pharmacologically active and available for distribution to tissues, metabolism by the liver, and excretion by the kidneys.

Protein Biosynthesis

Production of proteins by transcription of DNA into RNA followed by ribosomal translation of messenger RNA into an amino-acid chain.

Protein C

Natural anticoagulant. Inactivates factors Va and VIIIa. Warfarin can cause skin necrosis in deficiency.

Protein C Blood Test

A protein C blood test measures a natural anticoagulant and may help evaluate unexplained or recurrent blood clots or suspected inherited deficiency.

Protein C Deficiency

An inherited (autosomal dominant) thrombophilia causing increased VTE risk. Protein C is a vitamin K-dependent natural anticoagulant that inactivates factors Va and VIIIa and stimulates fibrinolysis. Heterozygous: 6-10 fold increased VTE risk.

Protein Deficiency

Inadequate protein intake, digestion, absorption, or use, causing loss of muscle, impaired immunity and wound healing, edema, or growth failure when severe or prolonged.

Protein Denaturation

Loss of a protein's normal three-dimensional structure from heat, chemicals, pH, or mechanical stress, often impairing its biological function without breaking its peptide backbone.

Protein Dephosphorylation

Removal of a phosphate group from a protein by a phosphatase, often reversing or modifying signaling, enzyme activity, localization, or cell-cycle control.

Protein Electrophoresis

A laboratory test that separates proteins in serum or urine according to their electrical properties, helping detect abnormal protein patterns.

Protein Electrophoresis - Serum

Serum protein electrophoresis separates blood proteins into patterns and helps evaluate abnormal antibodies, myeloma, inflammation, and some liver or kidney disorders.

Protein Engineering

Design or modification of proteins through genetic, biochemical, or computational methods to alter activity, stability, specificity, expression, or therapeutic usefulness.

Protein Expression

The process by which genetic information produces a protein, including transcription, RNA processing, translation, folding, modification, and cellular localization.

Protein Folding

The process by which a newly made protein changes from a chain of amino acids into its functional three-dimensional shape. The amino acid sequence provides the main instructions for folding, while the cellular environment and molecular chaperones may help the protein fold correctly. Incorrect folding can cause protein aggregation and contribute to disease.

Protein Glycosylation

Enzymatic attachment of carbohydrate chains to proteins, affecting folding, stability, trafficking, recognition, and function; inherited glycosylation defects can cause multisystem disease.

Protein Import

Transport of a protein into an organelle or across a cellular membrane using targeting signals, receptors, translocons, and energy-dependent folding or assembly mechanisms.

Protein In Diet

Dietary protein supplies amino acids needed for muscle, enzymes, hormones, immunity, and tissue repair; requirements vary with age, health, and kidney function.

Protein In Urine (Proteinuria)

An abnormal amount of protein in the urine, which may be temporary or indicate kidney disease, diabetes, high blood pressure, infection, or another condition.

Protein Isoform

One of several related protein forms produced from different genes, alternative splicing, or post-translational processing, with potentially different locations or functions.

Protein Kinase

An enzyme that transfers phosphate from ATP to a protein, changing its activity or interactions and regulating signaling, metabolism, growth, differentiation, and cell division.

Protein Malnutrition

Protein malnutrition refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Protein Metabolism

The synthesis, folding, modification, breakdown, and recycling of proteins and amino acids in cells and tissues.

Protein Methylation

Addition of methyl groups to selected amino-acid residues, regulating protein activity, chromatin, signaling, and gene expression.

Protein Overexpression

Production of an abnormally large amount of a protein, caused by gene amplification, transcriptional activation, altered degradation, or other mechanisms and potentially driving disease or serving as a biomarker.

Protein Phosphorylation

Addition of phosphate to a protein by a kinase, a reversible switch that regulates enzyme activity, signaling pathways, gene expression, transport, and cell division.

Protein Prenylation

Attachment of a hydrophobic isoprenoid group to a protein, helping target it to membranes and enabling signaling or trafficking functions.

Protein Purification

Separation of a desired protein from cells or a mixture using properties such as charge, size, solubility, affinity, or binding, while preserving its structure and activity.

Protein Refolding

Return of a denatured or newly synthesized protein to its functional three-dimensional structure, sometimes requiring chaperones or controlled laboratory conditions.

Protein S

Vitamin K-dependent plasma glycoprotein that acts as a cofactor for activated protein C in the inactivation of factors Va and VIIIa. Deficiency or dysfunction increases the risk of venous thromboembolism.

Protein S Blood Test

A protein S blood test measures a natural anticoagulant and may help evaluate suspected inherited or acquired tendency to form blood clots.

Protein S Deficiency

An inherited (autosomal dominant) thrombophilia. Protein S is a vitamin K-dependent natural anticoagulant that acts as a cofactor for activated protein C in inactivating factors Va and VIIIa. Heterozygous: 2-10 fold increased VTE risk. Homozygous: rare, purpura fulminans.

Protein Secretion

Transport of a protein out of a cell or into a specialized compartment, commonly through

the endoplasmic-reticulum and Golgi pathway or through regulated vesicle release.

Protein Sequence

The ordered chain of amino acids in a protein, which encodes information about folding, domains, binding sites, processing, and biological function.

Protein Splicing

Removal of an internal protein segment called an intein with ligation of the surrounding exteins, producing a mature functional protein.

Protein Subunit

One polypeptide component of a multi-subunit protein complex, whose interactions with other subunits create the complex's final structure and function.

Protein Synthesis

Translation of mRNA code into polypeptide chain. Initiation: ribosome assembly at start codon (AUG). Elongation: aminoacyl-tRNAs bring amino acids; peptide bonds form in peptidyl transferase center; ribosome translocates. Termination: stop codon (UAA, UAG, UGA) recognized by release factors.

Protein Translocation

Movement of a newly synthesized or mature protein across a membrane or to a different cellular compartment, directed by targeting signals and transport machinery.

Protein Transport

Cellular movement of proteins between compartments, across membranes, or outside the cell, ensuring that each protein reaches its functional location.

Protein Unfolding

Loss of a protein's native three-dimensional structure, caused by heat, chemicals, mutations, or cellular stress, which can impair function or promote aggregation.

Protein-Energy Malnutrition

Severe deficiency of protein and calories. Types:

marasmus (chronic, wasting), kwashiorkor (acute, edema, fatty liver), marasmic-kwashiorkor. Risk: famine, poverty, alcoholism, malignancy, malabsorption. Treatment: gradual refeeding, monitor refeeding syndrome.

Protein-Losing Enteropathy

Excessive loss of plasma proteins through the gastrointestinal tract, causing low albumin, swelling, diarrhea, or nutritional deficiency.

Proteinaceous Plug

A blockage made primarily of protein-rich material, mucus, cellular debris, or secretions that can obstruct an airway, duct, catheter, or other passage.

Proteinuria

Abnormally high protein excretion in urine, often indicating kidney disease but also occurring transiently.

Proteobacteria

A large bacterial group containing many gram-negative organisms, including *Escherichia*, *Salmonella*, *Pseudomonas*, and *Vibrio*.

Proteogenomics

Integration of genomic, transcriptomic, and proteomic data to identify proteins, discover coding variants, characterize tumors or microbes, and connect DNA changes with protein function.

Proteoglycan

A molecule made of a core protein with attached glycosaminoglycan chains, providing structure and signaling functions in connective tissue and cartilage.

Proteolipid Protein

Proteolipid protein refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Proteolipids

Proteins tightly associated with lipid molecules or

lipid membranes, important in myelin, mitochondrial membranes, ion transport, and other membrane structures.

Proteolysis

Breakdown of proteins into smaller peptides or amino acids.

Proteolytic

Pertaining to, promoting, or characterized by proteolysis—the hydrolytic cleavage of peptide bonds linking amino acids in peptides and proteins. Mediated by specialized enzymes known as proteases, peptidases, or proteinases. Essential for digestive enzyme activation (e.g., gastric pepsin, pancreatic trypsin, chymotrypsin), blood coagulation cascades (thrombin cleavage of fibrinogen to fibrin), the complement cascade, apoptotic cell death (caspases), and intracellular protein recycling via the ubiquitin-proteasome system. Pharmacologically targeted by protease inhibitors in HIV therapy (e.g., darunavir), hepatitis C therapy, and multiple myeloma treatment (e.g., bortezomib).

Proteome

The complete set of proteins produced by a cell, tissue, organism, or biological condition at a particular time.

Proteomic Profile

A proteomic profile is a pattern of proteins measured in a biological sample to study disease, treatment response, or biological processes.

Proteomics

The large-scale study of proteins, including their abundance, structure, modifications, interactions, and functions.

Proteus

A genus of gram-negative bacteria that can cause urinary and wound infections.

Proteus Hauseri

A gram-negative bacterium in the *Proteus* group

that is rarely isolated from human clinical specimens.

Proteus Mirabilis

A gram-negative bacterium that commonly causes urinary-tract infection and can produce urease-related kidney or bladder stones.

Proteus Penneri

A gram-negative *Proteus* species that can rarely cause urinary, wound, bloodstream, or other opportunistic infections.

Proteus Syndrome

A rare mosaic overgrowth disorder, usually caused by AKT1 variants, producing asymmetric enlargement of bones, skin, or other tissues and an increased risk of tumors.

Proteus Vulgaris

A gram-negative bacterium that can cause urinary, wound, bloodstream, and other infections and often produces urease.

Prothionamide

A thioamide antimycobacterial drug used as part of selected multidrug regimens for drug-resistant tuberculosis; gastrointestinal, hepatic, neurologic, and thyroid adverse effects require monitoring.

Prothrombin

A vitamin-K-dependent plasma protein converted to thrombin during blood coagulation.

Prothrombin Deficiency

Prothrombin deficiency is a rare bleeding disorder caused by insufficient or abnormal clotting factor II, leading to prolonged bleeding of variable severity.

Prothrombin Time

A blood test assessing the extrinsic and common coagulation pathways, commonly reported as INR.

Prothrombin Time

A blood test measuring the time required for clot

formation through the extrinsic and common coagulation pathways; it is commonly reported as the INR.

Alternate Names:

PT

Protista

Protista is a historical grouping for diverse mostly single-celled eukaryotes, including organisms formerly called protozoa and some algae.

Proto-Oncogene

A normal gene that regulates cell growth, division, differentiation, or survival and can become an oncogene after activating mutation, amplification, translocation, or overexpression.

Protocol

A predefined plan specifying procedures for a clinical study, examination, or treatment.

Protocol Agent

A protocol agent is a substance or intervention specified by a study or treatment protocol to produce a defined experimental or clinical effect.

Proton Pump Inhibitors

Proton pump inhibitors suppress gastric acid secretion and treat reflux, ulcers, and other acid-related disorders; prolonged or inappropriate use should be periodically reviewed.

Proton Therapy

Radiation using protons instead of photons. Bragg peak allows precise dose delivery. Used for pediatric tumors, skull base tumors.

Protozoa

Single-celled eukaryotic organisms, some of which cause human disease.

Protractile

Capable of being extended, protruded, or pushed forward from an anatomical resting position. In human anatomy, refers to structures capable of active

protrusion, such as the human tongue (protracted by bilateral contraction of the genioglossus muscle, innervated by the hypoglossal nerve [CN XII], whose integrity is clinically assessed by asking the patient to stick out their tongue) and the mandible (protracted primarily by the lateral pterygoid muscles).

Protractor

An instrument or muscle that moves a structure forward or outward.

Protrusion

A part that projects outward or a disc or tissue displaced beyond its normal boundary.

Protuberance

A protuberance is a structure or area that projects outward from a surface, such as a normal bone prominence, swelling, or mass.

Provisional Diagnosis

A working diagnosis made before all information is available and subject to revision.

Provisional Tic Disorder

A disorder with motor or vocal tics lasting less than one year, beginning in childhood and not better explained by another condition.

Provocation

An action or stimulus that triggers a symptom, response, or test finding.

Proximal

Closer to the point of attachment or to the center of the body than another structure.

Proximal Humerus Fractures

Proximal humerus fractures: Neer classification (4 parts: articular surface, greater tuberosity, lesser tuberosity, shaft; displaced >1 cm or angulated >45 degrees = part). 1-part (nondisplaced): sling, early motion. 2-part: surgical neck (ORIF plate, IM nail) or greater tuberosity (if displaced >5mm or rotator cuff involvement).

Proximal Neuropathy Diabetic (Diabetic Neuropathy)

Alternate index label for Diabetic Neuropathy.

Proximal Radioulnar Joint

Pivot synovial joint between radial head circumference and ulnar radial notch, held by the annular ligament. Allows pronation/supination with approximately 150 degrees range. Parallel bones in supination; radius crosses ulna in pronation. Significant in radial head fractures, Monteggia fracture-dislocations, and nursemaid's elbow.

Proximal Renal Tubular Acidosis

Proximal renal tubular acidosis occurs when the kidney cannot reclaim enough bicarbonate, causing excess acid in the blood and often low potassium.

Proximal Tubule

Segment of the nephron extending from Bowman capsule to the loop of Henle with a convoluted proximal straight tubule and convoluted proximal tubule. Responsible for reabsorbing approximately sixty-five percent of filtered sodium, water, bicarbonate, glucose, amino acids, and phosphate through an extensive brush border.

Proximal Tubule Reabsorption

The proximal convoluted tubule reabsorbs approximately 65 percent of filtered sodium, water, glucose, amino acids, bicarbonate, and phosphate.

Proximate Cause

The legally sufficient cause that directly contributes to an injury or outcome.

Prune Belly Syndrome

A congenital syndrome comprising the triad: deficient abdominal wall musculature (wrinkled, prune-like appearance), bilateral cryptorchidism (undescended testes), and urinary tract anomalies (dilated ureters, dilated and trabeculated bladder, and posterior urethral valves or urethral atresia). Almost exclusively affects boys.

Prurigo

A group of intensely itchy skin disorders with papules or nodules caused by inflammation and scratching.

Pruritic

Itchy or causing an urge to scratch; pruritus can result from skin inflammation, allergy, infection, systemic disease, medications, or neuropathic mechanisms.

Pruritic Eruption

An itchy skin eruption that may result from allergic, inflammatory, infectious, drug-related, or autoimmune disease; diagnosis depends on its morphology, distribution, timing, and associated findings.

Pruritis Ani

Alternate index label; see *Anal itching*.

Pruritus

Itching, an unpleasant skin sensation that creates the desire to scratch.

Pruritus (Itch)

Alternate index label for Itch.

Pruritus Ani

Persistent itching around the anus with multiple potential causes including hemorrhoids, fissures, infections such as pinworms and candidiasis, dermatologic conditions including psoriasis and lichen sclerosus, and idiopathic causes.

Pruritus In CKD

Severe itching in dialysis patients from uremic toxin accumulation, elevated calcium-phosphorus product, and secondary hyperparathyroidism. Difficult to treat. Include phosphate binders, dialysis optimization, gabapentin, and UV phototherapy.

Prussian Blue

A blue pigment used in laboratory staining and as an antidote for selected radioactive cesium or thallium poisoning.

PSA

Prostate-specific antigen, a protein measured in blood to help evaluate prostate conditions.

Alternate Names:

Prostate-specific antigen

PSA Velocity

The rate at which PSA concentration changes over time.

Psalterium

The body of the fornix in the brain, also called the hippocampal commissure; historically, a book of psalms.

Psammoma

A tumor or lesion containing concentrically laminated calcified bodies called psammoma bodies.

Psychiatry

A misspelling of psychiatry, the medical specialty concerned with mental disorders.

Pseudarthrosis

Failure of a fracture to unite, producing a false joint or persistent abnormal mobility.

Pseudoaneurysm

A contained arterial wall disruption in which blood communicates with the arterial lumen but is held by surrounding tissue rather than all layers of the arterial wall. It may follow catheterization, trauma, surgery, infection, or grafting and can rupture or embolize.

Pseudoangiomatous Stromal Hyperplasia

A benign breast stromal proliferation forming slit-like spaces within dense collagen that may mimic vascular channels. It can be incidental or present as a palpable or imaging-detected mass.

Pseudobulbar Affect

Involuntary episodes of laughing or crying disproportionate to mood, usually due to neurologic disease.

Alternate Names:

Pseudoclubbing

Broadening and flattening of the nail tips without the complete loss of the angle between the nail and nail fold that characterizes true clubbing. It is seen in conditions such as hyperparathyroidism, rheumatoid arthritis, and psoriasis.

Pseudocyesis

A false belief or physical state resembling pregnancy without a fetus.

Pseudodementia

Cognitive symptoms caused by depression or another potentially reversible condition rather than primary neurodegeneration.

Pseudoephedrine

A sympathomimetic (alpha-1 agonist, some beta effects) oral decongestant used for nasal congestion in colds and sinusitis. Increases nasal mucosal vasoconstriction.

Pseudogout

Calcium pyrophosphate deposition disease: deposition of calcium pyrophosphate dihydrate crystals in joints, causing acute arthritis resembling gout (pseudogout) or chronic osteoarthritis-like disease.

Alternate Names:

Arthritis Pseudogout (Pseudogout)

Pseudogout Flare

Acute crystal-induced arthritis caused by calcium pyrophosphate dihydrate deposition, often affecting the knee or wrist. Identification of positively birefringent rhomboid crystals in synovial fluid supports the diagnosis.

Pseudohypertrophic Muscular Dystrophy

An older name for Duchenne or related muscular dystrophy with enlarged-appearing muscles caused by fat and connective tissue.

Pseudohypoparathyroidism

A group of disorders in which target tissues resist parathyroid hormone, causing low calcium, high phosphate, and elevated PTH, sometimes with characteristic skeletal features or hormone resistance.

Pseudomelanosis Duodeni

Benign condition characterized by dark pigment deposits in the duodenal mucosa, visible as slate-gray or black patches on endoscopy, caused by accumulation of iron-containing pigment in macrophages within the lamina propria.

Pseudomembranous Colitis

Colonic inflammation caused by *C. difficile* infection showing yellow-white plaques on colonoscopy. The plaques consist of fibrin, inflammatory cells, and necrotic epithelial debris overlying mucosal ulceration. Presents with profuse watery diarrhea and abdominal pain.

Alternate Names:

C. Difficile Colitis Colitis, Pseudomembranous

Pseudomonas

A genus of gram-negative bacteria, including *Pseudomonas aeruginosa*, that can cause opportunistic infections.

Pseudomonas Aeruginosa

A gram-negative bacterium that can cause pneumonia, bloodstream, urinary, wound, and ear infections, especially in hospitalized or immunocompromised people; it often has antibiotic resistance.

Pseudomonas Aeruginosa Infection

Infection caused by *Pseudomonas aeruginosa*, an opportunistic gram-negative bacterium that can affect the lungs, wounds, urinary tract, blood, or other sites.

Pseudomonas Aeruginosa Resistance

Pseudomonas aeruginosa has intrinsic and acquired resistance mechanisms making it inherently resistant to many antibiotics. Intrinsic resistance includes low outer membrane permeability, efflux pumps, and

chromosomal beta-lactamase.

Pseudomyxoma Peritonei

Accumulation of mucinous ascites and mucin-producing tumor nodules in the peritoneal cavity, usually from a ruptured low-grade appendiceal or ovarian mucinous neoplasm; treated with cytoreductive surgery and hyperthermic intraperitoneal chemotherapy.

Pseudoneurosis

An obsolete term for symptoms resembling neurosis but caused by an underlying physical disorder.

Pseudophakia

The presence of an artificial intraocular lens after removal of the natural lens, usually during cataract surgery.

Pseudorabies

Aujeszky disease, a herpesvirus infection of animals; humans are rare accidental hosts.

Pseudothrombocytosis Due To Red Cell Fragments

Falsely elevated platelet count on automated hematology analyzers caused by fragmentation of red blood cells (schistocytes) being counted as platelets. Seen in microangiopathic hemolytic anemias such as TTP, HUS, DIC, and mechanical heart valves. Manual peripheral blood smear review confirms true platelet count.

Pseudotumor Cerebri

An older term for idiopathic intracranial hypertension, in which raised pressure around the brain causes headache, transient visual obscurations, pulsatile tinnitus, and possible papilledema without a mass lesion.

Pseudoxanthoma Elasticum

An inherited connective-tissue disorder with mineralization and fragmentation of elastic fibers, affecting skin, eyes, and blood vessels.

Alternate Names:

PXE (Pseudoxanthoma Elasticum)

Pseudoxanthoma Elasticum

Alternate index label for Pseudoxanthoma Elasticum.

Psittacosis

A zoonotic infection caused by *Chlamydia psittaci*, usually acquired from infected birds.

PSMA

Prostate-specific membrane antigen. Target for PET imaging and therapy in prostate cancer.

Psoas

A major hip-flexor muscle running from the lumbar spine to the femur.

Psoas Abscess

A collection of pus in or around the psoas muscle, caused by infection spreading through the blood or from nearby tissues. It may cause back or hip pain, fever, difficulty walking, or few symptoms and usually requires imaging and targeted treatment.

Psoriasis

A chronic immune-mediated skin disease producing sharply demarcated, often scaly plaques.

Psoriasis Of The Scalp (Scalp Psoriasis)

Alternate index label for Scalp Psoriasis.

Psoriatic Arthritis

An inflammatory arthritis associated with psoriasis of the skin or nails. It may affect the spine, peripheral joints, digits, or tendon insertions, and diagnosis uses the clinical pattern together with appropriate laboratory and imaging findings.

Alternate Names:

Arthritis Psoriatic (Psoriatic Arthritis)

PSP

PSP commonly means progressive supranuclear palsy, a neurodegenerative disorder causing early postural instability, vertical gaze impairment, rigid-

ity, and cognitive change.

Psychasthenia

An obsolete term for obsessive doubts, compulsive thoughts, or an inability to direct attention.

Psychedelic

Producing altered perception, mood, and cognition, as with certain hallucinogenic substances.

Psychiatric

Relating to the assessment, diagnosis, treatment, or prevention of disorders of mood, thought, perception, behavior, or emotional functioning.

Psychiatrist

Physician specializing in diagnosis and treatment of mental disorders; involved in pre-transplant psychological evaluation and post-transplant mental health management.

Psychiatry

The medical specialty concerned with mental health and the diagnosis and treatment of mental disorders.

Psychic

Relating to the mind or mental processes; claims of supernatural perception are not medical findings.

Psycho-Oncology

The clinical and research field addressing psychological, social, behavioral, and communication needs of people with cancer and their families across diagnosis, treatment, survivorship, and end of life.

Psychoacoustics

The study of how people perceive and interpret sound, including loudness, pitch, masking, localization, speech, and hearing thresholds.

Psychoactive

Affecting mood, thought, perception, behavior, or other mental processes.

Psychoactive Substance

A substance that alters mood, perception, cognition, behavior, or consciousness.

Psychoanalysis

A historical psychological theory and treatment approach emphasizing unconscious processes and conflicts.

Psychobiology

The study of interactions between biological processes and behavior or mental states, including brain circuits, hormones, genes, immune signals, and environmental effects.

Psychodermatology

Psychodermatology addresses the mind-skin connection. Stress exacerbates psoriasis, eczema, acne, and urticaria. Psychiatric disorders (depression, anxiety, body dysmorphic disorder) are associated with skin conditions.

Psychodidae

A family of small hairy flies that includes sand flies; some species transmit pathogens such as *Leishmania* and therefore have public-health importance.

Psychodrama

A therapeutic method using guided dramatic enactment to explore emotions, relationships, and behavior.

Psychodynamic Therapy

A form of depth psychology that explores how unconscious processes, early life experiences, and internal conflicts influence current thoughts, feelings, and behaviors. It involves exploration of transference and countertransference, defense mechanisms, and the therapeutic relationship as a vehicle for change.

Psychogenic

Caused or strongly influenced by psychological factors such as stress, trauma, or emotional conflict. The symptoms are real and not deliberately produced, and physical causes should still be consid-

ered.

Psychogenic Erectile Dysfunction

Erectile difficulty primarily related to psychological factors rather than a structural disease.

Psychogenic Non-Epileptic Seizures

Seizure-like episodes that are not caused by abnormal electrical activity in the brain. They are a form of conversion disorder with attacks or seizures. Events may include shaking, stiffening, unresponsiveness, and other movements.

Psycholinguistics

The study of how the brain and mind produce, understand, acquire, and remember language, including speech, reading, grammar, and meaning.

Psychological

Relating to mental processes, emotions, behavior, or their effects on health.

Psychological Adjustment

The process of adapting thoughts, emotions, and behavior to stress, illness, disability, loss, developmental change, or other life circumstances.

Psychological Dependence

A pattern in which a person feels a strong emotional or mental need for a substance or behavior, often with craving, preoccupation, or distress when it is unavailable.

Psychological Support

Counseling, therapy for patients with chronic illness, disability, pain. Address depression, anxiety, adjustment difficulties, coping strategies.

Psychological Transfer

The redirection of feelings, expectations, or relationship patterns from earlier experiences onto another person, especially the therapist in psychotherapy; interpretation requires clinical context.

Psychologist, Clinical

A licensed mental-health professional who assesses and treats psychological, emotional, behavioral, and cognitive problems using psychotherapy and other evidence-based interventions within local scope of practice.

Psychology

The scientific study of behavior, emotion, learning, and mental processes. Psychological care may include assessment, counseling, and behavioral therapies.

Alternate Names:

Mental Health (Psychology)

Psychometrics

The measurement of psychological characteristics using standardized tests and statistical methods.

Psychomotor

Relating to the interaction of mental processes and physical movement.

Psychomotor Agitation

Restless, excessive, or purposeless movement associated with inner tension, anxiety, mania, delirium, depression, substance effects, or medication reactions.

Psychomotor Disorder

A disorder affecting the speed, initiation, coordination, or expression of movement and mental activity.

Psychomotor Performance

The ability to coordinate mental processing with physical movement, assessed through tasks involving speed, precision, reaction time, balance, or sequencing.

Psychoneuroimmunology

The study of interactions among psychological processes, the nervous system, endocrine signals, and immune function in health and disease.

Psychoneurosis

An obsolete term for a nonpsychotic emotional disorder, historically including anxiety and phobic conditions.

Psychopathic

Relating to severe, persistent antisocial personality traits; the term is often used imprecisely.

Psychopathology

The study or description of abnormal mental states and behavior.

Psychopharmacology

The study of how drugs affect behavior, mood, cognition, and other mental processes.

Psychophysics

The quantitative study of relationships between physical stimuli and subjective sensory experience, such as thresholds, discrimination, and perceived intensity.

Psychophysiology

The study of links between psychological processes and measurable bodily functions such as heart rate, skin conductance, muscle activity, and brain signals.

Psychosexual

Relating to the interaction of psychological factors with sexual development, feelings, behavior, or functioning.

Psychosexual Development

Development of sexual identity, interest, relationships, and behavior across the lifespan as influenced by biology, cognition, attachment, culture, and experience.

Psychosis

A state involving impaired reality testing, such as delusions, hallucinations, or disorganized thought.

Psychosis ICU (ICU Psychosis)

Alternate index label for ICU Psychosis.

Psychosocial Rehabilitation

Support for mental illness recovery: skills training, supported employment, peer support.

Psychosocial Support

Cancer affects mental health. Depression, anxiety common. Support groups, counseling beneficial.

Psychosomatic

Relating to physical symptoms influenced by psychological processes or stress.

Psychosurgery

Surgery intended to alter brain circuits for severe, treatment-resistant psychiatric illness; it is rarely used and highly specialized.

Psychotherapy

Structured psychological treatment using therapeutic communication and evidence-based techniques.

Psychotic

Pertaining to, exhibiting, or characterized by psychosis—a profound psychiatric or neuropsychiatric syndrome defined by a gross loss of reality testing, accompanied by severe cognitive, behavioral, and perceptual impairments. Core clinical features include delusions (fixed, false idiosyncratic beliefs unshakeable by contradictory evidence), hallucinations (sensory perceptions in the absence of external stimuli, most commonly auditory in primary psychiatric illness), disorganized thinking and speech (loosening of associations, word salad), and grossly disorganized or catatonic behavior. Seen in primary psychotic disorders (schizophrenia, schizoaffective disorder), mood disorders with psychotic features (bipolar I manic episodes, severe major depression), substance-induced states, and secondary medical conditions (delirium, autoimmune encephalitis).

Psychotic Disorder

A mental disorder involving loss of contact with reality, such as delusions, hallucinations, disorganized thinking, or markedly disorganized behavior.

Psychotic Disorder Brief (Brief Psychotic Disorder)

Alternate index label for Brief Psychotic Disorder.

Psychotic Disorders

Psychotic disorders involve impaired reality testing, such as hallucinations, delusions, disorganized thought, or marked behavioral change. Causes include schizophrenia-spectrum illness, mood disorders, substances, medicines, neurologic disease, and medical illness; new psychosis requires prompt assessment and safety planning.

Psychrobacter

A genus of cold-tolerant gram-negative bacteria found in marine and other environments that rarely cause opportunistic infection.

Psychrobacter Immobilis

A cold-tolerant environmental bacterium rarely associated with human infection, including occasional bloodstream or wound infection.

PTCA

Percutaneous transluminal coronary angioplasty, a procedure that opens a narrowed coronary artery.

Alternate Names:

Percutaneous coronary intervention

PTCL

Alternate index label; see *Peripheral T-cell lymphoma*.

PTE

Alternate name; see *Pulmonary Thromboendarterectomy*.

Pteridine

A nitrogen-containing bicyclic chemical structure found in folates, pigments, and other biologically active compounds.

Pterygium

A wing-shaped growth of conjunctival tissue extending onto the cornea, usually related to ultraviolet

exposure.

PTH-Related Peptide

A peptide that activates the parathyroid-hormone receptor and can cause humoral hypercalcemia of malignancy. It increases bone resorption and renal calcium retention while reducing phosphate reabsorption.

Pthirus Pubis

The pubic louse, a parasitic insect that infests the pubic hair and occasionally other coarse body hair, causing pediculosis pubis. Transmission is primarily through sexual contact. It causes intense pruritus and may transmit *Borrelia recurrentis* causing relapsing fever. Diagnosis is by visual identification of the louse or nits.

Ptilosis

Abnormal growth, loss, or condition of hair or feathers; the exact meaning depends on context.

PTLD

Post-transplant lymphoproliferative disorder; a spectrum of lymphoid proliferations ranging from benign hyperplasia to lymphoma, occurring in immunosuppressed transplant recipients.

Ptosiss

Drooping of an eyelid or another body structure. Eyelid ptosis may result from muscle, nerve, tendon, aging, injury, or a neurologic disorder.

Ptosiss - Infants And Children

Ptosiss in an infant or child is drooping of the upper eyelid, which may obstruct vision or cause amblyopia and can reflect congenital, muscular, nerve, or other disease.

PTSD

Post-traumatic stress disorder, a condition involving persistent trauma-related memories, avoidance, negative mood or cognition, and heightened arousal after a severe traumatic event.

Ptyalin

An older name for salivary amylase, the enzyme that begins starch digestion in the mouth.

Ptyalism (Sialorrhoea)

Excessive salivation or drooling beyond normal physiological volumes (hyper-salivation). Can result from true salivary gland hypersecretion (induced by cholinergic medications, organophosphate poisoning, heavy metal toxicity, or gastroesophageal reflux [the "water brash" reflex]) or, far more commonly, from impaired swallowing (dysphagia) or oral neuromuscular incompetence with normal salivary production. Clinically prominent in advanced neurodegenerative disorders (Parkinson disease, amyotrophic lateral sclerosis [ALS], cerebral palsy) and acute pseudobulbar palsy; managed conservatively with oral motor therapy or pharmacologically with anticholinergic agents (glycopyrrolate, transdermal scopolamine) and intraglandular botulinum toxin injections.

Pubertal Failure

Failure to begin or progress through expected puberty because of hypothalamic, pituitary, gonadal, chromosomal, chronic-disease, nutritional, or other causes.

Puberty

The developmental period in which hormonal changes produce sexual maturation and reproductive capability.

Puberty In Boys

Puberty in boys is the hormone-driven development of testes, penis, body hair, voice, muscle, and fertility, usually progressing through predictable physical stages.

Puberty In Girls

Puberty in girls is the hormone-driven development of breasts, body hair, growth, menstrual cycles, and reproductive maturity, usually progressing through predictable stages.

Pubes

The anatomical anterior, lower hypogastric region

overlying the pubic bone (os pubis) and symphysis pubis, also referring to the coarse secondary sexual hair (pubic hair) that develops in this region during adrenarche and puberty under the influence of adrenal and gonadal androgens. Evaluated clinically in pediatric endocrinology using the standardized sexual maturity rating of Tanner stages (Stages 1 through 5).

Pubic

Relating to the pubis or the region over the front of the pelvis.

Pubic Lice

Pubic lice are parasitic insects infesting coarse body hair, causing intense itching and visible nits; treatment and assessment of close or sexual contacts are needed.

Pubic Lice (Crabs)

Infestation of coarse body hair by lice, usually causing intense itching and requiring treatment and contact precautions.

Alternate Names:

Crabs

Pubis

The front part of the pelvic bone, meeting its opposite side at the pubic symphysis.

Public Assistance

Government or community support providing eligible people with food, housing, income, healthcare, disability, or other essential services.

Public Exposure

Radiation exposure to public from nuclear medicine patients. Guidelines for discharge after therapy. I-131 therapy: radiation safety instructions for 3-7 days.

Public Health

The science and practice of protecting and improving the health of communities through education, policy-making, and research for disease and injury

prevention.

Public Health Emergency

Event threatening public health. Examples: pandemic, bioterrorism, natural disaster. Declaration enables special powers.

Public Health Research Bias

Systematic error in a population-health study, including selection bias, information bias, and confounding.

Publication

A formally distributed written, electronic, or multimedia work that communicates research, evidence, education, policy, or other information.

Puborectalis Muscle

A pelvic-floor muscle sling around the rectum that helps maintain continence.

Pudendal

Relating to the external genitalia, anus, or perineum. The pudendal nerve carries sensation from these areas and helps control several muscles involved in continence and sexual function.

Pudendal Arteries

Arteries supplying blood to the external genital and perineal region.

Pudendal Block

A local anesthetic block of the pudendal nerve (S2-S4) used for perineal anesthesia during the second stage of labor, outlet forceps delivery, or perineal repair. Technique: transvaginal approach — a needle guide is placed on the ischial spine, and the needle is advanced through the sacrospinous ligament.

Pudendal Nerve

The nerve carrying sensation and motor signals to much of the perineum and pelvic floor.

Puerperal

Relating to the period after childbirth, usually the

first six weeks.

Puerperal Disorder

A physical or mental health disorder occurring during the postpartum period, including infection, bleeding, hypertension, depression, or psychosis.

Puerperal Fever

Fever and infection occurring after childbirth, often from uterine or genital-tract infection.

Puerperal Infection

An infection after childbirth, commonly involving the uterus, surgical wound, urinary tract, breast, or bloodstream, often causing fever, pain, or foul discharge.

Puerperal Psychosis

A rare psychiatric emergency beginning soon after childbirth, characterized by rapidly changing mood, confusion, delusions, hallucinations, or disorganized behavior.

Puerperal Sepsis

A life-threatening systemic infection of the female reproductive and genital tract occurring between rupture of membranes or onset of labor and the 42nd day postpartum. Defined by the World Health Organization as pelvic pain, abnormal vaginal discharge, foul odor, and delayed uterine involution accompanied by fever ($\geq 38.5^{\circ}\text{C}$) and systemic inflammatory signs. It remains one of the primary worldwide causes of preventable maternal death, requiring emergency broad-spectrum parenteral antibiotics, supportive resuscitation, and search for retained tissue or pelvic abscesses.

Alternate Names:

Postpartum Sepsis; Puerperal Septicemia

Puerperium

The approximately six-week period after childbirth during which the reproductive organs return toward their nonpregnant state.

Puffy Eyes

Alternate index label; see *Bags under eyes*.

Pugilistic Stance

The characteristic body position adopted by fire victims—flexed arms, elbows bent, fists clenched, knees flexed—resulting from heat-induced contraction (denaturation) of muscles (mainly flexors) and tendons, not from conscious movement. It occurs with thermal injury and can be present regardless of cause (even if death preceded fire).

Pugilistica Dementia (Dementia)

Alternate index label for Dementia.

Pulex

A taxonomic genus of fleas belonging to the family Pulicidae, including *Pulex irritans* (the human flea) and related species. While not restricted exclusively to humans (also parasitizing swine, dogs, and rodents), *Pulex* fleas cause pruritic, erythematous papular urticaria (grouped "breakfast, lunch, and dinner" flea bites). Historically and epidemiologically significant as secondary vectors in the transmission of zoonotic pathogens, including *Yersinia pestis* (the etiologic agent of bubonic plague) and *Rickettsia typhi* (endemic/murine typhus), when sharing habitats with infected rodent reservoirs.

Pulex Irritans

The human flea, which can cause bites and occasionally transmit infections.

Pulled Hamstring (Hamstring Injury)

Alternate index label for Hamstring Injury.

Pulling A Patient Up In Bed

A safe repositioning technique using adequate staff, friction-reducing equipment, and coordinated movement to move a patient toward the head of the bed while preventing falls, shear, and staff injury.

Pulmonary

Pertaining to, situated within, or affecting the lungs (*pulmones*) or the respiratory gas exchange system.

Clinically central to thoracic medicine and cardiology, encompassing pulmonary circulation (low-pressure vascular circuit from the right ventricle through the pulmonary arterial tree to alveolar capillary networks and pulmonary veins into the left atrium), pulmonary function testing (spirometry, lung volumes, and diffusing capacity for carbon monoxide [D_LCO]), and major disease entities such as pulmonary hypertension, pulmonary embolism, pulmonary edema, chronic obstructive pulmonary disease, and pulmonary fibrosis.

Pulmonary Actinomycosis

Pulmonary actinomycosis is a chronic bacterial infection of the lungs caused by *Actinomyces* and may form mass-like lesions or spread across tissue planes.

Pulmonary Alveolar Hemorrhage

Bleeding into the alveolar spaces from disruption of the alveolar-capillary membrane. Causes include vasculitis (GPA, MPA, Goodpasture), connective tissue disease, medications, and coagulopathy. It presents with hemoptysis, falling hemoglobin, diffuse bilateral infiltrates, and serially bloody returns on BAL.

Pulmonary Alveolar Microlithiasis

A rare genetic disease caused by SLC34A2 mutations, leading to accumulation of calcium phosphate microliths within the alveoli. HRCT shows diffuse bilateral calcified nodules with a "sandstorm" appearance. It is usually diagnosed in the third to fourth decade.

Pulmonary Alveolar Proteinosis

A rare disease characterized by accumulation of surfactant-like material within the alveoli due to impaired clearance by alveolar macrophages. Most cases are autoimmune (anti-GM-CSF antibodies). Presents with progressive dyspnea and cough. HRCT shows bilateral ground-glass opacities with "crazy paving" pattern.

Pulmonary Angiography

Contrast imaging of the pulmonary arteries, usually by catheter. It can diagnose pulmonary vascular ab-

normalities but is now used less often for pulmonary embolism because CT pulmonary angiography is less invasive.

Pulmonary Arterial Hypertension

A precapillary pulmonary hypertension syndrome caused by increased pulmonary vascular resistance from pulmonary arteriolar disease. It may be idiopathic, heritable, drug-associated, or associated with conditions such as connective-tissue disease, congenital heart disease, portal hypertension, or HIV.

Pulmonary Arteries

The paired arteries that carry deoxygenated blood from the right ventricle to the lungs for gas exchange.

Alternate Names:

pulmonary artery

Pulmonary Arteriovenous Fistula

An abnormal direct connection between a pulmonary artery and vein that bypasses the capillary bed. It can cause hypoxemia, cyanosis, dyspnea, paradoxical emboli, or brain abscess.

Pulmonary Arteriovenous Malformation

An abnormal direct communication between a pulmonary artery and vein, causing a right-to-left shunt. It may be idiopathic or associated with hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu syndrome). It can cause hypoxemia, paradoxical emboli (stroke, brain abscess), and hemoptysis.

Pulmonary Artery

The artery carrying deoxygenated blood from the right ventricle to the lungs.

Pulmonary Artery Catheter

A catheter inserted through the right heart into the pulmonary artery to measure hemodynamic parameters (PA pressure, PAWP, cardiac output). Used for diagnosis and management of pulmonary hypertension, heart failure, and critical illness.

Pulmonary Artery Pulsatility Index

A novel, validated hemodynamic metric calculated during right heart catheterization to assess right ventricular pulsatile function and quantify the severity of right ventricular dysfunction. Defined as the ratio of pulmonary artery pulse pressure to right atrial pressure: $PAPi = (PASP - PADP)/RAP$, where PASP is pulmonary artery systolic pressure, PADP is pulmonary artery diastolic pressure, and RAP is mean right atrial pressure. A low PAPi (< 1.0 – 1.85) indicates impaired right ventricular systolic performance and strongly predicts the development of acute right ventricular failure and mortality following left ventricular assist device placement or acute inferior right ventricular myocardial infarction.

Alternate Names:

PAPi; PAP Index

Pulmonary Aspergilloma

A pulmonary aspergilloma is a ball of *Aspergillus* fungus and debris growing in a pre-existing lung cavity, sometimes causing coughing of blood.

Pulmonary Atresia

A congenital heart defect where the pulmonary valve is completely absent (atretic), obstructing right ventricular outflow. Two main types: pulmonary atresia with intact ventricular septum (PA-IVS) and pulmonary atresia with ventricular septal defect (PA-VSD, also called tetralogy of Fallot with pulmonary atresia).

Alternate Names:

tetralogy of Fallot with pulmonary atresia).

Pulmonary Blastoma

A rare malignant lung tumor composed of primitive epithelial and mesenchymal tissue, usually treated with specialist oncology care.

Pulmonary Capillary Wedge Pressure

An indirect estimate of left-atrial pressure obtained by briefly occluding a small pulmonary-artery branch with a balloon-tipped catheter. It reflects left-sided filling pressure when measured correctly and interpreted in the appropriate clinical and ventilatory context.

Pulmonary Circuit

The circulation in which blood travels from the right ventricle to the lungs for gas exchange and returns to the left atrium.

Pulmonary Circulation

The blood-flow circuit from the right ventricle through the pulmonary arteries and lung capillaries to the left atrium, where blood releases carbon dioxide and gains oxygen.

Pulmonary Compliance

The ease with which the lungs expand, calculated as change in lung volume divided by change in pressure; low compliance means stiff lungs.

Pulmonary Contusion

Bruising of the lung parenchyma from blunt chest trauma, causing hemorrhage and edema without lung laceration. It impairs gas exchange and can progress to ARDS. It is often associated with rib fractures and flail chest. Treatment is supportive with lung-protective ventilation.

Pulmonary Edema

A buildup of fluid in the lungs that can make breathing difficult. It is often caused by heart failure but may also result from kidney problems, lung injury, medicines, or high altitude. Sudden severe breathing difficulty is an emergency.

Alternate Names:

Edema, Pulmonary Edema Pulmonary (Pulmonary Edema)

Pulmonary Embolism

Blockage of a pulmonary artery, usually by a clot that traveled from a deep vein.

Alternate Names:

Blood Clot In The Lung (Pulmonary Embolism)
Embolism Pulmonary (Pulmonary Embolism)

Pulmonary Embolism Treatment

Anticoagulation: DOACs preferred (rivaroxaban, apixaban) or LMWH bridging to warfarin. Duration: 3 months for provoked (reversible risk factor),

indefinite for unprovoked or recurrent. Massive PE: systemic thrombolysis (alteplase 100 mg IV over 2 hours) or surgical catheter-directed embolectomy.

Pulmonary Embolus

A blockage of a pulmonary artery, usually by a blood clot that traveled from a deep vein, causing sudden breathlessness, chest pain, low oxygen, or collapse.

Pulmonary Emphysema

A chronic lung disease in which damage to the air sacs reduces the lungs' ability to exchange gases.

Pulmonary FibrosisIdiopathic (Idiopathic Pulmonary Fibrosis)

Alternate index label for Idiopathic Pulmonary Fibrosis.

Pulmonary Fibrosis

Scarring replacing normal alveolar architecture. Causes: IPF, connective tissue disease, asbestos, radiation, drugs (bleomycin, methotrexate). Features: progressive dyspnea, dry cough, digital clubbing.

Pulmonary Fistula

An abnormal passage connecting the lung or airways with another structure, such as the pleural space, skin, or esophagus.

Pulmonary Function Tests

Pulmonary-function testing may include spirometry, lung-volume measurement, and diffusing-capacity testing. Spirometry measures FVC (forced vital capacity), FEV1 (forced expiratory volume in one second), and the FEV1/FVC ratio. A low ratio supports an obstructive pattern; interpretation should use age-, sex-, height-, and population-appropriate reference equations and the lower limit of normal, although some guidelines use a fixed ratio such as less than 0.70 for selected COPD assessments. A low FVC with a preserved or high ratio may suggest restriction, but restriction requires confirmation by a reduced total lung capacity. Test quality, patient effort, and repeatability are essential.

Pulmonary Hemorrhage

Bleeding into the airways or lung tissue, which may cause coughing blood, low oxygen, anemia, or respiratory failure.

Pulmonary Hemosiderosis

Accumulation of iron-containing pigment in the lungs, usually after repeated bleeding into the air spaces. It may cause cough, coughing blood, anemia, shortness of breath, or diffuse lung changes and can result from immune-mediated or other disorders.

Pulmonary Hypertension

Elevated mean pulmonary artery pressure (>20 mmHg at rest) due to various etiologies. WHO Group 1: pulmonary arterial hypertension (idiopathic, heritable, drug-induced). Group 2: left heart disease. Group 3: lung disease. Group 4: chronic thromboembolic.

Alternate Names:

PH (Pulmonary Hypertension)

Pulmonary Hypertension

Alternate index label for Pulmonary Hypertension.

Pulmonary Hypertension - At Home

Home care for pulmonary hypertension includes taking prescribed medicines, monitoring symptoms, conserving energy, following oxygen advice, and seeking urgent help for severe breathlessness or chest pain.

Pulmonary Hypertension Classification

WHO Classification: Group 1 (PAH: idiopathic, heritable, drug-induced, connective tissue disease-associated). Group 2 (left heart disease: systolic/diastolic dysfunction, valvular). Group 3 (lung disease/hypoxia: COPD, ILD, sleep apnea). Group 4 (chronic thromboembolic). Group 5 (multifactorial: sarcoidosis, hematologic disorders).

Pulmonary Infarction

Lung necrosis from PE when bronchial supply is compromised. Causes pleuritic pain, hemoptysis, and wedge-shaped opacity. Most PE do not cause

infarction due to dual blood supply.

Pulmonary Infection

An infection of the lungs or lower respiratory tract caused by bacteria, viruses, fungi, or other organisms.

Pulmonary Langerhans Cell Histiocytosis

A rare smoking-related interstitial and cystic lung disease occurring predominantly in young adult smokers (aged 20 to 40 years). Driven by the clonal proliferation and accumulation of CD1a- and Langerin-positive dendritic Langerhans cells within peribronchial interstitial tissues, it forms cellular stellate granulomatous nodules that progressively cavitate and evolve into bizarre, bilobed, or cloverleaf-shaped thin-walled cysts distributed symmetrically across the mid-to-upper lung fields, with relative sparing of the lung bases and costophrenic angles. Smoking cessation is the cornerstone of therapy and often leads to stabilization or radiographic regression.

Alternate Names:

PLCH; Histiocytosis X of the Lung; Pulmonary Eosinophilic Granuloma

Pulmonary Lobes

The anatomical divisions of the lungs separated by interlobar fissures. The right lung has superior, middle, and inferior lobes; the left lung has superior and inferior lobes and a lingula.

Pulmonary Mass

A space-occupying lesion in or near the lung, which may represent infection, inflammation, benign growth, or cancer and requires appropriate imaging assessment.

Pulmonary Nocardiosis

A lung infection caused by *Nocardia* bacteria, usually affecting people with weakened immunity and sometimes spreading to the brain or skin.

Pulmonary Nodule

A rounded opacity in the lung, usually smaller than 3 cm, caused by scar, infection, inflammation, or tu-

mor and assessed according to size and risk factors.

Pulmonary Oedema

Accumulation of fluid in lung tissue or alveoli, causing breathlessness and impaired gas exchange.

Pulmonary Reactance

The elastic and inertial opposition to airflow and lung movement, measured during respiratory mechanics testing.

Pulmonary Rehabilitation

A supervised, multidisciplinary program combining exercise training, education, behavioral support, and self-management for chronic respiratory disease. It improves exercise capacity, dyspnea, functional status, and quality of life.

Pulmonary Sequestration

A congenital malformation of non-functioning lung tissue that lacks normal communication with the tracheobronchial tree and receives its blood supply from a systemic artery (usually the descending aorta). Intralobar sequestration is within the normal pleural investment; extralobar is outside.

Pulmonary Stenosis

Narrowing of the pulmonary valve or outflow tract, obstructing blood flow from the right ventricle.

Pulmonary Surfactant

A phospholipid-protein mixture produced by type II alveolar pneumocytes that reduces surface tension at the air-liquid interface within alveoli. The primary component is dipalmitoylphosphatidylcholine (DPPC, lecithin), which forms a monolayer that reduces surface tension to near zero during expiration.

Pulmonary Thromboendarterectomy

A complex, definitive surgical operation that provides a potentially curative treatment for operable chronic thromboembolic pulmonary hypertension. Performed through a median sternotomy with cardiopulmonary bypass and periods of deep hypothermic circulatory arrest (DHCA, at 18 to 20 degrees

Celsius), the surgeon establishes a meticulous dissection plane between the media and intima of the main, lobar, and segmental pulmonary arteries to extract organized, endothelialized, fibrotic chronic thromboembolic casts, thereby dramatically reducing pulmonary vascular resistance, reversing right ventricular strain, and curing pulmonary hypertension.

Alternate Names:

PTE; Pulmonary Endarterectomy; PEA (Pulmonary Endarterectomy)

Pulmonary Trunk

The large vessel leaving the right ventricle that divides into the right and left pulmonary arteries.

Pulmonary Tuberculosis

Tuberculosis infection involving the lungs, caused by *Mycobacterium tuberculosis*.

Pulmonary Valve

Semilunar valve between right ventricle and pulmonary trunk with three cusps. Most superiorly located cardiac valve with thinner cusps than the aortic valve. Pulmonic stenosis is a congenital defect causing right ventricular hypertrophy. Less commonly affected by endocarditis than mitral/aortic valves.

Pulmonary Valve Regurgitation

Incompetence or backflow of blood from the pulmonary artery across the pulmonary valve into the right ventricle during diastole. Most commonly caused by pulmonary arterial hypertension, infective endocarditis, or as a late sequela following surgical repair of congenital heart defects such as tetralogy of Fallot, potentially leading to progressive right ventricular dilation and dysfunction.

Alternate Names:

Pulmonic regurgitation; Pulmonary valve insufficiency

Pulmonary Valve Insufficiency

Leakage of blood backward through the pulmonary valve into the right ventricle during diastole.

Pulmonary Vascular Resistance

The resistance to blood flow through the pulmonary circulation, calculated approximately as the transpulmonary pressure gradient divided by cardiac output. Elevated resistance contributes to pulmonary hypertension and right-ventricular afterload.

Pulmonary Vasodilator Therapy

Medications that reduce pulmonary vascular resistance and pulmonary artery pressure. Used for WHO Group 1 PAH. Includes endothelin receptor antagonists (bosentan), PDE-5 inhibitors (sildenafil), prostacyclin analogs (epoprostenol), and soluble guanylate cyclase stimulators (riociguat).

Pulmonary Vein

The four veins (two from each lung) returning oxygenated blood from the lungs to the left atrium, a unique exception in the venous system (carrying oxygenated blood). Superior and inferior pulmonary veins enter the left atrium posteriorly.

Pulmonary Vein Isolation

An invasive electrophysiological catheter ablation procedure that serves as the cornerstone treatment for symptomatic, drug-refractory paroxysmal and persistent atrial fibrillation. Using fluoroscopic and 3D electroanatomical mapping guidance, circumferential transmural ablation lines are delivered around the antra of the four pulmonary veins using point-by-point radiofrequency energy, cryoballoon ablation, or pulsed-field ablation, electrically isolating ectopic pulmonary vein triggers from the left atrial myocardium and achieving entrance and exit block.

Alternate Names:

PVI; Antral Pulmonary Vein Isolation; AFib Catheter Ablation

Pulmonary Veins

Veins carrying oxygen-rich blood from the lungs to the left atrium.

Pulmonary Veno-Occlusive Disease

A rare progressive disease in which small pulmonary veins and venules become narrowed or blocked, caus-

ing pulmonary hypertension and possible pulmonary edema with vasodilator treatment.

Pulmonary Ventilation

The movement of air into and out of the lungs.

Pulmonary Ventilation/Perfusion Scan

A ventilation-perfusion scan compares air flow and blood flow in the lungs and is commonly used to evaluate suspected pulmonary embolism.

Pulmonary Wedge Pressure

The pressure measured by a balloon-tipped pulmonary artery catheter when the balloon is inflated, occluding a branch of the pulmonary artery. It approximates left atrial pressure. Normal is 6–12 mmHg. Elevated in left heart failure (PCWP > 18 mmHg). Low in hypovolemia.

Pulmonary-Artery Dissection

Rare but life-threatening tear in the pulmonary artery wall, most commonly associated with pulmonary hypertension, Marfan syndrome, or prior pulmonary artery catheterization. Presents with acute chest pain, dyspnea, and hemodynamic collapse. Diagnosis by CT angiography or echocardiography; management requires emergent surgical repair.

Pulmonic Valve Stenosis

Pulmonic valve stenosis is narrowing of the valve between the right ventricle and pulmonary artery, creating outflow obstruction and sometimes right-heart strain.

Pulmonologist

A pulmonologist is a physician specializing in diseases of the lungs and breathing system, including asthma, COPD, infections, and sleep-related breathing disorders.

Pulp

The soft vascular and nervous tissue inside a tooth or, in plants, soft internal tissue.

Pulp Canal

The pulp canal is the passage within a tooth root containing dental pulp, nerves, and blood vessels; infection may require root-canal treatment.

Pulp Disease

A disorder of the soft tissue inside a tooth, including inflammation, infection, loss of blood supply, or necrosis.

Pulp Stone

Calcified mass in pulp chamber or root canal. May be free, attached, or embedded. Incidental finding.

Pulpectomy

Complete removal of pulp from primary tooth. Followed by resorbable filling material.

Pulpitis

Inflammation of the dental pulp, usually caused by caries, trauma, cracks, or restorative treatment. Reversible pulpitis causes brief pain provoked by cold or sweet stimuli; irreversible pulpitis causes spontaneous or lingering pain and may progress to pulp necrosis, requiring endodontic treatment or extraction.

Pulposus

Relating to the nucleus pulposus, the gelatinous central portion of an intervertebral disk.

Pulpotomy

Removal of coronal pulp in primary tooth with reversible inflammation. Maintains radicular pulp vitality.

Pulse

The rhythmic expansion of an artery produced by each heartbeat.

Pulse - Bounding

A bounding pulse feels unusually strong or full and can occur with fever, anemia, hyperthyroidism, exercise, anxiety, aortic regurgitation, or high-output states.

Pulse Examination

Assessment of arterial pulses for rate, rhythm, strength, and symmetry to help evaluate circulation.

Pulse Irregular

An irregular pulse has uneven timing or varying beats and may reflect premature beats, atrial fibrillation, other arrhythmias, or measurement variation.

Pulse Oximeter

A noninvasive device that estimates peripheral oxygen saturation using light absorption. Readings can be affected by poor perfusion, motion, skin or nail characteristics, device limitations, and the clinical setting.

Pulse Oximetry

A noninvasive method for estimating peripheral oxygen saturation by measuring red and infrared light absorption through a pulsatile vascular bed. It is widely used for monitoring, but values must be interpreted with symptoms, perfusion, device limitations, and the individual's clinical targets.

Pulse Pressure

The systolic blood pressure minus the diastolic blood pressure. A widened or narrowed value may provide clues about circulation but must be interpreted with the full blood-pressure reading and clinical context.

Pulse Radiolysis

Pulse radiolysis is a research technique using short radiation pulses to create and study transient chemical species and reaction kinetics.

Pulse Rate

Pulse rate is the number of palpable arterial beats per minute and usually approximates heart rate, although some arrhythmias produce a pulse deficit.

Pulseless Disease

Takayasu arteritis, a large-vessel vasculitis that can reduce pulses in affected arteries.

Pulseless Electrical Activity

A cardiac arrest rhythm organized electrical activity without a detectable pulse. PEA has many reversible causes (5 H's and 5 T's): hypovolemia, hypoxia, hydrogen ion (acidosis), hypo/hyperkalemia, hypothermia; tension pneumothorax, tamponade (cardiac), toxins, thrombosis (pulmonary), thrombosis (coronary).

Pulsus

A Latin term for pulse, used in names describing abnormal pulse patterns.

Pulsus Paradoxus

A decrease in systolic blood pressure greater than 10 mmHg during inspiration, normally less than 10 mmHg. Exaggerated in cardiac tamponade (due to ventricular interdependence in the confined pericardial space), severe asthma, COPD exacerbation, and constrictive pericarditis.

Pulvinar

The pulvinar is the largest posterior nucleus of the thalamus and participates in visual attention, sensory integration, and connections with the cortex.

Pumice

A porous volcanic stone used as an abrasive; excessive dust inhalation can irritate the lungs.

Pump Failure

Severe weakening of the heart's pumping ability, causing inadequate blood flow to organs.

Alternate Names:

Cardiogenic pump failure

Pump-Oxygenator

A clinical or therapeutic concept involving pump-oxygenator. Its features, uses, and clinical significance depend on the relevant medical context.

Puna

A high-altitude region or, in some sources, a term related to altitude sickness; precise meaning depends on context.

Punch Biopsy

A circular, usually full-thickness skin specimen obtained with a punch instrument. It is useful for inflammatory, infectious, melanocytic, and other cutaneous lesions; small wounds may be closed with a suture.

Punch-Drunk

An obsolete term for chronic traumatic encephalopathy associated with repeated head injury, especially in boxing.

Punctate

Composed of or marked by small spots or points.

Punctate Hemorrhages (Tardieu Spots)

Small, pinpoint sub-ecchymotic hemorrhages (petechiae) resulting from the rupture of small capillaries under conditions of intense mechanical venous engorgement and asphyxia.

Alternate Names:

Tardieu Spots

Punctate Palmoplantar Keratoderma

A hereditary keratoderma characterised by multiple, small, firm, circumscribed keratotic papules and plaques scattered across the palms and soles, often with an autosomal dominant inheritance pattern. It may present in adolescence or adulthood and can be associated with malignant transformation in rare cases.

Puncture

A wound or procedure made by a pointed object or needle.

Puncture Biopsy

Alternate index label for needle biopsy: removal of cells or tissue through a percutaneous puncture for microscopic, molecular, or microbiologic examination; imaging may guide the target and route.

Puncture Wound

A wound produced when a pointed object penetrates the skin and underlying tissue, creating a small

surface opening with potentially deep tissue damage and infection risk.

Pungent

Having a strong, sharp smell or taste; pungent substances may irritate mucous membranes.

Punicic Acid

A conjugated polyunsaturated fatty acid found mainly in pomegranate seed oil and studied for lipid and inflammatory effects; clinical supplement benefits remain uncertain.

Punicin

A rare or historical term whose precise medical meaning is source-dependent and should be verified.

Punishment

Punishment is a consequence intended to reduce a behavior; in healthcare, coercive or harmful punishment is inappropriate and behavior support should prioritize safety and dignity.

Pupil

The central opening in the iris through which light enters the eye.

Pupil - White

A white pupil, or leukocoria, may result from reflection, cataract, retinal disease, infection, or retinoblastoma and requires prompt eye assessment, especially in a child.

Pupillary

Pertaining to the pupil, the central aperture of the iris that regulates the quantity of light entering the posterior segment of the eye. Modulated antagonistically by autonomic innervation: parasympathetic fibers traveling via the oculomotor nerve (CN III) constrict the pupil via the sphincter pupillae muscle (miosis), whereas sympathetic fibers originating from the superior cervical ganglion dilate the pupil via the dilator pupillae muscle (mydriasis). Examination of pupillary size, shape, and reactivity (light reflex and accommodation reflex) is critical

in neuro-ophthalmology and emergency medicine to assess optic nerve function, brainstem integrity, uncal herniation (fixed dilated "blown" pupil), and Horner syndrome.

Pure Red Cell Aplasia

A rare hematologic syndrome characterized by profound normocytic normochromic anemia, severe reticulocytopenia, and an isolated, selective absence of mature erythroid precursors in an otherwise normal bone marrow. May be congenital (Diamond-Blackfan anemia) or acquired (associated with thymoma, persistent parvovirus B19 infection, autoimmune disease, or medications).

Alternate Names:

PRCA; Pure red-cell aplasia

Pure Tone Audiometry

The standard behavioral test used to measure hearing sensitivity across the frequency range important for speech perception, typically from 250 to 8000 Hertz, by determining the lowest intensity level at which a pure tone can be detected approximately fifty percent of the time (the hearing threshold).

Pure-Tone Audiometry

A hearing test that measures the softest tones a person can hear at selected frequencies through air and bone conduction. The results are plotted on an audiogram and help distinguish conductive from sensorineural hearing loss.

Alternate Names:

Pure-tone threshold audiometry; pure-tone hearing test.

Purgative

A strong, fast-acting cathartic pharmacologic agent or substance that induces complete, vigorous evacuation and emptying of the bowels. Distinguished from gentle stool softeners or mild laxatives by its rapid onset and watery bowel purging effect. Clinically utilized in gastroenterology for bowel preparation prior to colonoscopy, barium enema, or colorectal surgery (e.g., high-volume polyethylene glycol [PEG] solutions, magnesium citrate, or sodium phosphate purgatives), and historically used in acute

poisoning management (though now largely abandoned due to electrolyte derangements, dehydration, and lack of proven efficacy).

Purge

To empty the bowel, stomach, or body of a substance; the term may also mean forceful vomiting or diarrhea.

Purging

Deliberate behaviors intended to compensate for eating, such as self-induced vomiting or misuse of laxatives, diuretics, or excessive exercise.

Purging Disorder

Recurrent purging behavior to influence weight or shape in the absence of binge eating. Self-induced vomiting, misuse of laxatives, diuretics, or other medications. It is classified under Other Specified Feeding or Eating Disorder (OSFED). It can lead to electrolyte imbalances, dental erosion, Russell's sign, and esophageal tears.

Purine

A nitrogen-containing compound forming part of nucleotides; its breakdown produces uric acid.

Purine Nucleosides

Nucleosides containing adenine or guanine bases, such as adenosine and guanosine, involved in energy transfer and genetic material.

Purine Nucleotides

Nucleotides containing purine bases, including ATP, GTP, and their metabolic precursors and products.

Purines

Nitrogenous bases containing a double-ring structure (a six-membered ring fused to a five-membered imidazole ring), which serve as essential components of nucleotides in DNA and RNA. The primary purines are adenine (A) and guanine (G).

Purinoceptor

A purinoceptor is a cell-surface receptor activated by

extracellular purines such as ATP, ADP, or adenosine and involved in signaling, inflammation, and vascular control.

Purkinje Cell

A large inhibitory neuron in the cerebellar cortex whose axons provide the sole output of the cerebellar cortex and regulate coordination, balance, and motor learning.

Purkinje Fibers

The terminal branches of the cardiac conduction system, distributed throughout the ventricular myocardium. They conduct impulses rapidly (up to 4 m/s) to ensure coordinated ventricular contraction from apex to base. Purkinje fibers have an intrinsic firing rate of 20-40 beats per minute.

Purkinje Fibres

Specialized cardiac conduction fibers that rapidly distribute impulses through the ventricular myocardium.

Puromycin

Puromycin is an antibiotic research compound that stops protein production in cells; it is not used as a routine human medicine.

Purpura

Purple or red nonblanching skin or mucosal discoloration caused by bleeding into tissue.

Purpura Fulminans

A rapidly progressive, life-threatening condition characterised by extensive, confluent, non-blanching purpuric skin necrosis with haemorrhagic bullae, resulting from acute intravascular thrombosis and disseminated intravascular coagulation.

Purulent

Consisting of, containing, producing, or resembling pus (*pus*)—a viscous inflammatory exudate composed of dead and dying neutrophils, liquefactive necrotic cellular debris, protein-rich serous fluid, and viable or non-viable microorganisms. A cardi-

nal hallmark of acute suppurative pyogenic infections, typically driven by bacterial pathogens such as *Staphylococcus aureus*, *Streptococcus pyogenes*, and *Pseudomonas aeruginosa*. Clinical identification of purulent drainage indicates active infection requiring diagnostic Gram stain/culture, possible surgical incision and drainage (I&D), and targeted antimicrobial therapy.

Purulent Pericarditis

Bacterial infection of the pericardial space, most commonly caused by *S. aureus*, *S. pneumoniae*, or gram-negative organisms. Often occurs as a complication of pneumonia, endocarditis, chest surgery, or immunosuppression. Presents with fever, pleuritic chest pain, and rapidly accumulating effusion with tamponade physiology.

Pus

A thick fluid containing dead cells, bacteria, inflammatory cells, and tissue debris.

Push Enteroscopy

Endoscopic technique using a longer endoscope to visualize and intervene in the proximal small bowel beyond the reach of standard upper endoscopy. Unlike capsule endoscopy, push enteroscopy allows direct visualization with the ability to obtain biopsies and perform hemostasis.

Pustule

A primary dermatologic lesion characterized by a small (< 1 cm), circumscribed, superficial, elevated collection of cloudy or turbid purulent fluid (pus) located within the epidermis or a hair follicle. Can be infectious or sterile: infectious pustules are seen in bacterial folliculitis, impetigo, and acne vulgaris (where *Cutibacterium acnes* colonizes occluded pilosebaceous units); non-infectious, sterile pustules characterize autoimmune or inflammatory dermatoses such as pustular psoriasis, acute generalized exanthematous pustulosis (AGEP, a severe cutaneous adverse drug reaction), and subcorneal pustular dermatosis.

Pustules

Pustules are small raised skin lesions filled with pus, occurring in acne, folliculitis, infections, inflammatory disorders, or drug reactions.

Pustulosis

A skin disorder characterized by sterile or infectious pustules, which may be localized or widespread and associated with inflammation, fever, psoriasis, drug reactions, or infection.

Putamen

The putamen is a basal-ganglia structure involved in movement selection, motor learning, and habit-related behavior.

Putrefaction

Decomposition of tissues after death by bacterial and enzymatic processes.

Putrescine

A foul-smelling diamine produced during breakdown of proteins in decaying tissue.

Putrid Throat

Putrid throat refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

PUVA

PUVA combines a psoralen medicine with ultraviolet-A exposure to treat selected skin diseases; eye protection and cancer-risk precautions are required.

PVC

A PVC, or premature ventricular contraction, is an early heartbeat arising in a ventricle; occasional PVCs are common, while frequent or symptomatic beats need evaluation.

PVD

Alternate index label; see *Peripheral artery disease*.

PVI

Alternate name; see *Pulmonary Vein Isolation*.

PVPS

Alternate index label; see *Post-vasectomy pain syndrome*.

Pyarthrosis

Infection of a joint with pus in the joint space, usually bacterial septic arthritis, causing acute pain, swelling, restricted movement, fever, and risk of cartilage destruction.

Pycnodysostosis

A rare autosomal-recessive bone dysplasia caused by CTSK deficiency, producing short stature, dense but fragile bones, acro-osteolysis, delayed closure of cranial sutures, and dental abnormalities.

Pyelitis

Inflammation of the renal pelvis, usually due to infection.

Pyelo

Pyelo is an informal shortening usually referring to pyelonephritis, an infection of the kidney and renal pelvis that can cause fever and flank pain.

Pyelogram

An imaging record of the renal pelvis and ureters after contrast administration.

Pyelography

Imaging of the renal pelvis and ureters using contrast.

Pyelolithotomy

Surgical removal of a stone from the renal pelvis.

Pyelonephritis

Infection and inflammation of the kidney and renal pelvis, usually caused by ascending bacteria.

Pyelotomy

An incision into the renal pelvis, usually made to remove an obstruction or kidney stone.

Pylephlebitis

Septic thrombophlebitis of the portal vein or its tributaries, most commonly arising from intra-abdominal infections such as appendicitis, diverticulitis, cholangitis, or hepatic abscess. Presents with high fever, rigors, right upper quadrant pain, jaundice, and sepsis.

Pyloric Antrum

The pyloric antrum is the lower stomach region that grinds food and regulates its passage toward the pyloric canal and duodenum.

Pyloric Sphincter

A muscular valve regulating passage of stomach contents into the duodenum.

Alternate Names:

Pylorus

Pyloric Stenosis

Narrowing of the pyloric outlet, classically causing projectile vomiting in infants or gastric-outlet symptoms in adults.

Alternate Names:

Stenosis, Pyloric

Pyloric Stenosis In Infants

Infantile pyloric stenosis is narrowing of the stomach outlet that causes forceful, non-bilious vomiting in young infants and usually requires surgical treatment.

Pyloroplasty

Surgical widening of the pylorus to improve gastric emptying.

Pylorus

The distal part of the stomach leading into the duodenum.

Pyoderma

A bacterial skin infection producing pus, crusts, erosions, or ulcers.

Pyoderma Gangrenosum

A neutrophilic inflammatory skin disorder that causes painful ulcers with undermined, violaceous borders. It may be associated with inflammatory bowel disease, arthritis, blood disorders, or other systemic conditions, and trauma can worsen the lesions.

Pyogenesis

The biological process of pus formation (suppuration). Initiated when pyogenic bacteria or inflammatory tissue insults trigger intense local chemotaxis and recruitment of polymorphonuclear neutrophils via chemokines (e.g., IL-8) and complement activation products (C5a). As recruited neutrophils engulf pathogens, they release reactive oxygen species and lysosomal enzymes (including myeloperoxidase and elastase), resulting in liquefactive necrosis of surrounding parenchyma, neutrophil apoptosis, and accumulation of viscous, proteinaceous purulent exudate within an enclosed cavity (abscess) or discharging across a mucosal or cutaneous surface.

Pyogenic

Producing pus or associated with a pus-forming infection. Pyogenic infections commonly cause redness, warmth, swelling, pain, and a collection of pus, although the cause and treatment vary.

Pyogenic Granuloma

A benign vascular proliferation presenting as a rapidly growing, friable, red papule or nodule that bleeds easily. It commonly occurs on fingers, face, and lips, and may be triggered by trauma or pregnancy (epulis gravidarum). Diagnosis is clinical.

Pyogenic Liver Abscess

A pyogenic liver abscess is a pus-filled infection in the liver, usually from biliary, abdominal, or bloodstream spread, causing fever, pain, and possible sepsis.

Pyometra

The abnormal accumulation of pus within the uterine cavity, resulting from infection of the endometrium combined with impaired cervical drainage (cervical stenosis). Most commonly encountered in elderly, postmenopausal women due to atrophic cervical canal stenosis, or secondary to cervical or endometrial malignancy occluding the internal os. Clinically presents with foul-smelling purulent vaginal discharge, lower abdominal pain, uterine tenderness, and fever, though some patients remain asymptomatic until acute spontaneous uterine rupture occurs, causing life-threatening purulent peritonitis. Managed by urgent cervical dilation, purulent drainage, endometrial curettage/biopsy to exclude occult malignancy, and broad-spectrum intravenous antibiotics.

Pyonephrosis

Pus in renal collecting system. Urological emergency. Requires drainage (stent or nephrostomy) and antibiotics.

Pyosalpinx

Accumulation of pus within a fallopian tube, usually due to pelvic infection.

Pyostomatitis Vegetans

A rare, chronic, inflammatory oral mucosal condition characterized by multiple, soft, yellowish, vegetative pustules on an erythematous base, predominantly affecting the buccal mucosa and gingiva. It is strongly associated with inflammatory bowel disease, particularly ulcerative colitis, and may precede gastrointestinal symptoms.

Pyramid

A pyramid-shaped structure; in neuroanatomy, the medullary pyramids contain corticospinal fibers.

Pyramidal Cell

A triangular-shaped excitatory neuron with a prominent apical dendrite, found especially in the cerebral cortex and hippocampus and involved in long-range brain signaling.

Pyramidal Tract

The pyramidal tract is a major motor pathway from the cerebral cortex to the brainstem and spinal cord that controls voluntary movement.

Pyramidal Tracts

Descending motor pathways from the cerebral cortex to the brainstem and spinal cord.

Pyramidon

An obsolete analgesic and antipyretic medicine, also called aminopyrine.

Pyrazinamide

An antituberculous medicine used in combination therapy, especially during the initial phase.

Pyrazines

Pyrazines are aromatic nitrogen-containing chemical compounds found in foods, flavors, medicines, and industrial materials; effects depend on the specific compound and exposure.

Pyrazole

Pyrazole is a five-membered chemical ring used as a building block in medicines and other compounds; it is not one specific therapeutic drug.

Pyrazoles

Pyrazoles are a family of chemical compounds containing a pyrazole ring, including some medicines and pesticides.

Pyrene

Pyrene is a four-ring polycyclic aromatic hydrocarbon produced by incomplete combustion and used as an environmental marker of complex combustion mixtures.

Pyrenochaeta

A genus of soil-associated fungi that can rarely cause subcutaneous or opportunistic human infection.

Pyrethrins

Natural insecticides derived from chrysanthemum

flowers; exposure can irritate skin and airways.

Pyrethrum

The dried flower heads of certain *Chrysanthemum* species, used as a natural source of pyrethrins, insecticidal compounds that affect insect sodium channels. Pyrethrum and pyrethrins differ from synthetic pyrethroids; exposure may irritate the skin or airways and can cause toxicity with substantial exposure.

Pyrexia (Fever)

An elevation of core body temperature above the normal circadian range (typically defined clinically as a core temperature $\geq 38.0^{\circ}\text{C}$ [$\geq 100.4^{\circ}\text{F}$]). A coordinated, cytokine-mediated physiological response driven by pyrogenic cytokines (interleukin-1 [IL-1], interleukin-6 [IL-6], and tumor necrosis factor- α [TNF- α]) acting on the organum vasculosum of the lamina terminalis in the preoptic area of the anterior hypothalamus to stimulate prostaglandin E₂ (PGE₂) synthesis. PGE₂ elevates the hypothalamic thermoregulatory set-point, triggering systemic heat conservation (peripheral vasoconstriction) and heat production (shivering thermogenesis). Distinguished from hyperthermia, where body temperature rises uncontrollably above the set-point due to failed thermoregulation (e.g., heat stroke, malignant hyperthermia).

Pyridine

A nitrogen-containing aromatic ring found in many medicines, vitamins, and biologically active compounds.

Pyridostigmine

An acetylcholinesterase inhibitor used mainly for myasthenia gravis and selected neuromuscular disorders.

Pyridostigmine Bromide

Pyridostigmine bromide increases communication between nerves and muscles and is used mainly for myasthenia gravis; it may cause abdominal cramps, diarrhea, or excess secretions.

Pyridoxal Phosphate

The active coenzyme form of vitamin B6, required for amino-acid metabolism, neurotransmitter synthesis, and heme production.

Pyridoxine

A form of vitamin B6 involved in metabolism and nervous-system function.

Alternate Names:

Vitamin B6

Pyridoxine (Vitamin B6)

A vitamin converted to pyridoxal phosphate, a coenzyme for transamination, decarboxylation, and other amino-acid reactions. Deficiency can cause neuropathy, dermatitis, seizures, and sideroblastic anemia; excessive supplementation can also cause neuropathy.

Alternate Names:

Vitamin B6

Pyrimethamine

An antiparasitic medicine that inhibits folate metabolism and is used for selected protozoal infections.

Pyrimidine

A six-membered nitrogen-containing ring that forms the bases cytosine, thymine, and uracil in nucleic acids.

Pyrimidine Analog

A medicine or chemical resembling a pyrimidine nucleotide that can interfere with DNA or RNA synthesis, often for cancer or antiviral treatment.

Pyrimidine Dimer

A DNA lesion formed when adjacent pyrimidine bases bond abnormally after ultraviolet exposure, potentially causing mutations if unrepaired.

Pyrimidine Nucleosides

Nucleosides containing cytosine, thymine, or uracil bases linked to a sugar, serving as components or precursors of DNA and RNA.

Pyrimidine Nucleotides

Nucleotides containing pyrimidine bases, including cytidine, uridine, thymidine, and their phosphorylated forms.

Pyrimidines

Nitrogenous bases containing a single six-membered heterocyclic ring, serving as structural building blocks for nucleic acids. The primary pyrimidines are cytosine (C), thymine (T, present in DNA), and uracil (U, present in RNA).

Pyrogen

A pyrogen is a substance that induces fever, commonly through immune signaling; bacterial endotoxin is an important example in contaminated medicines or fluids.

Pyroglyphidae

Pyroglyphidae is a mite family that includes house-dust mites whose allergens can trigger allergic rhinitis, asthma, or atopic dermatitis.

Pyronine

Pyronine is a red fluorescent dye used in histology and research to stain RNA and certain cellular components.

Pyrophosphatase

An enzyme that hydrolyzes inorganic pyrophosphate and helps drive many biosynthetic reactions forward.

Pyropoikilocytosis

A severe inherited red-cell membrane disorder combining marked poikilocytosis, small irregular erythrocytes, thermal instability, and hemolytic anemia.

Pyroptosis

A programmed inflammatory form of cell death involving gasdermin pores, cell swelling, membrane rupture, and release of cytokines, commonly triggered by intracellular infection or inflammasome activation.

Pyrosequencing

pyrosequencing is a laboratory test, imaging method, instrument, score, or measurement used to characterize a specimen or clinical finding; interpretation depends on the method and clinical context.

Pyrosis

Heartburn, a burning discomfort behind the breastbone caused commonly by reflux of gastric contents.

Pyroxamide

Pyroxamide is an investigational histone-deacetylase inhibitor studied for anticancer effects through altered gene expression and cell growth.

Pyrroles

A group of five-membered nitrogen-containing ring compounds found in heme, bile pigments, medicines, and other biologically active molecules.

Pyrrolidonecarboxylic Acid

A cyclic amino-acid derivative, also called pyroglutamic acid, found in proteins and metabolic pathways; abnormal accumulation may occur in selected metabolic or medication-related conditions.

Pyrrolnitrin

Pyrrolnitrin is an antimicrobial compound produced by some bacteria and studied for antifungal activity and biological control.

Pyruvate

The three-carbon alpha-keto acid produced at the end of glycolysis from phosphoenolpyruvate by pyruvate kinase.

Pyruvate Decarboxylase

Pyruvate decarboxylase is an enzyme that converts pyruvate to acetaldehyde and carbon dioxide during fermentation in some organisms.

Pyruvate Dehydrogenase

Converts pyruvate to acetyl-CoA. Regulated by phosphorylation (inactivated by PDH kinase). Requires thiamine, lipoate, CoA, FAD, NAD+.

Pyruvate Kinase

Pyruvate kinase catalyzes the final ATP-producing step of glycolysis; inherited deficiency causes hemolytic anemia because red cells cannot generate enough energy.

Pyruvate Kinase Blood Test

A pyruvate kinase blood test measures the enzyme in red blood cells and may help evaluate pyruvate kinase deficiency, an inherited cause of hemolytic anemia.

Pyruvate Kinase Deficiency

An inherited red-cell glycolytic defect causing chronic hemolytic anemia of variable severity. Manifestations include jaundice, splenomegaly, gallstones, fatigue, and complications of iron overload.

Pyuria

Presence of white blood cells or pus in urine, often indicating urinary-tract inflammation or infection.

Alternate Names:

Artificial pacemaker

PYY

PYY, or peptide YY, is a gut hormone released after eating that reduces appetite and slows gastrointestinal motility.

Q

Q Arm Of A Chromosome

The long arm of a chromosome, designated q in cytogenetic notation. It contains genes and regulatory regions, and deletions, duplications, or rearrangements of the q arm can cause genetic disease.

Q Fever

A zoonotic disease caused by *Coxiella burnetii*, an obligate intracellular Gram-negative bacterium. Q fever is transmitted to humans primarily by inhalation of contaminated aerosols from infected livestock (cattle, sheep, goats) and their products (placenta, birth fluids, urine, feces, milk). Incubation: 2–3 weeks.

Q Wave

The negative deflection that follows the P wave in some electrocardiographic leads, representing initial ventricular depolarization.

Q-T Interval

The ECG interval from the start of ventricular depolarization to the end of repolarization; it is corrected for heart rate when assessing arrhythmia risk.

QALY

Quality-adjusted life year, a measure combining the length of life with a value assigned to health-related quality of life during that time.

Qigong

Qigong is a traditional mind–body practice combining controlled movement, breathing, and attention; it may support relaxation or function, but evidence and safety vary and it should not replace indicated

medical care.

QRS Complex

The deflection representing ventricular depolarization, consisting of a Q wave (initial downward deflection), R wave (upward deflection), and S wave (downward deflection after R). Normal duration is 0.06–0.10 seconds (1.5–2.5 small boxes). Widened QRS (>0.12 s) indicates bundle branch block, ventricular rhythm, or hyperkalemia.

QT Interval

The time from the onset of the QRS complex to the end of the T wave, representing total ventricular depolarization and repolarization. Normal QT is 0.35–0.44 seconds. The corrected QT (QTc) adjusts for heart rate using Bazett's formula: $QTc = QT / \text{square root of RR interval}$.

QT Prolongation

An electrocardiographic finding characterized by prolongation of the QT interval, representing the time from ventricular depolarization to repolarization, which predisposes to the potentially fatal arrhythmia torsades de pointes. The corrected QT interval should be less than 440 milliseconds in men and 460 milliseconds in women.

Quad Screen (Maternal Serum Screening)

Second-trimester prenatal screening (15–22 weeks) for fetal aneuploidy and neural tube defects. Measures 4 markers: AFP (alpha-fetoprotein: elevated in NTD, low in Down syndrome), hCG (elevated in Down syndrome), uE3 (unconjugated estriol: low in Down syndrome), and inhibin A (elevated in Down

syndrome).

Alternate Names:

Maternal Serum Screening

Quadrant

One of four sections of the abdomen used for clinical description. RUQ: liver, gallbladder, duodenum, right kidney. LUQ: stomach, spleen, left kidney, pancreas tail. RLQ: appendix, cecum, right ovary. LLQ: sigmoid colon, left ovary.

Quadrantanopia

Loss of vision in one quarter of the visual field, usually caused by a lesion affecting specific portions of the optic pathways.

Quadrantectomy

Surgical removal of one quadrant of an organ, most commonly a breast-conserving operation removing a breast tumour with a surrounding margin of tissue. The exact procedure and need for additional treatment depend on pathology and clinical staging.

Quadriceps

The four muscles on the front of the thigh that extend the knee; the rectus femoris also assists hip flexion. The quadriceps tendon connects them to the patella and contributes to the extensor mechanism.

Quadriceps Femoris

The four-headed muscle group on the anterior thigh consisting of rectus femoris and the three vasti. It extends the knee; rectus femoris also flexes the hip, and the group contributes to standing, walking, and shock absorption.

Quadriceps Femoris Muscle

Four-muscle anterior thigh group: rectus femoris, vastus lateralis, vastus medialis, vastus intermedius, all inserting on the tibial tuberosity via the patellar ligament. Innervated by femoral nerve (L2-L4). Primary knee extensor for walking, running, rising from seated. VMO is important for patellar tracking.

Quadriceps Injury

Damage to the quadriceps muscle or tendon caused by sudden force, overuse, or direct trauma. It may cause front-of-thigh pain, bruising, weakness, or difficulty extending the knee. Severe pain, a palpable defect, or inability to bear weight requires assessment for a significant tear or tendon rupture.

Quadriceps Tendinitis

Overuse-related inflammation or degeneration of the quadriceps tendon, causing pain above the kneecap that worsens with jumping, running, or resisted knee extension.

Quadriceps Tendon Rupture

A tear of the tendon connecting the quadriceps muscle to the patella, causing pain, swelling, and inability to actively extend the knee.

Quadriceps Tendon Ruptures

A tear of the tendon connecting the quadriceps muscle to the patella, causing weakness or inability to extend the knee. Complete tears often require surgical repair; treatment depends on the extent of injury and the person's functional needs.

Quadripareisis

Weakness affecting all four limbs, usually resulting from cervical spinal-cord disease, brain injury, neuromuscular disease, or another disorder of the central or peripheral nervous system.

Quadriplegia

Paralysis of all four limbs and trunk, resulting from a spinal cord lesion at the cervical level or a bilateral brainstem lesion. Common causes include traumatic spinal cord injury, stroke, multiple sclerosis, spinal cord tumors, and demyelinating disease.

Quadruple Screen Test

The quadruple screen measures maternal blood markers to estimate risk of selected fetal chromosomal or neural-tube conditions; it is a screening test, not a diagnosis.

Quadruplet Pregnancy

A pregnancy involving four fetuses. It carries increased risks of preterm birth, fetal growth restriction, hypertensive and metabolic complications, and maternal morbidity and therefore requires specialized multidisciplinary antenatal care.

Quadruplets

Four offspring born from the same pregnancy or four infants born in a multiple birth.

Quality

In healthcare, quality is the degree to which care is safe, effective, timely, equitable, efficient, and centered on patient needs, measured against defined standards or outcomes.

Quality Assurance

Planned activities used to verify that healthcare services meet defined standards of safety, quality, and performance.

Quality Control

Daily, weekly, monthly tests ensure system performance. Uniformity, energy, sensitivity, spatial resolution, count rate performance.

Quality Control (Laboratory)

Procedures using control materials, calibration, equipment checks, and documented review to ensure that a laboratory method remains accurate and precise during routine testing.

Alternate Names:

Laboratory

Quality Improvement

Systematic efforts to improve healthcare. Methods: PDSA cycles, Lean, Six Sigma. Outcomes: safety, effectiveness, patient-centeredness.

Quality Of Life

A person's perceived physical, psychological, social, and functional well-being.

Quality-Adjusted Life Year

A health-economic measure combining length of life with health-related quality of life on a scale where 1 represents full health and 0 represents death. It is used to compare cost-effectiveness across interventions.

Quantification

Measurement of radiotracer uptake or clearance. SUV, clearance rates (renal), ejection fraction (cardiac). Essential for treatment monitoring.

Quantitative Bence-Jones Protein Test

A quantitative Bence-Jones protein test measures free monoclonal light chains in urine, supporting evaluation and monitoring of plasma-cell disorders such as multiple myeloma.

Quantitative Computed Tomography

CT-based imaging that measures bone mineral density as well as showing bone structure.

Quantitative Digital Radiography

Digital X-ray imaging that uses measured pixel data for assessment of anatomy, density, or change over time.

Quantitative Immunoglobulins

Quantitative immunoglobulin testing measures classes of antibodies in blood and helps evaluate immune deficiency, chronic inflammation, monoclonal gammopathy, or abnormal immune activation.

Quantitative Nephelometry Test

Quantitative nephelometry estimates specific proteins by measuring light scattered by immune complexes, supporting assessment of immunoglobulins, complement, and other serum proteins.

Quantitative PET

The measurement or estimation of radiotracer concentration and tissue kinetics from PET data, using measures such as standardized uptake value or compartmental modeling.

Quantum Dot

A quantum dot is a nanoscale semiconductor particle whose size-dependent electronic properties produce characteristic light emission; medical research uses include imaging probes, biosensors, and targeted delivery systems.

Quaque

Quaque is Latin for every or each and appears in historical prescription abbreviations; modern instructions should use plain language.

Quaque Die

Quaque die means every day or once daily; clear modern medication directions should spell this out.

Quarantine

Separation or restriction of movement of people who have been exposed to an infectious disease while monitoring whether they become ill. It differs from isolation of people who are already infected.

Quartan Fever

A fever pattern in which fever attacks recur approximately every 72 hours, classically associated with infection by *Plasmodium malariae*.

Quasidominant

Describing an inheritance pattern in which a recessive trait appears in successive generations because affected individuals reproduce with carriers or because of a high disease frequency. It can resemble dominant inheritance in a pedigree.

Quassins

Bitter triterpenoid compounds found in *Quassia* species and responsible for much of the plant's bitter taste; they have been studied experimentally for insecticidal and biological effects but have no established routine therapeutic use.

Quater In Die

Quater in die means four times a day; explicit timing is preferable to this Latin abbreviation.

Queckenstedt'S Test

A historical test in which jugular-vein compression was used during lumbar puncture to assess cerebrospinal-fluid pressure transmission; it is rarely used today.

Quercetin

A flavonoid polyphenol found in many fruits, vegetables, and teas; it is marketed in supplements and studied for antioxidant and anti-inflammatory effects, but clinical benefits vary and interactions with medicines are possible.

Questionnaire

A questionnaire is a structured set of written or electronic questions used to collect information about symptoms, exposures, behaviors, function, quality of life, or patient-reported outcomes; validity depends on its design and population.

Quetiapine Fumarate

The fumarate salt of quetiapine, an atypical antipsychotic used for schizophrenia, bipolar disorder, and selected adjunctive treatment of major depressive disorder. Sedation, metabolic effects, orthostatic hypotension, and movement-related adverse effects require monitoring.

Quick Relievers

Fast-acting medicines that relax airway muscle and rapidly ease bronchoconstriction.

Alternate Names:

Rescue medicines

Quickening

The first perception of fetal movement by the mother, typically felt between 16-20 weeks in multiparous women and 18-22 weeks in nulliparous women. Described initially as a subtle fluttering sensation that becomes stronger and more regular as pregnancy progresses.

Quiescence

A temporary state of cellular or physiologic inactivity or reduced activity. Some cells, including certain

stem cells and latent infectious agents, may remain quiescent and later become active.

Quiescent

Inactive or dormant state of disease. Example: quiescent ulcerative colitis (no symptoms but histologic inflammation may persist).

Quiet Sleep

Non-REM sleep, during which brain and body activity generally slow.

Alternate Names:

Non-REM sleep

Quinalphos

An organophosphate insecticide that inhibits acetylcholinesterase. Exposure can cause a cholinergic toxidrome with sweating, salivation, vomiting, muscle fasciculations, weakness, seizures, and respiratory failure and requires urgent toxicology care.

Quinapril

Quinapril is an angiotensin-converting-enzyme inhibitor used for hypertension and heart failure; cough, hyperkalemia, renal-function changes, angioedema, and fetal toxicity are important risks.

Quinazoline

A bicyclic nitrogen-containing heterocyclic chemical scaffold used in medicinal chemistry. Quinazoline derivatives include compounds with kinase-inhibitory and other pharmacologic activities.

Quincke'S Disease (Angioneurotic Oedema)

Sudden swelling of deeper layers of the skin or mucosa, commonly affecting the lips, face, tongue, throat, or bowel; airway involvement is an emergency.

Alternate Names:

Angioneurotic Oedema

Quincke'S Pulse

Visible capillary pulsation in the nail bed or skin, often associated with a wide pulse pressure and

severe aortic regurgitation.

Quincke'S Sign

A sign of severe aortic regurgitation in which alternating flushing and blanching of the nail bed or skin can be seen with each heartbeat.

Quinghaosu

Another name for artemisinin, an antimalarial compound derived from *Artemisia annua* and used in combination therapies.

Quinidine

A class IA antiarrhythmic medicine that blocks cardiac sodium channels and prolongs repolarization. It is used selectively because it can cause QT prolongation, torsades de pointes, thrombocytopenia, and other adverse effects.

Quinine

An alkaloid derived from cinchona bark, with antimalarial, antipyretic, and skeletal muscle relaxant properties. Quinine is a blood schizonticide with activity against all *Plasmodium* species, but resistance has emerged in *P. falciparum*.

Quinolinic Acid

Quinolinic acid is a neuroactive intermediate in the kynurenine pathway and an agonist at NMDA receptors; excessive accumulation may contribute to excitotoxic neuronal injury and occurs in some inflammatory or neurologic conditions.

Quinolone

A class of synthetic antibacterial drugs that inhibit bacterial DNA gyrase or topoisomerase IV. Examples include ciprofloxacin and levofloxacin.

Quinolones

A class of synthetic antibiotics that inhibit bacterial DNA gyrase or topoisomerase IV.

Quinone Metabolism

Quinone metabolism is the enzymatic reduction and oxidation of quinones, helping cells process

these compounds but sometimes generating reactive oxygen that can damage tissues.

Quinones

Oxidized aromatic compounds involved in electron-transfer reactions and redox chemistry. Some quinones have biological or pharmacologic roles, while others can generate reactive oxygen species or cause toxicity.

Quinsy

A peritonsillar abscess: a collection of pus between the tonsillar capsule and the superior pharyngeal constrictor muscle, usually developing as a complication of tonsillar infection.

Quintuplets

Five offspring born from the same pregnancy or five infants born in a multiple birth.

Alternate Names:

QCT

Quinuclidines

A group of bicyclic tertiary amines containing the quinuclidine ring system, found in some natural products and medicines. Their effects depend on the attached chemical groups and receptor targets.

Quinupristin-Dalfopristin

A streptogramin antibiotic combination that inhibits bacterial protein synthesis. It is used intravenously for selected serious gram-positive infections, including some resistant *Enterococcus faecium* infections, when appropriate alternatives are unavailable.

Quipazine

An experimental serotonergic compound used mainly in pharmacologic and neurobehavioral research. It is not an established routine clinical treatment.

Quit Smoking (Smoking And Quitting Smoking)

Alternate index label for Smoking and Quitting

Smoking.

Quorum Sensing

Cell-density-dependent bacterial communication system in which bacteria produce and detect small signaling molecules (autoinducers—acyl-homoserine lactones in gram-negative, autoinducing peptides in gram-positive, AI-2 inter-species) to coordinate population-wide behaviors.

Quotidian Fever

A fever pattern in which fever attacks occur approximately every 24 hours; it may occur in malaria and several other infections.

R

R-Factor

The ratio of a bacterium's minimum inhibitory concentration to the susceptibility breakpoint for an antibiotic; it is used in antimicrobial susceptibility interpretation, not as a diagnosis.

Rabeprazole

Rabeprazole is a proton-pump inhibitor that irreversibly suppresses gastric $H^+/K^+-ATPase$; it treats reflux and peptic-ulcer disease and supports *Helicobacter pylori* eradication.

Rabies

Rabies is a nearly always fatal viral encephalitis transmitted through saliva, usually by an animal bite, but prompt wound care and post-exposure prophylaxis can prevent disease. infected animals. After inoculation, the virus travels to the CNS via peripheral nerves. Symptoms include hydrophobia, aerophobia, and progressive neurological deterioration. Diagnosis is made by direct fluorescent antibody testing of brain tissue.

Rabies Vaccination

Immunization used before or after exposure to reduce the risk of rabies. Post-exposure management also includes wound care and, when indicated, rabies immunoglobulin.

Rabies Virus

The virus that causes rabies, a usually fatal viral infection of the central nervous system transmitted most often through the saliva of an infected animal.

Race

Race is a social classification that can influence lived experience and health through genetics in limited contexts, discrimination, environment, and access to care; it is not a precise proxy for biology.

Raclopride

Raclopride is a research radioligand that binds dopamine D2 receptors and is used in brain-imaging studies of dopamine signaling.

Radial

Radial means relating to the radius bone or the thumb side of the forearm, hand, or a structure arranged like radii from a center.

Radial Artery

The smaller terminal branch of the brachial artery, arising at the cubital fossa, running laterally along the radius to the wrist (palpable at the anatomical snuffbox) and hand. Supplies the lateral forearm and the thumb side of the hand; forms the deep palmar arch.

Radial Vein

Deep veins accompanying the radial artery that drain the lateral forearm and join the ulnar veins to form the brachial veins.

Radial Artery Thrombosis

Formation of a thrombus in the radial artery, often at the site of arterial catheterization or in the setting of a hypercoagulable state or vasculitis. May cause hand ischemia with pain, pallor, and decreased pulse. Treatment includes anticoagulation and surgical in-

tervention for limb-threatening ischemia.

Radial Club Hand

A congenital longitudinal deficiency of the radial ray, ranging from thumb hypoplasia to complete absence of the radius. The hand is radially deviated due to the absent radial support, with the carpus displaced proximally and radially.

Radial Head Subluxation

A common pediatric elbow injury (popularly known as nursemaid's elbow or pulled elbow) occurring primarily in toddlers and children aged 1 to 4 years. Caused by sudden axial traction on an extended, pronated forearm (such as when lifting or pulling a child by the hand), which allows the flexible annular ligament to slip over the head of the radius and become entrapped in the radiohumeral joint. The child typically holds the arm adducted and pronated, refusing to bend or use the extremity; reduction is achieved by gentle hyperpronation or supination-flexion maneuvers with immediate symptom relief.

Alternate Names:

Nursemaid's Elbow (Radial Head Subluxation); Pulled Elbow; Annular Ligament Displacement

Radial Nerve

From posterior cord (C5-T1), the plexus's largest branch. In the arm runs in the radial groove with the profunda brachii artery, innervating triceps and anconeus. At the elbow divides into superficial (sensory to dorsal hand) and deep (posterior interosseous) branches. Deep branch innervates forearm/hand extensors.

Radial Nerve Dysfunction

Impaired function of the radial nerve causing weakness of wrist or finger extension and sensory changes over part of the posterior arm or hand; causes include compression, trauma, and systemic neuropathy.

Radial Nerve Palsy

Dysfunction of the radial nerve causing weakness of wrist and finger extension, often producing wrist drop, with sensory loss over part of the back of the

hand in some cases.

Radial Tunnel Syndrome

Compression of the radial nerve near the elbow, causing arm pain, tingling, or weakness.

Radiation

Energy transmitted as particles or electromagnetic waves, including X-rays and gamma rays.

Radiation Biology

The study of how ionizing or nonionizing radiation affects cells, tissues, organisms, and populations.

Radiation Contamination

Unwanted radioactive material on a person, surface, object, or body site. Contamination may be removable or fixed and requires containment, monitoring, decontamination, and documentation according to radiation-safety procedures.

Radiation Dosage

The amount of radiation given during an imaging test or radiation treatment. The dose is planned to obtain the needed benefit while limiting unnecessary exposure.

Radiation Dose

The amount of ionizing radiation absorbed or received by a person or tissue. Dose varies with the examination, equipment, protocol, and patient size; imaging should be justified and optimized to obtain the needed information with the lowest reasonable exposure.

Radiation Emergency

An event involving actual or suspected harmful radiation exposure or contamination that requires urgent protective and medical measures.

Radiation Enteritis

Inflammation and injury of the small or large intestine caused by therapeutic ionizing radiation. It may be acute with diarrhea, cramping, and nausea or delayed with strictures, malabsorption, bleeding,

fistula, or obstruction.

Radiation Exposure

Contact with radiation from a medical test, treatment, workplace, or other source. The possible effects depend on the type, amount, and duration of exposure.

Radiation Fibrosis

Permanent scarring and stiffening of tissue caused by radiation injury, which may develop months or years after treatment. It can affect the skin, lungs, bowel, or other organs and may cause pain, reduced function, or shortness of breath.

Radiation Field

The area of the body targeted by a radiation treatment or exposed to a radiation beam.

Radiation Leukemia

Radiation leukemia is leukemia that develops after significant exposure to ionizing radiation; it is a late treatment- or exposure-related complication.

Radiation Monitoring

Measurement of radiation levels or individual exposure using devices such as personal dosimeters and area monitors.

Radiation Necrosis

Irreversible tissue injury caused by ionizing radiation, often developing months or years after treatment. In the brain it may produce edema, seizures, headache, or focal deficits and can resemble recurrent tumour on imaging.

Radiation Oncologist

A physician who plans and oversees radiation treatment for cancer and selected noncancerous conditions.

Radiation Oncology

A medical specialty that uses ionizing radiation to treat cancer. Treatment may aim to destroy or control tumors, prevent cancer from returning, or

relieve symptoms.

Radiation Physics

The study and application of physical principles governing radiation production, detection, imaging, therapy, and protection.

Radiation Pneumonitis

An inflammatory lung injury occurring weeks to months after thoracic radiation. It may cause dry cough, fever, dyspnea, and hypoxemia within the irradiated field and can progress to radiation fibrosis.

Radiation Poisoning

Illness caused by substantial exposure to ionizing radiation, potentially producing skin injury, bone-marrow suppression, gastrointestinal symptoms, or organ failure.

Radiation Pregnancy Screening

Assessment of pregnancy status before procedures involving ionizing radiation or radioactive materials when pregnancy is possible. The decision to proceed requires clinical justification, risk optimization, and consideration of gestational age and fetal dose.

Radiation Proctitis

Inflammation and injury of the rectum after pelvic radiation therapy, causing rectal pain, urgency, diarrhea, mucus, or bleeding.

Radiation Protection

The practice of keeping radiation exposure as low as reasonably achievable while obtaining the required diagnostic or therapeutic result. It relies on justification, optimization, time reduction, distance, shielding, and appropriate contamination control.

Radiation Recall

An inflammatory reaction that occurs in previously irradiated skin or mucosa after administration of certain chemotherapeutic agents, particularly antimetabolites and anthracyclines. It manifests as erythema, desquamation, or mucositis in the previously irradiated field, mimicking acute radiation

dermatitis.

Radiation Safety

Principles: time, distance, shielding. ALARA (As Low As Reasonably Achievable). Radioactive materials handled in designated areas. Waste disposal per regulations. Pregnancy screening for women of childbearing age.

Radiation Sickness

Acute radiation syndrome caused by high-dose whole-body or substantial partial-body exposure to ionizing radiation. It may produce nausea, vomiting, marrow failure, infection, bleeding, gastrointestinal injury, neurovascular collapse, and death depending on dose.

Radiation Therapy

Use of ionizing radiation to kill cancer cells. Types: external beam, brachytherapy, stereotactic. Mechanisms: DNA damage, cell death. Side effects: fatigue, skin reactions, site-specific effects.

Radiation Therapy - Skin Care

A medical procedure, treatment, device, or care topic related to radiation therapy - skin care; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Radiation Tolerance

The amount of radiation that a tissue or organ can receive before injury becomes likely. It varies with the tissue, dose, treatment schedule, and previous radiation.

Radiation-Induced Angiosarcoma

A rare, aggressive soft tissue sarcoma arising in previously irradiated tissue, typically with a long latency period. It carries a poor prognosis due to local recurrence and metastatic potential.

Radiation-Recall Dermatitis

An inflammatory skin reaction that occurs in a previously irradiated area after exposure to a triggering substance, often months or years later. The area

may become red, swollen, painful, or blistered and requires clinical evaluation.

Radical Cystoprostatectomy

Radical cystoprostatectomy removes the urinary bladder, prostate, and nearby structures, usually to treat invasive bladder cancer in men.

Radical Hysterectomy

An operation that removes the uterus and cervix together with surrounding parametrial tissues and the upper vagina, usually with pelvic lymph-node assessment or removal. It is used mainly for selected invasive cervical cancers and is more extensive than a simple or total hysterectomy.

Radical Lymphadenectomy

Radical lymphadenectomy is extensive surgical removal of lymph nodes and surrounding tissue to assess or treat cancer spread.

Radical Mastectomy

Radical mastectomy removes the breast, chest muscles, and many underarm lymph nodes; it is now rarely used because less extensive cancer surgery is often effective.

Radical Nephrectomy

Radical nephrectomy removes an entire kidney and sometimes nearby tissues, usually to treat a kidney tumor.

Radical Prostatectomy

Surgery that removes the entire prostate, usually with nearby seminal vesicles.

Radicular Cyst

An inflammatory odontogenic cyst arising at the apex of a nonvital tooth, usually from chronic pulpal infection and apical periodontitis.

Radiculitis

Inflammation or irritation of a spinal nerve root, causing radiating pain, paresthesia, sensory loss, weakness, or reflex change in a dermatomal or my-

otomal distribution. Causes include infection, inflammatory disease, disc pathology, and compression.

Radiculopathy

Dysfunction of a spinal nerve root, usually from disc herniation, narrowing, or inflammation. It may cause shooting pain, tingling, numbness, weakness, or altered reflexes along the affected nerve pathway.

Radio-Opaque

Blocking X-rays and appearing relatively white on a radiograph; radiopaque is the modern spelling.

Radioactive Iodine

Used for thyroid imaging and treatment. I-131 emits beta particles (therapy) and gamma rays (imaging). Half-life: 8 days. Used for hyperthyroidism (Graves, toxic nodules) and thyroid cancer ablation. I-123 for diagnostics (shorter half-life, pure gamma).

Radioactive Iodine Ablation

Treatment of differentiated thyroid carcinoma using iodine-131 to destroy residual thyroid tissue after total thyroidectomy. Indications include intermediate and high-risk disease for remnant ablation and treatment of residual or recurrent disease. Patients must be hypothyroid or receive recombinant TSH to maximize iodine uptake.

Radioactive Iodine Uptake

A test measuring the percentage of administered radioactive iodine taken up by the thyroid over time, used to evaluate hyperthyroidism and selected thyroid disorders.

Radioactive Iodine Uptake Test

A nuclear medicine test that measures how much radioactive iodine the thyroid gland absorbs. It helps evaluate hyperthyroidism and some thyroid disorders.

Radioactive Isotopes

Unstable forms of elements that emit radiation and are used in nuclear imaging, treatment, or research.

Radioactivity

Spontaneous decay of unstable atomic nuclei with emission of particles or electromagnetic radiation.

Radioallergosorbent Test

A blood test used to detect allergen-specific IgE antibodies.

Radiobiology

Study of radiation effects on biological systems. DNA damage, cell death, repair mechanisms.

Radiocarpal Joint

The wrist joint between the distal radius and articular disc proximally and the scaphoid, lunate, and triquetrum distally. It permits flexion, extension, radial deviation, ulnar deviation, and circumduction.

Radiochemical Purity

The fraction of total radioactivity present in the intended chemical form of a radiopharmaceutical. Impurities may alter biodistribution, reduce target uptake, and increase radiation dose to nontarget tissues.

Radiochemistry

The chemistry of radioactive substances and compounds. In medicine it includes making and labeling radiopharmaceuticals, studying radioactive decay, separating products, and checking radiochemical purity and safety.

Radiocontrast

A substance administered during imaging to make selected tissues, vessels, or body spaces more visible.

Radiodermatitis

Radiodermatitis is skin injury caused by ionizing radiation, ranging from redness and hair loss to chronic ulceration, fibrosis, or cancer risk.

Radioembolisation

Alternate index label; see *Radioembolization*.

Radioembolization

Intra-arterial delivery of radioactive microspheres (yttrium-90) to liver tumors through the hepatic artery, used for hepatocellular carcinoma and liver metastases.

Radiofrequency

Electromagnetic energy used in some diagnostic and therapeutic medical procedures.

Alternate Names:

RF energy

Radiofrequency Ablation

A procedure that uses high-frequency electrical energy to heat and destroy selected tissue, often under imaging guidance.

Radiofrequency Ablation For Chronic Pain

A procedure that applies radiofrequency energy to selected sensory nerves or other targets to interrupt pain signaling for a limited period in carefully selected chronic pain conditions.

Radiograph

A permanent visual diagnostic projection image produced by passing a collimated beam of ionizing X-radiation (X-rays) through an anatomical region onto a digital detector or photographic emulsion plate. Different biological tissues attenuate X-ray photons proportionally to their physical density and effective atomic number: dense cortical bone absorbs high amounts of radiation, appearing radiopaque (white), whereas aerated lung tissue absorbs minimal radiation, appearing radiolucent (black), with soft tissues, fluids, and fat presenting intermediate shades of gray. Serves as the foundational, rapid, first-line imaging modality in emergency trauma evaluation, chest pathology, and skeletal traumatology.

Alternate Names:

X-ray

Radiographer

A trained professional who performs diagnostic imaging examinations or delivers radiotherapy, de-

pending on specialty and jurisdiction.

Radiography

Production of images using X-rays or other ionizing radiation.

Radioimmunoassay

Sensitive immunoassay using radiolabeled antigens or antibodies. Competitive binding principle. Used for hormone measurement (thyroid, reproductive), tumor markers, drug levels. Largely replaced by non-radioactive methods (ELISA) but still used for some assays.

Radioimmunodetection

Detection of a substance using an antibody linked to a radioactive tracer, allowing localization or measurement with imaging or laboratory methods.

Radioimmunoguided Surgery

Radioimmunoguided surgery uses a radiolabeled antibody or tracer to help surgeons locate tumor tissue during an operation.

Radioimmunotherapy

Radioimmunotherapy attaches a radioactive substance to an antibody so radiation is delivered preferentially to cells carrying a specific target, often cancer cells.

Radioiodine Therapy

A medical procedure, treatment, device, or care topic related to radioiodine therapy; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Radioisotope

An unstable isotope that emits radiation as it decays; radioisotopes are used in imaging, treatment, research, and industrial applications.

Radioisotope (Radionuclide)

An unstable isotope that emits radiation as it decays; radioisotopes are used in medical imaging, diagnosis, and selected treatments.

Alternate Names:

Radionuclide

Radiologic Health

Radiologic health concerns protection from unnecessary ionizing radiation while preserving the diagnostic or therapeutic benefit of medical imaging.

Radiologic Response

The change in the size, number, or characteristics of lesions on imaging after treatment, assessed using defined criteria such as response, stability, or progression.

Radiologic Technologist

A trained health professional who performs selected medical-imaging examinations and helps protect patients from unnecessary radiation exposure.

Radiologic-Pathologic Correlation

Integration of imaging findings with gross and microscopic pathology to explain lesion location, extent, tissue composition, and sampling adequacy. It is important for discordant biopsy and imaging results.

Radiologist

A physician trained to interpret medical images and, in some specialties, perform image-guided procedures.

Radiology

The medical specialty using imaging to diagnose disease and guide procedures.

Radiology Report

Document describing imaging findings and interpretation. Structure: indication, technique, findings, impression. Clear, concise language essential. Critical findings require immediate communication.

Radiology-Patient Identification

Verification of the correct patient, examination, body site, indication, pregnancy status when relevant, allergies, renal function, and prior imaging before image acquisition or intervention. It is a core

diagnostic and procedural safety step.

Radiolucent

Describing a material that allows relatively greater passage of X-rays and therefore appears darker on a radiograph or CT image. Air is highly radiolucent; some stones, foreign bodies, and lesions may also be radiolucent.

Radiometry

Measurement of electromagnetic radiation, including its intensity, energy, or spectral distribution, for imaging, treatment, or laboratory purposes.

Radiomimetic

Radiomimetic describes a chemical or treatment that produces cellular or DNA damage resembling the effects of ionizing radiation.

Radionuclide

A radioactive isotope used as a tracer or therapeutic agent.

Radionuclide Angiography

A nuclear-medicine test that uses a radioactive tracer to show blood flow through the heart or blood vessels.

Radionuclide Cisternogram

A nuclear medicine study tracking a radiotracer in cerebrospinal fluid to evaluate CSF circulation and selected suspected leaks or hydrocephalus; it is now rarely used.

Radionuclide Cystogram

A nuclear medicine examination that tracks radio-labeled urine to detect vesicoureteral reflux or evaluate bladder emptying.

Radionuclide Generator

A device that produces a short-lived radioactive substance from a longer-lived parent substance for use in nuclear-medicine tests or treatment.

Radionuclide Heart Scan

A nuclear medicine study that uses a radioactive tracer to evaluate myocardial blood flow, heart muscle viability, or ventricular function.

Radionuclide Imaging

Imaging that uses a small amount of radioactive material to show the structure or function of organs and tissues.

Radionuclide Imaging Or Scan

Imaging that detects radiation from a tracer administered to the patient.

Alternate Names:

Nuclear scan; scintigraphy

Radionuclide Therapy

Internal radiation therapy using radiopharmaceuticals. Examples: I-131 (thyroid), Lu-177 DOTATATE (neuroendocrine tumors), Lu-177 PSMA (prostate cancer), Ra-223 (bone metastases), Y-90 microspheres (liver cancer).

Radionuclides

Radioactive atoms used as tracers in nuclear medicine imaging or treatment.

Radionuclidic Purity

The fraction of radioactivity arising from the intended radionuclide rather than contaminating radionuclides. It is important because contaminants may change radiation type, energy, half-life, dosimetry, and patient risk.

Radiopaque

Describing a material that attenuates X-rays strongly and therefore appears relatively white on a radiograph or CT image. Bone, metal, and some contrast agents are radiopaque.

Radiopharmaceutical

Radioactive compound used for diagnostic or therapeutic purposes. Components: radionuclide + targeting molecule. Selectivity determines biodistribution and image quality.

Radiopharmaceutical Extravasation

Inadvertent leakage of a radiopharmaceutical from the injection vessel into surrounding tissue. It can reduce diagnostic accuracy and, with therapeutic agents, cause local radiation injury; the event requires documentation and dose assessment.

Radiopharmaceutical Labeling Kit

A prepared set of reagents used to attach a radionuclide, commonly technetium-99m, to a compound for diagnostic imaging.

Radiopharmaceutical Quality Control

Testing performed to verify identity, radionuclidic purity, radiochemical purity, pH, sterility, endotoxin limits, and other release specifications before clinical administration. Requirements depend on the preparation and jurisdiction.

Radiopharmacy

Preparation and dispensing of radiopharmaceuticals. Good manufacturing practice required. Sterility, pyrogen testing, quality control.

Radioprotective Agent

A radioprotective agent reduces tissue injury from ionizing radiation by limiting DNA damage, scavenging radicals, or supporting repair.

Radiosensitization

Increased susceptibility of cells or tissues to injury from ionizing radiation.

Radiosurgery

Radiosurgery delivers highly focused radiation to a small target, usually in the brain or body, without making a conventional surgical incision.

Radiotherapy

Treatment using ionizing radiation to destroy or control abnormal cells, especially cancer.

Radiotherapy Bolus

A radiotherapy bolus is a material placed on the skin to bring the radiation dose closer to the surface

during selected treatments.

Radiotherapy Dosage

Radiotherapy dosage is the amount of ionizing radiation prescribed for a treatment target, usually divided into carefully planned fractions.

Radiotracer

A compound containing a radioactive nuclide that follows a biologic pathway and permits in-vivo imaging or measurement. Its distribution reflects the target physiology, receptor, enzyme, transporter, or tissue process.

Radium

A radioactive chemical element; radium exposure can cause radiation injury and cancer.

Radium-223

Alpha emitter, half-life 11.4 days. Used for bone metastases from prostate cancer. Mimics calcium, targets bone turnover sites.

Radius

Lateral forearm bone in the anatomical position. Proximal end: head, neck, radial tuberosity. Shaft. Distal end: styloid process, articular surfaces. The radial head articulates with the capitulum and ulnar notch, allowing pronation/supination.

Radius Fracture

A break in the radius, one of the two bones of the forearm, often occurring near the wrist after a fall onto an outstretched hand.

Radon

A naturally occurring radioactive gas that can accumulate indoors and increase lung-cancer risk when inhaled over time.

Raffinose

Raffinose is a plant oligosaccharide that humans digest poorly and that can be fermented by gut bacteria, causing gas in some people.

Ragged Red Fibers

A hallmark histopathologic finding in skeletal muscle tissue characteristic of mitochondrial myopathies (including MELAS, MERRF, Kearns-Sayre syndrome, and CPEO), visualized using modified Gömöri trichrome staining. Reflects the massive, compensatory subsarcolemmal and intermyofibrillar proliferation and accumulation of abnormal, structurally disordered mitochondria in response to defective oxidative phosphorylation, producing a distinct, irregularly contoured, bright reddish-purple rim and speckled granular appearance beneath the sarcolemma.

Alternate Names:

RRF; Ragged-Red Muscle Fibers

Rahnella

Rahnella is a genus of environmental gram-negative bacteria that rarely causes opportunistic human infection.

Rahnella Aquatilis

Rahnella aquatilis is an environmental water-associated bacterium that can rarely cause bloodstream or other opportunistic infection.

Raised Hemidiaphragm

A raised hemidiaphragm is elevation of one side of the diaphragm, caused by abdominal pressure, lung volume loss, phrenic-nerve dysfunction, or diaphragmatic disease.

Rale

A rale, or crackle, is an abnormal discontinuous breath sound produced by fluid, collapsed airways, or opening of small airways during breathing.

Rale (Rhonchus)

An abnormal sound heard through a stethoscope during breathing, caused by air moving through fluid, mucus, narrowed airways, or abnormal lung tissue.

Alternate Names:

Rhonchus

Raloxifene

A selective estrogen-receptor modulator used mainly to reduce fracture risk in selected postmenopausal people.

Raloxifene Hydrochloride

Raloxifene hydrochloride is a selective estrogen-receptor modulator used to prevent or treat osteoporosis and reduce breast-cancer risk in selected postmenopausal women.

Ralstonia

Ralstonia is a group of environmental bacteria that can contaminate fluids or medicines and rarely cause opportunistic infection.

Ralstonia Mannitolilytica

Ralstonia mannitolilytica is an environmental bacterium associated with occasional healthcare-associated bloodstream or respiratory infections.

Ralstonia Pickettii

Ralstonia pickettii is an environmental bacterium that can contaminate water or medical products and rarely cause infection in vulnerable patients.

Raltegravir Potassium

Raltegravir potassium is an HIV integrase inhibitor used with other antiretroviral medicines to suppress HIV replication.

Raltitrexed

Raltitrexed is an antimetabolite chemotherapy medicine used for selected colorectal cancers.

Ramipril

Ramipril is an angiotensin-converting-enzyme inhibitor used for hypertension, heart failure, and cardiovascular risk reduction; cough, hyperkalaemia, renal-function changes, angioedema, and fetal toxicity are important risks.

Rampant Caries

Widespread severe caries affecting multiple surfaces. Associated with sugary drinks, xerostomia, bulimia.

Ramsay Hunt Syndrome

Reactivation of varicella-zoster virus in the geniculate ganglion causing ipsilateral facial paralysis, severe ear pain, and vesicular eruption in or around the ear or oral cavity. Hearing loss and vertigo may occur; prompt treatment is important.

Ramucirumab

A monoclonal antibody that blocks vascular endothelial growth factor receptor 2 and is used in selected advanced cancers, including gastric, colorectal, lung, and liver cancers.

Ramus

A ramus is a branch-like extension of a bone, nerve, artery, or other structure, such as the mandibular ramus.

Random

Random means determined by chance rather than a predictable pattern; randomization in research helps balance known and unknown confounding factors.

Random Allocation

Assignment of participants to study groups by chance to reduce selection bias and balance known and unknown factors.

Randomised Controlled Trial

A study in which participants are assigned by chance to interventions or controls and outcomes are compared.

Randomization Validation

Checking that participants were assigned to study groups by chance as planned and that the assignment process was recorded correctly.

Randomized Controlled Trial

An experimental study in which participants are assigned by chance to different interventions or a control condition and followed for specified outcomes.

Range Of Motion

ROM: joint movement capacity. Goniometer measurement. Restrictions from contracture, pain, swelling.

Ranibizumab

An intravitreal monoclonal antibody fragment that inhibits vascular endothelial growth factor and treats neovascular macular degeneration, diabetic macular edema, retinal vein occlusion, and related retinal disease.

Ranitidine

An H₂-receptor antagonist formerly used to reduce gastric acid; availability has been withdrawn or restricted in many countries because of impurity concerns.

Ranolazine

Ranolazine is an antianginal drug that inhibits the late sodium current in cardiac cells and reduces chronic angina without substantially lowering heart rate or blood pressure; QT prolongation and drug interactions require caution.

Ranpirnase

An investigational ribonuclease-derived anticancer medicine studied for its ability to degrade RNA inside tumor cells.

Ranson Criteria

Clinical scoring system for acute pancreatitis severity incorporating eleven parameters at two time points. Admission parameters include age, white blood cell count, glucose, LDH, and AST. Forty-eight hour parameters include hematocrit drop, BUN increase, calcium, oxygen tension, base deficit, and fluid sequestration.

RANTES

RANTES, now called CCL5, is a chemokine that attracts immune cells and participates in inflammation, antiviral defense, and immune signaling.

Ranula

A mucocele (mucus extravasation cyst) of the sublingual salivary gland, presenting as a blue-translucent, fluctuant swelling in the floor of the mouth, typically lateral to the lingual frenulum. A ranula results from trauma or obstruction of a sublingual gland duct, causing mucus accumulation in the sublingual space.

Raoultella

Raoultella is a genus of gram-negative bacteria that can cause opportunistic urinary, respiratory, wound, or bloodstream infections.

Raoultella Planticola

A gram-negative environmental bacterium that can rarely cause urinary, bloodstream, respiratory, or other opportunistic infections.

Raoultella Terrigena

A gram-negative environmental bacterium that rarely causes opportunistic human infection.

Rape

Nonconsensual sexual penetration or sexual violence; it is a serious crime and a medical, safeguarding, and trauma-care concern.

Rape (Sexual Assault)

Alternate index label for Sexual Assault.

Raphe

A seam-like line or ridge where structures join, such as the median raphe or raphe nuclei of the brainstem.

Raphe Nuclei

The raphe nuclei are brainstem groups of neurons that produce serotonin and influence mood, pain, sleep, arousal, and autonomic functions.

Rapid Eye Movement Sleep

A sleep stage marked by rapid eye movements, vivid dreaming, and high brain activity.

Alternate Names:

REM sleep

Rapid Eye Movement Sleep Behavior Disorder

Alternate index label; see *REM sleep behavior disorder*.

Rapid Gastric Emptying

Abnormally rapid movement of food from the stomach into the small intestine, sometimes called dumping syndrome.

Rapid Heartbeat

Alternate index label; see *Tachycardia*.

Rapid Plasma Reagin Test

Rapid plasma reagin test is a medical test, procedure, treatment, or clinical measure used in patient care; its purpose and interpretation depend on the clinical question.

Rapid Sequence Induction

An intubation technique designed to minimize aspiration risk in patients with full stomach.

Rapid Shallow Breathing

Rapid shallow breathing is an increased breathing rate with small tidal volumes and may signal lung disease, pain, metabolic illness, fatigue, or respiratory failure.

Rapid Strep Test

A test performed on a throat swab to detect Group A Streptococcus. It provides a rapid result, but a negative result may require confirmatory testing depending on the patient's age, symptoms, and local guidance.

Rapid-Sequence Intubation

A planned emergency airway technique using a sedative-hypnotic and a neuromuscular blocker in rapid succession to facilitate tracheal intubation while minimizing aspiration risk. Preparation includes preoxygenation, backup airway plans, hemodynamic assessment, and confirmation of tube posi-

tion.

Rapidly Progressive Glomerulonephritis

A syndrome of rapidly worsening kidney function accompanied by glomerular inflammation, often with crescent formation on biopsy. Causes include anti-glomerular-basement-membrane disease, immune-complex disease, and pauci-immune vasculitis. It requires urgent evaluation and treatment.

Rare Disease

A condition affecting relatively few people in a defined population; many rare diseases are genetic, require specialized diagnosis, and may have limited evidence or treatment options.

Rarefaction

Reduction in density or pressure, as in the low-pressure phase of a sound wave.

Ras Peptide

A small protein fragment related to Ras proteins, which help control cell growth and division. Changes in Ras signaling can contribute to cancer.

Rasburicase

A recombinant urate-oxidase enzyme that converts uric acid to the more soluble allantoin. It is used in selected patients at risk of or experiencing tumorlysis-related hyperuricemia. It can cause serious hypersensitivity or hemolysis in people with glucose-6-phosphate dehydrogenase deficiency and requires appropriate screening and monitoring.

Rash

A general term for any visible skin eruption, change in skin color, texture, or appearance.

Alternate Names:

Allergy Rash (Rash)

Rash - Child Under 2 Years

A rash in a child under 2 can result from infection, allergy, heat, irritation, or inflammation; breathing difficulty, lethargy, fever, blistering, or rapid spread needs urgent care.

Rash Heat (Heat Rash)

Alternate index label for Heat Rash.

Rash, Poison Ivy

Alternate index label; see *Poison ivy rash*.

Rashes

Alternate index label; see *Rash*.

Rasp

A roughened surgical instrument used to file, shape, or smooth bone or other hard tissue.

RAST

An older name for a blood test that measures IgE antibodies directed against specific allergens. Modern laboratories usually use newer specific-IgE immunoassays; results show sensitization and must be interpreted with symptoms and exposure history.

Alternate Names:

Radioallergosorbent test

Rastelli Procedure

A definitive surgical repair for congenital heart anomalies characterized by ventricular septal defect, systemic ventricular-arterial alignment, and severe subpulmonary or pulmonary stenosis, most commonly D-transposition of the great arteries with VSD and LVOT obstruction, double outlet right ventricle, or truncus arteriosus. Involves constructing a wide intracardiac patch baffle that tunnels oxygenated left ventricular blood across the VSD to the posteriorly situated aorta, while an extracardiac valved conduit (homograft or bioprosthetic tube graft) is placed from an incision in the free wall of the anterior right ventricle to the pulmonary artery confluence, bypassing the obstructed natural outflow tract.

Alternate Names:

Rastelli Operation; Rastelli Repair

Rat Bite Fever

A systemic bacterial infection caused by *Streptobacillus moniliformis* or *Spirillum minus*, transmitted through rat bites or contact with rat-

contaminated food or water. It presents with relapsing fever, rash, polyarthritides, and lymphadenopathy. The characteristic rash is petechial or maculopapular on the extremities.

Rat-Bite Fever

Alternate index label; see *Rat Bite Fever*. exposure to infected rodents. Fever, migratory arthralgia, vomiting, and a maculopapular or petechial rash may occur.

Ratio

A ratio compares two quantities by division, such as a measurement of risk, concentration, or body proportions.

Rationalization

A psychological defense in which a person creates plausible explanations for behavior or feelings while avoiding the underlying motive or conflict.

RATS

Alternate name; see *Robotic-Assisted Thoracoscopic Surgery*.

Rauscher Virus

A mouse leukemia retrovirus used in research to study viral oncogenesis and blood-cell cancers.

Rauwolfia Alkaloid

Rauwolfia alkaloids are plant-derived compounds that deplete or reduce sympathetic neurotransmission and were historically used for hypertension.

Ravuconazole

Ravuconazole is a triazole antifungal medicine studied and used in some settings for invasive fungal infections.

Raw Egg

Raw eggs may contain *Salmonella* and are safest when properly refrigerated, handled hygienically, and cooked unless pasteurized products are used.

Raynaud Phenomenon

Exaggerated vasospastic response of digital arteries to cold or stress, causing well-demarcated color changes (white-blue-red) in fingers or toes. Primary Raynaud: idiopathic, symmetric, no tissue damage, normal capillaries.

Raynaud Syndrome

A vasospastic disorder causing episodic digital color changes (white -> blue -> red) triggered by cold or stress, due to exaggerated vasoconstriction of digital arteries. Primary Raynaud: benign, symmetric, in young women, no underlying disease.

Raynaud'S Disease

Primary episodic vasospasm of fingers or toes triggered by cold or stress, causing color changes and numbness.

Raynaud'S Phenomenon

Episodic constriction of small arteries in the fingers or toes, causing color changes, numbness, or pain after cold exposure or emotional stress.

Alternate Names:

Raynauds Phenomenon (Raynaud's Phenomenon)
Raynauds Syndrome (Raynaud's Phenomenon)

Raynauds Phenomenon (Raynaud'S Phenomenon)

Alternate index label for Raynaud's Phenomenon.

Raynauds Syndrome (Raynaud'S Phenomenon)

Alternate index label for Raynaud's Phenomenon.

Raynaud'S Syndrome

Episodic narrowing of small blood vessels in fingers or toes triggered by cold or stress.

Razor Bumps

Inflamed bumps caused by hairs curling back into the skin after shaving or other hair removal. They are more common with coarse or curly hair and may cause itching, pain, dark marks, or scarring; avoiding very close shaving and using proper shaving

technique can reduce recurrence.

Alternate Names:

Raynaud phenomenon

Razoxane

Razoxane is a topoisomerase-II inhibitor related to dexrazoxane and studied or used historically for selected anticancer or cardioprotective purposes.

RâLe

An older term for an abnormal crackling or bubbling respiratory sound, now generally described as a crackle.

RBC Count

A measurement of the number of red blood cells in a volume of blood, interpreted with hemoglobin, hematocrit, and red-cell indices to evaluate anemia or erythrocytosis.

RBC Indices

Calculated or measured red-cell characteristics, including mean corpuscular volume, mean corpuscular hemoglobin, mean corpuscular hemoglobin concentration, and red-cell distribution width.

RBC Nuclear Scan

A nuclear medicine scan using labeled red blood cells to localize active gastrointestinal bleeding or, in selected studies, evaluate cardiac blood-pool function.

RBC Urine Test

A urine examination detecting or counting red blood cells, used to evaluate hematuria from urinary-tract infection, stones, glomerular disease, tumors, trauma, or other causes.

RCT

Randomized controlled trial, a study comparing randomly assigned intervention and control groups.

Alternate Names:

Randomized controlled trial

RDA

Recommended dietary allowance, the daily nutrient intake sufficient for nearly all healthy people in a defined group.

Alternate Names:

Recommended dietary allowance

Reabsorption

The transport of a substance from a body cavity or tubular fluid back into tissue or the bloodstream, such as renal tubular recovery of water and filtered solutes.

Reaction

A response to a stimulus, event, medicine, treatment, or interaction; clinical meaning depends on the type and severity of the response.

Reactivation Tuberculosis

Reactivation of latent infection, typically years later. Characteristically affects apical and posterior upper lobe segments. Features include chronic cough, hemoptysis, night sweats, weight loss, and cavitation.

Reactive Arthritis

An inflammatory arthritis that develops after certain genitourinary or gastrointestinal infections. It may involve the joints, eyes, skin, or urinary tract; not every person has all of these features.

Alternate Names:

Arthritis Reactive (Reactive Arthritis) Arthritis Re-
iters (Reactive Arthritis)

Reactive Attachment Disorder

A pattern of inhibited, emotionally withdrawn behavior toward adult caregivers. The child rarely or minimally seeks comfort when distressed and rarely or minimally responds to comfort when distressed. The child shows a pattern of persistently irritable and sad mood that is not limited to interactions with caregivers.

Reactive Attachment Disorder Of Infancy

Or Early Childhood

A childhood disorder characterized by markedly disturbed and developmentally inappropriate attachment behavior associated with severe insufficient caregiving, including limited comfort-seeking or response to caregivers.

Reactive Hypoglycemia

Reactive hypoglycemia is low blood glucose occurring after eating, often several hours later, with shakiness, sweating, hunger, or weakness and requiring evaluation of causes.

Reactive Oxygen Species

Highly reactive oxygen-containing molecules, such as superoxide and hydrogen peroxide, formed during normal metabolism and inflammation. Excessive amounts can damage cells, proteins, and DNA.

Reactive Perforating Collagenosis

A perforating dermatosis in which dermal collagen is extruded through the epidermis, presenting as discrete, dome-shaped papules with a central keratotic plug, most commonly on the extensor extremities.

Read Codes

A clinical coding system historically used to represent diagnoses, findings, and procedures in electronic health records.

Reading Disability, Specific

Alternate index label; see *Dyslexia*.

Reagent

A reagent is a chemical or biological substance used to cause or detect a reaction in laboratory testing, research, or manufacturing.

Reality Testing

The ability to recognize what is real and distinguish it from thoughts, beliefs, or perceptions that are not based on reality.

Rebeccamycin

Rebeccamycin is an investigational antitumor com-

pound that inhibits DNA-related enzymes; it is not a routine clinical medicine.

Rebound Effect

Worsening or return of symptoms after stopping or reducing a treatment that had been suppressing them.

Rebound Headache

Alternate index label; see *Medication overuse headaches*.

Rebound Insomnia

Return or worsening of insomnia after stopping a sleep medicine.

Recalcitrant

Resistant to treatment, control, healing, or change; the term often describes a persistent disease or infection.

Receive

To receive means to be given or take in a treatment, substance, signal, transplant, specimen, or other intervention or material.

Receiver Operating Characteristic

ROC curve: plot of sensitivity vs. 1-specificity. Area under curve measures test accuracy.

Recent Memory

Recent memory is the ability to recall information or events from the near past and can be impaired by aging, delirium, brain disease, depression, or medicines.

response. Types: (1) Cell surface: ion channel (nicotinic AChR), G-protein coupled (GPCR: adrenergic, muscarinic, opioid), enzyme-linked (RTK: insulin receptor, cytokine receptors).

Receptor Binding

receptor binding is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute

to disease.

Receptor Desensitization

A rapid decrease in the response of a receptor to continuous or repeated agonist stimulation, occurring within minutes to hours.

Receptor Downregulation

A decrease in the number or sensitivity of receptors in response to chronic receptor stimulation by agonists. When receptors are chronically activated, the cells compensate by internalizing and degrading receptors, reducing receptor density on the cell surface, and decreasing receptor responsiveness.

Receptor Tyrosine Kinases

Single-pass transmembrane receptors with intrinsic tyrosine kinase activity in their intracellular domain. Ligand binding (e.g., insulin, EGF, PDGF, FGF) induces receptor dimerization and autophosphorylation of tyrosine residues.

Receptor Upregulation

An increase in the number or sensitivity of receptors in response to chronic receptor blockade or decreased neurotransmitter activity. When receptors are chronically blocked by antagonists, the cells compensate by synthesizing more receptors to maintain homeostatic signaling, leading to increased sensitivity to agonists.

Receptors

Cell proteins or structures that bind specific signaling molecules and trigger a response.

Recession Of Medial Rectus Muscle

Recession of Medial Rectus Muscle refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Recessive

Allele whose phenotype is masked in heterozygotes. Autosomal recessive disorders: cystic fibrosis, sickle cell anemia, Tay-Sachs disease. Requires two copies

for expression.

RECIST

Response Evaluation Criteria in Solid Tumors, standardized imaging-based criteria that categorize response to anticancer therapy as complete response, partial response, stable disease, or progressive disease.

Recognizing Medical Emergencies

Recognizing a medical emergency includes identifying severe breathing difficulty, chest pain, stroke signs, major bleeding, seizure, collapse, poisoning, or altered consciousness and calling emergency services.

Recognizing Teen Depression

Recognizing depression in adolescents involves looking for persistent low mood or irritability, loss of interest, sleep or appetite changes, poor concentration, hopelessness, or self-harm thoughts.

Recombinant

Produced by combining genetic material from different sources or by using engineered DNA in a cell or organism.

Recombinant DNA

DNA molecule formed by combining genetic material from different sources. Basis of biotechnology. Used to produce insulin, growth hormone, vaccines, gene therapy vectors.

Recombinant DNA Technology

Laboratory techniques that join DNA from different sources to produce a desired genetic construct or product.

Recombinant Protein

recombinant protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Recombinant Tissue Plasminogen Activator

A laboratory-made form of tissue plasminogen activator that helps convert plasminogen to plasmin and dissolve blood clots. It is used in selected emergencies, such as some ischemic strokes, heart attacks, or pulmonary emboli, when benefits outweigh bleeding risk.

Alternate Names:

Recombinant tPA

Recombinase

An enzyme that rearranges or exchanges DNA segments, supporting genetic recombination, repair, or programmed immune-gene assembly.

Recombinational Repair

A medical procedure, treatment, device, or care topic related to recombinational repair; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Recommended Daily Allowance

The Recommended Dietary Allowance is the average daily nutrient intake sufficient for nearly all healthy people in a specified age and sex group.

Recommended Dietary Allowance

The average daily intake expected to meet the nutrient needs of nearly all healthy people in a defined group.

Alternate Names:

RDA

Reconstructive (Plastic) Surgery

Surgery that restores form or function after congenital difference, injury, disease, or removal of tissue.

Record

A health record is a documented account of a person's history, findings, tests, diagnoses, treatments, decisions, and outcomes.

Recovered Memory Syndrome

A controversial and nonstandard label for memories

recalled after an alleged period of amnesia; recovered memories require careful, non-leading assessment.

Stroke Rehabilitation

Rehabilitation after a stroke helps improve movement, speech, swallowing, thinking, mood, and daily activities while reducing complications and preventing another stroke.

Recovery Position

A side-lying position used for an unconscious but normally breathing person to help keep the airway open and reduce aspiration risk.

Recreation Therapy

Recreation therapy uses adapted games, arts, exercise, and leisure activities to support physical, emotional, cognitive, or social functioning.

Recrudescence

Recurrence of symptoms or infection after apparent improvement, often from persistence of the original organism or disease process.

Rectal

Rectal means relating to the rectum, the final section of the large intestine before the anal canal.

Rectal (Adjective)

Rectal means relating to the rectum, the final part of the large intestine before the anal canal.

Rectal Biopsy

Removal of a small sample of rectal tissue for microscopic examination, used to evaluate Hirschsprung disease, inflammation, infection, neoplasia, or other rectal disorders.

Rectal Bleeding

Passage of bright-red or dark blood from the rectum, caused by hemorrhoids, fissures, polyps, inflammatory bowel disease, diverticular disease, or cancer. Persistent, heavy, or unexplained bleeding requires evaluation.

Alternate Names:

Blood In The Stool (Rectal Bleeding)

Rectal Bleeding (Blood In The Stool (Rectal Bleeding))

Alternate index label for Blood in the Stool (Rectal Bleeding).

Rectal Cancer

Cancer that develops in the tissues of the rectum.

Alternate Names:

Cancer, Rectal

Rectal Carcinoma

Rectal carcinoma is cancer arising in the rectum and may cause bleeding, altered bowel habits, pain, or anemia.

Rectal Culture

Laboratory culture of a rectal specimen to detect selected bacterial pathogens or colonization, depending on the clinical indication.

Rectal Disease

A disorder affecting the rectum, including inflammation, infection, bleeding, obstruction, prolapse, functional disease, or neoplasm.

Rectal Fistula

A rectal fistula is an abnormal tract connecting the rectum with skin or another organ, often causing drainage, infection, or passage of stool through an abnormal route.

Rectal Hemorrhage

Bleeding from the rectum, which may result from hemorrhoids, fissure, inflammation, ischemia, polyps, cancer, trauma, or other gastrointestinal disease.

Rectal Inflammation

Alternate index label; see *Proctitis*.

Rectal Itching

Alternate index label; see *Anal itching*.

Rectal Mucositis

Rectal mucositis is inflammation and injury of the rectal lining, often after radiation, chemotherapy, infection, inflammatory disease, or trauma.

Rectal Necrosis

Rectal necrosis is death of rectal tissue from inadequate blood flow, severe inflammation, infection, pressure, or another destructive process.

Rectal Neoplasm

A rectal neoplasm is an abnormal growth in the rectum that may be benign, precancerous, or malignant.

Rectal Obstruction

Impaired passage through the rectum caused by fecal impaction, stricture, tumor, foreign body, inflammation, or functional outlet disorder.

Rectal Perforation

A tear or hole through the rectal wall that can release bowel contents and cause peritonitis, sepsis, or severe bleeding.

Rectal Polyposis

Rectal polyposis is the presence of multiple polyps in the rectum, which may be sporadic or part of an inherited syndrome and can increase cancer risk.

Rectal Prolapse

Protrusion of rectal tissue through the anal opening. Complete prolapse involves all layers of the rectal wall and presents as a concentric cylindrical mass with visible mucosal folds. Partial prolapse involves only the mucosa. Associated with chronic straining, pelvic floor dysfunction, and neurological conditions.

Rectal Prolapse Repair

A surgical or procedural treatment that restores a prolapsed rectum to its normal position and secures or removes affected tissue according to the prolapse and patient factors.

Rectal Route

The rectal route administers a medicine or fluid through the anus into the rectum, where it may act locally or be absorbed systemically.

Rectal Stenosis

Abnormal narrowing of the rectal lumen caused by inflammation, scarring, surgery, radiation, ischemia, trauma, or neoplasm.

Rectal Tenesmus

Rectal tenesmus is a persistent or urgent feeling of needing to pass stool despite an empty or nearly empty rectum, caused by inflammation, infection, mass, or motility disorder.

Rectal Ulcer

Alternate index label; see *Solitary rectal ulcer syndrome*.

Rectocele

A weakening of the vaginal wall that allows the rectum to bulge into the vagina.

Rectosigmoid Neoplasm

A rectosigmoid neoplasm is an abnormal growth where the sigmoid colon joins the rectum and may be benign or malignant.

Rectouterine Pouch

The rectouterine pouch, or pouch of Douglas, is the peritoneal recess between the uterus and rectum where fluid or disease may collect.

Rectovaginal Fistula

Abnormal connection between the rectum and vagina causing passage of stool or gas through the vagina. Causes include obstetric injury particularly third and fourth degree perineal tears, Crohn disease, radiation therapy, and surgical complications. Presents with vaginal discharge of stool, flatus, or recurrent vaginal infections.

Rectum

The final straight section of the large intestine, ap-

proximately 12-15 cm long, extending from the sigmoid colon to the anal canal. The rectum stores feces before defecation.

Rectus Abdominis

The rectus abdominis is a paired abdominal muscle that flexes the trunk, stabilizes the pelvis, and helps compress abdominal contents.

Rectus Abdominis Muscle

Long flat vertical anterior abdominal wall muscle from pubic symphysis to xiphoid process/costal cartilages 5-7, divided by tendinous intersections ("six-pack"). Innervated by T7-T12. Flexes trunk, compresses abdomen, increases intra-abdominal pressure. Diastasis recti common in pregnancy/obesity.

Rectus Abdominis Sheath

A fibrous aponeurotic envelope formed by the abdominal wall muscles and surrounding the rectus abdominis muscle. It contains the superior and inferior epigastric vessels and related nerves.

Recur

To recur means to return or happen again, as symptoms, disease, infection, or a tumor after a period of improvement or treatment.

Recurrence

Return of cancer after treatment. Can be local, regional, or distant.

Recurrence Risk

The probability that a genetic condition will occur again in a future pregnancy or relative, estimated from the inheritance pattern, parental testing, penetrance, and observed or presumed pathogenic variant.

Recurrence-Free Survival

In clinical oncology trials and longitudinal post-treatment monitoring, the duration of time elapsed from the completion of primary curative-intent treatment (such as complete surgical resection with or without adjuvant systemic or radiation therapy) un-

til the first documented clinical, radiographic, or biochemical recurrence of the primary malignancy, or death from any cause. Frequently utilized as a primary or key secondary efficacy endpoint in adjuvant clinical trials for resected solid tumors (e.g., breast cancer, melanoma, colorectal carcinoma), serving as an objective measure of definitive cure or sustained disease suppression.

Recurrent

Happening again after improving or stopping. A recurrent disease may return after remission, such as recurrent urinary infection or recurrent herpes.

Recurrent Breast Cancer

Breast cancer that returns after a period in which it was undetectable.

Recurrent Caries

Caries under or around restoration. Due to microleakage, marginal breakdown.

Recurrent Disease

Return of a disease after a period of improvement, remission, or apparent absence, either at the original site or elsewhere.

Recurrent Fractures

Repeated bone fractures, suggesting persistent trauma, osteoporosis, bone fragility, metabolic disease, tumor, or an inherited skeletal disorder.

Recurrent Laryngeal Nerve

A branch of the vagus nerve supplying most intrinsic muscles of the larynx; injury can cause hoarseness.

Recurrent Oral Ulcers

Alternate index label; see *Canker sore*.

Recurrent Pregnancy Loss

A distinct reproductive pathology defined by major international obstetric societies as the consecutive loss of two or more clinical pregnancies prior to 20 to 24 weeks of gestation. Affecting 1–2% of couples, standard clinical evaluation encompasses parental

peripheral blood karyotyping, anatomic evaluation of the uterine cavity (for congenital septa, submucosal fibroids, or Asherman syndrome), screening for antiphospholipid antibody syndrome, assessment of parental endocrine factors (thyroid dysfunction, hyperprolactinemia, diabetes), and thrombophilia screening.

Alternate Names:

Recurrent Miscarriage; Habitual Abortion; RPL

Recurrent Prostate Cancer

Alternate index label; see *Prostate cancer recurrence*.

Red Birthmarks

Red birthmarks are vascular skin marks present at or soon after birth, ranging from harmless capillary marks to lesions requiring monitoring or specialist treatment.

Red Blood Cell

A blood cell that carries oxygen from the lungs to the body's tissues. It contains hemoglobin, has a flattened, indented shape, and lacks a nucleus, which helps it bend through small blood vessels. Red blood cells also carry some carbon dioxide back to the lungs and help keep the blood's acid level balanced. They are made mainly in the bone marrow and normally live for about 120 days.

Red Blood Cell Labeling

Nuclear medicine technique labeling RBCs with Tc-99m. Used for GI bleeding detection, cardiac blood pool imaging.

Red Eye

Redness of the white part of the eye caused by irritation, dryness, allergy, infection, injury, inflammation, or increased surface blood flow.

Red Flag

A symptom, sign, or test result suggesting a potentially serious condition that requires urgent assessment or escalation of care. Red flags are context-specific and should not be used as a substitute for clinical judgment.

Red Infarction

Red infarction, also known as hemorrhagic infarction, occurs in tissues with dual blood supply, loose tissue architecture, or venous obstruction, allowing blood to accumulate in the necrotic area and produce a dark red appearance.

Alternate Names:

hemorrhagic infarction, occurs in tissues with dual blood.

Red Nucleus

The red nucleus is a midbrain structure involved in motor coordination and connections with the cerebellum and spinal cord.

Reducing Agents

Reducing agents donate electrons in chemical reactions and can alter oxidation state; they are used in laboratory chemistry, biology, and manufacturing.

Reductase Deficiency

A disorder or laboratory finding caused by inadequate activity of a reductase enzyme, impairing a specific metabolic or hormone pathway.

Reduction

Restoration of a displaced fracture or joint to better alignment, or decrease in amount, size, or activity.

Reduction Division

Meiosis, a cell division that reduces chromosome number by half to form gametes.

Reduction Mammoplasty

Surgery that removes breast tissue, fat, and excess skin to reduce breast size and relieve symptoms such as neck, shoulder, or back discomfort; it may also reshape the breasts.

Redundant Mitral Valve

Excess leaflet tissue of the mitral valve, most commonly the posterior leaflet, which may prolapse into the left atrium during systole. Often associated with myxomatous degeneration and mitral valve prolapse. May be asymptomatic or cause mild mitral regurgi-

tation detected incidentally on echocardiography.

Reduviidae

A family of insects including assassin bugs; some species transmit pathogens such as the parasite causing Chagas disease.

Reed-Sternberg Cells

Large abnormal cells derived from B lymphocytes and characteristic of classic Hodgkin lymphoma.

Reemerging Disease

Previously controlled disease increasing. Examples: tuberculosis, measles, drug-resistant infections.

Reexpansion Pulmonary Edema

Noncardiogenic pulmonary edema developing rapidly after rapid reexpansion of a collapsed lung, typically following chest tube drainage of a large or long-standing pneumothorax.

Refeeding Syndrome

Life-threatening fluid and electrolyte shifts when nutrition restarted in malnourished patient. Hypophosphatemia, hypokalemia, hypomagnesemia, fluid overload, cardiac failure. Prevention: start low (25-30 kcal/kg), advance slowly, supplement electrolytes, monitor closely.

Reference Interval

The range of laboratory results expected in a defined reference population, usually encompassing the central 95 percent of values. A result outside the interval is not by itself diagnostic and must be interpreted with clinical context, biologic variation, and the laboratory's method-specific range.

Referral

A request for assessment or treatment by another healthcare professional or service.

Referred Pain

An unpleasant sensory and emotional experience associated with actual or potential tissue damage, serving as a protective alerting system but becoming

maladaptive when chronic (lasting >3 months). See pain.

Reflex

A reflex is a rapid, involuntary response to a stimulus mediated through a neural or physiologic pathway.

Reflex Action

An involuntary, rapid response to a stimulus mediated through a reflex arc.

Reflex Anoxic Seizure

A brief seizure-like episode caused by transient cerebral hypoperfusion after a sudden painful or emotional stimulus, usually with pallor, bradycardia, and rapid recovery.

Reflex Arc

The neural pathway that mediates a reflex action, bypassing conscious brain processing for rapid, automatic responses. A reflex arc consists of five components: (1) Receptor—sensory nerve ending detecting a stimulus (e.g., pain, stretch).

Reflex Epilepsy

Reflex epilepsy is epilepsy in which specific stimuli such as flashing light, sound, reading, or touch reliably trigger seizures.

Reflexes

Automatic, rapid responses to stimuli mediated through neural pathways called reflex arcs.

Reflux

Gastroesophageal reflux in infants: effortless regurgitation of gastric contents, very common (physiologic 'spitting up') and usually benign, resolving by 12 months with maturation of the lower esophageal sphincter.

Reflux Nephropathy

Renal scarring and chronic kidney damage caused by vesicoureteral reflux, recurrent urinary infection, or both. It may lead to hypertension, proteinuria,

impaired growth, and chronic kidney disease.

Reflux, Bile

Alternate index label; see *Bile reflux*.

Reflux, Vesicoureteral

Alternate index label; see *Vesicoureteral reflux*.

Refraction

The bending of light as it passes through the cornea, aqueous humour, lens, and vitreous humour. In an eye examination, refraction determines the optical power needed to focus a clear image on the retina; refractive power is measured in dioptres.

Refractive Error

A refractive error occurs when the eye does not focus light correctly on the retina, causing nearsightedness, farsightedness, astigmatism, or presbyopia.

Refractive Errors And Astigmatism

Refractive errors are vision disorders caused by the eye's inability to properly focus light onto the retina, resulting in blurred vision.

Refractive Surgery

Surgical procedures that reshape the cornea or alter the eye to correct refractive errors and reduce dependence on eyeglasses or contact lenses. Laser techniques include LASIK (corneal flap with excimer ablation), photorefractive keratectomy (surface ablation), and small-incision lenticule extraction (SMILE).

Refractometer

A refractometer measures how much light bends through a substance and is used for fluid concentration, optical properties, or eye refraction.

Refractometry

Measurement of the eye's refractive power to determine the optical correction needed for myopia, hyperopia, astigmatism, or presbyopia.

Refractory

Resistant to a treatment or not responding adequately to it. The term may refer to cancer or other diseases.

Refractory Anemia

Refractory anemia is an older term for anemia that persists despite usual treatment and may refer to a myelodysplastic syndrome or another underlying disorder.

Refractory Cancer

Refractory cancer is cancer that does not respond adequately to a treatment or begins progressing during treatment.

Refractory Disease

Disease that does not respond adequately to appropriate standard treatment or that progresses despite therapy.

Refractory Period

The period after an action potential during which a cell cannot (absolute refractory period) or can only partially (relative refractory period) generate a new action potential.

Refrigerant Poisoning

Refrigerant poisoning involves toxic or environmental exposure to refrigerant poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Refsum Disease

An autosomal-recessive peroxisomal disorder of phytanic-acid metabolism. It causes retinitis pigmentosa, anosmia, peripheral neuropathy, ataxia, hearing loss, ichthyosis, and shortened metacarpals or metatarsals with variable onset.

Regadenoson Pharmacological Stress

Selective A2A adenosine receptor agonist used as a pharmacological vasodilator for nuclear myocardial perfusion imaging. Single IV bolus injection produces coronary vasodilation with fewer side effects

than adenosine (less bronchospasm, AV block). Contraindicated in severe asthma or COPD requiring oxygen.

Regeneration

The growth or replacement of damaged or lost cells, tissues, or body parts.

Regimen

A planned schedule of medicines, treatments, diet, or other healthcare measures.

Regional

Relating to a particular body area, geographic area, or localized distribution rather than the whole body.

Regional Anaesthesia

Loss of sensation in a larger body region produced by blocking nerves or a nerve plexus while consciousness is usually preserved.

Regional Anesthesia

Regional anesthesia involves the administration of local anesthetic agents to specific nerves or nerve plexuses to provide surgical anesthesia and postoperative analgesia for a defined body region.

Regional Chemotherapy

Regional chemotherapy delivers anticancer medicine directly to a body region or organ to increase local exposure while limiting whole-body exposure.

Regional Enteritis

An older name for Crohn disease, a chronic inflammatory bowel disease that can affect any part of the digestive tract.

Regional Ileitis

An older name for Crohn disease affecting mainly the terminal ileum. It causes long-term inflammation that may lead to abdominal pain, diarrhea, weight loss, narrowing, fistulas, or nutritional problems.

Registrar

A doctor in postgraduate specialist training or a medical records professional, depending on context and country.

Regulation Of Health Professions

Legal and professional systems governing training, registration, scope, conduct, and accountability of healthcare workers.

Regulatory Approval

Authorization by a competent regulatory authority for a medicine, device, or procedure to be marketed or used for specified purposes after review of evidence.

Regulatory Authority

A regulatory authority is a government or official body that sets and enforces standards for medicines, devices, facilities, professionals, or public health.

Regulatory Gene

A gene that controls the activity of other genes or helps regulate cell growth, development, or body functions.

Regulatory T Cells

Tregs (CD4+ CD25+ FOXP3+): essential for peripheral tolerance. Thymic (tTreg) vs peripheral (pTreg) origin. FOXP3: master transcription factor; mutations cause IPEX.

Regurgitation

The backward flow of blood through a heart valve that fails to close properly (valvular incompetence or insufficiency).

Regurgitation, Aortic Valve

Alternate index label; see *Aortic valve regurgitation*.

Regurgitation, Mitral Valve

Alternate index label; see *Mitral valve regurgitation*.

Rehab

A structured rehabilitation program that helps re-

store function, independence, participation, or quality of life after illness or injury.

Rehabilitation

A coordinated process that helps a person restore or adapt physical, cognitive, psychological, and social function after illness, injury, surgery, or disability.

Rehabilitation Clinic

A specialized outpatient healthcare facility staffed by a multidisciplinary clinical team (physiatrists, physical therapists, occupational therapists, speech-language pathologists, and prosthetists/orthotists) dedicated to restoring, optimizing, or maintaining physical, cognitive, and functional independence in individuals recovering from injury, surgery, or chronic disabling disorders. Rehabilitation programs focus on gait retraining, neuromuscular re-education, stroke recovery, cardiac and pulmonary rehabilitation, musculoskeletal injury recovery, and chronic pain management, guided by objective functional metrics (e.g., the Functional Independence Measure [FIM] or 6-minute walk test).

Rehabilitation Engineering

Technology development for disability: prosthetics, orthotics, assistive devices, environmental control.

Rehabilitation Motivation

Patient factors that influence participation in rehabilitation, including intrinsic and extrinsic motivation and engagement with therapeutic goals.

Rehabilitation Potential

The expected capacity to improve function, independence, participation, or quality of life through rehabilitation. It depends on diagnosis, prognosis, cognition, motivation, comorbidity, social support, and environmental resources.

Rehabilitation Unit

A specialized, acute or subacute hospital-based inpatient medical ward designed to provide intensive, coordinated interdisciplinary rehabilitative care to patients recovering from catastrophic neurological, orthopedic, or medical illness (such as acute stroke,

traumatic brain injury, spinal cord injury, major polytrauma, or complex joint arthroplasty). Patients typically undergo at least 3 hours of intensive therapy per day across multiple modalities, supported by 24-hour medical and rehabilitation nursing supervision focused on functional restoration, preventing immobility complications, and safe community reintegration.

Rehydration

Replacement of water and electrolytes lost through dehydration, commonly using oral fluids or intravenous treatment when necessary.

Reifenstein Syndrome

An inherited disorder of androgen action, usually involving partial androgen insensitivity in an individual with a 46,XY chromosome pattern.

Reiki

Reiki is a complementary practice involving light touch or hands near the body; evidence for treating disease is limited and it should not replace effective medical care.

Reinfection

A new episode of infection by the same or a related pathogen after the original infection has resolved. Reinfection is distinct from relapse, in which the original infection persists or returns from an incompletely eradicated focus.

Reinnervation

Restoration of nerve supply to a muscle or tissue after nerve repair, regeneration, or collateral nerve growth.

Reiter'S Syndrome

An older name for reactive arthritis, an inflammatory arthritis occurring after certain infections.

Rejection

An immune response in which the body attacks transplanted tissue or an organ.

Alternate Names:

Transplant rejection

Relapse

Return or worsening of a disease after a period of improvement or remission. The term is used for cancer and many noncancerous disorders.

Relapse-Free Survival

The time from completion of treatment to recurrence of the cancer or death from any cause.

Relapsing Fever

An acute febrile illness caused by *Borrelia* species, characterized by recurrent episodes of fever separated by afebrile intervals.

Relapsing Polychondritis

A systemic autoimmune disorder causing inflammation and destruction of cartilage throughout the body. Airway involvement can cause tracheal and bronchial stenosis. Other features include auricular chondritis, nasal chondritis, ocular inflammation, and arthritis. Treatment is corticosteroids and immunosuppressants.

Relapsing-Remitting Multiple Sclerosis

The most prevalent clinical phenotype of multiple sclerosis, accounting for roughly 85% of initial presentations. Characterized by clearly defined acute inflammatory neurological relapses (attacks or flares) followed by periods of complete or partial recovery (remission) without noticeable clinical disease progression during the intervening remission phases.

Alternate Names:

RRMS; Relapsing-remitting MS

Relate Marriage Guidance

Counseling or relationship support provided by Relate or similar services; it is not a medical diagnosis.

Relative Afferent Pupillary Defect

An asymmetric reduction in the direct pupillary light response caused by unequal afferent input from the retina or optic nerve. It is detected with the

swinging-flashlight test and is also called a Marcus Gunn pupil.

Alternate Names:

a Marcus Gunn pupil.

Relative Biological Effectiveness

Ratio of dose of reference radiation to dose of test radiation for same effect. Alpha particles: RBE 10-20.

Relative Risk

Ratio of disease risk in exposed vs. unexposed. $RR = (a/(a+b)) / (c/(c+d))$. $RR > 1$: increased risk. $RR < 1$: protective.

Relaxation

A reduction in muscle tension, physiologic arousal, or psychological stress; techniques may include breathing, biofeedback, or progressive muscle relaxation.

Relaxation Response

Physiological changes produced by relaxation practices that counter the stress response.

Relaxation Techniques For Stress

Relaxation techniques for stress include slow breathing, progressive muscle relaxation, meditation, guided imagery, and other methods that reduce physical arousal.

Relaxation Therapy

Techniques such as breathing, muscle relaxation, or imagery used to reduce stress, arousal, or symptom burden.

Relaxin

Relaxin is a reproductive hormone that helps remodel connective tissue and may contribute to cervical, pelvic, and vascular changes during pregnancy.

Religion

A system of spiritual beliefs and practices that may influence health decisions, coping, diet, treatment preferences, and end-of-life care.

REM

Rapid eye movement, a sleep stage associated with vivid dreams and intense brain activity.

Alternate Names:

Rapid eye movement sleep

REM Behavior Disorder

A sleep disorder in which a person acts out dreams during REM sleep.

Alternate Names:

REM sleep behavior disorder

REM Rebound

Increased REM sleep after REM deprivation or withdrawal of a REM-suppressing medicine.

REM Sleep

Rapid-eye-movement sleep, a stage associated with vivid dreaming, active brain activity, and skeletal-muscle atonia.

REM Sleep Behavior Disorder

A sleep disorder in which the normal muscle paralysis of REM sleep is reduced or absent, allowing a person to move or vocalize while dreaming.

Alternate Names:

Rapid Eye Movement Sleep Behavior Disorder

Remedy

A treatment or measure intended to relieve a symptom or illness; safety and effectiveness depend on the specific remedy and condition.

Remifentanil

An ultra-short-acting synthetic opioid analgesic metabolized by non-specific plasma and tissue esterases, producing context-sensitive half-time of approximately 3 to 10 minutes regardless of infusion duration. This makes it ideal for total intravenous anesthesia (TIVA) combined with propofol, and for procedures requiring rapid emergence.

Remission

Decrease or disappearance of cancer signs. Complete remission: no detectable disease. Partial re-

mission: significant reduction but not complete.

Remittent Fever

Fever that varies during the day but does not return to normal temperature.

Remitting Seronegative Symmetrical Synovitis With Pitting Edema

RS3PE: an acute-onset, symmetric polyarthritis (usually in elderly) with pitting edema of the hands/feet, negative rheumatoid factor and anti-CCP, and marked response to low-dose corticosteroids.

Remodeling

Bone adaptation to implant stress. Wolff's law: bone remodels along lines of stress.

Removal Technique

A medical procedure, treatment, device, or care topic related to removal technique; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Renal

Renal means relating to the kidneys, which filter blood, regulate fluid and electrolytes, produce hormones, and excrete urine.

Renal Abscess

A localized collection of pus within or around the kidney, usually caused by infection. Diagnosis and treatment may require imaging, antibiotics, and drainage.

Renal Adenoma

A renal adenoma is a usually benign kidney tumor, often discovered incidentally; its size, imaging features, and growth determine whether follow-up is needed.

Renal Agenesis

Congenital absence of one (unilateral, 1 in 1,000) or both (bilateral, 1 in 4,000) kidneys. Unilateral renal agenesis: usually asymptomatic, discovered

incidentally — the solitary kidney undergoes compensatory hypertrophy. The contralateral kidney may have VUR, UPJ obstruction, or ectopia.

Renal And Urological Diseases

Renal and urological diseases affect the kidneys, urinary tract, bladder, urethra, or male reproductive urinary organs.

Renal Anemia

Anemia associated with chronic kidney disease, caused by reduced erythropoietin production and other factors such as iron deficiency, inflammation, and shortened red-cell survival. Evaluation and treatment are individualized; iron replacement and erythropoiesis-stimulating therapy may be considered when appropriate.

Renal Arteriography

Renal arteriography uses contrast X-rays to visualize kidney arteries and may identify narrowing, bleeding, aneurysm, or vascular anatomy during an intervention.

Renal Artery

A paired branch of the abdominal aorta supplying a kidney. It divides into segmental, interlobar, arcuate, and interlobular arteries within the renal parenchyma.

Renal Artery Aneurysm

A focal enlargement of a renal artery, often asymptomatic but capable of causing hypertension, thromboembolism, or rupture. Management depends on size, symptoms, anatomy, pregnancy, and growth, and may involve surveillance, endovascular treatment, or surgery.

Renal Artery Disease

Alternate index label; see *Renal artery stenosis*.

Renal Artery Doppler

Ultrasound assessing renal artery stenosis. Screening for renovascular hypertension.

Renal Artery Embolism

Acute occlusion of renal artery from cardiac emboli typically from atrial fibrillation or mural thrombus. Presents with acute flank pain and hematuria. CT angiography shows filling defect. Treat with anticoagulation and possibly thrombolysis or embolectomy if caught early.

Renal Artery Stenosis

Narrowing of renal artery causing renovascular hypertension. Atherosclerotic most common in elderly at ostium. Fibromuscular dysplasia in young women causing mid-distal stenosis with string-of-beads appearance. Flash pulmonary edema and declining function on ACE inhibitors indicate intervention.

Alternate Names:

Renal Artery Disease Renal Vascular Disease

Renal Biopsy

Alternate index label; see *Kidney biopsy*. It is used selectively for unexplained acute kidney injury, nephrotic syndrome, glomerular hematuria, or transplant dysfunction; bleeding is the main complication.

Renal Calculi

Renal calculi are kidney stones formed from crystallized minerals in urine and may cause colicky flank pain, blood in urine, obstruction, or infection.

Renal Calculi (Kidney Stones)

Crystalline mineral deposits that form in the kidneys or urinary tract. They may cause severe colicky pain, blood in the urine, urinary obstruction, or infection.

Alternate Names:

Kidney Stones

Renal Cancer

Alternate index label; see *Kidney cancer*.

Renal Carcinoma

Renal carcinoma is cancer arising from kidney tissue, most commonly renal cell carcinoma.

Renal Cell Carcinoma

A malignant tumor arising from renal tubular epithelium, most commonly clear-cell carcinoma.

Renal Changes In Pregnancy

Pregnancy increases renal blood flow and glomerular filtration. These changes lower usual serum creatinine values and may increase urinary loss of glucose and amino acids.

Renal Circulation

Blood flow through the kidneys, which supports filtration, oxygen delivery, hormone production, and regulation of fluid and electrolytes.

Renal Clearance

The volume of plasma completely cleared of a substance by the kidneys per unit time, measured in mL/min. Clearance of any substance X is calculated as: $CX = (Ux \times V)/Px$, where Ux is urine concentration, V is urine flow rate, and Px is plasma concentration.

Renal Colic

Severe, paroxysmal flank pain caused by ureteral obstruction, typically by a kidney stone, with radiation to the groin; often the worst pain experienced.

Renal Corpuscle

The filtration unit at the beginning of a nephron, consisting of the glomerulus enclosed by Bowman capsule.

Renal Cortex

Outer kidney portion between capsule and medulla. Granular tissue with glomeruli, convoluted tubules, and cortical collecting ducts. Extends between pyramids as renal columns. Receives approximately 90 percent of renal blood flow; primary filtration site.

Renal Cryptococcosis

Infection of the kidney by *Cryptococcus* species, usually as part of disseminated cryptococcosis in an immunocompromised person.

Renal Cyst

A renal cyst is a fluid-filled sac in or on the kidney, commonly benign but sometimes complex or associated with inherited cystic disease.

Renal Dialysis

Alternate term for dialysis: kidney replacement therapy using hemodialysis or peritoneal dialysis to remove excess fluid and dissolved waste products when kidney function is inadequate.

Renal Diet

Restricted protein, sodium, potassium, phosphorus for kidney disease. Individualized based on stage. Dialysis increases protein requirement.

Renal Diseases

Disorders affecting kidney tissue or function, including glomerular, tubular, vascular, infectious, obstructive, and neoplastic disease.

Renal Dose Adjustment

Modification of a medication dose, dosing interval, or both to account for reduced kidney clearance. The adjustment should use an appropriate estimate of renal function, drug-specific pharmacokinetics, treatment indication, and therapeutic monitoring when available.

Renal Failure

Decline in kidney function leading to accumulation of waste products (azotemia) and electrolyte abnormalities. Acute kidney injury (AKI): rapid decline over hours to days, potentially reversible. Chronic kidney disease (CKD): progressive decline over months to years.

Renal Failure (Kidney Failure)

Alternate index label for Kidney Failure.

Renal Function

The kidneys' ability to filter blood, remove waste, regulate fluids and electrolytes, maintain acid-base balance, and produce or activate hormones.

Renal Glomerulus

A renal glomerulus is a tuft of capillaries within a nephron that filters blood to begin urine formation.

Renal Hamartoma

A renal hamartoma is a benign disorganized overgrowth of mature tissues in the kidney, sometimes used to describe an angiomyolipoma-like lesion.

Renal Hypertension

High blood pressure caused by kidney disease or narrowing of the renal arteries.

Renal Hypertrophy

Renal hypertrophy is enlargement of kidney tissue, often compensating for loss or reduced function of the other kidney but sometimes caused by disease.

Renal Hypoperfusion

Reduced blood flow to the kidneys, causing decreased glomerular filtration and potentially acute kidney injury. Causes include dehydration, blood loss, shock, heart failure, renal artery disease, and severe hypotension. Management requires treating the cause and restoring adequate circulation while avoiding further kidney injury.

Renal Infarction

Renal infarction results from occlusion of the renal artery or its branches, causing coagulative necrosis of the renal parenchyma. The kidney has end-arterial supply with limited collateral circulation, making it susceptible to ischemic injury even from small vessel occlusion.

Renal Insufficiency

Renal insufficiency is reduced kidney function that impairs waste removal, fluid and electrolyte balance, acid handling, or hormone production.

Renal Leukemia Infiltration

Kidney involvement by chronic lymphocytic leukemia or other hematological malignancies causing bilateral renomegaly and renal dysfunction. Leukemic cells infiltrate the interstitium and per-

itubular capillaries. May cause AKI from compression of tubules and vessels or chronic kidney disease.

Renal Lithiasis

Alternate index label; see *Kidney stones*.

Renal Mass

Abnormality in kidney detected on imaging. Solid masses may be renal cell carcinoma. Cystic masses classified by Bosniak system.

Renal Medulla

The inner part of the kidney containing the renal pyramids, loops of Henle, vasa recta, and collecting ducts. Its structure and blood supply help establish the concentration gradient used to concentrate urine.

Renal Necrosis

Renal necrosis is death of kidney tissue caused by severe ischemia, toxins, infection, obstruction, or other injury and can cause acute kidney failure.

Renal Oncocytoma

A renal oncocytoma is a usually benign kidney tumor arising from oncocytic cells; imaging may not always distinguish it reliably from renal cancer.

Renal Osteodystrophy

Bone disease in CKD including osteitis fibrosa, osteomalacia, adynamic bone disease, and osteoporosis. Results from calcium-phosphate imbalance, secondary hyperparathyroidism, and vitamin D deficiency. Manage with phosphate binders, vitamin D analogues, and parathyroidectomy for refractory hyperparathyroidism.

Renal Papillary Necrosis

Renal papillary necrosis is death of the kidney papillae from ischemia, infection, obstruction, diabetes, sickle-cell disease, or analgesic injury and may cause blood or obstruction.

Renal Parenchyma

Renal parenchyma is the functional kidney tissue

containing nephrons and supporting structures that filter blood and produce urine.

Renal Pelvis

The funnel-shaped expansion of the ureter within the kidney, formed by the union of major calyces. It collects urine from the renal papillae and directs it into the ureter.

Renal Pelvis Or Ureter Cancer

Cancer of the renal pelvis or ureter usually arises from urothelial cells lining the urinary tract and may cause blood in the urine.

Renal Perfusion Scintiscan

A nuclear medicine study using a radiotracer to evaluate renal blood flow and relative perfusion of the kidneys.

Renal Pyramids

Cone-shaped structures in the renal medulla that contain tubules and collecting ducts and drain urine toward the renal papillae.

Renal Reabsorption

Movement of water and useful solutes from kidney tubules back into the blood, helping regulate urine composition and body chemistry.

Renal Replacement Therapy

Treatment that substitutes for inadequate kidney excretory and regulatory function. Modalities include intermittent hemodialysis, continuous kidney replacement therapy, peritoneal dialysis, and kidney transplantation.

Renal Replacement Therapy Indications

Urgent indications for dialysis include refractory hyperkalemia, severe metabolic acidosis, toxin exposure by a dialyzable agent, refractory fluid overload or pulmonary edema, and symptomatic uremia. The decision incorporates the patient's trajectory and overall goals of care.

Renal Sarcoma

Renal sarcoma is a rare malignant tumor arising from connective or supporting tissues of the kidney.

Renal Scan

Nuclear medicine imaging using Tc-99m compounds. Evaluates renal function, obstruction, blood flow. MAG3 for function, DTPA for GFR.

Renal Scintigraphy

Renal imaging using Tc-99m compounds. MAG3 (tubular secretion) for functional assessment, DTPA (glomerular filtration) for GFR measurement. Used for obstruction, function, transplant monitoring, renin-mediated hypertension.

Renal Stone

A renal stone is a solid crystal aggregate formed in the kidney that can pass through the urinary tract or obstruct urine flow.

Renal Transplant (Renal Transplantation)

Surgical placement of a functioning donor kidney into a person with advanced kidney failure. It can restore kidney function but requires suitability assessment and long-term immunosuppression.

Alternate Names:

Renal Transplantation

Renal Trauma

Kidney injury from trauma. CT staging guides management. Grades I-V based on injury severity.

Renal Tubular Acidosis

Group of disorders with normal anion gap metabolic acidosis from impaired renal acid excretion or bicarbonate reabsorption. Type 1 distal with impaired H⁺ secretion and urine pH greater than 5.5. Type 2 proximal with impaired bicarbonate reabsorption. Type 4 hyperkalemic from aldosterone deficiency or resistance.

Renal Tubule

The nephron segment that modifies filtrate by reab-

sorbing and secreting water and solutes.

Renal Tubule Acidosis

renal tubule acidosis is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Renal Ultrasound

Noninvasive imaging used to assess kidney size, structure, blood flow, and urinary obstruction. It is particularly useful when obstruction, cystic disease, or structural abnormality is suspected and is often used in the evaluation of acute or chronic kidney disease.

Renal Vascular Disease

Alternate index label; see *Renal artery stenosis*.

Renal Vein

A paired vein draining each kidney into the inferior vena cava. The left renal vein is longer and crosses anterior to the aorta, receiving the left gonadal and suprarenal veins.

Renal Vein Thrombosis

Renal vein thrombosis is a clot in a kidney vein that can cause flank pain, blood or protein in urine, kidney dysfunction, swelling, or pulmonary embolism.

Renal Venogram

Contrast imaging of the renal veins used selectively to evaluate venous anatomy, thrombosis, obstruction, or vascular abnormalities.

Renaut Body

A Renaut body is a specialized cluster of cells in the peripheral nerve endoneurium, associated with pressure sensitivity and sometimes nerve pathology.

Renin

An enzyme released mainly by juxtaglomerular kidney cells that initiates the renin-angiotensin-aldosterone system.

Renin Blood Test

A blood test measuring renin concentration or activity, interpreted with aldosterone, posture, sodium status, medications, and blood pressure to evaluate hypertension or mineralocorticoid disorders.

Renin-Angiotensin System

A hormone and enzyme system that regulates blood pressure, fluid balance, and vascular tone. Renin initiates formation of angiotensin II, which promotes vasoconstriction and aldosterone release; excessive activation contributes to hypertension and heart failure.

Renin-Angiotensin-Aldosterone System

A hormonal system in which renin initiates formation of angiotensin II, promoting vasoconstriction and aldosterone secretion. Aldosterone increases distal sodium reabsorption and potassium and hydrogen-ion secretion; RAAS dysregulation contributes to hypertension, heart failure, and CKD progression.

Renin-Angiotensin-Aldosterone System Physiology

The RAAS regulates blood pressure and fluid electrolyte balance. Renin released from juxtaglomerular cells cleaves angiotensinogen to angiotensin I. Angiotensin converting enzyme converts angiotensin I to angiotensin II which causes vasoconstriction and stimulates aldosterone secretion from the adrenal cortex.

Rennin

An older spelling that may refer to renin or to rennet enzyme; in humans, renin regulates blood pressure and fluid balance.

Renography

Nuclear medicine imaging of kidneys. Evaluates function, obstruction, blood flow.

Renovascular Hypertension

Hypertension due to renal artery stenosis. Diagnosis: Doppler ultrasound, CTA, MRA, renography. Treatment: angioplasty, stenting, surgery.

Renshaw Cell

A nerve cell in the spinal cord that helps limit and regulate the activity of motor neurons.

Reoviridae

A family of nonenveloped, segmented double-stranded RNA viruses that infect humans, animals, plants, or insects.

Reoviridae Infection

An infection caused by a reovirus; illness varies by virus and host and is often mild or asymptomatic in humans.

Rep

Rep is an abbreviation whose meaning depends on context, including repetition in exercise or representative in a professional role.

Repair Of Webbed Fingers Or Toes

A medical procedure, treatment, device, or care topic related to repair of webbed fingers or toes; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Repeat Surgery

Repeat surgery is a second or subsequent operation on the same or related area to treat persistent disease, recurrence, complications, or failure of prior treatment.

Reperfusion

Restoration of blood flow to tissue after an interruption.

Reperfusion Injury

Tissue damage following restoration of blood flow after ischemia. Mechanisms: oxidative stress, calcium overload, inflammation. Examples: myocardial reperfusion injury, stroke reperfusion injury. Mitigation: antioxidants, reperfusion protocols.

Reperfusion Therapy

Treatment that restores circulation to tissue deprived of blood, such as clot-dissolving treatment

or angioplasty.

Repetitive Motion Disorders (RMDs)

Alternate index label for RMDs.

Repetitive Motion Disorders (RMDs)

Alternate index label for Repetitive Motion Disorders (RMDs).

Repetitive Nerve Stimulation

A specialized neurophysiological electrodiagnostic study used to assess neuromuscular junction transmission by delivering supramaximal electrical stimuli to a motor nerve at standardized rates (low frequency: 2–3 Hz; high frequency: 20–50 Hz) while recording compound muscle action potentials (CMAPs) from an innervated muscle. A progressive decremental response > 10% in CMAP amplitude during low-frequency stimulation indicates postsynaptic transmission failure characteristic of myasthenia gravis, whereas a low baseline CMAP with marked incremental facilitation (> 100%) during high-frequency stimulation or following brief maximal isometric exercise indicates presynaptic impairment characteristic of Lambert-Eaton myasthenic syndrome.

Alternate Names:

RNS; Repetitive Stimulation Test

Repetitive Strain Injury

Pain or dysfunction caused by repeated forceful movements, sustained positions, or overuse of muscles, tendons, nerves, or other tissues.

Repetitive Strain Injury

Pain or dysfunction associated with repeated movements or sustained posture, commonly affecting tendons, nerves, and muscles.

Replantation Of An Avulsed Tooth

Replacement of an avulsed permanent tooth into its socket after trauma, followed by flexible stabilization and endodontic and periodontal follow-up when indicated. Prognosis depends strongly on extraoral dry time, storage conditions, root maturity, and contamination.

Replantation Of Digits

Microsurgical replacement of a completely amputated finger or toe, reconnecting bone, tendons, nerves, arteries, and veins when feasible.

Replication Initiation

The point and process at which copying of DNA or another genetic molecule begins.

Replicon

A unit of genetic material that can replicate from one origin, such as a chromosome segment, plasmid, or viral genome.

Repolarization

The return of the membrane potential toward the resting value after depolarization. In neurons and skeletal/cardiac muscle, repolarization is primarily caused by the efflux of potassium through voltage-gated potassium channels and inactivation of sodium channels.

Reportable Diseases

Reportable diseases are illnesses that must be notified to public-health authorities under applicable laws to support surveillance and outbreak control.

Reporter Gene

A gene added or used in research to produce an easily measured signal, such as light or color, showing whether another gene or treatment is active.

Reporting Standards

Standards for structured reporting. ACR guidelines, BI-RADS, LI-RADS, PI-RADS.

Repressed Memory Therapy

A controversial approach aimed at recovering allegedly repressed memories; suggestive techniques can create false memories and require caution.

Repression

Reduction or suppression of gene expression, cell activity, behavior, or a physiologic process.

Reproduction

The biological process by which organisms produce offspring through sexual or asexual mechanisms.

Reproduction Spores

Formation of spores for reproduction, dispersal, survival, or transmission by fungi, plants, algae, and some microorganisms.

Reproductive Behavior

Reproductive behavior includes behaviors involved in courtship, mating, pair bonding, pregnancy, parenting, or sexual response and is influenced by biology and environment.

Reproductive Cells

Cells used for reproduction, mainly sperm in males and eggs in females. Each normally carries half the usual number of chromosomes.

Reproductive History

Reproductive history records pregnancies, births, miscarriages, abortions, menstrual function, fertility, contraception, and related conditions relevant to health assessment.

Reproductive Organs

The organs involved in reproduction. Female: ovaries (produce eggs and hormones), fallopian tubes (transport eggs, site of fertilization), uterus (nurtures the developing fetus), cervix (lower part of uterus, dilates during labor), vagina (passage for sperm and baby), and vulva (external genitalia).

Reproductive Organs, Female

Female reproductive organs include the ovaries, fallopian tubes, uterus, cervix, vagina, and external genitalia, which support hormones, ovulation, menstruation, pregnancy, and birth.

Reproductive Process

A reproductive process is a biological event involved in gamete production, fertilization, implantation, pregnancy, birth, or development of offspring.

Reproductive System

The organs and tissues responsible for producing gametes, reproductive hormones, conception, and in some bodies pregnancy and birth.

Reproductive Technique

A reproductive technique is a medical or laboratory method used to assist conception, preserve fertility, diagnose reproductive disease, or manage pregnancy.

Reproductive Toxicity

Effects on reproduction: infertility, teratogenesis, fetal toxicity, developmental toxicity. Testing required for pharmaceuticals.

Reproductive Years

The period of a woman's life during which she is capable of conceiving, typically from menarche (first menstrual period, average age 12-13) to menopause (final menstrual period, average age 51).

Reprogram

To reprogram a cell means to alter its gene-expression state so it acquires a different developmental potential or function.

Reptiles

Cold-blooded vertebrate animals such as snakes, lizards, turtles, and crocodilians; some can transmit infections or cause bites.

Research

A systematic investigation designed to answer a question or generate reliable knowledge. Medical research may study causes, diagnosis, prevention, treatment, or outcomes and should follow ethical and scientific standards.

Research Activity

Research activity is systematic work that generates or evaluates evidence through observation, experimentation, analysis, or synthesis.

Research Design

Research design is the planned structure of a study,

including its question, participants, interventions, measurements, comparisons, and analysis.

Research Ethics

Principles that protect the rights, safety, welfare, privacy, and informed participation of people or animals involved in research.

Research Fraud And Misconduct

Fabrication, falsification, plagiarism, or other serious breaches of standards in planning, conducting, reporting, or reviewing research.

Research Personnel

Research personnel are people who design, conduct, coordinate, monitor, analyze, or support a research study according to its protocol and ethical requirements.

Research Study

A research study is a planned investigation that collects and analyzes data to answer a scientific or clinical question.

Research, Controlled

Controlled research compares an intervention or exposure with a specified control condition to estimate effects while reducing alternative explanations.

Resectable

Describing a tumor or lesion that can be removed completely by surgery with acceptable risk and an appropriate margin, based on its extent and relationship to vital structures.

Resection

Surgical removal of part or all of an organ, tissue, or lesion.

Alternate Names:

Resection, Intestinal

Resection, Intestinal

Alternate index label; see *Resection* and *Small bowel resection*.

Resectoscope

An endoscopic surgical instrument with a viewing system and working loop used to remove or cauterize tissue, especially during prostate or bladder procedures.

Reserpine

Reserpine depletes neuronal catecholamines and serotonin by inhibiting vesicular monoamine transport; it was formerly used for hypertension and psychosis but is now rarely used because of depression, parkinsonism, and hypotension.

Reserve Volume

The additional amount of air that can be inhaled or exhaled beyond a normal tidal breath.

Reservoir

A human, animal, or environmental habitat in which an infectious agent normally lives, multiplies, and persists. A reservoir may serve as the source from which the agent is transmitted to a susceptible host.

Reservoir Of Infection

A reservoir of infection is a person, animal, environment, or material in which an infectious organism normally lives and from which transmission can occur.

Resident

A resident is a physician or other trainee receiving supervised specialty education after initial professional qualification; the term can also mean a person living in a place.

Residential Mobility

Residential mobility is movement between homes or communities and can affect continuity of care, social support, exposure, and health access.

Residual

Residual means remaining after a process, treatment, measurement, or event, such as residual urine, disease, risk, or volume.

Residual Limb

The portion of a limb remaining after amputation. Care includes edema control, skin inspection, shaping, strengthening, desensitization, contracture prevention, and prosthetic preparation.

Residual Limb Pain

Pain in the remaining portion of a limb after amputation.

Alternate Names:

Stump Pain

Residual Neoplasm

A residual neoplasm is tumor tissue that remains after treatment intended to remove or destroy the original tumor.

Residual Risk

Residual risk is the chance of harm that remains after preventive measures, treatment, screening, or controls have been applied.

Residual Tumor

A residual tumor is cancer or other tumor tissue that remains after surgery, radiation, chemotherapy, or another treatment.

Residual Volume

The amount of air remaining in the lungs after maximal forced expiration.

Resin

A sticky plant or synthetic substance used in medicines, dental materials, adhesives, coatings, and other products.

Resiquimod

An investigational immune-response modifier that activates toll-like receptors 7 and 8 and has been studied for infections, cancer, and skin disease.

Resistance

The ability of a microorganism to withstand a medicine that previously inhibited or killed it.

Alternate Names:

Drug resistance

Resistance Training

Exercise in which muscles contract against external resistance to improve strength, power, or endurance.

Resistant Hypertension

High blood pressure that remains above goal despite appropriate treatment with multiple medicines.

Alternate Names:

Treatment-resistant hypertension

Resistant Starch

Starch that resists digestion in the small intestine and is fermented by gut bacteria in the colon.

Resistin

A signaling protein associated with adipose tissue and inflammation; its human clinical significance remains under study.

Resolution

The ability of an imaging system to distinguish separate structures, or the improvement or disappearance of a disease or symptom.

Resonance

Amplification or characteristic vibration produced when a system is driven at a suitable frequency; used in sound and MRI contexts.

Resorcinol

A phenolic compound used in selected topical dermatologic preparations; concentrated exposure can irritate or poison.

Resorption

The removal and absorption of tissue, especially the breakdown of bone by osteoclasts.

Resorption Atelectasis

Atelectasis caused by complete obstruction of an airway, leading to absorption of trapped air by the pulmonary capillaries. Common causes include mucus plugging, foreign body, tumor, and post-surgical

shallow breathing. The affected area appears as increased density on CXR with volume loss.

Respiration

The processes of breathing, gas exchange, and cellular energy production involving oxygen and carbon dioxide.

Respiratory

Respiratory means relating to breathing, the airways, lungs, gas exchange, or the organs and muscles that move air.

Respiratory Acidosis

Acid-base disturbance due to accumulation of CO₂ from hypoventilation, lowering pH (acidemia). Causes: respiratory depression (opioids, sedatives, CNS disease), airway obstruction, neuromuscular weakness (GBS, myasthenia), chest wall restriction, severe COPD/asthma, and ventilation-perfusion mismatch.

Respiratory Alkalosis

Acid-base disturbance from excessive elimination of CO₂ by hyperventilation, raising pH (alkalemia). Causes: anxiety/hyperventilation, hypoxia, pulmonary embolism, sepsis, fever, pain, salicylate toxicity, liver failure, pregnancy, high altitude. Compensation: renal bicarbonate excretion (days).

Respiratory Arrest

Cessation of effective breathing, requiring immediate assessment and resuscitation.

Respiratory Burst

A respiratory burst is rapid production of reactive oxygen species by activated phagocytes to help destroy engulfed microorganisms.

Respiratory Changes In Pregnancy

The growing uterus elevates the diaphragm, while hormonal changes increase ventilatory drive. Tidal volume and minute ventilation generally increase, while functional residual capacity decreases.

Respiratory Depression

Breathing that is abnormally slow or shallow, causing inadequate ventilation and possible carbon-dioxide retention or low oxygen. Medicines, brain disorders, and metabolic illness can cause it.

Respiratory Disorder

A disease affecting the airways, lungs, pleura, respiratory muscles, or control of breathing and causing impaired ventilation, oxygenation, or gas exchange.

Respiratory Distress

Difficulty breathing due to various causes. Assessment: work of breathing (retractions, nasal flaring, tripod position), oxygen saturation, breath sounds. Causes: asthma, COPD, pneumonia, pulmonary embolism, pneumothorax, anaphylaxis, heart failure.

Respiratory Distress Syndrome

Severe respiratory failure with diffuse lung injury; in adults it is ARDS, while neonatal RDS is surfactant deficiency.

Respiratory Distress Syndrome

A life-threatening acute neonatal respiratory disorder affecting premature infants (primarily < 32 weeks gestation), caused by developmental deficiency of pulmonary surfactant produced by type II alveolar pneumocytes. Surfactant deficiency increases alveolar surface tension, leading to widespread microatelectasis, ventilation-perfusion mismatch, intrapulmonary shunting, and severe hypoxemia. Chest radiography classically reveals diffuse reticulogranular "ground-glass" opacities with prominent air bronchograms. Managed with maternal antenatal corticosteroids for prevention, early nasal CPAP, and exogenous intratracheal surfactant replacement.

Respiratory Drive

The neural output that stimulates respiratory muscles and determines ventilation. It is generated by brainstem networks and modified by chemoreceptor input, mechanoreceptors, wakefulness, metabolic state, medications, and behavioral control.

Respiratory Failure

Inadequate gas exchange resulting in hypoxemia ($\text{PaO}_2 < 60 \text{ mmHg}$), hypercapnia ($\text{PaCO}_2 > 50 \text{ mmHg}$), or both. Type 1 (hypoxemic): V/Q mismatch, shunt, diffusion impairment. Type 2 (hypercapnic): hypoventilation from neuromuscular disease, COPD exacerbation, or opioid overdose.

Respiratory Gating

A radiation-treatment technique that monitors a patient's breathing and delivers the radiation beam during a selected part of the breathing cycle, or pauses it when the target moves outside a predefined window. It helps account for motion in tumors of the chest or upper abdomen and may reduce radiation exposure to nearby healthy tissue.

Respiratory Health

The proper functioning of the lungs and airways, enabling efficient gas exchange and oxygen delivery to the body. Maintaining respiratory health involves avoiding tobacco smoke, minimizing exposure to air pollutants and occupational hazards, and receiving vaccinations against respiratory infections like influenza and pneumococcal disease.

Respiratory Infection

An infection of the upper or lower respiratory tract caused by viruses, bacteria, fungi, or other organisms.

Respiratory Infections

Illnesses caused by pathogens including viruses, bacteria, and fungi that affect the airways and lungs.

Respiratory Insufficiency

Respiratory insufficiency is inadequate gas exchange or ventilatory function causing low oxygen, high carbon dioxide, or both; causes include lung, airway, neuromuscular, and central respiratory disorders.

Respiratory Muscle

A muscle used for breathing, especially the diaphragm and muscles between the ribs. These muscles expand and contract the chest to move air into and out of the lungs.

Respiratory Neoplasm

A respiratory neoplasm is an abnormal growth in the airways or lungs and may be benign or malignant.

Respiratory Paralysis

Respiratory paralysis is failure of the muscles needed for breathing because of nerve, muscle, brainstem, toxin, or spinal-cord dysfunction and is an emergency.

Respiratory Physiology

The study of pulmonary function encompassing ventilation (movement of air), gas exchange (diffusion of O₂ and CO₂ across the alveolar-capillary membrane), transport of gases in blood, and regulation of breathing.

Respiratory Protection

Measures and equipment used to reduce inhalation of harmful dusts, fumes, gases, aerosols, or infectious particles.

Respiratory Quotient

The ratio of carbon dioxide production to oxygen consumption for a particular substrate or metabolic state. It differs from the respiratory exchange ratio measured at the mouth and is used in metabolic and ventilatory physiology.

Respiratory Rate

The number of breaths taken per minute. The expected rate varies with age, activity, temperature, illness, and other factors.

Respiratory Rehabilitation

A program of exercise, education, breathing techniques, and self-management that improves function and quality of life in people with chronic respiratory disease.

Respiratory Syncytial Virus

A respiratory virus that commonly causes bronchiolitis and pneumonia in infants and young children. Infection is usually mild in adults but may be severe in infants, older adults, and people with chronic

heart or lung disease.

Alternate Names:

RSV (Respiratory Syncytial Virus)

Respiratory Syncytial Virus

A respiratory virus that commonly causes bronchiolitis and pneumonia in infants and can cause significant disease in older adults.

Respiratory System

The airways, lungs, respiratory muscles, and related structures that move air and exchange oxygen and carbon dioxide.

Respiratory Therapy

Respiratory therapy includes treatments and breathing support that improve ventilation, oxygenation, airway clearance, or lung function.

Respiratory Tract Infections

Infections affecting any part of the respiratory tract, classified by anatomical location into upper respiratory tract infections (URTIs) and lower respiratory tract infections (LRTIs).

Respiratory Transport

Respiratory transport is movement of oxygen and carbon dioxide between lungs, blood, and tissues, largely through hemoglobin and dissolved gas.

Respiratory Viruses

Viruses that cause respiratory tract infections. Major respiratory viruses include influenza, respiratory syncytial virus (RSV), parainfluenza virus, adenovirus, and human metapneumovirus. SARS-CoV-2 causes COVID-19. Symptoms range from mild upper respiratory illness to severe pneumonia.

Respiratory Viruses And Face Masks

Face masks can reduce spread of respiratory viruses by blocking droplets and some aerosols, especially in crowded or healthcare settings; fit and local guidance matter.

Respirologist

Physician specializing in diseases of the respiratory system, including lung transplantation evaluation and management.

Respirovirus

A genus of paramyxoviruses that includes human parainfluenza viruses, which can cause croup and other respiratory infections.

Respite Care

A medical procedure, treatment, device, or care topic related to respite care; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Response Assessment

Evaluation of treatment response. RECIST criteria for solid tumors. PET response criteria for lymphoma.

Response Evaluation Criteria

Standardized rules for judging whether a disease, especially a tumor, has responded to treatment, remained stable, or progressed. RECIST mainly uses tumor measurements on imaging; other systems may use metabolic or blood-based findings.

Responsible Drinking

Responsible drinking means limiting alcohol according to health guidance, avoiding driving or hazardous activities after drinking, and avoiding alcohol when it is unsafe or contraindicated.

Rest

Rest is reduced physical or mental activity used to recover, though excessive inactivity can impair strength and function in some conditions.

Restenosis

Re-narrowing of a blood vessel after it has been opened by angioplasty or another procedure.

Resting Energy Expenditure

The calories the body uses at rest to maintain es-

sential functions.

Alternate Names:

Resting metabolic rate

Resting Membrane Potential

The steady electrical potential difference across the cell membrane of an unstimulated cell, typically -70 mV in neurons, -80 to -90 mV in skeletal and cardiac muscle, and -50 to -60 mV in smooth muscle.

Resting Phase

The resting phase is a period of relative inactivity, such as G0 in the cell cycle or a physiologic interval between active events.

Restless Anal Syndrome

An uncommon term describing uncomfortable anal sensations and an urge to move or relieve the area, analogous to restless legs symptoms.

Restless Leg Syndrome

Restless legs syndrome causes an urge to move the legs with uncomfortable sensations that worsen at rest and in the evening and improve temporarily with movement. Iron deficiency, kidney disease, medicines, pregnancy, and genetics may contribute.

Alternate Names:

Legs Restless (Restless Leg Syndrome) Syndrome
Restless Legs (Restless Leg Syndrome)

Restless Legs

Restless-legs syndrome causes an urge to move the legs with uncomfortable sensations, beginning or worsening at rest and in the evening and improving temporarily with movement; iron deficiency and medicines may contribute.

Restless Legs Syndrome

An urge to move the legs with uncomfortable sensations, worse at rest and in the evening or night.

Restlessness

Restlessness is an uncomfortable need to move or inability to remain still, caused by anxiety, akathisia, restless-legs syndrome, pain, stimulants, withdrawal,

or medical illness.

Restorative Dentistry

Specialty of restoring damaged teeth. Procedures: fillings, inlays, onlays, crowns, bridges, implants. Goals: function, aesthetics, durability.

Restraint

Device or medication limiting movement. Physical restraints: vests, wrist restraints, lap traps. Chemical restraints: antipsychotics, benzodiazepines for behavior control. Should be last resort, time-limited, with monitoring. Alternatives: addressing underlying causes, environmental modification.

Restraint-Related Harm

Physical or psychological injury associated with using a device, medication, or environmental restriction to limit a person's movement. Older adults are at risk of falls, pressure injury, delirium, aspiration, and functional decline; the least restrictive alternatives should be prioritized.

Restriction Enzyme

A bacterial enzyme that cuts DNA at specific nucleotide sequences and is used in molecular biology.

Restriction Mapping

A laboratory method that uses restriction enzymes to cut DNA and infer the locations of specific recognition sites.

Restrictive Cardiomyopathy

A cardiomyopathy characterized by impaired ventricular relaxation and filling due to stiff, noncompliant myocardium, with preserved systolic function and normal or reduced ventricular volumes. Causes include amyloidosis, sarcoidosis, hemochromatosis, endomyocardial fibrosis, and radiation.

Resuscitation

Emergency care used to restore or support breathing and circulation, including airway support, chest compressions, defibrillation, medicines, and treatment of the cause.

Alternate Names:

CPR; revival

Resveratrol

A plant polyphenol found in grapes and some berries; it is studied in nutrition and aging research, but supplement benefits remain uncertain.

Retained Placenta

Failure of the placenta to deliver after birth within the interval expected for safe management. It may result from failure of separation, abnormal placental attachment, or trapping behind a closed cervix and can cause severe bleeding.

Retained Products Of Conception

Fragments of placental or fetal tissue retained in the uterus after delivery, miscarriage, or abortion. Causes: incomplete placental separation, succenturiate lobe, and placenta accreta. Presents: persistent or heavy bleeding, pain, and fever in the postpartum period.

Retainer

A dental appliance worn after orthodontic treatment to help keep teeth in their corrected positions.

Retardation

An outdated or context-dependent term for slowing or delay; modern clinical language should specify the affected function and cause.

Retching

Repeated involuntary effort to vomit without necessarily producing stomach contents.

Retention Of Urine

Inability to empty the bladder completely or at all despite urine production.

Retention Time

The time required for a compound to pass through a chromatographic system from injection to detection. In analytical toxicology, comparison with validated standards helps identify compounds, but retention

time alone is not definitive identification.

Reticular

Net-like or arranged in a network pattern, describing the appearance or distribution of a skin lesion, rash, or vascular pattern.

Reticular Activating System

A network involving the brainstem reticular formation and related structures that helps regulate wakefulness, alertness, attention, and sleep-wake transitions.

Reticular Dermis

The deeper layer of the dermis containing dense connective tissue, collagen, elastin, blood vessels, nerves, glands, and hair follicles.

Reticular Formation

A network of interconnected nuclei in the brainstem (midbrain, pons, medulla) that regulates arousal, consciousness, sleep-wake transitions, and alertness (ascending reticular activating system), as well as muscle tone, posture, cardiovascular and respiratory control (vasomotor and respiratory centers), pain modulation, and attention.

Reticulin

A network-forming type of collagen that supports bone marrow, liver, lymph nodes, and other organs.

Reticulo-Endothelial System

An older term for the mononuclear phagocyte system, including macrophages in blood, tissues, liver, spleen, and lymph nodes.

Reticulocyte

An immature red blood cell that still contains residual ribosomal RNA; its proportion or absolute count reflects marrow red-cell production.

Reticulocyte Count

The number or percentage of immature red blood cells in blood. It reflects recent bone-marrow red-cell production and helps assess anemia, especially

when interpreted with the hemoglobin and clinical context.

Reticulocytes

Immature red blood cells recently released from bone marrow; their proportion helps assess red-cell production.

Reticulocytopenia

An abnormally low reticulocyte count indicating reduced effective red-cell production, which may occur with marrow failure, nutrient deficiency, inflammation, or kidney disease.

Reticulocytosis

Reticulocytosis is an increased number of immature red blood cells in blood, usually indicating stimulated marrow production after blood loss or hemolysis.

Reticuloendothelial System

An older term for the mononuclear phagocyte system, including macrophages and related cells that remove particles and aged cells.

Reticulophagy

Selective autophagic degradation of endoplasmic reticulum, helping maintain cellular quality control.

Reticulosarcoma

Reticulosarcoma is an obsolete term historically used for certain lymphoid or reticular-cell cancers; modern diagnosis uses current lymphoma or sarcoma classifications.

Reticulum

A network-like structure; in anatomy or cell biology it may refer to a tissue network, endoplasmic reticulum, or ruminant stomach compartment.

Retina

The light-sensitive neural tissue lining the back of the eye that converts light into nerve signals.

Retina Disorder

A disease affecting the retina, including vascular, degenerative, inflammatory, infectious, traumatic, inherited, or neoplastic conditions that can impair vision.

Retina Hemorrhage

Bleeding within or beneath the retina caused by vascular disease, diabetes, hypertension, trauma, retinal vein occlusion, inflammation, or other disorders.

Retina, Disorders Of

Diseases affecting the retina, including vascular, degenerative, inflammatory, infectious, inherited, and neoplastic conditions.

Retinal Artery Occlusion

Blockage of a retinal artery causing sudden, usually painless loss of vision. It requires urgent ophthalmic and vascular evaluation because it may indicate a systemic embolic or vascular disorder.

Retinal Cone

A retinal cone is a light-sensitive photoreceptor specialized for color and detailed vision in brighter conditions.

Retinal Damage

Injury or disease affecting the retina, potentially causing blurred vision, field loss, flashes, floaters, or permanent visual impairment.

Retinal Degeneration

Gradual damage or loss of retinal cells, which can cause reduced vision, blind spots, or loss of the visual field.

Retinal Detachment

An eye emergency in which the light-sensitive neurosensory retina separates from the underlying retinal pigment epithelium. The main types are rhegmatogenous detachment caused by a retinal tear, tractional detachment caused by scar tissue pulling on the retina, and exudative detachment caused by

fluid collecting beneath it without a tear. Sudden new floaters, flashes of light, a curtain or shadow in the field of vision, or reduced vision require urgent eye assessment because untreated detachment can cause permanent vision loss.

Retinal Detachment Repair

Surgical or pneumatic treatment to close retinal breaks and reattach the retina. Techniques include vitrectomy, scleral buckling, and pneumatic retinopexy, selected according to the type and location of detachment.

Retinal Diseases

Disorders that damage the retina, the light-sensitive tissue at the back of the eye.

Retinal Dysplasia

Retinal dysplasia is abnormal development and organization of the retina, which can impair vision and occur with congenital, genetic, or prenatal disorders.

Retinal Dystrophy

Retinal dystrophy is progressive degeneration of retinal cells, often inherited, causing night blindness, visual-field loss, and eventual reduced vision.

Retinal Fold

A retinal fold is a crease or ridge in the retina caused by traction, developmental abnormality, detachment, swelling, or another ocular process.

Retinal Ganglion

A retinal ganglion cell that receives signals from retinal neurons and sends visual information through its axon in the optic nerve.

Retinal Hemorrhages

Bleeding within the retina, a hallmark finding in abusive head trauma (shaken baby syndrome). Found on dilated fundoscopic examination, often bilateral and multi-layered. Also seen in severe accidental trauma, bleeding disorders, and increased intracranial pressure.

Retinal Ischemia

Inadequate blood flow to the retina, often caused by a retinal artery occlusion or severe vascular disease. It may cause sudden, painless visual loss and requires emergency ophthalmic and vascular assessment; treatment depends on the cause and time of presentation.

Retinal Layers

The retina consists of ten distinct layers that process visual information through a complex neural circuit.

Retinal Macula

The macula is the central region of the retina responsible for detailed straight-ahead vision, reading, and color discrimination.

Retinal Neoplasm

A retinal neoplasm is an abnormal growth arising in or near the retina, such as retinoblastoma or a metastatic tumor.

Retinal Perforation

A full-thickness hole or tear involving the retina, which can allow fluid under the retina and lead to retinal detachment.

Retinal Pigment

Retinal pigment refers to melanin-containing pigment in the retinal pigment epithelium, which supports photoreceptors and absorbs excess light.

Retinal Prosthesis

Experimental device for blindness. Argus II system: camera to processor to electrode array stimulates retina. Indications: retinitis pigmentosa. Provides light perception, pattern recognition.

Retinal Tear

A full-thickness break in the neurosensory retina, often caused by vitreoretinal traction. It can permit fluid to enter the subretinal space and progress to rhegmatogenous retinal detachment; symptomatic tears are generally treated urgently with laser photocoagulation or cryotherapy.

Retinal Vasculitis

Retinal vasculitis is inflammation of retinal blood vessels, which can cause hemorrhage, leakage, ischemia, vision loss, and systemic inflammatory disease.

Retinal Vein Occlusion

Obstruction of a retinal vein producing retinal hemorrhages, edema, and variable visual loss. Branch and central forms may cause macular edema, retinal ischemia, and neovascular complications.

Retinaldehyde

An aldehyde form of vitamin A involved in the visual cycle and in conversion to retinal pigments or retinoic acid.

Retinitis

Inflammation or infection of the retina, the light-sensitive tissue at the back of the eye. It may cause blurred vision, floaters, flashes, or loss of sight and needs prompt eye assessment.

Retinitis Pigmentosa

A group of inherited retinal disorders in which rod and cone photoreceptors gradually degenerate. Symptoms may include night blindness, narrowing of the visual field, and progressive loss of vision.

Retinoblastoma

A malignant retinal tumor occurring mainly in young children, associated with RB1 gene abnormalities.

Alternate Names:

Cancer, Retinoblastoma

Retinoblastoma Protein

retinoblastoma protein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Retinoic Acid

An active metabolite of vitamin A that regulates

gene expression and cell differentiation; it is used in selected dermatologic and cancer treatments.

Retinoid

A vitamin-A-related compound used in some skin treatments and other medical applications.

Retinoids

Vitamin-A-related compounds used in selected dermatologic, ophthalmic, and oncologic treatments.

Retinol

Vitamin A alcohol, important for vision, epithelial health, and immune function.

Retinol (Vitamin A)

A fat-soluble vitamin required for vision, epithelial health, immune function, and normal growth; deficiency may cause night blindness and xerophthalmia.

Alternate Names:

Vitamin A

Retinopathy

Disease or damage of the retina, often due to diabetes, hypertension, prematurity, vascular disease, or inherited disorders.

Retinopathy Of Prematurity

Abnormal vascularization of the retina in premature infants, a leading cause of childhood blindness. Affects infants less than 32 weeks gestation and less than 1500 grams. Stages range from mild (avascular retina) to severe (retinal detachment). Screening began at 31-33 weeks corrected gestational age or 4 weeks chronological age.

Retinopathy, Diabetic

Alternate index label; see *Diabetic retinopathy*.

Retinoschisis

Splitting of the retinal layers, producing a separation between inner and outer retinal tissue. It may be degenerative and peripheral or inherited and juvenile, and can cause visual-field loss, reduced acuity,

or complications resembling retinal detachment.

Retinoscope

An instrument that shines light into the eye to help determine refractive error.

Retirement

Retirement is withdrawal from regular employment or a career; changes in income, activity, social contact, and health coverage can affect wellbeing.

Retortamonas Intestinalis

A usually harmless intestinal protozoan that may be found in stool, especially where sanitation is poor. Its presence does not always cause illness.

Retract (Vascular)

To pull a vessel or tissue away from an operative field using a surgical retractor.

Alternate Names:

Vascular

Retractile Testis

A testis that moves between the scrotum and groin because of a strong reflex but can be gently brought into the scrotum. It should be monitored because it may later become undescended.

Retraction (Self-Retaining)

Holding tissue away from an operative field with a retractor that remains in position without continuous manual support.

Alternate Names:

Self-Retaining

Retractor

An instrument or muscle that pulls a structure backward or holds tissue aside during examination or surgery.

Retractor (Abdominal)

A surgical instrument used to hold back the abdominal wall or organs to improve access and visibility during an operation.

Alternate Names:

Abdominal

Retractor (Hand)

A handheld surgical instrument used to hold tissue or an organ away from the operative field.

Alternate Names:

Hand

Retractor (Surgical)

An instrument used to separate or hold tissues aside during a surgical procedure.

Alternate Names:

Surgical

Retreatment

Retreatment is giving a therapy again after incomplete response, recurrence, treatment interruption, or later progression.

Retrieval

Process of organ or tissue recovery from a donor for transplantation; also called procurement.

Retrobulbar Hemorrhage

Bleeding behind the eyeball within the orbit, often after trauma or surgery, that can rapidly increase orbital pressure and threaten vision.

Retrobulbar Neuritis

Inflammation or demyelination of the optic nerve behind the eyeball, causing painful visual loss and often associated with multiple sclerosis.

Retroesophageal

Retroesophageal means located behind the esophagus, such as a mass, lymph node, vessel, or collection in the posterior mediastinum.

Retrognathia

Retrognathia is backward positioning of the lower jaw relative to the upper jaw and may affect facial structure, bite, airway, or sleep breathing.

Retrograde

Moving backward or against the normal direction,

such as retrograde amnesia or retrograde blood flow.

Retrograde Amnesia

Loss of memory for events that occurred before an injury, illness, or other triggering event, with variable preservation of more remote memories.

Retrograde Cystography

Retrograde cystography uses contrast placed through the urethra to outline the bladder and detect leaks, injury, or structural abnormalities.

Retrograde Degeneration

Damage or breakdown of a nerve cell or nerve fiber that spreads backward toward the cell body after the nerve is injured.

Retrograde Ejaculation

Semen enters bladder instead of exiting urethra. Causes: diabetes, surgery, medications. Dry orgasm. Treatment: medications, sperm retrieval for fertility.

Retrograde Intrarenal Surgery

RIRS is a minimally invasive procedure using a flexible scope passed through the urinary tract to inspect and fragment or remove kidney stones.

Retromingent

Retromingent describes urination directed backward, a term historically applied to unusual urinary stream direction or anatomy.

Retroperitoneal Fibrosis

Chronic fibroinflammatory tissue formation in the retroperitoneum, often encasing the ureters and great vessels. It can cause hydronephrosis, flank pain, reduced kidney function, hypertension, or lower-limb edema and may be idiopathic, immune-mediated, drug-related, or secondary to malignancy.

Retroperitoneal Hemorrhage

Bleeding into the retroperitoneal space, which may result from aortic rupture, anticoagulation therapy, retroperitoneal hematoma following femoral catheterization, trauma, or retroperitoneal neo-

plasms. Presents with flank pain, abdominal distension, hypotension, and anemia.

Retroperitoneal Inflammation

Retroperitoneal inflammation is inflammatory disease in the space behind the peritoneum, caused by infection, pancreatitis, urinary disease, trauma, medicines, or fibrosis.

Retroperitoneal Neoplasm

A retroperitoneal neoplasm is an abnormal growth in the tissue space behind the abdominal lining and may be benign or malignant.

Retroperitoneal Space

The retroperitoneal space is the area behind the peritoneal lining containing the kidneys, adrenal glands, pancreas, parts of the bowel, vessels, nerves, and lymphatics.

Retroperitoneum

The anatomical space posterior to the peritoneum. It contains the kidneys, adrenal glands, ureters, aorta, and inferior vena cava, as well as portions of the duodenum, pancreas, and colon. Some organs are secondarily retroperitoneal after development. It is divided into anterior pararenal, perirenal, and posterior pararenal spaces.

Retropharyngeal Abscess

A collection of pus behind the pharynx, potentially obstructing the airway or spreading into the chest.

Retrospective Studies

Retrospective studies examine existing records or past exposures and outcomes; they are efficient but vulnerable to missing data, bias, and confounding.

Retrospective Study

A study that examines past records or exposures to investigate disease causes or outcomes.

Retrospectoscope

A historical instrument or method for examining past events or records; the term is not a standard

current clinical procedure.

Retrosternal Thyroid Surgery

A medical procedure, treatment, device, or care topic related to retrosternal thyroid surgery; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Retrotracheal

Retrotracheal means located behind the trachea, such as a mass, lymph node, vessel, or collection in the posterior mediastinum.

Retrotransposition

Movement of a genetic element to a new genomic location through an RNA intermediate and reverse transcription.

Retroversion

Backward tilting or turning of an organ, especially a uterus tilted toward the spine.

Retroversion Of The Uterus

A uterine position in which the uterus tilts backward toward the spine; it is often a normal variant but can be associated with pain or endometriosis.

Retroviral Vector

A retroviral vector is a modified virus used to deliver genetic material into cells, especially in research and some gene therapies.

Retroviridae

A family of RNA viruses that use reverse transcriptase to make a DNA copy that integrates into the host genome; HIV is a member.

Retroviridae Infection

An infection caused by a retrovirus, which may persist through genomic integration and cause immune, neurologic, or malignant disease.

Retrovirus

An RNA virus that uses reverse transcriptase to make DNA from its RNA genome, including HIV.

Rett Syndrome

An X-linked dominant neurodevelopmental disorder affecting almost exclusively females, caused by mutations in the MECP2 gene.

Return Of Spontaneous Circulation

Restoration of sustained cardiac mechanical activity after cardiac arrest, evidenced by a palpable pulse or measurable blood pressure and signs of circulation. Post-arrest care addresses oxygenation, ventilation, blood pressure, temperature, neurologic injury, and the cause of arrest.

Reuptake

Reuptake is transport of a released neurotransmitter or other molecule back into the cell that released it or into nearby cells, limiting its signal.

Revalidation

Formal reassessment that a healthcare professional remains fit, competent, and up to date to practice.

Revascularisation

Restoration of blood flow to an organ or limb by a procedure such as angioplasty, stenting, or surgical bypass.

Revascularization

Restoration of blood flow to an area through bypass surgery, angioplasty, or another procedure.

Revascularization Surgery

Revascularization surgery restores blood flow around a blocked or narrowed vessel, such as in coronary artery bypass or limb bypass surgery.

Reverberation Artifact

Ultrasound artifact from multiple reflections. Reduced by changing angle, using tissue harmonic imaging.

Reversal Learning

Reversal learning is the ability to change a learned response when the previously rewarded or correct choice is no longer so.

Reverse Total Shoulder Replacement

A medical procedure, treatment, device, or care topic related to reverse total shoulder replacement; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Reverse Transcriptase

An enzyme that synthesizes DNA using RNA as a template.

Reverse Transcriptase Inhibitor

An antiviral medicine that blocks reverse transcriptase, used in combination treatment for HIV or selected infections.

Reverse Transcription

Synthesis of DNA from an RNA template by the enzyme reverse transcriptase.

Reverse Trendelenburg Position

A supine position in which the head and upper body are elevated above the feet.

Reversible Cell Injury

Cellular dysfunction caused by a harmful stimulus that can resolve when the stimulus is removed. Common changes include cellular swelling, fatty change, membrane blebbing, and reduced oxidative phosphorylation without irreversible membrane or nuclear destruction.

Revision Arthroplasty

Surgical replacement, exchange, or repair of a failed or painful joint prosthesis. Indications include infection, aseptic loosening, instability, periprosthetic fracture, wear, and recurrent dislocation.

Reye Johnson Syndrome (Reye Syndrome)

Alternate index label for Reye Syndrome.

Reye Syndrome

A rare acute encephalopathy with hepatic dysfunction, classically occurring in children recovering from a viral illness. It causes persistent vomiting, altered

mental status, cerebral edema, and elevated liver enzymes without primary inflammatory liver failure.

Alternate Names:

Reye Johnson Syndrome (Reye Syndrome)

Reye'S Syndrome

A rare, potentially fatal illness causing acute brain dysfunction and liver injury, associated particularly with viral illness and aspirin exposure in children.

Reynolds Pentad

Extension of the Charcot triad to include hypotension and altered mental status representing suppurative cholangitis, a severe life-threatening form of ascending cholangitis. Patients present with high fever, rigors, right upper quadrant pain, jaundice, hypotension, and confusion.

Rezafungin

A long-acting echinocandin antifungal medicine that inhibits fungal cell-wall synthesis. It is used for selected invasive candidiasis infections; its long duration permits an intermittent dosing schedule under specialist supervision.

Rgs Proteins

rgs proteins is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Rh Factor

The Rh blood-group antigen system, especially the D antigen.

Rh Incompatibility

An immune reaction that can occur when an Rh-negative pregnant person carries an Rh-positive fetus and forms antibodies against fetal red blood cells. Anti-D immune globulin can prevent sensitization.

Rh Isoimmunization

Maternal alloantibody production against fetal Rh D-positive red blood cells in an Rh D-negative mother, occurring when fetal RBCs enter the ma-

ternal circulation (fetomaternal hemorrhage).

Rhabdoid Tumor

A highly aggressive malignant tumor of infancy, occurring in the kidney (malignant rhabdoid tumor) or brain (atypical teratoid/rhabdoid tumor), characterized by loss of the SMARCB1 (INI1) gene.

Rhabdomyolysis

Rhabdomyolysis is rapid breakdown of skeletal muscle that releases myoglobin, potassium, and other intracellular contents into the bloodstream. Causes include crush injury, extreme exertion, seizures, heat illness, medicines, and metabolic disease. It may cause muscle pain, weakness, dark urine, hyperkalemia, acute kidney injury, arrhythmias, or compartment syndrome. Markedly elevated creatine kinase supports the diagnosis and requires prompt medical assessment.

Rhabdomyolysis And AKI

Severe muscle breakdown releases myoglobin and other intracellular contents, potentially causing acute kidney injury, electrolyte abnormalities, and cardiac arrhythmias. Findings may include muscle pain, weakness, dark urine, and elevated creatine kinase; treatment requires urgent assessment and cause-directed fluid and supportive therapy.

Rhabdomyoma

A benign cardiac tumour composed of striated muscle cells, occurring predominantly in infants and children and strongly associated with tuberous sclerosis. It commonly involves the ventricular myocardium and may regress spontaneously, although large or multiple tumours can cause obstruction or arrhythmias.

Rhabdomyosarcoma

Rhabdomyosarcoma is a malignant tumor showing skeletal-muscle differentiation, occurring most often in children but also in adults.

Rhabdoviridae

A family of bullet-shaped, negative-sense RNA viruses that includes rabies virus and related animal

or plant viruses.

Rhabdoviruses

A family of bullet-shaped RNA viruses including rabies virus and vesicular stomatitis virus.

Rhamnose

Rhamnose is a deoxy sugar found in plants and bacterial cell walls and used in biochemical and microbiologic research.

Rhegmatogenous Retinal Detachment

The most common type of retinal detachment, caused by a full-thickness break or tear in the retina that allows liquefied vitreous humor to enter the subretinal space, separating the neurosensory retina from the underlying retinal pigment epithelium.

Rhesus (Rh) Factor

A system of antigens on the surface of red blood cells, especially the D antigen. A person is Rh-positive when the D antigen is present and Rh-negative when it is absent. Rh incompatibility can affect pregnancy and transfusion safety.

Alternate Names:

Rh

Rhesus Factor

The Rh blood-group antigen, especially the D antigen, whose presence or absence is important in transfusion and pregnancy.

Rhesus Incompatibility

Rhesus incompatibility occurs when an RhD-negative pregnant person forms antibodies against RhD-positive fetal red cells, risking haemolytic disease in a later or affected pregnancy; anti-D immunoglobulin prevents sensitization.

Rhesus System

The Rh blood-group system, a group of inherited red-cell antigens including D, C, c, E, and e. RhD status is commonly reported as positive or negative and is important in transfusion compatibility and prevention of hemolytic disease of the fetus and

newborn.

Rheumatic Chorea

Rheumatic chorea is involuntary, irregular movement caused by an autoimmune reaction after group A streptococcal infection and is a manifestation of acute rheumatic fever.

Rheumatic Disease

A broad group of disorders affecting joints, muscles, bones, connective tissue, or immune function, including arthritis and lupus.

Rheumatic Fever

An inflammatory complication of untreated or inadequately treated group A streptococcal throat infection, affecting joints, heart, skin, or brain.

Rheumatic Heart Disease

Valvular damage resulting from acute rheumatic fever, an autoimmune response to Group A streptococcal pharyngitis. Most commonly affects the mitral valve (stenosis/regurgitation), followed by aortic. Prevention through prompt treatment of streptococcal pharyngitis.

Rheumatism

Pain, stiffness, or discomfort in joints and surrounding soft tissues.

Rheumatoid Arthritis

A chronic autoimmune inflammatory arthritis causing synovitis, pain, stiffness, deformity, and possible systemic disease.

Alternate Names:

RA

Rheumatoid Arthritis Cause Gastrointestinal Issues

Rheumatoid arthritis can affect the gastrointestinal system indirectly through inflammation, associated autoimmune disease, reduced mobility, or medicines such as nonsteroidal anti-inflammatory drugs. Abdominal pain, bleeding, persistent diarrhea, or weight loss needs evaluation rather than

attribution to arthritis alone.

Rheumatoid Arthritis Inflammation Of The Brain

Direct brain inflammation is uncommon in rheumatoid arthritis, but vasculitis, medication effects, infection, stroke risk, or another autoimmune process can cause neurologic symptoms. New confusion, seizure, weakness, severe headache, or vision change requires urgent assessment.

Rheumatoid Factor

An autoantibody, usually of the IgM class, directed against part of IgG. It may support a diagnosis of rheumatoid arthritis but can also occur in other autoimmune diseases, infections, and healthy people.

Rheumatoid Factor

An autoantibody, usually directed against the Fc portion of immunoglobulin G, found in many people with rheumatoid arthritis but also in other autoimmune, infectious, and healthy states.

Rheumatoid Lung Disease

Lung involvement associated with rheumatoid arthritis, including interstitial lung disease, pleural disease, nodules, or airway abnormalities.

Rheumatoid Nodule

A rheumatoid nodule is a firm inflammatory lump under the skin or near a joint in rheumatoid arthritis, often over pressure points.

Rheumatoid Pneumoconiosis

Rheumatoid pneumoconiosis is lung disease involving rheumatoid arthritis and dust-related lung nodules, historically called Caplan syndrome.

Rheumatoid Synovitis

Chronic autoimmune inflammation of synovial tissue in rheumatoid arthritis, producing pain, swelling, pannus formation, cartilage damage, and progressive joint destruction.

Rheumatologist

A physician specializing in inflammatory and autoimmune diseases of joints and connective tissue.

Rheumatology

The medical specialty concerned with inflammatory, autoimmune, and musculoskeletal disorders.

Rhinencephalon

The part of the brain concerned chiefly with the sense of smell and related limbic functions.

Rhinitis

Inflammation of the nasal lining causing congestion, discharge, sneezing, or itching.

Rhinitis, Allergic

Alternate index label; see *Hay fever*.

Rhinitis, Nonallergic

Alternate index label; see *Nonallergic rhinitis*.

Rhinitis, Vasomotor

Alternate index label; see *Nonallergic rhinitis*.

Rhinocephaly

Rhinocephaly is a rare congenital facial-development abnormality with a proboscis-like structure and severe forebrain malformation.

Rhinophyma

Progressive thickening and enlargement of the nose, usually associated with advanced rosacea.

Rhinoplasty

Surgery to reshape or reconstruct the nose for functional or cosmetic reasons.

Rhinorrhea

Excessive watery or mucoid discharge from the nose, commonly caused by infection, allergy, irritants, or changes in temperature.

Rhinorrhoea

Discharge of watery or mucoid fluid from the nose.

Rhinoscopy

Examination of the nasal cavity and nasopharynx using an instrument or endoscope to assess the mucosa, septum, turbinates, masses, bleeding, or obstruction.

Rhinovirus

A genus of non-enveloped, positive-sense RNA viruses in the Picornaviridae family. Rhinoviruses are a common cause of the common cold and spread mainly through respiratory secretions and contaminated hands or surfaces.

Rhinovirus A

A species group of rhinoviruses that commonly causes the common cold and can worsen asthma or trigger wheezing.

Rhinovirus C

A rhinovirus group associated with common-cold symptoms and, in some children, more severe wheezing or lower-respiratory illness.

Rhinoviruses

A group of viruses that commonly cause the common cold and can worsen asthma.

Rhizobiaceae

A family of soil bacteria that form symbiotic nitrogen-fixing relationships with plants; human infection is rare.

Rhizobium

A genus of soil bacteria that forms nitrogen-fixing root nodules on legumes and is not a usual human pathogen.

Rhizomucor

A genus of molds that can cause mucormycosis, especially in people with diabetes, neutropenia, or severe immune suppression.

Rhizomucor Pusillus

A mold that can cause invasive mucormycosis involving the lungs, sinuses, skin, or other organs in severely immunocompromised people.

Rhizopus

A genus of molds that includes important causes of mucormycosis, a rapidly invasive infection of sinuses, lungs, skin, or other tissues.

Rhizopus Oryzae

A mold, also called *Rhizopus arrhizus*, that is a common cause of invasive mucormycosis.

Rhizotomy

Surgical or radiofrequency interruption of selected nerve roots to reduce pain, spasticity, or abnormal reflexes.

Rhizoxin

Rhizoxin is a microtubule-disrupting compound produced by some fungi and studied as an experimental cytotoxic or anticancer agent.

Rhodium

A chemical element used in industrial and dental materials; its compounds can irritate tissues and may cause allergic reactions.

Rhodobacteraceae

A family of photosynthetic or metabolically diverse bacteria found in aquatic and soil environments; human disease is uncommon.

Rhodococcus

A genus of gram-positive bacteria that includes *Rhodococcus equi*, an opportunistic pathogen causing pneumonia mainly in immunocompromised people.

Rhodococcus Equi

A gram-positive bacterium that can cause cavitary pneumonia and disseminated infection, especially in people with advanced immune suppression.

Rhodopseudomonas

A genus of photosynthetic or metabolically versatile environmental bacteria rarely associated with human infection.

Rhodopsin

A light-sensitive protein in rod cells of the retina that enables vision in dim light.

Rhodospirillaceae

A family of metabolically diverse bacteria found in soil and water; some species carry out photosynthesis or nitrogen fixation.

Rhodospirillales

An order of diverse bacteria that includes Rhodospirillaceae and related environmental organisms.

Rhodospirillum

A genus of spiral-shaped aquatic bacteria capable of photosynthesis or nitrogen fixation; human infection is rare.

Rhodotorula Glutinis

A pigmented yeast found in the environment that can rarely cause bloodstream or catheter-related infection.

RhoGAM

Rh immune globulin containing anti-D antibodies, used to prevent Rh isoimmunization in Rh-negative mothers. Administered intramuscularly at 28 weeks gestation and within 72 hours after delivery of an Rh-positive infant. Also given after amniocentesis, chorionic villus sampling, miscarriage, ectopic pregnancy, or abdominal trauma.

Rhombencephalon

The rhombencephalon is the embryonic hindbrain, which develops into the pons, medulla, cerebellum, and related structures.

Rhonchi

Low-pitched continuous respiratory sounds caused by airflow through narrowed or mucus-filled large

airways.

Rhubarb Leaves Poisoning

Rhubarb leaves poisoning involves toxic or environmental exposure to rhubarb leaves poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Rhythm Method

A fertility-awareness method that estimates fertile days from menstrual-cycle timing to avoid or achieve pregnancy.

Rhytidectomy

Cosmetic surgery that removes excess facial skin and tightens supporting tissues.

Alternate Names:

Facelift

Rib

One of 12 paired, curved bones forming the thoracic cage, protecting the heart, lungs, and great vessels. True ribs (1–7) attach directly to the sternum via costal cartilage. False ribs (8–10) attach indirectly to the sternum via the costal cartilage of rib 7. Floating ribs (11–12) have no anterior attachment.

Alternate Names:

Ribs

Rib Cage

The rib cage consists of ribs, sternum, thoracic vertebrae, muscles, and connective tissues that protect chest organs and assist breathing.

Rib Fracture

A break in one or more ribs, usually causing localized chest pain worsened by breathing, coughing, or movement and sometimes associated with injury to underlying organs.

Ribavirin

An antiviral medicine used in selected viral infections, usually in combination and under specialist supervision.

Ribcage Pain

Ribcage pain may arise from muscles, ribs, joints, lungs, heart, or other organs; sudden severe pain or breathing difficulty requires urgent assessment.

Riboflavin

Vitamin B2, a precursor of FAD and FMN needed for cellular energy metabolism.

Riboflavin (Vitamin B2)

A water-soluble vitamin that forms the coenzymes FAD and FMN. These cofactors participate in oxidation-reduction reactions, including electron transfer, fatty-acid oxidation, and energy metabolism.

Alternate Names:

Vitamin B2

Riboflavin Assay

A laboratory test that measures riboflavin, or vitamin B2, in blood or other samples to assess vitamin status.

Riboflavin Deficiency

Deficiency of vitamin B2, which can cause inflammation or fissuring of the mouth and lips, sore throat, dermatitis, anemia, and other mucosal or systemic findings.

Ribonuclease

An enzyme that breaks down RNA into smaller fragments or nucleotides.

Ribonuclease H

An enzyme that degrades the RNA strand of an RNA-DNA hybrid, important in DNA replication and reverse transcription.

Ribonuclease Iii

An enzyme that cuts double-stranded RNA and helps process regulatory small RNAs.

Ribonucleic Acid

RNA, a nucleic acid involved in genetic information transfer, protein synthesis, regulation, and some

catalytic functions.

Ribonucleic Acid

RNA is a nucleic acid that carries genetic instructions, helps make proteins, and regulates cell activity; some viruses use RNA as their genetic material.

Ribonucleoprotein

ribonucleoprotein is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Ribonucleoprotein

A complex made of RNA and one or more proteins, involved in RNA processing, transport, translation, or regulation.

Ribonucleoside

A molecule made of ribose sugar linked to a nitrogenous base, such as adenosine, guanosine, cytidine, or uridine.

Ribonucleotide Reductase

Ribonucleotide reductase converts ribonucleotides to deoxyribonucleotides needed for DNA synthesis and repair.

Ribonucleotides

Nucleotides containing ribose sugar and RNA bases, serving as RNA building blocks and cellular energy or signaling molecules.

Ribophagy

Ribophagy is selective autophagic degradation of ribosomes, allowing cells to recycle ribosomal components when nutrients or growth conditions change.

Ribose

A five-carbon sugar found in RNA, ATP, and many other biologically important molecules.

Ribosomal Protein

ribosomal protein is a protein or protein-related

molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Ribosomal RNA

RNA component of ribosomes, catalyzing peptide bond formation (peptidyl transferase activity). Eukaryotic ribosome: 60S (28S, 5.8S, 5S rRNAs + proteins) + 40S (18S rRNA + proteins) = 80S. Prokaryotic: 50S (23S, 5S rRNAs) + 30S (16S rRNA) = 70S. rRNA genes are highly conserved, used in phylogenetics.

Ribosome

A cellular structure that translates messenger RNA into protein.

Ribosome Subunit

One of the RNA-protein components of a ribosome; the small and large subunits join to translate messenger RNA into protein.

Ribostamycin

Ribostamycin is an aminoglycoside antibiotic used mainly in research or selected settings; it can cause kidney and hearing toxicity.

Ribozyme

A catalytic ribonucleic acid molecule capable of facilitating, speeding up, or executing specific biochemical reactions in the absence of protein enzymes, functioning analogously to proteinaceous biocatalysts. Discovered by Thomas Cech and Sidney Altman, ribozymes operate through complex RNA tertiary folding that coordinates catalytic metal divalent cations (Mg^{2+}). Naturally occurring examples include the large ribosomal subunit (where 28S/23S ribosomal RNA acts as a peptidyl transferase synthesizing peptide bonds during protein translation), self-splicing group I and II introns, and small self-cleaving ribozymes (hammerhead and hairpin ribozymes); engineered ribozymes are investigated in molecular medicine and gene therapy to target and degrade disease-specific pathogenic messenger RNAs.

Richter's Syndrome

Transformation of chronic lymphocytic leukemia or small lymphocytic lymphoma into an aggressive lymphoma, most often diffuse large B-cell lymphoma.

Ricin

Ricin is a highly toxic plant protein that blocks protein synthesis; ingestion, inhalation, or injection can cause severe organ injury and death.

Rickets

Defective mineralization of growing bone, usually due to vitamin D, calcium, or phosphate deficiency.

Alternate Names:

Osteomalacia (Rickets)

Rickettsia

A group of small intracellular bacteria transmitted mainly by arthropods and causing diseases such as typhus and spotted fevers.

Rickettsia Felis

Rickettsia felis is a flea-associated bacterium that can cause a spotted-fever-like illness in humans.

Rickettsia Infections

Rickettsial infections are illnesses caused by intracellular bacteria spread mainly by ticks, fleas, lice, or mites and often causing fever and rash.

Rickettsia Japonica

Rickettsia japonica is a tick-borne bacterium associated with Japanese spotted fever.

Rickettsia Prowazekii

Rickettsia prowazekii causes epidemic typhus and is transmitted mainly by body lice.

Rickettsia Rickettsii

Rickettsia rickettsii causes Rocky Mountain spotted fever and is transmitted by infected ticks.

Rickettsia Typhi

Rickettsia typhi causes murine typhus and is transmitted mainly by fleas.

Rickettsiaceae

Rickettsiaceae is a family of small intracellular bacteria that includes *Rickettsia* and related tick-, flea-, louse-, or mite-borne pathogens.

Rickettsialpox

Rickettsialpox is a mite-borne infection caused by *Rickettsia akari*, producing an inoculation eschar followed by fever and a vesicular rash.

Ridaforolimus

Ridaforolimus is an investigational mTOR inhibitor studied for anticancer effects by reducing growth and proliferation signaling.

Ridge Preservation

Socket grafting after extraction to maintain bone volume. Reduces ridge resorption.

Ridged Nail

A ridged nail has longitudinal or transverse grooves in its surface, which may result from aging, trauma, inflammation, nutritional problems, or systemic disease.

Riedel's Thyroiditis

Riedel's thyroiditis is rare fibrosing inflammation that makes the thyroid hard and can extend into nearby tissues, causing compression or hypothyroidism.

Rieger Syndrome

A genetic disorder affecting development of the eyes, teeth, and sometimes the umbilicus or other organs; it is often related to PITX2 or FOXC1 variants.

Rifabutin

A rifamycin antibiotic used mainly for selected mycobacterial infections and tuberculosis regimens; it has important drug interactions and can discolor body fluids.

Rifampicin

A rifamycin antibiotic used in tuberculosis and selected other bacterial infections, usually in combi-

nation to prevent resistance.

Rifampin

Rifampin is a rifamycin antibiotic that inhibits bacterial DNA-dependent RNA polymerase and is a key component of tuberculosis treatment; it causes orange body fluids and has many drug interactions and hepatotoxicity risk.

Rifamycin

Rifamycin is an antibiotic that inhibits bacterial RNA polymerase; related rifamycin medicines are used for tuberculosis and other infections.

Rifaximin

Rifaximin is a minimally absorbed rifamycin antibiotic used for selected intestinal infections, prevention of recurrent hepatic encephalopathy, and some diarrhoeal syndromes; indication and resistance patterns guide use.

Rift Valley Fever

A mosquito-borne viral disease affecting animals and humans, usually causing fever but sometimes severe eye, brain, or hemorrhagic disease.

Right Atrium

Receives deoxygenated blood from superior vena cava, inferior vena cava, and coronary sinus. Smooth posterior part (sinus venosus) and rough anterior auricle with pectinate muscles. The fossa ovalis is the fetal foramen ovale remnant. The SA node pacemaker is near the superior vena cava opening.

Right Bundle Branch

A continuation of the bundle of His that travels along the right side of the interventricular septum to the right ventricle. It supplies the right ventricular myocardium and Purkinje fibers. Right bundle branch block (RBBB) occurs when this branch is damaged, causing delayed right ventricular activation.

Right Coronary Artery

Arises from the right aortic sinus and travels in the

right atrioventricular groove. It supplies the right atrium, right ventricle, SA node (60% of individuals), AV node (85% of individuals), and the inferior wall of the left ventricle.

Right Coronary Artery Occlusion

Blockage of the right coronary artery, reducing blood flow to areas of the heart it supplies and potentially causing myocardial infarction or rhythm disturbances.

Right Heart Failure

A syndrome in which the right ventricle cannot adequately deliver blood to the pulmonary circulation, usually causing elevated systemic venous pressure and congestion. It may occur with left-sided heart failure or independently. Causes include pulmonary hypertension, right-ventricular infarction, pulmonary embolism, cardiomyopathy, and constrictive pericarditis. Signs may include elevated jugular venous pressure, peripheral edema, hepatomegaly, ascites, and weight gain.

Right Heart Ventricular Angiography

Right-heart ventricular angiography uses injected contrast and X-ray imaging to show the right ventricle and its blood flow during selected cardiac evaluations.

Right Hypochondriac Region

One of nine abdominal regions in the upper right abdomen, bounded medially by the right midclavicular line and superiorly by the right costal margin. Contains the cecum, appendix, ascending colon, and inferior right liver lobe. Pain here may indicate appendicitis (McBurney's point), cecal pathology, or hepatic disease.

Right Iliac Region

Lower right abdomen containing the cecum, appendix, and terminal ileum. McBurney's point (junction of lateral and middle thirds from umbilicus to ASIS) is critical for appendicitis diagnosis. Also significant for evaluating inguinal and femoral hernias, and the right ureter's course.

Right Lymphatic Duct

Short lymphatic vessel (approximately 1.5 cm) draining lymph from the right upper quadrant: right arm, right side of the head and neck, and right thorax. Formed by the junction of right jugular, subclavian, and bronchomediastinal trunks, emptying into the right venous angle (junction of right internal jugular and subclavian veins).

Right Sixth Cranial Nerve Palsy

Weakness or paralysis of the right abducens nerve (CN VI), causing inability to abduct the right eye and resulting in horizontal diplopia worse at distance and on right gaze.

Right Ureterovesical Calculus

A stone lodged at the junction of the right ureter and bladder, causing flank pain, hematuria, and urinary urgency or frequency. May require endoscopic or lithotripsy intervention.

Right Ventricle

Receives deoxygenated blood from the right atrium through the tricuspid valve, pumping into the pulmonary trunk via the pulmonary valve. Thinner walls (3-5 mm) than the left ventricle due to low pulmonary resistance.

Righting Reflex

The righting reflex is an automatic response that restores head and body orientation when balance or position changes.

Rigid And Flexible Sigmoidoscopy

A sigmoidoscopy examines the rectum and lower colon with a flexible or rigid instrument. It can evaluate bleeding, pain, altered bowel habits, inflammation, polyps, or tumors; preparation, reach, comfort, and biopsy capability differ by technique.

Rigidity

Abnormally increased resistance to passive movement, classically seen in Parkinson disease or meningeal irritation.

Rigor

Marked stiffness or rigidity; rigor mortis is post-mortem muscle stiffening, while rigor may also mean a severe chill.

Rigor Mortis

Post-mortem stiffening of body muscles resulting from loss of ATP synthesis following death. Without ATP, actin and myosin filaments remain permanently cross-linked in a rigid state.

Rigosertib Sodium

Rigosertib sodium is an investigational anticancer agent studied for effects on cell-cycle regulation and survival signaling.

Rilotumumab

An investigational monoclonal antibody targeting hepatocyte growth factor, studied as an anticancer treatment.

Rilpivirine

Rilpivirine is a nonnucleoside reverse-transcriptase inhibitor used with other medicines to treat HIV infection.

Riluzole

A medicine used to modestly slow disease progression in amyotrophic lateral sclerosis and to treat selected forms of spinal muscular atrophy.

Rimantadine

An antiviral medicine formerly used against some influenza A strains; resistance and limited effectiveness have greatly reduced its role.

Rimmed Vacuoles

A pathognomonic histopathological feature of muscle biopsies in sporadic Inclusion Body Myositis (sIBM) and GNE myopathy (Nonaka myopathy), visualized on hematoxylin and eosin (H&E) and modified Gömöri trichrome staining. Consists of small, autophagic cytoplasmic or subsarcolemmal vacuoles that are lined and bordered by dense, basophilic or reddish granular material and membranous de-

bris; electron microscopy reveals tubulofilamentous inclusions measuring 15–18 nm in diameter, and immunohistochemistry demonstrates pathological accumulation of amyloid- β , hyperphosphorylated tau, ubiquitin, and p62 aggregates.

Alternate Names:

Autophagic Rimmed Vacuoles

Ring Block

Local anesthetic injected around the base of a finger or toe to numb the entire digit.

Ring Chromosome

A chromosome whose two ends have joined to form a ring, often with loss or disruption of genetic material. It may cause developmental or other health problems, depending on the chromosome and changes involved.

Ring Schatzki (Schatzki Ring)

Alternate index label for Schatzki Ring.

Ring Vaccination

A public-health strategy in which close contacts and contacts of contacts around an identified case are vaccinated to create a protective ring and limit transmission.

Ringling In The Ear

Alternate index label; see *Tinnitus*.

Ringling In The Ears

Perception of ringing, buzzing, hissing, or other sound without an external source, medically called tinnitus. It may be associated with hearing loss, loud-noise exposure, earwax, ear infection, medicines, or jaw disorders. Sudden hearing loss, one-sided symptoms, or pulsatile ringing requires medical evaluation.

Alternate Names:

Tinnitus (Ringling In The Ears)

Ringling In The Ears (Tinnitus (Ringling In The Ears))

Alternate index label for Tinnitus (Ringling in the

Ears).

Ringworm

A contagious superficial fungal infection of skin, hair, or nails; it is not caused by a worm.

Alternate Names:

Scalp And Body (Ringworm)

Ringworm (Tinea)

Ringworm is a superficial dermatophyte fungal infection of skin, scalp, or nails that produces scaly annular lesions or hair loss; treatment depends on the site and may require topical or oral antifungals.

Ringworm Of The Body

A contagious fungal infection causing an itchy, ring-shaped rash on the skin.

Ringworm Of The Foot

Alternate index label; see *Athlete's foot*.

Ringworm Of The Scalp

A fungal infection of the scalp and hair that can cause scaling, broken hairs, and hair loss.

Rinne Test

The Rinne test is a screening test that compares air conduction and bone conduction hearing using a vibrating tuning fork (typically 512 Hz) to help differentiate between conductive and sensorineural hearing loss.

Rinnes Test

Rinne's test compares air and bone conduction with a tuning fork to help assess conductive hearing loss.

RIPE Therapy

Standard 4-drug regimen for active TB: Rifampin, Isoniazid, Pyrazinamide, Ethambutol. Intensive phase 2 months, continuation phase 4 months (total 6 months). Pyridoxine prevents isoniazid-induced neuropathy.

Rippling Muscle Disease

A rare, benign myopathy characterized by

mechanically-induced muscle contractions that spread across the muscle as a visible wave (rippling), typically triggered by stretch, percussion, or movement.

Risedronate Sodium

The sodium salt of risedronate, a bisphosphonate used to prevent or treat osteoporosis and other disorders of excessive bone resorption.

Risk

Risk is the probability of an unwanted event or outcome, considered together with its severity and the circumstances that influence it.

Risk Adjustment

Risk adjustment accounts for differences in baseline patient characteristics so outcomes or payments can be compared more fairly across groups.

Risk Factor

A characteristic, exposure, or behavior associated with increased probability of disease or another outcome.

Risk Management

Identification, assessment, reduction, and monitoring of hazards that could cause harm.

Risk Register

A documented list of identified risks, their likelihood, impact, controls, owners, and review status.

Risk-Taking

Risk-taking is behavior involving deliberate or inadvertent exposure to possible harm, influenced by perception, reward, development, mental health, and circumstances.

Risks Of Hip And Knee Replacement

A medical procedure, treatment, device, or care topic related to risks of hip and knee replacement; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Risks Of Tobacco

Tobacco use increases risks of cancer, heart and blood-vessel disease, stroke, lung disease, pregnancy complications, addiction, and harm from second-hand smoke.

Risks Of Underage Drinking

Alcohol use by minors can impair brain development, judgment, learning, safety, and mental health and increases risks of injury, dependence, and unsafe sexual or substance-related behavior.

Risky Drinking

Alcohol use that increases the likelihood of health, safety, or social consequences.

Alternate Names:

Hazardous drinking

Risperidone

Risperidone is an atypical antipsychotic that blocks dopamine and serotonin receptors and treats schizophrenia, bipolar disorder, and selected severe behavioral symptoms; hyperprolactinaemia, metabolic effects, movement disorders, and sedation may occur.

Ristocetin

Ristocetin is an antibiotic used in laboratory testing of platelet aggregation and von Willebrand factor, not as a routine modern treatment.

Ristocetin-Induced Platelet Aggregation

A platelet-function assay that evaluates platelet agglutination in the presence of ristocetin, which depends on von Willebrand factor and platelet glycoprotein Ib.

Risus Sardonicus

A fixed, sardonic facial spasm classically associated with tetanus or strychnine poisoning.

Ritodrine

A beta-2 agonist formerly used as a tocolytic to delay premature labor; it has limited current use because of maternal adverse effects.

Ritonavir

Ritonavir is an HIV protease inhibitor mainly used at low doses to boost levels of other antiviral medicines by slowing their breakdown.

Ritual

A ritual is a repeated, structured behavior or ceremony; repetitive rituals may also occur as compulsions in obsessive-compulsive disorder.

Rituximab

Rituximab is a monoclonal antibody against CD20 that depletes B cells and is used for B-cell malignancies and selected autoimmune diseases; infusion reactions, infections, hepatitis-B reactivation, and rare PML are important risks.

Rivaroxaban

Rivaroxaban is an oral direct factor Xa inhibitor used to prevent or treat venous thromboembolism and reduce stroke risk in selected atrial fibrillation; bleeding and renal-function-dependent dosing are important considerations.

Rivastigmine

A cholinesterase inhibitor used for mild-to-moderate Alzheimer disease and Parkinson disease dementia.

River Blindness

Onchocerciasis, a parasitic infection caused by *Onchocerca volvulus* and transmitted by blackflies. It can cause severe itching, skin disease, and visual impairment.

RLS

Restless legs syndrome, an urge to move the legs with uncomfortable sensations, especially at night.

Alternate Names:

Restless legs syndrome

RMDs (Repetitive Motion Disorders (RMDs))

Alternate index label for Repetitive Motion Disorders (RMDs).

RNA

Ribonucleic acid, a nucleic acid involved in gene expression and regulation.

RNA Binding

The attachment of a protein or other molecule to RNA, affecting RNA stability, transport, folding, processing, or translation.

RNA Biosynthesis

The cellular production of RNA from a DNA template or, for some viruses, from an RNA template.

RNA Degradation

Enzymatic breakdown of RNA, which controls RNA lifespan, removes damaged transcripts, and recycles nucleotides.

RNA Folding

The process by which an RNA strand forms secondary and three-dimensional structures that influence its function.

RNA Helicase

An enzyme that uses energy to unwind or remodel RNA structures or RNA-protein complexes.

RNA Interference

A gene-silencing process in which small RNAs guide proteins to degrade or block specific messenger RNA molecules.

RNA Splicing

Removal of introns and joining of exons from a precursor RNA to produce mature messenger RNA or other functional RNA.

RNA Transport

Movement of RNA within a cell or between cells, often mediated by transport proteins and required for translation or signaling.

RNA Virus

A virus whose genetic material is RNA; its replication often involves an RNA-dependent polymerase

and may have a high mutation rate.

RNase P

An RNA-protein enzyme complex that processes precursor transfer RNA by removing its extra 5-prime sequence.

Road Rage

Alternate index label; see *Intermittent explosive disorder*.

Remote Surgery

Surgery performed when the surgeon is physically separated from the patient and controls robotic instruments through a telecommunications system. It may also include remote surgical assistance or guidance, with a qualified local team available to manage the patient and respond to emergencies.

Alternate Names:

Telesurgery Telepresence Surgery

Robinow Syndrome

A rare inherited skeletal-development disorder with distinctive facial features, short stature, vertebral or rib abnormalities, and genital differences.

Robot-Assisted Rehabilitation

Robotic devices for repetitive task training. Upper extremity, lower extremity, gait training. Objective measurement, high repetition.

Robotic Hysterectomy

Hysterectomy performed through minimally invasive abdominal access using a surgeon-controlled robotic platform. The extent of uterine, cervical, tubal, and ovarian removal depends on the indication and operative plan.

Robotic Surgery

A surgical technique in which a surgeon controls a computer-assisted system that moves a camera and small instruments. It is usually performed through small incisions and may allow precise movements, but the system does not operate independently.

Alternate Names:

Robotic-Assisted Thoracoscopic Surgery

An advanced minimally invasive thoracic surgical platform that uses a multi-arm robotic console system to perform complex pulmonary and mediastinal operations. The surgeon controls wristed instruments offering 7 degrees of freedom, tremor filtration, and stereoscopic three-dimensional high-definition visualization, which facilitates precise mediastinal and hilar lymph node dissection, segmentectomies, and sleeve resections through small intercostal port incisions without rib spreading.

Alternate Names:

RATS; Robotic Thoracic Surgery; Robotic Lobectomy

Robust

Robust means strong, resilient, or relatively insensitive to variation; in research, a robust result remains stable under reasonable analytic changes.

Rod

A rod is a long cylindrical structure; in the retina, a rod photoreceptor supports dim-light and peripheral vision.

Rod Photoreceptor

rod photoreceptor is a protein or protein-related molecule that performs a defined structural, signaling, transport, immune, or enzymatic function in cells; abnormal amounts or activity may contribute to disease.

Rodent Ulcer

An older name for basal cell carcinoma, a common locally invasive skin cancer.

Rods

Retinal photoreceptor cells that provide vision in dim light and contribute to peripheral vision but do not detect color.

Roentgenology

Roentgenology is the historical term for the use and

study of X-rays in medical diagnosis and imaging.

Rofecoxib

Rofecoxib is a selective COX-2 anti-inflammatory drug withdrawn in many countries because of increased cardiovascular risk.

Romberg Sign

Loss of balance when standing with the feet together and the eyes closed, suggesting impaired proprioception or vestibular function.

Romberg Test

The Romberg test is a clinical neurological test used to assess the integrity of the dorsal column-medial lemniscus pathway (proprioception) and the cerebellum in maintaining balance. The test is performed by asking the patient to stand with feet together and eyes open, and then to close the eyes.

Romberg's Sign

Increased unsteadiness when a person stands with the feet together and closes the eyes, suggesting impaired proprioception or vestibular function.

Rombergism

Unsteadiness or falling when standing with feet together and eyes closed, indicating impaired proprioception or vestibular function.

Romidepsin

A histone deacetylase inhibitor used for selected T-cell lymphomas; it can cause cytopenias, infection risk, and ECG-related effects.

Romiplostim

A thrombopoietin-receptor agonist that stimulates platelet production and is used mainly for chronic immune thrombocytopenia.

Rongeur

A strong, jawed surgical instrument used to bite and remove small pieces of bone or other hard tissue.

Ronnel

Ronnel is an organophosphate insecticide that can inhibit acetylcholinesterase and cause cholinergic toxicity after exposure.

Root (Dental Root)

The anatomical portion of a human tooth situated apical to the cemento-enamel junction, embedded and anchored firmly into the bony alveolar socket of the maxilla or mandible by the fibrous periodontal ligament. Composed internally of dentin enclosing the root canal space (which contains the neurovascular dental pulp) and covered externally by a specialized calcified avascular layer of cementum. Teeth possess one root (incisors, canines), two roots (mandibular molars), or three roots (maxillary molars), each terminating at the apical foramen where neurovascular bundles enter the pulp; susceptible to root resorption, radicular cysts, periapical abscesses, and vertical root fractures.

Root Canal

The channel inside a tooth root containing the pulp and its nerves and blood vessels.

Root Canal Therapy

Dental treatment in which infected or inflamed pulp is removed from a tooth, the canal is cleaned and sealed, and the tooth is restored.

Root Caries

Caries on exposed root surfaces. Common in elderly with recession. Fluoride effective.

Root Filling

Dental treatment in which the cleaned root canal is sealed to prevent reinfection.

Root Fracture

Vertical or horizontal fracture of root. Prognosis depends on location and direction. Vertical root fractures typically require extraction.

Rooting Reflex

A primitive reflex present from birth to 3-4 months,

triggered by stroking the infant's cheek or corner of the mouth. The infant turns toward the stimulus and opens the mouth, searching for the nipple. Facilitates breastfeeding. Absence may indicate neurologic impairment. Disappears as voluntary feeding develops.

Ropinirole

A non-ergot dopamine agonist (D2/D3) used for Parkinson disease and restless legs syndrome. Adverse effects: nausea, dizziness, somnolence (sudden sleep attacks), orthostatic hypotension, impulse control disorders (gambling, hypersexuality), hallucinations, and leg edema. Extended-release for once-daily Parkinson dosing.

Ropivacaine

A long-acting local anesthetic used for regional nerve blocks and epidural anesthesia; excessive blood levels can cause neurologic or cardiac toxicity.

Ropivacaine Hydrochloride

The hydrochloride salt of ropivacaine, a long-acting local anesthetic used for regional and epidural anesthesia.

Rorschach Test

A projective psychological test using inkblots; its diagnostic validity is limited and interpretation requires specialist training.

Rosacea

A chronic inflammatory facial skin disorder causing flushing, persistent redness, visible vessels, papules, pustules, or ocular symptoms.

Rose Bengal

Rose bengal is a dye used in eye-surface testing and laboratory or diagnostic procedures; it highlights damaged or poorly protected conjunctival cells.

Roseola

Roseola infantum is a common viral illness, usually caused by human herpesvirus 6, with several days of high fever followed by a pink trunk rash as the

fever resolves; febrile seizures may occur.

Alternate Names:

Sixth Disease (Roseola)

Roseola Infantum

A common childhood viral illness, usually caused by human herpesvirus 6, with high fever followed by a trunk rash as fever resolves.

Roseolovirus

A genus of herpesviruses that includes human herpesvirus 6 and 7, commonly associated with roseola and sometimes serious disease after reactivation.

Roseomonas

A genus of pink-pigmented gram-negative bacteria that can rarely cause opportunistic bloodstream or device-related infection.

Roseomonas Gilardii

A pink-pigmented gram-negative bacterium that can rarely cause bloodstream, wound, or other opportunistic infection.

Roseomonas Mucosa

A pink-pigmented environmental bacterium occasionally associated with opportunistic infection, including bloodstream infection.

Rosiglitazone

Rosiglitazone is a PPAR-gamma agonist that improves insulin sensitivity in type 2 diabetes; weight gain, oedema, heart-failure worsening, fractures, and cardiovascular risk require careful patient selection.

Rosiglitazone Maleate

The maleate salt of rosiglitazone, a thiazolidinedione that improves insulin sensitivity in type 2 diabetes but may cause weight gain, edema, fractures, and heart-failure worsening.

Ross Procedure

A complex cardiac surgical autograft operation used to treat severe aortic valve disease, particularly in

infants, children, and young adults. The patient's diseased native aortic valve is excised and replaced with their own native pulmonary valve (pulmonary autograft), and a cryopreserved human pulmonary homograft is implanted to reconstruct the right ventricular outflow tract. The living autograft retains capacity for somatic growth, demonstrates superior hemodynamics, and carries low thrombogenicity, completely avoiding the need for lifelong anticoagulation.

Alternate Names:

Pulmonary Autograft Procedure; Ross Operation

Ross River Virus

A mosquito-borne alphavirus that causes Ross River fever, with fever, rash, and often prolonged joint pain or swelling.

Rostaporfin

Rostaporfin is an investigational photosensitizing compound studied for photodynamic treatment of cancer or other abnormal tissue.

Rostral

Positioned toward the snout or front of the head; in the brain, it means toward the forehead.

Rosuvastatin Calcium

Rosuvastatin calcium is a statin medicine that lowers LDL cholesterol and reduces the risk of cardiovascular events.

Rotation

Movement around an axis; in anatomy it describes turning a body part inward, outward, or around its longitudinal axis.

Rotator Cuff

The group of four shoulder muscles and tendons that stabilize the glenohumeral joint and rotate the arm.

Alternate Names:

Tendinitis Rotator Cuff (Rotator Cuff)

Rotator Cuff Arthropathy

Glenohumeral joint degeneration associated with chronic, irreparable rotator-cuff deficiency. Superior migration of the humeral head, altered joint mechanics, pain, and loss of elevation may lead to consideration of reverse shoulder arthroplasty.

Rotator Cuff Exercises

Rotator cuff exercises strengthen and stretch shoulder muscles and are selected according to the injury, pain, range of motion, and rehabilitation plan.

Rotator Cuff Injury

Damage to one or more tendons or muscles that stabilize and move the shoulder.

Rotator Cuff Problems

Rotator cuff problems include tendinopathy, bursitis, and tears that cause shoulder pain, weakness, painful motion, and difficulty with overhead activity.

Rotator Cuff Repair

A medical procedure, treatment, device, or care topic related to rotator cuff repair; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Rotator Cuff Tear

A partial or complete tear of one or more tendons stabilizing the shoulder, causing pain, weakness, and difficulty raising or rotating the arm.

Rotavirus

A double-stranded RNA virus that is the leading cause of severe gastroenteritis in children worldwide. The virus has a segmented genome and infects enterocytes of the small intestine. Rotavirus causes profuse watery diarrhea, vomiting, and dehydration. Diagnosis is made by antigen detection in stool.

Rotavirus

Rotavirus is a highly contagious virus causing vomiting and watery diarrhea, especially in infants and young children; vaccination substantially reduces severe disease.

Rotavirus Antigen Test

A rotavirus antigen test detects rotavirus proteins in stool and can help diagnose viral gastroenteritis, especially in infants and young children.

Rotavirus Infection

A highly contagious viral gastroenteritis causing vomiting and watery diarrhea, especially in infants and young children; vaccination prevents most severe disease.

Rotaviruses

Viruses that cause acute gastroenteritis, especially severe diarrhea and dehydration in young children.

Rotenone

Rotenone is a naturally occurring pesticide that inhibits mitochondrial complex I and can be toxic to people and animals after exposure.

Roth Spots

Retinal hemorrhages with pale centers, seen on fundoscopic examination in infective endocarditis. They are a minor criterion in the Duke criteria. Roth spots are caused by immune complex-mediated vasculitis of retinal vessels.

Roth's Spots

Small retinal haemorrhages with pale centres, associated with conditions such as infective endocarditis, severe anaemia, leukaemia, and diabetes.

Rothia Dentocariosa

A gram-positive bacterium found in the mouth that can rarely cause dental, bloodstream, heart-valve, or other opportunistic infections.

Rothia Mucilaginosa

A gram-positive bacterium that colonizes the mouth and can rarely cause bloodstream or device-related infection.

Rothmund-Thomson Syndrome

A genetically heterogeneous disorder characterized by poikiloderma beginning in infancy, short

stature, sparse hair, skeletal abnormalities, juvenile cataracts in some forms, and increased risk of osteosarcoma or other cancers.

Rotifera

A phylum of microscopic aquatic animals called rotifers; they are mainly studied in biology and are not typical human pathogens.

Rotterdam Criteria

The internationally accepted 2003 consensus diagnostic criteria for Polycystic Ovary Syndrome. Diagnosis requires the presence of at least two of the following three features, provided other etiologies (such as congenital adrenal hyperplasia, hyperprolactinemia, and thyroid disorders) are excluded: (1) oligo- or anovulation; (2) clinical and/or biochemical signs of hyperandrogenism; and (3) polycystic ovarian morphology on ultrasound (≥ 20 follicles per ovary measuring 2–9 mm, or an ovarian volume > 10 mL).

Roughage

Dietary fiber, especially the indigestible plant material that adds bulk to stool and supports normal bowel function.

Rouleau

A stack or column of red blood cells, commonly seen in blood with increased plasma proteins.

Rouleaux

Stacking of red blood cells like coins, often caused by increased plasma proteins.

Round

Round means circular or approximately spherical; in anatomy or pathology it describes shape and must be interpreted with the structure being described.

Round Back

Alternate index label; see *Kyphosis*.

Round Ligament

A round ligament is a cord-like supporting structure,

including the uterine ligaments that pass toward the groin and help maintain uterine position.

Round Ligament Pain

Sharp or aching pain in the lower abdomen or groin caused by stretching of the round ligaments as the uterus enlarges during pregnancy. Persistent, severe, or bleeding-associated pain requires medical evaluation.

Round Pneumonia

Pneumonia appearing as a round opacity on CXR rather than typical consolidation. More common in children and elderly. May mimic lung mass.

Roundworm

Ascariasis: infection by *Ascaris lumbricoides*, the most common human soil-transmitted helminth, acquired by ingesting eggs from contaminated food/soil.

Roundworms

Nematode worms, some of which infect humans and cause intestinal or tissue disease.

Rous Sarcoma

A malignant connective-tissue tumor caused by Rous sarcoma virus in chickens; the historical term is important in cancer biology rather than routine human diagnosis.

Route Of Administration

The path by which a drug enters the body, influencing absorption, onset, bioavailability, and first-pass metabolism. Enteral routes: oral, sublingual, buccal, rectal. Parenteral: intravenous (100 percent bioavailability, immediate onset), intramuscular, subcutaneous, intradermal.

Rovsing Sign

Physical exam sign of acute appendicitis: palpation of the left lower quadrant elicits pain in the right lower quadrant due to displacement of gas stretching the inflamed appendix. See Appendicitis.

Roxithromycin

Roxithromycin is a macrolide antibiotic used in some countries for selected susceptible bacterial infections.

RPR Test

The RPR test is a blood screening test for syphilis that detects antibodies associated with tissue injury; reactive results require confirmatory testing.

RSI

Repetitive strain injury, a group of pain and dysfunction syndromes associated with repeated movement or sustained posture.

RSV

Respiratory syncytial virus, a common respiratory virus that can cause bronchiolitis and pneumonia.

RSV Antibody Test

An RSV antibody test detects antibodies to respiratory syncytial virus and may help assess past exposure or immune response, but is not usually the main test for acute infection.

RT-PCR

RT-PCR converts RNA to complementary DNA and amplifies target sequences, allowing detection or measurement of RNA viruses, genes, and transcripts.

Ruba Miliaria (Heat Rash)

Alternate index label for Heat Rash.

Rubber

Rubber is an elastic material used in gloves, devices, and equipment; natural-rubber latex can cause irritant or allergic reactions.

Rubber Cement Poisoning

Rubber cement poisoning involves toxic or environmental exposure to rubber cement poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Rubber Dams

Sheets used in dentistry to isolate teeth from saliva and the rest of the mouth during procedures.

Rubella

A contagious viral infection causing mild fever and rash but potentially severe congenital abnormalities when acquired during pregnancy.

Alternate Names:

Measles, German Three-Day Measles

Rubella Virus

A togavirus that causes rubella (German measles), a mild febrile illness with rash. The virus is transmitted by respiratory droplets. Rubella is most dangerous during pregnancy, causing congenital rubella syndrome (CRS) with deafness, cataracts, and cardiac defects. Diagnosis involves serology. Treatment is supportive.

Rubeola

Measles, a highly contagious viral illness caused by measles virus and characterized by fever, cough, conjunctivitis, and a widespread maculopapular rash.

Alternate Names:

Measles (Rubeola)

Rubeola (Measles (Rubeola))

Alternate index label for Measles (Rubeola).

Rubeosis Iridis

Rubeosis iridis is abnormal growth of new blood vessels on the iris, usually from retinal ischemia such as diabetes or vein occlusion, and may cause neovascular glaucoma.

Rubidium

A chemical element whose radioactive isotopes can be used in medical imaging; nonmedical rubidium compounds can affect potassium-related physiology.

Rubin Maneuver

An internal rotational maneuver for shoulder dystocia: the operator inserts a hand and applies pressure to the posterior aspect of the anterior fetal shoulder,

rotating it toward the fetal chest (adducting the shoulders). This reduces the bisacromial diameter and disimpacts the anterior shoulder from the pubic symphysis.

Rubinstein-Taybi Syndrome

A genetic developmental disorder associated with characteristic facial features, short stature, broad thumbs and toes, and intellectual disability. It is commonly caused by pathogenic variants in CREBBP or EP300.

Rubitecan

An investigational oral topoisomerase-I inhibitor studied for cancer treatment.

Rubivirus

A genus of viruses that includes rubella virus, which causes rubella and congenital rubella syndrome.

Rubredoxins

Small iron-containing proteins that transfer electrons in certain microorganisms and organelles.

Rubricyte Count

A rubricyte count measures rubricytes, an older term for immature red blood cells, and may help assess red-cell production in the bone marrow.

Rubulavirus

A genus of paramyxoviruses that includes mumps virus and related animal viruses.

Rule Of Nines

A method for estimating the percentage of body-surface area affected by burns.

Rule Out

Rule out means evaluate a possibility and determine whether it is present; it does not mean the condition has already been excluded.

Rumen

The rumen is the largest compartment of a ruminant animal's stomach where microbes ferment plant ma-

terial before digestion continues.

Rumination

Repeated regurgitation and rechewing of food after eating, without the usual vomiting process; persistent symptoms can cause weight loss and dental damage.

Rumination Disorder

Recurrent effortless regurgitation of recently ingested food, followed by rechewing, reswallowing, or spitting, without the retching typical of vomiting. It can cause weight loss, malnutrition, and dental complications.

Ruminococcus

A genus of anaerobic intestinal bacteria involved in digestion of complex carbohydrates and plant fiber.

Ruminococcus Gnavus

An intestinal bacterium involved in mucin and carbohydrate metabolism and studied in relation to gut inflammation.

Runner'S Knee

Alternate index label; see *Patellofemoral pain syndrome*.

Runny Nose

Runny nose refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Rupture

A tear or complete break in an organ, vessel, muscle, or other tissue.

Ruptured Abdominal Aortic Aneurysm

A tear through an abdominal aortic aneurysm causing internal bleeding, shock, and usually sudden abdominal or back pain. It requires emergency repair.

Ruptured Disk

A spinal disk whose outer layer has torn, allowing

inner material to bulge outward.

Alternate Names:

Herniated disk; slipped disk

Ruptured Eardrum

A ruptured eardrum is a tear in the tympanic membrane caused by infection, pressure, trauma, or foreign objects and may cause pain, drainage, hearing loss, or dizziness.

Ruptured Eardrum (Perforated Eardrum)

A hole or tear in the eardrum that may cause pain, drainage, hearing loss, or ringing and can result from infection, injury, or pressure changes.

Alternate Names:

Eardrum, Ruptured Perforated Eardrum

Ruptured Perforated Eardrum

A perforated eardrum is a tear in the tympanic membrane from infection, pressure change, trauma, or an inserted object. Pain, drainage, hearing loss, or ringing may occur; keeping the ear dry and avoiding unprescribed drops helps while healing or repair is assessed.

Rural Health

Rural health concerns health outcomes and care access in rural communities, where distance, workforce shortages, transport, resources, and social conditions may affect services.

Russell-Silver Syndrome

A congenital growth disorder involving prenatal and postnatal growth restriction, relative macrocephaly, triangular face, body asymmetry, and feeding difficulty. It is often associated with abnormalities of genomic imprinting on chromosomes 7 or 11.

Russell-Siver Syndrome

Likely a misspelling of Russell-Silver syndrome, a growth disorder characterized by poor fetal and postnatal growth, feeding difficulty, relative macrocephaly, and characteristic body asymmetry or facial features.

Ruthenium

A chemical element used in industry and research; some ruthenium compounds are studied as experimental anticancer agents.

Rutin

A flavonoid glycoside found in plants and foods; it is sold in supplements, but clinical benefits are not firmly established.

Rx

Prescription or treatment. Symbol derives from Latin "requirere" (to take). Also used as abbreviation for remedy. Rx symbol: Rx or %x;.

Ryanodine

A plant alkaloid that binds ryanodine receptors and alters calcium release from muscle-cell sarcoplasmic reticulum; it is mainly used in research.

Ryle'S Tube

A nasogastric tube used to drain stomach contents or deliver fluids, nutrition, or medicines.

Alternate Names:

RAST

S

16S Ribosomal RNA Sequencing

Molecular identification of bacteria and archaea using conserved and variable regions of the 16S ribosomal RNA gene. It is useful for difficult-to-culture organisms but may not distinguish closely related species or provide susceptibility data.

S Phase

S phase is the part of the cell cycle during which DNA is replicated so each daughter cell can receive a complete genome.

S-Adenosylmethionine

SAMe, a methyl-group donor involved in neurotransmitter, phospholipid, DNA, and other biochemical reactions and studied as a supplement for selected mood, liver, or pain conditions.

S3 Gallop Pathology

A third heart sound produced during rapid ventricular filling. In some adults it may indicate volume overload or ventricular dysfunction, although it can be physiologic in children, young adults, and pregnancy. Its significance depends on the complete clinical assessment.

S4 Gallop Pathology

A fourth heart sound indicates atrial contraction against a stiff ventricle (decreased compliance). Associated with left ventricular hypertrophy from hypertension, aortic stenosis, and hypertrophic cardiomyopathy. S4 is best heard at the apex with the bell, and is accentuated by exertion, expiration, and isorhythmic AV dissociation.

SA Node

The sinoatrial node is the heart's normal pacemaker, generating electrical impulses that initiate each sinus heartbeat.

SAA

SAA commonly means serum amyloid A, an acute-phase protein that rises with inflammation and can contribute to secondary amyloidosis when persistently elevated.

Sabarubicin

Sabarubicin is an investigational anthracycline-like antineoplastic drug studied for cytotoxic effects against cancer cells.

Sabin Vaccine

The oral polio vaccine containing live attenuated poliovirus strains.

Saccade

A saccade is a rapid, conjugate eye movement that shifts gaze from one target to another.

Saccharomyces

A genus of yeasts, including *S. cerevisiae*; some species are used industrially and can rarely cause opportunistic infection.

Saccharomyces Cerevisiae

Saccharomyces cerevisiae is a budding yeast used in baking, brewing, biotechnology, and some probiotic products; it can rarely cause invasive fungemia or other infection in vulnerable patients.

Saccular Aneurysm

A rounded pouch protruding from one side of an artery because of focal wall weakness, with rupture and clot formation risks determined by location, size, and growth.

Sachet Poisoning

Illness caused by swallowing, inhaling, or getting the contents of a sachet on the skin or in the eyes. Effects depend on the substance and amount; some sachets contain medicines or dangerous chemicals.

Sacral

Sacral means relating to the sacrum, the triangular bone at the base of the spine that connects with the pelvis.

Sacral Dimple

A small indentation in the skin over the lower spine that is present at birth.

Alternate Names:

Pilonidal Dimple

Sacral Fistula

A sacral fistula is an abnormal tract near the sacrum that may connect skin with deeper tissue and can reflect congenital disease, infection, or pilonidal disease.

Sacral Nerve Stimulation

Neuromodulation for urinary/fecal incontinence, constipation. Sacral nerve root (S3) stimulation. Implantable pulse generator + electrode. Indications: refractory urge incontinence, fecal incontinence, non-obstructive urinary retention.

Sacral Nerve Stimulator

An implanted device that delivers electrical stimulation to sacral nerves to alter bladder, bowel, or pelvic-floor function in selected patients.

Sacral Nerves

The spinal nerves arising from the sacral spinal cord and supplying the pelvis, pelvic floor, bladder, bowel, and lower limbs.

Sacral Region

The lower back and posterior pelvic region overlying the sacrum, a fused triangular bone that connects the lumbar spine with the pelvis.

Sacral Vertebra

One of the fused vertebrae forming the sacrum, which supports the spine, joins the pelvic bones, and transmits trunk weight to the lower limbs.

Sacral Vertebrae

Five vertebrae fused to form the sacrum. They transmit body weight from the vertebral column to the pelvic girdle and contain anterior and posterior sacral foramina for spinal nerves.

Sacral Vertebrae (S1-S5)

The five sacral vertebrae that fuse to form the sacrum, connect the spine to the pelvis, and provide foramina for sacral nerve roots.

Sacrococcygeal

Sacrococcygeal means relating to the sacrum and coccyx at the lower end of the vertebral column.

Sacrococcygeal Teratoma

A congenital tumor arising at the base of the spine and containing tissues from multiple germ layers; large fetal lesions can cause high-output heart failure or hydrops.

Sacrocolpopexy

An open, laparoscopic, or robotically assisted reconstructive surgical procedure considered the gold standard for durable correction of apical vaginal vault or uterine prolapse. Involves suspending the vaginal apex or cervix to the anterior longitudinal ligament of the sacral promontory (S_1) using a synthetic non-absorbable polypropylene Y-mesh graft, restoring anatomical vaginal axis and length while preserving coital function.

Sacroileitis

A misspelling or variant of sacroiliitis, inflammation of a sacroiliac joint.

Sacroiliac Joint

The synovial joint between the sacrum and the ilium (auricular surfaces), forming the connection between the axial skeleton and the pelvis. The sacroiliac joint is reinforced by strong ligaments (anterior, interosseous, and posterior sacroiliac ligaments, sacrotuberous and sacrospinous ligaments).

Sacroiliac Joint Dysfunction (Sacroiliac Joint Pain)

Alternate index label for Sacroiliac Joint Pain.

Sacroiliac Joint Dysfunction (SI Joint Pain)

Pain or abnormal movement from the joint between the lower spine and pelvis. It may cause pain in the lower back, buttock, or thigh and can follow injury, pregnancy, arthritis, or changes in walking.

Sacroiliac Joint Pain

Sacroiliac-joint pain arises near the junction of the lower spine and pelvis and may follow injury, pregnancy, arthritis, leg-length or gait changes, or inflammation. Pain can radiate to the buttock or thigh; examination distinguishes it from hip or spine disease.

Alternate Names:

Sacroiliac Joint Dysfunction (Sacroiliac Joint Pain)

Sacroiliitis

Inflammation of one or both sacroiliac joints, causing pain in the lower back, buttock, or posterior thigh that may worsen with prolonged rest. Causes include axial spondyloarthritis, psoriatic or enteropathic arthritis, infection, trauma, and degenerative disease.

Sacrospinous Ligament Fixation

A transvaginal surgical procedure designed to correct apical vaginal vault prolapse or uterine prolapse by securing the vaginal apex or posterior cervical remnant to the unilateral or bilateral sacrospinous ligament and coccygeus muscle complex. Offers an entirely vaginal approach avoiding abdominal incisions or peritoneal entry, while carrying low morbidity and rapid postoperative recovery.

Alternate Names:

Sacrospinous Colpopexy; Sacrospinous Suspension

Sacrum

Large triangular bone from 5 fused sacral vertebrae between the hip bones. Articulates with L5 superiorly and coccyx inferiorly. The sacral promontory is an obstetric landmark. Contains 4 pairs of sacral foramina. Forms the posterior pelvic wall.

SAD

Seasonal affective disorder, a recurring mood disorder linked to seasonal changes and reduced daylight.

Alternate Names:

Seasonal Affective Disorder (SAD)

Sad Mood

Sad mood is a feeling of unhappiness or low mood that may be a normal response or a symptom of depression, grief, illness, or another condition.

Saddle Pulmonary Embolism

A massive, life-threatening form of pulmonary embolism in which a large, non-fragmented thromboembolus lodges directly across the bifurcation of the main pulmonary artery trunk, extending into and partially or completely obstructing both the right and left pulmonary arteries. It precipitates sudden, severe elevation of right ventricular afterload, acute cor pulmonale, right ventricular dilatation and failure, cardiogenic shock, and cardiovascular arrest; emergency treatment requires rapid systemic thrombolysis, catheter-directed thrombectomy, or surgical pulmonary embolectomy.

Alternate Names:

Saddle Embolus; Bifurcation Pulmonary Embolism

Sadism

Sexual arousal or gratification from inflicting pain, humiliation, or suffering; clinical significance depends on distress, impairment, or nonconsent.

Safe Driving For Teens

Safe driving for teens means reducing crash risk by using seat belts, avoiding alcohol, cannabis, and

distracted driving, following speed limits, and using graduated licensing rules.

Safe Eating During Cancer Treatment

Food-safety measures for people receiving cancer treatment, including careful hand hygiene, avoiding unsafe or undercooked foods, preventing cross-contamination, and following individualized advice when immunity is low.

Safe Period

A fertility-awareness term for days believed to have a lower chance of conception; cycle variation makes it unreliable as sole contraception.

Safe Sex

Sexual practices that reduce the risk of sexually transmitted infections and unintended pregnancy, including condoms or other barriers, testing, vaccination, and informed consent.

Safety

Safety is protection from preventable harm and includes identifying hazards, reducing risk, using equipment correctly, and responding promptly to danger.

Safety Of Drugs

The assessment and management of medicine benefits, risks, interactions, contraindications, monitoring, and safe use.

Safety Plan

A collaboratively written, practical plan for responding to escalating suicidal thoughts or emotional crisis. It identifies warning signs, internal coping strategies, supportive contacts, professional resources, and methods to restrict access to lethal means.

Safety Study

A safety study evaluates whether a medicine, device, procedure, or treatment causes harmful effects and identifies risks before or during clinical use.

Safflower Oil

Safflower oil is a vegetable oil rich in unsaturated fatty acids, used as food or in products; its health effect depends on amount and dietary substitution.

Safingol

An investigational sphingolipid-analog drug that inhibits protein kinase C and has been studied in cancer treatment, usually in combination with other agents.

Safrole

Safrole is a natural aromatic compound found in some plants and oils; high or prolonged exposure is toxic and has carcinogenic concern.

Sagittal

Sagittal means oriented from front to back and dividing the body or a structure into right and left portions.

Sagittal Plane

A vertical plane that divides the body or an organ into right and left sides.

Sagopilone

An investigational epothilone-like anticancer drug that disrupts microtubules and has been studied for treatment of advanced solid tumors.

Salaam Attacks

Brief repeated flexor spasms or head drops in infants, historically associated with infantile spasms.

Salabrasion

A dermabrasion method using salt or an abrasive salt preparation to remove superficial skin layers; it carries risks of pain, infection, pigment change, and scarring.

Salads And Nutrients

Salads can provide fiber, vitamins, minerals, and beneficial plant compounds, but nutritional value depends on ingredients, portions, dressings, and overall dietary pattern.

Salbutamol (Albuterol)

A selective beta-2 adrenergic agonist bronchodilator used for acute relief and prevention of bronchospasm in asthma and COPD. Relaxes airway smooth muscle; short-acting (SABA), onset 5-15 minutes, duration 4-6 hours via inhaler or nebulizer. Adverse effects: tremor, tachycardia, palpitations, hypokalemia, and headache.

Alternate Names:

Albuterol

Salicylate

A group of compounds that includes aspirin and has analgesic, antipyretic, and anti-inflammatory effects.

Salicylate Toxicity

Poisoning caused by excessive aspirin or another salicylate, potentially producing vomiting, ringing in the ears, fever, abnormal breathing, confusion, and acid-base disturbances.

Salicylic Acid

A keratolytic and anti-inflammatory compound used topically for acne, psoriasis, warts, and other hyperkeratotic disorders.

Salient

Salient means prominent or especially relevant; a salient clinical finding is one that stands out and may influence diagnosis or management.

Saline

A sterile water solution containing sodium chloride, commonly used for fluids, irrigation, or drug delivery.

Alternate Names:

Normal saline

Saline Infusion Sonohysterography

Ultrasound evaluation of uterus using saline contrast. Evaluates intrauterine pathology.

Saline Nasal Washes

Rinsing the nasal passages with sterile, distilled, or

appropriately treated saline to loosen mucus and reduce nasal irritants; unsafe water can cause serious infection.

Saliva

Secretion of salivary glands (1-1.5 L/day). Functions: lubrication, digestion (amylase), buffering, antimicrobial, remineralization. Xerostomia (dry mouth) increases caries risk. Causes: medications, Sjogren syndrome, radiation, dehydration.

Salivary Duct Stones

Calcified deposits obstructing a salivary duct, most often the submandibular duct, causing recurrent painful swelling of a gland during meals and sometimes infection.

Salivary Gland

Any exocrine gland that secretes saliva into the oral cavity. The major paired glands are the parotid, submandibular, and sublingual glands; numerous minor glands occur throughout the oral mucosa.

Salivary Gland Biopsy

Removal of salivary-gland tissue or a needle sample for microscopic examination of suspected autoimmune disease, inflammation, infection, or neoplasm.

Salivary Gland Cancer

A rare malignancy most often arising in the parotid gland. Histologic types include mucoepidermoid carcinoma, adenoid cystic carcinoma, and acinic cell carcinoma; facial nerve preservation is a key surgical consideration.

Salivary Gland Infection

Infection of a salivary gland, usually bacterial and often involving the parotid or submandibular gland. Painful swelling, fever, dry mouth, and purulent drainage from the duct may occur.

Salivary Gland Infections

Infections of a salivary gland or its duct, commonly bacterial or viral, causing painful swelling, tenderness, fever, dry mouth, or purulent drainage.

Salivary Gland Neoplasm

A salivary gland neoplasm is an abnormal growth in a salivary gland that may be benign or malignant.

Salivary Gland Scintigraphy

Salivary gland function assessment using Tc-99m pertechnetate. Used for Sjogren syndrome, obstruction, radiation sialadenitis.

Salivary Gland Tumors

Tumors arising in the salivary glands. Many are benign, especially in the parotid gland; pleomorphic adenoma is a common benign type. Malignant types include mucoepidermoid, adenoid cystic, and acinic cell carcinoma.

Alternate Names:

Tumor, Salivary Gland

Salivary Glands

Exocrine glands that produce saliva, including the parotid, submandibular, and sublingual glands.

Salivation

Production and release of saliva by the salivary glands. Saliva moistens food, begins digestion, protects teeth, and helps swallowing; excessive or reduced salivation can have many causes.

Salk Vaccine

The inactivated poliovirus vaccine, given by injection to protect against poliomyelitis. It contains killed virus and cannot cause polio; it differs from the older oral live-attenuated vaccine.

Salmon Patches

Flat pink or red capillary birthmarks, commonly on the eyelids, forehead, or nape; many fade during infancy.

Salmonella

A genus of Gram-negative, facultatively anaerobic, rod-shaped bacteria in the Enterobacteriaceae family. It includes non-typhoidal strains that cause gastroenteritis and typhoidal strains that can cause enteric fever.

Salmonella Arizonae

A Salmonella serotype associated with reptiles and other animals that can cause gastroenteritis or invasive infection, especially in people with weakened immunity.

Salmonella Choleraesuis

A Salmonella serotype adapted to pigs that can cause severe, invasive infection in humans, including bloodstream infection and pneumonia.

Salmonella Enterica

The main Salmonella species causing human disease, including gastroenteritis, enteric fever, and invasive bloodstream infection.

Salmonella Enterocolitis

Salmonella enterocolitis is intestinal infection causing diarrhea, fever, abdominal cramps, and sometimes bacteremia after contaminated food, water, or animal exposure.

Salmonella Food Poisoning (Salmonellosis)

Alternate index label for Salmonellosis.

Salmonella Gallinarum

A host-adapted Salmonella bacterium that causes fowl typhoid in chickens and other poultry; it is not a usual cause of human disease.

Salmonella Infection

An intestinal infection caused by Salmonella bacteria, usually acquired through contaminated food or water.

Salmonella Infections

Infections caused by Salmonella bacteria, ranging from gastroenteritis to enteric fever or invasive bloodstream disease.

Salmonella Pullorum

A host-adapted Salmonella bacterium that causes pullorum disease in young poultry; it is not a usual cause of human disease.

Salmonella Typhi

The bacterium that causes typhoid fever, usually acquired through contaminated food or water and capable of spreading through the bloodstream.

Salmonella Typhimurium

A common Salmonella serotype that usually causes foodborne gastroenteritis but can cause invasive disease in vulnerable people.

Salmonella Typhisuis

A Salmonella serotype mainly associated with swine; human infection is uncommon but may cause enteric or invasive disease.

Salmonella Vaccines

Vaccines used mainly to prevent typhoid fever, including typhoid conjugate, Vi polysaccharide, and oral live-attenuated vaccines; routine vaccination does not prevent all Salmonella gastroenteritis.

Salmonellosis

Infection with Salmonella bacteria (non-typhoidal), typically acquired from contaminated food (eggs, poultry, meat). Presents with acute gastroenteritis: diarrhea, fever, and abdominal cramps within 6-72 hours of ingestion. Self-limited in immunocompetent patients. Bacteremia can occur, especially in immunocompromised.

Alternate Names:

Salmonella Food Poisoning (Salmonellosis)

Salpingectomy

Surgical removal of a fallopian tube. It may be used to treat a tubal ectopic pregnancy, hydrosalpinx, or other tubal disease. The approach and whether the tube is removed depend on the clinical condition, fertility plans, and the state of the opposite tube.

Salpingitis

Inflammation of the fallopian tubes, most commonly caused by ascending infection from the lower genital tract. Salpingitis is a component of pelvic inflammatory disease and is often bilateral.

Salpingitis Isthmica Nodosa

A pathological condition characterized by nodular muscular hypertrophy and benign diverticular herniation of the fallopian tube epithelium into the hypertrophied muscularis layer of the isthmus segment. Strongly linked to chronic pelvic inflammation and unrecognized subclinical salpingitis, it causes luminal narrowing and architectural distortion that drastically increases the risk of tubal ectopic pregnancy and tubal-factor infertility.

Alternate Names:

SIN

Salpingo-Oophorectomy

Surgical removal of one or both fallopian tubes and ovaries. Bilateral salpingo-oophorectomy treats and stages ovarian cancer and is used as risk-reducing surgery in BRCA mutation carriers.

Salpingo-OöPhorectomy

Surgical removal of a fallopian tube and ovary, also called salpingo-oophorectomy.

Salpingography

Contrast imaging of the fallopian tubes, usually performed as part of hysterosalpingography to assess whether the tubes are open and to examine the uterine cavity.

Salpingostomy (Salpingotomy)

A fertility-preserving surgical procedure for tubal ectopic pregnancy: a linear incision on the antimesenteric border of the fallopian tube to remove the ectopic gestation, leaving the tube in situ. Indications: unruptured ectopic, desire for future fertility, contralateral tubal damage or absence.

Alternate Names:

Salpingotomy

Salt

An ionic chemical compound; dietary sodium chloride is essential in small amounts but excess intake can increase blood pressure.

Salt Gland

A specialized gland in some marine animals and birds that excretes concentrated salt to maintain fluid and electrolyte balance; humans do not possess a comparable salt gland.

Saltatory Conduction

The rapid propagation of action potentials along myelinated nerve fibers, where the action potential appears to "jump" from one Node of Ranvier to the next. Myelin (produced by Schwann cells in the PNS and oligodendrocytes in the CNS) insulates the axon, preventing ion flow across the internodal membrane.

Salter-Harris Fracture

A fracture involving the growth plate of a child or adolescent. Types I through V describe progressively different patterns and risks to bone growth.

Salvage Therapy

Treatment given for cancer that has not responded to, or has returned after, an earlier course of treatment.

Sample Fixation

Preservation of a tissue or cell sample with a chemical or physical method, such as formalin fixation, to prevent decay and maintain microscopic structure for processing and staining.

Sample Size

Sample size is the number of observations or participants in a study or test and affects precision, statistical power, and generalizability.

Sandfly Fever

An acute viral illness transmitted by phlebotomine sandflies, causing fever, headache, muscle pain, and sometimes rash.

Sandhoff Disease

An autosomal-recessive lysosomal storage disorder caused by deficiency of beta-hexosaminidase A and B, leading to accumulation of GM2 ganglioside. It

causes progressive neurodegeneration, developmental regression, seizures, hypotonia, and sometimes organ enlargement.

Sandpaper Tooth

Rough enamel surface retains plaque. May indicate early demineralization or fluorosis.

Sanfilippo Syndrome

A clinically severe autosomal recessive lysosomal storage disease (Mucopolysaccharidosis Type III [MPS III]) caused by deficiency of any of four enzymes involved in the degradation of the glycosaminoglycan heparan sulfate: heparan *N*-sulfatase (type IIIA, *SGSH*), α -*N*-acetylglucosaminidase (type IIIB, *NAGLU*), acetyl-CoA: α -glucosaminide acetyltransferase (type IIIC, *HGSNAT*), or *N*-acetylglucosamine-6-sulfatase (type IIID, *GNS*). Distinctive among mucopolysaccharidoses for presenting with profound, devastating neurocognitive degeneration, extreme behavioral disturbances (hyperactivity, impulsivity, aggression), sleep-wake cycle inversion, and progressive loss of motor skills, with remarkably mild somatic and skeletal features.

Alternate Names:

Mucopolysaccharidosis Type III (MPS III)

Sanguine

Sanguine means blood-related or blood-red; in older usage it can also describe a temperament or complexion.

Sanguineous

Containing, consisting of, derived from, or resembling blood (*sanguis*). In surgical care and wound management, used specifically to characterize fresh, gross bloody wound drainage or bodily exudates, distinguished from serous drainage (clear, thin, yellowish watery fluid), serosanguineous drainage (thin, pale pink-to-red watery fluid containing red blood cells and serum), and purulent drainage (thick, cloudy, pus-filled fluid). Copious or rapidly expanding sanguineous wound drainage postoperatively raises urgent suspicion of active surgical site hemorrhage or disrupted vascular ligation.

Sanitation

Public-health measures that safely manage human waste, water, food, hygiene, and environmental contamination to prevent infection and disease transmission.

Sano Shunt

A surgical modification of the first-stage Norwood reconstruction for hypoplastic left heart syndrome, wherein pulmonary blood flow is provided by an anterior right ventricle-to-pulmonary artery non-valved polytetrafluoroethylene conduit, rather than a modified Blalock-Taussig (systemic-to-pulmonary) shunt. Because flow through the RV-PA conduit occurs predominantly during ventricular systole, the Sano shunt preserves systemic diastolic arterial blood pressure and eliminates the continuous diastolic coronary run-off (coronary steal) associated with systemic-to-pulmonary shunts, improving early post-operative coronary perfusion stability.

Alternate Names:

RV-PA Conduit; Norwood-Sano Modification

Saphenous

Relating to the superficial veins of the lower limb, especially the great or small saphenous vein.

Saphenous Vein

A superficial vein of the lower limb that drains the foot and leg. The great and small saphenous veins are clinically important in venous disease and may be used as graft material in selected vascular or cardiac bypass procedures.

Great Saphenous Vein

The longest superficial vein of the lower limb, ascending from the medial foot to drain into the femoral vein at the saphenofemoral junction.

Small Saphenous Vein

A superficial vein ascending along the posterior calf that usually drains into the popliteal vein.

SAPHO Syndrome

A rare inflammatory disorder involving synovitis,

acne, pustulosis, hyperostosis, and osteitis. It commonly affects the anterior chest wall, spine, or jaw and may resemble infection or malignancy. Diagnosis is clinical and treatment may include anti-inflammatory or specialist immunomodulatory therapy.

Sapovirus

A virus that can cause vomiting and watery diarrhea, especially in children. It spreads through contaminated food, water, surfaces, or close contact.

Sapphism

Sapphism is an historical term for sexual or romantic relationships between women; respectful modern language generally uses lesbian or same-sex relationships.

Sapporo Virus

An older name for sapovirus, a virus that can cause vomiting and watery diarrhea.

Saprophyte

An organism that obtains nutrients from dead organic material; some saprophytic organisms can become opportunistic pathogens.

Sapropterin

A synthetic form of tetrahydrobiopterin used for some people with phenylketonuria and certain tetrahydrobiopterin deficiencies; response and diet requirements vary.

Saquinavir

An HIV protease inhibitor used with other antiretroviral medicines; drug interactions, gastrointestinal effects, and cardiovascular concerns limit its use.

Saquinavir Mesylate

The mesylate salt of saquinavir, an HIV protease inhibitor used with other antiretroviral medicines.

Saracatinib

An investigational oral kinase inhibitor that blocks Src-family and related signaling pathways and has

been studied in cancer and other diseases.

Saralasin

Saralasin is an investigational angiotensin-receptor antagonist formerly used to study renin-angiotensin physiology and blood-pressure regulation.

Sarcocystis

A genus of protozoan parasites that cycle between animal hosts and can cause intestinal or muscular sarcocystosis in people.

Sarcocystis Hominis

A protozoan parasite acquired by eating undercooked beef; people may develop intestinal symptoms, while cattle serve as an intermediate host.

Sarcocystis Suihominis

A protozoan parasite acquired by eating undercooked pork that can cause intestinal sarcocystosis; pigs serve as an intermediate host.

Sarcocystosis

Parasitic infection by *Sarcocystis* species, acquired from undercooked meat or contaminated food and sometimes causing muscle pain, fever, diarrhea, or no symptoms.

Sarcoglycan Complex

A group of proteins in the muscle-cell membrane that supports muscle fibers; inherited defects can cause limb-girdle muscular dystrophy and cardiomyopathy.

Sarcoglycanopathy

A subgroup of four distinct autosomal recessive limb-girdle muscular dystrophies (LGMD R3 through R6, historically LGMD 2C, 2D, 2E, and 2F) caused by biallelic loss-of-function mutations in the genes encoding components of the transmembrane sarcoglycan subcomplex: γ -sarcoglycan (*SGCG*), α -sarcoglycan (*SGCA*), β -sarcoglycan (*SGCB*), and δ -sarcoglycan (*SGCD*). Loss of any single sarcoglycan subunit destabilizes the entire tetrameric complex and disrupts the linkage between dystrophin and

α -dystroglycan, presenting with a severe childhood Duchenne-like phenotype featuring early pelvic-girdle weakness, calf pseudohypertrophy, Gowers sign, high serum creatine kinase ($> 20\text{--}50\times$), and progressive cardiomyopathy.

Alternate Names:

Sarcoglycan-Deficient Limb-Girdle Muscular Dystrophy; LGMD R3-R6

Sarcoidosis

Sarcoidosis is multisystem granulomatous inflammation, commonly affecting lungs and lymph nodes but also skin, eyes, heart, nerves, and other organs. It commonly affects the lungs and lymph nodes but may involve the skin, eyes, heart, nervous system, or other organs.

Alternate Names:

Arthritis Sarcoid (Sarcoidosis)

Sarcoidosis Skin Manifestations

Cutaneous sarcoidosis affects 25 percent of patients with systemic sarcoidosis.

Sarcolemma

The specialized cell membrane surrounding a muscle fiber. It conducts electrical signals and helps maintain the muscle cell's structure and chemical balance.

Sarcoma

A malignant tumor arising from connective or supporting tissues, including bone, cartilage, fat, muscle, and blood vessels. Examples include osteosarcoma, liposarcoma, and rhabdomyosarcoma.

Sarcoma, Bone Cancer

Alternate index label; see *Bone cancer*.

Sarcoma, Soft Tissue

Alternate index label; see *Soft tissue sarcoma*.

Sarcomere

The basic contractile unit of skeletal and cardiac muscle, extending from one Z line to the next and containing actin and myosin filaments.

Sarcopenia

Age-related loss of muscle mass and function. Prevalence increases with age. Risk factors: reduced protein intake, decreased physical activity, chronic disease. Assessment: grip strength, gait speed, muscle mass measurement. Treatment: protein supplementation, resistance exercise, vitamin D, hormonal therapy if deficient.

Sarcopenic Obesity

The coexistence of excess adiposity with reduced skeletal-muscle mass, strength, or physical performance. It increases risks of disability, falls, metabolic disease, and poor surgical outcomes and requires assessment beyond body-mass index alone.

Sarcoplasm

The cytoplasm of a muscle cell, containing structures such as mitochondria, glycogen, myoglobin, and the contractile apparatus.

Sarcoplasmic Reticulum

A specialized calcium-storage network in muscle cells that releases calcium to trigger contraction and takes it back up for relaxation.

Sarcoptes

The mite genus containing *Sarcoptes scabiei*, the cause of scabies.

Sarcoptes Scabiei

A mite that causes scabies, a contagious skin infestation transmitted by prolonged skin-to-skin contact. The female mite burrows into the stratum corneum, laying eggs along the burrow, which appears as a greyish, serpiginous line. The life cycle from egg to adult takes approximately 10-14 days.

Sarcoptic Mange (Scabies)

Alternate index label for Scabies.

Sarcosine

Sarcosine is a naturally occurring amino-acid derivative involved in one-carbon metabolism and studied in neurologic and psychiatric research.

Sarcosinemia

An uncommon metabolic finding with elevated sarcosine in blood or urine, usually due to altered glycine metabolism and often without major clinical symptoms.

Sardomozide

Sardomozide is an investigational compound studied for anticancer activity through effects on cellular growth and survival.

Sargramostim

A recombinant granulocyte-macrophage colony-stimulating factor used to promote white-cell recovery after selected cancer treatments or transplantation; bone pain and fever may occur.

Sarnat Staging

A standardized clinical grading scale developed by Harvey and Margaret Sarnat used to assess the severity of neonatal hypoxic-ischemic encephalopathy within the first 24 to 72 hours of life. It stratifies encephalopathy into three clinical stages: Stage 1 (mild: hyperalertness, normal or increased muscle tone, uninhibited Moro reflex, dilated pupils, absence of seizures); Stage 2 (moderate: lethargy, hypotonia, suppressed primitive reflexes, miosis, frequent seizures); and Stage 3 (severe: stupor or coma, flaccidity, absent brainstem and primitive reflexes, refractory seizures, decerebrate posturing). Stage 2 and Stage 3 are clinical indicators for initiating therapeutic hypothermia in term and late-preterm infants.

Alternate Names:

Sarnat and Sarnat Grading; Sarnat Score; HIE Staging

SARS

Severe acute respiratory syndrome, a viral respiratory illness caused by SARS-CoV.

SARS Virus

The coronavirus that causes severe acute respiratory syndrome, a serious respiratory infection that can spread through respiratory droplets and close contact.

SARS-CoV-2

A coronavirus belonging to the Coronaviridae family that causes COVID-19. SARS-CoV-2 has a positive-sense single-stranded RNA genome and a spike glycoprotein that mediates attachment to ACE2 receptors. The virus replicates in the cytoplasm using its own RNA-dependent RNA polymerase.

Sassafras Oil Overdose

Poisoning from swallowing or otherwise being exposed to too much sassafras oil. It may cause nausea, vomiting, seizures, breathing problems, liver injury, or other serious effects and requires urgent medical advice.

Satellite DNA

Highly repetitive DNA sequences found mainly near centromeres and other chromosome regions; they help organize chromosome structure and can vary between individuals.

Satellite Virus

A virus-like infectious agent that can multiply only when another helper virus is present.

Satiation

The feeling of fullness that terminates a meal, produced by gastrointestinal stretch, nutrients, hormones, and brain signals regulating food intake.

Satiety

Feeling of fullness after eating. Affected by meal composition, volume, fiber, protein. Satiety hormones: CCK, PYY, GLP-1.

Satiety - Early

Feeling full sooner than expected while eating, sometimes after only a small amount of food. It may result from stomach or digestive disease and should be assessed if persistent or associated with weight loss.

Satisfaction

A person's evaluation of how well an experience, service, treatment, or outcome met expectations; in

healthcare it is one quality measure but does not replace clinical outcomes or safety.

Satisficer

A person who chooses an option that meets acceptable criteria without seeking the absolute best option.

Satraplatin

An investigational oral platinum-based anticancer drug that forms DNA cross-links; it has been studied mainly in advanced prostate and other cancers.

Saturated Fat

A dietary fat whose fatty-acid chains contain no carbon-carbon double bonds. It is found in foods such as fatty meat, butter, full-fat dairy, coconut oil, and palm oil; replacing some saturated fat with unsaturated fat can help lower LDL cholesterol and cardiovascular risk.

Saturated Phytosterol

A hydrogenated plant sterol with a saturated side chain; it is a dietary lipid studied for effects on cholesterol absorption and metabolism.

Saturation

Occupancy of available binding, metabolic, transport, or elimination capacity such that further increases in concentration or dose produce a nonlinear response. Saturable metabolism can cause disproportionate rises in plasma concentration and toxicity.

Sausage Digit

Diffuse swelling of an entire finger or toe, often seen with inflammatory arthritis.

Alternate Names:

Dactylitis

Saxitoxin

A powerful poison made by some marine algae that can build up in shellfish. Eating contaminated seafood may cause tingling, weakness, trouble breathing, or paralysis.

SBBO

Alternate index label; see *Small intestinal bacterial overgrowth*.

Scab

A crust of dried blood, serum, pus, or tissue fluid over a healing skin injury.

Scabies

Scabies is infestation by *Sarcoptes scabiei* mites, causing intense nighttime itching and a characteristic rash or burrows, with spread through close contact. mite *Sarcoptes scabiei* var. *hominis*. The hallmark symptom is severe, generalized pruritus that characteristically worsens at night, driven by a hypersensitivity reaction to the mite's proteins, eggs, and feces.

Alternate Names:

Sarcoptic Mange (Scabies)

SCAD

Alternate index label; see *Spontaneous coronary artery dissection*.

Scaffold

Three-dimensional structure for tissue engineering. Porous, biodegradable. Cells populate scaffold.

Scaffold Seeding

Placing living cells onto or into a three-dimensional biomaterial scaffold to begin tissue growth in tissue engineering or regenerative medicine.

Scalded Mouth Syndrome

Alternate index label; see *Burning mouth syndrome*.

Scalded Skin Syndrome

Alternate index label; see *Scalded-Skin Syndrome*.

Scalded-Skin Syndrome

Staphylococcal scalded-skin syndrome, a toxin-mediated superficial epidermal separation causing tender erythema and peeling, especially in children.

Scalds

A specific category of thermal burn injury caused by exposure to hot liquids (e.g., boiling water, hot oil, hot beverages) or pressurized steam. The most common mechanism of burn injury in infants and young children (often through accidental spills or tap water immersion) and vulnerable elderly adults. Assessed clinically by depth (superficial partial-thickness with blistering, or deep partial-thickness/full-thickness with epidermal loss) and total body surface area using the Wallace Rule of Nines or Lund-Browder charts; immersion scalds exhibiting sharp water-line demarcation and sparing of flexural creases warrant evaluation for intentional non-accidental trauma.

Scale

Visible flakes or plates of stratum corneum shed from the skin surface, commonly caused by abnormal keratinization, inflammation, infection, or dryness.

Scales

Scales are flakes or plates of shed skin on the surface, occurring with dryness, eczema, psoriasis, fungal infection, or other dermatologic conditions.

Scaling

Scaling is flaking or shedding of the outer skin layer and may occur with dryness, eczema, psoriasis, fungal infection, or other skin disease.

Scaling And Root Planing

Nonsurgical periodontal instrumentation that removes supra- and subgingival deposits and smooths contaminated root surfaces. It reduces inflammation and pocket depth in appropriate patients but does not regenerate lost periodontal attachment by itself.

Scalp

The skin and underlying tissues covering the skull. It contains hair follicles, glands, blood vessels, nerves, and connective tissue and can be affected by injury, infection, or skin disease.

Scalp And Body (Ringworm)

Alternate index label for Ringworm.

Scalp Psoriasis

Psoriasis affecting the scalp, causing sharply demarcated red plaques covered by silvery scale that may extend beyond the hairline. It can itch, crack, or bleed and is not contagious. Treatment may include medicated shampoos, topical corticosteroids, vitamin-D analogues, or systemic therapy for extensive disease.

Alternate Names:

Psoriasis Of The Scalp (Scalp Psoriasis)

Scalpel

Surgical knife with interchangeable blades. Blade numbers indicate size and shape. #10: common for incisions. #11: puncture. #15: precise work.

Scalpel Handle

The reusable handle that holds a detachable scalpel blade. Different handle and blade sizes are chosen for the type and precision of the incision.

Scan

Scan is a medical test, procedure, treatment, or clinical measure used in patient care; its purpose and interpretation depend on the clinical question.

Scan Gallium

A nuclear-medicine scan using radiolabeled gallium to identify areas of inflammation, infection, or some tumors; uptake is interpreted with anatomy and clinical context.

Scanning

Systematic examination of a body region or specimen using imaging, microscopy, or another detection method to locate, characterize, or monitor abnormalities.

Scanning Electron Microscopy

Microscopy that scans a focused electron beam across a specimen surface to produce a high-resolution image of its three-dimensional topography

and composition.

Scanning Speech

A slow, irregularly paced speech pattern with abnormal emphasis, classically associated with cerebellar or multiple-sclerosis disease.

Scanning Techniques

Imaging or measurement methods that systematically examine a body region, organ, or sample.

Scaphoid

Largest proximal carpal row bone on the radial wrist side. Most commonly fractured carpal bone from falls on an outstretched hand. Unique distal blood supply makes waist fractures prone to proximal fragment avascular necrosis. Anatomical snuffbox tenderness is characteristic.

Scaphoid Abdomen

A scaphoid abdomen is an unusually hollowed abdominal contour, classically seen with severe diaphragmatic hernia in a newborn but also possible with marked wasting.

Scaphoid Bone

The largest proximal-row carpal bone on the thumb side of the wrist; its waist is vulnerable to fracture and avascular necrosis because of its blood supply.

Scaphoid Fractures

Scaphoid fractures: most common carpal fracture (young males, FOOSH). Classification: Herbert: A (incomplete), B (complete: B1 distal waist, B2 proximal waist, B3 proximal pole, B4 distal tubercle, B5 transverse). Vascularity: retrograde from distal radius -> proximal pole at risk for AVN.

Scaphycephaly

Scaphycephaly is a long, narrow skull shape caused by premature fusion of the sagittal suture and may require evaluation for craniosynostosis.

Scapula

Flat triangular bone on the posterior thorax between

ribs 2-7. Key features: acromion, coracoid process, glenoid cavity, spine, supraspinous/infraspinous fossae. Stabilized by 17 muscles and serves as rotator cuff attachment. Has 3 borders, 3 angles, and 2 surfaces. Fractures are uncommon, usually from high-energy trauma.

Scapulothoracic Joint

The functional gliding connection between the scapula and the back of the chest wall.

Scar

Fibrous tissue formed during healing after injury, inflammation, or surgery.

Scar Excessive (Keloid)

Alternate index label for Keloid.

Scar Management

Measures used to reduce symptoms and abnormal scar formation. Options include silicone products, pressure therapy, massage, corticosteroid injections, or other specialist treatments, depending on the scar and its cause.

Scar Revision

A procedure that improves the appearance, contour, flexibility, or symptoms of a scar through excision, rearrangement, grafting, laser, injection, or other selected techniques.

Scarification

Intentional scratching, cutting, or abrasion of skin, historically used in some inoculation or cultural practices and medically used in limited procedures; it can cause infection and scarring.

Scarlatina

Scarlet fever, an illness caused by toxin-producing group A Streptococcus and characterized by sore throat, fever, and a fine rough-textured rash.

Alternate Names:

Scarlet Fever (Scarlatina)

Scarlatina (Scarlet Fever (Scarlatina))

Alternate index label for Scarlet Fever (Scarlatina).

Scarlet Fever

A toxin-mediated illness caused by Group A Streptococcus producing erythrogenic toxin. Presents with fever, pharyngitis, and a characteristic diffuse, erythematous, sandpaper-textured rash that blanches with pressure. Circumoral pallor, Pastia lines (linear petechiae in skin folds), and strawberry tongue are classic findings.

Scarlet Fever (Scarlatina)

Alternate index label for Scarlatina.

Scarring Alopecia

Permanent hair loss caused by destruction and fibrosis of the hair follicle. Causes include lichen planopilaris, discoid lupus, central centrifugal cicatricial alopecia, folliculitis decalvans, and severe infection; early diagnosis and biopsy can prevent progression.

Scars

A scar is fibrous tissue formed during healing after injury, surgery, inflammation, or burns. Scars may be flat, depressed, raised, or contractured and can itch, hurt, restrict movement, or affect appearance; treatment depends on maturity and type.

Scatter Correction

A method in medical imaging, especially nuclear medicine, that estimates and removes the effect of photons scattered away from their original path, improving the accuracy of the image or measured activity.

Scatter Events

Photons deflected from original path. Corrected by energy windowing and scatter correction algorithms.

Scavenger Receptor

A cell-surface or intracellular receptor that binds and removes modified lipids, debris, microbes, or dying cells and contributes to innate immunity and

atherosclerosis.

Scavenging System

The anesthesia gas scavenging system collects and removes waste anesthetic gases from the operating room environment to protect staff from chronic exposure. Active scavenging uses negative pressure from vacuum source; passive scavenging vents to exhaust through gravity.

Scedosporium

A genus of environmental molds that can cause difficult-to-treat lung, sinus, skin, bone, or disseminated infections, especially in immunocompromised people.

Scedosporium Apiospermum

A mold that can cause localized or disseminated infection, including mycetoma, pneumonia, sinusitis, and brain abscesses; treatment often requires specialist antifungal care.

Scedosporium Prolifican

An environmental mold, now commonly classified as *Lomentospora prolificans*, that causes severe disseminated infection and is often resistant to antifungal drugs.

Schaalia Radingae

A normally slow-growing anaerobic bacterium of the human mouth and skin that can rarely cause abscesses or other invasive infections.

Schaalia Turicensis

A filamentous anaerobic bacterium that may be part of normal flora but can cause actinomycosis-like abscesses and other opportunistic infections.

Schatzki Ring

Thin circumferential mucosal ring located at the squamocolumnar junction of the distal esophagus at the superior margin of a hiatal hernia. The ring is composed of mucosa and submucosa. May be asymptomatic or cause episodic dysphagia particularly to solid food and acute food bolus impaction.

Alternate Names:

Ring Schatzki (Schatzki Ring) Strictures Esophagus (Schatzki Ring)

Schedule I

A United States controlled-substance category for drugs with high misuse potential and no accepted medical use under federal law.

Schedule II

A United States controlled-substance category for drugs with accepted medical uses but high misuse potential.

Schedule III

A United States controlled-substance category for drugs with lower misuse potential than Schedule II drugs.

Schedule IV

A United States controlled-substance category for drugs with accepted medical uses and lower misuse potential than Schedule III drugs.

Schedule V

A United States controlled-substance category for drugs with the lowest level of misuse potential among the federal schedules.

Scheduled Notification

A planned alert or message delivered at a specified time to remind a person about an appointment, medicine, monitoring task, or health-related action.

Scheuermann Disease

A juvenile osteochondrosis of the spine in which vertebral end plates develop irregularities and anterior wedging during skeletal growth. It commonly produces a rigid thoracic kyphosis, although the thoracolumbar or lumbar spine may also be involved. Symptoms can include back pain, fatigue, limited spinal motion, and postural deformity that becomes more apparent during adolescence. Diagnosis is based on clinical examination and characteristic radiographs; management depends on deformity

severity, skeletal maturity, pain, and progression.

Scheuermann'S Disease

A juvenile spinal growth disorder causing wedging of vertebral bodies and structural kyphosis.

Schiff Base

A chemical compound formed when a primary amine reacts with an aldehyde or ketone; Schiff bases occur in biochemistry and drug chemistry.

Schiller Test

A cervical examination using iodine solution; glycogen-rich normal squamous epithelium stains dark, while iodine-negative areas require clinical correlation and sometimes biopsy.

Schilling Test

A historical test of vitamin B12 absorption, formerly used to investigate pernicious anaemia and intestinal malabsorption. It has largely been replaced by serum vitamin B12, methylmalonic acid, homocysteine, and intrinsic-factor antibody testing.

Schirmer Test

A test in which standardized filter paper strips measure tear production over a set time, commonly used to evaluate aqueous-deficient dry eye.

Schistocyte Count

Measurement of fragmented red blood cells on a blood smear, supporting evaluation of microangiopathic hemolysis, mechanical valve damage, disseminated intravascular coagulation, or severe burns.

Schistocytes

Fragmented, helmet-shaped, or triangular red blood cells produced by mechanical shearing as erythrocytes pass through obstructed or fibrin-laden microvessels.

Schistomicide

A therapeutic anti-parasitic pharmacological agent specifically formulated to kill or eradicate parasitic blood flukes (trematodes) of the genus *Schistosoma*

(most notably *S. mansoni*, *S. haematobium*, and *S. japonicum*), which cause schistosomiasis (bilharzia). The primary schistomicide of choice worldwide is praziquantel, which acts by increasing schistosome cell membrane permeability to calcium ions, causing rapid muscular contraction, paralysis, tegumental damage, and exposure of parasite antigens to host immune attack.

Schistosoma

A genus of parasitic blood flukes that cause schistosomiasis when larval forms from contaminated freshwater penetrate human skin.

Schistosoma Haematobium

A parasitic blood fluke that can cause urinary schistosomiasis, including hematuria and chronic injury of the urinary tract.

Schistosoma Intercalatum

A parasitic blood fluke that causes intestinal schistosomiasis, mainly in parts of Central and West Africa, after freshwater exposure.

Schistosomiasis

A parasitic disease caused by *Schistosoma* blood flukes acquired through freshwater exposure, potentially causing intestinal or urinary inflammation, bleeding, liver disease, or chronic organ damage.

Alternate Names:

bilharzia, caused by blood flukes of the genus. Bilharzia (Schistosomiasis)

Schizencephaly

A rare brain malformation characterized by a full-thickness, gray matter-lined cleft extending from the pial surface to the lateral ventricle. Caused by early focal neuronal migration abnormalities, often associated with prenatal vascular disruption or COL4A1 mutations. Closed-lip (Type I): the cleft walls are apposed.

Schizoaffective Disorder

A disorder characterized by an uninterrupted period of illness during which there is a major mood episode concurrent with criterion A of schizophre-

nia (delusions, hallucinations, disorganized speech, grossly disorganized or catatonic behavior, or negative symptoms).

Schizogony

A form of asexual reproduction in which one protozoan cell divides into many daughter cells.

Schizoid Personality Disorder

A pattern of detachment from social relationships with a restricted range of emotional expression.

Alternate Names:

Personality Disorder, Schizoid

Schizophrenia

Schizophrenia is a chronic psychotic disorder involving delusions, hallucinations, disorganized thought or behavior, negative symptoms, and impaired functioning. delusions, disorganized thinking) and negative symptoms (avolition, alogia, anhedonia, blunted affect), with cognitive impairment and functional decline. Diagnosis: at least two core symptoms for ≥ 1 month with continuous signs for ≥ 6 months.

Schizophrenia, Adolescent

Alternate index label; see *Childhood schizophrenia*.

Schizophrenia, Childhood

Alternate index label; see *Childhood schizophrenia*.

Schizophrenia, Juvenile

Alternate index label; see *Childhood schizophrenia*.

Schizotypal Personality Disorder

A pattern of social and interpersonal deficits marked by acute discomfort with and reduced capacity for close relationships, cognitive or perceptual distortions, and eccentricities of behavior.

Alternate Names:

Personality Disorder, Schizotypal

Schlemm'S Canal

A circular channel that drains aqueous humor from the eye.

Alternate Names:

Canal of Schlemm

Schnitzler Syndrome

A rare autoinflammatory syndrome characterized by chronic urticarial rash, intermittent fever, bone pain, arthralgia or arthritis, and elevated inflammatory markers, usually with a monoclonal IgM protein and less commonly IgG.

School Age Test Or Procedure Preparation

Preparing a school-age child for a test or procedure includes age-appropriate explanations, comfort measures, caregiver support, and following any fasting or medicine instructions.

School Sores (Impetigo)

Alternate index label for Impetigo.

School-Age Children Development

School-age development includes advances in growth, coordination, language, learning, independence, social relationships, and emotional regulation, with broad normal variation.

School-Based Rehabilitation

Therapy and other support services provided in a school setting to help a child participate in education and daily school activities.

Schwann Cell

A peripheral glial cell that forms myelin or supports axons.

Schwann Cells

Glial cells of the peripheral nervous system that form myelin around many axons and support the repair and maintenance of peripheral nerves.

Schwannoma

A benign peripheral nerve sheath tumor arising from Schwann cells. It presents as a solitary, slow-growing, painless nodule along a peripheral nerve course. It can be moved laterally but not longitudinally along the nerve. Vestibular schwannoma

(acoustic neuroma) arises from CN VIII. Diagnosis is by MRI.

Alternate Names:

Neurilemmoma

Schwannomatosis

A disorder in which multiple schwannomas, tumors of nerve-sheath cells, develop.

Alternate Names:

Neurinomatosis

SCI

Alternate index label; see *Spinal cord injury*.

Sciatic Nerve

Body's largest nerve (L4-S3), formed from the lumbosacral plexus, exiting via the greater sciatic foramen below the piriformis. Descends in the posterior thigh, dividing into tibial and common peroneal nerves at the popliteal fossa. Innervates hamstrings and provides lower leg/foot motor/sensory innervation via its branches.

Sciatic Neuritis (Sciatica)

Alternate index label for Sciatica.

Sciatica

Sciatica is radiating leg pain or sensory symptoms along a lumbosacral nerve-root distribution, commonly from disk herniation or spinal narrowing, through the buttock and down the posterior thigh to the foot. Most commonly caused by lumbar disc herniation compressing the nerve root (L4-L5 or L5-S1). Other causes: spinal stenosis, spondylolisthesis, piriformis syndrome, and epidural abscess.

Alternate Names:

Sciatic Neuritis (Sciatica)

Sciatica: Causes, Treatment, And Symptoms

Sciatica is radiating pain, tingling, numbness, or weakness along the sciatic-nerve distribution, commonly from lumbar disc herniation or spinal narrowing. Most cases improve with conservative care, but saddle numbness, bowel or bladder dysfunction,

or progressive weakness is urgent.

Scimitar Syndrome

A congenital cardiopulmonary anomaly involving partial or total anomalous pulmonary venous return from the right lung to the inferior vena cava, often with right-lung hypoplasia and dextroposition of the heart. It may cause recurrent respiratory infection, pulmonary hypertension, or heart failure.

Scintigraphy

Nuclear medicine imaging in which a radiotracer is administered and its distribution is recorded to evaluate the structure or function of an organ.

Scintillation

The emission of light by a material after it absorbs ionizing radiation. Scintillation detectors use this light to measure radiation.

Scintillation Crystal

Sodium iodide doped with thallium (NaI:Tl) crystal converts gamma rays to light photons. Coupled to photomultiplier tubes.

Scintillations

Perceived flashes or zigzag lines, often occurring during a migraine aura.

Scirrhus

A hard, fibrous form of carcinoma, historically used especially for breast or gastric cancer.

Scissors

Instruments with two blades used to cut tissue, sutures, dressings, or other material. Surgical types vary in length, blade shape, and sharpness; examples include Mayo and Metzenbaum scissors.

Scissors (Surgical)

Cutting instruments used to divide tissue, sutures, dressings, or other materials during medical procedures.

Alternate Names:

Surgical

Scissors (Vascular)

A surgical instrument with two blades used to cut tissue, sutures, dressings, or other materials during vascular procedures.

Alternate Names:

Vascular

Sclera

The sclera is the tough white outer coat of the eye that protects the globe and provides attachment for eye muscles.

Scleral Buckling

A retinal-detachment repair procedure in which a silicone band or buckle is placed around the outside of the eye to support the retinal break and reduce traction.

Scleredema Diabeticorum

Scleredema diabeticorum is thickening and hardening of the skin on the upper back and neck, usually associated with long-standing diabetes and obesity.

Scleritis

Scleritis is painful inflammation of the sclera, often associated with autoimmune disease and capable of threatening vision.

Alternate Names:

Inflammation Sclera (Scleritis)

Scleroderma

Scleroderma is a group of autoimmune diseases that cause the skin to become thick and hard. In systemic sclerosis, it can also affect blood vessels and internal organs such as the esophagus, lungs, kidneys, and heart.

Alternate Names:

Arthritis Scleroderma (Scleroderma) Sclerosis, Systemic

Scleroma

Scleroma is chronic granulomatous infection, usually of the upper airway, caused by *Klebsiella rhinoscleromatis* and potentially producing scarring and obstruction.

Scleromyxedema

A rare, chronic, progressive mucinosis characterised by generalised papular and sclerodermoid thickening of the skin with firm, waxy papules in a "peau d'orange" pattern, most prominently on the face, neck, and hands.

Sclerosing Cholangitis

Sclerosing cholangitis is inflammation and scarring that narrows bile ducts, causing cholestasis and sometimes cirrhosis; primary and secondary causes require different evaluation.

Sclerosing Mesenteritis

A rare inflammatory and fibrotic disorder affecting the mesentery, the tissue supporting the intestines.

Sclerosing Peritonitis

Thickening and scarring of the peritoneum that can restrict movement of the intestines.

Sclerosing Sialadenitis

A chronic fibrosing inflammation of the salivary glands characterized by progressive glandular enlargement and firmness, sometimes associated with IgG4-related disease.

Sclerosis

Abnormal hardening or scarring of tissue, often from excess connective tissue, chronic inflammation, degeneration, or neurologic plaques; the affected organ determines the specific disease.

Sclerosis, Multiple

Alternate index label; see *Multiple sclerosis*.

Sclerosis, Systemic

Alternate index label; see *Scleroderma*.

Sclerosteosis

A rare inherited sclerosing bone dysplasia characterized by progressive thickening of the skull and long bones, tall stature or syndactyly in some forms, and possible cranial-nerve compression. Hearing loss, facial palsy, headache, and raised intracranial

pressure may occur.

Sclerotherapy

A treatment in which an irritating or closing agent is injected into a blood vessel or body cavity to make it scar and seal, commonly used for varicose veins and selected vascular lesions.

Scolecobasidium Terreum

An environmental dematiaceous fungus rarely reported as an opportunistic human pathogen; clinical significance depends on isolation from a compatible infection.

Scoliosis

Scoliosis is a three-dimensional sideways curvature and rotation of the spine, which may be idiopathic, congenital, neuromuscular, or degenerative.

Alternate Names:

Spinal Curvature

Scopolamine

Scopolamine is a muscarinic antagonist used mainly to prevent motion sickness and postoperative nausea; dry mouth, blurred vision, urinary retention, confusion, and other anticholinergic effects may occur.

Scopolamine Hydrobromide

The hydrobromide salt of scopolamine, an anticholinergic medicine used mainly to prevent motion sickness and postoperative nausea.

Scopoletin

A plant-derived coumarin compound studied for antioxidant, anti-inflammatory, and antimicrobial effects; clinical benefits in humans are not established.

Scopulariopsis

Scopulariopsis is a mold genus that rarely causes opportunistic infection, especially in people with immune suppression or implanted devices.

Scorbutic

Pertaining to, affected by, or characteristic of scurvy—the clinical syndrome resulting from severe nutritional ascorbic acid (vitamin C) deficiency. Vitamin C is an indispensable cofactor for prolyl and lysyl hydroxylase enzymes required for post-translational collagen triple-helix cross-linking. Scorbutic pathology is characterized by extreme capillary fragility, defective wound healing, and musculoskeletal deterioration; classic physical findings include corkscrew hairs with surrounding perifollicular petechiae, purpura, bleeding spongy gums (scorbutic gingivitis), subperiosteal hematomas, and painful hemarthroses.

Scorpion Fish Sting

A painful sting from a venomous scorpion fish, causing severe local pain, swelling, and sometimes whole-body symptoms. The wound needs urgent medical assessment because venom may remain in the spine.

Scorpion Sting

A venomous injury caused by a scorpion, producing local pain and sometimes systemic toxicity.

Alternate Names:

Stings, Scorpion

Scorpions

Venomous arachnids whose stings can cause local pain and, depending on the species, sweating, muscle spasms, breathing problems, or other serious symptoms.

Scotoma

A focal area of diminished or absent vision within an otherwise intact visual field, classified by location and morphology.

Scrape

A scrape is a superficial abrasion in which friction removes part of the skin surface; cleaning, protection, and monitoring for infection support healing.

Scrapie

Scrapie is a fatal prion disease of sheep and goats

causing progressive neurologic dysfunction; it is not considered a usual human infection.

Screen Time And Children

Time children spend using televisions, computers, phones, tablets, or game devices; excessive or poorly timed use can affect sleep, activity, learning, mood, and family routines.

Screening

Testing people without symptoms to identify a disease or an increased risk of disease. The appropriate test, age range, interval, and follow-up depend on the condition, individual risk, and current guidance; screening benefits must be balanced against harms.

Screening (Geriatric)

Detection of disease in asymptomatic elderly. Balance benefits vs. harms, life expectancy, preferences. Examples: bone density, abdominal aortic aneurysm, colorectal cancer (individualized), depression, falls.

Alternate Names:

Geriatric

Screening And Diagnosis For HIV

Testing for human immunodeficiency virus using an antigen-antibody assay or nucleic-acid test, followed by confirmatory testing and assessment of immune status when positive.

Screening Guidelines

Recommendations describing who should be screened for a disease, which test to use, and when to begin or repeat testing.

Screening Procedure

A test or examination offered to people without symptoms to detect disease or risk early enough for effective prevention or treatment; eligibility and intervals depend on evidence and risk.

Screening Program

An organized public-health service that systematically invites a defined population to undergo testing

for an asymptomatic condition, with quality-assured follow-up, diagnosis, treatment, and evaluation of benefits and harms.

Screening Study

A planned investigation of an asymptomatic population to identify early disease, precursor lesions, or risk factors using a validated test and follow-up pathway.

Screening Test

A test applied to people without symptoms or a known diagnosis to identify those at increased risk of a condition. A screening result usually requires a diagnostic or confirmatory evaluation before treatment is initiated.

Screws

Threaded devices used to hold bone fragments together or attach implants to bone.

Script

A script is a written set of instructions or a prescription; in healthcare, a prescription specifies the medicine, dose, route, timing, and duration.

Scrivener'S Palsy

An older term for task-specific hand dystonia or cramp occurring during writing.

Scrofula

Tuberculous infection of cervical lymph nodes, producing chronic neck swellings that may suppurate or form draining sinuses; nontuberculous mycobacteria can cause a similar presentation.

Scrofulous

Scrofulous is an old term for tuberculous infection of lymph nodes, especially in the neck, which can cause chronic swelling and draining sinuses.

Scrombotoxin Poisoning

A misspelled or nonstandard form referring to scombrotoxin poisoning, caused by histamine in improperly stored fish.

Scrotal Calcinosis

The deposition of calcium within the scrotal skin, presenting as multiple, firm, whitish, painless nodules on the scrotum, which may be solitary or numerous and may ulcerate. The exact aetiology is debated and may involve dystrophic calcification of pre-existing epidermal inclusion cysts, dermal cysts, or eccrine duct abnormalities.

Scrotal Infection

Infection of the scrotal skin or underlying structures, which may cause redness, swelling, warmth, pain, or drainage. Rapidly worsening pain, fever, discoloration, or severe swelling is an emergency because serious soft-tissue infection must be excluded.

Scrotal Masses

Abnormal lumps or swellings in the tissues of the scrotum.

Scrotal Pain

Pain in the scrotum caused by infection, trauma, hernia, fluid collections, varicocele, or testicular torsion. Sudden severe pain, nausea, a high-riding testis, or abnormal testicular position requires immediate assessment for torsion.

Scrotal Swelling

Enlargement of the scrotum caused by fluid, inflammation, infection, hernia, varicocele, hydrocele, tumor, trauma, or testicular torsion; sudden pain is an emergency.

Scrotal Ultrasound

Ultrasound imaging of the scrotum and testes used to evaluate pain, swelling, masses, torsion, infection, trauma, and blood flow.

Scrotum

The scrotum is the external skin-and-muscle sac containing the testes and epididymides; its muscles help regulate testicular temperature, and swelling or pain may indicate infection, torsion, hernia, or tumor. spermatic cords. The scrotal wall has multiple layers reflecting the abdominal wall layers.

Scrub In

To scrub in is to perform surgical hand antisepsis and put on sterile clothing and gloves before participating in a sterile procedure.

Scrub Up

To scrub up means to clean and antiseptically prepare the hands and forearms before a sterile procedure.

Scrubs

Scrubs are protective work garments commonly worn by healthcare staff; they reduce clothing contamination but do not replace hand hygiene or other precautions.

Scrum-Pox

A nonstandard or historical term whose intended clinical meaning is unclear and should be verified from its source.

Scurf

Fine flakes or scales shed from the skin, especially the scalp.

Scurvy

A disease caused by severe vitamin C deficiency. It can cause bleeding or swollen gums, easy bruising, small skin bleeds, poor wound healing, anemia, bone or joint pain, and fatigue.

Sea-Blue Histiocyte

A sea-blue histiocyte is a macrophage containing blue-green lipid or ceroid pigment on staining, seen in some storage, hematologic, or metabolic disorders.

Sea-Sickness

Motion sickness caused by conflicting signals from the inner ear, eyes, and body during travel on a boat or ship. It may cause nausea, vomiting, dizziness, sweating, headache, and pallor.

Seasonal Affective Disorder

A specifier of recurrent major depressive episodes

that occur at a specific time of year, most commonly winter (with remission in spring/summer). Symptoms: hypersomnia, hyperphagia with carbohydrate craving, weight gain, fatigue, and low energy.

Seasonal Affective Disorder

Alternate index label for SAD.

Seasonal Affective Disorder

Alternate index label for Seasonal Affective Disorder.

Seasonal Affective Disorder Syndrome

Seasonal affective disorder, a recurring mood disorder with episodes linked to a particular season.

Seasonal Allergies (Hay Fever)

Alternate index label for Hay Fever.

Seasonal Allergy

Alternate index label; see *Hay fever*.

Sebaceous Adenoma

A sebaceous adenoma is a usually benign tumor of sebaceous glands that can appear as a yellowish skin nodule and may be associated with Muir-Torre syndrome.

Sebaceous Carcinoma

An uncommon skin cancer arising from oil-producing sebaceous glands.

Sebaceous Cyst

A common but imprecise term for an epidermoid cyst, a keratin-filled cyst beneath the skin.

Sebaceous Epithelioma

A benign sebaceous neoplasm, often presenting as a yellowish papule or nodule. Multiple or unusual lesions may warrant assessment for a DNA mismatch-repair syndrome such as Muir-Torre syndrome.

Sebaceous Gland

A small gland in the skin that secretes sebum (oil) into hair follicles to lubricate and waterproof the

skin and hair.

Sebaceous Gland Carcinoma

Sebaceous gland carcinoma is an uncommon skin cancer arising from oil glands, often appearing as a firm yellowish eyelid or skin lump.

Sebaceous Glands

Skin glands that secrete sebum, usually into hair follicles.

Sebaceous Hyperplasia

A benign enlargement of sebaceous glands, presenting as small yellowish papules with central umbilication, usually on the face of adults.

Seborrhea

Seborrhea is excessive or altered oil production that contributes to greasy scale and inflammation, commonly in seborrheic dermatitis of the scalp, face, or chest. Gentle cleansing, medicated shampoos, and topical anti-inflammatory or antifungal treatment may help.

Seborrheic Dermatitis

A chronic inflammatory skin condition affecting sebum-rich areas: scalp, face (nasolabial folds, eyebrows), chest, and intertriginous areas. Presents with erythematous patches with greasy, yellowish scales. Dandruff is a mild form. More common and severe in Parkinson disease, HIV/AIDS, and Down syndrome.

Alternate Names:

Dermatitis, Seborrheic

Seborrheic Keratoses

Common, benign, epidermal skin tumors presenting as waxy, stuck-on, pigmented plaques with a rough, verrucous surface, typically appearing on the trunk and face of middle-aged and older adults. They are the most common benign skin neoplasm and are caused by keratinocyte proliferation.

Seborrheic Keratosis

A benign epidermal tumor appearing as stuck-on,

waxy, hyperpigmented plaques with a verrucous surface. They are extremely common in older adults and increase in number with age. Seborrheic keratoses are not premalignant.

Alternate Names:

Keratosis, Seborrheic

Seborrhoea

A chronic inflammatory skin condition affecting sebum-rich areas: scalp, face (nasolabial folds, eyebrows), chest, and intertriginous areas. See Seborrheic Dermatitis.

Seborrhoeic

Relating to increased or altered activity of sebaceous glands or to areas rich in these glands, such as the scalp, face, chest, and upper back.

Seborrhoeic Keratosis

A common benign, usually pigmented, raised skin growth with a waxy or stuck-on appearance.

Sebum

An oily secretion produced by sebaceous glands that lubricates and protects skin and hair.

Seckel Syndrome

A rare recessive primordial dwarfism syndrome featuring severe proportionate growth restriction, microcephaly, characteristic facial features, and developmental or intellectual impairment. Skeletal, hematological, and other congenital abnormalities may occur.

Secobarbital Overdose

Poisoning from taking too much secobarbital, a sedative medicine. It can cause extreme sleepiness, confusion, slowed breathing, coma, or death and requires emergency care.

Second Malignancy

New cancer after initial cancer treatment. Risk from radiation, chemotherapy.

Second Messenger Systems

Intracellular signaling molecules that relay extracellular signals (first messengers: hormones, neurotransmitters, growth factors) to cellular responses.

Second Messengers

Intracellular signaling molecules generated in response to extracellular signals (first messengers like hormones or neurotransmitters). cAMP (cyclic adenosine monophosphate): produced from ATP by adenylyl cyclase, activated by Gs-protein coupled receptors, activates protein kinase A (PKA); degraded by phosphodiesterase.

Second Opinion

An independent medical assessment by another qualified clinician to help confirm or clarify a diagnosis, prognosis, or treatment plan.

Second Stage Of Labor

From full cervical dilatation (10 cm) to delivery of the baby. The mother feels the urge to push as the presenting part descends. Pushing: closed-glottis (Valsalva) or open-glottis. Duration: up to 3 hours in nulliparas with regional anesthesia (2 hours without), and 2 hours in multiparas (1 hour without).

Second Trimester

The period from 13 to 27 weeks gestational age. Fetal growth accelerates, mother begins to feel quickening (fetal movement) between 16-20 weeks. Anatomy ultrasound at 18-22 weeks screens for structural anomalies. Maternal serum screening offered. Nausea typically resolves.

Second-Degree AV Block

Intermittent failure of AV conduction. Mobitz Type I (Wenckebach): progressive PR prolongation then dropped QRS; usually benign (AV nodal, narrow QRS). Mobitz Type II: constant PR interval with sudden dropped QRS; more serious (infranodal, wide QRS), risk of progression to complete heart block and Stokes-Adams attacks.

Second-Degree Burn

Partial-thickness burn involving the epidermis and dermis: superficial partial-thickness (blisters, moist, painful, pink) vs. deep partial-thickness (blanching, painful, may have thrombosed capillaries). Causes: thermal (scalds, flames), chemical, electrical.

Second-Hand Smoke

Smoke inhaled from burning tobacco products or exhaled by a person who is smoking. It contains toxic substances and increases health risks in exposed people.

Second-Look Surgery

Second-look surgery is a planned or follow-up operation performed to inspect healing, remove remaining disease, assess treatment response, or manage complications.

Secondary Adrenal Insufficiency

Adrenocortical hypofunction caused by deficient secretion of adrenocorticotrophic hormone from the anterior pituitary or corticotropin-releasing hormone from the hypothalamus. Most frequently caused by prolonged exogenous corticosteroid therapy followed by abrupt withdrawal; unlike primary adrenal insufficiency (Addison disease), mineralocorticoid secretion is preserved and hyperpigmentation is absent.

Secondary Amenorrhea

Absence of menstruation for at least three months in someone with previously regular cycles or six months with previously irregular cycles, caused by pregnancy, endocrine, ovarian, uterine, or systemic factors.

Secondary Attack Rate

The proportion of susceptible contacts who develop infection after exposure to a primary case within a defined period. It helps estimate transmissibility in households, schools, healthcare settings, and other close-contact groups.

Secondary Brain Cancer

Alternate index label; see *Brain metastases*.

Secondary Hyperparathyroidism

Compensatory PTH increase in CKD from decreased phosphate excretion, low calcitriol, and hypocalcemia. Leads to renal osteodystrophy and vascular calcification. Managed with dietary phosphorus restriction, phosphate binders, active vitamin D, and calcimimetics. Refractory cases may require parathyroidectomy.

Secondary Hypertension

High blood pressure caused by another medical condition or by a medication.

Secondary Hypothyroidism

Low thyroid hormone production caused by inadequate pituitary TSH stimulation, usually from pituitary or hypothalamic disease; TSH may be low, normal, or inappropriately mildly high.

Secondary Immune Response

The rapid, enhanced immune response on re-exposure to an antigen, mediated by memory B cells and memory T cells generated during the primary response. Features: shorter lag phase, higher and faster antibody titers (predominantly high-affinity IgG), and more effective cell-mediated responses.

Secondary Immunodeficiency

Reduced immune function caused by an acquired condition or exposure rather than a primary genetic disorder. Causes include HIV infection, cancer, malnutrition, medicines, and chronic disease; the manifestations depend on which immune functions are impaired.

Secondary Infections

Infections that develop during or after another disease, injury, or treatment because of tissue damage, altered defenses, or immunosuppression.

Secondary Metabolism

Biochemical pathways that produce specialized compounds not required for basic growth but important for defense, signaling, pigmentation, or interactions with the environment.

Secondary Osteoporosis

Bone loss caused by another illness, medication, or prolonged immobility.

Secondary Parkinsonism

Secondary parkinsonism is slowed movement, rigidity, tremor, or postural instability caused by another condition, medicine, toxin, vascular injury, or structural brain disease.

Secondary Prevention

Early detection and treatment. Examples: screening (mammography, colonoscopy, blood pressure), self-examination.

Secondary Sexual Characteristics

Physical changes of puberty that distinguish mature males and females but are not directly part of the reproductive organs.

Secondary Progressive Multiple Sclerosis

A phase of multiple sclerosis that evolves from relapsing-remitting multiple sclerosis over 10 to 20 years, characterized by progressive, irreversible accumulation of neurological disability and neurodegeneration with or without occasional superimposed relapses. Characterized radiologically by cerebral atrophy and chronic black holes on MRI rather than active gadolinium-enhancing inflammatory lesions.

Alternate Names:

SPMS; Secondary-progressive MS

Secondary Syphilis

The systemic dissemination stage of *Treponema pallidum* infection, occurring weeks to months after the primary chancre, characterised by a generalised maculopapular rash (including the palms and soles), mucous patches, condylomata lata, lymphadenopathy, and systemic symptoms such as fever and malaise.

Secondary Systemic Amyloidosis

Secondary systemic amyloidosis results when chronic inflammation or infection drives production of serum amyloid A, which deposits in organs such as the kidneys, liver, and spleen.

Secondary Tumor

A tumor formed when cancer cells spread from the original site to another part of the body.

Secretin

A peptide hormone produced by S-cells in the duodenal and jejunal mucosa in response to acidic chyme (pH less than 4.5) entering the duodenum from the stomach.

Secretin Stimulation Test

A dynamic test that administers secretin and measures pancreatic or gastrointestinal responses, used selectively to evaluate pancreatic exocrine function or gastrinoma.

Secretion

Active movement of a substance from cells into a body fluid, lumen, or external environment. In toxicokinetics, renal tubular or biliary secretion can contribute to elimination of a toxin or metabolite.

Secretory Diarrhea

Diarrhea caused by active secretion or reduced absorption of electrolytes and water by the intestine. It persists despite fasting and may result from infections, medications, bile-acid disorders, endocrine tumors, or inflammatory disease.

Section

A defined part or subdivision of an organ, document, study, or procedure; in anatomy and imaging, the term identifies a specific slice or region examined.

Section, Cesarean

Cesarean section, an operation delivering a baby through incisions in the abdomen and uterus when vaginal birth is unsafe, impossible, or not advisable.

Sed Rate

A blood test measuring how quickly red blood cells settle, used as a nonspecific marker of inflammation.

Alternate Names:

Erythrocyte sedimentation rate; ESR

Sedation

The use of medicines to make a person relaxed, drowsy, or less aware during a procedure while maintaining appropriate monitoring and breathing.

Sedation Dentistry

Pharmacological techniques for anxiety management: nitrous oxide (laughing gas), oral sedation, IV sedation, general anesthesia. Indications: dental phobia, special needs, prolonged procedures. Monitoring required for deeper sedation.

Sedation Levels

A continuum ranging from minimal sedation, in which normal responses to verbal commands are preserved, to moderate or deep sedation, in which responses and airway or breathing support may be reduced. The level can change and requires appropriate monitoring.

Sedative-Hypnotic

CNS depressants. Benzodiazepines, barbiturates, Z-drugs. Overdose: respiratory depression, coma. Treatment: supportive, flumazenil (benzodiazepines only), airway management.

Sedative-Hypnotic Toxicdrome

A poisoning pattern caused by central nervous system depressants, producing drowsiness, slurred speech, impaired coordination, reduced reflexes, respiratory depression, coma, or loss of airway protection.

Sedatives

Medicines or substances that reduce central nervous system activity and produce calmness or sleepiness.

Sedimentation Rate

The erythrocyte sedimentation rate, a nonspecific blood test that may increase with inflammation, infection, anemia, or other conditions.

Segmental Pressures

Blood pressure measurements at different leg levels. Localizes level of obstruction.

Segmentation

A pattern of intestinal motility involving localized, rhythmic contractions of circular muscle that segment the intestinal contents, mixing them with digestive secretions and increasing contact with the absorptive mucosal surface. Unlike peristalsis, segmentation does not propel contents distally.

Segmentectomy

An anatomical sublobar surgical resection of one or more individual bronchopulmonary segments of a lung lobe. The procedure involves individual dissection, division, and ligation of the segmental pulmonary artery, segmental pulmonary vein, and segmental bronchus, along with identification of the intersegmental parenchymal plane (often using systemic indocyanine green fluorescence) and radical hilar/mediastinal lymph node dissection. Landmark clinical trials (JCOG0802 and CALGB 140503) established segmentectomy as non-inferior to lobectomy in overall survival for peripheral non-small cell lung cancer ≤ 2 cm with ≥ 50 percent consolidation, while significantly preserving postoperative pulmonary reserve.

Alternate Names:

Sublobar Resection; Anatomical Segmentectomy; Pulmonary Segmentectomy

SEID

Alternate index label; see *Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS)*.

Seizure

A transient occurrence of signs and/or symptoms due to abnormal, excessive, or synchronous neuronal activity in the brain. Provoked seizures have an identifiable acute cause (fever, electrolyte disturbance, drug toxicity, withdrawal). Unprovoked seizures occur without an acute trigger.

Seizure Disorder

Alternate index label; see *Epilepsy*.

Seizure Disorders

Conditions characterized by recurrent unprovoked seizures or an enduring predisposition to them, clas-

sified by seizure type and cause and treated according to epilepsy syndrome and risk.

Seizure Febrile (Febrile Seizures)

Alternate index label for Febrile Seizures.

Seizure Fever Induced (Febrile Seizures)

Alternate index label for Febrile Seizures.

Seizure, Absence

Alternate index label; see *Absence seizure*.

Seizure, Febrile

Alternate index label; see *Febrile seizure*.

Seizure, Grand Mal

Alternate index label; see *Tonic-clonic (grand mal) seizure*.

Seizure, Petit Mal

Alternate index label; see *Absence seizure*.

Seizure, Temporal Lobe

Alternate index label; see *Temporal lobe seizure*.

Seizures

Seizures are transient episodes of abnormal excessive brain electrical activity causing changes in awareness, movement, sensation, behavior, or autonomic function.

Selective Deficiency Of IgA

The most common primary antibody deficiency, characterized by very low or absent immunoglobulin A with normal or near-normal levels of other immunoglobulins; infections and autoimmune disease may occur.

Selective Estrogen Receptor Modulator

A class of medications that bind to estrogen receptors (ER-alpha and ER-beta) with tissue-specific agonist or antagonist activity, depending on the target tissue and differential co-regulator recruitment.

Selective Estrogen Receptor Modulators

Medicines that activate estrogen receptors in some tissues and block them in others.

Alternate Names:

SERMs

Selective IgA Deficiency

An immune disorder characterized by very low or absent immunoglobulin A levels.

Selective Antibody Deficiency With Normal Immunoglobulins

A primary antibody deficiency in which total immunoglobulin levels are normal but responses to specific vaccine antigens are inadequate, causing recurrent respiratory or other bacterial infections.

Selective Mutism

Consistent failure to speak in specific social situations where speaking is expected despite speaking in other situations. The disturbance interferes with educational or occupational achievement or social communication. Must last for at least one month and is not attributable to lack of knowledge of or comfort with the spoken language.

Selective Serotonin Reuptake Inhibitors

Antidepressants that increase serotonin signaling by reducing its reuptake into nerve cells.

Alternate Names:

SSRIs

Selective Serotonin-Reuptake Inhibitors (SSRIs)

Antidepressants that increase serotonin signaling by inhibiting its reuptake into nerve terminals.

Selegiline

A monoamine-oxidase-B inhibitor used in selected treatment of Parkinson disease and sometimes depression.

Selenium

Trace mineral, component of antioxidant enzymes. Deficiency: Keshan disease (cardiomyopathy),

Kashin-Beck disease (osteoarthropathy). Sources: Brazil nuts, seafood, meat.

Selenium In Diet

Selenium in diet provides a trace mineral needed for antioxidant enzymes, thyroid hormone metabolism, and reproduction. Excess supplements can cause toxicity, so intake should remain within recommended limits.

Selenium Sulphide

A topical antifungal and antiseborrheic medicine used for dandruff and seborrheic dermatitis.

Selenium-75

Calibration source. Half-life 119 days. Used for uniformity correction.

Self Catheterization - Female

Intermittent insertion and removal of a catheter through the female urethra to empty the bladder when normal urination is inadequate, reducing retention and protecting kidney function.

Self Catheterization - Male

Intermittent insertion and removal of a catheter through the male urethra to drain the bladder when neurologic, obstructive, or other conditions prevent adequate emptying.

Self Gratification Sexual (Masturbation)

Alternate index label for Masturbation.

Self Harming

Alternate index label; see *Self-injury/cutting*.

Self-Diagnosing Body Dysmorphic Disorder

Alternate index label for BDD.

Self-Efficacy

Patient belief in ability to manage condition. Influences adherence, outcomes.

Self-Experimentation

Research in which an investigator deliberately uses an intervention or procedure on themselves. It requires careful attention to consent, safety, and scientific validity.

Self-Harm

Intentional self-inflicted injury without suicidal intent, also termed non-suicidal self-injury; common methods include cutting, burning, and self-hitting. Prevalent in adolescents and in borderline personality disorder, depression, eating disorders, and trauma, serving affect regulation or self-punishment.

Self-Help Group

A peer group in which members support one another while changing a harmful behavior or coping with a problem.

Alternate Names:

Support group

Self-Injury/Cutting

Deliberate injury to one's own body, often used to cope with overwhelming emotions and requiring compassionate assessment of safety and mental health.

Alternate Names:

Self Harming

Self-Management

Actions a person takes to monitor and manage a health condition, including treatment use, symptom monitoring, and lifestyle measures.

Self-Poisoning

Intentional ingestion or exposure to a potentially harmful substance, often requiring urgent medical and mental-health assessment.

Self-Testing For COVID-19

Testing a sample collected at home, usually from the nose, to check for COVID-19 infection. A negative result does not always rule out infection, especially early in illness.

Sella Turcica

A saddle-shaped depression in the sphenoid bone containing the pituitary gland.

Semaglutide

Semaglutide is a glucagon-like peptide-1 receptor agonist used for selected patients with type 2 diabetes or chronic weight management. It slows gastric emptying and reduces appetite and glucose levels; nausea, gallbladder disease, pancreatitis risk, and contraindications require review.

Semen

The fluid ejaculated from the male reproductive tract, containing sperm (produced in the testes) suspended in seminal plasma (produced by the seminal vesicles, prostate, and bulbourethral glands). Semen provides nutrients (fructose, amino acids), protective agents (zinc, antioxidants), and a medium for sperm transport.

Semen Analysis

Laboratory assessment of ejaculate volume, sperm concentration, total count, motility, morphology, vitality, and other characteristics. It is a key initial test in male-factor infertility and should usually be repeated because results vary biologically.

Semen Blood (Blood In Semen)

Alternate index label for Blood In Semen.

Semen DNA Analysis

Semen DNA analysis evaluates sperm genetic material, often to investigate infertility, repeated pregnancy loss, or possible DNA damage.

Semi-Fowler Position

A semiupright position in which the head and upper body are elevated about 30 to 45 degrees.

Semicircular Canals

Three fluid-filled (endolymph) loops in each inner ear, oriented approximately orthogonally to detect angular (rotational) acceleration in three planes: horizontal (lateral), anterior (superior), and poste-

rior. Each has an ampulla with a crista containing hair cells embedded in a gelatinous cupula.

Semilunar Cartilages

The crescent-shaped fibrocartilages of the knee, more commonly called menisci.

Semilunar Valves

The aortic and pulmonary valves, which open during ventricular contraction and close during ventricular relaxation to prevent backflow into the ventricles.

Seminal Vesicle

One of a paired set of male accessory reproductive glands behind the urinary bladder. Its fructose-rich secretion contributes substantially to semen and joins the vas deferens through the ejaculatory duct.

Seminal Vesicles

Glands behind the bladder that produce much of the fluid in semen.

Seminiferous Tubules

Tightly coiled testicular tubules in which sperm are produced.

Seminoma

A malignant germ-cell tumor of the testis, usually sensitive to radiotherapy and chemotherapy.

Senescence

Biological changes and functional decline associated with aging. In cell biology, cellular senescence is a durable state of cell-cycle arrest in metabolically active cells; accumulation of senescent cells can contribute to tissue dysfunction and age-related disease.

Alternate Names:

Biological aging

Senile Dementia

An outdated term formerly used for dementia in older adults; current diagnosis specifies the cause.

Senile Lentigines

Flat, sharply defined brown or tan spots caused by chronic ultraviolet exposure, commonly appearing on sun-exposed skin with age.

Senility

An outdated, imprecise term for age-related decline; modern assessment identifies specific medical, cognitive, or functional causes.

Senna

A plant-derived stimulant laxative containing sennosides, used short term for constipation and sometimes to empty the bowel before a procedure. It increases intestinal activity and may cause abdominal cramps or diarrhea; prolonged or excessive use can cause dependence or fluid and electrolyte disturbances.

Sensate Focus Techniques

Structured touch exercises used in sex therapy to reduce performance pressure and improve intimacy.

Alternate Names:

Sensate focus

Sensation

Perception of stimuli such as touch, pain, temperature, sound, or position.

Sense

A sense is a capacity to detect and interpret stimuli such as touch, temperature, pain, sound, light, smell, taste, position, or internal body signals.

Sensitisation

The process by which repeated exposure increases an immune, neural, or behavioral response.

Sensitivity

The proportion of people with a condition who have a positive test result; it also means responsiveness to a stimulus.

Sensitivity Analysis

Sensitivity analysis tests how much a study's con-

clusion changes when assumptions, measurements, or input values are varied.

Sensitivity, Gluten (Nonceliac Gluten Sensitivity (Intolerance))

Alternate index label for Nonceliac Gluten Sensitivity (Intolerance).

Sensitized

State of having preformed antibodies against human leukocyte antigens; complicates transplant matching and increases rejection risk.

Sensorimotor Polyneuropathy

Sensorimotor polyneuropathy affects multiple peripheral nerves and causes combinations of numbness, pain, weakness, impaired reflexes, and distal muscle wasting.

Sensorineural Deafness

Sensorineural deafness is hearing loss from dysfunction of the inner ear, auditory nerve, or central pathways rather than blockage of sound conduction.

Sensorineural Hearing Loss

Hearing loss from pathology of the cochlea or auditory (eighth cranial) nerve, characterized on pure-tone audiometry by elevated bone conduction thresholds and no air-bone gap.

Sensorium

The state or level of consciousness and awareness of one's surroundings.

Sensory

Relating to sensation or to nerves and pathways that carry information such as touch, pain, temperature, position, hearing, or vision to the brain.

Sensory Cortex

Cerebral cortex that receives and interprets sensory information, especially the postcentral gyrus for body sensation.

Sensory Deprivation

Marked reduction of normal sensory input, which can affect mood, cognition, and perception.

Sensory Level

A sharply demarcated horizontal boundary below which sensation is altered on the trunk or limbs. A sensory level strongly suggests a spinal-cord lesion, although its localization requires correlation with motor, reflex, and autonomic findings.

Sensory Nerves

Nerves that transmit sensory information from peripheral receptors to the central nervous system. Sensory (afferent) neurons have their cell bodies in the dorsal root ganglia (spinal nerves) or cranial nerve ganglia (cranial nerves).

Sensory Neuron

A nerve cell that carries information from sensory receptors toward the central nervous system.

Sensory Processing Disorder

Difficulty detecting, interpreting, or responding to sensory input such as sound, touch, movement, light, or body position. It may affect behavior, learning, coordination, or daily activities and can occur with other neurodevelopmental conditions. Assessment and occupational-therapy strategies are individualized.

Sensory Receptors

Specialized cells or nerve endings that detect stimuli such as touch, temperature, pain, light, sound, or chemical changes.

Sentinel Event

Unexpected occurrence involving death or serious physical/psychological injury. Requires root cause analysis.

Sentinel Injury

A minor but potentially significant injury in a non-mobile infant, such as unexplained bruising, oral injury, or a fracture, that may precede more se-

vere abuse. It warrants careful history, complete examination, safeguarding assessment, and documentation.

Sentinel Lymph Node

First lymph node draining tumor site. Biopsy assesses nodal spread.

Sentinel Lymph Node Biopsy For Melanoma

Sentinel lymph-node biopsy identifies and removes the first draining lymph node or nodes from a melanoma site to detect microscopic spread. The result helps stage disease and guide further treatment; it does not remove all regional nodes routinely.

Sentinel Lymph-Node Biopsy

Surgical removal and pathologic examination of the first draining lymph node or nodes from a primary tumor. It is used for selected melanomas, breast cancers, and other tumors to assess regional metastatic risk and guide staging.

Sentinel Node Biopsy

Alternate index label for sentinel lymph-node biopsy: removal and examination of the first draining node or nodes to which a tumor is likely to spread, used to assess regional metastasis.

Sentinel Surveillance

Ongoing collection of health data from selected reporting sites, providers, or populations chosen to represent trends rather than enumerate every event. It can provide early warning and monitor changes in disease activity.

Separation Anxiety

Separation anxiety is developmentally excessive fear or distress when apart from an attachment figure, causing avoidance, physical complaints, sleep problems, or impairment. Treatment may include gradual exposure, family-based support, cognitive behavioral therapy, and selected medication.

Alternate Names:

Separation Anxiety Disorder

Developmentally inappropriate and excessive fear or anxiety concerning separation from attachment figures. The individual may have recurrent excessive distress when anticipating or experiencing separation from home or major attachment figures.

Separation Anxiety In Children

Developmentally excessive distress when separated from an attachment figure, with worry, refusal, physical symptoms, nightmares, or school avoidance that impairs daily life.

Separation Of The Iris

Disinsertion or dialysis of the iris root from its attachment at the ciliary body, typically caused by blunt ocular trauma. It appears as a peripheral iridodialysis visible on slit-lamp examination and may cause monocular double vision, glare, or secondary glaucoma if extensive.

Sepsis

Life-threatening organ dysfunction caused by a dysregulated response to infection.

Alternate Names:

Septicemia (Sepsis)

Sepsis Syndrome

The spectrum of the host's dysregulated response to infection: sepsis (infection + organ dysfunction, SOFA increase ≥ 2), septic shock (sepsis + hypotension requiring vasopressors with lactate > 2 mmol/L despite adequate fluid), and severe organ dysfunction/multi-organ failure.

Septal Defect

An abnormal opening between chambers of the heart or between other bodily compartments.

Septal Defect, Atrioventricular

Alternate index label; see *Atrioventricular canal defect*.

Septal Defect, Ventricular

Alternate index label; see *Ventricular septal defect*.

Septal Hematoma

Accumulation of blood between the perichondrium and the septal cartilage of the nose, typically following nasal trauma (nasal fracture). Presents with nasal obstruction, pain, and a bluish, boggy swelling of the nasal septum.

Septate Uterus

The most prevalent congenital uterine anomaly, accounting for approximately 55% of Müllerian malformations, resulting from failure of complete resorption of the midline longitudinal fibromuscular septum following fusion of the paired paramesonephric ducts. Characterized by a normal external convex uterine fundal contour with an internal septum dividing the cavity partially (incomplete) or completely down to the internal or external os (complete septate uterus), carrying the highest rates of spontaneous first- and second-trimester abortion, corrected via hysteroscopic metroplasty.

Alternate Names:

Uterine Septum

Septic

Relating to infection or contamination by pathogenic microorganisms; in "septic shock" it refers to life-threatening circulatory and metabolic dysfunction caused by an abnormal response to infection.

Septic Abortion

Any spontaneous or induced pregnancy loss complicated by ascending pelvic and intrauterine infection (endometritis, parametritis, or peritonitis). Characterized by fever, chills, tachycardia, malodorous purulent cervical discharge, marked uterine tenderness, and leukocytosis, with potential rapid progression to septic shock, pelvic thrombophlebitis, and disseminated intravascular coagulation. Management requires urgent high-dose intravenous broad-spectrum antibiotic therapy followed by expeditious surgical evacuation of the infected retained gestational tissue.

Alternate Names:

Septic Miscarriage; Infected Abortion

Septic Arthritis

Infection of a joint, usually bacterial, causing acute pain, swelling, restricted movement, and systemic illness.

Alternate Names:

Arthritis Infectious (Septic Arthritis)

Septic Embolism

Septic embolism occurs when infected material, typically thrombi containing bacteria, embolizes through the bloodstream to distant sites, causing metastatic infection. The most common source is infective endocarditis, where infected vegetations on heart valves break off and lodge in peripheral vessels.

Septic Pulmonary Infarction

Necrosis of lung tissue caused by infected emboli lodging in pulmonary arteries, often from right-sided infective endocarditis or infected venous thrombi. It may produce fever, pleuritic chest pain, cough, hemoptysis, and cavitating peripheral lung lesions.

Septic Shock

A life-threatening form of circulatory failure caused by sepsis, with persistent hypotension or tissue hypoperfusion despite fluid treatment.

Septicaemia

An older term for bloodstream infection, often used to describe bacteremia with systemic illness.

Septicemia

Septicemia is an older term for serious bloodstream infection; modern assessment distinguishes bacteremia from sepsis, which is life-threatening organ dysfunction caused by infection.

Septicemia (Sepsis)

Alternate index label for Sepsis.

Septo-Optic Dysplasia

A congenital disorder characterized by variable combinations of optic-nerve hypoplasia, midline brain abnormalities, and pituitary hormone deficiencies.

It may cause visual impairment, nystagmus, developmental differences, seizures, and growth or endocrine problems.

Septoplasty

Septoplasty is surgery to straighten or reshape the nasal septum to improve airflow or address structural obstruction.

Septum

A wall or partition separating two spaces or tissues.

Septum Deviated (Deviated Septum)

Alternate index label for Deviated Septum.

Septum, Heart

The wall separating the right and left chambers of the heart, consisting of atrial and ventricular portions and preventing normal mixing of oxygenated and deoxygenated blood.

Sequela

A lasting consequence or condition that remains after an earlier disease, injury, infection, or treatment has ended, such as paralysis after stroke or scarring after infection.

Sequelae

Persistent consequences or complications that remain after an illness or injury.

Sequential Therapy

Treatments given in sequence. Used when combination not suitable.

Sequester

To sequester means to isolate or set apart; in medicine, sequestered tissue or material is separated from its normal circulation or surroundings.

Sequestrum

A piece of dead bone or tissue separated from living tissue, commonly occurring in chronic osteomyelitis.

Ser

Ser is the standard abbreviation for serine, an amino acid incorporated into proteins and involved in metabolism and cell signaling.

Serendipity

Serendipity is an unexpected beneficial discovery; in research, it describes useful findings that arise while investigating another question.

Serine

Serine is a nonessential amino acid used in protein synthesis, one-carbon metabolism, membrane formation, and signaling.

Serious Diseases And Health Problems (Symptoms Of Serious Diseases And Health Problems)

Alternate index label for Symptoms of Serious Diseases and Health Problems.

SERMs

Selective estrogen receptor modulators that have estrogen-like effects in some tissues and blocking effects in others.

Alternate Names:

Selective estrogen receptor modulators

Seroconversion

Development of detectable antibodies in blood after infection, vaccination, or another immune stimulus.

Serodiagnostic

Relating to diagnosis by detecting antibodies or other markers in blood serum.

Serologic Test

A blood test that detects antibodies, antigens, or other immune markers to help diagnose infection, assess immunity, or monitor response to treatment.

Serology

The laboratory evaluation of serum for antibodies, antigens, or other immune markers. Results may indicate current infection, previous exposure, vacci-

nation, autoimmune reactivity, or passive antibody transfer and must be interpreted with timing and clinical context.

Serology For Brucellosis

Blood testing for antibodies to *Brucella* species, interpreted with exposure history and symptoms because cross-reactions and early false-negative results can occur.

Seroma

A pocket of clear fluid that develops near a surgical incision or injured tissue.

Seronegative Arthritis

Arthritis with negative rheumatoid factor and anti-CCP. Includes spondyloarthropathies, psoriatic arthritis, reactive arthritis.

Serosanguineous

Serosanguineous describes thin wound or body fluid containing both clear serum and a small amount of blood, often appearing pink or pale red.

Serositis

Inflammation of a serous membrane, such as the pleura, pericardium, or peritoneum, causing pain with breathing or movement and sometimes an inflammatory effusion.

Serotonin

A neurotransmitter involved in mood, sleep, appetite, and pain regulation.

Alternate Names:

5-hydroxytryptamine; 5-HT

Serotonin Blood Test

A blood test measuring serotonin, used selectively in evaluating suspected serotonin-producing neuroendocrine tumors; many medicines and specimen factors affect results.

Serotonin Syndrome

A potentially life-threatening condition caused by excessive serotonergic activity in the central and

peripheral nervous systems, typically resulting from drug interactions or overdose involving serotonergic medications.

Serotonin-Norepinephrine Reuptake Inhibitors

Antidepressants that block the reuptake of both serotonin and norepinephrine. They include venlafaxine, duloxetine, desvenlafaxine, and levomilnacipran. SNRIs are used for major depressive disorder, generalized anxiety disorder, diabetic neuropathy, fibromyalgia, and chronic pain.

Serotonin-Norepinephrine-Dopamine Reuptake Inhibitors

A class of antidepressants that inhibit reuptake of serotonin, norepinephrine, and dopamine. The primary agent is bupropion, which also acts as a norepinephrine-dopamine reuptake inhibitor. It is used for depression, smoking cessation, and as augmentation. It lacks sexual side effects and weight gain associated with SSRIs.

Serotype

A distinct variation of a microorganism or antigen defined by its immune reactivity.

Serous

Thin, watery, and usually clear; serous fluid resembles serum.

Serous Borderline Ovarian Tumor

An ovarian serous epithelial tumor with hierarchical papillae and cellular atypia but no destructive stromal invasion; extraovarian implants may occur.

Serous Cyst Adenomas Pancreas (Pancreatic Cysts)

Alternate index label for Pancreatic Cysts.

Serous Fluid Cytology

Cytologic examination of pleural, pericardial, or peritoneal fluid to assess for malignant cells, infection, and reactive inflammation. Cell-block preparation and ancillary tests may improve diagnostic

accuracy.

Serous Membranes

Smooth membranes lining closed body cavities and covering organs, such as pleura, pericardium, and peritoneum.

Serous Tubal Intraepithelial Carcinoma

A noninvasive high-grade serous lesion of the fallopian-tube fimbria, considered a precursor or early manifestation of many tubo-ovarian high-grade serous carcinomas.

Serpentine Supravenous Hyperpigmentation

A dermatological finding characterized by dark, wavy, serpiginous hyperpigmentation along the course of superficial veins, most commonly on the upper chest and neck. Associated with venous stasis, thrombophlebitis, and certain malignancies. The pigmentation results from hemosiderin deposition in the perivenous dermis.

Serrated Pathway

Alternative molecular pathway to colorectal carcinogenesis accounting for fifteen to thirty percent of cases. It involves serrated polyps as precursors including sessile serrated lesions with dysplasia. Molecular features include CpG island methylator phenotype, BRAF mutations, and microsatellite instability from MLH1 silencing.

Serrated Polyposis Syndrome

A disorder characterized by multiple serrated lesions or polyps in the colon and an increased risk of colorectal cancer, requiring colonoscopic surveillance and removal of significant lesions.

Serration

A row of small tooth-like projections or notches along an edge, used to describe anatomy, cells, instruments, or imaging appearances.

Serratus Anterior Muscle

Fan-shaped muscle from ribs 1-8/9 to the me-

dial scapular border. Innervated by long thoracic nerve (C5-C7). Protracts scapula, upwardly rotates glenoid for overhead elevation, stabilizes against thorax. Long thoracic nerve injury causes medial scapular winging, especially visible during wall push-ups. Commonly injured in breast surgery.

Sertoli-Cell Tumor

A sex-cord stromal tumor that may produce androgens or estrogens and can cause endocrine manifestations.

Sertoli-Leydig Cell Tumor

A rare ovarian tumor that may produce male hormones, causing increased facial hair, acne, a deeper voice, or irregular periods. Treatment depends on the tumor's stage and features.

Serum

The liquid portion of blood remaining after clotting, lacking fibrinogen and most clotting factors.

Serum Concentration

The concentration of a substance measured in the liquid component of clotted blood. Interpretation depends on sampling time, exposure history, distribution, metabolism, elimination, and the relationship between concentration and toxicity.

Serum Free Hemoglobin Test

A blood test measuring hemoglobin not contained within red cells, used to support evaluation of intravascular hemolysis, transfusion reactions, or mechanical red-cell destruction.

Serum Glutamic Oxaloacetic Transaminase

SGOT, now called AST, is an enzyme released by injured liver, heart, skeletal muscle, or other cells; its level must be interpreted with other findings.

Serum Glutamic Pyruvic Transaminase

SGPT, now called ALT, is an enzyme concentrated in liver cells and commonly measured as a marker of hepatocellular injury.

Serum Herpes Simplex Antibodies

Serum herpes simplex antibodies is a diagnostic or laboratory investigation; interpretation depends on the specimen, method, reference interval, timing, and clinical question.

Serum Iron Test

A blood test measuring circulating iron bound mainly to transferrin; it is interpreted with ferritin, transferrin saturation, and blood counts rather than used alone to diagnose iron deficiency.

Serum Osmolal Gap

The difference between measured serum osmolality and calculated osmolality. An increased gap suggests unmeasured osmoles, including toxic alcohols, but interpretation must consider timing, renal function, lactate, ketones, and laboratory variation.

Serum Phenylalanine Screening

A blood test measuring phenylalanine, used in newborn screening and follow-up of phenylketonuria or other disorders of phenylalanine metabolism.

Serum Progesterone

Serum progesterone is a diagnostic or laboratory investigation; interpretation depends on the specimen, method, reference interval, timing, and clinical question.

Serum Protein Electrophoresis

Separation of serum proteins by electrophoresis into 5 fractions: albumin, alpha-1, alpha-2, beta, and gamma. Monoclonal gammopathy: M-spike in gamma region (multiple myeloma, MGUS, Waldenstrom macroglobulinemia). Immunofixation: identifies heavy chain (IgG, IgA, IgM, IgD, IgE) and light chain (kappa, lambda) isotype.

Serum Sickness

A delayed immune reaction, usually occurring 1–3 weeks after exposure to certain medicines, anti-serum, or other injected proteins. It can cause fever, rash or hives, swollen painful joints, and sometimes swollen lymph nodes or organ problems.

Serum Therapy

Treatment using serum or serum-derived antibodies, historically including antitoxins and immunoglobulin preparations.

Serving Size

Serving size is the standardized amount used to describe the nutrients or calories in food and may differ from the amount a person actually eats.

Sesamoid Bones

Small bones embedded in tendons, such as the patella and bones beneath the big toe.

Sesamoiditis

Inflammation and pain around a small sesamoid bone, often beneath the big toe.

Sesquipedalian

Sesquipedalian means characterized by very long words; clear healthcare communication favors language that patients can understand.

Sessile

Attached by a broad base rather than a stalk, as in a sessile polyp.

Set

In exercise, a group of consecutive repetitions performed before a rest period. In medicine, “set” can also mean a prepared group of instruments, tests, or equipment.

Sever Disease

Calcaneal apophysitis in children/adolescents. Heel pain, worse with activity. Treatment: rest, heel cups, stretching.

Severe Acute Respiratory Syndrome

A serious contagious respiratory illness caused by the SARS coronavirus, with fever, dry cough, and breathing difficulty that may progress to pneumonia and respiratory failure. The term usually refers to the 2002–2004 SARS outbreak; SARS-CoV-2 causes COVID-19, a different disease.

Alternate Names:

Severe Acute Respiratory Syndrome (Severe Acute Respiratory Syndrome (SARS))

Severe Acute Respiratory Syndrome (Severe Acute Respiratory Syndrome)

Alternate index label for Severe Acute Respiratory Syndrome.

Severe Brain Injury, Coma

Alternate index label; see *Coma*.

Severe Combined Immunodeficiency

Severe combined immunodeficiency represents a group of genetic disorders characterized by profoundly defective T cell development and function, with variable B and NK cell involvement, resulting in severe susceptibility to life-threatening infections in infancy.

Severe Condition (Sever Condition)

Alternate index label for Sever Condition.

Sevoflurane

A volatile inhaled anesthetic with relatively low blood-gas solubility, permitting rapid changes in anesthetic depth and recovery. It can cause dose-dependent cardiovascular and respiratory depression and is a recognized trigger of malignant hyperthermia in susceptible patients.

Sex Change

A nontechnical term for gender transition or medical gender-affirming care; respectful, current terminology should be used.

Sex Chromosomes

The X and Y chromosomes, which contribute to sex development and carry many other genes.

Sex Education

Education about anatomy, relationships, consent, contraception, sexually transmitted infections, and sexual health.

Sex Headaches

Headaches brought on by sexual activity, including a gradually increasing ache or a sudden severe headache at orgasm. A first or thunderclap headache requires urgent assessment to exclude bleeding or other dangerous causes.

Sex Hormones

Hormones involved in reproductive development and function, including estrogens, progesterone, and androgens.

Sex Therapy

Structured psychological or behavioral therapy for sexual concerns or dysfunction.

Sex-Linked Dominant

Sex-linked dominant inheritance occurs when a disease-causing variant on a sex chromosome can produce the trait with one altered copy, with transmission patterns differing by sex.

Sex-Linked Inheritance

Inheritance of a trait caused by a gene on a sex chromosome, commonly the X chromosome.

Sex-Linked Recessive

Sex-linked recessive inheritance usually involves a variant on the X chromosome, affecting males more often while females may be carriers or variably affected.

Sexual (Sex) Problems In Men

Alternate index label for Sex) Problems in Me.

Sexual Abuse

Nonconsensual sexual contact, exploitation, or behavior; it is a safeguarding and health emergency when immediate danger or injury exists.

Compulsive Sexual Behavior Disorder

A persistent pattern of failure to control intense, repetitive sexual impulses or behaviors, resulting in marked distress or significant impairment and continuing despite adverse consequences. It is not de-

finied simply by a high sex drive or moral disapproval; assessment considers duration, control, distress or impairment, consent, and coexisting mental-health or substance-use conditions.

Alternate Names:

Sexual Addiction; Addiction Sexual (Sexual Addiction)

Sexual Assault

A broad legal category of any non-consensual sexual contact or penetration, encompassing rape, indecent assault, and drug-facilitated assault. Examination documents general and anogenital injuries, collects biological and trace samples, and screens for drug-facilitated assault by extended-window testing.

Alternate Names:

Rape (Sexual Assault)

Sexual Assault - Prevention

Sexual-assault prevention includes consent education, recognizing coercion, safer environments, bystander intervention, and access to confidential support; responsibility always lies with the person who commits the assault.

Sexual Deviation

An obsolete and stigmatizing term for atypical sexual interests; current assessment focuses on consent, distress, impairment, and harm.

Sexual Dysfunction

Persistent difficulty with sexual desire, arousal, orgasm, or pain that causes distress or impairment.

Sexual Dysfunction, Female

Alternate index label; see *Female sexual dysfunction*.

Sexual Health Overview

Sexual health includes physical, emotional, mental, and social well-being in relation to sexuality and relationships. It includes consent, contraception, sexually transmitted infection prevention and testing, fertility, sexual function, safety, and respectful communication.

Sexual Intercourse (Coitus)

The insertion of a penis into a partner's body for sexual pleasure, reproduction, or both. Most commonly vaginal intercourse (penile-vaginal penetration) but may also include anal intercourse. Sexual intercourse is a key component of human pair bonding and reproduction.

Alternate Names:

Coitus

Sexual Masochism Disorder

A paraphilic disorder characterized by recurrent, intense sexual arousal from the act of being humiliated, beaten, bound, or otherwise made to suffer, lasting ≥ 6 months and causing distress or impairment or involving acting on these urges.

Sexual Obsession

Alternate index label; see *Compulsive sexual behavior*.

Sexual Rehabilitation

Address sexual function after injury, illness. Counseling, devices, medications.

Sexual Sadism

Sexual sadism involves recurrent arousal from another person's suffering or humiliation; acts without consent are abuse, and safety-focused clinical care is required.

Sexual Sadism Disorder

A paraphilic disorder characterized by recurrent, intense sexual arousal from the physical or psychological suffering of another person, lasting ≥ 6 months and causing distress or impairment or involving acting on these urges with a nonconsenting person.

Sexual Violence

Sexual violence is any sexual act, attempt, contact, or coercion without freely given consent and can cause physical injury, infection, pregnancy, trauma, and long-term health effects.

Sexually Transmitted Disease Treatments

Alternate index label for STD) Treatment.

Sexually Transmitted Diseases

Infections transmitted primarily through sexual contact; also called sexually transmitted infections.

Alternate Names:

sexually transmitted infections.

Sexually Transmitted Diseases (STDs In Women)

Alternate index label for STDs In Women.

Sexually Transmitted Diseases (STDs In Women)

Alternate index label for Sexually Transmitted Diseases (STDs In Women).

Sexually Transmitted Diseases (STDs)

Infections transmitted mainly through sexual contact, including chlamydia, gonorrhea, syphilis, HIV, and HPV.

Alternate Names:

STDs

Sexually Transmitted Diseases And Pregnancy (STDs)

Alternate index label for STDs.

Sexually Transmitted Infection

STI: infection transmitted via sexual contact. Examples: syphilis, gonorrhea, chlamydia, HIV.

Sezary Disease

Alternate index label; see *Sezary syndrome*.

Sezary Syndrome

An uncommon form of cutaneous T-cell lymphoma involving widespread red skin and abnormal lymphocytes in the blood.

Alternate Names:

Sezary Disease

Sézary Syndrome

The leukemic variant of cutaneous T-cell lymphoma, characterized by erythroderma, lymphadenopathy, pruritus, and circulating atypical lymphocytes (Sézary cells).

Sgarbossa Criteria

A validated, standardized electrocardiographic diagnostic scoring system developed to identify acute myocardial infarction in the presence of baseline Left Bundle Branch Block or right ventricular paced rhythms, both of which normally cause secondary repolarization abnormalities that obscure ischemic ST changes. The original criteria award points for: (1) concordant ST-segment elevation ≥ 1 mm in leads with a positive QRS complex (5 points, highly specific); (2) concordant ST-segment depression ≥ 1 mm in leads V_1 , V_2 , or V_3 (3 points); and (3) excessively discordant ST-segment elevation ≥ 5 mm in leads with a negative QRS complex (2 points). The Smith-Modified Sgarbossa criteria replace the absolute 5 mm discordant rule with a proportional rule: ST elevation/S-wave depth ratio ≤ -0.25 (or $\geq 25\%$), greatly enhancing sensitivity for acute coronary occlusion.

Alternate Names:

Modified Sgarbossa Criteria; Smith-Modified Sgarbossa Rule

Shadowing Artifact

Ultrasound artifact from attenuating structures. Enhanced by harmonic imaging.

Shaken Baby Syndrome

A form of abusive head trauma caused by violent shaking, typically in infants under 1 year. The triad: subdural hemorrhage, retinal hemorrhages, and encephalopathy. May present with irritability, lethargy, vomiting, seizures, and apnea. CT shows subdural hematoma; MRI shows diffuse axonal injury.

Shaken Baby Syndrome (Abusive Head Trauma)

Alternate index label for Abusive Head Trauma.

Shaken Impact Syndrome

A term for abusive head trauma involving shaking with or without impact, especially in infants.

Shampoo - Swallowing

Accidental swallowing of a small amount of shampoo usually causes mouth or stomach irritation; larger or concentrated exposures may cause vomiting or breathing problems and warrant poison-control advice.

Shared Decision-Making

A collaborative process in which clinicians and patients combine best available evidence with patient values, preferences, goals, and circumstances to choose among reasonable options.

Shark Attack

A shark attack can cause severe lacerations, bleeding, tissue loss, drowning, and shock; emergency rescue, hemorrhage control, and trauma care are critical.

Sharp Force Trauma

Injury produced by an edged or pointed instrument that cuts or penetrates the skin, divided into incised wounds (longer than deep, clean margins) and stab wounds (deeper than wide). Wound length, depth, angle, hesitation marks, and defense injuries infer weapon type and manner.

Sharp Pains In Your Stomach During Pregnancy

Sharp abdominal pain in pregnancy may result from ligament stretching, gas, constipation, uterine contractions, urinary disease, ectopic pregnancy, placental complications, or other causes. Severe or persistent pain, bleeding, fever, faintness, vomiting, or reduced fetal movement needs urgent obstetric assessment.

Sharps Container

Puncture-resistant container for used needles and sharp instruments. Red, labeled biohazard.

Sharps Injury

A percutaneous or mucocutaneous injury caused by a needle or other sharp instrument that may expose a person to blood or body fluids. Immediate wound care, incident reporting, risk assessment, and pathogen-specific post-exposure management are required.

Shave Biopsy

Removal of a raised skin lesion by horizontal slicing with a scalpel at the level of the dermis; used for exophytic lesions such as seborrheic keratoses and superficial basal cell carcinomas.

Shaving Cream Poisoning

Shaving cream poisoning involves toxic or environmental exposure to shaving cream poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Sheehan Syndrome

Postpartum pituitary necrosis resulting from severe hemorrhage and hypotension during delivery. The pituitary gland is enlarged during pregnancy and particularly vulnerable to ischemic injury from blood loss. Presentation varies from fatigue and failure to lactate to panhypopituitarism. Growth hormone and prolactin deficiency occur first.

Sheehan Syndrome (Postpartum Pituitary Necrosis)

Ischemic necrosis of the anterior pituitary gland due to severe postpartum hemorrhage (PPH) and hypotension. The pituitary enlarges during pregnancy and is vulnerable to ischemia from hypotensive episodes.

Alternate Names:

Postpartum Pituitary Necrosis

Shellac Poisoning

Shellac poisoning involves toxic or environmental exposure to shellac poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Shellfish Allergy

An immune reaction to proteins in crustaceans or mollusks that can cause hives, digestive or respiratory symptoms, or anaphylaxis.

Shellfish Poisoning

Illness caused by toxins accumulated in shellfish, producing gastrointestinal, neurologic, or paralytic symptoms.

Shielding

Material placed between a person and a source of radiation to reduce exposure. The material depends on the radiation: lead or tungsten is used for many X-rays and gamma rays, while plastic or acrylic can be useful for beta radiation.

Shift Work Sleep Disorder

A circadian rhythm sleep-wake disorder caused by work schedules that overlap the usual sleep period, producing insomnia, excessive sleepiness, or both.

Shigella

A genus of bacteria causing shigellosis, an acute diarrheal illness that may include fever and bloody stools.

Shigellosis

Shigellosis is intestinal infection caused by Shigella bacteria, producing diarrhea that may be bloody, fever, cramps, and dehydration.

Shin Bone Fever

Shin bone fever refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Shin Splints

Pain along the inner border of the shin, often referring to medial tibial stress syndrome from repetitive loading.

Alternate Names:

Medial Tibial Stress Syndrome

Shin Spot

A shin spot is a nonspecific mark or lesion on the lower leg; its significance depends on appearance, duration, symptoms, injury, and associated disease.

Shingles

Herpes zoster, reactivation of varicella-zoster virus in sensory nerves causing a painful unilateral vesicular rash.

Shingles (Herpes Zoster)

Alternate index label for Herpes Zoster.

Shingles And Pregnancy

Shingles is reactivation of varicella-zoster virus and usually causes a painful one-sided blistering rash. It is not the same as primary chickenpox exposure, but pregnancy requires prompt clinical advice about treatment, pain, and contact with people at risk.

Ship Fever

An historical term for epidemic typhus, a louse-borne infection caused by *Rickettsia prowazekii* and associated with crowding, fever, rash, and severe systemic illness.

Shirodkar Cerclage

A transvaginal cervical cerclage technique that involves dissection of the bladder and rectum to place the suture (Mersilene tape) as high as possible at the level of the internal os. Technically more difficult than McDonald cerclage with higher risk of hemorrhage and bladder injury.

Shivering

Involuntary, rhythmic skeletal muscle contraction generating heat in response to cold exposure or fever, a key thermoregulatory response under hypothalamic (preoptic area) control. Shivering can increase heat production 3-5 fold.

Shock

A life-threatening state of circulatory failure in which inadequate tissue perfusion and oxygen delivery cause cellular and organ dysfunction. Major

mechanisms are hypovolaemia, cardiac pump failure, distributive vasodilation, and obstruction to blood flow; features may include hypotension, tachycardia, altered mental status, cold or abnormal skin perfusion, and reduced urine output.

Shock Lung

An older term for acute respiratory distress syndrome (ARDS), a life-threatening form of lung inflammation in which fluid and injury make it difficult for oxygen to enter the blood.

Shock Therapy

An imprecise term that may refer to electroconvulsive therapy or outdated shock treatments; the specific intervention should be named.

Shock Wave Lithotripsy

Non-invasive procedure using extracorporeal shock waves to fragment renal and ureteral stones. The patient is positioned over a water cushion coupling the shock wave generator to the body. Shock waves are focused on the stone under fluoroscopic or ultrasound guidance.

Shock, Secondary

Circulatory failure occurring as a consequence of another condition, such as infection, trauma, bleeding, heart failure, allergy, or endocrine crisis, with inadequate tissue perfusion.

Shone Complex

A group of congenital left-sided obstructive heart lesions, classically including supravulvular mitral ring, parachute mitral valve, subaortic stenosis, and coarctation of the aorta. The combination and severity vary and may lead to heart failure or systemic outflow obstruction.

Short Arm Of A Chromosome

The p arm, or shorter arm, of a chromosome, named from the French *petit*; cytogenetic bands on this arm locate genes and chromosomal abnormalities.

Short Bones

Small, roughly cube-shaped bones that provide stability and limited movement. The wrist and ankle contain most of the body's short bones.

Short Bowel Syndrome

A condition in which the small intestine is shortened or damaged and cannot absorb enough nutrients from food. It most often occurs after surgery to remove part of the small intestine and may cause diarrhea, weight loss, and nutritional deficiencies.

Short QT Syndrome

A rare inherited primary cardiac channelopathy inherited in an autosomal dominant pattern, caused by gain-of-function mutations in potassium channels (*KCNH2*, *KCNQ1*, *KCNJ2*) that accelerate cardiac repolarization. Characterized on electrocardiogram by a corrected QT interval consistently under 330–340 ms and tall, peaked T waves, conferring a high risk of atrial fibrillation, polymorphic ventricular tachycardia, and sudden cardiac death in young, structurally normal hearts.

Alternate Names:

SQTS

Short Philtrum

A short philtrum is reduced distance between the nose and upper lip and may be an isolated feature or part of a genetic, developmental, or prenatal-exposure syndrome.

Short Stature

Height substantially below the expected range for age and sex, requiring assessment of growth pattern and cause.

Short-Sight

Myopia, a refractive error in which distant objects are focused in front of the retina.

Short-Term Memory

The temporary storage and use of information over seconds to minutes.

Alternate Names:

Working memory

Shortness Of Breath

An uncomfortable awareness of difficult or insufficient breathing. See Dyspnea.

Shotty

Shotty describes small, firm, discrete lymph nodes, often reactive after infection; persistent, enlarging, hard, or generalized nodes require evaluation.

Shoulder

The region and joint connecting the upper limb to the trunk.

Shoulder Arthroscopy

A minimally invasive procedure using a camera and instruments through small incisions to diagnose and treat selected disorders of the shoulder joint and surrounding tissues.

Shoulder Bursitis

Shoulder bursitis is inflammation of a fluid-filled sac around the shoulder, often associated with rotator-cuff disease or repetitive overhead activity. Pain with elevation and lying on the affected side is common; activity modification, therapy, medicines, injection, or treatment of the underlying cause may help.

Alternate Names:

Bursitis Shoulder (Shoulder Bursitis)

Shoulder CT Scan

Computed tomography of the shoulder used to evaluate fractures, bone alignment, arthritis, tumors, infection, and complex bony anatomy.

Shoulder Dislocation

Complete displacement of the upper arm bone from the shoulder socket, usually caused by a fall, collision, or forceful twisting. It causes severe pain, deformity, and inability to move the arm; urgent reduction and assessment for associated fractures or nerve and blood-vessel injury are required.

Shoulder Dystocia

An obstetric emergency where the fetal head delivers but the anterior shoulder is impacted behind the maternal symphysis pubis. Defined as head-to-body delivery time exceeding 60 seconds or need for ancillary maneuvers. Risk factors include macrosomia, diabetes, prior shoulder dystocia, and operative vaginal delivery.

Shoulder Impingement

Pain associated with movement of the shoulder, often involving the rotator-cuff tendons or subacromial tissues. Symptoms include pain with overhead activity and sometimes weakness; assessment and treatment are individualized and may include exercise-based rehabilitation.

Shoulder Instability

Excessive or recurrent movement of the shoulder joint caused by injury, laxity, or altered joint anatomy. Associated lesions may include a Bankart tear, Hill-Sachs lesion, or injury to the joint capsule or ligaments.

Shoulder MRI Scan

Magnetic resonance imaging of the shoulder used to evaluate rotator-cuff tendons, labrum, cartilage, ligaments, muscles, bone marrow, and joint fluid.

Shoulder Pain

Pain in or around the shoulder caused by injury, tendon or bursa disorders, arthritis, nerve disease, or referred pain. Sudden weakness, deformity, chest symptoms, fever, or pain after major trauma requires prompt evaluation.

Shoulder Replacement

A medical procedure, treatment, device, or care topic related to shoulder replacement; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Shulmans Syndrome (Eosinophilic Fasciitis)

Alternate index label for Eosinophilic Fasciitis.

Shunt

A passage that diverts blood, cerebrospinal fluid, urine, or another fluid from one channel to another.

Shunt (Vascular)

A surgically created or implanted pathway that redirects blood or another fluid from one vessel or body space to another.

Alternate Names:

Vascular

Shwachman Syndrome

An inherited multisystem disorder usually involving exocrine pancreatic insufficiency, bone-marrow dysfunction, and skeletal abnormalities. Children may have steatorrhea, poor growth, recurrent infections, neutropenia, anemia, or increased risk of myeloid malignancy.

Shy-Drager Syndrome

Alternate index label; see *Multiple system atrophy*.

SI Units

The International System of Units used for standardized measurements, including meter, kilogram, second, mole, and pascal.

SIADH

Syndrome of Inappropriate Antidiuretic Hormone secretion characterized by hyponatremia (dilutional) with inappropriately concentrated urine (>100 mOsm/kg) and elevated urine sodium (>40 mEq/L). Euvolemic hyponatremia.

Sialadenitis

Inflammation of salivary gland. Acute (bacterial) or chronic (recurrent). Treatment: antibiotics, hydration, gland massage, surgery if chronic.

Sialagogues

Substances or medicines that increase saliva production. They may help some people with dry mouth, but treatment also considers hydration, medicines, oral care, and the cause of reduced saliva.

Sialogram

An imaging study of a salivary gland and its ducts after contrast is introduced through the duct opening, used to assess obstruction, stones, strictures, or chronic inflammation.

Sialography

Contrast imaging of salivary ducts. Evaluates obstruction, stricture, tumors. Rarely performed now.

Sialolithiasis

Formation of a stone in a salivary gland or its duct, most often the submandibular gland. It may cause recurrent swelling and pain, especially during meals, and can lead to infection or obstruction.

Sialorrhea

Excessive salivation. Neurological causes, medications, dental causes. Treatment: hydration, glycopyrrolate, botulinum toxin, surgery.

Sialorrhoea

The excessive pooling and involuntary flow of saliva from the oral cavity (drooling). Most frequently arises not from hypersecretion of salivary glands, but from oral motor dysfunction, impaired swallowing mechanics (dysphagia), or abnormal head posture. Highly prevalent in pediatric patients with cerebral palsy and adult patients with neurodegenerative disorders, such as Parkinson disease, amyotrophic lateral sclerosis, and progressive supranuclear palsy. Chronic sialorrhoea precipitates perioral dermatitis, social isolation, and life-threatening aspiration pneumonia; management options include behavioral swallowing retraining, anticholinergic pharmacotherapy (glycopyrrolate), salivary gland botulinum toxin A injections, or surgical duct relocation.

Sialuria

A rare inherited disorder of sialic-acid metabolism caused by abnormal regulation of the GNE enzyme. Excess free sialic acid accumulates in cells and urine and may cause developmental delay, coarse facial features, hepatosplenomegaly, and variable intellectual disability.

Sibling

A person's brother or sister; family history from siblings can inform genetic and disease risk assessment.

SIBO

Alternate index label; see *Small intestinal bacterial overgrowth*.

Sicca Syndrome

Dry eyes and dry mouth due to decreased tear and saliva production. Primary: Sjogren syndrome. Secondary: associated with RA, SLE, scleroderma, graft-versus-host disease, hepatitis C, sarcoidosis, HIV, medications (anticholinergics, antidepressants). Diagnosis: Schirmer test (<5 mm/5 min), rose bengal staining, salivary gland biopsy.

Sick Building Syndrome

Sick building syndrome describes nonspecific symptoms such as headache, fatigue, eye or throat irritation, cough, dizziness, or nausea associated with time in a building and improving away from it. Ventilation, humidity, pollutants, mold, and other environmental factors should be assessed.

Sick Sinus Syndrome

Sinus node dysfunction causing a spectrum of arrhythmias: sinus bradycardia, sinus arrest, sinoatrial block, and tachycardia-bradycardia syndrome. Most common in elderly patients. Presents with syncope, fatigue, palpitations, and exercise intolerance. Diagnosis: ECG, Holter monitor (pauses >3 seconds).

Alternate Names:

Bradycardia-Tachycardia Syndrome

Sickle Cell

See *Sickle Cell Disease*.

Sickle Cell Anemia

A form of sickle cell disease, usually HbSS, in which red blood cells are easily destroyed and can block small blood vessels. It can cause chronic anemia, severe pain episodes, acute chest syndrome, stroke,

jaundice, and serious infections.

Sickle Cell Anemia (Sickle Cell)

See *Sickle Cell Anemia*.

Sickle Cell Crisis

An acute clinical complication of sickle cell disease, most commonly a vaso-occlusive episode producing ischemic pain. Other acute crises include acute chest syndrome, splenic or hepatic sequestration, and aplastic crisis; treatment depends on the specific complication and severity.

Sickle Cell Dactylitis

Painful swelling of the hands or feet caused by blockage of small blood vessels during a sickle-cell episode, especially in infants and young children. It may be an early sign of sickle-cell disease and requires appropriate clinical evaluation.

Sickle Cell Disease

An inherited group of blood disorders caused by abnormal hemoglobin. Red blood cells can become stiff, curved, and easily destroyed, causing anemia. They may also block small blood vessels, causing episodes of severe pain and damage to organs. Complications can include serious infections, stroke, and acute chest syndrome. Common types include HbSS, HbSC, and sickle beta-thalassemia.

Sickle Cell Test

A laboratory test detecting hemoglobin S or assessing sickling, used in screening and diagnosis of sickle-cell trait or sickle-cell disease; definitive testing identifies hemoglobin variants.

Sickle Cell Trait

An inherited carrier state in which a person has one hemoglobin S allele and one normal beta-globin allele, usually producing both hemoglobin A and hemoglobin S. Most people are asymptomatic, but complications can occur in severe physiologic conditions.

Sickle-Cell Anaemia

See *Sickle Cell Anemia*.

Sickness

A general term for illness or nausea; clinical assessment should identify the specific symptom or disease.

Sickness Motion (Motion Sickness (Sea Sickness, Car Sickness))

Alternate index label for Motion Sickness (Sea Sickness, Car Sickness).

Side Effect

An unintended pharmacological effect of a drug occurring at therapeutic doses, usually predictable from the drug's mechanism of action (a Type A reaction).

Side Effects

Side effects are unintended effects of a medicine or treatment, ranging from mild symptoms to serious toxicity; risk depends on the product, dose, and patient.

Side-Effects

Unintended effects of a medicine or treatment, which may be mild, serious, predictable, or idiosyncratic.

Sideroblastic Anemia

An anemia caused by defective heme synthesis, in which iron accumulates in the mitochondria of developing erythroblasts and forms ring sideroblasts in the bone marrow. Causes may be inherited or acquired, including myelodysplastic syndromes, alcohol, certain medicines, toxins, or vitamin B6 deficiency.

Siderosis

Accumulation of iron in tissues, or lung disease caused by inhalation of iron-containing dust.

SIDS

Sudden infant death syndrome, the sudden unexplained death of an infant younger than one year

after thorough investigation.

Sievert (Sv)

The SI unit used for equivalent and effective dose, incorporating radiation type and, for effective dose, tissue sensitivity. It is used to estimate stochastic radiation risk rather than simply physical energy deposition.

Alternate Names:

Sv

Sight

The ability to see, involving the eyes and visual nervous system.

Sigmoid

Sigmoid means S-shaped; the sigmoid colon is the curved section of large intestine between the descending colon and rectum.

Sigmoid Colon

S-shaped segment from descending colon to rectum in the left iliac fossa, approximately 40-50 cm, with complete mesentery (intraperitoneal, mobile). Most common diverticular disease site from narrow lumen and high pressure. Common volvulus site, particularly in elderly/institutionalized patients. Second most common colorectal cancer site.

Sigmoid Volvulus

Axial twisting of the sigmoid colon around its mesentery, the most common cause of colonic volvulus, typically occurring in elderly, institutionalized, or neuropsychiatric patients with chronic constipation and a redundant sigmoid. Presents with acute large bowel obstruction, abdominal distension, and absent bowel sounds.

Sigmoidoscopy

Endoscopic examination of the rectum and lower colon using a flexible or rigid sigmoidoscope. It can identify inflammation, bleeding, polyps, or tumors and may allow biopsy or treatment.

Sign

Objective evidence of disease. Examples: rash, murmur, edema. Observed by examiner.

Sign Language

Sign language is a complete visual language using hand shape, movement, position, facial expression, and body movement rather than spoken sound.

Signal Peptide

signal peptide is a defined chemical or biologic substance involved in signal peptide and studied in physiology, pharmacology, laboratory medicine, or treatment.

Signal Transduction

The process by which an extracellular signal (hormone, neurotransmitter, growth factor, drug) binding to a cell-surface or intracellular receptor is converted into a specific intracellular response.

Signal-To-Noise Ratio

Ratio of true counts to statistical noise. Improved by longer acquisition, higher activity.

Signature

A signature is a person's identifying mark or electronic confirmation of approval, consent, authorship, or verification.

Signature Strengths

Personal character strengths that a person values and naturally uses.

Signet Ring Cell Carcinoma

A poorly differentiated adenocarcinoma in which intracellular mucin displaces the nucleus to the periphery, most often of the stomach or colon; associated with an aggressive course and diffuse growth pattern.

Signs Of An Asthma Attack

An asthma attack may cause worsening wheeze, cough, chest tightness, breathlessness, rapid breathing, difficulty speaking, or reduced peak flow; severe

symptoms need emergency care.

Sildenafil Citrate

The citrate salt of sildenafil, a phosphodiesterase-5 inhibitor used for erectile dysfunction and pulmonary arterial hypertension.

Silent Heart Attack

A myocardial infarction without recognized pain or typical warning symptoms.

Alternate Names:

Unrecognized myocardial infarction

Silent Ischemia

Inadequate blood flow to heart muscle without noticeable symptoms.

Alternate Names:

Asymptomatic ischemia

Silent Mutation

A DNA sequence change that does not alter the encoded amino acid, usually because of genetic-code redundancy, although it can sometimes affect splicing, RNA stability, or translation.

Silent Sinus Syndrome

Silent sinus syndrome is painless inward collapse of the maxillary sinus and floor of the orbit, often causing facial asymmetry, enophthalmos, or a sunken eye. Imaging confirms the diagnosis, and endoscopic treatment restores sinus ventilation when indicated.

Silent Thyroiditis

Silent thyroiditis is painless autoimmune inflammation of the thyroid that may briefly release excess hormone before causing temporary hypothyroidism and eventual recovery.

Silicate

A salt or ester containing silicon and oxygen, found in minerals and industrial materials; inhalation of some silicate dusts can irritate the lungs or contribute to pneumoconiosis.

Silicon Dioxide

Silicon dioxide is a mineral found in dust and industrial materials; inhalation of crystalline forms can cause silicosis and increase lung-cancer risk.

Silicone

A family of synthetic silicon-containing polymers used in medical devices, implants, sealants, and consumer products; medical risks depend on the product and exposure.

Silicone Elastomer

A flexible, rubber-like silicone material used in implants, tubing, dressings, and other devices; complications may include infection, leakage, allergy, or device failure.

Silicone Implants

Medical implants containing silicone gel or elastomer used for breast reconstruction or augmentation; complications may include rupture, capsular contracture, infection, malposition, or need for revision.

Silicone Mastitis

A granulomatous inflammatory reaction to silicone released from a breast implant or injected silicone. It may cause nodules, pain, lymphadenopathy, or imaging findings that mimic malignancy.

Silicone Oil Hyperoleon

Emulsification of silicone oil tamponade used in vitreoretinal surgery, where the oil breaks into tiny droplets that migrate into the anterior chamber or trabecular meshwork. It can cause elevated intraocular pressure, band keratopathy, and chronic inflammation requiring oil removal and anterior chamber washout.

Silicones

Synthetic polymers used in medical devices, implants, dressings, and other products; adverse effects depend on formulation and site.

Silicosis

A fibrotic lung disease caused by inhalation of respirable crystalline silica dust.

Silk

Silk is a protein fiber used in textiles and some medical sutures or biomaterials; reactions and tissue response depend on the product and processing.

Silk Road Disease (Behcet's Syndrome)

Alternate index label for Behcet's Syndrome.

Siltuximab

A monoclonal antibody that binds interleukin-6 and is used to treat adults with multicentric Castleman disease who are negative for HIV and human herpesvirus 8.

Silver

A metal used in some dressings, creams, and medical devices for antimicrobial activity; excessive exposure can cause permanent blue-gray skin discoloration called argyria.

Silver Bullet

An informal expression for a simple, seemingly definitive solution; it is not a specific medical treatment or diagnosis.

Silver Diamine Fluoride

Liquid that arrests caries and stains cavity black. Used in young children, special needs patients.

Silver Nitrate

A caustic antiseptic used topically to cauterize tissue, control minor bleeding, or treat some wounds; it can burn healthy skin and permanently stain tissue.

Silver Poisoning

Silver poisoning, or argyria, follows excessive silver exposure and can cause permanent blue-gray skin discoloration and, rarely, organ or neurologic effects.

Silver Sulfadiazine

A topical antimicrobial cream formerly used widely

for burns; it may delay healing in some wounds and is not appropriate for everyone, including some newborns and people with sulfonamide allergy.

Silver-Russell Syndrome

A growth disorder usually present from birth, marked by poor growth before and after birth, a small or asymmetrical body, and characteristic facial features. Most cases are linked to changes in gene activity rather than a change in the gene itself.

Silymarin

A mixture of flavonolignans from milk thistle marketed as a dietary supplement; evidence for treating liver disease is limited and product quality varies.

Simeprevir

An older hepatitis C virus NS3/4A protease inhibitor formerly used with other antivirals; it is no longer a usual first-line treatment because better direct-acting regimens are available.

Simethicone

Simethicone is a non-absorbed antifoaming agent that coalesces gas bubbles in the gastrointestinal tract and is used for symptomatic relief of bloating; it does not treat the underlying cause of pain.

Simian Crease

A simian crease is a single transverse crease across the palm, which may be a normal variant or occur with some chromosomal or developmental conditions.

Simian Virus 40

A polyomavirus first identified in monkeys. It has been studied in cancer research, but a clear role in causing human cancer has not been established.

Simmonds Disease

An uncommon form of hypopituitarism caused by inadequate pituitary hormone production, often resulting in fatigue, weight loss, low blood pressure, and impaired endocrine function.

Simmonds' Disease

An older term for hypopituitarism, especially pituitary failure causing wasting and endocrine deficiency.

Simple Febrile Seizure

A common, benign febrile seizure occurring in children between 6 months and 5 years of age during an acute febrile illness (temperature usually exceeding 38 degrees Celsius) without central nervous system infection, electrolyte imbalance, or history of prior afebrile seizures. Characterized by a primary generalized (non-focal) tonic-clonic convulsion lasting less than 15 minutes that does not recur within a 24-hour period or during the same illness. It carries an excellent prognosis with no elevated risk of subsequent neurocognitive deficit or chronic epilepsy compared with the general population.

Alternate Names:

Benign Febrile Seizure; Typical Febrile Seizure

Simple Goiter

A simple goiter is noncancerous enlargement of the thyroid without obvious thyroid-hormone excess or deficiency, caused by iodine imbalance, stimulation, or other factors.

Simple Mastectomy

Simple mastectomy removes the breast tissue and nipple while leaving most chest muscles and usually the underarm lymph nodes.

Simple Prostatectomy

Surgical removal of the enlarged central portion of the prostate to relieve bladder-outlet obstruction from benign prostatic hyperplasia; it does not remove the entire prostate.

Simple Pulmonary Eosinophilia

A biomedical term referring to simple pulmonary eosinophilia; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Simple, Heart-Smart Substitutions

Easy food changes that may support heart health, such as choosing unsaturated oils instead of butter, whole grains instead of refined grains, and less processed or salty food.

Simplexvirus

A genus of herpesviruses that includes herpes simplex virus types 1 and 2, which cause cold sores, genital herpes, and other infections.

Simuliidae

The family of black flies; bites can cause itchy skin reactions, and some species transmit *Onchocerca volvulus*, the parasite that causes river blindness.

Simultanagnosia

Simultanagnosia is inability to perceive more than one object or the overall visual scene at a time, usually from dysfunction of visual-association regions.

Simvastatin

A statin medicine that lowers LDL cholesterol by inhibiting HMG-CoA reductase.

Sin Nombre Virus Infection

Alternate index label; see *Hantavirus pulmonary syndrome*.

Sincalide

A synthetic form of cholecystokinin used during gallbladder imaging to stimulate contraction and measure bile flow; it may cause abdominal pain, nausea, or flushing.

Sindbis Virus

A mosquito-borne virus that can cause fever, rash, and joint pain. It occurs mainly in parts of Africa, Europe, Asia, and Australia.

Sinding-Larsen-Johansson Syndrome

Traction apophysitis of inferior patellar pole in adolescents. Similar to Osgood-Schlatter but at knee. Treatment: rest, activity modification.

Sinecatechins

A topical green-tea catechin ointment used for external genital and perianal warts caused by human papillomavirus; it can cause local irritation and should not be used internally.

Singer'S Nodule

A vocal-fold nodule caused by chronic voice overuse, producing persistent hoarseness.

Single Nucleotide Polymorphism

Single base pair variation occurring in >1% of population. Most common genetic variation. Used in GWAS, forensics, pharmacogenetics.

Single Palmar Crease

A single transverse palmar crease is one crease crossing the palm instead of the usual two; it can be a normal variant or a feature of genetic syndromes.

Single Photon Emission Computed Tomography

Nuclear medicine imaging using gamma-emitting isotopes (technetium-99m most common). Rotating gamma camera acquires images. Applications: myocardial perfusion, bone scans, thyroid imaging, renal scans, brain perfusion. Lower resolution than PET but more widely available and cost-effective.

Single Ventricle

A group of congenital heart defects where one functional ventricle supports both the systemic and pulmonary circulations. Includes tricuspid atresia, mitral atresia, double inlet left ventricle, hypoplastic left heart syndrome, and unbalanced AV canal.

Single-Dose Toxicity

Harm caused by one exposure to a substance or medicine, assessed according to dose, route, timing, and the exposed person's characteristics.

Single-Fiber Electromyography

The most sensitive diagnostic clinical electrophysiological examination for detecting disorders of the neuromuscular junction. Utilizes a specialized fine-

needle electrode with an ultra-small recording surface to record simultaneous action potentials from two individual muscle fibers belonging to the same motor unit. Measures the microscopic variability in neuromuscular transmission time between the two fibers, termed "jitter"; increased jitter and impulse blocking are the electrophysiological hallmarks of impaired neuromuscular safety margins, providing diagnostic confirmation in ocular and mild generalized myasthenia gravis when repetitive nerve stimulation is normal.

Alternate Names:

SFEMG; Single-Fiber EMG

Single-Photon Absorptiometry

An older technique using gamma rays to measure bone density, usually in an arm or peripheral bone.

Alternate Names:

SPA

Single-Photon Emission Computed Tomography

A nuclear medicine imaging technique that detects gamma rays from an injected radiotracer to create cross-sectional images of organ function and blood flow.

Singlet Oxygen

A highly reactive form of oxygen involved in oxidative reactions, immune defenses, photodynamic therapy, and tissue injury from oxidative stress.

Sinoatrial Node

The heart's natural pacemaker in the right atrium that normally initiates each heartbeat.

Alternate Names:

Pacemaker, natural

Sinuatrial Node

The sinoatrial node, a group of specialized cells in the right atrium that normally initiates the heartbeat and sets the cardiac rhythm.

Sinus

A cavity, channel, or tract; in clinical use it com-

monly refers to a paranasal sinus or an abnormal draining tract.

Sinus Arrhythmia

Sinus arrhythmia is normal variation in sinus-heart rate, commonly speeding during inhalation and slowing during exhalation, especially in young people.

Sinus Bradycardia

Sinus bradycardia is a sinus rhythm slower than the expected rate; it can be normal during sleep or athletic conditioning but may also reflect drugs or disease.

Sinus CT Scan

Computed tomography of the paranasal sinuses used to evaluate chronic or complicated sinusitis, anatomic obstruction, trauma, inflammatory disease, or tumors.

Sinus Headache

Pain attributed to a sinus headache is often migraine, but sinus inflammation can cause facial pressure, congestion, reduced smell, and pain that worsens with bending. Fever, purulent discharge, persistent symptoms, or eye or neurologic signs guide evaluation.

Sinus Histiocytosis

Expansion of lymph-node sinuses by histiocytes, often a reactive response to drainage of inflammation, infection, tumor, or tissue breakdown; interpretation depends on associated findings.

Sinus Lift

Procedure to increase bone height in posterior maxilla. Subperiosteal or transalveolar approach. Graft material placed.

Sinus MRI Scan

Magnetic resonance imaging of the paranasal sinuses and surrounding soft tissues used to evaluate tumors, invasive infection, orbital or intracranial extension, and selected inflammatory disease.

Sinus Node

The heart's natural pacemaker, located in the right atrium.

Alternate Names:

Sinoatrial node

Sinus Pain

Pressure or pain around the forehead, cheeks, nose, or eyes caused by sinus inflammation, infection, migraine, or other conditions. Severe headache, eye swelling, vision change, high fever, or neurologic symptoms requires urgent assessment.

Sinus Rhomboidalis

A diamond-shaped depression in the floor of the fourth ventricle of the brain.

Sinus Rhythm

The normal heart rhythm generated by electrical impulses from the sinoatrial node.

Sinus Tachycardia

A heart rhythm originating in the sinoatrial node with a faster-than-normal rate, often due to exercise, fever, pain, anemia, or stress.

Sinus X-Ray

Plain radiographic imaging of the paranasal sinuses; it has limited sensitivity and is often replaced by CT when detailed evaluation is needed.

Sinuses

Air-filled cavities within the skull bones lined by respiratory mucosa, connected to the nasal cavity via ostia.

Sinusitis

Inflammation of the paranasal sinuses, usually causing facial pressure, nasal obstruction, discharge, or reduced smell.

Sinusitis And Sinus Problems

Inflammation or infection of the paranasal sinuses that may cause nasal congestion, discharge, facial pressure, headache, or reduced smell. Symptoms

may also arise from other nasal or facial conditions and require assessment when persistent or severe.

Sinusitis, Acute

Alternate index label; see *Acute sinusitis*.

Sinusitis, Chronic

Alternate index label; see *Chronic sinusitis*.

Sinusoid

A small, highly permeable blood vessel, especially in the liver, spleen, and bone marrow.

Sinusoidal Fetal Heart Rate Pattern

A rare, ominous FHR pattern characterized by a smooth, sine-wave-like undulation with a fixed frequency (2-5 cycles/minute) and absent variability. Strongly associated with severe fetal anemia (Rh isoimmunization, fetomaternal hemorrhage, parvovirus B19), fetal hypoxia, and fetal acidosis.

Siplizumab

An investigational monoclonal antibody targeting CD2 on T cells and natural-killer cells; it has been studied for immune-mediated disease and cancer.

Sipuleucel-T

An autologous cellular immunotherapy used for certain asymptomatic or minimally symptomatic metastatic castration-resistant prostate cancers.

Sirenomelia

A rare congenital malformation in which the lower limbs are partially or completely fused. It is usually associated with severe abnormalities of the genitourinary, gastrointestinal, vascular, and spinal systems and is often fatal in the neonatal period.

Sirolimus

Sirolimus is an mTOR inhibitor used for immunosuppression after transplantation and selected vascular or lymphatic disorders; mouth ulcers, hyperlipidaemia, cytopenias, infection, impaired wound healing, and pneumonitis can occur.

Sisomicin

An aminoglycoside antibiotic related to gentamicin that has activity against some aerobic gram-negative bacteria; use is limited by kidney and hearing toxicity.

Sister Mary Joseph Nodule

A metastatic tumor deposit in or around the umbilicus, usually from an intra-abdominal or pelvic malignancy.

Sister Mary Joseph'S Nodule

A firm, hard nodule in the umbilicus resulting from metastatic malignancy, most commonly from intra-abdominal or pelvic cancers including gastric, ovarian, and colorectal carcinoma. It indicates advanced, disseminated disease with a poor prognosis. Diagnosis is by fine needle aspiration biopsy.

Sitagliptin Phosphate

Sitagliptin phosphate is a DPP-4 inhibitor that prolongs incretin activity and improves glucose-dependent insulin secretion in type 2 diabetes; dose adjustment in kidney disease and rare pancreatitis or severe joint pain are important considerations.

Site-Directed Mutagenesis

A laboratory method that introduces a planned nucleotide change at a selected DNA site to study gene function, protein structure, regulation, or disease variants.

Sitosterolemia

An autosomal-recessive disorder of plant-sterol transport caused by ABCG5 or ABCG8 variants. It can produce premature atherosclerosis, xanthomas, hemolytic anemia, macrothrombocytopenia, and elevated plasma plant sterols.

Sitosterols

Sitosterols are plant sterols that resemble cholesterol and reduce intestinal cholesterol absorption; rare inherited storage disorders cause excessive accumulation.

Sitting Position

An upright or semiupright position in which the patient's torso is elevated and the hips and knees are flexed; it is used for selected shoulder and neurosurgical procedures.

Situational Syncope

A reflex faint triggered by a specific situation such as coughing, urination, defecation, swallowing, laughter, or prolonged standing, caused by transient reduced cerebral perfusion.

Situs Ambiguus

Situs ambiguus is abnormal, inconsistent arrangement of internal organs along the left-right body axis and is often associated with congenital heart and splenic abnormalities.

Situs Inversus

A congenital condition in which the thoracic and abdominal organs are reversed or mirrored from their normal positions. Complete situs inversus (situs inversus totalis) involves all organs.

Situs Inversus Totalis

A congenital condition in which the major visceral organs are mirrored from their normal positions, with the liver on the left, heart on the right, and other viscera similarly reversed.

Sitz Bath

Immersion of the perineal and anal region in warm or prescribed water for symptom relief in conditions such as hemorrhoids, anal fissures, or postpartum discomfort.

Six Fingers Or Toes

Six fingers or toes is polydactyly, a congenital extra-digit difference that may occur alone or as part of a genetic syndrome.

Six-Minute Walk Test

A standardized submaximal functional exercise test used to measure the distance a patient can walk quickly on a flat, hard, straight 30-meter indoor

corridor in a period of 6 minutes (6-minute walk distance, 6MWD). It evaluates global functional exercise capacity and integrating responses of the pulmonary, cardiovascular, and neuromuscular systems; it is widely utilized to assess disease severity, monitor exertional oxygen desaturation, guide supplemental oxygen prescription, and evaluate therapeutic responsiveness in chronic obstructive pulmonary disease, idiopathic pulmonary fibrosis, and pulmonary arterial hypertension.

Alternate Names:

6MWT; Six-Minute Walk; 6-Minute Walk Test

Sixth Disease (Roseola)

Alternate index label for Roseola.

Size Perception

The sensory judgment of an object's dimensions, influenced by visual context, distance, depth cues, touch, and brain processing; neurologic or visual disease can distort it.

Sizofiran

Sizofiran is an investigational beta-glucan immunomodulator studied as an adjunct to cancer treatment.

Sjogren Syndrome

Primary Sjogren syndrome (pSS) is a chronic autoimmune exocrinopathy characterized by lymphocytic infiltration of lacrimal and salivary glands causing dry eyes (keratoconjunctivitis sicca) and dry mouth (xerostomia), with female predominance (9:1) and peak onset 40-60. Secondary SS occurs with RA, SLE, SSc.

Sjogren'S Syndrome

An autoimmune disease that commonly causes dry eyes and dry mouth by damaging tear- and saliva-producing glands.

Alternate Names:

Arthritis Sjogrens Syndrome (Sjogren's Syndrome)

Sjogren-Larsson Syndrome

An inherited disorder causing dry, scaly skin, muscle

stiffness, and problems with movement or development. It results from changes in a gene involved in breaking down certain fats.

Sjögren Syndrome

A systemic autoimmune disease characterized by lymphocytic destruction and dysfunction of lacrimal and salivary glands, producing dry eyes and dry mouth. It may involve multiple organs and increases the risk of salivary-gland lymphoma.

Sjögren's Disease

Alternate index label; see *Sjögren's Syndrome*.

Sjögren's Syndrome

An autoimmune disease causing dry eyes and mouth and sometimes systemic glandular, lung, kidney, nerve, or joint involvement.

Alternate Names:

Sjögren's Disease

SJS

Alternate index label; see *Stevens-Johnson syndrome*.

Skeletal Development

Skeletal development is formation, growth, remodeling, and maturation of bones and cartilage from embryonic life through adulthood.

Skeletal Dysplasia

A skeletal dysplasia is an inherited disorder of bone or cartilage growth causing disproportionate short stature, abnormal limbs, spine disease, or other skeletal features.

Skeletal Limb Abnormalities

Skeletal limb abnormalities are differences in the formation, length, alignment, or structure of an arm or leg caused by genetic, developmental, vascular, injury, or other factors.

Skeletal Muscle

Striated muscle attached to bones and generally under voluntary control.

Skeletal Muscle Tissue

Voluntary striated muscle attached to bones or other structures, composed of contractile fibers that generate movement, posture, heat, and joint stability.

Skeletal Muscles

Muscles attached to bones by tendons that produce movement, maintain posture, and generate heat. They are usually under voluntary control but also support automatic actions such as breathing.

Skeletal Myocytes

Skeletal myocytes are striated muscle cells responsible for voluntary movement, posture, heat production, and stabilization of joints.

Skeletal Survey

A series of radiographs obtained when child abuse is suspected. Includes AP and lateral views of the skull, chest (including oblique views), abdomen, spine, and bilateral upper and lower extremities. Identifies fractures of varying ages, which is highly suggestive of non-accidental trauma.

Skeletal System

The skeletal system consists of bones, cartilage, joints, and supporting ligaments that provide structure, protect organs, enable movement, and store minerals.

Skeleton

The framework of bones and cartilage supporting the body and protecting organs.

Skeleton, Bones Of The

The body's framework of bones and cartilage, providing support, protection, leverage for movement, mineral storage, and marrow-based blood-cell production.

Skeletonization

Final stage of postmortem decomposition where soft tissues, organs, and skin have completely decayed or been scavenged, leaving only clean osseous skeletal remains.

Skene Gland Cyst

A benign retention cyst formed in the anterior vaginal vestibule adjacent to the external urethral meatus, resulting from ductal obstruction and mucus accumulation within the paired paraurethral Skene glands. Usually asymptomatic when small, but can cause dysuria, urinary stream deviation, dyspareunia, or progress to an acute infected paraurethral abscess requiring marsupialization or excision.

Skilled Nursing Facility

Inpatient facility providing skilled nursing and therapy. For recovery after hospitalization.

Skilled Nursing Or Rehabilitation Facilities

Facilities that provide 24-hour nursing care, rehabilitation, medication management, and assistance with daily activities after illness, injury, surgery, or functional decline.

Skin

The body's outer covering, consisting mainly of epidermis and dermis with appendages such as hair and glands.

Skin Lesion

An abnormal area of skin that may result from infection, inflammation, injury, allergy, or a benign or malignant growth.

Skin - Clammy

Skin that feels cool, damp, or unusually sweaty. It may occur with heat, anxiety, pain, low blood sugar, shock, or other illness.

Skin Abscess

A painful, swollen pocket of pus under the skin, usually caused by a bacterial infection. It may need drainage and antibiotics.

Skin Absorption

Movement of a substance through the skin into underlying tissues or the bloodstream. Absorption depends on the substance, skin condition, contact

time, and exposed area.

Skin Aging

Skin aging has intrinsic (chronological) and extrinsic (photoaging) components. Photoaging causes wrinkle formation, telangiectasias, lentigines, leather-like texture, and actinic keratoses. Intrinsic aging causes thinning, wrinkling, and delayed wound healing. Prevention includes sun protection (broad-spectrum SPF 30 or greater daily).

Skin Aging And Appearance

Skin aging is a complex biological process influenced by both intrinsic factors, such as genetics and natural aging, and extrinsic factors, particularly ultraviolet radiation exposure.

Skin Allergies And Contact Dermatitis

Inflammatory skin reactions triggered by exposure to allergens or irritants that come into direct contact with the skin.

Skin Allergy Testing

Skin allergy testing includes prick testing (immediate hypersensitivity, IgE-mediated) and intradermal testing (more sensitive, used for drug and venom allergy). Prick testing introduces allergen extract into the epidermis; wheal and flare reaction at 15 minutes indicates sensitization. Intradermal testing injects dilute allergen into the dermis.

Skin Anatomy Picture Definition Function

Skin consists of epidermis, dermis, and subcutaneous tissue with appendages such as hair, nails, glands, nerves, vessels, and immune cells. It forms a barrier, regulates temperature, senses the environment, prevents fluid loss, and participates in immunity and vitamin D production.

Skin And Aging

Skin aging involves intrinsic and extrinsic changes. Intrinsic aging causes thinning, wrinkling, and fragility. Extrinsic (photo)aging causes wrinkles, telangiectasias, lentigines, and actinic keratoses. Collagen loss, elastin degradation, and decreased

cell turnover contribute to aging skin.

Skin And Environment

Environmental factors significantly affect skin health. Pollution causes oxidative stress, inflammation, and accelerated aging. Particulate matter penetrates skin, generating free radicals. Blue light from screens causes hyperpigmentation, particularly in darker skin tones. Indoor heating and air conditioning cause dryness.

Skin And Hair Changes During Pregnancy

Pregnancy can alter pigmentation, hair growth or shedding, acne, stretch marks, and vascular features through hormonal and circulatory changes; concerning lesions need assessment.

Skin Appendages

Skin structures include hair follicles, sebaceous glands, eccrine sweat glands, and apocrine sweat glands. Hair follicles cycle through anagen (growth), catagen (transition), and telogen (resting) phases. Sebaceous glands produce sebum, an oily substance lubricating skin and hair.

Skin Barrier Function

The skin barrier prevents water loss and protects against environmental insults. The stratum corneum acts as a physical barrier, while the acid mantle (pH 4 to 6) provides chemical defense. Lipids (ceramides, cholesterol, free fatty acids) in the intercellular space form a waterproof barrier.

Skin Biopsy Methods

Skin biopsy methods vary by lesion characteristics. Punch biopsy (2 to 6 mm circular blade) provides full-thickness specimen for most diagnoses. Shave biopsy (razor or scalpel) removes superficial lesions; deep shave (saucerization) includes subcutaneous fat for suspected melanoma.

Skin Biopsy Techniques

Skin biopsy is selected according to the suspected diagnosis and lesion depth. Punch biopsy samples full-thickness skin; shave biopsy samples superficial

lesions; excisional biopsy removes an entire small lesion; and incisional biopsy samples part of a large lesion.

Skin Blushing/Flushing

Temporary reddening or warming of the skin caused by widening of small blood vessels. It may result from heat, emotion, exercise, fever, medicines, or hormone changes.

Skin Cancer

Most common cancer. Types: basal cell, squamous cell, melanoma. Prevention: sun protection. Early detection critical.

Alternate Names:

Cancer, Skin

Skin Cancer Epidemiology

Skin cancers are common and include basal-cell carcinoma, squamous-cell carcinoma, and melanoma. Incidence and mortality vary by tumor type, geography, skin phenotype, ultraviolet exposure, immune status, and access to diagnosis and treatment.

Skin Cancer Overview

Skin cancer includes basal-cell carcinoma, squamous-cell carcinoma, melanoma, and less common tumors. Ultraviolet exposure, immune suppression, genetic susceptibility, and certain lesions increase risk; a changing, bleeding, nonhealing, or unusual lesion needs examination.

Alternate Names:

Melanoma (Skin Cancer Overview)

Skin Cancer Prevention

Prevention focuses on reducing ultraviolet exposure through shade, protective clothing, hats, sunglasses, and appropriately used broad-spectrum sunscreen. Avoiding tanning beds also reduces ultraviolet exposure.

Skin Cancer Screening

Skin cancer screening uses the ABCDE criteria for melanoma, the ugly duckling sign, and total body skin examination. Dermoscopy (dermatoscopy)

magnifies skin lesions revealing structures invisible to the naked eye, improving diagnostic accuracy. Skin self-examination teaches patients to monitor moles for changes.

Skin Cancer Staging

Skin cancer staging guides treatment and prognosis. Melanoma: Breslow thickness (most important prognostic factor), ulceration, mitotic rate, sentinel lymph node status (AJCC 8th edition).

Skin Cancer Surgery

Skin cancer surgical techniques maximize cure while minimizing morbidity. Wide local excision with appropriate margins (BCC: 4 mm, SCC: 4 to 6 mm, melanoma: based on Breslow thickness). Mohs micrographic surgery provides 99 percent cure rate for BCC and 97 percent for SCC with tissue preservation.

Skin Cancer, Melanoma

Alternate index label; see *Melanoma*.

Skin Carcinoma

Skin carcinoma is cancer arising from skin cells, most commonly basal-cell or squamous-cell carcinoma.

Skin Care

Measures that protect and maintain the skin, including gentle cleansing, moisturization, sun protection, wound care, avoidance of irritants, and treatment of identified disease.

Skin Care And Incontinence

Skin protection for people with urine or stool leakage, including gentle cleansing, thorough drying, absorbent products, barrier creams, regular changing, and treatment of the underlying cause.

Skin Color

Skin color refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Skin Erosion

A shallow loss of the epidermis that leaves a moist or crusted surface but does not extend deeply into the dermis, often caused by friction, scratching, blister rupture, or inflammation.

Skin Examination

Systematic inspection and palpation of the skin, hair, nails, and mucous membranes to assess color, lesions, texture, distribution, vascular signs, infection, and cancer warning features.

Skin Findings In Newborns

Common skin changes in newborns, such as milia, newborn rash, peeling, birthmarks, or bluish hands and feet. Most are harmless, but fever, blisters, widespread redness, or a sick appearance needs prompt assessment.

Skin Flaps

Skin flaps transfer tissue with its blood supply. Rotation flaps rotate adjacent tissue into defect. Advancement flaps slide tissue forward. Transposition flaps move tissue across intervening skin. Random pattern flaps rely on dermal plexus; axial pattern flaps have named vascular pedicle.

Skin Graft

Transfer of skin from a donor site to a recipient site to cover a wound. Split-thickness skin graft (STSG): epidermis + partial dermis (0.008-0.018 inches), harvested with dermatome, donor site heals spontaneously (2-3 weeks). Full-thickness skin graft (FTSG): epidermis + full dermis, donor site requires closure, better cosmetic outcome.

Skin Graft, Pedicle

Skin graft, pedicle refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Skin Immune System

The skin immune system (SALT: skin-associated lymphoid tissue) provides immunological surveillance. Langerhans cells (dendritic cells in epidermis) capture antigens and present them to T cells.

Dermal dendritic cells, mast cells, and resident T cells form an innate and adaptive immune network.

Skin Induration

Abnormal hardening or firmness of skin or underlying tissue caused by fibrosis, inflammation, edema, infiltrative disease, infection, or tumor.

Skin Infection

Infection of the skin or underlying soft tissue, commonly caused by bacteria, fungi, or viruses. Redness, warmth, swelling, pain, drainage, or fever may occur; rapidly spreading redness, severe pain, blisters, or systemic illness requires prompt medical assessment.

Skin Integrity

Localized damage to the skin and/or underlying soft tissue caused by prolonged pressure, shear, or friction, most commonly over bony prominences. See pressure injury.

Skin Irritation

Inflammation or discomfort caused by direct contact with an irritant, often producing redness, burning, itching, dryness, or scaling.

Skin Lesion Aspiration

Removal of fluid or cells from a skin lump with a needle for testing or drainage.

Skin Lesion Biopsy

Removal of part or all of a skin lesion for microscopic examination to diagnose infection, inflammatory disease, benign tumor, or skin cancer.

Skin Lesion KOH Exam

Microscopic examination of skin scrapings treated with potassium hydroxide to detect fungal hyphae or yeast in suspected superficial fungal infection.

Skin Lesion Of Blastomycosis

A blastomycosis skin lesion results from dissemination of *Blastomyces* fungal infection and may appear as verrucous or ulcerated lesions with raised borders.

Skin Lesion Removal

A medical procedure, treatment, device, or care topic related to skin lesion removal; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Skin Lumps

Skin lumps are localized bumps or masses caused by cysts, lipomas, inflamed follicles, infections, benign tumors, or less commonly cancer; persistence or change warrants assessment.

Skin Mastocytoma

A localized collection of mast cells in the skin, usually a yellow-brown or reddish plaque or nodule in a child that may swell, itch, or redden when rubbed.

Skin Microbiome

The skin microbiome is the community of microorganisms (bacteria, fungi, viruses) residing on the skin. *Staphylococcus aureus* colonization is increased in atopic dermatitis and psoriasis. *Cutibacterium acnes* plays a role in acne. Diversity is decreased in diseased skin.

Skin Necrosis

Death of skin tissue from ischemia, infection, pressure, toxins, trauma, vasculitis, or severe inflammation, producing discoloration, blisters, ulcers, or black eschar.

Skin Neoplasm

A skin neoplasm is an abnormal growth in the skin that may be benign, precancerous, or malignant.

Skin Nodules

Firm, raised lumps in or under the skin. They may result from inflammation, infection, injury, cysts, or benign or malignant growths.

Skin Of Color

Skin of color (Fitzpatrick IV to VI) has unique dermatologic considerations. It is more prone to hyperpigmentation disorders (post-inflammatory hyperpigmentation, melasma, acanthosis nigricans) and

keloid formation. It is less prone to skin cancer but presents with more advanced disease when it occurs.

Skin Or Nail Culture

Laboratory culture of skin, hair, or nail material to identify bacteria or fungi when infection is suspected; results require clinical correlation.

Skin Pigmentation

Determined by melanin produced by melanocytes in the stratum basale. Eumelanin (brown-black) and pheomelanin (red-yellow) are the two types. Melanin production is stimulated by UV radiation through melanocyte-stimulating hormone (MSH). Fitzpatrick skin type classification (I to VI) describes pigmentation and sun response.

Skin Resurfacing

Procedures that improve skin texture or pigmentation by removing or remodeling superficial skin layers.

Skin Sagging Treatment - Underarms

Treatment for loose skin beneath the upper arms. Options may include exercise, weight management, energy-based procedures, or surgery, depending on the amount of loose skin and the person's goals.

Skin Self-Exam

A skin self-exam is a regular check of the skin for new or changing moles, sores, lumps, or patches that may need medical assessment.

Skin Tag

A skin tag is a benign soft growth of connective tissue, commonly in body folds and associated with friction, age, pregnancy, obesity, or insulin resistance. Removal may be performed for irritation or appearance, but a changing or atypical lesion should be diagnosed first.

Alternate Names:

Acrochordon (Skin Tag)

Skin Tags

Benign fibroepithelial polyps occurring in intert-

triginous areas (neck, axillae, groin, under breasts). They are associated with obesity, insulin resistance, and pregnancy. Skin tags are asymptomatic but may become irritated or bleed. Treatment includes snipping, cryotherapy, electrocautery, and shave excision.

Skin Test

Alternate index label for diagnostic or immunologic skin tests, including prick, patch, and tuberculin testing.

Skin Turgor

The skin's ability to return to its normal position after being gently pinched. Reduced turgor may occur with dehydration, although it is less reliable in older adults.

Skin Writing

Alternate index label; see *Dermatographia* (*Dermatographism*).

Skin, Diseases Of

Conditions affecting skin, hair, nails, glands, or subcutaneous tissue, including infection, inflammation, neoplasia, and genetic disorders.

Skin, Itchy

Alternate index label; see *Itchy skin* (*pruritus*).

Skin-Grafting

Transfer of skin from a donor site to cover a wound or tissue defect.

Skipped Heartbeats

Alternate index label; see *Heart palpitations*.

Skull

The bony framework of the head, protecting the brain and supporting the face.

Skull Fracture

A break in one or more skull bones caused by trauma, potentially associated with intracranial bleeding, cerebrospinal-fluid leakage, or brain injury. linear,

depressed, basilar, or comminuted, with potential for intracranial injury and cerebrospinal fluid leak.

Skull X-Ray

A skull X-ray uses low-dose radiation to show skull bones and may help assess selected fractures, foreign bodies, or bone abnormalities.

Slanted Ear

Slanted ear refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

SLAP Lesion

Superior labrum anterior to posterior tear. Associated with overhead activities, trauma. Treatment: rest, physical therapy, surgical repair if symptomatic.

SLAP Tears

Tears of the superior glenoid labrum extending from anterior to posterior, sometimes involving the biceps tendon attachment. They may cause pain, clicking, or instability, especially with overhead activity.

Slapped Cheek Disease

Alternate index label; see *Parvovirus infection*.

Slapped Cheek Syndrome

Erythema infectiosum, a parvovirus B19 infection causing a slapped-cheek rash and sometimes joint or hematologic complications.

SLE

Systemic lupus erythematosus, a chronic autoimmune disease that can affect skin, joints, kidneys, blood, nervous system, and other organs.

Sleep

A recurring state of reduced awareness and responsiveness with characteristic brain and body physiology.

Alternate Names:

Disorders Sleep (Sleep)

Sleep (Geriatric)

Sleep patterns and sleep problems in older adults, including insomnia, sleep apnea, circadian changes, and medication-related disturbances.

Alternate Names:

Geriatric

Sleep And Sleep Disorders In Children And Teens

Children and teens need age-appropriate sleep, and disorders may involve insomnia, circadian delay, sleep apnea, parasomnias, restless legs, or insufficient opportunity. Snoring, pauses in breathing, unusual daytime sleepiness, mood change, or poor school performance warrants evaluation.

Alternate Names:

Sleep Apnea

Repeated pauses or reductions in breathing during sleep. It may be obstructive, central, or mixed and can disturb sleep and oxygenation.

Sleep Apnea (Dental)

Oral appliance for obstructive sleep apnea. Mandibular advancement device. Indications: mild to moderate OSA, CPAP intolerance.

Alternate Names:

Dental

Sleep Apnea (Obstructive Sleep Apnea - OSA)

A sleep disorder characterized by repeated episodes of partial or complete upper airway obstruction during sleep (obstructive sleep apnea—OSA, most common) or cessation of respiratory drive (central sleep apnea—CSA), causing intermittent hypoxia, hypercapnia, and sleep fragmentation.

Alternate Names:

Obstructive Sleep Apnea - OSA

Sleep Apnea, Central

Alternate index label; see *Central sleep apnea*.

Sleep Apnea, Obstructive

Alternate index label; see *Obstructive sleep apnea*.

Sleep Apnoea

Repeated pauses or reductions in breathing during sleep, most commonly because of upper-airway obstruction or less commonly inadequate respiratory drive.

Sleep Apnoeas

Repeated pauses or reductions in breathing during sleep, caused by airway obstruction, impaired respiratory drive, or both.

Sleep Architecture

The pattern and organization of sleep stages across a sleep period.

Sleep Bruxism

Repeated grinding or clenching of the teeth during sleep, which may cause tooth wear, jaw pain, or headaches.

Sleep Debt

The accumulated difference between the amount of sleep a person needs and the amount obtained over time; it may impair alertness, mood, cognition, and performance.

Sleep Diary

A daily record of sleep and wake times, awakenings, naps, medicines, caffeine, and related factors used to assess sleep patterns and treatment response.

Sleep Disorder Periodic Limb Movement (Periodic Limb Movement Disorder)

Alternate index label for Periodic Limb Movement Disorder.

Sleep Disorders

Conditions affecting quality, timing, and amount of sleep; includes insomnia, sleep apnea, narcolepsy, restless legs syndrome, and parasomnias.

Sleep Disorders In Older Adults

Sleep problems in later life, including insomnia, sleep apnea, restless legs, and circadian changes; medication effects, pain, depression, and medical illness should be assessed.

Sleep Efficiency

The percentage of time spent asleep while in bed, calculated as total sleep time divided by time in bed and multiplied by 100.

Sleep Fragmentation

Repeated brief awakenings or arousals that interrupt otherwise continuous sleep and may reduce restorative sleep.

Sleep Hygiene

A set of behavioral and environmental practices designed to promote consistent, uninterrupted, and restorative sleep. Key elements include maintaining a regular sleep schedule, creating a cool, dark, and quiet sleep environment, avoiding caffeine and electronic screens before bedtime, and engaging in relaxing pre-sleep routines.

Sleep Medications And Supplements

Pharmacological agents used to treat insomnia and other sleep disorders when behavioral interventions alone are insufficient.

Sleep Onset Latency

The time required to fall asleep after attempting to sleep; prolonged latency may occur with insomnia, anxiety, circadian misalignment, stimulants, or another sleep disorder.

Sleep Paralysis

Temporary inability to move or speak when falling asleep or waking, while awareness is partly or fully preserved. Episodes usually last seconds to minutes and may include frightening hallucinations. Recurrent episodes can be associated with sleep deprivation, irregular sleep, narcolepsy, or other sleep disorders.

Sleep Related Breathing Disorders

Sleep-related breathing disorders cause abnormal ventilation or oxygenation during sleep, including obstructive and central sleep apnea and sleep-related hypoventilation. Snoring, witnessed pauses, gasping, morning headache, or daytime sleepiness may prompt sleep testing and treatment.

Alternate Names:

Breathing Disorders Sleep Related (Sleep Related Breathing Disorders)

Sleep Spindles

Brief bursts of rhythmic brain activity during stage 2 non-REM sleep.

Sleep Studies (Polysomnography)

Sleep tests, most commonly polysomnography, are diagnostic studies used to evaluate sleep architecture and identify sleep disorders. Polysomnography records brain waves, eye movements, heart rate, breathing patterns, blood oxygen levels, and limb movements throughout the night in a controlled laboratory setting.

Alternate Names:

Polysomnography

Sleep, Disorders Of

Conditions that alter sleep quantity, timing, quality, breathing, movement, or behavior.

Sleep-Related Hypoventilation

Inadequate ventilation during sleep causing elevated carbon dioxide, oxygen desaturation, or both. Causes include obesity, neuromuscular disease, chest-wall disorders, chronic lung disease, medications, and central respiratory-control abnormalities.

Sleep-Related Hypoxemia

Abnormally low blood oxygen during sleep that may occur with sleep-disordered breathing, lung disease, neuromuscular weakness, hypoventilation, or other medical conditions.

Sleepiness, Daytime

Alternate index label; see *Narcolepsy*.

Sleeping Sickness

African trypanosomiasis, a tsetse-fly-borne infection caused by *Trypanosoma brucei* that can progress to neurologic disease.

Sleeplessness

Alternate index label; see *Insomnia*.

Sleepwalking

Recurrent walking or other complex behavior during sleep, usually with little memory afterward. It is more common in children and may be triggered by sleep deprivation, stress, fever, medicines, or other sleep disorders; injury prevention is important.

Sleepwalking And Children

Sleepwalking in children is usually benign and often resolves with age; regular sleep, hazard prevention, and evaluation of frequent, violent, injurious, or atypical episodes are important.

Sleeve Gastrectomy

Removal of major stomach, leaving sleeve. Restrictive procedure. Lower malabsorption risk than bypass. Effective weight loss.

Sleeve Lobectomy

A complex, parenchyma-sparing thoracic surgical procedure for centrally located bronchial tumors (such as non-small cell lung cancer or bronchial carcinoid) invading the main stem bronchus. The affected lung lobe is resected together with a circumferential cuff or sleeve of the adjacent main or intermediate bronchus, followed by an end-to-end bronchial anastomosis reconstructive procedure between the remaining distal bronchial tree and the main stem bronchus, thereby achieving complete oncologic resection while sparing the patient a complete pneumonectomy and preserving long-term pulmonary function.

Alternate Names:

Bronchial Sleeve Resection; Sleeve Resection of the

Lung

Sliding Filament

The mechanism of muscle contraction in which actin filaments slide past myosin filaments, shortening the sarcomere without changing filament lengths. Myosin cross-bridges bind actin, undergo a power stroke using ATP hydrolysis, detach, and reattach to new actin sites in a cyclic manner.

Sliding Hiatal Hernia

A sliding hiatal hernia occurs when the stomach and gastroesophageal junction move upward through the diaphragm, often contributing to reflux.

Sling

A supportive device or surgically placed strip used to support an injured limb or urinary structures.

Slipped Capital Femoral Epiphysis

Displacement of the femoral head relative to the femoral neck through the growth plate (physis) in adolescents. Most common hip disorder in adolescents (10-16 years). Risk factors: obesity, male sex, African American, and endocrine disorders (hypothyroidism, growth hormone deficiency).

Slipped Disk

A spinal disk that bulges or herniates through a tear in its outer layer.

Alternate Names:

Herniated disk

Slipped Vertebra

Forward displacement of one vertebra relative to the one below it.

Alternate Names:

Spondylolisthesis

Slipping Rib Syndrome

Pain caused by excessive movement of the lower ribs, usually from loosened cartilage. The pain may be sharp and can sometimes be reproduced by pressing or moving the rib.

Slit Lamp

An instrument using magnification and a narrow beam of light to examine the eye.

Alternate Names:

Biomicroscope

Slit Lamp Biomicroscopy

A fundamental ophthalmological examination technique that uses a high-intensity, adjustable slit beam of light combined with a binocular microscope to provide a magnified, stereoscopic view of the anterior segment and, with the use of auxiliary lenses, the posterior segment of the eye.

Slit-Lamp Biomicroscope

An instrument that provides a magnified, illuminated view of the eye, especially the cornea, anterior chamber, iris, lens, and conjunctiva.

Slit-Lamp Exam

A focused eye examination using a microscope and narrow beam of light to inspect the eyelids, conjunctiva, cornea, anterior chamber, iris, lens, and selected posterior structures.

SLL

Alternate index label; see *Small lymphocytic lymphoma*.

Slough

Dead or detached tissue separating from a wound, ulcer, or inflamed surface.

Slow Heart Rate - Bradycardia

A heart rate slower than expected, often below 60 beats per minute in an adult. It may be normal in some people but can cause dizziness, fainting, tiredness, or shortness of breath when it reduces blood flow.

Slow Heartbeat

Alternate index label; see *Bradycardia*.

Slow-Twitch Fiber

A muscle fiber adapted for sustained, aerobic activ-

ity and fatigue resistance.

Alternate Names:

Type I muscle fiber

Slow-Wave Sleep

Deep non-REM sleep characterized by slow brain waves and reduced responsiveness.

Alternate Names:

Deep sleep

Small Airway Disease

Disease affecting airways with diameter < 2 mm (bronchioles). It is the earliest site of damage in COPD and is often subclinical until advanced. It is characterized by narrowing, mucus plugging, and inflammation of small airways. PFTs may show air trapping (increased RV/TLC) before FEV1 decline.

Small Bowel Adenocarcinoma

A malignant gland-forming epithelial tumor of the small intestine. It is uncommon and may cause occult bleeding, obstruction, abdominal pain, or weight loss; risk is increased in selected inherited and inflammatory disorders.

Small Bowel Bacterial Overgrowth

Excessive or abnormal bacterial colonization of the small intestine, now often termed intestinal methanogen or bacterial overgrowth, which can cause bloating, diarrhea, malabsorption, and nutrient deficiency.

Small Bowel Follow-Through

Extended upper GI series tracking barium through small bowel.

Small Bowel Obstruction

Partial or complete blockage of the small intestine. Common causes include adhesions, hernias, tumors, and inflammatory narrowing. Symptoms may include abdominal pain, vomiting, distension, and inability to pass stool or gas.

Small Bowel Prolapse

Alternate index label; see *Small bowel prolapse (enterocele)*.

Small Bowel Resection

Surgical removal of a diseased segment of the small intestine followed by reconnection or stoma formation when needed.

Small Bowel Series

X-ray imaging of small intestine using barium. Evaluates Crohn disease, tumors, obstructions.

Small Bowel Tissue Smear/Biopsy

Microscopic or histologic examination of a small-intestinal tissue sample to evaluate malabsorption, celiac disease, inflammation, infection, or neoplasm.

Small Calorie (Calorie)

A small calorie is a unit of heat energy; food labels usually use a kilocalorie, written Calorie with a capital C, equal to 1,000 small calories.

Small Cell Carcinoma

A high-grade neuroendocrine carcinoma, most often of the lung, strongly associated with smoking. Highly responsive initially to chemotherapy and radiation but prone to rapid recurrence and early brain metastases.

Small Cell Carcinoma Of The Ovary, Hypercalcemic Type

A rare, highly aggressive ovarian tumor in young patients, often associated with alterations in SMARCA4 and sometimes presenting with hypercalcemia.

Small Cell Lung Cancer

Aggressive lung cancer, strongly associated with smoking. Early metastasis, responds to chemotherapy.

Small Eye

Small eye refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Small Fiber Neuropathy

Damage to small sensory and autonomic nerve fibers, often causing burning pain, tingling, altered temperature or pain sensation, abnormal sweating, or gastrointestinal and cardiovascular symptoms. Diabetes, autoimmune disease, vitamin deficiency, toxins, and inherited disorders may contribute; evaluation seeks an underlying cause.

Small For Gestational Age

Birth weight or size below the expected range for gestational age, caused by constitutional small size, placental insufficiency, infection, genetic disease, or maternal factors.

Small Intestinal Bacterial Overgrowth

An abnormal increase or change in bacteria within the small intestine, often causing bloating, abdominal pain, nausea, diarrhea, fullness, weight loss, or nutrient deficiency. Causes include impaired intestinal movement, structural problems, surgery, and some systemic diseases; persistent symptoms require medical evaluation.

Small Intestinal Bacterial Overgrowth

Alternate index label for SIBO.

Small Intestinal Bacterial Overgrowth (Small Intestinal Bacterial Overgrowth)

Alternate index label for Small Intestinal Bacterial Overgrowth.

Small Intestinal Ischemia And Infarction

Inadequate blood supply to the small intestine that can progress to irreversible tissue death, perforation, sepsis, and shock.

Small Intestine

Longest GI segment (approximately 6 meters) from pylorus to ileocecal valve. Primary nutrient digestion and absorption site. Three segments: duodenum (25 cm, chyme/bile/pancreatic enzymes), jejunum (2.5 m, prominent folds/villi for absorption), ileum (3.5 m, Peyer's patches for immunity).

Small Intestine Aspirate And Culture

Collection and culture of fluid aspirated from the small intestine to evaluate suspected bacterial overgrowth or selected infections; the procedure is invasive and results require clinical correlation.

Small Intestine Prolapse

Alternate index label; see *Small bowel prolapse (enterocele)*.

Small Round Blue Cell Tumor

Small round blue cell tumor refers to a defined body structure, tissue, or physiologic process; its normal location and function provide the basis for interpreting related disease or injury.

Small Vessel Disease

Disease affecting small blood vessels, which can impair blood flow and contribute to brain, heart, kidney, or other organ problems.

Alternate Names:

Coronary Microvascular Disease

Small-Bowel Transplantation

Surgical replacement of a diseased or nonfunctioning small intestine with donor bowel.

Small-Bowel Volvulus

Twisting of the small intestine around its mesentery, causing mechanical obstruction and potentially compromising mesenteric blood flow. It may be primary (idiopathic, more common in developing countries) or secondary to adhesions, malrotation, hernias, or intraluminal tumors.

Small-Cell Carcinoma

An aggressive neuroendocrine cancer, most commonly arising in the lung.

Smallpox

A severe contagious disease caused by variola virus; it has been eradicated globally.

SMART

A goal-setting framework emphasizing specific, mea-

surable, achievable, realistic, and time-bound goals.

Alternate Names:

SMART goals

Smashed Fingers

A smashed finger may injure skin, nail, bone, tendon, or nerve; deformity, severe pain, numbness, impaired movement, or blood under the nail may require urgent evaluation.

Smear Of Duodenal Fluid Aspirate

A smear of duodenal fluid aspirate examines fluid from the first part of the small intestine for selected parasites or other microscopic findings.

Smegma

A collection of shed cells, oils, and moisture around the genital folds, which may promote irritation if hygiene is poor.

Smell

The sense of detecting airborne chemicals through olfactory receptors.

Smell - Impaired

Reduced or absent ability to detect odors, called hyposmia or anosmia, caused by nasal obstruction, viral injury, head trauma, neurodegeneration, medications, or other disease.

SMILE

Small incision lenticule extraction, commonly known as SMILE, is a minimally invasive refractive surgical procedure that corrects myopia and myopic astigmatism by creating and removing a thin, disc-shaped piece of corneal stroma called a lenticule through a small incision, without creating a corneal flap.

Smile Design

Planning smile based on facial features, gum display, tooth proportions. Digital smile design software available.

Smith-Lemli-Opitz Syndrome

An autosomal recessive disorder of cholesterol

metabolism caused by mutations in DHCR7, encoding 7-dehydrocholesterol reductase, the final enzyme in the cholesterol biosynthetic pathway.

Smith-Magenis Syndrome

A genetic neurodevelopmental disorder, usually involving deletion of 17p11.2 or an RAI1 variant. It causes developmental delay, intellectual disability, distinctive facial features, sleep disturbance, behavioral dysregulation, and variable congenital abnormalities.

Smoke Evacuator

A device that captures and filters airborne surgical plume generated by electrosurgery, laser use, or ultrasonic energy. Evacuation reduces exposure to particulate matter, toxic gases, and potentially infectious biologic material.

Smoke Inhalation

Injury from breathing smoke, hot gases, or combustion products, potentially causing airway burns, poisoning, and lung injury.

Smoke Inhalation Injury

Pulmonary injury caused by inhalation of toxic products of combustion.

Smoke Secondhand (Secondhand Smoke)

Alternate index label for Secondhand Smoke.

Smokeless Tobacco

Smokeless tobacco delivers nicotine and causes dependence, oral lesions, gum recession, tooth loss, cardiovascular risk, and cancers of the mouth, throat, and pancreas. Quitting reduces harm, and counseling plus approved medicines improve cessation success.

Alternate Names:

Tobacco Chewing (Smokeless Tobacco)

Smokers Keratosis (Leukoplakia)

Alternate index label for Leukoplakia.

Smokers Lung (Smoker'S Lung:

Pathology Photo Essay)

Alternate index label for Smoker's Lung: Pathology Photo Essay.

Smoking

Inhalation of smoke from burning tobacco or other substances, associated with addiction, cancer, cardiovascular disease, and lung disease.

Smoking And Asthma

Smoking irritates and inflames airways, worsens asthma control, increases attacks, and reduces treatment response; stopping smoking and avoiding secondhand smoke improve outcomes.

Smoking And COPD

Smoking is the major preventable cause of COPD and accelerates lung-function loss; quitting reduces progression and exacerbations even after COPD is established.

Smoking And Surgery

A medical procedure, treatment, device, or care topic related to smoking and surgery; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Smoking Cessation

The process of stopping tobacco or nicotine use. It may involve counseling, behavioral support, medicines, or nicotine replacement.

Smoking Cessation Medications

Medicines that reduce nicotine withdrawal and cravings, including nicotine replacement, bupropion, and varenicline, used with behavioral support and individualized for health risks and preferences.

Smoldering Multiple Myeloma

An asymptomatic plasma-cell disorder with increased clonal plasma cells or monoclonal protein but no myeloma-defining organ damage; it requires risk-based monitoring for progression to active multiple myeloma.

Alternate Names:

Voluntary muscles

Smooth Muscle

Involuntary muscle in the walls of hollow organs, blood vessels, and airways.

Smothering

Mechanical obstruction of the external nose and mouth that prevents effective ventilation. It may be caused by a covering, compression, or an obstructing soft material; findings are often nonspecific and must be interpreted with the scene, history, and complete examination.

Snacking When You Have Diabetes

Planning snacks for diabetes by considering hunger, glucose pattern, activity, carbohydrate amount, medicine timing, and hypoglycemia risk rather than relying on unrestricted grazing.

Snacks And Sweetened Drinks - Children

Frequent sugary snacks and drinks increase children's risk of tooth decay and excess weight; water, nutritious foods, portions, and regular meals are healthier defaults.

Snacks For Adults

Healthy adult snacks emphasize nutrient-dense foods such as fruit, vegetables, legumes, nuts, yogurt, or whole grains while limiting excess sugar, salt, and highly processed calories.

Snake Bite

Snake bites can cause puncture wounds, tissue injury, bleeding, paralysis, or venom-related effects on clotting, nerves, muscles, kidneys, or the heart. Call emergency services, keep the person still, and do not cut, suck, ice, or tourniquet the wound.

Alternate Names:

Bite Snake (Snake Bite)

Snake Bites

Snake bites involves toxic or environmental exposure to snake bites; dose, route, timing, symptoms, and organ involvement determine urgency and treat-

ment.

Snake-Bite

A wound from a snake that may cause local injury, envenomation, infection, or systemic toxicity.

Snapping Finger

Trigger finger, painful catching or locking of a finger caused by stenosing inflammation around a flexor tendon.

Snapping Hip Syndrome

Hip pain/clicking from tendon snapping over bony prominence. Internal (iliopsoas), external (IT band), intra-articular (labral tear). Treatment: physical therapy, injection, surgery if refractory.

Sneddon Syndrome

A non-inflammatory thrombotic vasculopathy characterized by livedo racemosa and recurrent ischemic cerebrovascular events. Headache, hypertension, cognitive impairment, and cardiac-valve abnormalities may occur; evaluation addresses vascular risk and autoimmune or thrombophilic causes.

Sneezing

A forceful reflex expulsion of air through the nose and mouth, often triggered by nasal irritation.

Snellen Chart

A chart of letters or symbols used to measure visual acuity at a specified distance.

Snoring

Noisy breathing during sleep caused by vibration of narrowed upper-airway tissues.

Snoring - Adults

Noisy breathing during sleep caused by vibration of relaxed tissues in the nose, mouth, or throat; loud habitual snoring with witnessed pauses or daytime sleepiness may indicate obstructive sleep apnea.

Snoring And Sleep Apnea

Snoring is the sound produced by vibrating soft tis-

sues in the upper airway during sleep, often resulting from partial airway obstruction. Sleep apnea is a more serious condition characterized by repeated episodes of complete or partial airway collapse during sleep, leading to intermittent hypoxia, frequent arousal, and fragmented sleep.

Snort

A snort is a forceful nasal or respiratory sound; it may occur with congestion, sleep-disordered breathing, irritation, or inhalation of a substance.

Snot

Snot is an informal term for nasal mucus, which traps particles and helps protect the respiratory lining; its appearance changes with infection, allergy, and irritation.

Snow Blindness

Photokeratitis, painful temporary corneal injury from intense ultraviolet exposure, often reflected by snow or ice.

SNRIs

Serotonin-norepinephrine reuptake inhibitors are a class of antidepressant drugs that inhibit the reuptake of both serotonin and norepinephrine from the synaptic cleft, increasing both serotonergic and noradrenergic neurotransmission. Common SNRIs include venlafaxine, duloxetine, desvenlafaxine, and levomilnacipran.

Snuff

Snuff is finely ground tobacco inhaled through the nose or placed in the mouth and can cause nicotine dependence, oral disease, and cardiovascular harm.

Snuffles

A runny or congested nose; in newborns, persistent snuffles can be a sign of congenital syphilis.

Social And Psychological Causes Of Alcoholism

Alcohol-use disorder reflects interactions among genetics, brain reward pathways, stress, trauma, men-

tal illness, social environment, and alcohol availability—not a moral failure. Treatment may include withdrawal care, counseling, peer support, and medicines that reduce drinking or relapse.

Social Anxiety Disorder

Marked fear or anxiety about one or more social situations in which the individual may be scrutinized by others. The individual fears acting in a way or showing anxiety symptoms that will be negatively evaluated.

Social Anxiety Disorder (Social Phobia)

Intense fear and avoidance of social or performance situations because of concern about embarrassment, judgment, or scrutiny.

Alternate Names:

Phobia, Social

Social Isolation

Limited contact or interaction with other people. Persistent isolation may adversely affect mental and physical health.

Social Medicine

The study and practice of how social conditions affect health and healthcare.

Social Worker

Healthcare professional providing psychosocial support, resource coordination, and counseling for patients and families throughout the transplant process.

Sociophobia

An older term for social anxiety disorder, involving excessive fear of scrutiny, embarrassment, humiliation, or negative evaluation in social or performance situations.

Sodium

The main extracellular cation, essential for fluid balance, nerve function, and muscle activity.

Sodium Aurothiomalate

An injectable gold compound formerly used to treat rheumatoid arthritis and now rarely used.

Sodium Bicarbonate Therapy

Alkalinizing agent for metabolic acidosis in CKD. Target serum bicarbonate greater than 22 mEq/L. Slows CKD progression and improves muscle wasting. Dose titrated to serum levels.

Sodium Bisulfate Poisoning

Sodium bisulfate poisoning involves toxic or environmental exposure to sodium bisulfate poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Sodium Blood Test

A blood test measuring serum sodium, a major extracellular electrolyte involved in water balance and nerve and muscle function; results are interpreted with fluid status and other electrolytes.

Sodium Carbonate Poisoning

Sodium carbonate poisoning involves toxic or environmental exposure to sodium carbonate poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Sodium Chloride

The salt sodium chloride; sterile solutions are used for fluid replacement, irrigation, and drug dilution.

Sodium Cromoglycate

A mast-cell stabilizer used in selected allergy and asthma treatments; also called cromolyn.

Sodium Diatrizoate

An iodinated contrast medium used historically for radiographic imaging.

Sodium Hydroxide Poisoning

Sodium hydroxide poisoning involves toxic or environmental exposure to sodium hydroxide poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Sodium Hypochlorite

A chlorine-containing disinfectant and bleach; concentrated exposure can cause corrosive injury.

Sodium Hypochlorite Poisoning

Sodium hypochlorite poisoning involves toxic or environmental exposure to sodium hypochlorite poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Sodium In Diet

Dietary sodium helps regulate fluid balance and nerve and muscle function, but excess intake can raise blood pressure; processed foods are a major source and intake should follow individual health advice.

Sodium Low Levels In The Blood (Hyponatremia)

Alternate index label for Hyponatremia.

Sodium Polystyrene Sulfonate

Potassium-binding resin for hyperkalemia. Risk: colonic necrosis, especially with sorbitol.

Sodium Urine Test

A urine test measuring sodium excretion, used with blood tests and clinical findings to assess volume status, kidney handling of sodium, and causes of hyponatremia or hypertension.

Sodium Valproate

An anticonvulsant and mood stabilizer; it carries important pregnancy and reproductive risks and requires specialist prescribing.

Sodomy

A legal or historical term for certain sexual acts; it is not a medical diagnosis and should be replaced by precise, nonjudgmental language.

Soft Markers For Aneuploidy

Sonographic findings in the second trimester that are common in normal fetuses but have a statistically higher prevalence in aneuploid fetuses (especially

trisomy 21).

Soft Tissue Augmentation

Restoration or enhancement of volume and contour in skin or underlying soft tissue using fillers, fat transfer, implants, or other reconstructive techniques.

Soft Tissue Injury

Damage to muscles, tendons, ligaments, fascia, skin, or other non-bony tissues, often caused by trauma or overuse.

Soft Tissue Rheumatism

Pain conditions affecting muscles, tendons, ligaments, bursae. Examples: frozen shoulder, lateral epicondylitis, rotator cuff tendinopathy. Treatment: rest, physiotherapy, injections, surgery if refractory.

Soft Tissue Sarcoma

A group of malignant tumors arising in muscle, fat, fibrous tissue, blood vessels, or other connective tissues. Treatment depends on the subtype, location, stage, and relationship to nearby structures.

Alternate Names:

Cancer, Soft Tissue Sarcoma Sarcoma, Soft Tissue

Soft Tissue Tumors

Lipoma: most common benign soft tissue tumor, subcutaneous, encapsulated. Liposarcoma: most common malignant STS in adults, deep, subtypes (well-differentiated/dedifferentiated, myxoid/round cell, pleomorphic). Synovial sarcoma: t(X;18), SS18-SSX, young adults, deep extremity, "triple sign" on MRI.

Soft-Tissue Calcinosis

The deposition of insoluble calcium phosphate salts in soft tissues such as skin, subcutaneous tissue, muscles, and tendons, which may be metastatic (due to hypercalcaemia or hyperphosphataemia), dystrophic (in damaged or devitalised tissue), or idiopathic.

Soil Contamination

Environmental toxicology. Heavy metals, pesticides, industrial waste. Remediation: excavation, bioremediation, containment.

Solar Keratosis

Alternate index label; see *Actinic keratosis*.

Solar Keratosis (Actinic Keratosis)

Alternate index label for Actinic Keratosis.

Solar Lentigines

Alternate index label; see *Age spots (liver spots)*.

Solar Plexus

A network of autonomic nerves in the upper abdomen, centered around the celiac plexus.

Solar Urticaria

A rare form of inducible urticaria triggered by ultraviolet or visible light, producing pruritic erythema and wheals within minutes on exposed skin. Symptoms usually improve after removal from the light source. Diagnosis may involve phototesting.

Solder Poisoning

Solder poisoning involves toxic or environmental exposure to solder poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Sole Sweating Excessive (Hyperhidrosis)

Alternate index label for Hyperhidrosis.

Soleus Muscle

Broad flat deep posterior leg muscle from posterior tibia, fibula, and tendinous arch to the calcaneus via the Achilles tendon. Innervated by tibial nerve (S1-S2). Powerful ankle plantarflexor, the primary anti-gravity muscle for upright posture. One-joint muscle.

Solitary Fibrous Tumor

A solitary fibrous tumor is a usually slow-growing growth of connective tissue that can arise in the

pleura, meninges, or other sites and may rarely behave aggressively.

Solitary Pulmonary Nodule

Single, well-circumscribed lesion < 3 cm surrounded by normal lung. Differential includes granuloma, hamartoma, and carcinoma. PET-CT and tissue sampling help determine etiology.

Solution

A homogeneous mixture in which one or more substances are dissolved in a solvent; medical solutions include saline and antiseptics.

Solvent Abuse (Misuse)

Harmful inhalation or ingestion of volatile solvents, which can cause neurologic, cardiac, liver, kidney, or respiratory injury.

Solvent Exposure

Contact with organic solvents by inhalation, ingestion, or skin absorption. Acute exposure may cause irritation or central nervous system effects; repeated exposure may damage organs.

Somatic

Relating to the body, especially the body wall and musculoskeletal system, or to physical symptoms.

Somatic Mutation

A genetic change acquired by a body cell during life rather than inherited from a parent. It is passed to descendants of that cell and may contribute to cancer, but it usually is not passed to children.

Somatic Symptom Disorder

Characterized by one or more somatic symptoms that are distressing or result in significant disruption of daily life, accompanied by excessive thoughts, feelings, or behaviors related to the somatic symptoms.

Somatic Variant

A DNA variant acquired after conception in a non-germline cell lineage. It is not usually transmitted

to offspring and may produce mosaic disease or contribute to cancer when it affects growth-control pathways.

Somatization

The experience or expression of psychological distress through physical symptoms such as pain, fatigue, or stomach problems. The symptoms are real and not intentional; assessment should consider both physical and psychological causes.

Somatoform Diseases

An obsolete term for disorders with prominent physical symptoms or health concerns associated with psychological distress.

Somatostatin

A hormone that inhibits secretion of growth hormone, insulin, glucagon, and several gastrointestinal hormones.

Somatostatin Analog

A class of synthetic peptides that mimic the actions of native somatostatin, a hormone that inhibits the secretion of growth hormone, thyroid-stimulating hormone, insulin, glucagon, gastrin, and other GI hormones.

Somatostatinoma

Rare pancreatic or duodenal tumor producing somatostatin causing diabetes, gallstones, steatorrhea, and hypochlorhydria from suppression of insulin, gallbladder contraction, pancreatic enzymes, and gastric acid respectively. Diagnosed by elevated somatostatin levels. Treatment is surgical resection for localized disease.

Somatotropic Pituitary Adenoma

A benign neoplasm of the anterior pituitary arising from growth hormone-secreting cells, causing acromegaly in adults and gigantism in children if occurring before epiphyseal closure.

Somatotropin

Growth hormone, a pituitary hormone that pro-

motes growth and influences protein, fat, and carbohydrate metabolism.

Somatotype

A descriptive body-build classification; it is not a disease diagnosis.

Somnambulism

Sleepwalking, a parasomnia involving complex behavior during incomplete arousal from sleep.

Somniloquy (Sleep-Talking)

A parasomnia characterized by talking or producing vocalizations during sleep without conscious awareness. Occurs across all stages of non-rapid eye movement and rapid eye movement sleep, ranging from incomprehensible mumbling to fluent, emotionally charged discourse. While typically a benign, self-limiting developmental phenomenon in children with normal polysomnography, in adults it can occur in conjunction with other parasomnias, such as sleepwalking (somnambulism), sleep terrors, obstructive sleep apnea, or REM sleep behavior disorder (RBD, a prospective marker of neurodegenerative synucleinopathies).

Alternate Names:

Sleep talking

Somnolence

Abnormal sleepiness or a tendency to fall asleep during wakeful periods, caused by insufficient sleep, medicines, sleep disorders, metabolic disease, or neurologic illness.

Somnolent

Somnolent means unusually sleepy or drowsy, which may result from sleep deprivation, medicines, intoxication, metabolic illness, or neurologic disease.

Sonographer

A healthcare professional trained to perform ultrasound scans (sonography).

Soporifics

Medicines or substances that cause sleepiness or

promote sleep. They may impair alertness, coordination, and breathing, especially when combined with alcohol or other sedatives.

Sorbitol

A sugar alcohol used as a sweetener and, in larger amounts, as an osmotic laxative.

Sore

Painful or tender; a sore may also mean an open skin or mucosal lesion.

Sore Throat

Pain or irritation in the throat, commonly caused by viral infection but sometimes by streptococcal infection or other disease.

Sore Throat (Pharyngitis)

Alternate index label for Pharyngitis.

Sores Canker (Canker Sores)

Alternate index label for Canker Sores.

Sotalol

A medicine with beta-blocking and class III antiarrhythmic effects, used for selected cardiac rhythm disorders.

Sotos Syndrome

An autosomal dominant overgrowth syndrome caused in over 90% of cases by haploinsufficiency, point mutations, or microdeletions of the nuclear receptor-binding SET domain-containing protein 1 (*NSD1*) gene on chromosome 5q35. Classically defined by the cardinal triad of: (1) distinctive facial appearance (macrodolichocephaly, prominent forehead with frontotemporal hair thinning, downslanting palpebral fissures, long narrow face, pointed chin); (2) childhood overgrowth with height and head circumference ≥ 2 standard deviations above the mean; and (3) intellectual disability with developmental delay.

Alternate Names:

Cerebral Gigantism; NSD1-Related Overgrowth Syndrome

Sound

Mechanical energy transmitted as waves and perceived by hearing; clinical sound assessment includes speech and respiratory or cardiac sounds.

Source Control

A procedure or intervention that eliminates, drains, debrides, or otherwise controls an anatomic focus of infection or contamination. Examples include drainage of an abscess, removal of infected tissue or device, and repair of a perforation.

Soy

Soy is a legume providing protein and isoflavones; it is nutritious for many people but can cause allergy and may interact with some medicines or thyroid treatment timing.

Soy Protein Intolerance

Adverse gastrointestinal reaction to soy proteins, most often seen in infants and young children. It may cause vomiting, diarrhoea, abdominal pain, blood or mucus in the stool, feeding difficulty, or poor growth.

Soya

Soybean and products made from it; soy is a food source and can cause allergy in susceptible people.

Space Medicine

The medical specialty concerned with human health and performance in aviation and space environments.

Spacer

A chamber used with a metered-dose inhaler to improve delivery of medicine to the lungs.

Alternate Names:

Valved holding chamber

Spanish Fly

A historical name for cantharidin-containing blister beetles; ingestion is poisonous and it is not a safe aphrodisiac.

Spasm

An involuntary, often painful contraction of a muscle or group of muscles.

Spasmodic Dysphonia

Spasmodic dysphonia is a focal laryngeal dystonia causing involuntary voice-box spasms and strained, breathy, or interrupted speech.

Spasmodic Torticollis

Cervical dystonia causing involuntary contractions and abnormal twisting or posture of the neck.

Spasmus Nutans

An uncommon early-childhood syndrome of rapid shimmering eye movements, head nodding, and torticollis that usually resolves but requires evaluation to exclude neurologic or ocular disease.

Spastic

Characterized by increased muscle tone and velocity-dependent resistance to stretch.

Spastic Colon (Irritable Bowel Syndrome)

Alternate index label for Irritable Bowel Syndrome.

Spasticity

Velocity-dependent increase in muscle tone caused by an upper motor-neuron lesion, producing stiffness, spasms, abnormal reflexes, and restricted movement.

Spasticity Management

Assessment and treatment of abnormal muscle stiffness and spasms. Measures may include stretching, therapy, positioning, medicines, injections, or surgery.

Spatial Resolution

Ability to distinguish small objects. PET: 4-6 mm. SPECT: 8-12 mm.

Spatula

A flat instrument used to spread, mix, examine, or transfer material.

Special Care Baby Unit

A ward within a hospital that provides specialist care for newborn babies who need additional support but not intensive care.

Special Stain

A histochemical stain that highlights a particular tissue component, organism, or deposited substance not adequately demonstrated by routine hematoxylin and eosin. Examples include periodic acid-Schiff, Ziehl-Neelsen, Congo red, and Gomori methenamine silver stains.

Specific Absorbed Fraction

The fraction of radiation energy emitted by a source region that is absorbed in a specified target organ or tissue. It is used in internal dosimetry to estimate radiation dose.

Specific Activity

Radioactivity per unit mass of compound. High specific activity essential for receptor targeting (low receptor density).

Specific Gravity

Ratio of urine density to water density reflecting solute concentration. Normal 1.005 to 1.030. Less precise than osmolality but faster to measure. Elevated in dehydration, glycosuria, proteinuria, and contrast. Decreased in diabetes insipidus and fluid overload.

Specific Learning Disorder

A neurodevelopmental disorder causing persistent difficulty acquiring or using academic skills in reading, writing, or mathematics. The difficulties interfere with functioning and are not better explained by another condition or inadequate instruction.

Specific Phobia

A disorder involving marked, persistent fear or anxiety about a particular object or situation. The fear is excessive relative to the actual danger, leads to avoidance or intense distress, and interferes with daily functioning.

Specific Reading Disability

Alternate index label; see *Dyslexia*.

Specificity

The proportion of people without a condition who have a negative test result.

Specimen Adequacy

The degree to which a collected clinical specimen is sufficient in quantity, correctly identified, properly transported, and representative of the site being tested. Inadequacy can cause invalid, misleading, or falsely negative results.

Specimen Container

Container for collecting specimens. Types: urine cup, stool container, sputum cup, biohazard bag.

Specimen Cup

A sterile or non-sterile, wide-mouthed medical container fabricated from medical-grade polypropylene or polystyrene, equipped with a leak-proof screw cap or tamper-evident seal, used for the collection, transport, and containment of biological fluid and tissue specimens (most commonly midstream clean-catch urine, sputum, stool, or intraoperative biopsy specimens). Calibrated with volumetric volume markings and patient identification labeling surfaces to ensure chain-of-custody and prevent contamination during automated microbiological, cytological, or chemical laboratory analysis.

Specimen Transport Medium

A formulated medium that preserves the viability and diagnostic characteristics of microorganisms during transport from the collection site to the laboratory. The required medium, temperature, container, and transport time depend on the specimen and suspected pathogen.

SPECT Scan (Single Photon Emission CT)

Nuclear medicine imaging using gamma-emitting radioisotopes (Tc-99m, Tl-201, I-123) with a rotating gamma camera to create 3D images.

Alternate Names:

Single Photon Emission CT

Spectral Analysis

Energy discrimination to reject scatter. Energy window typically 20% centered on photopeak.

Speculum

An instrument used to gently separate or hold open the walls of a body passage so that the area can be visualized, examined, sampled, or treated. Common types include vaginal, nasal, ear, and rectal specula; design and size depend on the body site and intended procedure.

Speculum, Ear

A small cone-shaped tip fitted to an otoscope to direct light into the external ear canal and permit examination of the canal and tympanic membrane. Disposable and reusable types are available in different sizes.

Speech

Phonation and articulation. Teeth, tongue, lips, palate involved. Dental prostheses may affect speech initially.

Speech Audiometry

Hearing testing that measures detection or recognition of speech at different loudness levels. It commonly includes speech reception threshold and word-recognition testing, complementing pure-tone audiometry.

Alternate Names:

Speech testing; speech audiometric testing.

Speech Delay

A delay in language development compared to peers: by 12 months—no single words; 18 months—<10-20 words/no word combinations by 24 months; or regression.

Speech Discrimination

Speech discrimination testing, also known as word recognition scoring, measures the ability to under-

stand speech at a comfortable listening level, and it is an important component of the audiometric evaluation that provides information about cochlear and neural function beyond what pure tone audiometry alone can reveal.

Alternate Names:

word recognition scoring, measures the.

Speech Disorders

Conditions affecting production, fluency, voice, articulation, or motor control of speech.

Speech Impairment In Adults

Difficulty producing or understanding speech caused by aphasia, dysarthria, apraxia, hearing loss, neurologic disease, structural problems, or cognitive disorder; sudden onset requires stroke assessment.

Speech Therapy

Assessment and treatment of communication, speech, voice, language, and swallowing disorders.

Speech-Language Pathologist

A clinician who evaluates and treats disorders of speech, language, voice, communication, cognition, and swallowing across childhood, adulthood, and older age.

Speech-Language Pathology Assessment

An evaluation of speech, language, voice, communication, swallowing, and sometimes related cognitive skills. It combines interview, clinical observation, and standardized or functional tests to guide therapy.

Speed

Speed is an informal term for methamphetamine, a stimulant that can cause dependence, cardiovascular toxicity, psychosis, overheating, and other serious harms.

Sperm

Mature male reproductive cells that can fertilize an egg. Semen contains sperm plus fluids from the reproductive glands; sperm are produced in the

testes.

Spermarche

The onset of sperm production and the first ejaculation, marking an important stage of male puberty.

Spermatic

Relating to sperm or the structures that produce, transport, or support sperm.

Spermatic Cord

A cord-like structure containing the vas deferens, testicular vessels, nerves, and lymphatics.

Spermatocele

Benign, painless, fluid-filled cystic retention cavity arising from the epididymis (usually the head/caput), containing avascular fluid and non-viable spermatozoa. Etiology: obstruction of an efferent ductule of the epididymis.

Alternate Names:

Cyst, Spermatic

Spermatogenesis

Production and maturation of sperm in the seminiferous tubules.

Spermatogonium

A germ cell in the testis that divides and develops into primary spermatocytes.

Spermatorrhea

Involuntary leakage or discharge of semen, sometimes occurring during sleep or without orgasm. Occasional nocturnal emissions can be normal; persistent discharge, pain, blood, or urinary symptoms needs assessment.

Spermatorrhoea

An obsolete term for involuntary or excessive semen discharge.

Spermatozoa

Mature male reproductive cells, also called sperm. Each normally carries half the usual chromosome

number and has a head, middle section, and tail that helps it move toward an egg.

Spermatozoon

The singular form of spermatozoa: one mature male reproductive cell capable of carrying genetic material to an egg.

Spermicide

A substance that immobilizes or destroys sperm and is used as a contraceptive.

Sphenoid

Relating to the sphenoid bone at the base of the skull.

Sphenoid Bone

Butterfly-shaped bone at the skull base, with body (sphenoid sinuses), greater/lesser wings, and pterygoid processes. The sella turcica houses the pituitary gland.

Sphenoid Sinusitis

Inflammation or infection of the sphenoid sinus, which lies deep behind the nasal cavity. It may cause deep headache, facial pain, or visual symptoms and can rarely spread to nearby nerves or the brain; persistent or severe symptoms require medical assessment.

Sphenoidal Sinus

A paired air-filled sinus within the sphenoid bone at the base of the skull. It lies near important nerves and blood vessels, so inflammation or surgery in this area requires careful assessment.

Spherocytes

Small, dark, sphere-shaped red blood cells lacking central pallor. Formed when RBC membrane is lost without loss of volume, reducing the surface area-to-volume ratio.

Spherocytosis (Spherocytosis (Hereditary, HS))

Alternate index label for Spherocytosis (Hereditary,

HS).

Spherocytosis (Hereditary, HS)

Alternate index label for Hereditary, HS.

Sphincter

A ring-shaped muscle that regulates an opening by contracting and relaxing.

Sphincter Of Oddi

A muscular valve controlling the flow of bile and pancreatic secretions into the duodenum.

Sphygmograph

An instrument or sensor that records the arterial pulse waveform. Modern devices use pressure or optical sensors to analyze pulse timing and shape.

Sphygmomanometer

An instrument used to measure arterial blood pressure, typically consisting of an inflatable cuff and a pressure-measuring system. Manual devices use Korotkoff sounds, while automated devices generally use oscillometric sensing. Correct cuff size and technique are essential for reliable measurements.

Spider Angioma

A spider angioma is a central red blood vessel with radiating fine vessels that blanches with pressure and may occur normally or with liver or hormonal disease.

Spider Bite

A cutaneous envenomation resulting from the bite of a spider, ranging from minor local reactions to severe systemic envenomation. The brown recluse spider may cause loxoscelism with necrotic skin lesions, while the black widow spider may cause latrodectism with muscle cramps, hypertension, and autonomic dysfunction.

Spider Bites (Black Widow And Brown Recluse)

Alternate index label for Black Widow and Brown Recluse.

Spina Bifida

A birth defect in which the backbone and spinal cord do not form completely. It can be mild or severe and may cause weakness, loss of feeling, or problems controlling the bladder or bowel.

Spina Bifida Meningocele

A form of spina bifida where the meninges (but not neural tissue) herniate through a vertebral defect, forming a CSF-filled sac. The spinal cord remains in its normal position and is not involved. Presents as a visible, fluid-filled, translucent sac on the back, typically without neurologic deficits.

Spina Bifida Myelomeningocele

The most severe form of spina bifida, where both the meninges and neural tissue (spinal cord and nerve roots) herniate through a vertebral defect. Presents as a visible, flat or sac-like lesion on the back with neurologic deficits below the level of the lesion (paralysis, sensory loss, bowel/bladder dysfunction).

Spina Bifida Occulta

A usually mild form of spina bifida in which the vertebral arch does not fully close without herniation of the spinal cord or meninges. It is often asymptomatic but may be associated with a skin dimple, hair tuft, or other finding over the lower spine.

Spinal Anaesthesia

Injection of local anesthetic into cerebrospinal fluid in the lumbar spine to produce regional numbness.

Spinal Analgesia

Intrathecal administration of a local anesthetic and/or opioid primarily to provide analgesia, often with limited motor block. Intrathecal opioids can extend postoperative pain relief but may cause pruritus, urinary retention, respiratory depression, and other opioid effects.

Spinal And Bulbar Muscular Atrophy

An adult-onset, X-linked recessive neurodegenerative motor neuron disorder caused by an expanded

CAG trinucleotide repeat (> 38 repeats) encoding an elongated polyglutamine tract in exon 1 of the androgen receptor (*AR*) gene on chromosome Xq12. Characterized by selective degeneration of lower motor neurons in the anterior horns of the spinal cord and bulbar motor nuclei, presenting after age 30 with progressive muscle weakness, proximal limb atrophy, perioral and lingual fasciculations, dysarthria, and dysphagia, accompanied by systemic androgen insensitivity manifest as gynecomastia, testicular atrophy, and oligospermia.

Alternate Names:

Kennedy Disease; SBMA

Spinal And Epidural Anesthesia

Spinal and epidural anesthesia deliver local anesthetic near spinal nerves to numb the lower body; spinal is a single injection, while epidural uses a catheter for adjustable dosing.

Spinal Anesthesia

Injection of local anesthetic into the lumbar subarachnoid space to produce temporary sensory, motor, and sympathetic blockade below the level of injection. Complications include hypotension, bradycardia, urinary retention, post-dural puncture headache, high or total spinal block, infection, and bleeding.

Spinal Anesthesia For Cesarean Delivery

A neuraxial anesthetic technique used for elective and many emergency cesarean deliveries. A small dose of local anesthetic (hyperbaric bupivacaine 12-15 mg) with opioid (fentanyl 15-25 mcg, morphine 0.15-0.25 mg) is injected into the subarachnoid space. Produces rapid surgical anesthesia (T4 sensory level for cesarean).

Spinal Angiography

X-ray imaging of blood vessels supplying the spinal cord after contrast is injected. It is used to locate vascular malformations, fistulas, aneurysms, or sources of bleeding and may guide treatment.

Spinal Artery

An artery supplying the spinal cord and its cover-

ings, including the anterior spinal artery and paired posterior spinal arteries; blockage can cause characteristic neurologic deficits.

Spinal Block

A spinal block is injection of local anesthetic into cerebrospinal fluid in the lower back, producing temporary numbness and weakness for surgery or pain control.

Spinal Canal

The spinal canal is the bony passage through the vertebral column containing the spinal cord, its coverings, nerve roots, and blood vessels.

Spinal Cavity

The posterior body cavity formed by the vertebral column that contains and protects the spinal cord.

Spinal Column

The vertebral column, consisting of vertebrae, discs, ligaments, and joints that support and protect the spinal cord.

Spinal Cord

The cylindrical nervous tissue within the vertebral canal that carries signals between brain and body and mediates reflexes.

Spinal Cord Abscess

A spinal cord or epidural abscess is a collection of infection near the spinal cord that can compress neural tissue and cause pain, fever, weakness, paralysis, or bladder problems.

Spinal Cord Compression

Oncologic emergency. Back pain, weakness, sensory loss, bowel/bladder dysfunction. MRI diagnostic. Treatment: steroids, radiation, surgery.

Spinal Cord Disease

A disorder affecting the spinal cord, including compression, inflammation, vascular injury, infection, tumor, degeneration, or trauma, and potentially causing weakness, sensory loss, or autonomic dys-

function.

Spinal Cord Injury

Spinal cord injury: ASIA impairment scale (A-E). Complete (ASIA A) vs incomplete.

Alternate Names:
SCI

Spinal Cord Injury: Treatments And Rehabilitation

Spinal-cord injury can cause temporary or permanent weakness, sensory loss, autonomic dysfunction, and bowel or bladder problems. Acute care protects the spine and treats complications; rehabilitation addresses mobility, skin, breathing, function, equipment, sexuality, and community participation.

Spinal Cord Neoplasms

Spinal cord neoplasms are abnormal growths in or around the spinal cord that can cause pain, weakness, sensory changes, or bladder problems.

Spinal Cord Stimulation

An implanted treatment that delivers electrical impulses to the epidural space to alter pain signaling, used for selected chronic neuropathic or ischemic pain after careful assessment.

Spinal Cord Trauma

Spinal cord trauma is injury to the cord from force such as fracture, dislocation, or penetrating damage, potentially causing weakness, sensory loss, autonomic dysfunction, or paralysis.

Spinal Curvature

Alternate index label; see *Scoliosis*.

Spinal Disease

A disorder of the vertebrae, discs, spinal cord, nerve roots, or supporting tissues.

Spinal Dysraphism

A group of congenital malformations resulting from incomplete closure of the neural tube during embryonic development, affecting the spinal cord and

overlying vertebral structures. Includes open defects (myelomeningocele) and closed defects (spina bifida occulta, lipomyelomeningocele, dermal sinus).

Spinal Epidural Abscess

A collection of pus in the spinal epidural space, usually caused by haematogenous or contiguous infection. It may present with severe local back pain, fever, spinal tenderness, radicular pain, neurological deficit, or sphincter disturbance, although the classic triad is often incomplete.

Spinal Fluid Leak

Alternate index label; see *CSF leak (Cerebrospinal fluid leak)*.

Spinal Fracture

A fracture of one or more vertebrae caused by trauma, osteoporosis, tumor, or other bone-weakening disease.

Spinal Fusion

Surgical union of two or more vertebrae using bone graft or bone graft substitutes with instrumentation (rods, screws, cages). Approaches: posterior (posterolateral fusion, transforaminal lumbar interbody fusion TLIF), anterior (anterior lumbar interbody fusion ALIF), and lateral (extreme lateral interbody fusion XLIF).

Spinal Ganglion

A cluster of sensory neuron cell bodies, especially a dorsal-root ganglion, carrying signals from skin, muscles, and joints toward the spinal cord.

Spinal Injury

A clinical condition involving spinal Injury; its causes, manifestations, severity, and complications guide diagnosis and management.

Spinal Muscular Atrophy

An inherited disorder causing progressive loss of motor neurons and muscle weakness, often beginning in infancy or childhood but sometimes appearing in adulthood. Severity varies; breathing, swallowing,

and mobility may be affected. Early diagnosis allows genetic counseling and disease-modifying care.

Spinal Neoplasms

Spinal neoplasms are tumors arising in the spinal cord, its coverings, vertebrae, or nearby tissues.

Spinal Nerve

A mixed nerve formed by the union of dorsal and ventral roots as it exits the vertebral canal. There are 31 pairs, carrying sensory, motor, and autonomic fibers to and from the spinal cord.

Spinal Precautions

Immobilization of spine in trauma patients to prevent neurological injury. Indications: mechanism (fall >3 feet, MVC, gunshot), altered mental status, focal neurological deficit, midline spinal tenderness. Technique: cervical collar, backboard, log roll. Clearance: NEXUS criteria or Canadian C-spine rule.

Spinal Shock

A state of transient physiological areflexia and flaccid paralysis occurring immediately after acute spinal cord injury, below the level of the lesion.

Spinal Stenosis

Narrowing of the spinal canal causing compression of the spinal cord or nerve roots. Lumbar stenosis (most common): neurogenic claudication (leg pain/weakness with walking, relieved by sitting/leaning forward), often bilateral.

Alternate Names:

Stenosis, Spinal

Spinal Tap

Alternate index label; see *Lumbar puncture*.

Spinal Tumor

A spinal tumor is an abnormal growth in or near the spine that may be benign or malignant and may compress nerves or the spinal cord.

Spindle Pole

A spindle pole is one end of the mitotic spindle from which microtubules extend to separate duplicated chromosomes during cell division.

Spindle Pole Body

The spindle pole body is the microtubule-organizing structure in fungi that functions like a centrosome during nuclear division.

Spindle-Cell Sarcoma

A group of sarcomas characterized by elongated, spindle-shaped malignant cells. It includes entities like leiomyosarcoma, fibrosarcoma, and malignant peripheral nerve sheath tumor, with variable behavior.

Spine

The vertebral column, consisting of stacked vertebrae that support the body and protect the spinal cord.

Spine And Spinal Cord, Diseases And Injuries Of

Disorders or trauma affecting vertebrae, discs, ligaments, spinal cord, or nerve roots.

Spine Disorders

The spine, or vertebral column, is a complex bony and soft-tissue structure that provides axial support, protects the spinal cord, and enables flexible movement of the trunk. It consists of 33 vertebrae divided into cervical, thoracic, lumbar, sacral, and coccygeal regions, separated by intervertebral discs that act as shock absorbers.

Spine Injury

A clinical condition or finding involving spine injury. Its features, causes, evaluation, and management depend on the clinical context.

Spinocerebellar Ataxia

Spinocerebellar ataxias are inherited neurodegenerative disorders causing progressive incoordination, gait problems, dysarthria, abnormal eye movements,

and sometimes other neurologic signs.

Spinocerebellar Ataxias

A group of autosomal dominant neurodegenerative disorders characterized by progressive cerebellar ataxia, often with additional neurological features. Over forty different SCA types have been identified, with SCA1, SCA2, SCA3 (Machado-Joseph disease, the most common worldwide), SCA6, and SCA7 being the most prevalent.

Spinous Process

The posterior bony projection of a vertebra to which muscles and ligaments attach.

Spiradenocarcinoma

A rare malignant sweat-gland tumor that may arise in a long-standing spiradenoma, presenting as a rapidly enlarging, painful, or ulcerated nodule.

Spiradenoma

A usually benign painful sweat-gland tumor presenting as a dermal nodule; multiple spiradenomas may occur in Brooke-Spiegler syndrome.

Spiral Colon

A spiral arrangement of the proximal colon described in some animals, especially ruminants, that increases transit length and absorptive surface; it is not a normal human colonic segment.

Spiral Organ

The organ of Corti, a sensory structure in the cochlea containing hair cells that convert sound vibrations into nerve signals.

Spiramycin

A macrolide antibiotic used in selected infections, including some prenatal toxoplasmosis protocols.

Spirillum

A spiral-shaped bacterium; Spirillum minus can cause rat-bite fever.

Spirillum Minus

Spirillum minus is a spiral Gram-negative bacterium that causes rat-bite fever, characterized by fever, rash, and migratory joint pain after rodent exposure; culture is difficult and diagnosis may be clinical or molecular.

Spirit

A volatile liquid preparation or alcohol; the meaning depends on the medical, pharmaceutical, or general context.

Spiritual Pain

Distress involving meaning, purpose, beliefs, identity, or connection, often addressed in palliative and supportive care.

Spiritual Support

Support that helps a person address meaning, values, faith, hope, or existential distress during illness; it may be provided by chaplains, spiritual leaders, clinicians, or trusted people.

Spirochaetaceae

A family of spiral-shaped bacteria that includes medically important spirochetes such as *Borrelia*, *Treponema*, and *Leptospira*.

Spirochaetales Infection

Infection caused by spiral-shaped bacteria in the order Spirochaetales, including syphilis, Lyme disease, relapsing fever, and leptospirosis.

Spirochaete

A slender spiral-shaped bacterium, including *Treponema*, *Borrelia*, and *Leptospira*.

Spirochete

A thin, coiled gram-negative bacterium with internal periplasmic flagella conferring corkscrew motility, requiring darkfield or special stains. Pathogenic genera include *Treponema* (syphilis), *Borrelia* (Lyme disease, relapsing fever), and *Leptospira* (leptospirosis), diagnosed mainly by serology and darkfield microscopy.

Spirogram

A graphical record of air volumes and flow during pulmonary function testing.

Spirometer

An instrument that measures airflow and the volume of air moved during specified breathing maneuvers, especially forced expiration. It does not by itself measure all lung volumes; residual volume and total lung capacity require additional pulmonary-function testing.

Spirometry

A pulmonary-function test that measures airflow and exhaled volume during standardized breathing maneuvers. Key measurements include forced vital capacity, forced expiratory volume in one second (FEV1), and the FEV1/FVC ratio. Results can support the assessment of obstructive ventilatory impairment and may suggest restriction when FVC is reduced with a preserved ratio, but suspected restriction should be confirmed with lung-volume measurement. Interpretation requires appropriate reference equations, patient effort, acceptability, and repeatability; bronchodilator response may provide additional information when indicated.

Spiroinolactone

An aldosterone antagonist and potassium-sparing diuretic used for heart failure, hypertension, hyperaldosteronism, and other indications.

Spit

Spit is saliva expelled from the mouth; increased saliva or repeated spitting may reflect oral irritation, nausea, reflux, swallowing difficulty, or behavior.

Spitting

Spitting is forceful expulsion of saliva or other material from the mouth; new excessive spitting may reflect oral disease, reflux, nausea, swallowing difficulty, or behavior.

Spitz Nevus

A benign melanocytic nevus, often a pink or pigmented papule in children or young adults, that

may resemble melanoma clinically or histologically.

Splanchnic

Relating to the internal organs, especially abdominal organs and their blood vessels or nerves.

Splanchnic Nerve

A visceral autonomic nerve carrying sympathetic or parasympathetic fibers from the thorax or abdomen to prevertebral ganglia and internal organs.

Spleen

A lymphoid organ in the upper-left abdomen. Its red pulp filters blood and removes aged or abnormal blood cells; its white pulp supports immune responses. It also stores platelets and helps clear blood-borne microorganisms.

Spleen Development

Spleen development is the embryonic formation and postnatal maturation of the organ that filters blood, supports immune responses, and removes aged blood cells.

Spleen Disorder

A disease affecting the spleen, such as enlargement, infection, infarction, rupture, inflammation, cyst, or tumor, with possible effects on blood cells and immunity.

Spleen Enlarged (Enlarged Spleen)

Alternate index label for Enlarged Spleen.

Spleen Hemangioma

A spleen hemangioma is a usually benign blood-vessel tumor of the spleen, often incidental but occasionally associated with enlargement, anemia, or rupture risk.

Spleen Hilum

Concave medial surface where the tortuous splenic artery (celiac trunk branch) enters and splenic vein exits, with lymphatics and nerves. Artery divides into 5 segmental branches at the hilum (basis for partial splenectomy). Splenic vein joins the superior

mesenteric vein forming the portal vein.

Spleen Removal

A medical procedure, treatment, device, or care topic related to spleen removal; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Spleen, Diseases Of

Disorders affecting splenic structure or function, including splenomegaly, infarction, infection, cysts, abscess, rupture, vascular disease, and hematologic or infiltrative disorders.

Splenectomy

Surgical removal of the spleen, performed for selected traumatic, hematologic, malignant, vascular, or other disorders. It increases the risk of severe infection with encapsulated bacteria and requires preventive planning.

Splenic Abscess

A localized collection of pus in the spleen, usually from bloodstream infection, trauma, infarction, or spread from nearby disease and potentially causing fever, left-upper-quadrant pain, and sepsis.

Splenic Cyst

A fluid-filled lesion within the spleen, classified as true cysts with an epithelial lining, including congenital, epidermoid, and dermoid cysts, or pseudocysts without a lining, typically resulting from trauma, infection, or pancreatitis. Most are asymptomatic and discovered incidentally on imaging.

Splenic Flexure

The bend of the colon where the transverse colon becomes the descending colon, near the spleen.

Splenic Flexure Syndrome

Left-upper-abdominal pain caused by gas trapped at the bend between the transverse and descending colon.

Splenic Infarction

Ischemic necrosis of splenic tissue caused by obstruction of splenic arterial or venous blood flow. Causes include embolism, thrombosis, hematologic disease, hypercoagulability, infection, trauma, or vasculitis; it often causes sudden left-upper-quadrant pain and is evaluated with imaging.

Splenic Infection

An infectious process or toxic exposure involving splenic infection; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Splenic Neoplasm

A splenic neoplasm is an abnormal growth in the spleen that may be benign, malignant, or spread from another cancer.

Splenic Rupture

A tear of the spleen, usually after blunt abdominal trauma but sometimes spontaneous in an enlarged or diseased spleen. It can cause life-threatening internal bleeding; diagnosis and treatment depend on hemodynamic stability and imaging, with urgent surgery or embolization sometimes required.

Splenomegaly

Abnormal enlargement of the spleen, usually secondary to infection, liver or portal-venous disease, hemolysis, hematologic malignancy, storage disease, or inflammation. It may cause left-upper-quadrant fullness or pain and increases rupture risk.

Splenomegaly Gaucher (Gaucher Disease)

Alternate index label for Gaucher Disease.

Splice-Site Variant

A variant affecting the sequences required for accurate removal of introns and joining of exons during RNA processing. It may cause abnormal transcripts, exon skipping, intron retention, or reduced functional protein.

Spliceosome

The spliceosome is a nuclear RNA-protein complex that removes introns from pre-messenger RNA and joins exons to form mature RNA.

Splint

Immobilizes injured body part. Types: aluminum, plastic, canvas, pneumatic.

Splint (Dental)

Fixation of mobile teeth or fractured tooth. Wire and composite, orthodontic bracket splint.

Alternate Names:

Dental

Splinter Haemorrhages

Small linear bleeding beneath a nail, caused by trauma, endocarditis, vasculitis, or other conditions.

Splinter Hemorrhages

Linear hemorrhages under the nail bed, oriented longitudinally. In infective endocarditis, they are caused by septic microemboli to the nail bed capillaries. They are a minor Duke criterion. Splinter hemorrhages can also result from trauma, psoriasis, and systemic vasculitis.

Splinter Removal

A medical procedure, treatment, device, or care topic related to splinter removal; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Splints

Devices that support, immobilize, align, or protect an injured or diseased body part.

Split Brain

A state after surgical or pathological disruption of the corpus callosum in which information transfer between cerebral hemispheres is reduced, producing characteristic disconnection findings.

Split-Thickness Skin Graft

A graft containing the epidermis and part of the der-

mis, used to cover wounds or burns and allowing the donor site to heal from remaining skin structures.

Splitting

A psychological defense pattern in which a person or situation is viewed as entirely good or entirely bad, with difficulty integrating positive and negative qualities. It can occur in several mental-health conditions and is not diagnostic by itself.

Spondee

A two-syllable test word with equal emphasis on both syllables, used in speech audiometry.

Spondylarthritis

A family of inflammatory rheumatic diseases affecting the spine, sacroiliac joints, entheses, peripheral joints, skin, eyes, or bowel, including ankylosing spondylitis and psoriatic arthritis.

Spondylitis

Inflammation of one or more vertebrae or the spine.

Spondyloarthritis

Group of inflammatory arthritis associated with HLA-B27. Includes ankylosing spondylitis, psoriatic arthritis, reactive arthritis, enteropathic arthritis. Features: axial involvement, enthesitis, dactylitis.

Spondylolisthesis

Forward or backward displacement of one vertebra relative to another.

Spondylolysis

A stress fracture or defect in the pars interarticularis of a vertebral arch, most often in the lower lumbar spine.

Spondylosis

Degenerative changes of the spine involving vertebrae, discs, and facet joints.

Spondylosis, Cervical

Alternate index label; see *Cervical spondylosis*.

Sponge

A sponge is a porous material used to absorb fluid, clean or dress wounds, or deliver a product; retained surgical sponges are a preventable complication.

Sponge (Vascular)

A sterile absorbent material used during vascular procedures to control minor bleeding or protect tissue.

Alternate Names:

Vascular

Spongiform Encephalopathy

A fatal or progressive neurodegenerative disorder with sponge-like brain changes, including prion diseases.

Spongy Bone

Porous trabecular bone found inside many bones, especially near joints.

Alternate Names:

Cancellous bone; trabecular bone

Spontaneous Abortion

Pregnancy loss before fetal viability that occurs without intentional intervention, commonly caused by chromosomal abnormalities but also by uterine, endocrine, immune, infectious, or other factors.

Spontaneous Abortion (Miscarriage)

Unintentional loss of a pregnancy before fetal viability. Causes include chromosomal abnormalities, uterine or cervical problems, infection, and maternal medical conditions.

Alternate Names:

Miscarriage

Spontaneous Bacterial Peritonitis

An infection of ascitic fluid without an identifiable abdominal source, usually occurring in people with cirrhosis and ascites. It may cause fever, abdominal pain, tenderness, or worsening illness. Diagnosis is made by testing ascitic fluid, usually showing at least 250 neutrophils per mm³.

Spontaneous Coronary Artery Dissection

A non-atherosclerotic, non-traumatic separation of the coronary arterial wall layers, creating a false lumen that compresses the true lumen and causes acute myocardial infarction or sudden death. Predominantly affects young to middle-aged women without traditional cardiovascular risk factors, strongly associated with fibromuscular dysplasia, pregnancy, and hormonal or emotional triggers; conservative management is generally preferred over stenting when hemodynamically stable.

Alternate Names:
SCAD

Spontaneous Breathing Trial

A period of unsupported breathing to assess a patient's readiness for extubation. The patient breathes through a T-piece or with minimal pressure support for 30–120 minutes. Successful SBT predicts successful extubation.

Spontaneous Coronary Artery Dissection

A non-traumatic, non-iatrogenic separation within the wall of a coronary artery, usually caused by an intramural hematoma or an intimal tear. It can obstruct coronary blood flow and cause acute coronary syndrome, especially in younger or peripartum women; diagnosis and management require specialist cardiovascular assessment.

Alternate Names:
SCAD

Spontaneous Perforation

Unexpected rupture of a hollow organ without a clear external injury, allowing contents to escape and potentially causing peritonitis, mediastinitis, sepsis, or hemorrhage.

Spontaneous Pneumothorax

Pneumothorax without trauma. Primary in healthy individuals; secondary with underlying lung disease. Treatment depends on size.

Sporadic

Occurring irregularly or as an isolated case rather than in a predictable pattern or family cluster.

Spore

A resistant reproductive or survival structure produced by fungi, bacteria, or some plants.

Spores

Resistant reproductive or survival structures made by some fungi, bacteria, and plants; inhaled fungal spores can cause allergy or infection, while bacterial spores can persist in the environment.

Sporothrix

A genus of environmental fungi that can cause sporotrichosis, usually after inoculation through the skin and sometimes through inhalation or dissemination.

Sporothrix Schenckii

A group of fungi that causes sporotrichosis, a typically nodular skin and lymphatic infection acquired from plants, soil, or animal scratches.

Sporotrichosis

Sporotrichosis is a fungal infection caused by *Sporothrix* species, often introduced through plant matter and producing nodules along lymphatic channels.

Sporozoa

An older group name for spore-forming protozoan parasites, including *Plasmodium*.

Sporozoites

Motile infective forms of certain protozoa, such as the *Plasmodium* forms injected by mosquitoes.

Sports Cream Overdose

Sports cream overdose involves toxic or environmental exposure to sports cream overdose; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Sports Physical

A preparticipation health evaluation that reviews medical history, symptoms, medications, family risks, and examination findings to identify condi-

tions needing evaluation before sports.

Sports Rehabilitation

Rehabilitation that restores safe movement and performance after a sports-related injury.

Spots Before The Eyes

Visual floaters, flashes, or scotomata perceived in the visual field; sudden onset requires urgent assessment.

Spots In Front Of The Eyes

Floaters or visual spots caused by shadows from vitreous opacities, inflammation, bleeding, retinal tears, migraine, or other eye or neurologic conditions; sudden flashes or a curtain-like loss are urgent.

Spotted Fever

A group of tick-borne rickettsial illnesses causing fever, headache, and often a spotted rash.

Spousal Abuse

Spousal abuse is physical, sexual, emotional, financial, or coercive harm by an intimate partner; immediate safety planning and confidential support may be needed.

Spouse

A spouse is a person's legally or socially recognized partner; involving a spouse in healthcare requires the patient's consent unless law provides otherwise.

Sprain

An injury to a ligament (connective tissue connecting bone to bone) caused by excessive stretching or tearing. Grading: Grade I (mild stretching, microscopic tears), Grade II (partial tear with laxity), Grade III (complete tear with instability).

Sprain Neck (Whiplash)

Alternate index label for Whiplash.

Sprained Ankle

An ankle sprain is stretching or tearing of ligaments, usually after rolling the ankle inward. Pain, swelling,

bruising, and instability occur; protection, relative rest, compression, elevation, and progressive rehabilitation help, while inability to bear weight may require imaging.

Alternate Names:

Strains (Sprained Ankle)

Sprains

Injuries involving stretching or tearing of ligaments around a joint.

Spray Drying

Spray drying converts a liquid mixture into a dry powder by atomizing it into a heated gas stream, often used in pharmaceutical and food manufacturing.

Sprengel Deformity

A congenital elevation of the scapula due to failure of the scapula to descend from its embryonic cervical position to its normal thoracic position during weeks 9-12 of gestation. The scapula is higher, smaller, and rotated inferiorly compared to the normal side. Range of motion is limited, especially in shoulder abduction.

Spring Fever (Henoch-Schonlein Purpura)

Alternate index label for Henoch-Schonlein Purpura.

Sprue

A malabsorption syndrome, especially tropical sprue or celiac disease in older terminology.

Sprue, Nontropical

Nontropical sprue is an older term for celiac disease, an immune reaction to gluten that damages the small-intestinal lining and impairs absorption.

Spumavirus

A genus of retroviruses that can infect humans and other animals. Human infection is usually mild or causes no symptoms.

Spur Heel (Heel Spurs)

Alternate index label for Heel Spurs.

Spur-Cell Hemolytic Anemia

A severe hemolytic anemia seen in advanced liver disease where acanthocytosis develops due to altered membrane cholesterol, causing premature red cell destruction in the spleen.

Sputum

Mucus or other material coughed up from the lower respiratory tract.

Sputum Culture

Laboratory culture of expectorated or aspirated respiratory secretions to identify bacterial, fungal, or mycobacterial pathogens; specimen quality is essential.

Sputum Direct Fluorescent Antibody Test

A sputum DFA test uses fluorescent antibodies to detect selected respiratory pathogens in mucus coughed from the lungs.

Sputum Fungal Smear

Microscopic examination of sputum for fungal elements, used with culture, antigen, molecular testing, and clinical findings when pulmonary fungal infection is suspected.

Sputum Gram Stain

A Gram-stained microscopic examination of sputum to assess specimen quality and visualize bacteria or inflammatory cells before or alongside culture.

Sputum Stain For Mycobacteria

Microscopic staining of sputum for acid-fast bacilli, used in evaluation of suspected tuberculosis or non-tuberculous mycobacterial disease; a negative smear does not exclude infection.

Squalamine Lactate

An investigational aminosterol compound studied for antimicrobial, antiangiogenic, and neurologic

effects; established routine clinical uses are limited.

Squalene

A lipid precursor in cholesterol and steroid synthesis found in human tissues and some plant oils; it is also used in certain vaccine adjuvant formulations.

Squamous Cell

A thin, flattened, polygonal epithelial cell with an expanded surface area, resembling flat paving stones or scales. Forms the protective cellular architecture of simple squamous epithelium (specialized for passive filtration and gas diffusion in pulmonary alveoli, vascular endothelium, and mesothelium) and stratified squamous epithelium (specialized for resistance against mechanical abrasion, chemical friction, and microbial invasion in the epidermis, oral cavity, esophagus, anal canal, and cervix). Neoplastic transformation of these cells gives rise to squamous cell carcinoma, a major malignant tumor of the skin, head and neck, lungs, esophagus, and cervix.

Squamous Cell Carcinoma

Cutaneous squamous cell carcinoma is a malignant keratinocyte tumor commonly arising on chronically sun-exposed skin or in areas of chronic inflammation or immunosuppression. It may present as a scaly, indurated, hyperkeratotic papule, plaque, or ulcer.

Squamous Cell Carcinoma Of The Lung

A major histologic subtype of non-small cell lung cancer that arises from the bronchial respiratory epithelium following squamous metaplasia and dysplasia, strongly correlated with cigarette smoking. Typically presents as a central, cavitating hilar mass with bronchial obstruction, and histopathologically demonstrates keratinization, intercellular bridges, and squamous pearls.

Squamous Cell Carcinoma Of The Skin

A skin cancer arising from squamous cells, often appearing as a scaly, crusted, wart-like, or ulcerated growth.

Alternate Names:

Cancer, Squamous Cell

Squamous Cell Skin Cancer

Squamous cell skin cancer is a malignant keratinocyte tumor often arising on sun-exposed skin and capable of local invasion or metastasis, especially when high risk.

Squamous Cells

Flat epithelial cells with scale-like appearance that line surfaces such as skin, mouth, cervix, and esophagus; abnormal numbers or forms can reflect irritation, infection, dysplasia, or cancer.

Squamous Intraepithelial Lesion Cervical (Cervical Dysplasia)

Alternate index label for Cervical Dysplasia.

Square Foot

A square foot is a unit of area equal to a square measuring one foot on each side, used to describe surfaces, rooms, wounds, or treatment areas.

Square Inch

A square inch is a unit of area equal to a square measuring one inch on each side, used for small surfaces or wound measurements.

Square Metres

Square metres are units of area equal to the surface of a one-metre square, used in clinical, environmental, and dosage calculations.

Squill

A plant-derived cardiac glycoside source historically used as an expectorant or cardiac medicine; toxicity limits use.

Squint

Strabismus, misalignment of the eyes so they do not point in the same direction.

Sscp Analysis

Single-strand conformation polymorphism analysis, a laboratory method that detects some DNA sequence variants by differences in the migration of single-stranded fragments through a gel.

SSRI

An SSRI is a selective serotonin-reuptake inhibitor antidepressant used for depression and several anxiety-related disorders; effects and withdrawal symptoms require monitoring.

SSRIs

Selective serotonin reuptake inhibitors, a group of medicines that increase serotonin signaling in the brain. They are used for depression and several anxiety-related disorders; effects and side effects develop over time, not immediately.

St John'S Wort

A herbal preparation from *Hypericum perforatum* used for mood symptoms but capable of serious drug interactions.

ST Segment

The isoelectric period between ventricular depolarization and repolarization (T wave). It corresponds to the plateau phase of ventricular action potentials. ST segment elevation (>1 mm in limb leads or >2 mm in precordial leads) suggests acute myocardial infarction, pericarditis, or early repolarization.

St Vitus'S Dance

An older name for Sydenham chorea, involuntary movements following group A streptococcal infection.

ST-Segment Elevation

Elevation of the ECG ST segment above the isoelectric baseline, reflecting abnormal ventricular repolarization. It may indicate acute coronary occlusion but can also occur in pericarditis, early repolarization, ventricular aneurysm, or other conditions; interpretation requires clinical context and serial testing.

St. Anthony'S Fire

St. Anthony's fire is an historical name for ergot poisoning or, in some usage, erysipelas; modern diagnosis should identify the specific disease.

St. John'S Wort

Herbal supplement. CYP3A4 inducer, reduces efficacy of many drugs (oral contraceptives, warfarin, antiretrovirals).

Stab Wound

A penetrating sharp-force injury in which wound track depth exceeds skin defect length, produced by a pointed or sharp-edged implement. The skin defect approximates blade width only when perpendicular to Langer's lines, and a single-edged blade produces one sharp and one blunt end.

Stable Angina

Predictable exertional chest pain caused by temporary inadequate coronary blood flow.

Alternate Names:

Chronic stable angina

Stable Coronary Artery Disease

Coronary artery narrowing causing a consistent, predictable pattern of angina.

Alternate Names:

Stable ischemic heart disease

Stable Disease

An oncology response category in which a cancer has neither shrunk enough to meet criteria for partial response nor grown enough to meet criteria for progressive disease. It describes change over a specified assessment interval and does not mean that the cancer is cured or inactive.

Stabs

Brief, sharp, stabbing pains or injuries; the exact cause depends on site and context.

Stage 4 Prostate Cancer

Alternate index label; see *Metastatic (stage 4) prostate cancer*.

Stage I

A cancer stage that generally indicates limited disease, although the exact definition and prognosis depend on the cancer type and staging system.

Stage II

A cancer stage whose meaning depends on the cancer type and may indicate a larger primary tumor or limited local spread without distant metastasis.

Stage III

A cancer stage that often indicates more extensive local or regional disease, with the precise criteria depending on the cancer type.

Stage IV

A cancer stage that generally indicates spread to distant organs or sites. Prognosis and treatment vary by cancer type and individual factors.

Stages Of Change

A model describing gradual phases people pass through while changing behavior.

Alternate Names:

Transtheoretical model

Stages Of Labor

Four stages describing the progression from uterine contractions to delivery of the placenta. First stage: onset of regular contractions to full cervical dilatation (10 cm). Second stage: full dilatation to delivery of the baby. Third stage: delivery of the baby to delivery of the placenta.

Staghorn Calculi

Large branching kidney stones that fill part or all of the renal collecting system.

Staghorn Calculus

A branching kidney stone that fills part or all of the renal pelvis and calyces, often associated with infection.

Staging

Assessment of the extent of cancer, including the size of the primary tumor, lymph-node involvement, and distant metastases.

Staining

A laboratory technique applying dyes to microbial

smears or tissue sections to visualise microorganisms and reveal structural characteristics. Major methods include Gram, acid-fast, Giemsa, and fluorescent stains, providing rapid presumptive identification from clinical specimens and guiding initial empiric therapy.

Stainless Steel

A corrosion-resistant alloy containing iron and chromium, used for instruments, implants, orthodontic appliances, and some dental crowns. Its suitability depends on the grade, strength, and tissue exposure.

Stammering

Stuttering, a speech-fluency disorder involving repetitions, prolongations, or blocks.

Standard Deviation

A statistical measure of the spread of observations around their mean.

Standard Eye Exam

A comprehensive eye examination assessing visual acuity, refraction, pupils, eye movements, external structures, anterior eye, intraocular pressure, and ocular fundus as indicated.

Standard Of Care

The conduct, skill, diligence, and judgment a reasonably prudent practitioner of the same specialty would exercise under similar circumstances; the benchmark for medical negligence. It is established through guidelines, peer practice, and expert testimony rather than fixed statute.

Standard Precautions

Precautions for all patient care. Examples: hand hygiene, PPE, safe injection, respiratory hygiene.

Standardized Extract

An herbal extract adjusted to contain a specified amount of a selected constituent.

Standardized Uptake Value

Quantitative PET measure: tissue activity divided by injected dose per body weight. Normalizes for body size, injection dose. SUVmax: maximum pixel value in lesion. SUVmean: average lesion uptake. Background: blood pool, liver.

Stannosis

A usually mild form of pneumoconiosis caused by long-term inhalation of tin oxide dust. It may produce visible lung changes on imaging, often without major breathing impairment.

Stanozolol

A synthetic anabolic androgenic steroid with limited medical use and potential liver, lipid, and cardiovascular toxicity.

Stapedectomy

Surgical removal or replacement of the stapes, usually to treat otosclerosis and conductive hearing loss.

Stapedius

A small middle-ear muscle that helps limit movement of the stapes and may protect the inner ear from loud sounds.

Stapes

The smallest auditory ossicle, transmitting vibrations from the incus to the inner ear.

Staph Infection

Infections caused by *Staphylococcus aureus* (and coagulase-negative staphylococci), a gram-positive coccus in clusters.

Alternate Names:

Staphylococcus Infection (Staph Infection)

Staph Infections In The Hospital

Hospital staph infections are infections caused by *Staphylococcus* bacteria acquired or treated in healthcare settings and may be difficult to treat because of antibiotic resistance.

Staphylococcal Infection

An infection caused by *Staphylococcus aureus* or another *Staphylococcus* species. It may affect the skin, bone, joints, lungs, heart valves, blood, or other tissues. Diagnosis and antimicrobial treatment depend on the site, severity, and susceptibility of the organism.

Staphylococcal Meningitis

Staphylococcal meningitis is infection and inflammation of the meninges caused by *Staphylococcus* bacteria, often after surgery, trauma, bloodstream infection, or an invasive device.

Staphylococcal Pneumonia

Severe pneumonia from *S. aureus*, often post-influenza. May cause necrotizing pneumonia, lung abscess, and empyema.

Staphylococcal Scalded Skin

A toxin-mediated *Staphylococcus aureus* infection causing superficial separation and tenderness of the outer skin, usually in infants or young children. Widespread redness and fragile blisters may rupture to leave painful, moist areas resembling a scald burn and require prompt medical assessment.

Staphylococcal Scalded Skin Syndrome

A toxin-mediated skin disorder caused by *Staphylococcus aureus*, usually affecting infants or young children and producing widespread superficial blistering and peeling of the skin.

Staphylococcus

A genus of gram-positive cocci that includes *S. aureus* and common skin colonizers and pathogens.

Staphylococcus Aureus

A gram-positive, catalase-positive, coagulase-positive coccus that causes a wide range of infections including skin infections, pneumonia, endocarditis, and osteomyelitis. *S. aureus* produces numerous virulence factors including toxins, enzymes, and surface proteins. MRSA strains carry the *mecA* gene encoding PBP2a.

Staphylococcus Infection (Staph Infection)

Alternate index label for Staph Infection.

Staple (Vascular)

A small metal or polymer device used to close tissue or connect vascular structures.

Alternate Names:

Vascular

Stapling

Closing or joining tissue with metal or absorbable surgical staples. Staples may be used for skin, blood vessels, or parts of the digestive tract and must be selected and removed or absorbed appropriately.

Starch

A plant polysaccharide digested to glucose and an important dietary carbohydrate.

Starch Poisoning

Starch poisoning involves toxic or environmental exposure to starch poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Stargardt Disease

The most common inherited macular dystrophy, usually autosomal recessive and caused by *ABCA4* gene mutations. Presents in childhood or adolescence with progressive central vision loss, retinal flecks, and bull's-eye maculopathy; no curative treatment exists.

Starling Forces

Starling forces describe the balance of hydrostatic and oncotic pressures that govern fluid movement across capillary walls, determining the distribution of fluid between the intravascular and extravascular compartments.

Starvation

Severe prolonged deficiency of food or energy intake causing wasting, metabolic adaptation, and organ dysfunction.

Stasis

Slowing or stopping of normal flow, such as blood, lymph, or intestinal contents.

Stasis Dermatitis

Inflammation of the lower-leg skin caused by chronic venous hypertension and edema. It may cause itching, redness, scaling, brown discoloration, weeping, or ulcers around the ankles. Management focuses on treating venous disease, compression when safe, skin care, and evaluation for infection or arterial insufficiency.

Stasis Dermatitis And Ulcers

Stasis dermatitis and ulcers result from chronic venous hypertension in the lower legs, causing swelling, brown skin change, inflammation, itching, and slow-healing wounds.

Stasis Ulcer

A stasis ulcer is a chronic lower-leg wound caused by venous hypertension and poor return of blood, commonly near the ankle and associated with edema and skin changes.

STAT

A medical instruction meaning immediately or with urgent priority.

Statin

An inhibitor of HMG-CoA reductase that lowers hepatic cholesterol synthesis and increases hepatic LDL-receptor expression. Statins reduce atherosclerotic cardiovascular events; adverse effects include muscle symptoms, rare severe myopathy, and liver-enzyme abnormalities.

Statin-Induced Myopathy

The spectrum of toxic, non-autoimmune adverse muscle reactions caused by 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase inhibitors (statins). Encompasses common statin-associated myalgias (muscle aching, stiffness, or weakness without serum creatine kinase elevation, affecting up to 5–10% of patients), statin-induced toxic myopathy (proximal muscle weakness with

CK elevation 10 to 50 times the upper limit of normal), and rare acute rhabdomyolysis, all of which resolve completely within weeks to months following drug discontinuation, distinguishing them from autoantibody-mediated anti-HMGCR necrotizing myopathy.

Alternate Names:

Statin-Associated Muscle Symptoms (SAMS);
Statin Myotoxicity

Statins

A class of medicines that inhibit hepatic HMG-CoA reductase, lower low-density lipoprotein (LDL) cholesterol, and reduce the risk of heart attack and ischemic stroke in appropriate patients. Examples include atorvastatin, rosuvastatin, simvastatin, and pravastatin; muscle symptoms, drug interactions, and rare serious muscle injury require clinical consideration.

Alternate Names:

HMG-CoA reductase inhibitors

Statins In Diabetes

Use of HMG-CoA reductase inhibitors to reduce atherosclerotic cardiovascular risk in people with diabetes. The decision, intensity, and treatment target are individualized according to age, cardiovascular risk, kidney function, and other clinical factors.

Station

The relationship of the fetal presenting part to the level of the ischial spines, measured in centimeters.

Status Asthmaticus

A severe asthma exacerbation that fails to respond adequately to initial treatment and may cause respiratory failure.

Status Epilepticus

A prolonged seizure or recurrent seizures without recovery between them, requiring emergency treatment.

Status Migrainosus

A severe migraine attack that persists for more than 72 hours and may require urgent medical treatment, especially when dehydration, vomiting, or medication overuse is present.

Stay Active And Exercise When You Have Arthritis

Regular low-impact activity and strengthening can reduce stiffness and maintain function in arthritis when adjusted to pain, joint stability, and medical advice.

Stay Away From Asthma Triggers

Avoiding asthma triggers such as smoke, allergens, infections, fumes, and cold air can reduce symptoms and attacks, alongside prescribed controller and reliever treatment.

STDs

Alternate index label; see *Sexually transmitted diseases (STDs)*.

Steady State

The condition where the rate of drug administration equals the rate of elimination, resulting in constant average plasma concentrations with repeated dosing. During regular dosing at fixed intervals, steady state is achieved after approximately 4 to 5 half-lives, regardless of the dosing frequency or dose size.

Steal Syndrome

Ischemia of distal extremity due to blood flow diversion to bypass graft or AV fistula. Most common: upper extremity after AV fistula creation. Symptoms: hand pain, pallor, numbness, weakness. Treatment: banding, ligation, revision.

Steam Iron Cleaner Poisoning

Steam iron cleaner poisoning involves toxic or environmental exposure to steam iron cleaner poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Steatoma

A fatty or sebaceous mass; the term may refer to a lipoma or, in the ear, a cholesteatoma depending on context.

Steatorrhea

The passage of excess fat in the stool, causing greasy, pale, foul-smelling, floating stools—a hallmark of fat malabsorption.

Steatorrhea

Excess fat in stools, causing pale, bulky, greasy, foul-smelling stools and suggesting fat malabsorption.

Steele-Richardson-Olszewski Syndrome

Alternate index label; see *Progressive supranuclear palsy*.

Stein-Leventhal Syndrome

Alternate index label; see *Polyendocrine metabolic ovarian syndrome*.

Stem Cell

A cell that can make more stem cells and develop into one or more specialized cell types. Some stem cells produce blood cells; others help form or repair different tissues.

Stem Cell Donation

The collection of healthy blood-forming stem cells from a donor for transplantation into another person. Cells may be collected from the bloodstream by apheresis after medication increases their number in the blood, or directly from bone marrow under anesthesia. Donor and recipient compatibility, especially human leukocyte antigen matching, helps improve transplant success.

Stem Cell Transplantation

Hematopoietic stem cell transplantation replaces diseased or ablated bone marrow with healthy donor or autologous stem cells.

STEMI

ST-elevation myocardial infarction: transmural my-

ocardial necrosis from acute total occlusion of a coronary artery, an emergency requiring immediate reperfusion. ECG: ST-segment elevation in contiguous leads (or new LBBB); troponin elevated.

Stenosing Tenosynovitis

Alternate index label; see *Trigger finger*.

Stenosis

Abnormal narrowing of a passage, vessel, valve, or opening.

Stenosis Lumbar (Lumbar Stenosis)

Alternate index label for Lumbar Stenosis.

Stenosis Of The Left Anterior Descending Artery

Narrowing of the left anterior descending coronary artery, which can reduce blood flow to the front of the heart and cause ischemia.

Stenosis Spinal (Lumbar Stenosis)

Alternate index label for Lumbar Stenosis.

Stenosis, Aortic Valve

Alternate index label; see *Aortic valve stenosis*.

Stenosis, Mitral Valve

Alternate index label; see *Mitral valve stenosis*.

Stenosis, Pulmonary Valve

Alternate index label; see *Pulmonary valve stenosis*.

Stenosis, Pyloric

Alternate index label; see *Pyloric stenosis*.

Stenosis, Spinal

Alternate index label; see *Spinal stenosis*.

Stenotic

Stenotic means narrowed, as in a stenotic heart valve, blood vessel, airway, intestine, or other passage.

Stenotrophomonas Maltophilia Resistance

Resistance in *Stenotrophomonas maltophilia* results from intrinsic and acquired mechanisms that limit the activity of several antibiotics. Treatment should be guided by the site of infection, susceptibility testing, illness severity, and specialist advice.

Stent

A tube or scaffold placed inside a vessel or other passage to keep it open.

Stent (Vascular)

A small mesh or tube placed inside a narrowed or blocked vessel to help keep it open.

Alternate Names:

Vascular

Stent Balloon (Vascular)

A balloon catheter used to expand a stent against the wall of a blood vessel.

Alternate Names:

Vascular

Stent Graft

A fabric-covered stent placed inside a blood vessel to reinforce the vessel wall or exclude an aneurysm from blood flow.

Stent Placement

Image-guided deployment of stent. Vascular, biliary, ureteral, esophageal. Maintains patency.

Stent Restenosis

Re-narrowing of a blood vessel within or adjacent to a previously placed stent, usually caused by tissue growth, recurrent atherosclerosis, or thrombosis.

Stent-Graft (Vascular)

A fabric-covered vascular stent used to reinforce a vessel or exclude an aneurysm or leak.

Alternate Names:

Vascular

Stenting

Placement of mesh tube to keep vessel open. Drug-eluting stents reduce restenosis.

Stepping Reflex

A primitive reflex present from birth to 2 months, elicited by holding the infant upright with feet touching a flat surface. The infant makes stepping-like movements as if walking. Disappears by 2 months as voluntary motor control develops. Reappears around 9-10 months as voluntary walking emerges.

Steps To Take Before You Get Pregnant

Preconception care includes reviewing health conditions and medicines, folic acid, vaccinations, dental and genetic considerations, substance avoidance, and planning prenatal care.

Stereognosis

The ability to identify an object by touch without looking, requiring intact cortical sensory function.

Stereotactic

Stereotactic describes a technique using three-dimensional coordinates to target a precise location in the brain or another body structure.

Stereotactic Biopsy

Image-guided biopsy using three-dimensional coordinates to target a lesion that may be difficult to localize by direct palpation. It is used in the breast and, with specialized systems, the brain, lung, and other organs.

Stereotactic Body Radiation Therapy

A highly precise form of external radiation delivering high-dose ablative treatment in few fractions, used for early-stage lung cancer and selected liver, pancreatic, spinal, and kidney tumors.

SRT

Stereotactic radiotherapy is a highly precise form of external radiation therapy that uses three-dimensional imaging and careful patient positioning to deliver focused radiation to a small target.

Fractionated SRT is commonly given in two to five treatments, rather than a single session.

Stereotactic Radiosurgery

Single high-dose radiation to small target. Used for brain metastases, AVMs, acoustic neuroma.

Stereotactic Radiosurgery - CyberKnife

CyberKnife is a robotic system that delivers many precisely aimed radiation beams to a tumor or other target without an open operation.

Stereotactic Radiosurgery - Gamma Knife

Gamma Knife is a focused radiation system that delivers multiple precisely aimed beams, mainly to selected brain tumors and vascular or other brain lesions.

Stereotaxis

A technique using a three-dimensional coordinate system to target deep structures, especially in brain procedures.

Stereotypic Movement Disorder

A clinical condition involving stereotypic Movement Disorder; its causes, manifestations, severity, and complications guide diagnosis and management.

Sterile Field

A designated area maintained free of viable microorganisms during an invasive procedure. Only sterile equipment and materials may contact the field or the sterile portions of instruments.

Sterile Site

A body site or specimen type expected to contain no viable microorganisms in health, such as blood, cerebrospinal fluid, synovial fluid, or tissue from a closed space. Recovery of a pathogen from a sterile site is generally clinically significant, subject to contamination assessment.

Sterile Technique

Methods used to prevent introduction of microor-

ganisms into a sterile field, wound, tissue, device, or body site through hand hygiene, barriers, aseptic handling, and appropriate preparation.

Sterilisation

Destruction or removal of all forms of microbial life, or permanent contraception depending on context.

Sterility

Inability to reproduce, or complete absence of viable microorganisms from an object or preparation.

Sterilization

A process eliminating all microbial life including spores, achieving a sterility assurance level of at least 10^{-6} .

Sternal Exploration Or Closure

Surgical reopening or repair of the breastbone after median sternotomy, performed to treat infection, bleeding, instability, dehiscence, or other postoperative chest complications.

Sternoclavicular Joint

A saddle-type synovial joint between the clavicle and manubrium of the sternum. It is the only bony connection between the upper limb and axial skeleton and permits movement of the shoulder girdle.

Sternocleidomastoid Muscle

Strap-like anterior neck muscle from manubrium and medial clavicle to the mastoid process. Innervated by CN XI and C2-C3. Divides neck into anterior/posterior triangles. Unilaterally: laterally flexes to opposite side, rotates face to same side. Bilaterally: flexes neck, assists forced inspiration.

Sternotomy

Surgical division of the sternum, usually along the midline, to provide access to the heart or mediastinal structures.

Sternum

The breastbone forming the front of the central chest wall.

Sternutation

Sneezing, a reflex forceful expulsion of air through the nose and mouth triggered by irritation of nasal or upper-airway sensory receptors.

Steroid Myopathy

A toxic, non-inflammatory myopathy caused by prolonged administration of exogenous corticosteroids or by endogenous hypercortisolemia (Cushing syndrome). Characterized by the insidious, symmetrical development of painless proximal muscle weakness and atrophy, predominantly affecting the lower extremities, pelvic girdle, and quadriceps femoris, making rising from a chair or climbing stairs difficult. Serum creatine kinase and electromyography are characteristically normal; muscle biopsy demonstrates selective, non-necrotic atrophy of type II (fast-twitch glycolytic) muscle fibers without inflammatory infiltrates, reversing following corticosteroid dose reduction.

Alternate Names:

Corticosteroid-Induced Myopathy; Glucocorticoid Myopathy

Steroids

Hormone-like medicines or compounds with a four-ring chemical structure; corticosteroids are used to reduce inflammation.

Alternate Names:

Corticosteroids

Stertor

A low-pitched snoring or rattling breathing sound caused by upper-airway obstruction.

Stethoscope

An instrument for listening to sounds from the heart, lungs, blood vessels, and other body structures.

Stevens-Johnson Syndrome

A rare severe mucocutaneous reaction, usually to medicines or infection, with epidermal detachment and involvement of multiple mucosal surfaces.

Alternate Names:

SJS

STI

An STI is a sexually transmitted infection spread through sexual contact, including infections caused by bacteria, viruses, parasites, or fungi.

Stickler Syndrome

An autosomal dominant connective tissue disorder caused by mutations in collagen genes (COL2A1 most common, also COL11A1, COL11A2, COL9A1).

Stiff Baby Syndrome

A clinical condition involving stiff Baby Syndrome; its causes, manifestations, severity, and complications guide diagnosis and management.

Stiff-Person Syndrome

A rare, disabling autoimmune neurological and neuromuscular disorder characterized by progressive, fluctuating muscular rigidity and stiffness of axial, thoracic, and proximal limb muscles, with superimposed agonizing, violent episodic muscle spasms triggered by sudden auditory, tactile, or emotional stimuli. Pathophysiologically driven by autoimmune impairment of GABAergic neurotransmission, with > 80% of patients harboring high-titer autoantibodies against glutamic acid decarboxylase 65 (*GAD65*) in serum and cerebrospinal fluid (or less commonly anti-amphiphysin in paraneoplastic cases). Results in fixed lumbar hyperlordosis, an impaired “tin-man” gait, and frequent falls, responsive to GABA-enhancing therapies and immunomodulation.

Alternate Names:

SPS; Stiff-Man Syndrome; Moersch-Woltman Syndrome

Stiffness

Difficulty or discomfort with movement, often due to joint, muscle, neurologic, or inflammatory disease.

Stigma

A mark, characteristic, or social discrediting label; in medicine it may refer to a sign or to harmful social attitudes.

Stilboestrol

An older name for diethylstilbestrol, a synthetic estrogen no longer used routinely because of serious risks.

Stilet

A slender pointed instrument used to guide, clear, or stiffen a cannula or tube.

Still Disease

An inflammatory disease that can affect the whole body. In children it is called systemic juvenile idiopathic arthritis; in adults it is called adult-onset Still disease. It can cause fever, joint pain or swelling, and a rash.

Alternate Names:

Still's Disease; Arthritis Still (Still's Disease)

Stillbirth

The birth of a baby showing no signs of life after the locally defined gestational or birth-weight threshold.

Still's Murmur

The most common innocent (functional) heart murmur of childhood, typically identified between 2 and 7 years of age. Characterized on auscultation as a short, low-pitched, musical, vibratory, or twanging-string midsystolic ejection murmur heard loudest at the lower left sternal border or mid-precordium without radiation to the axilla or back. The murmur intensifies in high-output states (fever, exercise, anemia) and when lying supine, and softens or disappears when standing or during Valsalva maneuver; it is associated with a completely normal heart and disappears spontaneously by adolescence.

Alternate Names:

Innocent Vibratory Murmur; Normal Pediatric Murmur; Twanging-String Murmur

Stimulant

A substance that increases activity of the central nervous system or other body systems.

Stimulant Use Disorder

A problematic pattern of stimulant use (cocaine,

amphetamines, methamphetamine, MDMA) leading to significant impairment. Stimulants increase dopamine and norepinephrine activity. Withdrawal is characterized by fatigue, depression, increased appetite, hypersomnia, psychomotor retardation, and vivid unpleasant dreams.

Stimulants

Stimulants increase activity in the central nervous system or sympathetic system and may improve alertness but can cause insomnia, anxiety, dependence, or cardiovascular effects.

Stimulus

A stimulus is a physical, chemical, sensory, or social change that evokes a response from a cell, tissue, organism, or person.

Stingray

Stingray involves toxic or environmental exposure to stingray; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Stings

Injuries caused by venom injected through an insect, animal, or plant structure.

Stings, Bee

Alternate index label; see *Bee sting*.

Stings, Jellyfish

Alternate index label; see *Jellyfish stings*.

Stings, Scorpion

Alternate index label; see *Scorpion sting*.

Stippling (Tattooing)

Patterned dermal abrasions and punctate hemorrhages surrounding a firearm entrance wound, caused by unburned and partially burned gunpowder grains striking the skin at close range (15–75 cm).

Alternate Names:

Tattooing

Stitch

A suture used to close tissue, or a sharp transient pain, often in the side during exercise.

Stochastic Effects

Random radiation effects with no threshold. Cancer, genetic effects. Probability increases with dose.

Stoddard Solvent Poisoning

Stoddard solvent poisoning involves toxic or environmental exposure to stoddard solvent poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Stokes-Adams Syndrome

Transient loss of consciousness caused by an intermittent cardiac arrhythmia, usually profound bradycardia or heart block.

Stoma

A natural opening or a surgically created opening connecting an organ to the skin or another organ.

Stoma Complications

Problems involving an opening made from the intestine or urinary tract to the skin. They include skin irritation, bleeding, retraction, prolapse, narrowing, blockage, high output, infection, and a parastomal hernia; sudden severe pain, color change, or no output needs urgent care.

Stomach

A muscular digestive organ between the oesophagus and small intestine that stores and begins digestion of food.

Stomach Cancer

Malignant cancer arising in the stomach, most often an adenocarcinoma. Risk factors include *Helicobacter pylori* infection, smoking, certain dietary exposures, chronic gastric inflammation, and inherited syndromes. Symptoms, stage, and tumor type guide treatment.

Alternate Names:

Cancer, Gastric Cancer, Stomach

Stomach Cramps

Intermittent gripping or spasmodic upper-abdominal pain caused by gas, infection, indigestion, ulcer disease, gallbladder or pancreatic disease, obstruction, or other causes.

Stomach Fat (Belly Fat, Abdominal Fat)

Adipose tissue stored beneath the skin or around abdominal organs; excess visceral fat is metabolically active and associated with insulin resistance, fatty liver, and cardiovascular risk.

Stomach Flu

Alternate index label; see *Viral gastroenteritis (stomach flu)*.

Stomach Paralysis

Stomach paralysis, or gastroparesis, is delayed emptying of stomach contents without obstruction, causing early fullness, nausea, vomiting, and bloating.

Stomach Tube

A tube passed through the mouth or nose into the stomach for drainage, feeding, sampling, or treatment.

Stomach Ulcer

An open sore in the lining of the stomach, commonly associated with *Helicobacter pylori* infection or medicines such as NSAIDs.

Stomach Ulcer (Peptic Ulcer)

Alternate index label for Peptic Ulcer.

Stomach Washout

Gastric lavage, irrigation and suction of stomach contents in selected poisoning or other emergencies.

Stomach, Diseases Of

Disorders affecting the stomach, including gastritis, ulcers, motility disorders, obstruction, and tumors.

Stomatitis

Inflammation, soreness, redness, or ulcers of the lining of the mouth. Causes include infection, irri-

tation, nutritional deficiency, medicines, and cancer treatment; severe pain, dehydration, fever, or inability to eat or drink needs medical care.

Stomatodynia

Alternate index label; see *Burning mouth syndrome*.

Stone

A hard concretion formed in a body organ or duct, commonly from mineral salts.

Stonefish Sting

Stonefish sting involves toxic or environmental exposure to stonefish sting; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Stool

Fecal material expelled from the bowel, composed of water, undigested residue, bacteria, cells, mucus, and bile pigments; color, form, frequency, and blood can provide clinical clues.

Stool C Difficile Toxin

Stool C difficile toxin is a diagnostic or laboratory investigation; interpretation depends on the specimen, method, reference interval, timing, and clinical question.

Stool Examination

Laboratory analysis of a stool sample for blood, infection, parasites, inflammation, malabsorption, or other digestive abnormalities.

Alternate Names:

Stool test

Stool Gram Stain

Microscopic examination of a stool specimen after Gram staining; it may provide limited information about bacterial flora or inflammatory cells but is not a general test for infectious diarrhea.

Stool Guaiac Test

A chemical fecal occult blood test that detects the peroxidase activity of heme in stool. Dietary factors,

medicines, and specimen collection can affect results. It is a screening or detection test rather than a diagnosis; a positive result requires clinical follow-up and usually evaluation for a gastrointestinal source.

Stool Holding

Alternate index label; see *Encopresis*.

Stool Occult Blood Test

Alternate index label; see *Fecal occult blood test*.

Stool Ova And Parasites Exam

Microscopic examination of stool for parasite eggs, cysts, larvae, or other forms, sometimes requiring multiple specimens to improve detection.

Stools

Feces, the solid waste expelled from the digestive tract.

Stools - Foul Smelling

Strongly odorous feces that may result from diet, infection, bleeding, malabsorption, altered gut bacteria, or rapid transit; persistent change with weight loss or diarrhea warrants review.

Stools - Pale Or Clay-Colored

Light-colored stool caused by reduced bile pigment reaching the intestine, potentially indicating bile-duct obstruction, gallbladder disease, liver disease, or certain transient conditions.

Stools – Floating

Feces that remain on the water surface, often from trapped gas or diet but sometimes from excess fat in malabsorption; persistent greasy, pale, foul-smelling stool needs evaluation.

Stop Smoking Support Programs

Counseling, quit plans, telephone or digital coaching, and medicines such as nicotine replacement, varenicline, or bupropion used to help people stop tobacco use.

STOPP/START

Screening Tool of Older Person's Prescriptions and Potential Older Person's Prescriptions (START). Identifies inappropriate prescribing.

Storage Diseases

A group of inherited metabolic disorders characterized by accumulation of substrates within cells due to specific enzyme deficiencies. Lysosomal storage diseases result from defects in lysosomal hydrolytic enzymes, leading to accumulation of undigested macromolecules within lysosomes.

Storing Your Medicines

Medicines should be stored in their labeled containers at the specified temperature, protected from moisture and children, and checked for expiry or disposal instructions.

Stork Bite

A stork bite is a common harmless pink birthmark, usually a nevus simplex on the forehead, eyelids, nose, or neck, that often fades during childhood.

Strabismus

Misalignment of the eyes, causing them to point in different directions.

Strain

An injury from overstretching or tearing a muscle or tendon, or a force applied to tissue.

Strain Imaging

Echocardiographic or magnetic-resonance imaging that measures deformation of the myocardium during contraction and relaxation. It can detect abnormalities in myocardial function that may not be apparent from ejection fraction alone.

Strain Neck (Whiplash)

Alternate index label for Whiplash.

Strains

A strain is overstretching or tearing of a muscle or tendon, causing pain, tenderness, swelling, weakness,

or reduced function after force or overuse.

Strains (Sprained Ankle)

Alternate index label for Sprained Ankle.

Stranguary

An obsolete term for painful, difficult, or frequent urination.

Strangulated Hernia

A hernia in which the blood supply to the trapped tissue is cut off, causing ischaemia and possible tissue death; it is a surgical emergency.

Strangulation

Compression of a body structure, especially the neck or a hernia, that compromises blood flow or airway.

Strapping

Application of adhesive tape or bandage to support, protect, or restrict movement of a body part.

Strategies For Getting Through Labor

Practical approaches for coping with labor, including breathing, movement, position changes, support people, relaxation, analgesia, and communication with the maternity team.

Stratum Corneum

The outermost layer of the epidermis, composed mainly of dead keratinized cells.

Strawberry Hemangioma

Alternate index label; see *Hemangioma*.

Strength

The ability of a muscle or muscle group to exert force.

Strength Testing

Manual muscle testing, dynamometry, isokinetic testing. Quantifies weakness, tracks progress.

Strength Training

Exercise using resistance to increase muscle strength and capacity.

Alternate Names:

Resistance training; weight training

Strep Throat

A bacterial infection of the throat and tonsils caused by group A Streptococcus, also called Streptococcus pyogenes. It commonly causes a sudden sore throat, pain when swallowing, fever, and swollen or red tonsils, sometimes with white patches.

Strep Throat

Alternate index label for GAS.

Streptococcal Cellulitis With Lymphangitis

Streptococcal infection of the skin and subcutaneous tissue with inflammation of the draining lymphatic vessels.

Streptococcal Infection

Infections caused by Streptococcus pyogenes (Group A Strep, GAS). Common presentations: pharyngitis (strep throat), scarlet fever (pharyngitis with erythematous rash), impetigo, cellulitis, and necrotizing fasciitis.

Streptococcal Infections

Streptococcal bacteria can cause throat infection, scarlet fever, skin infection, pneumonia, invasive bloodstream disease, and post-infectious complications. Diagnosis and antibiotic choice depend on site and testing; severe pain, rapid spread, sepsis signs, or breathing difficulty is urgent.

Streptococcal Pharyngitis

Sore throat caused by infection with group A Streptococcus, often accompanied by fever and swollen cervical lymph nodes.

Streptococcal Pharyngitis (Strep Throat)

Alternate index label for Strep Throat.

Streptococcal Screen

A streptococcal screen looks for group A Streptococcus, usually from a throat swab, to help diagnose strep throat.

Streptococcus

A genus of gram-positive cocci causing infections such as pharyngitis, pneumonia, cellulitis, and endocarditis.

Streptococcus Agalactiae

A gram-positive, beta-hemolytic streptococcus (Group B Streptococcus) that is a leading cause of neonatal sepsis, meningitis, and pneumonia. The organism colonises the gastrointestinal and genital tracts. Transmission occurs during birth. Risk factors include prematurity, prolonged rupture of membranes, and maternal colonisation.

Streptococcus Pneumoniae

A gram-positive, encapsulated diplococcus that causes pneumonia, otitis media, sinusitis, meningitis, and sepsis.

Streptococcus Pneumoniae (Pneumococcus)

A gram-positive bacterium that can cause pneumonia, meningitis, sinusitis, otitis media, and blood-stream infection; vaccines reduce disease risk.

Streptococcus Pyogenes

A gram-positive, beta-hemolytic streptococcus (Group A Streptococcus) that causes pharyngitis, skin infections, and post-infectious complications including rheumatic fever and glomerulonephritis. The organism produces streptolysins, erythrogenic toxin, and streptokinase.

Streptokinase

A thrombolytic protein that activates plasminogen and dissolves clots; use is limited by bleeding risk and alternatives.

Streptomycin

An aminoglycoside antibiotic used in selected tuber-

culosis and other infections.

Stress

A physiological or psychological response to perceived demands or threat.

Stress And Your Heart

Psychological stress can increase heart rate and blood pressure and may worsen cardiovascular risk through sleep and behavior changes; chest pain or severe symptoms require urgent assessment.

Stress Cardiomyopathy

A temporary weakening of the heart muscle, often following intense emotional or physical stress, with symptoms that can resemble acute coronary syndrome.

Stress Echocardiography

Echocardiography during exercise or pharmacological stress. Detects coronary artery disease by identifying wall motion abnormalities.

Stress Fracture

A bone injury caused by repetitive submaximal loading that exceeds the rate of remodeling and repair. It often presents with gradually increasing, localized activity-related pain and may progress to a complete fracture if loading continues.

Stress Fractures

Small cracks in bone caused by repetitive loading that exceeds remodeling capacity.

Stress Incontinence

Involuntary leakage of urine during activities that increase intra-abdominal pressure, such as coughing, sneezing, lifting, or exercise.

Stress Injuries

Tissue injuries caused by repeated loading or overuse without sufficient recovery. Examples include stress fractures, tendinopathy, and muscle strain.

Stress Response

Physiological changes that prepare the body to respond to a perceived threat.

Alternate Names:

Fight-or-flight response

Stress Shielding

Reduction in physiologic loading of adjacent bone caused by an implant that carries a disproportionate share of mechanical force. Chronic stress shielding may lead to bone resorption and compromise fixation.

Stress Tests

Diagnostic tests that assess organ function under controlled physical, pharmacologic, or physiologic stress, commonly evaluating the heart for inducible ischemia or abnormal rhythm.

Stress Ulcer Prophylaxis

Prevention of stress-related mucosal disease in critically ill patients. Risk factors: mechanical ventilation >48 hours, coagulopathy, trauma, burns, sepsis. Agents: H2 blockers, proton pump inhibitors. Duration: while risk factors present.

Stress Urinary Incontinence

Involuntary leakage of urine, significantly affecting psychological, social, and physical well-being. See urinary incontinence.

Stressors

Events, conditions, or circumstances that trigger a stress response.

Stretch Marks

Linear bands of atrophic skin, also called striae, caused by stretching and hormonal or corticosteroid effects; they commonly occur during pregnancy, growth, weight change, or steroid exposure.

Stretch Reflex

A monosynaptic spinal reflex in which stretching a muscle (e.g., a tap on the patellar tendon) excites muscle spindles, generating an afferent signal (Ia

fibers) that directly excites the alpha motor neurons of the same muscle, causing contraction and resistance to stretch.

Stria

A line, streak, or band in tissue, such as a stretch mark.

Alternate Names:

Striae

Striae

Linear bands or streaks in skin, commonly stretch marks caused by dermal connective-tissue change.

Striae Atrophicae

Linear bands of thin, wrinkled skin caused by stretching and disruption of dermal connective tissue; commonly called stretch marks.

Alternate Names:

stretch marks.

Striae Gravidarum

Stretch marks that develop during pregnancy, commonly on the abdomen, breasts, thighs, or buttocks. They may begin as red or purple lines and gradually fade to lighter bands.

Stricture (Stenosis)

An abnormal, pathological localized narrowing, constriction, or stenosis of a hollow, tubular anatomical conduit, duct, or lumen. Typically initiated by chronic mucosal inflammation, mechanical trauma, ischemia, surgical anastomosis, or caustic injury, leading to circumferential fibroblastic proliferation and collagenous scarring. Prominent clinical forms include urethral strictures (secondary to catheter trauma or gonococcal urethritis), esophageal strictures (caused by chronic GERD with reflux peptic injury, radiation, or lye ingestion), biliary strictures, and benign or malignant intestinal strictures (common in fibrostenosing Crohn disease).

Strictures Esophagus (Schatzki Ring)

Alternate index label for Schatzki Ring.

Stridor

A harsh, high-pitched breathing sound caused by upper-airway narrowing.

String Test For Intestinal Parasites

A string test collects a sample from the upper digestive tract on a swallowed capsule and string to look for selected intestinal parasites.

Stripping Of Membranes (Membrane Sweep)

A procedure performed during a cervical exam where the examiner inserts a finger through the internal os and sweeps the membranes circumferentially, separating them from the lower uterine segment. Releases endogenous prostaglandins, promoting cervical ripening and potentially initiating labor.

Alternate Names:

Membrane Sweep

Stroke

A sudden neurologic deficit caused by interrupted blood flow to the brain or bleeding within or around it.

Stroke Heat (Heat Stroke)

Alternate index label for Heat Stroke.

Stroke Index

Stroke index is a laboratory test, imaging method, instrument, score, or measurement used to characterize a specimen or clinical finding; interpretation depends on the method and clinical context.

Stroke Risk Factors

Stroke risk factors include high blood pressure, smoking, diabetes, high cholesterol, atrial fibrillation, prior stroke, and certain blood or vascular disorders.

Alternate Names:

Stroke Volume

The volume of blood ejected by the left ventricle

with each heartbeat, normally 60-100 mL. It is determined by three factors: preload, afterload, and myocardial contractility. Stroke volume equals end-diastolic volume minus end-systolic volume. Reduced stroke volume occurs in heart failure, hypovolemia, and cardiogenic shock.

Stroma

The supporting connective tissue of an organ, surrounding its functional cells.

Strongyloides Stercoralis

A soil-transmitted nematode that can infect the human intestine and cause strongyloidiasis.

Strongyloidiasis

A parasitic infection caused by *Strongyloides stercoralis*, which can become life-threatening in immunosuppression.

Structured Reporting

Reporting that uses a standard format, headings, and required fields to organize findings. It can improve completeness, consistency, and communication in imaging, pathology, and other clinical reports, but does not replace professional judgment.

Struma Ovarii

A specialized mature ovarian teratoma composed predominantly of thyroid tissue; it may be hormonally active or develop malignant thyroid-type transformation.

Struvite Stones

Infection stones composed of magnesium ammonium phosphate. Form in alkaline urine with urease-producing organisms including *Proteus* and *Klebsiella*. Staghorn calculi fill renal pelvis almost exclusively from *Proteus*. Treatment requires complete stone removal with antibiotics and urinary acidification. High recurrence if infection persists.

Strychnine

A poisonous alkaloid causing painful generalized muscle spasms and convulsions with preserved con-

sciousness.

Student'S Elbow

Olecranon bursitis, an inflammation of the bursa over the point of the elbow, often related to repeated pressure or trauma.

Study Population

The defined group of people included in a research study, selected according to eligibility criteria and used to estimate effects, associations, or outcomes.

Stuffy Nose

Stuffy nose refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Stuffy Or Runny Nose - Children

A stuffy or runny nose in a child is usually caused by a viral infection or allergy; breathing difficulty, dehydration, prolonged fever, or symptoms in an infant need prompt assessment.

Stuffy Or Runny Nose – Adult

A stuffy or runny nose in an adult commonly results from viral infection, allergy, irritants, or sinus disease; persistent, severe, bloody, or one-sided symptoms need review.

Stump Pain

Alternate index label; see *Residual limb pain*.

Stupor

A state of markedly reduced responsiveness from which a person can be aroused only with vigorous stimulation.

Sturge-Weber Syndrome

A neurocutaneous syndrome (phakomatosis) characterized by a facial capillary malformation (port-wine stain) in the V1 distribution of the trigeminal nerve (forehead and upper eyelid) and ipsilateral leptomeningeal angiomas (venous malformations of the pia mater). Caused by a somatic activating mutation in GNAQ.

Stuttering

A speech-fluency disorder involving repetitions, prolongations, blocks, or struggle during speech.

Sty (Stye)

Alternate index label for Stye.

Stye

A tender localized infection or inflammation of an eyelid gland, also called a hordeolum.

Alternate Names:

Sty (Stye)

Stye (Sty (Stye))

Alternate index label for Sty (Stye).

Styptics

Agents that help stop minor bleeding by promoting local vasoconstriction, coagulation, or tissue contraction.

Subacute

Between acute and chronic in onset, duration, or severity.

Subacute Bacterial Endocarditis

A slowly progressive infection of the endocardium or heart valves, usually caused by less-virulent bacteria and often involving previously abnormal valves.

Subacute Combined Degeneration

Damage to parts of the spinal cord, usually from vitamin B12 deficiency. It can cause numbness, weakness, poor balance, and difficulty walking.

Subacute Combined Degeneration Of The Cord

Spinal-cord degeneration caused mainly by vitamin B12 deficiency, affecting posterior and lateral columns.

Subacute Rehabilitation

Less intensive than acute rehab. For patients needing continued therapy but not as intensive. Skilled nursing facility setting.

Subacute Sclerosing Panencephalitis

A rare progressive brain disease occurring years after measles infection, caused by persistent defective measles virus.

Subacute Thyroiditis

Inflammatory condition of the thyroid most likely viral in etiology presenting with painful tender thyroid enlargement, fever, and malaise. Associated with elevated erythrocyte sedimentation rate and C-reactive protein.

Subaortic Stenosis

Narrowing of the left-ventricular outflow tract just below the aortic valve, often caused by a fibrous ridge or tunnel and producing a pressure gradient, murmur, and possible aortic-valve injury.

Subarachnoid

Situated beneath the arachnoid membrane surrounding the brain and spinal cord.

Subarachnoid Haemorrhage

Bleeding into the space around the brain containing cerebrospinal fluid, often causing sudden severe headache.

Subarachnoid Hemorrhage

Bleeding into the subarachnoid space, the area between the arachnoid membrane and the pia mater surrounding the brain, most commonly caused by rupture of a cerebral (berry) aneurysm.

Subarachnoid Space

The space between the arachnoid and pia mater containing cerebrospinal fluid.

Subareolar Abscess

A pus-filled infection beneath the areola, often associated with periductal inflammation and smoking, causing a painful lump, redness, drainage, or recurrent fistula.

Subchorionic Hemorrhage

Bleeding between the chorion and the uterine wall

caused by partial separation of the gestational sac membranes. It may produce vaginal bleeding in early or mid-pregnancy and is assessed according to bleeding, gestational age, and ultrasound findings.

Subclavian

Relating to the vessels or structures beneath the clavicle.

Subclavian Artery

The artery supplying the upper limb, arising on the right from the brachiocephalic trunk and on the left directly from the aortic arch, passing over the first rib and becoming the axillary artery. Branches: vertebral, internal thoracic, thyrocervical trunk, costocervical trunk.

Subclavian Steal Syndrome

A constellation of neurological and vascular symptoms resulting from severe stenosis or occlusion of the proximal subclavian artery (prior to the vertebral artery origin). During ipsilateral upper-extremity exertion, blood flow reverses down the ipsilateral vertebral artery to supply the arm, "stealing" blood from the vertebrobasilar cerebral circulation and causing vertigo, syncope, visual disturbances, and arm claudication.

Subclavian Vein

The continuation of the axillary vein beyond the lateral border of the first rib. It joins the internal jugular vein behind the sternoclavicular joint to form the brachiocephalic vein.

Subclavian Vein Thrombosis

Formation of a blood clot in the subclavian vein, often associated with central venous catheterization, thoracic outlet syndrome, or Paget-Schroetter syndrome (effort-related thrombosis in young, active individuals). Presents with arm swelling, pain, and cyanosis.

Subclinical

Present without obvious symptoms or signs, often detectable only by testing.

Subclinical Disease

A clinical condition involving subclinical Disease; its causes, manifestations, severity, and complications guide diagnosis and management.

Subclinical Hypothyroidism

A condition with an elevated thyroid-stimulating hormone level and normal circulating free thyroid-hormone levels, often without symptoms or with only mild nonspecific symptoms.

Subclinical Infection

An infection that produces no noticeable symptoms or signs but may still be detectable by testing and may sometimes be transmissible.

Subconjunctival Hemorrhage

Bleeding beneath the conjunctiva that appears as a sharply demarcated red patch on the white of the eye; it is usually harmless but recurrent cases or trauma require assessment.

Subconjunctival Hemorrhage (Broken Blood Vessel In Eye)

A small blood vessel broken beneath the clear surface of the eye, causing a sharply demarcated red patch that usually resolves without treatment.

Alternate Names:

Broken Blood Vessel In Eye

Subconscious

Mental processes outside immediate awareness; it is a psychological term rather than a specific diagnosis.

Subcorneal Pustular Dermatoses

A chronic relapsing skin disorder characterized by superficial sterile pustules beneath the stratum corneum, often on the trunk and flexural areas. It may be associated with IgA or other monoclonal gammopathies and can resemble pustular psoriasis.

Subcu

Subcu is an abbreviation for subcutaneous, meaning beneath the skin; clear clinical instructions should spell out the route.

Subculture

The transfer of microorganisms from one culture to fresh culture medium, used to maintain pure cultures, isolate colonies, increase viability, prepare standardized inocula, and perform identification/susceptibility testing. On solid media, subculture aims to obtain isolated single colonies from a mixed or stock culture.

Subcutaneous

Subcutaneous means beneath the skin; injections by this route deliver medicine into the fatty tissue between skin and muscle.

Subcutaneous Injections

Administration of a medicine into the fatty tissue beneath the skin, commonly used for insulin, anticoagulants, and selected biologic medicines.

Subcutaneous Emphysema

Air in the subcutaneous tissues, often from pneumothorax, pneumomediastinum, or chest trauma. It causes a characteristic crepitus on palpation. It is usually benign and resolves when the underlying air leak is treated. Severe cases may cause skin necrosis and compartment syndrome.

Subcutaneous Implantable Cardioverter-Defibrillator

A medical procedure, treatment, device, or care topic related to subcutaneous implantable cardioverter-defibrillator; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Subcutaneous Layer

The layer of tissue beneath the dermis, composed mainly of fat and connective tissue. It insulates the body, stores energy, and contains larger vessels and nerves.

Subcutaneous Tissue

Subcutaneous tissue (hypodermis, superficial fascia) lies beneath the dermis and connects skin to underlying structures. It contains adipose tissue (fat), blood vessels, nerves, and the superficial muscu-

lar aponeurotic system (SMAS). Subcutaneous fat provides insulation, cushioning, and energy storage.

Subdural

Situated, occurring, or located in the anatomical space between the inner surface of the dura mater and the outer surface of the arachnoid mater. Clinically famous for subdural hematoma, an intracranial hemorrhage resulting from tearing of bridging cortical veins as they cross the subdural space into dural venous sinuses, typically precipitated by head trauma or brain atrophy in elderly patients and chronic alcoholics; subdural empyema (a purulent intracranial collection secondary to contiguous sinusitis or otitis media requiring emergent surgical evacuation); and subdural hygroma.

Subdural Effusion

Accumulation of usually clear fluid in the subdural space, often after trauma, surgery, infection, or cerebrospinal-fluid disturbance and sometimes causing headache or pressure effects.

Subdural Haematoma

Accumulation of blood between the dura mater and arachnoid membrane, usually caused by tearing of bridging veins after head injury and sometimes developing gradually.

Subdural Hematoma

A collection of blood between the dura mater and the arachnoid mater, caused by tearing of bridging veins that traverse the subdural space.

Subdural Hemorrhage

Accumulation of blood between the dura mater and arachnoid membrane, usually caused by tearing of bridging veins after head trauma. Acute hemorrhage may produce rapid neurological deterioration, whereas chronic hemorrhage can develop gradually, particularly in older adults or patients with cerebral atrophy. Clinical features include headache, confusion, drowsiness, focal neurological deficits, seizures, or loss of consciousness. Diagnosis is established with neuroimaging, usually computed tomography, and treatment ranges from observation to surgical

evacuation according to size, symptoms, and mass effect.

Subendocardial Myocytes

Heart-muscle cells in the inner layer of the myocardium, which are especially vulnerable to reduced blood flow.

Subependymal Giant Cell Astrocytoma

A usually low-grade astrocytic tumor arising near the foramen of Monro, strongly associated with tuberous sclerosis complex. It may obstruct cerebrospinal-fluid flow and cause hydrocephalus.

Subependymoma

A rare, usually benign slow-growing ependymal tumor, commonly arising in the ventricular system of adults. Symptoms result from obstruction of cerebrospinal-fluid pathways or incidental detection.

Subfertility

Alternate index label; see *Infertility*.

Subgaleal Hemorrhage

A potentially life-threatening accumulation of blood in the space beneath the scalp's epicranial aponeurosis. It can expand rapidly in a newborn, particularly after vacuum-assisted delivery, and requires urgent recognition and treatment.

Subinvolution

Failure of an organ, especially the uterus after childbirth, to return toward its normal size.

Subjacent

Subjacent means lying directly beneath or below another structure.

Subjective

Based on a person's own experience or report rather than directly observable measurement.

Sublimation

A psychological defense process in which unacceptable impulses are redirected into socially acceptable

activity; in chemistry it is solid-to-gas transition.

Sublingual

Sublingual means beneath the tongue; medicines given this way can be absorbed through oral mucosa and may bypass much first-pass liver metabolism.

Sublingual Epidermoid Cyst

A benign cyst lined by stratified squamous epithelium beneath the tongue mucosa, presenting as a slow-growing midline swelling that may require surgical excision if symptomatic.

Sublingual Gland

One of the paired major salivary glands, located beneath the tongue. It produces mainly mucous saliva that enters the mouth through several small ducts and helps lubricate the oral cavity.

Subluxation

Partial dislocation of a joint, with some contact between articular surfaces remaining.

Submandibular Gland

One of the three paired major salivary glands, located in the submandibular triangle beneath the mandible. Mixed gland (predominantly serous) secreting saliva via Wharton's duct opening at the sublingual caruncle (floor of mouth). Saliva contains amylase, mucins, lysozyme, IgA.

Submucosa

A layer of connective tissue beneath the mucosa. It may contain blood vessels, nerves, lymphatic vessels, and glands and helps anchor and support the mucosal lining, especially in hollow organs.

Subphrenic Abscess

A collection of pus beneath the diaphragm, usually after abdominal infection or surgery.

Subq

Subq is an abbreviation for subcutaneous, referring to tissue beneath the skin and the route used for some injections.

Subscapular Artery

An artery arising from the axillary artery that supplies the scapular region and adjacent muscles.

Substance

A substance is a material with defined or variable chemical composition; medical effects depend on identity, dose, route, duration, and host factors.

Substance Abuse

Legacy, nonspecific label; see *Substance Use Disorder*.

Substance Abuse Disorder

Legacy label; see *Substance Use Disorder*.

Substance Abuse Psychiatry

The subspecialty focused on the evaluation and treatment of substance use disorders and co-occurring psychiatric conditions. Dual diagnosis patients require integrated treatment addressing both disorders simultaneously.

Substance Dependence

A state in which repeated substance use produces physiologic adaptation, so that reducing or stopping the substance may cause withdrawal. It is distinct from, but may occur with, substance-use disorder.

Substance Intoxication

A reversible substance-specific syndrome caused by recent use that produces clinically significant behavioral or physiologic changes. Diagnosis requires assessing timing, dose or exposure, expected pharmacology, co-ingestants, and medical complications.

Substance Use

The consumption of alcohol, drugs, or other chemical substances, which can lead to health, social, and psychological issues if used excessively.

Substance Use - Amphetamines

Amphetamine use can cause dependence, insomnia, anxiety, psychosis, high blood pressure, arrhythmias, overheating, and overdose; urgent symptoms require

emergency care.

Substance Use - Cocaine

Use of cocaine, a stimulant that can cause dependence, agitation, seizures, heart attack, stroke, dangerous overheating, and severe psychiatric or cardiovascular complications.

Substance Use - Inhalants

Inhalant misuse involves breathing fumes or gases from household or industrial products and can cause sudden cardiac arrest, brain injury, organ damage, dependence, or death.

Substance Use - LSD

LSD is a potent hallucinogen that alters perception, mood, and thought and can cause panic, accidents, persistent perceptual symptoms, or psychosis in susceptible people.

Substance Use - Marijuana

Marijuana or cannabis can impair attention, memory, coordination, and driving and may cause dependence, anxiety, psychosis, or respiratory harm when smoked.

Substance Use - Phencyclidine

Phencyclidine is a dissociative drug that can cause agitation, hallucinations, loss of coordination, high blood pressure, seizures, dangerous behavior, or coma.

Substance Use - Prescription Drugs

Use of prescription medicines in a way not directed by a clinician, including taking someone else's medicine, taking extra doses, or using them for intoxication; dependence, overdose, and interactions may result.

Substance Use Disorder

A problematic pattern of substance use leading to clinically significant impairment or distress, manifested by at least two of eleven criteria within a twelve-month period.

Substance Use In Pregnancy

Use of alcohol, tobacco, cannabis, opioids, cocaine, or other substances during pregnancy. Effects depend on the substance, dose, timing, and other factors and may include maternal complications, impaired fetal growth, preterm birth, or neonatal withdrawal. Evaluation and treatment should be individualized.

Substance Withdrawal

A substance-specific syndrome occurring after reduction or cessation of heavy or prolonged use. Withdrawal can range from discomfort to life-threatening complications such as seizures, delirium tremens, severe autonomic instability, or respiratory depression depending on the substance.

Substernal Goiter

Extension of thyroid tissue into the mediastinum, typically arising from the thyroid gland in the neck and growing downward through the thoracic inlet. It may cause compressive symptoms including dyspnea, dysphagia, and superior vena cava syndrome. Most are multinodular goiters. Diagnosis is by CT or MRI.

Substrate

A substance on which an enzyme acts or a material used by a biochemical pathway.

Subungual Exostosis

A benign osteocartilaginous outgrowth arising beneath the nail, usually of a toe or finger. It causes a painful firm subungual mass and nail deformity and is treated by excision when symptomatic.

Succenturiat Lobe

An accessory lobe of the placenta, connected to the main placenta by blood vessels that traverse the membranes. If undetected after delivery, the succenturiat lobe may be retained in the uterus, causing postpartum hemorrhage or infection. Diagnosis: careful inspection of the placenta and membranes after delivery.

Succenturiate Placenta

A succenturiate placenta has one or more accessory lobes separate from the main placenta, which can increase risk of retained tissue or fetal-vessel injury.

Succinate Dehydrogenase

Complex II of electron transport chain. Also TCA cycle enzyme. Produces FADH₂.

Succinylcholine

A depolarizing neuromuscular-blocking drug that produces rapid, brief skeletal-muscle paralysis by activating nicotinic acetylcholine receptors at the neuromuscular junction. It can cause hyperkalemia, bradycardia, myalgia, masseter spasm, prolonged apnea in pseudocholinesterase deficiency, and malignant hyperthermia.

Succumb

To succumb means to die from or be overcome by a disease, injury, exposure, or other serious process.

Succussion

A shaking movement or splash-like sound produced by fluid and gas in a cavity, historically assessed over the chest or abdomen.

Sucking Blister

A blister or worn area of skin in a newborn, usually on a hand, finger, forearm, or lip, caused by repeated sucking before or during feeding. It is usually harmless and heals on its own.

Sucking Reflex

A primitive reflex present from birth, elicited by placing an object in the infant's mouth. The infant rhythmically sucks on the object. Coordinated sucking and swallowing typically develop by 34-36 weeks gestational age. Premature infants may lack coordinated suck-swallow, requiring gavage feeding.

Sucrose

Common table sugar, a disaccharide of glucose and fructose.

Sucrose Lysis Test

A historical screening test for paroxysmal nocturnal haemoglobinuria in which patient red blood cells are exposed to serum in a low-ionic-strength sucrose solution, allowing complement-mediated lysis to be detected.

Suction

Removal of fluid, secretions, or material using negative pressure.

Suction (Vascular)

A suction device used during a vascular procedure to remove blood or other fluid from the operative field.

Alternate Names:

Vascular

Suction Device

Creates negative pressure for fluid removal. Components: canister, tubing, suction catheter. Used for airway clearance, surgical field, wound drainage.

Suction Tip

The end attachment of a suction device used to remove blood, fluid, secretions, or debris. Common types include the Yankauer tip for oropharyngeal suction and surgical suction tips such as the Frink tip.

Sudden Cardiac Arrest

An abrupt loss of effective heart activity, causing unresponsiveness and absent or abnormal breathing; it is a medical emergency.

Alternate Names:

Cardiac Arrest, Sudden

Sudden Cardiac Death

Unexpected death from a cardiac cause, usually occurring rapidly after symptom onset or without witnessed symptoms. Causes include coronary disease, cardiomyopathy, inherited arrhythmia syndromes, and bradyarrhythmia; prevention depends on the underlying risk.

Alternate Names:

Arrest Cardiac (Sudden Cardiac Death) Death Sudden Cardiac (Sudden Cardiac Death)

Sudden Infant Death Syndrome

SIDS: the sudden, unexplained death of an infant <1 year (peak 2-4 months) that remains unexplained after a thorough investigation (autopsy, scene examination, and review of clinical history). Risk factors: prone sleep, bed-sharing, soft bedding, overheating, maternal smoking, prematurity/low birth weight, and co-sleeping.

Sudden Infant Death Syndrome

The unexplained sudden death of an apparently healthy infant, usually during sleep and before one year of age.

Alternate Names:

Crib Death

Sudden Sensorineural Hearing Loss

Rapid sensorineural hearing loss, conventionally at least 30 dB over three contiguous frequencies within 72 hours. It requires urgent audiologic and otologic assessment because early treatment may improve recovery in selected patients.

Sudden Unexpected Death In Infancy

Unexpected death of an infant, requiring investigation to determine whether it is explained or unexplained.

Sudden Unexpected Infant Death

The sudden and unexpected death of an infant younger than 1 year, whether or not the cause is known before investigation. Sudden infant death syndrome is the unexplained subtype after complete scene, history, and autopsy evaluation.

Sudek'S Atrophy

An older name for complex regional pain syndrome with pain, swelling, autonomic change, and regional bone loss.

Sudorifics (Diaphoretics)

Pharmacological agents, substances, or thermal

modalities that stimulate, induce, or accelerate the secretion of sweat by eccrine sweat glands. Physiologically mediated by postganglionic sympathetic fibers that release acetylcholine onto muscarinic cholinergic receptors (M_3) of sweat glands. Direct or indirect muscarinic agonists (such as pilocarpine, bethanechol, or acetylcholinesterase inhibitors) exert strong sudorific actions and are utilized diagnostically in cystic fibrosis sweat chloride testing (pilocarpine iontophoresis) or therapeutically to alleviate severe xerostomia in Sjögren syndrome.

Suffocation

Inability to breathe adequately because of airway obstruction, oxygen deprivation, compression, or another cause.

Sugammadex

A selective relaxant binding agent that reverses aminosteroid neuromuscular blockers (rocuronium and vecuronium) by encapsulating them in a cyclodextrin molecule, forming a 1:1 complex that is renally excreted. Onset of action is within 2 to 3 minutes.

Sugar-Water Hemolysis Test

An older laboratory test assessing red-cell resistance to hypotonic sugar solution, historically used in evaluating paroxysmal nocturnal hemoglobinuria and now largely replaced by flow-cytometric testing.

Suicidal Ideation

Thoughts about killing oneself, ranging from passive (wish to be dead, wishes to fall asleep and not wake) to active (thoughts of killing oneself with varying degrees of intent, plan, and means). The C-SSRS grades severity from 1 (wish to be dead) to 5 (active suicidal ideation with specific plan and intent).

Suicide

Death caused by intentional self-inflicted action; suicidal thoughts and behavior require urgent, compassionate assessment.

Suicide And Suicidal Behavior

Thoughts, plans, attempts, or acts intended to cause

one's own death; any current intent, plan, access to means, or inability to stay safe requires immediate emergency support.

Suicide Risk Assessment

A clinical evaluation of suicidal thoughts, intent, plan, access to means, past attempts, self-harm, psychiatric and medical illness, substance use, protective factors, and immediate safety. It informs but does not replace ongoing clinical judgment and safety planning.

Sulcus

A groove or furrow on an organ or surface, especially the brain or heart.

Sulfa Drugs

A class of bacteriostatic antibiotics that competitively inhibit dihydropteroate synthase, an enzyme essential for folic acid synthesis in bacteria. See sulfonamides.

Sulfamethoxazole

A sulfonamide antibiotic used commonly with trimethoprim in selected infections.

Sulfasalazine

An anti-inflammatory medicine used for inflammatory bowel disease and rheumatoid or other inflammatory arthritis.

Sulfatide

A sulfated glycolipid found particularly in myelin and nervous tissue.

Sulfinpyrazone

An older uricosuric medicine that increases urinary excretion of uric acid. It was used historically for gout but is uncommon now and may interact with other medicines or increase bleeding risk.

Sulfite Sensitivity

Sulfite sensitivity is an adverse reaction to sulfite preservatives, sometimes causing wheeze, flushing, headache, or other symptoms, especially in people

with asthma.

Sulfonamide

A class of antibiotics (sulfamethoxazole, sulfadiazine, sulfasalazine) that inhibit bacterial dihydropteroate synthase, blocking folate synthesis—bacteriostatic. Often combined with trimethoprim (co-trimoxazole: TMP-SMX). Used for UTIs, PCP (pneumocystis pneumonia), toxoplasmosis, Nocardia, and MRSA.

Sulfonamides

Sulfonamides are drugs containing a sulfonamide group, including antibacterial agents that inhibit bacterial folate synthesis; allergic reactions, severe skin reactions, cytopenias, renal effects, and crystalluria are possible.

Sulfonylurea

A sulfonylurea is an oral diabetes medicine that closes pancreatic beta-cell ATP-sensitive potassium channels and stimulates insulin release; hypoglycaemia and weight gain are the main class risks.

Sulfonylureas

Oral antidiabetic agents that stimulate insulin secretion from pancreatic beta cells by blocking ATP-sensitive potassium channels on the beta cell membrane. This causes membrane depolarization, opening of voltage-gated calcium channels, calcium influx, and exocytosis of insulin-containing granules.

Sulfur Colloid Scan

Nuclear medicine imaging using Tc-99m sulfur colloid. Evaluates liver, spleen, bone marrow, lymph nodes.

Sulfuric Acid Poisoning

Sulfuric acid poisoning involves toxic or environmental exposure to sulfuric acid poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Sulindac Overdose

Sulindac overdose involves toxic or environmental exposure to sulindac overdose; dose, route, timing,

symptoms, and organ involvement determine urgency and treatment.

Sulphonamides

A class of antimicrobial agents containing the sulfonamide group (SO₂NH₂). Sulphonamides were the first effective systemic antibacterial agents (Prontosil, 1935—Gerhard Domagk). Mechanism: competitive inhibition of dihydropteroate synthase, blocking folate synthesis (para-aminobenzoic acid analogue)—bacteriostatic.

Sulphonylureas

Oral medicines that stimulate pancreatic insulin release to lower blood glucose in type 2 diabetes.

Sumatriptan

A serotonin 5-HT_{1B}/1D agonist used for acute migraine and cluster headache attacks.

Summer Diarrhoea

An older term for seasonal diarrheal illness, often referring to infectious diarrhea in children.

Sun Poisoning

Alternate index label; see *Polymorphous light eruption*.

Sun Protection

Sun protection reduces ultraviolet exposure through shade, protective clothing, and broad-spectrum sunscreen and helps prevent sunburn, premature skin aging, and skin cancer.

Sun Protection (Photoprotection)

Sun protection involves measures taken to reduce exposure to ultraviolet radiation from the sun, which is the primary environmental risk factor for skin cancer and photoaging.

Alternate Names:

Photoprotection

Sun-Sensitive Drugs (Photosensitivity To Drugs)

Alternate index label for Photosensitivity to Drugs.

Sun-Sensitive Drugs (Sun-Sensitive Drugs (Photosensitivity To Drugs))

Alternate index label for Sun-Sensitive Drugs (Photosensitivity to Drugs).

Sunburn

An acute inflammatory injury of the skin caused by excessive ultraviolet radiation, resulting in redness, pain, warmth, and sometimes blistering or peeling.

Sundowning

Late-day or evening worsening of confusion, agitation, wandering, or behavioral symptoms in some people with dementia or delirium, influenced by fatigue, light, routines, and unmet needs.

Sunflower Syndrome

A clinical condition involving sunflower Syndrome; its causes, manifestations, severity, and complications guide diagnosis and management.

Sunken Chest

Alternate index label; see *Pectus excavatum*.

Sunstroke

A life-threatening form of heat illness in which body temperature rises dangerously high and the nervous system is impaired.

Superantigen

A protein that bypasses antigen processing and non-specifically activates large T-cell populations by binding MHC class II and the T-cell receptor V region, triggering massive cytokine release. Staphylococcal TSST-1 and enterotoxins, and streptococcal pyrogenic exotoxins, cause toxic shock and scarlet fever.

Superaspirin

Superaspirin is an informal term for selective COX-2 inhibitors or other anti-inflammatory drugs developed to reduce gastrointestinal effects; cardiovascular risks remain relevant.

Superbug

A superbug is an informal term for a microorganism resistant to multiple antimicrobial drugs, making infection harder to treat and prevent from spreading.

Superficial

In anatomical directional nomenclature, situated near, closer to, or positioned on the surface of the body or relative to the external boundary of an organ (the opposite of deep). Used across clinical specialties to describe anatomical planes (superficial fascia, superficial femoral artery, superficial venous system like the great saphenous vein), burn depths (superficial first-degree burns confined entirely to the epidermis, such as mild sunburn), and surgical wound infections (superficial incisional surgical site infection involving only skin and subcutaneous tissue).

Superficial Fascia

A layer of loose connective tissue beneath the dermis, also called subcutaneous tissue or the hypodermis.

Alternate Names:
subcutaneous tissue.

Superficial Spreading Melanoma

The most common cutaneous melanoma subtype, characterized initially by radial growth across the epidermis and an irregularly pigmented patch or plaque before invasion develops.

Superficial Thrombophlebitis

Superficial thrombophlebitis is inflammation and clotting in a vein near the skin, causing a tender red cord; extension into deep veins must be assessed.

Superimposed Preeclampsia

Preeclampsia developing in a person who already has chronic hypertension. It may present with new or worsening proteinuria, thrombocytopenia, abnormal liver or kidney tests, neurologic symptoms, or other features of maternal organ dysfunction.

Superinfection

A new infection that develops during or after treat-

ment of a previous infection, often involving an antimicrobial-resistant organism or an organism not susceptible to the original therapy, such as *Clostridioides difficile* after antibiotic exposure.

Superior

In standard anatomical directional terminology, positioned above, higher, or oriented toward the head or cephalic pole of the body relative to a reference structure (the opposite of inferior; synonymous with cranial or cephalic). For example, the cervical spine is superior to the thoracic spine, and the superior vena cava returns systemic deoxygenated blood from anatomical regions superior to the respiratory diaphragm (head, neck, upper limbs, and thorax) to the right atrium of the heart.

Superior Mesenteric Artery

Second major abdominal aortic branch at L1, the midgut artery. Supplies distal duodenum through proximal two-thirds of transverse colon. Branches: inferior pancreaticoduodenal, jejunal, ileal, ileocolic, right colic, middle colic. Arises anterior to the pancreatic uncinate process, crosses the third duodenal part.

Superior Mesenteric Artery Syndrome

Compression of the duodenum between the aorta and the superior mesenteric artery, causing proximal duodenal obstruction. Presents with postprandial nausea, vomiting, and epigastric pain, often after significant weight loss that narrows the SMA-aortic angle. Treatment includes nutritional rehabilitation and surgical duodenal mobilization.

Superior Semicircular Canal Dehiscence

A defect or marked thinning of the bony covering of the superior semicircular canal that can produce sound- or pressure-induced vertigo, autophony, and conductive-appearing hearing findings. Diagnosis combines characteristic symptoms, physiologic testing, and high-resolution temporal-bone CT.

Superior Vena Cava

Formed by brachiocephalic vein junction at the first right costal cartilage, returning upper body deoxy-

generated blood to the right atrium. Approximately 7 cm, valveless. Related to right lung/pleura, right phrenic nerve, and azygos vein.

Superior Vena Cava Syndrome

Obstruction of SVC by tumor. Facial swelling, dyspnea, dilated neck veins. Emergency: radiation, stenting.

Alternate Names:

SVCS (Superior Vena Cava Syndrome)

Superior Vena Cava Syndrome

Alternate index label for Superior Vena Cava Syndrome.

Superior Vena Caval Obstruction

Compression or thrombosis of the superior vena cava impairing venous return from the head, neck, and upper extremities. Most commonly caused by malignancy (lung cancer, lymphoma) or indwelling central venous devices. Presents with facial swelling, plethora, distended neck veins, and upper extremity edema.

Supernumerary Nipples

Supernumerary nipples are extra nipples along the embryonic milk line, usually harmless but occasionally associated with underlying breast or urinary-tract developmental anomalies.

Supernumerary Rib

An extra rib beyond the typical twelve pairs, most commonly a cervical rib above the first thoracic rib, which may compress the brachial plexus or subclavian vessel causing thoracic outlet syndrome.

Supernumerary Teeth

Extra teeth beyond normal complement. Most common: mesiodens between maxillary centrals. May cause impaction or malocclusion.

Supervised Exercise Therapy

A structured, monitored exercise program used to improve walking ability and reduce claudication symptoms in people with peripheral artery disease.

The exercise plan is adjusted to the person's symptoms and functional capacity.

Supination

Rotation of the forearm so the palm faces upward or forward, or a related foot movement.

Supine

An anatomical body posture in which an individual lies horizontally on their back with the face, chest, and palms facing upward toward the ceiling (the opposite of prone). Standard positioning for resting physical examination (cardiovascular, abdominal, and neurological exams), routine surgical procedures under general endotracheal anesthesia, and non-contrast and contrast CT and MRI imaging acquisitions. Prolonged supine immobility poses significant risks of occipital and sacral pressure injury, supine hypotensive syndrome in late pregnancy (aortocaval compression), and positional obstructive sleep apnea.

Supine Hypotensive Syndrome

Compression of the inferior vena cava by the gravid uterus when the pregnant woman lies flat on her back after 20 weeks gestation. Results in decreased venous return, reduced cardiac output, and maternal hypotension, dizziness, pallor, and syncope.

Supine Position

A position in which the patient lies on the back with the face upward; it is the most commonly used surgical position.

Supplemental Oxygen Therapy

Administration of oxygen at a concentration above that of room air to treat or prevent clinically important hypoxemia. Delivery method and oxygen target depend on the illness, baseline oxygenation, risk of carbon-dioxide retention, and monitoring results; excessive oxygen can also cause harm.

Supplementation

Use of a vitamin, mineral, protein, fiber, or other nutrient preparation to correct a deficiency or meet a clinically identified increased requirement. The

indication, dose, interactions, product quality, and risk of toxicity should be assessed before use.

Supportive Care

Treatment that maintains essential body functions, relieves symptoms, prevents complications, and supports recovery while the underlying cause is treated.

Suppository

A solid preparation inserted into the rectum, vagina, or urethra where it dissolves or melts.

Suppression

Reduction or inhibition of a function, response, symptom, or process.

Suppuration

The biological process of forming, producing, discharging, or exuding pus (*pus*). Triggered by acute pyogenic bacterial invasion (most frequently by *Staphylococcus aureus*, *Streptococcus pyogenes*, or *Pseudomonas aeruginosa*), which attracts massive numbers of polymorphonuclear neutrophils that phagocytose microbes, release lytic enzymes, and undergo apoptosis. The resulting liquefactive necrosis of native parenchymal tissue produces a viscous, protein-rich exudate that either organizes into a localized, walled-off abscess cavity or discharges through sinus tracts and mucosal surfaces.

Supracondylar Humerus Fracture

The most common type of pediatric elbow fracture, accounting for nearly 60 percent of all elbow injuries in children aged 5 to 10 years, typically sustained from a fall onto an outstretched hand with the elbow in extension (extension-type fracture). Classified using the Gartland system (Type I non-displaced, Type II displaced with intact posterior cortex, Type III completely displaced, Type IV multidirectionally unstable). Carries a high potential for severe neurovascular complications, notably injury to the anterior interosseous nerve, radial nerve, or brachial artery, and can lead to compartment syndrome of the forearm and Volkmann ischemic contracture if neglected.

Alternate Names:

Gartland Supracondylar Fracture; Pediatric Elbow Fracture

Supraglottic Airway Device

An airway device positioned above the glottis to maintain oxygenation and ventilation without passing through the vocal cords. It is used for selected anesthetized or unconscious patients and is not a definitive substitute for a cuffed tracheal tube when aspiration protection is required.

Supranuclear Ophthalmoplegia

Weakness or loss of control of eye movements caused by damage to brain pathways above the eye-movement nerves. It may make it difficult to look in certain directions.

Supranuclear Palsy

Alternate index label; see *Progressive supranuclear palsy*.

Suprapubic

Situated, located, or performed in the lower anterior abdominal wall immediately superior to the pubic bones and pubic symphysis. Clinically significant for suprapubic palpation and percussion to assess acute urinary retention with bladder distension; suprapubic aspiration (percutaneous needle puncture of the distended bladder under sterile conditions to obtain non-contaminated urine in infants); and suprapubic catheterization (surgical insertion of a percutaneous cystostomy tube directly into the bladder when urethral catheterization is contraindicated or technically impossible, such as in severe urethral strictures or urethral trauma).

Suprapubic Catheter Care

A medical procedure, treatment, device, or care topic related to suprapubic catheter care; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Suprarenal Glands (Adrenal Glands)

The paired, triangular endocrine glands perched retroperitoneally atop the superior pole of each kidney, enveloped by renal (Gerota) fascia. Anatom-

ically and functionally partitioned into two embryologically distinct zones: the outer adrenal cortex (mesodermal origin, divided into the zona glomerulosa producing mineralocorticoids like aldosterone, zona fasciculata producing glucocorticoids like cortisol, and zona reticularis producing adrenal androgens like DHEA) and the inner adrenal medulla (ectodermal neural crest origin, containing chromaffin cells that synthesize and secrete catecholamines, primarily epinephrine and norepinephrine). Pathologies include primary adrenal insufficiency (Addison disease), hypercortisolemia (Cushing syndrome), primary aldosteronism (Conn syndrome), and pheochromocytoma.

Suprascapular Nerve

A branch of the brachial plexus that supplies the supraspinatus and infraspinatus muscles.

Supraventricular Tachycardia

A rapid heart rhythm arising above the ventricles, often beginning and ending abruptly.

Alternate Names:

SVT

Supraventricular Tachycardia Paroxysmal (Paroxysmal Supraventricular Tachycardia)

Alternate index label for Paroxysmal Supraventricular Tachycardia.

Suramin

An antiparasitic medicine used mainly for selected African trypanosomiasis.

Surfactant

A substance that lowers surface tension; pulmonary surfactant prevents alveolar collapse.

Surgeon

A physician trained to perform operations and invasive procedures, and to evaluate and care for patients before and after surgery.

Surgery

Treatment in which a clinician operates on tissues or organs to diagnose, remove, repair, replace, or improve a body structure. Risks and recovery depend on the procedure and the patient's health.

Surgery For Pancreatic Cancer

A medical procedure, treatment, device, or care topic related to surgery for pancreatic cancer; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Surgery For Pilonidal Cyst

A medical procedure, treatment, device, or care topic related to surgery for pilonidal cyst; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Surgical Drain

A device placed to remove blood, pus, bile, urine, lymph, or other fluid from a surgical site or body cavity. Indications, output characteristics, duration, and removal criteria should be documented because drains can also cause infection, bleeding, or tissue injury.

Surgical Drill

A powered instrument used to cut, shape, or bore bone during surgery.

Surgical Excision

A medical procedure, treatment, device, or care topic related to surgical excision; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Surgical Fixation

Stabilization of a bone, joint, or other structure with devices such as plates, screws, rods, wires, or external frames.

Surgical Hemostasis

Control of bleeding during an operation using pressure, ligation, sutures, clips, electrosurgery, topical agents, vessel sealing, or other techniques. Effective

hemostasis requires identification of the bleeding source and correction of coagulopathy when present.

Surgical Instruments

Instruments for surgery. Categories: cutting (scalpel, scissors), grasping (forceps, clamps), retraction, hemostasis (hemostat, electrocautery).

Surgical Margin

The edge of tissue removed during an excision or resection, assessed microscopically for residual disease. A negative margin has no tumor at the examined edge; a positive margin indicates tumor extends to the inked or designated margin.

Surgical Repair

An operation that restores or reconstructs damaged tissue, an organ, or an anatomic function.

Surgical Risk Stratification

Estimation of the likelihood of perioperative complications using patient factors, urgency, procedure risk, functional capacity, and disease-specific assessment. It informs optimization, monitoring level, shared decisions, and postoperative planning but cannot eliminate uncertainty.

Surgical Safety Checklist

A structured checklist used at defined points before anesthesia, before incision, and before leaving the operating room to verify patient identity, procedure, site, consent, equipment, specimens, and team communication.

Surgical Site Infection

Infection involving the incision, organ, or operative space within the surveillance period after surgery. It may be superficial incisional, deep incisional, or organ-space and requires appropriate culture, antimicrobial therapy, drainage, debridement, or reoperation when indicated.

Surgical Stapler

A device that places metal or absorbable staples to close skin, tissue, or a hollow organ.

Surgical Time-Out

A deliberate team pause immediately before incision to verify patient identity, procedure, site, consent, allergies, antibiotics, imaging, implants, equipment, and anticipated critical events. It is a core surgical-safety intervention.

Surgical Wound Care - Closed

Care of an incision whose edges are closed with sutures, staples, adhesive, or strips, including keeping it clean and dry as instructed and watching for infection or separation.

Surgical Wound Care - Open

Care of a surgical wound left partly or completely open to drain or heal gradually, usually with dressings and scheduled assessment for infection, bleeding, and healthy tissue growth.

Surgical Wound Classification

Categorization of an operative wound as clean, clean-contaminated, contaminated, or dirty/infected according to the degree of microbial exposure and existing infection. The class helps estimate surgical-site infection risk and guide prophylaxis.

Surgical Wound Infection - Treatment

Management of infection in or around a surgical wound, including assessment of depth and severity, drainage or debridement when needed, wound care, and targeted antimicrobial therapy when indicated.

Surrogacy

Surrogacy is an arrangement in which a person carries a pregnancy for intended parent or parents; medical, legal, ethical, and psychosocial requirements vary by jurisdiction.

Surrogate

A substitute or proxy; in medicine, a surrogate endpoint is an intermediate measure used in place of a clinical outcome.

Surrogate Decision-Maker

A person authorized to make healthcare decisions for

someone who lacks decision-making capacity, guided by the person's known values and preferences.

Surveillance

Systematic collection, analysis, interpretation of health data. Types: passive (reporting), active (proactive case finding), sentinel (selected sites).

Surveillance Imaging

Repeat imaging to detect recurrence. Protocol varies by cancer type, stage, treatment.

Survival Rate

Percentage of patients surviving for specified time after diagnosis. 5-year survival commonly used.

Survivorship

Care for cancer survivors. Addresses late effects, surveillance, quality of life.

Survivorship Care Plan

Document summarizing treatment, follow-up, late effects. Provided to cancer survivors.

Susac Syndrome

An autoimmune microangiopathy affecting the brain, retina, and inner ear. The characteristic triad is encephalopathy, branch retinal-artery occlusions, and sensorineural hearing loss, though all features may not be present initially.

Susceptibility

The likelihood of being affected by a disease, exposure, drug, or microorganism.

Susceptibility Testing

Laboratory tests determining the sensitivity of a bacterial isolate to antimicrobials to guide therapy. Methods: disk diffusion (Kirby-Bauer—zone sizes), broth dilution (MIC—minimum inhibitory concentration, MBC—minimum bactericidal concentration), E-test (gradient strips), and automated systems (VITEK).

Suspected Cause

A condition or factor believed to explain a disease or symptom but not yet confirmed.

Suspension

A pharmaceutical formulation in which fine solid drug particles are dispersed (not dissolved) in a liquid vehicle, often with suspending agents and stabilizers. Suspensions are used when a drug is poorly soluble, for oral (antibiotic suspensions, antacids), topical, ophthalmic, and parenteral (depot) administration.

Suture

A stitch or material used to close or join tissue.

Suture (Perineal)

A thread or strand of material used to approximate tissues (bring wound edges together) and hold them in place until healing occurs. See Suture.

Alternate Names:

Perineal

Suture Anchor

A small implant that secures a tendon, ligament, or other soft tissue to bone. It may be made from metal, polymer, or bioabsorbable material and can fail through pullout, breakage, migration, or inflammatory reaction.

Sutures - Ridged

Ridged cranial sutures are raised skull seams in an infant and may reflect normal molding or premature suture fusion, which requires assessment for craniosynostosis.

Sutures - Separated

Separated cranial sutures in an infant can reflect normal growth or increased intracranial pressure, hydrocephalus, prematurity, or other illness and require clinical assessment.

SUV

Standardized uptake value, a semiquantitative measure of radiotracer concentration used mainly in

PET imaging.

SVC Obstruction

Obstruction of the superior vena cava that impairs venous drainage from the head, neck, upper limbs, and upper chest, causing facial or arm swelling, venous distension, cough, or breathing difficulty.

SVC Syndrome

Obstruction of the superior vena cava causing facial swelling, plethora, and distended neck veins. Commonly from lung cancer (SCLC, squamous) and lymphoma. Oncologic emergency.

SVT

Alternate index label; see *Supraventricular tachycardia*.

Swab

A sterile absorbent applicator or the specimen collected with it from a body surface or cavity.

Swallow Study

X-ray imaging of swallowing using barium. Evaluates dysphagia, aspiration, structural abnormalities.

Swallowing

Difficulty swallowing, classified as oropharyngeal (difficulty initiating a swallow, choking, coughing, nasal regurgitation) or esophageal (sensation of food sticking retrosternally). See dysphagia.

Alternate Names:

Difficulty Swallowing (Swallowing)

Swallowing Assessment

Clinical and, when indicated, instrumental evaluation of oral, pharyngeal, and esophageal swallowing to identify dysphagia, aspiration, and appropriate food or liquid consistency. Videofluoroscopic or endoscopic studies may be required when bedside assessment is insufficient.

Swallowing Chalk

Ingestion of chalk, which is usually low in toxicity in small amounts but can cause choking, constipa-

tion, or gastrointestinal upset; repeated eating may indicate pica.

Swallowing Difficulty

Dysphagia, or impaired movement of food or liquid from the mouth to the stomach, caused by oropharyngeal, esophageal, neuromuscular, structural, or functional disorders.

Swallowing Problems

Difficulty moving food, liquid, or saliva safely from the mouth to the stomach; dysphagia can cause choking, aspiration, weight loss, or dehydration and may need evaluation.

Swallowing Soap

Ingestion of soap, which commonly causes mouth or stomach irritation, nausea, or diarrhea; concentrated or large exposures can injure tissue or affect breathing.

Swallowing Sunscreen

Ingestion of sunscreen, usually causing mild gastrointestinal upset but potentially causing choking or aspiration depending on amount and product; breathing difficulty or persistent vomiting requires urgent help.

Swallowing Syncope

Fainting triggered by swallowing, usually through an abnormal reflex that slows the heart or lowers blood pressure; recurrent episodes require evaluation for rhythm and structural heart disease.

Swallowing Therapy

Assessment and treatment to improve the safety and efficiency of swallowing and reduce aspiration risk.

Swan-Ganz - Right Heart Catheterization

Insertion of a pulmonary-artery catheter through the right heart to measure central venous, pulmonary-artery, and wedge pressures and cardiac output in selected critically ill patients.

Swan-Ganz Catheter

A pulmonary-artery catheter used to measure pressures and cardiac output in selected critically ill patients.

Sweat

Fluid secreted by sweat glands, helping regulate body temperature.

Sweat Chloride Test

A test that measures chloride concentration in sweat after pilocarpine stimulation. It is an important diagnostic test for cystic fibrosis, but results must be interpreted with the clinical findings and, when needed, genetic or other testing. Some illnesses and technical factors can produce misleading results.

Sweat Electrolytes Test

A sweat electrolytes test measures salts, especially chloride, in sweat and is commonly used to help diagnose cystic fibrosis.

Sweat Gland

A skin gland that produces sweat, which contributes mainly to temperature regulation and skin surface function.

Sweat Gland Tumor

A biomedical term referring to sweat gland tumor; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Sweat Glands

Eccrine and apocrine glands that produce sweat or related secretions.

Sweating

Production of sweat by eccrine and apocrine glands for heat regulation and other responses; excessive, absent, or asymmetric sweating can indicate medication, autonomic, endocrine, or neurologic disease.

Sweating And Body Odor

Excess perspiration or noticeable odor produced

when skin bacteria break down sweat; it may be influenced by hygiene, diet, medicines, hormones, or disease.

Alternate Names:

Body Odor And Sweating

Sweating, Abnormally Excessive

Alternate index label; see *Hyperhidrosis*.

Sweating, Gustatory

Sweating triggered by eating or smelling food, usually over the cheek or temple after parotid-nerve injury or surgery and known as Frey syndrome.

Sweet Syndrome

An inflammatory skin disorder characterized by sudden fever and tender red or violaceous plaques or nodules containing many neutrophils. It may be associated with infection, medicines, inflammatory disease, or cancer.

Sweet'S Syndrome

Acute febrile neutrophilic dermatosis, characterized by fever, neutrophilia, and painful, erythematous plaques or nodules on the skin, histologically showing dense neutrophilic infiltration of the dermis. It may be idiopathic, drug-induced, or associated with malignancy, particularly hematological cancers.

Sweetened Beverages

Sweetened beverages contain added sugars and can increase calorie intake, dental caries, obesity, and diabetes risk when consumed frequently.

Sweeteners - Sugar Substitutes

Sugar substitutes provide sweetness with little or no sugar; types differ in energy, digestion, after-taste, and effects on glucose and gastrointestinal symptoms.

Sweeteners - Sugars

Sugars are simple carbohydrates that provide energy but can raise blood glucose and contribute to dental decay or excess calories when consumed in excess.

Swelling

Abnormal enlargement caused by fluid accumulation, inflammation, bleeding, infection, tumor, or tissue injury; location, onset, pain, and associated symptoms identify the likely cause.

Swimmer'S Ear

An infection or inflammation of the ear canal, often after water remains in the ear. It can cause ear pain, itching, redness, drainage, and pain when the outer ear is touched.

Alternate Names:

Ear Infection, Outer Ear Outer Ear Infection

Swimmer'S Itch

An itchy rash caused by an allergic reaction to microscopic parasites released by infected snails in freshwater or saltwater.

Alternate Names:

Cercarial Dermatitis Dermatitis, Cercarial

Swimmers Ear (Otitis Externa)

Alternate index label for Otitis Externa.

Swimming Pool Cleaner Poisoning

Swimming pool cleaner poisoning involves toxic or environmental exposure to swimming pool cleaner poisoning; dose, route, timing, symptoms, and organ involvement determine urgency and treatment.

Swimming Pool Granuloma

Swimming-pool granuloma is a skin infection caused by *Mycobacterium marinum* after exposure to contaminated water, producing persistent nodules along lymphatic channels.

Swine Flu

An informal term for infection with an influenza A virus that normally circulates in pigs or has swine-origin genes. In people, the term often refers to the 2009 H1N1 pandemic virus or to a swine-origin variant infection. Symptoms may resemble seasonal flu, including fever, cough, sore throat, headache, and muscle aches; the term does not refer to one single virus.

Swollen Ankles And Swollen Feet

Enlargement of the ankles or feet caused by injury, venous insufficiency, lymphatic disease, medication, pregnancy, or systemic fluid retention from heart, kidney, or liver disease. Sudden one-sided swelling with pain, redness, warmth, or breathlessness may indicate thrombosis or infection and requires urgent assessment.

Alternate Names:

Ankle Swollen (Swollen Ankles And Swollen Feet)

Swollen Lymph Nodes

Swollen lymph nodes are enlarged immune glands, commonly from infection but sometimes from inflammation, medication, or cancer; persistent, hard, or generalized nodes need evaluation.

Alternate Names:

Lymph Nodes, Swollen Lymph Swollen Glands (Swollen Lymph Nodes) Lymph Swollen Nodes (Swollen Lymph Nodes)

Swollen Tongue

Swollen tongue refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Sycosis Barbae

Chronic folliculitis of the beard area, usually caused by bacterial infection and ingrown hairs.

Sydenham Chorea

A delayed neurologic complication of group A streptococcal infection causing involuntary jerky movements, hypotonia, emotional lability, and coordination problems, usually in children.

Sydenham'S Chorea

A post-streptococcal autoimmune movement disorder causing involuntary, irregular movements and emotional or behavioral change.

Symblepharon

Adhesion of the palpebral conjunctiva (inner eyelid) to the bulbar conjunctiva or cornea, most commonly resulting from chemical burns, Stevens-Johnson syn-

drome, ocular cicatricial pemphigoid, or repeated surgical intervention. It restricts eye movement and may require surgical lysis with amniotic membrane or mucous membrane grafting.

Symbol (F)

A symbol that commonly means female, force, or Fahrenheit in medical records; the intended meaning depends on the surrounding context.

Symbol (R)

R is a symbol whose meaning depends on context, including right side, radius, resistance, or a statistical correlation coefficient.

Symbolism

The use of an image, object, or action to represent an idea or feeling; in clinical settings, symbolic meaning may be relevant to communication, culture, or psychotherapy.

Sympathectomy

Surgical interruption or destruction of sympathetic nerves to reduce abnormal vasoconstriction, sweating, or pain in selected conditions.

Sympathetic

Relating to the sympathetic division of the autonomic nervous system or to shared feeling, depending on context.

Sympathetic Ganglion

A cluster of sympathetic autonomic neuron cell bodies, located in paravertebral or prevertebral chains, that relays signals controlling vessels, sweat glands, viscera, and other targets.

Sympathetic Nervous System

The autonomic division that prepares the body for activity or stress by increasing heart rate and altering vascular, airway, and digestive function.

Sympatholytic

An agent that blocks the sympathetic nervous system, used to treat hypertension, sympathetic over-

activity, and catecholamine excess. Classes: beta-blockers (propranolol, metoprolol), alpha-blockers (prazosin, doxazosin, phenoxybenzamine), centrally acting agents (clonidine, methyldopa), and neuron blockers (reserpine, guanethidine).

Sympathomimetic Drugs

Medicines that mimic sympathetic nervous-system activity, often by stimulating alpha or beta adrenergic receptors.

Sympathomimetic Toxicidrome

A poisoning pattern caused by excessive adrenergic stimulation, producing agitation, sweating, dilated pupils, rapid heart rate, high blood pressure, elevated temperature, tremor, or seizures.

Symphysis

A joint in which bones are united by fibrocartilage, such as the pubic symphysis.

Symphysis-Fundal Height

A standardized clinical measurement performed during routine antenatal care, measured in centimeters using a non-elastic tape from the top of the maternal pubic symphysis to the uppermost border of the uterine fundus. Between 24 and 36 weeks of gestation, the fundal height in centimeters typically corresponds closely (± 2 to 3 cm) to gestational age in weeks. A discrepancy prompts diagnostic obstetric ultrasonography to evaluate for intrauterine growth restriction, oligohydramnios, multiple gestation, fetal macrosomia, or polyhydramnios.

Alternate Names:

Fundal Height Measurement; SFH

Symphysis Pubis

The fibrocartilaginous joint connecting the left and right pubic bones at the front of the pelvis.

Symptom

A subjective experience reported by a patient, such as pain, nausea, or breathlessness.

Symptomatic

Having symptoms caused by a disease or condition; symptomatic treatment relieves those symptoms without necessarily treating the underlying cause.

Symptomatic Treatment

Symptomatic treatment is a medical test, procedure, treatment, or clinical measure used in patient care; its purpose and interpretation depend on the clinical question.

Synapse

A junction where a neuron communicates with another neuron, muscle, or gland.

Synapse Assembly

The developmental process by which neurons form organized connections with other neurons or target cells, using cell-adhesion molecules and signaling proteins.

Synapsins

A family of neuronal proteins that help organize synaptic vesicles and regulate the release of neurotransmitters.

Synaptic Cleft

The very small gap between a presynaptic neuron and its target cell across which neurotransmitters travel.

Synaptic Membrane

The specialized membrane at a neuronal junction, including the neurotransmitter-releasing presynaptic membrane and receptor-containing postsynaptic membrane.

Synaptic Plasticity

Activity-dependent change in the strength or structure of synaptic transmission. Long-term potentiation and long-term depression are important cellular mechanisms underlying learning, memory, adaptation, and some forms of chronic pain.

Synaptic Transmission

The fundamental process by which neurons communicate with each other through specialized junctions called synapses.

Synaptic Vesicle

A small membrane-bound sac in a nerve ending that stores neurotransmitters and releases them into the synaptic cleft when the neuron is activated.

Synaptonemal Complex

A protein structure that pairs homologous chromosomes during meiosis and supports genetic recombination.

Synaptophysin

A protein in synaptic vesicle membranes and a commonly used marker of neuronal and neuroendocrine tissue in laboratory testing.

Synaptosome

A sealed nerve-terminal structure produced when brain tissue is mechanically disrupted; it is used to study neurotransmitter release and synaptic function.

Synaptotagmin

A calcium-sensing protein on synaptic vesicles that helps trigger rapid neurotransmitter release.

Synarthrosis

A joint with little or no movement, such as a skull suture or tooth anchored in its socket, in which bones are joined by fibrous tissue or cartilage.

Syncephalastrum

A genus of environmental molds in the order Mucorales; it rarely causes opportunistic mucormycosis, usually in people with major immune or metabolic risk factors.

Synchrotron

A synchrotron is a particle accelerator that produces intense beams of charged particles or radiation used in research, imaging, and materials analysis.

Syncope

A sudden, brief loss of consciousness and postural tone caused by temporary global cerebral hypoperfusion, followed by spontaneous, usually complete recovery. Common mechanisms are reflex syncope, orthostatic hypotension, and cardiac disease; prolonged confusion or seizure-like features require evaluation for another cause.

Syncope (Fainting)

A transient loss of consciousness and postural tone caused by temporary global cerebral hypoperfusion, followed by spontaneous recovery. Reflex, orthostatic, and cardiac causes are common; prolonged confusion suggests a different diagnosis. See Fainting.

Alternate Names:

Fainting

Syncope, Coughing

A brief faint caused by coughing, which raises chest pressure, reduces venous return and cardiac output, and transiently lowers cerebral blood flow.

Syndactyly

Congenital fusion or webbing of adjacent fingers or toes.

Syndecan

A family of cell-surface proteoglycans that bind growth factors and extracellular-matrix proteins and help regulate cell adhesion, signaling, and tissue repair.

Syndemic

A syndemic is the interaction of two or more diseases with social or environmental conditions that together worsen health outcomes.

Syndesmophyte

A thin, bony growth that forms along the edges of the vertebrae, often in ankylosing spondylitis, and may bridge adjacent vertebrae.

Syndrome

A recognizable pattern of symptoms, signs, or findings that occur together.

Syndrome Asperger (Asperger Syndrome)

Alternate index label for Asperger Syndrome.

Syndrome Cauda Equina (Cauda Equina Syndrome)

Alternate index label for Cauda Equina Syndrome.

Syndrome Compartment (Compartment Syndrome)

Alternate index label for Compartment Syndrome.

Syndrome Cyclic Vomiting (Cyclic Vomiting Syndrome)

Alternate index label for Cyclic Vomiting Syndrome.

Syndrome Loeys-Dietz (Loeys-Dietz Syndrome)

Alternate index label for Loeys-Dietz Syndrome.

Syndrome Of Inappropriate ADH Secretion

Excessive antidiuretic hormone activity causing eu-volemic hyponatremia with inappropriately concentrated urine and elevated urinary sodium.

Syndrome Of Inappropriate Antidiuretic Hormone

Excessive release or effect of antidiuretic hormone causing impaired free-water excretion, low serum sodium, and inappropriately concentrated urine in the absence of an appropriate physiologic stimulus.

Syndrome Of Inappropriate Antidiuretic Hormone Secretion

A clinical condition involving syndrome Of Inappropriate Antidiuretic Hormone Secretion; its causes, manifestations, severity, and complications guide diagnosis and management.

Syndrome Of Inappropriate Antidiuretic

Hormone Secretion

Excessive vasopressin activity causing impaired free-water excretion, euvolemichyponatremia, and inappropriately concentrated urine after exclusion of other causes. Treatment depends on severity and cause and may include fluid restriction or carefully monitored hypertonic saline for severe symptomatic cases.

Alternate Names:

Syndrome of Inappropriate Antidiuretic Hormone Secretion

Syndrome Pfeiffer (Pfeiffer Syndrome)

Alternate index label for Pfeiffer Syndrome.

Syndrome Restless Legs (Restless Leg Syndrome)

Alternate index label for Restless Leg Syndrome.

Syndrome Surveillance

The public health monitoring of pre-diagnostic clinical indicators (symptoms, syndromes—fever, rash, GI illness, influenza-like illness) in real time to detect outbreaks, bioterrorism, or emerging infections earlier than laboratory confirmation.

Syndrome X

Alternate index label; see *Metabolic syndrome*.

Syndrome, Nervous Colon

A clinical condition involving syndrome, Nervous Colon; its causes, manifestations, severity, and complications guide diagnosis and management.

Syndrome, Post-Polio

A clinical condition involving syndrome, Post-Polio; its causes, manifestations, severity, and complications guide diagnosis and management.

Syndrome, TAR

A clinical condition involving syndrome, TAR; its causes, manifestations, severity, and complications guide diagnosis and management.

Syndromic Surveillance

Monitoring of symptoms, clinical encounters, laboratory requests, or other near-real-time signals before definitive diagnoses are available. It supports early detection of outbreaks and public-health threats but may have lower specificity.

Synechia

An abnormal adhesion between adjacent tissues, especially between parts of the iris or between the iris and the lens or cornea.

Synechiae

Abnormal adhesions between tissues, especially within the eye, uterus, or nasal cavity.

Synergist

A muscle, drug, or factor that works together with another to produce or enhance an effect.

Synesthesia

Synesthesia is a stable perceptual phenomenon in which stimulation of one sensory pathway automatically produces an experience in another, such as seeing colors with sounds.

Synesthete

A synesthete is a person who experiences synesthesia, in which stimulation of one sense or cognitive pathway automatically evokes another perception.

Syngeneic

Genetically identical or nearly identical within the same species, as in tissues or cells exchanged between identical twins or inbred laboratory animals.

Synostosis

Fusion of two bones by bone tissue, congenital or acquired.

Synovectomy

Surgical removal of part or all of an inflamed synovial membrane.

Synovial Biopsy

A synovial biopsy removes a small sample of the joint lining to investigate infection, inflammatory arthritis, crystal disease, or an unusual joint lesion.

Synovial Bursa

A synovial bursa is a small fluid-filled sac that reduces friction between moving tissues near joints, such as tendons, skin, muscles, and bone.

Synovial Chondromatosis

A benign, metaplastic condition characterized by the formation of multiple cartilaginous loose bodies within joint synovium, bursae, or tendon sheaths. It presents with joint pain, swelling, locking, and restricted range of motion, most commonly affecting the knee, hip, and elbow.

Synovial Cyst

A synovial cyst is a fluid-filled sac arising from or near a joint or tendon sheath, sometimes compressing nearby nerves and causing pain or weakness.

Synovial Fluid

A viscous fluid within a synovial joint that lubricates articular surfaces and helps nourish cartilage.

Synovial Fluid Analysis

Laboratory examination of aspirated joint fluid for appearance, viscosity, leukocyte count and differential, crystals under polarized light, Gram stain, and culture. It helps distinguish noninflammatory, inflammatory, crystal-associated, and septic arthritis.

Synovial Joint

A freely movable joint in which the articulating bones are separated by a fluid-filled joint cavity enclosed by a capsule.

Synovial Membrane

The lining of a synovial joint that produces lubricating synovial fluid.

Synovial Reaction To Implanted

Prostheses

Inflammation and tissue response around an orthopedic implant caused by wear particles, loosening, corrosion, infection, or hypersensitivity. Evaluation distinguishes aseptic failure from periprosthetic infection.

Synovial Sarcoma

A soft tissue sarcoma typically arising near joints of the extremities in young adults, characterized by the t(X;18) chromosomal translocation.

Synoviocytes

Synoviocytes are cells lining the joint synovium that produce lubricating fluid and inflammatory mediators and help maintain or damage joint tissue.

Synovitis

Inflammation of a synovial membrane, causing swelling, pain, and restricted movement.

Synovium

The synovial membrane and its associated tissue lining a joint or tendon sheath.

Synovography

Imaging of a joint after contrast is injected into its cavity. It can show the joint lining, capsule, tears, or abnormal connections, although MRI and ultrasound are often used instead.

Syntax

Syntax is the arrangement of words, symbols, or genetic sequence elements according to rules that determine meaning and valid structure.

Synthesis

Synthesis is the biological or chemical production of a complex molecule or structure from simpler components, such as protein synthesis from amino acids or DNA synthesis from nucleotides.

Synthesis Module

Automated system for radiopharmaceutical preparation. PET tracer synthesis: FDG, Florbetapir.

Quality control: radiochemical purity, pH, sterility.

Synthetic

Produced by chemical or laboratory processes rather than directly obtained in its natural form.

Synthetic Androgen

A synthetic androgen is a manufactured compound that activates androgen receptors and may mimic testosterone; therapeutic and anabolic use can suppress gonadal function and cause acne, mood effects, infertility, liver injury, or cardiovascular harm.

Syphilis

A sexually transmitted infection caused by *Treponema pallidum*, progressing through primary, secondary, latent, and sometimes tertiary stages.

Syphilis In Women Overview

Syphilis is a sexually transmitted infection caused by *Treponema pallidum* and progresses through stages that may include painless sores, rash, neurologic or cardiovascular disease, or no symptoms. Penicillin treatment cures infection and is essential in pregnancy to prevent congenital disease.

Alternate Names:

STD (Syphilis In Women Overview)

Syphilis In Women Overview

Alternate index label for Syphilis in Women Overview.

Syphilis Serology (RPR/VDRL)

Nontreponemal blood tests used to screen for and monitor syphilis. Reactive results require interpretation with clinical findings and, usually, a treponemal confirmatory test.

Alternate Names:

Syphilis Serology

Syphilitic Alopecia

Non-scarring hair loss occurring as a manifestation of secondary syphilis, presenting with diffuse thinning or patchy, moth-eaten alopecia of the scalp, eyebrows, or beard. It is thought to result from

direct spirochaetal invasion of hair follicles during the spirochaetaemic phase.

Syphilitic Aseptic Meningitis

Syphilitic aseptic meningitis is inflammation of the meninges caused by *Treponema pallidum*, producing headache, neck stiffness, cranial-nerve symptoms, or other neurologic findings.

Syphilitic Gumma

A destructive, granulomatous lesion occurring in tertiary syphilis, characterized by a center of necrotic tissue surrounded by plasma cells, lymphocytes, and fibroblasts. Gummas may affect any organ, most commonly the skin, bones, liver, and central nervous system. They present as painless, firm nodules that may ulcerate.

Syringe

A device with a barrel and plunger used to withdraw or inject fluids.

Syringe Drivers

Small battery-powered pumps that deliver medicines continuously or at programmed intervals through a syringe.

Syringocystadenoma Papilliferum

A benign adnexal tumor with apocrine or eccrine differentiation, often presenting as a papule or plaque on the scalp or face and sometimes arising within a nevus sebaceus.

Syringoma

A syringoma is a benign tumor of sweat-duct cells, usually appearing as small skin-colored or yellowish papules around the eyelids or elsewhere.

Syringomyelia

A fluid-filled cavity within the spinal cord that can cause pain, sensory loss, weakness, or autonomic dysfunction.

Syrinx

A syrinx is a fluid-filled cavity within the spinal cord,

typically containing cerebrospinal fluid or a similar fluid, that develops as a result of abnormal CSF flow dynamics within the spinal cord. Syringomyelia: a syrinx in the central spinal cord. Hydromyelia: dilation of the central canal.

Syrup

A syrup is a viscous liquid preparation, often sweetened, used as a medicine vehicle or oral dosage form; concentration and ingredients determine safe dosing.

Systematic Review

Comprehensive summary of research on specific question. Uses explicit methods to minimize bias. Cochrane reviews example.

Systemic

Affecting the whole body or relating to an entire system rather than a localized site.

Systemic Antifungals

Antifungal medicines taken orally or given intravenously for infections that are extensive, invasive, refractory, or unsuitable for topical treatment. Choice and duration depend on the organism, site, severity, drug interactions, organ function, and local resistance patterns.

Systemic Chemotherapy

Systemic chemotherapy uses medicines that travel through the bloodstream to kill or slow cancer cells throughout the body.

Systemic Circuit

The circulation that carries oxygenated blood from the left ventricle through the body and returns deoxygenated blood to the right atrium.

Systemic Disease

A clinical condition or finding involving systemic disease. Its features, causes, evaluation, and management depend on the clinical context.

Systemic Effects

Whole-body effects of toxins. Cardiac, neurological,

renal, hepatic, hematological.

Systemic Exertional Intolerance Disease

Alternate index label; see *Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS)*.

Systemic Inflammatory Response Syndrome

A systemic inflammatory response that may occur with infection, trauma, burns, pancreatitis, surgery, or other insults. It can involve abnormal temperature, heart rate, respiratory rate, or white-cell count. Modern sepsis assessment uses broader clinical criteria rather than SIRS alone.

Systemic Juvenile Idiopathic Arthritis

A subtype of juvenile idiopathic arthritis with prominent systemic inflammation—daily spiking fevers, evanescent salmon-pink rash, arthritis, lymphadenopathy, hepatosplenomegaly, and elevated inflammatory markers (ferritin).

Systemic Lupus

Systemic lupus erythematosus is an autoimmune disease that can affect skin, joints, kidneys, blood, nervous system, lungs, and heart. Symptoms fluctuate, and treatment is tailored to organ involvement with sun protection, monitoring, and immunomodulatory medicines.

Alternate Names:

Arthritis SLE (Systemic Lupus) Lupus (Systemic Lupus) SLE (Systemic Lupus)

Systemic Lupus

Alternate index label for Systemic Lupus.

Systemic Lupus Erythematosus

Multisystem autoimmune disease, more common in women. Features: malar rash, photosensitivity, oral ulcers, arthritis, serositis, renal disorder, neurologic disorder, hematologic disorder, immunologic disorder. Diagnosis: ANA positive plus clinical/immunologic criteria.

Systemic Lupus Erythematosus

A chronic, multisystem autoimmune inflammatory disorder characterized by loss of immune tolerance to nuclear autoantigens, polyclonal B-cell hyperactivity, and tissue deposition of pathogenic immune complexes. Predominantly affects women of child-bearing age (9:1 female-to-male ratio). Clinical hallmarks include the classic malar rash, photosensitivity, discoid lesions, inflammatory polyarthritis, serositis, lupus nephritis (monitored by proteinuria and anti-dsDNA titers), cytopenias, and neuropsychiatric lupus. Serologically diagnosed by high-sensitivity antinuclear antibodies followed by disease-specific anti-double-stranded DNA (anti-dsDNA) and anti-Smith (anti-Sm) antibodies; managed with hydroxychloroquine as baseline therapy, glucocorticoids for flares, and immunosuppressants (mycophenolate mofetil, cyclophosphamide, belimumab).

Systemic Mastocytosis

A disorder in which excessive mast cells accumulate in one or more organs and release inflammatory chemicals.

Systemic Scleroderma

Systemic sclerosis, an autoimmune connective-tissue disease causing vascular dysfunction and fibrosis of skin and internal organs, commonly affecting the esophagus, lungs, kidneys, and heart.

Systemic Sclerosis

Systemic sclerosis, also known as scleroderma, is a chronic autoimmune disease characterized by progressive fibrosis of the skin and internal organs, vasculopathy, and immune dysfunction.

Alternate Names:

scleroderma, is a chronic autoimmune disease.

Systemic Therapy

Treatment that reaches the whole body, including chemotherapy, endocrine therapy, targeted therapy, and immunotherapy, as opposed to local therapies such as surgery and radiation.

Systemic Vascular Resistance

The resistance to blood flow offered by the systemic circulation, determined mainly by arteriolar tone. It is calculated from the pressure gradient across the systemic circulation and cardiac output and is a major determinant of arterial pressure.

Supracervical Hysterectomy

Hysterectomy in which the uterine body is removed while the cervix is left in place; it is also called subtotal or partial hysterectomy. The fallopian tubes and ovaries may be managed separately.

Systemic Venous

Systemic venous refers to veins returning deoxygenated blood from the body to the right side of the heart, excluding the pulmonary circulation.

Sustained Virologic Response

The absence of detectable hepatitis C virus RNA in the blood 12 or more weeks after completing antiviral treatment, indicating virologic cure. Reinfection remains possible.

Systole

The phase of the cardiac cycle during which a chamber contracts and ejects blood.

Systolic

Systolic means relating to contraction of the heart, especially the pressure in arteries when the ventricles eject blood.

Systolic Anterior Motion

An abnormal hemodynamic and anatomical motion of the mitral valve apparatus during ventricular systole, wherein the anterior mitral valve leaflet (or both anterior and posterior leaflets) is pulled and swept anteriorly into the left ventricular outflow tract toward the hypertrophied interventricular septum. Driven by hydrodynamic drag forces and the Venturi suction effect generated by high-velocity blood flow through a narrowed LVOT, it causes dynamic subaortic obstruction and concomitant eccentric, posteriorly directed mitral regurgitation; a defining pathophysiological hallmark of hyper-

trophic obstructive cardiomyopathy.

Alternate Names:

SAM; SAM of the Mitral Valve

Systolic Murmurs

Murmurs occurring between S1 and S2. Ejection systolic murmurs (crescendo-decrescendo) are caused by flow across semilunar valves (aortic stenosis, pulmonary stenosis) or increased flow across normal valves.

Systolic Pressure

The maximum arterial pressure during ventricular contraction, recorded as the upper number of a blood-pressure reading.

Alternate Names:

Seasonal affective disorder

Sz

Sz is an informal abbreviation for seizure; clinical documentation should identify the seizure type and relevant features clearly.

T

T Cell

Abbreviation for T lymphocyte, a type of white blood cell that is produced in the bone marrow and is part of the body's immune system.

T Wave

The electrocardiographic deflection representing ventricular repolarization. Its shape and direction vary by lead and clinical context. Abnormal T waves may occur with ischemia, electrolyte disorders, ventricular strain, or other conditions.

T-Cell Count

A laboratory measurement of T lymphocytes in blood, often including CD4 cells, used to assess immune function and monitor conditions such as HIV infection or immunosuppression.

T-Cell Lymphoma

A lymphoma arising from T lymphocytes, with multiple biologically distinct subtypes.

T-Cell Lymphomas

Lymphomas arising from mature or immature T lymphocytes. They include several biologically distinct diseases that may involve lymph nodes, skin, blood, or other organs; behavior and treatment depend on the subtype.

T-Cell Receptor

A clonally distributed receptor on T lymphocytes that recognizes a peptide antigen presented by a major histocompatibility complex molecule. TCR signaling requires accessory and co-stimulatory signals for effective T-cell activation.

T-Cells (T-Lymphocytes)

A type of lymphocyte that matures in the thymus and is the central effector of cell-mediated immunity. T-cells express the T-cell receptor (TCR) on their surface, which recognises peptide antigens presented by major histocompatibility complex (MHC) molecules on antigen-presenting cells.

Alternate Names:

T-Lymphocytes

T-Lymphocyte

A lymphocyte that matures in the thymus and mediates cellular immune responses.

T-Wave Alternans

T-wave alternans is a beat-to-beat change in the size or shape of the heart's T wave and may indicate increased risk of dangerous ventricular rhythms.

T3

Triiodothyronine, the most biologically active thyroid hormone, produced by deiodination of T4 in peripheral tissues (liver, kidney) and secreted by the thyroid. Regulates metabolism (BMR), protein/carbohydrate/lipid metabolism, growth, and development.

T3 Test

A blood test measuring triiodothyronine, used with thyroid-stimulating hormone and thyroxine to evaluate thyroid function, particularly suspected hyperthyroidism.

T3RU Test

An older test estimating the proportion of thyroid-

hormone-binding sites occupied in serum; it is used indirectly to assess thyroid-hormone binding and has largely been replaced by free-thyroxine measurements.

T4

Thyroxine, the main hormone secreted by the thyroid gland (thyroxine, 80-90 percent of secretion), converted to the active T3 in tissues. Bound largely to thyroxine-binding globulin (TBG); free T4 is the active unbound fraction. Regulates metabolism, growth, and development.

TAB Vaccine

An older combined vaccine against typhoid and paratyphoid A and B.

Tabes

An older term for tabes dorsalis, a late neurologic complication of untreated syphilis.

Tabes Dorsalis

A late manifestation of untreated syphilis in which degeneration of dorsal spinal-cord columns and roots causes sensory ataxia, lightning pains, loss of position sense, areflexia, and bladder or sexual dysfunction.

Tablet

A solid dosage form of compressed drug plus excipients (binders, fillers, disintegrants, lubricants, coatings). Types: immediate-release, enteric-coated (resistant to gastric acid), sustained/controlled-release, sublingual, buccal, effervescent, chewable, and orally disintegrating.

Tablet Splitting

Dividing a tablet to take a smaller dose; only tablets designed for splitting should be divided, because extended-release, enteric-coated, and some specialized tablets may not work safely when split.

TAC Regimen

A chemotherapy combination of docetaxel, doxorubicin, and cyclophosphamide, used for selected

breast cancers; it can cause infection risk, nausea, hair loss, heart effects, and other toxicities.

Tacalcitol

A vitamin-D analogue used topically for psoriasis and selected disorders of keratinization.

Tacedinaline

Tacedinaline is an investigational histone-deacetylase inhibitor studied for effects on gene expression and cancer-cell growth.

Tache Noire

Tache noire is a black eschar at the site of a tick bite, classically associated with Mediterranean spotted fever and related rickettsial infections.

Tachyarrhythmia

An abnormally fast heart rhythm, which may arise from the atria, ventricles, or the heart's conduction system. Symptoms can include palpitations, dizziness, chest discomfort, shortness of breath, or fainting; fainting or severe symptoms require urgent assessment.

Tachycardia

A faster-than-expected heart rate that may be a normal response or a sign of disease. See Fast Heart Rate.

Alternate Names:

Rapid Heartbeat

Tachycardia Paroxysmal Atrial (Paroxysmal Supraventricular Tachycardia)

Alternate index label for Paroxysmal Supraventricular Tachycardia.

Tachycardia Paroxysmal Supraventricular (Paroxysmal Supraventricular Tachycardia)

Alternate index label for Paroxysmal Supraventricular Tachycardia.

Tachycardia-Induced Cardiomyopathy

Weakening and dilation of the heart muscle caused by a persistent or frequently recurring rapid heart rhythm, often improving when the arrhythmia is controlled.

Tachykinin

A family of signaling peptides, including substance P, that influence pain, inflammation, smooth-muscle activity, and nerve communication.

Tachyphrenia

Tachyphrenia is rapid flow of thoughts or accelerated mental activity, often described in association with mania, stimulants, or other psychiatric states.

Tachyphylaxis

A rapid, acute decrease in response to a drug after repeated administration over a short period, distinct from tolerance which develops more gradually. Tachyphylaxis results from rapid depletion of neurotransmitter stores, receptor desensitization, or receptor downregulation.

Tachypnea

Abnormally rapid respiratory rate (adult >20 breaths/min). Physiologic (exercise, anxiety) or pathologic: hypoxia, fever, sepsis, metabolic acidosis (Kussmaul), pulmonary embolism, pneumonia, heart failure, pain, and compensation for metabolic derangement.

Tachypnoea

Abnormally rapid breathing, defined as a respiratory rate >20 breaths per minute in adults (age-dependent: >60 in neonates, >40 in 1–12 months, >30 in 1–5 years, >20 in >5 years). Tachypnoea is a non-specific sign of respiratory or metabolic distress.

Tachysystole (Uterine Hyperstimulation)

More than 5 contractions in 10 minutes averaged over 30 minutes, or contractions lasting ≥ 2 minutes. Tachysystole with fetal heart rate changes (late decelerations, prolonged deceleration, loss of variability) is more concerning. Causes: oxytocin

(most common), prostaglandins (misoprostol > dinoprostone), and cocaine.

Alternate Names:

Uterine Hyperstimulation

TACO

Transfusion-Associated Circulatory Overload, a complication of blood transfusion due to volume overload causing pulmonary edema. Risk factors: pre-existing cardiac or renal disease, extremes of age, large transfusion volume, rapid infusion rate.

Tacrine

An older acetylcholinesterase inhibitor formerly used for Alzheimer disease; liver toxicity and frequent dosing limited its use.

Tacrolimus

An immunosuppressant that inhibits calcineurin; used systemically after transplantation and topically for eczema.

Tactile

Pertaining to, perceived by, or relating to the sense of touch (*tactus*). Mediated by specialized cutaneous mechanoreceptors embedded within the dermis and epidermis: Meissner corpuscles (low-frequency vibration, light dynamic touch), Merkel discs (static pressure, fine tactile edge and form discrimination), Pacinian corpuscles (high-frequency vibration), and Ruffini endings (skin stretch). Sensory tactile signals ascend via large myelinated $A\beta$ primary afferents through the dorsal column-medial lemniscal pathway to the contralateral primary somatosensory cortex. Assessed clinically using two-point discrimination tests, monofilament testing in diabetic peripheral neuropathy, and tactile fremitus during thoracic lung examination.

Tactile Agnosia

Tactile agnosia is inability to recognize a familiar object by touch despite intact primary sensation, usually from parietal-cortex dysfunction.

Tactile Fremitus

Vibrations transmitted through the lung to the chest

wall, felt during palpation. It is increased in consolidation (pneumonia) due to better sound transmission through solid tissue. It is decreased in pleural effusion, pneumothorax, and atelectasis due to impaired transmission.

Tactile Sense

The ability to perceive touch, pressure, vibration, texture, and related mechanical stimuli through receptors in the skin and deeper tissues.

Tadalafil

Tadalafil is a phosphodiesterase-5 inhibitor used for erectile dysfunction, benign prostatic hyperplasia, and pulmonary hypertension; headache, flushing, hypotension, priapism, and dangerous nitrate interactions are important risks.

Taenia

A genus of tapeworms, including *T. solium* and *T. saginata*.

Taenia Saginata

Taenia saginata is the beef tapeworm acquired from undercooked infected beef; intestinal infection may cause abdominal symptoms or passage of motile segments but does not usually cause human cysticercosis.

Taenia Solium

Taenia solium is the pork tapeworm; eating undercooked pork causes intestinal taeniasis, while ingesting eggs can cause cysticercosis, including potentially serious neurocysticercosis.

Taeniasis

Intestinal infection caused by adult *Taenia* tapeworms, usually acquired by eating undercooked beef or pork containing larval cysts. It may cause abdominal symptoms or passage of worm segments; swallowing *Taenia solium* eggs causes cysticercosis in tissues, which is a different infection.

Taeniasis (Tapeworm Infection)

Intestinal infection caused by adult *Taenia* tape-

worms, usually acquired by eating undercooked meat containing larval cysts. *Taenia solium* and *Taenia saginata* are important species; *Taenia solium* eggs can also cause cysticercosis in tissues.

Alternate Names:

Tapeworm Infection

Tail

A tapering appendage extending from the end of an animal's body. In human anatomy, the term may refer to a tail-like congenital structure or the embryologic caudal end; evaluation depends on the specific finding.

Tailbone Trauma

A biomedical term referring to tailbone trauma; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Taint

A taint is contamination or an undesirable influence; in medicine or laboratory work it may refer to contamination that affects a specimen, product, or result.

Takayasu Arteritis

A chronic granulomatous vasculitis primarily affecting the aorta and its major branches, most commonly diagnosed in women under 50 years of age, particularly in Asian populations.

Takayasu Disease

A large-vessel vasculitis that primarily affects the aorta and its major branches. It may cause limb claudication, weak or unequal pulses, blood-pressure differences, bruits, fever, fatigue, or arterial narrowing and aneurysms. Imaging and inflammatory assessment guide immunosuppressive and vascular treatment.

Takayasu's Arteritis

A chronic granulomatous vasculitis of the aorta and its major branches, predominantly affecting young women of Asian descent. Causes arterial stenosis, occlusion, or aneurysm formation. Presents

with limb claudication, absent pulses, bruit, hypertension, and constitutional symptoms. Treatment: corticosteroids and immunosuppressives.

Takotsubo Cardiomyopathy

Transient left ventricular apical ballooning resembling a Japanese octopus trap (takotsubo), typically triggered by emotional or physical stress. ECG may show ST elevation mimicking STEMI. Coronary angiography shows no obstructive disease. Most patients recover normal ventricular function within weeks.

Talabostat

Talabostat is an investigational inhibitor of fibroblast activation protein and related enzymes studied for immune and anticancer effects.

Talactoferrin Alfa

Talactoferrin alfa is a recombinant lactoferrin-related immunomodulatory protein studied as a treatment or adjunct in cancer and infectious disease.

Talaporfin Sodium

Talaporfin sodium is a photosensitizer used or studied in photodynamic therapy, where light activation produces local cytotoxic effects.

Talc

A soft mineral; inhalation of talc dust can cause talcosis, and contaminated injected talc can injure vessels and lungs.

Talcosis

Pneumoconiosis from talc dust inhalation or IV injection of crushed oral medications. Causes upper lobe nodular fibrosis. HRCT may show “galaxy sign” of calcified nodules.

Talcum Powder Poisoning

Harmful effects from swallowing or inhaling talcum powder. Swallowing usually causes mild stomach upset, but inhaling it can injure the lungs and make breathing difficult.

Talipes

A congenital or acquired deformity of the foot and ankle, including clubfoot.

Talipes Calcaneovalgus

A foot deformity in which the heel is turned outward and the ankle is dorsiflexed so the foot points upward; flexible newborn cases often improve, while rigid cases need evaluation.

Talipes Equinovarus

Clubfoot, a congenital deformity combining ankle equinus, hindfoot varus, forefoot adduction, and cavus; early serial casting and bracing are standard initial treatment.

Tall Stature

Tall stature means height substantially above the expected range for age and sex and may be familial, endocrine, genetic, or associated with another condition.

Talotrexin

Talotrexin is an investigational antifolate antineoplastic drug designed to inhibit folate-dependent cellular metabolism.

Talus

Second largest tarsal bone articulating with tibia/fibula proximally at the ankle. Approximately 60 percent of its surface is articular cartilage with no muscular attachments. Transmits entire body weight to the calcaneus. Talar neck fractures may disrupt blood supply causing avascular necrosis in up to 50 percent.

Tamoxifen (Nolvadex)

A medicine, biologic, or pharmacologic term related to tamoxifen (nolvadex); its mechanism, indication, interactions, and adverse effects depend on the specific agent.

Tamoxifen Citrate

The citrate salt of tamoxifen, a selective estrogen-receptor modulator used to treat or prevent some

breast cancers; blood clots and endometrial cancer are important uncommon risks.

Tampon

A material inserted into the vagina to absorb menstrual blood; prolonged use can rarely lead to toxic shock syndrome.

Tamponade

Compression of an organ or vessel by accumulated fluid, blood, or gas; cardiac tamponade impairs cardiac filling.

Tamsulosin Hydrochloride

The hydrochloride salt of tamsulosin, an alpha-1 blocker that relaxes prostate and bladder-neck muscle to improve urinary symptoms from benign prostatic enlargement.

Tandutinib

An investigational tyrosine-kinase inhibitor studied mainly for myeloproliferative disorders and cancer.

Tanespimycin

An investigational derivative of 17-AAG that inhibits heat-shock protein 90 and has been studied as an anticancer drug.

Tangier Disease

An autosomal-recessive disorder of HDL metabolism caused by ABCA1 dysfunction. It produces very low HDL cholesterol, enlarged yellow-orange tonsils, hepatosplenomegaly, peripheral neuropathy, thrombocytopenia, and premature atherosclerosis in some patients.

Tanner Staging

A scale for assessing sexual maturity during puberty. Stage 1: prepubertal. Stage 2: breast buds (girls), testicular enlargement (boys). Stage 3: breast enlargement, pubic hair. Stage 4: areolar development, more pubic hair.

Tannic Acid

A plant-derived polyphenol with astringent and

protein-binding properties, used in some industrial or historical preparations; concentrated ingestion can irritate the gastrointestinal tract and affect absorption.

Tanomastat

Tanomastat is an investigational matrix-metalloproteinase inhibitor studied for reducing tumor invasion and tissue remodeling.

Tantalum

A corrosion-resistant metal used in some surgical implants and medical devices; exposure risk depends on the form, dose, and route.

Tantulum

A misspelling of tantalum, a metal used in selected implants and medical devices.

Tap, Joint (Aspiration)

Tap, joint (aspiration) refers to an anatomical structure, biological substance, or physiologic concept used in describing the human body and its function.

Tapentadol

Tapentadol is a centrally acting analgesic with mu-opioid agonist and norepinephrine-reuptake-inhibitory effects, used for moderate-to-severe pain; respiratory depression, dependence, seizures, and serotonin or sedative interactions are possible.

Tapeworm

A parasitic flatworm with a segmented body that may live in the intestine or form larval cysts in tissues. People acquire different species by eating infected meat or fish or by ingesting eggs from contaminated sources. Symptoms and complications depend on the species and site.

Tapeworm Infection

Infection caused by parasitic tapeworms. Taenia species may live as adult worms in the intestine, while some tapeworm larvae can form cysts in tissues.

Tapeworm Infection - Beef Or Pork

Intestinal infection with *Taenia saginata* or *Taenia solium* acquired from undercooked beef or pork; *T. solium* eggs can also cause cysticercosis in tissues.

Tapeworm Infection - Hymenolepis

Intestinal infection with *Hymenolepis nana* or *H. diminuta*, acquired by ingesting eggs or infected arthropods and causing variable gastrointestinal symptoms.

Tapotement

A massage technique using rhythmic tapping, patting, or percussion. It should be avoided over injured areas or in people at risk of bleeding unless a trained clinician recommends it.

Tapping

Repeated light striking or percussion used in examination, therapy, or testing.

TAPSE

Alternate name; see *Tricuspid Annular Plane Systolic Excursion*.

TAPVR

Alternate index label; see *Total anomalous pulmonary venous return*.

Taq Polymerase

A heat-stable DNA-copying enzyme from *Thermus aquaticus*, widely used in polymerase chain reaction testing and research.

Tar Remover Poisoning

Poisoning from swallowing, inhaling, or skin contact with tar-removing chemicals. Effects vary by product and may include irritation, dizziness, vomiting, or breathing problems; contact poison-control services promptly.

Tarantism

Tarantism is a historical term for a dancing or behavioral syndrome once attributed to spider bite and now understood as a cultural or neurologic

phenomenon rather than a specific disease.

Tarantula Spider Bite

A bite from a tarantula that usually causes local pain, redness, and swelling. Some species also release irritating hairs that can affect the skin or eyes.

Tardive Dyskinesia

Involuntary, repetitive, purposeless movements caused by prolonged use of antipsychotic medications, typically developing after months to years of exposure. Movements include choreoathetoid movements of the tongue, face, and extremities (lip smacking, tongue protrusion, grimacing, choreiform movements of limbs).

Tarenflurbil

Tarenflurbil is an investigational derivative of flurbiprofen studied for effects on amyloid processing and Alzheimer disease.

Target Cell

A cell specifically affected by or responding to a hormone, drug, toxin, antigen, or other signal; target cells may also describe abnormal erythrocytes.

Target Organ

An organ particularly affected by a disease, hormone, toxin, drug, or other exposure.

Target Population

The defined group of people for whom a medical intervention, screening program, or research study is intended. It is characterized by criteria such as age, diagnosis, risk, or other relevant features.

Target Tissue

A tissue containing receptors or other molecular targets on which a hormone, drug, toxin, or signaling molecule acts.

Target-Controlled Infusion

A computerized infusion pump system that maintains a predetermined drug concentration in plasma or effect site using pharmacokinetic modeling. TCI

systems use three-compartment pharmacokinetic models to calculate infusion rates that achieve and maintain target concentration.

Targeted Radiation Therapy

Radiation treatment aimed closely at a tumor or other abnormal area to damage it while limiting exposure to nearby healthy tissue.

Targeted Temperature Management

Controlled therapeutic hypothermia for comatose survivors of cardiac arrest. Target: 32-36 degrees C for 24-72 hours followed by gradual rewarming. Mechanism: reduces metabolic demand, limits reperfusion injury, and reduces free radical generation. TTM2 trial showed 33 degrees C was not superior to 36 degrees C.

Targeted Therapies For Cancer

Cancer medicines designed to act on specific features of cancer cells or their surroundings. They may be more selective than traditional chemotherapy, but can still cause important side effects.

Targeted Therapy

Drugs targeting specific molecular abnormalities in cancer. Examples: imatinib (BCR-ABL), trastuzumab (HER2), rituximab (CD20), tyrosine kinase inhibitors. Requires molecular testing.

Tariquidar

Tariquidar is an investigational inhibitor of the P-glycoprotein drug transporter studied to reverse multidrug resistance and alter medicine distribution.

Tarlov Cyst

A cerebrospinal-fluid-filled perineural cyst, usually located near the dorsal root ganglia in the sacral spine. Many are incidental, but larger or symptomatic cysts can cause sacral pain, radiculopathy, sensory change, bowel or bladder dysfunction, or sexual dysfunction.

Tars

A viscous black material derived from organic mat-

ter; tobacco tar contains harmful chemicals.

Tarsal

Relating to the tarsus or ankle and hindfoot bones.

Tarsal Bones

The seven bones of the hindfoot and midfoot that connect the ankle to the metatarsals and form the arches and load-bearing framework of the foot.

Alternate Names:

Tarsals

Tarsal Coalition

An abnormal connection between two bones in the rear or middle foot, usually present from birth. It can cause a stiff, painful flatfoot, recurrent ankle sprains, and limited foot motion, often becoming more noticeable during adolescence.

Tarsal Joints

The articulations among the tarsal bones and between the tarsals and metatarsals, providing stability and controlled movement for walking and adapting the foot to surfaces.

Tarsal Tunnel Syndrome

Tibial nerve compression at ankle. Symptoms: burning pain, tingling in foot sole. Tinel sign at ankle. Treatment: orthotics, corticosteroid injection, surgical decompression.

Tarsorrhaphy

A procedure that partially or completely closes the eyelids to protect the cornea or reduce exposure in conditions such as severe dry eye or facial nerve palsy.

Tarsus

The group of seven bones forming the hindfoot and midfoot.

Tartar

A hard deposit of mineralized dental plaque, also called calculus.

Tarui Disease

An autosomal recessive metabolic myopathy (Glycogen Storage Disease Type VII [GSD VII]) caused by deficiency of the muscle isoform of phosphofructokinase (*PFKM*), the key rate-limiting enzyme of glycolysis catalyzing the conversion of fructose-6-phosphate to fructose-1,6-bisphosphate. Impairs energy generation from both muscle glycogen and blood-borne glucose. Characterized clinically by exercise-induced muscle fatigue, weakness, painful cramping, and rhabdomyolysis, exacerbated by high-carbohydrate meals (which paradoxically lower exercise tolerance by suppressing fatty acid oxidation), accompanied by compensated non-spherocytic hemolytic anemia and hyperuricemia.

Alternate Names:

Glycogen Storage Disease Type VII (GSD VII); Muscle Phosphofructokinase Deficiency

Tassinari Syndrome

An older name associated with epileptic encephalopathy featuring frequent abnormal electrical activity during sleep, seizures, and possible effects on learning or development. The term is now usually described more specifically as electrical status epilepticus in sleep.

Taste

The sensation produced by the interaction of chemical substances with taste receptor cells (TRCs) on the tongue, palate, pharynx, epiglottis, and upper esophagus. Taste is one of the five primary senses (gustation).

Taste - Impaired

Impaired taste is reduced, distorted, or absent ability to taste and may result from infection, medicines, oral disease, nerve injury, or changes in smell.

Taste Bud

A taste bud is a cluster of specialized sensory cells in papillae of the tongue and some oral tissues that detects sweet, sour, salty, bitter, and umami stimuli and transmits signals through cranial nerves.

Taste Disorder

Abnormal reduction, loss, distortion, or unpleasant alteration of taste, caused by infection, medications, oral disease, nerve injury, neurologic disease, or other factors.

Tattooing

Insertion of pigment into the skin; complications include infection, allergy, scarring, and difficulty interpreting lesions.

Tattoos & Permanent Makeup

Tattoos and permanent makeup place pigment into the skin; risks include infection, allergy, scarring, blood-borne disease, and difficulty removing or changing the result.

Tau PET

Tracers: Flortaucipir, MK-6240. Detect tau tangles in neurodegenerative diseases.

Taurodontism

Enlarged pulp chamber, short roots. Molars most affected. Associated with genetic conditions.

Taussig-Bing Anomaly

A rare, complex cyanotic congenital heart malformation categorized as a form of double outlet right ventricle, characterized anatomically by both great arteries arising predominantly or entirely from the right ventricle, with the aorta positioned side-by-side or anterior to the pulmonary artery, accompanied by a committed, subpulmonic ventricular septal defect. Hemodynamically resembles D-transposition of the great arteries with a large VSD, wherein oxygenated pulmonary venous blood preferentially streams from the left ventricle across the subpulmonic VSD directly into the pulmonary artery, and systemic venous blood enters the right ventricle and aorta, resulting in severe early cyanosis, pulmonary arterial hypertension, and congestive heart failure, corrected surgically in infancy by an arterial switch operation and VSD closure.

Alternate Names:

Taussig-Bing Syndrome; DORV with Subpulmonic VSD

TAVR

Alternate name; see *Transcatheter Aortic Valve Replacement*.

Taxanes

A class of chemotherapy medicines, including paclitaxel and docetaxel, that stabilize microtubules.

Taxis

Directed movement toward or away from a stimulus; in surgery it may also mean orderly repositioning.

Tay-Sachs Disease

An inherited autosomal-recessive lysosomal storage disease caused by disease-causing variants in both copies of the HEXA gene. Reduced beta-hexosaminidase A activity causes GM2 ganglioside to accumulate in nerve cells. The infantile form causes developmental regression, exaggerated startle, weakness, seizures, and often a cherry-red spot on the retina, with severe progression and death usually in early childhood. Juvenile and adult forms can begin later and progress more slowly, with variable movement, coordination, speech, or psychiatric problems. Carrier frequency is higher in some populations, including Ashkenazi Jewish, French Canadian, and Cajun communities, but the disease can occur in any ancestry. There is currently no curative treatment; supportive care, genetic counseling, and appropriate carrier or diagnostic testing are important.

Tazarotene

A topical retinoid used for acne, psoriasis, and photorejuvenation.

TB

Alternate index label; see *Tuberculosis*.

TB Test

A tuberculosis test looks for infection with *Mycobacterium tuberculosis* using a skin test, blood test, or examination of respiratory specimens.

TBG Blood Test

A blood test measuring thyroxine-binding globulin, the principal thyroid-hormone transport protein; it helps interpret abnormal total thyroid-hormone levels when binding-protein changes are suspected.

TBI

Alternate index label; see *Traumatic brain injury*.

TCA Cycle

The tricarboxylic acid (Krebs) cycle occurs in the mitochondrial matrix, completing the oxidation of acetyl-CoA (from carbohydrates, fats, and proteins) to CO₂ while generating reducing equivalents (NADH, FADH₂) for the electron transport chain.

TCAs (Tricyclic Antidepressants)

Antidepressants that block reuptake of serotonin and norepinephrine, with additional anticholinergic, antihistaminic, and alpha-adrenergic blocking effects. Examples include amitriptyline, nortriptyline, imipramine, and desipramine. They are effective for depression, neuropathic pain, and migraine prophylaxis.

Alternate Names:

Tricyclic Antidepressants

Td

Td is a vaccine protecting against tetanus and diphtheria, generally used as a booster in adolescents and adults according to local schedules.

Td Immunization

Vaccination against tetanus and reduced-dose diphtheria, used for routine boosters, catch-up immunization, and wound-related protection according to age and vaccination history.

Tdap Vaccination In Pregnancy

Administration of tetanus, diphtheria, and acellular pertussis (Tdap) vaccine during each pregnancy, optimally at 27-36 weeks. The goal: boost maternal antibodies to protect the newborn from pertussis (whooping cough) before the infant can be vaccinated at 2 months.

Tear

A split or rupture in tissue, such as a muscle, tendon, ligament, meniscus, skin, or organ, caused by trauma, overuse, degeneration, or pressure.

Tearing

Excessive production or inadequate drainage of tears, commonly caused by irritation, allergy, dry-eye reflex tearing, infection, eyelid malposition, or lacrimal-duct obstruction.

Tears

Fluid secreted by the lacrimal glands that lubricates, nourishes, and protects the ocular surface.

Tears, Decreased Production

Alternate index label; see *Dry eyes*.

Technetium-99

A radioactive isotope related to technetium-99m; technetium isotopes are used in nuclear medicine and radioactive decay studies.

Technetium-99M

Most widely used radionuclide in nuclear medicine. Half-life: 6 hours. Emits 140 keV gamma rays. Produced from molybdenum-99/technetium-99m generator. Used with various ligands: MDP (bone), sestamibi (myocardium), DTPA (renal), nanocolloid (lymphatic), HMPAO (brain perfusion).

Technique

The method, equipment, and protocol used to perform a medical examination, procedure, or imaging study.

Tedizolid Pharmacokinetics

The study of how tedizolid is absorbed, distributed, metabolized, and eliminated. Tedizolid is available orally and intravenously and has pharmacokinetic properties that support once-daily treatment; its clinical use depends on the infection and patient factors.

TEE

TEE, or transesophageal echocardiography, uses an ultrasound probe in the esophagus to obtain detailed images of heart valves, chambers, clots, and nearby vessels.

Teen Depression

A depressive disorder in an adolescent that causes persistent low mood or loss of interest and impairs daily functioning.

Alternate Names:

Depression, Teen

Teenage Pregnancy

Pregnancy in adolescents (usually 10-19 years), associated with higher maternal and fetal risks: preterm birth, low birth weight, preeclampsia, anemia, and increased neonatal mortality; also socioeconomic and educational consequences. Causes: unprotected intercourse, lack of contraception/education, poverty, and peer/family factors.

Teenagers And Drugs

Drug use during adolescence can affect brain development, school and relationships, safety, and mental health; early, nonjudgmental screening and support can reduce harm.

Teenagers And Sleep

Teenagers commonly need about eight to ten hours of sleep, but schedules, screens, caffeine, mood, school demands, and sleep disorders can reduce sleep quality.

TEER

Alternate name; see *Transcatheter Edge-to-Edge Repair*.

Teeth

Hard, mineralized structures in the mouth used for cutting, tearing, and grinding food.

Teeth Grinding

Alternate index label; see *Teeth grinding (bruxism)*.

Teeth Grinding (Bruxism)

Alternate index label for Bruxism.

Teeth, Disorders Of

Conditions affecting teeth, including caries, developmental defects, infection, trauma, malocclusion, and tumors.

Teeth-Grinding

Bruxism, repetitive clenching or grinding of teeth, often during sleep and capable of causing tooth wear and jaw pain.

Teething

Teething is eruption of an infant's primary teeth through the gums and may cause gum discomfort, drooling, and chewing; high fever or severe illness suggests another cause.

Teichopsia

A visual disturbance involving shimmering or fortification spectra, commonly occurring as a migraine aura.

Teicoplanin

A glycopeptide antibiotic used for selected serious gram-positive infections.

Telangiectasia

A visible, permanently dilated small blood vessel near the skin or mucosal surface, occurring alone or with conditions such as rosacea, liver disease, or inherited vascular syndromes.

Telangiectasis

Dilation of small superficial blood vessels, producing visible red or purple lines or patches.

Tele Radiology (Teleradiology)

The medical practice and technology of transmitting digital radiological patient images (such as plain radiographs, CT scans, MRIs, and ultrasonograms) across telecommunication networks from the site of acquisition to off-site, remote radiologists for the purposes of primary interpretation, emer-

gency triage, or secondary expert subspecialty consultation. Utilizes standardized Picture Archiving and Communication Systems and Digital Imaging and Communications in Medicine data transmission protocols with secure encryption compliant with healthcare privacy regulations. Essential for providing continuous 24/7 overnight emergency coverage to community hospitals and rural trauma centers.

Tele-Rehabilitation

Remote rehabilitation via technology. Video consultations, remote monitoring, home exercise programs. Increases access, especially rural.

Telehealth

Delivery of healthcare through secure electronic communication or remote monitoring, including video visits, telephone care, messaging, and transmission of clinical data.

Telemedicine

Delivery of medical care at a distance using information and communication technology. It may include video or telephone consultations, secure messages, remote diagnosis, treatment, and review of clinical information.

Telogen Effluvium

A common, non-scarring form of diffuse hair shedding that occurs when more hairs than usual enter the resting (telogen) phase. Shedding often begins 2–4 months after illness, fever, surgery, childbirth, major stress, rapid weight loss, nutritional deficiency, or certain medicines. It is usually temporary, with regrowth after the trigger is corrected, although recovery may take several months.

Telomere

Protective DNA sequence at chromosome ends. Shortens with cell division. Associated with aging.

Temazepam

A benzodiazepine hypnotic used short term for insomnia; sedation, dependence, and respiratory depression are risks.

Temperature

A measurement of body heat. The result varies with the measurement site, time, activity, age, and illness.

Temperature (Body Temperature)

The balance between heat production (metabolism, muscle activity) and heat loss (radiation, conduction, convection, evaporation). Normal core body temperature: 36.5–37.5°C (97.7–99.5°F), regulated by the hypothalamus (preoptic anterior hypothalamus—thermoregulatory center).

Alternate Names:

Body Temperature

Temperature Measurement

Measurement of body or environmental temperature using a thermometer or sensor at a specified site, interpreted with method, age, timing, and clinical context.

Temperature Scales

Systems for expressing temperature, including Celsius, Fahrenheit, and Kelvin.

Temple

The region of the skull at the side of the head between the forehead and ear.

Temple Syndrome

A genomic-imprinting disorder involving chromosome 14q32 that causes prenatal and postnatal growth restriction, hypotonia, developmental delay, feeding difficulty, short stature, and abnormal pubertal development.

Temporal

Relating to time or to the side of the skull and temporal lobe.

Temporal Arteritis

A vasculitic disorder affecting medium and large arteries, particularly the temporal artery, predominantly occurring in adults over 50 years of age.

Temporal Artery

An artery on the side of the scalp, especially the superficial temporal artery.

Temporal Bones

Paired bones forming the cranial floor and lateral wall, with squamous, petrous, mastoid, tympanic, and styloid parts. Contain the external auditory canal, middle and inner ear, and foramina: carotid canal (internal carotid), jugular foramen (CN IX-XI, IJV), and foramen lacerum.

Temporal Lobe

One of the four major lobes of the cerebral hemisphere, located beneath the lateral fissure (Sylvian fissure) on the lateral surface of the brain, extending medially to the parahippocampal gyrus. The temporal lobe is involved in auditory processing, language comprehension, memory, and emotion.

Temporal Lobe Epilepsy

The most common form of focal (partial) epilepsy, arising from the medial temporal lobe structures (hippocampus, amygdala, parahippocampal gyrus).

Temporal Lobe Seizure

A seizure that begins in the temporal lobe of the brain and may cause altered awareness, unusual sensations, or repetitive movements.

Alternate Names:

Seizure, Temporal Lobe

Temporary Loss Of Consciousness

Temporary loss of consciousness is a brief episode of unresponsiveness with loss of postural tone, commonly from syncope but sometimes from seizure or another disorder.

Temporomandibular Disorders

Alternate index label; see *TMJ disorders*.

Temporomandibular Joint

Synovial ginglymoarthrodial joint between mandibular condyle and temporal bone's mandibular fossa, separated by an articular disc. Only bilateral body

joint, allowing hinge and sliding movements. Disc divides into superior/inferior compartments. Temporomandibular disorders cause pain, clicking, limited movement, common with bruxism.

Temporomandibular Joint Disorder

Pain and dysfunction of TMJ and masticatory muscles. Symptoms: jaw pain, clicking, limited opening, headache, ear pain. Etiology: multifactorial (bruxism, stress, trauma, occlusion). Treatment: conservative (reassurance, self-care, splint, NSAIDs), physical therapy, rare surgery.

Temporomandibular Joint Replacement

Total joint replacement for severe TMJ pathology. Indications: ankylosis, severe degenerative disease, failed previous surgery.

Temporomandibular Joint Syndrome

Pain or dysfunction involving the jaw joint and chewing muscles, often causing jaw pain, clicking, restricted opening, headache, or tooth-related symptoms. Causes include clenching, trauma, arthritis, and stress-related muscle tension. Persistent locking, severe pain, or inability to eat requires dental or medical evaluation.

Temporomandibular Joint Syndrome

Alternate index label for TMJ.

Temporomandibular Joint Syndrome

Alternate index label for Temporomandibular Joint Syndrome.

TEN

Alternate index label; see *Toxic epidermal necrolysis*.

Tenascin

A family of extracellular-matrix proteins involved in cell adhesion, tissue development, wound healing, and tumor biology.

Tender

Painful or sensitive when touched or pressed, a

finding that may indicate inflammation, infection, trauma, ischemia, or another underlying process.

Tenderness

Pain or discomfort produced by pressing a body part.

Tendinitis

Tendinitis is painful inflammation or degeneration of a tendon, often from repetitive loading, causing localized tenderness and pain with movement.

Tendinitis Rotator Cuff (Rotator Cuff)

Alternate index label for Rotator Cuff.

Tendinitis, Achilles

Alternate index label; see *Achilles tendinitis*.

Tendinitis, Patellar

Alternate index label; see *Patellar tendinitis*.

Tendinopathy

A disorder involving tendon pain and impaired tendon structure, usually related to overuse or degeneration.

Tendon

Dense regular connective tissue connecting muscle to bone, composed primarily of type I collagen fibers arranged in parallel bundles. Transmits muscle force to bone to produce movement. Poor vascularity (slower healing). Enclosed in tendon sheath (synovial) at friction points.

Tendon Injury

Damage to a tendon ranging from microscopic overuse injury to partial or complete rupture, causing localized pain, weakness, swelling, or loss of function.

Tendon Repair

A medical procedure, treatment, device, or care topic related to tendon repair; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Tendon Sheath

A thin lubricated covering around some tendons that reduces friction as the tendon moves; inflammation is called tenosynovitis.

Tendon Transfer

Surgical relocation of a functioning tendon to restore movement or balance when the original muscle-tendon unit is absent, paralyzed, or irreparable. Successful transfer requires an adequate donor, appropriate excursion, and a stable recipient joint.

Tendonitis

Tendinopathy: inflammation or degeneration of a tendon from overuse, repetitive strain, or trauma, causing localized pain, swelling, and tenderness with activity. Common sites: rotator cuff (supraspinatus), lateral elbow (tennis elbow), medial elbow (golfer's elbow), Achilles, patellar (jumper's knee), biceps, and De Quervain.

Tenecteplase

A modified tissue plasminogen activator (TNK-tPA) thrombolytic with longer half-life, higher fibrin specificity, and resistance to PAI-1. Used for acute ischemic stroke and STEMI. Given as a single IV bolus (weight-based), simplifying administration. Adverse effects: hemorrhage (intracranial—major concern), allergic reactions.

Tenericutes

A former bacterial group characterized by organisms that lack a typical cell wall, including *Mycoplasma* and related bacteria.

Tenesmus

A distressing sensation of needing to empty the bowel or bladder despite little output, commonly caused by rectal inflammation, infection, tumor, constipation, or urinary irritation.

Tenia

Tenia is an older or variant spelling related to *Taenia*, a genus of tapeworms that can infect humans or animals.

Teniposide

Teniposide is a podophyllotoxin-derived antineoplastic drug that inhibits topoisomerase II and is used in selected leukemia or lymphoma regimens.

Tennis Elbow

Lateral epicondylitis, a tendinopathy of the common extensor tendon at the lateral epicondyle (extensor carpi radialis brevis). Caused by repetitive wrist extension and overuse, not limited to tennis. Presents with lateral elbow pain worsened by wrist extension and grip. Tenderness over the lateral epicondyle.

Alternate Names:

Epicondylitis, Lateral Lateral Elbow Tendinopathy
Lateral Epicondylitis Lateral Epicondylitis

Tennis Elbow (Lateral Epicondylitis)

Alternate index label for Lateral Epicondylitis.

Tennis Elbow Surgery

A medical procedure, treatment, device, or care topic related to tennis elbow surgery; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Tenodesis

Surgical fixation of a tendon to bone or another tendon to restore function, stabilize a joint, or treat tendon imbalance.

Tenofovir

Tenofovir is a nucleotide reverse-transcriptase inhibitor used with other medicines for HIV and as treatment or prevention for hepatitis B; renal and bone toxicity and formulation-specific monitoring are important.

Tenosynovitis

Tenosynovitis is inflammation of the sheath surrounding a tendon, producing pain, swelling, and restricted gliding, commonly in the hand or wrist.

Tenosynovitis, De Quervain'S

Alternate index label; see *De Quervain tenosynovitis*.

Tenotomy

The surgical cutting, division, or transection of a tendon, performed through open, percutaneous, or endoscopic approaches. Utilized in orthopedic, pediatric, and ophthalmic surgery to relieve severe muscle contractures, eliminate pathological spastic deforming forces, or rebalance joint kinematics. Classic applications include percutaneous Achilles tenotomy (a key step in the Ponseti method for correcting congenital talipes equinovarus / clubfoot), extraocular muscle tenotomy (to adjust alignment in strabismus surgery), and release of contracted flexor tendons in cerebral palsy or stroke spasticity.

Tenovaginitis

Inflammation of a tendon and its surrounding sheath, also called tenosynovitis.

Tensile Strength

The ability of a tissue or material to resist being pulled apart; it is important in wound healing and the integrity of sutures, tendons, and other connective tissues.

Tensilon Test

An older pharmacologic test using edrophonium to look for brief improvement in muscle weakness in suspected myasthenia gravis; it is rarely used because safer tests are available.

Tension

A pulling force or state of being stretched; in medicine it may describe tissue tension, blood-vessel wall tension, muscle tightness, or psychological strain depending on context.

Tension Headache

The most common headache type, characterized by bilateral, pressing/tightening head pain of mild-moderate intensity (like a band around the head), lasting 30 min to days, without nausea or vomiting; photophobia or phonophobia may be present but not both.

Tension Pneumothorax

Life-threatening pneumothorax with mediastinal

shift. Emergency needle decompression (2nd intercostal space, midclavicular line) followed by chest tube. Signs: respiratory distress, hypotension, tracheal deviation, distended neck veins, unilateral absent breath sounds.

Tension-Type Headache

A usually bilateral, pressing or tightening headache of mild-to-moderate intensity without prominent nausea, often associated with stress, muscle tenderness, or poor sleep.

Tentorium

A fold of dura mater separating the cerebellum from the occipital lobes.

Tenuazonic Acid

A mycotoxin produced by some fungi that contaminates certain plants and foods and can inhibit protein synthesis; toxicity depends on exposure and the producing organism.

Teprotumumab

A monoclonal antibody targeting the insulin-like growth factor 1 receptor and used to treat thyroid eye disease in selected patients.

Ter In Die

Ter in die means three times a day; explicit plain-language timing is safer than relying on Latin abbreviations.

Terameprocol

Terameprocol is an investigational anticancer compound that affects transcription factors and cell-survival pathways.

Teratocarcinoma

A teratocarcinoma is a malignant germ-cell tumor containing tissues from more than one embryonic layer, often arising in the testis or ovary.

Teratogen

Substance causing birth defects. Examples: thalidomide, isotretinoin, alcohol, warfarin, valproate, ACE

inhibitors. Prenatal exposure critical.

Teratogenesis

The production of congenital abnormalities by genetic, environmental, infectious, nutritional, or drug-related influences.

Teratogenic

Capable of causing congenital malformations, growth impairment, functional deficits, or pregnancy loss when an embryo or fetus is exposed during a vulnerable developmental period.

Teratogenic Drugs

Medicines that can cause birth defects, pregnancy loss, or impaired fetal development; risk depends on the drug, dose, timing, and pregnancy.

Teratogenicity

The ability of a drug, chemical, or infectious agent to cause birth defects or developmental abnormalities when exposure occurs during pregnancy. The risk of teratogenic effects depends on the timing of exposure, with the embryonic period from 3 to 8 weeks after fertilization being the most critical for organogenesis.

Teratoma

A type of germ cell tumor composed of tissues derived from two or three germ layers (ectoderm, endoderm, mesoderm), reflecting pluripotent differentiation.

Teratozoospermia

Teratozoospermia is a high proportion of abnormally shaped sperm and may contribute to reduced fertility, though semen results must be interpreted as a whole.

Terazosin Hydrochloride

The hydrochloride salt of terazosin, an alpha-1 blocker used for urinary symptoms from benign prostatic enlargement and sometimes for hypertension; dizziness and first-dose fainting can occur.

Terbinafine

An allylamine antifungal agent that inhibits squalene epoxidase, an early enzyme in the ergosterol biosynthesis pathway, causing accumulation of squalene and depletion of ergosterol in the fungal cell membrane. The accumulation of squalene is fungicidal, while ergosterol depletion disrupts membrane function.

Terbufos

Terbufos is a highly toxic organophosphate pesticide that inhibits acetylcholinesterase and can cause severe cholinergic poisoning.

Terbutaline

A beta-2 agonist used as a bronchodilator; it has limited and specialist use as a tocolytic.

Terfenadine

An older antihistamine withdrawn or restricted in many places because interactions and excessive blood levels can cause dangerous heart-rhythm abnormalities.

Teriparatide For Osteoporosis

Recombinant human parathyroid hormone fragments one through thirty-four that stimulate osteoblast-mediated bone formation. Indicated for severe osteoporosis with fractures or very low T-scores and for patients who have failed or are intolerant to bisphosphonates. Daily subcutaneous injection for up to two years.

Terlipressin

Terlipressin is a vasopressin analogue that constricts splanchnic vessels and is used in selected cases of variceal bleeding or hepatorenal syndrome.

Term

The end of pregnancy at 37 completed weeks of gestation; births before this are classified as preterm.

Term Birth

Term birth occurs at or after 37 completed weeks of pregnancy; early term, full term, and late or post-

term categories describe more specific gestational ranges.

Terminal Care

Care focused on comfort, symptom relief, dignity, and support for a person with an advanced illness who is approaching the end of life.

Terminal Delirium

Delirium occurring near the end of life, often caused by advanced organ failure, medication effects, metabolic disturbance, infection, or reduced cerebral perfusion. Management prioritizes comfort, reversible contributors when consistent with goals, and support for the patient and family.

Terminal Ileitis

Inflammation of the distal ileum, commonly associated with Crohn disease but also caused by infection, ischemia, medication, or other inflammatory conditions and producing pain, diarrhea, or bleeding.

Terminal Illness

An advanced disease that is expected to lead to death, although the timing and response to available treatment may vary.

Terminal Insomnia

Waking too early and being unable to return to sleep, a pattern that can occur in depression, circadian-rhythm disorders, aging, or other insomnia conditions.

Terms Medical Abbreviations (Common Medical Abbreviations And Terms)

Alternate index label for Common Medical Abbreviations and Terms.

Terpene

A terpene is a hydrocarbon built from isoprene units and found in plant oils, fragrances, medicines, and other natural products.

Terry'S Nails

A nail change in which most of the nail plate appears

white with a narrow pink or brown distal band; it may be associated with liver disease, heart failure, diabetes, or ageing.

Tertian Fever

A fever pattern recurring approximately every 48 hours, classically associated with *Plasmodium vivax* or *P. falciparum* malaria.

Tertiary Hyperparathyroidism

Autonomous overproduction of parathyroid hormone after long-standing secondary hyperparathyroidism, usually in chronic kidney disease, causing high calcium and high PTH.

Tertiary Prevention

Reducing complications from established disease. Examples: rehabilitation, chronic disease management, palliative care.

Tertiary Protein

The three-dimensional folding of a single protein chain, created by interactions among amino-acid side chains and determining the protein's functional shape.

Tesetaxel

Tesetaxel is an investigational oral taxane that disrupts microtubules and has been studied as an anti-cancer treatment.

Test Meal

A standardized meal given before measuring a physiologic response, such as gastric acid or glucose handling.

Test Strips

Reagents for diagnostic testing. Types: urine dipstick, blood glucose, ketone.

Test-Tube

A laboratory glass or plastic vessel used to contain, mix, or test small samples.

Test-Tube Baby

A lay term for a baby conceived through in-vitro fertilization.

Testes

The paired male gonads that produce sperm and testosterone and are located in the scrotum.

Testicle

One of the male reproductive glands in the scrotum. It produces sperm and testosterone; sudden severe testicular pain is an emergency because it may indicate torsion.

Testicle Lump

A localized mass or enlargement in or around a testis caused by cyst, infection, inflammation, torsion, trauma, or tumor; a new intratesticular mass requires prompt evaluation.

Testicle Pain

Pain in one or both testes or the surrounding structures, with urgent causes including torsion, infection, trauma, and incarcerated hernia.

Testicle, Diseases Of

Disorders of the testis, including torsion, infection, trauma, tumors, vascular disease, and developmental abnormalities.

Testicle, Retractable

Alternate index label; see *Retractable testicle*.

Testicle, Undescended

Alternate index label; see *Undescended testicle*.

Testicles

The paired male reproductive glands in the scrotum that produce sperm and testosterone; pain, enlargement, a new lump, or sudden change requires examination.

Testicular Atrophy

Shrinkage and loss of functional tissue in a testicle caused by undescended testis, infection, torsion,

trauma, varicocele, hormones, aging, or other injury.

Testicular Biopsy

Removal of a small sample of testicular tissue for microscopic examination, used selectively to evaluate infertility, atypical lesions, or suspected testicular disease.

Testicular Cancer

Most common solid malignancy in young men 15-35 years. Types: germ cell tumors (seminoma, non-seminoma), sex cord-stromal tumors. Presentation: painless testicular mass, heaviness, ache. Diagnosis: scrotal ultrasound, tumor markers (AFP, beta-hCG, LDH).

Alternate Names:

Cancer, Testicular

Testicular Disorders

Conditions affecting the testes or surrounding structures, including torsion, infection, inflammation, trauma, tumors, and hormonal disorders. Symptoms may include pain, swelling, a mass, infertility, or changes in sexual development and require appropriate evaluation.

Alternate Names:

Infection Testicle (Testicular Disorders) Pain Scrotum (Testicular Disorders) Torsion Testicle (Testicular Disorders) Tumor Testicle (Testicular Disorders)

Testicular Gonadoblastoma

A rare mixed germ-cell and sex-cord tumor arising in a dysgenetic or undescended gonad, often associated with Y-chromosome material and risk of another germ-cell tumor.

Testicular Hydrocele

A collection of clear fluid around a testicle within the tunica vaginalis, causing painless scrotal enlargement that transilluminates; it may be congenital, reactive, or secondary to disease.

Testicular Mass

A lump or enlargement within or around a testicle caused by tumor, infection, inflammation, cyst,

hematoma, or torsion; a solid intratesticular mass requires urgent ultrasound and specialist assessment.

Testicular Microlithiasis

Multiple tiny calcifications within the testicular tissue seen on ultrasound. The finding alone does not establish cancer, and follow-up depends on other risk factors and clinical findings.

Testicular Self-Exam

A testicular self-exam is a gentle check for new lumps, swelling, heaviness, or changes in size or shape; a new abnormality should be medically assessed.

Testicular Torsion

Surgical emergency caused by twisting of the spermatic cord around its longitudinal axis, cutting off venous return and leading to arterial occlusion, ischemia, and infarction of the testicle.

Alternate Names:

Torsion, Testicular

Testicular Torsion Emergency

Rotation of the spermatic cord that compromises testicular blood flow, causing sudden unilateral scrotal pain, nausea, a high-riding testis, and an abnormal cremasteric reflex. Immediate surgical exploration and detorsion are required; imaging must not delay treatment when suspicion is high.

Testicular Torsion Repair

Urgent surgery to untwist a spermatic cord and restore testicular blood flow, followed by fixation of the affected and usually opposite testis to prevent recurrence.

Testis

The male reproductive gland that produces spermatozoa in seminiferous tubules and secretes testosterone from interstitial Leydig cells. It is normally located in the scrotum.

Testis (Testicle)

The male gonad, a paired ovoid organ (4–5 cm ×

2–3 cm, weight 15–20 g) located in the scrotum, responsible for sperm production (spermatogenesis) and testosterone secretion. Each testis is covered by the tunica vaginalis (peritoneal remnant) and tunica albuginea (dense fibrous capsule).

Alternate Names:

Testicle

Testis Disorder

A condition affecting a testicle, including torsion, infection, tumor, trauma, infertility-related damage, hormonal dysfunction, or abnormal development.

Testis Neoplasm

A tumor arising in testicular tissue, most commonly a germ-cell tumor in younger adults; painless enlargement, heaviness, or a solid mass requires prompt ultrasound and evaluation.

Testolactone

An aromatase inhibitor formerly used to reduce estrogen production in selected endocrine conditions; current use is limited by availability and newer alternatives.

Testosterone

A steroid sex hormone produced mainly by the testes and, in smaller amounts, by the ovaries and adrenal glands. It supports reproductive development, muscle and bone maintenance, red-blood-cell production, libido, and other physiologic functions; levels vary with age, sex, time of day, illness, and medicines.

Testosterone Deficiency

Alternate index label; see *Male hypogonadism*.

Testosterone Low (Low Testosterone (Low T))

Alternate index label for Low Testosterone (Low T).

Testosterone Propionate

A short-acting injectable testosterone ester used in selected hormone-replacement or androgen-

deficiency settings; risks include erythrocytosis, infertility, acne, mood effects, and prostate-related monitoring needs.

Tests And Visits Before Surgery

Preoperative tests and visits review health conditions, medicines, anesthesia risk, needed laboratory or imaging tests, and instructions for the operation.

Tests For H Pylori

Tests for *Helicobacter pylori* infection, including urea breath testing, stool antigen testing, biopsy-based testing, or selected blood tests; choice depends on treatment and clinical context.

Tet Spell

An acute, paroxysmal, life-threatening hypercyanotic episode occurring in infants and young children with unrepaired Tetralogy of Fallot. Triggered by crying, defecation, feeding, pain, or exertion, which causes catecholamine release and infundibular spasm of the right ventricular outflow tract, resulting in a dramatic increase in right-to-left intracardiac shunting through the ventricular septal defect, hypoxemia, metabolic acidosis, and profound cyanosis. Emergency management includes calming the child, placing the patient in a knee-to-chest position (which increases systemic vascular resistance and reduces right-to-left shunting), administering supplemental oxygen, intravenous fluids, and morphine or beta-blockers (propranolol).

Alternate Names:

Hypercyanotic Spell; Paroxysmal Dyspneic Attack; Blue Spell

Tetanus

Tetanus is toxin-mediated neurologic disease caused by *Clostridium tetani* entering a wound, producing painful muscle rigidity, spasms, autonomic instability, and respiratory failure. introduced through wounds. Characterized by generalized rigidity and painful muscle spasms (trismus/lockjaw, risus sardonicus, opisthotonos). Autonomic instability can occur in severe cases. Diagnosis: clinical.

Alternate Names:

Lockjaw

Tetanus Toxin

An infectious process or toxic exposure involving tetanus toxin; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Tetany

A clinical syndrome of neuromuscular hyperexcitability caused by hypocalcaemia (most common—total calcium <2.1 mmol/L, ionised calcium <1.1 mmol/L), but also caused by hypomagnesaemia, hypokalaemia, and alkalosis (respiratory or metabolic).

Tethered Cord Syndrome

A congenital or acquired condition where the spinal cord is abnormally attached (tethered) to surrounding tissues, restricting its normal upward movement within the spinal canal. Caused by a thickened filum terminale, lipomeningomyelocele, diastematomyelia, or previous myelomeningocele repair.

Tetrabenazine

A vesicular monoamine transporter 2 inhibitor used for selected hyperkinetic movement disorders.

Tetracaine

Tetracaine is a potent ester local anaesthetic used topically or for spinal anaesthesia; excessive dose can cause seizures, cardiovascular toxicity, and methemoglobinaemia.

Tetrachloroethylene

Tetrachloroethylene, or perchloroethylene, is an industrial solvent used in dry cleaning; substantial exposure can affect the nervous system, liver, and kidneys.

Tetrachlorvinphos

Tetrachlorvinphos is an organophosphate insecticide that can inhibit acetylcholinesterase and cause cholinergic toxicity after exposure.

Tetracycline

A medicine, biologic, or pharmacologic term related

to tetracycline; its mechanism, indication, interactions, and adverse effects depend on the specific agent.

Tetracycline Stain

Intrinsic discoloration from tetracycline antibiotics during tooth development. Gray or brown bands. Treatment: bleaching, veneers, crowns.

Tetracyclines

A class of bacteriostatic antibiotics that bind reversibly to the 30S ribosomal subunit and block the attachment of aminoacyl-tRNA to the acceptor site, inhibiting protein synthesis. They include tetracycline, doxycycline, and minocycline.

Tetrahydrozoline Poisoning

Poisoning from swallowing eye drops or nasal drops containing tetrahydrozoline. It can cause extreme sleepiness, slow heart rate, low blood pressure, and breathing problems, especially in children.

Tetralogy Of Fallot

Tetralogy of Fallot is a congenital heart defect combining ventricular septal defect, right-ventricular outflow obstruction, overriding aorta, and right-ventricular hypertrophy.

Tetralogy Of Fallot

A congenital heart defect comprising ventricular septal defect, right-ventricular outflow obstruction, overriding aorta, and right-ventricular hypertrophy.

Tetraplegia

Paralysis or severe weakness of all four limbs, usually from cervical spinal-cord disease or injury.

Tetraplegia (Quadriplegia)

Paralysis or severe weakness affecting all four limbs and usually the trunk, commonly caused by cervical spinal-cord injury or disease.

Alternate Names:

Quadriplegia

TGA

Alternate index label; see *Transient global amnesia*.

Th1/Th2/Th17/Tfh/Treg Balance

CD4+ T helper subsets defined by master transcription factors and signature cytokines: Th1 (Tbet, IFN-gamma, IL-12, STAT4): intracellular pathogens, autoimmunity (RA, MS, type 1 diabetes). Th2 (GATA3, IL-4/5/13, IL-4/STAT6): helminths, allergy, asthma.

Thalamus

Bilateral ovoid gray matter structure forming the third ventricle's lateral walls, the brain's relay station for nearly all sensory (except olfaction), motor, and limbic information to the cortex. Numerous nuclei: VPL (spinothalamic, dorsal column), VPM (trigeminal), LGN (visual), MGN (auditory), VL/VA (motor).

Thalassaemia

The British spelling of thalassemia, inherited reduced production of alpha- or beta-globin chains causing anemia.

Thalassemia

A group of inherited hemoglobin disorders caused by reduced or absent synthesis of one or more globin chains, leading to microcytic hypochromic anemia.

Alternate Names:

Cooley's Anemia Mediterranean Anemia

Thalassemia Major

Beta-thalassemia, an autosomal recessive hemoglobin disorder with absent or minimal beta-globin chain production. Presents in infancy with severe anemia, failure to thrive, hepatosplenomegaly, and skeletal deformities. Diagnosis by hemoglobin electrophoresis showing absent HbA and elevated HbF and HbA2.

Beta-Thalassemia Trait

The heterozygous carrier state for a beta-globin gene variant. It is usually asymptomatic or causes mild microcytosis with mild or no anemia; hemoglobin

electrophoresis often shows increased HbA2. It is also called beta-thalassemia minor.

Alternate Names:

Beta Thalassemia Trait Beta-Thalassemia Minor
Thalassemia Minor

Thalidomide

An immunomodulatory medicine used for selected cancers and inflammatory disorders; it is strongly teratogenic.

Thallium

Thallium is a toxic heavy metal that can cause gastrointestinal illness followed by painful peripheral neuropathy, hair loss, skin changes, kidney or liver injury, seizures, and cardiovascular or neurologic toxicity. Suspected exposure requires immediate poison-control or emergency guidance.

Alternate Names:

Poisoning Thallium (Thallium)

Thallium Scan

Nuclear medicine imaging using Tl-201. Myocardial perfusion, tumor detection. Largely replaced by Tc-99m agents.

Thallium-201

SPECT radionuclide, half-life 73 hours. Used for myocardial perfusion imaging. Mechanism: K⁺ analog, taken up by Na⁺/K⁺ ATPase.

Thanatology

The scientific study of death, dying, postmortem changes, and the psychological, social, and ethical aspects of bereavement. In forensic medicine, it also examines the circumstances and evidence surrounding death.

Thanatophoric Dysplasia

A severe skeletal dysplasia, usually caused by an FGFR3 variant, with markedly shortened limbs, narrow thorax, macrocephaly, and characteristic skull and vertebral abnormalities. The severe thoracic restriction commonly causes neonatal respiratory failure.

Theca

A connective-tissue sheath, especially the theca surrounding an ovarian follicle.

Theca Lutein Cyst

A benign, functional ovarian cyst characteristically occurring bilaterally and filled with clear straw-colored fluid, resulting from hyperstimulation and marked luteinization of thecal cells driven by abnormally high circulating levels of human chorionic gonadotropin (β -hCG). Strongly associated with gestational trophoblastic disease (hydatidiform mole and choriocarcinoma), multiple gestations, and clomiphene or gonadotropin therapy, typically resolving spontaneously once β -hCG levels normalize.

Thecoma

A usually benign ovarian sex-cord stromal tumor composed of theca-like cells, often producing estrogen and presenting with abnormal uterine bleeding or an incidental adnexal mass.

Thelarche

The onset of breast development, usually the first visible sign of female puberty. It commonly begins between ages 8 and 13, with substantial normal variation.

Thenar Eminence

The fleshy muscular prominence at the base of the thumb.

Theophylline

A methylxanthine bronchodilator with a narrow therapeutic index and substantial interaction and toxicity risk.

Theranostic Pair

Combined therapy and diagnostics using matched radiopharmaceutical pairs. See theranostics.

Theranostics

Combined therapy and diagnostics using matched radiopharmaceutical pairs. Diagnostic isotope (e.g., Ga-68) identifies target, therapeutic isotope (e.g.,

Lu-177) delivers treatment. Examples: Ga-68/Lu-177 DOTATATE (neuroendocrine tumors), PSMA theranostics (prostate cancer).

Therapeutic Abortion

Intentional termination of a pregnancy because of medical, fetal, or other legally recognized indications, according to applicable law and clinical guidance.

Therapeutic Alliance

The collaborative, trusting relationship between therapist and patient that is foundational to effective psychotherapy. It includes agreement on goals, tasks, and the bond between therapist and patient. It is the strongest predictor of psychotherapy outcome across all therapeutic modalities.

Therapeutic Cloning

Creating a genetically matched embryonic stem-cell line for research or potential tissue replacement; it is distinct from reproductive cloning and remains ethically and technically limited.

Therapeutic Exercise

Exercise prescribed to improve function, reduce pain, prevent disability. Types: stretching, strengthening, aerobic, balance, functional. Progression: low to high intensity.

Therapeutic Hypothermia

Controlled cooling of the body temperature used in psychiatric practice for severe catatonia and neuroleptic malignant syndrome. In NMS, cooling is a key treatment alongside discontinuation of the offending agent and dantrolene administration.

Therapeutic Index

The therapeutic index is a quantitative measure of the relative safety of a drug, defined as the ratio of the dose that produces toxicity in 50 percent of the population to the dose that produces the desired therapeutic effect in 50 percent of the population.

Therapeutic Window

The range of plasma drug concentrations between the minimum effective concentration (producing a therapeutic effect) and the minimum toxic concentration (producing toxicity), in which efficacy is achieved with acceptable toxicity.

Therapeutics

The branch of medicine concerned with the treatment of disease, encompassing all interventions used to prevent, manage, or cure illness.

Therapy

Treatment intended to prevent, relieve, or cure disease or to improve function and quality of life.

Therapy Monitoring

Repeated clinical assessment, laboratory testing, or imaging used to evaluate response, adverse effects, and ongoing need for treatment.

Thermal Injury

Tissue damage caused by heat, flame, hot liquid, steam, or cold. Burn severity depends on depth, total body surface area, location, inhalation injury, associated trauma, and patient factors; assessment should include airway and circulatory complications.

Thermal Injury (Burns)

Tissue damage caused by heat. Initial care includes stopping the exposure, removing loose constricting material, cooling briefly with clean water, covering the injury, and assessing depth and extent. Larger or deeper burns require specialized evaluation, fluid management, pain control, wound care, and tetanus assessment.

Alternate Names:

Burns

Thermal Therapy

Treatment using heat or cold to influence pain, swelling, muscle tension, or tissue temperature. Effects and safety depend on the condition and method used.

Thermogenesis

Heat production from food digestion, muscle activity, brown fat. Diet-induced thermogenesis 10% of TDEE.

Thermography

Imaging or recording of heat patterns from the body surface.

Thermoluminescent Dosimeter

A radiation-monitoring device that measures accumulated exposure by light released from an irradiated crystal when heated.

Thermometer

Device measuring body temperature. Types: mercury (phased out), digital, tympanic, temporal artery, infrared. Normal oral: 36.5-37.5°C.

Thermometer Scales

Temperature measurement systems such as Celsius, Fahrenheit, and Kelvin.

Thermoneutral Zone

The range of ambient temperatures over which body temperature is maintained by basal metabolic heat production alone, without additional thermoregulatory responses such as shivering or sweating. In humans the zone is approximately 27-32 degrees C for a resting nude adult. Within the zone, vasomotor tone adjusts heat loss.

Thermoplasty

A bronchoscopic procedure that uses radiofrequency energy to ablate smooth muscle in the airways, reducing bronchospasm. It is approved for severe, persistent asthma in adults not controlled with ICS/LABA. It reduces exacerbations and emergency department visits but does not improve lung function.

Thermoreceptor

A specialized free sensory nerve ending or receptive ending that detects physiological variations in local temperature and thermal gradients, classified into

cool/cold receptors and warm/heat receptors. Cold receptors are primarily mediated by transient receptor potential channels such as TRPM8 (activated at 10–25°C and by menthol) transmitting via thinly myelinated A δ fibers, whereas warm receptors are mediated by channels such as TRPV1 (activated by temperatures > 43°C and capsaicin) transmitting via unmyelinated C fibers. Afferent signals ascend through the spinothalamic tract to the preoptic area of the hypothalamus to regulate systemic autonomic thermoregulation (sweating, shivering, cutaneous vasodilation, and vasoconstriction).

Thermoregulation

The physiological maintenance of core body temperature near the set point (37 degrees C) via a balance of heat production and heat loss, controlled by the preoptic area of the hypothalamus.

Thermostat

A thermostat regulates temperature by sensing heat and controlling a heating or cooling system; accurate environmental temperature can matter for patients, specimens, and medicines.

Thiabendazole

An anthelmintic used for selected worm infections and, historically, some fungal infections.

Thiamin

Vitamin B1, a water-soluble vitamin required for carbohydrate metabolism and neural function; severe deficiency can cause beriberi or Wernicke encephalopathy.

Thiamine

Vitamin B1, a water-soluble vitamin required for carbohydrate metabolism and nerve function. Severe deficiency can cause beriberi or Wernicke encephalopathy.

Thiamine (Vitamin B1)

A water-soluble vitamin whose active coenzyme form, thiamine pyrophosphate, is required by pyruvate dehydrogenase, alpha-ketoglutarate dehydrogenase, transketolase, and branched-chain ketoacid

dehydrogenase. Deficiency causes beriberi and Wernicke encephalopathy.

Alternate Names:

Vitamin B1

Thiazide Diuretics

Thiazide diuretics act on the distal convoluted tubule by inhibiting the sodium-chloride cotransporter, reducing sodium and chloride reabsorption and producing natriuresis and diuresis. They are less potent than loop diuretics but have a longer duration of action, typically 6 to 12 hours.

Thiazide Overdose

Taking too much of a thiazide diuretic can cause dehydration, low blood pressure, abnormal sodium or potassium levels, confusion, or abnormal heart rhythms.

Thiazides

Diuretics that inhibit sodium-chloride reabsorption in the distal convoluted tubule.

Thiazolidinedione Drugs

A group of diabetes medicines that make body tissues more sensitive to insulin by activating a protein called PPAR-gamma. Pioglitazone is the main example; possible effects include weight gain, swelling, bone fractures, and worsening heart failure.

Thiazolidinediones

Insulin sensitizers activating peroxisome proliferator-activated receptors primarily in adipose tissue and skeletal muscle. Agents include pioglitazone and rosiglitazone. Mechanisms include enhanced insulin-mediated glucose uptake, altered adipokine secretion, and reduced hepatic glucose output.

Thigh

The part of the lower limb between the hip and knee.

ThinPrep Pap Test

A cervical cancer screening test in which cells collected from the cervix are placed in a liquid preser-

vative and examined for abnormal changes.

Alternate Names:

Thin Prep Pap Screen

Thiopental Sodium

An ultra-short-acting barbiturate, used for induction of general anesthesia (IV, onset 30–60 seconds, duration 5–10 minutes) and as a third-line agent for raised intracranial pressure (cerebroprotective effect: reduces cerebral metabolism, cerebral blood flow, and intracranial pressure).

Thiopentone Sodium

A short-acting barbiturate anesthetic used for induction in selected settings.

Thioridazine

An older phenothiazine antipsychotic with important cardiac and neurologic toxicity.

Thioridazine Overdose

Poisoning from taking too much thioridazine, an antipsychotic medicine. It can cause severe sleepiness, low blood pressure, seizures, and dangerous heart rhythms.

Thiotepa

An alkylating chemotherapy medicine used in selected cancers and conditioning regimens.

Third Stage Of Labor

From delivery of the baby to delivery of the placenta. The placenta separates from the uterine wall and is expelled spontaneously or with gentle cord traction. Normal duration: 5-30 minutes.

Third Trimester

The period from 28 weeks to delivery. Rapid fetal weight gain occurs, lungs mature with surfactant production. Fetal movements become regular and countable. Braxton Hicks contractions begin. Group B Streptococcus screening at 36 weeks.

Third-Degree AV Block

Complete heart block with no conduction between

atria and ventricles. Atria and ventricles beat independently (AV dissociation). Ventricular escape rhythm: junctional (40-60, narrow QRS) or ventricular (20-40, wide QRS).

Third-Degree Burn

A full-thickness burn that destroys the epidermis and dermis and may extend into subcutaneous tissue. The skin may appear white, brown, or charred and sensation may be reduced. Assessment includes burn depth, extent, inhalation injury, and associated trauma; deep or extensive burns require specialist treatment and may need grafting.

Third-Degree Perineal Tear

A severe perineal laceration sustained during vaginal birth that involves the perineal muscles and extends into the anal sphincter complex, classified under Obstetric Anal Sphincter Injuries. Subdivided internationally into 3a (<50% thickness of the external anal sphincter torn), 3b ($\geq 50\%$ thickness of the external anal sphincter torn), and 3c (both external and internal anal sphincters torn). It requires prompt identification, layered surgical repair in an operating room under adequate regional anesthesia, and proactive postpartum management with laxatives, pelvic floor therapy, and antibiotics to prevent anal incontinence and rectovaginal fistula.

Alternate Names:

Third-Degree Tear; Obstetric Anal Sphincter Tear

Thirst

The conscious desire to drink, regulated largely by osmolality and circulating volume.

Thirst - Absent

Reduced or absent desire to drink despite a potential need for fluids, which can occur with hypothalamic disease, aging, altered consciousness, medications, or severe illness.

Thirst - Excessive

Polydipsia, an abnormally strong or persistent desire to drink, commonly associated with dehydration, diabetes mellitus, diabetes insipidus, medicines, or psychiatric conditions.

Thomsen Disease

The classic autosomal dominant form of myotonia congenita caused by heterozygous mutations in the *CLCN1* gene on chromosome 7q35, which encodes the skeletal muscle voltage-gated chloride channel ClC-1. Characterized by delayed muscle relaxation following voluntary contraction (myotonia) with childhood onset, commonly affecting the hands, legs, eyelids, and jaw. A defining clinical feature is the “warm-up phenomenon,” wherein muscle stiffness significantly diminishes with repeated, consecutive muscle contractions; patients develop diffuse, athletic muscular hypertrophy and action/percussion myotonia, typically experiencing a benign, non-progressive course with normal life expectancy.

Alternate Names:

Autosomal Dominant Myotonia Congenita; Myotonia Congenita (Thomsen Type)

Thoracentesis

Insertion of a needle or catheter into the pleural space to remove fluid for diagnosis or to relieve respiratory symptoms caused by a pleural effusion.

Thoracic

Thoracic means relating to the chest, the thoracic spine, or structures within the thorax such as the heart and lungs.

Thoracic Aortic Aneurysm

Aneurysm of thoracic aorta. Associated with hypertension, Marfan syndrome, bicuspid aortic valve.

Thoracic Cage

The bony and cartilaginous framework formed by the thoracic vertebrae, ribs, costal cartilages, and sternum. It protects thoracic organs and assists respiratory movements.

Thoracic Cavity

The chest cavity, enclosed by the ribs and chest muscles, containing the lungs, heart, esophagus, and trachea.

Thoracic Duct

Body's largest lymphatic vessel, approximately 40 cm long, draining lymph from the entire body below the diaphragm and left side above it (approximately 75 percent of total lymph).

Thoracic Endometriosis

Endometrial tissue within the thoracic cavity causing catamenial pneumothorax, hemothorax, or lung nodules, recurring cyclically with menstruation and often requiring hormonal suppression.

Thoracic Outlet Syndrome

Compression of neurovascular bundle at thoracic outlet. Types: neurogenic (most common), venous, arterial. Symptoms: neck/shoulder/arm pain, paresthesia, weakness, vascular symptoms. Diagnosis: clinical, imaging, nerve conduction. Treatment: physical therapy, surgical decompression.

Alternate Names:

TOS (Thoracic Outlet Syndrome)

Thoracic Outlet Syndrome

Alternate index label for Thoracic Outlet Syndrome.

Thoracic Spine CT Scan

Computed tomography of the thoracic spine used to evaluate vertebral fractures, alignment, bone lesions, spinal canal narrowing, and other bony abnormalities.

Thoracic Spine X-Ray

Plain radiographic imaging of the thoracic spine used to assess vertebral alignment, fractures, curvature, degenerative change, and selected bone lesions.

Thoracic Surgeon

Surgeon specializing in operations on the chest, including lungs, heart, esophagus, and mediastinum.

Thoracic Vertebrae

Have costal facets on bodies and transverse processes for rib articulation. Heart-shaped bodies, long downward spinous processes, coronally-oriented facets allowing rotation and lateral flexion but lim-

iting flexion/extension. Small circular vertebral foramen.

Thoracic Vertebrae (T1-T12)

The twelve vertebrae of the chest region, each articulating with ribs and protecting the thoracic spinal cord while supporting the trunk and rib cage.

Thoracolumbar Spine Trauma

Injury to the thoracic or lumbar spine. Assessment includes fracture morphology, posterior ligamentous complex integrity, and neurologic status. Classification systems such as AO Spine and TLICS help guide management, which depends on stability and the clinical context.

Thoracoscopy

A minimally invasive procedure using a camera inserted through the chest wall to examine or treat the pleura, lungs, or mediastinum.

Thoracotomy

A surgical incision into the chest wall to access the thoracic cavity (pleural space, lungs, heart, great vessels, esophagus, diaphragm, mediastinum).

Thorax

The thorax is the chest region between the neck and abdomen, enclosed by ribs and containing the heart, lungs, major vessels, and mediastinal structures.

Thought Blocking

An abrupt interruption in the train of thought before it is completed, as if the thought was removed or pushed out of the patient's mind. The patient may pause mid-sentence, appear confused, and report that the thought was "taken out" or "blocked." It is a formal thought disorder seen in schizophrenia.

Thought Disorders

Disturbances of the form, flow, or content of thinking, occurring in psychosis and other conditions.

Threadworm

An intestinal infection or the worm *Enterobius ver-*

micularis, also called pinworm.

Threadworm (Enterobiasis)

An intestinal parasitic infection caused by *Enterobius vermicularis*, also called pinworm. Infection spreads when eggs are swallowed, often after contact with contaminated hands, bedding, food, or surfaces.

Alternate Names:

Enterobiasis

Threadworms (Pinworms)

Small intestinal worms, *Enterobius vermicularis*, whose females lay eggs around the anus and commonly cause nocturnal itching.

Alternate Names:

Pinworms

Threatened Miscarriage

Vaginal bleeding during early pregnancy with a closed cervix and no passage of pregnancy tissue; the pregnancy may remain viable. Evaluation commonly includes clinical assessment, serial human chorionic gonadotropin testing when indicated, and ultrasound.

Alternate Names:

Threatened abortion.

Three-Day Measles

Alternate index label; see *Rubella*.

Three-Dimensional Ultrasound

Ultrasound that combines many sound-wave measurements to create a three-dimensional image. It is used in obstetrics, gynecology, and other settings to show anatomy from several angles.

Threonine

An essential amino acid used to build proteins and other body molecules. The body cannot make enough of it, so it must come from food.

Threshold

The minimum stimulus or level required to produce a detectable response or effect.

Thrill

A palpable vibration over the chest or a blood vessel caused by turbulent blood flow.

Throat

The pharynx and adjacent structures behind the mouth and nose.

Throat Cancer

Cancer arising in the pharynx or nearby structures of the throat. Persistent hoarseness, difficulty or pain with swallowing, a neck lump, unexplained weight loss, or persistent sore throat requires medical evaluation, especially in people who use tobacco or alcohol.

Alternate Names:

Cancer, Throat

Throat Cancer (Larynx Cancer)

Alternate index label for Larynx Cancer.

Throat Or Larynx Cancer

Throat or larynx cancer is malignant growth in the pharynx or voice box and may cause persistent hoarseness, swallowing difficulty, pain, or a neck lump.

Throat Swab Culture

Culture of a throat specimen to detect bacterial pathogens, especially group A *Streptococcus*, when evaluating pharyngitis.

Alternate Names:

Throat Culture

Throbbing

A pulsating or beating quality of pain or sensation.

Thrombectomy

A surgical or interventional radiology procedure to remove a blood clot (thrombus) from a blood vessel, restoring blood flow. Types: (1) Endovascular thrombectomy (mechanical thrombectomy)—for acute ischemic stroke due to large vessel occlusion (LVO—internal carotid artery, middle cerebral artery M1 segment).

Thrombectomy (Vascular)

Removal of a blood clot from a blood vessel, using an open surgical or catheter-based procedure.

Alternate Names:

Vascular

Thrombin

A coagulation enzyme that converts fibrinogen to fibrin and activates platelets.

Thromboangiitis Obliterans

Buerger disease, an inflammatory tobacco-associated disease of small and medium arteries and veins.

Thromboangiitis Obliterans (Buerger Disease)

An inflammatory disease of small and medium blood vessels, strongly associated with tobacco use. It can cause pain, color changes, ulcers, or gangrene in the hands or feet. Complete tobacco cessation is essential; further treatment depends on ischemia, tissue loss, and other complications.

Alternate Names:

Buerger Disease

Thrombocytopenia

A misspelling of thrombocytopenia, an abnormally low platelet count.

Thrombocyte (Platelet)

A small (2–4 μm), anucleate, discoid cellular fragment shed from the cytoplasm of mature polyploid megakaryocytes in the bone marrow under the regulation of thrombopoietin. Normal circulating platelet counts range from 150,000 to 450,000/ μL , with a circulating lifespan of 7 to 10 days. Essential for primary hemostasis: upon vascular endothelial disruption, platelets adhere to exposed subendothelial collagen via von Willebrand factor (vWF) and glycoprotein Ib/IX receptors, undergo activation and shape change to release granule contents (ADP, thromboxane A_2), and aggregate via fibrinogen cross-linking of activated glycoprotein IIb/IIIa receptors to form a primary hemostatic plug. Quantitative or qualitative defects lead to

thrombocytopenia or platelet dysfunction, presenting with mucocutaneous bleeding, petechiae, and purpura.

Thrombocytes

Small, anucleate, disc-shaped cell fragments (2–4 μm diameter) derived from megakaryocytes in the bone marrow, circulating at $150\text{--}400 \times 10^9/\text{L}$. See Platelets (Thrombocytes).

Thrombocytopathy

A disorder of platelet function that impairs primary hemostasis despite a normal or near-normal platelet count. It may be inherited or acquired and can cause mucocutaneous bleeding, easy bruising, prolonged bleeding after procedures, or menorrhagia.

Thrombocytopenia

Thrombocytopenia is a low platelet count caused by reduced production, increased destruction or consumption, sequestration, or dilution and may increase bleeding risk.

Alternate Names:

Low Platelet Count

Thrombocytopenia (Low Platelet Count)

Alternate index label for Low Platelet Count.

Thrombocytopenic

Characterized by too few circulating platelets. It may cause easy bruising, nosebleeds, heavy menstrual bleeding, or other bleeding, although some people have no symptoms.

Thrombocytosis

An elevated platelet count ($>450,000/\mu\text{L}$). Reactive (secondary): infection, inflammation, iron deficiency, malignancy, post-splenectomy, hemorrhage, exercise—usually mild, no thrombotic risk.

Thromboelastography

A viscoelastic whole-blood assay that measures the initiation, formation, strength, and breakdown of a clot. It provides a functional overview of coagulation and fibrinolysis and may guide targeted

blood-component therapy in major bleeding.

Thromboembolism

Obstruction of a blood vessel by a clot that formed elsewhere and traveled through the circulation.

Thrombolysis

The dissolution of a blood clot using a medicine such as a fibrinolytic agent. It is used only in selected conditions because it can cause serious bleeding.

Thrombolysis For PE

Systemic thrombolytics (alteplase) for massive PE with hemodynamic instability. Rapid clot dissolution but significant bleeding risk.

Thrombolysis For Stroke

Thrombolysis with intravenous alteplase (tPA) is the primary acute treatment for ischemic stroke when administered within the time window of symptom onset. Alteplase is a recombinant tissue plasminogen activator that converts plasminogen to plasmin, which degrades fibrin in the thrombus, restoring blood flow to the ischemic brain.

Thrombolytic Agents

Medicines that activate fibrinolysis and dissolve an established blood clot; they are used selectively for time-critical conditions because they can cause serious bleeding.

Thrombolytic Therapy

A medical procedure, treatment, device, or care topic related to thrombolytic therapy; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Thrombophilia

A tendency to develop venous thromboembolism due to inherited or acquired hypercoagulable states. Inherited: factor V Leiden, prothrombin G20210A mutation, antithrombin deficiency, protein C deficiency, protein S deficiency.

Thrombophilia Testing

Thrombophilia testing evaluates for inherited and acquired hypercoagulable states in patients with venous thromboembolism, recurrent pregnancy loss, or strong family history of VTE.

Thrombophlebitis

Thrombophlebitis is inflammation of a vein associated with a blood clot and may be superficial or involve deep veins with risk of pulmonary embolism.

Thrombopoiesis

The production of platelets from megakaryocytes in the bone marrow. It is regulated principally by thrombopoietin and involves megakaryocyte maturation, proplatelet formation, and release of circulating platelets.

Thrombosed Intracranial Aneurysm

An intracranial arterial aneurysm that has undergone partial or complete thrombosis of its lumen, which may reduce rupture risk but can cause mass effect on adjacent brain structures. Presents with headache, cranial nerve palsies, or seizures depending on location. Diagnosis: CT, MRI, or conventional angiography.

Thrombosis

The formation of a blood clot within the cardiovascular system during life, potentially obstructing blood flow and causing tissue ischemia or infarction. Arterial thrombi typically form on ruptured atherosclerotic plaques, are composed primarily of platelets and fibrin, and appear pale and grossly attached to the vessel wall.

Thrombosis Of The Inferior Vena Cava

Formation of a blood clot within the inferior vena cava, which may occur in the setting of IVC filters, malignancy, hypercoagulable states, or extension from iliofemoral deep vein thrombosis. May cause bilateral lower extremity edema, abdominal pain, and hepatic or renal dysfunction. Requires anticoagulation and possible thrombectomy.

Thrombotic Microangiopathy

Endothelial damage with platelet-rich microthrombi causing MAHA with schistocytes, thrombocytopenia, and organ ischemia. TTP from ADAMTS13 deficiency treated with plasmapheresis. HUS from Shiga toxin managed supportively. Atypical HUS from complement dysregulation responds to eculizumab.

Thrombotic Thrombocytopenic Purpura

A life-threatening thrombotic microangiopathy usually caused by acquired or inherited deficiency of ADAMTS13, the enzyme that cleaves large von Willebrand factor multimers. It causes platelet consumption, red-cell fragmentation, and possible organ ischemia.

Thrombotic Thrombocytopenic Purpura

A life-threatening hematologic emergency and thrombotic microangiopathy caused by severe deficiency (< 10% activity) of the metalloprotease ADAMTS13, most often mediated by acquired autoantibodies. Loss of ADAMTS13 impairs cleavage of ultra-large von Willebrand factor (vWF) multimers, promoting widespread platelet-rich microvascular thrombosis in arterioles and capillaries. Clinically characterized by the classic pentad (seen completely in < 10% of patients): microangiopathic hemolytic anemia (MAHA with schistocytes on blood smear), severe consumptive thrombocytopenia, fluctuating neurologic deficits, renal impairment, and fever. Mandates immediate emergency therapeutic plasma exchange with corticosteroids and the anti-vWF nanobody caplacizumab; platelet transfusions are strictly contraindicated.

Thrombotic Vasculopathy

A spectrum of disorders characterized by thrombus formation within blood vessels of various sizes, often associated with antiphospholipid syndrome, disseminated intravascular coagulation, thrombotic thrombocytopenic purpura, or malignancy.

Thromboxane

A lipid mediator produced by platelets that promotes platelet aggregation and vasoconstriction.

Thrombus

A blood clot that forms within a blood vessel or heart and remains attached at its site of formation, potentially obstructing blood flow or embolizing.

Thrombus Organization

The process by which a thrombus is gradually replaced by granulation tissue and eventually fibrous scar tissue. The process begins when inflammatory cells, particularly macrophages, infiltrate the thrombus and begin digesting the fibrin and dead cells through fibrinolysis.

Thrush

A Candida infection, usually of the mouth or throat, causing creamy white plaques, soreness, altered taste, or discomfort with eating or swallowing. Vaginal candidiasis is a separate condition. Risk increases with antibiotics, inhaled corticosteroids, diabetes, or immune suppression.

Alternate Names:

Oral Thrush

Thrush - Children And Adults

Oral candidiasis caused by Candida overgrowth, producing removable white plaques, soreness, redness, altered taste, or cracks at the mouth corners; risk rises with antibiotics, inhaled steroids, diabetes, or immune suppression.

Thrush And Other Yeast Infections In Children

Thrush is Candida overgrowth in the mouth, producing white plaques, soreness, or feeding difficulty; Candida can also affect skin folds or diaper areas. Persistent, recurrent, severe, or immune-associated infection needs pediatric evaluation and appropriate antifungal treatment.

Thrush In Newborns

Oral Candida infection in an infant, causing white plaques on the tongue or mouth lining that may interfere with feeding and can be passed between infant and breastfeeding parent.

Thrush, Oral

Alternate index label; see *Oral thrush*.

Thumb Arthritis

Degeneration or inflammation of a thumb joint, especially the joint at the base of the thumb, causing pain, stiffness, and reduced grip.

Alternate Names:

Basal Joint Arthritis

Thumb CMC Arthritis

Thumb carpometacarpal arthritis: Eaton-Littler classification: I (normal), II (slight narrowing, osteophytes <2mm), III (narrowing, osteophytes >2mm, subluxation), IV (pantrapezial arthritis). Clinical: base of thumb pain, pinch weakness, thenar wasting, squaring (subluxation).

Thumb-Sucking

A common childhood habit of sucking the thumb; persistent habit can affect dental and facial development.

Thymic Hyperplasia

Thymic hyperplasia is enlargement of the thymus from increased lymphoid tissue or rebound growth and may be incidental. It can be associated with autoimmune disease, infection, endocrine conditions, or stress; imaging and clinical context distinguish it from a mass.

Thymocyte

An immature T lymphocyte developing in the thymus, where it undergoes selection before becoming a mature T cell.

Thymoma

A tumor of thymic epithelial cells, associated with myasthenia gravis, pure red cell aplasia, and hypogammaglobulinemia. Malignant behavior is determined by capsular invasion and stage.

Alternate Names:

Neoplasm Thymomic (Thymoma)

Thymus

Primary lymphoid organ in the anterior mediastinum, largest in childhood, involutes with age. Site of T-cell maturation (thymopoiesis): thymocytes undergo positive and negative selection to produce self-tolerant, MHC-restricted T cells. Epithelial cells produce thymic hormones (thymosin, thymopoietin).

Thymus Gland

A lymphoid organ in the chest where T lymphocytes mature.

Thyrocalcitonin

Another name for calcitonin, a hormone made by thyroid C cells that lowers blood calcium mainly by reducing bone resorption.

Thyroglobulin Surveillance

Thyroglobulin produced only by thyroid follicular cells serves as a tumor marker after total thyroidectomy for differentiated thyroid carcinoma. In patients who have undergone complete thyroidectomy and radioactive iodine ablation thyroglobulin should be undetectable.

Thyroglossal Duct Cyst

Congenital midline neck mass resulting from persistence of the thyroglossal duct the embryological tract of thyroid descent. Most common congenital neck mass presenting in childhood. Located at or near the hyoid bone and moves with swallowing and tongue protrusion. Diagnosis is by ultrasound demonstrating a cystic midline neck mass.

Thyroid

An endocrine gland in the front of the neck that produces thyroid hormones, which regulate metabolism, growth, and development.

Thyroid Cancer

A malignant tumor of the thyroid. Major types include papillary, follicular, medullary, and anaplastic carcinoma, which differ in behavior and treatment.

Alternate Names:

Thyroid Cancer - Medullary Carcinoma

Medullary thyroid carcinoma is cancer of thyroid C cells that produce calcitonin and may be inherited as part of a multiple endocrine neoplasia syndrome.

Thyroid Cancer - Papillary Carcinoma

Papillary thyroid carcinoma is the most common thyroid cancer and usually grows slowly, although it can spread to nearby lymph nodes.

Thyroid Cancer Staging

American Joint Committee on Cancer TNM staging for differentiated thyroid carcinoma incorporates tumor size, extent of invasion, lymph node metastases, and distant metastases. Age is a critical prognostic factor with patients under fifty having more favorable outcomes. Stage I is any tumor without metastases in patients under fifty.

Thyroid Cartilage

The largest cartilage of the larynx, formed by two plates that meet in front and create the laryngeal prominence. It protects the vocal folds and provides attachment for muscles and ligaments.

Thyroid Disorders

Conditions affecting the thyroid gland, a butterfly-shaped endocrine organ in the neck that produces hormones regulating metabolism, growth, and development.

Thyroid Eye Disease

Autoimmune inflammatory disorder of the orbital tissues associated with Graves disease characterized by expansion of extraocular muscles and orbital fat. Pathogenesis involves orbital fibroblasts expressing the TSH receptor targeted by the same thyroid-stimulating immunoglobulins affecting the thyroid.

Thyroid Fine-Needle Aspiration

Image-guided sampling of thyroid-nodule cells for cytologic classification. Results are commonly reported using the Bethesda system and guide surveil-

lance, molecular testing, surgery, or other management.

Thyroid Follicular Carcinoma

approximately 10 to 15 percent of thyroid cancers and is the second most common differentiated thyroid malignancy. Unlike papillary carcinoma, follicular carcinoma typically spreads hematogenously to distant sites including bone, lung, and liver, rather than to regional lymph nodes.

Thyroid Function Tests

Laboratory assessment of the hypothalamic-pituitary-thyroid axis. TSH is the most sensitive screening test with elevated levels indicating primary hypothyroidism and suppressed levels indicating hyperthyroidism. Free T4 measures the biologically active unbound thyroxine.

Thyroid Gland

An endocrine gland located in the anterior neck, consisting of two lobes (right and left) connected by an isthmus, lying anterior to the trachea between the cricoid cartilage and the suprasternal notch.

Thyroid Gland Enlargement

Alternate index label; see *Goiter*.

Thyroid Gland Removal

A medical procedure, treatment, device, or care topic related to thyroid gland removal; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Thyroid Gland, Diseases Of

Disorders of the thyroid, including hypothyroidism, hyperthyroidism, inflammation, nodules, and cancer.

Thyroid Hormone

T3 and T4. Increase basal metabolic rate. Required for normal development.

Thyroid Hormones

Thyroxine (T4) and triiodothyronine (T3), hor-

mones that regulate metabolic rate, growth, brain development, heat production, and cardiovascular and gastrointestinal activity.

Thyroid Nodule

A discrete lump within the thyroid gland that may be solid, cystic, benign, or malignant. Most are benign, but evaluation uses clinical assessment, thyroid testing, ultrasound, and biopsy when indicated.

Thyroid Nodule Evaluation

Systematic approach to evaluation of thyroid nodules detected by physical examination or imaging. Thyroid ultrasound characterizes nodule features including size, composition, echogenicity, margins, calcifications, and shape. TI-RADS classification stratifies malignancy risk and guides biopsy decisions.

Thyroid Nodules

A thyroid nodule is a discrete growth within the thyroid and is often benign. Ultrasound, thyroid-function testing, and selected needle biopsy assess cancer risk; rapid growth, hoarseness, swallowing difficulty, or breathing problems needs prompt review.

Alternate Names:

Nodule Thyroid (Thyroid Nodules)

Thyroid Papillary Carcinoma

The most common differentiated thyroid cancer, arising from thyroid follicular cells. It often grows slowly and has a favorable prognosis, although outcome depends on tumor features, spread, age, and response to treatment.

Thyroid Peroxidase Antibody

An autoantibody against thyroid peroxidase, an enzyme required for thyroid-hormone synthesis. It supports autoimmune thyroid disease, especially Hashimoto thyroiditis, but may be present in some euthyroid individuals.

Thyroid Peroxidase Test

Thyroid peroxidase test is a medical test, procedure, treatment, or clinical measure used in patient care;

its purpose and interpretation depend on the clinical question.

Thyroid Preparation Overdose

Taking too much thyroid hormone medicine can cause a fast or irregular heartbeat, shaking, sweating, anxiety, vomiting, fever, or, in severe cases, heart failure or a dangerous thyroid crisis.

Thyroid Scan

Nuclear medicine imaging using I-123 or Tc-99m. Evaluates thyroid function and nodules. Hot nodules rarely malignant. Cold nodules require further evaluation.

Thyroid Scintigraphy

Thyroid imaging using I-123, I-131, or Tc-99m pertechnetate. Assesses function, nodules, ectopic tissue. Hot nodules (functioning) rarely malignant; cold nodules require ultrasound/FNA.

Thyroid Stimulating Hormone

TSH (thyrotropin): the anterior pituitary hormone that stimulates the thyroid to produce and release T4 and T3, controlled by TRH (hypothalamus) and negative feedback from thyroid hormones.

Thyroid Storm

Life-threatening exacerbation of thyrotoxicosis presenting with hyperpyrexia, tachycardia, heart failure, altered mental status, and multi-organ dysfunction. Precipitated by infection, surgery, trauma, or iodine load in patients with untreated hyperthyroidism. Mortality rates range from twenty to thirty percent.

Thyroid Surgery

Operations that remove part or all of the thyroid gland. They may be performed for selected cancers, enlarging or compressive nodules, goiter, hyperthyroidism, or diagnostic uncertainty; the procedure depends on the disease and its extent.

Thyroid Ultrasound

High-frequency sonographic imaging used to eval-

uate thyroid size, nodules, cysts, cervical lymph nodes, and structural features associated with malignancy. Suspicious nodules may require ultrasound-guided fine-needle aspiration.

Thyroid, Overactive

Alternate index label; see *Hyperthyroidism (overactive thyroid)*.

Thyroid-Stimulating Hormone

A pituitary hormone that stimulates the thyroid to produce thyroid hormones.

Alternate Names:

TSH

Thyroidectomy

Surgical removal of part or all of the thyroid gland.

Thyroiditis

Inflammation of the thyroid gland with various etiologies. Hashimoto thyroiditis: autoimmune, most common cause of hypothyroidism in iodine-sufficient areas. Subacute (De Quervain) thyroiditis: post-viral, presents with painful thyroid and transient hyperthyroidism followed by hypothyroidism.

Thyroiditis Classification

Inflammatory conditions of the thyroid classified by clinical presentation and etiology. Hashimoto thyroiditis is chronic autoimmune with hypothyroidism. Graves disease is autoimmune with hyperthyroidism. Subacute thyroiditis is painful and likely viral.

Thyroiditis, Chronic Lymphocytic

Alternate index label; see *Hashimoto's disease*.

Thyrotoxic Myopathy

A neuromuscular complication of severe, uncontrolled hyperthyroidism, occurring in up to 60–80% of patients with thyrotoxicosis. Characterized by insidious, progressive, painless proximal muscle weakness and striking muscle wasting, predominantly involving the shoulder and pelvic girdles, accompanied by muscle fatigue and hyperreflexia. Serum creatine kinase levels are typically normal or mildly reduced;

muscle biopsy shows non-specific type II fiber atrophy with normal sarcolemma, and all symptoms resolve rapidly within weeks of medical restoration of the euthyroid state.

Alternate Names:

Hyperthyroid Myopathy

Thyrotoxic Periodic Paralysis

Recurrent episodes of transient flaccid weakness associated with thyrotoxicosis, usually accompanied by hypokalemia caused by intracellular potassium shift. Attacks often affect the proximal limbs and are triggered by rest after exercise or carbohydrate-rich meals.

Thyrotoxicosis

The clinical state caused by excessive circulating thyroid hormone, regardless of source.

Thyrotroph Adenoma

A thyrotroph adenoma is a pituitary tumor arising from thyroid-stimulating-hormone-producing cells and may cause excess TSH and abnormal thyroid function.

Thyrotrophin-Releasing Hormone

A hypothalamic hormone that stimulates pituitary release of TSH and prolactin.

Thyrotrophin-Stimulating Hormone

Thyroid-stimulating hormone, a pituitary hormone that stimulates thyroid hormone production.

Thyrotropin

Thyroid-stimulating hormone, a pituitary hormone that stimulates thyroid growth and thyroid-hormone production; it is also called TSH.

Thyrotropin Alfa

A recombinant form of thyroid-stimulating hormone used to stimulate residual thyroid tissue or differentiated thyroid cancer cells during selected diagnostic or treatment protocols.

Thyroxine

The thyroid hormone T₄, which contains four iodine atoms. The thyroid releases it into the blood, and much of it is converted to the more active hormone triiodothyronine (T₃).

Alternate Names:

T₄

Thyroxine-Binding Globulin

The principal transport protein for thyroid hormones in the blood, binding most circulating thyroxine and triiodothyronine.

TI-RADS

The Thyroid Imaging Reporting and Data System is an ultrasound-based system that assigns a risk category to thyroid nodules using features such as composition, echogenicity, shape, margins, and calcification. It helps guide follow-up and biopsy decisions.

TIA

A transient ischemic attack, a temporary episode of neurological dysfunction from reduced brain blood flow.

Alternate Names:

Transient ischemic attack; mini-stroke

Tiazofurin

Tiazofurin is an investigational nucleoside analogue that interferes with NAD synthesis and has been studied as an antineoplastic agent.

Tibia

The tibia is the larger medial bone of the lower leg, extending from the knee to the ankle; it bears much of the body's weight and forms parts of both joints. Proximal: tibial plateau (medial/lateral condyles) and tibial tuberosity (patellar ligament insertion). Shaft: sharp subcutaneous anterior border. Distal: medial malleolus articulating with the talus at the ankle.

Tibia Vara

Tibia vara is inward angulation of the upper shin-

bone causing bowing of the leg, classically associated with Blount disease or other growth disorders.

Tibial Artery

An artery of the lower leg, chiefly the anterior and posterior tibial arteries, supplying the leg, ankle, and foot and contributing to pedal pulses.

Tibial Fracture

A break in the tibia, the major weight-bearing bone of the lower leg, often caused by high-energy trauma, twisting injury, or repetitive stress.

Tibial Nerve

Sciatic division (L₄-S₃) continuing from posterior thigh through the popliteal fossa into the deep posterior leg, innervating gastrocnemius, soleus, tibialis posterior, flexor digitorum longus, and flexor hallucis longus. Passes behind the medial malleolus as the tarsal tunnel, dividing into medial/lateral plantar nerves for the foot.

Tibial Shaft Fractures

Tibial shaft fractures: most common long bone fracture. AO/OTA: 42-A (simple), 42-B (wedge), 42-C (complex). Soft tissue critical: Gustilo for open.

Tibialcrural Artery

A lower-leg arterial vessel or branching region supplying the ankle and foot, usually discussed in relation to tibial and fibular arterial anatomy.

Tibialis Anterior Muscle

Superficial anterior leg muscle from lateral tibial condyle and shaft to medial cuneiform and first metatarsal base. Innervated by deep peroneal nerve (L₄-L₅). Primary dorsiflexor for foot clearance during gait swing phase; inverts foot. Weakness causes foot drop with steppage gait.

Tibialis Cranialis

The animal-anatomy term corresponding broadly to the tibialis anterior muscle, which dorsiflexes the ankle and helps invert the foot.

Tibialis Posterior Muscle

Deepest posterior leg muscle from posterior tibia, fibula, and interosseous membrane to navicular tuberosity, medial cuneiform, and sustentaculum tali. Innervated by tibial nerve (L4-L5). Plantarflexes and inverts foot; primary dynamic medial longitudinal arch stabilizer.

Tic

A sudden, rapid, recurrent, nonrhythmic movement or sound that a person may feel an urge to make. Tics can be motor or vocal and may worsen with stress.

Tic Disorder

A neurologic disorder characterized by recurrent, sudden, rapid, nonrhythmic motor movements or vocalizations that may be temporarily suppressible.

Tic Douloureux

A chronic pain condition characterized by recurrent, sudden, severe, electric shock-like pain in the distribution of the trigeminal nerve (typically V2/V3). See Trigeminal Neuralgia.

Tic Douloureux (Trigeminal Neuralgia)

Alternate index label for Trigeminal Neuralgia.

Ticarcillin

A penicillin antibiotic with activity against many gram-negative bacteria, including *Pseudomonas*; it is usually given with clavulanate and can cause allergy or electrolyte disturbances.

Tick

A small blood-feeding arachnid that can attach to people or animals and transmit infections such as Lyme disease, rickettsial diseases, and babesiosis.

Tick Bite

A bite from a tick, which may cause local irritation and can sometimes spread infections such as Lyme disease, ehrlichiosis, or Rocky Mountain spotted fever. Remove the tick promptly and watch for fever or a spreading rash.

Tick Paralysis

Acute, progressive flaccid paralysis caused by neurotoxin released by an attached tick. Weakness usually begins in the legs and ascends; respiratory failure can occur and symptoms improve after prompt tick removal.

Tick Removal

A medical procedure, treatment, device, or care topic related to tick removal; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Tick Typhus

A rickettsial infection transmitted by ticks, usually causing fever, headache, rash, and sometimes severe systemic illness.

Tick-Borne Encephalitis

A viral infection transmitted mainly by ticks that can cause fever, meningitis, encephalitis, or lasting neurological complications.

Tickborne Relapsing Fever

A vector-borne illness caused by *Borrelia hermsii* and related species, transmitted by *Ornithodoros* ticks. It presents with recurrent episodes of fever, headache, myalgia, and arthralgia, separated by afebrile intervals. The relapsing pattern results from antigenic variation of the spirochete's surface lipoproteins.

Ticks

Small arachnids that attach to skin and may transmit infections or cause tick paralysis.

Alternate Names:

Ticks Bite (Ticks)

Ticks Bite (Ticks)

Alternate index label for Ticks.

Ticlopidine

An antiplatelet medicine that reduces platelet aggregation and clot formation.

Tics

Sudden, rapid, recurrent, nonrhythmic stereotyped movements (motor tics) or vocalizations (vocal tics) that are usually brief and suppressible. Transient tics (most common in childhood) vs. chronic tic disorder; Tourette syndrome: multiple motor + at least one vocal tic for >1 year beginning before age 18.

Tidal Volume

The volume of air inspired or expired during each normal breath, approximately 500 mL (6-8 mL/kg) in a resting adult. Tidal volume is measured by spirometry. Minute ventilation = tidal volume x respiratory rate (6 L/min). Only a portion reaches the alveoli (alveolar ventilation) after subtracting dead space (150 mL).

Tidy Wound

A clean, sharply incised wound with minimal contamination, crushing, or tissue devitalization and clearly defined viable edges. After appropriate assessment and cleaning, it may often be closed primarily.

Tietze Syndrome (Costochondritis And Tietze Syndrome)

Alternate index label for Costochondritis and Tietze Syndrome.

Tigecycline

Tigecycline is a glycylcycline antibiotic that binds the bacterial 30S ribosomal subunit and inhibits protein synthesis; it treats selected complicated infections but commonly causes nausea and has a warning for increased mortality in some severe infections.

Tight Junction

A cell-to-cell junction that seals the space between epithelial cells and controls paracellular movement while helping maintain distinct apical and basal cell surfaces.

Tilorone

Tilorone is a synthetic antiviral and immune-

modulating compound used or studied in some countries; evidence and indications vary by setting.

Tilt Table Test

Evaluation of unexplained syncope by reproducing orthostatic stress. Patient is strapped to a motorized table tilted from supine to 60-80 degrees for 20-45 minutes. Positive: reproduction of symptoms with hypotension and/or bradycardia (vasovagal syncope, delayed orthostatic hypotension).

Tilt Table Testing

A supervised test in which a patient is moved from a lying to an upright or head-up tilt position while blood pressure, heart rate, and symptoms are monitored. It may help evaluate reflex syncope, orthostatic hypotension, or selected autonomic disorders when the diagnosis remains uncertain; it does not replace assessment for cardiac or neurologic causes.

Tilted Disc Syndrome

A congenital appearance in which the optic disc is angled or rotated, often with an unusual shape of nearby retina and blood vessels. Vision may be normal, but some people have visual-field changes or other eye conditions.

Time In Range

Percentage of time that continuous glucose monitoring readings remain within the target glucose range of seventy to one hundred eighty milligrams per deciliter for most adults with diabetes. General target is greater than seventy percent corresponding to approximately seventeen hours per day.

Time Of Death

The moment or estimated time at which biological death occurred, a key medicolegal question. Estimated from postmortem changes: body cooling (algor mortis), rigor mortis, lividity (livor mortis), decomposition/putrefaction, insect colonization, and chemical changes (vitreous potassium, vitreous glucose, body temperature).

Time-Of-Flight PET

A PET acquisition and reconstruction technique

that uses the measured arrival-time difference of coincident annihilation photons to improve localization of the annihilation event. It can improve signal-to-noise performance, especially in larger patients.

Timed Up-And-Go Test

A functional mobility test measuring the time required to stand from a chair, walk a short distance, turn, return, and sit. Prolonged performance suggests impaired mobility and increased fall risk but must be interpreted with gait, vision, cognition, and environment.

TIMI Risk Score

A clinical prognostic scoring system developed by the Thrombolysis in Myocardial Infarction study group to risk-stratify patients presenting with acute coronary syndromes. For unstable angina and NSTEMI, the score assigns one point each to seven variables: age ≥ 65 years, ≥ 3 CAD risk factors, known CAD with stenosis ≥ 50 percent, aspirin use in past 7 days, severe angina (≥ 2 episodes in 24h), ST-segment deviation ≥ 0.5 mm, and elevated cardiac biomarkers; higher scores predict significantly elevated 14-day risks of all-cause mortality, new or recurrent myocardial infarction, or severe recurrent ischemia requiring urgent revascularization.

Alternate Names:

TIMI Score; TIMI Risk Score for UA/NSTEMI; TIMI Score for STEMI

Timing Of Breastfeeding

The timing and frequency of breastfeeds, usually guided by early hunger cues and effective milk transfer rather than a rigid clock schedule, with special considerations for newborns and preterm infants.

Timolol

Timolol is a non-selective beta blocker used in glaucoma eye drops and for selected cardiovascular conditions; topical ocular use can still cause bradycardia, hypotension, bronchospasm, or fatigue.

Timolol Maleate

The maleate salt of timolol, a beta blocker used in

glaucoma and selected cardiovascular conditions.

Timothy Syndrome

A rare multisystem disorder caused by pathogenic variants in CACNA1C, characterized by marked QT prolongation and congenital heart disease with syndactyly. Developmental, neurological, immune, and facial features may also occur.

Tincture

Alcohol extract of plant or chemical substance. Common concentration: 10-40% alcohol. Examples: tincture of iodine (antiseptic), tincture of benzoin (skin protector). Also used for liquid medications.

Tinea

Dermatophytosis: superficial fungal infection of keratinized tissue (skin, hair, nails) by dermatophytes (Trichophyton, Microsporum, Epidermophyton).

Tinea (Dermatophytosis)

Superficial fungal infection by dermatophytes. Tinea corporis (ringworm): annular, scaly, erythematous plaque with central clearing. Tinea pedis (athlete's foot): interdigital scaling, maceration, and fissuring. Tinea cruris (jock itch): erythematous, pruritic rash in groin with raised border.

Alternate Names:

Dermatophytosis

Tinea Barbae

Dermatophyte infection of the beard and mustache area, presenting with erythematous, scaly, follicular papulopustules or deep inflammatory nodules, often acquired from contact with infected animals or through grooming. It may be inflammatory (kerion-like) or superficial, and differential diagnosis includes bacterial folliculitis and acne.

Tinea Capitis

A dermatophyte infection of the scalp and hair shafts, most common in children. It may cause scaling, alopecia, broken hairs, black dots, or an inflammatory kerion.

Tinea Circinata

An older term for tinea corporis (ringworm) of the body, presenting as annular, erythematous, scaly plaques with active, raised borders and central clearing. It is caused by dermatophyte fungi and may be acquired through direct contact with infected humans, animals, or fomites.

Tinea Corporis

A dermatophyte infection of glabrous skin, presenting as annular, erythematous, scaly plaques with raised, active borders and central clearing. It is contagious through direct contact with infected persons, animals, or contaminated objects. Diagnosis is by KOH preparation showing branching septate hyphae.

Tinea Cruris

Alternate index label; see *Jock itch*.

Tinea Faciei

Dermatophyte infection of the facial skin, often presenting with erythematous, scaly, annular or ill-defined plaques that may mimic eczema or rosacea, leading to frequent misdiagnosis. It is commonly caused by *Trichophyton* or *Microsporum* species and may be acquired from pets, other humans, or self-inoculation from other body sites.

Tinea Favosa

A chronic dermatophyte infection of the scalp or skin, usually caused by *Trichophyton schoenleinii*, producing yellow cup-shaped crusts and possible scarring alopecia.

Tinea Imbricata

A chronic dermatophyte infection caused by *Trichophyton concentricum*, endemic in the South Pacific and Southeast Asia, characterised by widespread, concentric, overlapping, polycyclic scaly rings producing a "wood-grain" pattern on the trunk and limbs.

Tinea Incognito

A dermatophyte infection whose clinical appearance has been altered by inappropriate topical corticosteroid or immunosuppressive treatment.

It may become less scaly and more extensive or atypical; fungal testing is often needed before directed therapy.

Tinea Nigra

A superficial fungal infection of the stratum corneum caused by *Hortaea werneckii*, presenting as a painless, brown to black, non-scaly macule or patch, most commonly on the palms or soles. It is acquired in tropical and subtropical regions and is often confused with acral melanoma.

Tinea Pedis

A dermatophyte infection of the feet presenting in interdigital, moccasin, vesicobullous, or ulcerative patterns. It may coexist with tinea unguium. Diagnosis is clinical or mycologic when uncertain; treatment uses topical antifungals for limited disease and oral therapy for extensive, refractory, or nail-associated infection.

Tinea Pedis (Athlete's Foot)

Alternate index label for Athlete's Foot.

Tinea Unguium

A fungal infection of the nail plate, also called onychomycosis, causing thickening, discoloration, brittleness, or separation of the nail.

Tinea Versicolor

A superficial fungal infection caused by *Malassezia* species, presenting as hypo- or hyperpigmented macules with fine scale on the trunk and proximal extremities. It is more noticeable after sun exposure due to unaffected tanning of surrounding skin.

Alternate Names:

Pityriasis Versicolor

Tinel Sign

Tapping over nerve produces tingling in distribution. Indicates nerve irritation/regeneration. Used for carpal tunnel, cubital tunnel, tarsal tunnel.

Tinidazole

Tinidazole is a nitroimidazole antimicrobial effective against anaerobic bacteria and protozoa such as *Giardia* and *Trichomonas*; metallic taste, gastrointestinal effects, neuropathy, and an alcohol interaction may occur.

Tinnitus

Tinnitus is perception of sound without an external source, often related to hearing loss, noise exposure, ear disease, medications, vascular causes, or neurologic factors.

Alternate Names:

Ringing In The Ear

Tinnitus (Ringing In The Ears)

Alternate index label for Ringing in the Ears.

Tinzaparin

A low-molecular-weight heparin anticoagulant used for prevention or treatment of venous thromboembolism; bleeding and heparin-induced thrombocytopenia are important risks.

Tinzaparin Sodium

The sodium salt of tinzaparin, a low-molecular-weight heparin used to prevent or treat venous blood clots; bleeding and heparin-induced thrombocytopenia are important risks.

Tiopronin

A thiol-containing medicine used in selected cases of cystinuria to increase urinary cystine solubility; rash, proteinuria, and blood abnormalities may occur.

Tiotropium Bromide

A long-acting muscarinic antagonist bronchodilator used for maintenance therapy in COPD. Once-daily dosing provides sustained bronchodilation. It reduces exacerbations and improves lung function. It is often combined with a LABA for additive bronchodilation.

Tipifarnib

An investigational farnesyl-transferase inhibitor

studied for cancers and certain blood disorders.

TIPS

TIPS, or transjugular intrahepatic portosystemic shunt, connects portal and hepatic veins to reduce portal hypertension, often for variceal bleeding or refractory ascites.

Tirapazamine

Tirapazamine is an investigational prodrug activated in low-oxygen tissue and studied to target hypoxic tumor cells.

Tirbanibulin

A topical medicine used in short courses to treat actinic keratoses on the face or scalp; local redness, scaling, or irritation may occur.

Tired

Feeling a lack of energy or physical or mental stamina, which may result from sleep loss, exertion, infection, anemia, endocrine disease, medication, depression, or other illness.

Tiredness

Fatigue or reduced energy and endurance that can be physical, mental, or both; persistent unexplained fatigue warrants assessment for sleep, mood, systemic, medication, and metabolic causes.

Tiredness (Fatigue)

A subjective lack of energy or exhaustion that may result from inadequate sleep, illness, psychological stress, anaemia, or many other causes.

Alternate Names:

Fatigue

Tirofiban

An intravenous glycoprotein IIb/IIIa platelet inhibitor used in selected acute coronary syndromes or coronary interventions to reduce platelet aggregation.

Tissierella

A genus of anaerobic bacteria found in the environ-

ment and sometimes isolated from polymicrobial human infections.

Tissierella Praeacuta

An anaerobic bacterium rarely associated with human infection; significance depends on recovery from a compatible sterile-site infection rather than colonization.

Tissue

A group of similar cells and their surrounding materials that work together to perform a specific function.

Tissue Adhesions

Bands of fibrous scar tissue that abnormally connect surfaces or organs, often after surgery, inflammation, infection, or bleeding and potentially causing pain, infertility, or obstruction.

Tissue Banks

Facilities that recover, screen, process, preserve, store, and distribute donated human tissue for transplantation, research, or reconstructive treatment under safety and traceability standards.

Tissue Congestion

Accumulation of blood or fluid within tissue because of impaired venous drainage, inflammation, obstruction, or increased vascular flow, producing swelling, redness, or impaired function.

Tissue Culture

Growth of cells or tissue outside the body in a controlled nutrient medium for research, diagnosis, vaccine production, regenerative medicine, or drug testing.

Tissue Distribution

Reversible movement of a chemical from blood into tissues and organs. Distribution is influenced by perfusion, lipid solubility, protein binding, tissue affinity, barriers, body composition, and time after exposure.

Tissue Elastance

The change in pressure required to produce a given change in tissue or organ volume, the inverse of compliance and important in lung and chest-wall mechanics.

Tissue Engineering

The use of cells, supportive materials called scaffolds, and biologically active substances to create or repair tissues that restore, maintain, or improve body function.

Tissue Expander

Balloon implanted under skin/muscle, gradually filled with saline to expand tissue. Used for breast reconstruction, scalp reconstruction, burn coverage. Removed after expansion, replaced with permanent implant or flap.

Tissue Expansion

Gradual stretching of skin or soft tissue with an implanted or external expander so additional tissue can be created for reconstruction; it usually requires staged treatment and monitoring for infection, pain, or device problems.

Tissue Fixation

Preservation of tissue morphology and molecular components by chemical or physical methods before processing. Neutral buffered formalin is commonly used for routine histopathology, while fixation choice affects antigen detection, nucleic-acid testing, and tissue architecture.

Tissue Kallikrein

A serine protease that cleaves kininogen to release kinins such as bradykinin, influencing local blood flow, vascular permeability, pain, and inflammation.

Tissue Membrane

A thin sheet or boundary of tissue that separates compartments, covers surfaces, or surrounds structures, with composition and function varying by epithelial, connective, muscle, or biologic membrane type.

Tissue Plasminogen Activator

An enzyme that converts plasminogen to plasmin, helping dissolve fibrin in blood clots. Recombinant forms are used as thrombolytic medicines in selected emergencies.

Tissue Plasminogen Activator

An enzyme or recombinant medicine that converts plasminogen to plasmin and dissolves clots.

Tissue Plasminogen Activator-Associated Angioedema

Angioedema occurring as a complication of tissue plasminogen activator therapy for acute ischemic stroke or myocardial infarction, affecting approximately 1 to 5 percent of patients. It typically involves the face, tongue, and upper airway and may be life-threatening.

Tissue Processor

A laboratory instrument that moves tissue through fixation, dehydration, clearing, and paraffin infiltration before embedding and microscopic sectioning.

Tissue Scaffold

A natural or engineered three-dimensional framework that supports cell attachment, growth, organization, and tissue repair in regenerative medicine.

Tissue Transglutaminase (tTG)

An enzyme that cross-links proteins and serves as the principal autoantigen in celiac disease; IgA antibodies against tissue transglutaminase are commonly used in celiac disease evaluation.

Tissue Typing

Testing tissue antigens, especially HLA, to assess compatibility for transplantation.

Tissues Of The Body

Groups of similar cells and extracellular material performing a common function: epithelial, connective, muscle, and nervous tissue.

Titanium

A strong, lightweight, corrosion-resistant metal used in implants, prostheses, dental devices, and surgical instruments. Bone can form a stable connection with some titanium implant surfaces.

Titer

The concentration or dilution level at which a measurable antibody, antigen, or other activity remains detectable in a specimen. expressed as the highest dilution that still produces a detectable reaction (e.g., 1:128).

Titration

Gradual adjustment of a medicine dose or measurement of a substance using a reagent of known concentration.

Titre

The concentration or level of an antibody or other substance measured by serial dilution.

Titubation

A rhythmic nodding or tremulous movement, especially of the head or trunk, often associated with cerebellar disease.

Tizanidine

A centrally acting alpha-2 agonist used to reduce painful muscle spasm and spasticity.

Thin-Layer Chromatography

A laboratory method that separates substances on a thin layer of silica, alumina, or another material as a solvent moves across it. It can help identify compounds, compare samples, or measure radiochemical purity.

Alternate Names:

TLC

Tmax

The time after a dose when a medicine reaches its highest measured concentration in blood or plasma.

TMD

Alternate index label; see *TMJ disorders*.

TMJ Disorders

Temporomandibular joint disorders affect the jaw joints or chewing muscles and may cause jaw pain, clicking, stiffness, headaches, or limited movement.

Alternate Names:

Temporomandibular Disorders Temporomandibular Joint Disorders TMD

TNBC

Alternate index label; see *Triple-negative breast cancer*.

TNF Alpha

Tumor necrosis factor alpha is a potent pro-inflammatory cytokine produced primarily by activated macrophages, as well as by T cells, NK cells, mast cells, and endothelial cells. TNF-alpha exists as a 26-kilodalton transmembrane protein that is cleaved by the metalloprotease TACE to release a soluble 17-kilodalton active form.

TNM Classification

A cancer-staging system describing primary tumor size or extent, regional nodes, and distant metastasis.

TNM Staging

A cancer-staging system describing the primary tumor, regional lymph nodes, and distant metastases to estimate disease extent and prognosis.

Tobacco

The plant and its products containing nicotine and many toxic chemicals; use causes addiction and major disease risk.

Tobacco Chewing (Smokeless Tobacco)

Alternate index label for Smokeless Tobacco.

Tobacco Use

Tobacco use exposes the body to nicotine and toxic combustion or aerosol chemicals, causing addiction

and increased risks of cancer, vascular disease, and lung disease.

Tobramycin

An aminoglycoside antibiotic used for selected bacterial infections, including some gram-negative infections.

Tocolysis

The use of medication to delay preterm birth for a limited time, often to allow corticosteroid treatment or transfer to a facility with appropriate neonatal care. The drug, gestational-age range, and contraindications depend on local guidance and the clinical situation.

Tocolytic Agents

Tocolytic agents suppress uterine contractions temporarily to delay preterm birth, often allowing time for corticosteroids or transfer to specialist care.

Tocolytic Therapy

Medications used to inhibit uterine contractions temporarily to prolong pregnancy, allowing time for corticosteroid administration and transport to a tertiary care center. Indications: preterm labor with intact membranes, 24-34 weeks.

Tocolytics

Medications used to suppress preterm labor (contractions before 37 weeks gestation), to delay delivery for corticosteroid administration (fetal lung maturity) and in utero transport to a tertiary care center.

Tocopherol

A form of vitamin E, a fat-soluble antioxidant involved in protecting cell membranes; deficiency is uncommon but can cause neurologic or hematologic abnormalities.

Tocotrienols

Vitamin E compounds with unsaturated side chains found in some plant oils and grains; proposed cardiovascular and metabolic benefits remain under

study.

Toddler

A toddler is a young child, usually about one to three years old, developing rapidly in movement, language, behavior, feeding, and independence.

Toddler Test Or Procedure Preparation

Preparing a toddler for a test or procedure uses simple explanations, play, distraction, comfort objects, caregiver support, and any required fasting or medicine instructions.

Toddler'S Fracture

A subtle, non-displaced spiral or oblique hairline fracture of the distal third of the tibial shaft in ambulatory young children between 9 months and 3 years of age, typically occurring after a low-energy rotational or twisting injury (such as stumbling, catching the foot in a crib slat, or going down a slide on a parent's lap). Characterized by sudden refusal to bear weight or limp without outward signs of severe trauma, erythema, or deformity. Initial radiographs may appear normal or show an faint hairline fracture line, with subperiosteal bone formation and callus becoming visible on follow-up radiographs after 10 to 14 days.

Alternate Names:

Cast-Tibia Fracture; Childhood Spiral Tibial Fracture

Toe Broken (Broken Toe)

Alternate index label for Broken Toe.

Toe Joint

Any joint connecting the bones of a toe. Toe joints include the metatarsophalangeal and interphalangeal joints; they permit toe movement and may be affected by arthritis, gout, injury, or deformity.

Alternate Names:

toe articulation.

Toe Pressure

Pressure in toe digits. Useful when ankle pressures non-compressible (diabetes, renal disease).

Toe-Brachial Index

The ratio of systolic toe pressure to brachial systolic pressure. It can support diagnosis of peripheral arterial disease when ankle arteries are noncompressible because of medial arterial calcification.

Toenail

A toenail is the hard keratin plate covering the end of a toe and protecting its underlying tissue; injury, fungus, and ingrowth can cause disease.

Toenail Fungus

Alternate index label; see *Nail fungus*.

Toes (Digits Of The Foot)

The five terminal articulated digits of the human foot (digits I through V), consisting of the medial hallux (great toe) and four lateral lesser toes. The hallux contains two robust phalanges and two constant plantar sesamoid bones beneath the first metatarsal head, functioning as a critical biomechanical fulcrum during the push-off phase of normal bipedal gait. Lesser toes contain three phalanges each (proximal, middle, distal); prone to structural deformities including hallux valgus (bunions), hammer toes, claw toes, and mallet toes resulting from neuromuscular imbalances, poorly fitting footwear, or inflammatory arthritis.

Togaviridae

A family of viruses that includes some mosquito-borne viruses, such as chikungunya and eastern equine encephalitis virus, which can cause fever, rash, joint pain, or brain inflammation.

Togaviridae Infections

Infections caused by viruses in the Togaviridae family. Symptoms vary by virus and may include fever, rash, joint pain, or encephalitis.

Toilet

A fixture or place used for urination and defecation; in healthcare, toilet access and toileting assistance are important parts of continence and fall prevention.

Toilet Bowl Cleaners And Deodorizers Poisoning

Poisoning from swallowing or contacting toilet cleaners or deodorizers. Acidic or alkaline products can burn the mouth, throat, eyes, and skin; mixing cleaners may release dangerous gases.

Tolazamide

An older sulfonylurea diabetes medicine that increases insulin release; hypoglycemia and weight gain can occur, especially in older adults or people with kidney disease.

Tolazomide

An older sulfonylurea medicine used to lower blood glucose in type 2 diabetes.

Tolbutamide

An older first-generation sulfonylurea used for type 2 diabetes.

Tolcapone

A catechol-O-methyltransferase inhibitor used with levodopa for Parkinson disease; potentially fatal liver injury limits use and requires monitoring.

Tolerable Upper Intake Level

The highest usual daily nutrient intake unlikely to cause harm in nearly all healthy people.

Alternate Names:

UL

Tolerance

Reduced response to a drug or exposure after repeated use, requiring greater exposure for the same effect.

Toll-Like Receptor

A class of single-pass membrane-spanning pattern recognition receptors (PRRs) expressed on innate immune cells (macrophages, dendritic cells). Recognize evolutionary conserved microbial structures (PAMPs).

Toll-Like Receptors

Innate-immune receptors that detect characteristic microbial or tissue-danger signals.

Alternate Names:

TLRs

Tolmetin

A nonsteroidal anti-inflammatory drug used for pain and arthritis; gastrointestinal bleeding, kidney injury, and cardiovascular events are important risks.

Tolmetin Overdose

Taking too much tolmetin, a nonsteroidal anti-inflammatory medicine, can cause stomach bleeding, vomiting, kidney injury, drowsiness, seizures, or other serious effects.

Tolnaftate

A topical antifungal medicine used for some dermatophyte infections such as athlete's foot and ringworm; it is not effective for every type of fungal infection.

Tolonium Chloride

Tolonium chloride, or toluidine blue, is a dye used in laboratory and selected clinical procedures to highlight cells or tissue changes.

Tolterodine Tartrate

Tolterodine tartrate is an antimuscarinic medicine used for overactive bladder and urinary urgency; dry mouth, constipation, and urinary retention can occur.

Toluene

Toluene is a volatile solvent that can irritate tissues and, with substantial inhalation or ingestion, injure the nervous system, liver, kidneys, or heart.

Toluene And Xylene Poisoning

Poisoning from solvents containing toluene or xylene, often through swallowing or inhalation. It can cause dizziness, confusion, abnormal heart rhythms, breathing problems, or organ injury.

Tolvaptan

A vasopressin V2-receptor antagonist used in selected patients with hyponatremia or autosomal dominant polycystic kidney disease; thirst, rapid sodium correction, and liver injury require attention.

Tomogram

A tomogram is an image of a selected cross-sectional layer of the body or another object, produced by tomography such as CT or MRI.

Tomography

Imaging that produces cross-sectional views, or slices, through the body. Computed tomography uses X-rays, while other forms use ultrasound, magnetic fields, or radioactive tracers.

Tomosynthesis

An imaging technique that acquires multiple low-dose X-ray projections and reconstructs thin section images, commonly used for three-dimensional breast imaging.

Tomotherapy

Tomotherapy combines CT imaging with rotating, precisely shaped radiation beams to deliver radiation treatment while adjusting for the target and nearby tissues.

Tongue

A muscular organ involved in taste, speech, chewing, and swallowing.

Tongue Biopsy

A tongue biopsy removes a small tissue sample from a tongue lesion for microscopic examination of infection, inflammation, precancer, or cancer.

Tongue Carcinoma

Tongue carcinoma is cancer of the tongue, usually squamous-cell cancer, and may cause a persistent sore, lump, pain, or difficulty speaking or swallowing.

Tongue Disease

A clinical condition or finding involving tongue disease. Its features, causes, evaluation, and management depend on the clinical context.

Tongue Neoplasm

A tongue neoplasm is an abnormal growth in the tongue that may be benign, precancerous, or malignant.

Tongue Problems

Changes in the tongue, such as pain, swelling, sores, color change, numbness, or difficulty moving or tasting. Causes include injury, infection, inflammation, nutritional deficiency, and other disease.

Tongue Tie

A short or tight band of tissue under the tongue that limits tongue movement. It may interfere with breastfeeding, speech, or eating in some people.

Tongue, Disorders Of

Conditions affecting the tongue, including infection, inflammation, trauma, tumors, and developmental abnormalities.

Tongue-Tie (Ankyloglossia)

Restricted movement of the tongue caused by an unusually short or tight lingual frenulum, sometimes affecting feeding, speech, or oral hygiene.

Alternate Names:

Ankyloglossia

Tonic Neck Reflex

A primitive reflex present from birth to 6 months, also called fencer's pose. When the head is turned to one side, the arm on that side extends while the opposite arm flexes. Present in both directions. Asymmetry may indicate neurologic injury. Disappears as voluntary head control develops.

Alternate Names:

fencer's pose.

Tonic Pupil

A tonic pupil is a pupil with poor light response but

a slow constriction to near focus, often caused by injury to postganglionic parasympathetic fibers.

Tonic Seizures

Seizures that cause sustained stiffening of muscles, usually lasting seconds to minutes.

Tonic-Clonic Seizure

A generalized seizure with a tonic stiffening phase followed by rhythmic clonic jerking, impaired consciousness, and a postictal state.

Tonics

Preparations traditionally claimed to invigorate or strengthen; their contents and evidence vary.

Tonometer

An instrument used to measure intraocular pressure, an important part of glaucoma assessment.

Tonometry

Measurement of pressure inside the eye, usually to assess for glaucoma or monitor its treatment. Methods include applanation, noncontact air-puff, rebound, and electronic devices; results must be interpreted with corneal thickness and the clinical context.

Tonsil

A mass of lymphoid tissue at the entrance to the pharynx.

Alternate Names:

Tonsils

Tonsil Inflammation (Adenoids And Tonsils)

Alternate index label for Adenoids and Tonsils.

Tonsil Stones

Small, hard lumps formed when food particles, mucus, dead cells, and bacteria become trapped and calcify in the folds of the tonsils. They are often harmless but may cause bad breath, throat irritation, or a feeling that something is stuck in the throat. Good oral hygiene and gargling may help;

recurrent symptoms, bleeding, or one-sided tonsil enlargement requires examination.

Tonsillar Cancer

A malignancy arising from the palatine tonsils, most commonly squamous cell carcinoma associated with HPV infection. It often presents with throat pain, otalgia, and cervical lymphadenopathy.

Tonsillar Neoplasm

A tonsillar neoplasm is an abnormal growth in a tonsil that may be benign or malignant and can cause a persistent sore throat, swallowing difficulty, or neck swelling.

Tonsillectomy

Surgical removal of the palatine tonsils, performed for selected recurrent infection, obstructive sleep-disordered breathing, or other tonsillar disease. apnea, or other selected conditions after assessment of potential benefits and risks.

Tonsillitis

Inflammation or infection of the tonsils, commonly viral or streptococcal.

Tonus

The normal resting partial contraction of a muscle or the baseline tension of a tissue.

Tooth

A mineralized structure in the jaw used for cutting, tearing, and grinding food.

Tooth - Abnormal Colors

Abnormal tooth color can result from plaque, decay, trauma, fluorosis, medicines, enamel defects, or staining, and the cause determines treatment.

Tooth - Abnormal Shape

Abnormal tooth shape may result from developmental defects, genetic conditions, trauma, wear, or tumors and can affect function, appearance, and eruption.

Tooth Abrasion

Loss of tooth substance caused by external mechanical friction, such as aggressive brushing, abrasive products, or habits, producing grooves, notches, or sensitivity.

Tooth Abscess

A tooth abscess is a localized collection of pus from bacterial infection of the tooth or surrounding tissues, causing pain, swelling, and possible spreading infection.

Tooth Anatomy

Tooth anatomy includes enamel, dentin, pulp, cementum, root, crown, periodontal ligament, and supporting bone and gums.

Tooth Ankylosis

Fusion of a tooth root to alveolar bone through loss of the periodontal ligament, preventing normal eruption and making the tooth appear infraoccluded as neighboring teeth grow.

Tooth Attrition

Wear of tooth surfaces caused by tooth-to-tooth contact, commonly from chewing, bruxism, or malocclusion, producing flattened cusps and possible sensitivity.

Tooth Calcification

Deposition of mineral that forms and hardens enamel, dentin, cementum, and other tooth structures during development; abnormal timing or amount can affect tooth strength and appearance.

Tooth Component

Tooth components include enamel, dentin, cementum, pulp, and supporting tissues, each contributing to tooth strength, sensation, nutrition, and attachment.

Tooth Crowding

Tooth crowding occurs when teeth lack enough space to align normally, often from tooth size, jaw size, eruption pattern, or retained primary teeth.

Tooth Crown

The tooth crown is the visible portion above the gum, covered mainly by hard enamel and containing underlying dentin and pulp.

Tooth Decay

Destruction of tooth structure caused by acids produced by bacteria.

Alternate Names:

Dental caries; cavities

Tooth Decay - Early Childhood

Early-childhood tooth decay is bacterial acid destruction of primary teeth, promoted by frequent sugars, poor brushing, enamel defects, and prolonged bottle or nursing exposure.

Tooth Discoloration

Change in tooth color caused by surface stains, dental decay, trauma, medications, fluorosis, aging, or changes within enamel or dentin.

Tooth Disease

A clinical condition or finding involving tooth disease. Its features, causes, evaluation, and management depend on the clinical context.

Tooth Enamel

Tooth enamel is the highly mineralized outer covering of the crown that protects dentin and pulp from wear, acid, and bacteria.

Tooth Eruption

Process by which tooth moves from intraosseous position to functional occlusion. Primary (deciduous) dentition: 20 teeth, eruption 6-30 months. Permanent dentition: 32 teeth (including third molars), eruption 6-21 years. Delayed eruption may indicate systemic conditions, local obstruction, or normal variant.

Tooth Extraction

Removal of a tooth from its socket because of severe decay, infection, periodontal disease, fracture, impaction, orthodontic planning, or another clinical

indication.

Tooth Formation - Delayed Or Absent

Delayed or absent tooth formation means teeth develop or erupt later than expected or fail to form, due to local, genetic, endocrine, nutritional, or systemic causes.

Tooth Fracture

A clinical condition or finding involving tooth fracture. Its features, causes, evaluation, and management depend on the clinical context.

Tooth Infection

Infection of the tooth, its pulp, or surrounding tissues, commonly caused by untreated decay, cracks, or gum disease. It may cause toothache, sensitivity, swelling, fever, or drainage. Facial swelling, difficulty swallowing or breathing, or spreading infection requires emergency assessment.

Tooth Loss

Tooth loss is absence of a tooth due to decay, gum disease, trauma, developmental absence, or extraction and can affect chewing, speech, and oral health.

Tooth Luxation

Traumatic displacement of a tooth without complete avulsion. Concussion and subluxation involve periodontal injury with little or no displacement, whereas extrusive, lateral, and intrusive luxation involve increasingly marked displacement and require prompt dental evaluation.

Tooth Replantation

Replacement of an accidentally displaced or avulsed tooth into its socket, followed by stabilization and dental follow-up.

Tooth Resorption

Loss of dental hard tissue caused by odontoclast activity, occurring internally or on the root surface and associated with trauma, infection, orthodontics, or idiopathic disease.

Tooth Root

The tooth root anchors a tooth in its socket and contains dentin, cementum, pulp, nerves, and blood vessels beneath the gum.

Tooth Sensibility Testing

Clinical testing of a tooth's response to thermal, electric, or dentin stimulation. It assesses neural response rather than directly measuring pulp blood flow, so results must be interpreted with history, examination, and imaging.

Tooth Sensitivity

Pain in response to thermal, sweet, acidic, or tactile stimulus. Caused by dentin exposure (recession, erosion, abrasion, abfraction). Management: desensitizing toothpaste, fluoride varnish, restorative coverage, gum graft for recession.

Tooth, Supernumerary

An extra tooth beyond the normal number, which may cause crowding or impaction.

Toothache

Pain arising from a tooth or its supporting tissues.

Toothache Overview

Toothache usually reflects decay, pulp inflammation, fracture, gum disease, abscess, trauma, or referred pain. Dental treatment is needed to correct the cause; facial swelling, fever, difficulty swallowing, or breathing difficulty requires urgent care.

Alternate Names:

Pain Tooth (Toothache Overview)

Toothaches

Toothache is pain arising from decay, pulp inflammation, abscess, gum disease, trauma, cracking, or referred facial pain and requires dental assessment if persistent.

Toothpaste Overdose

Swallowing too much toothpaste can cause stomach upset, vomiting, or diarrhea. Large amounts of fluoride toothpaste can cause serious salt and calcium

disturbances, especially in children.

Tophaceous Gout

Chronic tophaceous gout represents advanced gout with deposition of large, visible tophi consisting of monosodium urate crystals surrounded by granulomatous inflammation. Tophi typically develop after years of inadequately treated gout and occur on the ears, fingers, toes, and olecranon bursa.

Tophus

A deposit of monosodium urate crystals in gout, often around joints or in soft tissues.

Topical

Applied directly to a body surface, usually the skin or a mucous membrane.

Topical Antibiotics

Antibiotics applied to the skin to treat selected superficial bacterial infections. The choice and duration depend on the organism, site, extent of infection, resistance patterns, and local guidance. Unnecessary or prolonged use can cause irritation and antimicrobial resistance.

Topical Antifungals

Topical antifungals treat superficial fungal infections. Allylamines (terbinafine, naftifine) inhibit squalene epoxidase, fungicidal. Azoles (clotrimazole, miconazole, ketoconazole, econazole) inhibit ergosterol synthesis, fungistatic. Ciclopirox has broad-spectrum activity. Tolnaftate is effective against dermatophytes.

Topical Calcineurin Inhibitors

Non-steroidal anti-inflammatory agents used for atopic dermatitis and facial eczema. They inhibit T-cell activation by blocking calcineurin, preventing cytokine production. Advantages over corticosteroids: no skin atrophy, safe for face and intertriginous areas. Side effects include transient burning and itching at application site.

Topical Corticosteroid Potency

Topical corticosteroids are classified into seven potency groups. Group 1 (ultra-high): clobetasol propionate 0.05 percent. Group 2 (high): betamethasone dipropionate 0.05 percent. Group 3 to 4 (medium): triamcinolone acetonide 0.1 percent. Group 5 to 6 (low): hydrocortisone 1 percent.

Topical Retinoids

Vitamin A derivatives used for acne, photoaging, and hyperpigmentation. Tretinoin (all-trans retinoic acid) is the most studied. Adapalene is more stable and less irritating. Tazarotene is the most potent. Mechanism: normalizes follicular keratinization, increases cell turnover, and stimulates collagen production.

Topiramate

Topiramate is an antiseizure medicine also used to prevent migraine; cognitive slowing, paresthesias, weight loss, kidney stones, metabolic acidosis, and fetal oral-cleft risk are important considerations.

Topoisomerase

An enzyme that temporarily cuts and rejoins DNA to control its twisting during replication, transcription, and chromosome separation.

Topoisomerase II

An enzyme that cuts both DNA strands, passes another DNA segment through the break, and rejoins the strands; it is targeted by some anticancer drugs.

Topoisomerase Inhibitors

Drugs that interfere with DNA topoisomerases; some are antibiotics and others are anticancer medicines, with toxicity depending on the drug and target enzyme.

Topotecan

A topoisomerase-I inhibitor used for selected ovarian, small-cell lung, and cervical cancers; bone-marrow suppression is a major toxicity.

Topotecan Hydrochloride

The hydrochloride salt of topotecan, a topoisomerase-I inhibitor used for selected cancers.

TORCH Panel

Serologic screening for congenital infections: Toxoplasma, Rubella, CMV, Herpes simplex (and others: syphilis, parvovirus B19, VZV, Zika). IgM: indicates recent/active infection (maternal or fetal). IgG: past infection or passive immunity. Paired maternal and neonatal testing.

TORCH Screen

Testing for selected congenital infections—classically Toxoplasma, Other infections, Rubella, Cytomegalovirus, and Herpes—when fetal or newborn findings suggest an in-utero infection; broad panels require careful interpretation.

Toremifene

A selective estrogen-receptor modulator used for metastatic breast cancer in postmenopausal women; hot flashes and, rarely, dangerous QT prolongation may occur.

Torn ACL

A torn anterior cruciate ligament is a knee injury commonly caused by sudden pivoting, landing, or contact, producing a pop, swelling, instability, and difficulty changing direction. Rehabilitation, bracing, or reconstruction is selected according to activity, associated injury, and goals.

Torn Meniscus

A torn meniscus is a split in the knee's shock-absorbing cartilage caused by twisting or degeneration. Pain, swelling, catching, or locking may occur; physical therapy or arthroscopic treatment depends on tear pattern, symptoms, age, and associated disease.

Alternate Names:

Meniscus Tear

Torpor

A state of greatly reduced activity, alertness, or responsiveness. In a patient, marked or new torpor may signal a serious illness, medicine effect, or metabolic problem.

Torsade De Pointes

Torsade de pointes is a potentially fatal polymorphic ventricular tachycardia associated with prolonged QT, electrolyte disturbance, medicines, or inherited channel disorders.

Torsades De Pointes

A specific form of polymorphic ventricular tachycardia with QRS complexes that twist around the baseline, occurring in the setting of QT prolongation. It may degenerate to ventricular fibrillation.

Torsemide

Torsemide is a loop diuretic that inhibits sodium-potassium-chloride reabsorption in the thick ascending limb and treats oedema from heart, kidney, or liver disease; dehydration and electrolyte loss may occur.

Torsion

Twisting of a structure around its axis, potentially cutting off blood supply.

Torsion Dystonia

A movement disorder causing sustained or repetitive muscle contractions that twist the body or limbs into abnormal postures; it may be inherited, acquired, focal, or generalized.

Torsion Of The Appendix Testis

Acute twisting and infarction of the appendix testis (hydatid of Morgagni), a small embryonic remnant of the paramesonephric (Müllerian) duct attached to the upper pole of the testis. It represents the single most common cause of acute scrotum in pre-pubertal boys, peaking between ages 7 and 12 years. Patients present with sudden, localized upper-pole scrotal tenderness with a normal-sized, non-tender lower testis and an intact cremasteric reflex. In early stages, a pathognomonic hard, blue-tinged nodule

visible through the thin scrotal skin (the “blue dot sign”) may be present. Scrotal color Doppler ultrasonography confirms normal testicular perfusion and visualizes an avascular, enlarged, hyperechoic nodule adjacent to the testis. It is benign and self-limiting, managed conservatively with rest, scrotal elevation, and nonsteroidal anti-inflammatory drugs.

Alternate Names:

Hydatid of Morgagni Torsion; Appendix Testis Torsion

Torsion Testicle (Testicular Disorders)

Alternate index label for Testicular Disorders.

Torsion, Testicular

Alternate index label; see *Testicular torsion*.

Torticollis

Abnormal contraction or posture of the neck causing the head to tilt or rotate.

Torticollis (Wryneck)

A condition in which the head is tilted to one side (ear toward the shoulder) and rotated to the opposite side (chin toward opposite shoulder), due to contraction or shortening of the sternocleidomastoid muscle.

Alternate Names:

Wryneck

Torticollis, Spasmodic

Alternate index label; see *Cervical dystonia*.

Torulaspora

A genus of yeasts used in food and fermentation; human infection is rare and usually opportunistic.

Torulaspora Delbrueckii

A yeast used in wine and food fermentation; it rarely causes infection in people with major immune compromise.

Torus Fracture

An incomplete, stable compression fracture of pe-

diatric bone (also known as a buckle fracture), representing the most frequent skeletal injury in childhood. Characterized by compaction and outward bulging of the porous, trabecular metaphyseal cortex (most commonly in the distal radius) in response to an axial compressive load on a fall onto an outstretched hand, with an intact periosteum and without disruption of the opposite cortex. It carries an excellent prognosis, heals rapidly, and is typically treated conservatively with a removable wrist splint or short-arm cast for 3 to 4 weeks.

Alternate Names:

Buckle Fracture; Metaphyseal Buckle Fracture; Torus Fracture of Radius

Torus Mandibularis

A bony outgrowth arising from the lingual surface of the mandible, typically bilateral and located in the premolar region above the mylohyoid ridge. It is a common, asymptomatic developmental anomaly, more prevalent in women and certain ethnic populations.

Torus Palatinus

A bony exostosis arising from the midline of the hard palate, typically presenting as a slow-growing, asymptomatic, hard, bony protuberance discovered incidentally. It is more common in women and certain ethnic groups.

Tositumomab

A monoclonal antibody formerly used with radioactive iodine-131 for some relapsed or refractory B-cell non-Hodgkin lymphomas; it is no longer routinely available in many settings.

Total Abdominal Colectomy

Surgical removal of the entire colon through an abdominal approach, with restoration of intestinal continuity or creation of an ileostomy as clinically appropriate.

Total Anomalous Pulmonary Venous Connection

Alternate index label; see *Total anomalous pulmonary venous return*.

Total Anomalous Pulmonary Venous Return

A congenital heart defect in which the pulmonary veins drain into the systemic venous circulation instead of the left atrium. The clinical effects depend on the anatomic type, the degree of obstruction, and associated defects; affected newborns require specialist cardiac evaluation.

Total Artificial Heart

A mechanical device that replaces the pumping function of both ventricles and is used in selected patients with severe biventricular heart failure, usually as a bridge to transplantation. It requires specialized center-based management and carries risks including bleeding, infection, thrombosis, and device malfunction.

Total Body Surface Area

The percentage of body surface involved by a burn or other extensive skin process. It is estimated using methods such as the rule of nines, Lund-Browder chart, or the patient's palm and is used to guide burn severity assessment and resuscitation.

Total Breech Extraction

A maneuver where the operator grasps the fetal feet and delivers the entire baby in breech presentation. The most common indication is delivery of the second twin (malpresentation). Contraindicated for term singleton breech due to high risk of head entrapment, nuchal arms, and birth trauma.

Total Cholesterol

Total cholesterol is the combined cholesterol carried in major blood lipoproteins; cardiovascular risk is assessed with HDL, LDL, triglycerides, and other factors.

Total Daily Energy Expenditure

Total calories burned in 24 hours: BMR + thermic effect of food + physical activity. Estimation guides dietary recommendations. Activity factor multiplies BMR.

Total Haemolytic Complement Test

A functional assay of the classical complement pathway, commonly reported as CH50.

Total Hysterectomy

Removal of the uterus and cervix. The fallopian tubes and ovaries are not necessarily removed and should be specified separately, such as salpingectomy or oophorectomy.

Total Hip Arthroplasty

Replacement of the hip joint with prosthetic components. Acetabular component: metal shell + polyethylene liner. Femoral component: metal stem + femoral head (cobalt-chrome or ceramic). Fixation: cemented (PMMA, for elderly, osteoporotic bone), cementless (porous-coated, for younger patients, biologic fixation), and hybrid.

Total Hip Joint Replacement - Revision

A medical procedure, treatment, device, or care topic related to total hip joint replacement - revision; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Total Intravenous Anesthesia

Total intravenous anesthesia uses intravenous agents exclusively for induction and maintenance of anesthesia, avoiding volatile inhalation agents. The most common TIVA regimen combines propofol infusion (50 to 200 mcg/kg/min) with remifentanyl infusion (0.1 to 0.5 mcg/kg/min).

Total Iron Binding Capacity

TIBC: a measure of the maximum iron-binding capacity of transferrin in serum, reflecting total iron-transport capacity. Calculated from transferrin or measured directly.

Total Knee Arthroplasty

Replacement of the knee joint with prosthetic components. Femoral component (cobalt-chrome), tibial component (titanium tray + polyethylene insert), and patellar component (polyethylene, optional). Fixation: cement (PMMA) most common in older adults; cementless in younger.

Total Knee Joint Replacement - Revision

A medical procedure, treatment, device, or care topic related to total knee joint replacement - revision; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Total Knee Replacement

A medical procedure, treatment, device, or care topic related to total knee replacement; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Total Laryngectomy

Total laryngectomy removes the entire voice box and creates a permanent opening in the neck for breathing, usually to treat advanced laryngeal cancer.

Total Lesion Glycolysis

Product of metabolic tumor volume and mean SUV. Prognostic marker.

Total Lung Capacity

The total volume of air in the lungs after maximal inspiration. It is measured by lung-volume testing, such as body plethysmography, and may be increased in hyperinflation or decreased in restrictive lung disease.

Total Pancreatectomy

Total pancreatectomy removes the entire pancreas and causes permanent loss of insulin and digestive-enzyme production, requiring lifelong replacement treatment.

Total Parenteral Nutrition

Intravenous administration of nutrients sufficient to meet essentially all nutritional requirements when the gastrointestinal tract cannot be used adequately.

Total Parenteral Nutrition - Infants

Intravenous nutrition for infants who cannot receive enough nutrients through the gastrointestinal tract; it requires careful monitoring of glucose, electrolytes, liver function, infection, and catheter complications.

Total Proctocolectomy And Ileal-Anal Pouch

Total proctocolectomy with an ileal-anal pouch removes the colon and rectum and creates a pouch from the ileum that is connected to the anus.

Total Proctocolectomy With Ileostomy

Total proctocolectomy with ileostomy removes the colon and rectum and diverts the end of the small intestine through an abdominal stoma.

Total Protein

The combined amount of albumin and other proteins in the blood. A blood test may help assess nutrition, liver or kidney disease, inflammation, or abnormal antibody production.

Total Sleep Time

The total amount of time spent asleep during a defined sleep period or sleep study, excluding periods of wakefulness.

Touch

The sensory perception of contact, pressure, texture, or vibration.

Tourette Disorder

A neurodevelopmental disorder characterized by multiple motor tics and one or more vocal tics lasting >1 year, with onset before age 18. Motor tics: eye blinking, head jerking, shoulder shrugging, facial grimacing. Vocal tics: throat clearing, sniffing, grunting, barking, coprolalia (obscene words, occurs in <10%).

Tourette Syndrome

A neurodevelopmental disorder characterized by multiple motor tics and at least one vocal tic occurring over at least one year. Tics are sudden, rapid, recurrent, stereotyped movements or vocalizations. They are often preceded by premonitory urges. Tics typically worsen in adolescence and may improve in adulthood.

Alternate Names:

Tourette's

Tourette'S

Alternate index label; see *Tourette syndrome*.

Tourette'S Syndrome

A neurodevelopmental disorder involving multiple motor tics and at least one vocal tic persisting for more than one year.

Tourniquet

A device or band used to compress a limb and temporarily reduce blood flow.

Towel Clamp

A clamp used to secure surgical drapes or hold them in position.

Townes Syndrome

A congenital disorder caused by pathogenic variants in *SALL1*, classically involving anal atresia and characteristic thumb abnormalities. Renal, cardiac, hearing, foot, and genital anomalies are common and variable.

Toxaemia

An older term for toxins in the blood or systemic illness; in pregnancy it was historically used for pre-eclampsia.

Toxaphene

An infectious process or toxic exposure involving toxaphene; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Toxbase

A clinical toxicology information service used to support assessment and management of poisoning.

Toxic

Poisonous or capable of causing injury to living tissue.

Toxic Dose

Dose causing toxicity. Varies by substance, individual, route, duration.

Toxic Effect

An infectious process or toxic exposure involving toxic effect; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Toxic Encephalopathy

Brain dysfunction caused by exposure to a toxic substance or medicine. Symptoms may include confusion, altered behavior, seizures, or reduced consciousness.

Toxic Epidermal Necrolysis

A severe, potentially fatal mucocutaneous reaction, usually to a medication, causing extensive epidermal necrosis and skin detachment. It is on the severe end of the Stevens–Johnson syndrome and toxic epidermal necrolysis spectrum and requires urgent specialist care.

Alternate Names:

TEN

Toxic Epidermal Necrolysis

Toxic epidermal necrolysis is a life-threatening severe skin reaction, usually to a medicine, causing widespread epidermal death and skin detachment with mucosal injury.

Toxic Megacolon

Acute life-threatening dilation of the colon associated with systemic toxicity, representing one of the most serious complications of inflammatory bowel disease, particularly ulcerative colitis. It can also occur in infectious colitis caused by *Clostridioides difficile* and cytomegalovirus.

Toxic Metabolic Encephalopathy

Diffuse brain dysfunction caused by systemic metabolic or toxic disturbances, producing confusion, altered alertness, agitation, seizures, or coma.

Toxic Nodular Goiter

An enlarged thyroid with nodules that produce excessive thyroid hormone.

Toxic Shock Syndrome

A severe toxin-mediated illness, usually caused by *Staphylococcus aureus* or *Streptococcus pyogenes*, with fever, rash, hypotension, and organ dysfunction.

Toxic Shock Syndrome

Alternate index label for TSS.

Toxic Shock Syndrome

Alternate index label for Toxic Shock Syndrome.

Toxic Synovitis

Temporary inflammation of the lining of the hip joint, usually in a child after a viral illness. It can cause hip pain and limping and must be distinguished from a serious joint infection.

Toxicity

Toxicity refers to harmful exposure to the named substance or condition; severity depends on dose, route, timing, symptoms, and patient factors, and urgent professional assessment may be required.

Toxicity Grade

An infectious process or toxic exposure involving toxicity grade; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Toxicity Test

toxicity test refers to harmful exposure to the named substance or condition; severity depends on dose, route, timing, symptoms, and patient factors, and urgent professional assessment may be required.

Toxicogenomics

An infectious process or toxic exposure involving toxicogenomics; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Toxicokinetic Description

An infectious process or toxic exposure involving toxicokinetic description; diagnosis considers the

cause, route of exposure, affected tissues, severity, and host response.

Toxicokinetics

The study of absorption, distribution, metabolism, and excretion of a toxic substance and its relationship to concentration over time. Toxicokinetics helps estimate exposure, timing, accumulation, and expected duration of toxicity.

Toxicology

The study of poisons, toxic exposures, their effects, prevention, and treatment.

Toxicology Screen

A test of urine, blood, or another sample for medicines, drugs, alcohol, or poisons. Results must be interpreted with the person's symptoms, timing, and prescribed medicines.

Toxicology Specimen Preservation

Collection, labeling, storage, and transport of biological specimens to preserve analytes and prevent contamination or degradation. Appropriate specimens may include peripheral blood, urine, vitreous humor, bile, liver, hair, or gastric contents depending on the suspected exposure and timing.

Toxidrome

A recognizable pattern of signs and symptoms produced by a class of toxins. Important patterns include opioid, sympathomimetic, anticholinergic, cholinergic, sedative-hypnotic, and serotonin toxidromes; treatment combines supportive care, targeted antidotes when available, and decontamination decisions.

Toxin

A poisonous substance made by a living organism, such as a bacterium, plant, animal, or fungus. Toxins can damage specific tissues or disrupt nerve, muscle, or cellular functions.

Alternate Names:

Biological poison

Toxin Ciguatera (Ciguatera Poisoning)

Alternate index label for Ciguatera Poisoning.

Toxins

Poisonous substances produced by living organisms, including bacteria, fungi, plants, and animals. They can injure cells or disrupt organ function; the term is usually distinguished from toxicants, which are harmful substances of nonbiological origin.

Toxocara

An infectious process or toxic exposure involving toxocara; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Toxocara Canis

An infectious process or toxic exposure involving toxocara canis; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Toxocara Cati

An infectious process or toxic exposure involving toxocara cati; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Toxocariasis

A zoonotic infection caused by Toxocara larvae, potentially affecting viscera or the eye.

Toxoid

A toxin rendered non-toxic but still antigenic, used in vaccines such as tetanus and diphtheria toxoids.

Toxoid Vaccines

Vaccines that protect against harmful bacterial toxins. These vaccines contain toxins that have been detoxified and rendered inactive.

Toxoplasma

An infectious process or toxic exposure involving toxoplasma; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Toxoplasma Blood Test

A blood test detecting antibodies or genetic material related to Toxoplasma gondii infection, interpreted with symptoms, immune status, pregnancy, and timing of exposure.

Toxoplasma Gondii

An obligate intracellular protozoan parasite that causes toxoplasmosis. The definitive host is the cat (family Felidae), which sheds environmentally resistant oocysts in feces.

Toxoplasmosis

Infection by Toxoplasma gondii, usually mild but potentially severe in pregnancy or immunosuppression.

Alternate Names:

Chorioretinitis Toxoplasma (Toxoplasmosis) Meningoencephalitis Toxoplasma (Toxoplasmosis)

Toxoplasmosis (Toxo)

Infection with Toxoplasma gondii, often mild in healthy people but potentially severe in pregnancy or in people with weakened immunity.

TPA

Tissue plasminogen activator, a natural clot-dissolving protein or a laboratory-made medicine based on it. In selected emergencies it helps break down clots, but it can cause serious bleeding.

Alternate Names:

Tissue plasminogen activator

TPMT Polymorphism

Thiopurine S-methyltransferase is an enzyme that metabolizes thiopurine drugs including azathioprine, 6-mercaptopurine, and thioguanine, which are used in the treatment of leukemia, inflammatory bowel disease, and autoimmune conditions.

TPN

TPN, or total parenteral nutrition, supplies nutrients intravenously when the gastrointestinal tract cannot be used adequately.

Tra Gene

A gene involved in sex determination and sexual differentiation in some organisms, especially insects; its function and sequence vary by species.

Trabectedin

Trabectedin is an antineoplastic drug that binds DNA and alters transcription and repair, used for selected advanced soft-tissue sarcomas and ovarian cancer.

Trabecula

A small beam, strut, or partition of tissue, especially in cancellous bone.

Trabecular Bone

Porous, lattice-like bone tissue found inside bones and near many joints.

Alternate Names:

Spongy bone; cancellous bone

Trabecular Meshwork

A porous tissue in the eye through which aqueous humor drains.

Trabeculectomy

A conventional glaucoma filtration surgery that creates a new drainage pathway for aqueous humor by removing a small piece of the trabecular meshwork and creating a partial-thickness scleral flap under which aqueous humor filters from the anterior chamber into the subconjunctival space, forming a filtering bleb.

Trabeculoplasty

Laser treatment that improves aqueous-humor drainage through the trabecular meshwork.

Trabulsiella

A genus of gram-negative bacteria related to *Enterobacter* that is rarely associated with opportunistic human infection.

Trabulsiella Guamensis

A gram-negative bacterium rarely isolated from hu-

man clinical specimens; disease significance depends on a compatible infection rather than colonization.

Trace Elements

Minerals required in very small amounts, including iron, zinc, copper, iodine, and selenium.

Trace Mineral

A mineral required by the body in very small amounts.

Tracer

A labeled substance used to follow a biological, chemical, or physical process.

Tracer Principle

Administered in amounts too small to produce pharmacological effect. Dose optimized for imaging, not therapeutic.

Trachea

The windpipe connecting the larynx with the main bronchi.

Tracheal Agenesis With Bronchoesophageal Fistula

Congenital malformation characterized by absent or severely hypoplastic trachea with an abnormal communication between the bronchus and esophagus, a rare and typically fatal condition presenting in neonates with respiratory distress, cyanosis, and inability to intubate.

Tracheal Foreign Body

A foreign object lodged in the trachea, causing severe respiratory distress, stridor, and complete airway obstruction. It is a medical emergency requiring immediate bronchoscopic removal. In complete obstruction, emergency cricothyrotomy may be needed.

Tracheal Rupture

Tracheal rupture is a tear in the windpipe caused by trauma, instrumentation, surgery, or pressure and can produce air leakage, breathing difficulty, and

life-threatening complications.

Tracheal Stenosis

Narrowing of the trachea, often a complication of prolonged endotracheal intubation or tracheostomy. It presents with progressive dyspnea, stridor, and wheezing that may be mistaken for asthma. Diagnosis is by bronchoscopy or CT. Treatment is balloon dilation, stenting, or surgical resection.

Tracheitis

Inflammation or infection of the mucous membranes of the trachea. While mild tracheitis frequently accompanies viral upper respiratory tract infections (tracheobronchitis), acute bacterial tracheitis (bacterial croup / pseudomembranous croup) is a life-threatening pediatric airway emergency. Typically caused by secondary bacterial superinfection (*Staphylococcus aureus*, *Streptococcus pneumoniae*, *Haemophilus influenzae*) following a viral prodrome, presenting with high fever, toxic appearance, barking cough, and severe inspiratory and expiratory stridor refractory to racemic epinephrine. Mandates urgent airway visualization and stabilization via endotracheal intubation, deep tracheal suctioning of thick mucopurulent pseudomembranes, and broad-spectrum intravenous antibiotic therapy.

Trachelectomy

Trachelectomy is surgical removal of the cervix, sometimes with nearby tissue, used in selected cervical cancers or other conditions.

Tracheobronchomalacia

Abnormal softening of the trachea and/or bronchi causing excessive airway collapse during breathing. It may be congenital or acquired. Symptoms include chronic cough, recurrent infections, and expiratory airflow obstruction. Diagnosis is by dynamic CT or bronchoscopy. Treatment includes CPAP, stenting, or surgical reconstruction.

Tracheobronchomegaly

Abnormal enlargement of the trachea and main bronchi, usually associated with connective-tissue weakness. It may cause recurrent lower-respiratory

infection, bronchiectasis, ineffective cough, dyspnea, and difficulty clearing airway secretions.

Tracheoesophageal Fistula

An abnormal connection between the trachea and esophagus, often associated with esophageal atresia. Presents with recurrent aspiration, coughing with feedings, and cyanosis during meals. The H-type fistula may present later in infancy with chronic aspiration and failure to thrive. Diagnosis by esophagram or bronchoscopy.

Tracheoesophageal Fistula And Esophageal Atresia Repair

A medical procedure, treatment, device, or care topic related to tracheoesophageal fistula and esophageal atresia repair; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Tracheomalacia

Softening of the tracheal cartilage causing excessive collapse of the trachea during expiration. It may be congenital or acquired (prolonged intubation, relapsing polychondritis). Symptoms include expiratory stridor and difficulty clearing secretions. Diagnosis is by dynamic CT or bronchoscopy.

Tracheomalacia - Acquired

Acquired tracheomalacia is weakening of the tracheal wall after injury, inflammation, compression, or prolonged ventilation, causing airway collapse and breathing symptoms.

Tracheomalacia - Congenital

Congenital tracheomalacia is weakness of the infant tracheal cartilage causing dynamic airway collapse, noisy breathing, cough, or recurrent respiratory symptoms.

Tracheostomy

A surgically created opening through the neck into the trachea, usually with insertion of a tube.

Tracheostomy Care

A medical procedure, treatment, device, or care topic related to tracheostomy care; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Tracheostomy Tube

A tube inserted through a tracheostomy opening to maintain access to the airway and permit breathing, suction, or mechanical ventilation.

Tracheostomy Tube - Eating

Eating and drinking with a tracheostomy tube requires swallowing assessment and individualized precautions because aspiration risk depends on airway protection, secretions, and the underlying illness.

Tracheostomy Tube - Speaking

Speech with a tracheostomy tube may be supported by cuff deflation, a speaking valve, or other methods when airflow and airway protection are adequate; specialist assessment is required.

Tracheotomy

An operation that creates an opening through the neck into the trachea. The opening is called a tracheostomy.

Trachoma

A chronic eye infection caused by *Chlamydia trachomatis* that can scar the conjunctiva and cornea.

Trachyonychia

Rough, thin, ridged nails with many tiny pits, sometimes called twenty-nail dystrophy when most or all nails are affected. It may occur on its own or with psoriasis, alopecia areata, eczema, or lichen planus.

Traction

A pulling force used to align, immobilize, or relieve pressure on tissue or a fracture.

Tractional Retinal Detachment

Separation of the neurosensory retina caused by contraction of fibrovascular membranes, commonly

associated with proliferative diabetic retinopathy.

Tractotomy

Surgical interruption of a nerve tract, usually to reduce severe pain or other neurologic symptoms.

Train-Of-Four Monitoring

Train-of-four monitoring assesses neuromuscular blockade by delivering four supramaximal electrical stimuli at 2 Hz to a peripheral nerve, commonly the ulnar nerve at the wrist. The ratio of the fourth twitch height to the first twitch height estimates recovery from non-depolarizing blockade.

Trained Immunity

Innate immune memory: epigenetic/metabolic reprogramming of monocytes/macrophages/NK cells after initial stimulus (BCG, beta-glucan, *Candida*, oxidized LDL) -> enhanced response to secondary challenge (heterologous protection).

Training

Structured practice intended to develop knowledge, skills, physical capacity, or safe performance of a task. Medical training combines instruction, supervised practice, and assessment.

Trait

A trait is a measurable characteristic of an organism influenced by genes, environment, development, or their interaction.

Trajectory

The path a projectile follows in flight and, in wound ballistics, the course it traverses through tissue from entrance to exit or rest. Internal trajectory is reconstructed from entrance and exit defects, the organ track, recovered fragments, and clothing damage; angles disclose shooter position and victim posture.

TRALI

Transfusion-Related Acute Lung Injury, a life-threatening complication of blood transfusion characterized by acute hypoxemia and bilateral pulmonary edema within 6 hours of transfusion, with-

out evidence of volume overload.

Tram-Track Appearance

Histological finding on silver stain of MPGN representing GBM duplication. Results from mesangial cell interposition into capillary wall with new basement membrane formation. Creates double-contour pattern on Jones methenamine silver stain.

Trance

An altered state of awareness; clinical interpretation depends on cause and context.

Tranexamic Acid

An antifibrinolytic medicine that reduces bleeding by inhibiting plasminogen activation.

Trans Fat

An unsaturated fat containing at least one trans-configured double bond. Industrial trans fats, formerly common in partially hydrogenated oils, can raise LDL cholesterol, lower HDL cholesterol, and increase cardiovascular risk; small amounts also occur naturally in some meat and dairy foods.

Trans Fatty Acid

An unsaturated fatty acid with trans configuration, often formed during industrial oil processing.

Alternate Names:

Trans fat

Transabdominal Cerclage

A surgical procedure in which a permanent, non-absorbable synthetic band or braided suture is placed around the uterine cervico-isthmic junction via laparotomy or laparoscopy. Indicated in high-risk pregnant or pre-conceptional women with severe cervical insufficiency in whom standard transvaginal cerclage (McDonald or Shirodkar) is technically impossible or has repeatedly failed due to congenital cervical hypoplasia, cervical amputation, or extensive prior trachelectomy.

Alternate Names:

TAC; Cervico-Isthmic Cerclage

Transabdominal Scan

An ultrasound scan performed with the transducer placed on the maternal abdomen. A full bladder improves visualization of pelvic structures. Used in early pregnancy to confirm viability and gestational age, and throughout pregnancy to assess fetal growth, anatomy, amniotic fluid volume, and placental location.

Transaminase, Serum Glutamic Pyruvic

SGPT, now called ALT, is a liver-associated enzyme measured in blood and often elevated when liver cells are injured.

Transaminitis

Elevation of blood levels of aminotransferase enzymes, usually alanine aminotransferase and aspartate aminotransferase, indicating possible liver-cell injury but not identifying its cause.

Transarterial Chemoembolization

A procedure in which chemotherapy and embolizing particles are infused selectively into tumor-feeding hepatic arteries, combining regional chemotherapy with ischemic tumor necrosis; used for unresectable hepatocellular carcinoma.

Transbronchial Biopsy

Bronchoscopic sampling of lung parenchyma through the bronchial wall. It may evaluate interstitial or granulomatous disease, infection, transplant rejection, or peripheral lesions; complications include bleeding and pneumothorax.

Transcatheter Aortic Valve Replacement

A minimally invasive catheter-based cardiac intervention in which a collapsible bioprosthetic aortic valve (balloon-expandable or self-expanding) is delivered percutaneously (most commonly via the transfemoral artery, or via transaxillary, carotid, or transapical routes) and deployed within the diseased native calcified aortic valve under fluoroscopic guidance. Established as the standard of care for severe, symptomatic aortic stenosis across patients at high, intermediate, and low surgical risk, it avoids median sternotomy, cardiopulmonary bypass, and

cardiac arrest, resulting in reduced recovery times and mortality comparable or superior to surgical aortic valve replacement.

Alternate Names:

TAVR; TAVI; Transcatheter Aortic Valve Implantation

Transcatheter Edge-To-Edge Repair

A minimally invasive percutaneous catheter-based intervention for patients with severe symptomatic mitral regurgitation who are at high or prohibitive surgical risk, or who have secondary (functional) mitral regurgitation despite guideline-directed medical therapy. Under real-time transesophageal echocardiographic guidance via transfemoral transseptal access, a mechanical implant (such as the MitraClip or PASCAL device) is navigated into the left atrium and deployed to grasp and approximate the middle scallops of the anterior and posterior mitral valve leaflets (Alfieri stitch equivalent), creating a permanent double-orifice mitral valve that significantly reduces regurgitant volume.

Alternate Names:

TEER; MitraClip Procedure; Percutaneous Mitral Valve Repair

Transcervical Foley Catheter For Cervical Ripening

A mechanical method for cervical ripening: a Foley catheter (16-30 Fr) is inserted through the cervix, the balloon is inflated with 30-60 mL of saline, and traction is applied (or the catheter is taped to the thigh with gentle tension). The balloon applies direct pressure to the internal os, stimulating endogenous prostaglandin release.

Transcervical Resection Of Endometrium

An endoscopic procedure that removes or ablates the endometrium to reduce heavy menstrual bleeding.

Transcranial Doppler

Ultrasound measuring blood flow velocity in cerebral arteries. Used for vasospasm detection, stroke risk assessment.

Transcranial Doppler Scanning

Ultrasound measurement of blood flow through major arteries at the base of the brain.

Alternate Names:

Transcranial Doppler; TCD

Transcranial Doppler Ultrasound

Transcranial Doppler ultrasound measures blood-flow speed and direction in major arteries inside the skull.

Transcription

Synthesis of RNA from DNA template by RNA polymerase. Eukaryotic: RNA Pol II transcribes mRNA, rRNA, snRNA; RNA Pol I transcribes rRNA; RNA Pol III transcribes tRNA, 5S rRNA. Steps: initiation (promoter recognition, transcription factors), elongation (5' to 3'), termination. Promoter elements: TATA box, CAAT box, GC box.

Transcription Factor

Protein binding to specific DNA sequences, regulating transcription. Activators enhance transcription; repressors inhibit. Examples: p53, NF-kB, steroid hormone receptors.

Transcutaneous Electrical Nerve Stimulation

Low-voltage electrical stimulation delivered through skin electrodes to reduce pain.

Alternate Names:

TENS

Transcutaneous Electrical Nerve Stimulation

Electrical stimulation delivered through skin electrodes to modulate pain.

Transdermal

A route of drug administration delivering drug through intact skin into the systemic circulation via a patch or gel, bypassing first-pass metabolism and providing steady plasma concentrations over hours to days. Examples: nitroglycerin, fentanyl, nicotine, estrogen, clonidine, rotigotine.

Transfer Training

Teaching safe transfer techniques: bed to chair, chair to toilet, floor to standing. Use of transfer boards, lift equipment.

Transferrase

An enzyme that transfers a functional group from one molecule to another.

Transference

A psychoanalytic concept in which feelings, attitudes, and expectations originally directed toward a significant figure from childhood are unconsciously redirected onto the therapist. In psychodynamic therapy, transference is a therapeutic tool for understanding the patient's relational patterns and resolving conflicts.

Transferrin

An iron transport protein synthesized by the liver, carrying iron in plasma (2 iron-binding sites). In iron deficiency: transferrin is elevated with low ferritin, giving high TIBC and low transferrin saturation; in iron overload/inflammation/chronic disease: transferrin is low (acute phase).

Transferrin Saturation

The percentage of transferrin iron-binding sites occupied by iron.

Alternate Names:

TSAT

Transfusion

Administration of donated blood or blood components into a person's circulation.

Transfusion Of Neonate

Administration of a blood product to a newborn for clinically significant anemia, bleeding, or another specific indication. The decision, product characteristics, volume, and rate depend on the infant's symptoms, laboratory results, gestational age, and local transfusion protocol.

Transfusion Reaction

An adverse reaction to transfusion of blood products. Acute hemolytic transfusion reaction: ABO incompatibility, immediate fever/chills, hypotension, back pain, DIC, renal failure; management: STOP transfusion, maintain BP/urine output, supportive care.

Transient Familial Hyperbilirubinemia

Transient familial hyperbilirubinemia is inherited temporary elevation of bilirubin, often from reduced conjugation, causing episodic jaundice without major liver injury.

Transient Global Amnesia

A temporary episode of sudden inability to form new memories, with repetitive questioning, that usually resolves within 24 hours.

Transient Hypogammaglobulinemia Of Infancy

A temporary delay in normal immunoglobulin production during infancy, causing low antibody levels and increased susceptibility to infections before immune function improves spontaneously.

Alternate Names:

TGA

Transient Ischaemic Attack

A brief episode of neurological dysfunction caused by temporary focal brain, spinal-cord, or retinal ischaemia without persistent tissue infarction.

Alternate Names:

TIA

Transient Ischaemic Attacks Or Episodes

Brief focal neurologic episodes caused by temporary ischemia without persistent infarction.

Transient Ischemic Attack

A transient ischemic attack is a temporary episode of neurological dysfunction caused by focal brain, spinal cord, or retinal ischemia without acute infarction.

Transient Ischemic Attack (TIA, Mini-Stroke)

Alternate index label for TIA, Mini-Stroke.

Transient Ischemic Attack (TIA, Mini-Stroke)

Alternate index label for Transient Ischemic Attack (TIA, Mini-Stroke).

Transient Pain

Pain that is brief and resolves rather than persisting.

Transient Synovitis

A benign, self-limited inflammation of the synovial lining of the hip joint, representing the most common cause of acute hip pain and limping in children aged 3 to 8 years. Typically follows a viral upper respiratory tract infection by 1 to 2 weeks and presents with acute or insidious groin, thigh, or knee pain, antalgic limp, and mild restriction of hip abduction and internal rotation, with normal or slightly elevated inflammatory markers. It must be carefully distinguished from septic arthritis of the hip using the Kocher criteria (fever, non-weight-bearing, ESR >40 mm/hr, WBC >12,000/uL); treatment is supportive with rest and oral nonsteroidal anti-inflammatory drugs (NSAIDs).

Alternate Names:

Toxic Synovitis; Irritable Hip; Transient Synovitis of the Hip

Transient Tachypnea - Newborn

Temporary rapid breathing in a newborn caused by delayed clearance of fetal lung fluid, usually beginning soon after birth and improving within one to three days with supportive care.

Transient Tachypnea Of The Newborn

Temporary rapid breathing after birth, usually caused by delayed clearance of fetal lung fluid. It is more common after caesarean birth and typically improves within one to three days with supportive care.

Transillumination

Transillumination shines light through a body part or structure to reveal fluid, cavities, or other differences in tissue density.

Transition To Adulthood

Support for youth with disabilities transitioning to adult services.

Transitional Care

Coordinated care provided when a person moves between healthcare settings or levels of care, such as from hospital to home.

Transitional Epithelium

Specialized stretchable epithelium lining the renal pelvis, ureters, bladder, and proximal urethra, allowing urinary organs to expand while maintaining a protective barrier.

Alternate Names:

Transjugular Intrahepatic Portosystemic Shunt

Transjugular Intrahepatic Portosystemic Shunt

A radiologically created channel between a portal vein branch and a hepatic vein that lowers portal pressure, used for complications such as variceal bleeding or refractory ascites; encephalopathy and shunt dysfunction can occur.

Translational Medicine

An interdisciplinary field that turns findings from laboratory, clinical, and community research into better methods of disease prevention, diagnosis, treatment, and public health. It connects research findings with patient care and community health.

Alternate Names:

Translational Medicines

Translation

Decoding of mRNA sequence into polypeptide chain by ribosomes. Occurs in cytoplasm. tRNAs carry specific amino acids and recognize codons via anticodons. Steps: initiation, elongation, termination. Energy: GTP hydrolysis.

Translation Initiation

The first stage of protein synthesis, when a ribosome recognizes messenger RNA, selects the start codon, and assembles the initiator transfer RNA and ribosomal subunits.

Translocation

Movement of a chromosome segment, cell, organ, or substance to a different position.

Transmembrane Transport

Movement of ions, nutrients, drugs, or other molecules across a cell membrane through channels, carriers, pumps, or vesicles, sometimes requiring energy.

Transmission

Passage of an infection, signal, genetic trait, force, or substance from one source to another; in infectious disease, route, dose, and host susceptibility determine risk.

Transmural Infarction

A heart attack involving the full thickness of part of the heart wall.

Transplacental

Passing from the pregnant person's blood across the placenta to the fetus, or in the opposite direction. Antibodies, medicines, chemicals, and some infections can cross the placenta, with effects that depend on the substance and timing.

Transplant

A medical procedure, treatment, device, or care topic related to transplant; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Transplant Immunology

Hyperacute rejection: preformed anti-donor antibodies -> immediate graft failure (minutes-hours); prevented by crossmatch.

Transplant Rejection

Immune-mediated destruction of a transplanted organ or tissue, classified by timing and mechanism. Hyperacute: preformed donor-specific antibodies, minutes to hours post-transplant, immediate graft thrombosis, irreversible; prevention by cross-matching.

Transplant Services

A medical procedure, treatment, device, or care topic related to transplant services; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Transplant Surgery

Transplant surgery involves replacing a failing organ with a healthy donor organ. Kidney transplant: living donors (laparoscopic nephrectomy, left kidney preferred) or deceased donors (standard criteria, expanded criteria, DCD).

Transplantation

Transfer of an organ, tissue, or cells from a donor to a recipient.

Alternate Names:

Transplant

Transplantation Conditioning

A medical procedure, treatment, device, or care topic related to transplantation conditioning; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Transplantation Genetics

A medical procedure, treatment, device, or care topic related to transplantation genetics; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Transplantation Immunology

A medical procedure, treatment, device, or care topic related to transplantation immunology; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Transport

Regulations for shipping radioactive materials.
Package types: excepted, A, B.

Transport Protein

A membrane or soluble protein that carries a molecule or ion across a membrane or through body fluids, often determining absorption, distribution, and cellular concentration.

Transport Vesicle

A transport vesicle is a membrane-bound carrier that moves proteins, lipids, or other cargo between cellular compartments.

Transposition Of Great Vessels

Alternate index label; see *Transposition of the great arteries*.

Transposition Of The Great Arteries

A congenital heart defect in which the aorta arises from the right ventricle and the pulmonary artery from the left ventricle. This creates parallel circulations, so survival after birth depends on adequate mixing through an associated opening or ductus until definitive treatment is provided.

Alternate Names:

Transposition Of Great Vessels

Transposons

Mobile genetic elements that can move between different locations within the bacterial genome or between the chromosome and plasmids. Transposons carry genes including antibiotic resistance genes and can facilitate the spread of resistance.

Transrectal Ultrasonography

Ultrasound imaging of the prostate or nearby structures using a rectal probe.

Alternate Names:

TRUS

Transtracheal

Transtracheal means passing through or relating to the trachea, such as a catheter, oxygen delivery

route, or diagnostic procedure.

Transthyretin Amyloidosis

A progressive, infiltrative systemic amyloidosis caused by the dissociation and misfolding of tetrameric transthyretin into amyloid fibrils that deposit in the myocardium and peripheral nervous system. Occurs in two forms: hereditary (variant ATTR, caused by pathogenic *TTR* mutations such as Val122Ile or Val30Met) and wild-type (senile systemic amyloidosis, predominantly causing restrictive cardiomyopathy in elderly men). Diagnosed by technetium-99m pyrophosphate scintigraphy or tissue biopsy and treated with TTR tetramer stabilizers (such as tafamidis) or TTR-gene-silencing agents.

Alternate Names:

ATTR amyloidosis; ATTR-CM; Transthyretin-related amyloidosis

Transtympanic

Transtympanic means passing through the tympanic membrane, such as medicine delivered into the middle ear or a procedure crossing the eardrum.

Transudate

A transudate is a type of edema fluid with low protein content and few cells, resulting from imbalances in hydrostatic and oncotic pressures across capillary walls rather than from inflammation or increased vascular permeability.

Transudation

Passage of low-protein fluid through a membrane because of pressure or osmotic imbalance.

Transudative Pleural Effusion

From systemic factors altering hydrostatic/oncotic pressures. Most commonly from heart failure. Low protein, low LDH. Treatment addresses the underlying cause.

Transurethral Incision Of The Prostate

An operation using small prostate incisions to relieve obstruction of urine flow.

Alternate Names:

TUIP

Transurethral Microwave Thermotherapy

Heat treatment using microwaves to destroy prostate tissue obstructing urine flow.

Alternate Names:

TUMT

Transurethral Needle Ablation

Treatment using radiofrequency heat delivered by needles to destroy obstructing prostate tissue.

Alternate Names:

TUNA

Transurethral

Performed through the urethra, usually using an instrument to examine, treat, or remove tissue from the urinary tract.

Transurethral Resection

Removal of tissue through an instrument passed along the urethra, commonly for prostate or bladder disease.

Transurethral Resection Of The Prostate

An endoscopic operation that removes obstructing prostate tissue to improve urine flow in selected people with benign prostatic enlargement. Alternatives and indications depend on symptoms, prostate size, complications, and patient preference.

Transvaginal

Performed through the vagina, commonly describing an ultrasound examination or a procedure involving the uterus, cervix, ovaries, or other pelvic structures.

Transvaginal Ultrasound

Transvaginal ultrasound uses a covered probe in the vagina to produce images of the uterus, ovaries, cervix, pregnancy, or nearby pelvic structures.

Transverse

Running across or perpendicular to the long axis or midline.

Transverse Colon

Longest, most mobile colonic segment from hepatic to splenic flexure, approximately 45 cm, with complete mesentery (transverse mesocolon). Crosses anterior to duodenum, pancreas, and major vessels. Splenic flexure is the highest colon point and common gas trapping site. Most common diverticular disease site along with the sigmoid.

Transverse Lie

A fetal lie where the long axis of the fetus is perpendicular to the long axis of the mother. The presenting part is a shoulder or arm. Occurs in approximately 1% of term pregnancies. Risk of cord prolapse and obstructed labor. Cesarean delivery required as vaginal delivery is not possible.

Transverse Myelitis

An inflammatory disorder of the spinal cord causing acute or subacute motor, sensory, and autonomic dysfunction.

Transverse Plane

A horizontal plane that divides the body or an organ into upper (superior) and lower (inferior) portions.

Transverse Vaginal Septum

A congenital obstructive anomaly caused by incomplete canalization and failure of fusion between the caudal end of the paired Müllerian ducts and the cranial evagination of the urogenital sinus. Creates a horizontal partition across the vagina (most commonly at the junction of the upper and middle thirds), which when complete causes primary amenorrhea, cyclic pelvic pain, cryptomenorrhea, hematocolpos, and hematometra upon menarche, treated by surgical septum resection.

Transversus Abdominis Muscle

Deepest flat abdominal muscle with horizontal fibers, from inguinal ligament, iliac crest, thoracolumbar fascia, and inner lower six costal cartilages. Innervated by T7-T12 and L1. Compresses abdomen without trunk movement, providing core stability and increased intra-abdominal pressure during coughing, lifting, Valsalva.

Transvestic Disorder

A paraphilic disorder involving recurrent sexual arousal from cross-dressing that causes marked distress, impairment, or harm, or is acted on without consent. It is distinct from gender dysphoria, which concerns gender identity rather than sexual arousal.

Tranylcypromine

A monoamine oxidase inhibitor antidepressant requiring careful interaction and dietary management.

TRAP Sequence

Twin Reversed Arterial Perfusion sequence, a complication of monochorionic twin gestations where one twin (the pump twin) perfuses the other (the acardiac twin) via arterioarterial anastomoses.

Trapezium (Multangulum Majus)

A carpal bone of the wrist situated on the radial (lateral) aspect of the distal carpal row. Articulates proximally with the scaphoid, medially with the trapezoid, and distally with the first and second metacarpals. Most notably forms the classic saddle-shaped first carpometacarpal joint with the base of the first metacarpal, conferring unique multidirectional mobility and opposition to the thumb. Because of high mechanical joint loading forces during gripping and pinching, the trapeziometacarpal articulation is the most common site of disabling degenerative osteoarthritis in the human hand (basal thumb arthritis / rhizarthrosis), frequently managed with splinting, corticosteroid injections, or surgical trapeziectomy.

Trapezius Muscle

Large triangular muscle from occipital bone and C7-T12 spinous processes to clavicle, acromion, and scapular spine. Innervated by CN XI and C3-C4. Upper fibers elevate scapula, middle fibers retract, lower fibers depress. Essential for shoulder stability and scapulothoracic motion.

Trapezoid

Smallest distal row carpal bone between trapezium and capitate, articulating with the second metacarpal. Wedge shape provides stability for the

firmly fixed second metacarpal. Fractures rare (less than 1 percent of carpal fractures). Contributes to the radial column wrist stability.

Trapezoid Bone

A small distal-row carpal bone between the trapezium and capitate that articulates mainly with the index metacarpal and helps stabilize the radial side of the hand.

Trastuzumab

Trastuzumab is a monoclonal antibody targeting HER2 used for HER2-positive breast, gastric, and other cancers; infusion reactions and potentially reversible cardiomyopathy require cardiac monitoring.

Trauma

Physical injury caused by external force, or the psychological response to a distressing event.

Trauma (Vascular)

Injury to a blood vessel caused by blunt force, penetrating injury, fracture, medical procedures, or other mechanisms.

Alternate Names:

Vascular

Trauma Center

Hospital designated to provide comprehensive care for injured patients. Levels: Level I (complete care, research, prevention), Level II (similar but limited resources), Level III (stabilization, transfer), Level IV (advanced trauma life support, transfer). Transfer criteria: need for higher level of care, specific injuries.

Trauma Surgery

Trauma surgery encompasses the acute care of injured patients, guided by ATLS principles: Airway maintenance with cervical spine protection, Breathing and ventilation, Circulation with hemorrhage control, Disability/neurologic assessment, and Exposure/environmental control.

Traumatic Amputation

Traumatic amputation is loss of a body part from injury and requires hemorrhage control, preservation of the amputated part when possible, urgent surgery, rehabilitation, and psychological support.

Traumatic Bleb

A filtering bleb formed after glaucoma filtration surgery (trabeculectomy) that develops complications from trauma, or a thin-walled, elevated bleb that leaks or is at risk of rupture. Traumatic bleb failure or overfiltering can lead to hypotony, choroidal detachment, and endophthalmitis risk.

Traumatic Brain Injury

Disruption of brain function caused by an external mechanical force. Severity: mild (GCS 13-15, concussion), moderate (GCS 9-12), severe (GCS 3-8). Primary injury: direct damage at moment of impact (contusion, diffuse axonal injury, intracranial hemorrhage).

Alternate Names:

TBI

Traumatic Events And Children

Traumatic events can affect children's safety, behavior, sleep, learning, and emotions; supportive caregivers, stable routines, and professional assessment help identify persistent trauma responses.

Traumatic Grief

Alternate index label; see *Complicated grief*.

Traumatic Injury

A clinical condition or finding involving traumatic injury. Its features, causes, evaluation, and management depend on the clinical context.

Traumatic Injury Of The Bladder And Urethra

Damage to the bladder or urethra caused by an accident, surgery, or penetrating injury. Blood in the urine, inability to urinate, pelvic pain, or shock requires urgent evaluation.

Traumatic Shock

A life-threatening state in which severe injury causes inadequate blood flow and oxygen delivery to tissues.

Traumatic Spondylolisthesis Of The Axis (Hangman'S Fracture)

A bilateral fracture of the pars interarticularis of the C2 vertebra (axis), typically caused by hyperextension of the head. Classically associated with judicial hanging but also occurs in motor vehicle accidents and falls. May present with neck pain, torticollis, or neurological deficits if the spinal cord is compressed.

Alternate Names:

hangman's fracture

Traumatology

The study and treatment of physical injury and its consequences.

Traumeel S

Traumeel S is a multi-ingredient complementary product marketed for pain and inflammation; evidence, formulation, and safety depend on the product and it should not replace needed care.

Travel Sickness (Motion Sickness / Kinetosis)

A common physiological neurosensory mismatch syndrome provoked by unfamiliar real or perceived passive motion during vehicular, sea, or air travel. Pathophysiologically driven by the sensory conflict theory, where vestibular inputs from the semicircular canals and otolith organs conflict directly with visual cues and proprioceptive inputs regarding body position in space. Triggers central autonomic activation in the vestibular nuclei and chemoreceptor trigger zone, manifesting with pallor, cold diaphoresis, gastric awareness, malaise, nausea, and vomiting. Prevented and managed by visual fixation on the horizon, ginger, antihistamines with anticholinergic properties (dimenhydrinate, meclizine), or transdermal scopolamine patches.

Travelers' Diarrhea

Acute diarrhea occurring during or shortly after travel, usually caused by contaminated food or water and commonly involving bacterial, viral, or parasitic infection.

Travoprost

Travoprost is a prostaglandin analogue eye drop that increases uveoscleral outflow and lowers intraocular pressure in glaucoma or ocular hypertension; iris pigmentation and eyelash changes may occur.

Trazodone

Trazodone is a serotonin antagonist and reuptake inhibitor used for depression and sometimes insomnia; sedation, orthostatic hypotension, serotonin syndrome, and rare priapism are important adverse effects.

Trazodone Overdose

Poisoning from taking too much trazodone. It can cause severe sleepiness, vomiting, low blood pressure, seizures, abnormal heart rhythms, or a prolonged painful erection.

Trb Gene

A gene involved in biological processes that vary by species; its precise identity and function must be interpreted from the organism and gene nomenclature used.

Trd Gene

The gene encoding the delta chain of the T-cell receptor in humans, contributing to gamma-delta T-cell antigen recognition.

Treacher Collins Syndrome

An autosomal dominant craniofacial disorder caused by mutations in TCOF1 (most common), POLR1C, or POLR1D genes, encoding ribosomal RNA transcription factors.

Treacle

Treacle is a thick sugar syrup; in older medical usage, related terms referred to antidotes or mixtures, so

context matters.

Treat-To-Target Strategy

A management approach in which treatment is adjusted at regular intervals toward a predefined goal such as remission or low disease activity. It requires validated disease-activity measures, shared decisions, monitoring, and timely escalation or modification.

Treatment

Medical intervention for disease. Includes pharmacological, surgical, lifestyle, supportive.

Treatment Epoch

A treatment epoch is a defined period in a clinical study or treatment plan during which a particular therapy or treatment phase is given.

Treatment-Resistant Depression

Depression that does not improve adequately after appropriate trials of standard treatments. It requires reassessment of diagnosis, adherence, comorbidity, and treatment options.

Trematoda

A class of parasitic flatworms called flukes; human infections include schistosomiasis and liver or lung fluke disease.

Trematosphaeria Grisea

An environmental dematiaceous fungus rarely associated with human infection; clinical importance depends on recovery from a compatible lesion or sterile site.

Tremelimumab

A monoclonal antibody that blocks CTLA-4 and enhances T-cell activity; it is used with durvalumab for selected unresectable hepatocellular carcinoma and other cancers, with immune-related adverse effects possible.

Tremor

An involuntary, rhythmic shaking of a body part.

It may occur at rest, while holding a position, or during movement and can result from medicines, caffeine, anxiety, essential tremor, Parkinson disease, or other neurologic or metabolic conditions.

Trench Fever

A louse-borne infection caused by *Bartonella quintana*, producing recurring fever and bone pain.

Trench Foot

Cold and wet injury to the feet causing numbness, swelling, pain, and tissue damage.

Trench Mouth

Trench mouth is acute necrotizing ulcerative gingivitis causing painful bleeding gums, foul breath, and ulceration, often associated with plaque, smoking, stress, or poor nutrition.

Trendelenberg Position

A position in which the patient lies supine with the feet higher than the head; clinical uses and risks depend on context.

Trendelenburg

Trendelenburg position places the body supine with the feet higher than the head; its use is procedure-specific and may affect breathing, pressure, and circulation.

Trendelenburg Position

A supine position in which the head and upper body are lower than the feet.

Treosulfan

An alkylating chemotherapy drug used in conditioning before allogeneic hematopoietic stem-cell transplantation; infection, mucositis, infertility, and organ toxicities may occur.

Trepanning

Drilling or cutting a circular opening in bone, especially the skull.

Trephine

A circular cutting instrument used to remove a disk of tissue or bone, such as during a bone-marrow biopsy or corneal procedure.

Trephining

Another term for trepanning, creating a circular opening in bone.

Trephining (Trepanation)

A surgical procedure in which a hole is drilled or scraped into the skull (cranium), exposing the dura mater. Historically, trephining dates back to Neolithic times (10,000 BCE) and was performed for ritual, therapeutic (releasing 'evil spirits'), and post-traumatic indications.

Alternate Names:

Trepanation

Treponema

A genus of spiral bacteria including *T. pallidum*, the cause of syphilis.

Treponema Pallidum

A spirochete that causes syphilis, a sexually transmitted infection. The organism is transmitted through sexual contact or transplacentally (congenital syphilis). Primary syphilis presents with a painless genital ulcer (chancre). Secondary syphilis presents with rash, mucous patches, and condylomata lata.

Treponema Pallidum Infection

Infection with the bacterium *Treponema pallidum*, which causes syphilis and can affect the skin, nervous system, cardiovascular system, and other organs.

Treponemal Infections

An infectious process or toxic exposure involving treponemal infections; diagnosis considers the cause, route of exposure, affected tissues, severity, and host response.

Trepopnea

Dyspnea that occurs in one lateral decubitus posi-

tion but not the other. It is typically worse when lying on the side of the affected lung. It may be caused by unilateral lung disease, pleural effusion, or mediastinal mass.

Tretinoin

Tretinoin is all-trans retinoic acid used topically for acne and photoaging and orally with arsenic or chemotherapy for acute promyelocytic leukaemia; skin irritation and systemic differentiation syndrome are important risks.

TRG Gene

A gene encoding the gamma chain of the T-cell receptor, contributing to gamma-delta T-cell development and antigen recognition.

Triiodothyronine

The thyroid hormone T3, which contains three iodine atoms and has stronger biological activity than T4. It helps regulate metabolism, heart rate, temperature, and growth.

Triage

Sorting patients by urgency and resource needs so those at greatest risk receive timely care.

Trial Indication

The disease, condition, symptom, or clinical purpose for which an investigational intervention is being studied in a clinical trial.

Trial Of Labor After Cesarean

An attempt to give birth vaginally after a previous cesarean delivery. Whether it is appropriate depends on the prior uterine incision, obstetric history, current pregnancy, facility resources, and the person's preferences. Counseling includes the possibility of successful VBAC, repeat cesarean delivery, and uterine rupture.

Trial Screening

The process of checking potential participants against a clinical trial's eligibility criteria, including medical history, examination, laboratory tests, and

informed consent.

Triamcinolone

A synthetic glucocorticoid used topically, by injection, or in other formulations for inflammation.

Triamcinolone Acetonide

A synthetic corticosteroid used topically, by injection, or in other preparations to reduce inflammation; prolonged or extensive use can cause local and systemic steroid effects.

Triamterene

Triamterene is a potassium-sparing diuretic that blocks epithelial sodium channels in the collecting duct and is used for hypertension or oedema; hyperkalaemia and kidney stones are important risks.

Triazenes

A group of chemical compounds that includes some anticancer agents; their effects and toxicity depend on the specific compound.

Triazine

A chemical ring structure found in some medicines, herbicides, and industrial compounds; health effects depend on the specific substance and exposure.

Triaziquone

An older alkylating anticancer drug that cross-links DNA; its clinical use is limited by toxicity and availability.

Triazolam

Triazolam is a short-acting benzodiazepine hypnotic for short-term insomnia; amnesia, falls, dependence, respiratory depression, and dangerous interactions with opioids or other sedatives can occur.

Triazole

A five-membered nitrogen-containing chemical ring found in several antifungal medicines and other drugs; clinical effects depend on the specific compound.

Triazophos

An organophosphate insecticide that can inhibit acetylcholinesterase and cause cholinergic poisoning after significant exposure.

Tributylin

A triglyceride that releases the short-chain fatty acid butyrate during digestion; it is studied experimentally for intestinal and metabolic effects.

Triceps

A three-headed muscle, especially the triceps brachii that extends the elbow.

Triceps Brachii Muscle

Three-headed posterior arm muscle. Long head from infraglenoid tubercle, lateral head from posterior humerus above radial groove, medial head below. All insert on the olecranon. Innervated by radial nerve (C7-C8). Primary elbow extensor; assists shoulder extension/adduction.

Trichiasis

Inward misdirection of eyelashes so they rub against the eye.

Trichilemmal Carcinoma

A rare malignant follicular tumor showing outer-root-sheath differentiation, usually occurring on sun-damaged skin of older adults as a papule or nodule; complete excision is generally required.

Trichilemmoma

A benign follicular tumor with outer-root-sheath differentiation, usually presenting as a small facial papule; multiple lesions may be associated with Cowden syndrome.

Trichiniasis

An infection caused by *Trichinella* larvae, acquired from undercooked meat.

Trichinosis

Infection with *Trichinella* worms, usually acquired by eating undercooked infected pork or wild game.

It may cause diarrhea, fever, muscle pain, and swelling around the eyes.

Trichobezoar

A gastrointestinal mass composed mainly of swallowed hair, usually located in the stomach.

Trichoblastoma

A rare benign follicular tumor of the scalp or face, usually presenting as a solitary dermal or subcutaneous nodule; it can resemble basal-cell carcinoma and is diagnosed histologically.

Trichoepithelioma

A rare benign follicular adnexal tumor that usually presents as a skin-coloured papule or nodule on the face or scalp; it can resemble basal-cell carcinoma and may require histopathologic confirmation.

Trichomonas

A flagellated protozoan, *Trichomonas vaginalis*, that infects the urogenital tract and causes trichomoniasis. It is transmitted mainly through sexual contact and may cause discharge, irritation, or no symptoms.

Trichomonas Vaginalis

A flagellated protozoan that causes trichomoniasis, a sexually transmitted infection.

Trichomoniasis

A sexually transmitted infection caused by the parasite *Trichomonas vaginalis*. It may cause genital irritation and discharge or no symptoms; testing and treatment of partners help prevent reinfection.

Trichophyton

A genus of dermatophyte fungi causing infections of skin, hair, and nails.

Trichorrhexis Nodosa

A hair-shaft defect in which weakened points split into brush-like ends, caused by inherited disease, trauma, chemicals, heat, or nutritional or metabolic disorders.

Trichothiodystrophy Syndrome

A group of inherited disorders characterized by brittle sulfur-deficient hair with a distinctive banded appearance, often accompanied by photosensitivity, ichthyosis, developmental delay, short stature, or recurrent infection.

Trichotillomania

Recurrent compulsive pulling out of one's hair, causing hair loss and distress or impairment.

Trichuriasis

Intestinal infection caused by the whipworm *Trichuris trichiura*, usually acquired by swallowing eggs from contaminated soil or food. Heavy infection can cause abdominal pain, diarrhea, anemia, or rectal prolapse.

Tricuspid Annular Plane Systolic Excursion

A quantitative echocardiographic measurement used as the primary, standard bedside index of right ventricular longitudinal systolic function. Obtained using M-mode echocardiography aligned along the lateral tricuspid annulus in the apical four-chamber view, it measures the maximum total systolic displacement of the tricuspid annulus toward the RV apex in millimeters. A TAPSE value of < 17 mm indicates right ventricular systolic dysfunction and carries strong prognostic significance in pulmonary hypertension, heart failure, and acute pulmonary embolism.

Alternate Names:

TAPSE; Tricuspid Annular Excursion; RV Longitudinal Systolic Function

Tricuspid Atresia

Absence of the tricuspid valve, resulting in no direct communication between the right atrium and right ventricle. Blood flow depends on an atrial septal defect and ventricular septal defect for mixing. Presents with cyanosis.

Tricuspid Incompetence

Failure of the tricuspid valve to close completely, causing regurgitation into the right atrium.

Tricuspid Regurgitation

Incomplete closure of the tricuspid valve allowing blood to flow backward from the right ventricle to the right atrium during systole. Causes include right ventricular dilation (from pulmonary hypertension or left heart failure), endocarditis, rheumatic disease, and carcinoid syndrome.

Tricuspid Stenosis

Narrowing of the tricuspid valve opening, obstructing flow from right atrium to right ventricle.

Tricuspid Valve

The right atrioventricular heart valve between the right atrium and right ventricle.

Tricyclic Antidepressant Drugs

Antidepressants that inhibit norepinephrine and serotonin reuptake and can cause anticholinergic, cardiac, and toxic effects.

Tricyclic Antidepressants

Older antidepressants that non-selectively inhibit the reuptake of serotonin and norepinephrine. They include amitriptyline, nortriptyline, imipramine, clomipramine, and desipramine.

Trifluoperazine

A phenothiazine antipsychotic used for selected psychotic disorders. It can cause drowsiness, movement disorders, and other serious effects, so treatment requires medical supervision.

Trigeminal Nerve

Cranial nerve V, carrying facial sensation and motor supply to muscles of mastication.

Trigeminal Neuralgia

Brief, severe, electric-shock-like facial pain in one or more trigeminal-nerve divisions.

Alternate Names:

Neuralgia, Trigeminal Tic Douloureux (Trigeminal Neuralgia)

Trigger

A trigger is an event, exposure, stimulus, or condition that initiates or worsens a symptom, disease episode, behavior, or physiologic response.

Trigger Finger

Stenosing tenosynovitis causing painful catching or locking of a finger during movement.

Alternate Names:

Stenosing Tenosynovitis Trigger Thumb (Trigger Finger)

Trigger Thumb

Stenosing tenosynovitis causing painful catching or locking of the thumb.

Trigger Thumb (Trigger Finger)

Alternate index label for Trigger Finger.

Triglyceride

A lipid composed of glycerol and three fatty acids, transported in blood and stored in adipose tissue.

Triglyceride Level

A triglyceride level measures a type of fat in the blood; persistently high levels can increase cardiovascular risk and, when very high, cause pancreatitis.

Triglycerides

Triglycerides are blood fats made of glycerol and three fatty acids; elevated levels contribute to cardiovascular risk and, when very high, pancreatitis risk.

Trigone

A triangular area, especially the smooth region at the base of the urinary bladder.

Trimester

One of the three approximately three-month periods of pregnancy.

Trimethoprim

An antibacterial medicine that inhibits bacterial folate metabolism, often combined with sulfamethox-

azole.

Trimethoprim-Sulfamethoxazole

A fixed-dose combination of trimethoprim and sulfamethoxazole that synergistically inhibits folate synthesis. TMP-SMX is indicated for urinary tract infections, *Pneumocystis jirovecii* pneumonia, and community-acquired pneumonia. The combination is bactericidal against many gram-positive and gram-negative organisms.

Trimethylaminuria

A metabolic disorder in which impaired oxidation of trimethylamine causes a characteristic fish-like body odor in sweat, urine, and breath. It may be inherited or acquired and can cause substantial psychosocial distress despite limited physical illness.

Trimipramine

A sedating tricyclic antidepressant used for depression in some countries. Its anticholinergic and cardiovascular effects can cause dry mouth, constipation, blurred vision, drowsiness, orthostatic hypotension, or dangerous toxicity in overdose; interactions and suicide risk require clinical monitoring.

Trinitrin

Another name for glyceryl trinitrate or nitroglycerin, a nitrate vasodilator.

Trinitrophenol

Picric acid, a corrosive and explosive chemical historically used in medicine and industry.

Trinucleotide Repeat

Expansion of three-nucleotide sequences causes diseases: Huntington, Fragile X (CGG), myotonic dystrophy (CTG), Friedreich ataxia (GAA). Anticipation common.

Trip

A trip is a journey or an accidental stumble; in drug-related language it can also mean an episode of altered perception after a hallucinogen.

Tripe Palms

Diffuse, velvety, thickened hyperkeratosis of the palmar skin with a cerebriform, corrugated appearance resembling the lining of a cow's stomach (tripe), which is strongly associated with underlying malignancy, most commonly gastric adenocarcinoma. It may occur alone or in conjunction with acanthosis nigricans.

Triple X Syndrome

A sex chromosome aneuploidy characterized by the presence of an additional X chromosome in females, resulting in a 47,XXX karyotype, arising from non-disjunction during maternal meiosis I or II. Phenotypically, affected females are typically tall with normal sexual development and preserved fertility; clinical features may include mild motor, speech, and language delays, auditory processing difficulties, increased prevalence of learning disabilities and anxiety, mild epicanthal folds, and clinodactyly.

Alternate Names:

Trisomy X; 47,XXX Syndrome

Triple-Negative Breast Cancer

Breast cancer lacking ER, PR, HER2. Aggressive, limited treatment options.

Triplets

Three offspring born from the same pregnancy or three infants born in a multiple birth.

Triploidy Syndrome

A chromosomal condition in which cells contain three complete haploid chromosome sets. It causes severe fetal growth restriction and multiple congenital anomalies and is usually incompatible with prolonged survival; partial or mosaic forms vary.

Triptan

A class of medications used for acute migraine treatment, acting as 5-HT_{1B}/1D receptor agonists. 5-HT_{1B} receptors are located on cranial blood vessels; activation causes vasoconstriction of dilated meningeal arteries.

Triptans (Triptan)

Triptans are serotonin-receptor agonists used to abort migraine attacks by constricting cranial vessels and reducing trigeminal signaling.

Triquetrum

Pyramidal proximal row carpal bone on the ulnar side, articulating with lunate, hamate, pisiform, and TFCC. Second most commonly fractured carpal bone. Contributes to proximal carpal row and ulnar wrist stability. Pisiform articulates with its volar surface.

Trismus

Restricted ability to open the mouth, commonly caused by tetanus, dental infection, or muscle spasm.

Trisodium Phosphate Poisoning

Poisoning from swallowing or contacting trisodium phosphate, a strong alkaline cleaner. It can burn the mouth, throat, eyes, and skin and may cause vomiting or breathing problems.

Trisomy

Presence of three copies of a chromosome instead of two. Trisomy 21 (Down syndrome): intellectual disability, characteristic facies, heart defects. Trisomy 18 (Edwards): severe intellectual disability, multiple anomalies. Trisomy 13 (Patau): severe intellectual disability, holoprosencephaly, cleft lip/palate.

Trisomy 13

Patau syndrome, a chromosomal disorder caused by an extra chromosome 13 and associated with severe developmental, brain, facial, heart, kidney, and limb abnormalities.

Trisomy 18

Edwards syndrome, a chromosomal disorder caused by an extra chromosome 18 and associated with growth restriction, characteristic craniofacial and limb findings, heart disease, and severe developmental impairment.

Trisomy 18 (Edwards Syndrome)

Alternate index label for Edwards Syndrome.

Trisomy 21

Down syndrome: the most common chromosomal disorder, caused by an extra chromosome 21 (trisomy 21; 95 percent meiotic nondisjunction, mosaicism, translocation).

Trisomy 21 (Down Syndrome Overview)

Alternate index label for Down Syndrome Overview.

Trisomy 21 Syndrome

Alternate index label; see *Down syndrome*.

Trisomy X

Alternate index label; see *Triple X syndrome*.

Trocar

A pointed instrument in a cannula used to enter a body cavity or tissue.

Trochanter

A bony prominence on the femur, especially the greater or lesser trochanter.

Trochanteric Bursitis (Hip Bursitis)

Alternate index label for Hip Bursitis.

Troche

A troche is a small medicated lozenge that dissolves slowly in the mouth or throat for local or systemic drug delivery.

Trochlear Nerve

Cranial nerve IV, supplying the superior oblique eye muscle.

Trophic

Relating to the nutrition, growth, development, or maintenance of tissue. Trophic changes may occur when nerve supply or blood flow is impaired.

Trophoblast

The outer layer of the early embryo that contributes to implantation and placenta formation.

Trophoderm

The outer cellular layer of a blastocyst that contributes to formation of the placenta.

Trophozoite

The active, feeding, and multiplying stage of a protozoan parasite.

Tropical Chronic Pancreatitis

A historically recognized form of chronic pancreatitis reported mainly in tropical and subtropical regions, often beginning at a younger age and associated with recurrent abdominal pain, pancreatic duct stones, and diabetes. Its causes are multifactorial and may include genetic susceptibility, nutritional factors, and environmental exposures; the term should not be applied solely because a patient lives in a tropical region.

Tropical Diseases

Diseases more common in tropical or subtropical regions, including malaria, dengue, and schistosomiasis.

Tropical Medicine

The specialty concerned with diseases prevalent in tropical and subtropical regions and travel-related illness.

Tropical Oil

Tropical oils are plant-derived oils from warm regions, such as coconut or palm oil; their nutritional effects depend on the specific oil and amount consumed.

Tropical Sprue

Chronic diarrhea and malabsorption syndrome of uncertain etiology occurring predominantly in tropical and subtropical regions. The condition is characterized by villous atrophy of the small intestinal mucosa leading to malabsorption of nutrients in-

cluding folate, iron, and vitamin B12.

Tropical Ulcer

A chronic ulcer occurring in tropical regions, often associated with infection, trauma, or poor wound healing.

Troponin

Cardiac-specific troponin I and T, the preferred biomarkers for myocardial injury. Highly sensitive and specific. Rise 2-4 hours after symptom onset, peak at 12-24 hours, remain elevated for 7-10 days. High-sensitivity troponin (hs-cTn) allows earlier detection (1-2 hours).

Troponin Test

A troponin test measures heart-muscle proteins in the blood and helps assess possible heart injury, especially during suspected heart attack.

Trouble Swallowing

Alternate index label; see *Dysphagia*.

Trousseau Sign Of Latent Tetany

A sensitive and specific clinical sign of hypocalcemia. Elicited by inflating a blood pressure cuff on the upper arm to a pressure greater than systemic systolic blood pressure for 3 minutes.

Trousseau Syndrome

Recurrent or migratory venous thrombosis associated with active malignancy, particularly mucin-producing adenocarcinoma. It reflects tumor-associated activation of coagulation and is distinct from Trousseau sign of latent tetany.

Trp

Trp is the standard abbreviation for tryptophan, an essential amino acid used in protein synthesis and serotonin and niacin pathways.

True Ribs

Ribs 1-7 articulating directly with the sternum via their own costal cartilages. Each has a head, neck, tubercle, and shaft. The first two are short and

strongly curved for neck/shoulder muscle attachment. Costal cartilages provide thoracic cage elasticity for respiratory expansion.

Truncation

A shortened or incomplete form caused by removal of part of a sequence, structure, chromosome, protein, or process; the effect depends on what was lost and where.

Truncus Arteriosus

A congenital heart defect where a single large vessel arises from both ventricles, receiving a mixed blood supply. Associated with a large ventricular septal defect and abnormal semilunar valve. Presents with cyanosis and heart failure early in life.

Trunk

The central part of the body excluding the head and limbs, or a major nerve or vessel trunk.

Truss

A supportive appliance used historically for a hernia; definitive management depends on the type and complications.

Trypanosoma

A genus of protozoan parasites causing African and American trypanosomiasis.

Trypanosoma Brucei

A protozoan parasite transmitted by tsetse flies that causes African trypanosomiasis, or sleeping sickness. The human-infecting subspecies differ in geographic distribution and typical disease course; infection can progress from fever and lymph-node enlargement to neurologic involvement.

Trypanosomiasis

Disease caused by Trypanosoma parasites, including African sleeping sickness and Chagas disease.

Trypanosomiasis, American

Alternate index label; see *Chagas disease*.

Trypsin

A pancreatic proteolytic enzyme secreted as trypsinogen and activated in the small intestine, where it helps digest dietary proteins and activates other pancreatic enzymes.

Trypsin And Chymotrypsin In Stool

Stool trypsin and chymotrypsin testing measures digestive enzymes and may help assess pancreatic enzyme production, although other tests are often preferred.

Trypsinogen Test

A trypsinogen test measures a pancreatic enzyme precursor and may help evaluate pancreatic function or newborn screening for cystic fibrosis.

Tryptophan

An essential amino acid and precursor of serotonin, melatonin, and niacin.

Tsetse Fly

A biting fly of the *Glossina* genus that transmits African trypanosomes.

TSH

Thyroid-stimulating hormone, a pituitary hormone that stimulates thyroid hormone production; its blood concentration is the main initial test for many thyroid disorders.

TSH Test

A TSH test measures thyroid-stimulating hormone in the blood and is a common initial test of thyroid function.

TSH-Receptor Antibody

An antibody that binds the thyrotropin receptor. Stimulating antibodies cause Graves disease, while blocking antibodies can cause hypothyroidism; measurement supports diagnosis and can help assess fetal or neonatal risk in pregnancy.

TSI Test

A TSI test detects thyroid-stimulating immunoglob-

ulins and can support diagnosis or monitoring of Graves disease.

Tsutsugamushi Fever

Scrub typhus, a mite-borne infection caused by *Orientia tsutsugamushi*.

Tubal Ligation

Surgical sterilization by occluding the fallopian tubes, preventing the ovum from reaching the uterus. Postpartum tubal ligation: after vaginal or cesarean delivery (Pomeroy technique, Parkland). Interval: laparoscopic (bands, clips, bipolar cautery) or mini-laparotomy (Hulka or Filshie clip).

Tubal Ligation Reversal

Microsurgical reconnection of previously divided or blocked fallopian-tube segments to restore a possible path for sperm and egg; success depends on remaining tube length, age, scarring, and original method.

Tubal Occlusion Procedure, Selective

Selective tubal occlusion blocks a fallopian tube to prevent sperm and egg from meeting and is a permanent contraceptive procedure.

Tubal Pregnancy

An ectopic pregnancy that implants in a fallopian tube instead of the uterus. It can rupture and cause internal bleeding; one-sided pain, vaginal bleeding, dizziness, or fainting needs urgent evaluation.

Tuber

A rounded swelling or growth, often an anatomical prominence or a plant storage organ; clinical meaning depends on the site and context.

Tubercle

A small rounded projection or nodule; it may refer to bone anatomy or a tuberculous lesion.

Tuberculide

A hypersensitivity skin eruption associated with underlying tuberculosis, usually without organisms

in the lesions.

Tuberculin

Protein material from *Mycobacterium tuberculosis* used in skin testing for immune sensitization.

Tuberculin Skin Test

Intradermal injection of purified protein derivative to test for *Mycobacterium tuberculosis* infection. Read at 48-72 hours measuring induration (not erythema). Positive: greater than 5 mm in HIV or close contacts; greater than 10 mm in high-risk groups; greater than 15 mm in low-risk populations.

Tuberculin Test

A skin test that assesses whether the immune system has reacted to tuberculosis proteins.

Tuberculoid Leprosy

The paucibacillary pole of the leprosy spectrum characterised by strong cell-mediated immunity to *Mycobacterium leprae*, resulting in well-formed granulomas and few or no organisms in tissue.

Tuberculosis

An infectious disease caused by *Mycobacterium tuberculosis*, most often affecting the lungs.

Alternate Names:

TB

Tuberculosis In Children

Infection with *Mycobacterium tuberculosis*, often from household contacts. Latent TB infection (LTBI) positive tuberculin skin test or interferon-gamma release assay with no symptoms or radiographic findings. Active TB presents with cough, fever, weight loss, and lymphadenopathy. Diagnosis by PPD, chest X-ray, and sputum culture.

Tuberculosis, MDR

Multidrug-resistant tuberculosis is TB caused by bacteria resistant to at least isoniazid and rifampin, requiring specialized prolonged treatment.

Tuberculous Meningitis

The most severe, life-threatening form of extrapulmonary tuberculosis, resulting from hematogenous dissemination of *Mycobacterium tuberculosis* and the rupture of subependymal Rich foci into the subarachnoid space. Causes thick gelatinous basilar exudate, cranial nerve palsies, communicating hydrocephalus, and cerebral vasculitis with stroke.

Tuberous Sclerosis

A rare genetic disorder, also called tuberous sclerosis complex, that causes growths in the brain and other organs, including the skin, kidneys, heart, and lungs. It may cause seizures, developmental or behavioral problems, and skin changes; the severity varies widely.

Tuberous Sclerosis Complex

An autosomal dominant genetic disorder caused by mutations in TSC1 (hamartin) or TSC2 (tuberin) genes, leading to mTOR pathway activation. Features: cortical tubers, subependymal nodules, cardiac rhabdomyomas, angiomyolipomas, facial angiofibromas, and seizures (often infantile spasms).

Tubes

Hollow structures that carry air, fluid, food, or instruments; the clinical meaning depends on the specific tubes being described.

Tubes Ear Problems (Eustachian Tube Problems)

Alternate index label for Eustachian Tube Problems.

Tubo-Ovarian Abscess

A complex infected mass involving the fallopian tube and ovary, usually arising as a complication of pelvic inflammatory disease. It may cause pelvic pain, fever, and sepsis and requires prompt antimicrobial and sometimes drainage treatment.

Tubocurarine

A historical muscle-paralyzing medicine derived from curare. It blocks nerve signals at skeletal-muscle junctions, including signals needed for breathing, but does not cause unconsciousness or re-

lieve pain; safer modern drugs have largely replaced it.

Tubography

Contrast imaging of the fallopian tubes, usually as part of hysterosalpingography, to see whether they are open. It may help assess infertility but can cause temporary cramping, bleeding, infection, or rarely injury.

Tubular Adenoma

A benign glandular polyp, most often in the colon, that can develop dysplastic changes and is considered a precursor lesion for some colorectal cancers.

Tubular Reabsorption

The process by which the renal tubules return filtered water and solutes from the tubular lumen to the blood, reclaiming 99 percent of the glomerular filtrate. Proximal tubule: reabsorbs 65 percent of sodium, most glucose, amino acids, water, bicarbonate, phosphate, and urate.

Tubular Secretion

The transport of substances from the peritubular capillary blood into the tubular lumen, a second mechanism (besides filtration) by which the kidney eliminates waste and regulates ions.

Tubule

A small tube or tubular structure, such as a renal tubule.

Tubuloglomerular Feedback

A feedback mechanism at the juxtaglomerular apparatus that adjusts GFR based on tubular fluid NaCl concentration. The macula densa cells (specialized tubular epithelial cells in the thick ascending limb) sense NaCl concentration via the NKCC2 cotransporter.

Tularaemia

A zoonotic infection caused by *Francisella tularensis*, acquired from animals, ticks, contaminated water, or aerosols.

Tularemia

Tularemia is infection with *Francisella tularensis* acquired from ticks, animals, contaminated water, or aerosols and causing ulcer-glandular, pulmonary, or other disease.

Tularemia Blood Test

A tularemia blood test looks for antibodies or other evidence of infection with *Francisella tularensis*; antibodies may not appear early in illness.

Tulle Gras

A paraffin-impregnated gauze dressing used to cover wounds and burns.

Tumescence

Swelling or enlargement caused by engorgement with blood, fluid, or tissue growth.

Tumor

A tumor is an abnormal mass of cells caused by excessive or uncontrolled growth; it may be benign or malignant.

Tumor Adrenal Gland (Pheochromocytoma)

Alternate index label for Pheochromocytoma.

Tumor Board

A multidisciplinary conference of oncologists, surgeons, radiologists, and pathologists who review individual cancer cases to coordinate diagnosis and treatment planning.

Tumor Brain Cancer (Brain Cancer)

Alternate index label for Brain Cancer.

Tumor Burden

The total amount of cancer in the body, estimated from the number, size, and extent of tumor deposits; higher burden is generally associated with worse prognosis.

Tumor Differentiation

The degree to which neoplastic cells morphologically

and functionally resemble their normal counterparts of origin.

Tumor Embolism

Tumor embolism occurs when malignant cells, tumor fragments, or tumor thrombi enter the venous circulation and travel to distant sites. The pulmonary vasculature is the most common site of embolization, as venous blood from most organs drains to the lungs through the pulmonary arteries.

Tumor Grade

A histologic measure of how much tumor cells resemble normal tissue; low-grade (well-differentiated) tumors behave less aggressively than high-grade (poorly differentiated) tumors. Grading is a prognostic, not staging, feature.

Tumor Grading

A description of a tumor based on how abnormal the cancer cells look under a microscope compared to normal cells; a lower grade indicates slower-growing cancer.

Tumor Heterogeneity

The presence of genetically, epigenetically, or phenotypically distinct subpopulations within the same tumor or between primary and metastatic sites. Heterogeneity can influence prognosis, sampling, treatment response, and resistance.

Tumor Lysis Syndrome

Metabolic emergency from rapid cell death. Hyperuricemia, hyperkalemia, hyperphosphatemia, hypocalcemia. Prevention: hydration, allopurinol, rasburicase.

Tumor Marker

Substance produced by cancer or body in response to cancer. Used for diagnosis, monitoring, prognosis. Examples: PSA (prostate), CA-125 (ovarian), CEA (colon), AFP (liver), CA 19-9 (pancreas).

Tumor Marker, CEA

CEA is a blood tumor marker used mainly to help

monitor some colorectal and other cancers; it is not specific enough to diagnose cancer alone.

Tumor Marker, NSE

NSE is a protein marker that may rise in some neuroendocrine tumors or nerve-cell injury but is not diagnostic by itself.

Tumor Microenvironment

Surrounding tissue influencing tumor growth. Includes immune cells, blood vessels, fibroblasts, signaling molecules.

Tumor Mutational Burden

The number of somatic mutations per unit of coding DNA in a tumor specimen. TMB may correlate with production of neoantigens and response to some immune checkpoint inhibitors, but interpretation requires a validated assay and disease-specific context.

Tumor Necrosis Factor

A proinflammatory cytokine (TNF-alpha) produced mainly by macrophages, acting on TNFR1/TNFR2 receptors. Functions: activation of endothelium and leukocytes, fever, cachexia, apoptosis, and coordination of inflammation and the acute-phase response.

Tumor Necrosis Factor Receptor-Associated Periodic Syndrome

A rare inherited autoinflammatory disorder caused by variants in the TNFRSF1A gene, producing recurrent prolonged fever, muscle pain, abdominal pain, rash, and inflammation; it is also called TRAPS.

Tumor Staging

A system of describing the size of a cancer and how far it has spread throughout the body from its original location (e.g., using the TNM system).

Tumor Suppressor

A gene or protein that helps limit cell growth, repair DNA damage, or promote removal of abnormal cells. Loss of its function can contribute to cancer.

Tumor Suppressor Gene

Gene that prevents uncontrolled cell growth. Examples: TP53, RB1, BRCA1, BRCA2, APC. Loss of function contributes to cancer.

Tumor Testicle (Testicular Disorders)

Alternate index label for Testicular Disorders.

Tumor, Salivary Gland

Alternate index label; see *Salivary gland tumors*.

Tumor, Spinal

Alternate index label; see *Spinal tumor and spinal mass*.

Tumor-Suppressor Gene

A gene whose normal product restrains cell proliferation, promotes DNA repair, induces apoptosis, or maintains genomic stability. Loss or inactivation of both functional copies is common in the development of many cancers, although mechanisms vary by gene.

Tumors Uterine (Uterine Fibroids)

Alternate index label for Uterine Fibroids.

Tumour

A mass produced by abnormal cell growth; it may be benign or malignant.

Tumour (Neoplasm)

An abnormal mass of tissue produced by uncontrolled or inappropriate cell growth; it may be benign or malignant.

Alternate Names:

Neoplasm

Tungiasis

A skin infestation caused by the sand flea *Tunga* penetrans, producing an itchy or painful lesion, usually on the feet.

Tunica Adventitia

The outer connective-tissue layer of a blood-vessel wall. It contains collagen and elastic fibers that

provide structural support, along with nerves and, in larger vessels, small vessels called vasa vasorum that supply the outer wall. It is relatively prominent in veins and helps anchor vessels to surrounding tissues.

Tunica Media

The middle layer of a blood-vessel wall, located between the tunica intima and tunica adventitia. It consists primarily of smooth muscle and elastic fibers; its contraction or relaxation changes vascular diameter, regulating blood pressure and tissue perfusion. It is thickest in arteries and thinner in veins.

Tunnel Vision

Loss of peripheral visual field leaving a narrowed central field, caused by glaucoma, retinal disease, optic-nerve or brain lesions, severe hypoxia, or other conditions.

Turbinate Surgery

A medical procedure, treatment, device, or care topic related to turbinate surgery; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Turbinates

Curved bony shelves on the lateral walls of the nasal cavity, covered by vascular mucosa. They increase surface area and help warm, humidify, and filter inhaled air.

Turbinates Hypertrophy

Chronic enlargement of the nasal turbinates, most commonly the inferior turbinates, resulting in nasal airway obstruction, which can be caused by chronic inflammation from allergic rhinitis, non-allergic rhinitis, chronic sinusitis, hormonal changes, or vasomotor dysfunction.

Turcot Syndrome

A clinical descriptor for the rare concurrent occurrence of primary colorectal polyposis or colorectal cancer and primary central nervous system neoplasms. Genetically classified into two distinct

molecular entities: Type 1 Turcot syndrome, associated with hereditary nonpolyposis colorectal cancer (Lynch syndrome) carrying germline DNA mismatch repair gene mutations (*MLH1*, *PMS2*, *MSH2*) and predisposing to high-grade glioblastoma multiforme; and Type 2 Turcot syndrome, associated with familial adenomatous polyposis caused by germline mutations in the *APC* gene and predisposing to cerebellar medulloblastomas.

Alternate Names:

Brain Tumor-Polyposis Syndrome

Turgor

The firmness of tissue produced by fluid pressure within cells; skin turgor can help assess dehydration.

Turmeric

Turmeric is a plant spice containing curcuminoids and used in foods and supplements; evidence varies by condition and interactions or liver injury are possible.

Turner Hypoplasia

Enamel defect on permanent tooth from infection of overlying primary tooth. Localized hypoplasia.

Turner Syndrome

A chromosomal condition, most commonly with a 45,X karyotype but also with mosaic or structurally abnormal X chromosomes. It typically causes short stature and gonadal dysgenesis with ovarian insufficiency; associated findings include webbed neck, broad chest, bicuspid aortic valve, coarctation of the aorta, renal anomalies, hearing problems, and autoimmune thyroid disease.

Turner'S Syndrome

A chromosomal condition, usually involving complete or partial monosomy X, that typically affects phenotypic females. Common features include short stature, ovarian insufficiency with delayed or absent puberty and infertility, webbed neck, broad chest, congenital heart defects such as bicuspid aortic valve or coarctation of the aorta, and increased risk of autoimmune thyroid disease.

Turning Patients Over In Bed

Repositioning a patient at planned intervals to protect skin, improve comfort and breathing, prevent pressure injuries and contractures, and reduce complications of immobility.

Turpentine Oil Poisoning

Poisoning from swallowing or inhaling turpentine oil. It can irritate or damage the lungs, stomach, and nervous system and may cause chemical pneumonia.

Tussis

Cough; a forceful reflex expiration that helps clear secretions, irritants, or foreign material from the airways.

TVS

TVS commonly means transvaginal sonography, an ultrasound examination using a probe in the vagina to assess pelvic organs and early pregnancy.

Twilight Sleep

Twilight sleep is an older term for a sedated, partly conscious state used during procedures; modern analgesia and anesthesia specify the medicines and monitoring used.

Twin

A twin is one of two offspring from the same pregnancy, either identical from one fertilized egg or fraternal from two eggs and two sperm.

Twin Placenta

Placental anatomy in a twin pregnancy, classified by chorionicity and amnionicity; these features determine risks such as twin-to-twin transfusion and guide antenatal monitoring.

Twin Pregnancy

A pregnancy involving two fetuses. Clinical risk depends on chorionicity and amnionicity, with increased risks including preterm birth, fetal growth problems, and hypertensive disorders.

Twin-To-Twin Transfusion Syndrome

A complication of monochorionic twin pregnancy in which shared placental vessels cause unequal blood flow between the twins.

Twin-Twin Transfusion Syndrome

A complication of monochorionic diamniotic twin pregnancies where placental vascular anastomoses (arteriovenous connections) cause unbalanced blood flow from one twin (the donor) to the other (the recipient). The donor twin becomes hypovolemic, anemic, and growth-restricted with oligohydramnios ("stuck twin").

Twins

Two offspring from the same pregnancy, either monozygotic or dizygotic.

Twitch

A brief involuntary muscle contraction or movement, which may be benign, medication-related, stress-related, or a sign of nerve, muscle, or brain disease.

Twitching

Repeated involuntary muscle contractions or fasciculations, commonly caused by fatigue, stimulants, electrolyte disturbance, nerve irritation, medicines, or neurologic disease.

Tympanic

Tympanic means relating to the eardrum or middle ear, or resembling a drum in shape or sound.

Tympanic Bulla

The tympanic bulla is a hollow or expanded bony chamber surrounding the middle ear, especially prominent in many mammals.

Tympanic Cavity

The air-filled space of the middle ear containing the malleus, incus, and stapes.

Tympanic Hyperectasis

Abnormal outward bulging or retraction of the tympanic membrane from chronic middle ear pressure

changes, potentially causing conductive hearing loss and requiring tympanostomy tube placement.

Tympanic Membrane

The eardrum, a thin membrane separating the external ear canal from the middle ear.

Tympanic Membrane Perforation

A tear or hole in the eardrum caused by infection, trauma, pressure changes, or instrumentation, potentially producing pain, drainage, or conductive hearing loss.

Tympanites

Abdominal swelling or discomfort caused by excessive gas in the stomach or intestines. Persistent, severe, or painful distension may signal obstruction or another illness.

Tympanometry

An objective test that varies air pressure in the ear canal while measuring movement of the tympanic membrane and middle-ear pressure. It helps assess middle-ear fluid, eustachian-tube dysfunction, perforation, and ossicular or tympanic-membrane abnormalities.

Tympanoplasty

Surgical repair or reconstruction of the tympanic membrane and, when necessary, the middle-ear ossicles to close a perforation or improve hearing.

Tympanosclerosis

Scarring and calcification of the tympanic membrane or middle-ear lining, often following recurrent infection or previous inflammation. It may be asymptomatic or cause conductive hearing loss when the ossicles are affected.

Alternate Names:

Tympanic-membrane sclerosis; myringosclerosis when limited to the tympanic membrane.

Tympanostomy Tubes

Small tubes inserted through the tympanic membrane to ventilate the middle ear and prevent fluid

accumulation. Indicated for recurrent acute otitis media (3 episodes in 6 months or 4 in 12 months) or persistent otitis media with effusion greater than 3 months with hearing loss. Performed as outpatient procedure under general anesthesia.

Tympanum

The middle-ear cavity or, in some contexts, the tympanic membrane.

Tympany

Tympany is a drumlike sound on percussion, often produced by gas in a hollow organ such as the stomach or intestine.

Type 1 Diabetes

Type 1 diabetes is autoimmune destruction of pancreatic beta cells causing absolute insulin deficiency, hyperglycemia, and risk of diabetic ketoacidosis without insulin replacement.

Alternate Names:

Diabetes, Type 1 Diabetes Type 1 (Type 1 Diabetes)

Type 1 Diabetes In Children

Type 1 diabetes diagnosed during childhood, causing high blood sugar and requiring insulin, monitoring, nutrition planning, and education.

Alternate Names:

Type 1 Diabetes Mellitus

Type 1 diabetes is an autoimmune disease in which the immune system attacks and destroys the insulin-producing beta cells of the pancreas, resulting in absolute insulin deficiency.

Type 2 Diabetes

Type 2 diabetes is chronic hyperglycemia caused by insulin resistance and progressive beta-cell dysfunction, with risks to the eyes, kidneys, nerves, heart, and blood vessels.

Alternate Names:

Diabetes, Type 2 Diabetes Type 2 (Type 2 Diabetes)

Type 2 Diabetes - Oral Medicines

Oral medicines for type 2 diabetes lower glucose through effects on insulin, liver glucose production, kidney reabsorption, or intestinal absorption; choice depends on health and treatment goals.

Type 2 Diabetes In Children

Type 2 diabetes beginning during childhood or adolescence, often involving insulin resistance and requiring lifestyle and medical management.

Alternate Names:

Type 2 Diabetes Mellitus

Type 2 diabetes is a chronic metabolic disorder characterized by insulin resistance and progressive pancreatic beta-cell dysfunction, leading to hyperglycemia.

Type 3B Diabetes

Type 3b diabetes is a form of diabetes caused by disease or removal of the pancreas, which can impair both insulin and digestive-enzyme production.

Type I Hypersensitivity

An immediate (anaphylactic) hypersensitivity reaction mediated by allergen-specific IgE bound to mast cells and basophils; re-exposure cross-links IgE, triggering degranulation and release of histamine and mediators within minutes. Manifestations: allergic rhinitis, asthma, urticaria, food allergy, and anaphylaxis.

Type II Hypersensitivity

An antibody-mediated (cytotoxic) hypersensitivity reaction in which IgG or IgM binds to cell-surface or matrix antigens, causing cell destruction by complement activation (MAC), opsonization/phagocytosis, or antibody-dependent cellular cytotoxicity.

Type III Hypersensitivity

An immune complex-mediated hypersensitivity reaction in which circulating antigen-antibody complexes deposit in tissues and vessels, activating complement and attracting neutrophils, causing inflam-

mation and tissue damage. Local forms: Arthus reaction, serum sickness.

Type III Hypersensitivity Reaction

Tissue injury caused by deposition of antigen-antibody complexes, activating complement and inflammation.

Type IV Hypersensitivity

A delayed-type hypersensitivity (cell-mediated) reaction mediated by sensitized T cells (not antibody), occurring 24-72 hours after antigen exposure.

Type V Glycogen Storage Disease

An inherited muscle disorder, also called McArdle disease, in which muscle cells cannot properly break down glycogen for energy. Exercise can cause muscle pain, cramps, and sometimes dark urine from muscle injury.

Types Of Health Care Providers

Health professionals and services that diagnose, treat, prevent, or support health needs, including physicians, nurses, pharmacists, therapists, dentists, mental-health professionals, and community services.

Types Of Ileostomy

Types of ileostomy include end, loop, and continent procedures, which divert stool through an opening in the abdominal wall when the colon or rectum cannot be used normally.

Typhlitis

Inflammation of the cecum in immunocompromised patients particularly those with neutropenia from chemotherapy. Also known as neutropenic enterocolitis. Presents with right lower quadrant pain, fever, and diarrhea. Diagnosis is by CT showing cecal wall thickening.

Alternate Names:
neutropenic enterocolitis.

Typhoid

A systemic infection caused by *Salmonella enterica*

serovar Typhi, usually acquired through contaminated food or water and characterized by prolonged fever and gastrointestinal or systemic symptoms.

Typhoid Fever

A systemic infection caused by *Salmonella* Typhi, transmitted through contaminated food or water.

Alternate Names:
Enteric Fever (Typhoid Fever)

Typhus

Typhus refers to several rickettsial infections transmitted by lice, fleas, or mites and characterized by fever, headache, rash, and variable systemic complications. arthropod vectors. Three main forms: (1) Epidemic (louse-borne) typhus—*Rickettsia prowazekii*, transmitted by the human body louse (*Pediculus humanus corporis*). Associated with overcrowding, poverty, cold climate, and refugees ('war typhus').

Typhus Fever

A group of infections caused by *Rickettsia* species and transmitted mainly by lice or fleas.

Typical Antipsychotics

Typical antipsychotics, also known as first-generation antipsychotics, primarily block dopamine D2 receptors in the mesolimbic pathway, reducing positive symptoms of schizophrenia including hallucinations, delusions, and disorganized thinking.

Alternate Names:
first-generation antipsychotics, primarily block.

Tyramine

A naturally occurring amine in aged or fermented foods that can interact dangerously with monoamine oxidase inhibitors.

Tyrosinase Peptide

A short protein fragment related to tyrosinase, the enzyme that helps make melanin, the pigment in skin, hair, and eyes. Such peptides are mainly used in research.

Tyrosine

An amino acid used to make proteins, thyroid hormones, melanin, and catecholamines.

Alternate Names:

Full term

Tyrosinemia

An inherited disorder of tyrosine metabolism that causes accumulation of tyrosine or its metabolites. Type I can cause severe liver and kidney disease and is associated with fumarylacetoacetate hydrolase deficiency.

Tzanck Smear

A microscopic examination of cells from the base of a skin blister or erosion, historically used to support diagnoses such as herpesvirus infection or pemphigus.

U

U Wave

A small deflection following the T wave, best seen in precordial leads V2-V3. U waves are thought to represent repolarization of the Purkinje fibers or papillary muscles. Prominent U waves are associated with hypokalemia, bradycardia, and certain drugs (digitalis, quinidine).

UA

UA commonly means urinalysis, laboratory examination of urine for blood, protein, cells, glucose, infection, and other findings.

Ubidecarenone

Coenzyme Q10, a lipid-soluble molecule involved in mitochondrial electron transport and cellular energy production; it is sold as a supplement and studied in selected cardiovascular and neuromuscular conditions.

Ubiquinone

A lipid-soluble electron carrier in the mitochondrial membrane. See Coenzyme Q.

Ubiquitin

A small regulatory protein that is covalently attached to other proteins to label them for degradation by the proteasome or to alter their cellular trafficking and signaling.

Ubiquitination

Ubiquitination is attachment of ubiquitin to a target protein through an E1–E2–E3 enzyme cascade, marking it for proteasomal degradation or altering its localization, activity, or signaling.

Ubiquitous

Present or found everywhere. In medical writing, the term is descriptive rather than a diagnosis and may refer to a substance, organism, or process that is widely distributed.

UBT

UBT commonly means urea breath test, which detects active *Helicobacter pylori* infection by measuring labeled carbon dioxide after ingested urea.

UDP-Glucuronosyltransferase

UDP-glucuronosyltransferase is an enzyme family that attaches glucuronic acid to bilirubin, medicines, hormones, and other compounds for elimination.

Uhl Anomaly

A rare congenital absence or severe deficiency of right-ventricular myocardium, leaving a thin-walled right ventricle. It may cause right-sided heart failure, arrhythmia, cyanosis, or sudden deterioration in infancy or childhood.

Ulcer

Full-thickness loss of epidermis extending into the dermis or deeper tissue, often healing with a scar. Causes include vascular disease, pressure, infection, neuropathy, inflammatory disease, trauma, and malignancy.

Ulcer (Vascular)

A persistent break in the skin caused or maintained by inadequate blood flow, venous hypertension, neuropathy, or related vascular disease.

Alternate Names:

Vascular

Ulcer Healing Drugs

Medicines used to promote healing of gastrointestinal ulcers, usually by suppressing acid secretion or protecting the mucosa; treatment also addresses causes such as *H. pylori* infection or NSAID use.

Ulcer, Aphthous

Alternate index label; see *Canker sore*.

Ulcer, Peptic

Alternate index label; see *Peptic ulcer*.

Ulcer, Rectal

Alternate index label; see *Solitary rectal ulcer syndrome*.

Ulceration

The formation of an open sore caused by loss of surface tissue, usually involving skin or a mucous membrane and sometimes extending into deeper tissue.

Ulcerative Colitis

A chronic inflammatory disease of the colon and rectum, usually causing continuous mucosal inflammation that begins in the rectum and extends proximally. Symptoms may include bloody diarrhea, urgency, abdominal pain, and extraintestinal manifestations.

Alternate Names:

Colitis, Ulcerative Colitis Ulcerative (Ulcerative Colitis)

Ulcerative Keratitis

An ulcerative infection or inflammatory defect of the cornea that can cause pain, redness, photophobia, and impaired vision and may threaten sight if untreated.

Ulcerative Proctitis

Inflammation and ulceration limited to the lining of the rectum, often causing rectal bleeding, urgency, tenesmus, mucus, and diarrhea.

Ullrich Congenital Muscular Dystrophy

A severe congenital muscular dystrophy belonging to the collagen VI deficiency spectrum, inherited in an autosomal recessive (or dominant de novo) pattern caused by mutations in *COL6A1*, *COL6A2*, or *COL6A3*. Characterized from birth or early infancy by profound generalized muscle weakness and hypotonia, accompanied by a pathognomonic combination of proximal joint contractures (involving the hips, knees, and elbows) and striking distal joint hyperlaxity (hyperextensibility of the fingers, wrists, and ankles), congenital torticollis, prominent calcaneal protrusion, and progressive early respiratory failure due to severe diaphragm weakness, often requiring nocturnal non-invasive ventilation before the second decade of life.

Alternate Names:

UCMD; Ullrich Scleroatonic Muscular Dystrophy

Ulna

Medial forearm bone, longer than the radius. Proximal end: trochlear notch, olecranon, coronoid process. Shaft. Distal end: head, styloid process. The trochlear notch articulates with the trochlea forming the primary elbow hinge.

Ulnar

Relating to the ulna bone of the forearm or the ulnar side (medial side) of the hand. Ulnar nerve compression causes paresthesia in medial one and a half fingers.

Ulnar Artery

The larger terminal branch of the brachial artery at the cubital fossa, running medially down the forearm to the wrist (palpable medial to the pisiform) and hand. Supplies the medial forearm and ulnar side of the hand; forms the superficial palmar arch. Provides the common interosseous artery.

Ulnar Vein

Deep veins accompanying the ulnar artery that drain the medial forearm and join the radial veins to form the brachial veins.

Ulnar Hypoplasia

Underdevelopment or partial absence of the ulna, usually presenting as a congenital forearm and hand anomaly. It may cause forearm shortening, radial deviation of the hand, limited elbow or wrist movement, and associated digital or systemic abnormalities.

Ulnar Nerve

From medial cord (C8-T1), passing behind the medial epicondyle (cubital tunnel), into the forearm innervating flexor carpi ulnaris and medial FDP half. In the hand innervates hypothenar muscles, interossei, medial two lumbricals, and adductor pollicis. Sensation to medial 1.5 digits.

Ulnar Nerve Dysfunction

Impairment of the ulnar nerve, commonly caused by compression at the elbow (cubital tunnel) or wrist (Guyon canal). Symptoms include numbness or tingling in the little and ring fingers, weakened grip, and hand-muscle wasting; severe lesions may produce an ulnar claw.

Ulnar Nerve Entrapment

Compression or irritation of the ulnar nerve, commonly at the elbow or wrist, causing numbness or tingling in the ring and little fingers and sometimes weakness of the hand.

Ulnar Paradox

The clinical observation that a lesion of the ulnar nerve at the wrist may cause more clawing of the fingers than a lesion higher in the forearm, because the ulnar-innervated finger flexors remain intact.

Ulnar Wrist Pain

Pain on the side of the wrist nearest the little finger, often involving the ulna or nearby tendons and ligaments.

Alternate Names:

Ulnar-Sided Wrist Pain

Ulnar-Sided Wrist Pain

Alternate index label; see *Ulnar wrist pain*.

Ulocladium

A genus of dematiaceous filamentous fungi found in soil and plant material. It is an uncommon opportunistic pathogen that can cause keratitis, sinusitis, or invasive infection in susceptible people.

Ultracentrifugation

Ultracentrifugation separates particles or macromolecules at very high centrifugal force according to size, shape, and density; it is used to study viruses, organelles, lipoproteins, and protein complexes.

Ultradian Rhythm

A biological rhythm that repeats more than once in 24 hours, such as pulsatile hormone secretion, sleep-stage cycling, or periodic changes in heart and breathing patterns.

Ultrafiltration

Pressure-driven movement of water and small solutes across a semipermeable membrane while retaining larger molecules or particles. It contributes to glomerular filtration and is used in dialysis and laboratory processing.

Ultralente Insulin

Ultralente insulin is a prolonged-acting zinc insulin suspension formerly used for basal insulin replacement; it has largely been replaced by newer long-acting analogues and can cause hypoglycaemia and dosing variability.

Ultrasonic Therapy

Use of high-frequency sound energy for therapeutic purposes such as selected physical-therapy applications or tissue ablation; benefits and risks depend on the device and treatment site.

Ultrasonography

Imaging that uses high-frequency sound waves to produce pictures of tissues and organs.

Ultrasound

Ultrasound uses high-frequency sound waves to create images of organs, vessels, tissues, or a devel-

oping fetus without ionizing radiation. quadrant (RUQ): gallbladder (gallstones, cholecystitis), liver (steatosis, cysts), kidneys, and pancreas. Pelvic: uterus, ovaries, adnexa. Renal: hydronephrosis, stones, cysts, tumors. Scrotal: testicular torsion, epididymitis, varicocele, hydrocele.

Ultrasound Biomicroscopy

High-frequency ultrasound imaging used to visualize the anterior segment of the eye, including the cornea, iris, ciliary body, lens, and angle when ordinary examination or imaging is insufficient.

Ultrasound Contrast

A substance administered during ultrasound to increase the visibility of blood flow or selected tissues.

Ultrasound Therapy

Use of sound waves at therapeutic intensity to heat or mechanically affect tissue. It is used in selected rehabilitation settings, but evidence and suitability vary by condition and it is not appropriate over every body area.

Ultrasound Transducer

A device that emits and receives high-frequency sound waves to generate diagnostic images or guide procedures. Linear, curvilinear, phased-array, and endocavitary transducers are selected according to depth and anatomic application.

Ultraviolet Microscopy

Microscopy that uses ultraviolet wavelengths to produce images or fluorescence from structures and substances that absorb or emit ultraviolet light. It has specialized laboratory and research applications.

Ultraviolet Rays

Invisible electromagnetic radiation with wavelengths shorter than visible violet light. Excess exposure can cause sunburn, photoaging, eye injury, and skin cancers; controlled ultraviolet exposure has selected therapeutic uses.

Ultraviolet Rays

Electromagnetic radiation beyond visible violet light. Excessive exposure can cause sunburn, photoaging, eye injury, and skin cancer.

Ultraviolet Therapy

Controlled exposure to ultraviolet light used for selected skin disorders such as psoriasis, eczema, and vitiligo; treatment requires eye and skin protection and monitoring for burns and long-term skin cancer risk.

Umbilical

Relating to the umbilicus or navel, including structures, lesions, discharge, or abnormalities located at or connected with it.

Umbilical Artery Doppler

Ultrasound assessment of blood flow in the umbilical arteries to evaluate placental resistance. Normal flow shows a pulsatility index decreasing throughout pregnancy with forward diastolic flow. Absent or reversed end-diastolic flow indicates severe placental dysfunction and is associated with increased perinatal morbidity and mortality.

Umbilical Catheters

Umbilical catheters provide vascular access through a newborn's umbilical vessels for fluids, medicines, monitoring, or blood sampling while complications are monitored.

Umbilical Cord

The connection between the fetus and the placenta, containing two umbilical arteries (carrying deoxygenated blood from the fetus to the placenta) and one umbilical vein (carrying oxygenated blood from the placenta to the fetus). The vessels are embedded in Wharton jelly, a mucopolysaccharide matrix that prevents compression.

Umbilical Cord Blood Gas

Analysis of arterial and venous cord blood collected after delivery. The umbilical artery best reflects fetal acid-base status, while the umbilical vein reflects placental blood. Results are interpreted using

gestational age, delivery circumstances, pH, carbon dioxide, and base deficit or excess.

Umbilical Cord Prolapse

Descent of the umbilical cord through the cervix alongside or past the presenting fetal part, occurring when the membranes rupture. Risk factors: malpresentation (transverse lie, breech), preterm/low birth weight, polyhydramnios, multiparity, and artificial ROM with high presenting part.

Umbilical Granuloma

The most common umbilical lesion in newborns and young infants, presenting as a soft, moist, vascular, beefy-red or pink pedunculated or sessile mass of excess granulation tissue at the base of the umbilicus after umbilical cord separation. It may produce persistent serous or serosanguineous discharge and mild local periumbilical skin irritation without signs of systemic infection or purulence (distinguishing it from omphalitis). Standard treatment is topical application of silver nitrate (with protection of surrounding skin) or topical salt application.

Alternate Names:

Granuloma Umbilicale; Neonatal Umbilical Granuloma

Umbilical Hernia

A hernia through the umbilical ring, common in infants (most umbilical hernias close spontaneously by age 4-5) and adults (risk factors: obesity, pregnancy, ascites, cirrhosis). Presents as a reducible bulge at the umbilicus, worse with increased intra-abdominal pressure. In adults: risk of incarceration and strangulation is higher.

Umbilical Region

Central region around the umbilicus containing small intestine, transverse colon, and great vessels. The umbilicus is a landmark for assessing ascites, caput medusae in portal hypertension, and umbilical hernias. Percussion and auscultation here are essential components of the abdominal examination.

Umbilical Vein

The umbilical vein carries oxygen- and nutrient-

rich blood from the placenta to the fetus during pregnancy.

Umbilication

Formation of a central depression or navel-like indentation in a lesion or structure.

Umbilicus

The navel, a scar on the anterior abdominal wall marking the former attachment of the umbilical cord. It can be affected by hernia, infection, discharge, or metastatic disease.

Umbilicus (Navel)

The scar on the anterior abdominal wall marking the site of attachment of the umbilical cord in fetal life, located at the level of the L3-L4 intervertebral disc, approximately midway between the xiphoid process and the pubic symphysis.

Alternate Names:

Navel

Uncertainty

Lack of complete certainty about a diagnosis, prognosis, measurement, or treatment effect. Clinical uncertainty is managed by combining available evidence with follow-up, additional testing, and shared decision-making.

Uncinate Fit

A nonstandard or unclear historical label in the source dictionary; its intended clinical meaning should be verified against the original reference.

Uncompetitive Inhibition

A type of enzyme inhibition where the inhibitor binds only to the enzyme-substrate complex at a site distinct from the active site, forming an inactive ESI complex. The inhibitor cannot bind to the free enzyme.

Unconscious

Not awake and not responsive to ordinary external stimuli. Unconsciousness may result from syncope, seizure, trauma, intoxication, metabolic dis-

ease, stroke, or other emergencies.

Unconsciousness

Loss of awareness and responsiveness, ranging from brief fainting to prolonged coma.

Unconsciousness, Temporary

A brief episode of loss of awareness and responsiveness, often caused by syncope but also possible with seizure, concussion, hypoglycemia, or other conditions requiring assessment.

Underventilation

Breathing that is too shallow or slow to remove enough carbon dioxide, causing hypercapnia and sometimes low oxygen; causes include sedatives, neuromuscular weakness, obesity hypoventilation, and lung disease.

Underweight

BMI < 18.5 kg/m². Associated with osteoporosis, infertility, anemia, weakened immunity, surgical complications. Causes: hypermetabolism, malabsorption, eating disorders, malignancy, chronic infection.

Undescended Testicle

A testicle that has not moved into the scrotum before birth.

Alternate Names:

Testicle, Undescended

Undescended Testicle Repair

Surgical relocation and fixation of a testis that has not descended into the scrotum, usually performed in childhood to preserve function and reduce complications.

Undescended Testis

Failure of one or both testes to descend into the scrotum. Some descend during early infancy, but persistent cryptorchidism can impair fertility and increase the risk of torsion, hernia, or testicular cancer.

Undescended Testis (Cryptorchidism)

Congenital failure of one or both testes to descend along the normal path into the scrotum. Persistent cases require evaluation because they may affect fertility and increase the risk of torsion, hernia, or testicular cancer.

Alternate Names:

Cryptorchidism

Undetermined

Not established or not able to be classified with the available clinical information. An undetermined result may require repeat assessment or additional testing.

Undifferentiated

Lacking recognizable specialized features. In pathology, an undifferentiated tumor has cells that cannot be assigned readily to a specific tissue of origin based on their appearance.

Undifferentiated Cancer

A cancer whose cells lack recognizable specialized features, making the tissue of origin or exact subtype difficult to identify; diagnosis requires pathology and staging.

Undifferentiated Carcinoma

A high-grade carcinoma whose cells lack recognizable features of a specific tissue or tumor subtype; diagnosis and treatment require pathology, immunostaining, and staging.

Undifferentiated Pleomorphic Sarcoma

An uncommon aggressive cancer of soft tissue or bone whose cells have no clearly identifiable line of differentiation.

Alternate Names:

Malignant Fibrous Histiocytoma

Undifferentiated Pleomorphic Sarcoma Of The Breast

A rare high-grade breast sarcoma formerly called malignant fibrous histiocytoma. It is a diagnosis of exclusion after other specific sarcomas and meta-

plastic carcinoma have been ruled out histologically.

Undulant Fever

A zoonotic infectious disease caused by gram-negative coccobacilli of the genus *Brucella*, transmitted to humans through contact with infected animals (cattle, goats, pigs, sheep) or consumption of unpasteurized dairy products. See Brucellosis.

Undulate

To move or have a wave-like contour. In a clinical description, it may refer to a palpable fluid wave, a wavy tissue margin, or a rhythmic motion.

Undulate Ribs

Ribs with a wavy or irregular contour, a descriptive anatomic finding that may reflect a developmental variation, deformity, prior injury, or underlying bone disease.

Unencoded Data

Health information that has not been converted into a defined coding system or standardized data format. It may require interpretation or data transformation before analysis or exchange.

Unequivocal

Clear and not open to more than one reasonable interpretation. An unequivocal clinical finding or test result provides strong evidence for the stated interpretation when considered in context.

Unerupted Tooth

A tooth that has not emerged into the oral cavity at the expected time. It may be normally developing, delayed, impacted, or prevented from erupting by obstruction or abnormal position.

Unevaluable

Unable to be assessed or interpreted reliably because of inadequate data, poor test quality, an unavailable endpoint, or another limiting circumstance.

Ungual

Pertaining to, situated near, or affecting the nail

(*unguis*) or the entire functional nail apparatus of a digit (comprising the hard keratinized nail plate, nail matrix, nail bed, hyponychium, eponychium, and lateral nail folds). Clinically utilized in dermatology and podiatry to describe anatomical landmarks (subungual space beneath the nail plate, periungual tissue surrounding the nail border) and pathological conditions, such as subungual hematoma (blood collection following crush injury), subungual melanoma (requiring biopsy of pigmented longitudinal melanonychia), and periungual verrucae (warts).

Unguentum

A semisolid, greasy preparation intended for application to the skin or, when formulated for that purpose, to mucous membranes or the eyes.

Unicornuate Uterus

A congenital uterine malformation resulting from the complete or near-complete failure of development of one Müllerian duct, producing an elongated, banana-shaped hemi-uterus that communicates with a single fallopian tube. Frequently accompanied by a contralateral rudimentary horn (which can be cavitory or non-cavitory, communicating or non-communicating); a non-communicating cavitory horn carries a severe risk of ectopic rudimentary horn pregnancy and rupture, necessitating elective laparoscopic resection.

Unilaminar Epithelium

An epithelium consisting of a single layer of cells. Its structure is adapted for functions such as diffusion, absorption, secretion, or filtration depending on the organ.

Unilateral

Affecting, occurring on, or involving one side of the body or one paired organ, such as unilateral weakness or unilateral eye disease.

Unilateral Gynecomastia

Enlargement of breast glandular tissue affecting one breast in a male patient; a new, firm, fixed, painful, or nipple-associated mass requires clinical

assessment.

Unilateral Microphthalmos

A congenital condition in which one eye is abnormally small. It may be associated with other ocular malformations and can cause substantial visual impairment.

Uninodular Goiter

A goiter containing one dominant thyroid nodule. Evaluation generally includes thyroid-function testing and imaging, with further assessment to determine whether the nodule is benign or malignant.

Uniparental Disomy

Inheritance of both copies of a chromosome or chromosome segment from one parent and none from the other. It can cause disease through abnormal genomic imprinting, homozygosity for a recessive variant, or chromosomal dosage effects.

Unipolar Neuron

A neuron with a single process extending from its cell body. True unipolar neurons occur in some invertebrates; many sensory neurons in humans are more accurately described as pseudounipolar.

Unit

A standardized quantity used to express measurement, dosage, concentration, or a healthcare service.

Unknown

Not known or not determined from the information currently available. In a medical record, an unknown value should be distinguished from a negative or normal result.

Unopposed Estrogen

Estrogen therapy given without a progestogen in a person with a uterus.

Unresectable

Describing a tumor or lesion that cannot be completely removed safely by surgery because of its location, extent, invasion of vital structures, or the

patient's operative risk.

Unresolved Losses

Significant losses that have not been fully acknowledged or emotionally processed.

Unresponsive Coma

A state of profound unresponsiveness in which the person cannot be awakened or follow commands.

Unsaturated Phytosterol

A plant sterol containing one or more carbon-carbon double bonds. Phytosterols can reduce intestinal cholesterol absorption, although clinical effects depend on intake, formulation, and patient context.

Unstable Angina

Myocardial ischemia at rest, new-onset severe angina, or a crescendo pattern (more frequent, longer, lower-threshold) without evidence of myocardial necrosis (normal troponin). Pathophysiology: acute plaque rupture with thrombosis causing transient severe coronary obstruction.

Unsteadiness

Impaired balance or gait stability, which may result from vestibular disease, neurologic disorders, weakness, medication effects, visual impairment, or musculoskeletal problems.

Untidy Wound

An irregular or contaminated wound with crushing, devitalized tissue, tissue loss, foreign material, or associated injury to vessels, nerves, tendons, or bone. It has increased infection risk and may require exploration, irrigation, debridement, and delayed or staged closure.

Untranslated Region

A part of an RNA molecule that is not translated into protein; the 5-prime and 3-prime untranslated regions regulate RNA stability, localization, and translation.

Unwell

A general description of feeling ill or having reduced health, without identifying a specific disease. Persistent or severe symptoms require assessment for an underlying cause.

Upamostat

An investigational small-molecule serine-protease inhibitor studied for inflammatory and infectious diseases. It is not an established routine treatment, and its clinical role depends on trial evidence and regulatory approval.

Upper Airway Biopsy

Removal of tissue from the nasal cavity, pharynx, larynx, or related upper airway for microscopic evaluation of suspected infection, inflammation, infiltrative disease, or tumor.

Upper Airway Obstruction

Narrowing of the airway above the carina, causing inspiratory stridor, dyspnea, and characteristic changes on flow-volume loops (flattening of the inspiratory limb). Causes include tumors, stenosis, goiter, and vocal cord dysfunction. It is a medical emergency when severe.

Upper Airway Resistance Syndrome

Increased effort to breathe through a narrowed upper airway, often causing sleep disruption and daytime sleepiness.

Alternate Names:

UARS

Upper Arm

The portion of the upper limb between the shoulder and elbow, containing the humerus and muscles, nerves, and vessels that serve the arm.

Upper Endoscopy

Upper endoscopy: flexible endoscopic examination of the esophagus, stomach, and duodenum. See Esophagogastroduodenoscopy (EGD).

Upper Esophageal Sphincter

A muscular valve at the top of the esophagus that opens during swallowing.

Alternate Names:

UES

Upper Extremity

The anatomic region extending from the shoulder through the arm, forearm, wrist, and hand. Disorders may involve its bones, joints, muscles, nerves, vessels, or skin.

Upper Extremity Venous Doppler

An ultrasound examination of the arm and related veins that assesses compressibility and blood flow, commonly to evaluate suspected venous thrombosis.

Upper Gastrointestinal Bleeding

Bleeding from esophagus, stomach, or duodenum. Causes: peptic ulcer, varices, Mallory-Weiss tear, gastritis. Presentation: hematemesis, melena, hematochezia. Management: ABCs, IV access, type and crossmatch, PPI infusion, endoscopy within 24 hours. Variceal bleeding: octreotide, antibiotics, band ligation.

Upper GI And Small Bowel Series

A fluoroscopic x-ray examination in which swallowed contrast outlines the esophagus, stomach, duodenum, and small intestine to evaluate structure and transit.

Upper GI Series

Fluoroscopic examination of the esophagus, stomach, and duodenum after oral barium contrast. Single-contrast: looks for gross abnormalities (obstruction, masses). Double-contrast (barium + gas): better mucosal detail for ulcers, erosions, and early malignancies.

Upper Leg

The portion of the lower limb between the hip and knee, containing the femur and the thigh muscles, nerves, and blood vessels.

Upper Limb Disorders

Conditions affecting the shoulder, arm, elbow, forearm, wrist, or hand, including musculoskeletal, nerve, vascular, and work-related disorders.

Upper Motor Neuron

A neuron in the brain or spinal cord that influences lower motor neurons controlling skeletal muscle. Damage can cause weakness, spasticity, increased reflexes, and pathologic reflexes.

Upper Motor Neuron Syndrome

A pattern caused by injury to corticospinal pathways above the anterior horn cell or cranial motor nucleus. Findings may include weakness with spasticity, hyperreflexia, clonus, and an extensor plantar response, although acute lesions can initially produce flaccidity.

Upper Respiratory Infection

An infection of the nose, sinuses, pharynx, or larynx, most often caused by a virus and producing symptoms such as congestion, sore throat, cough, or runny nose.

Upregulation

An increase in the activity, amount, or expression of a gene, receptor, enzyme, or signaling pathway in response to developmental, physiologic, or disease-related signals.

Upright Labor Positions

Maternal positions during labor, such as walking, standing, squatting, kneeling, or sitting upright. They may improve comfort and mobility and can be used when safe and acceptable to the laboring person and clinical team.

Uptake

Cellular accumulation of radiotracer. Depends on blood flow, membrane transport, metabolism, binding.

Urachal Carcinoma

A rare malignant epithelial tumor arising from remnants of the embryonic urachus, usually located at the bladder dome or along the urachal tract.

Most tumors are adenocarcinomas and may produce mucin, with local invasion and metastatic spread occurring in advanced disease. Clinical features can include hematuria, mucousuria, urinary symptoms, pelvic pain, or an incidental mass detected on imaging. Diagnosis requires cystoscopic and histopathologic evaluation with staging; treatment is based on resectability, local extension, and metastatic status.

Urachal Cyst

A cystic remnant of the embryonic urachus between the bladder and umbilicus. It may be asymptomatic or become infected, causing lower-abdominal pain, fever, umbilical discharge, or urinary symptoms.

Urachal Mucinous Cystic Tumor

A rare tumor arising from the urachal remnant between the bladder dome and umbilicus. It may be benign or malignant, with mucinous histology, and requires surgical excision.

Urachus

An embryologic connection between the fetal bladder and umbilicus that normally becomes the median umbilical ligament after birth. Persistent urachal remnants can form cysts, sinuses, or other complications.

Uracil

A pyrimidine nitrogenous base found principally in RNA. It pairs with adenine and is also a component of several metabolic intermediates and nucleotide medicines.

Uracil Mustard

An older nitrogen-mustard alkylating compound related to uracil and studied or used historically as an antineoplastic drug. Its clinical use is limited by toxicity and the availability of newer treatments.

Uracil Nucleotides

Nucleotides containing uracil, including UMP, UDP, and UTP. They participate in RNA synthesis, carbohydrate metabolism, and activation of sugars for

glycoconjugate production.

Uraemia

The clinical syndrome resulting from accumulation of urea and other nitrogenous waste products in the blood in advanced chronic kidney disease (CKD stage 4-5) and acute kidney injury.

Uraemic Syndrome

The collection of symptoms and biochemical abnormalities caused by severe accumulation of uraemic waste products in advanced kidney failure.

Uranium

A naturally occurring radioactive heavy metal. Internal exposure can damage the kidneys, while inhaled or retained radioactive material can also expose tissues to ionizing radiation.

Urate

The ionized form of uric acid, the end product of purine metabolism in humans. Excess serum urate can promote monosodium urate crystal deposition and gout.

Urate-Lowering Therapy

Long-term treatment that reduces serum urate to dissolve monosodium urate deposits and prevent gout flares and structural damage. Xanthine oxidase inhibitors and uricosuric medicines are selected according to renal function, comorbidity, and treatment goals.

Urates

Salts or ions of uric acid. Monosodium urate crystals can accumulate in joints in gout, and urate crystals may contribute to urinary stones.

Urea

The principal nitrogen-containing waste product of protein metabolism, formed in the liver and excreted mainly by the kidneys.

Urea Breath Test

A noninvasive test for active *Helicobacter pylori* in-

fection: the person drinks labeled urea, and labeled carbon dioxide in exhaled breath indicates bacterial urease activity.

Urea Cycle

A metabolic pathway occurring in both the mitochondria and cytoplasm of hepatocytes that converts toxic ammonia (NH₃), derived from amino acid catabolism, into nontoxic, water-soluble urea for excretion by the kidneys.

Urea Cycle Disorder

A group of inherited disorders caused by deficiency of an enzyme or transporter in the urea cycle, leading to impaired ammonia clearance. Ornithine transcarbamylase deficiency is an X-linked form; presentation and severity vary by the gene and residual enzyme activity.

Urea Nitrogen Urine Test

A urine test measuring urea nitrogen excretion, interpreted with blood urea nitrogen, creatinine, diet, and urine volume to assess protein metabolism and renal handling.

Ureaplasma

A genus of very small bacteria without a cell wall that can colonize the genital or urinary tract and is sometimes associated with urethritis or pregnancy complications.

Ureaplasma Urealyticum

A cell-wall-free bacterium that can colonize the genital or urinary tract and is sometimes associated with nongonococcal urethritis, pregnancy complications, or infection in vulnerable newborns.

Uremia

A clinical syndrome caused by advanced kidney failure and accumulation of waste products and other biologically active substances. Symptoms may include nausea, itching, fatigue, confusion, bleeding, or pericarditis; laboratory values and symptoms must be interpreted together.

Uremic Encephalopathy

Neurological syndrome in severe renal failure. Confusion, asterixis, seizures, coma. Results from uremic toxin accumulation. Diagnosis is clinical in context of advanced renal failure. Emergent dialysis indicated with rapid neurological improvement typically.

Uremic Frost

Urea crystallization on skin in severe untreated uremia. Rare finding indicating extremely advanced uremia. White crystalline deposits on skin. Indicates immediate need for dialysis. Extremely rare in developed countries due to early detection.

Uremic Pericarditis

Pericardial inflammation in advanced uremia. Pleuritic chest pain, friction rub, diffuse ST elevation. Indicates need for emergent dialysis. Does not respond to anti-inflammatory therapy. Complications include tamponade.

Uremic Platelet Dysfunction

Impaired platelet adhesion and aggregation in uremia causing bleeding tendency. Easy bruising, epistaxis, GI bleeding. Improves with dialysis and desmopressin. Cryoprecipitate provides temporary benefit before procedures.

Uremic Stomatitis

Oral mucosal inflammation occurring in advanced kidney failure, presenting as white plaques, ulceration, or bleeding of the gums and tongue due to accumulation of uremic toxins.

Ureter

The paired muscular tubes (25-30 cm) conveying urine from the renal pelvis to the urinary bladder, crossing the pelvic brim at the pelvic inlet. Three anatomical narrowings (pelviureteric junction, pelvic brim, vesicoureteric junction) are common sites of ureteric stone impaction.

Ureter Duplex

A congenital duplication of the ureter, complete or partial, draining one kidney. It may be asymp-

tomatic or associated with vesicoureteral reflux, ureterocele, obstruction, recurrent urinary infection, or impaired renal drainage.

Ureter Fissus

A congenital longitudinal splitting or duplication anomaly of the ureter. Its clinical importance depends on the anatomy and whether it causes reflux, obstruction, infection, or impaired drainage.

Ureter Obstruction

Blockage of urine flow from a kidney to the bladder, commonly caused by a stone, stricture, tumor, or external compression and potentially leading to hydronephrosis, infection, or kidney injury.

Ureteral Calculi

Stones that form in or pass through a ureter, the tube carrying urine from a kidney to the bladder.

Ureteral Catheters

Thin tubes passed into the ureter to drain urine, inject contrast, measure pressure, or guide procedures. Complications include infection, bleeding, perforation, migration, and obstruction.

Ureteral Obstruction

Blockage of urine flow causing proximal dilation and increased pressure. Acute causes renal colic with flank pain. Chronic leads to hydronephrosis and CKD. Emergent decompression with stent or nephrostomy for infected obstructed kidney. Diagnosis by CT or ultrasound showing hydronephrosis.

Alternate Names:

Blocked Ureter

Ureteral Perforation

A tear or hole in the ureter, sometimes caused by a stone, trauma, surgery, or an endoscopic procedure. Urine may leak into surrounding tissues, causing pain, infection, or inflammation; prompt imaging and urologic assessment are required.

Ureteral Polyp

Usually a benign fibroepithelial growth arising from

the ureter. It may cause hematuria, flank pain, or obstruction and must be distinguished from urothelial malignancy when evaluated.

Ureteral Retrograde Brush Biopsy

A brush passed through a ureteroscope to collect cells from the lining of the ureter or renal pelvis for cytologic evaluation of suspected urothelial cancer.

Ureteral Stenosis

Narrowing of the ureter caused by scar tissue, prior surgery, radiation, stones, inflammation, or congenital disease, impairing urine drainage and possibly causing hydronephrosis or recurrent infection.

Ureteral Stent

Double-J stent maintaining ureteral patency. Indications: obstruction (stone, stricture, malignancy), post-ureteroscopy. Materials: silicone, polyurethane. Sizes: 4.7-7 Fr, lengths 24-30 cm. Complications: irritation, hematuria, encrustation, migration.

Ureteral Stent (Double-J Stent)

A tube placed in the ureter from the renal pelvis to the bladder to maintain patency, with pigtail coils at both ends (hence double-J). Indications: ureteral obstruction (stones, stricture, malignancy), after ureteroscopy (prevent post-op obstruction), and as a pre-treatment for complex ureteral stones.

Alternate Names:

Double-J Stent

Ureteral Stricture

Narrowing of the ureter caused by fibrosis, surgery, instrumentation, radiation, infection, malignancy, or congenital disease. It can produce upstream hydronephrosis, pain, infection, and declining renal function.

Uretereroenterostomy

A urinary diversion procedure connecting a ureter to a segment of intestine; this spelling is a source variant of *ureteroenterostomy*.

Ureteric Perforation

A tear or hole in the ureter that can allow urine to leak into surrounding tissue, usually after instrumentation, surgery, trauma, or a stone; it needs prompt urologic assessment.

Ureteritis

Inflammation of the ureter, usually related to infection, obstruction, stone passage, instrumentation, or adjacent inflammatory disease. It may cause flank or abdominal pain, urinary symptoms, hematuria, and fever.

Ureterocele

A balloon-like dilation of the lower ureter where it enters the bladder, usually congenital; it can obstruct urine flow and cause infection, hydronephrosis, reflux, stones, or impaired renal function.

Ureterolithiasis

The presence of kidney stones (calculi) within the ureter, causing ureteral obstruction and renal colic. Stones form from calcium oxalate/phosphate (most), uric acid, struvite (infection), and cystine.

Ureteropelvic Junction Obstruction

Impaired drainage from the renal pelvis into the proximal ureter, caused by intrinsic narrowing, aberrant vessels, scarring, or functional peristaltic abnormality. It may cause hydronephrosis, flank pain, infection, stones, or loss of renal function.

Ureteroscope

A slender endoscope used to inspect and treat the ureter and, when advanced, the renal pelvis.

Ureteroscopes

Small endoscopes used to inspect the ureter and renal collecting system and to perform procedures such as stone removal, biopsy, or stent placement.

Ureteroscopy

Ureteroscopy uses a small endoscope passed through the urinary tract to inspect, biopsy, or treat ureteral or kidney stones and other lesions. the urethra, blad-

der, and into the ureter or renal collecting system. Used primarily for the diagnostic evaluation and therapeutic management of ureteral and renal calculi, ureteral strictures, and upper tract urothelial carcinoma (UTUC).

Ureterostomy Procedure

Surgery that creates an opening from a ureter to the skin or another urinary conduit to divert urine when normal drainage to the bladder is impossible or must be bypassed.

Urethan

An older name for ethyl carbamate, a chemical compound with carcinogenic potential in experimental and exposure settings. It is not a routine therapeutic medicine.

Urethra

The tube conveying urine from the bladder to the exterior. Male urethra (18-20 cm): prostatic, membranous, and spongy (penile) parts; carries urine and semen; longer and more tortuous than female. Female urethra (3-4 cm): short, straight, predisposing to ascending urinary tract infections.

Urethra Disorder

A condition affecting the urethra, such as infection, inflammation, stricture, trauma, congenital abnormality, or tumor, that may cause pain, discharge, weak stream, or retention.

Urethra, Diseases Of And Injury To

Disorders or trauma affecting the urethra, including infection, stricture, diverticulum, obstruction, and laceration.

Urethracele

A herniation or prolapse of the urethra into the anterior vaginal wall, often occurring together with a cystocele (bladder prolapse). Urethracele is caused by weakening of the pubocervical fascia, often from childbirth, aging, or increased intra-abdominal pressure.

Urethral Carcinoma

A rare cancer arising from the urethra that may cause bleeding, a lump, discharge, pain, or difficult urination; diagnosis requires examination, imaging, and biopsy.

Urethral Catheter

A tube inserted through the urethra into the bladder to drain urine, obtain a specimen, instill medication, or monitor output. Urinary infection and urethral trauma are important risks.

Urethral Diverticulum

A periurethral outpouching that communicates with the urethral lumen, usually arising from infected or obstructed periurethral glands. It may cause dysuria, post-void dribbling, recurrent urinary infection, dyspareunia, or a palpable anterior-vaginal-wall mass.

Urethral Fistula

An abnormal channel between the urethra and another epithelial surface, often the vagina or skin. It can cause continuous urine leakage, recurrent infection, and local irritation.

Urethral Hypermobility

Excessive movement of the urethra during increased abdominal pressure, contributing to stress incontinence.

Urethral Infection

Infection involving the urethra, commonly caused by bacteria, including sexually transmitted organisms. Symptoms may include burning urination, discharge, frequency, or pelvic discomfort; testing and appropriate treatment help prevent complications and transmission.

Urethral Meatus

External opening of urethra. Position abnormal in hypospadias, epispadias.

Urethral Meatus, Female

The external opening of the female urethra, located

in the vestibule anterior to the vaginal opening. Its position and appearance may be relevant in urinary symptoms, congenital variation, or examination.

Urethral Obstruction

Partial or complete blockage of urine leaving the bladder, often from stricture, stone, prostate disease, tumor, or a congenital valve; acute retention requires prompt treatment.

Urethral Pressure Profile

Measurement of urethral closure pressure along urethra. Used in incontinence evaluation.

Urethral Pressures

Measurements of pressure within or along the urethra used to assess urethral closure and urinary continence.

Urethral Sling

Synthetic or fascial tape supporting urethra for stress incontinence. Mid-urethral sling (TVT, TOT) standard. Minimally invasive. Complications: erosion, urgency, retention.

Urethral Sphincter

The muscular mechanism that helps maintain urinary continence by compressing the urethra. It includes smooth and striated muscle components controlled by autonomic and voluntary pathways.

Urethral Stenosis

Scar-related narrowing of the urethral channel that restricts urinary flow and can cause spraying, weak stream, recurrent infection, retention, or bladder damage.

Urethral Stricture

Narrowing of the urethra caused by scar tissue. It may produce a weak urinary stream, spraying, straining, incomplete emptying, infections, or urinary retention and can result from injury, instrumentation, infection, or inflammatory disease.

Urethral Syndrome

Urinary frequency, urgency, dysuria, or discomfort without a clear bacterial urinary-tract infection; causes may include irritation, pelvic-floor dysfunction, or inflammation.

Urethritis

Urethritis is inflammation of the urethra, commonly from sexually transmitted infection, causing dysuria, irritation, or discharge and requiring evaluation of partners and complications. (Chlamydia trachomatis, Neisseria gonorrhoeae) or non-infectious causes. Presents with dysuria, urethral discharge, and meatal itching. Diagnosis: urine nucleic acid amplification testing (NAAT).

Urethrocele

Prolapse or herniation of the urethra, usually into the anterior vaginal wall.

Urethrography

Contrast imaging of urethra. Evaluates urethral strictures, trauma, anomalies.

Urethroplasty

Surgical reconstruction or widening of the urethra, most commonly to treat a urethral stricture.

Urethroscopy

Endoscopic examination of the urethra using a urethroscope, sometimes combined with treatment of a stricture or other lesion.

Urge Urinary Incontinence

Involuntary leakage of urine, significantly affecting psychological, social, and physical well-being. See urinary incontinence.

Urgency

A sudden, compelling desire to urinate that is difficult to defer—a lower urinary tract symptom. Causes: overactive bladder (detrusor overactivity, often idiopathic), urinary tract infection (urgency + dysuria), bladder outlet obstruction (BPH), neurogenic bladder, interstitial cystitis, and pelvic floor

disorders.

URI

URI usually means upper respiratory infection, commonly a viral infection of the nose, throat, or upper airways causing congestion, cough, and sore throat.

Uric Acid

The end product of purine breakdown in humans, eliminated mainly in urine. Increased levels can contribute to gout or uric-acid kidney stones.

Uric Acid - Blood

Uric acid - blood is a fatty acid or other lipid-related molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Uric Acid Nephropathy

Acute kidney injury from uric acid crystal precipitation in tubules. Occurs in tumor lysis syndrome and severe hyperuricemia. Prevent with hydration and rasburicase. Alkalinization of urine may be harmful as uric acid precipitates at alkaline pH unlike other stone types.

Uric Acid Stones

Kidney stones composed primarily of uric acid. They form more readily in acidic urine and are associated with hyperuricemia, gout, high purine intake, or chronic diarrhea. Some can dissolve when urine is alkalinized.

Uric Acid Urine Test

A urine test measuring uric acid excretion, used in evaluation of gout, kidney stones, uric-acid overproduction, and selected metabolic disorders.

Uricosuric Drug

A medicine that increases renal excretion of uric acid, such as probenecid, used in selected patients with gout.

Uridine

A nucleoside composed of uracil linked to ribose. It is a building block of RNA and a precursor of

uridine nucleotides involved in carbohydrate and membrane metabolism.

Uridine Diphosphate

A nucleotide consisting of uridine and two phosphate groups. UDP participates in glycogen metabolism and transfers activated sugars during glycosylation reactions.

Uridine Monophosphate

A nucleotide consisting of uridine and one phosphate group. UMP is a component of RNA and an intermediate in pyrimidine-nucleotide metabolism.

Uridine Triacetate

An oral prodrug that supplies uridine and is used as an emergency antidote for severe or life-threatening toxicity from fluorouracil or related fluoropyrimidines, under specialist direction.

Uridine Triphosphate

A high-energy uridine nucleotide used in RNA synthesis and in the formation of activated sugar donors for glycogen and glycoprotein production.

Urinal

A specialized, portable, graduated medical receptacle made of molded medical-grade plastic, designed for bedbound, non-ambulatory, or mobility-impaired patients (particularly male patients) to void urine while remaining in bed or seated in a wheelchair without requiring full transfer to a commode or bathroom. Features an anatomically angled neck and wide opening to prevent spillage, volume graduation markings along the transparent plastic body for accurate fluid balance charting (intake and output [I&O] monitoring), and a fitted cap to control odors and reduce surface contamination.

Urinalysis

Urinalysis examines urine appearance, chemistry, and sediment for evidence of infection, kidney disease, bleeding, diabetes, dehydration, or other systemic conditions. (bacteria), protein, glucose, ketones, bilirubin, urobilinogen, blood, and hemoglobin.

Urinary

Relating to urine or to the organs and processes involved in producing and eliminating urine.

Urinary Bladder

A hollow, distensible muscular organ located in the pelvic cavity posterior to the pubic symphysis, serving as a reservoir for urine.

Urinary Bladder Disorder

A disease affecting bladder storage or emptying, including infection, overactivity, pain syndrome, stones, tumor, neurologic dysfunction, or outlet obstruction.

Urinary Bladder, Diseases Of

Disorders of the bladder, including infection, stones, tumors, neurogenic dysfunction, inflammation, obstruction, and impaired emptying.

Urinary Calculi

Stones formed from crystallized minerals in the urinary tract. They may cause colicky pain, hematuria, obstruction, infection, or kidney injury depending on their size and location.

Urinary Casts

Cylindrical structures formed in renal tubules and passed in urine; their type can help identify kidney disease, such as glomerular inflammation or tubular injury.

Urinary Catheter

A hollow, flexible tube inserted through the urethra into the bladder to drain urine; it may be used intermittently or left in place.

Urinary Catheterization

Insertion of a tube into the bladder to drain urine, measure output, obtain a specimen, or deliver treatment; infection, trauma, and blockage are possible complications.

Urinary Catheters

Urinary catheters drain the bladder when a person

cannot empty it normally or accurate urine measurement is needed, with risks including infection, trauma, blockage, and discomfort.

Urinary Diversion

Surgical or catheter-based rerouting of urine when the normal passage from the kidneys to the urethra cannot be used, often to a stoma or bowel reservoir.

Urinary Fistula

An abnormal connection between part of the urinary tract and another organ or the skin, allowing urine to leak outside its normal pathway. Causes include surgery, childbirth injury, inflammation, and cancer.

Urinary Frequency

The need to urinate more often than usual, which may result from infection, bladder irritation, increased urine production, or other urinary disorders.

Urinary Hesitation

Difficulty starting or maintaining urination, which may result from prostate enlargement, urethral narrowing, medications, nerve disorders, pain, or impaired bladder contraction.

Urinary Incontinence

Involuntary leakage of urine, significantly affecting psychological, social, and physical well-being.

Alternate Names:

Bladder Control, Loss Of Incontinence, Urinary Loss Of Bladder Control

Urinary Incontinence Products

Urinary-incontinence products such as pads, protective underwear, catheters, and mattress covers manage leakage but do not treat its underlying cause.

Urinary Lithiasis

The formation or presence of stones anywhere in the urinary tract, including the kidneys, ureters, bladder, or urethra.

Urinary Retention

Inability to completely empty the bladder, which can be acute (sudden, painful, inability to void) or chronic (gradual, often painless, with overflow incontinence). Causes: benign prostatic hyperplasia, urethral stricture, neurogenic bladder, medications (anticholinergics), and pelvic organ prolapse.

Urinary Retraining

Scheduled voiding and behavioral techniques used to improve bladder control and reduce urgency or incontinence.

Urinary Stasis

Slowing or stopping of normal urine flow, which can promote urinary infection, stone formation, or obstruction-related kidney injury.

Urinary System

The kidneys, ureters, bladder, and urethra, which produce, transport, store, and eliminate urine while regulating fluid, electrolytes, acid–base balance, and blood pressure.

Urinary Tract

The kidneys, ureters, bladder, and urethra, which produce, transport, store, and excrete urine.

Urinary Tract Disorder

Any disease of the kidneys, ureters, bladder, or urethra, including infection, stones, obstruction, congenital abnormalities, cancer, and functional or neurologic disorders.

Urinary Tract Infection

Infection of the urinary tract, classified as lower infection such as cystitis or urethritis, or upper infection such as pyelonephritis. Symptoms vary; dysuria, frequency, urgency, fever, flank pain, or nausea may occur. *Escherichia coli* is a common cause, but organisms vary by setting and patient factors.

Alternate Names:

Bladder Infection (Urinary Tract Infection)

Urinary Tract Infection In Pregnancy

A urinary infection or asymptomatic bacterial growth occurring during pregnancy. Pregnancy increases the risk of upper urinary infection, so screening and treatment follow pregnancy-specific clinical guidance.

Urinary Tract Neoplasm

A benign or malignant growth in the kidneys, ureters, bladder, or urethra; evaluation depends on its site, symptoms, tissue type, and extent of spread.

Urinary Urgency

A sudden, difficult-to-delay need to urinate. It may occur with urinary infection, bladder irritation, overactive bladder, stones, neurologic disease, or other conditions. Urgency with fever, flank pain, blood in the urine, or inability to urinate requires medical assessment.

Urination

The biological process of releasing urine from the bladder through the urethra out of the body; also known as micturition.

Alternate Names:

micturition.

Urination Disorder

Abnormal storage or emptying of urine causing frequency, urgency, dysuria, incontinence, hesitancy, weak stream, retention, or excessive or reduced urine output.

Urination, Difficulty

Trouble starting, maintaining, or emptying urine, often described as hesitancy, weak stream, straining, or incomplete emptying; causes include obstruction, infection, medicines, and nerve or muscle disorders.

Urine

The liquid waste produced by the kidneys, containing water and dissolved substances eliminated from the blood.

Urine 24-Hour Volume

Twenty-four-hour urine volume is the total urine produced in a day and helps assess hydration, kidney function, endocrine disease, or abnormal urine production.

Urine Albuminuria

Abnormal urinary loss of albumin resulting from increased glomerular permeability or impaired tubular reclamation. Persistent albuminuria is independently associated with CKD progression and cardiovascular risk, even when eGFR is preserved.

Urine Alkalinization

Deliberate increase of urine pH, usually with intravenous sodium bicarbonate, to enhance elimination of selected weak acids such as salicylate; it requires monitoring of pH, potassium, fluid status, and acid-base balance.

Urine Analysis

Laboratory examination of urine for cells, blood, proteins, glucose, microorganisms, and other substances.

Urine Chemistry

Measurement of chemical constituents in urine, such as glucose, protein, electrolytes, ketones, blood, and pH, to evaluate kidney function and systemic disease.

Urine Concentration Test

A test of the kidney's ability to concentrate urine, assessed by urine osmolality or specific gravity after controlled fluid conditions when appropriate.

Urine Conductivity

Urine conductivity measures how readily urine carries electrical current and reflects its dissolved electrolyte and solute concentration.

Urine Culture

A laboratory test that grows microorganisms from a urine specimen to identify urinary-tract infection and guide antimicrobial selection. Results may re-

port the organism, quantity of growth, and antimicrobial susceptibility. Proper collection is important because contamination can produce misleading growth; results must be interpreted with symptoms and urinalysis.

Urine Culture - Catheterized Specimen

A urine culture performed on a specimen collected through urinary catheterization, usually when a clean-catch sample cannot be obtained or when catheter-associated infection is being assessed. Collection technique and timing affect interpretation because colonization and contamination may cause bacterial growth without symptomatic infection.

Urine Cytology

Microscopic examination of cells shed into urine, mainly used to help detect high-grade urothelial carcinoma and some other abnormalities. A negative result does not exclude all urinary-tract cancers.

Urine Drainage Bags

Urine drainage bags collect urine from a catheter or urinary diversion and require correct positioning, emptying, hygiene, and monitoring for blockage or infection.

Urine Formation

The process by which the kidney produces urine through three steps: glomerular filtration (formation of 180 L/day ultrafiltrate), tubular reabsorption (return of 99 percent of water and most solutes), and tubular secretion (active addition of waste and ions).

Urine HCG (Pregnancy Test)

Rapid point-of-care test detecting human chorionic gonadotropin in urine. Sensitivity: detects as low as 25 mIU/mL. Positive as early as 12-14 days after conception (before missed period). Results in 3-5 minutes. False negative: dilute urine, very early pregnancy, ectopic pregnancy.

Alternate Names:

Pregnancy Test

Urine Hemoglobin

Hemoglobin detected in urine, which may reflect intravascular hemolysis or contamination with intact red blood cells. Interpretation requires comparison with urine microscopy and clinical findings.

Urine Osmolality

Concentration of urine relative to plasma normally 50 to 1200 mOsm/kg. High greater than 500 in prerenal states reflecting intact concentrating ability. Low less than 350 indicates impaired concentrating ability in diabetes insipidus, chronic kidney disease, and excessive fluid intake. More precise than specific gravity.

Urine PH

A measurement of urine acidity or alkalinity. It varies with diet, metabolism, medications, infection, and kidney function and can influence the formation of some urinary stones.

Urine PH Test

A test measuring urine acidity or alkalinity, useful in evaluating acid-base handling, urinary infection, kidney stones, and some metabolic conditions.

Urine Protein Dipstick Test

A screening test using a reagent strip to detect mainly albumin in urine; positive results require confirmation and quantitative assessment when clinically indicated.

Urine Protein Electrophoresis Test

A laboratory test that separates urine proteins by electrical charge and size to help detect abnormal proteins, including monoclonal light chains associated with plasma-cell disorders.

Urine Retention

Inability to empty the urinary bladder completely or at all. Acute retention can cause severe pain and requires prompt medical assessment.

Urine Sediment

Microscopic examination of centrifuged urine pro-

viding diagnostic information about renal pathology. Normal findings include few RBCs, WBCs, and hyaline casts.

Urine Sodium

Concentration of sodium in urine measured in mEq/L. Low less than 20 in prerenal states reflecting avid tubular reabsorption. High greater than 40 in intrinsic renal disease, diuretic use, and salt-wasting nephropathies. Should be interpreted in context of volume status, medications, and concurrent therapies.

Urine Specific Gravity Test

A test estimating urine density relative to water and reflecting solute concentration; results help assess hydration and renal concentrating ability.

Urine Toxicology

Laboratory analysis of urine for medicines, drugs, or their metabolites. Results require interpretation because detection does not always establish impairment or the timing of exposure.

Urinometer

A hydrometer used to measure the specific gravity of urine.

Urobilinogen

A colourless product formed in the intestine from bilirubin and partly reabsorbed and excreted in urine; abnormal levels may provide clues to liver disease or haemolysis.

Urocele

An uncommon term for a cystic dilation or protrusion involving a urinary-tract structure; the precise meaning depends on the anatomical context.

Urochordata

A subphylum of chordates, also called tunicates, whose members include sea squirts. They are primarily biological research organisms rather than human pathogens.

Urodynamic Testing

Suite of diagnostic procedures designed to evaluate the physiological function of the lower urinary tract (bladder and urethral sphincter) during filling, storage, and emptying phases.

Urodynamics

A set of tests that assess the function of the lower urinary tract (bladder and urethra).

Uroflow

Measurement of the rate and pattern of urine flow during voiding.

Uroflowmetry

A noninvasive test measuring the rate and pattern of urine flow during voiding to evaluate obstruction, weak detrusor function, or voiding dysfunction. and testing conditions; a low or abnormal flow may reflect obstruction, weak bladder contraction, or inadequate effort and requires clinical correlation.

Urogenital

Relating to the urinary and reproductive systems or to structures that involve both systems.

Urogenital Fistula

An abnormal connection between the urinary and genital tracts, such as a vesicovaginal or ureterovaginal fistula. It commonly causes continuous urinary leakage and requires specialist evaluation.

Urogenital System

The related urinary and reproductive organs and their associated ducts, vessels, and supporting tissues. The systems share developmental origins and anatomic relationships.

Urogenital Tract

The combined urinary and reproductive organs and passages, which develop from related embryologic structures.

Urography

Imaging of the urinary tract, traditionally us-

ing radiography after administration of a contrast medium.

Urokinase

A plasminogen activator produced by the body and also used as a thrombolytic medicine to help dissolve blood clots.

Urolithiasis

Formation of stones in the urinary tract, including the kidneys, ureters, bladder, or urethra. Stone composition varies by population and may include calcium oxalate, calcium phosphate, uric acid, struvite, or cystine.

Urologic Diagnostic Yield

The proportion of urologic diagnostic procedures or tests that provide a definitive and clinically useful result.

Urologist

A physician specializing in urinary-tract disorders and the male reproductive system.

Urology

Surgical and medical specialty concerned with the study, diagnosis, and management of diseases affecting the male and female urinary tract (kidneys, ureters, urinary bladder, urethra) and the male reproductive system (prostate, testes, epididymis, seminal vesicles, penis).

Urosepsis

Sepsis caused by infection of the urinary tract or urinary system, often associated with pyelonephritis, obstruction, catheterization, or an infected urinary calculus. The infection may progress from a localized urinary source to bacteremia, systemic inflammation, organ dysfunction, and septic shock. Clinical findings may include fever or hypothermia, flank pain, dysuria, hypotension, altered mental status, oliguria, or elevated lactate. Diagnosis combines clinical assessment with urine and blood cultures, laboratory testing, and imaging when obstruction or a complicated source is suspected.

Urostomy Leak

Escape of urine beneath or around a urostomy appliance or from a surgical connection. Persistent leakage can irritate the skin and may signal poor appliance fit, obstruction, infection, or a surgical complication.

Urostomy Procedure

Surgery that diverts urine through a stoma into an external pouch or conduit, commonly after bladder removal or when the lower urinary tract cannot safely transport urine.

Urostomy Prolapse

Protrusion of a segment of bowel used as a urostomy stoma beyond its usual level. It may cause bleeding, swelling, appliance problems, or impaired urine drainage and should be assessed clinically.

Urostomy Stenosis

Narrowing of a urinary stoma or conduit that impedes drainage, causing reduced urine output, leakage, infection, or kidney-pressure changes and sometimes requiring revision.

Urothelial Carcinoma

Urothelial carcinoma, previously called transitional cell carcinoma, is the most common malignancy of the urinary tract, arising from the urothelium that lines the renal pelvis, ureters, bladder, and proximal urethra.

Urothelium

The specialized transitional epithelium lining the renal pelvis, ureters, bladder, and part of the urethra.

Ursodeoxycholic Acid

A hydrophilic bile acid used for primary biliary cholangitis and selected cholestatic liver diseases; it may also dissolve certain small cholesterol gallstones when bile flow is preserved.

Urticaria

A skin reaction causing transient, itchy raised welts

that change location and usually resolve within 24 hours. It may occur with angioedema, which is deeper swelling often affecting the lips, eyelids, tongue, or extremities.

Urticaria Multiforme

A hypersensitivity reaction in children, often triggered by infections or medications, characterised by widespread, annular or polycyclic erythematous plaques with dusky, ecchymotic, or targetoid centres and pseudovesication, differing from true urticaria by the presence of dusky discolouration and prolonged lesion duration.

Urticaria Pigmentosa

A cutaneous form of mastocytosis in which excessive mast cells accumulate in the skin, producing tan or brown macules or papules that may swell and itch when rubbed. Flushing, abdominal symptoms, or systemic involvement can occur.

Urticarial Vasculitis

Inflammation of small blood vessels associated with hive-like lesions that often last longer than ordinary urticaria and may burn, bruise, or leave residual discoloration.

Usher Syndrome

An inherited disorder combining sensorineural hearing loss with retinitis pigmentosa, the most common cause of combined deaf-blindness. Three types are distinguished by severity and age of onset; management requires coordinated audiological, ophthalmological, and genetic care.

Ustekinumab

Ustekinumab is a monoclonal antibody blocking interleukin-12 and interleukin-23 signaling, used for psoriasis, psoriatic arthritis, inflammatory bowel disease, and other approved conditions; infection risk and hypersensitivity require monitoring.

Ustilago

A genus of smut fungi that infect plants, especially grasses and cereals. Human infection is rare and generally occurs as an opportunistic infection in

people with impaired immunity.

Usual Interstitial Pneumonia

The histopathological pattern in IPF with temporal heterogeneity, fibroblast foci, and honeycombing. HRCT shows basal-predominant, subpleural reticular opacities with traction bronchiectasis.

Ut Dict.

An abbreviation of the Latin phrase “ut dictum,” meaning “as stated” or “as directed.” In medical notes it refers the reader to information already given or to an instruction previously specified.

Uterine (Fallopian) Tubes

Paired muscular tubes extending from the uterus toward the ovaries. They transport the oocyte and are the usual site of fertilization; blockage can contribute to infertility or ectopic pregnancy.

Alternate Names:

fallopian tubes; uterine tubes

Uterine Adnexae

The uterine adnexa are structures beside the uterus, chiefly the fallopian tubes, ovaries, and supporting tissues.

Uterine Atony

Failure of the uterus to contract adequately after delivery, resulting in postpartum hemorrhage. The most common cause of PPH. Risk factors include prolonged labor, multiple gestation, macrosomia, oxytocin augmentation, and uterine fibroids. Diagnosis is clinical: boggy, soft uterus with excessive bleeding.

Uterine Balloon Tamponade

A minimally invasive technique for managing postpartum hemorrhage due to uterine atony when uterotonics fail. A balloon (Bakri Postpartum Balloon, 500-1000 mL capacity) is inserted transvaginally into the uterine cavity and inflated with saline until bleeding stops. The balloon provides mechanical compression against the uterine wall.

Uterine Bleeding

Uterine bleeding is bleeding from the uterus and may result from menstruation, pregnancy complications, fibroids, polyps, hormonal disorders, cancer, or medicines.

Uterine Cancer

Cancer arising in the uterus, most commonly from the endometrium. Abnormal vaginal bleeding, bleeding after menopause, pelvic pain, or unusual discharge requires evaluation; early diagnosis often permits effective treatment.

Uterine Cavity

The uterine cavity is the internal space of the uterus lined by endometrium and is where implantation and pregnancy occur.

Uterine Contraction

Tightening of the uterine muscle, occurring normally during menstruation and labor and sometimes abnormally with pain, pregnancy complications, or medication effects.

Uterine Fibroid (Leiomyoma)

A benign tumor arising from the smooth muscle of the uterus. Fibroids may be submucosal, intramural, or subserosal and may cause heavy menstrual bleeding, pelvic pressure, pain, urinary symptoms, or reproductive problems.

Alternate Names:

Leiomyoma

Uterine Fibroid Embolization

A minimally invasive procedure that blocks the arteries supplying uterine fibroids, causing them to shrink.

Uterine Fibroids

Benign tumors of uterine smooth muscle, also called leiomyomas. They may cause heavy menstrual bleeding, pelvic pressure, pain, infertility, or no symptoms.

Alternate Names:

leiomyomas. Polyps Uterus (Uterine Fibroids) Tu-

mors Uterine (Uterine Fibroids) Womb Growths (Uterine Fibroids)

Uterine Fistula

A uterine fistula is an abnormal passage connecting the uterus with another organ or the skin, often after surgery, childbirth injury, infection, or cancer.

Uterine Fornix

The uterine or vaginal fornix is a recess around the cervix where the upper vagina surrounds the projecting lower uterus.

Uterine Gland

A small gland in the lining of the uterus that produces fluid and helps support the early embryo. Its activity changes during the menstrual cycle and pregnancy.

Uterine Growths

Uterine growths may include fibroids, polyps, adenomyosis, or rarely cancer. They can cause heavy or irregular bleeding, pelvic pressure, pain, infertility, or no symptoms; examination and imaging determine the type and treatment.

Uterine Hemorrhage

Heavy or abnormal bleeding from the uterus, occurring during pregnancy, after childbirth, or at other times. Causes include pregnancy complications, fibroids, hormonal disorders, and malignancy; severe bleeding with dizziness or weakness is an emergency.

Uterine Inversion

A rare but life-threatening obstetric emergency where the uterus turns inside out, protruding through the cervix or vaginal introitus. Occurs in 1:2000-1:20,000 deliveries. Causes: fundal pressure, excessive cord traction (especially with fundal placenta), uterine atony, and fundal fibroids.

Uterine Leiomyosarcoma

A rare malignant tumor of uterine smooth muscle that may cause abnormal bleeding, pelvic pain, or an enlarging mass; diagnosis is based on tissue

examination and staging.

Uterine Monitoring

Assessment of uterine contractions and, in pregnancy, their frequency, duration, and strength to evaluate labor progress, induction response, or excessive uterine activity.

Uterine Myomectomy

Surgical removal of uterine fibroids while preserving the uterus; it may be performed through hysteroscopy, laparoscopy, or open surgery depending on fibroid size and location.

Uterine Obstruction

Blockage that prevents normal passage through the uterus or cervix, which may cause menstrual outflow obstruction, infertility, pregnancy complications, or retained blood or tissue.

Uterine Perforation

A full-thickness breach of the uterine wall, most commonly occurring during instrumentation such as dilation and curettage or hysteroscopy, potentially causing hemorrhage and requiring surgical repair.

Uterine Polyps

Growths arising from the inner lining of the uterus that may cause irregular bleeding, heavy periods, or fertility problems.

Alternate Names:

Polyps, Endometrial Polyps, Uterine

Uterine Prolapse

Descent of the uterus toward or through the vaginal opening because the supporting pelvic floor and ligaments have weakened. Risk factors include childbirth, aging, obesity, and conditions that increase chronic pressure in the abdomen.

Alternate Names:

Pelvic Support Problems, Uterine Prolapse Proapsed Uterus

Uterine Rupture

A full-thickness tear through the uterine wall, po-

tentially life-threatening to both mother and fetus. Risk factors include prior caesarean section (especially classical incision), oxytocin augmentation, and operative vaginal delivery. Presents with sudden abdominal pain, loss of fetal heart rate, and maternal hemorrhage.

Uterine Sarcoma

Uterine sarcoma is a rare malignant tumor arising from uterine muscle, connective tissue, or related cells and may cause bleeding, pain, or an enlarging mass.

Uterine Tachysystole

More than five uterine contractions in 10 minutes averaged over 30 minutes, whether or not accompanied by fetal heart-rate abnormalities. It can reduce uteroplacental perfusion, particularly when caused by oxytocin or prostaglandins.

Uterine Tube

A paired muscular tube that transports the oocyte from the ovary toward the uterus and is the usual site of fertilization.

Utero-Ovarian Ligament

A round, cord-like fibromuscular band extending from the lower, medial (uterine) pole of the ovary to the superolateral angle of the uterus (uterine cornu), immediately posterior and inferior to the insertion of the fallopian tube. Lies enclosed within the leaves of the broad ligament and anchors the ovary to the uterus without containing significant vascular structures.

Alternate Names:

Ovarian Ligament; Proper Ovarian Ligament

Uterosacral Ligament

Paired strong fibromuscular condensations of pelvic endopelvic fascia extending posterolaterally from the supravaginal cervix and upper vaginal fornices, curving around the rectum, to insert into the presacral fascia overlying the second to fourth sacral vertebrae (S_2 – S_4). Provides vital apical support to keep the cervix in its normal posterior position; represents an essential anatomical structure targeted

in surgical apical suspension for uterine prolapse, and a frequent site for painful deep infiltrating endometriosis nodules.

Uterus

Hollow muscular organ in the pelvis between the bladder and rectum, approximately 7.5 cm long in adult females, responsible for menstruation, implantation, and fetal development during pregnancy.

Common name: Womb

Uterus Didelphys

A congenital Müllerian anomaly resulting from the complete failure of fusion of the two paramesonephric ducts during early embryogenesis. Produces two completely separate, symmetrical uterine bodies (hemi-uteruses), each possessing its own cervix, and almost invariably accompanied by a longitudinal vaginal septum (double uterus and double vagina). Associated with an increased risk of preterm labor, malpresentation, and cesarean delivery, though fertile pregnancies can occur in either or both horns.

Alternate Names:

Double Uterus; Didelphic Uterus

Uterus Disorder

A condition affecting the uterus, including fibroids, adenomyosis, endometriosis, infection, abnormal bleeding, congenital anomalies, prolapse, and cancer.

Uterus Neoplasm

A benign or malignant growth arising in the uterus, including fibroids, endometrial cancer, and sarcomas; diagnosis depends on examination, imaging, and tissue sampling.

Uterus, Diseases Of

Disorders affecting the uterus, including fibroids, adenomyosis, endometriosis, infection, abnormal bleeding, prolapse, and cancer.

UTI

Infection of the urinary tract, classified as lower

(cystitis, urethritis) or upper (pyelonephritis). See Urinary Tract Infection.

Utricle

One of the otolith organs of the inner ear, detecting horizontal linear acceleration and head position relative to gravity.

UV Light

Ultraviolet radiation, an invisible part of sunlight or artificial light that can affect the skin and eyes and may cause sunburn, aging, or skin cancer.

Unsaturated Fat

A dietary fat whose fatty-acid chains contain one or more carbon-carbon double bonds. Monounsaturated and polyunsaturated fats are common in vegetable oils, nuts, seeds, and fish; replacing saturated fat with unsaturated fat generally supports healthier LDL cholesterol levels.

Uvea

The uvea is the vascular middle layer of the eye, comprising the iris, ciliary body, and choroid; it regulates light entry, supports the lens, and supplies the outer retina. focus, and blood supply to the eye.

Uveal Disease

A disorder of the iris, ciliary body, or choroid, including uveitis, tumors, vascular disease, and inflammation that may cause eye pain, photophobia, floaters, or visual loss.

Uveal Melanoma

The most common primary intraocular malignancy in adults, arising from melanocytes within the uveal tract (choroid, ciliary body, or iris), presenting with visual symptoms, a pigmented intraocular mass, or incidental finding on examination.

Uveal Neoplasm

A tumor arising in the uveal tract of the eye—the iris, ciliary body, or choroid; malignant melanoma may threaten vision and spread to distant organs.

Uveitis

Inflammation of the uveal tract—the iris, ciliary body, and choroid—and sometimes nearby eye tissues. It may result from infection, autoimmune or other inflammatory disease, trauma, or an unknown cause. Symptoms can include eye pain, redness, light sensitivity, floaters, blurred vision, or reduced vision; prompt eye assessment is important because untreated inflammation can damage vision.

Alternate Names:

Chorioretinitis (Uveitis)

Uveoparotid Fever

A manifestation of sarcoidosis characterized by the triad of bilateral parotid gland enlargement, anterior uveitis, and fever, sometimes associated with facial nerve palsy. It reflects granulomatous inflammation and is treated with systemic corticosteroids and potentially disease-modifying agents.

Uvula

A small, fleshy flap of tissue that hangs from the back of the throat over the root of the tongue.

Uvulitis

Uvulitis is inflammation and swelling of the uvula from infection, trauma, allergy, dehydration, or irritation and may interfere with swallowing or airway protection. dehydration, or irritation. It may cause throat pain, difficulty swallowing, gagging, or a foreign-body sensation; marked swelling with breathing difficulty requires urgent assessment.

Uvulopalatopharyngoplasty

Surgery that removes or reshapes the uvula, tonsils, and soft-palate tissue to treat obstructive sleep apnea.

Alternate Names:

UPPP

Uvulopalatopharyngoplasty

A surgical procedure that removes or reshapes selected tissue of the soft palate, uvula, and pharynx to enlarge the upper airway. It may be used in selected patients with obstructive sleep apnea, but effectiveness and risks depend on airway anatomy

and other treatments.

V

V-Fib

Alternate index label; see *Ventricular fibrillation*.

Vaccination

Administration of a vaccine to stimulate immune response and provide protection against infectious disease. Types: live attenuated, inactivated, subunit, mRNA, viral vector. Schedule varies by country. Herd immunity achieved when sufficient proportion immune.

Vaccination Schedule

A time-based plan for administering vaccines according to age, previous doses, medical conditions, exposure risks, and public-health guidance. Schedules vary by country and are updated as recommendations change.

Vaccines

Biological preparations that stimulate active adaptive immunity against specific infectious agents or their toxins. They reduce the risk of infection, severe disease, complications, and transmission.

Vaccines - Immunizations

Vaccines train the immune system to recognize specific pathogens or toxins and reduce the risk of infection, severe disease, complications, and transmission.

Vaccinia

The orthopoxvirus used in the smallpox vaccine; infection is uncommon and may cause a localized or disseminated lesion, especially in people with impaired immunity.

Vaccinia Virus

Vaccinia virus is a poxvirus used to make smallpox vaccines and studied in some experimental cancer treatments; accidental infection is uncommon but possible.

VACTERL Association

A pattern of congenital abnormalities involving vertebral, anal, cardiac, tracheoesophageal, renal, and limb development. The diagnosis is considered when several component abnormalities occur together and other syndromes have been excluded.

Vacuole

A membrane-bound compartment within a cell containing fluid, solutes, or cellular material. Vacuoles have specialized roles in storage, transport, and degradation and may enlarge or accumulate material during disease.

Vacuum Curettage

A procedure that removes tissue from the uterus using suction, commonly for uterine evacuation or endometrial sampling. Risks include bleeding, infection, retained tissue, and rarely uterine injury.

Vacuum Delivery

Assisted vaginal delivery using a vacuum extractor (cup applied to the fetal scalp with negative pressure) to expedite delivery for prolonged second stage, fetal distress, or maternal indications.

Vacuum Erection Devices

Pumps that draw blood into the penis to produce an erection.

Alternate Names:

Vacuum erection pump

Vacuum Extractor

A cup-shaped device applied to the fetal scalp to assist vaginal delivery when specific maternal or fetal indications are present.

Vacuum-Assisted Delivery

Operative vaginal delivery using a vacuum extractor applied to the fetal scalp. Types: soft cup (Kiwi OmniCup, Mityvac, for outlet and low deliveries) and rigid cup (metal Kiwi, for posterior positions and midpelvic deliveries).

Vadimezan

An investigational vascular-disrupting and tumor-necrosis-factor-inducing compound studied as an anticancer agent. It is not an established routine treatment and has no general clinical indication.

Vagal Paraganglioma

A usually slow-growing tumor arising from paraganglion cells near the vagus nerve in the neck.

Vagina

A muscular, elastic canal connecting the cervix to the vaginal opening. It allows menstrual blood to leave the body, receives a penis or other objects during vaginal sex, and serves as the birth canal.

Vagina Carcinoma

A malignant neoplasm arising from the vaginal epithelium; it may present with abnormal bleeding, discharge, or a visible lesion and requires evaluation for local and regional spread.

Vagina Neoplasm

A tumor arising in the vagina; diagnosis determines whether it is benign or malignant, its tissue type, extent, and appropriate management.

Vaginal Agenesis

A condition in which the vagina is absent or incompletely developed.

Vaginal Atresia

Congenital absence or complete obstruction of the vaginal canal. It may cause absent menstrual flow, cyclic pelvic pain, or difficulty with sexual activity and requires specialist evaluation.

Vaginal Atrophy

Thinning, drying, and inflammation of vaginal tissues caused by reduced estrogen.

Vaginal Bleeding

Unexpected, unusually heavy, or prolonged bleeding from the vagina, including bleeding between periods, after sexual intercourse, during pregnancy, or after menopause. Causes include hormonal changes, infection, pregnancy-related conditions, structural lesions, medications, and cancer.

Postcoital Bleeding

Bleeding from the vagina after sexual intercourse. Causes include infection, dryness, injury, polyps, pregnancy-related conditions, or cancer.

Vaginal Bleeding Between Periods

Bleeding between menstrual periods may result from contraception, infection, pregnancy, polyps, fibroids, hormonal changes, or cancer and requires evaluation if persistent or heavy.

Vaginal Bleeding During Early Pregnancy

Bleeding from the vagina during the first part of pregnancy can result from cervical changes, miscarriage, ectopic pregnancy, infection, or other conditions; a small amount of bleeding may occur without a serious cause, but it should not be assumed to be implantation bleeding. Heavy bleeding, severe or one-sided pain, shoulder pain, dizziness, or fainting requires urgent assessment.

Vaginal Bleeding During Late Pregnancy

Bleeding from the vagina later in pregnancy may be caused by placenta previa, placental abruption, labor, cervical disease, or other complications. Any significant late-pregnancy bleeding requires immediate obstetric assessment.

Vaginal Bleeding In Early Pregnancy

Alternate index label; see *Vaginal Bleeding During Early Pregnancy*.

Vaginal Bleeding In Late Pregnancy

Alternate index label; see *Vaginal Bleeding During Late Pregnancy*.

Vaginal Bleeding In Pregnancy

Broad index label for bleeding during pregnancy; see *Vaginal Bleeding During Early Pregnancy* and *Vaginal Bleeding During Late Pregnancy*. Any bleeding during pregnancy should be assessed clinically, especially if heavy or accompanied by pain, dizziness, fainting, or reduced fetal movement.

Vaginal Cancer

A rare gynecologic malignancy, most commonly squamous cell carcinoma linked to human papillomavirus. Treatment depends on stage and includes surgery, radiation, and chemotherapy.

Alternate Names:

Cancer, Vagina

Vaginal Candidiasis

Alternate index label; see *Yeast infection (vaginal)*.

Vaginal Cysts

Vaginal cysts are fluid-filled or tissue-lined sacs in or beside the vaginal wall, commonly benign but sometimes causing pain, pressure, infection, or obstruction.

Vaginal Delivery (Spontaneous)

The spontaneous passage of the fetus through the birth canal without the use of operative instruments (vacuum or forceps). Typically occurs in the second stage of labor with maternal pushing.

Alternate Names:

Spontaneous

Vaginal Dryness

Reduced vaginal lubrication, commonly associated with menopause, breastfeeding, hormonal changes, medications, or irritation. It may cause discomfort,

itching, painful intercourse, or minor bleeding.

Vaginal Dryness And Atrophy (Genitourinary Syndrome Of Menopause)

Vaginal dryness and atrophy, collectively termed genitourinary syndrome of menopause, result from declining estrogen levels that cause thinning, drying, and loss of elasticity of the vaginal tissues. Symptoms include vaginal dryness, irritation, burning, dyspareunia, and urinary symptoms such as frequency and recurrent infections.

Alternate Names:

Genitourinary Syndrome of Menopause

Vaginal Fistula

An abnormal opening connecting the vagina to another organ, such as the bladder or rectum.

Vaginal Foreign Body

An object retained in the vagina, most often a tampon or other inserted material. It may cause discharge, odor, bleeding, pain, or infection and should be removed safely with examination when necessary.

Vaginal Fornix

One of the recesses surrounding the cervix where the vagina meets the cervix. The posterior fornix is adjacent to the peritoneal cul-de-sac and is relevant in pelvic examination and some procedures.

Vaginal Hematoma

A collection of blood within the vaginal wall or surrounding tissues, usually caused by childbirth, trauma, or surgery and potentially producing severe pain, swelling, or blood loss.

Vaginal Hemorrhage

Heavy or clinically significant bleeding from the vagina. Causes include childbirth, pregnancy complications, trauma, infection, structural disease, hormonal disorders, and gynecologic malignancy; severe or pregnancy-related bleeding requires urgent evaluation.

Vaginal Hysterectomy

Surgical removal of the uterus through the vagina. It may be used for selected benign uterine conditions and usually avoids an abdominal incision, although suitability depends on anatomy and disease.

Vaginal Intercourse

The insertion of the penis into the vagina for sexual pleasure or reproduction. The most common form of sexual intercourse. Vaginal intercourse can lead to pregnancy if sperm are deposited in the vagina.

Vaginal Intraepithelial Neoplasia

Precancerous abnormal growth of squamous cells in the vaginal epithelium, usually associated with human papillomavirus and graded according to the severity of dysplasia.

Vaginal Introitus

The entrance to the vaginal canal, located in the vestibule of the vulva. Its appearance and tenderness may be relevant in examinations for infection, trauma, pain, or pelvic-floor disorders.

Vaginal Itching

Itching or irritation of the vulva or vagina caused by candidiasis, bacterial vaginosis, sexually transmitted infections, dermatitis, hormonal changes, or other conditions. It may occur with discharge, odor, soreness, or burning; recurrent or unexplained symptoms require examination and appropriate testing rather than self-diagnosis.

Vaginal Membrane

A thin mucosal or connective-tissue fold associated with the vaginal canal; the term may refer to the hymen at the vaginal opening or to an abnormal obstructing membrane, so the intended anatomical meaning should be specified.

Vaginal Microbiome

The community of microorganisms living in the vagina, commonly dominated by *Lactobacillus* species that help maintain an acidic environment. Changes in this community may contribute to bacterial vaginosis, candidiasis, or other symptoms.

Vaginal Odor

An unusual or unpleasant odor from vaginal discharge that may result from bacterial vaginosis, infection, retained material, hormonal changes, or other conditions.

Vaginal Opening

The external entrance to the vagina. It is part of the vulvar vestibule and may be affected by infection, inflammation, trauma, scarring, or congenital variation.

Vaginal Prolapse, Posterior

Alternate index label; see *Posterior vaginal prolapse (rectocele)*.

Vaginal Speculum

An instrument inserted into the vagina and gently opened to separate the vaginal walls and visualize the vagina and cervix. It may be used for pelvic examination, cervical cytology such as a Pap test, swab collection, assessment of bleeding or discharge, and selected procedures. Size and design are chosen according to anatomy, comfort, and the clinical purpose.

Vaginal Stenosis

Narrowing or shortening of the vaginal canal, which may result from surgery, radiation, scarring, inflammation, trauma, or congenital development. It can cause pain, difficult examination, or difficulty with penetration.

Vaginal Trauma

Injury to the vaginal tissues caused by childbirth, sexual activity, foreign objects, procedures, or blunt or penetrating trauma. Bleeding, pain, swelling, or urinary symptoms require assessment; severe bleeding or suspected deep injury is urgent.

Vaginal Vault Prolapse

The descent of the apex (cuff or top) of the vagina into or beyond the vaginal introitus following a total hysterectomy, resulting from the detachment or weakening of the cardinal and uterosacral ligament complexes and levator ani muscles. Symptoms in-

clude a sensation of pelvic heaviness, a visible or protruding vaginal bulge, lower back ache, and obstructive voiding or defecatory dysfunction. Management options include pelvic floor muscle training, supportive pessaries, or surgical reconstructive procedures such as sacrocolpopexy or uterosacral/sacrospinous ligament fixation.

Alternate Names:

Post-Hysterectomy Vault Prolapse; Apical Prolapse

Vaginal Vestibule

The area of the vulva enclosed by the labia minora that contains the openings of the urethra and vagina and the ducts of several glands.

Vaginal Yeast Infection

A vaginal yeast infection is usually *Candida* overgrowth causing itching, burning, soreness, and thick discharge without the odor typical of many bacterial infections.

Vaginectomy

Surgical removal of part or all of the vagina, usually to treat selected cancers or severe disease; effects may include changes in sexual function, fertility, urination, and bowel function.

Vaginismus

Involuntary tightening of the muscles around the vaginal opening that can make penetration painful or impossible. It may be associated with pelvic-floor dysfunction, fear, prior pain, or other factors and is treatable with a tailored multidisciplinary approach.

Vaginitis

Inflammation of the vaginal mucosa, one of the most common gynecologic complaints.

Vaginospasm

Involuntary contraction of the pelvic-floor muscles surrounding the vagina, causing pain or difficulty with vaginal penetration. It may be associated with fear of pain, prior trauma, pelvic-floor dysfunction, infection, hormonal factors, or other genitopelvic pain conditions.

Vagotomy

Surgical cutting of selected vagus-nerve branches, historically used to reduce stomach acid secretion.

Vagus

Relating to the vagus nerve, the tenth cranial nerve, which carries parasympathetic, sensory, and motor fibers between the brainstem and thoracic and abdominal organs.

Vagus Nerve

The tenth cranial nerve, carrying parasympathetic, sensory, and motor fibres between the brainstem and the heart, lungs, gastrointestinal tract, pharynx, and larynx.

Vagus Nerve (Cranial Nerve X)

The tenth cranial nerve (CN X), the longest and most widely distributed cranial nerve in the body. It contains sensory, motor, and parasympathetic fibers, exiting the skull via the jugular foramen.

Alternate Names:

Cranial Nerve X

Vagus Nerve Stimulation

A neuromodulation treatment involving electrical stimulation of the left vagus nerve via an implanted device. It is FDA-approved for treatment-resistant depression and epilepsy. It requires surgical implantation and delivers intermittent stimulation. Improvement may take months to develop.

Valacyclovir

Valacyclovir is the oral prodrug of acyclovir that inhibits herpesvirus DNA polymerase after activation; it treats herpes simplex and shingles, with renal dosing and neurotoxicity risk in accumulation.

Valganciclovir

Valganciclovir is the oral prodrug of ganciclovir used mainly to treat or prevent cytomegalovirus infection after transplantation; myelosuppression, renal toxicity, infertility, and teratogenicity are important risks.

Valgus

A deformity in which the distal part of a limb deviates away from the body's midline; examples include hallux valgus and genu valgum.

Validity

The extent to which a test or measurement accurately assesses what it is intended to assess.

Valine

An essential branched-chain amino acid required for protein synthesis and metabolism.

Valinemia

A rare inherited disorder of branched-chain amino-acid metabolism causing elevated plasma valine. It may present with developmental delay, feeding difficulty, hypotonia, or neurological symptoms, although some individuals are minimally affected.

Valinomycin

A potassium-carrying ionophore produced by some bacteria and used mainly in laboratory research; it is not a routine human medicine and can be toxic to cells.

Valley Fever

The common name for coccidioidomycosis, a fungal infection caused by *Coccidioides* species and acquired by inhaling airborne spores from soil. It usually causes a mild respiratory illness but can disseminate to skin, bone, joints, or meninges, especially in people at higher risk.

Valproate (Sodium Valproate)

Valproate is a broad-spectrum antiseizure medicine used for several generalized and focal seizure disorders and for selected mood disorders. Choice requires consideration of age, pregnancy potential, liver disease, interactions, and other safety risks.

Alternate Names:

Sodium Valproate

Valproic Acid

valproic acid is a fatty acid or other lipid-related

molecule whose chain structure influences membranes, energy storage, signaling, and metabolism.

Valrubicin

An anthracycline antineoplastic drug administered into the bladder for selected cases of carcinoma in situ that have not responded adequately to intravesical bacillus Calmette–Guérin. It can irritate the bladder and may cause systemic toxicity.

Valsalva

Forced expiration against a closed glottis, which raises intrathoracic pressure. The maneuver affects venous return and heart rate and is used in selected examinations and treatments, but may be unsafe in some conditions.

Valsalva Maneuver

Alternate index label; see *Valsalva's Manoeuvre* and *Valsalva*.

Valsalva'S Manoeuvre

Forced expiratory effort against a closed airway (glottis closed or mouth and nose pinched shut), typically performed by bearing down as if straining during a bowel movement. The manoeuvre has four phases: Phase I (onset of straining): increased intrathoracic pressure compresses the aorta, causing a transient rise in blood pressure.

Alternate Names:

Valsalva Maneuver

Valsartan

Valsartan is an angiotensin-II receptor blocker used for hypertension, heart failure, and selected cardiovascular indications; hyperkalaemia, renal-function changes, hypotension, and fetal toxicity are important risks.

Valves

Structures that permit fluid or blood to move predominantly in one direction and prevent backward flow; cardiac valves and venous valves are important examples.

Valvotomy

Surgical or catheter-based division of narrowed valve tissue to improve blood flow, most often performed for selected stenotic heart valves.

Valvular Disease

Disease affecting one or more heart valves, causing stenosis, regurgitation, or both and potentially leading to heart failure or arrhythmia.

Valvular Heart Disease

Valvular heart disease involves dysfunction of one or more heart valves, leading to stenosis, regurgitation, or a combination of both. The most common causes are degenerative calcification of aortic and mitral valves, rheumatic heart disease, infective endocarditis, and ischemic papillary muscle dysfunction.

Valvular Insufficiency

Incomplete closure of a heart valve allowing retrograde blood flow (regurgitation); affects mitral, aortic, tricuspid, or pulmonic valves.

Valvuloplasty

A procedure used to widen or repair a narrowed or abnormal heart valve, by balloon catheter or open surgery depending on the valve and condition.

Valvulotomy

Surgical incision of a stenosed heart valve to improve the opening through which blood flows.

Van Den Bergh Test

A historical test measuring the reaction of serum bilirubin with a diazo reagent to help distinguish conjugated from unconjugated hyperbilirubinemia.

Vancomycin

A glycopeptide antibiotic that inhibits Gram-positive bacterial cell-wall synthesis. It is given intravenously for serious resistant infections and orally for *C. difficile* colitis; infusion reactions, nephrotoxicity, and ototoxicity require caution.

Vancomycin-Resistant Enterococci -

Hospital

Vancomycin-resistant enterococci are enterococcal bacteria resistant to vancomycin and capable of causing healthcare-associated urinary, bloodstream, wound, or other infections.

Vancomycin-Resistant Enterococcus

Enterococcus bacteria that are resistant to vancomycin, usually because of altered cell-wall precursors. VRE can cause healthcare-associated urinary, bloodstream, wound, or other infections and may require susceptibility-guided treatment.

Vandetanib

Vandetanib is a tyrosine-kinase inhibitor targeting RET, VEGFR, and EGFR, used for selected medullary thyroid cancers; diarrhoea, hypertension, skin effects, and QT prolongation require monitoring.

Vaporiser

A device that converts a liquid medicine or volatile substance into a vapor for inhalation or controlled delivery, such as in anesthesia.

Variable Decelerations

Abrupt decreases in fetal heart rate that vary in timing, shape, and duration, commonly caused by umbilical-cord compression. Their significance depends on frequency, duration, associated baseline changes, variability, and the clinical setting.

Variance

A statistical measure of dispersion equal to the average squared deviation from the mean.

Variant Angina Pectoris

Chest pain caused by temporary coronary-artery spasm, often occurring at rest.

Alternate Names:

Prinzmetal angina; vasospastic angina

Variant CJD

Alternate index label; see *Creutzfeldt-Jakob disease*.

Variant Classification

The process of assigning a genetic variant to categories such as benign, likely benign, uncertain significance, likely pathogenic, or pathogenic using evidence from population data, laboratories, family studies, functional studies, and clinical correlation.

Varibaculum

A genus of bacteria in the family Lachnospiraceae, identified mainly in microbiologic and genomic research. Its clinical significance depends on the species and clinical specimen.

Varibaculum Cambriense

A bacterial species in the genus *Varibaculum* reported from human clinical specimens. Its role may be opportunistic and requires identification with the clinical context and susceptibility data.

Variceal Hemorrhage

Life-threatening complication of portal hypertension involving rupture of dilated esophageal or gastric varices. The risk is determined by variceal size, red color signs, liver disease severity, and portal hypertension degree.

Varicella

A primary infection with varicella-zoster virus that causes fever and a generalized itchy rash. See Chickenpox.

Alternate Names:

Chickenpox (*Varicella*)

Varicella (Chickenpox)

Alternate index label; see *Varicella*.

Varicella-Zoster Virus

A herpesvirus that causes chickenpox (*varicella*) and shingles (*herpes zoster*). VZV establishes latent infections in dorsal root ganglia and can reactivate decades later. Chickenpox is highly contagious and typically occurs in childhood. Reactivation as shingles causes painful dermatomal vesicular rash.

Varicellovirus

Varicellovirus is a genus of herpesviruses that includes varicella-zoster virus, the cause of chickenpox and shingles.

Varices

Abnormally dilated and tortuous veins, commonly occurring in the legs or in the oesophagus and stomach as a result of portal hypertension.

Varices, Esophageal

Alternate index label; see *Esophageal varices*.

Varicocele

A varicocele is enlargement of veins in the scrotum, often on the left, that may cause aching, testicular growth differences, infertility, or no symptoms. cord, resulting from incompetent or absent venous valves.

Varicose Ulcer

Impaired venous return from the lower extremities due to venous valve incompetence or obstruction. See chronic venous insufficiency.

Varicose Veins

Dilated, tortuous superficial veins, most commonly in the lower extremities. Caused by venous valve incompetence leading to venous reflux. Risk factors: family history, prolonged standing, obesity, pregnancy, and older age. Symptoms: leg heaviness, swelling, aching. Complications: skin changes, ulceration, thrombophlebitis.

Variegate Porphyria

An inherited hepatic porphyria caused by reduced protoporphyrinogen oxidase activity. It can cause acute neurovisceral attacks with abdominal pain and neuropathy and/or photosensitive blistering skin disease.

Variola

A highly contagious and deadly disease caused by variola virus, eradicated globally by 1980 through vaccination. See Smallpox.

Varix

An abnormally dilated, tortuous vein; multiple varices commonly occur in portal hypertension or chronic venous disease.

Varus

A deformity in which the distal part of a limb deviates toward the body's midline; examples include genu varum and talipes equinovarus.

Vas

A duct or vessel; in reproductive anatomy, the vas deferens transports sperm from the epididymis toward the ejaculatory duct.

Vas Deferens

Muscular duct (approximately 45 cm long) transporting sperm from the epididymal tail to the ejaculatory duct. It ascends through the spermatic cord, passes through the inguinal canal, enters the pelvis, crosses over the ureter, and joins the seminal vesicle duct to form the ejaculatory duct (in the prostate).

Vasa Deferentia

The paired muscular ducts that transport mature sperm from the epididymides toward the ejaculatory ducts during ejaculation.

Alternate Names:

vas deferens; ductus deferentes

Vasa Efferentia

Small ducts carrying sperm from the rete testis to the epididymal duct.

Vasa Previa

Fetal blood vessels traverse the membranes over or near the internal cervical os without protection from placental tissue or the umbilical cord. Rupture of membranes can cause rapid fetal exsanguination, so antenatal diagnosis and planned cesarean delivery are important.

Vascular

Pertaining to, composed of, containing, or relating to the blood vessels (arteries, arterioles, capillar-

ies, venules, and veins) or lymphatic channels of the circulatory system. Clinically central to every branch of medicine, encompassing vascular surgery (revascularization, endarterectomy, bypass grafting), vascular imaging (Doppler ultrasound, CT angiography, digital subtraction angiography), and vascular pathologies including systemic atherosclerosis, vasculitis, aneurysms, arteriovenous malformations, and peripheral vascular disease.

Vascular Access

A route into the bloodstream for treatment or monitoring. Hemodialysis access may be an arteriovenous fistula, arteriovenous graft, or central venous catheter; the choice depends on vascular anatomy, urgency, expected duration, and individual risks.

Vascular Access For Hemodialysis

A surgically created or catheter-based route that permits high-flow blood removal and return during hemodialysis. Arteriovenous fistulae are generally preferred when feasible, followed by grafts; central venous catheters have higher infection and thrombosis risks.

Vascular Bed

The network of blood vessels supplying a tissue or organ. Regional vascular beds regulate perfusion through vessel diameter, pressure, resistance, and local metabolic demands.

Vascular Claudication

Alternate index label; see *Claudication*.

Vascular Dementia

Cognitive impairment caused by cerebrovascular disease, such as strokes or chronic small-vessel injury. Symptoms may develop suddenly or gradually and can include problems with attention, executive function, memory, or walking. Management addresses the underlying vascular risks and functional needs.

Alternate Names:

Dementia, Vascular

Vascular Diagnostic Yield

The proportion of vascular diagnostic procedures or

tests that provide a clinically useful result.

Vascular Disease

Vascular disease affects arteries, veins, or lymphatic vessels and includes atherosclerosis, aneurysm, peripheral artery disease, venous thrombosis, varicose disease, and inflammatory disorders. Symptoms and treatment depend on the vessel and may involve risk-factor control, medicines, procedures, or surgery.

Vascular Ehlers-Danlos Syndrome

The most life-threatening subtype of Ehlers-Danlos syndrome (vEDS / Type IV EDS), inherited in an autosomal dominant manner caused by heterozygous pathogenic variants in the *COL3A1* gene on chromosome 2q32, which encodes pro-alpha-1 chains of type III procollagen. Produces severe structural fragility of vascular connective tissue and hollow internal organs. Classically features thin, translucent skin with prominent visible venous patterning, easy bruising, characteristic facial appearance (pinched nose, thin vermilion border, prominent eyes), and catastrophic vascular complications including spontaneous arterial dissection, aneurysm, or rupture, spontaneous intestinal perforation (typically sigmoid colon), and uterine rupture during pregnancy.

Alternate Names:

vEDS; Ehlers-Danlos Syndrome Type IV

Vascular Graft

Synthetic or biologic conduit for vascular reconstruction. PTFE, Dacron, saphenous vein.

Vascular Graft Patency

Continued openness and functional blood flow through a vascular graft or bypass. Patency is monitored clinically and with duplex or other imaging because thrombosis, stenosis, infection, intimal hyperplasia, and anastomotic problems can cause failure.

Vascular Imaging

Techniques for visualizing blood vessels: duplex ultrasound, CT angiography, MR angiography, con-

ventional angiography.

Vascular Malformation

Abnormal blood vessel development. Arteriovenous malformation, venous malformation, lymphatic malformation.

Vascular Ring

A congenital abnormality of the aortic arch or its branches that encircles and may compress the trachea, esophagus, or both. Symptoms include stridor, wheezing, recurrent respiratory infection, dysphagia, feeding difficulty, or respiratory distress.

Vascular Surgery

Surgery that repairs or reroutes blood vessels to improve circulation.

Vascular System

The network of arteries, veins, and capillaries that transports blood through the body.

Vascular Trauma

Injury to an artery or vein from blunt, penetrating, iatrogenic, or crush mechanisms. Assessment prioritizes hemorrhage control, distal perfusion, ischemia, expanding hematoma, bruit, thrill, and associated nerve or bone injury.

Vascularity

The amount and distribution of blood vessels supplying a tissue or lesion.

Vasculitis

Inflammation of blood vessels causing tissue ischemia and necrosis. Classification by vessel size. Large vessel: giant cell arteritis, Takayasu arteritis. Medium vessel: polyarteritis nodosa (PAN), Kawasaki disease.

Alternate Names:

Angiitis (Vasculitis)

Vasculitis (Systemic)

Inflammation of blood vessels that may involve large, medium, or small vessels and impair blood flow

or cause tissue injury. Examples include ANCA-associated vasculitis and IgA vasculitis.

Alternate Names:

Systemic

Vasculostatin

Vasculostatin is an experimental anti-angiogenic fragment related to angiostatin that has been studied for inhibition of new blood-vessel growth; it is not an established routine clinical medicine.

Vasectomy

A permanent contraceptive procedure in which the vas deferens are divided or occluded to prevent sperm from entering the ejaculate. It does not immediately provide sterility and does not protect against sexually transmitted infections.

Vasectomy Reversal

A microsurgical reconstructive operative procedure performed to re-establish patency of the male reproductive tract and restore fertility following a previous vasectomy. Operating under high-power optical magnification using 9-0 or 10-0 nylon sutures, the urological microsurgeon performs a two-layer end-to-end anastomosis reconnecting the severed vasal lumens (vasovasostomy). If intraoperative aspiration of fluid from the testicular vas end reveals absence of sperm and thick pasty fluid indicative of secondary epididymal obstruction from prolonged back-pressure, a microsurgical bypass connecting the abdominal vas lumen directly to an individual patent epididymal tubule (vasoepididymostomy) is performed instead. Patency rates exceed 85–95% depending on the interval since vasectomy and surgeon experience.

Alternate Names:

Vasovasostomy; Vasoepididymostomy

Vasoactive Intestinal Peptide Test

A blood test measuring vasoactive intestinal peptide, used mainly when a VIP-secreting neuroendocrine tumor is suspected in a patient with secretory diarrhea and electrolyte abnormalities.

Vasoconstriction

Source spelling variant of *vasoconstriction*, the narrowing of a blood vessel caused by contraction of its muscular wall.

Vasoconstriction

Narrowing of blood vessels caused by contraction of vascular smooth muscle, which can raise blood pressure or reduce blood flow to affected tissues.

Vasoconstrictor

A physiological mediator, autonomic nerve signal, or pharmacological agent that induces vasoconstriction—the narrowing of blood vessel lumens through the contraction of vascular smooth muscle cells within tunica media walls. Endogenous vasoconstrictors include endothelin-1, angiotensin II, vasopressin, and norepinephrine. Pharmacological vasoconstrictors (vasopressors, such as norepinephrine, epinephrine, phenylephrine, and vasopressin) act primarily via vascular α_1 -adrenergic or V_1 receptors, raising systemic vascular resistance and mean arterial pressure to restore tissue perfusion in distributive and septic shock, and are co-administered with local anesthetics to prolong anesthetic duration and limit surgical bleeding.

Vasoconstrictors

Substances or medicines that narrow blood vessels and usually increase vascular resistance and blood pressure.

Vasodepressor Syncope

Fainting caused predominantly by a sudden reduction in vascular tone and blood pressure, often as part of a vasovagal reflex. Warning symptoms may include nausea, sweating, warmth, and lightheadedness.

Vasodilatation

Widening of blood vessels caused by relaxation of vascular smooth muscle, increasing blood flow and potentially lowering blood pressure.

Vasodilation

The widening of blood vessels during inflammation,

resulting in increased blood flow to the affected area. This process is mediated primarily by histamine released from mast cell granules acting on H1 receptors, nitric oxide produced by endothelial nitric oxide synthase, and prostaglandins particularly PGE2.

Vasodilator

An endogenous biochemical signaling molecule or pharmacological agent that induces vasodilation—the relaxation and widening of the lumen of blood vessels, decreasing vascular resistance and systemic blood pressure while increasing regional blood flow. Endogenous vasodilators include endothelial nitric oxide, prostacyclin (PGI₂), adenosine, bradykinin, and histamine. Pharmacologic vasodilators are categorized by mechanism and vascular selectivity: arterial dilators (hydralazine, calcium channel blockers), venodilators (nitroglycerin, reducing cardiac preload and myocardial wall stress in acute coronary syndromes and heart failure), and mixed arterial-venous dilators (sodium nitroprusside, ACE inhibitors).

Vasodilators

Medicines or substances that relax vascular smooth muscle and widen blood vessels. They are used for conditions such as hypertension, angina, heart failure, or pulmonary hypertension depending on the agent.

Vasomotor

Relating to the control of blood-vessel diameter by nerves, hormones, and local chemical factors.

Vasomotor Centre

Brainstem networks, chiefly in the medulla, that regulate vascular tone and contribute to control of blood pressure.

Vasomotor Nerves

Autonomic nerves that alter blood-vessel diameter by causing vascular smooth-muscle contraction or relaxation.

Vasomotor Rhinitis

Nonallergic inflammation of the nasal lining causing

congestion, runny nose, or sneezing, often triggered by odors, temperature changes, or irritants.

Vasopressin

Antidiuretic hormone, also called arginine vasopressin, which promotes renal water reabsorption and can constrict blood vessels.

Vasopressin (Antidiuretic Hormone, ADH)

A hormone that increases water reabsorption in the kidney and can constrict blood vessels. It is released from the posterior pituitary.

Alternate Names:

Antidiuretic Hormone, ADH

Vasopressin Receptor

A protein on or inside a cell that responds to vasopressin. It helps control water balance, blood-vessel narrowing, and other body functions.

Vasopressor

An agent or physiological signal that raises blood pressure mainly by constricting blood vessels and sometimes by increasing cardiac activity.

Vasopressors In Anesthesia

Medicines that increase vascular tone or cardiac output to support blood pressure during anesthesia or critical illness. The choice and dose depend on the cause of hypotension, heart rate, cardiac function, and local protocol; monitoring is required.

Vasospasm

Sudden involuntary narrowing caused by contraction of a blood vessel.

Vasovagal Attack

A transient reflex episode caused by reduced sympathetic tone and increased vagal activity, producing hypotension and often bradycardia with fainting or near-fainting. Common triggers include prolonged standing, pain, fear, heat, dehydration, or emotional stress.

Vasovagal Reaction

A reflex that slows the heart and widens blood vessels, briefly lowering blood pressure and sometimes causing fainting; triggers include pain, fear, prolonged standing, heat, or blood exposure.

Vasovagal Syncope

Reflex syncope mediated by increased vagal tone and withdrawal of sympathetic activity, causing bradycardia and/or vasodilation resulting in transient cerebral hypoperfusion. Triggers: prolonged standing, heat, pain, emotional stress. Preceded by prodrome (lightheadedness, nausea, diaphoresis, visual tunneling).

Alternate Names:

Neurocardiogenic Syncope

Vatalanib

An investigational inhibitor of vascular endothelial growth factor receptors studied as an anticancer drug; it is not an established routine treatment.

VATS

Alternate name; see *Video-Assisted Thoracoscopic Surgery*.

Vaughan Williams Classification

The Vaughan Williams classification categorizes antiarrhythmic drugs into four classes based on their primary mechanism of action.

VCJD

Alternate index label; see *Creutzfeldt-Jakob disease*.

VDRL Test

A nontreponemal blood or cerebrospinal-fluid test for reagin antibodies used in syphilis screening and monitoring; reactive results require treponemal confirmation.

VEAL CHOP Mnemonic

A memory aid for interpreting fetal heart rate patterns. VEAL: Variable decelerations Cord compression, Early decelerations Head compression, Accelerations OK (fetal well-being), Late decelerations

Placental insufficiency.

Vector

A living carrier, commonly an arthropod, that transmits an infectious agent between hosts. Biological vectors support replication or development of the agent; mechanical vectors transfer it without such biologic change.

Vector Control

Reduction of vector populations. Examples: insecticides, larvicides, environmental modification, personal protection.

Vector-Borne Disease

Disease transmitted by vector (insect, tick). Examples: malaria, dengue, Lyme disease.

Vector-Borne Diseases

Diseases transmitted by arthropod vectors including mosquitoes, ticks, fleas, and sandflies. Major vector-borne diseases include malaria, dengue, Zika, chikungunya, Lyme disease, and leishmaniasis. Prevention through vector control, personal protection, and chemoprophylaxis is essential.

Veganism

A dietary or lifestyle pattern that excludes animal-derived foods; nutritional adequacy requires attention to nutrients such as vitamin B12, iron, calcium, iodine, and omega-3 fatty acids.

Vegetarianism

A dietary pattern that excludes meat, with inclusion or exclusion of eggs and dairy depending on the subtype; nutritional planning may be needed.

Vegetation

A vegetation is a mass composed of platelets, fibrin, and microorganisms that forms on a heart valve or endocardial surface in the setting of infective endocarditis. The vegetation typically attaches to the valve leaflet on the low-pressure side and may progressively destroy the underlying valve tissue, causing valvular regurgitation.

Vegetations

Abnormal masses, often composed of fibrin, platelets, and microorganisms, that form on heart valves in infective endocarditis; sterile vegetations can occur in other conditions.

Vehicle

A substance or medium that carries an active medicine, microorganism, or toxic agent; in pharmacology it may be the inactive base in which a drug is formulated.

Vein

A blood vessel that carries deoxygenated blood toward the heart (except pulmonary veins, which carry oxygenated blood). Veins have thinner walls and larger lumens than arteries, with less smooth muscle and elastic tissue.

Vein-Graft Aneurysms

Focal dilation of a venous bypass graft, most commonly a saphenous vein graft used in coronary artery bypass surgery. Rare but may cause graft thrombosis, distal embolization, compression of adjacent structures, or rupture. Diagnosis: coronary angiography or CT angiography. Treatment: re-intervention or medical management.

Veins

Blood vessels that return blood toward the heart. Most carry deoxygenated blood, while pulmonary and umbilical veins carry oxygenated blood.

Veins, Diseases Of

Disorders including thrombosis, varicose veins, chronic venous insufficiency, venous ulcers, inflammation, and congenital venous malformations.

VeIP Regimen

A combination chemotherapy regimen of vinblastine, ifosfamide, and cisplatin, used in selected germ-cell tumors and other oncology settings. Treatment requires specialist monitoring for marrow, kidney, nerve, and bladder toxicity.

Veisalgia

The medical term for a hangover: the cluster of headache, thirst, nausea, fatigue, poor concentration, and malaise that can follow alcohol use as blood alcohol falls.

Velafermin

A recombinant fibroblast-growth-factor analog studied to promote epithelial repair, including in ocular-surface and mucosal disease. It remains investigational in many settings.

Velamentous Cord Insertion

An abnormal structural variation of umbilical cord attachment in which the umbilical vessels separate and diverge prior to reaching the placental disc, coursing between the amnion and chorion across the fetal membranes unsupported by protective Wharton's jelly. It significantly increases the risk of vessel compression, thrombosis, fetal growth restriction, and vasa previa, wherein exposed fetal vessels cross or lie within 2 cm of the internal cervical os, risking catastrophic fetal exsanguination upon membrane rupture.

Alternate Names:

Velamentous Insertion of the Umbilical Cord

Veliparib

An investigational inhibitor of poly(ADP-ribose) polymerase studied in combination cancer treatment, particularly for tumors with impaired DNA repair. Its role depends on clinical-trial evidence and regulatory approval.

Velocardiofacial Syndrome

Alternate index label; see *DiGeorge syndrome (22q11.2 deletion syndrome)*.

Veltuzumab

An investigational monoclonal antibody directed against CD20 on B cells, studied for B-cell cancers and autoimmune disease.

Vemurafenib

Vemurafenib is a BRAF V600 kinase inhibitor used

for selected BRAF-mutated melanoma and other cancers; arthralgia, photosensitivity, QT prolongation, new skin tumors, and resistance are important concerns.

Vena Cava

The two largest systemic veins returning deoxygenated blood to the right atrium. Superior Vena Cava (SVC): formed by brachiocephalic veins, drains head, neck, upper limbs, thorax (no valves). Inferior Vena Cava (IVC): formed by common iliac veins at L5, drains abdomen, pelvis, lower limbs (receives hepatic, renal, gonadal, lumbar veins).

Venae Cavae

The superior and inferior venae cavae are the major veins returning systemic blood to the right atrium. The superior vena cava drains the head, neck, chest, and upper limbs; the inferior vena cava drains the abdomen, pelvis, and lower limbs.

Alternate Names:

venae cava; vena cava

Venepuncture

Puncture of a vein with a needle to obtain blood or establish intravenous access.

Venereal

Relating to sexual intercourse or to infection transmitted through sexual contact. The term is largely historical; sexually transmitted infection is the preferred modern expression.

Venereal Disease Research Laboratory Test

Venereal Disease Research Laboratory test is a medical test, procedure, treatment, or clinical measure used in patient care; its purpose and interpretation depend on the clinical question.

Venereal Diseases

Sexually transmitted infections are infections passed from one person to another through sexual contact, including vaginal, anal, and oral sex. See Sexually Transmitted Infections (STIs).

Venesection

The controlled removal of blood from a vein, used diagnostically or therapeutically, for example in selected cases of hemochromatosis or polycythemia.

Venesection (Phlebotomy)

The controlled removal of blood from a vein for therapeutic or diagnostic purposes.

Alternate Names:

Phlebotomy

Venin

A toxic secretion injected by an animal through a bite, sting, or other specialized apparatus. Its effects depend on the species, dose, route, and toxins involved.

Venipuncture

Puncture of a vein with a needle to obtain blood, administer fluids or medicines, or establish intravenous access. Bruising, infection, bleeding, and nerve or artery injury are uncommon complications.

Veno-Occlusive Disease

Obstruction of small veins, particularly in the liver after certain chemotherapy or hematopoietic stem-cell transplantation, causing painful hepatomegaly, fluid retention, jaundice, and weight gain.

Venogram

Contrast imaging of veins used to detect thrombosis, obstruction, abnormal anatomy, or selected venous flow disorders.

Venogram - Leg

Contrast imaging of the veins of a leg, used selectively to evaluate suspected deep-vein thrombosis or venous obstruction when noninvasive imaging is inadequate.

Venography

Imaging of veins, usually after contrast material is administered, to assess venous anatomy, obstruction, thrombosis, or malformation.

Venom

A biological poison produced by a microorganism, plant, or animal that damages host tissues at low concentrations. See toxin.

Venomous

Describing an animal that actively delivers toxins through a bite, sting, spine, or similar mechanism. This differs from poisonous, which generally refers to toxicity after ingestion or contact.

Venomous Bite

A bite that injects venom from an animal such as a snake, spider, or arthropod. Effects range from local pain and swelling to bleeding, paralysis, or systemic toxicity.

Venous

Pertaining to, situated within, originating from, or affecting the veins—the thin-walled, compliant capacitance blood vessels that transport deoxygenated blood and metabolic waste products from peripheral capillary beds back toward the right atrium of the heart (with the exception of pulmonary veins, which carry oxygenated blood to the left atrium). Governed by low luminal pressure and equipped with one-way intraluminal bicuspid valves in the extremities to prevent gravitational retrograde flow. Clinically critical in deep vein thrombosis, chronic venous insufficiency (CVI; with stasis dermatitis and venous ulcers), venous thromboembolism, and venous access procedures (peripheral cannulation, central venous catheters).

Venous Aneurysm

A localized abnormal dilatation of a vein. It may be asymptomatic or present as a soft mass, pain, thrombosis, embolism, or bleeding depending on its location and size.

Venous Hum

A benign, innocent vascular sound caused by turbulent blood flow in the internal jugular veins as blood drains rapidly into the superior vena cava. Common in healthy children aged 2 to 8 years, it is heard on auscultation as a continuous, soft, low-pitched

blowing or humming murmur over the supraclavicular and infraclavicular regions (more commonly on the right). It is loudest in the upright seated position during diastole, and uniquely diminishes or completely vanishes when the child lies supine, rotates the head, or when gentle digital compression is applied over the ipsilateral internal jugular vein.

Alternate Names:

Cervical Venous Hum; Innocent Venous Hum

Venous Insufficiency

Impaired venous return from the lower extremities due to venous valve incompetence or obstruction. See chronic venous insufficiency.

Venous Leak

Inadequate retention of blood in the penis during erection because blood escapes through veins.

Alternate Names:

Veno-occlusive dysfunction

Venous Pressure

The pressure within the venous circulation, influenced by blood volume, venous tone, and the function of the right side of the heart.

Venous Stasis Ulcer

A chronic leg wound caused by venous hypertension and chronic venous insufficiency. It often occurs near the ankle and may be accompanied by edema, varicose veins, skin darkening, and lipodermatosclerosis.

Venous Thromboembolism

Formation of a venous blood clot, usually deep-vein thrombosis, with or without embolization to the pulmonary arteries; pulmonary embolism is the major acute complication.

Venous Thromboembolism In Pregnancy

Pregnancy increases the risk of VTE (DVT and PE) by 5-10 fold due to a hypercoagulable state (increased fibrinogen, factors VII-XII, vWF, decreased protein S, and venous stasis from the gravid uterus).

Venous Thromboembolism Prophylaxis

Measures used to reduce perioperative deep-vein thrombosis and pulmonary embolism, including early mobilization, mechanical compression, and pharmacologic anticoagulation when bleeding risk permits.

Venous Thrombosis

The formation of a blood clot (thrombus) within a vein, most commonly in the deep veins of the lower extremities (deep vein thrombosis, DVT).

Venous Ulcer

A chronic wound caused by impaired venous drainage, usually occurring near the ankle in a limb with chronic venous insufficiency.

Ventilation

Movement of air into and out of the lungs; effective ventilation brings oxygen to the alveoli and removes carbon dioxide.

Ventilation, Artificial

Mechanical movement of air into and out of the lungs using a device when spontaneous breathing is inadequate or absent.

Ventilation-Perfusion Mismatch

Unequal matching of air reaching the alveoli and blood reaching the pulmonary capillaries. Low or high regional ratios can impair oxygen exchange and occur in conditions such as COPD, pneumonia, or pulmonary embolism.

Ventilation-Perfusion Ratio

The relationship between alveolar ventilation and pulmonary capillary perfusion. Regional mismatch is a major cause of impaired oxygenation; low V/Q produces venous admixture, while high V/Q reflects wasted ventilation.

Ventilation-Perfusion Scan

Lung imaging assessing ventilation and perfusion. Mismatch indicates pulmonary embolism. Alternative to CTPA in renal impairment or contrast

allergy.

Ventilation/Perfusion Scan

Pulmonary embolism diagnosis. Ventilation: inhaled Tc-99m DTPA aerosol. Perfusion: injected Tc-99m MAA. Mismatch (perfusion defect with preserved ventilation) suggests PE. High single-photon emission computed tomography (SPECT) sensitivity.

Ventilator

Mechanical device providing ventilatory support. Modes: assist-control, SIMV, CPAP, PRVC. Parameters: tidal volume, respiratory rate, FiO₂, PEEP.

Ventilator-Associated Pneumonia

Pneumonia developing after at least 48 hours of mechanical ventilation. Diagnosis is based on clinical findings and respiratory evaluation; cultures may help identify the organism but are not interpreted in isolation.

Ventilator-Induced Lung Injury

Lung injury caused or amplified by mechanical ventilation, including volutrauma, barotrauma, atelectrauma, and biotrauma. Lung-protective ventilation limits tidal volume and excessive airway pressures while maintaining appropriate oxygenation and carbon-dioxide targets.

Ventouse

A vacuum extractor used for selected assisted vaginal deliveries.

Ventral

Relating to the belly or front side of the body; synonym for anterior in human anatomy.

Ventral Hernia

Protrusion of abdominal tissue through a weakness in the front abdominal wall, including incisional and umbilical hernias. It may cause a bulge or pain and requires urgent assessment if incarceration or strangulation is suspected.

Ventricle

One of the two lower, thicker-walled chambers of the heart that pump blood out of the heart. The right ventricle pumps deoxygenated blood into the pulmonary artery (low-pressure system); the left ventricle pumps oxygenated blood into the aorta (high-pressure system, generating systemic blood pressure).

Ventricle (Brain)

The interconnected system of fluid-filled cavities within the brain producing and containing cerebrospinal fluid: two lateral ventricles (cerebrum), third ventricle (diencephalon), and fourth ventricle (pons/medulla, continuous with the central canal).

Alternate Names:

Brain

Ventricular

Relating to a ventricle, especially one of the heart or the brain. The meaning depends on the organ, such as ventricular tachycardia or a cerebral ventricular system.

Ventricular Arrhythmia

An abnormal heart rhythm arising from the ventricles. It may cause palpitations, dizziness, fainting, chest discomfort, or sudden cardiac arrest; fainting or severe symptoms require emergency care.

Ventricular Arrhythmias

Abnormal heart rhythms arising from the ventricles. They range from premature ventricular contractions to ventricular tachycardia and fibrillation, which can cause sudden cardiac arrest.

Ventricular Assist Device

A mechanical pump supporting circulation in advanced heart failure. Types include left-, right-, and biventricular devices. Uses may include temporary support, a bridge to transplantation, or long-term therapy; complications include bleeding, infection, stroke, and pump thrombosis.

Ventricular Fibrillation

Chaotic electrical activity in ventricles, causing no cardiac output. Immediate defibrillation is treatment. Causes: coronary artery disease, cardiomyopathy, electrolyte disturbances, drug toxicity. Post-resuscitation care: targeted temperature management, coronary angiography, ICU admission.

Alternate Names:

V-Fib

Ventricular Hypertrophy

Thickening of a heart-ventricle wall, usually from chronic pressure or volume overload, inherited cardiomyopathy, or athletic adaptation. It may cause diastolic dysfunction, arrhythmia, ischemia, or heart failure.

Ventricular Myocardium

The heart muscle forming the walls of the ventricles. Its contraction pumps blood to the lungs and the rest of the body.

Ventricular Premature Contraction

Alternate index label; see *Premature ventricular contractions (PVCs)*.

Ventricular Rupture

A tear in a ventricular wall, often a life-threatening complication of myocardial infarction.

Ventricular Septal Defect

An opening in the wall between the ventricles that can allow blood to flow from the left ventricle to the right. The effects depend on the size and location of the defect and may range from no symptoms to heart failure or pulmonary vascular disease.

Ventricular Tachycardia

A wide-complex tachycardia (QRS >0.12 s) originating from the ventricles, with a rate of 100-250 bpm. Monomorphic VT has uniform QRS complexes; polymorphic VT has varying QRS morphology. VT is life-threatening and may degenerate to ventricular fibrillation.

Ventriculoatrial

Relating to or connecting a ventricle of the brain or heart with an atrium; often describing a cerebrospinal-fluid shunt to the right atrium.

Ventriculography

Imaging ventricular function. Radionuclide ventriculography (MUGA scan) accurately measures ejection fraction. Cardiac MRI also accurate.

Ventriculoperitoneal Shunt

A surgically implanted system that drains cerebrospinal fluid from a brain ventricle into the abdominal cavity, where the fluid can be absorbed.

Ventrosuspension

Surgical fixation of a displaced or prolapsed uterus to the anterior abdominal wall or nearby supporting tissues.

Venturi Mask

A high-flow, fixed-performance oxygen delivery device designed to deliver a precise, constant, and reproducible fraction of inspired oxygen (FiO_2 ; typically ranging from 24% to 50%) irrespective of the patient's spontaneous respiratory rate, depth, or peak inspiratory flow demands. Operates on the Bernoulli principle of fluid dynamics and fluid entrainment: oxygen flows at high velocity through a narrow jet orifice, generating a localized low-pressure zone that draws in (entrains) a calibrated, fixed proportion of room air through color-coded, interchangeable base adapters. The clinical gold standard for acute oxygen therapy in patients with chronic obstructive pulmonary disease or chronic hypercapnic respiratory failure, where uncontrolled supplemental oxygen could blunt hypoxic respiratory drive and precipitate worsening respiratory acidosis.

Venule

A small vessel that drains blood from capillary beds into veins. Postcapillary venules are important sites for leukocyte migration and inflammatory fluid exchange.

Venules

Small blood vessels that collect blood from capillaries and join to form veins. Their walls also participate in inflammation by allowing white blood cells to leave the bloodstream.

Verapamil

A non-dihydropyridine calcium-channel blocker used for selected hypertension, angina, and supraventricular tachyarrhythmias; it can cause bradycardia and constipation.

Vermicides

Medicines or chemical agents that kill parasitic worms; the appropriate agent depends on the parasite and site of infection.

Vernal Keratoconjunctivitis

A recurrent allergic inflammation of the conjunctiva and cornea, usually affecting children or adolescents and causing itching, redness, and light sensitivity.

Vernix

A white, greasy protective coating on the skin of a fetus or newborn, composed mainly of water, lipids, and shed skin cells.

Vernix Caseosa

A white, greasy protective coating on the skin of a fetus or newborn, composed mainly of water, lipids, and shed skin cells.

Vero Cell

A laboratory cell line originally taken from monkey kidney tissue and used to grow viruses, make vaccines, and conduct biomedical research. It is not a normal human cell.

Verruca

A benign epidermal growth caused by infection with a human papillomavirus. Common warts, plantar warts, and flat warts are clinical types; treatment depends on site, symptoms, and immune status.

Verruciform Xanthoma

A rare benign lesion characterized by a verrucous epithelial surface with lipid-laden macrophages in the underlying connective tissue. It most often occurs in oral or genital mucosa and may resemble a wart or malignant lesion clinically.

Verrucose

Having a rough, thickened, wart-like, or finger-like surface. It is a descriptive term for a skin, mucosal, or other lesion and does not by itself identify whether the lesion is benign, infectious, or cancerous.

Verrucous

Having a rough, thickened, irregular, or wart-like surface. The appearance can occur in benign growths, infections, chronic inflammation, or some cancers.

Verrucous Carcinoma

A low-grade, well-differentiated variant of squamous cell carcinoma with a warty, exophytic appearance. It is locally invasive but rarely metastasizes, commonly affecting the oral cavity, skin, or genitalia.

Version

A term describing a change in the position or presentation of a body structure, especially the fetus during pregnancy; its meaning depends on the clinical context.

Vertebra

The individual bone of the vertebral column (33 vertebrae: 7 cervical, 12 thoracic, 5 lumbar, 5 sacral fused, 4 coccygeal). Parts: vertebral body, vertebral arch (pedicles, laminae), spinous and transverse processes, articular processes/facets, and vertebral foramen (spinal canal).

Vertebrae

Vertebrae are the bones of the spinal column that protect the spinal cord, support the body, allow movement, and provide attachment for muscles and ligaments.

Vertebral Artery

An artery ascending through the neck that supplies blood to the brainstem and posterior brain.

Vertebral Canal

The bony canal formed by the vertebrae that encloses and protects the spinal cord.

Vertebral Column

33 vertebrae: 7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral, 4 fused coccygeal. Protects the spinal cord, supports the head/trunk, provides muscle/rib attachment. Each typical vertebra has a body, vertebral arch, and 7 processes. Intervertebral discs (nucleus pulposus/annulus fibrosus) provide cushioning.

Vertebral Compression Fracture

Collapse of a vertebral body, often caused by osteoporosis, trauma, or malignancy, producing back pain, loss of height, or progressive kyphosis.

Vertebral Hemangioma

Alternate index label; see *Spinal hemangioma*.

Vertebral Tumor

A tumor arising in or near a vertebra of the spinal column.

Vertebrobasilar Circulatory Disorders

Conditions that reduce blood flow through the vertebral or basilar arteries supplying the brainstem, cerebellum, and posterior brain; symptoms may include vertigo, double vision, imbalance, weakness, or stroke.

Vertebrobasilar Insufficiency

A condition of reduced blood flow through the vertebral arteries or basilar artery to the posterior circulation of the brain, including the brainstem, cerebellum, occipital lobes, and thalami.

Vertebrobasilar Insufficiency

Source spelling variant of *vertebrobasilar insufficiency*, reduced blood flow in the vertebral or basi-

lar circulation supplying the brainstem, cerebellum, and posterior brain.

Vertical

Oriented from superior to inferior or perpendicular to a horizontal plane. In medicine, it may describe position, transmission from parent to child, or a body-oriented imaging or movement axis.

Vertical Talus

A congenital rigid flatfoot deformity in which the talus is positioned vertically and the midfoot is dorsiflexed. It may be isolated or associated with neuromuscular or genetic syndromes and usually requires early orthopedic assessment and treatment.

Vertical Transmission

Passage of an infection or genetic condition from parent to child during pregnancy, delivery, or breastfeeding.

Vertigo

A false sensation of rotational or linear movement of the self or the environment, and it is a symptom rather than a disease, requiring differentiation from dizziness, disequilibrium, and presyncope.

Vertigo Overview

Vertigo is the false sensation of movement, often caused by inner-ear disorders such as benign positional vertigo, vestibular neuritis, or Ménière disease, but sometimes by migraine or brain disease. New neurologic deficits, severe headache, or inability to walk is urgent.

Vertigo-Associated Disorders

Vertigo-associated disorders cause an illusion of movement from vestibular, neurologic, visual, medication-related, or systemic dysfunction and may be accompanied by imbalance, nausea, or hearing symptoms.

Alternate Names:

VLCD

Very Long-Chain Acyl-CoA

Dehydrogenase Deficiency

An autosomal recessive inborn error of mitochondrial long-chain fatty acid β -oxidation caused by pathogenic variants in the *ACADVL* gene on chromosome 17p13.1. Impairs the initial dehydrogenation step for fatty acids 14 to 20 carbons in length. Clinically manifests during periods of prolonged fasting or metabolic stress; the severe infantile-onset form presents with life-threatening hypoketotic hypoglycemia, severe hypertrophic or dilated cardiomyopathy, pericardial effusion, ventricular arrhythmias, and acute hepatic failure, whereas the later-onset myopathic form presents with recurrent exercise-induced rhabdomyolysis and myoglobinuria.

Alternate Names:

VLCAD Deficiency; VLCADD

Very-Low-Density Lipoprotein

A lipoprotein made by the liver that carries mainly triglycerides to body tissues. As it loses triglycerides, it can form remnant particles and contribute to LDL production; elevated levels are associated with atherosclerotic cardiovascular risk.

Alternate Names:

VLDL

Vesical

Relating to bladder. Vesical carcinoma, vesical diverticulum, vesical neck.

Vesicant

A substance that causes blistering and tissue injury on contact. In oncology, vesicant drugs can cause severe extravasation injury if they escape from a vein into surrounding tissue.

Vesicants

Substances that cause blistering and tissue injury on contact; examples include certain chemotherapy agents and chemical-warfare agents.

Vesicle

A small (<1 cm) fluid-filled blister within the epidermis, containing clear fluid (serum) or, if infected, pus. Causes: viral (herpes simplex, herpes zoster,

varicella, hand-foot-mouth disease), dermatitis (contact, dyshidrotic eczema), autoimmune (pemphigus, bullous pemphigoid early), and friction burns.

Vesicles

Small, elevated, fluid-filled blisters on the skin or mucous membrane, generally less than 1 cm in diameter. Causes include viral infection, dermatitis, burns, and autoimmune blistering disease.

Vesico-Ureteric Reflux

Backward flow of urine from the bladder into one or both ureters and sometimes the kidneys, increasing the risk of recurrent urinary infection and renal scarring.

Vesicoureteral Reflux

Retrograde flow of urine from the bladder into the ureters and potentially the kidneys, most commonly due to incompetent ureterovesical valve. Predisposes to pyelonephritis and renal scarring. More common in children, presenting with recurrent febrile UTIs. Diagnosis: voiding cystourethrogram (VCUG). Grading I-V.

Alternate Names:

Reflux, Vesicoureteral

Vesicoureteric Reflux

Backward flow of urine from the bladder into one or both ureters and sometimes the kidneys, increasing the risk of urinary infection and renal scarring.

Vesicovaginal Fistula

An abnormal connection between the bladder and vagina that causes continuous or frequent leakage of urine through the vagina. It may follow childbirth injury, pelvic surgery, radiation, or advanced infection and usually requires specialist repair.

Vesicular

Characterized by or containing vesicles. The term commonly describes a vesicular rash or a vesicle-related structure in cell biology and pathology.

Vesicular Breathing

The normal soft, rustling breath sound heard over most peripheral lung fields during auscultation.

Vesiculobullous Dermatoses

A group of skin disorders characterized by vesicles or bullae caused by autoimmune, genetic, infectious, inflammatory, drug-related, or traumatic processes. Diagnosis depends on lesion morphology, distribution, biopsy, and immunofluorescence when indicated.

Vesiculovirus

Vesiculovirus is a genus of rhabdoviruses that mainly infect animals and can rarely cause a flu-like illness in humans.

Vessel

A vessel is a tube carrying blood, lymph, air, urine, bile, or another fluid through the body, depending on context.

Vestibular

Relating to the vestibular system of the inner ear and brain, which detects head motion and position and helps maintain balance and gaze stability.

Vestibular Migraine

Episodic vestibular symptoms associated with migraine biology, with or without simultaneous headache. Episodes may include vertigo, motion sensitivity, visual aura, or sound sensitivity; diagnosis is clinical after exclusion of other vestibular and neurologic causes.

Vestibular Neuritis

Acute vestibular dysfunction due to inflammation of the vestibular nerve, typically viral in origin. Presents with acute onset of vertigo, nausea, vomiting, and nystagmus lasting days. Hearing is normal (unlike labyrinthitis). Diagnosis: clinical, HINTS exam (Head Impulse, Nystagmus, Test of Skew) to differentiate from central causes.

Vestibular Papillomatosis

A normal anatomical variant of the female vulva characterised by multiple, soft, finger-like or filiform papules on the labia minora and vestibule, often mistaken for genital warts (condylomata acuminata).

Vestibular Rehabilitation

Exercise-based therapy that improves balance, gaze stability, and adaptation in people with vestibular disorders or dizziness.

Vestibular Schwannoma

A benign, slow-growing tumor of the vestibular portion of the eighth cranial nerve. See Acoustic Neuroma.

Vestibular System

Inner-ear and brain structures responsible for balance and spatial orientation.

Vestibule

Central bony labyrinth between cochlea and semi-circular canals, containing utricle (horizontal linear acceleration and head tilt detection) and saccule (vertical linear acceleration detection). Each has a macula with hair cells covered by gelatinous membrane embedded with otoconia (calcium carbonate crystals).

Vestibule, Vaginal

The space of the vulva between the labia minora that contains the external openings of the urethra and vagina and the ducts of the greater and lesser vestibular glands.

Vestibulo-Ocular Reflex

A reflex that moves the eyes opposite to head movement, stabilizing visual fixation during motion through signals from the vestibular system.

Vestibulocochlear Nerve

The eighth cranial nerve, carrying information for hearing and balance from the inner ear to the brain.

Vestibulodynia

A localized subtype of vulvodynia characterized by persistent, sharp, burning pain strictly confined to the vulvar vestibule, provoked by light touch, pressure, vaginal entry, or tampon insertion (allodynia). Diagnosis is confirmed clinically using the cotton-swab test (Q-tip palpation eliciting acute point tenderness across the vestibular glands, especially at 4 to 8 o'clock). Etiology is multifactorial, involving peripheral neuroproliferation, pelvic floor hypertonicity, and altered central pain processing; management comprises topical anesthetics, pelvic floor physical therapy, neuromodulators, and in refractory cases, vestibulectomy.

Alternate Names:

Provoked Vestibulodynia; Vulvar Vestibulitis Syndrome; PVD

Vestigial

Describing a structure or feature that has become greatly reduced through evolution and has little or no apparent original function.

VEXAS Syndrome

VEXAS (Vacuoles, E1 ubiquitin-activating enzyme, X-linked, Autoinflammatory, Somatic) syndrome is an adult-onset autoinflammatory disorder caused by somatic UBA1 mutations in hematopoietic stem cells, affecting men >50.

Viable

Capable of surviving or developing under suitable conditions. In clinical contexts it may describe a viable pregnancy, tissue, organism, cell, or treatment option.

Vibration Problem

Symptoms or injury caused by repeated exposure to vibrating tools or equipment, which may affect nerves, blood vessels, muscles, and joints in the hands or arms.

Vibrator

A device that produces mechanical vibration and may be used for sexual stimulation, therapeutic massage, or selected rehabilitation purposes.

Vibrio

A genus of curved, Gram-negative bacteria; species include *V. cholerae*, which causes cholera, and *V. vulnificus*, which can cause severe wound or bloodstream infection.

Vibrio Cholerae

A gram-negative, curved rod that causes cholera, a severe diarrhoeal illness. The organism produces cholera toxin, which causes profuse watery diarrhea (rice-water stools). Transmission occurs through contaminated water and food. The infectious dose varies depending on stomach acidity.

Vibrio Infection

Infection caused by *Vibrio* bacteria, acquired through contaminated seafood or exposure of wounds to warm seawater. It may cause watery diarrhea, wound infection, or rapidly progressive sepsis, especially in people with liver disease or impaired immunity. Severe wound pain, blistering, fever, or shock requires emergency treatment.

Alternate Names:
vibriosis.

Vibrio Vulnificus Infections

Infections caused by *Vibrio vulnificus*, a gram-negative bacterium found in warm seawater and raw seafood. It can cause wound infection, bloodstream infection, or gastroenteritis, with particularly severe disease in people with chronic liver disease or other risk factors.

Vibrios

Comma-shaped bacteria, usually Gram-negative, that include species capable of causing cholera and wound or gastrointestinal infections.

Video Laryngoscope

An airway device with a camera near the blade tip that displays the laryngeal view on a monitor. It may improve visualization during tracheal intubation, particularly when direct laryngoscopy is difficult, but does not eliminate aspiration or airway-trauma risk.

Video Laryngoscopy

Video laryngoscopy uses a camera-equipped blade to display an indirect view of the glottis on a screen.

Video-Assisted Thoracoscopic Surgery

A minimally invasive thoracic surgical technique that utilizes a specialized rigid fiberoptic camera (thoracoscope) and long endoscopic instruments inserted into the hemithorax through 1 to 3 small incisions (ports) without rib spreading. It has become the standard surgical approach for diagnostic lung and pleural biopsies, anatomic lobectomy, segmentectomy, wedge resection, pleurodesis, thymectomy, and decortication, resulting in significantly reduced postoperative pain, shorter chest tube duration, reduced morbidity, and faster functional recovery compared with open posterolateral thoracotomy.

Alternate Names:

VATS; Video Thoracoscopy; Thoracoscopic Surgery

Videofluoroscopic Swallow Study

Real-time X-ray imaging of swallowing. Detailed assessment of oropharyngeal phase.

Videostroboscopy

A diagnostic technique that uses synchronized stroboscopic light to create a slow-motion view of vocal-fold vibration. It assesses the mucosal wave, amplitude, symmetry, and glottic closure.

Vildagliptin

An oral dipeptidyl-peptidase-4 inhibitor used in some countries for type 2 diabetes, alone or with other medicines. It improves glucose-dependent insulin secretion; liver function and pancreatitis symptoms may require attention.

Villi

Finger-like projections that increase the absorptive surface of the small-intestinal lining; similar structures occur in the placenta and other tissues.

Villous Atrophy

Flattening and loss of the normal small-intestinal villi, reducing absorptive surface area and occurring

in conditions such as coeliac disease.

Villus

A small finger-like projection, especially one of the intestinal mucosa that increases absorptive surface area; chorionic villi mediate exchange between placenta and fetus.

Vinca Alkaloids

Anticancer medicines, including vincristine and vinblastine, that inhibit microtubule formation and cell division.

Vincent Angina

An acute necrotizing infection of the gums and oropharynx associated with fusobacteria and spirochetes. It causes painful ulceration, bleeding, foul breath, and gingival necrosis and requires dental or medical treatment.

Vincent Disease

Acute necrotizing ulcerative gingivitis associated with fusospirochetal bacteria, causing painful bleeding gums, ulceration, and foul breath.

Vincent Infection

Vincent infection is acute necrotizing ulcerative gingivitis caused by mixed oral bacteria, producing painful bleeding gums, ulcers, and foul breath.

Vincent Stomatitis

A painful necrotizing infection of the gums caused by mixed anaerobic bacteria, producing foul breath, bleeding, ulceration, and gray necrotic tissue; it is also called acute necrotizing ulcerative gingivitis.

Vincent's Angina

Acute necrotizing ulcerative gingivitis or tonsillitis associated with fusiform bacteria and spirochetes, causing painful ulcers, bleeding, and fetid breath.

Vinorelbine

A semisynthetic vinca-alkaloid chemotherapy medicine used primarily for selected lung and breast cancers; myelosuppression is a major adverse effect.

Viomycin

An older aminoglycoside antibiotic related to capreomycin that was used against tuberculosis and other mycobacterial infections; kidney and hearing toxicity limit its use.

Viraemia

The presence of viruses in the bloodstream, which may be transient or persistent and may permit spread to distant tissues.

Viral

Pertaining to, caused by, characteristic of, or resembling a virus—an obligate intracellular submicroscopic infectious agent possessing a DNA or RNA genome enclosed within a protective protein capsid, sometimes surrounded by a host-derived lipid envelope. Because viruses lack autonomous metabolic and translational machinery, they must hijack host cell biochemical systems to replicate, often triggering cellular apoptosis, cytopathic lysis, or oncogenic transformation (e.g., HPV, HBV, HCV, EBV). Manifestations range from self-limiting viral exanthems and upper respiratory tract infections to life-threatening viral hemorrhagic fevers, viral encephalitis, and pandemic respiratory diseases (e.g., influenza, COVID-19); diagnosed by reverse transcription-polymerase chain reaction, viral antigen detection, or serology, and managed with supportive care, vaccines, and targeted antiviral therapies.

Viral Antibody

An immunoglobulin produced after infection or vaccination that recognizes a viral antigen; viral-antibody tests help assess previous exposure, immune response, or selected active infections, depending on the assay and timing.

Viral Arthritis

Viral arthritis is joint inflammation caused by or associated with a viral infection and may be acute, symmetric, and self-limited or part of a broader illness.

Viral Assembly

The process of packaging newly synthesised viral genomes and proteins into complete virions. Assembly typically occurs in the nucleus or cytoplasm, depending on the virus type. The process involves self-assembly of capsid proteins around the viral genome.

Viral Attachment

The initial step in viral infection where viral surface proteins bind to specific receptors on host cells. Attachment determines viral tropism and host range. The viral attachment protein is called the ligand, and the host cell receptor is typically a protein or glycan.

Viral Carcinogenesis

Viral carcinogenesis is cancer development driven or promoted by viral infection through oncogenic proteins, altered cell signaling, immune suppression, or chronic inflammation; examples include HPV, EBV, HBV, HCV, and HTLV-1.

Viral Conjunctivitis

The most common form of infectious conjunctivitis, caused predominantly by adenoviruses (serotypes 3, 4, 7, 8, 19, and 37), and it is highly contagious, spreading through direct contact with infected ocular secretions or contaminated surfaces.

Viral Culture

Growth of a virus in susceptible cell cultures to identify an infectious agent. It can be useful for selected viruses, but molecular and antigen-based tests are often faster or more sensitive; the appropriate method depends on the suspected infection and specimen.

Viral Cytopathic Effects

Observable changes in host cells caused by viral infection. CPE include cell rounding, detachment, syncytia formation (fusion of infected cells), inclusion bodies, and cell lysis. CPE are used in viral diagnosis and can be observed by light microscopy.

Viral Encephalitis

Inflammation of the brain caused by a viral infection, resulting in dysfunction of neuronal tissue and surrounding inflammatory responses. Important causes include herpes simplex virus, varicella-zoster virus, enteroviruses, arboviruses, and other regionally prevalent pathogens. Clinical manifestations may include fever, headache, altered mental status, seizures, focal neurological deficits, or behavioral changes. Diagnosis requires urgent clinical assessment, cerebrospinal-fluid analysis, neuroimaging, and targeted microbiologic testing; treatment depends on the suspected virus and severity.

Viral Entry

The process by which viruses cross the host cell membrane to deliver their genome into the cytoplasm. Entry occurs through membrane fusion (enveloped viruses) or endocytosis (enveloped and non-enveloped viruses). For membrane fusion, viral fusion proteins mediate merger of the viral envelope with the host cell membrane.

Viral Exanthem

A widespread skin eruption caused by a viral infection, often accompanied by fever or other systemic symptoms.

Viral Gastroenteritis

Inflammation of the stomach or intestines caused by a virus. Common symptoms include watery diarrhea, vomiting, abdominal cramps, and fever. Diagnosis is usually clinical; stool testing is reserved for selected situations. Treatment focuses on fluid replacement and symptom control.

Viral Gastroenteritis (Stomach Flu)

Viral gastroenteritis is infection of the stomach and intestines causing vomiting, watery diarrhea, cramps, and fever, with dehydration as the main acute risk.

Alternate Names:

Stomach Flu

Viral Haemorrhagic Fever

A severe illness caused by certain viruses that may

produce fever, vascular injury, thrombocytopenia, and bleeding; severity and transmission vary by virus.

Viral Hemorrhagic Fevers

Severe viral infections that can cause fever, vascular injury, bleeding, shock, and organ failure.

Alternate Names:

Bunyaviruses

Viral Hepatitis

Inflammation of the liver caused by viral infections. The major hepatitis viruses include A, B, C, D, and E. Hepatitis A and E are transmitted through the fecal-oral route and cause acute, self-limited infections.

Alternate Names:

Infectious Hepatitis

Viral Immune Evasion

Mechanisms by which viruses evade host immune responses. Strategies include antigenic variation, inhibition of interferon signalling, downregulation of MHC molecules, and encoding viral homologues of cytokines. Understanding viral immune evasion is essential for developing effective vaccines and antiviral therapies.

Viral Infection

A viral infection occurs when a virus enters and replicates in host cells, causing symptoms that vary with the virus, tissue, immune response, and host.

Viral Infections

Illnesses caused by viruses, which are intracellular pathogens that hijack host cellular machinery to replicate. They range from self-limited conditions such as the common cold and gastroenteritis to serious diseases including influenza, hepatitis, HIV, and COVID-19.

Viral Latency

A state in which viral genomes persist in host cells without producing infectious particles. Latent viruses can reactivate under certain conditions

including immunosuppression. Herpesviruses establish latent infections in specific cell types (HSV in neurons, EBV in B cells, CMV in myeloid cells).

Viral Latency Mechanisms

The strategies by which viruses establish and maintain latent infections. Herpesviruses maintain their genomes as episomes in the nucleus. HIV integrates into the host genome as a provirus. Latent viruses express minimal proteins to avoid immune detection. Reactivation can be triggered by immunosuppression, stress, or other stimuli.

Viral Load

The amount of virus or viral genetic material in a body fluid or tissue, measured by a laboratory assay; trends can help assess disease activity or response to antiviral treatment.

Viral Load (HIV RNA)

The amount of HIV RNA in plasma, measured by a laboratory assay. It is used to assess viral replication and response to antiretroviral therapy. An undetectable result indicates that the amount is below the assay's reporting limit; testing intervals and treatment decisions follow current clinical guidance.

Alternate Names:

HIV RNA

Viral Meningitis

Inflammation of the meninges caused by viruses, usually Enteroviruses. Presents similarly to bacterial meningitis but generally less severe. CSF shows lymphocytic pleocytosis, normal glucose, and mildly elevated protein. Self-limited in most cases; supportive treatment.

Viral Pathogenesis

The mechanisms by which viruses cause disease. Pathogenesis involves attachment, entry, replication, and spread within the host. Viral damage can be direct (cytopathic effects) or indirect (immune-mediated). Direct effects include cell death, altered cell function, and cell transformation.

Viral Pneumonia

Infection and inflammation of lung tissue caused by a virus, producing cough, fever, chills, breathlessness, chest discomfort, fatigue, and variable impairment of oxygenation. Severe disease can cause respiratory failure or secondary bacterial infection.

Viral Release

The process by which newly assembled virions exit the host cell to infect other cells. Release occurs through budding (enveloped viruses) or cell lysis (non-enveloped viruses). Budding allows the virus to acquire an envelope from host cell membranes without killing the cell.

Viral Replication

The process of synthesising new viral genomes and proteins within host cells. The mechanism varies by virus type. DNA viruses typically replicate in the nucleus using host cell DNA polymerases (except poxviruses). RNA viruses typically replicate in the cytoplasm using viral RNA-dependent RNA polymerase.

Viral Replication Cycle

The series of steps by which viruses reproduce within host cells.

Viral Structure

The basic components of viral particles (virions). Viruses consist of a nucleic acid genome (DNA or RNA) surrounded by a protein coat called a capsid. Some viruses have an outer lipid envelope derived from host cell membranes. The capsid is composed of structural units called capsomeres.

Viral Uncoating

The process by which the viral capsid is disassembled to release the viral genome into the host cell cytoplasm. Uncoating occurs after entry and before replication. The mechanism varies by virus type and can involve pH-dependent conformational changes, proteolytic cleavage, or disassembly of capsid proteins.

Virchow's Triad

Virchow's triad describes three broad categories of factors that contribute to thrombosis: endothelial injury or dysfunction, stasis or turbulent blood flow, and hypercoagulability. Endothelial injury exposes subendothelial collagen and tissue factor, initiating platelet adhesion and the coagulation cascade.

Viremia

The presence of virus in the bloodstream. Viremia may be transient or persistent and can enable dissemination to target organs; it may be detected by culture, antigen testing, or nucleic-acid amplification.

Virginal Membrane

A historical term usually referring to the hymen, a variable rim or fold of mucosal tissue at the vaginal opening; its appearance cannot reliably establish sexual history.

Virginiamycin

A streptogramin antibiotic combination used mainly in veterinary medicine; it inhibits bacterial protein synthesis and is not routinely used to treat people.

Viridans Streptococci

Viridans streptococci are a group of bacteria commonly found in the mouth that can cause dental infections and, after entering the blood, infect heart valves.

Virilisation

The development of male secondary sexual characteristics in females, resulting from excessive androgen production or exposure.

Virilism

Development of masculinizing features in a female, usually from excess androgen exposure or production, such as hirsutism, deep voice, acne, and menstrual disturbance.

Virilism Foetal

Prenatal or neonatal masculinization of a genetically

female fetus or infant due to excess androgen exposure. Causes include congenital adrenal hyperplasia, maternal androgen-producing disease, or placental and fetal steroidogenic disorders.

Virilization

Virilization is development of typically male secondary sexual features, such as hirsutism, deep voice, acne, increased muscle mass, or clitoromegaly, from androgen excess or androgen exposure.

Virilize

To virilize is to develop or cause typically male physical changes from androgen excess or treatment, including facial hair, voice deepening, acne, muscle change, or genital enlargement.

Virology

The branch of microbiology concerned with viruses: structure, classification (DNA/RNA, enveloped/nonenveloped, Baltimore scheme), replication, pathogenesis, epidemiology, diagnosis, treatment, and prevention. Diagnostic methods: cell culture, antigen detection, nucleic acid amplification (PCR), and serology (IgM/IgG).

Virtual Colonoscopy

Virtual colonoscopy is CT colonography, which uses bowel preparation and gas or carbon-dioxide insufflation to create images of the colon; it can detect polyps and masses but cannot remove lesions or biopsy them.

Virulence

The relative capacity of a microorganism to cause disease or tissue injury in a host. It depends on microbial factors such as toxins, adhesins, invasiveness, and immune evasion, together with host susceptibility.

Virulence Factor

A microbial product or characteristic that helps an organism colonize, evade host defenses, injure tissue, or otherwise cause disease. Examples include adhesins, toxins, capsules, enzymes, and mechanisms that alter immune recognition.

Virulent

Highly harmful, severe, or capable of causing serious disease.

Virus

An infectious agent consisting of genetic material enclosed in a protein coat, sometimes with a lipid envelope, that replicates only inside living host cells. bleeding. Newborns have limited vitamin K stores, so the recommended route and dose are chosen according to local guidance and the infant's clinical circumstances.

Virus Activation

Virus activation is the switch from a dormant or inactive viral state to production of new viral components and particles.

Virus Assembly

Virus assembly is the stage of the viral life cycle when viral genetic material and proteins are packaged into new virus particles.

Virus Diseases

Viral diseases are illnesses caused by viruses, ranging from mild localized infections to severe diseases affecting multiple organs.

Virus Integration

Virus integration is insertion of viral genetic material into the DNA of a host cell, a process used by some viruses and important in certain cancers.

Virus Protein

A virus protein is a protein made from viral genetic instructions that helps form the virus, enter cells, replicate, or evade immunity.

Virus Receptor

A virus receptor is a molecule on a host cell that a virus binds to in order to attach to and enter the cell.

Virus Replication

Virus replication is the process by which a virus

copies its genetic material and makes proteins and new virus particles inside a host cell.

Virus Uncoating

Virus uncoating is removal of the virus's protective outer layer after entry into a host cell, releasing its genetic material for replication.

Viruses

Viruses are infectious particles containing DNA or RNA within a protein coat, sometimes with a lipid envelope; they replicate only inside host cells and cause disease through direct injury or immune responses.

Viscera

The internal organs of the body, particularly those within the thoracic and abdominal cavities. The thoracic viscera include the heart, lungs, esophagus, trachea, and thymus.

Visceral

Relating to the internal organs, especially those within the thorax or abdomen, rather than the body wall or limbs.

Visceral Larva Migrants

A syndrome caused by migration of animal nematode larvae through human tissues, most often from *Toxocara* species. It may produce fever, eosinophilia, liver lesions, or ocular and neurologic disease.

Visceromegaly

Abnormal enlargement of one or more internal organs, such as the liver, spleen, or kidneys. It is a physical finding with causes that include infection, congestion, storage disease, inflammation, and cancer.

Viscus

An internal organ, particularly an organ in the thoracic or abdominal cavity; the plural is viscera.

Vision

The sensory ability to perceive light and interpret

visual information processed by the eyes and brain, enabling recognition of shapes, colors, depth, and motion. Normal vision depends on the proper function of the cornea, lens, retina, and optic nerve, as well as intact neural pathways to the visual cortex.

Vision - Night Blindness

Difficulty seeing in dim light or adapting from bright to dark conditions, also called nyctalopia; causes include vitamin-A deficiency, retinal disease, cataract, and some inherited disorders.

Vision And Hearing

The two primary sensory modalities through which humans perceive and interact with their environment. Age-related declines in both senses are common, with presbyopia and cataracts affecting vision and presbycusis affecting hearing, often beginning as early as the fourth decade of life.

Vision Impairment

Reduced visual acuity or visual-field function that affects daily activities. Causes include refractive error, cataract, glaucoma, retinal disease, optic-nerve injury, and neurologic disease. Evaluation and treatment depend on the cause; rehabilitation and visual aids may improve function.

Vision Loss

A decrease in the ability to see that cannot be fully corrected with eyeglasses, contact lenses, or medical treatment. It ranges from partial impairment to complete blindness and can result from conditions such as macular degeneration, glaucoma, diabetic retinopathy, cataracts, and traumatic injury.

Vision Loss, Acute

Sudden reduction or loss of vision in one or both eyes caused by retinal vascular occlusion, retinal detachment, optic-nerve disease, stroke, acute glaucoma, inflammation, trauma, or other emergency. It requires immediate medical or ophthalmic assessment.

Vision Problems

Changes in visual acuity, field, color, or clarity

caused by refractive error, eye disease, neurologic disease, or systemic illness. Sudden vision loss is an emergency.

Vision Receptors

Photoreceptors in the retina, mainly rods and cones, that convert light into neural signals.

Vision Screening

Screening for visual problems in children: newborn—red reflex examination (cataract, retinoblastoma); 3-5 years—visual acuity, strabismus screening (cover test, photostereopsis), and amblyopia risk factors.

Vision Test

An examination that measures visual acuity, fields, color, alignment, or eye structures to detect refractive error, retinal disease, optic-nerve dysfunction, glaucoma, or other causes of visual impairment.

Vision, Blurred

Reduced sharpness or clarity of vision that may result from refractive error, dry eye, cataract, retinal or optic-nerve disease, neurologic disease, medication, or other causes. Sudden, painful, one-sided, or progressive blurring requires prompt evaluation.

Visual Acuity

The resolving power of the eye, defined as the ability to distinguish two points as distinct, and the most common clinical measure of visual function.

Visual Acuity Test

A test measuring the smallest standardized detail a person can resolve at a specified distance, usually reported as a fraction or decimal.

Visual Agnosia

Inability to recognize or identify visually presented objects, faces, or symbols despite adequate vision and attention. It results from dysfunction of higher-order visual association cortex and may follow stroke, trauma, neurodegenerative disease, or other brain injury.

Visual Field

The entire area that can be seen while the eyes remain fixed in one position. Visual-field loss may result from retinal, optic-nerve, or brain lesions and is assessed by perimetry.

Visual Field Defect

Loss or alteration of a portion of the normally perceived visual field caused by dysfunction of the retina, optic nerve, chiasm, optic tract, radiations, or visual cortex. The pattern can help localize the lesion.

Visual Field Perimetry

A systematic method of measuring the sensitivity of the retina across the entire visual field using standardized light stimuli, providing a quantitative map of visual function that can detect, characterize, and monitor visual field loss caused by diseases of the retina, optic nerve, and visual pathways.

Visual Fields

The total area that can be seen while the eyes are fixed in one position; defects may indicate retinal, optic-nerve, or brain disease.

Visual Impairment

Reduced vision that cannot be fully corrected by ordinary glasses or contact lenses. Causes include refractive error, cataract, glaucoma, retinal disease, optic-nerve injury, and neurologic disease; new or sudden loss of vision requires urgent assessment.

Visualized Laser-Assisted Prostatectomy

Laser prostate surgery performed with direct visualization to remove obstructing tissue.

Alternate Names:

VLAP

Vital

Essential for life or relating to vital physiologic functions. Vital signs commonly include temperature, pulse, respiratory rate, blood pressure, and oxygen saturation.

Vital Capacity

The largest volume of air that can be exhaled after a maximal inhalation, equal to inspiratory reserve volume plus tidal volume plus expiratory reserve volume.

Vital Signs

Basic measurements of physiologic status, commonly including temperature, pulse, respiratory rate, blood pressure, and oxygen saturation. Values vary with age, activity, illness, and measurement conditions; trends and clinical context are important.

Vitamin A

A fat-soluble vitamin needed for vision, immune function, skin and mucosal health, and growth; deficiency causes night blindness, while excess can injure the liver and developing fetus.

Vitamin A Blood Test

A blood test measuring circulating retinol, used to assess suspected vitamin A deficiency or toxicity in appropriate clinical settings.

Vitamin A Deficiency

A nutritional disorder resulting from inadequate intake or absorption of retinol, causing xerophthalmia, night blindness, follicular hyperkeratosis, and impaired immunity. Severe deficiency leads to corneal ulceration and blindness, and is a leading cause of preventable childhood blindness worldwide.

Vitamin A: Retinal

Vitamin A (retinol) is a fat-soluble vitamin essential for vision, immune function, cell differentiation, and reproduction. In the eye, retinol is oxidized to retinal (retinaldehyde), which binds to opsin in the photoreceptor outer segments forming rhodopsin (in rods) and photopsins (in cones).

Vitamin B Complex

A group of eight water-soluble vitamins that support energy metabolism, blood-cell production, nervous-system function, and other biochemical processes. The group includes vitamins B1 through B7 and

B9, as well as vitamin B12.

Vitamin B1

A vitamin or nutritional compound called vitamin b1. It supports specific physiological processes, and deficiency or excess can cause clinical disease.

Vitamin B1 (Thiamine)

Water-soluble vitamin, coenzyme in carbohydrate metabolism. Deficiency: beriberi (wet, dry), Wernicke-Korsakoff. Risk: alcoholism, bariatric surgery, dialysis. Treatment: supplementation.

Alternate Names:

Thiamine

Vitamin B12

A vitamin or nutritional compound called vitamin b12. It supports specific physiological processes, and deficiency or excess can cause clinical disease.

Vitamin B12 (Cobalamin)

Required for nerve function, red blood cell formation, DNA synthesis. Deficiency: megaloblastic anemia, neurological damage. Risk: vegans, elderly, pernicious anemia, metformin use, bariatric surgery. Requires intrinsic factor for absorption.

Alternate Names:

Cobalamin

Vitamin B12 Deficiency

Megaloblastic anemia, neurological symptoms. Pernicious anemia, vegan diet. Treated with B12 supplementation.

Vitamin B12 Deficiency Anemia

Vitamin B12 deficiency anemia is megaloblastic anemia from inadequate B12, impaired absorption, or medication effects and may cause irreversible neurologic injury if prolonged.

Vitamin B12 Deficiency In Older Adults

Deficiency caused by inadequate intake, malabsorption, autoimmune intrinsic-factor deficiency, gastric surgery, or medications. It may cause megaloblastic anemia, neuropathy, cognitive changes, gait impair-

ment, and glossitis; neurologic injury can occur without anemia.

Vitamin B12 Level

The amount of vitamin B12 measured in blood; results are interpreted with symptoms and related tests because serum levels may not fully reflect tissue deficiency.

Vitamin B12: Cobalamin

A water-soluble vitamin needed for red-blood-cell formation, DNA synthesis, and normal nerve function. Deficiency can cause megaloblastic anemia and neurologic abnormalities.

Vitamin B1: Thiamine

A water-soluble vitamin that acts as a cofactor in carbohydrate and amino-acid metabolism and supports nerve function. Severe deficiency can cause beriberi or Wernicke-Korsakoff syndrome.

Vitamin B2

A vitamin or nutritional compound called vitamin b2. It supports specific physiological processes, and deficiency or excess can cause clinical disease.

Vitamin B2 (Riboflavin)

Precursor for FAD and FMN. Deficiency: cheilosis, glossitis, seborrheic dermatitis. Common with B-complex deficiency.

Alternate Names:

Riboflavin

Vitamin B3 (Niacin)

Precursor for NAD⁺ and NADP⁺. Deficiency: pellagra (diarrhea, dermatitis, dementia). Sources: meat, fish, poultry, grains.

Alternate Names:

Niacin

Vitamin B3: Niacin

A water-soluble vitamin that forms the coenzymes NAD and NADP, which participate in many oxidation-reduction and energy-producing reactions. Severe deficiency causes pellagra.

Vitamin B5 (Pantothenic Acid)

Component of coenzyme A. Deficiency rare. Involved in fatty acid metabolism.

Alternate Names:

Pantothenic Acid

Vitamin B6

A vitamin or nutritional compound called vitamin b6. It supports specific physiological processes, and deficiency or excess can cause clinical disease.

Vitamin B6 (Pyridoxine)

Coenzyme in amino acid metabolism. Deficiency: seizures, anemia, dermatitis. Risk: isoniazid therapy, alcoholism.

Alternate Names:

Pyridoxine

Vitamin B6: Pyridoxine

Pyridoxine (pyridoxal phosphate, PLP, the active form) is a water-soluble B vitamin essential as a cofactor for over 100 enzymatic reactions, particularly in amino acid metabolism.

Vitamin B7 (Biotin)

Coenzyme for carboxylation reactions. Deficiency rare, may occur with raw egg consumption (avidin binds biotin).

Alternate Names:

Biotin

Vitamin B9 (Folate)

Essential for DNA synthesis, red blood cell formation. Deficiency: megaloblastic anemia, neural tube defects in pregnancy. Sources: leafy greens, legumes, fortified grains. Supplementation recommended before and during pregnancy.

Alternate Names:

Folate

Vitamin B9: Folate

Folate (folic acid, tetrahydrofolate/THF) is a water-soluble B vitamin essential for one-carbon transfer reactions in nucleotide synthesis (thymidylate synthesis for DNA, purine synthesis), amino acid

metabolism (methionine synthesis from homocysteine), and methylation reactions.

Vitamin C

A water-soluble vitamin needed for collagen formation, wound healing, antioxidant activity, and iron absorption; severe deficiency causes scurvy.

Vitamin C (Ascorbic Acid)

A water-soluble vitamin and a powerful antioxidant, essential for collagen synthesis (required for proline and lysine hydroxylation in collagen triple helix formation), carnitine synthesis, neurotransmitter synthesis (dopamine to norepinephrine), and iron absorption (reduces ferric to ferrous iron in the gut).

Alternate Names:

Ascorbic Acid

Vitamin D

Vitamin D supports calcium and phosphate balance, bone mineralization, muscle function, and immune processes; deficiency causes rickets in children and osteomalacia or bone loss in adults.

Vitamin D Deficiency

Inadequate vitamin-D status causing impaired calcium and phosphate homeostasis and, in severe cases, defective bone mineralization. It may result from low exposure or intake, malabsorption, liver or kidney disease, or altered metabolism.

Vitamin D3

A vitamin or nutritional compound called vitamin d3. It supports specific physiological processes, and deficiency or excess can cause clinical disease.

Vitamin D: Calcitriol

Vitamin D is a fat-soluble vitamin (actually a pro-hormone) essential for calcium and phosphorus homeostasis and bone health. Cholecalciferol (D3) is synthesized in the skin from 7-dehydrocholesterol by UVB radiation, or obtained from diet (ergocalciferol/D2 from plants, D3 from animal sources).

Vitamin Deficiency Anemia

Anemia caused by inadequate vitamin B-12, folate, or other nutrients needed for normal red blood cell production.

Vitamin E: Tocopherol

Vitamin E (alpha-tocopherol) is a fat-soluble vitamin and the body's primary lipid-soluble antioxidant, protecting polyunsaturated fatty acids in cell membranes from lipid peroxidation by free radicals (reactive oxygen species). It donates a hydrogen atom to lipid peroxy radicals, breaking the chain reaction of lipid peroxidation.

Vitamin K

A fat-soluble vitamin required to activate several blood-clotting proteins and support bone metabolism; deficiency can cause bleeding and is treated or prevented with replacement.

Vitamin K Deficiency

Coagulopathy, bleeding. Newborns receive prophylaxis. Antibiotics can cause deficiency.

Vitamin K Deficiency Bleeding Of The Newborn

Bleeding in an infant caused by inadequate vitamin K-dependent clotting factors; preventive vitamin K given at birth greatly reduces the risk.

Vitamin K Prophylaxis

Administration of vitamin K to a newborn soon after birth to prevent vitamin K deficiency

Vitamin K: Phylloquinone

Vitamin K is a fat-soluble vitamin essential for the post-translational gamma-carboxylation of glutamate residues to gamma-carboxyglutamate (Gla) in specific clotting factors (II, VII, IX, X, protein C, protein S) and anticoagulant proteins (protein C, protein S, protein Z).

Vitamins

Organic nutrients needed in small amounts for normal metabolism, growth, and tissue maintenance;

some are water-soluble and others fat-soluble, and both deficiency and excess can cause disease.

Vitrectomy

Surgical removal of the vitreous gel, typically via the pars plana approach, using a cutting probe under microscopic visualization. Indications include vitreous hemorrhage, complicated retinal detachment, macular holes, epiretinal membranes, intraocular foreign bodies, and endophthalmitis.

Vitreous

Relating to the transparent gel filling most of the space between the lens and retina, or describing a glass-like appearance. Vitreous disorders can cause floaters, hemorrhage, detachment, or retinal traction.

Vitreous Body

The transparent gel filling the large space between the lens and retina. It helps maintain the shape of the eye and transmits light; age-related liquefaction can cause floaters or contribute to retinal traction.

Alternate Names:

vitreous humor; vitreous

Vitreous Chemistry

Analysis of the fluid from the eye after death for substances such as electrolytes, glucose, urea, creatinine, and ethanol. It may support assessment of postmortem interval or metabolic disturbance but is not independently diagnostic of cause or time of death.

Vitreous Detachment

Separation of the vitreous gel from the retina, usually related to aging and often causing new floaters or flashes. Sudden symptoms require assessment for retinal tears or detachment.

Vitreous Hemorrhage

Bleeding into the vitreous cavity that causes sudden painless floaters, cobwebs, haze, or loss of vision. Common causes include proliferative diabetic retinopathy, retinal tear or detachment, trauma, and posterior vitreous separation.

Vitreous Humor

The clear gel that fills the space between the lens and retina. It helps maintain the shape of the eye and transmits light to the retina.

VLDL Test

A measurement or estimate of very-low-density lipoprotein cholesterol, which carries endogenous triglycerides and is considered an atherogenic lipoprotein.

VO2 Max

The maximum rate at which the body can use oxygen during intense exercise.

Alternate Names:

Maximal oxygen consumption

Vocal Cord Nodules

Benign, usually paired thickened areas of the vocal folds caused by repeated voice strain or overuse. They can cause hoarseness, vocal fatigue, and a rough or breathy voice.

Vocal Cord Paralysis

Immobility of one or both vocal cords due to disruption of the recurrent laryngeal nerve (RLN). Unilateral vocal cord paralysis: hoarseness, breathy voice, aspiration, and ineffective cough.

Alternate Names:

Vocal Fold Paralysis

Vocal Cord Paresis

Partial paralysis or weakness of one or both vocal cords due to impaired function of the recurrent laryngeal nerve (RLN) or the vagus nerve. Unilateral paresis is more common and presents with hoarseness, breathy dysphonia, vocal fatigue, and sometimes aspiration.

Vocal Cord Polyp

Vocal cord polyps are benign, unilateral, sessile or pedunculated lesions that arise from the vocal cord mucosa, typically as a result of a single traumatic vocal event such as shouting, screaming, or coughing, causing hemorrhage and edema in the superficial

lamina propria (Reinke space).

Vocal Fold Paralysis

Alternate index label; see *Vocal cord paralysis*.

Vocal Folds

Paired mucosal folds within the larynx that vibrate as air passes between them to produce sound. They also close to protect the airway and assist in raising intrathoracic pressure.

Vocal Resonance

The transmission of vocal sounds to the chest wall, assessed by having the patient speak while auscultating. It is increased in consolidation and decreased in pleural effusion or pneumothorax. It correlates with bronchophony and whispered pectoriloquy.

Vocal-Cord Dysfunction

Episodic inappropriate adduction of the vocal folds, especially during inspiration, causing throat tightness, noisy breathing, and apparent asthma-like attacks. Diagnosis is supported by laryngoscopy during symptoms; speech therapy and trigger management are central treatments.

Voice

Sound produced by vibration of the vocal folds as air passes through the larynx, then shaped by the throat, mouth, and nose for speech.

Voice And Speech

The products of complex coordinated activity of the respiratory, laryngeal, and articulatory systems.

Voice Box

The larynx, an organ in the neck containing the vocal folds and forming part of the airway. It produces voice and helps protect the lower respiratory tract during swallowing.

Voice Disorders

Conditions affecting voice quality, pitch, loudness, or ease of speaking because of problems involving the vocal folds, airway, nerves, or voice use.

Alternate Names:

Difficulty Speaking

Void

To empty the urinary bladder by urinating; in other medical contexts, void may mean an empty space or absence of a substance.

Voiding Cystourethrogram

Fluoroscopic imaging of the bladder and urethra during voiding, after filling the bladder with contrast via a catheter. Indications: evaluation of vesicoureteral reflux (VUR) in children with recurrent UTIs, posterior urethral valves in males, and post-surgical assessment of the bladder/urethra.

Volar

Relating to the palm of the hand or sole of the foot, especially the surface opposite the dorsum.

Volatile Anesthetics

Halogenated inhaled anesthetic agents, including sevoflurane, desflurane, and isoflurane, that reversibly depress central nervous system function. They provide dose-dependent hypnosis and immobility but may cause hypotension, respiratory depression, postoperative nausea, and malignant hyperthermia in susceptible individuals.

Volatile Oils

Plant oils containing compounds that evaporate readily at room temperature.

Alternate Names:

Essential oils

Volkmann Contracture

A permanent flexion deformity of the hand, wrist, and fingers caused by ischemic injury and fibrosis of forearm muscles. It most often follows untreated or inadequately treated compartment syndrome after fracture, crush injury, vascular compromise, or severe swelling. The degree of functional loss depends on the muscles and nerves affected and may include pain, weakness, sensory impairment, and restricted finger extension. Prevention requires urgent recognition of limb ischemia and compartment pressure;

established contractures require reconstructive assessment and may need surgical release or tendon procedures.

Volume Of Distribution

A theoretical pharmacokinetic parameter that relates the total amount of drug in the body to the plasma concentration, calculated as V_d equals amount of drug in the body divided by plasma concentration. It represents the apparent volume into which a drug distributes and is not a true anatomical volume.

Volume Overload

Excess total body sodium and water, causing peripheral or pulmonary oedema; common causes include heart failure, kidney failure, liver disease, and excessive intravenous fluid, and treatment depends on the cause and severity.

Voluntary

Under conscious control or performed by choice, such as voluntary movement. Voluntary action is distinguished from reflex, autonomic, or involuntary activity.

Volutrauma

Lung injury caused by excessive mechanical stretch, usually from an inappropriately large tidal volume during ventilation. Risk is reduced by lung-protective ventilation tailored to the patient's predicted body weight, lung mechanics, and clinical condition.

Volvulus

Twisting of a loop of intestine around its mesenteric attachment point causing luminal obstruction and compromise of the mesenteric blood supply which can rapidly progress to bowel ischemia.

Volvulus - Childhood

Twisting of a loop of intestine around its mesentery, causing obstruction and potentially loss of blood supply. In children it is often related to intestinal malrotation and is a surgical emergency when acute.

Vomit

To eject stomach or upper gastrointestinal contents through the mouth; the expelled material may also be called vomitus. Persistent vomiting can cause dehydration and electrolyte abnormalities.

Vomiting

Reflex expulsion of gastric contents. Causes: GI, neurological, metabolic, toxic, psychological. Treatment: antiemetics, identify cause, prevent aspiration.

Vomiting Blood

The passage of blood or coffee-ground material in vomit, usually indicating upper gastrointestinal bleeding. It is a potentially serious symptom requiring prompt medical evaluation.

Vomitus

Material expelled from the stomach or upper gastrointestinal tract during vomiting. Its color and contents can provide clues but do not by themselves establish a diagnosis.

Von Gierke Disease

Glycogen storage disease type I, an inherited deficiency of glucose-6-phosphatase or its transport system that causes fasting hypoglycemia, lactic acidosis, enlarged liver, and excess uric acid or lipids.

Von Hippel-Lindau Disease

Autosomal dominant syndrome caused by mutations in the VHL tumor suppressor gene on chromosome three predisposing to hemangioblastomas of the central nervous system and retina, clear cell renal cell carcinoma, pheochromocytoma, pancreatic cysts and neuroendocrine tumors, and endolymphatic sac tumors.

Von Recklinghausen'S Disease

Alternate index label; see *Neurofibromatosis type 1*.

Von Recklinghausen'S Neurofibromatosis

Alternate index label; see *Neurofibromatosis type 1*.

Von Recklinghausen'S Phakomatosis

Alternate index label; see *Neurofibromatosis type 1*.

Von Willebrand Disease

An inherited or acquired bleeding disorder caused by deficient or abnormal von Willebrand factor, which helps platelets adhere and carries factor VIII. It commonly causes nosebleeds, easy bruising, heavy menstrual bleeding, or prolonged bleeding after procedures.

VOR Reflex

The vestibulo-ocular reflex stabilizes gaze by moving the eyes opposite to head movement through inner-ear, brainstem, and eye-muscle pathways. Damage can cause oscillopsia, vertigo, imbalance, and abnormal head-impulse testing.

Voriconazole

A triazole antifungal agent used in the treatment of invasive aspergillosis, candidemia, and other serious fungal infections. Voriconazole inhibits the fungal cytochrome P450 enzyme lanosterol 14-alpha demethylase, preventing ergosterol synthesis and disrupting fungal cell membrane integrity.

Voyeurism

A paraphilic disorder involving recurrent intense sexual arousal from observing an unsuspecting person who is naked, disrobing, or engaging in sexual activity, when it causes distress, impairment, or involves nonconsenting behavior.

Voyeuristic Disorder

A paraphilic disorder involving recurrent, intense sexual arousal from observing an unsuspecting person who is naked, disrobing, or engaging in sexual activity. The individual has acted on these urges or they cause marked distress or interpersonal difficulty. Duration ≥ 6 months. Most common in males with onset in adolescence.

VPCs

Alternate index label; see *Premature ventricular contractions (PVCs)*.

VRE

Enterococcus bacteria that are resistant to vancomycin, often because of acquired changes in cell-wall precursors. VRE can cause urinary, blood-stream, wound, or other healthcare-associated infections. Treatment requires susceptibility-guided selection of an effective antimicrobial and source control when needed.

VSD

Ventricular septal defect, a hole in the wall separating the heart's ventricles.

Alternate Names:

Ventricular septal defect

Vulnerable

A substance traditionally believed to help heal wounds. Such products may lack good evidence and can cause allergy or contamination, so wound care should follow appropriate medical advice.

Vulva

The external female genital structures, including the mons pubis, labia, clitoris, and vestibule. It protects the openings of the urinary and reproductive tracts and contains specialized sensory and glandular tissue.

Alternate Names:

female external genitalia

Vulvar

Relating to the vulva, the external female genital structures including the labia, clitoris, vestibule, and associated glands.

Vulvar Carcinoma

An invasive malignant neoplasm of the external female genitalia, predominantly squamous cell carcinoma (90% of cases). Occurs in two distinct clinical pathways: a high-risk HPV-associated type in younger patients and an HPV-independent, differentiated VIN-related type associated with chronic inflammatory dermatoses (such as lichen sclerosus) in postmenopausal women.

Alternate Names:

Vulvar cancer; Carcinoma of the vulva

Vulvar Intraepithelial Neoplasia

A premalignant or potentially premalignant squamous lesion of the vulva. Most usual-type lesions are related to high-risk HPV, whereas differentiated lesions are commonly associated with lichen sclerosus and have a higher risk of invasive carcinoma.

Vulvar Melanoma

A malignant melanocytic neoplasm arising in vulvar skin or mucosa, most often affecting the labia in older adults. It may present as a pigmented or amelanotic mass, plaque, ulcer, bleeding lesion, or persistent area of altered pigmentation. Diagnosis requires biopsy with assessment of tumor thickness, ulceration, mitotic activity, and regional or distant spread. Management is based on stage and resectability and may involve local excision, evaluation of regional lymph nodes, and systemic treatment for advanced disease.

Vulvar Squamous Cell Carcinoma

The most common invasive vulvar malignancy, arising through HPV-related or HPV-independent pathways. It usually presents as a persistent mass, ulcer, plaque, or wart-like lesion and requires biopsy for diagnosis.

Vulvectomy

Surgical removal of part or all of the vulva, performed for selected cancers, precancerous lesions, or severe persistent disease.

Vulvitis

Inflammation of the vulva (external female genitalia), often presenting with pruritus, burning, irritation, erythema, edema, and discharge. Vulvitis is rarely isolated and is most commonly part of vulvovaginitis (inflammation of both vulva and vagina).

Vulvitis, Yeast

Inflammation of the vulva associated with vulvovaginal candidiasis, usually caused by *Candida* species. It may cause itching, burning, redness, swelling, and soreness and should be distinguished from other

causes of vulvar symptoms.

Vulvodynia

Vulvodynia is chronic vulvar pain without an identifiable specific cause, often described as burning or rawness and triggered by touch, pressure, or occurring spontaneously.

Alternate Names:

Chronic Vulvar Pain

Vulvovaginal Candidiasis

Symptomatic infection of the vulva and vagina, usually caused by *Candida albicans*. Pruritus, soreness, erythema, and thick discharge are common; diagnosis is clinical with microscopy or culture when recurrent, severe, atypical, or treatment-resistant.

W

Waardenburg Syndrome

A group of inherited disorders characterized by sensorineural hearing loss and pigmentary abnormalities of the eyes, hair, or skin. Dystopia canthorum, limb abnormalities, or intestinal aganglionosis may occur in particular genetic forms.

Wafer

Thin, flat disc used in pharmaceutical formulations for controlled drug delivery, particularly in intravitreal implants for retinal conditions.

WAGR Syndrome

A contiguous-gene deletion syndrome involving Wilms tumor predisposition, aniridia, genitourinary anomalies, and variable developmental delay. It usually involves chromosome 11p13 and requires tumor surveillance and multidisciplinary care.

Waist Circumference

A measure of abdominal adiposity taken at a standardized site. Larger measurements are associated with increased cardiometabolic risk, but clinically used thresholds vary by sex, ancestry, and guideline; it complements, rather than replaces, body-mass index and other assessment.

Waist-Hip Ratio

Anthropometric measure calculated by dividing waist circumference by hip circumference; used to assess central obesity and cardiovascular risk.

Wait List

Registry of patients awaiting organ transplantation, maintained by national transplant organizations,

prioritized by medical urgency and compatibility.

Wakefulness

State of being conscious and alert; regulated by the reticular activating system and neurotransmitters including orexin, histamine, and acetylcholine.

Waldenstrom Macroglobulinemia

A rare B-cell lymphoplasmacytic lymphoma characterized by bone marrow infiltration by lymphoplasmacytic cells and production of monoclonal IgM paraprotein. It accounts for less than 2 percent of hematologic malignancies and typically affects older adults (median age 63-65 years).

Alternate Names:

Lymphoplasmacytic Lymphoma

Waldenstrom's Macroglobulinemia

A lymphoplasmacytic cancer of the bone marrow that produces excess IgM and may cause anemia, enlarged lymph nodes, or hyperviscosity symptoms.

Waldenström's Macroglobulinemia

An indolent B-cell lymphoplasmacytic lymphoma characterized by the production of a monoclonal IgM paraprotein, causing hyperviscosity syndrome with visual disturbances, bleeding, and neurological symptoms. It typically affects elderly patients and involves the bone marrow, lymph nodes, spleen, and liver.

Walker-Warburg Syndrome

The most severe, lethal phenotype among the congenital muscular dystrophy-dystroglycanopathies, inherited in an autosomal recessive manner and

caused by mutations in genes essential for the *O*-mannosyl glycosylation of α -dystroglycan (most frequently *POMT1* and *POMT2*). Clinically characterized by a classic triad of: (1) severe congenital muscular dystrophy with marked hypotonia, generalized weakness, and massive creatine kinase elevation; (2) severe structural central nervous system malformations, including cobblestone (type II) lissencephaly, complete agenesis of the corpus callosum, hydrocephalus, and cerebellar hypoplasia; and (3) complex congenital ocular malformations, including microphthalmia, congenital cataracts, buphthalmos, and retinal detachment, typically culminating in death during infancy.

Alternate Names:

WWS; Muscle-Eye-Brain Disease (Severe Type); HARD +/- E Syndrome

Walking Abnormalities

Walking abnormalities include changes in speed, balance, stride, posture, or coordination caused by pain, weakness, sensory loss, joint disease, or neurologic illness.

Walking Speed

The rate at which a person walks, often measured over a defined distance. It can provide information about mobility and function, but no single speed determines disability in every person.

Wall Stress

Force exerted on a heart chamber wall by pressure inside the chamber.

Wallerian Degeneration

The breakdown of the portion of a nerve fiber separated from its cell body after nerve injury.

Walleye

Walleye is an informal term for strabismus, in which the eyes are misaligned and do not point at the same target.

Walleyed

Walleyed describes eyes that appear misaligned or divergent; clinically, the condition may be strabis-

mus and requires eye assessment when persistent.

Wandering Spleen

Abnormal mobility of the spleen caused by laxity or absence of its supporting ligaments. It may be asymptomatic or cause abdominal pain, splenic enlargement, torsion, infarction, or acute abdomen.

Ward'S Triangle

Radiological landmark formed by the intersection of the iliopectineal line, ilioischial line, and teardrop line on pelvic radiographs; used to assess acetabular coverage.

Warfarin

An anticoagulant that reduces blood clot formation by inhibiting vitamin-K-dependent clotting factors.

Warfarin Syndrome

A pattern of fetal abnormalities caused by warfarin exposure during pregnancy, including nasal and skeletal abnormalities, eye or neurologic defects, and fetal or neonatal bleeding.

Warm Autoimmune Hemolytic Anemia

The most common form of autoimmune hemolytic anemia (70–80% of cases), mediated by warm-reactive IgG autoantibodies that bind red blood cell surface antigens with maximal avidity at 37°C. Antibody-coated erythrocytes are selectively removed and phagocytosed by splenic macrophages (extravascular hemolysis), resulting in spherocytosis and a positive direct antiglobulin (Coombs) test.

Alternate Names:

wAIHA; Warm antibody autoimmune hemolytic anemia

Warm Ischemia

Tissue oxygen deprivation at body temperature due to interrupted blood supply; occurs during organ procurement for transplantation and causes more cellular damage than cold ischemia.

Wart

A benign skin growth caused by infection with hu-

man papillomavirus.

Alternate Names:

Verruca

Wart Remover Poisoning

Toxic exposure to a wart-removal product, often involving salicylic acid or another keratolytic. Severity depends on the substance, amount, route, symptoms, and duration of exposure.

Warthin Tumor

A benign salivary-gland neoplasm composed of oncocytic epithelial and lymphoid tissue, usually arising in the parotid gland. It is strongly associated with tobacco exposure, may be multifocal or bilateral, and is generally managed according to symptoms, diagnosis, and growth.

Warty Dyskeratoma

A benign follicular keratinizing tumor presenting as a solitary umbilicated papule or nodule, often on the head or neck. Histology shows a cup-shaped invagination filled with keratin and acantholytic dyskeratotic cells.

Washout Period

Interval between discontinuation of a drug or treatment and the start of a subsequent treatment phase; allows clearance of prior agents to avoid confounding effects in clinical studies.

Wasp Sting

A venomous insect sting that may cause localized pain, redness, and swelling; severe reactions can include anaphylaxis or, rarely, systemic envenomation.

Wassermann Reaction

A historical complement-fixation blood test for syphilis. It has largely been replaced by modern treponemal and nontreponemal serologic tests.

Wasting

Abnormal loss of body weight, muscle, or other tissue, often associated with chronic disease, inad-

equately nutrition, inflammation, or cancer. Severe wasting may impair strength, immunity, and recovery.

Watchful Waiting

Planned observation without immediate active treatment, with clinical review and treatment if symptoms develop or the condition progresses. It may be used for selected cancers and other slowly progressive conditions.

Water Birth (Underwater Delivery)

Delivery of a baby while the birthing person remains in a tub or pool of warm water. It may reduce perceived labor pain for some people, but evidence about delivery-related benefits and risks is limited. Eligibility, monitoring, and facility policies determine whether it is offered.

Alternate Names:

Underwater Delivery

Water Blister

A fluid-filled blister, usually containing clear serum, caused by friction, pressure, burns, or some skin disorders. The blister should be protected from further trauma and assessed if infected or widespread.

Water Brash

A sudden increase in salty saliva associated with acid reflux.

Water Channel

Aquaporin membrane proteins that facilitate rapid transcellular water transport; found in renal collecting ducts, red blood cells, and other tissues involved in fluid homeostasis.

Water Consumption

Intake of water and other fluids to replace normal losses and maintain hydration. Requirements vary with body size, diet, activity, climate, pregnancy, illness, and kidney or heart function; thirst and urine concentration can help guide intake, while excessive intake can cause dangerous hyponatremia.

Water Intoxication

A potentially life-threatening dilutional hyponatremia caused by excessive water intake or retention relative to the body's ability to excrete free water. Severe cases can cause headache, confusion, seizures, or coma.

Water On The Brain

A nontechnical term usually referring to hydrocephalus, an abnormal accumulation or impaired circulation of cerebrospinal fluid that enlarges the ventricles and may increase intracranial pressure.

Water On The Knee

Alternate index label; see *Swollen knee*.

Water Pollution

Contamination of water bodies with pathogens, chemicals, or heavy metals; associated with gastrointestinal illness, developmental disorders, and chronic disease.

Water Retention

Excess accumulation of fluid in the body's tissues or cavities, commonly causing swelling and sometimes resulting from heart, kidney, liver, venous, hormonal, or medication-related disorders.

Water Soluble Toxins

Toxic substances that dissolve readily in water. Their elimination depends on the substance, kidney function, metabolism, protein binding, and other properties; some may be removed by dialysis.

Water Supply

Infrastructure for delivering potable water to populations; public health measures include chlorination, filtration, and monitoring for contamination.

Water-Borne Disease

An illness acquired through water contaminated with infectious organisms or harmful chemicals. Examples include cholera, typhoid fever, and hepatitis A; prevention includes safe drinking water, sanitation, hand hygiene, and appropriate treatment of

contaminated water.

Water-Deprivation Test

A supervised dynamic test assessing the kidney's ability to concentrate urine when free-water intake is withheld, followed when appropriate by desmopressin. It helps distinguish central diabetes insipidus, nephrogenic diabetes insipidus, and primary polydipsia but requires careful monitoring for hyponatremia and dehydration.

Water-Hammer Pulse (Corrigan Pulse)

A pulse with a rapid upstroke and collapse, classically associated with significant aortic regurgitation and other states of increased pulse pressure.

Alternate Names:

Corrigan Pulse

Water-Soluble Vitamin

Vitamin class dissolving in water and not stored in significant amounts; includes B-complex vitamins and vitamin C, requiring regular dietary intake.

Waterborne Bacterial Disease

An infection acquired from water contaminated with bacteria, such as cholera, typhoid fever, or leptospirosis; prevention depends on safe water, sanitation, and food hygiene.

Waterhouse-Friderichsen Syndrome

Waterhouse-Friderichsen syndrome is adrenal hemorrhage and acute adrenal failure associated with overwhelming sepsis, classically meningococcal infection, causing shock and coagulopathy.

Watery Eyes

Excessive tearing caused by irritation, allergy, infection, dry-eye reflex tearing, or blockage of the tear-drainage system. Persistent unilateral tearing or pain requires ocular assessment.

Wax

A water-resistant substance produced by ceruminous glands in the ear canal; earwax protects the canal and usually clears naturally.

Wax Poisoning

Toxic exposure to a wax product, usually causing little harm after a small accidental ingestion but potentially causing choking, aspiration, irritation, or toxicity when mixed with other chemicals.

Waxy Casts

Broad refractile cylindrical structures with smooth appearance indicating severe chronic kidney disease. Form from degeneration of granular casts in dilated tubules. Broad diameter reflects formation in dilated collecting ducts. Relatively specific for advanced CKD with significant tubular atrophy.

WBC Count

A measurement of the number of white blood cells in a volume of blood, used with the differential count to evaluate infection, inflammation, immune disorders, marrow disease, or treatment effects.

Weakness

Reduced ability to generate force, which may be generalized or limited to one body region. Causes include disorders of muscle, nerve, neuromuscular junction, brain, or spinal cord, as well as metabolic illness, infection, medication effects, and deconditioning; sudden one-sided weakness or weakness with breathing, swallowing, or speech difficulty requires emergency assessment.

Weakness, Generalized

Reduced strength affecting multiple body regions, caused by systemic illness, metabolic or electrolyte disturbance, medication, infection, muscle disease, nerve disease, neuromuscular-junction disease, or brain or spinal-cord disease. Sudden or progressive weakness, especially with speech, facial, breathing, or walking difficulty, requires urgent assessment.

Weaning

Gradual reduction and eventual discontinuation of breastfeeding, bottle-feeding, or a medicine or other treatment, according to the relevant clinical context.

Wear Debris

Particles generated by mechanical abrasion or degra-

dation of implant materials. Macrophage-mediated inflammation around debris can produce osteolysis, implant loosening, and failure of an arthroplasty.

Weaver Syndrome

A congenital overgrowth and neurodevelopmental disorder associated with pathogenic variants in EZH2. It may cause prenatal overgrowth, advanced bone age, characteristic facial features, hypotonia, developmental delay, and variable intellectual disability.

Weaver'S Bottom

Inflammation of the ischial bursa over the sitting bone, causing pain in the buttock that may worsen with prolonged sitting. It is also called ischial bursitis.

Webbed Fingers

Partial or complete joining of adjacent fingers by skin or soft tissue, known as syndactyly; it may be congenital or acquired.

Webbed Neck

Lateral cervical skin folds extending from the mastoid process to the acromion; characteristic of Turner syndrome and Noonan syndrome.

Webbing Of The Fingers Or Toes

Partial or complete joining of adjacent fingers or toes by skin or other tissue. It is called syndactyly and may be congenital or acquired.

Weber

Unit of measurement for the frequency of bone conduction; used in audiology testing (Weber test) to lateralize hearing loss.

Weber Test

The Weber test is a screening test for lateralization of hearing that uses a vibrating tuning fork (typically 512 Hz) placed on the midline of the skull (vertex or forehead) to determine whether the sound is heard equally in both ears or louder in one ear, providing information about the type and side of

hearing loss.

Weber-Christian Disease

Weber-Christian disease is an older, nonspecific term historically used for relapsing fever and inflammation of subcutaneous fat, now often classified as panniculitis. Painful subcutaneous nodules and systemic symptoms require modern evaluation for infection, autoimmune disease, pancreatic disease, or other causes.

Alternate Names:

Panniculitis Idiopathic Nodular (Weber-Christian Disease)

Wedge Excision

Surgical removal of a triangular or wedge-shaped portion of tissue; commonly performed for breast biopsies, cervical lesions, and skin tumors.

Wedge Resection

Surgical removal of a wedge-shaped piece of tissue, typically lung or breast. Used for diagnosis or treatment of localized lesions. Less extensive than lobectomy.

Weeksella

Genus of Gram-negative bacteria in the family Neisseriaceae; occasionally isolated from clinical specimens.

Weeksella Virosa

Species of Gram-negative, oxidase-positive bacteria; opportunistic pathogen rarely associated with human infection.

Weever Fish

A venomous marine fish whose dorsal spines can cause a painful sting, local swelling, and occasionally systemic symptoms.

Wegener's Granulomatosis

Wegener's granulomatosis, now called granulomatosis with polyangiitis, is a systemic vasculitis characterized by necrotizing granulomatous inflammation of the upper and lower respiratory tracts and pauci-

immune necrotizing glomerulonephritis.

Wegeners Granulomatosis (Granulomatosis With Polyangiitis)

Alternate index label for Granulomatosis With Polyangiitis.

Weight Gain

Increase in body weight; may be intentional (muscle building, pregnancy) or unintentional (fluid retention, endocrine disorders, medication effects).

Weight Gain In Pregnancy

Recommended gestational weight gain according to the Institute of Medicine guidelines, based on prepregnancy BMI. Underweight (BMI <18.5): 28-40 lbs (12.5-18 kg). Normal weight (BMI 18.5-24.9): 25-35 lbs (11.5-16 kg). Overweight (BMI 25-29.9): 15-25 lbs (7-11.5 kg). Obese (BMI ≥ 30): 11-20 lbs (5-9 kg).

Weight Gain – Unintentional

Increase in body weight without a planned change in diet or activity, often caused by fluid retention, medicines, endocrine disease, reduced activity, or changes in eating; rapid swelling or breathlessness needs prompt assessment.

Weight Loss

A decrease in body weight that may be intentional or unintentional. Intentional loss is usually achieved through sustainable changes in eating, physical activity, and behavior; unintentional or rapid loss can indicate illness, inadequate nutrition, or medication effects and may require medical evaluation.

Alternate Names:

Obesity (Weight Loss)

Weight Loss - Unintentional

Unintentional weight loss is reduction in body weight without planned dietary or activity change and may result from malignancy, endocrine disease, infection, gastrointestinal disease, medication, or mood disorder.

Weight Loss And Management

Weight loss and management encompasses the ongoing process of achieving and maintaining a healthy body weight through sustainable lifestyle modifications.

Weight Loss Medications

Pharmacologic agents approved for the treatment of obesity when used in conjunction with lifestyle modifications.

Weight-Loss Medicines

Alternate index label; see *Weight Loss Medications*.

Weight-Reducing Diets

Dietary plans intended to create a sustained energy deficit and reduce excess body weight, ideally individualized with attention to nutrition, health conditions, and long-term adherence.

Weil'S Disease

A zoonotic bacterial infection caused by spirochaetes of the genus *Leptospira*, transmitted through contact with water or soil contaminated with the urine of infected animals (rodents, cattle, dogs). See *Leptospirosis*.

Weils Disease (Leptospirosis)

Alternate index label for *Leptospirosis*.

Wellens Syndrome

A pre-infarction electrocardiographic syndrome strongly indicative of critical, high-grade (> 85–90%) stenosis of the proximal left anterior descending coronary artery. Typically identified during pain-free intervals in patients presenting with unstable angina, characterized by: (1) deeply and symmetrically inverted T waves in leads V_2 and V_3 (Type B, 75% of cases); or (2) biphasic T waves in V_2 and V_3 with initial positivity and terminal negativity (Type A, 25% of cases); alongside normal or minimally elevated cardiac biomarkers, absence of pathological Q waves, and preservation of precordial R-wave progression. Carries an imminent risk of extensive anterior wall myocardial infarction or sudden death if urgent coronary angiography and

revascularization are delayed; exercise stress testing is strictly contraindicated.

Alternate Names:

Wellens Pattern; LAD Coronary T-Wave Syndrome; Wellens Sign

Wells Score For Pulmonary Embolism

A widely validated clinical prediction score used in acute and emergency settings to estimate the pretest probability of acute pulmonary embolism. It assigns specific weighted points to seven clinical variables: clinical signs and symptoms of deep vein thrombosis (3.0 points), PE considered most likely or equally likely diagnosis (3.0 points), heart rate > 100 bpm (1.5 points), immobilization or surgery in past 4 weeks (1.5 points), prior documented DVT or PE (1.5 points), hemoptysis (1.0 point), and active malignancy (1.0 point). Patients stratified as low probability (score < 2) or PE unlikely (score $\leq$ 4) are evaluated with high-sensitivity D-dimer, whereas those stratified as moderate/high probability or PE likely (score > 4) proceed directly to computed tomography pulmonary angiography.

Alternate Names:

Wells Criteria for PE; Wells Score (PE)

Welt

A raised, often itchy area of skin caused by localized swelling, commonly seen in hives or after an insect bite. A welt may fade within hours, but persistent or severe reactions need evaluation.

Wen (Epidermoid Cyst)

A common benign cyst beneath the skin that is lined by squamous epithelium and contains keratin debris.

Alternate Names:

Epidermoid Cyst

Wenckebach Phenomenon (Mobitz Type I AV Block)

Progressive prolongation of the PR interval followed by a nonconducted atrial impulse and a dropped QRS complex. It reflects intermittent impairment of atrioventricular conduction.

Alternate Names:

Mobitz Type I AV Block

Wens

Alternate index label; see *Wen (Epidermoid Cyst)*.

Werdnig-Hoffmann Disease

The most severe, infantile-onset form of 5q-related spinal muscular atrophy (Spinal Muscular Atrophy Type 1 [SMA 1]), inherited in an autosomal recessive manner caused by homozygous deletion or loss-of-function mutations in the *SMN1* gene on chromosome 5q13, with patients carrying only 1 or 2 copies of the backup *SMN2* gene. Manifests before 6 months of age with profound, generalized, flaccid hypotonia (“floppy infant syndrome”), symmetrical proximal muscle weakness, absent deep tendon reflexes, fasciculations of the tongue, paradoxical bell-shaped chest breathing due to intercostal muscle paralysis with diaphragm sparing, weak cry, and inability to ever sit unsupported, historically leading to respiratory failure and death within the first two years of life without disease-modifying therapies.

Alternate Names:

Spinal Muscular Atrophy Type 1 (SMA 1); Severe Infantile Spinal Muscular Atrophy

Wermer Syndrome

Alternate index label; see *Multiple endocrine neoplasia, type 1 (MEN 1)*.

Wermer’S Syndrome

Alternate index label; see *Multiple endocrine neoplasia, type 1 (MEN 1)*.

Werner Syndrome

An autosomal-recessive progeroid disorder caused most often by pathogenic variants in the *WRN* gene. It typically presents after puberty with premature ageing, short stature, bilateral cataracts, scleroderma-like skin changes, thin limbs, diabetes, osteoporosis, vascular disease, and increased risk of malignancy.

Werner’S Syndrome

An autosomal-recessive premature-aging disorder caused by *WRN* gene defects, characterized after

childhood by short stature, cataracts, scleroderma-like skin, diabetes, atherosclerosis, and increased cancer risk.

Wernicke Area

Region of the left dominant cerebral hemisphere (Brodmann area 22) involved in language comprehension; damage causes Wernicke aphasia with fluent but nonsensical speech.

Wernicke Encephalopathy

An acute neurologic disorder caused by thiamine deficiency, classically producing confusion, impaired coordination, and abnormal eye movements. It is a medical emergency requiring prompt thiamine treatment.

Wernicke’S Encephalopathy

An acute neurologic disorder caused by thiamine deficiency, classically involving confusion, eye-movement abnormalities, and gait or coordination impairment. It requires prompt thiamine treatment.

Wernicke-Korsakoff Syndrome

A neurologic disorder caused by thiamine deficiency, combining acute or previous Wernicke encephalopathy with persistent memory impairment and confabulation in some patients. Causes include alcohol-use disorder, malnutrition, and prolonged vomiting.

Wernicke’S Area

A brain region important for understanding spoken and written language.

Alternate Names:

Wernicke area

Wertheim’S Hysterectomy

Radical hysterectomy involving removal of the uterus, cervix, surrounding tissues, and commonly the upper vagina and pelvic lymph nodes, historically used for cervical cancer.

West Nile Encephalitis

Inflammation of the brain caused by neuroinvasive West Nile virus, transmitted mainly by mosquitoes.

It may cause fever, headache, confusion, tremor, weakness, seizures, or coma. There is no specific routine antiviral treatment; severe illness requires hospital-based supportive care.

West Nile Virus

A mosquito-borne flavivirus that can cause asymptomatic infection, a febrile illness, or neuroinvasive disease such as meningitis, encephalitis, or acute flaccid paralysis. Risk of severe disease is higher in older adults and people with certain medical conditions.

West Nile Virus Infection

West Nile virus infection is a mosquito-borne illness ranging from fever and body aches to neuroinvasive meningitis, encephalitis, weakness, or paralysis.

West Syndrome

An infantile epileptic encephalopathy characterized by epileptic spasms, hypsarrhythmia on electroencephalography, and developmental regression or arrest. Causes include structural brain disease, genetic disorders, metabolic disease, or acquired brain injury.

Westermark Sign

A focal area of decreased pulmonary vascularity distal to an occluded pulmonary artery segment on a chest radiograph. It may occur with pulmonary embolism but is neither sensitive nor specific.

Western Blotting

Laboratory technique for detecting specific proteins in tissue samples using gel electrophoresis, membrane transfer, and antibody-based detection; used in HIV diagnosis and research.

Westley Croup Score

A standardized, validated clinical scoring system used in emergency departments and pediatric inpatient units to assess the severity of airway obstruction and respiratory distress in children with croup (acute laryngotracheobronchitis). It grades five clinical parameters: inspiratory stridor (0–2), retractions (0–3), air entry (0–2), cyanosis (0–5),

and level of consciousness (0–5). A total score of 0–2 indicates mild croup, 3–5 moderate croup, and 6–11 severe croup (with scores ≥ 12 indicating impending respiratory failure), guiding the use of oral dexamethasone versus nebulized racemic epinephrine and hospital admission.

Alternate Names:

Westley Clinical Scoring System for Croup; Croup Severity Score

Wet Age-Related Macular Degeneration

Alternate index label; see *Wet macular degeneration*.

Wet AMD

Alternate index label; see *Wet macular degeneration*.

Wet Dream

Involuntary ejaculation during sleep, usually occurring in adolescence or adulthood and generally a normal physiological event.

Wet Dry Drowning (Drowning)

Alternate index label for Drowning.

Wet Gangrene

Tissue death complicated by bacterial infection and tissue swelling, producing moist, discolored, rapidly spreading necrosis. It requires urgent medical assessment.

Wet Macular Degeneration

The neovascular form of age-related macular degeneration in which abnormal blood vessels grow beneath or within the retina and leak fluid or blood, causing rapid central-vision loss.

Alternate Names:

Macular Degeneration, Wet Wet Age-Related Macular Degeneration Wet AMD

Wharton Jelly

A gelatinous substance (mucopolysaccharide matrix) surrounding the umbilical cord vessels, composed primarily of hyaluronic acid and chondroitin sulfate. It protects the umbilical vessels from compression and kinking during fetal movement and uterine con-

tractions.

Wharton'S Jelly

Gelatinous connective tissue within the umbilical cord that surrounds and supports the umbilical blood vessels and helps protect them from compression.

Wheal

Raised, itchy swelling of skin (urticaria/hives). Caused by mast cell degranulation and histamine release. Examples: insect bite, allergic reaction, dermatographism.

Wheal And Flare

A transient raised area of dermal edema (wheal) surrounded by localized redness (flare), typically produced by histamine release in urticaria or another immediate hypersensitivity reaction.

Wheat Allergy

An immune reaction to proteins in wheat that may cause skin, digestive, or breathing symptoms.

Wheeled Stretcher

Mobile patient transport device with wheels; used in hospitals and emergency services for moving patients who cannot walk.

Wheeze

A high-pitched whistling sound produced by narrowed airways during breathing.

Wheezing

Wheezing is a musical sound from narrowed lower airways, commonly caused by asthma, infection, chronic lung disease, allergy, or an inhaled foreign body. airflow through narrowed or compressed lower airways during expiration (and occasionally inspiration). Pathophysiology: bronchospasm, mucosal edema, airway inflammation, or intraluminal secretions.

Whiplash

A neck injury caused by sudden acceleration and

deceleration of the head, often after a rear-end vehicle collision but also after a fall, sports injury, or assault. Symptoms may appear immediately or over the next several hours or days and can include neck pain and stiffness, reduced movement, headache, shoulder or upper-back pain, dizziness, or tingling in the arms. Most cases improve over several weeks, but persistent or worsening symptoms should be assessed; weakness, numbness, difficulty walking, or severe pain after significant trauma requires urgent medical evaluation.

Alternate Names:

Pain Whiplash (Whiplash) Sprain Neck (Whiplash) Strain Neck (Whiplash)

Whiplash Injury

Alternate index label; see *Whiplash*.

Whipple Disease

A rare systemic bacterial infection caused by *Tropheryma whippelii*, affecting the small intestine and multiple organ systems. Presents with arthralgias (prodromal, years before GI symptoms), weight loss, chronic diarrhea, steatorrhea, and abdominal pain.

Whipple Procedure

Pancreaticoduodenectomy for cancers of the pancreatic head, distal bile duct, or ampulla. Involves removal of the pancreatic head, duodenum, distal bile duct, and gallbladder with three anastomoses: hepaticojejunostomy, pancreaticojejunostomy, and gastrojejunostomy. The pylorus-preserving variant maintains the entire stomach.

Whipple Triad

A clinical framework supporting the diagnosis of symptomatic hypoglycemia: symptoms consistent with hypoglycemia, a documented low blood-glucose concentration, and improvement after the glucose level is raised. It is used when evaluating possible hypoglycemia, including suspected insulinoma.

Whipple'S Disease

A rare bacterial infection that can impair intestinal absorption and affect multiple organs.

Alternate Names:

Intestinal Lipodystrophy Lipodystrophy, Intestinal

Whipworm

A parasitic roundworm of the genus *Trichuris*, usually *Trichuris trichiura* in humans. Infection occurs by ingesting embryonated eggs and may cause abdominal pain, diarrhea, anemia, or rectal prolapse when severe.

Whipworm (Trichuriasis)

Intestinal infection caused by the soil-transmitted helminth *Trichuris trichiura* (whipworm), endemic in tropical and subtropical regions with poor sanitation. Humans acquire infection by ingesting embryonated eggs from contaminated soil, food, or water.

Alternate Names:

Trichuriasis

Whipworm Infection

Intestinal infection with *Trichuris trichiura* acquired by ingesting eggs from contaminated soil or food; heavy infection can cause abdominal symptoms, anemia, or rectal prolapse.

Whirlpool Bath

Hydrotherapy device providing agitated water for therapeutic bathing; used in wound care, physical therapy, and rehabilitation.

Whispered Pectoriloquy

The ability to hear whispered words clearly through the stethoscope over consolidated lung tissue. It is a sign of pulmonary consolidation. Normally, whispered words are barely audible over the lung fields. It is one of the classic signs of pneumonia on physical examination.

White Blood Cell

A cell of the immune system that helps protect the body from infections, foreign substances, and abnormal cells. White blood cells are made mainly in the bone marrow and include neutrophils, lymphocytes, monocytes, eosinophils, and basophils. Their number or function may change with infection, inflammation, immune disorders, or bone-marrow

disease.

White Blood Cell Count

WBC count: the total number of leukocytes per unit volume (normal $4.0\text{--}11.0 \times 10^9/\text{L}$), part of the CBC with differential (neutrophils, lymphocytes, monocytes, eosinophils, basophils). Leukocytosis: infection (neutrophilia), inflammation, stress, steroids, leukemia (marked), corticosteroids.

White Blood Cell Labeling

Nuclear medicine technique labeling WBCs with Tc-99m. Used for infection localization.

White Cell

The cells of the immune system that defend the body against infection and foreign substances. Produced in the bone marrow from hematopoietic stem cells.

White Cell Count

A laboratory measurement of the number of white blood cells (leukocytes) in a given volume of blood, part of the complete blood count (CBC). Normal range: 4,000-11,000 cells/ μL ($4\text{--}11 \times 10^9/\text{L}$). Differential white cell count measures the proportion of each leukocyte type: neutrophils, lymphocytes, monocytes, eosinophils, basophils.

White Finger

Episodic pallor or whitening of the fingers caused by vasospasm, commonly associated with Raynaud phenomenon or hand-arm vibration syndrome.

White Infarction

White infarction, also known as anemic infarction, occurs in solid organs with end-arterial circulation where collateral blood supply is limited, resulting in pale, well-demarcated areas of coagulative necrosis.

Alternate Names:

anemic infarction, occurs in solid organs with.

White Leg (Phlegmasia Alba Dolens)

Painful, marked swelling and pallor of a limb caused by extensive deep venous thrombosis, usually involving major venous outflow. It requires urgent

assessment.

Alternate Names:

Phlegmasia Alba Dolens

White Matter

Regions of the brain and spinal cord composed primarily of myelinated nerve fibers, responsible for transmitting signals between brain regions.

White Matter Of The Brain

The brain tissue composed mainly of myelinated axons and supporting glial cells, connecting different brain regions and the brain with the spinal cord; disease or injury can impair cognition, sensation, coordination, or movement.

White Of The Eye

The sclera, the tough, opaque outer coat of the eye surrounding the cornea and helping maintain the shape of the globe.

White Spot Disease

Alternate index label; see *Lichen sclerosus*.

White Tongue

A white coating or patches on the tongue caused by debris, dry mouth, irritation, infection, inflammation, or other oral conditions.

White-Coat Hypertension

High blood pressure in a medical setting with normal readings outside it.

Alternate Names:

White-coat effect

Whiteheads

Whiteheads are closed comedones formed when oil and dead skin block a hair follicle, producing small pale or skin-colored acne bumps.

Vaginal Discharge

Vaginal discharge may be normal or may indicate infection, inflammation, hormonal change, or another condition. Normal discharge is usually clear to whitish and odourless, and varies with the men-

strual cycle (increased at ovulation, pre-menstrually, during pregnancy, sexual arousal, and with oral contraceptive use).

Whitlow

An infection or inflammation of the tissues around a finger or toenail, often involving the nail fold; it may be bacterial, fungal, or herpetic.

Whitlow (Paronychia)

Acute or chronic infection and inflammation of the tissues around a fingernail or toenail, usually caused by bacteria, fungi, or repeated moisture and trauma.

Alternate Names:

Paronychia

Whole Breast Radiation Therapy

Whole breast radiation therapy treats remaining breast tissue after breast-conserving surgery to reduce local recurrence risk, with planned dosing and monitoring for skin and tissue effects.

Whole Exome Sequencing

Sequencing the protein-coding regions of the genome, which make up a small fraction of total DNA. It can identify variants associated with disease but does not detect all genetic causes; its yield depends on the disorder, test quality, and interpretation.

Whole Genome Sequencing

Laboratory analysis of nearly all of a person's nuclear DNA, including coding and noncoding regions, and sometimes mitochondrial DNA. It can identify many types of genetic variants but may produce uncertain or incidental findings and does not detect every cause of disease; results require appropriate clinical interpretation.

Whole-Body Irradiation

Radiotherapy technique delivering radiation to the entire body; used as conditioning before bone marrow transplantation or for treatment of systemic hematologic malignancies.

Whole-Body Scan

An imaging examination that surveys much or all of the body, using a technique such as PET/CT, CT, MRI, or a nuclear medicine scan. It may be used for selected cancer staging, infection localization, or evaluation of systemic disease; the coverage and usefulness depend on the test and clinical question.

Whole-Bowel Irrigation

Gastrointestinal decontamination using large volumes of an iso-osmotic polyethylene-glycol solution to flush intestinal contents; it is reserved for selected ingestions and requires airway and clinical assessment.

Whole-Brain Radiation Therapy

Radiation directed at the entire brain, used for multiple brain metastases, leptomeningeal disease, and prophylactic cranial irradiation in small cell lung cancer and acute lymphoblastic leukemia.

Whole-Genome Sequencing

Sequencing of nearly the complete nuclear genome, including coding and noncoding regions, together with mitochondrial DNA when covered by the assay. It can detect a broad range of variants but generates substantial interpretation and incidental-finding challenges.

Whooping Cough

A highly contagious respiratory infection caused by *Bordetella pertussis*. It causes prolonged bouts of rapid coughing that may be followed by a whooping sound or vomiting; infants may have pauses in breathing without the typical whoop. Also called pertussis.

Whooping-Cough

Alternate index label; see *Whooping Cough*.

Wickerhamiella Pararugosa

Species of basidiomatus yeast; rarely isolated from clinical specimens, generally considered a contaminant.

Widal Reaction

An agglutination test historically used to detect antibodies associated with *Salmonella Typhi* and paratyphoid infection. Its diagnostic accuracy is limited, and results require clinical interpretation.

Widal Test

An agglutination test historically used to detect antibodies associated with typhoid and paratyphoid fever; its limited accuracy means results require clinical interpretation.

Widely Spaced Teeth

Teeth separated by unusually large gaps, also called diastema when the gap is localized. Causes include normal development, missing teeth, tooth-size differences, habits, or jaw and gum conditions.

Williams Syndrome

A microdeletion syndrome involving the elastin gene on chromosome 7q11.23. Features: elfin facies, supravalvular aortic stenosis, hypercalcemia, intellectual disability with strong verbal skills, and overly friendly personality. Often associated with connective tissue abnormalities.

Wilm'S Tumour

Source spelling variant of *Wilms tumor*, a malignant embryonal kidney tumor occurring mainly in young children.

Wilms Tumor

A malignant embryonal tumor of the kidney in children. It may present as an abdominal mass, pain, hematuria, or an incidental finding and may be associated with syndromes such as WAGR or Beckwith-Wiedemann syndrome. Treatment and prognosis depend on stage and tumor biology.

Alternate Names:

Cancer, Wilms' Tumor

Wilms' Tumor (Nephroblastoma)

An embryonal malignant tumor of the kidney that usually occurs in young children. It may be associated with congenital syndromes, and treatment and

prognosis depend on tumor stage, histology, and molecular features.

Alternate Names:

Nephroblastoma

Wilson Disease

Autosomal recessive copper metabolism disorder (ATP7B gene). Copper accumulation in liver, brain, cornea. Kayser-Fleischer rings, hepatopathy, movement disorder, psychiatric symptoms. Treatment: chelation (penicillamine, trientine), zinc, liver transplant.

Wilson'S Disease

An autosomal recessive disorder of copper metabolism caused by mutations in the ATP7B gene, leading to impaired biliary copper excretion and toxic copper accumulation in the liver, brain, cornea, and other organs.

Windowing

Adjusting CT image brightness and contrast to optimize visualization of specific tissues. Bone window: wide window width, high level. Soft tissue window: narrow width, mid level. Lung window: wide width, low level.

Windpipe

The windpipe, a 10-12 cm fibrocartilaginous tube from the larynx (C6) to the carina (T4/T5) where it bifurcates into main bronchi. See Trachea.

Wing Bone

A bone forming the skeletal framework of a bird or bat wing; it is not a standard human anatomical term, though the bones correspond broadly to the upper limb.

Winter Vomiting Disease

An acute gastroenteritis, usually caused by norovirus, characterized by sudden vomiting, diarrhea, and abdominal cramps.

Wireless Motility Capsule

Ingestible capsule that measures pH, pressure, and

temperature throughout the gastrointestinal tract. Provides regional transit times through the stomach, small bowel, and colon in a single test. The pH sensor identifies transition points between segments based on characteristic pH changes.

Wisdom Teeth

Wisdom teeth are the third molars that commonly emerge in late adolescence or adulthood. Impaction, infection, decay, cysts, gum disease, or crowding can cause pain and swelling; dental examination and imaging determine whether monitoring or removal is appropriate.

Wisdom Teeth, Impacted

Alternate index label; see *Impacted wisdom teeth*.

Wisdom Tooth (Third Molar)

The last molar in each dental quadrant, usually erupting in late adolescence or early adulthood. Impaction may cause pain, infection, or damage to adjacent teeth.

Alternate Names:

Third Molar

Wiskott-Aldrich Syndrome

X-linked recessive disorder (WAS gene mutation, encodes WASP protein, regulates actin polymerization in hematopoietic cells). Triad: microthrombocytopenia (small platelets, <70 fl), eczema, recurrent infections (encapsulated bacteria, viruses). Combined immunodeficiency with progressive T-cell lymphopenia.

Witch'S Milk

A temporary milky breast discharge in a newborn caused by maternal hormones crossing the placenta; it usually resolves without squeezing or treatment, which can otherwise cause infection.

Witch'S Milk (Neonatal Galactorrhea)

A temporary milk-like breast secretion in a newborn caused by exposure to maternal hormones. It usually resolves without treatment; the breasts should not be squeezed.

Alternate Names:
Neonatal Galactorrhea

Withdrawal

Physical or psychological symptoms that occur when a person who has developed dependence reduces or stops using a medicine or psychoactive substance. Symptoms vary by substance and may include anxiety, agitation, sweating, tremor, nausea, pain, insomnia, cravings, or seizures; some forms require urgent medical care.

Alternate Names:
Withdrawal syndrome

Withdrawal Reflex

A protective polysynaptic spinal reflex in which a painful stimulus causes flexion (withdrawal) of the stimulated limb, removing it from the source of injury, with reciprocal extension of the contralateral limb (crossed extensor reflex) to maintain posture.

Withdrawal Symptoms

Physical or psychological symptoms that occur when a person stops or reduces a substance or medicine after the body has adapted to it. The symptoms depend on the substance, duration of use, and the person's health.

Withdrawal Syndrome

A predictable set of signs and symptoms that occur when a drug is discontinued or the dose is reduced after chronic use, reflecting the physiological rebound from drug adaptation.

Wolf-Hirschhorn Syndrome

A severe contiguous gene deletion syndrome caused by a terminal or interstitial microdeletion of the short arm of chromosome 4 (4p16.3), involving the critical genes *WHSC1* and *LETM1*. Characterized by a pathognomonic facial appearance described as a "Greek warrior helmet" facies (prominent glabella, hypertelorism, broad beaked nose continuing directly from the forehead), severe prenatal and postnatal growth restriction, microcephaly, profound intellectual disability, structural congenital cardiac defects (atrial and ventricular septal defects), and

early-onset intractable epilepsy.

Alternate Names:
4p- Syndrome; 4p Deletion Syndrome

Wolff Parkinson White Syndrome (Wolff-Parkinson-White Syndrome)

Alternate index label for Wolff-Parkinson-White Syndrome.

Wolff-Parkinson-White Accessory Pathway-Related Tachycardia

Alternate index label; see *Wolff-Parkinson-White syndrome*.

Wolff-Parkinson-White Syndrome

A pre-excitation syndrome caused by an accessory pathway (bundle of Kent) connecting atria and ventricles, bypassing the AV node. ECG shows short PR interval (<0.12 s), delta wave (slurred upstroke of QRS), and widened QRS.

Alternate Names:
Wolff Parkinson White Syndrome (Wolff-Parkinson-White Syndrome)

Wolff-Parkinson-White Syndrome

Wolff-Parkinson-White syndrome is pre-excitation through an extra atrioventricular pathway that can cause episodes of supraventricular tachycardia and rarely dangerous ventricular rhythms.

Wolffian Ducts

Embryonic mesonephric ducts that form parts of the male reproductive tract, including the epididymal duct, vas deferens, and ejaculatory ducts; most regress in typical female development.

Wolfram Syndrome

Autosomal recessive disorder caused by mutations in the *WFS1* gene encoding wolframin. Classic triad of diabetes mellitus, optic atrophy, and diabetes insipidus. Additional features include sensorineural deafness, neurodegeneration, and psychiatric illness. Onset of diabetes typically in the first decade.

Wolinella

Genus of anaerobic, Gram-negative bacteria found in the oral cavity and gastrointestinal tract; associated with periodontal disease.

Wolinella Succinogenes

An anaerobic gram-negative bacterium found in the mouth and gastrointestinal tract of some animals; it is mainly of microbiologic interest and is rarely linked to human infection.

Wolman Disease

A rare autosomal-recessive lysosomal storage disorder caused by lysosomal acid lipase deficiency, leading to accumulation of cholesteryl esters and triglycerides, adrenal calcification, hepatosplenomegaly, diarrhea, and severe early-onset disease.

Wolman'S Disease

A severe lysosomal acid-lipase deficiency presenting in infancy with hepatosplenomegaly, adrenal calcification, malabsorption, vomiting, diarrhea, and failure to thrive. It is a severe form of lysosomal acid-lipase deficiency and can be fatal early without treatment.

Womb

Common name for the uterus; the muscular organ where a fetus develops during pregnancy.

Alternate Names:

Uterus

Womb Growths (Uterine Fibroids)

Alternate index label for Uterine Fibroids.

Breast Lumps

Alternate index label; see *Breast Lumps In Women*.

Women'S Health

Medical specialty focusing on conditions unique to or more prevalent in females, including reproductive health, pregnancy, menopause, and gender-specific disease patterns.

Alternate Names:

Womens Health (Women's Health)

Women'S Hospital

A hospital or service that focuses on health needs common in women, such as gynecologic care, pregnancy, childbirth, breast health, and menopause; services vary by institution.

Womens Health (Women'S Health)

Alternate index label for Women's Health.

Wood Lamp Examination

Inspection of skin under long-wave ultraviolet A light in a darkened room. It can accentuate pigimentary differences and produce characteristic fluorescence in selected infections, but findings are not independently diagnostic.

Wood Naphtha

A historical term for methanol-containing wood distillate; ingestion or inhalation can cause severe metabolic acidosis, visual loss, and death.

Wood Spirit

An old name for methanol, a toxic alcohol whose ingestion can cause metabolic acidosis, blindness, and death.

Wood'S Light

Long-wave ultraviolet-A light used in a darkened room to enhance certain skin-color changes and aid assessment of selected dermatological conditions.

Woods Corkscrew Maneuver

An internal rotation maneuver for shoulder dystocia where the operator inserts a hand and rotates the fetal shoulders 180 degrees by applying pressure to the posterior aspect of the anterior shoulder. This disimpacts the shoulder from behind the symphysis pubis. Can be repeated if the shoulder re-impacts in the new position.

Woods Screw Maneuver

A rotational maneuver used during shoulder dystocia to rotate the fetal shoulders and assist delivery.

Woody Tongue

A rare chronic infection, usually caused by *Actinomyces*, producing a hard, swollen tongue and surrounding tissue inflammation.

Wool-Sorter'S Disease (Anthrax)

An occupational form of inhalational anthrax caused by inhalation of spores of *Bacillus anthracis*, historically associated with handling contaminated wool or animal products.

Alternate Names:

Anthrax

Wool-Sorters' Disease

An occupational form of inhalational anthrax caused by inhaling spores of *Bacillus anthracis* while handling contaminated wool or animal products.

Word Blindness

An older term for alexia, an acquired inability to read or recognize written words despite adequate vision and previously developed language ability, usually caused by brain dysfunction or injury.

Word Blindness (Alexia)

An acquired neurological disorder in which a person loses or markedly impairs the ability to read despite adequate vision and previously acquired language skills.

Alternate Names:

Alexia

Word Deafness (Auditory Verbal Agnosia)

An acquired inability to understand spoken words despite adequate hearing, often caused by dysfunction of auditory-language pathways.

Alternate Names:

Auditory Verbal Agnosia

Word Recognition Scoring

An audiologic test measuring how accurately a person repeats or identifies spoken words at a comfortable listening level.

Word Salad

Severe disorganized speech consisting of a jumble of words and phrases that lack meaning or logical connection. It is a hallmark of formal thought disorder in schizophrenia and represents the most severe form of disorganized speech. It reflects profound disruption of language processing.

Working Memory

The short-term holding and manipulation of information during a task.

Worm Infestation (Helminthiasis)

Infection by parasitic worms, including nematodes, cestodes, and trematodes. Manifestations depend on the species and the organs involved.

Alternate Names:

Helminthiasis

Worms Pinworms (Pinworm Infection)

Alternate index label for Pinworm Infection.

Wortmannin

Fungal metabolite and potent inhibitor of phosphatidylinositol 3-kinase (PI3K); used extensively in cell biology research.

Wound

An injury in which tissue is damaged by trauma, surgery, pressure, heat, chemicals, or other causes. Wounds may be open or closed and vary in depth, contamination, and effect on nearby structures.

Wound Care

Assessment and treatment of a wound through cleansing, bleeding control, removal of dead or contaminated tissue when indicated, appropriate closure or dressing, pain control, and follow-up. Care also includes tetanus prevention, pressure relief, and treatment of infection or underlying conditions; antibiotics are reserved for selected contaminated wounds, bites, or infections.

Wound Complication

A problem that develops during healing, such as

infection, separation of wound edges, bleeding, fluid collection, tissue death, or excessive scarring. Increasing pain, redness, swelling, drainage, fever, or wound opening should be assessed promptly.

Wound Dehiscence

Separation of the layers of a surgical wound after closure, most commonly occurring within 3-14 days postoperatively.

Wound Evisceration

Separation of a surgical wound with protrusion of internal organs or tissue through the opening. It is a surgical emergency requiring protection of the exposed tissue and urgent medical evaluation.

Wound Healing

The coordinated repair of injured tissue through overlapping phases of hemostasis, inflammation, new tissue formation, epithelial coverage, and remodeling. Healing is influenced by blood supply, infection, nutrition, diabetes, medications, smoking, mechanical stress, and the depth and location of the wound.

Wound Healing Phases

Wound healing involves overlapping stages of hemostasis, inflammation, tissue formation, epithelial coverage, and remodeling. The duration and result depend on the wound, blood supply, infection, nutrition, and overall health.

Wound Infection

Infection of a wound by microorganisms, causing findings such as increasing pain, redness, warmth, swelling, pus, fever, or delayed healing.

Wound Management

Assessment and treatment of acute or chronic wounds through cleansing, debridement when indicated, moisture and exudate control, infection management, pressure relief, nutrition, and correction of underlying disease.

Wound VAC

Vacuum-assisted closure for wound healing. Foam dressing connected to vacuum pump. Removes exudate, reduces edema, promotes granulation. Used for complex wounds, surgical sites, pressure ulcers.

Wounds

Injuries in which tissue is damaged by trauma, surgery, pressure, heat, chemicals, or other causes; wounds may be open or closed and may require cleansing, closure, or debridement.

Wright Stain

A blood-cell stain used to identify and classify cells in a blood smear.

Wrinkle

Alternate index label; see *Wrinkles*.

Wrinkles

Wrinkles are skin folds or lines produced by aging, sun exposure, repeated facial movement, loss of elasticity, smoking, or other factors.

Alternate Names:

Wrinkle

Wrisberg Ganglion

A sensory ganglion associated with the intermediate nerve of the facial nerve.

Wrist

Complex joint connecting the hand to the forearm; composed of eight carpal bones, ligaments, and tendons allowing flexion, extension, abduction, and adduction.

Wrist Arthroscopy

Wrist arthroscopy uses a small camera and instruments through portals to diagnose or treat cartilage, ligament, synovial, or other wrist problems.

Wrist Drop

Inability to extend the wrist and fingers, usually caused by radial nerve dysfunction.

Wrist Fracture

A break in one or more bones near the wrist, most commonly the distal radius or ulna, usually caused by a fall, direct trauma, or weakened bone.

Wrist Injury

Damage to the bones, ligaments, tendons, nerves, or vessels of the wrist from a fall, twist, impact, or overuse; fracture, instability, severe swelling, or neurovascular symptoms require assessment.

Wrist Joint

Condylloid synovial joint between distal radius and articular disc proximally and scaphoid/lunate/triquetrum distally. Allows flexion, extension, radial/ulnar deviation, circumduction. The triangular fibrocartilage complex absorbs ulnar side load.

Wrist Pain

Pain in or around the wrist, commonly caused by sprain, fracture, tendinitis, arthritis, or nerve compression such as carpal tunnel syndrome. Persistent pain, deformity, numbness, or weakness requires assessment.

Alternate Names:

Pain, wrist

Alternate Names:

Pain, Wrist

Wrist Sprain

Stretching or tearing of one or more ligaments supporting the wrist, causing pain, swelling, bruising, and reduced movement after injury.

Writer'S Cramp

A task-specific focal dystonia causing involuntary muscle contractions and abnormal postures during writing or closely related fine-motor activities.

Written Documentation

Formal recording of patient information, clinical decisions, and treatment plans; essential for continuity of care and medicolegal purposes.

Alternate Names:

Torticollis

Wry-Neck

Torticollis, an abnormal tilt or rotation of the head caused by involuntary contraction, shortening, or dysfunction of neck muscles.

Wryneck (Torticollis)

Abnormal tilting or rotation of the head caused by involuntary contraction, shortening, or dysfunction of neck muscles. It may be congenital, acquired, painful, or neurological.

Wt

A medical abbreviation commonly meaning weight. Its meaning should be confirmed from the surrounding documentation because abbreviations can be ambiguous.

Wuchereria

A genus of filarial nematodes that includes *Wucheria bancrofti*, a major cause of lymphatic filariasis. Infection is transmitted by mosquitoes and may cause lymphangitis, chronic lymphedema, hydrocele, and elephantiasis.

Wuchereria Bancrofti

A filarial nematode transmitted by mosquitoes and the major cause of lymphatic filariasis, which can produce lymphoedema and elephantiasis.

Wuhan Coronavirus (2019-NCoV)

Alternate index label for 2019-nCoV.

X

X Chromosome

One of the two human sex chromosomes, carrying genes involved in sex development and many non-sex-specific biological functions. Individuals typically have one or two X chromosomes, and dosage is partly regulated by X-chromosome inactivation.

X, Factor

Factor X is a vitamin-K-dependent coagulation protein activated in the common pathway to generate thrombin and form fibrin.

X-Inactivation

Process inactivating one X chromosome in female mammals, forming Barr body. Occurs early in development, random in most tissues. Ensures dosage compensation.

X-Linked

Relating to a gene or condition associated with the X chromosome; inheritance patterns differ between individuals with one or two X chromosomes.

X-Linked Adrenoleukodystrophy

An inherited disorder of peroxisomal fatty-acid metabolism, usually affecting boys, that can cause progressive damage to the adrenal glands, nervous system, and myelin.

X-Linked Agammaglobulinemia

An inherited immunodeficiency, usually caused by a pathogenic variant in the BTK gene, that prevents normal B-cell maturation. It causes very low immunoglobulin levels and recurrent bacterial or enteroviral infections, usually beginning in infancy

or early childhood.

X-Linked Hypophosphatemia

An inherited disorder of phosphate regulation, commonly caused by excess fibroblast growth factor 23, leading to renal phosphate wasting, rickets or osteomalacia, bone pain, and growth abnormalities.

X-Linked Lymphoproliferative Syndrome

An inherited immune-regulation disorder, usually caused by pathogenic variants in SH2D1A or XIAP, that can cause severe or persistent Epstein–Barr virus infection, hemophagocytic lymphohistiocytosis, dysgammaglobulinemia, lymphoma, or inflammatory bowel disease.

X-Linked Ichthyosis

An inherited, usually X-linked recessive skin disorder caused by a variant or deletion affecting the STS gene and deficiency of steroid sulfatase. It usually affects males and causes widespread dry, dark, plate-like scales that often begin in infancy or early childhood.

X-Linked Inheritance

Transmission of a pathogenic variant on the X chromosome. Phenotypes differ by sex, X-inactivation, and variant mechanism; fathers do not transmit an X-linked variant to their sons.

X-Linked Myotubular Myopathy

The most severe, life-threatening form of centronuclear myopathy, inherited in an X-linked recessive manner and caused by loss-of-function mutations in the myotubularin (*MTM1*) gene on chromosome

Xq28, which encodes a dual-specificity lipid phosphatase regulating endosomal trafficking. Affects male infants who present at birth with profound, generalized hypotonia, severe muscle weakness, bilateral ptosis, external ophthalmoplegia, facial diplegia, and life-threatening respiratory failure requiring immediate mechanical ventilation. Muscle biopsy characteristically demonstrates small, rounded myofibers with centrally placed vesicular nuclei resembling fetal myotubes, surrounded by a clear halo devoid of myofibrils.

Alternate Names:

XLMTM; Myotubularin Deficiency

X-Linked Retinoschisis

An inherited vitreoretinal dystrophy caused by mutations in the RS1 gene, characterized by splitting of the retinal nerve fibre layer producing a spoke-wheel pattern of cystic spaces at the macula and peripheral retinal schisis.

X-Linked Severe Combined Immunodeficiency

A severe inherited immune disorder, usually caused by defects affecting the interleukin-2 receptor gamma chain, resulting in profound T-cell dysfunction and vulnerability to serious infections.

X-Ray

An X-ray is an imaging study that passes a small amount of ionizing radiation through the body to create a projection image. Differences in tissue density help show bones, lungs, teeth, and selected soft-tissue abnormalities.

X-Ray (Plain Radiography)

An imaging method that uses X-rays to create a two-dimensional projection of internal structures. It is used to assess bones, the chest, abdomen, teeth, and other selected body regions.

Alternate Names:

Plain Radiography

X-Ray (Radiography)

Projectional imaging using ionizing radiation. Dense structures (bone, contrast) appear white; air

appears black. Fast, inexpensive, widely available. Two-dimensional projection of three-dimensional structures. Mammography: specialized breast X-ray for screening.

Alternate Names:

Radiography

X-Ray - Skeleton

A skeletal X-ray series uses radiographs of selected bones and joints to evaluate fractures, alignment, bone density patterns, arthritis, infection, tumors, and developmental abnormalities.

X-Ray Attenuation

Reduction in the intensity of an X-ray beam as it passes through tissue or another material.

X-Ray Crystallography

Analytical technique using X-ray diffraction patterns to determine three-dimensional atomic structure of crystals and proteins.

X-Ray Shield

Protective material, usually lead or another radiation-attenuating substance, used to reduce exposure to scattered X-rays during imaging; shielding is applied according to the procedure and current radiation-safety guidance.

X-Ray Therapy

Use of ionizing radiation from X-ray sources for treatment of malignant and benign conditions; encompasses external beam radiotherapy.

X-Rays (Röntgen Rays)

Alternate index label; see *X-Ray*.

Alternate Names:

Röntgen Rays

Xanthelasma

Xanthelasma palpebrarum are soft, yellow, lipid-filled papules or plaques on the medial canthus of the eyelids (upper more commonly than lower), the most common form of cutaneous xanthoma. They are composed of foam cells (lipid-laden macrophages)

in the superficial dermis.

Xanthine

Purine base intermediate in uric acid metabolism; accumulated in xanthinuria due to xanthine oxidase deficiency.

Xanthine Dehydrogenase

Enzyme converting hypoxanthine to xanthine and xanthine to uric acid; deficiency causes xanthinuria type I.

Xanthine Oxidase

Molybdenum-containing enzyme catalyzing oxidation of hypoxanthine to xanthine and xanthine to uric acid; target of allopurinol in gout treatment.

Xanthogranulomatous

Describing inflammation in which tissues contain many lipid-filled immune cells called macrophages. It is seen in conditions such as xanthogranulomatous pyelonephritis.

Xanthogranulomatous Oophoritis

A rare chronic inflammatory lesion of the ovary with lipid-laden macrophages, plasma cells, and fibrosis, often associated with infection or endometriosis and capable of mimicking ovarian malignancy.

Xanthogranulomatous Pyelonephritis

A chronic destructive kidney infection in which renal tissue is replaced by lipid-laden macrophages. It is often associated with obstruction or stones and may require antibiotics, drainage, surgery, or nephrectomy depending on the extent of damage.

Xanthoma

A xanthoma is a yellowish deposit of lipid-laden macrophages in the skin, subcutaneous tissue, or tendons. It may occur with hyperlipidemia, diabetes, cholestatic liver disease, or an inherited lipid disorder and warrants assessment of the underlying cause. presenting as yellow-orange papules, nodules, or plaques.

Xanthomata

Alternate index label; see *Xanthoma*.

Xanthomatosis

A disorder in which fatty deposits called xanthomas develop in the skin, tendons, or organs. It may be associated with inherited or acquired lipid disorders and can signal increased cardiovascular or organ-specific risk requiring medical evaluation.

Xanthomonadaceae

Family of Gram-negative, aerobic, rod-shaped bacteria; includes plant pathogens and rare human opportunistic isolates.

Xanthomonas

Genus of Gram-negative, aerobic bacteria in family Xanthomonadaceae; primarily plant pathogens, occasionally isolated from clinical specimens.

Xanthopsia

A visual disturbance in which objects appear yellow or yellowish. It can occur with certain eye disorders, medication toxicity such as digoxin toxicity, or other neurologic or retinal conditions and warrants evaluation when new or persistent.

Xanthosis

Yellow discoloration of tissues caused by deposition of carotenoid pigments or other yellow material, distinguished from jaundice by the absence of yellowing in the sclerae.

Xenobiotic

Foreign chemical substance in body. Drugs, pollutants, pesticides, industrial chemicals. Metabolized by liver enzymes.

Xenobiotic Metabolism

Biotransformation of foreign chemical compounds (drugs, toxins, pollutants) primarily by hepatic cytochrome P450 enzymes; Phase I and Phase II reactions.

Xenograft

Tissue or a biological material transplanted between different species, often used as a temporary graft or tissue substitute.

Xenograft (Heterograft)

Alternate index label; see *Xenograft*.

Alternate Names:

Heterograft

Xenograft Procedure

Transplantation of tissue between different species; used experimentally and clinically (e.g., porcine heart valves, skin grafts).

Xenon

Noble gas (Xe) with anesthetic properties; used in research imaging and experimental anesthesia.

Xenophagy

Cellular process of degrading engulfed foreign material or pathogens via autophagy machinery.

Xenophobia

An intense or persistent fear of unfamiliar people, cultures, or situations; when clinically significant, it may be considered a specific phobia.

Xenotransplant

Transplantation of cells, tissues, or organs between different species, such as experimental or clinical use of animal-derived biological material.

Xenotransplantation

The transplantation of living cells, tissues, or organs between members of different species, most commonly from an animal donor to a human recipient. It is being investigated as a potential source of organs and tissues when human donor supply is insufficient, but substantial immunologic and infectious risks remain. Major concerns include hyperacute or delayed rejection, graft dysfunction, transmission of animal pathogens, and physiologic incompatibility. Clinical use or research requires rigorous immunologic screening, infection surveil-

lance, ethical oversight, and specialized monitoring of recipients.

Xeroderma

Abnormally dry skin that may cause scaling, roughness, itching, or cracking. It can result from environmental exposure, skin disease, age, or medicines.

Xeroderma Pigmentosum

A rare autosomal recessive disorder of nucleotide-excision DNA repair that causes extreme sun sensitivity and a greatly increased risk of skin and eye damage and cancers. It is caused by pathogenic variants in genes involved in DNA repair.

Xerophthalmia

Dryness of the conjunctiva and cornea, often caused by vitamin A deficiency. Severe disease can damage the cornea and impair vision.

Xerosis

Abnormally dry skin, common in elderly patients and winter months. It results from decreased sebaceous and sweat gland activity, impaired skin barrier function, and environmental factors. Treatment includes regular emollient application (cream or ointment rather than lotion), gentle cleansing, and humidification.

Xerostomia

The sensation of dry mouth caused by reduced or altered saliva production. It may result from medicines, dehydration, salivary-gland disease, radiation, or systemic illness and can impair speaking, swallowing, taste, and dental health.

Xerostomia (Dry Mouth)

Alternate index label; see *Xerostomia*.

Alternate Names:

Dry Mouth

Xipamide

Thiazide-like diuretic used for hypertension and edema; acts on the distal convoluted tubule.

Xiphisternal Plane

The horizontal anatomical plane at the junction of the body of the sternum and the xiphoid process, approximately corresponding to the T9 vertebral level.

Xiphoid Process

The small inferior projection of the sternum. Its shape varies, and it serves as an attachment site for muscles and ligaments.

XX

XX is the usual human sex-chromosome complement associated with female development, though sex development has biologic variation.

XX Male Syndrome

XX male syndrome, also called de la Chapelle syndrome, is a disorder of sex development in which a person with an XX karyotype develops a typically male phenotype, usually from translocation of the SRY gene; infertility is common.

XXX Syndrome

Alternate index label; see *Triple X syndrome* (trisomy X), a sex-chromosome variation in which a female has an additional X chromosome.

XXXX Syndrome

XXXX syndrome is a rare sex-chromosome variation in which a female has four X chromosomes; it may cause developmental or learning difficulties, distinctive physical features, and ovarian dysfunction, with severity varying widely.

Xylitol

A sugar alcohol used as a sweetener and in some oral-care products. Large human ingestions may cause gastrointestinal symptoms; it is highly toxic to dogs.

Xylitol Toxicity

Poisoning caused by excessive xylitol exposure. It can cause marked hypoglycemia in dogs and may cause gastrointestinal symptoms in people.

Xylose

Five-carbon monosaccharide used in diagnostic tests for intestinal absorption (D-xylose test) and bacterial metabolism studies.

XYY Syndrome

A sex-chromosome variation in which a male has an extra Y chromosome. Many affected individuals have few or no obvious symptoms, though some have tall stature, learning difficulties, or developmental differences.

Y

Y Chromosome

The sex chromosome that typically carries the SRY gene and other genes involved in testicular development and male sex differentiation. It is present in most individuals with a typical 46,XY karyotype and is transmitted through the paternal line.

Y-Linked Inheritance

Y-linked inheritance involves a gene on the Y chromosome and typically passes from an affected father to his sons, not daughters.

YAG Capsulotomy

A laser procedure that creates an opening in an opaque posterior lens capsule after cataract surgery. It can improve vision when posterior capsule opacification is causing visual symptoms.

YAG Laser Capsulotomy

Alternate index label; see *YAG Capsulotomy*.

YAG Laser Surgery

YAG laser surgery uses a yttrium-aluminum-garnet laser to cut, vaporize, or treat tissue, including posterior-capsule clouding after cataract surgery.

Yarrowia

A genus of yeast that can rarely cause opportunistic infection, particularly in people who are immunocompromised or have indwelling medical devices. Identification and treatment depend on species and susceptibility testing.

Yarrowia Lipolytica

A yeast species in the genus *Yarrowia* that is primar-

ily environmental but has rarely been reported as an opportunistic human pathogen, including bloodstream or device-associated infection.

Yawn

An involuntary behavior consisting of a deep inhalation followed by a slower exhalation, often with stretching of the jaw and facial muscles. Yawning is common with fatigue but excessive or unusual yawning can occur with some medications or neurologic conditions.

Yawning

Alternate index label; see *Yawn*.

Yawning - Excessive

Excessive yawning is unusually frequent or difficult-to-control yawning. It may accompany sleep deprivation, excessive daytime sleepiness, medication effects, vasovagal reactions, or less commonly neurologic or cardiopulmonary disease; persistent unexplained episodes merit clinical evaluation.

Yaws

Yaws is a chronic infection caused by *Treponema pallidum* subsp. *pertenue* and spread mainly through close skin contact. It can cause an initial papilloma or ulcer and later destructive lesions of skin, bone, and cartilage; antibiotics can cure the infection.

Yaws (Frambesia)

Alternate index label; see *Yaws*.

Alternate Names:

Frambesia

Yd.

An abbreviation for yard, a unit of length equal to 0.9144 meter. In clinical documentation, the unit should be written clearly to avoid confusion with other abbreviations.

Yeast

Yeasts are unicellular fungi that reproduce mainly by budding or, in some species, fission; medically important examples include *Candida*, *Cryptococcus*, and *Malassezia*, which may colonize humans or cause opportunistic infection.

Yeast Infection

A *Candida* infection that may affect the mouth, skin, vagina, penis, or other tissues. Symptoms depend on the site and may include itching, soreness, redness, white plaques, discharge, or rash. Diagnosis and treatment depend on the affected site, severity, and whether the finding represents colonization or true infection.

Yeast Infection (Vaginal)

Alternate index label; see *Vulvovaginal Candidiasis*.

Alternate Names:

Candidiasis, Vaginal Vaginal Candidiasis

Yeast Infection In Women And Men

Genital *Candida* infection may affect the vulva or vagina in women and the glans penis or foreskin in men. Symptoms can include itching, soreness, redness, discharge, or rash. Recurrent, severe, atypical, or treatment-resistant symptoms need confirmation and evaluation for contributing conditions.

Yeast Oral (Thrush)

Alternate index label; see *Oral Thrush*.

Yeast Present

A laboratory report indicating that yeast forms were detected in a specimen. The finding may represent colonization, contamination, or infection and must be interpreted with symptoms, specimen type, and organism identification.

Yellow Fever

Yellow fever is an acute mosquito-borne infection caused by yellow fever virus. Fever, headache, muscle aches, nausea, and vomiting may be followed by jaundice, bleeding, shock, and organ failure; vaccination is the main preventive measure.

Yellow Fever Virus

A flavivirus transmitted mainly by infected mosquitoes that can cause yellow fever, ranging from a mild febrile illness to jaundice, bleeding, shock, and organ failure.

Yellow Nail Syndrome

Alternate index label; see *Yellow-Nail Syndrome*. Nail changes may include thickening, excessive curvature, loss of the cuticle, and separation of the nail plate from the nail bed. Respiratory manifestations can include recurrent sinusitis, chronic cough, pleural effusions, or bronchiectasis, although the complete triad is not always present. Diagnosis is clinical and supported by characteristic nail findings and evaluation for lymphatic or pulmonary complications; management is directed at the affected systems.

Yellow Tongue

Yellow discoloration or coating of the tongue that may result from oral hygiene, smoking, dry mouth, diet, medicines, or an underlying illness.

Yellow-Nail Syndrome

A rare triad of yellow, thickened, slow-growing nails, lymphedema, and chronic respiratory disease including bronchiectasis and pleural effusions. The nail changes result from impaired lymphatic drainage affecting nail growth. It typically affects middle-aged adults.

Yergason Test

A shoulder examination maneuver used to assess the long head of the biceps tendon and transverse humeral ligament by resisted supination and external rotation.

Yersinia

Yersinia is a genus of Gram-negative bacteria that includes *Y. pestis*, *Y. enterocolitica*, and *Y. pseudotuberculosis*; human disease ranges from enterocolitis and pseudoappendicitis to plague.

Yersinia Aldovae

An environmental *Yersinia* species rarely isolated from humans; its significance depends on recovery from a compatible infection rather than colonization or contamination.

Yersinia Bercovieri

An uncommon environmental or enteric *Yersinia* species with limited evidence of human pathogenicity; clinical significance depends on the specimen and symptoms.

Yersinia Enterocolitica

A gram-negative coccobacillus of the family Enterobacteriaceae, a common cause of enterocolitis and mesenteric lymphadenitis, especially in children and young adults in temperate climates.

Yersinia Frederiksenii

An environmental *Yersinia* species occasionally recovered from food, water, or clinical specimens; confirmed human infection is uncommon.

Yersinia Infection

Infection caused by *Yersinia* bacteria, commonly producing diarrhea, abdominal pain, fever, or vomiting. Some cases mimic appendicitis; severe illness, dehydration, bloody diarrhea, or persistent symptoms requires medical assessment.

Yersinia Intermedia

An environmental *Yersinia* species that is uncommon in human disease; interpretation of an isolate depends on the site, symptoms, and evidence of invasive infection.

Yersinia Kristensenii

An uncommon *Yersinia* species found in environmental or animal sources and rarely associated with

human infection.

Yersinia Mollaretii

An uncommon *Yersinia* species mainly identified in environmental or animal samples; human pathogenicity is not well established.

Yersinia Pestis

A gram-negative rod that causes plague, a severe and potentially fatal infection; see *Plague*. The organism is transmitted by flea bites or direct contact with infected animals. Bubonic plague presents with painful lymphadenopathy (buboes). Pneumonic plague involves the lungs and is transmitted by respiratory droplets.

Yersinia Pseudotuberculosis

Yersinia pseudotuberculosis is a Gram-negative bacterium acquired mainly from contaminated food or water that causes enterocolitis, fever, and mesenteric lymphadenitis resembling appendicitis; systemic infection is uncommon.

Yersinia Rohdei

An uncommon *Yersinia* species associated mainly with environmental or animal sources and rarely reported in human clinical specimens.

Yersinia Ruckeri

A bacterium that causes enteric redmouth disease in fish; human infection is extremely rare and should be assessed in the context of a compatible exposure and specimen.

Yersiniosis

Infection caused by *Yersinia* bacteria, most often *Y. enterocolitica* or *Y. pseudotuberculosis*, producing diarrhea, abdominal pain, fever, or pseudoappendicitis.

Yew Poisoning

Yew poisoning follows ingestion of yew plant material, especially leaves or seeds containing taxine compounds. Toxicity can cause nausea, abdominal symptoms, dizziness, seizures, dangerous slowing

or disruption of the heart rhythm, cardiovascular collapse, and death.

Yips

Sudden involuntary movements or muscle spasms that interfere with a skilled task, often in athletes or musicians.

Yoga

A mind-body practice combining physical postures, breathing techniques, and meditation or relaxation. It may support flexibility, balance, stress reduction, and symptom management, but exercises should be adapted to individual health conditions.

Yoga For Health

Yoga for health combines physical postures, breathing exercises, and meditation or relaxation. It may improve flexibility, balance, strength, stress, and some chronic symptoms when adapted to the person; instructors should account for injuries, pregnancy, and medical limitations.

Yohimbine

An alpha-2 adrenergic antagonist derived from yohimbe-related compounds and studied for erectile dysfunction; evidence is limited and adverse effects may include anxiety, high blood pressure, palpitations, and seizures.

Yokenella

A genus of Gram-negative bacteria in the family Enterobacteriales. *Yokenella* species are uncommon human pathogens but can cause opportunistic infections, including wound or bloodstream infections.

Yokenella Regensburgi

A Gram-negative bacterium that rarely causes human infection, including wound, urinary, or bloodstream infection, usually in people with underlying illness or impaired immunity. Treatment is guided by culture and antimicrobial susceptibility results.

Yolk Sac

An extraembryonic structure that supports the early

embryo before the placenta is fully functional. It contributes to early blood-cell formation and development of the primitive gut and is assessed on early pregnancy ultrasound.

Yolk Sac Tumor

A malignant ovarian germ-cell tumor that often occurs in children or young adults and produces alpha-fetoprotein; it requires prompt treatment.

Yolk Stalk

The embryonic connection between the yolk sac and the developing midgut, also called the vitelline duct. Persistence of this structure can result in a Meckel diverticulum or other vitelline duct anomaly.

Yolk-Sac Tumor

Alternate index label; see *Yolk Sac Tumor*.

Your Cancer Care Team

A cancer care team is the group of oncology professionals who coordinate diagnosis, treatment, symptom control, rehabilitation, psychosocial support, and follow-up. It may include medical, surgical, and radiation oncologists, nurses, pharmacists, social workers, dietitians, and palliative-care specialists.

Ytterbium

A chemical element with the symbol Yb and atomic number 70. Its compounds have specialized industrial and research uses; medical relevance depends on the specific compound and exposure.

Yttrium

A chemical element with the symbol Y and atomic number 39. The radioactive isotope yttrium-90 is used in selected radiation treatments, including some targeted therapies and radiosynovectomy procedures.

Yttrium-90

A radioactive beta-emitting isotope with a physical half-life of about 64 hours, used in selected targeted radiation treatments such as liver radioembolization and some radiosynovectomy procedures.

Z

Z-Plasty

A reconstructive surgical technique that uses interconnected triangular skin flaps in a Z-shaped arrangement to lengthen a contracted scar and change its direction. It is used in selected scar contractures and other soft-tissue deformities.

ZAP-70 Deficiency

A rare combined immunodeficiency caused by impaired ZAP-70 signaling in T lymphocytes, leading to abnormal T-cell development or function and recurrent, sometimes severe infections.

Zafirlukast

A leukotriene-receptor antagonist used for long-term control and prevention of asthma symptoms. It is taken orally and is not a rescue treatment for an acute asthma attack; liver injury and drug interactions are important safety considerations.

Zalcitabine

A nucleoside reverse-transcriptase inhibitor formerly used in combination antiretroviral treatment for HIV infection. It is rarely used because of toxicity and the availability of safer alternatives; peripheral neuropathy and pancreatitis are important adverse effects.

Zanamivir

An inhaled neuraminidase inhibitor used to treat or prevent influenza A and B when started appropriately. It may cause bronchospasm and is used cautiously in people with underlying airway disease.

Alternate Names:

Relenza

Zavanelli Maneuver

A rare emergency maneuver for unresolved shoulder dystocia in which the fetal head is flexed and replaced into the birth canal before emergency cesarean delivery. It is performed only by an experienced obstetric team when other measures have failed.

Alternate Names:

cephalic replacement

Zebrafish

A small freshwater fish (*Danio rerio*) widely used as a laboratory model in developmental biology, genetics, toxicology, and disease research. Its embryos are transparent and develop rapidly, but findings may not translate directly to humans.

Alternate Names:

Danio rerio

Zellweger Syndrome

A severe inherited peroxisome-biogenesis disorder, usually caused by pathogenic variants in a PEX gene. It presents in infancy with hypotonia, seizures, feeding difficulty, liver dysfunction, distinctive facial features, and impaired brain development.

Alternate Names:

Zellweger spectrum disorder

Zenker Diverticulum

A pouch that forms where the throat meets the esophagus when the inner lining pushes through a weak spot in a swallowing muscle. It can cause difficulty swallowing, regurgitation of undigested food, bad breath, cough, and aspiration; a barium swallow is commonly used for diagnosis.

Alternate Names:
Zenker's diverticulum

Zibotentan

An investigational selective endothelin-A-receptor antagonist studied mainly in oncology. It is not an established routine treatment, and its clinical role depends on evidence from clinical trials and regulatory approval.

Ziconotide

A synthetic analog of omega-conotoxin used by continuous intrathecal infusion for severe chronic pain when other treatments are inadequate. It can cause dizziness, confusion, hallucinations, and other serious neurologic or psychiatric adverse effects.

Alternate Names:
Prialt

Zidovudine

A nucleoside reverse-transcriptase inhibitor used in selected antiretroviral regimens for HIV infection and to reduce perinatal HIV transmission. Bone-marrow suppression, anemia, neutropenia, myopathy, and mitochondrial toxicity are important adverse effects.

Alternate Names:
azidothymidine; AZT

Ziehl-Neelsen Stain

An acid-fast staining method that helps detect organisms with waxy cell walls, especially mycobacteria. Acid-fast organisms retain carbol fuchsin after acid-alcohol decolorization and appear red against a contrasting background.

Alternate Names:
acid-fast stain

Zieve Syndrome

A complication of acute alcohol-associated liver injury characterized by hemolytic anemia, transient hyperlipidemia, and jaundice. Treatment focuses on stopping alcohol use and managing the associated liver disease and complications.

Alternate Names:

Zieve's syndrome

Zika

An illness caused by Zika virus, usually producing no symptoms or a mild fever, rash, conjunctivitis, arthralgia, and myalgia. Infection during pregnancy can cause congenital Zika syndrome, and infection has also been associated with Guillain-Barré syndrome.

Alternate Names:
Zika virus disease

Zika Virus

A positive-sense RNA flavivirus transmitted mainly by infected *Aedes* mosquitoes and also by sexual contact and other less common routes. Infection is often asymptomatic or mild, but infection during pregnancy can injure the developing fetus.

Alternate Names:
ZIKV

Zileuton

A 5-lipoxygenase inhibitor used for prevention and chronic control of asthma. It does not relieve acute bronchospasm; liver-enzyme elevations and interactions with other medicines require monitoring.

Alternate Names:
Zyflo

Zinc

An essential trace mineral required for numerous enzymes, gene-regulatory proteins, immune functions, growth, reproduction, and wound healing. Excess supplementation can cause gastrointestinal symptoms and copper deficiency.

Alternate Names:
Zn

Zinc Acetate

An oral zinc salt used to treat zinc deficiency and, in specific formulations, to reduce intestinal copper absorption in Wilson disease. Dose and monitoring depend on the indication; nausea and gastrointestinal irritation may occur.

Zinc Deficiency

A state of inadequate available zinc caused by insufficient intake, malabsorption, excessive losses, or increased requirements. Manifestations may include impaired growth, dermatitis, alopecia, diarrhea, altered taste, delayed wound healing, and increased infection susceptibility.

Alternate Names:

zinc depletion

Zinc Ointment (Zinc Oxide)

A topical preparation containing zinc oxide that forms a protective barrier over irritated or moist skin. It is commonly used for diaper dermatitis and minor skin irritation.

Alternate Names:

zinc oxide ointment

Zinc Oxide

An insoluble zinc compound used topically as a skin protectant and barrier, including in preparations for diaper dermatitis and minor irritation. It can reduce contact with moisture and irritants but does not treat every cause of a rash.

Alternate Names:

ZnO

Zinc Sulfate

An oral or parenteral zinc salt used to prevent or treat zinc deficiency when clinically indicated. Gastrointestinal upset is common with oral therapy, and prolonged excessive use can cause copper deficiency.

Zinc Toxicity

Adverse effects caused by excessive zinc ingestion or exposure. Acute toxicity may cause nausea, vomiting, abdominal pain, and diarrhea; chronic excess can impair copper absorption and lead to anemia or neurologic and immune abnormalities.

Alternate Names:

zinc poisoning

Zingiber Officinale

The botanical species commonly known as ginger. Rhizome preparations are used as a flavoring and

may provide modest relief of nausea in some settings; gastrointestinal irritation and drug interactions, including effects on bleeding risk, are possible.

Alternate Names:

ginger

Zinostatin

An antitumor antibiotic, also called neocarzinostatin, that damages DNA. It has been used or studied in oncology in some countries and settings; availability, indications, and toxicity depend on the specific formulation and regulatory status.

Alternate Names:

neocarzinostatin

Zirconia

A zirconium-oxide ceramic used in dentistry for crowns, bridges, implants, and other restorations. Its strength and tissue compatibility make it useful, although restoration design, surface treatment, and clinical indication affect performance.

Alternate Names:

zirconium dioxide

Zirconium

A metallic element used in some medical and dental materials. Certain zirconium-containing compounds have specialized clinical uses, while exposure to some compounds can irritate tissues or produce systemic toxicity.

Alternate Names:

Zr

Zoledronic Acid

A potent intravenous bisphosphonate used for osteoporosis and certain cancer-related conditions, including hypercalcemia of malignancy and skeletal complications of bone metastases. Renal injury, hypocalcemia, acute-phase reactions, and rare osteonecrosis of the jaw are important risks.

Alternate Names:

zoledronate; Reclast; Zometa

Zollinger-Ellison Syndrome

A disorder caused by one or more gastrin-secreting

neuroendocrine tumors, called gastrinomas, that produce marked gastric acid hypersecretion. Excess acid commonly causes recurrent or multiple peptic ulcers, gastroesophageal reflux, abdominal pain, and secretory diarrhea. Gastrinomas may arise in the duodenum or pancreas and can occur as part of multiple endocrine neoplasia type 1. Diagnosis uses fasting serum gastrin with assessment of gastric acidity, followed by localization and staging of the tumor when appropriate.

Alternate Names:

ddC

Zolpidem

A nonbenzodiazepine sedative-hypnotic used for short-term treatment of insomnia. It can cause next-day impairment, complex sleep behaviors, dependence, and respiratory or central-nervous-system depression, especially with other sedatives.

Alternate Names:

Ambien

Zolpidem Tartrate

The tartrate salt of zolpidem, formulated as an oral sedative-hypnotic for insomnia. It carries the same risks as zolpidem, including next-day impairment, abnormal sleep behaviors, dependence, and additive sedation.

Zona Pellucida

A glycoprotein-rich extracellular layer surrounding the mammalian oocyte. It participates in sperm binding and the acrosome reaction and helps prevent polyspermy after fertilization.

Alternate Names:

oocyte zona pellucida

Zonisamide

An antiseizure medicine used for focal-onset seizures and, in some countries, other seizure indications. It can cause somnolence, cognitive effects, metabolic acidosis, kidney stones, oligohidrosis, and serious hypersensitivity reactions.

Alternate Names:

Zonegran

Zonula Adherens

A belt-like cell junction that links the actin cytoskeletons of adjacent epithelial cells. It contributes to cell adhesion and helps maintain epithelial structure and barrier organization.

Alternate Names:

adherens junction; zonula adherente

Zonule

Fibrous suspensory fibers connecting the ciliary body to the lens capsule. They transmit forces from the ciliary muscle and help alter lens curvature during accommodation.

Alternate Names:

zonular fibers; ciliary zonule

Alternate Names:

zonular fibers

Zoon Balanitis

A chronic benign inflammation of the glans penis, usually seen in uncircumcised men. It commonly appears as a sharply demarcated shiny orange-red or rust-colored patch; diagnosis is clinical, with biopsy used when the diagnosis is uncertain.

Alternate Names:

plasma cell balanitis; balanitis circumscripta plasmacellularis

Zoonosis

An infection naturally transmissible between vertebrate animals and humans. Examples include rabies, brucellosis, leptospirosis, psittacosis, toxoplasmosis, and some influenza infections.

Alternate Names:

zoonotic disease

Zoonotic Disease

An infectious disease that can be naturally transmitted between vertebrate animals and humans. Transmission may occur through direct contact, contaminated food or water, animal products, or arthropod vectors.

Alternate Names:

zoonosis

Zoster

A painful, usually unilateral vesicular eruption caused by reactivation of latent varicella-zoster virus in a sensory nerve ganglion. It may be complicated by postherpetic neuralgia, ocular disease, or neurologic involvement.

Alternate Names:

shingles; herpes zoster

Zygoma (Zygomatic Bone)

A paired facial bone that forms the prominence of the cheek and part of the lateral orbital wall and floor. It articulates with the frontal, sphenoid, temporal, and maxillary bones.

Alternate Names:

zygomatic bone; cheekbone

Zygomatic Fracture

A fracture of the zygomatic bone or zygomaticomaxillary complex, usually caused by facial trauma. It may cause facial flattening, infraorbital numbness, diplopia, trismus, or orbital injury.

Alternate Names:

zygomaticomaxillary complex fracture

Zygomatic Gland

In veterinary anatomy, the zygomatic gland is an orbital gland that contributes to the tear film, especially in dogs and cats; it is distinct from the human lacrimal gland and should be interpreted according to species-specific anatomy.

Zygomycosis

An older term for invasive fungal infection caused by organisms formerly grouped as zygomycetes, particularly mucormycetes. Mucormycosis can affect the sinuses, orbit, lungs, skin, or gastrointestinal tract, especially in people with diabetes or impaired immunity.

Alternate Names:

mucormycosis; phycomycosis

Zygote

The diploid cell formed when a sperm fertilizes an oocyte. It undergoes early cell divisions and devel-

ops into an embryo.

Alternate Names:

fertilized egg

Zymogen

An inactive enzyme precursor that is converted into an active enzyme, often by proteolytic cleavage. Examples include pepsinogen and trypsinogen.

Alternate Names:

proenzyme

Drugs

Generic Medicines and Medication Terms

5-Alpha Reductase Inhibitor

A medicine that blocks conversion of testosterone to dihydrotestosterone. It is used mainly to treat benign prostatic hyperplasia and androgen-related hair loss.

Abortifacient

A drug or substance that induces abortion by ending a pregnancy.

Abortion - Medication

Medication abortion uses prescribed medicines, commonly mifepristone followed by misoprostol or misoprostol alone, to end an early intrauterine pregnancy; confirmation of completion and assessment for ectopic pregnancy are important.

Abuse Steroid (Anabolic Steroid Abuse)

Alternate index label for Anabolic Steroid Abuse.

Accident And Emergency Medicine (A&E Medicine)

The medical specialty concerned with the immediate assessment, resuscitation, diagnosis, and initial treatment of acute illness and injury.

Alternate Names:

A&E Medicine

ACE Inhibitor

A drug that inhibits angiotensin-converting enzyme,

reducing formation of angiotensin II and degradation of bradykinin.

Acitretin

An oral retinoid used mainly for severe psoriasis and selected keratinization disorders. It is highly teratogenic and requires strict pregnancy precautions and monitoring for adverse effects.

Adenovirus Vaccine

A vaccine used primarily to prevent adenovirus types 4 and 7 among selected military personnel. It is an oral, live vaccine and is not part of routine vaccination for the general public. Vaccine use and eligibility depend on current public-health recommendations.

Adverse Drug Event

Harm related to the use of a medication, including harm caused by an adverse drug reaction, medication error, overdose, or inappropriate use.

Adverse Drug Reaction

An adverse drug reaction is any noxious or unintended response to a drug that occurs at doses used for prophylaxis, diagnosis, or therapy.

Agent, Tocolytic

A medicine that reduces uterine contractions, used selectively to delay preterm birth long enough for corticosteroids, transfer, or other preparations to take effect.

Agonist-Antagonist

A drug that has mixed pharmacological activity,

acting as an agonist at one receptor type and an antagonist at another, or as a partial agonist at a single receptor.

AIDS Vaccine

A vaccine designed to prevent HIV infection or reduce its effects; no generally effective licensed vaccine currently exists.

Alendronate

A bisphosphonate that reduces bone resorption and is used to treat or prevent osteoporosis and selected other disorders of excessive bone turnover.

Alternate Names:

Alendronic acid; oral bisphosphonate therapy.

Allopathic Medicine

A historical term commonly used for conventional, evidence-based medical practice; it is not a precise scientific classification and may be used differently by different groups.

Alternative Medicine

A nonconventional health practice used instead of standard medical treatment; evidence, safety, and interactions vary widely, and replacing effective care can be harmful.

Alternative Medicine - Pain Relief

A clinical topic concerning alternative Medicine - Pain Relief; assessment and management depend on the specific clinical question, findings, and patient context.

Anabolic Steroid Abuse

Anabolic-androgenic steroid misuse can cause infertility, testicular shrinkage, acne, mood changes, high blood pressure, adverse cholesterol changes, liver injury, and cardiovascular events. Stopping may produce withdrawal symptoms and is safest with medical support.

Alternate Names:

Abuse Steroid (Anabolic Steroid Abuse)

Analgesic

A drug that relieves pain without producing loss of consciousness.

Analgesic Nephropathy

Chronic tubulointerstitial renal injury associated with prolonged or excessive use of analgesic medicines. It may cause impaired urinary concentration, sterile pyuria, haematuria, papillary necrosis, hypertension, and progressive chronic kidney disease.

Analgesics

Medicines used to relieve pain. They include nonopioid analgesics, opioid analgesics, and adjuvant medicines used for selected pain conditions.

Alternate Names:

Painkillers

Anesthetic

A medicine or technique that causes loss of sensation, with or without loss of consciousness.

Angiogenesis Inhibitor

A treatment or substance that prevents or slows formation of new blood vessels.

Alternate Names:

Antiangiogenic agent

Angiotensin Converting-Enzyme Inhibitor

A class of antihypertensive medications that block the conversion of angiotensin I to angiotensin II by inhibiting the angiotensin-converting enzyme, thereby reducing vasoconstriction and aldosterone secretion. Used to treat hypertension, heart failure, and diabetic nephropathy.

Angiotensin Receptor Blocker

A medicine that blocks angiotensin II at the AT1 receptor, reducing vasoconstriction and aldosterone effects; it is used for hypertension, heart failure, and selected kidney disease.

Anti-Infective

A medicine or other agent used to prevent or treat infection caused by microorganisms, including bacteria, viruses, fungi, or parasites.

Anti-Inflammatory

A medicine or other treatment that reduces inflammation.

Antiarrhythmic

A drug used to treat or prevent abnormal cardiac rhythms. Drug classes act mainly on sodium channels, sympathetic stimulation, potassium channels, or calcium channels; selection depends on the rhythm and the patient's clinical condition.

Antibiotic

A substance that kills or slows the growth of bacteria.

Antibiotic Cycling

A strategy in which the preferred antibiotic for a particular clinical use is changed at planned intervals. It is intended to limit selection of resistant organisms, although its benefit depends on local conditions and implementation.

Antibiotic De-Escalation

The process of narrowing antimicrobial spectrum based on microbiological results and clinical response. De-escalation aims to reduce selective pressure for resistance while maintaining therapeutic efficacy.

Antibiotic Mixing

A strategy in which different antibiotics are assigned concurrently to different patients with the same indication. It is intended to reduce uniform selection pressure for resistance in a patient population.

Antibiotic Resistance

The ability of bacteria to survive or grow despite an antibiotic that would normally kill them or stop their growth. It can result from genetic changes and the spread of resistance genes; unnecessary or

incorrect antibiotic use can accelerate it.

Alternate Names:

Resistance Antibiotic (Antibiotic Resistance)

Antibiotic Resistance Genes

Genes that encode mechanisms conferring resistance to antibiotics. These genes can be located on chromosomes, plasmids, or transposons. Major resistance genes include *mecA* (methicillin resistance in *S. aureus*), *vanA/vanB* (vancomycin resistance in *Enterococcus*), and *blaKPC* (carbapenem resistance).

Antibiotic Stewardship

Coordinated efforts to use antibiotics appropriately by selecting the right medicine, dose, route, and duration while reducing unnecessary treatment and resistance.

Antibiotic-Associated Colitis

Alternate Mayo Clinic label; see *Pseudomembranous colitis*.

Antibiotic-Associated Diarrhea

Loose or frequent stools that occur during or soon after antibiotic treatment.

Alternate Names:

Diarrhea, Antibiotic-Associated

Anticholinergic Cognitive Burden Scale

A medication-review tool that estimates the total anticholinergic load of a patient's medicines. It supports risk assessment and deprescribing but does not replace individualized clinical judgment.

Anticoagulant

A medicine or substance that reduces the blood's ability to form clots.

Anticoagulant Agent

A drug that reduces blood clot formation by interfering with the coagulation cascade; examples include heparin, warfarin, and direct oral anticoagulants.

Anticoagulant Monitoring

Laboratory assessment used to measure the effect or safety of an anticoagulant. The appropriate test depends on the drug and may include aPTT, anti-Xa activity, INR, or measurements of kidney function and blood counts.

Anticoagulant Rodenticides Poisoning

Toxic effects caused by excessive exposure to Anticoagulant rodenticides poisoning. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Anticonvulsant

A medicine used to prevent or control seizures; some are also used for nerve pain or mood disorders.

Alternate Names:

Antiseizure medicine

Anticonvulsant Mood Stabilizers

Antiepileptic drugs used as mood stabilizers in bipolar disorder. Valproate (divalproex) is effective for acute mania and rapid cycling. Carbamazepine is effective for acute mania but has numerous drug interactions via CYP3A4 induction. Lamotrigine is effective for bipolar depression prevention but not acute mania.

Antidepressant

A medicine used to treat depressive disorders and some other conditions involving mood, anxiety, pain, or sleep.

Antidiabetic Agent

A medicine that lowers blood glucose or improves glucose regulation in diabetes, such as insulin, metformin, sulfonylureas, or GLP-1 receptor agonists.

Antidiarrheal

A drug that reduces the frequency or volume of diarrhea. Some antidiarrheals should be avoided when there is bloody diarrhea, high fever, suspected invasive infection, or ileus.

Antidiarrheal Medicine Overdose

Toxic effects caused by excessive exposure to Antidiarrheal medicine. Severity depends on the dose, route, timing, coexisting conditions, and clinical findings; suspected poisoning requires prompt assessment.

Antiemetic

A medication that prevents or treats nausea and vomiting.

Antiestrogen

A drug that blocks estrogen effects or reduces estrogen production; it may be used in hormone-sensitive breast cancer or to alter reproductive hormone signaling.

Antifungal

A drug used to treat fungal infections.

Antifungal Agent

A drug or chemical that kills fungi or inhibits their growth, used to treat fungal infections of the skin, mucosa, or internal organs.

Antifungal Drug Classes

Major classes of antifungal agents include polyenes (amphotericin B), azoles (fluconazole, voriconazole), echinocandins (caspofungin, micafungin), and allylamines (terbinafine). Polyenes bind to ergosterol in fungal cell membranes. Azoles inhibit ergosterol synthesis. Echinocandins inhibit cell wall synthesis.

Antifungal Medication

A medicine used to treat fungal infection; the choice and route depend on the organism and site, and examples include azoles, terbinafine, polyenes, and echinocandins.

Antihistamine

A drug that blocks histamine receptors. H1 antihistamines treat allergic rhinitis, urticaria, pruritus, and (older sedating agents) insomnia and motion sickness. First-generation (diphenhydramine, chlorpheniramine): CNS-penetrant—sedating, anti-

cholinergic (dry mouth, urinary retention, confusion in elderly).

Antihypertensive

A medicine used to lower blood pressure. Classes include ACE inhibitors, angiotensin-receptor blockers, calcium-channel blockers, thiazide-type diuretics, beta-blockers, and other agents. Selection depends on blood pressure, kidney function, cardiovascular disease, pregnancy, and other clinical factors.

Antihypertensive Agent

A medicine or treatment that lowers elevated blood pressure.

Alternate Names:

Blood-pressure medicine

Antimalarial

A drug used to prevent or treat malaria caused by Plasmodium parasites; selection depends on species, disease severity, resistance, and geographic exposure.

Antimicrobial Drug

A medicine used to treat infection by bacteria, viruses, fungi, or parasites; the drug is selected according to the organism, infection site, and patient factors.

Antimicrobial Drug Interactions

Interactions between antimicrobial agents and other medications. Fluoroquinolones chelate divalent cations. Macrolides inhibit CYP3A4. Rifampin is a potent enzyme inducer. Azole antifungals inhibit CYP3A4.

Antineoplastic

A medicine or other treatment that inhibits the growth or survival of abnormal cells, especially cancer cells.

Antiparasitic

A drug used to treat infections caused by parasites (protozoa and helminths). Antiprotozoals: metronidazole (Giardia, amebiasis, trichomoniasis),

tinidazole, chloroquine and artemisinin combinations (malaria), pyrimethamine (toxoplasmosis), pentamidine, and nifurtimox (Chagas).

Antiparkinsonian

A drug used to treat Parkinson disease and drug-induced parkinsonism.

Antiplatelet Agent

A medicine that reduces platelet clumping and lowers the risk of arterial clots.

Alternate Names:

Antiplatelet medicine

Antiprotozoal Drug

A medicine used to prevent or treat disease caused by protozoa, including drugs directed against malaria, intestinal protozoa, or tissue parasites.

Antipruritic

A medication that relieves itching (pruritus).

Antipsychotic

A drug used to treat psychotic disorders (schizophrenia, schizoaffective disorder, psychosis in bipolar/manic states and delirium). Typical (first-generation): haloperidol, chlorpromazine—D2 blockade, high EPS/tardive dyskinesia risk, hyperprolactinemia.

Antipsychotic Agent

A medicine used to treat psychosis and, in some cases, bipolar disorder or severe behavioral symptoms.

Alternate Names:

Antipsychotic

Antipsychotic-Induced Parkinsonism

Drug-induced bradykinesia, rigidity, tremor, or postural instability caused by dopamine-receptor blockade, usually from antipsychotic treatment. Management involves reviewing the medication, dose, alternatives, and symptomatic treatment when appropriate.

Antipyretic

A drug that reduces fever by lowering the hypothalamic set point, typically by inhibiting prostaglandin synthesis in the preoptic hypothalamus. Acetaminophen and NSAIDs (ibuprofen, aspirin) are the principal antipyretics.

Antispasmodic

A medication that reduces smooth muscle spasm, primarily in the gastrointestinal tract. Anticholinergic agents: hyoscyamine, dicyclomine, atropine — block muscarinic M3 receptors on GI smooth muscle, reducing contractions and peristalsis, used for irritable bowel syndrome and functional dyspepsia.

Alternate Names:

Antispasmodics

Antithyroid

A medicine or treatment that reduces thyroid-hormone production or activity. Examples include thioamides, radioactive iodine, and medicines that reduce the effects of thyroid hormone; the choice depends on the cause and clinical situation.

Antithyroid Drug

A medicine that reduces thyroid hormone production or release, used to treat hyperthyroidism; examples include methimazole, propylthiouracil, and radioactive iodine.

Antitussive

A medication that suppresses the cough reflex, used for dry, non-productive cough.

Antiviral

A class of drugs that inhibit viral replication, used to treat viral infections.

Antiviral Drug Classes

Major classes of antiviral agents include nucleoside analogues (acyclovir, tenofovir), protease inhibitors (ritonavir, atazanavir), reverse transcriptase inhibitors (efavirenz, sofosbuvir), and neuraminidase inhibitors (oseltamivir). Each class targets a specific step in the viral replication cycle.

Anxiolytic

A medicine or other treatment that reduces anxiety. Choice depends on the anxiety disorder, urgency, coexisting conditions, adverse effects, and patient preference; psychological therapies and, when appropriate, antidepressant medicines are commonly used for persistent anxiety.

ARB-Neprilysin Inhibitor

Sacubitril/valsartan combines an ARB with a neprilysin inhibitor, increasing natriuretic peptide levels while blocking the renin-angiotensin system. The PARADIGM-HF trial demonstrated superior reduction in cardiovascular death and heart failure hospitalization compared to enalapril.

Arimidex

A brand name for anastrozole, an aromatase inhibitor used mainly to treat hormone-receptor-positive breast cancer in selected patients.

Aromatase Inhibitor

A class of medications that inhibit aromatase, the enzyme that converts androgens (androstenedione, testosterone) to estrogens (estrone, estradiol) in peripheral tissues (adipose, breast, muscle, bone). Used primarily in hormone receptor-positive breast cancer in postmenopausal women.

Aspirin

Acetylsalicylic acid, a salicylate that relieves pain and fever, reduces inflammation, and irreversibly inhibits platelet aggregation; it can cause gastrointestinal bleeding and is generally avoided in children with viral illness.

Asthma OTC Medication (Asthma: Over The Counter Treatment)

Alternate index label for Asthma: Over The Counter Treatment.

Attenuated Vaccine

A vaccine made with a weakened microorganism that retains the ability to stimulate immunity.

Alternate Names:

Live attenuated vaccine

Bacterial Vaccine

Bacterial vaccine is a biological organism or infectious agent; its clinical importance depends on the species, host, site, exposure, and evidence of infection or colonization.

BCG Vaccine

A live attenuated vaccine derived from *Mycobacterium bovis* that helps prevent severe forms of tuberculosis, especially in children. Its use depends on national tuberculosis-risk policies.

Behavioral Medicine

An interdisciplinary field integrating behavioral science with medical practice to prevent, diagnose, and treat illness. It addresses psychological factors such as stress, coping, adherence, and health behaviors that influence disease outcomes.

Benzodiazepines

A class of drugs that enhance the effect of the inhibitory neurotransmitter gamma-aminobutyric acid at the GABA-A receptor, increasing the frequency of chloride channel opening and producing anxiolytic, sedative, anticonvulsant, and muscle relaxant effects.

Beta Agonists

Drugs that activate beta-adrenergic receptors, producing effects such as relaxation of airway smooth muscle or increased cardiac stimulation depending on receptor selectivity.

Alternate Names:

Beta-adrenergic agonists; beta-agonist drugs.

Beta Blocker

A drug that blocks beta-adrenergic receptors, lowering sympathetic effects such as heart rate and contractility; beta blockers are used for conditions including hypertension, angina, arrhythmias, and heart failure.

Beta Blockers

A class of medicines that block the effects of adrenaline and related hormones at beta receptors. They usually slow the heart rate, reduce the force of heart contraction, and lower blood pressure. Depending on the medicine, they may be used for high blood pressure, angina, abnormal heart rhythms, heart failure, migraine prevention, tremor, or some symptoms of anxiety.

Beta-Adrenergic Blocker

A drug that antagonizes beta-adrenergic receptors, reducing heart rate, myocardial contractility, and renin release. It is used in selected patients with hypertension, angina, arrhythmias, post-myocardial infarction disease, and heart failure; abrupt withdrawal may worsen ischemia.

Beta-Agonist

A drug that stimulates beta-adrenergic receptors; beta-2 agonists relax airway smooth muscle and are used as bronchodilators.

Beta-Blocker Overdose

Toxic exposure to a beta-adrenergic blocker that may cause slow heart rate, low blood pressure, low blood glucose, bronchospasm, seizures, or altered consciousness. It requires urgent poison-control and medical management.

Beta-Lactamase Inhibitor Combinations

Combinations of beta-lactam antibiotics with beta-lactamase inhibitors that extend activity against beta-lactamase producing organisms. The major combinations include amoxicillin-clavulanate, ampicillin-sulbactam, piperacillin-tazobactam, cefoperazone-sulbactam, and ceftolozane-tazobactam.

Botox

A brand name for botulinum toxin type A, used to reduce muscle activity in selected neurologic, ophthalmic, dermatologic, cosmetic, and other conditions.

C1 Esterase Inhibitor

A plasma protein that regulates the complement, kallikrein-kinin, and coagulation pathways. Deficiency or dysfunction can cause hereditary angioedema.

Calcium Channel Blocker

A drug that inhibits calcium entry through L-type calcium channels, relaxing vascular smooth muscle and, for some agents, slowing cardiac conduction and reducing contractility.

Calcium Channel Blocker Overdose

Toxic exposure to a calcium-channel blocker that may cause low blood pressure, slow heart rate, conduction abnormalities, high blood glucose, or shock. It requires urgent medical and poison-control management.

Calcium Channel Blockers

Medicines that reduce the movement of calcium into heart and blood-vessel muscle cells. Dihydropyridines, such as amlodipine and nifedipine, mainly relax arteries and are used for high blood pressure and angina. Other calcium channel blockers, such as diltiazem and verapamil, also slow the heart rate and electrical conduction and may be used for certain abnormal heart rhythms.

Camptothecin Analogue

A drug structurally related to camptothecin that inhibits topoisomerase I, disrupting DNA replication and causing cancer-cell death; toxicity commonly includes marrow or gastrointestinal effects.

Cancer Vaccine

A vaccine intended either to prevent cancer-causing infections or to stimulate an immune response against cancer cells. Preventive examples target oncogenic infections, while therapeutic vaccines are designed for selected established cancers.

Cardiotonic Agent

A drug or substance that increases the force or efficiency of cardiac contraction, used selectively in heart failure or shock because benefits must be

balanced against arrhythmia and other risks.

Carubicin

A medicine or investigational pharmacologic compound; its mechanism, indication, dosing, interactions, and adverse effects depend on the specific agent and patient.

Carvedilol Phosphate

A medicine or investigational pharmacologic compound; its mechanism, indication, dosing, interactions, and adverse effects depend on the specific agent and patient.

Carzelesin

A medicine or investigational pharmacologic compound; its mechanism, indication, dosing, interactions, and adverse effects depend on the specific agent and patient.

Caspase Inhibitor

A biological or biochemical term referring to caspase inhibitor. Its role depends on the relevant tissue, pathway, or clinical context.

Cebocephaly

A medicine or investigational pharmacologic compound; its mechanism, indication, dosing, interactions, and adverse effects depend on the specific agent and patient.

Celecoxib

A medicine or investigational pharmacologic compound; its mechanism, indication, dosing, interactions, and adverse effects depend on the specific agent and patient.

Checkpoint Inhibitor

A medicine that blocks an inhibitory immune-checkpoint pathway, helping selected immune cells recognize and attack tumor cells. Examples target PD-1, PD-L1, or CTLA-4.

Cholesterol - Drug Treatment

A medicine or chemical compound identified by

cholesterol - drug treatment. Its uses, effects, and risks depend on the specific substance and clinical context.

Cholinesterase Inhibitor

A class of medications that inhibit acetylcholinesterase, the enzyme that breaks down acetylcholine in the synaptic cleft, thereby increasing cholinergic neurotransmission in the central nervous system and peripheral tissues.

Cold And Cough Medicine For Infants And Children

Many nonprescription cold and cough medicines are unsafe or ineffective for young children and can cause sedation, agitation, abnormal heart rate, overdose, or breathing problems. Age-appropriate fluids, saline, humidification, and clinician-guided treatment are safer approaches.

Complementary Medicine

Health practices used together with conventional medical treatment, such as selected mind-body practices or acupuncture; it is distinct from alternative medicine, which replaces conventional care.

Conjugate Vaccine

A vaccine in which a poorly immunogenic polysaccharide antigen is linked to a carrier protein to improve the immune response, especially in infants and young children.

Alternate Names:

Protein-conjugate vaccine; conjugated polysaccharide vaccine.

Conscious Sedation

A drug-induced depression of consciousness in which patients respond purposefully to verbal or tactile stimulation, maintain a patent airway, and retain protective reflexes. Typical agents include midazolam, fentanyl, or low-dose propofol. Indications include endoscopic and minor surgical procedures.

Controlled Substance

A drug or chemical whose manufacture, possession, and use are regulated by government authorities

due to its potential for abuse, dependence, or harm.

COX Inhibitor

A class of drugs that inhibit cyclooxygenase enzymes (COX-1 and COX-2), which catalyze the conversion of arachidonic acid to prostaglandins and thromboxanes. COX-1 is constitutively expressed in most tissues, producing prostaglandins that protect the gastric mucosa, maintain renal blood flow, and support platelet aggregation.

Cox-1 Inhibitor

A drug that inhibits cyclooxygenase-1 and reduces production of prostaglandins and thromboxanes; inhibition can contribute to analgesia but also gastrointestinal, renal, and platelet effects.

Crack (Drug)

A smokable, rapidly acting form of cocaine; it can cause dependence, severe cardiovascular effects, seizures, stroke, psychiatric symptoms, and overdose.

Cutaneous Drug Eruptions

a spectrum of reactions. Exanthematous (maculopapular) is the most common, appearing 4 to 14 days after drug initiation. Urticarial appears within hours. Photoallergic occurs at sun-exposed sites. Lichenoid resembles lichen planus.

Cyclooxygenase Inhibitor

A medicine that reduces production of prostaglandins and may relieve pain and inflammation.

Cymbalta

A brand name for duloxetine, a serotonin-norepinephrine reuptake inhibitor used for depression, anxiety, and selected neuropathic or chronic pain conditions.

Designer Steroid

A chemically modified anabolic-androgenic steroid created to produce performance-enhancing effects or evade detection; products may be unapproved,

contaminated, and medically dangerous.

Diarrhea, Antibiotic-Associated

Alternate index label; see *Antibiotic-associated diarrhea*.

Dietary Steroid

A dietary steroid is a steroid compound consumed in food or supplements; sterols and hormones can affect metabolism or interact with medicines.

Digitalis

A group of cardiac glycosides, including digoxin, that can increase cardiac contraction and affect heart rhythm.

Diphtheria Vaccine

Toxoid vaccine containing formalin-inactivated diphtheria toxin, providing immunity against the exotoxin produced by *Corynebacterium diphtheriae* rather than the bacterium itself.

Disease-Modifying Antirheumatic Drug

A medication that reduces inflammatory disease activity and helps prevent structural damage or disability rather than providing only symptomatic relief. Conventional synthetic, biologic, and targeted synthetic DMARDs are used according to disease and patient factors.

Disulfiram

A medicine used as an adjunct for alcohol-use disorder that inhibits aldehyde dehydrogenase and can cause an unpleasant reaction if alcohol is consumed.

Diuretic

A drug that increases urine production, used for edema (heart failure, cirrhosis, renal disease), hypertension, and electrolyte disorders.

Doctor Of Osteopathic Medicine

A physician who has completed an accredited osteopathic medical degree and residency training and is licensed to diagnose, prevent, and treat disease.

Drug Abuse

A nonpreferred term for harmful or nonmedical use of a drug; current clinical language generally uses substance use, misuse, or substance use disorder according to the situation.

Drug Addiction (Substance Use Disorder)

A chronic disorder involving compulsive substance use despite harmful consequences, with impaired control, cravings, tolerance, or withdrawal.

Alternate Names:

Chemical Dependency Substance Abuse Disorder

Drug Administration Route

The pathway by which a medicine enters the body, such as oral, intravenous, intramuscular, inhaled, topical, or subcutaneous administration.

Drug Allergies

Drug allergies are immune-mediated reactions to a medicine, ranging from rash to anaphylaxis; they must be distinguished from predictable side effects and nonimmune intolerance.

Drug Allergy

An adverse immune-mediated reaction to a medication, representing a hypersensitivity response that is distinct from side effects, toxicities, and idiosyncratic reactions.

Drug Antagonism

An interaction where one drug blocks or decreases the effectiveness of another, such as when an antidote blocks a poison.

Drug Bioavailability

The fraction of an administered drug that reaches systemic circulation unchanged. Oral bioavailability is affected by first-pass metabolism, absorption, and drug formulation. IV administration has 100% bioavailability. Bioavailability influences drug dosing and route selection.

Drug Classification Systems

Systems that group medicines according to proper-

ties such as their chemical structure, mechanism of action, therapeutic use, or effects on the body.

Drug Clearance

The volume of plasma completely cleared of drug per unit time. Hepatic clearance depends on hepatic blood flow and enzyme activity; renal clearance depends on glomerular filtration, tubular secretion, and reabsorption. Total clearance determines maintenance dose.

Drug Dependence

A state of physiological or psychological adaptation to a drug, characterized by a withdrawal syndrome upon discontinuation or dose reduction. Physical dependence: the body has adapted to the presence of the drug, and abrupt cessation produces a predictable withdrawal syndrome specific to the drug class.

Drug Emulsion

A pharmaceutical preparation in which one liquid is dispersed as droplets within another immiscible liquid, often stabilized with an emulsifier.

Drug Eruption

A skin eruption caused by a medication, ranging from a mild morbilliform rash to severe reactions such as Stevens–Johnson syndrome or toxic epidermal necrolysis.

Drug Hypersensitivity Reactions

Immune-mediated adverse reactions to medicines. They may be immediate or delayed and can range from a mild rash to severe systemic illness.

Drug Hypersensitivity Syndrome

A severe delayed drug reaction occurring 2 to 8 weeks after drug initiation. It presents with fever, rash (morbilliform, facial edema, exfoliative dermatitis), lymphadenopathy, eosinophilia, and internal organ involvement (hepatitis most common, also nephritis, pneumonitis, myocarditis).

Drug Implants

Implantable dosage forms or devices that release a medicine over time; selection and monitoring depend on the product and clinical indication.

Drug Induced Liver Disease

Drug-induced liver injury occurs when a medicine, supplement, or toxin damages liver cells or bile flow. It may cause fatigue, nausea, itching, dark urine, jaundice, or no symptoms; identifying and stopping the responsible exposure under medical guidance is central to care.

Drug Interaction

A change in the effect or concentration of one medicine caused by another medicine, food, substance, or disease.

Drug Metabolism

The enzymatic conversion of medicines into metabolites, mainly in the liver but also in the intestine, kidneys, lungs, and other tissues.

Drug Metabolism Pharmacogenomics

Pharmacogenomics of drug metabolism examines how genetic variations affect individual responses to drugs, enabling personalized medicine approaches to prescribing.

Drug Overdose

Excessive ingestion of medication. Common: opioids, benzodiazepines, acetaminophen, salicylates, tricyclic antidepressants. Management: ABCs, activated charcoal, specific antidotes (naloxone, flumazenil, N-acetylcysteine), supportive care.

Drug Overdose Management

Management of poisoning begins with assessment and support of airway, breathing, circulation, and neurologic status. Decontamination, enhanced elimination, and specific antidotes are considered only when appropriate for the substance, timing, and patient.

Drug Packaging

The container, labeling, and protective materials used to preserve a medicine and support correct identification, storage, and use.

Drug Reaction With Eosinophilia And Systemic Symptoms

A severe hypersensitivity syndrome characterized by fever, rash, lymphadenopathy, and multiorgan involvement triggered by medications, requiring prompt drug cessation and systemic corticosteroids.

Drug Resistance

The ability of microorganisms, cancer cells, or parasites to survive and proliferate despite exposure to drugs that were previously effective.

Drug Therapeutics

The use of medicines to prevent, diagnose, or treat disease while balancing expected benefit and potential harm.

Drug Tolerance

A decreased pharmacological response to a drug after repeated administration, requiring dose escalation to achieve the same effect.

Drug Toxicity

Harmful effects caused by a medicine or its metabolites. Toxicity may result from excessive exposure, accumulation, interactions, or an unpredictable individual reaction.

Drug Use First Aid

Immediate care for suspected drug intoxication, including assessment of breathing and responsiveness, emergency activation, overdose-specific support, and naloxone when opioid toxicity is suspected.

Drug Vehicle

An inactive substance or carrier used to deliver an active drug ingredient in a formulation.

Drug Withdrawal

A clinically significant syndrome or return of symp-

toms caused by abrupt reduction or cessation of a drug after physiologic adaptation or dependence has developed. Tapering is required for some medications to reduce withdrawal and rebound effects.

Drug, Prescription

A medicine legally supplied under the order of an authorized prescriber, with its indication, dose, duration, interactions, and risks specified for the patient.

Drug, Sulfa

A sulfonamide drug; some are antibacterial medicines, while others are used as diuretics or for other indications, and sulfonamide reactions vary by drug and patient.

Drug-Drug Interactions

When one drug alters the pharmacokinetics or pharmacodynamics of another. Pharmacokinetic interactions affect absorption, distribution, metabolism, or excretion (e.g., CYP inhibitors increasing warfarin levels).

Drug-Eluting Balloon (Vascular)

A vascular balloon catheter that delivers a medicine to the vessel wall during dilation, without leaving a permanent implant.

Alternate Names:

Vascular

Drug-Eluting Stent

A vascular stent that slowly releases a medicine to reduce tissue growth and the risk of restenosis after angioplasty.

Drug-Eluting Stent (Vascular)

A vascular stent coated with medicine that is released gradually to reduce tissue growth and recurrent narrowing.

Alternate Names:

Vascular

Drug-Facilitated Sexual Assault

Sexual contact occurring when a person's capacity to

consent or resist is impaired by alcohol, medication, or another drug, whether administered covertly or knowingly consumed. Medical care includes urgent safety assessment, treatment of injuries, toxicology sampling when indicated, prophylaxis, and trauma-informed support.

Drug-Induced Autoimmunity

Drug-induced lupus (DIL): procainamide, hydralazine, isoniazid, minocycline, TNF inhibitors, anti-TNF; anti-histone antibodies, positive ANA, negative anti-dsDNA/anti-Sm; symptoms resolve after drug cessation.

Drug-Induced Diarrhea

Diarrhea caused by a medicine or drug interaction. Common causes include antibiotics, laxatives, metformin, and medicines that alter intestinal motility.

Drug-Induced Immune Hemolytic Anemia

An uncommon immune-mediated destruction of red blood cells triggered by a medication; evaluation includes blood tests and review of exposures, with specialist management.

Drug-Induced Liver Injury

Liver damage caused by a medicine, herbal product, or dietary supplement. It may produce abnormal liver tests, hepatitis, cholestasis, or liver failure; evaluation requires considering the exposure, timing, competing causes, and severity.

Drug-Induced Low Blood Sugar

Low blood glucose caused by a medicine, most often insulin or an insulin secretagogue. Symptoms may include sweating, tremor, confusion, seizure, or loss of consciousness.

Drug-Induced Lupus Erythematosus

A lupus-like autoimmune syndrome caused by certain medicines. Symptoms commonly include joint pain, muscle pain, and fever and usually improve after the causative medicine is stopped.

Drug-Induced Pulmonary Disease

A pharmacologic agent associated with drug-Induced Pulmonary Disease; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

Drug-Induced Thrombocytopenia

A low platelet count caused by a medicine through immune destruction, reduced production, or other mechanisms, increasing bleeding risk.

Drug-Induced Tremor

Tremor caused or worsened by a medicine, stimulant, or toxin. Assessment considers timing, dose, other causes, and whether the medicine can be changed.

DTaP Vaccine

Diphtheria, Tetanus, and acellular Pertussis vaccine given at 2, 4, 6, and 15-18 months, and 4-6 years. Protects against three serious bacterial diseases. Diphtheria causes respiratory and cardiac complications. Tetanus causes lockjaw and muscle rigidity. Pertussis causes whooping cough, most severe in young infants.

Effornithine

A medicine that inhibits ornithine decarboxylase; topical use reduces unwanted facial hair, while intravenous use has been used against selected trypanosome infections.

Elavil

A brand name formerly associated with amitriptyline, a tricyclic antidepressant used for depression and selected pain, migraine, or sleep-related indications.

Enzyme Inhibitor

A substance that reduces enzyme activity by binding the active site or another regulatory site, thereby altering a biochemical pathway or therapeutic target.

EpiPen

A brand of epinephrine auto-injector used for emer-

gency treatment of anaphylaxis.

Estrogen Inhibitor

Estrogen inhibitor is a medical term whose precise meaning depends on the clinical context, associated findings, and source definition.

Etidronate

A bisphosphonate used in selected conditions involving abnormal bone remodeling.

Evidence-Based Medicine

Integration of best research evidence with clinical expertise and patient values. Foundation for clinical guidelines.

Expectorant

A medicine intended to help loosen or mobilize respiratory mucus so it can be coughed up; benefit varies by cause and product.

Extensively Drug-Resistant TB

TB resistant to isoniazid, rifampin, fluoroquinolones, and at least one injectable agent. Very limited treatment options and poor outcomes.

Family Medicine

A medical specialty providing comprehensive continuing care for people of all ages, including prevention, acute illness, chronic disease, and coordination of referrals.

Fentanyl

A potent synthetic opioid analgesic used for severe pain and anesthesia. It can cause respiratory depression, sedation, constipation, nausea, tolerance, dependence, and fatal overdose, especially when combined with alcohol or other sedatives.

Fertility Agent

A medicine or biologic intervention intended to improve the chance of conception by inducing ovulation, supporting sperm production, or treating a reproductive mechanism.

Fidaxomicin

A narrow-spectrum oral macrocyclic antibiotic used to treat *Clostridioides difficile* infection. It has minimal systemic absorption and reduces disruption of normal intestinal flora compared with many alternatives, but recurrence can still occur.

Filgrastim

A recombinant granulocyte colony-stimulating factor that increases production and release of neutrophils from bone marrow. It is used to reduce or treat selected forms of chemotherapy-related neutropenia; bone pain and rare splenic complications can occur.

Finasteride

A 5-alpha-reductase inhibitor used for benign prostatic enlargement and some types of hair loss.

Fixed Drug Eruption

A recurrent drug reaction appearing at the same site with each exposure to the offending drug. It presents as well-demarcated, round to oval, erythematous to violaceous plaques that may blister. Common culprits include sulfonamides, tetracyclines, NSAIDs, and barbiturates.

Fluticasone

A corticosteroid available in inhaled, nasal, topical, and other formulations to reduce inflammation. Adverse effects depend on route and may include local irritation, oral candidiasis, skin thinning, adrenal suppression, or growth effects with prolonged exposure.

Food-Drug Interaction

An alteration in drug absorption, metabolism, distribution, or effect caused by a food, beverage, nutrient, or supplement. Examples include vitamin K affecting warfarin response, grapefruit affecting selected CYP3A4 substrates, and calcium impairing absorption of some medicines.

Forensic Medicine

Application of medical knowledge to legal questions, including injury interpretation, cause of death, tox-

cology, identification, sexual assault evaluation, and documentation.

Forensic Medicine (Legal Medicine)

Branch of medicine that applies medical, biological, and scientific knowledge to legal questions and the administration of justice.

Alternate Names:

Legal Medicine

Foscarnet

An intravenous antiviral agent used for selected cytomegalovirus or herpesvirus infections, especially when resistance or intolerance limits other treatment. Important toxicities include kidney injury and disturbances of calcium, magnesium, potassium, or phosphate.

Furosemide

A loop diuretic that inhibits sodium and chloride reabsorption in the thick ascending limb of the loop of Henle, increasing urine production. It treats edema and some cases of hypertension but can cause dehydration, low electrolytes, kidney injury, and ototoxicity.

Ganglionic Blocker

A class of drugs that block nicotinic acetylcholine receptors at autonomic ganglia (both sympathetic and parasympathetic), inhibiting neurotransmission in both divisions of the autonomic nervous system.

Generic Drug

A medicine equivalent to a brand-name medicine in active ingredient, strength, dosage form, and intended use, usually sold under a nonproprietary name.

Alternate Names:

Generic medicine

Generic Medicine

A medicine containing the same active ingredient, strength, dosage form, route, and intended clinical effect as an approved reference product. Regulatory requirements generally include demonstration of

pharmaceutical equivalence and bioequivalence.

Generic Name, Drug

A medicine, biologic, or pharmacologic term related to generic name, drug; its mechanism, indication, interactions, and adverse effects depend on the specific agent.

Genomic Medicine

Use of a person's genetic or genomic information to guide disease diagnosis, risk assessment, prevention, or treatment.

Geriatric Medicine

The medical specialty focused on the health, function, diseases, medications, and goals of older adults, often using comprehensive and coordinated care.

Growth Inhibitor

growth inhibitor is a medical or biological concept whose definition and clinical significance depend on its characteristic features, setting, associated findings, and clinical context.

Haemophilus Vaccine

haemophilus vaccine is a bacterial taxon; disease risk and clinical significance depend on the strain, specimen source, host factors, and evidence of infection rather than colonization.

Heparin

An anticoagulant that reduces formation and extension of blood clots.

Hepatic Drug Metabolism

The liver chemically modifies medicines through reactions such as oxidation, reduction, hydrolysis, and conjugation. These reactions can inactivate a drug, activate a prodrug, or produce an active or toxic metabolite, and are influenced by genetics, disease, age, and other medicines.

Hepatitis B Vaccine

Recombinant vaccine given at birth, 1-2 months, and 6-18 months. Protects against hepatitis B virus,

which can cause chronic liver disease, cirrhosis, and hepatocellular carcinoma. Also prevents perinatal transmission when given to infants of HBsAg-positive mothers within 12 hours of birth, plus hepatitis B immune globulin.

Hepatitis B Vaccine (Birth Dose)

The first dose of the hepatitis B vaccine is recommended within 24 hours of birth for all medically stable infants. Administered IM in the vastus lateralis. Prevents perinatal transmission of HBV. For infants born to HBsAg-positive mothers: HBIG (0.5 mL IM) is given within 12 hours of birth along with the vaccine.

Alternate Names:

Birth Dose

HER2 Inhibitor

A targeted cancer treatment that blocks signaling through human epidermal growth factor receptor 2 in cancers with HER2 overexpression or amplification.

High Blood Pressure - Medicine-Related

Hypertension caused or worsened by a medicine or substance, such as NSAIDs, steroids, decongestants, stimulants, oral contraceptives, or immunosuppressants; review and substitution may improve control.

Histamine-2 Receptor Antagonist

A drug that blocks histamine H2 receptors on gastric parietal cells, reducing gastric acid secretion. Agents: cimetidine, ranitidine, famotidine, nizatidine. Used for GERD, peptic ulcer disease, and dyspepsia—less potent than PPIs. Adverse effects: headache, diarrhea, confusion (elderly, cimetidine), and rare thrombocytopenia.

Hormone Inhibitor

A medicine or biologic that blocks hormone synthesis, release, receptor binding, or downstream signaling, used in selected endocrine disorders and hormone-sensitive cancers.

HPV Vaccine

Human Papillomavirus vaccine given at 11-12 years

(can start at 9), with series completion by 15 years. Two-dose series if started before 15; three-dose series if started at 15 or older. Protects against HPV types causing cervical cancer, oropharyngeal cancer, and genital warts. Gardasil 9 covers nine HPV types.

Hypoglycemic Agent

A medicine that lowers blood glucose, including insulin, insulin secretagogues, insulin sensitizers, incretin-based drugs, and sodium-glucose cotransporter inhibitors; choice depends on diabetes type and patient risks.

Hypolipidemic Agent

A medication that lowers blood lipids such as cholesterol or triglycerides.

Imitrex

A brand name for sumatriptan, a serotonin receptor agonist used for acute treatment of migraine and cluster headache in selected patients.

Immunosuppressant

A drug that reduces immune system activity, used to prevent organ transplant rejection and to treat autoimmune and inflammatory diseases.

Immunosuppressant Drug

A medicine that reduces immune-system activity, often to prevent transplant rejection or treat autoimmune disease.

Immunosuppressive Drug

A drug that suppresses immune responses, used for organ transplantation, graft-versus-host disease, and autoimmune/inflammatory diseases.

Inactivated Polio Vaccine

Salk vaccine, a formalin-inactivated preparation containing all three poliovirus serotypes (1, 2, and 3), administered by intramuscular injection. The only polio vaccine used in the United States since 2000, and the recommended vaccine for immunocompromised individuals and pregnant women worldwide.

Influenza Vaccine

Inactivated influenza vaccine given annually starting at 6 months of age. Two doses needed in the first season for children under 9. Live attenuated nasal spray vaccine (FluMist) available for children over 2. Updated annually to match circulating strains. Reduces influenza complications, hospitalizations, and deaths.

Inotrope

A drug that changes myocardial contractility; positive inotropes increase the force of contraction. Dobutamine and milrinone may be used short term under specialist monitoring for cardiogenic shock or selected cases of acute decompensated heart failure with low output. Dopamine, epinephrine, and norepinephrine are primarily vasopressors or emergency agents with additional inotropic effects, while digoxin is a separate cardiac glycoside used in selected chronic heart-failure or rate-control settings. These medicines can cause arrhythmias, ischemia, or other serious effects and are not routine long-term treatment for chronic heart failure.

Integrative Medicine For Cancer Treatment

A medicine or chemical compound called integrative medicine for cancer treatment. Its uses, effects, interactions, and risks depend on the specific substance and clinical indication.

Intensive Care Medicine

Specialty managing critically ill patients. Components: monitoring, ventilatory support, hemodynamic support, infection control, nutrition, sedation, pain management. Common conditions: sepsis, ARDS, traumatic brain injury, post-cardiac arrest, multiorgan failure.

Internal Medicine

The medical specialty focused on prevention, diagnosis, and nonsurgical treatment of diseases in adults, including complex multisystem illness.

Japanese Encephalitis Vaccine

A vaccine that protects against Japanese encephali-

tis, a mosquito-borne viral infection of the central nervous system. Vaccine type, schedule, and eligibility depend on age, destination or exposure risk, and local recommendations.

Laxative

A drug that promotes bowel evacuation, used for constipation and bowel preparation. Classes: bulk-forming (psyllium, methylcellulose), osmotic (polyethylene glycol, lactulose, magnesium salts), stimulant (senna, bisacodyl), stool softener (docusate), and lubricant (mineral oil). Choice depends on constipation type and patient factors.

Legal Medicine

The application of medical knowledge to legal questions, including forensic examination, injury assessment, death investigation, and issues of capacity or consent.

Lipitor

A brand name for atorvastatin, an HMG-CoA reductase inhibitor used to reduce low-density lipoprotein cholesterol and cardiovascular risk.

Liquid Medication Administration

A medicine or pharmacologic agent called liquid medication administration. Its uses, contraindications, interactions, and adverse effects depend on the specific agent and patient.

Local Anesthetic

A medication that causes reversible loss of sensation in a specific, localized area of the body without affecting consciousness. Local anesthetics block voltage-gated sodium channels in nerve cell membranes, preventing action potential generation and propagation.

Luteinizing Hormone-Releasing Hormone Agonist

A medicine that initially stimulates and then suppresses reproductive-hormone production with continued use.

Alternate Names:

GnRH agonist; LHRH agonist

Macrolide Antibiotic

An antibiotic of the macrolide class that binds the bacterial 50S ribosomal subunit and usually inhibits protein synthesis; activity and resistance vary by organism and drug.

Malaria Vaccine

A vaccine intended to reduce malaria caused by *Plasmodium* parasites, especially *P. falciparum*; the RTS,S/AS01 and R21/Matrix-M vaccines are used in recommended childhood immunization programs in some endemic settings, alongside other preventive measures.

Manufactured Form

A drug, chemical, nutrient, or other product prepared by an industrial or laboratory process into a defined formulation, such as a tablet, capsule, solution, or injectable preparation.

MAO Inhibitor

A drug that inhibits monoamine oxidase, the enzyme that metabolizes monoamine neurotransmitters; some are used for depression or Parkinson disease and require attention to interactions and dietary precautions.

Me-Too Drug

A medicine, biologic, or pharmacologic term related to me-too drug; its mechanism, indication, interactions, and adverse effects depend on the specific agent.

Measles Vaccine

Measles vaccine is a live attenuated vaccine that induces active immunity against measles virus. It is usually given as part of the measles–mumps–rubella (MMR) or measles–mumps–rubella–varicella (MMRV) vaccine.

Measles-Rubella Vaccine

A combined live attenuated vaccine containing measles and rubella antigens without the mumps component, widely used in countries where mumps is not a significant public health burden, including India's Universal Immunization Schedule.

Medication

A medicine, biologic, or pharmacologic term related to medication; its mechanism, indication, interactions, and adverse effects depend on the specific agent.

Medication Adherence

The extent to which a patient's medication-taking behavior corresponds with the agreed treatment plan. Adherence is influenced by understanding, adverse effects, cost, access, regimen complexity, beliefs, and cognitive or social factors.

Medication Adverse-Effect Profile

The clinically relevant pattern of predictable pharmacologic effects and harmful reactions associated with a medication. It includes common dose-related effects, serious rare events, organ-specific toxicity, interactions, and monitoring requirements.

Medication Cascade

A prescribing sequence in which an adverse drug effect is misinterpreted as a new medical condition and another drug is prescribed to treat it. Recognizing cascades is an important goal of medication reconciliation and deprescribing.

Medication Deprescribing

A planned, supervised process of reducing the dose or stopping a medicine when its current or future harms outweigh its benefits or when treatment goals change. Tapering may be required to prevent withdrawal or rebound disease.

Medication Error

A preventable event that may cause or lead to inappropriate medication use or patient harm while the medication is in the control of a healthcare professional, patient, or consumer. Errors can occur during prescribing, transcribing, dispensing, administration, or monitoring.

Medication Management

The systematic process of optimizing pharmacotherapy to achieve therapeutic goals while minimizing adverse effects, encompassing prescribing, dispens-

ing, administration, monitoring, and deprescribing.

Medication Overuse Headaches

Frequent headaches caused or worsened by regular overuse of headache medicines.

Alternate Names:

Rebound Headache

Medication Reconciliation

A structured process of obtaining and comparing the most accurate medication list with current prescriptions at transitions of care. It identifies omissions, duplications, incorrect doses, interactions, and unintended changes.

Medication Review

Systematic examination of all medications for appropriateness. Components: indication, efficacy, safety, adherence, interactions. Tools: Beers Criteria, STOPP/START. Goals: deprescribing, optimizing therapy, reducing adverse effects.

Medication Safety

Practices that reduce medication errors and prevent avoidable harm during prescribing, dispensing, administration, and use.

Medication Side Effects

Unintended harmful responses to medications occurring at standard doses, ranging from mild (nausea, rash, dry mouth) to severe (anaphylaxis, Stevens-Johnson syndrome, QT prolongation, hepatotoxicity).

Medicine

A substance, preparation, or therapeutic practice used to prevent, diagnose, relieve, or treat disease.

Meningococcal Vaccine

Conjugate vaccine (MenACWY) and serogroup B vaccine (MenB) given at 11-12 years with booster at 16 years. Protects against *Neisseria meningitidis*, causing meningitis and septicemia. MenACWY booster at 16 years covers the high-risk period for serogroup C and Y disease.

Mineralocorticoid Receptor Antagonist

A drug that blocks aldosterone receptors, promoting natriuresis while limiting potassium loss and adverse cardiac remodeling. Spironolactone and eplerenone improve outcomes in selected heart-failure patients; hyperkalemia and renal dysfunction require monitoring.

MMR Vaccine

A live attenuated vaccine against measles, mumps, and rubella. It is given according to the immunization schedule and is generally avoided during pregnancy and in people with certain severe immune deficiencies.

Molecular Medicine

Molecular medicine applies molecular and cellular biology to understand, diagnose, prevent, and treat disease. It includes genomics, precision oncology, molecular biomarkers, targeted therapy, and other mechanism-based approaches.

Monoamine Oxidase Inhibitor

A medicine, biologic, or pharmacologic term related to monoamine oxidase inhibitor; its mechanism, indication, interactions, and adverse effects depend on the specific agent.

MRNA Vaccine

A vaccine delivering messenger RNA encoding a pathogen antigen that host cells translate into protein, triggering an immune response. Approved for COVID-19 (Pfizer-BioNTech, Moderna) and RSV; under study for influenza, cancer, and other infections.

Multi-Drug Resistance

Multidrug resistance is resistance of a microorganism, cancer, or other disease process to multiple drugs that would otherwise be active. Mechanisms include target changes, drug inactivation, reduced uptake, efflux, and altered repair or survival pathways.

Multidrug-Resistant Extensively

Drug-Resistant TB

Multidrug-resistant tuberculosis is resistant to key first-line drugs, while extensively drug-resistant TB has additional resistance that greatly limits options. Diagnosis requires molecular or culture testing, and treatment uses prolonged individualized combinations with public-health oversight.

Neuropathy Secondary To Drugs

Drug-induced neuropathy is peripheral nerve damage caused by a medicine or toxin, often producing numbness, burning pain, weakness, or altered reflexes; management may require dose reduction, withdrawal, and treatment of symptoms.

Nonsteroidal Anti-Inflammatory Drug

A medicine that inhibits cyclooxygenase enzymes and reduces prostaglandin production, producing analgesic, anti-inflammatory, and fever-reducing effects. NSAIDs can cause gastrointestinal, kidney, cardiovascular, or other adverse effects.

NSAIDs

Nonsteroidal anti-inflammatory drugs are a class of drugs that inhibit cyclooxygenase enzymes, COX-1 and COX-2, reducing the synthesis of prostaglandins and thromboxanes from arachidonic acid. They produce analgesic, anti-inflammatory, antipyretic, and antiplatelet effects.

Nuclear Medicine

A medical specialty that uses small amounts of radioactive substances (radiotracers) to diagnose, stage, and treat various diseases, including cancer and heart disease.

Nuclear Medicine Documentation

Record keeping for radioactive materials, including inventory, disposal, administered activity, and patient release.

Nuclear Medicine Reporting

Structured interpretation of nuclear medicine studies using standardized terminology and clinically relevant findings.

Nuclear Medicine Resolution

Ability of a PET or SPECT system to distinguish small details, limited by factors such as positron range, non-collinearity, and detector geometry.

Occupational Medicine

The medical specialty concerned with the effects of work on health, prevention of work-related illness and injury, and safe return to work.

Opioid Analgesics

A class of drugs that bind to opioid receptors in the central and peripheral nervous systems, producing analgesia and other effects.

Opioid Antagonist

A medicine that blocks opioid receptors and can reverse opioid effects such as respiratory depression. Naloxone is a commonly used example.

Opioids

A class of drugs that bind to opioid receptors in the central and peripheral nervous systems, producing analgesia and other effects. See Opioid Analgesics.

Oral Hypoglycemic

A class of drugs used to treat type 2 diabetes (with lifestyle).

Oral Medicine

Specialty diagnosing oral mucosal diseases, dental causes of facial pain, oral manifestations of systemic disease.

Oral Polio Vaccine

A live attenuated poliovirus vaccine administered by mouth. Its formulation and use depend on the public-health program and circulating poliovirus strains; it has played an important role in polio control.

Oriental Medicine (Traditional)

A biological or biochemical term involving oriental medicine (traditional). Its role depends on the relevant cells, molecules, and physiological process.

Orphan Drug

A pharmaceutical agent developed specifically to treat a rare medical condition (affecting fewer than 200,000 people in the United States, as defined by the Orphan Drug Act of 1983).

Over-The-Counter (OTC) Drug

A pharmacologic agent associated with over-The-Counter (OTC) Drug; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

Over-The-Counter Drug

A medication that can be purchased without a prescription, deemed safe and effective for self-medication when used according to label instructions. OTC drugs have established safety profiles, low toxicity, and clear dosing guidelines suitable for consumer use.

P2Y₁₂ Inhibitor

An antiplatelet drug that blocks the platelet P2Y₁₂ ADP receptor, reducing activation and aggregation. Clopidogrel, prasugrel, and ticagrelor are used in selected acute coronary syndrome and percutaneous coronary intervention settings, with bleeding and drug-specific contraindications considered.

Palifermin

A pharmacologic agent associated with palifermin; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

Palonosetron

A pharmacologic agent associated with palonosetron; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

Parasympatholytic

A drug that blocks muscarinic (cholinergic) receptors, inhibiting parasympathetic nervous system activity. Agents: atropine (bradycardia, organophosphate poisoning), ipratropium/tiotropium (COPD/asthma—bronchodilation), scopolamine

(motion sickness), oxybutynin/tolterodine (overactive bladder), and glycopyrrolate.

Parasympathomimetic

A drug that mimics or enhances parasympathetic (cholinergic) activity. Direct agonists: muscarinic (pilocarpine—glaucoma, dry mouth) and nicotinic agonists.

Paroxetine

Paroxetine is a selective serotonin-reuptake inhibitor used for depression, anxiety disorders, obsessive-compulsive disorder, PTSD, and some other conditions; sexual dysfunction, withdrawal symptoms, serotonin syndrome, and pregnancy-related risks require consideration.

Pediatric Nuclear Medicine

Special considerations for children: lower doses, sedation often needed, distraction techniques. Dose reduction: weight-based activity, optimized protocols.

Pefloxacin

A pharmacologic agent associated with pefloxacin; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

PEG

A pharmacologic agent associated with pEG; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

Pegargiminase

A pharmacologic agent associated with pegargiminase; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

Pegaspargase

A pharmacologic agent associated with pegaspargase; its mechanism, indications, interactions, contraindications, and adverse effects depend on the

specific product.

Pegfilgrastim

A pharmacologic agent associated with pegfilgrastim; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

Penicillamine

A chelating and immunomodulatory medicine used in selected copper overload and autoimmune disorders.

Pentavalent Vaccine

A WHO-prequalified five-in-one combination vaccine containing Diphtheria toxoid, Tetanus toxoid, whole-cell Pertussis antigen, Hepatitis B (HBsAg), and Haemophilus influenzae type b (Hib) conjugate, administered as a single intramuscular injection.

Pentobarbital

A barbiturate sedative and anticonvulsant with substantial respiratory-depression and overdose risk.

Pergolide

A pharmacologic agent associated with pergolide; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

Perioperative Antibiotic Prophylaxis

Administration of an appropriate antimicrobial within a defined interval before incision to reduce infection risk for selected procedures. Choice, timing, redosing, and discontinuation depend on procedure, organisms, patient factors, and local guidance.

Personalized Medicine

An approach that uses a person's biological, clinical, and sometimes genetic information to guide prevention, diagnosis, or treatment.

Pertussis Vaccine

A vaccine containing selected antigens from Bordetella pertussis, given in combination vaccines to

prevent whooping cough and reduce severe disease and transmission.

Pharmaceutical

Relating to medicines or the preparation and manufacture of medicines.

Pharmacodynamic Interaction

A drug interaction in which one drug changes the effect of another at its site of action without necessarily changing its concentration. Effects may be additive, synergistic, or antagonistic, such as combined respiratory depression or bleeding risk.

Pharmacokinetic Interaction

A drug interaction that changes absorption, distribution, metabolism, or excretion of another drug, thereby altering its concentration over time. Enzyme inhibition, enzyme induction, transporter effects, and altered gastric pH are common mechanisms.

Phosphodiesterase Inhibitor

A class of drugs that inhibit phosphodiesterase enzymes, which break down cyclic nucleotides (cAMP and cGMP), thereby increasing intracellular levels of these second messengers and potentiating their downstream effects.

Physical Medicine And Rehabilitation

A medical procedure, treatment, device, or care topic related to physical medicine and rehabilitation; indications, benefits, risks, and follow-up depend on the specific intervention and patient.

Pneumococcal Conjugate Vaccine

A vaccine that links pneumococcal polysaccharides to a carrier protein to produce protective immunity against selected Streptococcus pneumoniae serotypes. Product choice and schedule depend on age, vaccination history, medical risk, and local recommendations.

Pneumococcal Vaccine

A vaccine that protects against disease caused by

Streptococcus pneumoniae, including invasive infection, pneumonia, meningitis, and otitis media; product and schedule depend on age and risk.

Poliovirus Vaccine

poliovirus vaccine is a virus or viral group classified by its genetic and biologic features. Clinical importance depends on the strain, host, route of transmission, tissue tropism, and ability to cause disease.

Prazosin

An alpha-1 blocker that relaxes blood vessels and smooth muscle.

Pre-Medication

Medicine given before a procedure, treatment, or exposure to prevent a predictable problem such as pain, nausea, allergy, infection, or anxiety.

Precision Medicine

Treatment tailored to individual tumor characteristics. Requires molecular profiling.

Prescription Drug

A medication whose dispensing or use is subject to prescription requirements under the applicable law. It requires professional oversight because of its risks, complexity, or need for monitoring; regulatory approval and prescribing rules vary by country and product.

Procalcitonin-Guided Antibiotic Therapy

The use of procalcitonin levels to guide initiation and discontinuation of antibiotics. Procalcitonin is a biomarker that rises in bacterial infections and falls with effective treatment. Elevated levels support the need for antibiotic therapy, while declining levels support discontinuation.

Prostaglandin Antagonist

A drug that blocks a prostaglandin receptor or pathway, reducing selected prostaglandin-mediated effects such as inflammation, pain, uterine contraction, or gastric secretion.

Protease Inhibitor

A class of antiretroviral drugs that block the viral protease enzyme responsible for cleaving polyprotein precursors into functional proteins during HIV replication, preventing the formation of mature, infectious virions. Used in combination antiretroviral therapy for HIV infection.

Proteasome Inhibitor

proteasome inhibitor is a medicine or pharmacologic agent; its indication, dose, interactions, contraindications, and adverse effects depend on the specific product and patient.

Proton Pump Inhibitor

A class of medications that irreversibly inhibit the gastric H⁺/K⁺ ATPase (proton pump) in parietal cells, the final common pathway of gastric acid secretion, producing profound and sustained suppression of both basal and stimulated acid production.

Psychotropic Drug

psychotropic drug is a medicine or pharmacologic agent; its indication, dose, interactions, contraindications, and adverse effects depend on the specific product and patient.

Psychotropic Medication

A medicine, biologic, or pharmacologic term related to psychotropic medication; its mechanism, indication, interactions, and adverse effects depend on the specific agent.

Pulmonary Medicine

A biomedical term referring to pulmonary medicine; its classification, distinguishing features, and medical significance depend on the clinical or biologic context.

Rabies Vaccine

A vaccine containing inactivated rabies virus used for preexposure protection or postexposure prophylaxis after a possible rabies exposure, together with wound care and immune globulin when indicated.

Rehabilitation Medicine

Physician specialty focusing on function, quality of life for patients with impairment, disability. Also called physiatry.

Alternate Names:
physiatry.

Renal Drug Excretion

The elimination of drugs by the kidney through glomerular filtration, active tubular secretion, and passive tubular reabsorption. Drugs eliminated renally (e.g., aminoglycosides, vancomycin, lithium, metformin) require dose adjustment in renal impairment.

Renin Inhibitor

A class of antihypertensive drugs that directly inhibit renin, the first and rate-limiting enzyme in the renin-angiotensin-aldosterone system. Renin catalyzes the conversion of angiotensinogen to angiotensin I, which is then converted to angiotensin II by ACE. Aliskiren is the only direct renin inhibitor available.

Resistance Antibiotic (Antibiotic Resistance)

Alternate index label for Antibiotic Resistance.

Rock (Drug)

A medicine, biologic, or pharmacologic term related to rock (drug); its mechanism, indication, interactions, and adverse effects depend on the specific agent.

Rohypnol

A brand name for flunitrazepam, a benzodiazepine not approved in some countries and associated with sedation, dependence, misuse, and serious interaction with other central-nervous-system depressants.

Rotavirus Vaccine

An oral live attenuated vaccine that protects infants and young children against severe rotavirus gastroenteritis. The product, schedule, and age limits depend on the immunization program; mild fever

or diarrhea may occur.

Scabicide

A medicine that kills scabies mites.

Sedative

A medication that reduces central nervous system activity, producing calming effects, drowsiness, and sleep at higher doses. Sedatives are used for anxiety, procedural sedation, insomnia, and as adjuncts in anesthesia.

Selective Serotonin Reuptake Inhibitor

A medicine, biologic, or pharmacologic term related to selective serotonin reuptake inhibitor; its mechanism, indication, interactions, and adverse effects depend on the specific agent.

SGLT2 Inhibitor

A drug that reduces sodium-glucose cotransport in the proximal renal tubule, increasing urinary glucose and sodium excretion.

Siddha Medicine

Siddha medicine is a traditional South Asian healing system using dietary, herbal, mineral, and other practices; products may have contamination, toxicity, or interaction risks.

Skeletal Muscle Relaxant

A medication that reduces muscle spasm, spasticity, and associated pain.

Sleep Drug

A medicine, biologic, or pharmacologic term related to sleep drug; its mechanism, indication, interactions, and adverse effects depend on the specific agent.

SMART Goals For Antibiotic Stewardship

Stewardship objectives that are specific, measurable, achievable, relevant, and time-limited. They define a clear antimicrobial-use or patient-outcome target and a way to monitor progress.

Sports Medicine

The field concerned with prevention, diagnosis, treatment, and rehabilitation of activity-related injury and performance problems.

Steroid

A compound with a characteristic four-ring structure, including corticosteroids and sex hormones.

Steroid Abuse

Steroid misuse is non-prescribed or excessive use of corticosteroids or anabolic-androgenic steroids, which can cause endocrine suppression, psychiatric effects, infection, metabolic disease, liver injury, cardiovascular harm, or tissue damage.

Steroid Hormones

Lipid-soluble hormones derived from cholesterol, including cortisol, aldosterone, estrogen, progesterone, and testosterone.

Steroid Injections - Tendon, Bursa, Joint

Injection of a corticosteroid, often with local anesthetic, into or around a joint, bursa, or tendon sheath to reduce inflammation and pain; injections directly into tendons are generally avoided because of rupture risk.

Tamoxifen

A selective estrogen-receptor modulator used mainly to treat or prevent selected breast cancers.

Tamsulosin

Tamsulosin is a selective alpha-1A blocker that relaxes prostate and bladder-neck smooth muscle to improve urinary symptoms of benign prostatic hyperplasia; dizziness, postural hypotension, and ejaculatory dysfunction may occur.

TCA's

Tricyclic antidepressants are a class of drugs that inhibit the reuptake of serotonin and norepinephrine, similar to SNRIs, but also block histamine H1 receptors, muscarinic acetylcholine receptors, and alpha-1 adrenergic receptors, producing numerous side ef-

fects.

Tdap Vaccine

Tetanus, Diphtheria, and acellular Pertussis booster given at 11-12 years and during each pregnancy (27-36 weeks gestation). Replaces one dose of Td. Important for pertussis protection, particularly for protecting newborns through maternal antibody transfer. Also given to adolescents and adults to maintain tetanus and diphtheria immunity.

Tenormin

A brand name for atenolol, a beta-adrenergic blocker used for selected cardiovascular conditions.

Terazosin

A medicine or pharmacologic agent called terazosin. Its uses, contraindications, interactions, and adverse effects depend on the specific agent and patient.

Therapeutic Drug Levels

A medicine or pharmacologic agent called therapeutic drug levels. Its uses, contraindications, interactions, and adverse effects depend on the specific agent and patient.

Therapeutic Drug Monitoring

The clinical practice of measuring specific drug concentrations in blood or plasma to individualize drug dosing and ensure therapeutic effectiveness while minimizing toxicity. TDM is particularly important for drugs with narrow therapeutic indices, where small changes in concentration can lead to therapeutic failure or toxicity.

Thrombolytic

A drug that dissolves blood clots by activating plasminogen to plasmin, which degrades fibrin. Agents: alteplase (tPA), tenecteplase (TNK-tPA), reteplase, streptokinase, urokinase. Indications: acute ischemic stroke (within 4.5 hours), STEMI, massive pulmonary embolism, and acute limb ischemia.

Thrombolytic Drugs For Heart Attack

A pharmacologic agent associated with thrombolytic

Drugs For Heart Attack; its mechanism, indications, interactions, contraindications, and adverse effects depend on the specific product.

Tranquillisers

An older term for medicines that reduce anxiety or agitation, including benzodiazepines and some antipsychotics.

Transfusion Medicine

Blood components: PRBCs (leukoreduced, irradiated for at-risk), FFP (coagulation factors), platelets (apheresis pooled, leukoreduced), cryoprecipitate (fibrinogen, vWF, factor VIII, XIII). ABO/Rh compatibility.

Travel Medicine

Medical care concerned with prevention and treatment of health problems related to travel.

Tricyclic Antidepressant Overdose

Anticholinergic effects, cardiac toxicity (QRS prolongation, arrhythmias), seizures, hypotension. Treatment: sodium bicarbonate (QRS >100 ms), seizure control, supportive care.

Trimox

A brand name for amoxicillin, a penicillin-class antibiotic used for susceptible bacterial infections.

Tumor Necrosis Factor Inhibitor

TNF inhibitor (biologic): monoclonal antibodies/fusion proteins blocking TNF-alpha—infliximab, adalimumab (monoclonals), etanercept (soluble receptor), certolizumab, golimumab. Used in rheumatoid arthritis (after methotrexate failure), ankylosing spondylitis, psoriatic arthritis, psoriasis, IBD (Crohn, UC).

Tylenol

A brand name for acetaminophen, also called paracetamol, used to reduce pain and fever.

Tyrosine Kinase Inhibitor

A class of targeted cancer therapy drugs that in-

hibit tyrosine kinases, enzymes that phosphorylate tyrosine residues on proteins and activate signaling pathways involved in cell growth, proliferation, differentiation, and survival. Tyrosine kinase mutations or overexpression drive many cancers.

Uricosuric Agent

A medicine that increases renal excretion of uric acid, lowering serum urate in selected patients with gout. Adequate kidney function and hydration are important, and urinary uric-acid stones may be a concern.

Urine Drug Screen

Immunoassay-based screening for common drugs of abuse in urine. Detects: amphetamines, benzodiazepines, cocaine (benzoylecgonine), opiates (morphine, codeine, heroin), oxycodone, methadone, fentanyl, barbiturates, PCP, THC (cannabinoids), and alcohol. Pros: rapid (minutes at POC), inexpensive.

Uterotonic

A medication that stimulates uterine contraction, used for induction/augmentation of labor, prevention and treatment of postpartum hemorrhage, and medical termination of pregnancy.

Vaccine

A biological preparation that stimulates an immune response to a pathogen or its components. Vaccines are used mainly to prevent infection or reduce disease severity; a limited number are used in therapeutic settings.

Vaccine Adjuvant

A substance added to some vaccines to enhance and prolong the immune response to the antigen. Examples include aluminum salts and newer adjuvant systems.

Vaccine Development

The research and testing process used to create a vaccine, including antigen selection, laboratory studies, clinical trials, manufacturing, safety monitoring, and regulatory review.

Vaccine Immunology

The study of how vaccines stimulate innate and adaptive immune responses. Vaccine types include live attenuated, inactivated, subunit, conjugate, toxoid, viral-vector, and nucleic-acid vaccines; their immune responses, precautions, and schedules vary.

Vaccine Therapy

vaccine therapy is used in relation to vaccine therapy during assessment, diagnosis, treatment, monitoring, or patient support.

Varicella Vaccine

A live attenuated vaccine that prevents or reduces the severity of chickenpox. The number and timing of doses depend on age, vaccination history, immune status, pregnancy, and local recommendations. It is generally avoided during pregnancy and in people with certain immune deficiencies.

Wellcovorin

A brand name for leucovorin, the calcium folinate form of folinic acid, used in selected chemotherapy protocols and to reduce toxicity from certain folate antagonists.

Xanax

A brand name for alprazolam, a benzodiazepine used for selected anxiety and panic disorders; dependence, sedation, and respiratory depression are important risks.

Xanthine Oxidase Inhibitor

A class of medications that inhibit xanthine oxidase, the enzyme that catalyzes the final steps of purine metabolism: hypoxanthine to xanthine and xanthine to uric acid. By inhibiting this enzyme, XOIs reduce uric acid production and lower serum uric acid levels, used for management of chronic gout and tumor lysis syndrome.

Brand Medicines

Medical Prefix

Prefix	Meaning	Example of use in medical terms
Ab-	A medical prefix meaning away from, off, or separating from,	abduction (movement away from the midline) and ablation (removal or destruction of tissue)
Acou-	A medical combining form referring to hearing or sound,	acoustic nerve, audiology, and acoumeter
Acro-	Acro- is a medical word element meaning extremity, tip, or highest part,	acromegaly and acrocyanosis
Acu-	A combining form meaning sharp, pointed, or related to acuity; in medical terms it may refer to sharp sensation or vision,	acuity, acute, or acupuncture, depending on the word formed
Ad-	A medical prefix meaning toward, to, or near,	adduction (movement toward the midline) and adrenal (near the kidney)
Aden- Adeno-	A combining form meaning gland or glandular tissue, A combining form referring to a gland or glandular tissue,	adenitis and adenoma adenovirus, adenoid, and adenocarcinoma
Aer-, Aero-	A combining form meaning air or gas,	aerobic, aerosol, and aerophagia
Alb- Amyl-	A prefix derived from the Latin albus, meaning white, A combining form referring to starch or a starch-like substance,	albino and albinism amylase and amyloid
Amylo-	A combining form referring to starch-like material or amyloid protein deposits	amyloidosis and amyloidoma
An-	A prefix meaning without, lacking, or not,	anoxia, anesthesia, and anencephaly
Aniso-	A combining form meaning unequal or unequal in size, number, or degree	anisocytosis and anisometropia
Antero-	A combining form meaning anterior or toward the front	anteroposterior and anterolateral
Anti-	A prefix meaning against, opposed to, or preventing a specified process or condition	antibiotic and anticoagulant
Arthro- Bi-	A combining form meaning joint A prefix meaning two, twice, or involving two sides or components	arthritis and arthroscopy bilateral and bicuspid

Prefix	Meaning	Example of use in medical terms
Bio-	A combining form meaning life or living organisms	biochemistry and biopsy
Brachy-	A combining form meaning short or abnormally short	brachycephaly and brachydactyly
Cephal-	A combining form meaning head	cephalic and cephalalgia
Cholangi-	A combining form meaning bile duct or bile vessel	cholangitis and cholangiography
Circum-	A combining form meaning around or surrounding	circumcision and circumoral
Colpo-	A combining form meaning vagina	colposcopy and colpotomy
Cranio-	A combining form meaning skull or cranium	craniotomy and craniosynostosis
Cyto-	A combining form meaning cell	cytology and cytotoxic
Dacryo-	A combining form meaning tear or tear duct	dacryocystitis and dacryostenosis
Dactyl-, -Dactyl	A combining form or suffix meaning finger or toe	polydactyly and brachydactyly
Dextro-	A combining form meaning right, toward the right, or sometimes clockwise	dextrocardia and dextroposition
Di-	A combining form meaning two, twice, or double	diplopia and diplegia
Dia-	A combining form with meanings including through, across, apart, or complete, depending on the word	dialysis and diagnosis
Disto-	A combining form meaning distal, distant, or away from a point of attachment or origin	distal and distobuccal
Dys-	A prefix meaning abnormal, difficult, painful, or impaired,	dysphagia (difficulty swallowing) and dysuria (painful urination)
Entero-	A combining form meaning intestine or bowel,	enteropathy and enteroscopy
Epi-	A prefix meaning on, upon, over, or above,	epidermis and epigastric
Extra-	A clinical condition or finding involving extra-. Its defining features, causes, evaluation, and management depend on the clinical context	extraocular and extracellular
Gyne-	A combining form meaning woman or female,	gynecology
Haem- / Haema- / Haemato- / Haemo-	Prefixes derived from the Greek word haima (blood), used in medical terminology to denote blood-related conditions and procedures	hematology, hematoma, and hemoglobin
(Prefixes Relating To Blood)		
Hemi-	A prefix meaning half or one side,	hemiplegia and hemithorax
Hepat-	A combining form meaning liver,	hepatic and hepatocellular
Hepato-	A combining form meaning liver,	hepatobiliary and hepatomegaly
Hetero-	A prefix meaning different, other, or unequal,	heterozygous and heterogeneous
Histo-	A combining form meaning tissue,	histology and histopathology
Homo-	A prefix meaning same or alike,	homozygous and homologous
Hyper-	A prefix meaning excessive, above normal, or increased,	hyperglycemia and hypertension
Hypo-	A prefix meaning below, deficient, reduced, or less than normal,	hypoglycemia and hypothermia

Prefix	Meaning	Example of use in medical terms
Iatr-	A combining form meaning medical treatment or physician,	iatrogenic and iatrogeny
Infra-	A medical term referring to infra-. Its precise meaning and clinical importance depend on the context in which it is used	infrapatellar and infraorbital
Intra-	Intra- is an anatomical structure, tissue relationship, physiologic process, or body function; its location and mechanism explain its clinical significance	intraocular and intravenous
Iso-	A prefix meaning equal, same, or uniform,	isotonic and isometric
Juxta-	A prefix meaning next to, beside, or adjoining,	juxtaglomerular
Kerato-	A combining form relating to the cornea or to horny or keratinized tissue,	keratitis and keratosis
Leuk-	A combining form meaning white, especially in terms relating to white blood cells,	leukocyte
Leuko-	A combining form meaning white,	leukoderma or leukopenia
Levo-	A prefix meaning left or turning toward the left, often used in chemistry or to describe spatial orientation	levocardia and levodopa
Lieno-	A combining form meaning relating to the spleen	lienorenal and lienectomy
Lipo-	A combining form meaning relating to fat or lipids	lipoma and lipoprotein
Litho-	A combining form meaning stone,	lithotripsy and nephrolithiasis
Macro- (Prefix)	A prefix meaning large, abnormally large, or on a large scale,	macrocytic, meaning involving abnormally large cells
Medical Prefix	A prefix used in medical terminology to modify a word's meaning	hyper- (above or excessive), hypo- (below or deficient), macro- (large), and micro- (small)
Mega-	A combining form meaning large, great, or abnormally enlarged	megacolon and megakaryocyte
Mega- (Prefix)	A prefix derived from Greek meaning large, great, or abnormally enlarged,	megacolon, megacardia, and megaloccephaly
Melan- (Prefix)	A combining form meaning dark or black, derived from the Greek melas; it refers especially to melanin or dark pigmentation,	melanoma and melanocyte
Mes-	A combining form meaning middle	mesencephalon and mesoderm
Meso-	A combining form meaning middle	mesothelioma and mesocolon
Mi-	A combining form meaning small or relating to muscle, depending on the word in which it appears	micturition and miosis
Micro-	A combining form meaning small or millionth, depending on context	microscopy and microcephaly
Myc-	A combining form meaning fungus or pertaining to fungi, derived from the Greek mykes	mycosis and mycology
Myco-	The term Myco- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	mycosis and mycology
Myel-	A combining form relating to bone marrow or the spinal cord, depending on the word	myelitis and myelopathy

Prefix	Meaning	Example of use in medical terms
Myo- (Prefix)	A medical term referring to myo- (prefix). Its clinical meaning depends on the context in which it is used	myopathy and myocarditis
Naso-	The term Naso- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	nasogastric and nasopharyngeal
Neo- (Prefix)	Neo- is a prefix meaning new or recently formed	neoplasm and neonatal
Nephro-	The term Nephro- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	nephritis and nephrectomy
Oligo-	A combining form meaning few, scanty, or deficient	oliguria and oligospermia
Oligo- (Prefix)	Oligo- is a prefix meaning few, scanty, or small in number or amount	oliguria and oligospermia
Onco-	A combining form relating to tumors or cancer	oncology and oncogenesis
Onycho-	A combining form relating to the nail	onychomycosis and onycholysis
Onycho- (Prefix)	Onycho- is a prefix meaning relating to a nail or claw	onychomycosis and onycholysis
Ortho- (Prefix)	Ortho- is a prefix meaning straight, correct, upright, or related to orthopedic care depending on the term	orthopedics and orthostatic
Osteo- (Prefix)	Osteo- is a prefix meaning relating to bone	osteomyelitis and osteoporosis
Ot-	The term Ot- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	otitis and otalgia
Oto-	The term Oto- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	otology and otoscope
Oxy	Oxy- is a prefix relating to oxygen, sharpness, or acidity depending on the medical or chemical term	oxygen and oxyhemoglobin
Pachy- (Prefix)	A word-forming prefix meaning thick or thickness,	pachyonychia, pachyderma, and pachycephaly
Para	Para- is a prefix meaning beside, beyond, abnormal, or related to childbirth depending on the word; its exact meaning must be read in context	parathyroid and paranasal
Para-	A combining form meaning beside, beyond, abnormal, or closely related, depending on the word	paracentesis and paraesthesia
Para- (Prefix)	Para- is a medical prefix whose meaning varies by use, including beside, beyond, abnormal, or the number of births reaching a specified gestational age	parathyroid and parity in obstetric history
Patho-	The term Patho- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	pathology and pathogenesis
Peri-	The term Peri- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	pericardium and periosteum
Phago-	The term Phago- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	phagocytosis and phagocyte
Phlebo- (Prefix)	Phlebo- is a medical prefix meaning relating to a vein,	phlebitis, phlebectomy, or phlebography

Prefix	Meaning	Example of use in medical terms
Pneumo-	The term Pneumo- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	pneumothorax and pneumonia
Poly-	A combining form meaning many or much	polyuria and polyneuropathy
Post	Post- is a prefix meaning after,	post-operative, postpartum, or postprandial
Post-	A combining form meaning after or behind	postnatal and postprandial
Pre-	A combining form meaning before or in front of	prenatal and preoperative
Prefix	A prefix is a word element placed before a root to modify meaning; medical prefixes commonly indicate location, number, time, direction, or abnormality	hyper-, hypo-, peri-, and brady-
Prefixes, Medical	Word beginnings that modify medical terms by indicating location, number, time, speed, size, direction, deficiency, excess, or other meaning,	hyper-, hypo-, peri-, and brady-
Pro-	A combining form meaning before, forward, or in favor of	prolactin and prognosis
Pseudo-	A combining form meaning false or resembling but not truly	pseudocyst and pseudarthrosis
Retro-	A combining form meaning backward, behind, or posterior	retroperitoneal and retrosternal
Salpingo-	A combining form relating to a fallopian tube	salpingitis and salpingectomy
Sarco-	A combining form meaning flesh or soft tissue	sarcoma and sarcoidosis
Schizo-	A combining form relating to splitting or schizophrenia	schizophrenia and schizoid
Scler-	The term Scler- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	sclerosis and scleritis
Sclero-	The term Sclero- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	scleroderma and sclerotherapy
Sub-	A combining form meaning under, below, or less than normal	subcutaneous and subdural
Super-	A combining form meaning above, excessive, or superior	superinfection and supernumerary
Supra-	A combining form meaning above or beyond	suprarenal and suprapubic
Syn-	A combining form meaning together, with, or joined	syndactyly and synovitis
Teno-	A combining form relating to a tendon	tenosynovitis and tenodesis
Thermo-	A combining form relating to heat or temperature	thermoregulation and thermotherapy
Toc- (Prefix)	Toc- is a prefix used in some medical terms and often relates to childbirth or labor,	tocolysis; the complete word determines its meaning
Toco-	The term Toco- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	tocography and tocolysis
Toko- (Prefix)	Toko- is a prefix relating to childbirth or labor, derived from Greek terms for birth	tocolysis and tokodynamometry
Tom-	The term Tom- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	tomography and tomodensitometry

Prefix	Meaning	Example of use in medical terms
Tomo-	The term Tomo- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	tomography and tomographic
Trans- (Prefix)	The term Trans- (prefix) is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	transdermal and transcatheter
Trich- (Prefix)	The term Trich- (prefix) is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	trichiasis and trichology
Tricho-	A combining form relating to hair	trichoscopy and trichotillomania
Tricho- (Prefix)	The term Tricho- (prefix) is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	trichorrhexis and trichomycosis
Tympano-	The term Tympano- is a medical word element or prefix whose precise meaning depends on the word or clinical context in which it is used	tympanoplasty and tympanometry
Xanth- (Prefix)	Xanth- is a medical word element meaning yellow or yellowish,	xanthoma, xanthelasma, and xanthopsia
Xen- (Prefix)	A medical word element meaning foreign, strange, or originating outside the usual organism or system,	xenograft (a tissue transplanted between different species)
Xeno- (Prefix)	Xeno- is a medical word element meaning foreign, strange, or from another species,	xenograft and xenotransplantation
Xero-	Xero- is a medical word element meaning dry or dryness,	xeroderma, xerophthalmia, and xerostomia

Complete reference tables

These categorized tables reproduce the prefix reference structure from the Pressbooks resource cited below. Examples are retained as learning aids; the meaning of a word part can vary by term.

Prefix	Meaning	Example of use in medical terms
Common prefixes		
A-, An-	Without; lacking	Anemia
Andr/o-	Male	Androgen
Anti-	Against	Anticholinergic drugs
Auto-	Self	Autocrine
Bio-	Life	Biology
Chem/o-	Chemistry	Chemotherapy
Contra-	Against	Contraception
Cyt/o-	Cell	Cytokine
Dis-	Separation; taking apart	Dissection
Dys-	Difficult; abnormal	Dyspnea
Eu-	Good; well	Eupnea
Fibr/o-	Fiber	Fibrosis
Gluco-, Glyco-	Glucose; sugar	Glycogen

Prefix	Meaning	Example of use in medical terms
Gyn/o-, Gynec-	Female	Gynecology
Hydr/o-	Water	Hydrocephalus
Idio-	Self; one's own	Idiopathic
Lyso-, Lys-	Break down; destruction; dissolving	Lysosome
Mal-	Bad; abnormal	Malignant
Myc/o-	Fungus	Mycetoma
Necr/o-	Death	Necrosis
Neo-	New	Neonate
Oxy-	Sharp; acute; oxygen	Oxytocin
Pan-, Pant/o-	All or everywhere	Pancytopenia
Pharmaco-	Drug; medicine	Pharmacist
Re-	Again; backward	Rejuvenation
Somat/o-, Somatico-	Body; bodily	Somatic cell
Body-part prefixes		
Acous/o-	Hearing	Acoustic meatus
Aden/o-	Gland	Adenoid
Adip/o-	Fat	Adipocyte
Adren/o-	Gland	Adrenal cortex
Angi/o-	Blood vessel	Angioplasty
Arteri/o-	Artery	Arteriole
Arthr/o-	Joint	Arthroplasty
Bucc/o-	Cheek	Buccal cavity
Bronch/i-	Bronchus	Bronchioles
Burs/o-	Bursa	Bursa
Carcin/o-	Cancer	Basal cell carcinoma
Cardi/o-	Heart	Cardiology
Cephal/o-	Head	Cephalic flexure
Chol-	Bile	Cholesterol
Chondri-	Cartilage	Chondrosarcoma
Coron-	Heart	Coronary arteries
Cost-	Rib	Costal cartilage
Crani/o-	Brain	Cranium
Cutane-	Skin	Cutaneous
Cyst/o-, Cysti-	Bladder or sac	Cystoscopy
Derm-, Dermat/o-	Skin	Dermatologist
Duoden/o-	Duodenum	Duodenitis
Gastr-	Stomach	Gastrectomy
Gloss-	Tongue	Glossectomy
Hem-, Hema-, Hemat-, Hemo-, Hemat/o-	Blood	Hematopoiesis
Hepat/o-, Hepatico-	Liver	Hepatic portal system
Hist/o-, Histio-	Tissue	Histology
Hyster/o-	Uterus	Hysterectomy
Ileo-	Ileum	Ileostomy
Ischi/o-	Ischium	Ischial tuberosity
Kerat/o-	Cornea (eye or skin)	Keratin
Lacrim/o-	Tear	Lacrimal fluid
Lact/o-, Lacti-	Milk	Lactose
Laryng/o-	Larynx	Laryngitis
Lingu/o-	Tongue	Lingual tonsil

Prefix	Meaning	Example of use in medical terms
Lip/o-	Fat	Lipolysis
Lymph/o-	Lymph	Lymphocyte
Mamm-, Mast/o-	Breast	Mammary glands
Mening/o-	Meninges	Meningitis
Muscul/o-	Muscle	Musculoskeletal
My/o-	Muscle	Myocardium
Myel/o-	Spinal cord or bone marrow	Myelin
Nephr/o-	Kidney	Nephron
Neur/i-, Neur/o-	Nerve	Neuron
Oculo-	Eye	Oculomotor nerve
Onco-	Tumor; bulk; volume	Oncogene
Onych/o-	Fingernail; toenail	Onychodystrophy
Oo-	Egg; ovary	Oocyte
Oophor/o-	Ovary	Oophorectomy
Op-, Opt-	Vision	Optic nerve
Ophthalm/o-	Eye	Ophthalmic artery
Orchid/o-, Orchio-	Testis	Orchidectomy
Orth/o-	Straight; normal; correct	Orthostatic
Osseo-	Bony	Osseous tissue
Ossi-	Bone	Ossicles
Ost-, Oste/o-	Bone	Osteoporosis
Ot/o-	Ear	Otolaryngologist
Ovar/i-, Ovario-, Ovi-, Ovo-	Ovary	Ovarian follicle
Phalang-	Phalanx	Phalanges
Pharyng/o-	Pharynx; throat	Pharyngeal tonsil
Phleb/o-	Vein	Phlebotomist
Phren/i-, Phreno-, Phrenico-	Diaphragm	Phrenic nerve
Pleur-, Pleur/a-, Pleur/o-	Rib; pleura	Pleural cavity
Pneum/a-, Pneumat/o-	Air; lung	Pneumonia
Proct/o-	Anus; rectum	Proctoscopy
Prostat-	Prostate	Prostatectomy
Pseudo-	False	Pseudostratified
Psych/o-, Psyche-	Mind	Psychiatrist
Radio-	Radiation; radius	Radioisotopes
Ren/o-	Kidney	Renal cortex
Retin-	Retina	Retinitis pigmentosa
Rhin/o-	Nose	Rhinoscope
Salping/o-	Tube	Salpingo-oophorectomy
Sarco-	Muscular; flesh-like	Sarcomere
Schiz/o-	Split; cleft	Schizophrenia
Sclera-, Sclero-	Hardness	Sclerosis
Sigmoid/o-	Sigmoid colon	Sigmoidoscopy
Sperma-, Sperm-, Spermato-	Sperm	Spermatocyte
Splen/o-	Spleen	Splenomegaly
Sten/o-	Narrowed; blocked	Stenosis
Stern-	Sternum	Sternoclavicular joint

Prefix	Meaning	Example of use in medical terms
Stom/a-, Stomat/o-	Mouth	Stomatitis
Thorac/o-, Thoracico-	Chest	Thoracic cavity
Thromb/o-	Blood clot	Thrombolytic
Thyr/o-	Thyroid gland	Thyroiditis
Trache/o-	Trachea	Trachealis
Tympan/o-	Eardrum	Tympanic membrane
Ur/o-	Urine	Urologist
Vagin-	Vagina	Vaginal
Varic/o-	Duct; blood vessel	Varicose veins
Vasculo-	Blood vessel	Vasculitis
Ven/o-	Vein	Venae cavae
Vertbr-	Vertebra; spine	Vertebral column
Color prefixes		
Chlor/o-	Green	Chlorophyll
Chrom-, Chromato-	Color	Chromosome
Cyano-	Blue	Cyanosis
Erythr/o-	Red	Erythrocyte
Leuk/o-	White	Leukocyte
Melan/o-	Black	Melanin
Physical property and shape prefixes		
Cry/o-	Cold	Cryotherapy
Elect-	Electrical activity	Electrocardiogram
Kin/o-, Kine-, Kinesi/o-	Movement	Kinetic energy
Kyphy/o-	Humped	Kyphosis
Rhabd/o-	Rod-shaped; striated	Rhabdomyosarcoma
Phot/o-	Light	Photoreceptor
Reticul/o-	Net	Reticulocytes
Scoli/o-	Twisted	Scoliosis
Therm/o-	Heat	Thermotherapy
Direction and position prefixes		
Ab-, Abs-	Away from	Abductor
Ad-	Towards	Adductor
Ante-	Before; forward	Antenatal
Circum-	Around	Circumcision
Cycl-	Circle; cycle	Cyclic neutropenia
De-	Away from; ending	Dehydration
Dia-	Across; through	Diagnosis
Ect/o-, Exo-	Outer; outside	Exocrine gland
End/o-, Ent-, Enter/o-	Within; inner	Endocrine gland
Epi-	Upon; outside of	Epidermis
Ex-, Extra-	Beyond	Expiration
Infra-	Beneath; below	Infratemporal fossa
Inter-	Between	Interstitial fluid
Intra-	Within	Intracellular fluid
Meso-	Middle	Mesoderm
Meta-	Beyond; change	Metabolism
Para-	Alongside; abnormal	Parathyroid glands

Prefix	Meaning	Example of use in medical terms
Path/o-	Disease	Pathologist
Peri-	Around	Pericardium
Post-	Behind; after	Postpartum
Pre-	Before; in front	Precancerous
Retro-	Backward; behind	Retroperitoneum
Sub-	Under	Subcutaneous layer
Super-	Above	Superior
Supra-	Above; upon	Supraglottis
Sy-, Syl-, Sym-, Syn-, Sys-	Together	Syndrome
Trans-	Across; through	Transdermal
Quantity prefixes		
Bi-	Two	Biceps
Brady-	Slow	Bradycardia
Diplo-	Double	Diploid
Hemi-	Half	Hemihypertrophy
Hetero-	Other; different	Heterogeneous
Homo-	Same	Homozygous genotype
Hyper-	Above; beyond; excessive	Hypertension
Hypo-	Under; deficient	Hypotension
Iso-	Equal; like	Isointense
Macro-	Large; long; big	Macrophage
Mic-, Micro-	Small	Microglia
Mon-, Mono-	One	Monocyte
Olig/o-	Few; little	Oliguria
Poly-	Many; excessive	Polyuria
Quadri-	Four	Quadriceps
Semi-	Half	Semilunar valves
Tachy-	Fast	Tachycardia
Tetra-	Four	Tetralogy of Fallot
Tri-	Three	Triceps
Uni-	One	Unicellular

Source and license: Andrea Nelson and Katherine Greene, *Medical Terminology for Healthcare Professions*, “Prefixes and Suffixes,” University of West Florida Libraries, CC BY 4.0. <https://pressbooks.uwf.edu/medicalterminology/chapter/prefixes-and-suffixes/>

Medical Suffixes

Suffix	Meaning	Example of use in medical terms
-Stomy	A suffix meaning surgical creation of an opening,	colostomy or gastrostomy
-Tomy	A suffix meaning incision or cutting into	laparotomy or tracheotomy
Cidal	A suffix meaning killing or destroying,	bactericidal or fungicidal
Cide	A suffix meaning an agent or process that kills,	pesticide, bactericide, or homicide
Cyte	A suffix meaning cell,	erythrocyte or leukocyte
Cytosis	A suffix or term indicating a condition involving cells or, in some contexts, an increase in the number of a particular type of cell	leukocytosis or erythrocytosis
Dactyl-, -Dactyl	A combining form or suffix meaning finger or toe	polydactyly or brachydactyly
Ectomy	A suffix meaning surgical removal or excision,	appendectomy or mastectomy
Emia	A suffix meaning a condition of the blood or a substance present in the blood,	anemia or bacteremia
Iatrics	A suffix or word element referring to medical treatment or a medical specialty,	geriatrics and pediatrics
Iatry	A suffix meaning medical treatment or a medical specialty,	psychiatry and podiatry
Itis	A suffix meaning inflammation,	dermatitis, bronchitis, and arthritis
Kalemia	A suffix or term referring to the concentration of potassium in the blood,	hypokalemia or hyperkalemia
MAB	Abbreviation for monoclonal antibody; the suffix -mab in a generic drug name indicates a monoclonal antibody,	adalimumab or rituximab
Oma	-oma is a suffix commonly indicating a tumor or mass, although some words ending in -oma describe non-neoplastic conditions	lipoma or carcinoma
Pathy	A suffix meaning disease, disorder, or abnormality,	neuropathy, cardiomyopathy, and nephropathy
Plegia	-plegia is a suffix meaning paralysis,	paraplegia, hemiplegia, or quadriplegia

Complete reference tables

Suffix	Meaning	Example of use in medical terms
Common suffixes		
-ac	Pertaining to	Cardiac
-blast, -blasto, -blastic	Bud; germ	Myeloblast
-cyte, -cytic	Cell	Thrombocyte
-dynia	Pain; swelling	Thoracodynia
-eal, -ial	Pertaining to	Esophageal
-ectasis	Expansion; dilation	Atelectasis
-emia	Blood condition	Anemia
-ia	Condition	Hemophilia
-iasis	Condition; formation of	Psoriasis
-ism	Condition	Hypothyroidism
-ites, -itis	Inflammation	Arthritis
-ity	Pertaining to	Immunity
-ium	Structure or tissue	Epithelium
-lysis, -lytic	Breakdown; destruction; dissolving	Osteolytic
-malacia	Softening	Osteomalacia
-megaly	Enlargement	Acromegaly
-oid	Resembling	Arachnoid trabeculae
-oma	Tumor	Angiosarcoma
-osis	Condition; usually abnormal	Endometriosis
-ous	Pertaining to	Aqueous
-pathy	Disease	Lymphadenopathy
-penia	Deficiency; lack of	Thrombocytopenia
-phagia, -phagy	Eating; swallowing	Dysphagia
-phasia	Speech	Aphasia
-plasia, -plastic	Growth	Hyperplasia
-plegia	Paralysis	Hemiplegia
-pnea	Breathing	Sleep apnea
-poiesis	Production	Hemopoiesis
-ptosis	Falling; drooping	Apoptosis
-rrhage, -rrhagic	Bleeding	Hemorrhage
-rrhea	Flow or discharge	Diarrhea
-sclerosis	Hardening	Arteriosclerosis
-sis	Condition	Agranulocytosis
-stasis	Level; unchanging	Homeostasis
-trophy	Growth	Hypertrophy
-uria	In the urine	Anuria
Procedure suffixes		
-centesis	Surgical puncture to remove fluid	Thoracentesis
-desis	Surgical binding	Pleurodesis
-ectomy	Cut out; removal	Mastectomy
-gram	Record; picture	Electrocardiogram
-graph	Instrument used to create a record or picture	Electrocardiograph
-graphy	To record or take a picture	Echocardiography
-meter	Device used for measuring	Sphygmomanometer
-opsy	Visual examination	Biopsy

Suffix	Meaning	Example of use in medical terms
-ostomy	Opening	Colostomy
-otomy	Incision	Laparotomy
-pexy	Surgical fixation	Oophoropexy
-plasty	Surgical reconstruction	Vertebroplasty
-scope	For examining	Endoscope
-scopy	Examine	Endoscopy

Source and license: Andrea Nelson and Katherine Greene, *Medical Terminology for Healthcare Professions*, “Prefixes and Suffixes,” University of West Florida Libraries, CC BY 4.0. <https://pressbooks.uwf.edu/medicalterminology/chapter/prefixes-and-suffixes/>

Alternate Names

Primary term	Alternate names
ACE	angiotensin-converting enzyme
ACE Inhibitor	angiotensin-converting enzyme inhibitor
Acquired Immunodeficiency Syndrome	AIDS
Adenosine Triphosphate	ATP
AIDS	acquired immunodeficiency syndrome
Alcoholism	alcohol dependence: A chronic
Alimentary Canal	gastrointestinal
AMD	age-related macular degeneration
Angiotensin-Converting Enzyme	ACE
Angiotensin-Converting Enzyme Inhibitor	ACE
Amyloid Precursor Protein	APP
ARB	angiotensin II receptor blockers
ATP	adenosine triphosphate
AV Node	atrioventricular node
Balloon Dilation	Balloon dilatation
BMD	bone mineral density
BMI	body-mass index
BPH	benign prostatic hyperplasia
CABG	coronary artery bypass graft
CBC	complete blood count
	mdash
CCU	coronary care unit
Cerebral Hemorrhage	hemorrhagic stroke
CFS	chronic fatigue syndrome
CK	creatine kinase
COX-2 Inhibitors	cyclooxygenase-2 inhibitors
CPAP	continuous positive airway pressure
CPR	cardiopulmonary resuscitation
CRP	C-reactive protein
DNA	deoxyribonucleic acid
DNR	do-not-resuscitate order
DRI	dietary reference intake
EBCT	electron-beam computed tomography
ECG	electrocardiogram
ED	erectile dysfunction

Primary term	Alternate names
EEG	electroencephalogram
EKG	electrocardiogram
EMG	electromyography
	mdash
FSH	follicle-stimulating hormone
GERD	gastroesophageal reflux disease
GFR	glomerular filtration rate
Gingiva	gums: a form of oral tissue that covers the roots of teeth and surrounding bone
GnRH	gonadotropin-releasing hormone
HbA1C	hemoglobin A1c
HDL	high-density lipoprotein
HEPA Filter	high-efficiency particulate air
HIV	human immunodeficiency virus
HLA	human leukocyte antigen
HPA	hypothalamic-pituitary-adrenal axis
Human Immunodeficiency Virus	HIV
IBD	inflammatory bowel disease
ICD	implantable cardioverter defibrillator
IgE	immunoglobulin E
IU	international unit
LDL	low-density lipoprotein
LH	luteinizing hormone
LVAD	left-ventricular assist device
MAOIs	monoamine oxidase inhibitors
METs	metabolic equivalents
MHC Molecule	major histocompatibility complex molecule
Mitral Valve Prolapse	Barlow's Syndrome; Barlow Syndrome (Mitral Valve Prolapse)
MRI	magnetic resonance imaging—a scan that creates detailed pictures of internal organs
MUSE	medicated urethral system for erection
NK	natural killer cells
NMDA Receptor	N-methyl-D-aspartame receptor
NO	nitric oxide
NSAID	nonsteroidal anti-inflammatory drug
PAD	peripheral artery disease
PCOS	polycystic ovary syndrome
PDE5 Inhibitors	phosphodiesterase type 5 inhibitors
PET Scan	positron emission tomography
PSA	prostate-specific antigen
PTCA	percutaneous transluminal coronary angioplasty
PTSD	post-traumatic stress disorder
RAST	radioallergosorbent test
RCT	randomized controlled trial
RDA	recommended dietary allowance
REM	rapid eye movement
RLS	restless legs syndrome
SAD	seasonal affective disorder
SERMs	selective estrogen receptor modulators
SSRIs	selective serotonin reuptake inhibitors

Primary term	Alternate names
Steroids	corticosteroids
T Cell	mdash T lymphocyte
TDD	telecommunications device for the deaf
TENS	mdash transcutaneous electrical nerve stimulation
TIA	mdash transient ischemic attack
tPA	tissue plasminogen activator
TSH	thyroid-stimulating hormone
VLDL	very-low-density lipoprotein
VSD	ventricular septal defect
ABO Blood Group System	mdash ABO Blood Group; ABO Blood Groups; ABO Blood Types; ABO System
Absolute Neutrophil Count	ANC
Aarskog-Scott syndrome	Aarskog Syndrome
Banti Syndrome	Banti disease; Banti's syndrome
Barber itch	Barber's itch
Glycated Hemoglobin (A1C)	A1C; A1C test; Hemoglobin A1C; HbA1C

Common Medical Abbreviations

Abbreviation	Full form
ABG	Arterial blood gas
ACE	Angiotensin-converting enzyme
ACE inhibitor	Angiotensin-converting enzyme inhibitor
ACL	Anterior cruciate ligament
ACS	Acute coronary syndrome
ADHD	Attention-deficit/hyperactivity disorder
ADL	Activity of daily living
AED	Automated external defibrillator
AFib	Atrial fibrillation
AIDS	Acquired immunodeficiency syndrome
AKI	Acute kidney injury
ALP	Alkaline phosphatase
ALT	Alanine aminotransferase
AMI	Acute myocardial infarction
ALS	Amyotrophic lateral sclerosis; advanced life support, depending on context
ANC	Absolute neutrophil count
ANA	Antinuclear antibody
ANP	Atrial natriuretic peptide
APGAR	Appearance, pulse, grimace, activity, and respiration
ARDS	Acute respiratory distress syndrome
AST	Aspartate aminotransferase
ATN	Acute tubular necrosis
AV	Atrioventricular
AVM	Arteriovenous malformation
AVN	Atrioventricular node
BID	Twice a day; prescription shorthand that should be avoided when possible
BP	Blood pressure
BMI	Body mass index
BMR	Basal metabolic rate
BPH	Benign prostatic hyperplasia
BRCA	Breast cancer susceptibility gene; commonly referring to the BRCA1 or BRCA2 gene
BUN	Blood urea nitrogen
CABG	Coronary artery bypass graft
CA	Cancer antigen or calcium, depending on context
CA-125	Cancer antigen 125

Abbreviation	Full form
CAD	Coronary artery disease
CAT	Computed tomography; an older term for computerized axial tomography
CBC	Complete blood count
CBT	Cognitive behavioural therapy
CDC	Centers for Disease Control and Prevention
CEA	Carcinoembryonic antigen
CHF	Congestive heart failure
CHD	Congenital heart disease
CIN	Cervical intraepithelial neoplasia
CJD	Creutzfeldt–Jakob disease
CK	Creatine kinase
CMV	Cytomegalovirus
CNS	Central nervous system
COPD	Chronic obstructive pulmonary disease
CPAP	Continuous positive airway pressure
CPK	Creatine phosphokinase; an older name for creatine kinase
CPR	Cardiopulmonary resuscitation
CRF	Chronic renal failure; chronic kidney disease is preferred in current usage
CRP	C-reactive protein
CSF	Cerebrospinal fluid
CT	Computed tomography
CVS	Chorionic villus sampling
CVA	Cerebrovascular accident
CXR	Chest radiograph
D&C	Dilatation and curettage
DKA	Diabetic ketoacidosis
DJD	Degenerative joint disease
DM	Diabetes mellitus
DNA	Deoxyribonucleic acid
DTP	Diphtheria, tetanus, and pertussis
DVT	Deep-vein thrombosis
DX	Diagnosis
ECG	Electrocardiogram
EKG	Electrocardiogram
ECHO	Echocardiogram
EEG	Electroencephalogram
eGFR	Estimated glomerular filtration rate
EMG	Electromyography
ENT	Ear, nose, and throat
ERCP	Endoscopic retrograde cholangiopancreatography
ESR	Erythrocyte sedimentation rate
ESRD	End-stage renal disease
FBC	Full blood count
FEV1	Forced expiratory volume in the first second
FSH	Follicle-stimulating hormone
GABHS	Group A beta-haemolytic streptococcus
GERD	Gastroesophageal reflux disease
GFR	Glomerular filtration rate
GI	Gastrointestinal
GIST	Gastrointestinal stromal tumour
GU	Genitourinary
HbA1c	Glycated haemoglobin

Abbreviation	Full form
HDL	High-density lipoprotein
HAV	Hepatitis A virus
HBV	Hepatitis B virus
HCT	Hematocrit
HCV	Hepatitis C virus
HGB	Hemoglobin
HIV	Human immunodeficiency virus
HLA	Human leukocyte antigen
HPV	Human papillomavirus
HRT	Hormone replacement therapy
HTN	Hypertension
IBD	Inflammatory bowel disease
IBS	Irritable bowel syndrome
ICD	Implantable cardioverter-defibrillator
ICU	Intensive care unit
IDDM	Insulin-dependent diabetes mellitus; type 1 diabetes mellitus is preferred in current usage
IM	Intramuscular
IU	International unit
IUD	Intrauterine device
IV	Intravenous
IVP	Intravenous pyelography or intravenous pyelogram
LDL	Low-density lipoprotein
LH	Luteinizing hormone
LFT	Liver function test
MI	Myocardial infarction
MMR	Measles, mumps, and rubella
MRI	Magnetic resonance imaging
MRSA	Methicillin-resistant <i>Staphylococcus aureus</i>
MS	Multiple sclerosis; may have other meanings depending on context
NG	Nasogastric
NIDDM	Non-insulin-dependent diabetes mellitus; type 2 diabetes mellitus is preferred in current usage
NKDA	No known drug allergies
NSAID	Nonsteroidal anti-inflammatory drug
OCD	Obsessive-compulsive disorder
PAD	Peripheral arterial disease
PAP	Papanicolaou
PCOS	Polycystic ovary syndrome
PCR	Polymerase chain reaction
PE	Pulmonary embolism
PET	Positron emission tomography
PFT	Pulmonary function test
PID	Pelvic inflammatory disease
PMS	Premenstrual syndrome
PPD	Purified protein derivative
PRN	As needed (from the Latin <i>pro re nata</i>)
PSA	Prostate-specific antigen
PT	Prothrombin time
PTCA	Percutaneous transluminal coronary angioplasty
PTH	Parathyroid hormone
PTT	Partial thromboplastin time

Abbreviation	Full form
PUD	Peptic ulcer disease
PVC	Premature ventricular contraction
QID	Four times a day; prescription shorthand that should be avoided when possible
RA	Rheumatoid arthritis; may have other meanings depending on context
RAST	Radioallergosorbent test
RBC	Red blood cell
RCT	Randomized controlled trial
REM	Rapid eye movement
RLS	Restless legs syndrome
RNA	Ribonucleic acid
RSV	Respiratory syncytial virus
Rx	Prescription or treatment, depending on context
SaO ₂	Arterial oxygen saturation
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
SIDS	Sudden infant death syndrome
SLE	Systemic lupus erythematosus
SOB	Shortness of breath
SSRI	Selective serotonin reuptake inhibitor
STD	Sexually transmitted disease; sexually transmitted infection is preferred in current usage
STI	Sexually transmitted infection
T ₃	Triiodothyronine
T ₄	Thyroxine; commonly referring to tetraiodothyronine
TAH	Total abdominal hysterectomy
TB	Tuberculosis
TIA	Transient ischemic attack
TIBC	Total iron-binding capacity
TID	Three times a day; prescription shorthand that should be avoided when possible
TMJ	Temporomandibular joint
TORCH	Toxoplasmosis, other infections, rubella, cytomegalovirus, and herpes simplex; a group of congenital-infection evaluations
TSH	Thyroid-stimulating hormone
TURP	Transurethral resection of the prostate
URI	Upper respiratory infection
UTI	Urinary tract infection
VSD	Ventricular septal defect
WBC	White blood cell
WDWN	Well-developed, well-nourished
WNL	Within normal limits; a finding is within the expected reference range or clinical appearance
WHO	World Health Organization
XRT	Radiotherapy

A.C. A prescription abbreviation meaning “before meals,” from the Latin *ante cibum*.

ALA An abbreviation for alpha-linolenic acid, an essential omega-3 fatty acid, or aminolevulinic acid, a heme-pathway compound used topically in some photodynamic treatments; context determines the meaning.

AP An abbreviation commonly meaning anteroposterior, or directed from front to back; in some contexts it may have another medical expansion.

ATM An abbreviation that may mean the temporomandibular joint or ataxia-telangiectasia mutated, a protein kinase involved in DNA-damage repair; context determines the meaning.

-
- BF** Physician's shorthand for Black female; demographic abbreviations should be used cautiously and replaced with precise, patient-preferred language when possible.
- BPD** An abbreviation commonly meaning bronchopulmonary dysplasia, chronic lung disease after premature birth; it can also mean borderline personality disorder in psychiatric contexts.
- CA** A medical abbreviation that may mean cancer, carcinoma, or calcium; the intended meaning depends on the surrounding clinical context.
- CKD** Abbreviation for Chronic Kidney Disease: a progressive loss of kidney function over months to years, defined by a glomerular filtration rate (GFR) less than 60 mL/min/1.73m² or markers of kidney damage (albuminuria, haematuria, structural abnormalities) persisting for at least 3 months.
- CLA** An abbreviation that commonly means conjugated linoleic acid, a group of fatty-acid isomers; in other medical contexts it may have a different expansion.
- CTG** Abbreviation for cardiotocography; continuous electronic monitoring of fetal heart rate and uterine contractions during labour.
- CVA** An abbreviation for cerebrovascular accident, meaning a stroke caused by interruption of blood flow to the brain or bleeding within the brain. It may cause sudden neurologic deficits.
- CVS** An abbreviation that may mean cardiovascular system or chorionic-villus sampling, a prenatal diagnostic procedure; context determines the meaning.
- DDAVP** The abbreviation for desmopressin, a synthetic vasopressin-like medicine used for some bleeding disorders, central diabetes insipidus, and nighttime urination.
- DES** An abbreviation that may mean diethylstilbestrol, a synthetic estrogen, or drug-eluting stent; diethylstilbestrol exposure during pregnancy is associated with later reproductive and cancer risks.
- DHHS** The abbreviation for the U.S. Department of Health and Human Services.
- DNR** An abbreviation for do-not-resuscitate, indicating that cardiopulmonary resuscitation should not be attempted if breathing or heartbeat stops.
- DNS** An abbreviation that may mean deviated nasal septum or dextrose in normal saline; the intended expansion depends on the clinical context.
- DPT** An older abbreviation for the combined diphtheria, pertussis, and tetanus vaccine; current formulations include DTaP and Tdap.
- DT** An abbreviation that may mean delirium tremens or diphtheria–tetanus vaccine; the intended meaning depends on context.
- DTC** An abbreviation that may mean differentiated thyroid carcinoma or direct-to-consumer, such as direct-to-consumer genetic testing; the intended meaning depends on context.
- DTP** An older abbreviation for diphtheria, tetanus, and pertussis vaccine; current childhood formulations are generally called DTaP.
- DV** An abbreviation that may mean domestic violence, ductus venosus, or another context-specific term; the full expression should be specified.
- DYS** An abbreviation or word root related to dysplasia, dystonia, dyslexia, or another condition beginning with dys-; the intended meaning depends on context.
- EC** An abbreviation that may mean enteric-coated, emergency contraception, or endothelial cell; the intended meaning depends on context.
- EFT** An abbreviation whose medical meaning depends on context, such as endocrine-function testing or emotional freedom technique; the intended expansion should be specified.
- ENT** Abbreviation for ear, nose, and throat; it commonly refers to the otolaryngology specialty or an otolaryngologist.
-

EOS Commonly an abbreviation for eosinophils or an eosinophil count in laboratory reports; the intended meaning should be confirmed from the report's context.

EPA Abbreviation for eicosapentaenoic acid, an omega-3 polyunsaturated fatty acid found in oily fish and fish-oil supplements.

EPR Abbreviation for electrophrenic respiration, also called diaphragm pacing, in which electrical stimulation helps activate the diaphragm in selected patients with respiratory muscle paralysis.

EPT Abbreviation for estrogen/progestin therapy, treatment using both an estrogen and a progestin, commonly discussed in menopausal hormone therapy when the uterus is present.

ER Abbreviation for emergency room or emergency department, the hospital service that evaluates and treats urgent illness and injury.

FCC In medicine, an abbreviation for familial colorectal cancer; in other contexts it may have nonmedical meanings.

FFI Abbreviation for fatal familial insomnia, a rare inherited prion disease causing progressively severe insomnia, autonomic dysfunction, and neurologic decline.

FISH Abbreviation for fluorescence in situ hybridization, a test that uses fluorescent DNA probes to detect specific genes, chromosomes, or chromosomal rearrangements in cells.

FLAP Abbreviation for 5-lipoxygenase-activating protein, a protein encoded by *ALOX5AP* that helps produce leukotrienes involved in inflammation.

GALT Abbreviation for gut-associated lymphoid tissue, immune tissue in the gastrointestinal tract that samples intestinal contents and helps defend against pathogens.

GBV-C Abbreviation for GB virus C, also called human pegivirus, a blood-borne virus that can persist in people but is not established as a cause of chronic hepatitis.

GD In the RxList medical-dictionary context, abbreviation for Gaucher disease, an inherited lysosomal storage disorder caused by deficient glucocerebrosidase.

GI Abbreviation for gastrointestinal, relating to the digestive tract from the mouth through the anus and its associated organs.

GPO Abbreviation for group purchasing organization, an organization that combines buyers' purchasing power to negotiate prices and contracts for medicines, supplies, or services.

HAART Abbreviation for highly active antiretroviral therapy, an older term for combination antiretroviral treatment used to suppress HIV replication and preserve immune function.

HAPE Abbreviation for high-altitude pulmonary edema, a life-threatening accumulation of fluid in the lungs after rapid ascent; immediate descent and supplemental oxygen are urgent treatments.

HAV Abbreviation for hepatitis A virus, an RNA virus spread mainly by the fecal-oral route that causes acute hepatitis but usually does not cause chronic infection.

HBV Abbreviation for hepatitis B virus, a DNA virus transmitted through infected blood or body fluids that can cause acute or chronic hepatitis and cirrhosis or liver cancer.

HEPA Abbreviation for high-efficiency particulate air, describing a filter designed to remove a very high proportion of airborne particles of specified sizes from an air stream.

HGD Commonly an abbreviation for high-grade dysplasia, a marked abnormality of cell development that may be a precursor to invasive cancer but is not itself invasive cancer.

HNP Abbreviation for herniated nucleus pulposus, in which the inner material of an intervertebral disk protrudes through the outer ring and may compress a nerve root.

HPS Common medical abbreviations include hantavirus pulmonary syndrome, a severe respiratory illness, and Hermansky-Pudlak syndrome, an inherited disorder with oculocutaneous albinism and platelet-storage abnormalities; context determines the meaning.

-
- HS** Commonly an abbreviation for hereditary spherocytosis, an inherited hemolytic anemia; it can also mean half-strength, heart sounds, hamstrings, heparan sulfate, hernia sac, or house surgeon, so context is essential.
- HSA** Abbreviation that may mean a US health savings account or human serum albumin, the main plasma protein that helps maintain oncotic pressure; context determines the meaning.
- ICU** Abbreviation for intensive care unit, a hospital unit providing continuous monitoring and specialized support for critically ill patients.
- ID** A context-dependent abbreviation that may mean infectious disease, infectious diseases specialist, intellectual disability, or identification; the surrounding clinical text determines the meaning.
- IFA** Abbreviation for indirect fluorescent antibody, a laboratory method that detects antibodies or antigens using fluorescent labels, commonly in serologic testing.
- IHS** Abbreviation for the Indian Health Service, a US federal agency providing health services to American Indians and Alaska Native people and communities.
- IPT** Abbreviation for interpersonal therapy, a structured psychotherapy that focuses on relationships, grief, role transitions, and interpersonal difficulties.
- IPV** Abbreviation for inactivated polio vaccine, an injectable vaccine containing killed poliovirus and used to prevent poliomyelitis.
- IQ** Abbreviation for intelligence quotient, a standardized score from cognitive testing; it describes performance on specified tests and is not a complete measure of a person's abilities or worth.
- ITC** Abbreviation for in-the-canal, describing a hearing aid or other device positioned within the ear canal.
- ITN** Abbreviation for insecticide-treated bednet, a net treated with insecticide to reduce mosquito bites and malaria transmission.
- IVC** Commonly abbreviation for inferior vena cava, the large vein returning blood from the lower body to the heart; it can also mean intravenous cholangiogram, an X-ray study of the biliary system.
- JIP** Abbreviation for juvenile intestinal polyposis, an inherited condition in which children develop multiple hamartomatous polyps in the gastrointestinal tract and may have increased cancer risk.
- KB** In medical usage, an abbreviation for keratoderma blennorrhagicum or kilobase of DNA; lower-case *kb* is preferred for kilobase, so context is important.
- KUB** Abbreviation for kidney, ureter, and bladder, commonly referring to a plain abdominal X-ray used to assess urinary tract calculi, bowel gas patterns, or other abdominal findings.
- L** Abbreviation for liter, a metric unit of volume equal to 1,000 milliliters.
- LAVH** Abbreviation for laparoscopically assisted vaginal hysterectomy, removal of the uterus through the vagina with laparoscopic assistance.
- LL** Commonly an abbreviation for lower lid of the eye or lower limit; the intended meaning depends on the clinical context.
- LLL** Abbreviation for the left lower lobe of the lung.
- LLQ** Abbreviation for left lower quadrant, usually referring to the lower-left quarter of the abdomen.
- LMP** Abbreviation for last menstrual period, the first day of the most recent menstrual bleeding, used in estimating gestational age when the date is reliable.
- LPA** LPA abbreviates Little People of America and may also refer to the left pulmonary artery.
- LUL** Abbreviation for the left upper lobe of the lung.
- LUQ** Abbreviation for the left upper quadrant of the abdomen.
- LVF** Abbreviation for left ventricular failure, meaning failure of the left side of the heart; it may also mean left ventricular function, depending on context.
-

-
- MAOI** Abbreviation for monoamine oxidase inhibitor, a drug that inhibits monoamine oxidase and is used in selected treatment of depression and other disorders.
- MAP** Common abbreviation for *Mycobacterium avium* subspecies *paratuberculosis*, an organism of the *M. avium* complex; the abbreviation can have other meanings in different clinical contexts.
- MCD** An abbreviation with several medical meanings, including minimal change disease, mean corpuscular diameter, and medullary cystic disease; the intended meaning depends on context.
- MCP** Membrane cofactor protein, also called CD46, is a cell-surface protein that regulates complement activation and helps protect host cells from complement-mediated injury; the abbreviation can have other meanings in different contexts.
- MDI** An abbreviation that may mean metered-dose inhaler, multiple daily injections, or mental developmental index; the intended meaning depends on context.
- MEDVAC** Abbreviation for medical evacuation: the transport of ill or injured people to a medical facility or to a location providing a higher level of care.
- MI** Abbreviation for myocardial infarction, or heart attack: death of heart-muscle tissue caused by prolonged inadequate blood flow, usually from acute coronary artery blockage.
- MM** An abbreviation that may mean meningomyelocele, malignant melanoma, or multiple myeloma; in lowercase, *mm* commonly means millimeter. The intended meaning depends on context.
- MP** Abbreviation for mercaptopurine, a purine-antimetabolite drug, or for molecular pathology; the intended meaning depends on context.
- MPS** Abbreviation for mucopolysaccharidosis, a group of inherited lysosomal storage disorders caused by deficient breakdown of glycosaminoglycans.
- MSG** Abbreviation for monosodium glutamate, the sodium salt of glutamic acid used as a flavor enhancer and contributing umami taste.
- MTX** Abbreviation for methotrexate, a folate-antagonist drug used in selected cancers, autoimmune diseases, and other conditions.
- MVP** Abbreviation for mitral valve prolapse, in which one or both mitral-valve leaflets bow into the left atrium during ventricular systole; it is often mild but can sometimes cause mitral regurgitation or arrhythmias.
- MYH** MYH is a gene-family abbreviation involving myosin or DNA-repair proteins; the specific gene and variant determine the clinical meaning.
- ND** ND is a context-dependent abbreviation that may mean naturopathic doctor, neurodevelopmental, or another term; the full meaning should be specified.
- NEP** NEP is a context-dependent abbreviation that may mean neuroendocrine pancreatic tumor, nasoendotracheal procedure, or another clinical term.
- NF** NF is a context-dependent abbreviation, including neurofibromatosis, a group of genetic disorders causing nerve-sheath tumors and other features.
- NG** NG commonly means nasogastric, referring to a tube passed through the nose into the stomach, but the abbreviation has other uses.
- NS** NS is a context-dependent abbreviation, commonly normal saline, a sterile sodium-chloride solution, or not significant; the full term should be stated.
- OB** OB is an abbreviation commonly meaning obstetrics, the care of pregnancy, labor, birth, and the postpartum period, but it may have other meanings.
- OBS** OBS is a context-dependent abbreviation that may mean observation, obstetric service, or organic brain syndrome; the intended term should be written out.

-
- OCG** OCG is a context-dependent abbreviation that may refer to oral contraceptive guidance, oculocardiac testing, or another clinical term; the full term should be specified.
- OMS** An abbreviation for oral and maxillofacial surgery; the intended meaning depends on the clinical context.
- PAP** PAP is an abbreviation with context-dependent meanings, including Papanicolaou testing, pulmonary alveolar proteinosis, or prostatic acid phosphatase; the full term should be specified.
- PAS** PAS is an abbreviation that may mean periodic-acid–Schiff staining, para-aminosalicylic acid, or pulmonary-artery systolic pressure; context determines the meaning.
- PAT** PAT is a context-dependent abbreviation that may refer to paroxysmal atrial tachycardia, a rapid rhythm beginning in the atria, or another clinical term.
- PCA** PCA is a context-dependent abbreviation, commonly meaning patient-controlled analgesia, in which a patient activates prescribed doses from an infusion device.
- PCT** PCT is a context-dependent abbreviation that may mean procalcitonin, porphyria cutanea tarda, or patient-care technician; the full term should be specified.
- PDR** PDR commonly means Physicians’ Desk Reference, a drug-information reference, but the abbreviation has other uses and should be interpreted in context.
- PEP** PEP is a context-dependent abbreviation, including post-exposure prophylaxis, a preventive medicine course after a possible infection exposure.
- PHS** PHS is an abbreviation whose meaning depends on context, including public health service or primary hyperhidrosis; the full term should be specified.
- PI** PI is a context-dependent abbreviation, including principal investigator, pulmonary insufficiency, or protease inhibitor; the intended meaning must be specified.
- PLP** PLP commonly means pyridoxal 5′-phosphate, the active coenzyme form of vitamin B6 required by many enzymes, but the abbreviation has other uses.
- PMD** PMD is a context-dependent abbreviation, including Pelizaeus–Merzbacher disease, a rare inherited disorder of central myelination, or a professional medical designation.
- PON** PON is a context-dependent abbreviation that may refer to postoperative nausea, paraoxonase, or another clinical term; the full meaning should be stated.
- PTU** PTU is the abbreviation for propylthiouracil, an antithyroid medicine used selectively because of liver toxicity and other safety considerations.
- PV** PV is a context-dependent abbreviation, including polycythemia vera, plasma volume, or peripheral vision; the full term should be specified.
- PVB** PVB is a context-dependent abbreviation that may mean paravertebral block, a regional anesthetic technique, or another medical term.
- PX** PX is a context-dependent abbreviation that may mean prognosis, patient, prescription, or a genetic marker; clinical documentation should spell out the intended term.
- Q.2H** A prescription abbreviation meaning “every 2 hours.” Medication instructions should use clear wording because abbreviation misinterpretation can cause dosing errors.
- Q.3H.** A prescription abbreviation meaning “every 3 hours.” Medication instructions should use clear wording because abbreviation misinterpretation can cause dosing errors.
- Q.S.** A prescription abbreviation from the Latin “quantum sufficiat,” meaning “quantity sufficient” to make the required amount or concentration. The exact quantity must be specified by the formulation or prescription context.
- R/O** R/O is an abbreviation for rule out, indicating that a condition is being considered but has not been confirmed; clear documentation should state the actual diagnostic question.
-

RAD Acronym whose medical meaning depends on context, commonly radiation absorbed dose or right axis deviation on an ECG.

RBC RBC is the abbreviation for red blood cell, the cell that transports oxygen with hemoglobin and carries carbon dioxide back toward the lungs.

RF RF is a context-dependent abbreviation, including rheumatoid factor, radiofrequency, or risk factor; the complete term should be stated.

RLL RLL is the abbreviation for right lower lobe, usually referring to the lower lobe of the right lung.

RLQ RLQ is the abbreviation for right lower quadrant of the abdomen, containing structures including the appendix, cecum, and right adnexa.

RML RML is the abbreviation for right middle lobe, the middle lobe of the right lung.

RP RP is a context-dependent abbreviation, commonly retinitis pigmentosa, an inherited retinal degeneration causing night blindness and progressive visual-field loss.

RUQ RUQ is the abbreviation for right upper quadrant of the abdomen, containing the liver, gallbladder, duodenum, and portions of bowel and kidney.

SGOT SGOT is the former abbreviation for AST, an enzyme measured in blood that can rise with injury to liver, heart, muscle, or other tissues.

SGPT SGPT is the former abbreviation for ALT, a liver-associated enzyme that rises in many forms of hepatocellular injury.

SHP SHP is a context-dependent abbreviation that may refer to a healthcare professional, a gene or protein, or another clinical term; the full meaning should be specified.

SLV SLV is a context-dependent abbreviation that may mean single-lung ventilation, serum lithium value, or another term; the full meaning should be specified.

STD(S) Abbreviation for sexually transmitted disease(s), now often called sexually transmitted infections.

STS STS is a context-dependent abbreviation that may mean soft-tissue sarcoma, sodium thiosulfate, or another clinical term; the full term should be stated.

TMR TMR is a context-dependent abbreviation, including targeted muscle reinnervation, a surgical technique used to improve control of a prosthesis and reduce amputation pain.

TPA/TPA Abbreviation for tissue plasminogen activator.

TSD TSD is a context-dependent abbreviation that may mean total sleep deprivation, traumatic stress disorder, or another clinical term; the full term should be specified.

TSE TSE is a context-dependent abbreviation, including transmissible spongiform encephalopathy, the prion-disease group that includes Creutzfeldt-Jakob disease.

UL UL is a context-dependent abbreviation that may mean upper limit, ulnar or ulna-related, or another clinical term; the full meaning should be specified.

URG URG is a context-dependent abbreviation that may refer to urinary urgency or an urgent clinical priority; documentation should state the full term.

URTI Abbreviation for upper respiratory tract infection, commonly involving the nose, sinuses, pharynx, or larynx and often caused by viruses.

VD Abbreviation for venereal disease, an older term for a sexually transmitted infection.

WBS Abbreviation for Williams–Beuren syndrome, a genetic condition associated with characteristic facial features, developmental differences, and cardiovascular disease. In other clinical contexts, WBS may have a different meaning, so the full term should be specified.

Modern Terms

Old or obsolete term	Modern or preferred term
Acute renal failure	Acute kidney injury
Adult-onset diabetes mellitus	Type 2 diabetes mellitus
Ague	Malaria; a historical term for intermittent fever, usually associated with malaria
Apoplexy	Stroke or intracerebral haemorrhage, depending on the historical context
Arteriosclerosis obliterans	Peripheral arterial disease
Bright's disease	Glomerular kidney disease; the specific modern diagnosis should be stated when known
Chronic renal failure	Chronic kidney disease
Congestive heart failure	Heart failure
Consumption	Tuberculosis, especially pulmonary tuberculosis
Crippling arthritis	Disabling arthritis; describe the functional limitation specifically
Diabetic retinopathy, background	Mild or nonproliferative diabetic retinopathy, with severity stated
Dropsy	Oedema or fluid accumulation, with the affected site specified
Eclampsia toxemia	Preeclampsia or eclampsia, depending on the clinical condition
Essential hypertension	Primary hypertension
Fallopian tube	Uterine tube; fallopian tube remains an accepted anatomical term
Female hysteria	No single modern diagnosis; reassess using the presenting symptoms and an appropriate diagnosis
Gastric ulcer disease	Peptic ulcer disease, with the ulcer site specified
Heart attack	Myocardial infarction in formal medical usage
Herpes zoster	Shingles in common language; herpes zoster remains the preferred medical term
Hyperlipaemia	Hyperlipidaemia
Idiopathic thrombocytopenic purpura	Immune thrombocytopenia
Imbecile	Intellectual disability; the historical term is offensive and must not be used clinically
Insulin-dependent diabetes mellitus	Type 1 diabetes mellitus
Junkie's abscess	Injection-related abscess or skin and soft-tissue infection
Leprosy	Hansen disease; leprosy remains a recognized medical term, but person-first language is preferred

Old or obsolete term	Modern or preferred term
Manic depression	Bipolar disorder
Mongolism	Down syndrome; the historical term is unacceptable and must not be used
Multiple personality disorder	Dissociative identity disorder
Non-insulin-dependent diabetes mellitus	Type 2 diabetes mellitus
Old age dementia	Dementia or major neurocognitive disorder, with the cause specified when known
Palsy	Paralysis or the specific neurological disorder, depending on the context
Paralytic stroke	Stroke with paralysis; specify the neurological deficit and stroke type
Phthisis	Pulmonary tuberculosis; historically, progressive wasting disease
Senile dementia	Dementia or major neurocognitive disorder; age alone should not define the diagnosis
Senility	Cognitive impairment, dementia, or another specific diagnosis, after assessment
Spastic colon	Irritable bowel syndrome
Stomach flu	Acute gastroenteritis
Sunstroke	Heatstroke
Terminal illness	Advanced or life-limiting illness, with the diagnosis and prognosis specified
Toxemia of pregnancy	Preeclampsia or eclampsia
Venereal disease	Sexually transmitted infection
Wassermann reaction	A modern syphilis serologic test, such as a nontreponemal or treponemal test
Water on the brain	Hydrocephalus

Medical Specialties

Medical specialty	Principal area of practice
Allergy and Immunology	Allergic disease, asthma, immune deficiency, and immune-mediated disorders
Anesthesiology	Perioperative care, anesthesia, critical care, and pain medicine
Cardiology	Diseases of the heart and circulatory system
Cardiothoracic Surgery	Surgical treatment of diseases of the heart, lungs, and chest
Clinical Genetics	Diagnosis, counseling, and management of inherited and genomic disorders
Clinical Pharmacology	Effects, uses, interactions, and safe clinical use of medicines
Colon and Rectal Surgery	Surgical treatment of diseases of the colon, rectum, and anus
Critical Care Medicine	Assessment and treatment of life-threatening organ dysfunction
Dermatology	Diseases of the skin, hair, nails, and mucous membranes
Emergency Medicine	Immediate assessment and treatment of acute illness and injury
Endocrinology	Hormones, metabolism, diabetes, and endocrine glands
Family Medicine	Comprehensive continuing care for individuals and families of all ages
Forensic Medicine	Application of medical knowledge to legal and medicolegal questions
Gastroenterology	Diseases of the digestive tract, liver, gallbladder, and pancreas
General Practice	First-contact and continuing medical care across a broad range of conditions
General Surgery	Operative treatment of diseases involving multiple body systems
Geriatric Medicine	Health care of older adults, including multimorbidity and functional decline
Hematology	Blood cells, blood-forming organs, clotting, and blood disorders
Hepatology	Diseases of the liver and biliary system
Hospital Medicine	Comprehensive medical care of hospitalized adults
Infectious Disease Medicine	Diagnosis, treatment, and prevention of infections
Internal Medicine	Prevention, diagnosis, and treatment of adult medical disorders
Interventional Cardiology	Catheter-based diagnosis and treatment of cardiovascular disease
Interventional Radiology	Image-guided minimally invasive diagnosis and treatment
Medical Oncology	Systemic treatment and supportive care for cancer
Neonatology	Care of newborn infants, especially premature or critically ill infants
Nephrology	Diseases of the kidneys, fluid balance, electrolytes, and hypertension
Neurology	Diseases of the brain, spinal cord, peripheral nerves, and muscles

Medical specialty	Principal area of practice
Neurosurgery	Surgical treatment of disorders of the nervous system
Nuclear Medicine	Diagnosis and treatment using radioactive substances and molecular imaging
Obstetrics and Gynecology	Pregnancy, childbirth, and diseases of the female reproductive system
Occupational Medicine	Health, safety, and disease prevention in the workplace
Ophthalmology	Medical and surgical care of the eyes and visual system
Oral and Maxillofacial Surgery	Surgical treatment of diseases and injuries of the mouth, jaws, face, and neck
Orthopedic Surgery	Surgical and nonsurgical treatment of the musculoskeletal system
Otolaryngology–Head and Neck Surgery	Diseases and surgery of the ear, nose, throat, and head and neck
Pain Medicine	Assessment and treatment of acute, chronic, and cancer-related pain
Palliative Medicine	Relief of symptoms and support for people with serious or life-limiting illness
Pathology	Laboratory examination of tissues, cells, blood, and body fluids
Pediatric Surgery	Surgical care of infants, children, and adolescents
Pediatrics	Medical care of infants, children, and adolescents
Physical Medicine and Rehabilitation	Restoration of function and independence after illness or injury
Plastic and Reconstructive Surgery	Reconstruction, restoration, and alteration of body tissues
Preventive Medicine	Prevention of disease and promotion of health in individuals and populations
Psychiatry	Diagnosis and treatment of mental, emotional, and behavioral disorders
Public Health Medicine	Protection and improvement of population health
Pulmonary Medicine	Diseases of the lungs and respiratory system
Radiation Oncology	Use of ionizing radiation to treat cancer
Radiology	Diagnostic and image-guided evaluation of disease
Reproductive Endocrinology and Infertility	Hormonal and reproductive disorders and fertility care
Rheumatology	Autoimmune, inflammatory, and musculoskeletal disorders
Sleep Medicine	Sleep disorders and their effects on health
Sports Medicine	Prevention, diagnosis, treatment, and rehabilitation of activity-related conditions
Thoracic Surgery	Surgery of the chest, lungs, esophagus, and mediastinum
Transfusion Medicine	Collection, testing, processing, and clinical use of blood components
Trauma Surgery	Surgical management of serious and complex injuries
Urology	Diseases of the urinary tract and male reproductive system
Vascular Surgery	Surgical and endovascular treatment of blood-vessel disease

Lay Language Medical Terms

Lay term	Medical term	Lay-language explanation
Belly area	ABDOMEN/ABDOMINAL	The body cavity below the diaphragm containing the stomach, intestines, liver, and other organs.
Take in	ABSORB	To take up or take in a substance, such as fluid.
Too much acid in blood	ACIDOSIS	A condition in which the blood is more acidic than normal.
Sharpness or clarity	ACUITY	The clearness or keenness of a sense, especially vision.
Recent or sudden	ACUTE	New, recent, sudden, or requiring urgent attention.
Swollen glands	ADENOPATHY	Enlarged or swollen lymph nodes.
Helpful added treatment	ADJUVANT	Something that supports or assists the main treatment.
Added treatment	ADJUVANT TREATMENT	Treatment given in addition to the main or standard treatment.
Germ-killing drug	ANTIBIOTIC	A drug used to kill or control bacteria and some other germs.
Germ-killing treatment	ANTIMICROBIAL	A substance or treatment that acts against microorganisms such as bacteria, fungi, or parasites.
Antiviral drug	ANTIRETROVIRAL	A drug that slows or stops the growth of certain viruses.
Side effect	ADVERSE EFFECT	An unwanted or harmful response to a treatment or medicine.
Walking	AMBULATE/AMBULATION/ AMBULATORY	To walk, or to be able to walk.
Severe allergy	ANAPHYLAXIS	A serious, potentially life-threatening allergic reaction.
Low red-cell count	ANEMIA	A reduced number of red blood cells or a low red-cell blood count.
Numbing medicine	ANESTHETIC	A drug or agent that reduces or removes the feeling of pain, sometimes by causing sleep.
Heart pain	ANGINA	Chest pain caused by inadequate blood flow to the heart muscle.
Heart pain	ANGINA PECTORIS	Another medical term for chest pain caused by reduced blood flow to the heart.
Loss of appetite	ANOREXIA	Lack of appetite or, in some contexts, a disorder involving refusal to eat.

Lay term	Medical term	Lay-language explanation
Inner elbow area	ANTECUBITAL	Relating to the inner side of the elbow or forearm.
Protective protein	ANTIBODY	A protein made by the body in response to a foreign substance.
Seizure-prevention drug	ANTICONVULSANT	A medicine used to prevent or control seizures.
Cholesterol-lowering drug	ANTILIPEMIC	A drug that lowers fat or cholesterol levels in the blood.
Cough medicine	ANTITUSSIVE	A medicine used to reduce coughing.
Irregular heartbeat	ARRHYTHMIA	Any abnormal change in the normal heartbeat or rhythm.
Breathing material into the lungs	ASPIRATION	Entry of fluid, food, or other material into the lungs, often after vomiting.
Laboratory test	ASSAY	A laboratory test used to measure or identify something.
Check or evaluate	ASSESS	To examine, measure, evaluate, or look at something carefully.
Asthma	ASTHMA	A lung disease in which narrowed or tightened airways make breathing difficult.
Without symptoms	ASYMPTOMATIC	Having no noticeable symptoms.
Armpit	AXILLA	The armpit area.
Not cancerous	BENIGN	Not malignant and generally not cancerous.
Twice daily	BID	Prescription shorthand meaning twice a day.
Carried or stuck together	BINDING/BOUND	Carried by something, or made to stick or join together.
Drug availability	BIOAVAILABILITY	The amount or proportion of a drug that becomes available for the body to use.
Series of blood tests	BLOOD PROFILE	A group or panel of blood tests.
Large dose at once	BOLUS	A relatively large amount given all at one time.
Amount of mineral in bone	BONE MASS	The amount of calcium and other minerals in a given amount of bone.
Slow irregular heartbeats	BRADYARRHYTHMIAS	Slow heart rhythms that may also be irregular.
Slow heartbeat	BRADYCARDIA	A slower-than-usual heartbeat.
Narrowed airways	BRONCHOSPASM	Breathing difficulty caused by tightening of the airway muscles.
Cancer-causing	CARCINOGENIC	Capable of causing cancer.
Cancer type	CARCINOMA	A cancer that begins in lining or covering cells of the body.
Related to the heart	CARDIAC	Relating to the heart.
Resetting the heartbeat	CARDIOVERSION	Restoring a normal heart rhythm, often with a controlled electrical shock.
Tube	CATHETER	A thin tube used to give or remove fluids, or to perform certain medical procedures.
Epidural tube	CATHETER	A catheter placed near the spinal cord to deliver anesthesia during surgery.
Brain and spinal cord	CENTRAL NERVOUS SYSTEM (CNS)	The brain and spinal cord.
Brain injury	CEREBRAL TRAUMA	Damage to the brain caused by an injury.
Stopping	CESSATION	The act of stopping something.
Coronary heart disease	CHD	Disease of the blood vessels that supply the heart.
Cancer drug treatment	CHEMOTHERAPY	Treatment of disease, usually cancer, with chemical medicines.

Lay term	Medical term	Lay-language explanation
Long-lasting	CHRONIC	Continuing for a long time or ongoing.
Related to patient care	CLINICAL	Pertaining to medical care or the care of patients.
Human research study	CLINICAL TRIAL	A research study involving human participants.
Unconscious state	COMA	A prolonged state of unconsciousness in which a person cannot be awakened normally.
All disease gone	COMPLETE RESPONSE	Total disappearance of the detectable disease being treated.
Present from birth	CONGENITAL	Present before or at birth.
Pink, irritated eye	CONJUNCTIVITIS	Redness and irritation of the thin membrane covering the eye.
Treatment to keep improvement	CONSOLIDATION PHASE	A treatment phase intended to make a remission last, usually after initial treatment.
Comparison study	CONTROLLED TRIAL	A study comparing an experimental treatment with a standard or control treatment.
Research partnership	COOPERATIVE GROUP	An association of institutions that work together on clinical research.
Related to heart arteries	CORONARY	Relating to the heart or the blood vessels that supply it.
Computerized X-ray scan	CT SCAN (CAT)	A series of computer-processed X-ray images that show sections of the body.
Germ-growth test	CULTURE	A test that grows or detects organisms that may cause infection.
Added together	CUMULATIVE	Added up from the beginning.
Related to the skin	CUTANEOUS	Relating to the skin.
Stroke	CVA	A stroke, also called a cerebrovascular accident.
Related to the skin	DERMATOLOGIC	Pertaining to the skin.
Lower blood-pressure number	DIASTOLIC	Relating to the lower blood-pressure number, when the heart relaxes.
Farther away	DISTAL	Toward the end or away from the center of the body.
Water pill	DIURETIC	A drug that increases urination.
Sound-wave blood-flow test	DOPPLER	A device or test using sound waves to assess blood flow.
Masked study	DOUBLE BLIND	A study in which neither the participants nor investigators know which treatment is being given.
Improper function	DYSFUNCTION	A state in which something does not work normally.
Abnormal cells	DYSPLASIA	Abnormal changes in the size, shape, or organization of cells.
Heart sound-wave test	ECHOCARDIOGRAM	An ultrasound test that creates pictures of the heart.
Swelling from fluid	EDEMA	Excess fluid collecting in body tissues.
Brain-wave test	EEG	A recording of the brain's electrical activity.
Effectiveness	EFFICACY	How well a treatment works under specified conditions.
Heart electrical tracing	ELECTROCARDIOGRAM	A tracing of the heart's electrical activity, also called an ECG or EKG.
Mineral imbalance	ELECTROLYTE IMBALANCE	An abnormal level or balance of minerals in the blood.
Vomiting	EMESIS	The act of vomiting.

Lay term	Medical term	Lay-language explanation
Based on experience	EMPIRIC	Based on experience or likely cause rather than definitive test results.
Viewing inside the body	ENDOSCOPIC	Looking inside the body with a lighted tube or camera.
Through the intestines	ENTERAL	Given through or involving the digestive tract.
Outside the spinal cord	EPIDURAL	Located outside the spinal cord covering, or medicine given into that space.
Remove completely	ERADICATE	To get rid of a disease, infection, or other problem.
Examined	EVALUATED, ASSESSED	Examined or checked for a medical condition.
Fast ethics review	EXPEDITED REVIEW	A rapid research-protocol review allowed for certain lower-risk studies.
Outside the body	EXTERNAL	Located or occurring outside the body.
Leak out	EXTRAVASATE	To leak outside the intended area, such as from a blood vessel into nearby tissue.
U.S. drug agency	FDA	The U.S. Food and Drug Administration, which regulates medicines and medical products.
Scar-like	FIBROUS	Containing many fibers, often resembling scar tissue.
Irregular quivering	FIBRILLATION	Very rapid, uncoordinated activity of the heart or another muscle.
Being put fully to sleep	GENERAL ANESTHESIA	Use of medicines to cause unconsciousness and prevent pain during a procedure.
Related to pregnancy	GESTATIONAL	Pertaining to pregnancy.
Red-cell percentage	HEMATOCRIT	The proportion of blood made up of red blood cells.
Bruise or blood collection	HEMATOMA	A collection of blood under the skin or in a tissue, often appearing as a bruise.
Blood-flow measurement	HEMODYNAMIC MEASURING	Measuring blood flow and circulation.
Red-cell breakdown	HEMOLYSIS	Breakdown or destruction of red blood cells.
Heparin tube or port	HEPARIN LOCK	An intravenous access device kept open with a prescribed solution, sometimes containing heparin.
Liver tumor	HEPATOMA	A tumor of the liver, often referring to hepatocellular carcinoma.
Inherited disease	HERITABLE DISEASE	A disease that can be passed through genes to future children.
Tissue-disease examination	HISTOPATHOLOGIC	Relating to examination of tissues or cells for disease.
Portable heart monitor	HOLTER MONITOR	A portable device that records heartbeats, usually for a day or longer.
High blood calcium	HYPERCALCEMIA	An abnormally high calcium level in the blood.
High blood potassium	HYPERKALEMIA	An abnormally high potassium level in the blood.
High blood sodium	HYPERNATREMIA	An abnormally high sodium level in the blood.
High blood pressure	HYPERTENSION	Blood pressure that is higher than recommended.
Low blood calcium	HYPOCALCEMIA	An abnormally low calcium level in the blood.
Low blood potassium	HYPOKALEMIA	An abnormally low potassium level in the blood.
Low blood sodium	HYPONATREMIA	An abnormally low sodium level in the blood.
Low blood pressure	HYPOTENSION	Blood pressure that is lower than expected.
Low blood oxygen	HYPOXEMIA	A low level of oxygen in the blood.
Low tissue oxygen	HYPOXIA	Too little oxygen reaching body tissues.

Lay term	Medical term	Lay-language explanation
Removal of the uterus	HYSTERECTOMY	Surgical removal of the uterus; the ovaries and other structures may or may not also be removed.
Treatment-caused	IATROGENIC	Caused by medical treatment or a healthcare procedure.
Device research permission	IDE	Permission to study an unapproved medical device in people.
Cause unknown	IDIOPATHIC	Having no known cause.
Body protection	IMMUNITY	The body's defense against infection or other foreign substances.
Antibody protein	IMMUNOGLOBIN	A protein involved in making antibodies and immune defense; the standard spelling is immunoglobulin.
Immune-suppressing	IMMUNOSUPPRESSIVE	A drug or treatment that reduces the body's immune response.
Immune-based treatment	IMMUNOTHERAPY	Treatment that uses or changes the immune system, often against cancer or immune disease.
Abnormal function	IMPAIRED FUNCTION	Function that is reduced or not working normally.
Placed inside the body	IMPLANTED	Surgically or otherwise placed inside the body.
New-drug research permission	IND	Authorization to test an unapproved new drug in people.
Hardening	INDURATION	An area that has become unusually firm or hard.
Left in place	INDWELLING	Remaining in a particular location, such as a catheter.
Dead tissue from poor blood flow	INFARCT	Death of tissue caused by inadequate blood supply.
Disease that can spread	INFECTIOUS DISEASE	An illness caused by organisms that may spread between people or through other routes.
Swelling and irritation	INFLAMMATION	The body's response to injury or infection, often causing swelling, warmth, redness, or pain.
Slow delivery into the body	INFUSION	Slow delivery of a substance, usually through a vein or catheter.
Taking by mouth	INGESTION	Eating, drinking, or taking something into the body by mouth.
Antiviral immune protein	INTERFERON	A medicine or natural protein that helps the body act against viruses.
On and off	INTERMITTENT	Occurring repeatedly with pauses between episodes.
Inside the body	INTERNAL	Located or occurring within the body.
Inside	INTERIOR	Located on the inside.
Into a muscle	INTRAMUSCULAR	Given into a muscle.
Into the abdomen	INTRAPERITONEAL	Given into or located within the abdominal cavity.
Into spinal fluid	INTRATHECAL	Given into the fluid space around the spinal cord.
Into a vein	INTRAVENOUS (IV)	Given through a vein.
Into the bladder	INTRAVESICAL	Given into the bladder.
Place a breathing tube	INTUBATE	To place a tube into the airway.
Breaking the skin	INVASIVE PROCEDURE	A procedure that punctures, opens, or cuts the skin or enters the body.
Unapproved-drug study	INVESTIGATIONAL NEW DRUG (IND)	A new drug being studied before it has full approval for general use.

Lay term	Medical term	Lay-language explanation
Unproven treatment method	INVESTIGATIONAL METHOD	A treatment method still being studied and not established as standard care.
Reduced blood flow	ISCHEMIA	Too little blood and oxygen reaching a tissue, usually because blood flow is reduced.
Open abdominal surgery	LAPAROTOMY	Surgery using an incision in the abdominal wall to reach the organs inside.
Abnormal area	LESION	A wound, injury, or area of abnormal tissue.
Very tired or sleepy	LETHARGY	Sleepiness, tiredness, or reduced energy.
Low white-cell count	LEUKOPENIA	A lower-than-normal number of white blood cells.
Fat	LIPID	A fat or fat-like substance.
Amount of fat	LIPID CONTENT	The amount of fat in a substance, tissue, or blood sample.
Cholesterol and fat panel	LIPID PROFILE (PANEL)	Blood tests measuring fats and cholesterol.
Numbing a small area	LOCAL ANESTHESIA	Numbing medicine used to prevent pain in a limited area.
Limited to one area	LOCALIZED	Restricted to one area.
Inner space	LUMEN	The hollow space inside a tube or organ, such as a blood vessel.
Lymph-vessel X-ray	LYMPHANGIOGRAPHY	An X-ray study of lymph vessels or nodes after contrast dye is injected.
Immune white blood cell	LYMPHOCYTE	A type of white blood cell that helps protect against infection.
Lymph-node cancer	LYMPHOMA	A cancer of lymph nodes or related lymphatic tissue.
Feeling unwell	MALAISE	A vague feeling of illness or bodily discomfort.
Not working correctly	MALFUNCTION	A condition in which a device, organ, or system does not work properly.
Cancerous growth	MALIGNANCY	A cancer or other tumor capable of growing and spreading.
Brain tumor	MEDULLABLASTOMA	A type of tumor that begins in the brain.
Abnormal red-cell development	MEGALOBLASTOSIS	A change in blood-forming cells that produces unusually large, abnormal red-cell precursors.
Break down in the body	METABOLIZE	To chemically process or break down a substance in the body.
Cancer spread	METASTASIS	Spread of cancer cells from one body part to another.
Antibiotic medicine	METRONIDAZOLE	A medicine used for certain bacterial and parasitic infections.
Heart attack	MI	Myocardial infarction, or death of heart muscle from blocked blood flow.
Small or slight	MINIMAL	Very small or slight.
Reduce	MINIMIZE	To reduce something as much as possible.
Check or watch	MONITOR	To observe, measure, or keep track of something over time.
Ease of movement	MOBILITY	The ability to move easily.
Illness or complication	MORBIDITY	Illness, disability, or an unwanted result of disease or treatment.
Death rate or death	MORTALITY	Death, or the rate of death in a population.
Ability to move	MOTILITY	The ability of a body part, cell, or organism to move.
Magnetic body scan	MRI	Pictures of the inside of the body made with magnets and radio waves rather than X-rays.

Lay term	Medical term	Lay-language explanation
Moist body lining	MUCOSA, MUCOUS MEMBRANE	The moist lining of the digestive, respiratory, reproductive, and urinary passages.
Muscle aches	MYALGIA	Pain or aching in the muscles.
Related to heart muscle	MYOCARDIAL	Pertaining to the muscle of the heart.
Heart attack	MYOCARDIAL INFARCTION	Death of heart muscle caused by inadequate blood flow.
Nose-to-stomach tube	NASOGASTRIC TUBE	A tube passed through the nose into the stomach.
U.S. health institute	NCI	The National Cancer Institute.
Tissue death	NECROSIS	Death of cells or tissue.
Tumor	NEOPLASIA/NEOPLASM	An abnormal new growth that may be benign or malignant.
Nerve-tissue cancer	NEUROBLASTOMA	A cancer that develops from immature nerve tissue.
Related to the nervous system	NEUROLOGICAL	Pertaining to the brain, spinal cord, or nerves.
Low infection-fighting cells	NEUTROPENIA	A decrease in neutrophils, a major type of infection-fighting white blood cell.
U.S. health agency	NIH	The National Institutes of Health.
Not entering the body	NONINVASIVE	Not breaking, cutting, or entering the skin or body.
Hospital-acquired	NOSOCOMIAL	Acquired during a stay in a healthcare facility.
Blockage	OCCLUSION	Closing, blockage, or obstruction.
Cancer specialty	ONCOLOGY	The study and treatment of cancer and tumors.
Related to the eye	OPHTHALMIC	Pertaining to the eye.
Best choice	OPTIMAL	Best, most favorable, or most suitable.
By mouth	ORAL ADMINISTRATION	Giving a medicine through the mouth.
Related to bones	ORTHOPEDIC	Pertaining to bones, joints, muscles, or related structures.
Rare dense-bone disorder	OSTEOPETROSIS	A rare disorder in which bones become unusually dense but may be fragile.
Weak bones	OSTEOPOROSIS	A condition in which bones lose strength and become more likely to break.
Female sex glands	OVARIES	The organs that produce eggs and reproductive hormones.
Given by injection	PARENTERAL	Given by injection rather than through the digestive tract.
Open passage	PATENCY	The condition of being open and unobstructed.
How a disease develops	PATHOGENESIS	The process by which a disease or unhealthy condition develops.
Through the skin	PERCUTANEOUS	Done through or across the skin.
Away from the center	PERIPHERAL	At or toward the outer part rather than the center.
By mouth	PER OS (PO)	A prescription direction meaning by mouth.
How the body handles a drug	PHARMACOKINETICS	The study of how the body absorbs, distributes, changes, and removes a drug.
First drug-study phase	PHASE I	The first human study phase, usually focused on safety, effects, and dosing.
Early effectiveness study	PHASE II	A study phase examining safety and early evidence of benefit in people with a condition.
Large confirmation study	PHASE III	A larger study phase used to confirm safety and effectiveness and identify common side effects.

Lay term	Medical term	Lay-language explanation
After-approval study	PHASE IV	A study performed after approval to monitor safety or examine additional uses.
Vein irritation	PHLEBITIS	Irritation or inflammation of a vein.
Inactive treatment	PLACEBO	An inactive substance or treatment that looks like the real treatment.
Improvement from expectation	PLACEBO EFFECT	An improvement that occurs after an inactive treatment, often because of expectation or context.
Clotting blood particles	PLATELETS	Small blood components that help stop bleeding by forming clots.
Possible	POTENTIAL	Possible or capable of happening.
Make stronger	POTENTIATE	To increase or multiply the effect of a drug, treatment, or substance.
Effect booster	POTENTIATOR	An agent that helps another agent work more strongly.
Before birth	PRENATAL	Occurring before birth.
Preventive treatment	PROPHYLAXIS	A medicine or action used to prevent disease or infection.
As needed	PRN	Prescription shorthand meaning when needed.
Expected outlook	PROGNOSIS	The expected course and outcome of a disease.
Lying face down	PRONE	Lying on the stomach or front.
Study looking forward	PROSPECTIVE STUDY	A study that follows participants forward in time.
Artificial body part	PROSTHESIS	An artificial replacement for a body part.
Study plan	PROTOCOL	The written plan for a study or treatment.
Closer to the center	PROXIMAL	Closer to the center of the body or point of attachment.
Related to the lungs	PULMONARY	Pertaining to the lungs.
Every day	QD	Prescription shorthand meaning every day; clearer wording is preferred.
Four times daily	QID	Prescription shorthand meaning four times a day.
Radiation treatment	RADIATION THERAPY	Treatment using high-energy radiation to damage or destroy abnormal cells.
By chance	RANDOM	Selected or occurring by chance.
Chance assignment	RANDOMIZATION	Assigning participants to groups by chance.
Red blood cell	RBC	A red blood cell.
Genetically combined	RECOMBINANT	Made by combining genetic material from different sources.
Put back together	RECONSTITUTION	Restoring something by putting its parts back together or adding a liquid.
Happen again	RECUR	To happen again.
Not responding	REFRACTORY	Not responding adequately to treatment.
Regrowth or repair	REGENERATION	Growth again or replacement of lost or damaged tissue.
Treatment schedule	REGIMEN	The planned pattern, dose, and timing of treatment.
Return of disease	RELAPSE	Return of a disease after improvement or remission.
No detectable disease	REMISSION	Disappearance or reduction of evidence of disease; it may be partial or complete.
Related to kidneys	RENAL	Pertaining to the kidneys.
Can be repeated	REPLICABLE	Able to be repeated with similar results.

Lay term	Medical term	Lay-language explanation
Remove surgically	RESECT	To remove part or all of an organ or tissue by surgery.
Study looking back	RETROSPECTIVE STUDY	A study that examines past records or past events.
Connective-tissue cancer	SARCOMA	A cancer arising from connective, bone, muscle, fat, or related tissues.
Calming medicine	SEDATIVE	A medicine that calms or makes a person sleepy.
Testicular cancer	SEMINOMA	A type of cancer arising in the testicle.
In order	SEQUENTIALLY	Happening one after another in order.
Sleepiness	SOMNOLENCE	Unusual sleepiness or a tendency to fall asleep.
Air-measuring instrument	SPIROMETER	An instrument that measures the amount of air moved into and out of the lungs.
Extent of disease	STAGING	Assessment of how far a disease has progressed or spread.
Usual accepted treatment	STANDARD OF CARE	A treatment plan generally accepted by the medical community as appropriate.
Narrowing	STENOSIS	Narrowing of a passage, duct, or blood vessel.
Mouth inflammation	STOMATITIS	Sores or inflammation inside the mouth.
Group for comparison	STRATIFY	To arrange people or results into groups for analysis.
Barely responsive state	STUPOR	A greatly reduced level of alertness in which a person is difficult to awaken or respond to.
Under the collarbone	SUBCLAVIAN	Located beneath the collarbone.
Under the skin	SUBCUTANEOUS	Beneath the skin.
Lying on the back	SUPINE	Lying face up on the back.
Care focused on comfort	SUPPORTIVE CARE	General care that relieves symptoms and supports the person without directly curing the underlying disease.
Having symptoms	SYMPTOMATIC	Showing or having symptoms.
Group of related findings	SYNDROME	A condition identified by a characteristic group of symptoms and signs.
Upper blood-pressure number	SYSTOLIC	Relating to the upper blood-pressure number when the heart contracts.
Birth-defect-causing	TERATOGENIC	Capable of causing abnormalities in a developing fetus.
Male sex glands	TESTES/TESTICLES	The organs that produce sperm and male reproductive hormones.
Clot formation	THROMBOSIS	Formation of a blood clot inside a blood vessel or other location.
Blood clot	THROMBUS	A blood clot formed within a blood vessel or the heart.
Three times daily	TID	Prescription shorthand meaning three times a day.
Gradual dose adjustment	TITRATION	Gradually changing a dose or solution to reach a desired effect.
Immune white blood cells	T-LYMPHOCYTES	White blood cells that help coordinate or carry out immune responses.
On the surface	TOPICAL	Applied to the surface, usually the skin or a mucous membrane.
Surface numbing medicine	TOPICAL ANESTHETIC	A numbing medicine applied to a particular surface area.

Lay term	Medical term	Lay-language explanation
Harmful effects	TOXICITY	Harmful or unwanted effects caused by a drug, chemical, or treatment.
Through the skin	TRANSDERMAL	Delivered through the skin.
Temporarily	TRANSIENTLY	For a short time or temporarily.
Injury	TRAUMA	Physical injury or wound.
Walking exercise machine	TREADMILL	A moving walking surface used for exercise or heart testing.
Absorption into tissue	UPTAKE	Absorbing or taking in a substance by living tissue.
Heart-valve repair	VALVULOPLASTY	A procedure that repairs or widens a heart valve.
Enlarged veins	VARICES	Abnormally enlarged or twisted veins.
Blood-vessel spasm	VASOSPASM	Sudden narrowing of blood vessels.
Disease carrier	VECTOR	An organism or carrier that can transmit a disease-causing microorganism.
Blood draw	VENIPUNCTURE	Puncturing a vein with a needle to draw blood or give treatment.
Parent-to-child spread	VERTICAL TRANSMISSION	Spread of an infection from a parent to a child, often during pregnancy, birth, or breastfeeding.
White blood cell	WBC	A white blood cell involved in immune defense.

Source and adaptation: Terms and concepts were checked against the University of Iowa Human Subjects Office, “Medical Terms in Lay Language,” available at <https://hso.research.uiowa.edu/get-started/guides-and-standard-operating-procedures-sops/medical-terms-lay-language>. The explanations in this appendix are paraphrased for this dictionary and are not the only acceptable plain-language alternatives.

Bibliography and Sources

This is a selected bibliography of standard medical dictionaries and authoritative biomedical terminology resources used to guide terminology, spelling, synonyms, and general definitions in this dictionary. The source material is not mapped term-by-term, so this list should not be read as a claim that every entry was copied from every source. Clinical decisions should be based on current guidelines, specialist references, and primary literature.

BMA Illustrated Medical Dictionary

British Medical Association. *The British Medical Association Illustrated Medical Dictionary*. Dorling Kindersley; 2007. ISBN 978-1-4053-1997-3. Bibliographic record available from: <https://obnb.uk/p13578671-the-british-medical-association-illustrated-medical-dictionary>

Black's Medical Dictionary

Marcovitch H, editor. *Black's Medical Dictionary*. 43rd ed. Bloomsbury Publishing; 2023. ISBN 978-1-3994-1231-5. Available from: <https://www.bloomsbury.com/us/blacks-medical-dictionary-9781399412315/>

Dorland's Illustrated Medical Dictionary

Dorland. *Dorland's Illustrated Medical Dictionary*. 33rd ed. Elsevier; 2020. ISBN 978-1-4557-5643-8. Available from: <https://www.us.elsevierhealth.com/dorlands-illustrated-medical-dictionary-9781455756438.html>

Mosby's Medical Dictionary

Mosby. *Mosby's Medical Dictionary*. 11th ed. Elsevier; 2021. ISBN 978-0-323-63915-6. Available from: <https://shop.elsevier.com/books/mosbys-medical-dictionary/mosby/978-0-323-63915-6>

Stedman's Medical Dictionary

Stedman. *Stedman's Medical Dictionary*. 28th ed. Lippincott Williams & Wilkins; 2013. ISBN 978-0-7817-3390-8. Available from: <https://www.wolterskluwer.com/en/solutions/ovid/stedmans-medical-dictionary-13117>

Taber's Cyclopedic Medical Dictionary

Venes D, editor. *Taber's Cyclopedic Medical Dictionary*. 25th ed. F.A. Davis; 2025. Available from: https://www.tabers.com/tabersonline/view/Tabers-Dictionary/729000/all/Taber%27s_Cyclopedic_Medical_Dictionary

Merriam-Webster Medical Dictionary

Merriam-Webster. *Medical Dictionary*. Online reference. Available from: <https://www.merriam-webster.com/medical>

Medical Subject Headings (MeSH)

National Library of Medicine. *Medical Subject Headings (MeSH)*. Controlled and hierarchical biomedical terminology resource. Available from: <https://www.nlm.nih.gov/mesh/meshhome.html>

MeSH Data Downloads

National Library of Medicine. *MeSH Data*. Current production data and download information. Available from: <https://www.nlm.nih.gov/databases/download/mesh.html>

MedlinePlus

National Library of Medicine. *MedlinePlus*. Health information, medical encyclopedia, and patient-oriented reference resource. Available from: <https://medlineplus.gov/>

National Institutes of Health (NIH)

National Institutes of Health. *Health Information*. Federal biomedical research and health-information portal. Available from: <https://www.nih.gov/health-information>

National Cancer Institute (NCI)

National Cancer Institute. *Definition of oncology*. NCI Dictionary of Cancer Terms. Available from: <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/oncology>

National Cancer Institute. *Definition of image-guided radiation therapy*. NCI Dictionary of Cancer Terms. Available from: <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/image-guided-radiation-therapy>

National Cancer Institute. *Stereotactic radiotherapy*. Clinical trial description. Available from: <https://www.cancer.gov/research/participate/clinical-trials-search/v?id=NCI-2026-00405>

National Cancer Institute. *Definition of brachytherapy*. NCI Dictionary of Cancer Terms. Available from: <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/brachytherapy>

National Cancer Institute. *Radiation Therapy for Cancer*. Available from: <https://www.cancer.gov/about-cancer/treatment/types/radiation-therapy>

American Society for Radiation Oncology (ASTRO)

American Society for Radiation Oncology. *What is radiation oncology?* Medical Student FAQs. Available from: <https://www.astro.org/interest-groups/medical-students/faqs>

RadiologyInfo.org

RadiologyInfo.org. *Functional MRI (fMRI) of the brain*. Patient and health information reviewed by radiology and medical-imaging specialists. Available from: <https://www.radiologyinfo.org/en/info/fmribrain>

RadiologyInfo.org. *IGRT – Image-Guided Radiation Therapy*. Patient and health information reviewed by radiology and medical-imaging specialists. Available from: <https://www.radiologyinfo.org/en/info/igrt>

RadiologyInfo.org. *Stereotactic Radiosurgery (SRS) and Stereotactic Body Radiotherapy (SBRT)*. Patient and health information reviewed by radiology and medical-imaging specialists. Available from: <https://www.radiologyinfo.org/en/info/stereotactic>

RadiologyInfo.org. *Brachytherapy (Internal radiation therapy)*. Patient and health information reviewed by radiology and medical-imaging specialists. Available from: <https://www.radiologyinfo.org/en/info/brachy>

RadiologyInfo.org. *Radiation Therapy for Pain Relief*. Patient and health information reviewed by radiology and medical-imaging specialists. Available from: <https://www.radiologyinfo.org/en/info/radiation-therapy-for-pain-relief>

National Cancer Institute. *Head and Neck Cancers*. Available from: <https://www.cancer.gov/types/head-and-neck/head-neck-fact-sheet>

National Cancer Institute. *Surgery: Head and Neck Cancer*. SEER Training Modules. Available from: <https://training.seer.cancer.gov/head-neck/treatment/surgery/>

National Cancer Institute. *Breast Cancer Surgery*. Available from: <https://www.cancer.gov/types/breast/treatment/surgery>

National Cancer Institute. *Non-Small Cell Lung Cancer Treatment*. Patient Version. Available from: <https://www.cancer.gov/types/lung/patient/non-small-cell-lung-treatment-pdq>

National Institute on Deafness and Other Communication Disorders (NIDCD). *Smell (Olfactory) Disorders*. Available from: <https://www.nidcd.nih.gov/health/smell-disorders>

National Library of Medicine. *Adnexa Uteri*. Medical Subject Headings (MeSH). Available from: <https://meshb.nlm.nih.gov/record/ui?ui=D000290>

National Institute of Biomedical Imaging and Bioengineering (NIBIB). *Tissue Engineering and Regenerative Medicine Fact Sheet*. National Institutes of Health. Available from: <https://www.nibib.nih.gov/sites/default/files/2022-04/Fact-Sheet-Tissue-Engineering.pdf>

National Library of Medicine. *Robotic surgery*. MedlinePlus Medical Encyclopedia. Available from: <https://medlineplus.gov/ency/article/007339.htm>

National Library of Medicine. *Erythrocyte Sedimentation Rate (ESR)*. MedlinePlus Medical Test. Available from: <https://medlineplus.gov/lab-tests/erythrocyte-sedimentation-rate-esr/>

National Library of Medicine. *Throat Swab Culture*. MedlinePlus Medical Encyclopedia. Available from: <https://medlineplus.gov/ency/article/003746.htm>

National Cancer Institute. *Cervical Cancer Screening (PDQ)*. Available from: <https://www.cancer.gov/types/cervical/hp/cervical-screening-pdq>

National Cancer Institute. *Definition of liquid biopsy*. NCI Dictionary of Cancer Terms. Available from: <https://www.cancer.gov/publications/dictionaries/cancer-terms/def/liquid-biopsy>

Japan Surgical Society. *Clinical practice guidelines for telesurgery 2022*. National Library of Medicine, PubMed Central. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11266380/>

World Health Organization. *Scaling up telemedicine in the WHO European Region*. Available from: <https://www.who.int/europe/publications/i/item/WHO-EURO-2025-12185-51957-79682>

Aerospace Medical Association. *What is Aerospace Medicine?* Available from: <https://asma.org/about/what-is-aerospace-medicine/>

Centers for Disease Control and Prevention (CDC)

Centers for Disease Control and Prevention. *Health Topics*. Public health and disease information resource. Available from: <https://www.cdc.gov/health-topics.html>

MedicineNet

MedicineNet. *MedicineNet*. Doctor-authored health and medical information resource. Available from: <https://www.medicinenet.com/>

WebMD

WebMD. *WebMD*. Consumer health and medical information resource. Available from: <https://www.webmd.com/>

Medscape

Medscape. *Medscape*. Medical reference and professional health information resource. Available from: <https://www.medscape.com/>

Mayo Clinic

Mayo Clinic. *Health Information*. Diseases, symptoms, tests, procedures, and general health information resource. Available from: <https://www.mayoclinic.org/patient-consumer>

OpenMD Medical Dictionary

OpenMD. *OpenMD Medical Dictionary*. Online medical terminology reference with definitions, abbreviations, and word parts. Available from: <https://openmd.com/dictionary/>

RxList

RxList. *MedTerms Medical Dictionary*. Online medical dictionary and A–Z term index. Available from: <https://www.rxlist.com/drug-medical-dictionary/>

Online resources accessed 24 August 2026.